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# Preface

Regression is taught under at least five different descriptions, and each one is correct. In one, regression is least squares together with the distribution theory of quadratic forms. In another, it is orthogonal projection onto a subspace, and everything else follows as a corollary. A third starts from likelihood and treats the normal linear model as the Gaussian member of the generalized linear family. A fourth defines regression as estimating a conditional distribution, with linearity one structural assumption among many. A fifth treats it as a data-analytic procedure whose assumptions must be checked, one at a time, against the data.

Most textbooks pick one of these descriptions and stay with it. This book uses all of them, in order, as rungs of a single ladder. Parts I–III set up the classical linear model and its exact theory. After that, *each part relaxes exactly one assumption*, and asks what survives, what must be rebuilt, and what new tools the relaxation requires.

| Part | What changes                                                           | What we study                                                                                                                                                                                               |
|------|------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| I    | —                                                                      | The apparatus: matrices, random vectors, the normal distribution, quadratic forms                                                                                                                           |
| II   | —                                                                      | $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ , $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ , $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$ , as a projection problem |
| III  | adds $\boldsymbol{\varepsilon} \sim N(\mathbf{0}, \sigma^2\mathbf{I})$ | Exact finite-sample inference                                                                                                                                                                               |
| IV   | drops $\text{rank}(\mathbf{X}) = p$                                    | Estimability, analysis of variance, designed structure                                                                                                                                                      |
| V    | audits every assumption                                                | What breaks, how to see it, what inference survives                                                                                                                                                         |
| VI   | drops unbiasedness                                                     | Shrinkage, selection, regularization, $p > n$                                                                                                                                                               |
| VII  | drops $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$      | Generalized least squares, general Gauss–Markov, mixed models                                                                                                                                               |
| VIII | drops normality of $\mathbf{Y}$                                        | Exponential families and generalized linear models                                                                                                                                                          |
| IX   | drops the linear predictor                                             | Polynomials, splines, smoothing, additive models                                                                                                                                                            |
| X    | drops modeling only the mean                                           | Quantile regression and distributional models                                                                                                                                                               |

## Choices this book makes

**Geometry before distribution theory.** Least squares is introduced as a nearest-point problem ([Chapter 6](#)) before any sampling theory. Fitted values, residuals and sums of squares are projections. Coefficients are coordinates, and coordinates depend on the basis you choose. This is the only framing in which rank-deficient models need no special treatment.

**Estimability is an estimation idea.** It appears in Part II, not as a preliminary to the analysis of variance. Putting it there is what allows [Part IV](#) to treat ANOVA as ordinary subspace comparison.

**Computation is part of the theory.** Conditioning, QR and the singular value decomposition come immediately after least squares ([Chapter 10](#)), because numerical stability and statistical stability are closely linked.

**The Bayesian treatment is a thread, not a quarantined chapter.** Boxed “Bayesian thread” sections close Parts II, III, VII and VIII. Each develops the Bayesian counterpart of the result just proved.

**Departures have a theory.** [Part V](#) opens by deriving what omitted variables, wrong covariance, non-normality, outliers and collinearity actually do to the estimator. Diagnostics and remedies come after that.

**Material most regression texts omit.** This book adds chapters on bootstrap and permutation inference ([Chapter 23](#)), errors in variables ([Chapter 24](#)), the causal interpretation of coefficients ([Chapter 25](#)), high-dimensional regression ([Chapter 28](#)) and missing data ([Chapter 41](#)).

## Code

Every figure and every number obtained by computation is produced by a Python script in the book’s `code/` directory, and the scripts check their own claims with assertions. The code in the text is taken from those scripts. On this website each code block is a live cell: press **Run** (or Shift+Enter) to execute it in your browser, edit it, and run it again. Nothing needs to be installed; the first run downloads a Python environment with numpy, scipy, pandas and statsmodels. Cells on a page share their variables, and running a cell first runs the cells above it that have not run yet.

## Sources

The coverage of this book was planned against six standard texts: Rencher and Schaalje (2008), Seber and Lee (2003), Christensen (2020), Sen and Srivastava (1990), Agresti (2015) and Fahrmeir et al. (2021). Their chapters were mapped onto a single prerequisite graph, and the order of this book respects that graph. Each chapter’s *Notes and sources* section names the treatments it drew on and the primary literature. The exposition, proofs, examples and exercises are original to this book.

# How to Use This Book

## Parts and chapters

Parts I–VI (Chapters 1–30) are the classical linear model, complete in itself: estimation, exact inference, designed experiments, diagnostics, and the move from unbiased estimation to regularization. Parts VII–X (Chapters 31–45) climb the rest of the ladder: correlated errors, non-normal responses, nonlinear predictors, and models for entire conditional distributions. Chapters are published here as they are written; the ones not yet on the site are named in plain text.

## Reading paths

- **A first course on linear models.** Chapters 5–9, 11–12, 15, 19–20 and 29. Consult Part I as needed.
- **The theory spine.** Part I, then Chapters 6, 7, 8, 9, 11 and 31. This is the shortest route to a full structural understanding of the linear model.
- **For implementers.** Chapters 5, 6, 10 and 20, Part VI and Chapter 34.
- **A course on generalized models.** Chapters 5, 6 and 11, then Parts VII–X.

## Prerequisites

Each chapter opens with a list of the earlier chapters it uses, taken from the prerequisite graph the book was planned against. Part I is written as reference material. Read it quickly the first time and come back to it when a later chapter sends you there.

## Exercises

Exercises come at the end of each section in three groups: **A. Check your understanding** (routine: a definition or computation has been understood), **B. Practice** (core: the result is used later in the book) and **C. Going deeper** (hard: may need an idea not in the text). Many have a worked solution, folded under the exercise; try the exercise before opening it.

# Notation

| Symbol                                             | Meaning                                                                                                |
|----------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| $\mathbf{Y}, \mathbf{y}$                           | response vector (random, observed), $n \times 1$                                                       |
| $\mathbf{X}$                                       | model matrix, $n \times p$ ; columns $\mathbf{x}_1, \dots, \mathbf{x}_p$                               |
| $\boldsymbol{\beta}, \hat{\boldsymbol{\beta}}$     | coefficient vector and a least squares estimate                                                        |
| $\boldsymbol{\varepsilon}$                         | error vector, $\boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}$             |
| $\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}}$ | fitted values $\mathbf{X}\hat{\boldsymbol{\beta}}$ and residuals $\mathbf{Y} - \hat{\mathbf{Y}}$       |
| $\sigma^2$                                         | error variance                                                                                         |
| $\mathbb{R}^n$                                     | $n$ -dimensional real space with $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v}$ |
| $\mathcal{C}(\mathbf{A}), \mathcal{N}(\mathbf{A})$ | column space and null space of $\mathbf{A}$                                                            |
| $\mathcal{S}^\perp$                                | orthogonal complement of a subspace $\mathcal{S}$                                                      |
| $\mathbf{A}^\top, \mathbf{A}^-$                    | transpose; a generalized inverse ( $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ )                   |
| $\mathbf{M}$                                       | orthogonal projection onto $\mathcal{C}(\mathbf{X})$                                                   |
| $\mathbf{M}_0$                                     | orthogonal projection onto the column space of a reduced model                                         |

| Symbol                                       | Meaning                                                                                                                                                                                                                      |
|----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| rank, tr, diag                               | rank, trace, diagonal matrix                                                                                                                                                                                                 |
| <b>1</b> , <b>0</b> , <b>I</b>               | vector of ones, zero vector, identity                                                                                                                                                                                        |
| $\ \cdot\ $                                  | Euclidean norm unless stated otherwise                                                                                                                                                                                       |
| E, Var, Cov                                  | expectation, variance, covariance (matrix)                                                                                                                                                                                   |
| $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ | $n$ -variate normal distribution                                                                                                                                                                                             |
| $\chi^2(r, \gamma)$ , $F(r, s, \gamma)$      | noncentral distributions, noncentrality $\gamma$                                                                                                                                                                             |
| <b>V</b>                                     | error covariance: a known shape in<br>$\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$ ( <a href="#">Chapter 31</a> ), or<br>$\mathbf{V} = \mathbf{ZGZ}^\top + \mathbf{R}$ in a mixed model ( <a href="#">Chapters 32–33</a> ) |
| <b>Z</b> , <b>u</b> , <b>G</b> , <b>R</b>    | random-effect design matrix, random effects,<br>$\text{Cov}(\mathbf{u})$ and $\text{Cov}(\boldsymbol{\epsilon})$ ( <a href="#">Part VII</a> )                                                                                |
| $\theta$ , $b(\theta)$ , $\phi$ , $V(\mu)$   | natural parameter, cumulant function,<br>dispersion and variance function of an<br>exponential dispersion family ( <a href="#">Part VIII</a> )                                                                               |
| $\eta$ , $g$ , <b>W</b> , $D$                | linear predictor $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , link function, working<br>weights and deviance of a generalized linear<br>model ( <a href="#">Part VIII</a> )                                                  |
| <b>R<sub>i</sub></b> , <b>R</b>              | working correlation matrix of a cluster<br>( <a href="#">Chapter 40</a> ) and missingness indicator array<br>( <a href="#">Chapter 41</a> ): two local uses of <b>R</b> , each<br>announced where it starts                  |

Vectors are columns. Matrices and vectors are set in bold upright type and scalars in italic. A bare “projection” always means an orthogonal projection with respect to the Euclidean inner product. [Section 6.9](#) covers the other kinds.

# Chapter 1

## Matrix Algebra for Linear Models

A linear model is a statement about vectors and matrices, and nearly every argument in this book is an argument about them. This chapter collects the matrix algebra the book relies on: rank and the fundamental subspaces, inverses and generalized inverses, consistent systems of equations, orthogonal matrices, determinants and partitioned matrices, the spectral theorem and positive definiteness, the singular value decomposition, idempotent matrices, Kronecker products and the calculus of quadratic forms. It is a reference chapter, meant to be skimmed first and consulted later, but every central result is proved.

**What you need.** None beyond the Notation chapter. A first course in linear algebra helps, but is not assumed.

### 1.0.1 Roadmap

- [Vectors, matrices and the operations we use](#)
- [Rank, column space and null space](#)
- [Inverses and generalized inverses](#)
- [Systems of equations and consistency](#)
- [Orthogonality and orthogonal matrices](#)
- [Trace, determinants and partitioned matrices](#)
- [Eigenvalues, the spectral theorem and positive definiteness](#)
- [The singular value decomposition and conditioning](#)
- [Idempotent matrices](#)
- [Kronecker products and the vec operator](#)
- [Vector and matrix calculus](#)
- [Summary and notes](#)

## 1.1 Vectors, matrices and the operations we use

This chapter works in  $\mathbb{R}^n$  and with real matrices only. It fixes the conventions, states the results later chapters cite, and proves them. A reader who already knows linear algebra can go straight

to the results that are specific to statistics: generalized inverses (Section 1.3 and Section 1.4), Schur complements and the Woodbury identity (Section 1.6), nonnegative definite matrices (Section 1.7), conditioning (Section 1.8) and idempotent matrices (Section 1.9). Orthogonal projections, the geometric heart of least squares, get only the minimum here. Chapter 6 develops them.

### 1.1.1 Conventions

An  $m \times n$  matrix  $\mathbf{A} = (a_{ij})$  has  $m$  rows and  $n$  columns. We write  $\mathbf{A} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$  for its columns and  $\mathbf{a}_{(i)}^\top$  for its  $i$ th row. A **vector** is always a column, so  $\mathbf{x} \in \mathbb{R}^n$  is an  $n \times 1$  matrix and  $\mathbf{x}^\top$  is a row. Bold upright letters denote vectors and matrices, and italic letters denote scalars. The special matrices we use constantly are the identity  $\mathbf{I}_n$ , the zero matrix  $\mathbf{0}$ , the vector of ones  $\mathbf{1}_n$ , the matrix of ones  $\mathbf{J}_n = \mathbf{1}_n \mathbf{1}_n^\top$ , and the coordinate vectors  $\mathbf{e}_1, \dots, \mathbf{e}_n$ , the columns of  $\mathbf{I}_n$ . Subscripts giving sizes are dropped when the size is clear.  $\text{diag}(d_1, \dots, d_n)$  is the diagonal matrix with the given diagonal entries. A square matrix is **symmetric** if  $\mathbf{A}^\top = \mathbf{A}$ , *upper triangular* if  $a_{ij} = 0$  for  $i > j$ , and *lower triangular* if  $\mathbf{A}^\top$  is upper triangular.

The **transpose** satisfies  $(\mathbf{A}^\top)^\top = \mathbf{A}$ ,  $(\mathbf{A} + \mathbf{B})^\top = \mathbf{A}^\top + \mathbf{B}^\top$  and, most usefully,

$$(\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top. \quad (1.1.1)$$

### 1.1.2 Products as linear combinations

If  $\mathbf{A}$  is  $m \times n$  and  $\mathbf{B}$  is  $n \times q$ , the  $(i, j)$  entry of  $\mathbf{AB}$  is  $\sum_k a_{ik} b_{kj}$ . Three equivalent readings of the product carry most of the arguments in this book:

$$\mathbf{Ab} = \sum_{j=1}^n b_j \mathbf{a}_j, \quad \mathbf{AB} = [\mathbf{Ab}_1, \dots, \mathbf{Ab}_q], \quad \mathbf{AB} = \sum_{k=1}^n \mathbf{a}_k \mathbf{b}_{(k)}^\top. \quad (1.1.2)$$

The first says that  $\mathbf{Ab}$  combines the columns of  $\mathbf{A}$ , with weights taken from  $\mathbf{b}$ . The second says that each column of  $\mathbf{AB}$  is such a combination. The third writes a product as a sum of  $n$  rank-one matrices (outer products), and is the form in which spectral and singular value decompositions are usually used. Transposing the first reading shows that each row of  $\mathbf{AB}$  is a combination of the rows of  $\mathbf{B}$ .

Matrices partitioned into conformable blocks multiply as if the blocks were scalars, as long as the order of factors is kept:

$$\begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix} \begin{pmatrix} \mathbf{B}_{11} & \mathbf{B}_{12} \\ \mathbf{B}_{21} & \mathbf{B}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11}\mathbf{B}_{11} + \mathbf{A}_{12}\mathbf{B}_{21} & \mathbf{A}_{11}\mathbf{B}_{12} + \mathbf{A}_{12}\mathbf{B}_{22} \\ \mathbf{A}_{21}\mathbf{B}_{11} + \mathbf{A}_{22}\mathbf{B}_{21} & \mathbf{A}_{21}\mathbf{B}_{12} + \mathbf{A}_{22}\mathbf{B}_{22} \end{pmatrix}.$$

Multiplication is associative and distributive but not commutative, and  $\mathbf{AB} = \mathbf{0}$  does not force  $\mathbf{A} = \mathbf{0}$  or  $\mathbf{B} = \mathbf{0}$ . The **Hadamard product**  $\mathbf{A} \circ \mathbf{B} = (a_{ij} b_{ij})$  multiplies entrywise. It appears only occasionally.

### 1.1.3 Inner products, lengths and the Cauchy–Schwarz inequality

The **inner product** of  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$  is  $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v} = \sum_i u_i v_i$ , and the **Euclidean norm** is  $\|\mathbf{u}\| = (\mathbf{u}^\top \mathbf{u})^{1/2}$ . The entries of  $\mathbf{A}^\top \mathbf{A}$  are the inner products of the columns of  $\mathbf{A}$ , and those of  $\mathbf{A} \mathbf{A}^\top$  are the inner products of its rows. In particular

$$\sum_{i,j} a_{ij}^2 = \sum_j \|\mathbf{a}_j\|^2, \quad (1.1.3)$$

so  $\mathbf{A}^\top \mathbf{A} = \mathbf{0}$  forces every column to be zero, that is,  $\mathbf{A} = \mathbf{0}$ . The square root of Eq. 1.1.3 is the **Frobenius norm**  $\|\mathbf{A}\|_F$ .

#### Proposition 1.1.1 — Cauchy–Schwarz

For  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ ,  $(\mathbf{u}^\top \mathbf{v})^2 \leq (\mathbf{u}^\top \mathbf{u})(\mathbf{v}^\top \mathbf{v})$ , with equality iff one vector is a multiple of the other. Consequently  $\|\mathbf{u} + \mathbf{v}\| \leq \|\mathbf{u}\| + \|\mathbf{v}\|$ .

*Proof.* If  $\mathbf{v} = \mathbf{0}$  there is nothing to prove. Otherwise, the quadratic  $q(t) = \|\mathbf{u} - t\mathbf{v}\|^2 = \mathbf{u}^\top \mathbf{u} - 2t \mathbf{u}^\top \mathbf{v} + t^2 \mathbf{v}^\top \mathbf{v}$  is nonnegative for all  $t$ , so its discriminant  $4(\mathbf{u}^\top \mathbf{v})^2 - 4(\mathbf{u}^\top \mathbf{u})(\mathbf{v}^\top \mathbf{v})$  is at most zero. Equality holds iff  $q$  has a real root, that is, iff  $\mathbf{u} = t\mathbf{v}$  for some  $t$ . The triangle inequality follows by expanding  $\|\mathbf{u} + \mathbf{v}\|^2$  and bounding the cross term. ■

### 1.1.4 Quadratic forms

A **quadratic form** in  $\mathbf{x} \in \mathbb{R}^n$  is a function  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \sum_{i,j} a_{ij} x_i x_j$ . Since  $\mathbf{x}^\top \mathbf{A} \mathbf{x}$  is a scalar it equals its own transpose  $\mathbf{x}^\top \mathbf{A}^\top \mathbf{x}$ , so

$$\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \left( \frac{1}{2} (\mathbf{A} + \mathbf{A}^\top) \right) \mathbf{x}. \quad (1.1.4)$$

Every quadratic form therefore has a symmetric matrix, and that matrix is unique: if  $\mathbf{x}^\top \mathbf{S} \mathbf{x} = 0$  for all  $\mathbf{x}$  with  $\mathbf{S}$  symmetric, then taking  $\mathbf{x} = \mathbf{e}_i$  gives  $s_{ii} = 0$ , and  $\mathbf{x} = \mathbf{e}_i + \mathbf{e}_j$  gives  $2s_{ij} = 0$ . From now on we always take the matrix in a quadratic form to be symmetric.

#### Example. (Sums of squares as quadratic forms)

The sum of squared deviations of  $\mathbf{y} \in \mathbb{R}^n$  from its mean is

$$\sum_{i=1}^n (y_i - \bar{y})^2 = \mathbf{y}^\top \mathbf{y} - n\bar{y}^2 = \mathbf{y}^\top \left( \mathbf{I}_n - \frac{1}{n} \mathbf{J}_n \right) \mathbf{y},$$

because  $n\bar{y}^2 = (\mathbf{1}^\top \mathbf{y})^2 / n = \mathbf{y}^\top \mathbf{J}_n \mathbf{y} / n$ . The **centring matrix**  $\mathbf{I}_n - n^{-1} \mathbf{J}_n$  is symmetric and satisfies  $(\mathbf{I}_n - n^{-1} \mathbf{J}_n)^2 = \mathbf{I}_n - n^{-1} \mathbf{J}_n$ , because  $\mathbf{J}_n^2 = n \mathbf{J}_n$ . Every sum of squares



in an analysis of variance is a quadratic form whose matrix has these two properties (Section 1.9).

## 1.2 Rank, column space and null space

### 1.2.1 Subspaces and dimension

A nonempty set  $\mathcal{S} \subseteq \mathbb{R}^n$  is a **subspace** if  $\mathbf{u}, \mathbf{v} \in \mathcal{S}$  and  $a, b \in \mathbb{R}$  imply  $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$ . Vectors  $\mathbf{v}_1, \dots, \mathbf{v}_k$  are **linearly independent** if  $\sum_j c_j \mathbf{v}_j = \mathbf{0}$  only for  $c_1 = \dots = c_k = 0$ , that is, if the matrix  $[\mathbf{v}_1, \dots, \mathbf{v}_k]$  sends only the zero vector to zero. Their **span** is the set of all their linear combinations. A **basis** of  $\mathcal{S}$  is an independent set that spans it.

We use the following facts of elementary linear algebra without proof (Halmos 1958, chapter 1). Every subspace of  $\mathbb{R}^n$  has a basis. All bases of a subspace have the same number of elements, its **dimension**  $\dim \mathcal{S}$ . In an  $r$ -dimensional subspace, any  $r$  independent vectors form a basis, and any independent set can be extended to a basis. If  $\mathcal{S} \subseteq \mathcal{T}$  and  $\dim \mathcal{S} = \dim \mathcal{T}$ , then  $\mathcal{S} = \mathcal{T}$ . Finally, if  $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$ , the union of a basis of  $\mathcal{S}$  and a basis of  $\mathcal{T}$  is independent, so the sum  $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$  has dimension  $\dim \mathcal{S} + \dim \mathcal{T}$ . Such a sum is called **direct** and written  $\mathcal{S} \oplus \mathcal{T}$ .

### 1.2.2 The four subspaces of a matrix

An  $m \times n$  matrix  $\mathbf{A}$  carries two subspaces of  $\mathbb{R}^m$  and two of  $\mathbb{R}^n$ :

$$\begin{aligned} \mathcal{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^n\} \subseteq \mathbb{R}^m, & \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^n : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^n, \\ \mathcal{C}(\mathbf{A}^\top) &\subseteq \mathbb{R}^n, & \mathcal{N}(\mathbf{A}^\top) &\subseteq \mathbb{R}^m. \end{aligned}$$

By Eq. 1.1.2,  $\mathcal{C}(\mathbf{A})$  is the span of the columns, the **column space**, and  $\mathcal{C}(\mathbf{A}^\top)$  is the span of the rows, the **row space**.  $\mathcal{N}(\mathbf{A})$  is the **null space**. The **rank** of  $\mathbf{A}$  is  $\text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$ , the largest number of independent columns. The number of independent rows,  $\dim \mathcal{C}(\mathbf{A}^\top)$ , looks like a different quantity. It is not.

#### Theorem 1.2.1 — Row rank equals column rank

For every matrix  $\mathbf{A}$ ,  $\dim \mathcal{C}(\mathbf{A}^\top) = \dim \mathcal{C}(\mathbf{A})$ . That is,  $\text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A})$ . Moreover, if  $\text{rank}(\mathbf{A}) = r \geq 1$  there are an  $m \times r$  matrix  $\mathbf{B}$  and an  $r \times n$  matrix  $\mathbf{F}$ , both of rank  $r$ , with  $\mathbf{A} = \mathbf{B}\mathbf{F}$ .

*Proof.* Let  $r = \text{rank}(\mathbf{A})$  and let the columns of the  $m \times r$  matrix  $\mathbf{B}$  be a basis of  $\mathcal{C}(\mathbf{A})$ . Each column  $\mathbf{a}_j$  is a combination  $\mathbf{B}\mathbf{f}_j$  of these, so  $\mathbf{A} = \mathbf{B}\mathbf{F}$  with  $\mathbf{F} = [\mathbf{f}_1, \dots, \mathbf{f}_n]$ . By the transposed reading of Eq. 1.1.2, every row of  $\mathbf{A} = \mathbf{B}\mathbf{F}$  is a combination of the  $r$  rows of  $\mathbf{F}$ , so  $\dim \mathcal{C}(\mathbf{A}^\top) \leq r = \dim \mathcal{C}(\mathbf{A})$ . Applying this to  $\mathbf{A}^\top$  gives the reverse inequality. Finally  $\text{rank}(\mathbf{F}) \leq r$  because  $\mathbf{F}$  has  $r$  rows, and  $r = \dim \mathcal{C}(\mathbf{A}^\top) \leq \dim \mathcal{C}(\mathbf{F}^\top) = \text{rank}(\mathbf{F})$ , so  $\text{rank}(\mathbf{F}) = r$ . ■

The factorization  $\mathbf{A} = \mathbf{B}\mathbf{F}$  is a **rank factorization**:  $\mathbf{B}$  has **full column rank** (its columns are independent) and  $\mathbf{F}$  has **full row rank**. It is a convenient tool for proofs, as [Proposition 1.9.1](#) shows.

### Theorem 1.2.2 — Rank-nullity

For an  $m \times n$  matrix  $\mathbf{A}$ ,  $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = n$ .

*Proof.* Let  $\mathbf{v}_1, \dots, \mathbf{v}_k$  be a basis of  $\mathcal{N}(\mathbf{A})$  and extend it to a basis  $\mathbf{v}_1, \dots, \mathbf{v}_k, \mathbf{w}_1, \dots, \mathbf{w}_{n-k}$  of  $\mathbb{R}^n$ . Every  $\mathbf{A}\mathbf{b}$  is a combination of  $\mathbf{A}\mathbf{w}_1, \dots, \mathbf{A}\mathbf{w}_{n-k}$ , since  $\mathbf{A}\mathbf{v}_i = \mathbf{0}$ . These vectors are independent: if  $\sum_j c_j \mathbf{A}\mathbf{w}_j = \mathbf{0}$ , then  $\sum_j c_j \mathbf{w}_j \in \mathcal{N}(\mathbf{A})$  is a combination of the  $\mathbf{v}_i$ , and independence of the full basis forces every  $c_j = 0$ . So  $\text{rank}(\mathbf{A}) = n - k$ . ■

In a linear model with  $n \times p$  model matrix  $\mathbf{X}$ ,  $\dim \mathcal{N}(\mathbf{X}) = p - \text{rank}(\mathbf{X})$  counts the directions in which the coefficient vector can move without changing the mean  $\mathbf{X}\boldsymbol{\beta}$ .  $\mathbf{X}$  has full column rank iff  $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$ .

## 1.2.3 Rank of products and of $\mathbf{A}^\top \mathbf{A}$

### Proposition 1.2.3

Let  $\mathbf{A}$  be  $m \times n$  and  $\mathbf{B}$  be  $n \times q$ .

- (a)  $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$  and  $\mathcal{N}(\mathbf{B}) \subseteq \mathcal{N}(\mathbf{AB})$ , so  $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$ .
- (b) If  $\mathbf{B}$  has full row rank, then  $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A})$ . If  $\mathbf{A}$  has full column rank, then  $\mathcal{N}(\mathbf{AB}) = \mathcal{N}(\mathbf{B})$ . So  $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{A})$  in the first case and  $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B})$  in the second. In particular, multiplying by a nonsingular matrix does not change rank.
- (c)  $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$  and  $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$ , so  $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A})$ .
- (d)  $\text{rank}(\mathbf{A} + \mathbf{C}) \leq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{C})$  for  $\mathbf{C}$  of the same size as  $\mathbf{A}$ .

*Proof.*

- (a) is [Eq. 1.1.2](#), together with  $\text{rank}(\mathbf{AB}) = \text{rank}(\mathbf{B}^\top \mathbf{A}^\top) \leq \text{rank}(\mathbf{B}^\top)$  by the first inclusion and [Theorem 1.2.1](#). (b) If  $\mathbf{B}$  has full row rank,  $\mathcal{C}(\mathbf{B}) = \mathbb{R}^n$  because it is an  $n$ -dimensional subspace of  $\mathbb{R}^n$ , so every  $\mathbf{A}\mathbf{u}$  equals  $\mathbf{A}\mathbf{B}\mathbf{v}$  for some  $\mathbf{v}$ . If  $\mathbf{A}$  has full column rank,  $\mathbf{A}\mathbf{B}\mathbf{v} = \mathbf{0}$  forces  $\mathbf{B}\mathbf{v} = \mathbf{0}$ . The rank statements follow from rank-nullity. (c) If  $\mathbf{A}^\top \mathbf{A}\mathbf{v} = \mathbf{0}$  then  $\|\mathbf{A}\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}^\top \mathbf{A}\mathbf{v} = 0$ , so  $\mathbf{A}\mathbf{v} = \mathbf{0}$ . Thus the null spaces agree, and by [Theorem 1.2.2](#)  $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A})$ . Since  $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) \subseteq \mathcal{C}(\mathbf{A}^\top)$  and both have dimension  $\text{rank}(\mathbf{A})$ , they are equal. Applying this to  $\mathbf{A}^\top$  gives  $\text{rank}(\mathbf{AA}^\top) = \text{rank}(\mathbf{A})$ . (d)  $\mathcal{C}(\mathbf{A} + \mathbf{C}) \subseteq \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{C})$ , and the dimension of a sum is at most the sum of the dimensions. ■

Part (c) is the reason the normal equations  $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$  always have a solution: the right-hand side lies in  $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$  (Section 1.4).

## 1.2.4 Nonsingular submatrices

### Proposition 1.2.4

Let  $\mathbf{A}$  have rank  $r \geq 1$ . Choose any  $r$  linearly independent rows, indexed by  $I$ , and any  $r$  linearly independent columns, indexed by  $J$ . Then the  $r \times r$  submatrix  $\mathbf{A}_{IJ}$  at their intersection is nonsingular, and

$$\mathbf{A} = \mathbf{A}_{\cdot J} \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I\cdot}, \quad (1.2.1)$$

where  $\mathbf{A}_{\cdot J}$  holds the columns in  $J$  and  $\mathbf{A}_{I\cdot}$  the rows in  $I$ .

*Proof.* The columns in  $J$  are a basis of  $\mathcal{C}(\mathbf{A})$ , so  $\mathbf{A} = \mathbf{A}_{\cdot J} \mathbf{F}$  for some  $r \times n$  matrix  $\mathbf{F}$ . Keeping only the rows in  $I$  gives  $\mathbf{A}_{I\cdot} = \mathbf{A}_{IJ} \mathbf{F}$ . The left side has rank  $r$ , so by Proposition 1.2.3(a) the  $r \times r$  matrix  $\mathbf{A}_{IJ}$  has rank  $r$  and is nonsingular. Then  $\mathbf{F} = \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I\cdot}$ , which gives Eq. 1.2.1. ■

Conversely, a nonsingular  $k \times k$  submatrix makes the  $k$  columns of  $\mathbf{A}$  that contain it linearly independent, so  $\text{rank}(\mathbf{A}) \geq k$ . In particular, a matrix has rank  $r$  iff it has a nonsingular  $r \times r$  submatrix and no nonsingular larger one. This proposition gives an explicit generalized inverse in Theorem 1.3.2.

## 1.2.5 Exercises

### 1.2.5.1 A. Check your understanding

#### Exercise A1.

Show that  $\sum_{i < j} (y_i - y_j)^2 = n \sum_i (y_i - \bar{y})^2$ , and write both sides as  $\mathbf{y}^\top \mathbf{A} \mathbf{y}$  with  $\mathbf{A}$  symmetric. Find the symmetric matrix of the quadratic form  $2x_1^2 - x_2^2 + 6x_1x_2 - 4x_2x_3$ .

#### Exercise A2.

Show that  $\mathbf{X}^\top \mathbf{X} \mathbf{B} = \mathbf{X}^\top \mathbf{X} \mathbf{C}$  implies  $\mathbf{X} \mathbf{B} = \mathbf{X} \mathbf{C}$ . Deduce that  $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{0}$  iff  $\mathbf{X} \mathbf{b} = \mathbf{0}$ .

*Solution.* Put  $\mathbf{D} = \mathbf{B} - \mathbf{C}$ , so that  $\mathbf{X}^\top \mathbf{X} \mathbf{D} = \mathbf{0}$ . Then  $(\mathbf{X} \mathbf{D})^\top (\mathbf{X} \mathbf{D}) = \mathbf{D}^\top \mathbf{X}^\top \mathbf{X} \mathbf{D} = \mathbf{0}$ , and by Eq. 1.1.3 a matrix  $\mathbf{Z}$  with  $\mathbf{Z}^\top \mathbf{Z} = \mathbf{0}$  is zero. So  $\mathbf{X} \mathbf{D} = \mathbf{0}$ . The second statement is the case  $\mathbf{C} = \mathbf{0}$ , with the converse obvious.

**Exercise A3.**

For the additive two-way layout with two rows, three columns and one observation per cell, write the  $6 \times 6$  model matrix  $\mathbf{X} = [\mathbf{1}, \mathbf{R}, \mathbf{C}]$ , where  $\mathbf{R}$  and  $\mathbf{C}$  hold the row and column indicators. Find  $\text{rank}(\mathbf{X})$  and a basis of  $\mathcal{N}(\mathbf{X})$ . Compare  $[\mathbf{R}, \mathbf{C}]^\top$  with the matrix  $\mathbf{A}$  of [Example \(Filling a table from its margins\)](#).

**1.2.5.2 B. Practice****Exercise B1.**

Let  $\mathbf{A}$  be  $m \times n$  and  $\mathbf{B}$  be  $n \times q$ . Prove  $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{A}) + \text{rank}(\mathbf{B}) - n$ . *Hint:* apply rank-nullity to  $\mathbf{AB}_0$ , where the columns of  $\mathbf{B}_0$  are a basis of  $\mathcal{C}(\mathbf{B})$ .

*Solution.* Let the  $n \times s$  matrix  $\mathbf{B}_0$ ,  $s = \text{rank}(\mathbf{B})$ , have columns forming a basis of  $\mathcal{C}(\mathbf{B})$ . Then  $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{AB}_0)$ , so by rank-nullity  $\text{rank}(\mathbf{AB}) = s - \dim \mathcal{N}(\mathbf{AB}_0)$ . The map  $\mathbf{c} \mapsto \mathbf{B}_0 \mathbf{c}$  is one-to-one and sends  $\mathcal{N}(\mathbf{AB}_0)$  into  $\mathcal{N}(\mathbf{A})$ , so  $\dim \mathcal{N}(\mathbf{AB}_0) \leq \dim \mathcal{N}(\mathbf{A}) = n - \text{rank}(\mathbf{A})$ . Hence  $\text{rank}(\mathbf{AB}) \geq \text{rank}(\mathbf{B}) - n + \text{rank}(\mathbf{A})$ .

**Exercise B2.**

Show that if  $\mathbf{A} = \mathbf{B}_1 \mathbf{F}_1 = \mathbf{B}_2 \mathbf{F}_2$  are two rank factorizations, then  $\mathbf{B}_2 = \mathbf{B}_1 \mathbf{T}$  and  $\mathbf{F}_2 = \mathbf{T}^{-1} \mathbf{F}_1$  for a nonsingular  $\mathbf{T}$ .

## 1.3 Inverses and generalized inverses

### 1.3.1 The inverse of a square matrix

An  $n \times n$  matrix  $\mathbf{A}$  is **nonsingular** if there is a matrix  $\mathbf{A}^{-1}$  with  $\mathbf{AA}^{-1} = \mathbf{A}^{-1}\mathbf{A} = \mathbf{I}_n$ . The inverse is unique: if  $\mathbf{B}$  and  $\mathbf{C}$  both qualify, then  $\mathbf{B} = \mathbf{B}(\mathbf{AC}) = (\mathbf{BA})\mathbf{C} = \mathbf{C}$ .

**Proposition 1.3.1**

For an  $n \times n$  matrix  $\mathbf{A}$  the following are equivalent: (a)  $\mathbf{A}$  is nonsingular; (b)  $\text{rank}(\mathbf{A}) = n$ ; (c)  $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$ ; (d)  $\mathbf{BA} = \mathbf{I}$  for some  $\mathbf{B}$ ; (e)  $\mathbf{AC} = \mathbf{I}$  for some  $\mathbf{C}$ . When they hold,  $\mathbf{B} = \mathbf{C} = \mathbf{A}^{-1}$ , and

$$(\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top, \quad (\mathbf{AB})^{-1} = \mathbf{B}^{-1}\mathbf{A}^{-1}$$

for nonsingular  $\mathbf{A}, \mathbf{B}$  of the same size. A nonsingular symmetric matrix has a symmetric inverse.

*Proof.* (b)  $\Leftrightarrow$  (c) is rank-nullity. (a)  $\Rightarrow$  (d) and (a)  $\Rightarrow$  (e) are trivial. (d)  $\Rightarrow$  (c):  $\mathbf{Ax} = \mathbf{0}$  gives  $\mathbf{x} = \mathbf{BAx} = \mathbf{0}$ . (e)  $\Rightarrow$  (b):  $\mathbf{I} = \mathbf{AC}$  has rank  $n$ , and  $\text{rank}(\mathbf{AC}) \leq \text{rank}(\mathbf{A})$ . (b)  $\Rightarrow$  (a): since  $\mathcal{C}(\mathbf{A}) = \mathbb{R}^n$ , we can solve  $\mathbf{Ac}_j = \mathbf{e}_j$  for each  $j$ , giving  $\mathbf{AC} = \mathbf{I}$ . Then  $\mathbf{A}(\mathbf{CA} - \mathbf{I}) = \mathbf{0}$ ,

and because  $\mathcal{N}(\mathbf{A}) = \{\mathbf{0}\}$  each column of  $\mathbf{CA} - \mathbf{I}$  is zero. So  $\mathbf{CA} = \mathbf{I}$  as well. If  $\mathbf{BA} = \mathbf{I}$  then  $\mathbf{B} = \mathbf{BAA}^{-1} = \mathbf{A}^{-1}$ , and similarly for  $\mathbf{C}$ . The formulas are verified by multiplication:  $\mathbf{A}^\top (\mathbf{A}^{-1})^\top = (\mathbf{A}^{-1}\mathbf{A})^\top = \mathbf{I}$  and  $(\mathbf{AB})(\mathbf{B}^{-1}\mathbf{A}^{-1}) = \mathbf{I}$ . If  $\mathbf{A} = \mathbf{A}^\top$ , then  $\mathbf{A}^{-1} = (\mathbf{A}^\top)^{-1} = (\mathbf{A}^{-1})^\top$ .

■

A rectangular matrix has no two-sided inverse. If  $\mathbf{X}$  is  $n \times p$  with full column rank, then  $\mathbf{X}^\top \mathbf{X}$  is nonsingular by Proposition 1.2.3(c), and  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  is a *left inverse*: it multiplies  $\mathbf{X}$  on the left to give  $\mathbf{I}_p$ . This is the matrix that computes least squares coefficients.

## 1.3.2 Generalized inverses

When  $\mathbf{A}$  is singular or rectangular we still want a matrix that inverts  $\mathbf{A}$  on the vectors where inversion makes sense, namely those in  $\mathcal{C}(\mathbf{A})$ . The following weak requirement turns out to be exactly what solving equations needs.

### Definition 1.3.1 — Generalized inverse

A **generalized inverse** of an  $m \times n$  matrix  $\mathbf{A}$  is any  $n \times m$  matrix  $\mathbf{A}^-$  with

$$\mathbf{AA}^-\mathbf{A} = \mathbf{A}.$$

Read columnwise, the definition says that  $\mathbf{AA}^-$  acts as the identity on every column of  $\mathbf{A}$ , hence on all of  $\mathcal{C}(\mathbf{A})$ : if  $\mathbf{u} = \mathbf{Ab}$  then  $\mathbf{A}(\mathbf{A}^-\mathbf{u}) = \mathbf{AA}^-\mathbf{Ab} = \mathbf{u}$ . So  $\mathbf{A}^-\mathbf{u}$  is a solution of  $\mathbf{Ax} = \mathbf{u}$  whenever one exists. If  $\mathbf{A}$  is nonsingular, multiplying the definition by  $\mathbf{A}^{-1}$  on both sides gives  $\mathbf{A}^- = \mathbf{A}^{-1}$ . In general there are many generalized inverses. Chapter 6 uses them to write projections, and Chapter 8 uses them to describe estimable functions in models of less than full rank.

### Theorem 1.3.2 — Existence

Every matrix has a generalized inverse. Explicitly, if  $\text{rank}(\mathbf{A}) = r \geq 1$  and  $\mathbf{A}_{IJ}$  is the nonsingular  $r \times r$  submatrix of Proposition 1.2.4, the  $n \times m$  matrix  $\mathbf{G}$  that holds  $\mathbf{A}_{IJ}^{-1}$  in the rows indexed by  $J$  and the columns indexed by  $I$ , and zeros elsewhere, is a generalized inverse of  $\mathbf{A}$ .

*Proof.* For  $\mathbf{A} = \mathbf{0}$  any matrix of the right size works. Otherwise let  $\mathbf{E}_I$  and  $\mathbf{E}_J$  be the columns of  $\mathbf{I}_m$  and  $\mathbf{I}_n$  indexed by  $I$  and  $J$ . Then  $\mathbf{G} = \mathbf{E}_J \mathbf{A}_{IJ}^{-1} \mathbf{E}_I^\top$ ,  $\mathbf{AE}_J = \mathbf{A}_{.J}$  and  $\mathbf{E}_I^\top \mathbf{A} = \mathbf{A}_{I.}$ . So  $\mathbf{AGA} = \mathbf{A}_{.J} \mathbf{A}_{IJ}^{-1} \mathbf{A}_{I.} = \mathbf{A}$  by Eq. 1.2.1.

■

### Proposition 1.3.3 — Properties of generalized inverses

Let  $\mathbf{A}$  be  $m \times n$  of rank  $r$  and let  $\mathbf{G}$  be a generalized inverse of  $\mathbf{A}$ .

- (a)  $\mathbf{AG}$  and  $\mathbf{GA}$  are idempotent. Moreover  $\mathcal{C}(\mathbf{AG}) = \mathcal{C}(\mathbf{A})$  and  $\mathcal{N}(\mathbf{GA}) = \mathcal{N}(\mathbf{A})$ , and  $\text{rank}(\mathbf{AG}) = \text{rank}(\mathbf{GA}) = r \leq \text{rank}(\mathbf{G})$ .
- (b)  $\mathbf{G}^\top$  is a generalized inverse of  $\mathbf{A}^\top$ .
- (c) If  $\mathbf{G}_1, \mathbf{G}_2$  are generalized inverses of  $\mathbf{A}$ , so is  $\mathbf{G}_1\mathbf{AG}_2$ . In particular  $\mathbf{GAG}$  is a generalized inverse which also satisfies  $(\mathbf{GAG})\mathbf{A}(\mathbf{GAG}) = \mathbf{GAG}$ .
- (d) If  $\mathbf{A}$  is symmetric, then  $\mathbf{G}^\top$  and  $\frac{1}{2}(\mathbf{G} + \mathbf{G}^\top)$  are generalized inverses of  $\mathbf{A}$ , so a symmetric generalized inverse always exists.
- (e)  $\mathbf{G} + (\mathbf{I} - \mathbf{GA})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{AG})$  is a generalized inverse of  $\mathbf{A}$  for every  $\mathbf{U}$  and  $\mathbf{V}$  of size  $n \times m$ .

*Proof.*

- (a)  $(\mathbf{AG})^2 = (\mathbf{AGA})\mathbf{G} = \mathbf{AG}$ , and likewise for  $\mathbf{GA}$ . Next,  $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{AGA}) \subseteq \mathcal{C}(\mathbf{AG}) \subseteq \mathcal{C}(\mathbf{A})$ , and  $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{GA}) \subseteq \mathcal{N}(\mathbf{AGA}) = \mathcal{N}(\mathbf{A})$ . The ranks follow from these and rank-nullity, and  $r = \text{rank}(\mathbf{AG}) \leq \text{rank}(\mathbf{G})$ .
- (b) Transpose  $\mathbf{AGA} = \mathbf{A}$ . (c)  $\mathbf{AG}_1\mathbf{AG}_2\mathbf{A} = \mathbf{AG}_2\mathbf{A} = \mathbf{A}$ . The second statement is  $\mathbf{GAGAGAG} = \mathbf{G}(\mathbf{AGA})\mathbf{GAG} = \mathbf{GAGAG} = \mathbf{GAG}$ .
- (c) By (b),  $\mathbf{G}^\top$  is a generalized inverse of  $\mathbf{A}^\top = \mathbf{A}$ . The condition  $\mathbf{AGA} = \mathbf{A}$  is linear in  $\mathbf{G}$ , so it is preserved by averaging.
- (d) Multiply out, using  $\mathbf{A}(\mathbf{I} - \mathbf{GA}) = \mathbf{0}$  and  $(\mathbf{I} - \mathbf{AG})\mathbf{A} = \mathbf{0}$ .

■

A generalized inverse with the extra property in (c),  $\mathbf{GAG} = \mathbf{G}$ , is called **reflexive**. [Exercise \(B1\)](#) shows that the reflexive ones are exactly those of rank  $r$ . Part (e) in fact produces *every* generalized inverse from a single one as  $\mathbf{U}$  and  $\mathbf{V}$  vary. Idempotence in (a) means that  $\mathbf{AG}$  and  $\mathbf{GA}$  are projections, generally not orthogonal ones. [Chapter 6](#) shows that the particular combination  $\mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$  is symmetric, does not depend on the choice of generalized inverse, and is the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$ .

### 1.3.3 The Moore–Penrose inverse

One generalized inverse is singled out by symmetry requirements.

#### Definition 1.3.2 — Moore–Penrose inverse

The **Moore–Penrose inverse** of  $\mathbf{A}$  is the matrix  $\mathbf{A}^+$  satisfying

$$\mathbf{AA}^+\mathbf{A} = \mathbf{A}, \quad \mathbf{A}^+\mathbf{AA}^+ = \mathbf{A}^+, \quad (\mathbf{AA}^+)^\top = \mathbf{AA}^+, \quad (\mathbf{A}^+\mathbf{A})^\top = \mathbf{A}^+\mathbf{A}.$$

There is at most one such matrix. If  $\mathbf{G}$  and  $\mathbf{H}$  both satisfy the four conditions, then

$$\mathbf{G} = \mathbf{GAG} = \mathbf{G}(\mathbf{AG})^\top = \mathbf{GG}^\top(\mathbf{AHA})^\top = \mathbf{G}(\mathbf{AG})^\top(\mathbf{AH})^\top = \mathbf{GAGAH} = \mathbf{GAH},$$

where the steps use the second and third conditions for  $\mathbf{G}$ , the first for  $\mathbf{H}$ , and the third for both. The same argument with the fourth condition in place of the third gives

$$\mathbf{H} = (\mathbf{HA})^\top\mathbf{H} = \mathbf{A}^\top\mathbf{H}^\top\mathbf{H} = (\mathbf{AGA})^\top\mathbf{H}^\top\mathbf{H} = (\mathbf{GA})^\top(\mathbf{HA})^\top\mathbf{H} = \mathbf{GAHAH} = \mathbf{GAH},$$

so  $\mathbf{G} = \mathbf{H}$ . Existence follows from the singular value decomposition, which gives  $\mathbf{A}^+$  explicitly (Theorem 1.8.1).

## 1.4 Systems of equations and consistency

A system  $\mathbf{Ax} = \mathbf{b}$  with  $\mathbf{A}$  of size  $m \times n$  is **consistent** if it has at least one solution. Least squares leads to such systems in which  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  may be singular, so we need to know when solutions exist, what all of them look like, and which features are common to all of them.

### Theorem 1.4.1 — Consistency and the general solution

Let  $\mathbf{G}$  be any generalized inverse of  $\mathbf{A}$ . The following are equivalent:

- (a)  $\mathbf{Ax} = \mathbf{b}$  is consistent;
- (b)  $\mathbf{b} \in \mathcal{C}(\mathbf{A})$ ;
- (c)  $\text{rank}[\mathbf{A}, \mathbf{b}] = \text{rank}(\mathbf{A})$ ;
- (d)  $\mathbf{AGb} = \mathbf{b}$ .

If the system is consistent,  $\mathbf{x}_0 = \mathbf{Gb}$  is a solution, and the set of all solutions is

$$\{\mathbf{Gb} + (\mathbf{I} - \mathbf{GA})\mathbf{z} : \mathbf{z} \in \mathbb{R}^n\} = \mathbf{x}_0 + \mathcal{N}(\mathbf{A}). \quad (1.4.1)$$

*Proof.* (a) $\Leftrightarrow$ (b) is the first reading in Eq. 1.1.2. (b) $\Leftrightarrow$ (c): appending  $\mathbf{b}$  as a column leaves the column space unchanged iff  $\mathbf{b}$  is already in it, and a subspace of  $\mathcal{C}([\mathbf{A}, \mathbf{b}])$  with the same dimension equals it. (d) $\Rightarrow$ (a):  $\mathbf{Gb}$  is a solution. (b) $\Rightarrow$ (d): if  $\mathbf{b} = \mathbf{Au}$ , then  $\mathbf{AGb} = \mathbf{AGA}\mathbf{u} = \mathbf{Au} = \mathbf{b}$ .

The solutions of a consistent system are  $\mathbf{x}_0 + \mathcal{N}(\mathbf{A})$ , since  $\mathbf{Ax} = \mathbf{b}$  iff  $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$ . It remains to show  $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{I} - \mathbf{GA})$ . The inclusion  $\supseteq$  holds because  $\mathbf{A}(\mathbf{I} - \mathbf{GA}) = \mathbf{0}$ . Conversely, if  $\mathbf{Av} = \mathbf{0}$ , then  $\mathbf{v} = (\mathbf{I} - \mathbf{GA})\mathbf{v}$ .

■

Criterion (d) is useful in practice: it tests consistency with a single matrix product, for any generalized inverse. By Proposition 1.3.3(a) and rank-nullity,  $\text{rank}(\mathbf{I} - \mathbf{GA}) = n - r$ , so the solution set is an affine subspace of dimension  $n - r$ . The solution is unique iff  $\text{rank}(\mathbf{A}) = n$ , and then  $\mathbf{Gb}$  is the same for every  $\mathbf{G}$ .

*Idea.* Any generalized inverse does three jobs for  $\mathbf{Ax} = \mathbf{b}$ . It tests consistency ( $\mathbf{AGb} = \mathbf{b}$ ), produces one solution ( $\mathbf{Gb}$ ), and parametrizes all of them ( $\mathbf{Gb} + (\mathbf{I} - \mathbf{GA})\mathbf{z}$ ). Which  $\mathbf{G}$  is used changes the particular solution but never the solution set, and a linear function  $\mathbf{q}^\top \mathbf{x}$  is the same across solutions exactly when  $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$ .

### Corollary 1.4.2 — Invariant linear functions

Let  $\mathbf{Ax} = \mathbf{b}$  be consistent and let  $\mathbf{q} \in \mathbb{R}^n$ . The value  $\mathbf{q}^\top \mathbf{x}$  is the same for every solution

$\mathbf{x}$  iff  $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$ , that is, iff  $\mathbf{q}^\top = \mathbf{q}^\top \mathbf{G} \mathbf{A}$  for one (and then every) generalized inverse  $\mathbf{G}$ .

*Proof.* By Eq. 1.4.1,  $\mathbf{q}^\top \mathbf{x}$  is constant over solutions iff  $\mathbf{q}^\top (\mathbf{I} - \mathbf{G} \mathbf{A}) \mathbf{z} = 0$  for all  $\mathbf{z}$ , that is, iff  $\mathbf{q}^\top = \mathbf{q}^\top \mathbf{G} \mathbf{A}$ . That equation puts  $\mathbf{q}$  in  $\mathcal{C}(\mathbf{A}^\top)$ . Conversely, if  $\mathbf{q} = \mathbf{A}^\top \mathbf{w}$ , then  $\mathbf{q}^\top \mathbf{G} \mathbf{A} = \mathbf{w}^\top \mathbf{A} \mathbf{G} \mathbf{A} = \mathbf{w}^\top \mathbf{A} = \mathbf{q}^\top$ . ■

Applied to the normal equations  $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ , which are consistent by Proposition 1.2.3(c), the corollary says that  $\mathbf{q}^\top \hat{\boldsymbol{\beta}}$  is the same for all least squares solutions iff  $\mathbf{q} \in \mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$ . This is the algebra behind estimable functions (Theorem 8.3.1).

### Proposition 1.4.3 — Minimum-norm solution

If  $\mathbf{A} \mathbf{x} = \mathbf{b}$  is consistent, then  $\mathbf{A}^+ \mathbf{b}$  is its unique solution lying in  $\mathcal{C}(\mathbf{A}^\top)$ , and  $\|\mathbf{A}^+ \mathbf{b}\| < \|\mathbf{x}\|$  for every other solution  $\mathbf{x}$ .

*Proof.*  $\mathbf{A}^+ \mathbf{b}$  is a solution because  $\mathbf{A}^+$  is a generalized inverse. From the second and fourth Penrose conditions,  $\mathbf{A}^+ = (\mathbf{A}^+ \mathbf{A}) \mathbf{A}^+ = (\mathbf{A}^+ \mathbf{A})^\top \mathbf{A}^+ = \mathbf{A}^\top (\mathbf{A}^+)^{\top} \mathbf{A}^+$ , so  $\mathbf{A}^+ \mathbf{b} \in \mathcal{C}(\mathbf{A}^\top)$ . Every vector of  $\mathcal{C}(\mathbf{A}^\top)$  is orthogonal to every  $\mathbf{v} \in \mathcal{N}(\mathbf{A})$ , since  $(\mathbf{A}^\top \mathbf{w})^\top \mathbf{v} = \mathbf{w}^\top \mathbf{A} \mathbf{v} = 0$ . Any other solution is  $\mathbf{x} = \mathbf{A}^+ \mathbf{b} + \mathbf{v}$  with  $\mathbf{0} \neq \mathbf{v} \in \mathcal{N}(\mathbf{A})$ , and then  $\|\mathbf{x}\|^2 = \|\mathbf{A}^+ \mathbf{b}\|^2 + \|\mathbf{v}\|^2$ . The same computation shows that  $\mathbf{x} \notin \mathcal{C}(\mathbf{A}^\top)$ , because  $\mathbf{x}^\top \mathbf{v} = \|\mathbf{v}\|^2 \neq 0$ . ■

### Example. (Filling a table from its margins)

Suppose the six entries of a  $2 \times 3$  table are unknown, but its row totals (9, 6) and column totals (4, 5, 6) are known. With the cells ordered  $x_{11}, x_{12}, x_{13}, x_{21}, x_{22}, x_{23}$ , the equations are  $\mathbf{A} \mathbf{x} = \mathbf{b}$  with

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} 9 \\ 6 \\ 4 \\ 5 \\ 6 \end{pmatrix}.$$

The rows of  $\mathbf{A}$  satisfy one relation: the two row-total rows and the three column-total rows both add to  $\mathbf{1}_6^\top$ . So  $\text{rank}(\mathbf{A}) = 4$  and there is a  $6 - 4 = 2$ -dimensional family of solutions. The same relation must hold for  $\mathbf{b}$ . By Theorem 1.4.1(c), the system is consistent iff the row totals and the column totals have the same sum, and here both sums are 15.

The listing builds three generalized inverses:  $\mathbf{G}_1$  from the nonsingular submatrix in rows 1–4 and columns 1–4 (Theorem 1.3.2), the Moore–Penrose inverse  $\mathbf{G}_2 = \mathbf{A}^+$ , and  $\mathbf{G}_3 = \mathbf{G}_1 + (\mathbf{I} - \mathbf{G}_1 \mathbf{A}) \mathbf{Z}$  for an integer matrix  $\mathbf{Z}$  (Proposition 1.3.3(e)). They give the



solutions

$$\mathbf{G}_1 \mathbf{b} = (-2, 5, 6, 6, 0, 0)^\top, \quad \|\mathbf{G}_1 \mathbf{b}\| = 10.050,$$

$$\mathbf{G}_2 \mathbf{b} = (5/2, 3, 7/2, 3/2, 2, 5/2)^\top, \quad \|\mathbf{G}_2 \mathbf{b}\| = 6.325,$$

$$\mathbf{G}_3 \mathbf{b} = (21, -39, 27, -17, 44, -21)^\top, \quad \|\mathbf{G}_3 \mathbf{b}\| = 73.192.$$

The individual cell  $x_{11}$  takes the values  $-2$ ,  $5/2$  and  $21$ : margins do not determine cells. But  $x_{12} + x_{13} - x_{21}$  equals  $5$  for all three, as [Corollary 1.4.2](#) predicts, because its coefficient vector is the first row of  $\mathbf{A}$  minus the third. The minimum-norm solution has the additive form  $x_{ij} = r_i/3 + c_j/2 - 15/6$  in terms of the row totals  $r_i$  and column totals  $c_j$  ([Exercise \(A2\)](#)). If the last column total is changed to  $7$ , then  $\mathbf{A}\mathbf{G}\mathbf{b} \neq \mathbf{b}$  for each of the three matrices, and the script confirms that the test detects the inconsistency. The script also checks all 60 submatrices formed by 4 independent rows and 4 independent columns, and finds each one nonsingular, as [Proposition 1.2.4](#) requires.

```
from fractions import Fraction
from itertools import combinations
import numpy as np
m, n = A.shape
# cells ordered (1,1), (1,2), (1,3), (2,1), (2,2), (2,3)
A = np.array([[1, 1, 1, 0, 0, 0],      # row 1 total
              [0, 0, 0, 1, 1, 1],      # row 2 total
              [1, 0, 0, 1, 0, 0],      # column 1 total
              [0, 1, 0, 0, 1, 0],      # column 2 total
              [0, 0, 1, 0, 0, 1]],      # column 3 total
             dtype=float)
b = np.array([9, 6, 4, 5, 6], dtype=float) # both sets of totals add to 15
r = np.linalg.matrix_rank(A)
print("rank(A) =", r, " rank([A, b]) =", np.linalg.matrix_rank(np.column_stack([A, b]
)))

def ginverse_from_submatrix(A, rows, cols):
    """Place the inverse of the nonsingular block A[rows, cols] at [cols, rows]."""
    G = np.zeros(A.T.shape)
    G[np.ix_(cols, rows)] = np.linalg.inv(A[np.ix_(rows, cols)])
    return G

rows, cols = [0, 1, 2, 3], [0, 1, 2, 3] # r independent rows and
# columns
G1 = ginverse_from_submatrix(A, rows, cols)
G2 = np.linalg.pinv(A) # Moore-Penrose inverse
rng = np.random.default_rng(1)
Z = rng.integers(-2, 3, size=A.T.shape).astype(float)
G3 = G1 + (np.eye(n) - G1 @ A) @ Z # another, non-reflexive
# one

for G in (G1, G2, G3):
    assert np.allclose(A @ G @ A, A) # each is a g-inverse
    assert np.allclose(A @ G @ b, b) # consistency test passes
    x = G @ b
    print(np.round(x, 4), " A x - b =", np.abs(A @ x - b).max())
```

## 1.4.1 Exercises

### 1.4.1.1 A. Check your understanding

#### Exercise A1.

Find all generalized inverses of a nonzero vector  $\mathbf{a} \in \mathbb{R}^n$ , regarded as an  $n \times 1$  matrix. Which one is  $\mathbf{a}^+$ ?

#### Exercise A2.

Generalize [Example \(Filling a table from its margins\)](#) to an  $a \times b$  table with row totals  $r_i$ , column totals  $c_j$  and grand total  $T = \sum_i r_i = \sum_j c_j$ . Show that  $x_{ij} = r_i/b + c_j/a - T/(ab)$  is a solution lying in  $\mathcal{C}(\mathbf{A}^\top)$ , and hence is the minimum-norm solution.

*Solution.* Row  $i$  of the proposed table sums to  $r_i + \sum_j c_j/a - bT/(ab) = r_i$ , and column  $j$  sums to  $\sum_i r_i/b + c_j - T/b = c_j$ . The rows of  $\mathbf{A}$  are the row indicators and the column indicators, so  $\mathcal{C}(\mathbf{A}^\top)$  consists of the tables of the form  $x_{ij} = \alpha_i + \beta_j$ . The proposed solution has this form. By [Proposition 1.4.3](#), it is  $\mathbf{A}^+\mathbf{b}$ . For  $a = 2$ ,  $b = 3$  and  $T = 15$  this is the additive formula quoted in [Example \(Filling a table from its margins\)](#).

#### Exercise A3.

For which  $c$  and  $d$  is the system  $x_1 + x_2 + x_3 = 2$ ,  $x_1 - x_2 = c$ ,  $3x_1 + x_2 + 2x_3 = d$  consistent? When it is, give its general solution in the form [Eq. 1.4.1](#).

### 1.4.1.2 B. Practice

#### Exercise B1.

Let  $\mathbf{G}$  be a generalized inverse of  $\mathbf{A}$ . Show that  $\mathbf{GAG} = \mathbf{G}$  iff  $\text{rank}(\mathbf{G}) = \text{rank}(\mathbf{A})$ .

*Solution.* If  $\mathbf{GAG} = \mathbf{G}$ , then  $\text{rank}(\mathbf{G}) \leq \text{rank}(\mathbf{A})$  by [Proposition 1.2.3\(a\)](#), and  $\text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{G})$  by [Proposition 1.3.3\(a\)](#). Conversely, suppose  $\text{rank}(\mathbf{G}) = \text{rank}(\mathbf{A}) = r$ . Since  $\mathcal{C}(\mathbf{GA}) \subseteq \mathcal{C}(\mathbf{G})$  and  $\text{rank}(\mathbf{GA}) = r$ , the two column spaces coincide, so  $\mathbf{G} = \mathbf{GAK}$  for some  $\mathbf{K}$ . Then  $\mathbf{GAG} = \mathbf{GAGAK} = \mathbf{GAK} = \mathbf{G}$ , using  $\mathbf{AGA} = \mathbf{A}$ .

#### Exercise B2.

Let  $\mathbf{A} = \mathbf{BF}$  be a rank factorization. Show that  $\mathbf{A}^+ = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$ . Deduce that  $\mathbf{A}^+ = (\mathbf{A}^\top\mathbf{A})^{-1}\mathbf{A}^\top$  when  $\mathbf{A}$  has full column rank.

*Solution.* Write  $\mathbf{G}$  for the proposed matrix. Then  $\mathbf{AG} = \mathbf{B}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top$  and  $\mathbf{GA} = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}\mathbf{F}$ , both symmetric. Also  $\mathbf{AGA} = \mathbf{B}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top\mathbf{BF} = \mathbf{A}$  and  $\mathbf{GAG} = \mathbf{F}^\top(\mathbf{FF}^\top)^{-1}\mathbf{FF}^\top(\mathbf{FF}^\top)^{-1}(\mathbf{B}^\top\mathbf{B})^{-1}\mathbf{B}^\top = \mathbf{G}$ . By uniqueness  $\mathbf{G} = \mathbf{A}^+$ . If  $\mathbf{A}$  has full column rank, take  $\mathbf{B} = \mathbf{A}$  and  $\mathbf{F} = \mathbf{I}$ .

**Exercise B3.**

Let  $\mathbf{A}$  be  $m \times n$  of rank  $r$  and let  $\mathbf{Ax} = \mathbf{b}$  be consistent with  $\mathbf{b} \neq \mathbf{0}$ . Show that the system has  $n - r + 1$  linearly independent solutions and no more, and that every solution is an affine combination of them.

**1.4.1.3 C. Going deeper****Exercise C1.**

Show that  $(\mathbf{A}^+)^+ = \mathbf{A}$ ,  $(\mathbf{A}^\top)^+ = (\mathbf{A}^+)^^\top$ ,  $(\mathbf{A}^\top \mathbf{A})^+ = \mathbf{A}^+ (\mathbf{A}^+)^^\top$  and  $\mathbf{A}^+ = (\mathbf{A}^\top \mathbf{A})^+ \mathbf{A}^\top$ . Give an example with  $(\mathbf{AB})^+ \neq \mathbf{B}^+ \mathbf{A}^+$ .

## 1.5 Orthogonality and orthogonal matrices

### 1.5.1 Orthogonal vectors and complements

Vectors  $\mathbf{u}, \mathbf{v}$  are **orthogonal**, written  $\mathbf{u} \perp \mathbf{v}$ , if  $\mathbf{u}^\top \mathbf{v} = 0$ . For orthogonal vectors the Pythagorean identity  $\|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2$  holds. A set of nonzero, mutually orthogonal vectors is independent: if  $\sum_j c_j \mathbf{v}_j = \mathbf{0}$ , taking the inner product with  $\mathbf{v}_k$  leaves  $c_k \|\mathbf{v}_k\|^2 = 0$ . A set is **orthonormal** if, in addition, each vector has unit length. The columns of an  $n \times r$  matrix  $\mathbf{Q}$  are orthonormal iff  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$ .

The **orthogonal complement** of a subspace  $\mathcal{S} \subseteq \mathbb{R}^n$  is  $\mathcal{S}^\perp = \{\mathbf{v} : \mathbf{v}^\top \mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}$ . The next proposition is all that this chapter needs about complements. [Chapter 6](#) builds the theory of orthogonal projections on it.

**Proposition 1.5.1**

Let  $\mathbf{A}$  be  $m \times n$  and let  $\mathcal{S} \subseteq \mathbb{R}^n$  be a subspace.

- (a)  $\mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp$  and  $\mathcal{N}(\mathbf{A}^\top) = \mathcal{C}(\mathbf{A})^\perp$ .
- (b)  $\dim \mathcal{S}^\perp = n - \dim \mathcal{S}$ , and  $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp$ . In particular  $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$  and  $\mathbb{R}^m = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A}^\top)$ .
- (c)  $(\mathcal{S}^\perp)^\perp = \mathcal{S}$ .

*Proof.*

- (a)  $\mathbf{Av} = \mathbf{0}$  says that  $\mathbf{v}$  is orthogonal to every row of  $\mathbf{A}$ , hence to their span. Apply this to  $\mathbf{A}^\top$  for the second identity.
- (b) Put a basis of  $\mathcal{S}$  ( $k$  vectors) into the columns of  $\mathbf{B}$ . Then  $\mathcal{S}^\perp = \mathcal{N}(\mathbf{B}^\top)$ , which has dimension  $n - k$  by rank-nullity. If  $\mathbf{u} \in \mathcal{S} \cap \mathcal{S}^\perp$  then  $\mathbf{u}^\top \mathbf{u} = 0$ , so the sum is direct and has dimension  $n$ , so it is  $\mathbb{R}^n$ .
- (c)  $\mathcal{S} \subseteq (\mathcal{S}^\perp)^\perp$ , and both have dimension  $\dim \mathcal{S}$  by (b).

## 1.5.2 Gram–Schmidt and the QR factorization

### Theorem 1.5.2 — Gram–Schmidt; QR factorization

Let  $\mathbf{X} = [\mathbf{x}_1, \dots, \mathbf{x}_p]$  be  $n \times p$  with full column rank. Define recursively

$$\mathbf{w}_j = \mathbf{x}_j - \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|.$$

Then  $\mathbf{u}_1, \dots, \mathbf{u}_p$  are orthonormal and, for each  $j$ , span the same subspace as  $\mathbf{x}_1, \dots, \mathbf{x}_j$ . Equivalently,  $\mathbf{X} = \mathbf{Q}\mathbf{R}$  with  $\mathbf{Q} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$  satisfying  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$  and  $\mathbf{R}$  upper triangular with positive diagonal entries.

*Proof.* By induction on  $j$ . Suppose  $\mathbf{u}_1, \dots, \mathbf{u}_{j-1}$  are orthonormal and span  $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_{j-1})$ . Then  $\mathbf{w}_j \neq \mathbf{0}$ , since otherwise  $\mathbf{x}_j$  would lie in that span, contradicting independence. For  $i < j$ ,  $\mathbf{u}_i^\top \mathbf{w}_j = \mathbf{u}_i^\top \mathbf{x}_j - \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k^\top \mathbf{u}_i = 0$ . So  $\mathbf{u}_1, \dots, \mathbf{u}_j$  are orthonormal, and they span  $\text{span}(\mathbf{x}_1, \dots, \mathbf{x}_j)$  because  $\mathbf{x}_j = \|\mathbf{w}_j\| \mathbf{u}_j + \sum_{k < j} (\mathbf{u}_k^\top \mathbf{x}_j) \mathbf{u}_k$ . That last identity, read column by column, is  $\mathbf{X} = \mathbf{Q}\mathbf{R}$  with  $r_{jj} = \|\mathbf{w}_j\| > 0$  and  $r_{kj} = \mathbf{u}_k^\top \mathbf{x}_j$  for  $k < j$ . ■

The factorization is unique ([Exercise \(B2\)](#)). Numerical software computes it by a more stable route, but the object is the same, and [Chapter 6](#) and [Chapter 10](#) rely on it. Two consequences are used repeatedly. First, any orthonormal set  $\mathbf{u}_1, \dots, \mathbf{u}_r$  in  $\mathbb{R}^n$  extends to an orthonormal basis: apply Gram–Schmidt to  $\mathbf{u}_1, \dots, \mathbf{u}_r, \mathbf{e}_1, \dots, \mathbf{e}_n$ , discarding each vector that depends on its predecessors. Second, every subspace has an orthonormal basis.

## 1.5.3 Orthogonal matrices

A square matrix  $\mathbf{Q}$  is **orthogonal** if  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$ . By [Proposition 1.3.1](#),  $\mathbf{Q}^\top$  is then the inverse of  $\mathbf{Q}$ , so also  $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$ : the rows of an orthogonal matrix are orthonormal as well as its columns.

### Proposition 1.5.3

Let  $\mathbf{Q}$  and  $\mathbf{P}$  be  $n \times n$  orthogonal matrices.

- (a)  $(\mathbf{Q}\mathbf{x})^\top (\mathbf{Q}\mathbf{v}) = \mathbf{x}^\top \mathbf{v}$  and  $\|\mathbf{Q}\mathbf{x}\| = \|\mathbf{x}\|$  for all  $\mathbf{x}, \mathbf{v}$ .
- (b)  $\mathbf{Q}^\top$  and  $\mathbf{Q}\mathbf{P}$  are orthogonal.
- (c) Every entry satisfies  $|q_{ij}| \leq 1$ , and  $\det \mathbf{Q} = \pm 1$ .
- (d) If  $[\mathbf{Q}_1, \mathbf{Q}_2]$  is orthogonal with  $\mathbf{Q}_1$  of size  $n \times r$ , then  $\mathbf{Q}_1 \mathbf{Q}_1^\top + \mathbf{Q}_2 \mathbf{Q}_2^\top = \mathbf{I}_n$ , and  $\mathcal{C}(\mathbf{Q}_2) = \mathcal{C}(\mathbf{Q}_1)^\perp$ .

*Proof.*

- (a)  $(\mathbf{Q}\mathbf{x})^\top \mathbf{Q}\mathbf{v} = \mathbf{x}^\top \mathbf{Q}^\top \mathbf{Q}\mathbf{v} = \mathbf{x}^\top \mathbf{v}$ . (b)  $(\mathbf{Q}\mathbf{P})^\top \mathbf{Q}\mathbf{P} = \mathbf{P}^\top \mathbf{P} = \mathbf{I}$ .  
 (b) Each column has unit length. For the determinant, see [Proposition 1.6.2](#):  $1 = \det(\mathbf{Q}^\top \mathbf{Q}) = (\det \mathbf{Q})^2$ . (d) The first statement is  $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$  written in blocks. For the second,  $\mathbf{Q}_1^\top \mathbf{Q}_2 = \mathbf{0}$  gives  $\mathcal{C}(\mathbf{Q}_2) \subseteq \mathcal{C}(\mathbf{Q}_1)^\perp$ , and both have dimension  $n - r$ .

■

Part (a) says an orthogonal matrix is a rigid motion that fixes the origin: a rotation, a reflection, or a composition of the two. Changing coordinates by an orthogonal matrix changes neither lengths nor angles. This is why the distribution theory of [Chapter 4](#) can rotate a normal vector with independent components into a convenient coordinate system without changing its distribution.

### Example. (Helmert matrices)

The  $n \times n$  **Helmert matrix**  $\mathbf{H}$  has first row  $n^{-1/2}\mathbf{1}^\top$  and, for  $k = 2, \dots, n$ ,  $k$ th row

$$\mathbf{h}_k^\top = \frac{1}{\sqrt{k(k-1)}} \left( \underbrace{1, \dots, 1}_{k-1}, -(k-1), 0, \dots, 0 \right).$$

Each row has unit length, since  $(k-1) + (k-1)^2 = k(k-1)$ . Row  $k$  is orthogonal to row 1 because its entries sum to zero. For  $2 \leq j < k$ , the nonzero entries of  $\mathbf{h}_j$  sit in positions  $1, \dots, j$ , where  $\mathbf{h}_k$  is constant, so  $\mathbf{h}_j^\top \mathbf{h}_k$  is proportional to the sum of the entries of  $\mathbf{h}_j$ , which is zero. So  $\mathbf{H}$  is orthogonal. For  $\mathbf{y} \in \mathbb{R}^n$ ,  $(\mathbf{H}\mathbf{y})_1 = \sqrt{n}\bar{y}$  and, by [Proposition 1.5.3\(a\)](#),

$$\sum_{k=2}^n (\mathbf{h}_k^\top \mathbf{y})^2 = \|\mathbf{H}\mathbf{y}\|^2 - n\bar{y}^2 = \sum_{i=1}^n (y_i - \bar{y})^2.$$

The sum of squared deviations is a sum of  $n-1$  squares of orthonormal contrasts. With independent normal observations this gives a short proof that the sample mean and variance are independent ([Exercise \(B1\)](#)).

Two other families appear later. **Permutation matrices**, the identity with its rows reordered, are orthogonal. For a unit vector  $\mathbf{u}$ , the **Householder reflection**  $\mathbf{I} - 2\mathbf{u}\mathbf{u}^\top$  is symmetric and orthogonal and reflects  $\mathbb{R}^n$  in the hyperplane  $\mathbf{u}^\perp$ . Householder reflections are how QR factorizations are computed in practice ([Exercise \(B1\)](#)).

## 1.5.4 Exercises

### 1.5.4.1 A. Check your understanding

#### Exercise A1.

Show that every  $2 \times 2$  orthogonal matrix is either a rotation  $\begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$  (determinant 1) or a reflection  $\begin{pmatrix} \cos \theta & \sin \theta \\ \sin \theta & -\cos \theta \end{pmatrix}$  (determinant  $-1$ ). Find the line fixed by the reflection.

### 1.5.4.2 B. Practice

**Exercise B1.**

Let  $\mathbf{u}$  be a unit vector and  $\mathbf{H} = \mathbf{I} - 2\mathbf{u}\mathbf{u}^\top$ . Show that  $\mathbf{H}$  is symmetric and orthogonal, that  $\mathbf{H}\mathbf{u} = -\mathbf{u}$ , and that  $\mathbf{H}\mathbf{v} = \mathbf{v}$  for  $\mathbf{v} \perp \mathbf{u}$ . Given  $\mathbf{x}$  not a nonnegative multiple of  $\mathbf{e}_1$ , find  $\mathbf{u}$  with  $\mathbf{H}\mathbf{x} = \|\mathbf{x}\|\mathbf{e}_1$ . Explain how repeating this column by column produces a QR factorization.

*Solution.*  $\mathbf{H}^\top = \mathbf{H}$ , and  $\mathbf{H}^\top\mathbf{H} = \mathbf{I} - 4\mathbf{u}\mathbf{u}^\top + 4\mathbf{u}(\mathbf{u}^\top\mathbf{u})\mathbf{u}^\top = \mathbf{I}$ . Also  $\mathbf{H}\mathbf{u} = \mathbf{u} - 2\mathbf{u} = -\mathbf{u}$ , and  $\mathbf{H}\mathbf{v} = \mathbf{v}$  when  $\mathbf{u}^\top\mathbf{v} = 0$ . Let  $\mathbf{w} = \mathbf{x} - \|\mathbf{x}\|\mathbf{e}_1 \neq \mathbf{0}$  and  $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$ . Then  $\mathbf{w}^\top\mathbf{w} = 2\|\mathbf{x}\|^2 - 2\|\mathbf{x}\|x_1 = 2\mathbf{w}^\top\mathbf{x}$ , so  $\mathbf{H}\mathbf{x} = \mathbf{x} - 2\mathbf{w}(\mathbf{w}^\top\mathbf{x})/(\mathbf{w}^\top\mathbf{w}) = \mathbf{x} - \mathbf{w} = \|\mathbf{x}\|\mathbf{e}_1$ . Applying such a reflection  $\mathbf{H}_1$  to the first column of  $\mathbf{X}$ , then a reflection acting on coordinates  $2, \dots, n$  to the second column of  $\mathbf{H}_1\mathbf{X}$ , and so on, gives  $\mathbf{H}_p \cdots \mathbf{H}_1\mathbf{X} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$  with  $\mathbf{R}$  upper triangular. Hence  $\mathbf{X} = \mathbf{Q}\mathbf{R}$ , where  $\mathbf{Q}$  holds the first  $p$  columns of the orthogonal matrix  $\mathbf{H}_1 \cdots \mathbf{H}_p$ . In floating point one uses  $\mathbf{w} = \mathbf{x} + \text{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$  instead, which avoids cancellation. It gives  $\mathbf{H}\mathbf{x} = -\text{sgn}(x_1)\|\mathbf{x}\|\mathbf{e}_1$ , so diagonal entries of  $\mathbf{R}$  may come out negative; changing the sign of such a row of  $\mathbf{R}$  and of the matching column of  $\mathbf{Q}$  at the end restores the positive diagonal of [Theorem 1.5.2](#).

**Exercise B2.**

Show that the QR factorization of a matrix with full column rank, with  $\mathbf{R}$  having positive diagonal, is unique, and that  $\mathbf{R}^\top$  is the Cholesky factor of  $\mathbf{X}^\top\mathbf{X}$ .

*Solution.* If  $\mathbf{X} = \mathbf{Q}_1\mathbf{R}_1 = \mathbf{Q}_2\mathbf{R}_2$ , then  $\mathbf{X}^\top\mathbf{X} = \mathbf{R}_1^\top\mathbf{R}_1 = \mathbf{R}_2^\top\mathbf{R}_2$ . Each  $\mathbf{R}_k^\top$  is lower triangular with positive diagonal, so both are the Cholesky factor of  $\mathbf{X}^\top\mathbf{X}$ , which is unique by [Theorem 1.7.4\(d\)](#). So  $\mathbf{R}_1 = \mathbf{R}_2$  and  $\mathbf{Q}_1 = \mathbf{X}\mathbf{R}_1^{-1} = \mathbf{Q}_2$ .

## 1.6 Trace, determinants and partitioned matrices

### 1.6.1 Trace

The **trace** of a square matrix is the sum of its diagonal entries,  $\text{tr}(\mathbf{A}) = \sum_i a_{ii}$ . It is linear,  $\text{tr}(a\mathbf{A} + b\mathbf{B}) = a\text{tr}(\mathbf{A}) + b\text{tr}(\mathbf{B})$ , and  $\text{tr}(\mathbf{A}^\top) = \text{tr}(\mathbf{A})$ .

#### Theorem 1.6.1 — Cyclic property

If  $\mathbf{A}$  is  $m \times n$  and  $\mathbf{B}$  is  $n \times m$ , then  $\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{BA})$ .

*Proof.*  $\text{tr}(\mathbf{AB}) = \sum_{i=1}^m \sum_{k=1}^n a_{ik}b_{ki} = \sum_{k=1}^n \sum_{i=1}^m b_{ki}a_{ik} = \text{tr}(\mathbf{BA})$ . ■

The two products need not have the same size. Applied repeatedly, the theorem allows cyclic

permutations such as  $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{CAB}) = \text{tr}(\mathbf{BCA})$ , but not arbitrary reorderings. Three consequences occur again and again:

$$\text{tr}(\mathbf{A}^\top \mathbf{A}) = \sum_{i,j} a_{ij}^2 = \|\mathbf{A}\|_F^2, \quad \mathbf{x}^\top \mathbf{A} \mathbf{x} = \text{tr}(\mathbf{A} \mathbf{x} \mathbf{x}^\top), \quad \text{tr}(\mathbf{P}^{-1} \mathbf{A} \mathbf{P}) = \text{tr}(\mathbf{A}). \quad (1.6.1)$$

The middle one turns a scalar quadratic form into the trace of a matrix that is linear in  $\mathbf{x} \mathbf{x}^\top$ . Taking expectations then gives the mean of a random quadratic form ([Chapter 2](#)).

## 1.6.2 Determinants

The **determinant** of an  $n \times n$  matrix is

$$\det \mathbf{A} = \sum_{\pi} \text{sgn}(\pi) a_{1\pi(1)} a_{2\pi(2)} \cdots a_{n\pi(n)},$$

the sum over all permutations  $\pi$  of  $\{1, \dots, n\}$ , where  $\text{sgn}(\pi) = \pm 1$  according to whether  $\pi$  is a product of an even or odd number of transpositions. We collect the standard properties. Parts (a)–(d) are proved in any linear algebra text (Harville 1997, chapter 13), and the others follow from them.

### Proposition 1.6.2 — Determinants

Let  $\mathbf{A}, \mathbf{B}$  be  $n \times n$ .

- (a)  $\det \mathbf{A}^\top = \det \mathbf{A}$ .
- (b)  $\det \mathbf{A}$  is a linear function of each column separately. It changes sign when two columns are interchanged, and is unchanged when a multiple of one column is added to another. The same holds for rows.
- (c) A triangular matrix has determinant equal to the product of its diagonal entries.
- (d)  $\det(\mathbf{AB}) = \det \mathbf{A} \det \mathbf{B}$ .
- (e)  $\mathbf{A}$  is nonsingular iff  $\det \mathbf{A} \neq 0$ , and then  $\det \mathbf{A}^{-1} = 1/\det \mathbf{A}$ .
- (f)  $\det(c\mathbf{A}) = c^n \det \mathbf{A}$ , and  $\det \mathbf{Q} = \pm 1$  for orthogonal  $\mathbf{Q}$ .

*Proof* (Proof of (e) and (f)). If  $\mathbf{A}$  is nonsingular,  $1 = \det \mathbf{I} = \det \mathbf{A} \det \mathbf{A}^{-1}$  by (d). If  $\mathbf{A}$  is singular, some column is a combination of the others. Subtracting that combination leaves a zero column without changing the determinant, by (b), and the determinant of a matrix with a zero column is zero, since every term of the sum contains a factor from that column. For (f), apply (b) to each of the  $n$  columns, and use (a) and (d) on  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$ .

■

### 1.6.3 Schur complements and block determinants

Partitioned matrices arise whenever the regressors split into two groups, the model matrix  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$  giving

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} \mathbf{X}_1^\top \mathbf{X}_1 & \mathbf{X}_1^\top \mathbf{X}_2 \\ \mathbf{X}_2^\top \mathbf{X}_1 & \mathbf{X}_2^\top \mathbf{X}_2 \end{pmatrix},$$

and whenever a random vector splits into two blocks (Chapter 3). In what follows

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}, \quad (1.6.2)$$

with  $\mathbf{A}_{11}$  of size  $k \times k$  and  $\mathbf{A}_{22}$  of size  $(n - k) \times (n - k)$ .

#### Definition 1.6.1 — Schur complement

If  $\mathbf{A}_{11}$  is nonsingular, the **Schur complement** of  $\mathbf{A}_{11}$  in  $\mathbf{A}$  is  $\mathbf{A}_{22 \cdot 1} = \mathbf{A}_{22} - \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$ .  
If  $\mathbf{A}_{22}$  is nonsingular, the Schur complement of  $\mathbf{A}_{22}$  is  $\mathbf{A}_{11 \cdot 2} = \mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21}$ .

Everything in this subsection comes from one identity, block Gaussian elimination. If  $\mathbf{A}_{11}$  is nonsingular, multiplying out shows

$$\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22 \cdot 1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}. \quad (1.6.3)$$

The outer factors are block triangular with identity blocks on the diagonal, and are nonsingular with inverses obtained by changing the sign of the off-diagonal block.

#### Theorem 1.6.3 — Block determinants

Let  $\mathbf{A}$  be partitioned as in Eq. 1.6.2.

- (a) If  $\mathbf{A}_{12} = \mathbf{0}$  or  $\mathbf{A}_{21} = \mathbf{0}$ , then  $\det \mathbf{A} = \det \mathbf{A}_{11} \det \mathbf{A}_{22}$ .
- (b) If  $\mathbf{A}_{11}$  is nonsingular,  $\det \mathbf{A} = \det \mathbf{A}_{11} \det \mathbf{A}_{22 \cdot 1}$ . If  $\mathbf{A}_{22}$  is nonsingular,  $\det \mathbf{A} = \det \mathbf{A}_{22} \det \mathbf{A}_{11 \cdot 2}$ .
- (c) For  $m \times n$  matrices  $\mathbf{U}$  and  $n \times m$  matrices  $\mathbf{V}$ ,  $\det(\mathbf{I}_m + \mathbf{UV}) = \det(\mathbf{I}_n + \mathbf{VU})$ . If  $\mathbf{B}$  is  $m \times m$  and nonsingular,  $\det(\mathbf{B} + \mathbf{UV}) = \det \mathbf{B} \det(\mathbf{I}_n + \mathbf{VB}^{-1} \mathbf{U})$ .

*Proof.*

- (a) Suppose  $\mathbf{A}_{12} = \mathbf{0}$  (otherwise transpose). In the permutation expansion, a term that is nonzero cannot use an entry of the zero block  $\mathbf{A}_{12}$ . So each of the first  $k$  rows takes its entry from the first  $k$  columns, and  $\pi$  maps  $\{1, \dots, k\}$  onto itself. Such a permutation splits into a permutation of the first  $k$  indices and one of the rest. Its sign is the product of their signs, so the sum factors as  $\det \mathbf{A}_{11} \det \mathbf{A}_{22}$ .
- (b) Take determinants in Eq. 1.6.3, using (a) and Proposition 1.6.2(d). The outer factors have determinant 1. The second statement follows in the same way from the elimination that starts with  $\mathbf{A}_{22}$ .



- (c) Apply (b) in both ways to  $\begin{pmatrix} \mathbf{I}_m & -\mathbf{U} \\ \mathbf{V} & \mathbf{I}_n \end{pmatrix}$ , whose Schur complements are  $\mathbf{I}_m + \mathbf{V}\mathbf{U}$  and  $\mathbf{I}_n + \mathbf{U}\mathbf{V}$ . For the second statement, write  $\mathbf{B} + \mathbf{U}\mathbf{V} = \mathbf{B}(\mathbf{I}_m + \mathbf{B}^{-1}\mathbf{U}\mathbf{V})$  and apply the first. ■

Part (c) with  $n = 1$  is the **matrix determinant lemma**  $\det(\mathbf{B} + \mathbf{u}\mathbf{v}^\top) = \det \mathbf{B} (1 + \mathbf{v}^\top \mathbf{B}^{-1} \mathbf{u})$ .

## 1.6.4 Inverses of partitioned matrices

### Theorem 1.6.4 — Partitioned inverse

Let  $\mathbf{A}$  be partitioned as in Eq. 1.6.2 with  $\mathbf{A}_{11}$  nonsingular. Then  $\mathbf{A}$  is nonsingular iff  $\mathbf{A}_{22 \cdot 1}$  is, and in that case

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \\ -\mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{A}_{22 \cdot 1}^{-1} \end{pmatrix}. \quad (1.6.4)$$

If  $\mathbf{A}_{22}$  is also nonsingular, the leading block of  $\mathbf{A}^{-1}$  equals  $\mathbf{A}_{11 \cdot 2}^{-1}$ :

$$(\mathbf{A}_{11} - \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21})^{-1} = \mathbf{A}_{11}^{-1} + \mathbf{A}_{11}^{-1} \mathbf{A}_{12} \mathbf{A}_{22 \cdot 1}^{-1} \mathbf{A}_{21} \mathbf{A}_{11}^{-1}. \quad (1.6.5)$$

*Proof.* The nonsingularity statement is Theorem 1.6.3(b). Inverting the three factors of Eq. 1.6.3 and multiplying them in reverse order gives

$$\mathbf{A}^{-1} = \begin{pmatrix} \mathbf{I} & -\mathbf{A}_{11}^{-1} \mathbf{A}_{12} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} \begin{pmatrix} \mathbf{A}_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_{22 \cdot 1}^{-1} \end{pmatrix} \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ -\mathbf{A}_{21} \mathbf{A}_{11}^{-1} & \mathbf{I} \end{pmatrix},$$

which multiplies out to Eq. 1.6.4. Eliminating with  $\mathbf{A}_{22}$  first gives a second expression for  $\mathbf{A}^{-1}$ , whose leading block is  $\mathbf{A}_{11 \cdot 2}^{-1}$ . The inverse is unique, so the two leading blocks agree. ■

For a symmetric positive definite  $\mathbf{X}^\top \mathbf{X}$  (Section 1.7), the theorem says that the block of  $(\mathbf{X}^\top \mathbf{X})^{-1}$  belonging to  $\mathbf{X}_1$  is  $(\mathbf{X}_1^\top \mathbf{X}_1 - \mathbf{X}_1^\top \mathbf{X}_2 (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top \mathbf{X}_1)^{-1}$ . Chapter 6 derives the same formula geometrically, from the Frisch–Waugh–Lovell theorem.

Identity Eq. 1.6.5 is a disguised form of a result that deserves its own name. Rename  $\mathbf{A}_{11} = \mathbf{B}$ ,  $\mathbf{A}_{12} = \mathbf{U}$ ,  $\mathbf{A}_{21} = \mathbf{V}^\top$  and  $\mathbf{A}_{22} = -\mathbf{C}^{-1}$ .

### Theorem 1.6.5 — Sherman–Morrison–Woodbury

Let  $\mathbf{B}$  ( $n \times n$ ) and  $\mathbf{C}$  ( $k \times k$ ) be nonsingular, and let  $\mathbf{U}, \mathbf{V}$  be  $n \times k$ . If  $\mathbf{C}^{-1} + \mathbf{V}^\top \mathbf{B}^{-1} \mathbf{U}$  is nonsingular, then so is  $\mathbf{B} + \mathbf{U}\mathbf{C}\mathbf{V}^\top$ , and

$$(\mathbf{B} + \mathbf{UCV}^\top)^{-1} = \mathbf{B}^{-1} - \mathbf{B}^{-1}\mathbf{U}(\mathbf{C}^{-1} + \mathbf{V}^\top\mathbf{B}^{-1}\mathbf{U})^{-1}\mathbf{V}^\top\mathbf{B}^{-1}. \quad (1.6.6)$$

In particular, for vectors  $\mathbf{u}, \mathbf{v}$  with  $1 + \mathbf{v}^\top\mathbf{B}^{-1}\mathbf{u} \neq 0$ ,

$$(\mathbf{B} + \mathbf{uv}^\top)^{-1} = \mathbf{B}^{-1} - \frac{\mathbf{B}^{-1}\mathbf{uv}^\top\mathbf{B}^{-1}}{1 + \mathbf{v}^\top\mathbf{B}^{-1}\mathbf{u}}. \quad (1.6.7)$$

*Proof.* Apply [Theorem 1.6.4](#) to  $\mathbf{N} = \begin{pmatrix} \mathbf{B} & \mathbf{U} \\ \mathbf{V}^\top & -\mathbf{C}^{-1} \end{pmatrix}$ . Both diagonal blocks are nonsingular. The Schur complement of  $\mathbf{B}$  is  $-(\mathbf{C}^{-1} + \mathbf{V}^\top\mathbf{B}^{-1}\mathbf{U})$ , which is nonsingular by hypothesis, so  $\mathbf{N}$  is nonsingular. Then the Schur complement of the lower block,  $\mathbf{B} + \mathbf{UCV}^\top$ , is nonsingular by the same theorem with the roles of the blocks exchanged, and [Eq. 1.6.5](#) becomes [Eq. 1.6.6](#). For [Eq. 1.6.7](#) take  $k = 1$  and  $\mathbf{C} = 1$ . ■

*Idea.* Every result in this section comes from one block elimination, [Eq. 1.6.3](#): a partitioned matrix is a unit triangular matrix times  $\text{diag}(\mathbf{A}_{11}, \mathbf{A}_{22.1})$  times another. Taking determinants gives [Theorem 1.6.3](#), inverting the factors gives [Theorem 1.6.4](#), and eliminating in the other order and comparing gives the Woodbury identity. When a formula about a block is forgotten, redo the elimination.

The identity replaces the inversion of an  $n \times n$  matrix by that of a  $k \times k$  matrix. That matters when  $\mathbf{B}^{-1}$  is already known or cheap, as in the following two examples.

#### Example. (Deleting an observation)

Removing the  $i$ th row  $\mathbf{x}_i^\top$  from a model matrix changes  $\mathbf{X}^\top\mathbf{X}$  into  $\mathbf{X}^\top\mathbf{X} - \mathbf{x}_i\mathbf{x}_i^\top$ . With  $\mathbf{u} = -\mathbf{x}_i$ ,  $\mathbf{v} = \mathbf{x}_i$  and  $h_i = \mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i$ , [Eq. 1.6.7](#) gives

$$(\mathbf{X}^\top\mathbf{X} - \mathbf{x}_i\mathbf{x}_i^\top)^{-1} = (\mathbf{X}^\top\mathbf{X})^{-1} + \frac{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{x}_i\mathbf{x}_i^\top(\mathbf{X}^\top\mathbf{X})^{-1}}{1 - h_i},$$

valid whenever  $h_i < 1$ , and the determinant lemma gives  $\det(\mathbf{X}^\top\mathbf{X} - \mathbf{x}_i\mathbf{x}_i^\top) = (1 - h_i)\det(\mathbf{X}^\top\mathbf{X})$ . The quantity  $h_i$  is the leverage of observation  $i$  ([Chapter 6](#)), and these formulas are what make the deletion diagnostics of [Chapter 20](#) cheap ([Theorem 20.3.1](#)). For Brownlee's stack-loss data ( $n = 21$ ,  $p = 4$ ), the listing updates the inverse for each of the 21 deletions. The largest relative discrepancy from direct inversion is  $1 \times 10^{-13}$ . Observation 17 has the largest leverage, 0.412, and removing it multiplies  $\det(\mathbf{X}^\top\mathbf{X})$  by 0.588.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.stackloss.load_pandas().data
X = np.column_stack([np.ones(len(data)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape

A_inv = np.linalg.inv(X.T @ X)
h = np.einsum("ij,jk,ik->i", X, A_inv, X) # leverages x_i' (X'X)^{-1} x_i
```

```

worst = 0.0
for i in range(n):
    x = X[i]
    # Sherman-Morrison with u = -x, v = x: (X'X - x x')^{-1}
    B_update = A_inv + np.outer(A_inv @ x, x @ A_inv) / (1 - h[i])
    X_del = np.delete(X, i, axis=0)
    B_direct = np.linalg.inv(X_del.T @ X_del)
    worst = max(worst, np.abs(B_update - B_direct).max() / np.abs(B_direct).max())
print(f"largest relative discrepancy over all {n} deletions: {worst:.1e}")

```

### Example. (Diagonal plus low rank)

In factor models and in the linear mixed models of [Chapter 32](#), covariance matrices have the form  $\Sigma = \mathbf{D} + \mathbf{L}\mathbf{L}^\top$  with  $\mathbf{D}$  diagonal and positive and  $\mathbf{L}$  of size  $N \times k$ ,  $k \ll N$ . By [Eq. 1.6.6](#) and [Theorem 1.6.3\(c\)](#),

$$\Sigma^{-1} = \mathbf{D}^{-1} - \mathbf{D}^{-1}\mathbf{L}(\mathbf{I}_k + \mathbf{L}^\top\mathbf{D}^{-1}\mathbf{L})^{-1}\mathbf{L}^\top\mathbf{D}^{-1}, \quad \det \Sigma = \det \mathbf{D} \det(\mathbf{I}_k + \mathbf{L}^\top\mathbf{D}^{-1}\mathbf{L}).$$

Each needs only a  $k \times k$  factorization. The script checks this with  $N = 1500$  and  $k = 4$ . The solution of  $\Sigma\mathbf{w} = \mathbf{v}$  agrees with a direct solve to a relative error of  $8 \times 10^{-14}$ .

### Example. (Equicorrelation)

The matrix  $\mathbf{E} = (1 - \rho)\mathbf{I}_n + \rho\mathbf{J}_n$  has unit diagonal and every off-diagonal entry equal to  $\rho$ . It is the correlation matrix of exchangeable observations. Taking  $\mathbf{B} = (1 - \rho)\mathbf{I}$  and  $\mathbf{u} = \rho\mathbf{1}$ ,  $\mathbf{v} = \mathbf{1}$ , for  $\rho \neq 1$  and  $1 + (n - 1)\rho \neq 0$ ,

$$\mathbf{E}^{-1} = \frac{1}{1 - \rho} \left( \mathbf{I}_n - \frac{\rho}{1 + (n - 1)\rho} \mathbf{J}_n \right), \quad \det \mathbf{E} = (1 - \rho)^{n-1} (1 + (n - 1)\rho).$$

## 1.6.5 Exercises

### 1.6.5.1 A. Check your understanding

#### Exercise A1.

Show that  $\text{tr}(\mathbf{ABC}) = \text{tr}(\mathbf{ACB})$  when  $\mathbf{A}, \mathbf{B}, \mathbf{C}$  are symmetric, and find  $2 \times 2$  matrices for which the two traces differ.

*Solution.* For symmetric matrices,  $\text{tr}(\mathbf{ABC}) = \text{tr}((\mathbf{ABC})^\top) = \text{tr}(\mathbf{CBA}) = \text{tr}(\mathbf{ACB})$ , using transposition and then the cyclic property. For a counterexample take  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ ,  $\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{C} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$ . Then  $\mathbf{ABC} = \mathbf{C}$  has trace 1, while  $\mathbf{AC} = \mathbf{0}$  gives  $\text{tr}(\mathbf{ACB}) = 0$ .

#### Exercise A2.

Let  $\mathbf{A}$  be nonsingular,  $\mathbf{a}$  a vector and  $c$  a scalar with  $s = c - \mathbf{a}^\top \mathbf{A}^{-1} \mathbf{a} \neq 0$ . Write down the inverse and the determinant of  $\begin{pmatrix} \mathbf{A} & \mathbf{a} \\ \mathbf{a}^\top & c \end{pmatrix}$ . Use it to update  $(\mathbf{X}^\top \mathbf{X})^{-1}$  when a column

is added to  $\mathbf{X}$ .

### 1.6.5.2 B. Practice

#### Exercise B1.

Let  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ ,  $e_i = y_i - \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$  and  $h_i = \mathbf{x}_i^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i < 1$ . Show that the least squares coefficients computed without observation  $i$  are

$$\hat{\boldsymbol{\beta}}_{(i)} = \hat{\boldsymbol{\beta}} - \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_i e_i}{1 - h_i}.$$

*Solution.* Write  $\mathbf{K} = (\mathbf{X}^\top \mathbf{X})^{-1}$  and  $\mathbf{x} = \mathbf{x}_i$ . Deleting row  $i$  turns  $\mathbf{X}^\top \mathbf{y}$  into  $\mathbf{X}^\top \mathbf{y} - \mathbf{x} y_i$ , so by [Example \(Deleting an observation\)](#)

$$\hat{\boldsymbol{\beta}}_{(i)} = \left( \mathbf{K} + \frac{\mathbf{K} \mathbf{x} \mathbf{x}^\top \mathbf{K}}{1 - h_i} \right) (\mathbf{X}^\top \mathbf{y} - \mathbf{x} y_i) = \hat{\boldsymbol{\beta}} - \mathbf{K} \mathbf{x} y_i + \frac{\mathbf{K} \mathbf{x} (\mathbf{x}^\top \hat{\boldsymbol{\beta}} - h_i y_i)}{1 - h_i} = \hat{\boldsymbol{\beta}} - \mathbf{K} \mathbf{x} \frac{y_i - \mathbf{x}^\top \hat{\boldsymbol{\beta}}}{1 - h_i},$$

since  $(1 - h_i) y_i - \mathbf{x}^\top \hat{\boldsymbol{\beta}} + h_i y_i = y_i - \mathbf{x}^\top \hat{\boldsymbol{\beta}}$ .

#### Exercise B2.

For an  $n \times p$  matrix  $\mathbf{X}$  and  $\lambda > 0$ , show that  $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1}$ . Explain why the right-hand side is preferable when  $p$  is much larger than  $n$ , and derive the identity a second way from [Theorem 1.6.5](#).

*Solution.* Both  $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$  and  $\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}$  are positive definite. The identity  $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p) \mathbf{X}^\top = \mathbf{X}^\top \mathbf{X} \mathbf{X}^\top + \lambda \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)$  gives the result after multiplying by the two inverses. When  $p \gg n$ , the right-hand side inverts an  $n \times n$  matrix instead of a  $p \times p$  one. For the second derivation, apply [Theorem 1.6.5](#) with  $\mathbf{B} = \lambda \mathbf{I}_p$ ,  $\mathbf{U} = \mathbf{X}^\top$ ,  $\mathbf{C} = \mathbf{I}_n$  and  $\mathbf{V} = \mathbf{X}^\top$ :  $(\lambda \mathbf{I}_p + \mathbf{X}^\top \mathbf{X})^{-1} = \lambda^{-1} \mathbf{I}_p - \lambda^{-2} \mathbf{X}^\top \mathbf{W}^{-1} \mathbf{X}$  with  $\mathbf{W} = \mathbf{I}_n + \mathbf{X} \mathbf{X}^\top / \lambda$ . Multiplying on the right by  $\mathbf{X}^\top$  gives  $\lambda^{-1} \mathbf{X}^\top [\mathbf{I}_n - \mathbf{W}^{-1} (\mathbf{X} \mathbf{X}^\top / \lambda)] = \lambda^{-1} \mathbf{X}^\top \mathbf{W}^{-1} [\mathbf{W} - \mathbf{X} \mathbf{X}^\top / \lambda] = \lambda^{-1} \mathbf{X}^\top \mathbf{W}^{-1} = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1}$ .

## 1.7 Eigenvalues, the spectral theorem and positive definiteness

### 1.7.1 Eigenvalues

A scalar  $\lambda$  is an **eigenvalue** of the  $n \times n$  matrix  $\mathbf{A}$ , with **eigenvector**  $\mathbf{x} \neq \mathbf{0}$ , if  $\mathbf{A} \mathbf{x} = \lambda \mathbf{x}$ . Equivalently,  $\mathbf{A} - \lambda \mathbf{I}$  is singular, that is,  $\lambda$  is a root of the **characteristic polynomial**  $p(\lambda) = \det(\lambda \mathbf{I} - \mathbf{A})$ . This polynomial has degree  $n$  and leading coefficient one, so over the complex numbers it factors as  $\prod_{i=1}^n (\lambda - \lambda_i)$ . The eigenvalues  $\lambda_1, \dots, \lambda_n$  are listed with multiplicity and may be complex when  $\mathbf{A}$  is not symmetric.

### Proposition 1.7.1

Let  $\mathbf{A}$  be  $n \times n$  with eigenvalues  $\lambda_1, \dots, \lambda_n$ .

- (a)  $\det \mathbf{A} = \prod_i \lambda_i$  and  $\operatorname{tr} \mathbf{A} = \sum_i \lambda_i$ .
- (b) If  $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$ , then  $f(\mathbf{A})\mathbf{x} = f(\lambda)\mathbf{x}$  for every polynomial  $f$ . If  $\mathbf{A}$  is nonsingular,  $\mathbf{A}^{-1}\mathbf{x} = \lambda^{-1}\mathbf{x}$ .
- (c)  $\mathbf{A}$  and  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  have the same characteristic polynomial for nonsingular  $\mathbf{P}$ .
- (d) If  $\mathbf{C}$  is  $m \times n$  and  $\mathbf{D}$  is  $n \times m$ , then  $\lambda^n \det(\lambda \mathbf{I}_m - \mathbf{CD}) = \lambda^m \det(\lambda \mathbf{I}_n - \mathbf{DC})$ . So  $\mathbf{CD}$  and  $\mathbf{DC}$  have the same nonzero eigenvalues, with the same multiplicities.

*Proof.*

- (a) Setting  $\lambda = 0$  in  $\det(\lambda \mathbf{I} - \mathbf{A}) = \prod(\lambda - \lambda_i)$  gives  $(-1)^n \det \mathbf{A} = (-1)^n \prod \lambda_i$ . For the trace, compare coefficients of  $\lambda^{n-1}$ . On the right it is  $-\sum \lambda_i$ . On the left, in the permutation expansion of  $\det(\lambda \mathbf{I} - \mathbf{A})$ , only the product of diagonal entries  $\prod_i (\lambda - a_{ii})$  contains  $\lambda^{n-1}$ , because every other term omits at least two diagonal entries. Its coefficient is  $-\sum a_{ii}$ .
- (b)  $\mathbf{A}^k \mathbf{x} = \lambda^k \mathbf{x}$  by induction, and  $\mathbf{x} = \mathbf{A}^{-1} \mathbf{A} \mathbf{x} = \lambda \mathbf{A}^{-1} \mathbf{x}$  with  $\lambda \neq 0$ . (c)  $\det(\lambda \mathbf{I} - \mathbf{P}^{-1} \mathbf{A} \mathbf{P}) = \det(\mathbf{P}^{-1}(\lambda \mathbf{I} - \mathbf{A})\mathbf{P}) = \det(\lambda \mathbf{I} - \mathbf{A})$ .
- (d) For  $\lambda \neq 0$ , Theorem 1.6.3(c) gives  $\det(\mathbf{I}_m - \lambda^{-1} \mathbf{CD}) = \det(\mathbf{I}_n - \lambda^{-1} \mathbf{DC})$ . Multiplying by  $\lambda^{m+n}$  gives the identity for  $\lambda \neq 0$ . Both sides are polynomials, so it holds for all  $\lambda$ .

■

## 1.7.2 The spectral theorem

Symmetric matrices have the best possible eigenstructure, and almost every matrix whose eigenvalues matter in statistics is symmetric.

### Theorem 1.7.2 — Spectral theorem

Let  $\mathbf{A}$  be an  $n \times n$  symmetric matrix. Then all eigenvalues of  $\mathbf{A}$  are real, and there is an orthogonal matrix  $\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_n]$  with

$$\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top = \sum_{i=1}^n \lambda_i \mathbf{q}_i \mathbf{q}_i^\top, \quad \mathbf{\Lambda} = \operatorname{diag}(\lambda_1, \dots, \lambda_n). \quad (1.7.1)$$

The columns of  $\mathbf{Q}$  are an orthonormal basis of eigenvectors,  $\mathbf{A}\mathbf{q}_i = \lambda_i \mathbf{q}_i$ . Eigenvectors belonging to distinct eigenvalues are orthogonal.

*Proof. Real eigenvalues.* Let  $\mathbf{A}\mathbf{z} = \lambda\mathbf{z}$  with  $\mathbf{z} \in \mathbb{C}^n$  nonzero, and let  $\bar{\mathbf{z}}$  be the complex conjugate. The scalar  $s = \bar{\mathbf{z}}^\top \mathbf{A}\mathbf{z}$  is real, since its conjugate is  $\mathbf{z}^\top \mathbf{A}\bar{\mathbf{z}} = (\mathbf{z}^\top \mathbf{A}\bar{\mathbf{z}})^\top = \bar{\mathbf{z}}^\top \mathbf{A}\mathbf{z} = s$  by symmetry. Also  $s = \lambda \bar{\mathbf{z}}^\top \mathbf{z} = \lambda \sum_i |z_i|^2$ , with the sum real and positive. So  $\lambda$  is real. The real singular matrix  $\mathbf{A} - \lambda \mathbf{I}$  then has a real nonzero null vector, so there

is a real eigenvector.

*Diagonalization.* By induction on  $n$ , the case  $n = 1$  being trivial. Let  $\mathbf{q}_1$  be a real unit eigenvector for  $\lambda_1$  and extend it to an orthogonal matrix  $[\mathbf{q}_1, \mathbf{Q}_2]$  (Section 1.5). Since  $\mathbf{Q}_2^\top \mathbf{A} \mathbf{q}_1 = \lambda_1 \mathbf{Q}_2^\top \mathbf{q}_1 = \mathbf{0}$  and  $\mathbf{A}$  is symmetric,

$$[\mathbf{q}_1, \mathbf{Q}_2]^\top \mathbf{A} [\mathbf{q}_1, \mathbf{Q}_2] = \begin{pmatrix} \lambda_1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{Q}_2^\top \mathbf{A} \mathbf{Q}_2 \end{pmatrix}.$$

The  $(n-1) \times (n-1)$  block is symmetric, so by induction  $\mathbf{Q}_2^\top \mathbf{A} \mathbf{Q}_2 = \mathbf{P} \mathbf{\Lambda}_2 \mathbf{P}^\top$  with  $\mathbf{P}$  orthogonal. Then  $\mathbf{Q} = [\mathbf{q}_1, \mathbf{Q}_2 \mathbf{P}]$  is orthogonal (Proposition 1.5.3(b)) and  $\mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \text{diag}(\lambda_1, \mathbf{\Lambda}_2)$ .

*Orthogonality.* If  $\mathbf{A} \mathbf{u} = \lambda \mathbf{u}$  and  $\mathbf{A} \mathbf{v} = \mu \mathbf{v}$ , then  $\lambda \mathbf{u}^\top \mathbf{v} = \mathbf{u}^\top \mathbf{A} \mathbf{v} = \mu \mathbf{u}^\top \mathbf{v}$ , so  $\mathbf{u}^\top \mathbf{v} = 0$  when  $\lambda \neq \mu$ . ■

Unless stated otherwise, the eigenvalues of symmetric  $\mathbf{A}$  are ordered as  $\lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_n$ , and we write  $\lambda_{\max} = \lambda_1$  and  $\lambda_{\min} = \lambda_n$ . The following consequences are used without comment later on.

### Corollary 1.7.3

Let  $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$  be symmetric.

- (a)  $\text{rank}(\mathbf{A})$  counts the nonzero eigenvalues.  $\mathcal{C}(\mathbf{A})$  is spanned by the  $\mathbf{q}_i$  with  $\lambda_i \neq 0$ , and  $\mathcal{N}(\mathbf{A})$  by those with  $\lambda_i = 0$ .
- (b)  $\mathbf{A}^k = \mathbf{Q} \mathbf{\Lambda}^k \mathbf{Q}^\top$  for  $k = 1, 2, \dots$ . If  $\mathbf{A}$  is nonsingular,  $\mathbf{A}^{-1} = \mathbf{Q} \mathbf{\Lambda}^{-1} \mathbf{Q}^\top$ , and  $\mathbf{A}^+ = \mathbf{Q} \mathbf{\Lambda}^+ \mathbf{Q}^\top$  in general, where  $\mathbf{\Lambda}^+$  inverts the nonzero diagonal entries.
- (c) The eigenspace  $\{\mathbf{x} : \mathbf{A} \mathbf{x} = \lambda \mathbf{x}\}$  has dimension equal to the multiplicity of  $\lambda$  as a root of the characteristic polynomial.

*Proof.* Multiplying by the nonsingular  $\mathbf{Q}$  and  $\mathbf{Q}^\top$  changes neither rank (Proposition 1.2.3(b)) nor, after the change of variable  $\mathbf{x} = \mathbf{Q} \mathbf{c}$ , the structure of null spaces and eigenspaces. So each statement reduces to the diagonal matrix  $\mathbf{\Lambda}$ , where it is immediate. For (b),  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}$  collapses the middle factors, and the four Penrose conditions for  $\mathbf{Q} \mathbf{\Lambda}^+ \mathbf{Q}^\top$  reduce to those for  $\mathbf{\Lambda}^+$ . ■

### Example. (Eigenvalues of the equicorrelation matrix)

For  $\mathbf{E} = (1 - \rho) \mathbf{I} + \rho \mathbf{J}_n$  of Example (Equicorrelation),  $\mathbf{E} \mathbf{1} = (1 + (n-1)\rho) \mathbf{1}$ , and  $\mathbf{E} \mathbf{v} = (1 - \rho) \mathbf{v}$  for every  $\mathbf{v} \perp \mathbf{1}$ . So the eigenvalues are  $1 + (n-1)\rho$  once and  $1 - \rho$  with multiplicity  $n-1$ , with eigenvectors  $\mathbf{1}/\sqrt{n}$  and the last  $n-1$  rows of the Helmert matrix (Example (Helmert matrices)). The determinant and inverse found earlier follow at once from Proposition 1.7.1 and Corollary 1.7.3(b).

### 1.7.3 Positive definite and nonnegative definite matrices

#### Definition 1.7.1

A symmetric matrix  $\mathbf{A}$  is **positive definite** if  $\mathbf{x}^\top \mathbf{A} \mathbf{x} > 0$  for every  $\mathbf{x} \neq \mathbf{0}$ , and **nonnegative definite** if  $\mathbf{x}^\top \mathbf{A} \mathbf{x} \geq 0$  for every  $\mathbf{x}$ . For symmetric  $\mathbf{A}, \mathbf{B}$  of the same size we write  $\mathbf{A} \succeq \mathbf{B}$  if  $\mathbf{A} - \mathbf{B}$  is nonnegative definite, and  $\mathbf{A} \succ \mathbf{B}$  if it is positive definite (the **Loewner order**).

In this book, “nonnegative definite” includes positive definite. Some texts, Rencher and Schaalje among them, reserve “positive semidefinite” for matrices that are nonnegative definite but singular. We use the terms only for symmetric matrices. Covariance matrices are nonnegative definite (Chapter 2), and *Gram matrices*  $\mathbf{X}^\top \mathbf{X}$  are too, since  $\mathbf{x}^\top \mathbf{X}^\top \mathbf{X} \mathbf{x} = \|\mathbf{X} \mathbf{x}\|^2$ .

#### Theorem 1.7.4 — Characterizations

For a symmetric  $n \times n$  matrix  $\mathbf{A}$  the following are equivalent:

- (a)  $\mathbf{A}$  is positive definite;
- (b) every eigenvalue of  $\mathbf{A}$  is positive;
- (c)  $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$  for some matrix  $\mathbf{B}$  with  $n$  columns and full column rank;
- (d)  $\mathbf{A} = \mathbf{L} \mathbf{L}^\top$  with  $\mathbf{L}$  lower triangular with positive diagonal entries (the **Cholesky factorization**;  $\mathbf{L}$  is unique);
- (e) every leading principal minor  $\det \mathbf{A}_{[k]}$ ,  $k = 1, \dots, n$ , is positive, where  $\mathbf{A}_{[k]}$  is the upper left  $k \times k$  block.

Likewise, the following are equivalent: (a')  $\mathbf{A}$  is nonnegative definite; (b') every eigenvalue is nonnegative; (c')  $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$  for some  $\mathbf{B}$ , which can be taken of size  $\text{rank}(\mathbf{A}) \times n$ ; (e') every principal minor (determinant of a submatrix with the same row and column indices) is nonnegative.

*Proof.* Write  $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$ . (a) $\Rightarrow$ (b):  $\lambda_i = \mathbf{q}_i^\top \mathbf{A} \mathbf{q}_i > 0$ . (b) $\Rightarrow$ (c):  $\mathbf{B} = \mathbf{A}^{1/2} \mathbf{Q}^\top$  is nonsingular. (c) $\Rightarrow$ (a):  $\mathbf{x}^\top \mathbf{B}^\top \mathbf{B} \mathbf{x} = \|\mathbf{B} \mathbf{x}\|^2$ , which is positive for  $\mathbf{x} \neq \mathbf{0}$  by full column rank. (d) $\Rightarrow$ (c) is immediate, with  $\mathbf{B} = \mathbf{L}^\top$ .

(a) $\Rightarrow$ (d), by induction on  $n$ . Write  $\mathbf{A} = \begin{pmatrix} a & \mathbf{b}^\top \\ \mathbf{b} & \mathbf{A}_2 \end{pmatrix}$ . Then  $a = \mathbf{e}_1^\top \mathbf{A} \mathbf{e}_1 > 0$ . The Schur complement  $\mathbf{A}_2 - \mathbf{b} \mathbf{b}^\top / a$  is positive definite: for  $\mathbf{v} \neq \mathbf{0}$ , put  $\mathbf{x} = (-\mathbf{b}^\top \mathbf{v} / a, \mathbf{v}^\top)^\top$  and expand to get  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{v}^\top (\mathbf{A}_2 - \mathbf{b} \mathbf{b}^\top / a) \mathbf{v} > 0$ . By induction  $\mathbf{A}_2 - \mathbf{b} \mathbf{b}^\top / a = \mathbf{L}_2 \mathbf{L}_2^\top$ , and  $\mathbf{L} = \begin{pmatrix} \sqrt{a} & \mathbf{0}^\top \\ \mathbf{b} / \sqrt{a} & \mathbf{L}_2 \end{pmatrix}$  satisfies  $\mathbf{L} \mathbf{L}^\top = \mathbf{A}$ . For uniqueness, if  $\mathbf{L} \mathbf{L}^\top = \mathbf{K} \mathbf{K}^\top$ , then  $\mathbf{K}^{-1} \mathbf{L} = \mathbf{K}^\top (\mathbf{L}^\top)^{-1}$  is both lower and upper triangular, hence diagonal, say  $\mathbf{D}$ . From  $\mathbf{K}^{-1} \mathbf{L} = \mathbf{D}$  and  $\mathbf{L} \mathbf{L}^\top = \mathbf{K} \mathbf{K}^\top$  we get  $\mathbf{D} \mathbf{D}^\top = \mathbf{I}$ , so  $\mathbf{D} = \mathbf{I}$  because its diagonal is positive.

(a) $\Rightarrow$ (e):  $\mathbf{A}_{[k]} = \mathbf{E}^\top \mathbf{A} \mathbf{E}$  with  $\mathbf{E} = [\mathbf{e}_1, \dots, \mathbf{e}_k]$  is positive definite, since  $\mathbf{x}^\top \mathbf{A}_{[k]} \mathbf{x} = (\mathbf{E} \mathbf{x})^\top \mathbf{A} (\mathbf{E} \mathbf{x})$ . By (b) and Proposition 1.7.1(a) its determinant is positive. (e) $\Rightarrow$ (a), by induction on  $n$ . The block  $\mathbf{A}_{[n-1]}$  satisfies (e) and so is positive definite, so by (a) $\Rightarrow$ (d),

which is already proved,  $\mathbf{A}_{[n-1]} = \mathbf{L}\mathbf{L}^\top$ . Write  $\mathbf{A} = \begin{pmatrix} \mathbf{A}_{[n-1]} & \mathbf{c} \\ \mathbf{c}^\top & d \end{pmatrix}$  and  $s = d - \mathbf{c}^\top \mathbf{A}_{[n-1]}^{-1} \mathbf{c}$ . By Theorem 1.6.3(b),  $s = \det \mathbf{A} / \det \mathbf{A}_{[n-1]} > 0$ . With  $\mathbf{T} = \begin{pmatrix} \mathbf{I} & \mathbf{A}_{[n-1]}^{-1} \mathbf{c} \\ \mathbf{0}^\top & 1 \end{pmatrix}$ , a direct check gives  $\mathbf{A} = \mathbf{T}^\top \text{diag}(\mathbf{A}_{[n-1]}, s) \mathbf{T} = \mathbf{B}^\top \mathbf{B}$  with  $\mathbf{B} = \text{diag}(\mathbf{L}^\top, \sqrt{s}) \mathbf{T}$  nonsingular, so  $\mathbf{A}$  is positive definite by (c)  $\Rightarrow$  (a).

The primed statements: (a')  $\Leftrightarrow$  (b')  $\Leftrightarrow$  (c') exactly as above, taking for  $\mathbf{B}$  the rows of  $\mathbf{A}^{1/2} \mathbf{Q}^\top$  that belong to nonzero eigenvalues. (a')  $\Rightarrow$  (e') as above, with an arbitrary index set in place of  $\{1, \dots, k\}$ . For (e')  $\Rightarrow$  (a'), fix  $t > 0$ . Each column of  $\mathbf{A}_{[k]} + t\mathbf{I}$  is a column of  $\mathbf{A}_{[k]}$  plus  $t$  times a coordinate vector. Expanding the determinant by linearity in each column gives  $\det(\mathbf{A}_{[k]} + t\mathbf{I}) = \sum_S t^{k-|S|} \det \mathbf{A}_{SS}$ , the sum over all subsets  $S \subseteq \{1, \dots, k\}$ , with  $\det \mathbf{A}_{\emptyset\emptyset} = 1$ . Every term is nonnegative and the term for  $S = \emptyset$  is  $t^k > 0$ . So  $\mathbf{A} + t\mathbf{I}$  is positive definite by (e)  $\Rightarrow$  (a), and  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \lim_{t \rightarrow 0} \mathbf{x}^\top (\mathbf{A} + t\mathbf{I}) \mathbf{x} \geq 0$ . ■

Two warnings. Leading principal minors do not suffice for nonnegative definiteness:  $\text{diag}(0, -1)$  has leading minors 0 and 0. And (c') holds with many matrices  $\mathbf{B}$ . The Cholesky factor and the symmetric square root below are two convenient choices.

### Proposition 1.7.5

Let  $\mathbf{A}$  be  $n \times n$  nonnegative definite.

- (a)  $a_{ii} \geq 0$ , and if  $a_{ii} = 0$  then row  $i$  and column  $i$  of  $\mathbf{A}$  are zero.
- (b)  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = 0$  iff  $\mathbf{A} \mathbf{x} = \mathbf{0}$ .
- (c)  $\mathbf{B}^\top \mathbf{A} \mathbf{B}$  is nonnegative definite for any  $n \times q$  matrix  $\mathbf{B}$ . If  $\mathbf{A}$  is positive definite,  $\mathbf{B}^\top \mathbf{A} \mathbf{B}$  is positive definite iff  $\mathbf{B}$  has full column rank.
- (d) Every principal submatrix of  $\mathbf{A}$  is nonnegative definite, and positive definite if  $\mathbf{A}$  is.
- (e) If  $\mathbf{A}$  is positive definite, it is nonsingular and  $\mathbf{A}^{-1}$  is positive definite. If also  $\mathbf{A}$  is partitioned as in Eq. 1.6.2, both Schur complements  $\mathbf{A}_{11.2}$  and  $\mathbf{A}_{22.1}$  are positive definite.
- (f) If  $\mathbf{A} \succeq \mathbf{B} \succ \mathbf{0}$ , then  $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$ .

*Proof.*

- (b) Write  $\mathbf{A} = \mathbf{B}^\top \mathbf{B}$  by Theorem 1.7.4(c'). Then  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{B} \mathbf{x}\|^2$ , which vanishes iff  $\mathbf{B} \mathbf{x} = \mathbf{0}$ , which implies  $\mathbf{A} \mathbf{x} = \mathbf{0}$ .
- (c)  $a_{ii} = \mathbf{e}_i^\top \mathbf{A} \mathbf{e}_i \geq 0$ . If  $a_{ii} = 0$  then  $\mathbf{A} \mathbf{e}_i = \mathbf{0}$  by (b), so column  $i$  is zero, and row  $i$  by symmetry.
- (d)  $\mathbf{x}^\top \mathbf{B}^\top \mathbf{A} \mathbf{B} \mathbf{x} = (\mathbf{B} \mathbf{x})^\top \mathbf{A} (\mathbf{B} \mathbf{x})$  is nonnegative, and when  $\mathbf{A}$  is positive definite it is zero only if  $\mathbf{B} \mathbf{x} = \mathbf{0}$ .
- (e) is (c) with  $\mathbf{B}$  a set of columns of  $\mathbf{I}$ .
- (f) The eigenvalues of  $\mathbf{A}^{-1}$  are  $1/\lambda_i > 0$ . The inverse in Eq. 1.6.4 has  $\mathbf{A}_{22.1}^{-1}$  as a principal submatrix, which is positive definite by (d). The other Schur complement follows by symmetry of the argument.
- (g) By Theorem 1.7.6 below,  $\mathbf{B} = \mathbf{B}^{1/2} \mathbf{B}^{1/2}$ . Then  $\mathbf{C} = \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} \succeq \mathbf{I}$  by (c), so



every eigenvalue of  $\mathbf{C}$  is at least 1 and every eigenvalue of  $\mathbf{C}^{-1} = \mathbf{B}^{1/2}\mathbf{A}^{-1}\mathbf{B}^{1/2}$  is at most 1. So  $\mathbf{I} \succeq \mathbf{B}^{1/2}\mathbf{A}^{-1}\mathbf{B}^{1/2}$ , and multiplying on both sides by  $\mathbf{B}^{-1/2}$  gives  $\mathbf{B}^{-1} \succeq \mathbf{A}^{-1}$  by (c). ■

Part (f) is used in [Chapter 7](#): an estimator whose covariance matrix is smaller in the Loewner order has smaller variance for every linear combination of its components.

### Theorem 1.7.6 — Square root

A nonnegative definite matrix  $\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$  has exactly one nonnegative definite matrix  $\mathbf{B}$  with  $\mathbf{B}^2 = \mathbf{A}$ , namely  $\mathbf{A}^{1/2} = \mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^\top$ . It satisfies  $\mathcal{C}(\mathbf{A}^{1/2}) = \mathcal{C}(\mathbf{A})$ . If  $\mathbf{A}$  is positive definite,  $\mathbf{A}^{1/2}$  is positive definite and its inverse is  $\mathbf{A}^{-1/2} = \mathbf{Q}\mathbf{\Lambda}^{-1/2}\mathbf{Q}^\top$ .

*Proof.*  $\mathbf{A}^{1/2}$  is symmetric with nonnegative eigenvalues and  $(\mathbf{Q}\mathbf{\Lambda}^{1/2}\mathbf{Q}^\top)^2 = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^\top$ . Its column space is spanned by the  $\mathbf{q}_i$  with  $\lambda_i > 0$ , as is  $\mathcal{C}(\mathbf{A})$  ([Corollary 1.7.3\(a\)](#)). For uniqueness, let  $\mathbf{B}$  be nonnegative definite with  $\mathbf{B}^2 = \mathbf{A}$ , and write  $\mathbf{B} = \mathbf{P}\mathbf{\Delta}\mathbf{P}^\top$  with  $\mathbf{\Delta} = \text{diag}(\delta_i)$ ,  $\delta_i \geq 0$ . Then  $\mathbf{A} = \mathbf{P}\mathbf{\Delta}^2\mathbf{P}^\top$ . Let  $\mathbf{A}\mathbf{v} = \lambda\mathbf{v}$  and expand  $\mathbf{v} = \sum_i c_i \mathbf{p}_i$ . Comparing  $\mathbf{A}\mathbf{v} = \sum_i c_i \delta_i^2 \mathbf{p}_i$  with  $\lambda\mathbf{v}$  gives  $c_i(\delta_i^2 - \lambda) = 0$ , so  $c_i \neq 0$  only if  $\delta_i = \sqrt{\lambda}$ . Hence  $\mathbf{B}\mathbf{v} = \sum_i c_i \delta_i \mathbf{p}_i = \sqrt{\lambda}\mathbf{v}$ . Applying this to each  $\mathbf{q}_j$  gives  $\mathbf{B}\mathbf{Q} = \mathbf{Q}\mathbf{\Lambda}^{1/2}$ , that is,  $\mathbf{B} = \mathbf{A}^{1/2}$ . The last statement is [Corollary 1.7.3\(b\)](#). ■

## 1.7.4 Extremal properties of eigenvalues

### Theorem 1.7.7 — Rayleigh quotient

Let  $\mathbf{A}$  be symmetric with eigenvalues  $\lambda_1 \geq \dots \geq \lambda_n$  and orthonormal eigenvectors  $\mathbf{q}_1, \dots, \mathbf{q}_n$ . For every  $\mathbf{x} \neq \mathbf{0}$ ,

$$\lambda_n \leq \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} \leq \lambda_1, \quad (1.7.2)$$

with equality on the right at  $\mathbf{x} = \mathbf{q}_1$  and on the left at  $\mathbf{x} = \mathbf{q}_n$ . More generally, the maximum of  $\mathbf{x}^\top \mathbf{A} \mathbf{x} / \mathbf{x}^\top \mathbf{x}$  over nonzero  $\mathbf{x}$  orthogonal to  $\mathbf{q}_1, \dots, \mathbf{q}_{k-1}$  is  $\lambda_k$ , attained at  $\mathbf{q}_k$ .

*Proof.* Put  $\mathbf{c} = \mathbf{Q}^\top \mathbf{x}$ , so that  $\mathbf{x}^\top \mathbf{x} = \mathbf{c}^\top \mathbf{c}$  and

$$\frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}} = \frac{\sum_i \lambda_i c_i^2}{\sum_i c_i^2},$$

a weighted average of the eigenvalues with weights  $c_i^2 / \sum_j c_j^2$ . It lies between the smallest and largest eigenvalue, and equals  $\lambda_1$  when all weight is on  $c_1$ , that is, at  $\mathbf{x} = \mathbf{q}_1$ . The condition  $\mathbf{x} \perp \mathbf{q}_1, \dots, \mathbf{q}_{k-1}$  means  $c_1 = \dots = c_{k-1} = 0$ , so the average is then over

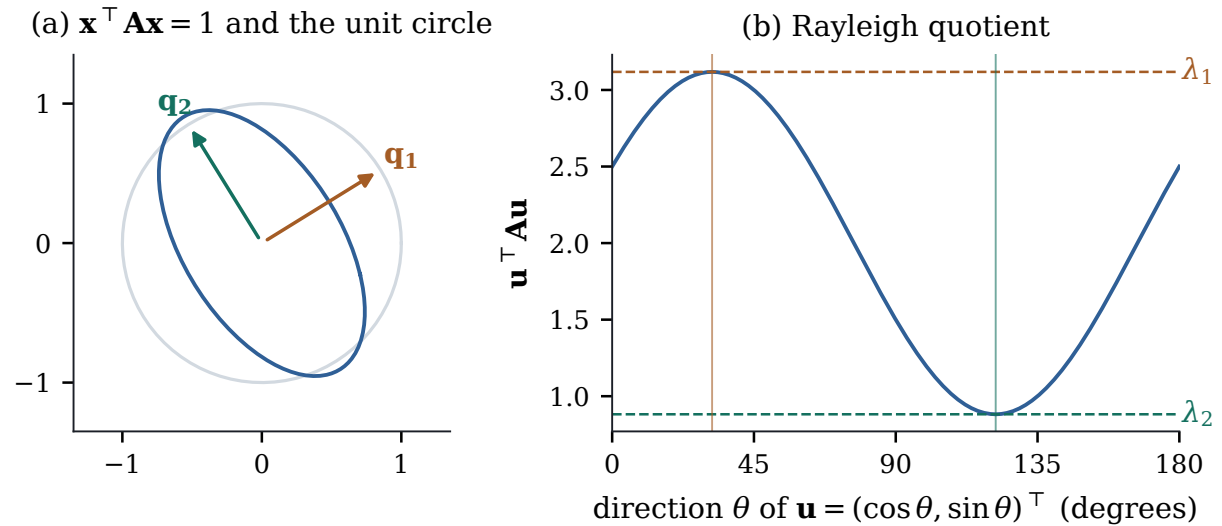


Figure 1.7.1: The Rayleigh quotient of a  $2 \times 2$  symmetric matrix. (a) The level set  $\mathbf{x}^T \mathbf{A} \mathbf{x} = 1$ , an ellipse with axes along the orthonormal eigenvectors  $\mathbf{q}_1, \mathbf{q}_2$ , and the unit circle. (b) The quotient  $\mathbf{u}^T \mathbf{A} \mathbf{u}$  along the unit circle. Its maximum  $\lambda_1$  and minimum  $\lambda_2$  are attained at the eigenvector directions (Theorem 1.7.7).

$\lambda_k, \dots, \lambda_n$ , whose largest member is  $\lambda_k$ .

■

Figure 1.7.1 shows the theorem for  $\mathbf{A} = \begin{pmatrix} 2.5 & 1 \\ 1 & 1.5 \end{pmatrix}$ . The eigenvalues are 3.118 and 0.882, with eigenvectors at angles  $31.7^\circ$  and  $121.7^\circ$ . As  $\mathbf{x}$  turns around the unit circle, the quotient swings between the two eigenvalues. The ellipse  $\mathbf{x}^T \mathbf{A} \mathbf{x} = 1$  has its principal axes along the eigenvectors, with semi-axis lengths  $\lambda_i^{-1/2}$ . Consequently the Rayleigh quotient is largest in the direction in which the ellipse is narrowest.

### Corollary 1.7.8 — Relative to a positive definite matrix

Let  $\mathbf{A}$  be symmetric and  $\mathbf{B}$  positive definite, both  $n \times n$ .

- (a)  $\max_{\mathbf{x} \neq \mathbf{0}} \mathbf{x}^T \mathbf{A} \mathbf{x} / \mathbf{x}^T \mathbf{B} \mathbf{x}$  is the largest eigenvalue of  $\mathbf{B}^{-1} \mathbf{A}$ , which has the same eigenvalues as the symmetric  $\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2}$ .
- (b) There is a nonsingular  $\mathbf{P}$  with  $\mathbf{P}^T \mathbf{B} \mathbf{P} = \mathbf{I}$  and  $\mathbf{P}^T \mathbf{A} \mathbf{P}$  diagonal.
- (c) For  $\mathbf{a} \in \mathbb{R}^n$ ,  $\max_{\mathbf{x} \neq \mathbf{0}} (\mathbf{a}^T \mathbf{x})^2 / \mathbf{x}^T \mathbf{B} \mathbf{x} = \mathbf{a}^T \mathbf{B}^{-1} \mathbf{a}$ , attained at  $\mathbf{x} = \mathbf{B}^{-1} \mathbf{a}$ .

*Proof.* Substitute  $\mathbf{x} = \mathbf{B}^{-1/2} \mathbf{z}$ . The quotient in (a) becomes  $\mathbf{z}^T \mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} \mathbf{z} / \mathbf{z}^T \mathbf{z}$ , to which Theorem 1.7.7 applies, and  $\mathbf{B}^{-1} \mathbf{A} = \mathbf{B}^{-1/2} (\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2}) \mathbf{B}^{1/2}$  is similar to it (Proposition 1.7.1(c)). For (b), if  $\mathbf{B}^{-1/2} \mathbf{A} \mathbf{B}^{-1/2} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^T$ , take  $\mathbf{P} = \mathbf{B}^{-1/2} \mathbf{Q}$ . For (c), the matrix  $\mathbf{A} = \mathbf{a} \mathbf{a}^T$  has  $\mathbf{B}^{-1/2} \mathbf{a} \mathbf{a}^T \mathbf{B}^{-1/2}$  of rank at most one, whose only nonzero eigenvalue is its trace  $\mathbf{a}^T \mathbf{B}^{-1} \mathbf{a}$ . Substituting  $\mathbf{x} = \mathbf{B}^{-1} \mathbf{a}$  attains it.

■

Part (c) is the Cauchy–Schwarz inequality in the inner product  $\mathbf{x}^T \mathbf{B} \mathbf{v}$ . It is the algebra behind simultaneous confidence intervals (Section 12.2), and part (b) is used for linear discriminants

and for comparing two covariance matrices.

## 1.7.5 Exercises

### 1.7.5.1 A. Check your understanding

#### Exercise A1.

Show that  $\begin{pmatrix} a & b \\ b & c \end{pmatrix}$  is positive definite iff  $a > 0$  and  $ac > b^2$ , and nonnegative definite iff  $a \geq 0$ ,  $c \geq 0$  and  $ac \geq b^2$ . Why is  $a \geq 0$  and  $ac \geq b^2$  not enough?

### 1.7.5.2 B. Practice

#### Exercise B1.

Let  $\mathbf{A} \succeq \mathbf{B} \succ \mathbf{0}$ . Show that  $\det \mathbf{A} \geq \det \mathbf{B}$  and  $\operatorname{tr} \mathbf{A} \geq \operatorname{tr} \mathbf{B}$ .

#### Exercise B2.

Let  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$  have full column rank. Show that the leading block of  $(\mathbf{X}^\top \mathbf{X})^{-1}$  satisfies  $[(\mathbf{X}^\top \mathbf{X})^{-1}]_{11} \succeq (\mathbf{X}_1^\top \mathbf{X}_1)^{-1}$ , with equality iff  $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$ . Interpret this for the variances of least squares coefficients when regressors are added to a model.

*Solution.* Write  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  in blocks. By [Theorem 1.6.4](#),  $[\mathbf{A}^{-1}]_{11} = \mathbf{A}_{11.2}^{-1}$ . Now  $\mathbf{A}_{11} - \mathbf{A}_{11.2} = \mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21}$  is nonnegative definite by [Proposition 1.7.5\(c,e\)](#), and  $\mathbf{A}_{11.2}$  is positive definite. So  $\mathbf{A}_{11} \succeq \mathbf{A}_{11.2} \succ \mathbf{0}$ , and [Proposition 1.7.5\(f\)](#) gives  $\mathbf{A}_{11.2}^{-1} \succeq \mathbf{A}_{11}^{-1}$ . Equality forces  $\mathbf{A}_{12} \mathbf{A}_{22}^{-1} \mathbf{A}_{21} = \mathbf{0}$ , hence  $\mathbf{A}_{21} \mathbf{x} = \mathbf{0}$  for all  $\mathbf{x}$  by [Proposition 1.7.5\(b\)](#), that is,  $\mathbf{X}_2^\top \mathbf{X}_1 = \mathbf{0}$ . When  $\operatorname{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , the covariance matrix of the coefficients of  $\mathbf{X}_1$  is  $\sigma^2 [\mathbf{A}^{-1}]_{11}$  in the larger model and  $\sigma^2 \mathbf{A}_{11}^{-1}$  in the smaller one. Adding regressors never decreases the variance of any linear combination of the original coefficients, and leaves it unchanged only when the new regressors are orthogonal to the old ones.

### 1.7.5.3 C. Going deeper

#### Exercise C1.

Let  $\mathbf{A}$  be symmetric with eigenvalues  $\lambda_1 \geq \dots \geq \lambda_n$ . Prove the Courant–Fischer theorem

$$\lambda_k = \max_{\dim S=k} \min_{\mathbf{0} \neq \mathbf{x} \in S} \frac{\mathbf{x}^\top \mathbf{A} \mathbf{x}}{\mathbf{x}^\top \mathbf{x}}.$$

Deduce that if  $\mathbf{B}$  is obtained from  $\mathbf{A}$  by deleting one row and the corresponding column, its eigenvalues  $\mu_1 \geq \dots \geq \mu_{n-1}$  interlace:  $\lambda_k \geq \mu_k \geq \lambda_{k+1}$ .

#### Exercise C2.

Let  $\mathbf{S}$  be positive definite of size  $p \times p$ . Show that  $g(\boldsymbol{\Sigma}) = -\log \det \boldsymbol{\Sigma} - \operatorname{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{S})$ , over positive definite  $\boldsymbol{\Sigma}$ , is maximized uniquely at  $\boldsymbol{\Sigma} = \mathbf{S}$ . *Hint:* write  $g$  in terms of the

eigenvalues of  $\mathbf{S}^{-1/2}\mathbf{\Sigma}\mathbf{S}^{-1/2}$ .

*Solution.* Let  $\mathbf{M} = \mathbf{S}^{-1/2}\mathbf{\Sigma}\mathbf{S}^{-1/2}$ , which is positive definite with eigenvalues  $\mu_1, \dots, \mu_p > 0$ . Then  $\det \mathbf{\Sigma} = \det \mathbf{S} \det \mathbf{M}$  and  $\operatorname{tr}(\mathbf{\Sigma}^{-1}\mathbf{S}) = \operatorname{tr}(\mathbf{S}^{1/2}\mathbf{\Sigma}^{-1}\mathbf{S}^{1/2}) = \operatorname{tr}(\mathbf{M}^{-1})$ . So

$$g(\mathbf{\Sigma}) = -\log \det \mathbf{S} - \sum_{i=1}^p \left( \log \mu_i + \frac{1}{\mu_i} \right).$$

The function  $\mu \mapsto \log \mu + 1/\mu$  has derivative  $(\mu - 1)/\mu^2$  and a unique minimum at  $\mu = 1$ . So  $g$  is maximized iff every  $\mu_i = 1$ , that is,  $\mathbf{M} = \mathbf{I}$  and  $\mathbf{\Sigma} = \mathbf{S}$ . The maximum is  $-\log \det \mathbf{S} - p$ .

## 1.8 The singular value decomposition and conditioning

The spectral theorem needs a symmetric matrix. A model matrix is rectangular, and the decomposition that fits it is the singular value decomposition.

### Theorem 1.8.1 — Singular value decomposition

Let  $\mathbf{A}$  be  $m \times n$  of rank  $r \geq 1$ . There are an  $m \times r$  matrix  $\mathbf{U}$  and an  $n \times r$  matrix  $\mathbf{V}$ , both with orthonormal columns, and numbers  $\sigma_1 \geq \dots \geq \sigma_r > 0$ , such that

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top, \quad \mathbf{D} = \operatorname{diag}(\sigma_1, \dots, \sigma_r). \quad (1.8.1)$$

The **singular values**  $\sigma_i$  are the positive square roots of the nonzero eigenvalues of  $\mathbf{A}^\top \mathbf{A}$  (equivalently of  $\mathbf{A}\mathbf{A}^\top$ ). The columns of  $\mathbf{V}$  and  $\mathbf{U}$  are eigenvectors of  $\mathbf{A}^\top \mathbf{A}$  and  $\mathbf{A}\mathbf{A}^\top$ , with  $\mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{A}^\top)$  and  $\mathcal{C}(\mathbf{U}) = \mathcal{C}(\mathbf{A})$ . Moreover  $\mathbf{A}^+ = \mathbf{V}\mathbf{D}^{-1}\mathbf{U}^\top$  is the Moore–Penrose inverse of  $\mathbf{A}$ .

*Proof.*  $\mathbf{A}^\top \mathbf{A}$  is nonnegative definite of rank  $r$  (Proposition 1.2.3(c)). By the spectral theorem its eigenvalues are  $\sigma_1^2 \geq \dots \geq \sigma_r^2 > 0$  and  $n-r$  zeros, with orthonormal eigenvectors  $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_r]$  for the nonzero ones and  $\mathbf{V}_0$  for the zeros. Define  $\mathbf{U} = \mathbf{A}\mathbf{V}\mathbf{D}^{-1}$ . Then  $\mathbf{U}^\top \mathbf{U} = \mathbf{D}^{-1}\mathbf{V}^\top \mathbf{A}^\top \mathbf{A}\mathbf{V}\mathbf{D}^{-1} = \mathbf{D}^{-1}\mathbf{D}^2\mathbf{D}^{-1} = \mathbf{I}_r$ . For each column  $\mathbf{v}$  of  $\mathbf{V}_0$ ,  $\|\mathbf{A}\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}^\top \mathbf{A}\mathbf{v} = 0$ , so  $\mathbf{A}\mathbf{V}_0 = \mathbf{0}$ . Since  $\mathbf{V}\mathbf{V}^\top + \mathbf{V}_0\mathbf{V}_0^\top = \mathbf{I}$  (Proposition 1.5.3(d)),

$$\mathbf{A} = \mathbf{A}\mathbf{V}\mathbf{V}^\top + \mathbf{A}\mathbf{V}_0\mathbf{V}_0^\top = \mathbf{U}\mathbf{D}\mathbf{V}^\top.$$

Then  $\mathbf{A}\mathbf{A}^\top \mathbf{U} = \mathbf{U}\mathbf{D}\mathbf{V}^\top \mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{U} = \mathbf{U}\mathbf{D}^2$ , so the columns of  $\mathbf{U}$  are eigenvectors of  $\mathbf{A}\mathbf{A}^\top$ .  $\mathcal{C}(\mathbf{U}) \subseteq \mathcal{C}(\mathbf{A})$  by definition of  $\mathbf{U}$ , and both have dimension  $r$ . The statement for  $\mathbf{V}$  is the same argument applied to  $\mathbf{A}^\top = \mathbf{V}\mathbf{D}\mathbf{U}^\top$ . The four Penrose conditions for  $\mathbf{V}\mathbf{D}^{-1}\mathbf{U}^\top$  follow from  $\mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_r$ . For instance,  $\mathbf{A}\mathbf{A}^+ = \mathbf{U}\mathbf{U}^\top$  and  $\mathbf{A}^+ \mathbf{A} = \mathbf{V}\mathbf{V}^\top$  are symmetric. ■

Completing  $\mathbf{U}$  and  $\mathbf{V}$  to orthogonal matrices gives the *full* SVD  $\mathbf{A} = \tilde{\mathbf{U}}\tilde{\mathbf{D}}\tilde{\mathbf{V}}^\top$ , with  $\tilde{\mathbf{D}}$  of size  $m \times n$  carrying  $\sigma_1, \dots, \sigma_r$  on its diagonal and zeros elsewhere. By convention,  $\sigma_{r+1} = \dots = \sigma_{\min(m,n)} = 0$ . Geometrically,  $\mathbf{A}$  maps the unit ball of  $\mathbb{R}^n$  onto a solid ellipsoid in  $\mathcal{C}(\mathbf{A})$  with semi-axes  $\sigma_i \mathbf{u}_i$ . The input direction  $\mathbf{v}_i$  is stretched by the factor  $\sigma_i$  and turned into  $\mathbf{u}_i$ . The Moore–Penrose inverse undoes the stretching on  $\mathcal{C}(\mathbf{A})$  and sends  $\mathcal{C}(\mathbf{A})^\perp$  to zero. The matrix  $\mathbf{U}\mathbf{U}^\top$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{A})$ , whose theory is in [Chapter 6](#).

### Proposition 1.8.2 — Norms and extremal properties

Let  $\mathbf{A}$  be  $m \times n$  with singular values  $\sigma_1 \geq \sigma_2 \geq \dots$ , and let  $\|\mathbf{A}\|_2 = \max_{\mathbf{x} \neq \mathbf{0}} \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\|$  be the **spectral norm**.

- (a)  $\|\mathbf{A}\|_2 = \sigma_1$  and  $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$ .
- (b) If  $\mathbf{A}$  has full column rank  $n$ , then  $\min_{\mathbf{x} \neq \mathbf{0}} \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\| = \sigma_n$ , attained at  $\mathbf{v}_n$ .
- (c) (Eckart–Young.) For  $k < r$ , let  $\mathbf{A}_k = \sum_{i \leq k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ . Then  $\|\mathbf{A} - \mathbf{B}\|_2 \geq \sigma_{k+1} = \|\mathbf{A} - \mathbf{A}_k\|_2$  for every  $\mathbf{B}$  with  $\text{rank}(\mathbf{B}) \leq k$ .

*Proof.*

- (a) and (b):  $\|\mathbf{A}\mathbf{x}\|^2/\|\mathbf{x}\|^2 = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x}/\mathbf{x}^\top \mathbf{x}$ , and the eigenvalues of  $\mathbf{A}^\top \mathbf{A}$  are the  $\sigma_i^2$  (padded with zeros), so [Theorem 1.7.7](#) applies. For the Frobenius norm,  $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^\top \mathbf{A}) = \sum \sigma_i^2$  by [Proposition 1.7.1\(a\)](#).
- (b)  $\mathbf{A} - \mathbf{A}_k = \sum_{i > k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$  has largest singular value  $\sigma_{k+1}$ . If  $\text{rank}(\mathbf{B}) \leq k$ , then  $\dim \mathcal{N}(\mathbf{B}) \geq n - k$ , and since  $(n - k) + (k + 1) > n$ , a basis of  $\mathcal{N}(\mathbf{B})$  together with  $\mathbf{v}_1, \dots, \mathbf{v}_{k+1}$  is dependent. So the two subspaces share a unit vector  $\mathbf{x} = \sum_{i \leq k+1} c_i \mathbf{v}_i$ . Then  $\|(\mathbf{A} - \mathbf{B})\mathbf{x}\|^2 = \|\mathbf{A}\mathbf{x}\|^2 = \sum_{i \leq k+1} \sigma_i^2 c_i^2 \geq \sigma_{k+1}^2$ .

■

The Frobenius version of (c),  $\|\mathbf{A} - \mathbf{B}\|_F^2 \geq \sum_{i > k} \sigma_i^2$ , also holds (Eckart and Young 1936). It is the basis of principal components.

## 1.8.1 The condition number

### Definition 1.8.1 — Condition number

The **condition number** of an  $m \times n$  matrix  $\mathbf{A}$  of full column rank is

$$\kappa(\mathbf{A}) = \frac{\sigma_{\max}(\mathbf{A})}{\sigma_{\min}(\mathbf{A})} = \frac{\sigma_1}{\sigma_n}.$$

For a nonsingular square matrix this equals  $\|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2$ . A matrix without full column rank has  $\kappa(\mathbf{A}) = \infty$  by convention.

By [Proposition 1.8.2](#),  $\kappa(\mathbf{A})$  is the ratio of the largest to the smallest stretching that  $\mathbf{A}$  applies to a unit vector. It is at least 1, is unchanged by scaling  $\mathbf{A}$  or multiplying it on the left by a matrix with orthonormal columns, and equals 1 iff  $\mathbf{A}^\top \mathbf{A} = c\mathbf{I}$ , that is, iff the columns are orthogonal with equal lengths. Two further facts explain its role in regression. First, the singular values of

$\mathbf{A}^\top \mathbf{A}$  are the  $\sigma_i^2$ , so

$$\kappa(\mathbf{A}^\top \mathbf{A}) = \kappa(\mathbf{A})^2. \quad (1.8.2)$$

Second,  $\kappa$  bounds how much relative errors in the data can grow when a linear system is solved. If  $\mathbf{A}$  is square and nonsingular,  $\mathbf{A}\mathbf{x} = \mathbf{b}$  and  $\mathbf{A}(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}$ , then  $\delta\mathbf{x} = \mathbf{A}^{-1}\delta\mathbf{b}$ , and  $\|\mathbf{b}\| \leq \sigma_1 \|\mathbf{x}\|$  gives

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\|\mathbf{A}^{-1}\|_2 \|\delta\mathbf{b}\|}{\|\mathbf{b}\|/\sigma_1} = \kappa(\mathbf{A}) \frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}. \quad (1.8.3)$$

In floating point, relative errors of about  $10^{-16}$  in the data can therefore become errors of about  $\kappa \times 10^{-16}$  in the solution. Solving the normal equations works with  $\mathbf{X}^\top \mathbf{X}$  and so, by [Eq. 1.8.2](#), squares the damage. [Chapter 6](#) and [Chapter 10](#) take this up.

A small  $\sigma_{\min}$  has a direct statistical reading. By [Proposition 1.8.2\(b\)](#),  $\|\mathbf{X}\mathbf{v}_n\| = \sigma_n$  with  $\|\mathbf{v}_n\| = 1$ . The coefficients  $\mathbf{v}_n$  combine the columns of  $\mathbf{X}$  into a vector that is nearly zero, which is a near-dependence among the regressors. This is the basis of the collinearity diagnostics of [Chapter 26](#), such as the condition indices of [Definition 26.3.1](#). The size of  $\kappa$  depends on the units of the columns, so columns are usually rescaled before it is interpreted.

#### Example. (A macroeconomic design)

Take the 203 quarterly US observations of real GDP, real investment, real government spending, real disposable income, population and unemployment, with an intercept, as a model matrix with  $p = 7$  columns. As recorded,  $\kappa(\mathbf{X}) = 5.10 \times 10^5$ . Most of that is units: GDP is measured in billions and unemployment in percent. Rescaling each column to unit length reduces it to 383, and centring the regressors before rescaling reduces it further to 124. By [Eq. 1.8.2](#) the corresponding cross-product matrix has condition number 15386.

What remains is not an artefact of units. [Figure 1.8.1\(b\)](#) shows the right singular vector for the smallest singular value,  $\sigma_{\min} = 0.0175$ , of the centred and scaled design. Its weight is concentrated on GDP (0.78) and disposable income ( $-0.61$ ). The combination of the scaled columns with these coefficients is a vector of length 0.0175, although every column has length one. The two series move almost in lockstep: their correlation is 0.9990. No rescaling can separate their coefficients well. Centring is not a neutral choice, however: it removes near-dependences that involve the intercept, and [Belsley et al. \(1980\)](#) recommend scaling without centring for that reason, so their diagnostics would use the unit-length figure 383. [Section 26.3](#) returns to the choice. The listing computes the singular values, and the script also checks [Eq. 1.8.2](#), the Penrose conditions for  $\mathbf{V}\mathbf{D}^{-1}\mathbf{U}^\top$ , and the Eckart–Young bound.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.macrodta.load_pandas().data
names = ["realgdp", "realinv", "realgovt", "realdpi", "pop", "unemp"]
labels = ["GDP", "investment", "government", "disp. income", "population", "unemployment"]
Z = data[names].to_numpy()
n = len(Z)
```

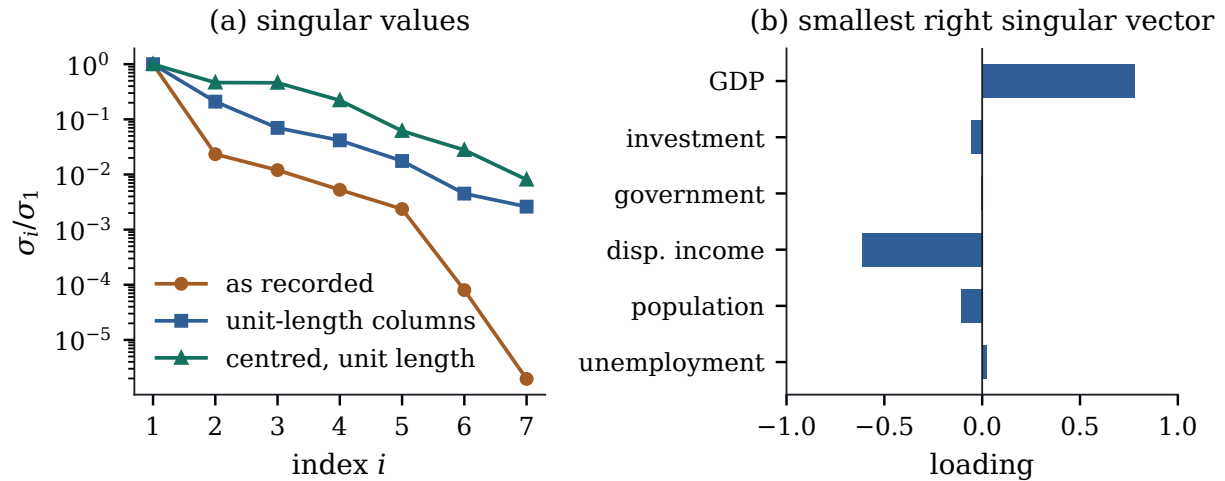


Figure 1.8.1: Singular value decomposition of a macroeconomic model matrix (Example (A macroeconomic design)). (a) Singular values relative to the largest, for the columns as recorded, rescaled to unit length, and centred then rescaled. The ratio of the first to the last is the condition number. (b) The right singular vector for the smallest singular value of the centred design. Its two large loadings reveal a near-dependence between GDP and disposable income.

```
X = np.column_stack([np.ones(n), Z]) # as recorded
X_unit = X / np.linalg.norm(X, axis=0) # unit-length columns
Xc = np.column_stack([np.ones(n), Z - Z.mean(axis=0)]) # regressors centred
X_std = Xc / np.linalg.norm(Xc, axis=0) # ... then length 1

for name, M in [("raw", X), ("unit length", X_unit), ("centred", X_std)]:
    U, s, Vt = np.linalg.svd(M, full_matrices=False) # M = U diag(s) V'
    print(f"{name:12s} kappa = {s[0] / s[-1]:10.1f} singular values / s_1:",
          np.array2string(s / s[0], precision=4))
```

## 1.8.2 Exercises

### 1.8.2.1 B. Practice

#### Exercise B1.

Let  $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2]$  with  $\|\mathbf{x}_1\| = \|\mathbf{x}_2\| = 1$  and  $\mathbf{x}_1^\top \mathbf{x}_2 = r$ ,  $|r| < 1$ . Show that  $\kappa(\mathbf{X}) = \sqrt{(1+|r|)/(1-|r|)}$ . How large must  $|r|$  be for  $\kappa(\mathbf{X}) > 30$ ?

*Solution.*  $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$  has eigenvalues  $1 \pm r$ , with eigenvectors  $(1, \pm 1)^\top$ . The singular values of  $\mathbf{X}$  are their square roots, so  $\kappa(\mathbf{X})^2 = (1+|r|)/(1-|r|)$ . For  $\kappa > 30$  we need  $(1+|r|)/(1-|r|) > 900$ , that is,  $|r| > 899/901 \approx 0.998$ . For two columns, even  $\kappa(\mathbf{X}) = 100$  requires  $|r| > 9999/10001 \approx 0.9998$ .

#### Exercise B2.

Show that  $\kappa(\mathbf{X}) \geq \max_j \|\mathbf{x}_j\| / \min_k \|\mathbf{x}_k\|$ , so that columns on very different scales force a large condition number. Show by example that rescaling the columns can also increase

$\kappa$ .

## 1.9 Idempotent matrices

A square matrix  $\mathbf{A}$  is **idempotent** if  $\mathbf{A}^2 = \mathbf{A}$ . Applying it twice has the same effect as applying it once, which is how a projection behaves. Examples are  $\mathbf{I}$ ,  $\mathbf{0}$ , the centring matrix of [Example \(Sums of squares as quadratic forms\)](#), the products  $\mathbf{AG}$  and  $\mathbf{GA}$  for a generalized inverse  $\mathbf{G}$  ([Proposition 1.3.3\(a\)](#)), and the non-symmetric  $\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ . In linear models, sums of squares are quadratic forms in *symmetric* idempotent matrices, and their degrees of freedom are ranks.

### Proposition 1.9.1 — General idempotent matrices

Let  $\mathbf{A}$  be  $n \times n$  and idempotent of rank  $r$ .

- (a) Every eigenvalue of  $\mathbf{A}$  is 0 or 1.
- (b)  $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$ .
- (c)  $\mathbf{I} - \mathbf{A}$  is idempotent,  $\mathbf{A}(\mathbf{I} - \mathbf{A}) = (\mathbf{I} - \mathbf{A})\mathbf{A} = \mathbf{0}$ ,  $\mathcal{C}(\mathbf{I} - \mathbf{A}) = \mathcal{N}(\mathbf{A})$ , and  $\text{rank}(\mathbf{A}) + \text{rank}(\mathbf{I} - \mathbf{A}) = n$ .
- (d) An idempotent matrix that is nonsingular equals  $\mathbf{I}$ .
- (e)  $\mathbf{P}^{-1}\mathbf{A}\mathbf{P}$  is idempotent for nonsingular  $\mathbf{P}$ , and  $\mathbf{A}^\top$  is idempotent.

*Proof.*

- (a) If  $\mathbf{A}\mathbf{x} = \lambda\mathbf{x}$  with  $\mathbf{x} \neq \mathbf{0}$ , then  $\lambda\mathbf{x} = \mathbf{A}\mathbf{x} = \mathbf{A}^2\mathbf{x} = \lambda^2\mathbf{x}$ , so  $\lambda^2 = \lambda$ .
- (b) If  $r = 0$  both sides vanish. Otherwise take a rank factorization  $\mathbf{A} = \mathbf{BF}$  ([Theorem 1.2.1](#)). Then  $\mathbf{BFBF} = \mathbf{BF}$ . Since  $\mathbf{B}$  has full column rank and  $\mathbf{F}$  has full row rank,  $\mathbf{B}$  has a left inverse and  $\mathbf{F}$  a right inverse ([Section 1.3](#)), and cancelling them gives  $\mathbf{FB} = \mathbf{I}_r$ . So  $\text{tr}(\mathbf{A}) = \text{tr}(\mathbf{BF}) = \text{tr}(\mathbf{FB}) = r$  by [Theorem 1.6.1](#).
- (c) The first two claims are one-line expansions.  $\mathcal{C}(\mathbf{I} - \mathbf{A}) \subseteq \mathcal{N}(\mathbf{A})$  because  $\mathbf{A}(\mathbf{I} - \mathbf{A}) = \mathbf{0}$ , and  $\mathbf{v} \in \mathcal{N}(\mathbf{A})$  gives  $\mathbf{v} = (\mathbf{I} - \mathbf{A})\mathbf{v}$ . The rank statement is then rank-nullity.
- (d) Multiply  $\mathbf{A}^2 = \mathbf{A}$  by  $\mathbf{A}^{-1}$ .
- (e)  $(\mathbf{P}^{-1}\mathbf{A}\mathbf{P})^2 = \mathbf{P}^{-1}\mathbf{A}^2\mathbf{P}$  and  $(\mathbf{A}^\top)^2 = (\mathbf{A}^2)^\top$ .

■

By (c), every  $\mathbf{x}$  splits as  $\mathbf{A}\mathbf{x} + (\mathbf{I} - \mathbf{A})\mathbf{x}$  with the first part in  $\mathcal{C}(\mathbf{A})$  and the second in  $\mathcal{N}(\mathbf{A})$ . So  $\mathbb{R}^n = \mathcal{C}(\mathbf{A}) \oplus \mathcal{N}(\mathbf{A})$ , and  $\mathbf{A}$  projects onto  $\mathcal{C}(\mathbf{A})$  along  $\mathcal{N}(\mathbf{A})$ . The projection is orthogonal exactly when the two subspaces are orthogonal, which happens iff  $\mathbf{A}$  is symmetric ([Exercise \(B1\)](#)).

### Theorem 1.9.2 — Symmetric idempotent matrices

Let  $\mathbf{A}$  be an  $n \times n$  symmetric matrix of rank  $r$ . Then  $\mathbf{A}$  is idempotent iff all its eigenvalues are 0 or 1. In that case



- (a)  $\mathbf{A} = \mathbf{Q}_1 \mathbf{Q}_1^\top$ , where the  $n \times r$  matrix  $\mathbf{Q}_1$  has orthonormal columns spanning  $\mathcal{C}(\mathbf{A})$ ;
- (b)  $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = r$ , the multiplicity of the eigenvalue 1;
- (c)  $\mathbf{A}$  and  $\mathbf{I} - \mathbf{A}$  are nonnegative definite, and  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{A} \mathbf{x}\|^2 \leq \|\mathbf{x}\|^2$ .

*Proof.* If  $\mathbf{A}$  is idempotent its eigenvalues are 0 or 1 by [Proposition 1.9.1\(a\)](#). Conversely, if  $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top$  with each  $\lambda_i \in \{0, 1\}$ , then  $\mathbf{A}^2 = \mathbf{Q} \mathbf{\Lambda}^2 \mathbf{Q}^\top = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top = \mathbf{A}$ . In that case, order the eigenvalues so that the  $r$  ones come first and let  $\mathbf{Q}_1$  hold the corresponding eigenvectors. Then  $\mathbf{A} = \mathbf{Q} \mathbf{\Lambda} \mathbf{Q}^\top = \mathbf{Q}_1 \mathbf{Q}_1^\top$ , and  $\mathcal{C}(\mathbf{A}) = \mathcal{C}(\mathbf{Q}_1)$  by [Corollary 1.7.3\(a\)](#). This is (a). The trace is the sum of the eigenvalues, which is (b). For (c), the eigenvalues of  $\mathbf{A}$  and of  $\mathbf{I} - \mathbf{A}$  are nonnegative, and  $\mathbf{x}^\top \mathbf{A} \mathbf{x} = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x}$ . Also  $\|\mathbf{x}\|^2 - \|\mathbf{A} \mathbf{x}\|^2 = \mathbf{x}^\top (\mathbf{I} - \mathbf{A}) \mathbf{x} \geq 0$ . ■

[Chapter 6](#) shows that symmetric idempotent matrices are exactly the orthogonal projections, and  $\mathbf{Q}_1 \mathbf{Q}_1^\top$  in (a) is the projection onto  $\mathcal{C}(\mathbf{A})$ . In [Chapter 4](#), part (b) becomes the degrees of freedom of a chi-squared distribution.

### Theorem 1.9.3 — Sums of symmetric idempotent matrices

Let  $\mathbf{A}_1, \dots, \mathbf{A}_k$  be symmetric  $n \times n$  matrices and  $\mathbf{A} = \sum_i \mathbf{A}_i$ . Consider

- (i)  $\mathbf{A}$  is idempotent;
- (ii) every  $\mathbf{A}_i$  is idempotent;
- (iii)  $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$  for all  $i \neq j$ .

Any two of these imply the third. Moreover, if  $\mathbf{A} = \mathbf{I}_n$  and  $\sum_i \text{rank}(\mathbf{A}_i) = n$ , then (ii) and (iii) hold.

*Proof.*

(ii) and (iii) imply (i):  $\mathbf{A}^2 = \sum_i \mathbf{A}_i^2 + \sum_{i \neq j} \mathbf{A}_i \mathbf{A}_j = \sum_i \mathbf{A}_i = \mathbf{A}$ .

(iii) and (ii) imply (iii): By [Theorem 1.9.2\(c\)](#),  $\|\mathbf{x}\|^2 \geq \mathbf{x}^\top \mathbf{A} \mathbf{x}$  for all  $\mathbf{x}$ . Fix  $j$  and take  $\mathbf{x} = \mathbf{A}_j \mathbf{y}$ , so that  $\mathbf{A}_j \mathbf{x} = \mathbf{x}$  and  $\mathbf{x}^\top \mathbf{A}_j \mathbf{x} = \|\mathbf{x}\|^2$ . Then

$$\|\mathbf{x}\|^2 \geq \mathbf{x}^\top \mathbf{A} \mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \mathbf{x}^\top \mathbf{A}_i \mathbf{x} = \|\mathbf{x}\|^2 + \sum_{i \neq j} \|\mathbf{A}_i \mathbf{x}\|^2.$$

So  $\mathbf{A}_i \mathbf{A}_j \mathbf{y} = \mathbf{A}_i \mathbf{x} = \mathbf{0}$  for every  $i \neq j$  and every  $\mathbf{y}$ .

(iv) and (iii) imply (ii): By (iii),  $\mathbf{A} \mathbf{A}_i = \mathbf{A}_i^2$ . Using  $\mathbf{A}^2 = \mathbf{A}$ ,  $\mathbf{A}_i^2 = \mathbf{A} \mathbf{A}_i = \mathbf{A}^2 \mathbf{A}_i = \mathbf{A} \mathbf{A}_i^2 = \mathbf{A}_i^3$ . The eigenvalues of the symmetric  $\mathbf{A}_i$  therefore satisfy  $\lambda^2 = \lambda^3$ , so they lie in  $\{0, 1\}$ , and  $\mathbf{A}_i$  is idempotent by [Theorem 1.9.2](#).

For the last statement, let  $r_i = \text{rank}(\mathbf{A}_i)$ . For each  $i$ ,  $\mathbf{I} - \mathbf{A}_i = \sum_{j \neq i} \mathbf{A}_j$  has rank at most  $\sum_{j \neq i} r_j = n - r_i$  ([Proposition 1.2.3\(d\)](#)). So the eigenvalue 1 of  $\mathbf{A}_i$ , whose eigenspace is  $\mathcal{N}(\mathbf{I} - \mathbf{A}_i)$ , has multiplicity at least  $r_i$ . But  $\mathbf{A}_i$  has exactly  $r_i$  nonzero eigenvalues, so all of them equal 1 and  $\mathbf{A}_i$  is idempotent. Now (i) holds trivially, and (iii) follows from the

second part of the proof. ■

The last statement is the algebraic core of Cochran's theorem (Chapter 4): if  $\mathbf{y}^\top \mathbf{y} = \sum_i \mathbf{y}^\top \mathbf{A}_i \mathbf{y}$  and the ranks add up to  $n$ , the quadratic forms are built from mutually orthogonal projections.

## 1.9.1 Exercises

### 1.9.1.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{A}, \mathbf{B}$  be idempotent of the same size. Show that  $\mathbf{AB}$  is idempotent if  $\mathbf{AB} = \mathbf{BA}$ , and that  $\mathbf{A} + \mathbf{B}$  is idempotent iff  $\mathbf{AB} + \mathbf{BA} = \mathbf{0}$ . Show that the latter forces  $\mathbf{AB} = \mathbf{0}$  (symmetry is not needed).

### 1.9.1.2 B. Practice

#### Exercise B1.

Let  $\mathbf{A}$  be idempotent. Show that  $\mathbf{A}$  is symmetric iff  $\mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$ .

*Solution.* Suppose  $\mathbf{A}$  is symmetric. For  $\mathbf{u} = \mathbf{Ax} \in \mathcal{C}(\mathbf{A})$  and  $\mathbf{v} \in \mathcal{N}(\mathbf{A})$ ,  $\mathbf{u}^\top \mathbf{v} = \mathbf{x}^\top \mathbf{Av} = 0$ . Conversely, suppose  $\mathcal{C}(\mathbf{A}) \perp \mathcal{N}(\mathbf{A})$ . For all  $\mathbf{x}, \mathbf{y}$ ,  $(\mathbf{I} - \mathbf{A})\mathbf{y} \in \mathcal{N}(\mathbf{A})$ , so  $\mathbf{x}^\top \mathbf{A}^\top (\mathbf{I} - \mathbf{A})\mathbf{y} = 0$ . Hence  $\mathbf{A}^\top (\mathbf{I} - \mathbf{A}) = \mathbf{0}$ , that is,  $\mathbf{A}^\top = \mathbf{A}^\top \mathbf{A}$ . The right side is symmetric, so  $\mathbf{A}^\top$  is symmetric, and so is  $\mathbf{A}$ .

## 1.10 Kronecker products and the vec operator

Balanced designs have model matrices with a product structure, and multivariate observations have covariance matrices with one. The Kronecker product expresses both.

### Definition 1.10.1

The **Kronecker product** of an  $m \times n$  matrix  $\mathbf{A}$  and a  $p \times q$  matrix  $\mathbf{B}$  is the  $mp \times nq$  block matrix  $\mathbf{A} \otimes \mathbf{B} = (a_{ij}\mathbf{B})$ . The **vec operator** places the  $n$  columns of an  $m \times n$  matrix one below another, giving a vector of length  $mn$ :  $\text{vec}(\mathbf{A}) = (\mathbf{a}_1^\top, \dots, \mathbf{a}_n^\top)^\top$ .

### Proposition 1.10.1

For matrices of compatible sizes:

- (a)  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{C} \otimes \mathbf{D}) = \mathbf{AC} \otimes \mathbf{BD}$  (mixed-product rule);
- (b)  $(\mathbf{A} \otimes \mathbf{B})^\top = \mathbf{A}^\top \otimes \mathbf{B}^\top$ , and  $(\mathbf{A} \otimes \mathbf{B})^{-1} = \mathbf{A}^{-1} \otimes \mathbf{B}^{-1}$  for nonsingular  $\mathbf{A}, \mathbf{B}$ . More generally  $\mathbf{A}^- \otimes \mathbf{B}^-$  is a generalized inverse of  $\mathbf{A} \otimes \mathbf{B}$ ;
- (c)  $\text{tr}(\mathbf{A} \otimes \mathbf{B}) = \text{tr}(\mathbf{A}) \text{tr}(\mathbf{B})$  and  $\text{rank}(\mathbf{A} \otimes \mathbf{B}) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$ ;
- (d) if  $\mathbf{A}$  ( $m \times m$ ) and  $\mathbf{B}$  ( $p \times p$ ) are symmetric with eigenvalues  $\lambda_i$  and  $\mu_j$ , the eigenvalues of  $\mathbf{A} \otimes \mathbf{B}$  are the products  $\lambda_i \mu_j$ , and  $\det(\mathbf{A} \otimes \mathbf{B}) = (\det \mathbf{A})^p (\det \mathbf{B})^m$ ;
- (e) Kronecker products of orthogonal, symmetric, idempotent, or nonnegative definite matrices are of the same kind;
- (f)  $\text{vec}(\mathbf{ABC}) = (\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$  and  $\text{tr}(\mathbf{A}^\top \mathbf{B}) = \text{vec}(\mathbf{A})^\top \text{vec}(\mathbf{B})$ .

*Proof.*

- (a) The  $(i, j)$  block of the left side is  $\sum_k a_{ik} c_{kj} \mathbf{B} \mathbf{D}$ , which is the  $(i, j)$  block of the right side. (b) Transposition is direct, and the rest follows from (a), for example  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-)(\mathbf{A} \otimes \mathbf{B}) = \mathbf{AA}^- \otimes \mathbf{BB}^-$ .
- (b) With  $\mathbf{A} = \mathbf{PAP}^\top$  and  $\mathbf{B} = \mathbf{QMQ}^\top$ , (a) and (b) give  $\mathbf{A} \otimes \mathbf{B} = (\mathbf{P} \otimes \mathbf{Q})(\mathbf{A} \otimes \mathbf{M})(\mathbf{P} \otimes \mathbf{Q})^\top$ , a spectral decomposition because  $\mathbf{P} \otimes \mathbf{Q}$  is orthogonal and  $\mathbf{A} \otimes \mathbf{M}$  is diagonal with entries  $\lambda_i \mu_j$ . The determinant is the product of these. (c) The trace statement is direct. For the rank, by (a)  $(\mathbf{A} \otimes \mathbf{B})(\mathbf{A}^- \otimes \mathbf{B}^-) = \mathbf{AA}^- \otimes \mathbf{BB}^-$  is idempotent with the same rank as  $\mathbf{A} \otimes \mathbf{B}$  (Proposition 1.3.3(a)). Its trace is  $\text{tr}(\mathbf{AA}^-) \text{tr}(\mathbf{BB}^-) = \text{rank}(\mathbf{A}) \text{rank}(\mathbf{B})$  by Proposition 1.9.1(b).
- (c) follows from (a), (b) and (d). (f) The  $j$ th column of  $\mathbf{ABC}$  is  $\mathbf{ABc}_j = \sum_k c_{kj} \mathbf{Ab}_k$ , which is the  $j$ th block of  $(\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$ . The trace identity is  $\sum_{ij} a_{ij} b_{ij}$  written two ways.

■

The determinant formula in (d) holds for all square  $\mathbf{A}$  and  $\mathbf{B}$ , not only symmetric ones. We need only the symmetric case.

### Example. (The balanced two-way layout)

Let observations  $y_{ij}$ ,  $i = 1, \dots, a$ ,  $j = 1, \dots, b$ , be stacked with  $j$  running inside  $i$ . The additive model  $E(y_{ij}) = \mu + \alpha_i + \beta_j$  has model matrix

$$\mathbf{X} = [\mathbf{1}_a \otimes \mathbf{1}_b, \mathbf{I}_a \otimes \mathbf{1}_b, \mathbf{1}_a \otimes \mathbf{I}_b],$$

because  $\mathbf{I}_a \otimes \mathbf{1}_b$  has a 1 in column  $i$  exactly for the  $b$  observations in row  $i$ . With  $\bar{\mathbf{J}}_k = k^{-1} \mathbf{J}_k$ , a symmetric idempotent averaging matrix, the mixed-product rule shows that  $\mathbf{I}_a \otimes \bar{\mathbf{J}}_b$ ,  $\bar{\mathbf{J}}_a \otimes \mathbf{I}_b$  and  $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$  replace each observation by its row mean, its column mean and the grand mean. The matrix  $\mathbf{M} = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b + \bar{\mathbf{J}}_a \otimes \mathbf{I}_b - \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$  is symmetric and idempotent, with trace  $a + b - 1 = \text{rank}(\mathbf{X})$ . It is the projection onto  $\mathcal{C}(\mathbf{X})$  (Theorem 16.1.2 in Chapter 16). The script checks this for  $a = 3$  and  $b = 4$ , together with each part of Proposition 1.10.1.

```
import numpy as np
rng = np.random.default_rng(11)
def vec(M):
    return M.reshape(-1, order="F")
```

# stack the columns

```

A, B, C = rng.normal(size=(3, 4)), rng.normal(size=(4, 2)), rng.normal(size=(2, 5))
assert np.allclose(vec(A @ B @ C), np.kron(C.T, A) @ vec(B))          # vec(ABC)

# balanced two-way layout: a = 3 rows, b = 4 columns, one observation per cell
a, b = 3, 4
one = lambda k: np.ones((k, 1))
X = np.hstack([np.kron(one(a), one(b)),          # intercept
               np.kron(np.eye(a), one(b)),      # row indicators
               np.kron(one(a), np.eye(b))])      # column indicators
print("rank(X) =", np.linalg.matrix_rank(X), "= a + b - 1 =", a + b - 1)

```

## 1.11 Vector and matrix calculus

### 1.11.1 Gradients of linear and quadratic functions

For a differentiable scalar function  $f$  of  $\mathbf{x} \in \mathbb{R}^n$ , the **gradient**  $\partial f / \partial \mathbf{x}$  is the column vector with entries  $\partial f / \partial x_i$ , and the **Hessian**  $\partial^2 f / \partial \mathbf{x} \partial \mathbf{x}^\top$  is the  $n \times n$  matrix of second derivatives. For a vector-valued  $\mathbf{g}(\mathbf{x}) \in \mathbb{R}^m$ , the  $m \times n$  *Jacobian*  $\partial \mathbf{g} / \partial \mathbf{x}^\top$  has  $(i, j)$  entry  $\partial g_i / \partial x_j$ .

#### Theorem 1.11.1 — Derivatives of linear and quadratic forms

Let  $\mathbf{a} \in \mathbb{R}^n$ , let  $\mathbf{A}$  be  $n \times n$  and  $\mathbf{B}$  be  $m \times n$ . Then

$$\frac{\partial(\mathbf{a}^\top \mathbf{x})}{\partial \mathbf{x}} = \mathbf{a}, \quad \frac{\partial(\mathbf{B}\mathbf{x})}{\partial \mathbf{x}^\top} = \mathbf{B}, \quad \frac{\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}, \quad \frac{\partial^2(\mathbf{x}^\top \mathbf{A}\mathbf{x})}{\partial \mathbf{x} \partial \mathbf{x}^\top} = \mathbf{A} + \mathbf{A}^\top.$$

In particular  $\partial(\mathbf{x}^\top \mathbf{A}\mathbf{x}) / \partial \mathbf{x} = 2\mathbf{A}\mathbf{x}$  when  $\mathbf{A}$  is symmetric.

*Proof.* The first two are immediate from  $\mathbf{a}^\top \mathbf{x} = \sum_i a_i x_i$  and  $(\mathbf{B}\mathbf{x})_i = \sum_j b_{ij} x_j$ . For the quadratic form, expand

$$(\mathbf{x} + \mathbf{h})^\top \mathbf{A}(\mathbf{x} + \mathbf{h}) = \mathbf{x}^\top \mathbf{A}\mathbf{x} + \mathbf{h}^\top (\mathbf{A} + \mathbf{A}^\top)\mathbf{x} + \mathbf{h}^\top \mathbf{A}\mathbf{h}.$$

The middle term is linear in  $\mathbf{h}$  and the last is  $O(\|\mathbf{h}\|^2)$ , so  $(\mathbf{A} + \mathbf{A}^\top)\mathbf{x}$  is the gradient. The gradient is linear in  $\mathbf{x}$  with matrix  $\mathbf{A} + \mathbf{A}^\top$ , which is the Hessian by the second formula. ■

#### Example. (The least squares criterion)

Let  $S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}^\top \mathbf{y} - 2\mathbf{y}^\top \mathbf{X}\mathbf{b} + \mathbf{b}^\top \mathbf{X}^\top \mathbf{X}\mathbf{b}$ . By Theorem 1.11.1,

$$\frac{\partial S}{\partial \mathbf{b}} = -2\mathbf{X}^\top \mathbf{y} + 2\mathbf{X}^\top \mathbf{X}\mathbf{b} = -2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}), \quad \frac{\partial^2 S}{\partial \mathbf{b} \partial \mathbf{b}^\top} = 2\mathbf{X}^\top \mathbf{X}.$$

Setting the gradient to zero gives the normal equations  $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ . The Hessian is nonnegative definite, so  $S$  is convex and every solution is a global minimizer. The next

result proves this without calculus, and covers the rank-deficient case.

### Proposition 1.11.2 — Minimizing a quadratic function

Let  $\mathbf{A}$  be  $n \times n$  nonnegative definite,  $\mathbf{b} \in \mathbb{R}^n$  and  $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x} - 2\mathbf{b}^\top \mathbf{x}$ . If  $\mathbf{b} \in \mathcal{C}(\mathbf{A})$ , the minimizers of  $f$  are exactly the solutions of  $\mathbf{A}\mathbf{x} = \mathbf{b}$ , and the minimum value is  $-\mathbf{b}^\top \mathbf{A}^- \mathbf{b}$  for any generalized inverse  $\mathbf{A}^-$ . If  $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$ , then  $f$  is unbounded below.

*Proof.* Let  $\mathbf{b} \in \mathcal{C}(\mathbf{A})$  and let  $\mathbf{x}_0$  solve  $\mathbf{A}\mathbf{x}_0 = \mathbf{b}$ . Completing the square,

$$f(\mathbf{x}) = (\mathbf{x} - \mathbf{x}_0)^\top \mathbf{A}(\mathbf{x} - \mathbf{x}_0) - \mathbf{x}_0^\top \mathbf{A} \mathbf{x}_0.$$

The first term is nonnegative, and by Proposition 1.7.5(b) it is zero iff  $\mathbf{A}(\mathbf{x} - \mathbf{x}_0) = \mathbf{0}$ , that is, iff  $\mathbf{A}\mathbf{x} = \mathbf{b}$ . The minimum is  $-\mathbf{x}_0^\top \mathbf{A} \mathbf{x}_0 = -\mathbf{x}_0^\top \mathbf{A} \mathbf{A}^- \mathbf{A} \mathbf{x}_0 = -\mathbf{b}^\top \mathbf{A}^- \mathbf{b}$ . If  $\mathbf{b} \notin \mathcal{C}(\mathbf{A})$ , write  $\mathbf{b} = \mathbf{b}_1 + \mathbf{b}_2$  with  $\mathbf{b}_1 \in \mathcal{C}(\mathbf{A})$  and  $\mathbf{0} \neq \mathbf{b}_2 \in \mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A})$  (Proposition 1.5.1). Then  $f(t\mathbf{b}_2) = -2t\|\mathbf{b}_2\|^2 \rightarrow -\infty$  as  $t \rightarrow \infty$ . ■

With  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  and  $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ , the proposition gives least squares for every model matrix: the minimizers are the solutions of the normal equations, and the minimum of  $S$  is  $\mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top \mathbf{y}$ .

## 1.11.2 Constrained minimization

To minimize a differentiable  $f(\mathbf{x})$  subject to linear constraints  $\mathbf{H}\mathbf{x} = \mathbf{h}$ , with  $\mathbf{H}$  of size  $q \times n$ , the method of **Lagrange multipliers** looks for stationary points of  $f(\mathbf{x}) + 2\boldsymbol{\theta}^\top (\mathbf{H}\mathbf{x} - \mathbf{h})$  in  $(\mathbf{x}, \boldsymbol{\theta})$ . The factor 2 is only for convenience. For quadratic  $f$  the stationary point can be written down and checked directly.

### Proposition 1.11.3

Let  $\mathbf{A}$  be positive definite,  $\mathbf{H}$  of size  $q \times n$  with full row rank, and  $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A} \mathbf{x} - 2\mathbf{b}^\top \mathbf{x}$ . The unique minimizer of  $f$  subject to  $\mathbf{H}\mathbf{x} = \mathbf{h}$  is

$$\mathbf{x}_* = \mathbf{x}_0 - \mathbf{A}^{-1} \mathbf{H}^\top (\mathbf{H} \mathbf{A}^{-1} \mathbf{H}^\top)^{-1} (\mathbf{H} \mathbf{x}_0 - \mathbf{h}), \quad \mathbf{x}_0 = \mathbf{A}^{-1} \mathbf{b}.$$

*Proof.*  $\mathbf{H} \mathbf{A}^{-1} \mathbf{H}^\top$  is positive definite by Proposition 1.7.5(c,e), so  $\mathbf{x}_*$  is well defined, and  $\mathbf{H}\mathbf{x}_* = \mathbf{h}$  by direct substitution. Stationarity of the Lagrangian gives  $2\mathbf{A}\mathbf{x} - 2\mathbf{b} + 2\mathbf{H}^\top \boldsymbol{\theta} = \mathbf{0}$ . Indeed  $\mathbf{A}\mathbf{x}_* - \mathbf{b} = -\mathbf{H}^\top \boldsymbol{\theta}_*$  with  $\boldsymbol{\theta}_* = (\mathbf{H} \mathbf{A}^{-1} \mathbf{H}^\top)^{-1} (\mathbf{H} \mathbf{x}_0 - \mathbf{h})$ . For any feasible  $\mathbf{x}$ , put  $\mathbf{d} = \mathbf{x} - \mathbf{x}_*$ , so that  $\mathbf{H}\mathbf{d} = \mathbf{0}$ . Then

$$f(\mathbf{x}) = f(\mathbf{x}_*) + 2\mathbf{d}^\top (\mathbf{A}\mathbf{x}_* - \mathbf{b}) + \mathbf{d}^\top \mathbf{A} \mathbf{d} = f(\mathbf{x}_*) - 2(\mathbf{H}\mathbf{d})^\top \boldsymbol{\theta}_* + \mathbf{d}^\top \mathbf{A} \mathbf{d} = f(\mathbf{x}_*) + \mathbf{d}^\top \mathbf{A} \mathbf{d},$$

which exceeds  $f(\mathbf{x}_*)$  unless  $\mathbf{d} = \mathbf{0}$ . ■

With  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  and  $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ , this is least squares subject to a linear hypothesis  $\mathbf{H}\boldsymbol{\beta} = \mathbf{h}$ , the starting point of [Chapter 11](#).

### 1.11.3 Derivatives with respect to matrices and scalars

Likelihoods for normal models involve  $\log \det \boldsymbol{\Sigma}$  and  $\text{tr}(\boldsymbol{\Sigma}^{-1} \mathbf{S})$ . For a scalar function  $f$  of an  $m \times n$  matrix  $\mathbf{W}$  with functionally independent entries,  $\partial f / \partial \mathbf{W}$  is the  $m \times n$  matrix with entries  $\partial f / \partial w_{ij}$ .

#### Proposition 1.11.4

- (a)  $\partial \text{tr}(\mathbf{A}\mathbf{W}) / \partial \mathbf{W} = \mathbf{A}^\top$ .
- (b) If  $\det \mathbf{W} > 0$ , then  $\partial \log \det \mathbf{W} / \partial \mathbf{W} = (\mathbf{W}^{-1})^\top$ .
- (c) If  $\mathbf{A}(t)$  is a nonsingular matrix whose entries are differentiable functions of a scalar  $t$ , with derivative  $\dot{\mathbf{A}} = \partial \mathbf{A} / \partial t$ , then

$$\frac{\partial \mathbf{A}^{-1}}{\partial t} = -\mathbf{A}^{-1} \dot{\mathbf{A}} \mathbf{A}^{-1}, \quad \frac{\partial \log |\det \mathbf{A}|}{\partial t} = \text{tr}(\mathbf{A}^{-1} \dot{\mathbf{A}}).$$

*Proof.*

- (a)  $\text{tr}(\mathbf{A}\mathbf{W}) = \sum_{i,j} a_{ji} w_{ij}$ . (c) Differentiate  $\mathbf{A}\mathbf{A}^{-1} = \mathbf{I}$  by the product rule to get  $\dot{\mathbf{A}}\mathbf{A}^{-1} + \mathbf{A}\partial \mathbf{A}^{-1} / \partial t = \mathbf{0}$ . For the determinant,  $\mathbf{A}(t+h) = \mathbf{A}(t)(\mathbf{I} + h\mathbf{A}^{-1}\dot{\mathbf{A}} + o(h))$ . In the permutation expansion of  $\det(\mathbf{I} + h\mathbf{C})$  only the diagonal product contributes terms of first order in  $h$ , so  $\det(\mathbf{I} + h\mathbf{C}) = 1 + h \text{tr} \mathbf{C} + O(h^2)$ . Hence  $\det \mathbf{A}(t+h) = \det \mathbf{A}(t)(1 + h \text{tr}(\mathbf{A}^{-1}\dot{\mathbf{A}}) + o(h))$ , and taking logarithms gives the formula. (b) Apply (c) with  $\mathbf{A}(t) = \mathbf{W} + t\mathbf{e}_i\mathbf{e}_j^\top$ . Then  $\text{tr}(\mathbf{W}^{-1}\mathbf{e}_i\mathbf{e}_j^\top) = \mathbf{e}_j^\top \mathbf{W}^{-1} \mathbf{e}_i$ , the  $(j, i)$  entry of  $\mathbf{W}^{-1}$ .

■

When  $\mathbf{W}$  is constrained to be symmetric,  $w_{ij}$  and  $w_{ji}$  are the same variable, so the derivatives with respect to off-diagonal entries double. Some books (Renchner and Schaalje 2008) report formulas of the type  $2\mathbf{W}^{-1} - \text{diag}(\mathbf{W}^{-1})$  for this reason. The simplest way to avoid the issue is to differentiate along a symmetric direction  $\mathbf{W} + t\mathbf{E}$ , as in (c). This is the convenient route for the normal likelihoods of [Chapter 32](#). The listing checks the gradient formulas against central finite differences, and the script also checks part (c) and [Proposition 1.11.2](#). The largest discrepancy in the listing is  $3 \times 10^{-9}$ , which is the size one expects from the finite-difference approximation.

```
import numpy as np
rng = np.random.default_rng(5)
p = 4
def num_grad(f, x, h=1e-6):
    """Central-difference gradient of a scalar function of a vector or matrix."""
    g = np.zeros_like(x)
    for idx in np.ndindex(x.shape):
        e = np.zeros_like(x)
        e[idx] = h
        g[idx] = (f(x + e) - f(x - e)) / (2 * h)
```

```

    return g

a = rng.normal(size=p)
A = rng.normal(size=(p, p))          # not symmetric
x = rng.normal(size=p)
checks = {
    "a'x": (num_grad(lambda v: a @ v, x), a),
    "x'Ax": (num_grad(lambda v: v @ A @ v, x), (A + A.T) @ x),
}
X = rng.normal(size=(20, p))
y = rng.normal(size=20)
b = rng.normal(size=p)
grad_ls = -2 * X.T @ (y - X @ b)
checks["||y - Xb||^2"] = (num_grad(lambda v: np.sum((y - X @ v) ** 2), b), grad_ls)

S = rng.normal(size=(p, p)); S = S @ S.T + p * np.eye(p)  # positive definite
checks["log det S"] = (num_grad(lambda W: np.linalg.slogdet(W)[1], S), np.linalg.inv(S).T)
checks["tr(AS)"] = (num_grad(lambda W: np.trace(A @ W), S), A.T)
for name, (numeric, formula) in checks.items():
    print(f"{name:14s} max error {np.abs(numeric - formula).max():.1e}")

```

## 1.11.4 Exercises

### 1.11.4.1 A. Check your understanding

#### Exercise A1.

Observations fall in  $a$  clusters of size  $b$ , and  $\Sigma = \sigma^2 \mathbf{I}_a \otimes \mathbf{E}_b$ , where  $\mathbf{E}_b = (1 - \rho)\mathbf{I}_b + \rho\mathbf{J}_b$ . Find the eigenvalues, determinant and inverse of  $\Sigma$  using [Proposition 1.10.1](#), [Example \(Equicorrelation\)](#) and [Example \(Eigenvalues of the equicorrelation matrix\)](#).

#### Exercise A2.

Use [Proposition 1.11.2](#) to show that  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda\|\mathbf{b}\|^2$  with  $\lambda > 0$  has the unique minimizer  $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ , whatever the rank of  $\mathbf{X}$ .

### 1.11.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{X}$  have full column rank,  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ , and let  $\mathbf{H}$  be  $q \times p$  with full row rank. Show that the minimizer of  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  subject to  $\mathbf{H}\mathbf{b} = \mathbf{h}$  is

$$\hat{\boldsymbol{\beta}}_H = \hat{\boldsymbol{\beta}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}), \quad \mathbf{W} = \mathbf{H} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top,$$

and that the increase in the residual sum of squares is  $(\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})$ .

*Solution.* Apply [Proposition 1.11.3](#) with  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  and  $\mathbf{b} = \mathbf{X}^\top \mathbf{y}$ . Since  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \mathbf{y}^\top \mathbf{y} + f(\mathbf{b})$  and  $\mathbf{x}_0 = \hat{\boldsymbol{\beta}}$ , this gives  $\hat{\boldsymbol{\beta}}_H$ . By completing the square as in [Proposition 1.11.2](#),  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + (\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}})$  for every  $\mathbf{b}$ . With

$\mathbf{b} - \hat{\boldsymbol{\beta}} = -(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})$ , the increase is

$$(\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} \mathbf{H} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{H}^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}) = (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h})^\top \mathbf{W}^{-1} (\mathbf{H} \hat{\boldsymbol{\beta}} - \mathbf{h}).$$

This quadratic form is the numerator of the  $F$  statistic in Chapter 11 (Theorem 11.3.2).

## 1.12 Summary and Notes

### 1.12.1 Summary

*Idea.*

1. Row rank equals column rank, and  $\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A})$  is the number of columns.  $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$ , so the normal equations are always consistent (Theorem 1.2.1, Theorem 1.2.2 and Proposition 1.2.3).
2. A generalized inverse satisfies  $\mathbf{A} \mathbf{A}^- \mathbf{A} = \mathbf{A}$ . Every matrix has one, built from any nonsingular  $r \times r$  submatrix. The Moore–Penrose inverse is the unique one that is also reflexive and makes  $\mathbf{A} \mathbf{A}^+$  and  $\mathbf{A}^+ \mathbf{A}$  symmetric (Definition 1.3.1, Theorem 1.3.2, Proposition 1.2.4 and Definition 1.3.2).
3.  $\mathbf{A} \mathbf{x} = \mathbf{b}$  is consistent iff  $\mathbf{A} \mathbf{A}^- \mathbf{b} = \mathbf{b}$ , and then its solutions are  $\mathbf{A}^- \mathbf{b} + (\mathbf{I} - \mathbf{A}^- \mathbf{A}) \mathbf{z}$ . A linear function  $\mathbf{q}^\top \mathbf{x}$  is the same for all solutions iff  $\mathbf{q} \in \mathcal{C}(\mathbf{A}^\top)$  (Theorem 1.4.1 and Corollary 1.4.2).
4. Orthogonal matrices preserve lengths and angles. Gram–Schmidt gives the QR factorization, and  $\mathbb{R}^n = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$  (Proposition 1.5.3, Theorem 1.5.2 and Proposition 1.5.1).
5.  $\text{tr}(\mathbf{A} \mathbf{B}) = \text{tr}(\mathbf{B} \mathbf{A})$ . Block elimination gives the determinant and inverse of a partitioned matrix through Schur complements, and the Sherman–Morrison–Woodbury identity as a special case (Theorem 1.6.1, Theorem 1.6.3, Theorem 1.6.4 and Theorem 1.6.5).
6. A symmetric matrix is  $\mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top$  with  $\mathbf{Q}$  orthogonal. Positive definiteness is equivalent to positive eigenvalues, to a Cholesky factorization, and to positive leading minors. Nonnegative definite matrices have a unique nonnegative definite square root (Theorem 1.7.2, Theorem 1.7.4 and Theorem 1.7.6).
7. The Rayleigh quotient  $\mathbf{x}^\top \mathbf{A} \mathbf{x} / \mathbf{x}^\top \mathbf{x}$  ranges over  $[\lambda_{\min}, \lambda_{\max}]$  (Theorem 1.7.7 and Corollary 1.7.8).
8. Every matrix is  $\mathbf{U} \mathbf{D} \mathbf{V}^\top$ . The condition number  $\kappa = \sigma_{\max} / \sigma_{\min}$  bounds error growth, is squared by forming  $\mathbf{X}^\top \mathbf{X}$ , and its smallest singular vector exhibits near-dependences among the columns (Theorem 1.8.1, Definition 1.8.1 and Proposition 1.8.2).
9. Idempotent matrices have eigenvalues 0 and 1 and rank equal to trace. Symmetric ones are  $\mathbf{Q}_1 \mathbf{Q}_1^\top$ . If symmetric matrices summing to  $\mathbf{I}$  have ranks summing to  $n$ , they are idempotent and mutually orthogonal (Proposition 1.9.1, Theorem 1.9.2 and Theorem 1.9.3).
10. Kronecker products factor balanced designs.  $\text{vec}(\mathbf{A} \mathbf{B} \mathbf{C}) = (\mathbf{C}^\top \otimes \mathbf{A}) \text{vec}(\mathbf{B})$  (Proposition 1.10.1).
11.  $\partial(\mathbf{x}^\top \mathbf{A} \mathbf{x}) / \partial \mathbf{x} = (\mathbf{A} + \mathbf{A}^\top) \mathbf{x}$ . Quadratic functions with nonnegative definite ma-



trices are minimized by solving a linear system, with or without linear constraints ([Theorem 1.11.1](#), [Proposition 1.11.2](#) and [Proposition 1.11.3](#)).

## 1.12.2 Notes and sources

**Coverage and framing.** The selection of results follows what the rest of the book uses. Rencher and Schaalje (2008, chapter 2) was the checklist for topics: its sections on positive definite matrices, generalized inverses, orthogonal matrices, idempotent matrices and matrix calculus correspond to [Section 1.7](#), [Section 1.3](#), [Section 1.5](#), [Section 1.9](#) and [Section 1.11](#). Several results that Rencher and Schaalje state without proof or with a reference, such as the general solution of a consistent system, the characterizations of positive definiteness and the sums of idempotent matrices, are proved here. The vector-space point of view, in which column spaces, null spaces and orthogonal complements come first and matrices second, is that of Christensen (2020, appendices A and B). Seber and Lee (2003, appendix A) is a compact list of further matrix facts. The Kronecker product results in [Section 1.10](#) are also in Christensen’s appendix B.

**Linear algebra for statisticians.** Harville (1997) is the most complete treatment written for statisticians, with full proofs of the determinant facts assumed in [Proposition 1.6.2](#) and a long development of generalized inverses. Searle (1982) is more elementary and has many worked examples from linear models. Horn and Johnson (2013) is the standard reference for eigenvalue inequalities, positive definite matrices and norms. Halmos (1958) gives the coordinate-free view of the vector-space facts used in [Section 1.2](#).

**Generalized inverses.** The four conditions of [Definition 1.3.2](#) are due to Penrose (1955), who proved existence and uniqueness. E.H. Moore had described an equivalent inverse decades earlier. Rao (1973) made the weaker inverse  $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$  the basic tool of linear-model theory, and it is his approach that [Section 1.4](#) and [Chapter 8](#) follow. The construction from a nonsingular submatrix in [Theorem 1.3.2](#) is a matrix-free way of stating an algorithm that is often given as a sequence of steps (Searle 1982).

**Schur complements and updating formulas.** The name “Schur complement” and the notation for it vary. Zhang (2005) collects the history and many applications. The rank-one formula [Eq. 1.6.7](#) is from Sherman and Morrison (1950), and the general identity [Eq. 1.6.6](#) from Woodbury (1950). Henderson and Searle (1981) trace earlier appearances and compare several derivations. Obtaining the identity by comparing two expressions for one block of a partitioned inverse, as in [Theorem 1.6.5](#), is a classical route.

**Singular values and conditioning.** Stewart (1993) tells the early history of the singular value decomposition, beginning with Beltrami and Jordan in the 1870s. The low-rank approximation theorem, [Proposition 1.8.2\(c\)](#) and its Frobenius form, is usually credited to Eckart and Young (1936), although Stewart points out that Schmidt had proved it earlier in the setting of integral equations. Golub and Van Loan (2013) is the reference for computing the decomposition and for perturbation bounds such as [Eq. 1.8.3](#), and Higham (2002) for their refinements. The use of scaled condition numbers and singular vectors to diagnose collinearity is due to Belsley et al. (1980). [Chapter 26](#) returns to it.

**Kronecker products and calculus.** Henderson and Searle (1979) survey the vec operator and its uses in multivariate statistics. Magnus and Neudecker (2019) develop matrix calculus systematically through differentials, which avoid the bookkeeping of the entrywise definitions used in [Section 1.11](#) and handle symmetric matrix arguments cleanly.

### 1.12.3 References

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# Chapter 2

## Random Vectors: Expectation and Covariance Operators

A linear model is a statement about the first two moments of a random vector: its mean lies in a subspace and its covariance matrix has a given form. This chapter builds the calculus for such statements. Expectation and covariance become matrix operators that commute with linear maps, covariance matrices turn out to be exactly the nonnegative definite matrices, and a singular covariance means the vector is confined to a flat. The mean of a quadratic form needs only two moments; its variance needs four. Partitioning a vector gives cross-covariances, Schur complements and the best linear predictor, which foreshadow conditional distributions under normality.

**What you need.** [Chapter 1](#) (trace, spectral theorem, nonnegative definite matrices and their square roots, generalized inverses, partitioned matrices), and probability at the level of expectations, variances and independence of random variables.

### 2.0.1 Roadmap

- [Mean vectors and covariance matrices](#)
- [Linear and affine transformations](#)
- [Expectation of quadratic forms](#)
- [Correlation matrices](#)
- [Moments of partitioned random vectors](#)
- [Moment generating functions and independence](#)
- [Sample moments and their computation](#)
- [Summary and notes](#)

## 2.1 Mean vectors and covariance matrices

Random vectors enter regression in two different ways. The first is the response vector. In the model

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon},$$

$\mathbf{Y} = (Y_1, \dots, Y_n)^\top$  has one entry per observational unit, and so does the unobservable error  $\boldsymbol{\varepsilon}$ . The assumptions that do the real work in [Chapter 5](#), [Chapter 6](#) and [Chapter 7](#) are not about the shape of any distribution. They are statements about moments:  $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ , and the errors have equal variances and are uncorrelated. The second way is the measurement vector. When several quantities are recorded on the same unit, such as income, age and years of schooling, the unit contributes a vector whose components are usually correlated and measured in different units. Random regressors, errors-in-variables models and multivariate responses are all built from such vectors.

Both uses need the same toolkit: expectation and covariance as operations on vectors and matrices, their behaviour under linear maps, and the expected values of quadratic forms. Throughout, a random vector *has finite second moments* if every component has finite variance.

## 2.1.1 Random matrices and their expectation

### Definition 2.1.1 — Expectation of a random matrix

A **random matrix**  $\mathbf{Z} = (Z_{ij})$  is an  $m \times k$  array of random variables. If every  $Z_{ij}$  has a finite expectation, the **expectation** of  $\mathbf{Z}$  is the  $m \times k$  matrix of expectations,

$$E(\mathbf{Z}) = (E(Z_{ij})).$$

A random vector is the case  $k = 1$ .

The definition works entry by entry, so every rule that is linear in the entries carries over at once.

### Proposition 2.1.1 — Linearity of expectation

Let  $\mathbf{Z}$  and  $\mathbf{Z}_2$  be  $m \times k$  random matrices with integrable entries, and let  $\mathbf{A}$  ( $l \times m$ ),  $\mathbf{B}$  ( $k \times s$ ) and  $\mathbf{C}$  ( $l \times s$ ) be constant matrices. Then

- (a)  $E(\mathbf{AZB} + \mathbf{C}) = \mathbf{A} E(\mathbf{Z}) \mathbf{B} + \mathbf{C}$ ;
- (b)  $E(\mathbf{Z} + \mathbf{Z}_2) = E(\mathbf{Z}) + E(\mathbf{Z}_2)$  and  $E(\mathbf{Z}^\top) = E(\mathbf{Z})^\top$ ;
- (c) if  $m = k$ , then  $E \operatorname{tr}(\mathbf{Z}) = \operatorname{tr} E(\mathbf{Z})$ .

*Proof.* The  $(i, j)$  entry of  $\mathbf{AZB} + \mathbf{C}$  is  $\sum_r \sum_t a_{ir} Z_{rt} b_{tj} + c_{ij}$ , a finite linear combination of integrable random variables with constant coefficients. Its expectation is  $\sum_r \sum_t a_{ir} E(Z_{rt}) b_{tj} + c_{ij}$ , which is the  $(i, j)$  entry of  $\mathbf{A} E(\mathbf{Z}) \mathbf{B} + \mathbf{C}$ . Parts (b) and (c) are the same argument applied to  $Z_{ij} + Z_{2,ij}$ , to  $Z_{ji}$ , and to  $\sum_i Z_{ii}$ . ■

Two warnings go with this proposition. The multipliers must be constant: if  $\mathbf{A}$  is random,  $E(\mathbf{AZ})$  is generally not  $E(\mathbf{A})E(\mathbf{Z})$ . And the expectation of a product of random matrices is generally not the product of expectations. It is when every entry of the first is independent of every entry of the second, because then  $E(Z_{ir}W_{rj}) = E(Z_{ir})E(W_{rj})$  for each term.

## 2.1.2 Mean vector and covariance matrix

### Definition 2.1.2 — Mean vector and covariance matrices

Let  $\mathbf{Y}$  be a  $p \times 1$  random vector with finite second moments. Its **mean vector** is  $\boldsymbol{\mu} = E(\mathbf{Y})$ , and its **covariance matrix** is the  $p \times p$  matrix

$$\text{Cov}(\mathbf{Y}) = E[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top],$$

whose  $(i, j)$  entry is  $\sigma_{ij} = \text{Cov}(Y_i, Y_j)$  and whose diagonal holds the variances  $\sigma_{ii} = \text{Var}(Y_i)$ . If  $\mathbf{U}$  ( $p \times 1$ ) and  $\mathbf{V}$  ( $q \times 1$ ) both have finite second moments, their **cross-covariance matrix** is the  $p \times q$  matrix

$$\text{Cov}(\mathbf{U}, \mathbf{V}) = E[(\mathbf{U} - E \mathbf{U})(\mathbf{V} - E \mathbf{V})^\top],$$

with  $(i, j)$  entry  $\text{Cov}(U_i, V_j)$ .

The expectations exist:  $|U_i V_j| \leq \frac{1}{2}(U_i^2 + V_j^2)$ , so finite second moments make every product integrable. We often write  $\boldsymbol{\Sigma}$  for  $\text{Cov}(\mathbf{Y})$ , and  $\text{Cov}(\mathbf{Y}) = \text{Cov}(\mathbf{Y}, \mathbf{Y})$ . Some books write  $\text{Var}(\mathbf{Y})$  or  $D(\mathbf{Y})$  for the covariance matrix. We keep  $\text{Var}$  for scalars. The symbol  $\text{Cov}$  then has three uses:  $\text{Cov}(Y_i, Y_j)$  is a number,  $\text{Cov}(\mathbf{Y})$  is a symmetric square matrix, and  $\text{Cov}(\mathbf{U}, \mathbf{V})$  is a rectangular matrix that is usually not symmetric even when  $p = q$ .

### Proposition 2.1.2 — Elementary properties

Let  $\mathbf{Y}$ ,  $\mathbf{U}$ ,  $\mathbf{V}$  have finite second moments, with  $\boldsymbol{\mu} = E(\mathbf{Y})$  and  $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$ , and let  $\mathbf{a}$ ,  $\mathbf{b}$  be constant vectors of matching sizes. Then:

- (a)  $\boldsymbol{\Sigma}$  is symmetric;
- (b)  $\text{Cov}(\mathbf{U}, \mathbf{V}) = E(\mathbf{UV}^\top) - E(\mathbf{U})E(\mathbf{V})^\top$ , and in particular  $\boldsymbol{\Sigma} = E(\mathbf{YY}^\top) - \boldsymbol{\mu}\boldsymbol{\mu}^\top$ ;
- (c)  $\text{Cov}(\mathbf{V}, \mathbf{U}) = \text{Cov}(\mathbf{U}, \mathbf{V})^\top$ ;
- (d)  $\text{Cov}(\mathbf{U} + \mathbf{a}, \mathbf{V} + \mathbf{b}) = \text{Cov}(\mathbf{U}, \mathbf{V})$ ;
- (e)  $E[(\mathbf{Y} - \mathbf{a})(\mathbf{Y} - \mathbf{a})^\top] = \boldsymbol{\Sigma} + (\boldsymbol{\mu} - \mathbf{a})(\boldsymbol{\mu} - \mathbf{a})^\top$ , and hence  $E\|\mathbf{Y} - \mathbf{a}\|^2 = \text{tr}(\boldsymbol{\Sigma}) + \|\boldsymbol{\mu} - \mathbf{a}\|^2$ .

*Proof.*

- (a) and (c) hold because  $(\mathbf{uv}^\top)^\top = \mathbf{vu}^\top$  and expectation commutes with transposition. For (b), expand  $(\mathbf{U} - \boldsymbol{\mu}_U)(\mathbf{V} - \boldsymbol{\mu}_V)^\top = \mathbf{UV}^\top - \mathbf{U}\boldsymbol{\mu}_V^\top - \boldsymbol{\mu}_U\mathbf{V}^\top + \boldsymbol{\mu}_U\boldsymbol{\mu}_V^\top$  and take expectations with Proposition 2.1.1. The two middle terms each give  $-\boldsymbol{\mu}_U\boldsymbol{\mu}_V^\top$ . (d) holds because adding a constant does not change the centred vectors. For (e), write  $\mathbf{Y} - \mathbf{a} = (\mathbf{Y} - \boldsymbol{\mu}) + (\boldsymbol{\mu} - \mathbf{a})$  and expand. The cross terms contain the factor  $E(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{0}$  and vanish. The trace identity follows from  $\|\mathbf{v}\|^2 = \text{tr}(\mathbf{vv}^\top)$  and Proposition 2.1.1(c). ■

Part (e) is the vector form of “mean squared error equals variance plus squared bias”. It shows that the mean is the constant vector closest to  $\mathbf{Y}$  in expected squared distance, and that the minimal value is  $\text{tr}(\boldsymbol{\Sigma}) = \sum_i \text{Var}(Y_i)$ . The matrix statement is stronger than the trace statement.

It says that the second-moment matrix about any point exceeds  $\Sigma$  by a nonnegative definite matrix of rank at most one.

### 2.1.3 Uncorrelated is not independent

Random vectors  $\mathbf{U}$  and  $\mathbf{V}$  are **uncorrelated** if  $\text{Cov}(\mathbf{U}, \mathbf{V}) = \mathbf{0}$ . If  $\mathbf{U}$  and  $\mathbf{V}$  are independent, then  $E(U_i V_j) = E(U_i) E(V_j)$  for all  $i, j$ , so they are uncorrelated. The converse is false, and it fails in the most innocent-looking cases.

#### Example. (A point on a circle)

Let  $\Theta$  be uniform on  $[0, 2\pi)$  and  $\mathbf{Y} = (\cos \Theta, \sin \Theta)^\top$ . Then  $E(\cos \Theta) = E(\sin \Theta) = 0$ ,  $E(\cos^2 \Theta) = E(\sin^2 \Theta) = \frac{1}{2}$ , and  $E(\cos \Theta \sin \Theta) = \frac{1}{2} E(\sin 2\Theta) = 0$ . So  $\boldsymbol{\mu} = \mathbf{0}$  and  $\text{Cov}(\mathbf{Y}) = \frac{1}{2} \mathbf{I}_2$ , exactly the moments of two independent coordinates. Yet  $Y_1^2 + Y_2^2 = 1$ : once  $Y_1$  is known,  $Y_2$  is known up to sign. Covariance measures *linear* association only, and the dependence here is entirely nonlinear.

For jointly normal vectors, zero covariance and independence coincide ([Theorem 3.4.1](#)). That equivalence rests on joint normality, not on normal marginals, and it is one of the main reasons normal theory is so tractable.

**Scalar summaries.** The **total variance**  $\text{tr}(\Sigma)$  ignores correlation; the **generalized variance**  $\det(\Sigma)$  does not. If  $\lambda_1, \dots, \lambda_p$  are the eigenvalues of  $\Sigma$  ([Theorem 1.7.2](#)), these are  $\sum_i \lambda_i$  and  $\prod_i \lambda_i$ , and both are unchanged by a rotation  $\mathbf{Y} \mapsto \mathbf{QY}$ . The determinant can be tiny even when every variance is large, if one eigenvalue is near zero. [Theorem 2.2.3](#) shows that a zero eigenvalue means some linear combination of  $\mathbf{Y}$  does not vary at all.

### 2.1.4 Exercises

#### 2.1.4.1 A. Check your understanding

##### Exercise A1.

Let  $\mathbf{Z}$  be an  $n \times p$  random matrix whose entries are independent, with mean 0 and variance  $\sigma^2$ . Show that  $E(\mathbf{Z}^\top \mathbf{Z}) = n\sigma^2 \mathbf{I}_p$  and  $E(\mathbf{Z}\mathbf{Z}^\top) = p\sigma^2 \mathbf{I}_n$ . For a constant  $p \times p$  matrix  $\mathbf{B}$ , find  $E(\mathbf{Z}\mathbf{B}\mathbf{Z}^\top)$ .

#### 2.1.4.2 B. Practice

##### Exercise B1.

Let  $X$  have a distribution symmetric about 0 with  $\text{Var}(X) = 1$  and  $E X^4 < \infty$ , let  $S$  be independent of  $X$  with  $\Pr(S = 1) = \Pr(S = -1) = \frac{1}{2}$ , and put  $\mathbf{Y} = (X, SX)^\top$ . Show that the two components have the same distribution and are uncorrelated, and that  $\text{Cov}(Y_1^2, Y_2^2) = \text{Var}(X^2)$ . Deduce that  $Y_1$  and  $Y_2$  are independent iff  $X$  is itself a random sign. ([Chapter 3](#) takes  $X$  standard normal.)

*Solution.* Since  $S$  is independent of  $X$  and  $X$  is symmetric,  $\Pr(SX \leq y) = \frac{1}{2} \Pr(X \leq y) + \frac{1}{2} \Pr(-X \leq y) = \Pr(X \leq y)$ . By independence,  $\text{Cov}(X, SX) = E(S)E(X^2) = 0$ . Both squares equal  $X^2$ , so  $\text{Cov}(Y_1^2, Y_2^2) = \text{Var}(X^2)$ . Functions of independent variables are independent, hence uncorrelated, so  $\text{Var}(X^2) > 0$  makes  $Y_1$  and  $Y_2$  dependent. If  $\text{Var}(X^2) = 0$ , then  $X^2 = E X^2 = 1$  with probability one, and by symmetry  $X = \pm 1$  with probability  $\frac{1}{2}$  each. Then  $\Pr(Y_1 = a, Y_2 = b) = \Pr(X = a) \Pr(S = ab) = \frac{1}{4}$  for all signs  $a, b$ , which is the product of the marginal probabilities, as in [Example \(Pairwise but not mutually independent\)](#).

### Exercise B2.

Show that  $E[(\mathbf{Y} - \mathbf{a})(\mathbf{Y} - \mathbf{a})^\top] - \text{Cov}(\mathbf{Y})$  is nonnegative definite for every constant  $\mathbf{a}$ , with equality iff  $\mathbf{a} = \boldsymbol{\mu}$ . Deduce that  $\mathbf{a} = \boldsymbol{\mu}$  minimizes  $E[(\mathbf{Y} - \mathbf{a})^\top \mathbf{N}(\mathbf{Y} - \mathbf{a})]$  for every positive definite  $\mathbf{N}$ .

## 2.2 Linear and affine transformations

Almost every statistic in a linear model is a linear function of the response: least squares coefficients, fitted values, residuals, contrasts. Their means and covariance matrices follow from one theorem.

### 2.2.1 The transformation rules

#### Theorem 2.2.1 — Moments of affine transformations

Let  $\mathbf{Y}$  be a  $p \times 1$  random vector with finite second moments,  $E(\mathbf{Y}) = \boldsymbol{\mu}$  and  $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$ . Let  $\mathbf{A}$  be  $k \times p$ ,  $\mathbf{B}$  be  $m \times p$ , and  $\mathbf{b}$  be  $k \times 1$ , all constant. Then

- (a)  $E(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$ ;
- (b)  $\text{Cov}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$ ;
- (c)  $\text{Cov}(\mathbf{A}\mathbf{Y}, \mathbf{B}\mathbf{Y}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top$ .

More generally, if  $\mathbf{U}$  and  $\mathbf{V}$  have finite second moments and  $\mathbf{A}, \mathbf{B}, \mathbf{b}, \mathbf{c}$  are constants of conformable sizes, then  $\text{Cov}(\mathbf{A}\mathbf{U} + \mathbf{b}, \mathbf{B}\mathbf{V} + \mathbf{c}) = \mathbf{A} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{B}^\top$ .

*Proof.*

- (a) is [Proposition 2.1.1\(a\)](#). For the general statement, the centred vectors are  $\mathbf{A}\mathbf{U} + \mathbf{b} - E(\mathbf{A}\mathbf{U} + \mathbf{b}) = \mathbf{A}(\mathbf{U} - E\mathbf{U})$  and similarly  $\mathbf{B}(\mathbf{V} - E\mathbf{V})$ . Hence

$$\text{Cov}(\mathbf{A}\mathbf{U} + \mathbf{b}, \mathbf{B}\mathbf{V} + \mathbf{c}) = E[\mathbf{A}(\mathbf{U} - E\mathbf{U})(\mathbf{V} - E\mathbf{V})^\top \mathbf{B}^\top] = \mathbf{A} E[(\mathbf{U} - E\mathbf{U})(\mathbf{V} - E\mathbf{V})^\top] \mathbf{B}^\top,$$

again by [Proposition 2.1.1\(a\)](#). Parts (c) and (b) are the cases  $\mathbf{U} = \mathbf{V} = \mathbf{Y}$ , and  $\mathbf{B} = \mathbf{A}$  in (b).

For a single linear combination,  $\mathbf{A} = \mathbf{a}^\top$  is a row vector, and (b) and (c) give

$$\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a} = \sum_{i=1}^p \sum_{j=1}^p a_i a_j \sigma_{ij}, \quad \text{Cov}(\mathbf{a}^\top \mathbf{Y}, \mathbf{b}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{b}. \quad (2.2.1)$$

Every variance and covariance enters  $\text{Var}(\mathbf{a}^\top \mathbf{Y})$ , weighted by the products of the coefficients. Conversely,  $\boldsymbol{\Sigma}$  is determined by the variances of linear combinations, because  $\sigma_{ij} = \frac{1}{2}[\text{Var}(Y_i + Y_j) - \text{Var}(Y_i) - \text{Var}(Y_j)]$ .

### Proposition 2.2.2 — Covariance of sums

Let  $\mathbf{U}, \mathbf{U}_2, \mathbf{V}$  have finite second moments, with  $\mathbf{U}$  and  $\mathbf{U}_2$  of the same size in (a), and  $\mathbf{U}$  and  $\mathbf{V}$  of the same size in (b). Then

- (a)  $\text{Cov}(\mathbf{U} + \mathbf{U}_2, \mathbf{V}) = \text{Cov}(\mathbf{U}, \mathbf{V}) + \text{Cov}(\mathbf{U}_2, \mathbf{V})$ ;
- (b)  $\text{Cov}(\mathbf{U} + \mathbf{V}) = \text{Cov}(\mathbf{U}) + \text{Cov}(\mathbf{U}, \mathbf{V}) + \text{Cov}(\mathbf{V}, \mathbf{U}) + \text{Cov}(\mathbf{V})$ ;
- (c) if  $\mathbf{Y}_1, \dots, \mathbf{Y}_n$  are pairwise uncorrelated  $p \times 1$  vectors and  $c_1, \dots, c_n$  are constants, then  $\text{Cov}(\sum_i c_i \mathbf{Y}_i) = \sum_i c_i^2 \text{Cov}(\mathbf{Y}_i)$ . In particular, if every  $\text{Cov}(\mathbf{Y}_i) = \boldsymbol{\Sigma}$ , then  $\text{Cov}(\bar{\mathbf{Y}}) = \boldsymbol{\Sigma}/n$ .

*Proof.*

- (a) Centre each vector and use linearity of the outer product and of E. (b) Apply (a) in each argument; note that the two cross terms are transposes of each other, not equal.
- (b) By (a) and (b), applied repeatedly,  $\text{Cov}(\sum_i c_i \mathbf{Y}_i) = \sum_i \sum_j c_i c_j \text{Cov}(\mathbf{Y}_i, \mathbf{Y}_j)$ , using  $\text{Cov}(c\mathbf{U}, \mathbf{V}) = c \text{Cov}(\mathbf{U}, \mathbf{V})$  from [Theorem 2.2.1](#). The terms with  $i \neq j$  vanish.

### Example. (Centring a sample)

Let  $\mathbf{Y}$  be  $n \times 1$  with  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$ , and write  $\mathbf{J} = \mathbf{1}\mathbf{1}^\top$ . The mean is  $\bar{Y} = n^{-1} \mathbf{1}^\top \mathbf{Y}$ , and the vector of deviations is  $\mathbf{d} = (\mathbf{I} - n^{-1} \mathbf{J})\mathbf{Y}$ . The matrix  $\mathbf{I} - n^{-1} \mathbf{J}$  is symmetric and idempotent, so [Theorem 2.2.1](#) gives

$$\text{Cov}(\mathbf{d}) = \sigma^2 (\mathbf{I} - n^{-1} \mathbf{J}), \quad \text{Cov}(\bar{Y}, \mathbf{d}) = \sigma^2 n^{-1} \mathbf{1}^\top (\mathbf{I} - n^{-1} \mathbf{J}) = \mathbf{0}^\top.$$

The deviations have variances  $\sigma^2(1 - 1/n)$  and are negatively correlated, with correlation  $-1/(n-1)$  between each pair, because they are forced to sum to zero. And each deviation is uncorrelated with the mean. Under normality, uncorrelated becomes independent ([Example \(Sample mean and deviations\)](#)), which is the first half of [Corollary 4.4.5](#).



## 2.2.2 Covariance matrices are nonnegative definite

A variance cannot be negative, and by Eq. 2.2.1 this simple fact constrains the whole matrix. Recall from Definition 1.7.1 that a symmetric  $\Sigma$  is nonnegative definite if  $\mathbf{a}^\top \Sigma \mathbf{a} \geq 0$  for all  $\mathbf{a}$ , and positive definite if in addition  $\mathbf{a}^\top \Sigma \mathbf{a} > 0$  for  $\mathbf{a} \neq \mathbf{0}$ .

### Theorem 2.2.3 — Covariance matrices

Let  $\mathbf{Y}$  have finite second moments, mean  $\boldsymbol{\mu}$  and covariance matrix  $\Sigma$ .

- (a)  $\Sigma$  is nonnegative definite.
- (b) Conversely, every  $p \times p$  nonnegative definite matrix is the covariance matrix of some random vector.
- (c) For a constant  $\mathbf{a}$ ,  $\Sigma \mathbf{a} = \mathbf{0}$  iff  $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top \boldsymbol{\mu}$  with probability one.
- (d)  $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma)) = 1$ . If  $\mathcal{S}$  is any subspace with  $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{S}) = 1$ , then  $\mathcal{C}(\Sigma) \subseteq \mathcal{S}$ .

In particular,  $\Sigma$  is singular iff some nontrivial linear combination of  $\mathbf{Y}$  is constant with probability one. Then  $\mathbf{Y}$  lies, with probability one, in the affine subspace  $\boldsymbol{\mu} + \mathcal{C}(\Sigma)$  of dimension  $\text{rank}(\Sigma) < p$ , and this is the smallest such flat.

*Proof.*

- (a)  $\Sigma$  is symmetric, and  $\mathbf{a}^\top \Sigma \mathbf{a} = \text{Var}(\mathbf{a}^\top \mathbf{Y}) \geq 0$  by Eq. 2.2.1.
- (b) Let  $\Sigma$  be nonnegative definite and let  $\Sigma^{1/2}$  be its symmetric nonnegative definite square root (Theorem 1.7.6). Let  $\mathbf{Z}$  have independent components with mean 0 and variance 1, for instance independent standard normal variables, so that  $\text{Cov}(\mathbf{Z}) = \mathbf{I}$ . Then  $\text{Cov}(\Sigma^{1/2} \mathbf{Z}) = \Sigma^{1/2} \mathbf{I} \Sigma^{1/2} = \Sigma$ .
- (c) If  $\Sigma \mathbf{a} = \mathbf{0}$ , the random variable  $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu})$  has mean zero and variance  $\mathbf{a}^\top \Sigma \mathbf{a} = 0$ , so it is zero with probability one. Conversely, if  $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0$  with probability one, then  $\Sigma \mathbf{a} = \mathbb{E}[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top \mathbf{a}] = \mathbf{0}$ .
- (d) Let  $\mathbf{a}_1, \dots, \mathbf{a}_s$  be a basis of  $\mathcal{N}(\Sigma)$ . By (c), each event  $\{\mathbf{a}_j^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0\}$  has probability one, and so does their intersection. On that intersection  $\mathbf{Y} - \boldsymbol{\mu}$  is orthogonal to  $\mathcal{N}(\Sigma)$ . Because  $\Sigma$  is symmetric,  $\mathcal{N}(\Sigma)$  is the orthogonal complement of  $\mathcal{C}(\Sigma)$ , and so  $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma)$ . Now let  $\mathcal{S}$  be a subspace with  $\Pr(\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{S}) = 1$ . For every  $\mathbf{a} \in \mathcal{S}^\perp$  we have  $\mathbf{a}^\top (\mathbf{Y} - \boldsymbol{\mu}) = 0$  with probability one, hence  $\mathbf{a} \in \mathcal{N}(\Sigma) = \mathcal{C}(\Sigma)^\perp$  by (c). So  $\mathcal{S}^\perp \subseteq \mathcal{C}(\Sigma)^\perp$ , and taking orthogonal complements gives  $\mathcal{C}(\Sigma) \subseteq \mathcal{S}$ . The final statements follow, with  $\dim \mathcal{C}(\Sigma) = \text{rank}(\Sigma)$ . For minimality among all flats, suppose  $\Pr(\mathbf{Y} \in \mathbf{c} + \mathcal{S}) = 1$  for a vector  $\mathbf{c}$  and a subspace  $\mathcal{S}$ . Every  $\mathbf{a} \in \mathcal{S}^\perp$  has  $\mathbf{a}^\top (\mathbf{Y} - \mathbf{c}) = 0$  with probability one, and taking expectations gives  $\mathbf{a}^\top (\boldsymbol{\mu} - \mathbf{c}) = 0$ . Hence  $\boldsymbol{\mu} - \mathbf{c} \in \mathcal{S}$ , so  $\mathbf{c} + \mathcal{S} = \boldsymbol{\mu} + \mathcal{S}$ , and the subspace case applies.

■

A covariance matrix is therefore positive definite exactly when no linear combination of the components is degenerate. Singular covariance matrices are not pathological. They arise whenever the components satisfy an exact linear constraint, as with the deviations of Example (Centring

a sample), whose covariance matrix has rank  $n - 1$  because  $\mathbf{1}^\top \mathbf{d} = 0$ . The residual vector of a regression is another case: it is orthogonal to every column of  $\mathbf{X}$ , and when  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$  its covariance matrix has rank  $n - \text{rank}(\mathbf{X})$  (Section 2.2, Chapter 6).

### Example. (Multinomial counts)

In  $m$  independent trials, each outcome falls into one of  $k$  categories with probabilities  $\boldsymbol{\pi} = (\pi_1, \dots, \pi_k)^\top$ , where  $\sum_j \pi_j = 1$ . Let  $\mathbf{Z}_t$  be the indicator vector of trial  $t$ , with a single 1 in the position of its category. Then  $E(\mathbf{Z}_t) = \boldsymbol{\pi}$ , and  $\mathbf{Z}_t \mathbf{Z}_t^\top = \text{diag}(\mathbf{Z}_t)$  because the indicator has only one nonzero entry. So  $\text{Cov}(\mathbf{Z}_t) = \text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi} \boldsymbol{\pi}^\top$ . The count vector  $\mathbf{Y} = \sum_t \mathbf{Z}_t$  is a sum of independent vectors, and Proposition 2.2.2(c) gives

$$\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y}) = m(\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi} \boldsymbol{\pi}^\top).$$

Because  $\boldsymbol{\Sigma} \mathbf{1} = m(\boldsymbol{\pi} - \boldsymbol{\pi} \boldsymbol{\pi}^\top \mathbf{1}) = \mathbf{0}$ , the matrix is singular, and Theorem 2.2.3(c) recovers the constraint  $\mathbf{1}^\top \mathbf{Y} = m$ . For  $m = 20$  and  $\boldsymbol{\pi} = (0.5, 0.3, 0.2)^\top$  the eigenvalues are 7.762, 4.638 and 0. All 5000 simulated count vectors lie in the plane  $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma}) = \{\mathbf{y} : \mathbf{1}^\top \mathbf{y} = m\}$ .

Linear maps cannot increase the rank of a covariance matrix, since  $\text{rank}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top) \leq \text{rank}(\boldsymbol{\Sigma})$  (Exercise B2). They can, however, discard degenerate directions. If  $\mathbf{Y} = (Z, Z)^\top$  with  $\text{Var}(Z) = 1$ , then  $\boldsymbol{\Sigma}$  has all entries equal to 1 and is singular, while  $\mathbf{A} = (1, 0)$  gives  $\text{Var}(\mathbf{A} \mathbf{Y}) = 1 > 0$ . By Theorem 2.2.1, the covariance matrix of  $\mathbf{A} \mathbf{Y}$  is  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top$ . If  $\boldsymbol{\Sigma}$  is positive definite and  $\mathbf{A}$  has full row rank  $k \leq p$ , then for  $\mathbf{b} \neq \mathbf{0}$  the vector  $\mathbf{A}^\top \mathbf{b}$  is nonzero and  $\mathbf{b}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top \mathbf{b} > 0$ . So  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top$  is positive definite. If  $k > p$ , then  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top$  has rank at most  $p$  and is singular. A vector of more than  $p$  linear combinations of  $p$  variables always satisfies a linear constraint.

## 2.2.3 Standardization and Mahalanobis distance

A scalar with mean  $\mu$  and variance  $\sigma^2 > 0$  is standardized by  $(Y - \mu)/\sigma$ . The vector version must remove correlation as well as scale.

### Proposition 2.2.4 — Whitening

Let  $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$  be positive definite, with  $E(\mathbf{Y}) = \boldsymbol{\mu}$ .

- (a)  $\mathbf{Z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{Y} - \boldsymbol{\mu})$  has  $E(\mathbf{Z}) = \mathbf{0}$  and  $\text{Cov}(\mathbf{Z}) = \mathbf{I}_p$ .
- (b) A  $p \times p$  matrix  $\mathbf{W}$  satisfies  $\mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{I}_p$  iff  $\mathbf{W} = \mathbf{Q} \boldsymbol{\Sigma}^{-1/2}$  for some orthogonal  $\mathbf{Q}$ . In particular, if  $\boldsymbol{\Sigma} = \mathbf{L} \mathbf{L}^\top$  is the Cholesky factorization,  $\mathbf{W} = \mathbf{L}^{-1}$  is such a matrix.
- (c) For every  $\mathbf{W}$  as in (b),  $\|\mathbf{W}(\mathbf{Y} - \boldsymbol{\mu})\|^2 = (\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$ .

*Proof.*

- (a) By Theorem 2.2.1,  $\text{Cov}(\mathbf{Z}) = \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\Sigma} \boldsymbol{\Sigma}^{-1/2} = \mathbf{I}$ , since  $\boldsymbol{\Sigma}^{-1/2}$  is symmetric and commutes with  $\boldsymbol{\Sigma}$ . (b) If  $\mathbf{W} = \mathbf{Q} \boldsymbol{\Sigma}^{-1/2}$ , then  $\mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{Q} \mathbf{Q}^\top = \mathbf{I}$ . Conversely, if  $\mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{I}$ , put  $\mathbf{Q} = \mathbf{W} \boldsymbol{\Sigma}^{1/2}$ . Then  $\mathbf{Q} \mathbf{Q}^\top = \mathbf{W} \boldsymbol{\Sigma} \mathbf{W}^\top = \mathbf{I}$ , so the square matrix  $\mathbf{Q}$  is orthogonal, and  $\mathbf{W} = \mathbf{Q} \boldsymbol{\Sigma}^{-1/2}$ . For the Cholesky factor,  $\mathbf{L}^{-1} \boldsymbol{\Sigma} (\mathbf{L}^\top)^{-1} = \mathbf{L}^{-1} \mathbf{L} \mathbf{L}^\top (\mathbf{L}^\top)^{-1} = \mathbf{I}$ . (c)  $\mathbf{W}^\top \mathbf{W} = \boldsymbol{\Sigma}^{-1/2} \mathbf{Q}^\top \mathbf{Q} \boldsymbol{\Sigma}^{-1/2} = \boldsymbol{\Sigma}^{-1}$ .



```

lam, U = np.linalg.eigh(Sigma)                # spectral decomposition
W_sym = U @ np.diag(lam ** -0.5) @ U.T        # Sigma^{-1/2}
L = np.linalg.cholesky(Sigma)                # Sigma = L L', L lower triangular
W_chol = np.linalg.inv(L)                    # L^{-1}

n = 400
Y = mu + rng.standard_normal((n, 2)) @ L.T    # any draws with Cov = Sigma will
do
Z_sym = (Y - mu) @ W_sym.T
Z_chol = (Y - mu) @ W_chol.T

Q = W_chol @ np.linalg.inv(W_sym)            # L^{-1} Sigma^{1/2}
print("Q'Q =", np.round(Q.T @ Q, 12))

```

Whitening runs in both directions. If  $\text{Cov}(\mathbf{Z}) = \mathbf{I}$ , then  $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$ , or  $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$ , has mean  $\boldsymbol{\mu}$  and covariance  $\boldsymbol{\Sigma}$ . This is how correlated vectors are simulated, and with  $\mathbf{Z}$  standard normal it is how [Chapter 3](#) constructs  $N_p(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ . The Mahalanobis distance then appears in the exponent of the density ([Theorem 3.2.2](#)). When  $\boldsymbol{\Sigma}$  is singular, no matrix whitens all of  $\mathbf{Y}$ , since a degenerate combination cannot be scaled to variance one. The best one can do is whiten within the support  $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$  using the Moore–Penrose inverse square root ([Exercise \(B4\)](#)).

## 2.2.4 Second moments in the linear model

Here is how the rules of this section enter regression. Suppose  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $\mathbf{X}$  an  $n \times p$  constant matrix of rank  $p$ ,  $E(\boldsymbol{\varepsilon}) = \mathbf{0}$  and  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$ . Adding the constant  $\mathbf{X}\boldsymbol{\beta}$  changes the mean but not the covariance, so  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$ . The least squares estimator  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$  is  $\mathbf{A}\mathbf{Y}$  with  $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ , so

$$E(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}.$$

No distributional assumption was used, and none is needed for the Gauss–Markov theorem ([Theorem 7.2.1](#)). Fitted values and residuals are also linear in  $\mathbf{Y}$ , and their covariance matrices are computed the same way in [Chapter 5](#) and [Chapter 6](#). If instead  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$  with  $\mathbf{V}$  positive definite, then  $\mathbf{V}^{-1/2} \mathbf{Y} = \mathbf{V}^{-1/2} \mathbf{X} \boldsymbol{\beta} + \mathbf{V}^{-1/2} \boldsymbol{\varepsilon}$  is a model with  $\text{Cov}(\mathbf{V}^{-1/2} \boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ . Whitening the data turns correlated errors back into the standard case. This is generalized least squares, which [Chapter 6](#) and [Chapter 31](#) develop.

## 2.2.5 Exercises

### 2.2.5.1 A. Check your understanding

#### Exercise A1.

Let  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_3$ . Find the covariance and correlation matrices of the successive differences  $(Y_2 - Y_1, Y_3 - Y_2)^\top$ , and of  $(Y_2 - Y_1, Y_3 - Y_2, Y_3 - Y_1)^\top$ . Which is singular, and which linear combination is degenerate?

**Exercise A2.**

For  $p = 2$  and  $\Sigma = \mathbf{L}\mathbf{L}^\top$ , show that the second component of  $\mathbf{L}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$  is a multiple of  $Y_2 - \mu_2 - (\sigma_{12}/\sigma_{11})(Y_1 - \mu_1)$ . Explain the connection with [Theorem 2.5.2](#).

**2.2.5.2 B. Practice****Exercise B1.**

Let  $\varepsilon_0, \dots, \varepsilon_n$  be uncorrelated with variance  $\sigma^2$ , and let  $Y_t = \varepsilon_t + \theta\varepsilon_{t-1}$  for  $t = 1, \dots, n$ . Find  $\text{Cov}(\mathbf{Y})$ , and show that it is positive definite for every real  $\theta$ , including  $\theta = \pm 1$ .

*Solution.*  $\mathbf{Y} = \mathbf{A}\boldsymbol{\varepsilon}$ , where  $\mathbf{A}$  is  $n \times (n+1)$  with  $\theta$  in column  $t$  and 1 in column  $t+1$  of row  $t$  (columns indexed by  $\varepsilon_0, \dots, \varepsilon_n$ ). So  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{A}\mathbf{A}^\top$  is tridiagonal, with  $\sigma^2(1 + \theta^2)$  on the diagonal and  $\sigma^2\theta$  next to it. The last  $n$  columns of  $\mathbf{A}$  form a lower bidiagonal matrix with unit diagonal, which is nonsingular. So  $\mathbf{A}$  has full row rank  $n$ , and  $\mathbf{A}\mathbf{A}^\top$  is positive definite ([Section 2.2](#)) whatever the value of  $\theta$ .

**Exercise B2.**

Show that  $\mathcal{C}(\mathbf{A}\Sigma\mathbf{A}^\top) = \mathcal{C}(\mathbf{A}\Sigma)$  for every nonnegative definite  $\Sigma$  and conformable  $\mathbf{A}$ . Deduce that  $\text{rank Cov}(\mathbf{A}\mathbf{Y}) \leq \text{rank Cov}(\mathbf{Y})$ .

**Exercise B3.**

Let  $\Sigma$  be positive definite. Among all  $\mathbf{W}$  with  $\mathbf{W}\Sigma\mathbf{W}^\top = \mathbf{I}$ , show that  $\mathbb{E}\|\mathbf{W}(\mathbf{Y} - \boldsymbol{\mu}) - (\mathbf{Y} - \boldsymbol{\mu})\|^2$  is minimized uniquely by  $\mathbf{W} = \Sigma^{-1/2}$ . *Hint:* use [Proposition 2.2.4\(b\)](#) and show that  $\text{tr}(\mathbf{Q}\Sigma^{1/2}) \leq \text{tr}(\Sigma^{1/2})$  for orthogonal  $\mathbf{Q}$ .

*Solution.* Write  $\mathbf{W} = \mathbf{Q}\Sigma^{-1/2}$ . By [Corollary 2.3.2\(a\)](#) with  $\mathbf{A} = (\mathbf{W} - \mathbf{I})^\top(\mathbf{W} - \mathbf{I})$ ,

$$\mathbb{E}\|(\mathbf{W} - \mathbf{I})(\mathbf{Y} - \boldsymbol{\mu})\|^2 = \text{tr}((\mathbf{W} - \mathbf{I})\Sigma(\mathbf{W} - \mathbf{I})^\top) = \text{tr}(\mathbf{W}\Sigma\mathbf{W}^\top) - 2\text{tr}(\mathbf{W}\Sigma) + \text{tr}(\Sigma) = p + \text{tr}(\Sigma) - 2\text{tr}(\mathbf{Q}\Sigma^{1/2}).$$

Let  $\Sigma^{1/2} = \mathbf{\Gamma}\mathbf{\Lambda}^{1/2}\mathbf{\Gamma}^\top$  and  $\mathbf{G} = \mathbf{\Gamma}^\top\mathbf{Q}\mathbf{\Gamma}$ , which is orthogonal, so each  $|g_{ii}| \leq 1$ . Then  $\text{tr}(\mathbf{Q}\Sigma^{1/2}) = \text{tr}(\mathbf{G}\mathbf{\Lambda}^{1/2}) = \sum_i g_{ii}\lambda_i^{1/2} \leq \sum_i \lambda_i^{1/2}$ . Since every  $\lambda_i > 0$ , equality forces every  $g_{ii} = 1$ . An orthogonal matrix with unit diagonal is  $\mathbf{I}$ , because each column has norm one. So  $\mathbf{G} = \mathbf{I}$ , which means  $\mathbf{Q} = \mathbf{I}$ .

**Exercise B4.**

Let  $\Sigma = \text{Cov}(\mathbf{Y})$  have rank  $r$ , with spectral decomposition  $\Sigma = \mathbf{\Gamma}_r\mathbf{\Lambda}_r\mathbf{\Gamma}_r^\top$ , where  $\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r = \mathbf{I}_r$  and  $\mathbf{\Lambda}_r$  holds the positive eigenvalues. Show that  $\mathbf{Z} = \mathbf{\Lambda}_r^{-1/2}\mathbf{\Gamma}_r^\top(\mathbf{Y} - \boldsymbol{\mu})$  has  $\text{Cov}(\mathbf{Z}) = \mathbf{I}_r$  and that  $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{\Gamma}_r\mathbf{\Lambda}_r^{1/2}\mathbf{Z}$  with probability one. Show that  $(\mathbf{Y} - \boldsymbol{\mu})^\top\Sigma^+(\mathbf{Y} - \boldsymbol{\mu}) = \|\mathbf{Z}\|^2$ , where  $\Sigma^+$  is the Moore–Penrose inverse, and that its expectation is  $r$ .

*Solution.* By [Theorem 2.2.1](#),  $\text{Cov}(\mathbf{Z}) = \mathbf{\Lambda}_r^{-1/2}\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r\mathbf{\Lambda}_r\mathbf{\Gamma}_r^\top\mathbf{\Gamma}_r\mathbf{\Lambda}_r^{-1/2} = \mathbf{I}_r$ . By [Theorem 2.2.3\(d\)](#),  $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma) = \mathcal{C}(\mathbf{\Gamma}_r)$  with probability one, and on that event  $\mathbf{Y} -$

$\boldsymbol{\mu} = \boldsymbol{\Gamma}_r \boldsymbol{\Gamma}_r^\top (\mathbf{Y} - \boldsymbol{\mu}) = \boldsymbol{\Gamma}_r \boldsymbol{\Lambda}_r^{1/2} \mathbf{Z}$ . The Moore–Penrose inverse is  $\boldsymbol{\Sigma}^+ = \boldsymbol{\Gamma}_r \boldsymbol{\Lambda}_r^{-1} \boldsymbol{\Gamma}_r^\top$ , so  $(\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^+ (\mathbf{Y} - \boldsymbol{\mu}) = \|\boldsymbol{\Lambda}_r^{-1/2} \boldsymbol{\Gamma}_r^\top (\mathbf{Y} - \boldsymbol{\mu})\|^2 = \|\mathbf{Z}\|^2$ , with expectation  $\text{tr Cov}(\mathbf{Z}) = r$ .

### 2.2.5.3 C. Going deeper

#### Exercise C1.

In  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $\mathbf{X}$  of full column rank, let  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$  with  $\mathbf{V}$  positive definite. Find  $\text{Cov}(\hat{\boldsymbol{\beta}})$  for  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$  and  $\text{Cov}(\tilde{\boldsymbol{\beta}})$  for  $\tilde{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$ . Show that  $\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\tilde{\boldsymbol{\beta}})$  is nonnegative definite. *Hint:* show that  $\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}$  and  $\tilde{\boldsymbol{\beta}}$  are uncorrelated.

*Solution.* By [Theorem 2.2.1](#),

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}, \quad \text{Cov}(\tilde{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1}.$$

Write  $\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}} = \mathbf{D}\mathbf{Y}$  with  $\mathbf{D} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top - (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1}$ , and note  $\mathbf{D}\mathbf{X} = \mathbf{I} - \mathbf{I} = \mathbf{0}$ . Then  $\text{Cov}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}}, \tilde{\boldsymbol{\beta}}) = \sigma^2 \mathbf{D} \mathbf{V} \mathbf{V}^{-1} \mathbf{X} (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} = \mathbf{0}$ , so by [Proposition 2.2.2\(b\)](#),  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \text{Cov}(\tilde{\boldsymbol{\beta}}) + \text{Cov}(\hat{\boldsymbol{\beta}} - \tilde{\boldsymbol{\beta}})$ , and the last term is nonnegative definite.

## 2.3 Expectation of quadratic forms

Sums of squares are quadratic forms in the response. The sample variance is  $(n-1)^{-1} \mathbf{Y}^\top (\mathbf{I} - n^{-1} \mathbf{J}) \mathbf{Y}$ . The residual sum of squares of a regression is  $\mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$ , where  $\mathbf{M}$  is the projection onto the column space of  $\mathbf{X}$  ([Chapter 6](#)). Test statistics are ratios of such forms. Their expectations follow from a short trace argument that needs no distributional assumption.

### 2.3.1 The basic formula

#### Theorem 2.3.1 — Mean of a quadratic form

Let  $\mathbf{Y}$  be a  $p \times 1$  random vector with finite second moments,  $E(\mathbf{Y}) = \boldsymbol{\mu}$  and  $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$ , and let  $\mathbf{A}$  be a constant  $p \times p$  matrix. Then

$$E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}.$$

*Proof.* A  $1 \times 1$  matrix equals its trace, and  $\text{tr}(\mathbf{a} \mathbf{b}^\top) = \mathbf{b}^\top \mathbf{a}$  ([Theorem 1.6.1](#)), so  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \text{tr}(\mathbf{A} \mathbf{Y} \mathbf{Y}^\top)$ . By [Proposition 2.1.1](#) and [Proposition 2.1.2\(b\)](#),

$$E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} E(\mathbf{Y} \mathbf{Y}^\top)) = \text{tr}(\mathbf{A} (\boldsymbol{\Sigma} + \boldsymbol{\mu} \boldsymbol{\mu}^\top)) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}.$$

■

The formula has two parts. The trace term is what the form picks up from random variation. The second term is the value of the form at the mean, which is what it would be without any

randomness. The theorem does not require independence, normality or symmetry of  $\mathbf{A}$ . Since  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \mathbf{Y}^\top \mathbf{A}_s \mathbf{Y}$  with  $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$ , and  $\text{tr}(\mathbf{A} \mathbf{\Sigma}) = \text{tr}(\mathbf{A}_s \mathbf{\Sigma})$  when  $\mathbf{\Sigma}$  is symmetric, we may assume  $\mathbf{A}$  symmetric whenever that is convenient.

### Corollary 2.3.2

Under the conditions of [Theorem 2.3.1](#):

- (a)  $E[(\mathbf{Y} - \mathbf{b})^\top \mathbf{A}(\mathbf{Y} - \mathbf{b})] = \text{tr}(\mathbf{A} \mathbf{\Sigma}) + (\boldsymbol{\mu} - \mathbf{b})^\top \mathbf{A}(\boldsymbol{\mu} - \mathbf{b})$  for any constant  $\mathbf{b}$ ;
- (b) if  $\mathbf{\Sigma} = \sigma^2 \mathbf{I}$ , then  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2 \text{tr}(\mathbf{A}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ ;
- (c) if  $\mathbf{\Sigma}$  is positive definite, the squared Mahalanobis distance [Eq. 2.2.2](#) has  $E \Delta^2(\mathbf{Y}) = p$ ;
- (d) for random vectors  $\mathbf{U}$  ( $p \times 1$ ) and  $\mathbf{V}$  ( $q \times 1$ ) with finite second moments and a constant  $p \times q$  matrix  $\mathbf{A}$ ,  $E(\mathbf{U}^\top \mathbf{A} \mathbf{V}) = \text{tr}(\mathbf{A} \text{Cov}(\mathbf{V}, \mathbf{U})) + E(\mathbf{U})^\top \mathbf{A} E(\mathbf{V})$ .

*Proof.*

- (a) applies the theorem to  $\mathbf{Y} - \mathbf{b}$ , which has covariance matrix  $\mathbf{\Sigma}$ . (b) is immediate.
- (c) is (a) with  $\mathbf{b} = \boldsymbol{\mu}$  and  $\mathbf{A} = \mathbf{\Sigma}^{-1}$ , so the trace is  $\text{tr}(\mathbf{I}_p) = p$ . For (d),  $\mathbf{U}^\top \mathbf{A} \mathbf{V} = \text{tr}(\mathbf{A} \mathbf{V} \mathbf{U}^\top)$  and  $E(\mathbf{V} \mathbf{U}^\top) = \text{Cov}(\mathbf{V}, \mathbf{U}) + E(\mathbf{V}) E(\mathbf{U})^\top$ .

■

#### Example. (Sample variance with unequal means and variances)

Let  $Y_1, \dots, Y_n$  be uncorrelated with means  $\mu_i$  and variances  $\sigma_i^2$ , and let  $S^2 = (n-1)^{-1} \mathbf{Y}^\top \mathbf{C} \mathbf{Y}$  with  $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{J}$ . Here  $\mathbf{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$ , so  $\text{tr}(\mathbf{C} \mathbf{\Sigma}) = (1 - 1/n) \sum_i \sigma_i^2$ , and  $\boldsymbol{\mu}^\top \mathbf{C} \boldsymbol{\mu} = \sum_i (\mu_i - \bar{\mu})^2$ . Therefore

$$E(S^2) = \overline{\sigma^2} + \frac{1}{n-1} \sum_{i=1}^n (\mu_i - \bar{\mu})^2, \quad \overline{\sigma^2} = \frac{1}{n} \sum_{i=1}^n \sigma_i^2.$$

With a common mean,  $S^2$  is unbiased for the *average* variance, even when the variances differ. Differences in the means inflate it. That is exactly why a residual mean square estimates  $\sigma^2$  only when the mean structure is right.

#### Example. (Residual sum of squares under a wrong mean)

Let  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$  and  $E(\mathbf{Y}) = \boldsymbol{\theta}$ , where  $\boldsymbol{\theta}$  need not lie in  $\mathcal{C}(\mathbf{X})$ . The residual sum of squares is  $\mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$ , where  $\mathbf{M}$  is symmetric and idempotent of rank  $r = \text{rank}(\mathbf{X})$ , so  $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$  ([Theorem 1.9.2](#)). By [Corollary 2.3.2](#)(b),

$$E(\text{RSS}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2. \quad (2.3.1)$$

If the model is right,  $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$ , the second term vanishes and  $\text{RSS}/(n - r)$  is unbiased for  $\sigma^2$ . If the model is wrong, the estimate is biased upwards by the squared distance from the true mean to the model space, divided by  $n - r$ . For instance, fit a straight

line at  $x = 1, \dots, 6$  when the true mean is the parabola  $\theta_i = \frac{1}{4}(x_i - 3.5)^2$  and  $\sigma^2 = 1$ . Then  $\sigma^2(n - r) = 4$ ,  $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2 = 2.333$ , and  $E(\text{RSS}) = 6.333$ . Chapter 19 studies such departures systematically (Theorem 19.1.2).

The theorem uses only two moments, so it holds equally for skewed and heavy-tailed data. The listing checks it for a correlated vector  $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{B}\mathbf{U}$ , where  $\mathbf{U}$  has independent, centred exponential components, which are strongly skewed. With the matrices in the listing,  $\text{tr}(\mathbf{A}\boldsymbol{\Sigma}) = 6.9$  and  $\boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu} = 17.0$ , so  $E(\mathbf{Y}^\top \mathbf{A}\mathbf{Y}) = 23.9$ . The average over  $10^6$  draws is 23.940, with Monte Carlo standard error 0.018. The script also verifies the formula exactly, by enumerating a discrete distribution with eight atoms.

```
import itertools
import numpy as np
rng = np.random.default_rng(20260917)

mu = np.array([1.0, -1.0, 2.0])
B = np.array([[1.0, 0.0, 0.0],
              [0.5, 1.0, 0.0],
              [-0.5, 0.8, 0.6]])
Sigma = B @ B.T                                     # Cov(mu + B U) when Cov(U) = I
A = np.array([[2.0, 1.0, 0.0],
              [1.0, 1.0, -1.0],
              [0.0, -1.0, 3.0]])

theory = np.trace(A @ Sigma) + mu @ A @ mu          # Theorem: no normality needed

N = 1_000_000
U = rng.exponential(size=(N, 3)) - 1.0             # skewed, mean 0, variance 1
Y = mu + U @ B.T
Q = np.einsum("ij,jk,ik->i", Y, A, Y)             # Y_i' A Y_i for every draw
print(f"theory {theory:.4f}    simulation {Q.mean():.4f}  {Q.std() / np.sqrt(N):.4f}")
```

## 2.3.2 The variance needs fourth moments

The mean of  $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$  involves only  $E(Y_i Y_j)$ . Its second moment involves  $E(Y_i Y_j Y_k Y_l)$ , so no formula for  $\text{Var}(\mathbf{Y}^\top \mathbf{A}\mathbf{Y})$  can be written in terms of  $\boldsymbol{\mu}$  and  $\boldsymbol{\Sigma}$  alone. Two random vectors with the same mean and covariance matrix can give the same quadratic form very different variances. The next theorem handles independent components, the case that matters for residual sums of squares under the classical error assumptions. Chapter 4 gives the normal case with a general  $\boldsymbol{\Sigma}$  (Theorem 4.2.3).

### Theorem 2.3.3 — Variance of a quadratic form

Let  $Y_1, \dots, Y_n$  be independent with means  $\theta_1, \dots, \theta_n$  and common central moments  $\sigma^2 = E(Y_i - \theta_i)^2$ ,  $\mu_3 = E(Y_i - \theta_i)^3$  and  $\mu_4 = E(Y_i - \theta_i)^4 < \infty$ . Let  $\mathbf{A}$  be a constant symmetric  $n \times n$  matrix, and let  $\mathbf{a} = (a_{11}, \dots, a_{nn})^\top$  be its diagonal. Then

$$\text{Var}(\mathbf{Y}^\top \mathbf{A}\mathbf{Y}) = (\mu_4 - 3\sigma^4) \mathbf{a}^\top \mathbf{a} + 2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta} + 4\mu_3 \boldsymbol{\theta}^\top \mathbf{A} \mathbf{a}.$$



*Proof.* Write  $\mathbf{Y} = \boldsymbol{\theta} + \boldsymbol{\varepsilon}$ , where the  $\varepsilon_i$  are independent with mean zero. Then

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = q + \ell + \boldsymbol{\theta}^\top \mathbf{A} \boldsymbol{\theta}, \quad q = \boldsymbol{\varepsilon}^\top \mathbf{A} \boldsymbol{\varepsilon}, \quad \ell = 2\mathbf{b}^\top \boldsymbol{\varepsilon}, \quad \mathbf{b} = \mathbf{A} \boldsymbol{\theta},$$

so  $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{Var}(q) + \text{Var}(\ell) + 2 \text{Cov}(q, \ell)$ . We compute the three terms.

*The linear term.*  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ , so  $\text{Var}(\ell) = 4\sigma^2 \mathbf{b}^\top \mathbf{b} = 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta}$ .

*The cross term.* Since  $E(\ell) = 0$ ,  $\text{Cov}(q, \ell) = E(q\ell) = 2 \sum_{i,j,k} a_{ij} b_k E(\varepsilon_i \varepsilon_j \varepsilon_k)$ . By independence and zero means,  $E(\varepsilon_i \varepsilon_j \varepsilon_k)$  vanishes unless  $i = j = k$ , when it equals  $\mu_3$ . So  $\text{Cov}(q, \ell) = 2\mu_3 \sum_i a_{ii} b_i = 2\mu_3 \boldsymbol{\theta}^\top \mathbf{A} \mathbf{a}$ .

*The quadratic term.*  $E(q^2) = \sum_{i,j,k,l} a_{ij} a_{kl} E(\varepsilon_i \varepsilon_j \varepsilon_k \varepsilon_l)$ . The fourth-order moment is  $\mu_4$  if all four indices agree. It is  $\sigma^4$  if the indices form two distinct pairs, which happens in three patterns:  $i = j \neq k = l$ ,  $i = k \neq j = l$  and  $i = l \neq j = k$ . Otherwise some index occurs exactly once, and the moment is zero. Using the symmetry of  $\mathbf{A}$ ,

$$E(q^2) = \mu_4 \sum_i a_{ii}^2 + \sigma^4 \left[ \sum_{i \neq k} a_{ii} a_{kk} + 2 \sum_{i \neq j} a_{ij}^2 \right] = \mu_4 \mathbf{a}^\top \mathbf{a} + \sigma^4 [(\text{tr } \mathbf{A})^2 - \mathbf{a}^\top \mathbf{a} + 2 \text{tr}(\mathbf{A}^2) - 2\mathbf{a}^\top \mathbf{a}],$$

since  $\sum_{i,j} a_{ij}^2 = \text{tr}(\mathbf{A}^2)$ . By [Theorem 2.3.1](#),  $E(q) = \sigma^2 \text{tr } \mathbf{A}$ , and subtracting its square gives  $\text{Var}(q) = (\mu_4 - 3\sigma^4) \mathbf{a}^\top \mathbf{a} + 2\sigma^4 \text{tr}(\mathbf{A}^2)$ . Adding the three terms proves the formula. ■

For normal errors,  $\mu_3 = 0$  and  $\mu_4 = 3\sigma^4$ , and the formula collapses to  $2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta}$ . Both extra terms describe how non-normality leaks into second-order behaviour. The kurtosis term depends on  $\mathbf{A}$  only through its diagonal. The skewness term needs both a skewed error distribution and a mean that the form does not annihilate. For the residual sum of squares of a correct model,  $\mathbf{A} \boldsymbol{\theta} = (\mathbf{I} - \mathbf{M}) \boldsymbol{\theta} = \mathbf{0}$ , so the skewness term disappears but the kurtosis term remains.

### Example. (Skewed errors and a misspecified line)

Continue [Example \(Residual sum of squares under a wrong mean\)](#), now with independent centred exponential errors, for which  $\sigma^2 = 1$ ,  $\mu_3 = 2$  and  $\mu_4 = 9$ . [Theorem 2.3.3](#) gives  $\text{Var}(\text{RSS}) = 29.80$ . Of this, the kurtosis term contributes 16.73 and the skewness term  $-4.27$ . Normal errors with the same mean and variance would give 17.33, so the exponential errors multiply the variance by 1.72 even though the mean of RSS is unchanged. Over  $10^6$  simulated data sets, the variance of RSS is 30.04. The script also confirms the formula exactly for a two-point error distribution.

```
se = Q.std() / np.sqrt(N)
vals, probs = np.array([-1.0, 2.0]), np.array([2 / 3, 1 / 3])
exact = 0.0
for idx in itertools.product(range(2), repeat=3):
    u = vals[list(idx)]
    y = mu + B @ u
    exact += np.prod(probs[list(idx)]) * (y @ A @ y)
x = np.arange(1.0, 7.0) # n = 6 design points
X = np.column_stack([np.ones_like(x), x])
M = X @ np.linalg.solve(X.T @ X, X.T)
A2 = np.eye(6) - M # RSS = Y'(I - M)Y for a straight
```

```

line
theta = 0.25 * (x - 3.5) ** 2          # true mean is curved: the line is
wrong
a = np.diag(A2)

def var_quadform(A, theta, m2, m3, m4):
    """Var(Y'AY) for independent Y_i with means theta_i and common central moments.
    """
    a = np.diag(A)
    return ((m4 - 3 * m2**2) * a @ a + 2 * m2**2 * np.trace(A @ A)
            + 4 * m2 * theta @ A @ A @ theta + 4 * m3 * theta @ A @ a)

# A2 = I - M for a straight line at x = 1..6; theta = true (curved) mean
mean_rss = np.trace(A2) + theta @ A2 @ theta          # sigma^2 = 1
var_expo = var_quadform(A2, theta, 1.0, 2.0, 9.0) # centred exponential errors
var_norm = var_quadform(A2, theta, 1.0, 0.0, 3.0) # what normal errors would give

E = rng.exponential(size=(N, 6)) - 1.0
R = np.einsum("ij,jk,ik->i", theta + E, A2, theta + E)
print(f"mean   {mean_rss:.3f} vs {R.mean():.3f}")
print(f"var    {var_expo:.3f} vs {R.var():.3f}   (normal-theory value {var_norm:.3f})")

```

## 2.3.3 Exercises

### 2.3.3.1 A. Check your understanding

#### Exercise A1.

For random vectors  $\mathbf{U}$  and  $\mathbf{V}$  of the same size, show that  $E\|\mathbf{U} - \mathbf{V}\|^2 = \text{tr Cov}(\mathbf{U}) + \text{tr Cov}(\mathbf{V}) - 2 \text{tr Cov}(\mathbf{U}, \mathbf{V}) + \|E\mathbf{U} - E\mathbf{V}\|^2$ .

### 2.3.3.2 B. Practice

#### Exercise B1.

Let  $E(Y_t) = \mu$  and  $\text{Cov}(Y_t, Y_s) = \sigma^2 \phi^{|t-s|}$  for  $t, s = 1, \dots, n$ , with  $|\phi| < 1$ . Show that the sample variance satisfies

$$E(S^2) = \sigma^2 \left[ 1 - \frac{2}{n(n-1)} \sum_{k=1}^{n-1} (n-k) \phi^k \right].$$

Is  $S^2$  biased up or down when  $\phi > 0$ ? What happens as  $n \rightarrow \infty$  with  $\phi$  fixed?

*Solution.* With  $\mathbf{C} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$  and  $E(\mathbf{Y}) = \mu\mathbf{1}$ ,  $\mathbf{C}E(\mathbf{Y}) = \mathbf{0}$ , so  $E(\mathbf{Y}^\top \mathbf{C} \mathbf{Y}) = \text{tr}(\mathbf{C}\Sigma) = \text{tr}(\Sigma) - n^{-1}\mathbf{1}^\top \Sigma \mathbf{1}$ . Here  $\text{tr}(\Sigma) = n\sigma^2$ , and  $\mathbf{1}^\top \Sigma \mathbf{1}$  sums all entries. There are  $n$  diagonal entries and  $2(n-k)$  entries at lag  $k$ , so  $\mathbf{1}^\top \Sigma \mathbf{1} = \sigma^2 [n + 2 \sum_{k=1}^{n-1} (n-k) \phi^k]$ . Dividing  $n\sigma^2 - n^{-1}\mathbf{1}^\top \Sigma \mathbf{1}$  by  $n-1$  gives the formula. For  $\phi > 0$  the sum is positive, so  $S^2$  is biased downwards: positively correlated observations cluster, and the sample looks

less variable than the process is. As  $n \rightarrow \infty$  the sum is  $O(n)$ , so the bias is  $O(1/n)$  and vanishes.

### Exercise B2.

Suppose  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$ . Show that  $E(\text{RSS}) = \sigma^2 \text{tr}((\mathbf{I} - \mathbf{M})\mathbf{V})$ . For the equicorrelated  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$  and a model with  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , show that  $E(\text{RSS})/(n - r) = \sigma^2(1 - \rho)$ .

### Exercise B3.

Let  $Y_1, \dots, Y_n$  be independent and identically distributed with variance  $\sigma^2$  and finite fourth central moment  $\mu_4$ . Use [Theorem 2.3.3](#) to show that  $\text{Var}(S^2) = (\mu_4 - 3\sigma^4)/n + 2\sigma^4/(n - 1)$ . Compare with the normal case.

*Solution.* Take  $\mathbf{A} = \mathbf{C}/(n - 1)$  and  $\boldsymbol{\theta} = \mu\mathbf{1}$ . Then  $\mathbf{A}\boldsymbol{\theta} = \mathbf{0}$ , so the last two terms of [Theorem 2.3.3](#) vanish. Every diagonal entry of  $\mathbf{A}$  is  $(1 - 1/n)/(n - 1) = 1/n$ , so  $\mathbf{a}^\top \mathbf{a} = 1/n$ , and  $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{C})/(n - 1)^2 = 1/(n - 1)$ . This gives the formula. Normal data have  $\mu_4 = 3\sigma^4$  and  $\text{Var}(S^2) = 2\sigma^4/(n - 1)$ . Heavy tails ( $\mu_4 > 3\sigma^4$ ) add a term of the same order  $1/n$ , so the normal-theory variance is wrong even in large samples.

### Exercise B4.

Under the conditions of [Theorem 2.3.3](#) with  $\boldsymbol{\theta} = \mathbf{0}$ , show that  $\text{Cov}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}, \mathbf{Y}^\top \mathbf{B} \mathbf{Y}) = (\mu_4 - 3\sigma^4)\mathbf{a}^\top \mathbf{b} + 2\sigma^4 \text{tr}(\mathbf{A}\mathbf{B})$ , where  $\mathbf{a}$  and  $\mathbf{b}$  are the diagonals of the symmetric matrices  $\mathbf{A}$  and  $\mathbf{B}$ . *Hint:* apply the theorem to  $\mathbf{A} + \mathbf{B}$ .

### Exercise B5.

Let  $Y_1, \dots, Y_n$  be independent  $N(0, \sigma^2)$ . Among symmetric  $\mathbf{A}$  with  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2$ , show that  $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$  is minimized by  $\mathbf{A} = n^{-1}\mathbf{I}$ . *Hint:*  $\text{tr}(\mathbf{A}^2) \geq (\text{tr} \mathbf{A})^2/n$ .

## 2.4 Correlation matrices

Covariances carry units: the covariance of income in dollars with age in years is in dollar-years. To compare the strength of linear association across pairs of variables, each covariance is divided by the two standard deviations.

### Definition 2.4.1 — Correlation matrix

Let  $\mathbf{Y}$  have covariance matrix  $\boldsymbol{\Sigma} = (\sigma_{ij})$  with every  $\sigma_{ii} > 0$ , and let  $\mathbf{D}_\sigma = \text{diag}(\sigma_{11}^{1/2}, \dots, \sigma_{pp}^{1/2})$  be the diagonal matrix of standard deviations. The **correlation matrix** of  $\mathbf{Y}$  is

$$\mathbf{R} = \mathbf{D}_\sigma^{-1} \boldsymbol{\Sigma} \mathbf{D}_\sigma^{-1}, \quad \text{with entries} \quad \rho_{ij} = \frac{\sigma_{ij}}{\sqrt{\sigma_{ii}\sigma_{jj}}} = \text{corr}(Y_i, Y_j).$$

Equivalently  $\Sigma = \mathbf{D}_\sigma \mathbf{R} \mathbf{D}_\sigma$ . For random vectors  $\mathbf{U}$  and  $\mathbf{V}$  with standard-deviation matrices  $\mathbf{D}_U$  and  $\mathbf{D}_V$ , the cross-correlation matrix is  $\mathbf{D}_U^{-1} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{D}_V^{-1}$ .

Some books write  $\mathbf{P}_\rho$  for the correlation matrix. We use  $\mathbf{R}$  in this chapter, and  $\hat{\mathbf{R}}$  for its sample counterpart in [Section 2.7](#).

### Proposition 2.4.1 — Properties of correlation matrices

Let  $\mathbf{R}$  be the correlation matrix of  $\mathbf{Y}$ .

- (a)  $\mathbf{R}$  is the covariance matrix of the standardized vector  $\mathbf{Z} = \mathbf{D}_\sigma^{-1}(\mathbf{Y} - \boldsymbol{\mu})$ . Hence  $\mathbf{R}$  is nonnegative definite with unit diagonal, and  $\text{tr}(\mathbf{R}) = p$ .
- (b)  $|\rho_{ij}| \leq 1$ , with  $|\rho_{ij}| = 1$  iff  $Z_j = \rho_{ij} Z_i$  with probability one.
- (c)  $\text{rank}(\mathbf{R}) = \text{rank}(\Sigma)$ . In particular,  $\mathbf{R}$  is positive definite iff  $\Sigma$  is.
- (d) If  $\mathbf{Y}^* = \mathbf{D}\mathbf{Y} + \mathbf{c}$  with  $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$  and every  $d_i \neq 0$ , then  $\rho_{ij}^* = \text{sign}(d_i d_j) \rho_{ij}$ . Positive rescaling and shifting leave  $\mathbf{R}$  unchanged.
- (e) Conversely, every nonnegative definite matrix with unit diagonal is the correlation matrix of some random vector.

*Proof.*

- (a) [Theorem 2.2.1](#) gives  $\text{Cov}(\mathbf{Z}) = \mathbf{D}_\sigma^{-1} \Sigma \mathbf{D}_\sigma^{-1} = \mathbf{R}$ , and [Theorem 2.2.3\(a\)](#) applies.
- (b) The  $2 \times 2$  matrix of rows and columns  $i, j$  of  $\mathbf{R}$  is the covariance matrix of  $(Z_i, Z_j)$ , so it is nonnegative definite and its determinant  $1 - \rho_{ij}^2$  is nonnegative. If  $\rho_{ij} = \pm 1$ , the vector  $(\rho_{ij}, -1)^\top$  is in the null space of that  $2 \times 2$  matrix, so  $\rho_{ij} Z_i - Z_j = 0$  with probability one by [Theorem 2.2.3\(c\)](#). Conversely,  $Z_j = c Z_i$  with  $\text{Var}(Z_i) = \text{Var}(Z_j) = 1$  forces  $c = \pm 1$  and  $\rho_{ij} = c$ .
- (c)  $\mathbf{D}_\sigma$  is nonsingular.
- (d)  $\text{Cov}(\mathbf{Y}^*) = \mathbf{D} \Sigma \mathbf{D}$  has entries  $d_i d_j \sigma_{ij}$  and standard deviations  $|d_i| \sigma_{ii}^{1/2}$ .
- (e) Such a matrix is a covariance matrix by [Theorem 2.2.3\(b\)](#), and its variances are one.

■

Part (d) is the reason correlation is the natural scale-free summary. Part (e) is less obvious than it looks. A symmetric matrix with unit diagonal and off-diagonal entries in  $[-1, 1]$  need not be a correlation matrix, because the correlations must be jointly consistent.

### Proposition 2.4.2 — Three correlations

Let  $|\rho_{12}| < 1$  and  $|\rho_{13}| < 1$ . The matrix

$$\mathbf{R} = \begin{pmatrix} 1 & \rho_{12} & \rho_{13} \\ \rho_{12} & 1 & \rho_{23} \\ \rho_{13} & \rho_{23} & 1 \end{pmatrix}$$

is a correlation matrix iff

$$\rho_{12}\rho_{13} - \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)} \leq \rho_{23} \leq \rho_{12}\rho_{13} + \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}.$$

*Proof.* By [Proposition 2.4.1\(e\)](#) we must decide when  $\mathbf{R}$  is nonnegative definite. Write  $\mathbf{r} = (\rho_{12}, \rho_{13})^\top$  and let  $\mathbf{R}_2$  be the lower right  $2 \times 2$  block. The nonsingular matrix  $\mathbf{T} = \begin{pmatrix} 1 & \mathbf{0}^\top \\ -\mathbf{r} & \mathbf{I}_2 \end{pmatrix}$  gives

$$\mathbf{T}\mathbf{R}\mathbf{T}^\top = \begin{pmatrix} 1 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{R}_2 - \mathbf{r}\mathbf{r}^\top \end{pmatrix},$$

and  $\mathbf{a}^\top \mathbf{R} \mathbf{a} = \mathbf{b}^\top (\mathbf{T}\mathbf{R}\mathbf{T}^\top) \mathbf{b}$  with  $\mathbf{b} = (\mathbf{T}^\top)^{-1} \mathbf{a}$  ranges over all vectors as  $\mathbf{a}$  does. So  $\mathbf{R}$  is nonnegative definite iff the Schur complement

$$\mathbf{R}_2 - \mathbf{r}\mathbf{r}^\top = \begin{pmatrix} 1 - \rho_{12}^2 & \rho_{23} - \rho_{12}\rho_{13} \\ \rho_{23} - \rho_{12}\rho_{13} & 1 - \rho_{13}^2 \end{pmatrix}$$

is. A symmetric  $2 \times 2$  matrix with positive diagonal is nonnegative definite iff its determinant is nonnegative, because its eigenvalues have positive sum and their product is the determinant. The determinant is  $(1 - \rho_{12}^2)(1 - \rho_{13}^2) - (\rho_{23} - \rho_{12}\rho_{13})^2$ , and it is nonnegative exactly on the stated interval. ■

If  $Y_2$  and  $Y_3$  both correlate 0.8 with  $Y_1$ , the interval is  $[0.28, 1]$ . Two variables that are strongly related to a third cannot be negatively correlated with each other, nor even uncorrelated. The quantity  $(\rho_{23} - \rho_{12}\rho_{13}) / \sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}$ , which the proof forces into  $[-1, 1]$ , is the correlation of  $Y_2$  and  $Y_3$  after the linear effect of  $Y_1$  has been removed from both. [Section 2.5](#) develops this residualization, and [Definition 3.6.1](#) names the result the partial correlation.

Correlation matrices do not transform simply under linear maps: the correlation matrix of  $\mathbf{A}\mathbf{Y}$  must be recomputed from  $\mathbf{A}\Sigma\mathbf{A}^\top$ . Theory is therefore done with covariance matrices, and correlation matrices are used for description.

## 2.4.1 Exercises

### 2.4.1.1 A. Check your understanding

#### Exercise A1.

Show that  $(1 - \rho)\mathbf{I}_p + \rho\mathbf{1}\mathbf{1}^\top$  is a correlation matrix iff  $-1/(p-1) \leq \rho \leq 1$ . Explain the lower bound through the variance of  $\sum_i Z_i$ .

*Solution.* The matrix sends  $\mathbf{1}$  to  $(1 + (p-1)\rho)\mathbf{1}$  and every  $\mathbf{u} \perp \mathbf{1}$  to  $(1 - \rho)\mathbf{u}$ . Its eigenvalues are  $1 + (p-1)\rho$  (once) and  $1 - \rho$  ( $p-1$  times). By [Proposition 2.4.1\(e\)](#) it is a correlation matrix iff both are nonnegative. If  $\mathbf{Z}$  has this correlation matrix,  $\text{Var}(\sum_i Z_i) = \mathbf{1}^\top \mathbf{R} \mathbf{1} = p[1 + (p-1)\rho]$ , which must be nonnegative. Many variables cannot all be strongly negatively correlated with one another, because their sum would then have negative variance.

### 2.4.1.2 B. Practice

#### Exercise B1.

Show that  $\det(\mathbf{\Sigma}) \leq \prod_i \sigma_{ii}$  for every covariance matrix, with equality for positive definite  $\mathbf{\Sigma}$  iff  $\mathbf{\Sigma}$  is diagonal. *Hint:* factor out  $\mathbf{D}_\sigma$  and use the eigenvalues of  $\mathbf{R}$ .

*Solution.* If some  $\sigma_{ii} = 0$ , the  $i$ th row of  $\mathbf{\Sigma}$  vanishes (apply Theorem 2.2.3(c) with  $\mathbf{a}$  the  $i$ th coordinate vector) and both sides are zero. Otherwise  $\det \mathbf{\Sigma} = \det(\mathbf{D}_\sigma)^2 \det \mathbf{R} = \prod_i \sigma_{ii} \cdot \det \mathbf{R}$ . The eigenvalues  $\lambda_i \geq 0$  of  $\mathbf{R}$  sum to  $\text{tr} \mathbf{R} = p$ , so by the inequality of arithmetic and geometric means  $\det \mathbf{R} = \prod_i \lambda_i \leq (p^{-1} \sum_i \lambda_i)^p = 1$ . Equality holds iff every  $\lambda_i = 1$ , that is, iff  $\mathbf{R} = \mathbf{I}$ , that is, iff  $\mathbf{\Sigma}$  is diagonal.

### 2.4.1.3 C. Going deeper

#### Exercise C1.

- (a) Show that a  $p \times p$  matrix is a correlation matrix iff it is the Gram matrix  $(\mathbf{u}_i^\top \mathbf{u}_j)$  of  $p$  unit vectors in  $\mathbb{R}^p$ . (b) For  $p = 3$ , write  $\rho_{ij} = \cos \vartheta_{ij}$  with  $\vartheta_{ij} \in [0, \pi]$ . Show that Proposition 2.4.2 is equivalent to the spherical triangle inequalities  $|\vartheta_{12} - \vartheta_{13}| \leq \vartheta_{23} \leq \min(\vartheta_{12} + \vartheta_{13}, 2\pi - \vartheta_{12} - \vartheta_{13})$ .

## 2.5 Moments of partitioned random vectors

Regression with random regressors studies a response vector  $\mathbf{Y}$  ( $p \times 1$ ) together with a regressor vector  $\mathbf{X}$  ( $q \times 1$ ) observed on the same unit. In this section  $\mathbf{X}$  is a random vector, not a model matrix. We stack the two into one vector and read off what the blocks of its moments mean.

### 2.5.1 Blocks of the mean and covariance

#### Theorem 2.5.1 — Partitioned moments

Let  $\mathbf{W} = (\mathbf{Y}^\top, \mathbf{X}^\top)^\top$  have finite second moments, and write

$$\mathbf{E}(\mathbf{W}) = \begin{pmatrix} \boldsymbol{\mu}_Y \\ \boldsymbol{\mu}_X \end{pmatrix}, \quad \text{Cov}(\mathbf{W}) = \begin{pmatrix} \boldsymbol{\Sigma}_{YY} & \boldsymbol{\Sigma}_{YX} \\ \boldsymbol{\Sigma}_{XY} & \boldsymbol{\Sigma}_{XX} \end{pmatrix}.$$

- (a)  $\boldsymbol{\mu}_Y = \mathbf{E}(\mathbf{Y})$ ,  $\boldsymbol{\mu}_X = \mathbf{E}(\mathbf{X})$ ,  $\boldsymbol{\Sigma}_{YY} = \text{Cov}(\mathbf{Y})$ ,  $\boldsymbol{\Sigma}_{XX} = \text{Cov}(\mathbf{X})$ ,  $\boldsymbol{\Sigma}_{YX} = \text{Cov}(\mathbf{Y}, \mathbf{X})$  is  $p \times q$ , and  $\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{YX}^\top$ .  
 (b) For constants  $\mathbf{A}$ ,  $\mathbf{B}$ ,  $\mathbf{c}$  of conformable sizes,  $\text{Cov}(\mathbf{AY}, \mathbf{BX}) = \mathbf{A}\boldsymbol{\Sigma}_{YX}\mathbf{B}^\top$ , and

$$\text{Cov}(\mathbf{AY} + \mathbf{BX} + \mathbf{c}) = \mathbf{A}\boldsymbol{\Sigma}_{YY}\mathbf{A}^\top + \mathbf{A}\boldsymbol{\Sigma}_{YX}\mathbf{B}^\top + \mathbf{B}\boldsymbol{\Sigma}_{XY}\mathbf{A}^\top + \mathbf{B}\boldsymbol{\Sigma}_{XX}\mathbf{B}^\top.$$

- (c)  $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$  and  $\mathcal{C}(\boldsymbol{\Sigma}_{YX}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{YY})$ . Equivalently, for every generalized inverse  $\boldsymbol{\Sigma}_{XX}^-$ ,

$$\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XY}, \quad \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX},$$

and similarly with the roles of  $\mathbf{Y}$  and  $\mathbf{X}$  exchanged.

*Proof.*

- (a) The outer product  $(\mathbf{W} - \mathbf{E}\mathbf{W})(\mathbf{W} - \mathbf{E}\mathbf{W})^\top$  has blocks  $(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{Y} - \boldsymbol{\mu}_Y)^\top$ ,  $(\mathbf{Y} - \boldsymbol{\mu}_Y)(\mathbf{X} - \boldsymbol{\mu}_X)^\top$ , its transpose, and  $(\mathbf{X} - \boldsymbol{\mu}_X)(\mathbf{X} - \boldsymbol{\mu}_X)^\top$ . Take expectations blockwise. (b)  $\mathbf{A}\mathbf{Y} + \mathbf{B}\mathbf{X} + \mathbf{c} = [\mathbf{A}, \mathbf{B}]\mathbf{W} + \mathbf{c}$ , and  $\mathbf{A}\mathbf{Y} = [\mathbf{A}, \mathbf{0}]\mathbf{W}$ ,  $\mathbf{B}\mathbf{X} = [\mathbf{0}, \mathbf{B}]\mathbf{W}$ . Apply [Theorem 2.2.1](#) and multiply out the blocks.
- (b) Let  $\mathbf{a} \in \mathcal{N}(\boldsymbol{\Sigma}_{XX})$ . By [Theorem 2.2.3\(c\)](#),  $\mathbf{a}^\top \mathbf{X}$  is constant with probability one, so its covariance with  $\mathbf{Y}$  vanishes:  $\boldsymbol{\Sigma}_{YX}\mathbf{a} = \text{Cov}(\mathbf{Y}, \mathbf{a}^\top \mathbf{X}) = \mathbf{0}$ . Thus  $\mathcal{N}(\boldsymbol{\Sigma}_{XX}) \subseteq \mathcal{N}(\boldsymbol{\Sigma}_{YX})$ . Taking orthogonal complements, and using  $\mathcal{N}(\mathbf{B})^\perp = \mathcal{C}(\mathbf{B}^\top)$  and the symmetry of  $\boldsymbol{\Sigma}_{XX}$ , gives  $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$ . The same argument with  $\mathbf{Y}$  and  $\mathbf{X}$  exchanged gives the second inclusion. A column inclusion  $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$  means  $\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XX}\mathbf{K}^\top$  for some  $\mathbf{K}$ . Then  $\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY} = \boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX}\mathbf{K}^\top = \boldsymbol{\Sigma}_{XY}$ . Conversely, that identity exhibits every column of  $\boldsymbol{\Sigma}_{XY}$  as a combination of columns of  $\boldsymbol{\Sigma}_{XX}$  ([Theorem 1.4.1](#)). Transposing gives  $\boldsymbol{\Sigma}_{YX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}$  and the second identity. ■

Part (c) looks technical, but it removes every difficulty with singular  $\boldsymbol{\Sigma}_{XX}$  in what follows. A direction in which  $\mathbf{X}$  does not vary cannot covary with anything.

## 2.5.2 Linear prediction and residual covariance

The most important use of the blocks is to split  $\mathbf{Y}$  into a part explained linearly by  $\mathbf{X}$  and a part uncorrelated with  $\mathbf{X}$ .

### Theorem 2.5.2 — Best linear predictor

In the setting of [Theorem 2.5.1](#), fix a generalized inverse  $\boldsymbol{\Sigma}_{XX}^-$  and put

$$\mathbf{B}_* = \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-, \quad \mathbf{E} = \mathbf{Y} - \boldsymbol{\mu}_Y - \mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X), \quad \boldsymbol{\Sigma}_{YY \cdot X} = \boldsymbol{\Sigma}_{YY} - \boldsymbol{\Sigma}_{YX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XY}.$$

- (a)  $\mathbf{E}(\mathbf{E}) = \mathbf{0}$  and  $\text{Cov}(\mathbf{E}, \mathbf{X}) = \mathbf{0}$ .
- (b)  $\text{Cov}(\mathbf{E}) = \boldsymbol{\Sigma}_{YY \cdot X}$ . This matrix is nonnegative definite and does not depend on the choice of generalized inverse. Neither does  $\mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X)$ , with probability one.
- (c) For every constant  $p \times q$  matrix  $\mathbf{B}$  and  $p \times 1$  vector  $\mathbf{c}$ ,

$$\mathbf{E}[(\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X})(\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X})^\top] = \boldsymbol{\Sigma}_{YY \cdot X} + (\mathbf{B} - \mathbf{B}_*)\boldsymbol{\Sigma}_{XX}(\mathbf{B} - \mathbf{B}_*)^\top + \mathbf{d}\mathbf{d}^\top,$$

where  $\mathbf{d} = \boldsymbol{\mu}_Y - \mathbf{c} - \mathbf{B}\boldsymbol{\mu}_X$ . So the mean squared error matrix is never smaller, in the nonnegative definite ordering, than  $\boldsymbol{\Sigma}_{YY \cdot X}$ . Equality holds iff  $\mathbf{c} = \boldsymbol{\mu}_Y - \mathbf{B}\boldsymbol{\mu}_X$  and  $\mathbf{B}\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$ .

*Proof.* By [Theorem 2.5.1\(c\)](#), write  $\boldsymbol{\Sigma}_{YX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}$ . Then  $\mathbf{B}_*\boldsymbol{\Sigma}_{XX} = \mathbf{K}\boldsymbol{\Sigma}_{XX}\boldsymbol{\Sigma}_{XX}^-\boldsymbol{\Sigma}_{XX} = \mathbf{K}\boldsymbol{\Sigma}_{XX} = \boldsymbol{\Sigma}_{YX}$ .

- (a) The mean is clear. By [Theorem 2.5.1\(b\)](#),  $\text{Cov}(\mathbf{E}, \mathbf{X}) = \boldsymbol{\Sigma}_{YX} - \mathbf{B}_*\boldsymbol{\Sigma}_{XX} = \mathbf{0}$ .
- (b) Again by [Theorem 2.5.1\(b\)](#),  $\text{Cov}(\mathbf{E}) = \boldsymbol{\Sigma}_{YY} - \mathbf{B}_*\boldsymbol{\Sigma}_{XY} - \boldsymbol{\Sigma}_{YX}\mathbf{B}_*^\top + \mathbf{B}_*\boldsymbol{\Sigma}_{XX}\mathbf{B}_*^\top$ . Us-

ing  $\mathbf{B}_* \Sigma_{XX} = \Sigma_{YX}$ , the last term is  $\Sigma_{YX} \mathbf{B}_*^\top$ , which cancels the third, so  $\text{Cov}(\mathbf{E}) = \Sigma_{YY} - \mathbf{B}_* \Sigma_{XY} = \Sigma_{YY.X}$ . Next,  $\Sigma_{YX} \Sigma_{XX}^{-1} \Sigma_{XY} = \mathbf{K} \Sigma_{XX} \Sigma_{XX}^{-1} \Sigma_{XX} \mathbf{K}^\top = \mathbf{K} \Sigma_{XX} \mathbf{K}^\top$ , which involves no generalized inverse. A covariance matrix is nonnegative definite. Finally, by [Theorem 2.2.3\(d\)](#),  $\mathbf{X} - \boldsymbol{\mu}_X = \Sigma_{XX} \mathbf{v}$  for some random  $\mathbf{v}$ , with probability one, and then  $\mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X) = \Sigma_{YX} \mathbf{v}$  for every choice of generalized inverse.

- (c) Decompose  $\mathbf{Y} - \mathbf{c} - \mathbf{B}\mathbf{X} = \mathbf{E} + (\mathbf{B}_* - \mathbf{B})(\mathbf{X} - \boldsymbol{\mu}_X) + \mathbf{d}$ . The three pieces are, respectively, mean zero, mean zero and constant, and the first two are uncorrelated by (a). So the cross products have zero expectation, and the expected outer product is  $\text{Cov}(\mathbf{E}) + (\mathbf{B}_* - \mathbf{B}) \Sigma_{XX} (\mathbf{B}_* - \mathbf{B})^\top + \mathbf{d} \mathbf{d}^\top$ . Both added matrices are nonnegative definite. They vanish iff  $\mathbf{d} = \mathbf{0}$  and  $(\mathbf{B} - \mathbf{B}_*) \Sigma_{XX}^{1/2} = \mathbf{0}$ , and the latter is equivalent to  $(\mathbf{B} - \mathbf{B}_*) \Sigma_{XX} = \mathbf{0}$ , that is, to  $\mathbf{B} \Sigma_{XX} = \Sigma_{YX}$ .

■

The affine function  $\boldsymbol{\mu}_Y + \mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X)$  is the **best linear predictor** of  $\mathbf{Y}$  from  $\mathbf{X}$ . Because the ordering in (c) is the nonnegative definite ordering, it is simultaneously best for every linear combination  $\mathbf{a}^\top \mathbf{Y}$  and for the total squared error. The matrix  $\Sigma_{YY.X}$  is the **partial covariance matrix** of  $\mathbf{Y}$  given  $\mathbf{X}$ . When  $\Sigma_{XX}$  is nonsingular,  $\mathbf{B}_* = \Sigma_{YX} \Sigma_{XX}^{-1}$  holds the population regression coefficients, and  $\Sigma_{YY.X}$  is a Schur complement. Two consequences follow from [Chapter 1](#). If  $\text{Cov}(\mathbf{W})$  is positive definite, the  $YY$  block of its inverse is  $\Sigma_{YY.X}^{-1}$  ([Theorem 1.6.4](#)). And  $\det \text{Cov}(\mathbf{W}) = \det(\Sigma_{XX}) \det(\Sigma_{YY.X})$  ([Theorem 1.6.3](#)), so the generalized variance factorizes into the part of  $\mathbf{X}$  and the part of  $\mathbf{Y}$  that  $\mathbf{X}$  leaves unexplained.

Nothing here says that  $E(\mathbf{Y} | \mathbf{X})$  is linear. The best linear predictor is best among affine functions, whatever the joint distribution. For jointly normal vectors, the conditional mean happens to be this affine function, and  $\Sigma_{YY.X}$  is the conditional covariance matrix ([Theorem 3.5.2](#)). The partial correlation of [Definition 3.6.1](#) is the correlation computed from  $\Sigma_{YY.X}$ . [Chapter 6](#) shows that least squares estimates  $\mathbf{B}_*$  in large samples, whether or not the regression function is linear. It also recovers the predictor as an orthogonal projection in the space of square-integrable random variables. For scalar  $Y$ , the coefficient vector used there is  $\mathbf{B}_*^\top$ .

### Proposition 2.5.3 — Multiple correlation

Let  $Y$  be scalar with  $\sigma_{YY} = \text{Var}(Y) > 0$ , let  $\Sigma_{XX}$  be positive definite, and write  $\sigma_{XY} = \text{Cov}(\mathbf{X}, Y)$ . Over all  $\mathbf{a} \neq \mathbf{0}$ ,

$$\max_{\mathbf{a}} \text{corr}(Y, \mathbf{a}^\top \mathbf{X})^2 = \rho_{Y.X}^2 = \frac{\sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY}}{\sigma_{YY}},$$

attained at  $\mathbf{a} = \Sigma_{XX}^{-1} \sigma_{XY}$ , the coefficients of the best linear predictor (if  $\sigma_{XY} = \mathbf{0}$ , every  $\mathbf{a}$  attains the maximum 0). Moreover  $1 - \rho_{Y.X}^2 = \sigma_{YY.X} / \sigma_{YY}$ .

*Proof.*  $\text{corr}(Y, \mathbf{a}^\top \mathbf{X})^2 = (\mathbf{a}^\top \sigma_{XY})^2 / (\sigma_{YY} \mathbf{a}^\top \Sigma_{XX} \mathbf{a})$ . Write  $\mathbf{a}^\top \sigma_{XY} = (\Sigma_{XX}^{1/2} \mathbf{a})^\top (\Sigma_{XX}^{-1/2} \sigma_{XY})$ . The Cauchy–Schwarz inequality gives  $(\mathbf{a}^\top \sigma_{XY})^2 \leq (\mathbf{a}^\top \Sigma_{XX} \mathbf{a}) (\sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY})$ , with equality iff  $\Sigma_{XX}^{1/2} \mathbf{a}$  is proportional to  $\Sigma_{XX}^{-1/2} \sigma_{XY}$ . Dividing gives the bound and the maximizer. The last identity is the definition of  $\sigma_{YY.X} = \sigma_{YY} - \sigma_{XY}^\top \Sigma_{XX}^{-1} \sigma_{XY}$ , divided by  $\sigma_{YY}$ .

■



$\rho_{Y \cdot X}^2$  is the population version of  $R^2$ : the fraction of the variance of  $Y$  that the best linear predictor accounts for. [Chapter 14](#) studies its sample version.

## 2.5.3 Conditioning: total expectation and total covariance

When the joint distribution is available, conditioning gives another decomposition of the covariance matrix. Conditional expectations and covariances are taken entry by entry, as in [Definition 2.1.1](#), and  $\text{Cov}(\mathbf{Y} \mid \mathbf{X}) = \mathbb{E}[(\mathbf{Y} - \mathbb{E}(\mathbf{Y} \mid \mathbf{X}))(\mathbf{Y} - \mathbb{E}(\mathbf{Y} \mid \mathbf{X}))^\top \mid \mathbf{X}]$ .

### Proposition 2.5.4 — Laws of total expectation and covariance

If  $\mathbf{Y}$  has finite second moments, then  $\mathbb{E}(\mathbf{Y}) = \mathbb{E}[\mathbb{E}(\mathbf{Y} \mid \mathbf{X})]$  and

$$\text{Cov}(\mathbf{Y}) = \mathbb{E}[\text{Cov}(\mathbf{Y} \mid \mathbf{X})] + \text{Cov}(\mathbb{E}(\mathbf{Y} \mid \mathbf{X})).$$

*Proof.* The first identity is the tower property of conditional expectation, applied to each component. For the second, let  $\mathbf{m}(\mathbf{X}) = \mathbb{E}(\mathbf{Y} \mid \mathbf{X})$  and write  $\mathbf{Y} - \boldsymbol{\mu} = (\mathbf{Y} - \mathbf{m}(\mathbf{X})) + (\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})$ . The outer product has two cross terms. The conditional expectation of  $(\mathbf{Y} - \mathbf{m}(\mathbf{X}))(\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})^\top$  given  $\mathbf{X}$  is  $[\mathbb{E}(\mathbf{Y} \mid \mathbf{X}) - \mathbf{m}(\mathbf{X})](\mathbf{m}(\mathbf{X}) - \boldsymbol{\mu})^\top = \mathbf{0}$ , so its expectation is zero, and likewise for the other. The expectation of  $(\mathbf{Y} - \mathbf{m})(\mathbf{Y} - \mathbf{m})^\top$  is  $\mathbb{E}[\text{Cov}(\mathbf{Y} \mid \mathbf{X})]$  by the tower property, and that of  $(\mathbf{m} - \boldsymbol{\mu})(\mathbf{m} - \boldsymbol{\mu})^\top$  is  $\text{Cov}(\mathbf{m}(\mathbf{X}))$  because  $\mathbb{E} \mathbf{m}(\mathbf{X}) = \boldsymbol{\mu}$ . ■

### Example. (A shared random effect)

Measurements  $Y_1, \dots, Y_k$  on the same subject share a subject effect:  $\mathbf{Y} = \alpha \mathbf{1} + \boldsymbol{\varepsilon}$ , where  $\alpha$  has variance  $\tau^2$ ,  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_k$ , and  $\boldsymbol{\varepsilon}$  is independent of  $\alpha$  with mean zero. Given  $\alpha$ , the mean is  $\alpha \mathbf{1}$  and the covariance is  $\sigma^2 \mathbf{I}$ . By [Proposition 2.5.4](#),

$$\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_k + \tau^2 \mathbf{1} \mathbf{1}^\top,$$

so every pair of measurements has correlation  $\tau^2/(\tau^2 + \sigma^2)$ . Independent errors combined with a shared random component produce exactly this *equicorrelation* structure. It is the starting point of [Chapter 32](#). Ordinary least squares ignores it, with consequences that [Chapter 33](#) examines ([Proposition 33.2.1](#)).

## 2.5.4 Exercises

### 2.5.4.1 A. Check your understanding

#### Exercise A1.

For scalar  $Y$  and  $X$  with  $\text{Var}(X) > 0$ , show that the best linear predictor has slope

$\sigma_{YX}/\sigma_{XX}$ , and that its mean squared error is  $\sigma_{YY}(1 - \rho^2)$ .

### Exercise A2.

Let  $X$  be uniform on  $(-1, 1)$  and  $Y = X^2 + \varepsilon$ , with  $\varepsilon$  independent of  $X$ , mean 0 and variance  $\sigma^2$ . Find the best linear predictor of  $Y$  from  $X$  and its mean squared error, and compare with the mean squared error of  $E(Y | X)$ .

## 2.5.4.2 B. Practice

### Exercise B1.

Let  $(Y_1, Y_2, Y_3)$  have correlation matrix  $\mathbf{R}$ . Compute the partial covariance matrix of  $(Y_2, Y_3)$  given  $Y_1$  from [Theorem 2.5.2](#), and show that the corresponding correlation is

$$\frac{\rho_{23} - \rho_{12}\rho_{13}}{\sqrt{(1 - \rho_{12}^2)(1 - \rho_{13}^2)}}.$$

### Exercise B2.

Let  $\mathbf{X} = (X_1, X_1)^\top$  with  $\text{Var}(X_1) = 1$ , and let  $\text{Cov}(Y, X_1) = c$ . Find two different generalized inverses of  $\Sigma_{XX}$  that give different  $\mathbf{B}_*$ , and show that both give the same predictor  $\mathbf{B}_*(\mathbf{X} - \boldsymbol{\mu}_X)$ .

### Exercise B3.

A random label  $K$  takes values  $1, \dots, g$  with probabilities  $\pi_k$ . Given  $K = k$ ,  $\mathbf{Y}$  has mean  $\mathbf{m}_k$  and covariance  $\Sigma_k$ . Show that  $\text{Cov}(\mathbf{Y}) = \sum_k \pi_k \Sigma_k + \sum_k \pi_k (\mathbf{m}_k - \bar{\mathbf{m}})(\mathbf{m}_k - \bar{\mathbf{m}})^\top$ , where  $\bar{\mathbf{m}} = \sum_k \pi_k \mathbf{m}_k$ . Interpret the two terms as within-group and between-group covariance.

## 2.5.4.3 C. Going deeper

### Exercise C1.

Let  $\mathbf{N}$  be the symmetric partitioned matrix

$$\mathbf{N} = \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix}.$$

Show that  $\mathbf{N}$  is nonnegative definite iff  $\Sigma_{XX}$  is nonnegative definite,  $\mathcal{C}(\Sigma_{XY}) \subseteq \mathcal{C}(\Sigma_{XX})$ , and  $\Sigma_{YY} - \Sigma_{YX}\Sigma_{XX}^- \Sigma_{XY}$  is nonnegative definite for some generalized inverse  $\Sigma_{XX}^-$  (equivalently, given the column-space condition, for every one). Show by a  $2 \times 2$  example that the column-space condition cannot be dropped.

*Solution.* If the matrix is nonnegative definite, it is the covariance matrix of some  $\mathbf{W} = (\mathbf{Y}^\top, \mathbf{X}^\top)^\top$  ([Theorem 2.2.3\(b\)](#)). Then  $\Sigma_{XX}$  is a covariance matrix, the column condition is [Theorem 2.5.1\(c\)](#), and the Schur complement is  $\text{Cov}(\mathbf{E})$  by [Theorem 2.5.2\(b\)](#).

Conversely, the column condition gives  $\Sigma_{YX} = \mathbf{K}\Sigma_{XX}$ , and then  $\Sigma_{YX}\Sigma_{XX}^{-}\Sigma_{XY} = \mathbf{K}\Sigma_{XX}\mathbf{K}^{\top}$ . With the nonsingular  $\mathbf{T} = \begin{pmatrix} \mathbf{I} & -\mathbf{K} \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$ ,

$$\mathbf{T} \begin{pmatrix} \Sigma_{YY} & \Sigma_{YX} \\ \Sigma_{XY} & \Sigma_{XX} \end{pmatrix} \mathbf{T}^{\top} = \begin{pmatrix} \Sigma_{YY} - \mathbf{K}\Sigma_{XX}\mathbf{K}^{\top} & \mathbf{0} \\ \mathbf{0} & \Sigma_{XX} \end{pmatrix},$$

which is nonnegative definite. So is the original matrix, since  $\mathbf{a}^{\top}\mathbf{N}\mathbf{a} = \mathbf{b}^{\top}(\mathbf{T}\mathbf{N}\mathbf{T}^{\top})\mathbf{b}$  with  $\mathbf{b} = (\mathbf{T}^{\top})^{-1}\mathbf{a}$ . For the example, take  $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ . Here  $\Sigma_{XX} = 0$  is nonnegative definite, and with the generalized inverse 0 the Schur complement is  $1 \geq 0$ . But the determinant is  $-1$ , so the matrix is not nonnegative definite. The column condition fails, because  $\mathcal{C}(1) \not\subseteq \mathcal{C}(0)$ . Without that condition the Schur complement also depends on the choice: every scalar  $t$  is a generalized inverse of 0, and the Schur complement is then  $1 - 1 \cdot t \cdot 1 = 1 - t$ , which is negative for  $t > 1$ . Under the column condition,  $\Sigma_{YX}\Sigma_{XX}^{-}\Sigma_{XY} = \mathbf{K}\Sigma_{XX}\mathbf{K}^{\top}$  for every choice, as shown above.

## 2.6 Moment generating functions and independence

Means and covariance matrices summarize a distribution. They do not determine it, and they cannot certify independence. For that we need a transform of the whole distribution. This section collects the facts that [Chapter 3](#) and [Chapter 4](#) use.

### Definition 2.6.1 — Moment generating function

The **moment generating function** of a  $p \times 1$  random vector  $\mathbf{Y}$  is

$$M_{\mathbf{Y}}(\mathbf{t}) = \mathbb{E} \exp(\mathbf{t}^{\top} \mathbf{Y}), \quad \mathbf{t} \in \mathbb{R}^p,$$

where the expectation may be  $+\infty$ . We say that  $M_{\mathbf{Y}}$  *exists* if it is finite for all  $\mathbf{t}$  in some neighbourhood  $\{\|\mathbf{t}\| < h\}$  of the origin,  $h > 0$ .

### Theorem 2.6.1 — Properties of moment generating functions

Suppose  $M_{\mathbf{Y}}$  exists.

- (a) *Uniqueness.* If  $M_{\mathbf{V}}$  also exists and  $M_{\mathbf{Y}} = M_{\mathbf{V}}$  on a neighbourhood of  $\mathbf{0}$ , then  $\mathbf{Y}$  and  $\mathbf{V}$  have the same distribution.
- (b) *Moments.* All moments of  $\mathbf{Y}$  are finite. The gradient of  $M_{\mathbf{Y}}$  at  $\mathbf{0}$  is  $\mathbb{E}(\mathbf{Y})$ , and its Hessian at  $\mathbf{0}$  is  $\mathbb{E}(\mathbf{Y}\mathbf{Y}^{\top})$ . The Hessian of  $\log M_{\mathbf{Y}}$  at  $\mathbf{0}$  is  $\text{Cov}(\mathbf{Y})$ .
- (c) *Affine maps.* For a constant  $k \times p$  matrix  $\mathbf{A}$  and  $k \times 1$  vector  $\mathbf{b}$ ,  $M_{\mathbf{A}\mathbf{Y}+\mathbf{b}}(\mathbf{s}) = \exp(\mathbf{s}^{\top} \mathbf{b}) M_{\mathbf{Y}}(\mathbf{A}^{\top} \mathbf{s})$ .
- (d) *Independence.* Partition  $\mathbf{Y} = (\mathbf{Y}_1^{\top}, \mathbf{Y}_2^{\top})^{\top}$  and  $\mathbf{t} = (\mathbf{t}_1^{\top}, \mathbf{t}_2^{\top})^{\top}$  conformably. Then  $\mathbf{Y}_1$  and  $\mathbf{Y}_2$  are independent iff  $M_{\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{Y}_1}(\mathbf{t}_1) M_{\mathbf{Y}_2}(\mathbf{t}_2)$  for all  $\mathbf{t}$  in a neighbourhood of  $\mathbf{0}$ .

*Proof.* For (a), the scalar case is a theorem of analysis: a moment generating function that is finite near the origin extends analytically, which pins down the characteristic function and hence the distribution (Billingsley 1995). For vectors, fix  $\mathbf{a}$ . For small  $|s|$ ,  $M_{\mathbf{a}^\top \mathbf{Y}}(s) = M_{\mathbf{Y}}(\mathbf{s}\mathbf{a}) = M_{\mathbf{V}}(\mathbf{s}\mathbf{a}) = M_{\mathbf{a}^\top \mathbf{V}}(s)$ , so  $\mathbf{a}^\top \mathbf{Y}$  and  $\mathbf{a}^\top \mathbf{V}$  have the same distribution by the scalar case, and Proposition 2.6.2 below finishes the argument. For (b), finiteness near  $\mathbf{0}$  implies  $E \exp(h'|Y_i|) < \infty$  for some  $h' > 0$  and each  $i$ , so every polynomial in the components is integrable and differentiation under the expectation is justified. Then  $\partial M / \partial t_i = E(Y_i e^{\mathbf{t}^\top \mathbf{Y}})$  and  $\partial^2 M / \partial t_i \partial t_j = E(Y_i Y_j e^{\mathbf{t}^\top \mathbf{Y}})$ . At  $\mathbf{t} = \mathbf{0}$  these are  $E(Y_i)$  and  $E(Y_i Y_j)$ . For  $K = \log M$ , the chain rule gives  $\nabla^2 K = M^{-1} \nabla^2 M - M^{-2} (\nabla M)(\nabla M)^\top$ , which at  $\mathbf{0}$ , where  $M = 1$ , is  $E(\mathbf{Y}\mathbf{Y}^\top) - \boldsymbol{\mu}\boldsymbol{\mu}^\top = \text{Cov}(\mathbf{Y})$ . For (c),  $E \exp(\mathbf{s}^\top (\mathbf{A}\mathbf{Y} + \mathbf{b})) = e^{\mathbf{s}^\top \mathbf{b}} E \exp((\mathbf{A}^\top \mathbf{s})^\top \mathbf{Y})$ . For (d), independence gives  $E(e^{\mathbf{t}_1^\top \mathbf{Y}_1} e^{\mathbf{t}_2^\top \mathbf{Y}_2}) = E(e^{\mathbf{t}_1^\top \mathbf{Y}_1}) E(e^{\mathbf{t}_2^\top \mathbf{Y}_2})$ . Conversely, let  $\mathbf{V}_1$  and  $\mathbf{V}_2$  be independent with the distributions of  $\mathbf{Y}_1$  and  $\mathbf{Y}_2$ . The vector  $(\mathbf{V}_1^\top, \mathbf{V}_2^\top)^\top$  has moment generating function  $M_{\mathbf{Y}_1}(\mathbf{t}_1) M_{\mathbf{Y}_2}(\mathbf{t}_2)$ , which by hypothesis equals  $M_{\mathbf{Y}}$  near  $\mathbf{0}$ . By (a) it has the same joint distribution as  $\mathbf{Y}$ , so  $\mathbf{Y}_1$  and  $\mathbf{Y}_2$  are independent. ■

The next fact explains why Chapter 3 can define the multivariate normal through linear combinations.

### Proposition 2.6.2 — Distributions are determined by linear combinations

If  $\mathbf{a}^\top \mathbf{Y}$  and  $\mathbf{a}^\top \mathbf{V}$  have the same distribution for every  $\mathbf{a} \in \mathbb{R}^p$ , then  $\mathbf{Y}$  and  $\mathbf{V}$  have the same distribution.

*Proof.* The characteristic function  $\phi_{\mathbf{Y}}(\mathbf{t}) = E \exp(i\mathbf{t}^\top \mathbf{Y})$  exists for every random vector and determines its distribution (Billingsley 1995). Since  $\phi_{\mathbf{Y}}(\mathbf{t}) = \phi_{\mathbf{t}^\top \mathbf{Y}}(1) = \phi_{\mathbf{t}^\top \mathbf{V}}(1) = \phi_{\mathbf{V}}(\mathbf{t})$  for every  $\mathbf{t}$ , the two distributions coincide. ■

If  $\mathbf{Y}_1$  and  $\mathbf{Y}_2$  are independent, so are  $g(\mathbf{Y}_1)$  and  $h(\mathbf{Y}_2)$  for measurable  $g$  and  $h$ . Independence of statistics is usually proved this way. For normal vectors, Theorem 3.4.1 reduces the independence of two blocks to a zero cross-covariance, and Theorem 4.4.1 and Theorem 4.4.2 build on that.

Independence of *several* vectors means that the joint distribution factors into all the marginals, not merely that each pair is independent.

### Example. (Pairwise but not mutually independent)

Let  $S_1$  and  $S_2$  be independent, each equal to  $\pm 1$  with probability  $\frac{1}{2}$ , and put  $S_3 = S_1 S_2$ . Each  $S_j$  is  $\pm 1$  with probability  $\frac{1}{2}$ . Each pair is independent: for instance,  $\Pr(S_1 = s, S_3 = u) = \Pr(S_1 = s, S_2 = su) = \frac{1}{4}$  for all signs  $s, u$ . So  $\text{Cov}(\mathbf{S}) = \mathbf{I}_3$ . But the three are not mutually independent, since  $S_3$  is a function of the other two. A third-order moment shows the failure directly:  $E(S_1 S_2 S_3) = 1$ , whereas mutual independence would force it to be 0. No condition on the covariance matrix can detect this, because covariances involve only pairs.

## 2.6.1 Exercises

### 2.6.1.1 A. Check your understanding

#### Exercise A1.

Find the moment generating function of the multinomial count vector of [Example \(Multinomial counts\)](#), and recover its mean and covariance matrix from [Theorem 2.6.1\(b\)](#).

### 2.6.1.2 B. Practice

#### Exercise B1.

Let  $S_1, \dots, S_k$  be independent random signs. For each nonempty  $A \subseteq \{1, \dots, k\}$  put  $T_A = \prod_{i \in A} S_i$ . Show that the  $2^k - 1$  variables  $T_A$  are pairwise independent with covariance matrix  $\mathbf{I}$ , and find a triple of them that is not mutually independent.

## 2.7 Sample moments and their computation

The population quantities of this chapter have sample analogues, and the operator rules give their expectations at once. Let  $\mathbf{y}_1, \dots, \mathbf{y}_n$  be  $p \times 1$  observations, and stack them as the rows of the  $n \times p$  data matrix  $\mathbf{Y} = [\mathbf{y}_1, \dots, \mathbf{y}_n]^\top$ . The script letter keeps the data matrix apart from a single random vector  $\mathbf{Y}$ . The rows keep the lowercase  $\mathbf{y}_i$  of observed data, but in [Proposition 2.7.1](#) they are treated as random. With  $\mathbf{C} = \mathbf{I}_n - n^{-1}\mathbf{1}\mathbf{1}^\top$ , the sample mean vector and the **sample covariance matrix** are

$$\bar{\mathbf{y}} = \frac{1}{n} \sum_{i=1}^n \mathbf{y}_i = \frac{1}{n} \mathbf{Y}^\top \mathbf{1}, \quad \mathbf{S} = \frac{1}{n-1} \sum_{i=1}^n (\mathbf{y}_i - \bar{\mathbf{y}})(\mathbf{y}_i - \bar{\mathbf{y}})^\top = \frac{1}{n-1} \mathbf{Y}^\top \mathbf{C} \mathbf{Y}.$$

The second form of  $\mathbf{S}$  holds because  $\mathbf{C}\mathbf{Y}$  is the data matrix with its column means subtracted, and  $\mathbf{C}$  is symmetric and idempotent. The sample correlation matrix is  $\hat{\mathbf{R}} = \hat{\mathbf{D}}^{-1} \mathbf{S} \hat{\mathbf{D}}^{-1}$ , where  $\hat{\mathbf{D}}$  holds the sample standard deviations.

### Proposition 2.7.1 — Moments of sample moments

Suppose the rows  $\mathbf{y}_i$  are pairwise uncorrelated random vectors with common mean  $\boldsymbol{\mu}$  and common covariance matrix  $\boldsymbol{\Sigma}$ . Then

- (a)  $E(\bar{\mathbf{y}}) = \boldsymbol{\mu}$  and  $\text{Cov}(\bar{\mathbf{y}}) = \boldsymbol{\Sigma}/n$ ;
- (b)  $E(\mathbf{S}) = \boldsymbol{\Sigma}$ ;
- (c)  $\mathbf{S}$  is nonnegative definite, and  $\text{rank}(\mathbf{S}) \leq \min(p, n-1)$  for every realization.

*Proof.*

- (a) is [Proposition 2.2.2\(c\)](#) with  $c_i = 1/n$ . For (b), let  $\mathbf{u}_j$  denote the  $j$ th column of  $\mathbf{Y}$ , the  $n$  observations of variable  $j$ . The  $(j, k)$  entry of  $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$  is the bilinear form  $\mathbf{u}_j^\top \mathbf{C} \mathbf{u}_k$ . Because different rows are uncorrelated,  $\text{Cov}(\mathbf{u}_k, \mathbf{u}_j) = \sigma_{kj} \mathbf{I}_n$ , and

$E(\mathbf{u}_j) = \mu_j \mathbf{1}$ . By [Corollary 2.3.2\(d\)](#),

$$E(\mathbf{u}_j^\top \mathbf{C} \mathbf{u}_k) = \sigma_{kj} \operatorname{tr}(\mathbf{C}) + \mu_j \mu_k \mathbf{1}^\top \mathbf{C} \mathbf{1} = (n-1) \sigma_{jk},$$

since  $\operatorname{tr}(\mathbf{C}) = n-1$  and  $\mathbf{C} \mathbf{1} = \mathbf{0}$ . (c) For any  $\mathbf{a}$ ,  $\mathbf{a}^\top \mathbf{S} \mathbf{a} = \|\mathbf{C} \mathbf{y} \mathbf{a}\|^2 / (n-1) \geq 0$ . The rank of  $\mathbf{y}^\top \mathbf{C} \mathbf{y} = (\mathbf{C} \mathbf{y})^\top (\mathbf{C} \mathbf{y})$  equals  $\operatorname{rank}(\mathbf{C} \mathbf{y})$ , which is at most  $\operatorname{rank}(\mathbf{C}) = n-1$  and at most  $p$ . ■

The divisor  $n-1$  is the trace of  $\mathbf{C}$ , the dimension of the space the centred data live in, exactly as  $n-r$  is for regression residuals in [Example \(Residual sum of squares under a wrong mean\)](#). Part (c) has a practical side. With  $n \leq p$ , as in genomics or finance, the sample covariance matrix is always singular, even when  $\mathbf{\Sigma}$  is positive definite. It has a linear combination with zero sample variance whether or not one exists in the population. [Chapter 28](#) returns to this.

**Computation.** In practice  $\mathbf{C}$  is never formed: one subtracts column means. The eigenvalues of  $\mathbf{S}$  are the squared singular values of  $\mathbf{C} \mathbf{y} / \sqrt{n-1}$  ([Theorem 1.8.1](#)), which can be computed without forming a cross-product matrix ([Chapter 10](#)).

#### Example. (Macroeconomic series)

Longley's data give  $n = 16$  annual observations (1947–1962) on  $p = 7$  US series: total employment, the GNP deflator, GNP, unemployment, armed forces, population and the year. The eigenvalues of  $\mathbf{S}$  run from  $9.94 \times 10^9$  down to  $1.07 \times 10^{-2}$ , a ratio of  $9.3 \times 10^{11}$ . That spread says little about the structure of the data. It mostly reflects units: the standard deviation of GNP is 99395, that of the deflator 10.79.

The correlation matrix removes the units ([Proposition 2.4.1\(d\)](#)). Its eigenvalues sum to  $p = 7$ . The largest is 5.533, so one direction carries about 79% of the standardized variation, and the smallest is  $2.59 \times 10^{-4}$ . By [Theorem 1.7.7](#), the unit vector  $\mathbf{v}$  belonging to the smallest eigenvalue minimizes the sample variance of  $\mathbf{v}^\top \mathbf{z}$  over standardized combinations. Its two largest weights are 0.681 on year and  $-0.674$  on GNP, and the combination has standard deviation 0.016, against 1 for each standardized variable. In these sixteen years GNP is almost an exact linear function of time (their correlation is 0.995). This is the sample version of [Theorem 2.2.3\(c\)](#): a nearly zero eigenvalue marks a nearly constant linear combination. For regression it signals collinearity, which [Chapter 26](#) studies. Finally, the first 5 rows alone give a sample covariance matrix of rank 4, as [Proposition 2.7.1\(c\)](#) requires.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
names = list(data.columns)

Y = data.to_numpy(dtype=float)           # n x p data matrix, one row per
    year
n, p = Y.shape
C = np.eye(n) - np.ones((n, n)) / n     # centring matrix I - J/n
S = Y.T @ C @ Y / (n - 1)               # sample covariance matrix
d = np.sqrt(np.diag(S))
R = S / np.outer(d, d)                  # D^{-1} S D^{-1}, D = diag(d)
```

```

lam_S = np.linalg.eigvalsh(S)
lam_R, V = np.linalg.eigh(R)                                # ascending eigenvalues
print("eigenvalues of S:", lam_S)
print("eigenvalues of R:", lam_R)
v = V[:, 0]                                                  # direction of least variance
Z = (Y - Y.mean(0)) / d                                      # standardized variables
print("sample variance of Z v:", np.var(Z @ v, ddof=1))

```

## 2.7.1 Exercises

### 2.7.1.1 B. Practice

#### Exercise B1.

Suppose the rows of  $\mathbf{Y}$  are observed in time order, with common mean  $\boldsymbol{\mu}$  and covariance  $\boldsymbol{\Sigma}$ , and that neighbouring rows are correlated:  $\text{Cov}(\mathbf{y}_i, \mathbf{y}_{i+1}) = \rho\boldsymbol{\Sigma}$ , while  $\text{Cov}(\mathbf{y}_i, \mathbf{y}_j) = \mathbf{0}$  for  $|i - j| \geq 2$ . Show that  $E(\mathbf{S}) = (1 - 2\rho/n)\boldsymbol{\Sigma}$  and  $\text{Cov}(\bar{\mathbf{y}}) = n^{-2}[n + 2(n - 1)\rho]\boldsymbol{\Sigma}$ . Which of the two effects disappears as  $n \rightarrow \infty$ ?

*Solution.* As in the proof of [Proposition 2.7.1](#),  $\text{Cov}(\mathbf{u}_k, \mathbf{u}_j) = \sigma_{kj}\mathbf{T}$ , where  $\mathbf{T}$  is the  $n \times n$  tridiagonal matrix with ones on the diagonal and  $\rho$  next to it. Since  $\mathbf{C}\mathbf{1} = \mathbf{0}$ , [Corollary 2.3.2\(d\)](#) gives  $E(\mathbf{u}_j^\top \mathbf{C}\mathbf{u}_k) = \sigma_{kj} \text{tr}(\mathbf{C}\mathbf{T}) = \sigma_{kj} [\text{tr}(\mathbf{T}) - n^{-1}\mathbf{1}^\top \mathbf{T}\mathbf{1}]$ . Here  $\text{tr}(\mathbf{T}) = n$  and  $\mathbf{1}^\top \mathbf{T}\mathbf{1} = n + 2(n - 1)\rho$ , so the bracket is  $(n - 1) - 2(n - 1)\rho/n$ , and dividing by  $n - 1$  gives  $E(\mathbf{S}) = (1 - 2\rho/n)\boldsymbol{\Sigma}$ . For the mean,  $\bar{\mathbf{y}} = n^{-1} \sum_i \mathbf{y}_i$  and [Proposition 2.2.2](#) gives  $n^{-2}$  times the sum of all  $\text{Cov}(\mathbf{y}_i, \mathbf{y}_j)$ :  $n$  diagonal terms  $\boldsymbol{\Sigma}$  and  $2(n - 1)$  neighbouring terms  $\rho\boldsymbol{\Sigma}$ . The bias of  $\mathbf{S}$  is  $O(1/n)$  and disappears, but  $n \text{Cov}(\bar{\mathbf{y}}) \rightarrow (1 + 2\rho)\boldsymbol{\Sigma}$ , not  $\boldsymbol{\Sigma}$ . For  $\rho > 0$  the naive standard error  $\sqrt{s_{jj}/n}$  stays too small by a fixed factor however large the sample.

## 2.8 Summary and Notes

### 2.8.1 Summary

*Idea.*

1. Expectation of a random matrix is taken entry by entry, and it commutes with constant linear maps, transposes and traces ([Proposition 2.1.1](#)).
2.  $\text{Cov}(\mathbf{Y}) = E[(\mathbf{Y} - \boldsymbol{\mu})(\mathbf{Y} - \boldsymbol{\mu})^\top] = E(\mathbf{Y}\mathbf{Y}^\top) - \boldsymbol{\mu}\boldsymbol{\mu}^\top$ , and  $\text{Cov}(\mathbf{U}, \mathbf{V})$  is the rectangular cross-covariance, with  $\text{Cov}(\mathbf{V}, \mathbf{U}) = \text{Cov}(\mathbf{U}, \mathbf{V})^\top$  ([Definition 2.1.2](#) and [Proposition 2.1.2](#)). Uncorrelated does not imply independent ([Example \(A point on a circle\)](#)).
3. Affine maps:  $E(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$ ,  $\text{Cov}(\mathbf{A}\mathbf{Y} + \mathbf{b}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$ ,  $\text{Cov}(\mathbf{A}\mathbf{U}, \mathbf{B}\mathbf{V}) = \mathbf{A} \text{Cov}(\mathbf{U}, \mathbf{V}) \mathbf{B}^\top$  ([Theorem 2.2.1](#)). In particular  $\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$ .
4. Covariance matrices are exactly the nonnegative definite matrices.  $\boldsymbol{\Sigma} \mathbf{a} = \mathbf{0}$  iff  $\mathbf{a}^\top \mathbf{Y}$  is constant with probability one, and  $\mathbf{Y}$  lives in the flat  $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$  ([Theorem 2.2.3](#)).
5. Whitening by  $\mathbf{W}$  with  $\mathbf{W}\boldsymbol{\Sigma}\mathbf{W}^\top = \mathbf{I}$  is unique up to an orthogonal factor. Ev-

- ery choice gives the same Mahalanobis distance  $(\mathbf{Y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \boldsymbol{\mu})$  (Proposition 2.2.4).
6.  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$  under second moments only (Theorem 2.3.1). Hence  $E(\text{RSS}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2$  (Example (Residual sum of squares under a wrong mean)). The variance of a quadratic form needs third and fourth moments (Theorem 2.3.3).
  7. Correlation matrices are the nonnegative definite matrices with unit diagonal. They are invariant to rescaling, and their entries must be jointly consistent (Proposition 2.4.1 and Proposition 2.4.2).
  8. For partitioned vectors,  $\mathcal{C}(\boldsymbol{\Sigma}_{XY}) \subseteq \mathcal{C}(\boldsymbol{\Sigma}_{XX})$ . The best linear predictor  $\boldsymbol{\mu}_Y + \boldsymbol{\Sigma}_{YX} \boldsymbol{\Sigma}_{XX}^{-1}(\mathbf{X} - \boldsymbol{\mu}_X)$  leaves an error uncorrelated with  $\mathbf{X}$  with covariance matrix given by the Schur complement  $\boldsymbol{\Sigma}_{YY.X}$  (Theorem 2.5.1 and Theorem 2.5.2).
  9. Moment generating functions determine distributions and factor exactly under independence (Theorem 2.6.1). Pairwise independence is weaker than mutual independence (Example (Pairwise but not mutually independent)).
  10. The sample covariance matrix is unbiased under uncorrelated rows and has rank at most  $n - 1$ . A tiny eigenvalue of the sample correlation matrix flags a nearly constant combination (Proposition 2.7.1 and Example (Macroeconomic series)).

## 2.8.2 Notes and sources

**The operator calculus.** Treating expectation and covariance as operators on random vectors and matrices is standard in linear-model texts. Rencher and Schaalje (2008, chapter 3) develops mean vectors, covariance and correlation matrices, partitioned vectors and linear functions in detail and at a gentle pace. Seber and Lee (2003, §§1.4–1.6) give a compact operator treatment, first and second moments of quadratic forms, and moment generating functions with their use in proving independence. This chapter covers the same ground. Rencher and Schaalje already introduce the standardized distance and the transformation by  $\boldsymbol{\Sigma}^{-1/2}$ . The chapter adds the characterization of singular covariance matrices through their column space (Theorem 2.2.3), the non-uniqueness of whitening and the comparison of symmetric and Cholesky whitening, the best linear predictor with generalized inverses, and the sample covariance matrix. Christensen (2020, chapter 1) states the covariance rules briefly in the notation of linear models. For the multivariate setting, Anderson (2003) remains the standard reference.

**Mahalanobis distance and whitening.** The distance Eq. 2.2.2 is due to Mahalanobis (1936), who introduced it to compare populations described by several correlated measurements. Kessy et al. (2018) compare whitening transformations. They show that the symmetric choice  $\boldsymbol{\Sigma}^{-1/2}$  is the one closest to the original data in mean squared error (Exercise (B3)), and set it against Cholesky and principal-component whitening.

**Quadratic forms.** The trace argument for Theorem 2.3.1 is folklore. Theorem 2.3.3 is stated without proof by Atiqullah (1962), who used it to study how non-normal errors affect the residual variance estimate and the  $F$ -test. The kurtosis term depends on  $\mathbf{A}$  only through its diagonal. For the residual sum of squares of a correct model it equals  $(\mu_4 - 3\sigma^4) \sum_i (1 - h_{ii})^2$ , which takes its simplest form,  $(\mu_4 - 3\sigma^4)(n - r)^2/n$ , when all leverages  $h_{ii}$  are equal. These are the quadratically balanced designs of Atiqullah's title. Seber and Lee (2003, theorem 1.6) give a proof by direct expansion. The proof here organizes the same expansion by splitting the form into quadratic, linear and constant parts. Mathai and Provost (1992) is a book-length treatment of moments and distributions of quadratic forms, including general covariance structures.



**Correlation, partial covariance and prediction.** The feasible range in [Proposition 2.4.2](#) is a special case of the fact that a symmetric matrix is nonnegative definite iff a Schur complement is, when the leading block is positive definite. Horn and Johnson (2013) give the general Schur-complement and Loewner-order results used in [Theorem 2.5.2](#). Stating the best linear predictor with a generalized inverse, so that nothing needs  $\Sigma_{XX}$  to be nonsingular, is in the spirit of Rao (1973), who uses generalized inverses throughout.

**Probability background.** Uniqueness of moment generating functions and characteristic functions, and the device of [Proposition 2.6.2](#), are proved in Billingsley (1995). The Longley data come from Longley (1967), who used them to test the numerical accuracy of regression programs. Their near-collinearity, visible in [Example \(Macroeconomic series\)](#), is the reason.

## 2.8.3 References

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# Chapter 3

## The Multivariate Normal Distribution

A random vector is normal when every linear combination of its entries is normal. From that definition come the moment generating function, the density, and closure under all linear maps, singular ones included. Two further properties make the normal distribution the working assumption of linear-model theory: zero covariance between jointly normal blocks means independence, and conditioning one block on another gives a linear regression with constant error variance. Partial correlation, the correlation that remains after linear adjustment, connects these facts to least squares.

**What you need.** [Chapter 1](#) (spectral theorem, square roots, generalized and partitioned inverses), [Chapter 2](#) (mean vectors and covariance matrices). Some examples refer forward to least squares ([Chapter 5](#) and [Chapter 6](#)).

### 3.0.1 Roadmap

- [Definition through linear combinations](#)
- [Moment generating function and density](#)
- [Linear transformations and marginals](#)
- [Independence and zero covariance](#)
- [Conditional distributions](#)
- [Partial correlation](#)
- [Summary and notes](#)

## 3.1 Definition through linear combinations

Nearly every exact distributional statement in the first half of this book has the same form. The response is  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with normal errors, and the object of interest is a linear function of  $\mathbf{Y}$  (a coefficient estimate, a fitted value, a residual) or a quadratic function of  $\mathbf{Y}$  (a sum of squares). A usable notion of normality for random vectors must therefore do two things. It must survive every linear map, including maps that collapse dimensions. And it must say when two linear functions of the same vector are independent.

The residual vector shows why the first requirement has to include singular cases. In a model with  $p$  linearly independent columns its covariance matrix has rank  $n - p$  (Chapter 6), so the residuals have no density on  $\mathbb{R}^n$ . Defining multivariate normality through a density would exclude it. This chapter uses a definition that does not mention densities at all: a random vector is normal when every linear combination of its entries is normal. The density, the moment generating function and the representation  $\boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$  all follow as theorems.

### 3.1.1 The univariate normal, including the degenerate case

For  $\mu \in \mathbb{R}$  and  $\sigma^2 > 0$ , a random variable  $X$  has the  $N(\mu, \sigma^2)$  distribution when it has density  $(2\pi\sigma^2)^{-1/2} \exp\{-(x - \mu)^2/(2\sigma^2)\}$ . We also allow  $\sigma^2 = 0$ , and take  $N(\mu, 0)$  to be the point mass at  $\mu$ . Treating constants as normal variables with zero variance costs nothing and removes a great many special cases later.

#### Lemma 3.1.1

Let  $X \sim N(\mu, \sigma^2)$  with  $\sigma^2 \geq 0$ .

- (a)  $E e^{sX} = \exp(\mu s + \frac{1}{2}\sigma^2 s^2)$  for every real  $s$ .
- (b) A random variable whose moment generating function is  $\exp(\mu s + \frac{1}{2}\sigma^2 s^2)$  on an interval around 0 has the  $N(\mu, \sigma^2)$  distribution.
- (c)  $aX + b \sim N(a\mu + b, a^2\sigma^2)$ , and if  $X_1 \sim N(\mu_1, \sigma_1^2)$  and  $X_2 \sim N(\mu_2, \sigma_2^2)$  are independent, then  $X_1 + X_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$ .

*Proof.*

- (a) For  $\sigma^2 = 0$  this is  $e^{\mu s}$ . Otherwise, completing the square,  $sx - (x - \mu)^2/(2\sigma^2) = \mu s + \sigma^2 s^2/2 - (x - \mu - \sigma^2 s)^2/(2\sigma^2)$ , and the remaining Gaussian integral equals  $(2\pi\sigma^2)^{1/2}$ .
- (b) is the uniqueness theorem for moment generating functions (Theorem 2.6.1(a)), in its scalar form.
- (c) The moment generating function of  $aX + b$  is  $e^{bs} E e^{(as)X} = \exp\{(a\mu + b)s + \frac{1}{2}a^2\sigma^2 s^2\}$ . For the sum, independence gives  $E e^{s(X_1 + X_2)} = E e^{sX_1} E e^{sX_2}$ , which is the normal moment generating function with the added parameters. Apply (b). ■

### 3.1.2 The definition

#### Definition 3.1.1 — Multivariate normal distribution

A random vector  $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$  has a **multivariate normal distribution** if  $\mathbf{a}^\top \mathbf{Y}$  has a univariate normal distribution (possibly degenerate) for every  $\mathbf{a} \in \mathbb{R}^n$ . Such a  $\mathbf{Y}$  is also said to be **jointly normal**, and its entries are said to be jointly normal random

variables.

Taking  $\mathbf{a}$  to be a coordinate vector shows that each  $Y_i$  is normal, so  $E Y_i^2 < \infty$ . By the Cauchy–Schwarz inequality every product  $Y_i Y_j$  is integrable, so the mean vector  $\boldsymbol{\mu} = E \mathbf{Y}$  and the covariance matrix  $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$  of Definition 2.1.2 exist. By Theorem 2.2.1,  $E(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\mu}$  and  $\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$ , and a normal variable is determined by its mean and variance. Hence

$$\mathbf{a}^\top \mathbf{Y} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}) \quad \text{for every } \mathbf{a} \in \mathbb{R}^n. \quad (3.1.1)$$

We write  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ . The notation claims that the joint distribution of  $\mathbf{Y}$  depends only on  $\boldsymbol{\mu}$  and  $\boldsymbol{\Sigma}$ . That is true, but it is not obvious: the definition fixes the law of each one-dimensional projection, and it is not yet clear that these laws pin down the joint law. The claim is proved in Theorem 3.2.1. Until then,  $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  should be read as “normal, with mean  $\boldsymbol{\mu}$  and covariance  $\boldsymbol{\Sigma}$ ”.

### 3.1.3 Existence

#### Theorem 3.1.2 — Existence and the representation $\boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$

Let  $Z_1, \dots, Z_m$  be independent  $N(0, 1)$  variables and  $\mathbf{Z} = (Z_1, \dots, Z_m)^\top$ .

- (a)  $\mathbf{Z} \sim N_m(\mathbf{0}, \mathbf{I}_m)$ .
- (b) For any  $n \times m$  matrix  $\mathbf{A}$  and  $\boldsymbol{\mu} \in \mathbb{R}^n$ ,  $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$  is multivariate normal with mean  $\boldsymbol{\mu}$  and covariance  $\mathbf{A}\mathbf{A}^\top$ .
- (c) For every  $\boldsymbol{\mu} \in \mathbb{R}^n$  and every nonnegative definite  $n \times n$  matrix  $\boldsymbol{\Sigma}$  there is a random vector with the  $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  distribution, for example  $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$  with  $m = n$ .

No other matrices occur: the covariance matrix of a normal vector is nonnegative definite.

*Proof.*

- (a)  $\mathbf{a}^\top \mathbf{Z} = \sum_i a_i Z_i$  is a sum of independent normal variables  $a_i Z_i \sim N(0, a_i^2)$ , which is normal by Lemma 3.1.1(c).
- (b)  $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top \boldsymbol{\mu} + (\mathbf{A}^\top \mathbf{a})^\top \mathbf{Z}$  is a constant plus a linear combination of  $\mathbf{Z}$ , hence normal by (a). Its moments come from Theorem 2.2.1:  $E \mathbf{Y} = \boldsymbol{\mu}$  and  $\text{Cov}(\mathbf{Y}) = \mathbf{A} \mathbf{I} \mathbf{A}^\top = \mathbf{A} \mathbf{A}^\top$ .
- (c) By Theorem 1.7.6 a nonnegative definite  $\boldsymbol{\Sigma}$  has a symmetric nonnegative definite square root with  $\boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}$ . Take  $\mathbf{A} = \boldsymbol{\Sigma}^{1/2}$  in (b). The last sentence is Theorem 2.2.3.

■

Any factorization  $\boldsymbol{\Sigma} = \mathbf{A} \mathbf{A}^\top$  serves in (b). The symmetric square root is convenient in proofs. The Cholesky factor, a lower triangular  $\mathbf{L}$  with  $\mathbf{L} \mathbf{L}^\top = \boldsymbol{\Sigma}$ , is cheaper to compute and is what software uses. A factor with fewer columns than rows is natural when  $\boldsymbol{\Sigma}$  is singular. Once Theorem 3.2.1 is proved, all of these give the same distribution. The first listing checks this by simulation. It draws 400,000 vectors from a three-dimensional normal distribution with each of

two factors. Both sample covariance matrices agree with  $\Sigma$  to within 0.016 in every entry, and the quantiles of a linear combination match the exact normal quantiles.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(301)
mu = np.array([1.0, -2.0, 0.5])
Sigma = np.array([[4.0, 1.2, -0.8],
                  [1.2, 2.0, 0.6],
                  [-0.8, 0.6, 1.5]])
N = 400_000

def draw_normal(mu, A, size, rng):
    """Rows are independent copies of mu + A Z with Z ~ N(0, I)."""
    Z = rng.standard_normal((size, A.shape[1]))
    return mu + Z @ A.T

L = np.linalg.cholesky(Sigma)                # lower triangular, L L^T = Sigma
lam, U = np.linalg.eigh(Sigma)
root = U @ np.diag(np.sqrt(lam)) @ U.T       # symmetric square root

Y_chol = draw_normal(mu, L, N, rng)
Y_root = draw_normal(mu, root, N, rng)
for Y in (Y_chol, Y_root):
    print(np.round(np.cov(Y, rowvar=False) - Sigma, 3))
```

### 3.1.4 Singular normal distributions

When  $\Sigma$  is singular, [Definition 3.1.1](#) still applies, and the distribution is called **singular** or degenerate. If  $\Sigma \mathbf{v} = \mathbf{0}$ , then  $\text{Var}(\mathbf{v}^\top \mathbf{Y}) = \mathbf{v}^\top \Sigma \mathbf{v} = 0$ , so  $\mathbf{v}^\top \mathbf{Y} = \mathbf{v}^\top \boldsymbol{\mu}$  with probability one. Every vector in  $\mathcal{N}(\Sigma)$  gives such a linear constraint, and together they say that  $\mathbf{Y}$  lies in the affine subspace  $\boldsymbol{\mu} + \mathcal{C}(\Sigma)$  with probability one ([Theorem 2.2.3](#)). A subspace of dimension  $\text{rank}(\Sigma) < n$  has Lebesgue measure zero, so no density on  $\mathbb{R}^n$  is possible.

#### Example. (Three overlapping sums)

Let  $\mathbf{Z} \sim N_3(\mathbf{0}, \mathbf{I})$  and  $\mathbf{U} = (Z_1 + Z_2, Z_2 - Z_3, Z_1 + Z_3)^\top$ . Then  $\mathbf{U} = \mathbf{A}\mathbf{Z}$  with rows  $(1, 1, 0)$ ,  $(0, 1, -1)$  and  $(1, 0, 1)$ , so  $\mathbf{U}$  is normal with mean zero and covariance

$$\mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 2 & 1 & 1 \\ 1 & 2 & -1 \\ 1 & -1 & 2 \end{pmatrix}.$$

The vector  $(1, -1, -1)^\top$  lies in  $\mathcal{N}(\mathbf{A}\mathbf{A}^\top)$ , and the rank is 2. Correspondingly  $U_1 - U_2 - U_3 = 0$  for every outcome, not merely with probability one. Each  $U_i$  is  $N(0, 2)$ , each pair  $(U_i, U_j)$  has a nonsingular covariance matrix (determinant 3) and a bivariate density, and yet the triple lives on a plane.

In regression the singular case is the rule. With  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ , the fitted values and the residuals are normal vectors of ranks  $\text{rank}(\mathbf{X})$  and  $n - \text{rank}(\mathbf{X})$  ([Example \(Least squares under normal errors\)](#)). The last part of the listing draws a normal vector in  $\mathbb{R}^3$  from a  $3 \times 2$  factor. Every draw lies on the plane  $\boldsymbol{\mu} + \mathcal{C}(\mathbf{A})$  to within  $2.7 \times 10^{-15}$ , which is rounding error.

### 3.1.5 Why linear combinations suffice

**Definition 3.1.1** describes a joint distribution entirely through its one-dimensional projections. That this loses no information is a general fact about random vectors, the Cramér–Wold device proved with characteristic functions in [Proposition 2.6.2](#): if  $\mathbf{a}^\top \mathbf{Y}$  and  $\mathbf{a}^\top \mathbf{W}$  have the same distribution for every  $\mathbf{a} \in \mathbb{R}^n$ , then  $\mathbf{Y}$  and  $\mathbf{W}$  have the same distribution. We do not need it in this generality. For normal vectors the moment generating function is finite everywhere, and the next section runs the same argument with it. The device explains why the definition is natural. It also warns that all directions  $\mathbf{a}$  matter. Normality of the  $n$  coordinate variables, which are only  $n$  directions, is not enough ([Section 3.3](#) and [Exercise \(B3\)](#)).

### 3.1.6 Exercises

#### 3.1.6.1 A. Check your understanding

##### Exercise A1.

Let  $Y_1, \dots, Y_n$  be independent with  $Y_i \sim N(\mu_i, \sigma_i^2)$ . Show directly from [Definition 3.1.1](#) that  $\mathbf{Y} = (Y_1, \dots, Y_n)^\top$  is multivariate normal, and find its mean and covariance. Show also that a constant vector  $\mathbf{c}$  is  $N_n(\mathbf{c}, \mathbf{0})$ .

##### Exercise A2.

For the bivariate normal of [Example \(The bivariate normal\)](#), find the Cholesky factor  $\mathbf{L}$  of  $\Sigma$  and write  $Y_1$  and  $Y_2$  in terms of two independent standard normals. Do the same with the symmetric square root when  $\sigma_1 = \sigma_2 = 1$ . Why do the two representations define the same distribution?

#### 3.1.6.2 B. Practice

##### Exercise B1.

Let  $Z \sim N(0, 1)$  and  $\mathbf{Y} = (Z, 2Z - 1, 3 - Z)^\top$ . Show that  $\mathbf{Y}$  is normal, find  $\boldsymbol{\mu}$  and  $\Sigma$ , and find the rank of  $\Sigma$ . Describe the set in which  $\mathbf{Y}$  lies, and write it as the solution set of two linear equations.

*Solution.*  $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{A}Z$  with  $\boldsymbol{\mu} = (0, -1, 3)^\top$  and  $\mathbf{A} = (1, 2, -1)^\top$ , so  $\mathbf{Y}$  is normal by [Theorem 3.1.2](#) with

$$\Sigma = \mathbf{A}\mathbf{A}^\top = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \\ -1 & -2 & 1 \end{pmatrix}, \quad \text{rank}(\Sigma) = 1.$$

$\mathbf{Y}$  lies on the line  $\{\boldsymbol{\mu} + s\mathbf{A} : s \in \mathbb{R}\}$ . The null space of  $\Sigma$  is  $\mathbf{A}^\perp$ , spanned by  $(2, -1, 0)^\top$  and  $(1, 0, 1)^\top$ . The corresponding constraints  $\mathbf{v}^\top \mathbf{Y} = \mathbf{v}^\top \boldsymbol{\mu}$  are  $2Y_1 - Y_2 = 1$  and  $Y_1 + Y_3 = 3$ , and both hold identically.

## 3.2 Moment generating function and density

### 3.2.1 Moment generating functions of random vectors

The moment generating function of a random vector  $\mathbf{Y} \in \mathbb{R}^n$  is  $M_{\mathbf{Y}}(\mathbf{t}) = \mathbb{E} \exp(\mathbf{t}^\top \mathbf{Y})$ , and [Theorem 2.6.1](#) proves the two properties we need when it is finite on a neighbourhood of  $\mathbf{0}$ . *Uniqueness*: two random vectors whose moment generating functions are finite and equal near  $\mathbf{0}$  have the same distribution. *Factorization*: two blocks  $\mathbf{Y}_1, \mathbf{Y}_2$  are independent iff  $M_{\mathbf{Y}}(\mathbf{t}) = M_{\mathbf{Y}_1}(\mathbf{t}_1)M_{\mathbf{Y}_2}(\mathbf{t}_2)$  near  $\mathbf{0}$ . The second extends to  $k$  blocks by induction. If  $M_{\mathbf{Y}}$  is the product of the  $k$  block functions, setting  $\mathbf{t}_1 = \mathbf{0}$  shows that  $(\mathbf{Y}_2, \dots, \mathbf{Y}_k)$  has the product of the last  $k-1$ , so  $\mathbf{Y}_1$  is independent of  $(\mathbf{Y}_2, \dots, \mathbf{Y}_k)$ , whose blocks are mutually independent by the induction hypothesis. The converse is immediate.

#### Theorem 3.2.1 — Moment generating function

A random vector  $\mathbf{Y}$  is  $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  iff

$$M_{\mathbf{Y}}(\mathbf{t}) = \exp(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}) \quad \text{for every } \mathbf{t} \in \mathbb{R}^n. \quad (3.2.1)$$

Consequently the distribution of a normal vector is determined by its mean vector and covariance matrix.

*Proof.* Suppose  $\mathbf{Y}$  is normal with mean  $\boldsymbol{\mu}$  and covariance  $\boldsymbol{\Sigma}$ . By [Eq. 3.1.1](#),  $\mathbf{t}^\top \mathbf{Y} \sim N(\mathbf{t}^\top \boldsymbol{\mu}, \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})$ , and  $M_{\mathbf{Y}}(\mathbf{t})$  is the moment generating function of this scalar evaluated at  $s = 1$ . [Lemma 3.1.1\(a\)](#) gives [Eq. 3.2.1](#).

Conversely, suppose [Eq. 3.2.1](#) holds for a symmetric  $\boldsymbol{\Sigma}$ . Fix  $\mathbf{a}$ . The moment generating function of  $\mathbf{a}^\top \mathbf{Y}$  at  $s$  is  $M_{\mathbf{Y}}(s\mathbf{a}) = \exp\{(\mathbf{a}^\top \boldsymbol{\mu})s + \frac{1}{2}(\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})s^2\}$ . The coefficient  $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$  is nonnegative, since otherwise this function would tend to zero as  $s \rightarrow \pm\infty$ , whereas every random variable  $X$  has  $\mathbb{E} e^{sX} + \mathbb{E} e^{-sX} \geq \mathbb{E} e^{|sX|} \geq 1$ . By [Lemma 3.1.1\(b\)](#),  $\mathbf{a}^\top \mathbf{Y} \sim N(\mathbf{a}^\top \boldsymbol{\mu}, \mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a})$ . So  $\mathbf{Y}$  is normal, and matching means and variances of all linear combinations shows  $\mathbb{E} \mathbf{Y} = \boldsymbol{\mu}$  and  $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$ .

Finally, two normal vectors with the same  $\boldsymbol{\mu}$  and  $\boldsymbol{\Sigma}$  have the same moment generating function [Eq. 3.2.1](#), which is finite everywhere, so they have the same distribution by the uniqueness property. ■

The last sentence is what makes  $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  a well-defined distribution. Combined with [Theorem 3.1.2](#), it says that  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  holds iff  $\mathbf{Y}$  has the same distribution as  $\boldsymbol{\mu} + \mathbf{A}\mathbf{Z}$  for some, and then every, matrix  $\mathbf{A}$  with  $\mathbf{A}\mathbf{A}^\top = \boldsymbol{\Sigma}$ . The number of columns of  $\mathbf{A}$ , and so the length of  $\mathbf{Z}$ , does not matter. Some books take this representation as the definition ([Christensen 2020](#); [Seber and Lee 2003](#)). The proof above shows the two definitions agree.

The moment generating function also encodes moments. Differentiating [Eq. 3.2.1](#) at  $\mathbf{t} = \mathbf{0}$  gives  $\partial M_{\mathbf{Y}}/\partial \mathbf{t} = \boldsymbol{\mu}$  and  $\partial^2 M_{\mathbf{Y}}/\partial \mathbf{t} \partial \mathbf{t}^\top = \boldsymbol{\Sigma} + \boldsymbol{\mu}\boldsymbol{\mu}^\top = \mathbf{E}(\mathbf{Y}\mathbf{Y}^\top)$ . Higher derivatives give the fourth moments needed for variances of quadratic forms in [Chapter 4](#) ([Exercise \(B1\)](#)).

## 3.2.2 The density

### Theorem 3.2.2 — Density

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ .

- (a) If  $\boldsymbol{\Sigma}$  is positive definite,  $\mathbf{Y}$  has the density

$$f(\mathbf{y}) = (2\pi)^{-n/2} \det(\boldsymbol{\Sigma})^{-1/2} \exp\left\{-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})\right\}, \quad \mathbf{y} \in \mathbb{R}^n. \quad (3.2.2)$$

- (b) Conversely, a random vector with density [Eq. 3.2.2](#), for some positive definite  $\boldsymbol{\Sigma}$ , is  $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ .
- (c) If  $\boldsymbol{\Sigma}$  is singular,  $\mathbf{Y}$  has no density on  $\mathbb{R}^n$ .

*Proof.*

- (a) By [Theorem 3.2.1](#),  $\mathbf{Y}$  has the same distribution as  $\boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{Z}$  with  $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$ , so it suffices to find the density of the latter. The entries of  $\mathbf{Z}$  are independent standard normals, so  $\mathbf{Z}$  has density  $\prod_i (2\pi)^{-1/2} e^{-z_i^2/2} = (2\pi)^{-n/2} \exp(-\frac{1}{2}\mathbf{z}^\top \mathbf{z})$ . Since  $\boldsymbol{\Sigma}$  is positive definite, so is  $\boldsymbol{\Sigma}^{1/2}$  (its eigenvalues are the positive square roots of those of  $\boldsymbol{\Sigma}$ ), and the map  $\mathbf{z} \mapsto \mathbf{y} = \boldsymbol{\mu} + \boldsymbol{\Sigma}^{1/2}\mathbf{z}$  is an affine bijection of  $\mathbb{R}^n$  with inverse  $\mathbf{z} = \boldsymbol{\Sigma}^{-1/2}(\mathbf{y} - \boldsymbol{\mu})$ . The change-of-variables formula multiplies the density of  $\mathbf{Z}$  at  $\mathbf{z}(\mathbf{y})$  by  $|\det \boldsymbol{\Sigma}^{-1/2}| = \det(\boldsymbol{\Sigma})^{-1/2}$ . Finally  $\mathbf{z}^\top \mathbf{z} = (\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\Sigma}^{-1/2}(\mathbf{y} - \boldsymbol{\mu}) = (\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})$ .
- (b) A density determines a distribution, and by (a) the  $N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  distribution has this density.
- (c)  $\mathbf{Y}$  lies with probability one in  $\boldsymbol{\mu} + \mathcal{C}(\boldsymbol{\Sigma})$ , an affine subspace of dimension less than  $n$ . It has Lebesgue measure zero, so it would receive probability zero under any distribution with a density. ■

A singular normal vector does have a density with respect to the natural measure on the affine subspace where it lives ([Proposition 3.3.4](#) and [Exercise \(B2\)](#)), but that is rarely needed. In this book, arguments are made with the moment generating function or with linear transformations, which work in every case, and densities are used only for pictures and likelihoods.

### Example. (The bivariate normal)

For  $n = 2$ , write  $\sigma_1^2, \sigma_2^2 > 0$  for the variances and  $\rho$  for the correlation. Then

$$\boldsymbol{\Sigma} = \begin{pmatrix} \sigma_1^2 & \rho\sigma_1\sigma_2 \\ \rho\sigma_1\sigma_2 & \sigma_2^2 \end{pmatrix}, \quad \det \boldsymbol{\Sigma} = \sigma_1^2\sigma_2^2(1 - \rho^2),$$



so  $\Sigma$  is positive definite iff  $|\rho| < 1$ . With standardized coordinates  $u_i = (y_i - \mu_i)/\sigma_i$ , the density Eq. 3.2.2 is

$$f(y_1, y_2) = \frac{1}{2\pi\sigma_1\sigma_2\sqrt{1-\rho^2}} \exp\left\{-\frac{u_1^2 - 2\rho u_1 u_2 + u_2^2}{2(1-\rho^2)}\right\}.$$

When  $\rho = 0$  it factors into the two univariate normal densities. When  $\rho = \pm 1$  there is no density:  $\text{Var}(u_2 \mp u_1) = 0$  in standardized units, so all the probability lies on a line.

### 3.2.3 Contours and the Mahalanobis distance

For positive definite  $\Sigma$ , the density Eq. 3.2.2 depends on  $\mathbf{y}$  only through

$$\Delta^2(\mathbf{y}) = (\mathbf{y} - \boldsymbol{\mu})^\top \Sigma^{-1}(\mathbf{y} - \boldsymbol{\mu}),$$

the squared Mahalanobis distance from  $\mathbf{y}$  to  $\boldsymbol{\mu}$  introduced in Section 2.2. The density is a decreasing function of  $\Delta^2$ , so its contours are the ellipsoids  $\{\mathbf{y} : \Delta^2(\mathbf{y}) = c^2\}$ . Their shape comes from the spectral decomposition  $\Sigma = \sum_i \lambda_i \mathbf{u}_i \mathbf{u}_i^\top$  (Theorem 1.7.2). In the coordinates  $w_i = \mathbf{u}_i^\top(\mathbf{y} - \boldsymbol{\mu})$  we have  $\Delta^2 = \sum_i w_i^2 / \lambda_i$ , so the ellipsoid has principal axes along the eigenvectors  $\mathbf{u}_i$ , with half-lengths  $c\sqrt{\lambda_i}$ . Large-variance directions are long axes. The Mahalanobis distance measures distance from the centre in units of standard deviation along each principal direction. The volume of  $\{\mathbf{y} : \Delta^2(\mathbf{y}) \leq c^2\}$  is proportional to  $\prod_i \sqrt{\lambda_i} = \det(\Sigma)^{1/2}$ , so the generalized variance  $\det \Sigma$  (Section 2.1) measures the size of the region the distribution occupies. The factor  $\det(\Sigma)^{-1/2}$  in Eq. 3.2.2 spreads the same total probability over that region: a larger generalized variance means a flatter density.

#### Proposition 3.2.3

If  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$  with  $\Sigma$  positive definite, then  $\Delta^2(\mathbf{Y})$  has the chi-squared distribution with  $n$  degrees of freedom,  $\chi^2(n)$ , which is  $\chi^2(n, 0)$  in the notation of Definition 4.1.1. So the ellipsoid  $\{\mathbf{y} : \Delta^2(\mathbf{y}) \leq q\}$ , where  $q$  is the  $1 - \alpha$  quantile of  $\chi^2(n)$ , has probability exactly  $1 - \alpha$ .

*Proof.* Put  $\mathbf{Z} = \Sigma^{-1/2}(\mathbf{Y} - \boldsymbol{\mu})$ . For every  $\mathbf{a}$ ,  $\mathbf{a}^\top \mathbf{Z} = (\Sigma^{-1/2}\mathbf{a})^\top \mathbf{Y} - \mathbf{a}^\top \Sigma^{-1/2}\boldsymbol{\mu}$  is a linear combination of  $\mathbf{Y}$  minus a constant, hence normal by Definition 3.1.1, so  $\mathbf{Z}$  is normal. Its mean is zero and its covariance is  $\mathbf{I}$  (Proposition 2.2.4(a)). By Theorem 3.2.1 its moment generating function is  $\exp(\frac{1}{2}\mathbf{t}^\top \mathbf{t}) = \prod_i \exp(\frac{1}{2}t_i^2)$ , which factorizes, so its entries are independent standard normals.  $\Delta^2(\mathbf{Y}) = \mathbf{Z}^\top \mathbf{Z} = \sum_i Z_i^2$  is a sum of  $n$  independent squared standard normals, which is the definition of  $\chi^2(n)$ . ■

Chapter 4 generalizes this considerably, to  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  with a nonzero mean and a general  $\mathbf{A}$  (Theorem 4.3.2). Because the density is largest inside these ellipsoids, they are the smallest regions with their probability content, which is why they reappear as confidence regions for  $\boldsymbol{\beta}$  in Chapter 12 (Theorem 12.2.1).

**Computing with the density.** Neither  $\Sigma^{-1}$  nor  $\det \Sigma$  should be formed explicitly. With the Cholesky factorization  $\Sigma = \mathbf{L}\mathbf{L}^\top$ , we have  $\Delta^2(\mathbf{y}) = \|\mathbf{L}^{-1}(\mathbf{y} - \boldsymbol{\mu})\|^2$ , computed by a triangular solve, and  $\log \det \Sigma = 2 \sum_i \log l_{ii}$ . The listing evaluates the log density this way, checks it

against `scipy`, and counts how many of the simulated vectors fall inside the 95% ellipsoid. The fraction is 0.9500. The average of  $\Delta^2$  over the draws is 3.004 and its variance is 5.994, against the  $\chi^2(3)$  values 3 and 6.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(301)
mu = np.array([1.0, -2.0, 0.5])
Sigma = np.array([[4.0, 1.2, -0.8],
                  [1.2, 2.0, 0.6],
                  [-0.8, 0.6, 1.5]])

N = 400_000
a = np.array([0.3, -1.0, 2.0])
q = [0.05, 0.25, 0.5, 0.75, 0.95]
exact = stats.norm.ppf(q, loc=a @ mu, scale=np.sqrt(a @ Sigma @ a))
cov_err = max(np.abs(np.cov(Y, rowvar=False) - Sigma).max() for Y in (Y_chol, Y_root))

def draw_normal(mu, A, size, rng):
    """Rows are independent copies of mu + A Z with Z ~ N(0, I)."""
    Z = rng.standard_normal((size, A.shape[1]))
    return mu + Z @ A.T

L = np.linalg.cholesky(Sigma)                # lower triangular, L L^T = Sigma
lam, U = np.linalg.eigh(Sigma)
root = U @ np.diag(np.sqrt(lam)) @ U.T       # symmetric square root

Y_chol = draw_normal(mu, L, N, rng)
Y_root = draw_normal(mu, root, N, rng)
for Y in (Y_chol, Y_root):
    print(np.round(np.cov(Y, rowvar=False) - Sigma, 3))

def normal_logpdf(y, mu, Sigma):
    """log density of N(mu, Sigma) at the rows of y, Sigma positive definite."""
    L = np.linalg.cholesky(Sigma)
    w = np.linalg.solve(L, (y - mu).T)       # w = L^{-1}(y - mu), so w^T w =
    Delta^2
    D2 = np.sum(w**2, axis=0)
    logdet = 2.0 * np.sum(np.log(np.diag(L)))
    return -0.5 * (len(mu) * np.log(2 * np.pi) + logdet + D2), D2

logf, D2 = normal_logpdf(Y_chol, mu, Sigma)
inside = np.mean(D2 <= stats.chi2.ppf(0.95, df=3))
print(f"fraction inside the 95% ellipsoid: {inside:.4f}")
```

## 3.2.4 Exercises

### 3.2.4.1 A. Check your understanding

#### Exercise A1.

Show that  $f(y_1, y_2) = k \exp\{-\frac{1}{2}(2y_1^2 + 2y_1y_2 + y_2^2 - 4y_1 - 2y_2)\}$  is a bivariate normal density for a suitable constant  $k$ . Find  $k$ ,  $\mu$ ,  $\Sigma$  and the correlation.

*Solution.* The quadratic part is  $\mathbf{y}^\top \mathbf{P} \mathbf{y}$  with  $\mathbf{P} = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$ , which is positive definite with  $\det \mathbf{P} = 1$ . Matching  $(\mathbf{y} - \boldsymbol{\mu})^\top \mathbf{P} (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{y}^\top \mathbf{P} \mathbf{y} - 2\boldsymbol{\mu}^\top \mathbf{P} \mathbf{y} + \boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu}$  with the linear terms requires  $\mathbf{P} \boldsymbol{\mu} = (2, 1)^\top$ , so  $\boldsymbol{\mu} = (1, 0)^\top$  and  $\boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu} = 2$ . The exponent is  $-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \mathbf{P} (\mathbf{y} - \boldsymbol{\mu}) + 1$ . By [Theorem 3.2.2\(b\)](#) this is the  $N_2(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  density with  $\boldsymbol{\Sigma} = \mathbf{P}^{-1} = \begin{pmatrix} 1 & -1 \\ -1 & 2 \end{pmatrix}$  provided  $ke = (2\pi)^{-1}(\det \boldsymbol{\Sigma})^{-1/2} = (2\pi)^{-1}$ . So  $k = e^{-1}/(2\pi)$ , and the correlation is  $-1/\sqrt{2}$ .

### Exercise A2.

Let  $\boldsymbol{\Sigma}$  have unit variances and correlation  $\rho$ , with  $|\rho| < 1$ . Find the principal axes and half-lengths of the ellipse  $\Delta^2 = c^2$ , show that its area is  $\pi c^2 \sqrt{1 - \rho^2}$ , and show that it contains probability  $1 - e^{-c^2/2}$ .

### 3.2.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  and  $\mathbf{X} = \mathbf{Y} - \boldsymbol{\mu}$ . Expand the moment generating function of  $\mathbf{X}$  in a power series to show that all odd-order moments of  $\mathbf{X}$  vanish and

$$E(X_i X_j X_k X_l) = \sigma_{ij} \sigma_{kl} + \sigma_{ik} \sigma_{jl} + \sigma_{il} \sigma_{jk}.$$

Deduce that  $\text{Var}(\mathbf{X}^\top \mathbf{A} \mathbf{X}) = 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma})$  for symmetric  $\mathbf{A}$ .

*Solution.* By [Theorem 3.2.1](#),  $M_{\mathbf{X}}(\mathbf{t}) = \exp(\frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t}) = \sum_{m \geq 0} (\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})^m / (2^m m!)$ , a series containing only terms of even degree in  $\mathbf{t}$ . Mixed moments of order  $d$  are the  $d$ th partial derivatives at  $\mathbf{0}$ , which come from the degree- $d$  terms, so odd-order moments vanish. The degree-four term is  $p(\mathbf{t}) = \frac{1}{8} (\mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t})^2 = T(\mathbf{t}, \mathbf{t}, \mathbf{t}, \mathbf{t})$ , where  $T(\mathbf{u}, \mathbf{v}, \mathbf{w}, \mathbf{x}) = \frac{1}{8} (\mathbf{u}^\top \boldsymbol{\Sigma} \mathbf{v})(\mathbf{w}^\top \boldsymbol{\Sigma} \mathbf{x})$  is 4-linear. Differentiating  $T(\mathbf{t}, \mathbf{t}, \mathbf{t}, \mathbf{t})$  once in each of the directions  $\mathbf{e}_i, \mathbf{e}_j, \mathbf{e}_k, \mathbf{e}_l$  gives the sum of  $T$  over the 24 ways of placing these four vectors in the four slots. This holds whether or not the indices are distinct, since it is the product rule applied to a multilinear form. A placement contributes  $\frac{1}{8} \sigma_{ab} \sigma_{cd}$ , where the first two slots receive the indices  $a, b$  and the last two receive  $c, d$ . Each of the three ways of splitting the four *directions* into two pairs,  $\{ij, kl\}$ ,  $\{ik, jl\}$ ,  $\{il, jk\}$ , arises from 8 placements (choose which pair goes first, and the order within each pair), which gives the formula. Then, for symmetric  $\mathbf{A}$ ,

$$E(\mathbf{X}^\top \mathbf{A} \mathbf{X})^2 = \sum_{i,j,k,l} a_{ij} a_{kl} E(X_i X_j X_k X_l) = (\text{tr} \mathbf{A} \boldsymbol{\Sigma})^2 + 2 \text{tr}(\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}),$$

and subtracting  $(E \mathbf{X}^\top \mathbf{A} \mathbf{X})^2 = (\text{tr} \mathbf{A} \boldsymbol{\Sigma})^2$  gives the variance.

#### Exercise B2.

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  with  $\text{rank}(\boldsymbol{\Sigma}) = r < n$ , and use the notation of [Proposition 3.3.4](#). Show that  $\mathbf{V} = \mathbf{Q}^\top (\mathbf{Y} - \boldsymbol{\mu})$  has a density on  $\mathbb{R}^r$ , and that this density can be written as  $(2\pi)^{-r/2} (\lambda_1 \cdots \lambda_r)^{-1/2} \exp\{-\frac{1}{2}(\mathbf{y} - \boldsymbol{\mu})^\top \boldsymbol{\Sigma}^+ (\mathbf{y} - \boldsymbol{\mu})\}$  with  $\mathbf{y} = \boldsymbol{\mu} + \mathbf{Q} \mathbf{v}$ , where  $\boldsymbol{\Sigma}^+ = \mathbf{Q} \boldsymbol{\Lambda}^{-1} \mathbf{Q}^\top$  is the Moore–Penrose inverse.

**Exercise B3.**

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  with  $\boldsymbol{\Sigma}$  positive definite and let  $\mathbf{v}$  be a fixed vector. Show that  $(\mathbf{Y} - \mathbf{v})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{Y} - \mathbf{v})$  has the noncentral chi-squared distribution  $\chi^2(n, \gamma)$  with  $\gamma = (\boldsymbol{\mu} - \mathbf{v})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\mu} - \mathbf{v})$ , in the convention of [Definition 4.1.1](#), and find its mean.

**3.2.4.3 C. Going deeper****Exercise C1.**

Let  $\mathbf{Y}$  have density [Eq. 3.2.2](#), and let  $E = \{\mathbf{y} : \Delta^2(\mathbf{y}) \leq q\}$ . Show that every Borel set  $A$  with  $\Pr(\mathbf{Y} \in A) \geq \Pr(\mathbf{Y} \in E)$  has volume at least that of  $E$ . *Hint:* compare  $\int_{A \setminus E} f$  with  $\int_{E \setminus A} f$ .

*Solution.* The density is a decreasing function of  $\Delta^2$ , so with  $c = (2\pi)^{-n/2} \det(\boldsymbol{\Sigma})^{-1/2} e^{-q/2} > 0$  we have  $f \geq c$  on  $E$  and  $f < c$  off  $E$ . We may assume  $A$  has finite volume. Subtracting  $\Pr(\mathbf{Y} \in A \cap E)$  from both sides of  $\Pr(\mathbf{Y} \in A) \geq \Pr(\mathbf{Y} \in E)$  gives  $\int_{A \setminus E} f \geq \int_{E \setminus A} f$ . Hence

$$c \operatorname{vol}(A \setminus E) \geq \int_{A \setminus E} f \geq \int_{E \setminus A} f \geq c \operatorname{vol}(E \setminus A),$$

so  $\operatorname{vol}(A \setminus E) \geq \operatorname{vol}(E \setminus A)$ . Adding  $\operatorname{vol}(A \cap E)$  to both sides gives  $\operatorname{vol}(A) \geq \operatorname{vol}(E)$ . The argument uses only that  $E$  is a set on which  $f$  is at least as large as anywhere outside it, so it applies to the level sets of any density.

## 3.3 Linear transformations and marginals

### 3.3.1 Affine maps

**Theorem 3.3.1 — Affine transformations**

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ , let  $\mathbf{A}$  be any  $m \times n$  matrix and  $\mathbf{b} \in \mathbb{R}^m$ . Then

$$\mathbf{A}\mathbf{Y} + \mathbf{b} \sim N_m(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}, \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top).$$

*Proof.* For  $\mathbf{c} \in \mathbb{R}^m$ ,  $\mathbf{c}^\top(\mathbf{A}\mathbf{Y} + \mathbf{b}) = (\mathbf{A}^\top \mathbf{c})^\top \mathbf{Y} + \mathbf{c}^\top \mathbf{b}$  is a linear combination of  $\mathbf{Y}$  plus a constant, which is normal by [Definition 3.1.1](#) and [Lemma 3.1.1\(c\)](#). So  $\mathbf{A}\mathbf{Y} + \mathbf{b}$  is normal, and its mean and covariance are given by [Theorem 2.2.1](#). ■

The proof is one line because the definition was chosen to make it one line. There is no rank condition on  $\mathbf{A}$ . If  $\mathbf{A}$  has more rows than  $\operatorname{rank}(\boldsymbol{\Sigma})$ , the result is a singular normal vector, and that is allowed. With a density-based definition, the same theorem needs  $\mathbf{A}$  to have full row rank and  $\boldsymbol{\Sigma}$  to be positive definite, and the most useful cases fall outside it. The moment generating function gives a second proof:  $E e^{\mathbf{t}^\top(\mathbf{A}\mathbf{Y} + \mathbf{b})} = e^{\mathbf{t}^\top \mathbf{b}} M_{\mathbf{Y}}(\mathbf{A}^\top \mathbf{t}) = \exp\{\mathbf{t}^\top(\mathbf{A}\boldsymbol{\mu} + \mathbf{b}) + \frac{1}{2} \mathbf{t}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A}^\top \mathbf{t}\}.$

### Corollary 3.3.2 — Marginal distributions

Every subvector of a normal vector is normal. If  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  and  $\mathbf{Y}_1$  consists of the entries with indices in a set  $I$ , then  $\mathbf{Y}_1 \sim N(\boldsymbol{\mu}_I, \boldsymbol{\Sigma}_{II})$ , where  $\boldsymbol{\mu}_I$  and  $\boldsymbol{\Sigma}_{II}$  keep the entries, rows and columns indexed by  $I$ . In particular  $Y_i \sim N(\mu_i, \sigma_{ii})$ .

*Proof.*  $\mathbf{Y}_1 = \mathbf{A}\mathbf{Y}$ , where the rows of  $\mathbf{A}$  are the coordinate vectors  $\mathbf{e}_i^\top$ ,  $i \in I$ . Then  $\mathbf{A}\boldsymbol{\mu} = \boldsymbol{\mu}_I$  and  $\mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top = \boldsymbol{\Sigma}_{II}$ . ■

### Corollary 3.3.3 — Standardization and rotation

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ .

- (a) If  $\boldsymbol{\Sigma}$  is positive definite and  $\boldsymbol{\Sigma} = \mathbf{A}\mathbf{A}^\top$  with  $\mathbf{A}$  square, then  $\mathbf{A}^{-1}(\mathbf{Y} - \boldsymbol{\mu}) \sim N_n(\mathbf{0}, \mathbf{I})$ . Its entries are independent standard normal variables.
- (b) If  $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$  and  $\mathbf{Q}$  is an  $n \times n$  orthogonal matrix, then  $\mathbf{Q}^\top\mathbf{Y} \sim N_n(\mathbf{Q}^\top\boldsymbol{\mu}, \sigma^2\mathbf{I})$ .

*Proof.* Both follow from [Theorem 3.3.1](#):  $\mathbf{A}^{-1}\mathbf{A}\mathbf{A}^\top(\mathbf{A}^{-1})^\top = \mathbf{I}$  and  $\mathbf{Q}^\top(\sigma^2\mathbf{I})\mathbf{Q} = \sigma^2\mathbf{I}$ . Independence in (a) was shown in the proof of [Proposition 3.2.3](#). ■

Part (b) is the probabilistic basis of the canonical form of the linear model. Spherical normal errors look the same in every orthonormal coordinate system, so one may rotate to coordinates adapted to  $\mathcal{C}(\mathbf{X})$ , in which fitted values and residuals occupy separate independent coordinates. [Chapter 4](#) uses this rotation to find the distributions of sums of squares.

## 3.3.2 The rank of a normal vector

[Theorem 3.1.2](#) writes a normal vector as an affine function of independent standard normals *in distribution*. When  $\boldsymbol{\Sigma}$  is singular the representation can be made exact, with as few standard normals as the rank.

### Proposition 3.3.4

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  with  $\text{rank}(\boldsymbol{\Sigma}) = r \geq 1$ , and let  $\boldsymbol{\Sigma} = \mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top$ , where  $\mathbf{Q}$  is  $n \times r$  with orthonormal columns and  $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_r)$  holds the positive eigenvalues. Put  $\mathbf{W} = \boldsymbol{\Lambda}^{-1/2}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu})$ . Then  $\mathbf{W} \sim N_r(\mathbf{0}, \mathbf{I}_r)$ , and

$$\mathbf{Y} = \boldsymbol{\mu} + \mathbf{Q}\boldsymbol{\Lambda}^{1/2}\mathbf{W} \quad \text{with probability one.}$$

*Proof.* By [Theorem 3.3.1](#),  $\mathbf{W}$  is normal with covariance  $\boldsymbol{\Lambda}^{-1/2}\mathbf{Q}^\top\mathbf{Q}\boldsymbol{\Lambda}\mathbf{Q}^\top\mathbf{Q}\boldsymbol{\Lambda}^{-1/2} = \mathbf{I}_r$ . Next,  $\mathbf{Q}\boldsymbol{\Lambda}^{1/2}\mathbf{W} = \mathbf{Q}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu})$ , and  $\mathbf{Q}\mathbf{Q}^\top$  is the symmetric idempotent matrix with range  $\mathcal{C}(\boldsymbol{\Sigma})$  (the orthogonal projection onto it, in the language of [Chapter 6](#)). Since

$\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\boldsymbol{\Sigma})$  with probability one (Theorem 2.2.3),  $\mathbf{Q}\mathbf{Q}^\top(\mathbf{Y} - \boldsymbol{\mu}) = \mathbf{Y} - \boldsymbol{\mu}$  with probability one. ■

So a normal vector of rank  $r$  is exactly  $r$  independent standard normals placed in  $\mathbb{R}^n$  by an injective affine map. Exercise (B4) gives the same construction for any random vector, with uncorrelated entries in place of independent normal ones. Singular normal vectors are lower-dimensional normal vectors in disguise.

**Example. (Least squares under normal errors)**

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$  with  $\mathbf{X}$  of full column rank  $p$ , and let  $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ , the symmetric idempotent matrix of rank  $p$  with  $\mathbf{M}\mathbf{X} = \mathbf{X}$  (Chapter 5 and Chapter 6). By Theorem 3.3.1:

- (a)  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y} \sim N_p(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1})$ , a nonsingular normal vector;
- (b)  $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\lambda}^\top\boldsymbol{\beta}, \sigma^2\boldsymbol{\lambda}^\top(\mathbf{X}^\top\mathbf{X})^{-1}\boldsymbol{\lambda})$  for every  $\boldsymbol{\lambda} \in \mathbb{R}^p$ ;
- (c) the fitted values  $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{M})$ , singular of rank  $p$ ;
- (d) the residuals  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$ , singular of rank  $n - p$ .

No density for  $\hat{\mathbf{Y}}$  or  $\hat{\boldsymbol{\varepsilon}}$  was needed, and none exists.

### 3.3.3 Independent normal vectors

**Lemma 3.3.5**

If  $\mathbf{Y}_1 \sim N_{n_1}(\boldsymbol{\mu}_1, \boldsymbol{\Sigma}_1)$  and  $\mathbf{Y}_2 \sim N_{n_2}(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_2)$  are independent, then  $(\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$  is jointly normal with mean  $(\boldsymbol{\mu}_1^\top, \boldsymbol{\mu}_2^\top)^\top$  and block-diagonal covariance  $\text{diag}(\boldsymbol{\Sigma}_1, \boldsymbol{\Sigma}_2)$ . Consequently, if also  $n_1 = n_2$ , then  $\mathbf{Y}_1 + \mathbf{Y}_2 \sim N(\boldsymbol{\mu}_1 + \boldsymbol{\mu}_2, \boldsymbol{\Sigma}_1 + \boldsymbol{\Sigma}_2)$ .

*Proof.*  $\mathbf{a}_1^\top\mathbf{Y}_1 + \mathbf{a}_2^\top\mathbf{Y}_2$  is a sum of independent normal variables, which is normal by Lemma 3.1.1(c). Independence makes the cross-covariance zero. The sum is  $[\mathbf{I}, \mathbf{I}]$  applied to the stacked vector. ■

Independence is essential in this lemma. Two normal vectors defined on the same probability space need not be jointly normal, as we now see.

### 3.3.4 Normal marginals are not enough

Corollary 3.3.2 cannot be reversed. A vector whose entries are each normal need not be jointly normal. Two examples make the point, one extreme and one realistic.

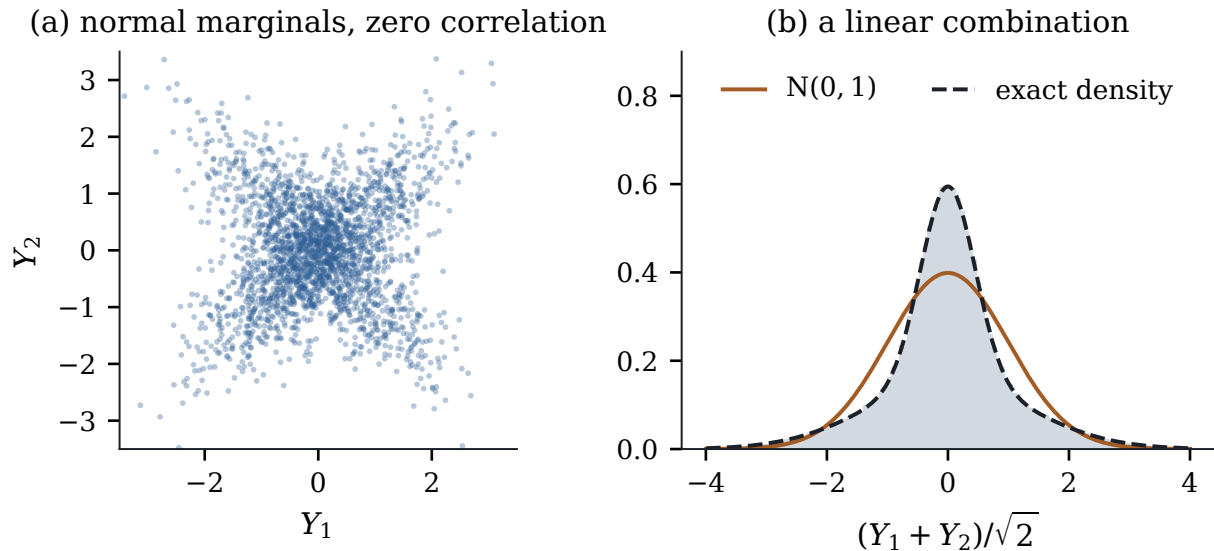


Figure 3.3.1: An equal mixture of bivariate normals with correlations  $\pm 0.8$ . (a) Both margins are exactly  $N(0, 1)$  and the correlation is zero, but the scatter is an X, not an ellipse. (b) The linear combination  $(Y_1 + Y_2)/\sqrt{2}$  has variance one but is sharply peaked and heavy-tailed compared with  $N(0, 1)$ . By Definition 3.1.1, one non-normal direction is enough to rule out joint normality.

#### Example. (A random sign)

Let  $Z \sim N(0, 1)$  and let  $S$  be independent of  $Z$  with  $\Pr(S = 1) = \Pr(S = -1) = \frac{1}{2}$ . Put  $W = SZ$ . By symmetry of  $Z$ ,  $W \sim N(0, 1)$ . Also  $\text{Cov}(Z, W) = E(S)E(Z^2) = 0$ . But  $Z + W = (1 + S)Z$  equals zero with probability  $\frac{1}{2}$  and is otherwise  $N(0, 4)$ . A variable with an atom at zero and a continuous part is not normal, so  $(Z, W)$  is not jointly normal. And  $Z$  and  $W$  are far from independent:  $|W| = |Z|$  always.

#### Example. (A mixture of two correlations)

Fix  $0 < \rho < 1$ . With probability  $\frac{1}{2}$  draw  $(Y_1, Y_2)$  from the bivariate normal with unit variances and correlation  $\rho$ , and otherwise from the one with correlation  $-\rho$ . Both components have  $N(0, 1)$  margins, so the mixture does too, exactly. The covariance is  $\frac{1}{2}\rho + \frac{1}{2}(-\rho) = 0$ . The pair is nevertheless dependent: since  $E(Y_1^2 Y_2^2) = 1 + 2\rho^2$  for a standard bivariate normal with correlation  $r$ , the squares have correlation  $\rho^2$ , not zero. The direction  $\mathbf{a} = (1, 1)^\top/\sqrt{2}$  exposes the failure of joint normality.  $S = (Y_1 + Y_2)/\sqrt{2}$  is an equal mixture of  $N(0, 1 + \rho)$  and  $N(0, 1 - \rho)$ . It has variance 1 but kurtosis  $E S^4 = \frac{3}{2}\{(1 + \rho)^2 + (1 - \rho)^2\} = 3(1 + \rho^2)$ , where a normal variable would have 3.

Figure 3.3.1 shows a simulation with  $\rho = 0.8$  and 200,000 draws. Kolmogorov–Smirnov tests of normality for the two margins give  $p$ -values 0.84 and 0.29. The sample correlation is  $-0.0033$ , the correlation of the squares is 0.640 (theory 0.64), and the kurtosis of  $S$  is 4.88 (theory 4.92). The Kolmogorov–Smirnov distance of  $S$  from  $N(0, 1)$  is 0.067, which with this many draws rejects normality at any conventional level.

```
import numpy as np
from scipy import integrate, stats
rng = np.random.default_rng(302)
```

```

rho, N = 0.8, 200_000

sign = rng.choice([-1.0, 1.0], size=N)          # which component each draw uses
Z1, Z2 = rng.standard_normal((2, N))
Y1 = Z1
Y2 = sign * rho * Z1 + np.sqrt(1 - rho**2) * Z2 # corr(Y1, Y2 | sign) = sign * rho

S = (Y1 + Y2) / np.sqrt(2)                      # a direction that is not normal
print("marginal KS p-values:", stats.kstest(Y1, "norm").pvalue, stats.kstest(Y2, "
    norm").pvalue)
print("correlation:", np.corrcoef(Y1, Y2)[0, 1])
print("corr of squares:", np.corrcoef(Y1**2, Y2**2)[0, 1])
print("kurtosis of S:", stats.kurtosis(S, fisher=False), " theory:", 3 * (1 + rho
    **2))
print("KS p-value of S:", stats.kstest(S, "norm").pvalue)

```

*Idea.* Normality of a vector is a property of *every direction*, not of the coordinate axes. Normal margins with zero correlation can hide strong dependence. The theorems of the next two sections, which turn zero covariance into independence and give linear conditional means, need *joint* normality. They are not about variables that merely happen to be normal one at a time.

### 3.3.5 Exercises

#### 3.3.5.1 A. Check your understanding

##### Exercise A1.

Let  $\mathbf{Y} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  with  $\boldsymbol{\mu} = (0, 1, 2)^\top$  and  $\boldsymbol{\Sigma} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$ . Find the joint distribution of  $U = Y_1 + Y_2 + Y_3$ ,  $V = Y_1 - Y_3$  and  $W = Y_1 - 2Y_2 + Y_3$ . Which of the pairs are independent? Is  $V$  independent of  $(U, W)$ ?

##### Exercise A2.

In [Example \(Least squares under normal errors\)](#), show that the  $i$ th residual is  $N(0, \sigma^2(1 - m_{ii}))$ , where  $m_{ii}$  is the  $i$ th diagonal entry of  $\mathbf{M}$ , and that it is zero with probability one iff  $m_{ii} = 1$ .

#### 3.3.5.2 B. Practice

##### Exercise B1.

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}\mathbf{1}, \sigma^2\mathbf{I})$  and let  $\mathbf{Q}$  be an orthogonal matrix whose first row is  $n^{-1/2}\mathbf{1}^\top$ . Show that the entries of  $\mathbf{V} = \mathbf{Q}\mathbf{Y}$  are independent, with  $V_1 \sim N(\sqrt{n}\boldsymbol{\mu}, \sigma^2)$  and  $V_i \sim N(0, \sigma^2)$  for  $i \geq 2$ . Show that  $\sum_i (Y_i - \bar{Y})^2 = \sum_{i \geq 2} V_i^2$ , and conclude again that  $\bar{Y}$  and  $S^2$  are independent.



**Exercise B2.**

Let  $Z \sim N(0, 1)$  and, for  $c > 0$ , let  $W = Z$  if  $|Z| \leq c$  and  $W = -Z$  otherwise. Show that  $W \sim N(0, 1)$ , that there is a  $c$  with  $\text{Cov}(Z, W) = 0$ , and that  $(Z, W)$  is not jointly normal for any  $c$ .

*Solution.* Because  $Z$  and  $-Z$  have the same distribution and the event  $\{|Z| > c\}$  is unchanged by the sign change,  $\Pr(W \leq w) = \Pr(Z \leq w, |Z| \leq c) + \Pr(-Z \leq w, |Z| > c) = \Pr(Z \leq w)$ . Next,  $\text{Cov}(Z, W) = E(Z^2; |Z| \leq c) - E(Z^2; |Z| > c) = 2E(Z^2; |Z| \leq c) - 1$ , which is continuous and increasing in  $c$ , tends to  $-1$  as  $c \rightarrow 0$  and to  $1$  as  $c \rightarrow \infty$ , and so vanishes at some  $c$ . Finally  $Z + W = 2Z \mathbf{1}\{|Z| \leq c\}$  equals 0 with probability  $\Pr(|Z| > c) \in (0, 1)$  but is not constant. A normal variable is either constant or has a density, so  $Z + W$  is not normal.

**Exercise B3.**

For the pair  $(Z, W)$  of [Example \(A random sign\)](#), find every  $\mathbf{a} \neq \mathbf{0}$  for which  $a_1 Z + a_2 W$  is normal. *Hint:* an equal mixture of  $N(0, v_1)$  and  $N(0, v_2)$  has fourth moment  $\frac{3}{2}(v_1^2 + v_2^2)$ . Conclude that  $(Z, W)$  is normal along exactly two lines through the origin, the coordinate axes.

## 3.4 Independence and zero covariance

For any random vectors with finite second moments, independence implies zero cross-covariance, because the expectation of a product of functions of independent variables factors. The converse fails in general, as [Example \(A random sign\)](#) and [Example \(A mixture of two correlations\)](#) show. For jointly normal vectors it holds. This is the single most used property of the normal distribution in linear-model theory. Every independence statement behind a  $t$  or  $F$  statistic reduces to it.

### Theorem 3.4.1 — Independence

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  be partitioned into blocks  $\mathbf{Y} = (\mathbf{Y}_1^\top, \dots, \mathbf{Y}_k^\top)^\top$ , with  $\boldsymbol{\mu}$  and  $\boldsymbol{\Sigma} = (\boldsymbol{\Sigma}_{ij})$  partitioned to match.

- (a) The blocks  $\mathbf{Y}_1, \dots, \mathbf{Y}_k$  are mutually independent iff  $\boldsymbol{\Sigma}_{ij} = \text{Cov}(\mathbf{Y}_i, \mathbf{Y}_j) = \mathbf{0}$  for all  $i \neq j$ .
- (b) In particular,  $\mathbf{Y}_1$  and  $\mathbf{Y}_2$  are independent iff  $\text{Cov}(\mathbf{Y}_1, \mathbf{Y}_2) = \mathbf{0}$ , and the scalar entries  $Y_1, \dots, Y_n$  are mutually independent iff  $\boldsymbol{\Sigma}$  is diagonal.

*Proof.* If the blocks are independent, every cross-covariance is zero. Conversely, suppose  $\boldsymbol{\Sigma}_{ij} = \mathbf{0}$  for  $i \neq j$ , and partition  $\mathbf{t}$  to match. Then

$$\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \boldsymbol{\Sigma} \mathbf{t} = \sum_{i=1}^k \left( \mathbf{t}_i^\top \boldsymbol{\mu}_i + \frac{1}{2} \mathbf{t}_i^\top \boldsymbol{\Sigma}_{ii} \mathbf{t}_i \right),$$

because all the cross terms  $\mathbf{t}_i^\top \boldsymbol{\Sigma}_{ij} \mathbf{t}_j$  vanish. Exponentiating, [Theorem 3.2.1](#) gives  $M_{\mathbf{Y}}(\mathbf{t}) = \prod_i M_i(\mathbf{t}_i)$ , where  $M_i$  is the moment generating function of  $N(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_{ii})$ . By [Corollary 3.3.2](#) that is the distribution of  $\mathbf{Y}_i$ , so  $M_i = M_{\mathbf{Y}_i}$ . The factorization property of [Section 3.2](#) gives mutual independence. Part (b) is the cases  $k = 2$  and  $k = n$ . ■

A consequence worth stating explicitly: for jointly normal blocks, *pairwise* independence implies *mutual* independence, because condition (a) involves only pairs. For general random vectors this is false, as the example of three random signs in [Example \(Pairwise but not mutually independent\)](#) shows. Joint normality of the whole vector matters here: joint normality of each pair is not enough ([Exercise \(C1\)](#)).

The theorem is usually applied to two linear functions of the same normal vector.

### Corollary 3.4.2

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$ , and let  $\mathbf{A}$  and  $\mathbf{B}$  be matrices with  $n$  columns. Then  $\mathbf{AY}$  and  $\mathbf{BY}$  are independent iff  $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top = \mathbf{0}$ . In that case  $g(\mathbf{AY})$  and  $h(\mathbf{BY})$  are independent for all measurable functions  $g$  and  $h$ .

*Proof.* The stacked vector  $((\mathbf{AY})^\top, (\mathbf{BY})^\top)^\top$  is a linear function of  $\mathbf{Y}$ , so it is normal by [Theorem 3.3.1](#). Its off-diagonal covariance block is  $\text{Cov}(\mathbf{AY}, \mathbf{BY}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top$  ([Theorem 2.2.1](#)). Apply [Theorem 3.4.1](#). Functions of independent random vectors are independent. ■

The condition  $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B}^\top = \mathbf{0}$  involves only second moments, but the conclusion is full independence. In particular, any quadratic form in  $\mathbf{BY}$ , such as  $\|\mathbf{BY}\|^2$ , is independent of  $\mathbf{AY}$ . [Chapter 4](#) extends this to quadratic forms that are not written as squared lengths ([Theorem 4.4.1](#)).

#### Example. (Sample mean and deviations)

Let  $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$ , so that  $Y_1, \dots, Y_n$  is a random sample from  $N(\mu, \sigma^2)$ . The mean is  $\bar{Y} = n^{-1}\mathbf{1}^\top\mathbf{Y}$ , and the vector of deviations is  $\mathbf{d} = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{Y}$ . As in [Example \(Centring a sample\)](#),

$$\text{Cov}(\bar{Y}, \mathbf{d}) = \sigma^2 n^{-1} \mathbf{1}^\top (\mathbf{I} - n^{-1} \mathbf{1}\mathbf{1}^\top) = \mathbf{0}^\top,$$

By [Corollary 3.4.2](#),  $\bar{Y}$  is independent of  $\mathbf{d}$ , and so of every function of the deviations: the sample variance  $S^2 = \|\mathbf{d}\|^2/(n-1)$ , the range, the sample skewness. The distribution of  $(n-1)S^2/\sigma^2$  is found in [Corollary 4.4.5](#). The computation used  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$  only through  $\boldsymbol{\Sigma}\mathbf{1} \in \text{span}(\mathbf{1})$ , which is the precise condition for the conclusion ([Exercise \(B1\)](#)).

#### Example. (Coefficients and residuals)

In the setting of [Example \(Least squares under normal errors\)](#),

$$\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\varepsilon}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\sigma^2 \mathbf{I}) (\mathbf{I} - \mathbf{M}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \{(\mathbf{I} - \mathbf{M})\mathbf{X}\}^\top = \mathbf{0},$$

since  $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$ . Hence  $\hat{\boldsymbol{\beta}}$  is independent of the residual vector, and so of the residual sum of squares  $\|\hat{\boldsymbol{\epsilon}}\|^2$  and of the estimate  $s^2 = \|\hat{\boldsymbol{\epsilon}}\|^2/(n-p)$ . The same argument shows that the fitted values  $\hat{\mathbf{Y}}$  and the residuals are independent. This independence is what allows a  $t$  statistic  $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/\{s^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}\}^{1/2}$  to have a  $t$  distribution (Theorem 11.6.1).

*Remark* (What joint normality buys). The two examples use only a covariance calculation. Without joint normality, zero covariance between  $\hat{\boldsymbol{\beta}}$  and  $\hat{\boldsymbol{\epsilon}}$  still holds under the second-moment assumptions of Definition 5.3.1, but independence does not follow, and neither do exact  $t$  and  $F$  distributions. Chapter 19 (Theorem 19.3.2) and Chapter 21 examine what survives approximately.

## 3.4.1 Exercises

### 3.4.1.1 A. Check your understanding

#### Exercise A1.

Let  $Y_1, Y_2$  be independent  $N(\mu, \sigma^2)$ . Show that  $Y_1 + Y_2$  and  $Y_1 - Y_2$  are independent. What if the variances differ?

#### Exercise A2.

Let  $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$  and let  $\mathbf{u}$  be a unit vector. Show that  $\mathbf{u}^\top \mathbf{Y}$  and  $\|\mathbf{Y} - (\mathbf{u}^\top \mathbf{Y})\mathbf{u}\|^2$  are independent.

### 3.4.1.2 B. Practice

#### Exercise B1.

Let  $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \boldsymbol{\Sigma})$  with  $\boldsymbol{\Sigma}$  arbitrary. Show that  $\bar{Y}$  is independent of the deviations  $Y_i - \bar{Y}$  iff  $\boldsymbol{\Sigma}\mathbf{1} \in \text{span}(\mathbf{1})$ , that is, iff all row sums of  $\boldsymbol{\Sigma}$  are equal. Check the condition for the equicorrelation matrix  $\sigma^2\{(1-\rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\}$ , and for a two-group matrix: the first  $m_1$  and the last  $m_2$  observations form groups, all variances are 1, the correlation is  $\rho$  within a group and  $\tau$  between groups.

*Solution.*  $\bar{Y} = n^{-1}\mathbf{1}^\top \mathbf{Y}$  and the deviations are  $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{Y}$ , so by Corollary 3.4.2 they are independent iff  $n^{-1}\mathbf{1}^\top \boldsymbol{\Sigma}(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top) = \mathbf{0}^\top$ , that is,  $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\boldsymbol{\Sigma}\mathbf{1} = \mathbf{0}$ . The null space of  $\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$  is  $\text{span}(\mathbf{1})$ , and  $\boldsymbol{\Sigma}\mathbf{1}$  is the vector of row sums. For the equicorrelation matrix,  $\boldsymbol{\Sigma}\mathbf{1} = \sigma^2(1-\rho + n\rho)\mathbf{1}$ , so independence holds for every admissible  $\rho$ . In the two-group matrix, a row in the first group sums to  $1 + (m_1 - 1)\rho + m_2\tau$  and a row in the second to  $1 + (m_2 - 1)\rho + m_1\tau$ . The difference is  $(m_1 - m_2)(\rho - \tau)$ , so independence holds iff the groups have equal size or  $\rho = \tau$ , and the latter is the equicorrelated case.

#### Exercise B2.

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$  with  $\text{rank}(\mathbf{X}) = r < p$ , let  $\mathbf{G}$  be a generalized inverse of  $\mathbf{X}^\top \mathbf{X}$

and  $\mathbf{M} = \mathbf{XGX}^\top$ . Show that the least squares estimate  $\mathbf{GX}^\top \mathbf{Y}$  and the residual vector  $(\mathbf{I} - \mathbf{M})\mathbf{Y}$  are independent, and that every estimable function  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  with  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$  has an estimate independent of the residual sum of squares.

### 3.4.1.3 C. Going deeper

#### Exercise C1.

Let  $\varphi$  be the  $N(0, 1)$  density,  $0 < \varepsilon \leq e^{3/2}$ , and

$$f(x, y, z) = \varphi(x)\varphi(y)\varphi(z)\{1 + \varepsilon xyz e^{-(x^2+y^2+z^2)/2}\}.$$

Show that  $f$  is a probability density, and that under  $f$  every pair of coordinates is  $N_2(\mathbf{0}, \mathbf{I})$ . So the three variables are pairwise jointly normal and pairwise independent. Show that they are not mutually independent and that  $(X, Y, Z)$  is not jointly normal. Which step of [Theorem 3.4.1](#) fails?

*Solution.* Since  $|t|e^{-t^2/2} \leq e^{-1/2}$  for all  $t$ , the perturbation satisfies  $|\varepsilon xyz e^{-(x^2+y^2+z^2)/2}| \leq \varepsilon e^{-3/2} \leq 1$ , so  $f \geq 0$ . The function  $z \mapsto z\varphi(z)e^{-z^2/2}$  is odd and integrable, so integrating  $f$  over  $z$  removes the perturbation and leaves  $\varphi(x)\varphi(y)$ . Integrating further shows that  $f$  integrates to 1, and by symmetry each pair has density  $\varphi \otimes \varphi$ , the  $N_2(\mathbf{0}, \mathbf{I})$  density. Mutual independence would force  $f = \varphi(x)\varphi(y)\varphi(z)$  almost everywhere, but the perturbation is nonzero off the coordinate planes. Directly,  $E(XYZ) = \varepsilon(\int t^2 \varphi(t) e^{-t^2/2} dt)^3 > 0$ , whereas independence with mean-zero margins gives 0. If  $(X, Y, Z)$  were jointly normal, its covariance matrix would be  $\mathbf{I}$  (unit variances from the margins, zero covariances from the pairs), and by [Theorem 3.4.1](#) the coordinates would be independent, which they are not. The theorem's hypothesis is joint normality of the whole vector. Here only the pairs are normal, and the moment generating function of the triple does not factor.

## 3.5 Conditional distributions

Throughout this section  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  is split into  $\mathbf{Y}_1 \in \mathbb{R}^{n_1}$  and  $\mathbf{Y}_2 \in \mathbb{R}^{n_2}$ , with

$$\boldsymbol{\mu} = \begin{pmatrix} \boldsymbol{\mu}_1 \\ \boldsymbol{\mu}_2 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\Sigma}_{11} & \boldsymbol{\Sigma}_{12} \\ \boldsymbol{\Sigma}_{21} & \boldsymbol{\Sigma}_{22} \end{pmatrix}, \quad \boldsymbol{\Sigma}_{21} = \boldsymbol{\Sigma}_{12}^\top,$$

as in [Theorem 2.5.1](#). The question is what  $\mathbf{Y}_1$  looks like once  $\mathbf{Y}_2$  has been observed. The answer is the probabilistic core of regression.

### 3.5.1 Partial covariance with a generalized inverse

The natural answer involves  $\boldsymbol{\Sigma}_{22}^{-1}$ , but  $\boldsymbol{\Sigma}_{22}$  may be singular, for instance when  $\mathbf{Y}_2$  contains a sum and its summands. A generalized inverse ([Definition 1.3.1](#)) removes the need for a separate treatment. The facts that make this work were proved for arbitrary random vectors in [Section 2.5](#). We collect them in the notation of this chapter.

**Lemma 3.5.1**

Let  $\Sigma$  be nonnegative definite and partitioned as above, and let  $\mathbf{G}$  be any generalized inverse of  $\Sigma_{22}$ . Then:

- (a)  $\mathcal{C}(\Sigma_{21}) \subseteq \mathcal{C}(\Sigma_{22})$ , so  $\Sigma_{21} = \Sigma_{22}\mathbf{H}$  for some matrix  $\mathbf{H}$ ;
- (b)  $\Sigma_{12}\mathbf{G}\Sigma_{22} = \Sigma_{12}$  and  $\Sigma_{22}\mathbf{G}\Sigma_{21} = \Sigma_{21}$ ;
- (c)  $\Sigma_{12}\mathbf{G}\Sigma_{21} = \mathbf{H}^\top \Sigma_{22}\mathbf{H}$ , which does not depend on the choice of  $\mathbf{G}$  and is nonnegative definite.

The matrix  $\Sigma_{11.2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^-\Sigma_{21}$ , the **Schur complement** of  $\Sigma_{22}$  in  $\Sigma$ , is therefore well defined. It is the partial covariance matrix of  $\mathbf{Y}_1$  given  $\mathbf{Y}_2$  in the sense of [Theorem 2.5.2](#).

*Proof.* By [Theorem 3.1.2](#),  $\Sigma$  is the covariance matrix of some random vector, so (a) and (b) are [Theorem 2.5.1\(c\)](#). For (c),  $\Sigma_{12}\mathbf{G}\Sigma_{21} = \mathbf{H}^\top \Sigma_{22}\mathbf{G}\Sigma_{22}\mathbf{H} = \mathbf{H}^\top \Sigma_{22}\mathbf{H}$ . ■

**3.5.2 The conditional distribution****Theorem 3.5.2 — Conditional distribution**

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$  be partitioned as above, let  $\mathbf{G}$  be a generalized inverse of  $\Sigma_{22}$ , and put  $\mathbf{B} = \Sigma_{12}\mathbf{G}$ . Then:

- (a)  $\mathbf{W} = \mathbf{Y}_1 - \boldsymbol{\mu}_1 - \mathbf{B}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$  is independent of  $\mathbf{Y}_2$  and  $\mathbf{W} \sim N_{n_1}(\mathbf{0}, \Sigma_{11.2})$ ;
- (b) the conditional distribution of  $\mathbf{Y}_1$  given  $\mathbf{Y}_2 = \mathbf{y}_2$  is

$$\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2 \sim N_{n_1}\left(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^-(\mathbf{y}_2 - \boldsymbol{\mu}_2), \Sigma_{11} - \Sigma_{12}\Sigma_{22}^-\Sigma_{21}\right), \quad (3.5.1)$$

and for  $(\mathbf{y}_2)$  in  $(\boldsymbol{\mu}_2 + \mathcal{C}(\Sigma_{22}))$ , where  $(\mathbf{Y}_2)$  lies with one, the conditional mean does not depend on the choice of  $(\mathbf{G})$ ;

- (c) if  $\Sigma$  is positive definite, then  $\Sigma_{22}^- = \Sigma_{22}^{-1}$ ,  $\Sigma_{11.2}$  is positive definite,  $\Sigma_{11.2}^{-1}$  is the leading  $n_1 \times n_1$  block of  $\Sigma^{-1}$ , and  $\det \Sigma = \det \Sigma_{22} \det \Sigma_{11.2}$ .

*Proof.*

- (a)  $\mathbf{W} = [\mathbf{I}, -\mathbf{B}](\mathbf{Y} - \boldsymbol{\mu})$ , so  $(\mathbf{W}^\top, \mathbf{Y}_2^\top)^\top$  is an affine function of  $\mathbf{Y}$  and is jointly normal ([Theorem 3.3.1](#)). The vector  $\mathbf{W}$  is the error of the best linear predictor of  $\mathbf{Y}_1$  from  $\mathbf{Y}_2$ , so by [Theorem 2.5.2\(a,b\)](#) it has mean zero,  $\text{Cov}(\mathbf{W}, \mathbf{Y}_2) = \mathbf{0}$  and  $\text{Cov}(\mathbf{W}) = \Sigma_{11.2}$ . These are moment calculations, valid for any random vector. What normality adds is that zero covariance becomes independence ([Theorem 3.4.1](#)), and that  $\mathbf{W}$ , a linear function of  $\mathbf{Y}$ , is normal.
- (b) Write  $\mathbf{Y}_1 = \boldsymbol{\mu}_1 + \mathbf{B}(\mathbf{Y}_2 - \boldsymbol{\mu}_2) + \mathbf{W}$ . The first two terms are a function of  $\mathbf{Y}_2$ , and  $\mathbf{W}$  is independent of  $\mathbf{Y}_2$ . For independent  $\mathbf{W}$  and  $\mathbf{Y}_2$ , the conditional distribution

of  $h(\mathbf{W}, \mathbf{Y}_2)$  given  $\mathbf{Y}_2 = \mathbf{y}_2$  is the distribution of  $h(\mathbf{W}, \mathbf{y}_2)$  (Billingsley 1995). With  $h(\mathbf{w}, \mathbf{y}_2) = \boldsymbol{\mu}_1 + \mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2) + \mathbf{w}$ , this is the normal distribution Eq. 3.5.1. If  $\mathbf{y}_2 - \boldsymbol{\mu}_2 = \boldsymbol{\Sigma}_{22}\mathbf{k}$ , then  $\mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2) = \boldsymbol{\Sigma}_{12}\mathbf{G}\boldsymbol{\Sigma}_{22}\mathbf{k} = \boldsymbol{\Sigma}_{12}\mathbf{k}$ , which does not involve  $\mathbf{G}$ .

- (c) If  $\boldsymbol{\Sigma}$  is positive definite, so is its principal submatrix  $\boldsymbol{\Sigma}_{22}$ , and its only generalized inverse is its inverse. By the block determinant formula (Theorem 1.6.3), which needs only  $\boldsymbol{\Sigma}_{22}$  nonsingular,  $\det \boldsymbol{\Sigma} = \det \boldsymbol{\Sigma}_{22} \det \boldsymbol{\Sigma}_{11.2}$ . The left side and  $\det \boldsymbol{\Sigma}_{22}$  are positive, so  $\boldsymbol{\Sigma}_{11.2}$  is nonsingular. It is nonnegative definite, being  $\text{Cov}(\mathbf{W})$  by (a), hence positive definite. With both  $\boldsymbol{\Sigma}_{22}$  and its Schur complement nonsingular, the partitioned-inverse formula (Theorem 1.6.4) gives  $\boldsymbol{\Sigma}_{11.2}^{-1}$  as the leading block of  $\boldsymbol{\Sigma}^{-1}$ .

When  $\boldsymbol{\Sigma}$  is positive definite, (b) can also be read off the densities. By (c),  $\boldsymbol{\Sigma}_{11.2}$  is positive definite, so  $\mathbf{W}$  has a density. The map  $(\mathbf{y}_1, \mathbf{y}_2) \mapsto (\mathbf{w}, \mathbf{y}_2)$  has unit Jacobian, so  $f_{\mathbf{Y}}(\mathbf{y}_1, \mathbf{y}_2) = f_{\mathbf{W}}(\mathbf{y}_1 - \boldsymbol{\mu}_1 - \mathbf{B}(\mathbf{y}_2 - \boldsymbol{\mu}_2)) f_{\mathbf{Y}_2}(\mathbf{y}_2)$  by independence, and dividing by  $f_{\mathbf{Y}_2}(\mathbf{y}_2)$  gives the conditional density.

■

*Idea.* Every jointly normal vector splits as

$$\mathbf{Y}_1 = \underbrace{\boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)}_{\text{function of } \mathbf{Y}_2} + \underbrace{\mathbf{W}}_{\text{independent of } \mathbf{Y}_2}.$$

The first term is a linear regression of  $\mathbf{Y}_1$  on  $\mathbf{Y}_2$ . The second is a normal error whose distribution is the same whatever value  $\mathbf{Y}_2$  takes.

### 3.5.3 What the theorem says

Three features of Eq. 3.5.1 are special to the normal distribution, and each is an assumption of the classical linear model.

- (i) *The regression is linear.*  $E(\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2)$  is an affine function of  $\mathbf{y}_2$ , with coefficient matrix  $\boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}$ .
- (ii) *The errors are homoscedastic.* The conditional covariance  $\boldsymbol{\Sigma}_{11.2}$  is the same for every  $\mathbf{y}_2$ . Observing  $\mathbf{Y}_2$  tells us where  $\mathbf{Y}_1$  is centred, but not how spread out it is.
- (iii) *Conditioning cannot increase uncertainty.*  $\boldsymbol{\Sigma}_{11} - \boldsymbol{\Sigma}_{11.2} = \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}\boldsymbol{\Sigma}_{21}$  is nonnegative definite (Lemma 3.5.1(c)), so every linear combination  $\mathbf{a}^\top \mathbf{Y}_1$  has conditional variance at most its unconditional variance. The reduction is zero iff  $\boldsymbol{\Sigma}_{12} = \mathbf{0}$ .

None of these holds in general. In the mixture of Example (A mixture of two correlations), given  $Y_1 = y_1$  the variable  $Y_2$  is an equal mixture of  $N(\rho y_1, 1 - \rho^2)$  and  $N(-\rho y_1, 1 - \rho^2)$ . Its conditional mean is 0 for every  $y_1$ , but its conditional variance is  $1 - \rho^2 + \rho^2 y_1^2$ , which grows with  $|y_1|$ . Linearity and constant variance are not consequences of normal margins or of zero correlation. They are consequences of joint normality.

**One response, several predictors.** The case used most in this book has a scalar response  $Y$  and a vector  $\mathbf{X}$  of  $k$  predictors, jointly normal, with  $\text{Var}(Y) = \sigma_Y^2$ ,  $\boldsymbol{\sigma}_{XY} = \text{Cov}(\mathbf{X}, Y)$  and  $\boldsymbol{\Sigma}_{XX} = \text{Cov}(\mathbf{X})$  positive definite. Theorem 3.5.2 says

$$Y = \alpha + \boldsymbol{\beta}^\top \mathbf{X} + e, \quad \boldsymbol{\beta} = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}, \quad \alpha = \mu_Y - \boldsymbol{\beta}^\top \boldsymbol{\mu}_X, \quad e \sim N(0, \sigma^2) \text{ independent of } \mathbf{X}, \quad (3.5.2)$$

with  $\sigma^2 = \sigma_Y^2 - \boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY} = \sigma_Y^2(1 - \rho_{Y \cdot X}^2)$ , where  $\rho_{Y \cdot X}^2 = \boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY} / \sigma_Y^2$  is the population squared multiple correlation ([Proposition 2.5.3](#)). So if observations  $(Y_i, \mathbf{X}_i)$  are drawn independently from a joint normal distribution, then conditionally on the predictors, the normal linear model with independent homoscedastic errors holds exactly. Inference that treats the predictors as fixed is valid for random predictors, as [Chapter 14](#) develops ([Theorem 14.1.2](#)).

**The bivariate case.** With  $k = 1$ , correlation  $\rho$  and standard deviations  $\sigma_X, \sigma_Y$ ,

$$E(Y | X = x) = \mu_Y + \rho \frac{\sigma_Y}{\sigma_X} (x - \mu_X), \quad \text{Var}(Y | X = x) = \sigma_Y^2(1 - \rho^2).$$

In standard units,  $(E(Y | X = x) - \mu_Y) / \sigma_Y = \rho(x - \mu_X) / \sigma_X$ . The predicted response is fewer standard deviations from its mean than the predictor is from its mean, whenever  $|\rho| < 1$ . This is **regression toward the mean**, Galton's observation about the heights of parents and children (Galton 1886), and the origin of the word “regression”. It is also a warning. Regressing  $X$  on  $Y$  is a different line,  $E(X | Y = y) = \mu_X + \rho(\sigma_X / \sigma_Y)(y - \mu_Y)$ , and when  $0 < |\rho| < 1$  neither line is the major axis of the density contours.

#### Example. (Two regression lines)

Take  $\mu_X = 1$ ,  $\mu_Y = 2$ ,  $\sigma_X = 1$ ,  $\sigma_Y = 1.5$  and  $\rho = 0.7$ . The regression of  $Y$  on  $X$  has slope 1.05 and residual standard deviation 1.071. Drawn in the same  $(x, y)$  plane, the regression of  $X$  on  $Y$  has slope 2.14, and the major axis of the ellipses lies between them, with slope 1.76. [Figure 3.5.1](#) shows the three lines. The line  $E(Y | X = x)$  passes through the points where each contour ellipse has a vertical tangent ([Exercise \(B2\)](#)): on a vertical line  $X = x$ , the density is highest at the conditional mean. A least squares line fitted to 60 draws has slope 1.12 (standard error 0.16). It estimates the regression of  $Y$  on  $X$ , not the major axis.

#### Example. (Holding a third variable fixed)

Let  $\mathbf{Y} \sim N_3(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  with

$$\boldsymbol{\mu} = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}, \quad \boldsymbol{\Sigma} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 3 & 2 \\ 2 & 2 & 2 \end{pmatrix}.$$

Conditioning  $Y_1$  on  $Y_2$  alone gives slope  $\sigma_{12} / \sigma_{22} = \frac{1}{3}$  and conditional variance  $\frac{11}{3}$ . Conditioning on  $(Y_2, Y_3)$ , the listing computes  $\mathbf{B} = \boldsymbol{\Sigma}_{12} \boldsymbol{\Sigma}_{22}^{-1}$  in exact arithmetic and finds

$$E(Y_1 | Y_2 = y_2, Y_3 = y_3) = 3 - y_2 + 2y_3, \quad \text{Var}(Y_1 | Y_2, Y_3) = 1.$$

The coefficient of  $y_2$  changes sign. Among outcomes with the same  $Y_3$ , larger  $Y_2$  goes with smaller  $Y_1$ . Across all outcomes, larger  $Y_2$  goes with larger  $Y_1$ , because both tend to be large when  $Y_3$  is large. The conditional variance 1 also equals  $\det \boldsymbol{\Sigma} / \det \boldsymbol{\Sigma}_{22} = 2/2$ , as [Theorem 3.5.2\(c\)](#) predicts.

```
from fractions import Fraction as F
```

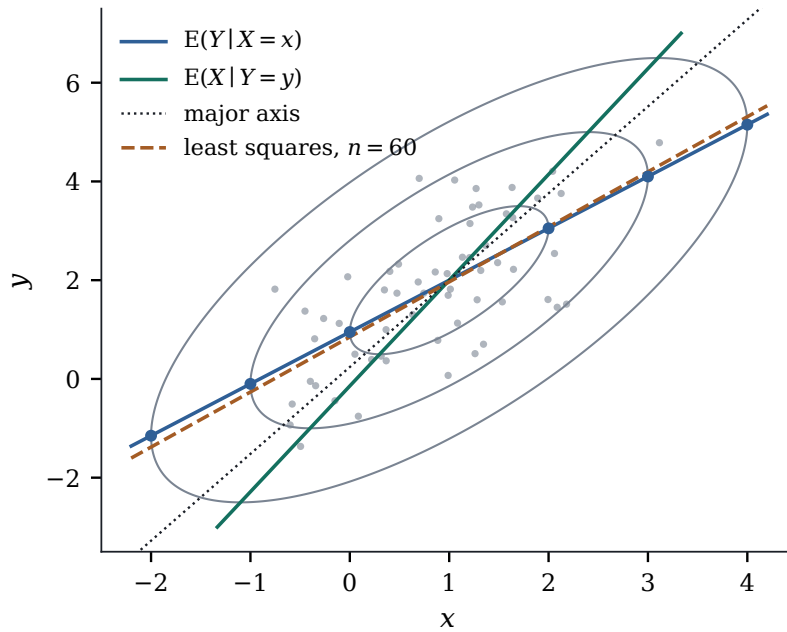


Figure 3.5.1: Contours  $\Delta^2 = 1, 4, 9$  of a bivariate normal density with  $\rho = 0.7$ . The conditional mean line  $E(Y | X = x)$  passes through the points of vertical tangency (dots). The line  $E(X | Y = y)$  passes through the points of horizontal tangency and is steeper, and the major axis lies between the two. Least squares on a sample of 60 points estimates  $E(Y | X = x)$ .

```
import numpy as np
rng = np.random.default_rng(305)
mx, my, sx, sy, rho = 1.0, 2.0, 1.0, 1.5, 0.7
Sigma = np.array([[sx**2, rho * sx * sy], [rho * sx * sy, sy**2]])
slope_y_on_x = rho * sy / sx # E(Y | X = x)
slope_x_on_y = rho * sx / sy # E(X | Y = y), as dx/dy
cond_sd = sy * np.sqrt(1 - rho**2)
n = 60
XY = rng.multivariate_normal([mx, my], Sigma, size=n)
b_ls, a_ls = np.polyfit(XY[:, 0], XY[:, 1], 1)
Lin = np.linalg.inv(np.linalg.cholesky(Sigma))
theta = np.linspace(0, 2 * np.pi, 200_001)
circle = np.vstack([np.cos(theta), np.sin(theta)])
L = np.linalg.cholesky(Sigma)
big = rng.multivariate_normal([mx, my], Sigma, size=4_000_000)
x0 = 2.0
s1 = big[np.abs(big[:, 0] - x0) < 0.01, 1]
resid = XY[:, 1] - a_ls - b_ls * XY[:, 0]
se = np.sqrt(resid @ resid / (n - 2) / np.sum((XY[:, 0] - XY[:, 0].mean())**2))
lam, U = np.linalg.eigh(Sigma)
slope_major = U[1, 1] / U[0, 1]
levels = [1, 4, 9]
gx, gy = np.meshgrid(np.linspace(-3, 5, 300), np.linspace(-4, 8, 300))
D = np.stack([gx - mx, gy - my], axis=-1)
D2 = np.einsum("...i,ij,...j->...", D, np.linalg.inv(Sigma), D)
xs = np.linspace(-2.2, 4.2, 2)
ys = np.linspace(-3.0, 7.0, 2)

mu = [F(1), F(0), F(-1)]
S = [[F(4), F(1), F(2)],
      [F(1), F(3), F(2)],
```



```

[F(2), F(2), F(2)]

def inv2(M):
    det = M[0][0] * M[1][1] - M[0][1] * M[1][0]
    return [[M[1][1] / det, -M[0][1] / det], [-M[1][0] / det, M[0][0] / det]]

# Y1 given (Y2, Y3): B = sigma_12 Sigma_22^{-1}, variance sigma_11 - B sigma_21
S22inv = inv2([[S[1][1], S[1][2]], [S[2][1], S[2][2]]])
s12 = [S[0][1], S[0][2]]
B = [s12[0] * S22inv[0][j] + s12[1] * S22inv[1][j] for j in range(2)]
var_1_23 = S[0][0] - (B[0] * s12[0] + B[1] * s12[1])
const = mu[0] - B[0] * mu[1] - B[1] * mu[2]
print("E(Y1 | y2, y3) =", const, "+", B[0], "y2 +", B[1], "y3; variance", var_1_23)

# (Y1, Y2) given Y3, and the partial correlation of Y1 and Y2 given Y3
C = [[S[i][j] - S[i][2] * S[2][j] / S[2][2] for j in range(2)] for i in range(2)]
print("Cov((Y1, Y2) | Y3) =", C)

```

### 3.5.4 The conditional mean is the best predictor

Section 6.11 defines the best linear predictor  $L(\mathbf{Y}_1 | \mathbf{Y}_2)$  as a projection, and shows that the conditional mean is the best predictor among *all* square-integrable functions of  $\mathbf{Y}_2$ . For a scalar response, the gap between the two is the approximation error of Proposition 6.11.2(c). Under joint normality the gap is zero.

#### Proposition 3.5.3

Let  $\mathbf{Y}$  be jointly normal and partitioned as above, and let  $\mathbf{m}(\mathbf{Y}_2) = \boldsymbol{\mu}_1 + \boldsymbol{\Sigma}_{12}\boldsymbol{\Sigma}_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$ . For every function  $\mathbf{g}$  with  $E\|\mathbf{g}(\mathbf{Y}_2)\|^2 < \infty$ ,

$$E\|\mathbf{Y}_1 - \mathbf{g}(\mathbf{Y}_2)\|^2 = \text{tr}(\boldsymbol{\Sigma}_{11.2}) + E\|\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2)\|^2.$$

So  $\mathbf{m}(\mathbf{Y}_2)$  minimizes the mean squared prediction error over all predictors, it equals the best linear predictor of Theorem 6.11.1 (entry by entry, when  $\boldsymbol{\Sigma}_{22}$  is positive definite), and the minimum is  $\text{tr}(\boldsymbol{\Sigma}_{11.2})$ .

*Proof.*  $\mathbf{Y}_1 - \mathbf{g}(\mathbf{Y}_2) = \mathbf{W} + (\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2))$  with  $\mathbf{W}$  as in Theorem 3.5.2. The cross term  $E[\mathbf{W}^\top(\mathbf{m}(\mathbf{Y}_2) - \mathbf{g}(\mathbf{Y}_2))]$  vanishes, because  $\mathbf{W}$  has mean zero and is independent of  $\mathbf{Y}_2$ . And  $E\|\mathbf{W}\|^2 = \text{tr Cov}(\mathbf{W}) = \text{tr}(\boldsymbol{\Sigma}_{11.2})$ . The minimizer  $\mathbf{m}$  is affine, so it is also the best affine predictor, which is the best linear predictor. ■

The proof is short because independence of  $\mathbf{W}$  and  $\mathbf{Y}_2$  is much stronger than the zero correlation that defines a projection. For non-normal vectors only zero correlation is available, and the best linear predictor is merely the best *linear* predictor.

### 3.5.5 Exercises

#### 3.5.5.1 A. Check your understanding

##### Exercise A1.

For  $\boldsymbol{\mu}$  and  $\boldsymbol{\Sigma}$  of [Example \(Holding a third variable fixed\)](#), find the conditional distribution of  $(Y_1, Y_2)$  given  $Y_3 = y_3$  and of  $Y_3$  given  $(Y_1, Y_2) = (y_1, y_2)$ . Verify that the conditional variances are smaller than the unconditional ones.

##### Exercise A2.

In the setting of [Eq. 3.5.2](#), let  $\boldsymbol{\Omega}$  be the inverse of  $\text{Cov}((Y, \mathbf{X}^\top)^\top)$ . Show that the row of  $\boldsymbol{\Omega}$  belonging to  $Y$  is  $(1, -\boldsymbol{\beta}^\top)/\sigma^2$ , where  $\sigma^2 = \sigma_Y^2(1 - \rho_{Y,X}^2)$ . So the regression coefficients and the error variance can be read off one row of the precision matrix.

*Solution.* Put  $Y$  first and apply [Exercise \(B5\)](#) with the scalar block  $Y$  and  $\mathbf{Y}_2 = \mathbf{X}$ . The conditional variance is  $\omega_{YY}^{-1} = \text{Var}(Y | \mathbf{X}) = \sigma^2$ , and the coefficient vector of the conditional mean is  $-\omega_{YY}^{-1}\boldsymbol{\Omega}_{YX} = \boldsymbol{\beta}^\top$ . Hence  $\omega_{YY} = 1/\sigma^2$  and  $\boldsymbol{\Omega}_{YX} = -\boldsymbol{\beta}^\top/\sigma^2$ . By [Eq. 3.5.2](#),  $\sigma^2 = \sigma_Y^2(1 - \rho_{Y,X}^2)$ .

#### 3.5.5.2 B. Practice

##### Exercise B1.

Suppose  $\mathbf{Y}_2 \sim N(\boldsymbol{\mu}_2, \boldsymbol{\Sigma}_{22})$  and that, given  $\mathbf{Y}_2 = \mathbf{y}_2$ ,  $\mathbf{Y}_1$  is  $N(\mathbf{a} + \mathbf{B}\mathbf{y}_2, \mathbf{S})$  with  $\mathbf{a}$ ,  $\mathbf{B}$  and  $\mathbf{S}$  not depending on  $\mathbf{y}_2$ . Show that  $(\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$  is jointly normal and find its mean and covariance. Explain why [Example \(A mixture of two correlations\)](#) does not contradict this.

*Solution.* Condition on  $\mathbf{Y}_2$ :

$$\mathbb{E} e^{\mathbf{t}_1^\top \mathbf{Y}_1 + \mathbf{t}_2^\top \mathbf{Y}_2} = \mathbb{E} \left[ e^{\mathbf{t}_2^\top \mathbf{Y}_2} e^{\mathbf{t}_1^\top (\mathbf{a} + \mathbf{B}\mathbf{Y}_2) + \frac{1}{2} \mathbf{t}_1^\top \mathbf{S} \mathbf{t}_1} \right] = e^{\mathbf{t}_1^\top \mathbf{a} + \frac{1}{2} \mathbf{t}_1^\top \mathbf{S} \mathbf{t}_1} M_{\mathbf{Y}_2}(\mathbf{t}_2 + \mathbf{B}^\top \mathbf{t}_1).$$

Inserting  $M_{\mathbf{Y}_2}$  from [Theorem 3.2.1](#), the exponent is linear plus  $\frac{1}{2} \mathbf{t}^\top \mathbf{C} \mathbf{t}$  with  $\mathbf{C} = \begin{pmatrix} \mathbf{S} + \mathbf{B}\boldsymbol{\Sigma}_{22}\mathbf{B}^\top & \mathbf{B}\boldsymbol{\Sigma}_{22} \\ \boldsymbol{\Sigma}_{22}\mathbf{B}^\top & \boldsymbol{\Sigma}_{22} \end{pmatrix}$ , so the pair is normal with mean  $((\mathbf{a} + \mathbf{B}\boldsymbol{\mu}_2)^\top, \boldsymbol{\mu}_2^\top)^\top$  and covariance  $\mathbf{C}$ . In [Example \(A mixture of two correlations\)](#), the conditional distribution of  $Y_2$  given  $Y_1$  is a mixture, not a normal distribution, and its variance depends on  $y_1$ .

##### Exercise B2.

Show that for a bivariate normal density with  $|\rho| < 1$ , the points of each contour ellipse at which the tangent is vertical lie on the line  $y = \mathbb{E}(Y | X = x)$ . Explain the result in terms of the conditional density of  $Y$  given  $X = x$ .

*Solution.* Write the ellipse as  $\{\boldsymbol{\mu} + \mathbf{u} : \mathbf{u}^\top \boldsymbol{\Sigma}^{-1} \mathbf{u} = c^2\}$ . At a point with vertical tangent, the normal vector  $\boldsymbol{\Sigma}^{-1} \mathbf{u}$  is horizontal, so  $\boldsymbol{\Sigma}^{-1} \mathbf{u} = \lambda(1, 0)^\top$  and  $\mathbf{u} = \lambda(\sigma_X^2, \rho\sigma_X\sigma_Y)^\top$ . The slope  $u_2/u_1 = \rho\sigma_Y/\sigma_X$  is that of the conditional mean line, and both such points ( $\lambda > 0$

and  $\lambda < 0$ ) lie on it. Along the vertical line  $X = x$ , the joint density is proportional to the conditional density of  $Y$ , which is highest at  $E(Y | X = x)$ . The contour that just touches the vertical line touches it at the point of highest density on that line.

### Exercise B3.

Let  $Z_1, \dots, Z_n$  be independent  $N(0, 1)$  and  $S_k = Z_1 + \dots + Z_k$ . Show that  $(S_1, \dots, S_n)$  is jointly normal with  $\text{Cov}(S_j, S_k) = \min(j, k)$ . Find the conditional distribution of  $S_j$  given  $S_n = s$ , and comment on its variance as a function of  $j$ .

*Solution.*  $(S_1, \dots, S_n)^\top = \mathbf{L}\mathbf{Z}$  with  $\mathbf{L}$  lower triangular and all entries on and below the diagonal equal to one, so the vector is normal with covariance  $\mathbf{L}\mathbf{L}^\top$ , whose  $(j, k)$  entry counts the common indices,  $\min(j, k)$ . By [Theorem 3.5.2](#) with  $\mathbf{Y}_1 = S_j$ ,  $\mathbf{Y}_2 = S_n$ ,

$$S_j | S_n = s \sim N\left(\frac{j}{n}s, j - \frac{j^2}{n}\right) = N\left(\frac{j}{n}s, \frac{j(n-j)}{n}\right).$$

The conditional mean interpolates linearly between 0 and  $s$ . The variance is zero at both ends, where the value is known, and largest in the middle.

### Exercise B4.

Let  $(Y, X_1, X_2)$  be normal with positive definite covariance, and let  $\mathbf{Y}_2 = (X_1, X_2, X_1 + X_2)^\top$ . Show that  $\Sigma_{22}$  is singular. Using a generalized inverse, show that the conditional distribution of  $Y$  given  $\mathbf{Y}_2$  from [Theorem 3.5.2](#) coincides with its conditional distribution given  $(X_1, X_2)$ .

### Exercise B5.

With  $\Omega = \Sigma^{-1}$  partitioned like  $\Sigma$ , show that  $E(\mathbf{Y}_1 | \mathbf{Y}_2 = \mathbf{y}_2) = \boldsymbol{\mu}_1 - \Omega_{11}^{-1}\Omega_{12}(\mathbf{y}_2 - \boldsymbol{\mu}_2)$  and  $\text{Cov}(\mathbf{Y}_1 | \mathbf{Y}_2) = \Omega_{11}^{-1}$ .

*Solution.* By [Theorem 1.6.4](#),  $\Omega_{11} = \Sigma_{11.2}^{-1}$  and  $\Omega_{12} = -\Sigma_{11.2}^{-1}\Sigma_{12}\Sigma_{22}^{-1}$ . Hence  $-\Omega_{11}^{-1}\Omega_{12} = \Sigma_{12}\Sigma_{22}^{-1}$  and  $\Omega_{11}^{-1} = \Sigma_{11.2}$ . Substitute into [Eq. 3.5.1](#).

### 3.5.5.3 C. Going deeper

#### Exercise C1.

Assume  $\Sigma$  is positive definite. Prove [Theorem 3.5.2\(b\)](#) by dividing the joint density by the marginal density of  $\mathbf{Y}_2$  directly, using the partitioned inverse and the block determinant formula.

## 3.6 Partial correlation

Two variables can be strongly correlated only because both depend on a third. [Example \(Holding a third variable fixed\)](#) showed a positive association between  $Y_1$  and  $Y_2$  turning negative once  $Y_3$  is held fixed. The conditional covariance matrix  $\Sigma_{11 \cdot 2}$  measures association that remains after conditioning, and its correlation form is the partial correlation.

### 3.6.1 Definition and interpretations

#### Definition 3.6.1 — Partial correlation

Let  $\mathbf{Y} = (\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$  be a random vector with covariance matrix  $\Sigma$ , partitioned as in [Section 3.5](#), and write  $\Sigma_{11 \cdot 2} = \Sigma_{11} - \Sigma_{12}\Sigma_{22}^{-1}\Sigma_{21} = (\sigma_{ij \cdot 2})$ . The **partial covariance** of  $Y_i$  and  $Y_j$  (two entries of  $\mathbf{Y}_1$ ) given  $\mathbf{Y}_2$  is  $\sigma_{ij \cdot 2}$ , and their **partial correlation** given  $\mathbf{Y}_2$  is

$$\rho_{ij \cdot 2} = \frac{\sigma_{ij \cdot 2}}{\sqrt{\sigma_{ii \cdot 2} \sigma_{jj \cdot 2}}},$$

provided  $\sigma_{ii \cdot 2}$  and  $\sigma_{jj \cdot 2}$  are positive. When the conditioning variables are  $Y_k, Y_l, \dots$  we also write  $\rho_{ij \cdot kl \dots}$ . The matrix of partial correlations is  $\mathbf{D}^{-1/2} \Sigma_{11 \cdot 2} \mathbf{D}^{-1/2}$  with  $\mathbf{D} = \text{diag}(\Sigma_{11 \cdot 2})$ , the correlation matrix ([Definition 2.4.1](#)) built from  $\Sigma_{11 \cdot 2}$ .

The definition uses only  $\Sigma$ , so partial correlations exist for any random vector with finite second moments. The next result gives them two meanings, one that always applies and one that needs normality.

#### Proposition 3.6.1

In the setting of [Definition 3.6.1](#):

- (a)  $\Sigma_{11 \cdot 2}$  is the covariance matrix of the prediction errors  $\mathbf{W} = \mathbf{Y}_1 - \boldsymbol{\mu}_1 - \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$ , and  $\mathbf{W}$  is uncorrelated with  $\mathbf{Y}_2$ . If  $\Sigma_{22}$  is positive definite, the entries of  $\mathbf{W}$  are the errors of the best linear predictors  $L(Y_i | \mathbf{Y}_2)$ , so  $\rho_{ij \cdot 2}$  is the correlation between  $Y_i - L(Y_i | \mathbf{Y}_2)$  and  $Y_j - L(Y_j | \mathbf{Y}_2)$ .
- (b) If  $\mathbf{Y}$  is jointly normal,  $\rho_{ij \cdot 2}$  is the correlation of  $Y_i$  and  $Y_j$  in their conditional distribution given  $\mathbf{Y}_2 = \mathbf{y}_2$ , the same for every  $\mathbf{y}_2$ . Moreover  $\rho_{ij \cdot 2} = 0$  iff  $Y_i$  and  $Y_j$  are conditionally independent given  $\mathbf{Y}_2$ .

*Proof.*

- (a) This is [Theorem 2.5.2](#), restated:  $\text{Cov}(\mathbf{W}, \mathbf{Y}_2) = \mathbf{0}$  and  $\text{Cov}(\mathbf{W}) = \Sigma_{11 \cdot 2}$  are the moment calculations in the proof of [Theorem 3.5.2\(a\)](#), which use no normality. When  $\Sigma_{22}$  is positive definite, the  $i$ th entry of  $\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^{-1}(\mathbf{Y}_2 - \boldsymbol{\mu}_2)$  is  $L(Y_i | \mathbf{Y}_2)$  by [Theorem 6.11.1](#).
- (b) By [Theorem 3.5.2](#), the conditional distribution of  $(Y_i, Y_j)$  given  $\mathbf{Y}_2 = \mathbf{y}_2$  is bivariate normal with covariance matrix the corresponding  $2 \times 2$  block of  $\Sigma_{11 \cdot 2}$ . Its correlation is  $\rho_{ij \cdot 2}$ , and by [Theorem 3.4.1](#) applied to this conditional normal distribution, the two are independent iff the off-diagonal entry is zero.

Part (a) is the definition to remember: *a partial correlation is the correlation of two residuals*. Remove from each variable the part that is linearly predictable from the conditioning variables, and correlate what is left. This is the population version of the Frisch–Waugh–Lovell construction ([Theorem 6.6.2](#)).

### 3.6.2 Computing partial correlations

Two routes avoid computing  $\Sigma_{11 \cdot 2}$  in full.

With a single conditioning variable,  $\Sigma_{11 \cdot 2}$  is a  $2 \times 2$  matrix with entries  $\sigma_{ij} - \sigma_{i3}\sigma_{j3}/\sigma_{33}$ , and in correlation units this is the familiar three-variable formula of [Exercise \(B1\)](#),

$$\rho_{12 \cdot 3} = \frac{\rho_{12} - \rho_{13}\rho_{23}}{\sqrt{(1 - \rho_{13}^2)(1 - \rho_{23}^2)}}, \quad |\rho_{13}| < 1, \quad |\rho_{23}| < 1. \quad (3.6.1)$$

The residual interpretation shows that the same formula can be applied one conditioning variable at a time.

#### Proposition 3.6.2 — Recursion

Let  $S$  be a set of indices and  $k \notin S \cup \{1, 2\}$  one more. Suppose  $\sigma_{11 \cdot S}$ ,  $\sigma_{22 \cdot S}$  and  $\sigma_{kk \cdot S}$  are positive and  $|\rho_{1k \cdot S}| < 1$ ,  $|\rho_{2k \cdot S}| < 1$ . Then

$$\sigma_{12 \cdot Sk} = \sigma_{12 \cdot S} - \frac{\sigma_{1k \cdot S} \sigma_{2k \cdot S}}{\sigma_{kk \cdot S}}, \quad \rho_{12 \cdot Sk} = \frac{\rho_{12 \cdot S} - \rho_{1k \cdot S} \rho_{2k \cdot S}}{\sqrt{(1 - \rho_{1k \cdot S}^2)(1 - \rho_{2k \cdot S}^2)}}.$$

With  $S$  empty this is [Eq. 3.6.1](#).

*Proof.* No normality is needed. For  $i \in \{1, 2, k\}$  let  $e_i$  be the error of the best linear predictor of  $Y_i$  from  $\mathbf{Y}_S$ , as in [Proposition 3.6.1\(a\)](#). Then each  $e_i$  has mean zero, is uncorrelated with  $\mathbf{Y}_S$ , and  $\text{Cov}(e_i, e_j) = \sigma_{ij \cdot S}$ . Put  $r_i = e_i - c_i e_k$  with  $c_i = \sigma_{ik \cdot S}/\sigma_{kk \cdot S}$ , for  $i = 1, 2$ .

We claim that  $r_i$  is the error of the best linear predictor of  $Y_i$  from  $(\mathbf{Y}_S, Y_k)$ . First,  $Y_i - r_i$  is an affine function of  $(\mathbf{Y}_S, Y_k)$ , because  $e_i$  and  $e_k$  differ from  $Y_i$  and  $Y_k$  by affine functions of  $\mathbf{Y}_S$ . Second,  $r_i$  has mean zero and is uncorrelated with  $\mathbf{Y}_S$  and with  $e_k$ , by the choice of  $c_i$ , hence with  $Y_k = e_k + (\text{affine in } \mathbf{Y}_S)$ . Now let  $r'_i$  be the error from [Proposition 3.6.1\(a\)](#) for predicting  $Y_i$  from  $(\mathbf{Y}_S, Y_k)$ , which has the same two properties. The difference  $d = r_i - r'_i$  is an affine function of  $(\mathbf{Y}_S, Y_k)$  with mean zero and is uncorrelated with  $(\mathbf{Y}_S, Y_k)$ , so  $\text{Var}(d) = \text{Cov}(d, d) = 0$ . Thus  $r_i = r'_i$  with probability one.

Therefore  $\sigma_{12 \cdot Sk} = \text{Cov}(r_1, r_2) = \sigma_{12 \cdot S} - c_2 \sigma_{1k \cdot S} - c_1 \sigma_{2k \cdot S} + c_1 c_2 \sigma_{kk \cdot S}$ , which simplifies to the first formula, and similarly  $\sigma_{ii \cdot Sk} = \sigma_{ii \cdot S}(1 - \rho_{ik \cdot S}^2)$ , which is positive by hypothesis. Dividing  $\sigma_{12 \cdot Sk}$  by  $\sqrt{\sigma_{11 \cdot Sk} \sigma_{22 \cdot Sk}}$  and then numerator and denominator by  $\sqrt{\sigma_{11 \cdot S} \sigma_{22 \cdot S}}$  gives the second.

**Proposition 3.6.3 — Partial correlations from the precision matrix**

Let  $\Sigma$  be positive definite and  $\Omega = \Sigma^{-1} = (\omega_{ij})$ . The partial correlation of  $Y_i$  and  $Y_j$  given all the other entries of  $\mathbf{Y}$  is

$$\rho_{ij \cdot \text{rest}} = -\frac{\omega_{ij}}{\sqrt{\omega_{ii}\omega_{jj}}},$$

and the conditional variance of  $Y_i$  given all the others is  $1/\omega_{ii}$ .

*Proof.* Permute  $\mathbf{Y}$  so that  $(Y_i, Y_j)$  comes first. By [Theorem 3.5.2\(c\)](#) (a statement about matrices, valid whether or not  $\mathbf{Y}$  is normal), the leading  $2 \times 2$  block of  $\Omega$  is  $\Sigma_{11 \cdot 2}^{-1}$ . Inverting,

$$\Sigma_{11 \cdot 2} = \begin{pmatrix} \omega_{ii} & \omega_{ij} \\ \omega_{ij} & \omega_{jj} \end{pmatrix}^{-1} = \frac{1}{\omega_{ii}\omega_{jj} - \omega_{ij}^2} \begin{pmatrix} \omega_{jj} & -\omega_{ij} \\ -\omega_{ij} & \omega_{ii} \end{pmatrix},$$

whose correlation is  $-\omega_{ij}/\sqrt{\omega_{ii}\omega_{jj}}$ . The same argument with a  $1 \times 1$  leading block gives the conditional variance. ■

[Proposition 3.6.3](#) has a striking consequence for normal vectors: *a zero entry  $\omega_{ij}$  of the inverse covariance matrix means that  $Y_i$  and  $Y_j$  are conditionally independent given all the other variables.* Zeros in  $\Sigma$  describe marginal independence, zeros in  $\Sigma^{-1}$  describe conditional independence. The pattern of zeros in  $\Sigma^{-1}$  defines a graph on the variables, which is the starting point of Gaussian graphical models (Dempster 1972; Lauritzen 1996).

**Example. (The trivariate example, continued)**

For  $\Sigma$  of [Example \(Holding a third variable fixed\)](#), the listing gives the covariance of  $(Y_1, Y_2)$  given  $Y_3$  as  $\begin{pmatrix} 2 & -1 \\ -1 & 1 \end{pmatrix}$ . So  $\rho_{12} = 0.289$  while  $\rho_{12 \cdot 3} = -0.707$ . The script confirms that [Eq. 3.6.1](#), [Proposition 3.6.3](#), and the correlation of simulated residuals all give this value.

**3.6.3 Sample partial correlations**

For data, replace  $\Sigma$  by the sample covariance matrix. Because the empirical distribution of the rows is a probability distribution in its own right ([Section 6.11](#)), all the identities above hold exactly for the sample versions. In particular, for two variables  $\mathbf{y}, \mathbf{x}$  and conditioning variables collected with an intercept in  $\mathbf{Z}$ , the following numbers are *identical*:

- (1) the correlation of the residuals of  $\mathbf{y}$  and of  $\mathbf{x}$  after least squares regression on  $\mathbf{Z}$ ;
- (2) the recursion [Eq. 3.6.1](#) applied to sample correlations (one conditioning variable) or repeatedly (several);
- (3)  $-w_{12}/\sqrt{w_{11}w_{22}}$  computed from the inverse of the sample covariance or correlation matrix of  $(\mathbf{y}, \mathbf{x}, \text{conditioning variables})$ ;
- (4)  $\text{sign}(t)\sqrt{t^2/(t^2 + n - p)}$ , where  $t$  is the  $t$  statistic of  $\mathbf{x}$  in the regression of  $\mathbf{y}$  on  $[\mathbf{Z}, \mathbf{x}]$  with  $p$  columns ([Exercise \(B3\)](#)).

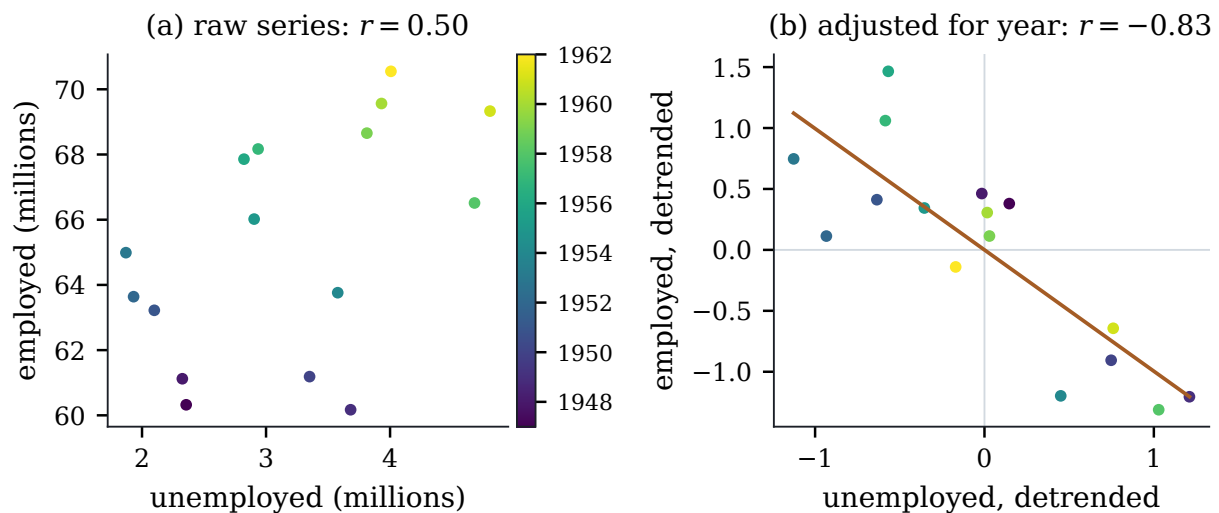


Figure 3.6.1: Employment and unemployment in the United States, 1947–1962, coloured by year. (a) The raw series are positively correlated because both trend upwards. (b) After removing the linear trend in year from each, the residuals are strongly negatively correlated. Their correlation is the partial correlation given year, and the fitted slope is the multiple-regression coefficient.

#### Example. (Employment and unemployment)

Longley’s macroeconomic series record, among other things, total employment and the number unemployed in the United States for the 16 years 1947–1962. Their correlation is 0.502: years with more people employed had more people unemployed. Both series grew with the population. Their correlations with the calendar year are 0.971 and 0.668. Adjusting for year, the partial correlation is  $-0.829$ . Once the common trend is removed, a year with unemployment above its trend is a year with employment below its trend, as one would expect. Figure 3.6.1 shows both scatterplots. The listing computes the partial correlation in all four ways above. They agree to ten decimal places. The slope in panel (b),  $-0.996$ , is the coefficient of unemployment in the regression of employment on unemployment and year, as Theorem 6.6.2 requires. Its  $t$  statistic is  $-5.34$ .

Removing a linear time trend before correlating two series is the problem that Frisch and Waugh (1933) solved, and it is a special case of partial correlation.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data

emp = data["TOTEMP"].to_numpy() / 1000      # thousands -> millions of persons
unemp = data["UNEMP"].to_numpy() / 1000
year = data["YEAR"].to_numpy()
n = len(emp)
Z = np.column_stack([np.ones(n), year])     # the variables we adjust for

def resid(v, Z):
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

# 1. correlation of residuals (detrended series)
r_resid = np.corrcoef(resid(emp, Z), resid(unemp, Z))[0, 1]
```

```

# 2. the one-variable recursion from ordinary correlations
R = np.corrcoef(np.vstack([emp, unemp, year]))
r12, r13, r23 = R[0, 1], R[0, 2], R[1, 2]
r_formula = (r12 - r13 * r23) / np.sqrt((1 - r13**2) * (1 - r23**2))

# 3. from the inverse of the correlation (or covariance) matrix
W = np.linalg.inv(R)
r_precision = -W[0, 1] / np.sqrt(W[0, 0] * W[1, 1])

# 4. from the t statistic of unemployment in the regression of emp on (1, unemp,
    year)
fit = sm.OLS(emp, np.column_stack([Z, unemp])).fit()
t = fit.tvalues[2]
r_t = np.sign(t) * np.sqrt(t**2 / (t**2 + n - 3))

print(f"marginal r = {r12:.4f}")
print(f"partial r:  {r_resid:.10f} {r_formula:.10f} {r_precision:.10f} {r_t:.10f}")

```

*Remark* (Adjusting is not always right). A partial correlation answers a precise question: how are the parts of two variables not linearly explained by  $\mathbf{Y}_2$  related? Whether that is the *right* question depends on what  $\mathbf{Y}_2$  is. Adjusting for a common cause, as the year stands in for population growth here, removes a spurious association. Adjusting for a common effect can *create* an association between variables that are independent. [Chapter 25](#) treats the difference ([Theorem 25.5.1](#)). Under normality, the sampling distribution of a sample partial correlation given  $q$  variables is that of an ordinary sample correlation from  $n - q$  observations, a result used for tests in [Chapter 14](#).

## 3.6.4 Exercises

### 3.6.4.1 A. Check your understanding

#### Exercise A1.

Show that  $\rho_{12 \cdot 3} = 0$  iff  $\rho_{12} = \rho_{13}\rho_{23}$ . Find  $\rho_{12 \cdot 3}$  when  $\rho_{12} = 0$  and  $\rho_{13} = \rho_{23} = 0.6$ , and interpret.

### 3.6.4.2 B. Practice

#### Exercise B1.

Let  $X$ ,  $Z$  and  $e$  be independent, with  $X, Z \sim N(0, 1)$  and  $e \sim N(0, \tau^2)$ , and let  $C = X + Z + e$ . Show that  $\rho_{XZ} = 0$  but  $\rho_{XZ \cdot C} = -1/(1 + \tau^2)$ . What happens as  $\tau \rightarrow 0$  and as  $\tau \rightarrow \infty$ ?

*Solution.* The covariance matrix of  $(X, Z, C)$  is  $\begin{pmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 + \tau^2 \end{pmatrix}$ . Conditioning on  $C$ ,

$$\sigma_{XZ \cdot C} = 0 - \frac{1}{2 + \tau^2}, \quad \sigma_{XX \cdot C} = \sigma_{ZZ \cdot C} = 1 - \frac{1}{2 + \tau^2} = \frac{1 + \tau^2}{2 + \tau^2},$$

so  $\rho_{XZ \cdot C} = -1/(1 + \tau^2)$ . As  $\tau \rightarrow 0$  it tends to  $-1$ : if  $X + Z$  is known exactly, a larger



$X$  forces a smaller  $Z$ . As  $\tau \rightarrow \infty$  it tends to 0, since  $C$  carries little information about  $X + Z$ . Two independent causes become dependent once their common effect is held fixed.

### Exercise B2.

Let  $\Sigma$  be positive definite. Show that the coefficient of  $Y_j$  in the best linear predictor of  $Y_1$  from all the other entries is  $-\omega_{1j}/\omega_{11}$ . Deduce that  $\rho_{1j \cdot \text{rest}}^2$  equals the product of the coefficient of  $Y_j$  for predicting  $Y_1$  and the coefficient of  $Y_1$  for predicting  $Y_j$ , each from all the other variables.

*Solution.* Put  $Y_1$  first. By [Exercise \(B5\)](#) with a scalar first block, the coefficient vector of the best linear predictor of  $Y_1$  from the rest is  $-\Omega_{12}/\omega_{11}$ , whose  $j$ th entry is  $-\omega_{1j}/\omega_{11}$ . Likewise the coefficient of  $Y_1$  in the predictor of  $Y_j$  is  $-\omega_{j1}/\omega_{jj}$ . Their product is  $\omega_{1j}^2/(\omega_{11}\omega_{jj})$ , which is  $\rho_{1j \cdot \text{rest}}^2$  by [Proposition 3.6.3](#).

### Exercise B3.

Let  $\mathbf{D}$  be an  $n \times k$  data matrix whose columns are centred, and let  $\mathbf{S} = \mathbf{D}^\top \mathbf{D}$ . Show, using [Theorem 6.6.2](#) and the partitioned inverse, that  $-w_{12}/\sqrt{w_{11}w_{22}}$ , computed from  $\mathbf{S}^{-1} = (w_{ij})$ , equals the correlation of the residuals of columns 1 and 2 after regression on the remaining columns.

### Exercise B4.

Using the Longley data of [Example \(Employment and unemployment\)](#), compute the partial correlation of employment and unemployment given both year and GNP, in at least two of the four ways. Draw the corresponding residual scatterplot. Does adjusting for GNP as well as year change the conclusion?

## 3.7 Summary and Notes

### 3.7.1 Summary

*Idea.*

1.  $\mathbf{Y}$  is multivariate normal when every linear combination  $\mathbf{a}^\top \mathbf{Y}$  is normal, constants included ([Definition 3.1.1](#)). The definition covers singular covariance matrices, which regression needs for fitted values and residuals.
2.  $\boldsymbol{\mu} + \mathbf{AZ}$ , with  $\mathbf{Z}$  standard normal, is  $N(\boldsymbol{\mu}, \mathbf{AA}^\top)$ . Every mean and every nonnegative definite covariance occurs ([Theorem 3.1.2](#)).
3. The moment generating function is  $\exp(\mathbf{t}^\top \boldsymbol{\mu} + \frac{1}{2} \mathbf{t}^\top \Sigma \mathbf{t})$ , so a normal distribution is determined by  $\boldsymbol{\mu}$  and  $\Sigma$  ([Theorem 3.2.1](#)). When  $\Sigma$  is positive definite there is a density whose contours are the ellipsoids of constant Mahalanobis distance, and  $\Delta^2(\mathbf{Y}) \sim \chi^2(n)$  ([Theorem 3.2.2](#) and [Proposition 3.2.3](#)).

4. Affine maps of normal vectors are normal, with no rank condition, and so are all subvectors ([Theorem 3.3.1](#) and [Corollary 3.3.2](#)). A normal vector of rank  $r$  is an exact affine image of  $r$  independent standard normals ([Proposition 3.3.4](#)).
5. Normal margins do not imply joint normality, and zero correlation between normal variables does not imply independence without it ([Example \(A random sign\)](#) and [Example \(A mixture of two correlations\)](#)).
6. For jointly normal blocks, zero cross-covariance is equivalent to independence.  $\mathbf{AY}$  and  $\mathbf{BY}$  are independent iff  $\mathbf{A}\Sigma\mathbf{B}^\top = \mathbf{0}$  ([Theorem 3.4.1](#) and [Corollary 3.4.2](#)).
7.  $\mathbf{Y}_1 \mid \mathbf{Y}_2 = \mathbf{y}_2 \sim N(\boldsymbol{\mu}_1 + \Sigma_{12}\Sigma_{22}^-(\mathbf{y}_2 - \boldsymbol{\mu}_2), \Sigma_{11.2})$ : linear mean, constant covariance, reduced variance ([Theorem 3.5.2](#)). The conditional mean is the best predictor of all, and coincides with the best linear predictor ([Proposition 3.5.3](#)).
8. The partial correlation is the correlation of the residuals from linear prediction on the conditioning variables. Under normality it is the conditional correlation, and it is zero iff the variables are conditionally independent ([Definition 3.6.1](#) and [Proposition 3.6.1](#)). It can be computed by a recursion, from the inverse covariance matrix, or by FWL residuals ([Proposition 3.6.2](#) and [Proposition 3.6.3](#)).

### 3.7.2 Notes and sources

**Three definitions.** Linear-model texts reach the multivariate normal by different routes, and comparing them explains the choices made here. Rencher and Schaalje (2008, chapter 4) start from the positive definite case, obtain the density by transforming independent standard normals with  $\Sigma^{1/2}$ , and prove the properties of linear functions and independence with moment generating functions, for full-rank transformations. Seber and Lee (2003, chapter 2) also begin with the density, then extend the definition to singular covariance matrices through the representation  $\mathbf{AZ} + \boldsymbol{\mu}$  and prove that normality of all linear combinations characterizes the family. Their moment-generating-function proof of the independence theorem is the model for [Section 3.4](#). Christensen (2020, §1.2) takes  $\mathbf{AZ} + \boldsymbol{\mu}$  as the definition from the outset, shows with characteristic functions that the law depends only on  $\boldsymbol{\mu}$  and  $\Sigma$ , and proves the independence theorem by building an explicit independent version. Defining the distribution through linear combinations, as this chapter does, goes back at least to Rao (1973, chapter 8). It has the advantage that the theorem on affine transformations needs no rank condition and a one-line proof.

**Cramér–Wold.** The device of reducing a vector to its linear combinations is from Cramér and Wold (1936), where it was used to reduce multivariate limit theorems to univariate ones. [Chapter 2](#) proves it, together with the uniqueness and factorization properties of moment generating functions used in [Section 3.2](#). Billingsley (1995) is the reference for these facts and for the conditioning property of independent variables used in the proof of [Theorem 3.5.2](#).

**Conditional distributions and partial correlation.** The proof of [Theorem 3.5.2](#) through the independent error  $\mathbf{W}$  borrows one idea from Rencher and Schaalje (2008) and Seber and Lee (2003): both pass through the transformed vector  $\mathbf{Y}_1 - \mathbf{BY}_2$ , which is independent of  $\mathbf{Y}_2$ , although both still obtain the conditional density as a ratio of densities in the positive definite case. The proof here uses only independence and a generalized inverse, so it covers singular  $\Sigma_{22}$ , an extension that is standard in multivariate analysis (Rao 1973). Anderson (2003) is the classical reference for the multivariate normal in statistics, including the distribution theory of sample partial and multiple correlation coefficients that [Chapter 14](#) uses. Rencher and Schaalje (2008, §4.5) define partial correlation as a conditional correlation, as in [Proposition 3.6.1\(b\)](#).

The residual interpretation (a) does not require normality and is the one that connects to least squares. Regression toward the mean was described by Galton (1886). The link between zeros of the inverse covariance matrix and conditional independence was made the basis of a modelling strategy by Dempster (1972), and Lauritzen (1996) develops the theory of graphical models that grew from it.

**Data.** [Example \(Employment and unemployment\)](#) uses the macroeconomic series published by Longley (1967) as a test of numerical accuracy, available in `statsmodels`. The same series appear in [Chapter 2](#), as an example of sample moments, and in [Chapter 6](#), to illustrate ill-conditioning.

### 3.7.3 References

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# Chapter 4

## Quadratic Forms and Their Distributions

Every sum of squares in a linear model is a quadratic form  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  in a normal vector, and every  $F$  and  $t$  statistic is a ratio built from such forms. This chapter works out their laws: the noncentral chi-squared,  $F$  and  $t$  distributions and the monotonicity of their tails; the mean, variance and a canonical representation of an arbitrary quadratic form; exactly when a quadratic form is chi-squared; when two forms, or a form and a linear function, are independent; and Cochran's theorem, which says when a sum of squares splits into independent chi-squared pieces.

**What you need.** [Chapter 1](#) (spectral theorem, symmetric idempotent matrices, trace), [Chapter 2](#) (means and covariances of random vectors) and [Chapter 3](#) (the multivariate normal distribution and its independence properties). Orthogonal projections appear here only as symmetric idempotent matrices. [Chapter 6](#) develops their geometry.

### 4.0.1 Roadmap

- [Noncentral chi-squared, t and F distributions](#)
- [Mean and variance of quadratic forms](#)
- [When a quadratic form is chi-squared](#)
- [Independence of quadratic forms and linear forms](#)
- [The Cochran theorem](#)
- [Summary and notes](#)

## 4.1 Noncentral chi-squared, t and F distributions

A regression analysis reports a handful of numbers whose sampling behaviour decides everything: the residual sum of squares, the reduction in it when a term is added, the ratio of the two. Each

is a function of the response vector of the form

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} Y_i Y_j,$$

a **quadratic form** in  $\mathbf{Y}$ . The sample variance is one, with  $\mathbf{A} = (n-1)^{-1}(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)$ . So is the squared length  $\|\mathbf{P}\mathbf{Y}\|^2 = \mathbf{Y}^\top \mathbf{P} \mathbf{Y}$  of any symmetric idempotent image of  $\mathbf{Y}$ , and this is how the sums of squares of [Chapter 6](#) will arise.

Since  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  is a scalar, it equals its own transpose  $\mathbf{Y}^\top \mathbf{A}^\top \mathbf{Y}$ , and so also  $\mathbf{Y}^\top \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top) \mathbf{Y}$ . Replacing  $\mathbf{A}$  by its symmetric part changes nothing, and *throughout this chapter the matrix of a quadratic form is symmetric*. Several formulas below, the variance formula among them, are false for a non-symmetric  $\mathbf{A}$  ([Exercise \(A1\)](#)).

When  $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$  and  $\mathbf{A}$  is symmetric idempotent,  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  has a chi-squared distribution, which is the null distribution of a test. Under an alternative the mean is not zero, and a slightly larger family appears. We start with that family.

### 4.1.1 Noncentral chi-squared

Recall that  $\chi^2(r)$ , the chi-squared distribution with  $r$  degrees of freedom, is the law of  $Z_1^2 + \dots + Z_r^2$  for independent standard normal  $Z_i$ . It is the gamma distribution with shape  $r/2$  and scale 2, with density

$$g_r(x) = \frac{x^{r/2-1} e^{-x/2}}{2^{r/2} \Gamma(r/2)}, \quad x > 0, \quad (4.1.1)$$

mean  $r$ , variance  $2r$  and moment generating function  $(1-2t)^{-r/2}$  for  $t < 1/2$ . By convention  $\chi^2(0)$  is the point mass at zero. Everything about the noncentral version follows from one computation.

#### Lemma 4.1.1

Let  $W \sim N(0, 1)$  and  $\nu \in \mathbb{R}$ . For  $t < 1/2$ ,

$$\mathbb{E}[e^{t(W+\nu)^2}] = (1-2t)^{-1/2} \exp\left(\frac{\nu^2 t}{1-2t}\right).$$

*Proof.* With  $u = w + \nu$  the expectation is  $(2\pi)^{-1/2} \int \exp\{tu^2 - (u-\nu)^2/2\} du$ . Completing the square in the exponent,

$$tu^2 - \frac{1}{2}(u-\nu)^2 = -\frac{1-2t}{2} \left(u - \frac{\nu}{1-2t}\right)^2 + \frac{\nu^2}{2} \left(\frac{1}{1-2t} - 1\right).$$

The last term equals  $\nu^2 t / (1-2t)$ . The remaining Gaussian integral is  $(2\pi)^{1/2} (1-2t)^{-1/2}$  when  $1-2t > 0$ . ■

**Definition 4.1.1 — Noncentral chi-squared**

Let  $\mathbf{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I}_r)$ . The distribution of  $\mathbf{Z}^\top \mathbf{Z} = \sum_i Z_i^2$  is the **noncentral chi-squared distribution** with  $r$  degrees of freedom and **noncentrality**  $\gamma = \boldsymbol{\mu}^\top \boldsymbol{\mu} = \|\boldsymbol{\mu}\|^2$ , written  $\chi^2(r, \gamma)$ .

The definition hides a claim: the law of  $\|\mathbf{Z}\|^2$  depends on the mean vector  $\boldsymbol{\mu}$  only through its length. [Theorem 4.1.2](#) proves it. Intuitively, a rotation  $\mathbf{Q}\mathbf{Z}$  of a spherical normal is again spherical normal, with mean  $\mathbf{Q}\boldsymbol{\mu}$ , and it has the same length as  $\mathbf{Z}$ , so every mean vector of a given length gives the same answer.

*Remark (Convention).* In this book the noncentrality is  $\gamma = \boldsymbol{\mu}^\top \boldsymbol{\mu}$ , with no factor  $\frac{1}{2}$ . Then  $E \chi^2(r, \gamma) = r + \gamma$ , which is easy to remember, and  $\gamma$  agrees with the parameter `nc` of `scipy.stats.ncx2`. Several texts, including Rencher and Schaalje (2008) and Christensen (2020), use  $\lambda = \boldsymbol{\mu}^\top \boldsymbol{\mu}/2$  instead. When reading them, replace their  $\lambda$  by  $\gamma/2$ . The same convention applies to the noncentral  $F$  distribution below.

**Theorem 4.1.2 — Properties of the noncentral chi-squared**

Let  $U \sim \chi^2(r, \gamma)$ .

- (a) The moment generating function is

$$M_U(t) = (1 - 2t)^{-r/2} \exp\left(\frac{\gamma t}{1 - 2t}\right), \quad t < \frac{1}{2}, \quad (4.1.2)$$

so the law depends on  $\boldsymbol{\mu}$  only through  $\gamma$ , and  $\chi^2(r, 0) = \chi^2(r)$ .

- (b)  $E(U) = r + \gamma$  and  $\text{Var}(U) = 2r + 4\gamma$ .  
 (c) If  $U_1, \dots, U_k$  are independent with  $U_i \sim \chi^2(r_i, \gamma_i)$ , then  $\sum_i U_i \sim \chi^2(\sum_i r_i, \sum_i \gamma_i)$ .  
 (d) (*Poisson mixture.*) Let  $K \sim \text{Poisson}(\gamma/2)$  and, given  $K = k$ , let  $V \sim \chi^2(r + 2k)$ . Then  $V \sim \chi^2(r, \gamma)$ . Consequently  $U$  has density

$$f_{r,\gamma}(x) = \sum_{k=0}^{\infty} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} g_{r+2k}(x), \quad x > 0. \quad (4.1.3)$$

*Proof.*

- (a) With  $\mathbf{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I})$ , the  $Z_i$  are independent, and  $Z_i = W_i + \mu_i$  with  $W_i \sim N(0, 1)$ . By [Lemma 4.1.1](#),

$$E e^{t\mathbf{Z}^\top \mathbf{Z}} = \prod_{i=1}^r (1 - 2t)^{-1/2} \exp\left(\frac{\mu_i^2 t}{1 - 2t}\right) = (1 - 2t)^{-r/2} \exp\left(\frac{\gamma t}{1 - 2t}\right).$$

A moment generating function that is finite on an open interval around zero determines the distribution, so the law depends only on  $(r, \gamma)$ . Setting  $\gamma = 0$  gives the central mgf.

- (b) The cumulant generating function is  $K(t) = \log M_U(t) = -\frac{r}{2} \log(1 - 2t) + \gamma t/(1 - 2t)$ . Differentiating,  $K'(t) = r(1 - 2t)^{-1} + \gamma(1 - 2t)^{-2}$  and  $K''(t) = 2r(1 - 2t)^{-2} + 4\gamma(1 - 2t)^{-3}$ . The mean and variance are  $K'(0)$  and  $K''(0)$ .

- (c) By independence the mgf of the sum is the product of the mgfs [Eq. 4.1.2](#), which has the same form with  $\sum r_i$  and  $\sum \gamma_i$ .
- (d) Conditioning on  $K$ ,

$$\mathbb{E} e^{tV} = \sum_{k=0}^{\infty} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} (1-2t)^{-(r+2k)/2} = (1-2t)^{-r/2} e^{-\gamma/2} \exp\left(\frac{\gamma/2}{1-2t}\right),$$

and  $-\frac{\gamma}{2} + \frac{\gamma}{2(1-2t)} = \frac{\gamma t}{1-2t}$ . This is [Eq. 4.1.2](#). The density of a mixture is the mixture of the densities. ■

The mixture in (d) is more than a formula. It says that noncentrality acts like a random number of extra degrees of freedom, two at a time, with  $\gamma/2$  of them on average. Since each extra pair of degrees of freedom pushes the distribution to the right, a larger  $\gamma$  should give a stochastically larger variable. That is the next result, and it is what makes tests based on these distributions sensible.

**Example. (A noncentral chi-squared with four degrees of freedom)**

Take  $r = 4$  and  $\gamma = 6$ . The Poisson(3) weights of the central densities  $g_4, g_6, g_8, g_{10}$  in [Eq. 4.1.3](#) are 0.050, 0.149, 0.224 and 0.224, and the mean and variance are 10 and 32. The listing checks the mixture against a library implementation of the density. The script also squares normal vectors with a mean of length  $\sqrt{6}$  in an arbitrary direction, and their average squared length is 10.002. A test that rejects when a  $\chi^2(4)$  statistic exceeds its upper 5% point 9.488 rejects with probability 0.470 when the statistic is in fact  $\chi^2(4, 6)$ . [Figure 4.1.1](#) shows how the density moves right and spreads out as  $\gamma$  grows.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(20240401)

def ncx2_pdf_mixture(x, r, gamma, terms=200):
    """Density of chi^2(r, gamma) as a Poisson(gamma/2) mixture of central densities
    . """
    k = np.arange(terms)
    weights = stats.poisson.pmf(k, gamma / 2)          # P(K = k)
    dens = stats.chi2.pdf(np.asarray(x)[: , None], r + 2 * k)
    return dens @ weights

r, gamma = 4, 6.0
x = np.linspace(0.05, 40, 400)
print(np.max(np.abs(ncx2_pdf_mixture(x, r, gamma) - stats.ncx2.pdf(x, r, gamma))))
```

**Proposition 4.1.3 — Monotonicity in the noncentrality**

For fixed  $r \geq 1$  and  $x > 0$ , the tail probability  $\gamma \mapsto \Pr\{\chi^2(r, \gamma) > x\}$  is continuous and strictly increasing on  $[0, \infty)$ , and tends to 1 as  $\gamma \rightarrow \infty$ .

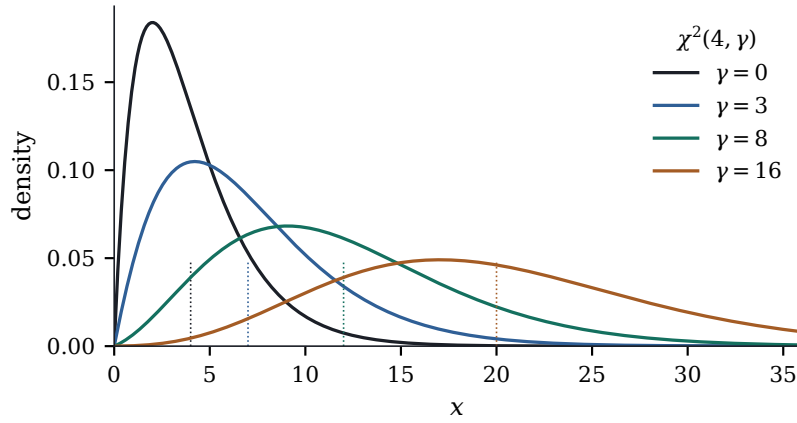


Figure 4.1.1: Densities of  $\chi^2(4, \gamma)$ . Dotted ticks mark the means  $4 + \gamma$ . The variance  $2r + 4\gamma$  grows with the noncentrality, so the distributions flatten as they move right.

*Proof.* Let  $h(k) = \Pr\{\chi^2(r + 2k) > x\}$  and  $p_k(\lambda) = e^{-\lambda}\lambda^k/k!$ . By Theorem 4.1.2(d), the tail probability is  $\phi(\lambda) = \sum_k p_k(\lambda)h(k)$  with  $\lambda = \gamma/2$ . First,  $h$  is strictly increasing: if  $V \sim \chi^2(r + 2k)$  and  $E \sim \chi^2(2)$  are independent, then  $V + E \sim \chi^2(r + 2k + 2)$  and  $h(k + 1) - h(k) = \Pr(V \leq x < V + E) > 0$ , since  $(V, E)$  has a positive joint density on  $(0, \infty)^2$ . Next,  $p'_k(\lambda) = p_{k-1}(\lambda) - p_k(\lambda)$  (with  $p_{-1} = 0$ ). Because  $0 \leq h \leq 1$ , the series of derivatives converges uniformly on bounded intervals, so term-by-term differentiation is valid and

$$\phi'(\lambda) = \sum_{k \geq 0} (p_{k-1}(\lambda) - p_k(\lambda))h(k) = \sum_{k \geq 0} p_k(\lambda)(h(k+1) - h(k)) > 0.$$

For the limit, fix  $\varepsilon > 0$ . Since  $\chi^2(r + 2m)$  has mean  $r + 2m$  and standard deviation  $\sqrt{2r + 4m}$ , Chebyshev's inequality gives  $h(m) > 1 - \varepsilon$  for some  $m$ . Then  $\phi(\lambda) \geq h(m) \Pr(K \geq m)$  for  $K \sim \text{Poisson}(\lambda)$ , and  $\Pr(K \geq m) \rightarrow 1$  as  $\lambda \rightarrow \infty$ . ■

## 4.1.2 Noncentral $F$ and $t$

Test statistics divide one quadratic form by another, to remove the unknown scale. The two ratio distributions that result are defined next.

### Definition 4.1.2 — Noncentral $F$

Let  $U \sim \chi^2(r, \gamma)$  and  $V \sim \chi^2(s)$  be independent,  $r, s \geq 1$ . The law of

$$F = \frac{U/r}{V/s}$$

is the **noncentral  $F$  distribution**  $F(r, s, \gamma)$ , with numerator and denominator degrees of freedom  $r$  and  $s$  and noncentrality  $\gamma$ . The central  $F(r, s) = F(r, s, 0)$  has upper  $\alpha$  point  $F_\alpha(r, s)$ .



**Definition 4.1.3 — Noncentral  $t$** 

Let  $X \sim N(\delta, 1)$  and  $V \sim \chi^2(s)$  be independent. The law of  $T = X/\sqrt{V/s}$  is the **noncentral  $t$  distribution**  $t(s, \delta)$ , and  $t(s) = t(s, 0)$ .

Only the numerator is allowed to be noncentral. A noncentral denominator gives a *doubly noncentral*  $F$ , which appears when the model used for the error sum of squares is itself wrong ([Exercise \(B3\)](#)). Two relations are immediate. First,  $X^2 \sim \chi^2(1, \delta^2)$ , so  $T^2 \sim F(1, s, \delta^2)$ : a two-sided  $t$  test is an  $F$  test with one numerator degree of freedom. Second, since  $E(V^{-1}) = 1/(s-2)$  for  $s > 2$ , independence gives

$$E(F) = \frac{r + \gamma}{r} \cdot \frac{s}{s - 2}, \quad s > 2.$$

The central  $F$  is the null distribution of the tests of [Chapter 11](#), and  $F(r, s, \gamma)$  is their distribution under an alternative. The power of such a test is therefore  $\Pr\{F(r, s, \gamma) > F_\alpha(r, s)\}$ , and the following fact is what guarantees that it behaves as a test should.

**Theorem 4.1.4 — Power of the  $F$  test increases with  $\gamma$** 

For fixed  $r, s \geq 1$  and  $c > 0$ ,  $\beta(\gamma) = \Pr\{F(r, s, \gamma) > c\}$  is continuous and strictly increasing in  $\gamma \geq 0$ , and  $\beta(\gamma) \rightarrow 1$  as  $\gamma \rightarrow \infty$ . In particular, the level- $\alpha$  test that rejects when  $F > F_\alpha(r, s)$  has power greater than  $\alpha$  at every  $\gamma > 0$ .

*Proof.* With  $U, V$  as in [Definition 4.1.2](#),  $F > c$  iff  $U > crV/s$ . Condition on  $V$ , which is independent of  $U$ :

$$\beta(\gamma) = E[\psi_\gamma(crV/s)], \quad \psi_\gamma(x) = \Pr\{\chi^2(r, \gamma) > x\}.$$

Since  $V > 0$  with probability one, [Proposition 4.1.3](#) gives, for  $\gamma < \gamma'$ ,  $\psi_\gamma(crV/s) < \psi_{\gamma'}(crV/s)$  almost surely, so  $\beta(\gamma) < \beta(\gamma')$ . Continuity and the limit follow from the same proposition by dominated convergence, since  $0 \leq \psi_\gamma \leq 1$ . ■

Two further monotonicity facts hold for fixed  $\gamma > 0$  and fixed level. The power decreases as the numerator degrees of freedom  $r$  grow, and increases as the denominator degrees of freedom  $s$  grow (Ghosh 1973). The first says that spreading a fixed amount of signal over more dimensions makes it harder to detect. The second says that estimating  $\sigma^2$  better helps. We use both only qualitatively, and [Figure 4.4.1](#) in [Section 4.4](#) shows the first. Because  $T^2 \sim F(1, s, \delta^2)$ , [Theorem 4.1.4](#) also shows that the power of the two-sided  $t$  test increases with  $|\delta|$  ([Exercise \(A2\)](#)).

**4.1.3 Exercises****4.1.3.1 A. Check your understanding**

**Exercise A1.**

Show that  $\chi^2(2)$  is the exponential distribution with mean 2. Use [Theorem 4.1.2\(d\)](#) to show that

$$\Pr\{\chi^2(2, \gamma) > x\} = \sum_{k \geq 0} e^{-\gamma/2} \frac{(\gamma/2)^k}{k!} e^{-x/2} \sum_{j=0}^k \frac{(x/2)^j}{j!}.$$

**Exercise A2.**

Show that  $T \sim t(s, \delta)$  implies  $T^2 \sim F(1, s, \delta^2)$ . Deduce that the power of the two-sided level- $\alpha$   $t$  test depends on  $\delta$  only through  $|\delta|$  and increases with it. Show directly, by conditioning on the denominator, that the power of the one-sided test that rejects for  $T > c$  increases with  $\delta$ .

*Solution.*  $T^2 = X^2/(V/s)$  with  $X^2 \sim \chi^2(1, \delta^2)$  independent of  $V$ , which is [Definition 4.1.2](#). The two-sided test rejects when  $|T| > t_{\alpha/2}(s)$ , that is, when  $T^2 > F_{\alpha}(1, s)$ . Its power is therefore  $\Pr\{F(1, s, \delta^2) > F_{\alpha}(1, s)\}$ , a strictly increasing function of  $\delta^2$  by [Theorem 4.1.4](#). For the one-sided test,  $\Pr(T > c) = \mathbb{E} \Pr\{X > c\sqrt{V/s} \mid V\} = \mathbb{E} \Phi(\delta - c\sqrt{V/s})$ , and  $\Phi$  is strictly increasing.

**4.1.3.2 B. Practice****Exercise B1.**

Show that the  $k$ th cumulant of  $\chi^2(r, \gamma)$  is  $2^{k-1}(k-1)!(r+k\gamma)$ . Deduce that its skewness tends to zero like  $3/\sqrt{\gamma}$  as  $\gamma \rightarrow \infty$  with  $r$  fixed.

*Solution.* Using  $-\frac{1}{2} \log(1-2t) = \sum_{k \geq 1} (2t)^k/(2k)$  and  $t/(1-2t) = \sum_{k \geq 1} 2^{k-1}t^k$ , the cumulant generating function is

$$K(t) = \sum_{k \geq 1} t^k 2^{k-1} \left( \frac{r}{k} + \gamma \right) = \sum_{k \geq 1} \frac{t^k}{k!} 2^{k-1} (k-1)! (r+k\gamma).$$

The  $k$ th cumulant is  $k!$  times the coefficient of  $t^k$ . In particular  $\kappa_2 = 2(r+2\gamma)$  and  $\kappa_3 = 8(r+3\gamma)$ , so the skewness is  $\kappa_3/\kappa_2^{3/2} = 8(r+3\gamma)/\{2(r+2\gamma)\}^{3/2}$ . As  $\gamma \rightarrow \infty$  this behaves like  $24\gamma/(4\gamma)^{3/2} = 3/\sqrt{\gamma}$ .

**Exercise B2.**

Show that  $\chi^2(1, \gamma)$  has density  $(2\pi x)^{-1/2} e^{-(x+\gamma)/2} \cosh \sqrt{\gamma x}$  for  $x > 0$ . Verify that expanding  $\cosh$  in a power series reproduces [Eq. 4.1.3](#). *Hint:*  $\Gamma(k+\frac{1}{2}) = \sqrt{\pi} (2k)!/(4^k k!)$ .

*Solution.* Let  $X = (W + \sqrt{\gamma})^2$  with  $W \sim N(0, 1)$ . For  $x > 0$ ,  $X \leq x$  iff  $-\sqrt{x} - \sqrt{\gamma} \leq W \leq \sqrt{x} - \sqrt{\gamma}$ . Differentiating with  $\phi$  the standard normal density,

$$f(x) = \frac{\phi(\sqrt{x} - \sqrt{\gamma}) + \phi(\sqrt{x} + \sqrt{\gamma})}{2\sqrt{x}} = \frac{e^{-(x+\gamma)/2}}{\sqrt{2\pi x}} \cdot \frac{e^{\sqrt{\gamma x}} + e^{-\sqrt{\gamma x}}}{2}.$$

Expanding,  $f(x) = (2\pi x)^{-1/2} e^{-(x+\gamma)/2} \sum_k (\gamma x)^k / (2k)!$ . The  $k$ th term of Eq. 4.1.3 is

$$\frac{e^{-\gamma/2} (\gamma/2)^k x^{k-1/2} e^{-x/2}}{k! 2^{k+1/2} \Gamma(k + \frac{1}{2})}.$$

By the hint the denominator is  $k! 2^{k+1/2} \sqrt{\pi} (2k)! / (4^k k!) = \sqrt{2\pi} 2^{-k} (2k)!$ , and the  $k$ th term becomes  $(2\pi x)^{-1/2} e^{-(x+\gamma)/2} (\gamma x)^k / (2k)!$ , as required.

### Exercise B3.

Let  $U_\gamma \sim \chi^2(r, \gamma)$  with  $r$  fixed. Show that  $(U_\gamma - r - \gamma) / \sqrt{2r + 4\gamma}$  converges in distribution to  $N(0, 1)$  as  $\gamma \rightarrow \infty$ . *Hint:*  $U_\gamma$  has the law of  $(W_1 + \sqrt{\gamma})^2 + W_2^2 + \cdots + W_r^2$  with  $W_1, \dots, W_r$  independent standard normal.

### Exercise B4.

Let  $U \sim \chi^2(a)$  and  $V \sim \chi^2(b)$  be independent. Show that  $B = U/(U + V)$  has the  $\text{Beta}(a/2, b/2)$  distribution and is independent of  $U + V$ , and that  $F = (U/a)/(V/b) = (b/a) B/(1 - B)$ . Deduce the null distribution of  $R^2$  asked for in Chapter 6.

*Solution.* The joint density of  $(U, V)$  is proportional to  $u^{a/2-1} v^{b/2-1} e^{-(u+v)/2}$ . Change variables to  $w = u/(u + v) \in (0, 1)$  and  $s = u + v > 0$ , so that  $u = ws$ ,  $v = (1 - w)s$  and the Jacobian is  $s$ . The density of  $(W, S)$  is proportional to

$$w^{a/2-1} (1 - w)^{b/2-1} \cdot s^{(a+b)/2-1} e^{-s/2},$$

a product of a function of  $w$  and a function of  $s$ . So  $B$  and  $U + V$  are independent,  $B \sim \text{Beta}(a/2, b/2)$  and  $U + V \sim \chi^2(a + b)$ . Also  $U/V = B/(1 - B)$ . For  $R^2$  in a model with intercept,  $R^2 = \text{SSR}/(\text{SSR} + \text{SSE})$ , where  $\text{SSR} = \|(\mathbf{M} - \mathbf{P}_1)\mathbf{Y}\|^2$  and  $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$  come from orthogonal projections of ranks  $r - 1$  and  $n - r$ . If  $\boldsymbol{\mu} = \mu\mathbf{1}$ , both noncentralities vanish, and Theorem 4.4.4 with  $a = r - 1$ ,  $b = n - r$  gives  $R^2 \sim \text{Beta}((r - 1)/2, (n - r)/2)$ .

### Exercise B5.

Show that  $f_{r,\gamma}(x)/g_r(x)$  is strictly increasing in  $x > 0$  for every  $\gamma > 0$ . Conclude that among tests based on one observation  $U \sim \chi^2(r, \gamma)$  of  $\gamma = 0$  against a fixed  $\gamma > 0$ , rejecting for large  $U$  is most powerful.

## 4.2 Mean and variance of quadratic forms

### 4.2.1 The mean needs no normality

Theorem 2.3.1 computed  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$  for any random vector with mean  $\boldsymbol{\mu}$  and covariance  $\boldsymbol{\Sigma}$ :

$$E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}. \quad (4.2.1)$$

The formula has two parts with distinct roles. The trace term is what the noise contributes. The term  $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$  is what the signal contributes. For nonnegative definite  $\mathbf{A}$  it vanishes exactly when  $\mathbf{A} \boldsymbol{\mu} = \mathbf{0}$ , which describes the mean vectors a well-designed estimator is meant to ignore.

**Example. (Two estimators of  $\sigma^2$  under a drifting mean)**

Let  $Y_1, \dots, Y_n$  be uncorrelated with common variance  $\sigma^2$ , and consider

$$S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2, \quad Q = \frac{1}{2(n-1)} \sum_{i=1}^{n-1} (Y_{i+1} - Y_i)^2,$$

the sample variance and the **mean square successive difference**. In matrix form  $S^2 = \mathbf{Y}^\top \mathbf{C} \mathbf{Y} / (n-1)$  with  $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$ , and  $Q = \mathbf{Y}^\top \mathbf{D}^\top \mathbf{D} \mathbf{Y} / \{2(n-1)\}$ , where  $\mathbf{D}$  is the  $(n-1) \times n$  first-difference matrix with rows  $(\dots, -1, 1, \dots)$ . Both matrices annihilate  $\mathbf{1}$ , so both estimators ignore a constant mean. Since  $\text{tr}(\mathbf{C}) = n-1$  and  $\text{tr}(\mathbf{D}^\top \mathbf{D}) = \text{tr}(\mathbf{D} \mathbf{D}^\top) = 2(n-1)$ , Eq. 4.2.1 with  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$  shows that both are unbiased when the mean is constant.

Now let the mean drift linearly,  $\mu_i = \mu_0 + \theta i$ . Then  $\mathbf{C} \boldsymbol{\mu}$  is the centred trend and  $\mathbf{D} \boldsymbol{\mu} = \theta \mathbf{1}_{n-1}$ , so

$$E(S^2) = \sigma^2 + \theta^2 \frac{n(n+1)}{12}, \quad E(Q) = \sigma^2 + \frac{\theta^2}{2}.$$

The bias of  $S^2$  grows quadratically with the length of the series, while that of  $Q$  does not grow at all. With  $n = 20$ ,  $\sigma = 1$  and  $\theta = 0.1$ , the means are 1.350 and 1.005. Differencing removes smooth trends, and this is why  $Q$  was proposed for series whose level may wander (von Neumann 1941). The price of robustness to drift is paid in variance, which we compute below.

## 4.2.2 A canonical representation

Higher moments and distributions need normality, and one representation answers all such questions at once.

### Theorem 4.2.1 — Canonical form of a normal quadratic form

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  and let  $\mathbf{A}$  be symmetric. Write  $\boldsymbol{\Sigma} = \mathbf{L} \mathbf{L}^\top$  with  $\mathbf{L}$  of size  $n \times k$ , and take a spectral decomposition  $\mathbf{L}^\top \mathbf{A} \mathbf{L} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top$  with  $\boldsymbol{\Lambda} = \text{diag}(\lambda_1, \dots, \lambda_k)$ . Put  $\mathbf{b} = \mathbf{Q}^\top \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}$ . Then

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + \sum_{j=1}^k (\lambda_j W_j^2 + 2b_j W_j), \quad \mathbf{W} \sim N_k(\mathbf{0}, \mathbf{I}_k). \quad (4.2.2)$$

For  $\lambda_j \neq 0$  the  $j$ th term equals  $\lambda_j (W_j + b_j/\lambda_j)^2 - b_j^2/\lambda_j$ . The nonzero  $\lambda_j$  are the nonzero

eigenvalues of  $\mathbf{A}\Sigma$ .

*Proof.* Such an  $\mathbf{L}$  exists. By [Theorem 1.7.2](#),  $\Sigma = \mathbf{U}\mathbf{D}\mathbf{U}^\top$  with  $\mathbf{U}$  orthogonal and  $\mathbf{D}$  diagonal with entries  $d_i \geq 0$ , and  $\mathbf{L} = \mathbf{U}\mathbf{D}^{1/2}$  works with  $k = n$ . Let  $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$ . The vector  $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$  is normal with mean  $\boldsymbol{\mu}$  and covariance  $\mathbf{L}\mathbf{L}^\top = \Sigma$  ([Theorem 3.3.1](#)). A normal distribution is determined by its mean and covariance ([Theorem 3.2.1](#)), so  $\mathbf{Y}$  and  $\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}$  have the same law, and so do  $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$  and

$$(\boldsymbol{\mu} + \mathbf{L}\mathbf{Z})^\top \mathbf{A}(\boldsymbol{\mu} + \mathbf{L}\mathbf{Z}) = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu} + 2\boldsymbol{\mu}^\top \mathbf{A}\mathbf{L}\mathbf{Z} + \mathbf{Z}^\top \mathbf{L}^\top \mathbf{A}\mathbf{L}\mathbf{Z}.$$

Put  $\mathbf{W} = \mathbf{Q}^\top \mathbf{Z}$ , again  $N_k(\mathbf{0}, \mathbf{I})$ . Then  $\mathbf{Z}^\top \mathbf{L}^\top \mathbf{A}\mathbf{L}\mathbf{Z} = \mathbf{W}^\top \boldsymbol{\Lambda} \mathbf{W} = \sum \lambda_j W_j^2$  and  $\boldsymbol{\mu}^\top \mathbf{A}\mathbf{L}\mathbf{Z} = \mathbf{b}^\top \mathbf{W}$ , which is [Eq. 4.2.2](#). The completed square is algebra. Finally,  $\mathbf{L}^\top \mathbf{A}\mathbf{L} = \mathbf{H}\mathbf{F}$  with  $\mathbf{F} = \mathbf{A}\mathbf{L}$  and  $\mathbf{H} = \mathbf{L}^\top$ , and  $\mathbf{F}\mathbf{H} = \mathbf{A}\Sigma$ . By [Proposition 1.7.1\(d\)](#) these two products have the same nonzero eigenvalues, with the same multiplicities. ■

So a normal quadratic form is a constant plus a sum of independent pieces. Each piece is either a scaled noncentral  $\chi^2(1)$  variable, with scale  $\lambda_j$ , or a normal variable with mean zero, when  $\lambda_j = 0$  but  $b_j \neq 0$ . Because the  $\lambda_j$  may be negative,  $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$  need not be positive. The linear pieces are special to singular covariances. If  $\Sigma$  is positive definite,  $\mathbf{L}$  can be taken square and nonsingular, and then  $\lambda_j = 0$  gives  $\mathbf{A}\mathbf{L}\mathbf{q}_j = \mathbf{0}$  for the  $j$ th column  $\mathbf{q}_j$  of  $\mathbf{Q}$ , so  $b_j = (\mathbf{A}\mathbf{L}\mathbf{q}_j)^\top \boldsymbol{\mu} = 0$ . A linear piece needs a direction in which  $\mathbf{A}$  responds to the mean while the noise is absent.

### Corollary 4.2.2 — Moment generating function

In the setting of [Theorem 4.2.1](#), for  $|t| < 1/(2 \max_j |\lambda_j|)$  (any  $t$  if all  $\lambda_j = 0$ ),

$$\mathbb{E} e^{t\mathbf{Y}^\top \mathbf{A}\mathbf{Y}} = e^{t\boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}} \prod_{j=1}^k (1 - 2\lambda_j t)^{-1/2} \exp\left(\frac{2b_j^2 t^2}{1 - 2\lambda_j t}\right).$$

*Proof.* The terms of [Eq. 4.2.2](#) are independent. If  $\lambda_j = 0$ , the term  $2b_j W_j$  has mgf  $e^{2b_j^2 t^2}$ . If  $\lambda_j \neq 0$ , apply [Lemma 4.1.1](#) to  $W_j + \nu_j$  with  $\nu_j = b_j/\lambda_j$  at the argument  $\lambda_j t$ , and multiply by  $e^{-tb_j^2/\lambda_j}$ . The exponent is  $b_j^2 t / \{\lambda_j(1 - 2\lambda_j t)\} - b_j^2 t / \lambda_j = 2b_j^2 t^2 / (1 - 2\lambda_j t)$ . ■

## 4.2.3 Variance and covariances

### Theorem 4.2.3 — Moments of normal quadratic forms

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ , let  $\mathbf{A}$  and  $\mathbf{B}$  be symmetric  $n \times n$  matrices, and let  $\mathbf{K}$  be an  $m \times n$  matrix. Then

- (a)  $\mathbb{E}(\mathbf{Y}^\top \mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\Sigma) + \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}$ ;
- (b)  $\text{Var}(\mathbf{Y}^\top \mathbf{A}\mathbf{Y}) = 2 \text{tr}\{(\mathbf{A}\Sigma)^2\} + 4\boldsymbol{\mu}^\top \mathbf{A}\Sigma \mathbf{A}\boldsymbol{\mu}$ ;

- (c)  $\text{Cov}(\mathbf{KY}, \mathbf{Y}^\top \mathbf{AY}) = 2\mathbf{K}\Sigma\mathbf{A}\boldsymbol{\mu}$ ;  
 (d)  $\text{Cov}(\mathbf{Y}^\top \mathbf{AY}, \mathbf{Y}^\top \mathbf{BY}) = 2\text{tr}(\mathbf{A}\Sigma\mathbf{B}\Sigma) + 4\boldsymbol{\mu}^\top \mathbf{A}\Sigma\mathbf{B}\boldsymbol{\mu}$ .

*Proof.*

- (a) is [Eq. 4.2.1](#). For (b), use [Eq. 4.2.2](#). Independent terms have additive variances, and for  $W \sim N(0, 1)$ ,  $\text{Var}(\lambda W^2 + 2bW) = 2\lambda^2 + 4b^2$ , because  $\text{Cov}(W^2, W) = E W^3 = 0$ . Hence

$$\text{Var}(\mathbf{Y}^\top \mathbf{AY}) = 2 \sum_j \lambda_j^2 + 4\|\mathbf{b}\|^2 = 2\text{tr}\{(\mathbf{L}^\top \mathbf{AL})^2\} + 4\boldsymbol{\mu}^\top \mathbf{AL}\mathbf{Q}\mathbf{Q}^\top \mathbf{L}^\top \mathbf{A}\boldsymbol{\mu}.$$

Now  $\text{tr}\{(\mathbf{L}^\top \mathbf{AL})^2\} = \text{tr}(\mathbf{L}^\top \mathbf{A}\Sigma\mathbf{AL}) = \text{tr}(\mathbf{A}\Sigma\mathbf{A}\Sigma)$  by [Theorem 1.6.1](#), and  $\mathbf{Q}\mathbf{Q}^\top = \mathbf{I}$ , so the second term is  $4\boldsymbol{\mu}^\top \mathbf{A}\Sigma\mathbf{A}\boldsymbol{\mu}$ . For (c), write  $\mathbf{Y} = \boldsymbol{\mu} + \mathbf{LZ}$  as in the proof of [Theorem 4.2.1](#). Then  $\mathbf{Y}^\top \mathbf{AY} - E(\mathbf{Y}^\top \mathbf{AY}) = \mathbf{Z}^\top \mathbf{L}^\top \mathbf{ALZ} - \text{tr}(\mathbf{A}\Sigma) + 2\boldsymbol{\mu}^\top \mathbf{ALZ}$  and  $\mathbf{KY} - E(\mathbf{KY}) = \mathbf{KLZ}$ . Every entry of  $E[\mathbf{Z}(\mathbf{Z}^\top \mathbf{GZ})]$  is a sum of third moments of a  $N_k(\mathbf{0}, \mathbf{I})$  vector, which vanish. So only the linear term contributes, and  $E[\mathbf{KLZ} \cdot 2\mathbf{Z}^\top \mathbf{L}^\top \mathbf{A}\boldsymbol{\mu}] = 2\mathbf{KLL}^\top \mathbf{A}\boldsymbol{\mu}$ . For (d), apply (b) to  $\mathbf{A}$ ,  $\mathbf{B}$  and  $\mathbf{A} + \mathbf{B}$ , and use  $\text{Var}(X + Y) = \text{Var} X + \text{Var} Y + 2\text{Cov}(X, Y)$ .

■

For  $\boldsymbol{\mu} = \mathbf{0}$ , part (b) can also be read off the fourth moments of a normal vector ([Exercise \(B1\)](#)). The variance has the same two-part structure as the mean: a pure-noise term, plus a term that is present only when  $\mathbf{A}$  responds to the mean. Part (c) shows that a quadratic form and a linear form are uncorrelated when  $\mathbf{K}\Sigma\mathbf{A} = \mathbf{0}$ . Under normality this will turn out to give independence ([Theorem 4.4.1](#)).

### Example.

#### Example (Two estimators of $\sigma^2$ under a drifting mean)

With  $\Sigma = \sigma^2 \mathbf{I}$  and a constant mean, [Theorem 4.2.3\(b\)](#) gives  $\text{Var}(S^2) = 2\sigma^4 \text{tr}(\mathbf{C}^2)/(n-1)^2 = 2\sigma^4/(n-1)$ . For  $Q$ , the matrix  $\mathbf{DD}^\top$  is tridiagonal with 2 on the diagonal and -1 beside it, so  $\text{tr}\{(\mathbf{D}^\top \mathbf{D})^2\} = \text{tr}\{(\mathbf{DD}^\top)^2\} = 4(n-1) + 2(n-2) = 6n-8$  and

$$\text{Var}(Q) = \frac{2\sigma^4(6n-8)}{4(n-1)^2} = \frac{\sigma^4(3n-4)}{(n-1)^2}.$$

The ratio  $\text{Var}(Q)/\text{Var}(S^2) = (3n-4)/\{2(n-1)\}$  tends to 3/2. With  $n = 20$  the two variances are 0.1551 and 0.1053, a ratio of 1.474. So with no drift,  $Q$  needs about half as many observations again to match  $S^2$ . Under the drift  $\theta = 0.1$ , the signal term  $4\boldsymbol{\mu}^\top \mathbf{A}\Sigma\mathbf{A}\boldsymbol{\mu}$  raises the variances to 0.1552 and 0.1789. Combining variance and bias, the root mean squared errors are 0.394 for  $Q$  and 0.549 for  $S^2$ . The listing evaluates the formulas, and the script confirms all eight moments with 400,000 simulated series.

```
import numpy as np
rng = np.random.default_rng(4402)
n, sigma, theta = 20, 1.0, 0.1
t = np.arange(1, n + 1)
mu = 5.0 + theta * t
Sigma = sigma**2 * np.eye(n)
```

# mean with a linear trend

```

def qf_moments(A, mu, Sigma):
    """Mean and variance of  $Y^T A Y$  for  $Y \sim N(\mu, \text{Sigma})$ ,  $A$  symmetric."""
    AS = A @ Sigma
    mean = np.trace(AS) + mu @ A @ mu
    var = 2 * np.trace(AS @ AS) + 4 * mu @ AS @ A @ mu
    return mean, var

C = np.eye(n) - np.ones((n, n)) / n          # centring matrix
D = np.diff(np.eye(n), axis=0)               # (n-1) x n first differences
A_s2 = C / (n - 1)
A_q = D.T @ D / (2 * (n - 1))
for name, A in [("S2", A_s2), ("Q", A_q)]:
    print(name, qf_moments(A, mu, Sigma), qf_moments(A, 0 * mu, Sigma))

```

Normality enters (b)–(d) through the third and fourth moments of  $\mathbf{Y}$ . For independent coordinates with excess kurtosis the variance acquires an extra term that involves only the diagonal of  $\mathbf{A}$  ([Theorem 2.3.3](#)). Variance formulas for sums of squares derived under normality can be badly wrong for heavy-tailed errors, a point taken up in [Chapter 19](#) ([Proposition 19.3.4](#), [Proposition 19.3.5](#)).

## 4.2.4 Exercises

### 4.2.4.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$  and  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ . Compute  $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$  directly, and compare with  $2 \text{tr}(\mathbf{A}^2)$  and with  $2 \text{tr}(\mathbf{A}_s^2)$ , where  $\mathbf{A}_s = \frac{1}{2}(\mathbf{A} + \mathbf{A}^\top)$ .

### 4.2.4.2 B. Practice

#### Exercise B1.

Let  $(U_i, W_i), i = 1, \dots, n$ , be independent copies of a bivariate normal pair with variances  $\sigma_U^2, \sigma_W^2$  and covariance  $\sigma_{UW}$ , and let  $s_{UW} = (n-1)^{-1} \sum_i (U_i - \bar{U})(W_i - \bar{W})$ . Write  $s_{UW} = \mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  for the stacked vector  $\mathbf{Y} = (\mathbf{U}^\top, \mathbf{W}^\top)^\top$  and a symmetric  $2n \times 2n$  matrix  $\mathbf{A}$ , and use [Theorem 4.2.3](#) to show that  $\text{Var}(s_{UW}) = (\sigma_U^2 \sigma_W^2 + \sigma_{UW}^2) / (n-1)$ . Check the answer against  $\text{Var}(S^2)$  when  $\mathbf{U} = \mathbf{W}$ .

#### Exercise B2.

Let  $Y_1, \dots, Y_n$  be independent with mean 0, variance  $\sigma^2$  and  $E Y_i^4 = \mu_4$ . By [Exercise \(B4\)](#), for symmetric  $\mathbf{A}$  and  $\mathbf{B}$ ,

$$\text{Cov}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}, \mathbf{Y}^\top \mathbf{B} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \sum_i a_{ii} b_{ii} + 2\sigma^4 \text{tr}(\mathbf{A} \mathbf{B}).$$

Show that in a one-way layout with  $g \geq 2$  groups, not all of size one, the between-group and within-group sums of squares are correlated unless  $\mu_4 = 3\sigma^4$ . Which sign does the

correlation have for heavy-tailed errors?

*Solution.* In the one-way layout the between-group matrix is  $\mathbf{A} = \mathbf{M} - \mathbf{P}_1$  and the within-group matrix is  $\mathbf{B} = \mathbf{I} - \mathbf{M}$ . They satisfy  $\mathbf{AB} = \mathbf{0}$ , so  $\text{tr}(\mathbf{AB}) = 0$  and the covariance reduces to  $(\mu_4 - 3\sigma^4) \sum_i a_{ii} b_{ii}$ . An observation in group  $\ell$  has  $a_{ii} = 1/n_\ell - 1/n$  and  $b_{ii} = 1 - 1/n_\ell$ , so

$$\sum_i a_{ii} b_{ii} = \sum_\ell n_\ell \left( \frac{1}{n_\ell} - \frac{1}{n} \right) \left( 1 - \frac{1}{n_\ell} \right) = \sum_\ell \left( 1 - \frac{n_\ell}{n} \right) \left( 1 - \frac{1}{n_\ell} \right).$$

With  $g \geq 2$  every first factor is positive, and the second is positive for any group with two or more observations. So the sum is positive, and the covariance vanishes iff  $\mu_4 = 3\sigma^4$ . For heavy-tailed errors,  $\mu_4 > 3\sigma^4$ , the two sums of squares are positively correlated: a large error inflates both.

### Exercise B3.

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ , and let  $\mathbf{M}$  be symmetric idempotent of rank  $r$  with  $\mathbf{M}\boldsymbol{\mu} \neq \boldsymbol{\mu}$ , as when a regression model omits a needed regressor. Show that  $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2 / \sigma^2 \sim \chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2 / \sigma^2)$ , and find the bias of  $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2 / (n - r)$  as an estimator of  $\sigma^2$ . In the setting of [Theorem 4.4.6](#) without the assumption  $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$ , which parts of the conclusion survive?

### Exercise B4.

Let  $\mathbf{Y} \sim N_n(\mathbf{0}, \boldsymbol{\Sigma})$ . Use [Corollary 4.2.2](#) and [Proposition 1.7.1\(d\)](#) to show that  $E e^{t\mathbf{Y}^\top \mathbf{A} \mathbf{Y}} = \det(\mathbf{I}_n - 2t\mathbf{A}\boldsymbol{\Sigma})^{-1/2}$  for  $|t|$  small.

#### 4.2.4.3 C. Going deeper

### Exercise C1.

Let  $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$  with  $\mu$  and  $\sigma^2$  unknown. Among symmetric  $\mathbf{A}$  with  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2$  for all  $\mu$  and  $\sigma^2$ , show that  $\mathbf{A} = \mathbf{C}/(n - 1)$ , which gives  $S^2$ , uniquely minimizes  $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y})$  at every parameter value. (Compare [Exercise \(B5\)](#), where the mean is known to be zero.)

*Solution.* Since  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \sigma^2 \text{tr} \mathbf{A} + \mu^2 \mathbf{1}^\top \mathbf{A} \mathbf{1}$ , unbiasedness means  $\text{tr} \mathbf{A} = 1$  and  $\mathbf{1}^\top \mathbf{A} \mathbf{1} = 0$ . By [Theorem 4.2.3](#),  $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2\sigma^4 \text{tr}(\mathbf{A}^2) + 4\mu^2 \sigma^2 \|\mathbf{A} \mathbf{1}\|^2$ . The map  $\mathbf{A} \mapsto \mathbf{C} \mathbf{A} \mathbf{C}$  is an orthogonal projection for the inner product  $\text{tr}(\mathbf{A} \mathbf{B})$  on symmetric matrices, so  $\text{tr}(\mathbf{A}^2) \geq \text{tr}\{(\mathbf{C} \mathbf{A} \mathbf{C})^2\}$ , with equality iff  $\mathbf{A} = \mathbf{C} \mathbf{A} \mathbf{C}$ . Also  $\text{tr}(\mathbf{C} \mathbf{A} \mathbf{C}) = \text{tr} \mathbf{A} - \mathbf{1}^\top \mathbf{A} \mathbf{1} / n = 1$ . The symmetric matrix  $\mathbf{C} \mathbf{A} \mathbf{C}$  has rank at most  $n - 1$ . If  $\lambda_1, \dots, \lambda_{n-1}$  are its eigenvalues on  $\mathbf{1}^\perp$ , the Cauchy-Schwarz inequality gives  $1 = (\sum \lambda_i)^2 \leq (n - 1) \sum \lambda_i^2$ . Hence  $\text{tr}(\mathbf{A}^2) \geq 1/(n - 1)$ , with equality iff  $\mathbf{A} = \mathbf{C} \mathbf{A} \mathbf{C}$  and all  $\lambda_i = 1/(n - 1)$ , that is, iff  $\mathbf{A} = \mathbf{C}/(n - 1)$ . That matrix also has  $\mathbf{A} \mathbf{1} = \mathbf{0}$ , so it minimizes both variance terms at once.



## 4.3 When a quadratic form is chi-squared

A quadratic form is a weighted sum of independent noncentral  $\chi^2(1)$  pieces, and the weights are eigenvalues. It is itself chi-squared when every nonzero weight equals one. This section turns that observation into checkable conditions on  $\mathbf{A}$  and  $\Sigma$ , and shows that for the spherical case the condition is necessary as well as sufficient.

### 4.3.1 Spherical normal vectors

#### Theorem 4.3.1 — Chi-squared forms in a spherical normal vector

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{I}_n)$ , let  $\mathbf{A}$  be symmetric, and let  $r \geq 1$  be an integer and  $\gamma \geq 0$ . Then  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \gamma)$  if and only if  $\mathbf{A}$  is idempotent with  $\text{rank}(\mathbf{A}) = r$  and  $\gamma = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ .

*Proof. Sufficiency.* By Theorem 1.9.2,  $\mathbf{A} = \mathbf{Q}_1 \mathbf{Q}_1^\top$ , where the  $n \times r$  matrix  $\mathbf{Q}_1$  has orthonormal columns and  $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A})$ . Then  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \|\mathbf{Q}_1^\top \mathbf{Y}\|^2$  and  $\mathbf{Q}_1^\top \mathbf{Y} \sim N_r(\mathbf{Q}_1^\top \boldsymbol{\mu}, \mathbf{Q}_1^\top \mathbf{Q}_1) = N_r(\mathbf{Q}_1^\top \boldsymbol{\mu}, \mathbf{I}_r)$ , so by definition  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \|\mathbf{Q}_1^\top \boldsymbol{\mu}\|^2)$ , and  $\|\mathbf{Q}_1^\top \boldsymbol{\mu}\|^2 = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ .

*Necessity.* Apply Theorem 4.2.1 with  $\mathbf{L} = \mathbf{I}$ , so that  $\mathbf{A} = \mathbf{Q} \boldsymbol{\Lambda} \mathbf{Q}^\top$ . Let  $\mathbf{v} = \mathbf{Q}^\top \boldsymbol{\mu}$ . Then  $\mathbf{b} = \mathbf{Q}^\top \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Lambda} \mathbf{v}$ , so  $b_j = \lambda_j \nu_j$ . The constant left after completing squares is  $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} - \sum_{\lambda_j \neq 0} \lambda_j \nu_j^2 = \mathbf{v}^\top \boldsymbol{\Lambda} \mathbf{v} - \sum_j \lambda_j \nu_j^2 = 0$ . Hence  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  has the law of  $\sum_{j \in J} \lambda_j (W_j + \nu_j)^2$ , where  $J$  indexes the nonzero eigenvalues.  $J$  is not empty, because a  $\chi^2(r, \gamma)$  variable with  $r \geq 1$  is not identically zero.

By Lemma 4.1.1, near  $t = 0$  the logarithm of the mgf of this sum is  $\sum_{j \in J} \{-\frac{1}{2} \log(1 - 2\lambda_j t) + \lambda_j \nu_j^2 t / (1 - 2\lambda_j t)\}$ . Expanding  $-\frac{1}{2} \log(1 - x) = \sum_{k \geq 1} x^k / (2k)$  and  $x / (1 - x) = \sum_{k \geq 1} x^k$ , the coefficient of  $t^k$  is  $(2^{k-1}/k) \sum_{j \in J} \lambda_j^k (1 + k \nu_j^2)$ . The same expansion of Eq. 4.1.2 gives  $(2^{k-1}/k)(r + k\gamma)$ . Equal distributions have equal mgfs, and two power series that agree on an interval have equal coefficients, so

$$\sum_{j \in J} \lambda_j^k (1 + k \nu_j^2) = r + k\gamma \quad \text{for every } k \geq 1. \quad (4.3.1)$$

Let  $\rho = \max_{j \in J} |\lambda_j|$ . For even  $k$  every term on the left is nonnegative and one of them is at least  $\rho^k$ . The right side grows only linearly in  $k$ , so  $\rho \leq 1$ . Split  $J$  into  $J_+ = \{\lambda_j = 1\}$ ,  $J_- = \{\lambda_j = -1\}$  and  $J_0 = \{0 < |\lambda_j| < 1\}$ , and write  $N_\pm = |J_\pm|$  and  $S_\pm = \sum_{j \in J_\pm} \nu_j^2$ . The terms over  $J_0$  tend to zero as  $k \rightarrow \infty$ , so

$$N_+ + (-1)^k N_- + k(S_+ + (-1)^k S_-) - r - k\gamma \rightarrow 0.$$

Dividing by  $k$  and letting  $k \rightarrow \infty$  through even and through odd values gives  $S_+ + S_- = \gamma = S_+ - S_-$ . Then the same limit without the factor  $k$  gives  $N_+ + N_- = r = N_+ - N_-$ . So  $N_- = S_- = 0$ ,  $N_+ = r$  and  $S_+ = \gamma$ . The terms over  $J_+$  in Eq. 4.3.1 now account for all of  $r + k\gamma$ , and with  $k = 2$  the remaining sum  $\sum_{j \in J_0} \lambda_j^2 (1 + 2\nu_j^2)$  is zero. Every term of that sum is positive, so  $J_0$  is empty. Thus every eigenvalue of  $\mathbf{A}$  is 0 or 1,  $\mathbf{A}$  is idempotent,  $\text{rank}(\mathbf{A}) = N_+ = r$ , and  $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} = \sum_{j \in J_+} \nu_j^2 = \gamma$ .

The necessity half is the one that makes the theorem a tool for *recognizing* distributions. If a quadratic form in a spherical normal vector is chi-squared at all, then it is the squared length of a projection. There is no other way.

### 4.3.2 General covariance matrices

#### Theorem 4.3.2 — When a quadratic form is chi-squared

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  and let  $\mathbf{A}$  be symmetric.

- (a) (*Projections.*) If  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$  with  $\sigma > 0$  and  $\mathbf{P}$  is symmetric and idempotent, then

$$\frac{\|\mathbf{PY}\|^2}{\sigma^2} = \frac{\mathbf{Y}^\top \mathbf{PY}}{\sigma^2} \sim \chi^2(\text{rank}(\mathbf{P}), \frac{\|\mathbf{P}\boldsymbol{\mu}\|^2}{\sigma^2}).$$

- (b) (*Positive definite  $\boldsymbol{\Sigma}$ .*) If  $\boldsymbol{\Sigma}$  is positive definite, then  $\mathbf{Y}^\top \mathbf{AY}$  has a  $\chi^2(r, \gamma)$  distribution for some  $r \geq 1$  and  $\gamma \geq 0$  iff  $\mathbf{A}\boldsymbol{\Sigma}$  is idempotent and  $\mathbf{A} \neq \mathbf{0}$ . In that case  $r = \text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$  and  $\gamma = \boldsymbol{\mu}^\top \mathbf{A}\boldsymbol{\mu}$ .

- (c) (*Any  $\boldsymbol{\Sigma}$ .*) If

$$(i) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}, \quad (ii) \ \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}, \quad (iii) \ \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu},$$

then  $\mathbf{Y}^\top \mathbf{AY} \sim \chi^2(\text{tr}(\mathbf{A}\boldsymbol{\Sigma}), \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu})$ . The three conditions hold in particular when  $\mathbf{A}\boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$ , and when  $\mathbf{A}\boldsymbol{\Sigma}$  is idempotent and  $\boldsymbol{\mu} \in \mathcal{C}(\boldsymbol{\Sigma})$ .

*Proof.*

- (a)  $\sigma^{-1} \mathbf{Y} \sim N_n(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$ . Apply Theorem 4.3.1 with  $\mathbf{A} = \mathbf{P}$ , using  $\mathbf{P}^\top \mathbf{P} = \mathbf{P}$  to write  $\mathbf{Y}^\top \mathbf{PY} = \|\mathbf{PY}\|^2$  and  $\boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu} = \|\mathbf{P}\boldsymbol{\mu}\|^2$ .
- (b) Let  $\boldsymbol{\Sigma}^{1/2}$  be the positive definite square root (Theorem 1.7.6). Then  $\mathbf{Z} = \boldsymbol{\Sigma}^{-1/2} \mathbf{Y} \sim N_n(\boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}, \mathbf{I})$  and  $\mathbf{Y}^\top \mathbf{AY} = \mathbf{Z}^\top \mathbf{BZ}$  with  $\mathbf{B} = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$  symmetric. By Theorem 4.3.1,  $\mathbf{Y}^\top \mathbf{AY}$  is noncentral chi-squared iff  $\mathbf{B}$  is idempotent and nonzero. Now  $\mathbf{B}^2 = \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$ , so  $\mathbf{B}^2 = \mathbf{B}$  iff  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} = \mathbf{A}$  (multiply by  $\boldsymbol{\Sigma}^{-1/2}$  on both sides) iff  $\mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\Sigma} = \mathbf{A} \boldsymbol{\Sigma}$  (multiply by  $\boldsymbol{\Sigma}$  or  $\boldsymbol{\Sigma}^{-1}$  on the right). Also  $\mathbf{B} \neq \mathbf{0}$  iff  $\mathbf{A} \neq \mathbf{0}$ . The degrees of freedom are  $\text{rank}(\mathbf{B}) = \text{rank}(\mathbf{A})$ , which equals  $\text{tr}(\mathbf{B}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma})$ , and the noncentrality is  $\boldsymbol{\mu}^\top \boldsymbol{\Sigma}^{-1/2} \mathbf{B} \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ .
- (c) Take  $\mathbf{L}$  of size  $n \times k$  with full column rank  $k = \text{rank}(\boldsymbol{\Sigma})$  and  $\mathbf{L} \mathbf{L}^\top = \boldsymbol{\Sigma}$ , for instance  $\mathbf{L} = \mathbf{U}_1 \mathbf{D}_1^{1/2}$  from the eigenvectors  $\mathbf{U}_1$  for the positive eigenvalues  $\mathbf{D}_1$  of  $\boldsymbol{\Sigma}$ . (If  $\boldsymbol{\Sigma} = \mathbf{0}$  the claim is trivial.) As in Theorem 4.2.1,  $\mathbf{Y} \stackrel{d}{=} \boldsymbol{\mu} + \mathbf{LZ}$  with  $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{I})$ . Since  $\mathbf{L}$  has full column rank,  $\mathbf{L}^\top$  has full row rank, so  $\mathbf{L} \mathbf{G} \mathbf{L}^\top = \mathbf{L} \mathbf{H} \mathbf{L}^\top$  implies  $\mathbf{G} = \mathbf{H}$ , and  $\mathbf{L} \mathbf{g} = \mathbf{L} \mathbf{h}$  implies  $\mathbf{g} = \mathbf{h}$ . Put  $\mathbf{B} = \mathbf{L}^\top \mathbf{A} \mathbf{L}$  and  $\mathbf{c} = \mathbf{L}^\top \mathbf{A} \boldsymbol{\mu}$ . Condition (i) reads  $\mathbf{L} \mathbf{B}^2 \mathbf{L}^\top = \mathbf{L} \mathbf{B} \mathbf{L}^\top$ , so  $\mathbf{B}^2 = \mathbf{B}$ . Condition (ii) reads  $\mathbf{L} \mathbf{B} \mathbf{c} = \mathbf{L} \mathbf{c}$ , so  $\mathbf{B} \mathbf{c} = \mathbf{c}$ . Condition (iii) reads  $\mathbf{c}^\top \mathbf{c} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ . Hence

$$(\boldsymbol{\mu} + \mathbf{LZ})^\top \mathbf{A} (\boldsymbol{\mu} + \mathbf{LZ}) = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu} + 2\mathbf{c}^\top \mathbf{Z} + \mathbf{Z}^\top \mathbf{BZ} = (\mathbf{Z} + \mathbf{c})^\top \mathbf{B} (\mathbf{Z} + \mathbf{c}),$$

because  $(\mathbf{Z} + \mathbf{c})^\top \mathbf{B} (\mathbf{Z} + \mathbf{c}) = \mathbf{Z}^\top \mathbf{BZ} + 2\mathbf{c}^\top \mathbf{BZ} + \mathbf{c}^\top \mathbf{B} \mathbf{c}$  and  $\mathbf{B} \mathbf{c} = \mathbf{c}$ . Now  $\mathbf{Z} +$

$\mathbf{c} \sim N_k(\mathbf{c}, \mathbf{I})$ , and [Theorem 4.3.1](#) gives the law  $\chi^2(\text{rank } \mathbf{B}, \mathbf{c}^\top \mathbf{B} \mathbf{c})$ , where  $\text{rank } \mathbf{B} = \text{tr } \mathbf{B} = \text{tr}(\mathbf{A}\Sigma)$  and  $\mathbf{c}^\top \mathbf{B} \mathbf{c} = \mathbf{c}^\top \mathbf{c} = \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ . If  $\mathbf{B} = \mathbf{0}$ , then  $\mathbf{c} = \mathbf{B} \mathbf{c} = \mathbf{0}$  and the form is identically 0, which is  $\chi^2(0)$ . If  $\mathbf{A}\Sigma\mathbf{A} = \mathbf{A}$ , all three conditions follow by substitution. If  $\mathbf{A}\Sigma$  is idempotent and  $\boldsymbol{\mu} = \Sigma \mathbf{v}$ , then (i) follows from multiplying  $\mathbf{A}\Sigma\mathbf{A}\Sigma = \mathbf{A}\Sigma$  by  $\Sigma$  on the left, and (ii) and (iii) from the same identity applied to  $\mathbf{v}$ .

■

Part (a) is the case used most often in this book. For the linear model  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$  and the projection  $\mathbf{M}$  onto  $\mathcal{C}(\mathbf{X})$ , [Chapter 6](#) shows that the residual sum of squares is  $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ . Since  $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ , part (a) gives

$$\text{SSE}/\sigma^2 \sim \chi^2(n - \text{rank}(\mathbf{X})),$$

a central distribution whatever the value of  $\boldsymbol{\beta}$ . The projections that define test statistics, by contrast, do not annihilate the mean under the alternative, and their noncentrality  $\|\mathbf{P}\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$  measures how far the truth is from the hypothesis in the metric that the test can see.

In part (c) idempotence of  $\mathbf{A}\Sigma$  alone is *not* enough when  $\Sigma$  is singular. A mean with a component outside  $\mathcal{C}(\Sigma)$  is a deterministic shift that the form may pick up as a constant or as a linear term ([Exercise \(B2\)](#)). Part (c) also covers the natural generalization of the Mahalanobis distance.

### Corollary 4.3.3 — Mahalanobis forms

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ .

- (a) If  $\Sigma$  is positive definite,  $(\mathbf{Y} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{Y} - \boldsymbol{\mu}) \sim \chi^2(n)$  and  $\mathbf{Y}^\top \Sigma^{-1} \mathbf{Y} \sim \chi^2(n, \boldsymbol{\mu}^\top \Sigma^{-1} \boldsymbol{\mu})$ .
- (b) If  $\mathbf{G}$  is a symmetric generalized inverse of  $\Sigma$  and  $\boldsymbol{\mu} \in \mathcal{C}(\Sigma)$ , then  $\mathbf{Y}^\top \mathbf{G} \mathbf{Y} \sim \chi^2(\text{rank}(\Sigma), \boldsymbol{\mu}^\top \mathbf{G} \boldsymbol{\mu})$ .

*Proof.*

- (a)  $\mathbf{A} = \Sigma^{-1}$  satisfies  $\mathbf{A}\Sigma\mathbf{A} = \mathbf{A}$ , and  $\mathbf{Y} - \boldsymbol{\mu} \sim N_n(\mathbf{0}, \Sigma)$ . The central statement was already obtained by whitening in [Proposition 3.2.3](#).
- (b)  $\mathbf{G}\Sigma$  is idempotent because  $\Sigma\mathbf{G}\Sigma = \Sigma$ , so part (c) of the theorem applies and the degrees of freedom are  $\text{tr}(\mathbf{G}\Sigma) = \text{tr}(\mathbf{L}^\top \mathbf{G} \mathbf{L})$ . With  $\mathbf{L}$  as in that proof,  $\mathbf{L}\mathbf{L}^\top \mathbf{G} \mathbf{L}\mathbf{L}^\top = \mathbf{L}\mathbf{L}^\top$  gives  $\mathbf{L}^\top \mathbf{G} \mathbf{L} = \mathbf{I}_k$ , whose trace is  $k = \text{rank}(\Sigma)$ .

■

A symmetric generalized inverse always exists ([Proposition 1.3.3\(d\)](#)). The value of  $\mathbf{Y}^\top \mathbf{G} \mathbf{Y}$  does not depend on which one is chosen, with probability one ([Exercise \(B3\)](#)).

### Example. (Correlated observations about a common level)

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$  with  $\Sigma$  positive definite, and suppose we want to measure how far  $\mathbf{Y}$  is from a common level  $m\mathbf{1}$ . The generalized least squares level is  $\hat{m} = \mathbf{1}^\top \Sigma^{-1} \mathbf{Y} / \mathbf{1}^\top \Sigma^{-1} \mathbf{1}$ ,

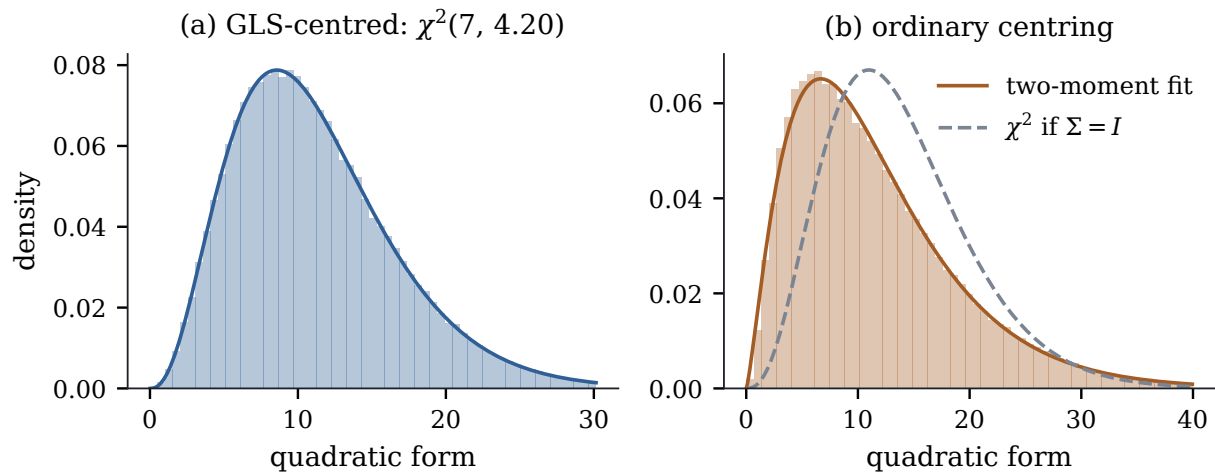


Figure 4.3.1: Two quadratic forms measuring spread about a common level, for  $n = 8$  autoregressive observations with a trending mean. (a) With generalized least squares centring,  $\mathbf{A}\Sigma$  is idempotent and the form is exactly noncentral chi-squared. (b) With ordinary centring it is a weighted sum of noncentral  $\chi^2(1)$  terms. A two-moment scaled chi-squared fits closely. The chi-squared law that would hold for independent observations does not.

and the natural discrepancy is  $(\mathbf{Y} - \hat{m}\mathbf{1})^\top \Sigma^{-1}(\mathbf{Y} - \hat{m}\mathbf{1}) = \mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  with

$$\mathbf{A} = \Sigma^{-1} - \frac{\Sigma^{-1} \mathbf{1} \mathbf{1}^\top \Sigma^{-1}}{\mathbf{1}^\top \Sigma^{-1} \mathbf{1}}.$$

A direct multiplication gives  $\mathbf{A}\Sigma\mathbf{A} = \mathbf{A}$ , and  $\text{tr}(\mathbf{A}\Sigma) = n - 1$ . By Theorem 4.3.2(b) the form is noncentral chi-squared with  $n - 1$  degrees of freedom and noncentrality  $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$ , which is zero exactly when  $\boldsymbol{\mu}$  is constant. The listing takes  $n = 8$ , an autoregressive correlation  $\Sigma = (\rho^{|i-j|})$  with  $\rho = 0.6$ , and a linear trend in the mean, for which  $\gamma = 4.200$ . Figure 4.3.1(a) compares 200,000 simulated values with the  $\chi^2(7, 4.200)$  density.

```
import numpy as np
from scipy import integrate, stats
rng = np.random.default_rng(31415)
n, rho = 8, 0.6
idx = np.arange(n)
Sigma = rho ** np.abs(idx[:, None] - idx[None, :]) # AR(1) correlation matrix
mu = 0.4 * (idx - idx.mean()) # linear trend, mean zero
one = np.ones(n)

W = np.linalg.inv(Sigma)
A = W - np.outer(W @ one, W @ one) / (one @ W @ one) # GLS-centred form
AS = A @ Sigma
print("A Sigma idempotent:", np.allclose(AS @ AS, AS))
df, gamma = np.trace(AS), mu @ A @ mu # chi^2(df, gamma)

L = np.linalg.cholesky(Sigma) # Y = mu + L Z
Z = rng.standard_normal((200_000, n))
Y = mu + Z @ L.T
q_gls = np.einsum("ij,jk,ik->i", Y, A, Y)
print(df, gamma, stats.kstest(q_gls, stats.ncx2(df, gamma).cdf).pvalue)
```

### 4.3.3 Forms that are not chi-squared

Most quadratic forms fail the conditions of [Theorem 4.3.2](#). The ordinary sum of squares about the mean,  $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$  with  $\mathbf{C} = \mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top$ , is an example as soon as the observations are correlated. Then  $\mathbf{C} \boldsymbol{\Sigma}$  is not idempotent, and by [Theorem 4.2.1](#) the law is that of  $\sum_j \lambda_j (W_j + \nu_j)^2$ , with weights  $\lambda_j$  equal to the nonzero eigenvalues of  $\mathbf{C} \boldsymbol{\Sigma}$ . In [Example \(Correlated observations about a common level\)](#) these range from 0.259 to 1.832. Two practical ways to handle such a law are the following.

**Exact computation.** The characteristic function of the weighted sum is known in closed form (put  $t = iu$  in [Corollary 4.2.2](#)), and a distribution function can be recovered from a characteristic function  $\varphi$  by the inversion formula of Gil-Pelaez (1951),

$$\Pr(X > x) = \frac{1}{2} + \frac{1}{\pi} \int_0^\infty \frac{\operatorname{Im}\{e^{-iux} \varphi(u)\}}{u} du.$$

For a weighted sum of  $k$  chi-squared terms the integrand decays like  $u^{-1-k/2}$ , so ordinary quadrature works well. Imhof (1961) gave the real-variable form of this integral that most software uses. The listing implements it directly and checks it against simulation.

**Two-moment approximation.** Match  $a\chi^2(\nu)$  to the mean  $m$  and variance  $v$  from [Theorem 4.2.3](#). Since  $a\chi^2(\nu)$  has mean  $a\nu$  and variance  $2a^2\nu$ , this gives  $a = v/(2m)$  and  $\nu = 2m^2/v$ , with  $\nu$  usually not an integer (Satterthwaite 1946; Patnaik 1949).

In [Example \(Correlated observations about a common level\)](#), the simulated 95th percentile of  $\mathbf{Y}^\top \mathbf{C} \mathbf{Y}$  is 26.51. The exact tail probability beyond it is 0.0490, and the approximation with  $a = 2.488$  and  $\nu = 4.68$  gives 0.0484. Treating the observations as independent, and so using  $\chi^2(7, 6.720)$ , gives 0.040. [Figure 4.3.1\(b\)](#) shows the three laws. Here the error from ignoring correlation is modest at the 5% point but visible in the shape. It is not a monotone function of  $\rho$ , but strong correlation makes it severe: with  $\rho = 0.9$  and the same mean, the naive tail probability at the true 95th percentile is 0.197.

```
def canonical(A, mu, L):
    """Y^T A Y = sum_j lam_j (W_j + nu_j)^2 + c (all lam_j nonzero here), W ~ N(0, I)"""
    lam, Qm = np.linalg.eigh(L.T @ A @ L)
    b = Qm.T @ L.T @ A @ mu
    keep = np.abs(lam) > 1e-10
    assert np.allclose(b[~keep], 0) # no linear terms in this example
    c = mu @ A @ mu - np.sum(b[keep] ** 2 / lam[keep])
    return lam[keep], b[keep] / lam[keep], c

def tail_prob(x, lam, nu, c=0.0):
    """P(sum lam_j (W_j + nu_j)^2 + c > x) by Gil-Pelaez inversion."""
    def integrand(u):
        z = 1 - 2j * lam * u
        phi = np.prod(z ** -0.5 * np.exp(1j * u * lam * nu**2 / z))
        return (np.exp(-1j * u * (x - c)) * phi).imag / u
    edges = np.linspace(0, 400, 401) # integrand decays like u
    # ^(-1-k/2)
    val = sum(integrate.quad(integrand, a, b)[0] for a, b in zip(edges[:-1], edges[1:]))
    return 0.5 + val / np.pi
```

## 4.3.4 Exercises

### 4.3.4.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$  with  $n \geq 2$ , and  $\mathbf{A} = \alpha \mathbf{I} + \beta \mathbf{1}\mathbf{1}^\top$ . Find all  $(\alpha, \beta)$  for which  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  has a chi-squared distribution, and identify each form.

*Solution.*  $\mathbf{A}$  has eigenvalue  $\alpha + n\beta$  on  $\mathbf{1}$  and  $\alpha$  on  $\mathbf{1}^\perp$ . By [Theorem 4.3.1](#), the form is chi-squared iff  $\mathbf{A} \neq \mathbf{0}$  and both eigenvalues lie in  $\{0, 1\}$ . The three solutions are  $(1, 0)$ , giving  $\|\mathbf{Y}\|^2 \sim \chi^2(n)$ ;  $(0, 1/n)$ , giving  $n\bar{Y}^2 \sim \chi^2(1)$ ; and  $(1, -1/n)$ , giving  $\sum (Y_i - \bar{Y})^2 \sim \chi^2(n-1)$ .

#### Exercise A2.

Two independent samples of sizes  $m$  and  $k$  come from  $N(\mu_1, \sigma^2)$  and  $N(\mu_2, \sigma^2)$ . Write the pooled within-sample sum of squares as  $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$  for a suitable projection, and show that the two-sample  $t$  statistic has the  $t(m+k-2, (\mu_1 - \mu_2)/(\sigma\sqrt{1/m+1/k}))$  distribution.

### 4.3.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\{(1-\rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\})$  with  $-1/(n-1) < \rho < 1$ , and let  $\bar{Y}$  and  $S^2$  be the sample mean and variance. Show that  $(n-1)S^2/\{\sigma^2(1-\rho)\} \sim \chi^2(n-1)$  and that  $S^2$  is independent of  $\bar{Y}$ . Deduce that  $\sqrt{n}(\bar{Y} - \mu)/S$  has the law of  $\{(1 + (n-1)\rho)/(1-\rho)\}^{1/2}$  times a  $t(n-1)$  variable, and describe what a positive  $\rho$  does to the coverage of the usual 95%  $t$  interval for  $\mu$ .

#### Exercise B2.

Let  $\Sigma = \text{diag}(1, 0)$ ,  $\mu = (0, 1)^\top$ ,  $\mathbf{A} = \mathbf{I}_2$  and  $\mathbf{Y} \sim N_2(\mu, \Sigma)$ . Show that  $\mathbf{A}\Sigma$  is idempotent but  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  is not chi-squared. Which of the conditions of [Theorem 4.3.2\(c\)](#) fail? What changes if  $\mu = (m, 0)^\top$ ? Repeat the first part with  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$ , where the shift enters through a linear term.

*Solution.*  $\mathbf{A}\Sigma = \Sigma$  is idempotent. But  $Y_2 = 1$  with probability one, so  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 + 1 \geq 1$ , while every  $\chi^2(r, \gamma)$  with  $r \geq 1$  gives positive probability to  $(0, 1)$ . Conditions (i) and (ii) hold, since  $\Sigma^3 = \Sigma$  and  $\Sigma^2\mu = \mathbf{0} = \Sigma\mu$ . Condition (iii) fails, since  $\mu^\top \Sigma\mu = 0 \neq 1 = \mu^\top \mu$ . The part of the mean outside  $\mathcal{C}(\Sigma)$  adds a constant. If  $\mu = (m, 0)^\top \in \mathcal{C}(\Sigma)$ , then  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 \sim \chi^2(1, m^2)$ , as [Theorem 4.3.2\(c\)](#) predicts. For  $\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mu = (0, 1)^\top$ ,  $\mathbf{A}\Sigma = \begin{pmatrix} 1 & 0 \\ 1 & 0 \end{pmatrix}$  is again idempotent, and  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = Y_1^2 + 2Y_1Y_2 = Y_1^2 + 2Y_1 = (Y_1 + 1)^2 - 1$ , which is negative with positive probability. Conditions (i) and (ii) hold, since  $\Sigma\mathbf{A}\Sigma = \Sigma$  and  $\Sigma\mathbf{A}\Sigma\mu = \Sigma\mathbf{A}\mu = (1, 0)^\top$ , but (iii) fails because  $\mu^\top \mathbf{A}\Sigma\mathbf{A}\mu = 1 \neq 0 = \mu^\top \mathbf{A}\mu$ . The shift now enters through the linear term  $2Y_1$ .

**Exercise B3.**

In [Corollary 4.3.3\(b\)](#), show that  $\mathbf{Y}^\top \mathbf{G} \mathbf{Y}$  takes the same value for every generalized inverse  $\mathbf{G}$  of  $\Sigma$ , with probability one. *Hint:* [Theorem 2.2.3](#).

*Solution.* By [Theorem 2.2.3](#),  $\mathbf{Y} - \boldsymbol{\mu} \in \mathcal{C}(\Sigma)$  with probability one. Since  $\boldsymbol{\mu} \in \mathcal{C}(\Sigma)$ , also  $\mathbf{Y} \in \mathcal{C}(\Sigma)$ , so  $\mathbf{Y} = \Sigma \mathbf{v}$  for some (random)  $\mathbf{v}$ . For any generalized inverse  $\mathbf{G}$ ,  $\mathbf{Y}^\top \mathbf{G} \mathbf{Y} = \mathbf{v}^\top \Sigma \mathbf{G} \Sigma \mathbf{v} = \mathbf{v}^\top \Sigma \mathbf{v}$ , which does not involve  $\mathbf{G}$ .

**Exercise B4.**

Let  $\mathbf{Y} \sim N_n(\mathbf{0}, \Sigma)$  with  $\Sigma$  positive definite, and let  $\mathbf{A} \neq \mathbf{0}$  be nonnegative definite. Show that the two-moment approximation  $a\chi^2(\nu)$  to  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  has  $\nu = \{\text{tr}(\mathbf{A}\Sigma)\}^2 / \text{tr}\{(\mathbf{A}\Sigma)^2\}$ , and that  $\nu \leq \text{rank}(\mathbf{A})$ , with equality iff all nonzero eigenvalues of  $\mathbf{A}\Sigma$  are equal.

**4.3.4.3 C. Going deeper****Exercise C1.**

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$  with  $\Sigma$  possibly singular, and suppose  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r, \gamma)$  with  $r \geq 1$ . Show that  $\Sigma \mathbf{A} \Sigma \mathbf{A} \Sigma = \Sigma \mathbf{A} \Sigma$ . *Hint:* in [Corollary 4.2.2](#), compare the coefficients of  $t^k$  for  $k \geq 3$  with those of [Eq. 4.1.2](#), and adapt the argument of [Theorem 4.3.1](#).

## 4.4 Independence of quadratic forms and linear forms

An  $F$  or  $t$  statistic is a ratio, and its distribution is known only if numerator and denominator are independent. For normal vectors, uncorrelated linear functions are independent ([Theorem 3.4.1](#)). Quadratic forms are not linear, but the following observation brings them within reach of that result.

*Idea.* A quadratic form is a function of a linear form: for any generalized inverse  $\mathbf{A}^-$  of  $\mathbf{A}$ ,

$$\mathbf{Y}^\top \mathbf{A} \mathbf{Y} = \mathbf{Y}^\top \mathbf{A} \mathbf{A}^- \mathbf{A} \mathbf{Y} = (\mathbf{A} \mathbf{Y})^\top \mathbf{A}^- (\mathbf{A} \mathbf{Y}).$$

So  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  is independent of anything that is independent of the vector  $\mathbf{A} \mathbf{Y}$ .

### 4.4.1 The independence theorems

#### **Theorem 4.4.1** — Linear and quadratic forms

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$ , let  $\mathbf{B}$  be  $m \times n$  and  $\mathbf{A}$  symmetric  $n \times n$ . If  $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$ , then  $\mathbf{B}\mathbf{Y}$  and  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  are independent. Conversely, if  $\mathbf{B}\mathbf{Y}$  and  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  are independent for every  $\boldsymbol{\mu} \in \mathbb{R}^n$ , with  $\Sigma$  fixed, then  $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$ .

*Proof.* The vector  $(\mathbf{BY}, \mathbf{AY})$  is a linear function of  $\mathbf{Y}$ , hence jointly normal (Theorem 3.3.1), and its cross-covariance is  $\text{Cov}(\mathbf{BY}, \mathbf{AY}) = \mathbf{B}\Sigma\mathbf{A}^\top = \mathbf{B}\Sigma\mathbf{A}$  (Theorem 2.2.1). If this is zero,  $\mathbf{BY}$  and  $\mathbf{AY}$  are independent by Theorem 3.4.1, and so are  $\mathbf{BY}$  and the function  $(\mathbf{AY})^\top \mathbf{A}^-(\mathbf{AY})$  of  $\mathbf{AY}$ , which is  $\mathbf{Y}^\top \mathbf{AY}$  by the key idea above. Conversely, independent variables with finite variances are uncorrelated, so Theorem 4.2.3(c) gives  $2\mathbf{B}\Sigma\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$  for every  $\boldsymbol{\mu}$ , that is,  $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$ . ■

### Theorem 4.4.2 — Two quadratic forms

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \Sigma)$  and let  $\mathbf{A}, \mathbf{B}$  be symmetric. If  $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$ , equivalently  $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$ , then  $\mathbf{Y}^\top \mathbf{AY}$  and  $\mathbf{Y}^\top \mathbf{BY}$  are independent.

*Proof.* The two conditions are transposes of each other. As before,  $\mathbf{AY}$  and  $\mathbf{BY}$  are jointly normal with  $\text{Cov}(\mathbf{AY}, \mathbf{BY}) = \mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$ , hence independent, and each form is a function of one of them. ■

The condition is sufficient for every  $\boldsymbol{\mu}$  and every  $\Sigma$ . When  $\Sigma$  is singular one might hope that the weaker requirement  $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$  suffices, since only directions in  $\mathcal{C}(\Sigma)$  carry noise. For nonnegative definite  $\mathbf{A}$  and  $\mathbf{B}$  the two requirements turn out to be equivalent (Exercise (B2)). In the other direction, the condition is necessary under mild assumptions, a result usually credited to Craig (1943). Its general proof is surprisingly delicate, and the history of incorrect published proofs is recounted by Driscoll and Gundberg (1986). The case that matters most for sums of squares has a short proof.

### Proposition 4.4.3 — Necessity for nonnegative definite forms

Let  $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$  and let  $\mathbf{A}, \mathbf{B}$  be nonnegative definite. If  $\mathbf{Y}^\top \mathbf{AY}$  and  $\mathbf{Y}^\top \mathbf{BY}$  are independent, then  $\mathbf{AB} = \mathbf{0}$ .

*Proof.* Independence makes the covariance zero, and by Theorem 4.2.3(d) the covariance is  $2\text{tr}(\mathbf{AB})$ . With the square roots of Theorem 1.7.6,

$$\text{tr}(\mathbf{AB}) = \text{tr}(\mathbf{A}^{1/2}\mathbf{A}^{1/2}\mathbf{B}^{1/2}\mathbf{B}^{1/2}) = \text{tr}\{(\mathbf{A}^{1/2}\mathbf{B}^{1/2})^\top(\mathbf{A}^{1/2}\mathbf{B}^{1/2})\},$$

the sum of squares of the entries of  $\mathbf{A}^{1/2}\mathbf{B}^{1/2}$ . So  $\mathbf{A}^{1/2}\mathbf{B}^{1/2} = \mathbf{0}$ , and  $\mathbf{AB} = \mathbf{A}^{1/2}(\mathbf{A}^{1/2}\mathbf{B}^{1/2})\mathbf{B}^{1/2} = \mathbf{0}$ . ■

For nonnegative definite forms in a standard normal vector, then, zero correlation and independence coincide, just as for jointly normal linear forms. This fails for general symmetric matrices (Exercise (B1)).



## 4.4.2 Orthogonal projections of a spherical normal vector

The situation in regression is the tidiest one: a spherical normal vector and several projections onto mutually orthogonal subspaces.

### Theorem 4.4.4 — Independent projections

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ , and let  $\mathbf{P}_1, \dots, \mathbf{P}_k$  be symmetric idempotent matrices with  $\mathbf{P}_i \mathbf{P}_j = \mathbf{0}$  for  $i \neq j$ , and ranks  $r_1, \dots, r_k$ . Then  $\mathbf{P}_1 \mathbf{Y}, \dots, \mathbf{P}_k \mathbf{Y}$  are mutually independent normal vectors,  $\mathbf{P}_i \mathbf{Y} \sim N_n(\mathbf{P}_i \boldsymbol{\mu}, \sigma^2 \mathbf{P}_i)$ , and the sums of squares

$$\frac{\|\mathbf{P}_i \mathbf{Y}\|^2}{\sigma^2} \sim \chi^2\left(r_i, \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{\sigma^2}\right), \quad i = 1, \dots, k,$$

are mutually independent.

*Proof.* Stack the vectors  $\mathbf{P}_i \mathbf{Y}$  into one  $nk$ -vector. It is a linear function of  $\mathbf{Y}$ , so it is normal, and the covariance between blocks  $i$  and  $j$  is  $\mathbf{P}_i(\sigma^2 \mathbf{I})\mathbf{P}_j^\top = \sigma^2 \mathbf{P}_i \mathbf{P}_j$ . This is  $\mathbf{0}$  for  $i \neq j$  and  $\sigma^2 \mathbf{P}_i$  for  $i = j$ . By the block form of [Theorem 3.4.1](#) the blocks are mutually independent, and functions of them, the squared lengths, are then mutually independent. Their laws are [Theorem 4.3.2\(a\)](#). ■

In geometric language, which [Chapter 6](#) develops: the coordinates of a spherical normal vector in mutually orthogonal subspaces are independent. It is the reason the rows of an analysis of variance table can be treated separately.

### Corollary 4.4.5 — Sample mean and sample variance

Let  $Y_1, \dots, Y_n$  ( $n \geq 2$ ) be independent  $N(\mu, \sigma^2)$ , with sample mean  $\bar{Y}$  and sample variance  $S^2 = (n-1)^{-1} \sum_i (Y_i - \bar{Y})^2$ . Then

- (a)  $\bar{Y}$  and  $S^2$  are independent, and  $\bar{Y} \sim N(\mu, \sigma^2/n)$ ;
- (b)  $(n-1)S^2/\sigma^2 \sim \chi^2(n-1)$ ;
- (c) for any  $\mu_0$ ,  $T = \sqrt{n}(\bar{Y} - \mu_0)/S \sim t(n-1, \delta)$  with  $\delta = \sqrt{n}(\mu - \mu_0)/\sigma$ . In particular  $T \sim t(n-1)$  when  $\mu = \mu_0$ .

*Proof.* Here  $\mathbf{Y} \sim N_n(\mu \mathbf{1}, \sigma^2 \mathbf{I})$ . Let  $\mathbf{P}_1 = n^{-1} \mathbf{1} \mathbf{1}^\top$  and  $\mathbf{C} = \mathbf{I} - \mathbf{P}_1$ . Both are symmetric and idempotent,  $\mathbf{P}_1 \mathbf{C} = \mathbf{0}$ , and  $\text{rank}(\mathbf{C}) = \text{tr}(\mathbf{C}) = n-1$ . Since  $\mathbf{P}_1 \mathbf{Y} = \bar{Y} \mathbf{1}$  and  $\|\mathbf{C} \mathbf{Y}\|^2 = (n-1)S^2$ , [Theorem 4.4.4](#) gives (a), which [Example \(Sample mean and deviations\)](#) obtained directly, and, because  $\mathbf{C} \mu \mathbf{1} = \mathbf{0}$ , also (b). For (c),  $X = \sqrt{n}(\bar{Y} - \mu_0)/\sigma \sim N(\delta, 1)$  and  $V = (n-1)S^2/\sigma^2 \sim \chi^2(n-1)$  are independent, and  $T = X/\sqrt{V/(n-1)}$ . Apply [Definition 4.1.3](#). ■

The independence in (a) characterizes the normal distribution: for  $n \geq 2$  independent identically distributed observations with finite variance,  $\bar{Y}$  and  $S^2$  are independent only under normality (Lukacs 1942). We will not need this, but it shows that normality is used essentially here. It is not a technicality that more careful arguments could remove.

### 4.4.3 Preview: the $F$ statistic for nested models

The main use of these results comes in [Chapter 11](#). Its core fits in a few lines.

#### Theorem 4.4.6 — $F$ statistic for nested subspaces

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ , and let  $\mathbf{P}_0$  and  $\mathbf{P}$  be symmetric idempotent matrices of ranks  $r_0 < r < n$  with  $\mathbf{P}\mathbf{P}_0 = \mathbf{P}_0$ . Suppose that  $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$ . Then

$$F = \frac{\mathbf{Y}^\top (\mathbf{P} - \mathbf{P}_0) \mathbf{Y} / (r - r_0)}{\mathbf{Y}^\top (\mathbf{I} - \mathbf{P}) \mathbf{Y} / (n - r)} \sim F\left(r - r_0, n - r, \frac{\|(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu}\|^2}{\sigma^2}\right),$$

and the noncentrality is zero iff  $\mathbf{P}_0\boldsymbol{\mu} = \boldsymbol{\mu}$ .

*Proof.* Transposing  $\mathbf{P}\mathbf{P}_0 = \mathbf{P}_0$  gives  $\mathbf{P}_0\mathbf{P} = \mathbf{P}_0$ . Hence  $(\mathbf{P} - \mathbf{P}_0)^2 = \mathbf{P} - 2\mathbf{P}_0 + \mathbf{P}_0 = \mathbf{P} - \mathbf{P}_0$ , so  $\mathbf{P} - \mathbf{P}_0$  is symmetric idempotent with rank  $\text{tr}(\mathbf{P}) - \text{tr}(\mathbf{P}_0) = r - r_0$ . Also  $(\mathbf{P} - \mathbf{P}_0)(\mathbf{I} - \mathbf{P}) = \mathbf{P} - \mathbf{P} - \mathbf{P}_0 + \mathbf{P}_0 = \mathbf{0}$ , and  $\mathbf{I} - \mathbf{P}$  is symmetric idempotent of rank  $n - r$ . By [Theorem 4.4.4](#) the numerator and denominator sums of squares are independent, with  $\sigma^{-2}$  times them distributed as  $\chi^2(r - r_0, \|(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu}\|^2/\sigma^2)$  and  $\chi^2(n - r, \|(\mathbf{I} - \mathbf{P})\boldsymbol{\mu}\|^2/\sigma^2)$ . The second noncentrality is 0 since  $\mathbf{P}\boldsymbol{\mu} = \boldsymbol{\mu}$ . The unknown  $\sigma^2$  cancels in the ratio, and [Definition 4.1.2](#) applies. Finally,  $(\mathbf{P} - \mathbf{P}_0)\boldsymbol{\mu} = \boldsymbol{\mu} - \mathbf{P}_0\boldsymbol{\mu}$ . ■

In a linear model,  $\mathbf{P} = \mathbf{M}$  projects onto the column space of the full model matrix  $\mathbf{X}$  and  $\mathbf{P}_0 = \mathbf{M}_0$  onto that of a submodel  $\mathbf{X}_0$ . The numerator is the drop in residual sum of squares from the submodel to the full model ([Chapter 6](#)). The theorem says three things. The test that rejects for large  $F$  has an exactly known null distribution. Its power depends on the unknown parameters only through  $\gamma = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2$ . And, by [Theorem 4.1.4](#), that power increases with  $\gamma$ .

#### Example. (Detecting curvature)

Let  $x_1, \dots, x_n$ ,  $n = 15$ , be equally spaced on  $[0, 1]$ , and suppose  $Y_i = 1 + 0.5x_i + \beta_2 x_i^2 + \varepsilon_i$  with independent  $N(0, 1)$  errors. To test for curvature, compare the straight line ( $\mathbf{X}_0 = [\mathbf{1}, \mathbf{x}]$ ) with the quadratic ( $\mathbf{X} = [\mathbf{1}, \mathbf{x}, \mathbf{x}^2]$ ). The  $F$  statistic has 1 and 12 degrees of freedom, and the 5% critical value is 4.747. For  $\beta_2 = 8.0$ , the listing finds  $\gamma = 6.873$  and power 0.673 from the noncentral  $F$ , and the rejection rate in 100,000 simulated data sets is 0.674.

The noncentrality has a clear meaning. It is  $\beta_2^2$  times the squared length of the part of  $\mathbf{x}^2$  orthogonal to  $\mathbf{1}$  and  $\mathbf{x}$ , divided by  $\sigma^2$ . It grows roughly in proportion to  $n$  for a fixed design pattern. Keeping the same design on  $[0, 1]$ , the smallest  $n$  with power at least 0.8 is 21, where  $\gamma = 8.97$ . [Figure 4.4.1](#) places the example on the power curve and shows the loss of power when the same noncentrality is spread over more numerator degrees

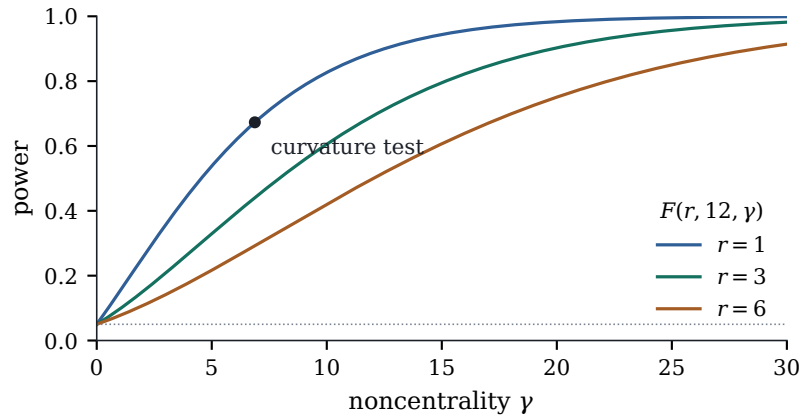


Figure 4.4.1: Power  $\Pr\{F(r, 12, \gamma) > F_{0.05}(r, 12)\}$  of the level-0.05  $F$  test as a function of the noncentrality. Every curve starts at 0.05 and increases to 1 (Theorem 4.1.4). For a fixed  $\gamma$ , power falls as the number  $r$  of numerator degrees of freedom grows. The dot is the curvature test of Example (Detecting curvature).

of freedom  $r$ . At  $\gamma = 10$  the power is 0.827, 0.606 and 0.421 for  $r = 1, 3, 6$ .

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(1104)
alpha, s = 0.05, 12
def power(gamma, q, s, alpha=0.05):
    crit = stats.f.ppf(1 - alpha, q, s)
    return stats.ncf.sf(crit, q, s, np.maximum(gamma, 1e-12))
gammas = np.linspace(0, 30, 301)
gamma_ref, rs = 10.0, (1, 3, 6) # r = numerator degrees of
                                # freedom
curves = {q: power(gammas, q, s) for q in rs}

n, sigma, beta0, beta1, beta2, reps = 15, 1.0, 1.0, 0.5, 8.0, 100_000
x = np.linspace(0, 1, n)
X0 = np.column_stack([np.ones(n), x]) # straight line
X = np.column_stack([X0, x**2]) # adds curvature

def proj(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T

M, M0 = proj(X), proj(X0)
mean = X @ np.array([beta0, beta1, beta2])
gamma = mean @ (M - M0) @ mean / sigma**2 # noncentrality
q, df_resid = 1, n - 3
crit = stats.f.ppf(0.95, q, df_resid)
exact_power = stats.ncf.sf(crit, q, df_resid, gamma)

Y = mean + sigma * rng.standard_normal((reps, n))
num = np.einsum("ij,jk,ik->i", Y, M - M0, Y) / q
den = np.einsum("ij,jk,ik->i", Y, np.eye(n) - M, Y) / df_resid
print(gamma, exact_power, np.mean(num / den > crit))
```

## 4.4.4 Exercises

### 4.4.4.1 A. Check your understanding

#### Exercise A1.

In [Example \(Two estimators of  \$\sigma^2\$  under a drifting mean\)](#) let  $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$  with  $n \geq 3$ . Use [Theorem 4.2.3\(d\)](#) to show that  $\text{Cov}(S^2, Q) = \text{Var}(S^2)$ , and deduce that  $S^2$  and  $Q$  are dependent with correlation  $\{2(n-1)/(3n-4)\}^{1/2}$ . Which linear forms  $\mathbf{a}^\top \mathbf{Y}$  are independent of  $Q$  for every mean vector  $\mu \in \mathbb{R}^n$ ?

*Solution.* With  $\Sigma = \sigma^2\mathbf{I}$ , part (d) of [Theorem 4.2.3](#) gives  $\text{Cov}(S^2, Q) = 2\sigma^4 \text{tr}(\mathbf{C}\mathbf{D}^\top \mathbf{D})/\{2(n-1)^2\}$ . Since  $\mathbf{D}\mathbf{1} = \mathbf{0}$ ,  $\mathbf{C}\mathbf{D}^\top \mathbf{D} = \mathbf{D}^\top \mathbf{D}$ , whose trace is  $2(n-1)$ , so  $\text{Cov}(S^2, Q) = 2\sigma^4/(n-1) = \text{Var}(S^2)$ . The mean terms vanish because  $\mathbf{C}\mu = \mathbf{D}\mu = \mathbf{0}$ . Using [Var\(Q\)](#) from [Example](#), the correlation is  $\{\text{Var}(S^2)/\text{Var}(Q)\}^{1/2} = \{2(n-1)/(3n-4)\}^{1/2}$ , which is nonzero, so the forms are dependent. By [Theorem 4.4.1](#),  $\mathbf{a}^\top \mathbf{Y}$  is independent of  $Q$  for every  $\mu$  iff  $\mathbf{a}^\top \mathbf{D}^\top \mathbf{D} = \mathbf{0}^\top$ . Multiplying by  $\mathbf{a}$  gives  $\|\mathbf{D}\mathbf{a}\|^2 = 0$ , so  $\mathbf{D}\mathbf{a} = \mathbf{0}$  and  $\mathbf{a}$  is a multiple of  $\mathbf{1}$ ; conversely such  $\mathbf{a}$  work. The only such linear forms are the multiples of  $\bar{Y}$ .

### 4.4.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$ . Show that  $Y_1^2 - Y_2^2$  and  $2Y_1Y_2$  are uncorrelated but not independent. Why does [Proposition 4.4.3](#) not apply?

*Solution.* With  $\mathbf{A} = \text{diag}(1, -1)$  and  $\mathbf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ,  $\text{tr}(\mathbf{AB}) = 0$ , so the covariance  $2\text{tr}(\mathbf{AB})$  is zero. In polar coordinates  $Y_1 = R \cos \Theta$ ,  $Y_2 = R \sin \Theta$ , with  $R^2 \sim \chi^2(2)$  independent of  $\Theta$  uniform on  $[0, 2\pi)$ , the two forms are  $R^2 \cos 2\Theta$  and  $R^2 \sin 2\Theta$ . Now  $E(R^4) = 8$  and  $E(R^8) = 384$ , so  $E[(R^2 \cos 2\Theta)^2 (R^2 \sin 2\Theta)^2] = 384/8 = 48$ , but the product of the expected squares is  $(8/2)^2 = 16$ . The forms are dependent. [Proposition 4.4.3](#) needs nonnegative definite matrices, and  $\mathbf{A}$  and  $\mathbf{B}$  are indefinite.

#### Exercise B2.

Let  $\mathbf{Y} \sim N_n(\mu, \Sigma)$  and let  $\mathbf{A}, \mathbf{B}$  be nonnegative definite with  $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$ . Show that  $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$ , so that  $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$  and  $\mathbf{Y}^\top \mathbf{B}\mathbf{Y}$  are independent. Show by an example that the implication can fail when  $\mathbf{A}$  is not nonnegative definite.

*Solution.* Write  $\mathbf{A} = \mathbf{R}\mathbf{R}^\top$  and  $\mathbf{B} = \mathbf{S}\mathbf{S}^\top$ , for instance with symmetric square roots, and  $\Sigma = \mathbf{L}\mathbf{L}^\top$  with  $\mathbf{L}$  of full column rank. Cancelling  $\mathbf{L}$  on both sides of  $\mathbf{L}(\mathbf{L}^\top \mathbf{A}\mathbf{L})(\mathbf{L}^\top \mathbf{B}\mathbf{L})\mathbf{L}^\top = \mathbf{0}$  gives  $\mathbf{L}^\top \mathbf{A}\mathbf{L}\mathbf{L}^\top \mathbf{B}\mathbf{L} = \mathbf{0}$ . The trace of this matrix is

$$\text{tr}(\mathbf{L}^\top \mathbf{R}\mathbf{R}^\top \mathbf{L}\mathbf{L}^\top \mathbf{S}\mathbf{S}^\top \mathbf{L}) = \text{tr}\{(\mathbf{R}^\top \Sigma \mathbf{S})(\mathbf{R}^\top \Sigma \mathbf{S})^\top\} = \|\mathbf{R}^\top \Sigma \mathbf{S}\|_F^2,$$

so  $\mathbf{R}^\top \Sigma \mathbf{S} = \mathbf{0}$ . Hence  $\mathbf{A}\Sigma\mathbf{B} = \mathbf{R}(\mathbf{R}^\top \Sigma \mathbf{S})\mathbf{S}^\top = \mathbf{0}$ , and [Theorem 4.4.2](#) applies. For nonnegative definite matrices the two conditions are therefore equivalent, since  $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$  trivially implies  $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$ . Without nonnegative definiteness the argument breaks

at the trace step. Take  $\Sigma = \text{diag}(1, 0)$ ,  $\mathbf{A} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$  and  $\mathbf{B} = \mathbf{I}_2$ . Then  $\Sigma\mathbf{A}\Sigma = \mathbf{0}$ , so  $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$ , but  $\mathbf{A}\Sigma\mathbf{B} = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix} \neq \mathbf{0}$ . With  $\mu = (0, m)^\top$  and  $m \neq 0$  we have  $Y_2 = m$  with probability one, so the forms are  $2mY_1$  and  $Y_1^2 + m^2$ . The second is a nonconstant function of the first, so they are dependent.

### Exercise B3.

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\beta, \sigma^2\mathbf{I})$  with  $\mathbf{X}$  of full column rank  $p < n$ , let  $\hat{\beta} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{Y}$ ,  $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$  and  $s^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/(n - p)$ . With  $v_{jj}$  the  $j$ th diagonal entry of  $(\mathbf{X}^\top\mathbf{X})^{-1}$ , show that  $(\hat{\beta}_j - b)/(s\sqrt{v_{jj}}) \sim t(n - p, (\beta_j - b)/(\sigma\sqrt{v_{jj}}))$ .

*Solution.* Here  $\mathbf{M}$  is symmetric and idempotent with  $\text{tr } \mathbf{M} = \text{tr}\{(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top\mathbf{X}\} = p$ , and  $\mathbf{M}\mathbf{X} = \mathbf{X}$ . So  $(n - p)s^2/\sigma^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/\sigma^2 \sim \chi^2(n - p)$  by [Theorem 4.3.2\(a\)](#), because  $(\mathbf{I} - \mathbf{M})\mathbf{X}\beta = \mathbf{0}$ . With  $\mathbf{B} = (\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top$ ,  $\mathbf{B}(\sigma^2\mathbf{I})(\mathbf{I} - \mathbf{M}) = \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1}(\mathbf{X}^\top - \mathbf{X}^\top\mathbf{M}) = \mathbf{0}$ , so  $\hat{\beta} = \mathbf{B}\mathbf{Y}$  is independent of  $s^2$  ([Theorem 4.4.1](#)). Since  $\hat{\beta} \sim N_p(\beta, \sigma^2(\mathbf{X}^\top\mathbf{X})^{-1})$ , the variable  $X = (\hat{\beta}_j - b)/(\sigma\sqrt{v_{jj}})$  is  $N((\beta_j - b)/(\sigma\sqrt{v_{jj}}), 1)$ . The statistic is  $X/\sqrt{V/(n - p)}$  with  $V = (n - p)s^2/\sigma^2$ , and [Definition 4.1.3](#) applies.

### Exercise B4.

Show that  $F_\alpha(r, s) \rightarrow \chi_\alpha^2(r)/r$  as  $s \rightarrow \infty$ , where  $\chi_\alpha^2(r)$  is the upper  $\alpha$  point of  $\chi^2(r)$ , and that the power  $\Pr\{F(r, s, \gamma) > F_\alpha(r, s)\}$  tends to  $\Pr\{\chi^2(r, \gamma) > \chi_\alpha^2(r)\}$ , the power of the test that knows  $\sigma^2$ .

## 4.5 The Cochran theorem

An analysis of variance starts from an identity

$$\mathbf{Y}^\top\mathbf{Y} = \mathbf{Y}^\top\mathbf{A}_1\mathbf{Y} + \cdots + \mathbf{Y}^\top\mathbf{A}_k\mathbf{Y}, \quad \mathbf{A}_1 + \cdots + \mathbf{A}_k = \mathbf{I},$$

that splits the total sum of squares into pieces attributed to sources of variation. [Theorem 4.4.4](#) says the pieces are independent chi-squared variables when the  $\mathbf{A}_i$  are mutually orthogonal projections. Checking that directly means checking  $k$  idempotence conditions and  $\binom{k}{2}$  orthogonality conditions. Cochran's theorem shows that one count of ranks does all of it.

### 4.5.1 The algebra

The matrix algebra behind this was done in [Chapter 1](#). For symmetric matrices, any two of the three properties “each  $\mathbf{A}_i$  is idempotent”, “ $\mathbf{A}_i\mathbf{A}_j = \mathbf{0}$  for  $i \neq j$ ” and “the sum is idempotent” imply the third, and symmetric matrices that add to  $\mathbf{I}$  with ranks adding to  $n$  are idempotent and mutually orthogonal ([Theorem 1.9.3](#)). The next theorem adds the rank count for a sum that is idempotent but not necessarily  $\mathbf{I}$ , which is what nested models need.

**Theorem 4.5.1 — Cochran, algebraic form**

Let  $\mathbf{A}_1, \dots, \mathbf{A}_k$  be symmetric  $n \times n$  matrices and  $\mathbf{A} = \sum_i \mathbf{A}_i$ . Consider

- (a) each  $\mathbf{A}_i$  is idempotent;
- (b)  $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$  for all  $i \neq j$ ;
- (c)  $\mathbf{A}$  is idempotent;
- (d)  $\text{rank}(\mathbf{A}) = \sum_i \text{rank}(\mathbf{A}_i)$ .

Then any two of (a), (b), (c) imply all four, and (c) together with (d) implies all four.

*Proof.* Any two of (a), (b), (c). By Theorem 1.9.3, any two of them imply the third. When all three hold, rank equals trace for symmetric idempotent matrices (Theorem 1.9.2), so  $\text{rank}(\mathbf{A}) = \text{tr}(\mathbf{A}) = \sum_i \text{tr}(\mathbf{A}_i) = \sum_i \text{rank}(\mathbf{A}_i)$ , which is (d).

(c) and (d) imply (a) and (b). Let  $r = \text{rank}(\mathbf{A})$  and  $r_i = \text{rank}(\mathbf{A}_i)$ . Every  $\mathbf{A}\mathbf{v} = \sum_i \mathbf{A}_i\mathbf{v}$  lies in  $\mathcal{C}(\mathbf{A}_1) + \dots + \mathcal{C}(\mathbf{A}_k)$ , whose dimension is at most  $\sum_i r_i = r = \dim \mathcal{C}(\mathbf{A})$ . So the two spaces coincide, and in particular  $\mathcal{C}(\mathbf{A}_i) \subseteq \mathcal{C}(\mathbf{A})$ . By Theorem 1.9.2(a),  $\mathbf{A} = \mathbf{Q}\mathbf{Q}^\top$  with  $\mathbf{Q}$  of size  $n \times r$  and  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_r$ . Since  $\mathbf{A}$  is symmetric and idempotent it fixes every vector of its column space, so  $\mathbf{A}\mathbf{A}_i = \mathbf{A}_i$ , and transposing,  $\mathbf{A}_i \mathbf{A} = \mathbf{A}_i$ . Hence  $\mathbf{A}_i = \mathbf{A}\mathbf{A}_i \mathbf{A} = \mathbf{Q}\mathbf{B}_i \mathbf{Q}^\top$  with  $\mathbf{B}_i = \mathbf{Q}^\top \mathbf{A}_i \mathbf{Q}$ . The  $r \times r$  matrices  $\mathbf{B}_i$  are symmetric, satisfy  $\sum_i \mathbf{B}_i = \mathbf{Q}^\top \mathbf{A} \mathbf{Q} = \mathbf{I}_r$ , and have  $\text{rank}(\mathbf{B}_i) = r_i$  because  $\mathbf{Q}$  has orthonormal columns. By the last statement of Theorem 1.9.3,  $\mathbf{B}_i^2 = \mathbf{B}_i$  and  $\mathbf{B}_i \mathbf{B}_j = \mathbf{0}$  for  $i \neq j$ , and multiplying by  $\mathbf{Q}$  on the left and  $\mathbf{Q}^\top$  on the right transfers both identities to the  $\mathbf{A}_i$ . ■

**4.5.2 The distributional statement****Theorem 4.5.2 — Cochran**

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ , and let  $\mathbf{A}_1, \dots, \mathbf{A}_k$  be symmetric with ranks  $r_1, \dots, r_k$ , whose sum  $\mathbf{A} = \sum_i \mathbf{A}_i$  is idempotent of rank  $r$  (for example  $\mathbf{A} = \mathbf{I}$ ,  $r = n$ ). The following are equivalent:

- (i) the forms  $\mathbf{Y}^\top \mathbf{A}_i \mathbf{Y} / \sigma^2$  are mutually independent, and  $\mathbf{Y}^\top \mathbf{A}_i \mathbf{Y} / \sigma^2 \sim \chi^2(r_i, \boldsymbol{\mu}^\top \mathbf{A}_i \boldsymbol{\mu} / \sigma^2)$  for each  $i$ ;
- (ii)  $r_1 + \dots + r_k = r$ ;
- (iii) each  $\mathbf{A}_i$  is idempotent;
- (iv)  $\mathbf{A}_i \mathbf{A}_j = \mathbf{0}$  for all  $i \neq j$ .

*Proof.* Condition (c) of Theorem 4.5.1 holds by assumption, so that theorem shows that (ii), (iii) and (iv) are equivalent, and each implies the other two. If (iii) and (iv) hold, Theorem 4.4.4 gives (i). If (i) holds, apply Theorem 4.3.1 to  $\sigma^{-1} \mathbf{Y} \sim N_n(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$ . Each  $\mathbf{A}_i$  with  $r_i \geq 1$  is idempotent, and an  $\mathbf{A}_i$  of rank zero is  $\mathbf{0}$ . This is (iii). ■

In words: *if the degrees of freedom add up, the sums of squares are independent chi-squared variables.* Independence in (i) is not needed to reach (iii). The marginal chi-squared laws alone force the whole structure.

**Example. (The one-way layout)**

Let observations fall into  $g$  groups of sizes  $n_1, \dots, n_g$ , with  $Y_{\ell j}$  independent  $N(\mu_\ell, \sigma^2)$ . Let  $\mathbf{M}$  replace each observation by its group mean, and  $\mathbf{P}_1 = n^{-1}\mathbf{1}\mathbf{1}^\top$  by the grand mean. Then

$$\mathbf{I} = \mathbf{P}_1 + (\mathbf{M} - \mathbf{P}_1) + (\mathbf{I} - \mathbf{M})$$

splits  $\|\mathbf{Y}\|^2$  into  $n\bar{Y}^2$ , the between-group sum of squares  $\sum_\ell n_\ell(\bar{Y}_\ell - \bar{Y})^2$  and the within-group sum of squares  $\sum_{\ell,j} (Y_{\ell j} - \bar{Y}_\ell)^2$ . The three matrices are symmetric with ranks 1,  $g - 1$  and  $n - g$ , which add to  $n$ . Cochran's theorem needs nothing more. The three sums of squares, divided by  $\sigma^2$ , are independent noncentral chi-squared variables. Their noncentralities are  $n\bar{\mu}^2/\sigma^2$ ,  $\sum_\ell n_\ell(\mu_\ell - \bar{\mu})^2/\sigma^2$  with  $\bar{\mu} = \sum_\ell n_\ell\mu_\ell/n$ , and 0. The last is zero because  $\mathbf{M}\boldsymbol{\mu} = \boldsymbol{\mu}$ . The ratio of the between and within mean squares is  $F(g - 1, n - g, \gamma)$  by [Theorem 4.4.6](#), and it is central iff all group means are equal.

The script uses  $g = 3$  groups of sizes 3, 5, 4, means 2, 3, 1.5, and  $\sigma = 1$ , and draws 200,000 data sets. [Table 4.5.1](#) compares the theoretical means  $r_i + \gamma_i$  and variances  $2r_i + 4\gamma_i$  with the simulated ones. The largest absolute correlation between the three sums of squares is 0.0013. As a check that goes beyond correlation, the proportion of data sets in which both the between and the within sums of squares exceed their medians is 0.2497, against 0.25 for independent variables.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(5150)

sizes, means, sigma = [3, 5, 4], [2.0, 3.0, 1.5], 1.0
groups = np.repeat(np.arange(3), sizes)
n = len(groups)
Z = (groups[:, None] == np.arange(3)).astype(float) # group indicators
M = Z @ np.diag(1 / np.array(sizes)) @ Z.T # group-mean projection
J = np.ones((n, n)) / n
pieces = [J, M - J, np.eye(n) - M]
ranks = [np.linalg.matrix_rank(P) for P in pieces]
print("ranks", ranks, "sum", sum(ranks), "n", n)

mu = np.array(means)[groups]
gammas = [mu @ P @ mu / sigma**2 for P in pieces]
Y = mu + sigma * rng.standard_normal((200_000, n))
Q = np.stack([np.einsum("ij,jk,ik->i", Y, P, Y) for P in pieces], axis=1) / sigma**2
print(np.corrcoef(Q, rowvar=False).round(3))
```

**Table 4.5.1.** Cochran decomposition for a one-way layout with  $n = 12$ : theory against 200,000 simulated data sets.

| Sum of<br>squares | $r_i$ | $\gamma_i$ | mean   | simulated | variance | simulated |
|-------------------|-------|------------|--------|-----------|----------|-----------|
| grand<br>mean     | 1     | 60.750     | 61.750 | 61.763    | 245.00   | 244.95    |

| Sum of squares | $r_i$ | $\gamma_i$ | mean  | simulated | variance | simulated |
|----------------|-------|------------|-------|-----------|----------|-----------|
| between groups | 2     | 5.250      | 7.250 | 7.230     | 25.00    | 24.85     |
| within groups  | 9     | 0.000      | 9.000 | 8.994     | 18.00    | 17.96     |

*Remark* (What the theorem does not cover). Two assumptions carry the weight. The first is normality. Without it, [Theorem 4.2.3\(d\)](#) fails, and with independent non-normal errors the between and within sums of squares are in general correlated, with a correlation driven by the fourth cumulant ([Exercise \(B2\)](#)). The second is the spherical covariance  $\sigma^2 \mathbf{I}$ . With  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$ ,  $\mathbf{V}$  positive definite, apply the theorem to  $\mathbf{V}^{-1/2} \mathbf{Y}$ , whose forms have matrices  $\mathbf{V}^{1/2} \mathbf{A}_i \mathbf{V}^{1/2}$ . Their ranks are those of the  $\mathbf{A}_i$ , so the rank count is unchanged. What changes is the condition on the sum:  $\mathbf{V}^{1/2} \mathbf{A} \mathbf{V}^{1/2}$  is idempotent iff  $\mathbf{A} \mathbf{V} \mathbf{A} = \mathbf{A}$  (for example  $\sum_i \mathbf{A}_i = \mathbf{V}^{-1}$ ), and conditions (iii) and (iv) become  $\mathbf{A}_i \mathbf{V} \mathbf{A}_i = \mathbf{A}_i$  and  $\mathbf{A}_i \mathbf{V} \mathbf{A}_j = \mathbf{0}$ . These are the generalized least squares decompositions of [Chapter 31](#) ([Theorem 31.1.1](#)).

Most analysis of variance tables in this book come from a chain of nested model spaces. The successive differences of their projections are mutually orthogonal projections whose ranks add up, so Cochran's theorem applies without further work ([Exercise \(B2\)](#), [Theorem 9.1.1](#)).

### 4.5.3 Exercises

#### 4.5.3.1 A. Check your understanding

##### Exercise A1.

Let  $\mathbf{A}_1 = \text{diag}(2, 0)$  and  $\mathbf{A}_2 = \text{diag}(-1, 1)$ , so that  $\mathbf{A}_1 + \mathbf{A}_2 = \mathbf{I}_2$ . Which hypotheses of [Theorem 4.5.2](#) fail, and what are the laws of the two forms when  $\mathbf{Y} \sim N_2(\mathbf{0}, \mathbf{I})$ ?

*Solution.* The sum  $\mathbf{I}_2$  is idempotent, but every one of the conditions (ii)–(iv) fails. The ranks are 1 and 2, which add to  $3 \neq 2$ . Neither matrix is idempotent, since  $\mathbf{A}_1^2 = \text{diag}(4, 0)$  and  $\mathbf{A}_2^2 = \mathbf{I}_2$ . And  $\mathbf{A}_1 \mathbf{A}_2 = \text{diag}(-2, 0) \neq \mathbf{0}$ . The forms are  $2Y_1^2$ , which is twice a  $\chi^2(1)$  variable, and  $Y_2^2 - Y_1^2$ , the difference of two independent  $\chi^2(1)$  variables, which takes negative values. Neither is chi-squared. They are dependent, because

$$\text{Cov}(2Y_1^2, Y_2^2 - Y_1^2) = 2 \text{Cov}(Y_1^2, Y_2^2) - 2 \text{Var}(Y_1^2) = 0 - 2 \cdot 2 = -4 \neq 0.$$

#### 4.5.3.2 B. Practice

##### Exercise B1.

A two-way layout has one observation  $Y_{ij}$  for each of  $a$  rows and  $b$  columns. Write  $\|\mathbf{Y}\|^2$  as the sum of the grand-mean, row, column and residual sums of squares, identify the four matrices, and use [Theorem 4.5.2](#) to show that under  $Y_{ij} \sim N(\mu + \alpha_i + \beta_j, \sigma^2)$



independently the four sums of squares are independent, with degrees of freedom 1,  $a - 1$ ,  $b - 1$  and  $(a - 1)(b - 1)$ .

### Exercise B2.

Let  $\mathbf{P}_1, \dots, \mathbf{P}_m$  be symmetric idempotent with  $\mathcal{C}(\mathbf{P}_1) \subset \mathcal{C}(\mathbf{P}_2) \subset \dots \subset \mathcal{C}(\mathbf{P}_m)$ , and put  $\mathbf{P}_0 = \mathbf{0}$ ,  $\mathbf{P}_{m+1} = \mathbf{I}$ . Show that the matrices  $\mathbf{P}_i - \mathbf{P}_{i-1}$ ,  $i = 1, \dots, m+1$ , satisfy the conditions of [Theorem 4.5.2](#), so that the sequential sums of squares  $\|(\mathbf{P}_i - \mathbf{P}_{i-1})\mathbf{Y}\|^2$  are independent noncentral chi-squared variables when  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ .

### 4.5.3.3 C. Going deeper

#### Exercise C1.

Let  $\mathbf{Y} \sim N_n(\mathbf{0}, \mathbf{I})$ , and  $Q = Q_1 + Q_2$  with  $Q = \mathbf{Y}^\top \mathbf{A} \mathbf{Y} \sim \chi^2(r)$ ,  $Q_1 = \mathbf{Y}^\top \mathbf{A}_1 \mathbf{Y} \sim \chi^2(r_1)$  and  $Q_2 = \mathbf{Y}^\top \mathbf{A}_2 \mathbf{Y}$  for a nonnegative definite  $\mathbf{A}_2$ . Show that  $Q_1$  and  $Q_2$  are independent and  $Q_2 \sim \chi^2(r - r_1)$ .

*Solution.* By [Theorem 4.3.1](#),  $\mathbf{A}$  and  $\mathbf{A}_1$  are idempotent, with ranks  $r$  and  $r_1$ . If  $\mathbf{A}\mathbf{v} = \mathbf{0}$ , then  $0 = \mathbf{v}^\top \mathbf{A}_1 \mathbf{v} + \mathbf{v}^\top \mathbf{A}_2 \mathbf{v}$  with both terms nonnegative, so  $\|\mathbf{A}_1 \mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}_1 \mathbf{v} = 0$ . Thus  $\mathcal{N}(\mathbf{A}) \subseteq \mathcal{N}(\mathbf{A}_1)$ , hence  $\mathcal{C}(\mathbf{A}_1) \subseteq \mathcal{C}(\mathbf{A})$  and  $\mathbf{A}\mathbf{A}_1 = \mathbf{A}_1 = \mathbf{A}_1 \mathbf{A}$ . Then  $\mathbf{A}_2^2 = (\mathbf{A} - \mathbf{A}_1)^2 = \mathbf{A} - 2\mathbf{A}_1 + \mathbf{A}_1 = \mathbf{A}_2$ , so  $\mathbf{A}_2$  is idempotent, and  $\mathbf{A}_1 \mathbf{A}_2 = \mathbf{A}_1 \mathbf{A} - \mathbf{A}_1 = \mathbf{0}$ . By [Theorem 4.4.4](#),  $Q_1$  and  $Q_2$  are independent and  $Q_2 \sim \chi^2(\text{tr } \mathbf{A}_2) = \chi^2(r - r_1)$ .

## 4.6 Summary and Notes

### 4.6.1 Summary

*Idea.*

1. If  $\mathbf{Z} \sim N_r(\boldsymbol{\mu}, \mathbf{I})$  then  $\|\mathbf{Z}\|^2 \sim \chi^2(r, \gamma)$  with  $\gamma = \|\boldsymbol{\mu}\|^2$ , mean  $r + \gamma$  and variance  $2r + 4\gamma$ . The law is a  $\text{Poisson}(\gamma/2)$  mixture of  $\chi^2(r + 2k)$  laws, and its tails increase strictly with  $\gamma$  ([Theorem 4.1.2](#) and [Proposition 4.1.3](#)).
2.  $F(r, s, \gamma)$  and  $t(s, \delta)$  put a noncentral numerator over an independent central denominator. The power of an  $F$  test increases strictly with  $\gamma$ , and  $T^2 \sim F(1, s, \delta^2)$  ([Theorem 4.1.4](#)).
3.  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A}\boldsymbol{\Sigma}) + \boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$  for any distribution. Under normality

$$\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2 \text{tr}\{(\mathbf{A}\boldsymbol{\Sigma})^2\} + 4\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}, \quad \text{Cov}(\mathbf{K} \mathbf{Y}, \mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = 2\mathbf{K} \boldsymbol{\Sigma} \mathbf{A} \boldsymbol{\mu}$$

([Theorem 4.2.3](#)).

4. A normal quadratic form is a constant plus independent terms  $\lambda_j(W_j + \nu_j)^2$ , and for singular  $\boldsymbol{\Sigma}$  possibly normal terms. The weights  $\lambda_j$  are the nonzero eigenvalues of  $\mathbf{A}\boldsymbol{\Sigma}$  ([Theorem 4.2.1](#)).
5. For  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \mathbf{I})$ ,  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  is chi-squared iff  $\mathbf{A}$  is idempotent. Then the degrees of freedom are  $\text{rank } \mathbf{A} = \text{tr } \mathbf{A}$  and the noncentrality is  $\boldsymbol{\mu}^\top \mathbf{A} \boldsymbol{\mu}$  ([Theorem 4.3.1](#)).

For positive definite  $\Sigma$  the condition is that  $\mathbf{A}\Sigma$  be idempotent. For projections,  $\|\mathbf{P}\mathbf{Y}\|^2/\sigma^2 \sim \chi^2(\text{rank } \mathbf{P}, \|\mathbf{P}\boldsymbol{\mu}\|^2/\sigma^2)$  (Theorem 4.3.2).

6.  $\mathbf{B}\mathbf{Y}$  and  $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$  are independent if  $\mathbf{B}\Sigma\mathbf{A} = \mathbf{0}$ , and  $\mathbf{Y}^\top \mathbf{A}\mathbf{Y}$  and  $\mathbf{Y}^\top \mathbf{B}\mathbf{Y}$  are independent if  $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$ , because a quadratic form is a function of  $\mathbf{A}\mathbf{Y}$  (Theorem 4.4.1 and Theorem 4.4.2).
7. Projections of a spherical normal vector onto mutually orthogonal subspaces are independent. This gives the independence of  $\bar{Y}$  and  $S^2$  and the exact  $F(r - r_0, n - r, \gamma)$  law of the nested-model  $F$  statistic (Theorem 4.4.4, Corollary 4.4.5 and Theorem 4.4.6).
8. Cochran: if symmetric matrices add to an idempotent matrix and their ranks add up, the corresponding sums of squares are independent chi-squared variables (Theorem 4.5.1 and Theorem 4.5.2).

## 4.6.2 Notes and sources

**Source texts.** Rencher and Schaalje (2008, chapter 5) cover the material of Section 4.2, Section 4.3, Section 4.4 and Section 4.5 for positive definite covariance matrices. They derive the mgf of a quadratic form by integrating against the normal density and obtain its variance by differentiating the logarithm of the mgf; the converse of the chi-squared theorem is referred to the literature. They use the noncentrality  $\boldsymbol{\mu}^\top \boldsymbol{\mu}/2$ . Seber and Lee (2003, section 2.4) reduce the general case to a spherical normal vector by an orthogonal diagonalization. They prove necessity of idempotence for central forms by factorizing the moment generating function, and give the eigenvalue criterion for singular covariance matrices. Christensen (2020, section 1.3) states the three conditions of Theorem 4.3.2(c) for arbitrary nonnegative definite  $\Sigma$ , and obtains independence by viewing quadratic forms as functions of linear ones, the approach followed in Section 4.4. The canonical representation (Theorem 4.2.1), the proof of necessity through the coefficients Eq. 4.3.1, the monotonicity proof through the Poisson mixture, and the treatment of the Cochran algebra are this book's own arrangements of standard material.

**Noncentral distributions.** The noncentral chi-squared distribution first appeared in Fisher (1928). Patnaik (1949) studied its approximation by a scaled central chi-squared variable, and Satterthwaite (1946) used the same two-moment idea for linear combinations of mean squares, which reappears in the mixed models of Chapter 32 (Proposition 32.5.1). Johnson et al. (1995, chapters 29–31) collect the properties of the noncentral chi-squared,  $F$  and  $t$  families. The monotonicity of power in the noncentrality and in the degrees of freedom is treated systematically by Ghosh (1973).

**General quadratic forms.** The exact distribution of a quadratic form that is not chi-squared can be computed by numerical inversion of its characteristic function (Gil-Pelaez 1951; Imhof 1961). Mathai and Provost (1992) give a book-length treatment of quadratic forms in normal and non-normal variables. The mean square successive difference of Example (Two estimators of  $\sigma^2$  under a drifting mean) goes back to von Neumann (1941), who derived the exact distribution of its ratio to  $S^2$ .

**Independence and Cochran's theorem.** Cochran (1934) proved the decomposition theorem that bears his name, in the context of the analysis of covariance. The necessity of  $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$  for independence of two forms is attributed to Craig (1943). Its full proof for arbitrary symmetric matrices and arbitrary means turned out to be subtle, and Driscoll and Gundberg (1986) trace the flawed and correct arguments. Searle (1971) gives general proofs of the independence and Cochran theorems in the style most linear-models texts follow. For nonnegative definite  $\mathbf{A}$  and

$\mathbf{B}$ , the condition  $\Sigma\mathbf{A}\Sigma\mathbf{B}\Sigma = \mathbf{0}$  of Christensen (2020, Theorem 1.3.8) is in fact equivalent to  $\mathbf{A}\Sigma\mathbf{B} = \mathbf{0}$  (Exercise (B2)), so that theorem adds nothing to Theorem 4.4.2 even for singular  $\Sigma$ , whereas his Theorem 1.3.9, for general symmetric  $\mathbf{A}$  and  $\mathbf{B}$ , does. That the independence of  $\bar{Y}$  and  $S^2$  characterizes the normal distribution was shown by Lukacs (1942).

### 4.6.3 References

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# Chapter 5

## The Model, Least Squares and the Normal Equations

A regression model is a claim about how the distribution of a response changes with a set of regressors, and the linear model is the simplest useful claim of that kind. It says that the mean of the response is a linear combination of known columns, that the errors around this mean have a common variance and are uncorrelated, and nothing more. This chapter sets the model up and draws out what follows from those assumptions alone. It starts with what the assumptions do and do not say, fits a straight line by calculus, and passes to the matrix form of the model, where polynomials, harmonics, indicator variables and interactions all turn out to be linear models. It then derives the least squares estimator and the normal equations, computes the mean and covariance of the estimator, estimates the error variance without bias, and asks what a fitted coefficient means. The last section surveys the models the rest of the book builds on this one. Throughout, the model matrix has full column rank. [Chapter 6](#) retells the same story geometrically and removes that condition.

**What you need.** [Chapter 1](#): rank ([Proposition 1.2.3](#)), inverses, nonnegative definite matrices ([Proposition 1.7.5](#)), traces ([Theorem 1.6.1](#)) and the derivatives of linear and quadratic forms ([Theorem 1.11.1](#)). [Chapter 2](#): means and covariances of linear transformations ([Theorem 2.2.1](#)), the mean and variance of a quadratic form ([Theorem 2.3.1](#), [Theorem 2.3.3](#)) and the laws of total expectation and covariance ([Proposition 2.5.4](#)). Normality ([Chapter 3](#)) appears only in remarks.

### 5.0.1 Roadmap

- [What a regression model claims](#)
- [Simple linear regression as a warm-up](#)
- [The multiple linear regression model](#)
- [Least squares and the normal equations](#)
- [Properties of the estimator under second-moment assumptions](#)
- [Estimating the error variance](#)
- [Interpreting coefficients](#)
- [A panorama of regression models to come](#)
- [Summary and notes](#)

## 5.1 What a regression model claims

A data set for regression consists of  $n$  cases. Each case records a response  $y_i$  and a few other measurements, the regressors, on the same unit, such as a patient or a month. The question is how the response depends on the regressors. Two cases with identical regressors almost always have different responses, so a regression model cannot be an equation linking  $y$  to the regressors. It is a statement about the *distribution* of the response given the regressors, and usually only about one feature of that distribution, its mean.

This section separates what is automatically true of any response and regressors from what a model actually assumes. The linear model's assumptions come in layers, and each layer buys a different part of the theory.

### 5.1.1 The regression function

Let  $Y$  be a response with finite variance and  $\mathbf{X}$  a random vector of regressors, defined on the same probability space. We need only elementary properties of conditional expectation: the tower property  $E[E(Y | \mathbf{X})] = E(Y)$ , and  $E[g(\mathbf{X})Y | \mathbf{X}] = g(\mathbf{X})E(Y | \mathbf{X})$  for functions  $g$  of the regressors.

#### Definition 5.1.1 — Regression function

The **regression function** of  $Y$  on  $\mathbf{X}$  is the conditional mean

$$m(\mathbf{x}) = E(Y | \mathbf{X} = \mathbf{x}).$$

The **error** is  $\varepsilon = Y - m(\mathbf{X})$ , so that  $Y = m(\mathbf{X}) + \varepsilon$ .

The decomposition  $Y = m(\mathbf{X}) + \varepsilon$  looks like a model, but it is not one. It holds for every pair  $(\mathbf{X}, Y)$  with  $EY^2 < \infty$ , and the error it defines has two properties that are often listed as “assumptions” even though they are automatic.

#### Proposition 5.1.1 — The error of the regression function

Let  $EY^2 < \infty$ , and let  $m$  and  $\varepsilon$  be as in Definition 5.1.1.

- (a)  $E(\varepsilon | \mathbf{X}) = 0$ , and hence  $E(\varepsilon) = 0$ .
- (b)  $E[\varepsilon g(\mathbf{X})] = 0$  for every function  $g$  with  $Eg(\mathbf{X})^2 < \infty$ . In particular  $\varepsilon$  is uncorrelated with every such function of the regressors.
- (c) For every such  $g$ ,

$$E[(Y - g(\mathbf{X}))^2] = E(\varepsilon^2) + E[(m(\mathbf{X}) - g(\mathbf{X}))^2].$$

So  $m(\mathbf{X})$  is the best predictor of  $Y$  from  $\mathbf{X}$  in mean squared error, and it is the only one up to events of probability zero.

*Proof.* (a)  $E(\varepsilon | \mathbf{X}) = E(Y | \mathbf{X}) - m(\mathbf{X}) = 0$ , and the tower property gives  $E(\varepsilon) = E[E(\varepsilon | \mathbf{X})] = 0$ . (b) The product  $\varepsilon g(\mathbf{X})$  is integrable by the Cauchy–Schwarz inequality,

because  $E\varepsilon^2 \leq 2EY^2 + 2Em(\mathbf{X})^2$  and  $Em(\mathbf{X})^2 \leq EY^2$  by Jensen's inequality for conditional expectations. Then  $E[\varepsilon g(\mathbf{X})] = E[g(\mathbf{X}) E(\varepsilon | \mathbf{X})] = 0$ . (c) Write  $Y - g(\mathbf{X}) = \varepsilon + (m(\mathbf{X}) - g(\mathbf{X}))$  and expand the square. The cross term is  $2E[\varepsilon(m(\mathbf{X}) - g(\mathbf{X}))]$ , which is zero by (b) applied to the function  $m - g$ . The second term on the right is zero only if  $g(\mathbf{X}) = m(\mathbf{X})$  with probability one. ■

Part (c) is the reason the conditional mean is the natural target. It is what an ideal forecaster would report under squared-error loss. It is also the reason regression is sometimes misread. The function  $m$  summarizes the conditional distribution by its centre. Two populations can have the same regression function and very different spreads, shapes and tails around it.

*Warning.* Part (b) says the error is uncorrelated with the regressors *by construction*. The lack of correlation between errors and regressors therefore cannot be tested from the data or defended as a scientific assumption about the regression function. The real assumption hides elsewhere: in the claim that  $m$  has a particular form, or that the quantity of interest is  $m$  at all. When someone says “the error is correlated with the regressor”, the error they have in mind is a different object, usually the deviation from a *causal* or *structural* relation rather than from the conditional mean. [Chapter 24](#) and [Chapter 25](#) take this up.

## 5.1.2 What the linear model adds

A linear regression model makes claims about  $m$  and about the errors that  $EY^2 < \infty$  does not guarantee. Write  $\mathbf{x}_{(i)}$  for the regressors of case  $i$ , already transformed into the columns the model uses (a constant 1, the regressors themselves, their squares, indicators of categories, and so on; [Section 5.3](#) gives many examples). The assumptions come in four layers.

- (L1) **Linearity.**  $E(Y_i) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  for an unknown  $\boldsymbol{\beta} \in \mathbb{R}^p$ . The regression function belongs to a fixed  $p$ -dimensional family of functions.
- (L2) **Constant variance.**  $\text{Var}(Y_i) = \sigma^2$  for every  $i$ , with  $\sigma^2$  unknown.
- (L3) **Uncorrelated errors.**  $\text{Cov}(Y_i, Y_j) = 0$  for  $i \neq j$ .
- (L4) **Normality.** The errors are jointly normal.

The first three layers are the **second-moment assumptions**. They mention only means, variances and covariances, and they are all that [Section 5.5](#) and [Section 5.6](#) use. Adding (L4) gives the normal linear model, and with it exact sampling distributions for estimates and test statistics, which [Chapter 7](#) and [Chapter 11](#) develop. Each layer is used by a specific part of the theory:

| Assumptions | What they deliver                                                                      | Where                                                                                    |
|-------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| (L1)        | the least squares estimator is unbiased                                                | <a href="#">Section 5.5</a>                                                              |
| (L1)–(L3)   | its covariance matrix; an unbiased estimate of $\sigma^2$ ;<br>Gauss–Markov optimality | <a href="#">Section 5.5</a> , <a href="#">Section 5.6</a> ,<br><a href="#">Chapter 7</a> |

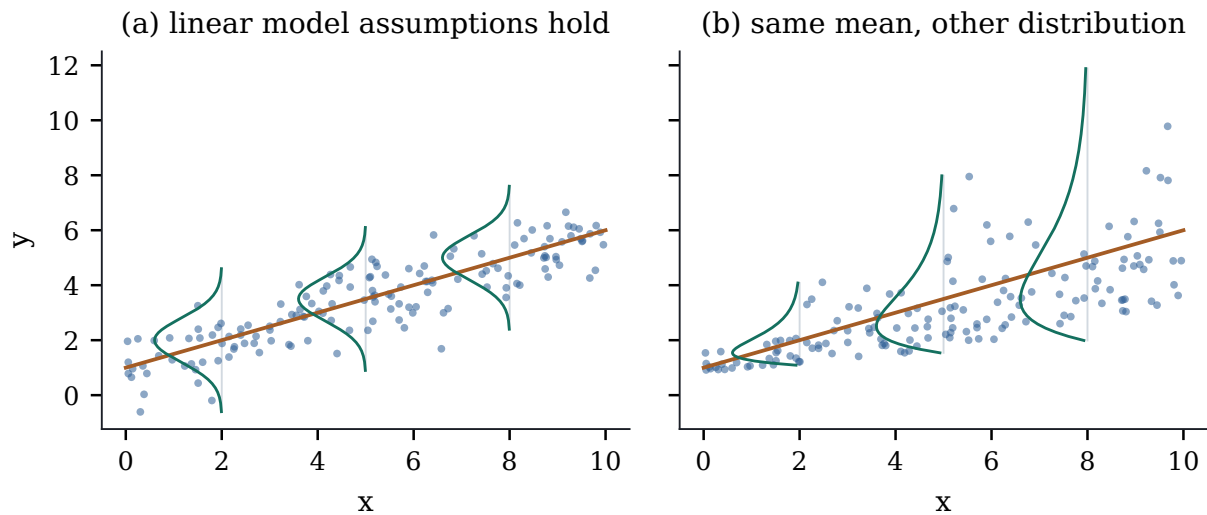


Figure 5.1.1: Two data-generating processes with the same regression function  $1 + 0.5x$  (brown line). The curves are conditional densities of  $Y$  given  $x$ , drawn sideways at  $x = 2, 5, 8$ . (a) Normal errors with constant variance: all four layers of assumptions hold. (b) Right-skewed errors whose spread grows with  $x$ : only linearity holds.

| Assumptions | What they deliver                                                          | Where                                                                                  |
|-------------|----------------------------------------------------------------------------|----------------------------------------------------------------------------------------|
| (L1)–(L4)   | exact $t$ and $F$ distributions;<br>maximum likelihood; exact<br>intervals | <a href="#">Chapter 7</a> , <a href="#">Chapter 11</a> ,<br><a href="#">Chapter 12</a> |

Least squares itself needs none of them. The estimator of [Section 5.4](#) is defined for any data at all. The assumptions only tell us how good it is.

**Example. (Same regression function, different models)**

[Figure 5.1.1](#) shows two simulated data sets with the same regression function  $m(x) = 1 + 0.5x$ . In panel (a) the errors are normal with constant standard deviation 0.8. In panel (b) they are skewed to the right, and their standard deviation grows linearly in  $x$ , from 0.15 at  $x = 0$  to 2.65 at  $x = 10$ . The curves show the conditional density of  $Y$  at three values of  $x$ . Both processes satisfy (L1), so least squares estimates the line without bias in both. Only process (a) satisfies (L2), so only there do the variance formulas of [Section 5.5](#) apply as stated. In process (b) most points lie below the line, because the median of a right-skewed distribution is below its mean. The line is still the conditional mean.

The assumptions concern the errors  $\varepsilon_i$ , which are never observed. What we see are residuals, the differences between the responses and a fitted function. Whether the assumptions are credible is judged from residuals, from the design of the study, and from knowledge of the subject. [Part V](#) of the book is devoted to the first of these and to what goes wrong when the assumptions fail.



### 5.1.3 Fixed and random regressors

The layers (L1)–(L4) were written with  $E(Y_i)$  and  $\text{Var}(Y_i)$ , as if the regressors were constants. In a designed experiment they are. The experimenter chooses the temperatures, doses or fertilizer levels, and only the responses are random. In an observational study the regressors are measured on units drawn from a population, so the pairs  $(\mathbf{x}_{(i)}, Y_i)$  are random together.

This book's convention, like that of most linear-model theory, is to treat the regressors as fixed, and to read (L1)–(L4) in the random case as statements *conditional on the regressors*:  $E(Y_i | \mathbf{X}) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ ,  $\text{Var}(Y_i | \mathbf{X}) = \sigma^2$ , and so on, where  $\mathbf{X}$  collects all the regressors. There are two reasons for this choice.

First, conditional statements imply unconditional ones where it matters. If the estimator is unbiased given every possible  $\mathbf{X}$ , it is unbiased when  $\mathbf{X}$  is random, by the tower property. [Section 5.5](#) makes this precise ([Corollary 5.5.5](#)). Second, conditioning describes the precision of the data actually in hand. A sample whose regressors happen to be spread out estimates a slope better than one whose regressors happen to be bunched together. The conditional variance says so, and the unconditional variance averages the two cases.

#### Example. (One slope, two sampling schemes)

Take  $Y = 1 + 0.5x + \varepsilon$  with standard normal errors and  $n = 15$  cases. In the *fixed design*, the regressor values are drawn once from the uniform distribution on  $[0, 10]$  and then kept. They have  $S_{xx} = \sum_i (x_i - \bar{x})^2 = 103.73$ . In the *random design*, a new set of 15 regressor values is drawn for every sample. Over 50,000 simulated samples, the least squares slope (formula in [Section 5.2](#)) averages 0.5007 under the fixed design and 0.5006 under the random one. Both are unbiased. Their variances differ in meaning. Under the fixed design the simulated variance is 0.00960, matching  $\sigma^2/S_{xx} = 0.00964$  for this particular design. Under the random design it is 0.00933, matching the *average*  $E(\sigma^2/S_{xx}) = 0.00923$  over designs, estimated from the same simulation. Replacing  $S_{xx}$  by its expectation 116.67 would give 0.00857, which is too small. By Jensen's inequality  $E(1/S_{xx}) \geq 1/E(S_{xx})$ , so random bunching of the regressors costs precision on average.

```
import numpy as np

rng = np.random.default_rng(2005)
n, beta0, beta1, sigma = 15, 1.0, 0.5, 1.0
reps = 50_000

def slope(x, y):
    """Least squares slope of y on x (with an intercept)."""
    xc = x - x.mean()
    return xc @ (y - y.mean()) / (xc @ xc)

x_fixed = rng.uniform(0, 10, size=n)          # drawn once, then held fixed
Sxx_fixed = np.sum((x_fixed - x_fixed.mean()) ** 2)
b_fixed = np.empty(reps)
b_random = np.empty(reps)
Sxx_random = np.empty(reps)
for r in range(reps):
    e = sigma * rng.standard_normal(n)
    b_fixed[r] = slope(x_fixed, beta0 + beta1 * x_fixed + e)
    x = rng.uniform(0, 10, size=n)            # a new design for every sample
```



```

Sxx_random[r] = np.sum((x - x.mean()) ** 2)
b_random[r] = slope(x, beta0 + beta1 * x + e)

print(f"fixed design: mean {b_fixed.mean():.4f} var {b_fixed.var():.5f}"
      f" sigma^2/Sxx = {sigma**2 / Sxx_fixed:.5f}")
print(f"random design: mean {b_random.mean():.4f} var {b_random.var():.5f}"
      f" E(sigma^2/Sxx) = {np.mean(sigma**2 / Sxx_random):.5f}")

```

Conditioning on the regressors is not always harmless. It fails when the regressors are measured with error, because then the regression on the *measured* values is not the one of scientific interest (Chapter 24, Theorem 24.1.1). It fails when a regressor is a lagged value of the response, as in autoregressive time series, because then  $\mathbf{X}$  is not determined before the errors are drawn. It also changes the question when the target is a population quantity, such as the best linear predictor of one variable from others in a population where the regression function is not linear (Section 6.11 and Chapter 14).

### 5.1.4 What regression is used for

The same fitted equation can serve different purposes, and the purpose decides which assumptions matter. For **prediction** of new cases only the accuracy of the fitted mean matters, and a regressor can be useful without any causal role. For **description** the fitted mean summarizes how the response varies across the population sampled. For **estimation of a parameter**, such as an elasticity or a dose-response slope, one coefficient and its uncertainty are the point. For **control**, regressor values are chosen to achieve a desired response, which requires the fitted relation to hold when the regressors are *set* rather than observed: a causal claim.

*Warning.* The regression function describes how the mean response differs between cases that *are observed* to have different regressor values. It does not by itself describe what would happen if a regressor were *changed*. Units with high values of one regressor usually differ in other ways too. Randomized assignment of the regressor, as in a designed experiment, is the standard way to make the two questions coincide. Without it, the step from association to effect needs assumptions that the data cannot check. Section 5.7 returns to this point, and Chapter 25 treats it in full.

### 5.1.5 Exercises

#### 5.1.5.1 A. Check your understanding

##### Exercise A1.

Let  $X$  take the values  $-1, 0, 1$  with probability  $1/3$  each, and let  $Y = X^2 + XZ$ , where  $Z \sim N(0, 1)$  is independent of  $X$ . Find the regression function  $m$  and the error  $\varepsilon$ . Show that  $E(\varepsilon | X) = 0$  and  $\text{Cov}(\varepsilon, X) = 0$ , but that  $\varepsilon$  and  $X$  are not independent. Which of the layers (L1)–(L4) fail for a sample from this distribution, with the columns  $1, x, x^2$ ?

*Solution.* Since  $E(XZ | X) = X E(Z) = 0$ ,  $m(x) = x^2$  and  $\varepsilon = XZ$ . Then  $E(\varepsilon | X) = 0$ , and  $\text{Cov}(\varepsilon, X) = E(X^2Z) = E(X^2)E(Z) = 0$ . But  $\text{Var}(\varepsilon | X) = X^2$ , which is 0 when  $X = 0$  and 1 otherwise. The conditional distribution of  $\varepsilon$  depends on  $X$ , so they are not independent. With columns  $1, x, x^2$ , (L1) holds with  $\beta = (0, 0, 1)^\top$  and (L3) holds

for independent cases. (L2) fails, and so does (L4) as stated, because the error of a case with  $x = 0$  is identically zero and the joint distribution is not normal with covariance  $\sigma^2 \mathbf{I}$ .

### Exercise A2.

Let  $(X, Y)$  be bivariate normal with means  $\mu_X, \mu_Y$ , standard deviations  $\sigma_X, \sigma_Y > 0$  and correlation  $\rho$ . Use [Theorem 3.5.2](#) to find  $m(x)$  and  $\text{Var}(Y \mid X = x)$ . Conclude that a random sample from this distribution satisfies (L1)–(L4) conditionally on the  $x_i$ , with columns 1,  $x$ . What are  $\beta$  and  $\sigma^2$ ?

### 5.1.5.2 B. Practice

#### Exercise B1.

Under the conditions of [Proposition 5.1.1](#), show that  $\text{Var}(Y) = \text{Var}(m(\mathbf{X})) + \text{E}[\text{Var}(Y \mid \mathbf{X})]$  and that  $\text{E}(\varepsilon^2) = \text{E}[\text{Var}(Y \mid \mathbf{X})]$ . The ratio  $\eta^2 = \text{Var}(m(\mathbf{X})) / \text{Var}(Y)$  is the proportion of the variance of  $Y$  that the regression function explains. Compute it for the distribution of [Exercise \(A1\)](#).

*Solution.* The first identity is the scalar case of [Proposition 2.5.4](#). For the second,  $\text{E}(\varepsilon^2 \mid \mathbf{X}) = \text{E}[(Y - m(\mathbf{X}))^2 \mid \mathbf{X}] = \text{Var}(Y \mid \mathbf{X})$ ; take expectations. In [Exercise \(A1\)](#),  $m(X) = X^2$  takes the value 1 with probability 2/3 and 0 with probability 1/3, so  $\text{Var}(m(X)) = 2/9$ . Also  $\text{E}[\text{Var}(Y \mid X)] = \text{E}(X^2) = 2/3$ . Hence  $\text{Var}(Y) = 8/9$  and  $\eta^2 = 1/4$ .

#### Exercise B2.

In [Example \(One slope, two sampling schemes\)](#) the regressors are independent and uniform on  $[0, 10]$ . Show that  $\text{E}(S_{xx}) = (n-1) \cdot 100/12$ . Use Jensen's inequality to show that  $\text{E}(\sigma^2/S_{xx}) \geq \sigma^2/\text{E}(S_{xx})$ , with strict inequality here. Why is the right side of no use for describing the precision of the slope from a given sample?

### 5.1.5.3 C. Going deeper

#### Exercise C1.

Let  $X \sim N(0, 1)$  and  $Y = X^2$ . Find the best linear predictor  $\beta_0 + \beta_1 X$  of  $Y$  ([Theorem 2.5.2](#)) and its error  $u$ . Show that  $\text{E}(u) = 0$  and  $\text{Cov}(u, X) = 0$ , but  $\text{E}(u \mid X) \neq 0$ . Explain why fitting a straight line to a large sample from this distribution estimates a well-defined population quantity even though (L1) fails for the columns 1,  $x$ , and why the second-moment theory of this chapter does not describe the sampling variability of the fitted slope.

## 5.2 Simple linear regression as a warm-up

With a single regressor  $x$  and an intercept, the linear model is

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n, \quad (5.2.1)$$

where the  $x_i$  are known numbers and the errors satisfy the second-moment assumptions (L1)–(L3) of [Section 5.1](#):  $E(\varepsilon_i) = 0$ ,  $\text{Var}(\varepsilon_i) = \sigma^2$ , and  $\text{Cov}(\varepsilon_i, \varepsilon_j) = 0$  for  $i \neq j$ . The parameter  $\beta_1$  is the **slope**, the difference in mean response between two values of  $x$  one unit apart. The **intercept**  $\beta_0$  is the mean response at  $x = 0$ , which may or may not be a value of  $x$  that makes sense. Everything in this section is a special case of [Section 5.4](#). We work it out directly because the formulas are short, because they show the structure of the general answer, and because they are the ones used every day.

### 5.2.1 The least squares line

For a candidate line  $b_0 + b_1 x$ , the **residual** of case  $i$  is  $y_i - b_0 - b_1 x_i$ , the vertical distance from the data point to the line. The method of least squares chooses the line that minimizes the sum of squared residuals,

$$S(b_0, b_1) = \sum_{i=1}^n (y_i - b_0 - b_1 x_i)^2.$$

Other criteria are possible, for example the sum of absolute residuals (Chapter 45). Squared error is chosen because it leads to linear equations, because it matches the conditional mean of [Proposition 5.1.1\(c\)](#), and because its statistical properties under (L1)–(L3) can be worked out exactly, which is the business of this chapter.

We use the standard abbreviations

$$\begin{aligned} \bar{x} &= \frac{1}{n} \sum_i x_i, & \bar{y} &= \frac{1}{n} \sum_i y_i, & S_{xx} &= \sum_i (x_i - \bar{x})^2, \\ S_{xy} &= \sum_i (x_i - \bar{x})(y_i - \bar{y}), & S_{yy} &= \sum_i (y_i - \bar{y})^2. \end{aligned} \quad (5.2.2)$$

Two identities are used repeatedly:  $\sum_i (x_i - \bar{x}) = 0$ , and consequently  $S_{xy} = \sum_i (x_i - \bar{x})y_i$  and  $S_{xx} = \sum_i (x_i - \bar{x})x_i$ . Note that  $S_{xx} > 0$  unless all the  $x_i$  are equal.

#### Theorem 5.2.1 — Least squares for a straight line

Suppose the  $x_i$  are not all equal. Then  $S(b_0, b_1)$  has a unique minimizer  $(\hat{\beta}_0, \hat{\beta}_1)$ , given by

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}, \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}. \quad (5.2.3)$$

It is the unique solution of the two **normal equations**

$$\sum_{i=1}^n (y_i - b_0 - b_1 x_i) = 0, \quad \sum_{i=1}^n x_i (y_i - b_0 - b_1 x_i) = 0. \quad (5.2.4)$$

Moreover, for all  $b_0, b_1$ ,

$$\begin{aligned} S(b_0, b_1) &= S(\hat{\beta}_0, \hat{\beta}_1) + n[(b_0 - \hat{\beta}_0) + (b_1 - \hat{\beta}_1)\bar{x}]^2 \\ &\quad + S_{xx}(b_1 - \hat{\beta}_1)^2. \end{aligned} \quad (5.2.5)$$

*Proof.*  $S$  is a differentiable function of  $(b_0, b_1)$ , with

$$\frac{\partial S}{\partial b_0} = -2 \sum_i (y_i - b_0 - b_1 x_i), \quad \frac{\partial S}{\partial b_1} = -2 \sum_i x_i (y_i - b_0 - b_1 x_i).$$

A minimizer must make both derivatives zero, which gives [Eq. 5.2.4](#). The first normal equation divided by  $n$  says  $\bar{y} - b_0 - b_1 \bar{x} = 0$ , so  $b_0 = \bar{y} - b_1 \bar{x}$ . Substituting into the second, and using  $y_i - \bar{y} - b_1(x_i - \bar{x})$  for the residual,

$$\sum_i x_i [(y_i - \bar{y}) - b_1(x_i - \bar{x})] = S_{xy} - b_1 S_{xx} = 0,$$

so  $b_1 = S_{xy}/S_{xx}$ . This is the only stationary point.

A stationary point need not be a minimum, so we check directly. Write  $\hat{\varepsilon}_i = y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i$ ,  $d_0 = b_0 - \hat{\beta}_0$  and  $d_1 = b_1 - \hat{\beta}_1$ . Then  $y_i - b_0 - b_1 x_i = \hat{\varepsilon}_i - (d_0 + d_1 x_i)$ , and

$$S(b_0, b_1) = \sum_i \hat{\varepsilon}_i^2 - 2d_0 \sum_i \hat{\varepsilon}_i - 2d_1 \sum_i x_i \hat{\varepsilon}_i + \sum_i (d_0 + d_1 x_i)^2.$$

The two middle sums vanish by the normal equations. For the last,  $d_0 + d_1 x_i = (d_0 + d_1 \bar{x}) + d_1(x_i - \bar{x})$ , and the cross term again sums to zero, so  $\sum_i (d_0 + d_1 x_i)^2 = n(d_0 + d_1 \bar{x})^2 + d_1^2 S_{xx}$ . This proves [Eq. 5.2.5](#). Both added terms are nonnegative, and since  $S_{xx} > 0$  they are both zero only when  $d_1 = 0$  and then  $d_0 = 0$ . So  $(\hat{\beta}_0, \hat{\beta}_1)$  is the unique global minimizer. ■

The decomposition [Eq. 5.2.5](#) says more than minimality. It shows the level sets of  $S$  are ellipses centred at  $(\hat{\beta}_0, \hat{\beta}_1)$ , whose shape is set by  $n$ ,  $\bar{x}$  and  $S_{xx}$ . When  $\bar{x}$  is far from zero the ellipses are long and thin, tilted along the direction where  $b_0 + b_1 \bar{x}$  stays constant. Many lines then fit almost equally well, provided they pass near the centre of the data. This is a first look at the correlation between the estimated intercept and slope ([Corollary 5.5.4](#)).

The **fitted values** are  $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$  and the **residuals** are  $\hat{\varepsilon}_i = y_i - \hat{y}_i$ . The minimum SSE =  $\sum_i \hat{\varepsilon}_i^2$  is the **residual sum of squares** (also *error sum of squares*).

### Proposition 5.2.2 — Properties of the fitted line

Under the conditions of [Theorem 5.2.1](#):

- (a)  $\sum_i \hat{\varepsilon}_i = 0$ ,  $\sum_i x_i \hat{\varepsilon}_i = 0$  and  $\sum_i \hat{y}_i \hat{\varepsilon}_i = 0$ ;

- (b) the line passes through  $(\bar{x}, \bar{y})$ , and the mean of the fitted values is  $\bar{y}$ ;  
 (c) with  $SST = S_{yy}$  and  $SSR = \sum_i (\hat{y}_i - \bar{y})^2$ ,

$$SST = SSR + SSE, \quad SSR = \hat{\beta}_1^2 S_{xx} = \frac{S_{xy}^2}{S_{xx}},$$

and so  $SSE = S_{yy} - S_{xy}^2/S_{xx}$ ;

- (d) if also  $S_{yy} > 0$ , the **coefficient of determination**  $R^2 = SSR/SST$  equals  $r^2$ , the square of the sample correlation  $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$ .

*Proof.* (a) The first two are the normal equations Eq. 5.2.4. The third follows because  $\hat{y}_i$  is a combination of 1 and  $x_i$ . (b) The first normal equation is  $\bar{y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{x}$ , and averaging  $\hat{y}_i = y_i - \hat{\varepsilon}_i$  gives  $\bar{y}$ . (c) By (b),  $\hat{y}_i - \bar{y} = \hat{\beta}_1(x_i - \bar{x})$ , so  $SSR = \hat{\beta}_1^2 S_{xx} = S_{xy}^2/S_{xx}$ . Also  $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{\varepsilon}_i$ , and the cross term  $\sum_i (\hat{y}_i - \bar{y})\hat{\varepsilon}_i$  vanishes by (a). (d)  $R^2 = S_{xy}^2/(S_{xx}S_{yy}) = r^2$ . ■

Part (c) is the simplest instance of the analysis of variance: the total variation of the response about its mean splits into a part along the fitted line and a part around it. Chapter 9 develops the general decomposition, and Section 6.7 interprets  $R^2$  as a squared cosine.

## 5.2.2 Regression toward the mean

Divide  $S_{xy}/S_{xx}$  above and below by  $n-1$ . With the sample standard deviations  $s_x = \sqrt{S_{xx}/(n-1)}$  and  $s_y = \sqrt{S_{yy}/(n-1)}$ , the slope is

$$\hat{\beta}_1 = r \frac{s_y}{s_x}, \quad \text{equivalently} \quad \frac{\hat{y} - \bar{y}}{s_y} = r \frac{x - \bar{x}}{s_x}. \quad (5.2.6)$$

In standard units, the fitted value is the regressor's standardized value multiplied by  $r$ . Since  $|r| \leq 1$  (the Cauchy–Schwarz inequality, Proposition 1.1.1), a case that is two standard deviations above average in  $x$  is predicted to be *less* than two standard deviations above average in  $y$ , unless the points lie exactly on a line. This is **regression toward the mean**, and it gave the subject its name: Galton observed it in 1886 for the heights of parents and their adult children.

The effect is purely statistical and involves no mechanism. Regressing  $x$  on  $y$  instead gives the slope  $r s_x/s_y$ , which predicts  $x$  from  $y$  with the same shrinkage toward the mean in the other direction. The two fitted lines differ unless  $r^2 = 1$ . They answer different questions: the mean of  $y$  at given  $x$ , and the mean of  $x$  at given  $y$ . It is a common error to read regression toward the mean as evidence that extreme units become more ordinary over time, or that an intervention given to the most extreme units has worked.

## 5.2.3 A worked example

**Example. (Forecasting December sea temperature)**

The Niño 1+2 index is the mean sea surface temperature in a region of the Pacific off

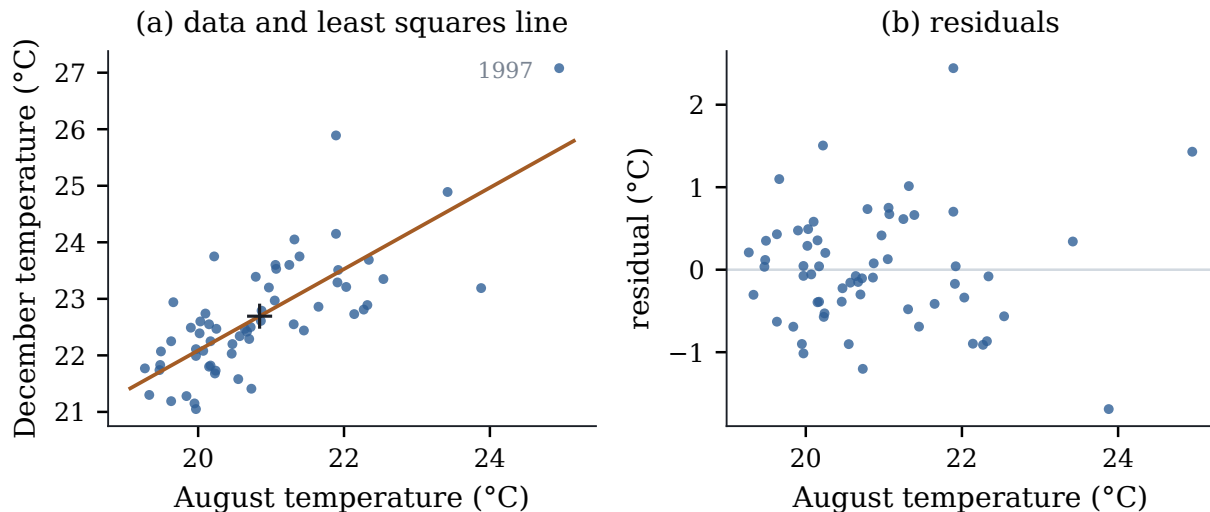


Figure 5.2.1: December against August sea surface temperature in the Niño 1+2 region, 1950–2010. (a) The data and the least squares line, which passes through the point of means (cross). (b) Residuals against the regressor. The El Niño year 1997 is the point at the far right of both panels.

Peru and Ecuador. It is high during El Niño events. We use the monthly values for the 61 years 1950–2010 and ask how well the August value predicts the December value four months later. Here  $x$  is the August temperature and  $y$  the December temperature, both in degrees Celsius. The summary statistics are

$$\begin{aligned}\bar{x} &= 20.843, & \bar{y} &= 22.693, \\ S_{xx} &= 77.801, & S_{xy} &= 56.008, & S_{yy} &= 70.380,\end{aligned}$$

so by [Theorem 5.2.1](#)

$$\hat{\beta}_1 = \frac{56.008}{77.801} = 0.7199, \quad \hat{\beta}_0 = 22.693 - 0.7199 \times 20.843 = 7.6887.$$

Each extra degree in August goes with 0.7199 of a degree more in the following December, on average. For an August temperature of 22°C the fitted December mean is 23.53°C. The intercept, the fitted December temperature after an August at 0°C, is meaningless: no August in the data is below 19°C.

The residual sum of squares is  $SSE = 30.061$ , and  $R^2 = 0.5729$ , the square of the correlation  $r = 0.7569$ . The slope is less than 1 partly because  $r < 1$  and partly because December temperatures vary a little less than August ones:  $s_y = 1.083$  against  $s_x = 1.139$ .

```
import numpy as np
import statsmodels.api as sm
sst = sm.datasets.elnino.load_pandas().data
x = sst["AUG"].to_numpy()           # August temperature
y = sst["DEC"].to_numpy()           # December temperature, same year
n = len(y)

xbar, ybar = x.mean(), y.mean()
Sxx = np.sum((x - xbar) ** 2)
Sxy = np.sum((x - xbar) * (y - ybar))
Syy = np.sum((y - ybar) ** 2)
```

```

b1 = Sxy / Sxx                # slope
b0 = ybar - b1 * xbar         # intercept
fitted = b0 + b1 * x
resid = y - fitted
print(f"n = {n}, slope = {b1:.4f}, intercept = {b0:.4f}")

```

```

print("sum of residuals      ", resid.sum())
print("sum of x * residuals  ", (x * resid).sum())
SSE = resid @ resid
R2 = 1 - SSE / Syy
r = Sxy / np.sqrt(Sxx * Syy)
print(f"R^2 = {R2:.4f}, r^2 = {r**2:.4f}")
print("numpy.polyfit agrees: ", np.polyfit(x, y, 1))

```

The residual plot in [Figure 5.2.1\(b\)](#) is the first diagnostic to draw after any fit. The residuals should show no pattern against  $x$ . Here there is no obvious curvature, but the spread is not uniform. The two warmest Augusts, in 1997 and 1983, both fell during strong El Niño events. The 1997 event was still building in December, while the 1982–83 event was over by then, and the two points sit well away from the line on opposite sides. The warmest of all, 1997 (August 24.95°C, December 27.08°C), has a residual of 1.430°C. Because it lies far from  $\bar{x}$ , it pulls on the slope: without it the slope would be 0.6214. Points with regressor values far from the mean have high *leverage*, a notion made precise in [Section 6.8](#) and [Chapter 20](#).

## 5.2.4 Regression through the origin

Sometimes theory says the mean response is zero when  $x = 0$ , and the model  $Y_i = \beta_1 x_i + \varepsilon_i$  is fitted without an intercept. Minimizing  $\sum_i (y_i - b_1 x_i)^2$  gives one normal equation,  $\sum_i x_i (y_i - b_1 x_i) = 0$ , with solution

$$\tilde{\beta}_1 = \frac{\sum_i x_i y_i}{\sum_i x_i^2},$$

valid whenever some  $x_i \neq 0$ . The residuals now satisfy  $\sum_i x_i \hat{\varepsilon}_i = 0$  but generally *not*  $\sum_i \hat{\varepsilon}_i = 0$ , and the decomposition of [Proposition 5.2.2\(c\)](#) about  $\bar{y}$  fails. It holds instead about zero, as  $\sum_i y_i^2 = \sum_i \hat{y}_i^2 + \sum_i \hat{\varepsilon}_i^2$ , which is why software reports an “uncentred”  $R^2$  for such fits. That number is not comparable with the  $R^2$  of a model with an intercept ([Exercise \(A2\)](#)).

Dropping the intercept is a strong assumption. In [Example \(Forecasting December sea temperature\)](#), forcing the line through the origin gives slope 1.0877, a fit much worse than the line with an intercept: the data are nowhere near  $x = 0$ , and the constraint has nothing to do with how ocean temperatures behave.

## 5.2.5 Exercises

### 5.2.5.1 A. Check your understanding

#### Exercise A1.

For the five points  $(x, y) = (0, 1), (1, 1), (2, 3), (3, 4), (4, 6)$ , compute  $\bar{x}, \bar{y}, S_{xx}, S_{xy}$ , the least squares line, the residuals, and  $R^2$ . Check the three identities of [Proposition 5.2.2\(a\)](#).

*Solution.*  $\bar{x} = 2$ ,  $\bar{y} = 3$ ,  $S_{xx} = 10$ ,  $S_{xy} = (-2)(-2) + (-1)(-2) + 0 + 1 \cdot 1 + 2 \cdot 3 = 13$ . So  $\hat{\beta}_1 = 1.3$  and  $\hat{\beta}_0 = 3 - 2.6 = 0.4$ . The fitted values are 0.4, 1.7, 3.0, 4.3, 5.6 and the residuals 0.6, -0.7, 0, -0.3, 0.4, which sum to zero. Also  $\sum_i x_i \hat{\epsilon}_i = 0 - 0.7 + 0 - 0.9 + 1.6 = 0$ .  $S_{yy} = 4 + 4 + 0 + 1 + 9 = 18$ ,  $SSE = 0.36 + 0.49 + 0 + 0.09 + 0.16 = 1.1$ , and  $R^2 = 1 - 1.1/18 = 0.939$ , which equals  $13^2/(10 \cdot 18)$ .

### Exercise A2.

For regression through the origin, show that  $\sum_i y_i^2 = \sum_i \hat{y}_i^2 + \sum_i \hat{\epsilon}_i^2$ , and give a small data set for which  $1 - \sum_i \hat{\epsilon}_i^2 / \sum_i y_i^2$  is close to 1 while  $1 - \sum_i \hat{\epsilon}_i^2 / S_{yy}$  is negative.

### 5.2.5.2 B. Practice

#### Exercise B1.

Show that  $\hat{\beta}_1 = \sum_i c_i y_i$  with  $c_i = (x_i - \bar{x})/S_{xx}$ , and that  $\sum_i c_i = 0$ ,  $\sum_i c_i x_i = 1$  and  $\sum_i c_i^2 = 1/S_{xx}$ . Write  $\hat{\beta}_0$  as  $\sum_i d_i y_i$  and find  $\sum_i d_i$ ,  $\sum_i d_i x_i$  and  $\sum_i d_i^2$ .

*Solution.*  $S_{xy} = \sum_i (x_i - \bar{x})y_i$ , so  $\hat{\beta}_1 = \sum_i c_i y_i$ . Then  $\sum_i c_i = 0$ ,  $\sum_i c_i x_i = \sum_i (x_i - \bar{x})x_i/S_{xx} = 1$ , and  $\sum_i c_i^2 = S_{xx}/S_{xx}^2 = 1/S_{xx}$ . For the intercept,  $\hat{\beta}_0 = \bar{y} - \bar{x}\hat{\beta}_1 = \sum_i d_i y_i$  with  $d_i = 1/n - \bar{x}c_i$ . Hence  $\sum_i d_i = 1$ ,  $\sum_i d_i x_i = \bar{x} - \bar{x} = 0$ , and  $\sum_i d_i^2 = \sum_i (1/n^2 - 2\bar{x}c_i/n + \bar{x}^2 c_i^2) = 1/n + \bar{x}^2/S_{xx}$ . These are the coefficients that give the moments in [Corollary 5.5.4](#).

#### Exercise B2.

Suppose all  $x_i$  are distinct. Show that  $\hat{\beta}_1$  is a weighted average of the slopes  $(y_j - y_i)/(x_j - x_i)$  of the lines through all pairs of points, with weights proportional to  $(x_j - x_i)^2$ . *Hint:* show that  $\sum_{i < j} (x_j - x_i)(y_j - y_i) = nS_{xy}$  and  $\sum_{i < j} (x_j - x_i)^2 = nS_{xx}$ .

### 5.2.5.3 C. Going deeper

#### Exercise C1.

The **least absolute deviations** line minimizes  $\sum_i |y_i - b_0 - b_1 x_i|$ . Show that some minimizing line passes through at least two of the data points (assume the  $x_i$  are not all equal). *Hint:* for fixed  $b_1$  the best  $b_0$  is a median of the  $y_i - b_1 x_i$ , and the objective is piecewise linear and convex in  $b_1$ . Compare the effect of the 1997 point in [Example \(Forecasting December sea temperature\)](#) on this line and on the least squares line.

## 5.3 The multiple linear regression model

A response rarely depends on one regressor alone. The money a country's residents hold depends on their income and on interest rates. The number of physician visits depends on health, age and insurance. Even with one measured regressor, a straight line is often too rigid: the curvature



of a trend or the shape of a seasonal cycle needs several columns to describe. Matrix notation handles all of these cases at once.

### 5.3.1 Matrix form

Suppose case  $i$  has regressor values that have been turned into  $p$  numbers  $x_{i1}, \dots, x_{ip}$ , and consider

$$Y_i = \beta_1 x_{i1} + \beta_2 x_{i2} + \dots + \beta_p x_{ip} + \varepsilon_i, \quad i = 1, \dots, n.$$

Stack the responses, the coefficients and the errors into vectors, and the regressor values into an  $n \times p$  matrix:

$$\mathbf{Y} = \begin{pmatrix} Y_1 \\ \vdots \\ Y_n \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{pmatrix},$$

$$\mathbf{X} = \begin{pmatrix} x_{11} & \cdots & x_{1p} \\ \vdots & & \vdots \\ x_{n1} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_{(1)}^\top \\ \vdots \\ \mathbf{x}_{(n)}^\top \end{pmatrix}.$$

The  $n$  equations become the single equation  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ . Row  $i$  of  $\mathbf{X}$ , written  $\mathbf{x}_{(i)}^\top$ , holds the regressor values of case  $i$ , its **design point**. Column  $j$ , written  $\mathbf{x}_j$ , holds the values of the  $j$ th regressor across the cases.

#### Definition 5.3.1 — The linear model

The **linear model** for a random vector  $\mathbf{Y} \in \mathbb{R}^n$  is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n, \quad (5.3.1)$$

where the **model matrix**  $\mathbf{X}$  is a known  $n \times p$  matrix of constants, and the **coefficient vector**  $\boldsymbol{\beta} \in \mathbb{R}^p$  and the **error variance**  $\sigma^2 > 0$  are unknown parameters. The conditions on  $\boldsymbol{\varepsilon}$  are the **second-moment assumptions**. If in addition  $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$ , equivalently  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$ , the model is the **normal linear model**. The model has **full rank** if  $\text{rank}(\mathbf{X}) = p$ .

The second-moment assumptions are exactly the layers (L1)–(L3) of [Section 5.1](#). By [Theorem 2.2.1](#) they say

$$\mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n. \quad (5.3.2)$$

The first equation is the whole content of (L1). It says the mean vector  $\mathbb{E}(\mathbf{Y})$  is a linear combination of the columns of  $\mathbf{X}$ , with unknown coefficients. The second combines constant variance and uncorrelated errors. When the regressors are random, both equations are read conditionally on  $\mathbf{X}$ , as explained in [Section 5.1](#). The normal version is a model for the full distribution of  $\mathbf{Y}$ , in the sense of [Chapter 3](#). Since a normal vector with covariance  $\sigma^2 \mathbf{I}$  has independent components ([Theorem 3.4.1](#)), the errors are then independent and not merely uncorrelated.

The definition does not require full rank; most results in this chapter assume it, which forces  $p \leq n$ , and [Chapter 6](#) and [Chapter 8](#) treat the general case. What must be *known* is  $\mathbf{X}$ , and

what must be *linear* is the dependence of  $E(\mathbf{Y})$  on  $\boldsymbol{\beta}$ . Nor does the model say that the errors are identically distributed, only that they have a common variance.

### 5.3.2 The intercept

Almost every model in practice includes a column of ones,  $\mathbf{x}_1 = \mathbf{1}$ , so that

$$E(Y_i) = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_k x_{ik}, \quad p = k + 1.$$

In this form it is conventional to number the coefficients from 0, and to call  $k$  the number of regressors. The coefficient of the column of ones is the **intercept**. Including it lets the fitted surface find its own level, rather than being forced through the origin, and it has two further consequences that [Section 5.4](#) proves: the residuals sum to zero, and the slopes can be computed from centred data. Leaving out the intercept is justified only when theory requires the mean response to vanish at the origin of every regressor, as in the regression through the origin at the end of [Section 5.2](#).

### 5.3.3 Linear in the parameters, not in the regressors

The word *linear* in “linear model” refers to  $\boldsymbol{\beta}$ . The columns of  $\mathbf{X}$  can be any known functions of the measured variables. This turns a remarkably wide class of curved relationships into linear models.

**Transformations.** If the mean response is thought to depend on  $\log z$ , the column  $\log z_i$  enters  $\mathbf{X}$ , and  $E(Y_i) = \beta_0 + \beta_1 \log z_i$  is linear in  $(\beta_0, \beta_1)$ . The response can be transformed too, as in the money demand model of [Section 5.4](#), where  $Y$  is the logarithm of money holdings. A transformation of the response changes the meaning of the error, and hence the assumptions; [Chapter 22](#) treats this.

**Polynomials.** A quadratic trend  $E(Y_i) = \beta_0 + \beta_1 t_i + \beta_2 t_i^2$  uses the columns  $\mathbf{1}$ ,  $\mathbf{t}$  and  $\mathbf{t}^2$  (squared entrywise). A polynomial of degree  $d$  in one variable needs  $d + 1$  columns.

**Periodic terms.** A seasonal cycle of period one year can be written as  $a \cos(2\pi t + \phi)$  with unknown amplitude  $a$  and phase  $\phi$ . This is not linear in  $\phi$ . But  $a \cos(2\pi t + \phi) = \gamma_1 \cos(2\pi t) + \gamma_2 \sin(2\pi t)$  with  $\gamma_1 = a \cos \phi$  and  $\gamma_2 = -a \sin \phi$ , and this *is* linear in  $(\gamma_1, \gamma_2)$ . The amplitude is recovered as  $a = (\gamma_1^2 + \gamma_2^2)^{1/2}$ .

#### Example. (A curved trend and a seasonal cycle)

The monthly mean concentration of carbon dioxide in the air at Mauna Loa, Hawaii, from March 1958 to December 2001, gives  $n = 521$  observations in parts per million. Let  $t$  be the time in years, at the middle of each month, and  $t_c = t - 1980$ . The model

$$E(Y) = \beta_0 + \beta_1 t_c + \beta_2 t_c^2 + \gamma_1 \cos 2\pi t + \gamma_2 \sin 2\pi t + \gamma_3 \cos 4\pi t + \gamma_4 \sin 4\pi t$$

has  $p = 7$  columns, and the model matrix has full rank. Its least squares fit ([Section 5.4](#)) has

$$\hat{\beta}_0 = 337.6163, \quad \hat{\beta}_1 = 1.3344, \quad \hat{\beta}_2 = 0.0118,$$

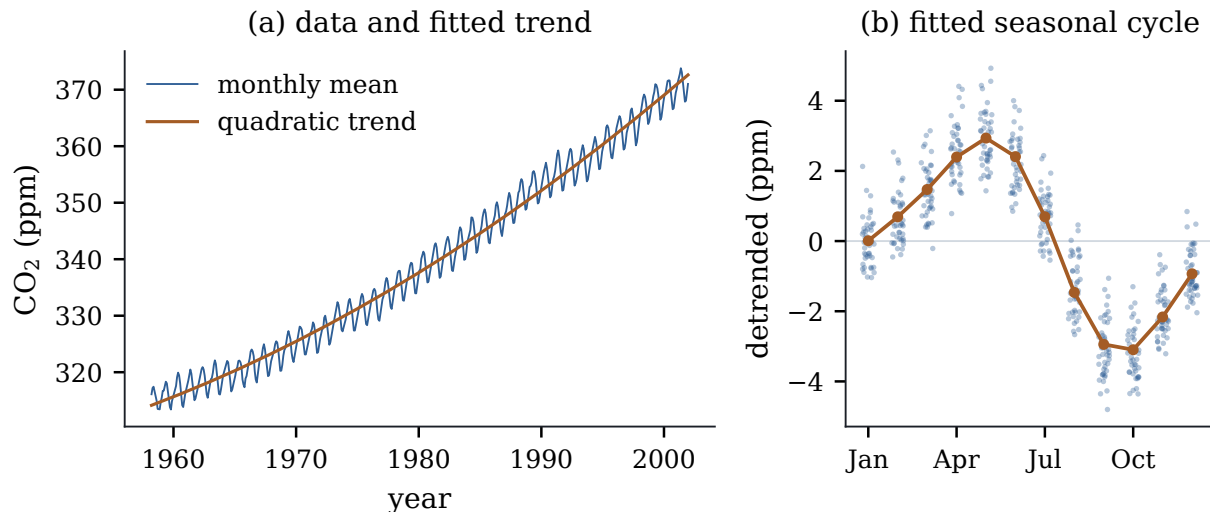


Figure 5.3.1: Monthly carbon dioxide at Mauna Loa, 1958–2001. (a) The data and the fitted quadratic trend. (b) The data with the fitted trend removed, by calendar month, and the fitted seasonal cycle made of two harmonics. The model is curved in time and in the calendar month, but linear in its seven coefficients.

$$\hat{\gamma}_1 = -1.0413, \quad \hat{\gamma}_2 = 2.5641, \quad \hat{\gamma}_3 = 0.6284, \quad \hat{\gamma}_4 = -0.3859.$$

The quadratic term says the rate of increase is itself increasing. The slope of the trend,  $\beta_1 + 2\beta_2 t_c$ , is estimated as 0.86 ppm per year in 1960 and 1.81 ppm per year in 2000. The annual harmonic has amplitude 2.77 ppm and the half-year harmonic 0.74 ppm. Together they make a seasonal swing of 6.0 ppm from the late-spring peak to the autumn trough, as the vegetation of the northern hemisphere takes up carbon over the summer. The residual standard deviation (estimated as in [Section 5.6](#)) is 0.760 ppm, against 2.743 ppm for a straight line in time alone. [Figure 5.3.1](#) shows the data, the trend and the seasonal cycle.

```
import numpy as np
import statsmodels.api as sm
co2 = sm.datasets.co2.load_pandas().data["co2"]
monthly = co2.resample("MS").mean().dropna()           # weekly values -> monthly means
y = monthly.to_numpy()
t = (monthly.index.year + (monthly.index.month - 0.5) / 12).to_numpy()
tc = t - 1980.0                                         # years since 1980

X = np.column_stack([
    np.ones_like(tc),                                  # intercept
    tc, tc**2,                                         # quadratic trend
    np.cos(2 * np.pi * t), np.sin(2 * np.pi * t),   # annual cycle
    np.cos(4 * np.pi * t), np.sin(4 * np.pi * t),   # half-year harmonic
])
n, p = X.shape
print("n =", n, " p =", p, " rank =", np.linalg.matrix_rank(X))
```

```
beta_hat = np.linalg.solve(X.T @ X, X.T @ y)           # normal equations (fine here)
fitted = X @ beta_hat
resid = y - fitted
print("coefficients:", np.round(beta_hat, 4))
print(f"residual standard deviation {np.sqrt(resid @ resid / (n - p)):.3f} ppm")
```

```
beta_line = np.linalg.solve(X[:, :2].T @ X[:, :2], X[:, :2].T @ y)
print(f"straight line only: slope {beta_line[1]:.3f} ppm per year")
```

The residuals of this model are strongly correlated from month to month: the correlation between consecutive residuals is 0.90, because departures from a smooth trend persist for many months. So (L3) fails and the standard errors of [Section 5.5](#) would be too optimistic. The *fit* is still a sensible description. [Chapter 21](#) treats serially correlated errors.

**Models that are not linear.** Some relations cannot be rearranged into linear form. Exponential decay toward an unknown asymptote,  $E(Y) = \beta_0 + \beta_1 e^{-\beta_2 x}$ , is nonlinear in  $\beta_2$ , and so is a cycle of unknown period. Such models are fitted by iterative nonlinear least squares, which this book does not treat. Others become linear after a transformation: if  $Y = \alpha x^\beta \eta$  with a positive multiplicative error  $\eta$ , then  $\log Y = \log \alpha + \beta \log x + \log \eta$  is a linear model for  $\log Y$ , provided the assumptions are made about  $\log \eta$  rather than  $\eta$ .

### 5.3.4 Categorical regressors and indicator variables

A regressor that records a category, such as treatment group, region, or a self-rating on a four-point scale, cannot enter  $\mathbf{X}$  as a single column of codes 1, 2, 3, 4. That would force the mean differences between consecutive categories to be equal, which is an arbitrary consequence of the labels. Instead, each category except one gets an **indicator variable** (or *dummy variable*): a column that is 1 for cases in that category and 0 otherwise. The omitted category is the **reference category**.

#### Example. (Self-rated health)

The RAND Health Insurance Experiment recorded, for 20,190 person-years, the number of outpatient visits to a physician and the person's own rating of their health: excellent, good, fair or poor. With excellent as the reference, the model

$$E(Y_i) = \beta_0 + \beta_1 g_i + \beta_2 f_i + \beta_3 q_i$$

uses three indicator columns **g, f, q** for good, fair and poor health. For a person in excellent health the mean is  $\beta_0$ ; in good health,  $\beta_0 + \beta_1$ ; in fair health,  $\beta_0 + \beta_2$ ; in poor health,  $\beta_0 + \beta_3$ . So  $\beta_1, \beta_2, \beta_3$  are the differences in mean visits between each category and the reference. [Proposition 5.4.4](#) shows that the least squares estimates are the corresponding differences of sample means. In these data they are 0.268, 1.058 and 3.160 visits a year, on top of  $\hat{\beta}_0 = 2.634$  for excellent health.

Why omit a category? With an indicator for *every* category the columns satisfy  $\mathbf{e} + \mathbf{g} + \mathbf{f} + \mathbf{q} = \mathbf{1}$ , where  $\mathbf{e}$  indicates excellent health. The model matrix  $[\mathbf{1}, \mathbf{e}, \mathbf{g}, \mathbf{f}, \mathbf{q}]$  then has five columns and rank four, so it does not have full rank. Such a model is not wrong. It describes the same set of mean vectors, but with one more parameter than it can identify, and the least squares coefficients are no longer unique. This situation is sometimes called the *dummy variable trap*. It is not a trap at all once the right theory is in place: [Section 6.4](#) shows that fitted values are unaffected, and [Chapter 8](#) shows which functions of the coefficients, such as the difference between two categories, are still estimated uniquely.

```
import numpy as np
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
```

```

y = rand["mdvis"].to_numpy(dtype=float)
D = rand[["hlthg", "hlthf", "hlthp"]].to_numpy(dtype=float)  # good, fair, poor
X = np.column_stack([np.ones(len(y)), D])                  # excellent =
  reference

excellent = 1.0 - D.sum(axis=1)
X_all = np.column_stack([X, excellent])                    # an indicator for every level, and
  1
print("columns:", X_all.shape[1], " rank:", np.linalg.matrix_rank(X_all))
print("1 - (sum of the four indicators) =", np.abs(X_all[:, 0] - X_all[:, 1:].sum(
  axis=1)).max())

```

Reference coding is one choice among several. **Effect coding** replaces the indicator of category  $k$  by a column that is 1 in category  $k$ ,  $-1$  in the reference category, and 0 elsewhere. Its coefficients measure deviations of category means from their unweighted average rather than from a reference. The fitted values are the same under either coding, because the two model matrices have the same column space ([Proposition 5.7.1](#)). Only the meaning of the coefficients changes.

### 5.3.5 Interactions

In the model  $E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$  the slope in  $x_1$  is  $\beta_1$  whatever the value of  $x_2$ : the effects are **additive**. An **interaction** lets the slope in one variable depend on another. Adding the product column  $x_1 x_2$ ,

$$E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_1 x_2 = (\beta_0 + \beta_2 x_2) + (\beta_1 + \beta_3 x_2) x_1, \quad (5.3.3)$$

gives a slope in  $x_1$  of  $\beta_1 + \beta_3 x_2$ . The model is still linear in the coefficients. When  $x_2$  is an indicator for a group, [Eq. 5.3.3](#) fits a separate straight line in each group: intercept  $\beta_0$  and slope  $\beta_1$  in the reference group, intercept  $\beta_0 + \beta_2$  and slope  $\beta_1 + \beta_3$  in the other. When both variables are categorical, products of indicators give each combination of categories its own mean, the two-way layout of [Chapter 16](#) ([Definition 16.2.1](#)).

Interactions change what the other coefficients mean. In [Eq. 5.3.3](#),  $\beta_1$  is the slope in  $x_1$  when  $x_2 = 0$ , which may be far outside the data. Centring  $x_2$  before forming the product makes  $\beta_1$  the slope at the average  $x_2$  instead. [Section 5.7](#) returns to this.

### 5.3.6 When is the model matrix of full rank?

Full rank means the columns of  $\mathbf{X}$  are linearly independent: no column is an exact linear combination of the others. By rank-nullity ([Theorem 1.2.2](#)), this holds iff  $\mathbf{X}\mathbf{b} = \mathbf{0}$  only for  $\mathbf{b} = \mathbf{0}$ , that is, iff two different coefficient vectors never give the same mean vector. It fails in three common ways:

1. **Too few cases:**  $n < p$ . [Chapter 28](#) deals with this setting.
2. **Redundant coding:** an indicator for every category together with an intercept, as above, or the same quantity measured in two units.
3. **Unlucky data:** a design that happens to make columns dependent, such as a quadratic in a variable observed at only two distinct values, or an interaction between two indicators

when some combination of categories never occurs.

Near-failures, where columns are almost dependent, are more common and more troublesome. They make coefficients imprecise ([Chapter 26](#)) and computations inaccurate ([Chapter 10](#)).

## 5.3.7 Exercises

### 5.3.7.1 A. Check your understanding

#### Exercise A1.

Write out the model matrix for each model, for the data  $x = (1, 2, 3, 4)$  and a group label  $(A, A, B, B)$ , and give its rank: (a) a quadratic in  $x$ ; (b) separate intercepts for the two groups and a common slope; (c) separate lines in the two groups, with reference coding; (d) an intercept, an indicator for  $A$  and an indicator for  $B$ .

*Solution.* With  $g$  the indicator of group  $B$ : (a) rows  $(1, x, x^2)$ :  $(1, 1, 1), (1, 2, 4), (1, 3, 9), (1, 4, 16)$ , rank 3. (b) rows  $(1, g, x)$ :  $(1, 0, 1), (1, 0, 2), (1, 1, 3), (1, 1, 4)$ , rank 3. (c) rows  $(1, g, x, gx)$ :  $(1, 0, 1, 0), (1, 0, 2, 0), (1, 1, 3, 3), (1, 1, 4, 4)$ , rank 4; it is square and nonsingular, so the two lines pass through the data exactly. (d) rows  $(1, 1, 0), (1, 1, 0), (1, 0, 1), (1, 0, 1)$ , rank 2: the first column is the sum of the other two.

#### Exercise A2.

Which of these mean functions define linear models in the stated parameters (after a change of parameters if necessary)? (a)  $\beta_0 + \beta_1\sqrt{x}$ ; (b)  $\beta_0 e^{\beta_1 x}$ ; (c)  $\beta_0 + \beta_1 x + \beta_2 \max(x - 5, 0)$ ; (d)  $\beta_0 + \beta_1 \sin(x - \beta_2)$ ; (e)  $1/(\beta_0 + \beta_1 x)$ .

### 5.3.7.2 B. Practice

#### Exercise B1.

For a factor with three levels and an intercept, write the model matrix rows under reference coding (level 3 as reference) and under effect coding. Find the nonsingular  $3 \times 3$  matrix  $\mathbf{K}$  with  $\mathbf{X}_{\text{effect}} = \mathbf{X}_{\text{ref}}\mathbf{K}$ , and express the effect-coding coefficients in terms of the three level means  $\mu_1, \mu_2, \mu_3$ .

*Solution.* Reference coding has rows  $(1, 1, 0), (1, 0, 1), (1, 0, 0)$  for levels 1, 2, 3. Effect coding has rows  $(1, 1, 0), (1, 0, 1), (1, -1, -1)$ . Each effect-coded row is the reference-coded row times  $\mathbf{K} = \begin{pmatrix} 1 & -1 & -1 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{pmatrix}$ : for instance  $(1, 0, 0)\mathbf{K} = (1, -1, -1)$  and  $(1, 1, 0)\mathbf{K} = (1, 1, 0)$ .  $\det \mathbf{K} = 3 \neq 0$ . Under effect coding the level means are  $\mu_1 = \alpha_0 + \alpha_1$ ,  $\mu_2 = \alpha_0 + \alpha_2$ ,  $\mu_3 = \alpha_0 - \alpha_1 - \alpha_2$ . Adding,  $\alpha_0 = (\mu_1 + \mu_2 + \mu_3)/3$ , and then  $\alpha_k = \mu_k - \alpha_0$  for  $k = 1, 2$ : deviations from the unweighted average of the means.

#### Exercise B2.

In [Eq. 5.3.3](#), replace  $x_2$  by  $x_2 - c$  in both the main-effect column and the product. Find the new coefficients in terms of the old, and show that the fitted mean function is

unchanged. For which  $c$  is the new coefficient of  $x_1$  the slope at the average value of  $x_2$ ?

### 5.3.7.3 C. Going deeper

#### Exercise C1.

Let  $t_i = i/m$  for  $i = 0, \dots, m-1$  (one full period sampled at  $m$  equally spaced points, with  $m \geq 5$ ). Show that the columns  $\mathbf{1}, \cos 2\pi\mathbf{t}, \sin 2\pi\mathbf{t}, \cos 4\pi\mathbf{t}, \sin 4\pi\mathbf{t}$  are mutually orthogonal, and find their squared lengths. Conclude that for a whole number of periods the harmonic coefficients are estimated separately from each other and from the mean. Why is this not exactly the case in [Example \(A curved trend and a seasonal cycle\)](#)?

## 5.4 Least squares and the normal equations

The straight-line calculation of [Section 5.2](#) extends to any model matrix. The algebra is shorter in matrix form than it was for two coefficients, because the derivatives of linear and quadratic forms were worked out once and for all in [Chapter 1](#). In this section  $\mathbf{y}$  is a fixed data vector. Nothing here uses probability. The statistical properties come in [Section 5.5](#).

### 5.4.1 The least squares estimate

#### Definition 5.4.1 — Least squares estimate

For an  $n \times p$  matrix  $\mathbf{X}$  and  $\mathbf{y} \in \mathbb{R}^n$ , the **least squares criterion** is

$$S(\mathbf{b}) = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_{i=1}^n (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2, \quad \mathbf{b} \in \mathbb{R}^p.$$

A **least squares estimate** of  $\boldsymbol{\beta}$  is any  $\hat{\boldsymbol{\beta}}$  that minimizes  $S$ . The corresponding **fitted values** are  $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ , the **residuals** are  $\hat{\boldsymbol{\varepsilon}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$ , and the **residual sum of squares** is  $\text{SSE} = S(\hat{\boldsymbol{\beta}}) = \|\hat{\boldsymbol{\varepsilon}}\|^2$ .

Following the book's notation, we write  $\hat{\mathbf{Y}}$  and  $\hat{\boldsymbol{\varepsilon}}$  in bold for the vectors, and  $\hat{y}_i$  and  $\hat{\varepsilon}_i$  for their entries, whether the data are regarded as fixed numbers or as random.

#### Theorem 5.4.1 — Least squares for a full-rank model

Let  $\mathbf{X}$  be an  $n \times p$  matrix of rank  $p$ . Then:

- (a)  $\mathbf{X}^\top \mathbf{X}$  is positive definite, and in particular nonsingular;
- (b) the **normal equations**

$$\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y} \tag{5.4.1}$$

have exactly one solution,

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}; \quad (5.4.2)$$

(c) for every  $\mathbf{b} \in \mathbb{R}^p$ ,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + (\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}}), \quad (5.4.3)$$

so  $\hat{\boldsymbol{\beta}}$  is the unique least squares estimate.

If instead  $\text{rank}(\mathbf{X}) < p$ , least squares estimates exist but are never unique.

*Proof.* (a)  $\mathbf{X}^\top \mathbf{X}$  is symmetric, and  $\mathbf{b}^\top \mathbf{X}^\top \mathbf{X} \mathbf{b} = \|\mathbf{X}\mathbf{b}\|^2 \geq 0$ , with equality iff  $\mathbf{X}\mathbf{b} = \mathbf{0}$ . Since  $\mathbf{X}$  has rank  $p$ , rank-nullity (Theorem 1.2.2) gives  $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$ , so the quadratic form is positive for  $\mathbf{b} \neq \mathbf{0}$ . A positive definite matrix is nonsingular (Proposition 1.7.5(e)).

(b) This is the computation of Example (The least squares criterion): by Theorem 1.11.1 the gradient of  $S$  is  $-2\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b})$  and its Hessian is  $2\mathbf{X}^\top \mathbf{X}$ . Every minimizer of the differentiable function  $S$  is a stationary point, and the stationary points are the solutions of Eq. 5.4.1. By (a) there is exactly one, Eq. 5.4.2.

(c) Put  $\mathbf{d} = \mathbf{b} - \hat{\boldsymbol{\beta}}$ . Then  $\mathbf{y} - \mathbf{X}\mathbf{b} = \hat{\boldsymbol{\varepsilon}} - \mathbf{X}\mathbf{d}$ , and

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\hat{\boldsymbol{\varepsilon}}\|^2 - 2\mathbf{d}^\top \mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} + \mathbf{d}^\top \mathbf{X}^\top \mathbf{X} \mathbf{d}.$$

The middle term vanishes because  $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{0}$  by the normal equations. The last term is positive for  $\mathbf{d} \neq \mathbf{0}$  by (a). So  $S(\mathbf{b}) > S(\hat{\boldsymbol{\beta}})$  whenever  $\mathbf{b} \neq \hat{\boldsymbol{\beta}}$ .

For the last statement, suppose  $\text{rank}(\mathbf{X}) < p$ . The vector  $\mathbf{X}^\top \mathbf{y}$  lies in  $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$  (Proposition 1.2.3(c)), so by Proposition 1.11.2, applied with  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  and  $S(\mathbf{b}) = \mathbf{y}^\top \mathbf{y} + (\mathbf{b}^\top \mathbf{X}^\top \mathbf{X} \mathbf{b} - 2\mathbf{b}^\top \mathbf{X}^\top \mathbf{y})$ , the minimizers of  $S$  are exactly the solutions of the normal equations, and they exist. But there is a  $\mathbf{v} \neq \mathbf{0}$  with  $\mathbf{X}\mathbf{v} = \mathbf{0}$ , and then  $S(\mathbf{b} + t\mathbf{v}) = S(\mathbf{b})$  for every  $t$ , so no minimizer is unique. ■

The calculus proof finds the stationary point and then needs a separate argument, Eq. 5.4.3, to show it is the global minimum. The Hessian  $2\mathbf{X}^\top \mathbf{X}$  tells the same story: it is positive definite, so  $S$  is a strictly convex quadratic bowl with a single lowest point. The name “normal equations” has nothing to do with the normal distribution. It refers to the orthogonality they express:  $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$  says the residual vector is *normal*, that is, perpendicular, to every column of  $\mathbf{X}$ . Chapter 6 makes that geometry the starting point, derives the normal equations again without calculus (Theorem 6.4.2), and treats the rank-deficient case in full.

*Remark* (The rank-deficient case). When  $\text{rank}(\mathbf{X}) < p$ , the proof shows that the least squares estimates form an affine set  $\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X})$  of dimension  $p - \text{rank}(\mathbf{X})$ . Different programs report different members of this set, according to which columns they drop or which constraints they impose. The fitted values  $\mathbf{X}\hat{\boldsymbol{\beta}}$  are the same for all of them, since  $\mathbf{X}(\hat{\boldsymbol{\beta}} + \mathbf{v}) = \mathbf{X}\hat{\boldsymbol{\beta}}$  for  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ . Which linear combinations of the coefficients share this invariance is the question of estimability, answered in Chapter 8 (Definition 8.2.1 and Theorem 8.3.1).

Least squares under known linear restrictions  $\mathbf{H}\mathbf{b} = \mathbf{h}$ , such as shares that add to one, has a



closed form found by Lagrange multipliers in [Exercise \(B1\)](#); the increase in the residual sum of squares it computes drives the test of the restrictions in [Chapter 11](#) ([Theorem 11.3.2](#)).

**Example. (The straight line again)**

For  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ ,

$$\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum_i x_i^2 \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{y} = \begin{pmatrix} n\bar{y} \\ \sum_i x_i y_i \end{pmatrix}.$$

The determinant is  $n \sum_i x_i^2 - n^2 \bar{x}^2 = nS_{xx}$ , positive iff the  $x_i$  are not all equal, which is exactly when  $\text{rank}(\mathbf{X}) = 2$ . Inverting,

$$(\mathbf{X}^\top \mathbf{X})^{-1} = \frac{1}{nS_{xx}} \begin{pmatrix} \sum_i x_i^2 & -n\bar{x} \\ -n\bar{x} & n \end{pmatrix} = \begin{pmatrix} \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} & -\frac{\bar{x}}{S_{xx}} \\ -\frac{\bar{x}}{S_{xx}} & \frac{1}{S_{xx}} \end{pmatrix}, \quad (5.4.4)$$

using  $\sum_i x_i^2 = S_{xx} + n\bar{x}^2$ . The second entry of  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$  is  $(\sum_i x_i y_i - n\bar{x}\bar{y})/S_{xx} = S_{xy}/S_{xx}$ , and the first is  $\bar{y} - \bar{x}S_{xy}/S_{xx}$ . This recovers [Theorem 5.2.1](#), and [Eq. 5.4.3](#) reduces to [Eq. 5.2.5](#).

## 5.4.2 Fitted values, residuals and the hat matrix

Substituting [Eq. 5.4.2](#), the fitted values are

$$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{H}\mathbf{y}, \quad \mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top, \quad (5.4.5)$$

and the residuals are  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{y}$ . The  $n \times n$  matrix  $\mathbf{H}$  is the **hat matrix**, so called because it puts the hat on  $\mathbf{y}$ . It depends only on the regressors, never on the response.

### Proposition 5.4.2 — Algebra of the fit

Let  $\mathbf{X}$  have rank  $p$ , and let  $\mathbf{H}$ ,  $\hat{\mathbf{Y}}$  and  $\hat{\boldsymbol{\varepsilon}}$  be as above.

- (a)  $\mathbf{H}$  is symmetric and idempotent,  $\mathbf{H}\mathbf{X} = \mathbf{X}$  and  $(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{0}$ , and  $\text{tr } \mathbf{H} = p$ ,  $\text{tr}(\mathbf{I} - \mathbf{H}) = n - p$ .
- (b)  $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$  and  $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$ .
- (c)  $\|\mathbf{y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2$ , and  $\text{SSE} = \mathbf{y}^\top (\mathbf{I} - \mathbf{H})\mathbf{y} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ .
- (d) If  $\mathbf{1} = \mathbf{X}\mathbf{c}$  for some  $\mathbf{c}$  (for example, if  $\mathbf{X}$  has an intercept column), then  $\sum_i \hat{\varepsilon}_i = 0$ , the mean of the fitted values is  $\bar{y}$ , and

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{\varepsilon}_i^2,$$

that is,  $\text{SST} = \text{SSR} + \text{SSE}$ .

- (e) If  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  for some  $\boldsymbol{\beta}$  and  $\boldsymbol{\varepsilon}$ , then  $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$  and  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$ .

*Proof.* (a)  $\mathbf{H}^\top = \mathbf{X}((\mathbf{X}^\top \mathbf{X})^{-1})^\top \mathbf{X}^\top = \mathbf{H}$ , since the inverse of a symmetric matrix is symmetric.  $\mathbf{H}\mathbf{X} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} = \mathbf{X}$ , and then  $\mathbf{H}^2 = \mathbf{H}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{H}$ . By the cyclic property (Theorem 1.6.1),  $\text{tr } \mathbf{H} = \text{tr}((\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}) = \text{tr } \mathbf{I}_p = p$ . (b)  $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{X}^\top (\mathbf{I} - \mathbf{H})\mathbf{y} = ((\mathbf{I} - \mathbf{H})\mathbf{X})^\top \mathbf{y} = \mathbf{0}$ , and  $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = 0$ . (c)  $\mathbf{y} = \hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}}$  with  $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$  gives the first identity. For the second,  $(\mathbf{I} - \mathbf{H})$  is symmetric and idempotent, so  $\|(\mathbf{I} - \mathbf{H})\mathbf{y}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{H})\mathbf{y}$ ; and  $\mathbf{y}^\top \mathbf{H}\mathbf{y} = \mathbf{y}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ . (d)  $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{c}^\top \mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = 0$  by (b). Then  $\sum_i \hat{y}_i = \sum_i y_i$ . Write  $y_i - \bar{y} = (\hat{y}_i - \bar{y}) + \hat{\varepsilon}_i$ . The cross term  $\sum_i (\hat{y}_i - \bar{y})\hat{\varepsilon}_i = \hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} - \bar{y} \mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$ . (e) Substitute  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  into Eq. 5.4.2 and into  $(\mathbf{I} - \mathbf{H})\mathbf{y}$ , using (a). ■

Part (e) will carry most of the statistical theory. The estimation error  $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$  is a fixed linear function of the unobserved errors, and the residual vector does not depend on  $\boldsymbol{\beta}$  at all. Part (b) says that the residuals satisfy  $p$  independent linear constraints. That is the origin of the divisor  $n - p$  in Section 5.6.

For a model with an intercept, part (d) motivates the **coefficient of determination**

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}},$$

the proportion of the variation of the response about its mean that the fitted values reproduce. It generalizes Proposition 5.2.2(d), and lies in  $[0, 1]$ . It can only increase when a column is added to  $\mathbf{X}$  (Exercise (B1)), which is why Chapter 9 introduces an adjusted version (Proposition 9.5.3). Section 6.7 interprets  $R$  as the cosine of an angle.

### 5.4.3 Centring the regressors

For a model with an intercept, write  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$ , where  $\mathbf{X}_1$  is  $n \times k$  with column means  $\bar{\mathbf{x}} = \mathbf{X}_1^\top \mathbf{1}/n$ . The **centred regressors** are the columns of

$$\mathbf{X}_c = \mathbf{X}_1 - \mathbf{1}\bar{\mathbf{x}}^\top = \left(\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\right)\mathbf{X}_1,$$

each of which sums to zero. The next result is the matrix version of the formulas  $\hat{\beta}_1 = S_{xy}/S_{xx}$  and  $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ .

#### Proposition 5.4.3 — Centred form

Let  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$  have rank  $p = k + 1$ , and partition  $\hat{\boldsymbol{\beta}} = (\hat{\beta}_0, \hat{\boldsymbol{\beta}}_1^\top)^\top$ . Then  $\mathbf{X}_c$  has rank  $k$ , and

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top \mathbf{y} = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top (\mathbf{y} - \bar{y}\mathbf{1}), \quad \hat{\beta}_0 = \bar{y} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_1.$$

*Proof.* For  $\mathbf{b} = (b_0, \mathbf{b}_1^\top)^\top$ ,

$$\mathbf{X}\mathbf{b} = b_0\mathbf{1} + \mathbf{X}_1\mathbf{b}_1 = (b_0 + \bar{\mathbf{x}}^\top \mathbf{b}_1)\mathbf{1} + \mathbf{X}_c\mathbf{b}_1 = [\mathbf{1}, \mathbf{X}_c] \mathbf{T}\mathbf{b}, \quad \mathbf{T} = \begin{pmatrix} 1 & \bar{\mathbf{x}}^\top \\ \mathbf{0} & \mathbf{I}_k \end{pmatrix}.$$

$\mathbf{T}$  is nonsingular (it is upper triangular with unit diagonal), so  $[\mathbf{1}, \mathbf{X}_c] = \mathbf{X}\mathbf{T}^{-1}$  has rank

$p$  (Proposition 1.2.3(b)), and hence  $\mathbf{X}_c$  has rank  $k$ . The criterion satisfies  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - [\mathbf{1}, \mathbf{X}_c]\mathbf{a}\|^2$  with  $\mathbf{a} = \mathbf{T}\mathbf{b}$ , and  $\mathbf{b} \mapsto \mathbf{T}\mathbf{b}$  is a bijection of  $\mathbb{R}^p$ . So  $\hat{\boldsymbol{\beta}}$  minimizes the left side iff  $\hat{\mathbf{a}} = \mathbf{T}\hat{\boldsymbol{\beta}}$  minimizes the right side. Because  $\mathbf{1}^\top \mathbf{X}_c = \mathbf{0}^\top$ , the normal equations for  $[\mathbf{1}, \mathbf{X}_c]$  are block diagonal,

$$\begin{pmatrix} n & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{X}_c^\top \mathbf{X}_c \end{pmatrix} \hat{\mathbf{a}} = \begin{pmatrix} n\bar{y} \\ \mathbf{X}_c^\top \mathbf{y} \end{pmatrix},$$

giving  $\hat{a}_0 = \bar{y}$  and  $\hat{\mathbf{a}}_1 = (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \mathbf{X}_c^\top \mathbf{y}$ . Undoing  $\mathbf{T}$ :  $\hat{\boldsymbol{\beta}}_1 = \hat{\mathbf{a}}_1$  and  $\hat{\beta}_0 = \hat{a}_0 - \bar{\mathbf{x}}^\top \hat{\mathbf{a}}_1$ . Finally  $\mathbf{X}_c^\top \mathbf{1} = \mathbf{0}$ , so  $\mathbf{X}_c^\top \mathbf{y} = \mathbf{X}_c^\top (\mathbf{y} - \bar{y}\mathbf{1})$ .

■

In words: to find the slopes, centre everything and fit without an intercept; the intercept then makes the fitted surface pass through the point of means  $(\bar{\mathbf{x}}, \bar{y})$ . The matrix  $\mathbf{X}_c^\top \mathbf{X}_c / (n-1)$  is the sample covariance matrix of the regressors, and  $\mathbf{X}_c^\top \mathbf{y} / (n-1)$  their sample covariances with the response. So the slopes are “covariance divided by variance”, as in the straight-line case, with a matrix inverse in place of the division. This is the simplest case of the Frisch–Waugh–Lovell theorem of Section 6.6, with the intercept playing the part of the regressors that are “partialled out”.

## 5.4.4 Indicator variables

The least squares estimates in a model with only indicator columns are simple averages.

### Proposition 5.4.4 — Coefficients of indicator variables

Suppose each case belongs to one of  $c \geq 2$  groups, group  $k$  has  $n_k \geq 1$  cases with mean response  $\bar{y}_k$ , and  $\mathbf{z}_k$  is the indicator column of group  $k$ . Let  $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_{c-1}]$ , so that group  $c$  is the reference. Then  $\mathbf{X}$  has full rank, and

$$\hat{\beta}_0 = \bar{y}_c, \quad \hat{\beta}_k = \bar{y}_k - \bar{y}_c \quad (k = 1, \dots, c-1).$$

The fitted value of every case is the mean of its group.

*Proof.* If  $\mathbf{X}\mathbf{b} = \mathbf{0}$ , the entries for a case in group  $c$  give  $b_0 = 0$ , and those for a case in group  $k < c$  give  $b_0 + b_k = 0$ , so  $\mathbf{b} = \mathbf{0}$ : the rank is  $p = c$ . Let  $\mathbf{b}$  be the proposed vector. Its fitted value for a case in group  $k < c$  is  $\bar{y}_c + (\bar{y}_k - \bar{y}_c) = \bar{y}_k$ , and for a case in group  $c$  it is  $\bar{y}_c$ . The residuals  $y_i - \bar{y}_{k(i)}$  therefore sum to zero within each group, so  $\mathbf{z}_k^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = 0$  for every  $k = 1, \dots, c$ . Every column of  $\mathbf{X}$  is a sum of group indicators, since  $\mathbf{1} = \mathbf{z}_1 + \dots + \mathbf{z}_c$ , so  $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$ : the normal equations hold. By Theorem 5.4.1,  $\mathbf{b} = \hat{\boldsymbol{\beta}}$ .

■

```
import numpy as np
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
D = rand[["hlthg", "hlthf", "hlthp"]].to_numpy(dtype=float) # good, fair, poor
```

```

X = np.column_stack([np.ones(len(y)), D])           # excellent =
    reference

beta_hat = np.linalg.solve(X.T @ X, X.T @ y)
level = D @ np.array([1, 2, 3])                   # 0 = excellent, ...,
    3 = poor
means = np.array([y[level == k].mean() for k in range(4)])
print("coefficients:      ", np.round(beta_hat, 4))
print("group means:      ", np.round(means, 4))
print("means minus reference: ", np.round(means - means[0], 4))

```

For the health data of [Example \(Self-rated health\)](#) the group sizes are 11,019 (excellent), 7,309 (good), 1,560 (fair) and 302 (poor), and the group means are 2.634, 2.902, 3.692 and 5.795 visits. The listing confirms that the coefficients are these means' differences from the reference.

### 5.4.5 A worked example

#### Example. (Money demand in Denmark)

Economic theory relates the amount of money people choose to hold to their income, which drives the volume of transactions, and to interest rates, which measure the return given up by holding money rather than bonds. A classic data set records, for  $n = 55$  quarters from 1974 to 1987 in Denmark, the logarithm of real money holdings ( $y$ ), the logarithm of real income, the interest rate on bonds and the interest rate on deposits (rates as fractions per year). The model has  $p = 4$  columns: an intercept and the three regressors. Solving the normal equations gives

$$\hat{\beta} = (4.3945, 1.2958, -2.6163, 0.6186)^T,$$

with residual sum of squares  $SSE = 0.09253$  and  $R^2 = 0.9262$ . The listing checks that  $\mathbf{X}^T \hat{\mathbf{e}}$  is zero up to rounding error, that the two formulas for SSE in [Proposition 5.4.2\(c\)](#) agree, and that the slopes computed from centred data ([Proposition 5.4.3](#)) are the same. [Section 5.6](#) attaches standard errors to these numbers, and [Section 5.7](#) interprets them.

```

import numpy as np
import statsmodels.api as sm
dk = sm.datasets.danish_data.load_pandas().data
y = dk["lrm"].to_numpy()
X = np.column_stack([np.ones(len(y)), dk["lry"], dk["ibo"], dk["ide"]])
n, p = X.shape
print("n =", n, " p =", p)

```

```

XtX = X.T @ X
Xty = X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)           # solve X^T X b = X^T y
fitted = X @ beta_hat
resid = y - fitted
SSE = resid @ resid

print("beta_hat =", np.round(beta_hat, 4))
print("X^T e     =", X.T @ resid)             # zero up to rounding
print("SSE two ways:", SSE, y @ y - beta_hat @ Xty)

```

```

Z = X[:, 1:] - X[:, 1:].mean(axis=0)          # centred regressors, no intercept column
slopes = np.linalg.solve(Z.T @ Z, Z.T @ (y - y.mean()))
intercept = y.mean() - X[:, 1:].mean(axis=0) @ slopes
print("slopes from centred data:", np.round(slopes, 4))
print("intercept recovered:      ", round(intercept, 4))

```

*Warning* (Formulas are not algorithms). The formula  $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$  is for reasoning, not for computing. Forming  $\mathbf{X}^\top \mathbf{X}$  squares the condition number of the problem, and inverting it wastes work and accuracy. The listings solve the normal equations with a linear solver, which is acceptable for small, well-conditioned problems like these. Good software computes  $\hat{\beta}$  from an orthogonal factorization of  $\mathbf{X}$  itself. [Section 6.10](#) shows the loss of accuracy in an example, and [Chapter 10](#) develops the algorithms ([Theorem 10.2.1](#), [Theorem 10.3.2](#)).

## 5.4.6 Exercises

### 5.4.6.1 A. Check your understanding

#### Exercise A1.

Suppose the columns of  $\mathbf{X}$  are nonzero and mutually orthogonal. Show that  $\hat{\beta}_j = \mathbf{x}_j^\top \mathbf{y} / \mathbf{x}_j^\top \mathbf{x}_j$  for each  $j$ , so that each coefficient is the same as in the regression of  $\mathbf{y}$  on  $\mathbf{x}_j$  alone. Show also that deleting a column does not change the other estimates.

### 5.4.6.2 B. Practice

#### Exercise B1.

Let  $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}]$  have full rank, and let  $\text{SSE}_1$  and  $\text{SSE}$  be the residual sums of squares from regressing  $\mathbf{y}$  on  $\mathbf{X}_1$  and on  $\mathbf{X}$ . Show that  $\text{SSE} \leq \text{SSE}_1$ , directly from [Definition 5.4.1](#). Deduce that  $R^2$  cannot decrease when a column is added to a model with an intercept.

*Solution.* Let  $\hat{\beta}_1$  be the least squares estimate for  $\mathbf{X}_1$ . The vector  $(\hat{\beta}_1^\top, 0)^\top$  is one candidate in the minimization over  $\mathbf{b} \in \mathbb{R}^p$ , and it gives the value  $\|\mathbf{y} - \mathbf{X}_1 \hat{\beta}_1\|^2 = \text{SSE}_1$ . The minimum over all candidates is at most this. With an intercept in  $\mathbf{X}_1$ ,  $R^2 = 1 - \text{SSE}/\text{SST}$  and  $\text{SST}$  does not depend on the model, so  $R^2$  cannot decrease. Equality holds iff the extra coefficient in the larger fit is zero, because by [Eq. 5.4.3](#) the candidate  $(\hat{\beta}_1^\top, 0)^\top$  is optimal only if it equals  $\hat{\beta}$ .

#### Exercise B2.

Let  $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$  with all  $w_i > 0$ , and  $\mathbf{X}$  of full rank. Show that  $\sum_i w_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$  has the unique minimizer  $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}$ , by applying [Theorem 5.4.1](#) to the rescaled data  $\mathbf{W}^{1/2} \mathbf{y}$  and  $\mathbf{W}^{1/2} \mathbf{X}$ . Which cases get most influence when the weights are the reciprocals of the error variances?

*Solution.*  $\sum_i w_i (y_i - \mathbf{x}_i^\top \mathbf{b})^2 = \|\mathbf{W}^{1/2} \mathbf{y} - \mathbf{W}^{1/2} \mathbf{X} \mathbf{b}\|^2$ . The matrix  $\mathbf{W}^{1/2} \mathbf{X}$  has rank  $p$ , because  $\mathbf{W}^{1/2}$  is nonsingular (Proposition 1.2.3(b)). By Theorem 5.4.1 the unique minimizer is  $((\mathbf{W}^{1/2} \mathbf{X})^\top \mathbf{W}^{1/2} \mathbf{X})^{-1} (\mathbf{W}^{1/2} \mathbf{X})^\top \mathbf{W}^{1/2} \mathbf{y} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}$ . With  $w_i = 1/\text{Var}(Y_i)$ , the least variable cases count most. This is weighted least squares, the subject of Chapter 21 and, with a general covariance matrix, of Section 6.9.

### Exercise B3.

Show that  $\hat{\boldsymbol{\beta}} = \mathbf{A} \mathbf{y}$  with  $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ , that  $\mathbf{A} \mathbf{X} = \mathbf{I}_p$ , and that  $\mathbf{A} \mathbf{A}^\top = (\mathbf{X}^\top \mathbf{X})^{-1}$ . Give an example of a different  $p \times n$  matrix  $\mathbf{B}$  with  $\mathbf{B} \mathbf{X} = \mathbf{I}_p$  when  $n > p$ . (Chapter 7 compares all such estimators  $\mathbf{B} \mathbf{y}$ .)

### 5.4.6.3 C. Going deeper

### Exercise C1.

Let  $\mathbf{X}$  have full rank and let a new case  $(\mathbf{x}_0, y_0)$  be appended. Using Theorem 1.6.5, show that the new estimate is

$$\hat{\boldsymbol{\beta}}_{\text{new}} = \hat{\boldsymbol{\beta}} + \frac{(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0 (y_0 - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}})}{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}.$$

Interpret: the estimate moves in proportion to the new case's prediction error. This is the basis of *recursive least squares*; Chapter 10 treats the numerically stable version.

## 5.5 Properties of the estimator under second-moment assumptions

Least squares was defined for a fixed data vector. Now let the data be random, generated by the linear model of Definition 5.3.1:

$$\mathbf{Y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n,$$

with  $\mathbf{X}$  of full rank  $p$ . The estimator  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$  is then a random vector, and so are  $\hat{\mathbf{Y}}$  and  $\hat{\boldsymbol{\varepsilon}}$ . This section computes their means and covariances. Only the second-moment assumptions are used, so the results hold for skewed, heavy-tailed or discrete errors alike, as long as they have finite variance.

### 5.5.1 Mean and covariance of the estimator

The key observation is that  $\hat{\boldsymbol{\beta}}$  is a *linear* function of  $\mathbf{Y}$ :  $\hat{\boldsymbol{\beta}} = \mathbf{A} \mathbf{Y}$  with the constant matrix  $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ . The moments of a linear transformation were worked out in Chapter 2.

### Theorem 5.5.1 — Moments of the least squares estimator

Let  $\mathbf{X}$  have full rank  $p$  and  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ .

- (a) If  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ , then  $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$ : the least squares estimator is unbiased.
- (b) If moreover  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$ , then

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (5.5.1)$$

*Proof.* Let  $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ . By Theorem 2.2.1(a),  $E(\hat{\boldsymbol{\beta}}) = \mathbf{A} E(\mathbf{Y}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta} = \boldsymbol{\beta}$ . By Theorem 2.2.1(b),

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \mathbf{A} (\sigma^2 \mathbf{I}_n) \mathbf{A}^\top = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1},$$

using the symmetry of  $(\mathbf{X}^\top \mathbf{X})^{-1}$ . ■

The proof is two lines, but the division of labour between the assumptions is worth stating. Unbiasedness needs only (L1): the errors may have unequal variances or be correlated, and  $\hat{\boldsymbol{\beta}}$  still centres on  $\boldsymbol{\beta}$ . The covariance formula needs (L2) and (L3) as well. Neither needs normality. Under normality  $\hat{\boldsymbol{\beta}}$  is itself normal, since it is a linear function of a normal vector (Theorem 3.3.1), so  $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$ . Chapter 7 builds exact inference on this (Theorem 7.5.1).

The matrix  $(\mathbf{X}^\top \mathbf{X})^{-1}$  carries all the information about how the design affects precision. Its diagonal entries give the variances of the individual coefficients, and its off-diagonal entries their covariances. When two columns are strongly positively correlated, their coefficients are typically estimated with a strong *negative* correlation: the data can tell how large the two effects are together, but not how to split the total between them.

### Corollary 5.5.2 — Linear functions of the coefficients

Under the assumptions of Theorem 5.5.1(b), let  $\mathbf{a} \in \mathbb{R}^p$  and let  $\mathbf{L}$  be a constant  $q \times p$  matrix. Then

$$E(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \mathbf{a}^\top \boldsymbol{\beta}, \quad \text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a},$$

and  $\text{Cov}(\mathbf{L}\hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{L} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{L}^\top$ . In particular, the estimated mean response at a design point  $\mathbf{x}_0$ ,  $\hat{\mu}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ , is unbiased for  $\mu_0 = \mathbf{x}_0^\top \boldsymbol{\beta}$  with  $\text{Var}(\hat{\mu}_0) = \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$ , and  $\text{Var}(\hat{\beta}_j) = \sigma^2 [(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}$ .

*Proof.* Apply Theorem 2.2.1 to  $\mathbf{L}\hat{\boldsymbol{\beta}}$ , with covariance  $\mathbf{L} \text{Cov}(\hat{\boldsymbol{\beta}}) \mathbf{L}^\top$ , and take  $\mathbf{L} = \mathbf{a}^\top$ ,  $\mathbf{L} = \mathbf{x}_0^\top$  and  $\mathbf{L} = \mathbf{e}_j^\top$ . ■

Linear functions of  $\boldsymbol{\beta}$  are what applications usually need: a single coefficient, the difference between two coefficients, the mean response at a new design point, or the change in mean response between two profiles. Chapter 7 shows that for every such function,  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  has the smallest variance among all linear unbiased estimators (the Gauss–Markov theorem, Theorem 7.2.1).

## 5.5.2 Fitted values and residuals

### Proposition 5.5.3 — Moments of fitted values and residuals

Under the linear model with  $\mathbf{X}$  of full rank and  $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ :

- (a)  $E(\hat{\mathbf{Y}}) = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{H}$ ;
- (b)  $E(\hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$  and  $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{H})$ ; in particular  $\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii})$ , where  $h_{ii}$  is the  $i$ th diagonal entry of  $\mathbf{H}$ ;
- (c)  $\text{Cov}(\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$  and  $\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\varepsilon}}) = \mathbf{0}$ ;
- (d)  $0 \leq h_{ii} \leq 1$  for every  $i$ , and  $\sum_i h_{ii} = p$ .

*Proof.*  $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$  and  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$ , with  $\mathbf{H}$  symmetric and idempotent and  $\mathbf{H}\mathbf{X} = \mathbf{X}$  (Proposition 5.4.2(a)). By Theorem 2.2.1,  $E(\hat{\mathbf{Y}}) = \mathbf{H}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{H}\mathbf{H}^\top = \sigma^2 \mathbf{H}$ ; similarly  $E(\hat{\boldsymbol{\varepsilon}}) = (\mathbf{I} - \mathbf{H})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$  and  $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{H})^2 = \sigma^2(\mathbf{I} - \mathbf{H})$ . The cross-covariances are  $\text{Cov}(\mathbf{H}\mathbf{Y}, (\mathbf{I} - \mathbf{H})\mathbf{Y}) = \sigma^2 \mathbf{H}(\mathbf{I} - \mathbf{H}) = \mathbf{0}$  and  $\text{Cov}(\mathbf{A}\mathbf{Y}, (\mathbf{I} - \mathbf{H})\mathbf{Y}) = \sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{H}) = \sigma^2(\mathbf{A} - \mathbf{A}\mathbf{H}) = \mathbf{0}$ , since  $\mathbf{A}\mathbf{H} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{A}$ . For (d),  $\sigma^2 h_{ii}$  and  $\sigma^2(1 - h_{ii})$  are variances, so both are nonnegative, and  $\sum_i h_{ii} = \text{tr } \mathbf{H} = p$ . ■

Two consequences deserve emphasis. First, residuals are *not* uncorrelated and *not* of equal variance, even when the errors are both. The residual of a case with  $h_{ii}$  near 1 has variance near zero: the fit is pulled so close to such a case that its residual says little about its error. The number  $h_{ii}$  is the **leverage** of case  $i$  (Section 6.8, Chapter 20). Second, the fitted values and the residuals are uncorrelated. Under normality they are then independent (Theorem 3.4.1), which is the key to the exact tests of Chapter 11.

## 5.5.3 The straight line

Specializing to  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ , the inverse Eq. 5.4.4 gives the classical formulas at once.

### Corollary 5.5.4 — Moments for a straight line

In the model Eq. 5.2.1 with  $S_{xx} > 0$ ,

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}, \quad \text{Var}(\hat{\beta}_0) = \sigma^2 \left( \frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right), \quad \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -\frac{\sigma^2 \bar{x}}{S_{xx}},$$

and the estimated mean response at  $x_0$  has

$$\text{Var}(\hat{\beta}_0 + \hat{\beta}_1 x_0) = \sigma^2 \left( \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}} \right). \quad (5.5.2)$$

Both estimates are unbiased, and  $\bar{Y}$  is uncorrelated with  $\hat{\beta}_1$ .



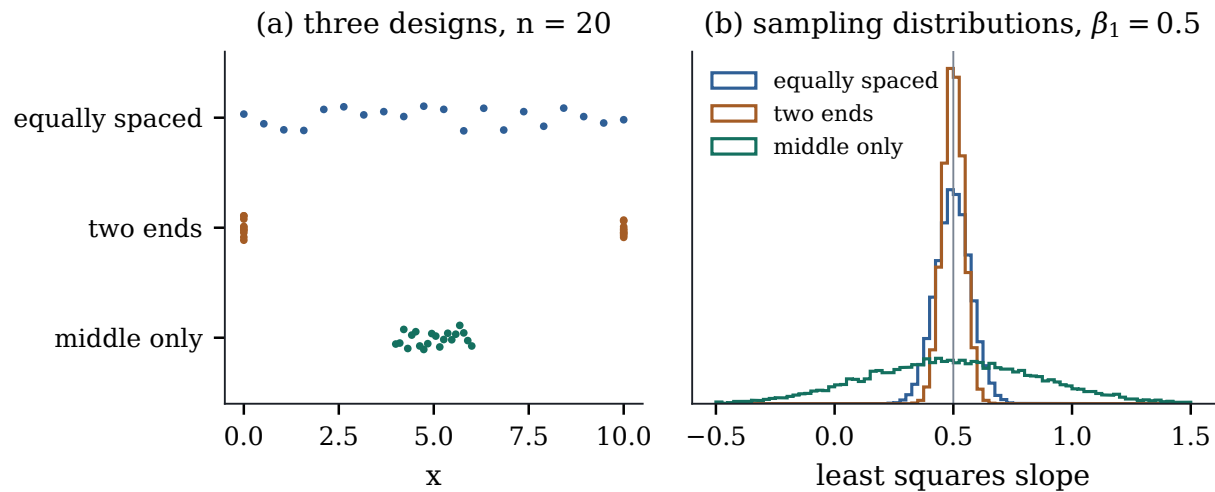


Figure 5.5.1: (a) Three designs with 20 points in  $[0, 10]$ . (b) Sampling distributions of the least squares slope under each design, with  $\beta_1 = 0.5$  and  $\sigma = 1$ . All three are centred on  $\beta_1$ ; their spreads are proportional to  $S_{xx}^{-1/2}$ .

*Proof.* The first three are the entries of  $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$  in Eq. 5.4.4. For the mean response, Corollary 5.5.2 with  $\mathbf{x}_0 = (1, x_0)^\top$  gives  $\sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} - \frac{2x_0\bar{x}}{S_{xx}} + \frac{x_0^2}{S_{xx}}\right)$ , which rearranges to Eq. 5.5.2. Finally  $\bar{Y} = \hat{\beta}_0 + \hat{\beta}_1\bar{x}$  (the line passes through the point of means), and  $\text{Cov}(\hat{\beta}_0 + \hat{\beta}_1\bar{x}, \hat{\beta}_1) = -\sigma^2\bar{x}/S_{xx} + \bar{x}\sigma^2/S_{xx} = 0$ . ■

Three design lessons can be read off. The slope is estimated precisely when the  $x_i$  are spread out, since its variance is inversely proportional to  $S_{xx}$ . The intercept is estimated precisely only near the data: its variance grows with  $\bar{x}^2$ , and the estimated intercept and slope are strongly negatively correlated when  $\bar{x}$  is large and positive, the tilting of the ellipses noted after Theorem 5.2.1. And the mean response is estimated best at  $x_0 = \bar{x}$ , with a variance that grows quadratically as  $x_0$  moves away. Extrapolation is doubly risky: the model may fail outside the range of the data, and even if it holds, the estimate is imprecise there.

#### Example. (Where to put the design points)

Suppose  $n = 20$  observations can be taken anywhere in  $[0, 10]$  to estimate a slope. Three designs are compared in Figure 5.5.1. Equally spaced points give  $S_{xx} = 184.21$  and a slope standard deviation of  $0.0737\sigma$ . Putting half the points at each end gives the largest possible  $S_{xx} = 500.00$  (Exercise B1) and  $0.0447\sigma$ . Bunching the points in the middle, over  $[4, 6]$ , gives  $S_{xx} = 7.37$  and  $0.3684\sigma$ , more than eight times worse than the two-ends design. The simulated sampling distributions in panel (b), each from 20,000 samples, match these standard deviations. The two-ends design has a price the variance formula does not show: with only two distinct values of  $x$ , the data cannot reveal curvature. The equally spaced design gives up some precision for the ability to check the model.

```
import numpy as np
n = 20
designs = {
    "equally spaced": np.linspace(0, 10, n),
```

```

    "two ends": np.repeat([0.0, 10.0], n // 2),
    "middle only": np.linspace(4, 6, n),
}
for name, xd in designs.items():
    Sxx = np.sum((xd - xd.mean()) ** 2)
    print(f"{name:15s} S_xx = {Sxx:7.2f}    sd(slope) = sigma * {1 / np.sqrt(Sxx):.4f}
    ")

```

### 5.5.4 Only two moments are needed

#### Example. (Skewed errors)

To see that normality plays no role in [Theorem 5.5.1](#), fit the quadratic  $E(Y) = \beta_0 + \beta_1 x + \beta_2 x^2$  at  $x = 1, \dots, 12$ , with  $\beta = (2, 0.5, -0.03)^\top$  and errors that are centred exponential variables scaled to  $\sigma = 1.5$ . These errors are strongly skewed. Over 100,000 simulated data sets, the averages of the three estimates are 1.9974, 0.5005 and  $-0.0300$ , and the simulated standard deviations are 1.5421, 0.5457 and 0.0409, against 1.5503, 0.5483 and 0.0411 from [Eq. 5.5.1](#). Every entry of the simulated covariance matrix is within 0.011 of the formula, measured as a correlation. The formula also predicts that  $\hat{\beta}_1$  and  $\hat{\beta}_2$  have correlation  $-0.973$ , because the columns  $x$  and  $x^2$  are nearly collinear on  $1, \dots, 12$ . A slightly larger linear coefficient can be almost exactly compensated by a slightly more negative quadratic one.

```

import numpy as np
rng = np.random.default_rng(5)
x = np.arange(1.0, 13.0)                # twelve design points
X = np.column_stack([np.ones_like(x), x, x**2])
beta = np.array([2.0, 0.5, -0.03])
sigma = 1.5
reps = 100_000

A = np.linalg.solve(X.T @ X, X.T)        # beta_hat = A y
E = sigma * (rng.exponential(size=(reps, len(x))) - 1.0) # skewed, mean 0, sd
sigma
B = (X @ beta + E) @ A.T                # one estimate per row

print("mean of estimates:", np.round(B.mean(axis=0), 4), " true:", beta)
print("Monte Carlo covariance:\n", np.round(np.cov(B.T), 5))
print("sigma^2 (X^T X)^{-1}:\n", np.round(sigma**2 * np.linalg.inv(X.T @ X), 5))

```

### 5.5.5 Random regressors

When the regressors are random, the assumptions are read conditionally on  $\mathbf{X}$  ([Section 5.1](#)). The conditional results then transfer.

#### Corollary 5.5.5 — Random regressors

Let  $(\mathbf{X}, \mathbf{Y})$  be random, with  $\text{rank}(\mathbf{X}) = p$  with probability one, and suppose  $E(\mathbf{Y} \mid$

$\mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y} \mid \mathbf{X}) = \sigma^2 \mathbf{I}_n$ . If  $E[(\mathbf{X}^\top \mathbf{X})^{-1}]$  is finite, then

$$E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}].$$

*Proof.* Conditionally on  $\mathbf{X}$ , [Theorem 5.5.1](#) gives  $E(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \boldsymbol{\beta}$  and  $\text{Cov}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ . By the laws of total expectation and covariance ([Proposition 2.5.4](#)),  $E(\hat{\boldsymbol{\beta}}) = E(\boldsymbol{\beta}) = \boldsymbol{\beta}$  and  $\text{Cov}(\hat{\boldsymbol{\beta}}) = E[\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}] + \text{Cov}(\boldsymbol{\beta}) = \sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}]$ , since  $\boldsymbol{\beta}$  is constant. The finiteness assumption guarantees  $\hat{\boldsymbol{\beta}}$  has finite second moments, which [Proposition 2.5.4](#) needs. ■

This is what [Example \(One slope, two sampling schemes\)](#) showed by simulation: under the random design, the variance of the slope was  $\sigma^2 E(1/S_{xx})$ , not  $\sigma^2/S_{xx}$  for any one design. For inference, the conditional covariance  $\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$  is the relevant one, because it describes the design actually observed.

## 5.5.6 Consistency

As the sample grows, a good estimator should approach the truth. For a fixed  $p$  and a sequence of model matrices  $\mathbf{X}_n$ , the mean squared error of  $\hat{\boldsymbol{\beta}}$  is

$$E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \text{tr Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \text{tr}(\mathbf{X}_n^\top \mathbf{X}_n)^{-1},$$

by [Theorem 5.5.1](#) and the fact that the trace of a covariance matrix is the sum of the variances.

### Proposition 5.5.6 — Consistency in mean square

Under the linear model with  $\mathbf{X}_n$  of full rank  $p$  for every  $n$ , if  $\text{tr}(\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \rightarrow 0$ , equivalently if the smallest eigenvalue of  $\mathbf{X}_n^\top \mathbf{X}_n$  tends to infinity, then  $E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 \rightarrow 0$ , and hence  $\Pr(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| > \delta) \rightarrow 0$  for every  $\delta > 0$ .

*Proof.* The first claim is the display above. The eigenvalues of  $(\mathbf{X}_n^\top \mathbf{X}_n)^{-1}$  are the reciprocals  $1/\lambda_1, \dots, 1/\lambda_p$  of those of  $\mathbf{X}_n^\top \mathbf{X}_n$ , all positive, so the trace is  $\sum_j 1/\lambda_j$ . It tends to zero iff every  $\lambda_j \rightarrow \infty$ , iff the smallest does. The last claim is Markov's inequality applied to  $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2$ :  $\Pr(\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| > \delta) \leq E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 / \delta^2$ . ■

For a straight line the condition is  $S_{xx} \rightarrow \infty$  together with  $\bar{x}^2/S_{xx} \rightarrow 0$ , which holds for any design that keeps spreading, but fails, for instance, for  $x_i = 1/i$ , where  $S_{xx}$  converges to a finite limit. Then even infinitely many observations do not pin down the slope ([Exercise \(C1\)](#)).

## 5.5.7 What if the assumptions fail?

The proof of [Theorem 5.5.1](#) shows exactly what each assumption contributes, so it also shows what goes wrong without it. Two cases recur throughout the book.

### Proposition 5.5.7 — Moments under departures

Let  $\mathbf{X}$  have full rank,  $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  and  $\hat{\boldsymbol{\beta}} = \mathbf{A}\mathbf{Y}$ .

- (a) If  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}$  for some  $\boldsymbol{\delta} \in \mathbb{R}^n$ , then  $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\delta}$ .
- (b) If  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}$  for some covariance matrix  $\boldsymbol{\Sigma}$ , then  $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$  and

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (5.5.3)$$

*Proof.* Both are [Theorem 2.2.1](#) applied to  $\mathbf{A}\mathbf{Y}$ :  $E(\mathbf{A}\mathbf{Y}) = \mathbf{A}\mathbf{X}\boldsymbol{\beta} + \mathbf{A}\boldsymbol{\delta}$  and  $\text{Cov}(\mathbf{A}\mathbf{Y}) = \mathbf{A}\boldsymbol{\Sigma}\mathbf{A}^\top$ . ■

Part (a) says that when the mean is misspecified, least squares estimates not  $\boldsymbol{\beta}$  but  $\boldsymbol{\beta}$  plus the coefficients from regressing the missing part  $\boldsymbol{\delta}$  on  $\mathbf{X}$ . The most important case is an omitted regressor, which [Section 5.7](#) examines. Part (b) says that with the wrong covariance, the estimator stays unbiased but the formula  $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$  is wrong. The “sandwich” [Eq. 5.5.3](#) is the correct covariance, and [Chapter 21](#) shows how to estimate it when  $\boldsymbol{\Sigma}$  is diagonal but unknown ([Theorem 21.4.2](#)). [Chapter 19](#) studies both departures systematically ([Theorem 19.1.2](#), [Theorem 19.2.2](#)).

## 5.5.8 Exercises

### 5.5.8.1 A. Check your understanding

#### Exercise A1.

In the model of [Proposition 5.4.4](#) with  $c = 2$  groups, show that  $\text{Var}(\hat{\beta}_1) = \sigma^2(1/n_1 + 1/n_2)$ , both from [Eq. 5.5.1](#) and directly from  $\hat{\beta}_1 = \bar{Y}_1 - \bar{Y}_2$ . For fixed  $n = n_1 + n_2$ , which split minimizes it?

#### Exercise A2.

Show that for a straight line  $h_{ii} = 1/n + (x_i - \bar{x})^2/S_{xx}$ . Which case in [Example \(Forecasting December sea temperature\)](#) has the largest leverage? Check that the leverages sum to 2.

### 5.5.8.2 B. Practice

#### Exercise B1.

Suppose  $n$  is even and the  $x_i$  may be chosen anywhere in  $[a, b]$ . Show that  $S_{xx} \leq n(b-a)^2/4$ , with equality iff half the points are at  $a$  and half at  $b$ . *Hint:*  $|x_i - c| \leq (b-a)/2$  with  $c = (a+b)/2$ , and  $S_{xx} = \sum_i (x_i - c)^2 - n(\bar{x} - c)^2$ .

*Solution.* With  $c$  the midpoint,  $S_{xx} = \sum_i (x_i - c)^2 - n(\bar{x} - c)^2 \leq \sum_i (x_i - c)^2 \leq n(b-a)^2/4$ . Equality in the second step needs every  $x_i \in \{a, b\}$ , and in the first step needs  $\bar{x} = c$ , that is, equally many points at each end. For  $[0, 10]$  and  $n = 20$  this gives  $S_{xx} = 20 \cdot 25 = 500$ , the value in [Example \(Where to put the design points\)](#).

### Exercise B2.

In [Example \(Money demand in Denmark\)](#), write the variances of  $\hat{\beta}_2 - \hat{\beta}_3$  and  $\hat{\beta}_2 + \hat{\beta}_3$ , the difference and the sum of the two interest-rate coefficients, in terms of the entries of  $\mathbf{G} = (\mathbf{X}^\top \mathbf{X})^{-1}$ . Show that if  $\hat{\beta}_2$  and  $\hat{\beta}_3$  are negatively correlated, the difference is estimated *less* precisely, and the sum *more* precisely, than they would be if the two estimates were uncorrelated. The two rate columns have sample correlation 0.80. Which combination would you expect to be better determined?

*Solution.* With  $\mathbf{a} = (0, 0, 1, \mp 1)^\top$  and indices counting the intercept as the first coefficient, [Corollary 5.5.2](#) gives  $\text{Var}(\hat{\beta}_2 \mp \hat{\beta}_3) = \sigma^2(g_{33} + g_{44} \mp 2g_{34})$ , and  $\text{Cov}(\hat{\beta}_2, \hat{\beta}_3) = \sigma^2 g_{34}$ . If  $g_{34} < 0$ , the variance of the difference exceeds  $\sigma^2(g_{33} + g_{44}) = \text{Var}(\hat{\beta}_2) + \text{Var}(\hat{\beta}_3)$ , and that of the sum falls below it. Positively correlated columns tend to produce negatively correlated estimates (as in [Example \(Skewed errors\)](#)), because the data can tell how large the combined effect of the two rates is more easily than how to divide it between them. So the sum should be the better determined combination.

### 5.5.8.3 C. Going deeper

#### Exercise C1.

Let  $x_i = 1/i$  for  $i = 1, 2, \dots$  in the straight-line model. Show that  $S_{xx}$  converges to a finite limit as  $n \rightarrow \infty$ , so that  $\text{Var}(\hat{\beta}_1)$  does not tend to zero. Then construct a sequence of designs, in which the points may change with  $n$  (for example, designs centred farther and farther from 0), with  $S_{xx} \rightarrow \infty$  but  $\bar{x}^2/S_{xx} \not\rightarrow 0$ , and decide whether the slope, the intercept, or both fail to be consistent.

*Solution.* With  $x_i = 1/i$ ,  $S_{xx} = \sum_{i \leq n} 1/i^2 - n\bar{x}^2$ . The sum increases to  $\pi^2/6$ , and  $n\bar{x}^2 = H_n^2/n \rightarrow 0$  because the harmonic number  $H_n = \sum_{i \leq n} 1/i$  grows like  $\log n$ . So  $S_{xx} \rightarrow \pi^2/6$  and  $\text{Var}(\hat{\beta}_1) = \sigma^2/S_{xx}$  stays above  $6\sigma^2/\pi^2$ .

For even  $n$  take  $x_i = n + (-1)^i$ ,  $i = 1, \dots, n$ : half the points at  $n-1$ , half at  $n+1$ . Then  $\bar{x} = n$  and  $S_{xx} = n$ , so  $\bar{x}^2/S_{xx} = n \rightarrow \infty$ . By [Corollary 5.5.4](#),  $\text{Var}(\hat{\beta}_1) = \sigma^2/n \rightarrow 0$ , so the slope is consistent (Chebyshev's inequality), while  $\text{Var}(\hat{\beta}_0) = \sigma^2(1/n + n) \rightarrow \infty$ . The intercept, the mean response at  $x = 0$ , ever farther from the data, is not consistent: with normal errors,  $\hat{\beta}_0 - \beta_0$  is normal with unbounded variance.

## 5.6 Estimating the error variance

The covariance formula  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$  is of little use until  $\sigma^2$  is estimated. Least squares does not estimate it: the criterion  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  involves only  $\boldsymbol{\beta}$ . But the size of the residuals clearly carries information about the size of the errors. This section finds the right way to turn one into

an estimate of the other, and shows why the obvious way is wrong.

## 5.6.1 The residual sum of squares

Since  $\sigma^2 = \text{Var}(\varepsilon_i) = E(\varepsilon_i^2)$ , a natural first idea is to replace each unobserved error by its residual and average:  $\text{SSE}/n$ . This underestimates  $\sigma^2$ . The fitted values were chosen to make the residuals as small as possible, so the residuals are systematically smaller than the errors. The next theorem measures exactly how much smaller.

### Theorem 5.6.1 — Unbiased estimation of the error variance

In the linear model [Definition 5.3.1](#) with  $\text{rank}(\mathbf{X}) = p < n$ ,

$$E(\text{SSE}) = \sigma^2(n - p).$$

Hence

$$s^2 = \frac{\text{SSE}}{n - p} = \frac{\|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2}{n - p} \quad (5.6.1)$$

is an unbiased estimator of  $\sigma^2$ .

*Proof.* By [Proposition 5.4.2\(e\)](#) the residual vector is  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$ , whatever  $\boldsymbol{\beta}$  is. Since  $\mathbf{I} - \mathbf{H}$  is symmetric and idempotent,  $\text{SSE} = \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}} = \boldsymbol{\varepsilon}^\top (\mathbf{I} - \mathbf{H})\boldsymbol{\varepsilon}$ , a quadratic form in the error vector, which has mean  $\mathbf{0}$  and covariance  $\sigma^2 \mathbf{I}_n$ . By [Corollary 2.3.2\(b\)](#),

$$E(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{H}) + \mathbf{0}^\top (\mathbf{I} - \mathbf{H})\mathbf{0} = \sigma^2(n - p),$$

using  $\text{tr}(\mathbf{I} - \mathbf{H}) = n - p$  from [Proposition 5.4.2\(a\)](#). ■

The computation is the case  $\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta}$  of [Example \(Residual sum of squares under a wrong mean\)](#) in [Chapter 2](#), specialized to full rank.

The same result follows from [Proposition 5.5.3](#) without quadratic forms. Each residual has mean zero and variance  $\sigma^2(1 - h_{ii})$ , so  $E(\text{SSE}) = \sum_i E(\hat{\varepsilon}_i^2) = \sigma^2 \sum_i (1 - h_{ii}) = \sigma^2(n - p)$ . This version shows *where* the shortfall comes from: each residual is less variable than its error by the factor  $1 - h_{ii}$ , and the leverages add up to  $p$ .

The divisor  $n - p$  is the number of **residual degrees of freedom**. The name reflects the constraints  $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$  of [Proposition 5.4.2\(b\)](#): the  $n$  residuals satisfy  $p$  independent linear equations, so only  $n - p$  of them can vary freely. [Chapter 6](#) makes this precise: the residual vector lies in a subspace of dimension  $n - p$ , and the degrees of freedom of a sum of squares is always the dimension of the space it lives in. For the straight line  $p = 2$ , so  $s^2 = \text{SSE}/(n - 2)$ . For the sample variance of a single sample, the model matrix is  $\mathbf{1}$ ,  $p = 1$ , and [Eq. 5.6.1](#) is the familiar divisor  $n - 1$ .

### Corollary 5.6.2 — Estimated covariance and standard errors

Under the conditions of [Theorem 5.6.1](#),  $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$  is an unbiased estimator of  $\text{Cov}(\hat{\boldsymbol{\beta}})$ , and  $s^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$  is an unbiased estimator of  $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})$ . The **standard error** of  $\hat{\beta}_j$  is

$$\text{se}(\hat{\beta}_j) = s \sqrt{[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}}.$$

*Proof.*  $(\mathbf{X}^\top \mathbf{X})^{-1}$  and  $\mathbf{a}$  are constants, so the expectations are  $E(s^2)(\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$  and similarly for the scalar. ■

The standard error is not unbiased for the standard deviation of  $\hat{\beta}_j$ . By Jensen's inequality  $E(s) \leq \sqrt{E(s^2)} = \sigma$ , with strict inequality unless  $s$  is constant. The bias is small when  $n - p$  is large, and it matters little, since the standard error is used mainly through the ratio  $(\hat{\beta}_j - \beta_j) / \text{se}(\hat{\beta}_j)$ , whose exact distribution under normality is the  $t$  distribution with  $n - p$  degrees of freedom ([Chapter 7](#), [Theorem 7.5.1](#)).

#### Example. (Standard errors for money demand)

For the money demand model of [Example \(Money demand in Denmark\)](#),  $n = 55$  and  $p = 4$ , so

$$s^2 = \frac{0.09253}{51} = 0.001814, \quad s = 0.04259.$$

Since the response is a logarithm,  $s$  says that money holdings typically deviate from the fitted relation by about 4 percent. The standard errors of the four coefficients are 0.5811, 0.0940, 0.3282 and 0.6911. The income coefficient 1.2958 is almost fourteen standard errors from zero and the bond rate coefficient  $-2.6163$  about eight, but the deposit rate coefficient 0.6186 is smaller than its standard error. Whether the deposit rate belongs in the model at all is a testing question for [Chapter 11](#).

These standard errors rest on (L2) and (L3). For a quarterly economic series (L3) is doubtful. The correlation between consecutive residuals is 0.62, which suggests that the errors are positively correlated in time, and then the formula  $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$  usually understates the true variability. [Chapter 21](#) shows how to detect and allow for this.

```
import numpy as np
import statsmodels.api as sm
dk = sm.datasets.danish_data.load_pandas().data
y = dk["lrm"].to_numpy()
X = np.column_stack([np.ones(len(y)), dk["lry"], dk["ibo"], dk["ide"]])
n, p = X.shape
XtX = X.T @ X
Xty = X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)          # solve X^T X b = X^T y
fitted = X @ beta_hat
resid = y - fitted
SSE = resid @ resid

s2 = SSE / (n - p)                            # unbiased estimate of sigma^2
cov_hat = s2 * np.linalg.inv(XtX)             # estimated Cov(beta_hat)
```

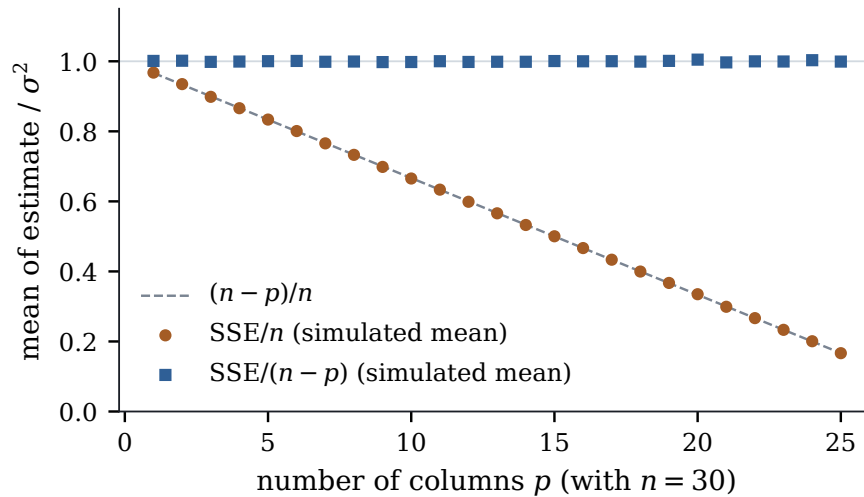


Figure 5.6.1: Average of two estimators of  $\sigma^2 = 1$  over 20,000 simulated data sets, with  $n = 30$  and  $p$  columns. Dividing the residual sum of squares by  $n$  underestimates  $\sigma^2$  by the factor  $(n-p)/n$ ; dividing by  $n-p$  is unbiased for every  $p$ .

```
se = np.sqrt(np.diag(cov_hat))
print(f"s^2 = {s2:.6f}, s = {np.sqrt(s2):.5f}")
print("standard errors:", np.round(se, 4))
print("SSE / n would give", SSE / n)
```

## 5.6.2 Dividing by $n$

The estimator  $\hat{\sigma}^2 = SSE/n$  is not unbiased, but it is not arbitrary either: under normality it is the maximum likelihood estimator (Theorem 7.3.1). Its expectation is

$$E(\hat{\sigma}^2) = \frac{n-p}{n} \sigma^2 = \sigma^2 - \frac{p}{n} \sigma^2.$$

The bias is negligible when  $p$  is a small fraction of  $n$ , and severe when it is not. For the money demand data the two estimates of  $\sigma$  are 0.04259 and 0.04102, a difference of about 4 percent. But a model with many columns relative to the number of cases, such as a two-way layout with a few replicates per cell or a regression with many indicator variables, can make  $SSE/n$  badly misleading.

### Example. (How bad is dividing by $n$ ?)

Take  $n = 30$  cases and model matrices with an intercept and  $p - 1$  further columns of standard normal numbers, for  $p = 1, \dots, 25$ . For each  $p$ , 20,000 error vectors are simulated with  $\sigma^2 = 1$ . Figure 5.6.1 plots the average of  $SSE/n$  and of  $SSE/(n-p)$ . The first follows the line  $(n-p)/n$ : at  $p = 5$  it averages 0.834, at  $p = 15$  it averages 0.500, half the true value, and at  $p = 25$  only 0.167. The unbiased estimator averages 1.000, 1.001 and 0.999 at the same values of  $p$ .

```
import numpy as np
rng = np.random.default_rng(506)
n, sigma2, reps = 30, 1.0, 20_000
Xfull = np.column_stack([np.ones(n), rng.normal(size=(n, 24))])
```



```

ps = np.arange(1, 26)
mean_n, mean_np = [], []
for p in ps:
    X = Xfull[:, :p]
    R = np.eye(n) - X @ np.linalg.solve(X.T @ X, X.T)      # I - H
    E = rng.standard_normal((reps, n))                      # errors, sigma^2 = 1
    SSE = np.einsum("ri,ij,rj->r", E, R, E)                 # e^T (I - H) e
    mean_n.append(SSE.mean() / n)
    mean_np.append(SSE.mean() / (n - p))
print("p      E(SSE/n)  E(SSE/(n-p))")
for p in (1, 5, 15, 25):
    print(f"{p:2d}      {mean_n[p - 1]:.3f}      {mean_np[p - 1]:.3f}")

```

### 5.6.3 The variance of $s^2$

Unbiasedness says where  $s^2$  is centred, not how much it varies. Its variance depends on the fourth moment of the errors, so the second-moment assumptions are not enough to find it. With independent errors, [Chapter 2](#) supplies the formula.

#### Proposition 5.6.3 — Variance of $s^2$

In the linear model with  $\text{rank}(\mathbf{X}) = p < n$ , suppose the errors  $\varepsilon_1, \dots, \varepsilon_n$  are independent with mean 0, variance  $\sigma^2$ , common third moment  $\mu_3 = E(\varepsilon_i^3)$  and common fourth moment  $\mu_4 = E(\varepsilon_i^4) < \infty$ . Then

$$\text{Var}(\text{SSE}) = 2\sigma^4(n - p) + (\mu_4 - 3\sigma^4) \sum_{i=1}^n (1 - h_{ii})^2.$$

For normal errors,  $\mu_4 = 3\sigma^4$  and  $\text{Var}(s^2) = 2\sigma^4/(n - p)$ .

*Proof.* Apply [Theorem 2.3.3](#) to the vector  $\boldsymbol{\varepsilon}$ , whose means  $\theta_i$  are all zero, with  $\mathbf{A} = \mathbf{I} - \mathbf{H}$ . The terms involving  $\boldsymbol{\theta}$  vanish (in particular  $\mu_3$  enters only multiplied by  $\boldsymbol{\theta} = \mathbf{0}$ , so its value is irrelevant). The diagonal of  $\mathbf{A}$  has entries  $1 - h_{ii}$ , and  $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{A}) = n - p$  because  $\mathbf{A}$  is idempotent. For the normal case, the fourth moment of a  $N(0, \sigma^2)$  variable is  $3\sigma^4$ , and  $\text{Var}(s^2) = \text{Var}(\text{SSE})/(n - p)^2$ . ■

The kurtosis term can be large. Heavy-tailed errors, with  $\mu_4$  much larger than  $3\sigma^4$ , make  $s^2$  much less reliable than the normal-theory formula suggests, even though it remains unbiased. Since  $0 \leq 1 - h_{ii} \leq 1$ , the sum in the kurtosis term is at most  $\sum_i (1 - h_{ii}) = n - p$ , which gives the following.

#### Corollary 5.6.4 — Consistency of $s^2$

Under the conditions of [Proposition 5.6.3](#),

$$\text{Var}(s^2) \leq \frac{2\sigma^4 + |\mu_4 - 3\sigma^4|}{n - p}.$$

So if  $n - p \rightarrow \infty$  along a sequence of models,  $s^2 \rightarrow \sigma^2$  in mean square and in probability.

*Proof.* Divide the formula of [Proposition 5.6.3](#) by  $(n - p)^2$  and bound the sum by  $n - p$ . Since  $E(s^2) = \sigma^2$ ,  $E(s^2 - \sigma^2)^2 = \text{Var}(s^2) \rightarrow 0$ , and Chebyshev's inequality gives convergence in probability. ■

### Example. (Skewed errors inflate the variance of $s^2$ )

With  $n = 30$  and the first  $p = 6$  columns of the model matrix of [Example \(How bad is dividing by  \$n\$ ?\)](#), the leverages give  $\sum_i (1 - h_{ii})^2 = 19.431$ . For normal errors with  $\sigma^2 = 1$ ,  $\text{Var}(\text{SSE}) = 2(n - p) = 48.00$ . For centred exponential errors, which have  $\sigma^2 = 1$  and  $\mu_4 = 9$ , [Proposition 5.6.3](#) gives  $48 + 6 \times 19.431 = 164.59$ , and a simulation with 200,000 data sets gives 165.19. The standard deviation of  $s^2$  is 0.289 under normal errors and 0.535 under exponential errors, nearly twice as large, although  $s^2$  is unbiased in both cases.

```
p = 6
X = Xfull[:, :p]
H = X @ np.linalg.solve(X.T @ X, X.T)
R = np.eye(n) - H
mu4 = 9.0                                     # fourth moment of a centred Exp(1)
variable
var_normal = 2 * (n - p)
var_exp = (mu4 - 3) * np.sum(np.diag(R) ** 2) + 2 * (n - p)
reps2 = 200_000
E = rng.exponential(size=(reps2, n)) - 1.0
SSE = np.einsum("ri,ij,rj->r", E, R, E)
print(f"Var(SSE): normal errors {var_normal:.2f}, exponential errors {var_exp:.2f},"
      f" simulated {SSE.var():.2f}")
```

Under normality much more is true:  $\text{SSE}/\sigma^2$  has the chi-squared distribution with  $n - p$  degrees of freedom and is independent of  $\hat{\beta}$  ([Theorem 7.5.1](#), using [Theorem 4.3.2](#) and [Theorem 4.4.1](#)). That is what turns  $\hat{\beta}$ ,  $s^2$  and the standard errors into exact confidence intervals and tests. Among unbiased estimators of  $\sigma^2$ ,  $s^2$  then has the smallest variance ([Theorem 7.4.6](#)).

## 5.6.4 When the mean is wrong

[Theorem 5.6.1](#) assumed that  $E(\mathbf{Y}) = \mathbf{X}\beta$ . If instead  $E(\mathbf{Y}) = \boldsymbol{\theta}$  is not of this form, [Example \(Residual sum of squares under a wrong mean\)](#) in [Chapter 2](#) shows that

$$E(\text{SSE}) = \sigma^2(n - p) + \|(\mathbf{I} - \mathbf{H})\boldsymbol{\theta}\|^2.$$

The extra term is the squared distance from the true mean vector to the nearest mean vector the model can produce. So an incomplete model *overestimates*  $\sigma^2$ : variation that the model fails

to capture is counted as error. This is one reason to prefer a model that includes all relevant regressors, and it is the basis of the lack-of-fit tests of [Chapter 14 \(Theorem 14.6.2\)](#), which compare  $s^2$  with an estimate of  $\sigma^2$  that does not depend on the form of the mean.

## 5.6.5 Exercises

### 5.6.5.1 A. Check your understanding

#### Exercise A1.

For regression through the origin,  $E(Y_i) = \beta x_i$ , what is the unbiased estimator of  $\sigma^2$ ? Verify  $E(\text{SSE}) = \sigma^2(n-1)$  directly, by writing  $\text{SSE} = \sum_i \varepsilon_i^2 - (\sum_i x_i \varepsilon_i)^2 / \sum_i x_i^2$ .

#### Exercise A2.

For [Example \(Forecasting December sea temperature\)](#), compute  $s$  and the standard error of the slope. Is the slope more than two standard errors away from 1, the value that would mean December simply tracks August one for one?

### 5.6.5.2 B. Practice

#### Exercise B1.

Show that  $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$ , and that for a model with an intercept  $\text{SSE} = S_{yy} - \hat{\boldsymbol{\beta}}_1^\top \mathbf{X}_c^\top \mathbf{X}_c \hat{\boldsymbol{\beta}}_1$  in the notation of [Proposition 5.4.3](#). Explain why the second form is numerically preferable when the responses are large and nearly equal, as in [Example \(A curved trend and a seasonal cycle\)](#).

*Solution.* By [Proposition 5.4.2\(c\)](#),  $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ , and  $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$  by the normal equations. For the centred form, the model with columns  $[\mathbf{1}, \mathbf{X}_c]$  has the same fitted values (the proof of [Proposition 5.4.3](#)), with coefficients  $(\bar{y}, \hat{\boldsymbol{\beta}}_1)$ . So  $\text{SSE} = \mathbf{y}^\top \mathbf{y} - n\bar{y}^2 - \hat{\boldsymbol{\beta}}_1^\top \mathbf{X}_c^\top \mathbf{X}_c \hat{\boldsymbol{\beta}}_1$ , because the cross-product matrix is block diagonal, and  $\mathbf{y}^\top \mathbf{y} - n\bar{y}^2 = S_{yy}$ . When the responses are about 340 with variation of a few units,  $\mathbf{y}^\top \mathbf{y}$  and  $\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$  are both near  $n \cdot 340^2$  and agree in their leading digits; their difference loses those digits to cancellation. The centred form subtracts numbers of the size of the variation itself.

#### Exercise B2.

Under normality  $\text{SSE}/\sigma^2$  has mean  $n-p$  and variance  $2(n-p)$ . Among estimators  $c \cdot \text{SSE}$ , find the  $c$  that minimizes the mean squared error  $E(c \text{SSE} - \sigma^2)^2$ . Compare with  $1/(n-p)$  and  $1/n$ .

*Solution.* Write  $\text{SSE} = \sigma^2 Q$  with  $E Q = m = n-p$  and  $E Q^2 = 2m + m^2$ . Then  $E(cQ - 1)^2 \sigma^4 = \sigma^4 (c^2(m^2 + 2m) - 2cm + 1)$ , minimized at  $c = m/(m^2 + 2m) = 1/(n-p+2)$ . The minimum-MSE estimator divides by  $n-p+2$ : it is biased downward but less variable. The ordering of divisors is  $n-p < n-p+2$  and, when  $p \leq 2$ ,  $n \leq n-p+2$ , so for small  $p$  the maximum likelihood divisor  $n$  lies between the unbiased and the minimum-MSE

choice. Unbiasedness is a convention, not an optimality property; it is kept because it makes  $s^2$  combine cleanly in the  $t$  and  $F$  statistics.

### 5.6.5.3 C. Going deeper

#### Exercise C1.

Show that a quadratic form  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  with  $\mathbf{A}$  symmetric is unbiased for  $\sigma^2$  for every  $\boldsymbol{\beta}$  and  $\sigma^2$  iff  $\mathbf{X}^\top \mathbf{A} \mathbf{X} = \mathbf{0}$  and  $\text{tr } \mathbf{A} = 1$ . Verify that  $\mathbf{A} = (\mathbf{I} - \mathbf{H})/(n - p)$  qualifies, and find another  $\mathbf{A}$  that does. (Under normality, [Theorem 7.4.6](#) shows that  $s^2$  has the smallest variance among all unbiased estimators.)

## 5.7 Interpreting coefficients

A fitted regression is usually reported as a table of coefficients. The short answer, “the change in the mean response per unit change in the regressor, holding the other regressors fixed”, is a useful slogan and a frequent source of error. This section makes the slogan precise within the model, shows how the meaning of a coefficient depends on units, centring, transformations and the other columns of the model, and marks where the slogan stops being true.

### 5.7.1 What a coefficient measures

#### Definition 5.7.1 — Interpretation of a coefficient

In a linear model  $E(Y | \mathbf{x}) = \beta_0 + \beta_1 x_1 + \cdots + \beta_k x_k$ , suppose the regressor  $x_j$  enters the model only through its own column, so that no other column is a function of  $x_j$ . For any two regressor profiles  $\mathbf{x}$  and  $\mathbf{x}'$  that agree in every regressor except the  $j$ th, and have  $x'_j = x_j + 1$ ,

$$E(Y | \mathbf{x}') - E(Y | \mathbf{x}) = \beta_j. \quad (5.7.1)$$

The coefficient  $\beta_j$  is the **partial effect** of  $x_j$  in this model: the difference in mean response per unit difference in  $x_j$ , between profiles that agree on all the other regressors in the model. Its estimate  $\hat{\beta}_j$  is the least squares estimate of this difference.

Three qualifications are built into the definition.

1. **It compares profiles, not interventions.** [Eq. 5.7.1](#) compares the mean response of two kinds of case that exist in the population the model describes. Whether changing  $x_j$  in a given case would change its response by  $\beta_j$  is a causal question, which the model alone cannot answer ([Chapter 25](#)).
2. **“The other regressors” means the ones in the model.** Adding or removing a regressor changes the comparison being made, and so changes  $\beta_j$ , unless the regressors are orthogonal. [Proposition 5.7.2](#) gives the exact relation.
3. **It needs  $x_j$  to vary alone.** If  $x_j$  also appears in other columns, as in a polynomial or

an interaction, there are no two profiles that differ only in the  $x_j$  column, and  $\beta_j$  alone does not measure the effect of  $x_j$ .

The geometric content of “holding the others fixed” in a data set, where nobody held anything fixed, is the Frisch–Waugh–Lovell theorem of [Section 6.6](#):  $\hat{\beta}_j$  is the slope of the response on the part of  $x_j$  that is not linearly explained by the other regressors.

## 5.7.2 Units, rescaling and reparameterization

The coefficient  $\beta_j$  is measured in units of the response per unit of  $x_j$ . Changing units rescales it. More generally, any invertible linear recoding of the columns changes the coefficients in a predictable way and leaves the fit untouched.

### Proposition 5.7.1 — Reparameterization

Let  $\mathbf{X}$  be  $n \times p$  of rank  $p$ , let  $\mathbf{K}$  be a nonsingular  $p \times p$  matrix, and put  $\mathbf{Z} = \mathbf{X}\mathbf{K}$ . Then  $\mathbf{Z}$  has rank  $p$ , and the least squares estimates from the two model matrices are related by

$$\hat{\beta}_Z = \mathbf{K}^{-1} \hat{\beta}_X.$$

The fitted values, residuals, SSE and  $s^2$  are the same for both, and the estimated covariance matrices satisfy  $s^2(\mathbf{Z}^\top \mathbf{Z})^{-1} = \mathbf{K}^{-1} [s^2(\mathbf{X}^\top \mathbf{X})^{-1}] (\mathbf{K}^{-1})^\top$ .

*Proof.*  $\text{rank}(\mathbf{Z}) = \text{rank}(\mathbf{X}) = p$  by [Proposition 1.2.3\(b\)](#). Then

$$\begin{aligned} \hat{\beta}_Z &= (\mathbf{K}^\top \mathbf{X}^\top \mathbf{X} \mathbf{K})^{-1} \mathbf{K}^\top \mathbf{X}^\top \mathbf{y} \\ &= \mathbf{K}^{-1} (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{K}^\top)^{-1} \mathbf{K}^\top \mathbf{X}^\top \mathbf{y} = \mathbf{K}^{-1} \hat{\beta}_X, \end{aligned}$$

and  $\mathbf{Z} \hat{\beta}_Z = \mathbf{X} \mathbf{K} \mathbf{K}^{-1} \hat{\beta}_X = \mathbf{X} \hat{\beta}_X$ . Equal fitted values give equal residuals, SSE and  $s^2$ . The covariance identity follows from  $(\mathbf{Z}^\top \mathbf{Z})^{-1} = \mathbf{K}^{-1} (\mathbf{X}^\top \mathbf{X})^{-1} (\mathbf{K}^{-1})^\top$ . ■

With  $\mathbf{K} = \text{diag}(1, c_1, \dots, c_k)$ , measuring  $x_j$  in units  $1/c_j$  times as large divides its coefficient and its standard error by  $c_j$ , and leaves everything else alone. Centring, effect coding of a factor ([Exercise \(B1\)](#)) and many other changes of coding are also of the form  $\mathbf{X}\mathbf{K}$ . [Section 6.7](#) shows that what really stays fixed is the column space of  $\mathbf{X}$ .

**The intercept and centring.** The intercept  $\beta_0$  is the mean response at the profile where every regressor is zero. That profile may be impossible (zero income) or far outside the data (an August sea temperature of  $0^\circ\text{C}$  in [Example \(Forecasting December sea temperature\)](#)). If the regressors are centred, the intercept becomes the mean response at the average profile  $\bar{\mathbf{x}}$ , and its estimate is  $\bar{y}$  ([Proposition 5.4.3](#)). Centring does not change the slopes. It makes the intercept interpretable, and it makes  $\hat{\beta}_0$  uncorrelated with the slopes ([Exercise \(B2\)](#)). With interactions or polynomials, centring also changes what the lower-order coefficients mean, as the next subsections show.

**Standardized coefficients.** Measuring every variable in units of its own sample standard deviation gives the **standardized coefficients**  $\hat{\beta}_j^* = \hat{\beta}_j s_{x_j}/s_y$ : the difference in the response, in standard deviations, per standard deviation of  $x_j$ . They make coefficients unit-free and are

sometimes used to rank the “importance” of regressors. That use is fragile. A standard deviation is a property of the sample, not of the relation: the same relation observed over a wider range of  $x_j$  gives a larger standardized coefficient.

### 5.7.3 Logarithms

When the response or a regressor is on a logarithmic scale, the coefficients describe proportional rather than absolute differences.

- **Log–log**,  $E(\log Y) = \beta_0 + \beta_1 \log x + \dots$ : a 1 percent difference in  $x$  goes with approximately a  $\beta_1$  percent difference in  $Y$  (exactly, a factor  $1.01^{\beta_1}$  in the geometric mean). Economists call  $\beta_1$  an **elasticity**.
- **Log–level**,  $E(\log Y) = \beta_0 + \beta_1 x + \dots$ : a unit difference in  $x$  goes with a factor  $e^{\beta_1}$  in the geometric mean of  $Y$ , a difference of  $100(e^{\beta_1} - 1)$  percent, approximately  $100\beta_1$  percent when  $\beta_1$  is small.
- **Level–log**,  $E(Y) = \beta_0 + \beta_1 \log x + \dots$ : a 1 percent difference in  $x$  goes with a difference of about  $\beta_1/100$  in  $Y$ .

The phrase “geometric mean” matters.  $E(\log Y)$  is not  $\log E(Y)$ , so a model for the mean of  $\log Y$  is not a model for the mean of  $Y$ . [Chapter 22](#) returns to back-transformation ([Proposition 22.5.1](#)).

#### Example. (Reading the money demand coefficients)

In [Example \(Money demand in Denmark\)](#) the response is log real money and the first regressor is log real income, so  $\hat{\beta}_1 = 1.2958$  is an income elasticity. Comparing quarters with the same interest rates, 1 percent more real income goes with about 1.3 percent more real money. The interest rates are measured as fractions, so the bond rate coefficient  $-2.6163$  is per unit of the fraction, that is, per 100 percentage points, an absurdly large change. Rescaling the rate to percentage points ([Proposition 5.7.1](#) with  $c = 100$ ) divides the coefficient by 100: a bond rate one percentage point higher, with income and the deposit rate the same, goes with money holdings lower by the factor  $e^{-0.026163}$ , a difference of  $-2.58$  percent. The deposit rate has a positive coefficient, which is plausible if deposits are part of the money measured, since a higher deposit rate makes holding them more attractive. But the coefficient is smaller than its standard error.

The standardized coefficients of income, bond rate and deposit rate are 0.619,  $-0.529$  and 0.057. The sample standard deviation of log income is 0.0728, about 7 percent, and that of the bond rate is 0.0308, about 3 percentage points; within those ranges of variation, income and the bond rate move money holdings by similar amounts.

```
import numpy as np
import statsmodels.api as sm
dk = sm.datasets.danish_data.load_pandas().data
y = dk["lrm"].to_numpy()
X = np.column_stack([np.ones(len(y)), dk["lry"], dk["ibo"], dk["ide"]])
XtX = X.T @ X
Xty = X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)          # solve X^T X b = X^T y

X_pct = X.copy()
```

```

X_pct[:, 2:] *= 100                                # rates in percentage points
beta_pct = np.linalg.solve(X_pct.T @ X_pct, X_pct.T @ y)
print("rates as fractions:", np.round(beta_hat, 4))
print("rates in percent:  ", np.round(beta_pct, 5))
print("same fitted values:", np.allclose(X @ beta_hat, X_pct @ beta_pct))

sd = X[:, 1:].std(axis=0, ddof=1)
standardized = beta_hat[1:] * sd / y.std(ddof=1)
print("standardized slopes:", np.round(standardized, 3))

```

### 5.7.4 Terms that move together

If  $x$  enters through both  $x$  and  $x^2$ , the mean response  $\beta_0 + \beta_1 x + \beta_2 x^2$  changes at the rate

$$\frac{\partial E(Y | x)}{\partial x} = \beta_1 + 2\beta_2 x,$$

which depends on  $x$ . The coefficient  $\beta_1$  alone is this rate at  $x = 0$ , and  $\beta_2$  is half the rate at which the slope itself changes. Neither is “the effect of  $x$ ”. In [Example \(A curved trend and a seasonal cycle\)](#) the growth rate of carbon dioxide was 0.86 ppm per year in 1960 and 1.81 ppm per year in 2000. The coefficient  $\hat{\beta}_1 = 1.3344$  is the growth rate in 1980, the year chosen as the origin of  $t_c$ , and would have been different had the origin been chosen differently.

Interactions behave the same way. In [Eq. 5.3.3](#) the slope in  $x_1$  is  $\beta_1 + \beta_3 x_2$ . Reporting “the effect of  $x_1$ ” requires choosing a value of  $x_2$ , and  $\beta_1$  is the choice  $x_2 = 0$ . Centring  $x_2$  at a representative value before forming the product makes  $\beta_1$  the slope at that value. The fitted surface does not change ([Exercise \(B2\)](#)).

### 5.7.5 Indicator variables

With reference coding, the coefficient of an indicator is the difference in mean response between its category and the reference category, among profiles that agree on the other regressors. Changing the reference changes every indicator coefficient but not the fit, since the reference coding for one reference category is a reparameterization of the coding for any other.

#### Example. (Self-rated health, with and without adjustment)

In [Example \(Self-rated health\)](#), people reporting poor health made 3.160 more physician visits a year, on average, than those reporting excellent health. Add two regressors: a chronic-disease score (the variable `disea`) and an indicator of a physical limitation. The coefficient of poor health falls to 1.554, and that of fair health from 1.058 to 0.202. The good-health coefficient changes sign, to  $-0.081$ . The adjusted comparison is between people with the same disease score and limitation status. Much of the difference in visits between health categories goes with differences in measured disease, and the self-rating adds less once disease is taken into account. Neither set of numbers is “the” effect of self-rated health. They answer different questions.

The data record the same people in several years, so the cases are not independent and the second-moment assumptions do not hold exactly; [Chapter 33](#) treats such clustered data.

```

import numpy as np
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
D = rand[["hlthg", "hlthf", "hlthp"]].to_numpy(dtype=float)  # good, fair, poor
X = np.column_stack([np.ones(len(y)), D])                  # excellent =
    reference

Xa = np.column_stack([X, rand["disea"], rand["physlm"]])
beta_a = np.linalg.solve(Xa.T @ Xa, Xa.T @ y)
print("adjusted for chronic-disease score and physical limitation:")
print(np.round(beta_a, 4))

```

## 5.7.6 Omitted regressors

How does a coefficient change when other regressors are added or removed? Partition the model matrix as  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ , with  $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top$  the estimate from the full (“long”) regression, and let  $\tilde{\boldsymbol{\beta}}_1 = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{y}$  be the estimate from the “short” regression on  $\mathbf{X}_1$  alone.

### Proposition 5.7.2 — Omitted regressors

Let  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$  have full rank, and let  $\boldsymbol{\Pi} = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2$  be the coefficient matrix from regressing each column of  $\mathbf{X}_2$  on  $\mathbf{X}_1$ . Then:

- (a) for every data vector  $\mathbf{y}$ ,  $\tilde{\boldsymbol{\beta}}_1 = \hat{\boldsymbol{\beta}}_1 + \boldsymbol{\Pi} \hat{\boldsymbol{\beta}}_2$ ;
- (b) if  $E(\mathbf{Y}) = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{X}_2 \boldsymbol{\beta}_2$ , then  $E(\tilde{\boldsymbol{\beta}}_1) = \boldsymbol{\beta}_1 + \boldsymbol{\Pi} \boldsymbol{\beta}_2$ .

In particular the short and long coefficients agree, for all  $\mathbf{y}$ , iff  $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$ .

*Proof.* (a) Write  $\mathbf{y} = \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 + \mathbf{X}_2 \hat{\boldsymbol{\beta}}_2 + \hat{\boldsymbol{\varepsilon}}$ , where  $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$  and so  $\mathbf{X}_1^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$ . Then

$$\begin{aligned} \tilde{\boldsymbol{\beta}}_1 &= (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{y} \\ &= \hat{\boldsymbol{\beta}}_1 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2 \hat{\boldsymbol{\beta}}_2 + (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \hat{\boldsymbol{\varepsilon}} = \hat{\boldsymbol{\beta}}_1 + \boldsymbol{\Pi} \hat{\boldsymbol{\beta}}_2. \end{aligned}$$

(b) Take expectations in (a), using  $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$  (Theorem 5.5.1), or apply Proposition 5.5.7(a) with  $\boldsymbol{\delta} = \mathbf{X}_2 \boldsymbol{\beta}_2$ . For the last claim, if  $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$  then  $\boldsymbol{\Pi} = \mathbf{0}$ . Conversely, if  $\boldsymbol{\Pi} \neq \mathbf{0}$ , take  $\mathbf{y} = \mathbf{X}_2 \mathbf{c}$  with  $\boldsymbol{\Pi} \mathbf{c} \neq \mathbf{0}$ : the long regression gives  $\hat{\boldsymbol{\beta}}_2 = \mathbf{c}$ , and the two coefficients differ by  $\boldsymbol{\Pi} \mathbf{c}$ .

■

Part (b) is the **omitted-variable bias** formula. The short regression does not estimate  $\boldsymbol{\beta}_1$  but  $\boldsymbol{\beta}_1 + \boldsymbol{\Pi} \boldsymbol{\beta}_2$ . Each omitted regressor contributes its own coefficient times its association with the included ones. The bias is zero if the omitted regressors have no effect or are uncorrelated with the included ones, in the sense  $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$ ; otherwise it is there, and it can have either sign. Part (a) says the same thing about the numbers actually computed from a data set, with no model assumptions at all.



**Example. (Income without interest rates)**

Regressing log money on log income alone, with an intercept, gives an income elasticity of 1.819, against 1.2958 when both interest rates are included. [Proposition 5.7.2\(a\)](#) accounts for the whole difference. Regressing each rate on log income gives slopes  $-0.2215$  (bond rate) and  $-0.0917$  (deposit rate): in these data, quarters with higher income had lower interest rates, and the correlation of log income with the bond rate is  $-0.52$ . The short-regression elasticity therefore absorbs

$$(-0.2215)(-2.6163) + (-0.0917)(0.6186) = 0.5229,$$

mostly the association of higher income with lower bond rates, which in turn go with more money. Which elasticity is the “right” one depends on the question. The long regression compares quarters with equal interest rates. The short one compares quarters as they came, with whatever rates accompanied a given income.

```
X1 = X[:, :2]                                # intercept and log income
X2 = X[:, 2:]                                # the two interest rates
b_short = np.linalg.solve(X1.T @ X1, X1.T @ y)
A = np.linalg.solve(X1.T @ X1, X1.T @ X2)    # regress each rate on (1, lry)
print("short regression slope on income:", round(b_short[1], 4))
print("long slope + A @ (rate coefs):", round(beta_hat[1] + A[1] @ beta_hat[2:],
4))
print("auxiliary slopes of the rates on income:", np.round(A[1], 4))
```

*Warning* (Adjustment is not a cure). [Proposition 5.7.2](#) suggests a tempting strategy: include every variable that might be relevant, so that nothing is omitted. Two problems remain. First, some variables should *not* be held fixed. If a treatment works partly by changing an intermediate variable, adjusting for that variable removes part of the effect one wants to measure. Adjusting for a common consequence of the regressor and the response can create an association where there is none. Second, adding regressors that are nearly collinear with  $x_j$  inflates the variance of  $\hat{\beta}_j$  ([Chapter 26, Definition 26.2.1](#)). Which variables to adjust for is a question about how the data were generated, not about the fit. [Chapter 25](#) develops the tools for answering it.

## 5.7.7 Exercises

### 5.7.7.1 A. Check your understanding

#### Exercise A1.

In [Example \(Forecasting December sea temperature\)](#), suppose both temperatures are converted to degrees Fahrenheit,  $F = 32 + 1.8C$ . What happens to the slope, the intercept,  $R^2$  and  $s$ ? Answer using [Proposition 5.7.1](#), noting that the response is also transformed.

### 5.7.7.2 B. Practice

**Exercise B1.**

Show that the standardized coefficients are the slopes obtained by regressing the standardized response  $(y_i - \bar{y})/s_y$  on the standardized regressors  $(x_{ij} - \bar{x}_j)/s_{x_j}$ , without an intercept. For a single regressor, show that the standardized slope is the correlation  $r$ .

*Solution.* Let  $\mathbf{X}_c$  be the centred regressors and  $\mathbf{D} = \text{diag}(s_{x_1}, \dots, s_{x_k})$ . The standardized regressors are  $\mathbf{X}_c \mathbf{D}^{-1}$  and the standardized response is  $(\mathbf{y} - \bar{y}\mathbf{1})/s_y$ . By [Proposition 5.4.3](#) the slopes of  $\mathbf{y}$  on  $\mathbf{X}_c$  are  $\hat{\boldsymbol{\beta}}_1$ . Dividing the response by  $s_y$  divides them by  $s_y$ , since least squares is linear in the response; replacing  $\mathbf{X}_c$  by  $\mathbf{X}_c \mathbf{D}^{-1}$  multiplies them by  $\mathbf{D}$  ([Proposition 5.7.1](#) with  $\mathbf{K} = \mathbf{D}^{-1}$ ). The result is  $\mathbf{D}\hat{\boldsymbol{\beta}}_1/s_y$ , whose entries are  $\hat{\beta}_j s_{x_j}/s_y$ . For one regressor this is  $(S_{xy}/S_{xx})\sqrt{S_{xx}/S_{yy}} = r$ , as in [Eq. 5.2.6](#).

**Exercise B2.**

In the centred parameterization  $E(\mathbf{Y}) = \alpha\mathbf{1} + \mathbf{X}_c\boldsymbol{\beta}_1$  of [Proposition 5.4.3](#), show that  $\text{Cov}(\hat{\alpha}, \hat{\boldsymbol{\beta}}_1) = \mathbf{0}$ ,  $\text{Var}(\hat{\alpha}) = \sigma^2/n$  and  $\text{Cov}(\hat{\boldsymbol{\beta}}_1) = \sigma^2(\mathbf{X}_c^\top \mathbf{X}_c)^{-1}$ . Deduce that  $\text{Var}(\hat{\beta}_0) = \sigma^2(1/n + \bar{\mathbf{x}}^\top (\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \bar{\mathbf{x}})$ , which generalizes [Corollary 5.5.4](#).

*Solution.* The matrix  $[\mathbf{1}, \mathbf{X}_c]$  has full rank and  $[\mathbf{1}, \mathbf{X}_c]^\top [\mathbf{1}, \mathbf{X}_c] = \text{diag}(n, \mathbf{X}_c^\top \mathbf{X}_c)$ , because  $\mathbf{1}^\top \mathbf{X}_c = \mathbf{0}^\top$ . By [Theorem 5.5.1](#) the covariance of  $(\hat{\alpha}, \hat{\boldsymbol{\beta}}_1)^\top$  is  $\sigma^2 \text{diag}(1/n, (\mathbf{X}_c^\top \mathbf{X}_c)^{-1})$ , which gives all three claims. Since  $\hat{\beta}_0 = \hat{\alpha} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_1$ , [Corollary 5.5.2](#) gives  $\text{Var}(\hat{\beta}_0) = \sigma^2/n + \bar{\mathbf{x}}^\top \sigma^2(\mathbf{X}_c^\top \mathbf{X}_c)^{-1} \bar{\mathbf{x}}$ , the cross term being zero. With one regressor,  $\mathbf{X}_c^\top \mathbf{X}_c = S_{xx}$ .

**Exercise B3.**

In a model with an intercept and two regressors, let the short regression omit  $x_2$ . Show that the bias of the slope on  $x_1$  is  $\beta_2 S_{12}/S_{11}$ , where  $S_{12} = \sum_i (x_{i1} - \bar{x}_1)(x_{i2} - \bar{x}_2)$  and  $S_{11} = \sum_i (x_{i1} - \bar{x}_1)^2$ . Tabulate the sign of the bias according to the signs of  $\beta_2$  and the correlation of the two regressors.

**5.7.7.3 C. Going deeper****Exercise C1.**

Construct a data set with  $n = 8$  cases, two regressors and an intercept, in which the slope of  $y$  on  $x_1$  is positive in the simple regression and negative in the regression on both  $x_1$  and  $x_2$ . Explain the reversal with [Proposition 5.7.2](#), and draw the scatterplot of  $y$  against  $x_1$  with the cases labelled by  $x_2$ .

## 5.8 A panorama of regression models to come

The linear model of this chapter makes three kinds of claim: about the *mean* of the response (a linear combination of known columns), about its *variability* (a common variance, no correlation) and, in the normal version, about its *distribution*. Every model in the rest of the book relaxes

one or more of these claims, or keeps them and asks harder questions about them. This section surveys the route.

## 5.8.1 Three components of a regression model

It helps to see any regression model as three separate choices.

1. **A distribution for the response given the regressors:** normal, binomial, Poisson, gamma, or something less specified, such as “any distribution with this mean and variance”.
2. **A predictor** that combines the regressors: the linear predictor  $\eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , or a sum of smooth functions, or a sum that includes random terms.
3. **A link** between the predictor and the mean,  $g(E(Y_i)) = \eta_i$ , where  $g$  is a known monotone function.

The normal linear model chooses the normal distribution with constant variance, the linear predictor, and the identity link  $g(\mu) = \mu$ . Changing one choice at a time generates the models below.

## 5.8.2 The ladder

| Model                                 | What changes                                                                                                                                              | Where          |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| Linear model, second moments          | the starting point: geometry, optimality, estimability, computation                                                                                       | Chapters 5–10  |
| Normal linear model                   | exact tests, intervals, multiple comparisons, designed experiments                                                                                        | Chapters 11–18 |
| Departures from the assumptions       | diagnostics, transformations, resampling, measurement error, causation                                                                                    | Chapters 19–25 |
| Many regressors                       | collinearity, shrinkage, $p > n$ , selection, boosting                                                                                                    | Chapters 26–30 |
| General covariance and random effects | $\text{Cov}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{V}$ ;<br>$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\epsilon}$ | Chapters 31–33 |
| Generalized linear models             | non-normal distribution and a link                                                                                                                        | Chapters 34–41 |
| Nonparametric and additive models     | smooth functions in the predictor                                                                                                                         | Chapters 42–44 |
| Distributional regression             | regressors act on more than the mean                                                                                                                      | Chapter 45     |

**The normal linear model and its designs.** With normal errors, the least squares estimator has an exact normal distribution and the residual sum of squares an exact chi-squared distribution ([Theorem 7.5.1](#)). This gives  $t$  and  $F$  tests and exact confidence regions (Chapters 11 and 12), and procedures for many comparisons at once ([Chapter 13](#)). Analysis of variance and designed experiments (Chapters 15–18) are linear models whose columns are indicator variables, often not of full rank; their theory rests on [Chapter 6](#) and [Chapter 8](#).

**When the assumptions fail, or there are many regressors.** Part V asks what happens when the mean is misspecified, the variance is not constant, the errors are correlated or not normal, the regressors are measured with error, or the question is causal; some remedies, such as robust standard errors based on Eq. 5.5.3 (Theorem 21.4.2), stay inside the linear model. Part VI treats large  $p$  and nearly dependent columns, where ridge regression, principal components and the lasso trade a little bias for less variance. Regression trees, random forests and neural networks also estimate  $m$ , judged by prediction error rather than by the meaning of coefficients; the book does not develop them, but Chapters 29 and 30 compare predictors and treat boosting.

**General covariance and mixed models.** The general Gauss–Markov model replaces  $\sigma^2\mathbf{I}$  by  $\sigma^2\mathbf{V}$  (Chapter 31), and generalized least squares is least squares in a different inner product (Section 6.9). Linear mixed models write the response as

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon},$$

where  $\mathbf{u}$  is a vector of random effects shared by related cases, such as repeated measurements on one person or pupils in one school (Chapters 32–33). The RAND data of Example (Self-rated health), with the same people observed in several years, are of this kind.

**Generalized linear models.** For a binary response the mean is a probability, and a linear predictor can leave  $[0, 1]$ . For a count the mean must be positive and the variance grows with the mean. Generalized linear models keep the linear predictor, choose a distribution from an exponential family, and connect the two by a link: the logit  $\log\{\mu/(1 - \mu)\}$  for binary responses (Chapter 35), the logarithm for counts (Chapter 37). Coefficients act additively on the scale of the link, so in a Poisson model with log link  $e^{\beta_j}$  is the factor by which the mean changes per unit of  $x_j$ . Chapters 34–41 develop the theory, overdispersion, correlated responses and missing data.

**Nonparametric and additive models.** Polynomials, as in Example (A curved trend and a seasonal cycle), force a global shape on a curve. Splines and smoothers let the data choose the shape locally (Chapters 42–43), and additive models replace each term  $\beta_j x_j$  by an unknown smooth function  $f_j(x_j)$  (Chapter 44). Much of this is still penalized least squares on a model matrix of basis functions.

**Beyond the mean.** Quantile regression models a chosen quantile of  $Y$ , such as the median or the tenth percentile, as a linear function of the regressors, and so describes how the whole conditional distribution shifts. Distributional regression lets the regressors act on the location, the scale and the shape of the response distribution through separate predictors, turning process (b) of Figure 5.1.1 into a model rather than a violation (Chapter 45).

**The Bayesian thread.** A prior distribution on  $(\boldsymbol{\beta}, \sigma^2)$  turns the likelihood into a posterior. For the normal linear model with a conjugate prior the posterior is available in closed form (Theorem 7.6.2), and it reappears for intervals (Theorem 12.6.2) and for generalized linear models (Chapter 39).

### 5.8.3 A first look beyond the linear model

#### Example. (Physician visits are counts)

The number of physician visits in the RAND data is a count. It is zero for 31.2 percent of person-years, and its variance is 7.09 times its mean. Group the person-years into ten groups by the chronic-disease score, and compute the mean and variance of visits in each (Figure 5.8.1). The variance rises with the mean, and in every group it is between

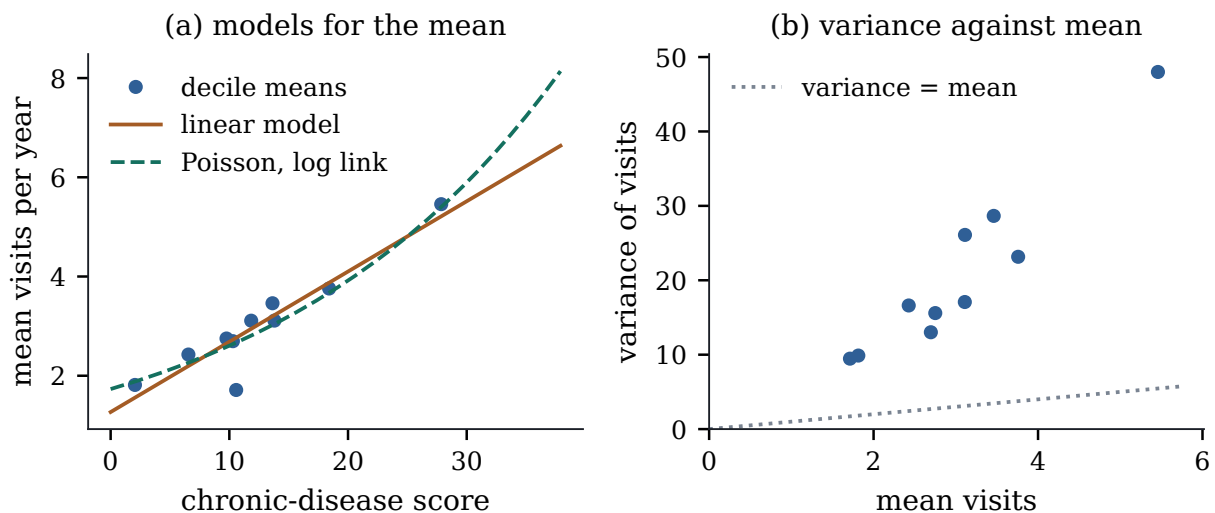


Figure 5.8.1: Annual physician visits in the RAND Health Insurance Experiment, grouped into ten groups by a chronic-disease score. (a) Group means with a linear model and a Poisson regression with log link. (b) Group variances against group means; the dotted line is the Poisson relation  $\text{variance} = \text{mean}$ .

4.8 and 8.8 times the mean. (L2) fails badly.

Two models for the mean are fitted to the disease score  $x$ . The linear model gives  $1.268 + 0.1416x$  visits. The Poisson regression with log link (Definition 37.1.1) gives  $\exp(0.549 + 0.0408x)$ : each unit of the index multiplies the mean by  $e^{0.0408}$ , 4.17 percent more visits. Over the bulk of the data the two curves are close, and both follow the group means. They differ in shape at high values of the index, where the log-linear mean grows ever faster and the linear one at a constant rate, and where the data are too sparse to decide between them. They differ more in what they assume about variability. The linear model assumes a constant variance; the Poisson model assumes variance equal to the mean. The data contradict both. Chapter 38 shows how to keep the log-linear mean and let the variance be a multiple of it.

The Poisson fit is not computed by ordinary least squares, but its estimating equations have the same form as the normal equations:  $\mathbf{X}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$ , where  $\hat{\boldsymbol{\mu}}$  is the vector of fitted means. The residuals are again orthogonal to every column of  $\mathbf{X}$ . The listing checks this.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
rand = sm.datasets.randhie.load_pandas().data
y = rand["mdvis"].to_numpy(dtype=float)
x = rand["disea"].to_numpy()
X = sm.add_constant(x)

bins = pd.qcut(x, 10, duplicates="drop") # ten groups by disease score
groups = pd.DataFrame({"y": y, "x": x}).groupby(bins, observed=True)
table = groups.agg(x=("x", "mean"), mean=("y", "mean"), var=("y", "var"))
print(table.round(2).to_string(index=False))
print(f"share of zeros {np.mean(y == 0):.3f}; variance/mean overall {y.var() / y.mean():.2f}")

linear = sm.OLS(y, X).fit()
```

```
poisson = sm.GLM(y, X, family=sm.families.Poisson()).fit()
print("linear model:      ", np.round(linear.params, 4))
print("Poisson, log link:", np.round(poisson.params, 4))
mu = poisson.fittedvalues
print("X^T (y - mu_hat) =", X.T @ (y - mu))
```

## 5.8.4 Exercises

### 5.8.4.1 A. Check your understanding

#### Exercise A1.

For each response, name the rung of the ladder you would start from, and say which of the three components (distribution, predictor, link) departs from the normal linear model: (a) whether a loan is repaid; (b) blood pressure measured monthly on the same patients; (c) the number of insurance claims per policy; (d) the 90th percentile of birth weight as a function of the mother's age; (e) crop yield in a randomized block experiment.

### 5.8.4.2 B. Practice

#### Exercise B1.

Let  $Y_i$  be binary with  $\Pr(Y_i = 1) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  (the *linear probability model*). Show that (L1) holds but (L2) fails, with  $\text{Var}(Y_i) = \mu_i(1 - \mu_i)$ . Show that least squares is still unbiased, find its covariance matrix from [Proposition 5.5.7\(b\)](#), and explain why fitted values outside  $[0, 1]$  are possible.

*Solution.*  $E(Y_i) = \Pr(Y_i = 1) = \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , which is (L1).  $\text{Var}(Y_i) = \mu_i - \mu_i^2$ , which varies with  $i$  unless all the  $\mu_i$  are equal or symmetric about  $1/2$ , so (L2) fails. Unbiasedness needs only (L1) ([Theorem 5.5.1\(a\)](#)). With independent cases,  $\boldsymbol{\Sigma} = \text{diag}(\mu_i(1 - \mu_i))$  and [Eq. 5.5.3](#) gives  $\text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$ . Nothing in least squares constrains  $\mathbf{x}^\top \hat{\boldsymbol{\beta}}$  to  $[0, 1]$ : the estimate is a linear function of  $\mathbf{y}$ , and at design points far from the centre of the data the fitted line can cross 0 or 1 even when every true probability is inside. The logit link of [Chapter 35 \(Definition 35.1.1\)](#) removes the problem.

#### Exercise B2.

For independent  $Y_i \sim \text{Poisson}(\mu_i)$  with  $\log \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , write the log-likelihood and show that its gradient is  $\mathbf{X}^\top (\mathbf{y} - \boldsymbol{\mu})$ . Deduce that when  $\mathbf{X}$  contains an intercept the fitted means sum to  $\sum_i y_i$ , and that for the intercept-only model the fitted mean is  $\bar{y}$ .

### 5.8.4.3 C. Going deeper

#### Exercise C1.

In process (b) of [Example \(Same regression function, different models\)](#) the error standard deviation is  $0.15 + 0.25x$ . Suppose instead  $\log \text{Var}(Y_i) = \gamma_0 + \gamma_1 x_i$  with unknown  $\gamma$ .

Propose a two-step procedure: fit the mean by least squares, then regress  $\log \hat{\varepsilon}_i^2$  on  $x_i$ . Show that, when the errors are normal and  $n \rightarrow \infty$  with  $p$  fixed, the slope of the second regression estimates  $\gamma_1$  consistently, while its intercept estimates  $\gamma_0 + E(\log \chi_1^2)$ . (Argue first with the errors  $\varepsilon_i$  in place of the residuals, then explain why the difference does not matter in the limit.) Compute this bias,  $-\gamma_E - \log 2 \approx -1.27$ , where  $\gamma_E$  is Euler's constant. (Chapter 45 treats models of this kind by likelihood.)

## 5.9 Summary and Notes

### 5.9.1 Summary

*Idea.*

1. The regression function  $m(\mathbf{x}) = E(Y \mid \mathbf{X} = \mathbf{x})$  is the best predictor in mean squared error, and its error is automatically uncorrelated with every function of the regressors (Proposition 5.1.1). A regression model is a claim about  $m$  and about the distribution around it, made in layers: linearity, constant variance, uncorrelated errors, normality (Section 5.1).
2. The least squares line has slope  $S_{xy}/S_{xx}$  and passes through the point of means (Theorem 5.2.1, Proposition 5.2.2). In standard units its slope is the correlation, which is regression toward the mean (Eq. 5.2.6).
3. The linear model  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ ,  $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ ,  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$ , is linear in  $\boldsymbol{\beta}$ , not in the regressors. Polynomials, harmonics, transformations, indicator variables and interactions are all linear models (Definition 5.3.1).
4. For a full-rank model matrix, least squares has the unique solution  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$  of the normal equations  $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$ , and the criterion exceeds its minimum by  $(\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}})$  (Theorem 5.4.1). The residuals are orthogonal to every column (Proposition 5.4.2). Slopes can be computed from centred data (Proposition 5.4.3), and indicator coefficients are differences of group means (Proposition 5.4.4).
5. Under the second-moment assumptions,  $\hat{\boldsymbol{\beta}}$  is unbiased with covariance  $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ , and so is every linear function  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  (Theorem 5.5.1, Corollary 5.5.2). Unbiasedness needs only the mean assumption; a wrong mean produces bias and a wrong covariance produces the sandwich formula (Proposition 5.5.7).
6.  $s^2 = \text{SSE}/(n - p)$  is unbiased for  $\sigma^2$ , because the residuals are the errors projected into a space with  $n - p$  degrees of freedom (Theorem 5.6.1). Its variance depends on the kurtosis of the errors (Proposition 5.6.3).
7. A coefficient is the difference in mean response per unit of its regressor between profiles that agree on the *other regressors in the model* (Definition 5.7.1). It changes with the units (Proposition 5.7.1) and with the set of other regressors, by exactly the amount given by the omitted-variable formula (Proposition 5.7.2). It is not by itself a causal effect.
8. Everything later in the book changes one of three components of this model: the distribution of the response, the form of the predictor, or the link between predictor and mean (Section 5.8).



## 5.9.2 Notes and sources

**History.** Least squares was published by Legendre (1805) as a method for fitting orbits, and claimed by Gauss (1809), who gave it a probabilistic justification through the normal distribution. Half a century earlier Boscovich had fitted a line to measurements of meridian arcs by making the absolute deviations as small as possible, an idea Laplace took up and that returns in this book as quantile regression. The priority dispute and the early history are told by Stigler (1981; 1986). The word *regression* comes from Galton (1886), whose study of the heights of parents and children found the phenomenon now called regression toward the mean. For Galton the slope of a fitted line measured an inherited tendency. Only later did the least squares machinery of the astronomers and the regression of the biometricians become one subject.

**The source books.** The chapter covers the material of Rencher and Schaalje (2008, chapter 1, chapter 6 and sections 7.1–7.7) and of Sen and Srivastava (1990, chapters 1 and 2), whose presentations of simple and multiple regression by calculus and matrix algebra are the closest to the one here. Sen and Srivastava’s emphasis on the Gauss–Markov conditions as a benchmark, and their discussion of centring and scaling, are reflected in [Section 5.1](#) and [Section 5.7](#). Rencher and Schaalje also treat the geometry, the normal model and  $R^2$  in their chapter 7; in this book those topics are in [Chapter 6](#), [Chapter 7](#) and [Chapter 9](#). The view of a model as a distribution, a linear predictor and a link, and the discussion of indicator variables and model matrices, follow Agresti (2015, chapter 1). The panorama of [Section 5.8](#) is organized like the survey in Fahrmeir, Kneib, Lang and Marx (2021, chapter 2), whose chapter 3 also discusses what the standard assumptions mean in applications and how covariates are coded. Seber and Lee (2003, chapter 3) and Christensen (2020, chapters 1–2) give terser, more general treatments that start from projections.

**Fixed and random regressors.** Conditioning on the regressors is standard in linear-model theory. Freedman (2009) is a careful discussion of what regression models assume and of the gap between association and causation that [Section 5.1](#) and [Section 5.7](#) point to.

**Standard errors under departures.** The sandwich formula [Eq. 5.5.3](#), with the unknown covariance matrix estimated from squared residuals, is due to White (1980) in econometrics; it is developed in [Chapter 21](#) ([Theorem 21.4.2](#)).

**Beyond the linear model.** Generalized linear models were introduced by Nelder and Wedderburn (1972) and quantile regression by Koenker and Bassett (1978). Additive models were developed and popularized by Hastie and Tibshirani (1986; 1990), and generalized additive models for location, scale and shape were proposed by Rigby and Stasinopoulos (2005). Each is the subject of a later part of the book.

**Data.** The Niño 1+2 sea surface temperatures are from the US National Oceanic and Atmospheric Administration. The carbon dioxide record is that of Keeling and Whorf (2004). The Danish money demand data are those analysed by Johansen and Juselius (1990). The RAND Health Insurance Experiment is described by Manning et al. (1987); the extract used here is the one analysed by Cameron and Trivedi (2005). All four data sets are distributed with the Python package statsmodels.

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# Chapter 6

## Projections: The Geometry of Least Squares

Least squares chooses the point of a subspace that is closest to the data. Everything else in this chapter follows from that one statement: the normal equations, why fitted values are unique when coefficients are not, the orthogonal decompositions behind every sum of squares, the Frisch–Waugh–Lovell theorem,  $R^2$  as a squared cosine, leverage, generalized least squares, and how the projection should actually be computed.

**What you need.** [Chapter 1](#) (rank, inverses, eigenvalues), [Chapter 5](#) (the linear model and least squares). Random vectors ([Chapter 2](#)) appear only in a few remarks.

### 6.0.1 Roadmap

- [Least squares as a nearest-point problem](#)
- [The subspaces we need](#)
- [Orthogonal projection](#)
- [Least squares via projection](#)
- [Nested models and sequential projection](#)
- [The Frisch–Waugh–Lovell theorem](#)
- [Reparameterization, centring and  \$R^2\$](#)
- [Leverage as geometry](#)
- [Other inner products](#)
- [Computation](#)
- [Projection in the population: best linear prediction](#)
- [Summary and notes](#)

## 6.1 Least squares as a nearest-point problem

[Chapter 5](#) introduced least squares as a calculus problem: write down the sum of squared errors as a function of the coefficients, differentiate, set the derivative to zero. That route gets the right answer, but it hides the structure of the answer. It treats the coefficient vector  $\beta$  as the basic unknown. It needs  $\mathbf{X}^\top \mathbf{X}$  to be invertible before it can say anything. And it gives no hint

of why some quantities, such as fitted values, residuals and sums of squares, behave well in every model, while others, such as individual coefficients, can be fragile or not defined at all.

This chapter starts from a different place. Instead of the  $p$ -dimensional space where  $\boldsymbol{\beta}$  lives, it works in the  $n$ -dimensional space where the data live.

### 6.1.1 Two pictures of the same data

A regression data set with  $n$  observations and a single regressor is usually drawn as a scatterplot:  $n$  points in the plane, one for each observation, with the regressor on one axis and the response on the other. Call this the **variable space** picture. It is the right picture for seeing the shape of a relationship, but it scales badly. With  $k$  regressors it needs  $k + 1$  dimensions, and it has no natural place for the model's coefficients.

The other picture turns this around. Stack the  $n$  responses into a single vector  $\mathbf{y} = (y_1, \dots, y_n)^\top$ , and regard it as *one* point in  $\mathbb{R}^n$ . Each column of the model matrix  $\mathbf{X}$  is also a vector in  $\mathbb{R}^n$ : the intercept column is  $\mathbf{1}$ , and the column for a regressor holds its  $n$  observed values. This is the **observation space** picture. Its dimension is the sample size, which is usually far more than three. But it contains only  $p + 1$  vectors of interest, and those always span a subspace of dimension at most  $p + 1$ . Most of the geometry happens in a subspace small enough to draw.

In observation space, the linear model

$$\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \boldsymbol{\beta} \in \mathbb{R}^p,$$

says one thing: the mean vector  $\mathbf{E}(\mathbf{Y})$  lies in the set

$$\mathcal{C}(\mathbf{X}) = \{\mathbf{X}\mathbf{b} : \mathbf{b} \in \mathbb{R}^p\},$$

the **column space** of  $\mathbf{X}$ . That set is a subspace of  $\mathbb{R}^n$ . The coefficients  $\boldsymbol{\beta}$  are the coordinates of  $\mathbf{E}(\mathbf{Y})$  with respect to the columns of  $\mathbf{X}$ , but the claim itself is only about *where*  $\mathbf{E}(\mathbf{Y})$  is. It does not depend on which spanning vectors are used to describe the subspace.

### 6.1.2 The least squares principle, restated

The observed  $\mathbf{y}$  is almost never in  $\mathcal{C}(\mathbf{X})$ , because noise moves it off the subspace. A natural estimate of  $\mathbf{E}(\mathbf{Y})$  is the point of  $\mathcal{C}(\mathbf{X})$  that is closest to  $\mathbf{y}$ . With Euclidean distance, this is exactly least squares:

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_{i=1}^n (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2,$$

where  $\mathbf{x}_{(i)}^\top$  is the  $i$ th row of  $\mathbf{X}$ . Minimizing over  $\mathbf{b}$  is the same as searching  $\mathcal{C}(\mathbf{X})$  for the point nearest to  $\mathbf{y}$ .

#### Definition 6.1.1 — Least squares

A vector  $\hat{\boldsymbol{\beta}} \in \mathbb{R}^p$  is a **least squares estimate** of  $\boldsymbol{\beta}$  if

$$\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\| \leq \|\mathbf{y} - \mathbf{X}\mathbf{b}\| \quad \text{for every } \mathbf{b} \in \mathbb{R}^p.$$

The vector  $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$  is the **fitted value** vector, and  $\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \hat{\mathbf{Y}}$  is the **residual** vector.

The definition does not assume that the minimizer is unique, and deliberately so. In [Section 6.4](#) we will see that the *nearest point*  $\hat{\mathbf{Y}}$  is always unique. The coefficient vector  $\hat{\boldsymbol{\beta}}$  that produces it is unique only when the columns of  $\mathbf{X}$  are linearly independent.

### 6.1.3 A picture with three observations

The smallest case that still shows all the geometry has  $n = 3$  and  $p = 2$ . Then  $\mathcal{C}(\mathbf{X})$  is a plane through the origin in  $\mathbb{R}^3$ , and  $\mathbf{y}$  is a point somewhere off the plane.

#### Example. (Three observations)

Take responses  $\mathbf{y} = (1, 4, 2)^\top$  at regressor values  $\mathbf{x} = (0, 1, 3)^\top$ , with an intercept, so that  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ . The least squares fit (computed by the listing below) is

$$\hat{\mathbf{Y}} = (2.143, 2.286, 2.571)^\top, \quad \hat{\boldsymbol{\epsilon}} = (-1.143, 1.714, -0.571)^\top,$$

with intercept 2.143 and slope 0.143. Two facts stand out. First, the residual is orthogonal to both columns of  $\mathbf{X}$ : the entries of  $\hat{\boldsymbol{\epsilon}}$  sum to zero, and  $\mathbf{x}^\top \hat{\boldsymbol{\epsilon}} = 0$ . Second, the squared lengths add:

$$\|\mathbf{y}\|^2 = 21.000 = 16.429 + 4.571 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\epsilon}}\|^2.$$

[Figure 6.1.1](#) shows the configuration. The fitted vector is the foot of the perpendicular dropped from  $\mathbf{y}$  to the plane.

```
import numpy as np

y = np.array([1.0, 4.0, 2.0])
X = np.column_stack([np.ones(3), [0.0, 1.0, 3.0]]) # intercept and one regressor

beta_hat, *_ = np.linalg.lstsq(X, y, rcond=None)
y_hat = X @ beta_hat # the point of C(X) nearest to y
e_hat = y - y_hat # the residual vector

print("fitted ", y_hat)
print("residual", e_hat)
print("X^T e ", X.T @ e_hat) # zero: the residual is orthogonal to C(X)
```

The two facts in [Example \(Three observations\)](#) are not coincidences of this data set. They are the whole theory in miniature. The residual is orthogonal to the subspace *because*  $\hat{\mathbf{Y}}$  is the nearest point, and the squared lengths add *because* the two pieces are orthogonal. The rest of the chapter makes these two statements precise and follows their consequences.

*Idea.* Fitted values are determined by the *subspace*  $\mathcal{C}(\mathbf{X})$ . Coefficients are determined by the *basis* used to describe it. Anything that depends only on  $\mathcal{C}(\mathbf{X})$  is unchanged by reparameterization, by rescaling or recoding regressors, and by rank deficiency. This covers fitted values, residuals, sums of squares,  $R^2$  and leverages.

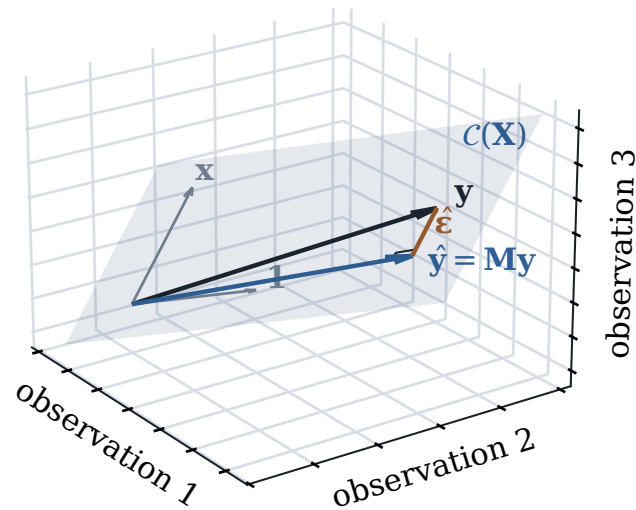


Figure 6.1.1: Least squares with  $n = 3$  observations. The model's column space  $\mathcal{C}(\mathbf{X})$  is the plane spanned by  $\mathbf{1}$  and  $\mathbf{x}$ . The fitted vector  $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$  is the point of the plane nearest to  $\mathbf{y}$ , and the residual  $\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \hat{\mathbf{Y}}$  is perpendicular to the plane.

### 6.1.4 Plan of the chapter

[Section 6.2](#) collects the facts about subspaces that the geometry needs. [Section 6.3](#) proves that nearest points exist and are unique, and identifies the matrices that compute them: the symmetric idempotent matrices, including  $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  for every generalized inverse. [Section 6.4](#) applies this to least squares. [Section 6.5](#) studies nested subspaces, which is where sums of squares come from. [Section 6.6](#) proves the Frisch–Waugh–Lovell theorem, which says what a single coefficient in a multiple regression measures. [Section 6.7](#) covers reparameterization and  $R^2$ , and [Section 6.8](#) covers leverage. [Section 6.9](#) replaces the Euclidean inner product with a general one, the step that leads to generalized least squares. [Section 6.10](#) shows how the projection should be computed in floating point.

## 6.2 The subspaces we need

This section gathers the vector-space facts used in the rest of the chapter. Readers who have worked through [Chapter 1](#) can skim the definitions. The results on orthogonal complements, especially [Lemma 6.2.1](#) and [Theorem 6.2.3](#), are used constantly afterwards and are proved in full.

### 6.2.1 Subspaces, spans and dimension

A nonempty set  $\mathcal{S} \subseteq \mathbb{R}^n$  is a **subspace** if it is closed under addition and scalar multiplication:  $\mathbf{u}, \mathbf{v} \in \mathcal{S}$  and  $a, b \in \mathbb{R}$  imply  $a\mathbf{u} + b\mathbf{v} \in \mathcal{S}$ . Every subspace contains  $\mathbf{0}$ . The **span** of vectors

$\mathbf{v}_1, \dots, \mathbf{v}_k$  is the set of all their linear combinations. It is the smallest subspace containing them. A set of vectors is *linearly independent* if the only linear combination equal to  $\mathbf{0}$  is the one with all coefficients zero. A **basis** for  $\mathcal{S}$  is a linearly independent set that spans  $\mathcal{S}$ . All bases of  $\mathcal{S}$  have the same number of elements, the **dimension**  $\dim \mathcal{S}$ .

For an  $n \times p$  matrix  $\mathbf{A}$  there are two subspaces we use constantly:

$$\begin{aligned}\mathcal{C}(\mathbf{A}) &= \{\mathbf{A}\mathbf{b} : \mathbf{b} \in \mathbb{R}^p\} \subseteq \mathbb{R}^n && \text{(column space),} \\ \mathcal{N}(\mathbf{A}) &= \{\mathbf{b} \in \mathbb{R}^p : \mathbf{A}\mathbf{b} = \mathbf{0}\} \subseteq \mathbb{R}^p && \text{(null space).}\end{aligned}$$

The **rank** of  $\mathbf{A}$  is  $\text{rank}(\mathbf{A}) = \dim \mathcal{C}(\mathbf{A})$ . Two standard facts from [Chapter 1](#): row rank equals column rank, so  $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{A}^\top)$ ; and the rank-nullity theorem

$$\text{rank}(\mathbf{A}) + \dim \mathcal{N}(\mathbf{A}) = p. \quad (6.2.1)$$

In regression,  $\mathcal{C}(\mathbf{X})$  is the **model space**.  $\mathcal{N}(\mathbf{X})$  is the set of directions in which  $\boldsymbol{\beta}$  can move without changing  $\mathbf{X}\boldsymbol{\beta}$ . It is  $\{\mathbf{0}\}$  exactly when  $\mathbf{X}$  has full column rank.

#### Example. (**A rank-deficient layout**)

Suppose  $n$  observations fall into  $g$  groups and  $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g]$ , where  $\mathbf{z}_k$  is the indicator of group  $k$ . Since  $\mathbf{z}_1 + \dots + \mathbf{z}_g = \mathbf{1}$ , the vector  $(1, -1, \dots, -1)^\top$  lies in  $\mathcal{N}(\mathbf{X})$ . The  $g$  indicators are linearly independent (their supports are disjoint), so  $\text{rank}(\mathbf{X}) = g$  while  $p = g + 1$ . By [Eq. 6.2.1](#),  $\mathcal{N}(\mathbf{X})$  is exactly the line spanned by  $(1, -1, \dots, -1)^\top$ . The model space  $\mathcal{C}(\mathbf{X}) = \mathcal{C}([\mathbf{z}_1, \dots, \mathbf{z}_g])$  is the set of vectors that are constant within groups.

## 6.2.2 Inner products, orthogonality and complements

Unless stated otherwise  $\mathbb{R}^n$  carries the Euclidean inner product  $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{v}$  and norm  $\|\mathbf{u}\| = (\mathbf{u}^\top \mathbf{u})^{1/2}$ . Vectors are **orthogonal**, written  $\mathbf{u} \perp \mathbf{v}$ , if  $\mathbf{u}^\top \mathbf{v} = 0$ . The Cauchy–Schwarz inequality  $|\mathbf{u}^\top \mathbf{v}| \leq \|\mathbf{u}\| \|\mathbf{v}\|$ , with equality iff one vector is a multiple of the other, lets us define the angle  $\theta \in [0, \pi]$  between nonzero vectors by

$$\cos \theta = \frac{\mathbf{u}^\top \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}. \quad (6.2.2)$$

The angle comes back in [Section 6.7](#), where  $R^2$  turns out to be a squared cosine. The Pythagorean identity

$$\mathbf{u} \perp \mathbf{v} \implies \|\mathbf{u} + \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 \quad (6.2.3)$$

follows by expanding  $(\mathbf{u} + \mathbf{v})^\top (\mathbf{u} + \mathbf{v})$ . The cross terms are the ones that vanish.

**Definition 6.2.1 — Orthogonal complement**

The **orthogonal complement** of a subspace  $\mathcal{S} \subseteq \mathbb{R}^n$  is

$$\mathcal{S}^\perp = \{\mathbf{v} \in \mathbb{R}^n : \mathbf{v}^\top \mathbf{u} = 0 \text{ for all } \mathbf{u} \in \mathcal{S}\}.$$

It is immediate that  $\mathcal{S}^\perp$  is a subspace and that  $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$ : a vector orthogonal to itself has zero length. To test whether  $\mathbf{v} \in \mathcal{S}^\perp$  it is enough to check orthogonality to a spanning set of  $\mathcal{S}$ . In particular,  $\mathbf{v} \in \mathcal{C}(\mathbf{A})^\perp$  iff  $\mathbf{A}^\top \mathbf{v} = \mathbf{0}$ .

**Lemma 6.2.1**

For any  $n \times p$  matrix  $\mathbf{A}$ ,

$$\mathcal{N}(\mathbf{A}^\top) = \mathcal{C}(\mathbf{A})^\perp \quad \text{and} \quad \mathcal{N}(\mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)^\perp.$$

*Proof.*  $\mathbf{v} \in \mathcal{N}(\mathbf{A}^\top)$  means  $\mathbf{A}^\top \mathbf{v} = \mathbf{0}$ , that is,  $\mathbf{v}$  is orthogonal to every column of  $\mathbf{A}$ , which is the same as  $\mathbf{v} \perp \mathcal{C}(\mathbf{A})$ . The second identity is the first applied to  $\mathbf{A}^\top$ . ■

**6.2.3 Orthonormal bases and Gram–Schmidt**

Vectors  $\mathbf{u}_1, \dots, \mathbf{u}_k$  are **orthonormal** if  $\mathbf{u}_i^\top \mathbf{u}_j$  is 1 for  $i = j$  and 0 otherwise. Orthonormal vectors are linearly independent: taking the inner product of  $\sum_j a_j \mathbf{u}_j = \mathbf{0}$  with  $\mathbf{u}_i$  gives  $a_i = 0$ . The coordinates of a vector  $\mathbf{v} \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_k)$  are simply the inner products  $a_i = \mathbf{u}_i^\top \mathbf{v}$ , with no linear system to solve. This is why orthonormal bases are so convenient.

**Proposition 6.2.2 — Gram–Schmidt**

Let  $\mathbf{v}_1, \dots, \mathbf{v}_k$  be linearly independent. Define recursively

$$\mathbf{w}_j = \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^\top \mathbf{v}_j) \mathbf{u}_i, \quad \mathbf{u}_j = \mathbf{w}_j / \|\mathbf{w}_j\|, \quad j = 1, \dots, k.$$

Then each  $\mathbf{w}_j \neq \mathbf{0}$ . The vectors  $\mathbf{u}_1, \dots, \mathbf{u}_k$  are orthonormal, and  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$  for every  $j$ .

*Proof.* Induction on  $j$ . Suppose the claims hold for  $j - 1$ . The vector  $\mathbf{w}_j$  is  $\mathbf{v}_j$  minus an element of  $\text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_{j-1})$ . If  $\mathbf{w}_j$  were zero,  $\mathbf{v}_j$  would lie in that span, contradicting independence. For  $l < j$ ,  $\mathbf{u}_l^\top \mathbf{w}_j = \mathbf{u}_l^\top \mathbf{v}_j - \sum_{i < j} (\mathbf{u}_i^\top \mathbf{v}_j) \mathbf{u}_l^\top \mathbf{u}_i = \mathbf{u}_l^\top \mathbf{v}_j - \mathbf{u}_l^\top \mathbf{v}_j = 0$ , so  $\mathbf{u}_j$  is orthogonal to its predecessors. Finally,  $\mathbf{u}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_{j-1}, \mathbf{v}_j) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_j)$  and  $\mathbf{v}_j \in \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_j)$ , so the two spans agree. ■

Two consequences are used repeatedly. Every subspace has an orthonormal basis (apply Gram–Schmidt to any basis). And any orthonormal basis of a subspace  $\mathcal{S}$  can be extended to an orthonormal basis of  $\mathbb{R}^n$ : extend it to a basis of  $\mathbb{R}^n$  in any way, then apply Gram–Schmidt, which leaves the first vectors unchanged. [Section 6.10](#) shows that Gram–Schmidt, written in matrix form, is the QR factorization.

## 6.2.4 Direct sums and the orthogonal decomposition

For subspaces  $\mathcal{S}, \mathcal{T}$ , the sum  $\mathcal{S} + \mathcal{T} = \{\mathbf{u} + \mathbf{v} : \mathbf{u} \in \mathcal{S}, \mathbf{v} \in \mathcal{T}\}$  is a subspace. When  $\mathcal{S} \cap \mathcal{T} = \{\mathbf{0}\}$  the sum is *direct*, written  $\mathcal{S} \oplus \mathcal{T}$ . Every element then decomposes as  $\mathbf{u} + \mathbf{v}$  in exactly one way: if  $\mathbf{u} + \mathbf{v} = \mathbf{u}' + \mathbf{v}'$  then  $\mathbf{u} - \mathbf{u}' = \mathbf{v}' - \mathbf{v}$  lies in both subspaces and so is zero. When in addition  $\mathcal{S} \perp \mathcal{T}$ , the sum is an *orthogonal* direct sum.

### Theorem 6.2.3 — Orthogonal decomposition

For every subspace  $\mathcal{S} \subseteq \mathbb{R}^n$ ,

$$\mathbb{R}^n = \mathcal{S} \oplus \mathcal{S}^\perp, \quad \dim \mathcal{S} + \dim \mathcal{S}^\perp = n, \quad (\mathcal{S}^\perp)^\perp = \mathcal{S}.$$

Hence every  $\mathbf{y} \in \mathbb{R}^n$  can be written uniquely as  $\mathbf{y} = \mathbf{u} + \mathbf{v}$  with  $\mathbf{u} \in \mathcal{S}$  and  $\mathbf{v} \perp \mathcal{S}$ .

*Proof.* Let  $\mathbf{u}_1, \dots, \mathbf{u}_r$  be an orthonormal basis of  $\mathcal{S}$ , extended to an orthonormal basis  $\mathbf{u}_1, \dots, \mathbf{u}_n$  of  $\mathbb{R}^n$ . A vector  $\mathbf{y} = \sum_{i=1}^n a_i \mathbf{u}_i$  is orthogonal to  $\mathcal{S}$  iff  $a_1 = \dots = a_r = 0$ , so  $\mathcal{S}^\perp = \text{span}(\mathbf{u}_{r+1}, \dots, \mathbf{u}_n)$  has dimension  $n - r$ . Writing  $\mathbf{y} = \sum_{i \leq r} a_i \mathbf{u}_i + \sum_{i > r} a_i \mathbf{u}_i$  exhibits the decomposition, and it is unique because  $\mathcal{S} \cap \mathcal{S}^\perp = \{\mathbf{0}\}$ . By the same argument with the roles of the two index sets swapped,  $(\mathcal{S}^\perp)^\perp = \text{span}(\mathbf{u}_1, \dots, \mathbf{u}_r) = \mathcal{S}$ . ■

Combining [Theorem 6.2.3](#) with [Lemma 6.2.1](#) gives a matrix fact used in almost every later chapter.

### Corollary 6.2.4

For any  $n \times p$  matrix  $\mathbf{A}$ :

- (a)  $\mathcal{N}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A})$ ;
- (b)  $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{C}(\mathbf{A}^\top)$  and  $\mathcal{C}(\mathbf{A} \mathbf{A}^\top) = \mathcal{C}(\mathbf{A})$ ;
- (c)  $\text{rank}(\mathbf{A}^\top \mathbf{A}) = \text{rank}(\mathbf{A} \mathbf{A}^\top) = \text{rank}(\mathbf{A})$ ;
- (d)  $\mathcal{C}(\mathbf{A})^\perp = \mathcal{N}(\mathbf{A}^\top)$  and  $\mathbb{R}^p = \mathcal{C}(\mathbf{A}^\top) \oplus \mathcal{N}(\mathbf{A})$ .

*Proof.*

- (a) If  $\mathbf{A}\mathbf{b} = \mathbf{0}$  then  $\mathbf{A}^\top \mathbf{A}\mathbf{b} = \mathbf{0}$ . Conversely,  $\mathbf{A}^\top \mathbf{A}\mathbf{b} = \mathbf{0}$  gives  $\mathbf{b}^\top \mathbf{A}^\top \mathbf{A}\mathbf{b} = \|\mathbf{A}\mathbf{b}\|^2 = 0$ , so  $\mathbf{A}\mathbf{b} = \mathbf{0}$ .
- (b) By [Lemma 6.2.1](#) applied to the symmetric matrix  $\mathbf{A}^\top \mathbf{A}$ , and then (a) and [Lemma 6.2.1](#)



again,  $\mathcal{C}(\mathbf{A}^\top \mathbf{A}) = \mathcal{N}(\mathbf{A}^\top \mathbf{A})^\perp = \mathcal{N}(\mathbf{A})^\perp = (\mathcal{C}(\mathbf{A}^\top)^\perp)^\perp = \mathcal{C}(\mathbf{A}^\top)$ , using  $(\mathcal{S}^\perp)^\perp = \mathcal{S}$  in the last step. The second identity is the first applied to  $\mathbf{A}^\top$ .

- (c) Take dimensions in (b) and use  $\text{rank}(\mathbf{A}^\top) = \text{rank}(\mathbf{A})$ .
- (d) This is [Lemma 6.2.1](#) and [Theorem 6.2.3](#).

■

Part (b) is the reason the normal equations  $\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$  can always be solved: the right-hand side lies in  $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ . We come back to this in [Section 6.4](#).

### Lemma 6.2.5

For subspaces  $\mathcal{S}_1, \mathcal{S}_2 \subseteq \mathbb{R}^n$ ,

$$(\mathcal{S}_1 + \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp \cap \mathcal{S}_2^\perp \quad \text{and} \quad (\mathcal{S}_1 \cap \mathcal{S}_2)^\perp = \mathcal{S}_1^\perp + \mathcal{S}_2^\perp.$$

*Proof.* A vector is orthogonal to  $\mathcal{S}_1 + \mathcal{S}_2$  iff it is orthogonal to every  $\mathbf{u}_1 + \mathbf{u}_2$ , which (taking  $\mathbf{u}_2 = \mathbf{0}$  or  $\mathbf{u}_1 = \mathbf{0}$ ) is the case iff it is orthogonal to both subspaces. For the second identity, apply the first to  $\mathcal{S}_1^\perp$  and  $\mathcal{S}_2^\perp$ :  $(\mathcal{S}_1^\perp + \mathcal{S}_2^\perp)^\perp = \mathcal{S}_1 \cap \mathcal{S}_2$ , and take complements of both sides.

■

[Lemma 6.2.5](#) is what we need to describe reduced models defined by linear constraints ([Theorem 6.5.2](#)).

## 6.2.5 Exercises

### 6.2.5.1 A. Check your understanding

#### Exercise A1.

Show that  $\mathcal{S}^\perp$  is a subspace for any set  $\mathcal{S}$  (not necessarily a subspace), and that  $\mathcal{S} \subseteq \mathcal{T}$  implies  $\mathcal{T}^\perp \subseteq \mathcal{S}^\perp$ . Show that  $\mathcal{S}^\perp = \text{span}(\mathcal{S})^\perp$ .

### 6.2.5.2 B. Practice

#### Exercise B1.

A one-way layout has three groups of sizes 2, 2, 1 and model matrix  $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$ . Find bases for  $\mathcal{C}(\mathbf{X})$ ,  $\mathcal{N}(\mathbf{X})$ ,  $\mathcal{C}(\mathbf{X}^\top)$  and  $\mathcal{C}(\mathbf{X})^\perp$ , and check that the dimensions agree with [Eq. 6.2.1](#) and [Theorem 6.2.3](#).

#### Exercise B2.

Show that  $\mathcal{C}(\mathbf{AB}) \subseteq \mathcal{C}(\mathbf{A})$ , and deduce  $\text{rank}(\mathbf{AB}) \leq \min\{\text{rank}(\mathbf{A}), \text{rank}(\mathbf{B})\}$ . Show that  $\mathcal{C}(\mathbf{AB}) = \mathcal{C}(\mathbf{A})$  whenever  $\mathbf{B}$  has full row rank.

### 6.2.5.3 C. Going deeper

#### Exercise C1.

For subspaces  $\mathcal{S}, \mathcal{T} \subseteq \mathbb{R}^n$ , prove  $\dim(\mathcal{S} + \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T} - \dim(\mathcal{S} \cap \mathcal{T})$ . Use it to show that two  $r$ -dimensional subspaces of  $\mathbb{R}^n$  with  $2r > n$  must share a nonzero vector.

## 6.3 Orthogonal projection

### 6.3.1 Nearest points

#### Theorem 6.3.1 — Projection theorem

Let  $\mathcal{S}$  be a subspace of  $\mathbb{R}^n$  and  $\mathbf{y} \in \mathbb{R}^n$ . There is exactly one  $\hat{\mathbf{y}} \in \mathcal{S}$  such that

$$\|\mathbf{y} - \hat{\mathbf{y}}\| \leq \|\mathbf{y} - \mathbf{u}\| \quad \text{for all } \mathbf{u} \in \mathcal{S}.$$

It is the unique  $\hat{\mathbf{y}} \in \mathcal{S}$  with  $\mathbf{y} - \hat{\mathbf{y}} \perp \mathcal{S}$ . Moreover, for every  $\mathbf{u} \in \mathcal{S}$ ,

$$\|\mathbf{y} - \mathbf{u}\|^2 = \|\mathbf{y} - \hat{\mathbf{y}}\|^2 + \|\hat{\mathbf{y}} - \mathbf{u}\|^2. \quad (6.3.1)$$

*Proof.* By Theorem 6.2.3 there is a unique decomposition  $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{v}$  with  $\hat{\mathbf{y}} \in \mathcal{S}$  and  $\mathbf{v} \in \mathcal{S}^\perp$ . For  $\mathbf{u} \in \mathcal{S}$ ,  $\mathbf{y} - \mathbf{u} = \mathbf{v} + (\hat{\mathbf{y}} - \mathbf{u})$ , where  $\mathbf{v} \perp \mathcal{S}$  and  $\hat{\mathbf{y}} - \mathbf{u} \in \mathcal{S}$ . The Pythagorean identity Eq. 6.2.3 then gives Eq. 6.3.1. The right-hand side is at least  $\|\mathbf{y} - \hat{\mathbf{y}}\|^2$ , with equality iff  $\mathbf{u} = \hat{\mathbf{y}}$ . So  $\hat{\mathbf{y}}$  is the unique nearest point. If some other  $\mathbf{w} \in \mathcal{S}$  also had  $\mathbf{y} - \mathbf{w} \perp \mathcal{S}$ , then  $\mathbf{y} = \mathbf{w} + (\mathbf{y} - \mathbf{w})$  would be a second decomposition, contradicting uniqueness. ■

The theorem has two halves, and it helps to keep them apart. The *variational* half says  $\hat{\mathbf{y}}$  minimizes a distance. The *orthogonality* half says the error  $\mathbf{y} - \hat{\mathbf{y}}$  is perpendicular to the subspace. The first half explains why least squares is a sensible estimation principle. The second is what we compute with, because orthogonality is a set of linear equations.

### 6.3.2 The projection as a matrix

Write  $\mathbf{P}_{\mathcal{S}}\mathbf{y}$  for the nearest point  $\hat{\mathbf{y}}$ . The map  $\mathbf{y} \mapsto \mathbf{P}_{\mathcal{S}}\mathbf{y}$  is linear: if  $\mathbf{y}_k = \hat{\mathbf{y}}_k + \mathbf{v}_k$  are the decompositions of  $\mathbf{y}_1$  and  $\mathbf{y}_2$ , then  $a\mathbf{y}_1 + b\mathbf{y}_2 = (a\hat{\mathbf{y}}_1 + b\hat{\mathbf{y}}_2) + (a\mathbf{v}_1 + b\mathbf{v}_2)$  is a decomposition of the same kind, and by uniqueness it is *the* decomposition. A linear map on  $\mathbb{R}^n$  is given by a unique  $n \times n$  matrix, since two matrices that agree on every vector agree on the standard basis vectors and are therefore equal.

### Definition 6.3.1 — Orthogonal projection matrix

The **orthogonal projection** onto a subspace  $\mathcal{S}$  is the unique  $n \times n$  matrix  $\mathbf{P}_{\mathcal{S}}$  such that, for every  $\mathbf{y}$ ,  $\mathbf{P}_{\mathcal{S}}\mathbf{y} \in \mathcal{S}$  and  $\mathbf{y} - \mathbf{P}_{\mathcal{S}}\mathbf{y} \perp \mathcal{S}$ . When  $\mathcal{S} = \mathcal{C}(\mathbf{X})$  for the model matrix of a linear model, we write  $\mathbf{M} = \mathbf{P}_{\mathcal{C}(\mathbf{X})}$ .

The most useful formula for  $\mathbf{P}_{\mathcal{S}}$  uses an orthonormal basis.

### Proposition 6.3.2

Let the columns of the  $n \times r$  matrix  $\mathbf{Q}$  be an orthonormal basis for  $\mathcal{S}$ , so that  $\mathbf{Q}^{\top}\mathbf{Q} = \mathbf{I}_r$ . Then

$$\mathbf{P}_{\mathcal{S}} = \mathbf{Q}\mathbf{Q}^{\top} = \sum_{i=1}^r \mathbf{u}_i\mathbf{u}_i^{\top},$$

where  $\mathbf{u}_1, \dots, \mathbf{u}_r$  are the columns of  $\mathbf{Q}$ .

*Proof.* For any  $\mathbf{y}$ ,  $\mathbf{Q}\mathbf{Q}^{\top}\mathbf{y} = \mathbf{Q}(\mathbf{Q}^{\top}\mathbf{y}) \in \mathcal{C}(\mathbf{Q}) = \mathcal{S}$ , and  $\mathbf{Q}^{\top}(\mathbf{y} - \mathbf{Q}\mathbf{Q}^{\top}\mathbf{y}) = \mathbf{Q}^{\top}\mathbf{y} - \mathbf{Q}^{\top}\mathbf{y} = \mathbf{0}$ , so the difference is orthogonal to every column of  $\mathbf{Q}$  and hence to  $\mathcal{S}$ . By uniqueness,  $\mathbf{Q}\mathbf{Q}^{\top} = \mathbf{P}_{\mathcal{S}}$ . ■

### Example. (Projections onto a line)

For  $\mathcal{S} = \text{span}(\mathbf{v})$  with  $\mathbf{v} \neq \mathbf{0}$ , the single vector  $\mathbf{v}/\|\mathbf{v}\|$  is an orthonormal basis, so

$$\mathbf{P}_{\mathcal{S}} = \frac{\mathbf{v}\mathbf{v}^{\top}}{\mathbf{v}^{\top}\mathbf{v}}, \quad \mathbf{P}_{\mathcal{S}}\mathbf{y} = \frac{\mathbf{v}^{\top}\mathbf{y}}{\mathbf{v}^{\top}\mathbf{v}} \mathbf{v}.$$

With  $\mathbf{v} = \mathbf{1}$  this gives  $\mathbf{P} = n^{-1}\mathbf{1}\mathbf{1}^{\top}$ , the matrix with every entry  $1/n$ , and  $\mathbf{P}\mathbf{y} = \bar{y}\mathbf{1}$ . *The sample mean is a projection:* fitting the intercept-only model replaces every observation by  $\bar{y}$ , and the residuals  $y_i - \bar{y}$  form a vector in  $\mathbf{1}^{\perp}$ , the set of vectors whose entries sum to zero.

## 6.3.3 Characterizing projection matrices

Which matrices are orthogonal projections? [Proposition 6.3.2](#) shows that every projection has the form  $\mathbf{Q}\mathbf{Q}^{\top}$ , so it is *symmetric*, and  $\mathbf{Q}\mathbf{Q}^{\top}\mathbf{Q}\mathbf{Q}^{\top} = \mathbf{Q}\mathbf{I}\mathbf{Q}^{\top} = \mathbf{Q}\mathbf{Q}^{\top}$ , so it is *idempotent*. The converse is the basic recognition theorem.

### Theorem 6.3.3

An  $n \times n$  matrix  $\mathbf{P}$  is the orthogonal projection onto some subspace iff  $\mathbf{P}^{\top} = \mathbf{P}$  and  $\mathbf{P}^2 = \mathbf{P}$ . In that case the subspace is  $\mathcal{C}(\mathbf{P})$ , and  $\mathbf{I} - \mathbf{P}$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{P})^{\perp}$ .

*Proof.* Necessity was shown above. For sufficiency, let  $\mathbf{P}$  be symmetric and idempotent and let  $\mathbf{y} \in \mathbb{R}^n$ . Certainly  $\mathbf{Py} \in \mathcal{C}(\mathbf{P})$ . For any  $\mathbf{u} = \mathbf{Pb} \in \mathcal{C}(\mathbf{P})$ ,

$$\mathbf{u}^\top (\mathbf{y} - \mathbf{Py}) = \mathbf{b}^\top \mathbf{P}^\top (\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{b}^\top (\mathbf{P} - \mathbf{P}^2)\mathbf{y} = 0,$$

so  $\mathbf{y} - \mathbf{Py} \perp \mathcal{C}(\mathbf{P})$  and  $\mathbf{P} = \mathbf{P}_{\mathcal{C}(\mathbf{P})}$ . For the last claim,  $\mathbf{I} - \mathbf{P}$  is symmetric and idempotent, since  $(\mathbf{I} - \mathbf{P})^2 = \mathbf{I} - 2\mathbf{P} + \mathbf{P}^2 = \mathbf{I} - \mathbf{P}$ . It sends  $\mathbf{y}$  to  $\mathbf{y} - \mathbf{Py} \in \mathcal{C}(\mathbf{P})^\perp$ , and  $\mathbf{y} - (\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{Py} \perp \mathcal{C}(\mathbf{P})^\perp$ . ■

Before this theorem, checking that a matrix computes a projection meant thinking about subspaces. Afterwards it is two matrix identities. Also, a projection is determined by its range: if  $\mathbf{P}_1, \mathbf{P}_2$  are symmetric idempotent with  $\mathcal{C}(\mathbf{P}_1) = \mathcal{C}(\mathbf{P}_2)$ , then  $\mathbf{P}_1 = \mathbf{P}_2$ , because both are the projection onto the same subspace.

### Proposition 6.3.4 — Spectrum, rank and trace

Let  $\mathbf{P}$  be the orthogonal projection onto an  $r$ -dimensional subspace  $\mathcal{S}$ . Then:

- (a) every eigenvalue of  $\mathbf{P}$  is 0 or 1, and 1 has multiplicity  $r$ ;
- (b)  $\text{rank}(\mathbf{P}) = \text{tr}(\mathbf{P}) = r$ ;
- (c)  $\mathbf{Pv} = \mathbf{v}$  iff  $\mathbf{v} \in \mathcal{S}$ , and  $\mathbf{Pv} = \mathbf{0}$  iff  $\mathbf{v} \in \mathcal{S}^\perp$ ;
- (d)  $\mathbf{v}^\top \mathbf{Pv} = \|\mathbf{Pv}\|^2 \leq \|\mathbf{v}\|^2$ , so  $\mathbf{P}$  and  $\mathbf{I} - \mathbf{P}$  are nonnegative definite.

*Proof.* Write  $\mathbf{P} = \mathbf{Q}\mathbf{Q}^\top$  as in Proposition 6.3.2, and extend  $\mathbf{Q}$  to an orthogonal matrix  $\mathbf{U} = [\mathbf{Q}, \mathbf{Q}_\perp]$ . Then  $\mathbf{P} = \mathbf{U} \text{diag}(\mathbf{I}_r, \mathbf{0}) \mathbf{U}^\top$  is a spectral decomposition, which gives (a). Rank and trace of a symmetric matrix are the number of nonzero eigenvalues and their sum, which gives (b). Directly,  $\text{tr}(\mathbf{Q}\mathbf{Q}^\top) = \text{tr}(\mathbf{Q}^\top \mathbf{Q}) = \text{tr}(\mathbf{I}_r) = r$ . Part (c) restates the projection theorem: the nearest point of  $\mathcal{S}$  to a vector already in  $\mathcal{S}$  is the vector itself, and  $\mathbf{v} \perp \mathcal{S}$  iff  $\mathbf{0}$  is its nearest point. For (d),  $\mathbf{v}^\top \mathbf{Pv} = \mathbf{v}^\top \mathbf{P}^\top \mathbf{Pv}$ , and  $\|\mathbf{v}\|^2 = \|\mathbf{Pv}\|^2 + \|(\mathbf{I} - \mathbf{P})\mathbf{v}\|^2$  by Pythagoras. ■

The identity  $\text{rank} = \text{tr}$  gets a lot of use. The dimension of a subspace, which is the “degrees of freedom” of every sum of squares in this book, can be read off as a sum of diagonal entries without finding a basis.

## 6.3.4 Generalized inverses

Proposition 6.3.2 needs an orthonormal basis. In statistics the subspace usually comes as the column space of a model matrix  $\mathbf{X}$  whose columns are not orthonormal and may not even be independent. To write  $\mathbf{M}$  directly in terms of  $\mathbf{X}$  we need a substitute for  $(\mathbf{X}^\top \mathbf{X})^{-1}$  that exists even when  $\mathbf{X}^\top \mathbf{X}$  is singular.

**Definition 6.3.2 — Generalized inverse**

A **generalized inverse** of an  $m \times k$  matrix  $\mathbf{A}$  is any  $k \times m$  matrix  $\mathbf{A}^-$  such that  $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$ .

If  $\mathbf{A}$  is square and nonsingular, multiplying  $\mathbf{A}\mathbf{A}^-\mathbf{A} = \mathbf{A}$  on both sides by  $\mathbf{A}^{-1}$  gives  $\mathbf{A}^- = \mathbf{A}^{-1}$ , so the generalized inverse is unique and equal to the inverse. When  $\mathbf{A}$  is singular there are infinitely many.

**Proposition 6.3.5 — Existence**

Every matrix has a generalized inverse. If  $\mathbf{A}$  has rank  $r$  and is permuted so that

$$\mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{22} \end{pmatrix}$$

with  $\mathbf{A}_{11}$  an  $r \times r$  nonsingular block, then  $\mathbf{A}^- = \begin{pmatrix} \mathbf{A}_{11}^{-1} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$  (permuted back) is a generalized inverse.

*Proof.* A rank- $r$  matrix has  $r$  independent rows and  $r$  independent columns, and the submatrix where they cross is nonsingular (Chapter 1). This guarantees the block form. Since the first  $r$  columns are a basis for the column space,  $\begin{pmatrix} \mathbf{A}_{12} \\ \mathbf{A}_{22} \end{pmatrix} = \begin{pmatrix} \mathbf{A}_{11} \\ \mathbf{A}_{21} \end{pmatrix} \mathbf{B}$  for some  $\mathbf{B}$ , so  $\mathbf{B} = \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$  and  $\mathbf{A}_{22} = \mathbf{A}_{21} \mathbf{A}_{11}^{-1} \mathbf{A}_{12}$ . Multiplying out,

$$\mathbf{A}\mathbf{A}^-\mathbf{A} = \begin{pmatrix} \mathbf{I} & \mathbf{0} \\ \mathbf{A}_{21}\mathbf{A}_{11}^{-1} & \mathbf{0} \end{pmatrix} \mathbf{A} = \begin{pmatrix} \mathbf{A}_{11} & \mathbf{A}_{12} \\ \mathbf{A}_{21} & \mathbf{A}_{21}\mathbf{A}_{11}^{-1}\mathbf{A}_{12} \end{pmatrix} = \mathbf{A}.$$

■

Among the many generalized inverses, one is singled out by extra symmetry conditions. The **Moore–Penrose inverse**  $\mathbf{A}^+$  is the unique matrix satisfying  $\mathbf{A}\mathbf{A}^+\mathbf{A} = \mathbf{A}$ ,  $\mathbf{A}^+\mathbf{A}\mathbf{A}^+ = \mathbf{A}^+$ , and both  $\mathbf{A}\mathbf{A}^+$  and  $\mathbf{A}^+\mathbf{A}$  symmetric (Penrose 1955). Section 6.10 computes it from the singular value decomposition. For projections we will need none of these extra conditions, and that is the point of the next result.

**Lemma 6.3.6**

Let  $\mathbf{X}$  be  $n \times p$  and let  $\mathbf{G}$  be any generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ . Then:

- (a)  $\mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{X} = \mathbf{X}$  and  $\mathbf{X}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top = \mathbf{X}^\top$ ;
- (b)  $\mathbf{X}\mathbf{G}\mathbf{X}^\top$  does not depend on the choice of generalized inverse;
- (c)  $\mathbf{X}\mathbf{G}\mathbf{X}^\top$  is symmetric, even when  $\mathbf{G}$  is not.

*Proof.* By Corollary 6.2.4(b),  $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ , so there is a  $p \times n$  matrix  $\mathbf{B}$  with

$\mathbf{X}^\top = \mathbf{X}^\top \mathbf{X} \mathbf{B}$ . Transposing,  $\mathbf{X} = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X}$ . Then

$$\mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X} = \mathbf{B}^\top (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X} = \mathbf{X},$$

which is the first identity in (a). The second follows the same way, using  $\mathbf{X}^\top = \mathbf{X}^\top \mathbf{X} \mathbf{B}$  on the right. For (b) and (c),

$$\mathbf{X} \mathbf{G} \mathbf{X}^\top = \mathbf{B}^\top (\mathbf{X}^\top \mathbf{X}) \mathbf{G} (\mathbf{X}^\top \mathbf{X}) \mathbf{B} = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X} \mathbf{B},$$

which involves no  $\mathbf{G}$  at all, and  $(\mathbf{B}^\top \mathbf{X}^\top \mathbf{X} \mathbf{B})^\top = \mathbf{B}^\top \mathbf{X}^\top \mathbf{X} \mathbf{B}$ . ■

### Theorem 6.3.7

For any  $n \times p$  matrix  $\mathbf{X}$  and any generalized inverse  $(\mathbf{X}^\top \mathbf{X})^-$ ,

$$\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top$$

is the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$ . In particular  $\text{rank}(\mathbf{M}) = \text{tr}(\mathbf{M}) = \text{rank}(\mathbf{X})$ . If  $\mathbf{X}$  has full column rank,  $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ .

*Proof.* By Lemma 6.3.6,  $\mathbf{M}$  is symmetric and  $\mathbf{M}^2 = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- [\mathbf{X}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top] = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top = \mathbf{M}$ . By Theorem 6.3.3,  $\mathbf{M}$  is the projection onto  $\mathcal{C}(\mathbf{M})$ . Clearly  $\mathcal{C}(\mathbf{M}) \subseteq \mathcal{C}(\mathbf{X})$ , and  $\mathbf{M}\mathbf{X} = \mathbf{X}$  (again by the lemma) shows  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{M})$ . The rank statement is Proposition 6.3.4(b). ■

It is worth pausing on how strong Theorem 6.3.7 is. The matrix  $(\mathbf{X}^\top \mathbf{X})^-$  can be almost anything: symmetric or not, reflexive or not, with entries of any size. The sandwich  $\mathbf{X}(\cdot)\mathbf{X}^\top$  removes all of that freedom. We will see in Section 6.4 that this is exactly the difference between coefficients and fitted values, made concrete.

#### Example. (The one-way layout)

In the setting of Example (A rank-deficient layout), sort the observations by group and let group  $k$  have  $n_k$  members. The model space is spanned by the orthogonal vectors  $\mathbf{z}_1, \dots, \mathbf{z}_g$ , so by Proposition 6.3.2 with  $\mathbf{u}_k = \mathbf{z}_k / \sqrt{n_k}$ ,

$$\mathbf{M} = \sum_{k=1}^g \frac{\mathbf{z}_k \mathbf{z}_k^\top}{n_k} = \begin{pmatrix} n_1^{-1} \mathbf{1}_{n_1} \mathbf{1}_{n_1}^\top & & \\ & \ddots & \\ & & n_g^{-1} \mathbf{1}_{n_g} \mathbf{1}_{n_g}^\top \end{pmatrix},$$

a block-diagonal matrix of within-group averaging operators.  $\mathbf{M}\mathbf{y}$  replaces each observation by its group mean. The intercept column adds nothing to  $\mathcal{C}(\mathbf{X})$ , so it does not appear, and  $\text{tr}(\mathbf{M}) = \sum_k n_k/n_k = g = \text{rank}(\mathbf{X})$ . We got here without a generalized inverse. Theorem 6.3.7 guarantees that any generalized inverse of the singular  $(g+1) \times (g+1)$  matrix  $\mathbf{X}^\top \mathbf{X}$  produces this same matrix. Section 6.4 checks this numerically.

### 6.3.5 Sums and products of projections

Complicated subspaces are often built from simpler ones, and their projections are built the same way.

#### Theorem 6.3.8

Let  $\mathbf{P}_1, \mathbf{P}_2$  be orthogonal projections. Then  $\mathbf{P}_1 + \mathbf{P}_2$  is an orthogonal projection iff  $\mathbf{P}_1\mathbf{P}_2 = \mathbf{0}$ , that is, iff  $\mathcal{C}(\mathbf{P}_1) \perp \mathcal{C}(\mathbf{P}_2)$ . In that case it projects onto  $\mathcal{C}(\mathbf{P}_1) \oplus \mathcal{C}(\mathbf{P}_2)$ .

*Proof.*  $\mathbf{P}_1 + \mathbf{P}_2$  is always symmetric, and  $(\mathbf{P}_1 + \mathbf{P}_2)^2 = \mathbf{P}_1 + \mathbf{P}_2 + \mathbf{P}_1\mathbf{P}_2 + \mathbf{P}_2\mathbf{P}_1$ . So it is idempotent iff  $\mathbf{P}_1\mathbf{P}_2 + \mathbf{P}_2\mathbf{P}_1 = \mathbf{0}$ . If  $\mathbf{P}_1\mathbf{P}_2 = \mathbf{0}$ , then transposing gives  $\mathbf{P}_2\mathbf{P}_1 = \mathbf{0}$  and the condition holds. Conversely, suppose  $\mathbf{P}_1\mathbf{P}_2 + \mathbf{P}_2\mathbf{P}_1 = \mathbf{0}$ . Multiplying on the left by  $\mathbf{P}_1$  gives  $\mathbf{P}_1\mathbf{P}_2 + \mathbf{P}_1\mathbf{P}_2\mathbf{P}_1 = \mathbf{0}$ , and on the right by  $\mathbf{P}_1$  gives  $\mathbf{P}_1\mathbf{P}_2\mathbf{P}_1 + \mathbf{P}_2\mathbf{P}_1 = \mathbf{0}$ . Subtracting,  $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$ , and then  $2\mathbf{P}_1\mathbf{P}_2 = \mathbf{0}$ . The equivalence with orthogonality of ranges holds because  $\mathbf{P}_1\mathbf{P}_2 = \mathbf{0}$  says exactly that every vector of  $\mathcal{C}(\mathbf{P}_2)$  is sent to  $\mathbf{0}$  by  $\mathbf{P}_1$ , that is (Proposition 6.3.4(c)), lies in  $\mathcal{C}(\mathbf{P}_1)^\perp$ . Finally,  $\mathcal{C}(\mathbf{P}_1 + \mathbf{P}_2) \subseteq \mathcal{C}(\mathbf{P}_1) + \mathcal{C}(\mathbf{P}_2)$ , and for  $\mathbf{u} = \mathbf{P}_1\mathbf{u} + \mathbf{P}_2\mathbf{u}$  in that sum we have  $(\mathbf{P}_1 + \mathbf{P}_2)\mathbf{u} = \mathbf{u}$ , so the ranges agree. ■

By induction, if  $\mathbf{P}_1, \dots, \mathbf{P}_k$  are projections with  $\mathbf{P}_i\mathbf{P}_j = \mathbf{0}$  for  $i \neq j$ , their sum projects onto the orthogonal direct sum of their ranges. When the sum is  $\mathbf{I}$ , we get an orthogonal decomposition

$$\mathbf{y} = \mathbf{P}_1\mathbf{y} + \dots + \mathbf{P}_k\mathbf{y}, \quad \|\mathbf{y}\|^2 = \|\mathbf{P}_1\mathbf{y}\|^2 + \dots + \|\mathbf{P}_k\mathbf{y}\|^2.$$

Every analysis of variance table in this book is a table of such a decomposition.

Products behave differently.  $\mathbf{P}_1\mathbf{P}_2$  need not be symmetric or idempotent, and it is generally not a projection. Exercise (B1) shows that it is one iff  $\mathbf{P}_1$  and  $\mathbf{P}_2$  commute, in which case it projects onto  $\mathcal{C}(\mathbf{P}_1) \cap \mathcal{C}(\mathbf{P}_2)$ . The most important instance is the nested case  $\mathcal{C}(\mathbf{P}_2) \subseteq \mathcal{C}(\mathbf{P}_1)$ , treated in Section 6.5.

### 6.3.6 Exercises

#### 6.3.6.1 A. Check your understanding

##### Exercise A1.

Let  $\mathcal{S} = \text{span}\{(1, 1, 0)^\top, (0, 1, 1)^\top\}$ . Compute  $\mathbf{P}_{\mathcal{S}}$  in two ways: by Gram-Schmidt and Proposition 6.3.2, and by Theorem 6.3.7. Verify symmetry, idempotence and  $\text{tr } \mathbf{P}_{\mathcal{S}} = 2$ . Find  $\mathcal{S}^\perp$  and check that  $\mathbf{I} - \mathbf{P}_{\mathcal{S}}$  projects onto it.

##### Exercise A2.

Show that a symmetric matrix whose eigenvalues all lie in  $\{0, 1\}$  is an orthogonal projection. Give a non-symmetric  $2 \times 2$  matrix with eigenvalues 0 and 1 that is not idempotent, and one that is idempotent.

**Exercise A3.**

Let  $\mathbf{P}$  be an orthogonal projection and  $a, b$  real numbers. Show that the eigenvalues of  $a\mathbf{P} + b\mathbf{I}$  are  $b$  and  $a + b$ . Show that when  $b \neq 0$  and  $a + b \neq 0$ ,

$$(a\mathbf{P} + b\mathbf{I})^{-1} = \frac{1}{b} \left( \mathbf{I} - \frac{a}{a+b} \mathbf{P} \right).$$

Use this to invert the equicorrelation matrix  $(1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ .

**Exercise A4.**

Show that if  $\mathbf{G}$  is a generalized inverse of the symmetric matrix  $\mathbf{X}^\top \mathbf{X}$ , then so is  $\mathbf{G}^\top$ , and so is  $\mathbf{G}\mathbf{X}^\top \mathbf{X}\mathbf{G}^\top$ . Which of the four Penrose conditions does the last one satisfy?

**6.3.6.2 B. Practice****Exercise B1.**

Let  $\mathbf{P}_1, \mathbf{P}_2$  be orthogonal projections. Show that  $\mathbf{P}_1\mathbf{P}_2$  is an orthogonal projection iff  $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$ , and that in this case it projects onto  $\mathcal{C}(\mathbf{P}_1) \cap \mathcal{C}(\mathbf{P}_2)$ . *Hint:* for the range, show  $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{u}$  iff  $\mathbf{u}$  lies in both subspaces.

*Solution.* If  $\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_2\mathbf{P}_1$ , then  $(\mathbf{P}_1\mathbf{P}_2)^\top = \mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_2$ , and  $(\mathbf{P}_1\mathbf{P}_2)^2 = \mathbf{P}_1\mathbf{P}_2\mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_2\mathbf{P}_1 = \mathbf{P}_1\mathbf{P}_1\mathbf{P}_2 = \mathbf{P}_1\mathbf{P}_2$ . So the product is symmetric and idempotent. Conversely, if  $\mathbf{P}_1\mathbf{P}_2$  is an orthogonal projection, it is symmetric, so  $\mathbf{P}_1\mathbf{P}_2 = (\mathbf{P}_1\mathbf{P}_2)^\top = \mathbf{P}_2\mathbf{P}_1$ . For the range: if  $\mathbf{u}$  lies in both subspaces, then  $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{P}_1\mathbf{u} = \mathbf{u}$ . If  $\mathbf{P}_1\mathbf{P}_2\mathbf{u} = \mathbf{u}$ , then  $\mathbf{u} \in \mathcal{C}(\mathbf{P}_1)$ , and also  $\mathbf{u} = \mathbf{P}_2\mathbf{P}_1\mathbf{u} \in \mathcal{C}(\mathbf{P}_2)$ . A projection's range is the set of vectors it fixes, so  $\mathcal{C}(\mathbf{P}_1\mathbf{P}_2) = \mathcal{C}(\mathbf{P}_1) \cap \mathcal{C}(\mathbf{P}_2)$ .

**Exercise B2.**

Let  $\mathbf{G}_0$  be a generalized inverse of  $\mathbf{A}$ .

- Show that  $\mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{A}\mathbf{G}_0)$  is a generalized inverse of  $\mathbf{A}$  for all conformable  $\mathbf{U}, \mathbf{V}$ .
- Show that every generalized inverse  $\mathbf{G}$  of  $\mathbf{A}$  has this form. *Hint:* take  $\mathbf{U} = \mathbf{G} - \mathbf{G}_0$  and  $\mathbf{V} = \mathbf{G}_0\mathbf{A}(\mathbf{G} - \mathbf{G}_0)$ , and use  $\mathbf{A}(\mathbf{G} - \mathbf{G}_0)\mathbf{A} = \mathbf{0}$ .
- Conclude that a matrix has a unique generalized inverse iff it is square and non-singular.

*Solution. Parts a,b.* For (a), multiply out:  $\mathbf{A}[\mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U} + \mathbf{V}(\mathbf{I} - \mathbf{A}\mathbf{G}_0)]\mathbf{A} = \mathbf{A}\mathbf{G}_0\mathbf{A} + (\mathbf{A} - \mathbf{A}\mathbf{G}_0\mathbf{A})\mathbf{U}\mathbf{A} + \mathbf{A}\mathbf{V}(\mathbf{A} - \mathbf{A}\mathbf{G}_0\mathbf{A}) = \mathbf{A}$ . For (b), put  $\mathbf{\Delta} = \mathbf{G} - \mathbf{G}_0$ , so that  $\mathbf{A}\mathbf{\Delta}\mathbf{A} = \mathbf{A} - \mathbf{A} = \mathbf{0}$ . With  $\mathbf{U} = \mathbf{\Delta}$  and  $\mathbf{V} = \mathbf{G}_0\mathbf{A}\mathbf{\Delta}$ ,

$$(\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{\Delta} + \mathbf{G}_0\mathbf{A}\mathbf{\Delta}(\mathbf{I} - \mathbf{A}\mathbf{G}_0) = \mathbf{\Delta} - \mathbf{G}_0\mathbf{A}\mathbf{\Delta}\mathbf{A} + \mathbf{G}_0\mathbf{A}\mathbf{\Delta}\mathbf{A} - \mathbf{G}_0(\mathbf{A}\mathbf{\Delta}\mathbf{A})\mathbf{G}_0 = \mathbf{\Delta},$$

so the formula reproduces  $\mathbf{G}_0 + \mathbf{\Delta} = \mathbf{G}$ .



**Exercise B3.**

Using the singular value decomposition, show that  $(\mathbf{X}^\top \mathbf{X})^+ \mathbf{X}^\top = \mathbf{X}^+$  and  $\mathbf{X} \mathbf{X}^+ = \mathbf{M}$ . Show also that  $\mathbf{X}^+ = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  when  $\mathbf{X}$  has full column rank.

**6.3.6.3 C. Going deeper****Exercise C1.**

Show that  $\mathbf{M}$  is the only  $n \times n$  matrix  $\mathbf{P}$  satisfying  $\mathbf{P}\mathbf{X} = \mathbf{X}$ ,  $\mathbf{P}^\top = \mathbf{P}$  and  $\text{rank}(\mathbf{P}) = \text{rank}(\mathbf{X})$ . Is the rank condition necessary? Is symmetry?

## 6.4 Least squares via projection

We now return to the linear model  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ , with  $\mathbf{X}$  an  $n \times p$  matrix of rank  $r \leq p$ , and  $\mathbf{M}$  the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$ . Nothing in this section uses probability. The results hold for every data vector  $\mathbf{y}$  and so apply whatever the distribution of the errors.

### 6.4.1 The characterization of least squares estimates

#### Theorem 6.4.1 — Least squares is projection

A vector  $\hat{\boldsymbol{\beta}}$  is a least squares estimate of  $\boldsymbol{\beta}$  iff

$$\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}\mathbf{y}.$$

Consequently the fitted values  $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$  and residuals  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$  are the same for every least squares estimate.

*Proof.* Apply the projection theorem (Theorem 6.3.1) with  $\mathcal{S} = \mathcal{C}(\mathbf{X})$  and  $\mathbf{u} = \mathbf{X}\mathbf{b}$ . By Eq. 6.3.1,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{M}\mathbf{y}\|^2 + \|\mathbf{M}\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 \quad \text{for every } \mathbf{b} \in \mathbb{R}^p.$$

The first term does not involve  $\mathbf{b}$ , and the second is zero iff  $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$ . Such a  $\mathbf{b}$  exists because  $\mathbf{M}\mathbf{y} \in \mathcal{C}(\mathbf{X})$ . ■

The orthogonality half of the projection theorem turns this into linear equations. The residual  $\mathbf{y} - \mathbf{X}\mathbf{b}$  is orthogonal to  $\mathcal{C}(\mathbf{X})$  iff it is orthogonal to every column of  $\mathbf{X}$ , that is, iff  $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$ .

#### Theorem 6.4.2 — Normal equations

The **normal equations**

$$\mathbf{X}^\top \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{y} \quad (6.4.1)$$

are consistent for every  $\mathbf{y}$ , and their solutions are exactly the least squares estimates. One solution is  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top \mathbf{y}$ , for any generalized inverse, and the set of all solutions is

$$\hat{\boldsymbol{\beta}} + \mathcal{N}(\mathbf{X}) = \{\hat{\boldsymbol{\beta}} + \mathbf{v} : \mathbf{X}\mathbf{v} = \mathbf{0}\}.$$

In particular, the least squares estimate is unique iff  $\text{rank}(\mathbf{X}) = p$ , and then  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ .

*Proof.* A vector  $\mathbf{b}$  solves Eq. 6.4.1 iff  $\mathbf{y} - \mathbf{X}\mathbf{b} \perp \mathcal{C}(\mathbf{X})$ . Since  $\mathbf{X}\mathbf{b} \in \mathcal{C}(\mathbf{X})$  automatically, this holds iff  $\mathbf{y} = \mathbf{X}\mathbf{b} + (\mathbf{y} - \mathbf{X}\mathbf{b})$  is the orthogonal decomposition of  $\mathbf{y}$ , that is, iff  $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$ . By Theorem 6.4.1 this is the same as  $\mathbf{b}$  being a least squares estimate. For the particular solution,  $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top \mathbf{y} = \mathbf{M}\mathbf{y}$  by Theorem 6.3.7. Finally,  $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}}$  iff  $\mathbf{X}(\mathbf{b} - \hat{\boldsymbol{\beta}}) = \mathbf{0}$ , and  $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$  iff  $\text{rank}(\mathbf{X}) = p$  by Eq. 6.2.1. ■

The logic of this proof runs opposite to the usual calculus derivation, and the difference matters. Setting a gradient to zero finds stationary points. One then has to argue separately that they are minima and that one exists. Here existence, minimality and the full solution set all come from the decomposition  $\mathbb{R}^n = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\mathbf{X})^\perp$ . The consistency of the normal equations, which can look surprising when  $\mathbf{X}^\top \mathbf{X}$  is singular, is just the statement  $\mathbf{X}^\top \mathbf{y} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{X})$  from Corollary 6.2.4.

## 6.4.2 Fitted values and residuals

Collecting the consequences for  $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$  and  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$ :

### Proposition 6.4.3

For any data vector  $\mathbf{y}$ :

- (a)  $\mathbf{X}^\top \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$  and  $\hat{\mathbf{Y}}^\top \hat{\boldsymbol{\varepsilon}} = 0$ ;
- (b) if  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  (for example, the model has an intercept), then  $\sum_i \hat{\varepsilon}_i = 0$  and the mean of the fitted values is  $\bar{y}$ ;
- (c)  $\|\mathbf{y}\|^2 = \|\hat{\mathbf{Y}}\|^2 + \|\hat{\boldsymbol{\varepsilon}}\|^2$ ;
- (d)  $\|\hat{\boldsymbol{\varepsilon}}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{y}$  and  $\|\hat{\mathbf{Y}}\|^2 = \mathbf{y}^\top \mathbf{M}\mathbf{y}$ ;
- (e) if  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  for some  $\boldsymbol{\beta}$ , then  $\hat{\mathbf{Y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{M}\boldsymbol{\varepsilon}$  and  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$ .

*Proof.*

- (a) is the orthogonality of  $\hat{\boldsymbol{\varepsilon}}$  to  $\mathcal{C}(\mathbf{X})$ , and  $\hat{\mathbf{Y}} \in \mathcal{C}(\mathbf{X})$ . For (b),  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  gives  $\mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$ , and then  $\mathbf{1}^\top \hat{\mathbf{Y}} = \mathbf{1}^\top \mathbf{y}$ . (c) is Pythagoras. (d) uses symmetry and idempotence:  $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})^\top (\mathbf{I} - \mathbf{M})\mathbf{y} = \mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{y}$ . For (e),  $\mathbf{M}\mathbf{X} = \mathbf{X}$  and  $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$ .

Part (e) is simple but important. The residual vector does not depend on  $\boldsymbol{\beta}$  at all. It is the projection of the unobservable error vector onto the  $(n-r)$ -dimensional space  $\mathcal{C}(\mathbf{X})^\perp$ . Everything about  $\sigma^2$  that the data can reveal must therefore come from  $(\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$ . If  $E(\boldsymbol{\varepsilon}) = \mathbf{0}$  and  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}$ , the expectation of a quadratic form (Chapter 2) gives

$$E\|\hat{\boldsymbol{\varepsilon}}\|^2 = E[\boldsymbol{\varepsilon}^\top (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}] = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2(n - r),$$

using Proposition 6.3.4. So  $\|\hat{\boldsymbol{\varepsilon}}\|^2/(n - r)$  is unbiased for  $\sigma^2$ . The divisor is the dimension of the space in which the residuals live, not the number of observations and not  $n - p$ . Chapter 7 develops the sampling theory, and Chapter 9 develops the decomposition.

### 6.4.3 Coefficients are not unique; fitted values are

When  $\text{rank}(\mathbf{X}) < p$ , Theorem 6.4.2 gives a whole affine subspace of least squares estimates, one for each generalized inverse and more besides. Statistical software has to pick one. Some programs drop columns that are linearly dependent on earlier ones. Others impose “side conditions” such as a zero coefficient for a reference level or effects that sum to zero. Others report the minimum-norm solution. These choices look very different in the printed coefficient table. The geometry says they are all describing the same fitted vector.

#### Example. (Four answers, one fit)

Nine observations fall in three groups of sizes 3, 2, 4 with group means 5.700, 7.800 and 4.300. The model matrix  $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$  has  $p = 4$  columns and rank 3, so  $\mathbf{X}^\top \mathbf{X}$  is singular. We compute  $\hat{\boldsymbol{\beta}} = \mathbf{G}\mathbf{X}^\top \mathbf{y}$  for four generalized inverses  $\mathbf{G}$  of  $\mathbf{X}^\top \mathbf{X}$ : the Moore–Penrose inverse; the inverse obtained by deleting the intercept row and column (Proposition 6.3.5); the one obtained by deleting the first group’s row and column; and a deliberately strange, non-symmetric one built from the general formula of Exercise (B2).

| Generalized<br>inverse | $\hat{\beta}_0$ | $\hat{\beta}_1$ | $\hat{\beta}_2$ | $\hat{\beta}_3$ | $\hat{\beta}_1 - \hat{\beta}_2$ | $\max \mathbf{M} - \mathbf{M}_{\text{MP}} $ |
|------------------------|-----------------|-----------------|-----------------|-----------------|---------------------------------|---------------------------------------------|
| Moore–<br>Penrose      | 4.450           | 1.250           | 3.350           | -0.150          | -2.100                          | 0                                           |
| drop<br>intercept      | 0.000           | 5.700           | 7.800           | 4.300           | -2.100                          | $1.1 \times 10^{-16}$                       |
| drop<br>group 1        | 5.700           | 0.000           | 2.100           | -1.400          | -2.100                          | $1.1 \times 10^{-16}$                       |
| random                 | -9.102          | 14.802          | 16.902          | 13.402          | -2.100                          | $1.8 \times 10^{-15}$                       |

The coefficient vectors disagree wildly, including in sign. Yet each produces the same projection matrix up to rounding error, and hence the same fitted values (the group means) and residuals. The difference  $\hat{\beta}_1 - \hat{\beta}_2$  is the same in every row, and so is  $\hat{\beta}_0 + \hat{\beta}_k$  for each  $k$ . Individual coefficients are not. The trace of  $\mathbf{M}$  is 3.000, the rank of  $\mathbf{X}$ .

```
import numpy as np
```

```

groups = np.array([0, 0, 0, 1, 1, 2, 2, 2, 2])    # three groups, sizes 3, 2, 4
y = np.array([5.1, 6.3, 5.7, 8.2, 7.4, 4.0, 4.9, 3.8, 4.5])
n, g = len(y), 3
X = np.column_stack([np.ones(n), np.eye(g)[groups]]) # [1, indicator columns]
A = X.T @ X
print("n =", n, " p =", X.shape[1], " rank(X) =", np.linalg.matrix_rank(X))

```

```

def g_drop(j):
    """Generalized inverse obtained by deleting row and column j of A."""
    keep = [k for k in range(A.shape[0]) if k != j]
    G = np.zeros_like(A)
    G[np.ix_(keep, keep)] = np.linalg.inv(A[np.ix_(keep, keep)])
    return G

rng = np.random.default_rng(6)
G0 = np.linalg.pinv(A)                    # Moore-Penrose inverse
U, V = rng.normal(size=A.shape), rng.normal(size=A.shape)
G_random = G0 + (np.eye(4) - G0 @ A) @ U + V @ (np.eye(4) - A @ G0)

ginverses = {
    "Moore-Penrose": G0,
    "drop intercept": g_drop(0),
    "drop group 1": g_drop(1),
    "random": G_random,
}

```

```

M_ref = X @ G0 @ X.T
for name, G in ginverses.items():
    assert np.allclose(A @ G @ A, A)        # G really is a g-inverse
    beta = G @ X.T @ y
    M = X @ G @ X.T
    print(f"{name:15s} beta = {np.round(beta, 3)}",
          f" max|M - M_ref| = {np.abs(M - M_ref).max():.1e}",
          f" group1-group2 = {beta[1] - beta[2]:.3f}")

```

Which linear functions of  $\hat{\beta}$  come out the same for every least squares estimate? The answer is a first look at estimability, the subject of [Chapter 8](#).

### Theorem 6.4.4

For  $\lambda \in \mathbb{R}^p$ , the value  $\lambda^\top \hat{\beta}$  is the same for every least squares estimate  $\hat{\beta}$  (and every  $\mathbf{y}$ ) iff  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ , that is, iff  $\lambda^\top = \rho^\top \mathbf{X}$  for some  $\rho \in \mathbb{R}^n$ . In that case  $\lambda^\top \hat{\beta} = \rho^\top \mathbf{M}\mathbf{y}$ .

*Proof.* By [Theorem 6.4.2](#) the least squares estimates form  $\hat{\beta} + \mathcal{N}(\mathbf{X})$ . So  $\lambda^\top \hat{\beta}$  is invariant iff  $\lambda^\top \mathbf{v} = 0$  for all  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ , that is, iff  $\lambda \in \mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$  ([Corollary 6.2.4](#)). If  $\lambda^\top = \rho^\top \mathbf{X}$  then  $\lambda^\top \hat{\beta} = \rho^\top \mathbf{X}\hat{\beta} = \rho^\top \mathbf{M}\mathbf{y}$ . ■

In [Example \(Four answers, one fit\)](#),  $\lambda = (0, 1, -1, 0)^\top$  is the difference between any row of  $\mathbf{X}$  from group 1,  $(1, 1, 0, 0)$ , and any row from group 2,  $(1, 0, 1, 0)$ . So  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$  and  $\hat{\beta}_1 - \hat{\beta}_2$  is invariant. For  $\lambda = (0, 1, 0, 0)^\top$ , note that  $\rho^\top \mathbf{X}$  has first entry  $\sum_i \rho_i$  and remaining entries equal

to the sums of  $\rho_i$  within each group. Matching  $(0, 1, 0, 0)$  would need the group sums to be  $1, 0, 0$  and their total to be  $0$ , which is impossible. So  $\hat{\beta}_1$  is not invariant.

*Remark* (Side conditions choose coordinates). Imposing  $\beta_0 = 0$  or  $\beta_1 = 0$  does not add information about the data. It picks one point from the affine set  $\hat{\beta} + \mathcal{N}(\mathbf{X})$ . A side condition  $\mathbf{c}^\top \beta = 0$  picks exactly one point iff  $\mathbf{c} \notin \mathcal{C}(\mathbf{X}^\top)$ . A condition inside  $\mathcal{C}(\mathbf{X}^\top)$  would constrain the fitted values and change the model rather than just its coordinates.

[Theorem 8.4.2](#) makes this precise.

## 6.4.4 Exercises

### 6.4.4.1 A. Check your understanding

#### Exercise A1.

For [Example \(Three observations\)](#), write out the  $3 \times 3$  matrix  $\mathbf{M}$  with exact fractional entries, verify  $\text{tr } \mathbf{M} = 2$ , and recover the fitted values and residuals quoted in the text.

#### Exercise A2.

Show that the residuals sum to zero whenever  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , even if no column of  $\mathbf{X}$  is constant. Give an example of a model without such a column in which this happens, and one in which the residuals do not sum to zero.

### 6.4.4.2 B. Practice

#### Exercise B1.

Suppose  $E(\mathbf{Y}) = \mathbf{X}\beta$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ . Show that  $E(\hat{\mathbf{Y}}) = \mathbf{X}\beta$ ,  $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{M}$ ,  $\text{Cov}(\hat{\epsilon}) = \sigma^2(\mathbf{I} - \mathbf{M})$  and  $\text{Cov}(\hat{\mathbf{Y}}, \hat{\epsilon}) = \mathbf{0}$ . Which of these statements survive when  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$  for a general positive definite  $\mathbf{V}$ ?

#### Exercise B2.

Let  $\text{rank}(\mathbf{X}) = r < p$  and let  $\mathbf{C}$  be a  $(p - r) \times p$  matrix. Show that the constrained problem “minimize  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|$  subject to  $\mathbf{C}\mathbf{b} = \mathbf{0}$ ” has a unique solution for every  $\mathbf{y}$ , and that its fitted values are  $\mathbf{M}\mathbf{y}$ , iff  $\text{rank}(\begin{smallmatrix} \mathbf{X} \\ \mathbf{C} \end{smallmatrix}) = p$ . Show that the condition implies  $\mathcal{C}(\mathbf{C}^\top) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$ , and interpret: the constraints must not restrict any estimable function.

### 6.4.4.3 C. Going deeper

#### Exercise C1.

Suppose  $\mathbf{X}^\top \mathbf{y} \neq \mathbf{0}$ . Show that every least squares estimate  $\mathbf{b}$  can be written as  $\mathbf{G}\mathbf{X}^\top \mathbf{y}$  for some generalized inverse  $\mathbf{G}$  of  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ . *Hint:* with  $\mathbf{c} = \mathbf{X}^\top \mathbf{y}$ , try  $\mathbf{G} = \mathbf{G}_0 + (\mathbf{b} - \mathbf{G}_0 \mathbf{c})\mathbf{c}^\top / (\mathbf{c}^\top \mathbf{c})$ . What goes wrong when  $\mathbf{X}^\top \mathbf{y} = \mathbf{0}$ ?

## 6.5 Nested models and sequential projection

Statistical questions about linear models are almost always comparisons. Does a regressor matter? Are several group means equal? Is a curve needed, or will a line do? Each comparison sets a full model against a *reduced* model whose mean space is a subspace of the full one. This section studies the geometry of such nested pairs. [Chapter 9](#) and [Chapter 11](#) build the analysis of variance and the  $F$  test on it.

### 6.5.1 Two nested subspaces

Let  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ , with ranks  $r_0 \leq r$ . Write  $\mathbf{M}_0$  and  $\mathbf{M}$  for the two projections.

#### Theorem 6.5.1 — Nested projections

If  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ , then:

- (a)  $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0\mathbf{M} = \mathbf{M}_0$ ;
- (b)  $\mathbf{M} - \mathbf{M}_0$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$ , the orthogonal complement of  $\mathcal{C}(\mathbf{X}_0)$  within  $\mathcal{C}(\mathbf{X})$ ;
- (c)  $\text{rank}(\mathbf{M} - \mathbf{M}_0) = r - r_0$ ;
- (d)  $(\mathbf{I} - \mathbf{M}_0) - (\mathbf{I} - \mathbf{M}) = \mathbf{M} - \mathbf{M}_0$ , and  $\mathbf{M}_0(\mathbf{M} - \mathbf{M}_0) = (\mathbf{M} - \mathbf{M}_0)(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ .

*Proof.*

- (a) Every column of  $\mathbf{M}_0$  lies in  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  and is left unchanged by  $\mathbf{M}$ , so  $\mathbf{M}\mathbf{M}_0 = \mathbf{M}_0$ . Transposing gives  $\mathbf{M}_0\mathbf{M} = \mathbf{M}_0$ .
- (b)  $\mathbf{M} - \mathbf{M}_0$  is symmetric, and by (a)  $(\mathbf{M} - \mathbf{M}_0)^2 = \mathbf{M} - \mathbf{M}\mathbf{M}_0 - \mathbf{M}_0\mathbf{M} + \mathbf{M}_0 = \mathbf{M} - \mathbf{M}_0$ . By [Theorem 6.3.3](#) it projects onto its column space. If  $\mathbf{v} \in \mathcal{C}(\mathbf{X})$  and  $\mathbf{v} \perp \mathcal{C}(\mathbf{X}_0)$ , then  $(\mathbf{M} - \mathbf{M}_0)\mathbf{v} = \mathbf{v} - \mathbf{0} = \mathbf{v}$ , so  $\mathbf{v} \in \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$ . Conversely, a vector  $\mathbf{v} = (\mathbf{M} - \mathbf{M}_0)\mathbf{u}$  lies in  $\mathcal{C}(\mathbf{X})$  and satisfies  $\mathbf{M}_0\mathbf{v} = (\mathbf{M}_0 - \mathbf{M}_0)\mathbf{u} = \mathbf{0}$ .
- (c)  $\text{tr}(\mathbf{M} - \mathbf{M}_0) = r - r_0$  and [Proposition 6.3.4](#).
- (d) Direct multiplication using (a).

■

The theorem gives a three-way orthogonal decomposition of  $\mathbb{R}^n$ ,

$$\mathbb{R}^n = \underbrace{\mathcal{C}(\mathbf{X}_0)}_{\dim r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})}_{\dim r - r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X})^\perp}_{\dim n - r},$$

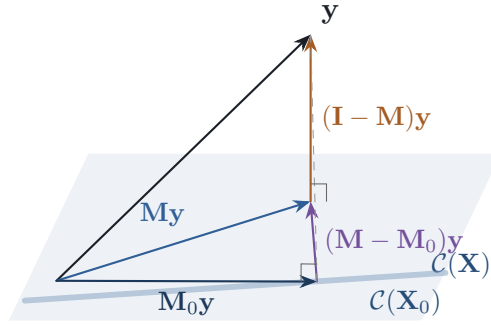
and correspondingly  $\mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$ , with the three terms mutually orthogonal projections. Applying the decomposition to  $\mathbf{y}$  and using Pythagoras,

$$\|\mathbf{y}\|^2 = \|\mathbf{M}_0\mathbf{y}\|^2 + \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2. \quad (6.5.1)$$

The middle term has a second reading that is the key to model comparison. Since  $(\mathbf{I} - \mathbf{M}_0) = (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$  is an orthogonal sum,

$$\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \|(\mathbf{I} - \mathbf{M}_0)\mathbf{y}\|^2 - \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \text{SSE}_0 - \text{SSE}. \quad (6.5.2)$$

The *drop in residual sum of squares* from the reduced to the full model is the squared length of a projection onto an  $(r - r_0)$ -dimensional space. That is why it has  $r - r_0$  degrees of freedom, and why it is independent of SSE under normality: the two lie in orthogonal subspaces (Chapter 4). It also shows that SSE can never increase when the model space grows, with equality iff  $\mathbf{M}\mathbf{y} = \mathbf{M}_0\mathbf{y}$ .



**Figure 6.5.1.** Nested projections. The fitted vector of the reduced model,  $\mathbf{M}_0\mathbf{y}$ , is also the projection of  $\mathbf{M}\mathbf{y}$  onto  $\mathcal{C}(\mathbf{X}_0)$ . The three coloured pieces  $\mathbf{M}_0\mathbf{y}$ ,  $(\mathbf{M} - \mathbf{M}_0)\mathbf{y}$  and  $(\mathbf{I} - \mathbf{M})\mathbf{y}$  are mutually orthogonal. The dashed segment is the reduced model's residual, whose squared length is the sum of the squared lengths of the two segments meeting at  $\mathbf{M}\mathbf{y}$ .

Part (a) also says  $\mathbf{M}_0\mathbf{y} = \mathbf{M}_0(\mathbf{M}\mathbf{y})$ . *Fitting the reduced model to the data gives the same result as fitting it to the full model's fitted values.* Projection can be done in stages (Figure 6.5.1).

**Example. (Intercept inside simple regression)**

Let  $\mathbf{X}_0 = \mathbf{1}$  and  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$  with  $\mathbf{x}$  not constant. Then  $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$  is spanned by the centred regressor  $\mathbf{x} - \bar{x}\mathbf{1} = (\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{x}$ , which lies in  $\mathcal{C}(\mathbf{X})$  and is orthogonal to  $\mathbf{1}$ . By Example (Projections onto a line),

$$\mathbf{M} - \mathbf{M}_0 = \frac{(\mathbf{x} - \bar{x}\mathbf{1})(\mathbf{x} - \bar{x}\mathbf{1})^\top}{S_{xx}}, \quad S_{xx} = \sum_i (x_i - \bar{x})^2,$$

and so

$$\mathbf{M}\mathbf{y} = \bar{y}\mathbf{1} + \frac{S_{xy}}{S_{xx}}(\mathbf{x} - \bar{x}\mathbf{1}), \quad S_{xy} = \sum_i (x_i - \bar{x})(y_i - \bar{y}).$$

Reading off coordinates, the slope is  $S_{xy}/S_{xx}$  and the intercept is  $\bar{y} - \bar{x} S_{xy}/S_{xx}$ . These are the familiar formulas, obtained without solving any equations, and the decomposition Eq. 6.5.1 becomes

$$\sum_i y_i^2 = n\bar{y}^2 + \frac{S_{xy}^2}{S_{xx}} + \text{SSE}.$$

## 6.5.2 Chains of subspaces

The argument extends to any chain  $\mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X}_2) \subseteq \cdots \subseteq \mathcal{C}(\mathbf{X}_k)$  with projections  $\mathbf{M}_1, \dots, \mathbf{M}_k$ . Setting  $\mathbf{M}_0 = \mathbf{0}$  and  $\mathbf{M}_{k+1} = \mathbf{I}$ , the differences  $\mathbf{M}_j - \mathbf{M}_{j-1}$  for  $j = 1, \dots, k+1$  are mutually orthogonal projections summing to  $\mathbf{I}$ . The corresponding lengths

$$\|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{y}\|^2 = \text{SSE}_{j-1} - \text{SSE}_j$$

are the **sequential sums of squares** of [Definition 9.3.1](#). Their values depend on the *order* in which the subspaces are nested. A regressor added early can take credit that it would not get if added last. The Frisch–Waugh–Lovell theorem of [Section 6.6](#) describes what the regressor added last actually contributes.

### Example. (Order matters)

For the state data of [Example \(Poverty and murder rates\)](#), enter the three regressors after the intercept in two different orders. The listing computes each squared length  $\|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{y}\|^2$  directly from the projections.

| Order A       |        | Order B       |        |
|---------------|--------|---------------|--------|
| poverty       | 102.49 | single-parent | 162.84 |
| single-parent | 80.04  | urban         | 3.59   |
| urban         | 0.27   | poverty       | 16.37  |
| residual      | 101.96 | residual      | 101.96 |

Entered first, poverty accounts for 102.49 of the corrected total sum of squares 284.76. Entered last, it accounts for only 16.37. The residual sum of squares, and the total explained by all three regressors together (182.80), do not depend on the order, because the final subspace does not. The individual pieces do depend on the order, because the intermediate subspaces differ. Poverty and single parenthood are correlated, so whichever enters first takes the variation they share. The piece for the *last* regressor is the only one with an order-free meaning. By [Theorem 6.6.2](#), it is the squared length of the projection of  $\mathbf{y}$  onto the part of that regressor orthogonal to all the others.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
cols = {"poverty": data["poverty"], "single": data["single"], "urban": data["urban"]}

def projection(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T
    # fine for n = 50; never do this for large n

def sequential_ss(y, columns):
    """Squared lengths ||(M_j - M_{j-1}) y||^2 as columns enter one at a time."""
    n = len(y)
    Z = np.ones((n, 1))
    M_prev = projection(Z)
```



```

pieces = {"mean": y @ M_prev @ y}
for name, x in columns:
    Z = np.column_stack([Z, x])
    M = projection(Z)
    pieces[name] = y @ (M - M_prev) @ y
    M_prev = M
pieces["residual"] = y @ (np.eye(n) - M_prev) @ y
return pieces

order_a = sequential_ss(y, [(k, cols[k]) for k in ["poverty", "single", "urban"]])
order_b = sequential_ss(y, [(k, cols[k]) for k in ["single", "urban", "poverty"]])
for name in ["poverty", "single", "urban", "residual"]:
    print(f"{name:9s}  order A: {order_a[name]:8.2f}  order B: {order_b[name]:8.2f}")
    )

```

### 6.5.3 Two factors and when the pieces are orthogonal

A sum of projections is a projection exactly when the pieces are orthogonal ([Theorem 6.3.8](#)). In designed experiments, orthogonality comes from balance, and when it fails, it fails in ways that later chapters have to face.

#### Example. (Additive two-factor model, one observation per cell)

Let factor  $A$  have  $a$  levels and factor  $B$  have  $b$  levels, with one observation  $y_{ij}$  for each of the  $n = ab$  combinations. Let  $\mathbf{P}_A$  project onto vectors that depend only on the  $A$  level, so  $\mathbf{P}_A \mathbf{y}$  has entries  $\bar{y}_{i\cdot}$ , and define  $\mathbf{P}_B$  similarly. Let  $\mathbf{P}_0 = n^{-1} \mathbf{1} \mathbf{1}^\top$ . Both  $\mathcal{C}(\mathbf{P}_A)$  and  $\mathcal{C}(\mathbf{P}_B)$  contain  $\mathbf{1}$ , so  $\mathbf{P}_A \mathbf{P}_0 = \mathbf{P}_B \mathbf{P}_0 = \mathbf{P}_0$  by [Theorem 6.5.1](#).

The crucial computation is  $\mathbf{P}_A \mathbf{P}_B$ . Averaging over  $j$  with  $i$  fixed a vector that depends only on  $j$  gives the overall average of that vector. So for every  $\mathbf{y}$  the entries of  $\mathbf{P}_A(\mathbf{P}_B \mathbf{y})$  are all equal to  $b^{-1} \sum_j \bar{y}_{\cdot j} = \bar{y}_{\cdot\cdot}$ . Therefore

$$\mathbf{P}_A \mathbf{P}_B = \mathbf{P}_0,$$

and so  $(\mathbf{P}_A - \mathbf{P}_0)(\mathbf{P}_B - \mathbf{P}_0) = \mathbf{P}_0 - \mathbf{P}_0 - \mathbf{P}_0 + \mathbf{P}_0 = \mathbf{0}$ . The two *centred* factor spaces are orthogonal. By [Theorem 6.3.8](#), the additive model space has projection  $\mathbf{P}_0 + (\mathbf{P}_A - \mathbf{P}_0) + (\mathbf{P}_B - \mathbf{P}_0) = \mathbf{P}_A + \mathbf{P}_B - \mathbf{P}_0$ , and

$$\mathbf{I} = \mathbf{P}_0 + (\mathbf{P}_A - \mathbf{P}_0) + (\mathbf{P}_B - \mathbf{P}_0) + (\mathbf{I} - \mathbf{P}_A - \mathbf{P}_B + \mathbf{P}_0)$$

is an orthogonal decomposition with ranks 1,  $a-1$ ,  $b-1$  and  $(a-1)(b-1)$ . The residual projection sends  $y_{ij}$  to  $y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot}$ .

The key step used the fact that every  $A$  level appears with every  $B$  level equally often. If cells are missing or unequally replicated,  $\mathbf{P}_A \mathbf{P}_B \neq \mathbf{P}_0$ , the centred factor spaces are no longer orthogonal, and the sum of squares for  $A$  depends on whether  $B$  was fitted first. [Chapter 17](#) is about that situation.

## 6.5.4 Reduced models defined by constraints

Reduced models often come as *constraints* on the parameters of the full model, for example  $\beta_2 = \beta_3$  or  $\beta_4 = 0$ , rather than as a smaller model matrix. In mean space, a constraint  $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{0}$  on an estimable function  $\mathbf{A}^\top = \mathbf{P}^\top \mathbf{X}$  restricts  $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$  to the subspace

$$\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{P}^\top) = \{\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}) : \mathbf{P}^\top \boldsymbol{\mu} = \mathbf{0}\}.$$

To test the constraint we need the piece of  $\mathcal{C}(\mathbf{X})$  that  $\mathcal{S}_0$  removes.

### Theorem 6.5.2

Let  $\mathbf{M}$  project onto  $\mathcal{C}(\mathbf{X})$  and let  $\mathbf{P}$  be an  $n \times q$  matrix. If  $\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{P}^\top)$ , then

$$\mathcal{S}_0^\perp \cap \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{MP}).$$

Hence the projection onto  $\mathcal{S}_0$  is  $\mathbf{M} - \mathbf{P}_{\mathcal{C}(\mathbf{MP})}$ , and  $\dim \mathcal{S}_0 = r - \text{rank}(\mathbf{MP})$ .

*Proof.* By Lemma 6.2.5 and Lemma 6.2.1,  $\mathcal{S}_0^\perp = \mathcal{C}(\mathbf{X})^\perp + \mathcal{N}(\mathbf{P}^\top)^\perp = \mathcal{C}(\mathbf{X})^\perp + \mathcal{C}(\mathbf{P})$ . If  $\mathbf{v} \in \mathcal{S}_0^\perp \cap \mathcal{C}(\mathbf{X})$ , write  $\mathbf{v} = \mathbf{w} + \mathbf{Pa}$  with  $\mathbf{w} \perp \mathcal{C}(\mathbf{X})$ . Applying  $\mathbf{M}$  gives  $\mathbf{v} = \mathbf{Mv} = \mathbf{MPa} \in \mathcal{C}(\mathbf{MP})$ . Conversely,  $\mathbf{MPa}$  lies in  $\mathcal{C}(\mathbf{X})$  and equals  $\mathbf{Pa} - (\mathbf{I} - \mathbf{M})\mathbf{Pa} \in \mathcal{C}(\mathbf{P}) + \mathcal{C}(\mathbf{X})^\perp = \mathcal{S}_0^\perp$ . The remaining claims follow from Theorem 6.5.1 with the roles  $\mathcal{C}(\mathbf{X}_0) = \mathcal{S}_0$ . ■

The subspace  $\mathcal{C}(\mathbf{MP})$  is the *test space* of the hypothesis. Its dimension  $\text{rank}(\mathbf{MP})$  counts how many independent restrictions the constraint really places on the mean. It can be smaller than  $q$  if some rows of the constraint are redundant. Chapter 11 uses exactly this space for the numerator of the  $F$  statistic.

## 6.5.5 Exercises

### 6.5.5.1 A. Check your understanding

#### Exercise A1.

Construct the projections  $\mathbf{P}_A$ ,  $\mathbf{P}_B$  and  $\mathbf{P}_0$  of Example (Additive two-factor model, one observation per cell) for  $a = 3$ ,  $b = 2$  and verify  $\mathbf{P}_A \mathbf{P}_B = \mathbf{P}_0$  numerically. Delete one observation and recompute. Show that the centred factor spaces are no longer orthogonal, and find the largest entry of  $(\mathbf{P}_A - \mathbf{P}_0)(\mathbf{P}_B - \mathbf{P}_0)$ .

### 6.5.5.2 B. Practice

#### Exercise B1.

With  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ , show that  $\text{SSE}_0 - \text{SSE} = \|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_0\|^2$ , the squared distance between the two fitted vectors.

*Solution.*  $\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_0 = (\mathbf{M} - \mathbf{M}_0)\mathbf{y}$ , and  $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \text{SSE}_0 - \text{SSE}$  by Eq. 6.5.2.

### Exercise B2.

In the model  $E(\mathbf{Y}) = \beta_0\mathbf{1} + \beta_1\mathbf{x}_1 + \beta_2\mathbf{x}_2 + \beta_3\mathbf{x}_3$  with full-rank  $\mathbf{X}$ , the hypothesis  $\beta_2 = \beta_3$  defines a reduced model with model matrix  $\mathbf{X}_0 = [\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2 + \mathbf{x}_3]$ . Show that the test space  $\mathcal{C}(\mathbf{X}_0)^\perp \cap \mathcal{C}(\mathbf{X})$  is spanned by  $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2$ , and that  $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2 = -(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_3$ . Verify that this agrees with Theorem 6.5.2 for a suitable  $\mathbf{P}$ .

*Solution.* The two model spaces differ in dimension by one, so the test space is a line. The vector  $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_2$  lies in  $\mathcal{C}(\mathbf{X})$  and is orthogonal to  $\mathcal{C}(\mathbf{X}_0)$ . It is nonzero, because  $\mathbf{x}_2 \in \mathcal{C}(\mathbf{X}_0)$  would make the columns of  $\mathbf{X}$  dependent. Since  $\mathbf{x}_2 + \mathbf{x}_3 \in \mathcal{C}(\mathbf{X}_0)$ ,  $(\mathbf{I} - \mathbf{M}_0)(\mathbf{x}_2 + \mathbf{x}_3) = \mathbf{0}$ . For Theorem 6.5.2, write the constraint as  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = 0$  with  $\boldsymbol{\lambda} = (0, 0, 1, -1)^\top$ , and take  $\mathbf{P} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ , so that  $\mathbf{P}^\top \mathbf{X} = \boldsymbol{\lambda}^\top$ . Then  $\mathbf{M}\mathbf{P} = \mathbf{P}$ . Each column of  $\mathbf{X}_0$  is  $\mathbf{X}\mathbf{k}$  with  $\boldsymbol{\lambda}^\top \mathbf{k} = 0$ , so  $\mathbf{P} \perp \mathcal{C}(\mathbf{X}_0)$  and  $\mathbf{P}$  spans the same line.

### 6.5.5.3 C. Going deeper

#### Exercise C1.

Extend Example (Additive two-factor model, one observation per cell) to  $m \geq 2$  observations in every cell and a model with interaction. Find five mutually orthogonal projections summing to  $\mathbf{I}$ , corresponding to the grand mean, factor  $A$ , factor  $B$ , the interaction, and within-cell error, and find their ranks. Describe the action of each on  $\mathbf{y}$  in terms of cell, row, column and grand means.

## 6.6 The Frisch–Waugh–Lovell theorem

A coefficient in a multiple regression is often described as the effect of its regressor “holding the other regressors fixed”. In observational data nobody held anything fixed, so the phrase needs a precise meaning. The Frisch–Waugh–Lovell (FWL) theorem supplies one. A coefficient in a multiple regression equals the slope of a simple regression in which the other regressors have been *projected out* of both the response and the regressor of interest.

### 6.6.1 Statement and proof

Partition the model matrix and coefficients as

$$\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2], \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix},$$

with  $\mathbf{X}_1$  of size  $n \times p_1$  and  $\mathbf{X}_2$  of size  $n \times p_2$ , and assume  $\mathbf{X}$  has full column rank. Let  $\mathbf{M}_2$  project onto  $\mathcal{C}(\mathbf{X}_2)$ , and write

$$\tilde{\mathbf{X}}_1 = (\mathbf{I} - \mathbf{M}_2)\mathbf{X}_1, \quad \tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_2)\mathbf{y}$$

for the parts of  $\mathbf{X}_1$  and  $\mathbf{y}$  that  $\mathbf{X}_2$  cannot explain linearly. Each column of  $\tilde{\mathbf{X}}_1$  is the residual vector from regressing the corresponding column of  $\mathbf{X}_1$  on  $\mathbf{X}_2$ .

**Lemma 6.6.1 — Orthogonalizing a partitioned model space**

Under the assumptions above:

- (a)  $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X}_2) \oplus \mathcal{C}(\tilde{\mathbf{X}}_1)$ , and the two summands are orthogonal;
- (b)  $\tilde{\mathbf{X}}_1$  has full column rank;
- (c)  $\mathbf{M} = \mathbf{M}_2 + \tilde{\mathbf{M}}_1$ , where  $\tilde{\mathbf{M}}_1 = \tilde{\mathbf{X}}_1(\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top$ .

*Proof.*

- (a) The columns of  $\tilde{\mathbf{X}}_1 = \mathbf{X}_1 - \mathbf{M}_2 \mathbf{X}_1$  lie in  $\mathcal{C}(\mathbf{X})$  and are orthogonal to  $\mathcal{C}(\mathbf{X}_2)$ . Conversely, for any  $\mathbf{b} = (\mathbf{b}_1^\top, \mathbf{b}_2^\top)^\top$ ,

$$\mathbf{X}\mathbf{b} = \mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2\mathbf{b}_2 = \tilde{\mathbf{X}}_1\mathbf{b}_1 + (\mathbf{M}_2\mathbf{X}_1\mathbf{b}_1 + \mathbf{X}_2\mathbf{b}_2),$$

with the second term in  $\mathcal{C}(\mathbf{X}_2)$ . So  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\tilde{\mathbf{X}}_1) + \mathcal{C}(\mathbf{X}_2)$ .

- (b) If  $\tilde{\mathbf{X}}_1\mathbf{a} = \mathbf{0}$ , then  $\mathbf{X}_1\mathbf{a} = \mathbf{M}_2\mathbf{X}_1\mathbf{a} = \mathbf{X}_2\mathbf{c}$  for some  $\mathbf{c}$ , so  $\mathbf{X}(\mathbf{a}^\top, -\mathbf{c}^\top)^\top = \mathbf{0}$ . Full column rank of  $\mathbf{X}$  forces  $\mathbf{a} = \mathbf{0}$ .
- (c) Combine (a) with [Theorem 6.3.8](#) and [Theorem 6.3.7](#).

■

**Theorem 6.6.2 — Frisch–Waugh–Lovell**

Let  $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top$  be the least squares estimate in the full model. Then:

- (a)  $\hat{\boldsymbol{\beta}}_1 = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{y}} = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \mathbf{y}$ . So  $\hat{\boldsymbol{\beta}}_1$  is the coefficient from regressing  $\tilde{\mathbf{y}}$  (or  $\mathbf{y}$  itself) on  $\tilde{\mathbf{X}}_1$ ;
- (b) the residual vector from regressing  $\tilde{\mathbf{y}}$  on  $\tilde{\mathbf{X}}_1$  equals the residual vector  $(\mathbf{I} - \mathbf{M})\mathbf{y}$  of the full model;
- (c)  $\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top (\mathbf{y} - \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1)$ .

*Proof.* Let  $\tilde{\mathbf{b}} = (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \mathbf{y}$ . Since  $\mathbf{I} - \mathbf{M}_2$  is symmetric and idempotent,  $\tilde{\mathbf{X}}_1^\top \mathbf{y} = \mathbf{X}_1^\top (\mathbf{I} - \mathbf{M}_2) \mathbf{y} = \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{y}}$ , so the two expressions in (a) agree. By [Lemma 6.6.1\(c\)](#),

$$\mathbf{M}\mathbf{y} = \mathbf{M}_2\mathbf{y} + \tilde{\mathbf{X}}_1\tilde{\mathbf{b}} = \mathbf{M}_2\mathbf{y} + \mathbf{X}_1\tilde{\mathbf{b}} - \mathbf{M}_2\mathbf{X}_1\tilde{\mathbf{b}} = \mathbf{X}_1\tilde{\mathbf{b}} + \mathbf{M}_2(\mathbf{y} - \mathbf{X}_1\tilde{\mathbf{b}}).$$

The last term lies in  $\mathcal{C}(\mathbf{X}_2)$ , so it equals  $\mathbf{X}_2\mathbf{c}$  with  $\mathbf{c} = (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top (\mathbf{y} - \mathbf{X}_1\tilde{\mathbf{b}})$ . Thus  $\mathbf{M}\mathbf{y} = \mathbf{X}_1\tilde{\mathbf{b}} + \mathbf{X}_2\mathbf{c}$ . Because  $\mathbf{X}$  has full column rank, the coefficient vector representing  $\mathbf{M}\mathbf{y}$  is unique, so  $\hat{\boldsymbol{\beta}}_1 = \tilde{\mathbf{b}}$  and  $\hat{\boldsymbol{\beta}}_2 = \mathbf{c}$ . This proves (a) and (c). For (b),  $\mathbf{y} - \mathbf{M}\mathbf{y} = (\mathbf{I} - \mathbf{M}_2)\mathbf{y} - \tilde{\mathbf{M}}_1\mathbf{y} = \tilde{\mathbf{y}} - \tilde{\mathbf{X}}_1\tilde{\mathbf{b}}$ , which is the residual of the regression of  $\tilde{\mathbf{y}}$  on  $\tilde{\mathbf{X}}_1$ , because  $\tilde{\mathbf{X}}_1\tilde{\mathbf{b}} = \tilde{\mathbf{M}}_1\mathbf{y} = \tilde{\mathbf{M}}_1\tilde{\mathbf{y}}$ .

■

The proof uses nothing beyond the orthogonal decomposition of the model space. Part (a) contains a small surprise. Residualizing  $\mathbf{y}$  is optional, because the part of  $\mathbf{y}$  lying in  $\mathcal{C}(\mathbf{X}_2)$  is orthogonal to  $\tilde{\mathbf{X}}_1$  and drops out of  $\tilde{\mathbf{X}}_1^\top \mathbf{y}$  anyway. Residualizing the regressor of interest is

essential.

*Idea.* In a multiple regression,  $\hat{\beta}_j$  measures how  $\mathbf{y}$  moves with *the part of  $\mathbf{x}_j$  that is orthogonal to the other regressors*. A regressor that is almost a linear combination of the others has a tiny orthogonal part, so its coefficient rests on very little information. That is collinearity, seen geometrically.

## 6.6.2 Consequences

**Variances.** The FWL formula writes  $\hat{\beta}_1$  as a linear function of  $\mathbf{y}$ . If  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , then

$$\text{Cov}(\hat{\beta}_1) = \sigma^2 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} = \sigma^2 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1}.$$

Comparing with  $\text{Cov}(\hat{\beta}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$  (Theorem 5.5.1) identifies the leading block of the inverse of a partitioned matrix,

$$[(\mathbf{X}^\top \mathbf{X})^{-1}]_{11} = (\mathbf{X}_1^\top (\mathbf{I} - \mathbf{M}_2) \mathbf{X}_1)^{-1} = (\mathbf{X}_1^\top \mathbf{X}_1 - \mathbf{X}_1^\top \mathbf{X}_2 (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} \mathbf{X}_2^\top \mathbf{X}_1)^{-1}, \quad (6.6.1)$$

a Schur-complement identity that Chapter 1 proves by algebra. For a single regressor  $\mathbf{x}_j$ , with all other columns (including the intercept) in  $\mathbf{X}_2$ ,

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2} = \frac{\sigma^2}{S_{jj}(1 - R_j^2)}, \quad (6.6.2)$$

where  $S_{jj} = \sum_i (x_{ij} - \bar{x}_j)^2$  and  $R_j^2$  is the coefficient of determination from regressing  $\mathbf{x}_j$  on the other regressors. The second equality is Exercise (B1). The factor  $1/(1 - R_j^2)$  is the *variance inflation factor* of Chapter 26 (Definition 26.2.1).

**Centring.** Take  $\mathbf{X}_2 = \mathbf{1}$ . Then  $\mathbf{M}_2 = n^{-1} \mathbf{1}\mathbf{1}^\top$  and  $\tilde{\mathbf{X}}_1, \tilde{\mathbf{y}}$  are the centred regressors and response. FWL says the slopes in a model with an intercept are the coefficients of a no-intercept regression of centred  $\mathbf{y}$  on centred regressors, and part (c) gives the intercept  $\bar{y} - \bar{\mathbf{x}}^\top \hat{\beta}_1$ . Example (Intercept inside simple regression) was the case  $p_1 = 1$ .

**Detrending and deseasonalizing.** The original application was economic time series. Frisch and Waugh (1933) showed that regressing on a time trend together with other variables gives the same slope as first detrending every series. Lovell (1963) proved the analogous statement for seasonal dummy variables, and noted a trap: if only the *response* is seasonally adjusted and the regressors are not, the result is generally wrong. Part (a) of the theorem shows why. What must be residualized is the regressor.

**A warning about standard errors.** If the residualized regression of  $\tilde{\mathbf{y}}$  on  $\tilde{\mathbf{X}}_1$  is fitted with ordinary software, the coefficient and residuals are right, but the software divides the residual sum of squares by  $n - p_1$  instead of  $n - p_1 - p_2$ . It does not know that  $p_2$  dimensions were used up in forming  $\tilde{\mathbf{y}}$ . The reported standard error is too small by the factor  $\sqrt{(n - p_1)/(n - p)}$ .

### Example. (Poverty and murder rates)

For the 50 US states in 2009, regress the murder rate (per 100,000) on the poverty rate, the percentage of single-parent households, and the percentage of the population living in urban areas, with an intercept. The coefficient of poverty is 0.2538. On its own,

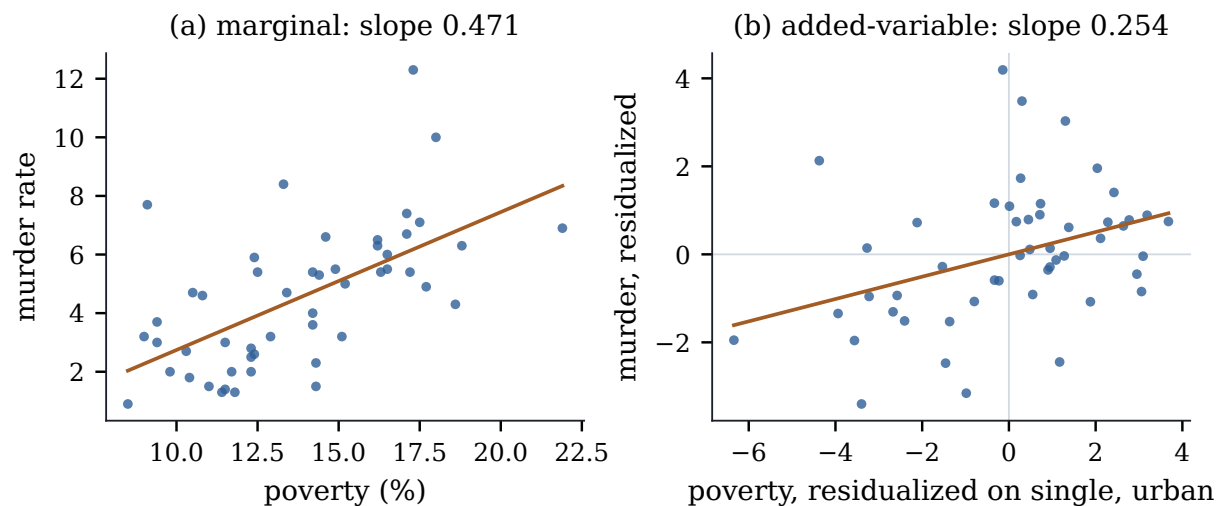


Figure 6.6.1: Murder rate and poverty for 50 US states. (a) The marginal relationship. (b) The added-variable plot: both variables residualized on the single-parent percentage, urban percentage and intercept. The slope in (b) is the multiple-regression coefficient of poverty (Theorem 6.6.2).

a simple regression of murder rate on poverty has slope 0.4706, nearly twice as large. Poverty and single parenthood are correlated, and the simple regression gives poverty credit for part of their shared association.

Figure 6.6.1 shows both regressions. Panel (b) is an **added-variable plot**: the residual of murder rate after regression on the other two regressors and the intercept, against the corresponding residual of poverty. By FWL its least squares slope is exactly the multiple-regression coefficient. The listing confirms the equality to machine precision.

The standard error of the coefficient in the full model is 0.0934. Fitting the residualized regression naively gives 0.0905, smaller by the factor 1.0321, which is  $\sqrt{(n-1)/(n-4)}$  as predicted.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data
data = data.drop(index="District of Columbia")      # an extreme point; see the
                                                    # exercises of Section 6.6

y = data["murder"].to_numpy()
x1 = data["poverty"].to_numpy()                    # the coefficient we care
                                                    # about
X2 = np.column_stack([np.ones(len(y)), data["single"], data["urban"]])
X = np.column_stack([x1, X2])

def resid(v, Z):
    """Residual of v after projecting onto C(Z)."""
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

beta_full, *_ = np.linalg.lstsq(X, y, rcond=None)  # regress y on everything

y_tilde = resid(y, X2)                            # (I - M2) y
x_tilde = resid(x1, X2)                            # (I - M2) x1
```

```

beta_fwl = (x_tilde @ y_tilde) / (x_tilde @ x_tilde) # simple regression, no
            intercept

print(f"full regression:  {beta_full[0]:.6f}")
print(f"FWL regression:   {beta_fwl:.6f}")

```

*Remark* (What FWL does not say). FWL is an algebraic identity about fitted coefficients. It says what the number  $\hat{\beta}_j$  is: an association with the unexplained part of  $\mathbf{x}_j$ . It does not say that changing  $x_j$  in the world, with the other regressors held fixed, would change the response by  $\hat{\beta}_j$ . That is a causal claim, and it needs assumptions about how the data arose. [Chapter 25](#) takes up this question, including cases where adding a regressor to  $\mathbf{X}_2$  makes the coefficient *less* interpretable rather than more.

### 6.6.3 Fixed effects are a projection

One of the most widely used estimators in applied statistics is the FWL theorem in disguise. Suppose units  $k = 1, \dots, K$  are observed repeatedly: firms over years, patients over visits, schools over cohorts. The analyst wants the effect of time-varying regressors  $\mathbf{X}_1$  while allowing each unit its own intercept:

$$E(\mathbf{Y}) = \mathbf{X}_1 \boldsymbol{\beta}_1 + \mathbf{D} \boldsymbol{\alpha},$$

where  $\mathbf{D}$  is the  $n \times K$  matrix of unit indicators. Fitting this directly means a regression with  $K$  extra columns, and  $K$  may be in the thousands.

FWL with  $\mathbf{X}_2 = \mathbf{D}$  removes that burden. The projection onto  $\mathcal{C}(\mathbf{D})$  replaces each entry by its unit's average ([Example \(The one-way layout\)](#)), so  $\mathbf{I} - \mathbf{M}_D$  subtracts unit means. Therefore  $\hat{\boldsymbol{\beta}}_1$  is the least squares coefficient from regressing the *within-unit deviations*  $y_{kt} - \bar{y}_k$  on  $\mathbf{x}_{kt} - \bar{\mathbf{x}}_k$ , with no indicators at all. This is the **within estimator** of panel-data econometrics. Its standard error needs the degrees-of-freedom correction described above, with  $p_2 = K$ , which is large.

#### Example. (Grunfeld's investment data)

Grunfeld's classic panel follows 11 US firms for 20 years, giving  $n = 220$  observations of gross investment, market value and capital stock. With a separate intercept for each firm, the coefficients of value and capital are 0.1101 and 0.3100. The same numbers come from demeaning every variable within firm and fitting two columns. Ignoring the firms altogether (one common intercept) gives 0.1145 and 0.2275 instead. That comparison uses variation *between* firms, of the sort "large firms invest more", which the within estimator deliberately discards.

```

import numpy as np
import pandas as pd
import statsmodels.api as sm
df = sm.datasets.grunfeld.load_pandas().data
y = df["invest"].to_numpy()
X1 = df[["value", "capital"]].to_numpy()

# (i) least squares with one indicator column per firm ("dummy variables")
D = pd.get_dummies(df["firm"]).to_numpy(dtype=float)
beta_dummies, *_ = np.linalg.lstsq(np.column_stack([X1, D]), y, rcond=None)

```

```

# (ii) FWL: projecting onto  $C(D)$  replaces each value by its firm mean,
#       so  $(I - M_D)$  subtracts firm means -- the "within" transformation
def demean_by(v, groups):
    v = pd.DataFrame(v)
    return (v - v.groupby(groups.to_numpy()).transform("mean")).to_numpy()

X1_within = demean_by(X1, df["firm"])
y_within = demean_by(y, df["firm"]).ravel()
beta_within, *_ = np.linalg.lstsq(X1_within, y_within, rcond=None)

print("with firm indicators:", beta_dummies[:2])
print("within transformation:", beta_within)

```

Whether discarding between-unit variation is wise depends on why units differ. That question is taken up for random-effects alternatives in [Chapter 32](#) and [Chapter 33](#).

Added-variable plots are among the most useful diagnostics for multiple regression. They show the evidence for a single coefficient in two dimensions, together with any points that dominate it. They reappear in [Section 20.1](#).

## 6.6.4 Exercises

### 6.6.4.1 A. Check your understanding

#### Exercise A1.

Refit the model of [Example \(Poverty and murder rates\)](#) with the District of Columbia included. How does the poverty coefficient change? Draw the added-variable plot and locate the District. Relate what you see to its leverage ([Example \(High-leverage states\)](#)).

### 6.6.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{x}_j$  be a column of  $\mathbf{X}$ , let  $\mathbf{X}_{(j)}$  be  $\mathbf{X}$  with that column removed, and assume  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$ . Show that  $\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2 = S_{jj}(1 - R_j^2)$ , where  $R_j^2$  is the  $R^2$  from regressing  $\mathbf{x}_j$  on  $\mathbf{X}_{(j)}$ . Deduce [Eq. 6.6.2](#).

*Solution.* Since  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$ , the vector  $(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$  is the residual from regressing  $\mathbf{x}_j$  on  $\mathbf{X}_{(j)}$ . Its squared length is that regression's residual sum of squares, which equals  $\text{SST}(1 - R_j^2)$  with  $\text{SST} = S_{jj}$  by the definition of  $R^2$ . Substituting into the variance formula from FWL gives [Eq. 6.6.2](#).

#### Exercise B2.

In [Theorem 6.6.2](#), regressing  $\mathbf{y}$  (not  $\tilde{\mathbf{y}}$ ) on  $\tilde{\mathbf{X}}_1$  gives the coefficient  $\hat{\beta}_1$ . Show that its residual vector is  $(\mathbf{I} - \mathbf{M})\mathbf{y} + \mathbf{M}_2\mathbf{y}$ , so the residuals are generally wrong even though the coefficient is right.



*Solution.* The fitted vector of the regression of  $\mathbf{y}$  on  $\tilde{\mathbf{X}}_1$  is  $\tilde{\mathbf{M}}_1\mathbf{y}$ . By Lemma 6.6.1,  $\mathbf{y} - \tilde{\mathbf{M}}_1\mathbf{y} = \mathbf{y} - (\mathbf{M} - \mathbf{M}_2)\mathbf{y} = (\mathbf{I} - \mathbf{M})\mathbf{y} + \mathbf{M}_2\mathbf{y}$ . The extra term  $\mathbf{M}_2\mathbf{y}$  is zero only if  $\mathbf{y} \perp \mathcal{C}(\mathbf{X}_2)$ .

### Exercise B3.

For a single regressor  $\mathbf{x}_j$ , the *partial correlation* of  $\mathbf{y}$  and  $\mathbf{x}_j$  given the other regressors is the sample correlation of  $\tilde{\mathbf{y}}$  and  $\tilde{\mathbf{x}}_j$  (both residualized on  $\mathbf{X}_{(j)}$ , which contains  $\mathbf{1}$ ). Let  $t = \hat{\beta}_j / \text{se}(\hat{\beta}_j)$  with  $\text{se}(\hat{\beta}_j)^2 = \text{SSE} / [(n - p) \|\tilde{\mathbf{x}}_j\|^2]$ . Show that the squared partial correlation equals  $t^2 / (t^2 + n - p)$ .

*Solution.* Write  $\tilde{\mathbf{x}} = \tilde{\mathbf{x}}_j$  and  $\tilde{\mathbf{y}} = \tilde{\mathbf{y}}$ . Both have mean zero because  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$ , so their correlation is  $r = \tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} / (\|\tilde{\mathbf{x}}\| \|\tilde{\mathbf{y}}\|)$ . By FWL,  $\hat{\beta}_j = \tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} / \|\tilde{\mathbf{x}}\|^2$ , and the full-model residual sum of squares equals that of the residualized regression:  $\text{SSE} = \|\tilde{\mathbf{y}}\|^2 - (\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}\|^2 = \|\tilde{\mathbf{y}}\|^2 (1 - r^2)$ . Hence

$$t^2 = \frac{\hat{\beta}_j^2 (n - p) \|\tilde{\mathbf{x}}\|^2}{\text{SSE}} = \frac{(n - p) r^2}{1 - r^2},$$

and solving gives  $r^2 = t^2 / (t^2 + n - p)$ .

#### 6.6.4.3 C. Going deeper

### Exercise C1.

Drop the full-rank assumption in Theorem 6.6.2. Show that  $\mathbf{b}_1$  solves the normal equations of the regression of  $\tilde{\mathbf{y}}$  on  $\tilde{\mathbf{X}}_1$  iff there exists  $\mathbf{b}_2$  such that  $(\mathbf{b}_1^\top, \mathbf{b}_2^\top)^\top$  is a least squares estimate in the full model.

## 6.7 Reparameterization, centring and $R^2$

### 6.7.1 Same subspace, same fit

The key idea of Section 6.1 now becomes a theorem.

#### Theorem 6.7.1 — Invariance under reparameterization

Let  $\mathbf{X}$  ( $n \times p$ ) and  $\mathbf{W}$  ( $n \times q$ ) satisfy  $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{W})$ . Then the two linear models  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  and  $E(\mathbf{Y}) = \mathbf{W}\boldsymbol{\gamma}$  have the same projection  $\mathbf{M}$ , the same fitted values and residuals for every  $\mathbf{y}$ , and the same residual sum of squares. There are a  $p \times q$  matrix  $\mathbf{T}$  with  $\mathbf{W} = \mathbf{X}\mathbf{T}$  and a  $q \times p$  matrix  $\mathbf{S}$  with  $\mathbf{X} = \mathbf{W}\mathbf{S}$ . If  $\hat{\boldsymbol{\gamma}}$  is a least squares estimate for  $\mathbf{W}$ , then  $\mathbf{T}\hat{\boldsymbol{\gamma}}$  is one for  $\mathbf{X}$ , and  $\mathbf{S}\hat{\boldsymbol{\beta}}$  is one for  $\mathbf{W}$  whenever  $\hat{\boldsymbol{\beta}}$  is one for  $\mathbf{X}$ . If both matrices have full column rank, then  $p = q$ ,  $\mathbf{T} = \mathbf{S}^{-1}$ , and  $\hat{\boldsymbol{\gamma}} = \mathbf{S}\hat{\boldsymbol{\beta}}$ .

*Proof.* A projection is determined by its range (Theorem 6.3.3), so both models have projection  $\mathbf{M}$ , and fitted values and residuals follow from Theorem 6.4.1. Each column

of  $\mathbf{W}$  lies in  $\mathcal{C}(\mathbf{X})$ , so  $\mathbf{W} = \mathbf{X}\mathbf{T}$  for some  $\mathbf{T}$ , and likewise  $\mathbf{X} = \mathbf{W}\mathbf{S}$ . If  $\mathbf{W}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y}$ , then  $\mathbf{X}(\mathbf{T}\hat{\boldsymbol{\gamma}}) = \mathbf{M}\mathbf{y}$ , so  $\mathbf{T}\hat{\boldsymbol{\gamma}}$  is a least squares estimate for  $\mathbf{X}$ . The other direction is symmetric. With full column rank,  $\mathbf{X} = \mathbf{X}\mathbf{T}\mathbf{S}$  gives  $\mathbf{X}(\mathbf{I} - \mathbf{T}\mathbf{S}) = \mathbf{0}$  and so  $\mathbf{T}\mathbf{S} = \mathbf{I}$ , and similarly  $\mathbf{S}\mathbf{T} = \mathbf{I}$ . Uniqueness of least squares estimates then gives  $\hat{\boldsymbol{\gamma}} = \mathbf{S}\hat{\boldsymbol{\beta}}$ . ■

Many operations that change the printed coefficient table leave the fit exactly unchanged:

- **Changing units** of a regressor (metres to centimetres, dollars to thousands of dollars) rescales one column:  $\mathbf{W} = \mathbf{X}\mathbf{D}$  with  $\mathbf{D}$  diagonal. The coefficient is divided by the scale factor, and nothing else changes.
- **Centring** regressors,  $\mathbf{x}_j \mapsto \mathbf{x}_j - \bar{x}_j\mathbf{1}$ , when the model has an intercept. The slopes are unchanged, and the intercept becomes the fitted value at the mean of the regressors instead of at zero, which is often far outside the data.
- **Recoding a factor**. Reference-level dummies, sum-to-zero effect coding, and cell-means coding without an intercept all span the space of vectors that are constant within levels.
- **Replacing polynomial terms by orthogonal polynomials** (Chapter 42), which span the same space with much better numerical behaviour (Section 6.10).

Operations that *do* change the fit are exactly those that change  $\mathcal{C}(\mathbf{X})$ . Examples are dropping the intercept when no other combination of columns produces  $\mathbf{1}$ , replacing a regressor by its logarithm, and adding an interaction.

## 6.7.2 Centring as a projection

Suppose  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ . By Theorem 6.5.1 with  $\mathbf{X}_0 = \mathbf{1}$ , the fitted vector splits as

$$\mathbf{M}\mathbf{y} = \bar{y}\mathbf{1} + (\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y},$$

and the second projection is onto  $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$ , the vectors in the model space with entries summing to zero. By Lemma 6.6.1 with  $\mathbf{X}_2 = \mathbf{1}$ , that subspace is spanned by the centred columns  $(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{X}$ . Applying Pythagoras to the orthogonal pieces  $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$  and  $\hat{\boldsymbol{\epsilon}}$  of  $\mathbf{y} - \bar{y}\mathbf{1}$  gives the *corrected* decomposition

$$\underbrace{\sum_i (y_i - \bar{y})^2}_{\text{SST}} = \underbrace{\sum_i (\hat{y}_i - \bar{y})^2}_{\text{SSR}} + \underbrace{\sum_i (y_i - \hat{y}_i)^2}_{\text{SSE}}, \quad (6.7.1)$$

with dimensions  $n - 1 = (r - 1) + (n - r)$ . Chapter 9 builds the analysis of variance on this identity.

## 6.7.3 $R^2$ is a squared cosine

The **coefficient of determination** of a model with  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  is

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}}.$$

It depends on the model only through  $\mathcal{C}(\mathbf{X})$ , so by Theorem 6.7.1 it is invariant under every reparameterization. Its geometric meaning is simple.

inside  $\mathbf{1}^\perp$ :  $R^2 = \cos^2 \theta = 0.617$

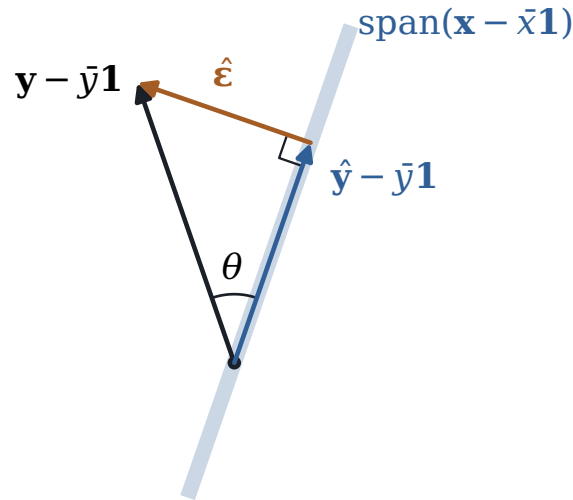


Figure 6.7.1:  $R^2$  as a squared cosine, for  $n = 3$  and one regressor. Centred vectors lie in the plane  $\mathbf{1}^\perp$ , so the whole configuration is two-dimensional. The centred fitted vector is the projection of the centred response onto the line spanned by the centred regressor.

### Theorem 6.7.2

Assume  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  and  $\mathbf{y} \notin \text{span}(\mathbf{1})$ . Let  $\theta$  be the angle between the centred response  $\mathbf{y} - \bar{y}\mathbf{1}$  and the centred fitted vector  $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ . Then

$$R^2 = \cos^2 \theta = \text{corr}(\mathbf{y}, \hat{\mathbf{Y}})^2,$$

where  $\text{corr}$  is the sample correlation. For simple regression on one regressor  $\mathbf{x}$ ,  $R^2 = \text{corr}(\mathbf{y}, \mathbf{x})^2$ .

*Proof.* Write  $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$  and  $\mathbf{v} = \hat{\mathbf{Y}} - \bar{y}\mathbf{1}$ . Then  $\mathbf{u} = \mathbf{v} + \hat{\mathbf{e}}$  with  $\mathbf{v} \perp \hat{\mathbf{e}}$ , so  $\mathbf{u}^\top \mathbf{v} = \|\mathbf{v}\|^2$  and

$$\cos^2 \theta = \frac{(\mathbf{u}^\top \mathbf{v})^2}{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2} = \frac{\|\mathbf{v}\|^2}{\|\mathbf{u}\|^2} = \frac{\text{SSR}}{\text{SST}}.$$

(If  $\mathbf{v} = \mathbf{0}$  the angle is undefined, but  $R^2 = 0$  and the correlation is taken to be zero.) Since  $\hat{\mathbf{Y}}$  has mean  $\bar{y}$  (Proposition 6.4.3(b)),  $\cos \theta$  is the sample correlation of  $\mathbf{y}$  and  $\hat{\mathbf{Y}}$ . In simple regression  $\mathbf{v} = (S_{xy}/S_{xx})(\mathbf{x} - \bar{x}\mathbf{1})$  by Example (Intercept inside simple regression), a scalar multiple of the centred regressor. Rescaling a vector by a nonzero constant does not change a squared cosine, so  $\cos^2 \theta$  is the squared correlation of  $\mathbf{y}$  and  $\mathbf{x}$ . When  $S_{xy} = 0$  both sides are zero. ■

So  $R^2$  measures how closely the model space lines up with the centred data. Figure 6.7.1 draws the case  $n = 3$ . For the three-regressor model of Example (Poverty and murder rates),  $R^2 = 0.6419$ , meaning the centred response makes an angle of  $36.8^\circ$  with the model space. The simple regression on poverty alone has  $R^2 = 0.3599$ , the square of the correlation 0.5999.

```

import numpy as np
import statsmodels.api as sm

def r_squared_as_cosine(y, X):
    """X must contain the intercept column. Returns (R^2 from sums of squares, cos^2
    of angle)."""
    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    y_hat = X @ beta
    yc, yhc = y - y.mean(), y_hat - y.mean()          # centre both vectors
    r2 = 1 - np.sum((y - y_hat) ** 2) / np.sum(yc ** 2)
    cos = (yc @ yhc) / (np.linalg.norm(yc) * np.linalg.norm(yhc))
    return r2, cos ** 2

```

Three consequences follow directly from the geometry.

1.  $R^2$  never decreases when the model space grows. If  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  and both contain  $\mathbf{1}$ , then  $\text{SSE}_0 - \text{SSE} = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 \geq 0$  by [Eq. 6.5.2](#). Adding a regressor of pure noise increases  $R^2$  almost surely. This is why  $R^2$  cannot be used for model selection ([Chapter 29](#)).
2.  $R^2 = 1$  once the model space contains  $\mathbf{y}$ , which always happens when  $\text{rank}(\mathbf{X}) = n$ . A model with as many free parameters as observations interpolates, and its  $R^2$  says nothing.
3. Without an intercept, the definition changes. If  $\mathbf{1} \notin \mathcal{C}(\mathbf{X})$ , the corrected decomposition [Eq. 6.7.1](#) fails, because  $\hat{\mathbf{Y}} - \bar{y}\mathbf{1}$  and  $\hat{\mathbf{e}}$  need not be orthogonal. Software then usually reports the *uncentred*  $R_0^2 = \|\hat{\mathbf{Y}}\|^2 / \|\mathbf{y}\|^2$ , the squared cosine of the angle between  $\mathbf{y}$  and  $\mathcal{C}(\mathbf{X})$  itself. It is typically much larger and cannot be compared with the centred  $R^2$  of a model with an intercept ([Exercise \(B1\)](#)).

## 6.7.4 Exercises

### 6.7.4.1 A. Check your understanding

#### Exercise A1.

A factor with three levels is coded with an intercept and indicators for levels 2 and 3. Write down the matrices  $\mathbf{T}$  and  $\mathbf{S}$  of [Theorem 6.7.1](#) that convert to the coding in which level 3, rather than level 1, is the omitted baseline, and to sum-to-zero effect coding. Express each new coefficient in terms of the old ones.

#### Exercise A2.

Show that, for a model with  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ ,  $R^2$  is unchanged when  $\mathbf{y}$  is replaced by  $a\mathbf{1} + b\mathbf{y}$  with  $b \neq 0$ , and when any column of  $\mathbf{X}$  is replaced by an affine function of itself. Is it unchanged when a regressor is replaced by its logarithm?

### 6.7.4.2 B. Practice

#### Exercise B1.

Show that if  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  then  $R_0^2 = \|\hat{\mathbf{Y}}\|^2 / \|\mathbf{y}\|^2 = (n\bar{y}^2 + \text{SSR}) / (n\bar{y}^2 + \text{SST}) \geq R^2$ .

Construct a data set and a model without an intercept for which  $R_0^2 > 0.95$  but the model's residual sum of squares exceeds that of the intercept-only model.

*Solution.* With  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , the fitted vector splits orthogonally as  $\bar{y}\mathbf{1} + (\hat{\mathbf{Y}} - \bar{y}\mathbf{1})$ , so  $\|\hat{\mathbf{Y}}\|^2 = n\bar{y}^2 + \text{SSR}$ , and likewise  $\|\mathbf{y}\|^2 = n\bar{y}^2 + \text{SST}$ . For  $a \geq 0$  and  $0 \leq \text{SSR} \leq \text{SST}$ ,  $(a + \text{SSR})/(a + \text{SST}) \geq \text{SSR}/\text{SST}$ . For the example, take responses near 100 with small noise and a regressor unrelated to them, and fit through the origin. The fitted line captures the large mean, so  $R_0^2$  is near 1, but its residuals exceed those around  $\bar{y}$ .

### 6.7.4.3 C. Going deeper

#### Exercise C1.

Suppose  $\mathbf{Y} \sim N_n(\mu\mathbf{1}, \sigma^2\mathbf{I})$  and  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  with  $\text{rank}(\mathbf{X}) = r$ . Using the results of Chapter 4, show that  $R^2 \sim \text{Beta}((r-1)/2, (n-r)/2)$ , so that  $E(R^2) = (r-1)/(n-1)$  even though no regressor is related to the response. Use this to motivate the adjusted  $R^2 = 1 - (1 - R^2)(n-1)/(n-r)$ .

## 6.8 Leverage as geometry

The projection  $\mathbf{M}$  is an  $n \times n$  matrix whose entries have a direct interpretation. Since  $\hat{y}_i = \sum_j m_{ij}y_j$ , the entry  $m_{ij}$  is the weight observation  $j$  receives in the fitted value for observation  $i$ . So

$$m_{ij} = \frac{\partial \hat{y}_i}{\partial y_j}.$$

The diagonal entries measure how strongly each observation pulls the fitted surface towards itself.

### Definition 6.8.1 — Leverage

The **leverage** of observation  $i$  is  $h_{ii} = m_{ii}$ , the  $i$ th diagonal entry of  $\mathbf{M}$ . In this context  $\mathbf{M}$  is often called the **hat matrix** and written  $\mathbf{H}$ , because it “puts the hat on”  $\mathbf{y}$ .

Leverage is a property of the design, not of the response. It depends on  $\mathbf{y}$  not at all, and on  $\mathbf{X}$  only through  $\mathcal{C}(\mathbf{X})$ , so it is unchanged by reparameterization.

### Proposition 6.8.1 — Properties of leverage

Let  $\text{rank}(\mathbf{X}) = r$ , and let  $\mathbf{e}_i$  denote the  $i$ th standard basis vector. Then:

- (a)  $h_{ii} = \|\mathbf{M}\mathbf{e}_i\|^2 = \sum_j m_{ij}^2$ , the squared length of the projection of  $\mathbf{e}_i$  onto  $\mathcal{C}(\mathbf{X})$ ;
- (b)  $0 \leq h_{ii} \leq 1$ , and  $\sum_i h_{ii} = r$ , so the average leverage is  $r/n$ ;
- (c) if  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , then  $h_{ii} \geq 1/n$ ;
- (d)  $h_{ii} = 1$  iff  $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$ . In that case  $\hat{y}_i = y_i$  for every  $\mathbf{y}$ , the  $i$ th residual is always zero, and  $m_{ij} = 0$  for  $j \neq i$ ;

- (e) if  $\mathbf{X} = \mathbf{QR}$  with  $\mathbf{Q}$  having orthonormal columns spanning  $\mathcal{C}(\mathbf{X})$ , then  $h_{ii}$  is the squared length of the  $i$ th row of  $\mathbf{Q}$ .

*Proof.*

- (a)  $h_{ii} = \mathbf{e}_i^\top \mathbf{M} \mathbf{e}_i = \mathbf{e}_i^\top \mathbf{M}^\top \mathbf{M} \mathbf{e}_i$ , and  $\mathbf{M} \mathbf{e}_i$  is the  $i$ th column of  $\mathbf{M}$ .  
 (b) By [Proposition 6.3.4\(d\)](#),  $0 \leq \|\mathbf{M} \mathbf{e}_i\|^2 \leq \|\mathbf{e}_i\|^2 = 1$ . The sum is  $\text{tr}(\mathbf{M}) = r$ .  
 (c) With  $\mathbf{M}_0 = n^{-1} \mathbf{1} \mathbf{1}^\top$ , the difference  $\mathbf{M} - \mathbf{M}_0$  is a projection ([Theorem 6.5.1](#)), hence nonnegative definite, so  $h_{ii} \geq (\mathbf{M}_0)_{ii} = 1/n$ .  
 (d) Equality holds in  $\|\mathbf{M} \mathbf{e}_i\| \leq \|\mathbf{e}_i\|$  iff  $(\mathbf{I} - \mathbf{M}) \mathbf{e}_i = \mathbf{0}$ , that is, iff  $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$ . Then  $\mathbf{M} \mathbf{e}_i = \mathbf{e}_i$ , so column  $i$  of  $\mathbf{M}$  is  $\mathbf{e}_i$ , and by symmetry so is row  $i$ . Hence  $\hat{y}_i = y_i$ .  
 (e)  $\mathbf{M} = \mathbf{Q} \mathbf{Q}^\top$ , so  $m_{ii} = \sum_k q_{ik}^2$ .

■

Part (d) describes the extreme case. An observation with leverage one has a coordinate direction of its own in the model space. This happens, for example, when it is the only member of a group in a one-way layout, or when a column of  $\mathbf{X}$  is an indicator for that single observation. The model then fits the observation exactly, whatever its value, and the data give no check on it. High leverage is a weaker form of the same thing. If  $h_{ii}$  is close to one, then  $\hat{y}_i \approx y_i$ , and a gross error in  $y_i$  barely shows up in its residual.

This shows up in the residual variances. If  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ , then by [Proposition 6.4.3\(e\)](#)

$$\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii}),$$

so residuals at high-leverage points are *less* variable than others, even though those are exactly the points where an error would do the most damage. [Chapter 20](#) standardizes residuals to correct for this ([Definition 20.1.1](#)).

## 6.8.1 Leverage as distance from the centre

When the model has an intercept, leverage has a second description in variable space: distance from the centroid of the regressors. Write  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$ , let  $\bar{\mathbf{x}}$  be the vector of column means of  $\mathbf{X}_1$ , and let  $\tilde{\mathbf{X}}_1 = (\mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top) \mathbf{X}_1$  be the centred regressors, assumed to have full column rank.

### Proposition 6.8.2

Under these assumptions,

$$h_{ii} = \frac{1}{n} + (\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} (\mathbf{x}_{(i)} - \bar{\mathbf{x}}),$$

where  $\mathbf{x}_{(i)}$  is the  $i$ th row of  $\mathbf{X}_1$ , written as a column vector.

*Proof.* By [Lemma 6.6.1](#) with  $\mathbf{X}_2 = \mathbf{1}$ ,  $\mathbf{M} = n^{-1} \mathbf{1} \mathbf{1}^\top + \tilde{\mathbf{X}}_1 (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} \tilde{\mathbf{X}}_1^\top$ . The  $i$ th row of  $\tilde{\mathbf{X}}_1$  is  $(\mathbf{x}_{(i)} - \bar{\mathbf{x}})^\top$ .

■

The quadratic form is  $(n-1)^{-1}$  times the squared Mahalanobis distance of  $\mathbf{x}_{(i)}$  from the centroid, measured in the metric of the sample covariance matrix of the regressors. Leverage is therefore high for points that are unusual *jointly*, even when no single coordinate is extreme. A state with moderate poverty and moderate single parenthood, but an unusual *combination* of the two, can have high leverage. For simple regression the formula reduces to  $h_{ii} = n^{-1} + (x_i - \bar{x})^2 / S_{xx}$ .

### Example. (High-leverage states)

Return to the state data, now with all 51 jurisdictions including the District of Columbia, and the model of [Example \(Poverty and murder rates\)](#). The average leverage is  $p/n = 0.078$ , and the smallest is 0.021, just above the bound  $1/n$ . The three largest are

| Jurisdiction         | $h_{ii}$ | multiple of average |
|----------------------|----------|---------------------|
| District of Columbia | 0.506    | 6.5                 |
| Mississippi          | 0.212    | 2.7                 |
| Alaska               | 0.177    | 2.3                 |

The District of Columbia is entirely urban, with by far the highest proportion of single-parent households. It sits at a corner of the regressor cloud that no state comes near, and its leverage is about half the maximum possible. It was omitted from [Example \(Poverty and murder rates\)](#). [Exercise \(A1\)](#) asks what happens to the poverty coefficient when it is put back.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data           # District of Columbia
# kept here
X = np.column_stack([np.ones(len(data)), data["poverty"], data["single"], data["urban"]])

Q, _ = np.linalg.qr(X)                                     # orthonormal basis for C(X)
h = np.sum(Q ** 2, axis=1)                                  # h_ii = m_ii = ||row i of Q||^2
n, p = X.shape
print("sum of leverages:", h.sum(), " (= p =", p, ")")
print("range:", h.min(), h.max(), " bounds:", 1 / n, 1)
top = np.argsort(h)[::-1][:3]
for i in top:
    print(f"{data.index[i]:22s} h = {h[i]:.3f}    ({h[i] / (p / n):.1f} x average)")
```

The listing computes leverages from a QR factorization rather than by forming  $\mathbf{M}$ . For large  $n$  the  $n \times n$  matrix  $\mathbf{M}$  should never be formed. [Section 6.10](#) explains why.

## 6.8.2 Exercises

### 6.8.2.1 A. Check your understanding

#### Exercise A1.

For simple regression, derive  $h_{ii} = n^{-1} + (x_i - \bar{x})^2 / S_{xx}$  directly from [Example \(Intercept](#)

inside simple regression), and show that the largest leverage belongs to the observation with  $x_i$  farthest from  $\bar{x}$ .

### 6.8.2.2 B. Practice

#### Exercise B1.

Show that  $m_{ij}^2 \leq m_{ii}(1 - m_{ii})$  for  $i \neq j$ , and hence  $|m_{ij}| \leq 1/2$ . When is the bound attained?

*Solution.* From [Proposition 6.8.1\(a\)](#),  $m_{ii} = \sum_k m_{ik}^2 \geq m_{ii}^2 + m_{ij}^2$ . So  $m_{ij}^2 \leq m_{ii}(1 - m_{ii}) \leq 1/4$ . Equality needs  $m_{ii} = 1/2$  and  $m_{ij}$  the only other nonzero entry of row  $i$ . The intercept-only model with  $n = 2$ , where every entry of  $\mathbf{M}$  is  $1/2$ , attains it.

#### Exercise B2.

Show that adding a column to  $\mathbf{X}$  cannot decrease any leverage. Show that the leverages in a one-way layout are  $1/n_k$  for members of group  $k$ , and explain why an observation forming a group on its own has leverage one.

## 6.9 Other inner products

Everything so far has used the Euclidean inner product  $\mathbf{u}^\top \mathbf{v}$ . That choice treats the  $n$  coordinates of observation space symmetrically: an error of a given size counts the same in every observation, and errors in different observations are unrelated. When observations have different variances or are correlated, Euclidean distance is the wrong way to measure how far  $\mathbf{y}$  is from the model space. This section shows that the whole theory carries over to any inner product, and that the result is generalized least squares.

### 6.9.1 Projection in a general inner product

Let  $\mathbf{A}$  be an  $n \times n$  symmetric positive definite matrix, and define

$$\langle \mathbf{u}, \mathbf{v} \rangle_{\mathbf{A}} = \mathbf{u}^\top \mathbf{A} \mathbf{v}, \quad \|\mathbf{u}\|_{\mathbf{A}} = (\mathbf{u}^\top \mathbf{A} \mathbf{u})^{1/2}.$$

This is an inner product: bilinear, symmetric, and positive on nonzero vectors. Say  $\mathbf{u}$  and  $\mathbf{v}$  are  $\mathbf{A}$ -orthogonal if  $\mathbf{u}^\top \mathbf{A} \mathbf{v} = 0$ . Nothing in the proofs of [Proposition 6.2.2](#), [Theorem 6.2.3](#) or [Theorem 6.3.1](#) used anything about  $\mathbf{u}^\top \mathbf{v}$  except these three properties. So those results hold verbatim with  $\langle \cdot, \cdot \rangle_{\mathbf{A}}$  in place of the Euclidean inner product. In particular, every  $\mathbf{y}$  has a unique  $\mathbf{A}$ -nearest point in a subspace  $\mathcal{S}$ , characterized by  $\mathbf{A}$ -orthogonality of the error to  $\mathcal{S}$ . The map sending  $\mathbf{y}$  to it is linear, and we write it  $\mathbf{P}_{\mathcal{S}, \mathbf{A}}$ .

#### Theorem 6.9.1

Let  $\mathbf{A}$  be positive definite and  $\mathbf{X}$  any  $n \times p$  matrix. The  $\mathbf{A}$ -orthogonal projection onto



$\mathcal{C}(\mathbf{X})$  is

$$\mathbf{P}_{\mathbf{X},\mathbf{A}} = \mathbf{X}(\mathbf{X}^\top \mathbf{A} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{A},$$

for any generalized inverse. It satisfies  $\mathbf{P}_{\mathbf{X},\mathbf{A}}^2 = \mathbf{P}_{\mathbf{X},\mathbf{A}}$  and  $\mathcal{C}(\mathbf{P}_{\mathbf{X},\mathbf{A}}) = \mathcal{C}(\mathbf{X})$ , and the matrix  $\mathbf{A}\mathbf{P}_{\mathbf{X},\mathbf{A}}$  is symmetric. The matrix  $\mathbf{P}_{\mathbf{X},\mathbf{A}}$  itself is symmetric iff  $\mathcal{A}\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , and in general it is not.

*Proof.* Let  $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$  with  $\mathbf{L}$  nonsingular (for instance the Cholesky factor), and put  $\mathbf{X}_* = \mathbf{L}^\top \mathbf{X}$  and  $\mathbf{y}_* = \mathbf{L}^\top \mathbf{y}$ . Then  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|_{\mathbf{A}}^2 = \|\mathbf{y}_* - \mathbf{X}_*\mathbf{b}\|^2$ , so  $\mathbf{b}$  minimizes the  $\mathbf{A}$ -distance iff it is an ordinary least squares estimate for  $(\mathbf{y}_*, \mathbf{X}_*)$ . By [Theorem 6.3.7](#) and [Theorem 6.4.2](#), one such estimate is  $(\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{y}_* = (\mathbf{X}^\top \mathbf{A} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{A} \mathbf{y}$ , and  $\mathbf{X}_*(\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top$  is invariant to the choice of generalized inverse. Multiplying on the left by  $\mathbf{L}^{-\top}$  and on the right by  $\mathbf{L}^\top$  shows that  $\mathbf{P}_{\mathbf{X},\mathbf{A}} = \mathbf{L}^{-\top} \mathbf{M}_* \mathbf{L}^\top$ , where  $\mathbf{M}_*$  projects onto  $\mathcal{C}(\mathbf{X}_*)$ . Idempotence and  $\mathcal{C}(\mathbf{P}_{\mathbf{X},\mathbf{A}}) = \mathbf{L}^{-\top} \mathcal{C}(\mathbf{X}_*) = \mathcal{C}(\mathbf{X})$  follow, and  $\mathbf{A}\mathbf{P}_{\mathbf{X},\mathbf{A}} = \mathbf{L}\mathbf{M}_*\mathbf{L}^\top$  is symmetric. For the last claim, see [Exercise \(B1\)](#). ■

The proof also gives the general principle:  *$\mathbf{A}$ -geometry is Euclidean geometry after the change of variables  $\mathbf{y} \mapsto \mathbf{L}^\top \mathbf{y}$ .* Any fact about orthogonal projections has an  $\mathbf{A}$ -version, obtained by transporting it through  $\mathbf{L}$ .

Symmetry is the Euclidean signature of an orthogonal projection. The  $\mathbf{A}$ -version of [Theorem 6.3.3](#) replaces it with self-adjointness.

### Proposition 6.9.2

An idempotent matrix  $\mathbf{P}$  is the  $\mathbf{A}$ -orthogonal projection onto  $\mathcal{C}(\mathbf{P})$  iff  $\mathbf{A}\mathbf{P}$  is symmetric, that is, iff  $\langle \mathbf{P}\mathbf{u}, \mathbf{v} \rangle_{\mathbf{A}} = \langle \mathbf{u}, \mathbf{P}\mathbf{v} \rangle_{\mathbf{A}}$  for all  $\mathbf{u}, \mathbf{v}$ .

*Proof.* For idempotent  $\mathbf{P}$ , the residual  $(\mathbf{I} - \mathbf{P})\mathbf{y}$  is  $\mathbf{A}$ -orthogonal to  $\mathcal{C}(\mathbf{P})$  for every  $\mathbf{y}$  iff  $\mathbf{P}^\top \mathbf{A}(\mathbf{I} - \mathbf{P}) = \mathbf{0}$ , that is, iff  $\mathbf{P}^\top \mathbf{A} = \mathbf{P}^\top \mathbf{A}\mathbf{P}$ . If  $\mathbf{A}\mathbf{P}$  is symmetric, then  $\mathbf{P}^\top \mathbf{A} = \mathbf{A}\mathbf{P}$ , and  $\mathbf{P}^\top \mathbf{A}\mathbf{P} = \mathbf{A}\mathbf{P}\mathbf{P} = \mathbf{A}\mathbf{P} = \mathbf{P}^\top \mathbf{A}$ . Conversely,  $\mathbf{P}^\top \mathbf{A} = \mathbf{P}^\top \mathbf{A}\mathbf{P}$  has a symmetric right-hand side, so  $\mathbf{P}^\top \mathbf{A}$  and hence its transpose  $\mathbf{A}\mathbf{P}$  are symmetric. ■

## 6.9.2 Generalized least squares

Suppose  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$  with  $\mathbf{V}$  known and positive definite. Observations with large variance should count for less, and correlated observations partly repeat each other. The natural distance is the one that makes the errors look uncorrelated:  $\|\cdot\|_{\mathbf{V}^{-1}}$ . If  $\mathbf{V} = \mathbf{L}\mathbf{L}^\top$ , then  $\mathbf{L}^{-1}\boldsymbol{\varepsilon}$  has covariance  $\sigma^2 \mathbf{I}$ , and  $\|\mathbf{u}\|_{\mathbf{V}^{-1}} = \|\mathbf{L}^{-1}\mathbf{u}\|$ .

### Definition 6.9.1 — Generalized least squares

A **generalized least squares** (GLS) estimate minimizes  $(\mathbf{y} - \mathbf{X}\mathbf{b})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\mathbf{b})$ . Equivalently,  $\mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}} = \mathbf{P}_{\mathbf{X},\mathbf{V}^{-1}}\mathbf{y}$ , and  $\hat{\boldsymbol{\beta}}_{\text{GLS}}$  solves the generalized normal equations

$$\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{y}.$$

**Theorem 6.9.1** with  $\mathbf{A} = \mathbf{V}^{-1}$  is the whole of the algebra. Computationally, GLS is ordinary least squares applied to the *whitened* data  $(\mathbf{L}^{-1}\mathbf{y}, \mathbf{L}^{-1}\mathbf{X})$ . Aitken (1935) showed that GLS gives the best linear unbiased estimator under this covariance, which is the Gauss–Markov theorem transported through  $\mathbf{L}$ . [Chapter 7](#) proves it as [Corollary 7.2.3](#), and [Chapter 31](#) handles singular  $\mathbf{V}$  and the more realistic case where  $\mathbf{V}$  must be estimated.

When is the ordinary least squares fit already the GLS fit? Geometrically, the two projections onto  $\mathcal{C}(\mathbf{X})$  coincide iff the Euclidean and the  $\mathbf{V}^{-1}$  notions of orthogonality agree on this particular subspace.

### Theorem 6.9.3 — Kruskal

For positive definite  $\mathbf{V}$ ,  $\mathbf{M}\mathbf{y} = \mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}\mathbf{y}$  for every  $\mathbf{y}$  iff  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ .

*Proof.* The two projections have the same range. They agree iff they have the same null space, since an idempotent matrix is determined by its range and null space ([Proposition 6.9.4](#)). The null space of  $\mathbf{M}$  is  $\mathcal{C}(\mathbf{X})^\perp = \mathcal{N}(\mathbf{X}^\top)$ . The null space of  $\mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}$  is the set of  $\mathbf{u}$  with  $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{u} = \mathbf{0}$ , that is,  $\mathbf{V} \mathcal{N}(\mathbf{X}^\top)$ . So the projections agree iff  $\mathbf{V} \mathcal{N}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X}^\top)$ , and since  $\mathbf{V}$  is nonsingular and the spaces have equal dimension, iff  $\mathbf{V} \mathcal{N}(\mathbf{X}^\top) \subseteq \mathcal{N}(\mathbf{X}^\top)$ . By symmetry of  $\mathbf{V}$ , this says  $\mathbf{X}^\top \mathbf{V} \mathbf{u} = \mathbf{0}$  whenever  $\mathbf{X}^\top \mathbf{u} = \mathbf{0}$ , that is,  $\mathcal{C}(\mathbf{V}\mathbf{X}) \perp \mathcal{C}(\mathbf{X})^\perp$ , which is  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ . ■

The condition holds, for example, for equicorrelated errors  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$  whenever  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  ([Exercise \(A1\)](#)). It also underlies the analysis of balanced split-plot designs in [Chapter 33](#) ([Theorem 33.1.2](#)). The result is due to Kruskal (1968), and the coordinate-free argument is his.

#### Example. (Autocorrelated errors)

Take eight equally spaced observations with a linear trend and error covariance  $v_{st} = 0.7^{|s-t|}$ . The  $\mathbf{V}^{-1}$ -projection onto  $\mathcal{C}(\mathbf{X})$  is idempotent and self-adjoint for  $\langle \cdot, \cdot \rangle_{\mathbf{V}^{-1}}$  but not symmetric: its largest asymmetry  $\max |p_{st} - p_{ts}|$  is 0.444. The ordinary and generalized slope estimates are 0.648 and 0.650. The listing checks each claim, including that the GLS fit is ordinary least squares on whitened data.

```
import numpy as np
rng = np.random.default_rng(9)
n = 8
X = np.column_stack([np.ones(n), np.arange(n, dtype=float)])
y = X @ np.array([1.0, 0.5]) + rng.normal(size=n)
t = np.arange(n)
V = 0.7 ** np.abs(t[:, None] - t[None, :])           # AR(1)-type covariance

Vinv = np.linalg.inv(V)
P_V = X @ np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv) # V^{-1}-orthogonal
               projection onto C(X)
```

```

print("idempotent:", np.allclose(P_V @ P_V, P_V))
print("symmetric: ", np.allclose(P_V, P_V.T))           # False: oblique in
               the usual geometry
print("self-adjoint for <u,v> = u'V^{-1}v:", np.allclose(Vinv @ P_V, (Vinv @ P_V).T)
      )

# GLS = ordinary least squares after whitening with a square root of V^{-1}
L = np.linalg.cholesky(V)                               # V = L L'
Xw, yw = np.linalg.solve(L, X), np.linalg.solve(L, y)
beta_whitened, *_ = np.linalg.lstsq(Xw, yw, rcond=None)
beta_gls = np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv @ y)
print("GLS via P_V equals OLS on whitened data:", np.allclose(beta_gls,
      beta_whitened))

```

### 6.9.3 Weighted least squares

The simplest non-Euclidean geometry is a diagonal one. If the errors are uncorrelated with variances  $\sigma^2/w_i$ , then  $\mathbf{V}^{-1} = \mathbf{W} = \text{diag}(w_1, \dots, w_n)$ , and GLS minimizes  $\sum_i w_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$ . Observations believed to be noisier get less weight in the distance. Whitening multiplies row  $i$  of  $\mathbf{X}$  and  $\mathbf{y}$  by  $\sqrt{w_i}$ . The fitted values are  $\mathbf{P}_{\mathbf{X}, \mathbf{W}} \mathbf{y}$ , and the natural leverages are the diagonal entries of  $\mathbf{P}_{\mathbf{X}, \mathbf{W}}$ . These equal the diagonal of the orthogonal projection onto  $\mathcal{C}(\mathbf{W}^{1/2} \mathbf{X})$  and still sum to  $\text{rank}(\mathbf{X})$  (Exercise (B3)).

#### Example. (Engel's food expenditure data)

Engel's 1857 survey of 235 Belgian working-class households records annual income and food expenditure. The spread of expenditure grows with income, as Figure 6.9.1(a) shows. Weighting by  $w_i = 1/\text{income}_i^2$  (standard deviation proportional to income) changes the fitted slope from 0.4852 to 0.5740. The change in geometry shows up more clearly in the leverages, panel (b). Under ordinary least squares the richest households have the most leverage, with a maximum of 0.255. In the weighted geometry their large variance shrinks their influence, and no household has leverage above 0.037. Whether this is the *right* geometry depends on whether the variance model holds. Chapter 21 discusses how to check it, and how to get valid standard errors when it fails.

```

import numpy as np
import statsmodels.api as sm
df = sm.datasets.engel.load_pandas().data
x, y = df["income"].to_numpy(), df["foodexp"].to_numpy()
X = np.column_stack([np.ones(len(x)), x])

w = 1.0 / x**2                                           # working model: sd proportional to
               income
sw = np.sqrt(w)
beta_ols, *_ = np.linalg.lstsq(X, y, rcond=None)
beta_wls, *_ = np.linalg.lstsq(X * sw[:, None], y * sw, rcond=None) # OLS on (W
               ^{1/2} X, W^{1/2} y)

# leverages in the weighted geometry: squared row norms of an orthonormal basis of C
               (W^{1/2} X)
Qw, _ = np.linalg.qr(X * sw[:, None])
h_wls = np.sum(Qw**2, axis=1)

```

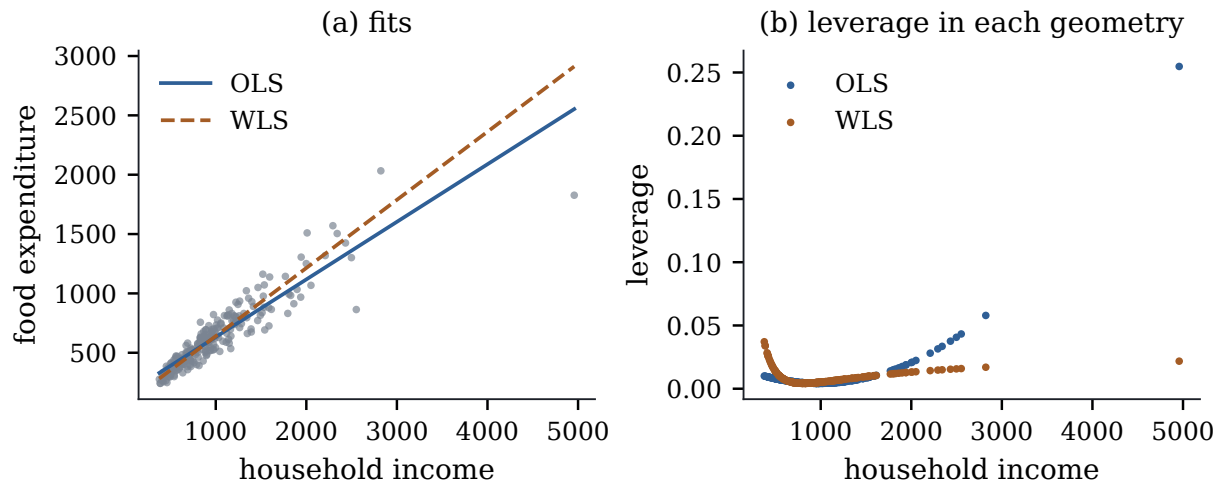


Figure 6.9.1: Engel's data. (a) Ordinary and weighted least squares fits, with weights proportional to  $1/\text{income}^2$ . (b) Leverages under each geometry. The weighted projection spreads influence away from the high-variance, high-income households.

```
Q, _ = np.linalg.qr(X)
h_ols = np.sum(Q**2, axis=1)
print("OLS:", beta_ols, " WLS:", beta_wls)
print("max leverage OLS %.3f, WLS %.3f" % (h_ols.max(), h_wls.max()))
```

## 6.9.4 Oblique projections

The matrices  $\mathbf{P}_{\mathbf{X}, \mathbf{A}}$  are idempotent without being symmetric. In the Euclidean geometry of  $\mathbb{R}^n$  they are **oblique projections**. Idempotent matrices in general have a clean description.

### Proposition 6.9.4

Let  $\mathbf{P}$  be idempotent. Then  $\mathbb{R}^n = \mathcal{C}(\mathbf{P}) \oplus \mathcal{N}(\mathbf{P})$ , and  $\mathbf{P}$  maps each  $\mathbf{y} = \mathbf{u} + \mathbf{v}$  ( $\mathbf{u} \in \mathcal{C}(\mathbf{P})$ ,  $\mathbf{v} \in \mathcal{N}(\mathbf{P})$ ) to  $\mathbf{u}$ . Conversely, for any decomposition  $\mathbb{R}^n = \mathcal{S} \oplus \mathcal{T}$  there is exactly one idempotent  $\mathbf{P}$  with  $\mathcal{C}(\mathbf{P}) = \mathcal{S}$  and  $\mathcal{N}(\mathbf{P}) = \mathcal{T}$ , the *projection onto  $\mathcal{S}$  along  $\mathcal{T}$* . It is an orthogonal projection iff  $\mathcal{T} = \mathcal{S}^\perp$ , that is, iff  $\mathbf{P}$  is symmetric.

*Proof.* Write  $\mathbf{y} = \mathbf{P}\mathbf{y} + (\mathbf{I} - \mathbf{P})\mathbf{y}$ . The first term lies in  $\mathcal{C}(\mathbf{P})$ , and  $\mathbf{P}(\mathbf{I} - \mathbf{P})\mathbf{y} = \mathbf{0}$  puts the second in  $\mathcal{N}(\mathbf{P})$ . If  $\mathbf{u} = \mathbf{P}\mathbf{w} \in \mathcal{N}(\mathbf{P})$  then  $\mathbf{u} = \mathbf{P}^2\mathbf{w} = \mathbf{P}\mathbf{u} = \mathbf{0}$ , so the sum is direct. For  $\mathbf{u} \in \mathcal{C}(\mathbf{P})$ ,  $\mathbf{P}\mathbf{u} = \mathbf{u}$  (write  $\mathbf{u} = \mathbf{P}\mathbf{w}$ ), so  $\mathbf{P}(\mathbf{u} + \mathbf{v}) = \mathbf{u}$ . Conversely, given  $\mathcal{S} \oplus \mathcal{T}$ , the map  $\mathbf{u} + \mathbf{v} \mapsto \mathbf{u}$  is well defined and linear by uniqueness of the decomposition, and it is idempotent with the required range and null space. It is unique because its values are forced on all of  $\mathbb{R}^n$ . If  $\mathcal{T} = \mathcal{S}^\perp$ , the map is the orthogonal projection, which is symmetric. If  $\mathbf{P}$  is symmetric, then  $\mathcal{N}(\mathbf{P}) = \mathcal{N}(\mathbf{P}^\top) = \mathcal{C}(\mathbf{P})^\perp$  by Lemma 6.2.1. ■

Figure 6.9.2 shows the difference. The nearest-point property belongs to the orthogonal projection alone. An oblique projection can move a vector *farther* away: its operator norm exceeds one unless it is orthogonal (Exercise (B2)). The GLS projection is still a nearest-point map,

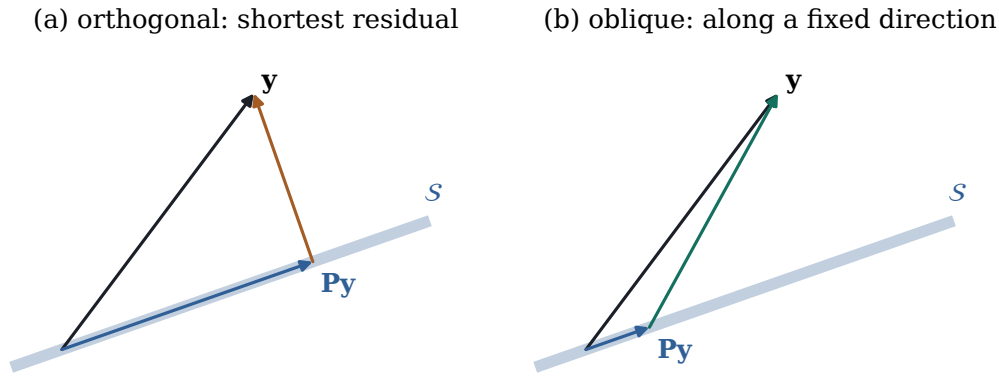


Figure 6.9.2: Projections onto a line  $\mathcal{S}$  in  $\mathbb{R}^2$ . (a) The orthogonal projection: the residual is perpendicular to  $\mathcal{S}$ , and  $\mathbf{Py}$  is the nearest point. (b) An oblique projection moves  $\mathbf{y}$  along a fixed direction that is not perpendicular to  $\mathcal{S}$ . The result is idempotent but is not the nearest point, and the residual is longer.

but in the  $\mathbf{V}^{-1}$  norm, not the Euclidean one. Which geometry is “right” is not a mathematical question. It is decided by the covariance of the errors.

## 6.9.5 Exercises

### 6.9.5.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$  with  $-1/(n - 1) < \rho < 1$ . Show that  $\mathbf{V}$  is positive definite and that  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  whenever  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ . Conclude that ordinary least squares is generalized least squares for equicorrelated errors in any model with an intercept.

*Solution.*  $\mathbf{V}\mathbf{1} = (1 + (n - 1)\rho)\mathbf{1}$  and  $\mathbf{V}\mathbf{u} = (1 - \rho)\mathbf{u}$  for  $\mathbf{u} \perp \mathbf{1}$ . So the eigenvalues are  $1 + (n - 1)\rho$  and  $1 - \rho$ , both positive exactly when  $-1/(n - 1) < \rho < 1$ . Then  $\mathbf{V}\mathbf{X} = (1 - \rho)\mathbf{X} + \rho\mathbf{1}(\mathbf{1}^\top\mathbf{X})$ , whose columns lie in  $\mathcal{C}(\mathbf{X})$  when  $\mathbf{1}$  does. [Theorem 6.9.3](#) applies.

### 6.9.5.2 B. Practice

#### Exercise B1.

Complete the proof of [Theorem 6.9.1](#). Show that  $\mathbf{P}_{\mathbf{X},\mathbf{A}}$  is symmetric iff it equals  $\mathbf{M}$ , and deduce from [Theorem 6.9.3](#) (with  $\mathbf{V} = \mathbf{A}^{-1}$ ) that this happens iff  $\mathcal{AC}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ .

#### Exercise B2.

Let  $\mathbf{P} \neq \mathbf{0}$  be idempotent. Show that the operator norm  $\|\mathbf{P}\|_2 = \max_{\|\mathbf{u}\|=1} \|\mathbf{P}\mathbf{u}\|$  is at least 1, with equality iff  $\mathbf{P}$  is symmetric. *Hint:* if  $\mathbf{P}$  is not symmetric, find  $\mathbf{u} \in \mathcal{C}(\mathbf{P})$  and  $\mathbf{v} \in \mathcal{N}(\mathbf{P})$  with  $\mathbf{u}^\top\mathbf{v} \neq 0$ , and consider  $\mathbf{u} + t\mathbf{v}$  for small  $t$ .

*Solution.* For  $\mathbf{u} \in \mathcal{C}(\mathbf{P})$ ,  $\mathbf{P}\mathbf{u} = \mathbf{u}$ , so  $\|\mathbf{P}\|_2 \geq 1$ . If  $\mathbf{P}$  is symmetric,  $\|\mathbf{P}\mathbf{u}\| \leq \|\mathbf{u}\|$  by Proposition 6.3.4(d). Suppose  $\mathbf{P}$  is not symmetric. Then  $\mathcal{N}(\mathbf{P}) \neq \mathcal{C}(\mathbf{P})^\perp$  by Proposition 6.9.4. The two spaces have the same dimension, so some  $\mathbf{v} \in \mathcal{N}(\mathbf{P})$  is not orthogonal to  $\mathcal{C}(\mathbf{P})$ . Choose a unit  $\mathbf{u} \in \mathcal{C}(\mathbf{P})$  with  $\mathbf{u}^\top \mathbf{v} > 0$ . For  $t > 0$ ,  $\mathbf{P}(\mathbf{u} - t\mathbf{v}) = \mathbf{u}$ , while  $\|\mathbf{u} - t\mathbf{v}\|^2 = 1 - 2t\mathbf{u}^\top \mathbf{v} + t^2\|\mathbf{v}\|^2 < 1$  for small  $t$ . So  $\mathbf{P}$  stretches  $\mathbf{u} - t\mathbf{v}$ , and  $\|\mathbf{P}\|_2 > 1$ .

### Exercise B3.

For weighted least squares,  $\mathbf{V} = \text{diag}(1/w_1, \dots, 1/w_n)$  with  $w_i > 0$ . Show that the diagonal entries of  $\mathbf{P}_{\mathbf{X}, \mathbf{V}^{-1}}$  equal those of the orthogonal projection onto  $\mathcal{C}(\mathbf{W}^{1/2}\mathbf{X})$ , where  $\mathbf{W} = \text{diag}(w_i)$ , and that they sum to  $\text{rank}(\mathbf{X})$ . Interpret them as leverages.

## 6.10 Computation

The formulas  $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  and  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$  are the right way to *reason* about least squares and the wrong way to *compute* it. There are two problems.  $\mathbf{M}$  is  $n \times n$ , so forming it costs  $O(n^2)$  memory for a quantity that is never needed in full. And forming  $\mathbf{X}^\top \mathbf{X}$  squares the condition number of the problem, which can throw away half of the available floating-point digits before any equation is solved. This section shows how the geometry itself suggests the fix, which is to build an orthonormal basis for  $\mathcal{C}(\mathbf{X})$ . Chapter 10 develops the numerical linear algebra in full.

### 6.10.1 QR: Gram–Schmidt as a factorization

Apply Gram–Schmidt (Proposition 6.2.2) to the columns  $\mathbf{x}_1, \dots, \mathbf{x}_p$  of a full-column-rank  $\mathbf{X}$ . Each  $\mathbf{x}_j$  is a combination of  $\mathbf{u}_1, \dots, \mathbf{u}_j$  with coefficients  $r_{ij} = \mathbf{u}_i^\top \mathbf{x}_j$  for  $i < j$  and  $r_{jj} = \|\mathbf{w}_j\| > 0$ . In matrix form,

$$\mathbf{X} = \mathbf{Q}\mathbf{R},$$

where  $\mathbf{Q} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$  is  $n \times p$  with  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$ , and  $\mathbf{R}$  is  $p \times p$  upper triangular with positive diagonal. This is the **thin QR factorization**. Since  $\mathcal{C}(\mathbf{Q}) = \mathcal{C}(\mathbf{X})$ , Proposition 6.3.2 gives  $\mathbf{M} = \mathbf{Q}\mathbf{Q}^\top$  at once, and the rest of least squares follows:

$$\hat{\mathbf{Y}} = \mathbf{Q}(\mathbf{Q}^\top \mathbf{y}), \quad \mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{Q}^\top \mathbf{y}, \quad \text{SSE} = \|\mathbf{y}\|^2 - \|\mathbf{Q}^\top \mathbf{y}\|^2, \quad h_{ii} = \sum_k q_{ik}^2. \quad (6.10.1)$$

The coefficient equation holds because the normal equations become  $\mathbf{R}^\top \mathbf{R}\mathbf{b} = \mathbf{R}^\top \mathbf{Q}^\top \mathbf{y}$ , and  $\mathbf{R}^\top$  is nonsingular. It is solved by back substitution, with no matrix inverse. Nothing of size  $n \times n$  is ever formed.

In floating point, Gram–Schmidt as written in Proposition 6.2.2 gradually loses orthogonality among the computed  $\mathbf{u}_j$ . Production software computes the same factorization with Householder reflections, which are backward stable (Golub and Van Loan 2013; Higham 2002). The geometry does not change:  $\mathbf{Q}$  is an orthonormal basis for the model space.

## 6.10.2 Conditioning

The sensitivity of least squares to rounding is governed by the **condition number**

$$\kappa(\mathbf{X}) = \frac{\sigma_{\max}(\mathbf{X})}{\sigma_{\min}(\mathbf{X})},$$

the ratio of the largest and smallest singular values of  $\mathbf{X}$  (Chapter 1). Geometrically,  $\kappa(\mathbf{X})$  is large when some combination of columns of unit coefficient norm is nearly zero. The columns are then close to linearly dependent, and the coordinates of a point in  $\mathcal{C}(\mathbf{X})$  are poorly determined, even though the point itself is not. The singular values of  $\mathbf{X}^\top \mathbf{X}$  are the squares of those of  $\mathbf{X}$ , so

$$\kappa(\mathbf{X}^\top \mathbf{X}) = \kappa(\mathbf{X})^2.$$

A useful rule of thumb from numerical analysis says that solving a linear system with condition number  $\kappa$  in double precision loses about  $\log_{10} \kappa$  of the roughly 16 available significant digits. The normal equations solve a system whose matrix is  $\mathbf{X}^\top \mathbf{X}$ , so they lose about  $2 \log_{10} \kappa(\mathbf{X})$  digits. QR solves the triangular system  $\mathbf{R}\mathbf{b} = \mathbf{Q}^\top \mathbf{y}$ , and  $\kappa(\mathbf{R}) = \kappa(\mathbf{X})$  because  $\mathbf{Q}$  has orthonormal columns, so it loses about  $\log_{10} \kappa(\mathbf{X})$ . The precise perturbation theory has one more subtlety. When the residual is large, the least squares *problem* itself has a sensitivity term proportional to  $\kappa(\mathbf{X})^2 \|\hat{\mathbf{e}}\|$ , and no algorithm can avoid it (Higham 2002, ch. 20). QR's advantage is largest when the model fits well.

### Example. (Polynomial regression)

The columns  $1, t, t^2, \dots, t^d$  evaluated at 200 equally spaced points in  $[0, 1]$  become nearly dependent as  $d$  grows. To isolate the numerical error, the listing sets  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$  exactly for a random  $\boldsymbol{\beta}$ , so that the true least squares solution is  $\boldsymbol{\beta}$  itself. For  $d = 8$  the condition number is  $6.8 \times 10^5$ . The normal equations return  $\boldsymbol{\beta}$  with relative error  $1.0 \times 10^{-5}$ , while QR achieves  $1.7 \times 10^{-11}$ . At  $d = 13$ , with  $\kappa(\mathbf{X}) = 4.1 \times 10^9$ , the normal-equation answer has relative error  $9.3 \times 10^{-1}$  and not one correct digit, while QR still gets  $4.4 \times 10^{-9}$ . Figure 6.10.1 plots the whole sequence against the reference lines  $\kappa\epsilon$  and  $\kappa^2\epsilon$ , where  $\epsilon \approx 2.2 \times 10^{-16}$  is machine precision.

```
import numpy as np
import statsmodels.api as sm

def fit_normal_equations(X, y):
    return np.linalg.solve(X.T @ X, X.T @ y)

def fit_qr(X, y):
    Q, R = np.linalg.qr(X)                # X = QR, Q has orthonormal columns
    return np.linalg.solve(R, Q.T @ y)    # triangular system R b = Q'y

rng = np.random.default_rng(10)
n = 200
t = np.linspace(0, 1, n)
rows = []

for degree in range(1, 14):
    X = np.vander(t, degree + 1, increasing=True)    # 1, t, t^2, ..., t^degree
    beta = rng.normal(size=degree + 1)
```

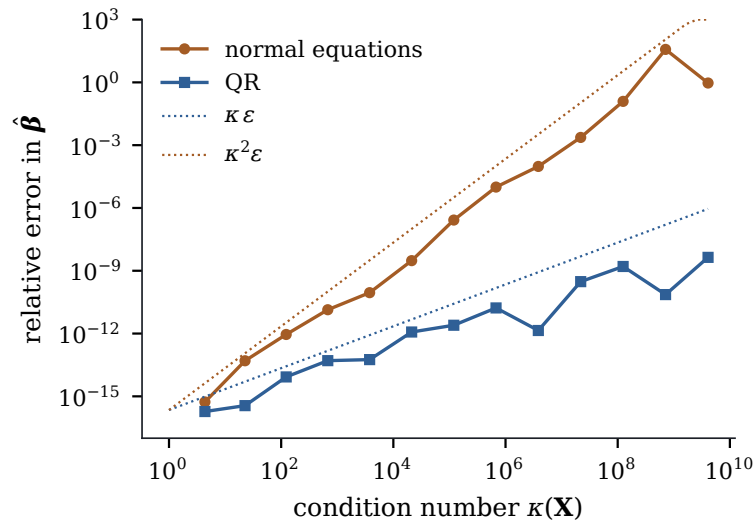


Figure 6.10.1: Accuracy of least squares coefficients for polynomial regressions of degree 1 to 13, against the condition number of the model matrix. The error of the normal equations tracks  $\kappa^2\epsilon$  until it reaches order one. The error of QR tracks  $\kappa\epsilon$ .

```

y = X @ beta                                     # exact fit: the true
answer is beta
kappa = np.linalg.cond(X)
err_ne = np.linalg.norm(fit_normal_equations(X, y) - beta) / np.linalg.norm(beta)
)
err_qr = np.linalg.norm(fit_qr(X, y) - beta) / np.linalg.norm(beta)
rows.append((degree, kappa, err_ne, err_qr))
print(f"degree {degree:2d}   cond(X) = {kappa:9.2e}   normal eq. {err_ne:8.1e}
      QR {err_qr:8.1e}")

```

The polynomial example is extreme by design, but ill-conditioning is common in real data. The macroeconomic series of Longley's classic test problem (in `statsmodels.datasets.longley`) give a model matrix with  $\kappa(\mathbf{X}) \approx 4.86 \times 10^9$ . Then  $\kappa(\mathbf{X}^\top \mathbf{X}) \approx 2.36 \times 10^{19}$ , which is larger than  $1/\epsilon$ . The rule of thumb predicts about 6 correct digits from QR and about 0 from the normal equations. Part of that condition number is only a matter of units: one column is a year, another is population in thousands. Rescaling columns ([Theorem 6.7.1](#)) changes  $\kappa$  without changing the fit ([Exercise \(B1\)](#)). The part that remains after rescaling reflects genuine near-dependence among the regressors, the subject of [Chapter 26](#).

Two practical lessons follow. First, never compute  $(\mathbf{X}^\top \mathbf{X})^{-1}$  just to obtain  $\hat{\beta}$ . Solve with QR, or call a least squares routine that does. Second, reparameterizations that leave  $\mathcal{C}(\mathbf{X})$  unchanged, such as centring, scaling and orthogonal polynomials, are statistically neutral but can be numerically decisive.

### 6.10.3 Rank deficiency and the singular value decomposition

QR with positive diagonal needs full column rank. For rank-deficient or nearly rank-deficient  $\mathbf{X}$ , the most informative tool is the **singular value decomposition**

$$\mathbf{X} = \mathbf{U} \mathbf{D} \mathbf{V}^\top = \mathbf{U}_r \mathbf{D}_r \mathbf{V}_r^\top,$$



where  $\mathbf{U}$  ( $n \times n$ ) and  $\mathbf{V}$  ( $p \times p$ ) are orthogonal,  $\mathbf{D}$  is  $n \times p$  with nonnegative diagonal entries  $\sigma_1 \geq \dots \geq \sigma_r > 0$ , and the subscript  $r$  keeps only the first  $r$  columns. Then:

- $\mathbf{U}_r$  is an orthonormal basis for  $\mathcal{C}(\mathbf{X})$ , so  $\mathbf{M} = \mathbf{U}_r \mathbf{U}_r^\top$ ;
- the last  $p - r$  columns of  $\mathbf{V}$  form an orthonormal basis for  $\mathcal{N}(\mathbf{X})$ ;
- the Moore–Penrose inverse is  $\mathbf{X}^+ = \mathbf{V}_r \mathbf{D}_r^{-1} \mathbf{U}_r^\top$ , and  $\mathbf{M} = \mathbf{X} \mathbf{X}^+$ .

### Proposition 6.10.1 — Minimum-norm least squares

$\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^+ \mathbf{y}$  is a least squares estimate, and it has the smallest Euclidean norm among all least squares estimates.

*Proof.*  $\mathbf{X} \mathbf{X}^+ \mathbf{y} = \mathbf{U}_r \mathbf{D}_r \mathbf{V}_r^\top \mathbf{V}_r \mathbf{D}_r^{-1} \mathbf{U}_r^\top \mathbf{y} = \mathbf{U}_r \mathbf{U}_r^\top \mathbf{y} = \mathbf{M} \mathbf{y}$ , so  $\hat{\boldsymbol{\beta}}^+$  is a least squares estimate. It lies in  $\mathcal{C}(\mathbf{V}_r) = \mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$ . Every other least squares estimate is  $\hat{\boldsymbol{\beta}}^+ + \mathbf{v}$  with  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$  (Theorem 6.4.2), and  $\|\hat{\boldsymbol{\beta}}^+ + \mathbf{v}\|^2 = \|\hat{\boldsymbol{\beta}}^+\|^2 + \|\mathbf{v}\|^2$ . ■

In Example (Four answers, one fit), the Moore–Penrose row is this minimum-norm solution. In floating point, “ $\sigma_j > 0$ ” becomes “ $\sigma_j$  larger than a tolerance”, and choosing the tolerance decides the numerical rank. That choice is a statistical decision in disguise. Chapter 10 discusses it, together with pivoted QR (Theorem 10.7.2), which gives most of the SVD’s rank-revealing ability at lower cost.

## 6.10.4 Exercises

### 6.10.4.1 A. Check your understanding

#### Exercise A1.

Prove the four identities in Eq. 6.10.1.

### 6.10.4.2 B. Practice

#### Exercise B1.

Compute  $\kappa(\mathbf{X})$  for the Longley data (with intercept) as given, after scaling every non-intercept column to unit Euclidean norm, and after centring and scaling. Confirm that the fitted values agree to the precision you expect in all three cases. Which part of the original condition number is “just units”?

#### Exercise B2.

Implement classical Gram–Schmidt as in Proposition 6.2.2, and the *modified* version that subtracts each projection from the working vector before computing the next inner

product. For the degree-10 polynomial design of [Example \(Polynomial regression\)](#), compare  $\|\mathbf{Q}^\top \mathbf{Q} - \mathbf{I}\|$  for the two and for `numpy.linalg.qr`.

### 6.10.4.3 C. Going deeper

#### Exercise C1.

Let  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$  with  $\bar{x} = c$  and  $S_{xx} = s^2 n$ . Find  $\kappa(\mathbf{X})$  as a function of  $c$ ,  $s$  and  $n$ , and show that it grows without bound as  $c \rightarrow \infty$  with  $s$  fixed. Show that centring the regressor, a reparameterization, reduces the condition number to  $\max(1, s)/\min(1, s)$ .

*Solution.* Write  $\mathbf{x} = c\mathbf{1} + \mathbf{d}$  with  $\mathbf{d} \perp \mathbf{1}$  and  $\|\mathbf{d}\|^2 = ns^2$ . Then  $\mathbf{X}^\top \mathbf{X} = n \begin{pmatrix} 1 & c \\ c & c^2 + s^2 \end{pmatrix}$ . The matrix in parentheses has trace  $T = 1 + c^2 + s^2$  and determinant  $s^2$ , so its eigenvalues are  $\lambda_{\pm} = (T \pm \sqrt{T^2 - 4s^2})/2$ , with  $\lambda_+ \lambda_- = s^2$  and  $\lambda_+ \geq T/2$ . Hence

$$\kappa(\mathbf{X})^2 = \frac{\lambda_+}{\lambda_-} = \frac{\lambda_+^2}{s^2} \geq \frac{T^2}{4s^2} \rightarrow \infty \quad \text{as } c \rightarrow \infty.$$

After centring,  $\mathbf{X} = [\mathbf{1}, \mathbf{d}]$  has orthogonal columns of lengths  $\sqrt{n}$  and  $s\sqrt{n}$ , so its singular values are these lengths and  $\kappa = \max(1, s)/\min(1, s)$ .

## 6.11 Projection in the population: best linear prediction

So far the geometry has lived in observation space  $\mathbb{R}^n$ , and every statement was about one fixed data vector. There is a parallel geometry one level up, in which the vectors are *random variables* and the inner product is an expectation. In that geometry, least squares estimates something even when the linear model is false, namely the population projection of  $Y$  onto a span of regressors. The conditional expectation turns out to be a projection too. This section sets up the correspondence. [Chapter 14](#) develops prediction theory from it, and [Chapter 19](#) uses it to study what least squares estimates under misspecification.

### 6.11.1 Random variables as vectors

Consider random variables with finite second moments on a common probability space. They form a vector space, and

$$\langle U, V \rangle = \mathbb{E}(UV), \quad \|U\| = (\mathbb{E}U^2)^{1/2}$$

is an inner product, provided we identify variables that are equal with probability one, so that  $\|U\| = 0$  forces  $U = 0$ . This space is usually written  $L^2$ . Two random variables are orthogonal when  $\mathbb{E}(UV) = 0$ . For variables with mean zero, this is the same as being uncorrelated. The Cauchy–Schwarz inequality [Eq. 6.2.2](#) becomes the statement that correlations lie in  $[-1, 1]$ . The *cosine* of the angle between two centred random variables is their correlation.

For a random  $k$ -vector  $\mathbf{X}$  with finite second moments, the set  $\mathcal{L} = \{a + \mathbf{b}^\top \mathbf{X} : a \in \mathbb{R}, \mathbf{b} \in \mathbb{R}^k\}$  is a subspace of dimension at most  $k + 1$ . The proofs of [Proposition 6.2.2](#), [Theorem 6.2.3](#) and

**Theorem 6.3.1** used only the axioms of an inner product and a finite spanning set, so they apply unchanged. Every  $Y$  with  $EY^2 < \infty$  has a unique nearest point in  $\mathcal{L}$ , characterized by orthogonality of the error to  $\mathcal{L}$ .

### Definition 6.11.1 — Best linear predictor

The **best linear predictor** of  $Y$  given  $\mathbf{X}$ , written  $L(Y | \mathbf{X})$ , is the projection of  $Y$  onto  $\mathcal{L}$ . It is the  $a + \mathbf{b}^\top \mathbf{X}$  that minimizes the mean squared error  $E(Y - a - \mathbf{b}^\top \mathbf{X})^2$ .

### Theorem 6.11.1

Let  $\boldsymbol{\mu}_X = E\mathbf{X}$ ,  $\mu_Y = EY$ ,  $\boldsymbol{\Sigma}_{XX} = \text{Cov}(\mathbf{X})$  and  $\boldsymbol{\sigma}_{XY} = \text{Cov}(\mathbf{X}, Y)$ , and suppose  $\boldsymbol{\Sigma}_{XX}$  is positive definite. Then

$$L(Y | \mathbf{X}) = \alpha^* + \boldsymbol{\beta}^{*\top} \mathbf{X}, \quad \boldsymbol{\beta}^* = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}, \quad \alpha^* = \mu_Y - \boldsymbol{\beta}^{*\top} \boldsymbol{\mu}_X.$$

The prediction error  $U = Y - L(Y | \mathbf{X})$  has mean zero and is uncorrelated with every component of  $\mathbf{X}$ . Moreover

$$\text{Var}(Y) = \text{Var}(L(Y | \mathbf{X})) + E U^2, \quad \frac{\text{Var}(L(Y | \mathbf{X}))}{\text{Var}(Y)} = \frac{\boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}}{\text{Var}(Y)}. \quad (6.11.1)$$

*Proof.* Orthogonality of  $U = Y - a - \mathbf{b}^\top \mathbf{X}$  to the spanning set  $\{1, X_1, \dots, X_k\}$  of  $\mathcal{L}$  gives the *population normal equations*

$$E(U) = 0, \quad E(\mathbf{X}U) = \mathbf{0}.$$

The first gives  $a = \mu_Y - \mathbf{b}^\top \boldsymbol{\mu}_X$ . Substituting into the second and subtracting  $\boldsymbol{\mu}_X E(U) = \mathbf{0}$  gives  $\text{Cov}(\mathbf{X}, Y) - \text{Cov}(\mathbf{X})\mathbf{b} = \mathbf{0}$ , so  $\mathbf{b} = \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$ . By the projection theorem the solution is the unique minimizer. Since  $U$  has mean zero and is uncorrelated with  $\mathbf{X}$ , it is uncorrelated with  $L(Y | \mathbf{X})$ , so the variances add. Finally,  $\text{Var}(\boldsymbol{\beta}^{*\top} \mathbf{X}) = \boldsymbol{\beta}^{*\top} \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}^* = \boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}$ . ■

The ratio in Eq. 6.11.1 is the population *squared multiple correlation*, the population counterpart of  $R^2$ . By the same cosine argument as Theorem 6.7.2, it is the squared correlation between  $Y$  and  $L(Y | \mathbf{X})$ .

## 6.11.2 Conditional expectation is also a projection

The linear span  $\mathcal{L}$  is one subspace of  $L^2$ . A much larger one is  $\mathcal{G} = \{g(\mathbf{X}) : E g(\mathbf{X})^2 < \infty\}$ , all square-integrable functions of  $\mathbf{X}$ . It is infinite-dimensional, but it is closed, and the projection theorem still holds in it. We accept that fact from measure-theoretic probability and only use its conclusion.

### Proposition 6.11.2

Let  $m(\mathbf{X}) = E(Y | \mathbf{X})$  with  $EY^2 < \infty$ . Then:

- (a)  $m(\mathbf{X})$  is the projection of  $Y$  onto  $\mathcal{G}$ . It minimizes  $E(Y - g(\mathbf{X}))^2$  over  $g$ , and  $E[(Y - m(\mathbf{X}))g(\mathbf{X})] = 0$  for all  $g(\mathbf{X}) \in \mathcal{G}$ ;
- (b)  $L(Y | \mathbf{X}) = L(m(\mathbf{X}) | \mathbf{X})$ : the best linear predictor of  $Y$  equals the best linear predictor of the regression function;
- (c)  $E(Y - L(Y | \mathbf{X}))^2 = E(Y - m(\mathbf{X}))^2 + E(m(\mathbf{X}) - L(Y | \mathbf{X}))^2$ ;
- (d) if  $m$  is affine,  $m(\mathbf{X}) = L(Y | \mathbf{X})$ .

*Proof.*

- (a) The orthogonality property follows from the tower rule:  $E[(Y - m(\mathbf{X}))g(\mathbf{X})] = E[g(\mathbf{X}) E(Y - m(\mathbf{X}) | \mathbf{X})] = 0$ . Orthogonality to the subspace characterizes the nearest point, by the argument of Theorem 6.3.1, which needs only Pythagoras.
- (b) This is Theorem 6.5.1(a) in  $L^2$ . Since  $\mathcal{L} \subseteq \mathcal{G}$ , projecting onto  $\mathcal{L}$  can be done in two stages.
- (c)  $Y - L = (Y - m) + (m - L)$ , and  $m - L \in \mathcal{G}$  is orthogonal to  $Y - m$  by (a).
- (d) If  $m \in \mathcal{L}$ , the second term in (c) is zero.

■

Part (c) is the population version of Figure 6.5.1. The error of the best linear predictor splits into two orthogonal parts. One is irreducible noise, which no function of  $\mathbf{X}$  can remove. The other is *approximation error*, the price of insisting on linearity. Chapters in Part IX reduce the second part by enlarging  $\mathcal{L}$  towards  $\mathcal{G}$ .

### 6.11.3 Least squares estimates the population projection

Here is the connection with the rest of the chapter. Given data  $(\mathbf{x}_{(i)}, y_i)$ , put probability  $1/n$  on each row. For this *empirical distribution*, the  $L^2$  inner product of two variables is  $n^{-1} \sum_i u_i v_i = n^{-1} \mathbf{u}^\top \mathbf{v}$ , which is the Euclidean inner product of observation space up to the factor  $1/n$ . So the best linear predictor under the empirical distribution is the least squares fit, and the population normal equations become the normal equations Eq. 6.4.1. Observation-space geometry is  $L^2$  geometry for the empirical distribution.

As the sample grows, the empirical distribution approaches the population distribution, and the projections converge with it.

#### Proposition 6.11.3 — Consistency for the projection

Let  $(\mathbf{X}_i, Y_i)$ ,  $i = 1, 2, \dots$ , be independent copies of  $(\mathbf{X}, Y)$  with finite second moments and  $\Sigma_{XX}$  positive definite. Let  $(\hat{\alpha}_n, \hat{\beta}_n)$  be the least squares coefficients from the first  $n$  observations, with an intercept. Then  $(\hat{\alpha}_n, \hat{\beta}_n) \rightarrow (\alpha^*, \beta^*)$  with probability one. No

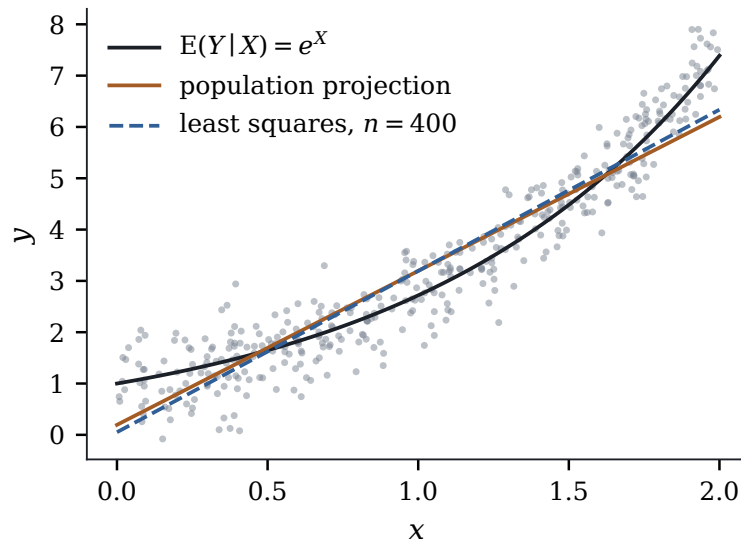


Figure 6.11.1: When the regression function is curved, least squares estimates its best linear approximation in the  $L^2$  sense, the population projection. The approximation error is largest where  $e^x$  bends away from the line.

assumption about the form of  $E(Y | \mathbf{X})$ , the distribution of the errors, or their variance is needed.

*Proof.* For large  $n$ ,  $\hat{\beta}_n = \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}$ , where  $\mathbf{S}_{XX}$  and  $\mathbf{s}_{XY}$  are sample covariances with divisor  $n$  (Section 6.6, centring). By the strong law of large numbers the sample means, second moments and cross-moments converge almost surely to their population values. So  $\mathbf{S}_{XX} \rightarrow \Sigma_{XX}$  and  $\mathbf{s}_{XY} \rightarrow \sigma_{XY}$ . Matrix inversion is continuous at the nonsingular  $\Sigma_{XX}$ , so eventually  $\mathbf{S}_{XX}$  is invertible and  $\hat{\beta}_n \rightarrow \beta^*$ . The intercept follows the same way. ■

### Example. (A curved mean)

Let  $X \sim \text{Uniform}(0, 2)$  and  $Y = e^X + \varepsilon$  with independent  $\varepsilon \sim N(0, 0.25)$ . The regression function is not linear, but Theorem 6.11.1 still identifies a best line. A short calculation gives  $\text{Cov}(X, Y) = 1$  and  $\text{Var}(X) = 1/3$ , so the population projection has slope 3.000 and intercept 0.195. The listing fits least squares lines to samples of increasing size. The estimates settle on these values, not on any feature of  $e^x$ . Figure 6.11.1 shows the regression function, the population projection, and a least squares fit.

```
import numpy as np
rng = np.random.default_rng(611)

# X ~ Uniform(0, 2), E(Y | X) = exp(X), Var(Y | X) = 0.25.
# Population projection of Y onto span{1, X}: slope Cov(X, Y)/Var(X), intercept E Y
# - slope * E X.
EX, VarX = 1.0, 4.0 / 12.0
EY = (np.exp(2) - 1) / 2
EXY = (np.exp(2) + 1) / 2
slope_pop = (EXY - EX * EY) / VarX
```

```

intercept_pop = EY - slope_pop * EX

for n in [20, 200, 2000, 20000]:
    x = rng.uniform(0, 2, n)
    y = np.exp(x) + rng.normal(0, 0.5, n)
    b, *_ = np.linalg.lstsq(np.column_stack([np.ones(n), x]), y, rcond=None)
    print(f"n = {n:6d}    intercept {b[0]:.3f}    slope {b[1]:.3f}")
print(f"population    intercept {intercept_pop:.3f}    slope {slope_pop:.3f}")

```

*Remark* (What changes under misspecification). [Proposition 6.11.3](#) is reassuring about the *target*. A least squares line always estimates a well-defined population quantity. It says nothing about the *uncertainty* of the estimate. When  $E(Y | \mathbf{X})$  is not linear, or the variance is not constant, the approximation error  $m(\mathbf{X}) - L(Y | \mathbf{X})$  behaves like extra noise that depends on  $\mathbf{X}$ . The textbook formula  $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$  for the covariance of  $\hat{\boldsymbol{\beta}}$  is then wrong, even in large samples. The “sandwich” covariance estimators of [Chapter 21](#) ([Theorem 21.4.2](#)) are designed for exactly this situation.

## 6.12 Summary and Notes

### 6.12.1 Summary

*Idea.*

1. Least squares is the nearest-point problem in observation space. The fitted vector  $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$  is the orthogonal projection of  $\mathbf{y}$  onto  $\mathcal{C}(\mathbf{X})$ , and the residual is orthogonal to  $\mathcal{C}(\mathbf{X})$  ([Theorem 6.4.1](#)).
2. Orthogonal projections are exactly the symmetric idempotent matrices. Their rank equals their trace ([Theorem 6.3.3](#) and [Proposition 6.3.4](#)).
3.  $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top$  for *every* generalized inverse ([Theorem 6.3.7](#)). Fitted values, residuals and every function of them are unique, whatever the rank. Coefficients are unique only when  $\text{rank}(\mathbf{X}) = p$ , and  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  is invariant iff  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$  ([Theorem 6.4.2](#) and [Theorem 6.4.4](#)).
4. Nested model spaces give orthogonal decompositions  $\mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$ . The drop in residual sum of squares is  $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2$ , with  $r - r_0$  degrees of freedom ([Theorem 6.5.1](#)).
5. A multiple-regression coefficient is the slope on the part of its regressor orthogonal to the others. Fixed-effects estimators and added-variable plots are applications ([Theorem 6.6.2](#)).
6. Anything that depends only on  $\mathcal{C}(\mathbf{X})$  is invariant to reparameterization.  $R^2$  is the squared cosine between centred data and centred fit ([Theorem 6.7.1](#) and [Theorem 6.7.2](#)).
7. Leverage  $h_{ii} = m_{ii}$  is the squared length of the projection of a coordinate vector. It lies in  $[1/n, 1]$  with an intercept and averages  $r/n$  ([Proposition 6.8.1](#)).
8. Replacing the inner product by  $\mathbf{u}^\top \mathbf{V}^{-1} \mathbf{v}$  turns least squares into generalized least squares. The projection is then idempotent and  $\mathbf{V}^{-1}$ -self-adjoint, but not symmetric ([Theorem 6.9.1](#) and [Theorem 6.9.3](#)).

9. Compute projections with orthonormal bases (QR, SVD), never by inverting  $\mathbf{X}^\top \mathbf{X}$ , which squares the condition number (Section 6.10).
10. In the population, least squares estimates the  $L^2$  projection of  $Y$  onto linear functions of  $\mathbf{X}$ , whether or not the regression function is linear (Theorem 6.11.1 and Proposition 6.11.3).

## 6.12.2 Notes and sources

**The geometric tradition.** Seeing least squares as projection is as old as least squares itself, but it became the organizing principle of linear-model theory only in the twentieth century. Herr (1980) traces the history, from Fisher’s geometric derivations of sampling distributions through the coordinate-free treatments of the 1960s. Book-length geometric developments at different levels include Saville and Wood (1991), written for a first course, and Wickens (1995), for multivariate methods. Christensen (2020) is the most sustained vector-space treatment of linear model theory. Its chapter 2 and appendices A–B cover the material of Section 6.2, Section 6.3 and Section 6.4 for the rank-deficient case throughout, and influenced this chapter’s choice to state results for general rank from the start. Seber and Lee (2003, appendix B) gives a compact account of projections, including the constraint-space result that appears here as Theorem 6.5.2. Agresti (2015, chapter 2) develops projection matrices through worked null-model, one-way and two-way examples, and is a gentle companion to Section 6.5. For the underlying linear algebra, Halmos (1958) remains a model of coordinate-free exposition.

**Generalized inverses.** The four-condition inverse is due to Penrose (1955). Its statistical uses, and the theory of general  $\mathbf{A}^-$  including the characterization in Exercise (B2), are developed by Rao (1973). The invariance of  $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top$  (Lemma 6.3.6) is the fact that makes generalized inverses harmless in linear models. Every quantity of statistical interest is sandwiched in this way.

**Frisch–Waugh–Lovell.** The theorem is due to Frisch and Waugh (1933) for time trends and Lovell (1963) for seasonal dummies and general partitions. It is also known under Lovell’s name alone, and in econometrics as the “partialling-out” result. It underlies added-variable plots, double/debiased machine-learning estimators, and the fixed-effects (“within”) estimator for panel data.

**Leverage.** Hoaglin and Welsch (1978) introduced the name “hat matrix” into common use and established the basic properties in Proposition 6.8.1. Diagnostic use of leverage is taken up in Chapter 20.

**Other inner products.** Generalized least squares is due to Aitken (1935). The coordinate-free condition for equality of ordinary and generalized least squares (Theorem 6.9.3) is from Kruskal (1968). Christensen (2020, chapter 10) develops the consequences for the general Gauss–Markov model, which Chapter 31 follows.

**Computation.** Golub and Van Loan (2013) is the standard reference for QR, the SVD and their perturbation theory. Higham (2002) gives the precise statement of the conditioning of least squares problems, including the  $\kappa^2 \|\hat{\mathbf{e}}\|$  term mentioned in Section 6.10. Seber and Lee (2003, chapter 11) surveys algorithms from a statistician’s perspective, and Chapter 10 builds on it.

### 6.12.3 References

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# Chapter 7

## Optimality: Gauss–Markov, Maximum Likelihood and Minimum Variance

Least squares was introduced as a way of fitting, and [Chapter 6](#) showed that it is a projection. Neither fact says it is a *good* way of estimating. This chapter asks in what precise senses least squares is the best one can do, and where those senses end. With only means and covariances assumed, least squares has the smallest variance among linear unbiased estimators (Gauss–Markov), and the proof is one more application of the projection theorem. With normal errors three more things hold. Least squares is maximum likelihood. It has the smallest variance among *all* unbiased estimators (Lehmann–Scheffé). And its sampling distribution is known exactly, which is what Part III builds tests and intervals on. The same assumptions also show where optimality stops. Nonlinear estimators can win when the errors are not normal, and biased estimators can win on mean squared error. A Bayesian analysis with a conjugate prior is one principled biased estimator, and it recovers least squares as the prior becomes flat.

**What you need.** Covariances of linear functions and the mean of a quadratic form ([Chapter 2: Theorem 2.2.1](#) and [Theorem 2.3.1](#)). The multivariate normal and its density ([Chapter 3: Theorem 3.3.1](#) and [Theorem 3.2.2](#)). Chi-squared quadratic forms and their independence ([Chapter 4: Theorem 4.3.2](#) and [Theorem 4.4.1](#)). The linear model and its first two moments ([Chapter 5: Definition 5.3.1](#) and [Theorem 5.5.1](#)). Least squares as projection, including the rank-deficient case ([Chapter 6: Theorem 6.4.1](#), [Theorem 6.4.2](#) and [Theorem 6.4.4](#)).

### 7.0.1 Roadmap

- [Linear unbiased estimation](#)
- [The Gauss–Markov theorem](#)
- [Maximum likelihood under normality](#)
- [Sufficiency, completeness and minimum variance](#)
- [Sampling distributions of the estimates](#)
- [Bayesian thread I: the conjugate linear model](#)
- [Summary and notes](#)

## 7.1 Linear unbiased estimation

To call an estimator the best, we have to say what it is compared with and by what criterion. The comparison can't be made against every estimator. The constant estimator  $\tilde{\beta}_1 \equiv 0.5$  of a slope has variance zero, and it is perfect when the true slope happens to be 0.5 and useless otherwise. No estimator has smaller variance, yet nobody would call it good. Some restriction on the class of competitors is unavoidable. The classical restriction has two parts. The estimator must be *unbiased*, which rules out estimators that buy precision by ignoring the data. And it must be *linear* in the response, which makes its variance computable from the first two moments of the errors alone. This section describes the class these two conditions carve out. The description is geometric, and the Gauss–Markov theorem of [Section 7.2](#) will follow from it in a few lines.

### 7.1.1 The setting

Throughout this chapter the response satisfies the linear model of [Chapter 5](#) ([Definition 5.3.1](#)),

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}, \quad (7.1.1)$$

with  $\mathbf{X}$  a fixed  $n \times p$  matrix of rank  $r \leq p$ ,  $\boldsymbol{\beta} \in \mathbb{R}^p$  unknown and  $\sigma^2 > 0$  unknown. These are the **second-moment assumptions**. The errors have mean zero, a common variance, and are uncorrelated. Nothing is assumed about their distribution beyond that. From [Section 7.3](#) on we add normality, and we will say so each time. As in [Chapter 6](#),  $\mathbf{M}$  denotes the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$ ,  $\hat{\boldsymbol{\beta}}$  is any least squares estimate, and  $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$  is the residual sum of squares.

The quantities we want to estimate are linear functions  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ , for fixed  $\boldsymbol{\lambda} \in \mathbb{R}^p$ . This covers a single coefficient ( $\boldsymbol{\lambda}$  a coordinate vector), a difference of two coefficients, and the mean response  $\mathbf{x}_0^\top \boldsymbol{\beta}$  at a new point  $\mathbf{x}_0$ . Several such functions at once are collected as  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}$ , where  $\boldsymbol{\Lambda}$  is a  $p \times q$  matrix.

### 7.1.2 Linear and unbiased

#### **Definition 7.1.1** — Linear unbiased and best linear unbiased estimators

Let  $\boldsymbol{\lambda} \in \mathbb{R}^p$ .

- (a) A **linear estimator** is a statistic of the form  $c + \mathbf{a}^\top \mathbf{Y}$ , with a constant  $c \in \mathbb{R}$  and a constant vector  $\mathbf{a} \in \mathbb{R}^n$  (constant means not depending on  $\mathbf{Y}$  or on the parameters).
- (b) A linear estimator is a **linear unbiased estimator** (LUE) of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  if  $\mathbf{E}(c + \mathbf{a}^\top \mathbf{Y}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$  for every  $\boldsymbol{\beta} \in \mathbb{R}^p$  and every  $\sigma^2 > 0$ .
- (c) An LUE  $c + \mathbf{a}^\top \mathbf{Y}$  of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is a **best linear unbiased estimator** (BLUE) if  $\text{Var}(c + \mathbf{a}^\top \mathbf{Y}) \leq \text{Var}(d + \mathbf{b}^\top \mathbf{Y})$  for every LUE  $d + \mathbf{b}^\top \mathbf{Y}$  of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ , for every  $\boldsymbol{\beta}$  and every  $\sigma^2 > 0$ .
- (d) For  $q$  functions  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}$ , a **linear estimator** is  $\mathbf{c} + \mathbf{A}\mathbf{Y}$  with a constant  $q \times n$  matrix  $\mathbf{A}$

and  $\mathbf{c} \in \mathbb{R}^q$ . It is an LUE of  $\mathbf{\Lambda}^\top \boldsymbol{\beta}$  if each coordinate is an LUE of the corresponding coordinate of  $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ , and a BLUE if  $\text{Cov}(\mathbf{BY} + \mathbf{d}) - \text{Cov}(\mathbf{AY} + \mathbf{c})$  is nonnegative definite for every LUE  $\mathbf{BY} + \mathbf{d}$  (Definition 1.7.1).

The words “for every  $\boldsymbol{\beta}$ ” in (b) carry all the weight. An estimator that is unbiased at one parameter value only, like the constant 0.5 when  $\beta_1 = 0.5$ , does not qualify. The first result (Proposition 7.1.1) turns unbiasedness into linear equations.

### Proposition 7.1.1 — Unbiasedness as a linear equation

The linear estimator  $c + \mathbf{a}^\top \mathbf{Y}$  is an LUE of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  iff  $c = 0$  and  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ . Consequently  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  has an LUE iff  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ .

*Proof.* By Theorem 2.2.1,  $E(c + \mathbf{a}^\top \mathbf{Y}) = c + \mathbf{a}^\top \mathbf{X}\boldsymbol{\beta}$ . This equals  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  for all  $\boldsymbol{\beta}$  iff the affine function  $\boldsymbol{\beta} \mapsto c + (\mathbf{X}^\top \mathbf{a} - \boldsymbol{\lambda})^\top \boldsymbol{\beta}$  vanishes identically. Putting  $\boldsymbol{\beta} = \mathbf{0}$  gives  $c = 0$ . Putting  $\boldsymbol{\beta}$  equal to each coordinate vector in turn then gives  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ . The converse is immediate. An LUE therefore exists iff the equation  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$  has a solution, that is, iff  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ . ■

So the constant  $c$  is always zero, and from now on we drop it. A linear function  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  with  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$  is called **estimable**. For these functions the data carry linear information. For the others no linear combination of the observations is centred on the target. Chapter 8 takes this as the definition of estimability (Definition 8.2.1) and develops its many equivalent forms. For now we need only Proposition 7.1.1 and one comparison. Theorem 6.4.4 of Chapter 6 found the same condition  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$  as the answer to a different question: when is  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  the same for every least squares estimate  $\hat{\boldsymbol{\beta}}$ ? The functions that least squares estimates unambiguously are exactly the functions that have linear unbiased estimators at all. When  $\mathbf{X}$  has full column rank,  $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}^p$ , so every linear function of  $\boldsymbol{\beta}$  is estimable.

## 7.1.3 The geometry of the unbiased class

Fix an estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  and write  $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$  for some  $\boldsymbol{\rho} \in \mathbb{R}^n$ . The coefficient vectors of its LUEs form the set

$$\mathcal{A}_\lambda = \{\mathbf{a} \in \mathbb{R}^n : \mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}\}.$$

This set is the key object of the chapter’s first half.

### Proposition 7.1.2 — The set of unbiased coefficient vectors

Let  $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$ . Then:

- (a)  $\mathcal{A}_\lambda = \mathbf{M}\boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$ , an affine subspace of  $\mathbb{R}^n$  of dimension  $n - r$ , parallel to  $\mathcal{C}(\mathbf{X})^\perp$ ;
- (b) every  $\mathbf{a} \in \mathcal{A}_\lambda$  has the same projection  $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$ , and  $\mathbf{M}\boldsymbol{\rho}$  is the only element of  $\mathcal{A}_\lambda$  that lies in  $\mathcal{C}(\mathbf{X})$ ;

- (c)  $\mathbf{M}\boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$  for every generalized inverse, and  $(\mathbf{M}\boldsymbol{\rho})^\top \mathbf{Y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  for every least squares estimate  $\hat{\boldsymbol{\beta}}$ .

*Proof.*

- (a) By [Proposition 7.1.1](#),  $\mathbf{a} \in \mathcal{A}_\lambda$  means  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ . Now  $\mathbf{X}^\top \mathbf{a} = \mathbf{X}^\top \boldsymbol{\rho}$  iff  $\mathbf{X}^\top (\mathbf{a} - \boldsymbol{\rho}) = \mathbf{0}$ , iff  $\mathbf{a} - \boldsymbol{\rho} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$  ([Lemma 6.2.1](#)). So  $\mathcal{A}_\lambda = \boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$ . Since  $\boldsymbol{\rho} - \mathbf{M}\boldsymbol{\rho} = (\mathbf{I} - \mathbf{M})\boldsymbol{\rho} \in \mathcal{C}(\mathbf{X})^\perp$ , the same set is  $\mathbf{M}\boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$ . Its dimension is  $\dim \mathcal{C}(\mathbf{X})^\perp = n - r$ .
- (b) If  $\mathbf{a} = \mathbf{M}\boldsymbol{\rho} + \mathbf{v}$  with  $\mathbf{v} \in \mathcal{C}(\mathbf{X})^\perp$ , then  $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho} + \mathbf{M}\mathbf{v} = \mathbf{M}\boldsymbol{\rho}$ . If moreover  $\mathbf{a} \in \mathcal{C}(\mathbf{X})$ , then  $\mathbf{a} = \mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$ .
- (c) By [Theorem 6.3.7](#),  $\mathbf{M}\boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ . By [Theorem 6.4.4](#),  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y} = (\mathbf{M}\boldsymbol{\rho})^\top \mathbf{Y}$ . ■

Part (c) says that least squares is itself one of the linear unbiased estimators. Its coefficient vector is  $\mathbf{a}_* = \mathbf{M}\boldsymbol{\rho}$ , the one member of the class that lies in the model space. It does not depend on which  $\boldsymbol{\rho}$  was used to represent  $\boldsymbol{\lambda}$ , because every such  $\boldsymbol{\rho}$  has the same projection. That  $\mathbf{a}_*^\top \mathbf{Y}$  is unbiased can also be seen directly:  $E(\mathbf{a}_*^\top \mathbf{Y}) = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ .

The variance of any linear estimator follows from [Theorem 2.2.1](#):

$$\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \mathbf{a}^\top (\sigma^2 \mathbf{I}) \mathbf{a} = \sigma^2 \|\mathbf{a}\|^2. \quad (7.1.2)$$

Under the second-moment assumptions, then, the variance of a linear estimator is  $\sigma^2$  times the squared length of its coefficient vector. Finding the BLUE means finding the shortest vector in the affine set  $\mathcal{A}_\lambda$ .

*Idea.* The linear unbiased estimators of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  correspond to the points of a flat  $\mathcal{A}_\lambda = \mathbf{a}_* + \mathcal{C}(\mathbf{X})^\perp$  in  $\mathbb{R}^n$ . The variance of each is  $\sigma^2$  times its squared distance from the origin. The question “which LUE is best?” is a nearest-point problem, the same kind of problem that least squares solves in [Chapter 6](#), now posed in the space of coefficient vectors.

A second description of the class is useful too. Every  $\mathbf{a} \in \mathcal{A}_\lambda$  is  $\mathbf{a}_* + (\mathbf{I} - \mathbf{M})\mathbf{w}$  for some  $\mathbf{w}$ , so every LUE has the form

$$\mathbf{a}^\top \mathbf{Y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{w}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{w}^\top \hat{\boldsymbol{\epsilon}}. \quad (7.1.3)$$

An LUE is the least squares estimate plus a linear function of the residuals. The linear functions of the residuals are exactly the linear unbiased estimators of zero ([Exercise \(B1\)](#)). Adding one to an estimator keeps it unbiased and can only add noise. [Section 7.2](#) makes “can only add noise” precise.

## 7.1.4 Three estimators of a slope

**Example. (Three linear unbiased estimators of a slope)**

A straight line  $E(Y_i) = \beta_0 + \beta_1 x_i$  is fitted to  $n = 12$  observations at the regressor values  $x = 1, 2, 2.5, 3, 4, 5, 5.5, 7, 8, 9, 10.5, 12$ , for which  $S_{xx} = \sum_i (x_i - \bar{x})^2 = 137.229$ . Here  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$  has full rank and the target is  $\boldsymbol{\lambda} = (0, 1)^\top$ . Three natural estimators of the slope are linear in  $\mathbf{Y}$ .

- (i) *Least squares*,  $\hat{\beta}_1 = \sum_i (x_i - \bar{x}) Y_i / S_{xx}$  ([Theorem 5.2.1](#)), with  $a_i = (x_i - \bar{x}) / S_{xx}$ .
- (ii) *End points*, the slope of the line through the first and last observations,  $(Y_{12} - Y_1) / (x_{12} - x_1)$ .
- (iii) *Group means*, which splits the data into the six smallest and six largest  $x$  values and takes  $(\bar{Y}_{\text{high}} - \bar{Y}_{\text{low}}) / (\bar{x}_{\text{high}} - \bar{x}_{\text{low}})$ .

Each coefficient vector satisfies  $\mathbf{1}^\top \mathbf{a} = 0$  and  $\mathbf{x}^\top \mathbf{a} = 1$ , which is the condition  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$  of [Proposition 7.1.1](#). So all three are unbiased, whatever the distribution of the errors. By [Eq. 7.1.2](#) their variances are  $\sigma^2 \|\mathbf{a}\|^2$ :

$$0.00729 \sigma^2, \quad 0.01653 \sigma^2, \quad 0.01008 \sigma^2.$$

Least squares has the smallest, and the other two have efficiencies 0.441 and 0.723 relative to it. The listing also confirms what [Proposition 7.1.2\(b\)](#) predicts: the three coefficient vectors differ, but their projections onto  $\mathcal{C}(\mathbf{X})$  all equal the least squares vector.

```
import numpy as np

x = np.array([1.0, 2.0, 2.5, 3.0, 4.0, 5.0, 5.5, 7.0, 8.0, 9.0, 10.5, 12.0])
n = len(x)
X = np.column_stack([np.ones(n), x])
lam = np.array([0.0, 1.0]) # target: the slope beta_1
M = X @ np.linalg.solve(X.T @ X, X.T) # projection onto C(X)

# coefficient vectors a of three linear estimators a^T y of the slope
a_ls = X @ np.linalg.solve(X.T @ X, lam) # least squares: a = X (X^T X)^{-1}
    lambda
a_end = np.zeros(n)
a_end[[0, -1]] = [-1.0, 1.0]
a_end /= x[-1] - x[0] # slope through the two end points
low, high = np.arange(n) < n // 2, np.arange(n) >= n // 2
a_grp = (high / high.sum() - low / low.sum()) / (x[high].mean() - x[low].mean()) #
    group means

for name, a in [("least squares", a_ls), ("end points", a_end), ("group means",
    a_grp)]:
    print(f"{name:14s} X^T a = {X.T @ a}    Var/sigma^2 = {a @ a:.5f}    Ma = a_ls: {
    np.allclose(M @ a, a_ls)}")
```

The end-point estimator throws away ten of the twelve observations, so its poor showing is no surprise. The group-means estimator uses every observation, and it is still clearly worse than least squares. It weights all observations in a half equally, while least squares weights each observation in proportion to its distance from  $\bar{x}$ . Estimators of this grouping type were proposed by Wald (1940) for regressions in which  $x$  itself is measured with error. [Chapter 24](#) returns to that problem.

## 7.1.5 Why these restrictions?

Linearity and unbiasedness are restrictions of convenience as well as of principle, and it is worth being clear about what each one does.

**Unbiasedness** removes estimators that are good only near one parameter value. It is not sacred. [Section 7.2](#) shows biased estimators with smaller mean squared error, and [Section 7.6](#) derives a whole family of them from prior information. Unbiasedness can also be too demanding: a nonestimable  $\lambda^\top \beta$  has no linear unbiased estimator, and under normality no unbiased estimator of any kind ([Exercise \(C1\)](#)).

**Linearity** is what makes the second-moment assumptions enough. The variance of a nonlinear estimator, such as a median or a trimmed mean, depends on the whole error distribution. Its variance can't be compared with that of least squares until that distribution is specified. For linear estimators [Eq. 7.1.2](#) needs only  $\sigma^2$ . The restriction therefore has a cost: optimality among linear estimators says nothing about nonlinear ones. [Section 7.2](#) gives an error distribution under which a simple nonlinear estimator is far better than least squares. [Section 7.4](#) shows that under normality this cannot happen.

## 7.1.6 Exercises

### 7.1.6.1 A. Check your understanding

#### Exercise A1.

In the location model  $Y_i = \mu + \varepsilon_i$  ( $\mathbf{X} = \mathbf{1}$ ), show that the LUEs of  $\mu$  are the weighted averages  $\sum_i a_i Y_i$  with  $\sum_i a_i = 1$ . Identify  $\mathcal{A}_\lambda$  and  $\mathbf{a}_*$  and find the LUE of smallest variance directly.

#### Exercise A2.

In the one-way layout  $E(Y_{ij}) = \mu + \alpha_i$  with three groups and an intercept ([Example \(Four answers, one fit\)](#)), show that  $\alpha_1$  has no linear unbiased estimator. Write down two different LUEs of  $\alpha_1 - \alpha_2$ , and verify that their coefficient vectors have the same projection onto  $\mathcal{C}(\mathbf{X})$ .

### 7.1.6.2 B. Practice

#### Exercise B1.

A linear estimator  $\mathbf{a}^\top \mathbf{Y}$  is an **unbiased estimator of zero** if  $E(\mathbf{a}^\top \mathbf{Y}) = 0$  for every  $\beta$ . Show that this holds iff  $\mathbf{a} \in \mathcal{C}(\mathbf{X})^\perp$ , iff  $\mathbf{a}^\top \mathbf{Y}$  is a linear function of the residual vector  $\hat{\varepsilon}$ . Show that every unbiased estimator of zero is uncorrelated with  $\lambda^\top \hat{\beta}$  for every estimable  $\lambda^\top \beta$ .

*Solution.* By [Proposition 7.1.1](#) with  $\lambda = \mathbf{0}$ ,  $\mathbf{a}^\top \mathbf{Y}$  is unbiased for zero iff  $\mathbf{X}^\top \mathbf{a} = \mathbf{0}$ , iff  $\mathbf{a} \in \mathcal{C}(\mathbf{X})^\perp$ . If so,  $\mathbf{a} = (\mathbf{I} - \mathbf{M})\mathbf{a}$  and  $\mathbf{a}^\top \mathbf{Y} = \mathbf{a}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{a}^\top \hat{\varepsilon}$ . Conversely,  $\mathbf{w}^\top \hat{\varepsilon} =$

$((\mathbf{I} - \mathbf{M})\mathbf{w})^\top \mathbf{Y}$  has coefficient vector in  $\mathcal{C}(\mathbf{X})^\perp$ . For the last claim, write  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y}$ . By [Theorem 2.2.1](#),  $\text{Cov}(\boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y}, \mathbf{a}^\top \mathbf{Y}) = \sigma^2 \boldsymbol{\rho}^\top \mathbf{M}\mathbf{a} = 0$ , because  $\mathbf{M}\mathbf{a} = \mathbf{0}$ .

### Exercise B2.

Let  $\mathbf{A}$  be  $p \times q$  with  $\mathcal{C}(\mathbf{A}) \subseteq \mathcal{C}(\mathbf{X}^\top)$ . Show that  $\mathbf{A}\mathbf{Y}$  is an LUE of  $\mathbf{A}^\top \boldsymbol{\beta}$  iff  $\mathbf{A}\mathbf{X} = \mathbf{A}^\top$ , and that the set of such  $\mathbf{A}$  is  $\{\mathbf{A}^\top (\mathbf{X}^\top \mathbf{X})^- \mathbf{X}^\top + \mathbf{W}(\mathbf{I} - \mathbf{M}) : \mathbf{W} \in \mathbb{R}^{q \times n}\}$ .

### 7.1.6.3 C. Going deeper

### Exercise C1.

Assume  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$  and  $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top)$ . Show that no statistic  $\delta(\mathbf{Y})$ , linear or not, satisfies  $E \delta(\mathbf{Y}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$  for all  $\boldsymbol{\beta}$ . *Hint:* the distribution of  $\mathbf{Y}$  depends on  $\boldsymbol{\beta}$  only through  $\mathbf{X}\boldsymbol{\beta}$ .

*Solution.* Since  $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$ , there is  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$  with  $\boldsymbol{\lambda}^\top \mathbf{v} \neq 0$ . The parameters  $\boldsymbol{\beta}$  and  $\boldsymbol{\beta} + \mathbf{v}$  give the same mean  $\mathbf{X}\boldsymbol{\beta}$ , hence the same distribution of  $\mathbf{Y}$ , hence the same value of  $E \delta(\mathbf{Y})$ . An unbiased  $\delta$  would need  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \boldsymbol{\lambda}^\top (\boldsymbol{\beta} + \mathbf{v})$ , which is false. The argument uses only that the law of  $\mathbf{Y}$  is a function of  $\mathbf{X}\boldsymbol{\beta}$  and  $\sigma^2$ . [Chapter 8](#) calls this *nonidentifiability* ([Definition 8.1.1](#)).

### Exercise C2.

For the group-means estimator of [Example \(Three linear unbiased estimators of a slope\)](#) with general  $x_1 \leq \dot{s} \leq x_n$  ( $n = 2m$ ), find its variance in terms of  $\bar{x}_{\text{high}} - \bar{x}_{\text{low}}$ . Show that its efficiency relative to least squares is  $(\bar{x}_{\text{high}} - \bar{x}_{\text{low}})^2 n / (4S_{xx})$ , and that this is 1 iff the  $x$  values take only two distinct values, with half of the observations at each.

## 7.2 The Gauss–Markov theorem

[Section 7.1](#) turned the search for a best linear unbiased estimator into a geometric problem. The unbiased coefficient vectors form the flat  $\mathcal{A}_{\boldsymbol{\lambda}} = \mathbf{a}_* + \mathcal{C}(\mathbf{X})^\perp$ , and the variance of each is  $\sigma^2$  times its squared length. The shortest vector in a flat parallel to  $\mathcal{C}(\mathbf{X})^\perp$  is the one in  $\mathcal{C}(\mathbf{X})$ , and that vector is the least squares coefficient vector  $\mathbf{a}_*$ . The Gauss–Markov theorem says exactly this, together with its matrix form and its consequences.

### 7.2.1 The theorem

We keep the second-moment assumptions [Eq. 7.1.1](#), with  $\mathbf{X}$  of any rank  $r$ , and write  $(\mathbf{X}^\top \mathbf{X})^-$  for any generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ .

**Theorem 7.2.1 — Gauss–Markov**

Assume [Eq. 7.1.1](#). Let  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ , say  $\lambda = \mathbf{X}^\top \rho$ , and let  $\hat{\beta}$  be any least squares estimate.

- (a)  $\lambda^\top \hat{\beta}$  is a linear unbiased estimator of  $\lambda^\top \beta$ , with

$$\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda.$$

- (b) For every linear unbiased estimator  $\mathbf{a}^\top \mathbf{Y}$  of  $\lambda^\top \beta$ ,

$$\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \text{Var}(\lambda^\top \hat{\beta}) + \sigma^2 \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2. \quad (7.2.1)$$

- (c) Consequently  $\lambda^\top \hat{\beta}$  is a BLUE of  $\lambda^\top \beta$ , and it is the only one: an LUE  $\mathbf{a}^\top \mathbf{Y}$  is a BLUE iff  $\mathbf{a} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \lambda$ , that is, iff  $\mathbf{a}^\top \mathbf{y} = \lambda^\top \hat{\beta}$  for every  $\mathbf{y}$ .

- (d) (*Matrix form.*) Let  $\mathbf{A}$  be  $p \times q$  with  $\mathcal{C}(\mathbf{A}) \subseteq \mathcal{C}(\mathbf{X}^\top)$ . For every linear unbiased estimator  $\mathbf{A}\mathbf{Y}$  of  $\mathbf{A}^\top \beta$ ,

$$\text{Cov}(\mathbf{A}\mathbf{Y}) = \text{Cov}(\mathbf{A}^\top \hat{\beta}) + \sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top, \quad (7.2.2)$$

where  $\text{Cov}(\mathbf{A}^\top \hat{\beta}) = \sigma^2 \mathbf{A}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{A}$ . So  $\text{Cov}(\mathbf{A}\mathbf{Y}) - \text{Cov}(\mathbf{A}^\top \hat{\beta})$  is nonnegative definite, and it is zero iff  $\mathbf{A}\mathbf{Y} = \mathbf{A}^\top \hat{\beta}$  for every  $\mathbf{y}$ .

*Proof.* Put  $\mathbf{a}_* = \mathbf{M}\rho = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \lambda$ . By [Proposition 7.1.2\(c\)](#),  $\mathbf{a}_*^\top \mathbf{Y} = \lambda^\top \hat{\beta}$ . Since  $\mathbf{a}_* \in \mathcal{A}_\lambda$ , it is an LUE by [Proposition 7.1.1](#). By [Eq. 7.1.2](#) its variance is

$$\sigma^2 \|\mathbf{M}\rho\|^2 = \sigma^2 \rho^\top \mathbf{M}\rho = \sigma^2 \rho^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \rho = \sigma^2 \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda,$$

using [Theorem 6.3.7](#). This proves (a).

For (b), let  $\mathbf{a}^\top \mathbf{Y}$  be any LUE. Then  $\mathbf{a} \in \mathcal{A}_\lambda$ , so  $\mathbf{M}\mathbf{a} = \mathbf{a}_*$  by [Proposition 7.1.2\(b\)](#). The decomposition  $\mathbf{a} = \mathbf{M}\mathbf{a} + (\mathbf{I} - \mathbf{M})\mathbf{a} = \mathbf{a}_* + (\mathbf{I} - \mathbf{M})\mathbf{a}$  is orthogonal, so  $\|\mathbf{a}\|^2 = \|\mathbf{a}_*\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2$ . Multiplying by  $\sigma^2$  and using [Eq. 7.1.2](#) twice gives [Eq. 7.2.1](#).

For (c), the second term of [Eq. 7.2.1](#) is nonnegative, so  $\lambda^\top \hat{\beta}$  is a BLUE. Because  $\sigma^2 > 0$ , equality holds iff  $(\mathbf{I} - \mathbf{M})\mathbf{a} = \mathbf{0}$ , that is, iff  $\mathbf{a} \in \mathcal{C}(\mathbf{X})$ . By [Proposition 7.1.2\(b\)](#) the only point of  $\mathcal{A}_\lambda$  in  $\mathcal{C}(\mathbf{X})$  is  $\mathbf{a}_*$ . Two linear statistics  $\mathbf{a}^\top \mathbf{y}$  and  $\mathbf{a}_*^\top \mathbf{y}$  agree for every  $\mathbf{y}$  iff  $\mathbf{a} = \mathbf{a}_*$ .

For (d), write  $\mathbf{A} = \mathbf{X}^\top \mathbf{P}$  for an  $n \times q$  matrix  $\mathbf{P}$ , and put  $\mathbf{A}_* = \mathbf{P}^\top \mathbf{M}$ , so that  $\mathbf{A}_* \mathbf{Y} = \mathbf{A}^\top \hat{\beta}$  by [Proposition 7.1.2\(c\)](#) applied column by column. Each row  $\mathbf{a}_k^\top$  of an LUE matrix  $\mathbf{A}$  has  $\mathbf{a}_k \in \mathcal{A}_{\lambda_k}$ , where  $\lambda_k$  is the  $k$ th column of  $\mathbf{A}$ . So  $\mathbf{M}\mathbf{a}_k = \mathbf{M}\mathbf{p}_k$ , and hence  $\mathbf{A}\mathbf{M} = \mathbf{A}_*$  (the  $k$ th row of  $\mathbf{A}\mathbf{M}$  is  $(\mathbf{M}\mathbf{a}_k)^\top$ , since  $\mathbf{M}$  is symmetric). Now split the identity:

$$\begin{aligned} \mathbf{A}\mathbf{A}^\top &= \mathbf{A}\mathbf{M}\mathbf{A}^\top + \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top \\ &= (\mathbf{A}\mathbf{M})(\mathbf{A}\mathbf{M})^\top + \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top = \mathbf{A}_* \mathbf{A}_*^\top + \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top, \end{aligned}$$

using  $\mathbf{M} = \mathbf{M}\mathbf{M}^\top$ . Multiplying by  $\sigma^2$  and using [Theorem 2.2.1](#) gives  $\text{Cov}(\mathbf{A}\mathbf{Y}) = \text{Cov}(\mathbf{A}_* \mathbf{Y}) + \sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top$ . Also  $\mathbf{A}_* \mathbf{A}_*^\top = \mathbf{P}^\top \mathbf{M}\mathbf{P} = \mathbf{A}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{A}$ , as in (a). The difference  $\sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top = \sigma^2 [\mathbf{A}(\mathbf{I} - \mathbf{M})][\mathbf{A}(\mathbf{I} - \mathbf{M})]^\top$  is nonnegative definite. It is zero iff  $\mathbf{A}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ , iff  $\mathbf{A} = \mathbf{A}\mathbf{M} = \mathbf{A}_*$ . ■



Three features of the theorem deserve comment.

**It is distribution-free.** The proof used  $E(\boldsymbol{\varepsilon}) = \mathbf{0}$  and  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ , and nothing else. The errors may be skewed, heavy-tailed, discrete, or dependent in any way that leaves them uncorrelated. They need not even be identically distributed, as long as their variances are equal.

**It holds in every rank.** When  $\mathbf{X}$  is rank deficient,  $\hat{\boldsymbol{\beta}}$  is not unique and neither is the generalized inverse. But for estimable  $\boldsymbol{\lambda}$ , both  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  and  $\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^- \boldsymbol{\lambda}$  are the same for every choice (Theorem 6.4.4 and Lemma 6.3.6). A nonestimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  has no LUE at all, so the theorem covers every function for which the question makes sense.

**The proof is a projection.** Equation Eq. 7.2.1 is Pythagoras in  $\mathbb{R}^n$ , applied to the coefficient vector and not to the data.

*Idea.* The least squares coefficient vector  $\mathbf{a}_*$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$  of the coefficient vector of *any* linear unbiased estimator. Projecting a coefficient vector onto  $\mathcal{C}(\mathbf{X})$  keeps the estimator unbiased and removes the part that only adds noise. That part,  $(\mathbf{I} - \mathbf{M})\mathbf{a}$ , multiplies the residuals, and the residuals carry no information about  $\boldsymbol{\beta}$ .

The decomposition Eq. 7.1.3 gives the same result in the language of estimators. Every LUE is  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{w}^\top \hat{\boldsymbol{\varepsilon}}$ , and the two terms are uncorrelated (Exercise (B1)), so their variances add. An estimator that uses the residuals only adds uncorrelated noise to least squares.

## 7.2.2 What the matrix form buys

The matrix form Eq. 7.2.2 is stronger than the scalar statement applied coordinate by coordinate, and it is the form usually quoted for vector estimators.

### Corollary 7.2.2 — Consequences of the Gauss–Markov theorem

Assume Eq. 7.1.1, let  $\mathcal{C}(\boldsymbol{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$ , and let  $\tilde{\boldsymbol{\theta}} = \mathbf{A}\mathbf{Y}$  be any LUE of  $\boldsymbol{\theta} = \boldsymbol{\Lambda}^\top \boldsymbol{\beta}$ , with  $\hat{\boldsymbol{\theta}} = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}$ .

- (a)  $\text{Var}(\mathbf{b}^\top \tilde{\boldsymbol{\theta}}) \geq \text{Var}(\mathbf{b}^\top \hat{\boldsymbol{\theta}})$  for every  $\mathbf{b} \in \mathbb{R}^q$ ; in particular each coordinate of  $\hat{\boldsymbol{\theta}}$  has the smallest variance.
- (b)  $E\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2 \geq E\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2$ .
- (c)  $\det \text{Cov}(\tilde{\boldsymbol{\theta}}) \geq \det \text{Cov}(\hat{\boldsymbol{\theta}})$ .
- (d) If  $\text{rank}(\mathbf{X}) = p$ , then  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$  is the BLUE of  $\boldsymbol{\beta}$ , and  $\text{Cov}(\hat{\boldsymbol{\beta}}) - \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$  is nonnegative definite for every LUE  $\tilde{\boldsymbol{\beta}}$  of  $\boldsymbol{\beta}$ .
- (e) The fitted vector  $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y}$  is the BLUE of  $\mathbf{X}\boldsymbol{\beta}$ , and the fitted value  $\hat{y}_i$  is the BLUE of  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , with variance  $\sigma^2 h_{ii}$ , where  $h_{ii}$  is the leverage.

*Proof.* Let  $\mathbf{D} = \text{Cov}(\tilde{\boldsymbol{\theta}}) - \text{Cov}(\hat{\boldsymbol{\theta}}) \succeq \mathbf{0}$  (Theorem 7.2.1(d)). (a)  $\text{Var}(\mathbf{b}^\top \tilde{\boldsymbol{\theta}}) - \text{Var}(\mathbf{b}^\top \hat{\boldsymbol{\theta}}) = \mathbf{b}^\top \mathbf{D} \mathbf{b} \geq 0$  by Definition 1.7.1. The coordinates are the case  $\mathbf{b} = \mathbf{e}_k$ . (b) Both estimators are unbiased, so the mean squared errors are the traces of the covariance matrices, and  $\text{tr } \mathbf{D} = \sum_k \mathbf{e}_k^\top \mathbf{D} \mathbf{e}_k \geq 0$ . (c) Write  $\mathbf{C} = \text{Cov}(\hat{\boldsymbol{\theta}})$ . If  $\mathbf{C}$  is singular, its determinant is zero and there is nothing to prove, because covariance matrices are nonnegative definite (Theorem 2.2.3). Otherwise  $\mathbf{C}$  is positive definite, with a symmetric positive definite

square root (Theorem 1.7.6). The matrix  $\mathbf{C}^{-1/2} \text{Cov}(\tilde{\boldsymbol{\theta}}) \mathbf{C}^{-1/2} = \mathbf{I} + \mathbf{C}^{-1/2} \mathbf{D} \mathbf{C}^{-1/2}$  has all eigenvalues at least 1: the matrix  $\mathbf{C}^{-1/2} \mathbf{D} \mathbf{C}^{-1/2}$  is nonnegative definite by Proposition 1.7.5(c), so its eigenvalues are nonnegative by Theorem 1.7.4, and adding  $\mathbf{I}$  adds 1 to each of them. Its determinant  $\det \text{Cov}(\tilde{\boldsymbol{\theta}}) / \det \mathbf{C}$  is therefore at least 1. (d) Take  $\boldsymbol{\Lambda} = \mathbf{I}_p$ , which is allowed because  $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}^p$ , and use  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$  (Theorem 5.5.1). (e) Take  $\boldsymbol{\Lambda} = \mathbf{X}^\top$ . Then  $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{M} \mathbf{Y}$  (Theorem 6.4.1), with covariance  $\sigma^2 \mathbf{M}$ . Its  $i$ th diagonal entry is  $\sigma^2 h_{ii}$  (Definition 6.8.1).

■

Part (b) says least squares minimizes the total mean squared error among unbiased linear estimators. Part (c) has a geometric meaning that Chapter 12 will use. For normal errors, the confidence ellipsoid for  $\boldsymbol{\theta}$  has volume proportional to  $\sqrt{\det \text{Cov}}$ , so least squares gives the smallest ellipsoids.

### Example. (The slope estimators under skewed errors)

Return to the three slope estimators of Example (Three linear unbiased estimators of a slope) and simulate 200,000 data sets with  $\beta_0 = 2$ ,  $\beta_1 = 0.5$  and errors  $\varepsilon_i = E_i - 1$ , where the  $E_i$  are independent exponential with mean one. These errors have mean zero and variance one, but they are strongly skewed: most are slightly negative and a few are large and positive. The average estimates are 0.5000, 0.4998 and 0.5001, and the simulated variances are 0.00724, 0.01643 and 0.01004, against the exact values 0.00729, 0.01653 and 0.01008. Figure 7.2.1(a) shows the three sampling distributions. All three are centred on the truth, and least squares is the most concentrated: its  $\pm 2$  standard deviation bar is the shortest. The end-point estimator, a difference of just two skewed errors, has a sharp peak but long tails, so the height of a density is a poor guide to variance. The errors are far from normal and the theorem does not care.

```
import numpy as np

x = np.array([1.0, 2.0, 2.5, 3.0, 4.0, 5.0, 5.5, 7.0, 8.0, 9.0, 10.5, 12.0])
n = len(x)
X = np.column_stack([np.ones(n), x])
lam = np.array([0.0, 1.0]) # target: the slope beta_1
a_ls = X @ np.linalg.solve(X.T @ X, lam) # least squares: a = X (X^T X)^{-1}
    lambda
a_end = np.zeros(n)
a_end[[0, -1]] = [-1.0, 1.0]
a_end /= x[-1] - x[0] # slope through the two end points
low, high = np.arange(n) < n // 2, np.arange(n) >= n // 2
a_grp = (high / high.sum() - low / low.sum()) / (x[high].mean() - x[low].mean()) #
    group means

estimators = {"ls": a_ls, "end": a_end, "grp": a_grp}
rng = np.random.default_rng(7)
beta, sigma, reps = np.array([2.0, 0.5]), 1.0, 200_000
E = sigma * (rng.exponential(size=(reps, n)) - 1.0) # skewed errors, mean 0,
    variance sigma^2
Y = X @ beta + E # one simulated data set per
    row
sims = {name: Y @ a for name, a in estimators.items()}
for name, est in sims.items():
    print(f"{name:4s} mean {est.mean():.4f}    variance {est.var():.4f}")
```

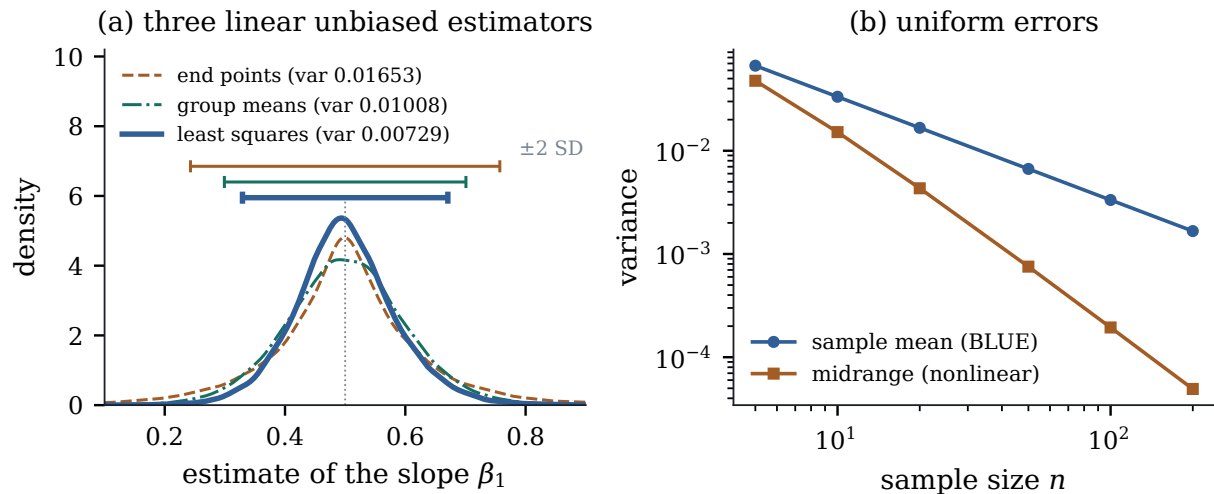


Figure 7.2.1: (a) Sampling distributions of three linear unbiased estimators of a slope, with skewed errors (200,000 simulated data sets), as smoothed densities, with bars marking the true value  $\pm 2$  standard deviations. Least squares has the smallest variance, as the Gauss–Markov theorem requires. (b) In the location model with uniform errors, the midrange, which is unbiased but not linear, has far smaller variance than the sample mean, which is the BLUE.

### 7.2.3 Correlated errors: Aitken’s theorem

If  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$  with  $\mathbf{V}$  known and positive definite, the theorem applies after a change of coordinates, and the BLUE becomes the generalized least squares estimate of Definition 6.9.1.

#### Corollary 7.2.3 — Aitken

Assume  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$ , with  $\mathbf{V}$  known and positive definite. For  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ , the BLUE of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\text{GLS}}$ , where  $\hat{\boldsymbol{\beta}}_{\text{GLS}}$  is any solution of  $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$ . Its variance is  $\sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \boldsymbol{\lambda}$ .

*Proof.* Let  $\mathbf{V}^{1/2}$  be the symmetric positive definite square root of  $\mathbf{V}$  (Theorem 1.7.6). Put  $\mathbf{Z} = \mathbf{V}^{-1/2} \mathbf{Y}$  and  $\mathbf{X}_* = \mathbf{V}^{-1/2} \mathbf{X}$ . By Theorem 2.2.1,  $E(\mathbf{Z}) = \mathbf{X}_* \boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Z}) = \sigma^2 \mathbf{I}$ , so  $\mathbf{Z}$  satisfies Eq. 7.1.1 with model matrix  $\mathbf{X}_*$ . Since  $\mathbf{V}^{-1/2}$  is nonsingular,  $\mathcal{C}(\mathbf{X}_*^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{V}^{-1/2}) = \mathcal{C}(\mathbf{X}^\top)$ , so the same functions are estimable. The map  $\mathbf{a}^\top \mathbf{Y} = (\mathbf{V}^{1/2} \mathbf{a})^\top \mathbf{Z}$  is a bijection between linear estimators based on  $\mathbf{Y}$  and those based on  $\mathbf{Z}$ . It preserves means and variances, since both sides are the same random variable. So a BLUE in one model is a BLUE in the other. By Theorem 7.2.1 applied to  $\mathbf{Z}$ , the BLUE is  $\boldsymbol{\lambda}^\top \mathbf{b}$  for any least squares solution  $\mathbf{b}$  of the  $\mathbf{Z}$  problem. That is,  $\mathbf{b}$  minimizes  $\|\mathbf{Z} - \mathbf{X}_* \mathbf{b}\|^2 = (\mathbf{Y} - \mathbf{X} \mathbf{b})^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X} \mathbf{b})$ , which is the generalized least squares criterion. Its normal equations (Theorem 6.4.2) are  $\mathbf{X}_*^\top \mathbf{X}_* \mathbf{b} = \mathbf{X}_*^\top \mathbf{Z}$ , that is,  $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$ . The variance is part (a) of the theorem with  $\mathbf{X}_*$  in place of  $\mathbf{X}$ . ■

Ordinary least squares remains unbiased when  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$ , but in general it is no longer

best. [Theorem 6.9.3](#) characterized the models in which the two estimators coincide, and [Exercise \(C3\)](#) shows that exactly these models keep ordinary least squares optimal. [Chapter 31](#) develops the general theory ([Theorem 31.2.2](#)), including singular  $\mathbf{V}$  ([Theorem 31.3.2](#)).

The whitening in the proof carries the rest of the least squares toolkit with it. The fitted vector  $\mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}} = \mathbf{V}^{1/2}\mathbf{M}_*\mathbf{Z}$ , where  $\mathbf{M}_*$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{X}_*)$ , is the generalized least squares fit of [Definition 6.9.1](#): it is idempotent but not symmetric, an oblique projection onto  $\mathcal{C}(\mathbf{X})$  in the sense of [Proposition 6.9.4](#). The residual sum of squares of the whitened problem is  $\|(\mathbf{I} - \mathbf{M}_*)\mathbf{Z}\|^2 = (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}})^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}})$ , and since  $\text{rank}(\mathbf{X}_*) = r$ , [Theorem 2.3.1](#) and [Proposition 6.3.4](#) give it expectation  $\sigma^2(n - r)$ . So

$$s_{\text{GLS}}^2 = \frac{(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}})^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\text{GLS}})}{n - r}$$

is unbiased for  $\sigma^2$ .

## 7.2.4 Design: when is a coefficient estimated best?

The theorem fixes  $\mathbf{X}$  and optimizes over estimators. The variance it leaves,  $\sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ , still depends on  $\mathbf{X}$ . When the regressor values can be chosen, one can ask which designs make it small. The Frisch–Waugh–Lovell theorem gives a clean answer for single coefficients.

### Proposition 7.2.4 — Orthogonal columns are best

Let  $\mathbf{X}$  have full column rank, let  $\mathbf{x}_j$  be its  $j$ th column and  $\mathbf{X}_{(j)}$  the other columns, with projection  $\mathbf{M}_{(j)}$  onto  $\mathcal{C}(\mathbf{X}_{(j)})$ . Under [Eq. 7.1.1](#),

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2} \geq \frac{\sigma^2}{\|\mathbf{x}_j\|^2},$$

with equality iff  $\mathbf{x}_j$  is orthogonal to every other column. If all columns are mutually orthogonal, then  $\hat{\beta}_j = \mathbf{x}_j^\top \mathbf{Y} / \|\mathbf{x}_j\|^2$ , and it is unchanged when any other columns are deleted from the model.

*Proof.* Put  $\tilde{\mathbf{x}}_j = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$ . By [Theorem 6.6.2\(a\)](#), with  $\mathbf{X}_1 = \mathbf{x}_j$  and  $\mathbf{X}_2 = \mathbf{X}_{(j)}$ ,  $\hat{\beta}_j = \tilde{\mathbf{x}}_j^\top \mathbf{Y} / \|\tilde{\mathbf{x}}_j\|^2$ . So  $\text{Var}(\hat{\beta}_j) = \sigma^2 / \|\tilde{\mathbf{x}}_j\|^2$  by [Theorem 2.2.1](#). By [Proposition 6.3.4\(d\)](#) applied to  $\mathbf{I} - \mathbf{M}_{(j)}$ ,  $\|\tilde{\mathbf{x}}_j\| \leq \|\mathbf{x}_j\|$ , with equality iff  $\mathbf{M}_{(j)}\mathbf{x}_j = \mathbf{0}$ , that is, iff  $\mathbf{x}_j \perp \mathcal{C}(\mathbf{X}_{(j)})$ . If all columns are orthogonal,  $\tilde{\mathbf{x}}_j = \mathbf{x}_j$ , and the formula for  $\hat{\beta}_j$  involves no other column, so deleting columns does not change it. ■

So among designs whose columns have given lengths, orthogonal designs minimize the variance of every coefficient at once. This is one reason designed experiments are built from orthogonal contrasts ([Part IV](#)), and why collinearity, meaning near-dependence among columns, inflates variances ([Theorem 26.1.1](#)). [Exercise \(C1\)](#) applies the proposition to a problem of weighing objects on a balance.

## 7.2.5 What the theorem does not say

The Gauss–Markov theorem is often quoted as “least squares is optimal”. The theorem is narrower than that. It compares least squares only with estimators that are linear and unbiased, and it assumes the model’s mean and covariance are right. Relax any of these and least squares can lose.

**Nonlinear estimators.** When the errors are not normal, a nonlinear unbiased estimator can beat least squares by a wide margin.

### Example. (Uniform errors and the midrange)

In the location model  $Y_i = \mu + \varepsilon_i$ , let the errors be independent and uniform on  $(-1, 1)$ , so  $\sigma^2 = 1/3$ . The BLUE of  $\mu$  is  $\bar{Y}$ , with variance  $1/(3n)$ . The **midrange**  $(Y_{(1)} + Y_{(n)})/2$ , the average of the smallest and largest observation, is not linear. It is unbiased, because the error distribution is symmetric about zero, and its variance is  $2/\{(n+1)(n+2)\}$  ([Exercise \(C2\)](#)). For  $n = 20$  the two variances are 0.0167 and 0.00433, a ratio of 3.9. For  $n = 100$  they are 0.0033 and 0.00019, a ratio of 17.2. The variance of the mean falls like  $1/n$  and that of the midrange like  $1/n^2$  ([Figure 7.2.1\(b\)](#)).

```
ns = np.array([5, 10, 20, 50, 100, 200])
var_mean = 1 / (3 * ns)                                # errors uniform on (-1, 1):
    variance 1/3
var_mid = 2 / ((ns + 1) * (ns + 2))                    # exact variance of the midrange
for m, v1, v2 in zip(ns, var_mean, var_mid):
    print(f"n = {m:3d}    Var(mean) = {v1:.5f}    Var(midrange) = {v2:.6f}    ratio = {
        v1 / v2:.1f}")
```

Uniform errors have sharp edges, and the extreme observations locate those edges very precisely. A linear estimator must give every observation a fixed weight whatever the sample looks like, so it cannot exploit this. Heavy-tailed errors produce the opposite phenomenon: there the extremes are unreliable, and estimators that downweight them, such as medians and M-estimators, beat least squares ([Exercise \(C1\)](#); [Chapter 20](#) and [Chapter 45](#) return to such estimators). [Section 7.4](#) shows that under *normal* errors neither phenomenon occurs, and least squares is best among all unbiased estimators.

**Biased estimators.** Mean squared error is variance plus squared bias. Accepting a little bias can reduce variance by more. For an estimator  $\tilde{\beta}$  of  $\beta$ , the natural matrix criterion is

$$E[(\tilde{\beta} - \beta)(\tilde{\beta} - \beta)^\top] = \text{Cov}(\tilde{\beta}) + (E\tilde{\beta} - \beta)(E\tilde{\beta} - \beta)^\top,$$

and the Gauss–Markov theorem ranks estimators by it only within the unbiased class. The ridge estimator  $(\mathbf{X}^\top \mathbf{X} + k\mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}$  is linear but biased, and some  $k > 0$  always gives a smaller total mean squared error than least squares ([Exercise \(B1\)](#)). Stein (1956) and James and Stein (1961) showed more: in the model  $\mathbf{Y} \sim N_n(\beta, \sigma^2 \mathbf{I})$  with known  $\sigma^2$  and  $n \geq 3$ , a nonlinear shrinkage estimator has smaller total mean squared error than  $\mathbf{Y}$  itself *for every*  $\beta$  ([Theorem 27.5.2](#)). [Chapter 27](#) develops shrinkage. [Section 7.6](#) derives a family of biased linear estimators from prior information.

**Wrong assumptions.** If  $\text{Cov}(\varepsilon) \neq \sigma^2 \mathbf{I}$ , [Corollary 7.2.3](#) replaces least squares by generalized least squares. If the mean is misspecified, so that  $E(\mathbf{Y}) \notin \mathcal{C}(\mathbf{X})$ , then  $E(\hat{\mathbf{Y}}) = \mathbf{M}E(\mathbf{Y})$  is the projection of the true mean. Least squares then estimates the best approximation within the

model, not the truth (compare [Theorem 6.11.1](#)). [Part V](#) is about diagnosing and repairing these departures.

## 7.2.6 Exercises

### 7.2.6.1 A. Check your understanding

#### Exercise A1.

Show that  $\sum_i \text{Var}(\hat{y}_i) = \sigma^2 r$  under [Eq. 7.1.1](#). Using [Corollary 7.2.2\(e\)](#), deduce that no linear unbiased estimator of  $\mathbf{X}\boldsymbol{\beta}$  has total variance less than  $\sigma^2 r$ . Interpret this as “each estimated dimension of the mean costs one  $\sigma^2$ ”.

### 7.2.6.2 B. Practice

#### Exercise B1.

Let  $\mathbf{X}$  have full column rank and, for  $k \geq 0$ , let  $\tilde{\boldsymbol{\beta}}_k = (\mathbf{X}^\top \mathbf{X} + k\mathbf{I})^{-1} \mathbf{X}^\top \mathbf{Y}$ . Under [Eq. 7.1.1](#), find the bias and covariance matrix of  $\tilde{\boldsymbol{\beta}}_k$ . Using a spectral decomposition  $\mathbf{X}^\top \mathbf{X} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top$ , show that the total mean squared error is

$$f(k) = \sum_{j=1}^p \frac{\sigma^2 d_j + k^2 \gamma_j^2}{(d_j + k)^2}, \quad \boldsymbol{\gamma} = \mathbf{Q}^\top \boldsymbol{\beta}, \quad \mathbf{D} = \text{diag}(d_1, \dotscots, d_p),$$

and that  $f'(0) < 0$ . Conclude that some  $k > 0$  beats least squares in total mean squared error, for every  $\boldsymbol{\beta}$  and  $\sigma^2$ , although the best  $k$  depends on them.

*Solution.* Write  $\mathbf{S} = \mathbf{X}^\top \mathbf{X}$  and  $\mathbf{W}_k = (\mathbf{S} + k\mathbf{I})^{-1}$ . Then  $\mathbb{E} \tilde{\boldsymbol{\beta}}_k = \mathbf{W}_k \mathbf{S} \boldsymbol{\beta} = \boldsymbol{\beta} - k \mathbf{W}_k \boldsymbol{\beta}$ , so the bias is  $-k \mathbf{W}_k \boldsymbol{\beta}$ , and  $\text{Cov}(\tilde{\boldsymbol{\beta}}_k) = \sigma^2 \mathbf{W}_k \mathbf{S} \mathbf{W}_k$ . With  $\mathbf{S} = \mathbf{Q}\mathbf{D}\mathbf{Q}^\top$ ,  $\mathbf{W}_k = \mathbf{Q}(\mathbf{D} + k\mathbf{I})^{-1} \mathbf{Q}^\top$ , so  $\text{tr Cov} = \sigma^2 \sum_j d_j / (d_j + k)^2$  and  $\|k \mathbf{W}_k \boldsymbol{\beta}\|^2 = k^2 \sum_j \gamma_j^2 / (d_j + k)^2$ . Adding gives  $f(k)$ . Differentiating,

$$f'(k) = \sum_j \frac{2d_j(k\gamma_j^2 - \sigma^2)}{(d_j + k)^3}, \quad f'(0) = -2\sigma^2 \sum_j d_j^{-2} < 0.$$

Since  $f$  is differentiable at 0 with negative derivative,  $f(k) < f(0)$  for all small  $k > 0$ , and  $f(0)$  is the total mean squared error of  $\hat{\boldsymbol{\beta}}$ . This does not contradict [Theorem 7.2.1](#), because  $\tilde{\boldsymbol{\beta}}_k$  is biased for  $k > 0$ .

#### Exercise B2.

Suppose the design grows with  $n$  and  $\mathbf{X}^\top \mathbf{X}$  has full rank for all large  $n$ . Show that  $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) \leq \sigma^2 \|\boldsymbol{\lambda}\|^2 / \lambda_{\min}(\mathbf{X}^\top \mathbf{X})$ . Deduce that if the smallest eigenvalue of  $\mathbf{X}^\top \mathbf{X}$  tends to infinity, then  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \rightarrow \boldsymbol{\lambda}^\top \boldsymbol{\beta}$  in mean square, for every  $\boldsymbol{\lambda}$ .

#### Exercise B3.

In the location model with  $\text{Cov}(\mathbf{Y}) = \sigma^2 \text{diag}(v_1, \dotscots, v_n)$ , compare the variance of the

ordinary mean  $\bar{Y}$  with that of the BLUE given by [Corollary 7.2.3](#). Show that the ratio is at least one, with equality iff all  $v_i$  are equal.

### 7.2.6.3 C. Going deeper

#### Exercise C1.

Each of  $k$  objects with unknown weights  $\beta_1, \dots, \beta_k$  is weighed in  $n$  weighings on a two-pan balance. In weighing  $i$  every object is put on the left pan or the right pan, and the reading  $Y_i$  is the difference of the pan loads plus an error. So  $\mathbf{X}$  is  $n \times k$  with entries  $\pm 1$ , and [Eq. 7.1.1](#) holds.

- (a) Show that  $\text{Var}(\hat{\beta}_j) \geq \sigma^2/n$  for every  $j$ , with equality for all  $j$  iff  $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$ .
- (b) Take  $k = 3$ ,  $n = 4$ , and the two designs whose rows are

$$\mathbf{X}_1 : (1, 1, 1), (1, -1, -1), (-1, 1, -1), (-1, -1, 1); \quad \mathbf{X}_2 : (1, 1, 1), (1, 1, -1), (1, -1, 1), (1, 1, 1).$$

For each, compute  $\text{Cov}(\hat{\boldsymbol{\beta}})/\sigma^2$ , its trace and its determinant. Which design is better by each criterion, and what do the criteria measure (compare [Corollary 7.2.2\(b\)](#) and (c))?

- (c) Show that if a  $\pm 1$  matrix with  $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$  exists with  $k = n \geq 3$ , then  $n$  is a multiple of 4. (Whether every multiple of 4 occurs is the Hadamard conjecture, which is open.)

*Solution.*

- (a) Every column has  $\|\mathbf{x}_j\|^2 = n$ , so [Proposition 7.2.4](#) gives  $\text{Var}(\hat{\beta}_j) \geq \sigma^2/n$ , with equality iff  $\mathbf{x}_j$  is orthogonal to the other columns. Equality for all  $j$  means all columns are mutually orthogonal, that is,  $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$ .
- (b)  $\mathbf{X}_1^\top \mathbf{X}_1 = 4\mathbf{I}$ , so  $\text{Cov}(\hat{\boldsymbol{\beta}})/\sigma^2 = \mathbf{I}/4$ , with trace  $3/4$  and determinant  $1/64$ . For  $\mathbf{X}_2$ , the columns are  $(1, 1, 1, 1)$ ,  $(1, 1, -1, 1)$ ,  $(1, -1, 1, 1)$ , so

$$\mathbf{X}_2^\top \mathbf{X}_2 = \begin{pmatrix} 4 & 2 & 2 \\ 2 & 4 & 0 \\ 2 & 0 & 4 \end{pmatrix}, \quad (\mathbf{X}_2^\top \mathbf{X}_2)^{-1} = \frac{1}{32} \begin{pmatrix} 16 & -8 & -8 \\ -8 & 12 & 4 \\ -8 & 4 & 12 \end{pmatrix},$$

with trace  $40/32 = 5/4$  and determinant  $1/32$ .  $\mathbf{X}_1$  wins on both counts, and on every single variance ( $1/4$  against  $1/2$ ,  $3/8$ ,  $3/8$ ). The trace is the total mean squared error of  $\hat{\boldsymbol{\beta}}$  divided by  $\sigma^2$ , and the determinant governs the volume of confidence ellipsoids.

- (c) Here  $\mathbf{X}$  is square and  $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$  gives  $\mathbf{X}\mathbf{X}^\top = n\mathbf{I}$ , so the rows are mutually orthogonal too. Changing the sign of a column keeps this property, so we may take the first row to be all ones. For the first three rows, let  $a, b, c, d$  count the columns in which rows 2 and 3 have signs  $(+, +)$ ,  $(+, -)$ ,  $(-, +)$ ,  $(-, -)$ . Then  $a + b + c + d = n$ , orthogonality of row 1 to rows 2 and 3 gives  $a + b - c - d = 0$  and  $a - b + c - d = 0$ , and orthogonality of rows 2 and 3 gives  $a - b - c + d = 0$ . Adding the four equations gives  $4a = n$ .



**Exercise C2.**

Let  $U_1, \dots, U_n$  be independent and uniform on  $(0, 1)$ . The  $k$ th smallest has a  $\text{Beta}(k, n - k + 1)$  distribution, and  $\text{Cov}(U_{(1)}, U_{(n)}) = 1/\{(n+1)^2(n+2)\}$ . Derive the variance  $2/\{(n+1)(n+2)\}$  of the midrange in [Example \(Uniform errors and the midrange\)](#). Show also that the midrange is unbiased for  $\mu$  whenever the errors have a distribution symmetric about zero with finite mean.

*Solution.*  $\text{Var}(U_{(1)}) = \text{Var}(U_{(n)}) = n/\{(n+1)^2(n+2)\}$  from the beta variance. Hence

$$\text{Var}\left(\frac{U_{(1)} + U_{(n)}}{2}\right) = \frac{1}{4} \cdot \frac{2n+2}{(n+1)^2(n+2)} = \frac{1}{2(n+1)(n+2)}.$$

Errors uniform on  $(-1, 1)$  are  $2U_i - 1$ , which multiplies variances by 4 and gives  $2/\{(n+1)(n+2)\}$ . For unbiasedness, if  $\varepsilon$  and  $-\varepsilon$  have the same distribution, then  $\varepsilon_{(1)} + \varepsilon_{(n)}$  and  $-(\varepsilon_{(n)} + \varepsilon_{(1)})$  have the same distribution. So the midrange minus  $\mu$  is symmetric about zero, and it has mean zero because  $|\varepsilon_{(1)} + \varepsilon_{(n)}| \leq 2 \max_i |\varepsilon_i|$  is integrable.

**Exercise C3.**

Let  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$  with  $\mathbf{V}$  positive definite. Show that an LUE  $\mathbf{a}^\top \mathbf{Y}$  of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is a BLUE iff it is uncorrelated with every linear unbiased estimator of zero. Deduce that  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  (ordinary least squares) is the BLUE for every estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  iff  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , in agreement with [Theorem 6.9.3](#).

*Solution.* Every LUE is  $\mathbf{a}^\top \mathbf{Y} + \mathbf{u}^\top \mathbf{Y}$  with  $\mathbf{u}^\top \mathbf{Y}$  an unbiased estimator of zero, and  $\text{Var}(\mathbf{a}^\top \mathbf{Y} + t\mathbf{u}^\top \mathbf{Y}) = \text{Var}(\mathbf{a}^\top \mathbf{Y}) + 2t \text{Cov}(\mathbf{a}^\top \mathbf{Y}, \mathbf{u}^\top \mathbf{Y}) + t^2 \text{Var}(\mathbf{u}^\top \mathbf{Y})$  for every real  $t$ . If  $\mathbf{a}^\top \mathbf{Y}$  is a BLUE, this quadratic in  $t$  is minimized at  $t = 0$ , which forces the covariance to vanish. Conversely, if every such covariance vanishes, the variance of every competitor is  $\text{Var}(\mathbf{a}^\top \mathbf{Y}) + \text{Var}(\mathbf{u}^\top \mathbf{Y}) \geq \text{Var}(\mathbf{a}^\top \mathbf{Y})$ . By [Exercise \(B1\)](#), the unbiased estimators of zero are  $\mathbf{w}^\top (\mathbf{I} - \mathbf{M})\mathbf{Y}$ . So ordinary least squares  $\boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y}$  is a BLUE for every  $\boldsymbol{\rho}$  iff  $\sigma^2 \boldsymbol{\rho}^\top \mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M})\mathbf{w} = 0$  for all  $\boldsymbol{\rho}, \mathbf{w}$ , that is, iff  $\mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ . Transposing, this says  $(\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{M} = \mathbf{0}$ , so every column of  $\mathbf{V}\mathbf{M}$  lies in  $\mathcal{C}(\mathbf{X})$ . Since  $\mathcal{C}(\mathbf{V}\mathbf{M}) = \mathcal{C}(\mathbf{V}\mathbf{X})$ , this is  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ .

## 7.3 Maximum likelihood under normality

The Gauss–Markov theorem needs only two moments. That is its strength, and also the reason it can say nothing about  $\sigma^2$ , nothing about nonlinear estimators, and nothing about the distribution of the estimates. From now on we assume more: the errors are normal. The linear model becomes a fully specified family of distributions,

$$\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}), \quad \boldsymbol{\beta} \in \mathbb{R}^p, \quad \sigma^2 > 0, \quad (7.3.1)$$

the normal linear model of [Definition 5.3.1](#). The first general-purpose method of estimation in such a family is maximum likelihood. This section shows that it gives back least squares for  $\boldsymbol{\beta}$ , and gives a slightly different estimate of  $\sigma^2$ .



### 7.3.1 The likelihood

By [Theorem 3.2.2](#) with  $\Sigma = \sigma^2 \mathbf{I}$ , which is positive definite with  $\det(\sigma^2 \mathbf{I}) = \sigma^{2n}$ , the density of  $\mathbf{Y}$  at the observed  $\mathbf{y}$  is

$$L(\boldsymbol{\beta}, \sigma^2) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}{2\sigma^2}\right\}.$$

As a function of the parameters with  $\mathbf{y}$  held fixed, this is the **likelihood**. Its logarithm is

$$\ell(\boldsymbol{\beta}, \sigma^2) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}{2\sigma^2}. \quad (7.3.2)$$

The data enter the log-likelihood only through the sum of squares  $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ . That is the first sign that normality and least squares belong together. The Gaussian density penalizes a residual by its square, so the most likely parameter values are those with the smallest squared residuals.

One technical point matters. If  $\mathbf{y} \in \mathcal{C}(\mathbf{X})$ , the residual sum of squares at the least squares fit is zero, and  $\ell \rightarrow \infty$  as  $\sigma^2 \rightarrow 0$ , so no maximum exists. This happens with probability zero when  $n > r$ . The subspace  $\mathcal{C}(\mathbf{X})$  has dimension  $r < n$ , so it has Lebesgue measure zero, and  $\mathbf{Y}$  has a density. We therefore assume  $n > r$  throughout, and  $\text{SSE} > 0$  for the observed data.

### 7.3.2 The maximum likelihood estimates

#### **Theorem 7.3.1 — Maximum likelihood in the normal linear model**

Assume [Eq. 7.3.1](#) with  $n > r$ , and let  $\mathbf{y}$  be such that  $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 > 0$ . Then  $(\tilde{\boldsymbol{\beta}}, \tilde{\sigma}^2)$  maximizes the likelihood iff  $\tilde{\boldsymbol{\beta}}$  is a least squares estimate and

$$\tilde{\sigma}^2 = \hat{\sigma}_{\text{ML}}^2 = \frac{\text{SSE}}{n}.$$

The maximized log-likelihood is  $\ell_{\max} = -\frac{n}{2} \{\log(2\pi \text{SSE}/n) + 1\}$ . The maximum likelihood estimate of  $\mathbf{X}\boldsymbol{\beta}$  is  $\mathbf{M}\mathbf{y}$ , and for estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  it is  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ . Both are unique.

*Proof.* Fix  $\sigma^2 > 0$ . By [Eq. 7.3.2](#),  $\ell(\mathbf{b}, \sigma^2)$  is a decreasing function of  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ . So it is maximized over  $\mathbf{b}$  exactly at the least squares estimates, and there  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE}$  ([Theorem 6.4.1](#)). This gives

$$\begin{aligned} \ell(\mathbf{b}, \sigma^2) &\leq \ell(\hat{\boldsymbol{\beta}}, \sigma^2) = g(\sigma^2) \\ &:= -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log \sigma^2 - \frac{\text{SSE}}{2\sigma^2}, \end{aligned}$$

with equality iff  $\mathbf{b}$  is a least squares estimate. Now put  $v_0 = \text{SSE}/n$  and  $u = v_0/\sigma^2 > 0$ . Then  $\text{SSE}/(2\sigma^2) = nu/2$  and  $\log \sigma^2 = \log v_0 - \log u$ , so

$$g(v_0) - g(\sigma^2) = \frac{n}{2}(u - 1 - \log u) \geq 0,$$

because  $\log u \leq u - 1$  for all  $u > 0$ , with equality only at  $u = 1$ . Hence  $\ell(\mathbf{b}, \sigma^2) \leq g(\sigma^2) \leq g(v_0)$ , with equality throughout iff  $\mathbf{b}$  is a least squares estimate and  $\sigma^2 = v_0$ . The value

$g(v_0)$  is the stated maximum. The least squares estimates all give the fitted vector  $\mathbf{My}$  (Theorem 6.4.1), and all give the same value of  $\lambda^\top \hat{\beta}$  for  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$  (Theorem 6.4.4). ■

The proof finds a global maximum without solving any equations. Calculus gives the same answer and is the usual route in a first course: set the gradient  $\partial\ell/\partial\beta = \mathbf{X}^\top(\mathbf{y} - \mathbf{X}\beta)/\sigma^2$  to zero, which gives the normal equations, and  $\partial\ell/\partial\sigma^2 = -n/(2\sigma^2) + \|\mathbf{y} - \mathbf{X}\beta\|^2/(2\sigma^4)$  to zero. But the stationarity argument still has to be completed by showing that the stationary point is a maximum (Exercise (B1)). The inequality  $\log u \leq u - 1$  handles existence, maximality and uniqueness together.

When  $\mathbf{X}$  is rank deficient, the likelihood is constant along the affine set of least squares estimates  $\hat{\beta} + \mathcal{N}(\mathbf{X})$ , so the MLE of  $\beta$  is not unique. This is a general fact about nonidentified parameters. If  $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$ , the two parameter values give the same distribution of  $\mathbf{Y}$  and hence the same likelihood for every data set. Maximum likelihood cannot separate what the model itself does not separate. What it does determine uniquely,  $\mathbf{X}\beta$  and the estimable functions, is exactly what least squares determines uniquely.

#### Example. (Stack loss)

Brownlee's stack loss data record  $n = 21$  days of operation of a plant that oxidizes ammonia to nitric acid. The response is the percentage of ammonia lost (times ten). The regressors are air flow, the temperature of the cooling water, and the acid concentration. With an intercept,  $p = 4$ , and least squares gives

$$\hat{\beta} = (-39.9197, 0.7156, 1.2953, -0.1521)^\top, \quad \text{SSE} = 178.830.$$

By Theorem 7.3.1 these are also the maximum likelihood estimates of  $\beta$ . The MLE of  $\sigma^2$  is  $178.830/21 = 8.516$ , while the unbiased estimate is  $s^2 = 178.830/17 = 10.519$ . The maximized log-likelihood is  $-52.288$ . A general-purpose numerical optimizer, started far from the answer and knowing nothing about least squares, arrives at the same point.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, stats

data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape

beta_hat, *_ = np.linalg.lstsq(X, y, rcond=None)
sse = np.sum((y - X @ beta_hat) ** 2)
sigma2_mle, s2 = sse / n, sse / (n - p)
loglik_max = -n / 2 * (np.log(2 * np.pi * sigma2_mle) + 1)
print("beta_hat", np.round(beta_hat, 4))
print(f"SSE = {sse:.3f}    MLE of sigma^2 = {sigma2_mle:.3f}    s^2 = {s2:.3f}")
print(f"maximized log-likelihood = {loglik_max:.3f}")
```

```
def negloglik(theta):
    """Minus the normal log-likelihood; theta = (beta, log sigma^2)."""
    b, log_s2 = theta[:p], theta[p]
    r = y - X @ b
```

```

    return n / 2 * (np.log(2 * np.pi) + log_s2) + r @ r / (2 * np.exp(log_s2))

start = np.r_[np.zeros(p), np.log(np.var(y))]
opt = optimize.minimize(negloglik, start, method="BFGS", options={"gtol": 1e-8})
print("numerical MLE of beta  ", np.round(opt.x[:p], 4))
print("numerical MLE sigma^2  ", round(float(np.exp(opt.x[p])), 3))

```

### 7.3.3 The bias of the variance estimate

Unlike least squares for  $\beta$ , the MLE of  $\sigma^2$  is biased. Since  $E(\text{SSE}) = \sigma^2(n - r)$  (Theorem 5.6.1 in full rank, and Section 6.4 in general),

$$E \hat{\sigma}_{\text{ML}}^2 = \frac{n - r}{n} \sigma^2.$$

The bias is downward, by the fraction  $r/n$ . The residual vector lives in the  $(n - r)$ -dimensional space  $\mathcal{C}(\mathbf{X})^\perp$ , and the MLE divides its squared length by  $n$  as if it had  $n$  dimensions. The  $r$  dimensions spent on fitting the mean are the dimensions in which the fit has pulled the data toward itself. With a few regressors and many observations the bias is negligible. It is not negligible when the number of parameters grows with the sample size.

#### Example. (Many means, one variance)

Suppose  $m$  laboratories each make two measurements,  $Y_{i1}, Y_{i2} \stackrel{\text{iid}}{\sim} N(\mu_i, \sigma^2)$ , with a different  $\mu_i$  for each laboratory and a common measurement variance. This is a linear model with  $n = 2m$  and  $r = m$ , one indicator column per laboratory. The fitted value for each pair is its mean, and

$$\begin{aligned} \hat{\sigma}_{\text{ML}}^2 &= \frac{1}{2m} \sum_{i=1}^m \sum_{j=1}^2 (Y_{ij} - \bar{Y}_{i\cdot})^2 = \frac{1}{4m} \sum_{i=1}^m (Y_{i1} - Y_{i2})^2, \\ E \hat{\sigma}_{\text{ML}}^2 &= \frac{\sigma^2}{2}. \end{aligned}$$

The bias does not disappear as  $m \rightarrow \infty$ . By the law of large numbers,  $\hat{\sigma}_{\text{ML}}^2 \rightarrow \sigma^2/2$  with probability one, so the MLE is *inconsistent*. The unbiased  $s^2 = \text{SSE}/(n - r)$  is exactly twice as large and is consistent. This is the example of Neyman and Scott (1948). It shows that the good large-sample behaviour of maximum likelihood can fail when the number of nuisance parameters grows with the data. Restricted maximum likelihood, which removes the mean before estimating the variance (Exercise (C2)), repairs it, and Chapter 32 builds variance-component estimation on that idea (Definition 32.4.1).

Unbiasedness is not the only possible criterion for choosing a divisor. Under normality we can compare estimators of the form  $\text{SSE}/d$  by mean squared error.

#### Proposition 7.3.2 — The best multiple of SSE

Assume Eq. 7.3.1 with  $k = n - r \geq 1$ . Among estimators  $\text{SSE}/d$  with  $d > 0$ ,

$$\frac{E(\text{SSE}/d - \sigma^2)^2}{\sigma^4} = \frac{2k}{d^2} + \left(\frac{k}{d} - 1\right)^2,$$

which is minimized at  $d = k + 2$ , with minimum value  $2/(k + 2)$ . The unbiased choice  $d = k$  gives  $2/k$ , and the maximum likelihood choice  $d = n$  gives  $(2k + r^2)/n^2$ .

*Proof.* Write  $\text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$ . By [Theorem 4.2.3](#) with  $\mathbf{A} = \mathbf{I} - \mathbf{M}$ ,  $\Sigma = \sigma^2 \mathbf{I}$  and  $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ , for which  $\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$ ,

$$E(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2 k, \quad \text{Var}(\text{SSE}) = 2\sigma^4 \text{tr}((\mathbf{I} - \mathbf{M})^2) = 2\sigma^4 k.$$

The mean squared error is variance plus squared bias:  $2\sigma^4 k/d^2 + (k/d - 1)^2 \sigma^4$ . As a function of  $t = 1/d$ , this is  $\sigma^4 \{(2k + k^2)t^2 - 2kt + 1\}$ , a convex quadratic minimized at  $t = k/(2k + k^2) = 1/(k + 2)$ . Substituting  $d = k + 2$ ,  $k$  and  $n = k + r$  gives the three values. For the last,  $2k/n^2 + (k - n)^2/n^2 = (2k + r^2)/n^2$ . ■

For the stack loss data,  $k = 17$ , and the three relative mean squared errors are 0.1176 for  $s^2$ , 0.1134 for the MLE and 0.1053 for  $\text{SSE}/(k + 2)$ . The unbiased estimator is the worst of the three by this criterion. Comparing  $(2k + r^2)/n^2$  with  $2/k$  shows that the MLE beats  $s^2$  iff  $(r - 4)k < 2r$ . So it always does when  $r \leq 4$ , and it loses once many parameters are fitted relative to the residual degrees of freedom, as in [Example \(Many means, one variance\)](#). In practice  $s^2$  is used anyway, because it is what makes the  $t$  and  $F$  statistics of [Section 7.5](#) exact. The choice of divisor matters for point estimation of  $\sigma^2$  and for nothing else. Any fixed multiple of SSE leads to the same tests and intervals once the matching distribution is used.

## 7.3.4 Profile likelihood

In most problems only some parameters are of interest. The others, called **nuisance parameters**, still appear in the likelihood. A simple and general way to eliminate them is to maximize them out.

### Definition 7.3.1 — Profile likelihood

Let the parameter be  $\boldsymbol{\theta} = (\psi, \boldsymbol{\eta})$ , where  $\psi$  is of interest and  $\boldsymbol{\eta}$  is a nuisance parameter. The **profile log-likelihood** of  $\psi$  is  $\ell_p(\psi) = \sup_{\boldsymbol{\eta}} \ell(\psi, \boldsymbol{\eta})$ .

The profile log-likelihood is maximized at the MLE of  $\psi$ , with maximum  $\ell_{\max}$ . The drop  $\ell_{\max} - \ell_p(\psi)$  measures how much less plausible the value  $\psi$  is than the best value. In the linear model the profiles have closed forms. The proof of [Theorem 7.3.1](#) computed two of them. With  $\sigma^2$  as nuisance, the profile of  $\boldsymbol{\beta}$  is

$$\ell_p(\boldsymbol{\beta}) = -\frac{n}{2} \left\{ \log \left( \frac{2\pi \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2}{n} \right) + 1 \right\},$$

a decreasing function of  $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2$ . Its level sets are the level sets of the residual sum of squares. When  $\mathbf{X}$  has full rank they are ellipsoids centred at  $\hat{\boldsymbol{\beta}}$ , and they are the basis of the confidence ellipsoids of [Chapter 12](#) ([Theorem 12.2.1](#)). With  $\boldsymbol{\beta}$  as nuisance, the profile of  $\sigma^2$  is the function  $g$  in the proof.

The most useful case is a single estimable function. Here the profile likelihood turns out to be a function of the familiar  $t$  statistic.

**Proposition 7.3.3 — Profile likelihood of an estimable function**

Assume Eq. 7.3.1 with  $n > r$ , and let  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ ,  $\lambda \neq \mathbf{0}$ . For  $\psi \in \mathbb{R}$ , put

$$t(\psi) = \frac{\lambda^\top \hat{\beta} - \psi}{s \sqrt{\lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda}}, \quad s^2 = \frac{\text{SSE}}{n - r}.$$

Then the smallest residual sum of squares subject to  $\lambda^\top \beta = \psi$  is

$$\text{SSE}(\psi) = \text{SSE} + \frac{(\lambda^\top \hat{\beta} - \psi)^2}{\lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda}, \quad (7.3.3)$$

and the profile log-likelihood of  $\psi = \lambda^\top \beta$ , with  $(\beta, \sigma^2)$  otherwise free, satisfies

$$2\{\ell_{\max} - \ell_p(\psi)\} = n \log \left( 1 + \frac{t(\psi)^2}{n - r} \right). \quad (7.3.4)$$

*Proof.* Write  $\lambda = \mathbf{X}^\top \rho$  and  $\mathbf{c} = \mathbf{M}\rho$ . Then  $\mathbf{c} \neq \mathbf{0}$ , since  $\mathbf{X}^\top \mathbf{c} = \mathbf{X}^\top \rho = \lambda \neq \mathbf{0}$ , and  $\|\mathbf{c}\|^2 = \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda$  as in the proof of Theorem 7.2.1(a). For any  $\mathbf{b}$ , Eq. 6.3.1 gives  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE} + \|\mathbf{X}\hat{\beta} - \mathbf{X}\mathbf{b}\|^2$ . Put  $\mathbf{u} = \mathbf{X}(\hat{\beta} - \mathbf{b}) \in \mathcal{C}(\mathbf{X})$ . The constraint  $\lambda^\top \mathbf{b} = \psi$  reads  $\rho^\top \mathbf{X}(\hat{\beta} - \mathbf{b}) = \lambda^\top \hat{\beta} - \psi =: \delta$ , that is,  $\rho^\top \mathbf{u} = \delta$ . Since  $\mathbf{u} \in \mathcal{C}(\mathbf{X})$ ,  $\rho^\top \mathbf{u} = \rho^\top \mathbf{M}\mathbf{u} = \mathbf{c}^\top \mathbf{u}$ . By the Cauchy–Schwarz inequality,  $|\delta| = |\mathbf{c}^\top \mathbf{u}| \leq \|\mathbf{c}\| \|\mathbf{u}\|$ , so  $\|\mathbf{u}\|^2 \geq \delta^2 / \|\mathbf{c}\|^2$ . Equality holds for  $\mathbf{u} = \delta \mathbf{c} / \|\mathbf{c}\|^2$ , which lies in  $\mathcal{C}(\mathbf{X})$  and so equals  $\mathbf{X}(\hat{\beta} - \mathbf{b})$  for some  $\mathbf{b}$ . That  $\mathbf{b}$  satisfies the constraint. This proves Eq. 7.3.3. Maximizing over  $\sigma^2$  as in the proof of Theorem 7.3.1, with SSE replaced by SSE( $\psi$ ), gives  $\ell_p(\psi) = -\frac{n}{2} \{\log(2\pi \text{SSE}(\psi)/n) + 1\}$ . Hence

$$\begin{aligned} 2\{\ell_{\max} - \ell_p(\psi)\} &= n \log \frac{\text{SSE}(\psi)}{\text{SSE}} = n \log \left( 1 + \frac{\delta^2}{\|\mathbf{c}\|^2 \text{SSE}} \right) \\ &= n \log \left( 1 + \frac{t(\psi)^2}{n - r} \right), \end{aligned}$$

since  $\delta^2 / (\|\mathbf{c}\|^2 \text{SSE}) = t(\psi)^2 / (n - r)$ . ■

Two consequences show up immediately. First, any procedure based on the profile likelihood of  $\lambda^\top \beta$  is a procedure based on  $|t|$ , because Eq. 7.3.4 is increasing in  $t^2$ . The likelihood ratio test of  $\lambda^\top \beta = \psi_0$  is therefore the  $t$  test, a fact Chapter 11 extends to general hypotheses (Theorem 11.4.1). Second, the large-sample calibration of the likelihood ratio, “reject when  $2\{\ell_{\max} - \ell_p\}$  exceeds a  $\chi^2(1)$  quantile”, is only approximate. The exact calibration uses the  $t(n - r)$  distribution of Section 7.5.

**Example. (Profile likelihood for the air-flow coefficient)**

For the stack loss model, Figure 7.3.1 plots  $\ell_p(\beta_1) - \ell_{\max}$  for the air-flow coefficient  $\beta_1$ , computed by refitting the model with  $\beta_1$  fixed on a grid. The listing confirms Eq. 7.3.4 to rounding error. The standard error of  $\hat{\beta}_1$  is 0.1349. The set of  $\beta_1$  with

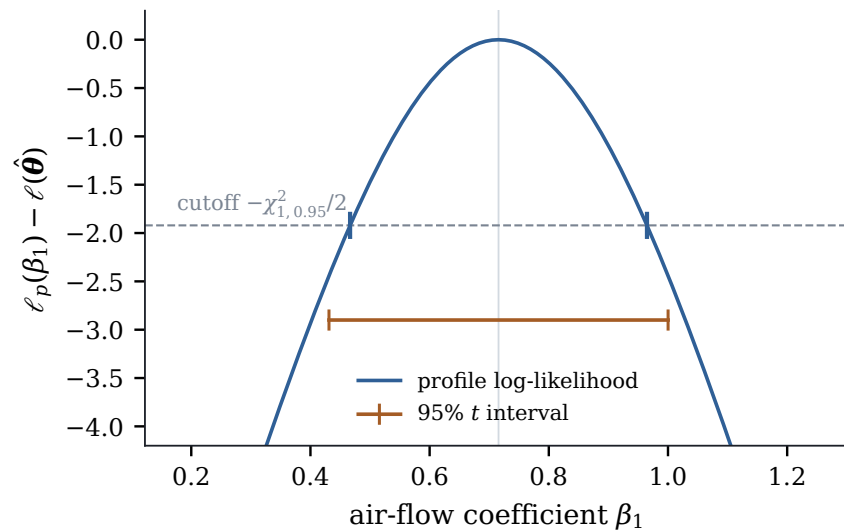


Figure 7.3.1: Profile log-likelihood of the air-flow coefficient in the stack loss regression, relative to its maximum. The ticks on the dashed cutoff line mark the  $\chi^2$ -calibrated likelihood interval, and the bar below it marks the exact 95%  $t$  interval, which is wider.

$2\{\ell_{\max} - \ell_p\} \leq 3.841$ , the 95% point of  $\chi^2(1)$ , is the likelihood interval (0.467, 0.965). By Eq. 7.3.4 it consists of the values with  $|t| \leq 1.847$ . The exact 95% interval uses the  $t(17)$  quantile 2.110 instead, and is (0.431, 1.000). With only 17 residual degrees of freedom, the  $\chi^2$  calibration makes the interval about 12% too short.

```
def profile_loglik(b1):
    """Profile log-likelihood of the air-flow coefficient: maximize over everything
    else."""
    others = np.delete(X, 1, axis=1)
    z = y - b1 * X[:, 1]
    coef, *_ = np.linalg.lstsq(others, z, rcond=None)
    sse_b = np.sum((z - others @ coef) ** 2)
    with beta_1 = b1
    return -n / 2 * (np.log(2 * np.pi * sse_b / n) + 1)

se1 = np.sqrt(s2 * np.linalg.inv(X.T @ X)[1, 1])
grid = np.linspace(beta_hat[1] - 4 * se1, beta_hat[1] + 4 * se1, 201)
lp = np.array([profile_loglik(b) for b in grid])
t = (beta_hat[1] - grid) / se1
print("max |2(l_max - l_p) - n log(1 + t^2/(n-p))| =",
      np.abs(2 * (loglik_max - lp) - n * np.log1p(t ** 2 / (n - p))).max())
```

*Remark* (Other error distributions). The agreement between maximum likelihood and least squares is special to normal errors. If the errors have the Laplace density  $(2\tau)^{-1}e^{-|\varepsilon|/\tau}$ , the log-likelihood is  $-n \log(2\tau) - \sum_i |y_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}|/\tau$ , and maximum likelihood minimizes the sum of *absolute* residuals; Exercise (C1) works out the location case. For errors with a  $t$  distribution it downweights large residuals. For each error distribution the likelihood picks its own loss function. Least squares is the one that goes with the Gaussian. Section 20.6 uses robust fitting as a diagnostic, and Chapter 45 treats median and quantile regression.

## 7.3.5 Exercises

### 7.3.5.1 A. Check your understanding

#### Exercise A1.

For  $Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2)$ , write the model in the form [Eq. 7.3.1](#) and read off the MLEs from [Theorem 7.3.1](#). Show that  $\hat{\sigma}_{\text{ML}}^2 = (n-1)s^2/n$ , where  $s^2$  is the sample variance.

#### Exercise A2.

Let  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  with residual sums of squares  $\text{SSE}_0 \geq \text{SSE}$ . Show that the ratio of maximized likelihoods is  $L_{0,\text{max}}/L_{\text{max}} = (\text{SSE}/\text{SSE}_0)^{n/2}$ , and write it as a decreasing function of  $(\text{SSE}_0 - \text{SSE})/\text{SSE}$ .

*Solution.* By [Theorem 7.3.1](#),  $L_{\text{max}} = (2\pi \text{SSE}/n)^{-n/2} e^{-n/2}$ , and similarly for the smaller model. The ratio is  $(\text{SSE}_0/\text{SSE})^{-n/2} = \{1 + (\text{SSE}_0 - \text{SSE})/\text{SSE}\}^{-n/2}$ , which decreases in  $(\text{SSE}_0 - \text{SSE})/\text{SSE}$ . That quantity is, up to degrees of freedom, the  $F$  statistic of [Theorem 4.4.6](#). So the likelihood ratio test for nested normal linear models is the  $F$  test. [Chapter 11](#) develops this ([Theorem 11.4.1](#)).

### 7.3.5.2 B. Practice

#### Exercise B1.

Assume  $\text{rank}(\mathbf{X}) = p$ . Compute the gradient and the Hessian of  $\ell(\boldsymbol{\beta}, \sigma^2)$ , show that the only stationary point is  $(\hat{\boldsymbol{\beta}}, \text{SSE}/n)$ , and show that the Hessian there is negative definite. Why does this not by itself prove that the stationary point is a global maximum? What additional argument completes the proof?

#### Exercise B2.

In [Example \(Many means, one variance\)](#), verify the identity  $\sum_j (Y_{ij} - \bar{Y}_{i\cdot})^2 = (Y_{i1} - Y_{i2})^2/2$ , compute  $E \hat{\sigma}_{\text{ML}}^2$  and  $\text{Var} \hat{\sigma}_{\text{ML}}^2$ , and prove that  $\hat{\sigma}_{\text{ML}}^2 \rightarrow \sigma^2/2$  in probability as  $m \rightarrow \infty$ .

*Solution.*  $Y_{i1} - \bar{Y}_{i\cdot} = (Y_{i1} - Y_{i2})/2 = -(Y_{i2} - \bar{Y}_{i\cdot})$ , so the sum of squares is  $2 \cdot (Y_{i1} - Y_{i2})^2/4$ . Now  $D_i = Y_{i1} - Y_{i2} \sim N(0, 2\sigma^2)$  independently, so  $D_i^2/(2\sigma^2) \sim \chi^2(1)$ , with mean 1 and variance 2. Hence  $\hat{\sigma}_{\text{ML}}^2 = \frac{1}{4m} \sum_i D_i^2$  has mean  $2\sigma^2/4 = \sigma^2/2$  and variance  $m \cdot 4\sigma^4 \cdot 2/(16m^2) = \sigma^4/(2m)$ . The variance tends to zero, so by Chebyshev's inequality  $\hat{\sigma}_{\text{ML}}^2 \rightarrow \sigma^2/2$  in probability.

#### Exercise B3.

Suppose  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{V})$  with  $\mathbf{V}$  known and positive definite and  $\mathbf{X}$  of full rank. Find the MLEs of  $\boldsymbol{\beta}$  and  $\sigma^2$ .

## 7.3.5.3 C. Going deeper

**Exercise C1.**

In the location model  $Y_i = \mu + \varepsilon_i$ ,  $i = 1, \dots, n$ , let the errors be independent with the Laplace density  $(2\tau)^{-1}e^{-|\varepsilon|/\tau}$ , so  $\text{Var}(\varepsilon_i) = 2\tau^2$ .

- Show that every sample median maximizes the likelihood in  $\mu$ , whatever  $\tau$  is, and that the MLE of  $\tau$  is the mean absolute deviation  $n^{-1} \sum_i |Y_i - \tilde{\mu}|$  about a median  $\tilde{\mu}$ .
- The sample median of  $n$  observations from a density  $f$  that is positive and continuous at its median  $\mu$  has asymptotic variance  $1/\{4nf(\mu)^2\}$ . Compute this for the Laplace density and compare it with  $\text{Var}(\bar{Y})$ . Which is larger, and by what factor?
- $\bar{Y}$  is the BLUE of  $\mu$  (Corollary 7.2.2). Explain why (b) does not contradict Theorem 7.2.1, and contrast the answer with Example (Uniform errors and the midrange), where the winning nonlinear estimator used the extremes rather than the middle of the sample.

*Solution.*

- The log-likelihood is  $-n \log(2\tau) - \tau^{-1} \sum_i |Y_i - \mu|$ . For fixed  $\tau$  it is maximized by minimizing  $g(\mu) = \sum_i |Y_i - \mu|$ . The function  $g$  is convex and piecewise linear, with slope  $\#\{i : Y_i < \mu\} - \#\{i : Y_i > \mu\}$  between data points. The slope is  $\leq 0$  to the left of every median and  $\geq 0$  to the right, so the minimizers are exactly the sample medians. Substituting  $\tilde{\mu}$  and maximizing  $-n \log \tau - g(\tilde{\mu})/\tau$  over  $\tau > 0$  gives  $\hat{\tau} = g(\tilde{\mu})/n$ .
- $f(\mu) = 1/(2\tau)$ , so the median has asymptotic variance  $\tau^2/n$ , while  $\text{Var}(\bar{Y}) = 2\tau^2/n$ . The mean has twice the variance of the median.
- The median is not a linear function of  $\mathbf{Y}$ : its weights depend on the ordering of the sample. The Gauss–Markov theorem compares  $\bar{Y}$  only with linear unbiased estimators. With Laplace errors, which have heavier tails than the normal, the extremes are unreliable and the median gains by ignoring them; with uniform errors the extremes are the most informative observations and the midrange gains by using only them.

**Exercise C2.**

Let  $\mathbf{Q}_2$  be an  $n \times (n-r)$  matrix whose columns are an orthonormal basis of  $\mathcal{C}(\mathbf{X})^\perp$ , and let  $\mathbf{Z} = \mathbf{Q}_2^\top \mathbf{Y}$  (“error contrasts”). Show that  $\mathbf{Z} \sim N_{n-r}(\mathbf{0}, \sigma^2 \mathbf{I})$ , whatever  $\boldsymbol{\beta}$  is, and that  $\|\mathbf{Z}\|^2 = \text{SSE}$ . Show that the MLE of  $\sigma^2$  based on  $\mathbf{Z}$  alone is  $\text{SSE}/(n-r)$ . This is **restricted** (or residual) maximum likelihood.

*Solution.*  $\mathbf{Q}_2^\top \mathbf{X} = \mathbf{0}$ , so  $\mathbf{Z} = \mathbf{Q}_2^\top \boldsymbol{\varepsilon} \sim N_{n-r}(\mathbf{0}, \sigma^2 \mathbf{Q}_2^\top \mathbf{Q}_2) = N_{n-r}(\mathbf{0}, \sigma^2 \mathbf{I})$  by Theorem 3.3.1. Also  $\mathbf{Q}_2 \mathbf{Q}_2^\top = \mathbf{I} - \mathbf{M}$  (Proposition 6.3.2), so  $\|\mathbf{Z}\|^2 = \mathbf{Y}^\top \mathbf{Q}_2 \mathbf{Q}_2^\top \mathbf{Y} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y} = \text{SSE}$ . The log-likelihood of  $\mathbf{Z}$  is  $-\frac{n-r}{2} \log(2\pi\sigma^2) - \|\mathbf{Z}\|^2/(2\sigma^2)$ , which is the function  $g$  of the proof of Theorem 7.3.1 with  $n$  replaced by  $n-r$ . It is maximized at  $\|\mathbf{Z}\|^2/(n-r) = \text{SSE}/(n-r)$ . In Example (Many means, one variance) this gives  $s^2$ , which is consistent.



## 7.4 Sufficiency, completeness and minimum variance

The Gauss–Markov theorem compares least squares only with *linear* unbiased estimators, and [Example \(Uniform errors and the midrange\)](#) showed that the restriction matters: with uniform errors a nonlinear estimator wins easily. This section shows that under normality the restriction can be dropped. In the normal linear model,  $\lambda^\top \hat{\beta}$  has the smallest variance among *all* unbiased estimators of an estimable  $\lambda^\top \beta$ , and  $s^2$  has the smallest variance among all unbiased estimators of  $\sigma^2$ . The argument is the classical one through sufficiency and completeness. It also produces best unbiased estimators of nonlinear functions, such as  $\sigma$  itself. At the end we compare the variances with the Cramér–Rao lower bound, which least squares attains for  $\beta$  and  $s^2$  does not attain for  $\sigma^2$ .

Two facts from mathematical statistics are quoted without proof: the factorization criterion for sufficiency and the completeness of full-rank exponential families. Both are proved in standard texts such as Lehmann and Casella (1998) and Lehmann and Romano (2005). Everything built on them is proved here.

### 7.4.1 Sufficiency

#### Definition 7.4.1 — Sufficient and complete statistics

Let  $\mathbf{Y}$  have a distribution  $P_\theta$  from a family indexed by  $\theta \in \Theta$ , and let  $\mathbf{T} = \mathbf{T}(\mathbf{Y})$  be a statistic.

- (a)  $\mathbf{T}$  is **sufficient** for  $\theta$  if the conditional law of  $\mathbf{Y}$  given  $\mathbf{T}$  is the same for every  $\theta$ .
- (b)  $\mathbf{T}$  is **complete** if, for every function  $h$  with  $E_\theta |h(\mathbf{T})| < \infty$  and  $E_\theta h(\mathbf{T}) = 0$  for all  $\theta \in \Theta$ , we have  $\Pr_\theta\{h(\mathbf{T}) = 0\} = 1$  for all  $\theta$ .

A sufficient statistic carries all the information about  $\theta$  in the data. Once  $\mathbf{T}$  is known, the rest of  $\mathbf{Y}$  is noise whose distribution does not involve the parameter. Completeness says that  $\mathbf{T}$  carries no redundant information either: no nontrivial function of  $\mathbf{T}$  has expectation zero for every parameter value. Equivalently, an unbiased estimator that is a function of  $\mathbf{T}$  is unique.

#### Lemma 7.4.1 — Factorization criterion

Suppose  $\mathbf{Y}$  has a density  $f_\theta(\mathbf{y})$  on  $\mathbb{R}^n$ . Then  $\mathbf{T}(\mathbf{Y})$  is sufficient iff there are functions  $g_\theta$  and  $h$ , with  $h$  not depending on  $\theta$ , such that  $f_\theta(\mathbf{y}) = g_\theta(\mathbf{T}(\mathbf{y})) h(\mathbf{y})$  for all  $\mathbf{y}$  and  $\theta$ .

#### Lemma 7.4.2 — Completeness in exponential families

Suppose  $\mathbf{Y}$  has a density of the form

$$f_\eta(\mathbf{y}) = c(\eta) h(\mathbf{y}) \exp\{\eta^\top \mathbf{T}(\mathbf{y})\}, \quad \eta \in \mathcal{E} \subseteq \mathbb{R}^k,$$

with  $\mathbf{T}(\mathbf{y}) \in \mathbb{R}^k$ . If  $\mathcal{E}$  contains a nonempty open subset of  $\mathbb{R}^k$ , then  $\mathbf{T}(\mathbf{Y})$  is complete

and sufficient.

Sufficiency follows from [Lemma 7.4.1](#) with  $g_{\boldsymbol{\eta}}(\mathbf{T}) = c(\boldsymbol{\eta})e^{\boldsymbol{\eta}^\top \mathbf{T}}$ . Completeness rests on the uniqueness of Laplace transforms:  $E_{\boldsymbol{\eta}} h(\mathbf{T}) = 0$  on an open set of  $\boldsymbol{\eta}$  forces a certain two-sided Laplace transform to vanish on an open set, and hence  $h(\mathbf{T}) = 0$  almost surely.

For the normal linear model, expand the exponent of the density:

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \mathbf{y}^\top \mathbf{y} - 2\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}.$$

So the density of [Eq. 7.3.1](#) is

$$f(\mathbf{y}) = (2\pi\sigma^2)^{-n/2} \exp\left\{-\frac{\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}}{2\sigma^2}\right\} \exp\left\{\frac{\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y}}{\sigma^2} - \frac{\mathbf{y}^\top \mathbf{y}}{2\sigma^2}\right\}, \quad (7.4.1)$$

and the data enter only through  $\mathbf{X}^\top \mathbf{y}$  and  $\mathbf{y}^\top \mathbf{y}$ .

### Proposition 7.4.3 — A complete sufficient statistic

In the normal linear model [Eq. 7.3.1](#), let  $\mathbf{Q}$  be an  $n \times r$  matrix whose columns are an orthonormal basis of  $\mathcal{C}(\mathbf{X})$ . Then  $\mathbf{T} = (\mathbf{Q}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$  is complete and sufficient for  $(\boldsymbol{\beta}, \sigma^2)$ . The statistics  $(\mathbf{X}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$  and  $(\mathbf{MY}, \text{SSE})$  are one-to-one functions of  $\mathbf{T}$ , so they are complete and sufficient too.

*Proof.* Since  $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X})$ ,  $\mathbf{X}\boldsymbol{\beta} = \mathbf{Q}\boldsymbol{\gamma}$  with  $\boldsymbol{\gamma} = \mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta} \in \mathbb{R}^r$ , and  $\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{y} = \boldsymbol{\gamma}^\top \mathbf{Q}^\top \mathbf{y}$ . By [Eq. 7.4.1](#) the density has the form of [Lemma 7.4.2](#) with  $k = r + 1$ ,

$$\mathbf{T}(\mathbf{y}) = (\mathbf{Q}^\top \mathbf{y}, \mathbf{y}^\top \mathbf{y}), \quad \boldsymbol{\eta} = (\boldsymbol{\gamma}/\sigma^2, -1/(2\sigma^2)).$$

As  $\boldsymbol{\beta}$  ranges over  $\mathbb{R}^p$ ,  $\boldsymbol{\gamma} = \mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta}$  ranges over all of  $\mathbb{R}^r$ : the  $r \times p$  matrix  $\mathbf{Q}^\top \mathbf{X}$  has rank  $r$ , because  $\mathbf{Q}\mathbf{Q}^\top \mathbf{X} = \mathbf{M}\mathbf{X} = \mathbf{X}$  has rank  $r$ . Hence  $\mathcal{E} = \mathbb{R}^r \times (-\infty, 0)$ , which is open, and [Lemma 7.4.2](#) applies. For the equivalent statistics:  $\mathbf{MY} = \mathbf{Q}\mathbf{Q}^\top \mathbf{Y}$  ([Proposition 6.3.2](#)) and  $\text{SSE} = \mathbf{Y}^\top \mathbf{Y} - \|\mathbf{Q}^\top \mathbf{Y}\|^2$  are functions of  $\mathbf{T}$ , and conversely  $\mathbf{Q}^\top \mathbf{Y} = \mathbf{Q}^\top \mathbf{MY}$  and  $\mathbf{Y}^\top \mathbf{Y} = \text{SSE} + \|\mathbf{MY}\|^2$ . Also  $\mathbf{X}^\top \mathbf{Y} = \mathbf{X}^\top \mathbf{Q}(\mathbf{Q}^\top \mathbf{Y})$ , and  $\mathbf{Q} = \mathbf{X}\mathbf{K}$  for some  $\mathbf{K}$ , since the columns of  $\mathbf{Q}$  lie in  $\mathcal{C}(\mathbf{X})$ , so  $\mathbf{Q}^\top \mathbf{Y} = \mathbf{K}^\top \mathbf{X}^\top \mathbf{Y}$ . A one-to-one function of a complete sufficient statistic is again complete and sufficient, since both definitions refer only to the information (the  $\sigma$ -field) the statistic generates. ■

The reduction is dramatic. An  $n$ -dimensional data vector is replaced by  $r+1$  numbers: the fitted values (equivalently, the least squares estimates of the estimable functions) and the residual sum of squares. Under normality the residual *vector*, apart from its length, carries no information about the parameters. Its direction is uniformly distributed over the unit sphere of  $\mathcal{C}(\mathbf{X})^\perp$  whatever  $(\boldsymbol{\beta}, \sigma^2)$  is ([Exercise \(C1\)](#)). This is the precise sense in which the residuals are “just noise”, and it is also why they are useful for checking the model. Their pattern can reveal departures from normality precisely because, under normality, it has a known and parameter-free distribution.

## 7.4.2 Rao–Blackwell and Lehmann–Scheffé

**Theorem 7.4.4 — Rao–Blackwell**

Let  $\mathbf{T}$  be sufficient for  $\boldsymbol{\theta}$ , and let  $\delta(\mathbf{Y})$  be an unbiased estimator of  $g(\boldsymbol{\theta})$  with  $\text{Var}_{\boldsymbol{\theta}} \delta < \infty$  for all  $\boldsymbol{\theta}$ . Then  $\delta^* = E(\delta \mid \mathbf{T})$  is a statistic, it is unbiased for  $g(\boldsymbol{\theta})$ , and

$$\begin{aligned} \text{Var}_{\boldsymbol{\theta}}(\delta) &= \text{Var}_{\boldsymbol{\theta}}(\delta^*) + E_{\boldsymbol{\theta}}[(\delta - \delta^*)^2] \\ &\geq \text{Var}_{\boldsymbol{\theta}}(\delta^*) \end{aligned}$$

for every  $\boldsymbol{\theta}$ , with equality iff  $\delta = \delta^*$  with probability one.

*Proof.* The law of  $\mathbf{Y}$  given  $\mathbf{T}$  does not depend on  $\boldsymbol{\theta}$ , so neither does  $E(\delta(\mathbf{Y}) \mid \mathbf{T})$ . It is therefore computable from the data, which makes it a statistic. By the law of total expectation (Proposition 2.5.4),  $E \delta^* = E \delta = g(\boldsymbol{\theta})$ . By the law of total variance in the same result,  $\text{Var} \delta = E\{\text{Var}(\delta \mid \mathbf{T})\} + \text{Var}(\delta^*)$ , and  $E\{\text{Var}(\delta \mid \mathbf{T})\} = E\{E[(\delta - \delta^*)^2 \mid \mathbf{T}]\} = E(\delta - \delta^*)^2$ . This last term is zero iff  $\delta = \delta^*$  almost surely. ■

Conditioning on a sufficient statistic averages out the part of an estimator that responds to parameter-free noise. It never hurts, and it helps unless the estimator was already a function of  $\mathbf{T}$ .

**Theorem 7.4.5 — Lehmann–Scheffé**

Let  $\mathbf{T}$  be complete and sufficient for  $\boldsymbol{\theta}$ . If  $\phi(\mathbf{T})$  is unbiased for  $g(\boldsymbol{\theta})$  and has finite variance, then  $\phi(\mathbf{T})$  is a **uniformly minimum variance unbiased estimator** (UMVUE) of  $g(\boldsymbol{\theta})$ :  $\text{Var}_{\boldsymbol{\theta}} \phi(\mathbf{T}) \leq \text{Var}_{\boldsymbol{\theta}} \delta$  for every unbiased  $\delta$  and every  $\boldsymbol{\theta}$ . It is the only unbiased function of  $\mathbf{T}$  and, up to sets of probability zero, the only UMVUE.

*Proof.* If  $\phi_1(\mathbf{T})$  and  $\phi_2(\mathbf{T})$  are both unbiased, then  $E_{\boldsymbol{\theta}}\{\phi_1(\mathbf{T}) - \phi_2(\mathbf{T})\} = 0$  for all  $\boldsymbol{\theta}$ , so  $\phi_1(\mathbf{T}) = \phi_2(\mathbf{T})$  almost surely by completeness. Now let  $\delta$  be any unbiased estimator with finite variance. By Theorem 7.4.4,  $E(\delta \mid \mathbf{T})$  is an unbiased function of  $\mathbf{T}$ , so it equals  $\phi(\mathbf{T})$  almost surely, and  $\text{Var} \phi(\mathbf{T}) \leq \text{Var} \delta$ , with equality iff  $\delta = \phi(\mathbf{T})$  almost surely. (If  $\delta$  has infinite variance there is nothing to prove.) ■

## 7.4.3 Best unbiased estimators in the normal linear model

**Theorem 7.4.6 — Minimum variance unbiased estimation**

Assume the normal linear model Eq. 7.3.1 with  $n > r$ , and let  $s^2 = \text{SSE}/(n - r)$ .

- (a) For every  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ ,  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  is the UMVUE of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ .
- (b)  $s^2$  is the UMVUE of  $\sigma^2$ .

- (c) More generally, every estimator of the form  $\phi(\mathbf{MY}, \text{SSE})$  with finite variance is the UMVUE of its expectation. In particular, for  $k = n - r$ ,

$$b_k = \sqrt{\frac{2}{k}} \frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2})},$$

$s/b_k$  is the UMVUE of  $\sigma$ , and  $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + z s/b_k$  is the UMVUE of  $\mathbf{x}_0^\top \boldsymbol{\beta} + z\sigma$  for every constant  $z$  and every  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$ .

*Proof.* By Proposition 7.4.3,  $(\mathbf{MY}, \text{SSE})$  is complete and sufficient, so by Theorem 7.4.5 it suffices to show that each estimator is an unbiased function of it with finite variance.

(a) Write  $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$ . Then  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{MY}$  is a function of  $\mathbf{MY}$ , it is unbiased by Theorem 7.2.1(a), and its variance is finite. (b)  $s^2$  is a function of SSE, it is unbiased because  $E(\text{SSE}) = \sigma^2(n - r)$  (Theorem 2.3.1), and it has finite variance. (c) The first statement is Theorem 7.4.5 again. For  $\sigma$ , note that  $\text{SSE}/\sigma^2 = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y} / \sigma^2 \sim \chi^2(k)$  by Theorem 4.3.2(a), since  $\mathbf{I} - \mathbf{M}$  is a projection of rank  $k$  and  $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ . If  $W \sim \chi^2(k)$ , with density  $w^{k/2-1}e^{-w/2}/\{2^{k/2}\Gamma(k/2)\}$ , then

$$\begin{aligned} E\sqrt{W} &= \frac{1}{2^{k/2}\Gamma(k/2)} \int_0^\infty w^{(k+1)/2-1} e^{-w/2} dw \\ &= \frac{2^{(k+1)/2}\Gamma(\frac{k+1}{2})}{2^{k/2}\Gamma(\frac{k}{2})} = \sqrt{2} \frac{\Gamma(\frac{k+1}{2})}{\Gamma(\frac{k}{2})}. \end{aligned}$$

Hence  $Es = \sigma E\sqrt{W/k} = b_k\sigma$ , and  $s/b_k$  is unbiased for  $\sigma$ , with variance at most  $E s^2 / b_k^2 < \infty$ . The last estimator is a sum of two unbiased functions of  $(\mathbf{MY}, \text{SSE})$ . ■

Part (a) is the promised strengthening of Gauss–Markov. Under normality no unbiased estimator of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ , linear or not, beats least squares. The midrange of Example (Uniform errors and the midrange) cannot help when the errors are normal. Part (c) shows the method at work beyond linear estimators. The obvious estimate  $s$  of  $\sigma$  is biased, because  $Es < \sqrt{E s^2}$  by Jensen’s inequality, and dividing by  $b_k$  removes the bias.

#### Example. (Estimating $\sigma$ without bias)

The constant  $b_k = E(s)/\sigma$  depends only on the residual degrees of freedom. For the stack loss regression,  $k = 17$  and  $b_{17} = 0.9854$ , so the UMVUE of  $\sigma$  is  $s/b_{17} = 1.0148 s$ . The correction is 1.5% here. It is large for very small  $k$ , with  $b_2 = 0.8862$ , and negligible for large  $k$ , with  $b_{200} = 0.9988$ .

```
import numpy as np
from scipy import special

def mean_s_factor(k):
    """b_k = E(s)/sigma when k s^2 / sigma^2 ~ chi^2(k)."""
    return np.sqrt(2 / k) * np.exp(special.gammaln((k + 1) / 2) - special.gammaln(k / 2))

for k in [2, 5, 10, 17, 50, 200]:
    b = mean_s_factor(k)
```

```
print(f"k = {k:3d}    E(s)/sigma = {b:.4f}    UMVUE of sigma = s / {b:.4f}")
```

## 7.4.4 Rao–Blackwellization by reflection

The Rao–Blackwell theorem is not only a device for proofs. In the normal linear model we can carry out the conditioning explicitly, and it takes every linear unbiased estimator to least squares.

### Example. (Conditioning on the sufficient statistic)

Let  $\mathbf{a}^\top \mathbf{Y}$  be any LUE of an estimable  $\lambda^\top \beta$ , such as the end-point slope of [Example \(Three linear unbiased estimators of a slope\)](#). Let  $\mathbf{R} = 2\mathbf{M} - \mathbf{I}$ , the reflection that fixes  $\mathcal{C}(\mathbf{X})$  and reverses  $\mathcal{C}(\mathbf{X})^\perp$ . It is symmetric and orthogonal ( $\mathbf{R}^2 = \mathbf{I}$ ) and  $\mathbf{R}\mathbf{X} = \mathbf{X}$ . Consequently:

- $\mathbf{R}\mathbf{Y} \sim N_n(\mathbf{R}\mathbf{X}\beta, \sigma^2 \mathbf{R}\mathbf{R}^\top) = N_n(\mathbf{X}\beta, \sigma^2 \mathbf{I})$  by [Theorem 3.3.1](#), so  $\mathbf{R}\mathbf{Y}$  has the same distribution as  $\mathbf{Y}$ ;
- the sufficient statistic is unchanged:  $\mathbf{X}^\top \mathbf{R}\mathbf{Y} = \mathbf{X}^\top \mathbf{Y}$  and  $\|\mathbf{R}\mathbf{Y}\| = \|\mathbf{Y}\|$ .

So the pairs  $(\mathbf{a}^\top \mathbf{Y}, \mathbf{T}(\mathbf{Y}))$  and  $(\mathbf{a}^\top \mathbf{R}\mathbf{Y}, \mathbf{T}(\mathbf{Y}))$  have the same joint distribution, where  $\mathbf{T} = (\mathbf{X}^\top \mathbf{Y}, \mathbf{Y}^\top \mathbf{Y})$ . Hence  $E(\mathbf{a}^\top \mathbf{Y} \mid \mathbf{T}) = E(\mathbf{a}^\top \mathbf{R}\mathbf{Y} \mid \mathbf{T})$ , and averaging,

$$E(\mathbf{a}^\top \mathbf{Y} \mid \mathbf{T}) = E\left(\mathbf{a}^\top \frac{\mathbf{Y} + \mathbf{R}\mathbf{Y}}{2} \mid \mathbf{T}\right) = E(\mathbf{a}^\top \mathbf{M}\mathbf{Y} \mid \mathbf{T}) = \mathbf{a}^\top \mathbf{M}\mathbf{Y} = \lambda^\top \hat{\beta},$$

because  $\mathbf{a}^\top \mathbf{M}\mathbf{Y}$  is a function of  $\mathbf{T}$  and  $\mathbf{M}\mathbf{a} = \mathbf{a}_*$  ([Proposition 7.1.2](#)). The Rao–Blackwell improvement of *every* LUE is the least squares estimate. The listing checks the identity  $\frac{1}{2}(\mathbf{a}^\top \mathbf{y} + \mathbf{a}^\top \mathbf{R}\mathbf{y}) = \lambda^\top \hat{\beta}$  for simulated data and the end-point slope.

```
import numpy as np

x = np.array([1.0, 2.0, 2.5, 3.0, 4.0, 5.0, 5.5, 7.0, 8.0, 9.0, 10.5, 12.0])
n = len(x)
X = np.column_stack([np.ones(n), x])
M = X @ np.linalg.solve(X.T @ X, X.T)
R = 2 * M - np.eye(n)                                # reflection: keeps C(X), flips C(X)
                                                        )-perp

rng = np.random.default_rng(71)
y = X @ np.array([2.0, 0.5]) + rng.normal(size=n)
y_ref = R @ y
print("X^T y unchanged:", np.allclose(X.T @ y_ref, X.T @ y))
print("y^T y unchanged:", np.isclose(y_ref @ y_ref, y @ y))

a_end = np.zeros(n)
a_end[[0, -1]] = [-1.0, 1.0]
a_end /= x[-1] - x[0]                                # the end-point slope estimator
a_ls = X @ np.linalg.solve(X.T @ X, [0.0, 1.0])
print("average of a^T y over the two points:", (a_end @ y + a_end @ y_ref) / 2)
print("least squares slope:", a_ls @ y)
```

The reflection argument uses the normal distribution only through its invariance under orthogo-

nal maps that fix the mean. It makes concrete what “the residuals are noise” means. Reversing the residual vector changes neither the likelihood nor the sufficient statistic, so an estimator should not depend on the residuals’ direction. An LUE that uses the residuals is replaced by the average of its value at  $\mathbf{y}$  and at the reflected data, and that average no longer involves the residuals.

## 7.4.5 The Cramér–Rao bound

A different route to optimality bounds the variance of every unbiased estimator from below and checks whether the bound is attained. We state the bound for the full-rank normal linear model, where every ingredient can be computed.

Write  $\boldsymbol{\theta} = (\boldsymbol{\beta}^\top, v)^\top$  with  $v = \sigma^2$ . The **score** is the gradient of the log-likelihood Eq. 7.3.2, evaluated at the random  $\mathbf{Y}$ :

$$\mathbf{S} = \begin{pmatrix} \partial \ell / \partial \boldsymbol{\beta} \\ \partial \ell / \partial v \end{pmatrix} = \begin{pmatrix} \mathbf{X}^\top \boldsymbol{\varepsilon} / v \\ -\frac{n}{2v} + \frac{\boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}}{2v^2} \end{pmatrix}, \quad \boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\boldsymbol{\beta}.$$

### Proposition 7.4.7 — Score and information

In Eq. 7.3.1 with  $\text{rank}(\mathbf{X}) = p$ ,  $\mathbb{E} \mathbf{S} = \mathbf{0}$  and

$$\mathcal{I}(\boldsymbol{\theta}) = \text{Cov}(\mathbf{S}) = \begin{pmatrix} \mathbf{X}^\top \mathbf{X} / \sigma^2 & \mathbf{0} \\ \mathbf{0}^\top & n / (2\sigma^4) \end{pmatrix}.$$

*Proof.*  $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, v\mathbf{I})$ . Hence  $\mathbb{E}(\mathbf{X}^\top \boldsymbol{\varepsilon} / v) = \mathbf{0}$  and  $\text{Cov}(\mathbf{X}^\top \boldsymbol{\varepsilon} / v) = \mathbf{X}^\top \mathbf{X} / v$  by Theorem 2.2.1. Next,  $\boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon} / v \sim \chi^2(n)$  by Theorem 4.3.2(a) with  $\mathbf{P} = \mathbf{I}$ , so it has mean  $n$  and variance  $2n$ . So the last coordinate has mean  $-n/(2v) + nv/(2v^2) = 0$  and variance  $2n/(4v^2) = n/(2v^2)$ . Finally, by Theorem 4.2.3(c) with  $\mathbf{K} = \mathbf{X}^\top$ ,  $\mathbf{A} = \mathbf{I}$  and  $\boldsymbol{\mu} = \mathbf{0}$ ,  $\text{Cov}(\mathbf{X}^\top \boldsymbol{\varepsilon}, \boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}) = \mathbf{0}$ . ■

### Theorem 7.4.8 — Cramér–Rao bound in the normal linear model

Assume Eq. 7.3.1 with  $\text{rank}(\mathbf{X}) = p$ . Let  $\delta(\mathbf{Y})$  have finite variance and satisfy  $\mathbb{E}_{\boldsymbol{\theta}} \delta = g(\boldsymbol{\theta})$  for all  $\boldsymbol{\theta}$ , with  $g$  differentiable. Then

$$\begin{aligned} \text{Var}_{\boldsymbol{\theta}}(\delta) &\geq \nabla g(\boldsymbol{\theta})^\top \mathcal{I}(\boldsymbol{\theta})^{-1} \nabla g(\boldsymbol{\theta}) \\ &= \sigma^2 \mathbf{g}_{\boldsymbol{\beta}}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{g}_{\boldsymbol{\beta}} + \frac{2\sigma^4}{n} g_v^2, \end{aligned}$$

where  $\mathbf{g}_{\boldsymbol{\beta}} = \partial g / \partial \boldsymbol{\beta}$  and  $g_v = \partial g / \partial v$ .

*Proof.* Write  $f_{\boldsymbol{\theta}}$  for the density. Differentiating  $g(\boldsymbol{\theta}) = \int \delta(\mathbf{y}) f_{\boldsymbol{\theta}}(\mathbf{y}) d\mathbf{y}$  under the integral sign, which is justified because the normal family is an exponential family (Lehmann and

Casella 1998, chapter 1), gives

$$\begin{aligned}\nabla g(\boldsymbol{\theta}) &= \int \delta(\mathbf{y}) \nabla \log f_{\boldsymbol{\theta}}(\mathbf{y}) f_{\boldsymbol{\theta}}(\mathbf{y}) d\mathbf{y} \\ &= E(\delta \mathbf{S}) = \text{Cov}(\delta, \mathbf{S}),\end{aligned}$$

the last step because  $E\mathbf{S} = \mathbf{0}$ . For any constant vector  $\mathbf{c}$ ,  $0 \leq \text{Var}(\delta - \mathbf{c}^\top \mathbf{S}) = \text{Var} \delta - 2\mathbf{c}^\top \nabla g + \mathbf{c}^\top \mathcal{I} \mathbf{c}$ . Choosing  $\mathbf{c} = \mathcal{I}^{-1} \nabla g$  gives  $\text{Var} \delta \geq \nabla g^\top \mathcal{I}^{-1} \nabla g$ . The block-diagonal form of [Proposition 7.4.7](#) gives the right-hand side.

■

For  $g(\boldsymbol{\theta}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$  the bound is  $\sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$ , exactly the variance of  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ . Least squares **attains** the bound, which gives a second proof of [Theorem 7.4.6\(a\)](#) in full rank. For  $g = \sigma^2$  the bound is  $2\sigma^4/n$ , while

$$\text{Var}(s^2) = \frac{\sigma^4}{(n-p)^2} \text{Var}\{\chi^2(n-p)\} = \frac{2\sigma^4}{n-p} > \frac{2\sigma^4}{n}.$$

So  $s^2$  does not attain the bound, although by [Theorem 7.4.6\(b\)](#) no unbiased estimator does better. The bound is simply not sharp here. Its ratio to the attained minimum, the **efficiency** of  $s^2$ , is  $(n-p)/n$ . That is 0.810 for the stack loss regression, and it tends to one as  $n \rightarrow \infty$  with  $p$  fixed. The same happens for  $\sigma$ : the UMVUE  $s/b_k$  has variance  $0.0298 \sigma^2$  when  $k = 17$ , against the bound  $\sigma^2/(2n) = 0.0238 \sigma^2$ , an efficiency of 0.798. The equality condition in the proof explains why. The bound is attained iff  $\text{Var}(\delta - \mathbf{c}^\top \mathbf{S}) = 0$ , that is, iff  $\delta = g(\boldsymbol{\theta}) + \nabla g^\top \mathcal{I}^{-1} \mathbf{S}$  with probability one. For  $g = \sigma^2$  the right-hand side is  $\sigma^2 + (2\sigma^4/n)\{-n/(2\sigma^2) + \boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon}/(2\sigma^4)\} = \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2/n$ . This is not a statistic, because it involves the unknown  $\boldsymbol{\beta}$ , so no unbiased estimator of  $\sigma^2$  attains the bound.

*Warning.* The Cramér–Rao bound is a lower bound, not a target. An estimator that fails to attain it may still be the best unbiased estimator, as  $s^2$  is. When an estimator does attain it, the bound certifies optimality without any appeal to completeness. The Lehmann–Scheffé route is stronger in one respect: it needs no differentiability and covers every estimable function in every rank.

## 7.4.6 What changes without normality

Every result of this section uses the normal density. Without it the conclusions can fail. Under uniform errors the midrange beats  $\bar{Y}$  ([Example \(Uniform errors and the midrange\)](#)). Under Laplace errors the sample median has asymptotic variance  $\sigma^2/(2n)$ , half that of  $\bar{Y}$ . Under the second-moment assumptions alone, the best that can be said for  $s^2$  is a result in the spirit of Gauss–Markov. Among quadratic unbiased estimators  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  of  $\sigma^2$  with  $\mathbf{A}$  nonnegative definite,  $s^2$  has the smallest variance when the errors have zero excess kurtosis, or when all leverages are equal. [Exercise \(C2\)](#) outlines this result of Atiqullah (1962).

## 7.4.7 Exercises

### 7.4.7.1 A. Check your understanding

**Exercise A1.**

Assume  $\text{rank}(\mathbf{X}) = p$ . Use [Lemma 7.4.1](#) and the identity  $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \text{SSE} + (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$  to show directly that  $(\hat{\boldsymbol{\beta}}, \text{SSE})$  is sufficient for  $(\boldsymbol{\beta}, \sigma^2)$ . Is  $\hat{\boldsymbol{\beta}}$  alone sufficient?

**Exercise A2.**

Suppose  $\sigma^2 = \sigma_0^2$  is known. Show that  $\mathbf{X}^\top \mathbf{Y}$  alone is complete and sufficient for  $\boldsymbol{\beta}$ , and that  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  is still the UMVUE of every estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ . Why is  $\mathbf{X}^\top \mathbf{Y}$  not sufficient when  $\sigma^2$  is unknown?

**7.4.7.2 B. Practice****Exercise B1.**

For estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ , find the UMVUE of  $(\boldsymbol{\lambda}^\top \boldsymbol{\beta})^2$  in the normal linear model. Can it be negative? Comment.

*Solution.*  $E(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})^2 = (\boldsymbol{\lambda}^\top \boldsymbol{\beta})^2 + \sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$  and  $E s^2 = \sigma^2$ , so  $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})^2 - s^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$  is unbiased. It is a function of  $(\mathbf{M}\mathbf{Y}, \text{SSE})$ , so it is the UMVUE by [Theorem 7.4.6\(c\)](#). It is negative whenever  $|t| < 1$  for the  $t$  statistic of [Proposition 7.3.3](#) at  $\psi = 0$ , which has positive probability. An estimator of a nonnegative quantity that can be negative is clearly improvable, for example by its positive part. The positive part is biased but has smaller mean squared error. Unbiasedness can be a costly requirement.

**Exercise B2.**

Compute the observed information  $-\nabla^2 \ell$  at  $(\boldsymbol{\beta}, v)$ , take its expectation, and confirm [Proposition 7.4.7](#) by this second route.

**Exercise B3.**

For the stack loss model, the 90% quantile of the response at a new operating condition  $\mathbf{x}_0$  is  $\mathbf{x}_0^\top \boldsymbol{\beta} + 1.2816 \sigma$ . Write down its UMVUE, and explain why  $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + 1.2816 s$  is biased. By how much, in units of  $\sigma$ ?

**7.4.7.3 C. Going deeper****Exercise C1.**

In the normal linear model with  $n - r \geq 2$ , let  $\mathbf{U} = (\mathbf{I} - \mathbf{M})\mathbf{Y} / \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|$ . Show that  $\mathbf{U}$  is independent of  $(\mathbf{M}\mathbf{Y}, \text{SSE})$ , and that its distribution (uniform on the unit sphere of  $\mathcal{C}(\mathbf{X})^\perp$ ) does not depend on  $(\boldsymbol{\beta}, \sigma^2)$ . *Hint:* write  $(\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{Q}_2 \mathbf{Z}$  as in [Exercise \(C2\)](#), and use the rotation invariance of  $N(\mathbf{0}, \sigma^2 \mathbf{I})$ .



*Solution.* With  $\mathbf{Q}_2$  as in [Exercise \(C2\)](#),  $(\mathbf{I} - \mathbf{M})\mathbf{Y} = \mathbf{Q}_2\mathbf{Z}$  with  $\mathbf{Z} \sim N_{n-r}(\mathbf{0}, \sigma^2\mathbf{I})$ , and  $\mathbf{Z}$  is independent of  $\mathbf{MY}$  by [Theorem 3.4.1](#), since  $\text{Cov}(\mathbf{MY}, \mathbf{Z}) = \sigma^2\mathbf{MQ}_2 = \mathbf{0}$ . Then  $\text{SSE} = \|\mathbf{Z}\|^2$  and  $\mathbf{U} = \mathbf{Q}_2\mathbf{Z}/\|\mathbf{Z}\|$ . For any orthogonal  $\mathbf{H}$ ,  $\mathbf{HZ}$  has the same law as  $\mathbf{Z}$ , so the law of  $\mathbf{Z}/\|\mathbf{Z}\|$  is invariant under all rotations. The only such law on the sphere is the uniform one. The independence of the direction  $\mathbf{Z}/\|\mathbf{Z}\|$  and the length  $\|\mathbf{Z}\|$  follows from the polar form of the density of  $\mathbf{Z}$ , which is a function of  $\|\mathbf{z}\|$  only. Hence  $\mathbf{U}$  is independent of  $(\mathbf{MY}, \text{SSE})$ , and its law involves neither  $\boldsymbol{\beta}$  nor  $\sigma^2$ .

### Exercise C2.

Assume only that the errors are independent with common variance  $\sigma^2$ , zero third moment and fourth moment  $\mu_4$ . Let  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  be an unbiased estimator of  $\sigma^2$  with  $\mathbf{A}$  symmetric nonnegative definite. (a) Show that  $\mathbf{A}\mathbf{X} = \mathbf{0}$  and  $\text{tr } \mathbf{A} = 1$ . (b) Using [Theorem 2.3.3](#), show that  $\text{Var}(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = (\mu_4 - 3\sigma^4) \sum_i a_{ii}^2 + 2\sigma^4 \text{tr}(\mathbf{A}^2)$ . (c) Show that  $\text{tr}(\mathbf{A}^2) \geq 1/(n-r)$ , with equality iff  $\mathbf{A} = (\mathbf{I} - \mathbf{M})/(n-r)$ . (d) Conclude that  $s^2$  has the smallest variance in this class when  $\mu_4 = 3\sigma^4$ . *Hint for (c):*  $\mathbf{A} = (\mathbf{I} - \mathbf{M})\mathbf{A}(\mathbf{I} - \mathbf{M})$ , so the eigenvalues of  $\mathbf{A}$  live on  $\mathcal{C}(\mathbf{X})^\perp$ .

## 7.5 Sampling distributions of the estimates

A point estimate is of little use without a measure of its uncertainty. Under the second-moment assumptions we know the mean and variance of  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  and the mean of  $s^2$ , but not their distributions. Under normality we know everything. The estimates of the estimable functions are jointly normal, the residual sum of squares is a scaled chi-squared variable, and the two are independent. These three facts are what the whole of Part III rests on. Tests, confidence intervals, confidence ellipsoids and prediction intervals are all built from them. The machinery was assembled in [Chapter 3](#) and [Chapter 4](#). Here we apply it.

### 7.5.1 The main theorem

#### Theorem 7.5.1 — Sampling distributions in the normal linear model

Assume [Eq. 7.3.1](#) with  $r = \text{rank}(\mathbf{X}) < n$ . Let  $\boldsymbol{\Lambda}$  be  $p \times q$  with  $\mathcal{C}(\boldsymbol{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$ , let  $(\mathbf{X}^\top \mathbf{X})^-$  be any generalized inverse, and let  $s^2 = \text{SSE}/(n-r)$ .

- (a)  $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} \sim N_q(\boldsymbol{\Lambda}^\top \boldsymbol{\beta}, \sigma^2 \boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^- \boldsymbol{\Lambda})$ . In particular  $\hat{\mathbf{Y}} = \mathbf{MY} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{M})$ , and if  $\text{rank}(\mathbf{X}) = p$ ,  $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1})$ .
- (b)  $\text{SSE}/\sigma^2 = (n-r)s^2/\sigma^2 \sim \chi^2(n-r)$ .
- (c)  $\mathbf{MY}$  and  $\text{SSE}$  are independent. Hence  $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}$ , and every function of the fitted values, is independent of  $s^2$ .
- (d)  $E(s^2) = \sigma^2$  and  $\text{Var}(s^2) = 2\sigma^4/(n-r)$ .

*Proof.* Write  $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{P}$  for an  $n \times q$  matrix  $\mathbf{P}$ . As in the proof of [Theorem 7.2.1\(d\)](#),  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{P}^\top \mathbf{M} \mathbf{Y}$  for every least squares estimate  $\hat{\boldsymbol{\beta}}$ .

- (a) By [Theorem 3.3.1](#),  $\mathbf{P}^\top \mathbf{M} \mathbf{Y}$  is normal with mean  $\mathbf{P}^\top \mathbf{M} \mathbf{X} \boldsymbol{\beta} = \mathbf{P}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{\Lambda}^\top \boldsymbol{\beta}$  and covariance  $\sigma^2 \mathbf{P}^\top \mathbf{M} \mathbf{M}^\top \mathbf{P} = \sigma^2 \mathbf{P}^\top \mathbf{M} \mathbf{P} = \sigma^2 \mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{\Lambda}$ , by [Theorem 6.3.7](#). The special cases are  $\mathbf{\Lambda} = \mathbf{X}^\top$ , with  $\mathbf{P} = \mathbf{I}$ , and  $\mathbf{\Lambda} = \mathbf{I}_p$  in full rank.
- (b)  $\text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$ , and  $\mathbf{I} - \mathbf{M}$  is symmetric and idempotent of rank  $n - r$  ([Theorem 6.3.3](#) and [Proposition 6.3.4](#)). By [Theorem 4.3.2\(a\)](#),  $\text{SSE}/\sigma^2 \sim \chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2)$ , and the noncentrality is zero because  $(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$ .
- (c) Apply [Theorem 4.4.1](#) with  $\mathbf{B} = \mathbf{M}$ ,  $\mathbf{A} = \mathbf{I} - \mathbf{M}$  and  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ . Then  $\mathbf{B} \boldsymbol{\Sigma} \mathbf{A} = \sigma^2 \mathbf{M}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ , so  $\mathbf{M} \mathbf{Y}$  and  $\mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$  are independent. Functions of independent random vectors are independent.
- (d) A  $\chi^2(k)$  variable has mean  $k$  and variance  $2k$  ([Theorem 4.1.2\(b\)](#) with  $\gamma = 0$ ). So  $\text{E SSE} = \sigma^2(n - r)$  and  $\text{Var}(\text{SSE}) = 2\sigma^4(n - r)$ . Divide by  $n - r$  and  $(n - r)^2$ .

■

The proof is short because the work was done earlier. It is worth seeing where each ingredient comes from geometrically. The fitted vector  $\mathbf{M} \mathbf{Y}$  and the residual vector  $(\mathbf{I} - \mathbf{M}) \mathbf{Y}$  are projections of  $\mathbf{Y}$  onto orthogonal subspaces. For a spherical normal vector, orthogonal projections are independent ([Theorem 4.4.4](#)). Everything about  $\boldsymbol{\beta}$  lives in the first projection. The second is pure noise, and its squared length, measured in units of  $\sigma^2$ , counts  $n - r$  independent squared standard normals.

The independence in (c) has no analogue under the second-moment assumptions. If the errors are independent with zero third moment,  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$  and  $s^2$  are still uncorrelated, but in general they are not independent. The simulation at the end of the section shows the difference.

## 7.5.2 Standardizing: the $t$ statistic

The distribution in (a) involves the unknown  $\sigma$ . Since  $\text{E}(s^2) = \sigma^2$ , the matrix  $s^2 \mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{\Lambda}$  is an unbiased estimate of  $\text{Cov}(\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}})$ , and in full rank  $s^2 (\mathbf{X}^\top \mathbf{X})^{-1}$  is unbiased for  $\text{Cov}(\hat{\boldsymbol{\beta}})$ ; this needs only the second-moment assumptions. The square roots of its diagonal entries are the standard errors reported by regression software. Replacing  $\sigma$  by  $s$  changes the normal distribution into a  $t$  distribution, and independence is what makes the change exact.

### Corollary 7.5.2 — The $t$ statistic for an estimable function

Assume [Eq. 7.3.1](#) with  $r < n$ , let  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$  with  $\boldsymbol{\lambda} \neq \mathbf{0}$ , and put  $\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = s \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}}$ . For every constant  $d$ ,

$$T_d = \frac{\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - d}{\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})} \sim t(n - r, \delta), \quad \delta = \frac{\boldsymbol{\lambda}^\top \boldsymbol{\beta} - d}{\sigma \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}}}.$$

In particular  $T = (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) \sim t(n - r)$ , and

$$\Pr\{|\lambda^\top \hat{\beta} - \lambda^\top \beta| \leq t_{n-r, \alpha/2} \text{se}(\lambda^\top \hat{\beta})\} = 1 - \alpha, \quad (7.5.1)$$

where  $t_{n-r, \alpha/2}$  is the upper  $\alpha/2$  point of  $t(n-r)$ .

*Proof.* Put  $c^2 = \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda$ . Then  $c > 0$ , since  $c^2 = \|\mathbf{M}\rho\|^2$  with  $\mathbf{X}^\top \mathbf{M}\rho = \lambda \neq \mathbf{0}$ . By Theorem 7.5.1(a),  $Z = (\lambda^\top \hat{\beta} - d)/(\sigma c) \sim N(\delta, 1)$ . By (b),  $V = (n-r)s^2/\sigma^2 \sim \chi^2(n-r)$ , and by (c)  $Z$  and  $V$  are independent. Now  $T_d = Z/\sqrt{V/(n-r)}$ , which is  $t(n-r, \delta)$  by Definition 4.1.3. With  $d = \lambda^\top \beta$ ,  $\delta = 0$ , and Eq. 7.5.1 restates  $\Pr\{|T| \leq t_{n-r, \alpha/2}\} = 1 - \alpha$ . ■

Eq. 7.5.1 is the familiar confidence interval  $\lambda^\top \hat{\beta} \pm t_{n-r, \alpha/2} \text{se}$ , and  $T_d$  is the statistic for testing  $\lambda^\top \beta = d$ . Its noncentrality  $\delta$  governs the power. Chapter 11 and Chapter 12 develop both. For the air-flow coefficient in the stack loss regression this interval was (0.431, 1.000) (Example (Profile likelihood for the air-flow coefficient)). Two features of the corollary are easy to overlook. The degrees of freedom are  $n-r$ , the dimension of the residual space. They are not  $n$ , and when  $\mathbf{X}$  is rank deficient they are not  $n-p$  either. And the interval is exact for every sample size, which no large-sample argument delivers.

### 7.5.3 Several functions at once

For a vector of estimable functions, the natural summary of Theorem 7.5.1(a) is a quadratic form. The covariance matrix  $\Lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \Lambda$  may be singular when the columns of  $\Lambda$  are linearly dependent, so a generalized inverse is needed.

#### Corollary 7.5.3 — A chi-squared pivot for several functions

In the setting of Theorem 7.5.1, let  $\mathbf{W} = \Lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \Lambda$ , let  $\mathbf{W}^-$  be any generalized inverse of  $\mathbf{W}$ , and put  $\mathbf{D} = \Lambda^\top \hat{\beta} - \Lambda^\top \beta$ . Then

$$Q = \frac{\mathbf{D}^\top \mathbf{W}^- \mathbf{D}}{\sigma^2} \sim \chi^2(\text{rank}(\Lambda)),$$

$Q$  does not depend on the choice of  $\mathbf{W}^-$  (with probability one), and  $Q$  is independent of SSE. Consequently

$$\frac{\mathbf{D}^\top \mathbf{W}^- \mathbf{D} / \text{rank}(\Lambda)}{s^2} \sim F(\text{rank}(\Lambda), n-r).$$

*Proof.* By Theorem 7.5.1(a),  $\mathbf{D} \sim N_q(\mathbf{0}, \sigma^2 \mathbf{W})$ . The quadratic form is unchanged if  $\mathbf{W}^-$  is replaced by its symmetric part  $\frac{1}{2}(\mathbf{W}^- + \mathbf{W}^{-\top})$ , which is again a generalized inverse of the symmetric matrix  $\mathbf{W}$  (Proposition 1.3.3(d)). So we may assume  $\mathbf{W}^-$  symmetric. Apply Theorem 4.3.2(c) with  $\Sigma = \sigma^2 \mathbf{W}$ ,  $\mu = \mathbf{0}$  and  $\mathbf{A} = \mathbf{W}^-/\sigma^2$ . Conditions (ii) and (iii) hold trivially because  $\mu = \mathbf{0}$ . Condition (i) is  $\Sigma \mathbf{A} \Sigma \mathbf{A} \Sigma = \sigma^2 \mathbf{W} \mathbf{W}^- \mathbf{W} \mathbf{W}^- \mathbf{W} = \sigma^2 \mathbf{W} = \Sigma \mathbf{A} \Sigma$ , by two uses of  $\mathbf{W} \mathbf{W}^- \mathbf{W} = \mathbf{W}$ . So  $Q \sim \chi^2(\text{tr}(\mathbf{A} \Sigma))$ , where  $\text{tr}(\mathbf{A} \Sigma) = \text{tr}(\mathbf{W}^- \mathbf{W}) = \text{rank}(\mathbf{W})$ , because  $\mathbf{W}^- \mathbf{W}$  is idempotent with the same rank as  $\mathbf{W}$  (Proposition 1.3.3(a)), and the trace of an idempotent matrix is its rank (Proposition 1.9.1).

It remains to show  $\text{rank}(\mathbf{W}) = \text{rank}(\mathbf{\Lambda})$ . With  $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{P}$ ,  $\mathbf{W} = \mathbf{P}^\top \mathbf{M} \mathbf{P} = (\mathbf{M} \mathbf{P})^\top (\mathbf{M} \mathbf{P})$ , so  $\text{rank} \mathbf{W} = \text{rank}(\mathbf{M} \mathbf{P})$ . Also  $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{P} = \mathbf{X}^\top \mathbf{M} \mathbf{P}$ . If  $\mathbf{X}^\top \mathbf{M} \mathbf{P} \mathbf{u} = \mathbf{0}$ , then  $\mathbf{M} \mathbf{P} \mathbf{u} \in \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X})^\perp = \{\mathbf{0}\}$ , so  $\mathbf{X}^\top$  is one-to-one on  $\mathcal{C}(\mathbf{M} \mathbf{P})$  and  $\text{rank} \mathbf{\Lambda} = \text{rank}(\mathbf{M} \mathbf{P})$ .

For invariance:  $\mathbf{D} \in \mathcal{C}(\mathbf{W})$  with probability one by [Theorem 2.2.3\(d\)](#). If  $\mathbf{D} = \mathbf{W} \mathbf{u}$ , then  $\mathbf{D}^\top \mathbf{W}^- \mathbf{D} = \mathbf{u}^\top \mathbf{W} \mathbf{W}^- \mathbf{W} \mathbf{u} = \mathbf{u}^\top \mathbf{W} \mathbf{u}$ , whatever  $\mathbf{W}^-$  is. Independence follows from [Theorem 7.5.1\(c\)](#), since  $\mathbf{D}$  is a function of  $\mathbf{M} \mathbf{Y}$ . The  $F$  ratio is then a ratio of independent chi-squared variables divided by their degrees of freedom ([Definition 4.1.2](#)). ■

The set of  $\mathbf{\Lambda}^\top \boldsymbol{\beta}$  values for which this  $F$  ratio stays below an  $F$  quantile is a confidence ellipsoid. [Chapter 12](#) studies these regions, and [Chapter 11](#) turns the same statistic into the test of the general linear hypothesis. Here we note only that the degrees of freedom in the numerator are the number of *linearly independent* functions being estimated, not the number of rows of  $\mathbf{\Lambda}^\top$ .

### Example. (A rank-deficient model, simulated)

Consider a one-way layout with three groups of sizes 4, 5, 6, an intercept, one indicator column per group, and a covariate. So  $n = 15$  and  $p = 5$ , but the rank is  $r = 4$ , because the indicators add up to the intercept. The contrast  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \alpha_1 - \alpha_2$ , with  $\boldsymbol{\lambda} = (0, 1, -1, 0, 0)^\top$ , is estimable. With  $\sigma = 2$ , [Theorem 7.5.1](#) gives  $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^- \boldsymbol{\lambda} = 1.9007$ , computed with the Moore–Penrose inverse. In 100,000 simulated data sets the variance of the estimated contrast is 1.9017, and the average of  $\text{SSE}/\sigma^2$  is 11.001, against  $n - r = 11$ . The sample correlation between the contrast estimate and  $s^2$  is 0.0036. The standardized contrast  $T$  falls within  $\pm 2.201$ , the 97.5% point of  $t(11)$ , in a proportion 0.9498 of the samples. [Figure 7.5.1](#) compares the simulated distributions with the theory. Near the centre the  $t(11)$  and  $N(0, 1)$  densities are hard to tell apart, so panel (b) uses a logarithmic density scale, on which the tails separate: the simulated points follow  $t(11)$ , and the normal density falls away too fast. The tail mass says the same in numbers. A proportion 0.0758 of the samples have  $|T| > 1.96$ , against 0.05 under  $N(0, 1)$  and 0.0758 under  $t(11)$ . A nominal 95% interval built with the normal point 1.96 would therefore undercover.

```
import numpy as np
from scipy import stats

groups = np.repeat([0, 1, 2], [4, 5, 6])
z = np.array([2.1, 3.4, 1.8, 4.0, 2.9, 3.3, 5.1, 2.2, 3.8, 1.5, 4.4, 2.7, 3.9, 5.0,
              3.1])
n = len(z)
X = np.column_stack([np.ones(n), np.eye(3)[groups], z]) # p = 5 columns, rank 4
r = np.linalg.matrix_rank(X)
G = np.linalg.pinv(X.T @ X) # one generalized inverse
M = X @ G @ X.T
lam = np.array([0.0, 1.0, -1.0, 0.0, 0.0]) # group 1 minus group 2:
    estimable
beta, sigma = np.array([1.0, 0.5, -0.5, 0.0, 0.8]), 2.0
var_lam = sigma ** 2 * lam @ G @ lam # Var(lambda^T beta_hat)
print(f"n = {n}, p = {X.shape[1]}, rank = {r}, Var(lambda^T beta_hat) = {var_lam:.4f}
      ")
```

```
rng = np.random.default_rng(75)
```

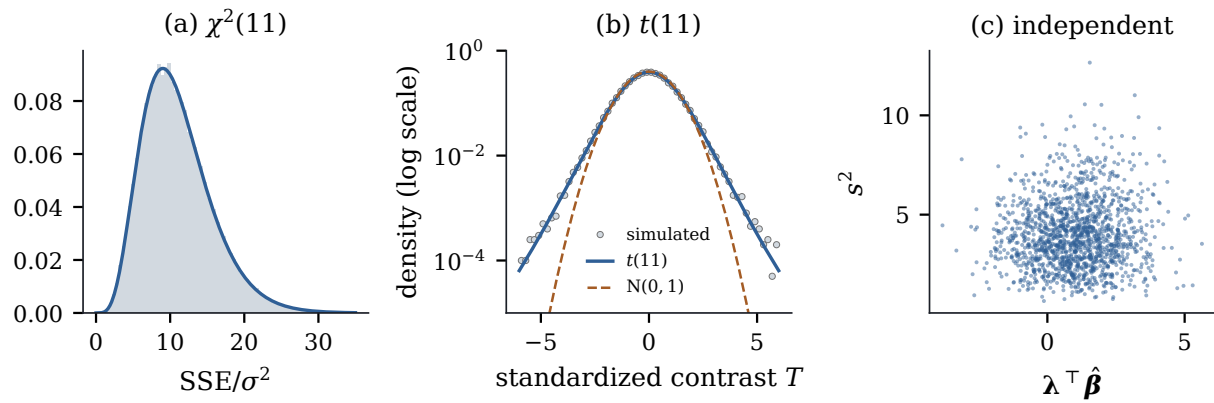


Figure 7.5.1: Sampling distributions in a rank-deficient normal linear model ( $n = 15$ ,  $r = 4$ ; 100,000 simulated data sets). (a)  $\text{SSE}/\sigma^2$  with the  $\chi^2(11)$  density. (b) The standardized contrast (binned density, points) with the  $t(11)$  and standard normal densities, on a logarithmic scale. (c) The contrast estimate against  $s^2$ : no relationship, as independence requires.

```

reps = 100_000
Y = X @ beta + sigma * rng.normal(size=(reps, n))           # one data set per row
est = Y @ (X @ G @ lam)                                     # lambda^T beta_hat =
    lambda^T G X^T y
sse = np.sum((Y - Y @ M) ** 2, axis=1)                       # y^T (I - M) y
s2 = sse / (n - r)
t = (est - lam @ beta) / np.sqrt(s2 * (lam @ G @ lam))
print(f"mean {est.mean():.4f}, variance {est.var():.4f}")
print(f"E(SSE)/sigma^2 = {np.mean(sse) / sigma**2:.3f} (n - r = {n - r})")
print(f"corr(estimate, s^2) = {np.corrcoef(est, s2)[0, 1]:.4f}")
print("KS p-value, SSE/sigma^2 vs chi^2(n-r):", round(stats.kstest(sse / sigma**2, "chi2", args=(n - r,)).pvalue, 3))
print("KS p-value, t vs t(n-r):", round(stats.kstest(t, "t", args=(n - r,)).pvalue, 3))

```

## 7.5.4 Without normality

Which parts of [Theorem 7.5.1](#) survive when the errors are merely uncorrelated with common variance? The first two moments of  $\lambda^\top \hat{\beta}$  do, by [Theorem 7.2.1](#), and so does  $E(s^2) = \sigma^2$ . The normality of  $\lambda^\top \hat{\beta}$  is lost, but it often returns approximately.  $\lambda^\top \hat{\beta}$  is a weighted sum of independent errors, and a central limit theorem applies when no single observation dominates the weights. [Chapter 19](#) makes this precise ([Theorem 19.3.2](#)). The distribution of SSE is less forgiving, because its variance depends on the fourth moment of the errors.

### Proposition 7.5.4 — Variance of SSE without normality

Let the errors be independent with mean zero, variance  $\sigma^2$ , third moment  $\mu_3$  and fourth moment  $\mu_4$ , and let  $h_{ii}$  be the leverages. Then

$$\text{Var}(\text{SSE}) = 2\sigma^4(n - r) + (\mu_4 - 3\sigma^4) \sum_{i=1}^n (1 - h_{ii})^2.$$

*Proof.* Apply [Theorem 2.3.3](#) to  $\text{SSE} = \mathbf{Y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$ , with  $\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta}$  and  $\mathbf{A} = \mathbf{I} - \mathbf{M}$ , whose diagonal entries are  $1 - h_{ii}$ . The terms  $4\sigma^2 \boldsymbol{\theta}^\top \mathbf{A}^2 \boldsymbol{\theta}$  and  $4\mu_3 \boldsymbol{\theta}^\top \mathbf{A} \mathbf{a}$  vanish because  $\mathbf{A}\boldsymbol{\theta} = (\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ , and  $\text{tr}(\mathbf{A}^2) = \text{tr}(\mathbf{A}) = n - r$ . ■

The factor  $\mu_4 - 3\sigma^4$  is  $\sigma^4$  times the excess kurtosis. For normal errors it vanishes and we recover  $2\sigma^4(n - r)$ . For heavy-tailed errors it is positive, and SSE is more variable than the chi-squared theory says. Intervals for  $\sigma^2$  built from [Theorem 7.5.1\(b\)](#) are then too short, and this does not improve with sample size: the ratio of the true variance to the nominal one tends to  $1 + \frac{1}{2}(\mu_4/\sigma^4 - 3)$  when the leverages are small. The  $t$  interval for  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is much less affected, because  $s^2$  enters only through  $s$ , and  $s$  converges to  $\sigma$  whatever the kurtosis.

### Example. (Laplace errors)

Rerun the simulation of [Example \(A rank-deficient model, simulated\)](#) with Laplace (double-exponential) errors of the same variance  $\sigma^2 = 4$ . For these  $\mu_4 = 6\sigma^4$ . The contrast still has variance 1.9007. For SSE,  $\sum_i (1 - h_{ii})^2 = 8.138$ , and [Proposition 7.5.4](#) gives  $\text{Var}(\text{SSE}) = 742.6$ , more than twice the normal-theory value 352.0. The simulated variance is 741.5. Independence fails too. The contrast and SSE are still uncorrelated, because the errors are symmetric, but the *squared* error of the contrast has correlation 0.211 with SSE. Samples with a few large errors inflate both.

```
E_lap = rng.laplace(scale=sigma / np.sqrt(2), size=(reps, n)) # variance sigma^2,
mu_4 = 6 * sigma^4
Y_lap = X @ beta + E_lap
est_lap = Y_lap @ (X @ G @ lam)
sse_lap = np.sum((Y_lap - Y_lap @ M) ** 2, axis=1)
h = np.diag(np.eye(n) - M)
var_sse_lap = (6 - 3) * sigma**4 * h @ h + 2 * sigma**4 * (n - r)
print(f"Var(SSE): normal theory {2 * sigma**4 * (n - r):.1f}, Laplace formula {
    var_sse_lap:.1f}, "
      f"simulated {sse_lap.var():.1f}")
print(f"corr(squared error of estimate, SSE) under Laplace errors: "
      f"{np.corrcoef((est_lap - lam @ beta) ** 2, sse_lap)[0, 1]:.3f}")
```

In the listing,  $\mathbf{h}$  holds the diagonal of  $\mathbf{I} - \mathbf{M}$ , that is,  $1 - h_{ii}$ .

## 7.5.5 Exercises

### 7.5.5.1 A. Check your understanding

#### Exercise A1.

Under [Eq. 7.3.1](#), find the joint distribution of  $(\hat{\mathbf{Y}}, \hat{\boldsymbol{\varepsilon}})$ . Show that  $\hat{\mathbf{Y}}$  and  $\hat{\boldsymbol{\varepsilon}}$  are independent, and that  $\hat{\boldsymbol{\varepsilon}} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$  whatever  $\boldsymbol{\beta}$  is. Is  $\hat{\boldsymbol{\varepsilon}}$  independent of SSE?

**Exercise A2.**

In simple linear regression, write out  $T$  of [Corollary 7.5.2](#) for the slope, in terms of  $S_{xx}$  and  $s$ , and state its distribution. What is  $r$ ?

**7.5.5.2 B. Practice****Exercise B1.**

Let  $\lambda_1, \lambda_2 \in \mathcal{C}(\mathbf{X}^\top)$ . Show that  $\text{Cov}(\lambda_1^\top \hat{\boldsymbol{\beta}}, \lambda_2^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \lambda_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda_2$ , and that under normality the two estimates are independent iff  $\lambda_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda_2 = 0$ . In a one-way layout with  $a$  groups of  $m$  observations each and cell-means parameterization  $E(Y_{ij}) = \mu_i$ , show that the estimates of  $\sum_i c_i \mu_i$  and  $\sum_i d_i \mu_i$  are independent iff  $\sum_i c_i d_i = 0$ .

*Solution.* The covariance is the off-diagonal entry of [Theorem 7.5.1\(a\)](#) with  $\boldsymbol{\Lambda} = [\lambda_1, \lambda_2]$ . Under normality the pair is bivariate normal, so zero covariance is equivalent to independence ([Theorem 3.4.1](#)). In the cell-means model  $\mathbf{X}^\top \mathbf{X} = m \mathbf{I}_a$ , so  $\lambda_1^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda_2 = \sum_i c_i d_i / m$ , which vanishes iff  $\sum_i c_i d_i = 0$ .

**Exercise B2.**

A new observation  $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$  will be taken, with  $\varepsilon_0 \sim N(0, \sigma^2)$  independent of  $\mathbf{Y}$  and  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$ . Show that

$$\frac{Y_0 - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}}{s \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}} \sim t(n - r),$$

and deduce a prediction interval for  $Y_0$ . Why is the extra 1 in the denominator never negligible?

*Solution.*  $Y_0$  is independent of  $\mathbf{Y}$ , so  $Y_0 - \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$  is normal with mean  $\mathbf{x}_0^\top \boldsymbol{\beta} - \mathbf{x}_0^\top \boldsymbol{\beta} = 0$  and variance  $\sigma^2 + \sigma^2 \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$ . It is a function of  $(Y_0, \mathbf{M}\mathbf{Y})$ , which is independent of SSE. Dividing by  $\sigma \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}$  gives a standard normal, independent of  $(n - r)s^2/\sigma^2 \sim \chi^2(n - r)$ , and the ratio is  $t(n - r)$  by [Definition 4.1.3](#). The interval is  $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \pm t_{n-r, \alpha/2} s \sqrt{1 + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0}$ . The 1 is the variance of the new error itself, which no amount of data about  $\boldsymbol{\beta}$  can reduce. The second term tends to zero as the design grows. The first does not.

**Exercise B3.**

Use [Theorem 7.5.1\(b\)](#) to construct an exact  $1 - \alpha$  confidence interval for  $\sigma^2$ . Then, using [Proposition 7.5.4](#) with small leverages, explain why its coverage does not tend to  $1 - \alpha$  as  $n \rightarrow \infty$  when the errors have nonzero excess kurtosis.

**7.5.5.3 C. Going deeper**



**Exercise C1.**

Give a second proof of [Theorem 7.5.1](#)(b) and (c) using [Theorem 4.4.4](#) with the two projections  $\mathbf{M}$  and  $\mathbf{I} - \mathbf{M}$ . Extend it: if  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ , show that  $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}\|^2$  and SSE are independent, and find the distribution of  $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}\|^2/\sigma^2$  when  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  is arbitrary.

**Exercise C2.**

For the test that rejects  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = d$  when  $|T_d| > t_{n-r, \alpha/2}$ , show that the power depends on the parameters only through  $|\delta|$  of [Corollary 7.5.2](#). For a fixed design, how does the power change when the sample is replicated  $m$  times ( $\mathbf{X}$  replaced by  $m$  stacked copies)?

## 7.6 Bayesian thread I: the conjugate linear model

The previous sections judged estimators by their behaviour over repeated samples, for each fixed value of the parameters. The Bayesian approach treats  $(\boldsymbol{\beta}, \sigma^2)$  as uncertain quantities with a probability distribution. It combines a **prior** distribution, expressing what is known before the data, with the likelihood of [Section 7.3](#), and the result is a **posterior** distribution. For the normal linear model there is a family of priors for which the posterior is available in closed form, and it connects directly to the rest of the chapter. The posterior mean is a matrix-weighted combination of the prior mean and the least squares estimate. It is biased in the frequentist sense, which is how it escapes the Gauss–Markov bound. And as the prior becomes flat, the posterior reproduces least squares and the  $t$  intervals of [Section 7.5](#) exactly. This is the first of several Bayesian threads in the book. [Section 12.6](#) continues it with credible regions and predictive distributions, [Chapter 27](#) with shrinkage priors, and [Chapter 39](#) with generalized linear models.

### 7.6.1 The normal-inverse-gamma family

#### **Definition 7.6.1** — Inverse gamma and normal-inverse-gamma distributions

- (a) A positive random variable  $V$  has the **inverse gamma** distribution  $\text{IG}(a, b)$ , with  $a, b > 0$ , if  $1/V$  has the gamma distribution with shape  $a$  and rate  $b$ . Its density is

$$\pi(v) = \frac{b^a}{\Gamma(a)} v^{-a-1} e^{-b/v}, \quad v > 0.$$

- (b) The pair  $(\boldsymbol{\beta}, \sigma^2)$  has the **normal-inverse-gamma** distribution  $\text{NIG}(\mathbf{m}, \mathbf{V}, a, b)$ , with  $\mathbf{m} \in \mathbb{R}^p$ ,  $\mathbf{V}$  positive definite and  $a, b > 0$ , if  $\sigma^2 \sim \text{IG}(a, b)$  and, given  $\sigma^2$ ,  $\boldsymbol{\beta} \sim N_p(\mathbf{m}, \sigma^2 \mathbf{V})$ . Its joint density is



$$\pi(\boldsymbol{\beta}, \sigma^2) = \frac{b^a}{\Gamma(a)(2\pi)^{p/2} \det(\mathbf{V})^{1/2}} (\sigma^2)^{-(a+p/2)-1} \times \exp\left\{-\frac{2b + (\boldsymbol{\beta} - \mathbf{m})^\top \mathbf{V}^{-1}(\boldsymbol{\beta} - \mathbf{m})}{2\sigma^2}\right\}. \quad (7.6.1)$$

The prior on  $\boldsymbol{\beta}$  is scaled by  $\sigma^2$ : prior uncertainty about the coefficients is measured in units of the noise level. This is a modelling choice, and it is what makes the family conjugate. It also makes  $\mathbf{V}$  dimensionless relative to the likelihood's  $(\mathbf{X}^\top \mathbf{X})^{-1}$ , which is the comparison that matters below. The marginal distribution of  $\boldsymbol{\beta}$  is a multivariate  $t$ .

### Definition 7.6.2 — Multivariate $t$

Let  $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{S})$  and  $W \sim \chi^2(\nu)$  be independent, with  $\mathbf{S}$  positive definite. The law of  $\mathbf{m} + \mathbf{Z}/\sqrt{W/\nu}$  is the **multivariate  $t$  distribution**  $t_\nu(\mathbf{m}, \mathbf{S})$ .

For  $p = 1$  and  $\mathbf{S} = 1$ , this is the  $t(\nu)$  distribution of Definition 4.1.3 shifted by  $\mathbf{m}$ . For every  $\mathbf{c} \neq \mathbf{0}$ ,  $\mathbf{c}^\top \mathbf{u}$  with  $\mathbf{u} \sim t_\nu(\mathbf{m}, \mathbf{S})$  is  $\mathbf{c}^\top \mathbf{m} + \sqrt{\mathbf{c}^\top \mathbf{S} \mathbf{c}} T$  with  $T \sim t(\nu)$ , because  $\mathbf{c}^\top \mathbf{Z}/\sqrt{\mathbf{c}^\top \mathbf{S} \mathbf{c}} \sim N(0, 1)$  is independent of  $W$ .

### Lemma 7.6.1 — Marginals of the normal-inverse-gamma

If  $(\boldsymbol{\beta}, \sigma^2) \sim \text{NIG}(\mathbf{m}, \mathbf{V}, a, b)$ , then  $2b/\sigma^2 \sim \chi^2(2a)$ ,  $E(\sigma^2) = b/(a - 1)$  for  $a > 1$ , and

$$\boldsymbol{\beta} \sim t_{2a}(\mathbf{m}, (b/a)\mathbf{V}).$$

In particular, for every  $\boldsymbol{\lambda} \neq \mathbf{0}$ ,  $(\boldsymbol{\lambda}^\top \boldsymbol{\beta} - \boldsymbol{\lambda}^\top \mathbf{m})/\sqrt{(b/a)\boldsymbol{\lambda}^\top \mathbf{V} \boldsymbol{\lambda}} \sim t(2a)$ .

*Proof.* Put  $W = 2b/\sigma^2$ . Changing variables  $v = 2b/w$  in the inverse gamma density, with  $|dv/dw| = 2b/w^2$ , gives the density of  $W$ :

$$\frac{b^a}{\Gamma(a)} \left(\frac{2b}{w}\right)^{-a-1} e^{-w/2} \frac{2b}{w^2} = \frac{w^{a-1} e^{-w/2}}{2^a \Gamma(a)}, \quad w > 0,$$

which is the  $\chi^2(2a)$  density (Eq. 4.1.1). Then  $E\sigma^2 = 2b E(1/W) = 2b/(2a - 2)$ , since  $E(1/W) = 1/(\nu - 2)$  for  $W \sim \chi^2(\nu)$ ,  $\nu > 2$ . Next let  $\mathbf{Z}_0 = (\boldsymbol{\beta} - \mathbf{m})/\sigma$ . Given  $\sigma^2$ ,  $\mathbf{Z}_0 \sim N_p(\mathbf{0}, \mathbf{V})$ , a law that does not depend on  $\sigma^2$ . So  $\mathbf{Z}_0$  is independent of  $\sigma^2$ , and hence of  $W$ , and

$$\boldsymbol{\beta} = \mathbf{m} + \sigma \mathbf{Z}_0 = \mathbf{m} + \sqrt{\frac{2b}{W}} \mathbf{Z}_0 = \mathbf{m} + \frac{\sqrt{b/a} \mathbf{Z}_0}{\sqrt{W/(2a)}}.$$

Since  $\sqrt{b/a} \mathbf{Z}_0 \sim N_p(\mathbf{0}, (b/a)\mathbf{V})$ , this is  $t_{2a}(\mathbf{m}, (b/a)\mathbf{V})$  by Definition 7.6.2. The last statement is the remark after Definition 7.6.2. ■

## 7.6.2 The conjugate update

### Theorem 7.6.2 — Conjugate posterior

Let  $\mathbf{y}$  follow the normal linear model [Eq. 7.3.1](#), with  $\mathbf{X}$  of any rank, and let the prior be  $(\boldsymbol{\beta}, \sigma^2) \sim \text{NIG}(\mathbf{m}_0, \mathbf{V}_0, a_0, b_0)$ . Then the posterior is  $\text{NIG}(\mathbf{m}_n, \mathbf{V}_n, a_n, b_n)$ , where

$$\begin{aligned} \mathbf{V}_n &= (\mathbf{V}_0^{-1} + \mathbf{X}^\top \mathbf{X})^{-1}, & \mathbf{m}_n &= \mathbf{V}_n(\mathbf{V}_0^{-1} \mathbf{m}_0 + \mathbf{X}^\top \mathbf{y}), \\ a_n &= a_0 + \frac{n}{2}, \end{aligned} \quad (7.6.2)$$

$$\begin{aligned} b_n &= b_0 + \frac{1}{2}(\mathbf{y}^\top \mathbf{y} + \mathbf{m}_0^\top \mathbf{V}_0^{-1} \mathbf{m}_0 - \mathbf{m}_n^\top \mathbf{V}_n^{-1} \mathbf{m}_n) \\ &= b_0 + \frac{1}{2}\{\|\mathbf{y} - \mathbf{X}\mathbf{m}_n\|^2 + (\mathbf{m}_n - \mathbf{m}_0)^\top \mathbf{V}_0^{-1}(\mathbf{m}_n - \mathbf{m}_0)\}. \end{aligned} \quad (7.6.3)$$

Consequently  $\boldsymbol{\beta} \mid \mathbf{y} \sim t_{2a_n}(\mathbf{m}_n, (b_n/a_n)\mathbf{V}_n)$ ,  $\sigma^2 \mid \mathbf{y} \sim \text{IG}(a_n, b_n)$ , and  $E(\boldsymbol{\beta} \mid \mathbf{y}) = \mathbf{m}_n$ .

*Proof.* Write  $\mathbf{P}_0 = \mathbf{V}_0^{-1}$  and  $\mathbf{P}_n = \mathbf{P}_0 + \mathbf{X}^\top \mathbf{X}$ . The matrix  $\mathbf{P}_n$  is positive definite, because  $\mathbf{P}_0$  is and  $\mathbf{X}^\top \mathbf{X}$  is nonnegative definite. This holds whatever the rank of  $\mathbf{X}$ . Expanding both sides and using  $\mathbf{P}_n \mathbf{m}_n = \mathbf{P}_0 \mathbf{m}_0 + \mathbf{X}^\top \mathbf{y}$  gives the completion of the square

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + (\boldsymbol{\beta} - \mathbf{m}_0)^\top \mathbf{P}_0(\boldsymbol{\beta} - \mathbf{m}_0) = (\boldsymbol{\beta} - \mathbf{m}_n)^\top \mathbf{P}_n(\boldsymbol{\beta} - \mathbf{m}_n) + C,$$

with  $C = \mathbf{y}^\top \mathbf{y} + \mathbf{m}_0^\top \mathbf{P}_0 \mathbf{m}_0 - \mathbf{m}_n^\top \mathbf{P}_n \mathbf{m}_n$ . Indeed, the left side is  $\boldsymbol{\beta}^\top \mathbf{P}_n \boldsymbol{\beta} - 2\boldsymbol{\beta}^\top (\mathbf{X}^\top \mathbf{y} + \mathbf{P}_0 \mathbf{m}_0) + \mathbf{y}^\top \mathbf{y} + \mathbf{m}_0^\top \mathbf{P}_0 \mathbf{m}_0$ , and  $(\boldsymbol{\beta} - \mathbf{m}_n)^\top \mathbf{P}_n(\boldsymbol{\beta} - \mathbf{m}_n) = \boldsymbol{\beta}^\top \mathbf{P}_n \boldsymbol{\beta} - 2\boldsymbol{\beta}^\top \mathbf{P}_n \mathbf{m}_n + \mathbf{m}_n^\top \mathbf{P}_n \mathbf{m}_n$ . Putting  $\boldsymbol{\beta} = \mathbf{m}_n$  in the identity shows that  $C = \|\mathbf{y} - \mathbf{X}\mathbf{m}_n\|^2 + (\mathbf{m}_n - \mathbf{m}_0)^\top \mathbf{P}_0(\mathbf{m}_n - \mathbf{m}_0) \geq 0$ , which is the second form of [Eq. 7.6.3](#).

By Bayes' theorem the posterior density is proportional, as a function of  $(\boldsymbol{\beta}, \sigma^2)$ , to prior times likelihood. Using [Eq. 7.6.1](#) and dropping factors free of the parameters,

$$\begin{aligned} \pi(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{y}) &\propto (\sigma^2)^{-(a_0+p/2)-1} e^{-\{2b_0 + (\boldsymbol{\beta} - \mathbf{m}_0)^\top \mathbf{P}_0(\boldsymbol{\beta} - \mathbf{m}_0)\}/(2\sigma^2)} \\ &\quad \times (\sigma^2)^{-n/2} e^{-\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2/(2\sigma^2)}. \end{aligned}$$

By the completion of the square, the exponent is  $-\{2b_0 + C + (\boldsymbol{\beta} - \mathbf{m}_n)^\top \mathbf{P}_n(\boldsymbol{\beta} - \mathbf{m}_n)\}/(2\sigma^2)$ , and  $2b_0 + C = 2b_n$ . The power of  $\sigma^2$  is  $-(a_0 + n/2 + p/2) - 1 = -(a_n + p/2) - 1$ . So the posterior is proportional to the  $\text{NIG}(\mathbf{m}_n, \mathbf{V}_n, a_n, b_n)$  density [Eq. 7.6.1](#). Two densities that are proportional are equal, since both integrate to one. The marginals follow from [Lemma 7.6.1](#). The posterior mean of  $\boldsymbol{\beta}$  is  $\mathbf{m}_n$ : by the tower property,  $E(\boldsymbol{\beta} \mid \mathbf{y}) = E\{E(\boldsymbol{\beta} \mid \sigma^2, \mathbf{y}) \mid \mathbf{y}\} = \mathbf{m}_n$ . ■

The update has a simple structure. **Precisions add:** the posterior precision (per unit  $\sigma^2$ ) is the prior precision  $\mathbf{V}_0^{-1}$  plus the data precision  $\mathbf{X}^\top \mathbf{X}$ . **Shape parameters count observations:** each observation adds  $1/2$  to  $a$ . **Scale parameters accumulate squared discrepancies:**  $b_n$  adds half of the residual sum of squares at the posterior mean and half of the prior-weighted distance between the posterior and prior means. Because  $\mathbf{P}_n$  is positive definite even when  $\mathbf{X}$  is rank deficient, the posterior is proper for every design. The prior supplies information about

the directions of  $\mathcal{N}(\mathbf{X})$  that the data cannot, and there the posterior equals the prior ([Exercise \(B2\)](#)).

### 7.6.3 The posterior mean as shrinkage

#### Proposition 7.6.3 — Posterior mean as a combination

In [Theorem 7.6.2](#), let  $\mathbf{X}$  have full column rank and put  $\mathbf{W} = \mathbf{V}_n \mathbf{V}_0^{-1}$ . Then:

- (a)  $\mathbf{m}_n = \mathbf{W}\mathbf{m}_0 + (\mathbf{I} - \mathbf{W})\hat{\boldsymbol{\beta}}$ , with  $\mathbf{I} - \mathbf{W} = \mathbf{V}_n \mathbf{X}^\top \mathbf{X}$ . Equivalently  $\mathbf{V}_n^{-1} \mathbf{m}_n = \mathbf{V}_0^{-1} \mathbf{m}_0 + (\mathbf{X}^\top \mathbf{X})\hat{\boldsymbol{\beta}}$ : the posterior mean is the precision-weighted average of the prior mean and the least squares estimate.
- (b)  $b_n = b_0 + \frac{1}{2} \{ \text{SSE} + (\hat{\boldsymbol{\beta}} - \mathbf{m}_0)^\top (\mathbf{V}_0 + (\mathbf{X}^\top \mathbf{X})^{-1})^{-1} (\hat{\boldsymbol{\beta}} - \mathbf{m}_0) \}$ .
- (c) (*g-prior.*) If  $\mathbf{V}_0 = g(\mathbf{X}^\top \mathbf{X})^{-1}$  with  $g > 0$ , then  $\mathbf{W} = \mathbf{I}/(1 + g)$  and  $\mathbf{m}_n = (\mathbf{m}_0 + g\hat{\boldsymbol{\beta}})/(1 + g)$ .
- (d) (*Ridge.*) If  $\mathbf{m}_0 = \mathbf{0}$  and  $\mathbf{V}_0 = k^{-1}\mathbf{I}$  with  $k > 0$ , then  $\mathbf{m}_n = (\mathbf{X}^\top \mathbf{X} + k\mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$ .

*Proof.*

- (a)  $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$  by the normal equations ([Theorem 6.4.2](#)), so  $\mathbf{m}_n = \mathbf{V}_n \mathbf{V}_0^{-1} \mathbf{m}_0 + \mathbf{V}_n \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$ , and  $\mathbf{V}_n \mathbf{V}_0^{-1} + \mathbf{V}_n \mathbf{X}^\top \mathbf{X} = \mathbf{V}_n \mathbf{V}_n^{-1} = \mathbf{I}$ .
- (b) Write  $\mathbf{P}_0 = \mathbf{V}_0^{-1}$  and  $\mathbf{K} = \mathbf{X}^\top \mathbf{X}$ . The completion of the square in the proof of [Theorem 7.6.2](#) shows that  $2(b_n - b_0)$  is the minimum over  $\boldsymbol{\beta}$  of  $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + (\boldsymbol{\beta} - \mathbf{m}_0)^\top \mathbf{P}_0 (\boldsymbol{\beta} - \mathbf{m}_0)$ , because the remaining term  $(\boldsymbol{\beta} - \mathbf{m}_n)^\top \mathbf{P}_n (\boldsymbol{\beta} - \mathbf{m}_n)$  is nonnegative and vanishes at  $\mathbf{m}_n$ . By [Eq. 6.3.1](#),  $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \text{SSE} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{K} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})$ . Put  $\mathbf{d} = \hat{\boldsymbol{\beta}} - \mathbf{m}_0$  and  $\boldsymbol{\beta} = \mathbf{m}_0 + \mathbf{u}$ . The quantity to minimize becomes  $\text{SSE} + (\mathbf{u} - \mathbf{d})^\top \mathbf{K} (\mathbf{u} - \mathbf{d}) + \mathbf{u}^\top \mathbf{P}_0 \mathbf{u}$ , whose minimum over  $\mathbf{u}$ , attained at  $\mathbf{u} = (\mathbf{K} + \mathbf{P}_0)^{-1} \mathbf{K} \mathbf{d}$ , is  $\text{SSE} + \mathbf{d}^\top \{ \mathbf{K} - \mathbf{K}(\mathbf{K} + \mathbf{P}_0)^{-1} \mathbf{K} \} \mathbf{d}$ . By [Theorem 1.6.5](#),  $\mathbf{K} - \mathbf{K}(\mathbf{K} + \mathbf{P}_0)^{-1} \mathbf{K} = (\mathbf{K}^{-1} + \mathbf{P}_0^{-1})^{-1} = (\mathbf{V}_0 + (\mathbf{X}^\top \mathbf{X})^{-1})^{-1}$ .
- (c) With  $\mathbf{V}_0^{-1} = \mathbf{X}^\top \mathbf{X}/g$ ,  $\mathbf{V}_n^{-1} = (1 + 1/g)\mathbf{X}^\top \mathbf{X}$ , so  $\mathbf{W} = \mathbf{V}_n \mathbf{V}_0^{-1} = \{g/(1 + g)\}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}/g = \mathbf{I}/(1 + g)$ . (d) Substitute into [Eq. 7.6.2](#).

■

Part (a) is the Bayesian counterpart of Gauss–Markov. The posterior mean is linear in  $\mathbf{y}$ , and it is biased unless  $\mathbf{m}_0 = \boldsymbol{\beta}$ :  $E(\mathbf{m}_n) = \mathbf{W}\mathbf{m}_0 + (\mathbf{I} - \mathbf{W})\boldsymbol{\beta}$ . So the Gauss–Markov theorem does not apply to it, and it can have smaller mean squared error than least squares when the prior mean is close to the truth. Part (d) identifies ridge regression, whose mean squared error was studied in [Exercise \(B1\)](#), as a posterior mean. [Chapter 27](#) develops the connection in [Theorem 27.2.2](#). When the prior is weak relative to the data,  $\mathbf{W}$  is close to  $\mathbf{0}$  and the posterior mean is close to  $\hat{\boldsymbol{\beta}}$ . When it is strong,  $\mathbf{W}$  is close to  $\mathbf{I}$  and the data barely move it.

In general  $\mathbf{W}$  is a full matrix, not a multiple of the identity. Shrinkage then happens along the eigenvectors of the problem, not coordinate by coordinate. A single coefficient’s posterior mean need not lie between its prior mean and its least squares estimate, as the next example shows. Only the *g*-prior of part (c) shrinks every coordinate by the same factor. Its prior covariance copies the shape of the sampling covariance of  $\hat{\boldsymbol{\beta}}$ , and that makes it convenient for model comparison. Zellner (1986) introduced it.

## 7.6.4 The flat-prior limit

What happens when the prior carries no information? Letting  $\mathbf{V}_0^{-1} \rightarrow \mathbf{0}$  in Eq. 7.6.2 gives  $\mathbf{m}_n \rightarrow \hat{\boldsymbol{\beta}}$  when  $\mathbf{X}$  has full rank. The shape parameter needs more care. The NIG density Eq. 7.6.1 carries the factor  $(\sigma^2)^{-(a_0+p/2)-1}$ , so letting only  $a_0, b_0 \rightarrow 0$  leads to the prior  $(\sigma^2)^{-p/2-1}$ , with  $a_n = n/2$  and  $t$  marginals on  $n$  degrees of freedom rather than  $n - p$ . The improper prior  $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$ , uniform in  $\boldsymbol{\beta}$  and in  $\log \sigma$ , is the limit  $\mathbf{V}_0^{-1} \rightarrow \mathbf{0}$ ,  $a_0 \rightarrow -p/2$ ,  $b_0 \rightarrow 0$ . A negative shape  $a_0 = -p/2$  lies outside the proper NIG family, so this prior cannot be reached inside Theorem 7.6.2 and has to be handled directly. The result is clean.

### Corollary 7.6.4 — The flat prior reproduces least squares

Let  $\text{rank}(\mathbf{X}) = p < n$ , let  $\text{SSE} > 0$ , and take the improper prior  $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$ . Then the posterior is proper and equals  $\text{NIG}(\hat{\boldsymbol{\beta}}, (\mathbf{X}^\top \mathbf{X})^{-1}, (n-p)/2, \text{SSE}/2)$ . In particular, for every  $\boldsymbol{\lambda} \neq \mathbf{0}$ ,

$$\frac{\boldsymbol{\lambda}^\top \boldsymbol{\beta} - \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}}{s \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}}} \mid \mathbf{y} \sim t(n-p),$$

so the  $1 - \alpha$  equal-tailed credible interval for  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is the  $t$  interval Eq. 7.5.1.

*Proof.* By Eq. 6.3.1,  $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \text{SSE} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})$ . So prior times likelihood is proportional to

$$\begin{aligned} & (\sigma^2)^{-1-n/2} \exp\left\{-\frac{\text{SSE} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})}{2\sigma^2}\right\} \\ &= (\sigma^2)^{-(a+p/2)-1} \exp\left\{-\frac{2b + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})}{2\sigma^2}\right\}, \end{aligned}$$

with  $a = (n-p)/2 > 0$  and  $b = \text{SSE}/2 > 0$ . This is proportional to the density Eq. 7.6.1 of  $\text{NIG}(\hat{\boldsymbol{\beta}}, (\mathbf{X}^\top \mathbf{X})^{-1}, a, b)$ , which is integrable. So the posterior is proper and equal to it. By Lemma 7.6.1 with  $2a = n - p$  and  $b/a = \text{SSE}/(n-p) = s^2$ , the stated ratio has the  $t(n-p)$  distribution. ■

The frequentist and Bayesian statements now read alike, but they say different things. The  $t$  interval of Section 7.5 has probability  $1 - \alpha$  of covering the fixed  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  *before the data are seen*. The credible interval has posterior probability  $1 - \alpha$  of containing the random  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  *given the observed data*. That the two coincide is a special property of this model and this prior. It is one reason why  $1/\sigma^2$  is regarded as the natural “noninformative” prior for the normal linear model.

## 7.6.5 A worked example

### Example. (A prior for the stack loss plant)

Return to the stack loss regression of Example (Stack loss). Suppose an engineer, before seeing these 21 days, believed from the chemistry of the process that stack loss should rise roughly one for one with air flow and with cooling water temperature, and should

not depend on acid concentration. This belief is hypothetical and is used only for illustration. We encode it as

$$\mathbf{m}_0 = (0, 1, 1, 0)^\top, \quad \mathbf{V}_0 = \text{diag}(100^2, 0.1^2, 0.2^2, 0.1^2), \quad a_0 = 3, \quad b_0 = 18.$$

So the prior standard deviations of the slopes are  $0.1\sigma$ ,  $0.2\sigma$  and  $0.1\sigma$ , the intercept is left essentially free, and the prior mean of  $\sigma^2$  is  $b_0/(a_0 - 1) = 9$ .

The posterior has  $a_n = 13.5$ , so the marginal posteriors are  $t$  with 27 degrees of freedom, and  $b_n = 112.523$ . The posterior means of the coefficients are

$$\mathbf{m}_n = (-41.1490, 0.7616, 1.1606, -0.1371)^\top,$$

against the least squares estimates  $(-39.9197, 0.7156, 1.2953, -0.1521)$ . Each slope has moved toward its prior mean. The intercept has moved too, though its prior is vague, because it is strongly correlated with the slopes when the regressors are not centred. The 95% credible interval for the air-flow coefficient, using the  $t(27)$  quantile 2.052, is  $(0.550, 0.973)$ , narrower than the least squares  $t$  interval  $(0.431, 1.000)$ , because the prior adds information. For water temperature and acid concentration the intervals are  $(0.594, 1.727)$  and  $(-0.392, 0.118)$ . The posterior mean of  $\sigma^2$  is 9.002, almost exactly the prior mean 9, although  $s^2 = 10.519$ . The two parts of [Eq. 7.6.3](#) explain why. The lack of fit at the posterior mean is  $\|\mathbf{y} - \mathbf{X}\mathbf{m}_n\|^2 = 180.667$ , and on its own, with  $b_0$ , it would give  $(b_0 + 180.667/2)/(a_n - 1) = 8.667$ : the shape  $a_n = a_0 + n/2$  counts all  $n$  observations, while the residuals have spent some of them on  $\boldsymbol{\beta}$ . The prior–data conflict term  $(\mathbf{m}_n - \mathbf{m}_0)^\top \mathbf{V}_0^{-1}(\mathbf{m}_n - \mathbf{m}_0) = 8.378$  adds to the scale and makes up the difference, so the data barely move  $\sigma^2$  from where the prior put it.

[Figure 7.6.1\(a\)](#) shows the three marginal densities for the air-flow coefficient. Panel (b) traces the posterior means of the slopes as the prior precision of the slopes is multiplied by a factor  $\tau$ , from  $10^{-3}$  (essentially least squares) to  $10^3$  (essentially the prior mean). The water-temperature coefficient does not move monotonically. It overshoots its prior mean of 1, down to 0.946, before returning. Matrix shrinkage acts on combinations of coefficients, and a single coordinate can pass beyond its prior mean on the way.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape
XtX, Xty = X.T @ X, X.T @ y
beta_hat = np.linalg.solve(XtX, Xty)
sse = np.sum((y - X @ beta_hat) ** 2)
```

```
def nig_posterior(m0, V0, a0, b0):
    """Conjugate update of the normal-inverse-gamma prior NIG(m0, V0, a0, b0)."""
    P0 = np.linalg.inv(V0) # prior precision (per unit sigma
    ~2)
    Vn = np.linalg.inv(P0 + XtX)
    mn = Vn @ (P0 @ m0 + Xty)
    an = a0 + n / 2
    bn = b0 + (y @ y + m0 @ P0 @ m0 - mn @ (P0 + XtX) @ mn) / 2
    return mn, Vn, an, bn
```

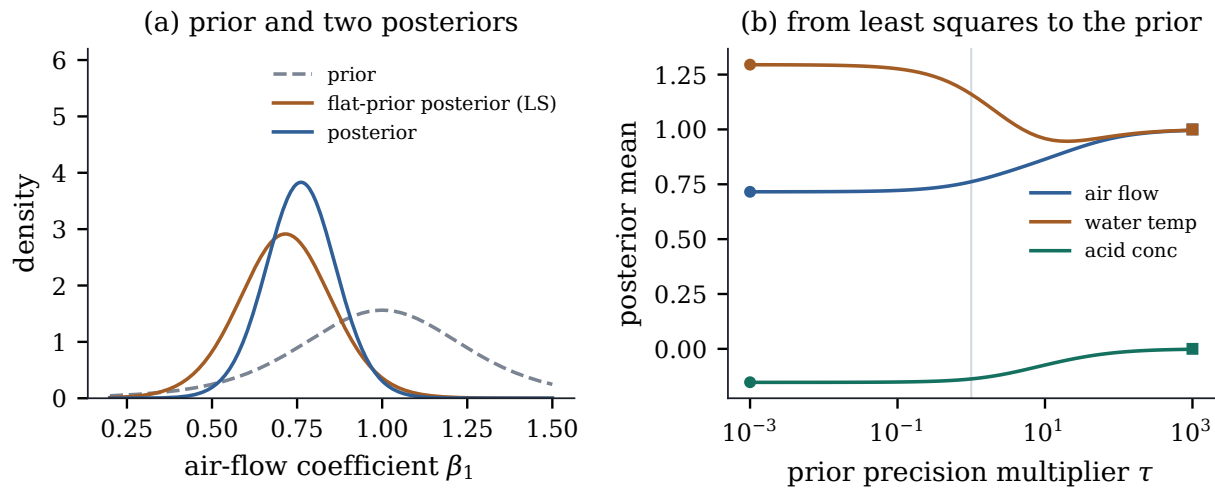


Figure 7.6.1: Conjugate analysis of the stack loss regression. (a) Marginal prior, flat-prior posterior (which is the least squares  $t$  distribution) and posterior for the air-flow coefficient. (b) Posterior means of the three slopes as the prior precision of the slopes is scaled by  $\tau$ . Circles mark least squares, squares mark the prior means, and the grey line marks the prior of the example ( $\tau = 1$ ).

```
# prior: vague intercept; slopes centred on 1, 1, 0 with sd 0.1, 0.2, 0.1 in units
# of sigma
m0 = np.array([0.0, 1.0, 1.0, 0.0])
V0 = np.diag([100.0**2, 0.1**2, 0.2**2, 0.1**2])
a0, b0 = 3.0, 18.0 # prior mean of sigma^2 is b0/(a0
                    # -1) = 9
mn, Vn, an, bn = nig_posterior(m0, V0, a0, b0)
scale = np.sqrt(np.diag(Vn) * bn / an) # marginal posterior: t with 2
a_n d.f.
q = stats.t.ppf(0.975, 2 * an)
for j, name in enumerate(["intercept", "air flow", "water temp", "acid conc"]):
    print(f"{name:10s} LS {beta_hat[j]:8.4f} posterior mean {mn[j]:8.4f} "
          f"95% interval ({mn[j] - q * scale[j]:.3f}, {mn[j] + q * scale[j]:.3f})")
print(f"posterior mean of sigma^2 = {bn / (an - 1):.3f}")
```

The script behind this example also checks the theorem numerically. The log posterior computed from Eq. 7.6.2 differs from log prior plus log-likelihood by the same constant at 50 random parameter values. The two forms of  $b_n$  agree. A Monte Carlo sample drawn from  $\sigma^2$  and then  $\beta \mid \sigma^2$  reproduces the marginal  $t$  quantiles. And a nearly flat prior reproduces least squares and the  $t$  interval.

*Remark* (How the Bayesian answer relates to the optimality results). Seen from the frequentist side, the posterior mean is a linear estimator with bias  $\mathbf{W}(\mathbf{m}_0 - \beta)$  and covariance  $\sigma^2(\mathbf{I} - \mathbf{W})(\mathbf{X}^\top \mathbf{X})^{-1}(\mathbf{I} - \mathbf{W})^\top$ , which is smaller than that of  $\hat{\beta}$  in the non-negative definite order. Whether its mean squared error is smaller depends on how far  $\mathbf{m}_0$  is from  $\beta$ , in the metric that  $\mathbf{W}$  defines. Averaged over the prior, the posterior mean is the estimator with the smallest expected squared error, because a conditional mean minimizes mean squared error. Different criteria single out different estimators. The Gauss–Markov theorem, the Lehmann–Scheffé theorem and the conjugate posterior (Theorem 7.6.2) answer three different questions, and in the flat-prior limit their answers agree.

## 7.6.6 Exercises

### 7.6.6.1 A. Check your understanding

#### Exercise A1.

Show that the  $\text{IG}(a, b)$  distribution has mean  $b/(a - 1)$  for  $a > 1$  and mode  $b/(a + 1)$ . What are the mean and mode of  $\sigma^2 \mid \mathbf{y}$  under the flat prior of [Corollary 7.6.4](#), and how do they compare with  $s^2$  and with the MLE of [Theorem 7.3.1](#)?

#### Exercise A2.

Suppose  $\sigma^2$  is known and the prior is  $\boldsymbol{\beta} \sim N_p(\mathbf{m}_0, \sigma^2 \mathbf{V}_0)$ . Show that the posterior is  $N_p(\mathbf{m}_n, \sigma^2 \mathbf{V}_n)$  with  $\mathbf{m}_n$  and  $\mathbf{V}_n$  as in [Eq. 7.6.2](#).

### 7.6.6.2 B. Practice

#### Exercise B1.

Under the  $g$ -prior of [Proposition 7.6.3\(c\)](#) with  $\mathbf{m}_0 = \mathbf{0}$ , show that  $b_n = b_0 + \frac{1}{2}\{\text{SSE} + \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} / (1 + g)\}$ . Show that as  $g \rightarrow \infty$  the posterior mean tends to  $\hat{\boldsymbol{\beta}}$ , and that for finite  $g$  the frequentist mean squared error of  $\mathbf{m}_n$  is  $\{g^2 \sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1} + \|\boldsymbol{\beta}\|^2\} / (1 + g)^2$ .

*Solution.* By [Proposition 7.6.3\(b\)](#), with  $\mathbf{V}_0 + (\mathbf{X}^\top \mathbf{X})^{-1} = (1 + g)(\mathbf{X}^\top \mathbf{X})^{-1}$ , the quadratic term is  $\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}} / (1 + g)$ . As  $g \rightarrow \infty$ ,  $\mathbf{m}_n = g\hat{\boldsymbol{\beta}} / (1 + g) \rightarrow \hat{\boldsymbol{\beta}}$ . For the mean squared error,  $\mathbf{m}_n - \boldsymbol{\beta} = \{g(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - \boldsymbol{\beta}\} / (1 + g)$ , and  $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}$  has mean zero and covariance  $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ . So the expected squared norm is  $\{g^2 \sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1} + \|\boldsymbol{\beta}\|^2\} / (1 + g)^2$ . It is smaller than  $\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$ , the value for  $\hat{\boldsymbol{\beta}}$ , iff  $\|\boldsymbol{\beta}\|^2 < (1 + 2g)\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$ .

#### Exercise B2.

Let  $\mathbf{X}$  be rank deficient,  $\mathbf{m}_0 = \mathbf{0}$  and  $\mathbf{V}_0 = \tau^{-1} \mathbf{I}$ . For  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ , show that  $\mathbf{v}^\top \mathbf{m}_n = 0$  and  $\mathbf{v}^\top \mathbf{V}_n \mathbf{v} = \|\mathbf{v}\|^2 / \tau$ . So the conditional posterior of  $\mathbf{v}^\top \boldsymbol{\beta}$  given  $\sigma^2$  equals its prior: the data do not update nonidentified directions. What happens to  $\boldsymbol{\lambda}^\top \mathbf{m}_n$  for  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$  as  $\tau \rightarrow 0$ ?

*Solution.*  $\mathbf{V}_n^{-1} \mathbf{v} = (\tau \mathbf{I} + \mathbf{X}^\top \mathbf{X}) \mathbf{v} = \tau \mathbf{v}$ , so  $\mathbf{V}_n \mathbf{v} = \mathbf{v} / \tau$ . Then  $\mathbf{v}^\top \mathbf{m}_n = \mathbf{v}^\top \mathbf{V}_n \mathbf{X}^\top \mathbf{y} = (\mathbf{V}_n \mathbf{v})^\top \mathbf{X}^\top \mathbf{y} = (\mathbf{X} \mathbf{v})^\top \mathbf{y} / \tau = 0$ , and  $\mathbf{v}^\top \mathbf{V}_n \mathbf{v} = \|\mathbf{v}\|^2 / \tau$ . These are the prior mean and variance (per unit  $\sigma^2$ ) of  $\mathbf{v}^\top \boldsymbol{\beta}$ . For the second question,  $\mathbf{m}_n = (\mathbf{X}^\top \mathbf{X} + \tau \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$  converges as  $\tau \rightarrow 0$  to the minimum-norm least squares estimate  $\mathbf{X}^+ \mathbf{y}$  ([Proposition 6.10.1](#)). This follows from the spectral decomposition of  $\mathbf{X}^\top \mathbf{X}$ : eigenvectors with eigenvalue  $d > 0$  get the factor  $1/(d + \tau) \rightarrow 1/d$ , and those with eigenvalue 0 get zero. Hence  $\boldsymbol{\lambda}^\top \mathbf{m}_n \rightarrow \boldsymbol{\lambda}^\top \mathbf{X}^+ \mathbf{y} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ , the least squares estimate of the estimable function.

### 7.6.6.3 C. Going deeper



**Exercise C1.**

Under the posterior of [Theorem 7.6.2](#), let  $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$  be a future observation, with  $\varepsilon_0 \sim N(0, \sigma^2)$  independent of everything given  $\sigma^2$ . Show that the posterior predictive distribution of  $Y_0$  is  $t_{2a_n}(\mathbf{x}_0^\top \mathbf{m}_n, (b_n/a_n)(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0))$ . Compare it with the frequentist prediction interval of [Exercise \(B2\)](#) under the flat prior.

*Solution.* Given  $\sigma^2$  and  $\mathbf{y}$ ,  $\mathbf{x}_0^\top \boldsymbol{\beta} \sim N(\mathbf{x}_0^\top \mathbf{m}_n, \sigma^2 \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0)$ , and  $\varepsilon_0$  is independent of it. So  $Y_0 \mid \sigma^2, \mathbf{y} \sim N(\mathbf{x}_0^\top \mathbf{m}_n, \sigma^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0))$ . Thus  $(Y_0, \sigma^2) \mid \mathbf{y}$  is normal-inverse-gamma with  $p = 1$ , mean  $\mathbf{x}_0^\top \mathbf{m}_n$  and  $\mathbf{V} = 1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$ , and [Lemma 7.6.1](#) gives the  $t_{2a_n}$  law. Under the flat prior,  $\mathbf{m}_n = \hat{\boldsymbol{\beta}}$ ,  $\mathbf{V}_n = (\mathbf{X}^\top \mathbf{X})^{-1}$ ,  $2a_n = n - p$  and  $b_n/a_n = s^2$ . The equal-tailed predictive interval is then exactly the prediction interval of [Exercise \(B2\)](#).

**Exercise C2.**

Split the data into two batches  $(\mathbf{y}_1, \mathbf{X}_1)$  and  $(\mathbf{y}_2, \mathbf{X}_2)$ . Show that updating the prior with the first batch, and then using that posterior as the prior for the second batch, gives the same posterior as a single update with all the data. Which quantities in [Eq. 7.6.2](#) make this obvious?

## 7.7 Summary and Notes

### 7.7.1 Summary

*Idea.*

1. A linear estimator  $\mathbf{a}^\top \mathbf{Y}$  is unbiased for  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  iff  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ . Such estimators exist iff  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ , the same condition under which least squares estimates  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  uniquely. Their coefficient vectors form the flat  $\mathbf{M}\boldsymbol{\rho} + \mathcal{C}(\mathbf{X})^\perp$  ([Proposition 7.1.1](#) and [Proposition 7.1.2](#)).
2. **Gauss–Markov.** Under  $E(\boldsymbol{\varepsilon}) = \mathbf{0}$  and  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ , in any rank,  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  is the unique BLUE of every estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ . For any LUE,  $\text{Var}(\mathbf{a}^\top \mathbf{Y}) = \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) + \sigma^2 \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2$ . In matrix form,  $\text{Cov}(\mathbf{A}\mathbf{Y}) - \text{Cov}(\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{A}^\top \succeq \mathbf{0}$  ([Theorem 7.2.1](#) and [Corollary 7.2.2](#)). With  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ , generalized least squares is the BLUE ([Corollary 7.2.3](#)).
3. The theorem does not cover nonlinear estimators (the midrange beats the mean under uniform errors, [Example \(Uniform errors and the midrange\)](#)) or biased ones (ridge, [Exercise \(B1\)](#)). Orthogonal columns minimize coefficient variances ([Proposition 7.2.4](#)).
4. Under normality, the MLE of  $\boldsymbol{\beta}$  is least squares and the MLE of  $\sigma^2$  is  $\text{SSE}/n$ , biased by the factor  $(n - r)/n$  and inconsistent when  $r$  grows with  $n$  ([Theorem 7.3.1](#) and [Example \(Many means, one variance\)](#)). The profile likelihood of an estimable function is a decreasing function of  $t^2$  ([Proposition 7.3.3](#)).
5.  $(\mathbf{M}\mathbf{Y}, \text{SSE})$  is complete and sufficient. By Lehmann–Scheffé,  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  and  $s^2$  are the UMVUEs, with no linearity restriction ([Proposition 7.4.3](#) and [Theorem 7.4.6](#)). Least squares attains the Cramér–Rao bound for  $\boldsymbol{\beta}$ , and  $s^2$  has efficiency  $(n - p)/n$ .



(Theorem 7.4.8).

6. Under normality,  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} \sim N(\mathbf{\Lambda}^\top \boldsymbol{\beta}, \sigma^2 \mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{\Lambda})$ ,  $\text{SSE}/\sigma^2 \sim \chi^2(n-r)$ , the two are independent, and standardized estimable functions are  $t(n-r)$  (Theorem 7.5.1, Corollary 7.5.2 and Corollary 7.5.3). Without normality the variance of SSE depends on the kurtosis (Proposition 7.5.4).
7. The normal-inverse-gamma prior is conjugate. The posterior mean is a precision-weighted combination of the prior mean and  $\hat{\boldsymbol{\beta}}$ , and the flat prior  $1/\sigma^2$  reproduces least squares and the  $t$  intervals (Theorem 7.6.2, Proposition 7.6.3 and Corollary 7.6.4).

## 7.7.2 Notes and sources

**Gauss and Markov.** Gauss (1823) proved that least squares gives the linear unbiased combination of observations with the smallest variance, without assuming normal errors. It was his second justification of the method, after the normal-theory argument of 1809. Markov's lectures on probability (1900 and later editions) restated the result, and Neyman (1934) attached Markov's name to it. The history is traced by Plackett (1949, 1972) and Stigler (1986). The projection proof of Section 7.2 splits the coefficient vector of a competing estimator by Pythagoras, so that its excess variance is  $\sigma^2 \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2$ . This is the proof of Seber and Lee (2003, section 3.2, Theorem 3.2), stated there for linear functions of the mean vector; we apply it to estimable functions in the rank-deficient case. Christensen (2020, section 2.3) argues instead through the covariance with  $\mathbf{p}^\top \mathbf{M}\mathbf{Y}$ . Rencher and Schaalje (2008, section 7.3) give the matrix form for full-rank models, and Agresti (2015, section 2.7) discusses its scope and its extension to generalized least squares. The generalized least squares version (Corollary 7.2.3) is due to Aitken (1935). The characterization of models in which ordinary least squares remains best (Exercise C3) is due to Zyskind (1967) and, in coordinate-free form, to Kruskal (1968).

**What Gauss–Markov does not cover.** That shrinkage can improve on unbiased estimation in three or more dimensions was discovered by Stein (1956), and James and Stein (1961) gave the explicit estimator. Hoerl and Kennard (1970) introduced ridge regression. The existence of a better ridge estimator (Exercise B1) is their theorem. Wald (1940) proposed the group-means slope estimator of Example (Three linear unbiased estimators of a slope) for regressions with errors in the regressor. The optimality of orthogonal weighing designs goes back to Hotelling (1944).

**Likelihood.** The inconsistency of the maximum likelihood estimate of a variance in the presence of many nuisance means is the example of Neyman and Scott (1948). Restricted maximum likelihood (Exercise C2) was developed by Patterson and Thompson (1971) for variance components. Christensen (2020, section 2.4), Seber and Lee (2003, section 3.5) and Rencher and Schaalje (2008, section 7.6) derive the normal-theory MLEs. The maximization through  $\log u \leq u - 1$  used in Theorem 7.3.1 follows Seber and Lee (2003, section 3.5).

**Minimum variance.** The Rao–Blackwell theorem is due to Rao (1945) and Blackwell (1947). Completeness and the uniqueness theorem are due to Lehmann and Scheffé (1950). The two results we quoted, the factorization criterion and the completeness of full-rank exponential families, are proved in Lehmann and Casella (1998) and Lehmann and Romano (2005). The same variance bound was found independently by Rao (1945) and Cramér (1946). Christensen (2020, section 2.5) outlines the minimum variance unbiased argument for the linear model by reparameterizing to full rank. Seber and Lee (2003, section 3.5) compare the Cramér–Rao bound with the variance of  $s^2$ . The optimality of  $s^2$  among nonnegative quadratic unbiased estimators

(Exercise (C2)) is due to Atiqullah (1962). That the divisor  $k + 2$  minimizes the mean squared error among multiples of SSE (Proposition 7.3.2) is due to Theil and Schweitzer (1961). The reflection argument of Example (Conditioning on the sufficient statistic) is our own illustration.

**Sampling distributions.** The distribution theory of Section 7.5 follows Christensen (2020, section 2.6), Seber and Lee (2003, section 3.4) and Rencher and Schaalje (2008, section 7.6.3), with the rank-deficient forms throughout. The effect of kurtosis on inference about variances, and the relative robustness of inference about means, were emphasized by Box (1953).

**Bayesian thread.** The conjugate normal-inverse-gamma analysis goes back to Raiffa and Schlaifer (1961) and is developed for regression by Zellner (1971) and Box and Tiao (1973). O’Hagan and Forster (2004) give a modern account. Zellner (1986) introduced the  $g$ -prior. Seber and Lee (2003, section 3.12), Christensen (2020, section 2.10) and Rencher and Schaalje (2008, chapter 11) treat Bayesian estimation in the linear model, and Fahrmeir, Kneib, Lang and Marx (2021, section 4.4) present the conjugate analysis as the starting point for richer priors. For an applied account of the Gauss–Markov conditions and what follows from them, see Sen and Srivastava (1990, section 2.5).

**Data.** The stack loss data are from Brownlee (1965) and are distributed with statsmodels.

## 7.7.3 References

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# Chapter 8

## Estimability, Identifiability and Indicator Variables

When the columns of the model matrix are linearly dependent, the data cannot tell some coefficient vectors apart, because different vectors produce the same mean. This chapter asks which questions about  $\beta$  still have answers. The answer is the same from every direction. A linear function  $\lambda^\top \beta$  can be estimated without bias exactly when it is determined by the mean vector  $\mathbf{X}\beta$ , exactly when  $\lambda$  lies in the row space of  $\mathbf{X}$ , and exactly when every least squares solution gives it the same value. The rest of the chapter uses this criterion. Reparameterizations and side conditions turn out to be two ways of choosing coordinates, and neither can change what is estimable. Factors enter a regression through indicator variables, and each coding of a factor amounts to a choice of which linear functions of the cell means the coefficients report. Contrasts are the estimable functions of a factor, and orthogonal sets of contrasts split the factor's sum of squares into separate pieces.

**What you need.** [Chapter 1](#) (rank, generalized inverses and the consistency of linear systems, especially [Theorem 1.4.1](#) and [Corollary 1.4.2](#)), [Chapter 2](#) (means and covariances of linear functions, [Theorem 2.2.1](#)), [Chapter 4](#) (for the distribution of contrast sums of squares under normality, [Theorem 4.4.4](#)), [Chapter 5](#) (the linear model, [Definition 5.3.1](#)), and [Chapter 6](#), above all [Section 6.2](#) and [Section 6.4](#), where rank-deficient least squares and the invariance of  $\lambda^\top \hat{\beta}$  for  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$  ([Theorem 6.4.4](#)) first appeared. A few remarks use the Gauss–Markov theorem of [Chapter 7](#).

### 8.0.1 Roadmap

- [Identifiability](#)
- [Estimable functions](#)
- [Characterizations of estimability](#)
- [Reparameterization and side conditions](#)
- [Indicator variables and the coding of a factor](#)
- [Several factors, interactions and covariates](#)
- [Contrasts](#)
- [Summary and notes](#)

## 8.1 Identifiability

Section 6.4 ended with a table of four coefficient vectors that disagreed in every entry, some even in sign, and still produced the same fitted values. Nothing had gone wrong with the computation. The model matrix had linearly dependent columns, and so the least squares problem had a whole affine subspace of solutions. This chapter asks what can be learned about  $\beta$  in that situation. We begin with a question that comes before estimation and does not involve data at all. Suppose we knew the distribution of  $\mathbf{Y}$  exactly, as if we had infinitely many replications. Would we then know  $\beta$ ? If not, which features of  $\beta$  would we know?

### 8.1.1 What the distribution of the data can reveal

Throughout,  $\mathbf{Y} = \mathbf{X}\beta + \epsilon$  is the linear model of Definition 5.3.1:  $\mathbf{X}$  is a fixed  $n \times p$  matrix of rank  $r$ , and  $E(\epsilon) = \mathbf{0}$ . We do not assume  $r = p$ . The coefficient vector enters the model only through the product  $\mathbf{X}\beta$ . Suppose the distribution of the error vector  $\epsilon$  is allowed to depend on other parameters, such as  $\sigma^2$ , but not on  $\beta$ . Then the distribution of  $\mathbf{Y}$  is the distribution of  $\epsilon$  shifted by the vector  $\mathbf{X}\beta$ . Two coefficient vectors with the same product  $\mathbf{X}\beta$  therefore give the same distribution of  $\mathbf{Y}$ , with the other parameters held fixed. Conversely, the distribution of  $\mathbf{Y}$  determines its mean  $E(\mathbf{Y}) = \mathbf{X}\beta$ . So the most we can learn about  $\beta$  from the distribution is the value of  $\mathbf{X}\beta$ . The mean vector

$$\mu = \mathbf{X}\beta \in \mathcal{C}(\mathbf{X})$$

is the object the data speak about. A value of  $\beta$  is one way of writing  $\mu$  in coordinates.

This observation reduces the question to linear algebra. The map  $\beta \mapsto \mathbf{X}\beta$  from  $\mathbb{R}^p$  to  $\mathbb{R}^n$  is linear. Two coefficient vectors have the same image iff their difference lies in the null space  $\mathcal{N}(\mathbf{X})$ . So

$$\{\mathbf{b} \in \mathbb{R}^p : \mathbf{X}\mathbf{b} = \mathbf{X}\beta\} = \beta + \mathcal{N}(\mathbf{X}), \quad (8.1.1)$$

an affine subspace of dimension  $p - r$  by rank-nullity (Eq. 6.2.1). All vectors in Eq. 8.1.1 are observationally equivalent: no experiment with this model matrix can distinguish between them.

#### Definition 8.1.1 — Identifiability

In the linear model  $E(\mathbf{Y}) = \mathbf{X}\beta$ ,  $\beta \in \mathbb{R}^p$ , the parameter  $\beta$  is **identifiable** if  $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$  implies  $\beta_1 = \beta_2$ . A function  $g$  defined on  $\mathbb{R}^p$ , with values in any set, is **identifiable** if  $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$  implies  $g(\beta_1) = g(\beta_2)$ .

The definition is phrased through the mean, not the whole distribution. By the argument above the two versions agree whenever the error distribution does not involve  $\beta$ , which is the situation throughout this book until the generalized linear models of Part VIII. There the mean is  $h(\mathbf{X}\beta)$  for a known one-to-one function  $h$ , and the same definition applies unchanged. It says that  $g(\beta)$  is a well-defined function of the point  $\mu$ , whichever coordinates were used to describe that point.

**Theorem 8.1.1 — Identifiable means “a function of the mean”**

A function  $g$  on  $\mathbb{R}^p$  is identifiable iff there is a function  $h$  on  $\mathcal{C}(\mathbf{X})$  with

$$g(\boldsymbol{\beta}) = h(\mathbf{X}\boldsymbol{\beta}) \quad \text{for every } \boldsymbol{\beta} \in \mathbb{R}^p.$$

Equivalently,  $g$  is constant on each affine subspace  $\boldsymbol{\beta} + \mathcal{N}(\mathbf{X})$ .

*Proof.* If  $g = h \circ \mathbf{X}$  and  $\mathbf{X}\boldsymbol{\beta}_1 = \mathbf{X}\boldsymbol{\beta}_2$ , then  $g(\boldsymbol{\beta}_1) = h(\mathbf{X}\boldsymbol{\beta}_1) = h(\mathbf{X}\boldsymbol{\beta}_2) = g(\boldsymbol{\beta}_2)$ . Conversely, let  $g$  be identifiable. For  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$  choose any  $\boldsymbol{\beta}$  with  $\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\mu}$  and put  $h(\boldsymbol{\mu}) = g(\boldsymbol{\beta})$ . The value does not depend on the choice, because any two choices have the same image under  $\mathbf{X}$ . By construction  $g(\boldsymbol{\beta}) = h(\mathbf{X}\boldsymbol{\beta})$ . The last sentence restates the definition using [Eq. 8.1.1](#). ■

The theorem makes identifiability a statement about information. A function that can be computed from  $\boldsymbol{\mu}$  is one that could, in principle, be estimated consistently, provided that  $\boldsymbol{\mu}$  itself can be estimated. A function that cannot be computed from  $\boldsymbol{\mu}$  is one about which the model is silent. Its value can be anything along the affine subspace, and more data from the same design cannot help.

**8.1.2 When the parameter is identifiable****Proposition 8.1.2**

The following are equivalent: (a)  $\boldsymbol{\beta}$  is identifiable; (b)  $\mathcal{N}(\mathbf{X}) = \{\mathbf{0}\}$ ; (c)  $\text{rank}(\mathbf{X}) = p$ ; (d)  $\mathbf{X}^\top \mathbf{X}$  is nonsingular. In that case the function  $h$  of [Theorem 8.1.1](#) for  $g(\boldsymbol{\beta}) = \boldsymbol{\beta}$  is  $h(\boldsymbol{\mu}) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\mu}$ .

*Proof.* By [Eq. 8.1.1](#),  $\boldsymbol{\beta}$  is identifiable iff every affine subspace  $\boldsymbol{\beta} + \mathcal{N}(\mathbf{X})$  is a single point, which is (b). (b) $\Leftrightarrow$ (c) is rank-nullity, and (c) $\Leftrightarrow$ (d) is [Corollary 6.2.4\(c\)](#). If  $\mathbf{X}^\top \mathbf{X}$  is nonsingular and  $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ , then  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\mu} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}$ . ■

When  $\text{rank}(\mathbf{X}) = r < p$  the parameter is not identifiable, but parts of it may be. The simplest parts are the individual coordinates.

**Proposition 8.1.3 — Identifiability of a single coefficient**

Write  $\mathbf{x}_1, \dots, \mathbf{x}_p$  for the columns of  $\mathbf{X}$ . For each  $j$  the following are equivalent:

- (a)  $\beta_j$  is identifiable;
- (b) every  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$  has  $v_j = 0$ ;

- (c)  $\mathbf{x}_j$  is not a linear combination of the other columns of  $\mathbf{X}$ ;
- (d) deleting column  $j$  from  $\mathbf{X}$  lowers the rank.

*Proof.* (a) $\Leftrightarrow$ (b): by Eq. 8.1.1, the  $j$ th coordinates of the vectors observationally equivalent to  $\boldsymbol{\beta}$  are  $\beta_j + v_j$ ,  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ . They all equal  $\beta_j$  iff  $v_j = 0$  for every null vector. (b) $\Leftrightarrow$ (c): a relation  $\mathbf{x}_j = \sum_{k \neq j} a_k \mathbf{x}_k$  is the same as a null vector with  $j$ th entry  $-1$ , and any null vector with  $v_j \neq 0$  can be rescaled to have  $v_j = -1$ . (c) $\Leftrightarrow$ (d): the column space loses a dimension when  $\mathbf{x}_j$  is removed iff  $\mathbf{x}_j$  is not in the span of the remaining columns. ■

Part (c) gives a practical test. A coefficient is lost exactly when its regressor is *aliased*, that is, when it can be rebuilt from the others. In a model that is only partly identifiable, some coefficients can still be perfectly well identified. If  $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}_j]$  and  $\mathbf{x}_j \notin \mathcal{C}(\mathbf{X}_1)$ , then  $\beta_j$  is identifiable however badly  $\mathbf{X}_1$  is parameterized. A typical case is a numeric covariate added to an overparameterized factor model: the covariate's coefficient is identifiable as long as the covariate is not constant within every level of the factor (Section 8.6).

### 8.1.3 Where rank deficiency comes from

Linear dependence among the columns of  $\mathbf{X}$  arises in three quite different ways, and it helps to distinguish them.

*By construction.* A factor with  $g$  levels entered as an intercept together with all  $g$  indicator columns gives  $g+1$  columns spanning a  $g$ -dimensional space (Example (A rank-deficient layout)). Such *overparameterized* models are written down deliberately, because they treat the levels symmetrically. The dependence is harmless once it is understood, and most of this chapter is about understanding it.

*By the logic of the variables.* Some sets of variables satisfy an identity that holds for every observation. Examples are shares of a whole that add to one, a total entered together with all its parts, and, the best-known case, age, period and birth cohort.

*By the design.* Two regressors that are logically unrelated may be exactly collinear in the sample at hand. A factor level may never be observed. The design may have fewer distinct rows than parameters. In each case another data set with the same model could identify the parameter, but this one cannot.

Identifiability is therefore a property of the model matrix, and through it of the design, not of the population being studied. The next example shows the second kind of dependence.

#### Example. (Age, period and cohort)

A survey is fielded every five years from 2000 to 2020, and its respondents are grouped in eleven five-year age bands from 20 to 70. For each of the 55 combinations we have one average response. A demographer wants to separate three linear trends: in age (people change as they grow older), in period (everyone changes with the times), and in birth cohort (generations differ). With age, period and cohort centred at 45, 2010 and 1965, the model is

$$E(y) = \beta_0 + \beta_A \text{age} + \beta_P \text{period} + \beta_C \text{cohort}.$$



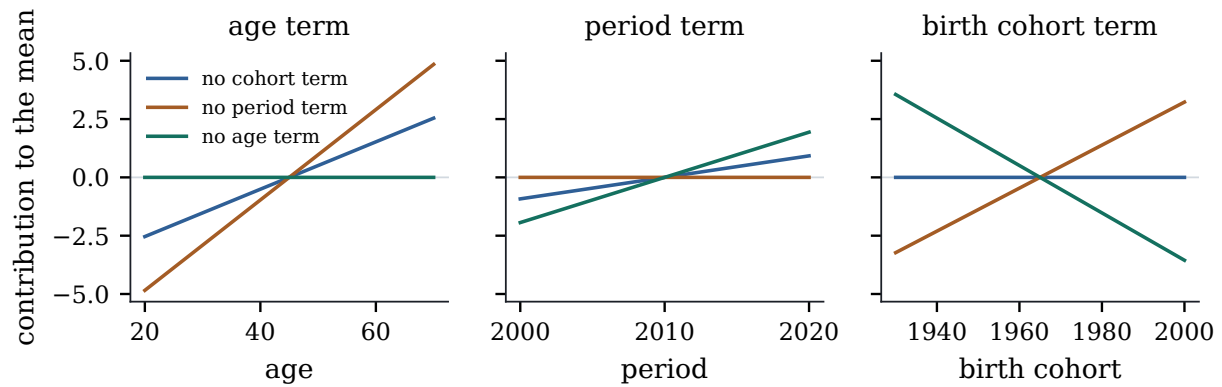


Figure 8.1.1: Three equally good fits of the age–period–cohort model. Each panel shows the contribution of one term to the fitted mean under the three fits of [Example \(Age, period and cohort\)](#). The sum of the three contributions is the same for every respondent, so the data cannot choose between the fits.

But cohort is period minus age, for every respondent. The four columns satisfy age – period + cohort =  $\mathbf{0}$ , so  $\mathbf{v} = (0, 1, -1, 1)^\top$  spans  $\mathcal{N}(\mathbf{X})$  and  $\text{rank}(\mathbf{X}) = 3$ . By [Proposition 8.1.3](#), none of the three slopes is identifiable.

The listing fits the model three times, each time dropping one of the three columns. All three fits have residual sum of squares 157.00, because they have the same column space. They tell three different stories:

| Fit            | age slope | period slope | cohort slope |
|----------------|-----------|--------------|--------------|
| no cohort term | 0.102     | 0.092        | 0.000        |
| no period term | 0.194     | 0.000        | 0.092        |
| no age term    | 0.000     | 0.194        | -0.102       |

One analyst reports a steady ageing effect and no generational change. Another reports a strong ageing effect and a positive cohort trend. A third reports no ageing at all, a strong secular trend, and younger generations scoring lower. The data were in fact simulated with slopes 0.30,  $-0.10$  and 0.20. That vector lies on the parallel line  $\boldsymbol{\beta} + t\mathbf{v}$  of equally valid truths, and the data cannot prefer it. Only the identifiable combinations are recovered: the fitted 0.194 and 0.092 estimate the true 0.20 and 0.10.

What the three fits share are the linear functions orthogonal to  $\mathbf{v} = (0, 1, -1, 1)^\top$ :

$$\beta_A + \beta_P = 0.194, \quad \beta_P + \beta_C = 0.092, \quad \beta_A - \beta_C = 0.102,$$

in every row of the table, and  $\beta_0$ . Any statement about the separate trends requires information from outside these data, for instance a constraint justified by subject knowledge, and the conclusions then rest on that constraint ([Exercise \(C1\)](#)). [Figure 8.1.1](#) shows how each fit divides the same fitted surface among the three terms.

```
import numpy as np

ages = np.arange(20, 75, 5)           # 11 age bands: 20, 25, ..., 70
periods = np.arange(2000, 2025, 5)   # 5 survey waves: 2000, ..., 2020
A, P = np.meshgrid(ages, periods, indexing="ij")
age, period = A.ravel() - 45.0, P.ravel() - 2010.0 # centred at 45 and 2010
```



```
cohort = period - age                                     # birth year minus 1965, exactly
X = np.column_stack([np.ones(age.size), age, period, cohort])
print("n =", X.shape[0], " p =", X.shape[1], " rank =", np.linalg.matrix_rank(X))
print("X @ (0, 1, -1, 1) =", np.abs(X @ [0, 1, -1, 1]).max())
```

```
rng = np.random.default_rng(8)
truth = np.array([50.0, 0.30, -0.10, 0.20])
y = X @ truth + rng.normal(scale=1.5, size=age.size)

def fit_without(j):
    """Least squares after deleting column j (one of age, period, cohort)."""
    keep = [k for k in range(4) if k != j]
    b = np.zeros(4)
    b[keep] = np.linalg.lstsq(X[:, keep], y, rcond=None)[0]
    return b

stories = {"no cohort term": fit_without(3),
          "no period term": fit_without(2),
          "no age term": fit_without(1)}
for name, b in stories.items():
    sse = np.sum((y - X @ b) ** 2)
    print(f"{name:15s} b = {np.round(b, 3)}   SSE = {sse:.3f}",
          f" age+period = {b[1] + b[2]:.3f}   period+cohort = {b[2] + b[3]:.3f}")
```

## 8.1.4 Linear and nonlinear identifiable functions

In the example the identifiable linear functions were exactly those whose coefficient vectors are orthogonal to the null space. That is true in general.

### Proposition 8.1.4

A linear function  $\lambda^\top \beta$  is identifiable iff  $\lambda \perp \mathcal{N}(\mathbf{X})$ , that is, iff  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ . The identifiable linear functions form a subspace of dimension  $r = \text{rank}(\mathbf{X})$ .

*Proof.* By [Theorem 8.1.1](#),  $\lambda^\top \beta$  is identifiable iff  $\lambda^\top (\beta + \mathbf{v}) = \lambda^\top \beta$  for all  $\beta$  and all  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ , that is, iff  $\lambda^\top \mathbf{v} = 0$  for all  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ . By [Corollary 6.2.4\(d\)](#),  $\mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$ , which has dimension  $\text{rank}(\mathbf{X}^\top) = r$ . ■

Identifiable functions need not be linear. If  $\lambda_1^\top \beta$  and  $\lambda_2^\top \beta$  are identifiable, so are their ratio (where defined), their product, and  $\max\{\lambda_1^\top \beta, 0\}$ . Every function of  $\mu$  is identifiable, for example  $\|\mu\|^2$  or the largest mean. For instance, in a quadratic regression  $E(y) = \beta_0 + \beta_1 x + \beta_2 x^2$  with a full-rank design, the location  $-\beta_1/(2\beta_2)$  of the turning point is identifiable. It is not linear, and [Section 12.5](#) has to estimate it with care. The next section shows that among the *linear* functions, identifiability coincides with a much more concrete property, the existence of a linear unbiased estimator.

*Idea.* The data determine the mean vector  $\mu = \mathbf{X}\beta$  and nothing else about  $\beta$ . A function of  $\beta$  is identifiable exactly when it is a function of  $\mu$ . A *linear* function  $\lambda^\top \beta$  is identifiable exactly when  $\lambda$  lies in the row space  $\mathcal{C}(\mathbf{X}^\top)$ , the orthogonal complement of

the invisible directions  $\mathcal{N}(\mathbf{X})$ .

## 8.1.5 Exercises

### 8.1.5.1 A. Check your understanding

#### Exercise A1.

In [Example \(Age, period and cohort\)](#), show directly that  $\beta_0$ ,  $\beta_A + \beta_P$  and  $\beta_P + \beta_C$  are identifiable, that every identifiable linear function is a combination of these three, and that  $\beta_A - \beta_C$  is one of them. Is  $\beta_A + \beta_C$  identifiable?

#### Exercise A2.

A household's spending is split into food, housing and other, with shares  $s_1 + s_2 + s_3 = 1$ . A regression of savings on an intercept and all three shares is proposed. Find  $\mathcal{N}(\mathbf{X})$  (assuming the shares vary enough that there is no other relation among the columns), decide which coefficients are identifiable, and describe two standard ways to make the model identifiable.

### 8.1.5.2 B. Practice

#### Exercise B1.

Let  $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_g]$  be the indicator matrix of a factor and  $\mathbf{x}$  a numeric covariate, and consider  $\mathbf{X} = [\mathbf{1}, \mathbf{Z}, \mathbf{x}]$ . Show that the slope of  $\mathbf{x}$  is identifiable iff  $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$ , that is, iff  $\mathbf{x}$  is not constant within every level. Show that no intercept-type parameter  $\beta_0$  or  $\alpha_k$  is identifiable.

*Solution.* The columns  $\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g$  satisfy  $\mathbf{1} - \sum_k \mathbf{z}_k = \mathbf{0}$ , so they are dependent, and  $\mathcal{C}([\mathbf{1}, \mathbf{Z}]) = \mathcal{C}(\mathbf{Z})$ . By [Proposition 8.1.3\(c\)](#), the slope is identifiable iff  $\mathbf{x}$  is not a combination of  $\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g$ , that is, iff  $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$ . A vector lies in  $\mathcal{C}(\mathbf{Z})$  iff it is constant within each level. For the other coefficients,  $\mathbf{v} = (1, -1, \dots, -1, 0)^\top$  is a null vector with nonzero entries in positions  $0, 1, \dots, g$ , so by [Proposition 8.1.3\(b\)](#) none of  $\beta_0, \alpha_1, \dots, \alpha_g$  is identifiable.

#### Exercise B2.

Let  $\mathbf{P}$  be the orthogonal projection onto  $\mathcal{C}(\mathbf{X}^\top)$ . Show that  $g$  is identifiable iff  $g(\boldsymbol{\beta}) = g(\mathbf{P}\boldsymbol{\beta})$  for every  $\boldsymbol{\beta}$ . Deduce that the identifiable functions are exactly the functions of  $\mathbf{P}\boldsymbol{\beta}$ , and that  $\mathbf{P}\boldsymbol{\beta}$  is the unique coefficient vector in  $\mathcal{C}(\mathbf{X}^\top)$  representing the mean  $\mathbf{X}\boldsymbol{\beta}$ .

#### Exercise B3.

A design with model matrix  $\mathbf{X}$  of rank  $r < p$  is augmented by one more observation at a point  $\mathbf{x}_0$ . Show that the rank increases iff  $\mathbf{x}_0 \notin \mathcal{C}(\mathbf{X}^\top)$ . Show that, in the age–period–cohort design, no additional observation can make the slopes identifiable, and explain

why in words.

### 8.1.5.3 C. Going deeper

#### Exercise C1.

Replace the three linear terms of [Example \(Age, period and cohort\)](#) by three factors: an indicator for each age band, each period and each cohort (with an intercept). Show that the null space of the resulting model matrix has dimension 3 plus exactly one extra dimension. Identify the extra null vector: it assigns the linear sequences  $a - \bar{a}$ ,  $-(t - \bar{t})$  and  $c - \bar{c}$  to the age, period and cohort effects. Deduce that every second difference of age effects, such as  $\alpha_{a+1} - 2\alpha_a + \alpha_{a-1}$ , is identifiable, while no linear trend in any single one of the three factors is.

*Solution.* Each factor's indicators sum to  $\mathbf{1}$ , which gives the usual three null vectors (intercept minus all indicators of one factor). The identity age – period + cohort = const holds for every observation. Writing the age of an observation as  $\sum_a a z_a^A$ , and similarly for period and cohort, it becomes a relation  $\sum_a a \mathbf{z}_a^A - \sum_t t \mathbf{z}_t^P + \sum_c c \mathbf{z}_c^C - \kappa \mathbf{1} = \mathbf{0}$  among the columns. Subtracting suitable multiples of the three standard relations centres the three sequences, and this gives the stated vector. It is not a combination of the other three, because those are constant on each factor's effects and this one is linear and nonconstant. To see that there are no more, check that the matrix has rank  $1 + (A - 1) + (P - 1) + (C - 1) - 1$ . This rank check can be done directly or numerically for the design of [Example \(Age, period and cohort\)](#). A function  $\sum_a d_a \alpha_a$  is identifiable iff  $\mathbf{d}$  is orthogonal to all four null vectors, that is, iff  $\sum_a d_a = 0$  and  $\sum_a d_a a = 0$ . A second difference satisfies both. A linear trend  $\sum_a (a - \bar{a}) \alpha_a$  fails the second condition.

## 8.2 Estimable functions

Identifiability asks whether a quantity is determined by the distribution of the data. Estimation asks for more. We want a statistic whose distribution is centred on the quantity, however the parameter is set. For linear functions of  $\boldsymbol{\beta}$ , the natural candidates are linear functions of  $\mathbf{Y}$ , and the natural centring requirement is unbiasedness. This leads to the classical notion of an estimable function.

### 8.2.1 Linear unbiased estimators

Here and below the model is  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  with  $\mathbf{X}$  an  $n \times p$  matrix of rank  $r$ ; no assumption on  $\text{Cov}(\mathbf{Y})$  is needed until variances are computed.

#### Definition 8.2.1 — Estimable function

A linear function  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ ,  $\boldsymbol{\lambda} \in \mathbb{R}^p$ , is **estimable** if there is a vector  $\mathbf{a} \in \mathbb{R}^n$  such that

$$E(\mathbf{a}^\top \mathbf{Y}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta} \quad \text{for every } \boldsymbol{\beta} \in \mathbb{R}^p.$$

A vector  $\Lambda^\top \beta$  of linear functions, with  $\Lambda$  a  $p \times q$  matrix, is estimable if each of its  $q$  components is. Equivalently, there is an  $n \times q$  matrix  $\mathbf{A}$  with  $E(\mathbf{A}^\top \mathbf{Y}) = \Lambda^\top \beta$  for all  $\beta$ .

The statistic  $\mathbf{a}^\top \mathbf{Y}$  is a **linear unbiased estimator** of  $\lambda^\top \beta$ . The requirement “for every  $\beta$ ” is essential. For a single fixed  $\beta$ , almost any statistic can be made unbiased by adding a constant. What the definition demands is one statistic that is centred correctly whatever the true coefficients are. Chapter 7 asked which linear unbiased estimator is best (Definition 7.1.1). Here we ask only whether there is one.

One might allow an *affine* estimator  $a_0 + \mathbf{a}^\top \mathbf{Y}$ . This gains nothing, as Chapter 7 already showed.

### Lemma 8.2.1

The statistic  $a_0 + \mathbf{a}^\top \mathbf{Y}$  is unbiased for  $\lambda^\top \beta$ , for every  $\beta$ , iff  $a_0 = 0$  and  $\mathbf{X}^\top \mathbf{a} = \lambda$ .

*Proof.* This is Proposition 7.1.1, proved in Chapter 7 by evaluating the expectation  $a_0 + \mathbf{a}^\top \mathbf{X}\beta$  at  $\beta = \mathbf{0}$  and at the coordinate vectors. ■

So the linear unbiased estimators of  $\lambda^\top \beta$  correspond exactly to the solutions  $\mathbf{a}$  of the linear system  $\mathbf{X}^\top \mathbf{a} = \lambda$ . There is at least one iff the system is consistent, and when it is, the solutions form the affine subspace  $\mathbf{a}_0 + \mathcal{N}(\mathbf{X}^\top)$  (Theorem 1.4.1).

## 8.2.2 Estimable is the same as identifiable

### Theorem 8.2.2

For  $\lambda \in \mathbb{R}^p$  the following are equivalent:

- (a)  $\lambda^\top \beta$  is estimable;
- (b)  $\lambda^\top \beta$  is identifiable;
- (c)  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ , that is,  $\lambda^\top = \rho^\top \mathbf{X}$  for some  $\rho \in \mathbb{R}^n$ .

For a  $p \times q$  matrix  $\Lambda$ ,  $\Lambda^\top \beta$  is estimable iff  $\Lambda = \mathbf{X}^\top \mathbf{P}$  for some  $n \times q$  matrix  $\mathbf{P}$ , that is, iff  $\mathcal{C}(\Lambda) \subseteq \mathcal{C}(\mathbf{X}^\top)$ .

*Proof.* (a)  $\Leftrightarrow$  (c) is Lemma 8.2.1: an unbiased  $\mathbf{a}^\top \mathbf{Y}$  exists iff  $\mathbf{X}^\top \mathbf{a} = \lambda$  has a solution, which is the statement  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ . (b)  $\Leftrightarrow$  (c) is Proposition 8.1.4. The matrix version applies the vector version to each column of  $\Lambda$  and collects the vectors  $\rho$  as the columns of  $\mathbf{P}$ . ■

The theorem connects two different ideas. Identifiability is about what the model could reveal with unlimited data, and it involves no estimator. Estimability is about a concrete property of a concrete statistic. For linear functions they are the same thing, and both reduce to membership

of  $\lambda$  in the row space of  $\mathbf{X}$ . The row space is spanned by the rows  $\mathbf{x}_{(1)}^\top, \dots, \mathbf{x}_{(n)}^\top$ , so the theorem says:

*A linear function of  $\beta$  is estimable iff it is a linear combination of the expected responses  $E(y_i) = \mathbf{x}_{(i)}^\top \beta$ .*

This reading is the most useful one in practice. To show that  $\lambda^\top \beta$  is estimable, write it as a combination of the means of observations that were actually made. To show that it is not, find a direction  $\mathbf{v}$  with  $\mathbf{X}\mathbf{v} = \mathbf{0}$  and  $\lambda^\top \mathbf{v} \neq 0$ .

### Corollary 8.2.3 — The estimable space

The set of  $\lambda$  for which  $\lambda^\top \beta$  is estimable is the subspace  $\mathcal{C}(\mathbf{X}^\top)$  of dimension  $r = \text{rank}(\mathbf{X})$ . In particular:

- (a) every expected response  $\mathbf{x}_{(i)}^\top \beta$ , and hence the whole mean vector  $\mathbf{X}\beta$ , is estimable;
- (b) every entry of  $\mathbf{X}^\top \mathbf{X}\beta$  is estimable;
- (c) at most  $r$  estimable functions can be linearly independent, and a set of  $r$  linearly independent ones determines all the others as linear combinations;
- (d) if  $r = p$ , every linear function of  $\beta$  is estimable.

*Proof.* The first sentence is [Theorem 8.2.2](#). For (a) take  $\rho = \mathbf{e}_i$ , the  $i$ th coordinate vector. For (b),  $\mathbf{X}^\top \mathbf{X}\beta = \mathbf{X}^\top (\mathbf{X}\beta)$  and the rows of  $\mathbf{X}^\top \mathbf{X}$  lie in  $\mathcal{C}(\mathbf{X}^\top)$ . Part (c) is the dimension count, and (d) is the case  $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}^p$ . ■

## 8.2.3 The mean at a new point

A frequent use of a fitted model is to estimate the mean response at a setting  $\mathbf{x}_0$  of the regressors, which may or may not have been observed. The quantity  $\mathbf{x}_0^\top \beta$  is a linear function of  $\beta$ , so the theorem applies. It is estimable iff  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$ . In a full-rank model this is automatic, and the only danger in predicting at a new point is extrapolation, meaning that the model may not hold there. In a rank-deficient model there is a sharper limit. Some settings are *algebraically* out of reach. The data contain no information about the mean there, even if the model is exactly true.

### Example. (A fertilizer trial with a fixed ratio)

Ten plots receive nitrogen at 0, 20, 40, 60, 80 kg/ha (two plots each), and phosphorus is always applied at half the nitrogen rate, as in a commercial blend. The model

$$E(y) = \beta_0 + \beta_1 N + \beta_2 P$$

is natural agronomically, but in this design  $\mathbf{x}_P = \frac{1}{2}\mathbf{x}_N$ , so  $\mathbf{v} = (0, 1, -2)^\top$  spans  $\mathcal{N}(\mathbf{X})$  and  $\text{rank}(\mathbf{X}) = 2$ . A linear function is estimable iff its coefficient vector satisfies  $\lambda_1 - 2\lambda_2 = 0$ . So:

- $\beta_0$ , the mean yield without fertilizer, is estimable. Plots with  $N = P = 0$  were observed.

- The mean yield at  $(N, P) = (40, 20)$  is estimable, since  $40 - 2 \cdot 20 = 0$ , and so is the mean at any dose of the blend, observed or not, such as  $(50, 25)$ .
- The mean yield at  $(40, 0)$ , nitrogen alone, is not estimable, since  $40 - 0 \neq 0$ .
- Neither  $\beta_1$  nor  $\beta_2$  is estimable, but  $\beta_1 + \frac{1}{2}\beta_2$ , the gain per kilogram of nitrogen when it comes with its phosphorus, is.

The listing computes three least squares solutions: the minimum-norm one, the one that omits  $P$ , and the one that omits  $N$ . They give intercepts 2.9579, 2.9579 and 2.9579, all equal as they must be. The pairs of slopes  $(0.0296, 0.0148)$ ,  $(0.0370, 0.0000)$  and  $(0.0000, 0.0740)$  all disagree. At the blend dose  $(40, 20)$  all three predict 4.439 t/ha. At the nitrogen-only dose  $(40, 0)$  they predict 4.143, 4.439 and 2.958. The third value is just  $\hat{\beta}_0$ , because that solution puts the whole fertilizer effect on phosphorus. Each of these is a legitimate least squares answer, and none of them estimates anything.

The two kinds of prediction differ in kind, not in degree. Nothing in this experiment says what nitrogen does without phosphorus. A larger experiment of the same design would not help. What is needed is at least one plot off the line  $P = N/2$  ([Exercise \(B3\)](#)).

```
import numpy as np

N = np.array([0, 0, 20, 20, 40, 40, 60, 60, 80, 80], dtype=float) # kg/ha nitrogen
P = N / 2 # phosphorus,
    always half
X = np.column_stack([np.ones(N.size), N, P])
rng = np.random.default_rng(80)
y = 3.0 + 0.020 * N + 0.030 * P + rng.normal(scale=0.25, size=N.size) # yield, t/
    ha
print("rank(X) =", np.linalg.matrix_rank(X), " null vector (0, 1, -2):", X @ [0, 1,
    -2])
```

```
b_min = np.linalg.pinv(X) @ y # minimum-norm
    solution
b_noP = np.append(np.linalg.lstsq(X[:, :2], y, rcond=None)[0], 0.0) # drop P
b_noN = np.insert(np.linalg.lstsq(X[:, [0, 2]], y, rcond=None)[0], 1, 0.0) # drop N
new_points = {"N=40, P=20 (on the line)": [1, 40, 20],
    "N=40, P=0 (off the line)": [1, 40, 0]}
for name, x0 in new_points.items():
    print(name, [round(float(np.dot(x0, b)), 3) for b in (b_min, b_noP, b_noN)])
```

## 8.2.4 Many unbiased estimators, one least squares estimator

An estimable function usually has many linear unbiased estimators. In [Example \(A fertilizer trial with a fixed ratio\)](#), the mean at the blend dose  $(40, 20)$  is estimated without bias by the average of the two plots that received exactly that dose,  $\mathbf{a}_1^\top \mathbf{Y}$  with  $\mathbf{a}_1$  putting weight  $\frac{1}{2}$  on each. It is also estimated without bias by the value of any least squares fit at that dose. If  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , the first estimator has variance  $\sigma^2 \|\mathbf{a}_1\|^2 = 0.500 \sigma^2$ , while the least squares estimator has variance  $0.100 \sigma^2$ , five times smaller, because it uses all ten plots.

The least squares estimator can be written in the same form. If  $\mathbf{a}$  is any solution of  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ ,

then [Theorem 6.4.4](#) gives

$$\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{a}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{a}^\top \mathbf{M} \mathbf{y} = (\mathbf{M} \mathbf{a})^\top \mathbf{y}, \quad (8.2.1)$$

for every least squares solution  $\hat{\boldsymbol{\beta}}$ . So the least squares estimator is the linear unbiased estimator whose coefficient vector is the projection  $\mathbf{M} \mathbf{a}$  of any one coefficient vector onto  $\mathcal{C}(\mathbf{X})$ . It is unbiased, because  $\mathbf{X}^\top \mathbf{M} \mathbf{a} = \mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ . Every solution of  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$  has the form  $\mathbf{M} \mathbf{a} + \mathbf{w}$  with  $\mathbf{w} \in \mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$ , and Pythagoras gives  $\|\mathbf{M} \mathbf{a} + \mathbf{w}\|^2 = \|\mathbf{M} \mathbf{a}\|^2 + \|\mathbf{w}\|^2$ . This is the Gauss–Markov theorem in its simplest form: among linear unbiased estimators of an estimable function, least squares has the smallest variance when  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ . [Chapter 7](#) states and proves it in general ([Theorem 7.2.1](#)).

## 8.2.5 What estimability does not depend on

Three remarks keep the concept in its place.

*It does not depend on the covariance.* Estimability involves only  $E(\mathbf{Y}) = \mathbf{X} \boldsymbol{\beta}$ , so it is the same whether the errors are uncorrelated, correlated, heteroscedastic or normal, provided  $\text{Cov}(\mathbf{Y})$  is free of  $\boldsymbol{\beta}$ . Generalized least squares in Part VII ([Chapter 31](#)) has the same estimable functions as ordinary least squares.

*It is a property of the design, not the response.* Whether  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is estimable can be decided before any response is measured, from  $\mathbf{X}$  alone. This makes estimability a planning tool. If a comparison matters, the design must contain the rows that make it estimable.

*It is a statement about linear functions.* A turning point  $-\beta_1/(2\beta_2)$  or a quotient of two estimable quantities is not “estimable” in the technical sense, and usually has no unbiased estimator of any kind. Yet when it is identifiable it is a perfectly sensible target, to be estimated by other means. The line that must not be crossed is identifiability, not linearity. In this chapter “nonestimable” is used only for linear functions, and for them it means the same as “not identifiable”.

*Warning.* Software fitted to a rank-deficient model reports a number for every coefficient and every requested combination, estimable or not. For a nonestimable function the number is an artefact of the particular solution the software chose, as the three predictions at  $(40, 0)$  show. It comes with a standard error that is equally arbitrary ([Section 8.3](#)).

## 8.2.6 Exercises

### 8.2.6.1 A. Check your understanding

#### Exercise A1.

In [Example \(A fertilizer trial with a fixed ratio\)](#), decide which of the following are estimable:  $\beta_1 - 2\beta_2$ ;  $2\beta_1 + \beta_2$ ; the mean yield at  $(N, P) = (100, 50)$ ; the mean yield at  $(0, 10)$ ; the difference between the mean yields at  $(60, 30)$  and  $(20, 10)$ . For the estimable ones give a vector  $\mathbf{a}$  with  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ .

**Exercise A2.**

In simple linear regression  $E(y_i) = \beta_0 + \beta_1 x_i$ , show that  $\beta_1$  is estimable iff the  $x_i$  are not all equal. If all  $x_i = c$ , which linear functions are estimable, and what is their least squares estimator?

**8.2.6.2 B. Practice****Exercise B1.**

Let  $\lambda^\top \beta$  be estimable and let  $\mathbf{a}^* = \mathbf{M}\mathbf{a}$  as in Eq. 8.2.1. Show that  $\mathbf{a}^*$  does not depend on which solution  $\mathbf{a}$  of  $\mathbf{X}^\top \mathbf{a} = \lambda$  is used, that  $\mathbf{a}^* = \mathbf{X}\mathbf{G}\lambda$  for any generalized inverse  $\mathbf{G}$  of  $\mathbf{X}^\top \mathbf{X}$ , and that  $\|\mathbf{a}^*\|^2 = \lambda^\top \mathbf{G}\lambda$ .

*Solution.* Two solutions differ by a vector of  $\mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$ , which  $\mathbf{M}$  sends to  $\mathbf{0}$ , so  $\mathbf{M}\mathbf{a}$  is the same for both. With  $\mathbf{M} = \mathbf{X}\mathbf{G}\mathbf{X}^\top$  (Theorem 6.3.7),  $\mathbf{M}\mathbf{a} = \mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{a} = \mathbf{X}\mathbf{G}\lambda$ . Finally,  $\|\mathbf{M}\mathbf{a}\|^2 = \mathbf{a}^\top \mathbf{M}\mathbf{a} = \mathbf{a}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{a} = \lambda^\top \mathbf{G}\lambda$ .

**Exercise B2.**

Show that  $\Lambda^\top \beta$  is estimable iff  $\Lambda^\top \mathbf{b}_1 = \Lambda^\top \mathbf{b}_2$  whenever  $\mathbf{X}\mathbf{b}_1 = \mathbf{X}\mathbf{b}_2$ , and that in that case  $\Lambda^\top \hat{\beta} = \mathbf{P}^\top \mathbf{M}\mathbf{y}$  for any  $\mathbf{P}$  with  $\mathbf{X}^\top \mathbf{P} = \Lambda$ . Show also that  $\mathbf{P}^\top \mathbf{M}$  does not depend on the choice of  $\mathbf{P}$ .

**Exercise B3.**

A study will compare three treatments,  $A$ ,  $B$  and  $C$ , in the additive model  $E(y) = \mu + \tau_{\text{trt}} + \delta_{\text{site}}$ . Two sites are available, and each site can host two of the three treatments. Which allocations make every difference  $\tau_s - \tau_t$  estimable? Show that putting  $\{A, B\}$  at both sites leaves  $\tau_A - \tau_C$  nonestimable, while  $\{A, B\}$  at one site and  $\{B, C\}$  at the other makes all three differences estimable.

**8.2.6.3 C. Going deeper****Exercise C1.**

Suppose the distribution of  $\varepsilon$  does not depend on  $\beta$ . Show that if some statistic  $f(\mathbf{Y})$ , linear or not, satisfies  $E f(\mathbf{Y}) = g(\beta)$  for every  $\beta$ , then  $g$  is identifiable. Conclude that no statistic whatever is unbiased for a nonestimable linear function.

*Solution.* The distribution of  $\mathbf{Y} = \mathbf{X}\beta + \varepsilon$  depends on  $\beta$  only through  $\mathbf{X}\beta$ . If  $\mathbf{X}\beta_1 = \mathbf{X}\beta_2$ , then  $\mathbf{Y}$  has the same distribution under both, so  $g(\beta_1) = E_{\beta_1} f(\mathbf{Y}) = E_{\beta_2} f(\mathbf{Y}) = g(\beta_2)$ . A linear function with an unbiased estimator is therefore identifiable, hence estimable by Theorem 8.2.2.



## 8.3 Characterizations of estimability

[Theorem 8.2.2](#) reduced estimability to a single condition,  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ . In practice that condition appears in several forms, depending on what is at hand. We may have the rows of  $\mathbf{X}$ , the matrix  $\mathbf{X}^\top \mathbf{X}$  and one of its generalized inverses, a collection of least squares solutions, or an SVD. This section collects the equivalent forms in one theorem, derives the estimator of an estimable function with its variance, and then turns to the practical question of deciding estimability in floating-point arithmetic, where exact rank is not available.

### 8.3.1 Eight equivalent conditions

#### Theorem 8.3.1 — Characterizations of estimability

Let  $\mathbf{X}$  be  $n \times p$ , let  $\mathbf{G}$  denote a generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ , and let  $\lambda \in \mathbb{R}^p$ . The following are equivalent:

- (a)  $\lambda^\top \beta$  is estimable;
- (b)  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ : there is  $\rho \in \mathbb{R}^n$  with  $\lambda^\top = \rho^\top \mathbf{X}$ ;
- (c)  $\lambda \perp \mathcal{N}(\mathbf{X})$ :  $\lambda^\top \mathbf{v} = 0$  whenever  $\mathbf{X}\mathbf{v} = \mathbf{0}$ ;
- (d)  $\lambda \in \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ : the system  $\mathbf{X}^\top \mathbf{X}\mathbf{r} = \lambda$  has a solution;
- (e)  $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \lambda^\top$  for some generalized inverse  $\mathbf{G}$ ;
- (f)  $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \lambda^\top$  for every generalized inverse  $\mathbf{G}$ ;
- (g) for every  $\mathbf{y}$ ,  $\lambda^\top \mathbf{b}$  has the same value for all solutions  $\mathbf{b}$  of the normal equations  $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$  ([Theorem 6.4.2](#));
- (h)  $\text{rank} \begin{pmatrix} \mathbf{X} \\ \lambda^\top \end{pmatrix} = \text{rank}(\mathbf{X})$ .

*Proof.* (a) $\Leftrightarrow$ (b) is [Theorem 8.2.2](#). (b) $\Leftrightarrow$ (c) holds because  $\mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$  ([Corollary 6.2.4\(d\)](#)). (b) $\Leftrightarrow$ (d) holds because  $\mathcal{C}(\mathbf{X}^\top \mathbf{X}) = \mathcal{C}(\mathbf{X}^\top)$  ([Corollary 6.2.4\(b\)](#)). (d) $\Rightarrow$ (f): if  $\lambda = \mathbf{X}^\top \mathbf{X}\mathbf{r}$ , then for every generalized inverse  $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \mathbf{r}^\top (\mathbf{X}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top \mathbf{X}) = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} = \lambda^\top$ . (f) $\Rightarrow$ (e) holds because generalized inverses exist ([Proposition 6.3.5](#)). (e) $\Rightarrow$ (d): transposing  $\lambda^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \lambda^\top$  gives  $\mathbf{X}^\top \mathbf{X}(\mathbf{G}^\top \lambda) = \lambda$ , so  $\mathbf{r} = \mathbf{G}^\top \lambda$  solves the system. (b) $\Leftrightarrow$ (g) is [Theorem 6.4.4](#). (b) $\Leftrightarrow$ (h): the rank of the stacked matrix is the dimension of its row space,  $\mathcal{C}([\mathbf{X}^\top, \lambda])$ . This contains  $\mathcal{C}(\mathbf{X}^\top)$ , and it has the same dimension iff it equals  $\mathcal{C}(\mathbf{X}^\top)$ , that is, iff  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ . ■

Conditions (e) and (f) are [Corollary 1.4.2](#) applied to the normal equations. Condition (g) is the one that matters for interpretation: an estimable function is one whose estimate does not depend on the arbitrary choice of solution. Condition (c) is the one to use when proving that something is *not* estimable, since a single null vector  $\mathbf{v}$  with  $\lambda^\top \mathbf{v} \neq 0$  settles the question. Condition (h) is the most direct computational test, but in floating point it inherits all the difficulties of deciding a rank.

### 8.3.2 The estimator and its variance

For an estimable function, every least squares solution produces the same estimate, and that estimate has a clean description.

#### Proposition 8.3.2 — Least squares estimation of estimable functions

Let  $\lambda^\top \beta$  be estimable, with  $\lambda = \mathbf{X}^\top \rho = \mathbf{X}^\top \mathbf{X} \mathbf{r}$ , let  $\hat{\beta}$  be any least squares solution and  $\mathbf{G}$  any generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ . Then:

- (a)  $\lambda^\top \hat{\beta} = \rho^\top \mathbf{M} \mathbf{y} = \mathbf{r}^\top \mathbf{X}^\top \mathbf{y} = \lambda^\top \mathbf{G} \mathbf{X}^\top \mathbf{y}$ ;
- (b)  $E(\lambda^\top \hat{\beta}) = \lambda^\top \beta$ ;
- (c) if  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , then

$$\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top \mathbf{G} \lambda = \sigma^2 \mathbf{r}^\top \lambda = \sigma^2 \|\mathbf{M} \rho\|^2,$$

the same for every  $\mathbf{G}$ ;

- (d) more generally, if  $\Lambda^\top \beta$  is estimable, then  $\text{Cov}(\Lambda^\top \hat{\beta}) = \sigma^2 \Lambda^\top \mathbf{G} \Lambda$  for every  $\mathbf{G}$ .

*Proof.*

- (a) The first equality is [Theorem 6.4.4](#). For the second, the normal equations give  $\mathbf{r}^\top \mathbf{X}^\top \mathbf{y} = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} \hat{\beta} = \lambda^\top \hat{\beta}$ . The last is the second with the solution  $\hat{\beta} = \mathbf{G} \mathbf{X}^\top \mathbf{y}$  and  $\mathbf{r} = \mathbf{G}^\top \lambda$  (see the proof of [Theorem 8.3.1](#)). (b)–(d) were proved in [Chapter 7](#): unbiasedness and  $\text{Var}(\lambda^\top \hat{\beta}) = \sigma^2 \lambda^\top \mathbf{G} \lambda$  are [Theorem 7.2.1\(a\)](#), and the matrix form  $\text{Cov}(\Lambda^\top \hat{\beta}) = \sigma^2 \Lambda^\top \mathbf{G} \Lambda$  is part (d) of the same theorem. Neither depends on  $\mathbf{G}$ , because  $\mathbf{M} = \mathbf{X} \mathbf{G} \mathbf{X}^\top$  for every  $\mathbf{G}$  ([Theorem 6.3.7](#)). The second form in (c) follows from  $\lambda^\top \mathbf{G} \lambda = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X} \mathbf{r} = \mathbf{r}^\top \mathbf{X}^\top \mathbf{X} \mathbf{r} = \mathbf{r}^\top \lambda$ . The third is  $\rho^\top \mathbf{M} \rho = \|\mathbf{M} \rho\|^2$ . ■

The formula  $\sigma^2 \lambda^\top \mathbf{G} \lambda$  is the rank-deficient version of the familiar  $\sigma^2 \lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda$  ([Theorem 5.5.1](#)). The generalized inverse is arbitrary, but the quadratic form in an estimable  $\lambda$  is not. The condition  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$  under which [Chapter 7](#) proved optimality is exactly estimability, so the Gauss–Markov theorem and the normal theory ([Theorem 7.5.1](#)) apply to every estimable  $\lambda^\top \hat{\beta}$  unchanged.

### 8.3.3 What a generalized inverse reports for a non-estimable function

Software that solves the normal equations with a particular generalized inverse will print  $\lambda^\top \mathbf{G} \mathbf{X}^\top \mathbf{y}$  and  $\hat{\sigma}^2 \lambda^\top \mathbf{G} \lambda$  for any  $\lambda$  whatever. To see what these numbers mean when  $\lambda$  is not estimable, look at the matrix

$$\mathbf{H}_\mathbf{G} = \mathbf{G} \mathbf{X}^\top \mathbf{X}. \quad (8.3.1)$$

**Proposition 8.3.3 — The matrix  $\mathbf{GX}^\top \mathbf{X}$** 

For any generalized inverse  $\mathbf{G}$  of  $\mathbf{X}^\top \mathbf{X}$ , with  $\mathbf{H}_\mathbf{G}$  as in Eq. 8.3.1:

- (a)  $\mathbf{H}_\mathbf{G}$  is idempotent,  $\mathcal{N}(\mathbf{H}_\mathbf{G}) = \mathcal{N}(\mathbf{X})$ ,  $\text{rank}(\mathbf{H}_\mathbf{G}) = r$ , and the rows of  $\mathbf{H}_\mathbf{G}$  span  $\mathcal{C}(\mathbf{X}^\top)$ ;
- (b)  $\lambda^\top \beta$  is estimable iff  $\lambda^\top \mathbf{H}_\mathbf{G} = \lambda^\top$ ;
- (c) the solution  $\hat{\beta}_\mathbf{G} = \mathbf{GX}^\top \mathbf{Y}$  has  $E(\hat{\beta}_\mathbf{G}) = \mathbf{H}_\mathbf{G} \beta$ . So for every  $\lambda$ ,  $\lambda^\top \hat{\beta}_\mathbf{G}$  is unbiased for the estimable function  $\lambda^\top \mathbf{H}_\mathbf{G} \beta$ , which equals  $\lambda^\top \beta$  iff  $\lambda^\top \beta$  is estimable;
- (d) for the Moore–Penrose inverse  $\mathbf{G} = (\mathbf{X}^\top \mathbf{X})^+$ ,  $\mathbf{H}_\mathbf{G}$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{X}^\top)$ .

*Proof.*

- (a)  $\mathbf{H}_\mathbf{G}^2 = \mathbf{G}(\mathbf{X}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X}) = \mathbf{GX}^\top \mathbf{X} = \mathbf{H}_\mathbf{G}$ . By Proposition 1.3.3(a),  $\mathcal{N}(\mathbf{GX}^\top \mathbf{X}) = \mathcal{N}(\mathbf{X}^\top \mathbf{X})$ , which is  $\mathcal{N}(\mathbf{X})$  by Corollary 6.2.4(a). Rank-nullity gives rank  $r$ . The row space of  $\mathbf{H}_\mathbf{G}$  is  $\mathcal{C}(\mathbf{H}_\mathbf{G}) = \mathcal{N}(\mathbf{H}_\mathbf{G})^\perp = \mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$ .
- (b) is Theorem 8.3.1(e),(f).
- (c)  $E(\mathbf{GX}^\top \mathbf{Y}) = \mathbf{GX}^\top \mathbf{X} \beta$ . The coefficient vector  $\mathbf{H}_\mathbf{G}^\top \lambda$  lies in the row space of  $\mathbf{H}_\mathbf{G}$ , which is  $\mathcal{C}(\mathbf{X}^\top)$ , so the function is estimable. It equals  $\lambda^\top \beta$  for all  $\beta$  iff  $\mathbf{H}_\mathbf{G}^\top \lambda = \lambda$ , which is (b).
- (d) By the Penrose conditions (Definition 1.3.2),  $(\mathbf{X}^\top \mathbf{X})^+ \mathbf{X}^\top \mathbf{X}$  is symmetric and idempotent. It is therefore an orthogonal projection (Theorem 6.3.3), and its null space is  $\mathcal{N}(\mathbf{X})$  by (a). So it projects onto  $\mathcal{N}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{X}^\top)$ .

■

Part (c) answers the question. For a nonestimable  $\lambda$ , the printed number is a perfectly good unbiased estimate, but of a *different* function,  $\lambda^\top \mathbf{H}_\mathbf{G} \beta$ , and which function depends on the software's choice of  $\mathbf{G}$ . The printed “variance”  $\sigma^2 \lambda^\top \mathbf{G} \lambda$  is the variance of that statistic when  $\mathbf{G}$  is symmetric and reflexive, as the Moore–Penrose inverse is, but in general it need not be the variance of anything (Exercise (C1)). Part (d) gives the cleanest picture. With the Moore–Penrose inverse,  $\beta = \mathbf{H} \beta + (\mathbf{I} - \mathbf{H}) \beta$  splits the coefficient vector into its component in the estimable space, which the data determine, and its component in  $\mathcal{N}(\mathbf{X})$ , which they cannot see. The minimum-norm solution estimates the first component and sets the second to zero.

**Example. (Checking estimability in a one-way layout)**

Eighteen observations fall into four groups of sizes 4, 6, 3, 5, with  $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3, \mathbf{z}_4]$  and  $\beta = (\mu, \alpha_1, \dots, \alpha_4)^\top$ , so  $\text{rank}(\mathbf{X}) = 4$  and  $\mathcal{N}(\mathbf{X})$  is spanned by  $(1, -1, -1, -1, -1)^\top$ . By condition (c),  $\lambda^\top \beta$  is estimable iff  $\lambda_0 = \lambda_1 + \lambda_2 + \lambda_3 + \lambda_4$ . The listing tests five vectors in two ways: condition (c) through the SVD of  $\mathbf{X}$ , and condition (e) with three generalized inverses (Moore–Penrose, the one that deletes the intercept row and column, and a non-symmetric one built as in Proposition 1.3.3(e)). The two tests always agree, and condition (e) holds for all three inverses or for none, as (f) requires.

| $\lambda^\top \beta$                        | estimable | $\lambda^\top \mathbf{G} \lambda$ :<br>Moore–Penrose | drop intercept | non-symmetric |
|---------------------------------------------|-----------|------------------------------------------------------|----------------|---------------|
| $\mu + \alpha_1$                            | yes       | 0.250                                                | 0.250          | 0.250         |
| $\alpha_1 - \alpha_2$                       | yes       | 0.417                                                | 0.417          | 0.417         |
| $\alpha_1 + \alpha_2 - 2\alpha_3$           | yes       | 1.750                                                | 1.750          | 1.750         |
| $\alpha_1$                                  | no        | 0.188                                                | 0.250          | 0.630         |
| $\alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$ | no        | 0.038                                                | 0.950          | 2.767         |

For the estimable rows the quadratic form is the same for all three and matches the direct computation:  $\text{Var}(\bar{y}_1 - \bar{y}_2) = \sigma^2(\frac{1}{4} + \frac{1}{6})$ , for instance. For  $\alpha_1$  the “variances” disagree. With the drop-intercept inverse, the solution is  $\hat{\beta}_{\mathbf{G}} = (0, \bar{y}_1, \dots, \bar{y}_4)^\top$ , so the reported “ $\hat{\alpha}_1$ ” is  $\bar{y}_1$ , which estimates  $\mu + \alpha_1$ . Its reported variance  $0.250 \sigma^2$  is the variance of  $\bar{y}_1$ , which is correct, but  $\bar{y}_1$  estimates  $\mu + \alpha_1$ , not  $\alpha_1$ .

```
import numpy as np

groups = np.repeat([0, 1, 2, 3], [4, 6, 3, 5])          # four groups, sizes 4, 6, 3,
5
X = np.column_stack([np.ones(groups.size), np.eye(4)[groups]]) # [1, z1, z2, z3, z4
]
A = X.T @ X

def is_estimable(lam, X, tol=None):
    """lambda is estimable iff it is orthogonal to N(X); use the SVD of X."""
    U, s, Vt = np.linalg.svd(X)
    if tol is None:
        tol = max(X.shape) * np.finfo(float).eps * s[0]
    r = int(np.sum(s > tol))
    V0 = Vt[r:].T                                     # orthonormal basis of N(X)
    return np.linalg.norm(V0.T @ lam) <= 1e-8 * max(1.0, np.linalg.norm(lam))

G_mp = np.linalg.pinv(A)                             # Moore-Penrose inverse of X^T X
G_ref = np.zeros((5, 5))
G_ref[1:, 1:] = np.linalg.inv(A[1:, 1:])             # delete the intercept row and
column
rng = np.random.default_rng(3)
G_odd = G_mp + (np.eye(5) - G_mp @ A) @ rng.normal(size=(5, 5)) # non-symmetric g-
inverse

candidates = {"mu + alpha_1": [1, 1, 0, 0, 0],
              "alpha_1 - alpha_2": [0, 1, -1, 0, 0],
              "alpha_1 + alpha_2 - 2 alpha_3": [0, 1, 1, -2, 0],
              "alpha_1": [0, 1, 0, 0, 0],
              "alpha_1 + ... + alpha_4": [0, 1, 1, 1, 1]}
for name, lam in candidates.items():
    lam = np.array(lam, dtype=float)
    H_checks = [np.allclose(lam @ G @ A, lam) for G in (G_mp, G_ref, G_odd)]
    variances = [lam @ G @ lam for G in (G_mp, G_ref, G_odd)]
    print(f"{name:30s} SVD test: {is_estimable(lam, X)!s:5s}",
          f" lambda'GX'X = lambda': {H_checks}",
          f" lambda'G lambda: {np.round(variances, 3)}")
```

### 8.3.4 Deciding estimability in floating point

Every condition in [Theorem 8.3.1](#) is a statement of exact linear algebra. On a computer, the rank of  $\mathbf{X}$  must be decided with a tolerance, and the answer to “is  $\lambda$  estimable?” inherits that tolerance. The most reliable test works with  $\mathbf{X}$  itself, not with  $\mathbf{X}^\top \mathbf{X}$ . Compute the singular value decomposition  $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$ . Declare the singular values below a threshold  $\tau$  to be zero. With the usual choice  $\tau = \max(n, p) \epsilon d_1$ , where  $d_1$  is the largest singular value and  $\epsilon \approx 2.2 \times 10^{-16}$  is the machine precision, the columns  $\mathbf{V}_0$  of  $\mathbf{V}$  that belong to the discarded singular values form an orthonormal basis of the numerical null space. Then  $\lambda$  is declared estimable iff  $\|\mathbf{V}_0^\top \lambda\|$  is negligible relative to  $\|\lambda\|$ . This is condition (c), and it is what the function `is_estimable` in the listing does. [Chapter 10](#) treats the SVD and its rank decisions in detail ([Theorem 10.4.1](#)).

Two things can go wrong, and both matter.

*Exact dependence can be blurred by rounding.* If a column is computed, say as a sum of other columns, rounding may leave a singular value of order  $10^{-15}$  rather than zero. The tolerance exists to absorb this.

*Near dependence is not dependence.* If the columns are nearly but not exactly dependent, every  $\lambda$  is estimable in the exact sense, but some are estimable only with enormous variance. Estimability is then a matter of degree, not a yes-or-no property.

#### Example. (Estimable in principle, not in practice)

Take twelve equally spaced values  $t_i$  on  $[-1, 1]$  and the model matrix with columns  $\mathbf{1}$ ,  $\mathbf{t}$  and  $\mathbf{t} + \delta \mathbf{w}$ , where  $w_i = \cos(\pi t_i)$  and  $\delta$  is a small perturbation. At  $\delta = 0$  the last two columns coincide. Then  $\beta_1 + \beta_2$  is estimable and  $\beta_1$  is not. For  $\delta > 0$  the matrix has full rank, so everything is estimable. But [Figure 8.3.1](#) shows how the variance of  $\hat{\beta}_1$  grows as  $\delta \rightarrow 0$ : in proportion to  $1/\delta^2$  (the fitted slope on the log-log scale is  $-2.00$ ). At  $\delta = 10^{-4}$ , the smallest singular value of  $\mathbf{X}$  is  $1.8 \times 10^{-4}$ , and  $\text{Var}(\hat{\beta}_1) = 1.56 \times 10^7 \sigma^2$ , while  $\text{Var}(\hat{\beta}_1 + \hat{\beta}_2) = 0.212 \sigma^2$  is unaffected. The function that is estimable at  $\delta = 0$  stays well estimated, and the one that is not becomes useless long before it becomes nonestimable.

The listing adds a warning about computation. At  $\delta = 10^{-8}$ , the SVD of  $\mathbf{X}$  still sees three nonzero singular values, so  $\beta_1$  is (barely) estimable. But the matrix  $\mathbf{X}^\top \mathbf{X}$  has eigenvalues equal to the *squared* singular values. Its smallest eigenvalue is about  $3 \times 10^{-16}$ , below any sensible tolerance, and `pinv(X.T @ X)` silently treats  $\mathbf{X}^\top \mathbf{X}$  as having rank 2. It then reports 0.0529 as the “variance” of  $\hat{\beta}_1$ , which is the Moore–Penrose quadratic form of a nonestimable function. The true variance, computed from the SVD of  $\mathbf{X}$  itself, is more than  $10^{16}$  times larger. Forming  $\mathbf{X}^\top \mathbf{X}$  squares the condition number, as [Section 6.10](#) showed, and here that turns a nearly nonestimable function into a numerically nonestimable one.

```
t = np.linspace(-1, 1, 12)
w = np.cos(np.pi * t)                                     # a fixed direction to perturb
                        along
def near_design(delta):
    """Columns 1, t and t + delta*w: exactly rank deficient at delta = 0."""
    return np.column_stack([np.ones_like(t), t, t + delta * w])

for delta in [1.0, 1e-2, 1e-4, 1e-8, 0.0]:
    Xd = near_design(delta)
```

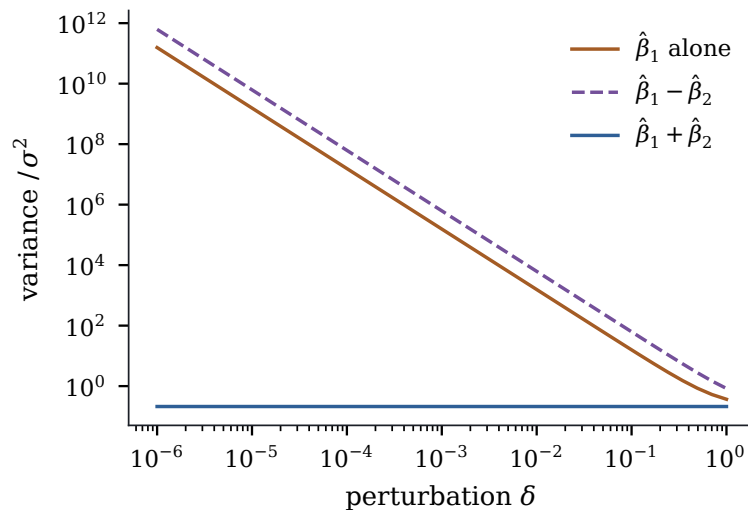


Figure 8.3.1: Variances of three linear functions of  $\hat{\beta}$  in the design with columns  $\mathbf{1}, \mathbf{t}, \mathbf{t} + \delta \mathbf{w}$ . As  $\delta \rightarrow 0$ , the functions that are not estimable at  $\delta = 0$  ( $\beta_1$  alone and  $\beta_1 - \beta_2$ ) have variances growing like  $\delta^{-2}$ , while  $\beta_1 + \beta_2$ , which is estimable at  $\delta = 0$ , is unaffected.

```
s = np.linalg.svd(Xd, compute_uv=False)
G = np.linalg.pinv(Xd.T @ Xd)
v_sum = np.array([0, 1, 1]) @ G @ [0, 1, 1]      # beta_1 + beta_2
v_one = np.array([0, 1, 0]) @ G @ [0, 1, 0]      # beta_1 alone
print(f"delta={delta:7.0e}  smallest singular value {s[-1]:8.1e}",
      f" Var(b1+b2)/s2 = {v_sum:8.3f}  Var(b1)/s2 = {v_one:9.2e}",
      f" beta_1 estimable? {is_estimable(np.array([0., 1, 0]), Xd)}")
```

The practical rule has two parts. Decide estimability with the SVD of  $\mathbf{X}$ , never with  $\mathbf{X}^\top \mathbf{X}$ . Then look at the variance, not only the verdict. A function that passes the test with a variance of  $10^7 \sigma^2$  has, for practical purposes, not been estimated. Collinearity, in [Chapter 26](#), is the study of this middle ground.

## 8.3.5 Exercises

### 8.3.5.1 A. Check your understanding

#### Exercise A1.

In [Example \(Checking estimability in a one-way layout\)](#), decide which of the following are estimable, using condition (c):  $2\mu + \alpha_1 + \alpha_2$ ;  $\alpha_1 + \alpha_2 - \alpha_3 - \alpha_4$ ;  $\mu$ ;  $4\mu + \alpha_1 + \alpha_2 + \alpha_3 + \alpha_4$ . For each estimable one give its least squares estimator in terms of the group means and its variance.

### 8.3.5.2 B. Practice

#### Exercise B1.

In [Example \(Checking estimability in a one-way layout\)](#), compute  $\mathbf{H}_G$  for the drop-intercept generalized inverse, and identify the estimable function  $\lambda^\top \mathbf{H}_G \beta$  that the soft-

ware is really estimating when it reports “ $\hat{\alpha}_k$ ”. Do the same for the generalized inverse that deletes the row and column of  $\alpha_4$ .

*Solution.* With  $\mathbf{G}$  holding  $\text{diag}(1/n_1, \dots, 1/n_4)$  in its lower right block,  $\mathbf{GX}^\top\mathbf{X}$  has first row zero and row  $k+1$  equal to  $(1, \mathbf{e}_k^\top)$ , because the row of  $\mathbf{X}^\top\mathbf{X}$  for  $\alpha_k$  is  $n_k(1, \mathbf{e}_k^\top)$ . So  $E(\hat{\alpha}_k) = \mu + \alpha_k$ : the reported “effect” estimates the group mean. If instead  $\alpha_4$  is deleted, the solution sets  $\hat{\alpha}_4 = 0$ ,  $\hat{\mu} = \bar{y}_4$  and  $\hat{\alpha}_k = \bar{y}_k - \bar{y}_4$ , so  $E(\hat{\mu}) = \mu + \alpha_4$  and  $E(\hat{\alpha}_k) = \alpha_k - \alpha_4$  for  $k \leq 3$ .

### Exercise B2.

Let  $\mathbf{\Lambda}^\top\boldsymbol{\beta}$  be estimable with  $\mathbf{\Lambda}$  of full column rank  $q$ , and suppose  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$  with  $\sigma^2 > 0$ . Show that  $\text{Cov}(\mathbf{\Lambda}^\top\hat{\boldsymbol{\beta}}) = \sigma^2\mathbf{\Lambda}^\top\mathbf{G}\mathbf{\Lambda}$  is positive definite, whichever generalized inverse is used. *Hint:* write  $\mathbf{\Lambda} = \mathbf{X}^\top\mathbf{P}$  and show  $\mathbf{MP}$  has full column rank.

### 8.3.5.3 C. Going deeper

#### Exercise C1.

Let  $\mathbf{\Lambda}^\top\boldsymbol{\beta}$  be nonestimable. Show that as  $\mathbf{G}$  ranges over all generalized inverses of  $\mathbf{X}^\top\mathbf{X}$ , the number  $\boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda}$  takes *every* real value, including negative ones. So the “variance” that software attaches to a nonestimable function can be anything at all.

*Solution.* Let  $\mathbf{A} = \mathbf{X}^\top\mathbf{X}$  and  $\mathbf{G}_0$  be one generalized inverse. By Proposition 1.3.3(e),  $\mathbf{G} = \mathbf{G}_0 + (\mathbf{I} - \mathbf{G}_0\mathbf{A})\mathbf{U}$  is a generalized inverse for every  $\mathbf{U}$ , and

$$\boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda} = \boldsymbol{\lambda}^\top\mathbf{G}_0\boldsymbol{\lambda} + \mathbf{u}^\top\mathbf{U}\boldsymbol{\lambda}, \quad \mathbf{u} = (\mathbf{I} - \mathbf{G}_0\mathbf{A})^\top\boldsymbol{\lambda}.$$

Because  $\boldsymbol{\lambda}$  is not estimable,  $\boldsymbol{\lambda}^\top\mathbf{G}_0\mathbf{A} \neq \boldsymbol{\lambda}^\top$  (by (a) $\Leftrightarrow$ (e) of Theorem 8.3.1), so  $\mathbf{u} \neq \mathbf{0}$ . Also  $\boldsymbol{\lambda} \neq \mathbf{0}$ , since  $\mathbf{0}$  is in  $\mathcal{C}(\mathbf{X}^\top)$ . Take  $\mathbf{U} = t\mathbf{u}\boldsymbol{\lambda}^\top/(\|\mathbf{u}\|^2\|\boldsymbol{\lambda}\|^2)$ . Then  $\boldsymbol{\lambda}^\top\mathbf{G}\boldsymbol{\lambda} = \boldsymbol{\lambda}^\top\mathbf{G}_0\boldsymbol{\lambda} + t$ , which takes every real value as  $t$  varies.

#### Exercise C2.

Let  $\mathbf{P}$  be the orthogonal projection onto  $\mathcal{C}(\mathbf{X}^\top)$ . Show that the minimum-norm least squares solution  $\hat{\boldsymbol{\beta}}^+$  satisfies  $E(\hat{\boldsymbol{\beta}}^+) = \mathbf{P}\boldsymbol{\beta}$ , and that for any  $\boldsymbol{\lambda}$ ,  $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}}^+$  is the least squares estimator of the estimable function  $(\mathbf{P}\boldsymbol{\lambda})^\top\boldsymbol{\beta}$ . Interpret  $\mathbf{P}\boldsymbol{\lambda}$  as the estimable function “closest” to  $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$ .

## 8.4 Reparameterization and side conditions

A rank-deficient model can be handled in three ways. One can work with it as it stands, estimating only estimable functions, which is what the last two sections did. One can *reparameterize*, replacing  $\mathbf{X}$  by a full-rank matrix with the same column space. Or one can impose *side conditions*, linear restrictions on  $\boldsymbol{\beta}$  that select a single coefficient vector from each set  $\boldsymbol{\beta} + \mathcal{N}(\mathbf{X})$ . This section shows that the second and third are two descriptions of the same act: choosing

coordinates for the mean vector. Neither changes any estimable quantity. It also records what the most common software does by default.

### 8.4.1 Full-rank reparameterizations

By [Corollary 8.2.3](#), at most  $r$  estimable functions are linearly independent. A set of exactly  $r$  of them can serve as a new parameter vector.

#### Theorem 8.4.1 — Reparameterizing to full rank

Let  $\text{rank}(\mathbf{X}) = r$ . Let  $\mathbf{U}$  be an  $r \times p$  matrix of rank  $r$  whose rows lie in  $\mathcal{C}(\mathbf{X}^\top)$ , so that  $\boldsymbol{\gamma} = \mathbf{U}\boldsymbol{\beta}$  is a vector of  $r$  linearly independent estimable functions, and put

$$\mathbf{Z} = \mathbf{X}\mathbf{U}^\top(\mathbf{U}\mathbf{U}^\top)^{-1}. \quad (8.4.1)$$

Then:

- (a)  $\mathbf{X} = \mathbf{Z}\mathbf{U}$ ,  $\mathbf{Z}$  has full column rank  $r$ , and  $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X})$ ;
- (b) the models  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ ,  $\boldsymbol{\beta} \in \mathbb{R}^p$ , and  $E(\mathbf{Y}) = \mathbf{Z}\boldsymbol{\gamma}$ ,  $\boldsymbol{\gamma} \in \mathbb{R}^r$ , describe the same set of mean vectors, with  $\boldsymbol{\gamma} = \mathbf{U}\boldsymbol{\beta}$ ;
- (c) the least squares estimate in the new model is  $\hat{\boldsymbol{\gamma}} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top\mathbf{y} = \mathbf{U}\hat{\boldsymbol{\beta}}$  for every least squares solution  $\hat{\boldsymbol{\beta}}$  of the old one;
- (d)  $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$  is estimable iff  $\boldsymbol{\lambda} = \mathbf{U}^\top\mathbf{c}$  for some  $\mathbf{c} \in \mathbb{R}^r$ , which is then unique, and in that case  $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} = \mathbf{c}^\top\hat{\boldsymbol{\gamma}}$ ;
- (e) conversely, if  $\mathbf{Z}$  is any  $n \times r$  matrix of full column rank with  $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X})$ , then  $\mathbf{X} = \mathbf{Z}\mathbf{U}$  for a unique  $\mathbf{U}$ , and  $\mathbf{U}$  has rank  $r$  and estimable rows.

*Proof.* Since  $\mathcal{C}(\mathbf{U}^\top) \subseteq \mathcal{C}(\mathbf{X}^\top)$  and both have dimension  $r$ ,  $\mathcal{C}(\mathbf{U}^\top) = \mathcal{C}(\mathbf{X}^\top)$ . The matrix  $\mathbf{Q} = \mathbf{U}^\top(\mathbf{U}\mathbf{U}^\top)^{-1}\mathbf{U}$  is symmetric and idempotent with column space  $\mathcal{C}(\mathbf{U}^\top)$ , so it is the orthogonal projection onto  $\mathcal{C}(\mathbf{X}^\top)$  ([Theorem 6.3.3](#)). Hence  $\mathbf{I} - \mathbf{Q}$  projects onto  $\mathcal{C}(\mathbf{X}^\top)^\perp = \mathcal{N}(\mathbf{X})$ , and  $\mathbf{X}(\mathbf{I} - \mathbf{Q}) = \mathbf{0}$ , that is,  $\mathbf{X} = \mathbf{X}\mathbf{Q} = \mathbf{Z}\mathbf{U}$ . (a) Now  $r = \text{rank}(\mathbf{X}) = \text{rank}(\mathbf{Z}\mathbf{U}) \leq \text{rank}(\mathbf{Z}) \leq r$ , so  $\mathbf{Z}$  has full column rank.  $\mathcal{C}(\mathbf{Z}) \subseteq \mathcal{C}(\mathbf{X})$  because  $\mathbf{Z} = \mathbf{X}(\cdot)$ , and  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$  because  $\mathbf{X} = \mathbf{Z}\mathbf{U}$ . (b)  $\mathbf{X}\boldsymbol{\beta} = \mathbf{Z}(\mathbf{U}\boldsymbol{\beta})$ , and  $\boldsymbol{\beta} \mapsto \mathbf{U}\boldsymbol{\beta}$  maps  $\mathbb{R}^p$  onto  $\mathbb{R}^r$  because  $\mathbf{U}$  has full row rank. So both models describe  $\mathcal{C}(\mathbf{X})$ . (c)  $\mathbf{Z}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{Z}(\mathbf{U}\hat{\boldsymbol{\beta}})$  by [Theorem 6.4.1](#), and  $\mathbf{Z}$  has full column rank. (d)  $\boldsymbol{\lambda}$  is estimable iff  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{U}^\top)$ . The vector  $\mathbf{c}$  is unique because  $\mathbf{U}^\top$  has full column rank, and  $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} = \mathbf{c}^\top\mathbf{U}\hat{\boldsymbol{\beta}} = \mathbf{c}^\top\hat{\boldsymbol{\gamma}}$ . (e) Each column of  $\mathbf{X}$  lies in  $\mathcal{C}(\mathbf{Z})$ , so  $\mathbf{X} = \mathbf{Z}\mathbf{U}$  with  $\mathbf{U} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top\mathbf{X}$ , unique because  $\mathbf{Z}$  has full column rank. Then  $\mathbf{U}\boldsymbol{\beta} = (\mathbf{Z}^\top\mathbf{Z})^{-1}\mathbf{Z}^\top(\mathbf{X}\boldsymbol{\beta})$  is a linear function of  $\mathbf{X}\boldsymbol{\beta}$ , so each row of  $\mathbf{U}$  is estimable. Finally  $r = \text{rank}(\mathbf{X}) \leq \text{rank}(\mathbf{U}) \leq r$ . ■

So reparameterization is the same thing as choosing a basis for the estimable space. [Theorem 6.7.1](#) of [Chapter 6](#) already showed that fitted values and residuals do not care which basis is chosen. Part (d) adds that estimable functions do not care either: each is a fixed linear combination of the new parameters, and its estimate is the same combination of their estimates. Everything in the full-rank theory of [Chapter 5](#) now applies to  $\hat{\boldsymbol{\gamma}}$ , including [Theorem 5.5.1](#) and



the variance estimator of [Theorem 5.6.1](#), with  $p$  replaced by  $r$ .

For the one-way layout with  $\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_g)^\top$ , two choices of  $\mathbf{U}$  stand out. The rows  $(1, \mathbf{e}_k^\top)$ ,  $k = 1, \dots, g$ , give  $\boldsymbol{\gamma} = (\mu + \alpha_1, \dots, \mu + \alpha_g)^\top$ , the cell means, and  $\mathbf{Z}$  is the indicator matrix. The rows  $(1, \mathbf{e}_1^\top)$  and  $(0, \mathbf{e}_k^\top - \mathbf{e}_1^\top)$ ,  $k = 2, \dots, g$ , give the first group's mean and the differences from it. This is the reference coding of [Section 8.5](#).

## 8.4.2 Side conditions

The other classical device leaves  $\mathbf{X}$  alone and adds equations. A **side condition** is a set of linear restrictions  $\mathbf{T}\mathbf{b} = \mathbf{0}$ , with  $\mathbf{T}$  a  $q \times p$  matrix, imposed on the solutions of the normal equations. The aim is to single out one solution. Examples in the one-way layout are  $\alpha_1 = 0$ ,  $\sum_k \alpha_k = 0$  and  $\mu = 0$ . The theorem says exactly which restrictions achieve this.

### Theorem 8.4.2 — Side conditions

Let  $\mathbf{X}$  be  $n \times p$  of rank  $r$  and  $\mathbf{T}$  be  $q \times p$ . The following are equivalent:

- (a) for every  $\mathbf{y} \in \mathbb{R}^n$  there is exactly one  $\mathbf{b}$  with  $\mathbf{X}^\top \mathbf{X}\mathbf{b} = \mathbf{X}^\top \mathbf{y}$  and  $\mathbf{T}\mathbf{b} = \mathbf{0}$ ;
- (b) for every  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$  there is exactly one  $\mathbf{b}$  with  $\mathbf{X}\mathbf{b} = \boldsymbol{\mu}$  and  $\mathbf{T}\mathbf{b} = \mathbf{0}$ ;
- (c)  $\mathbb{R}^p = \mathcal{N}(\mathbf{X}) \oplus \mathcal{N}(\mathbf{T})$ ;
- (d)  $\text{rank} \begin{pmatrix} \mathbf{X} \\ \mathbf{T} \end{pmatrix} = p$  and  $\mathcal{C}(\mathbf{T}^\top) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$ ;
- (e)  $\text{rank}(\mathbf{T}) = p - r$ , and no nonzero linear combination of the rows of  $\mathbf{T}$  is estimable.

When these hold,  $\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T}$  is nonsingular, the solution in (a) is

$$\hat{\boldsymbol{\beta}}_{\mathbf{T}} = (\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1} \mathbf{X}^\top \mathbf{y}, \quad (8.4.2)$$

and  $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1}$  is a generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ .

*Proof.* (a) $\Leftrightarrow$ (b). The solutions of the normal equations for  $\mathbf{y}$  are the  $\mathbf{b}$  with  $\mathbf{X}\mathbf{b} = \mathbf{M}\mathbf{y}$  ([Theorem 6.4.2](#)), and  $\mathbf{M}\mathbf{y}$  runs over all of  $\mathcal{C}(\mathbf{X})$  as  $\mathbf{y}$  does.

(b) $\Leftrightarrow$ (c). The set  $\{\mathbf{b} : \mathbf{X}\mathbf{b} = \boldsymbol{\mu}\}$  is a coset  $\mathbf{b}_0 + \mathcal{N}(\mathbf{X})$ , and every  $\mathbf{b}_0 \in \mathbb{R}^p$  arises (take  $\boldsymbol{\mu} = \mathbf{X}\mathbf{b}_0$ ). So (b) says that every coset of  $\mathcal{N}(\mathbf{X})$  meets  $\mathcal{N}(\mathbf{T})$  in exactly one point. Some point  $\mathbf{b}_0 + \mathbf{v}$  of the coset lies in  $\mathcal{N}(\mathbf{T})$  iff  $\mathbf{b}_0 \in \mathcal{N}(\mathbf{T}) + \mathcal{N}(\mathbf{X})$ , so existence for every coset means  $\mathbb{R}^p = \mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T})$ . Two points of the same coset that both lie in  $\mathcal{N}(\mathbf{T})$  differ by an element of  $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T})$ , and a nonzero element of that intersection added to one such point gives a second. So uniqueness means  $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) = \{\mathbf{0}\}$ . Together, these say the sum is direct.

(c) $\Leftrightarrow$ (d). By [Lemma 6.2.1](#) and [Lemma 6.2.5](#),

$$(\mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T}))^\perp = \mathcal{C}(\mathbf{X}^\top) \cap \mathcal{C}(\mathbf{T}^\top), \quad (\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}))^\perp = \mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top).$$

So  $\mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T}) = \mathbb{R}^p$  iff  $\mathcal{C}(\mathbf{X}^\top) \cap \mathcal{C}(\mathbf{T}^\top) = \{\mathbf{0}\}$ , and  $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) = \{\mathbf{0}\}$  iff  $\mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top) = \mathbb{R}^p$ . The last space is the row space of the stacked matrix, so this holds iff that matrix has rank  $p$ .

(d) $\Leftrightarrow$ (e). A vector in  $\mathcal{C}(\mathbf{T}^\top)$  is a combination of the rows of  $\mathbf{T}$ , and it is estimable iff it lies in  $\mathcal{C}(\mathbf{X}^\top)$ . So the second halves of (d) and (e) say the same thing. When they hold, the sum  $\mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top)$  is direct, with dimension  $r + \text{rank}(\mathbf{T})$ . This equals  $p$  iff  $\text{rank}(\mathbf{T}) = p - r$ .

For the formula, suppose (d). If  $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})\mathbf{v} = \mathbf{0}$ , then  $0 = \mathbf{v}^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} + \mathbf{v}^\top \mathbf{T}^\top \mathbf{T} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2 + \|\mathbf{T} \mathbf{v}\|^2$ , so  $\mathbf{v}$  lies in  $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) = \{\mathbf{0}\}$ . Hence the matrix is nonsingular. The solution  $\mathbf{b}$  of (a) satisfies  $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})\mathbf{b} = \mathbf{X}^\top \mathbf{y} + \mathbf{T}^\top \mathbf{0}$ , which gives Eq. 8.4.2. Finally, write  $\mathbf{G}_\mathbf{T} = (\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1}$ . For every  $\mathbf{y}$ ,  $\mathbf{G}_\mathbf{T} \mathbf{X}^\top \mathbf{y}$  solves the normal equations, so  $\mathbf{X}^\top \mathbf{X} \mathbf{G}_\mathbf{T} \mathbf{X}^\top = \mathbf{X}^\top$ . Multiplying on the right by  $\mathbf{X}$  gives  $\mathbf{X}^\top \mathbf{X} \mathbf{G}_\mathbf{T} \mathbf{X}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{X}$ . ■

Condition (e) is the familiar rule of thumb, made exact: a side condition must supply exactly as many restrictions as the rank deficiency, and none of them, singly or in combination, may be estimable. The reason for the second requirement is clear from (d). An estimable restriction constrains the mean vector itself, and so it changes the model instead of merely choosing coordinates.

*Remark* (Existence and uniqueness separately). The proof shows more than the theorem states. The condition  $\mathcal{C}(\mathbf{T}^\top) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$  alone guarantees that the restricted normal equations always have *at least* one solution. The condition  $\text{rank}\left(\begin{smallmatrix} \mathbf{X} \\ \mathbf{T} \end{smallmatrix}\right) = p$  alone guarantees *at most* one. A constraint that is too weak, such as a single restriction when  $p - r = 2$ , leaves many solutions. One with an estimable part leaves none for most  $\mathbf{y}$  (Exercise (C2)).

### 8.4.3 What the constrained coefficients mean

Under a side condition every coefficient becomes identifiable, and it acquires a meaning.

#### Proposition 8.4.3 — Constrained coefficients are estimable functions

Suppose  $\mathbf{T}$  satisfies the conditions of Theorem 8.4.2, and let  $\mathbf{P}_\mathbf{T} = (\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1} \mathbf{X}^\top \mathbf{X}$ . For each  $\boldsymbol{\beta}$ ,  $\mathbf{P}_\mathbf{T} \boldsymbol{\beta}$  is the unique vector  $\mathbf{b}$  with  $\mathbf{X} \mathbf{b} = \mathbf{X} \boldsymbol{\beta}$  and  $\mathbf{T} \mathbf{b} = \mathbf{0}$ . The matrix  $\mathbf{P}_\mathbf{T}$  is idempotent, with range  $\mathcal{N}(\mathbf{T})$  and null space  $\mathcal{N}(\mathbf{X})$ . Every row of  $\mathbf{P}_\mathbf{T}$  lies in  $\mathcal{C}(\mathbf{X}^\top)$ . So each constrained coefficient  $\mathbf{e}_j^\top \mathbf{P}_\mathbf{T} \boldsymbol{\beta}$  is an estimable function of the original parameters, and  $\hat{\boldsymbol{\beta}}_\mathbf{T}$  estimates it without bias.

*Proof.* With  $\mathbf{y} = \mathbf{X} \boldsymbol{\beta}$ , the vector  $\mathbf{P}_\mathbf{T} \boldsymbol{\beta} = \mathbf{G}_\mathbf{T} \mathbf{X}^\top (\mathbf{X} \boldsymbol{\beta})$  is the unique solution of Theorem 8.4.2(a) for that  $\mathbf{y}$ . It satisfies  $\mathbf{X} \mathbf{b} = \mathbf{M} \mathbf{X} \boldsymbol{\beta} = \mathbf{X} \boldsymbol{\beta}$  and  $\mathbf{T} \mathbf{b} = \mathbf{0}$ . The matrix  $\mathbf{P}_\mathbf{T}$  is  $\mathbf{H}_\mathbf{G}$  of Eq. 8.3.1 for the generalized inverse  $\mathbf{G}_\mathbf{T}$ , so Proposition 8.3.3 gives idempotence, the null space  $\mathcal{N}(\mathbf{X})$  and rows in  $\mathcal{C}(\mathbf{X}^\top)$ . Its range lies in  $\mathcal{N}(\mathbf{T})$ . The range has dimension  $r$ , and so does  $\mathcal{N}(\mathbf{T})$ , because  $\text{rank}(\mathbf{T}) = p - r$ . So the two coincide. Unbiasedness is Proposition 8.3.3(c). ■

In words,  $\mathbf{P}_\mathbf{T}$  is the projection of  $\mathbb{R}^p$  onto  $\mathcal{N}(\mathbf{T})$  along  $\mathcal{N}(\mathbf{X})$ , generally an oblique one. It sends

the true  $\beta$  to the unique representative of  $\beta + \mathcal{N}(\mathbf{X})$  that satisfies the side condition. That representative's coordinates are functions of the mean, so they can be estimated. Their *meaning* is fixed by  $\mathbf{T}$ . In the one-way layout, the condition  $\sum_k \alpha_k = 0$  makes  $\mu$  the unweighted average of the  $g$  cell means and  $\alpha_k$  the deviation of cell  $k$  from it. The condition  $\sum_k n_k \alpha_k = 0$  makes  $\mu$  the size-weighted average instead. The condition  $\alpha_1 = 0$  makes  $\mu$  the first cell mean. The data do not choose among these. The analyst chooses, by deciding which question the coefficients should answer.

One side condition deserves special mention, because it is the one many programs impose without saying so.

### Proposition 8.4.4 — Minimum norm as a side condition

Let the columns of the  $p \times (p-r)$  matrix  $\mathbf{V}_0$  be a basis of  $\mathcal{N}(\mathbf{X})$ . Then  $\mathbf{T} = \mathbf{V}_0^\top$  satisfies the conditions of [Theorem 8.4.2](#), and the resulting solution  $\hat{\beta}_{\mathbf{T}}$  is the minimum-norm least squares solution  $\mathbf{X}^+\mathbf{y}$ .

*Proof.*  $\mathcal{C}(\mathbf{T}^\top) = \mathcal{C}(\mathbf{V}_0) = \mathcal{N}(\mathbf{X})$ , which has dimension  $p-r$  and meets  $\mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$  only in  $\mathbf{0}$ . So (d) holds. The least squares solutions are  $\hat{\beta}_{\mathbf{T}} + \mathcal{N}(\mathbf{X})$ , and  $\hat{\beta}_{\mathbf{T}} \in \mathcal{N}(\mathbf{T}) = \mathcal{C}(\mathbf{V}_0)^\perp = \mathcal{N}(\mathbf{X})^\perp$ . For  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ , Pythagoras gives  $\|\hat{\beta}_{\mathbf{T}} + \mathbf{v}\|^2 = \|\hat{\beta}_{\mathbf{T}}\|^2 + \|\mathbf{v}\|^2$ , so  $\hat{\beta}_{\mathbf{T}}$  has the smallest norm. By [Proposition 6.10.1](#) this is  $\mathbf{X}^+\mathbf{y}$ . ■

In the one-way layout the null space is spanned by  $(1, -1, \dots, -1)^\top$ , so the minimum-norm solution is the one satisfying  $\mu = \alpha_1 + \dots + \alpha_g$ . No one would choose this restriction on purpose, and its coefficients have no natural interpretation.

### Example. (Five side conditions, one fit)

Eighteen simulated observations fall into four groups of sizes 5, 3, 6, 4, with group means 11.111, 15.204, 10.558 and 13.624. In the model  $E(y_{kj}) = \mu + \alpha_k$ ,  $p-r=1$ , so a side condition is a single nonestimable restriction. The listing solves [Eq. 8.4.2](#) for five of them and computes the minimum-norm solution.

| Side condition                           | $\hat{\mu}$ | $\hat{\alpha}_1$ | $\hat{\alpha}_2$ | $\hat{\alpha}_3$ | $\hat{\alpha}_4$ |
|------------------------------------------|-------------|------------------|------------------|------------------|------------------|
| $\alpha_1 = 0$                           | 11.111      | 0.000            | 4.093            | -0.553           | 2.513            |
| $\alpha_4 = 0$                           | 13.624      | -2.513           | 1.580            | -3.066           | 0.000            |
| $\sum_k \alpha_k = 0$                    | 12.624      | -1.513           | 2.580            | -2.066           | 1.000            |
| $\sum_k n_k \alpha_k = 0$                | 12.167      | -1.056           | 3.037            | -1.609           | 1.457            |
| $\mu = 0$                                | 0.000       | 11.111           | 15.204           | 10.558           | 13.624           |
| minimum norm ( $\mu = \sum_k \alpha_k$ ) | 10.100      | 1.012            | 5.105            | 0.459            | 3.525            |

Every row solves the normal equations. In every row  $\hat{\mu} + \hat{\alpha}_k$  is the  $k$ th group mean, and  $\hat{\alpha}_1 - \hat{\alpha}_2$  is the same. The value of  $\hat{\mu}$  is, in turn, the first group mean, the last group mean, the average of the four group means (12.624), the overall mean  $\bar{y} = 12.167$ , zero, and an uninterpretable 10.100.

Compare a restriction that *is* estimable,  $\alpha_1 = \alpha_2$ . It is not a side condition. Imposing it (by solving the Lagrange system for least squares under a linear restriction) forces the first two group means to be equal. The residual sum of squares rises from 14.665 to 46.078. The fit has changed, which is what testing the hypothesis  $\alpha_1 = \alpha_2$  in [Chapter 11](#) measures.

```
import numpy as np

sizes = [5, 3, 6, 4]
groups = np.repeat(np.arange(4), sizes)
rng = np.random.default_rng(84)
y = np.array([12.0, 15.0, 11.0, 14.0])[groups] + rng.normal(scale=1.0, size=groups.size)
X = np.column_stack([np.ones(groups.size), np.eye(4)[groups]]) # (mu, alpha_1..
                      alpha_4)
A, c = X.T @ X, X.T @ y
```

```
n_k = np.array(sizes, dtype=float)
side = {"alpha_1 = 0": [0, 1, 0, 0, 0],
        "alpha_4 = 0": [0, 0, 0, 0, 1],
        "sum alpha_k = 0": [0, 1, 1, 1, 1],
        "sum n_k alpha_k = 0": [0, *n_k],
        "mu = 0": [1, 0, 0, 0, 0]}
solutions = {}
for name, t in side.items():
    T = np.array([t], dtype=float)
    G = np.linalg.inv(A + T.T @ T) # (X'X + T'T)^{-1}: a g-inverse of X'X
    b = G @ c
    solutions[name] = b
    print(f"{name:20s} b = {np.round(b, 3)}   T b = {(T @ b)[0]:.1e}")
b_mn = np.linalg.pinv(X) @ y
print(f"{'minimum norm':20s} b = {np.round(b_mn, 3)}",
      f"   mu - sum(alpha) = {b_mn[0] - b_mn[1:].sum():.1e}")
```

```
# "alpha_1 = alpha_2" is a restriction on an estimable function: it changes the
  model
T_bad = np.array([[0, 1, -1, 0, 0]], dtype=float)
K = np.block([[A, T_bad.T], [T_bad, np.zeros((1, 1))]]) # Lagrange (bordered)
                      system
b_bad = np.linalg.lstsq(K, np.append(c, 0.0), rcond=None)[0][:5]
sse_free = np.sum((y - X @ solutions["alpha_1 = 0"]) ** 2)
sse_bad = np.sum((y - X @ b_bad) ** 2)
print(f"SSE without restriction {sse_free:.3f};   with alpha_1 = alpha_2 {sse_bad:.3f}
      ")
```

Reparameterization and side conditions are two languages for the same choice. Deleting the column of  $\alpha_1$  from  $\mathbf{X}$  and fitting the full-rank remainder gives exactly the solution with  $\alpha_1 = 0$ . More generally, deleting a set of columns whose removal leaves a full-rank matrix with the same column space is the side condition that sets those coefficients to zero, and it satisfies [Theorem 8.4.2\(c\)](#) ([Exercise \(B1\)](#)).

## 8.4.4 What software does

Programs treat a rank-deficient  $\mathbf{X}$  differently.

- *Code the factor at full rank before fitting.* Formula interfaces such as R's `lm()` and statsmodels' formula API expand a factor with  $g$  levels into an intercept and  $g - 1$  columns, by default the reference coding of [Section 8.5](#). The overparameterized matrix never appears. This is a reparameterization in the sense of [Theorem 8.4.1](#).
- *Drop columns found to be dependent.* If the columns supplied are dependent anyway, R's `lm()` computes a QR decomposition in which a column found to be (numerically) a combination of earlier ones is moved to the end and left out. Its coefficient is reported as NA. This is the side condition setting that coefficient to zero, and which column is dropped depends on the order of the columns. SAS's GLM procedure similarly solves the normal equations with a generalized inverse that zeroes the coefficients of dependent columns, and it flags the resulting estimates as not unique.
- *Return the minimum-norm solution.* `numpy.linalg.lstsq` returns  $\mathbf{X}^+\mathbf{y}$  silently (it reports the numerical rank only as a second output), and statsmodels' OLS, whose `fit` uses the pseudoinverse by default, returns the same vector with a `SingularMatrixWarning`. This is the side condition of [Proposition 8.4.4](#), and it does not depend on the order of the columns.
- *Pivot greedily.* A QR decomposition with column pivoting ([Chapter 10](#), [Theorem 10.7.2](#)) chooses at each step the remaining column with the largest component orthogonal to those already chosen, and the columns left at the end are dropped. In [Example \(Five side conditions, one fit\)](#) it takes the intercept, then  $\alpha_3$  (the largest group),  $\alpha_1$  and  $\alpha_4$ , and drops  $\alpha_2$ , the smallest group, a choice that depends on the group sizes.

```
import scipy.linalg
import statsmodels.api as sm
b_sm = sm.OLS(y, X).fit().params          # statsmodels default: method="pinv"
"
b_np = np.linalg.lstsq(X, y, rcond=None)[0]  # LAPACK gelsd: minimum-norm
solution
_, R, piv = scipy.linalg.qr(X, mode="economic", pivoting=True)
print("statsmodels:", np.round(b_sm, 3))
print("numpy lstsq:", np.round(b_np, 3))
print("pivoted QR column order:", piv, " |R_kk| =", np.round(np.abs(np.diag(R)), 3))
```

The lesson of this section is that all these outputs are correct, and none is more correct than the others. They are different coordinate systems for one fitted mean vector. Any conclusion that changes when the software changes is a conclusion about a nonestimable function.

*Warning.* Before interpreting a coefficient from a rank-deficient fit, find out which side condition produced it. A coefficient reported as “the effect of level 3” may be a difference from level 1, a deviation from an average of cell means, a deviation from the overall mean, or, under the minimum-norm convention, a quantity with no interpretation at all.

## 8.4.5 Exercises

### 8.4.5.1 A. Check your understanding

**Exercise A1.**

In the one-way model with four groups and  $\boldsymbol{\beta} = (\mu, \alpha_1, \dots, \alpha_4)^\top$ , decide which of the following are side conditions in the sense of [Theorem 8.4.2](#): (i)  $\alpha_2 = 0$ ; (ii)  $\mu = \alpha_3$ ; (iii)  $\alpha_1 - \alpha_2 = 0$ ; (iv)  $\alpha_1 = 0$  together with  $\alpha_2 = 0$ ; (v)  $2\mu + \alpha_1 + \alpha_2 = 0$ .

**8.4.5.2 B. Practice****Exercise B1.**

Let  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ , where  $\mathbf{X}_1$  has full column rank and  $\mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X})$ . Show that the least squares fit using  $\mathbf{X}_1$  alone, padded with zeros for the coefficients of  $\mathbf{X}_2$ , is the solution of the side condition  $\mathbf{T} = [\mathbf{0}, \mathbf{I}]$ . Check the conditions of [Theorem 8.4.2](#) directly.

*Solution.* The padded vector  $(\hat{\boldsymbol{\beta}}_1^\top, \mathbf{0}^\top)^\top$  satisfies  $\mathbf{X}(\hat{\boldsymbol{\beta}}_1^\top, \mathbf{0}^\top)^\top = \mathbf{X}_1 \hat{\boldsymbol{\beta}}_1 = \mathbf{M}\mathbf{y}$ , because  $\mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X})$ . So it solves the normal equations and satisfies  $\mathbf{T}\mathbf{b} = \mathbf{0}$ . For the conditions:  $\text{rank}(\mathbf{T}) = p_2$ , and  $\text{rank}(\mathbf{X}) = \text{rank}(\mathbf{X}_1) = p_1$ , so  $\text{rank}(\mathbf{T}) = p - r$ . If  $\mathbf{T}^\top \mathbf{c} = (\mathbf{0}^\top, \mathbf{c}^\top)^\top$  were in  $\mathcal{C}(\mathbf{X}^\top)$ , say  $\mathbf{X}_1^\top \boldsymbol{\rho} = \mathbf{0}$  and  $\mathbf{X}_2^\top \boldsymbol{\rho} = \mathbf{c}$ , then  $\boldsymbol{\rho} \perp \mathcal{C}(\mathbf{X}_1) = \mathcal{C}(\mathbf{X}) \supseteq \mathcal{C}(\mathbf{X}_2)$ , so  $\mathbf{c} = \mathbf{0}$ . Condition (e) holds.

**Exercise B2.**

For the additive model  $E(y_{ij}) = \mu + \alpha_i + \beta_j$  with  $i = 1, 2, j = 1, 2, 3$  and one observation per cell, take  $\mathbf{U}$  with rows giving  $\mu + \alpha_1 + \beta_1, \alpha_2 - \alpha_1, \beta_2 - \beta_1$  and  $\beta_3 - \beta_1$ . Compute  $\mathbf{Z}$  from [Eq. 8.4.1](#), and check that it is the reference-coded model matrix.

**Exercise B3.**

Let  $\mathbf{T}$  satisfy the conditions of [Theorem 8.4.2](#) and have full row rank  $p - r$ . Show that the bordered matrix  $\begin{pmatrix} \mathbf{X}^\top \mathbf{X} & \mathbf{T}^\top \\ \mathbf{T} & \mathbf{0} \end{pmatrix}$  is nonsingular, that the solution of the bordered system with right-hand side  $(\mathbf{X}^\top \mathbf{y}, \mathbf{0})$  has first block  $\hat{\boldsymbol{\beta}}_{\mathbf{T}}$  and second block  $\mathbf{0}$ , and interpret the zero Lagrange multiplier. Relate this to [Exercise \(B2\)](#).

**8.4.5.3 C. Going deeper****Exercise C1.**

Replace  $\mathbf{X}$  by  $\mathbf{X}\mathbf{D}$  for a nonsingular diagonal  $\mathbf{D}$  (a change of units). Show that the minimum-norm solution for  $\mathbf{X}\mathbf{D}$ , converted back to the original units, is the solution for the side condition  $\mathbf{V}_0^\top \mathbf{D}^{-2} \mathbf{b} = \mathbf{0}$ , which differs in general from  $\mathbf{V}_0^\top \mathbf{b} = \mathbf{0}$ . Conclude that the minimum-norm convention depends on the units of measurement, while no estimable function does.

**Exercise C2.**

Suppose  $\text{rank}(\mathbf{T}) = p - r$  but  $\mathcal{C}(\mathbf{T}^\top) \cap \mathcal{C}(\mathbf{X}^\top) \neq \{\mathbf{0}\}$ . Show that for some  $\mathbf{y}$  the restricted normal equations of [Theorem 8.4.2\(a\)](#) have no solution, and that whenever they have

one they have infinitely many.

*Solution.* By the dimension count,  $\mathcal{C}(\mathbf{X}^\top) + \mathcal{C}(\mathbf{T}^\top)$  has dimension less than  $r + (p - r) = p$ . So  $\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}) \neq \{\mathbf{0}\}$ , and any solution can be moved along a nonzero vector of that intersection. For existence, the proof of [Theorem 8.4.2](#) shows that a solution exists for every  $\mathbf{y}$  iff  $\mathcal{N}(\mathbf{X}) + \mathcal{N}(\mathbf{T}) = \mathbb{R}^p$ , which fails because its orthogonal complement  $\mathcal{C}(\mathbf{X}^\top) \cap \mathcal{C}(\mathbf{T}^\top)$  is nonzero. So some coset  $\mathbf{b}_0 + \mathcal{N}(\mathbf{X})$  misses  $\mathcal{N}(\mathbf{T})$ , and the response  $\mathbf{y} = \mathbf{X}\mathbf{b}_0$  has no solution.

## 8.5 Indicator variables and the coding of a factor

Most regressors in practice are not measurements on a continuous scale. They record a category: treatment arm, region, occupation, party, a yes-or-no answer. Such a variable is a **factor**, and its possible values are its **levels**. A factor enters a linear model through *indicator variables* (also called dummy variables), columns of zeros and ones that mark membership of a level. There are many ways to build these columns, and they give quite different coefficient tables. This section shows that they all describe the same fitted model. Each choice of coding is a choice of which linear functions of the level means the coefficients report. It gives a single formula that says exactly which ones.

### 8.5.1 One binary regressor

Suppose each observation belongs to one of two groups, and let  $d_i = 1$  if observation  $i$  is in group 1 and  $d_i = 0$  if it is in group 0. The model

$$E(y_i) = \beta_0 + \beta_1 d_i$$

says that the mean is  $\mu_0 = \beta_0$  in group 0 and  $\mu_1 = \beta_0 + \beta_1$  in group 1. So  $\beta_1 = \mu_1 - \mu_0$  is the difference between the two group means. The least squares fit sets the fitted value in each group to that group's mean (the model space is the set of vectors constant within groups, [Example \(The one-way layout\)](#)), so  $\hat{\beta}_0 = \bar{y}_0$  and  $\hat{\beta}_1 = \bar{y}_1 - \bar{y}_0$ . The  $t$  statistic for  $\beta_1 = 0$  is the pooled two-sample  $t$  statistic ([Exercise \(A1\)](#)). The familiar two-sample comparison is a regression on one indicator.

It is tempting to add a second indicator,  $1 - d_i$ , for group 0. Then  $\mathbf{X} = [\mathbf{1}, \mathbf{d}, \mathbf{1} - \mathbf{d}]$  has three columns, but they satisfy  $\mathbf{1} - \mathbf{d} - (\mathbf{1} - \mathbf{d}) = \mathbf{0}$ , so  $\text{rank}(\mathbf{X}) = 2$ . This is the **dummy variable trap**: an intercept together with an indicator for every level produces an exactly collinear model matrix. By [Section 8.1](#), none of the three coefficients is then identifiable. The trap is not a mistake in the model, which still describes two means, but in the coordinates.

### 8.5.2 A factor with $g$ levels

Let a factor have levels  $1, \dots, g$ , with  $n_k \geq 1$  observations at level  $k$ . Its **indicator matrix** is the  $n \times g$  matrix  $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_g]$  with  $z_{ik} = 1$  if observation  $i$  is at level  $k$  and 0 otherwise. Each row of  $\mathbf{Z}$  contains a single 1, so



$$\mathbf{Z}\mathbf{1}_g = \mathbf{1}_n, \quad \mathbf{Z}^\top \mathbf{Z} = \mathbf{D} = \text{diag}(n_1, \dots, n_g), \quad \mathbf{Z}^\top \mathbf{y} = (y_{1\cdot}, \dots, y_{g\cdot})^\top, \quad (8.5.1)$$

where  $y_{k\cdot}$  is the total of the responses at level  $k$ . The columns have disjoint supports, so  $\mathbf{Z}$  has full column rank  $g$ , and  $\mathcal{C}(\mathbf{Z})$  is the set of vectors that are constant within levels.

The *cell-means model*  $E(\mathbf{Y}) = \mathbf{Z}\boldsymbol{\mu}$ ,  $\boldsymbol{\mu} = (\mu_1, \dots, \mu_g)^\top$ , gives each level its own mean. It has full rank, and by [Eq. 8.5.1](#)

$$\hat{\boldsymbol{\mu}} = \mathbf{D}^{-1} \mathbf{Z}^\top \mathbf{y} = (\bar{y}_1, \dots, \bar{y}_g)^\top,$$

the vector of level means. Any other model containing  $\mathcal{C}(\mathbf{Z})$  and nothing more has the same fit. The overparameterized model  $[\mathbf{1}, \mathbf{Z}]$  of [Example \(A rank-deficient layout\)](#) is one example. The full-rank codings below are others.

### Definition 8.5.1 — Coding of a factor

A **coding** of a factor with  $g$  levels is a nonsingular  $g \times g$  matrix  $\mathbf{K}$ . The coded model matrix is  $\mathbf{ZK}$ , and the model is  $E(\mathbf{Y}) = \mathbf{ZK}\boldsymbol{\gamma}$ . When the first column of  $\mathbf{K}$  is  $\mathbf{1}_g$ , so that  $\mathbf{K} = [\mathbf{1}_g, \mathbf{C}]$ , the coded matrix is  $[\mathbf{1}_n, \mathbf{ZC}]$ : an intercept and the  $g - 1$  columns  $\mathbf{ZC}$ . The  $g \times (g - 1)$  matrix  $\mathbf{C}$  is then called the **coding matrix** of the factor.

Row  $k$  of  $\mathbf{K}$  lists the values that the coded columns take for an observation at level  $k$ . In the software of [Section 8.4](#) the matrix  $\mathbf{C}$  is what one specifies (in R, the “contrasts” attribute of a factor), and the program builds the columns  $\mathbf{ZC}$  from it.

### Proposition 8.5.1 — What a coding estimates

Let  $\mathbf{K}$  be a coding and  $\hat{\boldsymbol{\mu}} = \mathbf{D}^{-1} \mathbf{Z}^\top \mathbf{y}$  the vector of level means. Then:

- (a)  $\mathcal{C}(\mathbf{ZK}) = \mathcal{C}(\mathbf{Z})$ , so every coding has the same fitted values, residuals and residual sum of squares;
- (b) the coefficients are  $\boldsymbol{\gamma} = \mathbf{K}^{-1}\boldsymbol{\mu}$ , and their least squares estimates are  $\hat{\boldsymbol{\gamma}} = \mathbf{K}^{-1}\hat{\boldsymbol{\mu}}$ ;
- (c)  $[\mathbf{1}_g, \mathbf{C}]$  is nonsingular iff the columns of  $\mathbf{C}$  are linearly independent and  $\mathbf{1}_g \notin \mathcal{C}(\mathbf{C})$ ;
- (d) to make the coefficients estimate prescribed linear functions  $\mathbf{L}\boldsymbol{\mu}$  of the level means, with  $\mathbf{L}$  a nonsingular  $g \times g$  matrix, use the coding  $\mathbf{K} = \mathbf{L}^{-1}$ .

*Proof.*

- (a) The columns of  $\mathbf{ZK}$  lie in  $\mathcal{C}(\mathbf{Z})$ , and  $\text{rank}(\mathbf{ZK}) = \text{rank}(\mathbf{Z}) = g$  because  $\mathbf{K}$  is nonsingular ([Proposition 1.2.3\(b\)](#)). A subspace of  $\mathcal{C}(\mathbf{Z})$  of the same dimension is  $\mathcal{C}(\mathbf{Z})$  itself. The fit depends only on the column space ([Theorem 6.7.1](#)).
- (b) The two models describe the same means iff  $\mathbf{ZK}\boldsymbol{\gamma} = \mathbf{Z}\boldsymbol{\mu}$ , and since  $\mathbf{Z}$  has full column rank this means  $\mathbf{K}\boldsymbol{\gamma} = \boldsymbol{\mu}$ . For the estimates,  $\mathbf{ZK}\hat{\boldsymbol{\gamma}} = \mathbf{M}\mathbf{y} = \mathbf{Z}\hat{\boldsymbol{\mu}}$ , and the same cancellation gives  $\mathbf{K}\hat{\boldsymbol{\gamma}} = \hat{\boldsymbol{\mu}}$ .
- (c) A square matrix is nonsingular iff its columns are linearly independent. The columns  $\mathbf{1}_g, \mathbf{c}_1, \dots, \mathbf{c}_{g-1}$  are independent iff the  $\mathbf{c}_j$  are independent and  $\mathbf{1}_g$  is not in their span.
- (d) With  $\mathbf{K} = \mathbf{L}^{-1}$ , (b) gives  $\boldsymbol{\gamma} = \mathbf{L}\boldsymbol{\mu}$ .



Part (b) is the whole story. Whatever the coding, the coefficients are fixed linear combinations of the level means, and these combinations are the *rows of*  $\mathbf{K}^{-1}$ , not the columns of  $\mathbf{K}$ . Each such combination is estimable in the overparameterized model, since it is a function of the means. Part (d) reverses the logic, and in practice it is the useful direction. Decide which questions the coefficients should answer, write them as the rows of  $\mathbf{L}$ , and invert.

### 8.5.3 The standard codings

For a factor with  $g$  levels, the codings in common use are the following. For each we give the meaning of the intercept  $\gamma_0$  and of the other coefficients  $\gamma_1, \dots, \gamma_{g-1}$ , that is, the rows of  $\mathbf{K}^{-1}$ .

- **Reference (treatment) coding.**  $\mathbf{C} = \begin{pmatrix} \mathbf{0}^\top \\ \mathbf{I}_{g-1}^\top \end{pmatrix}$ : the columns are the indicators of levels  $2, \dots, g$ . Then  $\gamma_0 = \mu_1$ , and  $\gamma_j = \mu_{j+1} - \mu_1$  is the difference between level  $j+1$  and the *reference level* 1. The level chosen as reference is arbitrary.
- **Sum-to-zero (deviation or effect) coding.**  $\mathbf{C} = \begin{pmatrix} \mathbf{I}_{g-1}^\top \\ -\mathbf{1}^\top \end{pmatrix}$ . Level  $k < g$  has mean  $\gamma_0 + \gamma_k$  and level  $g$  has mean  $\gamma_0 - \sum_{j < g} \gamma_j$ . Summing over the  $g$  levels gives  $\gamma_0 = \bar{\mu} = g^{-1} \sum_k \mu_k$ , the unweighted average of the level means. Then  $\gamma_k = \mu_k - \bar{\mu}$  is the deviation of level  $k$  from that average. The deviation of the last level is  $-\sum_{j < g} \gamma_j$ .
- **Weighted effect coding.**  $\mathbf{C} = \begin{pmatrix} \mathbf{I}_{g-1}^\top \\ -(n_1, \dots, n_{g-1})/n_g \end{pmatrix}$ . Now  $\sum_k n_k \mu_k = n\gamma_0$ , so  $\gamma_0 = \sum_k n_k \mu_k / n$ , the size-weighted average, estimated by the overall mean  $\bar{y}$ . The coefficient  $\gamma_k = \mu_k - \gamma_0$  is the deviation from it.
- **Successive differences.** With  $\mathbf{L}$  having first row  $g^{-1}\mathbf{1}^\top$  and rows  $\mathbf{e}_{j+1}^\top - \mathbf{e}_j^\top$ , part (d) gives a coding in which  $\gamma_0 = \bar{\mu}$  and  $\gamma_j = \mu_{j+1} - \mu_j$ . This is natural for ordered levels.
- **Helmert and orthogonal polynomial codings**, whose columns are contrasts, are treated in [Section 8.7](#).

The first three are the side conditions of [Example \(Five side conditions, one fit\)](#) in another language. In the overparameterized model  $\mu + \alpha_k$ , reference coding is the side condition  $\alpha_1 = 0$ , sum-to-zero coding is  $\sum_k \alpha_k = 0$ , and weighted effect coding is  $\sum_k n_k \alpha_k = 0$ . In each case the constrained  $(\mu, \alpha_2, \dots, \alpha_g)$  or  $(\mu, \alpha_1, \dots, \alpha_{g-1})$  equals  $\boldsymbol{\gamma}$  ([Exercise \(C1\)](#)).

#### Example. (Party identification and ideology)

An extract of the 1996 American National Election Study, distributed with statsmodels as a public-domain data set, contains 944 respondents who placed themselves on a seven-point scale from extremely liberal (1) to extremely conservative (7). It also records their party identification in seven levels from strong Democrat to strong Republican. The level means and sizes are

| party<br>identifi-<br>cation | strong<br>Dem | weak<br>Dem | ind.<br>Dem | ind.<br>independenRep | weak<br>Rep | strong<br>Rep |
|------------------------------|---------------|-------------|-------------|-----------------------|-------------|---------------|
| $n_k$                        | 200           | 180         | 108         | 37                    | 94          | 150           |
| $\bar{y}_k$                  | 3.335         | 3.617       | 3.685       | 4.054                 | 4.957       | 5.691         |

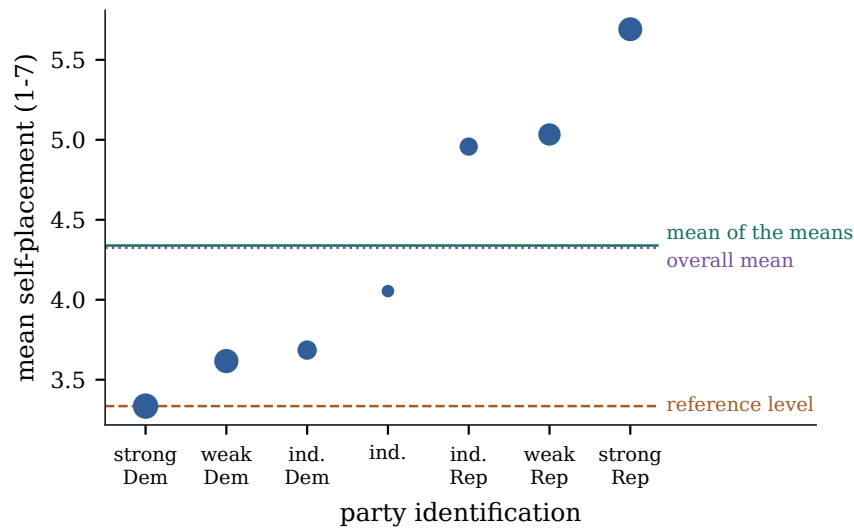


Figure 8.5.1: Mean self-placement by party identification in the 1996 ANES (point area proportional to group size), with the three baselines used by the codings of [Example \(Party identification and ideology\)](#): the reference level, the unweighted mean of the seven means, and the overall mean.

Here are the coefficients of four codings of the same factor ( $\gamma_1, \dots, \gamma_6$  refer to the columns of  $\mathbf{C}$  in the order defined above; for the two deviation codings they belong to the first six levels):

| coding                    | $\hat{\gamma}_0$ | $\hat{\gamma}_1$ | $\hat{\gamma}_2$ | $\hat{\gamma}_3$ | $\hat{\gamma}_4$ | $\hat{\gamma}_5$ | $\hat{\gamma}_6$ |
|---------------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| reference<br>(strong Dem) | 3.335            | 0.282            | 0.350            | 0.719            | 1.622            | 1.698            | 2.356            |
| sum to zero               | 4.339            | -1.004           | -0.722           | -0.654           | -0.285           | 0.618            | 0.694            |
| weighted effect           | 4.325            | -0.990           | -0.709           | -0.640           | -0.271           | 0.632            | 0.708            |
| successive differences    | 4.339            | 0.282            | 0.069            | 0.369            | 0.903            | 0.076            | 0.658            |

Each row is  $\mathbf{K}^{-1}\hat{\boldsymbol{\mu}}$  for its own  $\mathbf{K}$ , as [Proposition 8.5.1\(b\)](#) says, and all four have residual sum of squares 1178.3. The intercepts are the reference mean 3.335, the unweighted mean of the seven means 4.339, the overall mean  $\bar{y} = 4.325$  and again 4.339. [Figure 8.5.1](#) shows the three reference points. The successive differences show that the largest step in ideology, 0.903, falls between pure independents and independents leaning Republican, and the step between weak and strong Republicans is 0.658. The reference coding cannot show this directly. The two deviation codings nearly agree here because the groups are of similar size, apart from the small group of pure independents.

```
import numpy as np
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].to_numpy()      # 1 = very liberal ... 7 = very conservative
```

```

level = anes["PID"].to_numpy().astype(int)    # 0 = strong Dem. ... 6 = strong Rep.
g = 7
Z = np.eye(g)[level]                        # n x 7 indicator (cell-means) matrix
n_k = Z.sum(axis=0)
cell_means = (Z.T @ y) / n_k
print("group sizes:", n_k.astype(int))
print("cell means: ", np.round(cell_means, 3))

```

```

def coding_matrices(n_k):
    """Each coding is  $X = ZK$  for a nonsingular  $g \times g$   $K$ ; the coefficients are  $K^{-1}\mu$ ."""
    g = len(n_k)
    one = np.ones((g, 1))
    reference = np.vstack([np.zeros(g - 1), np.eye(g - 1)])    # level 0 is baseline
    deviation = np.vstack([np.eye(g - 1), -np.ones(g - 1)])    # sum-to-zero
    weighted = np.vstack([np.eye(g - 1), -n_k[:-1] / n_k[-1]])    # sum  $n_k \times$ 
    effect = 0
    successive = np.array([(j + 1 - g if i <= j else j + 1) / g for j in range(g - 1)]
                           for i in range(g))    # backward differences
    return {"cell means": np.eye(g),
            "reference": np.hstack([one, reference]),
            "sum to zero": np.hstack([one, deviation]),
            "weighted effect": np.hstack([one, weighted]),
            "successive differences": np.hstack([one, successive])}

for name, K in coding_matrices(n_k).items():
    Xc = Z @ K
    gamma = np.linalg.lstsq(Xc, y, rcond=None)[0]
    print(f"{name:23s}", np.round(gamma, 3))

```

## 8.5.4 Choosing a coding

Since every coding gives the same fit, the choice is about communication and, to a lesser extent, precision. Four points are worth keeping in mind.

*Anything that depends on the fit is invariant.* This includes fitted values, residuals,  $R^2$ ,  $\hat{\sigma}^2$ , and the  $F$  test that all levels have the same mean. That test compares  $\mathcal{C}(\mathbf{Z})$  with  $\mathcal{C}(\mathbf{1})$ , and the coding does not enter it ([Theorem 11.1.1](#)).

*Coefficient tests answer different questions.* The  $t$  test for a reference-coded coefficient tests whether level  $k$  differs from the reference level. The  $t$  test for a sum-coded coefficient tests whether level  $k$  differs from the average of all levels, a question that changes if a level is added. Neither question is wrong, but they are different questions. A table of coefficients from an unfamiliar coding should never be read as if it came from the familiar one.

*Precision depends on the reference.* In reference coding  $\text{Var}(\hat{\gamma}_j) = \sigma^2(1/n_{j+1} + 1/n_1)$ . A small reference group inflates the variance of every coefficient. In [Example \(Party identification and ideology\)](#), comparing each of the other five levels with the 37 pure independents instead of the 200 strong Democrats would multiply the standard errors by factors between 1.55 and 1.76. A common default is to use the largest group, or a natural control condition, as the reference.

With other regressors, coefficients are adjusted comparisons. When the model also contains other variables, a reference-coded coefficient compares level  $k$  with the reference level at equal values of those variables. That is the “holding the others fixed” reading of [Definition 5.7.1](#), with all its caveats. [Section 8.6](#) gives an example in which adjustment changes the answer a great deal.

*Idea.* A coding is a choice of coordinates for the  $g$ -dimensional space of level means. The coefficients are  $\mathbf{K}^{-1}\hat{\boldsymbol{\mu}}$ . To get the comparisons you want, write them down first as the rows of a nonsingular matrix  $\mathbf{L}$ , then code with  $\mathbf{K} = \mathbf{L}^{-1}$ .

For an ordered factor, a useful instance of this principle is the *cumulative* coding, with columns  $1\{\text{level} \geq k\}$  for  $k = 2, \dots, g$ . Then  $\mathbf{K}$  is lower triangular with ones on and below the diagonal. Its inverse takes successive differences, so the intercept is  $\mu_1$  and the coefficient of the  $k$ th column is  $\mu_k - \mu_{k-1}$  ([Exercise \(B2\)](#)). An observation’s mean is built up as a sum of steps, which is often how an ordered factor is best described.

## 8.5.5 Exercises

### 8.5.5.1 A. Check your understanding

#### Exercise A1.

In the model  $E(y_i) = \beta_0 + \beta_1 d_i$ , with  $n_0$  observations in group 0 (level 1) and  $n_1$  in group 1 (level 2), write down the coding matrix  $\mathbf{K}$  with  $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\beta}$ . Use [Proposition 8.5.1\(b\)](#), not an inverse of  $\mathbf{X}^\top \mathbf{X}$ , to find  $\hat{\boldsymbol{\beta}}$ . Then show that  $\text{Var}(\hat{\beta}_1) = \sigma^2(1/n_0 + 1/n_1)$  and that  $\hat{\beta}_1 / \text{se}(\hat{\beta}_1)$  is the pooled two-sample  $t$  statistic, where  $s^2 = \text{SSE}/(n - 2)$  replaces  $\sigma^2$ .

*Solution.* The two means are  $\mu_0 = \beta_0$  and  $\mu_1 = \beta_0 + \beta_1$ , so  $\mathbf{K} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$  and  $\mathbf{K}^{-1} = \begin{pmatrix} 1 & 0 \\ -1 & 1 \end{pmatrix}$ . By [Proposition 8.5.1\(b\)](#),  $\hat{\boldsymbol{\beta}} = \mathbf{K}^{-1}\hat{\boldsymbol{\mu}} = (\bar{y}_0, \bar{y}_1 - \bar{y}_0)^\top$ . The level means are the estimates of the full-rank cell-means model, with  $\text{Cov}(\hat{\boldsymbol{\mu}}) = \sigma^2 \mathbf{D}^{-1} = \sigma^2 \text{diag}(1/n_0, 1/n_1)$  ([Theorem 5.5.1](#)), so  $\text{Var}(\hat{\beta}_1) = (-1, 1) \sigma^2 \mathbf{D}^{-1} (-1, 1)^\top = \sigma^2(1/n_0 + 1/n_1)$ . The residuals are deviations from the group means ([Proposition 8.5.1\(a\)](#)), so  $s^2$  is the pooled variance, and the ratio is  $(\bar{y}_1 - \bar{y}_0) / \{s\sqrt{1/n_0 + 1/n_1}\}$ .

#### Exercise A2.

For a factor with three levels, write down the null space of  $[\mathbf{1}, \mathbf{z}_1, \mathbf{z}_2, \mathbf{z}_3]$ . Show that adding a second factor’s full set of indicators,  $[\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B]$ , creates a null space of dimension at least two, and give two independent null vectors.

### 8.5.5.2 B. Practice

#### Exercise B1.

Let  $\mathbf{X}$  have full column rank, let  $\mathbf{e}_i$  be the  $i$ th coordinate vector with  $\mathbf{e}_i \notin \mathcal{C}(\mathbf{X})$ , and fit  $[\mathbf{X}, \mathbf{e}_i]$  with coefficients  $(\boldsymbol{\beta}, \delta)$ . Write  $\hat{e}_i$  and  $h_{ii}$  for the residual and the leverage ([Definition 6.8.1](#)) of observation  $i$  in the fit of  $\mathbf{X}$  alone.

- (a) Use [Theorem 6.6.2](#), with  $\mathbf{e}_i$  as the regressor of interest and  $\mathbf{X}$  as the one partialled

out, to show that  $\hat{\delta} = \hat{e}_i/(1 - h_{ii})$ , and explain why  $h_{ii} < 1$ .

- (b) Show that the enlarged fit reproduces  $y_i$  exactly, and that its  $\hat{\boldsymbol{\beta}}$  is the estimate  $\hat{\boldsymbol{\beta}}_{(i)}$  without observation  $i$ .
- (c) Deduce the deletion identity  $y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)} = \hat{e}_i/(1 - h_{ii})$ : the error in predicting  $y_i$  from the other observations is the ordinary residual inflated by  $1/(1 - h_{ii})$ .

*Solution.*

- (a) Apply [Theorem 6.6.2\(a\)](#) with  $\mathbf{X}_1 = \mathbf{e}_i$  and  $\mathbf{X}_2 = \mathbf{X}$ , and let  $\mathbf{M}$  project onto  $\mathcal{C}(\mathbf{X})$ . Then  $\hat{\delta}$  is the coefficient in the regression of  $\mathbf{y}$  on  $(\mathbf{I} - \mathbf{M})\mathbf{e}_i$ :  $\hat{\delta} = \mathbf{e}_i^\top (\mathbf{I} - \mathbf{M})\mathbf{y} / \mathbf{e}_i^\top (\mathbf{I} - \mathbf{M})\mathbf{e}_i = \hat{e}_i/(1 - h_{ii})$ . The denominator is  $\|(\mathbf{I} - \mathbf{M})\mathbf{e}_i\|^2$ , which is positive because  $\mathbf{e}_i \notin \mathcal{C}(\mathbf{X})$ .
- (b) The residual vector of the enlarged fit is orthogonal to every column, in particular to  $\mathbf{e}_i$ , so its  $i$ th entry is zero. For fixed  $\boldsymbol{\beta}$ , the residual sum of squares  $\sum_{j \neq i} (y_j - \mathbf{x}_{(j)}^\top \boldsymbol{\beta})^2 + (y_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta} - \delta)^2$  is minimized over  $\delta$  by making the last term zero. What remains is the deletion criterion, and its minimizer  $\hat{\boldsymbol{\beta}}_{(i)}$  is unique: if  $\mathbf{X}\mathbf{b} = c\mathbf{e}_i$  with  $\mathbf{b} \neq \mathbf{0}$ , then  $c \neq 0$  by full rank, and  $\mathbf{e}_i \in \mathcal{C}(\mathbf{X})$ . So  $\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}_{(i)}$ .
- (c) By (b),  $\hat{\delta} = y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)}$ ; compare with (a). The  $t$  statistic for  $\delta$  is the externally studentized residual of [Chapter 20](#).

### Exercise B2.

For an ordered factor with  $g$  levels, let the coded columns be the indicators of  $\{\text{level} \geq k\}$ ,  $k = 2, \dots, g$ , together with an intercept. Show that  $\mathbf{K}$  is lower triangular with ones on and below the diagonal, that  $\mathbf{K}^{-1}$  is the first-difference matrix, and hence that the coefficients are  $\mu_1$  and  $\mu_k - \mu_{k-1}$ .

### Exercise B3.

On each of  $m$  days a warehouse runs  $N_{1i}$  machines of type 1 and  $N_{2i}$  of type 2, and meters its total energy use  $V_i$ . Suppose each running machine of type  $k$  uses an amount of energy with mean  $\theta_k$  and variance  $\tau_k^2$ , independently of the other machines and the same on every day. Show that  $E(V_i) = \theta_1 N_{1i} + \theta_2 N_{2i}$ , a linear model *without* intercept, and that  $(\theta_1, \theta_2)$  is identifiable iff the vectors  $(N_{1i})$  and  $(N_{2i})$  are not proportional. Find  $\text{Cov}(\mathbf{V})$ ; why is it not a multiple of  $\mathbf{I}$ ? Why is it essential that  $\theta_k$  does not vary by day?

#### 8.5.5.3 C. Going deeper

### Exercise C1.

Show that fitting the reference-coded model is the same as solving the normal equations of the overparameterized one-way model under the side condition  $\alpha_1 = 0$ , with  $\gamma_0 = \mu$  and  $\gamma_j = \alpha_{j+1}$ . State and prove the corresponding facts for the sum-to-zero and weighted effect codings.

## 8.6 Several factors, interactions and covariates

With one factor, rank deficiency is a nuisance, and a coding removes it once and for all. With two or more factors, or a factor together with a numeric regressor, it becomes informative. The rank of the model matrix now depends on which combinations of levels were observed. Whether a comparison is estimable depends on how the design links the levels involved. This section works out the three cases that occur most often: two factors without interaction, two factors with interaction, and a factor with a numeric covariate.

### 8.6.1 Two factors without interaction

Let factor  $A$  have levels  $i = 1, \dots, a$  and factor  $B$  levels  $j = 1, \dots, b$ . Call a pair  $(i, j)$  a **cell**. A cell is *occupied* if at least one observation has that combination, and we assume every level of each factor occurs in some occupied cell. The *additive* model is

$$E(y) = \mu + \alpha_i + \beta_j \quad \text{for an observation in cell } (i, j), \quad (8.6.1)$$

with model matrix  $\mathbf{X} = [\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B]$  built from the two indicator matrices. The number of parameters is  $p = 1 + a + b$ . Since  $\mathbf{Z}_A \mathbf{1}_a = \mathbf{Z}_B \mathbf{1}_b = \mathbf{1}$ , the vectors  $(1, -\mathbf{1}_a^\top, \mathbf{0}^\top)^\top$  and  $(1, \mathbf{0}^\top, -\mathbf{1}_b^\top)^\top$  always lie in  $\mathcal{N}(\mathbf{X})$ , so  $\text{rank}(\mathbf{X}) \leq a + b - 1$ . Whether equality holds, and which differences  $\alpha_i - \alpha_k$  are estimable, depends on the pattern of occupied cells.

Picture the design as a graph. The vertices are the  $a$  levels of  $A$  and the  $b$  levels of  $B$ , and each occupied cell  $(i, j)$  is an edge joining level  $i$  of  $A$  to level  $j$  of  $B$ . The design is **connected** if this graph is connected, that is, if one can walk from any level to any other through occupied cells.

#### Theorem 8.6.1 — Connectedness and estimability in the additive model

In the additive model Eq. 8.6.1, let the design graph have  $c$  connected components. Then:

- (a)  $\text{rank}(\mathbf{X}) = a + b - c$ ;
- (b)  $\alpha_i - \alpha_k$  is estimable iff levels  $i$  and  $k$  of  $A$  lie in the same component, and likewise for  $\beta_j - \beta_l$ ;
- (c) if the design is connected, the estimable functions are exactly the linear functions of the cell means  $\mu + \alpha_i + \beta_j$  over *all* cells, occupied or not. In particular every difference between levels of the same factor is estimable.

*Proof.* A vector  $(m, \mathbf{u}^\top, \mathbf{v}^\top)^\top$  lies in  $\mathcal{N}(\mathbf{X})$  iff  $m + u_i + v_j = 0$  for every occupied cell  $(i, j)$ . Fix such a vector. If  $(i, j)$  and  $(k, j)$  are both occupied, then  $u_i = u_k$ . If  $(i, j)$  and  $(i, l)$  are both occupied, then  $v_j = v_l$ . Following edges,  $u$  is constant on the  $A$ -levels of each component, say  $u_i = t_C$  for all  $A$ -levels  $i$  in component  $C$ , and then  $v_j = -m - t_C$  for the  $B$ -levels of  $C$ . Conversely, any choice of  $m$  and of numbers  $t_1, \dots, t_c$ , one for each component, defines in this way a vector satisfying all the equations. So  $\mathcal{N}(\mathbf{X})$

is parameterized linearly and injectively by  $(m, t_1, \dots, t_c)$ , and it has dimension  $c + 1$ . Rank-nullity gives  $\text{rank}(\mathbf{X}) = 1 + a + b - (c + 1)$ , which is (a).

For (b), by [Theorem 8.3.1\(c\)](#),  $\alpha_i - \alpha_k$  is estimable iff  $u_i - u_k = 0$  for every null vector, that is, iff  $t_{C(i)} = t_{C(k)}$  for all choices of the  $t$ 's. This holds iff  $C(i) = C(k)$ . The argument for  $B$  is the same.

For (c), if  $c = 1$ , then (b) makes every  $\alpha_i - \alpha_k$  and  $\beta_j - \beta_l$  estimable. Pick an occupied cell  $(i_0, j_0)$ . Then  $\mu + \alpha_i + \beta_j = (\mu + \alpha_{i_0} + \beta_{j_0}) + (\alpha_i - \alpha_{i_0}) + (\beta_j - \beta_{j_0})$  is estimable for every  $(i, j)$ . Conversely, every estimable function is a combination of the means of occupied cells ([Theorem 8.2.2](#)), and these are among the cell means.

■

Part (c) contains a point worth stating separately. In a connected additive design, the mean of an *empty* cell is estimable. The data never observed that combination, but additivity lets it be pieced together from cells that were observed:  $\mu + \alpha_i + \beta_j$  is the mean of an occupied cell  $(i, j')$ , plus the difference  $\beta_j - \beta_{j'}$  learned elsewhere. The estimate is only as good as the additivity assumption, which cannot be checked at the empty cell. This is extrapolation of a subtle kind.

### Example. (Two islands)

Take  $a = 4, b = 5$  and seven occupied cells:  $(1, 1), (1, 2), (2, 2), (2, 3)$  and  $(3, 4), (4, 4), (4, 5)$  (levels numbered from 1). The design graph has two components: rows 1–2 with columns 1–3, and rows 3–4 with columns 4–5. By [Theorem 8.6.1](#),  $\text{rank}(\mathbf{X}) = 4 + 5 - 2 = 7$ , one less than  $a + b - 1 = 8$ . The difference  $\alpha_1 - \alpha_2$  is estimable, since rows 1 and 2 are linked through column 2, but  $\alpha_1 - \alpha_3$  is not. The two islands could sit at any vertical offset from each other without changing any fitted value. Occupying a single further cell,  $(3, 3)$ , joins the islands. The rank rises to 8, and  $\alpha_1 - \alpha_3$  becomes estimable. The listing checks each statement with the rank test [Theorem 8.3.1\(h\)](#). The script also checks the rank formula on 200 random designs.

```
import numpy as np

def additive_matrix(cells, a, b):
    """Model matrix [1, row indicators, column indicators] for occupied (i, j) cells
    . """
    rows = np.array([i for i, j in cells])
    cols = np.array([j for i, j in cells])
    return np.column_stack([np.ones(len(cells)), np.eye(a)[rows], np.eye(b)[cols]])

def estimable(lam, X):
    """lambda is estimable iff appending it to the rows of X does not raise the rank
    . """
    return np.linalg.matrix_rank(np.vstack([X, lam])) == np.linalg.matrix_rank(X)

a, b = 4, 5
cells = [(0, 0), (0, 1), (1, 1), (1, 2), (2, 3), (3, 3), (3, 4)] # two separate
    blocks
X = additive_matrix(cells, a, b)
row_diff = lambda i, k: np.r_[0, np.eye(a)[i] - np.eye(a)[k], np.zeros(b)]
print("rank:", np.linalg.matrix_rank(X), " (a + b - 1 =", a + b - 1, ")")
print("alpha_1 - alpha_2 estimable:", estimable(row_diff(0, 1), X))
print("alpha_1 - alpha_3 estimable:", estimable(row_diff(0, 2), X))
```



```

X_linked = additive_matrix(cells + [(2, 2)], a, b)                                # one more
      occupied cell
print("after adding cell (3, 3): rank", np.linalg.matrix_rank(X_linked),
      " alpha_1 - alpha_3 estimable:", estimable(row_diff(0, 2), X_linked))

```

A complete design, with every cell occupied, is connected. So is any design in which one level of  $B$  appears with every level of  $A$  and one level of  $A$  with every level of  $B$ . Disconnected designs arise in practice more often than one might expect: in observational data with many sparse levels (workers and firms, students and schools, players and teams), and in experiments where blocks were filled carelessly. The estimable functions of a disconnected design are characterized in [Exercise \(C1\)](#).

## 8.6.2 Two factors with interaction

The additive model assumes that the difference between two levels of  $A$  is the same at every level of  $B$ . The *interaction model* drops that assumption:

$$E(y) = \mu + \alpha_i + \beta_j + \gamma_{ij} \quad \text{for an observation in cell } (i, j), \quad (8.6.2)$$

with an interaction parameter  $\gamma_{ij}$  for each of the  $ab$  cells. Write for the  $n \times ab$  indicator matrix of the cells. The column of an empty cell is zero. The model matrix is  $\mathbf{X} = [\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B, \gamma]$ , with  $p = 1 + a + b + ab$  columns, and  $m$  denotes the number of occupied cells.

### Proposition 8.6.2 — Estimability in the interaction model

In the interaction model [Eq. 8.6.2](#) with  $m$  occupied cells, write  $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$  for the mean of an occupied cell.

- (a)  $\mathcal{C}(\mathbf{X}) = \mathcal{C}()$  and  $\text{rank}(\mathbf{X}) = m$ : the model is the cell-means model in disguise.
- (b) The estimable functions are exactly the linear combinations  $\sum c_{ij}\mu_{ij}$  over occupied cells.
- (c) An interaction contrast  $\sum_{i,j} d_{ij}\gamma_{ij}$  whose coefficient table  $(d_{ij})$  has zero row and column sums is estimable iff  $d_{ij} = 0$  for every empty cell. In particular,  $\gamma_{ij} - \gamma_{il} - \gamma_{kj} + \gamma_{kl}$  is estimable iff all four cells are occupied.

*Proof.*

- (a) Every column of  $\mathbf{1}$ ,  $\mathbf{Z}_A$  and  $\mathbf{Z}_B$  is a sum of columns of  $\gamma$ . For example, the indicator of level  $i$  of  $A$  is the sum of the indicators of the cells in row  $i$ . So  $\mathcal{C}(\mathbf{X}) = \mathcal{C}()$ . The nonzero columns of  $\gamma$  are the  $m$  indicators of occupied cells, which have disjoint supports.
- (b) By [Theorem 8.2.2](#), the estimable functions are the combinations of the expected responses, and these are the  $\mu_{ij}$ .
- (c) If  $d_{ij} = 0$  off the occupied cells, then  $\sum_{\text{occ}} d_{ij}\mu_{ij} = \sum d_{ij}\mu + \sum_i \alpha_i \sum_j d_{ij} + \sum_j \beta_j \sum_i d_{ij} + \sum d_{ij}\gamma_{ij} = \sum d_{ij}\gamma_{ij}$ , because the row and column sums vanish. So the contrast is estimable. Conversely, if  $d_{i_0j_0} \neq 0$  for an empty cell  $(i_0, j_0)$ , the column of  $\gamma_{i_0j_0}$  in  $\mathbf{X}$  is zero, so the coordinate vector  $\mathbf{v}$  of that parameter lies in  $\mathcal{N}(\mathbf{X})$ . The contrast's coefficient vector has inner product  $d_{i_0j_0} \neq 0$  with  $\mathbf{v}$ , so it



is not estimable ([Theorem 8.3.1\(c\)](#)).

■

In the interaction model the fitted value of each occupied cell is its own mean, and an empty cell has no estimable mean at all. The two models treat empty cells in opposite ways. The additive model fills them by assumption, while the interaction model declines to say anything about them. Which is appropriate is a substantive question, and [Chapter 17](#) returns to it ([Theorem 17.4.1](#)).

### Example. (Empty cells in a survey)

Cross-classify the 944 ANES respondents of [Example \(Party identification and ideology\)](#) by party identification (seven levels) and education (seven levels, labelled in the statsmodels documentation from “grades 1–8” to “PhD”). Two of the 49 cells are empty: no pure independent and no weak Republican in the sample has only an elementary-school education. Other cells are thin: only 3 pure independents report some high school. The overparameterized interaction model has  $p = 64$  columns and rank 47, the number of occupied cells, as [Proposition 8.6.2\(a\)](#) says. The interaction contrast comparing strong Democrats with pure independents across “some high school” and “high-school graduate” involves four occupied cells and is estimable. It rests on only 3 respondents in one cell, so it is imprecise, but estimable. The same comparison across “grades 1–8” and “high-school graduate” involves an empty cell and is not estimable. No amount of computation will produce it, and software that prints a number for it is printing an artefact of its side condition. The additive model, with rank  $7 + 7 - 1 = 13$ , treats both comparisons as zero by assumption.

```
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
pid = anes["PID"].to_numpy().astype(int)          # 7 levels
educ = anes["educ"].to_numpy().astype(int) - 1    # 7 levels, now 0..6
counts = np.zeros((7, 7), dtype=int)
np.add.at(counts, (pid, educ), 1)
print("empty (PID, educ) cells:",
      [(int(i), int(j) + 1) for i, j in zip(*np.nonzero(counts == 0))])

n = len(pid)
cell = pid * 7 + educ
X_int = np.column_stack([np.ones(n), np.eye(7)[pid], np.eye(7)[educ], np.eye(49)[
    cell]])
print("interaction model: p =", X_int.shape[1], " rank =", np.linalg.matrix_rank(
    X_int),
      " occupied cells =", int((counts > 0).sum()))

def interaction_contrast(i, k, j, l):
    """gamma_ij - gamma_il - gamma_kj + gamma_kl, as a coefficient vector on X_int.
    """
    lam = np.zeros(X_int.shape[1])
    for (r, c), s in [((i, j), 1), ((i, l), -1), ((k, j), -1), ((k, l), 1)]:
        lam[15 + r * 7 + c] += s
    return lam

print("PID 0 vs 3, educ 2 vs 3:", estimable(interaction_contrast(0, 3, 1, 2), X_int)
      )
```

```
print("PID 0 vs 3, educ 1 vs 3:", estimable(interaction_contrast(0, 3, 0, 2), X_int)
)
```

### 8.6.3 A factor and a numeric covariate

Now let a factor with  $g$  levels, with indicator matrix  $\mathbf{Z}$ , share the model with a numeric covariate  $\mathbf{x}$ . The two basic models are the **parallel lines** model

$$E(y) = \mu_k + \theta x \quad \text{for an observation at level } k, \quad (8.6.3)$$

with model matrix  $[\mathbf{Z}, \mathbf{x}]$ , and the **separate lines** model

$$E(y) = \mu_k + \theta_k x, \quad (8.6.4)$$

with model matrix  $[\mathbf{Z}, \mathbf{Z} \circ \mathbf{x}]$ , where  $\mathbf{Z} \circ \mathbf{x}$  multiplies each indicator column entrywise by  $\mathbf{x}$ , so its  $k$ th column is  $\mathbf{x}$  on level  $k$  and zero elsewhere. The same models can be written with an intercept and any coding of the factor. For example, the separate lines model in reference coding has columns  $\mathbf{1}$ ,  $\mathbf{ZC}$ ,  $\mathbf{x}$ ,  $(\mathbf{ZC}) \circ \mathbf{x}$ . The coefficients of the last block are then differences of slopes from the reference level, and the column space is unchanged.

By [Exercise \(B1\)](#), the common slope  $\theta$  is identifiable iff  $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$ , that is, iff  $x$  varies within at least one level. In the separate lines model, the slope  $\theta_k$  is identifiable iff  $x$  varies within level  $k$ , since the  $k$ th column of  $\mathbf{Z} \circ \mathbf{x}$  is a multiple of  $\mathbf{z}_k$  exactly when  $x$  is constant on level  $k$ . When the slopes are identifiable, the separate lines model is simply  $g$  simple regressions fitted at once. Its coefficients are those of the separate fits, because the columns for different levels have disjoint supports.

The parallel lines model is where adjustment happens.

#### Proposition 8.6.3 — Adjusted differences

In the parallel lines model [Eq. 8.6.3](#) with  $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$ , let  $\bar{y}_k$  and  $\bar{x}_k$  be the level means. Then

$$\hat{\theta} = \frac{\sum_k \sum_{i \in k} (x_i - \bar{x}_k)(y_i - \bar{y}_k)}{\sum_k \sum_{i \in k} (x_i - \bar{x}_k)^2}, \quad \hat{\mu}_k - \hat{\mu}_l = (\bar{y}_k - \bar{y}_l) - \hat{\theta}(\bar{x}_k - \bar{x}_l).$$

*Proof.* The model matrix  $[\mathbf{Z}, \mathbf{x}]$  has full column rank because  $\mathbf{x} \notin \mathcal{C}(\mathbf{Z})$ . Apply [Theorem 6.6.2](#) with  $\mathbf{X}_1 = \mathbf{x}$  and  $\mathbf{X}_2 = \mathbf{Z}$ . Projecting onto  $\mathcal{C}(\mathbf{Z})$  replaces each entry by its level mean ([Example \(The one-way layout\)](#)), so  $(\mathbf{I} - \mathbf{M}_Z)\mathbf{x}$  and  $(\mathbf{I} - \mathbf{M}_Z)\mathbf{y}$  are the within-level deviations, and  $\hat{\theta}$  is the no-intercept slope of one on the other. This is the first formula. By [Theorem 6.6.2\(c\)](#),  $\hat{\boldsymbol{\mu}} = (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top (\mathbf{y} - \hat{\theta} \mathbf{x})$ , whose  $k$ th entry is  $\bar{y}_k - \hat{\theta} \bar{x}_k$ . Subtracting gives the second formula. ■

The difference  $\bar{y}_k - \bar{y}_l$  compares the levels as they happen to be. The adjusted difference compares them at equal values of the covariate, correcting for the fact that the groups differ in  $x$ . The

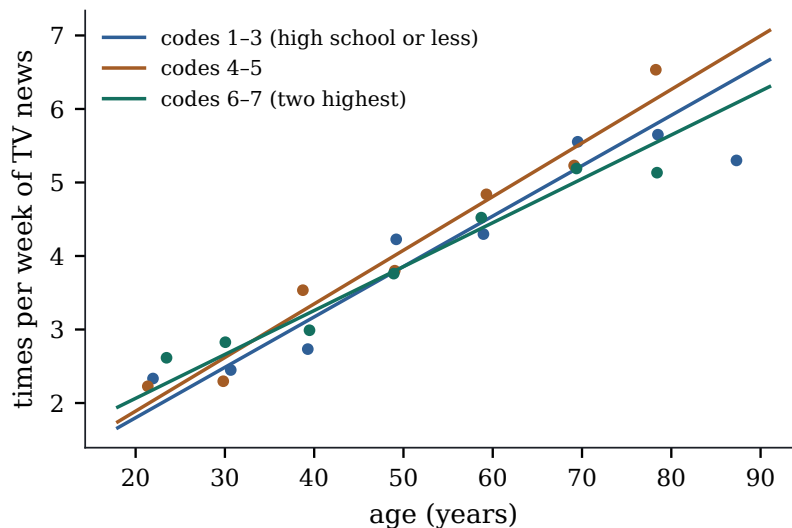


Figure 8.6.1: Times per week of television news against age for three education groups in the 1996 ANES: separate least squares lines (Eq. 8.6.4) and the group means within ten-year age bins (points).

correction is the slope times the difference in covariate means. It is large when the groups are unbalanced on a covariate that matters. The estimated slope uses only variation *within* levels, so that differences between levels in both  $x$  and  $y$  cannot be mistaken for an effect of  $x$ .

#### Example. (Education and television news, adjusted for age)

In the ANES data, respondents report how many times per week they watch television news. Coding education with seven levels and “high-school graduate” as the reference, respondents with some high school but no diploma watch 1.005 times a week more than high-school graduates. They are also older, with mean ages 59.8 and 48.2, and news watching rises with age. Adding age to the model, the within-level slope is 0.0674 times per week per year of age, and the adjusted difference shrinks to 0.224. More than three quarters of the raw difference is an age difference, and Proposition 8.6.3 accounts for it exactly. Whether the adjusted figure is the “effect of education” is a different question. Here age plausibly influences both schooling (older cohorts had less of it) and viewing habits, which argues for adjusting, but Chapter 25 shows that adjustment is not always appropriate.

Allowing separate age slopes for three education groups (codes 1–3, high school or less, 313 respondents; codes 4–5, 277; and the two highest levels, codes 6–7, 354) gives slopes 0.0686, 0.0729 and 0.0597, close to the common slope 0.0672 of the parallel lines model with the same three groups. The residual sum of squares falls only from 5623.6 to 5616.7 for the two extra parameters. Figure 8.6.1 shows the three fitted lines with binned means. The lines are nearly parallel, and the highest education group starts higher and rises a little more slowly.

```
tv = anes["TVnews"].to_numpy()           # times per week watching TV
news
age = anes["age"].to_numpy()
E = np.eye(7)[educ][:, [0, 1, 3, 4, 5, 6]] # reference: high-school
graduate
unadjusted = np.linalg.lstsq(np.column_stack([np.ones(n), E]), tv, rcond=None)[0]
```

```
adjusted = np.linalg.lstsq(np.column_stack([np.ones(n), E, age]), tv, rcond=None)[0]
print("some high school minus high-school graduate:",
      f"unadjusted {unadjusted[2]:.3f}, adjusted for age {adjusted[2]:.3f}")
print("mean age: some high school", age[educ == 1].mean().round(1),
      " high-school graduate", age[educ == 2].mean().round(1),
      " common age slope", adjusted[-1].round(4))
```

```
grp = np.select([educ <= 2, educ <= 4], [0, 1], 2) # education codes 1-3 / 4-5 /
6-7
D = np.eye(3)[grp]
X_par = np.column_stack([D, age]) # parallel lines: 3 intercepts,
one slope
X_sep = np.column_stack([D, D * age[:, None]]) # separate lines: 3 intercepts,
3 slopes
b_par = np.linalg.lstsq(X_par, tv, rcond=None)[0]
b_sep = np.linalg.lstsq(X_sep, tv, rcond=None)[0]
sse_par = np.sum((tv - X_par @ b_par) ** 2)
sse_sep = np.sum((tv - X_sep @ b_sep) ** 2)
print("separate slopes:", b_sep[3:].round(4),
      f" SSE parallel {sse_par:.1f}, separate {sse_sep:.1f}")
```

*Warning.* In a separate lines model the difference between two levels is  $(\mu_k - \mu_l) + (\theta_k - \theta_l)x$ , which depends on  $x$ . The coefficient that software labels as the “main effect” of level  $k$  is this difference at  $x = 0$ . For a covariate like age, that is a comparison of newborns, far outside the data. Centre the covariate at a meaningful value before reading such coefficients, or report the difference at several values of  $x$ .

## 8.6.4 Broken lines

Indicators can also switch a slope on. Let  $c$  be a known value of  $x$  and put  $(x-c)_+ = \max(x-c, 0)$ , which is  $x - c$  times the indicator of  $x > c$ . The model

$$E(y) = \beta_0 + \beta_1 x + \beta_2 (x - c)_+$$

is a continuous line with a bend at  $c$ : slope  $\beta_1$  to the left of  $c$  and  $\beta_1 + \beta_2$  to the right. Adding the indicator  $\mathbf{1}\{x > c\}$  itself allows a jump at  $c$  as well. With the bend point known, these are ordinary linear models, and all of this chapter applies. For example,  $\beta_2$  is identifiable iff the data contain values of  $x$  on both sides of  $c$  (strictly below and strictly above) and at least three distinct values of  $x$  in all ([Exercise \(B2\)](#)). When  $c$  is unknown the model is nonlinear in  $c$ . Such piecewise linear terms are the simplest splines, the subject of Part IX.

## 8.6.5 Exercises

### 8.6.5.1 A. Check your understanding

#### Exercise A1.

In the additive model with  $a = b = 2$ , suppose cells  $(1, 1)$ ,  $(2, 1)$  and  $(2, 2)$  are occupied and  $(1, 2)$  is empty. Find the rank of  $\mathbf{X}$  and a basis of  $\mathcal{N}(\mathbf{X})$ . Show that  $\mu + \alpha_1 + \beta_2$ , the mean of the empty cell, is estimable by writing it as a combination of the means of

occupied cells.

### Exercise A2.

A factor has three levels, and the covariate  $x$  is constant on level 2 but varies on levels 1 and 3. Find the rank of the separate lines model matrix  $[\mathbf{Z}, \mathbf{Z} \circ \mathbf{x}]$ , and say which of  $\mu_1, \mu_2, \mu_3, \theta_1, \theta_2, \theta_3$  are identifiable.

### 8.6.5.2 B. Practice

#### Exercise B1.

In the separate lines model, show that the mean at level  $k$  and covariate value  $x_0$ ,  $\mu_k + \theta_k x_0$ , is estimable iff either  $x$  varies within level  $k$ , or  $x_0$  equals the common value of  $x$  on level  $k$ . Interpret this as a statement about extrapolation.

*Solution.* The mean is  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  with  $\boldsymbol{\lambda}$  having 1 in the position of  $\mu_k$  and  $x_0$  in the position of  $\theta_k$ . The columns for  $\mu_k$  and  $\theta_k$  are  $\mathbf{z}_k$  and  $\mathbf{z}_k \circ \mathbf{x}$ , and they involve only level  $k$ . If  $x$  varies on level  $k$ , both coefficients are identifiable ([Proposition 8.1.3](#)), so  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is estimable. If  $x \equiv x^*$  on level  $k$ , then  $\mathbf{z}_k \circ \mathbf{x} = x^* \mathbf{z}_k$ , and the null space contains the vector with  $-x^*$  at  $\mu_k$  and 1 at  $\theta_k$ . Then  $\boldsymbol{\lambda}^\top \mathbf{v} = x_0 - x^*$ , which vanishes iff  $x_0 = x^*$ . A single covariate value within a level supports no statement about other values, however many observations share it.

#### Exercise B2.

Write the broken-line model  $E(y) = \beta_0 + \beta_1 x + \beta_2 (x - c)_+$  in matrix form and prove the identifiability condition stated in the text. Show that the continuous broken line is the same as the model with two separate lines, one on each side of  $c$ , constrained to meet at  $c$ , and count parameters to explain the difference of one.

#### Exercise B3.

In the interaction model, suppose all cells are occupied except  $(1, 1)$ . Show that the  $(a - 1)(b - 1)$  contrasts  $\gamma_{ij} - \gamma_{i1} - \gamma_{1j} + \gamma_{11}$  are no longer estimable, but that  $\gamma_{ij} - \gamma_{i2} - \gamma_{1j} + \gamma_{12}$  is estimable for  $i \geq 2, j \geq 3$ . How many linearly independent estimable interaction contrasts remain?

### 8.6.5.3 C. Going deeper

#### Exercise C1.

In the additive model, let the design graph have components  $C_1, \dots, C_c$ . Show that  $\lambda_0 \mu + \sum_i \lambda_i \alpha_i + \sum_j \kappa_j \beta_j$  is estimable iff  $\lambda_0 = \sum_j \kappa_j$  and, for every component  $C$ ,  $\sum_{i \in C} \lambda_i = \sum_{j \in C} \kappa_j$ . Deduce [Theorem 8.6.1\(b\)](#).

*Solution.* From the proof of [Theorem 8.6.1](#), the null vectors are  $(m, \mathbf{u}, \mathbf{v})$  with  $u_i = t_{C(i)}$  and  $v_j = -m - t_{C(j)}$ , for arbitrary  $m, t_1, \dots, t_c$ . The inner product with the coefficient vector is

$$\lambda_0 m + \sum_i \lambda_i t_{C(i)} - \sum_j \kappa_j (m + t_{C(j)}) = m \left( \lambda_0 - \sum_j \kappa_j \right) + \sum_C t_C \left( \sum_{i \in C} \lambda_i - \sum_{j \in C} \kappa_j \right).$$

This vanishes for all  $m, t$  iff every bracket is zero, which is the stated condition ([Theorem 8.3.1\(c\)](#)). For  $\alpha_i - \alpha_k$ ,  $\lambda_0 = 0$ , all  $\kappa_j = 0$ , and the component sums of  $\lambda$  are zero iff  $i$  and  $k$  lie in the same component.

## 8.7 Contrasts

In the one-way model the individual effects  $\alpha_k$  are not estimable, but their differences are. The general form of such a difference is a contrast. Contrasts are the natural targets of inference about a factor. They are estimable, they do not depend on the side condition or coding, and they are what a scientific question about a factor usually asks: does treatment beat control, is there a trend across doses, do two groups of levels differ? This section characterizes them, shows how an orthogonal set of contrasts splits the factor's sum of squares, and constructs the Helmert and polynomial families.

### 8.7.1 Definition and estimability

Throughout, the one-way model is  $E(y_{kj}) = \mu + \alpha_k$  for observation  $j$  at level  $k$ , with  $n_k \geq 1$  observations at level  $k = 1, \dots, g$ , level means  $\mu_k = \mu + \alpha_k$  and sample means  $\bar{y}_k$ .

#### Definition 8.7.1 — Contrast

In a factor model, a **contrast** in the effects of a factor is a linear function  $\sum_k c_k \alpha_k$  whose coefficients satisfy  $\sum_k c_k = 0$ , not all zero. In the one-way model it equals  $\sum_k c_k \mu_k$ , and we call either form a contrast in the level means. Two contrasts are **orthogonal** (for the given design) if  $\sum_k c_k d_k / n_k = 0$ .

The two forms agree because  $\sum_k c_k \mu_k = \mu \sum_k c_k + \sum_k c_k \alpha_k$ . This is where the condition  $\sum_k c_k = 0$  comes from: it makes the unidentified  $\mu$  drop out. Simple differences  $\alpha_k - \alpha_l$ , comparisons of one level with the average of others such as  $\alpha_1 - \frac{1}{2}(\alpha_2 + \alpha_3)$ , and trend coefficients are all contrasts.

#### Theorem 8.7.1 — Estimable functions in the one-way model

In the one-way model with every  $n_k \geq 1$ , the function  $\lambda_0 \mu + \sum_k \lambda_k \alpha_k$  is estimable iff  $\lambda_0 = \sum_k \lambda_k$ . In particular, a linear function of the effects alone,  $\sum_k c_k \alpha_k$ , is estimable iff it is a contrast (or zero). For a contrast, the least squares estimator and its variance under  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$  are

$$\hat{\psi} = \sum_k c_k \bar{y}_k, \quad \text{Var}(\hat{\psi}) = \sigma^2 \sum_k \frac{c_k^2}{n_k}. \quad (8.7.1)$$

*Proof.* The null space of  $\mathbf{X} = [\mathbf{1}, \mathbf{Z}]$  is spanned by  $\mathbf{v} = (1, -1, \dots, -1)^\top$  (Example (A rank-deficient layout)). By Theorem 8.3.1(c), the function is estimable iff  $\lambda_0 - \sum_k \lambda_k = 0$ . For an estimable function,  $\lambda_0 \mu + \sum_k \lambda_k \alpha_k = \sum_k \lambda_k (\mu + \alpha_k)$ . Every least squares solution has  $\hat{\mu} + \hat{\alpha}_k = \bar{y}_k$ , since the fitted values are the level means (Example (The one-way layout)). So the estimate is  $\sum_k \lambda_k \bar{y}_k$ . The level means are uncorrelated with variances  $\sigma^2/n_k$ , by Theorem 2.2.1, which gives the variance. ■

The estimator in Eq. 8.7.1 is the same whatever coding or side condition was used to fit the model. Proposition 8.3.2 guarantees this, and it is the reason contrasts, rather than coefficients, should be reported.

## 8.7.2 Sums of squares and orthogonal contrasts

Each contrast defines a direction in the observation space, and its estimate is the component of  $\mathbf{y}$  along that direction. With  $\mathbf{D} = \text{diag}(n_1, \dots, n_g)$  and  $\mathbf{Z}$  the indicator matrix, put

$$\mathbf{u}_c = \mathbf{ZD}^{-1}\mathbf{c}, \quad (8.7.2)$$

the vector whose entries equal  $c_k/n_k$  for the observations at level  $k$ .

### Proposition 8.7.2 — Contrasts as projections

Let  $\mathbf{c}$  and  $\mathbf{d}$  be contrasts, and write  $\hat{\psi}_c = \mathbf{c}^\top \hat{\boldsymbol{\mu}}$ . Then:

- (a)  $\hat{\psi}_c = \mathbf{u}_c^\top \mathbf{y}$ , and  $\mathbf{u}_c$  is a nonzero vector in  $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$ ;
- (b) the **sum of squares for the contrast**,

$$\text{SS}(\mathbf{c}) = \frac{\hat{\psi}_c^2}{\sum_k c_k^2/n_k},$$

equals  $\|\mathbf{P}\mathbf{y}\|^2$ , where  $\mathbf{P}$  is the orthogonal projection onto the line spanned by  $\mathbf{u}_c$ ;

- (c)  $\text{Cov}(\hat{\psi}_c, \hat{\psi}_d) = \sigma^2 \sum_k c_k d_k / n_k = \sigma^2 \mathbf{u}_c^\top \mathbf{u}_d$  when  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , so orthogonal contrasts have uncorrelated estimates and orthogonal vectors  $\mathbf{u}$ ;
- (d) if  $\mathbf{c}_1, \dots, \mathbf{c}_{g-1}$  are pairwise orthogonal contrasts, then

$$\sum_{j=1}^{g-1} \text{SS}(\mathbf{c}_j) = \sum_{k=1}^g n_k (\bar{y}_k - \bar{y})^2,$$

the sum of squares between levels.

*Proof.*

- (a)  $\hat{\boldsymbol{\mu}} = \mathbf{D}^{-1}\mathbf{Z}^\top \mathbf{y}$  (Eq. 8.5.1), so  $\mathbf{c}^\top \hat{\boldsymbol{\mu}} = (\mathbf{ZD}^{-1}\mathbf{c})^\top \mathbf{y}$ . Clearly  $\mathbf{u}_{\mathbf{c}} \in \mathcal{C}(\mathbf{Z})$ , and  $\mathbf{1}^\top \mathbf{u}_{\mathbf{c}} = \mathbf{1}^\top \mathbf{ZD}^{-1}\mathbf{c} = (n_1, \dots, n_g)\mathbf{D}^{-1}\mathbf{c} = \sum_k c_k = 0$ . It is nonzero because  $\mathbf{ZD}^{-1}$  has full column rank and  $\mathbf{c} \neq \mathbf{0}$ .
- (b) By Example (Projections onto a line),  $\|\mathbf{Py}\|^2 = (\mathbf{u}_{\mathbf{c}}^\top \mathbf{y})^2 / \|\mathbf{u}_{\mathbf{c}}\|^2$ , and  $\|\mathbf{u}_{\mathbf{c}}\|^2 = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{Z}^\top \mathbf{ZD}^{-1}\mathbf{c} = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{c} = \sum_k c_k^2 / n_k$ .
- (c) Theorem 2.2.1 gives  $\text{Cov}(\mathbf{u}_{\mathbf{c}}^\top \mathbf{Y}, \mathbf{u}_{\mathbf{d}}^\top \mathbf{Y}) = \sigma^2 \mathbf{u}_{\mathbf{c}}^\top \mathbf{u}_{\mathbf{d}}$ , and the same computation as in (b) gives  $\mathbf{u}_{\mathbf{c}}^\top \mathbf{u}_{\mathbf{d}} = \mathbf{c}^\top \mathbf{D}^{-1}\mathbf{d}$ .
- (d) The space  $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$  has dimension  $g - 1$  (Theorem 6.5.1(c), with  $\mathcal{C}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{Z})$ ). By (a) and (c) the vectors  $\mathbf{u}_{\mathbf{c}_1}, \dots, \mathbf{u}_{\mathbf{c}_{g-1}}$  are  $g - 1$  nonzero, mutually orthogonal vectors in it, so they form an orthogonal basis. By Theorem 6.3.8 the sum of the projections onto the lines they span is the projection onto that space, which is  $\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top$  (Theorem 6.5.1(b)). Applying it to  $\mathbf{y}$  and using Pythagoras gives  $\sum_j \text{SS}(\mathbf{c}_j) = \|(\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}\|^2$ . The vector  $(\mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}$  has entry  $\bar{y}_k - \bar{y}$  at every observation of level  $k$ .

■

So a complete set of orthogonal contrasts decomposes the between-level sum of squares into  $g - 1$  one-dimensional pieces. Under normality the pieces are independent, each is  $\sigma^2$  times a noncentral  $\chi^2$  with one degree of freedom, and each is independent of the residual sum of squares (Theorem 4.4.4). Each contrast can therefore be tested on its own. Chapter 9 places this decomposition in the general theory of orthogonal sums of squares, and Chapter 15 uses it for the one-way analysis of variance (Theorem 15.4.1). In a *balanced* design, with all  $n_k$  equal, orthogonality reduces to the ordinary  $\sum_k c_k d_k = 0$ . In an unbalanced design, contrasts that look orthogonal on paper need not be orthogonal for the data.

### 8.7.3 Helmert contrasts

The Helmert matrix of Example (Helmert matrices) has rows proportional to

$$\mathbf{h}_k = (\underbrace{1, \dots, 1}_{k-1}, -(k-1), 0, \dots, 0), \quad k = 2, \dots, g.$$

The contrast  $\mathbf{h}_k^\top \boldsymbol{\mu}$  compares level  $k$  with the average of the levels before it, since it is  $(k - 1)$  times  $\frac{1}{k-1} \sum_{l < k} \mu_l - \mu_k$ . The  $\mathbf{h}_k$  are mutually orthogonal in the ordinary sense, so in a balanced design they are orthogonal contrasts. They suit factors whose levels are added in a meaningful order, such as successive refinements of a treatment, where each new level is compared with everything before it.

### 8.7.4 Polynomial contrasts

When the levels of a factor are ordered and equally spaced, such as doses 0, 10, 20, 30 or the seven points of a party scale, the natural questions are about the *shape* of the sequence of means. Is there a linear trend? Curvature? Give the levels scores  $s_k = k$ . Apply Gram–Schmidt (Proposition 6.2.2) to the vectors  $\mathbf{1}, \mathbf{s}, \mathbf{s}^2, \dots, \mathbf{s}^{g-1}$  in  $\mathbb{R}^g$ , where  $\mathbf{s}^j$  has entries  $s_k^j$ . This gives orthogonal vectors  $\mathbf{p}_0, \dots, \mathbf{p}_{g-1}$ , and  $\mathbf{p}_j$  is a polynomial of degree  $j$  in the score. For  $j \geq 1$ ,



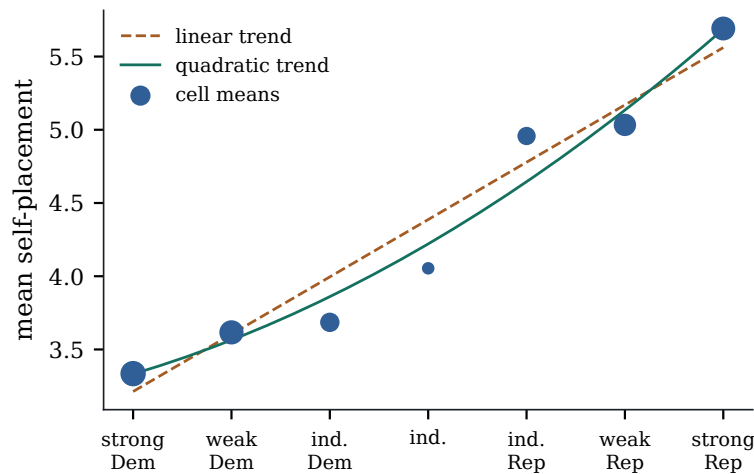


Figure 8.7.1: Mean ideological self-placement by party identification (point area proportional to group size), with the weighted least squares linear and quadratic trends in the score. The linear contrast accounts for almost all of the variation between groups.

$\mathbf{p}_j \perp \mathbf{1}$ , so  $\mathbf{p}_j$  is a contrast. Rescaled to integers, the first three for  $g = 7$  are

linear  $(-3, -2, -1, 0, 1, 2, 3)$ , quadratic  $(5, 0, -3, -4, -3, 0, 5)$ , cubic  $(-1, 1, 1, 0, -1, -1, 1)$ .

By construction  $\mathbf{p}_j$  is orthogonal to every polynomial of degree less than  $j$ . So if the level means follow a polynomial of degree  $d$  in the score, every contrast of degree above  $d$  is zero. The linear contrast measures the trend, the quadratic the curvature beyond it, and so on. This is the reason for their use.

With unequal group sizes the classical integer contrasts are no longer orthogonal for the design. The fix follows [Proposition 8.7.2\(c\)](#). Orthogonality is needed in the inner product  $\sum_k c_k d_k / n_k$ , which suggests setting  $\mathbf{c}_j = \mathbf{D}\mathbf{p}_j$ , where now the  $\mathbf{p}_j$  are orthogonal polynomials in the weighted inner product  $\sum_k n_k p_k q_k$ . Then  $\sum_k c_{jk} = \sum_k n_k p_{jk} = 0$ , and  $\sum_k c_{jk} c_{lk} / n_k = \sum_k n_k p_{jk} p_{lk} = 0$  for  $j \neq l$ . In observation space,  $\mathbf{u}_{\mathbf{c}_j} = \mathbf{Z}\mathbf{p}_j$  is the polynomial of degree  $j$  in each observation's score, orthogonalized over the observations. The weighted linear contrast gives exactly the regression sum of squares of  $\mathbf{y}$  on the score ([Exercise \(C1\)](#)).

#### Example. (Trend in ideology across party identification)

Return to the seven party-identification groups of [Example \(Party identification and ideology\)](#). The between-group sum of squares is 772.87. The classical linear contrast has estimate  $\hat{\psi} = \sum_k (k-4)\bar{y}_k = 11.175$ , variance  $0.1652 \sigma^2$  by [Eq. 8.7.1](#), and sum of squares 755.85, which is 97.8 percent of the between-group sum of squares. The quadratic and cubic contrasts have sums of squares 10.15 and 2.58. Ideology rises almost linearly across the seven-point party scale, with some upward curvature ([Figure 8.7.1](#)).

The group sizes range from 37 to 200, so the six classical contrasts are not orthogonal for this design. Their sums of squares add to 784.02, which exceeds the between-group total. Part of the variation is counted twice. The weighted orthogonal polynomial contrasts decompose the total exactly: 746.57 for the linear trend and 26.30 for the five higher-degree contrasts together.

```
import numpy as np
import statsmodels.api as sm
```

```

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].to_numpy()
level = anes["PID"].to_numpy().astype(int)          # 0 = strong Dem. ... 6 = strong
Rep.
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k

```

```

def orthogonal_polynomials(scores, weights):
    """Columns 1..g-1: polynomials in the scores, orthogonal in sum_k w_k u_k v_k.
    """
    V = np.vander(scores, len(scores), increasing=True)      # 1, s, s^2, ...
    W = np.sqrt(weights)[: , None]
    Q, _ = np.linalg.qr(W * V)                               # weighted Gram-Schmidt
    P = Q / W                                                  # back to the original
    scale
    P = P / np.abs(P).max(axis=0)                             # cosmetic rescaling
    return P[:, 1:]                                           # drop the constant
    column

def contrast_ss(c, means, n_k):
    """Estimate c'μ-hat and its sum of squares (c'μ-hat)^2 / sum(c_k^2 / n_k)."""
    psi = c @ means
    return psi, psi ** 2 / np.sum(c ** 2 / n_k)

scores = np.arange(g, dtype=float)
C_unw = orthogonal_polynomials(scores, np.ones(g))           # the classical table (
    equal n)
C_wtd = n_k[:, None] * orthogonal_polynomials(scores, n_k)   # c_k = n_k p(k)
ss_between = np.sum(n_k * (means - y.mean()) ** 2)
for name, C in [("unweighted", C_unw), ("weighted", C_wtd)]:
    ss = [contrast_ss(C[:, j], means, n_k)[1] for j in range(g - 1)]
    print(f"{name:10s} SS by degree: {np.round(ss, 2)} total {sum(ss):.2f}",
          f"(between groups {ss_between:.2f})")

```

## 8.7.5 Coding matrices are not contrast matrices

Section 8.5 coded a factor by an intercept and the columns  $\mathbf{ZC}$ . When the columns of  $\mathbf{C}$  are contrasts, as for Helmert or polynomial coding, it is tempting to read the coefficient of column  $j$  as the estimate of the contrast  $\mathbf{c}_j^\top \hat{\boldsymbol{\mu}}$ . That reading is wrong in general. By Proposition 8.5.1(b) the coefficients are the rows of  $\mathbf{K}^{-1}$ , with  $\mathbf{K} = [\mathbf{1}, \mathbf{C}]$ , applied to  $\hat{\boldsymbol{\mu}}$ . The rows of  $\mathbf{K}^{-1}$  and the columns of  $\mathbf{K}$  are related simply only when the columns are orthogonal.

### Proposition 8.7.3 — Orthogonal codings

If the columns of  $\mathbf{K} = [\mathbf{1}_g, \mathbf{c}_1, \dots, \mathbf{c}_{g-1}]$  are mutually orthogonal, then the coefficients of the coded model are

$$\hat{\gamma}_0 = \frac{1}{g} \sum_k \bar{y}_k, \quad \hat{\gamma}_j = \frac{\mathbf{c}_j^\top \hat{\boldsymbol{\mu}}}{\mathbf{c}_j^\top \mathbf{c}_j}, \quad j = 1, \dots, g-1,$$

whatever the group sizes.

*Proof.* With orthogonal columns,  $\mathbf{K}^\top \mathbf{K} = \text{diag}(g, \mathbf{c}_1^\top \mathbf{c}_1, \dots, \mathbf{c}_{g-1}^\top \mathbf{c}_{g-1})$ , so  $\mathbf{K}^{-1} = (\mathbf{K}^\top \mathbf{K})^{-1} \mathbf{K}^\top$ . Apply [Proposition 8.5.1\(b\)](#). ■

The orthogonality required here is the plain one,  $\mathbf{c}_j^\top \mathbf{c}_l = 0$ , not the design orthogonality of [Definition 8.7.1](#), because  $\hat{\boldsymbol{\gamma}} = \mathbf{K}^{-1} \hat{\boldsymbol{\mu}}$  is pure algebra and does not involve the  $n_k$ . Helmert coding, in the usual software convention, uses the columns  $\mathbf{c}_j = -\mathbf{h}_{j+1}$ , which compare level  $j+1$  with the levels before it, and  $\mathbf{c}_j^\top \mathbf{c}_j = j(j+1)$ . In the party data the first three Helmert coefficients are 0.1408, 0.0698 and 0.1271, while the contrasts themselves are 0.2817, 0.4187 and 1.5253, larger by the factors 2, 6 and 12. For a non-orthogonal coding there is no such shortcut. Reference coding has  $\mathbf{c}_j = \mathbf{e}_{j+1}$ , yet its coefficients are  $\mu_{j+1} - \mu_1$ , not  $\mu_{j+1}$ .

```
def helmert(g):
    """Coding matrix whose column k compares level k+1 with the mean of levels 1..k.
    """
    C = np.zeros((g, g - 1))
    for k in range(1, g):
        C[:k, k - 1] = -1.0
        C[k, k - 1] = k
    return C

H = helmert(g)
K = np.column_stack([np.ones(g), H])
gamma = np.linalg.solve(K, means)           # Helmert-coded coefficients
psi = H.T @ means                          # the contrasts c_k' mu-hat
print("coefficients:", np.round(gamma[1:], 4))
print("contrasts:    ", np.round(psi, 4))
print("ratio:        ", np.round(psi / gamma[1:], 1))  # = c_k' c_k = k(k+1)
```

*Warning.* A coefficient from a contrast-coded regression estimates a contrast only up to a scale factor, and only when the coding columns are orthogonal. To test or report a contrast, compute  $\mathbf{c}^\top \hat{\boldsymbol{\mu}}$  and its variance [Eq. 8.7.1](#) directly, or read it from the coded fit using  $\mathbf{K}^{-1}$ . Do not read it off the coefficient table.

## 8.7.6 Contrasts in the two-way additive model

In the additive model  $\mu + \alpha_i + \beta_j$  of [Section 8.6](#), a **row contrast** is  $\sum_i c_i \alpha_i$  with  $\sum_i c_i = 0$ . It is a combination of the differences  $\alpha_i - \alpha_1$ , so in a connected design it is estimable ([Theorem 8.6.1](#)). In a balanced design, with  $m \geq 1$  observations in every one of the  $ab$  cells, the least squares fit is  $\bar{y}_{i..} + \bar{y}_{.j.} - \bar{y}_{...}$ . This is shown in [Example \(Additive two-factor model, one observation per cell\)](#) for  $m = 1$ , and the averaging argument there is unchanged for  $m > 1$ . Hence

$$\widehat{\sum_i c_i \alpha_i} = \sum_i c_i (\bar{y}_{i..} + \bar{y}_{.j.} - \bar{y}_{...}) = \sum_i c_i \bar{y}_{i..}, \quad \text{Var} = \frac{\sigma^2}{bm} \sum_i c_i^2,$$

since the terms not depending on  $i$  are multiplied by  $\sum_i c_i = 0$ . The row contrast is estimated from the row means alone, as if factor  $B$  were absent. In an unbalanced design this is false. The row means are then contaminated by the column effects ([Exercise \(B2\)](#)), and the least squares

estimate has to adjust for them. That adjustment is the subject of [Chapter 17](#). Interaction contrasts, and their estimability when cells are empty, were treated in [Proposition 8.6.2](#).

## 8.7.7 Exercises

### 8.7.7.1 A. Check your understanding

#### Exercise A1.

Find the linear, quadratic and cubic orthogonal polynomial contrasts for  $g = 4$  equally spaced levels, scaled to integers. Check that they are mutually orthogonal and that the quadratic contrast vanishes when  $\mu_k = a + bk$ .

#### Exercise A2.

Show that the Helmert contrasts  $\mathbf{h}_2, \dots, \mathbf{h}_g$  are orthogonal for the design iff  $n_1 = n_2 = \dots = n_{g-1}$ , whatever  $n_g$  is. *Hint:* compute  $\sum_k h_{jk}h_{lk}/n_k$  for  $j < l$ .

### 8.7.7.2 B. Practice

#### Exercise B1.

Show that for every contrast  $\mathbf{c}$ ,  $\text{SS}(\mathbf{c}) \leq \sum_k n_k (\bar{y}_k - \bar{y})^2$ , with equality when  $c_k = n_k(\bar{y}_k - \bar{y})$ . This fact is the basis of Scheffé's method in [Chapter 13](#) ([Theorem 13.2.2](#)).

*Solution.* By [Proposition 8.7.2\(b\)](#),  $\text{SS}(\mathbf{c}) = \|\mathbf{P}\mathbf{y}\|^2$  with  $\mathbf{P}$  projecting onto a line inside  $\mathcal{S} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$ . Let  $\mathbf{P}_{\mathcal{S}} = \mathbf{M} - n^{-1}\mathbf{1}\mathbf{1}^\top$ . Because the line lies in  $\mathcal{S}$ ,  $\mathbf{P}\mathbf{y} = \mathbf{P}\mathbf{P}_{\mathcal{S}}\mathbf{y}$  ([Theorem 6.5.1\(a\)](#)), and so  $\|\mathbf{P}\mathbf{y}\|^2 \leq \|\mathbf{P}_{\mathcal{S}}\mathbf{y}\|^2 = \sum_k n_k (\bar{y}_k - \bar{y})^2$ . Equality holds when the line contains  $\mathbf{P}_{\mathcal{S}}\mathbf{y}$ , whose entries are  $\bar{y}_k - \bar{y}$  at level  $k$ . That vector is  $\mathbf{u}_{\mathbf{c}}$  for  $c_k = n_k(\bar{y}_k - \bar{y})$ , and this  $\mathbf{c}$  is a contrast because  $\sum_k n_k(\bar{y}_k - \bar{y}) = 0$ .

#### Exercise B2.

In the additive two-way model with  $n_{ij}$  observations in cell  $(i, j)$ , show that  $E(\bar{y}_{i..}) = \mu + \alpha_i + \sum_j n_{ij}\beta_j/n_{i..}$ . Deduce that  $\sum_i c_i \bar{y}_{i..}$  is unbiased for every row contrast and every  $\beta$  iff the proportions  $n_{ij}/n_{i..}$  do not depend on  $i$ , that is, iff the cell counts are *proportional*,  $n_{ij} = n_{i..}n_{.j}/n$ .

### 8.7.7.3 C. Going deeper

#### Exercise C1.

Let  $\mathbf{x}$  be the  $n$ -vector giving each observation's score  $s_k$ . Show that the sum of squares of the weighted linear contrast  $\mathbf{c}_1 = \mathbf{D}\mathbf{p}_1$  equals the regression sum of squares  $\|(\mathbf{M}_1 - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{y}\|^2$  of the simple regression of  $\mathbf{y}$  on  $\mathbf{x}$  with intercept, where  $\mathbf{M}_1$  projects onto

$\mathcal{C}([\mathbf{1}, \mathbf{x}])$ . Conclude that the remaining weighted contrasts together measure the lack of fit of the straight line to the level means.

*Solution.* In the weighted inner product,  $\mathbf{p}_1$  is  $\mathbf{s}$  minus its weighted mean,  $\mathbf{s} - \bar{s}_w \mathbf{1}$  with  $\bar{s}_w = \sum_k n_k s_k / n$ , up to scale. Then  $\mathbf{u}_{\mathbf{c}_1} = \mathbf{ZD}^{-1}\mathbf{Dp}_1 = \mathbf{Zp}_1$ , whose entries are  $s_k - \bar{s}_w$ . That is  $\mathbf{x} - \bar{x}\mathbf{1}$ , the centred score vector. By [Proposition 8.7.2\(b\)](#),  $SS(\mathbf{c}_1)$  is the squared length of the projection of  $\mathbf{y}$  onto the line spanned by  $\mathbf{x} - \bar{x}\mathbf{1}$ . By [Example \(Intercept inside simple regression\)](#) this line is  $\mathcal{C}([\mathbf{1}, \mathbf{x}]) \cap \mathbf{1}^\perp$ , and the projection onto it is  $\mathbf{M}_1 - n^{-1}\mathbf{1}\mathbf{1}^\top$ . The remaining weighted contrasts span the orthogonal complement of that line in  $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$ , which is  $\mathcal{C}(\mathbf{Z}) \cap \mathcal{C}([\mathbf{1}, \mathbf{x}])^\perp$ . Their total sum of squares is  $\|(\mathbf{M} - \mathbf{M}_1)\mathbf{y}\|^2$ , the reduction in residual sum of squares from the straight-line model to the model with a separate mean for each level.

## 8.8 Summary and Notes

### 8.8.1 Summary

*Idea.*

1. The data determine the mean vector  $\mathbf{X}\boldsymbol{\beta}$  and nothing more. A function of  $\boldsymbol{\beta}$  is identifiable iff it is a function of  $\mathbf{X}\boldsymbol{\beta}$ , and  $\boldsymbol{\beta}$  itself is identifiable iff  $\text{rank}(\mathbf{X}) = p$  ([Definition 8.1.1](#), [Theorem 8.1.1](#) and [Proposition 8.1.2](#)).
2. A single coefficient is identifiable iff its column is not a combination of the others ([Proposition 8.1.3](#)). Rank deficiency comes from construction, from identities among variables (age, period and cohort), or from the design ([Example \(Age, period and cohort\)](#)).
3. A linear function  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  has a linear unbiased estimator iff it is identifiable iff  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$  ([Definition 8.2.1](#) and [Theorem 8.2.2](#)). The estimable functions form a space of dimension  $\text{rank}(\mathbf{X})$  ([Corollary 8.2.3](#)).
4. Eight equivalent tests of estimability, including  $\boldsymbol{\lambda} \perp \mathcal{N}(\mathbf{X})$ ,  $\boldsymbol{\lambda}^\top \mathbf{G}\mathbf{X}^\top \mathbf{X} = \boldsymbol{\lambda}^\top$ , and invariance over least squares solutions ([Theorem 8.3.1](#)). The estimate is  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  for any solution, with variance  $\sigma^2 \boldsymbol{\lambda}^\top \mathbf{G}\boldsymbol{\lambda}$  for any generalized inverse ([Proposition 8.3.2](#)).
5. For a nonestimable  $\boldsymbol{\lambda}$ , a generalized-inverse solution estimates a different, estimable function  $\boldsymbol{\lambda}^\top \mathbf{G}\mathbf{X}^\top \mathbf{X}\boldsymbol{\beta}$ , and the reported variance can be any number ([Proposition 8.3.3](#) and [Exercise \(C1\)](#)). Decide estimability with the SVD of  $\mathbf{X}$ , and treat near-nonestimable functions as unestimated ([Example \(Estimable in principle, not in practice\)](#)).
6. Full-rank reparameterizations are choices of basis for the estimable space ([Theorem 8.4.1](#)). Side conditions  $\mathbf{T}\mathbf{b} = \mathbf{0}$  select a unique solution iff  $\text{rank}(\mathbf{T}) = p - r$  and no combination of them is estimable ([Theorem 8.4.2](#)). The constrained coefficients are then particular estimable functions ([Proposition 8.4.3](#)). The minimum-norm solution is itself a side condition ([Proposition 8.4.4](#)).
7. A coding of a factor is a nonsingular matrix  $\mathbf{K}$ , and its coefficients estimate  $\mathbf{K}^{-1}\boldsymbol{\mu}$ , the rows of  $\mathbf{K}^{-1}$  applied to the level means ([Definition 8.5.1](#) and [Proposition 8.5.1](#)).
8. In the additive two-way model the rank is  $a + b$  minus the number of connected components of the design, and a difference of levels is estimable iff the levels are

connected ([Theorem 8.6.1](#)). In the interaction model, exactly the combinations of occupied cell means are estimable ([Proposition 8.6.2](#)).

9. With a covariate, adjusted differences between levels subtract the slope times the difference in covariate means, the slope being estimated within levels ([Proposition 8.6.3](#)).
10. Contrasts are the estimable functions of a factor's effects ([Definition 8.7.1](#) and [Theorem 8.7.1](#)). Orthogonal contrasts, in the design inner product  $\sum_k c_k d_k / n_k$ , decompose the between-level sum of squares ([Proposition 8.7.2](#)). Coding coefficients equal scaled contrasts only for orthogonal codings ([Proposition 8.7.3](#)).

## 8.8.2 Notes and sources

**Identifiability and estimability.** Christensen (2020, §2.1) develops estimability from identifiability, as this chapter does, and stresses that the estimable functions are exactly the identifiable linear ones. Agresti (2015, §1.4) gives a short treatment aimed at generalized linear models, where the same definitions apply to the linear predictor. Rencher and Schaalje (2008, ch. 12) is a detailed textbook account of the non-full-rank model: estimable functions and their tests, the estimators  $\mathbf{r}^\top \mathbf{X}^\top \mathbf{y}$  and  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ , reparameterization and side conditions, with small worked layouts. The notion of an estimable function goes back to R. C. Bose's work on the design of experiments in the 1940s. Scheffé (1959) and Searle (1971) made it standard, and Rao (1973) treats it through generalized inverses. The characterization [Theorem 8.3.1](#) collects conditions that are scattered through these sources. That a generalized-inverse solution estimates the estimable function  $\boldsymbol{\lambda}^\top \mathbf{G} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}$  ([Proposition 8.3.3](#)) follows at once from  $E(\mathbf{G} \mathbf{X}^\top \mathbf{Y}) = \mathbf{G} \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}$ , but its consequence for software output deserves more emphasis than it usually gets.

**Side conditions and software.** The exact conditions of [Theorem 8.4.2](#), and the formula  $(\mathbf{X}^\top \mathbf{X} + \mathbf{T}^\top \mathbf{T})^{-1} \mathbf{X}^\top \mathbf{y}$ , are standard (see Rencher and Schaalje 2008, §12.6, and Searle 1971). The coding matrices, called “contrasts”, that S and R use for factors are described by Chambers and Hastie (1992), and Venables and Ripley (2002) discuss the treatment, sum, Helmert and polynomial codings of R. The minimum-norm solution rests on the Moore–Penrose inverse (Penrose 1955), and column-pivoted QR is described in Golub and Van Loan (2013).

**Indicator variables.** Sen and Srivastava (1990, ch. 4) is a practical account of indicator variables: the two-sample comparison as a regression, polychotomous factors, hierarchical codings, factors with continuous covariates, broken-line regression and aggregated indicators as responses. Where this chapter meets those themes it takes its own route: the two-sample comparison through the coding proposition ([Exercise \(A1\)](#)), the indicator of a single observation through the Frisch–Waugh–Lovell theorem ([Exercise \(B1\)](#)), and count regressors without intercept in a new setting ([Exercise \(B3\)](#)).

**Connectedness, empty cells and cohorts.** Connectedness of two-way designs, and its role in estimability, is treated at length by Searle (1971, 1987). Searle (1987) is also the standard reference for unbalanced data and empty cells, which [Chapter 17](#) takes up. The age–period–cohort problem of [Example \(Age, period and cohort\)](#) is a well-known instance of a dependency among variables. Holford (1983) showed which functions of the age, period and cohort effects are estimable, among them the second differences of [Exercise \(C1\)](#).

**Data.** The survey examples use the extract of the 1996 American National Election Study distributed with statsmodels (Seabold and Perktold 2010) as a public-domain data set. The other examples use simulated data, generated by the scripts in `code/ch08/`.

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# Chapter 9

## Sums of Squares and Orthogonal Decomposition

An analysis of variance table looks like bookkeeping. Behind every row there is a subspace, and behind the whole table an orthogonal decomposition of the space in which the data live: sums of squares are squared lengths of projections, and degrees of freedom are dimensions. Sequential and partial sums of squares, and the Type I, II and III sums of squares printed by software, differ only in which subspaces they project onto, and so in which hypotheses they address. When the subspaces are orthogonal, as in balanced designs and with orthogonal polynomials, the distinctions disappear. The coefficient of determination is a ratio of two of these squared lengths, and we mark what it does not measure. Finally, the expectation of each sum of squares shows what an  $F$  ratio compares, preparing the tests of Part III.

**What you need.** Chapter 6 throughout, especially projections and their ranks ([Theorem 6.3.8](#), [Proposition 6.3.4](#)), nested models ([Theorem 6.5.1](#), [Theorem 6.5.2](#)), the Frisch–Waugh–Lovell theorem ([Theorem 6.6.2](#)) and  $R^2$  as a squared cosine ([Theorem 6.7.2](#)). The linear model and the moments of least squares estimates from [Chapter 5](#) ([Definition 5.3.1](#), [Theorem 5.5.1](#)). The mean of a quadratic form ([Theorem 2.3.1](#)) and the distribution theory of [Chapter 4](#) ([Theorem 4.4.4](#), [Theorem 4.5.1](#)). Estimable functions and contrasts from [Chapter 8](#) appear in [Sections 9.2 to 9.4](#).

### 9.0.1 Roadmap

- [Decomposing the total sum of squares](#)
- [Extra sums of squares](#)
- [Sequential and partial sums of squares](#)
- [Orthogonal designs](#)
- [The coefficient of determination and its limits](#)
- [Expected mean squares](#)
- [Summary and notes](#)



## 9.1 Decomposing the total sum of squares

Every sum of squares in a linear model is the squared length of a projection of the data. [Chapter 6](#) showed this for the residual sum of squares,  $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2$ , and for the drop in residual sum of squares between nested models,  $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2$  ([Eq. 6.5.2](#)). This section turns these observations into a general framework. An analysis of variance is a choice of mutually orthogonal subspaces that together fill the model space. Once that choice is made, the whole table follows from Pythagoras: the sums of squares add up because the pieces are orthogonal, and the degrees of freedom add up because dimensions of orthogonal subspaces add.

Throughout the chapter the model is  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  with  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$  ([Definition 5.3.1](#)).  $\mathbf{X}$  is  $n \times p$  of rank  $r$ , which may be less than  $p$ , and  $\mathbf{M}$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$ . The identities of the chapter hold for every data vector  $\mathbf{y}$ . Probability enters only where a covariance, an expectation or a distribution is stated, and systematically in [Section 9.6](#).

### 9.1.1 Orthogonal decompositions of the model space

#### Definition 9.1.1 — Analysis of variance decomposition

An **analysis of variance decomposition** of the model space is a list of subspaces  $\mathcal{V}_1, \dots, \mathcal{V}_k$  of  $\mathcal{C}(\mathbf{X})$  that are mutually orthogonal and whose sum is  $\mathcal{C}(\mathbf{X})$ :

$$\mathcal{C}(\mathbf{X}) = \mathcal{V}_1 \oplus \mathcal{V}_2 \oplus \dots \oplus \mathcal{V}_k, \quad \mathcal{V}_i \perp \mathcal{V}_j \ (i \neq j).$$

The subspace  $\mathcal{V}_{k+1} = \mathcal{C}(\mathbf{X})^\perp$  is the **error space**. Writing  $\mathbf{P}_i$  for the orthogonal projection onto  $\mathcal{V}_i$ , the **sum of squares** for  $\mathcal{V}_i$  is  $\|\mathbf{P}_i\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{P}_i \mathbf{y}$ , its **degrees of freedom** are  $r_i = \dim \mathcal{V}_i$ , and its **mean square** is  $\|\mathbf{P}_i\mathbf{y}\|^2/r_i$ .

The same space can be decomposed in many ways, and the choice is the statistician's: the constants, a regressor of interest, a block of nuisance regressors, the linear and quadratic parts of a trend. The theorem collects what is true of all decompositions.

#### Theorem 9.1.1 — The analysis of variance identity

Let  $\mathcal{V}_1, \dots, \mathcal{V}_k$  be an analysis of variance decomposition of  $\mathcal{C}(\mathbf{X})$ , with projections  $\mathbf{P}_1, \dots, \mathbf{P}_k$  and dimensions  $r_1, \dots, r_k$ . Then:

- (a)  $\mathbf{P}_i\mathbf{P}_j = \mathbf{0}$  for  $i \neq j$ ,  $\mathbf{P}_1 + \dots + \mathbf{P}_k = \mathbf{M}$ , and

$$\mathbf{I} = \mathbf{P}_1 + \mathbf{P}_2 + \dots + \mathbf{P}_k + (\mathbf{I} - \mathbf{M}),$$

a sum of  $k + 1$  mutually orthogonal projections;

- (b) for every  $\mathbf{y} \in \mathbb{R}^n$  the vectors  $\mathbf{P}_1\mathbf{y}, \dots, \mathbf{P}_k\mathbf{y}, (\mathbf{I} - \mathbf{M})\mathbf{y}$  are mutually orthogonal and

$$\|\mathbf{y}\|^2 = \|\mathbf{P}_1\mathbf{y}\|^2 + \dots + \|\mathbf{P}_k\mathbf{y}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2; \quad (9.1.1)$$

- (c)  $r_i = \text{rank}(\mathbf{P}_i) = \text{tr}(\mathbf{P}_i)$ ,  $r_1 + \dots + r_k = r$ , and the error space has dimension  $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$ ;

- (d) if the columns of  $\mathbf{Q}_i$  ( $n \times r_i$ ) form an orthonormal basis of  $\mathcal{V}_i$ , then  $\|\mathbf{P}_i \mathbf{y}\|^2 = \|\mathbf{Q}_i^\top \mathbf{y}\|^2$  is a sum of  $r_i$  squares, and together with an orthonormal basis  $\mathbf{Q}_{k+1}$  of the error space the matrix  $\mathbf{U} = [\mathbf{Q}_1, \dots, \mathbf{Q}_{k+1}]$  is orthogonal;
- (e) the subspaces  $\mathcal{W}_j = \mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_j$  form a chain  $\mathcal{W}_1 \subseteq \dots \subseteq \mathcal{W}_k = \mathcal{C}(\mathbf{X})$  with projections  $\mathbf{N}_j = \mathbf{P}_1 + \dots + \mathbf{P}_j$ , and  $\mathbf{P}_j = \mathbf{N}_j - \mathbf{N}_{j-1}$  (with  $\mathbf{N}_0 = \mathbf{0}$ ). Conversely, every chain  $\{\mathbf{0}\} = \mathcal{W}_0 \subseteq \mathcal{W}_1 \subseteq \dots \subseteq \mathcal{W}_k = \mathcal{C}(\mathbf{X})$  gives an analysis of variance decomposition with  $\mathcal{V}_j = \mathcal{W}_{j-1}^\perp \cap \mathcal{W}_j$ .

*Proof.*

- (a) For any  $\mathbf{u}$ ,  $\mathbf{P}_j \mathbf{u} \in \mathcal{V}_j \subseteq \mathcal{V}_i^\perp$ , so  $\mathbf{P}_i \mathbf{P}_j \mathbf{u} = \mathbf{0}$  by Proposition 6.3.4(c). Applying Theorem 6.3.8 repeatedly, the sum  $\mathbf{P}_1 + \dots + \mathbf{P}_k$  is the orthogonal projection onto  $\mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_k = \mathcal{C}(\mathbf{X})$ , and a projection is determined by its range, so the sum is  $\mathbf{M}$ . Each  $\mathbf{P}_i$  satisfies  $\mathbf{M} \mathbf{P}_i = \mathbf{P}_i$ , because its columns lie in  $\mathcal{C}(\mathbf{X})$ , and therefore  $(\mathbf{I} - \mathbf{M}) \mathbf{P}_i = \mathbf{0}$ . The projection  $\mathbf{I} - \mathbf{M}$  onto  $\mathcal{C}(\mathbf{X})^\perp$  (Theorem 6.3.3) is thus orthogonal to every  $\mathbf{P}_i$ .
- (b) Multiply  $\mathbf{I} = \sum_i \mathbf{P}_i + (\mathbf{I} - \mathbf{M})$  by  $\mathbf{y}$ . For two different pieces,  $(\mathbf{P}_i \mathbf{y})^\top (\mathbf{P}_j \mathbf{y}) = \mathbf{y}^\top \mathbf{P}_i \mathbf{P}_j \mathbf{y} = 0$ , using symmetry and (a). Pythagoras gives the identity.
- (c) Rank equals trace for projections (Proposition 6.3.4(b)), and the trace is additive, so  $\sum_i r_i = \text{tr}(\mathbf{M}) = r$  by Theorem 6.3.7, and  $\text{tr}(\mathbf{I} - \mathbf{M}) = n - r$ .
- (d) By Proposition 6.3.2,  $\mathbf{P}_i = \mathbf{Q}_i \mathbf{Q}_i^\top$ , so  $\mathbf{y}^\top \mathbf{P}_i \mathbf{y} = \|\mathbf{Q}_i^\top \mathbf{y}\|^2$ . Columns taken from different  $\mathbf{Q}_i$  are orthogonal because the subspaces are, so  $\mathbf{U}$  has orthonormal columns. It has  $r_1 + \dots + r_k + (n - r) = n$  of them, so it is square and hence orthogonal.
- (e) The  $\mathbf{N}_j$  are projections onto  $\mathcal{W}_j$  by Theorem 6.3.8, and the chain is nested by construction. Conversely, given a chain, Theorem 6.5.1(b) shows that  $\mathbf{N}_j - \mathbf{N}_{j-1}$  is the projection onto  $\mathcal{V}_j = \mathcal{W}_{j-1}^\perp \cap \mathcal{W}_j$ . For  $i < j$  we have  $\mathcal{V}_i \subseteq \mathcal{W}_i \subseteq \mathcal{W}_{j-1}$ , which is orthogonal to  $\mathcal{V}_j$ . The projections telescope to  $\mathbf{N}_k = \mathbf{M}$ , so by Theorem 6.3.8 the orthogonal sum of the  $\mathcal{V}_j$  is  $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$ .

■

Part (e) says that decompositions and chains are the same thing. Listing the rows of an analysis of variance table from the top is the same as fitting a sequence of ever larger models and recording what each step adds. Section 9.3 builds on this.

## 9.1.2 Degrees of freedom are dimensions

“Degrees of freedom” can seem like a rule of thumb (“subtract one for every parameter estimated”). Part (d) of Theorem 9.1.1 gives the phrase an exact meaning. Rotate the data into the orthonormal coordinates  $\mathbf{z} = \mathbf{U}^\top \mathbf{y}$ . The rotation preserves length, and it splits  $\mathbf{z}$  into consecutive blocks  $\mathbf{Q}_1^\top \mathbf{y}, \dots, \mathbf{Q}_{k+1}^\top \mathbf{y}$  of sizes  $r_1, \dots, r_k, n - r$ . The sum of squares for  $\mathcal{V}_i$  is the sum of the squares of the  $r_i$  coordinates in its block. *A sum of squares has  $r_i$  degrees of freedom because it is a sum of  $r_i$  squares of orthonormal linear combinations of the data, and it cannot be written with fewer.*

The rotation also explains why degrees of freedom govern the size of a sum of squares. If  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , the coordinates  $\mathbf{U}^\top \mathbf{Y}$  are uncorrelated with variance  $\sigma^2$  (Theorem 2.2.1), so a block of  $r_i$  of them has expected sum of squares  $r_i \sigma^2$  plus the squared length of its mean

(Section 9.6). The divisor  $n - 1$  of the sample variance is the case  $\mathbb{R}^n = \text{span}(\mathbf{1}) \oplus \mathbf{1}^\perp$ : the  $n$  deviations  $y_i - \bar{y}$  lie in an  $(n - 1)$ -dimensional subspace, and the Helmert matrix of [Example \(Helmert matrices\)](#) writes their sum of squares as  $n - 1$  squared orthonormal contrasts.

### 9.1.3 The regression analysis of variance table

The most common decomposition comes from the chain  $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$ , available whenever  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , which is the case for every model with an intercept. Write  $\mathbf{P}_1 = n^{-1}\mathbf{1}\mathbf{1}^\top$  for the projection onto the constant vectors ([Example \(Projections onto a line\)](#)). The chain gives the three-way decomposition of [Eq. 6.5.1](#),

$$\mathbf{I} = \mathbf{P}_1 + (\mathbf{M} - \mathbf{P}_1) + (\mathbf{I} - \mathbf{M}),$$

with ranks 1,  $r - 1$  and  $n - r$ . The squared lengths are

$$\|\mathbf{P}_1\mathbf{y}\|^2 = n\bar{y}^2, \quad \text{SSR} = \|(\mathbf{M} - \mathbf{P}_1)\mathbf{y}\|^2 = \sum_i (\hat{y}_i - \bar{y})^2, \quad \text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \sum_i (y_i - \hat{y}_i)^2.$$

They are usually arranged as follows.

| Source                      | df      | Sum of squares               | Mean square                |
|-----------------------------|---------|------------------------------|----------------------------|
| Mean                        | 1       | $n\bar{y}^2$                 | $n\bar{y}^2$               |
| Regression (after the mean) | $r - 1$ | SSR                          | $\text{SSR}/(r - 1)$       |
| Residual                    | $n - r$ | SSE                          | $s^2 = \text{SSE}/(n - r)$ |
| Total                       | $n$     | $\mathbf{y}^\top \mathbf{y}$ |                            |

This is the **uncorrected** table. Its total is the raw sum of squares  $\sum_i y_i^2$ , with  $n$  degrees of freedom. The row for the mean is rarely of interest, because it records only that the average response is not zero. Deleting it gives the **corrected** table, whose total is  $\text{SST} = \sum_i (y_i - \bar{y})^2 = \|(\mathbf{I} - \mathbf{P}_1)\mathbf{y}\|^2$  on  $n - 1$  degrees of freedom. The corrected identity  $\text{SST} = \text{SSR} + \text{SSE}$  is [Eq. 6.7.1](#). “Correcting for the mean” is thus a projection. We first project the data onto  $\mathbf{1}^\perp$  and then decompose what is left.

The ratio of the two mean squares,

$$F = \frac{\text{SSR}/(r - 1)}{\text{SSE}/(n - r)},$$

compares the variation along the regression directions with the variation in the error space. Under normality and when no regressor matters, it has the  $F(r - 1, n - r)$  distribution ([Theorem 4.4.6](#)). [Section 9.6](#) explains why this ratio is the right comparison, and [Chapter 11](#) develops the test.

A few formulas make the table easy to compute from any least squares solution.

#### Proposition 9.1.2 — Computing the sums of squares

Let  $\hat{\boldsymbol{\beta}}$  be any least squares estimate. Then:

- (a)  $\|\mathbf{M}\mathbf{y}\|^2 = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$  and  $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ ;
- (b) if  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , then  $\text{SSR} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y} - n\bar{y}^2$ ;

- (c) if  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$  has full column rank and  $\tilde{\mathbf{X}}_1 = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_1$  is the matrix of centred regressors, then  $\text{SSR} = \hat{\boldsymbol{\beta}}_1^\top \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1 = \hat{\boldsymbol{\beta}}_1^\top \tilde{\mathbf{X}}_1^\top \mathbf{y}$ , where  $\hat{\boldsymbol{\beta}}_1$  holds the slopes.

*Proof.*

- (a)  $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}\mathbf{y}$  (Theorem 6.4.1), so  $\|\mathbf{M}\mathbf{y}\|^2 = \mathbf{y}^\top \mathbf{M}\mathbf{y} = \mathbf{y}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$ . The second formula follows from  $\text{SSE} = \mathbf{y}^\top \mathbf{y} - \mathbf{y}^\top \mathbf{M}\mathbf{y}$  (Proposition 6.4.3). (b)  $\mathbf{M} - \mathbf{P}_1$  is a projection (Theorem 6.5.1), so  $\text{SSR} = \mathbf{y}^\top \mathbf{M}\mathbf{y} - \mathbf{y}^\top \mathbf{P}_1 \mathbf{y} = \|\mathbf{M}\mathbf{y}\|^2 - n\bar{y}^2$ ; now use (a). (c) By Lemma 6.6.1 with  $\mathbf{X}_2 = \mathbf{1}$ ,  $\mathbf{M} - \mathbf{P}_1$  is the projection onto  $\mathcal{C}(\tilde{\mathbf{X}}_1)$ , and by Theorem 6.6.2 the slopes are the coefficients of  $(\mathbf{M} - \mathbf{P}_1)\mathbf{y}$  in terms of the columns of  $\tilde{\mathbf{X}}_1$ . So  $(\mathbf{M} - \mathbf{P}_1)\mathbf{y} = \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1$  and  $\text{SSR} = \|\tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1\|^2$ . The last equality uses  $\tilde{\mathbf{X}}_1^\top \mathbf{y} = \tilde{\mathbf{X}}_1^\top (\mathbf{M} - \mathbf{P}_1)\mathbf{y} = \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1 \hat{\boldsymbol{\beta}}_1$ . ■

In (a),  $\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y}$  is the same for every solution of the normal equations, even though  $\hat{\boldsymbol{\beta}}$  is not unique when  $r < p$ . That is no accident. It equals  $\|\mathbf{M}\mathbf{y}\|^2$ , which depends only on the model space.

#### Example. (An analysis of variance table for murder rates)

Return to the regression of the 2009 murder rate of the 50 US states on poverty, single parenthood and urbanization, with an intercept (Example (Poverty and murder rates)). Here  $r = 4$  and  $\bar{y} = 4.514$ . The listing computes every sum of squares from the coordinates of  $\mathbf{y}$  in an orthonormal basis of  $\mathcal{C}(\mathbf{X})$ , as in Theorem 9.1.1(d): the first coordinate gives the mean, the next three the regression.

| Source              | df | Sum of squares | Mean square |
|---------------------|----|----------------|-------------|
| Mean                | 1  | 1018.81        | 1018.81     |
| Regression          | 3  | 182.80         | 60.93       |
| Residual            | 46 | 101.96         | 2.217       |
| Total (uncorrected) | 50 | 1303.57        |             |
| Total (corrected)   | 49 | 284.76         |             |

The regression mean square is 27.49 times the residual mean square. If none of the three regressors were related to the murder rate and the errors were normal, a ratio this large would occur with probability about  $2.5 \times 10^{-10}$ . The proportion of the corrected total that the regression accounts for is  $182.80/284.76 = 0.6419$ , the  $R^2$  of Theorem 6.7.2.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), data["poverty"], data["single"], data["urban"]])
r = np.linalg.matrix_rank(X)

Q, _ = np.linalg.qr(X)                                # orthonormal basis of C(X); first column
```

```

    spans 1
z = Q.T @ y                                # coordinates of My in that basis
ss_mean = z[0] ** 2                         # n * ybar^2
ss_reg = np.sum(z[1:] ** 2)                 # ||(M - P1) y||^2, r - 1 squared coordinates
ss_res = y @ y - np.sum(z ** 2)             # ||(I - M) y||^2

rows = [("mean", 1, ss_mean), ("regression", r - 1, ss_reg),
        ("residual", n - r, ss_res), ("total", n, y @ y)]
for name, df, ss in rows:
    print(f"{name:11s} df = {df:3d}    SS = {ss:9.2f}    MS = {ss / df:8.3f}")
F = (ss_reg / (r - 1)) / (ss_res / (n - r))
print(f"corrected total = {ss_reg + ss_res:.2f},    F = {F:.2f}")

```

The QR factorization supplies the orthonormal basis of [Theorem 9.1.1\(d\)](#); its first column is  $\pm \mathbf{1}/\sqrt{n}$  because Gram–Schmidt starts with the intercept ([Proposition 6.2.2](#)). No  $n \times n$  projection matrix is formed.

## 9.1.4 Models without an intercept

The corrected table depends on  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ . Without it, the chain  $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$  is not available, and the corrected “identity” fails.

### Proposition 9.1.3

For any model matrix  $\mathbf{X}$ , with  $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{y}$  and  $\hat{\mathbf{e}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$ ,

$$\sum_i (y_i - \bar{y})^2 = \sum_i (\hat{y}_i - \bar{y})^2 + \sum_i \hat{e}_i^2 - 2\bar{y} \sum_i \hat{e}_i.$$

The last term vanishes for every  $\mathbf{y}$  iff  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ .

*Proof.* Write  $\mathbf{y} - \bar{y}\mathbf{1} = (\hat{\mathbf{Y}} - \bar{y}\mathbf{1}) + \hat{\mathbf{e}}$  and expand the squared length. The cross term is  $2(\hat{\mathbf{Y}} - \bar{y}\mathbf{1})^\top \hat{\mathbf{e}} = 2\hat{\mathbf{Y}}^\top \hat{\mathbf{e}} - 2\bar{y}\mathbf{1}^\top \hat{\mathbf{e}} = -2\bar{y}\mathbf{1}^\top \hat{\mathbf{e}}$ , since  $\hat{\mathbf{Y}} \perp \hat{\mathbf{e}}$ . If  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , then  $\mathbf{1}^\top \hat{\mathbf{e}} = 0$  ([Proposition 6.4.3\(b\)](#)). If  $\mathbf{1} \notin \mathcal{C}(\mathbf{X})$ , take  $\mathbf{y} = \mathbf{1}$ . Then  $\bar{y} = 1$  and  $\mathbf{1}^\top \hat{\mathbf{e}} = \mathbf{1}^\top (\mathbf{I} - \mathbf{M})\mathbf{1} = \|(\mathbf{I} - \mathbf{M})\mathbf{1}\|^2 > 0$ , so the last term is not zero. ■

Without an intercept the valid decomposition is the uncorrected one,  $\|\mathbf{y}\|^2 = \|\mathbf{M}\mathbf{y}\|^2 + \text{SSE}$ , whose regression row has  $r$  degrees of freedom. That is appropriate only when the mean response is known to vanish when every regressor is zero.

### Example. (Forcing the murder regression through the origin)

Refit the model of [Example \(An analysis of variance table for murder rates\)](#) without the intercept. The residual sum of squares rises from 101.96 to 177.73, so the fit is much worse. Yet the residuals now sum to  $-8.332$ , and the would-be corrected identity is off by  $-2\bar{y} \sum_i \hat{e}_i = 75.22$ : the corrected total 284.76 is not  $31.81 + 177.73 = 209.54$ . The uncentred ratio  $\|\hat{\mathbf{Y}}\|^2/\|\mathbf{y}\|^2$ , which some software reports as  $R^2$  for such models, is 0.8637, much larger than the 0.6419 of the better model with an intercept ([Exercise](#)

(B1)).

```

X0 = X[:, 1:]                                # drop the intercept column
b0, *_ = np.linalg.lstsq(X0, y, rcond=None)
e0 = y - X0 @ b0
ybar = y.mean()
sst = np.sum((y - ybar) ** 2)
ssr_star = np.sum((X0 @ b0 - ybar) ** 2)
sse0 = e0 @ e0
print(f"SST = {sst:.2f} but SSR* + SSE = {ssr_star + sse0:.2f}")
print(f"sum of residuals = {e0.sum():.3f}, -2 ybar * sum = {-2 * ybar * e0.sum():.2f}")

```

### 9.1.5 Coarsening and refining

**Coarsening** merges rows: replacing  $\mathcal{V}_1$  and  $\mathcal{V}_2$  by  $\mathcal{V}_1 \oplus \mathcal{V}_2$  gives another decomposition, whose sums of squares and degrees of freedom are sums of the old ones. The regression row is a coarsening of the table that gives each regressor its own row (Section 9.3). **Refining** splits a row: any subspace of dimension  $r_i$  can be cut into  $r_i$  orthogonal lines, each carrying one degree of freedom. A refinement is useful only when its lines answer questions of interest. Section 9.4 chooses them with orthogonal contrasts and orthogonal polynomials.

*Idea.* An analysis of variance table is a list of mutually orthogonal subspaces that fill  $\mathbb{R}^n$ . Its sums of squares are the squared lengths of the projections of  $\mathbf{y}$ , its degrees of freedom are the dimensions, and both columns add up because of Pythagoras.

### 9.1.6 Exercises

#### 9.1.6.1 A. Check your understanding

##### Exercise A1.

For simple regression,  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$  with  $\mathbf{x}$  not constant, show that  $\text{SSR} = \hat{\beta}_1^2 S_{xx} = S_{xy}^2 / S_{xx}$  and that the  $F$  ratio of the corrected table equals  $t^2$ , where  $t = \hat{\beta}_1 \sqrt{S_{xx}} / s$ .

##### Exercise A2.

A regression with an intercept has a corrected table in which the regression row has 3 degrees of freedom and sum of squares 120, the residual mean square is 2.5, and the corrected total sum of squares is 172.5. Find  $n$ , the residual degrees of freedom, the  $F$  ratio and  $R^2$ .

#### 9.1.6.2 B. Practice

##### Exercise B1.

Let  $\mathbf{U} = [\mathbf{Q}_1, \dots, \mathbf{Q}_{k+1}]$  be the orthogonal matrix of Theorem 9.1.1(d) and  $\mathbf{z} = \mathbf{U}^\top \mathbf{Y}$ . Suppose  $E(\mathbf{Y}) = \boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ . Show that the coordinates of  $\mathbf{z}$  are

uncorrelated with variance  $\sigma^2$ , that the last  $n - r$  of them have mean zero, and deduce  $E(\text{SSE}) = (n - r)\sigma^2$ . Show that if in addition  $\mathbf{Y}$  is normal, the coordinates are independent.

*Solution.* By [Theorem 2.2.1](#),  $\text{Cov}(\mathbf{z}) = \mathbf{U}^\top(\sigma^2\mathbf{I})\mathbf{U} = \sigma^2\mathbf{I}$  and  $E(\mathbf{z}) = \mathbf{U}^\top\boldsymbol{\mu}$ . The last block is  $\mathbf{Q}_{k+1}^\top\boldsymbol{\mu}$ , and  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$  is orthogonal to the error space, so it is zero. Then  $\text{SSE} = \|\mathbf{Q}_{k+1}^\top\mathbf{Y}\|^2$  is a sum of  $n - r$  squares of variables with mean zero and variance  $\sigma^2$ , so its expectation is  $(n - r)\sigma^2$ . Under normality  $\mathbf{z}$  is normal ([Theorem 3.3.1](#)) with diagonal covariance, and uncorrelated jointly normal variables are independent ([Theorem 3.4.1](#)).

### Exercise B2.

For a model without an intercept, the quantity  $1 - \text{SSE}/\text{SST}$  is sometimes reported as  $R^2$ . Show that it is negative iff the model fits worse than the intercept-only model. Construct a data set with  $n = 4$  and one regressor for which it is below  $-1$ .

#### 9.1.6.3 C. Going deeper

### Exercise C1.

Let  $\mathcal{V}$  be a subspace with projection  $\mathbf{P}$ . Show that

$$\|\mathbf{Py}\|^2 = \max_{\mathbf{v} \in \mathcal{V}, \mathbf{v} \neq \mathbf{0}} \frac{(\mathbf{v}^\top \mathbf{y})^2}{\mathbf{v}^\top \mathbf{v}},$$

with the maximum attained at  $\mathbf{v} = \mathbf{Py}$  when  $\mathbf{Py} \neq \mathbf{0}$ . Interpret: the sum of squares of a row with  $r_i$  degrees of freedom is the largest one-degree-of-freedom sum of squares obtainable by refining that row. [Chapter 13](#) uses this fact to build Scheffé's simultaneous intervals.

*Solution.* For  $\mathbf{v} \in \mathcal{V}$ ,  $\mathbf{v}^\top \mathbf{y} = \mathbf{v}^\top \mathbf{Py}$ , so by Cauchy–Schwarz ([Proposition 1.1.1](#))  $(\mathbf{v}^\top \mathbf{y})^2 \leq \mathbf{v}^\top \mathbf{v} \|\mathbf{Py}\|^2$ , with equality iff  $\mathbf{v}$  is a multiple of  $\mathbf{Py}$ . Taking  $\mathbf{v} = \mathbf{Py} \in \mathcal{V}$  attains the bound. A line  $\text{span}(\mathbf{v})$  inside  $\mathcal{V}$  carries the one-degree-of-freedom sum of squares  $(\mathbf{v}^\top \mathbf{y})^2 / \mathbf{v}^\top \mathbf{v}$  ([Example \(Projections onto a line\)](#)), so no refinement of the row gives a line with a larger sum of squares than the whole row.

## 9.2 Extra sums of squares

Most questions asked of a regression compare two models. Each question names a reduced model inside the full one, and the natural measure of what the full model adds is how much it lowers the residual sum of squares. This section develops that measure, its computation when regressors are added to a fitted model, and a closed form when the reduced model is defined by linear constraints.

### 9.2.1 Notation

**Definition 9.2.1 — Extra sum of squares**

Let  $\mathbf{X}_1$  and  $\mathbf{X}_2$  be matrices with  $n$  rows, let  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$ , and let  $\mathbf{M}_1$  and  $\mathbf{M}$  be the projections onto  $\mathcal{C}(\mathbf{X}_1)$  and  $\mathcal{C}(\mathbf{X})$ . The **extra sum of squares** of  $\mathbf{X}_2$  given  $\mathbf{X}_1$  is

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_1) \mathbf{y} = \text{SSE}(\mathbf{X}_1) - \text{SSE}(\mathbf{X}),$$

with  $\text{rank}(\mathbf{X}) - \text{rank}(\mathbf{X}_1)$  degrees of freedom. Here  $\text{SSE}(\cdot)$  is the residual sum of squares of the model with the given model matrix.

The two expressions agree by [Eq. 6.5.2](#). One also writes  $\text{SS}(\boldsymbol{\beta}_2 \mid \boldsymbol{\beta}_1)$ , or  $\text{SS}(B \mid \mu, A)$  in factor models; some books write  $R$ , for “reduction”. Four properties follow from [Theorem 6.5.1](#) and [Theorem 6.7.1](#).

1. *It depends only on the two column spaces.* Rescaling, recoding, centring, replacing  $\mathbf{X}_2$  by  $\mathbf{X}_2 + \mathbf{X}_1 \mathbf{B}$ , and rank deficiency change nothing.
2. *It is nonnegative*, and zero iff  $\mathbf{M} \mathbf{y} = \mathbf{M}_1 \mathbf{y}$ , that is, iff adding  $\mathbf{X}_2$  leaves the fitted values unchanged.
3. *It obeys a chain rule:*  $\text{SS}(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1) = \text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) + \text{SS}(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2)$ , because the projections telescope.
4. *It is the sum of squares of a residualized regression.* The next lemma makes this precise.

**Lemma 9.2.1 — Extra sums of squares by residualizing**

For any  $\mathbf{X}_1$  and  $\mathbf{X}_2$ , let  $\tilde{\mathbf{X}}_2 = (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2$ . Then  $\mathcal{C}([\mathbf{X}_1, \mathbf{X}_2]) = \mathcal{C}(\mathbf{X}_1) \oplus \mathcal{C}(\tilde{\mathbf{X}}_2)$  with the two summands orthogonal, so  $\mathbf{M} = \mathbf{M}_1 + \mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)}$  and

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \|\mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)} \mathbf{y}\|^2 = \mathbf{y}^\top \tilde{\mathbf{X}}_2 (\tilde{\mathbf{X}}_2^\top \tilde{\mathbf{X}}_2)^{-1} \tilde{\mathbf{X}}_2^\top \mathbf{y}.$$

*Proof.* The argument for part (a) of [Lemma 6.6.1](#) uses no rank condition, so it applies here: the columns of  $\tilde{\mathbf{X}}_2$  lie in  $\mathcal{C}(\mathbf{X})$  and are orthogonal to  $\mathcal{C}(\mathbf{X}_1)$ , and every  $\mathbf{X}_1 \mathbf{b}_1 + \mathbf{X}_2 \mathbf{b}_2$  equals  $\tilde{\mathbf{X}}_2 \mathbf{b}_2 + (\mathbf{X}_1 \mathbf{b}_1 + \mathbf{M}_1 \mathbf{X}_2 \mathbf{b}_2)$ . By [Theorem 6.3.8](#),  $\mathbf{M} = \mathbf{M}_1 + \mathbf{P}_{\mathcal{C}(\tilde{\mathbf{X}}_2)}$ . The formula is [Theorem 6.3.7](#) applied to  $\tilde{\mathbf{X}}_2$ . ■

**9.2.2 Adding regressors to a fitted model**

Suppose  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  has been fitted and we add the columns of an  $n \times t$  matrix  $\mathbf{Z}$ , giving  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma}$  with model matrix  $\mathbf{W} = [\mathbf{X}, \mathbf{Z}]$ . Only a  $t \times t$  matrix has to be inverted.

**Theorem 9.2.2 — Adding regressors**

Let  $\mathbf{W} = [\mathbf{X}, \mathbf{Z}]$  have full column rank  $p + t$ , let  $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}$ ,  $\hat{\boldsymbol{\beta}} = \mathbf{C} \mathbf{X}^\top \mathbf{y}$ , and  $\mathbf{R} = \mathbf{I} - \mathbf{X} \mathbf{C} \mathbf{X}^\top$ . Then:



- (a)  $\mathbf{Z}^\top \mathbf{R} \mathbf{Z}$  is positive definite;
- (b) the least squares coefficients of the enlarged model are

$$\hat{\gamma} = (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R} \mathbf{y}, \quad \hat{\beta}_W = \hat{\beta} - \mathbf{L} \hat{\gamma}, \quad \mathbf{L} = \mathbf{C} \mathbf{X}^\top \mathbf{Z};$$

- (c) the residual sum of squares falls by

$$\text{SS}(\mathbf{Z} \mid \mathbf{X}) = \mathbf{y}^\top \mathbf{R} \mathbf{Z} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R} \mathbf{y} = \hat{\gamma}^\top \mathbf{Z}^\top \mathbf{R} \mathbf{y} = \hat{\gamma}^\top (\mathbf{Z}^\top \mathbf{R} \mathbf{Z}) \hat{\gamma};$$

- (d) if  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , then  $\text{Cov}(\hat{\gamma}) = \sigma^2 (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1}$ ,  $\text{Cov}(\hat{\beta}_W, \hat{\gamma}) = -\mathbf{L} \text{Cov}(\hat{\gamma})$  and

$$\text{Cov}(\hat{\beta}_W) = \sigma^2 [\mathbf{C} + \mathbf{L} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{L}^\top].$$

*Proof.*

- (a)  $\mathbf{R}$  is the symmetric idempotent residual projection of the model  $\mathbf{X}$ , so  $\mathbf{a}^\top \mathbf{Z}^\top \mathbf{R} \mathbf{Z} \mathbf{a} = \|\mathbf{R} \mathbf{Z} \mathbf{a}\|^2 \geq 0$ . If  $\mathbf{R} \mathbf{Z} \mathbf{a} = \mathbf{0}$ , then  $\mathbf{Z} \mathbf{a} \in \mathcal{C}(\mathbf{X})$ , say  $\mathbf{Z} \mathbf{a} = \mathbf{X} \mathbf{b}$ , so  $\mathbf{W}(-\mathbf{b}^\top, \mathbf{a}^\top)^\top = \mathbf{0}$  and full column rank forces  $\mathbf{a} = \mathbf{0}$ .
- (b) Apply the Frisch–Waugh–Lovell theorem ([Theorem 6.6.2](#)) with  $\mathbf{X}_1 = \mathbf{Z}$  and  $\mathbf{X}_2 = \mathbf{X}$ . Then  $\mathbf{M}_2 = \mathbf{I} - \mathbf{R}$  and  $\tilde{\mathbf{X}}_1 = \mathbf{R} \mathbf{Z}$ , so part (a) of that theorem gives  $\hat{\gamma} = (\mathbf{Z}^\top \mathbf{R} \mathbf{R} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R} \mathbf{y}$ , which is the stated formula because  $\mathbf{R}$  is symmetric and idempotent. Part (c) of the theorem gives  $\hat{\beta}_W = \mathbf{C} \mathbf{X}^\top (\mathbf{y} - \mathbf{Z} \hat{\gamma}) = \hat{\beta} - \mathbf{L} \hat{\gamma}$ .
- (c) By [Lemma 9.2.1](#), the extra sum of squares is the squared length of the projection of  $\mathbf{y}$  onto  $\mathcal{C}(\mathbf{R} \mathbf{Z})$ , which is  $\mathbf{y}^\top \mathbf{R} \mathbf{Z} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R} \mathbf{y}$  by [Theorem 6.3.7](#). Substituting (b) gives the other two forms.
- (d) Both  $\hat{\gamma}$  and  $\hat{\beta}$  are linear in  $\mathbf{y}$ , and by [Theorem 2.2.1](#)

$$\text{Cov}(\hat{\gamma}) = \sigma^2 (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{R} \mathbf{R} \mathbf{Z} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} = \sigma^2 (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1}, \quad \text{Cov}(\hat{\beta}, \hat{\gamma}) = \sigma^2 \mathbf{C} \mathbf{X}^\top \mathbf{R} \mathbf{Z} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1},$$

since  $\mathbf{X}^\top \mathbf{R} = \mathbf{0}$ . With  $\text{Cov}(\hat{\beta}) = \sigma^2 \mathbf{C}$  ([Theorem 5.5.1](#)), the formula  $\hat{\beta}_W = \hat{\beta} - \mathbf{L} \hat{\gamma}$  gives  $\text{Cov}(\hat{\beta}_W, \hat{\gamma}) = -\mathbf{L} \text{Cov}(\hat{\gamma})$  and  $\text{Cov}(\hat{\beta}_W) = \text{Cov}(\hat{\beta}) + \mathbf{L} \text{Cov}(\hat{\gamma}) \mathbf{L}^\top$ .

■

Since  $\mathbf{L} (\mathbf{Z}^\top \mathbf{R} \mathbf{Z})^{-1} \mathbf{L}^\top$  is nonnegative definite, part (d) shows that *adding regressors never decreases the variances of the coefficients already in the model*, when both models are correct (compare [Exercise \(B2\)](#), and see [Exercise \(C2\)](#)). [Chapter 26](#) measures this price of adjustment as variance inflation ([Definition 26.2.1](#)). In computation one updates a QR factorization instead of using these formulas ([Chapter 10](#)).

### Example. (Adding poverty and urbanization)

In the state data, fit the murder rate on single parenthood first:  $\hat{y} = -8.2146 + 0.5147 x_{\text{single}}$ , with residual sum of squares 121.92. Now add  $\mathbf{Z} = [\mathbf{x}_{\text{poverty}}, \mathbf{x}_{\text{urban}}]$ . The listing computes  $\hat{\gamma} = (0.2538, 0.0046)$  from the old residual projection alone, and corrects the old coefficients to  $-9.0946$  and  $0.3980$ . These agree with a fresh fit of the enlarged model. The residual sum of squares falls by 19.96, on 2 degrees of freedom, to 101.96. Single parenthood's coefficient drops by almost a quarter once poverty is allowed to take its share, the pattern seen in [Example \(Order matters\)](#).

```

import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
n = len(y)
one = np.ones(n)
cols = {"poverty": data["poverty"].to_numpy(), "single": data["single"].to_numpy(),
        "urban": data["urban"].to_numpy()}

X = np.column_stack([one, cols["single"]])           # the model already
        fitted
Z = np.column_stack([cols["poverty"], cols["urban"]]) # the regressors to add
XtX_inv = np.linalg.inv(X.T @ X)
beta_X = XtX_inv @ X.T @ y
R = np.eye(n) - X @ XtX_inv @ X.T                 # residual projection of
        the old model

gamma = np.linalg.solve(Z.T @ R @ Z, Z.T @ R @ y)   # new coefficients
beta_new = beta_X - XtX_inv @ X.T @ Z @ gamma       # old coefficients,
        corrected
extra_ss = gamma @ Z.T @ R @ y                     # SS(Z | X) = SSE_X -
        SSE_W
print("gamma =", gamma.round(4), " corrected beta =", beta_new.round(4))
print(f"SSE_X = {y @ R @ y:.2f}, extra SS = {extra_ss:.2f}")

```

### 9.2.3 One column at a time

With  $t = 1$ , [Theorem 9.2.2](#) links the extra sum of squares of a single regressor to its coefficient and standard error.

#### Corollary 9.2.3 — The extra sum of squares of one column

Let  $\mathbf{X}$  have full column rank  $p$ , write  $c_{jj}$  for the  $j$ th diagonal entry of  $(\mathbf{X}^\top \mathbf{X})^{-1}$ , and let  $\mathbf{X}_{(j)}$  be  $\mathbf{X}$  without its  $j$ th column. Then

$$SS(\mathbf{x}_j | \mathbf{X}_{(j)}) = \frac{\hat{\beta}_j^2}{c_{jj}} = t_j^2 s^2, \quad t_j = \frac{\hat{\beta}_j}{s\sqrt{c_{jj}}}, \quad s^2 = \frac{SSE}{n-p}.$$

Consequently, the ratio of this one-degree-of-freedom sum of squares to the residual mean square is  $t_j^2$ .

*Proof.* Permute the columns so that  $\mathbf{x}_j$  comes last; this carries  $c_{jj}$  to the last diagonal entry of the inverse. Write  $\mathbf{X} = [\mathbf{X}_{(j)}, \mathbf{x}_j]$  and  $\mathbf{R} = \mathbf{I} - \mathbf{X}_{(j)}(\mathbf{X}_{(j)}^\top \mathbf{X}_{(j)})^{-1} \mathbf{X}_{(j)}^\top$  for the residual projection of  $\mathbf{X}_{(j)}$ . By [Theorem 1.6.4](#) the last diagonal entry of  $(\mathbf{X}^\top \mathbf{X})^{-1}$  is the inverse of the Schur complement  $\mathbf{x}_j^\top \mathbf{x}_j - \mathbf{x}_j^\top \mathbf{X}_{(j)}(\mathbf{X}_{(j)}^\top \mathbf{X}_{(j)})^{-1} \mathbf{X}_{(j)}^\top \mathbf{x}_j = \mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j$ , so  $\mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j = 1/c_{jj}$ . Now apply [Theorem 9.2.2](#) with  $\mathbf{X}_{(j)}$  as the old model and  $\mathbf{Z} = \mathbf{x}_j$ : then  $\hat{\gamma} = \hat{\beta}_j$ , and part (c) gives  $SS(\mathbf{x}_j | \mathbf{X}_{(j)}) = \hat{\beta}_j^2 \mathbf{x}_j^\top \mathbf{R} \mathbf{x}_j = \hat{\beta}_j^2 / c_{jj}$ . The definition of  $t_j$  gives the last form. ■

For poverty in the three-regressor model,  $\hat{\beta} = 0.2538$  and  $c_{jj} = 0.003935$ , so the extra sum of squares is  $0.2538^2 / 0.003935 = 16.37$ . This is exactly the last entry for poverty in Order B of [Example \(Order matters\)](#). The  $t$  statistic is 2.718, and  $t^2 s^2 = 7.386 \times 2.217 = 16.37$ .

```
Xf = np.column_stack([one, cols["poverty"], cols["single"], cols["urban"]])
fit = sm.OLS(y, Xf).fit()
C = np.linalg.inv(Xf.T @ Xf)
j = 1 # poverty
partial_ss = fit.params[j] ** 2 / C[j, j]
print(f"beta = {fit.params[j]:.4f}, c_jj = {C[j, j]:.6f}, beta^2/c_jj = {partial_ss:.2f}")
print(f"t = {fit.tvalues[j]:.3f}, t^2 * s^2 = {fit.tvalues[j] ** 2 * fit.scale:.2f}")
```

## 9.2.4 Sums of squares for linear hypotheses

Reduced models often come as constraints: two coefficients are equal, a set of coefficients vanishes, a contrast among group means is zero. [Section 6.5](#) identified the test space of such a constraint ([Theorem 6.5.2](#)). The next theorem gives its sum of squares in terms of the least squares estimate, with no reduced model fitted.

### Theorem 9.2.4 — Sum of squares of a linear hypothesis

Let  $\mathbf{\Lambda}$  be a  $p \times q$  matrix of rank  $q$  such that each entry of  $\mathbf{\Lambda}^\top \boldsymbol{\beta}$  is estimable, so that  $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$  for some  $n \times q$  matrix  $\mathbf{T} = [\boldsymbol{\rho}_1, \dots, \boldsymbol{\rho}_q]$  ([Definition 8.2.1](#), [Theorem 6.4.4](#)). Let  $\mathbf{G}$  be any generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ , let  $\hat{\boldsymbol{\beta}}$  be any least squares estimate, and let  $\mathbf{d} \in \mathbb{R}^q$ . Then:

- (a)  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$  and  $\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$  do not depend on the choices of  $\hat{\boldsymbol{\beta}}$  and  $\mathbf{G}$ , and  $\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$  is positive definite;
- (b) the subspace  $\mathcal{S}_0 = \{\mathbf{Xb} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$  of  $\mathcal{C}(\mathbf{X})$  has orthogonal complement in  $\mathcal{C}(\mathbf{X})$  equal to  $\mathcal{C}(\mathbf{MT})$ , of dimension  $q$ , and the projection onto  $\mathcal{C}(\mathbf{MT})$  is  $\mathbf{MT}(\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{T}^\top \mathbf{M}$ ;
- (c) the minimum of  $\|\mathbf{y} - \mathbf{Xb}\|^2$  over  $\{\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}\}$  exceeds SSE by

$$SS_H = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}). \quad (9.2.1)$$

*Proof.*

- (a)  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{T}^\top \mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{T}^\top \mathbf{M} \mathbf{y}$  and  $\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda} = \mathbf{T}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{T} = \mathbf{T}^\top \mathbf{M} \mathbf{T}$  by [Theorem 6.3.7](#), and neither expression involves the choices. Since  $\mathbf{X}^\top \mathbf{M} = \mathbf{X}^\top$ , we have  $\mathbf{X}^\top (\mathbf{MT}) = \mathbf{\Lambda}$ , which has rank  $q$ , so the  $q$  columns of  $\mathbf{MT}$  are linearly independent. Hence  $\mathbf{T}^\top \mathbf{M} \mathbf{T} = (\mathbf{MT})^\top (\mathbf{MT})$  is positive definite.
- (b)  $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{T}^\top \mathbf{Xb}$ , so  $\mathcal{S}_0 = \mathcal{C}(\mathbf{X}) \cap \mathcal{N}(\mathbf{T}^\top)$ . [Theorem 6.5.2](#) identifies its orthogonal complement in  $\mathcal{C}(\mathbf{X})$  as  $\mathcal{C}(\mathbf{MT})$ . The columns of  $\mathbf{MT}$  are independent, so [Theorem 6.3.7](#) gives the projection  $\mathbf{MT}(\mathbf{T}^\top \mathbf{M} \mathbf{T})^{-1} \mathbf{T}^\top \mathbf{M}$ , and  $\mathbf{T}^\top \mathbf{M} \mathbf{T} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$  by (a).
- (c) First let  $\mathbf{d} = \mathbf{0}$ . Minimizing over  $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}$  is minimizing  $\|\mathbf{y} - \boldsymbol{\mu}\|^2$  over  $\boldsymbol{\mu} \in \mathcal{S}_0$ , so by [Eq. 6.5.2](#) the excess over SSE is the squared length of the projection of  $\mathbf{y}$  onto  $\mathcal{C}(\mathbf{X}) \ominus \mathcal{S}_0 = \mathcal{C}(\mathbf{MT})$ , where  $\ominus$  denotes the orthogonal complement within

$\mathcal{C}(\mathbf{X})$ . By (b) this is  $\mathbf{y}^\top \mathbf{M} \mathbf{T} (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{T}^\top \mathbf{M} \mathbf{y}$ , and  $\mathbf{T}^\top \mathbf{M} \mathbf{y} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$ . For general  $\mathbf{d}$ , the rows of  $\mathbf{\Lambda}^\top$  are independent, so there is a  $\mathbf{b}_0$  with  $\mathbf{\Lambda}^\top \mathbf{b}_0 = \mathbf{d}$ . Substituting  $\mathbf{b} = \mathbf{b}_0 + \mathbf{b}_*$  turns the problem into minimizing  $\|\mathbf{y}_* - \mathbf{X} \mathbf{b}_*\|^2$  subject to  $\mathbf{\Lambda}^\top \mathbf{b}_* = \mathbf{0}$ , with  $\mathbf{y}_* = \mathbf{y} - \mathbf{X} \mathbf{b}_0$ . The data  $\mathbf{y}_*$  have the same residual sum of squares as  $\mathbf{y}$ , and  $\hat{\boldsymbol{\beta}} - \mathbf{b}_0$  is a least squares estimate for them, since  $\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}_0) = \mathbf{M} \mathbf{y}_*$ . Applying the case  $\mathbf{d} = \mathbf{0}$  to  $\mathbf{y}_*$  gives Eq. 9.2.1, because  $\mathbf{\Lambda}^\top (\hat{\boldsymbol{\beta}} - \mathbf{b}_0) = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$ .

■

In Eq. 9.2.1,  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$  measures how far the estimate is from the hypothesis, and  $\sigma^2 \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$  is the covariance of  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$  (Chapter 7), so  $\text{SS}_H / \sigma^2$  is the squared Mahalanobis length of the discrepancy (Proposition 3.2.3). No reduced model has to be fitted.

In the full-rank case the constrained minimizer itself has a closed form, which recovers the classical restricted least squares estimator.

### Theorem 9.2.5 — Restricted least squares

Let  $\mathbf{X}$  have full column rank,  $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1}$ , and let  $\mathbf{\Lambda}$  ( $p \times q$ ) have rank  $q$ . For any  $\mathbf{d} \in \mathbb{R}^q$ , the unique minimizer of  $\|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2$  subject to  $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$  is

$$\hat{\boldsymbol{\beta}}_H = \hat{\boldsymbol{\beta}} - \mathbf{C} \mathbf{\Lambda} (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}),$$

and  $\|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}_H\|^2 = \text{SSE} + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2$ , where the last term equals  $\text{SS}_H$  of Eq. 9.2.1 with  $\mathbf{G} = \mathbf{C}$ .

*Proof.*  $\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda}$  is positive definite because  $\mathbf{C}$  is and  $\mathbf{\Lambda}$  has full column rank, and  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}_H = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}) = \mathbf{d}$ . For any  $\mathbf{b}$ , Eq. 6.3.1 with  $\mathbf{u} = \mathbf{X} \mathbf{b}$  gives  $\|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2 = \text{SSE} + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b})\|^2$ . Every feasible  $\mathbf{b}$  has the form  $\hat{\boldsymbol{\beta}}_H + \mathbf{v}$  with  $\mathbf{\Lambda}^\top \mathbf{v} = \mathbf{0}$ , and then

$$\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b})\|^2 = \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 - 2(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} + \|\mathbf{X} \mathbf{v}\|^2.$$

The cross term vanishes:  $(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)^\top \mathbf{X}^\top \mathbf{X} \mathbf{v} = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{C} \mathbf{X}^\top \mathbf{X} \mathbf{v} = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{v} = 0$ . So the objective is  $\text{SSE} + \|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 + \|\mathbf{X} \mathbf{v}\|^2$ , minimized exactly when  $\mathbf{X} \mathbf{v} = \mathbf{0}$ , that is,  $\mathbf{v} = \mathbf{0}$ . Finally  $\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \mathbf{C} \mathbf{X}^\top \mathbf{X} \mathbf{C} \mathbf{\Lambda} (\mathbf{\Lambda}^\top \mathbf{C} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})$ , which simplifies to  $\text{SS}_H$ .

■

The same estimator comes out of Lagrange multipliers (Proposition 1.11.3 with  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ ). The proof above shows instead that the constrained fit is the feasible point nearest to  $\hat{\boldsymbol{\beta}}$  in the geometry of  $\mathbf{X}^\top \mathbf{X}$ .

#### Example. (Equal effects of poverty and single parenthood)

Both poverty and single parenthood are percentages of the population. Is one percentage point of either associated with the same change in the murder rate? The hypothesis is  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$  with  $\boldsymbol{\lambda} = (0, 1, -1, 0)^\top$ . Formula Eq. 9.2.1 gives  $\text{SS}_H = 1.807$  on one degree of freedom, identical to the drop in residual sum of squares when the two regressors are replaced by their sum. Its ratio to the residual mean square is 0.815, so the data are entirely compatible with equal effects. For a hypothesis with a nonzero right-hand side,

$\beta_{\text{single}} = 0.5$ , [Theorem 9.2.5](#) gives the constrained coefficients 0.1803 for poverty and  $-0.0044$  for urbanization, and a sum of squares of 3.357, both confirmed by a direct fit of the reduced model.

```
def hypothesis_ss(X, y, Lam, d=None):
    """(L'b - d)' [L' (X'X)^- L]^-1 (L'b - d) for estimable L'beta = d."""
    G = np.linalg.pinv(X.T @ X)
    b = G @ X.T @ y
    u = Lam.T @ b - (0 if d is None else d)
    return u @ np.linalg.solve(Lam.T @ G @ Lam, u)

def sse(Z):
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    e = y - Z @ b
    return e @ e

Lam_equal = np.array([[0.0, 1.0, -1.0, 0.0]]).T          # beta_poverty =
beta_single
ss_equal = hypothesis_ss(Xf, y, Lam_equal)
X_equal = np.column_stack([one, cols["poverty"] + cols["single"], cols["urban"]])
print(f"H: equal slopes    formula {ss_equal:.3f}    SSE_0 - SSE {sse(X_equal) - sse(
Xf):.3f}")
```

## 9.2.5 Constraints that are not estimable

Without estimability, the reduced model  $\mathcal{S}_0 = \{\mathbf{X}\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$  is still defined, but it may be larger than the number of constraints suggests.

### Proposition 9.2.6 — How much a constraint restricts

For any  $p \times q$  matrix  $\mathbf{\Lambda}$ , the subspace  $\mathcal{S}_0 = \{\mathbf{X}\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$  has dimension  $r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$ . So the extra sum of squares for the constraint has  $\dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$  degrees of freedom. It is identically zero iff no nonzero linear combination of the constrained functions is estimable.

*Proof.*  $\mathcal{S}_0$  is the image of  $\mathcal{N}(\mathbf{\Lambda}^\top)$  under  $\mathbf{b} \mapsto \mathbf{X}\mathbf{b}$ , and the kernel of this map restricted to  $\mathcal{N}(\mathbf{\Lambda}^\top)$  is  $\mathcal{N}(\mathbf{\Lambda}^\top) \cap \mathcal{N}(\mathbf{X})$ , the null space of the stacked matrix  $[\mathbf{\Lambda}, \mathbf{X}^\top]^\top$ . By rank-nullity ([Theorem 1.2.2](#)),

$$\dim \mathcal{S}_0 = \dim \mathcal{N}(\mathbf{\Lambda}^\top) - \dim(\mathcal{N}(\mathbf{\Lambda}^\top) \cap \mathcal{N}(\mathbf{X})) = (p - \text{rank } \mathbf{\Lambda}) - (p - \text{rank}[\mathbf{\Lambda}, \mathbf{X}^\top]).$$

The dimension formula for a sum of subspaces ([Section 6.2](#)) gives  $\text{rank}[\mathbf{\Lambda}, \mathbf{X}^\top] = \dim(\mathcal{C}(\mathbf{\Lambda}) + \mathcal{C}(\mathbf{X}^\top)) = \text{rank } \mathbf{\Lambda} + r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$ . Substituting,  $\dim \mathcal{S}_0 = r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top))$ . The degrees of freedom of the extra sum of squares are  $r - \dim \mathcal{S}_0$ . A combination  $\mathbf{\Lambda}\mathbf{a}$  is estimable iff it lies in  $\mathcal{C}(\mathbf{X}^\top)$  ([Theorem 6.4.4](#)), which gives the last statement. ■

In the one-way layout  $E(Y_{ij}) = \mu + \alpha_i$ , the constraint  $\alpha_1 = \alpha_2$  has one degree of freedom. The constraint  $\alpha_1 = 0$  has none: it does not restrict the mean vector, and only picks one of the least squares solutions ([Theorem 8.4.2](#)). Software asked to “test” a nonestimable hypoth-

esis either refuses or silently tests its estimable part. [Chapter 17](#) meets this with empty cells ([Theorem 17.4.1](#)).

## 9.2.6 Exercises

### 9.2.6.1 A. Check your understanding

#### Exercise A1.

In simple regression, compute  $SS_H$  for  $\beta_1 = 0$  from [Eq. 9.2.1](#) and show that it equals  $\hat{\beta}_1^2 S_{xx}$ , in agreement with [Exercise \(A1\)](#).

### 9.2.6.2 B. Practice

#### Exercise B1.

Prove the chain rule  $SS(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1) = SS(\mathbf{X}_2 \mid \mathbf{X}_1) + SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2)$ , and deduce  $SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2) \leq SS(\mathbf{X}_2, \mathbf{X}_3 \mid \mathbf{X}_1)$ . Show by an example from [Example \(Adding poverty and urbanization\)](#) or its listing that  $SS(\mathbf{X}_3 \mid \mathbf{X}_1, \mathbf{X}_2) \leq SS(\mathbf{X}_3 \mid \mathbf{X}_1)$  can fail.

*Solution.* Let  $\mathbf{M}_1$ ,  $\mathbf{M}_{12}$  and  $\mathbf{M}_{123}$  be the projections onto the nested column spaces. Then  $\mathbf{M}_{123} - \mathbf{M}_1 = (\mathbf{M}_{12} - \mathbf{M}_1) + (\mathbf{M}_{123} - \mathbf{M}_{12})$ , and the chain rule follows by taking  $\mathbf{y}^\top(\cdot)\mathbf{y}$ . Both terms are nonnegative, which gives the inequality. For the failure, the listing of [Section 9.3](#) reports  $SS(\mathbf{x}_{\text{urban}} \mid \mathbf{1}) = 6.30$  but  $SS(\mathbf{x}_{\text{urban}} \mid \mathbf{1}, \mathbf{x}_{\text{poverty}}) = 29.26$ . Adjusting for poverty more than quadruples the sum of squares for urbanization.

#### Exercise B2.

In [Theorem 9.2.5](#) with  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , show that

$$\text{Cov}(\hat{\boldsymbol{\beta}}_H) = \sigma^2 [\mathbf{C} - \mathbf{C}\mathbf{A}(\mathbf{A}^\top \mathbf{C}\mathbf{A})^{-1} \mathbf{A}^\top \mathbf{C}],$$

that  $\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\hat{\boldsymbol{\beta}}_H)$  is nonnegative definite, and that  $E(\hat{\boldsymbol{\beta}}_H) = \boldsymbol{\beta}$  when the constraint is true. What is  $E(\hat{\boldsymbol{\beta}}_H)$  when it is false?

*Solution.* Write  $\mathbf{K} = \mathbf{C}\mathbf{A}(\mathbf{A}^\top \mathbf{C}\mathbf{A})^{-1}$ , so  $\hat{\boldsymbol{\beta}}_H = (\mathbf{I} - \mathbf{K}\mathbf{A}^\top)\hat{\boldsymbol{\beta}} + \mathbf{K}\mathbf{d}$ . By [Theorem 2.2.1](#) and  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{C}$ ,  $\text{Cov}(\hat{\boldsymbol{\beta}}_H) = \sigma^2 (\mathbf{I} - \mathbf{K}\mathbf{A}^\top)\mathbf{C}(\mathbf{I} - \mathbf{A}\mathbf{K}^\top)$ . Expanding and using  $\mathbf{A}^\top \mathbf{C}\mathbf{A}(\mathbf{A}^\top \mathbf{C}\mathbf{A})^{-1} = \mathbf{I}$ , the two cross terms and the last term all equal  $\mathbf{K}\mathbf{A}^\top \mathbf{C} = \mathbf{C}\mathbf{A}(\mathbf{A}^\top \mathbf{C}\mathbf{A})^{-1} \mathbf{A}^\top \mathbf{C}$ , with signs  $-, -, +$ , which gives the formula. The difference  $\sigma^2 \mathbf{C}\mathbf{A}(\mathbf{A}^\top \mathbf{C}\mathbf{A})^{-1} \mathbf{A}^\top \mathbf{C}$  is of the form  $\mathbf{B}\mathbf{A}\mathbf{B}^\top$  with  $\mathbf{A}$  positive definite, hence nonnegative definite. Finally  $E(\hat{\boldsymbol{\beta}}_H) = \boldsymbol{\beta} - \mathbf{K}(\mathbf{A}^\top \boldsymbol{\beta} - \mathbf{d})$ , which is  $\boldsymbol{\beta}$  iff  $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{d}$ . A false constraint buys the smaller variance at the price of a bias.

#### Exercise B3.

In the one-way layout with model matrix  $\mathbf{X} = [\mathbf{1}, \mathbf{z}_1, \dots, \mathbf{z}_g]$  and group sizes  $n_1, \dots, n_g$ , use the generalized inverse  $\mathbf{G} = \text{diag}(0, 1/n_1, \dots, 1/n_g)$  in [Eq. 9.2.1](#) to show that the sum of squares for  $\alpha_1 = \alpha_2$  is  $(\bar{y}_1 - \bar{y}_2)^2 / (1/n_1 + 1/n_2)$ . Verify with [Proposition 9.2.6](#)

that the constraint  $\alpha_1 = 0$  has zero degrees of freedom.

*Solution.*  $\mathbf{G}$  is a generalized inverse because  $\mathbf{X}^\top \mathbf{X}$  has the block form with first row  $(n, n_1, \dots, n_g)$  and lower right block  $\text{diag}(n_1, \dots, n_g)$ , and a direct multiplication gives  $\mathbf{X}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{X}$ . With  $\boldsymbol{\lambda} = (0, 1, -1, 0, \dots, 0)^\top$ ,  $\mathbf{G} \mathbf{X}^\top \mathbf{y} = (0, \bar{y}_1, \dots, \bar{y}_g)^\top$ , so  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \bar{y}_1 - \bar{y}_2$  and  $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda} = 1/n_1 + 1/n_2$ . For  $\alpha_1 = 0$ ,  $\boldsymbol{\lambda} = (0, 1, 0, \dots, 0)^\top$  is not in  $\mathcal{C}(\mathbf{X}^\top)$  (Example (Four answers, one fit)), and  $\mathcal{C}(\boldsymbol{\lambda})$  is a line, so  $\mathcal{C}(\boldsymbol{\lambda}) \cap \mathcal{C}(\mathbf{X}^\top) = \{\mathbf{0}\}$ .

### 9.2.6.3 C. Going deeper

#### Exercise C1.

Using Exercise (C1) and Lemma 9.2.1, show that

$$\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \max_{\mathbf{a}} \frac{(\mathbf{a}^\top \tilde{\mathbf{X}}_2^\top \mathbf{y})^2}{\mathbf{a}^\top \tilde{\mathbf{X}}_2^\top \tilde{\mathbf{X}}_2 \mathbf{a}},$$

the maximum being over  $\mathbf{a}$  with  $\tilde{\mathbf{X}}_2 \mathbf{a} \neq \mathbf{0}$ . Deduce that the extra sum of squares of a block is at least that of any single column of the block, given  $\mathbf{X}_1$ , and that the sum of squares of an estimable hypothesis  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$  equals  $\max_{\mathbf{a}} (\mathbf{a}^\top \boldsymbol{\Lambda}^\top \boldsymbol{\beta})^2 / (\mathbf{a}^\top \boldsymbol{\Lambda}^\top \mathbf{G} \boldsymbol{\Lambda} \mathbf{a})$ .

#### Exercise C2.

Exercise (B2) gives the matrix inequality behind the remark after Theorem 9.2.2. Using part (d) of that theorem instead, show that  $\text{Var}(\hat{\beta}_{W,j}) = \text{Var}(\hat{\beta}_j)$  iff the  $j$ th row of  $\mathbf{L}$  is zero, and that this holds for every  $j$  iff  $\mathbf{X}^\top \mathbf{Z} = \mathbf{0}$ . Show that when  $\mathbf{X}^\top \mathbf{Z} = \mathbf{0}$  the estimates  $\hat{\boldsymbol{\beta}}$  do not change at all.

#### Exercise C3.

In the additive two-way model  $E(Y_{ijk}) = \mu + \alpha_i + \beta_j$  with all cells occupied, use Proposition 9.2.6 to find the degrees of freedom of the extra sums of squares for the constraints (i)  $\sum_i \alpha_i = 0$  and  $\sum_j \beta_j = 0$ ; (ii)  $\alpha_1 = \alpha_2$ ; (iii)  $\alpha_1 = 0$  and  $\alpha_1 = \alpha_2$ .

## 9.3 Sequential and partial sums of squares

How should the variation explained by a model be shared among its terms? A term is a block of columns representing one idea: a regressor, the coding columns of a factor, or an interaction. One answer enters the terms one at a time and credits each with what it adds to those already present. Another credits each term with what it adds when entered last. This section defines both, explains the three “types” of sums of squares that software prints, and identifies the hypothesis that each of them addresses.

## 9.3.1 Sequential sums of squares

Write the model matrix as a sequence of terms,

$$\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k],$$

where  $\mathbf{X}_0$  is usually the intercept column  $\mathbf{1}$ . Let  $\mathbf{M}_j$  be the projection onto  $\mathcal{C}(\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_j)$ , so that  $\mathbf{M}_k = \mathbf{M}$ .

### Definition 9.3.1 — Sequential sums of squares

The **sequential** (or **Type I**) sum of squares of the term  $\mathbf{X}_j$ , for the order  $\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k$ , is

$$\text{SS}(\mathbf{X}_j \mid \mathbf{X}_0, \dots, \mathbf{X}_{j-1}) = \mathbf{y}^\top (\mathbf{M}_j - \mathbf{M}_{j-1}) \mathbf{y}, \quad j = 1, \dots, k,$$

with  $d_j = \text{rank}(\mathbf{M}_j) - \text{rank}(\mathbf{M}_{j-1})$  degrees of freedom.

The sequential sums of squares are the analysis of variance decomposition of [Theorem 9.1.1\(e\)](#) for the chain  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_0, \mathbf{X}_1) \subseteq \dots \subseteq \mathcal{C}(\mathbf{X})$ , after the first piece  $\mathbf{M}_0$  has been set aside. They add up to the regression sum of squares  $\text{SS}(\mathbf{X}_1, \dots, \mathbf{X}_k \mid \mathbf{X}_0)$ , their degrees of freedom add up to  $r - \text{rank}(\mathbf{X}_0)$ , and under normality they are independent of each other and of the residual sum of squares ([Theorem 4.4.4](#)). A term linearly dependent on earlier ones receives zero degrees of freedom. The price is that each piece depends on the order:  $\mathbf{X}_j$  is credited only with the part of its span orthogonal to *the terms that happen to precede it*. For the murder rates of [Section 6.5](#), poverty received 102.49 first and 16.37 last ([Example \(Order matters\)](#)).

The order dependence can be more surprising than “entering first gives more credit”. The listing computes  $\text{SS}(\mathbf{x} \mid \mathbf{1}, S)$  for each of the three regressors and every set  $S$  of the other two.

| Regressor | given $\mathbf{1}$ only | given $\mathbf{1}$ and one other (named) | given $\mathbf{1}$ and the remaining other (named) | given $\mathbf{1}$ and both others |
|-----------|-------------------------|------------------------------------------|----------------------------------------------------|------------------------------------|
| poverty   | 102.49                  | 19.69 (single)                           | 125.45 (urban)                                     | 16.37                              |
| single    | 162.84                  | 80.04 (poverty)                          | 160.13 (urban)                                     | 51.06                              |
| urban     | 6.30                    | 29.26 (poverty)                          | 3.59 (single)                                      | 0.27                               |

Adjusting for urbanization *increases* the sum of squares of poverty from 102.49 to 125.45, and adjusting for poverty more than quadruples that of urbanization. Urban states have somewhat less poverty but more murder, so holding one fixed sharpens the association of the other; such a regressor is called a **suppressor** ([Exercise \(B1\)](#)). A sequential sum of squares is not a property of a regressor, and not even a monotone function of the set adjusted for.

```
from itertools import combinations
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
n = len(y)
one = np.ones(n)
cols = {"poverty": data["poverty"].to_numpy(), "single": data["single"].to_numpy(),
        "urban": data["urban"].to_numpy()}
```



```

def sse(Z):
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    e = y - Z @ b
    return e @ e

table = {}
for name, x in cols.items():
    others = [k for k in cols if k != name]
    for size in range(3):
        for S in combinations(others, size):
            base = np.column_stack([one] + [cols[k] for k in S])
            table[name, S] = sse(base) - sse(np.column_stack([base, x]))
            print(f"SS({name} | 1{''.join(', ' + k for k in S)}) = {table[name, S]
                  }:7.2f}")

```

## 9.3.2 Partial sums of squares: Types II and III

The second answer credits a term with what it adds last. For models whose terms are factors and their interactions, “last” needs care, because a main effect is part of every interaction that contains it. Say that a term  $\mathbf{X}_i$  **contains** a term  $\mathbf{X}_j$  if the factors of  $\mathbf{X}_j$  are a proper subset of the factors of  $\mathbf{X}_i$ . For example the interaction  $AB$  contains the main effects  $A$  and  $B$ .

### Definition 9.3.2 — Partial sums of squares; Types II and III

For a model with terms  $\mathbf{X}_0, \mathbf{X}_1, \dots, \mathbf{X}_k$ , write  $\mathbf{X}_{(-j)}$  for  $\mathbf{X}$  with the columns of  $\mathbf{X}_j$  removed.

- (a) The **partial** (or **last-in**) sum of squares of  $\mathbf{X}_j$  is  $SS(\mathbf{X}_j \mid \mathbf{X}_{(-j)}) = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_{(-j)}) \mathbf{y}$ , where  $\mathbf{M}_{(-j)}$  projects onto  $\mathcal{C}(\mathbf{X}_{(-j)})$ .
- (b) The **Type II** sum of squares of  $\mathbf{X}_j$  is its extra sum of squares given all terms that do not contain it:  $SS(\mathbf{X}_j \mid \mathbf{X}_i : \mathbf{X}_i \text{ does not contain } \mathbf{X}_j, i \neq j)$ .
- (c) The **Type III** sum of squares of  $\mathbf{X}_j$  is its partial sum of squares (a) computed when every factor is coded by sum-to-zero contrast columns and every interaction by the products of the columns of its factors.

When no term contains another, as in a regression on numerical regressors without interactions, the three coincide, and they equal the sequential sum of squares of  $\mathbf{X}_j$  when it is entered last. They also coincide for a term that is not contained in any other, such as the highest-order interaction. They differ for main effects in models with interactions.

Partial sums of squares are not a decomposition. The subspaces  $\mathcal{C}(\mathbf{X}) \ominus \mathcal{C}(\mathbf{X}_{(-j)})$  are generally not orthogonal, so the partial sums of squares can add up to more or less than the regression sum of squares ([Exercise \(C1\)](#)), and they need not be independent of each other, though each is independent of the residual sum of squares under normality.

Why does the definition of Type III name a coding? Because in models with interactions the partial sum of squares of a main effect depends on the coding. Deleting the columns of  $A$  while keeping those of  $AB$  leaves the reduced space  $\mathcal{C}(\mathbf{1}, \mathbf{X}_B, \mathbf{X}_{AB})$ , which depends on which columns represent  $AB$ , although the full space does not. Different codings give different reduced models and so test different hypotheses.

### 9.3.3 What each sum of squares tests

A sum of squares  $\mathbf{y}^\top \mathbf{P} \mathbf{y}$  with  $\mathbf{P}$  a projection of rank  $d$  has expectation

$$\mathbf{E}(\mathbf{Y}^\top \mathbf{P} \mathbf{Y}) = \sigma^2 d + \|\mathbf{P} \boldsymbol{\mu}\|^2, \quad \boldsymbol{\mu} = \mathbf{E}(\mathbf{Y}),$$

by [Theorem 2.3.1](#) with  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ . So the hypothesis a sum of squares addresses is  $\mathbf{P} \boldsymbol{\mu} = \mathbf{0}$ : *the mean has no component in the subspace onto which it projects*. Under normality this is exactly the condition for its  $F$  ratio to be central ([Section 9.6](#)). We translate it into statements about parameters.

#### Proposition 9.3.1 — Hypotheses of sequential and partial sums of squares

Let  $\mathbf{X} = [\mathbf{X}_0, \dots, \mathbf{X}_k]$  have full column rank and  $\boldsymbol{\mu} = \sum_l \mathbf{X}_l \beta_l$ .

- (a) The partial sum of squares of  $\mathbf{X}_j$  addresses  $\beta_j = \mathbf{0}$ :  $(\mathbf{M} - \mathbf{M}_{(-j)})\boldsymbol{\mu} = \mathbf{0}$  iff  $\beta_j = \mathbf{0}$ .
- (b) The sequential sum of squares of  $\mathbf{X}_j$  addresses

$$\beta_j + (\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \sum_{l>j} \mathbf{X}_l \beta_l = \mathbf{0}, \quad \tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{M}_{j-1})\mathbf{X}_j.$$

This is  $\beta_j = \mathbf{0}$  whenever the later terms have zero coefficients, or whenever  $\tilde{\mathbf{X}}_j^\top \mathbf{X}_l = \mathbf{0}$  for all  $l > j$ .

*Proof.*

- (a) For  $l \neq j$ ,  $\mathbf{X}_l \beta_l$  lies in  $\mathcal{C}(\mathbf{X}_{(-j)}) \subseteq \mathcal{C}(\mathbf{X})$ , so  $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_l \beta_l = \mathbf{0}$ . Let  $\tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{M}_{(-j)})\mathbf{X}_j$ . By [Lemma 9.2.1](#),  $\mathbf{M} - \mathbf{M}_{(-j)}$  projects onto  $\mathcal{C}(\tilde{\mathbf{X}}_j)$ , so  $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_j \beta_j$  is the projection of  $\mathbf{X}_j \beta_j$  onto that space. Since  $\mathbf{X}_j \beta_j - \tilde{\mathbf{X}}_j \beta_j = \mathbf{M}_{(-j)}\mathbf{X}_j \beta_j$  is orthogonal to  $\mathcal{C}(\tilde{\mathbf{X}}_j)$ , the projection is  $\tilde{\mathbf{X}}_j \beta_j$ . The columns of  $\tilde{\mathbf{X}}_j$  are independent ([Lemma 6.6.1\(b\)](#)), so it vanishes iff  $\beta_j = \mathbf{0}$ .
- (b) Terms with  $l < j$  are annihilated by  $\mathbf{M}_j - \mathbf{M}_{j-1}$ , which projects onto  $\mathcal{C}(\tilde{\mathbf{X}}_j)$  by [Lemma 9.2.1](#). As in (a),  $\mathbf{X}_j \beta_j$  projects to  $\tilde{\mathbf{X}}_j \beta_j$ . A later term projects to  $\tilde{\mathbf{X}}_j (\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \mathbf{X}_l \beta_l$  by [Theorem 6.3.7](#). Adding,  $(\mathbf{M}_j - \mathbf{M}_{j-1})\boldsymbol{\mu} = \tilde{\mathbf{X}}_j \mathbf{v}$ , where  $\mathbf{v}$  is the vector in the display, and this is zero iff  $\mathbf{v} = \mathbf{0}$  because  $\tilde{\mathbf{X}}_j$  has independent columns. ■

So a sequential sum of squares answers the question “does  $\mathbf{X}_j$  matter?” only if the terms fitted after it do not matter, or are orthogonal to it after adjusting for the terms before it. Otherwise the effects of later terms leak into it, through the coefficients  $(\tilde{\mathbf{X}}_j^\top \tilde{\mathbf{X}}_j)^{-1} \tilde{\mathbf{X}}_j^\top \mathbf{X}_l$  of the regressions of later terms on  $\tilde{\mathbf{X}}_j$ .

For factors, the natural parameters are the cell means. Consider a two-way layout with factor  $A$  at  $a$  levels and  $B$  at  $b$  levels,  $n_{ij} \geq 1$  observations in cell  $(i, j)$ , row totals  $n_{i.}$ , and cell means  $\mu_{ij}$ . The full model with interaction has model space  $\mathcal{C}(\mathbf{X}) =$  all vectors that are constant within

cells, of dimension  $ab$ . Define the **weighted** and **unweighted** row means

$$\bar{\mu}_i^w = \frac{1}{n_{i.}} \sum_j n_{ij} \mu_{ij}, \quad \bar{\mu}_{i.} = \frac{1}{b} \sum_j \mu_{ij}.$$

### Theorem 9.3.2 — What the sums of squares for a main effect test

In the two-way layout with all  $n_{ij} \geq 1$ :

- (a) the sequential sum of squares  $SS(A \mid \mu)$  of  $A$  entered first addresses  $\bar{\mu}_1^w = \cdots = \bar{\mu}_a^w$ ;
- (b) the Type III sum of squares of  $A$  addresses  $\bar{\mu}_{1.} = \cdots = \bar{\mu}_{a.}$ ;
- (c) the partial sum of squares of the columns of  $A$  under reference-level coding (indicators of levels  $2, \dots, a$  of  $A$  and  $2, \dots, b$  of  $B$ , and their products for  $AB$ ) addresses  $\mu_{11} = \mu_{21} = \cdots = \mu_{a1}$ , the equality of the  $A$  means at the first level of  $B$ ;
- (d) the Type II sum of squares  $SS(A \mid \mu, B)$  addresses a hypothesis that, when the cell means are additive ( $\mu_{ij} = \alpha_i + \beta_j$ ), is equivalent to  $\alpha_1 = \cdots = \alpha_a$ .

Each sum of squares equals the sum of squares Eq. 9.2.1 of the hypothesis it addresses, in the cell-means model.

*Proof.* Let  $\mathbf{P}_0$ ,  $\mathbf{P}_A$ ,  $\mathbf{P}_B$  and  $\mathbf{P}_{A+B}$  be the projections onto the constant vectors, the vectors constant within rows, those constant within columns, and their sum. Each hypothesis below defines a subspace  $\mathcal{S}_0$  of  $\mathcal{C}(\mathbf{X})$ , and each sum of squares is  $\|\mathbf{P}\mathbf{y}\|^2$  with  $\mathbf{P}$  the projection onto  $\mathcal{C}(\mathbf{X}) \ominus \mathcal{S}_0$ . Then  $\mathbf{P}\boldsymbol{\mu} = \mathbf{0}$  iff  $\boldsymbol{\mu} \in \mathcal{S}_0$ , and the sum of squares equals  $SSE_0 - SSE$  for  $\mathcal{S}_0$ , which is the sum of squares of the hypothesis by Theorem 9.2.4.

- (a)  $\mathbf{P}_A$  replaces each entry of a vector by the average of its row, so  $\mathbf{P}_A\boldsymbol{\mu}$  has entry  $\bar{\mu}_i^w$  in row  $i$ , and  $\mathbf{P}_0\boldsymbol{\mu}$  has every entry equal to the overall weighted mean  $\sum_i n_{i.} \bar{\mu}_i^w / n$ . So  $(\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu} = \mathbf{0}$  iff all  $\bar{\mu}_i^w$  equal their weighted average, that is, iff they are all equal. The relevant  $\mathcal{S}_0$  is  $\{\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}) : (\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu} = \mathbf{0}\}$ , whose orthogonal complement in  $\mathcal{C}(\mathbf{X})$  is  $\mathcal{C}(\mathbf{P}_A - \mathbf{P}_0)$ , because  $\mathcal{C}(\mathbf{P}_A - \mathbf{P}_0) \subseteq \mathcal{C}(\mathbf{X})$ .
- (b) Any  $a \times b$  array of cell means decomposes uniquely as

$$\mu_{ij} = \bar{\mu}_{..} + (\bar{\mu}_{i.} - \bar{\mu}_{..}) + (\bar{\mu}_{.j} - \bar{\mu}_{..}) + (\mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{..}),$$

where bars denote unweighted averages. The four arrays lie in the mutually orthogonal subspaces (for the unweighted inner product on arrays) of constants, row effects summing to zero, column effects summing to zero, and arrays with all row and column sums zero, of dimensions  $1, a-1, b-1, (a-1)(b-1)$ . Sum-to-zero coding spans exactly these: the columns  $\mathbf{u}_i - \mathbf{u}_1$  ( $i = 2, \dots, a$ ), where  $\mathbf{u}_i$  indicates row  $i$ , give all row effects  $\alpha_i$  with  $\sum_i \alpha_i = 0$ ; similarly for  $B$ ; and the  $(a-1)(b-1)$  products are linearly independent arrays with zero row and column sums, so they span that subspace. Deleting the columns of  $A$  removes exactly the row-effect component, so the reduced model is  $\mathcal{S}_0 = \{\boldsymbol{\mu} : \bar{\mu}_{i.} = \bar{\mu}_{..} \text{ for all } i\}$ .

- (c) With reference coding, every cell-mean array is uniquely  $\mu_{ij} = m + \alpha_i + \beta_j + \gamma_{ij}$  with  $\alpha_1 = \beta_1 = 0$  and  $\gamma_{1j} = \gamma_{i1} = 0$ ; explicitly  $m = \mu_{11}$ ,  $\alpha_i = \mu_{i1} - \mu_{11}$ ,  $\beta_j = \mu_{1j} - \mu_{11}$  and  $\gamma_{ij} = \mu_{ij} - \mu_{i1} - \mu_{1j} + \mu_{11}$ . Deleting the columns of  $A$  forces  $\alpha_i = 0$ , that is,  $\mu_{i1} = \mu_{11}$  for all  $i$ .
- (d) The Type II sum of squares is  $\mathbf{y}^\top (\mathbf{P}_{A+B} - \mathbf{P}_B) \mathbf{y}$ . If  $\boldsymbol{\mu}$  is additive, then  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{P}_{A+B})$  and  $(\mathbf{P}_{A+B} - \mathbf{P}_B)\boldsymbol{\mu} = \boldsymbol{\mu} - \mathbf{P}_B\boldsymbol{\mu}$ , which is zero iff  $\boldsymbol{\mu}$  is constant within columns,

that is, iff  $\alpha_i$  does not depend on  $i$ .

■

These are four different questions. Part (a) averages over  $B$  with the design's weights, so if  $B$  matters and the rows have different mixes of  $B$  levels, it can detect “an effect of  $A$ ” that is entirely an effect of  $B$  ([Exercise B3](#)). Part (b) averages with equal weights. Part (c) compares the rows at one level of  $B$ : a simple effect, not a main effect, and one that changes with the reference level. Part (d) is the natural test of  $A$  without interaction; with interaction its hypothesis has weights that depend on all the  $n_{ij}$ .

*Warning* (“Type III” with the default coding). Many regression programs code factors by reference levels unless told otherwise. Asking them for last-in sums of squares of main effects in a model with interactions then produces the hypotheses of [Theorem 9.3.2\(c\)](#), which concern a single level of the other factor. They are not Type III sums of squares, although they are sometimes reported under that name. Use sum-to-zero coding, or better, write down the hypothesis and use [Eq. 9.2.1](#).

## 9.3.4 An unbalanced example

### Example. (Party, education and political self-placement)

In the 1996 American National Election Study, 944 respondents placed themselves on a seven-point scale from extremely liberal (1) to extremely conservative (7). Classify them by party identification (Democrat, Independent, Republican, counting those who only lean to a party as Independents) and by education (high school or less, college, graduate study). The cell counts and means are:

|             | high school | college   | graduate  |
|-------------|-------------|-----------|-----------|
| Democrat    | 146, 3.97   | 108, 3.43 | 126, 2.92 |
| Independent | 80, 4.39    | 64, 4.28  | 95, 4.09  |
| Republican  | 87, 5.40    | 105, 5.36 | 133, 5.40 |

The design is unbalanced, and education is associated with party: 38% of the Democrats but only 27% of the Republicans have at most a high school education. The pattern of means shows an interaction. Among Democrats self-placement moves left with education by more than a scale point, while among Republicans it does not move at all. The sums of squares for the terms are:

| Term        | df  | Type I,<br>party first | Type I,<br>education<br>first | Type II | Type III | reference<br>coding |
|-------------|-----|------------------------|-------------------------------|---------|----------|---------------------|
| party       | 2   | 647.46                 | 674.42                        | 674.42  | 650.41   | 112.27              |
| education   | 2   | 43.26                  | 16.31                         | 43.26   | 32.03    | 75.12               |
| interaction | 4   | 35.81                  | 35.81                         | 35.81   | 35.81    | 35.81               |
| residual    | 935 | 1224.62                | 1224.62                       | 1224.62 | 1224.62  | 1224.62             |

The two Type I columns each add up to the model sum of squares 726.54, up to rounding. The other columns do not: the Type II entries add up to more. Education receives 16.31 when entered first and 43.26 after party. Since the least educated group contains the most Democrats, who are the most liberal, the marginal comparison mixes the leftward drift of educated Democrats with the partisan composition of each education group, and the two partly cancel. The Type III sum of squares 32.03 tests whether the unweighted averages over the three parties of the education means are equal. The last column, computed with reference coding, answers different questions: 75.12 for education measures the education gradient among *Democrats only* (the first party level), which is the steepest, and 112.27 for party compares the parties among respondents with a high school education only. The interaction row is the same in every column because the interaction is the highest-order term.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
party = pd.cut(d["PID"], [-1, 1.5, 4.5, 6], labels=["Dem", "Ind", "Rep"]).rename("party")
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"]).rename("educ")
y = d["selfLR"].to_numpy()
n = len(y)
print(pd.crosstab(party, educ))
print(d.groupby([party, educ], observed=True)["selfLR"].mean().unstack().round(3))
```

```
def proj(*blocks):
    """Orthogonal projection onto the span of the given column blocks."""
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]] # orthonormal basis of C(Z), any
    rank # rank
    return U @ U.T

one = np.ones((n, 1))
A = pd.get_dummies(party).to_numpy(float) # indicators of party
B = pd.get_dummies(educ).to_numpy(float) # indicators of education
AB = np.column_stack([A[:, [i]] * B for i in range(3)]) # the nine cells
PO, PA, PB = proj(one), proj(one, A), proj(one, B)
```

```

PAB, M = proj(one, A, B), proj(AB)                                # additive model, cell-means
                        model
I = np.eye(n)

def ss(P):
    return y @ P @ y

type1_AB = {"party": ss(PA - P0), "educ": ss(PAB - PA), "inter": ss(M - PAB)}
type1_BA = {"educ": ss(PB - P0), "party": ss(PAB - PB), "inter": ss(M - PAB)}
type2 = {"party": ss(PAB - PB), "educ": ss(PAB - PA), "inter": ss(M - PAB)}
sse = ss(I - M)

def coded(labels, kind):
    """Columns for a factor: sum-to-zero ("sum") or reference-level ("treatment")
    coding."""
    D = pd.get_dummies(labels).to_numpy(float)
    return D[:, 1:] - D[:, [0]] if kind == "sum" else D[:, 1:]

type3 = {}
for kind in ["sum", "treatment"]:
    Ca, Cb = coded(party, kind), coded(educ, kind)
    Cab = np.column_stack([Ca[:, [i]] * Cb for i in range(2)])
    full = proj(one, Ca, Cb, Cab)                                # always the cell-means space
    type3[kind] = {"party": ss(full - proj(one, Cb, Cab)),
                  "educ": ss(full - proj(one, Ca, Cab))}
    assert np.allclose(full, M)
for name, table in [("Type I, party first", type1_AB), ("Type I, education first",
type1_BA),
                    ("Type II", type2), ("Type III, sum coding", type3["sum"]),
                    ("Type III, treatment coding", type3["treatment"])] :
    print(f"{name:30s}", {k: round(float(v), 2) for k, v in table.items()})

```

The script behind these cells also checks every entry against statsmodels, and it confirms [Theorem 9.3.2](#) numerically: the party entries of the first, fourth and fifth columns equal [Eq. 9.2.1](#) for the hypotheses of equal weighted row means, equal unweighted row means, and equal means in the high-school column.

## 9.3.5 Which one to use

There is no single right sum of squares for a main effect in an unbalanced design with interaction, because there is no single main-effect hypothesis. Decide the hypothesis first, from the four above, and compute its sum of squares with [Eq. 9.2.1](#). If  $A$  and  $B$  interact, a single “main effect of  $A$ ” summarizes a heterogeneous comparison with arbitrary weights; Nelder (1977) argued that such tests are rarely meaningful, and Type II sums of squares respect this marginality by adjusting  $A$  for  $B$  but not for  $AB$ . Without interaction the Type II and Type III hypotheses coincide, and Type II, which does not adjust for the interaction columns, generally gives the more powerful test. In regression without interactions all partial types agree; the distinction matters only for terms contained in other terms.

[Chapter 17](#) treats empty cells, where even Type III hypotheses may be nonestimable ([Exercise \(B4\)](#), [Theorem 17.4.1](#)).

## 9.3.6 Exercises

### 9.3.6.1 A. Check your understanding

#### Exercise A1.

For a two-way model with terms  $A$ ,  $B$  and  $AB$ , write each of the Type I sums of squares (both orders), the Type II sums of squares and the partial sums of squares in the notation  $SS(\cdot | \cdot)$ . Which of them are equal for every data set, whatever the design?

*Solution.* Type I,  $A$  first:  $SS(A | \mu)$ ,  $SS(B | \mu, A)$ ,  $SS(AB | \mu, A, B)$ . Type I,  $B$  first:  $SS(B | \mu)$ ,  $SS(A | \mu, B)$ ,  $SS(AB | \mu, A, B)$ . Type II:  $SS(A | \mu, B)$ ,  $SS(B | \mu, A)$ ,  $SS(AB | \mu, A, B)$ . Partial:  $SS(A | \mu, B, AB)$ ,  $SS(B | \mu, A, AB)$ ,  $SS(AB | \mu, A, B)$ , where the first two depend on the coding. Always equal: the interaction entries of all four lists; the Type II entry for  $A$  and the Type I entry for  $A$  with  $B$  first; the Type II entry for  $B$  and the Type I entry for  $B$  with  $A$  first.

### 9.3.6.2 B. Practice

#### Exercise B1.

Let  $\mathbf{X} = [\mathbf{1}, \mathbf{x}_1, \mathbf{x}_2]$ , and let  $r_{y1}$ ,  $r_{y2}$  and  $r_{12}$  be the sample correlations among  $\mathbf{y}$ ,  $\mathbf{x}_1$  and  $\mathbf{x}_2$ , with  $|r_{12}| < 1$ . Show that

$$\frac{SS(\mathbf{x}_2 | \mathbf{1})}{SST} = r_{y2}^2, \quad \frac{SS(\mathbf{x}_2 | \mathbf{1}, \mathbf{x}_1)}{SST} = \frac{(r_{y2} - r_{y1}r_{12})^2}{1 - r_{12}^2}.$$

Show that adjusting for  $\mathbf{x}_1$  increases the sum of squares of  $\mathbf{x}_2$  whenever  $r_{y2} = 0$  and  $r_{y1}r_{12} \neq 0$ , and find the condition for it in general.

*Solution.* Work in  $\mathbf{1}^\perp$  with the centred vectors  $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$ ,  $\mathbf{v}_1$ ,  $\mathbf{v}_2$ , scaled to unit length, so that inner products are correlations and  $SST = \|\mathbf{y} - \bar{y}\mathbf{1}\|^2$ . By Lemma 9.2.1,  $SS(\mathbf{x}_2 | \mathbf{1}) = SST(\mathbf{u}^\top \mathbf{v}_2)^2 = SST r_{y2}^2$ . Adjusting for  $\mathbf{x}_1$  replaces  $\mathbf{v}_2$  by  $\mathbf{w} = \mathbf{v}_2 - r_{12}\mathbf{v}_1$ , with  $\|\mathbf{w}\|^2 = 1 - r_{12}^2$  and  $\mathbf{u}^\top \mathbf{w} = r_{y2} - r_{y1}r_{12}$ , so  $SS(\mathbf{x}_2 | \mathbf{1}, \mathbf{x}_1) = SST(\mathbf{u}^\top \mathbf{w})^2 / \|\mathbf{w}\|^2$ . If  $r_{y2} = 0$  the first ratio is zero and the second is  $r_{y1}^2 r_{12}^2 / (1 - r_{12}^2) > 0$ . In general the sum of squares increases iff  $(r_{y2} - r_{y1}r_{12})^2 > r_{y2}^2(1 - r_{12}^2)$ .

#### Exercise B2.

Show that the hypotheses of Theorem 9.3.2(a) and (b) are the same subspace of cell means iff  $n_{ij} = n_{i.}/b$  for all  $i, j$ , that is, iff within every row all cells have the same count. Show that under proportional frequencies,  $n_{ij} = n_{i.}n_{.j}/n$  (Theorem 9.4.3), the hypothesis (a) compares row means weighted by the column proportions  $n_{.j}/n$ , the same weights in every row.

*Solution.* Write  $w_{ij} = n_{ij}/n_{i.}$ , so  $\sum_j w_{ij} = 1$ , and let  $f_i(\boldsymbol{\mu}) = \sum_j w_{ij}\mu_{ij}$  and  $g_i(\boldsymbol{\mu}) = \sum_j \mu_{ij}/b$ . Hypothesis (a) is the null space of the linear functionals  $\sum_i c_i f_i$  with  $\sum_i c_i = 0$ , and (b) that of the functionals  $\sum_i d_i g_i$  with  $\sum_i d_i = 0$ . Two subspaces defined as null spaces coincide iff the spaces of functionals coincide. The functional  $f_i - f_1$  has coefficient

$w_{ij}$  on cell  $(i, j)$  and  $-w_{1j}$  on cell  $(1, j)$ . If it equals some  $\sum_i d_i g_i$ , whose coefficients are constant across each row, then  $w_{ij}$  does not depend on  $j$ , so  $w_{ij} = 1/b$ ; likewise  $w_{1j} = 1/b$ . Hence coincidence forces  $n_{ij} = n_{i\cdot}/b$ . Conversely, if all  $w_{ij} = 1/b$ , then  $f_i = g_i$ . Under proportional frequencies  $w_{ij} = n_{\cdot j}/n$ , which does not depend on  $i$ , so (a) compares  $\sum_j (n_{\cdot j}/n) \mu_{ij}$  across rows.

### Exercise B3.

Take  $a = b = 2$ , counts  $n_{11} = n_{22} = 9$  and  $n_{12} = n_{21} = 1$ , and additive cell means  $\mu_{ij} = \beta_j$  with  $\beta_1 = 0$  and  $\beta_2 = 1$ , so that  $A$  has no effect. Compute the weighted row means  $\bar{\mu}_i^w$  and show that the hypothesis addressed by the Type I sum of squares of  $A$  entered first is false. What do the Type II and Type III hypotheses say here?

### Exercise B4.

Suppose that in a  $2 \times 2$  layout the cell  $(2, 2)$  is empty. Show that  $\bar{\mu}_{1\cdot} - \bar{\mu}_{2\cdot}$  is not estimable. Compute what the last-in sum of squares of  $A$  under sum-to-zero coding tests in this case. *Hint:* with a cell missing, the interaction columns restricted to the observed cells are no longer independent of the main-effect columns.

#### 9.3.6.3 C. Going deeper

### Exercise C1.

With the notation of [Exercise \(B1\)](#), show that

$$\frac{SS(\mathbf{x}_1 \mid \mathbf{1}, \mathbf{x}_2) + SS(\mathbf{x}_2 \mid \mathbf{1}, \mathbf{x}_1) - SSR}{SST} = \frac{r_{12}[r_{12}(r_{y1}^2 + r_{y2}^2) - 2r_{y1}r_{y2}]}{1 - r_{12}^2}.$$

Deduce that the partial sums of squares add up to the regression sum of squares when  $r_{12} = 0$ , that they add up to less when  $r_{y1} = r_{y2} \neq 0$  and  $0 < r_{12} < 1$ , and give correlations for which they add up to more.

## 9.4 Orthogonal designs

[Section 9.3](#) showed how much the sums of squares for a term can depend on the other terms. This section asks when they do not. The answer is geometric: after the mean is removed, the terms must span mutually orthogonal subspaces. Balanced experiments achieve this by design and observational data essentially never do, but orthogonality can be manufactured inside a term by a change of basis, as with orthogonal polynomials and orthogonal contrasts.



### 9.4.1 When the order does not matter

Let the model matrix be  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$  with an intercept and  $k$  terms. Write  $\mathbf{P}_0 = n^{-1}\mathbf{1}\mathbf{1}^\top$ , and let

$$\tilde{\mathbf{X}}_j = (\mathbf{I} - \mathbf{P}_0)\mathbf{X}_j, \quad \mathcal{V}_j = \mathcal{C}(\tilde{\mathbf{X}}_j),$$

be the centred columns of term  $j$  and the subspace they span. By [Lemma 9.2.1](#) with  $\mathbf{X}_1 = \mathbf{1}$ ,  $\mathcal{C}(\mathbf{1}, \mathbf{X}_j) = \text{span}(\mathbf{1}) \oplus \mathcal{V}_j$ , so  $\text{SS}(\mathbf{X}_j | \mathbf{1}) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$ , the sum of squares of the term fitted on its own after the mean. Call the design **orthogonal** (for these terms) if the subspaces  $\mathcal{V}_1, \dots, \mathcal{V}_k$  are mutually orthogonal, that is, if  $\tilde{\mathbf{X}}_i^\top \tilde{\mathbf{X}}_j = \mathbf{0}$  for  $i \neq j$ . For numerical regressors this says that every pair of regressors from different terms has sample correlation zero.

#### Theorem 9.4.1 — Orthogonal designs

For a model  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$ , the following are equivalent:

- (a) the centred subspaces  $\mathcal{V}_1, \dots, \mathcal{V}_k$  are mutually orthogonal;
- (b)  $\mathbf{M} = \mathbf{P}_0 + \mathbf{P}_{\mathcal{V}_1} + \dots + \mathbf{P}_{\mathcal{V}_k}$ ;
- (c) for every  $j$  and every  $\mathbf{y}$ , the marginal and partial sums of squares agree:  $\text{SS}(\mathbf{X}_j | \mathbf{1}) = \text{SS}(\mathbf{X}_j | \mathbf{X}_{(-j)})$ ;
- (d) for every  $\mathbf{y}$ , the marginal sums of squares add up to the regression sum of squares:  $\sum_j \text{SS}(\mathbf{X}_j | \mathbf{1}) = \text{SS}(\mathbf{X}_1, \dots, \mathbf{X}_k | \mathbf{1})$ .

When they hold,  $\text{SS}(\mathbf{X}_j | \mathbf{1}, \mathbf{X}_S) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$  for every set  $S$  of other terms. In particular the sequential sum of squares of  $\mathbf{X}_j$  is the same in every order, and it equals the partial and the Type II sums of squares of  $\mathbf{X}_j$ .

*Proof.* (a) implies (b) and the final statement. Let  $S$  be any set of terms. The subspaces  $\text{span}(\mathbf{1})$  and  $\mathcal{V}_l$ ,  $l \in S$ , are mutually orthogonal, because each  $\mathcal{V}_l \subseteq \mathbf{1}^\perp$ , and they sum to  $\mathcal{C}(\mathbf{1}, \mathbf{X}_S)$ , because each  $\mathcal{C}(\mathbf{1}, \mathbf{X}_l) = \text{span}(\mathbf{1}) \oplus \mathcal{V}_l$ . By [Theorem 6.3.8](#) the projection onto  $\mathcal{C}(\mathbf{1}, \mathbf{X}_S)$  is  $\mathbf{P}_0 + \sum_{l \in S} \mathbf{P}_{\mathcal{V}_l}$ . With  $S$  all terms this is (b). With  $S$  and  $S \cup \{j\}$ , the difference of the two projections is  $\mathbf{P}_{\mathcal{V}_j}$ , which gives  $\text{SS}(\mathbf{X}_j | \mathbf{1}, \mathbf{X}_S) = \|\mathbf{P}_{\mathcal{V}_j}\mathbf{y}\|^2$ . Sequential, partial and Type II sums of squares of  $\mathbf{X}_j$  are all of this form.

(b) implies (d):  $\text{SS}(\mathbf{X}_1, \dots, \mathbf{X}_k | \mathbf{1}) = \mathbf{y}^\top (\mathbf{M} - \mathbf{P}_0) \mathbf{y} = \sum_j \mathbf{y}^\top \mathbf{P}_{\mathcal{V}_j} \mathbf{y}$ .

(d) implies (a). The two sides of (d) are the quadratic forms of the symmetric matrices  $\sum_j \mathbf{P}_{\mathcal{V}_j}$  and  $\mathbf{M} - \mathbf{P}_0$ . Two symmetric matrices with the same quadratic form are equal, because  $2\mathbf{u}^\top \mathbf{A} \mathbf{v} = (\mathbf{u} + \mathbf{v})^\top \mathbf{A} (\mathbf{u} + \mathbf{v}) - \mathbf{u}^\top \mathbf{A} \mathbf{u} - \mathbf{v}^\top \mathbf{A} \mathbf{v}$ . So  $\sum_j \mathbf{P}_{\mathcal{V}_j} = \mathbf{M} - \mathbf{P}_0$ , which is idempotent ([Theorem 6.5.1](#)). By the algebraic form of Cochran's theorem ([Theorem 4.5.1](#)), projections whose sum is idempotent satisfy  $\mathbf{P}_{\mathcal{V}_i} \mathbf{P}_{\mathcal{V}_j} = \mathbf{0}$  for  $i \neq j$ , which is (a).

(a) implies (c) is part of the final statement.

(c) implies (a). As before, equality of the quadratic forms gives  $\mathbf{P}_{\mathcal{V}_j} = \mathbf{M} - \mathbf{M}_{(-j)}$ . The right-hand side projects onto  $\mathcal{C}(\mathbf{X}) \ominus \mathcal{C}(\mathbf{X}_{(-j)})$  ([Theorem 6.5.1](#)), which is orthogonal to  $\mathcal{C}(\mathbf{X}_{(-j)})$ . The space  $\mathcal{C}(\mathbf{X}_{(-j)})$  contains  $\mathcal{V}_i$  for every  $i \neq j$ , since  $\tilde{\mathbf{X}}_i = \mathbf{X}_i - \mathbf{P}_0 \mathbf{X}_i$  has columns in  $\mathcal{C}(\mathbf{1}, \mathbf{X}_i)$ . So  $\mathcal{V}_j \perp \mathcal{V}_i$ . ■

*Idea.* Orthogonality is a property of the *centred* subspaces. The raw columns of two indicator variables are never orthogonal to the intercept, and a regressor and its square are rarely orthogonal. What matters is whether the parts of the terms that vary around their means point in perpendicular directions.

Orthogonality also stabilizes the coefficients.

### Corollary 9.4.2 — Coefficients in orthogonal designs

Suppose  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1, \dots, \mathbf{X}_k]$  has full column rank and satisfies the conditions of [Theorem 9.4.1](#). Then the least squares coefficients of the columns of  $\mathbf{X}_j$  are the same in every model that contains  $\mathbf{1}$  and  $\mathbf{X}_j$ , whichever other terms it contains. Only the intercept changes.

*Proof.* In the model with terms  $S \ni j$ , the fitted vector is  $\mathbf{P}_0 \mathbf{y} + \sum_{l \in S} \mathbf{P}_{\mathbf{X}_l} \mathbf{y}$  by the proof of the theorem, and  $\mathbf{P}_{\mathbf{X}_l} \mathbf{y} = \tilde{\mathbf{X}}_l \mathbf{b}_l$  with  $\mathbf{b}_l = (\tilde{\mathbf{X}}_l^\top \tilde{\mathbf{X}}_l)^{-1} \tilde{\mathbf{X}}_l^\top \mathbf{y}$ , which does not depend on  $S$ . Since  $\tilde{\mathbf{X}}_l \mathbf{b}_l = \mathbf{X}_l \mathbf{b}_l - \mathbf{P}_0 \mathbf{X}_l \mathbf{b}_l$ , the fitted vector equals  $\sum_{l \in S} \mathbf{X}_l \mathbf{b}_l$  plus a multiple of  $\mathbf{1}$ . The coefficients are unique by full rank, so the coefficient vector of  $\mathbf{X}_j$  is  $\mathbf{b}_j$  in every such model. ■

With uncorrelated regressors, dropping one leaves the others' estimates as they were, and each regressor has an order-free sum of squares. This is one reason factorial experiments are laid out on orthogonal arrays (Chapters 16 and 18).

## 9.4.2 Balanced and proportional layouts

For two factors the orthogonality condition reduces to a condition on the cell counts.

### Theorem 9.4.3 — Proportional frequencies

In a two-way layout with factor  $A$  at  $a$  levels, factor  $B$  at  $b$  levels, cell counts  $n_{ij}$ , and all row totals  $n_{i\cdot}$  and column totals  $n_{\cdot j}$  positive, the centred subspaces of  $A$  and  $B$  are orthogonal iff

$$n_{ij} = \frac{n_{i\cdot} n_{\cdot j}}{n} \quad \text{for all } i, j. \quad (9.4.1)$$

*Proof.* A vector in the centred subspace of  $A$  takes a value  $\alpha_i$  on every observation in row  $i$ , with  $\sum_i n_{i\cdot} \alpha_i = 0$ . A vector in that of  $B$  takes the value  $\beta_j$  in column  $j$ , with  $\sum_j n_{\cdot j} \beta_j = 0$ . Their inner product is  $\sum_{i,j} n_{ij} \alpha_i \beta_j = \boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta}$ , with  $\mathbf{N} = (n_{ij})$ . Write  $\mathbf{r} = (n_{1\cdot}, \dots, n_{a\cdot})^\top$  and  $\mathbf{c} = (n_{\cdot 1}, \dots, n_{\cdot b})^\top$ . The subspaces are orthogonal iff  $\boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta} = 0$  whenever  $\boldsymbol{\alpha}^\top \mathbf{r} = 0$  and  $\mathbf{c}^\top \boldsymbol{\beta} = 0$ .

If [Eq. 9.4.1](#) holds,  $\mathbf{N} = \mathbf{r} \mathbf{c}^\top / n$  and  $\boldsymbol{\alpha}^\top \mathbf{N} \boldsymbol{\beta} = (\boldsymbol{\alpha}^\top \mathbf{r})(\mathbf{c}^\top \boldsymbol{\beta}) / n = 0$ . Conversely, let  $\mathbf{E} = \mathbf{N} - \mathbf{r} \mathbf{c}^\top / n$ . Its row sums are  $\mathbf{r} - \mathbf{r}(\mathbf{c}^\top \mathbf{1}) / n = \mathbf{0}$  and its column sums are  $\mathbf{c}^\top - (\mathbf{1}^\top \mathbf{r}) \mathbf{c}^\top / n = \mathbf{0}^\top$ , since  $\mathbf{1}^\top \mathbf{r} = \mathbf{c}^\top \mathbf{1} = n$ . So  $\boldsymbol{\alpha}^\top \mathbf{E} \boldsymbol{\beta}$  does not change when  $\boldsymbol{\alpha}$  or  $\boldsymbol{\beta}$  is shifted by a multiple

of **1**. Given any  $\alpha$  and  $\beta$ , shift them to  $\alpha' = \alpha - (\alpha^\top \mathbf{r}/n)\mathbf{1}$  and  $\beta' = \beta - (\mathbf{c}^\top \beta/n)\mathbf{1}$ , which satisfy  $\alpha'^\top \mathbf{r} = 0$  and  $\mathbf{c}^\top \beta' = 0$ . Then  $\alpha^\top \mathbf{E}\beta = \alpha'^\top \mathbf{E}\beta' = \alpha'^\top \mathbf{N}\beta' - 0 = 0$  by assumption. As  $\alpha$  and  $\beta$  are arbitrary,  $\mathbf{E} = \mathbf{0}$ .

■

Equal replication,  $n_{ij} = m$ , is the common case ([Example \(Additive two-factor model, one observation per cell\)](#), [Exercise \(C1\)](#)). If the interaction subspace is defined as the orthogonal complement of the additive model within the cell-constant vectors, then under [Eq. 9.4.1](#) the subspaces for the mean,  $A$ ,  $B$  and  $AB$  form a decomposition whatever the order.

Orthogonality settles Type I against Type II, but not Type III. Under proportional frequencies the Type I and Type II sums of squares of  $A$  coincide and address equality of row means weighted by the column proportions  $n_{.j}/n$  ([Exercise \(B2\)](#)), while Type III weights the columns equally. They agree with equal replication or without interaction.

### Example. (Firms and years)

Grunfeld's investment panel ([Example \(Grunfeld's investment data\)](#)) records 11 firms in each of 20 years, one observation per firm and year. As a two-way layout with factors firm and year, it has  $n_{ij} = 1$ , so [Eq. 9.4.1](#) holds, and the additive model's sums of squares for the logarithm of investment are order-free:

| Source          | df  | Sum of squares |
|-----------------|-----|----------------|
| firm            | 10  | 473.482        |
| year            | 19  | 26.749         |
| residual        | 190 | 14.667         |
| corrected total | 219 | 514.898        |

Firms differ enormously in the scale of their investment, and years matter much less. Adding the logarithm of market value as a covariate destroys the orthogonality, because value varies with both firm and year. The sum of squares for firm then depends on what else is in the model: it is 90.701 after year and value, and 88.443 after value alone. Most of the variation between firms is variation in size, which value already measures. This is the setting of analysis of covariance ([Chapter 18](#), [Theorem 18.1.1](#)).

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

def proj(*blocks):
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]]
    return U @ U.T

g = sm.datasets.grunfeld.load_pandas().data
y = np.log(g["invest"].to_numpy())
n = len(y)
one = np.ones((n, 1))
F = pd.get_dummies(g["firm"]).to_numpy(float)      # 11 firms
T = pd.get_dummies(g["year"]).to_numpy(float)      # 20 years
```

```

P0, PF, PT, PFT = proj(one), proj(one, F), proj(one, T), proj(one, F, T)

print("centred spaces orthogonal:", np.abs((PF - P0) @ (PT - P0)).max() < 1e-10)
print(f"firm:  first {y @ (PF - P0) @ y:8.3f}    after year {y @ (PFT - PT) @ y:8.3f}"
      ")
print(f"year:  first {y @ (PT - P0) @ y:8.3f}    after firm {y @ (PFT - PF) @ y:8.3f}"
      ")

v = np.log(g["value"].to_numpy())[:, None] # log market value
PFv, PTv, PFTv = proj(one, F, v), proj(one, T, v), proj(one, F, T, v)
print(f"with log value: firm after year, value {y @ (PFTv - PTv) @ y:8.3f}"
      f"    firm after value {y @ (PFv - proj(one, v)) @ y:8.3f}")

```

### 9.4.3 Orthogonal polynomials

A polynomial regression in a single variable  $x$ , with distinct values  $x_1, \dots, x_n$ , has a built-in chain of models,

$$\mathcal{P}_0 \subseteq \mathcal{P}_1 \subseteq \dots \subseteq \mathcal{P}_d, \quad \mathcal{P}_k = \text{span}(\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \dots, \mathbf{x}^k),$$

where  $\mathbf{x}^k$  is the vector with entries  $x_i^k$ . The decomposition of interest is that of the chain ([Theorem 9.1.1\(e\)](#)). Its  $k$ th subspace,  $\mathcal{P}_{k-1}^\perp \cap \mathcal{P}_k$ , is a line. Applying Gram–Schmidt ([Proposition 6.2.2](#)) to  $\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \dots$  produces unit vectors  $\mathbf{q}_0, \mathbf{q}_1, \dots, \mathbf{q}_d$  with  $\text{span}(\mathbf{q}_0, \dots, \mathbf{q}_k) = \mathcal{P}_k$  for every  $k$ . So  $\mathbf{q}_k$  spans that line. The vectors  $\mathbf{q}_k$ , evaluated at the design points, are the **orthogonal polynomials** of the design. Three consequences follow at once.

1. The sequential sum of squares for degree  $k$ , the drop in residual sum of squares from degree  $k-1$  to degree  $k$ , is  $(\mathbf{q}_k^\top \mathbf{y})^2$ , with one degree of freedom.
2. The columns  $\mathbf{q}_1, \dots, \mathbf{q}_d$  are orthogonal and centred, so as separate terms they form an orthogonal design ([Theorem 9.4.1](#)), and their sums of squares do not depend on the order of entry.
3. The coefficient of  $\mathbf{q}_k$  is  $\mathbf{q}_k^\top \mathbf{y}$  in every fit that includes it ([Corollary 9.4.2](#)). Raising the degree never changes the lower coefficients.

For equally spaced  $x$  the orthogonal polynomials have integer multiples that are widely tabulated. For three points they are  $(-1, 0, 1)$  and  $(1, -2, 1)$ . For four points they are  $(-3, -1, 1, 3)$ ,  $(1, -1, -1, 1)$  and  $(-1, 3, -3, 1)$ . They serve as contrasts for a quantitative factor with equally spaced levels, splitting the treatment sum of squares into linear, quadratic and higher components. Their numerical merits and the recurrence that generates them belong to Chapter 42.

#### Example. (A trend in the Nile)

The annual flow of the Nile at Aswan was recorded for the 100 years 1871–1970. How much of its variation is a smooth trend? The listing builds the orthonormal polynomials  $\mathbf{q}_0, \dots, \mathbf{q}_4$  for the years by a QR factorization of the matrix of powers, and reads off the sequential sums of squares (in units of  $(10^8 \text{ m}^3)^2$ ):

| Degree                    | 1      | 2      | 3    | 4      |
|---------------------------|--------|--------|------|--------|
| Sequential sum of squares | 613893 | 309415 | 1894 | 134142 |

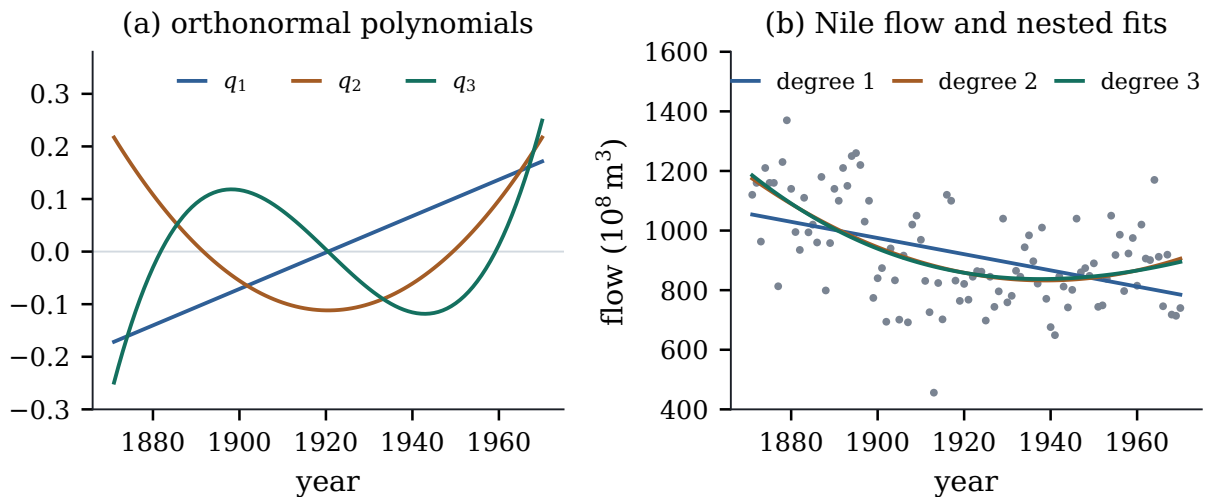


Figure 9.4.1: (a) The orthonormal polynomials  $\mathbf{q}_1, \mathbf{q}_2, \mathbf{q}_3$  for the years 1871–1970. (b) The Nile flow with the nested least squares fits of degrees 1, 2 and 3. The quadratic and cubic fits nearly coincide, because the cubic sum of squares is tiny.

The corrected total is 2835157. The linear term accounts for a fraction 0.217 of it, and the first three terms together for 0.326. Against the residual mean square of the cubic fit, 19895 on 96 degrees of freedom, the quadratic component has  $F = 15.55$  and the cubic only 0.10. The quartic sum of squares is large again, a sign that the pattern is not a gentle polynomial at all. The flow dropped abruptly around the end of the nineteenth century (Cobb 1978), and a low-degree polynomial can only approximate a step (Figure 9.4.1).

With the raw powers of the centred and scaled year, the sums of squares depend on the order: the linear column entered after the quadratic and cubic ones receives 74751 instead of 613893. The raw linear coefficient is  $-134.4$  in the linear and quadratic fits, but  $-117.3$  in the cubic and quartic ones. It survives the quadratic term because the years are symmetric about their mean, which makes odd and even centred powers orthogonal (Exercise (C1)), but not the cubic term. The coefficient of  $\mathbf{q}_1$  is  $-783.5$  in every fit.

```
import numpy as np
import statsmodels.api as sm
nile = sm.datasets.nile.load_pandas().data
year = nile["year"].to_numpy()
y = nile["volume"].to_numpy()

t = year - year.mean()
V = np.vander(t / t.max(), 5, increasing=True)
Q, R = np.linalg.qr(V)
Q = Q * np.sign(np.diag(R))
    with t^k
coef = Q.T @ y
    q_4
seq_ss = coef[1:] ** 2
    quartic
for k, s in enumerate(seq_ss, start=1):
    print(f"degree {k}: sequential SS = {s:10.0f}")
print(f"residual after degree 4: {y @ y - np.sum(coef ** 2):10.0f}")
```

# centring only helps rounding  
# 1, t, t^2, t^3, t^4 (scaled)  
# Gram-Schmidt in matrix form  
# make each  $q_k$  agree in sign  
  
# coordinates of  $y$  on  $q_0, \dots$   
  
# sequential SS: linear,  $\dots$

### 9.4.4 Single-degree-of-freedom decompositions

Orthogonal polynomials split a sum of squares into one-dimensional pieces along directions chosen for their meaning. The same can be done for any hypothesis, by choosing single estimable functions whose estimates are uncorrelated.

#### Theorem 9.4.4 — Orthogonal single-degree-of-freedom sums of squares

Let  $\lambda_1, \dots, \lambda_s$  be nonzero vectors with  $\lambda_k^\top \beta$  estimable, so that  $\lambda_k = \mathbf{X}^\top \rho_k$ , and let  $\mathbf{G}$  be a generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ .

- (a) The sum of squares for  $\lambda_k^\top \beta = 0$  is

$$SS_k = \frac{(\lambda_k^\top \hat{\beta})^2}{\lambda_k^\top \mathbf{G} \lambda_k} = \|\mathbf{P}_{\mathbf{v}_k} \mathbf{y}\|^2, \quad \mathbf{v}_k = \mathbf{M} \rho_k,$$

the squared length of the projection of  $\mathbf{y}$  onto the line spanned by  $\mathbf{v}_k$ .

- (b)  $\mathbf{v}_h^\top \mathbf{v}_k = \lambda_h^\top \mathbf{G} \lambda_k$ . If  $\lambda_h^\top \mathbf{G} \lambda_k = 0$  for all  $h \neq k$ , the lines are mutually orthogonal, and  $\sum_k SS_k$  is the sum of squares for the joint hypothesis  $\lambda_1^\top \beta = \dots = \lambda_s^\top \beta = 0$ , on  $s$  degrees of freedom.
- (c) Conversely, every orthonormal basis  $\mathbf{u}_1, \dots, \mathbf{u}_s$  of a subspace of  $\mathcal{C}(\mathbf{X})$  arises in this way, with  $\lambda_k = \mathbf{X}^\top \mathbf{u}_k$ .

*Proof.*

- (a) is Theorem 9.2.4 with  $q = 1$  and  $\mathbf{d} = 0$ : the test space is spanned by  $\mathbf{M} \rho_k = \mathbf{v}_k$ , and the projection of  $\mathbf{y}$  onto it has squared length  $(\mathbf{v}_k^\top \mathbf{y})^2 / \mathbf{v}_k^\top \mathbf{v}_k$ , where  $\mathbf{v}_k^\top \mathbf{y} = \rho_k^\top \mathbf{M} \mathbf{y} = \lambda_k^\top \hat{\beta}$  and  $\mathbf{v}_k^\top \mathbf{v}_k = \lambda_k^\top \mathbf{G} \lambda_k$ . (b)  $\mathbf{v}_h^\top \mathbf{v}_k = \rho_h^\top \mathbf{M} \rho_k = \rho_h^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \rho_k = \lambda_h^\top \mathbf{G} \lambda_k$  by Theorem 6.3.7. Orthogonal nonzero vectors are independent, so  $\mathbf{\Lambda} = [\lambda_1, \dots, \lambda_s]$  has rank  $s$  (because  $\mathbf{X}^\top \mathbf{V} = \mathbf{\Lambda}$  with  $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_s]$ , and  $\mathbf{X}^\top$  is one-to-one on  $\mathcal{C}(\mathbf{X})$ ). The test space of the joint hypothesis is  $\mathcal{C}(\mathbf{M} \mathbf{T}) = \mathcal{C}(\mathbf{V})$ , with  $\mathbf{T} = [\rho_1, \dots, \rho_s]$ , by Theorem 9.2.4(b), and the projection onto a span of orthogonal lines is the sum of the projections onto the lines (Theorem 6.3.8). (c) With  $\rho_k = \mathbf{u}_k$ ,  $\mathbf{v}_k = \mathbf{M} \mathbf{u}_k = \mathbf{u}_k$ , and  $\lambda_h^\top \mathbf{G} \lambda_k = \mathbf{u}_h^\top \mathbf{u}_k$ .

■

In the one-way layout with group sizes  $n_1, \dots, n_g$ , take the generalized inverse of Exercise (B3). A contrast  $\sum_i c_i \alpha_i$  with  $\sum_i c_i = 0$  (Definition 8.7.1) is estimated by  $\sum_i c_i \bar{y}_i$ , and  $\lambda^\top \mathbf{G} \lambda = \sum_i c_i^2 / n_i$ . So two contrasts  $\mathbf{c}$  and  $\mathbf{c}'$  give orthogonal sums of squares iff

$$\sum_{i=1}^g \frac{c_i c'_i}{n_i} = 0,$$

and  $g - 1$  such contrasts split the between-groups sum of squares into single degrees of freedom. With equal group sizes this is ordinary orthogonality,  $\sum_i c_i c'_i = 0$ ; with unequal sizes, contrasts that look orthogonal need not be.

**Example. (Splitting the education sum of squares)**

Consider the 944 self-placements of [Example \(Party, education and political self-placement\)](#) by education alone. The group sizes are 313, 277 and 354, and the group means are 4.476, 4.357 and 4.167. The between-groups sum of squares is 16.306 on two degrees of freedom. The contrast of college against graduate,  $\mathbf{c}_2 = (0, 1, -1)$ , has sum of squares 5.653. The contrast of high school against the other two, weighted by their sizes,  $\mathbf{c}_1 = (1, -0.439, -0.561)$ , is orthogonal to  $\mathbf{c}_2$  in the sense above, and has sum of squares 10.652. The two add up to the between-groups sum of squares, up to rounding. The more obvious contrast  $(1, -\frac{1}{2}, -\frac{1}{2})$ , high school against the plain average of the others, has sum of squares 9.534, and together with  $\mathbf{c}_2$  it gives 15.187: with unequal group sizes it is not orthogonal to  $\mathbf{c}_2$ .

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"])
y = d["selfLR"].to_numpy()

counts = educ.value_counts(sort=False).to_numpy().astype(float)
means = d.groupby(educ, observed=True)["selfLR"].mean().to_numpy()
grand = y.mean()
ss_treat = np.sum(counts * (means - grand) ** 2)          # SS(educ | 1), 2 df

def contrast_ss(c):
    """(c' ybar)^2 / sum(c_i^2 / n_i): the SS for the hypothesis c' mu = 0."""
    return (c @ means) ** 2 / np.sum(c ** 2 / counts)

def orthogonal(c, h):
    return np.isclose(np.sum(c * h / counts), 0.0)

n2, n3 = counts[1], counts[2]
c1 = np.array([1.0, -n2 / (n2 + n3), -n3 / (n2 + n3)]) # HS against the rest,
    weighted
c2 = np.array([0.0, 1.0, -1.0])                       # College against Graduate
c1_plain = np.array([1.0, -0.5, -0.5])                 # HS against the plain
    average
print("orthogonal:", orthogonal(c1, c2), orthogonal(c1_plain, c2))
print(f"treatment SS {ss_treat:.3f} = {contrast_ss(c1):.3f} + {contrast_ss(c2):.3f}"
      )
print(f"with the plain contrast: {contrast_ss(c1_plain):.3f} + {contrast_ss(c2):.3f}"
      )
```

Single-degree-of-freedom sums of squares reappear in [Chapter 13](#), where many of them are tested at once, and in [Chapter 15](#), where orthogonal contrasts organize the one-way analysis of variance ([Theorem 15.4.1](#)).

## 9.4.5 Exercises

### 9.4.5.1 A. Check your understanding

**Exercise A1.**

With  $x = 1, 2, 3, 4$  and  $\mathbf{y} = (3, 5, 4, 8)^\top$ , use the orthogonal polynomial contrasts for four equally spaced points to compute the linear, quadratic and cubic sums of squares, and check that they add up to the corrected total sum of squares.

**Exercise A2.**

Which of the  $2 \times 2$  count tables  $\begin{pmatrix} 2 & 4 \\ 3 & 6 \end{pmatrix}$ ,  $\begin{pmatrix} 2 & 4 \\ 4 & 2 \end{pmatrix}$  and  $\begin{pmatrix} 1 & 3 \\ 3 & 9 \end{pmatrix}$  give orthogonal centred factor spaces? For the one that does not, compute the inner product of the two centred indicator vectors of the first levels.

**Exercise A3.**

Show that if the regressors  $\mathbf{x}_1, \dots, \mathbf{x}_k$  of a model with an intercept are pairwise uncorrelated, then  $R^2 = \sum_j r_{yj}^2$ , where  $r_{yj}$  is the sample correlation of  $\mathbf{y}$  and  $\mathbf{x}_j$ .

**9.4.5.2 B. Practice****Exercise B1.**

In a two-way layout with interaction and  $n_{ij} = m \geq 1$  for all cells, show that the Type I sums of squares of  $A$  in either order, and its Type II and Type III sums of squares, are all equal to  $bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2$ .

*Solution.* Equal counts satisfy [Eq. 9.4.1](#), so by [Theorem 9.4.1](#) the Type I sums of squares of  $A$  in both orders and its Type II sum of squares equal  $\text{SS}(A \mid \mu) = \|(\mathbf{P}_A - \mathbf{P}_0)\mathbf{y}\|^2$ . Each row has  $bm$  observations, and  $\mathbf{P}_A - \mathbf{P}_0$  replaces an observation in row  $i$  by  $\bar{y}_{i..} - \bar{y}_{...}$ , which gives the formula. For Type III, the hypothesis of [Theorem 9.3.2\(b\)](#) is that of (a), because with equal counts weighted and unweighted row means coincide. A sum of squares for a hypothesis depends only on the hypothesis (it is  $\text{SSE}_0 - \text{SSE}$ ), so the Type III sum of squares equals  $\text{SS}(A \mid \mu)$  too.

**Exercise B2.**

Take counts  $n_{11} = 1$ ,  $n_{12} = 2$ ,  $n_{21} = 2$ ,  $n_{22} = 4$ . Check [Eq. 9.4.1](#). Find cell means for which the Type III hypothesis for  $A$  holds but the Type I (and Type II) hypothesis fails.

**Exercise B3.**

In a one-way layout with  $g$  groups of equal size  $m$ , show that the Helmert contrasts  $\mathbf{c}_k = (1, \dots, 1, -(k-1), 0, \dots, 0)$ , with  $k-1$  leading ones,  $k = 2, \dots, g$ , are mutually orthogonal, and write the between-groups sum of squares as a sum of  $g-1$  single-degree-of-freedom sums of squares. What goes wrong with unequal group sizes?



## 9.4.5.3 C. Going deeper

**Exercise C1.**

Suppose the design points are symmetric about their mean: after sorting,  $x_i - \bar{x} = -(x_{n+1-i} - \bar{x})$ . Show that every odd power of the centred variable is orthogonal to every even power (including the zeroth), that  $\mathbf{q}_k$  is an even or odd vector according to the parity of  $k$ , and explain the pattern of raw coefficients in [Example \(A trend in the Nile\)](#).

*Solution.* Let  $\mathbf{t}$  have entries  $t_i = x_i - \bar{x}$  and let  $\mathbf{J}$  be the reversal permutation, so that  $\mathbf{J}\mathbf{t} = -\mathbf{t}$ . A vector  $\mathbf{u}$  is even if  $\mathbf{J}\mathbf{u} = \mathbf{u}$  and odd if  $\mathbf{J}\mathbf{u} = -\mathbf{u}$ . Powers  $\mathbf{t}^k$  have the parity of  $k$ , and an even vector  $\mathbf{u}$  and an odd vector  $\mathbf{v}$  are orthogonal because  $\mathbf{u}^\top \mathbf{v} = (\mathbf{J}\mathbf{u})^\top (\mathbf{J}\mathbf{v}) = -\mathbf{u}^\top \mathbf{v}$ . Gram–Schmidt on  $\mathbf{1}, \mathbf{t}, \mathbf{t}^2, \dots$  subtracts from  $\mathbf{t}^k$  only its projections on the earlier  $\mathbf{q}_j$  of the same parity, by induction, so  $\mathbf{q}_k$  has the parity of  $k$ . Hence the column  $\mathbf{t}$  is orthogonal to the column  $\mathbf{t}^2$  and to  $\mathbf{1}$ , and adding the quadratic term does not change the linear coefficient ([Corollary 9.4.2](#), applied to the terms  $\mathbf{t}$  and  $\mathbf{t}^2 - \bar{\mathbf{t}^2}\mathbf{1}$ ). The cubic column is odd and not orthogonal to  $\mathbf{t}$ , so it changes the linear coefficient. The quartic column is even. By [Theorem 9.2.2\(b\)](#), adding it changes the cubic fit’s coefficients by  $-\mathbf{L}\hat{\gamma}$  with  $\mathbf{L} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{t}^4$ , where  $\mathbf{X} = [\mathbf{1}, \mathbf{t}, \mathbf{t}^2, \mathbf{t}^3]$ . Inner products between odd and even columns vanish, so  $\mathbf{X}^\top \mathbf{X}$  and its inverse do not link the odd and even columns, and the entries of  $\mathbf{X}^\top \mathbf{t}^4$  for the odd columns are zero. Hence the rows of  $\mathbf{L}$  for  $\mathbf{t}$  and  $\mathbf{t}^3$  vanish, and the linear coefficient is unchanged.

## 9.5 The coefficient of determination and its limits

The coefficient of determination is the most widely reported summary of a regression, and probably the most widely misread. It is a ratio of two rows of the corrected table,

$$R^2 = \frac{\text{SSR}}{\text{SST}} = 1 - \frac{\text{SSE}}{\text{SST}},$$

defined when  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  and  $\mathbf{y}$  is not constant. [Section 6.7](#) showed that it is a squared cosine ([Theorem 6.7.2](#)) that cannot decrease as the model space grows. This section adds  $R^2$  as a maximal correlation, its null distribution, the adjusted and partial versions, and what it does not measure.

### 9.5.1 A maximal correlation

The square root  $R = \sqrt{R^2}$  is called the **multiple correlation coefficient**. The name comes from the following property, the sample counterpart of the population result [Proposition 2.5.3](#).

**Theorem 9.5.1** —  $R$  is the largest correlation with a linear combination

Suppose  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  and  $\mathbf{y}$  is not constant. For every  $\mathbf{b}$  such that  $\mathbf{Xb}$  is not constant,

$$\text{corr}(\mathbf{y}, \mathbf{Xb})^2 \leq R^2.$$

If  $R^2 > 0$ , equality holds iff  $\mathbf{Xb} = a\hat{\mathbf{Y}} + c\mathbf{1}$  for some  $a \neq 0$  and some  $c$ , and the correlation  $\text{corr}(\mathbf{y}, \mathbf{Xb})$  is maximized, with value  $R$ , by the fitted values themselves. If  $R^2 = 0$ , every such correlation is zero.

*Proof.* Let  $\mathbf{u} = \mathbf{y} - \bar{y}\mathbf{1}$  and  $\mathcal{V} = \mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$ , whose projection is  $\mathbf{M} - \mathbf{P}_1$  (Theorem 6.5.1), and let  $\mathbf{v} = \mathbf{Xb} - \bar{\mathbf{Xb}}\mathbf{1}$ . Since  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ ,  $\mathbf{v} \in \mathcal{V}$ , so  $\mathbf{u}^\top \mathbf{v} = \mathbf{u}^\top (\mathbf{M} - \mathbf{P}_1)\mathbf{v} = ((\mathbf{M} - \mathbf{P}_1)\mathbf{u})^\top \mathbf{v}$ . The sample correlation is  $\mathbf{u}^\top \mathbf{v} / (\|\mathbf{u}\| \|\mathbf{v}\|)$ , and by Cauchy–Schwarz (Proposition 1.1.1)

$$\text{corr}(\mathbf{y}, \mathbf{Xb})^2 = \frac{((\mathbf{M} - \mathbf{P}_1)\mathbf{u})^\top \mathbf{v})^2}{\|\mathbf{u}\|^2 \|\mathbf{v}\|^2} \leq \frac{\|(\mathbf{M} - \mathbf{P}_1)\mathbf{u}\|^2}{\|\mathbf{u}\|^2} = \frac{\text{SSR}}{\text{SST}},$$

with equality iff  $\mathbf{v}$  and  $(\mathbf{M} - \mathbf{P}_1)\mathbf{u} = \hat{\mathbf{Y}} - \bar{y}\mathbf{1}$  are linearly dependent. If  $R^2 > 0$  the second vector is nonzero, and since  $\mathbf{v} \neq \mathbf{0}$  dependence means  $\mathbf{v} = a(\hat{\mathbf{Y}} - \bar{y}\mathbf{1})$  with  $a \neq 0$ , which is the stated condition; for  $a > 0$  the correlation itself is  $+R$ . If  $R^2 = 0$ , then  $(\mathbf{M} - \mathbf{P}_1)\mathbf{u} = \mathbf{0}$  and  $\mathbf{u}^\top \mathbf{v} = 0$ . ■

Among all linear combinations of the regressors, the fitted values are the one most correlated with the response. When the regressors are pairwise uncorrelated,  $R^2$  is the sum of the squared simple correlations (Exercise A3). In general it can be smaller or larger than that sum (Exercise C2).

## 9.5.2 $R^2$ when nothing is going on

Since  $R^2$  never decreases as regressors are added, a model with many regressors has a sizeable  $R^2$  even if none of them matters. The theory of Chapter 4 says how sizeable.

### Theorem 9.5.2 — The distribution of $R^2$

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$  with  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ ,  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  and  $1 < r = \text{rank}(\mathbf{X}) < n$ . Let  $F = [\text{SSR}/(r-1)]/[\text{SSE}/(n-r)]$  and  $\gamma = \|(\mathbf{M} - \mathbf{P}_1)\boldsymbol{\mu}\|^2/\sigma^2$ . Then:

- (a)  $R^2 = \frac{(r-1)F}{(r-1)F + n-r}$ , and  $F \sim F(r-1, n-r, \gamma)$ ;
- (b) if  $\boldsymbol{\mu} \in \text{span}(\mathbf{1})$ , so that no regressor matters, then  $R^2 \sim \text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$  and  $E(R^2) = \frac{r-1}{n-1}$ .

*Proof.*

- (a)  $R^2/(1-R^2) = \text{SSR}/\text{SSE} = (r-1)F/(n-r)$ , and solving for  $R^2$  gives the formula. The projections  $\mathbf{M} - \mathbf{P}_1$  and  $\mathbf{I} - \mathbf{M}$  are orthogonal with ranks  $r-1$  and  $n-r$ , and  $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \mathbf{0}$ , so by Theorem 4.4.4  $\text{SSR}/\sigma^2 \sim \chi^2(r-1, \gamma)$  and  $\text{SSE}/\sigma^2 \sim \chi^2(n-r)$ ,

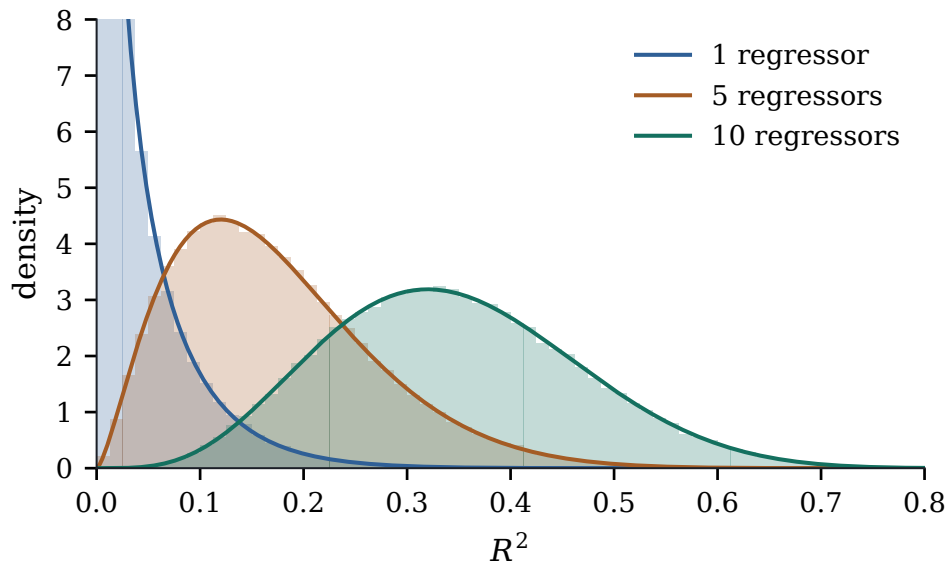


Figure 9.5.1: Null distribution of  $R^2$  for  $n = 30$  observations and 1, 5 or 10 regressors unrelated to the response. Histograms: 100,000 simulated data sets each. Curves: the Beta densities of [Theorem 9.5.2](#).

independently. The ratio  $F$  is therefore  $F(r-1, n-r, \gamma)$  ([Definition 4.1.2](#)).

- (b) Now  $\gamma = 0$ , and  $R^2 = U/(U+V)$  with  $U = \text{SSR}/\sigma^2 \sim \chi^2(r-1)$  and  $V = \text{SSE}/\sigma^2 \sim \chi^2(n-r)$  independent. By [Exercise \(B4\)](#) this is  $\text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$ , whose mean is  $\frac{r-1}{2} / \frac{n-1}{2} = (r-1)/(n-1)$ .

■

Part (b) answers the exercise [Exercise \(C1\)](#) of [Chapter 6](#). With  $k = r - 1$  regressors of pure noise,  $R^2$  is on average  $k/(n-1)$ . A model with 10 useless regressors fitted to 30 observations has expected  $R^2$  about one third. [Figure 9.5.1](#) compares the Beta densities with 100,000 simulated data sets with  $n = 30$  and  $k = 1, 5, 10$  regressors drawn at random. The simulated means are 0.0345, 0.1725 and 0.3445, against the exact 0.0345, 0.1724 and 0.3448. The upper 5% points of the three null distributions are 0.130, 0.353 and 0.556, so with ten regressors an  $R^2$  of one half is unremarkable. Part (a) shows that  $R^2$  and  $F$  carry the same information: testing  $R^2$  against its null distribution is the overall  $F$  test of the regression.

```
import numpy as np
rng = np.random.default_rng(20260919)
n, reps = 30, 100_000
results = {}
for k in (1, 5, 10):
    # number of regressors
    besides 1
    X = np.column_stack([np.ones(n), rng.normal(size=(n, k))])
    r = k + 1
    Q, _ = np.linalg.qr(X)
    Y = 3.0 + rng.normal(size=(n, reps))
    Z = Q.T @ Y
    sst = np.sum(Y ** 2, axis=0) - Z[0] ** 2
    ssr = np.sum(Z[1:] ** 2, axis=0)
    R2 = ssr / sst
    R2adj = 1 - (1 - R2) * (n - 1) / (n - r)
    # no regressor matters
    print(f"k = {k:2d}: mean R^2 = {R2.mean():.4f} (theory {k / (n - 1):.4f}),"
```

```
f" mean adjusted R^2 = {R2adj.mean():+.4f}"
results[k] = R2
```

### 9.5.3 Adjusted $R^2$

The bias of  $R^2$  under the null suggests a correction. Write  $s^2 = \text{SSE}/(n-r)$  and  $s_y^2 = \text{SST}/(n-1)$  for the unbiased estimators of the error variance in the fitted model and in the intercept-only model. Replacing sums of squares by these mean squares defines the **adjusted coefficient of determination**

$$\bar{R}^2 = 1 - \frac{s^2}{s_y^2} = 1 - (1 - R^2) \frac{n-1}{n-r}. \quad (9.5.1)$$

#### Proposition 9.5.3 — Properties of the adjusted $R^2$

Let  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ ,  $1 \leq r = \text{rank}(\mathbf{X}) < n$ , and let  $F$  be the overall  $F$  ratio of [Theorem 9.5.2](#).

- (a)  $\bar{R}^2 \leq R^2$ , with equality iff  $r = 1$  or  $R^2 = 1$ .
- (b) For  $r > 1$ ,  $\bar{R}^2 < 0$  iff  $F < 1$ .
- (c) If a model with rank  $r_0$  is enlarged to one of rank  $r = r_0 + q$  containing it, then  $\bar{R}^2$  increases iff the  $F$  ratio of the added terms,  $[\text{SS}(\text{added} \mid \text{old})/q]/s^2$ , exceeds 1. For a single added column this means  $|t| > 1$ .
- (d) Among models for the same response,  $\bar{R}^2$  is largest for the model with the smallest residual mean square  $s^2$ .
- (e) Under the normal model with  $\boldsymbol{\mu} \in \text{span}(\mathbf{1})$ ,  $E(\bar{R}^2) = 0$ .

*Proof.*

- (a)  $1 - \bar{R}^2 = (1 - R^2)(n-1)/(n-r)$ , and  $(n-1)/(n-r) \geq 1$  with equality iff  $r = 1$ , so  $1 - \bar{R}^2 \geq 1 - R^2$ , with equality iff  $r = 1$  or  $1 - R^2 = 0$ .
- (b)  $\bar{R}^2 < 0$  iff  $s^2 > s_y^2$ , that is,  $\text{SSE}(n-1) > (\text{SSR} + \text{SSE})(n-r)$ , that is,  $\text{SSE}(r-1) > \text{SSR}(n-r)$ , which for  $r > 1$  is  $F < 1$ .
- (c)  $s_y^2$  does not depend on the model, so  $\bar{R}^2$  increases iff  $s^2$  decreases. Write  $\text{SSE}_0 = \text{SSE} + D$  with  $D$  the extra sum of squares. Then  $\text{SSE}/(n-r) < (\text{SSE} + D)/(n-r_0)$  iff  $\text{SSE}(n-r_0) < (\text{SSE} + D)(n-r)$ , iff  $q\text{SSE} < D(n-r)$ , iff  $(D/q)/s^2 > 1$ . For one column the ratio is  $t^2$  ([Corollary 9.2.3](#)).
- (d) is immediate from [Eq. 9.5.1](#).
- (e) By [Theorem 9.5.2\(b\)](#),  $E(1 - R^2) = 1 - (r-1)/(n-1) = (n-r)/(n-1)$ , so  $E(1 - \bar{R}^2) = 1$ .

■

Part (e) is the point of the adjustment. In the simulation the averages of  $\bar{R}^2$  are 0.0000, 0.0001 and  $-0.0005$ , but  $\bar{R}^2$  is negative in a proportion 0.674 of the data sets with one regressor: it is not a proportion of anything. Part (c) shows what selection by  $\bar{R}^2$  amounts to: a regressor is

admitted when  $|t| > 1$ , far below the threshold near two of a 5% test, so the chosen models tend to be too large (Proposition 29.2.3).

**Example. (Adjusted  $R^2$  for the state data)**

For all seven models built from poverty, single parenthood and urbanization (each with an intercept):

| Regressors      | $R^2$  | $\bar{R}^2$ |
|-----------------|--------|-------------|
| poverty         | 0.3599 | 0.3466      |
| single          | 0.5719 | 0.5629      |
| urban           | 0.0221 | 0.0017      |
| poverty, single | 0.6410 | 0.6257      |
| poverty, urban  | 0.4626 | 0.4398      |
| single, urban   | 0.5844 | 0.5668      |
| all three       | 0.6419 | 0.6186      |

$R^2$  is largest for the full model, as it must be.  $\bar{R}^2$  prefers poverty and single parenthood without urbanization. Adding urbanization to that model raises  $R^2$  slightly but lowers  $\bar{R}^2$ , because its  $t$  statistic in the full model is only 0.349, so its  $F$  ratio is  $0.121 < 1$ .

## 9.5.4 Partial $R^2$

Applied to what a reduced model leaves unexplained, the idea of  $R^2$  gives a measure for a term.

**Definition 9.5.1 — Partial coefficient of determination**

Let  $\mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X})$  with  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$ . The **partial coefficient of determination** of  $\mathbf{X}$  over  $\mathbf{X}_1$  is

$$R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{\text{SSE}(\mathbf{X}_1) - \text{SSE}(\mathbf{X})}{\text{SSE}(\mathbf{X}_1)} = \frac{\text{SS}(\mathbf{X} | \mathbf{X}_1)}{\text{SSE}(\mathbf{X}_1)},$$

the proportion of the residual variation of the reduced model that the larger model explains.

**Theorem 9.5.4 — Partial  $R^2$ ,  $F$  and  $t$**

With the notation of Definition 9.5.1, let  $r_1 = \text{rank}(\mathbf{X}_1)$ ,  $r = \text{rank}(\mathbf{X})$ ,  $q = r - r_1 > 0$ ,  $s^2 = \text{SSE}(\mathbf{X})/(n - r)$ , and let  $F = [\text{SS}(\mathbf{X} | \mathbf{X}_1)/q]/s^2$ . Then:

- (a)  $1 - R^2 = (1 - R_1^2)(1 - R_{\mathbf{X}|\mathbf{X}_1}^2)$ , where  $R^2$  and  $R_1^2$  belong to the two models;
- (b)  $R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{qF}{qF + n - r}$ ;
- (c) if  $\mathbf{X} = [\mathbf{X}_1, \mathbf{x}_j]$  adds one column, then  $R_{\mathbf{X}|\mathbf{X}_1}^2 = \frac{t_j^2}{t_j^2 + n - r}$ , and it equals the squared sample correlation between the residuals  $(\mathbf{I} - \mathbf{M}_1)\mathbf{y}$  and  $(\mathbf{I} - \mathbf{M}_1)\mathbf{x}_j$ ;

(d) for a chain  $\mathcal{C}(\mathbf{1}) = \mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_1) \subseteq \cdots \subseteq \mathcal{C}(\mathbf{X}_k)$ ,  $1 - R_k^2 = \prod_{j=1}^k (1 - R_{\mathbf{X}_j|\mathbf{X}_{j-1}}^2)$ .

*Proof.*

- (a)  $1 - R^2 = \text{SSE}(\mathbf{X})/\text{SST} = (\text{SSE}(\mathbf{X}_1)/\text{SST})(\text{SSE}(\mathbf{X})/\text{SSE}(\mathbf{X}_1))$ , and the two factors are  $1 - R_1^2$  and  $1 - R_{\mathbf{X}|\mathbf{X}_1}^2$ .
- (b) Write  $D = \text{SS}(\mathbf{X} | \mathbf{X}_1) = qFs^2$  and  $\text{SSE}(\mathbf{X}) = (n - r)s^2$ . Then  $R_{\mathbf{X}|\mathbf{X}_1}^2 = D/(D + \text{SSE}(\mathbf{X})) = qF/(qF + n - r)$ .
- (c) With  $q = 1$ ,  $F = t_j^2$  by [Corollary 9.2.3](#), and (b) gives the first formula. For the correlation, let  $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_1)\mathbf{y}$  and  $\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{M}_1)\mathbf{x}_j$ . Then  $\text{SSE}(\mathbf{X}_1) = \|\tilde{\mathbf{y}}\|^2$ , and by [Lemma 9.2.1](#)  $\text{SS}(\mathbf{X} | \mathbf{X}_1) = (\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}\|^2$ . Their ratio is the squared cosine of the angle between  $\tilde{\mathbf{y}}$  and  $\tilde{\mathbf{x}}$ . Both have mean zero because  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$ , so the cosine is their sample correlation ([Exercise \(B3\)](#)).
- (d) Apply (a) repeatedly along the chain, with  $R_0^2 = 0$  for the intercept-only model.

■

Part (c) identifies the partial  $R^2$  of one regressor with a squared **sample partial correlation**, the sample counterpart of [Definition 3.6.1](#); [Chapter 14](#) studies its sampling distribution. Within one model, ranking regressors by  $|t|$  and by partial  $R^2$  is the same thing.

#### Example. (Partial $R^2$ in the state data)

In the three-regressor model, with  $n - r = 46$ :

| Regressor | $t$   | partial $R^2$ | $R^2$ without it |
|-----------|-------|---------------|------------------|
| poverty   | 2.718 | 0.1383        | 0.5844           |
| single    | 4.799 | 0.3337        | 0.4626           |
| urban     | 0.349 | 0.0026        | 0.6410           |

Each partial  $R^2$  equals  $t^2/(t^2 + 46)$ , and each row satisfies part (a): for poverty,  $1 - 0.6419 = (1 - 0.5844)(1 - 0.1383)$ . Poverty explains 14% of the variation that single parenthood and urbanization leave unexplained, although on its own it explains 36% of the total ( $R^2 = 0.3599$ ). For the pair poverty and urbanization over single parenthood, the  $F$  ratio on 2 and 46 degrees of freedom is 4.502 and the partial  $R^2$  is 0.1637.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
names = ["poverty", "single", "urban"]
n = len(y)

def fit(cols):
    X = np.column_stack([np.ones(n)] + [data[c].to_numpy() for c in cols])
    return sm.OLS(y, X).fit()

full = fit(names)
```

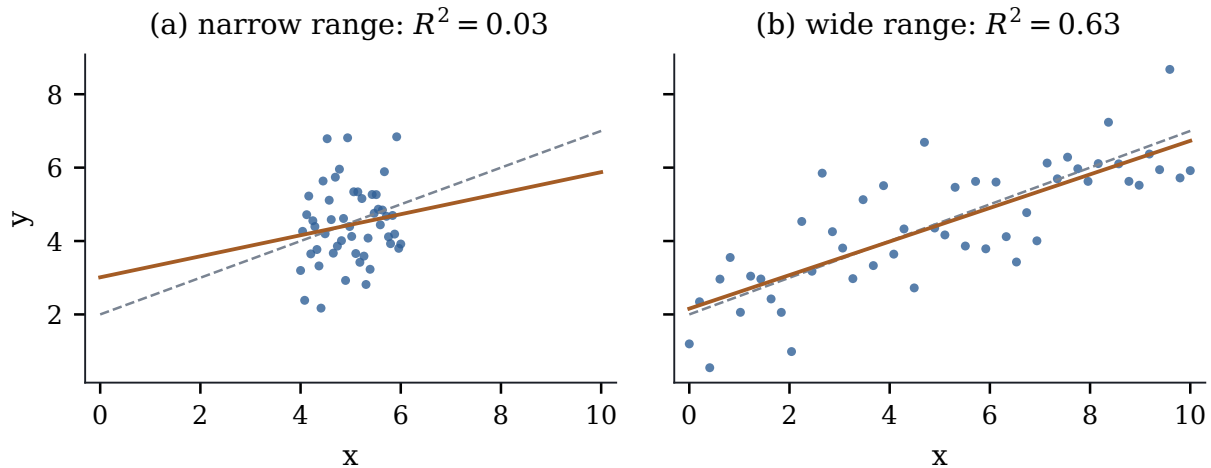


Figure 9.5.2: One line, two designs. The dashed line is the true mean  $2 + 0.5x$ , the solid line the least squares fit. The errors and the residual mean square are the same in both panels.

```
df_res = full.df_resid
for j, name in enumerate(names, start=1):
    reduced = fit([c for c in names if c != name])
    partial_r2 = (reduced.ssr - full.ssr) / reduced.ssr
    t = full.tvalues[j]
    print(f"{name:8s} partial R^2 = {partial_r2:.4f}, t = {t:6.3f},"
          f" t^2/(t^2 + df) = {t ** 2 / (t ** 2 + df_res):.4f}")
print(f"R^2 = {full.rsquared:.4f}, adjusted R^2 = {full.rsquared_adj:.4f}")
```

### 9.5.5 What $R^2$ does not measure

A high  $R^2$  is often read as “a good model” and a low one as “a useless model”; neither reading is safe.

**It is not a property of the relationship alone.**  $R^2$  depends on the spread of the regressors, which is a property of the design. In simple regression with fixed  $x$  values, the expectations of the two sums of squares are ([Exercise \(B1\)](#))

$$E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}, \quad E(\text{SST}) = (n-1)\sigma^2 + \beta_1^2 S_{xx},$$

so  $R^2$  is roughly  $\beta_1^2 S_{xx} / (\beta_1^2 S_{xx} + (n-1)\sigma^2)$ . In [Figure 9.5.2](#) the line  $E(y) = 2 + 0.5x$  and the same 50 errors with  $\sigma = 1$  are combined with  $x$  values spread evenly over  $[4, 6]$  and over  $[0, 10]$ . The two designs span the same  $\mathcal{C}(\mathbf{X})$ , so the residuals are identical and the residual mean square is 1.101 in both fits, but  $R^2$  is 0.026 in one case and 0.632 in the other. The approximate values from the formula are 0.081 and 0.689. Restricting the range of a regressor lowers  $R^2$  without changing the relationship.

```
import numpy as np
rng = np.random.default_rng(9)
n, beta0, beta1, sigma = 50, 2.0, 0.5, 1.0
eps = sigma * rng.normal(size=n)
# the same errors in both designs

def fit(x):
    y = beta0 + beta1 * x + eps
```

```

X = np.column_stack([np.ones(n), x])
b, *_ = np.linalg.lstsq(X, y, rcond=None)
sse = np.sum((y - X @ b) ** 2)
r2 = 1 - sse / np.sum((y - y.mean()) ** 2)
return y, b, sse / (n - 2), r2

for label, x in [("narrow", np.linspace(4, 6, n)), ("wide", np.linspace(0, 10, n))]:
    y, b, s2, r2 = fit(x)
    Sxx = np.sum((x - x.mean()) ** 2)
    expected = beta1 ** 2 * Sxx / (beta1 ** 2 * Sxx + (n - 1) * sigma ** 2)
    print(f"{label:6s}: slope {b[1]:.3f}, s^2 {s2:.3f}, R^2 {r2:.3f}, design value {
    expected:.3f}")

```

**It is not a test of the model.** A line fitted to points on a smooth curve can have a high  $R^2$  while missing the curvature; residual plots and lack-of-fit tests ([Theorem 14.6.2](#)) reveal it. Conversely, a correct model has a low  $R^2$  whenever the error variance is large, and its coefficients can still be precisely estimated and important.

**It is not comparable across responses.** Models for  $y$  and  $\log y$ , or for individuals and for group averages, have different denominators SST. Averaging replicates inflates  $R^2$  ([Exercise \(B2\)](#)).

**It is not comparable without an intercept** ([Proposition 9.1.3](#), [Example \(Forcing the murder regression through the origin\)](#)).

**It is not a measure of prediction or of cause.** Computed on the fitting data, it overstates predictive accuracy for new data ([Chapter 29](#)), and variation “explained” is linear association, not the effect of changing a regressor ([Chapter 25](#)).

## 9.5.6 Exercises

### 9.5.6.1 A. Check your understanding

#### Exercise A1.

A regression with an intercept,  $n = 20$  and  $r = 6$  has  $R^2 = 0.40$ . Compute  $\bar{R}^2$  and the overall  $F$  ratio. Is  $\bar{R}^2$  positive? Is it consistent with part (b) of [Proposition 9.5.3](#)?

### 9.5.6.2 B. Practice

#### Exercise B1.

In simple regression with fixed  $x$  values,  $E(\mathbf{Y}) = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , use [Theorem 2.3.1](#) to show that  $E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}$  and  $E(\text{SST}) = (n - 1)\sigma^2 + \beta_1^2 S_{xx}$ . Show that the ratio of these expectations tends to 1 as  $S_{xx} \rightarrow \infty$  with  $n$  fixed, and to  $1/(n - 1)$  as  $S_{xx} \rightarrow 0$ , and interpret the second limit.

*Solution.*  $\text{SSR} = \mathbf{y}^\top (\mathbf{M} - \mathbf{P}_1) \mathbf{y}$ , where  $\mathbf{M} - \mathbf{P}_1$  projects onto the line spanned by  $\mathbf{x} - \bar{x} \mathbf{1}$  ([Example \(Intercept inside simple regression\)](#)) and has trace 1. With  $\boldsymbol{\mu} = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$ ,  $(\mathbf{M} - \mathbf{P}_1) \boldsymbol{\mu} = \beta_1 (\mathbf{x} - \bar{x} \mathbf{1})$ , whose squared length is  $\beta_1^2 S_{xx}$ . [Theorem 2.3.1](#) gives  $E(\text{SSR}) = \sigma^2 + \beta_1^2 S_{xx}$ . Similarly  $\text{SST} = \mathbf{y}^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{y}$ , with trace  $n - 1$  and  $(\mathbf{I} - \mathbf{P}_1) \boldsymbol{\mu} = \beta_1 (\mathbf{x} - \bar{x} \mathbf{1})$ .



The ratio  $(\sigma^2 + \beta_1^2 S_{xx})/((n-1)\sigma^2 + \beta_1^2 S_{xx})$  tends to 1 as  $S_{xx} \rightarrow \infty$  and to  $1/(n-1)$  as  $S_{xx} \rightarrow 0$ . The second limit is the null value of  $E(R^2)$  for  $r = 2$  ([Theorem 9.5.2](#)): with almost no spread in  $x$ , the slope contributes almost nothing and the regression sum of squares is almost pure noise.

### Exercise B2.

Suppose each of  $g$  distinct values  $x_1, \dots, x_g$  carries  $m$  observations. Show that the least squares line fitted to all  $gm$  observations is the same as the line fitted to the  $g$  group means, and that its  $R^2$  for the individual data is at most its  $R^2$  for the means, with equality iff every observation equals its group mean.

*Solution.* The model space for the individual data is spanned by  $\mathbf{1}$  and  $\mathbf{x}$ , both constant within groups, so it lies inside the space of group-constant vectors. By [Theorem 6.5.1](#), projecting  $\mathbf{y}$  onto it equals projecting the vector of group means (repeated  $m$  times) onto it, which is the fit to the means with each point weighted equally. Write  $\text{SSR}_g$  and  $\text{SST}_g$  for the sums over the  $g$  means, and  $W$  for the within-group sum of squares. For the individual data  $\text{SSR} = m \text{SSR}_g$  and  $\text{SST} = m \text{SST}_g + W$ . Hence  $R^2 = m \text{SSR}_g / (m \text{SST}_g + W) \leq \text{SSR}_g / \text{SST}_g$ , with equality iff  $W = 0$ .

### Exercise B3.

Using part (d) of [Theorem 9.5.4](#) and the sequential sums of squares of [Example \(Order matters\)](#) (order A), express  $1 - R^2$  for the three-regressor model as a product of three factors, and identify each factor as one minus a partial  $R^2$ .

#### 9.5.6.3 C. Going deeper

### Exercise C1.

Under the conditions of [Theorem 9.5.2](#), show that for every  $c \in (0, 1)$  the probability  $\Pr(R^2 > c)$  is a strictly increasing function of the noncentrality  $\gamma$ . *Hint:* part (a) of the theorem and [Theorem 4.1.4](#).

### Exercise C2.

For two regressors with sample correlations  $r_{y1} = 0.5$ ,  $r_{y2} = 0$  and  $r_{12} = 0.6$ , compute  $R^2$  and compare it with  $r_{y1}^2 + r_{y2}^2$ . Use the formulas of [Exercise \(B1\)](#). Explain geometrically why a regressor that is uncorrelated with the response can raise  $R^2$ .

## 9.6 Expected mean squares

This section computes the expectations of sums of squares when the data are random, using only two moments, and their distributions under normality. Each mean square estimates  $\sigma^2$  plus a term that vanishes exactly when its hypothesis is true, while the residual mean square estimates  $\sigma^2$  alone. That is the structure of the  $F$  ratio, on which Part III builds.

## 9.6.1 Expected sums of squares for a decomposition

In this section  $E(\mathbf{Y}) = \boldsymbol{\mu}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ . We do not assume at first that  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ , because the effect of a wrong model on the mean squares is part of the story.

### Theorem 9.6.1 — Expected mean squares

Let  $\mathcal{V}_1, \dots, \mathcal{V}_k$  be an analysis of variance decomposition of  $\mathcal{C}(\mathbf{X})$ , with projections  $\mathbf{P}_i$  and dimensions  $r_i$ , and let  $E(\mathbf{Y}) = \boldsymbol{\mu}$ ,  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ .

- (a)  $E(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = \sigma^2 r_i + \|\mathbf{P}_i \boldsymbol{\mu}\|^2$ , so the mean square of  $\mathcal{V}_i$  has expectation

$$E(\text{MS}_i) = \sigma^2 + \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{r_i}. \quad (9.6.1)$$

- (b)  $E(\text{SSE}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2$ , so  $E(s^2) = \sigma^2$  iff  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ .  
 (c) If  $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$  with  $\mathbf{X} = [\mathbf{X}_0, \dots, \mathbf{X}_k]$  partitioned into terms, the partial sum of squares of  $\mathbf{X}_j$  has

$$E \text{ SS}(\mathbf{X}_j \mid \mathbf{X}_{(-j)}) = \sigma^2 d_j + \boldsymbol{\beta}_j^\top \mathbf{X}_j^\top (\mathbf{I} - \mathbf{M}_{(-j)}) \mathbf{X}_j \boldsymbol{\beta}_j,$$

with  $d_j = \text{rank}(\mathbf{M} - \mathbf{M}_{(-j)})$ , and the sequential sum of squares of  $\mathbf{X}_j$  has  $E \text{ SS}(\mathbf{X}_j \mid \mathbf{X}_0, \dots, \mathbf{X}_{j-1}) = \sigma^2 d'_j + \|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{X}\boldsymbol{\beta}\|^2$ , with  $d'_j = \text{rank}(\mathbf{M}_j - \mathbf{M}_{j-1})$ .

- (d) If moreover  $\mathbf{Y}$  is normal, then  $\text{Var}(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = 2\sigma^4 r_i + 4\sigma^2 \|\mathbf{P}_i \boldsymbol{\mu}\|^2$ , and the sums of squares of different subspaces, including the error space, are uncorrelated.

*Proof.*

- (a) By Theorem 2.3.1 with  $\mathbf{A} = \mathbf{P}_i$  and  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ ,  $E(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = \sigma^2 \text{tr}(\mathbf{P}_i) + \boldsymbol{\mu}^\top \mathbf{P}_i \boldsymbol{\mu}$ . The trace is  $r_i$  (Proposition 6.3.4), and  $\boldsymbol{\mu}^\top \mathbf{P}_i \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{P}_i^\top \mathbf{P}_i \boldsymbol{\mu} = \|\mathbf{P}_i \boldsymbol{\mu}\|^2$ . Divide by  $r_i$ .  
 (b) The same with  $\mathbf{A} = \mathbf{I} - \mathbf{M}$ , of trace  $n - r$ . The second term vanishes iff  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ .  
 (c) For the partial sum of squares,  $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_l \boldsymbol{\beta}_l = \mathbf{0}$  for  $l \neq j$ , because  $\mathbf{X}_l \boldsymbol{\beta}_l$  lies in  $\mathcal{C}(\mathbf{X}_{(-j)})$  and is fixed by both projections. And  $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_j \boldsymbol{\beta}_j = \mathbf{X}_j \boldsymbol{\beta}_j - \mathbf{M}_{(-j)}\mathbf{X}_j \boldsymbol{\beta}_j = (\mathbf{I} - \mathbf{M}_{(-j)})\mathbf{X}_j \boldsymbol{\beta}_j$ . Its squared length is  $\boldsymbol{\beta}_j^\top \mathbf{X}_j^\top (\mathbf{I} - \mathbf{M}_{(-j)}) \mathbf{X}_j \boldsymbol{\beta}_j$ , since  $\mathbf{I} - \mathbf{M}_{(-j)}$  is symmetric and idempotent. The sequential case is (a) for the chain decomposition (Definition 9.3.1).  
 (d) By Theorem 4.2.3(b) with  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ ,  $\text{Var}(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = 2 \text{tr}\{(\sigma^2 \mathbf{P}_i)^2\} + 4\sigma^2 \boldsymbol{\mu}^\top \mathbf{P}_i \mathbf{P}_i \boldsymbol{\mu} = 2\sigma^4 r_i + 4\sigma^2 \|\mathbf{P}_i \boldsymbol{\mu}\|^2$ . By part (d) of the same theorem the covariance of two forms is  $2\sigma^4 \text{tr}(\mathbf{P}_i \mathbf{P}_j) + 4\sigma^2 \boldsymbol{\mu}^\top \mathbf{P}_i \mathbf{P}_j \boldsymbol{\mu} = 0$  when  $\mathbf{P}_i \mathbf{P}_j = \mathbf{0}$ .

■

Parts (a) to (c) use nothing beyond second moments. They hold for skewed or heavy-tailed errors, provided the errors are uncorrelated with a common variance. Part (d) uses normality, and without it both the variances and the zero correlations fail in general (Exercise (C1)).

Part (c) confirms Proposition 9.3.1. For full-rank  $\mathbf{X}$  the term  $\|(\mathbf{I} - \mathbf{M}_{(-j)})\mathbf{X}_j \boldsymbol{\beta}_j\|^2$  is positive

unless  $\beta_j = \mathbf{0}$ , and it is large only if  $\mathbf{X}_j$  has a large part orthogonal to the other terms. A nearly collinear term has a small expected partial sum of squares even when its coefficients are substantial.

Under normality the expectations become full distributions.

### Corollary 9.6.2 — Distributions of the mean squares

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$  with  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ , and let  $\mathcal{V}_1, \dots, \mathcal{V}_k$  be an analysis of variance decomposition of  $\mathcal{C}(\mathbf{X})$ . Then the sums of squares  $\mathbf{Y}^\top \mathbf{P}_1 \mathbf{Y}, \dots, \mathbf{Y}^\top \mathbf{P}_k \mathbf{Y}$  and SSE are mutually independent,

$$\frac{\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}}{\sigma^2} \sim \chi^2(r_i, \gamma_i), \quad \gamma_i = \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{\sigma^2}, \quad \frac{\text{SSE}}{\sigma^2} \sim \chi^2(n - r),$$

and  $F_i = \text{MS}_i / s^2 \sim F(r_i, n - r, \gamma_i)$ , which is central iff  $\mathbf{P}_i \boldsymbol{\mu} = \mathbf{0}$ . The same holds for a single projection  $\mathbf{P}$  onto any subspace of  $\mathcal{C}(\mathbf{X})$ , such as a partial sum of squares, paired with SSE.

*Proof.* The projections  $\mathbf{P}_1, \dots, \mathbf{P}_k, \mathbf{I} - \mathbf{M}$  are mutually orthogonal (Theorem 9.1.1(a)), so Theorem 4.4.4 gives independence and the noncentral chi-squared laws, with  $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2 = 0$  for the error space. The ratio of two independent chi-squared variables, each divided by its degrees of freedom and the numerator noncentral, is the noncentral  $F$  (Definition 4.1.2). A single projection  $\mathbf{P}$  onto a subspace of  $\mathcal{C}(\mathbf{X})$  satisfies  $\mathbf{P}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ , and the same argument applies to the pair. ■

The residual part is the familiar Theorem 7.5.1; the rest applies the same argument to the other pieces.

## 9.6.2 The logic of the $F$ ratio

By Eq. 9.6.1 every mean square estimates  $\sigma^2$  plus  $\|\mathbf{P}_i \boldsymbol{\mu}\|^2 / r_i$ , the squared size per dimension of the mean's component in  $\mathcal{V}_i$ , while under a correct model the residual mean square estimates  $\sigma^2$  alone. So the ratio

$$F_i = \frac{\text{MS}_i}{s^2}$$

compares two estimates of  $\sigma^2$  that agree in expectation exactly when  $\mathbf{P}_i \boldsymbol{\mu} = \mathbf{0}$ . This is the logic of the analysis of variance: *choose a denominator whose expectation equals that of the numerator when the hypothesis holds*. Independence gives the distribution exactly, and the noncentrality  $\gamma_i = \|\mathbf{P}_i \boldsymbol{\mu}\|^2 / \sigma^2$  governs the power. The probability that  $F_i$  exceeds any fixed critical value increases strictly with  $\gamma_i$  (Theorem 4.1.4). The mean of the ratio follows from independence and  $E(1/V) = 1/(\nu - 2)$  for  $V \sim \chi^2(\nu)$ ,  $\nu > 2$ :

$$E(F_i) = \frac{n - r}{n - r - 2} \cdot \frac{r_i + \gamma_i}{r_i}. \quad (9.6.2)$$

The principle is more general than the choice of  $s^2$ . With random effects the expected mean squares contain variance components, and the right denominator may be another mean square

(Exercise (B1); Chapters 32 and 33, Proposition 32.1.1). Also, when the model is wrong, Theorem 9.6.1(b) shows that the denominator itself is inflated, by  $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/(n - r)$  (compare Example (Residual sum of squares under a wrong mean)). Omitted terms inflate the denominator of every  $F$  ratio, which pushes the ratios towards zero, and they can also shift the numerators, because the omitted part of  $\boldsymbol{\mu}$  may have components in the  $\mathcal{V}_i$  (Exercise (B2)).

Chapter 11 derives the  $F$  test from the likelihood ratio principle (Theorem 11.4.1) and studies its power (Theorem 11.5.1).

### 9.6.3 Expected mean squares in an unbalanced layout

In the two-way layout of Section 9.3, Eq. 9.6.1 makes the hypotheses of Theorem 9.3.2 concrete. For the sequential sum of squares of  $A$  entered first,  $(\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu}$  has entry  $\bar{\mu}_i^w - \bar{\mu}^w$  on each of the  $n_i$  observations of row  $i$ , where  $\bar{\mu}^w = \sum_i n_i \bar{\mu}_i^w / n$ . So

$$\text{E MS}(A \mid \mu) = \sigma^2 + \frac{1}{a-1} \sum_i n_i (\bar{\mu}_i^w - \bar{\mu}^w)^2.$$

For the Type III sum of squares, the noncentrality is the sum of squares Eq. 9.2.1 evaluated at the true means, divided by  $\sigma^2$  (Exercise (B3)):

$$\gamma = \frac{(\boldsymbol{\Lambda}^\top \boldsymbol{\mu}_c)^\top (\boldsymbol{\Lambda}^\top \mathbf{N}^{-1} \boldsymbol{\Lambda})^{-1} (\boldsymbol{\Lambda}^\top \boldsymbol{\mu}_c)}{\sigma^2},$$

where  $\boldsymbol{\mu}_c$  is the vector of  $ab$  cell means,  $\mathbf{N} = \text{diag}(n_{ij})$ , and the columns of  $\boldsymbol{\Lambda}$  express the  $a-1$  contrasts of unweighted row means. In the first, the design enters through the weights  $n_{ij}/n_i$ , which also define the hypothesis. In the second, the hypothesis does not depend on the design, which enters only through the precision matrix  $\mathbf{N}^{-1}$  of the cell means.

#### Example. (Checking expected mean squares by simulation)

Take the design of Example (Party, education and political self-placement):  $n = 944$  respondents in the nine cells of party by education, with the observed counts. Set the true cell means equal to the observed cell means, and  $\sigma^2 = 1.21$ . The script generates 40,000 data sets with normal errors and 40,000 with centred exponential errors of the same variance, which are strongly skewed. For each it computes the sequential decomposition (party, then education, then interaction), the residual, and the Type III sum of squares for party, and averages the mean squares:

| Source      | df  | $\gamma$ | E(MS)     | normal    | skewed    | Var(MS)  | simulated |
|-------------|-----|----------|-----------|-----------|-----------|----------|-----------|
| mean        | 1   | 14594.91 | 17661.050 | 17660.423 | 17661.259 | 85476.55 | 86357.17  |
| party       | 2   | 535.09   | 324.942   | 324.915   | 324.787   | 784.89   | 786.39    |
| education   | 2   | 35.75    | 22.840    | 22.836    | 22.891    | 53.81    | 53.13     |
| given       |     |          |           |           |           |          |           |
| party       |     |          |           |           |           |          |           |
| interaction | 4   | 29.60    | 10.163    | 10.174    | 10.179    | 11.57    | 11.75     |
| residual    | 935 | 0.00     | 1.210     | 1.210     | 1.210     | 0.0031   | 0.0031    |

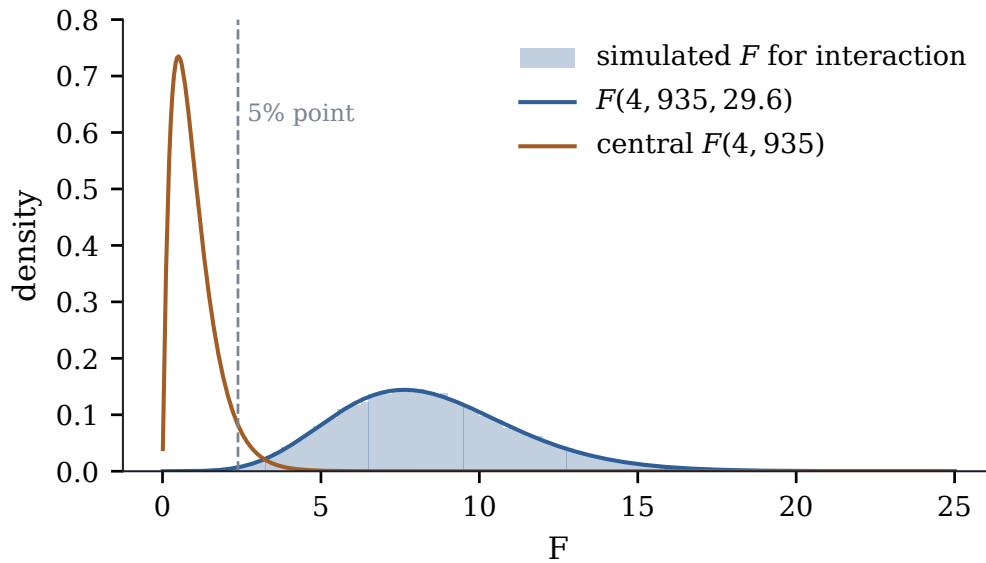


Figure 9.6.1: The  $F$  ratio for the interaction in [Example \(Checking expected mean squares by simulation\)](#) over 40,000 simulated data sets with normal errors, with the noncentral  $F(4, 935, 29.6)$  density of [Corollary 9.6.2](#) and the central null density  $F(4, 935)$  with its upper 5% point.

|          |   |        |         |         |         |        |        |
|----------|---|--------|---------|---------|---------|--------|--------|
| party,   | 2 | 537.53 | 326.414 | 326.362 | 326.259 | 788.46 | 789.47 |
| Type III |   |        |         |         |         |        |        |

The columns  $E(\text{MS})$  and  $\text{Var}(\text{MS})$  come from [Eq. 9.6.1](#) and [Theorem 9.6.1\(d\)](#), with  $\text{Var}(\text{MS}_i) = \sigma^4(2r_i + 4\gamma_i)/r_i^2$ ; the last column is the simulated variance under normal errors. The averages agree with the expectations under both error distributions, as parts (a) and (b) promise. The variances are another matter. Under skewed errors the variance of the residual mean square is 0.0124, 3.97 times the normal-theory value, because the variance of a quadratic form involves the fourth moment of the errors ([Theorem 2.3.3](#)). And the correlation between the interaction and residual mean squares, 0.008 under normal errors (zero in theory), is 0.020 under skewed errors, against a theoretical 0.025 from the third and fourth moments: the two mean squares are no longer independent.

For the interaction, the noncentrality  $\gamma = 29.60$  is large. By [Eq. 9.6.2](#), the  $F$  ratio has mean 8.417, and the simulated mean is 8.427. Its 5% critical value is 2.381, and the power is 0.9967 in theory and 0.9967 in the simulation. [Figure 9.6.1](#) shows the simulated distribution with the noncentral  $F$  density.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
party = pd.cut(d["PID"], [-1, 1.5, 4.5, 6], labels=["Dem", "Ind", "Rep"])
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"])
cell = party.cat.codes.to_numpy() * 3 + educ.cat.codes.to_numpy()
n = len(cell)

def basis(*blocks):
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
```

```

    return U[:, s > 1e-10 * s[0]]

one = np.ones((n, 1))
A = np.eye(3)[party.cat.codes.to_numpy()]
B = np.eye(3)[educ.cat.codes.to_numpy()]
Cell = np.eye(9)[cell]
U0, UA, UAB, UC = basis(one), basis(one, A), basis(one, A, B), basis(Cell)

def complement(U_big, U_small):
    """Orthonormal basis of  $C(U\_small)$ -perp within  $C(U\_big)$ ."""
    R = U_big - U_small @ (U_small.T @ U_big)
    return basis(R)

pieces = {"mean": U0, "party": complement(UA, U0),
          "educ | party": complement(UAB, UA), "interaction": complement(UC, UAB)}
Ub = basis(one, B, np.column_stack([(A[:, [i]] - A[:, [0]]) * (B[:, [j]] - B[:,
    [0]])
                                     for i in (1, 2) for j in (1, 2)]))
pieces["party (Type III)"] = complement(UC, Ub) # unweighted-means hypothesis

sigma = 1.1
mu = d.groupby(cell)["selfLR"].mean().to_numpy()[cell] # true means: observed cell
means
for name, U in pieces.items():
    r = U.shape[1]
    gamma = np.sum((U.T @ mu) ** 2) / sigma ** 2 # noncentrality  $\|P \mu\|^2 / \sigma^2$ 
    print(f"{name:17s} rank {r}  gamma = {gamma:9.2f}"
          f"  E(MS) = {sigma ** 2 * (1 + gamma / r):9.3f}")

rng = np.random.default_rng(99)
reps, chunk = 40_000, 4_000
dfe = n - 9
sums = {"normal": [], "skewed": []}
for kind in sums:
    out = []
    for _ in range(reps // chunk):
        if kind == "normal":
            E = sigma * rng.normal(size=(n, chunk))
        else:
            # centred exponential,
            # variance  $\sigma^2$ 
            E = sigma * (rng.exponential(size=(n, chunk)) - 1.0)
        Y = mu[:, None] + E
        ms = {name: np.sum((U.T @ Y) ** 2, axis=0) / U.shape[1] for name, U in
              pieces.items()}
        ms["residual"] = (np.sum(Y ** 2, axis=0) - np.sum((UC.T @ Y) ** 2, axis=0))
        / dfe
        out.append(pd.DataFrame(ms))
    sums[kind] = pd.concat(out, ignore_index=True)
print(pd.DataFrame({k: v.mean() for k, v in sums.items()}).round(3))

```

The simulation never forms an  $n \times n$  matrix: each sum of squares comes from the coordinates of  $\mathbf{Y}$  in an orthonormal basis of its subspace (Theorem 9.1.1(d)).

## 9.6.4 Exercises

### 9.6.4.1 A. Check your understanding

#### Exercise A1.

In simple regression with  $E(\mathbf{Y}) = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$ , show that the regression mean square has expectation  $\sigma^2 + \beta_1^2 S_{xx}$ , and that the  $F$  ratio of the corrected table has noncentrality  $\beta_1^2 S_{xx} / \sigma^2$  under normality.

#### Exercise A2.

In a one-way layout with  $g$  groups, group sizes  $n_i$  and means  $\mu_i$ , show that the between-groups mean square has expectation  $\sigma^2 + \sum_i n_i (\mu_i - \bar{\mu})^2 / (g - 1)$ , where  $\bar{\mu} = \sum_i n_i \mu_i / n$ . Compare with [Example \(The one-way layout\)](#).

### 9.6.4.2 B. Practice

#### Exercise B1.

In a balanced one-way layout with  $g$  groups of  $m$  observations, let  $Y_{ij} = \mu + a_i + \varepsilon_{ij}$ , where the  $a_i$  and  $\varepsilon_{ij}$  are uncorrelated random variables with mean zero and variances  $\sigma_a^2$  and  $\sigma^2$ . Show that  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$ , with  $\mathbf{Z}$  the matrix of group indicators, and use [Theorem 2.3.1](#) to show that

$$E(\text{MS}_{\text{between}}) = \sigma^2 + m\sigma_a^2, \quad E(\text{MS}_{\text{within}}) = \sigma^2.$$

Which ratio tests  $\sigma_a^2 = 0$ , and what unbiased estimator of  $\sigma_a^2$  do the two mean squares suggest?

*Solution.*  $\mathbf{Y} = \mu \mathbf{1} + \mathbf{Z}\mathbf{a} + \boldsymbol{\varepsilon}$ , so  $E(\mathbf{Y}) = \mu \mathbf{1}$  and  $\text{Cov}(\mathbf{Y}) = \mathbf{Z}(\sigma_a^2 \mathbf{I}_g) \mathbf{Z}^\top + \sigma^2 \mathbf{I}$  by [Theorem 2.2.1](#). The between-groups projection is  $\mathbf{M} - \mathbf{P}_1$ , where  $\mathbf{M} = m^{-1} \mathbf{Z}\mathbf{Z}^\top$  averages within groups, and the within-groups projection is  $\mathbf{I} - \mathbf{M}$ . The mean  $\mu \mathbf{1}$  is annihilated by both. By [Theorem 2.3.1](#),  $E(\mathbf{Y}^\top (\mathbf{M} - \mathbf{P}_1) \mathbf{Y}) = \text{tr}\{(\mathbf{M} - \mathbf{P}_1)(\sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top)\}$ . Now  $\mathbf{Z}\mathbf{Z}^\top = m\mathbf{M}$ , and  $(\mathbf{M} - \mathbf{P}_1)\mathbf{M} = \mathbf{M} - \mathbf{P}_1$ , so the trace is  $\sigma^2(g - 1) + \sigma_a^2 m(g - 1)$ . For the within-groups projection,  $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$ , so the trace is  $\sigma^2(n - g)$ . Dividing by the degrees of freedom gives the two expectations. The ratio  $\text{MS}_{\text{between}}/\text{MS}_{\text{within}}$  has expectations that agree iff  $\sigma_a^2 = 0$ , and  $(\text{MS}_{\text{between}} - \text{MS}_{\text{within}})/m$  is unbiased for  $\sigma_a^2$  (though it can be negative).

#### Exercise B2.

Suppose  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{z}\delta$  but the model  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  is fitted, with  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ . Show that

$$E(\text{SSE}) = \sigma^2(n - r) + \delta^2 \|(\mathbf{I} - \mathbf{M})\mathbf{z}\|^2, \quad E(\text{SSR}) = \sigma^2(r - 1) + \|(\mathbf{M} - \mathbf{P}_1)(\mathbf{X}\boldsymbol{\beta} + \mathbf{z}\delta)\|^2.$$

Give an example in which  $\boldsymbol{\beta}$  has all slopes zero but the regression mean square is inflated.

**Exercise B3.**

In the setting of [Theorem 9.2.4](#) with  $\mathbf{d} = \mathbf{0}$ , show that the expected sum of squares of the hypothesis  $\Lambda^\top \beta = \mathbf{0}$  is  $\sigma^2 q + (\Lambda^\top \beta)^\top (\Lambda^\top \mathbf{G} \Lambda)^{-1} \Lambda^\top \beta$ , so the noncentrality is obtained by evaluating [Eq. 9.2.1](#) at the true parameter.

*Solution.* By [Theorem 9.2.4](#)(b), the sum of squares is  $\mathbf{Y}^\top \mathbf{K} \mathbf{Y}$  with  $\mathbf{K} = \mathbf{M} \mathbf{T} (\Lambda^\top \mathbf{G} \Lambda)^{-1} \mathbf{T}^\top \mathbf{M}$ , a projection of rank  $q$ , where  $\mathbf{T}$  is the  $n \times q$  matrix with  $\Lambda = \mathbf{X}^\top \mathbf{T}$ . By [Theorem 9.6.1](#)(a) its expectation is  $\sigma^2 q + \beta^\top \mathbf{X}^\top \mathbf{K} \mathbf{X} \beta$ . Since  $\mathbf{T}^\top \mathbf{M} \mathbf{X} \beta = \mathbf{T}^\top \mathbf{X} \beta = \Lambda^\top \beta$ , the second term is  $(\Lambda^\top \beta)^\top (\Lambda^\top \mathbf{G} \Lambda)^{-1} \Lambda^\top \beta$ .

**9.6.4.3 C. Going deeper****Exercise C1.**

Let the errors be independent with mean zero, variance  $\sigma^2$  and fourth moment  $\mu_4 = E(\varepsilon_i^4)$ , and let  $\mu \in \mathcal{C}(\mathbf{X})$ . Use [Theorem 2.3.3](#) to show that

$$\text{Var}(\text{SSE}) = (\mu_4 - 3\sigma^4) \sum_i (1 - h_{ii})^2 + 2\sigma^4(n - r),$$

where  $h_{ii}$  are the leverages. For the centred exponential errors of [Example \(Checking expected mean squares by simulation\)](#),  $\mu_4 = 9\sigma^4$ . Show that  $\text{Var}(s^2) \approx 8\sigma^4/(n - r)$  when all leverages are small, and compare with the simulated variance of the residual mean square.

*Solution.* With  $\mathbf{A} = \mathbf{I} - \mathbf{M}$  and  $\theta = \mu$ , the terms of [Theorem 2.3.3](#) involving  $\theta$  vanish because  $(\mathbf{I} - \mathbf{M})\mu = \mathbf{0}$ . The diagonal of  $\mathbf{I} - \mathbf{M}$  is  $1 - h_{ii}$ , and  $\text{tr}\{(\mathbf{I} - \mathbf{M})^2\} = n - r$ , which gives the formula. If the leverages are small,  $\sum_i (1 - h_{ii})^2 \approx n - r$  (it equals  $n - 2r + \sum_i h_{ii}^2$ ), so  $\text{Var}(\text{SSE}) \approx (6 + 2)\sigma^4(n - r)$  and  $\text{Var}(s^2) \approx 8\sigma^4/(n - r)$ , four times the normal value  $2\sigma^4/(n - r)$ . This is the factor seen in the simulation.

**Exercise C2.**

For the Type I sum of squares of  $A$  entered first in a two-way layout, show that its noncentrality can be positive when  $\mu_{ij} = \beta_j$  depends on  $B$  only, and find the condition on the counts  $n_{ij}$  under which it is zero for all such  $\mu$ . Relate your answer to [Theorem 9.4.3](#).

## 9.7 Summary and notes

### 9.7.1 Summary

*Idea.*

1. An analysis of variance is a choice of mutually orthogonal subspaces filling the model space. Sums of squares are squared lengths of the projections of  $\mathbf{y}$ , degrees



- of freedom are dimensions, and both add up by Pythagoras. Decompositions and chains of nested models are the same thing (Definition 9.1.1, Theorem 9.1.1).
2. The regression table comes from the chain  $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{X})$ . The corrected identity  $\text{SST} = \text{SSR} + \text{SSE}$  needs  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ ; without it an extra term  $-2\bar{y} \sum_i \hat{\varepsilon}_i$  appears (Proposition 9.1.2, Proposition 9.1.3).
  3. The extra sum of squares  $\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_1) \mathbf{y}$  depends only on column spaces. Adding regressors to a fitted model needs only a  $t \times t$  inverse, and never lowers the variances of the old coefficients. For one column it equals  $t_j^2 s^2$  (Definition 9.2.1, Theorem 9.2.2, Corollary 9.2.3).
  4. The sum of squares of an estimable hypothesis  $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{d}$  is  $(\mathbf{A}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{A}^\top \mathbf{G} \mathbf{A})^{-1} (\mathbf{A}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})$ , whatever the rank. A nonestimable constraint restricts only through its estimable part (Theorem 9.2.4, Theorem 9.2.5, Proposition 9.2.6).
  5. Sequential (Type I) sums of squares decompose the regression sum of squares but depend on the order. Partial, Type II and Type III sums of squares do not decompose anything, and in models with interactions they address different hypotheses: weighted row means, additive main effects, unweighted row means, or, with reference coding, means at a single level of the other factor (Definition 9.3.1, Definition 9.3.2, Proposition 9.3.1, Theorem 9.3.2).
  6. Order stops mattering exactly when the centred subspaces of the terms are orthogonal. For two factors this is the proportional-frequency condition. Orthogonal polynomials and orthogonal contrasts manufacture orthogonality inside a single term (Theorem 9.4.1, Theorem 9.4.3, Theorem 9.4.4).
  7.  $R$  is the largest correlation of  $\mathbf{y}$  with a linear combination of the regressors. Under the null  $R^2 \sim \text{Beta}(\frac{r-1}{2}, \frac{n-r}{2})$ , with mean  $(r-1)/(n-1)$ . The adjusted  $R^2$  has mean zero under the null and rises exactly when an added term has  $F > 1$ . Partial  $R^2$  equals  $t^2/(t^2 + n - r)$  (Theorem 9.5.1, Theorem 9.5.2, Proposition 9.5.3, Theorem 9.5.4).
  8.  $R^2$  depends on the design, and it is not a test of the model, a comparison across responses, or a measure of predictive accuracy or of cause (Section 9.5).
  9. Every mean square has expectation  $\sigma^2 + \|\mathbf{P}\boldsymbol{\mu}\|^2 / \text{rank}(\mathbf{P})$ , using only second moments. Under normality the pieces of a decomposition are independent noncentral chi-squared variables, and an  $F$  ratio compares two estimates of  $\sigma^2$  that agree exactly when its hypothesis  $\mathbf{P}\boldsymbol{\mu} = \mathbf{0}$  holds (Theorem 9.6.1, Corollary 9.6.2).

## 9.7.2 Notes and sources

**Analysis of variance.** The decomposition of a total sum of squares into components with their own degrees of freedom is Fisher's. An analysis of variance already appears in Fisher and Mackenzie (1923), and Fisher set out the analysis of variance table for general use in *Statistical Methods for Research Workers* (Fisher 1925), after making the notion of degrees of freedom precise in Fisher (1922). The distribution theory that makes the components independent chi-squared variables is Cochran's theorem (Cochran 1934), proved in Chapter 4. The coordinate-free view, in which a table is a list of orthogonal subspaces, follows Christensen (2020). His section 3.6, which splits a test space into orthogonal lines, is the source of Theorem 9.4.4, and his section 7.4 treats proportional numbers. Scheffé (1959) remains a clear account of the classical theory, including expected mean squares for random effects.

**Extra sums of squares, adding regressors and restrictions.** Seber and Lee (2003, sections

3.7–3.9) give the updating formulas of [Theorem 9.2.2](#), restricted least squares, and their rank-deficient versions. Our proofs go through the Frisch–Waugh–Lovell theorem and Pythagoras instead of Lagrange multipliers. [Proposition 9.2.6](#) makes precise the remark, common to the sources, that only the estimable part of a hypothesis can be tested.

**Types of sums of squares.** Agresti (2015, sections 2.3–2.4) presents sequential and partial sums of squares through projection matrices, with the decomposition of the regression sum of squares when the regressors are uncorrelated, and notes the Type I, II and III terminology. The problem of main effects in unbalanced layouts with interaction goes back to Yates (1934), whose “weighted squares of means” is closely related to what software now calls Type III. The names of the types come from statistical software, and programs have used them inconsistently, which is why this chapter defines each type by its projection. Nelder (1977) argued that tests of main effects in the presence of interactions violate the marginality of the model terms; his position underlies the preference for Type II sums of squares when interactions are absent. Speed, Hocking and Hackney (1978) identify the cell-mean hypotheses tested by the various sums of squares, and Searle (1987) treats unbalanced data exhaustively in the same cell-mean terms.

**The coefficient of determination.** The distribution of the sample multiple correlation coefficient, under the null and in general for random normal regressors, is due to Fisher (1928); [Theorem 9.5.2](#) is the simpler fixed-regressor version. Kvralseth (1985) surveys the many different formulas that software has used for  $R^2$ , especially for models without an intercept, and Anscombe (1973) gives four data sets with the same  $R^2$  and very different relationships, a standing warning that  $R^2$  is not a test of the model.

**Data.** The state crime, election study, Grunfeld and Nile data are public-domain data sets distributed with statsmodels. The Nile series is the one analysed for a change point by Cobb (1978).

**What is new here.** The organization of the chapter around subspaces, the proofs of [Theorem 9.4.1](#) and [Theorem 9.4.3](#), the reference-coding case of [Theorem 9.3.2](#), and the numerical examples and simulations are this book’s.

### 9.7.3 References

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# Chapter 10

## Computation: QR, SVD and Conditioning

Every least squares estimate in this book is written as  $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$  or as  $\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ . These are the right formulas to reason with. They are not how a computer should produce the numbers. This chapter explains what a computer should do instead, and how accurate the result is. It separates two questions: how sensitive the least squares problem is to small changes in the data, which depends on  $\kappa(\mathbf{X})$  and the size of the residual, and whether an algorithm adds more error than that. The answers give a practical ranking. The Cholesky factorization of  $\mathbf{X}^\top \mathbf{X}$  is fast and is enough when  $\mathbf{X}$  is well conditioned. Householder or Givens QR is the default. The singular value decomposition is the tool for rank deficiency and near dependence. Pivoted QR decides the rank cheaply, which is how statistical software chooses what to do with aliased columns.

**What you need.** [Chapter 1](#): the Cholesky factorization ([Theorem 1.7.4](#)), QR ([Theorem 1.5.2](#)), the singular value decomposition ([Theorem 1.8.1](#)), the Eckart–Young inequality ([Proposition 1.8.2](#)) and the condition number ([Definition 1.8.1](#)). [Chapter 5](#): the linear model and the moments of  $\hat{\beta}$  ([Theorem 5.5.1](#)). [Chapter 6](#): least squares as projection, and the first look at computation in [Section 6.10](#). A few remarks use the sampling theory of [Chapter 7](#) and the sums of squares of [Chapter 9](#).

### 10.0.1 Roadmap

- [Why not invert the cross-product matrix](#)
- [The Cholesky factorization](#)
- [Householder and Givens QR](#)
- [The singular value decomposition](#)
- [Conditioning and perturbation bounds](#)
- [Updating and downdating](#)
- [Rank-revealing factorizations](#)
- [Summary and notes](#)

## 10.1 Why not invert the cross-product matrix

Chapter 5 found the least squares estimate by setting a gradient to zero, which gave  $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$  when  $\mathbf{X}$  has full column rank (Theorem 5.4.1). Read as a recipe, the formula says: form the  $p \times p$  matrix  $\mathbf{X}^\top \mathbf{X}$ , invert it, and multiply. Section 6.10 already showed that the recipe can lose twice as many digits as necessary. This section explains why.

### 10.1.1 Floating-point numbers

A double-precision number is zero or a number  $\pm m \times 2^{e-52}$ , where  $m$  is an integer with  $2^{52} \leq m < 2^{53}$  and  $e$  is an integer in a fixed range ( $-1022 \leq e \leq 1023$ ). Between  $2^e$  and  $2^{e+1}$  the doubles are therefore equally spaced,  $2^{e-52}$  apart. Just above 1 the spacing is  $\epsilon_M = 2^{-52} \approx 2.22 \times 10^{-16}$ , the **machine epsilon**. Rounding a real number  $x$  in the range of the doubles to the nearest double gives a number  $\text{fl}(x)$  with

$$\text{fl}(x) = x(1 + \delta), \quad |\delta| \leq u = 2^{-53} \approx 1.11 \times 10^{-16}.$$

The **unit roundoff**  $u = \epsilon_M/2$  is half the gap, because the nearest double is at most half a gap away. Rounding errors are *relative*: a double represents large and small numbers with the same number of significant digits, about sixteen.

The IEEE 754 standard, which essentially all current hardware follows, requires the basic operations to be correctly rounded. The computed result of  $x + y$  is the exact sum, rounded. That is the only property of the hardware that the analysis below uses.

#### Definition 10.1.1 — Standard model of floating-point arithmetic

For floating-point numbers  $x, y$  and  $\circ \in \{+, -, \times, /\}$ , the computed result satisfies

$$\text{fl}(x \circ y) = (x \circ y)(1 + \delta), \quad |\delta| \leq u,$$

and  $\text{fl}(\sqrt{x}) = \sqrt{x}(1 + \delta)$  with  $|\delta| \leq u$ , provided no result overflows or underflows.

Two consequences are worth seeing at once. The number  $1 + u$  lies exactly halfway between the doubles 1 and  $1 + \epsilon_M$ , and the tie is broken towards 1, so adding  $u$  to 1 changes nothing. And most decimal fractions are not doubles, so familiar identities fail at the level of  $u$ :

```
import numpy as np

eps = np.finfo(float).eps          # spacing of the doubles just above 1: 2^-52
u = eps / 2                        # unit roundoff: 2^-53
print("eps =", eps, " u =", u)
print("1 + u == 1:", 1.0 + u == 1.0, " 1 + eps == 1:", 1.0 + eps == 1.0)
print("0.1 + 0.2 == 0.3:", 0.1 + 0.2 == 0.3)
```

Trouble comes from accumulation and above all from subtraction. The difference of nearly equal numbers is computed exactly but exposes the errors they already carried: if  $a$  and  $b$  agree in

ten digits, a rounding error in either becomes a large relative error in  $a - b$ . This **cancellation** is behind every numerical difficulty in this chapter.

## 10.1.2 Accumulating rounding errors

Products of many factors  $1 + \delta_i$  appear whenever errors accumulate. The following lemma packages them.

### Lemma 10.1.1

Let  $|\delta_i| \leq u$  and  $\rho_i \in \{1, -1\}$  for  $i = 1, \dots, k$ , and suppose  $ku < 1$ . Then

$$\prod_{i=1}^k (1 + \delta_i)^{\rho_i} = 1 + \theta_k, \quad |\theta_k| \leq \gamma_k = \frac{ku}{1 - ku}. \quad (10.1.1)$$

*Proof.* Each factor lies between  $1 - u$  and  $1/(1 - u)$ : for  $\rho_i = 1$  because  $1 + u \leq 1/(1 - u)$ , and for  $\rho_i = -1$  because  $1/(1 + \delta_i) \geq 1/(1 + u) \geq 1 - u$ . So the product lies between  $(1 - u)^k$  and  $(1 - u)^{-k}$ . By Bernoulli's inequality  $(1 - u)^k \geq 1 - ku$ , which gives the lower end  $1 - ku \geq 1 - \gamma_k$ , and also the upper end  $(1 - u)^{-k} \leq 1/(1 - ku) = 1 + \gamma_k$ . ■

The constant  $\gamma_k$  is essentially  $ku$ : for  $k = 10^6$  it is about  $1.1 \times 10^{-10}$ . This is a worst case; in practice errors of different signs partly cancel and grow more like  $\sqrt{k}u$ .

The basic step of every least squares algorithm is an inner product.

### Proposition 10.1.2 — Rounding error in an inner product

Let  $\mathbf{x}, \mathbf{y} \in \mathbb{R}^n$  have floating-point entries and  $nu < 1$ . The inner product computed by recursive summation,  $s_1 = \text{fl}(x_1 y_1)$  and  $s_k = \text{fl}(s_{k-1} + \text{fl}(x_k y_k))$ , satisfies

$$s_n = \sum_{i=1}^n x_i y_i (1 + \theta_i), \quad |\theta_i| \leq \gamma_n.$$

Consequently  $|s_n - \mathbf{x}^\top \mathbf{y}| \leq \gamma_n \sum_i |x_i y_i|$ , and  $s_n = (\mathbf{x} + \Delta \mathbf{x})^\top \mathbf{y}$  for a vector  $\Delta \mathbf{x}$  with  $|\Delta x_i| \leq \gamma_n |x_i|$ .

*Proof.* By the standard model,  $\text{fl}(x_k y_k) = x_k y_k (1 + \mu_k)$  and  $s_k = (s_{k-1} + x_k y_k (1 + \mu_k))(1 + \sigma_k)$  for  $k \geq 2$ , with all  $|\mu_k|, |\sigma_k| \leq u$ . Unrolling the recursion,

$$s_n = \sum_{i=1}^n x_i y_i (1 + \mu_i) \prod_{k=\max(i,2)}^n (1 + \sigma_k).$$

The  $i$ th term carries at most  $n$  factors of the form  $1 + \delta$ , so by Lemma 10.1.1 it equals  $x_i y_i (1 + \theta_i)$  with  $|\theta_i| \leq \gamma_n$  (since  $\gamma_k \leq \gamma_n$  for  $k \leq n$ ). The two consequences follow by taking  $\Delta x_i = \theta_i x_i$ .

The first conclusion is a *forward* bound, which can be large relative to a result whose terms cancel. The second is a *backward* statement: the computed number is the exact inner product of slightly perturbed data. Over 200 random inner products of length up to 400, checked in rational arithmetic, the largest ratio of error to bound is 0.149.

```
from fractions import Fraction

def gamma(k, u=2.0 ** -53):
    return k * u / (1 - k * u)

def dot_recursive(x, y):
    """Inner product by recursive summation, one rounding per operation."""
    s = 0.0
    for a, b in zip(x, y):
        s = s + a * b
    return s

rng = np.random.default_rng(101)
worst = 0.0
for trial in range(200):
    n = int(rng.integers(2, 400))
    x = rng.normal(size=n) * 10.0 ** rng.integers(-3, 4, size=n)
    y = rng.normal(size=n)
    exact = sum(Fraction(a) * Fraction(b) for a, b in zip(x, y))      # exact
    rational value
    error = abs(Fraction(dot_recursive(x, y)) - exact)
    bound = Fraction(gamma(n)) * sum(abs(Fraction(a) * Fraction(b)) for a, b in zip(
        x, y))
    worst = max(worst, float(error / bound))
print(f"largest error / bound over 200 inner products: {worst:.3f}")
```

### 10.1.3 Conditioning and stability

A computation turns data  $d$  into a result  $f(d)$ . The computed result can be inaccurate because the *problem* is sensitive, or because the *algorithm* adds more error than the data justify. The first is measured by a **condition number**, the largest factor by which a small relative change in  $d$  can be magnified in  $f(d)$ ; the second by the following standard.

#### Definition 10.1.2 — Backward stability

An algorithm that computes  $f(d)$  is **backward stable** if, for every input  $d$ , its computed output  $\hat{f}(d)$  is the exact result for nearby data:  $\hat{f}(d) = f(d + \Delta d)$  with  $\|\Delta d\| \leq cu\|d\|$ , where  $c$  is a modest constant depending only on the dimensions of the problem.

Proposition 10.1.2 says that recursive summation is a backward stable way to compute inner products, with  $c = \gamma_n/u \approx n$ . Backward stability is the most one can ask, because storing the data as doubles already perturbs them by  $u$ . The two notions combine, to first order in  $u$ , as

$$\text{relative forward error} \lesssim \text{condition number} \times \text{backward error}. \quad (10.1.2)$$

For least squares the condition number, which involves  $\kappa(\mathbf{X})$  and sometimes  $\kappa(\mathbf{X})^2$ , is the subject of [Section 10.5](#). Sections 10.2 to 10.4 ask whether algorithms are backward stable *for the least squares problem*, with perturbations of  $\mathbf{X}$  and  $\mathbf{y}$ .

### 10.1.4 What forming the cross-product matrix loses

The normal-equations recipe is not backward stable for least squares, and the damage is done in its very first step. Here is what rounding does to  $\mathbf{X}^\top \mathbf{X}$ .

#### Proposition 10.1.3 — Rounding in the cross-product matrix

Let  $\mathbf{X}$  be  $n \times p$  with full column rank and let  $\hat{\mathbf{A}}$  be  $\mathbf{X}^\top \mathbf{X}$  computed entry by entry as in [Proposition 10.1.2](#). Then  $\hat{\mathbf{A}} = \mathbf{X}^\top \mathbf{X} + \Delta \mathbf{A}$  with  $|\Delta \mathbf{A}| \leq \gamma_n |\mathbf{X}|^\top |\mathbf{X}|$  entrywise, and

$$\|\Delta \mathbf{A}\|_2 \leq p \gamma_n \|\mathbf{X}\|_2^2.$$

In particular  $\hat{\mathbf{A}}$  is positive definite whenever  $p \gamma_n \kappa(\mathbf{X})^2 < 1$ .

*Proof.* The entrywise bound is [Proposition 10.1.2](#) applied to each pair of columns. The spectral norm is at most the Frobenius norm, the Frobenius norm is monotone in the absolute values of the entries and submultiplicative, and  $\|\mathbf{X}\|_F^2 = \sum_i \sigma_i^2 \leq p \sigma_1^2$  by [Proposition 1.8.2\(a\)](#). Hence

$$\|\Delta \mathbf{A}\|_2 \leq \gamma_n \|\mathbf{X}\|^\top \|\mathbf{X}\|_F \leq \gamma_n \|\mathbf{X}\|_F^2 \leq p \gamma_n \sigma_1^2.$$

For a unit vector  $\mathbf{v}$ ,  $\mathbf{v}^\top \hat{\mathbf{A}} \mathbf{v} = \|\mathbf{X} \mathbf{v}\|^2 + \mathbf{v}^\top \Delta \mathbf{A} \mathbf{v} \geq \sigma_p^2 - p \gamma_n \sigma_1^2$ , which is positive when  $p \gamma_n \sigma_1^2 / \sigma_p^2 < 1$ . ■

The information about  $\mathbf{X}$  that sits in its smallest singular value  $\sigma_p$  appears in  $\mathbf{X}^\top \mathbf{X}$  as an eigenvalue  $\sigma_p^2$ . Rounding perturbs  $\mathbf{X}^\top \mathbf{X}$  by an amount of order  $u \sigma_1^2$ . Once  $\sigma_p^2$  is below that level, the computed cross-product matrix may not determine the direction  $\mathbf{v}_p$  at all. Even rounding the *exact*  $\mathbf{X}^\top \mathbf{X}$  to doubles does this much damage. The loss becomes total around  $\kappa(\mathbf{X}) \approx u^{-1/2} = 9.5 \times 10^7$ , and it is already serious well before that: the normal equations are then a problem with condition number  $\kappa(\mathbf{X}^\top \mathbf{X}) = \kappa(\mathbf{X})^2$  ([Eq. 1.8.2](#)), and [Eq. 10.1.2](#) predicts relative errors of order  $\kappa(\mathbf{X})^2 u$  in  $\hat{\boldsymbol{\beta}}$  whatever method is then used to solve them.

#### Example. (A cross-product matrix that rounds to a singular one)

Take  $\delta = 10^{-9}$  and

$$\mathbf{X} = \begin{pmatrix} 1 & 1 \\ \delta & 0 \\ 0 & \delta \end{pmatrix}, \quad \mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 1 + \delta^2 & 1 \\ 1 & 1 + \delta^2 \end{pmatrix}.$$

The columns of  $\mathbf{X}$  are independent, and  $\kappa(\mathbf{X}) = \sqrt{2 + \delta^2} / \delta = 1.41 \times 10^9$ . But  $\delta^2 = 10^{-18}$  is smaller than  $u$ , so  $1 + \delta^2$  rounds to 1 and the computed  $\mathbf{X}^\top \mathbf{X}$  is the singular matrix of ones. The Cholesky factorization of [Section 10.2](#) fails, and no method applied to the computed normal equations can recover the solution. For  $\mathbf{y} = \mathbf{X}(1, 1)^\top$ , the exact least



squares estimate is  $(1, 1)^\top$  with zero residual. QR, which never forms  $\mathbf{X}^\top \mathbf{X}$ , returns it with relative error  $3.9 \times 10^{-16}$ .

```
delta = 1e-9
X = np.array([[1.0, 1.0],
              [delta, 0.0],
              [0.0, delta]])
y = X @ np.array([1.0, 1.0])          # exact data: the solution is b = (1, 1),
                                     residual 0

A = X.T @ X                          # 1 + delta^2 rounds to 1
print("computed X'X =\n", A)
print("cond(X) =", np.linalg.cond(X))
try:
    np.linalg.cholesky(A)
except np.linalg.LinAlgError as err:
    print("Cholesky of the computed X'X fails:", err)

Q, R = np.linalg.qr(X)
b_qr = np.linalg.solve(R, Q.T @ y)
print("QR solution:", b_qr)
```

## 10.1.5 Why not the inverse either

Even when  $\mathbf{X}^\top \mathbf{X}$  is safely positive definite, the inverse costs  $p^3$  flops against  $p^3/3$  for a Cholesky factorization plus  $2p^2$  for two triangular solves, and it is less accurate. Statistics needs only the diagonal of  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$  (Theorem 5.5.1): if  $\mathbf{X}^\top \mathbf{X} = \mathbf{R}^\top \mathbf{R}$ , its  $j$ th entry is  $\sigma^2$  times the squared length of row  $j$  of  $\mathbf{R}^{-1}$ .

## 10.1.6 Counting the work

Cost is traditionally counted in floating-point operations (**flops**), each addition, subtraction, multiplication and division counting one. Flop counts ignore memory traffic and parallelism, so they are only a first guide.

| Task                                                                                | Leading flop count | Where                        |
|-------------------------------------------------------------------------------------|--------------------|------------------------------|
| Form $\mathbf{X}^\top \mathbf{X}$ and $\mathbf{X}^\top \mathbf{y}$ (using symmetry) | $np^2$             | this section                 |
| Cholesky factorization of $\mathbf{X}^\top \mathbf{X}$                              | $p^3/3$            | <a href="#">Section 10.2</a> |
| Two triangular solves                                                               | $2p^2$             | <a href="#">Section 10.2</a> |
| Householder QR of $\mathbf{X}$ , with $\mathbf{Q}^\top \mathbf{y}$                  | $2np^2 - 2p^3/3$   | <a href="#">Section 10.3</a> |
| Givens QR of $\mathbf{X}$                                                           | $3np^2 - p^3$      | <a href="#">Section 10.3</a> |
| Modified Gram–Schmidt                                                               | $2np^2$            | <a href="#">Section 10.3</a> |
| Singular values and $\mathbf{V}$ , QR first (R-SVD)                                 | $2np^2 + 11p^3$    | <a href="#">Section 10.4</a> |

When  $n \gg p$ , as usual in regression, the normal equations cost about  $np^2$  and Householder

QR about  $2np^2$ : the accurate method is at most twice as expensive. And the one-pass,  $O(p^2)$ -memory accumulation of cross-products has a QR counterpart in row-wise Givens QR (Section 10.3).

*Idea.* Forming  $\mathbf{X}^\top \mathbf{X}$  squares the condition number before any equation is solved, and inverting it adds cost and error. Factor  $\mathbf{X}$  itself, by an orthogonal transformation, and read everything off the factors.

## 10.1.7 Exercises

### 10.1.7.1 A. Check your understanding

#### Exercise A1.

Show that the doubles in  $[1, 2)$  are exactly the numbers  $1 + k 2^{-52}$ ,  $k = 0, 1, \dots, 2^{52} - 1$ . What is the smallest double greater than 1? What is the largest integer  $N$  such that every integer from 0 to  $N$  is a double?

#### Exercise A2.

For  $x = 10^{-10}$ , the computer evaluates  $(1 + x) - 1$ . Use the standard model to show that the result is  $x + \delta(1 + x)$  for some  $|\delta| \leq u$ , where the subtraction itself is exact. Estimate the relative error, and check it numerically. Which of the two operations is responsible?

*Solution.* The addition gives  $\text{fl}(1 + x) = (1 + x)(1 + \delta)$ . This number is within a factor of two of 1, and the difference of two doubles within a factor of two of each other is itself a double, so the subtraction is exact:  $\text{fl}(1 + x) - 1 = x + \delta(1 + x)$ . The relative error is  $|\delta|(1 + x)/x \approx u/x \approx 10^{-6}$ . All of it comes from rounding  $1 + x$ , which kept only about six significant digits of  $x$ . The subtraction then exposed that loss.

### 10.1.7.2 B. Practice

#### Exercise B1.

Show that recursive summation of  $x_1, \dots, x_n$  returns  $\sum_i x_i(1 + \theta_i)$  with  $|\theta_i| \leq \gamma_{n-1}$ , and that pairwise summation returns such a sum with  $|\theta_i| \leq \gamma_{\lceil \log_2 n \rceil}$ .

*Solution.* With  $s_1 = x_1$  and  $s_k = (s_{k-1} + x_k)(1 + \sigma_k)$ ,  $s_n = \sum_i x_i \prod_{k=\max(i,2)}^n (1 + \sigma_k)$ . Term  $i$  has at most  $n - 1$  factors, so Lemma 10.1.1 gives  $|\theta_i| \leq \gamma_{n-1}$ . In pairwise summation each  $x_i$  passes through one addition per level of the tree, and a balanced tree over  $n$  leaves has  $\lceil \log_2 n \rceil$  levels. The same unrolling gives at most  $\lceil \log_2 n \rceil$  factors per term.

#### Exercise B2.

Let  $\mathbf{A}$  be  $m \times n$  and  $\mathbf{b} \in \mathbb{R}^n$ . Show that the computed product  $\text{fl}(\mathbf{A}\mathbf{b})$ , each entry an inner product, equals  $(\mathbf{A} + \Delta\mathbf{A})\mathbf{b}$  with  $|\Delta\mathbf{A}| \leq \gamma_n |\mathbf{A}|$ . Deduce  $\|\text{fl}(\mathbf{A}\mathbf{b}) - \mathbf{A}\mathbf{b}\| \leq$

$$\gamma_n \sqrt{n} \|\mathbf{A}\|_2 \|\mathbf{b}\|.$$

### 10.1.7.3 C. Going deeper

#### Exercise C1.

For  $0 < \delta \leq 2^{-27}$  let  $\mathbf{X}_\delta$  be the matrix of [Example \(A cross-product matrix that rounds to a singular one\)](#), and let  $\mathbf{y} = \mathbf{X}_\delta(3, 1)^\top$  and  $\mathbf{y}' = \mathbf{X}_\delta(1, 3)^\top$ . Show that the computed cross-product matrices of  $[\mathbf{X}_\delta, \mathbf{y}]$  and  $[\mathbf{X}_\delta, \mathbf{y}']$  are identical (and do not depend on  $\delta$ ), although the exact least squares estimates are  $(3, 1)^\top$  and  $(1, 3)^\top$ . Conclude that no algorithm that starts from the computed cross-product matrix can solve these problems. Is anything special about  $2^{-27}$ ?

## 10.2 The Cholesky factorization

If the cross-product matrix is to be used at all, the normal equations should be solved by factoring it as a triangular matrix times its transpose. One factorization of the augmented cross-product matrix then delivers every standard regression quantity.

### 10.2.1 The algorithm

[Chapter 1](#) proved that a positive definite matrix has a unique factorization  $\mathbf{A} = \mathbf{L}\mathbf{L}^\top$  with  $\mathbf{L}$  lower triangular and positive on the diagonal ([Theorem 1.7.4\(d\)](#)). Numerical work usually writes it with the upper triangular factor  $\mathbf{R} = \mathbf{L}^\top$ .

#### Theorem 10.2.1 — Cholesky factorization

Let  $\mathbf{A}$  be a  $p \times p$  positive definite matrix.

- (a) There is exactly one upper triangular  $\mathbf{R}$  with positive diagonal entries such that  $\mathbf{A} = \mathbf{R}^\top \mathbf{R}$ .
- (b) Its rows are given, for  $j = 1, \dots, p$  in turn, by

$$\begin{aligned} r_{jj} &= \left( a_{jj} - \sum_{k < j} r_{kj}^2 \right)^{1/2}, \\ r_{ji} &= \frac{1}{r_{jj}} \left( a_{ji} - \sum_{k < j} r_{kj} r_{ki} \right), \quad i > j, \end{aligned} \tag{10.2.1}$$

and every radicand in it is positive.

- (c) The computation takes  $p^3/3 + O(p^2)$  flops.
- (d) If  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  and  $\mathbf{X} = \mathbf{Q}\mathbf{R}_0$  is a QR factorization with  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{I}_p$  and  $\mathbf{R}_0$  upper triangular with positive diagonal, then  $\mathbf{R}_0 = \mathbf{R}$ .

*Proof.*

- (a) is [Theorem 1.7.4\(d\)](#) with  $\mathbf{R} = \mathbf{L}^\top$ .
- (b) Entry  $(j, i)$ ,  $i \geq j$ , of  $\mathbf{R}^\top \mathbf{R}$  is  $\sum_{k \leq j} r_{kj} r_{ki}$ , because  $r_{kj} = 0$  for  $k > j$ . Setting it equal to  $a_{ji}$  and isolating the  $k = j$  term gives the recursion [Eq. 10.2.1](#). So the entries of the unique factor of (a) satisfy [Eq. 10.2.1](#), and [Eq. 10.2.1](#) determines row  $j$  from rows  $1, \dots, j-1$ . By induction on  $j$ , the rows computed by [Eq. 10.2.1](#) are those of the factor in (a). In particular the  $j$ th radicand is  $r_{jj}^2 > 0$ .
- (c) Row  $j$  needs about  $2(j-1)$  flops for each of its  $p-j+1$  entries, so the total is  $\sum_j 2(j-1)(p-j+1) = p^3/3 + O(p^2)$ .
- (d)  $\mathbf{X}^\top \mathbf{X} = \mathbf{R}_0^\top \mathbf{Q}^\top \mathbf{Q} \mathbf{R}_0 = \mathbf{R}_0^\top \mathbf{R}_0$ , and uniqueness in (a).

■

By part (d), the triangle of a QR factorization of  $\mathbf{X}$  and the Cholesky factor of  $\mathbf{X}^\top \mathbf{X}$  are the same matrix (see also [Exercise \(B2\)](#)). The two routes of this chapter compute the same object, one from  $\mathbf{X}$ , the other from  $\mathbf{X}^\top \mathbf{X}$  after the small singular values may already have been damaged ([Proposition 10.1.3](#)).

The algorithm itself is backward stable *for the factorization*. The computed  $\hat{\mathbf{R}}$  satisfies  $\hat{\mathbf{R}}^\top \hat{\mathbf{R}} = \mathbf{A} + \Delta \mathbf{A}$  with  $|\Delta \mathbf{A}| \leq \gamma_{p+1} |\hat{\mathbf{R}}|^\top |\hat{\mathbf{R}}|$  entrywise, and it runs to completion whenever  $\mathbf{A}$  is not too close to singular (Higham 2002, chapter 10), with no pivoting. The weakness of the normal equations lies entirely in forming  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$ , not in factoring it.

```
import numpy as np
import statsmodels.api as sm

def cholesky_upper(A):
    """Upper triangular R with positive diagonal and R'R = A (A positive definite)."""
    A = np.array(A, dtype=float)
    p = A.shape[0]
    R = np.zeros_like(A)
    for j in range(p):
        d = A[j, j] - R[:j, j] @ R[:j, j]
        if d <= 0:
            raise np.linalg.LinAlgError(f"not positive definite at step {j + 1}")
        R[j, j] = np.sqrt(d)
        R[j, j + 1:] = (A[j, j + 1:] - R[:j, j] @ R[:j, j + 1:]) / R[j, j]
    return R

def solve_upper(R, b):
    """Back substitution for R x = b, R upper triangular."""
    x = np.zeros(np.shape(b))
    for i in range(len(b) - 1, -1, -1):
        x[i] = (b[i] - R[i, i + 1:] @ x[i + 1:]) / R[i, i]
    return x

def solve_lower(L, b):
    """Forward substitution for L x = b, L lower triangular."""
    x = np.zeros(np.shape(b))
    for i in range(len(b)):
        x[i] = (b[i] - L[i, :i] @ x[:i]) / L[i, i]
    return x
```

## 10.2.2 Everything from one factorization

Solving the normal equations  $\mathbf{R}^\top \mathbf{R} \mathbf{b} = \mathbf{X}^\top \mathbf{y}$  takes two triangular solves:  $\mathbf{R}^\top \mathbf{z} = \mathbf{X}^\top \mathbf{y}$  for  $\mathbf{z}$ , then  $\mathbf{R} \mathbf{b} = \mathbf{z}$ . Each costs  $p^2$  flops. It is neater to factor the cross-product matrix of the augmented matrix  $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$ . The factor then contains the solution of the first triangular system and the residual sum of squares.

### Proposition 10.2.2 — The augmented factorization

Let  $\mathbf{X}$  be  $n \times p$  of full column rank, suppose  $\mathbf{y} \notin \mathcal{C}(\mathbf{X})$ , and let

$$\mathbf{Z}^\top \mathbf{Z} = \begin{pmatrix} \mathbf{X}^\top \mathbf{X} & \mathbf{X}^\top \mathbf{y} \\ \mathbf{y}^\top \mathbf{X} & \mathbf{y}^\top \mathbf{y} \end{pmatrix} = \mathbf{T}^\top \mathbf{T}, \quad \mathbf{T} = \begin{pmatrix} \mathbf{R} & \mathbf{z} \\ \mathbf{0}^\top & d \end{pmatrix},$$

be the Cholesky factorization (Theorem 10.2.1) of the cross-product matrix of  $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$ . Then:

- (a)  $\mathbf{R}$  is the Cholesky factor of  $\mathbf{X}^\top \mathbf{X}$ , the least squares estimate solves  $\mathbf{R} \hat{\boldsymbol{\beta}} = \mathbf{z}$ , and  $d^2 = \text{SSE} = \|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}\|^2$ ;
- (b) for each  $k \leq p$ , the residual sum of squares of the regression of  $\mathbf{y}$  on the first  $k$  columns of  $\mathbf{X}$  is  $d^2 + \sum_{j>k} z_j^2$ . The drop in residual sum of squares when column  $k$  joins columns  $1, \dots, k-1$  is  $z_k^2$ ;
- (c)  $\det(\mathbf{X}^\top \mathbf{X}) = \prod_j r_{jj}^2$ ,  $(\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{R}^{-\top}$ , and the leverage of row  $\mathbf{x}_{(i)}^\top$  is  $h_{ii} = \|\mathbf{R}^{-\top} \mathbf{x}_{(i)}\|^2$ .

*Proof.*  $\mathbf{Z}$  has full column rank because  $\mathbf{y} \notin \mathcal{C}(\mathbf{X})$ , so  $\mathbf{Z}^\top \mathbf{Z}$  is positive definite and  $\mathbf{T}$  exists. Multiplying out  $\mathbf{T}^\top \mathbf{T}$  and comparing blocks with  $\mathbf{Z}^\top \mathbf{Z}$  gives

$$\mathbf{R}^\top \mathbf{R} = \mathbf{X}^\top \mathbf{X}, \quad \mathbf{R}^\top \mathbf{z} = \mathbf{X}^\top \mathbf{y}, \quad \mathbf{z}^\top \mathbf{z} + d^2 = \mathbf{y}^\top \mathbf{y}.$$

The first identity and uniqueness make  $\mathbf{R}$  the Cholesky factor of  $\mathbf{X}^\top \mathbf{X}$ . The second gives  $\mathbf{R}^\top \mathbf{R} \hat{\boldsymbol{\beta}} = \mathbf{R}^\top \mathbf{z}$  for the solution of the normal equations, so  $\mathbf{R} \hat{\boldsymbol{\beta}} = \mathbf{z}$ .

For (b), and for the rest of (a), let  $\mathbf{Q} = \mathbf{Z} \mathbf{T}^{-1}$ . Then  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{T}^{-\top} \mathbf{Z}^\top \mathbf{Z} \mathbf{T}^{-1} = \mathbf{I}$ , so the columns  $\mathbf{q}_1, \dots, \mathbf{q}_{p+1}$  of  $\mathbf{Q}$  are orthonormal, and  $\mathbf{Z} = \mathbf{Q} \mathbf{T}$ . Because  $\mathbf{T}$  is upper triangular with nonzero diagonal, the first  $k$  columns of  $\mathbf{X}$  span the same subspace as  $\mathbf{q}_1, \dots, \mathbf{q}_k$ . The last column of  $\mathbf{Z} = \mathbf{Q} \mathbf{T}$  reads  $\mathbf{y} = \sum_{j \leq p} z_j \mathbf{q}_j + d \mathbf{q}_{p+1}$ . By Proposition 6.3.2 the projection of  $\mathbf{y}$  onto the span of the first  $k$  columns is  $\sum_{j \leq k} z_j \mathbf{q}_j$ , and the residual sum of squares is  $\|\mathbf{y}\|^2 - \sum_{j \leq k} z_j^2 = d^2 + \sum_{j>k} z_j^2$ . With  $k = p$  this is  $d^2$ , which completes (a). Taking differences gives the rest of (b).

- (c) The determinant of a triangular matrix is the product of its diagonal. The inverse formula follows from  $\mathbf{X}^\top \mathbf{X} = \mathbf{R}^\top \mathbf{R}$ . Finally  $h_{ii} = \mathbf{x}_{(i)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} = \mathbf{x}_{(i)}^\top \mathbf{R}^{-1} \mathbf{R}^{-\top} \mathbf{x}_{(i)} = \|\mathbf{R}^{-\top} \mathbf{x}_{(i)}\|^2$ .

■

Part (b) says that the squares  $z_1^2, \dots, z_p^2$  are the *sequential sums of squares* of the regressors in the order of the columns (Definition 9.3.1). A single factorization gives the whole nested

sequence of fits. This is the content of [Theorem 6.5.1](#), now in coordinates. Part (c) costs about  $np^2$  flops for all  $n$  leverages, each a triangular solve of  $p^2$  flops.

### Example. (Murder rates by the augmented factorization)

For the state data of [Example \(Poverty and murder rates\)](#) ( $n = 50$ , with an intercept and the percentages in poverty, in single-parent households and urban, so  $p = 4$ ), the augmented factorization ([Proposition 10.2.2](#)) gives the coefficients  $-9.0946$ ,  $0.2538$ ,  $0.3980$  and  $0.0046$ , and the corner entry  $d = 10.0976$ , so  $\text{SSE} = d^2 = 101.96$ . The condition number of  $\mathbf{X}$  is only 509, and the solution agrees with a QR solution to a relative difference of  $2 \times 10^{-14}$ . The largest leverage is 0.260, and  $\log_{10} \det(\mathbf{X}^\top \mathbf{X}) = 11.14$ . The script also checks the standard errors  $s \| \mathbf{e}_j^\top \mathbf{R}^{-1} \|$  against statsmodels.

```
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
```

```
Z = np.column_stack([X, y])          # augmented matrix [X, y]
S = Z.T @ Z                          # (p+1) x (p+1) cross-product matrix
T = cholesky_upper(S)                # T = [[R, z], [0, d]]
R, z, d = T[:p, :p], T[:p, p], T[p, p]

beta = solve_upper(R, z)              # R b = z
sse = d ** 2                          # the last diagonal entry is sqrt(SSE)
K = solve_upper(R, np.eye(p))        # R^{-1}
XtX_inv = K @ K.T                    # (X'X)^{-1} = R^{-1} R^{-T}
h = np.array([np.sum(solve_lower(R.T, x) ** 2) for x in X]) # leverages
print("coefficients:", np.round(beta, 4))
print(f"SSE = {sse:.4f}, d = {d:.4f}")
```

## 10.2.3 When the normal equations are good enough

[Proposition 10.1.3](#) and [Eq. 10.1.2](#) predict a relative error of order  $\kappa^2 u$  in  $\hat{\beta}$  from the normal equations, where, as [Section 10.5](#) shows,  $\kappa$  is the condition number of  $\mathbf{X}$  after its columns are scaled to equal length. With  $\kappa \leq 10^4$  about eight digits survive, more than any coefficient is known statistically. The normal equations are therefore sound for well-conditioned designs, when only cross-products are available, and inside iterations such as the reweighted least squares of [Chapter 34](#) ([Theorem 34.4.1](#)). Nearly collinear observational data are where the  $\kappa^2$  loss bites.

## 10.2.4 The sweep operator

Before QR became standard, regression programs worked on the augmented cross-product matrix with an operation that adds one variable at a time to the fit. It is still used for fitting many subsets of variables.

**Definition 10.2.1 — Sweep**

Let  $\mathbf{A}$  be symmetric with  $a_{kk} \neq 0$ . **Sweeping**  $\mathbf{A}$  on pivot  $k$  produces the symmetric matrix  $\mathbf{B}$  with

$$b_{kk} = -\frac{1}{a_{kk}}, \quad b_{ik} = b_{ki} = \frac{a_{ik}}{a_{kk}}, \quad b_{ij} = a_{ij} - \frac{a_{ik}a_{kj}}{a_{kk}} \quad (i, j \neq k).$$

**Proposition 10.2.3 — What sweeping computes**

Let  $\mathbf{S} = \mathbf{Z}^\top \mathbf{Z}$  be the cross-product matrix of  $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_m]$ . Let  $K$  be a set of column indices such that  $\mathbf{Z}_K$  has full column rank, and let  $L$  be the remaining indices. Sweeping  $\mathbf{S}$  on the pivots in  $K$ , one at a time and in any order, meets only positive pivots and produces the matrix  $\mathbf{W}$  with blocks

$$\mathbf{W}_{KK} = -(\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1}, \quad \mathbf{W}_{KL} = \mathbf{W}_{LK}^\top = (\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1} \mathbf{Z}_K^\top \mathbf{Z}_L, \quad \mathbf{W}_{LL} = \mathbf{Z}_L^\top (\mathbf{I} - \mathbf{M}_K) \mathbf{Z}_L,$$

where  $\mathbf{M}_K$  is the projection onto  $\mathcal{C}(\mathbf{Z}_K)$ . Column  $j \in L$  of  $\mathbf{W}_{KL}$  holds the least squares coefficients of  $\mathbf{z}_j$  regressed on  $\mathbf{Z}_K$ , and  $\mathbf{W}_{LL}$  holds the residual cross-products. With  $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$  and  $K$  the columns of  $\mathbf{X}$ , the last column of  $\mathbf{W}$  contains  $\hat{\boldsymbol{\beta}}$  and, in its corner, the residual sum of squares.

*Proof.* We induct on the size of  $K$ ; the case  $K = \emptyset$  is  $\mathbf{W} = \mathbf{S}$ . Add a pivot  $m \in L$  with  $\mathbf{Z}_{K \cup \{m\}}$  of full column rank, and write  $\mathbf{G} = (\mathbf{Z}_K^\top \mathbf{Z}_K)^{-1}$ ,  $\mathbf{B}_j = \mathbf{G} \mathbf{Z}_K^\top \mathbf{z}_j$  and  $\mathbf{w} = (\mathbf{I} - \mathbf{M}_K) \mathbf{z}_m$ . The pivot is  $\mathbf{z}_m^\top (\mathbf{I} - \mathbf{M}_K) \mathbf{z}_m = \|\mathbf{w}\|^2 > 0$ . For  $i, j \in L \setminus \{m\}$  the new entries are  $\mathbf{z}_i^\top (\mathbf{I} - \mathbf{M}_K - \mathbf{w} \mathbf{w}^\top / \|\mathbf{w}\|^2) \mathbf{z}_j = \mathbf{z}_i^\top (\mathbf{I} - \mathbf{M}_{K \cup \{m\}}) \mathbf{z}_j$  by Lemma 6.6.1. The new entries  $\mathbf{w}^\top \mathbf{z}_j / \|\mathbf{w}\|^2$  in row  $m$ , and  $\mathbf{B}_j - \mathbf{B}_m \mathbf{w}^\top \mathbf{z}_j / \|\mathbf{w}\|^2$  in the rows of  $K$ , are the coefficients of  $\mathbf{z}_j$  on  $[\mathbf{Z}_K, \mathbf{z}_m]$  by Theorem 6.6.2. The partitioned inverse (Theorem 1.6.4), with Schur complement  $\|\mathbf{w}\|^2$ , shows that  $(\mathbf{Z}_{K \cup \{m\}}^\top \mathbf{Z}_{K \cup \{m\}})^{-1}$  has blocks  $\mathbf{G} + \mathbf{B}_m \mathbf{B}_m^\top / \|\mathbf{w}\|^2$ ,  $-\mathbf{B}_m / \|\mathbf{w}\|^2$  and  $1 / \|\mathbf{w}\|^2$ , whose negatives are what the sweep produces:  $-\mathbf{G} - \mathbf{B}_m \mathbf{B}_m^\top / \|\mathbf{w}\|^2$ ,  $\mathbf{B}_m / \|\mathbf{w}\|^2$  and  $-1 / \|\mathbf{w}\|^2$ . The result depends only on the set  $K$ , so the order of the sweeps does not matter. ■

Sweeping a pivot costs about  $(p+1)^2$  flops, so a new variable enters a fitted model for  $O(p^2)$  work. The price is the same as for the normal equations: the sweep works with cross-products and inherits their  $\kappa^2$  sensitivity. For the state data, sweeping all four columns reproduces the coefficients and SSE of Example (Murder rates by the augmented factorization). Sweeping only the intercept and the single-parent column puts the SSE of that two-column model, 121.92, in the corner.

```
def sweep(A, k):
    """Sweep the symmetric matrix A on pivot k."""
    A = np.array(A, dtype=float)
    a = A[k, k]
    B = A - np.outer(A[:, k], A[k, :]) / a
    B[k, :] = A[k, :] / a
    B[:, k] = A[:, k] / a
    B[k, k] = -1 / a
    return B
```

```

W = S.copy()
for k in range(p):
    W = sweep(W, k)
print("coefficients from the sweep:", np.round(W[:p, p], 4))
print(f"SSE from the sweep: {W[p, p]:.4f}")

```

## 10.2.5 Forming cross-products: the centring problem

With an intercept in the model, the cross-product matrix is usually computed in centred form,  $\sum_i (x_{ij} - \bar{x}_j)(x_{ik} - \bar{x}_k)$ , which by [Theorem 6.7.1](#) changes the parameterization but not the fit. How it is computed matters a great deal. The “textbook” formula  $\sum_i x_i^2 - n\bar{x}^2$  subtracts two large, nearly equal numbers whenever the mean is large compared with the spread. The loss is governed by the ratio

$$\kappa_{\mathbf{x}} = \frac{\|\mathbf{x}\|}{\|\mathbf{x} - \bar{x}\mathbf{1}\|} = \left(1 + \frac{n\bar{x}^2}{\sum_i (x_i - \bar{x})^2}\right)^{1/2},$$

which is the condition number of the column, and it is large when the coefficient of variation is small. Three algorithms are in common use:

- *textbook*: accumulate  $\sum x_i$  and  $\sum x_i^2$  in one pass, then subtract;
- *two-pass*: compute  $\bar{x}$  in a first pass, then sum  $(x_i - \bar{x})^2$  in a second;
- *updating*: one pass, maintaining the running mean and the running centred sum by the recursions below.

### Proposition 10.2.4 — Updating means and centred cross-products

For vectors  $\mathbf{x}_1, \mathbf{x}_2, \dots$  in  $\mathbb{R}^q$  let  $\mathbf{m}_k$  be the mean of the first  $k$  and  $\mathbf{C}_k = \sum_{i \leq k} (\mathbf{x}_i - \mathbf{m}_k)(\mathbf{x}_i - \mathbf{m}_k)^\top$ . With  $\mathbf{d}_k = \mathbf{x}_k - \mathbf{m}_{k-1}$ ,

$$\mathbf{m}_k = \mathbf{m}_{k-1} + \frac{\mathbf{d}_k}{k}, \quad \mathbf{C}_k = \mathbf{C}_{k-1} + \frac{k-1}{k} \mathbf{d}_k \mathbf{d}_k^\top.$$

*Proof.* The first identity is  $k\mathbf{m}_k = (k-1)\mathbf{m}_{k-1} + \mathbf{x}_k$ . For the second, write  $\mathbf{C}_k = \sum_{i \leq k} \mathbf{x}_i \mathbf{x}_i^\top - k\mathbf{m}_k \mathbf{m}_k^\top$  and put  $\mathbf{m} = \mathbf{m}_{k-1}$ ,  $\mathbf{d} = \mathbf{d}_k$ . Then

$$\mathbf{C}_k - \mathbf{C}_{k-1} = \mathbf{x}_k \mathbf{x}_k^\top - k\mathbf{m}_k \mathbf{m}_k^\top + (k-1)\mathbf{m} \mathbf{m}^\top.$$

Substituting  $\mathbf{x}_k = \mathbf{m} + \mathbf{d}$  and  $k\mathbf{m}_k \mathbf{m}_k^\top = k\mathbf{m} \mathbf{m}^\top + \mathbf{m} \mathbf{d}^\top + \mathbf{d} \mathbf{m}^\top + \mathbf{d} \mathbf{d}^\top / k$ , all the terms in  $\mathbf{m}$  cancel and  $\mathbf{d} \mathbf{d}^\top (1 - 1/k)$  remains. ■

The update adds a *centred* rank-one term, so it never subtracts large quantities. It is the cross-product analogue of the row updates of [Section 10.6](#). Chan, Golub and LeVeque (1983) give error bounds of roughly  $n\kappa_{\mathbf{x}}^2 u$  for the textbook formula,  $n\kappa_{\mathbf{x}} u$  for updating and  $nu + n^2 \kappa_{\mathbf{x}}^2 u^2$  for the two-pass algorithm.



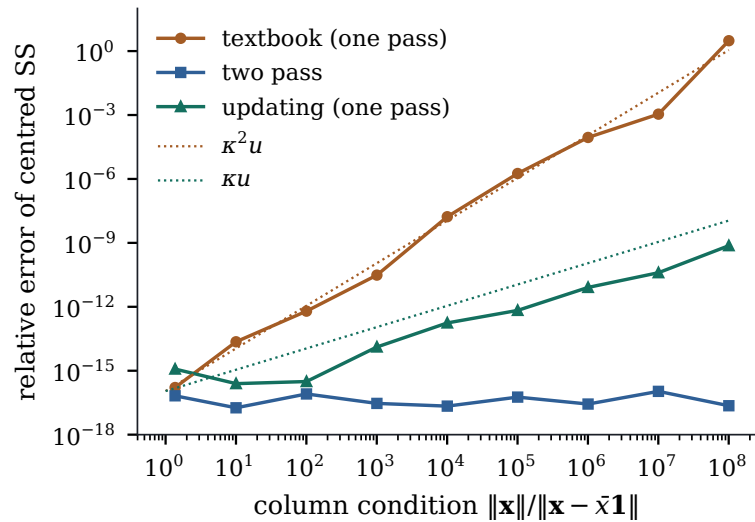


Figure 10.2.1: Relative error of the centred sum of squares of a column of 1000 values with standard deviation one and mean up to  $10^8$ , against the column condition number  $\kappa_{\mathbf{x}}$ . The one-pass textbook formula loses accuracy like  $\kappa^2 u$ , the one-pass updating formula like  $\kappa u$ , and the two-pass algorithm not at all.

#### Example. (Centring a column with a large mean)

A sample of 1000 standard normal values is shifted by  $c = 1, 10, \dots, 10^8$ , and the centred sum of squares is computed by the three algorithms and compared with its exact value (found in rational arithmetic). At  $c = 10^6$ , where  $\kappa_{\mathbf{x}} = 1.0 \times 10^6$ , the relative errors are  $8.9 \times 10^{-5}$  for the textbook formula,  $8.1 \times 10^{-12}$  for updating and  $2.7 \times 10^{-17}$  for two passes. At  $c = 10^8$  the textbook formula has relative error 3.1: not a single correct digit, and possibly a negative “sum of squares”. Updating gives  $7.4 \times 10^{-10}$  and two passes  $2.3 \times 10^{-17}$ . Figure 10.2.1 shows the three growth rates  $\kappa^2 u$ ,  $\kappa u$  and  $u$ .

```
def ss_textbook(x):
    return np.sum(x * x) - len(x) * np.mean(x) ** 2      # one pass, then subtract

def ss_two_pass(x):
    m = np.mean(x)                                       # first pass: the mean
    return np.sum((x - m) ** 2)                         # second pass: deviations

def ss_updating(x):
    m, s = 0.0, 0.0
    for k, xk in enumerate(x, start=1):                # one pass, no
        # cancellation
        d = xk - m
        m = m + d / k
        s = s + (k - 1) / k * d * d
    return s
```

Calendar years, population totals and price indices have large means and small spread. A program that forms cross-products should centre them with two passes (or update), solve for the slopes, and recover the intercept as  $\bar{y} - \bar{\mathbf{x}}^\top \hat{\boldsymbol{\beta}}_{\text{slopes}}$ .

## 10.2.6 Exercises

### 10.2.6.1 A. Check your understanding

#### Exercise A1.

Compute the Cholesky factor of  $\mathbf{A} = \begin{pmatrix} 4 & 2 & -2 \\ 2 & 10 & 2 \\ -2 & 2 & 6 \end{pmatrix}$  by Eq. 10.2.1, and use it to solve  $\mathbf{A}\mathbf{b} = (2, 14, 6)^\top$  and to find  $\det \mathbf{A}$ .

*Solution.* Row 1:  $r_{11} = 2$ ,  $r_{12} = 1$ ,  $r_{13} = -1$ . Row 2:  $r_{22} = \sqrt{10-1} = 3$ ,  $r_{23} = (2 - (1)(-1))/3 = 1$ . Row 3:  $r_{33} = \sqrt{6-1-1} = 2$ . So  $\mathbf{R} = \begin{pmatrix} 2 & 1 & -1 \\ 0 & 3 & 1 \\ 0 & 0 & 2 \end{pmatrix}$  and  $\det \mathbf{A} = (2 \cdot 3 \cdot 2)^2 = 144$ . Forward substitution in  $\mathbf{R}^\top \mathbf{z} = (2, 14, 6)^\top$  gives  $z_1 = 1$ ,  $z_2 = (14-1)/3 = 13/3$ ,  $z_3 = (6 + 1 - 13/3)/2 = 4/3$ . Back substitution in  $\mathbf{R}\mathbf{b} = \mathbf{z}$  gives  $b_3 = 2/3$ ,  $b_2 = (13/3 - 2/3)/3 = 11/9$  and  $b_1 = (1 - 11/9 + 2/3)/2 = 2/9$ . Check:  $\mathbf{A}\mathbf{b} = (8/9 + 22/9 - 12/9, 4/9 + 110/9 + 12/9, -4/9 + 22/9 + 36/9) = (2, 14, 6)$ .

#### Exercise A2.

For simple regression,  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$ , find the Cholesky factor (Theorem 10.2.1) of  $\mathbf{X}^\top \mathbf{X}$  and of the augmented matrix, and show that it reproduces the slope  $S_{xy}/S_{xx}$  of Theorem 5.2.1. Where does the centring happen?

*Solution.*  $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum x_i^2 \end{pmatrix}$  gives  $r_{11} = \sqrt{n}$ ,  $r_{12} = \sqrt{n}\bar{x}$  and  $r_{22} = (\sum x_i^2 - n\bar{x}^2)^{1/2} = S_{xx}^{1/2}$ . The augmented column gives  $z_1 = n\bar{y}/\sqrt{n} = \sqrt{n}\bar{y}$  and  $z_2 = (\sum x_i y_i - n\bar{x}\bar{y})/S_{xx}^{1/2} = S_{xy}/S_{xx}^{1/2}$ . Back substitution gives the slope  $b_2 = z_2/r_{22} = S_{xy}/S_{xx}$  and the intercept  $b_1 = (z_1 - r_{12}b_2)/r_{11} = \bar{y} - \bar{x}b_2$ . The centring happens in the subtraction  $a_{22} - r_{12}^2$ , which is the textbook formula of Example (Centring a column with a large mean). So the Cholesky factorization of an uncentred cross-product matrix inherits its cancellation.

### 10.2.6.2 B. Practice

#### Exercise B1.

Show that the standard error of  $\hat{\beta}_j$  is  $s \|\mathbf{e}_j^\top \mathbf{R}^{-1}\|$ , where  $s^2 = \text{SSE}/(n-p)$  (Theorem 5.6.1). Show that all  $p$  standard errors cost  $p^3/3 + O(p^2)$  flops, and all  $n$  leverages  $np^2 + O(np)$  flops. Why is computing  $\mathbf{M}$  itself a bad idea?

*Solution.* By Theorem 5.5.1 the estimated variance of  $\hat{\beta}_j$  is  $s^2[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj}$ , and  $[(\mathbf{X}^\top \mathbf{X})^{-1}]_{jj} = \mathbf{e}_j^\top \mathbf{R}^{-1} \mathbf{R}^{-\top} \mathbf{e}_j = \|\mathbf{e}_j^\top \mathbf{R}^{-1}\|^2$ . The inverse of a triangular matrix is triangular, and column  $j$  of  $\mathbf{R}^{-1}$  is found by back substitution on its leading  $j \times j$  block, at about  $j^2$  flops. The total is  $\sum_j j^2 \approx p^3/3$ . Each leverage is a triangular solve with  $\mathbf{R}^\top$  ( $p^2$  flops) and a squared norm. The projection  $\mathbf{M}$  has  $n^2$  entries, which for  $n = 10^6$  is too many to store, and only its diagonal is ever needed.

**Exercise B2.**

Two groups of observations have sizes  $n_a, n_b$ , means  $\mathbf{m}_a, \mathbf{m}_b$  and centred cross-product matrices  $\mathbf{C}_a, \mathbf{C}_b$ . Show that the pooled data have centred cross-product matrix

$$\mathbf{C} = \mathbf{C}_a + \mathbf{C}_b + \frac{n_a n_b}{n_a + n_b} (\mathbf{m}_a - \mathbf{m}_b)(\mathbf{m}_a - \mathbf{m}_b)^\top.$$

Show that [Proposition 10.2.4](#) is the special case  $n_b = 1$ , and explain how the formula allows cross-products to be computed in parallel on separate blocks of data.

**10.2.6.3 C. Going deeper****Exercise C1.**

Let  $\mathbf{A}$  be nonnegative definite of rank  $r$ . Show that the Cholesky recursion with *symmetric pivoting*, which at step  $j$  moves the largest remaining diagonal entry of the current Schur complement to position  $j$ , produces a permutation  $\mathbf{\Pi}$  and  $\mathbf{\Pi}^\top \mathbf{A} \mathbf{\Pi} = \mathbf{R}^\top \mathbf{R}$  with  $\mathbf{R} = \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$  and  $\mathbf{R}_{11}$  an  $r \times r$  nonsingular upper triangle. (Use [Proposition 1.7.5\(a\)](#).) How is this related to the pivoted QR of [Section 10.7](#)?

## 10.3 Householder and Givens QR

Orthogonal matrices preserve lengths ([Proposition 1.5.3\(a\)](#)). For any orthogonal  $\mathbf{Q}$ ,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\| = \|\mathbf{Q}^\top \mathbf{y} - \mathbf{Q}^\top \mathbf{X}\mathbf{b}\| \quad \text{for every } \mathbf{b}.$$

So a least squares problem can be replaced by an equivalent one in which  $\mathbf{Q}^\top \mathbf{X}$  is upper triangular, without ever forming  $\mathbf{X}^\top \mathbf{X}$ . Such a  $\mathbf{Q}$  is built from reflections, which zero a whole column below the diagonal at once, or from rotations, which zero one entry at a time.

### 10.3.1 Householder reflections

A reflection  $\mathbf{H} = \mathbf{I} - 2\mathbf{v}\mathbf{v}^\top/(\mathbf{v}^\top \mathbf{v})$  is symmetric and orthogonal, and it maps  $\mathbf{v}$  to  $-\mathbf{v}$  and fixes every vector orthogonal to  $\mathbf{v}$  ([Exercise \(B1\)](#)). The whole numerical content lies in choosing  $\mathbf{v}$  so that a given vector is mapped onto a coordinate axis, and in choosing its sign so that no cancellation occurs.

#### Lemma 10.3.1 — Householder vector

Let  $\mathbf{x} \in \mathbb{R}^m$ ,  $\mathbf{x} \neq \mathbf{0}$ , let  $\alpha = -\text{sign}(x_1)\|\mathbf{x}\|$  (with  $\text{sign}(0) = 1$ ), and let  $\mathbf{v} = \mathbf{x} - \alpha\mathbf{e}_1$ . Then  $\mathbf{v} \neq \mathbf{0}$ , and  $\mathbf{H} = \mathbf{I} - 2\mathbf{v}\mathbf{v}^\top/(\mathbf{v}^\top \mathbf{v})$  satisfies  $\mathbf{H}\mathbf{x} = \alpha\mathbf{e}_1$ . The first entry  $v_1 = x_1 - \alpha$  is a sum of two numbers of the same sign, so  $|v_1| = |x_1| + \|\mathbf{x}\|$ . Applying  $\mathbf{H}$  to a vector  $\mathbf{w}$  as  $\mathbf{w} - 2\mathbf{v}(\mathbf{v}^\top \mathbf{w})/(\mathbf{v}^\top \mathbf{v})$  costs about  $4m$  flops, and  $\mathbf{H}$  is never formed.

*Proof.*  $\alpha$  and  $x_1$  have opposite signs (or  $x_1 = 0$ ), so  $v_1 = x_1 - \alpha$  adds two numbers of the same sign and  $|v_1| = |x_1| + \|\mathbf{x}\| > 0$ . Next,  $\mathbf{v}^\top \mathbf{x} = \|\mathbf{x}\|^2 - \alpha x_1$  and  $\mathbf{v}^\top \mathbf{v} = \|\mathbf{x}\|^2 - 2\alpha x_1 + \alpha^2 = 2(\|\mathbf{x}\|^2 - \alpha x_1)$ , using  $\alpha^2 = \|\mathbf{x}\|^2$ . Hence  $2\mathbf{v}^\top \mathbf{x} / \mathbf{v}^\top \mathbf{v} = 1$  and  $\mathbf{H}\mathbf{x} = \mathbf{x} - \mathbf{v} = \alpha \mathbf{e}_1$ . ■

The opposite sign choice,  $\alpha = +\text{sign}(x_1)\|\mathbf{x}\|$ , also gives a reflection onto the axis. But then  $v_1 = x_1 - \alpha$  subtracts two nearly equal numbers whenever  $\mathbf{x}$  is nearly parallel to  $\mathbf{e}_1$ . This is the cancellation of [Section 10.1](#), and it can make  $\mathbf{v}$  inaccurate in every digit.

### Theorem 10.3.2 — Householder QR

Let  $\mathbf{X}$  be  $n \times p$  with  $n \geq p$ .

- (a) There are reflections  $\mathbf{H}_1, \dots, \mathbf{H}_p$ , with  $\mathbf{H}_j$  acting only on coordinates  $j, \dots, n$  (or equal to  $\mathbf{I}$ ), such that

$$\mathbf{H}_p \cdots \mathbf{H}_1 \mathbf{X} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$$

with  $\mathbf{R}$  upper triangular of size  $p \times p$ . Thus  $\mathbf{X} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$  with  $\mathbf{Q} = \mathbf{H}_1 \cdots \mathbf{H}_p$  orthogonal, and  $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$  where  $\mathbf{Q}_1$  holds the first  $p$  columns of  $\mathbf{Q}$ . If  $\text{rank } \mathbf{X} = p$ , then  $\mathbf{R}$  is nonsingular, and changing the signs of the rows of  $\mathbf{R}$  with negative diagonal entries gives the Cholesky factor of  $\mathbf{X}^\top \mathbf{X}$ .

- (b) The reduction takes  $2np^2 - 2p^3/3$  flops to leading order. Applying  $\mathbf{Q}^\top$  (or  $\mathbf{Q}$ ) to a vector takes  $4np - 2p^2$ , and forming  $\mathbf{Q}_1$  explicitly takes another  $2np^2 - 2p^3/3$ .
- (c) (Backward stability.) In floating-point arithmetic, the computed  $\hat{\mathbf{R}}$  is the exact triangular factor of  $\mathbf{X} + \Delta \mathbf{X}$ , where each column satisfies  $\|\Delta \mathbf{x}_j\| \leq c_{np} u \|\mathbf{x}_j\|$ . The least squares solution  $\hat{\mathbf{b}}$  computed from  $\hat{\mathbf{R}}$  and the computed  $\mathbf{Q}^\top \mathbf{y}$  is the exact least squares solution for data  $(\mathbf{X} + \Delta \mathbf{X}, \mathbf{y} + \Delta \mathbf{y})$  with perturbations of the same kind,  $\|\Delta \mathbf{y}\| \leq c_{np} u \|\mathbf{y}\|$ . Here  $c_{np}$  is a modest multiple of  $np$ .

*Proof.*

- (a) Induction on the column. Suppose  $\mathbf{H}_{j-1} \cdots \mathbf{H}_1 \mathbf{X}$  has zeros below the diagonal in its first  $j-1$  columns. Let  $\mathbf{x}$  be the part of column  $j$  in rows  $j, \dots, n$ . If  $\mathbf{x} = \mathbf{0}$ , take  $\mathbf{H}_j = \mathbf{I}$ . Otherwise take the reflection of [Lemma 10.3.1](#) for  $\mathbf{x}$ , acting on rows  $j, \dots, n$ . It zeroes column  $j$  below the diagonal. It leaves rows  $1, \dots, j-1$  alone. It maps the first  $j-1$  columns, which are zero in rows  $j, \dots, n$ , to themselves. After  $p$  steps the matrix is  $\begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ , and  $\mathbf{Q}$  is a product of orthogonal matrices. Orthogonal factors do not change rank, so  $\text{rank } \mathbf{R} = \text{rank } \mathbf{X}$ . If  $\mathbf{D}$  is the diagonal matrix of signs of the diagonal of  $\mathbf{R}$ , then  $\mathbf{X} = (\mathbf{Q}_1 \mathbf{D})(\mathbf{D} \mathbf{R})$  is a QR factorization with positive diagonal, and  $\mathbf{D} \mathbf{R}$  is the Cholesky factor by [Theorem 10.2.1\(d\)](#).
- (b) Step  $j$  builds  $\mathbf{v}$  (about  $2(n-j)$  flops) and applies the reflection to the  $p-j$  remaining columns of length  $n-j+1$ , at about  $4(n-j)(p-j)$  flops. Summing,

$$\sum_{j=1}^p 4(n-j)(p-j) \approx 4 \int_0^p (n-t)(p-t) dt = 2np^2 - \frac{2}{3}p^3.$$

Applying the  $p$  reflections to a vector costs  $\sum_j 4(n-j+1) \approx 4np - 2p^2$ . Forming

$\mathbf{Q}_1$  means applying the reflections in reverse order to the first  $p$  columns of  $\mathbf{I}_n$ . Reflection  $j$  then acts on columns  $j, \dots, p$  only, which gives the same integral again.

- (c) This is a theorem of rounding-error analysis, not of linear algebra, and we state it without proof. Its essence is that a reflection applied in floating point is an exact reflection applied to a slightly perturbed matrix, that orthogonal matrices do not amplify perturbations, and that the  $p$  steps add their perturbations. See Higham (2002, chapters 19 and 20) and Björck (1996, chapter 2).

■

Part (c) is why QR is the default. Its backward error is small relative to each column separately, which [Section 10.5](#) uses to explain why column scaling does not change the accuracy of QR. Software never forms  $\mathbf{Q}$ ; it stores the vectors  $\mathbf{v}_j$  in the zeroed part of  $\mathbf{X}$ .

```
import numpy as np
import statsmodels.api as sm

def house(x):
    """Unit vector v with (I - 2 v v') x = alpha e_1, alpha = -sign(x_1) ||x||."""
    alpha = -np.copysign(np.linalg.norm(x), x[0])
    v = x.astype(float).copy()
    v[0] -= alpha                                # x_1 - alpha adds two numbers of the same
    sign                                         sign
    return v / np.linalg.norm(v), alpha

def householder_qr(X):
    """Return the reflection vectors and the p x p triangle R of X = Q [R; 0]."""
    A = np.array(X, dtype=float)
    n, p = A.shape
    vs = []
    for j in range(p):
        v, alpha = house(A[j:, j])
        A[j:, j:] -= 2.0 * np.outer(v, v @ A[j:, j:])    # apply I - 2vv' to the
        trailing block
        vs.append(v)
    return vs, np.triu(A[:p, :])

def apply_Qt(vs, y):
    """Q'y, applying the reflections H_1, ..., H_p in turn (Q itself is never formed)."""
    z = np.array(y, dtype=float)
    for j, v in enumerate(vs):
        z[j:] -= 2.0 * v * (v @ z[j:])
    return z

def apply_Q(vs, z):
    """Q z, applying the reflections in reverse order."""
    w = np.array(z, dtype=float)
    for j in range(len(vs) - 1, -1, -1):
        v = vs[j]
        w[j:] -= 2.0 * v * (v @ w[j:])
    return w
```

## 10.3.2 Regression quantities from the factorization

Section 6.10 derived the regression quantities from the thin factorization  $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$  (Eq. 6.10.1). The full factorization, with  $\mathbf{Q}$  square, adds one thing: the part of  $\mathbf{Q}^\top \mathbf{y}$  that the model cannot reach, which is the residual in rotated coordinates.

### Proposition 10.3.3 — Least squares from QR

Let  $\mathbf{X} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$  with  $\mathbf{Q}$  orthogonal and  $\text{rank } \mathbf{X} = p$ , and partition  $\mathbf{c} = \mathbf{Q}^\top \mathbf{y}$  as  $(\mathbf{c}_1^\top, \mathbf{c}_2^\top)^\top$  with  $\mathbf{c}_1 \in \mathbb{R}^p$ . Then:

- (a)  $\hat{\boldsymbol{\beta}}$  is the solution of the triangular system  $\mathbf{R}\mathbf{b} = \mathbf{c}_1$ , and  $\text{SSE} = \|\mathbf{c}_2\|^2$ ;
- (b)  $\hat{\mathbf{Y}} = \mathbf{Q} \begin{pmatrix} \mathbf{c}_1 \\ \mathbf{0} \end{pmatrix}$  and  $\hat{\boldsymbol{\varepsilon}} = \mathbf{Q} \begin{pmatrix} \mathbf{0} \\ \mathbf{c}_2 \end{pmatrix}$ ;
- (c)  $(\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{R}^{-\top}$  and  $h_{ii} = \|\mathbf{e}_i^\top \mathbf{Q}_1\|^2$ ;
- (d)  $c_k^2$  is the drop in residual sum of squares when column  $k$  is added to columns  $1, \dots, k-1$ , the sequential sum of squares of Definition 9.3.1.

*Proof.*

- (a) With  $\mathbf{Q}_1$  the first  $p$  columns of  $\mathbf{Q}$ ,  $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$  is a thin QR factorization and  $\mathbf{c}_1 = \mathbf{Q}_1^\top \mathbf{y}$ , so  $\mathbf{R}\hat{\boldsymbol{\beta}} = \mathbf{c}_1$  and  $\text{SSE} = \|\mathbf{y}\|^2 - \|\mathbf{c}_1\|^2$  are Eq. 6.10.1. What the full  $\mathbf{Q}$  adds is  $\|\mathbf{c}\| = \|\mathbf{y}\|$ , hence  $\text{SSE} = \|\mathbf{c}_2\|^2$ .
- (b)  $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{Q} \begin{pmatrix} \mathbf{R}\hat{\boldsymbol{\beta}} \\ \mathbf{0} \end{pmatrix} = \mathbf{Q} \begin{pmatrix} \mathbf{c}_1 \\ \mathbf{0} \end{pmatrix}$ , and  $\mathbf{y} = \mathbf{Q}\mathbf{c}$ .
- (c)  $\mathbf{X}^\top \mathbf{X} = \mathbf{R}^\top \mathbf{Q}_1^\top \mathbf{Q}_1 \mathbf{R} = \mathbf{R}^\top \mathbf{R}$ , and the leverage formula is Eq. 6.10.1.
- (d) The first  $k$  columns of  $\mathbf{X}$  equal the first  $k$  columns of  $\mathbf{Q}$  times the leading  $k \times k$  block of  $\mathbf{R}$ , which is nonsingular. So they span the same subspace as  $\mathbf{q}_1, \dots, \mathbf{q}_k$ , and the squared length of the projection of  $\mathbf{y}$  onto it is  $\sum_{j \leq k} (\mathbf{q}_j^\top \mathbf{y})^2 = \sum_{j \leq k} c_j^2$  (Proposition 6.3.2).

■

Part (d) is the QR form of Proposition 10.2.2(b): by uniqueness of the triangle,  $\mathbf{c}_1$  equals the vector  $\mathbf{z}$  of that proposition up to signs. The vector  $\mathbf{c}_2$  replaces the  $n$  correlated residuals by  $n - p$  uncorrelated ones with the same sum of squares (Exercise (B1)).

### Example. (Sequential sums of squares from one factorization)

For the state data of Example (Murder rates by the augmented factorization), Householder QR (Theorem 10.3.2) gives  $\mathbf{R}$  with diagonal  $-7.071, 21.511, 21.609$  and  $113.413$ . The first entry is  $-\sqrt{50}$ : the sign convention of Lemma 10.3.1 makes it negative. The squares of the first four entries of  $\mathbf{Q}^\top \mathbf{y}$  are the sequential sums of squares: 1018.81 for the intercept (which is  $n\bar{y}^2$ ), 102.49 for poverty, 80.04 for single-parent households given poverty, and 0.27 for urbanization given the other two. The remaining 46 entries have squared length  $\text{SSE} = 101.96$ . The urbanization variable adds almost nothing once the other two are in the model, which is what its small coefficient in Example (Poverty and murder rates) suggested. The script checks every number against nested fits, and checks the leverages against the squared row lengths of  $\mathbf{Q}_1$ .

```

data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape

vs, R = householder_qr(X)
c = apply_Qt(vs, y)                                # c = Q'y = (c1, c2)
c1, c2 = c[:p], c[p:]
beta = np.linalg.solve(R, c1)                       # triangular system R b = c1
sse = c2 @ c2                                       # SSE = ||c2||^2
resid = apply_Q(vs, np.concatenate([np.zeros(p), c2]))
seq_ss = c1 ** 2                                   # sequential sums of squares, in column
                                                order
print("R diagonal:", np.round(np.diag(R), 4))
print("coefficients:", np.round(beta, 4), f"   SSE = {sse:.4f}")
print("sequential SS (intercept, poverty, single, urban):", np.round(seq_ss, 3))

```

### 10.3.3 Givens rotations

A reflection zeroes a whole column at once. Sometimes it is better to zero one entry at a time, touching only two rows. This is the case when  $\mathbf{X}$  is sparse, when rows arrive one by one, or when an existing factorization must be modified.

#### Theorem 10.3.4 — Givens rotations

- (a) For  $(a, b) \neq (0, 0)$  let  $r = (a^2 + b^2)^{1/2}$ ,  $c = a/r$  and  $s = b/r$ . The rotation  $\mathbf{G} = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}$  is orthogonal and  $\mathbf{G}(a, b)^\top = (r, 0)^\top$ . Embedded in rows  $i$  and  $k$  of the identity, it changes only rows  $i$  and  $k$  of a matrix it multiplies, at 6 flops per column.
- (b) Zeroing the subdiagonal of  $\mathbf{X}$  column by column, each entry against the row above it, produces  $\mathbf{GX} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$  with  $\mathbf{G}$  orthogonal. It takes  $np - p(p+1)/2$  rotations and  $3np^2 - p^3$  flops to leading order.
- (c) (Row-wise QR.) Let  $\mathbf{R}$  be upper triangular and  $\mathbf{x} \in \mathbb{R}^p$ . There are  $p$  rotations, the  $k$ th acting on row  $k$  of  $\mathbf{R}$  and on the new row, that reduce  $\begin{pmatrix} \mathbf{R} \\ \mathbf{x}^\top \end{pmatrix}$  to  $\begin{pmatrix} \mathbf{R}_+ \\ \mathbf{0}^\top \end{pmatrix}$  with  $\mathbf{R}_+$  upper triangular and  $\mathbf{R}_+^\top \mathbf{R}_+ = \mathbf{R}^\top \mathbf{R} + \mathbf{x}\mathbf{x}^\top$ . The cost is  $3p^2$  flops. If  $\mathbf{R}$  and  $\mathbf{x}$  are augmented by the response, as in Proposition 10.2.2, the new row ends as  $(0, \dots, 0, t)$ , and the residual sum of squares increases by  $t^2$ .
- (d) A rotation of rows  $i$  and  $k$  creates a nonzero in column  $j$  only if row  $i$  or row  $k$  already had one there.

*Proof.*

- (a)  $c^2 + s^2 = 1$  gives  $\mathbf{G}^\top \mathbf{G} = \mathbf{I}$ . Also  $ca + sb = (a^2 + b^2)/r = r$  and  $-sa + cb = 0$ . Each column of the product needs four multiplications and two additions.
- (b) When entry  $(i, j)$  is zeroed against row  $i - 1$ , working upwards from row  $n$ , both rows are already zero in columns  $1, \dots, j - 1$ , so no earlier zero is destroyed. Column  $j$  needs  $n - j$  rotations, each acting on  $p - j + 1$  columns, so the count is  $\sum_j 6(n - j)(p - j + 1) \approx 6 \int_0^p (n - t)(p - t) dt = 3np^2 - p^3$ .

- (c) Before rotation  $k$ , the new row is zero in columns  $1, \dots, k-1$ , and so is row  $k$  of  $\mathbf{R}$ . The rotation chosen by (a) to zero the  $k$ th entry of the new row keeps both rows zero in those columns. After  $p$  rotations the new row is zero, and row  $k$  of  $\mathbf{R}$  has been replaced by a row that is zero before column  $k$ . With  $\mathbf{G}$  the product of the rotations,

$$\mathbf{R}_+^\top \mathbf{R}_+ = \begin{pmatrix} \mathbf{R} \\ \mathbf{x}^\top \end{pmatrix}^\top \mathbf{G}^\top \mathbf{G} \begin{pmatrix} \mathbf{R} \\ \mathbf{x}^\top \end{pmatrix} = \mathbf{R}^\top \mathbf{R} + \mathbf{x} \mathbf{x}^\top.$$

Rotation  $k$  acts on  $p-k+1$  columns, which gives  $\sum_k 6(p-k+1) \approx 3p^2$ . In the augmented case, apply  $p$  rotations to the augmented  $(p+1) \times (p+1)$  triangle  $\mathbf{T}$  and the row  $(\mathbf{x}^\top, y)$ . The new row becomes  $(\mathbf{0}^\top, t)$ , and  $\mathbf{T}_+^\top \mathbf{T}_+ = \mathbf{T}^\top \mathbf{T} + (\mathbf{x}^\top, y)^\top (\mathbf{x}^\top, y) - (\mathbf{0}^\top, t)^\top (\mathbf{0}^\top, t)$ . The corner entry of  $\mathbf{T}_+$  is still  $d$ , so the augmented triangle with the new row included is obtained by one more rotation, of  $d$  against  $t$ . This gives a corner  $(d^2 + t^2)^{1/2}$ , and the claim follows from [Proposition 10.2.2\(a\)](#).

- (d) Each new row is a combination of the two old ones. ■

Part (c) turns QR into a streaming algorithm (Gentleman 1973): feed the rows of  $[\mathbf{X}, \mathbf{y}]$  one at a time into  $\mathbf{R} = \mathbf{0}$ . Memory is  $O(p^2)$ , one pass suffices, and the result is as accurate as Householder QR, so the one practical advantage of accumulating cross-products is kept without the  $\kappa^2$  loss. Part (d) keeps  $\mathbf{R}$  sparse for sparse designs, such as factors with many levels.

```
def givens(a, b):
    """c, s, r with [[c, s], [-s, c]] @ [a, b] = [r, 0]."""
    if b == 0.0:
        return 1.0, 0.0, a
    r = np.hypot(a, b) # sqrt(a^2 + b^2) without overflow
    return a / r, b / r, r
```

```
def add_row(R, z, x, y):
    """Rotate one observation (x, y) into the triangle R and the vector z = (Q'y)[:p].

    Returns the new R, z and the entry that leaves: its square is the increase in SSE.
    """
    R, z, x, y = R.copy(), z.copy(), np.array(x, dtype=float), float(y)
    for k in range(len(x)):
        c, s, r = givens(R[k, k], x[k])
        Rk, xk = R[k, k:].copy(), x[k:].copy()
        R[k, k:], x[k:] = c * Rk + s * xk, -s * Rk + c * xk
        z[k], y = c * z[k] + s * y, -s * z[k] + c * y
    return R, z, y

def rowwise_ls(X, y):
    """Least squares with one pass over the rows and O(p^2) memory."""
    p = X.shape[1]
    R, z, sse = np.zeros((p, p)), np.zeros(p), 0.0
    for xi, yi in zip(X, y):
        R, z, leftover = add_row(R, z, xi, yi)
        sse += leftover ** 2
    return R, z, sse
```



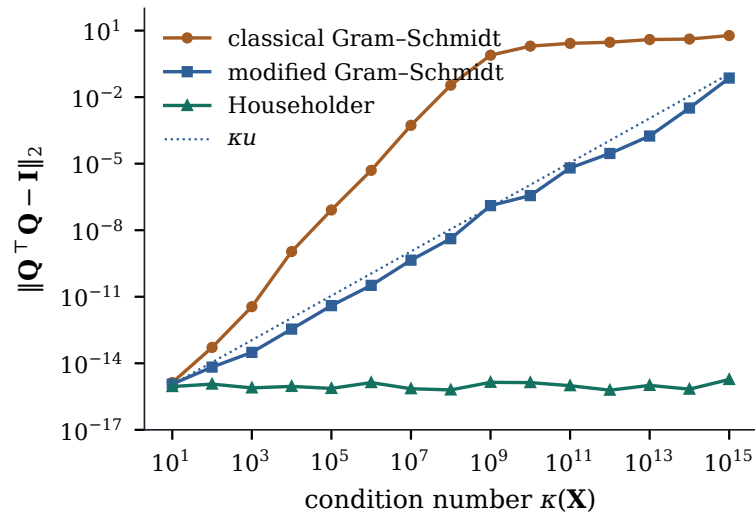


Figure 10.3.1: Departure from orthogonality of the computed  $\mathbf{Q}$  against the condition number of  $\mathbf{X}$ , for classical and modified Gram–Schmidt and for Householder QR. The dotted line is  $\kappa u$ .

```
R_row, z_row, sse_row = rowwise_ls(X, y)
print("coefficients (row-wise Givens):", np.round(np.linalg.solve(R_row, z_row), 4))
print(f"SSE (row-wise Givens): {sse_row:.4f}")
```

The rotation in `givens` uses `hypot`, which computes  $(a^2 + b^2)^{1/2}$  without overflow or underflow. Square-root-free “fast” variants (Gentleman 1973) matter less on modern hardware.

### 10.3.4 Gram–Schmidt revisited

Gram–Schmidt (Theorem 1.5.2) also computes a QR factorization, at  $2np^2$  flops, and forms  $\mathbf{Q}_1$  explicitly, which gives leverages directly. The classical form computes  $r_{kj} = \mathbf{q}_k^\top \mathbf{x}_j$  from the *original* column; the modified form subtracts each new direction from all remaining columns at once. The two agree in exact arithmetic and differ greatly in floating point.

#### Example. (Loss of orthogonality)

For  $100 \times 12$  matrices with prescribed condition numbers from 10 to  $10^{15}$ , Figure 10.3.1 plots  $\|\hat{\mathbf{Q}}^\top \hat{\mathbf{Q}} - \mathbf{I}\|_2$  for the computed  $\hat{\mathbf{Q}}$ . At  $\kappa = 10^8$  classical Gram–Schmidt gives  $3.5 \times 10^{-2}$ , and its columns are no longer an orthonormal basis in any useful sense. Modified Gram–Schmidt gives  $4.2 \times 10^{-9}$ , in line with the bound  $O(\kappa u)$  of Björck (1967). The  $\mathbf{Q}$  accumulated from Householder reflections gives  $6.3 \times 10^{-16}$ , orthogonal to working precision whatever the condition number.

```
def cgs(X):
    """Classical Gram-Schmidt: inner products with the original column."""
    n, p = X.shape
    Q = np.zeros((n, p))
    for j in range(p):
        w = X[:, j] - Q[:, :j] @ (Q[:, :j].T @ X[:, j])
        Q[:, j] = w / np.linalg.norm(w)
    return Q
```

```
def mgs(X):
    """Modified Gram-Schmidt: subtract each direction from the working vector at
    once."""
    Q = np.array(X, dtype=float)
    n, p = Q.shape
    for j in range(p):
        Q[:, j] /= np.linalg.norm(Q[:, j])
        Q[:, j + 1:] -= np.outer(Q[:, j], Q[:, j] @ Q[:, j + 1:])
    return Q
```

Loss of orthogonality does not make modified Gram–Schmidt useless. It is numerically equivalent to Householder QR applied to  $\mathbf{X}$  with  $p$  rows of zeros on top (Björck and Paige 1992), so applied to  $[\mathbf{X}, \mathbf{y}]$ , with  $\mathbf{Q}_1^\top \mathbf{y}$  taken from the last column, it gives a backward stable least squares solution. Leverages computed from the rows of  $\hat{\mathbf{Q}}_1$  are not reliable when  $\kappa$  is large.

*Idea.* Orthogonal transformations reduce  $\mathbf{X}$  to a triangle while preserving every residual length. Householder reflections do it fastest; Givens rotations do it one entry at a time, which suits streaming, updating and sparse data.

## 10.3.5 Exercises

### 10.3.5.1 A. Check your understanding

#### Exercise A1.

Find the Householder vector and  $\alpha$  of [Lemma 10.3.1](#) for  $\mathbf{x} = (3, 4)^\top$ , and verify  $\mathbf{H}\mathbf{x} = \alpha \mathbf{e}_1$ . For  $\mathbf{x} = (1, 10^{-9})^\top$ , compute  $v_1$  with both sign choices in double precision, and explain what goes wrong with the unstable one.

*Solution.*  $\|\mathbf{x}\| = 5$ ,  $\alpha = -5$ ,  $\mathbf{v} = (8, 4)^\top$ ,  $\mathbf{v}^\top \mathbf{v} = 80$  and  $\mathbf{v}^\top \mathbf{x} = 40$ , so  $\mathbf{H}\mathbf{x} = \mathbf{x} - 2 \cdot \frac{40}{80} \mathbf{v} = (3, 4) - (8, 4) = (-5, 0)$ . For  $\mathbf{x} = (1, 10^{-9})^\top$ ,  $\|\mathbf{x}\| = (1 + 10^{-18})^{1/2}$  rounds to 1. The stable choice gives  $v_1 = 1 + 1 = 2$ . The unstable one gives  $v_1 = 1 - 1 = 0$  in floating point, while the true value is  $1 - \|\mathbf{x}\| \approx -5 \times 10^{-19}$ . With  $\mathbf{v} = (0, 10^{-9})$  the “reflection” swaps the sign of the second coordinate and maps  $\mathbf{x}$  to  $(1, -10^{-9})$ , not onto the axis. An algebraically equivalent formula,  $v_1 = -(x_2^2 + \cdots + x_m^2)/(x_1 + \|\mathbf{x}\|)$ , rescues that choice.

### 10.3.5.2 B. Practice

#### Exercise B1.

In the setting of [Proposition 10.3.3](#), let  $\mathbf{Q} = [\mathbf{Q}_1, \mathbf{Q}_2]$ . Show that under the linear model with  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ , the vector  $\mathbf{c}_2 = \mathbf{Q}_2^\top \mathbf{y}$  has mean  $\mathbf{0}$  and covariance  $\sigma^2 \mathbf{I}_{n-p}$ , and that  $\hat{\boldsymbol{\varepsilon}} = \mathbf{Q}_2 \mathbf{c}_2$ . Under normality, deduce from  $\mathbf{c}_2$  alone that  $\text{SSE}/\sigma^2 \sim \chi^2(n-p)$  independently of  $\hat{\boldsymbol{\beta}}$  (compare [Theorem 7.5.1](#)). Is  $\mathbf{c}_2$  unique?

*Solution.*  $\mathbf{Q}_2^\top \mathbf{X} = \mathbf{0}$ , because  $\mathbf{Q}^\top \mathbf{X} = \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$ . So  $\mathbf{c}_2 = \mathbf{Q}_2^\top \mathbf{X}\boldsymbol{\beta} + \mathbf{Q}_2^\top \boldsymbol{\varepsilon} = \mathbf{Q}_2^\top \boldsymbol{\varepsilon}$ , with mean  $\mathbf{0}$  and covariance  $\sigma^2 \mathbf{Q}_2^\top \mathbf{Q}_2 = \sigma^2 \mathbf{I}$ . By [Proposition 10.3.3\(b\)](#),  $\hat{\boldsymbol{\varepsilon}} = \mathbf{Q} \begin{pmatrix} \mathbf{0} \\ \mathbf{c}_2 \end{pmatrix} = \mathbf{Q}_2 \mathbf{c}_2$ . Under normality,  $\mathbf{Q}^\top \boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$  ([Theorem 3.3.1](#)), so  $\mathbf{c}_1 - \mathbf{R}\boldsymbol{\beta} = \mathbf{Q}_1^\top \boldsymbol{\varepsilon}$  and  $\mathbf{c}_2 = \mathbf{Q}_2^\top \boldsymbol{\varepsilon}$  are independent normal vectors with identity covariance times  $\sigma^2$ . Then  $\text{SSE}/\sigma^2 =$

$\|\mathbf{c}_2/\sigma\|^2 \sim \chi^2(n-p)$ , and  $\hat{\boldsymbol{\beta}} = \mathbf{R}^{-1}\mathbf{c}_1$  is a function of  $\mathbf{c}_1$  alone.  $\mathbf{c}_2$  is not unique:  $\mathbf{Q}_2$  can be replaced by  $\mathbf{Q}_2\mathbf{O}$  for any orthogonal  $\mathbf{O}$ , which changes  $\mathbf{c}_2$  but not its length or its distribution.

### Exercise B2.

Weighted least squares with a diagonal weight matrix  $\mathbf{W}$  of positive weights is ordinary least squares for  $\mathbf{W}^{1/2}\mathbf{X}$  and  $\mathbf{W}^{1/2}\mathbf{y}$ . Show that  $\kappa(\mathbf{W}^{1/2}\mathbf{X}) \leq \kappa(\mathbf{W}^{1/2})\kappa(\mathbf{X})$ . Give an example with two very different weights in which  $\kappa(\mathbf{W}^{1/2}\mathbf{X})$  is large although  $\kappa(\mathbf{X})$  is small, and one in which it is small although  $\kappa(\mathbf{W}^{1/2})$  is large.

*Solution.*  $\sigma_{\max}(\mathbf{W}^{1/2}\mathbf{X}) \leq \sigma_{\max}(\mathbf{W}^{1/2})\sigma_{\max}(\mathbf{X})$  and  $\|\mathbf{W}^{1/2}\mathbf{X}\mathbf{v}\| \geq \sigma_{\min}(\mathbf{W}^{1/2})\|\mathbf{X}\mathbf{v}\| \geq \sigma_{\min}(\mathbf{W}^{1/2})\sigma_{\min}(\mathbf{X})$  for unit  $\mathbf{v}$ , by Proposition 1.8.2. Divide. For the first example take  $\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$  (with  $\kappa = 1$ ) and  $\mathbf{W} = \text{diag}(1, w^2)$  with  $w$  large. Then  $\mathbf{W}^{1/2}\mathbf{X}$  has rows  $(1, 1)$  and  $(w, -w)$ , whose singular values are  $\sqrt{2}$  and  $\sqrt{2}w$ , so  $\kappa = w$ . For the second take  $\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & 1/w \end{pmatrix}$  with the same  $\mathbf{W}$ . Then  $\mathbf{W}^{1/2}\mathbf{X} = \mathbf{I}$ .

### 10.3.5.3 C. Going deeper

#### Exercise C1.

Show that, in exact arithmetic, Householder QR (Theorem 10.3.2) applied to  $\begin{pmatrix} \mathbf{0}_{p \times p} \\ \mathbf{X} \end{pmatrix}$  produces the same  $\mathbf{R}$  as modified Gram–Schmidt applied to  $\mathbf{X}$  up to the signs of its rows, and that with the sign convention of Lemma 10.3.1 its reflection vectors are  $\mathbf{v}_j = \begin{pmatrix} \mathbf{e}_j \\ \mathbf{q}_j \end{pmatrix} / \sqrt{2}$ . This is the observation of Björck and Paige (1992), who take  $\alpha = +\|\mathbf{x}\|$  and so obtain  $\begin{pmatrix} -\mathbf{e}_j \\ \mathbf{q}_j \end{pmatrix} / \sqrt{2}$ .

## 10.4 The singular value decomposition

QR is the workhorse when  $\mathbf{X}$  has full column rank. When it does not, or nearly does not, the singular value decomposition tells the whole story: every least squares solution, the one of minimum norm, and, in the smallest singular values, the combinations of the columns that the data cannot resolve. Theorem 1.8.1 established the decomposition, and Proposition 6.10.1 previewed its use for least squares.

### 10.4.1 Least squares through the SVD

Throughout,  $\mathbf{X}$  is  $n \times p$  of rank  $r \geq 1$ , with singular value decomposition  $\mathbf{X} = \sum_{i \leq r} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$  (Theorem 1.8.1). We complete  $\mathbf{u}_1, \dots, \mathbf{u}_r$  and  $\mathbf{v}_1, \dots, \mathbf{v}_r$  to orthonormal bases  $\mathbf{u}_1, \dots, \mathbf{u}_n$  of  $\mathbb{R}^n$  and  $\mathbf{v}_1, \dots, \mathbf{v}_p$  of  $\mathbb{R}^p$ , and write  $c_i = \mathbf{u}_i^\top \mathbf{y}$  for the coordinates of the data in the left singular basis.

**Theorem 10.4.1 — Least squares via the SVD**

- (a)  $\mathbf{b}$  is a least squares estimate iff  $\mathbf{v}_i^\top \mathbf{b} = c_i/\sigma_i$  for  $i = 1, \dots, r$ . The coordinates  $\mathbf{v}_i^\top \mathbf{b}$  for  $i > r$  are arbitrary, and the set of least squares estimates is  $\mathbf{X}^+ \mathbf{y} + \mathcal{N}(\mathbf{X})$  with  $\mathcal{N}(\mathbf{X}) = \text{span}(\mathbf{v}_{r+1}, \dots, \mathbf{v}_p)$ .
- (b)  $\mathbf{X}^+ \mathbf{y} = \sum_{i \leq r} (c_i/\sigma_i) \mathbf{v}_i$  is the unique least squares estimate of smallest Euclidean norm.
- (c)  $\hat{\mathbf{Y}} = \sum_{i \leq r} c_i \mathbf{u}_i$  and  $\text{SSE} = \sum_{i > r} c_i^2 = \|\mathbf{y}\|^2 - \sum_{i \leq r} c_i^2$ .
- (d) If  $r = p$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , the estimate  $\hat{\boldsymbol{\beta}} = \mathbf{X}^+ \mathbf{Y}$  has  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \sum_{i \leq p} \sigma_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$ . In particular  $\text{Var}(\mathbf{v}_i^\top \hat{\boldsymbol{\beta}}) = \sigma^2/\sigma_i^2$ , and the total variance is  $\text{tr Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \sum_i \sigma_i^{-2}$ .

*Proof.* Let  $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_n]$ , an orthogonal matrix. For any  $\mathbf{b}$ ,  $\mathbf{X}\mathbf{b} = \sum_{i \leq r} \sigma_i (\mathbf{v}_i^\top \mathbf{b}) \mathbf{u}_i$ , so the coordinates of  $\mathbf{X}\mathbf{b}$  in the basis  $\mathbf{u}_1, \dots, \mathbf{u}_n$  are  $\sigma_i \mathbf{v}_i^\top \mathbf{b}$  for  $i \leq r$  and zero for  $i > r$ . By orthogonal invariance,

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{U}^\top \mathbf{y} - \mathbf{U}^\top \mathbf{X}\mathbf{b}\|^2 = \sum_{i \leq r} (c_i - \sigma_i \mathbf{v}_i^\top \mathbf{b})^2 + \sum_{i > r} c_i^2. \quad (10.4.1)$$

The second sum does not involve  $\mathbf{b}$ , and the first vanishes exactly when  $\mathbf{v}_i^\top \mathbf{b} = c_i/\sigma_i$  for  $i \leq r$ . This proves (a) and (c), since  $\mathbf{b} = \sum_i (\mathbf{v}_i^\top \mathbf{b}) \mathbf{v}_i$  and  $\mathcal{N}(\mathbf{X})$  is spanned by the  $\mathbf{v}_i$  with  $i > r$  (Theorem 1.8.1). Part (b) is Proposition 6.10.1. In coordinates,  $\|\mathbf{b}\|^2 = \sum_{i \leq p} (\mathbf{v}_i^\top \mathbf{b})^2$ , so the minimum-norm estimate is the one whose free coordinates are zero,  $\sum_{i \leq r} (c_i/\sigma_i) \mathbf{v}_i$ . For (d),  $\hat{\boldsymbol{\beta}} = \sum_i \sigma_i^{-1} \mathbf{v}_i \mathbf{u}_i^\top \mathbf{Y}$ , and  $\text{Cov}(\mathbf{u}_i^\top \mathbf{Y}, \mathbf{u}_j^\top \mathbf{Y}) = \sigma^2 \mathbf{u}_i^\top \mathbf{u}_j$  is  $\sigma^2$  or 0. By Theorem 2.2.1,  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \sum_i \sigma_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$ . Its trace is  $\sigma^2 \sum_i \sigma_i^{-2}$  because  $\|\mathbf{v}_i\| = 1$ . ■

Part (d) is the statistical face of ill-conditioning. The combination  $\mathbf{v}_p^\top \boldsymbol{\beta}$  of the coefficients, along the direction in which the columns of  $\mathbf{X}$  are nearly dependent, is estimated with standard deviation  $\sigma/\sigma_p$ . When  $\sigma_p$  is tiny, the data carry almost no information about that combination, and no algorithm can change this.

```
import numpy as np

def svd_ls(X, y, tol=None):
    """Minimum-norm least squares solution, treating singular values <= tol as zero.
    """
    U, s, Vt = np.linalg.svd(X, full_matrices=False)
    if tol is None:
        tol = max(X.shape) * np.finfo(float).eps * s[0] # the usual default
    r = int(np.sum(s > tol)) # numerical rank
    c = U[:, :r].T @ y # coordinates of the fit
    b = Vt[:r].T @ (c / s[:r])
    sse = y @ y - c @ c
    return b, r, sse, s
```

## 10.4.2 Computing the SVD

The SVD is not computed from the eigenvalues of  $\mathbf{X}^\top \mathbf{X}$ , which has already lost the small singular values (Section 10.1). The standard method (Golub and Kahan 1965; Golub and Reinsch 1970) first applies Householder reflections alternately from the left and the right to reduce  $\mathbf{X}$  to upper bidiagonal form, at about twice the cost of QR, and then drives the superdiagonal to zero by an implicitly shifted QR-type iteration that never forms a cross-product. When  $n \gg p$ , it pays to compute  $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$  first and then an SVD  $\mathbf{R} = \mathbf{U}_R \mathbf{\Sigma} \mathbf{V}^\top$  of the triangle, since  $\mathbf{X} = (\mathbf{Q}_1 \mathbf{U}_R) \mathbf{\Sigma} \mathbf{V}^\top$  (Exercise (B2)). Least squares needs only  $\mathbf{U}^\top \mathbf{y}$ , never  $\mathbf{U}$  itself, which gives the count  $2np^2 + 11p^3$  of Section 10.1 (Golub and Van Loan 2013).

The computed singular values are those of a nearby matrix  $\mathbf{X} + \mathbf{E}$  with  $\|\mathbf{E}\|_2 \leq cu\|\mathbf{X}\|_2$ . The next result turns this into accuracy.

### Proposition 10.4.2 — Singular values are well conditioned

For any matrices  $\mathbf{A}$  and  $\mathbf{E}$  of the same size and every  $k$ ,

$$|\sigma_k(\mathbf{A} + \mathbf{E}) - \sigma_k(\mathbf{A})| \leq \|\mathbf{E}\|_2.$$

*Proof.* For any matrix  $\mathbf{C}$  and any  $\mathbf{B}$  of rank at most  $k - 1$ ,  $\sigma_k(\mathbf{C}) \leq \|\mathbf{C} - \mathbf{B}\|_2$ . When  $k - 1 < \text{rank } \mathbf{C}$  this is the Eckart–Young inequality (Proposition 1.8.2(c)), and otherwise  $\sigma_k(\mathbf{C}) = 0$ . Let  $\mathbf{A}_{k-1}$  be the truncated SVD of  $\mathbf{A}$  with  $k - 1$  terms, or  $\mathbf{A}$  itself if  $\text{rank } \mathbf{A} < k$ , so that  $\|\mathbf{A} - \mathbf{A}_{k-1}\|_2 = \sigma_k(\mathbf{A})$ . Then

$$\sigma_k(\mathbf{A} + \mathbf{E}) \leq \|\mathbf{A} + \mathbf{E} - \mathbf{A}_{k-1}\|_2 \leq \sigma_k(\mathbf{A}) + \|\mathbf{E}\|_2.$$

Exchanging the roles of  $\mathbf{A}$  and  $\mathbf{A} + \mathbf{E}$  gives the other inequality. ■

So a backward stable SVD computes every singular value with *absolute* error of order  $u\sigma_1$ , and a singular value of that size or below is indistinguishable from zero. An exactly rank-deficient design, such as a one-way layout with an intercept and all its indicators, therefore shows a smallest singular value near  $10^{-16}\sigma_1$  rather than zero:  $5 \times 10^{-17}$  times the largest in the script's example.

## 10.4.3 Numerical rank

In floating point, and with data measured to finite precision, the meaningful question is how many singular values are *clearly* nonzero.

### Definition 10.4.1 — Numerical rank

For a tolerance  $\tau > 0$ , the **numerical rank** of  $\mathbf{X}$  is  $r_\tau = \#\{i : \sigma_i > \tau\}$ . The **truncated SVD** with  $k$  terms is  $\mathbf{X}_k = \sum_{i \leq k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ .

By the Eckart–Young inequality,  $r_\tau$  is the least rank of any matrix within spectral distance

$\tau$  of  $\mathbf{X}$  (Exercise (B1)), so  $\tau$  should reflect how well  $\mathbf{X}$  is known. The default in numpy and LAPACK-based software,  $\tau = \max(n, p) \epsilon_M \sigma_1$ , accounts for rounding only; data recorded to six significant figures cannot resolve singular values below about  $10^{-6} \sigma_1$ . The choice matters, because the minimum-norm solution is not a continuous function of  $\mathbf{X}$ .

### Example. (The pseudoinverse jumps)

Let  $\mathbf{x}_1 = (1, 2, 3, 4)^\top$ , let  $\mathbf{d} = (1, -1, -1, 1)^\top$  be orthogonal to it, and let  $\mathbf{X}_t = [\mathbf{x}_1, \mathbf{x}_1 + t\mathbf{d}]$ , with  $\mathbf{y} = (1, 3, 2, 5)^\top$ . For every  $t \neq 0$  the column space is the same plane  $\text{span}(\mathbf{x}_1, \mathbf{d})$ , so the fitted values do not depend on  $t$ . The coefficients do. At  $t = 10^{-6}$  they are  $-249999$  and  $250000$ , because the coefficients must reproduce a fixed multiple of  $\mathbf{d}$  using the small difference  $t\mathbf{d}$  of the two columns. At  $t = 0$  the rank drops to one. The minimum-norm solution jumps to  $0.550$  for both columns, and the fitted values jump too, because the  $\mathbf{d}$ -direction has disappeared from the model. Treating singular values below a tolerance as zero makes the two answers agree. Whether it should is a statistical question: is the difference between the columns signal, or recording error?

```
x1 = np.array([1.0, 2.0, 3.0, 4.0])
d = np.array([1.0, -1.0, -1.0, 1.0])          # orthogonal to x1
yd = np.array([1.0, 3.0, 2.0, 5.0])
for t in [1e-2, 1e-6, 1e-10, 0.0]:
    Xt = np.column_stack([x1, x1 + t * d])    # second column tends to the first
    b = np.linalg.pinv(Xt, rcond=1e-15) @ yd
    print(f"t = {t:7.0e}    b = {np.round(b, 3)}    fit = {np.round(Xt @ b, 3)}")
```

## 10.4.4 Truncated SVD as an estimator

The truncated solution  $\mathbf{b}_k = \sum_{i \leq k} (c_i / \sigma_i) \mathbf{v}_i$  is more than a numerical safeguard. It is a statistical estimator in its own right, and it trades bias for variance.

### Proposition 10.4.3 — Bias and variance of truncated SVD

Suppose  $\text{rank } \mathbf{X} = p$ ,  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , and let  $\mathbf{b}_k = \sum_{i \leq k} (\mathbf{u}_i^\top \mathbf{Y} / \sigma_i) \mathbf{v}_i$  for  $k \leq p$ . Then

$$E(\mathbf{b}_k) = \sum_{i \leq k} (\mathbf{v}_i^\top \boldsymbol{\beta}) \mathbf{v}_i, \quad \text{Cov}(\mathbf{b}_k) = \sigma^2 \sum_{i \leq k} \frac{\mathbf{v}_i \mathbf{v}_i^\top}{\sigma_i^2},$$

$$E\|\mathbf{b}_k - \boldsymbol{\beta}\|^2 = \sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta})^2 + \sigma^2 \sum_{i \leq k} \frac{1}{\sigma_i^2}. \quad (10.4.2)$$

Hence  $\mathbf{b}_k$  has smaller mean squared error than the least squares estimate  $\mathbf{b}_p$  iff  $\sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta})^2 < \sigma^2 \sum_{i > k} \sigma_i^{-2}$ .

*Proof.*  $E(\mathbf{u}_i^\top \mathbf{Y}) = \mathbf{u}_i^\top \mathbf{X}\boldsymbol{\beta} = \sigma_i \mathbf{v}_i^\top \boldsymbol{\beta}$ , which gives the mean. The covariance is part (d) of Theorem 10.4.1 with the sum stopped at  $k$ . For the mean squared error, write  $E\|\mathbf{b}_k - \boldsymbol{\beta}\|^2 = \|E\mathbf{b}_k - \boldsymbol{\beta}\|^2 + \text{tr Cov}(\mathbf{b}_k)$ . The bias  $E\mathbf{b}_k - \boldsymbol{\beta}$  is  $-\sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta}) \mathbf{v}_i$ , with squared length  $\sum_{i > k} (\mathbf{v}_i^\top \boldsymbol{\beta})^2$ . Comparing Eq. 10.4.2 for  $k$  and  $p$  gives the last claim.

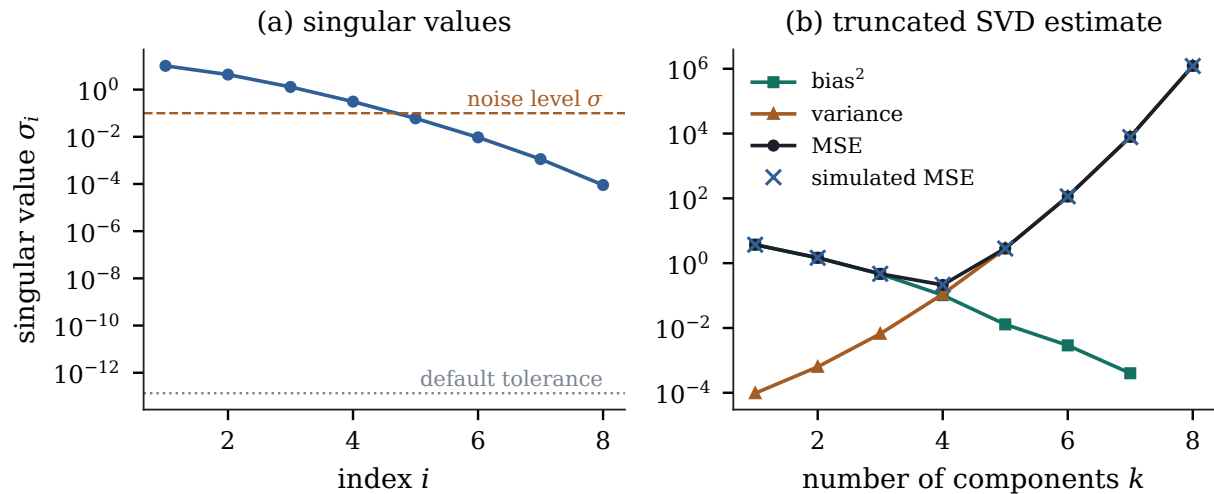


Figure 10.4.1: Truncated SVD for a degree-7 polynomial design. (a) Singular values, with the default numerical-rank tolerance and the noise level  $\sigma$ . (b) Squared bias, variance and mean squared error of the truncated estimate  $\mathbf{b}_k$  (Eq. 10.4.2), with simulated values. Least squares is  $k = 8$ .

Each dropped component costs its squared coefficient  $(\mathbf{v}_i^\top \boldsymbol{\beta})^2$  in bias and saves  $\sigma^2/\sigma_i^2$  in variance. It is worth dropping exactly when the signal along  $\mathbf{v}_i$  is weaker than the noise in its estimate, that is, when  $\sigma_i |\mathbf{v}_i^\top \boldsymbol{\beta}| < \sigma$ . The estimator  $\mathbf{b}_k$  is principal components regression in its uncentred form. Ridge regression replaces the hard cut-off by weights  $\sigma_i^2/(\sigma_i^2 + \lambda)$ . Both are developed in Chapter 27 (Theorem 27.2.2 and Definition 27.3.1), and choosing  $k$  from data is a model-selection problem (Chapter 29).

#### Example. (Truncating a polynomial design)

The powers  $1, t, \dots, t^7$  at 60 equally spaced points in  $[0, 1]$  form a design with  $\kappa(\mathbf{X}) = 1.1 \times 10^5$  and smallest singular value  $9.0 \times 10^{-5}$ . Take  $\sigma = 0.1$  and a coefficient vector whose components along  $\mathbf{v}_1, \dots, \mathbf{v}_8$  decay  $(2, -1.5, 1, -0.6, 0.3, -0.1, 0.05, -0.02)$ , as they do for smooth regression functions. By Eq. 10.4.2, least squares has mean squared error 1231055. The best truncation keeps 4 components and has mean squared error 0.21, made up of squared bias 0.10 and variance 0.11. A simulation with 4000 replicates agrees with the formula at every  $k$  (Figure 10.4.1). The default numerical tolerance treats the design as having full rank, which it does to working precision. The statistical “effective rank” is set by the noise level instead.

```
n, p, sigma = 60, 8, 0.1
t = np.linspace(0, 1, n)
X = np.vander(t, p, increasing=True) # 1, t, ..., t^7: badly conditioned
U, s, Vt = np.linalg.svd(X, full_matrices=False)
a = np.array([2.0, -1.5, 1.0, -0.6, 0.3, -0.1, 0.05, -0.02]) # decaying components
# v_i' beta
beta = Vt.T @ a
print("condition number:", s[0] / s[-1])
print("singular values:", np.array2string(s, precision=2))
```

```

bias2 = np.array([np.sum((Vt[k:] @ beta) ** 2) for k in range(1, p + 1)]) # //(I -
                V_k V_k') beta//2
var = np.array([sigma ** 2 * np.sum(1 / s[:k] ** 2) for k in range(1, p + 1)])
mse = bias2 + var
print("k      bias^2      variance      MSE")
for k in range(p):
    print(f"{k + 1}  {bias2[k]:10.3e}  {var[k]:10.3e}  {mse[k]:10.3e}")

```

*Idea.* The SVD gives every least squares solution, the precision  $\sigma/\sigma_i$  of each combination  $\mathbf{v}_i^\top \boldsymbol{\beta}$ , and, relative to a tolerance, how many combinations are determined at all.

## 10.4.5 Exercises

### 10.4.5.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{X} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 0 \end{pmatrix}$  and  $\mathbf{y} = (1, 3, 5)^\top$ . Find the SVD of  $\mathbf{X}$ , the set of least squares estimates, the minimum-norm estimate and the SSE.

*Solution.*  $\mathbf{X} = 2\mathbf{u}_1\mathbf{v}_1^\top$  with  $\mathbf{u}_1 = (1, 1, 0)^\top/\sqrt{2}$ ,  $\mathbf{v}_1 = (1, 1)^\top/\sqrt{2}$ ,  $\sigma_1 = 2$  and  $r = 1$ . Here  $c_1 = \mathbf{u}_1^\top \mathbf{y} = 4/\sqrt{2}$ , so the least squares estimates are the  $\mathbf{b}$  with  $\mathbf{v}_1^\top \mathbf{b} = c_1/\sigma_1 = \sqrt{2}$ , that is,  $b_1 + b_2 = 2$ . The minimum-norm estimate is  $\sqrt{2}\mathbf{v}_1 = (1, 1)^\top$ . The fitted vector is  $c_1\mathbf{u}_1 = (2, 2, 0)^\top$ , and  $\text{SSE} = \|\mathbf{y}\|^2 - c_1^2 = 35 - 8 = 27$ , which is  $(1-2)^2 + (3-2)^2 + 5^2$ .

### 10.4.5.2 B. Practice

#### Exercise B1.

Show that if  $\tau$  is not a singular value of  $\mathbf{X}$ , then  $r_\tau = \min\{\text{rank } \mathbf{B} : \|\mathbf{X} - \mathbf{B}\|_2 \leq \tau\}$ , and that the minimum is attained by the truncated SVD  $\mathbf{X}_{r_\tau}$ .

*Solution.*  $\|\mathbf{X} - \mathbf{X}_k\|_2 = \sigma_{k+1}$ . For  $k = r_\tau$ ,  $\sigma_{k+1} \leq \tau$  (with  $\sigma_{k+1} = 0$  if  $k = \text{rank } \mathbf{X}$ ), so  $\mathbf{X}_{r_\tau}$  is within  $\tau$  and has rank  $r_\tau$ . If  $\text{rank } \mathbf{B} = k < r_\tau$ , then by Proposition 1.8.2(c)  $\|\mathbf{X} - \mathbf{B}\|_2 \geq \sigma_{k+1} \geq \sigma_{r_\tau} > \tau$ . So no matrix of smaller rank is within  $\tau$ .

#### Exercise B2.

Let  $\mathbf{X} = \mathbf{Q}_1\mathbf{R}$  be a thin QR and  $\mathbf{R} = \mathbf{U}_R\boldsymbol{\Sigma}\mathbf{V}^\top$  an SVD of the triangle. Show that  $\mathbf{X} = (\mathbf{Q}_1\mathbf{U}_R)\boldsymbol{\Sigma}\mathbf{V}^\top$  is an SVD of  $\mathbf{X}$ , and that the minimum-norm least squares solution can be computed from  $\mathbf{R}$  and  $\mathbf{Q}_1^\top \mathbf{y}$  alone. Why is this attractive when  $n$  is much larger than  $p$ ?

#### Exercise B3.

Show that  $\mathbf{b}_k$  minimizes  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|$  over  $\mathbf{b} \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , and that it is the minimum-norm least squares estimate for the matrix  $\mathbf{X}_k$ . Interpret  $\mathbf{b}_k$  as least squares after



deleting the directions in which  $\mathbf{X}$  is nearly degenerate.

### 10.4.5.3 C. Going deeper

#### Exercise C1.

Show that the ridge estimate  $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$  equals  $\sum_i \frac{\sigma_i}{\sigma_i^2 + \lambda} c_i \mathbf{v}_i$ . Derive its bias and variance in the form of Proposition 10.4.3, and show that for every  $\boldsymbol{\beta}$  there is a  $\lambda > 0$  (depending on  $\boldsymbol{\beta}$  and  $\sigma^2$ ) for which its mean squared error is below that of least squares.

## 10.5 Conditioning and perturbation bounds

Sections 10.2 to 10.4 were about algorithms. This section is about the problem itself: if the data  $(\mathbf{X}, \mathbf{y})$  change a little, how much can the least squares solution and the residuals change? The answer does not depend on the algorithm. Combined with backward stability through Eq. 10.1.2, it predicts the accuracy of any computed solution.

### 10.5.1 The perturbation theorem

Throughout this section  $\mathbf{X}$  is  $n \times p$  with full column rank,  $\hat{\boldsymbol{\beta}} = \mathbf{X}^+ \mathbf{y}$  is its least squares solution (assumed nonzero) and  $\hat{\boldsymbol{\epsilon}} = \mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}$  its residual. Write  $\kappa = \kappa(\mathbf{X}) = \sigma_1 / \sigma_p$ . The perturbed data are  $\tilde{\mathbf{X}} = \mathbf{X} + \mathbf{E}$  and  $\tilde{\mathbf{y}} = \mathbf{y} + \mathbf{f}$ , with

$$\|\mathbf{E}\|_2 \leq \epsilon \|\mathbf{X}\|_2, \quad \|\mathbf{f}\| \leq \epsilon \|\mathbf{y}\|. \quad (10.5.1)$$

The solution and residual for the perturbed data are  $\tilde{\mathbf{b}}$  and  $\tilde{\boldsymbol{\epsilon}}$ . A single number measures how badly the model fits relative to the size of the problem:

$$\eta = \frac{\|\hat{\boldsymbol{\epsilon}}\|}{\|\mathbf{X}\|_2 \|\hat{\boldsymbol{\beta}}\|}.$$

Since  $\|\mathbf{X} \hat{\boldsymbol{\beta}}\| \leq \|\mathbf{X}\|_2 \|\hat{\boldsymbol{\beta}}\|$ ,  $\eta \leq \tan \theta$ , where  $\theta$  is the angle between  $\mathbf{y}$  and  $\mathcal{C}(\mathbf{X})$ . A model that fits well has small  $\eta$ .

#### Theorem 10.5.1 — Perturbation of least squares

Under Eq. 10.5.1 with  $\kappa \epsilon < 1$ , the matrix  $\tilde{\mathbf{X}}$  has full column rank, and

$$\frac{\|\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}}\|}{\|\hat{\boldsymbol{\beta}}\|} \leq \frac{\kappa \epsilon}{1 - \kappa \epsilon} (2 + \eta) + \frac{\kappa^2 \epsilon}{(1 - \kappa \epsilon)^2} \eta, \quad (10.5.2)$$

$$\frac{\|\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}}\|}{\|\mathbf{y}\|} \leq \epsilon \left( 1 + \frac{2\kappa}{1 - \kappa \epsilon} \right). \quad (10.5.3)$$

To first order in  $\epsilon$ , the coefficient bound is  $\epsilon(2\kappa + \kappa(\kappa + 1)\eta)$ .

*Proof.* By [Proposition 10.4.2](#),  $\sigma_p(\tilde{\mathbf{X}}) \geq \sigma_p - \|\mathbf{E}\|_2 \geq \sigma_p(1 - \kappa\epsilon) > 0$ , so  $\tilde{\mathbf{X}}$  has full column rank and  $\tilde{\mathbf{X}}^+ \tilde{\mathbf{X}} = \mathbf{I}$ . Hence

$$\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ \tilde{\mathbf{y}} - \tilde{\mathbf{X}}^+ \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ (\tilde{\mathbf{y}} - \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}}) = \tilde{\mathbf{X}}^+ (\hat{\boldsymbol{\epsilon}} + \mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}).$$

The residual term simplifies because  $\mathbf{X}^\top \hat{\boldsymbol{\epsilon}} = \mathbf{0}$ :  $\tilde{\mathbf{X}}^+ \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} (\mathbf{X} + \mathbf{E})^\top \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}$ . So

$$\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}} = \tilde{\mathbf{X}}^+ (\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}) + (\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}. \quad (10.5.4)$$

Now  $\|\tilde{\mathbf{X}}^+\|_2 = 1/\sigma_p(\tilde{\mathbf{X}})$  and  $\|(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}\|_2 = 1/\sigma_p(\tilde{\mathbf{X}})^2$  ([Theorem 1.8.1](#)). Also  $\|\mathbf{y}\| \leq \|\mathbf{X} \hat{\boldsymbol{\beta}}\| + \|\hat{\boldsymbol{\epsilon}}\| \leq \sigma_1 \|\hat{\boldsymbol{\beta}}\| + \|\hat{\boldsymbol{\epsilon}}\|$ , so  $\|\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}\| \leq \epsilon(\|\mathbf{y}\| + \sigma_1 \|\hat{\boldsymbol{\beta}}\|) \leq \epsilon \sigma_1 \|\hat{\boldsymbol{\beta}}\| (2 + \eta)$ . Therefore

$$\|\tilde{\mathbf{b}} - \hat{\boldsymbol{\beta}}\| \leq \frac{\epsilon \sigma_1 \|\hat{\boldsymbol{\beta}}\| (2 + \eta)}{\sigma_p(1 - \kappa\epsilon)} + \frac{\epsilon \sigma_1 \|\hat{\boldsymbol{\epsilon}}\|}{\sigma_p^2(1 - \kappa\epsilon)^2}.$$

Dividing by  $\|\hat{\boldsymbol{\beta}}\|$  and using  $\|\hat{\boldsymbol{\epsilon}}\| = \eta \sigma_1 \|\hat{\boldsymbol{\beta}}\|$  gives [Eq. 10.5.2](#).

For the residual,  $\tilde{\boldsymbol{\epsilon}} = (\mathbf{I} - \tilde{\mathbf{M}}) \tilde{\mathbf{y}}$ , where  $\tilde{\mathbf{M}}$  projects onto  $\mathcal{C}(\tilde{\mathbf{X}})$ . Write  $\tilde{\mathbf{y}} = \tilde{\mathbf{X}} \hat{\boldsymbol{\beta}} + (\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}) + \hat{\boldsymbol{\epsilon}}$ . The first term is annihilated by  $\mathbf{I} - \tilde{\mathbf{M}}$ , and  $\tilde{\mathbf{M}} \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^+)^T \tilde{\mathbf{X}}^\top \hat{\boldsymbol{\epsilon}} = (\tilde{\mathbf{X}}^+)^T \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}$  as before. So

$$\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}} = (\mathbf{I} - \tilde{\mathbf{M}})(\mathbf{f} - \mathbf{E} \hat{\boldsymbol{\beta}}) - (\tilde{\mathbf{X}}^+)^T \mathbf{E}^\top \hat{\boldsymbol{\epsilon}}.$$

The first term has norm at most  $\epsilon(\|\mathbf{y}\| + \sigma_1 \|\hat{\boldsymbol{\beta}}\|) \leq \epsilon(1 + \kappa)\|\mathbf{y}\|$ , because  $\|\hat{\boldsymbol{\beta}}\| = \|\mathbf{X}^+ \mathbf{y}\| \leq \|\mathbf{y}\|/\sigma_p$ . The second has norm at most  $\epsilon \sigma_1 \|\hat{\boldsymbol{\epsilon}}\|/(\sigma_p(1 - \kappa\epsilon)) \leq \epsilon \kappa \|\mathbf{y}\|/(1 - \kappa\epsilon)$ . Adding and using  $1 + \kappa + \kappa/(1 - \kappa\epsilon) \leq 1 + 2\kappa/(1 - \kappa\epsilon)$  gives the bound [Eq. 10.5.3](#). ■

The exact identity [Eq. 10.5.4](#) shows where each term comes from. Perturbing the data along  $\mathcal{C}(\mathbf{X})$  moves the solution through  $\tilde{\mathbf{X}}^+$ , with amplification  $1/\sigma_p$ . This is the familiar  $\kappa$  of linear systems ([Eq. 1.8.3](#)). The second term is new. A perturbation  $\mathbf{E}$  that tilts a column of  $\mathbf{X}$  towards the residual direction changes the subspace, and the residual, however large, then “leaks” into the solution through  $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$ , with amplification  $1/\sigma_p^2$ . The residuals themselves, and hence the fitted values, are only  $\kappa$ -sensitive.

The  $\kappa^2$  term is not an artefact of the proof.

#### Example. (The squared condition number is attained)

Let  $\delta = 10^{-3}$  and

$$\mathbf{X} = \begin{pmatrix} 1 & 0 \\ 0 & \delta \\ 0 & 0 \end{pmatrix}, \quad \mathbf{y} = \begin{pmatrix} 0 \\ \delta \\ 1 \end{pmatrix},$$

so that  $\hat{\boldsymbol{\beta}} = (0, 1)^\top$ ,  $\hat{\boldsymbol{\epsilon}} = (0, 0, 1)^\top$ ,  $\kappa = 1000$  and  $\eta = 1$ . Perturb only the  $(3, 2)$  entry of  $\mathbf{X}$ , by  $\epsilon = 10^{-10}$ . This tilts the weak second column towards the residual. Its coefficient becomes  $(\delta^2 + \epsilon)/(\delta^2 + \epsilon^2)$ , and the relative change in  $\hat{\boldsymbol{\beta}}$  is  $1.00 \times 10^{-4}$ , equal

to  $\kappa^2\epsilon = 1.00 \times 10^{-4}$ . A perturbation of relative size  $10^{-10}$  has changed the answer in the fourth digit.

```
import numpy as np
import statsmodels.api as sm

def ls(X, y):
    return np.linalg.lstsq(X, y, rcond=None)[0]

delta, eps = 1e-3, 1e-10
X = np.array([[1.0, 0.0], [0.0, delta], [0.0, 0.0]])
y = np.array([0.0, delta, 1.0])          # b = (0, 1), residual e = (0, 0, 1)
E = np.array([[0.0, 0.0], [0.0, 0.0], [0.0, eps]]) # ||E|| = eps ||X||
b = ls(X, y)
b_tilde = ls(X + E, y)
print("kappa =", np.linalg.cond(X))
print("relative change in b:", np.linalg.norm(b_tilde - b) / np.linalg.norm(b))
print("kappa^2 * eps      :", np.linalg.cond(X) ** 2 * eps)
```

The script also draws 300 random problems with random perturbations and checks [Eq. 10.5.2](#) on each. The worst ratio of the actual change to the bound is 0.31.

## 10.5.2 What the theorem says about algorithms

A backward stable algorithm (Householder QR, Givens QR, modified Gram–Schmidt on the augmented matrix, the SVD) returns the exact solution for data perturbed as in [Eq. 10.5.1](#) with  $\epsilon \approx cu$ , where  $c$  grows modestly with the dimensions. Its relative error is therefore about

$$cu(\kappa + \kappa^2\eta).$$

The normal equations incur a relative error of about  $cu\kappa^2$  whatever  $\eta$  is, because forming  $\mathbf{X}^\top \mathbf{X}$  already commits it ([Proposition 10.1.3](#)). Two regimes follow:

- **Small residual** ( $\eta \ll 1/\kappa$ ): QR gains a full factor  $\kappa$  over the normal equations. This is the regime of the experiment in [Section 6.10](#), where  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$  exactly.
- **Large residual** ( $\eta$  of order one): every method, however stable, loses about  $2\log_{10}\kappa$  digits, because the problem is that sensitive. The normal equations are then worse only by a modest factor.

[Figure 10.5.1](#) shows both regimes on test problems with prescribed condition numbers. With zero residual and  $\kappa = 10^6$ , the normal equations give relative error  $1.1 \times 10^{-5}$  and QR  $7.8 \times 10^{-12}$ . With  $\eta = 1$ , that is  $\|\hat{\mathbf{e}}\| = \|\mathbf{X}\|_2 \|\boldsymbol{\beta}\|$ , QR gives  $1.9 \times 10^{-7}$  and the SVD  $1.9 \times 10^{-7}$ . The  $\kappa^2$  term has caught up with the stable methods.

```
def by_cholesky(X, y):
    L = np.linalg.cholesky(X.T @ X)
    return np.linalg.solve(L.T, np.linalg.solve(L, X.T @ y))

def by_qr(X, y):
    Q, R = np.linalg.qr(X)
    return np.linalg.solve(R, Q.T @ y)

def by_svd(X, y):
    U, s, Vt = np.linalg.svd(X, full_matrices=False)
```

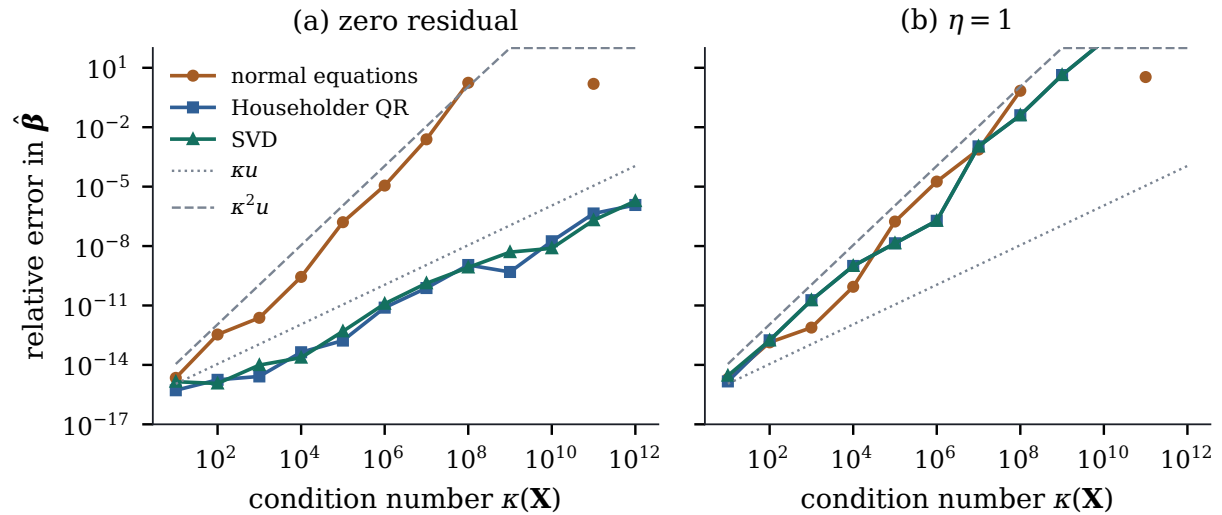


Figure 10.5.1: Relative error of the coefficients computed by the normal equations (Cholesky), Householder QR and the SVD, against  $\kappa(\mathbf{X})$ . (a) Consistent data,  $\hat{\varepsilon} = \mathbf{0}$ : QR and SVD track  $\kappa u$ , the normal equations track  $\kappa^2 u$ . (b) A large residual,  $\eta = 1$ : all three track  $\kappa^2 u$ , as Theorem 10.5.1 predicts.

```
return Vt.T @ ((U.T @ y) / s)
```

In regression  $\eta$  is rarely tiny, but QR keeps an advantage: its  $\kappa^2 \eta$  term is a property of the problem, not of the method.

### 10.5.3 Column scaling

The normwise condition number depends on the units of the columns: measuring one regressor in millimetres instead of metres can multiply  $\kappa(\mathbf{X})$  by up to 1000 (Exercise (B2)), although the fit is unchanged. Equalizing the column lengths is essentially the best diagonal scaling.

#### Proposition 10.5.2 — Equilibrating the columns

Let  $\mathbf{X}$  have full column rank and let  $\mathbf{D}_0 = \text{diag}(1/\|\mathbf{x}_1\|, \dots, 1/\|\mathbf{x}_p\|)$ . For every nonsingular diagonal  $\mathbf{D}$ ,

$$\kappa(\mathbf{X}\mathbf{D}_0) \leq \sqrt{p} \kappa(\mathbf{X}\mathbf{D}).$$

*Proof.* The columns of  $\mathbf{X}\mathbf{D}_0$  have unit length, so  $\sigma_1(\mathbf{X}\mathbf{D}_0) \leq \|\mathbf{X}\mathbf{D}_0\|_F = \sqrt{p}$ . For the smallest singular value, let  $\mathbf{z}$  be any nonzero vector and put  $\mathbf{w} = \mathbf{D}^{-1}\mathbf{D}_0\mathbf{z}$ . Then  $\|\mathbf{X}\mathbf{D}_0\mathbf{z}\| = \|\mathbf{X}\mathbf{D}\mathbf{w}\| \geq \sigma_p(\mathbf{X}\mathbf{D})\|\mathbf{w}\|$  and  $\|\mathbf{w}\| \geq \min_j (1/(|d_j|\|\mathbf{x}_j\|))\|\mathbf{z}\|$ . The largest column length of  $\mathbf{X}\mathbf{D}$ ,  $\max_j |d_j|\|\mathbf{x}_j\|$ , is at most  $\sigma_1(\mathbf{X}\mathbf{D})$ , because the columns are images of unit vectors. So  $\sigma_p(\mathbf{X}\mathbf{D}_0) \geq \sigma_p(\mathbf{X}\mathbf{D})/\sigma_1(\mathbf{X}\mathbf{D}) = 1/\kappa(\mathbf{X}\mathbf{D})$ , and  $\kappa(\mathbf{X}\mathbf{D}_0) \leq \sqrt{p} \kappa(\mathbf{X}\mathbf{D})$ . ■

This is due to van der Sluis (1969). What makes it more than a remark about units is the form of the backward errors. Householder QR has a *columnwise* backward error (Theorem 10.3.2(c)). Rescaling the columns rescales the perturbations with them, so the computed solution is as

accurate as if the columns had been equilibrated first. The rounding errors in  $\mathbf{X}^\top \mathbf{X}$  are entrywise small relative to  $|\mathbf{X}|^\top |\mathbf{X}|$  (Proposition 10.1.3), which has the same property. Both methods therefore behave according to the condition number of the *scaled* matrix,  $\kappa(\mathbf{XD}_0)$ , whatever the units. The rule of thumb should be applied to  $\kappa(\mathbf{XD}_0)$ , not to  $\kappa(\mathbf{X})$  as recorded.

### Example. (Longley's data, exactly)

Longley (1967) built his macroeconomic data to test regression programs. With an intercept and six regressors,  $\kappa(\mathbf{X}) = 4.86 \times 10^9$  (Section 6.10; the data are described in Example (Macroeconomic series)). The data are published as decimals, so the exact least squares solution can be computed in rational arithmetic. The script does so and counts the correct significant digits of the worst coefficient for each method:

| Method                                                    | Correct digits |
|-----------------------------------------------------------|----------------|
| Normal equations (Cholesky)                               | 7.2            |
| Householder QR                                            | 10.9           |
| SVD                                                       | 10.8           |
| statsmodels OLS                                           | 10.8           |
| Normal equations after scaling the columns to unit length | 7.0            |

The raw  $\kappa$  predicts about 6 digits from QR and none from the normal equations (Section 6.10). Both predictions are far too pessimistic. After scaling the columns to unit length,  $\kappa(\mathbf{XD}_0) = 4.33 \times 10^4$ . Then  $\kappa u \approx 5 \times 10^{-12}$  predicts about eleven digits from QR, and  $\kappa^2 u \approx 2 \times 10^{-7}$  about seven from the normal equations. That is what is observed. Scaling the columns explicitly before forming the normal equations changes nothing, as the argument above predicts. Centring as well reduces the condition number to  $1.11 \times 10^2$ . That is a reparameterization rather than a scaling (Theorem 6.7.1), and what remains is genuine collinearity among the regressors (Chapter 26).

```
lon = sm.datasets.longley.load_pandas()
XL = sm.add_constant(lon.exog.to_numpy())
yL = lon.endog.to_numpy()
names = ["const"] + list(lon.exog.columns)
print("kappa(X) as recorded:", f"{np.linalg.cond(XL):.3e}")
```

## 10.5.4 Numerical error and statistical error

The perturbation theorem applies to any perturbation of the data, including the rounding of recorded values, which is far above  $u$ . Longley's GNP deflator is given to one decimal place and the other series to whole units. Perturbing each regressor entry (the year is exact) uniformly within half a unit of its last recorded digit, 1000 times, the coefficients vary with standard deviations between 0.005 and 0.024 of their standard errors. The numerical error of the normal equations, the least accurate method, is at most  $4 \times 10^{-8}$  standard errors. Sampling error dominates, then comes the rounding of the published data, and the arithmetic is last by far. Beaton, Rubin and Barone (1976) made this point about Longley's data. A problem can be numerically delicate and statistically ordinary, or the reverse.

```
rng = np.random.default_rng(106)
```

```

half_unit = np.array([0.0, 0.05, 0.5, 0.5, 0.5, 0.5, 0.0]) # half a unit in the
    last digit
fitL = sm.OLS(yL, XL).fit() # year and constant are
    exact
reps = 1000
shifts = []
for _ in range(reps):
    Xp = XL + rng.uniform(-1, 1, size=XL.shape) * half_unit
    shifts.append(ls(Xp, yL) - fitL.params)
spread = np.std(shifts, axis=0)
ratio = spread / fitL.bse
for name, r in zip(names, ratio):
    print(f"{name:8s} sd of coefficient under rounding / standard error = {r:.2f}")

```

*Idea.* Least squares has a  $\kappa$  sensitivity, and a  $\kappa^2\eta$  one when the residual is large. Stable algorithms add nothing beyond this; the normal equations add  $\kappa^2$  regardless. Measure  $\kappa$  after equilibrating the columns.

## 10.5.5 Exercises

### 10.5.5.1 A. Check your understanding

#### Exercise A1.

In [Example \(The squared condition number is attained\)](#), find  $\tilde{\mathbf{b}}$  exactly and show that the relative change is  $\epsilon(1 - \epsilon)/(\delta^2 + \epsilon^2)$ . For which  $\epsilon$  is it close to  $\kappa^2\epsilon$ ?

### 10.5.5.2 B. Practice

#### Exercise B1.

Deduce from [Eq. 10.5.3](#) that the fitted values satisfy  $\|\tilde{\mathbf{X}}\tilde{\mathbf{b}} - \mathbf{X}\hat{\boldsymbol{\beta}}\| \leq \epsilon(2 + 2\kappa/(1 - \kappa\epsilon))\|\mathbf{y}\|$ . Why are fitted values never  $\kappa^2$ -sensitive, even though coefficients can be?

*Solution.*  $\tilde{\mathbf{X}}\tilde{\mathbf{b}} = \tilde{\mathbf{y}} - \tilde{\boldsymbol{\epsilon}}$  and  $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{y} - \hat{\boldsymbol{\epsilon}}$ , so the difference is  $\mathbf{f} - (\tilde{\boldsymbol{\epsilon}} - \hat{\boldsymbol{\epsilon}})$ , with norm at most  $\epsilon\|\mathbf{y}\| + \epsilon(1 + 2\kappa/(1 - \kappa\epsilon))\|\mathbf{y}\|$ . The  $\kappa^2$  term in the coefficients comes from  $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1} \mathbf{E}^\top \tilde{\boldsymbol{\epsilon}}$ , a change of coordinates *within* the slightly rotated column space. Multiplying by  $\tilde{\mathbf{X}}$  turns the factor  $1/\sigma_p^2$  into  $1/\sigma_p$ . A direction that the columns barely span is a direction in which large coefficient changes produce only small changes in the fit.

#### Exercise B2.

Show that for consistent data ( $\hat{\boldsymbol{\epsilon}} = \mathbf{0}$ ) the bound reduces to  $2\kappa\epsilon/(1 - \kappa\epsilon)$ . Construct a perturbation  $\mathbf{f}$  of  $\mathbf{y}$  alone that changes  $\hat{\boldsymbol{\beta}}$  by a relative amount  $\kappa\epsilon$ , so that the bound is attained up to a factor of two.

### 10.5.5.3 C. Going deeper

**Exercise C1.**

Take data with  $\hat{\beta} = \mathbf{v}_p$  and  $\hat{\epsilon} \neq \mathbf{0}$ , and the perturbation  $\mathbf{E} = \epsilon \sigma_1 \hat{\epsilon} \mathbf{v}_p^\top / \|\hat{\epsilon}\|$ , with  $\mathbf{f} = \mathbf{0}$ . Use Eq. 10.5.4 to show that, to first order in  $\epsilon$ ,  $\tilde{\mathbf{b}} - \hat{\beta} = \epsilon \sigma_1 \|\hat{\epsilon}\| \sigma_p^{-2} \mathbf{v}_p$ , so that the relative change is  $\epsilon \kappa^2 \eta$ . Compare with the first-order bound  $\epsilon(2\kappa + \kappa(\kappa + 1)\eta)$ .

## 10.6 Updating and downdating

A fitted regression often has to be changed slightly: new observations arrive, a suspicious case is set aside, a window slides along a time series, a variable is added or removed. Refitting from scratch costs  $O(np^2)$  each time. Modifying the factorization costs  $O(p^2)$  for a change of one row, and  $O(np)$  for a change of one column. This section develops those modifications and uses them to compute every leave-one-out quantity from a single fit.

### 10.6.1 Adding and deleting observations

Adding a row is part (c) of Theorem 10.3.4:  $p$  rotations fold the new row into the triangle. Removing a row is harder, because orthogonal transformations can only combine rows that are present. The trick is to run the addition backwards.

#### Theorem 10.6.1 — Updating and downdating a triangular factor

Let  $\mathbf{T}$  be an  $m \times m$  nonsingular upper triangular matrix with  $\mathbf{T}^\top \mathbf{T} = \mathbf{A}$ , and let  $\mathbf{z} \in \mathbb{R}^m$ .

- (a) (*Adding a row.*) There are  $m$  rotations, the  $k$ th acting on row  $k$  of  $\mathbf{T}$  and on an extra row, that turn  $\begin{pmatrix} \mathbf{T} \\ \mathbf{z}^\top \end{pmatrix}$  into  $\begin{pmatrix} \mathbf{T}_+ \\ \mathbf{0}^\top \end{pmatrix}$ , with  $\mathbf{T}_+$  upper triangular and  $\mathbf{T}_+^\top \mathbf{T}_+ = \mathbf{A} + \mathbf{z}\mathbf{z}^\top$ .
- (b) (*Deleting a row.*) Let  $\mathbf{a}$  solve  $\mathbf{T}^\top \mathbf{a} = \mathbf{z}$ . The matrix  $\mathbf{A} - \mathbf{z}\mathbf{z}^\top$  is positive definite iff  $\|\mathbf{a}\| < 1$ . In that case put  $\alpha = (1 - \|\mathbf{a}\|^2)^{1/2}$ . Choose rotations  $\mathbf{G}_m, \dots, \mathbf{G}_1$ , with  $\mathbf{G}_k$  acting on coordinates  $k$  and  $m+1$ , such that  $\mathbf{G}_1 \cdots \mathbf{G}_m (\mathbf{a}^\top, \alpha)^\top = (\mathbf{0}^\top, 1)^\top$ . Then

$$\mathbf{G}_1 \cdots \mathbf{G}_m \begin{pmatrix} \mathbf{T} \\ \mathbf{0}^\top \end{pmatrix} = \begin{pmatrix} \mathbf{T}_- \\ \mathbf{z}^\top \end{pmatrix}$$

with  $\mathbf{T}_-$  upper triangular and  $\mathbf{T}_-^\top \mathbf{T}_- = \mathbf{A} - \mathbf{z}\mathbf{z}^\top$ .

- (c) Each operation costs  $O(m^2)$  flops:  $3m^2$  for (a), and  $3m^2$  plus a triangular solve of  $m^2$  for (b).

*Proof.*

- (a) is Theorem 10.3.4(c).
- (b)  $\mathbf{A} - \mathbf{z}\mathbf{z}^\top = \mathbf{T}^\top (\mathbf{I} - \mathbf{a}\mathbf{a}^\top) \mathbf{T}$ . The matrix  $\mathbf{I} - \mathbf{a}\mathbf{a}^\top$  has eigenvalue  $1 - \|\mathbf{a}\|^2$  on  $\mathbf{a}$  and 1 on  $\mathbf{a}^\perp$ . So it is positive definite iff  $\|\mathbf{a}\| < 1$ , and then so is  $\mathbf{A} - \mathbf{z}\mathbf{z}^\top$  (Proposition 1.7.5(c)). The rotations exist:  $\mathbf{G}_m$  combines  $a_m$  with  $\alpha$  and leaves 0 in position  $m$  and  $(a_m^2 + \alpha^2)^{1/2}$  in position  $m+1$ . Then  $\mathbf{G}_{m-1}$  does the same with

$a_{m-1}$ , and so on (Theorem 10.3.4(a)). The final vector has norm  $(\|\mathbf{a}\|^2 + \alpha^2)^{1/2} = 1$  and is zero except in the last position. That entry is positive, so it is 1.

Apply the same rotations, in the same order, to  $\begin{pmatrix} \mathbf{T} \\ \mathbf{0}^\top \end{pmatrix}$ . Before  $\mathbf{G}_k$  is applied, the extra row is zero in columns  $1, \dots, k$ , because it has only been mixed with rows  $k+1, \dots, m$  of  $\mathbf{T}$ , which are zero there. Row  $k$  of  $\mathbf{T}$  is zero in columns  $1, \dots, k-1$ . So  $\mathbf{G}_k$  produces a new row  $k$  that is zero before column  $k$ , and the result is  $\begin{pmatrix} \mathbf{T}_- \\ \mathbf{w}^\top \end{pmatrix}$  with  $\mathbf{T}_-$  upper triangular. Write  $\mathbf{G} = \mathbf{G}_1 \cdots \mathbf{G}_m$ , which is orthogonal. Then

$$\mathbf{z} = \mathbf{T}^\top \mathbf{a} = \begin{pmatrix} \mathbf{T} \\ \mathbf{0}^\top \end{pmatrix}^\top \begin{pmatrix} \mathbf{a} \\ \alpha \end{pmatrix} = \left( \mathbf{G} \begin{pmatrix} \mathbf{T} \\ \mathbf{0}^\top \end{pmatrix} \right)^\top \mathbf{G} \begin{pmatrix} \mathbf{a} \\ \alpha \end{pmatrix} = \begin{pmatrix} \mathbf{T}_- \\ \mathbf{w}^\top \end{pmatrix}^\top \begin{pmatrix} \mathbf{0} \\ 1 \end{pmatrix} = \mathbf{w},$$

and in the same way  $\mathbf{T}^\top \mathbf{T} = \mathbf{T}_-^\top \mathbf{T}_- + \mathbf{w}\mathbf{w}^\top$ . Hence  $\mathbf{T}_-^\top \mathbf{T}_- = \mathbf{A} - \mathbf{z}\mathbf{z}^\top$ .

(c) The rotations act on rows of length at most  $m$ , and there are  $m$  of them. ■

If  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  and  $\mathbf{z} = \mathbf{x}_{(i)}$ , then  $\|\mathbf{a}\|^2 = \mathbf{z}^\top \mathbf{A}^{-1} \mathbf{z}$  is the leverage  $h_{ii}$  (Definition 6.8.1): a case can be deleted exactly when the others still determine every coefficient. DOWNDATING the augmented triangle of  $[\mathbf{X}, \mathbf{y}]$  (Proposition 10.2.2) gives the fit without case  $i$ , with the square root of the new SSE in the corner. DOWNDATING is less stable than updating:  $1 - \|\mathbf{a}\|^2$  cancels when  $h_{ii} \approx 1$ , and the relative error can reach about  $u/(1 - h_{ii})$  (Stewart 1979).

```
import numpy as np
import statsmodels.api as sm

def givens(a, b):
    """c, s, r with [[c, s], [-s, c]] @ [a, b] = [r, 0]."""
    if b == 0.0:
        return 1.0, 0.0, a
    r = np.hypot(a, b)
    return a / r, b / r, r

def update(T, z):
    """Triangle of [A; z'] from the triangle T of A: rotate the new row into T."""
    T, z = T.copy(), np.array(z, dtype=float)
    for k in range(len(z)):
        c, s, r = givens(T[k, k], z[k])
        Tk, zk = T[k, k:].copy(), z[k:].copy()
        T[k, k:], z[k:] = c * Tk + s * zk, -s * Tk + c * zk
    return T

def downdate(T, z):
    """Triangle of A with the row z' removed, from the triangle T of A (T'T = A'A)."""
    m = len(z)
    a = np.linalg.solve(T.T, z)          # T'a = z; ||a||^2 is the leverage of z
    alpha2 = 1.0 - a @ a
    if alpha2 <= 0:
        raise np.linalg.LinAlgError("the row cannot be removed (leverage >= 1)")
    alpha = np.sqrt(alpha2)
    T, w = T.copy(), np.zeros(m)        # w: the extra row, initially zero
    for k in range(m - 1, -1, -1):      # rotate (a_k, alpha) into (0, new
        alpha)                          alpha)
        c, s, r = givens(alpha, a[k])
```



```

alpha = r
Tk = T[k, k:].copy()
T[k, k:], w[k:] = c * Tk - s * w[k:], s * Tk + c * w[k:]
return T                                     # on exit w equals z

```

The alternative, **recursive least squares** (Plackett 1950), updates  $\mathbf{K} = (\mathbf{X}^\top \mathbf{X})^{-1}$  by the Sherman–Morrison formula ([Theorem 1.6.5](#), [Example \(Deleting an observation\)](#)): adding  $(\mathbf{x}, y)$  gives

$$\mathbf{K}_+ = \mathbf{K} - \frac{\mathbf{K} \mathbf{x} \mathbf{x}^\top \mathbf{K}}{1 + \mathbf{x}^\top \mathbf{K} \mathbf{x}}, \quad \hat{\boldsymbol{\beta}}_+ = \hat{\boldsymbol{\beta}} + \frac{\mathbf{K} \mathbf{x} (y - \mathbf{x}^\top \hat{\boldsymbol{\beta}})}{1 + \mathbf{x}^\top \mathbf{K} \mathbf{x}}.$$

It also costs  $O(p^2)$ , but it carries an explicit inverse of the cross-product matrix, with its  $\kappa^2$  sensitivity, and nothing corrects its accumulated errors ([Example \(Ten-year regressions of consumption\)](#)).

## 10.6.2 Leave-one-out quantities from one fit

Many diagnostics ask what would happen if observation  $i$  were left out. Computing them by  $n$  separate fits costs  $O(n^2 p^2)$ , but all of them follow from one fit.

### Proposition 10.6.2 — Deletion formulas

Let  $\mathbf{X}$  have full column rank, let  $h_{ii} < 1$ , and let  $\hat{\boldsymbol{\beta}}_{(i)}$  and  $\text{SSE}_{(i)}$  denote the least squares estimate and residual sum of squares computed without observation  $i$ . Then

- (a)  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} \hat{\varepsilon}_i / (1 - h_{ii})$ ;
- (b)  $y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)} = \hat{\varepsilon}_i / (1 - h_{ii})$ ;
- (c)  $\text{SSE}_{(i)} = \text{SSE} - \hat{\varepsilon}_i^2 / (1 - h_{ii})$ .

*Proof.*

- (a) is [Exercise \(B1\)](#), which follows from the deletion formula of [Example \(Deleting an observation\)](#).
- (b) Multiply (a) by  $\mathbf{x}_{(i)}^\top$ :  $\mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)} = \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}} - h_{ii} \hat{\varepsilon}_i / (1 - h_{ii})$ , so  $y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)} = \hat{\varepsilon}_i + h_{ii} \hat{\varepsilon}_i / (1 - h_{ii}) = \hat{\varepsilon}_i / (1 - h_{ii})$ .
- (c) Put  $g = \hat{\varepsilon}_i / (1 - h_{ii})$  and let  $h_{ji} = \mathbf{x}_{(j)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)}$  be the entries of  $\mathbf{M}$ . By (a),  $y_j - \mathbf{x}_{(j)}^\top \hat{\boldsymbol{\beta}}_{(i)} = \hat{\varepsilon}_j + h_{ji} g$  for every  $j$ , including  $j = i$ , where it gives  $g$  by (b). Summing squares over all  $j$ ,

$$\sum_j (\hat{\varepsilon}_j + h_{ji} g)^2 = \text{SSE} + 2g (\mathbf{M} \hat{\boldsymbol{\varepsilon}})_i + g^2 (\mathbf{M}^2)_{ii} = \text{SSE} + g^2 h_{ii},$$

because  $\mathbf{M} \hat{\boldsymbol{\varepsilon}} = \mathbf{0}$  and  $\mathbf{M}^2 = \mathbf{M}$ . Removing the  $j = i$  term, which is  $g^2$ , leaves  $\text{SSE}_{(i)} = \text{SSE} - g^2 (1 - h_{ii}) = \text{SSE} - \hat{\varepsilon}_i^2 / (1 - h_{ii})$ . ■

With a QR factorization the leverages cost  $O(np)$  ([Proposition 10.3.3\(c\)](#)), and the rest  $O(n)$ .

This gives the PRESS residuals  $\hat{\varepsilon}_i/(1-h_{ii})$ , whose sum of squares is leave-one-out cross-validation (Allen 1974; [Theorem 29.3.2](#)), and the deleted variances  $s_{(i)}^2 = \text{SSE}_{(i)}/(n-p-1)$  behind the externally studentized residuals  $t_i = \hat{\varepsilon}_i/(s_{(i)}\sqrt{1-h_{ii}})$  of [Chapter 20](#) ([Theorem 20.3.3](#)).

### Example. (Leaving out the District of Columbia)

Refit the state murder-rate regression with all 51 jurisdictions, including the District of Columbia, which [Chapter 6](#) set aside. The residual sum of squares is 142.53 and the PRESS statistic is 230.52. The largest PRESS residual belongs to the District of Columbia. Its ordinary residual is 4.475 and its leverage 0.506, so its prediction error when it is left out is  $\hat{\varepsilon}_i/(1-h_{ii}) = 9.066$ . Its externally studentized residual is 4.28. By [Proposition 10.6.2\(c\)](#), deleting it reduces the residual sum of squares to 101.96, the value found for the 50 states in [Example \(Murder rates by the augmented factorization\)](#). The script confirms all 51 deletions three ways: by the formulas, by downdating the augmented triangle, and by refitting.

```
data = sm.datasets.statecrime.load_pandas().data           # all 50 states and DC
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape

Q, R = np.linalg.qr(X)
beta = np.linalg.solve(R, Q.T @ y)
e = y - X @ beta
sse = e @ e
h = np.sum(Q ** 2, axis=1)                                # leverages
press_resid = e / (1 - h)                                  # y_i minus its prediction
                                without case i
sse_loo = sse - e ** 2 / (1 - h)                          # SSE after deleting case i
s_loo = np.sqrt(sse_loo / (n - 1 - p))
t_ext = e / (s_loo * np.sqrt(1 - h))                      # externally studentized
                                residuals
i = int(np.argmax(np.abs(press_resid)))
print(f"largest PRESS residual: {data.index[i]}, e = {e[i]:.3f}, h = {h[i]:.3f}, "
      f" e/(1-h) = {press_resid[i]:.3f}, t = {t_ext[i]:.2f}")
print(f"PRESS = {np.sum(press_resid ** 2):.2f}, SSE = {sse:.2f}")
```

## 10.6.3 Adding and deleting variables

Changing the columns of  $\mathbf{X}$  is a Gram–Schmidt step, or its reverse.

### Proposition 10.6.3 — Adding a column

Let  $\mathbf{X} = \mathbf{Q}_1\mathbf{R}$  be a thin QR factorization of a full-column-rank  $\mathbf{X}$ , and let  $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$ . Put  $\mathbf{r} = \mathbf{Q}_1^\top \mathbf{z}$ ,  $\mathbf{w} = \mathbf{z} - \mathbf{Q}_1\mathbf{r}$ ,  $\rho = \|\mathbf{w}\|$  and  $\mathbf{q} = \mathbf{w}/\rho$ . Then

$$[\mathbf{X}, \mathbf{z}] = [\mathbf{Q}_1, \mathbf{q}] \begin{pmatrix} \mathbf{R} & \mathbf{r} \\ \mathbf{0}^\top & \rho \end{pmatrix}$$

is a thin QR factorization. The residual sum of squares falls by  $(\mathbf{q}^\top \mathbf{y})^2$  when  $\mathbf{z}$  is added,

and the coefficient of  $\mathbf{z}$  in the enlarged model is  $\mathbf{q}^\top \mathbf{y} / \rho = \mathbf{w}^\top \mathbf{y} / \mathbf{w}^\top \mathbf{w}$ .

*Proof.*  $\mathbf{w} = (\mathbf{I} - \mathbf{M})\mathbf{z} \neq \mathbf{0}$  because  $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$ , and  $\mathbf{q}$  is a unit vector orthogonal to  $\mathcal{C}(\mathbf{Q}_1)$ . The product of the factors is  $[\mathbf{Q}_1 \mathbf{R}, \mathbf{Q}_1 \mathbf{r} + \rho \mathbf{q}] = [\mathbf{X}, \mathbf{z}]$ . By Proposition 10.3.3(d), applied to the enlarged factorization, the new column's sequential sum of squares is the square of its coordinate  $\mathbf{q}^\top \mathbf{y}$ . The last row of the triangular system of Proposition 10.3.3(a) reads  $\rho b_z = \mathbf{q}^\top \mathbf{y}$ . This is also the Frisch–Waugh–Lovell coefficient  $\mathbf{w}^\top \mathbf{y} / \mathbf{w}^\top \mathbf{w}$  of Theorem 6.6.2. ■

If  $\mathbf{z}$  is nearly in  $\mathcal{C}(\mathbf{X})$ , the computed  $\mathbf{w}$  loses orthogonality to  $\mathbf{Q}_1$ ; repeating the step once restores it (Björck 1996, chapter 2). Deleting column  $j$  of  $\mathbf{R}$  leaves an upper Hessenberg block that  $p-j$  rotations restore to triangular form, at  $O(p^2)$  cost (Exercise (B2)). So stepwise procedures move between models at  $O(p^2)$  or  $O(np)$  cost per step.

```
def add_column(Q1, R, x):
    """Thin QR of [X, x] from the thin QR of X (one Gram-Schmidt step, repeated once)."""
    r = Q1.T @ x
    w = x - Q1 @ r
    r2 = Q1.T @ w                                # reorthogonalize: guards against
    cancellation
    w, r = w - Q1 @ r2, r + r2
    rho = np.linalg.norm(w)
    Rn = np.block([[R, r[:, None]], [np.zeros((1, R.shape[1])), np.array([[rho]])]])
    return np.column_stack([Q1, w / rho], Rn)
```

## 10.6.4 A moving window

### Example. (Ten-year regressions of consumption)

Quarterly US real consumption is regressed on an intercept, real disposable income, real investment, population and unemployment (the public-domain macroeconomic series of Example (A macroeconomic design)), in a window of 40 quarters that slides forward one quarter at a time. Each step adds one row and deletes one. Over the 163 steps, whose windows have condition numbers up to  $2.9 \times 10^6$ , the fit is maintained in two ways. The first applies two Sherman–Morrison updates to  $(\mathbf{X}^\top \mathbf{X})^{-1}$ . The second updates and then downdates the augmented triangle. After each step both are compared with a fresh QR fit of the window (Figure 10.6.1). Inverse updating reaches a relative error of  $6.2 \times 10^{-7}$  and ends at  $8.4 \times 10^{-8}$ . Updating the triangle stays below  $1.1 \times 10^{-10}$  and ends at  $3.6 \times 10^{-11}$ . Both are cheap. Only one keeps QR's accuracy.

```
macro = sm.datasets.macrodta.load_pandas().data
yw = macro["realcons"].to_numpy()
Xw = np.column_stack([np.ones(len(yw)), macro[["realdpi", "realinv", "pop", "unemp"]]])
w = 40                                # ten years of quarterly data
print("kappa of the full design:", f"{np.linalg.cond(Xw):.2e}")
```

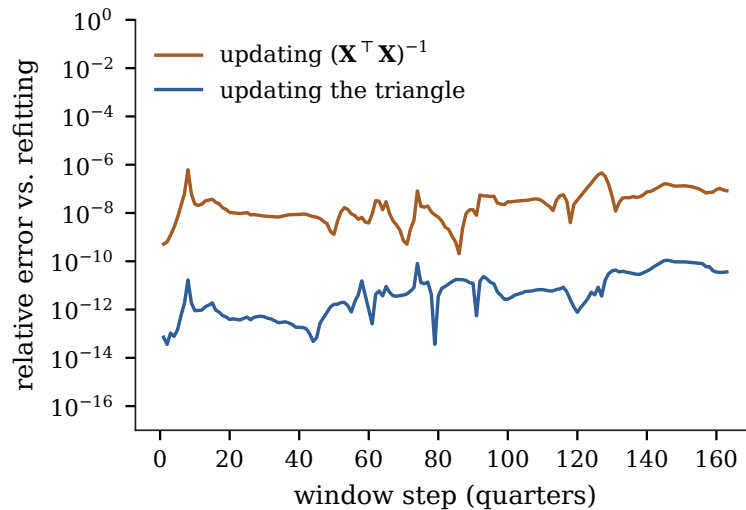


Figure 10.6.1: A moving-window regression maintained by rank-one updates of  $(\mathbf{X}^T \mathbf{X})^{-1}$  and by Givens updating and downdating of the augmented triangle. The vertical axis is the relative difference in the coefficients from a fresh QR fit of each window.

```

K = np.linalg.inv(Xw[:w].T @ Xw[:w])          # (a) rank-one updates of the
inverse
g = Xw[:w].T @ yw[:w]
Zw = np.column_stack([Xw, yw])
Tw = np.linalg.qr(Zw[:w], mode="r")          # (b) triangle of [X, y]
p1 = Xw.shape[1]
err_sm, err_giv = [], []
for t in range(w, len(yw)):
    new, old = Xw[t], Xw[t - w]
    Kx = K @ new                             # add the new quarter (Sherman-
Morrison)
    K = K - np.outer(Kx, Kx) / (1 + new @ Kx)
    Kx = K @ old                             # delete the oldest quarter
    K = K + np.outer(Kx, Kx) / (1 - old @ Kx)
    g = g + new * yw[t] - old * yw[t - w]
    b_sm = K @ g
    Tw = downdate(update(Tw, Zw[t]), Zw[t - w])
    b_giv = np.linalg.solve(Tw[:p1, :p1], Tw[:p1, p1])
    b_ref = np.linalg.lstsq(Xw[t - w + 1:t + 1], yw[t - w + 1:t + 1], rcond=None)[0]
    # (c) refit
    err_sm.append(np.linalg.norm(b_sm - b_ref) / np.linalg.norm(b_ref))
    err_giv.append(np.linalg.norm(b_giv - b_ref) / np.linalg.norm(b_ref))
print(f"final relative error: inverse updating {err_sm[-1]:.1e}, Givens {err_giv
[-1]:.1e}")

```

*Idea.* A change of one row or column costs  $O(p^2)$  or  $O(np)$  on the triangular factor, not  $O(np^2)$ . All  $n$  leave-one-out fits cost one, and the leverage says whether, and how accurately, a case can be removed.

## 10.6.5 Exercises

### 10.6.5.1 A. Check your understanding

**Exercise A1.**

Show that  $\text{PRESS} = \sum_i \hat{\varepsilon}_i^2 / (1 - h_{ii})^2 \geq \text{SSE}$ . Show that if all leverages equal  $p/n$ , then  $\text{PRESS} = \text{SSE } n^2 / (n - p)^2$ . Evaluate this for the state data with  $n = 51$  and  $p = 4$ , and compare with the actual ratio in [Example \(Leaving out the District of Columbia\)](#).

**10.6.5.2 B. Practice****Exercise B1.**

Show that the leverage of row  $i$  of the augmented matrix  $\mathbf{Z} = [\mathbf{X}, \mathbf{y}]$  is  $h_{ii} + \hat{\varepsilon}_i^2 / \text{SSE}$ . Deduce the condition under which the augmented triangle can be downdated, and show that it fails only if  $h_{ii} = 1$  or deleting case  $i$  leaves a model that fits the remaining data exactly.

*Solution.*  $\mathcal{C}(\mathbf{Z}) = \mathcal{C}(\mathbf{X}) \oplus \text{span}(\hat{\mathbf{e}})$ , with the two summands orthogonal, so by [Theorem 6.3.8](#) the projection onto  $\mathcal{C}(\mathbf{Z})$  is  $\mathbf{M} + \hat{\mathbf{e}}\hat{\mathbf{e}}^\top / \|\hat{\mathbf{e}}\|^2$ . Its  $i$ th diagonal entry is  $h_{ii} + \hat{\varepsilon}_i^2 / \text{SSE}$ . Downdating requires this to be below 1. By [Proposition 10.6.2\(c\)](#),  $1 - h_{ii} - \hat{\varepsilon}_i^2 / \text{SSE} = (1 - h_{ii}) \text{SSE}_{(i)} / \text{SSE}$ , which vanishes iff  $h_{ii} = 1$  or  $\text{SSE}_{(i)} = 0$ .

**Exercise B2.**

Let  $\mathbf{X} = \mathbf{Q}_1 \mathbf{R}$  and let  $\mathbf{X}_{-j}$  be  $\mathbf{X}$  without column  $j$ . Show that  $\mathbf{X}_{-j} = \mathbf{Q}_1 \mathbf{H}$ , where  $\mathbf{H}$  is  $\mathbf{R}$  without column  $j$ , which is upper Hessenberg in its last  $p - j$  columns. Show that  $p - j$  rotations of adjacent rows reduce  $\mathbf{H}$  to  $\begin{pmatrix} \mathbf{R}_{-j} \\ \mathbf{0}^\top \end{pmatrix}$ , and that  $\mathbf{R}_{-j}$  is the triangle of  $\mathbf{X}_{-j}$ . How much does the residual sum of squares rise?

*Solution.* Columns of  $\mathbf{X}$  are  $\mathbf{Q}_1$  times columns of  $\mathbf{R}$ , so deleting a column of both gives  $\mathbf{X}_{-j} = \mathbf{Q}_1 \mathbf{H}$ . Column  $k \geq j$  of  $\mathbf{H}$  is column  $k + 1$  of  $\mathbf{R}$ , which has nonzeros down to row  $k + 1$ : one below the diagonal. Rotate rows  $j, j + 1$  to zero entry  $(j + 1, j)$ . This mixes two rows that are zero before column  $j$ , so nothing earlier fills in. Continue with rows  $(j + 1, j + 2)$  and so on. After  $p - j$  rotations,  $\mathbf{G}\mathbf{H} = \begin{pmatrix} \mathbf{R}_{-j} \\ \mathbf{0}^\top \end{pmatrix}$  with  $\mathbf{R}_{-j}$  upper triangular, and  $\mathbf{X}_{-j} = (\mathbf{Q}_1 \mathbf{G}^\top) \begin{pmatrix} \mathbf{R}_{-j} \\ \mathbf{0}^\top \end{pmatrix}$ , where  $\mathbf{Q}_1 \mathbf{G}^\top$  has orthonormal columns. Apply the same rotations to  $\mathbf{c}_1 = \mathbf{Q}_1^\top \mathbf{y}$ . The last entry of the rotated vector is the coordinate of  $\mathbf{y}$  along the direction that has dropped out of the model, and its square is the rise in the residual sum of squares. This is the partial sum of squares of column  $j$  ([Definition 9.3.2](#)).

**10.6.5.3 C. Going deeper****Exercise C1.**

Feed the rows of  $[\mathbf{X}, \mathbf{y}]$  one at a time into the augmented triangle, as in [Theorem 10.3.4\(c\)](#). For  $k > p$ , let  $t_k$  be the entry left in the new row when case  $k$  is added. Show that

$$t_k = \pm \frac{y_k - \mathbf{x}_{(k)}^\top \hat{\boldsymbol{\beta}}_{k-1}}{(1 + \mathbf{x}_{(k)}^\top (\mathbf{X}_{k-1}^\top \mathbf{X}_{k-1})^{-1} \mathbf{x}_{(k)})^{1/2}},$$

where  $\hat{\beta}_{k-1}$  and  $\mathbf{X}_{k-1}$  use the first  $k-1$  cases. Show that under the normal linear model the  $t_k$  are independent  $N(0, \sigma^2)$ . These are the recursive residuals of Brown, Durbin and Evans (1975), used to test the constancy of a regression over time.

## 10.7 Rank-revealing factorizations

A model matrix with an intercept and indicators for every level of a factor has dependent, or **aliased**, columns. Software must notice them and decide which to set aside, but in floating point it can only ask whether a column is *nearly* a combination of others, relative to a tolerance. The SVD answers that (Section 10.4) but does not say which columns to drop, and it costs more. The standard tool is QR with column pivoting.

### 10.7.1 QR with column pivoting

At step  $k$  of Householder QR, the trailing columns (rows  $k$  to  $n$ , columns  $k$  to  $p$  of the partly reduced matrix) are what remains of the original columns after the parts explained by the first  $k-1$  chosen columns have been removed. **Column pivoting** (Businger and Golub 1965) moves the trailing column of largest norm, the one least explained by the columns already chosen, into position  $k$  before reflecting. The result is

$$\mathbf{X}\mathbf{\Pi} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}, \quad \mathbf{R} = \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{R}_{22} \end{pmatrix},$$

with  $\mathbf{\Pi}$  a permutation matrix,  $\mathbf{R}_{11}$  the leading  $k \times k$  block and  $\mathbf{R}_{22}$  the trailing  $(p-k) \times (p-k)$  block, for any  $k$ . We need one fact about submatrices.

#### Lemma 10.7.1 — Singular values of a submatrix

If  $\mathbf{X}_1$  is formed from some of the columns of  $\mathbf{X}$ , then  $\sigma_i(\mathbf{X}_1) \leq \sigma_i(\mathbf{X})$  for every  $i$  up to the number of columns of  $\mathbf{X}_1$ . The same holds for a submatrix formed from some of the rows.

*Proof.* Write  $\mathbf{X}_1 = \mathbf{X}\mathbf{S}$ , where  $\mathbf{S}$  holds the corresponding columns of the identity, so  $\|\mathbf{S}\|_2 = 1$ . Let  $\mathbf{X}_{i-1}$  be the truncated SVD with  $i-1$  terms, or  $\mathbf{X}$  itself if  $\text{rank } \mathbf{X} < i$ , so that  $\|\mathbf{X} - \mathbf{X}_{i-1}\|_2 = \sigma_i(\mathbf{X})$ . The matrix  $\mathbf{X}_{i-1}\mathbf{S}$  has rank at most  $i-1$ , so by the Eckart–Young inequality, in the form used in the proof of Proposition 10.4.2,

$$\sigma_i(\mathbf{X}_1) \leq \|\mathbf{X}\mathbf{S} - \mathbf{X}_{i-1}\mathbf{S}\|_2 \leq \|\mathbf{X} - \mathbf{X}_{i-1}\|_2 \|\mathbf{S}\|_2 = \sigma_i(\mathbf{X}).$$

For rows, apply this to  $\mathbf{X}^\top$ , which has the same singular values. ■

#### Theorem 10.7.2 — Column-pivoted QR

Let  $\mathbf{X}$  be  $n \times p$  with  $n \geq p$ , and let  $\mathbf{X}\mathbf{\Pi} = \mathbf{Q} \begin{pmatrix} \mathbf{R} \\ \mathbf{0} \end{pmatrix}$  be computed by Householder QR (Theorem 10.3.2) with column pivoting.

- (a) For every  $k \leq j$ ,  $r_{kk}^2 \geq \sum_{i=k}^j r_{ij}^2$ . In particular  $|r_{11}| \geq |r_{22}| \geq \cdots \geq |r_{pp}|$ .
- (b) For every  $k < p$ ,

$$\sigma_{k+1}(\mathbf{X}) \leq \|\mathbf{R}_{22}\|_2 \leq \sqrt{p-k} |r_{k+1,k+1}| \quad \text{and} \quad \sigma_{\min}(\mathbf{R}_{11}) \leq \sigma_k(\mathbf{X}).$$

- (c) In exact arithmetic, if  $\text{rank } \mathbf{X} = r$ , then  $r_{kk} \neq 0$  for  $k \leq r$  and  $\mathbf{R}_{22} = \mathbf{0}$  for  $k = r$ . The first  $r$  columns of  $\mathbf{X}\mathbf{\Pi}$  are a basis of  $\mathcal{C}(\mathbf{X})$ , and  $\mathbf{X}\mathbf{\Pi} = \mathbf{Q} \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$  with  $\mathbf{R}_{11}$  nonsingular.
- (d) The cost is that of the unpivoted factorization,  $2np^2 - 2p^3/3$  flops, plus  $O(np)$  for keeping track of the column norms.

*Proof.*

- (a) Reflections preserve the length of each column segment they act on. When column  $k$  is chosen, its segment in rows  $k, \dots, n$  has the largest norm among the trailing columns, and the reflection maps it to  $(r_{kk}, 0, \dots, 0)^\top$ , so  $|r_{kk}|$  equals that norm. Every column that ends in a position  $j > k$  was among the trailing columns at step  $k$ , so its segment in rows  $k, \dots, n$  had norm at most  $|r_{kk}|$ . The reflections at steps  $k, \dots, p$  act on rows  $k, \dots, n$  only, so they preserve this norm, and at the end the segment is  $(r_{kj}, \dots, r_{jj}, 0, \dots, 0)^\top$ . Hence  $\sum_{i=k}^j r_{ij}^2 \leq r_{kk}^2$ . With  $j = k+1$  this gives  $r_{k+1,k+1}^2 \leq r_{kk}^2$ .
- (b)  $\mathbf{X}\mathbf{\Pi}$  has the same singular values as  $\mathbf{X}$ , because  $\mathbf{\Pi}$  is orthogonal. The matrix  $\mathbf{Q} \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$  has rank at most  $k$  and differs from  $\mathbf{X}\mathbf{\Pi}$  by  $\mathbf{Q} \begin{pmatrix} \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{R}_{22} \end{pmatrix}$ , whose norm is  $\|\mathbf{R}_{22}\|_2$ . The Eckart–Young inequality (Proposition 1.8.2(c)) gives  $\sigma_{k+1}(\mathbf{X}) \leq \|\mathbf{R}_{22}\|_2$ . Next,  $\|\mathbf{R}_{22}\|_2 \leq \|\mathbf{R}_{22}\|_F \leq \sqrt{p-k} \max_j \|\text{column } j \text{ of } \mathbf{R}_{22}\|$ , and by (a) with  $k+1$  in place of  $k$ , each column of  $\mathbf{R}_{22}$  has norm at most  $|r_{k+1,k+1}|$ . Finally, the first  $k$  columns of  $\mathbf{X}\mathbf{\Pi}$  equal  $\mathbf{Q}_1 \begin{pmatrix} \mathbf{R}_{11} \\ \mathbf{0} \end{pmatrix}$ , so their singular values are those of  $\mathbf{R}_{11}$ . Then Lemma 10.7.1 gives  $\sigma_k(\mathbf{R}_{11}) \leq \sigma_k(\mathbf{X})$ .
- (c) Because  $\mathbf{R}_{11}$  is triangular,  $\text{rank } \mathbf{R} = k + \text{rank } \mathbf{R}_{22}$  whenever  $r_{11}, \dots, r_{kk}$  are nonzero (subtract multiples of the first  $k$  columns from the others to clear  $\mathbf{R}_{12}$ ). Also  $\text{rank } \mathbf{R} = \text{rank } \mathbf{X} = r$ . So while  $k < r$ ,  $\mathbf{R}_{22} \neq \mathbf{0}$ , some trailing column is nonzero, and the next pivot  $r_{k+1,k+1}$  is nonzero. At  $k = r$ ,  $\text{rank } \mathbf{R}_{22} = 0$ . The first  $r$  columns of  $\mathbf{X}\mathbf{\Pi}$  are  $\mathbf{Q}_1$  times the independent columns  $\begin{pmatrix} \mathbf{R}_{11} \\ \mathbf{0} \end{pmatrix}$ , so they are independent, and there are  $r = \dim \mathcal{C}(\mathbf{X})$  of them.
- (d) The norms of the trailing column segments are updated at each step by subtracting  $r_{kj}^2$ , at  $O(p)$  flops per step. When cancellation makes the update inaccurate, the norm is recomputed.

■

Part (b) is the rank-revealing property, and it is one-sided. A small pivot  $|r_{k+1,k+1}|$  certifies a small singular value  $\sigma_{k+1}$ : pivoted QR never reports a near dependence that is not there. The converse can fail. A matrix can have a tiny singular value although no pivot is small.

#### Example. (Kahan's matrix)

Kahan (1966) gave the upper triangular matrix with diagonal  $1, s, s^2, \dots, s^{m-1}$  and entries  $-cs^{i-1}$  above the diagonal in row  $i$ , where  $c^2 + s^2 = 1$ . Its columns are already in pivoted order (a tiny perturbation of the column scaling breaks ties in favour of that

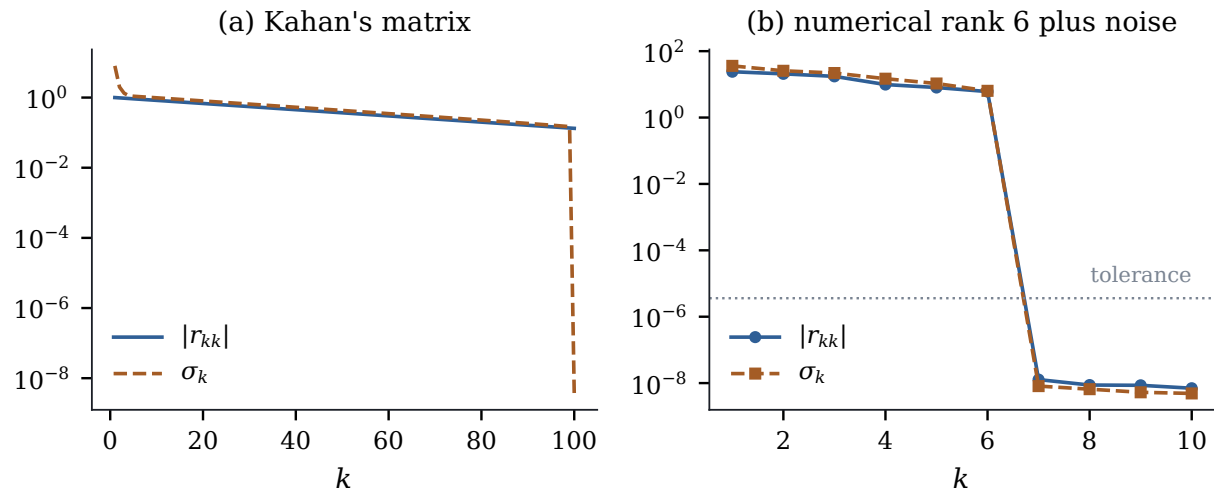


Figure 10.7.1: Diagonal of the pivoted triangle,  $|r_{kk}|$ , and singular values  $\sigma_k$ . (a) Kahan's matrix: the diagonal decreases slowly while the last singular value collapses. (b) A matrix of numerical rank six: both drop at  $k = 7$ , well below the tolerance.

order), so pivoted QR leaves it unchanged. With  $m = 100$  and  $c = 0.2$ , the smallest diagonal entry is 0.133, yet the smallest singular value is  $3.7 \times 10^{-9}$ . No diagonal entry hints at this (Figure 10.7.1(a)). Such matrices are rare in practice. Chan (1987) gave the first rank-revealing QR algorithm; factorizations whose bounds in both directions are polynomial in  $p$  exist (Hong and Pan 1992) and can be computed efficiently (Gu and Eisenstat 1996). Panel (b) shows the typical case: a  $50 \times 10$  matrix of rank six plus noise of relative size  $10^{-9}$ . The pivoted diagonal drops by eight orders of magnitude exactly where the singular values do.

```
import numpy as np
import statsmodels.api as sm

def pivoted_qr(X):
    """Householder QR with column pivoting:  $X[:, \text{perm}] = Q R$ ,  $|r_{11}| \geq |r_{22}| \geq \dots$ """
    A = np.array(X, dtype=float)
    n, p = A.shape
    perm = np.arange(p)
    for k in range(min(n, p)):
        norms = np.sum(A[k:, k:] ** 2, axis=0) # squared norms of the
        trailing columns
        j = k + int(np.argmax(norms))
        A[:, [k, j]], perm[[k, j]] = A[:, [j, k]], perm[[j, k]]
        x = A[k:, k]
        alpha = -np.copysign(np.linalg.norm(x), x[0])
        if alpha == 0.0:
            break # the trailing block is
        exactly zero
        v = x.copy()
        v[0] -= alpha
        v /= np.linalg.norm(v)
        A[k:, k:] -= 2.0 * np.outer(v, v @ A[k:, k:])
    return np.triu(A[:p, :]), perm

def numerical_rank(R, tol):
    """Number of diagonal entries of the pivoted triangle above  $\text{tol} * |r_{11}|$ ."""
```



```
d = np.abs(np.diag(R))
return int(np.sum(d > tol * d[0]))
```

```
m, c = 100, 0.2
sn = np.sqrt(1 - c ** 2)
K = np.diag(sn ** np.arange(m)) @ (np.eye(m) - c * np.triu(np.ones((m, m)), 1))
K = K @ np.diag(1 - 1e-10 * np.arange(m))          # tiny tilt so that pivoting
                                                    keeps the order
RK, permK = pivoted_qr(K)
sK = np.linalg.svd(K, compute_uv=False)
print("pivot order unchanged:", np.array_equal(permK, np.arange(m)))
print(f"smallest |r_kk| = {np.abs(RK[-1, -1]):.3f}, smallest singular value = {sK
      [-1]:.2e}")
```

## 10.7.2 Basic solutions and aliased columns

Suppose pivoted QR finds numerical rank  $r < p$ , so that  $\mathbf{X}\mathbf{\Pi} \approx \mathbf{Q} \begin{pmatrix} \mathbf{R}_{11} & \mathbf{R}_{12} \\ \mathbf{0} & \mathbf{0} \end{pmatrix}$ . Write  $\mathbf{Q}^\top \mathbf{y} = (\mathbf{c}_1^\top, \mathbf{c}_2^\top)^\top$  with  $\mathbf{c}_1 \in \mathbb{R}^r$ , and split the permuted coefficients as  $(\mathbf{b}_1, \mathbf{b}_2)$ . As in [Proposition 10.3.3](#), the least squares estimates are the solutions of

$$\mathbf{R}_{11}\mathbf{b}_1 + \mathbf{R}_{12}\mathbf{b}_2 = \mathbf{c}_1,$$

with  $\mathbf{b}_2$  free. Setting  $\mathbf{b}_2 = \mathbf{0}$  gives the **basic solution**, which simply drops the last  $p - r$  pivoted columns. It has at most  $r$  nonzero coefficients. It differs from the minimum-norm solution of [Theorem 10.4.1](#). That solution can also be computed at QR cost, by a second orthogonal reduction from the right that turns  $[\mathbf{R}_{11}, \mathbf{R}_{12}]$  into  $[\mathbf{T}, \mathbf{0}]$  (a *complete orthogonal decomposition*; Golub and Van Loan 2013). Every least squares solution gives the same fitted values. By [Theorem 6.4.4](#), and in the language of [Chapter 8](#) ([Theorem 8.3.1](#)), every estimable function  $\lambda^\top \beta$  has the same estimate under every convention. The coefficients of aliased columns are not estimable.

Software differs in the choice. R's `lm` takes the columns in order, moves to the end any column whose remaining norm falls below a relative tolerance (by default  $10^{-7}$ ), and reports its coefficient as NA. statsmodels' OLS uses the pseudoinverse and so reports the minimum-norm solution. Programs built on the sweep operator ([Proposition 10.2.3](#)), such as SAS's GLM procedure, skip numerically zero pivots and set those coefficients to zero, another basic solution.

### Example. (Firm effects in investment data)

Grunfeld's investment data ( $n = 220$ : 11 US firms over 20 years, public domain in statsmodels) are modelled with an intercept, the firm's market value and capital stock, and indicators for all 11 firms:  $p = 14$  columns of rank 13. Pivoted QR with a tolerance of  $10^{-10}$  finds rank 13 and leaves out the indicator for General Motors. Three conventions give three different intercepts:  $-50.67$  for the minimum-norm solution (as statsmodels reports),  $-20.58$  for the in-order rule that marks the last indicator as aliased (as R would), and  $-70.30$  for the basic solution from full pivoting. They agree to every printed digit on everything estimable: the coefficient 0.1101 of market value, 0.3100 of capital, the difference  $-172.20$  between the General Motors and US Steel effects, and the residual sum of squares 523719.

```

gr = sm.datasets.grunfeld.load_pandas().data
firms = list(dict.fromkeys(gr["firm"]))
D = np.column_stack([(gr["firm"] == f).to_numpy(float) for f in firms])    # 11
    indicators
y = gr["invest"].to_numpy()
X = np.column_stack([np.ones(len(y)), gr[["value", "capital"]], D])        # 14
    columns, rank 13
n, p = X.shape
R, perm = pivoted_qr(X)
r = numerical_rank(R, 1e-10)
print("p =", p, " numerical rank =", r, " column left out:", perm[r:])

```

```

def basic_solution(X, y, tol=1e-10):
    """Drop the columns that pivoted QR puts last; fit the rest (zeros for dropped
    columns)."""
    R, perm = pivoted_qr(X)
    r = numerical_rank(R, tol)
    b = np.zeros(X.shape[1])
    b[perm[:r]] = np.linalg.lstsq(X[:, perm[:r]], y, rcond=None)[0]
    return b

def in_order_solution(X, y, tol=1e-7):
    """Keep a column only if it is not (nearly) a combination of the columns kept
    before it."""
    kept = []
    for j in range(X.shape[1]):
        Z = X[:, kept + [j]]
        Q, R = np.linalg.qr(Z)
        if abs(R[-1, -1]) > tol * np.linalg.norm(X[:, j]):
            kept.append(j)
    b = np.full(X.shape[1], np.nan)    # aliased coefficients reported
    # as missing
    b[kept] = np.linalg.lstsq(X[:, kept], y, rcond=None)[0]
    return b

solutions = {
    "minimum norm (pinv)": np.linalg.pinv(X) @ y,
    "pivoted QR, basic": basic_solution(X, y),
    "in order, aliased = NA": in_order_solution(X, y),
}
for name, b in solutions.items():
    fitted = X @ np.nan_to_num(b)
    print(f"{name:24s} value {b[1]:.5f} capital {b[2]:.5f} const {b[0]:9.3f}"
          f" GM - US Steel {b[3] - b[4]:9.3f} SSE {np.sum((y - fitted) ** 2):.1f}"
          )

```

### 10.7.3 Choosing a tolerance

Exact aliasing, as in the example, is easy: the offending pivot is at the level of rounding error, far below any reasonable tolerance. The hard case is a column that is nearly, but not exactly, a combination of others. The numerical rank then depends on the tolerance, and so do the coefficients, dramatically.

**Example. (A nearly aliased column)**

Add to the Grunfeld regressors (intercept, value, capital) a fourth column equal to value plus capital plus noise of relative size  $10^{-9}$ . Its smallest singular value, relative to the largest, is  $4.0 \times 10^{-10}$ . The default tolerance of `numpy.linalg.lstsq`,  $\max(n, p) \epsilon_M$  relative to  $\sigma_1$ , treats the matrix as having full rank. It returns coefficients as large as  $8.1 \times 10^5$ , which spend one more degree of freedom to reduce the residual sum of squares from 1768678.4 to 1768252.5, by 0.02 per cent. With the tolerance set to  $10^{-8}$ , the fourth column is treated as aliased, and the largest coefficient is 38.41.

```
rng = np.random.default_rng(108)
z = X[:, 1] + X[:, 2] + 1e-9 * np.linalg.norm(X[:, 1]) * rng.normal(size=n) / np.
    sqrt(n)
Xz = np.column_stack([X[:, :3], z])          # constant, value, capital, value +
    capital + tiny
s = np.linalg.svd(Xz, compute_uv=False)
print("relative singular values:", np.array2string(s / s[0], precision=2))
for tol in [None, 1e-8]:
    b, _, rank, _ = np.linalg.lstsq(Xz, y, rcond=tol)
    print(f"rcond = {tol}: rank {rank}, largest |b_j| = {np.abs(b).max():.3g}, "
          f" SSE = {np.sum((y - Xz @ b) ** 2):.1f}")
```

Neither answer is wrong as arithmetic. But a column that differs from a combination of the others only at the level of  $10^{-9}$  of its size carries no information a real measurement could support. Three rules follow. Scale the columns to comparable lengths before deciding the rank ([Proposition 10.5.2](#)); R's relative test does this implicitly. Tie the tolerance to the accuracy of the data, not of the machine: regressors recorded to six significant figures cannot resolve near dependences below about  $10^{-6}$  of the column lengths ([Proposition 10.4.2](#), applied to the recording error). And look for a gap, as in [Figure 10.7.1\(b\)](#). Without one the rank decision is fragile, the data are showing collinearity ([Chapter 26](#)), and shrinkage is often a better response than a sharp cut-off ([Chapter 27](#)).

*Idea.* Pivoted QR brings the most independent columns to the front. A small trailing pivot proves a near dependence, and setting those coefficients aside changes nothing estimable. The tolerance that decides “small” is a statement about the accuracy of the data.

## 10.7.4 Exercises

### 10.7.4.1 A. Check your understanding

#### Exercise A1.

Apply column-pivoted QR ([Theorem 10.7.2](#)) by hand to the matrix with columns  $\mathbf{x}_1 = (1, 1, 0, 0)^\top$ ,  $\mathbf{x}_2 = (0, 0, 1, 1)^\top$  and  $\mathbf{x}_3 = (1, 1, 1, 1)^\top$  (use Gram–Schmidt on the pivoted columns, which gives the same  $|r_{ij}|$ ). Find the pivot order,  $\mathbf{R}$ , and the rank. Give the basic solution for  $\mathbf{y} = (1, 2, 3, 6)^\top$ .

*Solution.* The norms are  $\sqrt{2}, \sqrt{2}, 2$ , so  $\mathbf{x}_3$  is chosen first, with  $|r_{11}| = 2$  and  $\mathbf{q}_1 = \mathbf{x}_3/2$ . Then  $\mathbf{q}_1^\top \mathbf{x}_1 = \mathbf{q}_1^\top \mathbf{x}_2 = 1$ , and the remainders are  $\mathbf{x}_1 - \mathbf{q}_1 = (\frac{1}{2}, \frac{1}{2}, -\frac{1}{2}, -\frac{1}{2})^\top$  and  $\mathbf{x}_2 - \mathbf{q}_1$ ,

its negative. Both have norm 1, so the tie goes to  $\mathbf{x}_1$ , with  $|r_{22}| = 1$ . The remainder of  $\mathbf{x}_2$  is then zero, so  $r_{33} = 0$  and the rank is 2. The pivot order is  $(3, 1, 2)$ , and up to signs  $\mathbf{R} = \begin{pmatrix} 2 & 1 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$ . The basic solution regresses  $\mathbf{y}$  on  $\mathbf{x}_3$  and  $\mathbf{x}_1$  with  $b_2 = 0$ . The fit is the pair of group means, 1.5 for the first two observations and 4.5 for the last two. So  $b_3 = 4.5$  and  $b_1 = 1.5 - 4.5 = -3$ .

#### 10.7.4.2 B. Practice

##### Exercise B1.

In the setting of this section, show that the basic solution and the minimum-norm solution have the same fitted values. Show that the minimum-norm solution is orthogonal to  $\mathcal{N}(\mathbf{X})$ , while the basic solution in general is not. For the one-way layout of [Example \(Four answers, one fit\)](#), which generalized inverse in that example gives the basic solution that drops the intercept?

*Solution.* Both satisfy  $\mathbf{R}_{11}\mathbf{b}_1 + \mathbf{R}_{12}\mathbf{b}_2 = \mathbf{c}_1$ , so both are least squares estimates ([Proposition 10.3.3](#)), and by [Theorem 6.4.1](#) both have fitted values  $\mathbf{M}\mathbf{y}$ . The minimum-norm solution lies in  $\mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$  ([Theorem 10.4.1\(b\)](#)). The basic solution has zeros in the positions of the dropped columns. A null vector with a nonzero entry in one of those positions is generally not orthogonal to it. For the one-way layout, dropping the intercept column and fitting the three indicators is the “drop intercept” generalized inverse of [Example \(Four answers, one fit\)](#), which gives the group means as coefficients and a zero intercept.

##### Exercise B2.

Show that  $|r_{11}| = \max_j \|\mathbf{x}_j\|$ , and deduce  $|r_{11}| \leq \sigma_1(\mathbf{X}) \leq \sqrt{p}|r_{11}|$ . Combine with [Theorem 10.7.2\(b\)](#) to show that  $\kappa(\mathbf{X}) \geq |r_{11}/r_{pp}|$ , so that a large ratio of pivots proves ill-conditioning.

#### 10.7.4.3 C. Going deeper

##### Exercise C1.

For Kahan’s matrix  $\mathbf{K}_m(c)$ , find a vector  $\mathbf{x}$  with  $\|\mathbf{x}\| = 1$  and  $\|\mathbf{K}_m\mathbf{x}\|$  exponentially small in  $m$  when  $c$  is fixed. (Solve  $\mathbf{K}_m\mathbf{x} = \mathbf{e}_m s^{m-1}$  by back substitution and examine the growth of the entries.) Why does column pivoting not detect this?

## 10.8 Summary and Notes

### 10.8.1 Summary

*Idea.*

1. Floating-point arithmetic commits a relative error of at most  $u = 2^{-53}$  per opera-

- tion (Definition 10.1.1). Inner products are backward stable (Proposition 10.1.2), but forming  $\mathbf{X}^\top \mathbf{X}$  destroys the information in singular values below about  $u^{1/2}\sigma_1$  (Proposition 10.1.3, Example (A cross-product matrix that rounds to a singular one)).
2. The Cholesky factorization of the augmented cross-product matrix gives  $\hat{\boldsymbol{\beta}}$ , the SSE, the sequential sums of squares, the leverages and the standard errors in  $np^2 + p^3/3$  flops (Theorem 10.2.1, Proposition 10.2.2), accurately enough when the column-scaled condition number is modest. Sweeping fits nested models (Proposition 10.2.3); centring needs two passes or updating (Proposition 10.2.4).
  3. Householder QR reduces  $\mathbf{X}$  to a triangle by reflections in  $2np^2 - 2p^3/3$  flops and is backward stable with columnwise errors (Theorem 10.3.2, Proposition 10.3.3). Givens rotations work one row at a time, for streaming and sparse data (Theorem 10.3.4).
  4. The SVD gives every least squares solution, the minimum-norm one, and the precision  $\sigma/\sigma_i$  of each combination  $\mathbf{v}_i^\top \boldsymbol{\beta}$  (Theorem 10.4.1). Singular values are perturbed by at most  $\|\mathbf{E}\|_2$  (Proposition 10.4.2), so rank is meaningful only relative to a tolerance (Definition 10.4.1). Truncated SVD trades bias for variance (Proposition 10.4.3).
  5. The least squares solution has sensitivity  $\kappa + \kappa^2\eta$ , where  $\eta$  measures the residual relative to the fit; residuals have sensitivity  $\kappa$  only (Theorem 10.5.1). The normal equations add  $\kappa^2$  regardless. Equilibrating the columns is nearly optimal among scalings (Proposition 10.5.2, Example (Longley's data, exactly)).
  6. A triangular factor can be updated or downdated by one row in  $O(p^2)$  (Theorem 10.6.1), and by one column in  $O(np)$  (Proposition 10.6.3). All leave-one-out residuals, coefficients and sums of squares follow from a single fit (Proposition 10.6.2).
  7. QR with column pivoting orders the columns by how much each adds to the others, and a small trailing pivot certifies a small singular value (Theorem 10.7.2). Basic and minimum-norm solutions agree on everything estimable (Example (Firm effects in investment data)); the tolerance should reflect the accuracy of the data (Example (A nearly aliased column)).

## 10.8.2 Notes and sources

**Source coverage.** The chapter covers Seber and Lee (2003, chapter 11) on solving the normal equations, QR by Gram–Schmidt, Householder and Givens, the SVD, centring and leverages (sections 11.2–11.4, 11.7 and 11.10), with parts of sections 11.6, 11.8 and 11.9 on updating, comparing methods and rank deficiency.

**Floating point and stability.** The standard model and the  $\gamma_n$  notation follow Higham (2002), whose chapters 3, 10, 19 and 20 are the definitive treatment of inner products, Cholesky, QR and least squares. Goldberg (1991) is a readable introduction to IEEE arithmetic, now specified by IEEE Std 754-2019. The distinction between a problem's condition and an algorithm's backward stability goes back to Wilkinson (1963). Trefethen and Bau (1997) give a short modern account. Läuchli (1961) introduced the matrix of Example (A cross-product matrix that rounds to a singular one) to show the loss in forming normal equations.

**Algorithms.** Householder (1958) introduced the reflections and Golub (1965) applied them to least squares; Givens (1958) used plane rotations. Gentleman (1973) developed square-root-

free rotations and the row-wise algorithm, turned into widely used routines by Miller (1992). Björck (1967) analysed modified Gram–Schmidt, and Björck and Paige (1992) explained its good behaviour for least squares. The SVD algorithm is due to Golub and Kahan (1965) and Golub and Reinsch (1970), column pivoting to Businger and Golub (1965), and the counterexample of [Example \(Kahan’s matrix\)](#) to Kahan (1966). Chan (1987) gave the first rank-revealing QR algorithm, Hong and Pan (1992) proved that factorizations with polynomial bounds exist, and Gu and Eisenstat (1996) showed how to compute them efficiently. Goodnight (1979) describes the sweep operator for statisticians. Welford (1962) gave the updating recursion, and Chan, Golub and LeVeque (1983) compare variance algorithms. Downdating appears in LINPACK (Dongarra et al. 1979) and was analysed by Stewart (1979). The standard monographs are Björck (1996), Lawson and Hanson (1974) and Golub and Van Loan (2013); Thisted (1988) and Kennedy and Gentle (1980) take the statistician’s side.

**Perturbation theory.** Golub and Wilkinson (1966) first observed that the least squares solution can be sensitive to  $\kappa^2$  when the residual is large. Wedin (1973) gave sharp perturbation bounds for pseudoinverses and least squares. The equilibration result of [Proposition 10.5.2](#) is from van der Sluis (1969). The inequality of [Proposition 10.4.2](#) is the singular-value form of Weyl’s (1912) inequality for eigenvalues. Beaton, Rubin and Barone (1976) argued that errors from rounding the data should be the yardstick for computational accuracy, the point made at the end of [Section 10.5](#).

**Statistics.** Recursive least squares via the Sherman–Morrison formula goes back to Plackett (1950). The PRESS statistic is due to Allen (1974). The recursive residuals of [Exercise \(C1\)](#) are those of Brown, Durbin and Evans (1975). Deletion diagnostics are developed in [Chapter 20](#), collinearity in [Chapter 26](#), and ridge and principal components regression in [Chapter 27](#).

### 10.8.3 References

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# Chapter 11

## Testing Models: The General Linear Hypothesis

Almost every question put to a linear model asks whether a simpler model would do: whether a group of regressors can be dropped, whether two coefficients are equal, whether regions differ once other variables are accounted for. Each such question names a subspace of the model space. The whole theory of testing it rests on one picture. The part of the data that the simpler model cannot explain splits orthogonally into a piece in the *test space* and a piece in the residual space. The  $F$  statistic compares the squared lengths of the two pieces, per dimension. Normality turns the comparison into an exact test. This chapter derives that test in the three forms met in practice, as a comparison of fitted models, as a statement about estimable functions  $\mathbf{A}^\top \boldsymbol{\beta} = \mathbf{d}$ , and as a likelihood ratio. It shows that the Wald and score tests are the same test in disguise, and proves that no invariant test is more powerful. It then studies power, which depends on the unknown parameters only through one noncentrality, and ends with the  $t$  test, the overall  $F$  test and tests of groups of coefficients, including the ways a group test and the individual tests within it can disagree. Warnings about what a nonsignificant  $F$  does and does not show run through the chapter.

**What you need.** The normal linear model and its sampling distributions from [Chapter 7](#) (Eq. 7.3.1, [Theorem 7.5.1](#), [Corollary 7.5.2](#), [Theorem 7.3.1](#)). The noncentral  $\chi^2$ ,  $F$  and  $t$  distributions and the  $F$  statistic for nested subspaces from [Chapter 4](#) ([Definition 4.1.2](#), [Theorem 4.1.2](#), [Theorem 4.1.4](#), [Theorem 4.4.6](#)). Nested projections and the constraint space from [Chapter 6](#) ([Theorem 6.5.1](#), [Theorem 6.5.2](#), [Theorem 6.3.7](#)). Estimable functions from [Chapter 8](#) ([Definition 8.2.1](#), [Theorem 8.3.1](#)). Extra sums of squares and the sum of squares of a linear hypothesis from [Chapter 9](#) ([Definition 9.2.1](#), [Theorem 9.2.4](#), [Theorem 9.2.5](#), [Proposition 9.2.6](#)).

### 11.0.1 Roadmap

- [Testing reduced models](#)
- [The F statistic as a comparison of projections](#)
- [Testable hypotheses](#)
- [The likelihood ratio test and its relatives](#)
- [Power and noncentrality](#)
- [Testing single coefficients and groups of coefficients](#)
- [Summary and notes](#)

## 11.1 Testing reduced models

Part II fitted linear models without asking whether a fitted term was needed. Part III adds normal errors, and with them exact answers to that question. Does the murder rate of a state depend on its region once poverty and family structure are accounted for? Is a straight line enough, or is there curvature? Do two groups share a regression line? Is a coefficient equal to the value a theory predicts? Each asks whether a *reduced model*, a special case of the model already fitted, is adequate. This section sets up that comparison and derives its test. The rest of the chapter looks at the same test from other directions.

### 11.1.1 Hypotheses are subspaces

Throughout the chapter the data follow the normal linear model

$$\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}), \quad \boldsymbol{\beta} \in \mathbb{R}^p, \sigma^2 > 0,$$

of Eq. 7.3.1, with  $r = \text{rank}(\mathbf{X}) < n$ . We call it the **full model**. Its content is that the mean vector  $\boldsymbol{\mu} = E(\mathbf{Y})$  lies in the  $r$ -dimensional subspace  $\mathcal{C}(\mathbf{X})$  (Section 6.1). A reduced model says more: the mean lies in a smaller subspace.

#### Definition 11.1.1 — Reduced model and linear hypothesis

Let  $\mathbf{X}_0$  be an  $n \times p_0$  matrix with  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  and  $r_0 = \text{rank}(\mathbf{X}_0) < r$ . The **reduced model** is  $E(\mathbf{Y}) = \mathbf{X}_0\mathbf{c}$ ,  $\mathbf{c} \in \mathbb{R}^{p_0}$ . Inside the full model it corresponds to the **linear hypothesis**

$$H_0 : \mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0) \quad \text{against} \quad H_1 : \mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}), \mathbf{X}\boldsymbol{\beta} \notin \mathcal{C}(\mathbf{X}_0).$$

The number  $q = r - r_0$  is the number of **degrees of freedom of the hypothesis**, and we write  $\mathbf{M}$  and  $\mathbf{M}_0$  for the orthogonal projections onto  $\mathcal{C}(\mathbf{X})$  and  $\mathcal{C}(\mathbf{X}_0)$ .

Three features of the definition deserve comment. First, the hypothesis is a statement about the mean vector, not about coefficients. Coefficients enter only through the column spaces, so recoding the regressors of either model changes nothing (Theorem 6.7.1). Statements about coefficients, such as  $\beta_2 = \beta_3$ , are translated into subspaces in Section 11.3. Second, the models must be *nested*. Two models neither of which contains the other, say one with poverty and one with urbanization, are not compared by the tests of this chapter. Third, both models share the same  $\sigma^2$  and the same error law. The alternative is not “anything other than the reduced model”. It is “the full model, but not the reduced one”.

The most common reduced models come from three operations on  $\mathbf{X}$ :

- *Deleting columns.* Dropping a block  $\mathbf{X}_2$  from  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$  tests whether those regressors are needed given the others. With a single column this is the familiar question about one coefficient. With all columns except the intercept it is the “overall” test of Section 11.6.
- *Merging columns.* Replacing  $\mathbf{x}_2$  and  $\mathbf{x}_3$  by their sum  $\mathbf{x}_2 + \mathbf{x}_3$  forces equal coefficients. Merging the indicator columns of several groups forces those groups to share a mean.
- *Fixing a component.* Moving a known multiple of a column into an offset, for example testing that a slope equals 1, gives a reduced model of the form  $E(\mathbf{Y}) = \mathbf{X}_0\mathbf{c} + \mathbf{X}\mathbf{b}_0$  with  $\mathbf{b}_0$  known. It is handled by subtracting  $\mathbf{X}\mathbf{b}_0$  from the data (Section 11.3).

## 11.1.2 The $F$ statistic

If the reduced model is adequate, forcing the fit into  $\mathcal{C}(\mathbf{X}_0)$  should cost little: the residual sum of squares  $\text{SSE}_0 = \|(\mathbf{I} - \mathbf{M}_0)\mathbf{y}\|^2$  should exceed  $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2$  by no more than noise would produce. The difference is the extra sum of squares of [Section 9.2](#), and by [Eq. 6.5.2](#) it is a squared length,

$$\text{SSE}_0 - \text{SSE} = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \mathbf{y}^\top (\mathbf{M} - \mathbf{M}_0)\mathbf{y},$$

the squared length of the projection of  $\mathbf{y}$  onto the orthogonal complement of  $\mathcal{C}(\mathbf{X}_0)$  within  $\mathcal{C}(\mathbf{X})$ . How large is “more than noise”? Noise contributes  $\sigma^2$  per dimension ([Theorem 9.6.1](#)), and  $\mathbf{M} - \mathbf{M}_0$  has rank  $q$ , so we compare the **hypothesis mean square**  $(\text{SSE}_0 - \text{SSE})/q$  with the unbiased estimate  $s^2 = \text{SSE}/(n - r)$  of  $\sigma^2$  from the full model. The ratio

$$F = \frac{(\text{SSE}_0 - \text{SSE})/(r - r_0)}{\text{SSE}/(n - r)} = \frac{\mathbf{y}^\top (\mathbf{M} - \mathbf{M}_0)\mathbf{y}/q}{\mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{y}/(n - r)} \quad (11.1.1)$$

is the  $F$  **statistic** for the reduced model. It is scale-free: multiplying  $\mathbf{y}$  by a constant, or changing the units of the response, does not change it. Its exact distribution is the content of the first theorem of the chapter.

### Theorem 11.1.1 — The $F$ test of a reduced model

Assume the normal linear model [Eq. 7.3.1](#) and the setting of [Definition 11.1.1](#), so that  $0 \leq r_0 < r < n$ . Then:

- (a)  $\text{SSE}_0 - \text{SSE}$  and  $\text{SSE}$  are independent, with

$$\frac{\text{SSE}_0 - \text{SSE}}{\sigma^2} \sim \chi^2(r - r_0, \gamma), \quad \frac{\text{SSE}}{\sigma^2} \sim \chi^2(n - r);$$

- (b) the statistic [Eq. 11.1.1](#) has the noncentral  $F$  distribution

$$F \sim F(r - r_0, n - r, \gamma), \quad \gamma = \frac{\|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2}{\sigma^2}; \quad (11.1.2)$$

- (c)  $\gamma = \|(\mathbf{I} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2 = \min_{\mathbf{c}} \|\mathbf{X}\boldsymbol{\beta} - \mathbf{X}_0\mathbf{c}\|^2/\sigma^2$ , the squared distance from the mean vector to the reduced model space in units of  $\sigma^2$ . In particular  $\gamma = 0$  iff  $H_0$  holds;

- (d) under  $H_0$ ,  $F \sim F(r - r_0, n - r)$ , the central  $F$  distribution, whatever the values of  $\boldsymbol{\beta}$  and  $\sigma^2$ .

*Proof.* By [Theorem 6.5.1](#),  $\mathbf{M} - \mathbf{M}_0$  and  $\mathbf{I} - \mathbf{M}$  are symmetric idempotent with ranks  $r - r_0$  and  $n - r$ , and  $(\mathbf{M} - \mathbf{M}_0)(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ . [Theorem 4.4.4](#), applied with  $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ , shows that  $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}\|^2/\sigma^2$  and  $\|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2/\sigma^2$  are independent with distributions  $\chi^2(r - r_0, \|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2)$  and  $\chi^2(n - r, \|(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta}\|^2/\sigma^2)$ . The second noncentrality is zero because  $\mathbf{M}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}$ . With [Eq. 6.5.2](#) this is (a). Part (b) follows from (a) and [Definition 4.1.2](#), since  $\sigma^2$  cancels in the ratio. (It is also [Theorem 4.4.6](#) with  $\mathbf{P} = \mathbf{M}$  and  $\mathbf{P}_0 = \mathbf{M}_0$ .) For (c),  $\mathbf{M}\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}$  gives  $(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta} = (\mathbf{I} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}$ , which is the difference between  $\mathbf{X}\boldsymbol{\beta}$  and its nearest point in  $\mathcal{C}(\mathbf{X}_0)$  ([Theorem 6.3.1](#)). Its length is therefore the distance from  $\mathbf{X}\boldsymbol{\beta}$  to  $\mathcal{C}(\mathbf{X}_0)$ , which is zero iff  $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$ . Part (d) is (b) with  $\gamma = 0$ .

Part (d) is what makes a test possible. The reduced model is a *composite* hypothesis: it leaves  $\mathbf{c}$  and  $\sigma^2$  unspecified. Yet the null distribution of  $F$  is the same at every point of it. The unknown coefficients cancel because  $(\mathbf{M} - \mathbf{M}_0)\mathbf{X}_0\mathbf{c} = \mathbf{0}$  and  $(\mathbf{I} - \mathbf{M})\mathbf{X}_0\mathbf{c} = \mathbf{0}$ , and the unknown scale cancels in the ratio. A statistic with this property is called *pivotal* under the hypothesis.

Part (c) says what the test detects. The alternative is measured by one number, the squared distance of the true mean from the reduced model, relative to the noise variance. Two alternatives with the same  $\gamma$  are equally hard to detect, however different their coefficients. [Section 11.5](#) is built on this.

### 11.1.3 The test and its p-value

Large values of  $F$  are evidence against  $H_0$ : under the alternative the numerator mean square estimates  $\sigma^2 + \sigma^2\gamma/q$  while the denominator still estimates  $\sigma^2$  ([Eq. 9.6.1](#)). The  $F$  test at level  $\alpha$  rejects  $H_0$  when

$$F > F_\alpha(r - r_0, n - r),$$

where  $F_\alpha(a, b)$  is the upper  $\alpha$  point of the central  $F(a, b)$  distribution. By [Theorem 11.1.1\(d\)](#) its probability of rejecting is exactly  $\alpha$  at every point of the reduced model. It rejects with probability greater than  $\alpha$  at every point of the alternative, by (c) and [Theorem 4.1.4](#). Instead of a fixed level, one usually reports the **p-value**

$$p(\mathbf{y}) = \Pr\{F(r - r_0, n - r) \geq F_{\text{obs}}\},$$

the probability under the reduced model of a statistic at least as extreme as the one observed.

#### Proposition 11.1.2 — The p-value is uniform under the hypothesis

Under  $H_0$ , the p-value  $p(\mathbf{Y})$  has the uniform distribution on  $(0, 1)$ . For every  $\alpha \in (0, 1)$ , rejecting when  $p(\mathbf{Y}) \leq \alpha$  is the  $F$  test at level  $\alpha$ .

*Proof.* Let  $\Phi_F$  be the distribution function of  $F(q, n - r)$ . It is continuous and strictly increasing on  $(0, \infty)$ , since the distribution has a positive density there, and  $p(\mathbf{y}) = 1 - \Phi_F(F_{\text{obs}})$ . Under  $H_0$  the statistic  $F$  has distribution function  $\Phi_F$ , so for  $u \in (0, 1)$ ,  $\Pr\{p(\mathbf{Y}) \leq u\} = \Pr\{\Phi_F(F) \geq 1 - u\} = \Pr\{F \geq \Phi_F^{-1}(1 - u)\} = u$ . Finally,  $p(\mathbf{y}) \leq \alpha$  iff  $\Phi_F(F_{\text{obs}}) \geq 1 - \alpha$  iff  $F_{\text{obs}} \geq F_\alpha(q, n - r)$ , and the event  $F = F_\alpha$  has probability zero.

The p-value is a statement about the data under  $H_0$ . It is *not* the probability that  $H_0$  is true, and it is not the probability that the result is “due to chance”. A small p-value says that data like these would be unusual if the reduced model held.

#### Example. (Do regions matter?)

In the 2009 data on 50 US states used in [Chapter 6](#) and [Chapter 9](#), regress the murder rate on the poverty rate, the percentage of single-parent households and the percentage living in urbanized areas, with an intercept. This is the reduced model, with  $r_0 = 4$ .

The full model adds indicators of three of the four Census regions (Northeast, South, West, with the Midwest as reference), so  $r = 7$ . The question is whether the regions differ in murder rate once the three covariates are accounted for. The listing gives

| Source                        | Degrees of freedom | Sum of squares | Mean square |
|-------------------------------|--------------------|----------------|-------------|
| Regions, given the covariates | 3                  | 14.83          | 4.944       |
| Residual, full model          | 43                 | 87.13          | 2.026       |
| Residual, reduced model       | 46                 | 101.96         |             |

so  $F = 4.944/2.026 = 2.440$  on 3 and 43 degrees of freedom. The 5% critical value is 2.822 and the p-value is 0.077. At the conventional level the reduced model is not rejected. The data do not show convincingly that the regions differ, given the covariates.

Yet in the full model the Northeast coefficient is  $-1.736$  murders per 100,000, with a  $t$  statistic of  $-2.519$  and a p-value of 0.016. One of the three contrasts looks clearly nonzero while the joint test does not reject. There is no contradiction: [Section 11.6](#) shows how a joint test can dilute one strong effect among weak ones, and [Chapter 13](#) explains why a contrast singled out after looking at the estimates cannot be judged by its own p-value.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statecrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
# Census region of each state, in the data's alphabetical order (N, M, S, W)
codes = "SWWSWWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNNMSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates]) # reduced model
D = np.column_stack([(region == g).astype(float) for g in "NSW"]) # Midwest is the
# reference
X = np.column_stack([X0, D]) # full model

def sse(Z):
    """Residual sum of squares of the least squares fit of y on the columns of Z."""
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    return np.sum((y - Z @ b) ** 2)

r, r0 = np.linalg.matrix_rank(X), np.linalg.matrix_rank(X0)
sse_full, sse_reduced = sse(X), sse(X0)
F = ((sse_reduced - sse_full) / (r - r0)) / (sse_full / (n - r))
p_value = stats.f.sf(F, r - r0, n - r)
crit = stats.f.ppf(0.95, r - r0, n - r)
print(f"SSE0 = {sse_reduced:.2f}, SSE = {sse_full:.2f}, df = ({r - r0}, {n - r})")
print(f"F = {F:.3f}, p = {p_value:.4f}, 5% critical value = {crit:.3f}")
```

## 11.1.4 What the test assumes

[Theorem 11.1.1](#) rests on three assumptions.

*The full model must hold.* The denominator  $s^2$  estimates  $\sigma^2$  only if  $E(\mathbf{Y}) \in \mathcal{C}(\mathbf{X})$ . If the full model omits something,  $E(s^2)$  exceeds  $\sigma^2$  by  $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/(n - r)$  ([Theorem 9.6.1\(b\)](#)), the denominator is inflated, and  $F$  is pushed towards zero. The test then loses power, and a nonsignificant result says even less than usual. The test of a reduced model is a test *within* the full model. It cannot certify the full model itself. [Chapter 14](#) tests the full model against a nonparametric alternative when there are replicate observations.

*The errors must be independent with a common variance.* Correlated or heteroscedastic errors change both numerator and denominator, and the null distribution is no longer  $F$ . [Chapter 21](#) and [Chapter 31](#) treat these cases.

*Normality* gives the exact distribution, and it is the least critical of the three. The null distribution is exactly  $F$  for every spherically symmetric error law ([Section 11.2](#)), and approximately so for independent non-normal errors unless the sample is small or some points have high leverage ([Theorem 19.3.2](#)).

*Warning* (What a nonsignificant  $F$  does not show). Failing to reject the reduced model is *not* evidence that the reduced model is true. The test controls the probability of wrongly rejecting  $H_0$ . It says nothing on its own about the probability of wrongly accepting it, which depends on the unknown noncentrality  $\gamma$  and may be large. In [Example \(Do regions matter?\)](#),  $p = 0.077$  is compatible with no regional differences. It is equally compatible with sizeable regional differences. [Section 11.5](#) computes that the test had only about an even chance of detecting a Northeast shortfall of 1.5 murders per 100,000, about the size estimated in [Section 11.3](#) (1.379 below the other three regions). To support “the reduced model is adequate”, one must show that every alternative of practical importance would have been rejected. That calls for a power calculation made before the data were seen, or better, a confidence region for the omitted effects that excludes all important values ([Chapter 12](#)), or a test of equivalence in which the roles of hypothesis and alternative are reversed. Nor does a nonsignificant  $F$  license dropping the terms and then reporting the reduced fit as if it had been specified in advance.

*Warning* (What a significant  $F$  does not show). A significant  $F$  says that the mean vector is not in  $\mathcal{C}(\mathbf{X}_0)$ , assuming the full model holds. It does not say which of the  $q$  dimensions of the test space carries the departure, nor that every omitted coefficient is nonzero. It does not say that the departure is large enough to matter: with large  $n$ , a tiny  $\gamma/n$  is detected with certainty ([Section 11.5](#)). And it does not say that the full model is correct. The statistic only compares two models, and a third, better model may exist.

## 11.1.5 Exercises

### 11.1.5.1 A. Check your understanding

#### Exercise A1.

Show that the  $F$  statistic [Eq. 11.1.1](#) does not change when  $\mathbf{y}$  is replaced by  $a\mathbf{y} + \mathbf{X}_0\mathbf{c}$

for any  $a \neq 0$  and any vector  $\mathbf{c}$ . Interpret the result for the region test of [Example \(Do regions matter?\)](#) when murder rates are expressed per million instead of per 100,000.

### Exercise A2.

In the model  $E(Y_i) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \beta_3 x_{i3}$ , write down a matrix  $\mathbf{X}_0$  (or an offset and an  $\mathbf{X}_0$ ) for each reduced model and give  $q$ : (i)  $\beta_2 = \beta_3 = 0$ ; (ii)  $\beta_1 = \beta_2 = \beta_3$ ; (iii)  $\beta_1 = 2$ ; (iv)  $\beta_2 + \beta_3 = 0$ .

### 11.1.5.2 B. Practice

#### Exercise B1.

Suppose  $\sigma^2 = \sigma_0^2$  is known. Show that  $X^2 = (\text{SSE}_0 - \text{SSE})/\sigma_0^2$  has the  $\chi^2(q, \gamma)$  distribution and that the test rejecting when  $X^2$  exceeds the upper  $\alpha$  point of  $\chi^2(q)$  has level  $\alpha$ . Show that  $qF/X^2 \rightarrow 1$  in probability as  $n \rightarrow \infty$  with  $q$  fixed and the full model true, and explain why the  $F$  test and this test then behave alike.

*Solution.* The distribution is [Theorem 11.1.1\(a\)](#) with  $\sigma^2 = \sigma_0^2$ ; under  $H_0$  it is  $\chi^2(q)$ , so the level is exact. Now  $qF/X^2 = \sigma_0^2/s^2$ , and  $s^2 \rightarrow \sigma^2 = \sigma_0^2$  in probability, since  $E(s^2) = \sigma^2$  and  $\text{Var}(s^2) = 2\sigma^4/(n-r) \rightarrow 0$  ([Theorem 7.5.1\(d\)](#)) and Chebyshev's inequality applies. So  $qF$  and  $X^2$  differ by a factor tending to one, and  $qF_\alpha(q, n-r)$  tends to the upper  $\alpha$  point of  $\chi^2(q)$  for the same reason. With few residual degrees of freedom the two tests differ, and only the  $F$  test has the right level when  $\sigma^2$  must be estimated.

#### Exercise B2.

Suppose the full model is wrong:  $E(\mathbf{Y}) = \boldsymbol{\mu}$  with  $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \boldsymbol{\delta} \neq \mathbf{0}$ , but  $(\mathbf{M} - \mathbf{M}_0)\boldsymbol{\mu} = \mathbf{0}$ . Show that the numerator and denominator of [Eq. 11.1.1](#) are still independent, that  $E(\text{numerator}) = \sigma^2$  and  $E(\text{denominator}) = \sigma^2 + \|\boldsymbol{\delta}\|^2/(n-r)$ , and that  $\Pr(F \leq c) \geq \Pr\{F(q, n-r) \leq c\}$  for every  $c > 0$ . Explain why an  $F$  statistic far below one, with a p-value near one, is a hint that the full model is inadequate.

*Solution.* Independence and the distributions follow from [Theorem 4.4.4](#) as in the proof of [Theorem 11.1.1](#): the numerator is  $\sigma^2 \chi^2(q)/q$ , and  $\text{SSE}/\sigma^2 \sim \chi^2(n-r, \|\boldsymbol{\delta}\|^2/\sigma^2)$ , with mean  $n-r + \|\boldsymbol{\delta}\|^2/\sigma^2$  ([Theorem 4.1.2\(b\)](#)). For the probability, condition on the numerator  $U$ :  $F \leq c$  iff  $\text{SSE}/\sigma^2 \geq (n-r)U/(qc\sigma^2)$ , and a noncentral  $\chi^2$  exceeds any fixed value with at least the central probability ([Proposition 4.1.3](#)). Averaging over  $U$  gives the inequality. So omitted structure in the residual space makes small values of  $F$  more likely. Under the full model,  $\Pr(F \leq c)$  is small for small  $c$ , and an observed  $F$  in the far lower tail is surprising.

#### Exercise B3.

Let  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X}_1) \subseteq \mathcal{C}(\mathbf{X})$  with ranks  $r_0 < r_1 < r < n$ , and let  $s^2 = \text{SSE}/(n-r)$  come

from the largest model. Show that

$$F_1 = \frac{\mathbf{y}^\top (\mathbf{M}_1 - \mathbf{M}_0) \mathbf{y} / (r_1 - r_0)}{s^2}$$

has the  $F(r_1 - r_0, n - r, \gamma_1)$  distribution with  $\gamma_1 = \|(\mathbf{M}_1 - \mathbf{M}_0) \mathbf{X} \boldsymbol{\beta}\|^2 / \sigma^2$  whenever the largest model holds, and that its numerator is independent of that of the test of  $\mathbf{X}_1$  within  $\mathbf{X}$ . Why is this preferable to using the residual mean square of  $\mathbf{X}_1$ ?

#### Exercise B4.

Two samples of sizes  $n_1$  and  $n_2$  follow  $E(\mathbf{Y}_k) = \mathbf{X}_k \boldsymbol{\beta}_k$ ,  $k = 1, 2$ , with a common  $\sigma^2$  and  $\mathbf{X}_k$  of full column rank  $p$ . Write the model for both samples together as a full model with block-diagonal model matrix, and the hypothesis  $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_2$  as a reduced model. Show that

$$F = \frac{(\text{SSE}_{\text{pooled}} - \text{SSE}_1 - \text{SSE}_2)/p}{(\text{SSE}_1 + \text{SSE}_2)/(n_1 + n_2 - 2p)},$$

where  $\text{SSE}_k$  comes from fitting sample  $k$  alone and  $\text{SSE}_{\text{pooled}}$  from one fit to all the data.

Apply the test to the question whether two firms of the Grunfeld panel (`sm.datasets.grunfeld`) share their investment equation, and comment on the assumption of a common  $\sigma^2$ .

#### 11.1.5.3 C. Going deeper

#### Exercise C1.

In [Exercise \(B4\)](#) suppose instead that  $n_2 \leq p$  and  $\text{rank}(\mathbf{X}_2) = n_2$ , so that  $\boldsymbol{\beta}_2$  cannot be estimated from the second sample. Show that the full model then fits the second sample exactly, identify the test space, and show that the  $F$  statistic of [Theorem 11.1.1](#) becomes

$$F = \frac{(\text{SSE}_{\text{pooled}} - \text{SSE}_1)/n_2}{\text{SSE}_1/(n_1 - p)},$$

a test of whether the second sample is predicted by the fit to the first, on  $n_2$  and  $n_1 - p$  degrees of freedom.

## 11.2 The F statistic as a comparison of projections

The formula [Eq. 11.1.1](#) mentions two residual sums of squares, but the test is really about three subspaces and the way the data vector sits among them. Seen this way, the  $F$  statistic is a function of a single angle, its rejection region is a cone, and its null distribution survives far beyond normal errors. The same view gives a canonical form in which every linear hypothesis looks the same. [Section 11.4](#) needs that form to prove that the  $F$  test is optimal.



### 11.2.1 Three orthogonal pieces

By [Theorem 6.5.1](#), nested model spaces split  $\mathbb{R}^n$  into three mutually orthogonal subspaces:

$$\mathbb{R}^n = \underbrace{\mathcal{C}(\mathbf{X}_0)}_{\dim r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X}_0)^\perp}_{\dim q=r-r_0} \oplus \underbrace{\mathcal{C}(\mathbf{X})^\perp}_{\dim n-r}, \quad \mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M}). \quad (11.2.1)$$

The middle subspace,  $\mathcal{C}(\mathbf{M} - \mathbf{M}_0)$ , is the **test space** of the hypothesis. It consists of the directions in which the full model can move the mean and the reduced model cannot. Every data vector splits accordingly,

$$\mathbf{y} = \mathbf{M}_0\mathbf{y} + (\mathbf{M} - \mathbf{M}_0)\mathbf{y} + (\mathbf{I} - \mathbf{M})\mathbf{y},$$

and so does the mean vector,  $\mathbf{X}\boldsymbol{\beta} = \mathbf{M}_0\mathbf{X}\boldsymbol{\beta} + (\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta} + \mathbf{0}$ . The hypothesis says that the middle component of the *mean* is zero. The middle component of the *data*,  $(\mathbf{M} - \mathbf{M}_0)\mathbf{Y}$ , is its unbiased estimate, with covariance  $\sigma^2(\mathbf{M} - \mathbf{M}_0)$ . The residual component  $(\mathbf{I} - \mathbf{M})\mathbf{Y}$  has mean zero whether or not the hypothesis holds, and it serves as a yardstick for noise. The first component  $\mathbf{M}_0\mathbf{Y}$  estimates the part of the mean that both models allow. It carries no information about the hypothesis, and the  $F$  statistic ignores it.

What the statistic does use is  $(\mathbf{I} - \mathbf{M}_0)\mathbf{y}$ , the residual vector of the reduced model. It is split into a piece in the test space and a piece in the residual space, and

$$F = \frac{\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2/q}{\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2/(n-r)}$$

compares their squared lengths per dimension. Under the hypothesis both pieces are pure noise, and each dimension carries on average  $\sigma^2$ . Under the alternative the first piece also carries the signal  $(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}$ .

The test space can be described by a spanning set. Since  $\mathbf{M} - \mathbf{M}_0 = (\mathbf{I} - \mathbf{M}_0)\mathbf{M}$  by [Theorem 6.5.1\(a\)](#), and  $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$ ,

$$\mathcal{C}(\mathbf{M} - \mathbf{M}_0) = \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}).$$

The test space is spanned by the columns of  $\mathbf{X}$  after the reduced model has been regressed out of them. When  $\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_2]$  adds a block  $\mathbf{X}_2$ , this is  $\mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2)$ , the space of [Lemma 9.2.1](#). For a hypothesis on coefficients it is the constraint space  $\mathcal{C}(\mathbf{M}\mathbf{T})$  of [Theorem 6.5.2](#), as [Section 11.3](#) shows.

### 11.2.2 One angle

Because the two pieces of  $(\mathbf{I} - \mathbf{M}_0)\mathbf{y}$  are orthogonal, their lengths are the legs of a right triangle whose hypotenuse is the reduced model's residual vector.

#### Proposition 11.2.1 — $F$ as a function of an angle

Let  $\mathbf{w} = (\mathbf{I} - \mathbf{M}_0)\mathbf{y} \neq \mathbf{0}$  and let  $\theta \in [0, \pi/2]$  be the angle between  $\mathbf{w}$  and the test space, defined by  $\cos \theta = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|/\|\mathbf{w}\|$ . Then

$$\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2 = \|\mathbf{w}\|^2 \cos^2 \theta, \quad \text{SSE} = \|\mathbf{w}\|^2 \sin^2 \theta, \quad F = \frac{n-r}{q} \cot^2 \theta.$$

The partial coefficient of determination of [Definition 9.5.1](#) is  $\cos^2 \theta$ , and the  $F$  test rejects exactly when  $\theta < \theta_\alpha$ , where  $\tan^2 \theta_\alpha = (n - r)/\{q F_\alpha(q, n - r)\}$ .

*Proof.*  $\mathbf{w} = (\mathbf{M} - \mathbf{M}_0)\mathbf{y} + (\mathbf{I} - \mathbf{M})\mathbf{y}$  with orthogonal summands, since  $(\mathbf{M} - \mathbf{M}_0)(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ . Pythagoras gives  $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \|\mathbf{w}\|^2 - \|\mathbf{w}\|^2 \cos^2 \theta = \|\mathbf{w}\|^2 \sin^2 \theta$ . Since  $(\mathbf{M} - \mathbf{M}_0)\mathbf{y}$  is the projection of  $\mathbf{w}$  onto the test space,  $\theta$  is also the angle between  $\mathbf{w}$  and its projection, the smallest angle between  $\mathbf{w}$  and any vector of the test space. The ratio of the two squared lengths is  $\cot^2 \theta$ , which gives  $F$ . The partial coefficient of determination is  $(\text{SSE}_0 - \text{SSE})/\text{SSE}_0 = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2/\|\mathbf{w}\|^2 = \cos^2 \theta$ . Finally  $\cot^2$  is strictly decreasing on  $(0, \pi/2]$ , so  $F > F_\alpha$  iff  $\cot^2 \theta > qF_\alpha/(n - r)$  iff  $\theta < \theta_\alpha$ . ■

So the  $F$  test asks a geometric question: *does the reduced model's residual vector point suspiciously close to the test space?* Under the hypothesis,  $\mathbf{w}$  is noise spread over  $n - r_0$  dimensions, only  $q$  of which belong to the test space, so it typically points well away from it. The rejection region  $\{\mathbf{y} : \theta < \theta_\alpha\}$  is a circular cone around the test space inside  $\mathcal{C}(\mathbf{X}_0)^\perp$ , extended by arbitrary components in  $\mathcal{C}(\mathbf{X}_0)$ . That this is the same as the partial  $R^2$  test of [Theorem 9.5.4\(b\)](#) is no coincidence. The partial  $R^2$  is a squared cosine, just as [Theorem 6.7.2](#) showed for the ordinary  $R^2$ .

#### Example. (The region test as an angle)

In [Example \(Do regions matter?\)](#), the reduced model's residual vector has a component of length 3.851 in the three-dimensional test space and a component of length 9.334 in the 43-dimensional residual space. The angle between the residual vector and the test space is  $\theta = 67.6^\circ$ , and the partial  $R^2$  is  $\cos^2 \theta = 0.145$ . The 5% test rejects when  $\theta < 66.1^\circ$ , so the data fall just outside the cone.

[Figure 11.2.1](#) plots the two lengths for the data and for 4000 simulated data sets. With the reduced model true, 0.053 of them fall in the rejection region. With the Northeast lowered by two murders per 100,000, and all else as fitted, the fraction is 0.774.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statecrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNNSSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates])
D = np.column_stack([(region == g).astype(float) for g in "NSW"])
X = np.column_stack([X0, D])

def proj(Z):
    """Orthogonal projection onto C(Z), from an orthonormal basis."""
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T
```

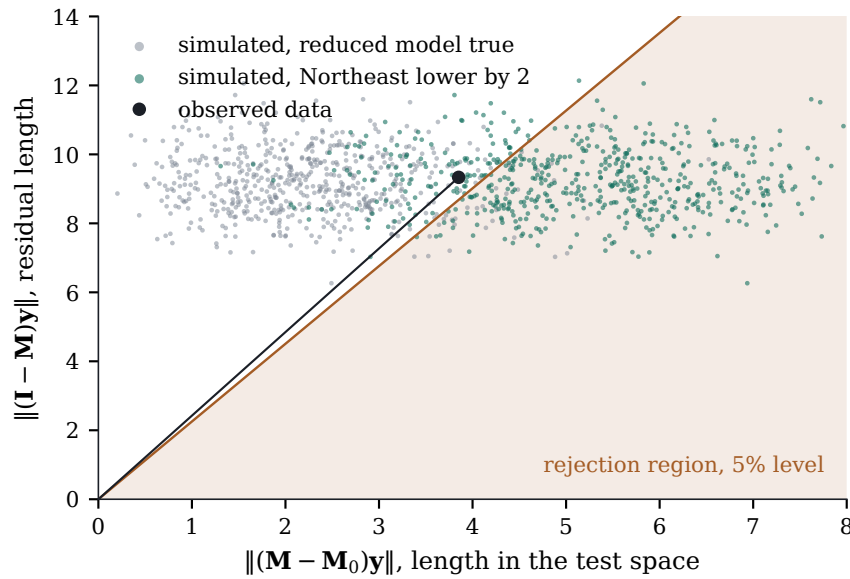


Figure 11.2.1: The region test of [Example \(Do regions matter?\)](#) in two coordinates: the length of  $(\mathbf{M} - \mathbf{M}_0)\mathbf{y}$  in the test space and the length of the residual vector  $(\mathbf{I} - \mathbf{M})\mathbf{y}$ . The  $F$  statistic depends only on the angle of the point, and the test rejects inside the shaded wedge. The grey points are simulated with the reduced model true, the green points with the Northeast lowered by two murders per 100,000. The observed data are the black point, just above the wedge.

```
M, M0 = proj(X), proj(X0)
q, nu = 3, n - 7
u = (M - M0) @ y
e = y - M @ y
w = y - M0 @ y
# r - r0 and n - r
# component in the test space
# residual vector
# what the reduced model leaves
unexplained
theta = np.arccos(np.linalg.norm(u) / np.linalg.norm(w))
F = (nu / q) / np.tan(theta) ** 2
theta_crit = np.arctan(np.sqrt(nu / (q * stats.f.ppf(0.95, q, nu))))
print(f"angle = {np.degrees(theta):.1f} degrees, F = {F:.3f}")
print(f"reject when the angle is below {np.degrees(theta_crit):.1f} degrees")
```

### 11.2.3 What the statistic ignores

Since  $F$  depends on  $\mathbf{y}$  only through the angle  $\theta$ , many transformations of the data leave it unchanged. They form the symmetry group of the testing problem, and [Section 11.4](#) uses them to characterize the  $F$  test.

#### Proposition 11.2.2 — Invariance of the $F$ statistic

The statistic [Eq. 11.1.1](#) depends on  $\mathbf{X}$  and  $\mathbf{X}_0$  only through  $\mathcal{C}(\mathbf{X})$  and  $\mathcal{C}(\mathbf{X}_0)$ . Where it is defined, that is, when  $(\mathbf{I} - \mathbf{M})\mathbf{y} \neq \mathbf{0}$ , it is unchanged when  $\mathbf{y}$  is replaced by

$$a\mathbf{H}\mathbf{y} + \mathbf{v}, \quad a \neq 0, \quad \mathbf{v} \in \mathcal{C}(\mathbf{X}_0),$$

where  $\mathbf{H}$  is any orthogonal matrix that maps  $\mathcal{C}(\mathbf{X}_0)$  onto itself and  $\mathcal{C}(\mathbf{X})$  onto itself.

*Proof.* The first claim holds because  $\mathbf{M}$  and  $\mathbf{M}_0$  are determined by the column spaces. For the second, the transformations can be applied one at a time. Adding  $\mathbf{v}$  changes nothing, because  $\mathbf{M} - \mathbf{M}_0$  and  $\mathbf{I} - \mathbf{M}$  annihilate  $\mathcal{C}(\mathbf{X}_0)$ . The factor  $a^2$  cancels in the ratio. For  $\mathbf{H}$ : the matrix  $\mathbf{HMH}^\top$  is symmetric and idempotent with column space  $\mathbf{HC}(\mathbf{X}) = \mathcal{C}(\mathbf{X})$ , so it equals  $\mathbf{M}$  (Theorem 6.3.3). Hence  $\mathbf{HM} = \mathbf{MH}$ , and likewise  $\mathbf{HM}_0 = \mathbf{M}_0\mathbf{H}$ . Then  $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{H}\mathbf{y}\| = \|\mathbf{H}(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\| = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|$ , because  $\mathbf{H}$  preserves lengths, and the same holds for  $\mathbf{I} - \mathbf{M}$ . ■

The orthogonal matrices allowed here are exactly those that rotate the test space within itself and the residual space within itself, and fix  $\mathcal{C}(\mathbf{X}_0)$  as a set. The invariance says that no direction within the test space is favoured. A departure of given size  $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|$  is equally detectable in every direction. When some directions matter more than others, for instance when one contrast is of primary interest, a test directed at them can have more power there. It pays for this with less power elsewhere (Section 11.6).

## 11.2.4 Canonical form

Orthonormal bases adapted to Eq. 11.2.1 turn every linear hypothesis into the same problem about independent normal coordinates.

### Proposition 11.2.3 — Canonical form of a linear hypothesis

Let  $\mathbf{Q} = [\mathbf{Q}_0, \mathbf{Q}_1, \mathbf{Q}_2]$  be an orthogonal  $n \times n$  matrix whose blocks of  $r_0$ ,  $q$  and  $n - r$  columns are orthonormal bases of  $\mathcal{C}(\mathbf{X}_0)$ , of the test space and of  $\mathcal{C}(\mathbf{X})^\perp$ . Put  $\mathbf{z}_k = \mathbf{Q}_k^\top \mathbf{Y}$  and  $\boldsymbol{\eta}_k = \mathbf{Q}_k^\top \mathbf{X}\boldsymbol{\beta}$ ,  $k = 0, 1, 2$ . Under Eq. 7.3.1:

- (a)  $\mathbf{z}_0, \mathbf{z}_1, \mathbf{z}_2$  are independent,  $\mathbf{z}_k \sim N(\boldsymbol{\eta}_k, \sigma^2 \mathbf{I})$ , and  $\boldsymbol{\eta}_2 = \mathbf{0}$ ;
- (b)  $\|\mathbf{z}_1\|^2 = \text{SSE}_0 - \text{SSE}$  and  $\|\mathbf{z}_2\|^2 = \text{SSE}$ , so  $F = \{\|\mathbf{z}_1\|^2/q\} / \{\|\mathbf{z}_2\|^2/(n - r)\}$ ;
- (c)  $\|\boldsymbol{\eta}_1\|^2 = \sigma^2\gamma$ , and  $H_0$  holds iff  $\boldsymbol{\eta}_1 = \mathbf{0}$ .

*Proof.*

- (a)  $\mathbf{Q}^\top \mathbf{Y} \sim N_n(\mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{Q}^\top \mathbf{Q}) = N_n(\mathbf{Q}^\top \mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$  by Theorem 3.3.1. Its blocks are uncorrelated, hence independent (Theorem 3.4.1). The columns of  $\mathbf{Q}_2$  are orthogonal to  $\mathcal{C}(\mathbf{X})$ , so  $\boldsymbol{\eta}_2 = \mathbf{Q}_2^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ . (b) By Proposition 6.3.2,  $\mathbf{Q}_1 \mathbf{Q}_1^\top = \mathbf{M} - \mathbf{M}_0$  and  $\mathbf{Q}_2 \mathbf{Q}_2^\top = \mathbf{I} - \mathbf{M}$ , so  $\|\mathbf{z}_1\|^2 = \mathbf{Y}^\top \mathbf{Q}_1 \mathbf{Q}_1^\top \mathbf{Y} = \mathbf{Y}^\top (\mathbf{M} - \mathbf{M}_0) \mathbf{Y}$  and similarly for  $\mathbf{z}_2$ . (c) In the same way  $\|\boldsymbol{\eta}_1\|^2 = \|(\mathbf{M} - \mathbf{M}_0)\mathbf{X}\boldsymbol{\beta}\|^2$ , and Theorem 11.1.1(c) applies. ■

In canonical form the problem is stripped to its essentials. One observes a  $q$ -vector  $\mathbf{z}_1$ , normal with unknown mean  $\boldsymbol{\eta}_1$  and covariance  $\sigma^2 \mathbf{I}$ , and wishes to test  $\boldsymbol{\eta}_1 = \mathbf{0}$ . An independent vector  $\mathbf{z}_2$  of pure noise, of dimension  $n - r$ , measures  $\sigma^2$ . The vector  $\mathbf{z}_0$  has an unknown mean that neither hypothesis restricts, so it is of no use. Every hypothesis of this chapter, whether about coefficients, contrasts or whole models, is an instance of this problem with particular  $q$  and  $n - r$ .

## 11.2.5 Beyond normal errors

The angle picture shows why normality is less important for the null distribution than it first appears. Under  $H_0$ ,  $(\mathbf{I} - \mathbf{M}_0)\mathbf{Y} = (\mathbf{I} - \mathbf{M}_0)\boldsymbol{\varepsilon}$ , so  $F$  is a function of the error vector alone, and it is a function of its *direction* only: replacing  $\boldsymbol{\varepsilon}$  by  $c\boldsymbol{\varepsilon}$ ,  $c > 0$ , leaves every angle unchanged. Any error law under which the direction of  $\boldsymbol{\varepsilon}$  behaves as it does for normal errors therefore gives the same null distribution. We call  $\mathbf{U}$  **uniformly distributed on the unit sphere** if it has the distribution of  $\mathbf{Z}/\|\mathbf{Z}\|$  for  $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$ .

### Proposition 11.2.4 — Spherical errors

Let  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$ , where  $\boldsymbol{\varepsilon} = R\mathbf{U}$  for a random variable  $R > 0$  and a vector  $\mathbf{U}$  uniformly distributed on the unit sphere. The variables  $R$  and  $\mathbf{U}$  need not be independent. Then  $F \sim F(r - r_0, n - r)$ .

*Proof.* Write  $g(\mathbf{v}) = \{\|(\mathbf{M} - \mathbf{M}_0)\mathbf{v}\|^2/q\} / \{\|(\mathbf{I} - \mathbf{M})\mathbf{v}\|^2/(n - r)\}$  for vectors with  $(\mathbf{I} - \mathbf{M})\mathbf{v} \neq \mathbf{0}$ . Under  $H_0$ ,  $(\mathbf{M} - \mathbf{M}_0)\mathbf{Y} = (\mathbf{M} - \mathbf{M}_0)\boldsymbol{\varepsilon}$  and  $(\mathbf{I} - \mathbf{M})\mathbf{Y} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$ , since both projections annihilate  $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$ . So  $F = g(\boldsymbol{\varepsilon})$ . Because  $g(c\mathbf{v}) = g(\mathbf{v})$  for  $c > 0$ ,  $F = g(R\mathbf{U}) = g(\mathbf{U})$ . The law of  $g(\mathbf{U})$  is the law of  $g(\mathbf{Z}/\|\mathbf{Z}\|) = g(\mathbf{Z})$  with  $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$ . By Theorem 11.1.1(d) with  $\sigma = 1$ , that law is  $F(r - r_0, n - r)$ . (The events  $\mathbf{Z} = \mathbf{0}$  and  $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$  have probability zero.)

■

The class of error laws covered is large. If  $\mathbf{Z} \sim N_n(\mathbf{0}, \mathbf{I})$  and  $S > 0$  is any random scale independent of  $\mathbf{Z}$ , or not, then  $\boldsymbol{\varepsilon} = S\mathbf{Z} = (S\|\mathbf{Z}\|)(\mathbf{Z}/\|\mathbf{Z}\|)$  qualifies. Taking  $S^{-2}$  proportional to a  $\chi^2(\nu)$  variable gives the spherical multivariate  $t$  distribution. Its components are uncorrelated and heavy-tailed but *not* independent: one common scale inflates or shrinks them all. The  $F$  test is exactly valid for all such laws. What the proposition does not cover are independent non-normal errors, whose direction is not uniform. For them the null distribution is only approximately  $F$ .

### Example. (Four error laws)

Keep the design of Example (Do regions matter?) and simulate 20000 data sets with the reduced model true, each with errors from one of four laws: independent normal; spherical  $t$  with three degrees of freedom; independent  $t$  with three degrees of freedom; independent centred exponential. The spherical errors are uncorrelated, but their sizes move together: the rank correlation between the absolute values of two components is 0.20. The rejection rates of the 5% test are

| Errors         | normal | spherical $t_3$ | independent $t_3$ | exponential |
|----------------|--------|-----------------|-------------------|-------------|
| Rejection rate | 0.0502 | 0.0502          | 0.0444            | 0.0474      |

with Monte Carlo standard error about 0.0015. The first two agree exactly, not just approximately: the spherical errors were made by rescaling the normal ones, and each rescaled data set gives the same  $F$  as its normal original. The two independent non-normal laws give rates slightly below 0.05. For this design, with 43 residual degrees of freedom and no extreme leverage, the independent  $t_3$  errors make the test slightly conservative, the exponential rate is within Monte Carlo error of 0.05, and both are close to the nominal level.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

rng = np.random.default_rng(1103)

data = sm.datasets.statcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNNSSWNSWSMW"
region = np.array(list(codes))
n = len(data)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates])
D = np.column_stack([(region == g).astype(float) for g in "NSW"])
X = np.column_stack([X0, D])

def proj(Z):
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T

M, M0 = proj(X), proj(X0)
q, nu = 3, n - 7
crit = stats.f.ppf(0.95, q, nu)

def F_stat(E):
    """F statistic for each row of E, used as a data vector under the reduced model.
    """
    num = np.sum((E @ (M - M0)) ** 2, axis=1) / q
    den = np.sum((E - E @ M) ** 2, axis=1) / nu
    return num / den

reps = 20000
Z = rng.standard_normal((reps, n))
scale = 1 / np.sqrt(rng.chisquare(3, size=(reps, 1)) / 3)  # one scale per data
set
errors = {
    "normal": Z,
    "spherical t3": Z * scale,  # dependent, heavy
    -tailed
    "independent t3": rng.standard_t(3, size=(reps, n)),
    "exponential": rng.exponential(size=(reps, n)) - 1.0,  # skewed
}
for name, E in errors.items():
    print(f"{name:15s} rejection rate {np.mean(F_stat(E) > crit):.4f}")
```

*Remark* (Only the null distribution). [Proposition 11.2.4](#) concerns the level of the test. Under an alternative,  $F$  depends on the error length  $R$  as well as the direction, and the

power under spherical  $t$  errors differs from the power under normal errors with the same covariance. Nor does the proposition say that the  $F$  test is a *good* test for heavy-tailed errors. Its power can be poor compared with robust methods. Those questions belong to [Chapter 19](#) and [Chapter 21](#).

## 11.2.6 Exercises

### 11.2.6.1 A. Check your understanding

#### Exercise A1.

Use [Proposition 11.2.1](#) to express  $\cos^2 \theta$  in terms of  $F$ ,  $q$  and  $n - r$ . Check the value 0.145 of [Example \(The region test as an angle\)](#) from  $F = 2.440$ ,  $q = 3$  and  $n - r = 43$ , and compute  $\theta_\alpha$  from  $F_{0.05}(3, 43) = 2.822$ .

### 11.2.6.2 B. Practice

#### Exercise B1.

Show that under  $H_0$  the squared cosine  $\cos^2 \theta$  of [Proposition 11.2.1](#) has the  $\text{Beta}(q/2, (n-r)/2)$  distribution, and that it is independent of  $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{Y}\|^2$ . What is its expectation?

*Solution.* By [Proposition 11.2.3](#),  $\cos^2 \theta = U/(U + V)$  with  $U = \|\mathbf{z}_1\|^2/\sigma^2 \sim \chi^2(q)$  and  $V = \|\mathbf{z}_2\|^2/\sigma^2 \sim \chi^2(n - r)$  independent under  $H_0$ , and  $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{Y}\|^2 = \sigma^2(U + V)$ . [Exercise \(B4\)](#) gives both the Beta law and the independence of  $U/(U + V)$  from  $U + V$ . The mean of  $\text{Beta}(a, b)$  is  $a/(a + b)$ , here  $q/(n - r_0)$ : under the hypothesis the residual vector of the reduced model puts, on average, a share of its squared length in the test space equal to that space's share of the dimensions.

#### Exercise B2.

Prove that  $\mathcal{C}(\mathbf{M} - \mathbf{M}_0) = \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X})$ , and that for  $\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_2]$  this equals  $\mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2)$ . Show that  $\dim \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2) = \text{rank}(\mathbf{X}) - \text{rank}(\mathbf{X}_0)$ .

*Solution.*  $\mathbf{M} - \mathbf{M}_0 = (\mathbf{I} - \mathbf{M}_0)\mathbf{M}$  by [Theorem 6.5.1\(a\)](#), so  $\mathcal{C}(\mathbf{M} - \mathbf{M}_0) \subseteq \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{M}) = (\mathbf{I} - \mathbf{M}_0)\mathcal{C}(\mathbf{M}) = (\mathbf{I} - \mathbf{M}_0)\mathcal{C}(\mathbf{X}) = \mathcal{C}((\mathbf{I} - \mathbf{M}_0)\mathbf{X})$ . Conversely each column  $(\mathbf{I} - \mathbf{M}_0)\mathbf{x}_j = \mathbf{x}_j - \mathbf{M}_0\mathbf{x}_j$  lies in  $\mathcal{C}(\mathbf{X})$  and is orthogonal to  $\mathcal{C}(\mathbf{X}_0)$ , so it lies in  $\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X}_0)^\perp = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$ . For  $\mathbf{X} = [\mathbf{X}_0, \mathbf{X}_2]$ ,  $(\mathbf{I} - \mathbf{M}_0)\mathbf{X} = [\mathbf{0}, (\mathbf{I} - \mathbf{M}_0)\mathbf{X}_2]$ . The dimension is  $\text{rank}(\mathbf{M} - \mathbf{M}_0) = r - r_0$  ([Theorem 6.5.1\(c\)](#)).

#### Exercise B3.

Which of the following error vectors satisfy the hypothesis of [Proposition 11.2.4](#)? (i)  $\boldsymbol{\varepsilon} = \sigma\mathbf{Z}$  with  $\sigma$  fixed; (ii)  $\boldsymbol{\varepsilon} = S\mathbf{Z}$  with  $S$  uniform on  $(1, 2)$  and independent of  $\mathbf{Z}$ ; (iii)  $\boldsymbol{\varepsilon}$  with independent Laplace components; (iv)  $\boldsymbol{\varepsilon} = \mathbf{D}\mathbf{Z}$  with  $\mathbf{D}$  a fixed diagonal matrix with unequal entries; (v)  $\boldsymbol{\varepsilon} = \mathbf{Z}/\|\mathbf{Z}\|^2$ .

## 11.2.6.3 C. Going deeper

**Exercise C1.**

In the canonical form, let two data vectors  $\mathbf{y}$  and  $\mathbf{y}'$  with  $\mathbf{z}_2, \mathbf{z}'_2 \neq \mathbf{0}$  have the same value of  $F$ . Construct  $a > 0$ ,  $\mathbf{v} \in \mathcal{C}(\mathbf{X}_0)$  and an orthogonal  $\mathbf{H}$  of the kind allowed in [Proposition 11.2.2](#) such that  $\mathbf{y}' = a\mathbf{H}\mathbf{y} + \mathbf{v}$ . (So  $F$  is a *maximal invariant*: any statistic unchanged by these transformations is a function of  $F$ .)

**Exercise C2.**

Take a one-way layout with two groups of sizes  $n_1 = 5$  and  $n_2 = 20$ , independent normal errors with standard deviations 3 in the small group and 1 in the large one, and equal means. Compute, by simulation or numerical integration, the true level of the nominal 5%  $F$  test of equal means. Explain the direction of the error using the angle picture: which way does the unequal variance tilt the residual vector?

## 11.3 Testable hypotheses

Scientific questions are usually phrased in terms of coefficients rather than subspaces: “the coefficients of poverty and single parenthood are equal”, “the four regions have the same intercept”, “the elasticity is one”. Such statements have the form

$$H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d},$$

with a known  $p \times m$  matrix  $\mathbf{\Lambda}$  and a known  $\mathbf{d} \in \mathbb{R}^m$ . This is the **general linear hypothesis**. This section decides when such a statement is a hypothesis about the distribution of  $\mathbf{Y}$  at all, gives its  $F$  statistic in terms of the unrestricted fit alone, shows that the statistic agrees with the reduced-model statistic of [Section 11.1](#), and describes least squares under the hypothesis. Nothing here assumes that  $\mathbf{X}$  has full column rank.

### 11.3.1 Which constraints are hypotheses

In the full model the distribution of  $\mathbf{Y}$  depends on  $\boldsymbol{\beta}$  only through  $\mathbf{X}\boldsymbol{\beta}$ . A statement about  $\boldsymbol{\beta}$  that cannot be read off from  $\mathbf{X}\boldsymbol{\beta}$  is therefore not a statement about the data-generating distribution, and no data can test it.

#### Definition 11.3.1 — Testable hypothesis

Let  $\mathbf{\Lambda}$  be a  $p \times m$  matrix and  $\mathbf{d} \in \mathbb{R}^m$ . The hypothesis  $H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  is **testable** if

- (i)  $\mathbf{\Lambda}^\top \boldsymbol{\beta}$  is estimable, that is,  $\mathcal{C}(\mathbf{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$  ([Definition 8.2.1](#) and [Theorem 8.3.1](#)), so that  $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$  for some  $n \times m$  matrix  $\mathbf{T}$ ; and
- (ii) the equations  $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$  are consistent, that is,  $\mathbf{d} \in \mathcal{C}(\mathbf{\Lambda}^\top)$  ([Theorem 1.4.1](#)).

Its number of degrees of freedom is  $q = \text{rank}(\mathbf{\Lambda})$ . We assume  $q \geq 1$ .



Condition (ii) is automatic when  $\mathbf{A}$  has full column rank, for then  $\mathcal{C}(\mathbf{A}^\top) = \mathbb{R}^m$ . It matters when the columns of  $\mathbf{A}$  are redundant, as when all six pairwise differences of four group effects are set to zero. Redundant constraints are harmless if they agree with each other. If they do not, as in  $\alpha_1 - \alpha_2 = 1$ ,  $\alpha_2 - \alpha_3 = 1$ ,  $\alpha_1 - \alpha_3 = 0$ , the “hypothesis” is false for every  $\boldsymbol{\beta}$ , and no test is possible.

Condition (i) is the substantive one. Without it the constraint cannot be settled by any amount of data.

### Proposition 11.3.1 — Nonestimable constraints cannot be tested

Let  $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top)$  and  $c \in \mathbb{R}$ .

- (a) For every  $\boldsymbol{\beta} \in \mathbb{R}^p$  there is  $\boldsymbol{\beta}'$  with  $\mathbf{X}\boldsymbol{\beta}' = \mathbf{X}\boldsymbol{\beta}$  and  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}' = c$ .
- (b) Let  $\phi$  be any test of  $H : \boldsymbol{\lambda}^\top \boldsymbol{\beta} = c$  in the model Eq. 7.3.1, with rejection probability  $\pi(\boldsymbol{\beta}, \sigma^2) = E_{\boldsymbol{\beta}, \sigma^2} \phi(\mathbf{Y})$ . For every point  $(\boldsymbol{\beta}, \sigma^2)$  of the alternative there is a point  $(\boldsymbol{\beta}', \sigma^2)$  of the hypothesis with  $\pi(\boldsymbol{\beta}', \sigma^2) = \pi(\boldsymbol{\beta}, \sigma^2)$ . So no test has power exceeding its size anywhere.

*Proof.*

- (a) By Theorem 8.3.1(c),  $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top)$  means that  $\boldsymbol{\lambda}$  is not orthogonal to  $\mathcal{N}(\mathbf{X})$ : there is  $\mathbf{v}$  with  $\mathbf{X}\mathbf{v} = \mathbf{0}$  and  $\boldsymbol{\lambda}^\top \mathbf{v} \neq 0$ . Put  $\boldsymbol{\beta}' = \boldsymbol{\beta} + t\mathbf{v}$  with  $t = (c - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/\boldsymbol{\lambda}^\top \mathbf{v}$ . Then  $\mathbf{X}\boldsymbol{\beta}' = \mathbf{X}\boldsymbol{\beta}$  and  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}' = c$ . (b) The law of  $\mathbf{Y}$ , and hence  $\pi$ , depends on  $(\boldsymbol{\beta}, \sigma^2)$  only through  $(\mathbf{X}\boldsymbol{\beta}, \sigma^2)$ . Apply (a).

■

For a matrix  $\mathbf{A}$ , Proposition 9.2.6 gives the general statement: the constraint  $\mathbf{A}^\top \mathbf{b} = \mathbf{0}$  restricts the mean vector only through its estimable part  $\mathcal{C}(\mathbf{A}) \cap \mathcal{C}(\mathbf{X}^\top)$ , and the extra sum of squares has  $\dim(\mathcal{C}(\mathbf{A}) \cap \mathcal{C}(\mathbf{X}^\top))$  degrees of freedom. In the one-way layout  $E(Y_{ij}) = \mu + \alpha_i$ , the constraint  $\alpha_1 = 0$  is a *side condition* (Theorem 8.4.2). It selects one of the equivalent parameterizations and says nothing about the data. Programs that report a “test” of  $\alpha_1 = 0$  are testing something else, usually a contrast such as  $\alpha_1 - \alpha_g$  that the program’s coding makes estimable.

## 11.3.2 The F statistic of a testable hypothesis

The sum of squares of a testable hypothesis was computed in Theorem 9.2.4 for  $\mathbf{A}$  of full column rank. The next theorem removes that restriction, identifies the reduced model, and supplies the distribution.

### Theorem 11.3.2 — The $F$ test of a testable hypothesis

Assume Eq. 7.3.1 with  $r < n$ , and let  $H : \mathbf{A}^\top \boldsymbol{\beta} = \mathbf{d}$  be testable with  $q = \text{rank}(\mathbf{A})$ . Let  $\mathbf{G}$  be any generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ ,  $\hat{\boldsymbol{\beta}}$  any least squares estimate,  $\mathbf{W} = \mathbf{A}^\top \mathbf{G} \mathbf{A}$  and  $\mathbf{W}^-$  any generalized inverse of  $\mathbf{W}$ . Put

$$SS_H = (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top \mathbf{W}^- (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}), \quad F_H = \frac{SS_H/q}{SSE/(n-r)}. \quad (11.3.1)$$

- (a)  $SS_H$  does not depend on the choices of  $\hat{\boldsymbol{\beta}}$ ,  $\mathbf{G}$  and  $\mathbf{W}^-$ .  
 (b) Let  $\mathbf{b}_0$  be any solution of  $\mathbf{\Lambda}^\top \mathbf{b}_0 = \mathbf{d}$  and  $\mathcal{S}_0 = \{\mathbf{Xb} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{0}\}$ . Then  $\mathcal{S}_0$  is a subspace of  $\mathcal{C}(\mathbf{X})$  of dimension  $r - q$ , the set of mean vectors allowed by  $H$  is  $\mathbf{Xb}_0 + \mathcal{S}_0$ , and

$$SS_H = \min_{\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}} \|\mathbf{y} - \mathbf{Xb}\|^2 - SSE = \|(\mathbf{M} - \mathbf{P}_{\mathcal{S}_0})(\mathbf{y} - \mathbf{Xb}_0)\|^2.$$

- (c)  $F_H \sim F(q, n - r, \gamma)$  with

$$\gamma = \frac{(\mathbf{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})^\top \mathbf{W}^- (\mathbf{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})}{\sigma^2}, \quad (11.3.2)$$

and  $\gamma = 0$  iff  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ . In particular  $F_H$  is the  $F$  statistic of [Theorem 11.1.1](#) for the data  $\mathbf{y} - \mathbf{Xb}_0$  and the reduced model space  $\mathcal{S}_0$ .

*Proof. Reduction to full column rank.* Choose  $q$  linearly independent columns of  $\mathbf{\Lambda}$ , forming a  $p \times q$  matrix  $\mathbf{\Lambda}_1$ . Every column of  $\mathbf{\Lambda}$  is a combination of them, so  $\mathbf{\Lambda} = \mathbf{\Lambda}_1 \mathbf{K}$  for a  $q \times m$  matrix  $\mathbf{K}$ , which has rank  $q$  because  $\text{rank } \mathbf{\Lambda} = q$ . Hence  $\mathbf{K}\mathbf{K}^\top$  is nonsingular. Put  $\mathbf{d}_1 = \mathbf{\Lambda}_1^\top \mathbf{b}_0$ , so that  $\mathbf{d} = \mathbf{K}^\top \mathbf{d}_1$ . Since  $\mathbf{K}^\top$  has full column rank it is one-to-one, so  $\mathbf{d}_1$  is determined by  $\mathbf{d}$  and does not depend on the choice of  $\mathbf{b}_0$ . Moreover  $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d} \iff \mathbf{K}^\top (\mathbf{\Lambda}_1^\top \mathbf{b} - \mathbf{d}_1) = \mathbf{0} \iff \mathbf{\Lambda}_1^\top \mathbf{b} = \mathbf{d}_1$ . The same argument with  $\mathbf{d}$  replaced by  $\mathbf{0}$  shows that  $\mathcal{S}_0$  is also  $\{\mathbf{Xb} : \mathbf{\Lambda}_1^\top \mathbf{b} = \mathbf{0}\}$ . The columns of  $\mathbf{\Lambda}_1$  are columns of  $\mathbf{\Lambda}$ , so  $\mathbf{\Lambda}_1^\top \boldsymbol{\beta}$  is estimable.

(a). By [Theorem 9.2.4\(a\)](#),  $\mathbf{u}_1 = \mathbf{\Lambda}_1^\top \hat{\boldsymbol{\beta}} - \mathbf{d}_1$  and the positive definite matrix  $\mathbf{W}_1 = \mathbf{\Lambda}_1^\top \mathbf{G} \mathbf{\Lambda}_1$  do not depend on  $\hat{\boldsymbol{\beta}}$  and  $\mathbf{G}$ . Now  $\mathbf{u} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d} = \mathbf{K}^\top \mathbf{u}_1$  and  $\mathbf{W} = \mathbf{K}^\top \mathbf{W}_1 \mathbf{K}$ . Let  $\mathbf{v} = \mathbf{K}(\mathbf{K}\mathbf{K}^\top)^{-1} \mathbf{W}_1^{-1} \mathbf{u}_1$ , so that  $\mathbf{Kv} = \mathbf{W}_1^{-1} \mathbf{u}_1$ . Then  $\mathbf{Wv} = \mathbf{K}^\top \mathbf{W}_1 \mathbf{Kv} = \mathbf{K}^\top \mathbf{u}_1 = \mathbf{u}$ , and for every generalized inverse,

$$\mathbf{u}^\top \mathbf{W}^- \mathbf{u} = \mathbf{v}^\top \mathbf{W} \mathbf{W}^- \mathbf{Wv} = \mathbf{v}^\top \mathbf{Wv} = (\mathbf{Kv})^\top \mathbf{W}_1 (\mathbf{Kv}) = \mathbf{u}_1^\top \mathbf{W}_1^{-1} \mathbf{u}_1.$$

The right side involves none of the choices.

(b).  $\mathcal{S}_0$  is the image of the subspace  $\mathcal{N}(\mathbf{\Lambda}^\top)$  under  $\mathbf{b} \mapsto \mathbf{Xb}$ , so it is a subspace of  $\mathcal{C}(\mathbf{X})$ . Its dimension is  $r - \dim(\mathcal{C}(\mathbf{\Lambda}) \cap \mathcal{C}(\mathbf{X}^\top)) = r - q$  by [Proposition 9.2.6](#), because  $\mathcal{C}(\mathbf{\Lambda}) \subseteq \mathcal{C}(\mathbf{X}^\top)$ . The solutions of  $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$  are  $\mathbf{b}_0 + \mathcal{N}(\mathbf{\Lambda}^\top)$ , so the allowed means are  $\mathbf{Xb}_0 + \mathcal{S}_0$ . For the minimum, write  $\mathbf{y}_* = \mathbf{y} - \mathbf{Xb}_0$ . As  $\mathbf{b}$  ranges over the constraint set,  $\mathbf{Xb} - \mathbf{Xb}_0$  ranges over  $\mathcal{S}_0$ , so

$$\min_{\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}} \|\mathbf{y} - \mathbf{Xb}\|^2 = \min_{\mathbf{m} \in \mathcal{S}_0} \|\mathbf{y}_* - \mathbf{m}\|^2 = \|(\mathbf{I} - \mathbf{P}_{\mathcal{S}_0})\mathbf{y}_*\|^2$$

by [Theorem 6.3.1](#). Also  $SSE = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{y}_*\|^2$ , since  $\mathbf{Xb}_0 \in \mathcal{C}(\mathbf{X})$ . Because  $\mathcal{S}_0 \subseteq \mathcal{C}(\mathbf{X})$ , [Eq. 6.5.2](#) turns the difference into  $\|(\mathbf{M} - \mathbf{P}_{\mathcal{S}_0})\mathbf{y}_*\|^2$ . By [Theorem 9.2.4\(c\)](#) applied to  $\mathbf{\Lambda}_1$  and  $\mathbf{d}_1$ , the same difference equals  $\mathbf{u}_1^\top \mathbf{W}_1^{-1} \mathbf{u}_1$ , which is  $SS_H$  by (a).

(c).  $\mathbf{Y} - \mathbf{Xb}_0 \sim N_n(\mathbf{X}(\boldsymbol{\beta} - \mathbf{b}_0), \sigma^2 \mathbf{I})$  is a normal linear model with mean in  $\mathcal{C}(\mathbf{X})$ , and by (b)  $F_H$  is its  $F$  statistic for the reduced model space  $\mathcal{S}_0$ , which has dimension  $r - q$ . [Theorem 11.1.1](#) gives  $F_H \sim F(q, n - r, \gamma)$  with  $\sigma^2 \gamma = \|(\mathbf{M} - \mathbf{P}_{\mathcal{S}_0})(\mathbf{X}\boldsymbol{\beta} - \mathbf{Xb}_0)\|^2$ . To evaluate this, note that the identity in (b) holds for every data vector. Apply it to the

vector  $\mathbf{X}\boldsymbol{\beta}$ , for which  $\boldsymbol{\beta}$  itself is a least squares estimate:  $\sigma^2\gamma = (\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})^\top \mathbf{W}^{-1}(\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})$ . Finally, by (a) this equals  $\mathbf{u}_1(\boldsymbol{\beta})^\top \mathbf{W}_1^{-1} \mathbf{u}_1(\boldsymbol{\beta})$  with  $\mathbf{u}_1(\boldsymbol{\beta}) = \boldsymbol{\Lambda}_1^\top \boldsymbol{\beta} - \mathbf{d}_1$ . It is zero iff  $\boldsymbol{\Lambda}_1^\top \boldsymbol{\beta} = \mathbf{d}_1$ , that is, iff  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ . ■

Formula Eq. 11.3.1 needs only the unrestricted fit. Its ingredients have a statistical meaning:  $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$  is the estimated departure from the hypothesis, and  $\sigma^2 \mathbf{W}$  is its covariance matrix (Theorem 7.5.1(a)). So  $\text{SS}_H/\sigma^2$  is the squared Mahalanobis length of the estimated departure, and the noncentrality Eq. 11.3.2 is the same length for the true departure. Under  $H$  the quadratic form  $\text{SS}_H/\sigma^2$  is  $\chi^2(q)$ , which is Corollary 7.5.3. The theorem adds the reduced model, the distribution under alternatives, and the fact that a redundant  $\boldsymbol{\Lambda}$  can be used as it stands, with any generalized inverse of  $\mathbf{W}$ .

The test space of the hypothesis is the orthogonal complement of  $\mathcal{S}_0$  within  $\mathcal{C}(\mathbf{X})$ . By Theorem 6.5.2 it is  $\mathcal{C}(\mathbf{MT})$ , where  $\boldsymbol{\Lambda} = \mathbf{X}^\top \mathbf{T}$ : the hypothesis constrains the mean vector to be orthogonal to the fitted values of the columns of  $\mathbf{T}$ .

### 11.3.3 Least squares under the hypothesis

The minimum in Theorem 11.3.2(b) is attained, and its minimizers have a closed form. The following extends Theorem 9.2.5, which assumed full rank, to every model matrix.

#### Proposition 11.3.3 — Least squares under a testable hypothesis

Let  $H : \boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  be testable with  $\boldsymbol{\Lambda}$  of full column rank  $q$ , and write  $\boldsymbol{\Lambda} = \mathbf{X}^\top \mathbf{T}$ . Let  $\mathbf{G}$  be a generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ ,  $\hat{\boldsymbol{\beta}} = \mathbf{GX}^\top \mathbf{y}$ ,  $\mathbf{W} = \boldsymbol{\Lambda}^\top \mathbf{G} \boldsymbol{\Lambda}$  and  $\mathbf{u} = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$ . Then  $\mathbf{W}$  is positive definite, and:

- (a)  $\hat{\boldsymbol{\beta}}_H = \hat{\boldsymbol{\beta}} - \mathbf{G} \boldsymbol{\Lambda} \mathbf{W}^{-1} \mathbf{u}$  satisfies  $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}_H = \mathbf{d}$  and minimizes  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  subject to  $\boldsymbol{\Lambda}^\top \mathbf{b} = \mathbf{d}$ . The set of all minimizers is  $\hat{\boldsymbol{\beta}}_H + (\mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\boldsymbol{\Lambda}^\top))$ .
- (b) The restricted fitted vector is unique:  $\mathbf{X}\hat{\boldsymbol{\beta}}_H = \mathbf{M}\mathbf{y} - \mathbf{MTW}^{-1} \mathbf{u} = \mathbf{X}\mathbf{b}_0 + \mathbf{P}_{\mathcal{S}_0}(\mathbf{y} - \mathbf{X}\mathbf{b}_0)$ , and  $\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_H\|^2 = \text{SSE} + \text{SS}_H$ , with  $\text{SS}_H = \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}$ .
- (c) With  $\hat{\boldsymbol{\theta}} = \mathbf{W}^{-1} \mathbf{u}$ , the restricted estimate solves the constrained normal equations (compare Theorem 6.4.2)

$$\mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}_H + \boldsymbol{\Lambda} \hat{\boldsymbol{\theta}} = \mathbf{X}^\top \mathbf{y}, \quad \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}_H = \mathbf{d},$$

$$\text{and } \text{SS}_H = \hat{\boldsymbol{\theta}}^\top \mathbf{W} \hat{\boldsymbol{\theta}}.$$

*Proof.*  $\mathbf{W} = \mathbf{T}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{T} = \mathbf{T}^\top \mathbf{MT}$  by Theorem 6.3.7. It is symmetric, and positive definite by Theorem 9.2.4(a). Also  $\mathbf{X} \mathbf{G} \boldsymbol{\Lambda} = \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{T} = \mathbf{MT}$ .

- (a)  $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}}_H = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{W} \mathbf{W}^{-1} \mathbf{u} = \mathbf{d}$ . For any feasible  $\mathbf{b}$ , Eq. 6.3.1 with  $\mathbf{u} = \mathbf{X}\mathbf{b}$  gives  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE} + \|\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\mathbf{b}\|^2$ . Write  $\mathbf{b} = \hat{\boldsymbol{\beta}}_H + \mathbf{v}$  with  $\boldsymbol{\Lambda}^\top \mathbf{v} = \mathbf{0}$ . Then  $\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\mathbf{b} = \mathbf{MTW}^{-1} \mathbf{u} - \mathbf{X}\mathbf{v}$ , and the cross term vanishes:

$$(\mathbf{MTW}^{-1} \mathbf{u})^\top \mathbf{X}\mathbf{v} = \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{T}^\top \mathbf{X}\mathbf{v} = \mathbf{u}^\top \mathbf{W}^{-1} \boldsymbol{\Lambda}^\top \mathbf{v} = 0,$$

using  $\mathbf{MX} = \mathbf{X}$ . Hence  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE} + \|\mathbf{MTW}^{-1}\mathbf{u}\|^2 + \|\mathbf{X}\mathbf{v}\|^2$ , which is smallest exactly when  $\mathbf{X}\mathbf{v} = \mathbf{0}$ . The minimizers are  $\hat{\boldsymbol{\beta}}_H + \mathbf{v}$  with  $\mathbf{v} \in \mathcal{N}(\mathbf{X}) \cap \mathcal{N}(\mathbf{\Lambda}^\top)$ .

- (b)  $\mathbf{X}\hat{\boldsymbol{\beta}}_H = \mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{XG}\mathbf{\Lambda}\mathbf{W}^{-1}\mathbf{u} = \mathbf{M}\mathbf{y} - \mathbf{MTW}^{-1}\mathbf{u}$ , and both terms are unique by [Theorem 9.2.4\(a\)](#). The minimum value is  $\text{SSE} + \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{T}^\top \mathbf{MTW}^{-1} \mathbf{u} = \text{SSE} + \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}$ , and  $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} = \text{SS}_H$  by [Theorem 11.3.2\(a\)](#). The fitted vector  $\mathbf{X}\hat{\boldsymbol{\beta}}_H$  is the point of  $\mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$  nearest to  $\mathbf{y}$ , which is  $\mathbf{X}\mathbf{b}_0 + \mathbf{P}_{\mathcal{S}_0}(\mathbf{y} - \mathbf{X}\mathbf{b}_0)$  by the argument in the proof of [Theorem 11.3.2\(b\)](#).
- (c)  $\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}}_H = \mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}^\top \mathbf{XG}\mathbf{\Lambda}\mathbf{W}^{-1}\mathbf{u} = \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{MT}\hat{\boldsymbol{\theta}} = \mathbf{X}^\top \mathbf{y} - \mathbf{\Lambda}\hat{\boldsymbol{\theta}}$ , because  $\mathbf{X}^\top \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$  and  $\mathbf{X}^\top \mathbf{M} = \mathbf{X}^\top$ . The last identity is  $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} = \hat{\boldsymbol{\theta}}^\top \mathbf{W}\hat{\boldsymbol{\theta}}$ .

■

Part (c) identifies  $\hat{\boldsymbol{\theta}}$  as a vector of Lagrange multipliers (compare [Proposition 1.11.3](#)). Each multiplier measures how hard the data push against its constraint, and the hypothesis sum of squares is their squared Mahalanobis length. This is the “Lagrange multiplier” form of the test met again in [Section 11.4](#).

### 11.3.4 Nonzero right-hand sides: offsets

When  $\mathbf{d} \neq \mathbf{0}$  the hypothesis restricts the mean to the *affine* set  $\mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$ , not to a subspace. [Theorem 11.3.2\(c\)](#) shows how to handle it: subtract a known point of the set from the data. The reduced model becomes

$$\mathbf{E}(\mathbf{Y} - \mathbf{X}\mathbf{b}_0) \in \mathcal{S}_0,$$

a linear model for the adjusted response  $\mathbf{Y} - \mathbf{X}\mathbf{b}_0$  with a **known offset**  $\mathbf{X}\mathbf{b}_0$ . Any solution  $\mathbf{b}_0$  of  $\mathbf{\Lambda}^\top \mathbf{b}_0 = \mathbf{d}$  may be used. Two solutions differ by an element of  $\mathcal{N}(\mathbf{\Lambda}^\top)$ , so their offsets differ by an element of  $\mathcal{S}_0$ , which the reduced model absorbs. In practice one fits  $\mathbf{y} - \mathbf{X}\mathbf{b}_0$  on a basis of  $\mathcal{S}_0$ , for example the columns of  $\mathbf{X}\mathbf{U}$  where the columns of  $\mathbf{U}$  span  $\mathcal{N}(\mathbf{\Lambda}^\top)$ , and takes the difference of residual sums of squares. The formula [Eq. 11.3.1](#) gives the same number with no second fit.

#### Example. (Regional hypotheses with an over-parameterized model)

Return to the state murder data, now with an intercept *and* an indicator for each of the four regions, so that  $\boldsymbol{\beta} = (\mu, \alpha_N, \alpha_M, \alpha_S, \alpha_W, \beta_{\text{pov}}, \beta_{\text{sin}}, \beta_{\text{urb}})^\top$ . The model matrix is  $50 \times 8$  of rank 7. No region effect is estimable on its own, but every contrast among them is.

*All four regions equal.* Write the hypothesis with all six pairwise differences  $\alpha_i - \alpha_j$  as the columns of  $\mathbf{\Lambda}$ . They are consistent and have rank  $q = 3$ . With the Moore–Penrose inverses of  $\mathbf{X}^\top \mathbf{X}$  and of  $\mathbf{W}$ , [Eq. 11.3.1](#) gives  $\text{SS}_H = 14.83$  and  $F = 2.440$ , exactly the reduced-model test of [Example \(Do regions matter?\)](#), as [Theorem 11.3.2\(b\)](#) requires. A second generalized inverse, made by deleting the row and column of  $\alpha_W$  from  $\mathbf{X}^\top \mathbf{X}$  before inverting, gives a different  $\hat{\boldsymbol{\beta}}$  but the same  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}$ ,  $\mathbf{W}$  and  $F$ .

*Midwest, South and West equal.* With  $\mathbf{\Lambda}$  made of  $\alpha_M - \alpha_S$  and  $\alpha_S - \alpha_W$ , the sum of squares is 3.881 on 2 degrees of freedom,  $F = 0.958$ ,  $p = 0.392$ . The restricted fit of [Proposition 11.3.3](#) coincides with the fit of the model in which the three regions are merged. Under the constraint, the Northeast lies 1.379 below the common level of the

other three, and the coefficient of single parenthood moves from 0.428 to 0.441.

*A nonzero right-hand side.* The contrast  $\alpha_N - (\alpha_M + \alpha_S + \alpha_W)/3$ , the Northeast against the average of the other regions, is estimated as  $\hat{\psi} = -1.426$  with standard error 0.595. To test whether it equals  $-1$ , take the offset  $\mathbf{b}_0 = (0, -1, 0, \dots, 0)^\top$  and fit  $\mathbf{y} - \mathbf{X}\mathbf{b}_0$  on a basis of  $\mathcal{S}_0$ . The drop in residual sum of squares is 1.040, equal to  $(\hat{\psi} + 1)^2 / \lambda^\top \mathbf{G} \lambda$  as Eq. 11.3.1 predicts, and  $F = 0.513$  with  $p = 0.478$ . Any other solution of  $\lambda^\top \mathbf{b}_0 = -1$  gives the same answer.

*A constraint that is not a hypothesis.* The constraint  $\alpha_N = 0$  is not estimable. Minimizing the residual sum of squares subject to it gives exactly SSE: the constrained model spans the whole of  $\mathcal{C}(\mathbf{X})$ , in agreement with Proposition 11.3.1.

```
import numpy as np
import statsmodels.api as sm
from scipy import linalg, stats

data = sm.datasets.statcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWSWNNSSSWMMMMSSNSNMMSMWNNWNMSMSSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
D = np.column_stack([(region == g).astype(float) for g in "NMSW"]) # all four
                                regions
X = np.column_stack([np.ones(n), D, data["poverty"], data["single"], data["urban"]])
r = np.linalg.matrix_rank(X) # 7, not 8

G = np.linalg.pinv(X.T @ X) # one generalized inverse of X'X
b = G @ X.T @ y # one least squares solution
s2 = np.sum((y - X @ b) ** 2) / (n - r)

def glh_test(Lam, d):
    """F test of the testable hypothesis Lam' beta = d, via any g-inverse of Lam' G
    Lam. """
    u = Lam.T @ b - d
    W = Lam.T @ G @ Lam
    q = np.linalg.matrix_rank(Lam)
    ss_h = u @ np.linalg.pinv(W) @ u
    F = ss_h / q / s2
    return ss_h, q, F, stats.f.sf(F, q, n - r)

def contrast(pairs):
    """Columns e_i - e_j on the region coefficients (positions 1..4 = N, M, S, W).
    """
    L = np.zeros((8, len(pairs)))
    for k, (i, j) in enumerate(pairs):
        L[1 + i, k], L[1 + j, k] = 1.0, -1.0
    return L

all_pairs = contrast([(0, 1), (0, 2), (0, 3), (1, 2), (1, 3), (2, 3)]) # rank 3,
                                redundant
three_equal = contrast([(1, 2), (2, 3)]) # M = S = W
for name, L in (("four regions equal", all_pairs), ("M = S = W", three_equal)):
    ss_h, q, F, p = glh_test(L, np.zeros(L.shape[1]))
    print(f"{name:20s} SS_H = {ss_h:.3f} q = {q} F = {F:.3f} p = {p:.4f}")
```

The offset computation fits the reduced model directly, as a check on the formula.

```
def sse(Z, target):
    coef, *_ = np.linalg.lstsq(Z, target, rcond=None)
    return np.sum((target - Z @ coef) ** 2)

sse_full = sse(X, y)
lam = np.array([0, 1, -1 / 3, -1 / 3, -1 / 3, 0, 0, 0]) # Northeast minus the
other three
d = -1.0 # hypothesized difference
b0 = np.array([0, -1.0, 0, 0, 0, 0, 0, 0]) # one solution of lam' b0
    = d
U = linalg.null_space(lam[None, :]) # beta - b0 ranges over C(
    U)
sse_H = sse(X @ U, y - X @ b0) # reduced model, offset X
    b0
est = lam @ b
se = np.sqrt(s2 * lam @ G @ lam)
print(f"estimate {est:.3f} (se {se:.3f}); SSE_H - SSE = {sse_H - sse_full:.3f}")
print(f"formula: {(est - d) ** 2 / (lam @ G @ lam):.3f}")
```

The restricted estimates, from two different generalized inverses:

```
keep = [0, 1, 2, 3, 5, 6, 7]
G2 = np.zeros((8, 8))
G2[np.ix_(keep, keep)] = np.linalg.inv((X.T @ X)[np.ix_(keep, keep)])

def restricted(Lam, d, Ginv):
    """Least squares under the testable constraint Lam' b = d (Lam of full column
    rank)."""
    bh = Ginv @ X.T @ y
    W = Lam.T @ Ginv @ Lam
    return bh - Ginv @ Lam @ np.linalg.solve(W, Lam.T @ bh - d)

bH = restricted(three_equal, np.zeros(2), G)
bH2 = restricted(three_equal, np.zeros(2), G2)
print("constraint holds:", np.allclose(three_equal.T @ bH, 0))
print("fitted values agree:", np.allclose(X @ bH, X @ bH2))
```

*Warning* (Hypotheses chosen after seeing the data). The same Northeast contrast, tested against zero, gives  $t = -2.398$  and  $p = 0.021$ . That p-value would be valid had the contrast been specified before the data were examined. It was not: it was suggested by the estimates, after the joint test of all regional differences had failed to reject. Among the many contrasts one could have tested, the one that looks largest is, by construction, the one most likely to look significant by chance. [Chapter 13](#) gives procedures, notably Scheffé's, that remain valid when the contrast is chosen by looking at the data.

## 11.3.5 Exercises

### 11.3.5.1 A. Check your understanding

**Exercise A1.**

In the one-way layout  $E(Y_{ij}) = \mu + \alpha_i$ ,  $i = 1, 2, 3$ , with all groups nonempty, decide which of the following are testable hypotheses: (i)  $\alpha_1 = \alpha_2$ ; (ii)  $\alpha_1 = 0$ ; (iii)  $\alpha_1 + \alpha_2 + \alpha_3 = 0$ ; (iv)  $\mu + \alpha_1 = 5$ ; (v)  $\alpha_1 - \alpha_2 = 1$ ,  $\alpha_2 - \alpha_3 = 2$ ,  $\alpha_1 - \alpha_3 = 3$ ; (vi)  $\alpha_1 - \alpha_2 = 1$ ,  $\alpha_2 - \alpha_3 = 2$ ,  $\alpha_1 - \alpha_3 = 4$ . Give  $q$  for those that are.

*Solution.* The estimable functions are the combinations  $c_0\mu + \sum_i c_i\alpha_i$  with  $c_0 = \sum_i c_i$  (Example (Four answers, one fit) and Section 8.2). (i) is testable,  $q = 1$ . (ii) and (iii) are not estimable: their coefficient of  $\mu$  is 0 while the  $\alpha$  coefficients sum to 1 and 3. They are side conditions (Theorem 8.4.2). (iv) is testable,  $q = 1$ . (v) is testable with  $q = 2$ : the three differences are estimable, have rank 2, and the right-hand sides are consistent because  $1 + 2 = 3$ . (vi) is estimable but inconsistent, since  $(\alpha_1 - \alpha_2) + (\alpha_2 - \alpha_3) = \alpha_1 - \alpha_3$  would need  $1 + 2 = 4$ . No parameter satisfies it.

**11.3.5.2 B. Practice****Exercise B1.**

Let  $\mathbf{\Lambda}$  be  $p \times m$  and  $\mathbf{A}$  a nonsingular  $m \times m$  matrix. Show that the hypotheses  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  and  $(\mathbf{\Lambda}\mathbf{A})^\top \boldsymbol{\beta} = \mathbf{A}^\top \mathbf{d}$  are the same, and that Eq. 11.3.1 gives the same  $SS_H$  for both. Conclude that the test does not depend on how a set of constraints is written.

*Solution.*  $(\mathbf{\Lambda}\mathbf{A})^\top \mathbf{b} = \mathbf{A}^\top \mathbf{d} \iff \mathbf{A}^\top (\mathbf{\Lambda}^\top \mathbf{b} - \mathbf{d}) = \mathbf{0} \iff \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$ , since  $\mathbf{A}^\top$  is nonsingular. So the constraint sets, and by Theorem 11.3.2(b) the sums of squares, agree. Directly: with  $\mathbf{u} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$ , the new departure is  $\mathbf{A}^\top \mathbf{u}$  and the new middle matrix is  $\mathbf{A}^\top \mathbf{W} \mathbf{A}$ , and one generalized inverse of it is  $\mathbf{A}^{-1} \mathbf{W}^{-1} \mathbf{A}^{-\top}$ , which gives  $\mathbf{u}^\top \mathbf{A} \mathbf{A}^{-1} \mathbf{W}^{-1} \mathbf{A}^{-\top} \mathbf{A}^\top \mathbf{u} = \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}$ . The choice of generalized inverse does not matter by Theorem 11.3.2(a).

**Exercise B2.**

In Proposition 11.3.3, show that  $E(\mathbf{X}\hat{\boldsymbol{\beta}}_H) = \mathbf{X}\boldsymbol{\beta} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\mathbf{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})$  and  $\text{Cov}(\mathbf{X}\hat{\boldsymbol{\beta}}_H) = \sigma^2 \mathbf{P}_{S_0}$ , where  $\mathbf{P}_{S_0} = \mathbf{M} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}\mathbf{T}^\top \mathbf{M}$ . Conclude that restricted fitted values are unbiased when  $H$  holds, have smaller variance than  $\mathbf{M}\mathbf{Y}$  in every direction, and are biased when  $H$  fails.

*Solution.*  $\mathbf{X}\hat{\boldsymbol{\beta}}_H = \mathbf{M}\mathbf{Y} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\mathbf{T}^\top \mathbf{M}\mathbf{Y} - \mathbf{d})$ , using  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{T}^\top \mathbf{M}\mathbf{Y}$ . Its mean is  $\mathbf{X}\boldsymbol{\beta} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\mathbf{T}^\top \mathbf{X}\boldsymbol{\beta} - \mathbf{d})$ , and  $\mathbf{T}^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{\Lambda}^\top \boldsymbol{\beta}$ . The random part is  $(\mathbf{M} - \mathbf{M}\mathbf{T}\mathbf{W}^{-1}\mathbf{T}^\top \mathbf{M})\mathbf{Y}$ . The matrix in parentheses is the projection onto  $\mathcal{C}(\mathbf{X})$  minus the projection onto  $\mathcal{C}(\mathbf{M}\mathbf{T})$  (Theorem 6.3.7, since  $\mathbf{W} = \mathbf{T}^\top \mathbf{M}\mathbf{T}$ ), which is  $\mathbf{P}_{S_0}$  by Theorem 6.5.2. It is symmetric idempotent, so the covariance is  $\sigma^2 \mathbf{P}_{S_0}$ . The difference  $\sigma^2(\mathbf{M} - \mathbf{P}_{S_0})$  from  $\text{Cov}(\mathbf{M}\mathbf{Y})$  is a projection times  $\sigma^2$ , so it is nonnegative definite. The bias is  $-\mathbf{M}\mathbf{T}\mathbf{W}^{-1}(\mathbf{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})$ , zero iff  $H$  holds, because  $\mathbf{M}\mathbf{T}$  has full column rank.

**Exercise B3.**

In simple regression  $E(Y_i) = \beta_0 + \beta_1 x_i$ , derive the  $F$  statistic for  $\beta_1 = 1$  in two ways:

from Eq. 11.3.1, and by regressing  $y_i - x_i$  on an intercept and comparing with the full fit. Show that both give  $F = (\hat{\beta}_1 - 1)^2 S_{xx}/s^2$ .

#### Exercise B4.

Show that the hypothesis  $\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\boldsymbol{\beta}_0$ , with  $\boldsymbol{\beta}_0$  known, is testable, that  $q = r$ , and that its  $F$  statistic is  $\{\|\mathbf{M}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}_0)\|^2/r\}/s^2$ . What is the reduced model?

### 11.3.5.3 C. Going deeper

#### Exercise C1.

In the one-way layout with two groups of sizes 2 and 3, take  $\boldsymbol{\Lambda} = (0, 1, 0)^\top$ , so that the “hypothesis” is  $\alpha_1 = 0$ . Evaluate  $(\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}})^2/(\boldsymbol{\Lambda}^\top \mathbf{G} \boldsymbol{\Lambda})$  for two different generalized inverses  $\mathbf{G}$  of  $\mathbf{X}^\top \mathbf{X}$  and show that the values differ. Explain which step of the proof of Theorem 11.3.2 fails.

#### Exercise C2.

Let  $\mathbf{X}$  have full column rank and write  $\boldsymbol{\Lambda}^\top = [\boldsymbol{\Lambda}_1^\top, \boldsymbol{\Lambda}_2^\top]$  with  $\boldsymbol{\Lambda}_2^\top$  a nonsingular  $q \times q$  block, after permuting the coefficients so that  $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^\top, \boldsymbol{\beta}_2^\top)^\top$  conformably. Show that  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$  is equivalent to  $\boldsymbol{\beta}_2 = -\boldsymbol{\Lambda}_2^{-\top} \boldsymbol{\Lambda}_1^\top \boldsymbol{\beta}_1$ , that the reduced model matrix is  $\mathbf{X}_1 - \mathbf{X}_2 \boldsymbol{\Lambda}_2^{-\top} \boldsymbol{\Lambda}_1^\top$ , and that this matrix has full column rank  $p - q$ . This is how the reduced model is fitted when no offset or null-space basis is at hand.

## 11.4 The likelihood ratio test and its relatives

The  $F$  test was motivated in Section 11.1 by a heuristic: compare the extra sum of squares with the noise level. General statistical theory offers other routes to a test, chiefly the likelihood ratio, the Wald statistic and the score statistic. In the normal linear model all three routes lead back to  $F$ . They differ only when they are calibrated by their common large-sample  $\chi^2$  approximation instead of the exact distribution, and then they can disagree about the same data. This section derives all three, shows how different the approximate versions can be, and then proves that the  $F$  test is the best test among those that respect the symmetries of Section 11.2.

### 11.4.1 The likelihood ratio

For a hypothesis  $H_0$  about the parameter of a family of densities  $L(\boldsymbol{\theta}; \mathbf{y})$ , the **likelihood ratio** is

$$\Lambda(\mathbf{y}) = \frac{\sup_{\boldsymbol{\theta} \in H_0} L(\boldsymbol{\theta}; \mathbf{y})}{\sup_{\boldsymbol{\theta}} L(\boldsymbol{\theta}; \mathbf{y})},$$



and the likelihood ratio test rejects when  $\Lambda$  is small, that is, when the best explanation the hypothesis can offer is much worse than the best explanation overall. In the linear model both suprema are available in closed form from [Theorem 7.3.1](#), and [Exercise \(A2\)](#) already carried out the computation for reduced models. The theorem records it for testable hypotheses, including the calibration.

### Theorem 11.4.1 — The $F$ test is the likelihood ratio test

Assume [Eq. 7.3.1](#) with  $r < n$ , and let  $H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  be testable with  $q = \text{rank}(\mathbf{\Lambda})$ . Let  $\text{SSE}_H = \min_{\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  and suppose  $\text{SSE} > 0$ , which holds with probability one. Then:

- (a) under  $H$  the likelihood is maximized at any minimizer  $\hat{\boldsymbol{\beta}}_H$  of  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  subject to  $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$  (for  $\mathbf{\Lambda}$  of full column rank, the estimate of [Proposition 11.3.3](#); compare [Theorem 9.2.5](#)), with  $\tilde{\sigma}_H^2 = \text{SSE}_H/n$ ;
- (b) the likelihood ratio is

$$\Lambda = \left( \frac{\text{SSE}}{\text{SSE}_H} \right)^{n/2} = \left( 1 + \frac{q F_H}{n - r} \right)^{-n/2}, \quad (11.4.1)$$

a strictly decreasing function of the statistic  $F_H$  of [Eq. 11.3.1](#);

- (c) the likelihood ratio test of size  $\alpha$  rejects exactly when  $F_H > F_\alpha(q, n - r)$ .

The same holds for a reduced model  $E(\mathbf{Y}) \in \mathcal{C}(\mathbf{X}_0)$ , with  $q = r - r_0$  and the statistic of [Theorem 11.1.1](#).

*Proof.*

- (a) By [Theorem 11.3.2\(b\)](#),  $H$  holds iff  $E(\mathbf{Y}) \in \mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$ . Maximizing the likelihood over  $H$  is therefore maximizing it in the linear model  $\mathbf{Y} - \mathbf{X}\mathbf{b}_0 \sim N_n(\mathbf{m}, \sigma^2 \mathbf{I})$ ,  $\mathbf{m} \in \mathcal{S}_0$ . By [Theorem 7.3.1](#) applied to that model (with any matrix whose columns span  $\mathcal{S}_0$ ), the maximizing mean is the least squares fit and the maximizing variance is its residual sum of squares divided by  $n$ . That fit is  $\mathbf{P}_{\mathcal{S}_0}(\mathbf{y} - \mathbf{X}\mathbf{b}_0)$ , so the maximizing mean of  $\mathbf{Y}$  is  $\mathbf{X}\mathbf{b}_0 + \mathbf{P}_{\mathcal{S}_0}(\mathbf{y} - \mathbf{X}\mathbf{b}_0)$ , the point of  $\mathbf{X}\mathbf{b}_0 + \mathcal{S}_0$  nearest to  $\mathbf{y}$ . As shown in the proof of [Theorem 11.3.2\(b\)](#), no rank condition on  $\mathbf{\Lambda}$  is needed for this. That point is  $\mathbf{X}\mathbf{b}$  exactly for the minimizers  $\mathbf{b}$  of  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  subject to  $\mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}$ , so it is  $\mathbf{X}\hat{\boldsymbol{\beta}}_H$ , and its residual sum of squares is  $\text{SSE}_H$ . This needs  $\text{SSE}_H > 0$ , which follows from  $\text{SSE}_H \geq \text{SSE} > 0$ .
- (b) By [Theorem 7.3.1](#) the two maxima are  $(2\pi e \text{SSE}/n)^{-n/2}$  and  $(2\pi e \text{SSE}_H/n)^{-n/2}$ . Their ratio is  $(\text{SSE}/\text{SSE}_H)^{n/2}$ , and  $\text{SSE}_H = \text{SSE} + \text{SS}_H = \text{SSE}\{1 + qF_H/(n - r)\}$  by [Theorem 11.3.2\(b\)](#).
- (c) By (b),  $\Lambda \leq k$  iff  $F_H \geq c$  for a constant  $c$  determined by  $k$ . By [Theorem 11.3.2\(c\)](#) the null distribution of  $F_H$  is  $F(q, n - r)$  at every point of  $H$ , so size  $\alpha$  requires  $c = F_\alpha(q, n - r)$ .

■

The large-sample theory of likelihood ratios (Wilks's theorem) would calibrate  $-2\log \Lambda = n \log\{1 + qF_H/(n - r)\}$  by the  $\chi^2(q)$  distribution. That calibration is only approximate. In the normal linear model the exact distribution is available through  $F_H$  and should be used.

## 11.4.2 Wald and score statistics

Two other large-sample recipes are in wide use. The **Wald statistic** measures the estimated departure  $\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$  against its estimated covariance, both computed at the unrestricted maximum likelihood estimate. The **score statistic**, also called the Lagrange multiplier statistic, measures the gradient of the log-likelihood at the *restricted* maximum likelihood estimate against the information there. In the linear model, with  $\mathbf{\Lambda}$  of full column rank  $q$  and the maximum likelihood variance estimates  $\hat{\sigma}^2 = \text{SSE}/n$  and  $\tilde{\sigma}_H^2 = \text{SSE}_H/n$ , they are

$$\text{Wald} = \frac{(\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})}{\hat{\sigma}^2}$$

and

$$\text{Score} = \mathbf{U}^\top \left( \frac{\mathbf{X}^\top \mathbf{X}}{\tilde{\sigma}_H^2} \right)^{-} \mathbf{U} = \tilde{\sigma}_H^2 \mathbf{U}^\top (\mathbf{X}^\top \mathbf{X})^{-} \mathbf{U},$$

where  $\mathbf{U} = \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}_H) / \tilde{\sigma}_H^2$ . Here  $\mathbf{U}$  is the gradient of the log-likelihood Eq. 7.3.1 with respect to  $\boldsymbol{\beta}$  at  $(\hat{\boldsymbol{\beta}}_H, \tilde{\sigma}_H^2)$ , and  $\mathbf{X}^\top \mathbf{X} / \tilde{\sigma}_H^2$  is the corresponding block of the Fisher information. The gradient with respect to  $\sigma^2$  vanishes at the restricted maximum, and the information matrix is block diagonal, because  $\mathbf{E}\{\mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})\} = \mathbf{0}$ . So the  $\sigma^2$  coordinates contribute nothing, and the formula is the full score statistic. The generalized inverse is harmless because  $\mathbf{U} \in \mathcal{C}(\mathbf{X}^\top \mathbf{X})$ .

### Proposition 11.4.2 — Wald, likelihood ratio and score

In the setting above, let  $u = \text{SS}_H / \text{SSE} = q F_H / (n - r)$ . Then

$$\text{Wald} = n u, \quad -2 \log \Lambda = n \log(1 + u), \quad \text{Score} = \frac{n u}{1 + u}.$$

Each is a strictly increasing function of  $F_H$ , so the three tests, calibrated exactly, coincide with the  $F$  test. For every sample with  $\text{SS}_H > 0$ ,

$$\text{Wald} > -2 \log \Lambda > \text{Score}.$$

If each is compared with the upper  $\alpha$  point  $c$  of  $\chi^2(q)$ , the exact sizes are

$$\Pr\left\{F(q, n - r) > \frac{n - r}{q} u_c\right\}, \quad u_c = \frac{c}{n}, \quad e^{c/n} - 1, \quad \frac{c}{n - c} \quad (n > c)$$

for the Wald, likelihood ratio and score tests respectively.

*Proof.* With  $\hat{\sigma}^2 = \text{SSE}/n$ ,  $\text{Wald} = n \text{SS}_H / \text{SSE} = n u$  by Eq. 11.3.1. For the score,  $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}$ , so  $\mathbf{X}^\top (\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}_H) = \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)$  and

$$\mathbf{U}^\top (\mathbf{X}^\top \mathbf{X})^{-} \mathbf{U} \cdot \tilde{\sigma}_H^2 = \frac{(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)^\top \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-} \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)}{\tilde{\sigma}_H^2} = \frac{\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2}{\tilde{\sigma}_H^2}.$$

By the proof of Proposition 11.3.3,  $\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_H)\|^2 = \text{SS}_H$ , so  $\text{Score} = n \text{SS}_H / \text{SSE}_H = n u / (1 + u)$ . The likelihood ratio is Eq. 11.4.1. All three are increasing in  $u$ , which is proportional to  $F_H$ . For  $u > 0$ ,  $\log(1 + u) = \int_0^u (1 + t)^{-1} dt$  lies strictly between  $u / (1 + u)$  and  $u$ , since the integrand lies strictly between  $(1 + u)^{-1}$  and 1 on  $(0, u)$ . Finally, each statistic exceeds  $c$  iff  $u > u_c$ , with  $u_c$  as stated (for the score,  $n u / (1 + u) > c$  iff  $u(n - c) > c$ , and it never exceeds  $c$  when  $n \leq c$ ), and  $u > u_c$  iff  $F_H > (n - r) u_c / q$ .

The inequality  $\text{Wald} \geq \text{LR} \geq \text{Score}$  is a general feature of linear restrictions in normal regression, and it has a practical consequence: with  $\chi^2$  critical values, the Wald test rejects most often and the score test least often, *on the same data*.

**Example. (Three verdicts from one data set)**

For the region test of [Example \(Do regions matter?\)](#),  $n = 50$ ,  $q = 3$  and  $n - r = 43$ . The three statistics are

| Statistic                           | Value | p-value from $\chi^2(3)$ |
|-------------------------------------|-------|--------------------------|
| Wald                                | 8.512 | 0.0365                   |
| Likelihood ratio, $-2 \log \Lambda$ | 7.861 | 0.0490                   |
| Score                               | 7.274 | 0.0637                   |
| $F = 2.440$ , exact                 |       | 0.0773                   |

With the 5% critical value 7.815 of  $\chi^2(3)$ , the Wald test rejects, the likelihood ratio test rejects barely, and the score test does not. The exact test, which all three approximate, does not reject. The disagreement is not a paradox. The approximate tests are different tests, and their true sizes differ from 0.05. With  $q = 3$  and  $r = 7$ , the formulas of [Proposition 11.4.2](#) give

| $n$              | 15    | 30    | 50    | 100   | 400   |
|------------------|-------|-------|-------|-------|-------|
| Wald             | 0.315 | 0.143 | 0.097 | 0.071 | 0.055 |
| Likelihood ratio | 0.221 | 0.106 | 0.079 | 0.063 | 0.053 |
| Score            | 0.102 | 0.069 | 0.060 | 0.055 | 0.051 |

At the sample size of the example the nominal 5% Wald test has true size close to 10%. All three sizes approach 0.05 as  $n$  grows, but slowly when  $r$  is not small compared with  $n$ .

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statcrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWWNSSSWMMMMSSNSNMMSMWNNWNSMMSWNNMSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
covariates = np.column_stack([data["poverty"], data["single"], data["urban"]])
X0 = np.column_stack([np.ones(n), covariates])
D = np.column_stack([(region == g).astype(float) for g in "NSW"])
X = np.column_stack([X0, D])
r, q = 7, 3
```

```

def fit(Z):
    b, *_ = np.linalg.lstsq(Z, y, rcond=None)
    return b, np.sum((y - Z @ b) ** 2)

b, sse = fit(X)
b0, sse0 = fit(X0)
F = (sse0 - sse) / q / (sse / (n - r))

u = (sse0 - sse) / sse          # the ratio behind every statistic
wald = n * u                  # n (SSE0 - SSE) / SSE
lr = n * np.log1p(u)           # n log(SSE0 / SSE)
score = n * u / (1 + u)        # n (SSE0 - SSE) / SSE0
crit = stats.chi2.ppf(0.95, q)
for name, stat in (("Wald", wald), ("LR", lr), ("score", score)):
    print(f"{name:5s} {stat:.3f}  chi2 p = {stats.chi2.sf(stat, q):.4f}")
print(f"F      {F:.3f}  exact p = {stats.f.sf(F, q, n - r):.4f}")

```

The lesson for other models is double. In the normal linear model none of the three approximations is needed, because the exact  $F$  test is available. In models without an exact theory, generalized linear models among them (Chapter 34 onwards), the three statistics are the standard tools (Proposition 34.5.3). Their disagreement in small samples is a warning that the  $\chi^2$  approximation is poor. The linear model suggests a remedy: dividing by  $q$  and referring to an  $F$  distribution with finite denominator degrees of freedom often calibrates better.

### 11.4.3 Invariance and optimality

Is the  $F$  test a *good* test? For a single constraint and a one-sided alternative, the  $t$  test of Section 11.6 has strong optimality properties. For  $q \geq 2$  no test is uniformly most powerful: the most powerful test against alternatives in one direction of the test space rejects for large values of the estimated departure *in that direction*, and it is a different test for each direction (Exercise C1). Some restriction on the class of tests is needed. The natural one is invariance. By Proposition 11.2.2, the problem does not change if the data are rescaled, shifted within the reduced model, or rotated within the test space and within the residual space. A test whose verdict changes under such operations would be favouring particular directions for no reason given by the problem. We show that among tests that do not, the  $F$  test is the most powerful, simultaneously against every alternative.

We need one property of the noncentral  $F$  family.

#### Lemma 11.4.3 — Monotone likelihood ratio of the noncentral $F$

For  $q, \nu \geq 1$  let  $f_\gamma$  be the density of  $F(q, \nu, \gamma)$ . For every  $\gamma > 0$ , the ratio  $f_\gamma(x)/f_0(x)$  is a strictly increasing function of  $x > 0$ .

*Proof.* Let  $V \sim \chi^2(\nu)$ , and for  $k = 0, 1, \dots$  let  $h_k$  be the density of  $(U_k/q)/(V/\nu)$ , where  $U_k \sim \chi^2(q + 2k)$  is independent of  $V$ . By Exercise (B4),  $B = U_k/(U_k + V) \sim \text{Beta}(a_k, b)$  with  $a_k = q/2 + k$  and  $b = \nu/2$ . The map  $B \mapsto x = \nu B/\{q(1 - B)\}$  is increasing, with inverse  $B = t(x) := qx/(\nu + qx)$ , and  $1 - t(x) = \nu/(\nu + qx)$ . Changing variables,

$$h_k(x) = \frac{t(x)^{a_k-1} \{1 - t(x)\}^{b-1}}{B(a_k, b)} \cdot \frac{q\nu}{(\nu + qx)^2}, \quad \text{so} \quad \frac{h_k(x)}{h_0(x)} = \frac{B(q/2, \nu/2)}{B(q/2 + k, \nu/2)} t(x)^k = c_k t(x)^k,$$

where  $B$  is the beta function and  $c_k > 0$ . By [Theorem 4.1.2\(d\)](#),  $U \sim \chi^2(q, \gamma)$  has the law of  $U_K$  with  $K \sim \text{Poisson}(\gamma/2)$  independent of the  $U_k$  and  $V$ . Conditioning on  $K$ ,  $F = (U/q)/(V/\nu)$  has density  $f_\gamma(x) = \sum_{k \geq 0} p_k h_k(x)$  with  $p_k = e^{-\gamma/2}(\gamma/2)^k/k!$ , and  $f_0 = h_0$ . Hence

$$\frac{f_\gamma(x)}{f_0(x)} = \sum_{k \geq 0} p_k c_k t(x)^k.$$

Every coefficient is positive when  $\gamma > 0$ , so the series is strictly increasing in  $t$ , and  $t(x)$  is strictly increasing in  $x$ . ■

A test is a measurable function  $\phi$  of the data with values in  $[0, 1]$ , the probability of rejecting. It is **invariant** if  $\phi(g(\mathbf{y})) = \phi(\mathbf{y})$  for all  $\mathbf{y}$  and all  $g$  in the set  $\mathcal{G}$  of maps

$$g(\mathbf{y}) = a\mathbf{H}\mathbf{y} + \mathbf{v}, \quad a > 0, \quad \mathbf{v} \in \mathcal{C}(\mathbf{X}_0), \quad \mathbf{H} \text{ orthogonal with } \mathbf{H}\mathcal{C}(\mathbf{X}_0) = \mathcal{C}(\mathbf{X}_0), \quad \mathbf{H}\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X}).$$

For a testable hypothesis with  $\mathbf{d} \neq \mathbf{0}$ , the same applies to the data  $\mathbf{y} - \mathbf{X}\mathbf{b}_0$ , with  $\mathcal{C}(\mathbf{X}_0)$  replaced by  $\mathcal{S}_0$ .

### Theorem 11.4.4 — The $F$ test is uniformly most powerful invariant

Assume [Eq. 7.3.1](#) and the setting of [Definition 11.1.1](#). Let  $\phi$  be an invariant test with  $\mathbb{E}\phi(\mathbf{Y}) \leq \alpha$  at every point of  $H_0$ . Then at every point  $(\boldsymbol{\beta}, \sigma^2)$  of the full model,

$$\mathbb{E}_{\boldsymbol{\beta}, \sigma^2} \phi(\mathbf{Y}) \leq \Pr\{F(q, n-r, \gamma) > F_\alpha(q, n-r)\},$$

with  $\gamma$  given by [Eq. 11.1.2](#). The right side is the power of the  $F$  test, which is itself invariant and of size  $\alpha$ .

*Proof.* Use the canonical coordinates  $\mathbf{z}_k = \mathbf{Q}_k^\top \mathbf{y}$  of [Proposition 11.2.3](#), and let  $\mathcal{Y} = \{\mathbf{y} : \mathbf{z}_2 \neq \mathbf{0}\}$ , a set of probability one on which  $F$  is defined.

*Step 1: an invariant test depends on  $\mathbf{y}$  only through  $F$ .* Let  $\mathbf{y}, \mathbf{y}' \in \mathcal{Y}$  with  $F(\mathbf{y}) = F(\mathbf{y}')$ . Put  $a = \|\mathbf{z}'_2\|/\|\mathbf{z}_2\| > 0$ . Equality of the  $F$  values gives  $\|\mathbf{z}'_1\| = a\|\mathbf{z}_1\|$ . Choose orthogonal matrices  $\mathbf{O}_1$  ( $q \times q$ ) and  $\mathbf{O}_2$   $((n-r) \times (n-r))$  with  $\mathbf{O}_1(a\mathbf{z}_1) = \mathbf{z}'_1$  and  $\mathbf{O}_2(a\mathbf{z}_2) = \mathbf{z}'_2$ . This is possible because the vectors in each pair have equal lengths, and if  $\mathbf{z}_1 = \mathbf{0}$  then  $\mathbf{z}'_1 = \mathbf{0}$  and  $\mathbf{O}_1 = \mathbf{I}$  will do. Put

$$\mathbf{H} = \mathbf{Q}_0\mathbf{Q}_0^\top + \mathbf{Q}_1\mathbf{O}_1\mathbf{Q}_1^\top + \mathbf{Q}_2\mathbf{O}_2\mathbf{Q}_2^\top, \quad \mathbf{v} = \mathbf{Q}_0(\mathbf{z}'_0 - a\mathbf{z}_0).$$

$\mathbf{H}$  is orthogonal, maps  $\mathcal{C}(\mathbf{X}_0) = \mathcal{C}(\mathbf{Q}_0)$  and  $\mathcal{C}(\mathbf{X}) = \mathcal{C}([\mathbf{Q}_0, \mathbf{Q}_1])$  onto themselves, and  $\mathbf{v} \in \mathcal{C}(\mathbf{X}_0)$ . The canonical coordinates of  $a\mathbf{H}\mathbf{y} + \mathbf{v}$  are  $a\mathbf{z}_0 + \mathbf{z}'_0 - a\mathbf{z}_0 = \mathbf{z}'_0$ ,  $a\mathbf{O}_1\mathbf{z}_1 = \mathbf{z}'_1$  and  $a\mathbf{O}_2\mathbf{z}_2 = \mathbf{z}'_2$ , so  $a\mathbf{H}\mathbf{y} + \mathbf{v} = \mathbf{y}'$  and  $\phi(\mathbf{y}') = \phi(\mathbf{y})$ . Now fix, for each  $f \geq 0$ , the point  $\mathbf{y}_f = \sqrt{qf/(n-r)}\mathbf{Q}_1\mathbf{e}_1 + \mathbf{Q}_2\mathbf{e}_1$ , where  $\mathbf{e}_1$  denotes a first coordinate vector, and put  $\psi(f) = \phi(\mathbf{y}_f)$ . Then  $F(\mathbf{y}_f) = f$ , so  $\phi(\mathbf{y}) = \psi(F(\mathbf{y}))$  for every  $\mathbf{y} \in \mathcal{Y}$ . The map  $f \mapsto \mathbf{y}_f$  is continuous, so  $\psi$  is measurable.

*Step 2: reduction to one dimension.* By Step 1 and [Theorem 11.1.1\(b\)](#),  $\mathbb{E}\phi(\mathbf{Y}) = \mathbb{E}\psi(F) = \int_0^\infty \psi(x)f_\gamma(x)dx$ , with  $f_\gamma$  the density of  $F(q, n-r, \gamma)$ . The hypothesis is  $\gamma = 0$  ([Theorem 11.1.1\(c\)](#)), so the size condition says  $\int \psi f_0 \leq \alpha$ .

*Step 3: Neyman–Pearson.* Fix  $\gamma > 0$ , let  $c = F_\alpha(q, n-r)$ ,  $\phi^*(x) = 1\{x > c\}$  and  $k = f_\gamma(c)/f_0(c)$ . By [Lemma 11.4.3](#),  $f_\gamma(x) - kf_0(x)$  is positive for  $x > c$  and negative

for  $x < c$ . Since  $\phi^* - \psi \geq 0$  where  $x > c$  and  $\phi^* - \psi \leq 0$  where  $x < c$ , the product  $(\phi^* - \psi)(f_\gamma - kf_0)$  is nonnegative everywhere. Integrating,

$$\int (\phi^* - \psi)f_\gamma \geq k \int (\phi^* - \psi)f_0 = k\left(\alpha - \int \psi f_0\right) \geq 0.$$

So  $E\phi(\mathbf{Y}) = \int \psi f_\gamma \leq \int \phi^* f_\gamma$ , which is the power of the  $F$  test. For  $\gamma = 0$  the inequality is the size condition. The  $F$  test is invariant by [Proposition 11.2.2](#). ■

The theorem says more than “the  $F$  test is good”. It says that once one agrees that no direction in the test space and no scale deserves special treatment, the choice of test is forced: every invariant test is a function of  $F$  (Step 1), and among functions of  $F$  the upper tail is best against every alternative at once. The likelihood ratio principle arrives at the same test without any appeal to optimality, which is reassuring but not a proof. In other models the likelihood ratio test need not be optimal in any sense.

*Remark* (What invariance costs). Invariance is a restriction, and the  $F$  test is not the most powerful test against every alternative. If one contrast in the test space is of special interest, the  $t$  test for that contrast is more powerful against departures in its direction and less powerful against others ([Section 11.6](#)). A Bayesian analysis with a prior concentrated on some directions behaves similarly. Invariance expresses the absence of such preferences. It is appropriate for an omnibus question like “do the regions differ?”, and inappropriate when the scientific question singles out a direction.

## 11.4.4 Exercises

### 11.4.4.1 A. Check your understanding

#### Exercise A1.

From  $F = 2.440$ ,  $q = 3$ ,  $n = 50$  and  $r = 7$ , compute the three statistics of [Proposition 11.4.2](#) and check them against [Example \(Three verdicts from one data set\)](#).

### 11.4.4.2 B. Practice

#### Exercise B1.

Fix  $q$  and  $r$ . Show that under  $H$ , as  $n \rightarrow \infty$ ,  $qF_H$  converges in distribution to  $\chi^2(q)$ , and that the differences Wald  $- qF_H$ ,  $-2\log \Lambda - qF_H$  and Score  $- qF_H$  converge to zero in probability. Deduce that the sizes in [Example \(Three verdicts from one data set\)](#) tend to 0.05.

*Solution.* Write  $qF_H = (SS_H/\sigma^2)/(s^2/\sigma^2)$ . The numerator is  $\chi^2(q)$  for every  $n$ , and  $s^2/\sigma^2 \rightarrow 1$  in probability ([Exercise \(B1\)](#)), so  $qF_H \rightarrow \chi^2(q)$  in distribution. With  $u = qF_H/(n-r)$ , Wald  $- qF_H = qF_H\{n/(n-r) - 1\} = qF_H r/(n-r) \rightarrow 0$  in probability. For the others,  $nu = O_p(1)$ , so  $u \rightarrow 0$ , and  $n \log(1+u) - nu = nu\{\log(1+u)/u - 1\}$  and  $nu/(1+u) - nu = -nu^2/(1+u)$  both tend to zero. Each statistic therefore converges in distribution to  $\chi^2(q)$ , whose upper 5% point is the critical value used, so the sizes

converge to 0.05.

### Exercise B2.

Let  $\hat{\epsilon}_H = \mathbf{y} - \mathbf{X}\hat{\beta}_H$  be the residual vector of the restricted fit. Show that the score statistic equals  $n\|\mathbf{M}\hat{\epsilon}_H\|^2/\|\hat{\epsilon}_H\|^2$ , which is  $n$  times the uncentred  $R^2$  from regressing  $\hat{\epsilon}_H$  on  $\mathbf{X}$ . Explain why this “ $nR^2$ ” recipe needs only the restricted fit.

*Solution.*  $\mathbf{M}\hat{\epsilon}_H = \mathbf{M}\mathbf{y} - \mathbf{X}\hat{\beta}_H = \mathbf{X}(\hat{\beta} - \hat{\beta}_H)$ , whose squared length is  $\text{SS}_H$  (proof of Proposition 11.4.2), and  $\|\hat{\epsilon}_H\|^2 = \text{SSE}_H$ . So  $n\|\mathbf{M}\hat{\epsilon}_H\|^2/\|\hat{\epsilon}_H\|^2 = n\text{SS}_H/\text{SSE}_H = \text{Score}$ . The uncentred  $R^2$  of a regression of  $\hat{\epsilon}_H$  on  $\mathbf{X}$  is  $\|\mathbf{M}\hat{\epsilon}_H\|^2/\|\hat{\epsilon}_H\|^2$ . Computing it requires the restricted residuals and one auxiliary regression, but not  $\hat{\beta}$  or the unrestricted model’s residual sum of squares as such. This is why score tests are popular when the full model is hard to fit, as in tests for omitted nonlinear terms or for heteroscedasticity (Chapter 21).

### Exercise B3.

Suppose  $\sigma^2 = \sigma_0^2$  is known. Show that the likelihood ratio test of a reduced model rejects for large  $(\text{SSE}_0 - \text{SSE})/\sigma_0^2$ , the statistic of Exercise (B1), and that  $-2\log \Lambda$  is then exactly  $\chi^2(q)$  under  $H_0$ .

#### 11.4.4.3 C. Going deeper

### Exercise C1.

In the canonical form with  $\sigma^2$  known and  $q \geq 2$ , consider testing  $\eta_1 = \mathbf{0}$  against the single alternative  $\eta_1 = \mathbf{a} \neq \mathbf{0}$ . Show by the Neyman–Pearson lemma that the most powerful level- $\alpha$  test rejects when  $\mathbf{a}^\top \mathbf{z}_1 > \sigma\|\mathbf{a}\|z_\alpha$ , where  $z_\alpha$  is the upper  $\alpha$  point of  $N(0, 1)$ . Show that this test has power below  $\alpha$  against  $-\mathbf{a}$ , and conclude that no uniformly most powerful test of  $\eta_1 = \mathbf{0}$  exists.

*Solution.* The likelihood ratio of  $N(\mathbf{a}, \sigma^2\mathbf{I})$  to  $N(\mathbf{0}, \sigma^2\mathbf{I})$  at  $\mathbf{z}_1$  is  $\exp\{(\mathbf{a}^\top \mathbf{z}_1 - \|\mathbf{a}\|^2/2)/\sigma^2\}$ , increasing in  $\mathbf{a}^\top \mathbf{z}_1$ . Under the null,  $\mathbf{a}^\top \mathbf{z}_1 \sim N(0, \sigma^2\|\mathbf{a}\|^2)$ , which gives the critical value, and the Neyman–Pearson argument of Step 3 of the proof of Theorem 11.4.4 shows that this test is most powerful. Against  $-\mathbf{a}$ ,  $\mathbf{a}^\top \mathbf{z}_1 \sim N(-\|\mathbf{a}\|^2, \sigma^2\|\mathbf{a}\|^2)$ , so the power is  $\Pr\{Z > z_\alpha + \|\mathbf{a}\|/\sigma\} < \alpha$ . A uniformly most powerful test would have to be most powerful against both  $\mathbf{a}$  and  $-\mathbf{a}$ . The most powerful tests against these two alternatives are essentially unique and different, so no single test can be both.

### Exercise C2.

Extend Lemma 11.4.3: show that for  $0 \leq \gamma_1 < \gamma_2$  the ratio  $f_{\gamma_2}(x)/f_{\gamma_1}(x)$  is increasing in  $x$ . (Write both densities as power series in  $t(x)$  with coefficients  $a_k(\gamma)$ , and show that  $a_k(\gamma_2)/a_k(\gamma_1)$  increases with  $k$ .) Deduce that the  $F$  test is also uniformly most powerful invariant for testing  $\gamma \leq \gamma_1$  against  $\gamma > \gamma_1$ , with a noncentral critical value. Such tests appear in equivalence testing.



## 11.5 Power and noncentrality

A test is judged by two probabilities. Its size, the probability of rejecting a true hypothesis, is fixed by the critical value. Its **power**, the probability of rejecting a false one, depends on how false the hypothesis is. For the  $F$  test the whole dependence runs through the single number  $\gamma$ , and that makes power calculations unusually transparent. This section states the power function, interprets the noncentrality, uses it to plan the size of a study, and asks how sensitive the region test of [Section 11.1](#) actually was. It ends with a popular calculation that should not be made.

### 11.5.1 The power function

#### Theorem 11.5.1 — Power of the $F$ test

Assume [Eq. 7.3.1](#) with  $r < n$ , and let  $H : \Lambda^\top \beta = \mathbf{d}$  be testable with  $q = \text{rank}(\Lambda)$  (or let  $H$  be a reduced model with  $q = r - r_0$ ). The probability that the level- $\alpha$   $F$  test rejects is

$$\pi(\gamma) = \Pr\{F(q, n - r, \gamma) > F_\alpha(q, n - r)\}, \quad (11.5.1)$$

where  $\gamma$  is the noncentrality of [Eq. 11.3.2](#), or of [Eq. 11.1.2](#) for a reduced model. Moreover:

- (a)  $\pi$  depends on  $(\beta, \sigma^2)$  only through  $\gamma$ . It is continuous and strictly increasing, with  $\pi(0) = \alpha$  and  $\pi(\gamma) \rightarrow 1$  as  $\gamma \rightarrow \infty$ . In particular the test is *unbiased*: its power exceeds  $\alpha$  at every alternative.
- (b)  $\sigma^2\gamma$  is the squared distance from the mean vector  $\mathbf{X}\beta$  to the set of mean vectors allowed by  $H$ :

$$\sigma^2\gamma = \min_{\Lambda^\top \mathbf{b} = \mathbf{d}} \|\mathbf{X}\beta - \mathbf{X}\mathbf{b}\|^2 = \min_{\Lambda^\top \mathbf{b} = \mathbf{d}} \sum_{i=1}^n (\mathbf{x}_{(i)}^\top \beta - \mathbf{x}_{(i)}^\top \mathbf{b})^2.$$

- (c) If further observations are added, with the same  $\beta, \sigma^2$  and hypothesis, the noncentrality does not decrease. If the whole design is replicated  $k$  times,  $\gamma$  is multiplied by  $k$ .

*Proof.* The distribution of  $F_H$  is [Theorem 11.3.2\(c\)](#), which gives [Eq. 11.5.1](#). (a) is [Theorem 4.1.4](#) with  $c = F_\alpha(q, n - r)$ . For (b), [Theorem 11.3.2\(b\)](#) applied to the data vector  $\mathbf{X}\beta$ , for which  $\beta$  is a least squares estimate and the residual sum of squares is zero, shows that  $\min_{\Lambda^\top \mathbf{b} = \mathbf{d}} \|\mathbf{X}\beta - \mathbf{X}\mathbf{b}\|^2 = (\Lambda^\top \beta - \mathbf{d})^\top \mathbf{W}^- (\Lambda^\top \beta - \mathbf{d}) = \sigma^2\gamma$ . For (c), adding rows to  $\mathbf{X}$  enlarges  $\mathcal{C}(\mathbf{X}^\top)$ , so  $H$  remains testable (the constraint set  $\{\mathbf{b} : \Lambda^\top \mathbf{b} = \mathbf{d}\}$  is unchanged), and the sum in (b) acquires extra nonnegative terms for every  $\mathbf{b}$ . The minimum of a larger function over the same set is at least as large. Replicating the design  $k$  times multiplies every sum of squares in (b) by  $k$ , and hence the minimum. ■

Part (b) is the reading of  $\gamma$  to remember. The noncentrality is the smallest total squared



discrepancy, over all observations, between the true mean and a mean the hypothesis allows, in units of the error variance. It is large when the hypothesis is badly wrong, when  $\sigma^2$  is small, and when there are many observations at which the discrepancy shows. Part (c) makes the last point precise. Note that adding observations also increases  $n - r$ , which helps as well, and that the sum in (b) does *not* depend on the parameterization.

Two consequences follow at once. First, for a fixed alternative,  $\gamma$  grows in proportion to the number of replicates, so the power tends to one: *every fixed departure from the hypothesis is detected with certainty in large enough samples*. A significant result from a very large sample may therefore reflect a departure too small to matter, and effect sizes, not p-values, answer the question “how much?”. Second, the ratio  $\gamma/n$  is a natural *effect size*, the average squared discrepancy per observation relative to  $\sigma^2$ . It is closely related to the effect size  $f^2$  of Cohen’s power tables, which are widely used in the behavioural sciences.

## 11.5.2 Planning a study

Before the data are collected, Eq. 11.5.1 turns the question “how many observations?” into arithmetic. One specifies the smallest departure worth detecting, a guess at  $\sigma$ , the level and the desired power, and solves for  $n$ . The difficulty is the first ingredient. A departure is a whole vector  $\mathbf{A}^\top \boldsymbol{\beta} - \mathbf{d}$ , and different vectors of the same apparent size give different  $\gamma$ . For the comparison of group means there is a clean way out.

### Proposition 11.5.2 — Least and most favourable configurations

In the balanced one-way layout with  $g \geq 2$  groups of  $m$  observations, group means  $\mu_1, \dots, \mu_g$  and common variance  $\sigma^2$ , the noncentrality of the  $F$  test of equal means is

$$\gamma = \frac{m}{\sigma^2} \sum_{i=1}^g (\mu_i - \bar{\mu})^2, \quad \bar{\mu} = \frac{1}{g} \sum_i \mu_i.$$

Among configurations whose range  $\max_i \mu_i - \min_i \mu_i$  equals  $\Delta$ ,

$$\frac{m\Delta^2}{2\sigma^2} \leq \gamma \leq \frac{mg\Delta^2}{4\sigma^2}.$$

The lower bound is attained exactly when one mean is at each extreme and all others are at the midpoint. For even  $g$ , the upper bound is attained exactly when half the means are at each extreme.

*Proof.* The full model has one indicator per group. The reduced model is  $\mathcal{C}(\mathbf{1})$ , so by Theorem 11.1.1(c),  $\sigma^2\gamma = \|(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\boldsymbol{\mu}\|^2$ , where  $\boldsymbol{\mu}$  repeats  $\mu_i$  on the  $m$  observations of group  $i$ . This is  $m \sum_i (\mu_i - \bar{\mu})^2$ . Let  $\mu_{\max}$  and  $\mu_{\min}$  be two groups at the extremes and put  $a = \mu_{\max} - \bar{\mu} \geq 0$ ,  $b = \bar{\mu} - \mu_{\min} \geq 0$ , so  $a + b = \Delta$ . Dropping the other groups from the sum,

$$\sum_i (\mu_i - \bar{\mu})^2 \geq a^2 + b^2 \geq \frac{1}{2}(a + b)^2 = \frac{1}{2}\Delta^2,$$

with equality in the first step iff every other mean equals  $\bar{\mu}$ , and in the second iff  $a = b$ . If all other means equal  $\bar{\mu}$ , then  $\bar{\mu}$  is also the average of the two extremes, so  $a = b$  holds automatically, and the equality case is the stated configuration. For the upper bound, let  $c$  be the midpoint of the range. Since  $\bar{\mu}$  minimizes  $t \mapsto \sum_i (\mu_i - t)^2$ ,  $\sum_i (\mu_i - \bar{\mu})^2 \leq$

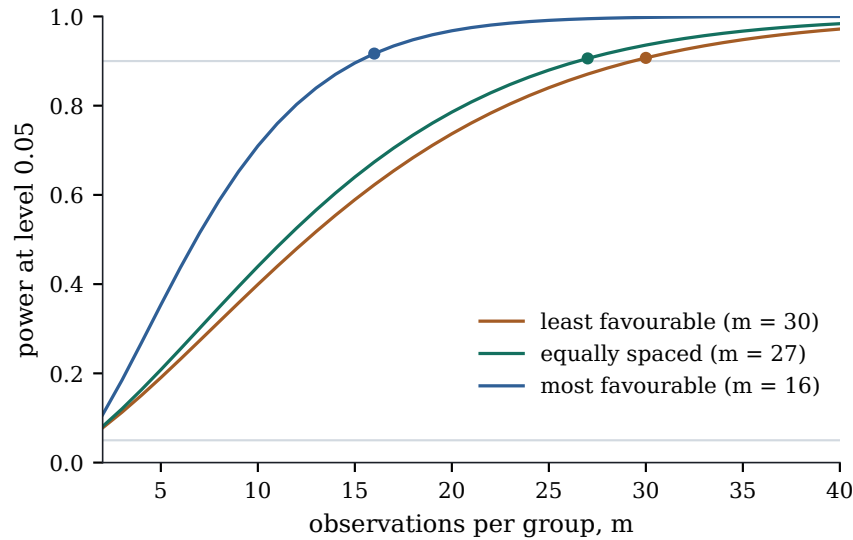


Figure 11.5.1: Power of the level-0.05  $F$  test of equal means for four groups of  $m$  observations, when the group means span one error standard deviation. The three curves are the least favourable configuration of Proposition 11.5.2, equally spaced means, and the most favourable configuration. Dots mark the smallest  $m$  giving power 0.9.

$\sum_i (\mu_i - c)^2 \leq g\Delta^2/4$ , because every  $|\mu_i - c| \leq \Delta/2$ . Equality requires every mean at an extreme and  $\bar{\mu} = c$ , which for even  $g$  means half at each end. ■

A design that has power 0.9 at the least favourable configuration has at least that power at every configuration with range  $\Delta$  or more. That is the guarantee a planner usually wants.

#### Example. (How many observations per group?)

Four treatments are to be compared at level 0.05, and a spread of one standard deviation between the best and the worst treatment ( $\Delta/\sigma = 1$ ) should be detected with probability at least 0.9. The noncentrality per observation per group is 0.500 for the least favourable configuration, 0.556 for equally spaced means, and 1.000 for the most favourable one (two means at each end). Searching over  $m$  with Eq. 11.5.1 gives

| Configuration    | $\gamma/m$ | smallest $m$ with<br>power $\geq 0.9$ | power at that $m$ |
|------------------|------------|---------------------------------------|-------------------|
| least favourable | 0.500      | 30                                    | 0.907             |
| equally spaced   | 0.556      | 27                                    | 0.906             |
| most favourable  | 1.000      | 16                                    | 0.917             |

To be safe against every configuration with range at least  $\sigma$ , the study needs 30 observations per group. A simulation of 20000 data sets at the least favourable configuration with  $m = 30$  rejects in a proportion 0.909 of them, against the computed 0.907. Figure 11.5.1 shows the whole power curves.

```
import numpy as np
from scipy import stats
```

```

def power(gamma, q, nu, alpha=0.05):
    """P{F(q, nu, gamma) > F_alpha(q, nu)}."""
    crit = stats.f.ppf(1 - alpha, q, nu)
    return stats.ncf.sf(crit, q, nu, gamma) if gamma > 0 else alpha

g, delta_over_sigma = 4, 1.0
patterns = {
    "least favourable": np.array([0, 0.5, 0.5, 1]),
    "equally spaced": np.array([0, 1 / 3, 2 / 3, 1]),
    "most favourable": np.array([0, 0, 1, 1]),
}

def gamma_oneway(m, means):
    """Noncentrality m * sum (mu_i - mu_bar)^2 / sigma^2 for m observations per
    group."""
    dev = (means - means.mean()) * delta_over_sigma
    return m * np.sum(dev ** 2)

for name, means in patterns.items():
    m = 2
    while power(gamma_oneway(m, means), g - 1, g * (m - 1)) < 0.9:
        m += 1
    print(f"{name:17s} m = {m} per group, power {power(gamma_oneway(m, means), g -
    1, g * (m - 1)):.3f}")

```

The same logic applies to any linear hypothesis. For a regression coefficient  $\beta_j$ , with the other regressors in the model,  $\sigma^2\gamma = \beta_j^2 \|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2$ , where  $\mathbf{M}_{(j)}$  projects onto the other columns. The design enters through the length of the part of  $\mathbf{x}_j$  that the other regressors cannot explain (Theorem 6.6.2). Spreading the values of  $x_j$  increases power, and collinearity with the other regressors destroys it. Adding a covariate that is unrelated to  $\mathbf{x}_j$  but explains part of the response reduces  $\sigma^2$ , and so increases  $\gamma$ , with little loss in the length of the residualized  $\mathbf{x}_j$ .

### 11.5.3 How sensitive was the region test?

Power calculations are also useful *after* a nonsignificant result, provided the alternatives are fixed by the scientific question and not by the data. For the region test of Example (Do regions matter?), consider the alternative in which the Northeast differs from the other three regions by  $\delta$  murders per 100,000, with everything else as in the reduced model. The mean vector is then  $\mathbf{X}_0\mathbf{c} + \delta\mathbf{d}_N$ , where  $\mathbf{d}_N$  is the Northeast indicator, and by Theorem 11.1.1(c),

$$\gamma = \frac{\delta^2 \|(\mathbf{I} - \mathbf{M}_0)\mathbf{d}_N\|^2}{\sigma^2}.$$

There are 9 Northeast states, but  $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{d}_N\|^2 = 5.76$  only: the covariates already account for part of the Northeast's distinctiveness, and only the rest counts as evidence about regions. Setting  $\sigma$  at  $s = 1.423$  as a planning value, the power of the 5% test is

| $\delta$ | 0.5   | 1.0   | 1.5   | 2.0   |
|----------|-------|-------|-------|-------|
| power    | 0.091 | 0.241 | 0.506 | 0.778 |

and power 0.8 needs  $\delta = 2.05$ . A Northeast shortfall of 1.5 murders per 100,000, about the size estimated in Example (Regional hypotheses with an over-parameterized model) (1.379 below the

other three regions), would have been missed about half the time. The nonsignificant verdict of [Example \(Do regions matter?\)](#) therefore says little about whether regional differences of that size exist.

```
import statsmodels.api as sm

data = sm.datasets.statocrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
codes = "SWWSWNSSSSWMMMMSSNSNMMSMWNNWNSMMSWNSMSSWNSWSMW"
region = np.array(list(codes))
y = data["murder"].to_numpy()
n = len(y)
X0 = np.column_stack([np.ones(n), data["poverty"], data["single"], data["urban"]])
D = np.column_stack([(region == c).astype(float) for c in "NSW"])
X = np.column_stack([X0, D])

b0, *_ = np.linalg.lstsq(X0, D[:, 0], rcond=None)
k2 = np.sum((D[:, 0] - X0 @ b0) ** 2)      #  $\|(I - M_0) d_N\|^2$ :  $\gamma = \delta^2 k2$ 
    / sigma^2
bf, *_ = np.linalg.lstsq(X, y, rcond=None)
sigma = np.sqrt(np.sum((y - X @ bf) ** 2) / (n - 7))  # planning value for sigma
for delta in (0.5, 1.0, 1.5, 2.0):
    print(f"Northeast lower by {delta}:  $\gamma = \{\delta^2 * k2 / \sigma^2\}$ , "
          f"power =  $\{\text{power}(\delta^2 * k2 / \sigma^2, 3, n - 7)\}$ ")
```

The calculation uses the estimate  $s$  in place of  $\sigma$ , which is itself uncertain. A more careful version would repeat it for a range of plausible values of  $\sigma$ , or replace the whole exercise by a confidence region for the regional effects, which shows directly which differences the data rule out ([Chapter 12](#)).

## 11.5.4 Observed power

A common calculation takes the observed data, estimates the noncentrality from them, and reports the power “the test had” at that estimate. It is meant to show whether a nonsignificant result was due to low power. It cannot show that.

### Proposition 11.5.3 — Observed power is a function of the p-value

Fix  $q$ ,  $n - r$  and  $\alpha$ , and define the **observed power** as  $\pi(\hat{\gamma})$ , with  $\pi$  as in [Eq. 11.5.1](#) and  $\hat{\gamma} = qF_{\text{obs}}$ . Then the observed power is a strictly decreasing function of the p-value of the  $F$  test. The same holds for any estimate  $\hat{\gamma} = h(F_{\text{obs}})$  with  $h$  strictly increasing and positive.

*Proof.* The p-value is  $1 - \Phi_F(F_{\text{obs}})$ , with  $\Phi_F$  the continuous, strictly increasing distribution function of  $F(q, n - r)$ , so  $F_{\text{obs}}$  is a strictly decreasing function of the p-value. The map  $F_{\text{obs}} \mapsto h(F_{\text{obs}})$  is strictly increasing, and  $\pi$  is strictly increasing by [Theorem 11.5.1\(a\)](#). The composition is strictly decreasing in the p-value. ■

So observed power adds nothing to the p-value. In the setting of the region test ( $q = 3$ ,

$n - r = 43$ ), a p-value of 0.2 always corresponds to an observed power of 0.394, a p-value of exactly 0.05 to 0.637, and 0.01 to 0.830. “The test was not significant and its observed power was low” says the same thing twice. The informative calculations are the ones above: power at alternatives chosen for their scientific meaning, or a confidence region.

*Warning* (Reading a nonsignificant  $F$ ). A nonsignificant  $F$  statistic is weak evidence for the reduced model unless the test had high power against every alternative that matters. Whether it had is a question about the design and about which alternatives matter, answered by [Eq. 11.5.1](#) at alternatives fixed in advance. It is not answered by the observed power, which merely restates the p-value, and it is not answered by the p-value being large. A p-value of 0.9 is no stronger evidence for  $H_0$  than one of 0.3 when the test has low power.

## 11.5.5 Exercises

### 11.5.5.1 A. Check your understanding

#### Exercise A1.

For the test of  $\beta_1 = 0$  in simple regression, show that  $\gamma = \beta_1^2 S_{xx} / \sigma^2$ , with  $S_{xx} = \sum_i (x_i - \bar{x})^2$ . With  $n = 20$  design points in  $[0, 1]$ , compare  $S_{xx}$  for equal spacing and for ten points at each end. What does the second design give up?

#### Exercise A2.

For the one-way layout with group sizes  $n_1, \dots, n_g$ , show that the noncentrality of the test of equal means is  $\gamma = \sum_i n_i (\mu_i - \bar{\mu}_w)^2 / \sigma^2$ , with  $\bar{\mu}_w = \sum_i n_i \mu_i / n$ .

### 11.5.5.2 B. Practice

#### Exercise B1.

Extend [Proposition 11.5.2](#) to unequal group sizes: show that among configurations with range  $\Delta$ ,

$$\gamma \geq \frac{\Delta^2}{\sigma^2} \frac{1}{1/n_{(1)} + 1/n_{(2)}},$$

where  $n_{(1)} \leq n_{(2)}$  are the two smallest group sizes, and describe the configuration attaining it.

*Solution.* Let groups  $a$  and  $b$  carry the largest and smallest means. Dropping the other terms of [Exercise \(A2\)](#),  $\sigma^2 \gamma \geq n_a (\mu_a - \bar{\mu}_w)^2 + n_b (\mu_b - \bar{\mu}_w)^2 \geq \min_t \{n_a (\mu_a - t)^2 + n_b (\mu_b - t)^2\}$ . The minimum is at  $t = (n_a \mu_a + n_b \mu_b) / (n_a + n_b)$  and equals  $\Delta^2 n_a n_b / (n_a + n_b) = \Delta^2 / (1/n_a + 1/n_b)$ . This is smallest when  $a$  and  $b$  are the two smallest groups. It is attained when those two groups carry the extremes and all other means equal  $(n_a \mu_a + n_b \mu_b) / (n_a + n_b)$ , which is then also  $\bar{\mu}_w$ . A departure is hardest to detect when it is confined to the groups with the fewest observations.

### Exercise B2.

Take the straight-line model with design points 0, 1, 2, 3, one observation each, and the hypothesis  $\beta_1 = 0$ . Compute  $\gamma$  as a multiple of  $\beta_1^2/\sigma^2$  and the power at  $\beta_1/\sigma = 1$ , level 0.05. Then compare (i) replicating the design twice, (ii) adding four observations at  $x = 3$ , and (iii) adding four observations at  $x = 1.5$ . Verify [Theorem 11.5.1\(c\)](#) in each case and explain the ranking.

### Exercise B3.

For  $q = 1$  and large  $n - r$ , the  $F$  test is approximately the test that rejects when  $|Z| > z_{\alpha/2}$ , with  $Z \sim N(\delta, 1)$ . Show that if the observed  $|z|$  equals  $z_{\alpha/2}$  exactly, the observed power  $\Pr\{|Z| > z_{\alpha/2}\}$  at  $\delta = z_{\alpha/2}$  is just above 1/2. Conclude that a result on the edge of significance always has observed power about one half, whatever the design.

*Solution.* At  $\delta = z_{\alpha/2}$ ,  $\Pr\{|Z| > z_{\alpha/2}\} = \Pr\{Z > z_{\alpha/2}\} + \Pr\{Z < -z_{\alpha/2}\} = \frac{1}{2} + \Pr\{Z - \delta < -2z_{\alpha/2}\}$ , and the last probability is  $\Phi(-2z_{\alpha/2})$ , which is  $\Phi(-3.92) \approx 4 \times 10^{-5}$  for  $\alpha = 0.05$ . The value depends only on  $\alpha$ , not on  $n$ ,  $\sigma$  or the design, as [Proposition 11.5.3](#) predicts.

#### 11.5.5.3 C. Going deeper

### Exercise C1.

In the sensitivity calculation, let  $n_N$  be the number of Northeast states and  $R_N^2$  the coefficient of determination from regressing the Northeast indicator on the reduced model's columns (which include  $\mathbf{1}$ ). Show that  $\|(\mathbf{I} - \mathbf{M}_0)\mathbf{d}_N\|^2 = n_N(1 - n_N/n)(1 - R_N^2)$ . Interpret each factor, and relate the formula to [Eq. 6.6.2](#).

## 11.6 Testing single coefficients and groups of coefficients

The tests most often printed by regression software are instances of the general theory: a  $t$  test for each coefficient, an  $F$  test that all slopes vanish, and on request an  $F$  test for a group of coefficients. This section derives each from the preceding sections, explains what each one tests, and studies how a group test and the individual tests inside it can disagree, in both directions.

### 11.6.1 The $t$ test

When the hypothesis consists of a single estimable function,  $q = 1$ , the  $F$  statistic is the square of a statistic with a sign, and the sign allows one-sided alternatives.

### Theorem 11.6.1 — The $t$ test for an estimable function

Assume [Eq. 7.3.1](#) with  $r < n$ . Let  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$ ,  $\lambda \neq \mathbf{0}$ , let  $d \in \mathbb{R}$ , let  $\mathbf{G}$  be a generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ , and put

$$T = \frac{\lambda^\top \hat{\boldsymbol{\beta}} - d}{s \sqrt{\lambda^\top \mathbf{G} \lambda}}, \quad s^2 = \frac{\text{SSE}}{n - r}. \quad (11.6.1)$$

- (a)  $T \sim t(n - r, \delta)$  with  $\delta = (\lambda^\top \boldsymbol{\beta} - d) / (\sigma \sqrt{\lambda^\top \mathbf{G} \lambda})$ .
- (b)  $T^2$  equals the statistic  $F_H$  of [Eq. 11.3.1](#) for  $H : \lambda^\top \boldsymbol{\beta} = d$ , and  $\gamma = \delta^2$ . The two-sided test that rejects when  $|T| > t_{n-r, \alpha/2}$  is the  $F$  test of  $H$ , with the same p-value.
- (c) The one-sided test that rejects when  $T > t_{n-r, \alpha}$  has size exactly  $\alpha$  for the hypothesis  $\lambda^\top \boldsymbol{\beta} \leq d$ : its rejection probability is at most  $\alpha$  whenever  $\lambda^\top \boldsymbol{\beta} \leq d$ , with equality when  $\lambda^\top \boldsymbol{\beta} = d$ . Its power increases strictly with  $\lambda^\top \boldsymbol{\beta}$ .

*Proof.*

- (a) is [Corollary 7.5.2](#). (b) With  $\mathbf{A} = \lambda$ ,  $\mathbf{W} = \lambda^\top \mathbf{G} \lambda > 0$  is a scalar, so  $F_H = (\lambda^\top \hat{\boldsymbol{\beta}} - d)^2 / (\lambda^\top \mathbf{G} \lambda s^2) = T^2$ , and [Eq. 11.3.2](#) gives  $\gamma = \delta^2$ . Under  $H$ ,  $T \sim t(n - r)$  is symmetric about zero, so  $\Pr\{|T| > t\} = \Pr\{T^2 > t^2\}$  for every  $t > 0$ . Hence  $t_{n-r, \alpha/2}^2 = F_\alpha(1, n - r)$ , the two rejection regions coincide, and the two-sided  $t$  p-value  $\Pr\{|T| \geq |T_{\text{obs}}|\}$  equals the  $F$  p-value  $\Pr\{T^2 \geq T_{\text{obs}}^2\}$ . (c) By [Exercise \(A2\)](#), the probability that a  $t(n - r, \delta)$  variable exceeds a fixed constant is strictly increasing in  $\delta$ , and  $\delta$  is an increasing function of  $\lambda^\top \boldsymbol{\beta}$ . At  $\delta = 0$  the probability is  $\alpha$ . ■

For a single coefficient  $\beta_j$  in a full-rank model, [Eq. 11.6.1](#) becomes the familiar  $t_j = \hat{\beta}_j / \{s \sqrt{c_{jj}}\}$ , with  $c_{jj}$  the  $j$ th diagonal entry of  $(\mathbf{X}^\top \mathbf{X})^{-1}$ . By [Corollary 9.2.3](#),  $t_j^2$  is the  $F$  statistic for dropping column  $j$  from the model. The  $t$  test of  $\beta_j = 0$  therefore compares the model with all regressors to the model with all *except*  $\mathbf{x}_j$ . It asks whether  $\mathbf{x}_j$  adds anything *given the others*. By the Frisch–Waugh–Lovell theorem ([Theorem 6.6.2](#)), what it measures is the association of  $\mathbf{y}$  with the part of  $\mathbf{x}_j$  orthogonal to the other columns. The same coefficient can be “significant” in one model and not in another, and there is no contradiction: the two tests answer different questions.

The one-sided test is appropriate only when the direction was specified before seeing the data, for example when theory says a coefficient cannot be negative, or when only an increase would lead to action. Choosing the direction after seeing the sign of the estimate doubles the actual size.

## 11.6.2 Tests and intervals are one computation

The two-sided test of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = d$  at level  $\alpha$  *accepts* exactly when  $|\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - d| \leq t_{n-r, \alpha/2} s \sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}}$ . Collecting the values of  $d$  that are accepted gives the interval

$$\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \pm t_{n-r, \alpha/2} s \sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}},$$

which is the  $1 - \alpha$  confidence interval [Eq. 7.5.1](#). Every interval of this kind is the set of hypothesized values that a test does not reject, and every test can be read off an interval: reject  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = d$  iff  $d$  lies outside. The interval reports more. It shows which values are ruled out, and so how precisely the data determine  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ , which the verdict of a single test does not. For several functions at once, inverting the  $F$  test of [Theorem 11.3.2](#) gives an ellipsoid. [Chapter 12](#) develops both and the duality between them.

## 11.6.3 The overall $F$ test

The most common reduced model is the one with no regressors at all. Let  $\mathbf{X}$  contain the intercept column  $\mathbf{1}$  and have rank  $r$ . The reduced model  $E(\mathbf{Y}) = \beta_0 \mathbf{1}$  has  $\text{SSE}_0 = \text{SST} = \sum_i (y_i - \bar{y})^2$ , and  $\text{SST} - \text{SSE} = \text{SSR}$  is the regression sum of squares. By [Theorem 11.1.1](#) and [Theorem 9.5.2](#),

$$F = \frac{\text{SSR}/(r-1)}{\text{SSE}/(n-r)} = \frac{R^2/(r-1)}{(1-R^2)/(n-r)} \sim F(r-1, n-r, \gamma), \quad \gamma = \frac{\|(\mathbf{M} - n^{-1} \mathbf{1} \mathbf{1}^\top) \mathbf{X} \boldsymbol{\beta}\|^2}{\sigma^2}.$$

The **overall  $F$  test** asks whether any linear combination of the regressors helps to predict the response. The answer is almost always yes in observational data with a sensible choice of regressors, and a significant overall  $F$  is correspondingly weak information. It does not show that any particular regressor matters, that the model fits well, or that it predicts usefully: with  $n$  large, a model with  $R^2 = 0.02$  has a highly significant overall  $F$ . A nonsignificant overall  $F$  is more informative. It says that the data give no evidence that the regressors, taken together, relate to the response at all.

## 11.6.4 Groups of coefficients

Between one coefficient and all of them lies the test of a group: whether the terms of a factor, the powers of a polynomial beyond the linear, or a set of related regressors can be dropped together. The statistic is the extra sum of squares of the group, divided by its degrees of freedom and by  $s^2$  ([Theorem 11.1.1](#)). How does it relate to the  $t$  statistics of its members?

### Proposition 11.6.2 — The group statistic is the largest single statistic

Let  $H : \boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  be testable with  $\boldsymbol{\Lambda}$  of full column rank  $q$ , let  $F_H$  be its statistic [Eq. 11.3.1](#), and for  $\mathbf{a} \in \mathbb{R}^q$ ,  $\mathbf{a} \neq \mathbf{0}$ , let  $T(\mathbf{a})$  be the  $t$  statistic [Eq. 11.6.1](#) for the single function  $\mathbf{a}^\top \boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{a}^\top \mathbf{d}$ . Then

$$qF_H = \max_{\mathbf{a} \neq \mathbf{0}} T(\mathbf{a})^2,$$



and the maximum is attained at  $\mathbf{a} \propto \mathbf{W}^{-1}(\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})$ , where  $\mathbf{W} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$ . In particular  $t_j^2 \leq qF_H$  for each coefficient  $\beta_j$  in a tested group  $\beta_j = 0$ ,  $j \in J$ , with  $q = |J|$ .

*Proof.* Write  $\mathbf{u} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$ . The function  $\mathbf{a}^\top \mathbf{\Lambda}^\top \boldsymbol{\beta}$  is estimable with  $\boldsymbol{\lambda} = \mathbf{\Lambda} \mathbf{a} \neq \mathbf{0}$  (the columns of  $\mathbf{\Lambda}$  are independent), and  $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda} = \mathbf{a}^\top \mathbf{W} \mathbf{a}$ , so  $T(\mathbf{a})^2 = (\mathbf{a}^\top \mathbf{u})^2 / (s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a})$ . The matrix  $\mathbf{W}$  is positive definite (Theorem 9.2.4(a)) and has a positive definite square root (Theorem 1.7.6). By the Cauchy–Schwarz inequality (Proposition 1.1.1) applied to  $\mathbf{W}^{1/2} \mathbf{a}$  and  $\mathbf{W}^{-1/2} \mathbf{u}$ ,

$$(\mathbf{a}^\top \mathbf{u})^2 \leq (\mathbf{a}^\top \mathbf{W} \mathbf{a})(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}),$$

with equality iff  $\mathbf{W}^{1/2} \mathbf{a}$  is proportional to  $\mathbf{W}^{-1/2} \mathbf{u}$ , that is,  $\mathbf{a} \propto \mathbf{W}^{-1} \mathbf{u}$  (or  $\mathbf{u} = \mathbf{0}$ ). Dividing by  $s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a}$  gives  $T(\mathbf{a})^2 \leq \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} / s^2 = qF_H$ , with equality at the stated  $\mathbf{a}$ . For the coefficient  $\beta_j$ , take  $\mathbf{a}$  to be the corresponding coordinate vector. ■

The proposition is the geometric heart of Scheffé’s method for multiple comparisons, and it explains the two ways in which a group test and its members can disagree.

*Dilution.* A single large  $t_j$  need not make the group significant. The group test rejects when  $qF_H > qF_\alpha(q, n-r)$ , and  $qF_\alpha(q, n-r)$  grows with  $q$  (Exercise (B2)). A  $t_j^2$  just above  $F_\alpha(1, n-r)$  can therefore leave  $qF_H$  below the group’s critical value, especially when the other members of the group contribute nothing. This is what happened in Example (Do regions matter?): the Northeast contrast had  $|t| = 2.519$ , but the three-degree-of-freedom test of all regional differences was not significant.

*Collinearity.* Conversely, the group can be highly significant while every member has a small  $t$ . The direction  $\mathbf{a}$  that attains the maximum is generally not a coordinate direction. When the estimates are strongly correlated, the data can say clearly that *some* combination of the coefficients is nonzero without being able to attribute it to any single one.

#### Example. (Five regressors for the murder rate)

Regress the murder rate of the 50 states on poverty, high-school graduation, single-parent households, percentage white and urbanization, with an intercept ( $p = 6$ ,  $n - p = 44$ ).

| Regressor                | Estimate | Standard error | $t$    | p-value |
|--------------------------|----------|----------------|--------|---------|
| intercept                | −10.0586 | 17.6785        | −0.569 | 0.572   |
| poverty                  | 0.2378   | 0.1412         | 1.684  | 0.099   |
| high-school graduates    | −0.0047  | 0.1530         | −0.030 | 0.976   |
| single-parent households | 0.4280   | 0.1372         | 3.120  | 0.003   |
| white                    | 0.0108   | 0.0270         | 0.398  | 0.692   |
| urban                    | 0.0046   | 0.0145         | 0.315  | 0.754   |

The overall test gives  $R^2 = 0.643$  and  $F = 15.87$  on 5 and 44 degrees of freedom,  $p = 6.4 \times 10^{-9}$ . Only single parenthood has a  $t$  statistic beyond the critical value 2.015. The group of three demographic and educational regressors, high-school graduation, percentage white and urbanization, has  $F = 0.096$  on 3 and 44 degrees of freedom,  $p = 0.962$ : the data give no evidence that these three add anything once poverty and single parenthood are in the model. Every  $t_j^2$  of the group is at most  $3F = 0.29$ , as [Proposition 11.6.2](#) requires.

This group was chosen here because its members have small  $t$  statistics. A group selected that way will tend to have a small  $F$ , and its large p-value makes dropping the group look safer than it is. Groups should be defined by the scientific question, before the individual results are seen.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statocrime.load_pandas().data
data.index = data.index.str.strip()
data = data.drop(index="District of Columbia")
names = ["poverty", "hs_grad", "single", "white", "urban"]
y = data["murder"].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), data[names]])
p = X.shape[1]

C = np.linalg.inv(X.T @ X)
b = C @ X.T @ y
sse = np.sum((y - X @ b) ** 2)
s2 = sse / (n - p)
se = np.sqrt(s2 * np.diag(C))
t = b / se
pv = 2 * stats.t.sf(np.abs(t), n - p)
for name, bj, sj, tj, pj in zip(["intercept"] + names, b, se, t, pv):
    print(f"{name:10s} {bj:9.4f} {sj:8.4f} {tj:7.3f} {pj:7.4f}")

sst = np.sum((y - y.mean()) ** 2)
R2 = 1 - sse / sst
F_all = (R2 / (p - 1)) / ((1 - R2) / (n - p))

def extra_F(keep):
    """F test that the coefficients not in `keep` (columns of X) are zero."""
    Xr = X[:, keep]
    br, *_ = np.linalg.lstsq(Xr, y, rcond=None)
    sse_r = np.sum((y - Xr @ br) ** 2)
    q = p - len(keep)
    F = ((sse_r - sse) / q) / s2
    return F, stats.f.sf(F, q, n - p)

F_group, p_group = extra_F([0, 1, 3]) # drop hs_grad, white, urban
print(f"overall: R2 = {R2:.3f}, F = {F_all:.2f} on ({p - 1}, {n - p})")
print(f"hs_grad, white, urban: F = {F_group:.3f}, p = {p_group:.3f}")
```

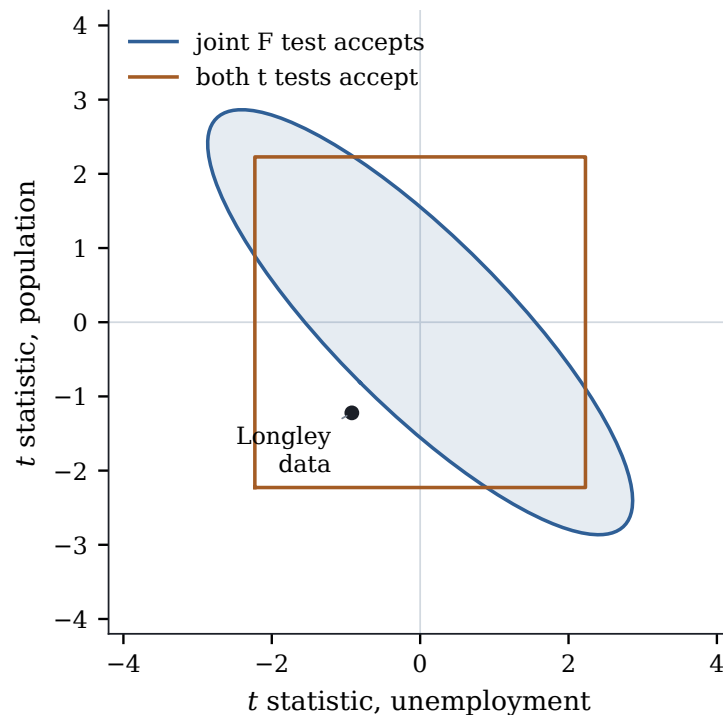


Figure 11.6.1: Acceptance regions for the hypothesis that the unemployment and population coefficients in Longley’s regression are both zero, drawn in the plane of the two  $t$  statistics. The square accepts when both individual  $t$  tests accept at level 0.05. The ellipse accepts when the joint  $F$  test does. The black point is the data.

#### Example. (Jointly significant, individually not)

Longley’s data record US employment for the 16 years 1947 to 1962, with five economic series: the GNP deflator, GNP, unemployment, the size of the armed forces and population. Regress employment on the five series and an intercept, leaving out the time trend, so that  $n - p = 10$ . For unemployment and population, the  $t$  statistics are  $-0.921$  and  $-1.222$ , with  $p$ -values 0.379 and 0.250. Neither comes near the critical value 2.228. The test that *both* coefficients are zero gives  $F = 7.175$  on 2 and 10 degrees of freedom,  $p = 0.0117$ .

Figure 11.6.1 shows why. The two estimates have correlation  $-0.840$ . Under the hypothesis, their  $t$  statistics are then likely to have opposite signs, and the  $F$  test’s acceptance region is an ellipse stretched along the line  $t_1 = -t_2$ . The observed pair has both statistics negative, a combination the ellipse excludes although each coordinate on its own is unremarkable. The data are clear that unemployment and population *together* matter, and unclear about how to divide the credit. With 16 consecutive years of trending series, the errors are also unlikely to be independent, a caution taken up in Chapter 21.

```
lg = sm.datasets.longley.load_pandas()
Z = sm.add_constant(lg.exog.drop(columns="YEAR"))
res = sm.OLS(lg.endog, Z).fit()
joint = res.f_test("UNEMP = 0, POP = 0")
print(res.tvalues[["UNEMP", "POP"]].round(3).to_dict(), res.pvalues[["UNEMP", "POP"]].round(3).to_dict())
print(f"joint F = {float(joint.fvalue):.3f}, p = {float(joint.pvalue):.4f}")
```

The figure also shows the opposite disagreement. Points inside the ellipse but outside the square have one significant  $t$  and a nonsignificant  $F$ . Neither verdict is wrong: the individual tests ask about the coordinate directions, and the joint test asks about all directions at once, paying for that breadth with a larger critical value.

*Warning* (Several  $t$  tests are not a test of the group). Rejecting the group hypothesis “all  $\beta_j = 0, j \in J$ ” whenever at least one  $|t_j|$  exceeds  $t_{n-r, \alpha/2}$  is not a level- $\alpha$  test: when the  $k$  coefficient estimates are independent and  $n - r$  is large, its size is about  $1 - (1 - \alpha)^k$ , about 0.23 for  $k = 5$  at  $\alpha = 0.05$ . (The  $t$  statistics share the estimate  $s$ , so they are not exactly independent.) Reporting the smallest of several p-values as if it were the only one overstates the evidence in the same way. The  $F$  test of the group, or a multiple-comparison procedure designed for the family (Chapter 13), controls the familywise error rate. Choosing regressors by their  $t$  statistics and then testing the chosen model on the same data invalidates all the p-values of the final fit. Section 29.5 returns to this problem.

## 11.6.5 Exercises

### 11.6.5.1 A. Check your understanding

#### Exercise A1.

Using the table of Example (Five regressors for the murder rate), compute the  $F$  statistic for dropping poverty from the five-regressor model and its p-value, without refitting.

#### Exercise A2.

Show that if the direction of a one-sided  $t$  test is chosen after seeing the sign of  $T$ , and the test then rejects when  $|T| > t_{n-r, \alpha}$ , its size is  $2\alpha$ .

### 11.6.5.2 B. Practice

#### Exercise B1.

In a full-rank model, show that the  $t$  statistic for  $\beta_j = \beta_k$  is

$$T = \frac{\hat{\beta}_j - \hat{\beta}_k}{s \sqrt{c_{jj} + c_{kk} - 2c_{jk}}},$$

where  $c_{jk}$  are the entries of  $(\mathbf{X}^\top \mathbf{X})^{-1}$ . Show that it equals the  $t$  statistic of the coefficient of  $\mathbf{x}_j$  in the reparameterized model with regressors  $\mathbf{x}_j$  and  $\mathbf{x}_j + \mathbf{x}_k$  in place of  $\mathbf{x}_j$  and  $\mathbf{x}_k$ .

#### Exercise B2.

Show that  $qF_\alpha(q, \nu)$  is strictly increasing in  $q$  for fixed  $\nu$  and  $\alpha$ . Deduce that a single  $t$  statistic with  $t^2$  just above  $F_\alpha(1, \nu)$  does not force the rejection of a group containing it.

*Solution.*  $qF(q, \nu)$  has the law of  $U_q/(V/\nu)$  with  $U_q \sim \chi^2(q)$  independent of  $V \sim \chi^2(\nu)$ . Taking  $U_{q+1} = U_q + W$  with  $W \sim \chi^2(1)$  independent of  $U_q$  and  $V$ , we have  $U_{q+1}/(V/\nu) > U_q/(V/\nu)$  with probability one, so  $\Pr\{U_{q+1}/(V/\nu) > c\} > \Pr\{U_q/(V/\nu) > c\}$  for every  $c > 0$  (strictly, since  $\Pr\{U_q/(V/\nu) \leq c < U_{q+1}/(V/\nu)\} > 0$ ). Taking  $c = qF_\alpha(q, \nu)$ , where the right side equals  $\alpha$ , shows that the upper  $\alpha$  point of  $U_{q+1}/(V/\nu)$  exceeds  $qF_\alpha(q, \nu)$ . For the deduction, if  $t_j^2$  lies between  $F_\alpha(1, \nu)$  and  $qF_\alpha(q, \nu)$  and the other members contribute nothing, then  $qF_H$  can equal  $t_j^2 < qF_\alpha(q, \nu)$ , and the group is not rejected.

### Exercise B3.

For two coefficients with  $t$  statistics  $t_1, t_2$  whose estimates have correlation  $\rho$ , show that the  $F$  statistic for both being zero is

$$F = \frac{t_1^2 + t_2^2 - 2\rho t_1 t_2}{2(1 - \rho^2)}.$$

Check the value 7.175 of [Example \(Jointly significant, individually not\)](#) from  $t_1 = -0.921$ ,  $t_2 = -1.222$  and  $\rho = -0.840$ , and find the value of  $F$  had the correlation been  $+0.840$ .

*Solution.* Let  $\mathbf{D} = \text{diag}(\text{se}_1, \text{se}_2)$  and  $\mathbf{R}$  the correlation matrix of the estimates, so that the estimated covariance is  $\mathbf{DRD}$ . With  $\mathbf{A}$  selecting the two coefficients,  $2F = \hat{\boldsymbol{\beta}}_J^\top (\mathbf{DRD})^{-1} \hat{\boldsymbol{\beta}}_J = \mathbf{t}^\top \mathbf{R}^{-1} \mathbf{t}$ , where  $\mathbf{t} = \mathbf{D}^{-1} \hat{\boldsymbol{\beta}}_J$ , and  $\mathbf{R}^{-1} = (1 - \rho^2)^{-1} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$ . Numerically,  $t_1^2 + t_2^2 = 2.341$ ,  $-2\rho t_1 t_2 = 1.890$  and  $2(1 - \rho^2) = 0.589$ , so  $F = 7.175$ , as in the example, up to the rounding of these three terms. With  $\rho = +0.840$  the middle term changes sign and  $F = 0.765$ , about  $(2.341 - 1.890)/0.589$ : the same two  $t$  statistics would then be entirely unremarkable.

### 11.6.5.3 C. Going deeper

#### Exercise C1.

Show that  $\max T(\boldsymbol{\lambda})^2$  over all nonzero estimable  $\boldsymbol{\lambda}$ , for the hypotheses  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = 0$ , equals  $rF_*$ , where  $F_*$  is the statistic for the reduced model  $E(\mathbf{Y}) = \mathbf{0}$ . Deduce that  $\Pr\{T(\boldsymbol{\lambda})^2 \leq rF_\alpha(r, n-r) \text{ for all estimable } \boldsymbol{\lambda}\} = 1 - \alpha$  when  $\boldsymbol{\beta} = \mathbf{0}$ . What changes if  $\boldsymbol{\beta} \neq \mathbf{0}$  and  $T(\boldsymbol{\lambda})$  is centred at  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ ? ([Chapter 13](#) builds simultaneous inference on this.)

#### Exercise C2.

Find the likelihood ratio test of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} \leq d$  against  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} > d$  for estimable  $\boldsymbol{\lambda}$ , and show that it rejects for large values of  $T$  in [Eq. 11.6.1](#). (Maximize the likelihood under the inequality constraint: if  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \leq d$  the unrestricted maximum is feasible; otherwise the maximum is on the boundary.)

*Solution.* If  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \leq d$ , the unrestricted maximizer satisfies the constraint and  $\Lambda = 1$ . Otherwise the log-likelihood, maximized over  $\sigma^2$ , is a decreasing function of  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ , which is a convex quadratic in  $\mathbf{b}$  with minimum at  $\hat{\boldsymbol{\beta}}$ . Its minimum over the half-space  $\boldsymbol{\lambda}^\top \mathbf{b} \leq d$  therefore lies on the boundary  $\boldsymbol{\lambda}^\top \mathbf{b} = d$ , where it equals  $\text{SSE} + (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - d)^2 / \boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$

(Eq. 7.3.3). So  $\Lambda = \{1 + T^2/(n - r)\}^{-n/2}$  when  $T > 0$  and  $\Lambda = 1$  when  $T \leq 0$ . This is a nonincreasing function of  $T$ , strictly decreasing for  $T > 0$ , so  $\Lambda \leq k < 1$  iff  $T \geq c$  for some  $c > 0$ , which is the one-sided  $t$  test.

## 11.7 Summary and notes

### 11.7.1 Summary

*Idea.*

1. A linear hypothesis says that the mean vector lies in a subspace  $\mathcal{C}(\mathbf{X}_0)$  of the model space. Its  $F$  statistic compares the extra sum of squares  $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{y}\|^2$ , per dimension of the test space, with  $s^2$ . Under normal errors it is  $F(r - r_0, n - r, \gamma)$ , central under the hypothesis, with  $\gamma$  the squared distance from the mean to the reduced model in units of  $\sigma^2$  (Definition 11.1.1, Theorem 11.1.1, Proposition 11.1.2).
2. Geometrically,  $F$  is a function of the angle between the reduced model's residual vector and the test space. The rejection region is a cone, the statistic is invariant under shifts within the reduced model, rescaling and rotations of the test and residual spaces, and in canonical coordinates every hypothesis is a test that a  $q$ -dimensional normal mean is zero. The null distribution holds for every spherically symmetric error law (Proposition 11.2.1, Proposition 11.2.2, Proposition 11.2.3, Proposition 11.2.4).
3. A constraint  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  is a hypothesis about the data only if it is estimable and consistent. Nonestimable constraints cannot be tested at all. For a testable hypothesis,  $(\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top [\mathbf{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-} \mathbf{\Lambda}]^{-} (\mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})$  is the extra sum of squares of the reduced model with offset  $\mathbf{X}\mathbf{b}_0$ , whatever the rank of  $\mathbf{X}$  or  $\mathbf{\Lambda}$ , and restricted least squares has a closed form through any generalized inverse (Definition 11.3.1, Proposition 11.3.1, Theorem 11.3.2, Proposition 11.3.3).
4. The  $F$  test is the likelihood ratio test. The Wald, likelihood ratio and score statistics are  $nu$ ,  $n \log(1 + u)$  and  $nu/(1 + u)$  with  $u = qF/(n - r)$ , so they always satisfy Wald > LR > Score. Calibrated by  $\chi^2$ , they disagree and have inflated sizes in small samples (Theorem 11.4.1, Proposition 11.4.2).
5. The noncentral  $F$  family has monotone likelihood ratio, and every test invariant under the symmetries of the problem is a function of  $F$ . Hence the  $F$  test is uniformly most powerful among invariant tests (Lemma 11.4.3, Theorem 11.4.4).
6. Power depends only on  $\gamma$ , increases with it, and grows with replication. In the one-way layout the least favourable configuration for a given range puts two means at the extremes and the rest at the midpoint. Observed power is a function of the p-value and adds nothing to it (Theorem 11.5.1, Proposition 11.5.2, Proposition 11.5.3).
7. For one estimable function the  $F$  test is the two-sided  $t$  test, and the one-sided version has exact size for one-sided hypotheses. A group statistic  $qF$  is the largest squared  $t$  statistic over all combinations in the group, which explains why group and individual tests can disagree in both directions (Theorem 11.6.1, Proposition 11.6.2).

## 11.7.2 Notes and sources

**Origins.** The ratio of mean squares as a test statistic is Fisher's. It grew out of his analysis of variance in the early 1920s, and Fisher (1925) tabulated it on the logarithmic scale, as  $z = \frac{1}{2} \log F$ . Snedecor (1934) introduced the letter  $F$  in Fisher's honour and published tables of the ratio itself. The exact  $t$  distribution of a regression coefficient, with  $n - p$  degrees of freedom, is also due to Fisher (1922). The geometry of [Section 11.2](#) follows the coordinate-free tradition described in the notes to [Chapter 6](#). Christensen (2020, chapter 3) develops testing entirely in that language, with test spaces, offsets and nonestimable constraints. Rencher and Schaalje (2008, chapters 8 and 12) and Seber and Lee (2003, chapter 4) give the matrix treatment for full-rank and rank-deficient models, including the canonical elimination of [Exercise \(C2\)](#). Sen and Srivastava (1990, chapter 3) and Agresti (2015, sections 3.1–3.2) emphasize applied examples. Note that Rencher and Schaalje and Christensen define the noncentrality as half of our  $\gamma$  ([Section 4.1](#)).

**Testable hypotheses.** The insistence that only estimable functions can be tested goes back at least to Searle (1971), whose treatment of “testable hypotheses” in models not of full rank is the standard reference; [Proposition 11.3.1](#) is a minimal version of his argument. Christensen (2020, section 3.3) shows how a nonestimable constraint reduces to its estimable part, which in our treatment is [Proposition 9.2.6](#).

**Likelihood ratio, Wald and score.** Neyman and Pearson (1933) introduced the most powerful test and the likelihood ratio principle for composite hypotheses. The large-sample  $\chi^2$  calibration of the likelihood ratio is due to Wilks (1938), the Wald statistic to Wald (1943), and the score statistic to Rao (1948). The inequality  $\text{Wald} \geq \text{LR} \geq \text{score}$  for linear restrictions in normal regression, and the resulting possibility of conflicting verdicts, were pointed out by Berndt and Savin (1977) and discussed further by Breusch (1979). The “ $nR^2$ ” form of the score test in [Exercise \(B2\)](#) is widely used in econometrics.

**Invariance.** Lehmann and Romano (2005) develop the theory of invariant tests and prove that the  $F$  test is uniformly most powerful invariant; [Theorem 11.4.4](#) follows their approach in outline. Scheffé (1959) treats the power of  $F$  tests in the analysis of variance. The robustness of the null distribution to spherically symmetric errors ([Proposition 11.2.4](#)) is part of the theory of spherical and elliptical distributions surveyed by Fang, Kotz and Ng (1990).

**Power.** The noncentral  $F$  distribution was tabulated for power calculations by Tang (1938). The least favourable configuration of [Proposition 11.5.2](#) is the basis of the classical sample-size tables for the analysis of variance. Cohen (1988) popularized effect-size based power analysis. Hoenig and Heisey (2001) explain why “observed power” computed from the data is uninformative, the point of [Proposition 11.5.3](#). Wasserstein and Lazar (2016) discuss common misreadings of p-values. Equivalence tests, which can support a claim of “no important difference”, go back to Schuirmann (1987).

**Other.** The test of equality of two regressions in [Exercise \(B4\)](#) and [Exercise \(C1\)](#) is due to Chow (1960). Longley (1967) assembled his data to test the numerical accuracy of regression programs ([Chapter 10](#)). Monotonicity properties of the  $\chi^2$ ,  $F$  and  $t$  distributions in their degrees of freedom, of the kind behind the dilution effect of [Exercise \(B2\)](#), are studied systematically by Ghosh (1973).

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# Chapter 12

## Confidence Intervals, Regions and Prediction Bands

A test asks whether one value of a parameter is compatible with the data; an interval reports all such values, and how precisely they are pinned down. Under normal errors the answers are exact for every sample size, and they come from three facts: the fitted vector is normal, the residual sum of squares is an independent chi-squared variable, and their ratio gives  $t$  and  $F$  pivots. One estimable function gives a  $t$  interval; several give an ellipsoid, the dual of the  $F$  test. A future observation gives a prediction interval, whose width never falls below the noise and whose validity rests on the normality of a single error. The whole regression surface gives the Working–Hotelling band, from the Cauchy–Schwarz inequality. Inverting a prediction band gives a calibration set, sometimes an interval, sometimes two rays, sometimes the whole line, and no honest procedure can avoid the last two. Finally, in the conjugate model of [Chapter 7](#), credible and predictive intervals have the same shapes, and under the flat prior they are the frequentist ones.

**What you need.** The sampling distributions of the normal linear model ([Chapter 7: Theorem 7.5.1](#), [Corollary 7.5.2](#) and [Corollary 7.5.3](#)) and the conjugate posterior ([Theorem 7.6.2](#), [Lemma 7.6.1](#), [Corollary 7.6.4](#)). Projections, leverage and the Frisch–Waugh–Lovell theorem ([Chapter 6: Theorem 6.3.7](#), [Proposition 6.8.1](#), [Theorem 6.6.2](#)). Estimability ([Chapter 8: Definition 8.2.1](#), [Theorem 8.3.1](#)) and the sum of squares of a linear hypothesis ([Chapter 9: Theorem 9.2.4](#)). Chi-squared and  $F$  distributions of quadratic forms ([Chapter 4: Theorem 4.3.2](#), [Definition 4.1.2](#), [Definition 4.1.3](#)). The tests of [Chapter 11](#) are the duals of the intervals here, but the proofs below do not depend on them.

### 12.0.1 Roadmap

- [Intervals for estimable functions](#)
- [Confidence ellipsoids](#)
- [Prediction intervals](#)
- [Simultaneous bands for the regression line and surface](#)
- [Inverse prediction and calibration](#)
- [Bayesian thread II: credible intervals and predictive distributions](#)
- [Summary and notes](#)

## 12.1 Intervals for estimable functions

A point estimate needs a statement of its precision. A confidence interval gives one exactly: a range computed from the data that contains the unknown quantity with a probability fixed in advance. This section builds the interval for one estimable function, relates it to the  $t$  test, and records two facts that are easy to overlook: an interval for a non-estimable function is useless, and the interval for  $\sigma^2$ , unlike the one for a coefficient, depends delicately on normality.

Throughout the chapter we work in the normal linear model

$$\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}), \quad \boldsymbol{\beta} \in \mathbb{R}^p, \sigma^2 > 0,$$

of Eq. 7.3.1, with  $r = \text{rank}(\mathbf{X}) < n$ . We write  $\mathbf{G}$  for any generalized inverse of  $\mathbf{X}^\top \mathbf{X}$ ,  $\hat{\boldsymbol{\beta}}$  for any least squares estimate,  $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$  and  $s^2 = \text{SSE}/(n - r)$ . For  $0 < \gamma < 1$ ,  $t_{\nu, \gamma}$ ,  $F_\gamma(q, \nu)$  and  $\chi_\gamma^2(\nu)$  denote the upper  $\gamma$  points of  $t(\nu)$ ,  $F(q, \nu)$  and  $\chi^2(\nu)$ .

### 12.1.1 Confidence sets

#### Definition 12.1.1 — Confidence set

Let  $\theta = g(\boldsymbol{\beta}, \sigma^2)$  be a quantity with values in a set  $\Theta$ . A **confidence set** for  $\theta$  with **confidence level**  $1 - \alpha$  is a rule that assigns to each data vector  $\mathbf{y}$  a subset  $C(\mathbf{y}) \subseteq \Theta$ , such that

$$\Pr_{\boldsymbol{\beta}, \sigma^2} \{\theta \in C(\mathbf{Y})\} \geq 1 - \alpha$$

for every  $\boldsymbol{\beta} \in \mathbb{R}^p$  and  $\sigma^2 > 0$ . The left side is the **coverage probability** at  $(\boldsymbol{\beta}, \sigma^2)$ . The set is **exact** if the coverage equals  $1 - \alpha$  at every parameter value. When  $C(\mathbf{y})$  is an interval it is a **confidence interval**.

The probability is taken over  $\mathbf{Y}$  with the parameters fixed: the set is random and the target is not. A realized interval either contains  $\theta$  or it does not;  $1 - \alpha$  describes the procedure.

The standard way to build an exact set is a **pivot**: a function  $Q(\mathbf{Y}, \theta)$  whose distribution is the same at every parameter value. If  $\Pr\{Q(\mathbf{Y}, \theta) \in A\} = 1 - \alpha$  for a fixed set  $A$ , then  $C(\mathbf{y}) = \{\theta_0 : Q(\mathbf{y}, \theta_0) \in A\}$  is exact, because the events  $\{\theta \in C(\mathbf{Y})\}$  and  $\{Q(\mathbf{Y}, \theta) \in A\}$  are the same event. All the sets of this chapter are built this way, from pivots supplied by Chapter 7.

### 12.1.2 The $t$ interval

#### Theorem 12.1.1 — The $t$ interval for an estimable function

Let  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ ,  $\boldsymbol{\lambda} \neq \mathbf{0}$ , and put  $\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = s\sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}}$ . Let

$$I(\mathbf{Y}) = [\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - t_{n-r, \alpha/2} \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}), \quad \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + t_{n-r, \alpha/2} \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})]. \quad (12.1.1)$$

- (a)  $I(\mathbf{Y})$  does not depend on the choice of  $\hat{\boldsymbol{\beta}}$  or of  $\mathbf{G}$ .
- (b)  $I(\mathbf{Y})$  is an exact  $1 - \alpha$  confidence interval for  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ .

- (c) Its length  $L = 2t_{n-r,\alpha/2} s \sqrt{\lambda^\top \mathbf{G} \lambda}$  has expectation  $E L = 2t_{n-r,\alpha/2} \kappa_{n-r} \sigma \sqrt{\lambda^\top \mathbf{G} \lambda}$ , where

$$\kappa_\nu = E \sqrt{V/\nu} = \sqrt{\frac{2}{\nu}} \frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)} < 1, \quad V \sim \chi^2(\nu).$$

- (d) The one-sided intervals  $(-\infty, \lambda^\top \hat{\beta} + t_{n-r,\alpha} \text{se}]$  and  $[\lambda^\top \hat{\beta} - t_{n-r,\alpha} \text{se}, \infty)$  are exact  $1 - \alpha$  confidence intervals for  $\lambda^\top \beta$ .

*Proof.*

- (a) Write  $\lambda = \mathbf{X}^\top \rho$ . Then  $\lambda^\top \hat{\beta} = \rho^\top \mathbf{X} \hat{\beta} = \rho^\top \mathbf{M} \mathbf{Y}$  and  $\lambda^\top \mathbf{G} \lambda = \rho^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \rho = \rho^\top \mathbf{M} \rho$  by Theorem 6.3.7. Neither involves the choices, and neither does  $s$ .
- (b) By Corollary 7.5.2,  $T = (\lambda^\top \hat{\beta} - \lambda^\top \beta) / \text{se}(\lambda^\top \hat{\beta}) \sim t(n-r)$  at every  $(\beta, \sigma^2)$ . The event  $\lambda^\top \beta \in I(\mathbf{Y})$  is the event  $|T| \leq t_{n-r,\alpha/2}$ , whose probability is  $1 - \alpha$ . So part (b) restates Eq. 7.5.1 as a confidence statement.
- (c)  $L$  is a constant multiple of  $s = \sigma \sqrt{V/(n-r)}$  with  $V = \text{SSE}/\sigma^2 \sim \chi^2(n-r)$  (Theorem 7.5.1(b)). With the  $\chi^2(\nu)$  density of Eq. 4.1.1,

$$E \sqrt{V} = \int_0^\infty \frac{v^{1/2} v^{\nu/2-1} e^{-v/2}}{2^{\nu/2} \Gamma(\nu/2)} dv = \frac{2^{(\nu+1)/2} \Gamma((\nu+1)/2)}{2^{\nu/2} \Gamma(\nu/2)},$$

because the integrand is a multiple of the  $\chi^2(\nu+1)$  density. Dividing by  $\sqrt{\nu}$  gives  $\kappa_\nu$ . By Jensen's inequality  $E \sqrt{V/\nu} < \sqrt{E(V/\nu)} = 1$ , with strict inequality since  $V$  is not constant.

- (d) The events are  $\{T \geq -t_{n-r,\alpha}\}$  and  $\{T \leq t_{n-r,\alpha}\}$ , each of probability  $1 - \alpha$ .

■

Part (a) matters when  $\mathbf{X}$  is rank deficient: the interval is a property of the model, not of the software's choice of generalized inverse. The degrees of freedom are  $n - r$ , not  $n - p$ .

Part (c) shows what the interval costs. Its expected length is proportional to  $\sigma \sqrt{\lambda^\top \mathbf{G} \lambda}$ , the standard deviation of the estimate. Estimating  $\sigma$  adds the factor  $t_{n-r,\alpha/2} \kappa_{n-r}$  in place of the normal point  $z_{\alpha/2}$  that would be used if  $\sigma$  were known. For  $\alpha = 0.05$  that factor is 2.45 at  $n - r = 5$  and 2.06 at  $n - r = 20$ , against 1.96. Beyond about twenty residual degrees of freedom, precision improves only through  $\lambda^\top \mathbf{G} \lambda$ , that is, through the design.

### Example. (Murder rates: coefficients and a mean)

Return to the regression of the 2009 murder rate of the 50 states on the poverty rate, the percentage of single-parent households and the percentage urban, with an intercept (Example (Poverty and murder rates)). Here  $n = 50$ ,  $r = 4$ ,  $s^2 = 2.217$ , and  $t_{46,0.025} = 2.013$ . The 95% intervals for the coefficients are

| Coefficient   | Estimate | Standard error | 95% interval      |
|---------------|----------|----------------|-------------------|
| intercept     | -9.0946  | 1.5555         | (-12.226, -5.964) |
| poverty       | 0.2538   | 0.0934         | (0.066, 0.442)    |
| single parent | 0.3980   | 0.0829         | (0.231, 0.565)    |
| urban         | 0.0046   | 0.0131         | (-0.022, 0.031)   |

The data are consistent with anything from 0.066 to 0.442 murders per 100,000 for each point of poverty, with the other regressors held fixed in the sense of [Theorem 6.6.2](#). The urban interval straddles zero but is narrow, which says more than “not significant”: any such association is at most 0.031 per percentage point.

A mean response is an estimable function too. For a hypothetical state with poverty 18%, single-parent households 28% and 70% urban,  $\mathbf{x}_0 = (1, 18, 28, 70)^\top$ , the estimated mean murder rate is 6.937 with standard error 0.412, and the 95% interval is (6.107, 7.767). Here  $\mathbf{x}_0^\top \mathbf{G} \mathbf{x}_0 = 0.0767$ . [Section 12.3](#) interprets this number as the leverage of the new point.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statelcrime.load_pandas().data
data = data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
r = np.linalg.matrix_rank(X)
G = np.linalg.pinv(X.T @ X)                    # any generalized inverse will do
beta_hat = G @ X.T @ y
s2 = np.sum((y - X @ beta_hat) ** 2) / (n - r)
q = stats.t.ppf(0.975, n - r)                  # upper 2.5% point of t(n - r)

def t_interval(lam):
    """95% interval for the estimable function lam^T beta."""
    est = lam @ beta_hat
    se = np.sqrt(s2 * lam @ G @ lam)
    return est, se, est - q * se, est + q * se

for j, name in enumerate(["intercept", "poverty", "single", "urban"]):
    est, se, lo, hi = t_interval(np.eye(p)[j])
    print(f"{name:9s} {est:8.4f} se {se:.4f} 95% interval ({lo:.3f}, {hi:.3f})")
```

## 12.1.3 What the standard error measures

For a coefficient in a full-rank model with an intercept, [Eq. 6.6.2](#) gives

$$\text{se}(\hat{\beta}_j) = \frac{s}{\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|} = \frac{s}{\sqrt{S_{jj}(1 - R_j^2)}},$$

where  $\mathbf{M}_{(j)}$  projects onto the span of the other columns. The interval is short when  $\mathbf{x}_j$  has a large component orthogonal to the other regressors; collinearity lengthens it through  $1 - R_j^2$ .

In general  $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda} = \|\mathbf{M} \boldsymbol{\rho}\|^2$ , with  $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$ , is the smallest squared length of a vector  $\mathbf{a}$  with  $E(\mathbf{a}^\top \mathbf{Y}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$  ([Theorem 7.2.1](#)).

## 12.1.4 Duality with tests

An interval and a family of tests carry the same information, in any model.

### Proposition 12.1.2 — Tests and confidence sets

For each  $\theta_0 \in \Theta$ , let  $A(\theta_0)$  be the acceptance region of a test of  $H : \theta = \theta_0$  with  $\Pr_{\beta, \sigma^2}\{\mathbf{Y} \in A(\theta_0)\} \geq 1 - \alpha$  whenever  $g(\beta, \sigma^2) = \theta_0$ . Then

$$C(\mathbf{y}) = \{\theta_0 \in \Theta : \mathbf{y} \in A(\theta_0)\}$$

is a  $1 - \alpha$  confidence set for  $\theta$ , and it is exact if every test has size exactly  $\alpha$  at every parameter value in its hypothesis. Conversely, if  $C$  is a  $1 - \alpha$  confidence set, the rule “reject  $H : \theta = \theta_0$  iff  $\theta_0 \notin C(\mathbf{y})$ ” is a level- $\alpha$  test for every  $\theta_0$ .

*Proof.* For every parameter value, with  $\theta = g(\beta, \sigma^2)$ , the events  $\{\theta \in C(\mathbf{Y})\}$  and  $\{\mathbf{Y} \in A(\theta)\}$  coincide, so they have the same probability. Both directions follow. ■

For an estimable function, the two-sided  $t$  test of  $\lambda^\top \beta = d$  accepts iff  $|\lambda^\top \hat{\beta} - d| \leq t_{n-r, \alpha/2} \text{se}$ , which is the condition  $d \in I(\mathbf{y})$ . So the interval Eq. 12.1.1 is exactly the set of values  $d$  that the  $t$  test of Theorem 11.6.1 does not reject. The  $p$ -value of the test of  $d$  is the smallest  $\alpha$  for which the  $1 - \alpha$  interval excludes  $d$ .

The interval is the more informative report. A test of  $\beta_j = 0$  reports one bit, confounded with the sample size; the interval shows location and resolution. Exercise (B2) uses it to answer “is the effect negligible?”, which a test of  $\beta_j = 0$  cannot answer at all.

## 12.1.5 Functions that are not estimable

What if  $\lambda \notin \mathcal{C}(\mathbf{X}^\top)$ ? The recipe above fails because  $\lambda^\top \hat{\beta}$  depends on the choice of  $\hat{\beta}$ . The failure is not a defect of the recipe. No procedure can do better.

### Proposition 12.1.3 — No useful interval for a non-estimable function

Let  $\lambda \notin \mathcal{C}(\mathbf{X}^\top)$  and let  $C$  be a  $1 - \alpha$  confidence set for  $\lambda^\top \beta$  with  $\alpha < 1$ . Then at every parameter value  $(\beta, \sigma^2)$  and for every real  $v$ ,  $\Pr_{\beta, \sigma^2}\{v \in C(\mathbf{Y})\} \geq 1 - \alpha$ . If the sets  $C(\mathbf{y})$  are such that  $\{(\mathbf{y}, v) : v \in C(\mathbf{y})\}$  is measurable, the expected length of  $C(\mathbf{Y})$  is infinite at every parameter value.

*Proof.* By Theorem 8.3.1 there is  $\mathbf{w} \in \mathcal{N}(\mathbf{X})$  with  $\lambda^\top \mathbf{w} \neq 0$ . Fix  $(\beta, \sigma^2)$  and  $v$ , and put  $\beta' = \beta + c\mathbf{w}$  with  $c = (v - \lambda^\top \beta)/\lambda^\top \mathbf{w}$ . Then  $\mathbf{X}\beta' = \mathbf{X}\beta$ , so  $\mathbf{Y}$  has the same distribution under  $(\beta', \sigma^2)$  as under  $(\beta, \sigma^2)$ , while  $\lambda^\top \beta' = v$ . Hence  $\Pr_{\beta, \sigma^2}\{v \in C(\mathbf{Y})\} =$

$\Pr_{\beta', \sigma^2} \{\lambda^\top \beta' \in C(\mathbf{Y})\} \geq 1 - \alpha$ . For the length, Fubini's theorem gives, for every  $a > 0$ ,

$$E[\text{length}(C(\mathbf{Y}) \cap [-a, a])] = \int_{-a}^a \Pr\{v \in C(\mathbf{Y})\} dv \geq 2a(1 - \alpha),$$

and the left side is at most the expected length of  $C(\mathbf{Y})$ . Let  $a \rightarrow \infty$ . ■

Thus the algebraic fact of [Chapter 8](#) becomes a statement about inference.

## 12.1.6 An interval for $\sigma^2$

### Proposition 12.1.4 — Interval for the error variance

The interval

$$\left[ \frac{\text{SSE}}{\chi_{\alpha/2}^2(n-r)}, \frac{\text{SSE}}{\chi_{1-\alpha/2}^2(n-r)} \right]$$

is an exact  $1 - \alpha$  confidence interval for  $\sigma^2$ , and the square roots of its endpoints give one for  $\sigma$ .

*Proof.*  $\text{SSE}/\sigma^2 \sim \chi^2(n-r)$  at every parameter value ([Theorem 7.5.1\(b\)](#)), so it is a pivot. The event  $\chi_{1-\alpha/2}^2(n-r) \leq \text{SSE}/\sigma^2 \leq \chi_{\alpha/2}^2(n-r)$  has probability  $1 - \alpha$ , and it is the event that  $\sigma^2$  lies in the interval, which is the interval of [Exercise \(B3\)](#). Taking square roots is monotone. ■

For the murder-rate regression,  $\text{SSE} = 46s^2$  with  $s^2 = 2.217$ , the chi-squared points are 29.160 and 66.617, and the 95% interval for  $\sigma^2$  is (1.531, 3.497). The interval is not symmetric about  $s^2$ , and it is not the shortest one with this coverage ([Exercise \(C2\)](#)).

The two pivots of this section react differently to non-normal errors. The  $t$  statistic needs normality of a weighted average of errors, which a central limit theorem restores, and  $s \rightarrow \sigma$ , which holds for any error law. The chi-squared pivot needs the whole error distribution, because the variance of SSE depends on the fourth moment ([Proposition 7.5.4](#)). Take the fitted murder-rate model as the truth, with  $\sigma^2 = s^2$ , and simulate 40000 data sets. With normal errors the two 95% intervals cover in proportions 0.9514 (poverty coefficient) and 0.9491 ( $\sigma^2$ ). With errors from a  $t(5)$  distribution rescaled to the same variance, the coefficient interval still covers 0.9498 of the time, but the interval for  $\sigma^2$  covers only 0.8165. More data do not repair this ([Exercise \(B3\)](#)).

```
x0 = np.array([1.0, 18.0, 28.0, 70.0])           # a state with high poverty and
single_parenthood
est0, se0, lo0, hi0 = t_interval(x0)
print(f"mean murder rate at x0: {est0:.3f}, se {se0:.3f}, interval ({lo0:.3f}, {hi0:.3f})")

sse = s2 * (n - r)
lo_s2 = sse / stats.chi2.ppf(0.975, n - r)       # interval for sigma^2 from SSE/
sigma^2 ~ chi^2(n-r)
```

```
hi_s2 = sse / stats.chi2.ppf(0.025, n - r)
print(f"s^2 = {s2:.3f}; 95% interval for sigma^2 ({lo_s2:.3f}, {hi_s2:.3f})")
```

## 12.1.7 Exercises

### 12.1.7.1 A. Check your understanding

#### Exercise A1.

A design is replicated  $m$  times:  $\mathbf{X}$  is replaced by  $m$  stacked copies of itself. For a fixed estimable  $\lambda^\top \beta$ , how does  $\lambda^\top \mathbf{G} \lambda$  change? Roughly how many replicates are needed to halve the expected length of the 95% interval when  $n - r$  is already large, and why is the answer slightly less than 4 when  $n - r$  is small?

*Solution.* The stacked matrix has  $\mathbf{X}^\top \mathbf{X}$  replaced by  $m\mathbf{X}^\top \mathbf{X}$ , and  $\mathbf{G}/m$  is a generalized inverse of  $m\mathbf{X}^\top \mathbf{X}$ , so  $\lambda^\top \mathbf{G} \lambda$  is divided by  $m$ . By [Theorem 12.1.1\(c\)](#) the expected length is proportional to  $t_{\nu, \alpha/2} \kappa_\nu / \sqrt{m}$ , with  $\nu = mn - r$ . For large  $\nu$  the factor  $t\kappa$  is nearly constant and  $m = 4$  halves the length; for small  $\nu$ , replication also lowers  $t\kappa$ , so slightly less suffices.

### 12.1.7.2 B. Practice

#### Exercise B1.

Show that  $\kappa_\nu \rightarrow 1$  as  $\nu \rightarrow \infty$ , and that  $s/\kappa_{n-r}$  is an unbiased estimator of  $\sigma$ . Compute  $\kappa_\nu$  for  $\nu = 1$  and  $\nu = 2$ .

*Solution.*  $V/\nu \rightarrow 1$  in probability by the law of large numbers, and  $\sqrt{V/\nu}$  is bounded in mean square, hence uniformly integrable, so  $E \sqrt{V/\nu} \rightarrow 1$ . Unbiasedness:  $E s = \sigma E \sqrt{V/(n-r)} = \sigma \kappa_{n-r}$ . For  $\nu = 1$ ,  $\kappa_1 = \sqrt{2} \Gamma(1)/\Gamma(1/2) = \sqrt{2/\pi} \approx 0.798$ . For  $\nu = 2$ ,  $\kappa_2 = \Gamma(3/2)/\Gamma(1) = \sqrt{\pi}/2 \approx 0.886$ .

#### Exercise B2.

Let  $\delta > 0$  be a margin below which  $|\lambda^\top \beta|$  is considered negligible. The *two one-sided tests* procedure declares equivalence when both  $H_1 : \lambda^\top \beta \leq -\delta$  and  $H_2 : \lambda^\top \beta \geq \delta$  are rejected by one-sided  $t$  tests at level  $\alpha$ . Show that this happens iff the  $1 - 2\alpha$  interval [Eq. 12.1.1](#) lies inside  $(-\delta, \delta)$ . Show that the procedure has level  $\alpha$ : when  $|\lambda^\top \beta| \geq \delta$ , the probability of declaring equivalence is at most  $\alpha$ .

*Solution.*  $H_1$  is rejected iff  $(\lambda^\top \hat{\beta} + \delta)/se > t_{n-r, \alpha}$ , that is iff the lower end of the  $1 - 2\alpha$  interval,  $\lambda^\top \hat{\beta} - t_{n-r, \alpha} se$ , exceeds  $-\delta$ . Symmetrically  $H_2$  is rejected iff the upper end is below  $\delta$ . For the level: if  $\lambda^\top \beta \leq -\delta$ , declaring equivalence requires rejecting  $H_1$ . Write  $\tau = \sigma \sqrt{\lambda^\top \mathbf{G} \lambda}$ ,  $Z = (\lambda^\top \hat{\beta} - \lambda^\top \beta)/\tau \sim N(0, 1)$  and  $V = SSE/\sigma^2 \sim \chi^2(\nu)$ ,  $\nu = n - r$ , independent of  $Z$  ([Theorem 7.5.1](#)). The test statistic is  $(Z + \eta)/\sqrt{V/\nu}$  with  $\eta = (\lambda^\top \beta + \delta)/\tau$ . For every realization of  $(Z, V)$  it increases in  $\eta$ , so the probability that it exceeds  $t_{\nu, \alpha}$  increases in  $\eta$ . At  $\eta = 0$  that probability is  $\alpha$ , so it is at most  $\alpha$  whenever



$\lambda^\top \beta \leq -\delta$ . The case  $\lambda^\top \beta \geq \delta$  is symmetric.

### Exercise B3.

Suppose  $\mathbf{Y} \sim N_n(\mathbf{X}\beta, \sigma^2\mathbf{V})$  with  $\mathbf{V}$  known and positive definite and  $\mathbf{X}$  of full column rank. Using the generalized least squares estimator  $\tilde{\beta} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$  and  $\tilde{s}^2 = (\mathbf{Y} - \mathbf{X}\tilde{\beta})^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X}\tilde{\beta}) / (n - p)$ , derive an exact  $1 - \alpha$  interval for  $\lambda^\top \beta$ . Write it out for a straight line  $E Y_i = \beta_0 + \beta_1 x_i$  with  $\mathbf{V} = \text{diag}(1/w_i)$ .

*Solution.* Let  $\mathbf{V} = \mathbf{L}\mathbf{L}^\top$  (Cholesky) and  $\mathbf{Z} = \mathbf{L}^{-1}\mathbf{Y} \sim N_n(\mathbf{L}^{-1}\mathbf{X}\beta, \sigma^2\mathbf{I})$ . The ordinary least squares estimate for  $(\mathbf{Z}, \mathbf{L}^{-1}\mathbf{X})$  is  $\tilde{\beta}$ , and its residual sum of squares is  $(n - p)\tilde{s}^2$ . Applying [Theorem 12.1.1](#) to the transformed model gives  $\lambda^\top \tilde{\beta} \pm t_{n-p, \alpha/2} \tilde{s} \sqrt{\lambda^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \lambda}$ . For the weighted straight line, with  $\bar{x}_w = \sum w_i x_i / \sum w_i$ ,  $\bar{y}_w$  defined likewise and  $S_w = \sum w_i (x_i - \bar{x}_w)^2$ , the slope estimate is  $\tilde{\beta}_1 = \sum w_i (x_i - \bar{x}_w)(y_i - \bar{y}_w) / S_w$  and its interval is  $\tilde{\beta}_1 \pm t_{n-2, \alpha/2} \tilde{s} / \sqrt{S_w}$ .

### Exercise B4.

In a one-way layout with  $a$  groups of sizes  $n_1, \dots, n_a$  and the overparameterized model  $E Y_{ij} = \mu + \alpha_i$ , show that the interval for  $\alpha_1 - \alpha_2$  is  $\bar{y}_1 - \bar{y}_2 \pm t_{n-a, \alpha/2} s \sqrt{1/n_1 + 1/n_2}$ , with  $n = \sum n_i$ . What does [Proposition 12.1.3](#) say about an interval for  $\alpha_1$  alone?

*Solution.* The rank is  $r = a$ . With  $\lambda = \mathbf{e}_2 - \mathbf{e}_3$  (coordinates  $\mu, \alpha_1, \dots, \alpha_a$ ),  $\lambda^\top \beta = \alpha_1 - \alpha_2$  is estimable, and  $\lambda^\top \hat{\beta} = \bar{y}_1 - \bar{y}_2$  for every least squares estimate. Its variance is  $\sigma^2(1/n_1 + 1/n_2)$  since the group means are independent, so  $\lambda^\top \mathbf{G}\lambda = 1/n_1 + 1/n_2$ . Substitute in [Eq. 12.1.1](#) with  $n - r = n - a$ . The function  $\alpha_1$  is not estimable (adding  $c$  to every  $\alpha_i$  and subtracting it from  $\mu$  leaves  $\mathbf{X}\beta$  unchanged), so every 95% confidence set for it covers every real number with probability at least 0.95.

### 12.1.7.3 C. Going deeper

#### Exercise C1.

Among intervals  $[\lambda^\top \hat{\beta} - a \text{ se}, \lambda^\top \hat{\beta} + b \text{ se}]$  with  $a, b \geq 0$  and coverage exactly  $1 - \alpha$ , show that the expected length is smallest when  $a = b = t_{n-r, \alpha/2}$ . (Use that the  $t$  density is symmetric and strictly decreasing in  $|x|$ .)

*Solution.* Coverage is  $F(b) - F(-a)$ , with  $F$  the  $t(n - r)$  distribution function, and the expected length is proportional to  $a + b$ . If  $a < b$ , shifting  $[-a, b]$  slightly left gains density  $f(a)$  and loses  $f(b) < f(a)$ , raising the coverage, so a strictly shorter interval has coverage  $1 - \alpha$ . Hence  $a = b$ .

#### Exercise C2.

Intervals for  $\sigma^2$  of the form  $[\text{SSE}/c_2, \text{SSE}/c_1]$  with  $\Pr(c_1 \leq \chi^2(\nu) \leq c_2) = 1 - \alpha$  all have exact coverage. Show that the one with the smallest expected length satisfies  $c_1^2 f(c_1) = c_2^2 f(c_2)$ , where  $f$  is the  $\chi^2(\nu)$  density, and that the equal-tailed choice of

[Proposition 12.1.4](#) does not satisfy this condition.

## 12.2 Confidence ellipsoids

Two 95% intervals, one for each of two coefficients, do not make a 95% statement about the pair, and they ignore the correlation between the estimates. The natural joint region is an ellipsoid: the set of values the  $F$  test does not reject, shaped like the covariance matrix of the estimates, with shadows that are wider than the  $t$  intervals. This geometry explains an  $F$  test that rejects while every  $t$  test accepts, and the reverse.

### 12.2.1 The ellipsoid

Let  $\mathbf{\Lambda}$  be a  $p \times q$  matrix of rank  $q$  whose columns lie in  $\mathcal{C}(\mathbf{X}^\top)$ , so that every entry of  $\boldsymbol{\phi} = \mathbf{\Lambda}^\top \boldsymbol{\beta}$  is estimable. Put

$$\hat{\boldsymbol{\phi}} = \mathbf{\Lambda}^\top \hat{\boldsymbol{\beta}}, \quad \mathbf{W} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}.$$

By [Theorem 9.2.4\(a\)](#), neither depends on the choices of  $\hat{\boldsymbol{\beta}}$  and  $\mathbf{G}$ , and  $\mathbf{W}$  is positive definite. By [Theorem 7.5.1\(a\)](#),  $\hat{\boldsymbol{\phi}} \sim N_q(\boldsymbol{\phi}, \sigma^2 \mathbf{W})$ .

#### Theorem 12.2.1 — Confidence ellipsoid for $\mathbf{\Lambda}^\top \boldsymbol{\beta}$

With the notation above, let  $F_\alpha = F_\alpha(q, n - r)$  and

$$E(\mathbf{Y}) = \{\boldsymbol{\phi}_0 \in \mathbb{R}^q : (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0)^\top \mathbf{W}^{-1} (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0) \leq q s^2 F_\alpha\}. \quad (12.2.1)$$

- (a)  $E(\mathbf{Y})$  is an exact  $1 - \alpha$  confidence set for  $\mathbf{\Lambda}^\top \boldsymbol{\beta}$ .
- (b)  $E(\mathbf{Y})$  is a solid ellipsoid centred at  $\hat{\boldsymbol{\phi}}$ . If  $\mathbf{W} = \sum_{k=1}^q w_k \mathbf{u}_k \mathbf{u}_k^\top$  is a spectral decomposition, its principal axes point along  $\mathbf{u}_1, \dots, \mathbf{u}_q$  and have half-lengths  $\sqrt{q s^2 F_\alpha w_k}$ . Its volume is

$$\frac{\pi^{q/2}}{\Gamma(q/2 + 1)} (q s^2 F_\alpha)^{q/2} \det(\mathbf{W})^{1/2}.$$

- (c) (*Duality with the  $F$  test.*) For  $\mathbf{d} \in \mathbb{R}^q$ , let  $\text{SSE}_{\mathbf{d}}$  be the minimum of  $\|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2$  over  $\{\mathbf{b} : \mathbf{\Lambda}^\top \mathbf{b} = \mathbf{d}\}$  and let

$$F_{\mathbf{d}} = \frac{(\text{SSE}_{\mathbf{d}} - \text{SSE})/q}{s^2} = \frac{(\hat{\boldsymbol{\phi}} - \mathbf{d})^\top \mathbf{W}^{-1} (\hat{\boldsymbol{\phi}} - \mathbf{d})}{q s^2}.$$

Then  $\mathbf{d} \in E(\mathbf{Y})$  iff  $F_{\mathbf{d}} \leq F_\alpha$ , that is, iff the level- $\alpha$   $F$  test of  $H : \mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  does not reject.

*Proof.*

- (a) Since  $\mathbf{W}$  is invertible,  $\mathbf{W}^{-1}$  is its only generalized inverse, and [Corollary 7.5.3](#) with

$\text{rank}(\mathbf{\Lambda}) = q$  gives

$$\frac{(\hat{\boldsymbol{\Phi}} - \boldsymbol{\Phi})^\top \mathbf{W}^{-1}(\hat{\boldsymbol{\Phi}} - \boldsymbol{\Phi})}{q s^2} \sim F(q, n - r)$$

at every parameter value. This is a pivot, and  $\boldsymbol{\Phi} \in E(\mathbf{Y})$  is the event that it is at most  $F_\alpha$ .

- (b) Put  $c = q s^2 F_\alpha$  and write  $\boldsymbol{\Phi}_0 - \hat{\boldsymbol{\Phi}} = \sum_k c_k \mathbf{u}_k$ . Since  $\mathbf{W}^{-1} = \sum_k w_k^{-1} \mathbf{u}_k \mathbf{u}_k^\top$ , the defining inequality becomes  $\sum_k c_k^2 / w_k \leq c$ , the standard form of an ellipsoid with half-axes  $\sqrt{c w_k}$  along  $\mathbf{u}_k$ . Equivalently  $E(\mathbf{Y}) = \hat{\boldsymbol{\Phi}} + \mathbf{W}^{1/2} B$ , where  $B$  is the ball of radius  $\sqrt{c}$  in  $\mathbb{R}^q$ . A linear map multiplies volume by the absolute value of its determinant, so the volume is  $\det(\mathbf{W})^{1/2} c^{q/2}$  times the volume  $\pi^{q/2} / \Gamma(q/2 + 1)$  of the unit ball.
- (c) The two expressions for  $F_d$  agree by [Theorem 9.2.4\(c\)](#), and the equivalence is the definition of  $E(\mathbf{Y})$ . ■

The statistic  $F_d$  is the  $F$  statistic of the general linear hypothesis in both of its forms (see [Theorem 11.3.2](#)), and part (c) is [Proposition 12.1.2](#) in concrete form. In particular the test of  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$  rejects iff the ellipsoid misses the origin.

In the full-rank case with  $\mathbf{\Lambda} = \mathbf{I}_p$ , the region for the whole coefficient vector is

$$\begin{aligned} \{\mathbf{b} : (\hat{\boldsymbol{\beta}} - \mathbf{b})^\top \mathbf{X}^\top \mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b}) &\leq p s^2 F_\alpha(p, n - p)\} \\ &= \{\mathbf{b} : \|\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\mathbf{b}\|^2 \leq p s^2 F_\alpha(p, n - p)\}. \end{aligned}$$

The second form says that  $\mathbf{b}$  is in the region iff its fitted vector  $\mathbf{X}\mathbf{b}$  is close to the least squares fit  $\hat{\mathbf{Y}}$ . This suggests looking at the mean vector directly.

## 12.2.2 A ball in the column space

### Proposition 12.2.2 — Confidence ball for the mean vector

Let  $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$  and  $c_r = r s^2 F_\alpha(r, n - r)$ . The set

$$B(\mathbf{Y}) = \{\boldsymbol{\mu}_0 \in \mathcal{C}(\mathbf{X}) : \|\hat{\mathbf{Y}} - \boldsymbol{\mu}_0\|^2 \leq c_r\}$$

is an exact  $1 - \alpha$  confidence set for  $\boldsymbol{\mu}$ : a ball of radius  $\sqrt{c_r}$  in the  $r$ -dimensional space  $\mathcal{C}(\mathbf{X})$ , centred at the fitted vector. Moreover, if  $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$  for an  $n \times q$  matrix  $\mathbf{T}$  (such a  $\mathbf{T}$  exists by [Theorem 8.3.1](#)), the image of  $B(\mathbf{Y})$  under  $\boldsymbol{\mu}_0 \mapsto \mathbf{T}^\top \boldsymbol{\mu}_0$  is the set

$$\{\boldsymbol{\Phi}_0 : (\hat{\boldsymbol{\Phi}} - \boldsymbol{\Phi}_0)^\top \mathbf{W}^{-1}(\hat{\boldsymbol{\Phi}} - \boldsymbol{\Phi}_0) \leq c_r\},$$

which contains the ellipsoid [Eq. 12.2.1](#).

*Proof.*  $\hat{\mathbf{Y}} - \boldsymbol{\mu} = \mathbf{M}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$ , and  $\mathbf{M}$  is symmetric idempotent of rank  $r$ , so  $\|\hat{\mathbf{Y}} - \boldsymbol{\mu}\|^2 / \sigma^2 \sim \chi^2(r)$  by [Theorem 4.3.2\(a\)](#), independently of SSE ([Theorem 7.5.1\(c\)](#)). Dividing by  $r s^2 / \sigma^2$  gives an  $F(r, n - r)$  pivot, and the first claim follows as before.

For the image, put  $\mathbf{A} = \mathbf{T}^\top \mathbf{M}$ , a  $q \times n$  matrix. For  $\boldsymbol{\mu}_0 \in \mathcal{C}(\mathbf{X})$ ,  $\mathbf{T}^\top \boldsymbol{\mu}_0 = \mathbf{A}\boldsymbol{\mu}_0$ , and  $\mathbf{A}\mathbf{A}^\top =$

$\mathbf{T}^\top \mathbf{M} \mathbf{T} = \mathbf{W}$  (Theorem 9.2.4(a)). Also  $\hat{\boldsymbol{\phi}} = \mathbf{T}^\top \hat{\mathbf{Y}} = \mathbf{A} \hat{\mathbf{Y}}$ . So the image consists of the points  $\hat{\boldsymbol{\phi}} - \mathbf{A} \mathbf{u}$  with  $\mathbf{u} \in \mathcal{C}(\mathbf{X})$  and  $\|\mathbf{u}\|^2 \leq c_r$ . If  $\mathbf{z} = \mathbf{A} \mathbf{u}$  with  $\|\mathbf{u}\|^2 \leq c_r$ , then  $\mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} = \mathbf{u}^\top \mathbf{A}^\top \mathbf{W}^{-1} \mathbf{A} \mathbf{u} \leq \|\mathbf{u}\|^2 \leq c_r$ , because  $\mathbf{A}^\top \mathbf{W}^{-1} \mathbf{A}$  is symmetric and idempotent, hence a projection, and a projection does not increase length (Proposition 6.3.4(d)). Conversely, if  $\mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} \leq c_r$ , then  $\mathbf{u} = \mathbf{A}^\top \mathbf{W}^{-1} \mathbf{z}$  lies in  $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$ , satisfies  $\mathbf{A} \mathbf{u} = \mathbf{z}$ , and has  $\|\mathbf{u}\|^2 = \mathbf{z}^\top \mathbf{W}^{-1} \mathbf{z} \leq c_r$ .

For the containment it remains to show  $q F_\alpha(q, \nu) \leq r F_\alpha(r, \nu)$  when  $q \leq r$ . Let  $U_q \sim \chi^2(q)$ ,  $U' \sim \chi^2(r-q)$  and  $V \sim \chi^2(\nu)$  be independent. Then  $U_q + U' \sim \chi^2(r)$ , so for every  $c$ ,  $\Pr\{(U_q + U')/(V/\nu) \leq c\} \leq \Pr\{U_q/(V/\nu) \leq c\}$ . The upper  $\alpha$  point of the variable on the left is  $r F_\alpha(r, \nu)$ , and that of the one on the right is  $q F_\alpha(q, \nu)$ .

■

So all confidence ellipsoids are views of one ball around  $\hat{\mathbf{Y}}$  in the column space. Each linear summary  $\mathbf{A}^\top \boldsymbol{\beta}$  maps the ball to an ellipsoid, and since the ball covers  $\boldsymbol{\mu}$  with probability  $1 - \alpha$ , all the images cover their targets *simultaneously* with probability at least  $1 - \alpha$ , at the price of the constant  $r F_\alpha(r, \nu)$  in place of  $q F_\alpha(q, \nu)$ . Section 12.4 takes  $q = 1$  and all directions at once.

### 12.2.3 Shadows

One inequality gives the extent of an ellipsoid in every direction.

#### Lemma 12.2.3 — Extent of an ellipsoid

Let  $\mathbf{W}$  be a  $q \times q$  positive definite matrix and  $c > 0$ . For every  $\mathbf{a} \in \mathbb{R}^q$ ,

$$\max\{\mathbf{a}^\top \mathbf{u} : \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c\} = \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}},$$

attained at  $\mathbf{u} = \sqrt{c/(\mathbf{a}^\top \mathbf{W} \mathbf{a})} \mathbf{W} \mathbf{a}$  when  $\mathbf{a} \neq \mathbf{0}$ . Consequently  $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c$  iff  $|\mathbf{a}^\top \mathbf{u}| \leq \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}}$  for every  $\mathbf{a} \in \mathbb{R}^q$ .

*Proof.* The bound is Corollary 1.7.8(c) with  $\mathbf{B} = \mathbf{W}^{-1}$ ; directly, write  $\mathbf{a}^\top \mathbf{u} = (\mathbf{W}^{1/2} \mathbf{a})^\top (\mathbf{W}^{-1/2} \mathbf{u})$ . By the Cauchy–Schwarz inequality,  $(\mathbf{a}^\top \mathbf{u})^2 \leq (\mathbf{a}^\top \mathbf{W} \mathbf{a})(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}) \leq c \mathbf{a}^\top \mathbf{W} \mathbf{a}$ , with equality when  $\mathbf{W}^{-1/2} \mathbf{u}$  is a nonnegative multiple of  $\mathbf{W}^{1/2} \mathbf{a}$  and  $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} = c$ , which is the stated  $\mathbf{u}$ . For the last statement, “only if” follows from the maximum applied to  $\mathbf{a}$  and  $-\mathbf{a}$ . For “if”, take  $\mathbf{a} = \mathbf{W}^{-1} \mathbf{u}$ : then  $(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u})^2 \leq c \mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}$ , so  $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq c$ .

■

#### Corollary 12.2.4 — Shadows of the confidence ellipsoid

For every  $\mathbf{a} \in \mathbb{R}^q$ , the projection of  $E(\mathbf{Y})$  onto the line spanned by  $\mathbf{a}$  is the interval

$$\mathbf{a}^\top \hat{\boldsymbol{\phi}} \pm \sqrt{q F_\alpha(q, n-r)} \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}}), \quad \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}}) = s \sqrt{\mathbf{a}^\top \mathbf{W} \mathbf{a}},$$

and  $E(\mathbf{Y})$  is the intersection of the slabs  $\{\boldsymbol{\phi}_0 : |\mathbf{a}^\top \hat{\boldsymbol{\phi}} - \mathbf{a}^\top \boldsymbol{\phi}_0| \leq \sqrt{q F_\alpha} \text{se}(\mathbf{a}^\top \hat{\boldsymbol{\phi}})\}$  over all  $\mathbf{a}$ . Hence, with probability exactly  $1 - \alpha$ , every one of these intervals contains  $\mathbf{a}^\top \boldsymbol{\Lambda}^\top \boldsymbol{\beta}$ .

*Proof.* Apply [Lemma 12.2.3](#) to  $\mathbf{u} = \boldsymbol{\phi}_0 - \hat{\boldsymbol{\phi}}$  with  $c = q s^2 F_\alpha$ . The extreme values of  $\mathbf{a}^\top \boldsymbol{\phi}_0$  over  $E(\mathbf{Y})$  are  $\mathbf{a}^\top \hat{\boldsymbol{\phi}} \pm \sqrt{c \mathbf{a}^\top \mathbf{W} \mathbf{a}}$ , and the set of values in between is attained because  $E(\mathbf{Y})$  is convex. The intersection statement is the last part of the lemma. The final claim holds because the event “all intervals cover” is the event  $\boldsymbol{\phi} \in E(\mathbf{Y})$ . ■

A shadow is a  $t$  interval with  $t_{n-r, \alpha/2}$  replaced by  $\sqrt{q F_\alpha(q, n-r)}$ . The multipliers agree when  $q = 1$  ([Exercise \(A1\)](#)) and the shadow is wider when  $q > 1$ , as it must be: the shadows hold simultaneously for all directions  $\mathbf{a}$ , including directions chosen after seeing the data. This is Scheffé’s method, developed in [Chapter 13](#) ([Theorem 13.2.2](#)).

The box of one-at-a-time  $t$  intervals undercovers, and it disagrees with the ellipsoid in both directions: a corner of the box can lie outside the ellipsoid, and the ellipsoid pokes out of the box along its long axis.

### Example. (A joint region for two coefficients)

In the murder-rate regression of [Example \(Murder rates: coefficients and a mean\)](#), take  $\mathbf{\Lambda}$  to pick out the coefficients of poverty and single parenthood, so  $q = 2$ ,  $\hat{\boldsymbol{\phi}} = (0.2538, 0.3980)^\top$ , and the correlation of the two estimates is  $-0.639$ . The negative correlation reflects the positive correlation of the regressors: the data can credit murder to either, but not to both at once. With  $F_{0.05}(2, 46) = 3.200$ , the 95% ellipse is drawn in [Figure 12.2.1](#). The  $F$  statistic for  $\boldsymbol{\phi} = \mathbf{0}$  is 39.81, so the origin is far outside.

The shadows of the ellipse on the two axes are (0.018, 0.490) and (0.188, 0.608), wider than the  $t$  intervals (0.066, 0.442) and (0.231, 0.565) by the factor  $\sqrt{2 \times 3.200}/2.013 = 1.257$ . Two points show how the ellipse and the box disagree.

- $\mathbf{A} = (0.442, 0.565)$ , the upper corner of the box. Neither  $t$  test rejects, but  $\mathbf{A}$  raises both coefficients at once, against the correlation, and the joint test rejects:  $F = 11.230$ ,  $p = 0.0001$ .
- $\mathbf{B} = (0.463, 0.224)$ , on the long axis of the ellipse. Both  $t$  statistics exceed 2.013 (2.242 and 2.098), but  $\mathbf{B}$  trades one coefficient for the other, which the data cannot resolve, and  $F = 2.888 < 3.200$ .

The ellipse and the box have similar areas (0.1197 and 0.1255) but different shapes. In 40000 simulated data sets from the fitted model the ellipse covered the true pair in a proportion 0.9500 and the box in only 0.9164.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statelcrime.load_pandas().data
data = data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
C = np.linalg.inv(X.T @ X)
beta_hat = C @ X.T @ y
s2 = np.sum((y - X @ beta_hat) ** 2) / (n - p)
```

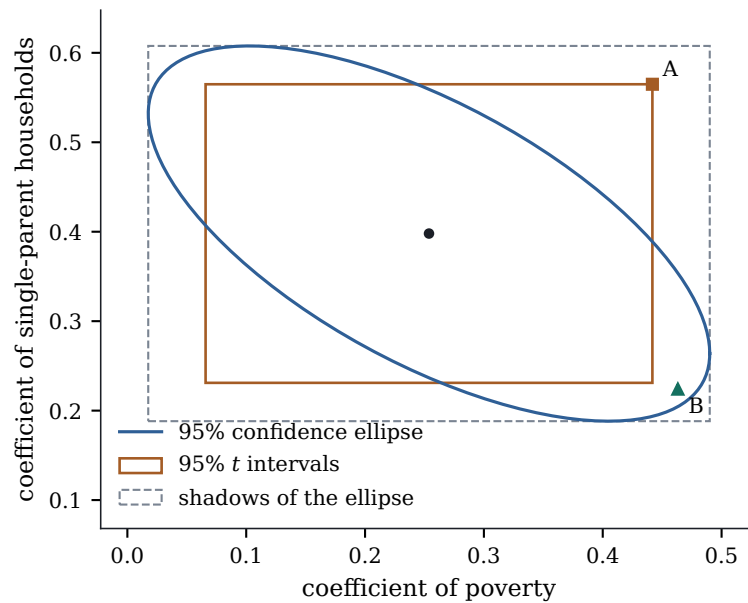


Figure 12.2.1: The 95% confidence ellipse for the poverty and single-parent coefficients, the box of 95%  $t$  intervals, and the ellipse's shadows (dashed). A is inside the box but outside the ellipse; B the reverse.

```

L = np.eye(p)[: , [1, 2]]                                # Lambda: the poverty and single
    coefficients
q = L.shape[1]
phi_hat = L.T @ beta_hat
W = L.T @ C @ L                                           # Cov(phi_hat) / sigma^2
Fq = stats.f.ppf(0.95, q, n - p)

def F_stat(d):
    """F statistic for the hypothesis  $\Lambda^T \beta = d$ ."""
    u = phi_hat - d
    return u @ np.linalg.solve(W, u) / (q * s2)

def in_ellipse(d):
    return F_stat(d) <= Fq

corr = W[0, 1] / np.sqrt(W[0, 0] * W[1, 1])
print(f"estimates {phi_hat}, correlation of the estimates {corr:.3f}")
print(f"F({q}, {n - p}) 95% point {Fq:.3f}; F statistic for (0, 0): {F_stat(np.zeros
(2)):.2f}")

tq = stats.t.ppf(0.975, n - p)
se = np.sqrt(s2 * np.diag(W))
t_box = np.column_stack([phi_hat - tq * se, phi_hat + tq * se])    # one-at-
    a-time t intervals
k = np.sqrt(q * Fq)
shadow = np.column_stack([phi_hat - k * se, phi_hat + k * se])    #
    projections of the ellipse
print("t intervals:\n", t_box.round(3))
print("shadows of the ellipse:\n", shadow.round(3), f"\nwider by the factor {k / tq
:.3f}")

```

## 12.2.4 Two cautions

**The ellipsoid is for a question fixed in advance.** If the coefficients are chosen because they look interesting, the ball of [Proposition 12.2.2](#) or the methods of [Chapter 13](#) are the honest tools.

**Intervals after a test.** If intervals are reported only when a preliminary test of  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$  rejects, their coverage *given* the rejection is no longer  $1 - \alpha$ , and can be larger or smaller (Olshen 1973).

## 12.2.5 Exercises

### 12.2.5.1 A. Check your understanding

#### Exercise A1.

Show that for  $q = 1$  the ellipsoid [Eq. 12.2.1](#) is the  $t$  interval [Eq. 12.1.1](#). Which fact about the  $F(1, \nu)$  distribution does this use?

*Solution.* With  $q = 1$ ,  $\mathbf{W} = w$  is a positive number and the region is  $(\hat{\phi} - \phi_0)^2 \leq s^2 w F_\alpha(1, n - r)$ . If  $T \sim t(\nu)$ , then  $T^2 \sim F(1, \nu)$  ([Definition 4.1.3](#) and the remark after it), so  $\Pr\{T^2 \leq t_{\nu, \alpha/2}^2\} = \Pr\{|T| \leq t_{\nu, \alpha/2}\} = 1 - \alpha$  and  $F_\alpha(1, \nu) = t_{\nu, \alpha/2}^2$ . The region is  $|\hat{\phi} - \phi_0| \leq t_{\nu, \alpha/2} s \sqrt{w}$ .

#### Exercise A2.

For the straight line  $E Y_i = \beta_0 + \beta_1 x_i$ , write the 95% confidence ellipse for  $(\beta_0, \beta_1)$  in terms of  $n$ ,  $\bar{x}$  and  $\sum x_i^2$ . Show that in the centred parameterization  $E Y_i = \gamma_0 + \beta_1(x_i - \bar{x})$  the axes of the ellipse for  $(\gamma_0, \beta_1)$  are parallel to the coordinate axes. When is it a circle?

*Solution.*  $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} n & n\bar{x} \\ n\bar{x} & \sum x_i^2 \end{pmatrix}$ , and the region is  $n(\hat{\beta}_0 - b_0)^2 + 2n\bar{x}(\hat{\beta}_0 - b_0)(\hat{\beta}_1 - b_1) + \sum x_i^2(\hat{\beta}_1 - b_1)^2 \leq 2s^2 F_\alpha(2, n - 2)$ . In the centred parameterization the cross-product matrix is  $\text{diag}(n, S_{xx})$ , so the quadratic form has no cross term, and the ellipse has axes along the coordinates with half-lengths  $\sqrt{2s^2 F_\alpha/n}$  and  $\sqrt{2s^2 F_\alpha/S_{xx}}$ . It is a circle iff  $S_{xx} = n$ , that is, iff the  $x_i$  have mean square deviation one.

### 12.2.5.2 B. Practice

#### Exercise B1.

Take  $q = 2$  and let  $\rho$  be the correlation of  $\hat{\phi}_1$  and  $\hat{\phi}_2$ . Show that the area of the ellipse divided by the area of the box of the two  $t$  intervals is

$$\frac{\pi F_\alpha(2, n - r)}{2 t_{n-r, \alpha/2}^2} \sqrt{1 - \rho^2},$$

so that it depends on the design only through  $\rho$ . Check it against [Example \(A joint](#)

region for two coefficients), where the areas are 0.1197 and 0.1255 at  $\rho = -0.639$ . For which  $|\rho|$  do the two regions have equal area there, and what happens to the ratio as the two regressors become collinear?

*Solution.* By [Theorem 12.2.1\(b\)](#) the ellipse has area  $\pi \cdot 2F_\alpha s^2 \det(\mathbf{W})^{1/2}$ , and writing  $\omega_j$  for the diagonal entries of  $\mathbf{W}$ ,  $\det \mathbf{W} = \omega_1 \omega_2 (1 - \rho^2)$ . The box has sides  $2ts\sqrt{\omega_j}$ , so area  $4t^2 s^2 \sqrt{\omega_1 \omega_2}$ ; divide. In the example  $F_{0.05}(2, 46) = 3.200$  and  $t = 2.013$  give 1.240 at  $\rho = 0$ , and  $1.240\sqrt{1 - \rho^2} = 0.954$  at  $\rho = -0.639$ , which is  $0.1197/0.1255$ . The areas agree when  $\sqrt{1 - \rho^2} = 1/1.240$ , that is  $|\rho| = 0.592$ . With uncorrelated estimates the ellipse is the larger region; as the regressors become collinear,  $|\rho| \rightarrow 1$ , the ratio tends to 0 and the ellipse becomes a thin needle across a large box, which the joint region mostly excludes.

### Exercise B2.

In [Example \(A joint region for two coefficients\)](#), the hypothesis that one point of poverty and one point of single parenthood have the same coefficient is the line  $\phi_1 = \phi_2$ . Using [Lemma 12.2.3](#), show that this line meets the ellipse iff  $|\hat{\phi}_1 - \hat{\phi}_2| \leq \sqrt{2F_\alpha(2, n - r)} \text{se}(\hat{\phi}_1 - \hat{\phi}_2)$ . Compare this with the  $t$  test of  $\phi_1 = \phi_2$ . Which is appropriate if the hypothesis was formulated before seeing the data?

*Solution.* The line is  $\{\boldsymbol{\phi}_0 : \mathbf{a}^\top \boldsymbol{\phi}_0 = 0\}$  with  $\mathbf{a} = (1, -1)^\top$ . By [Corollary 12.2.4](#), the values of  $\mathbf{a}^\top \boldsymbol{\phi}_0$  over the ellipse fill the interval  $\mathbf{a}^\top \hat{\boldsymbol{\phi}} \pm \sqrt{2F_\alpha} s \sqrt{\mathbf{a}^\top \mathbf{W} \mathbf{a}}$ , so the line meets the ellipse iff this interval contains 0. The  $t$  test uses the smaller multiplier  $t_{n-r, \alpha/2}$ . It is the right test for a hypothesis fixed in advance. The ellipse criterion is the right one for a contrast chosen after looking at the estimates, because it holds simultaneously for all  $\mathbf{a}$ .

### Exercise B3.

Let the columns of  $\mathbf{\Lambda}$  lie in  $\mathcal{C}(\mathbf{X}^\top)$  but be linearly dependent, with  $\text{rank}(\mathbf{\Lambda}) = k < q$ . Show that  $\mathbf{W} = \mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda}$  is singular, that  $\hat{\boldsymbol{\phi}} - \boldsymbol{\phi} \in \mathcal{C}(\mathbf{W})$  with probability one, and that

$$\left\{ \boldsymbol{\phi}_0 \in \hat{\boldsymbol{\phi}} + \mathcal{C}(\mathbf{W}) : (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0)^\top \mathbf{W}^- (\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}_0) \leq k s^2 F_\alpha(k, n - r) \right\}$$

is an exact  $1 - \alpha$  confidence set, a  $k$ -dimensional ellipsoid inside a  $k$ -dimensional affine subspace of  $\mathbb{R}^q$ .

*Solution.*  $\mathbf{W} = (\mathbf{M}\mathbf{T})^\top (\mathbf{M}\mathbf{T})$  has rank  $\text{rank}(\mathbf{M}\mathbf{T}) = \text{rank}(\mathbf{\Lambda}) = k$  (proof of [Corollary 7.5.3](#)). The vector  $\hat{\boldsymbol{\phi}} - \boldsymbol{\phi}$  is  $N_q(\mathbf{0}, \sigma^2 \mathbf{W})$ , so it lies in  $\mathcal{C}(\mathbf{W})$  almost surely ([Theorem 2.2.3](#)). By [Corollary 7.5.3](#) the quadratic form over  $k s^2$  is  $F(k, n - r)$  and does not depend on the generalized inverse. So the displayed set covers  $\boldsymbol{\phi}$  with probability  $1 - \alpha$ . Restricted to the affine subspace, write  $\mathbf{W} = \mathbf{U}\mathbf{D}\mathbf{U}^\top$  with  $\mathbf{U}$  ( $q \times k$ ) orthonormal and  $\mathbf{D}$  positive diagonal. For  $\boldsymbol{\phi}_0 - \hat{\boldsymbol{\phi}} = \mathbf{U}\mathbf{c}$ , the form is  $\mathbf{c}^\top \mathbf{D}^{-1} \mathbf{c}$ , an ellipsoid in the coordinates  $\mathbf{c}$ .

#### 12.2.5.3 C. Going deeper



**Exercise C1.**

Let  $\mathbf{X}$  have full column rank. Show that the confidence ellipsoid for  $\boldsymbol{\beta}$  is a region of the form  $\{\mathbf{b} : \ell_p(\mathbf{b}) \geq \ell_p(\hat{\boldsymbol{\beta}}) - k\}$ , where  $\ell_p(\mathbf{b}) = \max_{\sigma^2} \ell(\mathbf{b}, \sigma^2)$  is the profile log-likelihood of Chapter 7, and find  $k$  in terms of  $F_\alpha(p, n-p)$ . How does  $k$  compare with the value  $\frac{1}{2}\chi_\alpha^2(p)$  suggested by the asymptotic theory of likelihood ratios?

*Solution.* The profile log-likelihood is  $\ell_p(\mathbf{b}) = \text{const} - \frac{n}{2} \log \|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2$ , and  $\|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2 = \text{SSE} + (\hat{\boldsymbol{\beta}} - \mathbf{b})^\top \mathbf{X}^\top \mathbf{X} (\hat{\boldsymbol{\beta}} - \mathbf{b})$  by Eq. 6.3.1. So  $\ell_p(\hat{\boldsymbol{\beta}}) - \ell_p(\mathbf{b}) = \frac{n}{2} \log\{1 + pF_{\mathbf{b}}/(n-p)\}$ , where  $F_{\mathbf{b}}$  is the statistic of Theorem 12.2.1(c). This is increasing in  $F_{\mathbf{b}}$ , so the ellipsoid is the region with  $k = \frac{n}{2} \log\{1 + pF_\alpha(p, n-p)/(n-p)\}$ . As  $n \rightarrow \infty$ ,  $pF_\alpha(p, n-p) \rightarrow \chi_\alpha^2(p)$  and  $k \rightarrow \frac{1}{2}\chi_\alpha^2(p)$ . For finite  $n$ ,  $k$  is larger, and the exact region is larger than the asymptotic likelihood region.

## 12.3 Prediction intervals

Often the quantity of interest is an observation not yet made: tomorrow's stock loss, or the reading of the next specimen. An interval for it must allow for the error in the estimated mean, which shrinks as data accumulate, and the error of the new observation itself, which does not. The first depends on where the new point lies relative to the data; the second makes prediction intervals depend on normality in a way that confidence intervals do not.

### 12.3.1 Predicting a new observation

Let the data follow the normal linear model of Section 12.1, and let

$$Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0, \quad \varepsilon_0 \sim N(0, \sigma^2), \quad \varepsilon_0 \text{ independent of } \mathbf{Y}, \quad (12.3.1)$$

be a future observation at a regressor vector  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$ . The requirement  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$  makes the mean  $\mathbf{x}_0^\top \boldsymbol{\beta}$  estimable. Without it, Proposition 12.1.3 shows that nothing useful can be said. We call

$$h_0 = \mathbf{x}_0^\top \mathbf{G} \mathbf{x}_0$$

the **leverage** of the new point. When  $\mathbf{x}_0$  is the  $i$ th row of  $\mathbf{X}$ ,  $h_0$  is the diagonal entry  $h_{ii}$  of  $\mathbf{M}$  (Proposition 6.8.1), and  $\sigma^2 h_0$  is always the variance of  $\hat{Y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$  (Theorem 7.5.1(a)).

#### Theorem 12.3.1 — Prediction interval

Under Eq. 12.3.1, put  $\hat{Y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ .

- (a)  $Y_0 - \hat{Y}_0 \sim N(0, \sigma^2(1 + h_0))$ , it is independent of  $s^2$ , and

$$\frac{Y_0 - \hat{Y}_0}{s\sqrt{1 + h_0}} \sim t(n - r).$$

- (b) The interval  $\hat{Y}_0 \pm t_{n-r, \alpha/2} s\sqrt{1 + h_0}$  contains  $Y_0$  with probability exactly  $1 - \alpha$ , the

probability being computed over  $\mathbf{Y}$  and  $Y_0$  jointly, at every  $(\boldsymbol{\beta}, \sigma^2)$ .

- (c) More generally, if  $\bar{Y}_0$  is the mean of  $m$  independent new observations at  $\mathbf{x}_0$ , each distributed as in [Eq. 12.3.1](#), then  $\hat{Y}_0 \pm t_{n-r, \alpha/2} s \sqrt{1/m + h_0}$  contains  $\bar{Y}_0$  with probability exactly  $1 - \alpha$ . As  $m \rightarrow \infty$  this interval becomes the confidence interval  $\hat{Y}_0 \pm t_{n-r, \alpha/2} s \sqrt{h_0}$  for  $\mathbf{x}_0^\top \boldsymbol{\beta}$ .

*Proof.* We prove (c), of which (a) and (b) are the case  $m = 1$ . The new observations are independent of  $\mathbf{Y}$ , and  $\bar{Y}_0 \sim N(\mathbf{x}_0^\top \boldsymbol{\beta}, \sigma^2/m)$ . The vector  $(\bar{Y}_0, \mathbf{Y}^\top)^\top$  is normal with independent blocks, so  $\bar{Y}_0 - \hat{Y}_0$ , a linear function of it, is normal ([Theorem 3.3.1](#)) with mean  $\mathbf{x}_0^\top \boldsymbol{\beta} - \mathbf{x}_0^\top \boldsymbol{\beta} = 0$  and variance  $\sigma^2/m + \sigma^2 h_0$ , the two terms adding because  $\bar{Y}_0$  and  $\hat{Y}_0$  are independent. For independence from  $s^2$ :  $\hat{Y}_0$  is a function of  $\mathbf{MY}$ , and  $\mathbf{MY}$  and SSE are independent ([Theorem 7.5.1\(c\)](#)). Since  $\bar{Y}_0$  is independent of  $\mathbf{Y}$ , for any events,

$$\begin{aligned} \Pr(\bar{Y}_0 \in A, \mathbf{MY} \in B, \text{SSE} \in D) \\ = \Pr(\bar{Y}_0 \in A) \Pr(\mathbf{MY} \in B) \Pr(\text{SSE} \in D), \end{aligned}$$

so  $(\bar{Y}_0, \mathbf{MY})$  is independent of SSE, and so is  $\bar{Y}_0 - \hat{Y}_0$ . Dividing the standardized normal variable by  $\sqrt{(\text{SSE}/\sigma^2)/(n-r)}$  gives  $t(n-r)$  ([Definition 4.1.3](#)), and the interval statement is  $\Pr\{|T| \leq t_{n-r, \alpha/2}\} = 1 - \alpha$ . The limit as  $m \rightarrow \infty$  is immediate, and the limiting interval is [Eq. 12.1.1](#) with  $\boldsymbol{\lambda} = \mathbf{x}_0$ . ■

Part (a) was sketched in [Exercise \(B2\)](#). A *prediction* interval is for a random variable, and its probability refers to the joint experiment: collect the data, compute the interval, observe  $Y_0$ . Given the data, the probability that the interval contains  $Y_0$  is random,

$$\begin{aligned} \Pr(Y_0 \in \text{interval} \mid \mathbf{Y}) &= \Phi\left(\frac{\mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) + t s \sqrt{1 + h_0}}{\sigma}\right) \\ &\quad - \Phi\left(\frac{\mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) - t s \sqrt{1 + h_0}}{\sigma}\right), \end{aligned}$$

with  $t = t_{n-r, \alpha/2}$  and  $\Phi$  the standard normal distribution function, and it is only its expectation that equals  $1 - \alpha$ .

In the variance  $\sigma^2(1 + h_0)$ , the leverage  $h_0$  is the price of not knowing  $\boldsymbol{\beta}$ , of order  $r/n$  at a typical point ([Proposition 6.8.1](#)). The 1 is the new error, which no amount of data removes: the prediction interval never becomes shorter than about  $2z_{\alpha/2}\sigma$ , while the interval for the mean shrinks to a point.

### Example. (Predicting a state's murder rate)

In the murder-rate regression, consider a state at the centroid of the regressors. Its leverage is  $h_0 = 1/n = 0.0200$  (see [Proposition 12.3.2](#)), the fitted rate is 4.514, and the 95% intervals are

$$\text{mean response: } (4.090, 4.938), \quad \text{new state: } (1.487, 7.541).$$

The prediction interval is 7.14 times as long as the interval for the mean, and nearly all of its length, 6.053 per 100,000, comes from  $s$ .

```

import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statecrime.load_pandas().data
data = data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
C = np.linalg.inv(X.T @ X)
beta_hat = C @ X.T @ y
s = np.sqrt(np.sum((y - X @ beta_hat) ** 2) / (n - p))
q = stats.t.ppf(0.975, n - p)
h = np.einsum("ij,jk,ik->i", X, C, X)          # leverages of the 50 states

def intervals(x0):
    """Leverage h0, 95% interval for the mean response, 95% prediction interval."""
    h0 = x0 @ C @ x0
    fit = x0 @ beta_hat
    ci = (fit - q * s * np.sqrt(h0), fit + q * s * np.sqrt(h0))
    pi = (fit - q * s * np.sqrt(1 + h0), fit + q * s * np.sqrt(1 + h0))
    return h0, fit, ci, pi

x_centre = X.mean(axis=0)          # the average state
x_hidden = np.array([1.0, 20.0, 19.0, 60.0])  # each coordinate inside its
                                         observed range
for name, x0 in [("centre", x_centre), ("hidden", x_hidden)]:
    h0, fit, ci, pi = intervals(x0)
    print(f"{name}: h0 = {h0:.4f}, fit {fit:.3f}, mean ({ci[0]:.3f}, {ci[1]:.3f}), "
          f"new state ({pi[0]:.3f}, {pi[1]:.3f})")
print(f"largest leverage in the data: {h.max():.4f}")

```

## 12.3.2 Leverage and extrapolation

### Proposition 12.3.2 — Leverage of a new point

Let  $\mathbf{x}_0 = \mathbf{X}^\top \boldsymbol{\rho} \in \mathcal{C}(\mathbf{X}^\top)$  and  $h_0 = \mathbf{x}_0^\top \mathbf{G} \mathbf{x}_0$ .

- (a)  $h_0 = \min\{\|\mathbf{a}\|^2 : \mathbf{X}^\top \mathbf{a} = \mathbf{x}_0\}$ , and the minimum is attained only at  $\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$ .
- (b) If  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$  has full column rank,  $\mathbf{x}_0 = (1, \mathbf{x}_{01}^\top)^\top$ ,  $\bar{\mathbf{x}}$  is the vector of column means of  $\mathbf{X}_1$  and  $\tilde{\mathbf{X}}_1 = \mathbf{X}_1 - \mathbf{1}\bar{\mathbf{x}}^\top$ , then

$$h_0 = \frac{1}{n} + (\mathbf{x}_{01} - \bar{\mathbf{x}})^\top (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1} (\mathbf{x}_{01} - \bar{\mathbf{x}}).$$

- (c) If further columns are added,  $\mathbf{X}_+ = [\mathbf{X}, \mathbf{Z}]$ , and  $\mathbf{x}_{0+} = (\mathbf{x}_0^\top, \mathbf{z}_0^\top)^\top \in \mathcal{C}(\mathbf{X}_+^\top)$ , then the leverage of  $\mathbf{x}_{0+}$  in the enlarged model is at least  $h_0$ .

*Proof.*

- (a) Let  $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$ . Then  $\mathbf{X}^\top \mathbf{M} \mathbf{a} = \mathbf{X}^\top \mathbf{a} = \mathbf{x}_0 = \mathbf{X}^\top \mathbf{M} \boldsymbol{\rho}$ , so  $\mathbf{M} \mathbf{a} - \mathbf{M} \boldsymbol{\rho}$  lies in  $\mathcal{C}(\mathbf{X})$

and is orthogonal to  $\mathcal{C}(\mathbf{X})$ ; hence  $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$ . By Pythagoras,  $\|\mathbf{a}\|^2 = \|\mathbf{M}\boldsymbol{\rho}\|^2 + \|(\mathbf{I} - \mathbf{M})\mathbf{a}\|^2 \geq \|\mathbf{M}\boldsymbol{\rho}\|^2$ , with equality iff  $\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$ , and  $\|\mathbf{M}\boldsymbol{\rho}\|^2 = \boldsymbol{\rho}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top \boldsymbol{\rho} = h_0$  by Theorem 6.3.7.

- (b)  $\mathbf{X} = [\mathbf{1}, \tilde{\mathbf{X}}_1]\mathbf{A}$  with  $\mathbf{A} = \begin{pmatrix} 1 & \bar{\mathbf{x}}^\top \\ \mathbf{0} & \mathbf{I} \end{pmatrix}$ , which is invertible. Since  $\mathbf{1}^\top \tilde{\mathbf{X}}_1 = \mathbf{0}^\top$ ,  $\mathbf{X}^\top \mathbf{X} = \mathbf{A}^\top \text{diag}(n, \tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1) \mathbf{A}$ , and so  $h_0 = \mathbf{w}^\top \text{diag}(1/n, (\tilde{\mathbf{X}}_1^\top \tilde{\mathbf{X}}_1)^{-1}) \mathbf{w}$  with  $\mathbf{w} = \mathbf{A}^{-\top} \mathbf{x}_0$ . Solving  $\mathbf{A}^\top \mathbf{w} = \mathbf{x}_0$  gives  $\mathbf{w} = (1, (\mathbf{x}_{01} - \bar{\mathbf{x}})^\top)^\top$ .
- (c) Every  $\mathbf{a}$  with  $\mathbf{X}_+^\top \mathbf{a} = \mathbf{x}_{0+}$  also satisfies  $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$ . A minimum over a smaller set is at least as large, so (a), applied to both models, gives the claim. ■

Part (a) is the Gauss–Markov theorem in disguise. Part (b) makes leverage a distance: the quadratic form is  $(n - 1)^{-1}$  times the squared Mahalanobis distance of  $\mathbf{x}_{01}$  from the centroid of the regressors, as for the observed points (Proposition 6.8.2). So the interval for the mean is shortest at the centroid, where  $h_0 = 1/n$ , and widens along ellipsoidal contours. Part (c) says a larger model never predicts more precisely at a given point, though it may predict with less bias; Chapter 29 weighs the two.

With several correlated regressors, a point can lie inside the range of every regressor and still be far from the data, because it breaks their correlation. This **hidden extrapolation** is detected by the leverage, not by the separate ranges.

#### Example. (Hidden extrapolation)

Poverty ranges from 8.5% to 21.9% over the 50 states, and single-parent households from 17.9% to 32.8%. A state with poverty 20%, single-parent households 19% and 60% urban is inside every range, but poor with few single parents, a combination absent from the data. Its leverage is  $h_0 = 0.435$ , far above the largest leverage among the actual states, 0.260 (Mississippi). Figure 12.3.1 shows the contours of  $h_0$  over the two regressors, with urbanization held at its mean.

The fitted rate is 3.817. The 95% interval for the mean is (1.840, 5.794), longer than at the centroid by the factor  $\sqrt{0.435/0.0200} = 4.66$ . The prediction interval, (0.227, 7.407), is only 1.186 times as long as at the centroid, because it is dominated by  $s$ . The prediction interval barely registers the extrapolation, and the real danger, that the model may be wrong where there are no data to check it, is in neither formula. If urbanization is dropped from the model the leverage of this point falls to 0.287, as Proposition 12.3.2(c) requires.

### 12.3.3 When the errors are not normal

The estimate  $\hat{Y}_0$  is a weighted sum of  $n$  errors, approximately normal by a central limit theorem when no weight dominates (Theorem 19.3.2). The new error  $\varepsilon_0$  is a single draw, and its distribution enters the prediction interval unchanged, however large  $n$  is.

#### Proposition 12.3.3 — Prediction intervals without normality

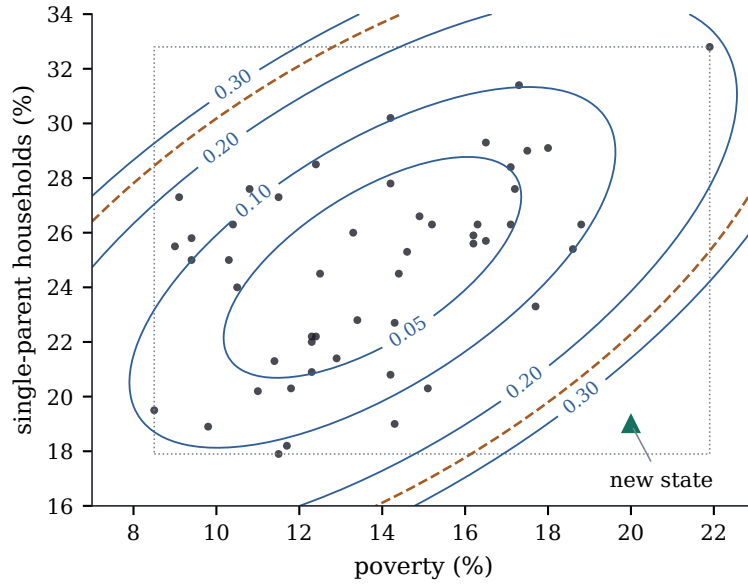


Figure 12.3.1: Contours of the leverage  $h_0$  of a new state (urbanization at its mean), with the 50 states, the box of observed ranges (dotted) and the largest observed leverage (dashed). The marked state is inside the box but far from the data.

Consider a sequence of full-rank models  $\mathbf{Y}_n = \mathbf{X}_n \boldsymbol{\beta} + \boldsymbol{\varepsilon}_n$  with  $p$  fixed and  $n \rightarrow \infty$ , and a fixed  $\mathbf{x}_0$  whose leverage  $h_{0,n} = \mathbf{x}_0^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{x}_0$  tends to zero. Let  $\varepsilon_1, \dots, \varepsilon_n$  and  $\varepsilon_0$  be independent and identically distributed with mean 0, variance  $\sigma^2$ , finite fourth moment and a continuous distribution function, and let  $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$ . Then the coverage of the nominal  $1 - \alpha$  prediction interval  $\hat{Y}_0 \pm t_{n-p, \alpha/2} s \sqrt{1 + h_{0,n}}$  satisfies

$$\begin{aligned} \Pr\{|Y_0 - \hat{Y}_0| \leq t_{n-p, \alpha/2} s \sqrt{1 + h_{0,n}}\} \\ \longrightarrow \Pr\{|\varepsilon_0| \leq z_{\alpha/2} \sigma\}, \end{aligned}$$

where  $z_{\alpha/2}$  is the upper  $\alpha/2$  point of  $N(0, 1)$ . Likewise the probabilities of missing above and below tend to  $\Pr\{\varepsilon_0 > z_{\alpha/2} \sigma\}$  and  $\Pr\{\varepsilon_0 < -z_{\alpha/2} \sigma\}$ .

*Proof.* Write  $T_n = (Y_0 - \hat{Y}_0)/(s \sqrt{1 + h_{0,n}})$ ,  $t_n = t_{n-p, \alpha/2}$  and  $E = \varepsilon_0/\sigma$ . First,  $Y_0 - \hat{Y}_0 = \varepsilon_0 - \mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ , and  $\mathbf{x}_0^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$  has mean zero and variance  $\sigma^2 h_{0,n} \rightarrow 0$  (Theorem 5.5.1), so it tends to zero in probability by Chebyshev's inequality. Second,  $s \rightarrow \sigma$  in probability (Corollary 5.6.4), and  $t_n \rightarrow z_{\alpha/2}$ . Hence  $D_n = T_n - E \rightarrow 0$  in probability. Fix  $\eta > 0$ . On the event  $\{|D_n| \leq \eta/2, |t_n - z_{\alpha/2}| \leq \eta/2\}$ , whose probability tends to one,

$$|E| \leq z_{\alpha/2} - \eta \implies |T_n| \leq t_n \implies |E| \leq z_{\alpha/2} + \eta.$$

So  $\Pr\{|E| \leq z_{\alpha/2} - \eta\} \leq \liminf \Pr\{|T_n| \leq t_n\} \leq \limsup \Pr\{|T_n| \leq t_n\} \leq \Pr\{|E| \leq z_{\alpha/2} + \eta\}$ . The distribution function of  $E$  is continuous, so letting  $\eta \rightarrow 0$  gives the limit. The one-sided statements are proved the same way. ■

The limit is the nominal  $1 - \alpha$  only if  $\Pr\{|\varepsilon_0| \leq z_{\alpha/2} \sigma\} = 1 - \alpha$ , a property of the normal distribution, not of large samples. A simulation shows both effects: a straight line with  $n$  equally spaced design points on  $[0, 1]$ , prediction at  $x_0 = 0.8$ , and errors with mean zero and

variance one that are normal, centred exponential (skewed) or uniform (short-tailed). With 20000 data sets for each case, the coverages of the nominal 80% and 95% intervals are:

| Errors      | $n$ | mean, 80% | mean, 95% | new obs.,<br>80% | new obs.,<br>95% |
|-------------|-----|-----------|-----------|------------------|------------------|
| normal      | 20  | 0.799     | 0.949     | 0.797            | 0.949            |
| normal      | 400 | 0.805     | 0.953     | 0.802            | 0.953            |
| exponential | 20  | 0.774     | 0.931     | 0.863            | 0.939            |
| exponential | 400 | 0.799     | 0.949     | 0.896            | 0.949            |
| uniform     | 20  | 0.797     | 0.948     | 0.787            | 0.983            |
| uniform     | 400 | 0.800     | 0.950     | 0.741            | 1.000            |

At  $n = 400$  the intervals for the mean are near nominal for every law, while the prediction intervals approach the limits of [Proposition 12.3.3](#): 0.898 and 0.948 (exponential), 0.740 and 1.000 (uniform). The exponential 95% interval looks fine only by accident: at  $n = 400$  it misses above in a proportion 0.051 of the samples and below in 0.0000, since a centred exponential error is never below  $-1$ .

```

rng = np.random.default_rng(1203)
laws = {
    "normal": lambda size: rng.normal(size=size),
    "exponential": lambda size: rng.exponential(size=size) - 1.0,          #
    "uniform": lambda size: rng.uniform(-np.sqrt(3), np.sqrt(3), size=size), #
    "skewed": lambda size: rng.exponential(size=size) - 1.0,             #
    "short tails": lambda size: rng.uniform(-np.sqrt(3), np.sqrt(3), size=size), #
}
all: mean 0, variance 1
x0_sim = 0.8

def simulate(law, n_sim, reps=20_000):
    x = np.linspace(0.0, 1.0, n_sim)
    Xs = np.column_stack([np.ones(n_sim), x])
    Hs = np.linalg.solve(Xs.T @ Xs, Xs.T)                                # beta_hat = Hs y
    z0 = np.array([1.0, x0_sim])
    h0 = z0 @ np.linalg.solve(Xs.T @ Xs, z0)
    E = laws[law](reps, n_sim)                                           # true beta = 0,
    sigma = 1
    fit = E @ (Hs.T @ z0)
    s_sim = np.sqrt(np.sum((E - (E @ Hs.T) @ Xs.T) ** 2, axis=1) / (n_sim - 2))
    new = laws[law](reps)                                               # the new
    observation's error
    out = {}
    for level in (0.80, 0.95):
        t = stats.t.ppf(0.5 + level / 2, n_sim - 2)
        out[("mean", level)] = np.mean(np.abs(fit) <= t * s_sim * np.sqrt(h0))
        out[("new", level)] = np.mean(np.abs(new - fit) <= t * s_sim * np.sqrt(1 +
h0))
    return out

table = {(law, m): simulate(law, m) for law in laws for m in (20, 400)}
for (law, m), out in table.items():
    print(f"{law:12s} n = {m:3d}: " + ", ".join(f"{k[0]} {k[1]:.2f}: {v:.3f}" for k,
v in out.items()))

```

Remedies, transformation ([Chapter 22](#)) and resampling ([Chapter 23](#)), come later. Checking normality matters much more for prediction than for estimation.

*Remark* (Prediction is not intervention). A prediction interval describes the next observation from the *same process*. If the regressors are set by intervention, say a policy that changes a state's poverty rate, the observed association need not describe what happens ([Chapter 25](#)).

## 12.3.4 Exercises

### 12.3.4.1 A. Check your understanding

#### Exercise A1.

Show that the ratio of the length of the prediction interval to that of the confidence interval for the mean at  $\mathbf{x}_0$  is  $\sqrt{(1 + h_0)/h_0}$ . What is it in [Example \(Predicting a state's murder rate\)](#), and what happens to it as  $n \rightarrow \infty$  with the design growing regularly?

*Solution.* Both intervals use the multiplier  $t_{n-r, \alpha/2} s$ , with  $\sqrt{1 + h_0}$  and  $\sqrt{h_0}$  respectively. At the centroid,  $h_0 = 0.0200$  and the ratio is  $\sqrt{51} = 7.14$ . As the design grows,  $h_0 \rightarrow 0$  and the ratio tends to infinity: the confidence interval shrinks to a point while the prediction interval does not.

### 12.3.4.2 B. Practice

#### Exercise B1.

Two new observations will be made, at  $\mathbf{x}_{01}$  and  $\mathbf{x}_{02}$ , with independent errors. Find an exact  $1 - \alpha$  prediction interval for their difference  $Y_{01} - Y_{02}$ . Show that when  $\mathbf{x}_{01} = \mathbf{x}_{02}$  it does not depend on the data except through  $s$ .

*Solution.*  $Y_{01} - Y_{02} - (\mathbf{x}_{01} - \mathbf{x}_{02})^\top \hat{\boldsymbol{\beta}}$  is normal with mean zero and variance  $\sigma^2\{2 + (\mathbf{x}_{01} - \mathbf{x}_{02})^\top \mathbf{G}(\mathbf{x}_{01} - \mathbf{x}_{02})\}$ , independent of  $s^2$  by the argument of [Theorem 12.3.1](#). The interval is  $(\mathbf{x}_{01} - \mathbf{x}_{02})^\top \hat{\boldsymbol{\beta}} \pm t_{n-r, \alpha/2} s \sqrt{2 + (\mathbf{x}_{01} - \mathbf{x}_{02})^\top \mathbf{G}(\mathbf{x}_{01} - \mathbf{x}_{02})}$ . When the two points coincide it is  $\pm t_{n-r, \alpha/2} s \sqrt{2}$ : the difference of two new errors, for which the fitted model contributes only its estimate of  $\sigma$ .

#### Exercise B2.

In [Proposition 12.3.2\(c\)](#), let  $\mathbf{X}$  have full column rank and add a single column  $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$ . Show that the new leverage is

$$h_{0+} = h_0 + \frac{(z_0 - \mathbf{k}^\top \mathbf{x}_0)^2}{\|(\mathbf{I} - \mathbf{M})\mathbf{z}\|^2}, \quad \mathbf{k} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{z}.$$

So the leverage is unchanged iff  $z_0$  equals the value  $\mathbf{k}^\top \mathbf{x}_0$  predicted for it by the other regressors.

*Solution.* Use [Proposition 12.3.2\(a\)](#) for the enlarged model. Write  $\mathbf{x}_0 = \mathbf{X}^\top \boldsymbol{\rho}$  and split any feasible  $\mathbf{a}$  as  $\mathbf{a} = \mathbf{M}\mathbf{a} + \mathbf{b}$  with  $\mathbf{b} \perp \mathcal{C}(\mathbf{X})$ . The constraint  $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$  forces  $\mathbf{M}\mathbf{a} = \mathbf{M}\boldsymbol{\rho}$ , as in the proof of the proposition. The constraint  $\mathbf{z}^\top \mathbf{a} = z_0$  then reads  $\mathbf{z}^\top \mathbf{M}\boldsymbol{\rho} + \mathbf{z}^\top \mathbf{b} = z_0$ , and  $\mathbf{z}^\top \mathbf{M}\boldsymbol{\rho} = \mathbf{z}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\rho} = \mathbf{k}^\top \mathbf{x}_0$ . Since  $\mathbf{b} \perp \mathcal{C}(\mathbf{X})$ ,  $\mathbf{z}^\top \mathbf{b} = ((\mathbf{I} - \mathbf{M})\mathbf{z})^\top \mathbf{b}$ , and by the Cauchy–Schwarz inequality the shortest  $\mathbf{b}$  with  $((\mathbf{I} - \mathbf{M})\mathbf{z})^\top \mathbf{b} = z_0 - \mathbf{k}^\top \mathbf{x}_0$  is a multiple of  $(\mathbf{I} - \mathbf{M})\mathbf{z}$ , with  $\|\mathbf{b}\|^2 = (z_0 - \mathbf{k}^\top \mathbf{x}_0)^2 / \|(\mathbf{I} - \mathbf{M})\mathbf{z}\|^2$ . Adding  $\|\mathbf{M}\boldsymbol{\rho}\|^2 = h_0$  gives the formula.

### Exercise B3.

For the line through the origin,  $E Y_i = \beta x_i$  with  $\sum x_i^2 > 0$ , write the  $1 - \alpha$  confidence interval for the mean at  $x_0$  and the prediction interval for  $Y_0$ . Show that the confidence interval for the mean has length proportional to  $|x_0|$ , and is a point at  $x_0 = 0$ . Why is that reasonable, and what does it assume?

*Solution.* Here  $r = 1$ ,  $\hat{\beta} = \sum x_i y_i / \sum x_i^2$ ,  $s^2 = \sum (y_i - \hat{\beta} x_i)^2 / (n - 1)$  and  $h_0 = x_0^2 / \sum x_i^2$ . The intervals are  $\hat{\beta} x_0 \pm t_{n-1, \alpha/2} s |x_0| / \sqrt{\sum x_i^2}$  and  $\hat{\beta} x_0 \pm t_{n-1, \alpha/2} s \sqrt{1 + x_0^2 / \sum x_i^2}$ . At  $x_0 = 0$  the model asserts that the mean is exactly zero, so there is nothing to estimate. The interval is only as good as that assertion: if the true line has a small intercept, the interval near  $x_0 = 0$  is too short.

### Exercise B4.

Suppose the new regressor vector  $\mathbf{X}_0$  is random, independent of the pair  $(\mathbf{Y}, \varepsilon_0)$ , and always lies in  $\mathcal{C}(\mathbf{X}^\top)$ . Show that the prediction interval of [Theorem 12.3.1](#), computed at the realized  $\mathbf{X}_0$ , still covers  $Y_0$  with probability exactly  $1 - \alpha$ .

*Solution.* Condition on  $\mathbf{X}_0 = \mathbf{x}_0$ . By independence, given this event  $(\mathbf{Y}, \varepsilon_0)$  has its unconditional distribution, so [Theorem 12.3.1](#) applies and the conditional coverage is  $1 - \alpha$  for every  $\mathbf{x}_0$ . Averaging over the distribution of  $\mathbf{X}_0$  gives  $1 - \alpha$ .

#### 12.3.4.3 C. Going deeper

### Exercise C1.

Let  $k$  new observations be made at the rows of a  $k \times p$  matrix  $\mathbf{X}_0$  whose rows lie in  $\mathcal{C}(\mathbf{X}^\top)$ , with errors independent of each other and of  $\mathbf{Y}$ . Show that  $\mathbf{D} = \mathbf{Y}_0 - \mathbf{X}_0 \hat{\boldsymbol{\beta}} \sim N_k(\mathbf{0}, \sigma^2(\mathbf{I}_k + \mathbf{X}_0 \mathbf{G} \mathbf{X}_0^\top))$ , independent of  $s^2$ , and deduce that the intervals

$$\mathbf{x}_{0j}^\top \hat{\boldsymbol{\beta}} \pm \sqrt{k F_\alpha(k, n - r)} s \sqrt{1 + h_{0j}}, \quad j = 1, \dots, k,$$

contain all  $k$  new observations simultaneously with probability at least  $1 - \alpha$ .

*Solution.* Normality, the mean and the covariance follow as in [Theorem 12.3.1](#), and  $\mathbf{S} = \mathbf{I}_k + \mathbf{X}_0 \mathbf{G} \mathbf{X}_0^\top$  is positive definite. So  $\mathbf{D}^\top \mathbf{S}^{-1} \mathbf{D} / (ks^2) \sim F(k, n - r)$  ([Corollary 4.3.3](#) and [Definition 4.1.2](#)). By [Lemma 12.2.3](#) with  $\mathbf{a} = \mathbf{e}_j$ , the event  $\mathbf{D}^\top \mathbf{S}^{-1} \mathbf{D} \leq ks^2 F_\alpha$  implies  $|D_j| \leq \sqrt{k F_\alpha} s \sqrt{S_{jj}}$  for every  $j$ , and  $S_{jj} = 1 + h_{0j}$ . [Chapter 13](#) compares this with Bonferroni's method, which is shorter when  $k$  is small.



**Exercise C2.**

A  $(P, 1 - \alpha)$  **tolerance interval** at  $\mathbf{x}_0$  is a random interval that, with probability at least  $1 - \alpha$  over  $\mathbf{Y}$ , contains at least a proportion  $P$  of the distribution of  $Y_0$ . Show that the prediction interval of [Theorem 12.3.1](#) is not a  $(1 - \alpha, 1 - \alpha)$  tolerance interval, by showing that its conditional coverage given  $\mathbf{Y}$  is below  $1 - \alpha$  with probability bounded away from zero. Construct a conservative tolerance interval from a lower confidence bound for the conditional coverage, using a  $1 - \alpha/2$  upper confidence bound for  $\sigma$  and a  $1 - \alpha/2$  confidence interval for  $\mathbf{x}_0^\top \boldsymbol{\beta}$ .

## 12.4 Simultaneous bands for the regression line and surface

A fitted line is usually drawn inside a shaded region. If the region is made of pointwise intervals, each vertical slice covers the true mean with probability  $1 - \alpha$ , but the region does not cover the whole true line with that probability. A reader who asks “which lines are consistent with the data?” needs a simultaneous band, and for a linear model the Cauchy–Schwarz inequality gives one with exact coverage.

### 12.4.1 Pointwise is not simultaneous

**Definition 12.4.1 — Simultaneous confidence band**

Let  $\mathcal{S} \subseteq \mathcal{C}(\mathbf{X}^\top)$  be a set of regressor vectors. Random functions  $\ell(\mathbf{x}) = \ell(\mathbf{x}; \mathbf{Y})$  and  $u(\mathbf{x}) = u(\mathbf{x}; \mathbf{Y})$  form a  $1 - \alpha$  **simultaneous confidence band** for the regression function over  $\mathcal{S}$  if

$$\Pr_{\boldsymbol{\beta}, \sigma^2} \{ \ell(\mathbf{x}) \leq \mathbf{x}^\top \boldsymbol{\beta} \leq u(\mathbf{x}) \text{ for every } \mathbf{x} \in \mathcal{S} \} \geq 1 - \alpha$$

for every  $(\boldsymbol{\beta}, \sigma^2)$ .

Pointwise  $t$  intervals fail this: for the stack loss line of [Example \(A band for the stack loss line\)](#), the pointwise 95% intervals cover the true line over the observed range in only 0.8705 of simulated samples.

### 12.4.2 The Working–Hotelling band

**Theorem 12.4.1 — Working–Hotelling band**

Let  $\mathcal{L}$  be a linear subspace of  $\mathcal{C}(\mathbf{X}^\top)$  of dimension  $k \geq 1$ , and let  $w_k = \sqrt{k F_\alpha(k, n - r)}$ . Then

$$\Pr \{ |\mathbf{x}^\top \hat{\boldsymbol{\beta}} - \mathbf{x}^\top \boldsymbol{\beta}| \leq w_k s \sqrt{\mathbf{x}^\top \mathbf{G} \mathbf{x}} \text{ for all } \mathbf{x} \in \mathcal{L} \} = 1 - \alpha. \quad (12.4.1)$$

In particular:

- (a) with  $\mathcal{L} = \mathcal{C}(\mathbf{X}^\top)$  and  $k = r$ , the functions  $\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm w_r s \sqrt{\mathbf{x}^\top \mathbf{G} \mathbf{x}}$  form an exact  $1 - \alpha$  simultaneous band for the regression surface over all estimable  $\mathbf{x}$ ;
- (b) for a straight line  $E Y_i = \beta_0 + \beta_1 x_i$  with the  $x_i$  not all equal, the band

$$\hat{\beta}_0 + \hat{\beta}_1 x \pm \sqrt{2F_\alpha(2, n-2)} s \sqrt{\frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}}}, \quad x \in \mathbb{R},$$

contains the true line  $\beta_0 + \beta_1 x$  for every real  $x$  with probability exactly  $1 - \alpha$ .

*Proof. Step 1: from  $\mathcal{L}$  to a subspace of  $\mathcal{C}(\mathbf{X})$ .* The map  $\mathbf{u} \mapsto \mathbf{X}^\top \mathbf{u}$  is one to one on  $\mathcal{C}(\mathbf{X})$ , since  $\mathbf{X}^\top \mathbf{u} = \mathbf{0}$  with  $\mathbf{u} \in \mathcal{C}(\mathbf{X})$  forces  $\mathbf{u} \in \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{X})^\perp = \{\mathbf{0}\}$ . It maps  $\mathcal{C}(\mathbf{X})$  onto  $\mathcal{C}(\mathbf{X}^\top)$ , because  $\mathbf{X}^\top \boldsymbol{\rho} = \mathbf{X}^\top \mathbf{M} \boldsymbol{\rho}$ . So  $\mathcal{U} = \{\mathbf{u} \in \mathcal{C}(\mathbf{X}) : \mathbf{X}^\top \mathbf{u} \in \mathcal{L}\}$  is a subspace of  $\mathcal{C}(\mathbf{X})$  of dimension  $k$ , and every  $\mathbf{x} \in \mathcal{L}$  is  $\mathbf{X}^\top \mathbf{u}$  for exactly one  $\mathbf{u} \in \mathcal{U}$ . For such a pair,

$$\begin{aligned} \mathbf{x}^\top \hat{\boldsymbol{\beta}} - \mathbf{x}^\top \boldsymbol{\beta} &= \mathbf{u}^\top (\mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{X} \boldsymbol{\beta}) = \mathbf{u}^\top \mathbf{M} (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta}) = \mathbf{u}^\top \mathbf{v}, \\ \mathbf{x}^\top \mathbf{G} \mathbf{x} &= \mathbf{u}^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \mathbf{u} = \mathbf{u}^\top \mathbf{M} \mathbf{u} = \|\mathbf{u}\|^2, \end{aligned}$$

where  $\mathbf{v} = \mathbf{P}_\mathcal{U}(\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})$ . The middle equality uses  $\mathbf{X} \hat{\boldsymbol{\beta}} = \mathbf{M} \mathbf{Y}$  and  $\mathbf{M} \mathbf{X} = \mathbf{X}$ . The last step of the first chain uses  $\mathbf{u} = \mathbf{M} \mathbf{u} = \mathbf{P}_\mathcal{U} \mathbf{u}$ , and the second chain uses [Theorem 6.3.7](#).

*Step 2: the supremum.* By the Cauchy–Schwarz inequality, for  $\mathbf{u} \in \mathcal{U}$ ,  $\mathbf{u} \neq \mathbf{0}$ ,

$$\frac{(\mathbf{x}^\top \hat{\boldsymbol{\beta}} - \mathbf{x}^\top \boldsymbol{\beta})^2}{\mathbf{x}^\top \mathbf{G} \mathbf{x}} = \frac{(\mathbf{u}^\top \mathbf{v})^2}{\|\mathbf{u}\|^2} \leq \|\mathbf{v}\|^2,$$

with equality at  $\mathbf{u} = \mathbf{v}$  when  $\mathbf{v} \neq \mathbf{0}$ , and  $\mathbf{v} \in \mathcal{U}$ . So the supremum of the left side over  $\mathbf{x} \in \mathcal{L} \setminus \{\mathbf{0}\}$  equals  $\|\mathbf{v}\|^2$  and is attained.

*Step 3: its distribution.*  $\mathbf{Y} - \mathbf{X} \boldsymbol{\beta} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ , and  $\mathbf{P}_\mathcal{U}$  is symmetric idempotent of rank  $k$ , so  $\|\mathbf{v}\|^2 / \sigma^2 \sim \chi^2(k)$  ([Theorem 4.3.2\(a\)](#)). Since  $\mathcal{U} \subseteq \mathcal{C}(\mathbf{X})$ ,  $\mathbf{P}_\mathcal{U} = \mathbf{P}_\mathcal{U} \mathbf{M}$ , so  $\mathbf{v}$  is a function of  $\mathbf{M} \mathbf{Y}$  and is independent of SSE ([Theorem 7.5.1\(c\)](#)). Hence  $\|\mathbf{v}\|^2 / (k s^2) \sim F(k, n-r)$  ([Definition 4.1.2](#)), and the event in [Eq. 12.4.1](#), which is  $\|\mathbf{v}\|^2 \leq w_k^2 s^2$  by Step 2, has probability  $1 - \alpha$ . (At  $\mathbf{x} = \mathbf{0}$  both sides of the inequality are zero.)

*Part (a)* is the case  $\mathcal{L} = \mathcal{C}(\mathbf{X}^\top)$ ,  $\mathcal{U} = \mathcal{C}(\mathbf{X})$ .

*Part (b).* Here  $r = k = 2$  and  $\mathcal{L} = \mathbb{R}^2$ , but the band uses only the vectors  $\mathbf{x} = (1, x)^\top$ . The ratio in Step 2 is unchanged when  $\mathbf{x}$  is multiplied by a nonzero scalar, so its supremum over  $\{(1, x)^\top : x \in \mathbb{R}\}$  equals its supremum over all  $\mathbf{x} = (a, b)^\top$  with  $a \neq 0$ . By Step 2 the supremum over all of  $\mathbb{R}^2$  is attained at  $\mathbf{x}^* = \mathbf{X}^\top \mathbf{v}$ , whose first coordinate is  $\mathbf{1}^\top \mathbf{v}$ . The random variable  $\mathbf{1}^\top \mathbf{v}$  is normal with variance  $\sigma^2 \|\mathbf{P}_\mathcal{U} \mathbf{1}\|^2 = \sigma^2 n > 0$ , because  $\mathbf{1} \in \mathcal{C}(\mathbf{X}) = \mathcal{U}$ . So with probability one  $\mathbf{x}^*$  has a nonzero first coordinate, the two suprema are equal, and the band event has the probability computed in Step 3. The formula for  $\mathbf{x}^\top \mathbf{G} \mathbf{x}$  is [Eq. 5.5.2](#) divided by  $\sigma^2$ . ■

The proof has one idea: every linear function of the fitted mean is an inner product with  $\mathbf{v}$ , and the worst case over all directions is  $\|\mathbf{v}\|$ . With  $\mathcal{L} = \mathcal{C}(\mathbf{X}^\top)$  the band holds exactly when the confidence ball of [Proposition 12.2.2](#) contains  $\mathbf{X} \boldsymbol{\beta}$ : the band is that ball, viewed one linear

functional at a time. Applied to general families of estimable functions, the argument gives Scheffé's intervals, which [Chapter 13](#) develops ([Theorem 13.2.2](#)) and compares with Bonferroni and Tukey.

The straight-line band is hyperbolic, narrowest at  $\bar{x}$ . Although  $x$  varies along a line, the multiplier uses  $k = 2$ , because the band must hold for intercept and slope together.

### 12.4.3 How much wider?

The band is wider than the pointwise intervals by the factor  $w_k/t_{n-r,\alpha/2}$ , the same at every  $\mathbf{x}$ . The multipliers  $w_k = \sqrt{kF_{0.05}(k, \nu)}$  for 95% coverage are:

| $k$ | $\nu = 5$ | $\nu = 10$ | $\nu = 20$ | $\nu = 60$ | $\nu = \infty$ |
|-----|-----------|------------|------------|------------|----------------|
| 1   | 2.571     | 2.228      | 2.086      | 2.000      | 1.960          |
| 2   | 3.402     | 2.865      | 2.643      | 2.510      | 2.448          |
| 3   | 4.028     | 3.335      | 3.049      | 2.876      | 2.795          |
| 5   | 5.025     | 4.078      | 3.682      | 3.441      | 3.327          |
| 10  | 6.881     | 5.457      | 4.845      | 4.464      | 4.279          |

The row  $k = 1$  is the  $t$  multiplier; the column  $\nu = \infty$  is  $\sqrt{\chi_{0.05}^2(k)}$ , for known  $\sigma$ . For a straight line the band is about a quarter wider than the pointwise intervals, for ten coefficients more than twice as wide: for large  $k$  simultaneity costs roughly a factor  $\sqrt{k}$ .

#### Example. (A band for the stack loss line)

Regress stack loss on air flow for the 21 days of Brownlee's plant data ([Example \(Stack loss\)](#)), ignoring the other two regressors. The fitted line is  $-44.132 + 1.0203x$  with  $s = 4.098$ ,  $\bar{x} = 60.43$  and  $S_{xx} = 1681.1$ . The pointwise multiplier is  $t_{19,0.025} = 2.093$ , and the Working–Hotelling multiplier is  $\sqrt{2F_{0.05}(2, 19)} = 2.654$ , larger by the factor 1.268. [Figure 12.4.1](#) shows the data, the band, the pointwise intervals, and the 95% prediction intervals of [Section 12.3](#).

Near the centre, at  $x = 60$ , the fitted mean is 17.09, and the half-widths are 1.87 (pointwise), 2.38 (band) and 8.78 (prediction). At  $x = 80$ , the edge of the data, the fitted mean is 37.49 and the half-widths are 4.50, 5.71 and 9.69.

In 40000 data sets simulated from the fitted line, the band contained the true line over the whole real line in a proportion 0.9500 of the samples and over the observed range  $50 \leq x \leq 80$  in 0.9537. The pointwise intervals covered at  $x = 60$  in 0.9517 of the samples, but covered the line over the whole observed range in only 0.8705.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.stackloss.load_pandas().data
x = data["AIRFLOW"].to_numpy()
y = data["STACKLOSS"].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), x])
```

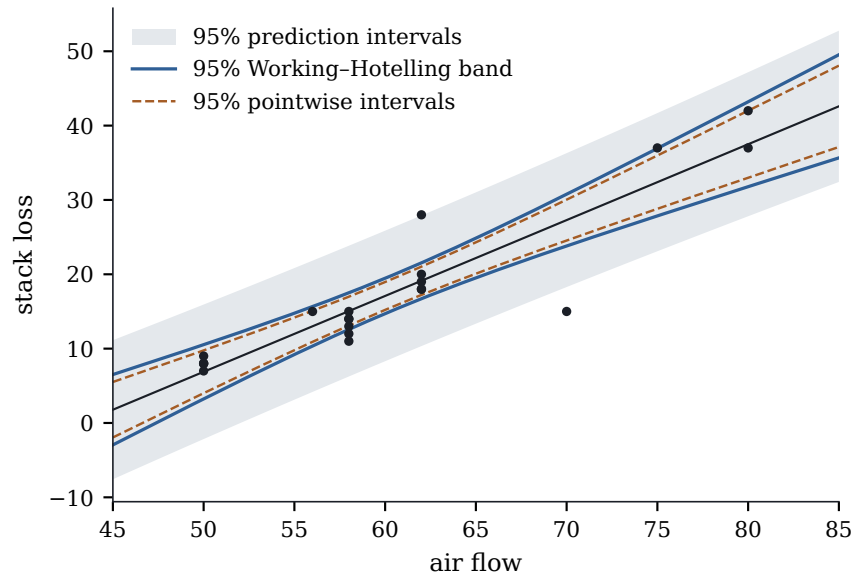


Figure 12.4.1: Stack loss against air flow: least squares line, 95% Working–Hotelling band (solid), pointwise 95% intervals for the mean (dashed) and pointwise 95% prediction intervals (shaded).

```
C = np.linalg.inv(X.T @ X)
beta_hat = C @ X.T @ y
s = np.sqrt(np.sum((y - X @ beta_hat) ** 2) / (n - 2))

grid = np.linspace(45, 85, 401)
Z = np.column_stack([np.ones_like(grid), grid])
fit = Z @ beta_hat
se_mean = s * np.sqrt(np.einsum("ij,jk,ik->i", Z, C, Z)) # s * sqrt(1/n + (x -
xbar)^2 / Sxx)
t_pt = stats.t.ppf(0.975, n - 2) # pointwise multiplier
w_wh = np.sqrt(2 * stats.f.ppf(0.95, 2, n - 2)) # Working-Hotelling
multiplier
pointwise = (fit - t_pt * se_mean, fit + t_pt * se_mean)
band = (fit - w_wh * se_mean, fit + w_wh * se_mean)
se_new = np.sqrt(s**2 + se_mean**2)
prediction = (fit - t_pt * se_new, fit + t_pt * se_new)
print(f"slope {beta_hat[1]:.4f}, s = {s:.3f}; multipliers: t {t_pt:.3f}, Working-
Hotelling {w_wh:.3f}")
```

### 12.4.4 Bands over part of the space, and other shapes

The band holds over the whole line. If only  $x \in [a, b]$  matters it is conservative (the simulated 0.9537), and an exact band over  $[a, b]$  uses a smaller multiplier (Wynn and Bloomfield 1971). The hyperbolic shape is also a choice: bands of constant width, or with straight edges, come from replacing the Euclidean norm in Step 2 by another norm (Gafarian 1964; Liu 2010). And if only some regressors vary,  $\mathcal{L}$  is smaller, and so is  $k$ .

## 12.4.5 Exercises

### 12.4.5.1 A. Check your understanding

#### Exercise A1.

For a straight line with many residual degrees of freedom, compute the ratio of the Working–Hotelling multiplier to the pointwise one at  $\alpha = 0.05$ . How does the ratio change at  $\alpha = 0.01$ ?

*Solution.* As  $\nu \rightarrow \infty$ ,  $2F_\alpha(2, \nu) \rightarrow \chi_\alpha^2(2) = -2 \log \alpha$ , since  $\chi^2(2)$  is exponential with mean 2. At  $\alpha = 0.05$  the ratio is  $\sqrt{2 \log 20} / z_{0.025} = 2.448 / 1.960 = 1.249$ . At  $\alpha = 0.01$  it is  $\sqrt{2 \log 100} / z_{0.005} = 3.035 / 2.576 = 1.178$ . The relative cost of simultaneity falls as the level becomes more demanding. As  $\alpha \rightarrow 0$ , both squared multipliers,  $z_{\alpha/2}^2$  and  $\chi_\alpha^2(2) = 2 \log(1/\alpha)$ , grow like  $2 \log(1/\alpha) + O(\log \log(1/\alpha))$ , so the ratio of the multipliers tends to 1.

#### Exercise A2.

For the line through the origin  $EY_i = \beta x_i$ , show that a simultaneous band for  $\beta x$  over all real  $x$  is  $\hat{\beta}x \pm t_{n-1, \alpha/2} s|x| / \sqrt{\sum x_i^2}$ , with no widening at all. Explain this with the dimension  $k$  of [Theorem 12.4.1](#).

*Solution.* Here  $\mathcal{C}(\mathbf{X}^\top) = \mathbb{R}$ , so  $k = 1$  and  $w_1 = t_{n-1, \alpha/2}$ . Directly: the band holds at every  $x$  iff it holds at  $x = 1$ , since both sides scale with  $|x|$ .

### 12.4.5.2 B. Practice

#### Exercise B1.

For the straight line, show that the Working–Hotelling band contains the line  $b_0 + b_1x$  for every  $x$  iff  $(b_0, b_1)$  lies in the  $1 - \alpha$  confidence ellipse for  $(\beta_0, \beta_1)$  of [Theorem 12.2.1](#). Deduce that the band is the envelope of all lines whose coefficients lie in the ellipse.

*Solution.* Put  $\mathbf{u} = (\hat{\beta}_0 - b_0, \hat{\beta}_1 - b_1)^\top$  and  $\mathbf{W} = (\mathbf{X}^\top \mathbf{X})^{-1}$ . The line  $b_0 + b_1x$  is inside the band at  $x$  iff  $|(1, x)\mathbf{u}| \leq w_2 s \sqrt{(1, x)\mathbf{W}(1, x)^\top}$ . By scale invariance this holds for all  $x$  iff it holds for all  $\mathbf{a} = (a_1, a_2)^\top$  with  $a_1 \neq 0$ , and by continuity iff it holds for all  $\mathbf{a}$ . By [Lemma 12.2.3](#) that is  $\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u} \leq w_2^2 s^2$ , which is the ellipse. At each  $x$ , the upper edge of the band is the maximum of  $b_0 + b_1x$  over the ellipse ([Corollary 12.2.4](#)), so the band is the envelope.

#### Exercise B2.

For quadratic regression  $EY_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$ , the natural band over all  $x$  uses  $\mathbf{x} = (1, x, x^2)^\top$ , a curve in  $\mathbb{R}^3$  whose directions do not fill  $\mathbb{R}^3$ . Show that the band with multiplier  $w_3$  has coverage at least  $1 - \alpha$  over all  $x$ , and argue that its coverage is strictly

larger than  $1 - \alpha$ . (Hint: when is the maximizing direction  $\mathbf{X}^\top \mathbf{v}$  of Step 2 a multiple of some  $(1, x, x^2)^\top$ ?)

*Solution.* The band event over all  $\mathbf{x} \in \mathbb{R}^3$  has probability  $1 - \alpha$  and is contained in the event over the curve, so the coverage is at least  $1 - \alpha$ . Write  $S = \|\mathbf{v}\|^2$  and  $S_c$  for the supremum of the ratio of Step 2 over the curve. The ratio depends on  $\mathbf{v}$  only through  $S$  and the direction  $\mathbf{v}/\|\mathbf{v}\|$ , so  $S_c = \rho S$ , where  $\rho \leq 1$  is a continuous function of the direction alone. Equality  $\rho = 1$  requires either that  $\mathbf{X}^\top \mathbf{v}$  be a multiple of some  $(1, x, x^2)^\top$ , that is, its coordinates  $(c_0, c_1, c_2)$  satisfy  $c_0 c_2 = c_1^2$ , or that  $\mathbf{X}^\top \mathbf{v}$  be a multiple of  $(0, 0, 1)^\top$ , the limit of  $(1, x, x^2)^\top/x^2$  as  $x \rightarrow \infty$ , where the supremum is approached but not attained. Both are events of probability zero. For a spherical normal vector in  $\mathcal{U}$ , the length and the direction are independent, and both are independent of  $s^2$ . So the coverage over the curve exceeds  $1 - \alpha$  by  $\Pr\{w_3^2 s^2 < S \leq w_3^2 s^2 / \rho\}$ , which is positive because  $S/\sigma^2$  has a positive density on  $(0, \infty)$  and  $\rho < 1$  almost surely.

### 12.4.5.3 C. Going deeper

#### Exercise C1.

A single new observation  $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$  will be taken at a regressor vector  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$  that is not known in advance. Show that the band  $\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm \sqrt{(r+1)F_\alpha(r+1, n-r)} s \sqrt{1 + \mathbf{x}^\top \mathbf{G} \mathbf{x}}$ ,  $\mathbf{x} \in \mathcal{C}(\mathbf{X}^\top)$ , contains  $Y_0$  with probability at least  $1 - \alpha$ , whatever  $\mathbf{x}_0$  turns out to be. (Hint: apply Step 2 of the proof to the  $(r+1)$ -dimensional vector  $(\mathbf{v}, \varepsilon_0)$ .)

*Solution.* For  $\mathbf{x} = \mathbf{X}^\top \mathbf{u}$  with  $\mathbf{u} \in \mathcal{C}(\mathbf{X})$ ,  $Y_0 - \mathbf{x}^\top \hat{\boldsymbol{\beta}} = \varepsilon_0 - \mathbf{u}^\top \mathbf{v}$ , with  $\mathbf{v} = \mathbf{M}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$ , and  $1 + \mathbf{x}^\top \mathbf{G} \mathbf{x} = 1 + \|\mathbf{u}\|^2$ . By Cauchy–Schwarz in  $\mathcal{C}(\mathbf{X}) \times \mathbb{R}$ ,  $(\varepsilon_0 - \mathbf{u}^\top \mathbf{v})^2 \leq (1 + \|\mathbf{u}\|^2)(\varepsilon_0^2 + \|\mathbf{v}\|^2)$ . The variable  $(\varepsilon_0^2 + \|\mathbf{v}\|^2)/\sigma^2$  is  $\chi^2(r+1)$ , independent of SSE, so  $(\varepsilon_0^2 + \|\mathbf{v}\|^2)/((r+1)s^2) \sim F(r+1, n-r)$ , and on the event that it is at most  $F_\alpha(r+1, n-r)$ , which has probability  $1 - \alpha$ , the band contains  $Y_0$  at every  $\mathbf{x}$ , in particular at  $\mathbf{x}_0$ .

#### Exercise C2.

Two straight lines are fitted in one model, with separate intercepts and slopes and a common  $\sigma^2$ . Construct simultaneous bands for both lines with joint coverage at least  $1 - \alpha$  in two ways: (i) [Theorem 12.4.1](#) with  $\mathcal{L} = \mathbb{R}^4$ , (ii) a band for each line with  $\mathcal{L} = \mathbb{R}^2$  at level  $\alpha/2$ , combined by the Bonferroni inequality. Compare the multipliers for  $n - r = 20$  and explain which is better and why.

## 12.5 Inverse prediction and calibration

Prediction runs from  $x$  to  $y$ . Calibration runs the other way: an instrument is calibrated on specimens of known value, a new specimen is measured, and its value is wanted. The estimate reads the fitted line backwards; the interval reads the prediction band backwards, and it is not always an interval.

## 12.5.1 Ratios of estimable functions

Inverse prediction belongs to a wider problem: a confidence set for a *ratio* of linear functions of the coefficients, such as the zero crossing  $-\beta_0/\beta_1$  of a line or the turning point  $-\beta_1/(2\beta_2)$  of a quadratic. A delta-method interval ignores that the denominator is uncertain and may be near zero. Fieller's approach tests, for each candidate value, a hypothesis that is linear in the coefficients.

### Proposition 12.5.1 — Fieller's confidence set for a ratio

Let  $\lambda_1, \lambda_2 \in \mathcal{C}(\mathbf{X}^\top)$  be linearly independent, put  $a = \lambda_1^\top \beta$ ,  $b = \lambda_2^\top \beta$ , their estimates  $\hat{a} = \lambda_1^\top \hat{\beta}$ ,  $\hat{b} = \lambda_2^\top \hat{\beta}$ , and  $v_{ij} = \lambda_i^\top \mathbf{G} \lambda_j$ , so that  $\mathbf{V} = (v_{ij})$  is positive definite. Suppose  $b \neq 0$  and let  $\theta = a/b$ . With  $t = t_{n-r, \alpha/2}$ , let

$$\mathcal{F}(\mathbf{Y}) = \{\theta_0 \in \mathbb{R} : (\hat{a} - \theta_0 \hat{b})^2 \leq t^2 s^2 (v_{11} - 2\theta_0 v_{12} + \theta_0^2 v_{22})\}. \quad (12.5.1)$$

- (a)  $\mathcal{F}(\mathbf{Y})$  is an exact  $1 - \alpha$  confidence set for  $\theta$ , at every parameter value with  $b \neq 0$ .
- (b) Let  $Q = (\hat{a}, \hat{b})\mathbf{V}^{-1}(\hat{a}, \hat{b})^\top$ . Outside an event of probability zero,  $\mathcal{F}(\mathbf{Y})$  is
  - a bounded closed interval containing  $\hat{\theta} = \hat{a}/\hat{b}$ , if  $\hat{b}^2 > t^2 s^2 v_{22}$ ;
  - the union of two disjoint closed rays, if  $\hat{b}^2 < t^2 s^2 v_{22}$  and  $Q > t^2 s^2$ ;
  - the whole real line, if  $\hat{b}^2 < t^2 s^2 v_{22}$  and  $Q < t^2 s^2$ .

*Proof.*

- (a) Fix  $\theta_0$ . The vector  $\lambda_1 - \theta_0 \lambda_2$  lies in  $\mathcal{C}(\mathbf{X}^\top)$  and is not zero, so  $(\lambda_1 - \theta_0 \lambda_2)^\top \beta = a - \theta_0 b$  is estimable, with estimate  $\hat{a} - \theta_0 \hat{b}$  and  $(\lambda_1 - \theta_0 \lambda_2)^\top \mathbf{G} (\lambda_1 - \theta_0 \lambda_2) = v_{11} - 2\theta_0 v_{12} + \theta_0^2 v_{22}$ . The inequality in Eq. 12.5.1 says that the  $t$  interval of Theorem 12.1.1 for  $a - \theta_0 b$  contains 0. At  $\theta_0 = \theta$ ,  $a - \theta b = 0$ , so the probability of this event is exactly  $1 - \alpha$ .
- (b) The defining inequality is  $q(\theta_0) = A\theta_0^2 - 2B\theta_0 + K \leq 0$ , with

$$A = \hat{b}^2 - t^2 s^2 v_{22}, \quad B = \hat{a}\hat{b} - t^2 s^2 v_{12}, \quad K = \hat{a}^2 - t^2 s^2 v_{11}.$$

Expanding and using  $(\hat{a}, \hat{b})\mathbf{V}^{-1}(\hat{a}, \hat{b})^\top = (v_{22}\hat{a}^2 - 2v_{12}\hat{a}\hat{b} + v_{11}\hat{b}^2)/\det \mathbf{V}$ ,

$$\begin{aligned} B^2 - AK &= t^2 s^2 (v_{22}\hat{a}^2 - 2v_{12}\hat{a}\hat{b} + v_{11}\hat{b}^2 - t^2 s^2 \det \mathbf{V}) \\ &= t^2 s^2 \det(\mathbf{V}) (Q - t^2 s^2). \end{aligned}$$

If  $A > 0$ , then  $q(\hat{\theta}) = -t^2 s^2 (v_{11} - 2\hat{a}\hat{b} + \hat{b}^2 v_{22}) < 0$ , so the upward parabola  $q$  has two real roots, and  $\{q \leq 0\}$  is the closed interval between them, which contains  $\hat{\theta}$ . If  $A < 0$ , the parabola opens downward. When  $B^2 - AK < 0$  it is negative everywhere and  $\mathcal{F}(\mathbf{Y}) = \mathbb{R}$ . When  $B^2 - AK > 0$  it is nonpositive exactly outside the open interval between its roots. The events  $A = 0$  and  $Q = t^2 s^2$  have probability zero, because  $\hat{b}$  and  $Q/s^2$  have continuous distributions. ■

The first case says that the  $t$  test rejects  $b = 0$ : the denominator is away from zero and the

ratio is confined. Otherwise very large  $|\theta|$  cannot be excluded, and  $Q/(2s^2)$ , the  $F$  statistic for  $a = b = 0$ , decides between two rays and the whole line. [Exercise \(C1\)](#) gives a picture:  $\mathcal{F}(\mathbf{Y})$  is the set of slopes of lines through the origin that meet a certain confidence ellipse for  $(a, b)$ .

## 12.5.2 The calibration problem

A calibration experiment measures  $n$  standards with known values  $x_1, \dots, x_n$ , not all equal, and records readings

$$Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \dots, n.$$

A new specimen with unknown value  $x_0$  is then measured  $m \geq 1$  times,  $Y_{0j} = \beta_0 + \beta_1 x_0 + \varepsilon_{0j}$ , and all  $n + m$  errors are independent  $N(0, \sigma^2)$ . Let  $\bar{Y}_0$  be the mean of the new readings, and let  $\hat{\beta}_0$ ,  $\hat{\beta}_1$  and  $s^2$  (with  $n - 2$  degrees of freedom) come from the standards alone. The **classical estimator** reads the fitted line backwards:

$$\hat{x}_0 = \frac{\bar{Y}_0 - \hat{\beta}_0}{\hat{\beta}_1} = \bar{x} + \frac{\bar{Y}_0 - \bar{Y}}{\hat{\beta}_1}.$$

It is the maximum likelihood estimator ([Exercise \(A2\)](#)) and, being a ratio, inherits a pathology of ratios.

### Proposition 12.5.2 — The classical estimator has no mean

If  $\Pr(\bar{Y}_0 \neq \bar{Y}) > 0$ , which always holds here, then  $E|\hat{x}_0| = \infty$ .

*Proof.*  $\bar{Y}$  and  $\hat{\beta}_1$  are jointly normal and uncorrelated ([Corollary 5.5.4](#)), hence independent ([Theorem 3.4.1](#)), and  $\bar{Y}_0$  is independent of both. So  $g = \bar{Y}_0 - \bar{Y}$  is independent of  $\hat{\beta}_1$  and  $E|\hat{x}_0 - \bar{x}| = E|g| E|1/\hat{\beta}_1|$ , with  $E|g| > 0$ . The density  $f$  of  $\hat{\beta}_1$  is continuous and positive at 0, so  $f \geq c > 0$  on some  $[-\epsilon, \epsilon]$  and  $E|1/\hat{\beta}_1| \geq c \int_{-\epsilon}^{\epsilon} |u|^{-1} du = \infty$ . Finally  $|\hat{x}_0| \geq |\hat{x}_0 - \bar{x}| - |\bar{x}|$ . ■

So a delta-method “standard error” of  $\hat{x}_0$  approximates a quantity that does not exist. That is harmless when  $\hat{\beta}_1$  is many standard errors from zero, and harmful exactly when the slope is poorly determined.

### Theorem 12.5.3 — Calibration set

In the calibration model, let  $t = t_{n-2, \alpha/2}$  and

$$\begin{aligned} \mathcal{C} = \left\{ x \in \mathbb{R} : (\bar{Y}_0 - \hat{\beta}_0 - \hat{\beta}_1 x)^2 \right. \\ \left. \leq t^2 s^2 \left( \frac{1}{m} + \frac{1}{n} + \frac{(x - \bar{x})^2}{S_{xx}} \right) \right\}. \end{aligned} \quad (12.5.2)$$

- (a)  $\mathcal{C}$  is an exact  $1 - \alpha$  confidence set for  $x_0$ , at every  $(\beta_0, \beta_1, \sigma^2, x_0)$ , including  $\beta_1 = 0$ .
- (b)  $\mathcal{C}$  is the set of  $x$  at which  $\bar{Y}_0$  lies inside the  $1 - \alpha$  prediction interval of [Theorem 12.3.1\(c\)](#) for the mean of  $m$  new readings at  $x$ .
- (c) Let  $T_1 = \hat{\beta}_1 \sqrt{S_{xx}}/s$  be the  $t$  statistic for the slope and  $g = \bar{Y}_0 - \bar{Y}$ . Outside an



event of probability zero,  $\mathcal{C}$  is

- a bounded interval containing  $\hat{x}_0$ , if  $T_1^2 > t^2$ ;
- the union of two disjoint closed rays, if  $T_1^2 < t^2$  and  $T_1^2 + g^2 / \{s^2(1/m + 1/n)\} > t^2$ ;
- the whole real line, if  $T_1^2 + g^2 / \{s^2(1/m + 1/n)\} < t^2$ .

*Proof.*

- (a) and (b). For fixed  $x$ ,  $\bar{Y}_0 - \hat{\beta}_0 - \hat{\beta}_1 x$  is the difference between the mean of the new readings and the estimated mean response at  $x$ . At  $x = x_0$ , [Theorem 12.3.1\(c\)](#), with  $\mathbf{x}_0 = (1, x_0)^\top$  and  $h_0 = 1/n + (x_0 - \bar{x})^2 / S_{xx}$  ([Eq. 5.5.2](#)), shows that

$$\frac{\bar{Y}_0 - \hat{\beta}_0 - \hat{\beta}_1 x_0}{s\sqrt{1/m + 1/n + (x_0 - \bar{x})^2 / S_{xx}}} \sim t(n-2),$$

whatever the parameters are. This is a pivot, and  $x_0 \in \mathcal{C}$  is the event that its absolute value is at most  $t$ . Nothing in the argument requires  $\beta_1 \neq 0$ . Part (b) restates the definition.

- (b) Put  $d = x - \bar{x}$  and  $c = 1/m + 1/n$ . Since  $\bar{Y}_0 - \hat{\beta}_0 - \hat{\beta}_1 x = g - \hat{\beta}_1 d$ , the defining inequality is  $Ad^2 - 2\hat{\beta}_1 g d + (g^2 - t^2 s^2 c) \leq 0$  with  $A = \hat{\beta}_1^2 - t^2 s^2 / S_{xx}$ . Its discriminant is

$$\begin{aligned} \hat{\beta}_1^2 g^2 - A(g^2 - t^2 s^2 c) &= t^2 s^2 \left( \frac{g^2}{S_{xx}} + cA \right) \\ &= \frac{t^2 s^4 c}{S_{xx}} \left( T_1^2 + \frac{g^2}{s^2 c} - t^2 \right). \end{aligned}$$

The sign of  $A$  is that of  $T_1^2 - t^2$ . If  $A > 0$ , the quadratic is negative at  $d = g/\hat{\beta}_1$ , that is at  $x = \hat{x}_0$ , where it equals  $-t^2 s^2 (c + d^2 / S_{xx})$ ; so it has two real roots and  $\mathcal{C}$  is the interval between them. If  $A < 0$ , the argument of [Proposition 12.5.1\(b\)](#) applies: two rays if the discriminant is positive and the whole line if it is negative. Equalities have probability zero. ■

Part (b) is the practical recipe ([Figure 12.5.1](#)): draw the prediction band for the mean of  $m$  readings and the horizontal line at  $\bar{Y}_0$ , and read off where the line is inside the band. A steep calibration line crosses the band in a short interval. A flat one lets both hyperbolic edges pass the horizontal line, giving two rays or everything. The set is bounded iff the slope test rejects, as it does for any useful instrument.

### Proposition 12.5.4 — Unbounded calibration sets cannot be avoided

Let  $\mathcal{D}$  be any confidence set for  $x_0$  with coverage at least  $1 - \alpha > 0$  at every parameter value, including those with  $\beta_1 = 0$ . Then at every parameter value with  $\beta_1 = 0$ ,  $\Pr\{v \in \mathcal{D}\} \geq 1 - \alpha$  for every real  $v$ , and if  $\mathcal{D}$  is measurable in the sense of [Proposition 12.1.3](#), its expected length is infinite.

*Proof.* When  $\beta_1 = 0$ , the joint distribution of the standards and the new readings does not involve  $x_0$ . So for every  $v$ , the probability that  $v \in \mathcal{D}$  is the same as it would be if  $x_0$  were  $v$ , and that probability is at least  $1 - \alpha$ . The rest is the Fubini argument in the

proof of [Proposition 12.1.3](#).

■

By continuity the same happens approximately for small  $\beta_1$ . A procedure that always reports a bounded interval, like the delta method, pays with coverage below  $1 - \alpha$  somewhere. The Fieller set's unbounded answers are the honest answer to a question the data cannot settle.

### Example. (Calibrating an assay)

In a simulated assay, nine standards of concentration  $0, 1, \dots, 8$  are each measured three times ( $n = 27$ ). The fitted line is  $0.0338 + 0.1161x$ , with  $s = 0.0131$  and slope  $t$  statistic 118.7. A new specimen is read  $m = 3$  times, with mean reading 0.6521. The classical estimate is  $\hat{x}_0 = 5.327$ , and the 95% calibration interval is  $(5.184, 5.471)$  ( $t_{25,0.025} = 2.060$ ). For an assay this precise the delta-method interval,  $\hat{x}_0 \pm t(s/|\hat{\beta}_1|)\sqrt{1/m + 1/n + (\hat{x}_0 - \bar{x})^2/S_{xx}}$ , agrees to three decimals:  $(5.184, 5.471)$ .

Now consider a weak assay with the same design, in which the fitted slope is 0.0021 and its  $t$  statistic is only 1.67. The mean reading of the standards is 0.0475. A specimen with mean reading 0.0525, close to that mean, gives the whole real line: nothing about its concentration can be concluded. A specimen with mean reading 0.0775 gives two rays,  $(-\infty, -55.3] \cup [7.6, \infty)$ : the middle of the scale is excluded, but not extreme concentrations of either sign. [Figure 12.5.1](#) shows both assays.

In 20000 simulated experiments at  $x_0 = 5.3$ , both intervals cover in 0.9517 for the precise assay. With the slope three standard errors from zero ( $\beta_1 = 0.00335$ ) the Fieller set again covers in 0.9517, since its event involves only the errors, and is bounded in 0.817 of the experiments, close to the power 0.822 of the slope  $t$  test; the delta-method interval covers in 0.9827, wastefully. With the slope two standard errors from zero and  $x_0 = 12$ , beyond the standards, the Fieller set covers in 0.9528 and the delta-method interval in only 0.9337.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(1206)
x = np.repeat(np.arange(9.0), 3)           # standards, in triplicate
n = len(x)
y = 0.04 + 0.115 * x + 0.015 * rng.normal(size=n) # absorbance readings
m = 3
y0 = 0.04 + 0.115 * 5.3 + 0.015 * rng.normal(size=m) # the unknown sample, true x0
    = 5.3

def calibrate(x, y, y0bar, m, level=0.95):
    """Fieller set {x0 : |y0bar - b0 - b1 x0| <= t s sqrt(1/m + 1/n + (x0 - xbar)^2 / Sxx)}. """
    n = len(x)
    xbar, Sxx = x.mean(), np.sum((x - x.mean()) ** 2)
    b1 = np.sum((x - xbar) * (y - y.mean())) / Sxx
    s2 = np.sum((y - y.mean() - b1 * (x - xbar)) ** 2) / (n - 2)
    t2 = stats.t.ppf(0.5 + level / 2, n - 2) ** 2
    g = y0bar - y.mean()
    # quadratic in d = x0 - xbar: A d^2 - 2 B d + K <= 0
```

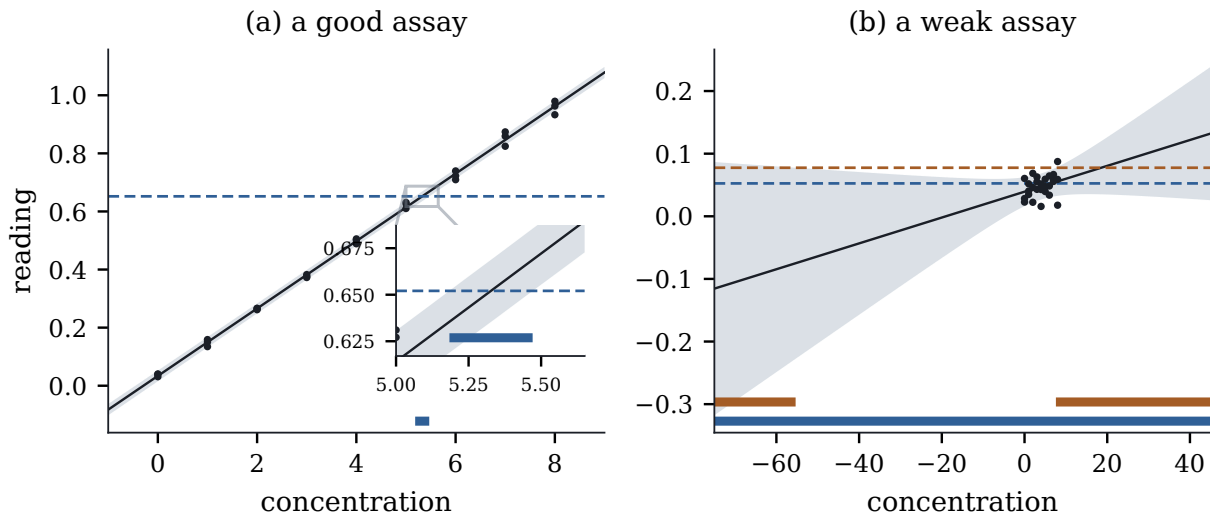


Figure 12.5.1: Calibration inverts the 95% prediction band for the mean of  $m = 3$  readings (shaded); bars mark the calibration sets for the observed readings (horizontal lines). (a) A precise assay, magnified in the inset. (b) A weak assay: whole line (lower bar) and two rays (upper bar).

```

A = b1**2 - t2 * s2 / Sxx
B = b1 * g
K = g**2 - t2 * s2 * (1 / m + 1 / n)
disc = B**2 - A * K
if disc < 0:
    return "whole line", (-np.inf, np.inf), xbar + g / b1
r1, r2 = sorted([(B - np.sqrt(disc)) / A, (B + np.sqrt(disc)) / A])
if A > 0:
    return "interval", (xbar + r1, xbar + r2), xbar + g / b1
return "two rays", (xbar + r1, xbar + r2), xbar + g / b1  # (-inf, first] and [
second, inf)

shape, (lo, hi), x0_hat = calibrate(x, y, y0.mean(), m)
print(f"estimate {x0_hat:.3f}; 95% calibration set: {shape} ({lo:.3f}, {hi:.3f})")

```

*Remark* (Classical or inverse estimation?). The *inverse* estimator regresses the  $x_i$  on the readings and predicts from  $\bar{Y}_0$ :  $\tilde{x}_0 = \bar{x} + (S_{xy}/S_{yy})(\bar{Y}_0 - \bar{Y})$ , with  $S_{xy} = \sum(x_i - \bar{x})(Y_i - \bar{Y})$  and  $S_{yy} = \sum(Y_i - \bar{Y})^2$ . Since  $\bar{Y}_0 - \bar{Y} = \hat{\beta}_1(\hat{x}_0 - \bar{x})$  and  $\hat{\beta}_1 = S_{xy}/S_{xx}$ ,

$$\tilde{x}_0 - \bar{x} = \frac{S_{xy}^2}{S_{xx}S_{yy}}(\hat{x}_0 - \bar{x}) = R^2(\hat{x}_0 - \bar{x}),$$

where  $R$  is the sample correlation of the calibration pairs. So  $\tilde{x}_0$  shrinks  $\hat{x}_0$  toward  $\bar{x}$  by the factor  $R^2 \in [0, 1]$ . It has finite moments and can beat  $\hat{x}_0$  in mean squared error near the centre of the standards (Krutchhoff 1967); away from the centre it does worse (Exercise (B3)). Hoadley (1970) showed that it is a Bayes estimator under a particular prior for  $x_0$ .

## 12.5.3 Exercises

### 12.5.3.1 A. Check your understanding

#### Exercise A1.

For a straight line, write down Fieller's  $1 - \alpha$  confidence set for the point  $\xi = -\beta_0/\beta_1$  where the line crosses zero, and state the condition under which it is a bounded interval. Show that it is the calibration set of [Theorem 12.5.3](#) with  $\bar{Y}_0$  replaced by 0 and  $1/m$  replaced by 0.

*Solution.* With  $\lambda_1 = (1, 0)^\top$  and  $\lambda_2 = (0, 1)^\top$ ,  $\xi = -a/b$ . Apply [Proposition 12.5.1](#) to  $\theta = a/b$  and change sign, or directly:  $\xi_0$  is in the set iff the  $t$  interval for  $\beta_0 + \beta_1\xi_0$  contains 0, that is iff  $(\hat{\beta}_0 + \hat{\beta}_1\xi_0)^2 \leq t^2 s^2 \{1/n + (\xi_0 - \bar{x})^2/S_{xx}\}$ . This is [Eq. 12.5.2](#) with  $\bar{Y}_0 = 0$  and the  $1/m$  term absent, because 0 is a known number rather than an average of new readings. It is bounded iff  $T_1^2 > t^2$ .

#### Exercise A2.

Show that  $\hat{x}_0$ , together with the least squares estimates from the standards, maximizes the joint likelihood of the  $n + m$  readings over  $(\beta_0, \beta_1, \sigma^2, x_0)$  when  $\hat{\beta}_1 \neq 0$ . What is the maximum likelihood estimate of  $\sigma^2$ ?

*Solution.* For fixed  $\sigma^2$  the likelihood is maximized by minimizing  $\sum_i (Y_i - \beta_0 - \beta_1 x_i)^2 + \sum_j (Y_{0j} - \beta_0 - \beta_1 x_0)^2$ . For any  $\beta_1 \neq 0$  the second sum is minimized over  $x_0$  by choosing  $\beta_0 + \beta_1 x_0 = \bar{Y}_0$ , leaving  $\sum_j (Y_{0j} - \bar{Y}_0)^2$ , which is free of the parameters. The first sum is then minimized by least squares on the standards, and  $x_0 = (\bar{Y}_0 - \hat{\beta}_0)/\hat{\beta}_1 = \hat{x}_0$ . The maximum likelihood estimate of  $\sigma^2$  is  $\{\text{SSE} + \sum_j (Y_{0j} - \bar{Y}_0)^2\}/(n + m)$ .

### 12.5.3.2 B. Practice

#### Exercise B1.

When  $m \geq 2$ , the new readings carry their own information about  $\sigma^2$ . Let  $s_p^2 = \{\text{SSE} + \sum_j (Y_{0j} - \bar{Y}_0)^2\}/(n + m - 3)$ . Show that replacing  $s$  by  $s_p$  and  $t_{n-2, \alpha/2}$  by  $t_{n+m-3, \alpha/2}$  in [Eq. 12.5.2](#) gives another exact  $1 - \alpha$  set. Why is it shorter on average, and why is it more often bounded?

*Solution.* The within-specimen sum of squares is  $\sigma^2 \chi^2(m - 1)$ , independent of  $\bar{Y}_0$  (the sample mean and sample variance of a normal sample are independent) and of the standards. So  $\{\text{SSE} + \sum_j (Y_{0j} - \bar{Y}_0)^2\}/\sigma^2 \sim \chi^2(n + m - 3)$ , independent of the numerator of the pivot, and the pivot with  $s_p$  is  $t(n + m - 3)$ . More degrees of freedom give a smaller multiplier and a less variable estimate of  $\sigma$ , so the set is shorter on average. The bounded case requires  $\hat{\beta}_1^2 S_{xx} > t^2 s_p^2$ , and a smaller  $t$  makes this easier.

#### Exercise B2.

In the quadratic regression  $EY_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2$ , use [Proposition 12.5.1](#) to give

a confidence set for the location  $x^* = -\beta_1/(2\beta_2)$  of the turning point. When is it a bounded interval, and what does that condition mean for the curvature?

*Solution.* Take  $\lambda_1 = (0, 1, 0)^\top$ ,  $\lambda_2 = (0, 0, 2)^\top$  and  $\theta = \beta_1/(2\beta_2)$ , so  $x^* = -\theta$ . The set is  $\{x_0 : (\hat{\beta}_1 + 2\hat{\beta}_2 x_0)^2 \leq t^2 s^2 (c_{11} + 4c_{12}x_0 + 4c_{22}x_0^2)\}$ , where  $c_{ij}$  are the entries of  $(\mathbf{X}^\top \mathbf{X})^{-1}$  belonging to  $\beta_1, \beta_2$ . It is the set of  $x_0$  at which the estimated derivative  $\beta_1 + 2\beta_2 x_0$  is not significantly different from zero. It is bounded iff  $\hat{\beta}_2^2 > t^2 s^2 c_{22}$ , that is iff the curvature is significant: without significant curvature, the fitted curve may be nearly straight and its turning point anywhere.

### Exercise B3.

Study the two estimators of the Remark “Classical or inverse estimation?” in an idealized large calibration experiment: the standards are so numerous that  $\hat{\beta}_0, \hat{\beta}_1$  and  $R^2$  may be replaced by constants  $\beta_0, \beta_1 \neq 0$  and  $\rho^2 \in (0, 1)$ , while  $\bar{Y}_0$  is still the mean of  $m$  new readings. Let  $v = \sigma^2/(m\beta_1^2)$ .

1. Show that  $\hat{x}_0$  is unbiased with mean squared error  $v$ , and that  $\tilde{x}_0 = \bar{x} + \rho^2(\hat{x}_0 - \bar{x})$  has mean squared error  $\rho^4 v + (1 - \rho^2)^2(x_0 - \bar{x})^2$ .
2. Show that  $\tilde{x}_0$  has the smaller mean squared error iff  $(x_0 - \bar{x})^2 < v(1 + \rho^2)/(1 - \rho^2)$ .
3. Let  $\tau^2 = S_{xx}/n$ , so that in this limit  $\rho^2 = \beta_1^2 \tau^2 / (\beta_1^2 \tau^2 + \sigma^2)$ . Show that the condition of part 2 reads  $(x_0 - \bar{x})^2 < v + 2\tau^2/m$ . With  $m = 1$  and a precise instrument, roughly how far from  $\bar{x}$ , in units of the spread  $\tau$  of the standards, does the inverse estimator stay ahead?

*Solution.*

1. Here  $\hat{x}_0 = x_0 + \bar{\varepsilon}_0/\beta_1$  with  $\bar{\varepsilon}_0 \sim N(0, \sigma^2/m)$ , so it is unbiased with variance  $v$ . Then  $\tilde{x}_0 - x_0 = \rho^2(\hat{x}_0 - x_0) - (1 - \rho^2)(x_0 - \bar{x})$ : variance  $\rho^4 v$  plus squared bias  $(1 - \rho^2)^2(x_0 - \bar{x})^2$ .
2. The inequality  $\rho^4 v + (1 - \rho^2)^2(x_0 - \bar{x})^2 < v$  is  $(1 - \rho^2)^2(x_0 - \bar{x})^2 < (1 - \rho^2)(1 + \rho^2)v$ ; divide by  $(1 - \rho^2)^2 > 0$ .
3.  $(1 + \rho^2)/(1 - \rho^2) = (2\beta_1^2 \tau^2 + \sigma^2)/\sigma^2$ , so  $v(1 + \rho^2)/(1 - \rho^2) = v + 2\tau^2/m$ . When  $\sigma$  is small,  $v \approx 0$  and the inverse estimator wins only for  $|x_0 - \bar{x}| < \sqrt{2}\tau$ , a little more than one spread from the centre; beyond that its bias dominates.

### 12.5.3.3 C. Going deeper

#### Exercise C1.

In the setting of [Proposition 12.5.1](#), let  $E_t = \{\boldsymbol{\phi} \in \mathbb{R}^2 : ((\hat{a}, \hat{b})^\top - \boldsymbol{\phi})^\top \mathbf{V}^{-1}((\hat{a}, \hat{b})^\top - \boldsymbol{\phi}) \leq t^2 s^2\}$ , an ellipse with the  $t$  constant in place of  $2F_\alpha(2, n - r)$ . Show that  $\theta_0 \in \mathcal{F}(\mathbf{Y})$  iff the line  $\{(\phi_1, \phi_2) : \phi_1 = \theta_0 \phi_2\}$  meets  $E_t$ . Use the picture to explain the three cases: the set is the whole line iff  $E_t$  contains the origin, and it is bounded iff  $E_t$  does not meet the axis  $\phi_2 = 0$ .

*Solution.* The line is  $\{\boldsymbol{\phi} : \mathbf{c}^\top \boldsymbol{\phi} = 0\}$  with  $\mathbf{c} = (1, -\theta_0)^\top$ . By [Corollary 12.2.4](#) (with the constant  $t^2 s^2$ ), the values of  $\mathbf{c}^\top \boldsymbol{\phi}$  over  $E_t$  form the interval  $\hat{a} - \theta_0 \hat{b} \pm ts\sqrt{\mathbf{c}^\top \mathbf{V} \mathbf{c}}$ , and  $\mathbf{c}^\top \mathbf{V} \mathbf{c} = v_{11} - 2\theta_0 v_{12} + \theta_0^2 v_{22}$ . So the line meets  $E_t$  iff this interval contains 0, which is [Eq. 12.5.1](#). Every line through the origin meets  $E_t$  iff the origin is in  $E_t$ ,

which is  $Q \leq t^2 s^2$ . Lines of large slope approach the axis  $\phi_2 = 0$  (the  $\phi_1$ -axis in these coordinates); the set is unbounded iff lines of arbitrarily large slope meet  $E_t$ , iff  $E_t$  meets that axis, which by [Corollary 12.2.4](#) with  $\mathbf{c} = (0, 1)^\top$  is  $\hat{b}^2 \leq t^2 s^2 v_{22}$ .

### Exercise C2.

Two straight lines  $EY = \alpha_k + \beta_k x$ ,  $k = 1, 2$ , are fitted to two independent samples in one model with a common  $\sigma^2$ , as in a two-phase regression. Use [Proposition 12.5.1](#) to construct a confidence set for the  $x$ -coordinate  $(\alpha_1 - \alpha_2)/(\beta_2 - \beta_1)$  of their intersection. When is it bounded? Interpret the unbounded cases in terms of parallel lines.

## 12.6 Bayesian thread II: credible intervals and predictive distributions

[Section 7.6](#) derived the posterior distribution of  $(\boldsymbol{\beta}, \sigma^2)$  under the conjugate normal-inverse-gamma prior, and showed that the flat prior  $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$  reproduces least squares. Here the posterior is summarized by credible regions and used to predict. In the conjugate model these have the shapes of the frequentist sets, and under the flat prior they coincide with them. We end by asking how often a credible interval covers in repeated sampling.

### 12.6.1 Credible sets

#### Definition 12.6.1 — Credible set

Let  $\theta$  be a function of the parameters and let  $\pi(\cdot | \mathbf{y})$  be its posterior distribution. A set  $C(\mathbf{y})$  with  $\Pr\{\theta \in C(\mathbf{y}) | \mathbf{y}\} \geq 1 - \alpha$  is a  $1 - \alpha$  **credible set** for  $\theta$ . If the posterior has a density  $\pi(\theta | \mathbf{y})$ , a set of the form  $\{\theta : \pi(\theta | \mathbf{y}) \geq k\}$  with posterior probability  $1 - \alpha$  is a **highest posterior density** (HPD) set.

Here the statement concerns the realized set. An HPD set has the smallest volume among sets with its posterior probability. We need three facts about the multivariate  $t$  distribution of [Definition 7.6.2](#).

#### Lemma 12.6.1 — Multivariate $t$

Let  $\mathbf{u} \sim t_\nu(\mathbf{m}, \mathbf{S})$  in  $\mathbb{R}^p$ , with  $\mathbf{S}$  positive definite.

- (a) If  $\mathbf{A}$  is  $q \times p$  of rank  $q$ , then  $\mathbf{A}\mathbf{u} \sim t_\nu(\mathbf{A}\mathbf{m}, \mathbf{A}\mathbf{S}\mathbf{A}^\top)$ .
- (b)  $(\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m})/p \sim F(p, \nu)$ .
- (c)  $\mathbf{u}$  has density proportional to  $\{1 + (\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m})/\nu\}^{-(\nu+p)/2}$ .

*Proof.* By Definition 7.6.2,  $\mathbf{u} = \mathbf{m} + \mathbf{Z}/\sqrt{W/\nu}$  with  $\mathbf{Z} \sim N_p(\mathbf{0}, \mathbf{S})$  and  $W \sim \chi^2(\nu)$  independent.

- (a)  $\mathbf{A}\mathbf{u} = \mathbf{A}\mathbf{m} + \mathbf{A}\mathbf{Z}/\sqrt{W/\nu}$ , and  $\mathbf{A}\mathbf{Z} \sim N_q(\mathbf{0}, \mathbf{A}\mathbf{S}\mathbf{A}^\top)$  (Theorem 3.3.1) is independent of  $W$ , with  $\mathbf{A}\mathbf{S}\mathbf{A}^\top$  positive definite.
- (b)  $(\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m}) = \mathbf{Z}^\top \mathbf{S}^{-1} \mathbf{Z}/(W/\nu)$ , and  $\mathbf{Z}^\top \mathbf{S}^{-1} \mathbf{Z} \sim \chi^2(p)$  independently of  $W$  (Corollary 4.3.3(a)). Divide by  $p$  and use Definition 4.1.2.
- (c) Given  $W = w$ ,  $\mathbf{u} \sim N_p(\mathbf{m}, (\nu/w)\mathbf{S})$ . Write  $D = (\mathbf{u} - \mathbf{m})^\top \mathbf{S}^{-1}(\mathbf{u} - \mathbf{m})$ . Integrating the normal density (Theorem 3.2.2) against the  $\chi^2(\nu)$  density of  $W$ , and dropping factors free of  $\mathbf{u}$ ,

$$\int_0^\infty w^{p/2} e^{-wD/(2\nu)} w^{\nu/2-1} e^{-w/2} dw = \int_0^\infty w^{(\nu+p)/2-1} e^{-w(1+D/\nu)/2} dw \\ \propto (1 + D/\nu)^{-(\nu+p)/2},$$

by the substitution  $w' = w(1 + D/\nu)$ . ■

## 12.6.2 Credible sets and predictive distributions in the conjugate model

### Theorem 12.6.2 — Credible sets and posterior predictive distribution

Let the prior be  $\text{NIG}(\mathbf{m}_0, \mathbf{V}_0, a_0, b_0)$ , so that by Theorem 7.6.2 the posterior is  $\text{NIG}(\mathbf{m}_n, \mathbf{V}_n, a_n, b_n)$ . Put  $\nu = 2a_n$  and  $\tau^2 = b_n/a_n$ .

- (a) For  $\boldsymbol{\lambda} \neq \mathbf{0}$ , the interval

$$\boldsymbol{\lambda}^\top \mathbf{m}_n \pm t_{\nu, \alpha/2} \tau \sqrt{\boldsymbol{\lambda}^\top \mathbf{V}_n \boldsymbol{\lambda}}$$

is the  $1 - \alpha$  HPD credible interval for  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ ; it is also equal-tailed.

- (b) For a  $p \times q$  matrix  $\boldsymbol{\Lambda}$  of rank  $q$ , the set

$$\{\boldsymbol{\Phi}_0 : (\boldsymbol{\Phi}_0 - \boldsymbol{\Lambda}^\top \mathbf{m}_n)^\top (\boldsymbol{\Lambda}^\top \mathbf{V}_n \boldsymbol{\Lambda})^{-1} (\boldsymbol{\Phi}_0 - \boldsymbol{\Lambda}^\top \mathbf{m}_n) \\ \leq q \tau^2 F_\alpha(q, \nu)\}$$

is the  $1 - \alpha$  HPD credible region for  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}$ .

- (c) Let  $\mathbf{Y}_0 = \mathbf{X}_0 \boldsymbol{\beta} + \boldsymbol{\varepsilon}_0$  be  $k$  future observations, with  $\boldsymbol{\varepsilon}_0 \sim N_k(\mathbf{0}, \sigma^2 \mathbf{I}_k)$  independent of  $\mathbf{Y}$  and of  $\boldsymbol{\beta}$  given  $\sigma^2$ . The posterior predictive distribution of  $\mathbf{Y}_0$  is

$$\mathbf{Y}_0 \mid \mathbf{y} \sim t_\nu(\mathbf{X}_0 \mathbf{m}_n, \tau^2(\mathbf{I}_k + \mathbf{X}_0 \mathbf{V}_n \mathbf{X}_0^\top)).$$

In particular, for one future observation at  $\mathbf{x}_0$ , the interval  $\mathbf{x}_0^\top \mathbf{m}_n \pm t_{\nu, \alpha/2} \tau \sqrt{1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0}$  has posterior predictive probability  $1 - \alpha$ .

(d)  $[2b_n/\chi_{\alpha/2}^2(\nu), 2b_n/\chi_{1-\alpha/2}^2(\nu)]$  is an equal-tailed  $1 - \alpha$  credible interval for  $\sigma^2$ .

*Proof.* By Theorem 7.6.2 and Lemma 7.6.1,  $\boldsymbol{\beta} \mid \mathbf{y} \sim t_\nu(\mathbf{m}_n, \tau^2 \mathbf{V}_n)$  and  $2b_n/\sigma^2 \mid \mathbf{y} \sim \chi^2(\nu)$ .

- (a) By Lemma 12.6.1(a) with  $\mathbf{A} = \boldsymbol{\lambda}^\top$ ,  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \boldsymbol{\lambda}^\top \mathbf{m}_n + \tau \sqrt{\boldsymbol{\lambda}^\top \mathbf{V}_n \boldsymbol{\lambda}} T$  with  $T \sim t(\nu)$ . The  $t(\nu)$  density is symmetric and decreasing in  $|x|$ , so the symmetric interval is both equal-tailed and HPD.
- (b) By Lemma 12.6.1(a),  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} \mid \mathbf{y} \sim t_\nu(\boldsymbol{\Lambda}^\top \mathbf{m}_n, \tau^2 \boldsymbol{\Lambda}^\top \mathbf{V}_n \boldsymbol{\Lambda})$ . By (b) of the lemma the quadratic form divided by  $q\tau^2$  is  $F(q, \nu)$ , so the set has posterior probability  $1 - \alpha$ . By (c) the density is a decreasing function of the quadratic form, so the set has the form  $\{\text{density} \geq k\}$ .
- (c) Given  $\sigma^2$  and  $\mathbf{y}$ ,  $\boldsymbol{\beta} \sim N_p(\mathbf{m}_n, \sigma^2 \mathbf{V}_n)$  and  $\boldsymbol{\varepsilon}_0 \sim N_k(\mathbf{0}, \sigma^2 \mathbf{I}_k)$  independently, so  $\mathbf{Y}_0 \sim N_k(\mathbf{X}_0 \mathbf{m}_n, \sigma^2(\mathbf{X}_0 \mathbf{V}_n \mathbf{X}_0^\top + \mathbf{I}_k))$  (Theorem 3.3.1). Together with  $\sigma^2 \mid \mathbf{y} \sim \text{IG}(a_n, b_n)$ , this says that  $(\mathbf{Y}_0, \sigma^2) \mid \mathbf{y}$  is NIG( $\mathbf{X}_0 \mathbf{m}_n, \mathbf{I}_k + \mathbf{X}_0 \mathbf{V}_n \mathbf{X}_0^\top, a_n, b_n$ ) (Definition 7.6.1), the matrix being positive definite. Lemma 7.6.1 gives the  $t_\nu$  law, and (a) of the present theorem, applied to it with  $k = 1$ , gives the interval.
- (d)  $\Pr\{\chi_{1-\alpha/2}^2(\nu) \leq 2b_n/\sigma^2 \leq \chi_{\alpha/2}^2(\nu) \mid \mathbf{y}\} = 1 - \alpha$ ; invert the inequalities.

■

The formulas mirror the frequentist ones:  $\hat{\boldsymbol{\beta}}$  becomes  $\mathbf{m}_n$ ,  $\mathbf{G}$  becomes  $\mathbf{V}_n = (\mathbf{V}_0^{-1} + \mathbf{X}^\top \mathbf{X})^{-1}$ ,  $s^2$  becomes  $\tau^2$ , and  $n - r$  becomes  $\nu = 2a_0 + n$ , which counts every observation plus  $2a_0$  “prior observations” about  $\sigma^2$ . The 1 in  $1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$  is again the new error.

### Corollary 12.6.3 — The flat prior reproduces the frequentist sets

Let  $\text{rank}(\mathbf{X}) = p < n$ ,  $\text{SSE} > 0$ , and let the prior be  $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$ . Then the credible interval (a), the region (b), the predictive interval (c) and the interval for  $\sigma^2$  (d) of Theorem 12.6.2 are, respectively, the  $t$  interval of Theorem 12.1.1, the confidence ellipsoid of Theorem 12.2.1, the prediction interval of Theorem 12.3.1 and the interval of Proposition 12.1.4.

*Proof.* By Corollary 7.6.4 the posterior is NIG( $\hat{\boldsymbol{\beta}}, (\mathbf{X}^\top \mathbf{X})^{-1}, (n - p)/2, \text{SSE}/2$ ). The proof of Theorem 12.6.2 used only that the posterior is normal-inverse-gamma, so its conclusions hold with  $\mathbf{m}_n = \hat{\boldsymbol{\beta}}$ ,  $\mathbf{V}_n = (\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{G}$ ,  $\nu = n - p = n - r$ ,  $\tau^2 = \text{SSE}/(n - p) = s^2$  and  $2b_n = \text{SSE}$ . Substituting gives the four frequentist sets.

■

The same set then carries two readings: “this procedure covers the fixed truth in 95% of samples” and “given these data, the parameter lies here with probability 0.95”. The agreement is special; it fails for proper priors and for most other models.

## 12.6.3 A worked example



**Example. (Predicting a day of stack loss)**

Take the stack loss regression with the engineer's prior of [Example \(A prior for the stack loss plant\)](#), and consider a new day with air flow 65, water temperature 23 and acid concentration 87. The posterior has  $a_n = 13.5$ , so  $\nu = 27$  and  $t_{27,0.025} = 2.052$ . By [Theorem 12.6.2](#), the 95% credible interval for the mean stack loss at these conditions is  $23.12 \pm 1.51$ , and the 95% posterior predictive interval for the day's stack loss is  $23.12 \pm 6.11$ . The least squares answers are  $23.15 \pm 1.77$  for the mean and  $23.15 \pm 7.07$  for the new day, with  $t_{17,0.025} = 2.110$ . The Bayesian intervals are shorter because the prior informs the slopes, shrinking  $\mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$ , and  $\sigma^2$ , raising the degrees of freedom from 17 to 27.

The 2.5% and 97.5% points of 400000 posterior predictive draws are 17.00 and 29.21, matching the  $t$  interval within Monte Carlo error. The simulated posterior probability of the credible ellipse of (b) for the air-flow and water-temperature coefficients is 0.9503.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]]])
n, p = X.shape
XtX, Xty = X.T @ X, X.T @ y

def nig_posterior(m0, P0, a0, b0, y=y):
    """Conjugate update; P0 is the prior precision V0^{-1} (per unit sigma^2)."""
    Vn = np.linalg.inv(P0 + XtX)
    mn = Vn @ (P0 @ m0 + X.T @ y)
    an = a0 + n / 2
    bn = b0 + (y @ y + m0 @ P0 @ m0 - mn @ (P0 + XtX) @ mn) / 2
    return mn, Vn, an, bn

m0 = np.array([0.0, 1.0, 1.0, 0.0]) # the prior of Chapter 7
P0 = np.linalg.inv(np.diag([100.0**2, 0.1**2, 0.2**2, 0.1**2]))
a0, b0 = 3.0, 18.0
mn, Vn, an, bn = nig_posterior(m0, P0, a0, b0)

x0 = np.array([1.0, 65.0, 23.0, 87.0]) # a new operating condition
qb = stats.t.ppf(0.975, 2 * an)
centre = x0 @ mn
half_mean = qb * np.sqrt(bn / an * (x0 @ Vn @ x0)) # credible interval for x0^T
beta
half_new = qb * np.sqrt(bn / an * (1 + x0 @ Vn @ x0)) # posterior predictive
interval for Y0

beta_hat = np.linalg.solve(XtX, Xty) # frequentist intervals, for
comparison
s2 = np.sum((y - X @ beta_hat) ** 2) / (n - p)
qf = stats.t.ppf(0.975, n - p)
h0 = x0 @ np.linalg.solve(XtX, x0)
print(f"Bayes: mean response {centre:.2f} +- {half_mean:.2f}; new day {centre:.2f}")
```

```

+- {half_new:.2f}")
print(f"least squares: mean response {x0 @ beta_hat:.2f} +- {qf * np.sqrt(s2 * h0)
:.2f}; "
      f"new day +- {qf * np.sqrt(s2 * (1 + h0)):.2f}")

```

## 12.6.4 How often does a credible interval cover?

One coverage statement holds for every proper prior.

### Proposition 12.6.4 — Coverage averaged over the prior

Let the prior  $\pi$  be proper, and let  $C(\mathbf{y})$  be a  $1 - \alpha$  credible set for  $\theta$  for every  $\mathbf{y}$ . If  $(\boldsymbol{\beta}, \sigma^2)$  is drawn from  $\pi$  and then  $\mathbf{Y}$  from the model,  $\Pr\{\theta \in C(\mathbf{Y})\} \geq 1 - \alpha$ , with equality if every  $C(\mathbf{y})$  has posterior probability exactly  $1 - \alpha$ . Equivalently, the frequentist coverage  $\Pr_{\boldsymbol{\beta}, \sigma^2}\{\theta \in C(\mathbf{Y})\}$ , averaged over  $\pi$ , is at least  $1 - \alpha$ .

*Proof.* Under the joint distribution of parameters and data, the conditional distribution of the parameters given  $\mathbf{Y} = \mathbf{y}$  is the posterior. By the tower property,  $\Pr\{\theta \in C(\mathbf{Y})\} = \mathbb{E}[\Pr\{\theta \in C(\mathbf{Y}) \mid \mathbf{Y}\}] \geq 1 - \alpha$ . Conditioning instead on the parameters gives the second form. ■

The average hides variation: coverage is high where the prior is right and low where it is wrong. Consider the credible interval for the air-flow coefficient in [Example \(Predicting a day of stack loss\)](#), and simulate its frequentist coverage when the true air-flow coefficient is varied, the other coefficients are held at their least squares values, and  $\sigma^2 = s^2$ . [Figure 12.6.1](#) shows the result. With the prior of Chapter 7, the coverage is 0.959 when the truth is the prior mean 1, and 0.936 at 0.7 and at 1.5. It ranges from 0.877 to 0.965 over the plotted range. This prior is mild, and prior–data conflict inflates  $b_n$ , which widens the interval. A prior with slope standard deviations five times smaller behaves very differently: its coverage is 1.000 at the prior mean but 0.007 at 0.7, close to the least squares estimate, and 0.000 at 1.5. Averaged over the prior of Chapter 7, the simulated coverage is 0.9512 (40000 draws), as [Proposition 12.6.4](#) requires. The flat-prior interval is the  $t$  interval and covers exactly 95% at every parameter value.

An informative prior makes intervals shorter, and better when the prior is sound; when it is not, they are confidently wrong. The frequentist interval makes no bet and pays in length. [Chapter 27](#) meets the same trade-off as bias against variance.

## 12.6.5 Exercises

### 12.6.5.1 A. Check your understanding

#### Exercise A1.

Is the equal-tailed interval of [Theorem 12.6.2\(d\)](#) the HPD interval for  $\sigma^2$ ? Describe the HPD interval and explain the difference.

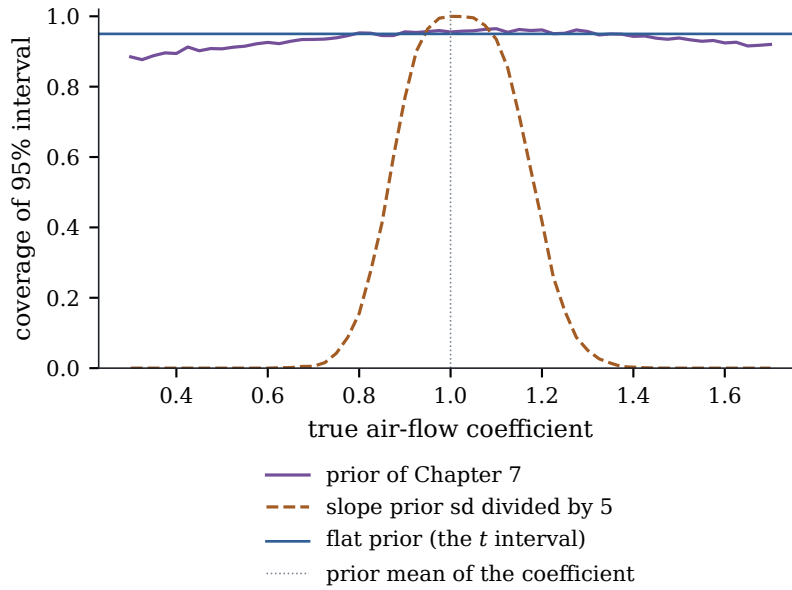


Figure 12.6.1: Frequentist coverage of the 95% credible interval for the air-flow coefficient against the true coefficient, for the prior of Chapter 7 and for a prior with slope standard deviations divided by five (4000 data sets per point). The flat-prior interval covers 0.95 everywhere.

*Solution.* No. The inverse gamma density is skewed to the right, so the HPD interval  $[l, u]$ , with  $\pi(l | \mathbf{y}) = \pi(u | \mathbf{y})$ , is shifted toward smaller values. It also depends on the parameterization: the HPD interval for  $\log \sigma^2$  does not transform into the one for  $\sigma^2$ , while equal-tailed intervals do.

### Exercise A2.

For  $\nu > 2$ , find the posterior predictive mean and variance of a single future observation at  $\mathbf{x}_0$ . Why is the variance larger than  $\tau^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0)$ ?

*Solution.* From Theorem 12.6.2(c),  $Y_0 | \mathbf{y} = \mathbf{x}_0^\top \mathbf{m}_n + \tau \sqrt{1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0} T$  with  $T \sim t(\nu)$ , so the mean is  $\mathbf{x}_0^\top \mathbf{m}_n$  and the variance is  $\tau^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0) \nu/(\nu - 2)$ . The extra factor comes from the uncertainty about  $\sigma^2$ : given  $\sigma^2$  the predictive variance is  $\sigma^2(1 + \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0)$ , and averaging over the posterior gives  $E(\sigma^2 | \mathbf{y}) = b_n/(a_n - 1)$ , which exceeds  $\tau^2 = b_n/a_n$  by the factor  $a_n/(a_n - 1) = \nu/(\nu - 2)$ .

### 12.6.5.2 B. Practice

#### Exercise B1.

Under the  $g$ -prior of Proposition 7.6.3(c) with  $\mathbf{m}_0 = \mathbf{0}$ , show that the credible interval for  $\lambda^\top \boldsymbol{\beta}$  is centred at  $\{g/(1+g)\} \lambda^\top \hat{\boldsymbol{\beta}}$  and has half-width  $t_{\nu, \alpha/2} \tau \sqrt{g/(1+g)} \sqrt{\lambda^\top (\mathbf{X}^\top \mathbf{X})^{-1} \lambda}$ . Compare it with the  $t$  interval as  $g \rightarrow \infty$ , paying attention to  $\nu$  and  $\tau^2$ .

*Solution.* With  $\mathbf{V}_0 = g(\mathbf{X}^\top \mathbf{X})^{-1}$ ,  $\mathbf{V}_n = \{g/(1+g)\}(\mathbf{X}^\top \mathbf{X})^{-1}$  and  $\mathbf{m}_n = \{g/(1+g)\} \hat{\boldsymbol{\beta}}$  (Proposition 7.6.3(c)). Substitute in Theorem 12.6.2(a). As  $g \rightarrow \infty$  the centre and the matrix tend to those of least squares, but  $\nu = 2a_0 + n$  stays larger than  $n - p$  and

$\tau^2 = b_n/a_n$  tends to  $(2b_0 + \text{SSE})/(2a_0 + n)$ , not to  $s^2$ . The interval does not become the  $t$  interval: that needs the flat prior on  $\sigma^2$  as well, with  $a_0 = -p/2$  and  $b_0 = 0$  (Corollary 7.6.4).

### Exercise B2.

Two future observations are taken at the same  $\mathbf{x}_0$ . Show that, under the posterior predictive distribution, they are correlated, and find the correlation. Why are they correlated when their errors are independent? Are they independent under the frequentist model?

*Solution.* By Theorem 12.6.2(c) with  $\mathbf{X}_0$  having two equal rows  $\mathbf{x}_0^\top$ , the scale matrix is  $\tau^2(\mathbf{I}_2 + h\mathbf{1}\mathbf{1}^\top)$  with  $h = \mathbf{x}_0^\top \mathbf{V}_n \mathbf{x}_0$ , so the correlation is  $h/(1+h)$ . Both observations share the unknown  $\boldsymbol{\beta}$  (and  $\sigma^2$ ), and uncertainty about the shared mean correlates them. Under the frequentist model with fixed parameters the two observations are independent, but the two prediction errors  $Y_{0j} - \hat{Y}_0$  share  $\hat{Y}_0$  and have the same correlation  $h_0/(1+h_0)$  under the flat prior.

### 12.6.5.3 C. Going deeper

### Exercise C1.

Let  $\mathbf{X}$  have rank  $r < p$  and take the flat prior  $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$ . Show that the posterior of  $\boldsymbol{\beta}$  is improper, but that for estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  the posterior of  $(\boldsymbol{\lambda}^\top \boldsymbol{\beta}, \sigma^2)$  can be defined by reparameterizing with  $\boldsymbol{\gamma} = \mathbf{U}^\top \boldsymbol{\beta}$  and  $\boldsymbol{\delta} = \mathbf{N}^\top \boldsymbol{\beta}$ , where the columns of  $\mathbf{U}$  and  $\mathbf{N}$  are orthonormal bases of  $\mathcal{C}(\mathbf{X}^\top)$  and  $\mathcal{N}(\mathbf{X})$ , and integrating over nothing but  $\boldsymbol{\gamma}$ . Show that the resulting credible interval is the  $t$  interval of Theorem 12.1.1 with  $n - r$  degrees of freedom, provided the flat prior is taken as  $1/\sigma^2$  in  $(\boldsymbol{\gamma}, \sigma^2)$  alone.

*Solution.* The likelihood depends on  $\boldsymbol{\beta}$  only through  $\mathbf{X}\boldsymbol{\beta} = \mathbf{X}\mathbf{U}\boldsymbol{\gamma}$ , since  $\mathbf{X}\mathbf{N} = \mathbf{0}$ . So the posterior is flat in  $\boldsymbol{\delta}$ , which is not integrable: the posterior of  $\boldsymbol{\beta}$  is improper. In the model  $\mathbf{Y} \sim N_n(\mathbf{X}\mathbf{U}\boldsymbol{\gamma}, \sigma^2\mathbf{I})$  the matrix  $\mathbf{X}\mathbf{U}$  has full column rank  $r$ , and Corollary 7.6.4 with  $p$  replaced by  $r$  gives a proper posterior with  $t(n - r)$  marginals. An estimable  $\boldsymbol{\lambda}$  lies in  $\mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{U})$ , so  $\boldsymbol{\lambda} = \mathbf{U}\mathbf{U}^\top \boldsymbol{\lambda}$  and  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = (\mathbf{U}^\top \boldsymbol{\lambda})^\top \boldsymbol{\gamma}$  involves  $\boldsymbol{\gamma}$  only. Its credible interval is centred at  $(\mathbf{U}^\top \boldsymbol{\lambda})^\top \hat{\boldsymbol{\gamma}} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  with scale  $s^2(\mathbf{U}^\top \boldsymbol{\lambda})^\top (\mathbf{U}^\top \mathbf{X}^\top \mathbf{X} \mathbf{U})^{-1} (\mathbf{U}^\top \boldsymbol{\lambda})$ , and  $\mathbf{U}(\mathbf{U}^\top \mathbf{X}^\top \mathbf{X} \mathbf{U})^{-1} \mathbf{U}^\top$  is a generalized inverse of  $\mathbf{X}^\top \mathbf{X}$  (check  $\mathbf{X}^\top \mathbf{X} \mathbf{U}(\mathbf{U}^\top \mathbf{X}^\top \mathbf{X} \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{X}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{X}$  using  $\mathbf{X} = \mathbf{X}\mathbf{U}\mathbf{U}^\top$ ), so the scale is  $s^2 \boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$ .

### Exercise C2.

In the calibration model of Section 12.5, give  $x_0$  a normal prior independent of the regression parameters, and use the flat prior  $1/\sigma^2$  for  $(\beta_0, \beta_1, \sigma^2)$ . Write down the posterior density of  $x_0$  up to a constant, and show that it is proper even when the slope is not significant. Compare the behaviour of the resulting credible interval with the Fieller set in the weak assay of Example (Calibrating an assay).

## 12.7 Summary and Notes

### 12.7.1 Summary

*Idea.*

1. Confidence sets come from pivots, and are the sets of values that a family of level- $\alpha$  tests accepts (Definition 12.1.1, Proposition 12.1.2).
2. For estimable  $\lambda^\top \beta$ ,  $\lambda^\top \hat{\beta} \pm t_{n-r, \alpha/2} s \sqrt{\lambda^\top \mathbf{G} \lambda}$  is exact and free of the generalized inverse (Theorem 12.1.1). A non-estimable function has no confidence set of finite expected length (Proposition 12.1.3). The interval for  $\sigma^2$  is exact under normality and fragile without it (Proposition 12.1.4).
3. The confidence ellipsoid for  $\Lambda^\top \beta$  is the acceptance region of the  $F$  test, shaped by the covariance of the estimates (Theorem 12.2.1); all such ellipsoids are images of one ball around  $\hat{\mathbf{Y}}$  (Proposition 12.2.2). Its shadows are Scheffé-type intervals (Lemma 12.2.3, Corollary 12.2.4).
4. The prediction interval  $\hat{Y}_0 \pm t_{n-r, \alpha/2} s \sqrt{1 + h_0}$  never shrinks below the noise (Theorem 12.3.1). The leverage  $h_0$  is a Mahalanobis distance that reveals hidden extrapolation (Proposition 12.3.2). Without normality prediction intervals stay wrong as  $n \rightarrow \infty$  (Proposition 12.3.3).
5. The Working–Hotelling band, with multiplier  $\sqrt{k F_\alpha(k, n-r)}$ , covers the regression surface over a  $k$ -dimensional family exactly;  $k = 2$  for a straight line (Theorem 12.4.1).
6. Fieller sets for ratios, and calibration sets, are exact, and are intervals, two rays or the whole line (Proposition 12.5.1, Theorem 12.5.3). The classical calibration estimator has no mean (Proposition 12.5.2), and unbounded sets cannot be avoided (Proposition 12.5.4).
7. Conjugate credible sets and predictive intervals have the frequentist shapes (Lemma 12.6.1, Theorem 12.6.2) and equal them under the flat prior (Corollary 12.6.3). Their coverage is nominal on average over the prior, not pointwise (Proposition 12.6.4).

### 12.7.2 Notes and sources

**Confidence intervals and duality.** Confidence intervals, and their construction by inverting tests, are due to Neyman (1937). Lehmann and Romano (2005) develop the duality and its optimality consequences. The two one-sided tests procedure of Exercise (B2) is due to Schuirman (1987). The shortest interval for a normal variance (Exercise (C2)) was worked out by Tate and Klett (1959).

**The source books.** The chapter covers Seber and Lee (2003, chapter 5 and sections 6.1–6.5), Rencher and Schaalje (2008, sections 8.6–8.7) and Sen and Srivastava (1990, section 3.8). These books state the intervals for full-rank models; here they are stated for estimable functions in models of any rank, as in Christensen (2020). Rencher and Schaalje (2008, section 8.7) derive the  $F$  test as a likelihood ratio test, which Chapter 11 covers (Theorem 11.4.1). Hahn and Meeker (1991) is a practical guide to confidence, prediction and tolerance intervals.

**Bands.** The band for a straight line is due to Working and Hotelling (1929), and the general

projection argument to Scheffé (1953). Exact bands over finite intervals are due to Wynn and Bloomfield (1971) and constant-width bands to Gafarian (1964). Miller (1981) surveys the early literature and Liu (2010) gives a modern book-length treatment.

**Normality and prediction.** The warning about conditional coverage after a preliminary test is from Olshen (1973).

**Ratios and calibration.** Fieller’s method came from bioassay and is set out in general form in Fieller (1954). Krutchkoff (1967) proposed the inverse estimator and Hoadley (1970) gave its Bayesian interpretation. Osborne (1991) reviews calibration, and Brown (1993) treats the multivariate case. Gleser and Hwang (1987) proved the general form of [Proposition 12.5.4](#).

**Bayesian intervals.** Box and Tiao (1973) and Gelman et al. (2013) treat credible and predictive intervals for the normal linear model under conjugate and flat priors.

**Data.** The murder-rate data are the 2009 US state data of statsmodels; the stack loss data are from Brownlee (1965); the assay is simulated.

### 12.7.3 References

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# Chapter 13

## Multiplicity: Scheffé, Tukey, Bonferroni and the False Discovery Rate

A test with error rate 0.05 keeps that promise only when it is the only inference drawn from the data. A regression analysis rarely stops at one: it compares every pair of groups, examines each of many coefficients, and follows a significant  $F$  test with the contrast that looks most interesting. This chapter measures the error of a whole family of inferences and controls it in four ways. Scheffé's method covers every estimable function in a subspace, even functions chosen after seeing the data, and is exactly the  $F$  test read one direction at a time. Tukey's studentized range is exact for all pairwise differences of balanced means. Bonferroni's inequality, and Holm's sharper step-down version, need only the level of each test. The Benjamini–Hochberg procedure controls the expected proportion of false discoveries, which is what screening many hypotheses needs. The chapter ends with a guide to choosing among them.

**What you need.** The normal linear model and the sampling distributions of [Chapter 7](#) ([Theorem 7.5.1](#), [Corollary 7.5.2](#), [Corollary 7.5.3](#)), independence of projections from [Chapter 4](#) ([Theorem 4.4.4](#)), and orthogonal projections onto subspaces from [Chapter 6](#) ([Theorem 6.5.1](#), [Theorem 6.5.2](#)). Estimable functions and contrasts in the one-way layout come from [Chapter 8](#) ([Theorem 8.7.1](#), [Proposition 8.7.2](#)), and the sum of squares of a linear hypothesis from [Chapter 9](#) ([Theorem 9.2.4](#)). The  $F$  test of [Chapter 11](#) ([Theorem 11.1.1](#), [Theorem 11.3.2](#)) and the confidence intervals and ellipsoids of [Chapter 12](#) ([Theorem 12.1.1](#), [Theorem 12.2.1](#)) are the single-inference versions of what this chapter does for families.

### 13.0.1 Roadmap

- [The multiple testing problem](#)
- [Scheffé's projection method](#)
- [Tukey's studentized range](#)
- [Bonferroni and Holm](#)
- [False discovery rate control](#)
- [Choosing a procedure](#)
- [Summary and notes](#)



## 13.1 The multiple testing problem

Seven groups give 21 pairs, and a two-sided  $t$  test at level 0.05 for each pair is a correct test. But an analyst who runs all 21 and reports the significant ones is not using a procedure with error rate 0.05: even if all seven means are equal, about one pair is expected to be significant, and the reader cannot tell it from a real difference. This section defines the error rates of a family of inferences, measures by simulation how large they become without adjustment, and separates strong control from a weak form that protects only against the situation in which nothing is going on.

### 13.1.1 A family of hypotheses

Let  $H_1, \dots, H_m$  be null hypotheses about the parameters of a model, tested on the same data. The model's true parameter makes some of them true and the others false. Write  $I_0 \subseteq \{1, \dots, m\}$  for the set of true hypotheses and  $m_0 = |I_0|$  for their number. A **multiple testing procedure** maps the data to a set  $\mathcal{R} \subseteq \{1, \dots, m\}$  of rejected hypotheses. The outcome can be summarized by three counts:

|                  | not rejected  | rejected | total     |
|------------------|---------------|----------|-----------|
| true hypotheses  | $m_0 - V$     | $V$      | $m_0$     |
| false hypotheses | $m - m_0 - S$ | $S$      | $m - m_0$ |
| total            | $m - R$       | $R$      | $m$       |

Here  $R = |\mathcal{R}|$  is the number of rejections,  $V = |\mathcal{R} \cap I_0|$  the number of **false rejections** (Type I errors), and  $S = R - V$  the number of correct ones. The counts  $R$ ,  $V$  and  $S$  are random, and  $m_0$  is unknown. The analyst sees  $R$  but not how it splits into  $V$  and  $S$ .

With one true hypothesis, a level- $\alpha$  test makes  $\Pr(V = 1) \leq \alpha$ . For a family there are several reasonable ways to summarize the distribution of  $V$ , and they lead to different procedures.

#### Definition 13.1.1 — Error rates of a family

For a multiple testing procedure and a given value of the parameters:

- (a) the **familywise error rate** is  $\text{FWER} = \Pr(V \geq 1)$ , the probability of at least one false rejection;
- (b) the **per-family error rate** is  $\text{PFER} = \mathbb{E}(V)$ , the expected number of false rejections, and the **per-comparison error rate** is  $\text{PCER} = \mathbb{E}(V)/m$ ;
- (c) the **false discovery proportion** is  $Q = V/\max(R, 1)$ , which is  $V/R$  when something is rejected and 0 otherwise, and the **false discovery rate** is its expectation,  $\text{FDR} = \mathbb{E}(Q)$ .

A procedure **controls** one of these rates at level  $\alpha$  if the rate is at most  $\alpha$  for every value of the parameters.

The familywise error rate treats one false claim as a failure of the whole family; it suits conclusions that will each be acted on. The false discovery rate accepts some errors in a long list of

discoveries and bounds their expected share. The unadjusted procedure, testing each hypothesis at level  $\alpha$ , controls only the per-comparison rate.

### Proposition 13.1.1 — How the error rates compare

For every procedure and every value of the parameters,

$$\text{PCER} \leq \text{FWER} \leq \text{PFER}, \quad \text{FDR} \leq \text{FWER}.$$

If all the hypotheses are true ( $m_0 = m$ ), then  $\text{FDR} = \text{FWER}$ . Consequently a procedure that controls the per-family error rate at  $\alpha$  controls the familywise rate at  $\alpha$ , and a procedure that controls the familywise rate controls the false discovery rate.

*Proof.* All four rates are expectations of functions of  $V$  and  $R$ , so it is enough to compare the functions pointwise. Since  $0 \leq V \leq m$ , we have  $V/m \leq 1_{\{V \geq 1\}} \leq V$ , which gives the first chain after taking expectations. Since  $V \leq R$ , the ratio  $V/\max(R, 1)$  is at most 1, and it is 0 when  $V = 0$ ; so  $V/\max(R, 1) \leq 1_{\{V \geq 1\}}$ , which gives  $\text{FDR} \leq \text{FWER}$ . If every hypothesis is true, every rejection is false, so  $V = R$  and  $V/\max(R, 1) = 1_{\{V \geq 1\}}$  exactly. ■

In particular, a procedure that controls the false discovery rate is a valid level- $\alpha$  test of the hypothesis that *all* the  $H_j$  are true, but not in general a procedure with familywise control (Section 13.5).

## 13.1.2 How often something is significant by chance

If the  $m$  tests are independent and each has exact level  $\alpha$ , then when all hypotheses are true,

$$\text{FWER} = 1 - \Pr(\text{no rejection}) = 1 - (1 - \alpha)^m. \quad (13.1.1)$$

With  $\alpha = 0.05$  this is 0.401 for ten tests and 0.994 for a hundred. In a linear model the tests are rarely independent. They share the estimate  $s$  of  $\sigma$ , and comparisons involving a common group share that group's mean. The next example measures the effect of that dependence for the most common family of all.

### Example. (All pairs of equal means)

Take  $g$  groups of 5 observations each, with all group means equal, and compare every pair with an unadjusted two-sided  $t$  test at level 0.05, using the pooled variance estimate on  $g(5 - 1)$  degrees of freedom. The listing simulates 20000 data sets for each  $g$ . For  $g = 3, 5, 7, 10$  (so 3, 10, 21, 45 pairs), the proportion of data sets with at least one significant pair is

$$0.116, \quad 0.265, \quad 0.412, \quad 0.589.$$

Independent tests would give 0.143, 0.401, 0.659 and 0.901. The average number of significant pairs is 0.15, 0.50, 1.08 and 2.23, in each case 0.05 times the number of pairs up to simulation error. Figure 13.1.1 plots the familywise rate against the number of tests.

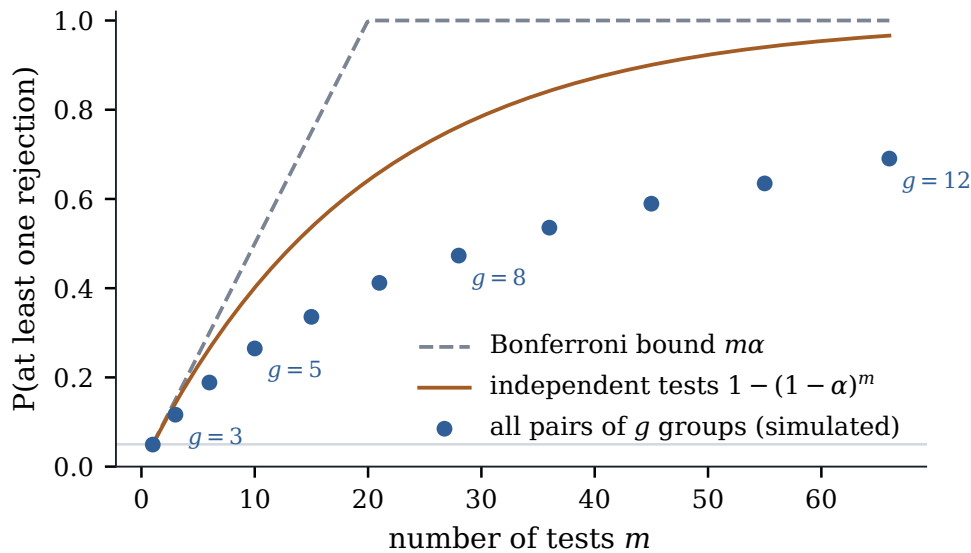


Figure 13.1.1: Probability of at least one false rejection among  $m$  unadjusted level-0.05 tests: all pairs of  $g = 2, \dots, 12$  equal means (dots, simulated), independent tests (curve, Eq. 13.1.1) and the Bonferroni bound  $m\alpha$  (dashed).

```
import itertools
import numpy as np
from scipy import stats

alpha, reps, m_rep = 0.05, 20_000, 5                                # 5 observations per group

def any_significant(g, m_rep, reps, alpha):
    """Share of data sets with at least one unadjusted pairwise rejection, and mean
    count."""
    rng = np.random.default_rng([20260919, g])                      # one fixed stream per g
    nu = g * (m_rep - 1)                                           # error degrees of freedom
    y = rng.normal(size=(reps, g, m_rep))                          # all group means equal
    means = y.mean(axis=2)
    s2 = ((y - means[:, :, None]) ** 2).sum(axis=(1, 2)) / nu
    se = np.sqrt(s2 * 2 / m_rep)                                    # standard error of a
    difference
    crit = stats.t.ppf(1 - alpha / 2, nu)
    count = np.zeros(reps)
    for k, l in itertools.combinations(range(g), 2):
        count += np.abs(means[:, k] - means[:, l]) / se > crit
    return np.mean(count > 0), np.mean(count)

for g in (3, 5, 10):
    pairs = g * (g - 1) // 2
    fwer, mean_count = any_significant(g, m_rep, reps, alpha)
    print(f"g = {g:2d}, {pairs:2d} pairs: P(some rejection) = {fwer:.3f}, "
          f"mean number = {mean_count:.2f} (alpha x pairs = {alpha * pairs:.2f}), "
          f"independent tests would give {1 - (1 - alpha) ** pairs:.3f}")
```

The expected number of false rejections is additive whatever the dependence, which is why the Bonferroni bound of Section 13.4 needs no assumptions. The probability of at least one grows more slowly than for independent tests, because false rejections cluster when  $s$  is small or one mean is extreme. Still, with ten groups some pair is “significant” in well over half of all

experiments in which nothing differs.

### 13.1.3 Simultaneous intervals and the tests they induce

Let  $\psi_j$ ,  $j \in J$ , be real parameters, where the index set  $J$  may be infinite. Intervals  $[L_j, U_j]$  computed from the data are **simultaneous confidence intervals** at level  $1 - \alpha$  if

$$\Pr(L_j \leq \psi_j \leq U_j \text{ for every } j \in J) \geq 1 - \alpha$$

for every value of the parameters. Scheffé's and Tukey's methods and the Bonferroni intervals all produce them, and they yield tests with the strongest guarantees.

#### Proposition 13.1.2 — Tests from simultaneous intervals

Let  $[L_j, U_j]$ ,  $j \in J$ , be simultaneous confidence intervals at level  $1 - \alpha$ , and for given constants  $d_j$  reject  $H_j : \psi_j = d_j$  exactly when  $d_j \notin [L_j, U_j]$ . Then:

- (a) the probability of rejecting at least one true  $H_j$  is at most  $\alpha$ , for every value of the parameters;
- (b) if, whenever  $H_j$  is rejected, one also declares  $\psi_j > d_j$  or  $\psi_j < d_j$  according to the side of  $d_j$  on which the interval lies, the probability of at least one false rejection or wrong declaration of sign is at most  $\alpha$ .

*Proof.* (a) If a true  $H_j$  is rejected, then  $\psi_j = d_j \notin [L_j, U_j]$ , so some interval fails to cover. That event has probability at most  $\alpha$ . (b) A wrong declaration, say  $\psi_j > d_j$  when in fact  $\psi_j \leq d_j$ , happens only if  $L_j > d_j \geq \psi_j$ ; again an interval fails to cover. Both kinds of error are contained in the event that some interval misses, whose probability is at most  $\alpha$ . ■

Part (b) matters because few parameters are exactly equal to their hypothesized values; the real question is which way each difference goes. Some good procedures, such as Holm's, give tests but no simple companion intervals.

### 13.1.4 Weak control is not enough

The error rates depend on which hypotheses are true. The strongest requirement bounds the rate in every configuration; a much weaker one only when every hypothesis is true.

#### Definition 13.1.2 — Strong and weak control

A procedure controls the familywise error rate **strongly** at level  $\alpha$  if  $\Pr(V \geq 1) \leq \alpha$  for every value of the parameters, and **weakly** if  $\Pr(V \geq 1) \leq \alpha$  whenever all of  $H_1, \dots, H_m$  are true.

Weak control is easy to obtain. Let  $H_0 = \bigcap_j H_j$  be the **global null hypothesis** that every  $H_j$  holds.

### Proposition 13.1.3 — A gate gives weak control

Let  $\phi_0$  be a level- $\alpha$  test of the global null  $H_0$ . A procedure that rejects no  $H_j$  unless  $\phi_0$  rejects  $H_0$  controls the familywise error rate weakly at level  $\alpha$ .

*Proof.* When all  $H_j$  are true,  $V \geq 1$  requires a rejection by  $\phi_0$ , which has probability at most  $\alpha$  because  $H_0$  is true. ■

The best-known gated procedure is Fisher's **protected least significant difference** (LSD): in a one-way layout, test equality of all  $g$  means with the  $F$  test at level  $\alpha$ , and only if it rejects, compare each pair with an unadjusted  $t$  test at level  $\alpha$ . Weak control is all it has.

#### Example. (The protected LSD without strong control)

With  $g = 10$  groups of 5 observations and all means equal, the simulation below gives a familywise error rate of 0.049, as Proposition 13.1.3 promises. Now move one mean far from the other nine, which remain equal. The  $F$  test rejects in essentially every data set, the gate is always open, and the 36 pairwise comparisons among the nine equal means are made without adjustment. The familywise error rate rises to 0.539.

```
import itertools
import numpy as np
from scipy import stats

alpha, reps, m_rep = 0.05, 20_000, 5

def protected_lsd_fwer(mu, m_rep, reps, alpha, seed):
    """Share of data sets in which the protected LSD rejects some true equality."""
    rng = np.random.default_rng(seed)
    g = len(mu)
    nu = g * (m_rep - 1)
    y = np.asarray(mu)[None, :, None] + rng.normal(size=(reps, g, m_rep))
    means = y.mean(axis=2)
    s2 = ((y - means[:, :, None]) ** 2).sum(axis=(1, 2)) / nu
    between = m_rep * ((means - means.mean(axis=1, keepdims=True)) ** 2).sum(axis=1)
    gate = between / (g - 1) / s2 > stats.f.ppf(1 - alpha, g - 1, nu)  # the F test
    crit = stats.t.ppf(1 - alpha / 2, nu) * np.sqrt(2 * s2 / m_rep)
    false = np.zeros(reps, dtype=bool)
    for k, l in itertools.combinations(range(g), 2):
        if mu[k] == mu[l]:  # a true null hypothesis
            false |= gate & (np.abs(means[:, k] - means[:, l]) > crit)
    return false.mean()

all_equal = protected_lsd_fwer([0.0] * 10, m_rep, reps, alpha, seed=1)
one_apart = protected_lsd_fwer([0.0] * 9 + [10.0], m_rep, reps, alpha, seed=2)
print(f"10 groups, all means equal: FWER = {all_equal:.3f}")
print(f"10 groups, one mean far from rest: FWER = {one_apart:.3f}")
```

Weak control protects only against the one configuration in which there is nothing to find. Every procedure recommended in this chapter controls its error rate strongly.

## 13.1.5 Exercises

### 13.1.5.1 A. Check your understanding

#### Exercise A1.

Ten and twenty independent tests of true hypotheses are made, each at level 0.01. Find the familywise error rate in each case. What common level for twenty independent tests gives a familywise rate of exactly 0.05?

*Solution.* By [Eq. 13.1.1](#) the rates are  $1 - 0.99^{10} = 0.0956$  and  $1 - 0.99^{20} = 0.1821$ . The level  $a$  with  $1 - (1 - a)^{20} = 0.05$  is  $a = 1 - 0.95^{1/20} = 0.00256$ , slightly more than  $0.05/20 = 0.00250$ . This is the Šidák level of [Section 13.4](#).

#### Exercise A2.

Give a configuration of true and false hypotheses, and a procedure, for which the familywise error rate is close to one while the false discovery rate is close to zero.

### 13.1.5.2 B. Practice

#### Exercise B1.

Show that with  $g = 3$  groups the protected LSD controls the familywise error rate strongly. *Hint:* consider separately the configurations with three, one and no true equalities among the three pairwise hypotheses.

*Solution.* The three hypotheses are  $\mu_1 = \mu_2$ ,  $\mu_1 = \mu_3$ ,  $\mu_2 = \mu_3$ . If two of them are true, the third is true as well, so the possible numbers of true hypotheses are 3, 1 and 0. With three, all means are equal and weak control ([Proposition 13.1.3](#)) applies. With none, no false rejection is possible. With exactly one, say  $\mu_1 = \mu_2 \neq \mu_3$ , a false rejection needs the unadjusted  $t$  test of  $\mu_1 = \mu_2$  to reject, an event of probability  $\alpha$  whether or not the gate opens. So the familywise rate is at most  $\alpha$  in every configuration. With four or more groups, a configuration can hold two disjoint true equalities, and the argument fails.

#### Exercise B2.

Let  $m_0$  true hypotheses have independent uniform  $p$ -values, and let  $m_1$  false hypotheses have  $p$ -values equal to 0. The unadjusted procedure rejects when  $p_j \leq \alpha$ . Find its familywise error rate, show that its false discovery rate is at most  $m_0\alpha/m_1$ , and conclude that the false discovery rate tends to 0 as  $m_1 \rightarrow \infty$  with  $m_0$  fixed.

### 13.1.5.3 C. Going deeper

**Exercise C1.**

In a one-way layout with  $g \geq 3$  groups of  $m$  observations and known  $\sigma$ , run the protected LSD with  $z$  tests: the gate is the level- $\alpha$  test that rejects when  $\sum_k m(\bar{y}_k - \bar{y})^2/\sigma^2$  exceeds the upper  $\alpha$  point of  $\chi^2(g-1)$ , and a pair is declared different when  $|\bar{y}_k - \bar{y}_l| > z_{\alpha/2} \sigma \sqrt{2/m}$ , where  $z_{\alpha/2}$  is the upper  $\alpha/2$  point of  $N(0, 1)$ . Let  $\mu_1 = \dots = \mu_{g-1} = 0$  and  $\mu_g = \Delta$ . Show that as  $\Delta \rightarrow \infty$  the familywise error rate tends to  $\Pr(W_{g-1} > \sqrt{2} z_{\alpha/2})$ , where  $W_{g-1}$  is the range of  $g-1$  independent  $N(0, 1)$  variables. Show that this limit is at least  $1 - (1 - \alpha)^{\lfloor (g-1)/2 \rfloor}$  and that it tends to 1 as  $g \rightarrow \infty$ .

*Solution.* With  $Z_k = \sqrt{m} \bar{y}_k / \sigma$ ,  $k < g$ , independent  $N(0, 1)$  and free of  $\Delta$ , some pair  $k, l < g$  is declared different iff  $W_{g-1} = \max_{k < g} Z_k - \min_{k < g} Z_k > \sqrt{2} z_{\alpha/2}$ , so the rate is  $\Pr(G \cap \{W_{g-1} > \sqrt{2} z_{\alpha/2}\})$ , with  $G$  the event that the gate opens. The gate statistic is at least  $m(\bar{y}_g - \bar{y})^2/\sigma^2$ , where  $\bar{y}_g - \bar{y}$  is  $\frac{g-1}{g}\Delta$  plus a variable free of  $\Delta$ ; so  $\Pr(G) \rightarrow 1$ , giving the limit. The disjoint pairs  $(1, 2), (3, 4), \dots$  give  $\lfloor (g-1)/2 \rfloor$  independent statistics  $|Z_{2i-1} - Z_{2i}|/\sqrt{2}$ , each exceeding  $z_{\alpha/2}$  with probability  $\alpha$ , and  $W_{g-1}$  is at least  $\sqrt{2}$  times each of them; hence the lower bound, which tends to 1.

## 13.2 Scheffé's projection method

Scheffé's method answers the hardest version of the problem: the family is every estimable function in a subspace, and the analyst may choose which ones to look at after seeing the data. The intervals have simultaneous coverage exactly  $1 - \alpha$ , because the largest standardized error in the family is the length of a projection, and Scheffé's method *is* the  $F$  test, read one direction at a time.

Throughout this section the model is the normal linear model [Eq. 7.3.1](#),  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ , with  $r = \text{rank}(\mathbf{X}) < n$ . Write  $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$ ,  $\nu = n - r$  for the error degrees of freedom,  $s^2 = \text{SSE}/\nu$ ,  $\mathbf{G} = (\mathbf{X}^\top \mathbf{X})^-$  for any generalized inverse, and  $F_\alpha(q, \nu)$  for the upper  $\alpha$  point of  $F(q, \nu)$ . For an estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  the standard error is  $\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = s\sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}}$  ([Corollary 7.5.2](#)).

### 13.2.1 Linear functions of the mean as vectors

An estimable function is a linear function of the mean vector, and a linear function of  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$  can be represented by a vector of  $\mathcal{C}(\mathbf{X})$ . If  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is estimable, then  $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$  for some vector  $\boldsymbol{\rho} \in \mathbb{R}^n$  ([Theorem 8.3.1](#)), and with  $\mathbf{v} = \mathbf{M}\boldsymbol{\rho}$ ,

$$\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{v}^\top \boldsymbol{\mu}, \quad \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{M}\mathbf{Y} = \mathbf{v}^\top \mathbf{Y}, \quad \boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda} = \boldsymbol{\rho}^\top \mathbf{X}\mathbf{G}\mathbf{X}^\top \boldsymbol{\rho} = \boldsymbol{\rho}^\top \mathbf{M}\boldsymbol{\rho} = \|\mathbf{v}\|^2. \quad (13.2.1)$$

The second identity is [Theorem 6.4.4](#), and the third uses  $\mathbf{M} = \mathbf{X}\mathbf{G}\mathbf{X}^\top$  ([Theorem 6.3.7](#)). The vector  $\mathbf{v}$  does not depend on the choice of  $\boldsymbol{\rho}$ , since two choices differ by some  $\boldsymbol{\delta}$  with  $\mathbf{X}^\top \boldsymbol{\delta} = \mathbf{0}$ , hence  $\mathbf{M}\boldsymbol{\delta} = \mathbf{0}$ . So the estimate is the inner product of  $\mathbf{v}$  with the data, the estimand its inner product with the mean, and the standard error is  $s\|\mathbf{v}\|$ .

A *family* of estimable functions closed under linear combination is then a subspace  $\mathcal{V} \subseteq \mathcal{C}(\mathbf{X})$ .

It is usually described in one of two ways. If the family is  $\{\lambda^\top \beta : \lambda \in \mathcal{C}(\Lambda)\}$  for a  $p \times q$  matrix  $\Lambda = \mathbf{X}^\top \mathbf{T}$ , then  $\mathbf{v} = \mathbf{M}\mathbf{T}\mathbf{a}$  and  $\mathcal{V} = \mathcal{C}(\mathbf{M}\mathbf{T})$ , the test space of  $\Lambda^\top \beta = \mathbf{0}$  (Theorem 6.5.2); if  $\Lambda$  has rank  $q$ , so does  $\mathbf{M}\mathbf{T}$ , because  $\mathbf{X}^\top \mathbf{M}\mathbf{T} = \Lambda$ . If instead the family is the set of functions about which a reduced model  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  makes a claim, those vanishing on  $\mathcal{C}(\mathbf{X}_0)$ , then  $\mathcal{V} = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$ , of dimension  $r - r_0$  (Theorem 6.5.1).

### 13.2.2 The theorem

#### Lemma 13.2.1 — The largest standardized component

Let  $\mathcal{V}$  be a subspace of  $\mathbb{R}^n$  with orthogonal projection  $\mathbf{P}_\mathcal{V}$ , and let  $\mathbf{u} \in \mathbb{R}^n$ . Then

$$\max_{\mathbf{v} \in \mathcal{V}, \mathbf{v} \neq \mathbf{0}} \frac{(\mathbf{v}^\top \mathbf{u})^2}{\|\mathbf{v}\|^2} = \|\mathbf{P}_\mathcal{V} \mathbf{u}\|^2,$$

and if  $\mathbf{P}_\mathcal{V} \mathbf{u} \neq \mathbf{0}$  the maximum is attained exactly at the nonzero multiples of  $\mathbf{P}_\mathcal{V} \mathbf{u}$ .

*Proof.* For  $\mathbf{v} \in \mathcal{V}$ ,  $\mathbf{v} = \mathbf{P}_\mathcal{V} \mathbf{v}$ , and since  $\mathbf{P}_\mathcal{V}$  is symmetric,  $\mathbf{v}^\top \mathbf{u} = \mathbf{v}^\top \mathbf{P}_\mathcal{V} \mathbf{u}$ . By Cauchy–Schwarz (Proposition 1.1.1),  $(\mathbf{v}^\top \mathbf{P}_\mathcal{V} \mathbf{u})^2 \leq \|\mathbf{v}\|^2 \|\mathbf{P}_\mathcal{V} \mathbf{u}\|^2$ , with equality iff  $\mathbf{v}$  is a multiple of  $\mathbf{P}_\mathcal{V} \mathbf{u}$ . That multiple lies in  $\mathcal{V}$ , so the bound is attained. If  $\mathbf{P}_\mathcal{V} \mathbf{u} = \mathbf{0}$  both sides are zero. ■

The lemma is Exercise (C1) of Chapter 9, where it described the sum of squares of a row of an analysis of variance table as the largest one-degree-of-freedom sum of squares inside it. Applied to  $\mathbf{u} = \mathbf{Y} - \boldsymbol{\mu}$ , it describes the worst standardized error in a family.

#### Theorem 13.2.2 — Scheffé's simultaneous intervals

Assume Eq. 7.3.1 with  $r < n$ , and let  $0 < \alpha < 1$ .

(a) Let  $\mathcal{V} \subseteq \mathcal{C}(\mathbf{X})$  be a subspace of dimension  $q \geq 1$ . Then

$$\max_{\mathbf{v} \in \mathcal{V}, \mathbf{v} \neq \mathbf{0}} \frac{(\mathbf{v}^\top \mathbf{Y} - \mathbf{v}^\top \boldsymbol{\mu})^2}{q s^2 \|\mathbf{v}\|^2} = \frac{\|\mathbf{P}_\mathcal{V}(\mathbf{Y} - \boldsymbol{\mu})\|^2 / q}{s^2} \sim F(q, \nu),$$

and consequently

$$\Pr\left(|\mathbf{v}^\top (\mathbf{Y} - \boldsymbol{\mu})| \leq \sqrt{q F_\alpha(q, \nu)} s \|\mathbf{v}\| \quad \forall \mathbf{v} \in \mathcal{V}\right) = 1 - \alpha.$$

(b) Let  $\Lambda$  be  $p \times q$  of rank  $q$  with  $\mathcal{C}(\Lambda) \subseteq \mathcal{C}(\mathbf{X}^\top)$ . With probability exactly  $1 - \alpha$ , for every  $\lambda \in \mathcal{C}(\Lambda)$  simultaneously,

$$\lambda^\top \beta \in \left[ \lambda^\top \hat{\beta} - \sqrt{q F_\alpha(q, \nu)} \text{se}(\lambda^\top \hat{\beta}), \quad \lambda^\top \hat{\beta} + \sqrt{q F_\alpha(q, \nu)} \text{se}(\lambda^\top \hat{\beta}) \right]. \quad (13.2.2)$$

Moreover, with  $\boldsymbol{\theta} = \Lambda^\top \beta$ ,  $\hat{\boldsymbol{\theta}} = \Lambda^\top \hat{\beta}$  and  $\mathbf{W} = \Lambda^\top \mathbf{G} \Lambda$ ,



$$\max_{\lambda \in \mathcal{C}(\Lambda), \lambda \neq \mathbf{0}} \frac{(\lambda^\top \hat{\beta} - \lambda^\top \beta)^2}{\text{se}(\lambda^\top \hat{\beta})^2} = \frac{(\hat{\theta} - \theta)^\top \mathbf{W}^{-1}(\hat{\theta} - \theta)}{s^2}, \quad (13.2.3)$$

attained at  $\lambda = \Lambda \mathbf{W}^{-1}(\hat{\theta} - \theta)$ .

*Proof.* (a) The first equality is Lemma 13.2.1 with  $\mathbf{u} = \mathbf{Y} - \boldsymbol{\mu}$ . The map  $\mathbf{v} \mapsto \mathbf{X}^\top \mathbf{v}$  is one to one on  $\mathcal{C}(\mathbf{X})$  (Step 1 of the proof of Theorem 12.4.1), so  $\mathcal{L} = \{\mathbf{X}^\top \mathbf{v} : \mathbf{v} \in \mathcal{V}\}$  is a subspace of  $\mathcal{C}(\mathbf{X}^\top)$  of dimension  $q$ . For  $\mathbf{x} = \mathbf{X}^\top \mathbf{v}$ , Eq. 13.2.1 with  $\boldsymbol{\rho} = \mathbf{v}$  gives  $\mathbf{x}^\top \hat{\beta} - \mathbf{x}^\top \beta = \mathbf{v}^\top (\mathbf{Y} - \boldsymbol{\mu})$  and  $\mathbf{x}^\top \mathbf{G} \mathbf{x} = \|\mathbf{v}\|^2$ . So the event in the display is the event Eq. 12.4.1 with  $k = q$ , and it has probability exactly  $1 - \alpha$  by Theorem 12.4.1. That event is also the event that the maximum is at most  $F_\alpha(q, \nu)$ . Since this holds for every  $\alpha \in (0, 1)$ , the maximum has the distribution function of  $F(q, \nu)$ .

(b) Write  $\Lambda = \mathbf{X}^\top \mathbf{T}$  and  $\mathcal{V} = \mathcal{C}(\mathbf{M}\mathbf{T})$ , of dimension  $q$ . Every  $\lambda \in \mathcal{C}(\Lambda)$  is  $\Lambda \mathbf{a} = \mathbf{X}^\top (\mathbf{T} \mathbf{a})$ , and by Eq. 13.2.1  $\lambda^\top \hat{\beta} - \lambda^\top \beta = \mathbf{v}^\top \mathbf{Y} - \mathbf{v}^\top \boldsymbol{\mu}$  and  $\text{se}(\lambda^\top \hat{\beta}) = s \|\mathbf{v}\|$  with  $\mathbf{v} = \mathbf{M} \mathbf{T} \mathbf{a} \in \mathcal{V}$ . As  $\mathbf{a}$  ranges over  $\mathbb{R}^q$ ,  $\mathbf{v}$  ranges over all of  $\mathcal{V}$ . So the event in (b) is the event in (a). Finally, Eq. 13.2.3 is Proposition 11.6.2 for the hypothesis  $\Lambda^\top \beta = \theta$ , which is testable because  $\theta$  is the true value: its statistic Eq. 11.3.1 is  $(\hat{\theta} - \theta)^\top \mathbf{W}^{-1}(\hat{\theta} - \theta)/(q s^2)$ , and each  $T(\mathbf{a})^2$  is the ratio on the left of Eq. 13.2.3 at  $\lambda = \Lambda \mathbf{a}$ , so the maximum is attained at  $\mathbf{a} \propto \mathbf{W}^{-1}(\hat{\theta} - \theta)$ . ■

The coverage is exact, not a bound: some function in the family, depending on the data, always lies on the edge of the event. Only the subspace matters, so dependent columns of  $\Lambda$  may be dropped first. For  $q = 1$  the multiplier is  $t_{\nu, \alpha/2}$ , since  $F_\alpha(1, \nu) = t_{\nu, \alpha/2}^2$ , and the interval is that of Theorem 12.1.1.

Part (b) is the observation-space form of a fact met twice already. Section 11.6 proved Eq. 13.2.3, with  $\mathbf{d}$  in place of  $\theta$ , as the statement that a group  $F$  statistic is the largest single  $t$  statistic in the group, and Section 12.2 showed that the intervals Eq. 13.2.2 are the *shadows* of the confidence ellipsoid for  $\theta = \Lambda^\top \beta$  (Theorem 12.2.1). The ellipsoid covers  $\theta$  iff every shadow covers the corresponding  $\mathbf{a}^\top \theta$ , which is why an infinite family costs no more than one ellipsoid. The coverage statement itself is the Working–Hotelling band of Theorem 12.4.1 for the subspace  $\mathcal{L} = \mathcal{C}(\Lambda)$ : a band for a regression surface and a family of intervals for estimable functions are one object, written in the coordinates of  $\mathcal{C}(\mathbf{X}^\top)$  or of  $\mathcal{C}(\mathbf{X})$ . Figure 13.2.1 extends Figure 12.2.1, which drew the same ellipse for two slopes of the crime regression, by adding the Bonferroni rectangle and the tangent lines that bound the Scheffé interval for a combination  $a_1 \beta_1 + a_2 \beta_2$ .

### Example. (Three slopes of a crime regression)

Return to the regression of the murder rate on poverty, single parenthood and urbanization for the 50 states of Example (Poverty and murder rates), with  $\nu = 46$  error degrees of freedom. The poverty coefficient is 0.2538 with standard error 0.0934, so  $t = 2.718$ . Suppose the family of interest is every linear combination of the three slopes,  $q = 3$ . The 95% intervals for the poverty coefficient are:

| method                   | multiplier | interval       |
|--------------------------|------------|----------------|
| unadjusted $t$           | 2.013      | [0.066, 0.442] |
| Bonferroni, three slopes | 2.485      | [0.022, 0.486] |

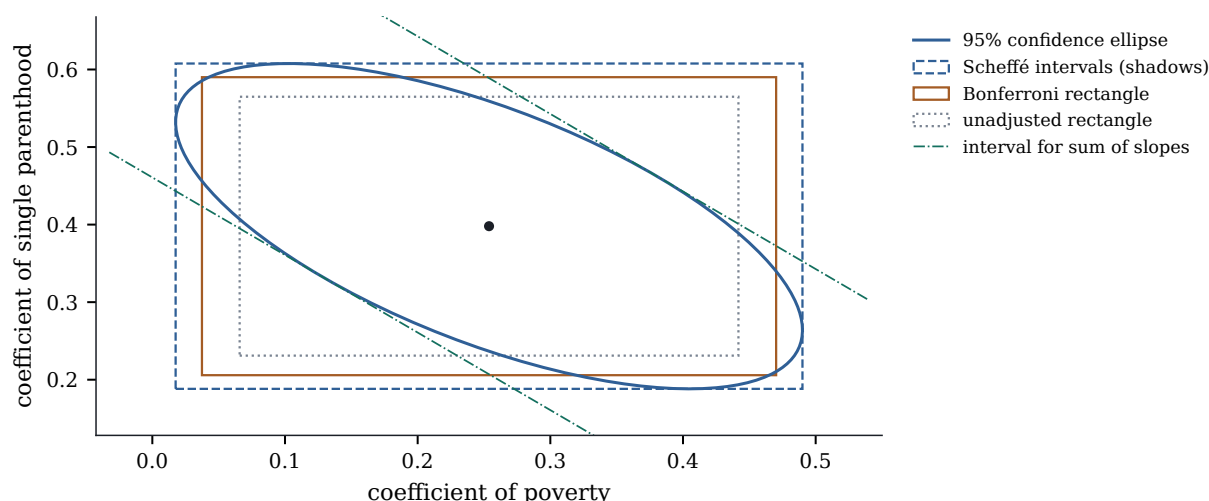


Figure 13.2.1: The coefficients of poverty and single parenthood in the crime regression. The Scheffé intervals (dashed box, multiplier 2.530) are the shadows of the 95% confidence ellipse; the Bonferroni rectangle (multiplier 2.317) covers only the two coefficients. The tangent lines bound the Scheffé interval for the sum of the slopes. The ellipse is that of Figure 12.2.1.

Scheffé, all combinations    2.902     $[-0.017, 0.525]$

The poverty coefficient is significant on its own and among three planned comparisons, but not once every combination of the slopes is admitted. The  $F$  statistic for the hypothesis that all three slopes are zero is 27.49, and the largest  $|t|$  over all combinations is  $9.081 = \sqrt{3 \times 27.49}$ , in agreement with Eq. 13.2.3.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statcrime.load_pandas().data
data = data.drop(index="District of Columbia")

y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
nu = n - p                                # error degrees of freedom
XtX_inv = np.linalg.inv(X.T @ X)
beta_hat = XtX_inv @ X.T @ y
s2 = np.sum((y - X @ beta_hat) ** 2) / nu
se = np.sqrt(s2 * np.diag(XtX_inv))
```

```
alpha, slopes = 0.05, [1, 2, 3]           # poverty, single, urban
k = len(slopes)
mult = {
    "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
    "Bonferroni": stats.t.ppf(1 - alpha / (2 * k), nu),
    "Scheffe": np.sqrt(k * stats.f.ppf(1 - alpha, k, nu)),    # q = 3 slopes
}
for name, c in mult.items():
    lo, hi = beta_hat[slopes] - c * se[slopes], beta_hat[slopes] + c * se[slopes]
    print(f"{name:10s} c = {c:.3f} poverty [{lo[0]:.3f}, {hi[0]:.3f}]")
```

```

L = np.eye(p)[: , slopes]                                # Lambda: the three slopes
W = L.T @ XtX_inv @ L
d = L.T @ beta_hat
F = d @ np.linalg.solve(W, d) / (k * s2)                # F test that all slopes are 0
a_star = np.linalg.solve(W, d)                           # the most significant
combination
t_star = (a_star @ d) / np.sqrt(s2 * a_star @ W @ a_star)
print(f"F = {F:.2f}; largest |t| over all combinations = {t_star:.3f} = sqrt(3F)")

```

### 13.2.3 Scheffé's method is the F test

Let  $\mathbf{d} \in \mathbb{R}^q$  and consider the hypothesis  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ , with  $\mathbf{\Lambda}$  as in [Theorem 13.2.2\(b\)](#). Because  $\mathbf{\Lambda}^\top$  has rank  $q$ , the hypothesis is consistent for every  $\mathbf{d}$ . By [Theorem 11.3.2](#) its  $F$  statistic is

$$F = \frac{(\hat{\boldsymbol{\theta}} - \mathbf{d})^\top \mathbf{W}^{-1} (\hat{\boldsymbol{\theta}} - \mathbf{d})}{q s^2}.$$

The hypothesis implies a single-degree-of-freedom statement for every  $\boldsymbol{\lambda} = \mathbf{\Lambda} \mathbf{a}$  in the family, namely  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \mathbf{a}^\top \mathbf{d}$ . Call such a statement **Scheffé-rejected** if its hypothesized value lies outside the Scheffé interval [Eq. 13.2.2](#).

#### Corollary 13.2.3 — The $F$ test and Scheffé's intervals

In the setting of [Theorem 13.2.2\(b\)](#), for every  $\mathbf{d} \in \mathbb{R}^q$ :

- (a)  $\max_{\mathbf{a} \neq \mathbf{0}} \frac{(\mathbf{a}^\top \hat{\boldsymbol{\theta}} - \mathbf{a}^\top \mathbf{d})^2}{s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a}} = qF$ , attained at  $\mathbf{a} = \mathbf{W}^{-1}(\hat{\boldsymbol{\theta}} - \mathbf{d})$ ;
- (b) the level- $\alpha$   $F$  test rejects  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  iff at least one statement  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \mathbf{a}^\top \mathbf{d}$ ,  $\boldsymbol{\lambda} = \mathbf{\Lambda} \mathbf{a}$ , is Scheffé-rejected;
- (c) if  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  is true, the probability that any of these statements is Scheffé-rejected is exactly  $\alpha$ . For any  $\boldsymbol{\beta}$ , the probability of Scheffé-rejecting at least one *true* statement  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = c$ ,  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{\Lambda})$ , is at most  $\alpha$ .

*Proof.* (a) For  $\boldsymbol{\lambda} = \mathbf{\Lambda} \mathbf{a}$ , the ratio is  $(\mathbf{a}^\top \mathbf{w})^2 / (s^2 \mathbf{a}^\top \mathbf{W} \mathbf{a})$  with  $\mathbf{w} = \hat{\boldsymbol{\theta}} - \mathbf{d}$ , and its maximum is  $\mathbf{w}^\top \mathbf{W}^{-1} \mathbf{w} / s^2 = qF$ , attained at  $\mathbf{a} = \mathbf{W}^{-1} \mathbf{w}$ , by [Corollary 1.7.8\(c\)](#) with  $\mathbf{W}$  positive definite ([Theorem 9.2.4\(a\)](#)); this is the maximum identity of [Section 11.6](#). (b) The statement for  $\mathbf{a}$  is Scheffé-rejected iff its squared  $t$  statistic exceeds  $qF_\alpha(q, \nu)$ , so some statement is rejected iff the maximum in (a) exceeds  $qF_\alpha(q, \nu)$ , that is, iff  $F > F_\alpha(q, \nu)$ . (c) The first claim follows from (b) and the exact level of the  $F$  test ([Theorem 11.3.2](#)). The second is [Proposition 13.1.2\(a\)](#) applied to the simultaneous intervals of [Theorem 13.2.2](#). ■

When the  $F$  test rejects, *some* function in the family is Scheffé-significant, though perhaps not one of interest; conversely, no function is Scheffé-significant unless the  $F$  test rejects, so no separate protecting  $F$  test is needed. For a reduced model the same holds with  $\mathcal{V} = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$ : by [Lemma 13.2.1](#) the  $F$  statistic of [Theorem 11.1.1](#) is the maximum over  $\mathbf{v} \in \mathcal{V}$  of  $(\mathbf{v}^\top \mathbf{Y})^2 / ((r - r_0)s^2 \|\mathbf{v}\|^2)$  ([Exercise \(B1\)](#)). Taking  $\mathcal{C}(\mathbf{\Lambda}) = \mathcal{C}(\mathbf{X}^\top)$ , so that  $q = r$  and the family contains

the mean response  $\mathbf{x}^\top \boldsymbol{\beta}$  at every  $\mathbf{x} \in \mathcal{C}(\mathbf{X}^\top)$ , gives the band  $\mathbf{x}^\top \hat{\boldsymbol{\beta}} \pm \sqrt{rF_\alpha(r, \nu)} s \sqrt{\mathbf{x}^\top \mathbf{G} \mathbf{x}}$ , the Working–Hotelling band of [Theorem 12.4.1](#).

### 13.2.4 Smaller families give shorter intervals

The Scheffé multiplier depends on the family only through its dimension  $q$ , and  $qF_\alpha(q, \nu)$  is strictly increasing in  $q$  for fixed  $\nu$  and  $\alpha$ , as a  $\chi^2$  coupling shows (an exercise of [Section 11.6](#)). So one should use the smallest subspace containing every function of interest, whose dimension can be smaller than the length of a list spanning it ([Exercise \(A2\)](#)). For a few planned functions, Bonferroni's method ([Section 13.4](#)) is usually shorter still, as in [Example \(Three slopes of a crime regression\)](#); Scheffé's multiplier is earned when functions are picked after the estimates are seen.

### 13.2.5 All contrasts in a one-way layout

In the one-way model  $E(y_{kj}) = \mu + \alpha_k$  with  $g$  levels,  $n_k \geq 1$  observations at level  $k$  and  $n = \sum_k n_k$ , the estimable functions of the effects alone are the contrasts  $\psi = \sum_k c_k \mu_k$  with  $\sum_k c_k = 0$  ([Theorem 8.7.1](#)). They form a space of dimension  $g - 1$ , the test space of the hypothesis of equal means. The estimate is  $\hat{\psi} = \sum_k c_k \bar{y}_k$  with standard error  $s(\sum_k c_k^2/n_k)^{1/2}$ , and  $\nu = n - g$ . Scheffé's intervals are

$$\sum_k c_k \bar{y}_k \pm \sqrt{(g-1)F_\alpha(g-1, n-g)} s \left( \sum_k \frac{c_k^2}{n_k} \right)^{1/2} \quad \text{for every contrast } \mathbf{c}, \quad (13.2.4)$$

with simultaneous coverage exactly  $1 - \alpha$ , balanced or not. The most significant contrast has  $c_k = n_k(\bar{y}_k - \bar{y})$  ([Exercise \(B1\)](#)), and its squared  $t$  statistic is  $(g-1)F$ , where  $F$  is the one-way  $F$  statistic.

#### Example. (Education and the perceived position of a candidate)

The 1996 American National Election Study asked respondents to place Bill Clinton on a seven-point scale from extremely liberal (1) to extremely conservative (7). Group the  $n = 944$  respondents by education, coded from 1 (least) to 7 (most). The group sizes are 13, 52, 248, 187, 90, 227, 127, and the mean placements are

$$4.08, 3.58, 3.21, 2.95, 2.79, 2.66, 2.65.$$

More educated respondents place Clinton further to the left. The pooled standard deviation is  $s = 1.354$  on  $\nu = 937$  degrees of freedom, and the one-way  $F$  statistic is 7.88, with  $p$ -value  $2.7 \times 10^{-8}$ . The between-groups sum of squares is 86.72.

Having seen the means, one might ask how the two least educated groups compare with the two most educated, the contrast  $\psi = \frac{1}{2}(\mu_1 + \mu_2) - \frac{1}{2}(\mu_6 + \mu_7)$ . Its estimate is 1.176 with standard error 0.223, so  $t = 5.27$ . Because the contrast was suggested by the data, the unadjusted  $t$  interval is not legitimate. The Scheffé interval [Eq. 13.2.4](#), with multiplier  $\sqrt{6F_{0.05}(6, 937)} = 3.557$ , is  $[0.383, 1.969]$ , and it is legitimate however the contrast was found. The most significant contrast of all has  $t = 6.88 = \sqrt{6 \times 7.88}$ , and its sum of squares is the whole between-groups sum of squares.

```

import numpy as np
import statsmodels.api as sm
from scipy import stats
anes = sm.datasets.anes96.load_pandas().data

y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1      # 0, ..., 6
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g      # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu      # pooled variance estimate
s = np.sqrt(s2)
between = np.sum(n_k * (means - y.mean()) ** 2)
F = between / (g - 1) / s2
print("group sizes", n_k.astype(int))
print("means", means.round(3))
print(f"s = {s:.4f} on {nu} df; F = {F:.3f}, p = {stats.f.sf(F, g - 1, nu):.2g}")

alpha = 0.05
c_sch = np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu))      # Scheffe multiplier,
q = g - 1

def contrast(c):
    """Estimate, standard error and sum of squares of the contrast sum_k c_k mu_k.
    """
    est = c @ means
    v = np.sum(c ** 2 / n_k)
    return est, s * np.sqrt(v), est ** 2 / v

c_max = n_k * (means - y.mean())      # the most significant
contrast
c_lohi = np.array([0.5, 0.5, 0, 0, 0, -0.5, -0.5])      # levels 1-2 against levels
6-7
for name, c in (("maximal", c_max), ("1-2 vs 6-7", c_lohi)):
    est, se, ss = contrast(c)
    print(f"{name:11s} estimate {est:8.3f} t = {est / se:6.2f} SS = {ss:8.2f}"
          f" Scheffe interval [{est - c_sch * se:.3f}, {est + c_sch * se:.3f}]")
print(f"between-groups SS = {between:.2f}; Scheffe multiplier = {c_sch:.3f}")

```

## 13.2.6 Exercises

### 13.2.6.1 A. Check your understanding

#### Exercise A1.

Show that for  $q = 1$  Scheffé's interval Eq. 13.2.2 is the  $t$  interval of Theorem 12.1.1, and explain why no multiplicity adjustment appears.

**Exercise A2.**

In [Example \(Education and the perceived position of a candidate\)](#), an analyst decides before seeing the data to examine the three contrasts  $\frac{1}{2}(\mu_1 + \mu_2) - \frac{1}{2}(\mu_6 + \mu_7)$ ,  $\mu_1 - \mu_7$  and  $\mu_2 - \mu_6$ . Show that they span a space of dimension 2, and state the Scheffé multiplier for this family.

**13.2.6.2 B. Practice****Exercise B1.**

Let  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  with ranks  $r_0 < r$ , and  $\mathcal{V} = \mathcal{C}(\mathbf{M} - \mathbf{M}_0)$ . Show that a vector  $\mathbf{v} \in \mathcal{C}(\mathbf{X})$  lies in  $\mathcal{V}$  iff  $\mathbf{v}^\top \boldsymbol{\mu} = 0$  for every  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}_0)$ . Show that the maximum over  $\mathbf{v} \in \mathcal{V}$  of the squared  $t$  statistic for  $\mathbf{v}^\top \boldsymbol{\mu} = 0$  equals  $(r - r_0)F$ , where  $F$  is the statistic of [Theorem 11.1.1](#).

**13.2.6.3 C. Going deeper****Exercise C1.**

Show that Scheffé's multiplier cannot be reduced: if  $c < \sqrt{qF_\alpha(q, \nu)}$ , the intervals  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} \pm c \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})$ ,  $\boldsymbol{\lambda} \in \mathcal{C}(\boldsymbol{\Lambda})$ , have simultaneous coverage less than  $1 - \alpha$ . So among intervals of the form estimate  $\pm$  constant  $\times$  standard error, valid for the whole family, Scheffé's are the shortest.

*Solution.* By [Eq. 13.2.3](#), all intervals cover iff the maximal squared  $t$  statistic, which has  $q$  times an  $F(q, \nu)$  distribution, is at most  $c^2$ . So the coverage is  $\Pr\{F(q, \nu) \leq c^2/q\}$ , which is less than  $1 - \alpha$  when  $c^2/q < F_\alpha(q, \nu)$ , because the  $F$  distribution has a positive density.

## 13.3 Tukey's studentized range

For the family of all pairwise differences  $\mu_k - \mu_l$ , Tukey saw that the largest standardized error is the range of the estimated means divided by an estimate of their common standard deviation. When the means are estimated independently and with equal precision, as in a balanced one-way layout, this ratio is pivotal, and its upper  $\alpha$  point gives intervals for all pairs with simultaneous coverage exactly  $1 - \alpha$ , shorter than Scheffé's or Bonferroni's.

### 13.3.1 The studentized range distribution

**Definition 13.3.1 — Studentized range**

Let  $Z_1, \dots, Z_g$  be independent  $N(0, 1)$  variables,  $g \geq 2$ , and let  $V \sim \chi^2(\nu)$  be indepen-

dent of them. The law of

$$Q = \frac{\max_k Z_k - \min_k Z_k}{\sqrt{V/\nu}}$$

is the **studentized range distribution** with parameters  $g$  and  $\nu$ . Its upper  $\alpha$  point is written  $q_\alpha(g, \nu)$ .

The numerator is the range of a normal sample of size  $g$ . Dividing by  $\sqrt{V/\nu}$  replaces the known standard deviation 1 by an independent estimate of it with  $\nu$  degrees of freedom, just as a  $t$  variable is a studentized normal. For  $g = 2$  the range is  $|Z_1 - Z_2|$ , a  $N(0, 2)$  variable in absolute value, so  $Q/\sqrt{2}$  is the absolute value of a  $t(\nu)$  variable and

$$q_\alpha(2, \nu) = \sqrt{2} t_{\nu, \alpha/2}. \quad (13.3.1)$$

For larger  $g$  the distribution function is a double integral: conditioning on the smallest  $Z_k$  gives  $\Pr(\max_k Z_k - \min_k Z_k \leq w) = g \int \varphi(z) [\Phi(z + w) - \Phi(z)]^{g-1} dz$ , and one further integration over  $V$  gives the distribution of  $Q$ . SciPy evaluates it as `scipy.stats.studentized_range`. For  $g = 5$  and  $\nu = 20$  it gives  $q_{0.05}(5, 20) = 4.232$ , and 200000 simulated values of  $Q$  put the upper 5% point at 4.230.

```
import itertools
import numpy as np
from scipy import stats

alpha, reps = 0.05, 40_000

g, nu = 5, 20
rng = np.random.default_rng(20260919)
Z = rng.normal(size=(200_000, g))
V = rng.chisquare(nu, size=200_000)
Q = (Z.max(axis=1) - Z.min(axis=1)) / np.sqrt(V / nu)    # studentized range
q_sim = np.quantile(Q, 1 - alpha)
q_exact = stats.studentized_range.ppf(1 - alpha, g, nu)
print(f"upper 5% point of the studentized range, g = {g}, nu = {nu}:"
      f" simulated {q_sim:.3f}, scipy {q_exact:.3f}")
```

### 13.3.2 Tukey's intervals for a balanced layout

Consider the one-way model with  $g$  levels and the same number  $m$  of observations at every level, so  $n = gm$ , and assume normal errors. The level means  $\bar{y}_1, \dots, \bar{y}_g$  are the least squares estimates of  $\mu_1, \dots, \mu_g$ , the pooled variance is  $s^2 = \text{SSE}/\nu$  with  $\nu = g(m - 1)$ , and the standard error of every difference  $\bar{y}_k - \bar{y}_l$  is  $s\sqrt{2/m}$ .

#### Lemma 13.3.1 — A contrast is bounded by the range

If  $\sum_k c_k = 0$ , then for all real  $x_1, \dots, x_g$ ,

$$\left| \sum_k c_k x_k \right| \leq \frac{1}{2} \sum_k |c_k| \left( \max_k x_k - \min_k x_k \right),$$

with equality when  $\mathbf{c}$  is a multiple of a pairwise difference  $\mathbf{e}_k - \mathbf{e}_l$  and  $x_k, x_l$  are the

largest and smallest of the  $x$ 's.

*Proof.* Let  $a = \frac{1}{2}(\max_k x_k + \min_k x_k)$ . Because  $\sum_k c_k = 0$ ,  $\sum_k c_k x_k = \sum_k c_k (x_k - a)$ , and  $|x_k - a| \leq \frac{1}{2}(\max_k x_k - \min_k x_k)$  for every  $k$ . The triangle inequality gives the bound. For  $\mathbf{c} = \mathbf{e}_k - \mathbf{e}_l$ , the left side is  $|x_k - x_l|$  and the right side is the range, which agree when  $x_k, x_l$  are the extremes. ■

### Theorem 13.3.2 — Tukey's simultaneous intervals

In the balanced one-way model with normal errors,  $g \geq 2$  and  $\nu = g(m-1) \geq 1$ , let  $c_T = q_\alpha(g, \nu)$ . Then:

(a) with probability exactly  $1 - \alpha$ ,

$$\mu_k - \mu_l \in \left[ \bar{y}_k - \bar{y}_l - c_T \frac{s}{\sqrt{m}}, \bar{y}_k - \bar{y}_l + c_T \frac{s}{\sqrt{m}} \right] \quad \text{for all } k \neq l; \quad (13.3.2)$$

(b) on the same event, for every contrast  $\psi = \sum_k c_k \mu_k$ ,

$$\left| \sum_k c_k \bar{y}_k - \psi \right| \leq c_T \frac{s}{\sqrt{m}} \cdot \frac{1}{2} \sum_k |c_k|,$$

so these intervals for all contrasts also have simultaneous coverage exactly  $1 - \alpha$ .

*Proof.* By Theorem 7.5.1, the means  $\bar{y}_k$  are independent  $N(\mu_k, \sigma^2/m)$ , and they are independent of  $s^2$ , with  $\nu s^2/\sigma^2 \sim \chi^2(\nu)$ . Put  $Z_k = \sqrt{m}(\bar{y}_k - \mu_k)/\sigma$  and  $V = \nu s^2/\sigma^2$ . These satisfy the assumptions of Definition 13.3.1, whatever the  $\mu_k$  are. Now

$$\max_{k \neq l} |(\bar{y}_k - \bar{y}_l) - (\mu_k - \mu_l)| = \max_{k, l} [(\bar{y}_k - \mu_k) - (\bar{y}_l - \mu_l)] = \frac{\sigma}{\sqrt{m}} (\max_k Z_k - \min_k Z_k),$$

so all intervals Eq. 13.3.2 cover iff  $(\max_k Z_k - \min_k Z_k)/\sqrt{V/\nu} \leq c_T$ , an event of probability  $1 - \alpha$ , since the studentized range has a continuous distribution. For (b), apply Lemma 13.3.1 with  $x_k = \bar{y}_k - \mu_k$ : on that event the range of the  $x_k$  is at most  $c_T s/\sqrt{m}$ . The pairwise differences belong to the family of contrasts, so the coverage of the enlarged family cannot exceed that of the pairs, and it equals  $1 - \alpha$ . ■

### Proposition 13.3.3 — Tukey's intervals are the shortest for all pairs

In the balanced one-way model, suppose intervals  $\bar{y}_k - \bar{y}_l \pm c s \sqrt{2/m}$  with a common multiplier  $c$  have simultaneous coverage at least  $1 - \alpha$  for all pairs. Then  $c \geq q_\alpha(g, \nu)/\sqrt{2}$ . In particular the Bonferroni, Šidák and Scheffé intervals for all pairs are at least as long as Tukey's.



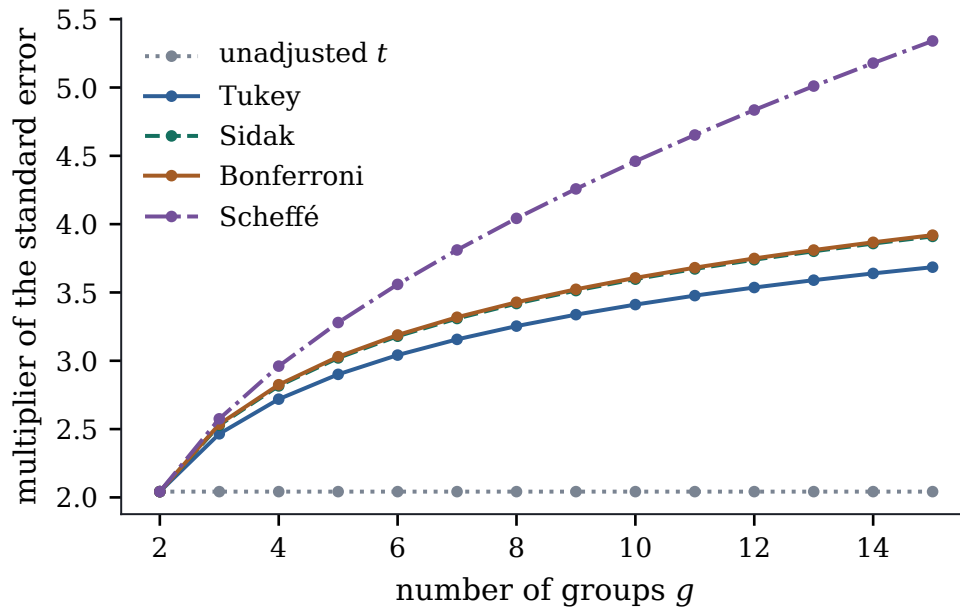


Figure 13.3.1: Multipliers of the standard error for 95% simultaneous intervals for all pairwise differences of  $g$  balanced means,  $\nu = 30$  error degrees of freedom. Tukey's multiplier is  $q_{0.05}(g, 30)/\sqrt{2}$ . The Šidák and Bonferroni curves nearly coincide.

*Proof.* By the proof of [Theorem 13.3.2](#), the simultaneous coverage of such intervals is  $\Pr(Q \leq c\sqrt{2})$ , where  $Q$  has the studentized range distribution. This is at least  $1 - \alpha$  only if  $c\sqrt{2} \geq q_\alpha(g, \nu)$ . The other three methods give intervals of this form with coverage at least  $1 - \alpha$  ([Theorem 13.2.2](#) and [Section 13.4](#)).

■

[Figure 13.3.1](#) compares the multipliers of the standard error of a difference for  $\nu = 30$ . With  $g = 5$  groups they are 2.901 for Tukey, 3.021 for Šidák, 3.030 for Bonferroni and 3.280 for Scheffé. With  $g = 10$  they are 3.411, 3.597, 3.607 and 4.461. The unadjusted multiplier is 2.042 throughout. For  $g = 2$  all methods coincide, by [Eq. 13.3.1](#). Scheffé's method falls further behind as  $g$  grows, because it covers a space of dimension  $g - 1$  while the family of pairs is finite.

```
import numpy as np
from scipy import stats

alpha, nu = 0.05, 30

def pairwise_multipliers(g, nu, alpha=0.05):
    m = g * (g - 1) // 2                                # number of pairs
    return {
        "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
        "Tukey": stats.studentized_range.ppf(1 - alpha, g, nu) / np.sqrt(2),
        "Sidak": stats.t.ppf((1 + (1 - alpha) ** (1 / m)) / 2, nu),
        "Bonferroni": stats.t.ppf(1 - alpha / (2 * m), nu),
        "Scheffe": np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu)),
    }

for g in (2, 3, 5, 10):
    row = pairwise_multipliers(g, nu)
    print(f"g = {g:2d}: " + " ".join(f"{k} {v:.3f}" for k, v in row.items()))
```

For contrasts other than pairs the comparison can reverse. Part (b) of [Theorem 13.3.2](#) gives the half-width  $q_\alpha(g, \nu) (s/\sqrt{m}) \frac{1}{2} \sum_k |c_k|$ , while Scheffé's is  $\sqrt{(g-1)F_\alpha(g-1, \nu)} (s/\sqrt{m}) (\sum_k c_k^2)^{1/2}$ . With  $g = 6$  and  $\nu = 30$ , in units of  $s/\sqrt{m}$ , a pairwise difference has half-width 4.301 by Tukey's method and 5.033 by Scheffé's; for  $c = \frac{1}{3}(1, 1, 1, -1, -1, -1)$  the half-widths are 4.301 and 2.906. Tukey's bound is sharp for pairs and crude for contrasts spread over many means.

### 13.3.3 Unbalanced layouts: the Tukey–Kramer intervals

With unequal group sizes  $n_k$  the maximum standardized difference is no longer a studentized range. Tukey and Kramer proposed keeping the multiplier and using the standard error of each difference:

$$\bar{y}_k - \bar{y}_l \pm \frac{q_\alpha(g, \nu)}{\sqrt{2}} s \sqrt{\frac{1}{n_k} + \frac{1}{n_l}}, \quad \nu = n - g. \quad (13.3.3)$$

Tukey conjectured, and Hayter (1984) proved, that the simultaneous coverage of these intervals is never less than  $1 - \alpha$ , whatever the group sizes. The proof is beyond this chapter. How conservative the intervals are is a separate question; the simulation below estimates the coverage of 95% intervals in three designs from 40000 data sets each (standard error about 0.0011):

| design                                  | coverage |
|-----------------------------------------|----------|
| balanced, five groups of 5              | 0.9511   |
| the education example (sizes 13 to 248) | 0.9562   |
| four groups of 2 and four of 40         | 0.9614   |

The balanced design is exact; the unbalanced ones overcover only modestly, even with very different group sizes.

```
import itertools
import numpy as np
from scipy import stats

alpha, reps = 0.05, 40_000

def tukey_kramer_coverage(n_k, reps, alpha, seed):
    """Share of data sets in which every Tukey-Kramer interval covers its difference
    ."""
    rng = np.random.default_rng(seed)
    n_k = np.asarray(n_k)
    g, nu = len(n_k), n_k.sum() - len(n_k)
    q = stats.studentized_range.ppf(1 - alpha, g, nu)
    # group means (true means 0) and an independent pooled variance, as in the model
    means = rng.normal(size=(reps, g)) / np.sqrt(n_k)
    s = np.sqrt(rng.chisquare(nu, size=reps) / nu)
    ok = np.ones(reps, dtype=bool)
    for k, l in itertools.combinations(range(g), 2):
        half = q / np.sqrt(2) * s * np.sqrt(1 / n_k[k] + 1 / n_k[l])
        ok &= np.abs(means[:, k] - means[:, l]) <= half
    return ok.mean()
```

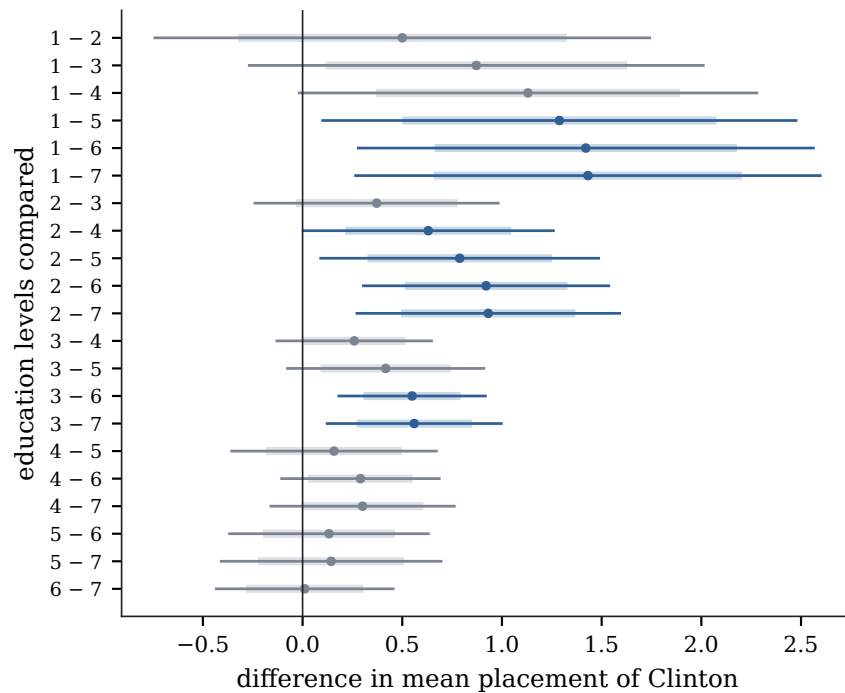


Figure 13.3.2: Differences in mean placement of Clinton between education levels, with 95% Tukey–Kramer simultaneous intervals (thin lines) and unadjusted  $t$  intervals (thick pale bands). Differences whose Tukey–Kramer interval excludes zero are drawn in blue.

```
balanced = tukey_kramer_coverage([5] * 5, reps, alpha, seed=1)
education = tukey_kramer_coverage([13, 52, 248, 187, 90, 227, 127], reps, alpha,
    seed=2)
lopsided = tukey_kramer_coverage([2, 2, 2, 2, 40, 40, 40, 40], reps, alpha, seed=3)
print(f"balanced (5 x 5):           {balanced:.4f}")
print(f"education design:          {education:.4f}")
print(f"four groups of 2, four of 40: {lopsided:.4f}")
```

#### Example. (All pairs of education levels)

In [Example \(Education and the perceived position of a candidate\)](#) there are  $g = 7$  education levels, so 21 pairs, and  $\nu = 937$ . The Tukey–Kramer multiplier is  $q_{0.05}(7, 937)/\sqrt{2} = 4.179/\sqrt{2} = 2.955$ . It declares 9 of the 21 differences nonzero, against 14 for unadjusted  $t$  intervals ([Section 13.6](#) compares all the methods). [Figure 13.3.2](#) shows the intervals. The significant differences all separate a lower education level from a higher one, and the three most educated levels do not differ from one another. The smallest group, with only 13 respondents, has the highest mean, but its intervals are so wide that only its differences from levels 5, 6 and 7 are established.

```
import itertools
import numpy as np
import statsmodels.api as sm
from scipy import stats
anes = sm.datasets.anes96.load_pandas().data
y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1      # 0, ..., 6
```

```

g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g                                # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu  # pooled variance estimate
s = np.sqrt(s2)
alpha = 0.05
c_sch = np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu))  # Scheffe multiplier,
    q = g - 1

pairs = list(itertools.combinations(range(g), 2))
m = len(pairs)                            # 21 comparisons
diff = np.array([means[k] - means[l] for k, l in pairs])
se_d = np.array([s * np.sqrt(1 / n_k[k] + 1 / n_k[l]) for k, l in pairs])
t = diff / se_d
pval = 2 * stats.t.sf(np.abs(t), nu)

mult = {                                # half-width = multiplier x se
    "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
    "Bonferroni": stats.t.ppf(1 - alpha / (2 * m), nu),
    "Sidak": stats.t.ppf((1 + (1 - alpha) ** (1 / m)) / 2, nu),
    "Tukey-Kramer": stats.studentized_range.ppf(1 - alpha, g, nu) / np.sqrt(2),
    "Scheffe": c_sch,
}
for name, c in mult.items():
    print(f"{name:13s} multiplier {c:.3f}    pairs declared different: {np.sum(np.abs
(t) > c)}")

```

### 13.3.4 Beyond the one-way layout

The proof of [Theorem 13.3.2](#) used only that the  $g$  estimates are independent, with a common variance  $\sigma^2 c$  for a known  $c$ , and independent of  $s^2$ . That also holds for the levels of one factor in a balanced multi-way design ([Exercise B1](#)), the subject of [Chapters 15](#) and [16](#). For correlated estimates, such as regression coefficients or adjusted means, neither form is exact; the exact critical value for any planned family then comes from a multivariate  $t$  distribution ([Exercise C1](#)), which standard software computes.

### 13.3.5 Exercises

#### 13.3.5.1 A. Check your understanding

##### Exercise A1.

Prove [Eq. 13.3.1](#), and deduce that Tukey's intervals for  $g = 2$  are the ordinary  $t$  intervals.

#### 13.3.5.2 B. Practice

##### Exercise B1.

In a two-way additive model  $E(y_{ijk}) = \mu + \alpha_i + \beta_j$  with  $a$  rows,  $b$  columns and  $N$

observations in every cell, show that the column means  $\bar{y}_{\cdot j}$  are independent, each with variance  $\sigma^2/(aN)$ , and independent of the residual mean square. Deduce Tukey intervals for all differences  $\beta_j - \beta_{j'}$ , and state their degrees of freedom.

*Solution.* Each  $\bar{y}_{\cdot j}$  averages the  $aN$  observations of column  $j$ , and different columns use disjoint sets of independent observations, so the column means are independent with variance  $\sigma^2/(aN)$ . Their expectations are  $\mu + \bar{\alpha} + \beta_j$ , with  $\bar{\alpha}$  the average row effect, so differences estimate  $\beta_j - \beta_{j'}$ . The column means are functions of the fitted values (the fitted value of cell  $(i, j)$  is  $\bar{y}_{i\cdot} + \bar{y}_{\cdot j} - \bar{y}$ , and averaging it over  $i$  gives  $\bar{y}_{\cdot j}$ ), so they are independent of  $s^2$  by [Theorem 7.5.1](#). The residual degrees of freedom are  $\nu = abN - a - b + 1$ , and the intervals are  $\bar{y}_{\cdot j} - \bar{y}_{\cdot j'} \pm q_\alpha(b, \nu) s / \sqrt{aN}$ , exact for all pairs of columns.

### Exercise B2.

Treatments  $1, \dots, g-1$  are to be compared with a control, level 0, in a balanced layout. Show that the exact simultaneous multiplier for the  $g-1$  differences  $\mu_k - \mu_0$  is the upper  $\alpha$  point of  $\max_{k \geq 1} |Z_k - Z_0| / \sqrt{V/\nu}$ , and that it is smaller than  $q_\alpha(g, \nu)$ .

*Solution.* As in the proof of [Theorem 13.3.2](#), the largest standardized error among the differences with the control is  $(\sigma/\sqrt{m}) \max_{k \geq 1} |Z_k - Z_0|$ , and the statistic  $D = \max_{k \geq 1} |Z_k - Z_0| / \sqrt{V/\nu}$  is pivotal. Let  $Q$  be the studentized range of  $Z_0, \dots, Z_{g-1}$ , so that  $D \leq Q$ . The event  $\{D \leq q_\alpha(g, \nu) < Q\}$  has positive probability: it contains an open set of values of  $(Z_0, \dots, Z_{g-1}, V)$ , for instance those with  $Z_0$  near the middle of the others and the range just above the critical value, and these variables have a positive joint density. Hence  $\Pr(D > q_\alpha(g, \nu)) = \alpha - \Pr(D \leq q_\alpha(g, \nu) < Q) < \alpha$ , and since  $D$  has a continuous distribution its upper  $\alpha$  point is smaller than  $q_\alpha(g, \nu)$ . This is Dunnett's many-to-one procedure.

### Exercise B3.

In the balanced one-way layout, the **studentized range test** rejects  $\mu_1 = \dots = \mu_g$  when  $Q_{\text{obs}} = (\max_k \bar{y}_k - \min_k \bar{y}_k) / (s/\sqrt{m}) > q_\alpha(g, \nu)$ . Show that it has exact level  $\alpha$ , and that it rejects iff some Tukey interval [Eq. 13.3.2](#) excludes 0. Show that the  $F$  test rejects iff  $\sum_k (\bar{y}_k - \bar{y})^2 > (g-1)F_\alpha(g-1, \nu) s^2/m$ , and describe, for  $g=3$ , configurations of the means that one test rejects and the other does not.

*Solution.* Under the hypothesis  $Q_{\text{obs}}$  has the studentized range distribution (proof of [Theorem 13.3.2](#) with all  $\mu_k$  equal), so the level is exactly  $\alpha$ , and  $Q_{\text{obs}} > q_\alpha(g, \nu)$  says that the extreme pair's interval excludes 0; every other pair has a smaller difference. The  $F$  statistic is  $m \sum_k (\bar{y}_k - \bar{y})^2 / \{(g-1)s^2\}$ . For  $g=3$  and a fixed range  $2a$ , the sum of squares is  $2a^2$  for means  $(-a, 0, a)$  and  $8a^2/3$  for  $(-a, -a, a)$ . In the limit  $\nu \rightarrow \infty$ , in units of  $s/\sqrt{m}$ , the range test rejects when  $2a > 3.314$  and the  $F$  test when the sum of squares exceeds 5.991: so  $(-a, 0, a)$  with  $1.657 < a < 1.731$  is rejected only by the range test, and  $(-a, -a, a)$  with  $1.499 < a < 1.657$  only by the  $F$  test. The range test is the natural gate for pairs, the  $F$  test for contrasts in general.

#### 13.3.5.3 C. Going deeper

**Exercise C1.**

Let  $\lambda_1, \dots, \lambda_k \in \mathcal{C}(\mathbf{X}^\top)$  be chosen in advance, with  $t$  statistics  $T_j = (\lambda_j^\top \hat{\beta} - \lambda_j^\top \beta) / \text{se}(\lambda_j^\top \hat{\beta})$ . Show that  $(T_1, \dots, T_k) = \mathbf{Z} / (s/\sigma)$ , where  $\mathbf{Z} \sim N_k(\mathbf{0}, \mathbf{R})$  is independent of  $s$ , and  $\mathbf{R}$  is the correlation matrix with entries  $\lambda_i^\top \mathbf{G} \lambda_j / \sqrt{\lambda_i^\top \mathbf{G} \lambda_i \cdot \lambda_j^\top \mathbf{G} \lambda_j}$ . Deduce that the constant  $c$  with  $\Pr(\max_j |T_j| \leq c) = 1 - \alpha$  depends only on  $\mathbf{R}$  and  $\nu$ , and that it is at most the Bonferroni multiplier and at most Scheffé's multiplier for the span of the  $\lambda_j$ .

## 13.4 Bonferroni and Holm

Scheffé's and Tukey's methods need an exact distribution for the worst error in a family. The methods of this section need only the level of each separate test and a probability inequality, so they apply to any finite list of hypotheses in any model, whatever the dependence. Holm's step-down procedure recovers part of the resulting conservatism at no cost in generality.

### 13.4.1 The Bonferroni inequality

#### Theorem 13.4.1 — Bonferroni

- (a) For any events  $A_1, \dots, A_m$ ,  $\Pr(\bigcup_j A_j) \leq \sum_j \Pr(A_j)$ .
- (b) Let  $\alpha_1, \dots, \alpha_m > 0$  with  $\sum_j \alpha_j = \alpha$ . If  $[L_j, U_j]$  is a confidence interval for  $\psi_j$  with coverage at least  $1 - \alpha_j$ , for each  $j$ , then the  $m$  intervals have simultaneous coverage at least  $1 - \alpha$ .
- (c) Let  $p_j$  be a  $p$ -value for  $H_j$ , valid in the sense that  $\Pr(p_j \leq u) \leq u$  for  $0 \leq u \leq 1$  whenever  $H_j$  is true. The procedure that rejects  $H_j$  when  $p_j \leq \alpha_j$  has  $\text{PFER} \leq \sum_{j \in I_0} \alpha_j \leq \alpha$ , and hence controls the familywise error rate strongly at level  $\alpha$ .

No assumption is made about the dependence among the events, intervals or  $p$ -values.

*Proof.* (a) Pointwise, the indicator of a union is at most the sum of the indicators:  $1_{\bigcup_j A_j} \leq \sum_j 1_{A_j}$ , since the right side is at least 1 wherever the left side is. Take expectations. (b) Apply (a) to the events  $A_j = \{\psi_j \notin [L_j, U_j]\}$ . (c) The number of false rejections is  $V = \sum_{j \in I_0} 1_{\{p_j \leq \alpha_j\}}$ , so  $E(V) = \sum_{j \in I_0} \Pr(p_j \leq \alpha_j) \leq \sum_{j \in I_0} \alpha_j$ . Strong control follows from Proposition 13.1.1. ■

Part (c) bounds even the expected *number* of false rejections, which is why the procedure is conservative when false rejections tend to occur together.

In the linear model, the **Bonferroni intervals** for  $k$  estimable functions  $\lambda_1^\top \beta, \dots, \lambda_k^\top \beta$  chosen in advance are

$$\lambda_j^\top \hat{\beta} \pm t_{\nu, \alpha/(2k)} \text{se}(\lambda_j^\top \hat{\beta}), \quad j = 1, \dots, k, \quad (13.4.1)$$

each the  $t$  interval of [Theorem 12.1.1](#) at level  $1 - \alpha/k$ . The equal split is conventional; any split fixed before the data are seen is valid.

*Warning.* The Bonferroni guarantee is for a list of  $k$  hypotheses fixed in advance. If the analyst looks at the estimates and then chooses which  $k$  to test, the hypotheses actually tested are a data-dependent selection from a larger family, and the relevant  $k$  is the size of that larger family. For data-driven choices within a subspace, use Scheffé's method.

## 13.4.2 Independent tests: Šidák's improvement

If the tests of the true hypotheses are independent and each has exact level  $\alpha$ , then by [Eq. 13.1.1](#) the familywise error rate is  $1 - (1 - \alpha)^m$ . Choosing

$$\alpha = 1 - (1 - \alpha)^{1/m} \quad (13.4.2)$$

makes this at most  $1 - (1 - \alpha)^m = \alpha$ , with equality when all hypotheses are true. This **Šidák level** is slightly larger than  $\alpha/m$ , because  $(1 - \alpha/m)^m > 1 - \alpha$  for  $m \geq 2$  (Bernoulli's inequality), so the Šidák procedure is slightly less conservative than Bonferroni's.

In a linear model, however,  $t$  statistics are never independent: they share  $s$ . The following result shows that the sharing works in the right direction.

### Lemma 13.4.2 — Chebyshev's association inequality

Let  $U$  be a real random variable and  $f_1, \dots, f_k$  nonnegative, nondecreasing functions with  $E f_j(U)^k < \infty$ . Then  $E[\prod_j f_j(U)] \geq \prod_j E f_j(U)$ .

*Proof.* For two functions  $f, h$ , let  $U'$  be an independent copy of  $U$ . Since  $f$  and  $h$  are both nondecreasing,  $(f(U) - f(U'))(h(U) - h(U')) \geq 0$ . Taking expectations gives  $2E[f(U)h(U)] - 2E f(U) E h(U) \geq 0$ . For  $k$  functions, argue by induction: the product  $f_1 \cdots f_{k-1}$  of nonnegative nondecreasing functions is nonnegative and nondecreasing, so  $E[f_1 \cdots f_k(U)] \geq E[f_1 \cdots f_{k-1}(U)] E f_k(U)$ , and the induction hypothesis applies to the first factor. The moment condition makes every expectation finite, by Hölder's inequality. ■

### Proposition 13.4.3 — Šidák intervals

Let  $Z_1, \dots, Z_k$  be independent  $N(0, 1)$  variables and  $V \sim \chi^2(\nu)$  independent of them, and put  $T_j = Z_j/\sqrt{V/\nu}$ . Then for every  $c > 0$ ,

$$\Pr(|T_j| \leq c \text{ for all } j) \geq \prod_{j=1}^k \Pr(|T_j| \leq c).$$

Consequently, if  $\lambda_1^\top \hat{\beta}, \dots, \lambda_k^\top \hat{\beta}$  are uncorrelated, the intervals  $\lambda_j^\top \hat{\beta} \pm c \text{se}(\lambda_j^\top \hat{\beta})$  with  $c$  the upper  $\frac{1}{2}(1 - (1 - \alpha)^{1/k})$  point of  $t(\nu)$  have simultaneous coverage at least  $1 - \alpha$ .

*Proof.* Conditionally on  $V = v$ , the events  $\{|Z_j| \leq c\sqrt{v/\nu}\}$  are independent, each with probability  $f(v) = 2\Phi(c\sqrt{v/\nu}) - 1$ . So  $\Pr(|T_j| \leq c \ \forall j) = E[f(V)^k]$ , while  $\Pr(|T_j| \leq c) = E f(V)$ . The function  $f$  is nonnegative, bounded and nondecreasing, and Lemma 13.4.2 gives  $E[f(V)^k] \geq (E f(V))^k$ . For the intervals, the estimates are jointly normal (Theorem 7.5.1), so uncorrelated estimates are independent (Theorem 3.4.1), and they are independent of  $s$ . The standardized errors are therefore of the form  $T_j$ , and each  $\Pr(|T_j| \leq c) = (1 - \alpha)^{1/k}$ . ■

The proposition covers orthogonal contrasts (Proposition 8.7.2(c)) and regressions with orthogonal columns. Šidák (1967) proved the corresponding inequality  $\Pr(|Z_j| \leq c_j \ \forall j) \geq \prod_j \Pr(|Z_j| \leq c_j)$  for *any* centred jointly normal  $(Z_1, \dots, Z_k)$ ; with the conditioning argument above it makes Šidák's intervals valid for any finite family of estimable functions. We use that theorem without proof. The gain over Bonferroni is small: for  $k = 10$  and  $\nu = 29$  the multipliers are 3.0289 and 3.0380, and with independent numerators and a shared  $s$  the exact coverage of the nominal 95% intervals, computed by integrating over  $s$ , is 0.9520 and 0.9530.

### 13.4.3 The closure principle

Bonferroni's procedure keeps the threshold  $\alpha/m$  even after rejections, although a rejected false hypothesis can no longer be falsely rejected. The closure principle exploits this. For each nonempty  $I \subseteq \{1, \dots, m\}$  let  $H_I = \bigcap_{i \in I} H_i$  be the **intersection hypothesis**, and let  $\phi_I$  be a level- $\alpha$  **local test** of it.

#### Theorem 13.4.4 — Closed testing

The procedure that rejects  $H_j$  iff  $\phi_I$  rejects  $H_I$  for every  $I$  containing  $j$  controls the familywise error rate strongly at level  $\alpha$ .

*Proof.* Fix the parameters and suppose  $I_0 \neq \emptyset$ ; otherwise there is nothing to prove. The intersection hypothesis  $H_{I_0}$  is true, so  $\Pr(\phi_{I_0} \text{ rejects}) \leq \alpha$ . If some true  $H_j$  is rejected, then  $j \in I_0$ , so by the definition of the procedure  $\phi_{I_0}$  rejects. Hence  $\Pr(V \geq 1) \leq \Pr(\phi_{I_0} \text{ rejects}) \leq \alpha$ . ■

Rejecting one hypothesis requires rejecting every intersection containing it, in general  $2^{m-1}$  tests. Holm's procedure is a case where this collapses to  $m$  steps.

### 13.4.4 Holm's step-down procedure

Order the  $p$ -values as  $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$ , breaking ties arbitrarily, and let  $H_{(i)}$  be the hypothesis with the  $i$ th smallest  $p$ -value.

**Holm's procedure at level  $\alpha$ .** For  $i = 1, 2, \dots, m$  in turn: if  $p_{(i)} \leq \alpha/(m - i + 1)$ , reject  $H_{(i)}$  and continue; otherwise stop, and reject none of  $H_{(i)}, \dots, H_{(m)}$ .



The first threshold is Bonferroni's,  $\alpha/m$ . Each rejection relaxes the next threshold: to  $\alpha/(m-1)$ , then  $\alpha/(m-2)$ , and so on, up to  $\alpha$  for the last hypothesis.

### Theorem 13.4.5 — Holm

Let each  $p_j$  be valid for  $H_j$ , as in [Theorem 13.4.1\(c\)](#), with any dependence among them. Then:

- (a) Holm's procedure controls the familywise error rate strongly at level  $\alpha$ ;
- (b) it rejects every hypothesis that Bonferroni's procedure (all  $\alpha_j = \alpha/m$ ) rejects, and possibly more.

*Proof.* (a) Suppose  $m_0 \geq 1$ , and let  $j^*$  be the rank, in the ordering of all  $m$   $p$ -values, of the smallest  $p$ -value belonging to a true hypothesis, so that  $p_{(j^*)} = \min_{i \in I_0} p_i$ . The hypotheses ranked before  $j^*$  are all false, and there are only  $m - m_0$  false hypotheses, so  $j^* \leq m - m_0 + 1$  and  $\alpha/(m - j^* + 1) \leq \alpha/m_0$ . Holm's procedure rejects an initial segment  $H_{(1)}, \dots, H_{(i)}$  of the ordering. If that segment contains any true hypothesis, it contains  $H_{(j^*)}$ , so  $p_{(j^*)} \leq \alpha/(m - j^* + 1) \leq \alpha/m_0$ . Therefore

$$\Pr(V \geq 1) \leq \Pr\left(\min_{i \in I_0} p_i \leq \frac{\alpha}{m_0}\right) \leq \sum_{i \in I_0} \Pr\left(p_i \leq \frac{\alpha}{m_0}\right) \leq \alpha,$$

by [Theorem 13.4.1\(a\)](#) and the validity of the  $p$ -values. (b) If Bonferroni rejects  $H_{(i)}$ , then  $p_{(j)} \leq p_{(i)} \leq \alpha/m \leq \alpha/(m - j + 1)$  for every  $j \leq i$ , so Holm's procedure does not stop before step  $i$  and rejects  $H_{(i)}$ . ■

Bonferroni's  $\alpha/m$  pays for  $m$  potential errors when only  $m_0$  are possible. Holm's procedure is closed testing with the Bonferroni local tests ([Exercise \(B3\)](#)). It gives no simple simultaneous intervals, since the threshold for a hypothesis depends on the other  $p$ -values; its output is a list of rejections, or **adjusted  $p$ -values**

$$\tilde{p}_{(i)} = \max_{j \leq i} \min\{1, (m - j + 1)p_{(j)}\}, \quad (13.4.3)$$

with  $H_{(i)}$  rejected at level  $\alpha$  iff  $\tilde{p}_{(i)} \leq \alpha$  ([Exercise \(B1\)](#)).

#### Example. (Holm for the education pairs)

For the 21 pairwise comparisons of [Example \(All pairs of education levels\)](#) the smallest  $p$ -value is  $1.1 \times 10^{-5}$ . Bonferroni's threshold is  $0.05/21 = 0.00238$ , and 8 of the  $p$ -values fall below it. Holm's procedure starts at the same threshold but relaxes it after each rejection. It rejects 10 hypotheses: the tenth smallest  $p$ -value is 0.0037, below its threshold  $0.05/12$ , and the eleventh, 0.0126, exceeds  $0.05/11 = 0.0045$ , so the procedure stops.

```
import itertools
import numpy as np
import statsmodels.api as sm
```

```

from scipy import stats
anes = sm.datasets.anes96.load_pandas().data
y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1          # 0, ..., 6
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g                                               # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu                # pooled variance estimate
s = np.sqrt(s2)
alpha = 0.05
pairs = list(itertools.combinations(range(g), 2))
diff = np.array([means[k] - means[l] for k, l in pairs])
se_d = np.array([s * np.sqrt(1 / n_k[k] + 1 / n_k[l]) for k, l in pairs])
t = diff / se_d
pval = 2 * stats.t.sf(np.abs(t), nu)

def holm(p, alpha):
    """Holm's step-down procedure: reject the i smallest p-values, i maximal with
    p_(j) <= alpha / (m - j + 1) for all j <= i."""
    m = len(p)
    order = np.argsort(p)
    reject = np.zeros(m, dtype=bool)
    for j, idx in enumerate(order):                      # j = 0, 1, ...
        if p[idx] > alpha / (m - j):
            break
        reject[idx] = True
    return reject

def benjamini_hochberg(p, q):
    """Reject the k smallest p-values, k the largest index with p_(k) <= k q / m."""
    m = len(p)
    order = np.argsort(p)
    below = np.nonzero(p[order] <= q * np.arange(1, m + 1) / m)[0]
    reject = np.zeros(m, dtype=bool)
    if below.size:
        reject[order[: below[-1] + 1]] = True
    return reject

print("Holm rejects", holm(pval, alpha).sum(), " BH (q = 0.05) rejects",
      benjamini_hochberg(pval, alpha).sum())

```

### 13.4.5 How much does Holm gain?

If all hypotheses are true, Holm and Bonferroni make a false rejection in the same data sets. The more hypotheses are false and rejected early, the larger the later thresholds and the gain.

#### Example. (Ten regression coefficients)

A regression has an intercept and ten centred, mutually orthogonal regressors, with  $n = 40$ . The ten  $t$  statistics for the slopes are then  $T_j = Z_j / \sqrt{V/29}$  with independent  $Z_j \sim N(\delta_j, 1)$  and a shared  $V \sim \chi^2(29)$  (Theorem 7.5.1). Let  $k$  of the slopes be nonzero, each with  $\delta_j = 3.5$ , and test all ten at familywise level 0.05. Figure 13.4.1

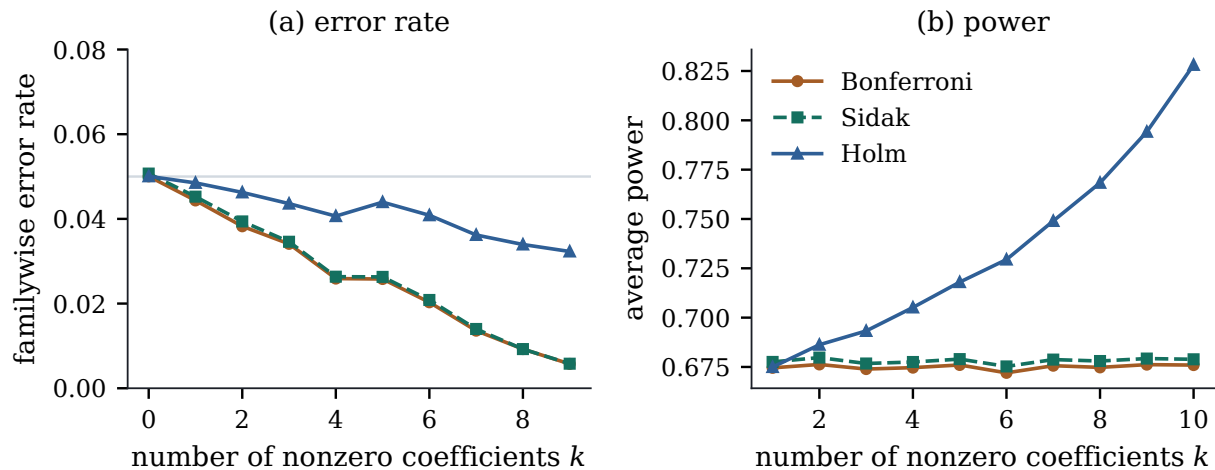


Figure 13.4.1: Ten orthogonal regression coefficients,  $k$  of them nonzero, tested at familywise level 0.05: (a) familywise error rate, (b) average power, the share of nonzero coefficients detected.

shows the result of 20000 simulated data sets for each  $k$ . All three procedures keep the familywise error rate at or below 0.05, up to simulation error. The average power of Bonferroni's procedure, the proportion of nonzero slopes it detects, hardly depends on  $k$ : it is 0.675 with one nonzero slope and 0.676 with five or nine. Šidák's is larger by a few thousandths. Holm's power is 0.675 with one nonzero slope, 0.718 with five and 0.794 with nine.

```
import numpy as np
from scipy import stats

alpha, m, nu, delta, reps = 0.05, 10, 29, 3.5, 20_000

def simulate(k, reps, seed):
    """FWER and average power of three procedures when k of m coefficients are
    nonzero."""
    rng = np.random.default_rng(seed)
    shift = np.r_[np.full(k, delta), np.zeros(m - k)]
    T = (rng.normal(size=(reps, m)) + shift) / np.sqrt(rng.chisquare(nu, (reps, 1))
    / nu)
    p = 2 * stats.t.sf(np.abs(T), nu)
    rules = {
        "Bonferroni": p <= alpha / m,
        "Sidak": p <= 1 - (1 - alpha) ** (1 / m),
    }
    # Holm: sort each row, compare p_(j) with alpha / (m - j + 1), stop at the first
    failure
    order = np.argsort(p, axis=1)
    ps = np.take_along_axis(p, order, axis=1)
    passed = np.cumprod(ps <= alpha / (m - np.arange(m)), axis=1).astype(bool)
    holm = np.zeros_like(passed)
    np.put_along_axis(holm, order, passed, axis=1)
    rules["Holm"] = holm
    out = {}
    for name, rej in rules.items():
        fwer = np.mean(rej[:, k:].any(axis=1)) if k < m else np.nan
        power = rej[:, :k].mean() if k > 0 else np.nan
        out[name] = (fwer, power, rej)
    return out
```

```
res = simulate(5, reps, seed=5)
for name, (fwer, power, _) in res.items():
    print(f"{name:10s} FWER = {fwer:.3f}    average power = {power:.3f}")
```

There is no reason to use Bonferroni's *tests* rather than Holm's, though Bonferroni's *intervals* remain useful. With independent  $p$ -values, Hochberg (1988) proposed the *step-up* version: reject  $H_{(1)}, \dots, H_{(i)}$  for the largest  $i$  with  $p_{(i)} \leq \alpha/(m - i + 1)$ . It rejects everything Holm's procedure does ([Exercise \(C1\)](#)), but its validity rests on an inequality of Simes (1986) that holds under independence and some positive dependence, not in general. The step-up idea returns in the next section, controlling a different error rate.

## 13.4.6 Exercises

### 13.4.6.1 A. Check your understanding

#### Exercise A1.

Five hypotheses have  $p$ -values 0.001, 0.011, 0.012, 0.04, 0.2. At familywise level 0.05, which are rejected by Bonferroni's procedure and which by Holm's? Compute Holm's adjusted  $p$ -values [Eq. 13.4.3](#).

*Solution.* Bonferroni's threshold is 0.01, so only the first is rejected. Holm compares  $0.001 \leq 0.05/5$ ,  $0.011 \leq 0.05/4$ ,  $0.012 \leq 0.05/3$ , and then  $0.04 > 0.05/2$ , so it rejects the first three and stops. The adjusted  $p$ -values are  $5 \times 0.001 = 0.005$ ,  $\max(0.005, 4 \times 0.011) = 0.044$ ,  $\max(0.044, 3 \times 0.012) = 0.044$ ,  $\max(0.044, 2 \times 0.04) = 0.08$  and  $\max(0.08, 0.2) = 0.2$ . The three below 0.05 are the rejected hypotheses.

### 13.4.6.2 B. Practice

#### Exercise B1.

Show that Holm's procedure rejects  $H_{(i)}$  iff the adjusted  $p$ -value [Eq. 13.4.3](#) satisfies  $\tilde{p}_{(i)} \leq \alpha$ , and that the adjusted  $p$ -values are nondecreasing in  $i$ .

#### Exercise B2.

All  $m \geq 2$  hypotheses are true and the  $p$ -values are independent and uniform. Show that the per-family error rate of Bonferroni's procedure is exactly  $\alpha$ , while that of Holm's procedure is larger than  $\alpha$ . So Holm's procedure controls the familywise rate but not the per-family rate.

#### Exercise B3.

Show that Holm's procedure is closed testing ([Theorem 13.4.4](#)) with the Bonferroni local tests  $\phi_I$ : reject  $H_I$  when  $\min_{i \in I} p_i \leq \alpha/|I|$ .

*Solution.* Suppose Holm rejects  $H_{(1)}, \dots, H_{(i)}$  and stops at step  $i + 1$ . Take  $j \leq i$  and any  $I \ni (j)$ , writing  $(j)$  for the index of  $H_{(j)}$ . If  $I$  contains some  $(l)$  with  $l \leq j$ , let  $l$  be the smallest such; then  $I$  contains no  $(1), \dots, (l - 1)$ , so  $|I| \leq m - l + 1$  and  $\min_I p \leq p_{(l)} \leq \alpha / (m - l + 1) \leq \alpha / |I|$ :  $\phi_I$  rejects. So closed testing rejects  $H_{(j)}$ . Conversely, the set  $I = \{(i + 1), \dots, (m)\}$  has  $m - i$  elements and minimum  $p_{(i+1)} > \alpha / (m - i)$ , so  $\phi_I$  does not reject, and no hypothesis in  $I$  is rejected by closed testing. The two procedures therefore reject the same hypotheses.

### 13.4.6.3 C. Going deeper

#### Exercise C1.

Show that Hochberg's step-up procedure rejects every hypothesis rejected by Holm's step-down procedure with the same thresholds. Give an example with  $m = 2$  where Hochberg rejects both hypotheses and Holm rejects neither.

## 13.5 False discovery rate control

With hundreds of hypotheses, as when a regression screens many candidate predictors, family-wise control becomes self-defeating: the threshold  $\alpha/m$  shrinks with  $m$ , and power disappears exactly when there is most to find. The false discovery rate of [Definition 13.1.1](#) asks instead that, on average, only a small *proportion* of the rejections be false. This section proves that the procedure of Benjamini and Hochberg controls it for independent  $p$ -values, and describes what is known under dependence.

### 13.5.1 The Benjamini–Hochberg procedure

As before,  $p_{(1)} \leq \dots \leq p_{(m)}$  are the ordered  $p$ -values,  $H_{(i)}$  the corresponding hypotheses, and  $q \in (0, 1)$  is the target false discovery rate.

**Benjamini–Hochberg (BH) procedure at level  $q$ .** Let

$$\hat{k} = \max \left\{ k \in \{1, \dots, m\} : p_{(k)} \leq \frac{kq}{m} \right\},$$

and reject  $H_{(1)}, \dots, H_{(\hat{k})}$ . If no  $k$  qualifies, reject nothing.

The thresholds rise linearly from Bonferroni's  $q/m$  to  $q$ . The procedure is a **step-up** procedure: it rejects everything up to the *last*  $p$ -value on or below the line, including smaller  $p$ -values that lie above the line at their own rank ([Figure 13.5.1](#)).

Why is the line the right one? Suppose  $k$  hypotheses are rejected, all with  $p$ -values at most  $t = kq/m$ . The  $m_0$  true hypotheses have uniform  $p$ -values, so about  $m_0 t \leq mt$  of them fall below  $t$  by chance, and the proportion of false ones among the  $k$  rejections is about  $mt/k = q$  at most. The BH procedure chooses the largest threshold at which this estimate of the false discovery proportion is at most  $q$ . The theorem makes the heuristic exact.

Two simple descriptions of the rejected set are useful. Write  $N(t) = \#\{j : p_j \leq t\}$  for the number of  $p$ -values at most  $t$ . Since  $p_{(k)} \leq t$  iff  $N(t) \geq k$ ,

$$\hat{k} = \max\left\{k : N(kq/m) \geq k\right\}. \quad (13.5.1)$$

And the rejected hypotheses are exactly those with  $p_j \leq \hat{k}q/m$ . Indeed, every rejected  $p$ -value is at most  $p_{(\hat{k})} \leq \hat{k}q/m$ . Conversely, if some  $p_j$  with  $j$  not rejected were at most  $\hat{k}q/m$ , then  $N(\hat{k}q/m) \geq \hat{k} + 1$ , hence  $N((\hat{k} + 1)q/m) \geq \hat{k} + 1$ , and  $\hat{k} + 1$  would qualify, contradicting the maximality of  $\hat{k}$ .

## 13.5.2 Control under independence

### Lemma 13.5.1 — Replacing one $p$ -value by zero

Fix  $i$ , and let  $\hat{k}^{(i)}$  be the number of rejections of the BH procedure applied to the  $p$ -values with  $p_i$  replaced by 0. Then  $\hat{k}^{(i)}$  is a function of  $(p_j)_{j \neq i}$  alone, and for each  $k = 1, \dots, m$ ,

$$\{H_i \text{ is rejected and } \hat{k} = k\} = \{p_i \leq kq/m \text{ and } \hat{k}^{(i)} = k\}.$$

*Proof.* The first claim is clear. Let  $N^{(i)}$  count the modified  $p$ -values. Then  $N^{(i)}(t) = N(t)$  for  $t \geq p_i$ , because at such  $t$  the  $p$ -value  $p_i$  is counted in both. By the second description above,  $H_i$  is rejected and  $\hat{k} = k$  iff  $p_i \leq kq/m$  and  $\hat{k} = k$ . Assume  $p_i \leq kq/m$ . By Eq. 13.5.1,  $\hat{k} = k$  iff  $N(kq/m) \geq k$  and  $N(k'q/m) < k'$  for every  $k' > k$ . These conditions involve  $N$  only at points  $t \geq kq/m \geq p_i$ , where  $N$  and  $N^{(i)}$  agree, so they hold iff the same conditions hold for  $N^{(i)}$ , that is, iff  $\hat{k}^{(i)} = k$ . ■

### Theorem 13.5.2 — Benjamini–Hochberg

Suppose that for each true hypothesis  $H_i$ ,  $i \in I_0$ , the  $p$ -value  $p_i$  is independent of the other  $m - 1$   $p$ -values and satisfies  $\Pr(p_i \leq u) \leq u$  for  $0 \leq u \leq 1$ . Then the BH procedure at level  $q$  has

$$\text{FDR} \leq \frac{m_0}{m} q \leq q,$$

with equality in the first inequality when every  $p_i$ ,  $i \in I_0$ , is uniform on  $(0, 1)$ .

*Proof.* The false discovery proportion is  $Q = \sum_{i \in I_0} 1_{\{H_i \text{ rejected}\}} / \max(\hat{k}, 1)$ . A rejection forces  $\hat{k} \geq 1$ , so

$$\text{FDR} = \sum_{i \in I_0} \sum_{k=1}^m \frac{1}{k} \Pr(H_i \text{ rejected}, \hat{k} = k) = \sum_{i \in I_0} \sum_{k=1}^m \frac{1}{k} \Pr\left(p_i \leq \frac{kq}{m}\right) \Pr(\hat{k}^{(i)} = k),$$

by Lemma 13.5.1 and the independence of  $p_i$  from  $\hat{k}^{(i)}$ . Now  $\Pr(p_i \leq kq/m) \leq kq/m$ , so the inner sum is at most  $(q/m) \sum_{k=1}^m \Pr(\hat{k}^{(i)} = k) = q/m$ , because  $\hat{k}^{(i)} \geq 1$  always (the replaced  $p$ -value 0 lies below the first threshold). Summing over the  $m_0$  true hypotheses gives  $\text{FDR} \leq m_0 q/m$ . With uniform null  $p$ -values every inequality is an equality. ■

The false hypotheses'  $p$ -values may depend on each other in any way. With uniform null  $p$ -values the false discovery rate is exactly  $\pi_0 q$ ,  $\pi_0 = m_0/m$ . With  $m = 100$  independent  $z$ -tests, 90 of them null and the rest shifted by 3, 20000 simulated repetitions estimate it at 0.0439, against the exact 0.0450.

When all hypotheses are true the false discovery rate is the familywise error rate ([Proposition 13.1.1](#)), so BH is a level- $q$  test of the global null. It rejects everything Bonferroni's and Holm's procedures at level  $q$  reject ([Exercise \(B2\)](#)). But it does *not* control the familywise error rate strongly: once many false hypotheses are rejected, the thresholds for the rest approach  $q$ .

### 13.5.3 Dependent tests

In a regression the  $t$  statistics share  $s$ , and their numerators are correlated unless  $\mathbf{X}$  has orthogonal columns. Two results, stated here without proof, cover much of what is needed.

**Positive dependence.** Benjamini and Yekutieli (2001) showed that  $\text{FDR} \leq m_0 q/m$  remains true if the  $p$ -values are *positively regression dependent on each one from the subset of true hypotheses* (PRDS): for every true  $H_i$  and every increasing set  $D$  of  $p$ -value vectors,  $\Pr((p_1, \dots, p_m) \in D \mid p_i = u)$  is nondecreasing in  $u$ . One-sided tests based on jointly normal statistics with nonnegative correlations are an example; two-sided tests of correlated statistics need not be.

**Arbitrary dependence.** They also showed that under *any* dependence, BH at level  $q/c_m$  with  $c_m = \sum_{i=1}^m 1/i \approx \log m + 0.577$  has false discovery rate at most  $m_0 q/m$  ([Exercise \(C1\)](#)). For  $m = 100$ ,  $c_m = 5.187$ , so a target of 0.05 becomes an effective level of 0.0096.

BH is routinely applied to regression  $t$  tests, and simulations, including the one below, find its false discovery rate below  $q$ . That is evidence, not a theorem; when a guarantee is essential, the Benjamini–Yekutieli version is the safe choice.

### 13.5.4 Adjusted $p$ -values and $q$ -values

As for Holm's procedure, the BH decisions for all levels at once are summarized by **adjusted  $p$ -values**

$$\tilde{p}_{(i)} = \min_{j \geq i} \min \left\{ 1, \frac{m p_{(j)}}{j} \right\}, \quad (13.5.2)$$

with  $H_{(i)}$  rejected at level  $q$  iff  $\tilde{p}_{(i)} \leq q$  ([Exercise \(B1\)](#)). Since BH controls the rate at  $\pi_0 q$ , it is conservative when many hypotheses are false. **Adaptive** procedures estimate  $\pi_0$ , for instance by twice the proportion of  $p$ -values above one half, and run BH at level  $q/\hat{\pi}_0$ . Storey (2002) developed this approach and defined the  **$q$ -value** of a hypothesis, the smallest false discovery rate at which it would be rejected; with  $\hat{\pi}_0 = 1$  the  $q$ -values are the adjusted values [Eq. 13.5.2](#). The gain is largest when  $\pi_0$  is well below one.

### 13.5.5 Screening regression coefficients

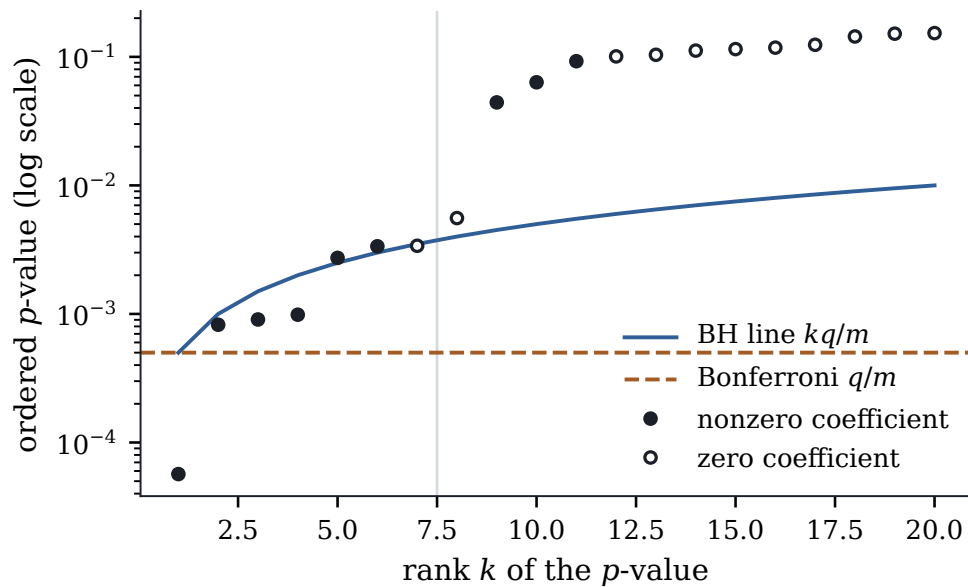


Figure 13.5.1: The twenty smallest of 100 ordered  $p$ -values (log scale), with the BH line  $kq/m$  and Bonferroni's threshold  $q/m$ ,  $q = 0.05$ . Filled points are nonzero coefficients; BH rejects the seven left of the vertical line.

#### Example. (One hundred candidate regressors)

A simulated regression has  $n = 250$  observations, an intercept and 100 regressors with independent standard normal entries, leaving  $\nu = 149$  error degrees of freedom. The first 10 slopes are nonzero, each chosen so that the noncentrality of its  $t$  statistic is 3; the other 90 are zero. The 100 two-sided  $t$  tests are run on one data set at level 0.05:

| procedure                 | rejections | of which false |
|---------------------------|------------|----------------|
| unadjusted, $p \leq 0.05$ | 9          | 2              |
| Bonferroni                | 1          | 0              |
| Holm                      | 1          | 0              |
| BH, $q = 0.05$            | 7          | 1              |

Bonferroni's threshold is  $0.05/100 = 0.0005$ , and only one  $p$ -value is below it. The BH line reaches  $7 \times 0.05/100 = 0.0035$  at rank 7, and the seventh smallest  $p$ -value is 0.0034 (Figure 13.5.1). Six of the seven BH discoveries are real.

```
import numpy as np
from scipy import stats

q = 0.05

rng = np.random.default_rng(20260919)
n, m = 250, 100
X = np.column_stack([np.ones(n), rng.normal(size=(n, m))])
Qx, Rx = np.linalg.qr(X)
nu = n - m - 1
c_diag = np.diag(np.linalg.inv(Rx.T @ Rx))[1:]
slopes
```

# 149 error degrees of freedom  
# [(X'X)^{-1}]\_{jj} for the



```

def t_statistics(Y):
    """t statistics of the 100 slopes for each column of Y."""
    coef = np.linalg.solve(Rx, Qx.T @ Y)
    resid = Y - X @ coef
    s2 = np.sum(resid ** 2, axis=0) / nu
    return coef[1:] / np.sqrt(np.outer(c_diag, s2))

def bh(p, q):
    """Benjamini-Hochberg step-up at level q; returns a boolean rejection vector."""
    m = len(p)
    order = np.argsort(p)
    below = np.nonzero(p[order] <= q * np.arange(1, m + 1) / m)[0]
    reject = np.zeros(m, dtype=bool)
    if below.size:
        reject[order[: below[-1] + 1]] = True
    return reject

def holm(p, alpha):
    m = len(p)
    order = np.argsort(p)
    ok = np.cumprod(p[order] <= alpha / (m - np.arange(m))).astype(bool)
    reject = np.zeros(m, dtype=bool)
    reject[order] = ok
    return reject

procedures = {
    "unadjusted": lambda p: p <= q,
    "Bonferroni": lambda p: p <= q / m,
    "Holm": lambda p: holm(p, q),
    "BH": lambda p: bh(p, q),
}

```

```

m1 = 10                                # the first 10 slopes are
    nonzero
beta = np.zeros(m + 1)
beta[1:m1 + 1] = 3.0 * np.sqrt(c_diag[:m1])    # noncentrality 3 when sigma =
    1
y = X @ beta + rng.normal(size=n)
p = 2 * stats.t.sf(np.abs(t_statistics(y[:, None]))[:, 0]), nu)
for name, rule in procedures.items():
    rej = rule(p)
    print(f"{name:10s} rejects {rej.sum():3d}, of which {rej[m1:].sum()} are false
    discoveries")

```

A simulation with the same design,  $m_1 = 0, 5, 10, 25, 50$  nonzero slopes and 2000 responses for each, shows the average behaviour (Figure 13.5.2). With  $m_1 = 10$ , the false discovery rate is 0.327 without adjustment, 0.014 for Bonferroni and Holm and 0.046 for BH, and the average power is 0.845, 0.289, 0.291 and 0.476. With  $m_1 = 50$ , BH's power rises to 0.731 and its false discovery rate falls to 0.024, in line with  $\pi_0 q$ , while Bonferroni's and Holm's power stay near 0.3. BH pays with familywise errors: at least one of its rejections is false in 0.238 of the data sets with  $m_1 = 10$  and 0.570 with  $m_1 = 50$ . With all slopes zero, its false discovery rate and familywise error rate coincide, at 0.050.

For the 21 education comparisons of Example (Holm for the education pairs), BH at  $q = 0.05$  rejects 13 hypotheses: the thirteenth smallest  $p$ -value is  $0.0303 \leq 13 \times 0.05/21 = 0.0310$ , and no

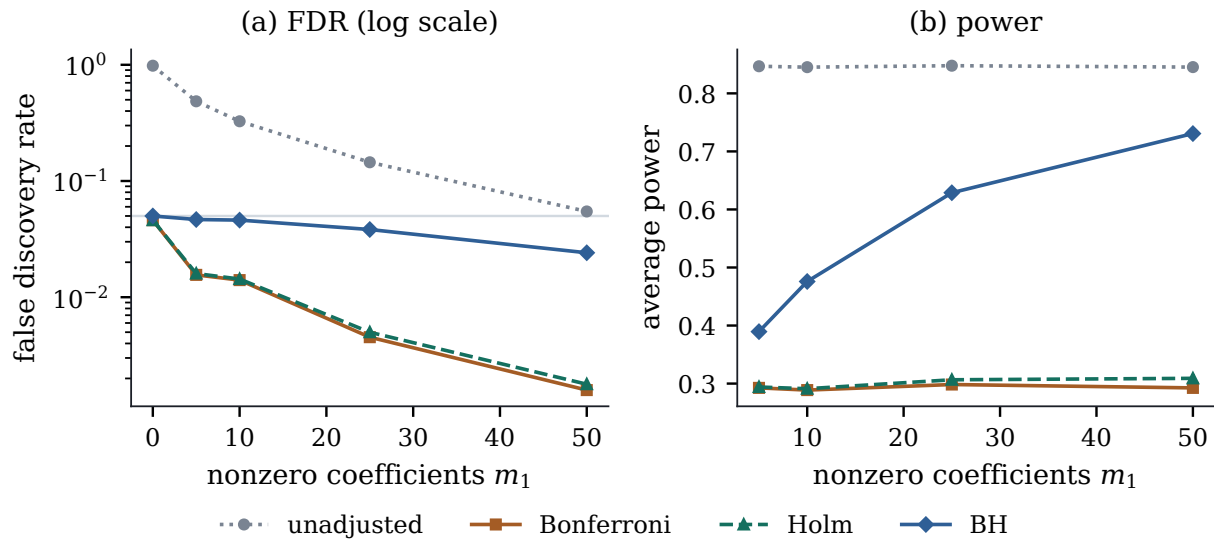


Figure 13.5.2: False discovery rate (a, log scale) and average power (b) of four procedures when  $m_1$  of 100 regression slopes are nonzero, 2000 simulated data sets per point, nominal level 0.05. The pale line in (a) marks 0.05.

later one comes back below the line. The next section discusses how to choose between the two error rates.

## 13.5.6 Exercises

### 13.5.6.1 A. Check your understanding

#### Exercise A1.

Apply the BH procedure at  $q = 0.05$  to the five  $p$ -values of [Exercise \(A1\)](#), and compute the adjusted  $p$ -values [Eq. 13.5.2](#).

*Solution.* The thresholds are 0.01, 0.02, 0.03, 0.04, 0.05. From the top:  $0.2 > 0.05$ , but  $0.04 \leq 0.04$ , so  $\hat{k} = 4$  and the first four hypotheses are rejected, one more than Holm's procedure rejects. The adjusted values are  $\min(5 \times 0.001, \dots) = 0.005$ , then  $\min(5 \times 0.011/2, 5 \times 0.012/3, 5 \times 0.04/4, 1) = 0.02, 0.02, 0.05$  and 0.2.

#### Exercise A2.

A report says: "Using BH at  $q = 0.1$  we made 40 discoveries, so at most 4 of them are false." What is wrong with this statement? What would be a correct one?

### 13.5.6.2 B. Practice

#### Exercise B1.

Prove that BH at level  $q$  rejects  $H_{(i)}$  iff the adjusted value [Eq. 13.5.2](#) satisfies  $\tilde{p}_{(i)} \leq q$ .

**Exercise B2.**

Show that the BH procedure at level  $q$  rejects every hypothesis that Holm's procedure at level  $q$  rejects. *Hint:*  $q/(m - i + 1) \leq iq/m$  for  $1 \leq i \leq m$ .

*Solution.* The hint holds because  $i(m - i + 1) \geq m$  for  $1 \leq i \leq m$ : the product of two positive integers summing to  $m + 1$  is at least  $m$ . If Holm rejects  $H_{(1)}, \dots, H_{(i)}$ , then  $p_{(i)} \leq q/(m - i + 1) \leq iq/m$ , so  $\hat{k} \geq i$ .

**Exercise B3.**

With  $m_0 = 2$  true hypotheses with independent uniform  $p$ -values and  $m - 2$  false hypotheses whose  $p$ -values are 0, find the familywise error rate of BH at level  $q$ . Compare it with the false discovery rate given by [Theorem 13.5.2](#).

**13.5.6.3 C. Going deeper****Exercise C1.**

Prove the Benjamini–Yekutieli bound: under arbitrary dependence, BH at level  $q/c_m$  has false discovery rate at most  $m_0 q/m$ . *Hint:* write the false discovery rate as  $\sum_{i \in I_0} \sum_k \Pr(p_i \in (\frac{(j-1)q'}{m}, \frac{j q'}{m}], \hat{k} = k)/k$  summed over  $j \leq k$ , with  $q' = q/c_m$ , and exchange the order of summation.

## 13.6 Choosing a procedure

None of the procedures of this chapter is best in general, because they answer different questions. This section compares them on one data set, collects the comparison into a guide, and explains why several procedures common in software are not developed here.

### 13.6.1 Decide the family first

Every guarantee in this chapter is for a *family* of inferences, and is only as good as the honesty with which the family is defined. A short list of functions written down before the data arrive is a **planned** family. All  $g(g - 1)/2$  pairwise differences form a family with its own exact method. A function chosen *because* the data make it look interesting belongs to the family of everything the analyst might have chosen, usually a whole subspace; this is **post hoc** inference, or data snooping. And a long list screened for leads worth following up calls for a different error rate altogether.

*Warning.* The family must be fixed before the data are examined, and reported. Choosing the adjustment after seeing the results, for example reporting Bonferroni's intervals for the three comparisons that turned out largest, reproduces the problem that adjustment was meant to solve. So does splitting one analysis into several “families” so that each is small.

## 13.6.2 One data set, five sets of intervals

The education example of [Example \(Education and the perceived position of a candidate\)](#) and [Example \(All pairs of education levels\)](#) makes the differences concrete. The table gives, for each method, the multiplier of the standard error for all 21 pairwise differences, the resulting half-width for two pairs with very different precision, and the number of pairs declared different at familywise level 0.05 (false discovery rate 0.05 for BH). The pair of levels 1 and 2 involves the two smallest groups, with standard error 0.420; the pair of levels 3 and 6 involves two large groups, with standard error 0.124.

| method                 | multiplier | ratio to $t$ | half-width, 1<br>vs 2 | half-width, 3<br>vs 6 | pairs<br>declared<br>different |
|------------------------|------------|--------------|-----------------------|-----------------------|--------------------------------|
| unadjusted $t$         | 1.962      | 1.00         | 0.824                 | 0.244                 | 14                             |
| Tukey–<br>Kramer       | 2.955      | 1.51         | 1.241                 | 0.368                 | 9                              |
| Šidák                  | 3.039      | 1.55         | 1.276                 | 0.378                 | 8                              |
| Bonferroni             | 3.046      | 1.55         | 1.279                 | 0.379                 | 8                              |
| Scheffé                | 3.557      | 1.81         | 1.494                 | 0.442                 | 6                              |
| Holm                   |            |              |                       |                       | 10                             |
| Benjamini–<br>Hochberg |            |              |                       |                       | 13                             |

The ordering is the one the theory predicts: the method designed for pairs is shortest, the general finite-family methods are a few percent longer, and the method covering every contrast is much longer. Holm’s procedure rejects more by stepping down, and BH by controlling a weaker error rate. The Šidák row relies on Šidák’s inequality for correlated normal variables, stated without proof in [Section 13.4](#).

```
import itertools
import numpy as np
import statsmodels.api as sm
from scipy import stats
anes = sm.datasets.anes96.load_pandas().data
y = anes["ClinLR"].to_numpy()
level = anes["educ"].to_numpy().astype(int) - 1      # 0, ..., 6
g = 7
n_k = np.bincount(level).astype(float)
means = np.bincount(level, weights=y) / n_k
n = len(y)
nu = n - g                                           # error degrees of freedom
s2 = np.sum((y - means[level]) ** 2) / nu           # pooled variance estimate
s = np.sqrt(s2)
alpha = 0.05
c_sch = np.sqrt((g - 1) * stats.f.ppf(1 - alpha, g - 1, nu))  # Scheffe multiplier,
q = g - 1
pairs = list(itertools.combinations(range(g), 2))
m = len(pairs)                                     # 21 comparisons
diff = np.array([means[k] - means[l] for k, l in pairs])
se_d = np.array([s * np.sqrt(1 / n_k[k] + 1 / n_k[l]) for k, l in pairs])
t = diff / se_d
mult = {
    "unadjusted": stats.t.ppf(1 - alpha / 2, nu),
    # half-width = multiplier x se
```

```

    "Bonferroni": stats.t.ppf(1 - alpha / (2 * m), nu),
    "Sidak": stats.t.ppf((1 + (1 - alpha) ** (1 / m)) / 2, nu),
    "Tukey-Kramer": stats.studentized_range.ppf(1 - alpha, g, nu) / np.sqrt(2),
    "Scheffe": c_sch,
}
for name, c in mult.items():
    print(f"{name:13s} multiplier {c:.3f}    pairs declared different: {np.sum(np.abs
(t) > c)}")

```

### 13.6.3 A decision guide

The choice can be put as a short sequence of questions.

- (1) *Is there one question?* Then use the ordinary  $t$  or  $F$  procedure of [Chapter 11](#) or [Chapter 12](#). No adjustment is needed.
- (2) *Will the functions be chosen after seeing the data?* If they lie in a known subspace, use Scheffé's intervals for that subspace, choosing the smallest subspace that contains every function one might pick ([Section 13.2](#)).
- (3) *Is the family all pairs of means?* Use Tukey's intervals if the design is balanced and the means are independent, Tukey–Kramer if the one-way layout is unbalanced, and the exact multivariate  $t$  computation otherwise.
- (4) *Is the family a short list fixed in advance?* Use Bonferroni's (or Šidák's) intervals, or Holm's tests if intervals are not needed, or the exact multivariate  $t$  value if software is at hand.
- (5) *Is the family long, and the goal a list of leads for further study?* Control the false discovery rate with the Benjamini–Hochberg procedure, or with the Benjamini–Yekutieli version if dependence among the tests is strong and of unknown form.

Where intervals are available they are preferable to tests alone, because they report the size of each difference as well as its sign ([Proposition 13.1.2](#)). Adjusted  $p$ -values, [Eq. 13.4.3](#) and [Eq. 13.5.2](#), let readers apply a different level. And a report should say how many comparisons were made in all, not only which were significant.

### 13.6.4 Planned versus post hoc

A post hoc comparison is chosen because it looks large, and its  $t$  statistic has the distribution of a maximum. In the education example the most extreme contrast has  $t = 6.88$ . Under equal means its square is not an  $F(1, 937)$  variable but 6 times an  $F(6, 937)$  variable ([Corollary 13.2.3](#)), and treating it as planned would overstate the evidence by exactly the amount Scheffé's multiplier corrects. The same happens at a larger scale when regressors are selected by their  $t$  statistics; [Section 29.5](#) takes up inference after model selection.

### 13.6.5 Procedures not developed in this book

Several procedures widely available in software are deliberately left out. They control the familywise error rate only weakly, or not at all.

**Fisher's protected LSD** controls the familywise error rate only weakly, as [Example \(The protected LSD without strong control\)](#) and [Exercise \(C1\)](#) show, except with three groups ([Exercise \(B1\)](#)).

**The Newman–Keuls procedure** orders the means and tests the range of each set of  $w$  adjacent means against  $q_\alpha(w, \nu) s / \sqrt{m}$ , stepping down from  $w = g$  and never testing inside a set whose range was not significant. Under the global null it rejects exactly when Tukey's test does, so its control is weak; with means in well-separated groups, each small group is tested nearly unprotected.

**Example. (Newman–Keuls without strong control)**

Take  $g = 10$  groups of 5 observations, with the true means equal in pairs and the five pairs far apart, and  $\nu = 40$ . The larger ranges are always significant, so the procedure reaches every pair and tests it with  $q_\alpha(2, \nu)$ , the critical value of a single unadjusted comparison. In 10000 simulated data sets, the Newman–Keuls procedure declares some pair of equal means different in a proportion 0.219 of them. Five independent level-0.05 tests would err with probability 0.226; the shared  $s$  lowers the rate slightly, as [Proposition 13.4.3](#) implies. Tukey's intervals err in 0.011 of the data sets.

```
import numpy as np
from scipy import stats

alpha, g, m_rep, reps = 0.05, 10, 5, 10_000
nu = g * (m_rep - 1)
mu = np.repeat(np.arange(5) * 20.0, 2)           # pairs 0,0, 20,20, 40,40, ...

def newman_keuls(means, s, m_rep, q_crit):
    """Pairs (k, l) declared different by the Newman-Keuls step-down procedure;
    q_crit[w] is the upper-alpha point of the studentized range of w means."""
    g = len(means)
    order = np.argsort(means)
    ybar = means[order]
    se = s / np.sqrt(m_rep)
    kept = []                                     # windows found homogeneous
    different = set()
    for width in range(g, 1, -1):                 # ranges of width, width-1,
        ..., 2
        crit = q_crit[width] * se
        for i in range(g - width + 1):
            j = i + width - 1
            if any(a <= i and j <= b for a, b in kept):
                continue                         # inside a homogeneous window
            if ybar[j] - ybar[i] > crit:
                different.add((min(order[i], order[j]), max(order[i], order[j])))
            else:
                kept.append((i, j))
    return different

q_crit = {w: stats.studentized_range.ppf(1 - alpha, w, nu) for w in range(2, g + 1)}
rng = np.random.default_rng(20260919)
q_tukey = stats.studentized_range.ppf(1 - alpha, g, nu)
nk_false = tukey_false = 0
for r in range(reps):
    y = mu[:, None] + rng.normal(size=(g, m_rep))
```

```

means = y.mean(axis=1)
s = np.sqrt(np.sum((y - means[:, None]) ** 2) / nu)
true_pairs = [(2 * i, 2 * i + 1) for i in range(5)]
nk = newman_keuls(means, s, m_rep, q_crit)
nk_false += any(pair in nk for pair in true_pairs)
tukey_false += any(abs(means[a] - means[b]) > q_tukey * s / np.sqrt(m_rep)
                    for a, b in true_pairs)
print(f"P(some false difference): Newman-Keuls {nk_false / reps:.3f}, "
      f"Tukey {tukey_false / reps:.3f}")

```

With  $2J$  means in  $J$  separated pairs, the familywise error rate of Newman–Keuls approaches  $1 - (1 - \alpha)^J$  as the degrees of freedom grow ([Exercise \(B2\)](#)). No bound below one holds uniformly in  $g$ .

**Duncan’s multiple range test.** Duncan’s procedure has the structure of Newman–Keuls but raises the level used for a set of  $w$  means to  $1 - (1 - \alpha)^{w-1}$ . Its first step, the range of all  $g$  means, is therefore a test at level  $1 - (1 - \alpha)^{g-1}$ , and if it rejects, at least the most extreme pair is declared different. Under the global null its familywise error rate is exactly  $1 - (1 - \alpha)^{g-1}$ , which is 0.370 for  $g = 10$  and  $\alpha = 0.05$ . It does not control the rate even weakly.

These procedures buy power over Tukey’s by relaxing the error rate when some hypotheses are false. If that trade is wanted, it is better made openly, by controlling the false discovery rate, whose meaning is clear. Closed testing ([Theorem 13.4.4](#)) with range or  $F$  local tests gives step-down procedures that are more powerful than Tukey’s and still control the familywise error rate strongly; the books cited in the notes describe them.

A Bayesian response to multiplicity is a hierarchical model that treats the group means as draws from a common distribution and shrinks the estimated differences toward each other ([Exercise \(C1\)](#)). [Chapter 32](#) develops random-effects models of this kind ([Proposition 32.6.1](#)), and the conjugate analysis of [Theorem 7.6.2](#) is their simplest case.

## 13.6.6 Exercises

### 13.6.6.1 A. Check your understanding

#### Exercise A1.

Choose a procedure for each situation, and say what family its guarantee refers to: (a) a trial with three doses and a placebo, where the protocol specifies the three comparisons with placebo; (b) eight fertilizers compared in a balanced field trial, with interest in every pair; (c) 5000 genes screened for association with a trait, to select candidates for a second study; (d) a contrast suggested by a plot of the group means.

### 13.6.6.2 B. Practice

#### Exercise B1.

Show that under the global null the familywise error rate of Duncan’s procedure is  $1 - (1 - \alpha)^{g-1}$  in a balanced one-way layout.

**Exercise B2.**

In [Example \(Newman–Keuls without strong control\)](#) with  $J$  pairs, let the separation between pairs tend to infinity. Show that the familywise error rate of Newman–Keuls converges to  $1 - \Pr(|T_j| \leq q_\alpha(2, \nu)/\sqrt{2}, j = 1, \dots, J)$ , where the  $T_j$  are  $t$  statistics with independent numerators and a common  $s$ . Using [Proposition 13.4.3](#), show that the limit is at most  $1 - (1 - \alpha)^J$ , and that it tends to this value as  $\nu \rightarrow \infty$ .

**13.6.6.3 C. Going deeper****Exercise C1.**

In the balanced one-way layout with known  $\sigma^2$ , give the group means the prior  $\mu_k \stackrel{\text{iid}}{\sim} N(\mu_0, \tau^2)$ , with  $\mu_0$  and  $\tau^2$  known. Find the posterior distribution of  $\mu_k - \mu_l$  and compare the width and centre of its central 95% posterior interval with the unadjusted confidence interval. Explain in what sense the shrinkage addresses the multiplicity of comparisons, and in what sense it does not.

## 13.7 Summary and Notes

### 13.7.1 Summary

*Idea.*

1. The error rates of a family, familywise, per-family and false discovery, are ordered ([Definition 13.1.1](#), [Proposition 13.1.1](#)). Control should be strong; a gate such as the protected LSD gives only weak control ([Definition 13.1.2](#), [Proposition 13.1.3](#)). Simultaneous intervals give strong control, including of sign errors ([Proposition 13.1.2](#)).
2. Scheffé: a family of estimable functions closed under linear combination is a subspace of  $\mathcal{C}(\mathbf{X})$ , and its largest squared  $t$  statistic is  $q$  times an  $F(q, n - r)$  variable. The intervals with multiplier  $\sqrt{qF_\alpha(q, \nu)}$  are exact, even for functions chosen after seeing the data ([Theorem 13.2.2](#)); the  $F$  test rejects iff some Scheffé interval excludes its hypothesized value ([Corollary 13.2.3](#)); smaller subspaces give shorter intervals ([Section 13.2](#)).
3. Tukey: in a balanced one-way layout the intervals  $\bar{y}_k - \bar{y}_l \pm q_\alpha(g, \nu)s/\sqrt{m}$  are exact for all pairs, extend to all contrasts, and are the shortest of their form ([Definition 13.3.1](#), [Theorem 13.3.2](#), [Proposition 13.3.3](#)). The Tukey–Kramer intervals [Eq. 13.3.3](#) are conservative.
4. Bonferroni needs no assumption on dependence and bounds the expected number of false rejections ([Theorem 13.4.1](#)); Šidák’s level is valid for independent estimates, and a shared  $s$  only helps ([Proposition 13.4.3](#)). Closed testing gives strong control ([Theorem 13.4.4](#)), and Holm’s step-down procedure, its Bonferroni case, rejects everything Bonferroni does and possibly more ([Theorem 13.4.5](#)).
5. The Benjamini–Hochberg procedure controls the false discovery rate at  $\pi_0 q$  for independent null  $p$ -values ([Theorem 13.5.2](#)), and Benjamini and Yekutieli’s results



extend it to positively dependent (PRDS) and arbitrarily dependent  $p$ -values.

6. Fix the family first: planned lists call for Bonferroni, Holm or exact multivariate  $t$  values, all pairs for Tukey, data snooping for Scheffé, screening for the false discovery rate ([Section 13.6](#)).

## 13.7.2 Notes and sources

**Sources for this chapter.** Christensen (2020, chapter 5) treats multiple comparisons within the projection framework used here, describing Scheffé’s method through a test space and showing that the  $F$  test rejects exactly when some single-degree-of-freedom hypothesis in that space does. [Section 13.2](#) follows that view. His chapter also compares the least significant difference, Bonferroni, Tukey, Newman–Keuls and Duncan procedures and discusses the Fisherian and Neyman–Pearson views of multiple testing. Agresti (2015, section 3.5) gives a concise account of Bonferroni’s and Tukey’s methods, the Tukey–Kramer extension and the Benjamini–Hochberg procedure. The examples, simulations and the treatment of error rates, Šidák’s inequality and closed testing are this book’s own.

**Scheffé and Tukey.** Scheffé (1953) introduced the method for all contrasts, and his book (Scheffé 1959) develops it for general linear hypotheses and relates it to the  $F$  test and the confidence ellipsoid. Tukey’s 1953 manuscript circulated for decades before it appeared in his collected works (Tukey 1994). Kramer (1956) proposed the extension to unequal group sizes, and Hayter (1984) proved Tukey’s conjecture that these intervals are conservative. The case of estimates with general correlation, where neither form is exact, is handled by computing multivariate  $t$  probabilities; Hothorn, Bretz and Westfall (2008) describe a general framework and Bretz, Hothorn and Westfall (2010) a practical guide. Dunnett (1955) gave the exact procedure for comparisons with a control.

**Bonferroni, Šidák and Holm.** The name commemorates Bonferroni’s (1936) work on inequalities for probabilities of unions. Dunn (1961) popularized its use for simultaneous confidence intervals. Šidák (1967) proved the inequality for rectangular regions of the multivariate normal distribution that underlies [Proposition 13.4.3](#) for correlated estimates. Holm (1979) introduced the step-down procedure, and Marcus, Peritz and Gabriel (1976) the closure principle, of which it is a case. Hochberg (1988) proposed the step-up version, whose validity rests on the inequality of Simes (1986).

**False discovery rate.** Benjamini and Hochberg (1995) defined the false discovery rate and proved [Theorem 13.5.2](#). The leave-one-out proof given here is a standard later simplification. Benjamini and Yekutieli (2001) proved control under positive regression dependence and gave the version valid under arbitrary dependence. Storey (2002) introduced estimation of  $\pi_0$  and the  $q$ -value, and Benjamini, Krieger and Yekutieli (2006) gave adaptive procedures with proven control.

**Books.** Miller (1981) is the classic account of simultaneous inference, and Hochberg and Tamhane (1987) and Hsu (1996) are comprehensive monographs. All three treat the procedures of Duncan (1955) and of Newman (1939) and Keuls (1952), and the closed step-down procedures that improve on Tukey’s. Gelman, Hill and Yajima (2012) give a Bayesian view of why hierarchical models reduce the need for multiplicity adjustments.

### 13.7.3 References

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# Chapter 14

## Correlation, Lack of Fit and Prediction Theory

So far the regressors have been fixed numbers. In most observational studies they are random, drawn along with the response. This chapter shows that the exact tests and intervals of Part III survive that change, because their conditional laws do not depend on the design, while the laws of the estimates, and the power of tests, do change. With random rows a regression is a prediction rule, and the chapter measures how well an estimated predictor predicts new cases and how much its in-sample error flatters it. It then studies the correlations that measure linear association: the multiple correlation, whose sample version  $R^2$  is the overall  $F$  statistic in disguise; the simple correlation and Fisher's transformation; and the partial correlation, whose test is the  $t$  test of a coefficient. The last two sections test whether a mean function is adequate: pure error gives an exact test when rows are replicated, and several exact tests, none dominating the others, replace it when they are not.

**What you need.** The normal linear model and its sampling distributions from [Chapter 7](#) ([Theorem 7.5.1](#), [Corollary 7.5.2](#), [Corollary 7.5.3](#)), nested projections and the Frisch–Waugh–Lovell theorem from [Chapter 6](#) ([Theorem 6.5.1](#), [Theorem 6.6.2](#)), best linear prediction in the population ([Theorem 2.5.2](#), [Theorem 6.11.1](#), [Proposition 6.11.3](#)) and the multiple correlation ([Proposition 2.5.3](#)) from [Chapter 2](#) and [Chapter 6](#), normal conditional distributions and partial correlations from [Chapter 3](#) ([Theorem 3.5.2](#), [Definition 3.6.1](#)), the noncentral distributions of [Chapter 4](#) ([Theorem 4.4.4](#), [Theorem 4.1.4](#)), and  $R^2$  and partial  $R^2$  from [Chapter 9](#) ([Theorem 9.5.2](#), [Theorem 9.5.4](#)). The  $F$  test of a reduced model ([Chapter 11](#), [Theorem 11.1.1](#)) is the template for every lack-of-fit test.

### 14.0.1 Roadmap

- [Random regressors and conditional inference](#)
- [Best linear prediction](#)
- [Multiple correlation](#)
- [The correlation coefficient and Fisher's  \$z\$](#)
- [Partial correlation](#)
- [Testing lack of fit with replicates](#)
- [Near replicates and nonparametric lack-of-fit tests](#)
- [Summary and notes](#)

## 14.1 Random regressors and conditional inference

The theory of Part III was built for a model matrix  $\mathbf{X}$  of known constants. In observational data the regressors are drawn with the response, and a second sample would bring a different  $\mathbf{X}$ . [Section 5.1](#) already argued that the assumptions of the linear model should then be read *conditionally on  $\mathbf{X}$* , and [Corollary 5.5.5](#) carried unbiasedness over to random regressors. This section does the same for the exact distribution theory. Tests and intervals stay exact; the estimates acquire heavier-tailed distributions, and power becomes a random quantity that depends on the design the data deliver.

### 14.1.1 The model with random regressors

#### Definition 14.1.1 — Normal linear model with random regressors

Let  $\mathbf{X}$  be a random  $n \times p$  matrix and  $\mathbf{Y}$  a random  $n$ -vector. They follow the **normal linear model with random regressors** if  $\text{rank}(\mathbf{X}) = p$  with probability one and, conditionally on  $\mathbf{X}$ ,

$$\mathbf{Y} \mid \mathbf{X} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}_n)$$

for constants  $\boldsymbol{\beta} \in \mathbb{R}^p$  and  $\sigma^2 > 0$ . Equivalently,  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}_n)$  independent of  $\mathbf{X}$ .

The leading case has independent and identically distributed rows  $(\mathbf{x}_{(i)}^\top, Y_i)$  with  $Y_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i$ , where  $\varepsilon_i \sim N(0, \sigma^2)$  is independent of  $\mathbf{x}_{(i)}$ . Independence of the rows then makes the errors independent given the whole of  $\mathbf{X}$ . Jointly normal rows satisfy the definition ([Eq. 3.5.2](#)), but the regressors may equally be skewed, discrete or dependent on one another; only the errors must be normal, homoscedastic and independent of the regressors.

### 14.1.2 Conditioning on the design

#### Lemma 14.1.1 — Pivots given the design

Let  $S$  be a random vector and  $U$  any random element. Suppose there is a distribution  $Q$  such that, for almost every value  $u$ , the conditional distribution of  $S$  given  $U = u$  is  $Q$ . Then  $S \sim Q$ , and  $S$  is independent of  $U$ .

*Proof.* For a Borel set  $B$  and an event  $\{U \in A\}$ , the tower rule gives

$$\begin{aligned} \Pr(S \in B, U \in A) &= E[\mathbf{1}\{U \in A\} \Pr(S \in B \mid U)] \\ &= E[\mathbf{1}\{U \in A\} Q(B)] = Q(B) \Pr(U \in A). \end{aligned}$$

Taking  $A$  to be the whole space gives  $\Pr(S \in B) = Q(B)$ , and then the display says that the joint probability factorizes, which is independence.

A statistic whose conditional law given  $\mathbf{X}$  is the same for every design is a *pivot given the design*. The  $t$  and  $F$  statistics of Part III are of this kind.

### Theorem 14.1.2 — Conditional inference is valid unconditionally

Assume the model of Definition 14.1.1, with  $n > p$ . Let  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$  and  $s^2 = \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 / (n - p)$ .

- (a) For every fixed  $\boldsymbol{\lambda} \neq \mathbf{0}$ , the statistic  $T = (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta}) / (s \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}})$  has the  $t(n - p)$  distribution and is independent of  $\mathbf{X}$ .
- (b) Let  $\boldsymbol{\Lambda}$  be a fixed  $p \times q$  matrix of rank  $q$ , let  $\mathbf{d} \in \mathbb{R}^q$ , and put

$$F = \frac{(\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top [\boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda}]^{-1} (\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})}{q s^2}.$$

Given  $\mathbf{X}$ ,  $F \sim F(q, n - p, \gamma(\mathbf{X}))$  with

$$\gamma(\mathbf{X}) = \frac{(\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})^\top [\boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda}]^{-1} (\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d})}{\sigma^2}. \quad (14.1.1)$$

If  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ , then  $F \sim F(q, n - p)$  unconditionally, independently of  $\mathbf{X}$ .

- (c) A confidence set whose coverage probability given  $\mathbf{X}$  equals  $1 - \alpha$  for almost every  $\mathbf{X}$  has unconditional coverage  $1 - \alpha$ . A test whose rejection probability under the null hypothesis equals  $\alpha$  given almost every  $\mathbf{X}$  has unconditional size  $\alpha$ .
- (d) The unconditional power of the level- $\alpha$  test that rejects when  $F > F_\alpha(q, n - p)$  is  $E[\pi(\gamma(\mathbf{X}))]$ , where  $\pi(\gamma) = \Pr\{F(q, n - p, \gamma) > F_\alpha(q, n - p)\}$ .

*Proof.* Fix a value of  $\mathbf{X}$  of rank  $p$ . Given this value,  $\mathbf{Y}$  follows the normal linear model with a fixed model matrix, so the results of Chapter 7 apply to the conditional law given  $\mathbf{X}$ .

- (a) By Corollary 7.5.2 with  $d = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ , the conditional law of  $T$  is  $t(n - p)$ . It is the same for every design, so Lemma 14.1.1 gives the unconditional law and the independence.
- (b) Write  $\mathbf{W} = \boldsymbol{\Lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Lambda}$ , which is positive definite because  $\boldsymbol{\Lambda}$  has rank  $q$ . By Theorem 7.5.1(a), given  $\mathbf{X}$ ,  $\mathbf{D} = \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d} \sim N_q(\boldsymbol{\Lambda}^\top \boldsymbol{\beta} - \mathbf{d}, \sigma^2 \mathbf{W})$ , so by Theorem 4.3.2(b) with  $\mathbf{A} = \mathbf{W}^{-1}/\sigma^2$  (for which  $\mathbf{A} \cdot \sigma^2 \mathbf{W} = \mathbf{I}$  is idempotent) the quadratic form  $\mathbf{D}^\top \mathbf{W}^{-1} \mathbf{D} / \sigma^2$  is  $\chi^2(q, \gamma(\mathbf{X}))$ . It is independent of  $(n - p)s^2 / \sigma^2 \sim \chi^2(n - p)$  by Theorem 7.5.1(b) and (c). Hence the conditional law of  $F$  is  $F(q, n - p, \gamma(\mathbf{X}))$  (Definition 4.1.2). Under  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  the noncentrality is zero for every design, and Lemma 14.1.1 applies.
- (c) Let  $A$  be the event that the set covers the true value, or that the test rejects. Then  $\Pr(A) = E[\Pr(A | \mathbf{X})] = E(1 - \alpha) = 1 - \alpha$ , and likewise for the size.
- (d) Given  $\mathbf{X}$ , the rejection probability is  $\pi(\gamma(\mathbf{X}))$  by (b). Take expectations.

The statistic in (b) is the  $F$  statistic of the general linear hypothesis ([Theorem 11.3.2](#)) for a full-rank design. So every test of [Chapter 11](#) and every interval of [Chapter 12](#) that is exact for fixed  $\mathbf{X}$  is exact for random  $\mathbf{X}$  too, whatever the distribution of the regressors, even without finite moments.

### Example. (Skewed regressors)

Take  $n = 10$  cases with an intercept and two regressors drawn afresh for every sample, one exponential and one lognormal, and  $E(Y) = 1 + 0 \cdot x_1 + 0.5x_2$ . The  $t$  test of  $\beta_1 = 0$  at level 0.05 uses the  $t(7)$  quantile. Over 100000 samples with standard normal errors independent of the regressors, the test rejects in a proportion 0.0508 of them, as [Theorem 14.1.2](#) requires; panel (b) of [Figure 14.1.1](#) shows the simulated statistics against the  $t(7)$  density. If instead the error of each case is multiplied by its own value of  $x_1$ , so that the spread depends on the regressor, the same test rejects in a proportion 0.2798. The conditional model no longer holds, and nothing in the theorem protects the test.

```
import numpy as np
from scipy import stats
reps = 100_000

rng = np.random.default_rng(1402)
n2, p2 = 10, 3
beta = np.array([1.0, 0.0, 0.5])          # H0: beta_1 = 0 is true
tq = stats.t.ppf(0.975, n2 - p2)

def t_stats(errors_from):
    """t statistics for beta_1 = 0 over many samples with skewed random regressors.
    """
    out = np.empty(reps)
    for s in range(reps):
        X = np.column_stack([np.ones(n2), rng.exponential(size=n2), rng.lognormal(
            size=n2)])
        y = X @ beta + errors_from(X)
        XtX_inv = np.linalg.inv(X.T @ X)
        b = XtX_inv @ X.T @ y
        s2 = np.sum((y - X @ b) ** 2) / (n2 - p2)
        out[s] = b[1] / np.sqrt(s2 * XtX_inv[1, 1])
    return out

t_indep = t_stats(lambda X: rng.normal(size=n2))          # errors independent of X
t_hetero = t_stats(lambda X: X[:, 1] * rng.normal(size=n2)) # spread grows with x_1
for label, t in [("independent errors", t_indep), ("spread depends on X", t_hetero)]:
    print(f"{label:20s} rejection rate {np.mean(np.abs(t) > tq):.4f}")
```

*Warning.* The conditional model can fail in ways no amount of data repairs. If the error variance depends on the regressors, the  $t$  and  $F$  statistics lose their exact laws ([Chapter 21](#)). If the regressors are correlated with the errors, as with measurement error ([Theorem 24.1.1](#)), an omitted common cause ([Chapter 25](#)) or a lagged response, least squares estimates something other than  $\beta$ . And if the mean is not linear, the population projection of [Section 6.11](#) is still estimated consistently, but its errors depend on the regressors and the usual standard errors are wrong.

### 14.1.3 What changes: the law of the estimates

Pivots keep their laws, but estimates do not. Given the design,  $\hat{\beta} \sim N_p(\beta, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})$ . Unconditionally  $\hat{\beta}$  is a mixture of normal distributions over the random covariance  $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ . Its mean is still  $\beta$ , and its covariance is  $\sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}]$  by [Corollary 5.5.5](#), but it is not normal. For a single normal regressor the mixture can be written down exactly. We need one moment of the chi-squared distribution.

#### Lemma 14.1.3 — Mean of an inverse chi-squared variable

If  $V \sim \chi^2(\nu)$  with  $\nu > 2$ , then  $E(1/V) = 1/(\nu - 2)$ , the fact used without proof for [Eq. 9.6.2](#). For  $\nu \leq 2$  the mean is infinite.

*Proof.* With the density [Eq. 4.1.1](#),

$$E\left(\frac{1}{V}\right) = \int_0^\infty \frac{v^{\nu/2-2} e^{-v/2}}{2^{\nu/2} \Gamma(\nu/2)} dv = \frac{2^{\nu/2-1} \Gamma(\nu/2 - 1)}{2^{\nu/2} \Gamma(\nu/2)} = \frac{1}{2(\nu/2 - 1)} = \frac{1}{\nu - 2},$$

using  $\Gamma(a + 1) = a\Gamma(a)$ . The integral converges at zero iff  $\nu/2 - 2 > -1$ , that is,  $\nu > 2$ . ■

#### Proposition 14.1.4 — The slope under a random normal regressor

Let  $(X_i, Y_i)$ ,  $i = 1, \dots, n$  with  $n \geq 3$ , be independent draws from a bivariate normal distribution with  $\text{Var}(X_i) = \sigma_X^2$ , regression slope  $\beta_1$  and conditional variance  $\sigma^2 = \text{Var}(Y_i | X_i)$ . Let  $\hat{\beta}_1$  be the least squares slope. Then

$$\frac{\sigma_X \sqrt{n-1}}{\sigma} (\hat{\beta}_1 - \beta_1) \sim t(n-1),$$

so  $\hat{\beta}_1$  has heavier tails than any normal distribution, and for  $n > 3$

$$\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sigma_X^2 (n-3)}. \quad (14.1.2)$$

*Proof.* Given the  $X$  values,  $\hat{\beta}_1 \sim N(\beta_1, \sigma^2/S_{xx})$  with  $S_{xx} = \sum_i (X_i - \bar{X})^2$  ([Theorem 7.5.1\(a\)](#) and [Corollary 5.5.4](#)). So  $Z = \sqrt{S_{xx}}(\hat{\beta}_1 - \beta_1)/\sigma$  is standard normal given every design, and by [Lemma 14.1.1](#) it is standard normal and independent of the  $X$  values, hence of  $V = S_{xx}/\sigma_X^2$ . The  $X_i$  are independent  $N(\mu_X, \sigma_X^2)$ , so  $V \sim \chi^2(n-1)$  by [Corollary 4.4.5\(b\)](#). Therefore

$$\frac{\sigma_X \sqrt{n-1}}{\sigma} (\hat{\beta}_1 - \beta_1) = \frac{Z}{\sqrt{V/(n-1)}} \sim t(n-1)$$

by [Definition 4.1.3](#). For the variance,  $E(Z^2/V) = E(Z^2) E(1/V) = 1/(n-3)$  by independence and [Lemma 14.1.3](#), so  $\text{Var}(\hat{\beta}_1) = \sigma^2 \sigma_X^{-2}/(n-3)$ . ■



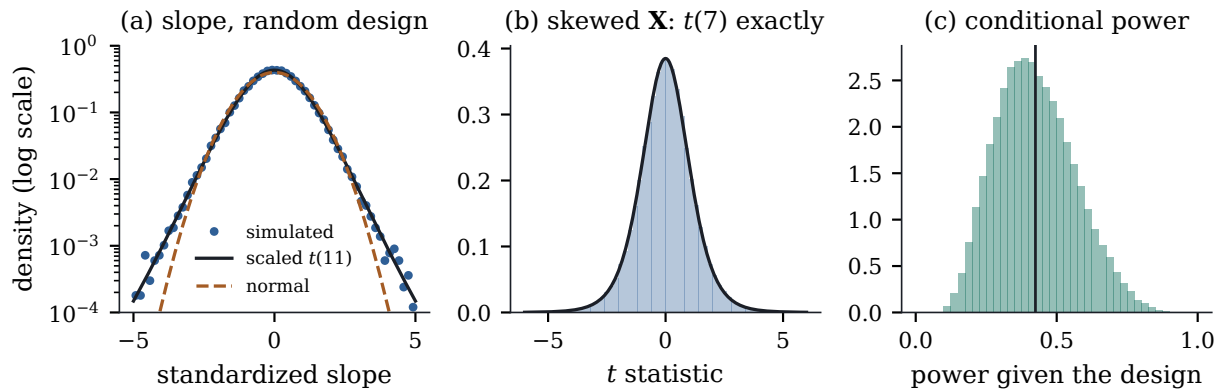


Figure 14.1.1: Random regressors. (a) The standardized slope with a random normal regressor ( $n = 12$ ) follows the scaled  $t(11)$  law of Proposition 14.1.4. (b) With skewed regressors the  $t$  statistic is exactly  $t(7)$ . (c) Conditional power varies with the design; the line marks the unconditional power.

The naive guess, the fixed-design variance  $\sigma^2/S_{xx}$  with  $S_{xx}$  replaced by its mean  $(n-1)\sigma_X^2$ , is too small by the factor  $(n-3)/(n-1)$ . This is Jensen's inequality at work, as in Example (One slope, two sampling schemes): designs that happen to bunch the regressor values together cost more precision than spread-out designs gain. Section 14.2 extends Eq. 14.1.2 to several regressors.

#### Example. (A heavy-tailed slope)

With  $n = 12$ ,  $\sigma_X = 2$ ,  $\sigma = 1$  and slope 0.5, the variance of the slope over 100000 simulated samples, each with its own design, is 0.02751, against 0.02778 from Eq. 14.1.2 and 0.02273 from the naive guess. The excess kurtosis of the simulated slopes is 0.85, close to the value 0.86 of the  $t(11)$  distribution. Panel (a) of Figure 14.1.1 shows the tails on a logarithmic scale. The scaled  $t(11)$  density follows the simulated slopes, and the normal density with the same variance falls away too fast.

```
rng = np.random.default_rng(1401)
n, beta1, sigma, sigma_x = 12, 0.5, 1.0, 2.0
x = sigma_x * rng.normal(size=(reps, n))           # a new design for every sample
y = 1.0 + beta1 * x + sigma * rng.normal(size=(reps, n))
xc = x - x.mean(axis=1, keepdims=True)
Sxx = np.sum(xc ** 2, axis=1)
b1 = np.sum(xc * y, axis=1) / Sxx                  # least squares slope, one per
sample

scale = sigma / (sigma_x * np.sqrt(n - 1))          # b1 - beta1 = scale * t(n-1)
print(f"variance of the slope {b1.var():.5f}")
print(f"sigma^2/(sigma_x^2 (n-3)) {sigma**2 / (sigma_x**2 * (n - 3)):.5f}")
print(f"sigma^2/E(Sxx) {sigma**2 / (sigma_x**2 * (n - 1)):.5f}")
print("KS p-value against the scaled t(n-1):",
      round(stats.kstest((b1 - beta1) / scale, "t", args=(n - 1,)).pvalue, 3))
```

Which of the two distributions should a report describe? For the data in hand, the conditional one. The design is *ancillary*: its distribution does not involve  $(\beta, \sigma^2)$ , and it tells us how precise this particular sample is. A slope estimated from widely spread regressor values deserves a shorter interval than one estimated from bunched values, which the conditional standard error

$s/\sqrt{S_{xx}}$  delivers, and by [Theorem 14.1.2](#) the interval is also exact on average over designs. The unconditional law answers questions asked before the data arrive, such as the power of a planned study.

### 14.1.4 Power depends on the design

Part (d) of [Theorem 14.1.2](#) says that power, unlike size, is not free of the design. For a single normal regressor the noncentrality of the test of a zero slope is  $\gamma(\mathbf{X}) = \beta_1^2 S_{xx} / \sigma^2$ , which is proportional to a  $\chi^2(n-1)$  variable.

#### Example. (How much power has this design?)

Take  $n = 12$ ,  $\sigma = \sigma_X = 1$  and slope 0.6. The test of a zero slope at level 0.05 has conditional power between 0.321 and 0.518 for the middle half of the designs, with median 0.414 (panel (c) of [Figure 14.1.1](#)). The unconditional power, the average over designs, is 0.425; a direct simulation of the whole procedure gives 0.426. Plugging in the average design,  $S_{xx} = n - 1$ , gives 0.436, which overstates the power a study should expect.

```
rng = np.random.default_rng(1403)
n3, b3, df3 = 12, 0.6, 10                                # slope 0.6, sigma = sigma_x =
1                                                         #
Sxx3 = stats.chi2.rvs(n3 - 1, size=reps, random_state=rng) # Sxx / sigma_x^2 ~ chi
~2(n-1)
gamma = b3 ** 2 * Sxx3                                    # noncentrality given the
design
fq = stats.f.ppf(0.95, 1, df3)
cond_power = stats.ncf.sf(fq, 1, df3, gamma)              # power of the F = t^2 test
given X
print(f"conditional power: quartiles {np.percentile(cond_power, [25, 50, 75]).round
(3)}")
print(f"unconditional power {cond_power.mean():.3f}")
print(f"power at the average design {stats.ncf.sf(fq, 1, df3, b3**2 * (n3 - 1)):.3f}
")
```

### 14.1.5 Exercises

#### 14.1.5.1 A. Check your understanding

##### Exercise A1.

In the setting of [Proposition 14.1.4](#) take  $n = 20$ ,  $\sigma = 2$  and  $\sigma_X = 1$ . Compute the unconditional variance of  $\hat{\beta}_1$  and compare it with  $\sigma^2 / E(S_{xx})$ . Which quantile would you use for a 95% interval for  $\beta_1$  computed from one sample?

*Solution.*  $\text{Var}(\hat{\beta}_1) = 4/17 = 0.235$ , against  $4/19 = 0.211$  for the naive value. For an interval from one sample, use the conditional standard error  $s/\sqrt{S_{xx}}$  and the  $t(18)$  quantile, as for a fixed design; by [Theorem 14.1.2](#) its coverage is exactly 95% over random designs too. The unconditional  $t(19)$  law of the proposition involves the unknown  $\sigma/\sigma_X$

and is not used for intervals.

### Exercise A2.

Under [Definition 14.1.1](#), which of the following hold unconditionally: (i)  $E(s^2) = \sigma^2$ ; (ii)  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ ; (iii)  $(n-p)s^2/\sigma^2 \sim \chi^2(n-p)$ , independent of  $\mathbf{X}$ ; (iv) the 95% confidence ellipsoid for  $\boldsymbol{\beta}$  has coverage 0.95?

### 14.1.5.2 B. Practice

#### Exercise B1.

In the setting of [Proposition 14.1.4](#), let  $x_0$  be a fixed value and  $\hat{\mu}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$  the fitted mean there. Show that

$$\text{Var}(\hat{\mu}_0) = \sigma^2 \left[ \frac{1}{n} + \frac{(x_0 - \mu_X)^2 + \sigma_X^2/n}{\sigma_X^2(n-3)} \right].$$

*Hint:*  $\bar{X}$  and  $S_{xx}$  are independent.

*Solution.* Given the design,  $\hat{\mu}_0$  is unbiased with variance  $\sigma^2[1/n + (x_0 - \bar{X})^2/S_{xx}]$  ([Eq. 5.5.2](#)). By [Proposition 2.5.4](#), the unconditional variance is the expectation of this, since the conditional mean  $\beta_0 + \beta_1 x_0$  is constant. By [Corollary 4.4.5](#),  $\bar{X} \sim N(\mu_X, \sigma_X^2/n)$  is independent of  $S_{xx} = \sigma_X^2 V$  with  $V \sim \chi^2(n-1)$ . Hence  $E[(x_0 - \bar{X})^2/S_{xx}] = E(x_0 - \bar{X})^2 E(1/V)/\sigma_X^2 = [(x_0 - \mu_X)^2 + \sigma_X^2/n]/[\sigma_X^2(n-3)]$  by [Lemma 14.1.3](#).

#### Exercise B2.

Suppose a pilot sample is used to choose the regressor values of the main study, so that the distribution of  $\mathbf{X}$  in the main study depends on  $\boldsymbol{\beta}$ . Assume that, given  $\mathbf{X}$ , the main-study responses follow the normal linear model with errors independent of the pilot. Show that the  $t$  test of [Theorem 14.1.2\(a\)](#), computed from the main study alone, still has exact size. Why is the design no longer ancillary, and what information about  $\boldsymbol{\beta}$  does the  $t$  test ignore?

### 14.1.5.3 C. Going deeper

#### Exercise C1.

In the setting of [Proposition 14.1.4](#), with  $\sigma$  known, consider the interval  $\hat{\beta}_1 \pm 1.96 \sigma / \sqrt{(n-1)\sigma_X^2}$ , which uses the average design instead of the observed one. Show that its coverage given the design is  $2\Phi(1.96\sqrt{V/(n-1)}) - 1$  with  $V \sim \chi^2(n-1)$ , and that its unconditional coverage is below 0.95. Which designs make it badly wrong, and in which direction?

## 14.2 Best linear prediction

With random rows, a regression is a rule for predicting the response of a new case from its regressors. The population theory is in place: the conditional mean  $m(\mathbf{X}) = E(Y | \mathbf{X})$  is the best predictor ([Proposition 6.11.2](#)), the best *linear* predictor  $L(Y | \mathbf{X}) = \alpha^* + \boldsymbol{\beta}^{*\top} \mathbf{X}$  is a projection ([Theorem 6.11.1](#), [Theorem 2.5.2](#)), the two coincide for jointly normal variables ([Proposition 3.5.3](#)), and least squares estimates  $L$  consistently ([Proposition 6.11.3](#)). This section asks the finite-sample questions: how large is the error of the estimated predictor on a new case, and how misleading is its error on the data used to fit it?

### 14.2.1 Characterizing the best predictor

The next result gives a test for whether a predictor is the best one, and a correlation that measures how well the best predictor can do.

#### Proposition 14.2.1 — The best predictor and the correlation ratio

Let  $EY^2 < \infty$ ,  $\text{Var}(Y) > 0$ , and  $m(\mathbf{X}) = E(Y | \mathbf{X})$ .

- (a) A predictor  $g(\mathbf{X})$  with  $Eg(\mathbf{X})^2 < \infty$  equals  $m(\mathbf{X})$  with probability one iff  $Eg(\mathbf{X}) = EY$  and  $\text{Cov}(h(\mathbf{X}), Y - g(\mathbf{X})) = 0$  for every square-integrable  $h$ .
- (b) For every such  $g$  with  $\text{Var } g(\mathbf{X}) > 0$ ,

$$\text{corr}(Y, g(\mathbf{X}))^2 \leq \eta^2 = \frac{\text{Var } m(\mathbf{X})}{\text{Var } Y}, \quad (14.2.1)$$

with equality for  $g = m$  when  $\text{Var } m(\mathbf{X}) > 0$ . The **correlation ratio**  $\eta^2$  exceeds the squared multiple correlation  $\rho_{Y \cdot X}^2$  of [Proposition 2.5.3](#) by  $E(m(\mathbf{X}) - L(Y | \mathbf{X}))^2 / \text{Var } Y$ , so  $\eta^2 = \rho_{Y \cdot X}^2$  iff  $m$  is affine.

*Proof.*

- (a) If  $g = m$ , then  $Em(\mathbf{X}) = EY$  by the tower rule, and  $E[(Y - m)h] = 0$  by [Proposition 6.11.2\(a\)](#), so the covariance is zero. Conversely, let  $d = g - m$ . Then  $Ed = 0$  and, taking  $h = d$ ,  $\text{Var } d = \text{Cov}(d, Y - m) - \text{Cov}(d, Y - g) = 0 - 0$ , so  $d = 0$  with probability one.
- (b) By (a) with  $h = g$ ,  $\text{Cov}(Y, g) = \text{Cov}(m, g)$ . Cauchy–Schwarz for covariances gives  $\text{Cov}(m, g)^2 \leq \text{Var } m \text{ Var } g$ , hence  $\text{corr}(Y, g)^2 = \text{Cov}(m, g)^2 / (\text{Var } Y \text{ Var } g) \leq \text{Var } m / \text{Var } Y$ . For  $g = m$  the covariance is  $\text{Var } m$  and equality holds. For the last claim write  $L = L(Y | \mathbf{X}) = L(m | \mathbf{X})$  ([Proposition 6.11.2\(b\)](#)). The error  $m - L$  has mean zero and is uncorrelated with the affine function  $L$ , so  $\text{Var } m = \text{Var } L + E(m - L)^2$ . Divide by  $\text{Var } Y$  and use  $\text{Var } L / \text{Var } Y = \rho_{Y \cdot X}^2$  ([Eq. 6.11.1](#)). ■

Part (a) is the population basis of residual plots. If a fitted model reproduces the conditional mean, its residuals are uncorrelated with *every* function of the regressors, not only with the regressors themselves. A residual plot that shows a trend against  $x_j^2$ , against a product  $x_j x_k$  or

against the fitted values is evidence that the fit is not  $m$ . Sections 14.6 and 14.7 turn this into tests. Part (b) says that no transformation of the regressors predicts better, in the correlation sense, than the conditional mean. The gap  $\eta^2 - \rho_{Y \cdot X}^2$  is the share of the variance of  $Y$  that a linear predictor leaves on the table.

## 14.2.2 The error of an estimated predictor

In practice  $\alpha^*$  and  $\beta^*$  are replaced by least squares estimates from a training sample  $(\mathbf{X}_i, Y_i)$ ,  $i = 1, \dots, n$ . By centring and Theorem 6.6.2 these are the plug-in estimates  $\hat{\beta} = \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}$  and  $\hat{\alpha} = \bar{Y} - \hat{\beta}^\top \bar{\mathbf{X}}$ , built from the sample means and covariances. Their performance is judged on a new case  $(\mathbf{X}_0, Y_0)$  drawn from the same distribution, independently of the training data.

### Proposition 14.2.2 — Excess error of a fitted linear predictor

Let  $(\mathbf{X}_0, Y_0)$  have finite second moments,  $\text{Cov}(\mathbf{X}_0) = \Sigma_{XX}$ , mean  $\mu_X$ , and best linear predictor  $L_0 = \alpha^* + \beta^{*\top} \mathbf{X}_0$  with error variance  $\sigma_*^2 = E(Y_0 - L_0)^2$ . Let  $\hat{Y}_0 = a + \mathbf{b}^\top \mathbf{X}_0$ , where  $(a, \mathbf{b})$  depend only on training data independent of  $(\mathbf{X}_0, Y_0)$ . Then, given the training data,

$$E[(Y_0 - \hat{Y}_0)^2 \mid \text{training data}] = \sigma_*^2 + (\alpha^* - a + (\beta^* - \mathbf{b})^\top \mu_X)^2 + (\beta^* - \mathbf{b})^\top \Sigma_{XX} (\beta^* - \mathbf{b}).$$

*Proof.* Condition on the training data, so that  $a$  and  $\mathbf{b}$  are constants. Write  $Y_0 - \hat{Y}_0 = (Y_0 - L_0) + (L_0 - \hat{Y}_0)$ . The second term is an affine function of  $\mathbf{X}_0$ , and by Theorem 6.11.1 the first has mean zero and is uncorrelated with every affine function of  $\mathbf{X}_0$ . So the cross term vanishes, and the mean square of the affine term is its squared mean plus its variance  $(\beta^* - \mathbf{b})^\top \Sigma_{XX} (\beta^* - \mathbf{b})$ . ■

The first term is the error of the best linear predictor, which no amount of data can remove. The other two are the price of estimation, and by Proposition 6.11.3 they vanish as  $n \rightarrow \infty$  whatever the joint distribution. To see how large they are for moderate  $n$ , we need the distribution of the regressors. Under joint normality the calculation can be done exactly. It uses one new distribution.

### Definition 14.2.1 — Wishart distribution

Let the rows of the  $m \times k$  matrix  $\mathbf{Z}$  be independent  $N_k(\mathbf{0}, \Sigma)$  vectors. The law of  $\mathbf{W} = \mathbf{Z}^\top \mathbf{Z} = \sum_{i=1}^m \mathbf{z}_i \mathbf{z}_i^\top$  is the **Wishart distribution**  $W_k(m, \Sigma)$  with  $m$  degrees of freedom. For  $k = 1$ ,  $W_1(m, \sigma^2)$  is the law of  $\sigma^2 \chi^2(m)$ .

### Lemma 14.2.3 — Centred cross-products

Let  $\mathbf{X}_1, \dots, \mathbf{X}_n$  ( $n \geq 2$ ) be independent  $N_k(\mu, \Sigma)$  vectors with mean  $\bar{\mathbf{X}}$ , and let  $\mathbf{W} = \sum_i (\mathbf{X}_i - \bar{\mathbf{X}})(\mathbf{X}_i - \bar{\mathbf{X}})^\top$ . Then  $\mathbf{W} \sim W_k(n-1, \Sigma)$ , and  $\mathbf{W}$  is independent of  $\bar{\mathbf{X}}$ .

*Proof.* Let  $\mathbf{X}_1$  be the  $n \times k$  matrix with rows  $\mathbf{X}_i^\top$ . Write  $\mathbf{X}_1 = \mathbf{1}\boldsymbol{\mu}^\top + \mathbf{Z}_0\boldsymbol{\Sigma}^{1/2}$ , where  $\boldsymbol{\Sigma}^{1/2}$  is the symmetric square root (Theorem 1.7.6) and  $\mathbf{Z}_0$  has independent standard normal entries. Let  $\mathbf{O}$  be an orthogonal  $n \times n$  matrix with first row  $n^{-1/2}\mathbf{1}^\top$ , for instance the Helmert matrix of Example (Helmert matrices), and let  $\mathbf{K}$  consist of its last  $n-1$  rows. From  $\mathbf{O}^\top\mathbf{O} = \mathbf{I}$ ,  $\mathbf{K}^\top\mathbf{K} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$ , so  $\mathbf{W} = \mathbf{X}_1^\top\mathbf{K}^\top\mathbf{K}\mathbf{X}_1 = (\mathbf{K}\mathbf{X}_1)^\top(\mathbf{K}\mathbf{X}_1)$ , and  $\mathbf{K}\mathbf{1} = \mathbf{0}$  gives  $\mathbf{K}\mathbf{X}_1 = (\mathbf{K}\mathbf{Z}_0)\boldsymbol{\Sigma}^{1/2}$ .

Each column of  $\mathbf{O}\mathbf{Z}_0$  is  $N_n(\mathbf{0}, \mathbf{O}\mathbf{O}^\top) = N_n(\mathbf{0}, \mathbf{I})$  by Theorem 3.3.1, and the columns are independent, so  $\mathbf{O}\mathbf{Z}_0$  again has independent standard normal entries. Its first row is  $\sqrt{n}(\bar{\mathbf{X}} - \boldsymbol{\mu})^\top\boldsymbol{\Sigma}^{-1/2}$ , and its other rows form  $\mathbf{K}\mathbf{Z}_0$ . Hence the  $n-1$  rows of  $\mathbf{K}\mathbf{X}_1$  are independent  $N_k(\mathbf{0}, \boldsymbol{\Sigma})$  vectors, independent of  $\bar{\mathbf{X}}$ , and  $\mathbf{W} \sim W_k(n-1, \boldsymbol{\Sigma})$ . ■

### Lemma 14.2.4 — Mean of an inverse Wishart matrix

Let  $\mathbf{W} \sim W_k(m, \boldsymbol{\Sigma})$  with  $\boldsymbol{\Sigma}$  positive definite and  $m \geq k+2$ . Then  $\mathbf{W}$  is nonsingular with probability one, and

$$\mathbb{E}(\mathbf{W}^{-1}) = \frac{\boldsymbol{\Sigma}^{-1}}{m - k - 1}.$$

*Proof. Reduction.* Write  $\mathbf{Z} = \mathbf{Z}_0\boldsymbol{\Sigma}^{1/2}$  with  $\mathbf{Z}_0$  as in the previous proof. Then  $\mathbf{W} = \boldsymbol{\Sigma}^{1/2}\mathbf{W}_0\boldsymbol{\Sigma}^{1/2}$  with  $\mathbf{W}_0 = \mathbf{Z}_0^\top\mathbf{Z}_0 \sim W_k(m, \mathbf{I})$ , and  $\mathbf{W}^{-1} = \boldsymbol{\Sigma}^{-1/2}\mathbf{W}_0^{-1}\boldsymbol{\Sigma}^{-1/2}$ . It suffices to show  $\mathbb{E}(\mathbf{W}_0^{-1}) = \mathbf{I}/(m-k-1)$ .

*Nonsingularity.*  $\mathbf{W}_0$  is singular iff some column  $\mathbf{z}_j$  of  $\mathbf{Z}_0$  lies in the span of the others. Given the other columns, that span has dimension at most  $k-1 < m$ , and  $\mathbf{z}_j \sim N_m(\mathbf{0}, \mathbf{I})$ , which has a density, falls in it with probability zero.

*Diagonal entries.* Partition  $\mathbf{Z}_0 = [\mathbf{z}_1, \mathbf{Z}_2]$ . By Theorem 1.6.4, the  $(1,1)$  entry of  $\mathbf{W}_0^{-1}$  is  $1/(\mathbf{z}_1^\top\mathbf{z}_1 - \mathbf{z}_1^\top\mathbf{Z}_2(\mathbf{Z}_2^\top\mathbf{Z}_2)^{-1}\mathbf{Z}_2^\top\mathbf{z}_1) = 1/\mathbf{z}_1^\top(\mathbf{I} - \mathbf{P}_2)\mathbf{z}_1$ , where  $\mathbf{P}_2$  projects onto  $\mathcal{C}(\mathbf{Z}_2)$ , of dimension  $k-1$ . Given  $\mathbf{Z}_2$ ,  $\mathbf{z}_1$  is still  $N_m(\mathbf{0}, \mathbf{I})$ , so  $\mathbf{z}_1^\top(\mathbf{I} - \mathbf{P}_2)\mathbf{z}_1 \sim \chi^2(m-k+1)$  by Theorem 4.3.2(a). This law does not depend on  $\mathbf{Z}_2$ , so it holds unconditionally (Lemma 14.1.1), and Lemma 14.1.3 gives mean  $1/(m-k-1)$ , finite because  $m-k+1 > 2$ . The same argument applies to every diagonal entry.

*Off-diagonal entries.* Let  $\mathbf{D}_i$  be the identity with its  $i$ th diagonal entry replaced by  $-1$ . Then  $\mathbf{Z}_0\mathbf{D}_i$  has the same law as  $\mathbf{Z}_0$ , so  $\mathbf{D}_i\mathbf{W}_0^{-1}\mathbf{D}_i = (\mathbf{D}_i\mathbf{W}_0\mathbf{D}_i)^{-1}$  has the same law as  $\mathbf{W}_0^{-1}$ . The mean of an off-diagonal entry  $(i,j)$  therefore equals its own negative, and is zero. It exists because  $|w^{ij}| \leq \frac{1}{2}(w^{ii} + w^{jj})$  for the entries of the positive definite  $\mathbf{W}_0^{-1}$ . ■

With these two lemmas the random-design analogue of Theorem 5.5.1 follows at once.

### Corollary 14.2.5 — Covariance of the slopes under normal regressors

Let the rows  $(\mathbf{X}_i^\top, Y_i)$  be independent draws from a  $(k+1)$ -variate normal distribution with  $\boldsymbol{\Sigma}_{XX}$  positive definite and conditional variance  $\sigma^2 = \text{Var}(Y | \mathbf{X})$ , and let  $\hat{\boldsymbol{\beta}}$  be the

least squares slopes from a fit with an intercept. If  $n \geq k + 3$ ,

$$\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \frac{\sigma^2}{n - k - 2} \boldsymbol{\Sigma}_{XX}^{-1}.$$

*Proof.* Given the regressors, the slopes are unbiased with covariance  $\sigma^2 \mathbf{W}^{-1}$ , where  $\mathbf{W}$  is the centred cross-product matrix (Eq. 6.6.1 with  $\mathbf{X}_2 = \mathbf{1}$ ). By Lemma 14.2.3  $\mathbf{W} \sim W_k(n - 1, \boldsymbol{\Sigma}_{XX})$ , and by Proposition 2.5.4 and Lemma 14.2.4 the unconditional covariance is  $\sigma^2 \mathbb{E}(\mathbf{W}^{-1}) = \sigma^2 \boldsymbol{\Sigma}_{XX}^{-1} / (n - k - 2)$ . ■

For  $k = 1$  this is Eq. 14.1.2. The factor  $1/(n - k - 2)$ , against  $1/(n - 1)$  for a fixed design with the average spread, grows quickly with  $k$ .

### Theorem 14.2.6 — Prediction error under normal regressors

In the setting of Corollary 14.2.5, let  $(\mathbf{X}_0, Y_0)$  be a further independent draw and  $\hat{Y}_0 = \hat{\alpha} + \hat{\boldsymbol{\beta}}^\top \mathbf{X}_0$  the fitted predictor. If  $n \geq k + 3$ , then

$$\mathbb{E}(Y_0 - \hat{Y}_0)^2 = \sigma^2 \left(1 + \frac{1}{n}\right) \frac{n - 2}{n - k - 2}. \quad (14.2.2)$$

*Proof.* Write  $\hat{Y}_0 = \bar{Y} + \hat{\boldsymbol{\beta}}^\top (\mathbf{X}_0 - \bar{\mathbf{X}})$ , and use the model  $Y_i = \alpha + \boldsymbol{\beta}^\top \mathbf{X}_i + e_i$  of Eq. 3.5.2 for  $i = 0, \dots, n$ . Since  $\bar{Y} = \alpha + \boldsymbol{\beta}^\top \bar{\mathbf{X}} + \bar{e}$ ,

$$Y_0 - \hat{Y}_0 = e_0 - \bar{e} - (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top (\mathbf{X}_0 - \bar{\mathbf{X}}).$$

Condition on  $\mathbf{X}_0, \dots, \mathbf{X}_n$ . Then  $e_0$ ,  $\bar{e}$  and  $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \mathbf{W}^{-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}}) e_i$  are normal with mean zero, and they are uncorrelated:  $e_0$  is independent of the training errors, and  $\text{Cov}(\bar{e}, \hat{\boldsymbol{\beta}}) = \sigma^2 n^{-1} \mathbf{W}^{-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}}) = \mathbf{0}$ . So the conditional mean square is

$$\sigma^2 \left[ 1 + \frac{1}{n} + (\mathbf{X}_0 - \bar{\mathbf{X}})^\top \mathbf{W}^{-1} (\mathbf{X}_0 - \bar{\mathbf{X}}) \right].$$

Now  $\mathbf{u} = \mathbf{X}_0 - \bar{\mathbf{X}} \sim N_k(\mathbf{0}, (1 + 1/n) \boldsymbol{\Sigma}_{XX})$ , and  $\mathbf{u}$  is independent of  $\mathbf{W}$ , because  $\mathbf{X}_0$  is independent of the training sample and  $\bar{\mathbf{X}}$  is independent of  $\mathbf{W}$  (Lemma 14.2.3). Hence  $\mathbb{E}(\mathbf{u}^\top \mathbf{W}^{-1} \mathbf{u}) = \text{tr}[\mathbb{E}(\mathbf{W}^{-1}) \mathbb{E}(\mathbf{u} \mathbf{u}^\top)] = (1 + 1/n) k / (n - k - 2)$  by Lemma 14.2.4, and

$$\mathbb{E}(Y_0 - \hat{Y}_0)^2 = \sigma^2 \left(1 + \frac{1}{n}\right) \left(1 + \frac{k}{n - k - 2}\right) = \sigma^2 \left(1 + \frac{1}{n}\right) \frac{n - 2}{n - k - 2}. \quad \blacksquare$$

The conditional mean square in the proof,  $\sigma^2 [1 + 1/n + (\mathbf{X}_0 - \bar{\mathbf{X}})^\top \mathbf{W}^{-1} (\mathbf{X}_0 - \bar{\mathbf{X}})]$ , is the variance behind the prediction interval of Theorem 12.3.1, so that interval is exact for random regressors as well (Theorem 14.1.2). What the theorem adds is the average over new cases and over training designs.

### 14.2.3 The optimism of the in-sample error

The residual mean square  $\text{SSE}/n$  is the error the fitted predictor makes on the cases used to fit it. It is a biased guide to the error on new cases, in both the fixed and the random design.

#### Proposition 14.2.7 — In-sample and out-of-sample error

- (a) (*Fixed design.*) Let  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n$  with  $\text{rank}(\mathbf{X}) = r$ , and let  $\mathbf{Y}^*$  be a vector of new responses at the same design, uncorrelated with  $\mathbf{Y}$ , with the same mean and covariance. Then

$$E \frac{\|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2}{n} = \sigma^2 \frac{n-r}{n}, \quad E \frac{\|\mathbf{Y}^* - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2}{n} = \sigma^2 \frac{n+r}{n},$$

so the in-sample error understates the error on new responses by  $2r\sigma^2/n$  on average.

- (b) (*Normal random design.*) In the setting of [Theorem 14.2.6](#), the in-sample error has  $E(\text{SSE}/n) = \sigma^2(n-k-1)/n$ , and the error on a new case is [Eq. 14.2.2](#).

*Proof.*

- (a) The first identity is  $E(\text{SSE}) = \sigma^2(n-r)$  ([Theorem 5.6.1](#), or [Eq. 2.3.1](#) with a mean in  $\mathcal{C}(\mathbf{X})$ ). For the second,  $\mathbf{Y}^* - \mathbf{X}\hat{\boldsymbol{\beta}} = (\mathbf{Y}^* - \mathbf{X}\boldsymbol{\beta}) - \mathbf{M}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$ , and the two terms are uncorrelated, so the expected squared length is  $n\sigma^2 + \sigma^2 \text{tr}(\mathbf{M}) = \sigma^2(n+r)$  by [Theorem 2.3.1](#).
- (b) Given the regressors,  $E(\text{SSE} \mid \mathbf{X}) = \sigma^2(n-k-1)$ , which does not depend on  $\mathbf{X}$ ; take expectations. ■

A new random case brings its own, possibly outlying, regressor values, where the fitted plane is least reliable, so the random design is worse. [Figure 14.2.1](#) shows the three expected errors for  $n = 30$ . With  $k = 10$  regressors the in-sample error is  $0.633\sigma^2$ , the error on new responses at the same design is  $1.367\sigma^2$ , and the error on a new random case is  $1.607\sigma^2$ : the in-sample figure flatters the fit by a factor of about 2.5. At  $k = 20$  the three values are 0.300, 1.700 and 3.617, in units of  $\sigma^2$ . The simulated averages over 20000 training samples per point, for instance 1.604 at  $k = 10$  and 3.612 at  $k = 20$ , agree with [Eq. 14.2.2](#). Estimating the out-of-sample error honestly, by penalties such as Mallows'  $C_p$  or by cross-validation, is the subject of [Chapter 29](#).

```
import numpy as np
rng = np.random.default_rng(1422)
n, sigma2, reps = 30, 1.0, 20_000

def simulate(k):
    """Average in-sample and out-of-sample squared error over many training samples.
    """
    ins, out = np.empty(reps), np.empty(reps)
    beta = np.full(k, 0.5)
    for s in range(reps):
        X = np.column_stack([np.ones(n), rng.normal(size=(n, k))])
        y = X @ np.r_[1.0, beta] + rng.normal(size=n)
```



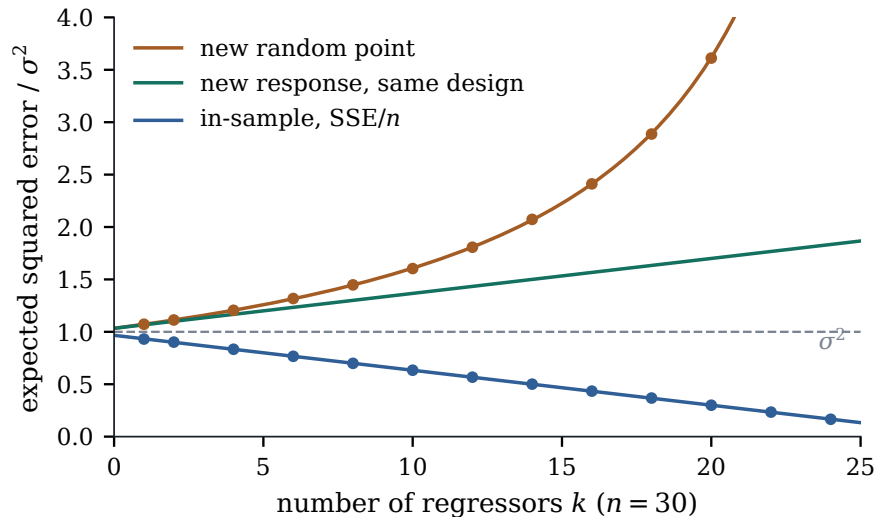


Figure 14.2.1: Expected squared error of a least squares fit,  $n = 30$ : in-sample (bottom), new responses at the same design (middle) and a new case with normal regressors (top, Eq. 14.2.2). Points: simulation.

```

b, *_ = np.linalg.lstsq(X, y, rcond=None)
ins[s] = np.mean((y - X @ b) ** 2)          # SSE / n
# new point  $X_0 \sim N(0, I)$ ,  $Y_0 = 1 + \beta^T X_0 + e_0$ : exact conditional error
out[s] = sigma2 + (b[0] - 1.0) ** 2 + np.sum((b[1:] - beta) ** 2)
return ins.mean(), out.mean()

results = {}
for k in (1, 5, 10, 20):
    ins, out = simulate(k)
    results[k] = (ins, out)
    theory = sigma2 * (1 + 1 / n) * (n - 2) / (n - k - 2)
    print(f"k = {k:2d}: in-sample {ins:.3f} (theory {sigma2 * (n - k - 1) / n:.3f}),",
          "
          f" new point {out:.3f} (theory {theory:.3f})")

```

### Example. (Predicting doctor visits)

The RAND Health Insurance Experiment records, for 20190 person-years, the number of outpatient visits and characteristics of the insurance plan and of the person. Treat these rows as the population, and predict  $\log(1 + \text{visits})$  linearly from 6 of the characteristics (coinsurance rate, deductible and participation-incentive terms, maximum expenditure, physical limitation and a chronic-disease index). The population projection has mean squared error 0.6330 and explains a fraction 0.094 of the variance. Draw training samples of 40 rows, fit by least squares, and evaluate each fit exactly on the whole population. Over 4000 training samples the average in-sample error is 0.5182 and the average error on new cases is 0.8201, against normal-theory values 0.5223 and 0.7705 from Proposition 14.2.7(b). The normal-theory value understates the error on new cases: the regressors are far from normal (two are indicators, one skewed), the linear predictor need not be the conditional mean, and the error variance need not be constant; the formula assumes none of this. It is a first guide to the optimism, not a substitute for checking it.

```

import statsmodels.api as sm
data = sm.datasets.randhie.load_pandas().data
cols = ["lncoins", "idp", "lpi", "fmde", "physlm", "disea"]
Xpop = np.column_stack([np.ones(len(data)), data[cols].to_numpy()])
ypop = np.log1p(data["mdvis"].to_numpy()) # log(1 + number of visits)
b_pop, *_ = np.linalg.lstsq(Xpop, ypop, rcond=None) # the population projection
sigma2_pop = np.mean((ypop - Xpop @ b_pop) ** 2) # its mean squared error

rng = np.random.default_rng(1423)
n_train, draws = 40, 4000
k_rand = len(cols)
ins, out = np.empty(draws), np.empty(draws)
for s in range(draws):
    idx = rng.integers(0, len(ypop), n_train) # iid draws from the population
    b, *_ = np.linalg.lstsq(Xpop[idx], ypop[idx], rcond=None)
    ins[s] = np.mean((ypop[idx] - Xpop[idx] @ b) ** 2)
    out[s] = np.mean((ypop - Xpop @ b) ** 2) # exact error over the
    population
print(f"population projection error {sigma2_pop:.4f}")
print(f"in-sample {ins.mean():.4f}, new case {out.mean():.4f}")
print(f"normal-theory values: {sigma2_pop * (n_train - k_rand - 1) / n_train:.4f},"
      f" {sigma2_pop * (1 + 1 / n_train) * (n_train - 2) / (n_train - k_rand - 2):.4f}")

```

## 14.2.4 Best linear unbiased prediction

So far the new response was independent of the training data. When it is not, as for the next value of a time series or the yield at an unsampled site near sampled ones, the training residuals carry information about the new error, and a good predictor should use it.

### Theorem 14.2.8 — Best linear unbiased predictor

Let  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  and  $y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + e_0$ , with  $E(\boldsymbol{\varepsilon}) = \mathbf{0}$ ,  $E(e_0) = 0$ ,  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$  for a known positive definite  $\mathbf{V}$ ,  $\text{Cov}(\boldsymbol{\varepsilon}, e_0) = \sigma^2 \mathbf{v}_0$  and  $\text{Var}(e_0) = \sigma^2 v_{00}$ . Suppose  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$ . A predictor  $\mathbf{a}^\top \mathbf{Y} + a_0$  is **unbiased** if  $E(\mathbf{a}^\top \mathbf{Y} + a_0 - y_0) = 0$  for every  $\boldsymbol{\beta}$ . Among unbiased linear predictors, the mean squared error  $E(y_0 - \mathbf{a}^\top \mathbf{Y} - a_0)^2$  is minimized by

$$\hat{y}_0 = \mathbf{x}_0^\top \tilde{\boldsymbol{\beta}} + \mathbf{v}_0^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\tilde{\boldsymbol{\beta}}), \quad (14.2.3)$$

where  $\tilde{\boldsymbol{\beta}}$  solves the generalized least squares equations  $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$ . With  $\mathbf{c} = \mathbf{x}_0 - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{v}_0$ , its mean squared error is  $\sigma^2 [v_{00} - \mathbf{v}_0^\top \mathbf{V}^{-1} \mathbf{v}_0 + \mathbf{c}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{c}]$ .

*Proof.* Put  $\boldsymbol{\delta} = \mathbf{V}^{-1} \mathbf{v}_0$  and split the prediction error of any linear predictor:

$$y_0 - \mathbf{a}^\top \mathbf{Y} - a_0 = \underbrace{(e_0 - \boldsymbol{\delta}^\top \boldsymbol{\varepsilon})}_u + \underbrace{(\mathbf{x}_0^\top \boldsymbol{\beta} - \boldsymbol{\delta}^\top \mathbf{X} \boldsymbol{\beta} - (\mathbf{a} - \boldsymbol{\delta})^\top \mathbf{Y} - a_0)}_w.$$

The term  $u$  is uncorrelated with  $\boldsymbol{\varepsilon}$ , since  $\text{Cov}(\boldsymbol{\varepsilon}, e_0 - \boldsymbol{\delta}^\top \boldsymbol{\varepsilon}) = \sigma^2 (\mathbf{v}_0 - \mathbf{V} \boldsymbol{\delta}) = \mathbf{0}$ . The term  $w$  is a constant plus a linear function of  $\mathbf{Y}$ , hence of  $\boldsymbol{\varepsilon}$ , so  $E(uw) = 0$  and  $E(y_0 - \mathbf{a}^\top \mathbf{Y} -$

$a_0)^2 = E u^2 + E w^2$ . Only  $E w^2$  depends on the predictor.

Unbiasedness means  $\mathbf{a}^\top \mathbf{X} \boldsymbol{\beta} + a_0 = \mathbf{x}_0^\top \boldsymbol{\beta}$  for all  $\boldsymbol{\beta}$ , that is,  $a_0 = 0$  and  $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$ . Equivalently,  $(\mathbf{a} - \boldsymbol{\delta})^\top \mathbf{Y}$  is a linear unbiased estimator of  $\mathbf{c}^\top \boldsymbol{\beta}$ , which is estimable because  $\mathbf{x}_0$  and  $\mathbf{X}^\top \boldsymbol{\delta}$  lie in  $\mathcal{C}(\mathbf{X}^\top)$ . For such predictors  $E w = 0$ , so  $E w^2$  is the variance of  $(\mathbf{a} - \boldsymbol{\delta})^\top \mathbf{Y}$ . By [Corollary 7.2.3](#) this variance is smallest when  $(\mathbf{a} - \boldsymbol{\delta})^\top \mathbf{Y} = \mathbf{c}^\top \tilde{\boldsymbol{\beta}}$ , the generalized least squares estimator, with variance  $\sigma^2 \mathbf{c}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{c}$ . The predictor is then  $\boldsymbol{\delta}^\top \mathbf{Y} + \mathbf{c}^\top \tilde{\boldsymbol{\beta}} = \mathbf{x}_0^\top \tilde{\boldsymbol{\beta}} + \boldsymbol{\delta}^\top (\mathbf{Y} - \mathbf{X} \tilde{\boldsymbol{\beta}})$ , which is [Eq. 14.2.3](#). Finally  $E u^2 = \sigma^2 (v_{00} - 2\boldsymbol{\delta}^\top \mathbf{v}_0 + \boldsymbol{\delta}^\top \mathbf{V} \boldsymbol{\delta}) = \sigma^2 (v_{00} - \mathbf{v}_0^\top \mathbf{V}^{-1} \mathbf{v}_0)$ . ■

The predictor is the estimated mean at the new point plus a prediction of the new error  $e_0$  from the residuals, by the best linear predictor of [Theorem 2.5.2](#). In the ordinary model,  $\mathbf{V} = \mathbf{I}$  and  $\mathbf{v}_0 = \mathbf{0}$ , the correction vanishes and [Eq. 14.2.3](#) is the fitted value  $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ , with mean squared error  $\sigma^2 [v_{00} + \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0]$ . The same theorem, with  $\mathbf{v}_0$  built from a spatial covariance function, is the kriging predictor of geostatistics, and with random effects in place of  $e_0$  it gives the predictors of mixed models ([Chapter 32](#), [Theorem 32.3.1](#)).

## 14.2.5 Exercises

### 14.2.5.1 A. Check your understanding

#### Exercise A1.

With normal regressors,  $n = 50$  and  $k = 10$ , compute the expected in-sample error  $E(\text{SSE}/n)$  and the expected error on a new case, in units of  $\sigma^2$ . By what factor does the first understate the second?

### 14.2.5.2 B. Practice

#### Exercise B1.

For any predictor  $g(\mathbf{X})$  with  $0 < \text{Var } g < \infty$ , let  $\tilde{g} = EY + b(g - EY)$  with  $b = \text{Cov}(Y, g) / \text{Var } g$ . Show that  $\tilde{g}$  predicts at least as well as  $g$  and that  $E(Y - \tilde{g})^2 = \text{Var}(Y)(1 - \text{corr}(Y, g)^2)$ . Conclude that among such *linearized* predictors, a higher squared correlation with  $Y$  means a smaller mean squared error, and that [Eq. 14.2.1](#) identifies the best of them.

*Solution.*  $\tilde{g}$  is the best linear predictor of  $Y$  from the single variable  $g$  ([Theorem 6.11.1](#) with  $k = 1$ ), and  $g$  itself is one such linear function (intercept 0, slope 1), so  $E(Y - \tilde{g})^2 \leq E(Y - g)^2$ . By [Eq. 6.11.1](#) the error of  $\tilde{g}$  is  $\text{Var } Y - \text{Cov}(Y, g)^2 / \text{Var } g = \text{Var } Y (1 - \text{corr}(Y, g)^2)$ , which is decreasing in the squared correlation. The largest squared correlation is  $\eta^2$ , attained by  $g = m$ , whose linearization is  $m$  itself.

#### Exercise B2.

Let the errors of a regression follow a stationary autoregression,  $e_t = \phi e_{t-1} + u_t$ ,  $|\phi| < 1$ , with white noise  $u_t$ , so that  $\text{Cov}(e_s, e_t) \propto \phi^{|s-t|}$ . Observe  $t = 1, \dots, n$  and predict

$y_{n+1} = \mathbf{x}_{n+1}^\top \boldsymbol{\beta} + e_{n+1}$ . Show that  $\mathbf{V}^{-1}\mathbf{v}_0 = \phi \mathbf{e}_n$ , the last standard basis vector scaled by  $\phi$ , and hence that the BLUP is  $\mathbf{x}_{n+1}^\top \tilde{\boldsymbol{\beta}} + \phi(y_n - \mathbf{x}_n^\top \tilde{\boldsymbol{\beta}})$ .

*Solution.* With  $\mathbf{V}_{st} = \phi^{|s-t|}$  (the common factor cancels in  $\mathbf{V}^{-1}\mathbf{v}_0$ ), the covariance of  $e_s$  with  $e_{n+1}$  is  $\phi^{n+1-s} = \phi \cdot \phi^{n-s} = \phi \mathbf{V}_{sn}$ . So  $\mathbf{v}_0 = \phi \mathbf{V} \mathbf{e}_n$  and  $\mathbf{V}^{-1}\mathbf{v}_0 = \phi \mathbf{e}_n$ . Substituting into Eq. 14.2.3, the correction term is  $\phi$  times the last residual  $y_n - \mathbf{x}_n^\top \tilde{\boldsymbol{\beta}}$ . Only the most recent residual matters, because given  $e_n$  the earlier errors carry no further linear information about  $e_{n+1}$ .

### Exercise B3.

In the setting of Corollary 14.2.5, let  $s^2 = \text{SSE}/(n - k - 1)$  and let  $\mathbf{W}$  be the centred cross-product matrix. Show that  $s^2 \mathbf{W}^{-1}$ , the usual estimate of the covariance of the slopes, is an unbiased estimator of their *unconditional* covariance  $\sigma^2 \boldsymbol{\Sigma}_{XX}^{-1}/(n - k - 2)$ .

*Solution.* Given the regressors,  $(n - k - 1)s^2/\sigma^2 \sim \chi^2(n - k - 1)$  (Theorem 7.5.1(b)), a law free of the design, so by Lemma 14.1.1  $s^2$  is independent of the regressors and hence of  $\mathbf{W}$ . Therefore  $E(s^2 \mathbf{W}^{-1}) = E(s^2) E(\mathbf{W}^{-1}) = \sigma^2 \boldsymbol{\Sigma}_{XX}^{-1}/(n - k - 2)$  by Lemma 14.2.3 and Lemma 14.2.4.

#### 14.2.5.3 C. Going deeper

### Exercise C1.

Let the rows  $(\mathbf{X}_i^\top, Y_i)$  be jointly normal with  $k$  regressors, and fit only the first  $k_1 < k$  of them. Show that the expected error on a new case is  $\sigma_1^2(1 + 1/n)(n - 2)/(n - k_1 - 2)$ , where  $\sigma_1^2$  is the error variance of the best linear predictor from the first  $k_1$  regressors. *Hint:* under joint normality the smaller model is itself a correct normal model with random regressors. For  $n = 30$ ,  $k = 10$ ,  $k_1 = 4$  and  $\sigma_1^2 = 1.2\sigma^2$ , is the smaller model better for prediction? What is the largest  $k_1$  for which it is?

*Solution.* The pairs  $(\mathbf{X}_{1i}, Y_i)$ , with  $\mathbf{X}_{1i}$  the first  $k_1$  regressors, are jointly normal, so given  $\mathbf{X}_1$  the response follows a normal linear model with  $k_1$  regressors and error variance  $\sigma_1^2$  (Eq. 3.5.2). The fitted predictor uses only these pairs, so Theorem 14.2.6 applies with  $k_1$  and  $\sigma_1^2$  in place of  $k$  and  $\sigma^2$ . With the numbers given, the full model has expected error  $1.607\sigma^2$  and the smaller one  $1.2 \cdot (31/30) \cdot 28/24 \sigma^2 = 1.447\sigma^2$ , so the smaller model predicts better despite its larger population error. Since the error of the smaller model increases with  $k_1$ , it wins as long as  $1.2 \cdot 28/(28 - k_1) < 28/18$ , that is  $k_1 < 6.4$ : at  $k_1 = 6$  its error is  $1.578\sigma^2$ , at  $k_1 = 7$  it is  $1.653\sigma^2$ . The largest such  $k_1$  is 6.

## 14.3 Multiple correlation

The population squared multiple correlation

$$\rho_{Y \cdot X}^2 = \frac{\boldsymbol{\sigma}_{XY}^\top \boldsymbol{\Sigma}_{XX}^{-1} \boldsymbol{\sigma}_{XY}}{\sigma_Y^2}$$

is the largest squared correlation between  $Y$  and a linear combination of  $\mathbf{X}$  (Proposition 2.5.3). Section 9.5 studied its sample counterpart  $R^2$  for a fixed design (Theorem 9.5.1, Theorem 9.5.2). With random rows,  $R^2$  estimates  $\rho_{Y \cdot X}^2$ . This section finds its exact distribution under multivariate normality, and uses it to test  $\rho_{Y \cdot X}^2 = 0$ , to build an interval and to measure the bias.

Throughout,  $(\mathbf{X}_i^\top, Y_i)$ ,  $i = 1, \dots, n$ , are independent draws from a  $(k+1)$ -variate normal distribution with  $\Sigma_{XX}$  positive definite and  $\sigma_Y^2 > 0$ , and  $n \geq k+2$ . By Eq. 3.5.2,  $Y_i = \alpha + \boldsymbol{\beta}^\top \mathbf{X}_i + e_i$  with independent  $e_i \sim N(0, \sigma^2)$  independent of the regressors, where

$$\sigma^2 = \sigma_Y^2(1 - \rho_{Y \cdot X}^2), \quad \boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta} = \sigma_Y^2 \rho_{Y \cdot X}^2, \quad \text{so} \quad \frac{\boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta}}{\sigma^2} = \frac{\rho_{Y \cdot X}^2}{1 - \rho_{Y \cdot X}^2}. \quad (14.3.1)$$

The ratio on the right is the population *signal-to-noise ratio*. We write  $\rho^2$  for  $\rho_{Y \cdot X}^2$  when no confusion can arise.

### 14.3.1 The sample multiple correlation as an estimator

Let  $\mathbf{S}_{XX}$ ,  $\mathbf{s}_{XY}$  and  $s_Y^2$  be the sample covariances, with any common divisor. The fit with an intercept has  $\text{SSR} = \hat{\boldsymbol{\beta}}^\top \mathbf{W} \hat{\boldsymbol{\beta}}$ , where  $\mathbf{W}$  is the centred cross-product matrix of the regressors and  $\hat{\boldsymbol{\beta}} = \mathbf{W}^{-1} \sum_i (\mathbf{X}_i - \bar{\mathbf{X}})(Y_i - \bar{Y})$  (Theorem 6.6.2 with  $\mathbf{X}_2 = \mathbf{1}$ ). Dividing by SST gives

$$R^2 = \frac{\mathbf{s}_{XY}^\top \mathbf{S}_{XX}^{-1} \mathbf{s}_{XY}}{s_Y^2}, \quad (14.3.2)$$

the population formula with sample moments in place of population moments. Under normality it is more than a plug-in.

#### Proposition 14.3.1 — Maximum likelihood with normal rows

Suppose the sample covariance matrix of the rows is nonsingular. The maximum likelihood estimates of  $(\alpha, \boldsymbol{\beta}, \sigma^2)$  are the least squares estimates and  $\text{SSE}/n$ , and the maximum likelihood estimate of  $\rho_{Y \cdot X}^2$  is  $R^2$ .

*Proof.* The joint density of a row factorizes as  $f(\mathbf{x}; \boldsymbol{\mu}_X, \Sigma_{XX}) f(y | \mathbf{x}; \alpha, \boldsymbol{\beta}, \sigma^2)$ , a normal density for the regressors times the conditional normal density of Theorem 3.5.2. The map from  $(\boldsymbol{\mu}, \Sigma)$  to  $(\boldsymbol{\mu}_X, \Sigma_{XX}, \alpha, \boldsymbol{\beta}, \sigma^2)$  is one-to-one onto the set where  $\Sigma_{XX}$  is positive definite and  $\sigma^2 > 0$ , with inverse  $\boldsymbol{\sigma}_{XY} = \Sigma_{XX} \boldsymbol{\beta}$ ,  $\sigma_Y^2 = \sigma^2 + \boldsymbol{\beta}^\top \Sigma_{XX} \boldsymbol{\beta}$ ,  $\mu_Y = \alpha + \boldsymbol{\beta}^\top \boldsymbol{\mu}_X$ . So the log-likelihood is a sum of two terms with separate parameters, and each can be maximized on its own. The conditional term is the likelihood of a normal linear model with model matrix  $[\mathbf{1}, \mathbf{X}_1]$ , maximized by least squares and  $\text{SSE}/n$  (Theorem 7.3.1).

In the marginal term the quadratic form is minimized over  $\boldsymbol{\mu}_X$  at  $\bar{\mathbf{x}}$ , leaving  $-\frac{n}{2} [\log \det \Sigma_{XX} + \text{tr}(\Sigma_{XX}^{-1} \mathbf{S})]$  with  $\mathbf{S} = \mathbf{W}/n$  positive definite. If  $\lambda_1, \dots, \lambda_k > 0$  are the eigenvalues of

$\mathbf{S}^{1/2}\boldsymbol{\Sigma}_{XX}^{-1}\mathbf{S}^{1/2}$ , then

$$\log \det \boldsymbol{\Sigma}_{XX} + \text{tr}(\boldsymbol{\Sigma}_{XX}^{-1}\mathbf{S}) - \log \det \mathbf{S} - k = \sum_{j=1}^k (\lambda_j - \log \lambda_j - 1) \geq 0,$$

with equality iff every  $\lambda_j = 1$ , that is, iff  $\boldsymbol{\Sigma}_{XX} = \mathbf{S}$ . By invariance of maximum likelihood under reparameterization, the estimate of  $\rho^2 = \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta} / (\sigma^2 + \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta})$  is  $\hat{\boldsymbol{\beta}}^\top \mathbf{S} \hat{\boldsymbol{\beta}} / (\text{SSE}/n + \hat{\boldsymbol{\beta}}^\top \mathbf{S} \hat{\boldsymbol{\beta}}) = \text{SSR}/(\text{SSE} + \text{SSR}) = R^2$ . ■

The rest of the section concerns the sampling distribution of this estimator.

## 14.3.2 The distribution of R-squared

### Theorem 14.3.2 — Distribution of $R^2$ with normal rows

Under the assumptions above, let  $F = \frac{n-k-1}{k} \frac{R^2}{1-R^2}$  and  $\lambda = \rho^2/(1-\rho^2)$ .

- (a) Given the regressors,  $F \sim F(k, n-k-1, \gamma)$  with  $\gamma = \boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} / \sigma^2$ , where  $\mathbf{W}$  is the centred cross-product matrix of the regressors.
- (b) If  $\rho^2 = 0$ , then  $F \sim F(k, n-k-1)$  and  $R^2 \sim \text{Beta}(\frac{k}{2}, \frac{n-k-1}{2})$ , independently of the regressors. The overall  $F$  test of the regression is an exact size- $\alpha$  test of  $\rho^2 = 0$ , which under joint normality is the hypothesis that  $Y$  is independent of  $\mathbf{X}$ .
- (c) If  $\rho^2 > 0$ , then  $\gamma = \lambda V$  with  $V \sim \chi^2(n-1)$ , and for  $0 < c < 1$

$$\Pr(R^2 \leq c) = \mathbb{E} \left[ \Pr \left\{ F(k, n-k-1, \lambda V) \leq \frac{(n-k-1)c}{k(1-c)} \right\} \right]. \quad (14.3.3)$$

- (d) For each  $c \in (0, 1)$ ,  $\Pr(R^2 > c)$  is a strictly increasing function of  $\rho^2 \in [0, 1)$ .

*Proof.*

- (a) The centred regressor matrix has rank  $k$  with probability one, because  $\mathbf{W} \sim W_k(n-1, \boldsymbol{\Sigma}_{XX})$  (Lemma 14.2.3) is nonsingular when  $n-1 \geq k$  (the nonsingularity step in the proof of Lemma 14.2.4 needs only  $m \geq k$ ). Given the regressors, the model is a normal linear model with  $r = k+1$  and mean  $\alpha \mathbf{1} + \mathbf{X}_1 \boldsymbol{\beta}$ . By Theorem 9.5.2(a),  $F \sim F(k, n-k-1, \gamma)$  with  $\gamma = \|(\mathbf{M} - \mathbf{P}_1)(\alpha \mathbf{1} + \mathbf{X}_1 \boldsymbol{\beta})\|^2 / \sigma^2$ . Since  $(\mathbf{M} - \mathbf{P}_1)\mathbf{1} = \mathbf{0}$  and  $(\mathbf{M} - \mathbf{P}_1)\mathbf{X}_1 \boldsymbol{\beta} = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_1 \boldsymbol{\beta}$ , this is  $\boldsymbol{\beta}^\top \mathbf{X}_1^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{X}_1 \boldsymbol{\beta} / \sigma^2 = \boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} / \sigma^2$ .
- (b)  $\rho^2 = 0$  means  $\boldsymbol{\sigma}_{XY} = \mathbf{0}$ , so  $\boldsymbol{\beta} = \mathbf{0}$  and  $\gamma = 0$  for every design. By Lemma 14.1.1,  $F \sim F(k, n-k-1)$  independently of the regressors, and so is  $R^2 = kF/(kF + n-k-1)$ , which is  $\text{Beta}(k/2, (n-k-1)/2)$  by Theorem 9.5.2(b). For jointly normal variables, zero covariance is independence (Theorem 3.4.1).
- (c) By the proof of Lemma 14.2.3,  $\mathbf{W} = (\mathbf{K}\mathbf{X}_1)^\top (\mathbf{K}\mathbf{X}_1)$ , where the  $n-1$  rows of  $\mathbf{K}\mathbf{X}_1$  are independent  $N_k(\mathbf{0}, \boldsymbol{\Sigma}_{XX})$ . So  $\mathbf{K}\mathbf{X}_1 \boldsymbol{\beta}$  has independent  $N(0, \boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta})$  entries, and  $\boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta} = \|\mathbf{K}\mathbf{X}_1 \boldsymbol{\beta}\|^2 = (\boldsymbol{\beta}^\top \boldsymbol{\Sigma}_{XX} \boldsymbol{\beta}) V$  with  $V \sim \chi^2(n-1)$ . By Eq. 14.3.1,  $\gamma = \lambda V$ . Since  $R^2 \leq c$  iff  $F \leq (n-k-1)c/(k(1-c))$ , taking expectations of the

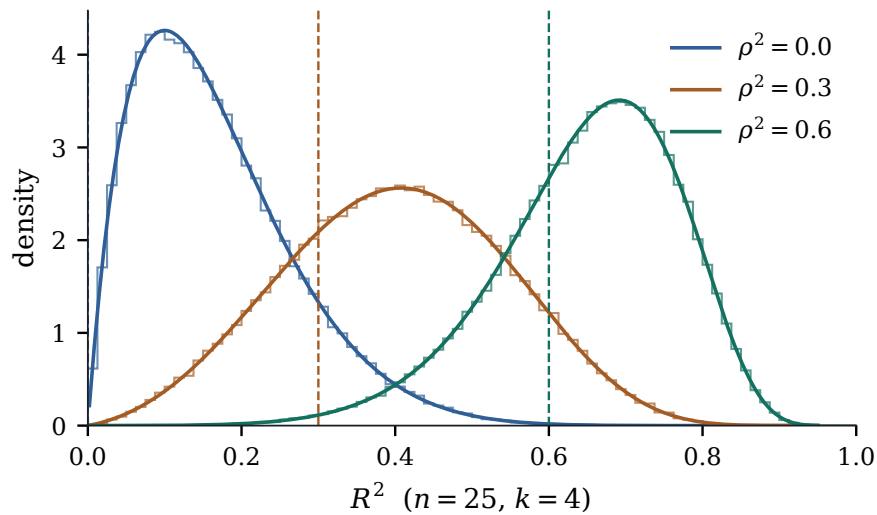


Figure 14.3.1: The law of  $R^2$  with normal rows,  $n = 25$ ,  $k = 4$ , for  $\rho^2 = 0, 0.3$  and  $0.6$  (dashed). Step outlines: simulation. Curves: Eq. 14.3.3.

conditional probability from (a) gives Eq. 14.3.3.

- (d) Fix  $v > 0$  and  $f > 0$ . By Theorem 4.1.4,  $\Pr\{F(k, n - k - 1, \lambda v) > f\}$  is strictly increasing in  $\lambda$ , and  $\lambda = \rho^2/(1 - \rho^2)$  is strictly increasing in  $\rho^2$ . If  $\rho_1^2 < \rho_2^2$ , the conditional tail probabilities therefore differ by a positive amount for every value of  $V > 0$ , and so do their expectations.

■

Part (c) is Fisher's (1928) distribution of the multiple correlation coefficient, reached here without any integration. Given the design,  $R^2$  is a transformed noncentral  $F$  variable, and the design enters only through the single number  $\boldsymbol{\beta}^\top \mathbf{W} \boldsymbol{\beta}$ , which is a scaled chi-squared variable because the regressors are normal. Figure 14.3.1 shows the resulting densities for  $n = 25$  and  $k = 4$ , computed from Eq. 14.3.3 by averaging over a grid of quantiles of  $V$ .

#### Example. (How biased is R-squared?)

With  $n = 25$ ,  $k = 4$  and 100000 simulated samples, the averages of  $R^2$  are 0.1669, 0.4026 and 0.6500 for  $\rho^2 = 0, 0.3$  and  $0.6$ , against exact values 0.1667, 0.4026 and 0.6498. The exact values come from the formula  $E(R^2) = 1 - \frac{n-k-1}{n-1}(1 - \rho^2) {}_2F_1(1, 1; \frac{n+1}{2}; \rho^2)$ , which follows from Fisher's (1928) distribution of  $R^2$  (see Olkin and Pratt 1958), where  ${}_2F_1$  is the hypergeometric function; it can also be computed from Eq. 14.3.3. The bias shrinks as  $\rho^2$  grows. The adjusted  $R^2$  of Eq. 9.5.1 averages 0.0003, 0.2831 and 0.5800: unbiased under the null (Proposition 9.5.3(e)), but biased *downwards* when  $\rho^2 > 0$ . Neither is unbiased everywhere. The standard deviations of  $R^2$  are 0.104, 0.146 and 0.118: with 25 cases, a single  $R^2$  says little about  $\rho^2$ .

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats

def r2_cdf(r, rho2, n, k, m=4000):
    """P(R^2 <= r) for iid normal rows: average the conditional noncentral F over V.
    """
```

```

df2 = n - k - 1
f = df2 * r / (k * (1 - r))
if rho2 == 0:
    return stats.f.cdf(f, k, df2)
V = stats.chi2.ppf((np.arange(m) + 0.5) / m, n - 1)      # equal-probability
nodes
return np.mean(stats.ncf.cdf(f, k, df2, rho2 / (1 - rho2) * V))

```

Part (d) says that the overall  $F$  test is unbiased and has power increasing in  $\rho^2$ , as a test of  $\rho^2 = 0$  against random-regressor alternatives. For  $n = 25$  and  $k = 4$  it rejects when  $R^2 > 0.364$ , and its power is 0.182, 0.377, 0.599 and 0.922 at  $\rho^2 = 0.1, 0.2, 0.3$  and  $0.5$ . A population in which the regressors account for a fifth of the variance of the response is detected less than half the time.

### 14.3.3 An interval for the squared multiple correlation

Part (d) also makes an exact confidence interval possible. For observed  $R^2 = r$ , let  $\rho_L^2$  be the value at which  $\Pr(R^2 \leq r) = 0.975$  and  $\rho_U^2$  the value at which it equals 0.025 (with  $\rho_L^2 = 0$  if  $\Pr(R^2 \leq r) < 0.975$  already at  $\rho^2 = 0$ ). Because the probability decreases in  $\rho^2$ ,  $[\rho_L^2, \rho_U^2]$  covers the true  $\rho^2$  exactly when  $R^2$  lies between the 2.5% and 97.5% points of its law at that  $\rho^2$ , which has probability 0.95. This is the usual inversion of a family of tests ([Proposition 12.1.2](#)).

#### Example. (Violent crime in the states)

For the 50 states in 2009, regress the violent crime rate on the high-school graduation rate, the poverty rate, the percentage of single-parent households, the percentage of white residents and the urbanization rate. Then  $R^2 = 0.4607$ , the adjusted value is 0.3994, and  $F = 7.52$  on 5 and 44 degrees of freedom, with  $p$ -value  $3.6 \times 10^{-5}$ . Read with random regressors, the 95% interval for  $\rho^2$  is (0.175, 0.610). The interval is wide: five regressors and fifty cases leave much room for chance. It also treats the states as a random sample, which they are not; it describes hypothetical replications of a process generating rows like these, and here, as in [Example \(Poverty and graduation\)](#), no more can be claimed.

```

data = sm.datasets.statecrime.load_pandas().data.drop(index="District of Columbia")
regs = ["hs_grad", "poverty", "single", "white", "urban"]
y = data["violent"].to_numpy()
X = np.column_stack([np.ones(len(y))] + [data[c].to_numpy() for c in regs])
fit = sm.OLS(y, X).fit()
n_s, k_s = len(y), len(regs)
R2_obs = fit.rsquared
print(f"R^2 = {R2_obs:.4f}, adjusted {fit.rsquared_adj:.4f}, F = {fit.fvalue:.2f},"
      f" p = {fit.f_pvalue:.2g}")

# 95% interval for rho^2: the values at which the observed R^2 is not extreme
def limit(prob):
    g = lambda rho2: r2_cdf(R2_obs, rho2, n_s, k_s) - prob
    return 0.0 if g(1e-12) < 0 else optimize.brentq(g, 1e-12, 0.999)

lower, upper = limit(0.975), limit(0.025)

```



```
print(f"95% interval for rho^2: ({lower:.3f}, {upper:.3f})")
```

*Remark* (Fixed or random?). Under a fixed design the expectation of  $R^2$  depends on the chosen regressor values ([Exercise \(B1\)](#)). With random rows  $R^2$  estimates  $\rho_{Y \cdot X}^2$ , which still depends on the regressor distribution: a narrower range gives a smaller  $\rho^2$ . The null law of the  $F$  test is the same in both readings.

## 14.3.4 Exercises

### 14.3.4.1 A. Check your understanding

#### Exercise A1.

A regression on  $k = 3$  random regressors from  $n = 40$  cases has  $R^2 = 0.25$ . Compute the overall  $F$  statistic and its degrees of freedom. Under the null hypothesis  $\rho^2 = 0$ , what are the mean and the distribution of  $R^2$ ?

### 14.3.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{R}$  be the sample correlation matrix of  $(Y, X_1, \dots, X_k)$ , assumed nonsingular, and let  $r^{YY}$  be the  $(1, 1)$  entry of  $\mathbf{R}^{-1}$ . Show that  $R^2 = 1 - 1/r^{YY}$ , and that  $1 - R^2 = \det \mathbf{R} / \det \mathbf{R}_{XX}$ , where  $\mathbf{R}_{XX}$  is the correlation matrix of the regressors.

*Solution.* [Eq. 14.3.2](#) is unchanged when every variable is standardized, so  $R^2 = \mathbf{r}_{XY}^\top \mathbf{R}_{XX}^{-1} \mathbf{r}_{XY}$ . By [Theorem 1.6.4](#), the leading entry of  $\mathbf{R}^{-1}$  is the inverse of the Schur complement  $1 - \mathbf{r}_{XY}^\top \mathbf{R}_{XX}^{-1} \mathbf{r}_{XY} = 1 - R^2$ , so  $r^{YY} = 1/(1 - R^2)$ . By [Theorem 1.6.3\(b\)](#),  $\det \mathbf{R} = \det \mathbf{R}_{XX} (1 - R^2)$ .

#### Exercise B2.

Show that  $R^2$  is unchanged if  $Y$  is replaced by  $aY + b$  with  $a \neq 0$ , or the regressors by  $\mathbf{B}^\top \mathbf{X} + \mathbf{c}$  with  $\mathbf{B}$  nonsingular. Show that  $\rho_{Y \cdot X}^2$  has the same invariance.

#### Exercise B3.

Show that  $R^2 \rightarrow \rho_{Y \cdot X}^2$  with probability one as  $n \rightarrow \infty$ , without assuming normality, provided the rows are independent and identically distributed with finite second moments and  $\Sigma_{XX}$  positive definite.

### 14.3.4.3 C. Going deeper

#### Exercise C1.

With normal rows, show that the likelihood ratio statistic for  $\rho_{Y \cdot X}^2 = 0$  against an unre-

stricted alternative is  $(1 - R^2)^{n/2}$ , so the likelihood ratio test is the overall  $F$  test. *Hint:* under the null the covariance matrix is block diagonal; use the proof of [Proposition 14.3.1](#) and [Exercise \(B1\)](#).

*Solution.* With  $\mathbf{S}$  the covariance matrix of the rows with divisor  $n$ , the maximized log-likelihood of a  $(k + 1)$ -variate normal sample is  $-\frac{n}{2}[\log \det \mathbf{S} + (k + 1)(1 + \log 2\pi)]$ , by the argument in the proof of [Proposition 14.3.1](#) applied to all  $k + 1$  variables. Under the null,  $Y$  and  $\mathbf{X}$  are independent, the likelihood factorizes, and the maximum is the same expression with  $\log(s_Y^2 \det \mathbf{S}_{XX})$  in place of  $\log \det \mathbf{S}$ . The likelihood ratio is therefore  $(\det \mathbf{S} / (s_Y^2 \det \mathbf{S}_{XX}))^{n/2} = (1 - R^2)^{n/2}$ , by [Theorem 1.6.3\(b\)](#) as in [Exercise \(B1\)](#). It is a decreasing function of  $R^2$ , hence of  $F$ .

## 14.4 The correlation coefficient and Fisher's Z

With a single regressor, the multiple correlation reduces to the ordinary correlation  $\rho$  of the pair  $(X, Y)$ , and  $R^2$  to the square of the sample correlation

$$r = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sqrt{\sum_i (X_i - \bar{X})^2 \sum_i (Y_i - \bar{Y})^2}}.$$

Keeping the sign, this section develops two routes to inference about  $\rho$ : the exact distribution of  $r$ , and Fisher's transformation to a nearly normal variable whose variance does not depend on  $\rho$ . Throughout,  $(X_i, Y_i)$ ,  $i = 1, \dots, n$ , are independent draws from a bivariate normal distribution with correlation  $|\rho| < 1$ , and  $n \geq 3$ .

### 14.4.1 The exact distribution of r

#### Proposition 14.4.1 — Exact law of the sample correlation

Let  $T = r\sqrt{n-2}/\sqrt{1-r^2}$  and  $\delta = \rho/\sqrt{1-\rho^2}$ .

- (a) Given the  $X$  values,  $T \sim t(n-2, \delta\sqrt{V})$ , where  $V = \sum_i (X_i - \bar{X})^2 / \sigma_X^2$ .
- (b)  $V \sim \chi^2(n-1)$ , and for  $-1 < c < 1$ ,

$$\Pr(r \leq c) = \mathbb{E}\left[\Pr\{t(n-2, \delta\sqrt{V}) \leq c\sqrt{n-2}/\sqrt{1-c^2}\}\right]. \quad (14.4.1)$$

- (c) If  $\rho = 0$ , then  $T \sim t(n-2)$  and  $r^2 \sim \text{Beta}(\frac{1}{2}, \frac{n-2}{2})$ , independently of the  $X$  values.
- (d) The law of  $r$  is stochastically increasing in  $\rho$ : for each  $c \in (-1, 1)$ ,  $\Pr(r \leq c)$  is a strictly decreasing function of  $\rho$ .

*Proof.*

- (a) Let  $\hat{\beta}_1$  be the least squares slope of  $Y$  on  $X$  and  $t_1$  its  $t$  statistic for the value zero. By [Theorem 9.5.4\(c\)](#) (see also [Exercise \(B3\)](#)), with only the intercept as the other

column,  $r^2 = t_1^2/(t_1^2 + n - 2)$ , and  $t_1$  has the sign of  $\hat{\beta}_1$ , which is the sign of  $r$ . Solving,  $t_1 = T$ . Given the  $X$  values the pairs follow a normal simple regression with slope  $\beta_1 = \rho\sigma_Y/\sigma_X$  and error variance  $\sigma^2 = \sigma_Y^2(1 - \rho^2)$  (Eq. 3.5.2), so by Corollary 7.5.2 with  $d = 0$ ,  $T$  is noncentral  $t$  with  $n - 2$  degrees of freedom and noncentrality  $\beta_1\sqrt{S_{xx}}/\sigma = \rho\sqrt{S_{xx}}/(\sigma_X\sqrt{1 - \rho^2}) = \delta\sqrt{V}$ .

- (b)  $V \sim \chi^2(n - 1)$  by Corollary 4.4.5. Since  $T$  is an increasing function of  $r$ ,  $r \leq c$  iff  $T \leq c\sqrt{n - 2}/\sqrt{1 - c^2}$ , and Eq. 14.4.1 is the expectation of the conditional probability.
- (c) With  $\delta = 0$  the conditional law is  $t(n - 2)$  for every design; apply Lemma 14.1.1. Then  $r^2 = T^2/(T^2 + n - 2)$  is the  $k = 1$  case of Theorem 14.3.2(b).
- (d) Write a  $t(\nu, \theta)$  variable as  $(Z + \theta)/\sqrt{W/\nu}$  (Definition 4.1.3). For  $\theta_1 < \theta_2$  and any  $u$ ,  $\Pr\{t(\nu, \theta_1) \leq u\} - \Pr\{t(\nu, \theta_2) \leq u\} = \Pr\{u\sqrt{W/\nu} - \theta_2 < Z \leq u\sqrt{W/\nu} - \theta_1\} > 0$ . Since  $\delta = \rho/\sqrt{1 - \rho^2}$  is strictly increasing in  $\rho$ , for every  $c$  and every  $V > 0$  the conditional probability in Eq. 14.4.1 strictly decreases in  $\rho$ , and so does its expectation.

■

Fisher (1915) found the density of  $r$  as an infinite series; for computation the mixture Eq. 14.4.1, an average of noncentral  $t$  probabilities over quantiles of  $V$ , is enough. Part (c) is the familiar test of  $\rho = 0$ , the  $t$  test of a zero slope, symmetric in the two variables. Part (d) makes exact intervals possible, by the inversion used for  $\rho^2$  in Section 14.3.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, stats

def r_cdf(c, rho, n, m=4000):
    """Exact P(r <= c) for n iid bivariate normal pairs with correlation rho."""
    if c < 0:
        # law of -r is that for
        -rho
        return 1 - r_cdf(-c, -rho, n, m)
    t = c * np.sqrt(n - 2) / np.sqrt(1 - c ** 2)
    # increasing in c
    if rho == 0:
        return stats.t.cdf(t, n - 2)
    V = stats.chi2.ppf((np.arange(m) + 0.5) / m, n - 1)
    # equal-probability
    nodes
    delta = rho / np.sqrt(1 - rho ** 2)
    return np.mean(stats.nct.cdf(t, n - 2, delta * np.sqrt(V)))
```

When  $\rho \neq 0$  the distribution of  $r$  is skewed towards zero, because  $r$  cannot pass  $\pm 1$ . For  $\rho = 0.8$  and  $n = 15$ , 200000 simulated samples give a mean of 0.7886, a median of 0.811, a skewness of  $-1.40$  and a variance of 0.01195 (panel (a) of Figure 14.4.1). The variance depends strongly on  $\rho$ : the standard deviation of  $r$  is 0.267 at  $\rho = 0$  and 0.032 at  $\rho = 0.95$  (panel (c)). An interval  $r \pm \text{constant} \times \text{standard error}$  would inherit both problems.

## 14.4.2 Fisher's transformation

Fisher (1921) proposed to work instead with

$$z = \tanh^{-1} r = \frac{1}{2} \log \frac{1+r}{1-r}, \quad \zeta = \tanh^{-1} \rho. \quad (14.4.2)$$

To see why this helps we need the large-sample behaviour of  $r$ , and for that three standard tools from probability theory, used here without proof (see van der Vaart 1998, chapters 2 and 3).

*Remark* (Three limit theorems).

- (i) *Central limit theorem.* If  $\mathbf{W}_1, \mathbf{W}_2, \dots$  are independent copies of a random vector with mean  $\boldsymbol{\mu}$  and covariance matrix  $\boldsymbol{\Gamma}$ , then  $\sqrt{n}(\bar{\mathbf{W}}_n - \boldsymbol{\mu}) \rightarrow N(\mathbf{0}, \boldsymbol{\Gamma})$  in distribution.
- (ii) *Slutsky's lemma.* If  $\mathbf{U}_n \rightarrow \mathbf{U}$  in distribution and  $\mathbf{V}_n \rightarrow \mathbf{c}$  in probability, then  $\mathbf{U}_n + \mathbf{V}_n \rightarrow \mathbf{U} + \mathbf{c}$ , and  $V_n U_n \rightarrow cU$  for scalars. (iii) *Delta method.* If  $\sqrt{n}(\mathbf{T}_n - \boldsymbol{\theta}) \rightarrow N(\mathbf{0}, \boldsymbol{\Gamma})$  and  $g$  is differentiable at  $\boldsymbol{\theta}$  with gradient  $\nabla g$ , then  $\sqrt{n}(g(\mathbf{T}_n) - g(\boldsymbol{\theta})) \rightarrow N(0, \nabla g^\top \boldsymbol{\Gamma} \nabla g)$ .

The limiting variance of  $r$  needs fourth moments of the bivariate normal distribution.

### Lemma 14.4.2 — Fourth moments of a standard bivariate normal pair

Let  $(X, Y)$  be bivariate normal with means 0, variances 1 and correlation  $\rho$ . Then  $E X^4 = 3$ ,  $E X^3 Y = 3\rho$  and  $E X^2 Y^2 = 1 + 2\rho^2$ . Consequently the covariance matrix of  $(X^2, Y^2, XY)$  is

$$\boldsymbol{\Gamma} = \begin{pmatrix} 2 & 2\rho^2 & 2\rho \\ 2\rho^2 & 2 & 2\rho \\ 2\rho & 2\rho & 1 + \rho^2 \end{pmatrix}.$$

*Proof.* Write  $Y = \rho X + \sqrt{1 - \rho^2} U$  with  $U \sim N(0, 1)$  independent of  $X$ ; the pair has the right means, variances and covariance, and it is normal by [Theorem 3.3.1](#). Using  $E X^2 = 1$ ,  $E X^4 = 3$  (the fourth moment of a standard normal) and the vanishing of odd moments,  $E X^3 Y = \rho E X^4 = 3\rho$  and  $E X^2 Y^2 = \rho^2 E X^4 + (1 - \rho^2) E X^2 E U^2 = 3\rho^2 + 1 - \rho^2 = 1 + 2\rho^2$ . The covariances follow by subtracting products of means,  $E X^2 = E Y^2 = 1$  and  $E XY = \rho$ : for instance  $\text{Var}(XY) = 1 + 2\rho^2 - \rho^2$  and  $\text{Cov}(X^2, XY) = E X^3 Y - \rho = 2\rho$ . ■

### Theorem 14.4.3 — Fisher's z transformation

As  $n \rightarrow \infty$ ,

$$\sqrt{n}(r - \rho) \rightarrow N(0, (1 - \rho^2)^2) \quad \text{and} \quad \sqrt{n}(z - \zeta) \rightarrow N(0, 1)$$

in distribution. The same limits hold with  $\sqrt{n-3}$  in place of  $\sqrt{n}$ . The limiting variance of  $z$  does not depend on  $\rho$ :  $\tanh^{-1}$  is a variance-stabilizing transformation for  $r$ .

*Proof.* The correlation is unchanged by the maps  $X \mapsto a + bX$ ,  $Y \mapsto c + dY$  with  $b, d > 0$ , so we may assume  $E X = E Y = 0$  and  $\text{Var } X = \text{Var } Y = 1$ . Let  $\bar{\mathbf{W}}_n$  be the average of  $\mathbf{W}_i = (X_i^2, Y_i^2, X_i Y_i)$ , whose mean is  $\boldsymbol{\theta} = (1, 1, \rho)$  and whose covariance

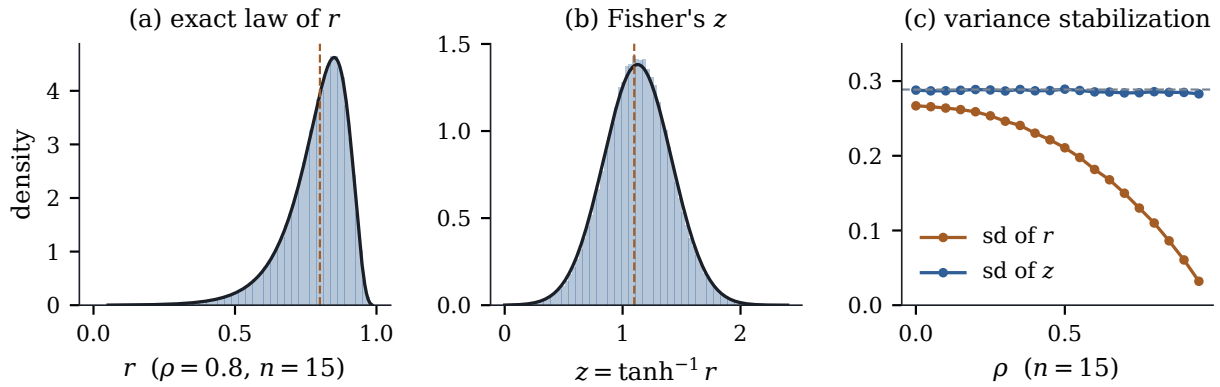


Figure 14.4.1: The sample correlation of  $n = 15$  normal pairs. (a)  $r$  for  $\rho = 0.8$ , with the exact density, obtained by differentiating Eq. 14.4.1. (b) Fisher's  $z$ , with the normal approximation. (c) The standard deviations of  $r$  and  $z$  across  $\rho$ ; dashed:  $1/\sqrt{n-3}$ .

is  $\mathbf{\Gamma}$  of Lemma 14.4.2. With divisor  $n$ , the sample variances and covariance are  $\bar{\mathbf{W}}_n - (\bar{X}^2, \bar{Y}^2, \bar{X}\bar{Y})$ . Since  $\sqrt{n}\bar{X}$  converges in distribution and  $\bar{X} \rightarrow 0$  in probability,  $\sqrt{n}\bar{X}^2 \rightarrow 0$  in probability, and similarly for the other two products, so by Slutsky's lemma the centring does not affect the limit. The divisor cancels in  $r = g(\text{sample moments})$ , where  $g(a, b, c) = c/\sqrt{ab}$ , and  $\nabla g(\boldsymbol{\theta}) = (-\rho/2, -\rho/2, 1)$ . By the central limit theorem and the delta method,  $\sqrt{n}(r - \rho)$  is asymptotically normal with variance

$$\nabla g^\top \mathbf{\Gamma} \nabla g = \frac{\rho^2}{4}(2+2+4\rho^2) - 2 \cdot \frac{\rho}{2}(2\rho+2\rho) + (1+\rho^2) = \rho^2 + \rho^4 - 4\rho^2 + 1 + \rho^2 = (1-\rho^2)^2.$$

The derivative of  $\tanh^{-1}$  at  $\rho$  is  $1/(1-\rho^2)$ , so a second application of the delta method gives variance 1 for  $z$ . Finally  $\sqrt{n-3}/\sqrt{n} \rightarrow 1$ , and Slutsky's lemma allows the substitution.

■

The transformation solves  $g'(\rho) = 1/(1-\rho^2)$ , the condition for a limiting standard deviation of one (Exercise B1). Why the approximation is so good for small  $n$  is another matter: Fisher (1921) showed that the mean of  $z$  exceeds  $\zeta$  by about  $\rho/(2(n-1))$  and that its variance is close to  $1/(n-3)$ , and the distribution of  $z$  is nearly symmetric even when that of  $r$  is strongly skewed. In the simulation with  $\rho = 0.8$  and  $n = 15$ ,  $z$  has mean 1.1295 against  $\zeta = 1.0986$  and  $\zeta + \rho/(2(n-1)) = 1.1272$ , variance 0.0811 against  $1/(n-3) = 0.0833$ , and skewness 0.01. Its standard deviation is 0.288 at  $\rho = 0$  and 0.283 at  $\rho = 0.95$  (panel (c)).

### 14.4.3 Intervals and tests for a correlation

The approximation  $z \approx N(\zeta, 1/(n-3))$  gives the interval

$$\tanh\left(z - \frac{z_{\alpha/2}}{\sqrt{n-3}}\right) \leq \rho \leq \tanh\left(z + \frac{z_{\alpha/2}}{\sqrt{n-3}}\right), \quad (14.4.3)$$

with  $z_{\alpha/2}$  the upper  $\alpha/2$  point of  $N(0, 1)$ . It is asymmetric about  $r$ , as it should be, and it never leaves  $(-1, 1)$ . The exact interval inverts Eq. 14.4.1, using part (d) of Proposition 14.4.1 exactly as for  $\rho^2$ : its limits are the values of  $\rho$  at which the observed  $r$  is the 97.5% and the 2.5% point.

How good is the approximation? For  $n = 10$  pairs, the coverage of the 95% interval [Eq. 14.4.3](#) over 200000 samples is 0.949, 0.949 and 0.952 at  $\rho = 0, 0.5$  and  $0.9$ . The interval  $r \pm 1.96(1 - r^2)/\sqrt{n}$ , which uses the asymptotic standard error of [Theorem 14.4.3](#) directly, covers only 0.839, 0.838 and 0.834.

```
def sample_r(rho, n, reps, rng):
    x = rng.normal(size=(reps, n))
    y = rho * x + np.sqrt(1 - rho ** 2) * rng.normal(size=(reps, n))
    xc = x - x.mean(axis=1, keepdims=True)
    yc = y - y.mean(axis=1, keepdims=True)
    return np.sum(xc * yc, axis=1) / np.sqrt(np.sum(xc ** 2, axis=1) * np.sum(yc **
    2, axis=1))

rng = np.random.default_rng(1443)
n_c, reps_c, zq = 10, 200_000, stats.norm.ppf(0.975)
for rho_c in (0.0, 0.5, 0.9):
    r_c = sample_r(rho_c, n_c, reps_c, rng)
    z_c = np.arctanh(r_c)
    fisher = np.abs(z_c - np.arctanh(rho_c)) <= zq / np.sqrt(n_c - 3)
    naive = np.abs(r_c - rho_c) <= zq * (1 - r_c ** 2) / np.sqrt(n_c)
    print(f"rho = {rho_c}: Fisher z covers {fisher.mean():.4f}, naive covers {naive.
    mean():.4f}")
```

Because  $z$  has an approximately known variance, it also compares correlations. With independent samples of  $n_1$  and  $n_2$  cases, the hypothesis  $\rho_1 = \rho_2$  is tested by referring

$$\frac{z_1 - z_2}{\sqrt{1/(n_1 - 3) + 1/(n_2 - 3)}}$$

to the standard normal distribution. The  $t$  statistic of [Proposition 14.4.1\(c\)](#) remains the exact test of  $\rho = 0$ ; for any other null value  $\rho_0$ , either  $(z - \tanh^{-1} \rho_0)\sqrt{n - 3}$  or the exact law [Eq. 14.4.1](#) is used.

### Example. (Poverty and graduation)

Across the 50 states, the correlation between the poverty rate and the high-school graduation rate is  $r = -0.7613$ . Its  $t$  statistic for  $\rho = 0$  is  $-8.13$  on 48 degrees of freedom, far out in the tail. The Fisher interval [Eq. 14.4.3](#) is  $(-0.858, -0.613)$  and the exact interval is  $(-0.855, -0.609)$ , almost the same, and both are noticeably asymmetric about  $r$ .

```
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
x, y = data["poverty"].to_numpy(), data["hs_grad"].to_numpy()
n_s = len(x)
r_s = np.corrcoef(x, y)[0, 1]
t_s = r_s * np.sqrt(n_s - 2) / np.sqrt(1 - r_s ** 2)
lo_z, hi_z = np.tanh(np.arctanh(r_s) + np.array([-1, 1]) * zq / np.sqrt(n_s - 3))
# exact interval: the rho at which the observed r sits at the 97.5% and 2.5% points
lo_x = optimize.brentq(lambda rh: r_cdf(r_s, rh, n_s) - 0.975, -0.99, 0.99)
hi_x = optimize.brentq(lambda rh: r_cdf(r_s, rh, n_s) - 0.025, -0.99, 0.99)
print(f"r = {r_s:.4f}, t = {t_s:.2f} on {n_s - 2} df")
print(f"Fisher z interval ({lo_z:.3f}, {hi_z:.3f}); exact interval ({lo_x:.3f}, {
    hi_x:.3f})")
```

*Warning.* Everything in this section uses bivariate normality, and not only through the conditional linear model. The asymptotic variance  $(1 - \rho^2)^2/n$  comes from the fourth moments of [Lemma 14.4.2](#). For other distributions it can be quite different, and then neither  $z$  nor the exact interval is reliable ([Exercise \(B2\)](#)). The  $t$  test of  $\rho = 0$  is more robust: by [Theorem 14.1.2](#) it needs only that  $Y$  given  $X$  follow a normal linear model with constant variance, and when  $X$  and  $Y$  are independent this holds as soon as  $Y$  is normal.

## 14.4.4 Exercises

### 14.4.4.1 A. Check your understanding

#### Exercise A1.

A sample of  $n = 28$  pairs has  $r = 0.62$ . Compute the 95% Fisher interval for  $\rho$ .

*Solution.*  $z = \tanh^{-1} 0.62 = 0.7250$  and  $1.96/\sqrt{25} = 0.392$ , so  $\zeta$  lies in  $(0.333, 1.117)$ , and  $\rho$  in  $(\tanh 0.333, \tanh 1.117) = (0.321, 0.806)$ . The interval is longer below  $r$  than above.

#### Exercise A2.

Two independent samples give  $r_1 = 0.45$  with  $n_1 = 60$  and  $r_2 = 0.20$  with  $n_2 = 80$ . Test  $\rho_1 = \rho_2$  at level 0.05.

### 14.4.4.2 B. Practice

#### Exercise B1.

Suppose  $\sqrt{n}(T_n - \theta) \rightarrow N(0, s(\theta)^2)$  for every  $\theta$  in an interval, with  $s$  positive and continuous. Show that if  $g'(\theta) = 1/s(\theta)$ , then  $\sqrt{n}(g(T_n) - g(\theta)) \rightarrow N(0, 1)$ . Solve  $g'(\rho) = 1/(1 - \rho^2)$  and recover [Eq. 14.4.2](#).

*Solution.* By the delta method the limiting variance is  $g'(\theta)^2 s(\theta)^2 = 1$ . With partial fractions,  $1/(1 - \rho^2) = \frac{1}{2}[1/(1 + \rho) + 1/(1 - \rho)]$ , whose antiderivative vanishing at zero is  $\frac{1}{2}[\log(1 + \rho) - \log(1 - \rho)] = \tanh^{-1} \rho$ .

#### Exercise B2.

Let  $X$  and  $U$  be independent standard normal variables and  $Y = XU$ . Show that  $X$  and  $Y$  are uncorrelated but not independent, and that  $\sqrt{n}r \rightarrow N(0, 3)$ . What happens to the level of the test that rejects  $\rho = 0$  when  $|T| > t_{n-2, 0.025}$  in large samples? Which assumption of [Theorem 14.1.2](#) fails?

*Solution.*  $EXY = EX^2EU = 0$ , but  $E(X^2Y^2) = EX^4EU^2 = 3 \neq EX^2EY^2 = 1$ , so they are dependent. Repeat the proof of [Theorem 14.4.3](#) at  $\rho = 0$ : the gradient is  $(0, 0, 1)$ , so the limiting variance of  $\sqrt{n}r$  is  $\text{Var}(XY)/(\text{Var } X \text{Var } Y) = 3$ . Since  $T \approx \sqrt{n}r$

for large  $n$ , the test rejects with probability tending to  $2\{1 - \Phi(1.96/\sqrt{3})\} = 0.258$ , not 0.05. Given  $X, Y$  is normal with mean zero but variance  $X^2$ : the errors are not homoscedastic, so the conditional model of [Definition 14.1.1](#) fails.

#### 14.4.4.3 C. Going deeper

##### Exercise C1.

For a general pair  $(X, Y)$  with finite fourth moments, standardized to means 0 and variances 1, show that  $\sqrt{n}(r - \rho)$  is asymptotically normal with variance  $(1 + \frac{\rho^2}{2}) E X^2 Y^2 + \frac{\rho^2}{4} (E X^4 + E Y^4) - \rho(E X^3 Y + E X Y^3)$ . Check that it reduces to  $(1 - \rho^2)^2$  under normality, and explain why no transformation of  $r$  stabilizes the variance across all distributions.

## 14.5 Partial correlation

A partial correlation measures linear association after the linear effects of conditioning variables have been removed from both variables ([Definition 3.6.1](#), [Proposition 3.6.1](#)). [Section 3.6](#) computed the sample version and [Theorem 9.5.4](#) identified its square with a partial  $R^2$ ; the sampling distribution is still missing. This section shows that a partial correlation from  $n$  cases with  $q$  conditioning variables behaves exactly like an ordinary correlation from  $n - q$  cases, and that its test is the  $t$  test of a regression coefficient.

### 14.5.1 The sample partial correlation

#### Definition 14.5.1 — Sample partial correlation

Let  $\mathbf{y}, \mathbf{x} \in \mathbb{R}^n$ , let  $\mathbf{Z}$  be an  $n \times q$  matrix, and let  $\mathbf{D} = [\mathbf{1}, \mathbf{Z}]$  have rank  $q + 1 < n$ . With  $\mathbf{M}_D$  the projection onto  $\mathcal{C}(\mathbf{D})$ , put  $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_D)\mathbf{y}$  and  $\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{M}_D)\mathbf{x}$ , and suppose both are nonzero. The **sample partial correlation** of  $\mathbf{y}$  and  $\mathbf{x}$  given  $\mathbf{Z}$  is

$$r_{yx \cdot Z} = \frac{\tilde{\mathbf{y}}^\top \tilde{\mathbf{x}}}{\|\tilde{\mathbf{y}}\| \|\tilde{\mathbf{x}}\|}.$$

Both residual vectors have mean zero, because  $\mathbf{1} \in \mathcal{C}(\mathbf{D})$ , so  $r_{yx \cdot Z}$  is the ordinary sample correlation of the two residual vectors. With  $q = 0$  it is the ordinary correlation of  $\mathbf{y}$  and  $\mathbf{x}$ .

### 14.5.2 Its distribution

The key is that residualizing on  $q + 1$  columns leaves  $n - q - 1$  dimensions, exactly as centring alone leaves  $n - 1$ .

#### Theorem 14.5.1 — Distribution of a sample partial correlation

Let  $\mathbf{Z}$  be a random  $n \times q$  matrix with  $\text{rank}[\mathbf{1}, \mathbf{Z}] = q + 1$  with probability one, and



$n \geq q + 3$ . Suppose that, given  $\mathbf{Z}$ , the rows  $(y_i, x_i)$  are independent bivariate normal vectors with means  $\mathbf{a} + \mathbf{B}\mathbf{z}_{(i)}$ , affine in the  $i$ th row  $\mathbf{z}_{(i)}$  of  $\mathbf{Z}$ , and a common covariance matrix with correlation  $\rho$ . Then  $r_{yx \cdot \mathbf{Z}}$  has the same distribution as the sample correlation of  $n - q$  independent bivariate normal pairs with correlation  $\rho$ , and it is independent of  $\mathbf{Z}$ .

In particular, if the rows  $(y_i, x_i, \mathbf{z}_{(i)}^\top)$  are independent draws from a multivariate normal distribution with a nonsingular covariance matrix, the conclusion holds with  $\rho = \rho_{yx \cdot \mathbf{Z}}$ , the population partial correlation.

*Proof.* Fix  $\mathbf{Z}$  and write  $\mathbf{D} = [\mathbf{1}, \mathbf{Z}]$ . Let  $\mathbf{K}$  be an  $(n - q - 1) \times n$  matrix whose rows form an orthonormal basis of  $\mathcal{C}(\mathbf{D})^\perp$ , so that  $\mathbf{KD} = \mathbf{0}$  and  $\mathbf{K}^\top \mathbf{K} = \mathbf{I} - \mathbf{M}_D$ . Then  $\tilde{\mathbf{y}}^\top \tilde{\mathbf{x}} = \mathbf{y}^\top (\mathbf{I} - \mathbf{M}_D) \mathbf{x} = (\mathbf{K}\mathbf{y})^\top (\mathbf{K}\mathbf{x})$ , and similarly for the squared lengths, so

$$r_{yx \cdot \mathbf{Z}} = \frac{(\mathbf{K}\mathbf{y})^\top (\mathbf{K}\mathbf{x})}{\|\mathbf{K}\mathbf{y}\| \|\mathbf{K}\mathbf{x}\|}.$$

Stack the pairs as the  $n \times 2$  matrix  $[\mathbf{y}, \mathbf{x}]$ . Given  $\mathbf{Z}$ ,  $[\mathbf{y}, \mathbf{x}] = \mathbf{DA} + \mathbf{E}$  for a  $(q + 1) \times 2$  matrix of coefficients  $\mathbf{A}$ , where  $\mathbf{E}$  has independent  $N_2(\mathbf{0}, \mathbf{C})$  rows. Hence  $\mathbf{K}[\mathbf{y}, \mathbf{x}] = \mathbf{KE}$ . Writing  $\mathbf{E} = \mathbf{E}_0 \mathbf{C}^{1/2}$  with independent standard normal entries in  $\mathbf{E}_0$ , each column of  $\mathbf{KE}_0$  is  $N(\mathbf{0}, \mathbf{K}\mathbf{K}^\top) = N(\mathbf{0}, \mathbf{I}_{n-q-1})$  (Theorem 3.3.1), so the  $n - q - 1$  rows of  $\mathbf{KE}$  are independent  $N_2(\mathbf{0}, \mathbf{C})$  vectors. Thus, given  $\mathbf{Z}$ ,  $r_{yx \cdot \mathbf{Z}}$  is the *uncentred* correlation of  $m = n - q - 1$  independent  $N_2(\mathbf{0}, \mathbf{C})$  pairs.

The same argument with  $q = 0$  and  $\mathbf{Z}$  absent shows that the ordinary sample correlation of  $m + 1$  bivariate normal pairs is the uncentred correlation of  $m$  independent  $N_2(\mathbf{0}, \mathbf{C})$  pairs. So the conditional law of  $r_{yx \cdot \mathbf{Z}}$  is that of an ordinary correlation from  $m + 1 = n - q$  pairs, whatever  $\mathbf{Z}$  is, and Lemma 14.1.1 gives the unconditional statement.

For the multivariate normal case, Theorem 3.5.2 says that given the conditioning variables the pair is bivariate normal with mean affine in them and covariance  $\Sigma_{11 \cdot 2}$ , whose correlation is  $\rho_{yx \cdot \mathbf{Z}}$  by Definition 3.6.1. ■

This is the result promised in Section 3.6: each conditioning variable costs one case. Everything in Section 14.4 transfers with  $n$  replaced by  $n - q$ : the exact law Eq. 14.4.1, the  $t$  test with  $n - q - 2$  degrees of freedom, and Fisher's interval with  $\sqrt{n - q - 3}$  (Fisher 1924). The conditioning variables need not be normal.

#### Example. (Twelve cases, two skewed conditioning variables)

Take  $n = 12$ ,  $q = 2$  conditioning variables drawn as squares of exponential variables, which are very skewed, a pair whose conditional means depend linearly on them, and conditional correlation  $\rho = 0.6$ . Over 100000 samples, the 10%, 50% and 90% points of the sample partial correlation are 0.257, 0.624 and 0.834. For ordinary correlations from  $n - q = 10$  normal pairs they are 0.256, 0.625 and 0.835. A two-sample Kolmogorov–Smirnov test finds no difference between the two sets of simulated values ( $p$ -value 0.09).

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, stats
```

```

rng = np.random.default_rng(1451)
n, q, rho, reps = 12, 2, 0.6, 100_000
B = np.array([[1.0, -2.0], [0.5, 3.0]])          # dependence of the pair on Z
C = np.array([[1.0, rho], [rho, 1.0]])          # conditional covariance of the
pair
Lc = np.linalg.cholesky(C)

partial = np.empty(reps)
for s in range(reps):
    Zc = rng.exponential(size=(n, q)) ** 2        # skewed conditioning variables
    pair = Zc @ B.T + rng.normal(size=(n, 2)) @ Lc.T
    D = np.column_stack([np.ones(n), Zc])
    coef, *_ = np.linalg.lstsq(D, pair, rcond=None)
    e = pair - D @ coef                          # residuals of both variables
    partial[s] = e[:, 0] @ e[:, 1] / np.sqrt((e[:, 0] @ e[:, 0]) * (e[:, 1] @ e[:, 1]))

# ordinary correlations from n - q pairs with correlation rho
x = rng.normal(size=(reps, n - q))
y = rho * x + np.sqrt(1 - rho ** 2) * rng.normal(size=(reps, n - q))
ordinary = np.array([np.corrcoef(a, b)[0, 1] for a, b in zip(x, y)])
print("quantiles, partial :", np.quantile(partial, [0.1, 0.5, 0.9]).round(3))
print("quantiles, ordinary:", np.quantile(ordinary, [0.1, 0.5, 0.9]).round(3))
print("two-sample KS p-value:", round(stats.ks_2samp(partial, ordinary).pvalue, 3))

```

### 14.5.3 The test of a partial correlation is a coefficient test

In the population, a zero partial correlation and a zero regression coefficient are the same thing.

#### Proposition 14.5.2 — Partial correlation and the regression coefficient

Let  $(Y, X, \mathbf{Z})$  have finite second moments, with  $\text{Cov}((\mathbf{Z}^\top, X)^\top)$  positive definite, and let  $\beta_X$  be the coefficient of  $X$  in the best linear predictor of  $Y$  from  $(\mathbf{Z}, X)$ . Then

$$\beta_X = \frac{\sigma_{yx \cdot Z}}{\sigma_{xx \cdot Z}} = \rho_{yx \cdot Z} \sqrt{\frac{\sigma_{yy \cdot Z}}{\sigma_{xx \cdot Z}}},$$

so  $\beta_X = 0$  iff  $\rho_{yx \cdot Z} = 0$ .

*Proof.* Let  $e_X = X - L(X | \mathbf{Z})$  and  $e_Y = Y - L(Y | \mathbf{Z})$  be the prediction errors of [Proposition 3.6.1\(a\)](#), with  $\text{Var}(e_X) = \sigma_{xx \cdot Z} > 0$  and  $\text{Cov}(e_Y, e_X) = \sigma_{yx \cdot Z}$ . The affine span of  $(\mathbf{Z}, X)$  is the orthogonal sum, in  $L^2$ , of the affine span of  $\mathbf{Z}$  and the line spanned by  $e_X$ , which is orthogonal to it. By the population version of [Theorem 6.3.8](#), projecting  $Y$  onto the sum projects it onto each piece:  $L(Y | \mathbf{Z}, X) = L(Y | \mathbf{Z}) + (\text{Cov}(Y, e_X) / \text{Var}(e_X))e_X$ . Since  $L(Y | \mathbf{Z})$  is orthogonal to  $e_X$ ,  $\text{Cov}(Y, e_X) = \text{Cov}(e_Y, e_X) = \sigma_{yx \cdot Z}$ . The coefficient of  $X$  in this expression comes only from the last term, because  $e_X = X -$

(affine function of  $\mathbf{Z}$ ); it is  $\sigma_{yx \cdot Z} / \sigma_{xx \cdot Z}$ .

■

This is the Frisch–Waugh–Lovell theorem in the population, and the sample version is [Theorem 6.6.2](#) itself. Combining the two with [Theorem 14.5.1](#):

### **Theorem 14.5.3 — The partial correlation test is the coefficient $t$ test**

Regress  $\mathbf{y}$  on the  $p = q + 2$  columns  $[\mathbf{1}, \mathbf{Z}, \mathbf{x}]$ , assumed of full rank, and let  $t$  be the  $t$  statistic of the coefficient of  $\mathbf{x}$  for the value zero. Then:

- (a) for every data set,  $t = \sqrt{n - p} r_{yx \cdot Z} / \sqrt{1 - r_{yx \cdot Z}^2}$ ;
- (b) under the assumptions of [Theorem 14.5.1](#) with  $\rho = 0$ ,  $t \sim t(n - p)$ , whatever the distribution of  $\mathbf{Z}$ ;
- (c) when the rows are multivariate normal, the  $t$  test of the coefficient of  $\mathbf{x}$  and the test of  $\rho_{yx \cdot Z} = 0$  are the same test, and it has exact size.

*Proof.*

- (a) By [Theorem 9.5.4\(c\)](#) (see also [Exercise \(B3\)](#)),  $r_{yx \cdot Z}^2 = t^2 / (t^2 + n - p)$ . By [Theorem 6.6.2](#) the coefficient of  $\mathbf{x}$  is  $\tilde{\mathbf{x}}^\top \tilde{\mathbf{y}} / \|\tilde{\mathbf{x}}\|^2$ , which has the sign of  $r_{yx \cdot Z}$ , so  $t$  and  $r_{yx \cdot Z}$  have the same sign. Solving for  $t$  gives (a).
- (b) By [Theorem 14.5.1](#),  $r_{yx \cdot Z}$  has the law of an ordinary correlation from  $n - q$  pairs with  $\rho = 0$ , and by [Proposition 14.4.1\(c\)](#) the statistic in (a) is then  $t((n - q) - 2) = t(n - p)$ .
- (c) Under multivariate normality the best linear predictor ([Theorem 2.5.2](#)) is the conditional mean ([Proposition 3.5.3](#)), so the regression coefficient of  $\mathbf{x}$  in the conditional model is  $\beta_X$  of [Proposition 14.5.2](#), which vanishes iff  $\rho_{yx \cdot Z} = 0$ . By (a) the two test statistics are monotone functions of each other, and by (b) or by [Theorem 14.1.2](#) the null distribution is exact.

■

So the test of a single coefficient in [Chapter 11](#) ([Theorem 11.6.1](#)) and the test of a partial correlation are one and the same, under either reading of the regressors. What the partial correlation adds is a scale-free measure of effect size, with its own confidence interval, which a coefficient does not provide.

#### **Example. (Violent crime and urbanization)**

Across the 50 states, the violent crime rate and the urbanization rate have correlation 0.345. Given the percentage of single-parent households and the poverty rate ( $q = 2$ ), their partial correlation is  $r = 0.1863$ . In the regression of violent crime on the three variables and an intercept, the coefficient of urbanization is 1.388 with  $t = 1.286$  on 46 degrees of freedom ( $p$ -value 0.2049), and indeed  $\sqrt{46} r / \sqrt{1 - r^2} = 1.286$ . With  $n - q = 48$ , the Fisher interval for  $\rho_{yx \cdot Z}$  is  $(-0.103, 0.447)$  and the exact interval is  $(-0.103, 0.443)$ . Much of the marginal association is shared with single parenthood and

poverty, and what remains is compatible with none.

```
def r_cdf(c, rho, n, m=4000):
    """Exact  $P(r \leq c)$  for an ordinary sample correlation of  $n$  bivariate normal
    pairs."""
    if c < 0:
        return 1 - r_cdf(-c, -rho, n, m)
    t = c * np.sqrt(n - 2) / np.sqrt(1 - c ** 2)
    if rho == 0:
        return stats.t.cdf(t, n - 2)
    V = stats.chi2.ppf((np.arange(m) + 0.5) / m, n - 1)
    return np.mean(stats.nct.cdf(t, n - 2, rho / np.sqrt(1 - rho ** 2) * np.sqrt(V))
    )

data = sm.datasets.statecrime.load_pandas().data.drop(index="District of Columbia")
y = data["violent"].to_numpy()
x = data["urban"].to_numpy()
Z = np.column_stack([np.ones(len(y)), data["single"], data["poverty"]])
n_s, q_s = len(y), Z.shape[1] - 1          # q = 2 conditioning variables

def resid(v, Z):
    coef, *_ = np.linalg.lstsq(Z, v, rcond=None)
    return v - Z @ coef

ey, ex = resid(y, Z), resid(x, Z)
r_p = ey @ ex / np.sqrt((ey @ ey) * (ex @ ex))          # sample partial correlation
fit = sm.OLS(y, np.column_stack([Z, x])).fit()          # p = 4 columns
t = fit.tvalues[-1]
df = n_s - 4
print(f"partial r = {r_p:.4f}, t = {t:.3f}, sqrt(df) r/sqrt(1-r^2) = "
      f"{np.sqrt(df) * r_p / np.sqrt(1 - r_p**2):.3f}")
zq = stats.norm.ppf(0.975)
lo_z, hi_z = np.tanh(np.arctanh(r_p) + np.array([-1, 1]) * zq / np.sqrt(n_s - q_s -
3))
lo_x = optimize.brentq(lambda rh: r_cdf(r_p, rh, n_s - q_s) - 0.975, -0.99, 0.99)
hi_x = optimize.brentq(lambda rh: r_cdf(r_p, rh, n_s - q_s) - 0.025, -0.99, 0.99)
print(f"Fisher z interval ({lo_z:.3f}, {hi_z:.3f}); exact ({lo_x:.3f}, {hi_x:.3f})")
```

*Remark* (Which variables to condition on). Conditioning can shrink an association, as in the example, reverse its sign, as in [Example \(Employment and unemployment\)](#) and Simpson's paradox, or create one where none exists, when the conditioning variable is a common effect of the two ([Exercise \(C1\)](#)). The distribution theory is silent on the choice of conditioning set; [Chapter 25](#) takes it up.

## 14.5.4 Exercises

### 14.5.4.1 A. Check your understanding

#### Exercise A1.

From  $n = 40$  cases, the partial correlation of  $y$  and  $x$  given  $q = 3$  other variables is

−0.30. Compute the  $t$  statistic, its degrees of freedom, and the 95% Fisher interval for the population partial correlation.

### Exercise A2.

Explain, in terms of dimensions of subspaces, why the  $t$  test of a partial correlation given  $q$  variables has  $n - q - 2$  degrees of freedom, and why this equals  $n - p$  for the corresponding regression.

#### 14.5.4.2 B. Practice

### Exercise B1.

Let  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$  with  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_1)$ ,  $\text{rank } \mathbf{X}_1 = r_1$ ,  $\text{rank } \mathbf{X} = r$  almost surely, and suppose that, given the regressors,  $\mathbf{Y}$  follows the normal linear model with mean  $\mathbf{X}_1\boldsymbol{\beta}_1$ . Show that the partial  $R^2$  of Definition 9.5.1 has the  $\text{Beta}(\frac{r-r_1}{2}, \frac{n-r}{2})$  distribution, whatever the distribution of the regressors.

*Solution.* By Theorem 9.5.4(b),  $R^2_{\mathbf{X}|\mathbf{X}_1} = qF/(qF+n-r)$  with  $q = r - r_1$ , where  $F$  is the  $F$  ratio for the added block. Given the regressors, the null hypothesis holds, so  $F \sim F(q, n-r)$  for every design (Theorem 14.1.2(b)), and by Lemma 14.1.1 unconditionally. Writing  $F = (U/q)/(V/(n-r))$  with independent chi-squared variables,  $R^2_{\mathbf{X}|\mathbf{X}_1} = U/(U+V)$ , which is  $\text{Beta}(q/2, (n-r)/2)$  by Exercise (B4).

### Exercise B2.

The *semipartial* correlation of  $\mathbf{y}$  and  $\mathbf{x}$  given  $\mathbf{Z}$  is the correlation between  $\mathbf{y}$  itself and the residual  $\tilde{\mathbf{x}} = (\mathbf{I} - \mathbf{M}_D)\mathbf{x}$ . Show that its square equals  $R^2_{\text{full}} - R^2_{\text{reduced}}$ , the increase in  $R^2$  when  $\mathbf{x}$  is added to  $[\mathbf{1}, \mathbf{Z}]$ . How does it relate to the squared partial correlation?

*Solution.* By Lemma 9.2.1, the extra sum of squares of  $\mathbf{x}$  is  $(\tilde{\mathbf{x}}^\top \mathbf{y})^2 / \|\tilde{\mathbf{x}}\|^2$ . Dividing by  $\text{SST} = \|(\mathbf{I} - \mathbf{P}_1)\mathbf{y}\|^2$  gives  $R^2_{\text{full}} - R^2_{\text{reduced}}$ . Since  $\tilde{\mathbf{x}}$  has mean zero,  $\tilde{\mathbf{x}}^\top \mathbf{y} = \tilde{\mathbf{x}}^\top (\mathbf{I} - \mathbf{P}_1)\mathbf{y}$ , so the quotient is the squared correlation of  $\mathbf{y}$  and  $\tilde{\mathbf{x}}$ . The squared partial correlation divides the same extra sum of squares by  $\text{SSE}_{\text{reduced}}$  instead of  $\text{SST}$ , so it equals the squared semipartial correlation divided by  $1 - R^2_{\text{reduced}}$ , and is never smaller.

#### 14.5.4.3 C. Going deeper

### Exercise C1.

Let  $X$  and  $Y$  be independent  $N(0, 1)$  and  $C = X + Y + U$  with  $U \sim N(0, 1)$  independent of both. Compute  $\rho_{XY \cdot C}$ . With  $n = 200$  cases, what does the  $t$  test of the coefficient of  $X$  in the regression of  $Y$  on  $C$  and  $X$  tend to conclude? Is the conclusion about the relation between  $X$  and  $Y$  wrong, or is the question wrong?

## 14.6 Testing lack of fit with replicates

Every test and interval so far has assumed that the mean function is right, that  $E(\mathbf{Y}) \in \mathcal{C}(\mathbf{X})$ . When it is wrong, the residual mean square overestimates the error variance: [Example \(Residual sum of squares under a wrong mean\)](#) showed that  $E(\text{SSE}) = \sigma^2(n-r) + \|(\mathbf{I}-\mathbf{M})\boldsymbol{\theta}\|^2$  for a mean vector  $\boldsymbol{\theta}$  outside the model space. Separating noise from systematic error needs an estimate of  $\sigma^2$  free of the form of the mean, and repeated rows provide one: observations at identical settings differ only by noise. The idea, and the test built on it, go back to Fisher (1922).

### 14.6.1 Replicates and the cell-means model

Suppose the model matrix has  $c$  distinct rows  $\mathbf{x}_1^\top, \dots, \mathbf{x}_c^\top$ , the  $i$ th occurring  $n_i \geq 1$  times, with  $n = \sum_i n_i$ . Index the observations by the distinct row and the replicate,  $y_{ij}$  with  $j = 1, \dots, n_i$ , so that the model is

$$E(y_{ij}) = \mathbf{x}_i^\top \boldsymbol{\beta}, \quad i = 1, \dots, c, \quad j = 1, \dots, n_i.$$

Let  $\mathbf{Z}$  be the  $n \times c$  matrix of group indicators, the model matrix of the one-way layout in which each distinct row has its own mean  $\mu_i$ . Call two matrices with  $n$  rows *alike in row structure* as  $\mathbf{X}$  if their rows are equal whenever the corresponding rows of  $\mathbf{X}$  are equal.

#### Lemma 14.6.1 — The largest model with the same replicates

- (a)  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ . More generally, every matrix  $\mathbf{W}$  with  $n$  rows whose rows are constant within the groups satisfies  $\mathcal{C}(\mathbf{W}) \subseteq \mathcal{C}(\mathbf{Z})$ .
- (b)  $\mathbf{M}_{\mathbf{Z}\mathbf{Y}}$  replaces each observation by its group mean  $\bar{y}_i$ , and  $\mathbf{M}\mathbf{y}$  is constant within groups.

*Proof.*

- (a) Let  $\bar{\mathbf{W}}$  be the  $c \times s$  matrix whose  $i$ th row is the common row of  $\mathbf{W}$  in group  $i$ . Then  $\mathbf{W} = \mathbf{Z}\bar{\mathbf{W}}$ , so  $\mathcal{C}(\mathbf{W}) \subseteq \mathcal{C}(\mathbf{Z})$ . The matrix  $\mathbf{X}$  is a special case.
- (b) The first claim is [Example \(The one-way layout\)](#). For the second,  $\mathbf{M}\mathbf{y} \in \mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ , and the vectors of  $\mathcal{C}(\mathbf{Z})$  are constant within groups. ■

So the cell-means model is the largest linear model that treats identical rows identically. Any mean function of the regressors at all, linear or not, produces a mean vector in  $\mathcal{C}(\mathbf{Z})$ .

#### Definition 14.6.1 — Pure error and lack of fit

With  $\mathbf{M}_{\mathbf{Z}}$  the projection onto  $\mathcal{C}(\mathbf{Z})$ , the **pure error** and **lack-of-fit** sums of squares are

$$\begin{aligned} \text{SSPE} &= \|(\mathbf{I} - \mathbf{M}_Z)\mathbf{y}\|^2 = \sum_{i=1}^c \sum_{j=1}^{n_i} (y_{ij} - \bar{y}_i)^2, \\ \text{SSLF} &= \|(\mathbf{M}_Z - \mathbf{M})\mathbf{y}\|^2 = \sum_{i=1}^c n_i (\bar{y}_i - \hat{y}_i)^2, \end{aligned} \quad (14.6.1)$$

with  $n - c$  and  $c - r$  degrees of freedom, where  $\hat{y}_i = \mathbf{x}_i^\top \hat{\boldsymbol{\beta}}$  is the common fitted value of group  $i$  and  $r = \text{rank}(\mathbf{X})$ .

The two expressions for each sum of squares agree by [Lemma 14.6.1\(b\)](#):  $(\mathbf{I} - \mathbf{M}_Z)\mathbf{y}$  has entries  $y_{ij} - \bar{y}_i$ , and  $(\mathbf{M}_Z - \mathbf{M})\mathbf{y}$  has entries  $\bar{y}_i - \hat{y}_i$ , repeated  $n_i$  times.

## 14.6.2 The test

### Theorem 14.6.2 — Pure-error lack-of-fit test

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$  with  $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$ , that is,  $E(y_{ij}) = \mu_i$  depends only on the row of  $\mathbf{X}$ ; assume  $r < c < n$ . Then:

- (a)  $\text{SSE} = \text{SSLF} + \text{SSPE}$ , and  $\text{SSLF}$  and  $\text{SSPE}$  are independent;
- (b)  $\text{SSPE}/\sigma^2 \sim \chi^2(n - c)$  for every  $\mu_1, \dots, \mu_c$ ;
- (c)  $\text{SSLF}/\sigma^2 \sim \chi^2(c - r, \gamma)$  with

$$\gamma = \frac{\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{i=1}^c n_i (\mu_i - \tilde{\mu}_i)^2, \quad (14.6.2)$$

where  $\tilde{\mu}_i$  is the  $i$ th group value of  $\mathbf{M}\boldsymbol{\theta}$ , the least squares fit to the true means;

- (d) the ratio

$$F = \frac{\text{SSLF}/(c - r)}{\text{SSPE}/(n - c)} \sim F(c - r, n - c, \gamma),$$

and  $\gamma = 0$  iff  $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$ , that is, iff the model is correct. The test that rejects the model when  $F > F_\alpha(c - r, n - c)$  has size  $\alpha$ , and its power increases strictly with  $\gamma$ .

*Proof.* By [Lemma 14.6.1](#),  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ , so by [Theorem 6.5.1](#)  $\mathbf{M}_Z - \mathbf{M}$  is the projection onto the orthogonal complement of  $\mathcal{C}(\mathbf{X})$  in  $\mathcal{C}(\mathbf{Z})$ , of rank  $c - r$ , and  $\mathbf{I} - \mathbf{M} = (\mathbf{M}_Z - \mathbf{M}) + (\mathbf{I} - \mathbf{M}_Z)$  splits into two orthogonal projections. This gives the identity in (a). By [Theorem 4.4.4](#),  $\text{SSLF}/\sigma^2$  and  $\text{SSPE}/\sigma^2$  are independent noncentral chi-squared variables with  $c - r$  and  $n - c$  degrees of freedom and noncentralities  $\|(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2$  and  $\|(\mathbf{I} - \mathbf{M}_Z)\boldsymbol{\theta}\|^2/\sigma^2$ . The second is zero because  $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$ , which proves (b). In the first,  $\mathbf{M}_Z\boldsymbol{\theta} = \boldsymbol{\theta}$ , so  $(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta} = (\mathbf{I} - \mathbf{M})\boldsymbol{\theta}$ , whose entries are  $\mu_i - \tilde{\mu}_i$ , repeated  $n_i$  times; this is (c). Part (d) follows from [Definition 4.1.2](#), the fact that  $(\mathbf{I} - \mathbf{M})\boldsymbol{\theta} = \mathbf{0}$  iff  $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$ , and [Theorem 4.1.4](#). ■

The test is the  $F$  test of the reduced model  $\mathcal{C}(\mathbf{X})$  inside the full model  $\mathcal{C}(\mathbf{Z})$  ([Theorem 11.1.1](#),

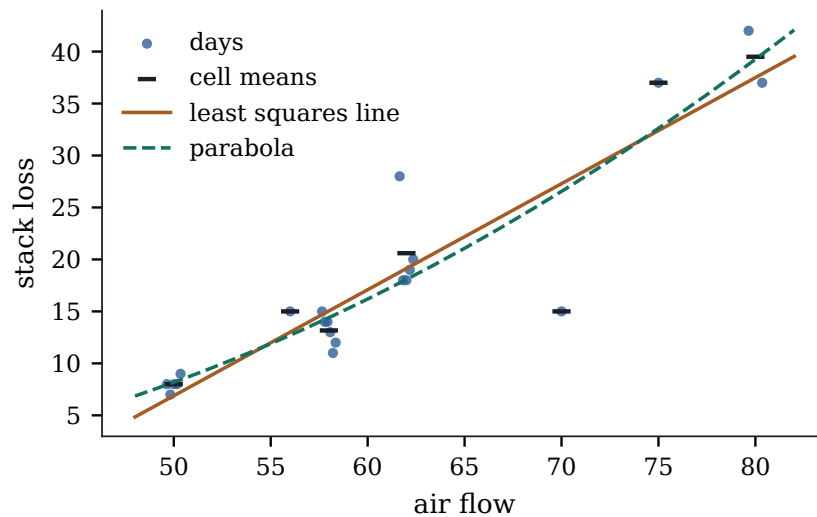


Figure 14.6.1: Stack loss against air flow on 21 days (replicates spread sideways), the seven group means (bars), the least squares line and parabola.

[Theorem 4.4.6](#)). What makes it special is the full model, which is chosen not by the analyst but by the replication in the data, and which contains every mean function of the regressors. The residual sum of squares of the working model is split into two rows of an analysis of variance:

| Source      | Degrees of freedom | Sum of squares                         | Mean square           |
|-------------|--------------------|----------------------------------------|-----------------------|
| Lack of fit | $c - r$            | $\sum_i n_i (\bar{y}_i - \hat{y}_i)^2$ | $\text{SSLF}/(c - r)$ |
| Pure error  | $n - c$            | $\sum_i \sum_j (y_{ij} - \bar{y}_i)^2$ | $\text{SSPE}/(n - c)$ |
| Residual    | $n - r$            | SSE                                    | $s^2$                 |

The noncentrality [Eq. 14.6.2](#) is the weighted squared distance between the true group means and their best approximation by the model, in units of  $\sigma^2$ . It grows with the replication, so replicates buy power twice: through  $\gamma$  and through the pure-error degrees of freedom.

#### Example. (Stack loss and air flow)

Brownlee's stack loss data record  $n = 21$  days of operation of a plant that oxidizes ammonia. Air flow, the rate of operation, takes only  $c = 7$  distinct values, so a straight-line regression of stack loss on air flow can be tested for lack of fit with 14 degrees of freedom for pure error. The line has intercept  $-44.13$  and slope  $1.020$ , and  $R^2 = 0.846$ . Its residual sum of squares  $319.12$  splits into  $\text{SSPE} = 96.53$  and  $\text{SSLF} = 222.58$  on 5 degrees of freedom. The mean squares are  $44.52$  and  $6.90$ , so  $F = 6.46$  with  $p$ -value  $0.0026$ ; the 5% point of  $F(5, 14)$  is  $2.96$ . The residual mean square of the line,  $16.80$ , is more than twice the pure-error estimate of  $\sigma^2$ .

[Figure 14.6.1](#) shows why. The group means do not lie on any smooth curve: the single day at air flow 70 has stack loss  $15.0$ , below the mean  $20.6$  of the five days at air flow 62. A parabola does not help, with  $F = 7.22$  on 4 and 14 degrees of freedom ( $p$ -value  $0.0023$ ). The mean of stack loss is not a function of air flow alone, which points to the other operating variables, taken up in [Example \(Is the three-variable model adequate?\)](#).

```
import numpy as np
import statsmodels.api as sm
```



```

from scipy import stats
data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
x = data["AIRFLOW"].to_numpy()
n = len(y)
levels, groups = np.unique(x, return_inverse=True)  # c distinct rows of X
c = len(levels)
Z = np.eye(c)[groups]  # cell-means model matrix

def sse(A):
    coef, *_ = np.linalg.lstsq(A, y, rcond=None)
    return np.sum((y - A @ coef) ** 2)

ss_pe = sse(Z)  # pure error, n - c df
for name, X in [("line", np.column_stack([np.ones(n), x])),
                ("parabola", np.column_stack([np.ones(n), x, x ** 2]))]:
    r = np.linalg.matrix_rank(X)
    ss_lf = sse(X) - ss_pe  # lack of fit, c - r df
    F = (ss_lf / (c - r)) / (ss_pe / (n - c))
    print(f"{name:8s}: SSE {sse(X):7.2f} = SSLF {ss_lf:7.2f} ({c - r} df)"
          f" + SSPE {ss_pe:6.2f} ({n - c} df); F = {F:.2f}, p = {stats.f.sf(F, c -
r, n - c):.4f}")

```

### 14.6.3 When pure error is not pure

The theorem assumes  $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$ : the mean depends on the observation only through its row of  $\mathbf{X}$ . If a variable that is not in the model affects the response and varies within the groups of replicates, as water temperature does among the days with equal air flow in the stack loss data, then observations in a group do not share a mean. By [Theorem 9.6.1\(b\)](#), applied to the cell-means model,

$$E\left(\frac{\text{SSPE}}{n - c}\right) = \sigma^2 + \frac{\|(\mathbf{I} - \mathbf{M}_Z)\boldsymbol{\theta}\|^2}{n - c},$$

so the “pure” error estimate is inflated, and  $F$  becomes a ratio of two noncentral chi-squared variables, a *doubly noncentral F*, which rejects less often than it would with a pure estimate of  $\sigma^2$ . Two further cautions: the pooled estimate assumes a common variance across groups ([Exercise \(C2\)](#)), and a non-significant result with few degrees of freedom is weak evidence that the model is adequate.

*Remark* (Designing for a lack-of-fit test). The test needs  $c > r$  distinct rows and  $n > c$  observations. The design that estimates a straight line most precisely puts half the observations at each end of the range; it has  $c = r = 2$  and cannot detect any curvature. A few intermediate settings buy a check of the model at a small cost in precision. Replicates also cap what any model of the regressors can achieve:  $R^2 \leq 1 - \text{SSPE}/\text{SST}$  ([Exercise \(B2\)](#)).

### 14.6.4 Exercises

#### 14.6.4.1 A. Check your understanding

**Exercise A1.**

A straight line is fitted to  $n = 24$  observations at  $c = 6$  distinct values of  $x$ , each replicated four times. The residual sum of squares is 50.0 and the pure-error sum of squares is 36.0. Complete the analysis of variance table for lack of fit and compute  $F$ . Is there evidence of lack of fit at level 0.05, given that  $F_{0.05}(4, 18) = 2.93$ ?

**14.6.4.2 B. Practice****Exercise B1.**

Show that the least squares estimate  $\hat{\boldsymbol{\beta}}$  from all  $n$  observations is also the weighted least squares estimate from the  $c$  group means, with weights  $n_i$ , and that SSLF is the weighted residual sum of squares of that fit.

*Solution.* By [Theorem 6.5.1](#),  $\mathbf{M}\mathbf{y} = \mathbf{M}\mathbf{M}_Z\mathbf{y}$ : projecting  $\mathbf{y}$  onto  $\mathcal{C}(\mathbf{X})$  is the same as projecting the vector of group means  $\mathbf{M}_Z\mathbf{y}$ . For a vector constant within groups,  $\|\mathbf{M}_Z\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \sum_i n_i(\bar{y}_i - \mathbf{x}_i^\top \mathbf{b})^2$ , so minimizing it over  $\mathbf{b}$  is weighted least squares on the  $c$  means with weights  $n_i$ . Its minimum is  $\|(\mathbf{M}_Z - \mathbf{M})\mathbf{y}\|^2 = \text{SSLF}$ .

**Exercise B2.**

Suppose  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ . Show that every model whose model matrix has the row structure of  $\mathbf{X}$  has  $R^2 \leq 1 - \text{SSPE}/\text{SST}$ , with equality for the cell-means model.

*Solution.* By [Lemma 14.6.1\(a\)](#) the column space  $\mathcal{V}$  of such a model lies in  $\mathcal{C}(\mathbf{Z})$ , so its residual sum of squares  $\|(\mathbf{I} - \mathbf{P}_\mathcal{V})\mathbf{y}\|^2$  is at least  $\|(\mathbf{I} - \mathbf{M}_Z)\mathbf{y}\|^2 = \text{SSPE}$  (projection onto a smaller space leaves a longer residual). Hence  $R^2 = 1 - \text{SSE}/\text{SST} \leq 1 - \text{SSPE}/\text{SST}$ , with equality when  $\mathcal{V} = \mathcal{C}(\mathbf{Z})$ .

**Exercise B3.**

Show that rows  $i$  and  $j$  of  $\mathbf{M}$  are equal iff rows  $i$  and  $j$  of  $\mathbf{X}$  are equal. So replicates can be read off the hat matrix.

**14.6.4.3 C. Going deeper****Exercise C1.**

A straight line is to be tested for lack of fit at the four values  $x = 0, 1, 2, 3$ , each replicated  $m$  times, when the true mean is  $\beta_0 + \beta_1 x + \beta_2 x^2$  with  $\beta_2/\sigma = 0.5$ . Show that  $\gamma = m\beta_2^2 \|(\mathbf{I} - \mathbf{M}_4)\mathbf{q}\|^2/\sigma^2$ , where  $\mathbf{q} = (0, 1, 4, 9)^\top$  and  $\mathbf{M}_4$  is the projection for a line on the four points, and find the smallest  $m$  for which the test at level 0.05 has power at least 0.8.

**Exercise C2.**

Suppose the groups have different variances  $\sigma_i^2$ . Compute  $E(\text{SSPE})$  and  $E(\text{SSLF})$  under

a correct mean model, and show by an example with two large groups of very different sizes and variances that the  $F$  test can then reject far more or far less often than  $\alpha$ .

## 14.7 Near replicates and nonparametric lack-of-fit tests

Exact replicates are a luxury of designed experiments. With continuous regressors, rows of the model matrix are rarely identical: stack loss with all three operating variables has 21 rows and a single repeated pair, one degree of freedom for pure error. The pure-error test survives in a more general form: any larger model chosen without looking at the response gives an exact test, and so do regressors built from the fitted values. What is lost is the guarantee that the larger model contains every alternative: each test looks in some directions and is blind in others.

### 14.7.1 A general recipe

#### Proposition 14.7.1 — Testing against any larger model

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ , let  $\mathbf{X}$  have rank  $r$ , and let  $\mathbf{Z}$  be a fixed matrix, chosen without reference to  $\mathbf{Y}$ , with  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$  and  $\text{rank}(\mathbf{Z}) = s$ ,  $r < s < n$ . Let  $\mathbf{M}$  and  $\mathbf{M}_Z$  be the two projections and

$$F = \frac{[\text{SSE}(\mathbf{X}) - \text{SSE}(\mathbf{Z})]/(s - r)}{\text{SSE}(\mathbf{Z})/(n - s)}.$$

- (a) If  $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{X})$ , then  $F \sim F(s - r, n - s)$ .
- (b) If  $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$ , then  $F \sim F(s - r, n - s, \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2)$ .
- (c) In general  $F = \frac{U/(s - r)}{V/(n - s)}$  with independent  $U \sim \chi^2(s - r, \gamma_1)$  and  $V \sim \chi^2(n - s, \gamma_2)$ , where  $\gamma_1 = \|(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2$  and  $\gamma_2 = \|(\mathbf{I} - \mathbf{M}_Z)\boldsymbol{\theta}\|^2/\sigma^2$ . For every  $c > 0$ ,  $\Pr(F > c)$  is strictly increasing in  $\gamma_1$  and strictly decreasing in  $\gamma_2$ .

*Proof.* The decomposition  $\mathbf{I} - \mathbf{M} = (\mathbf{M}_Z - \mathbf{M}) + (\mathbf{I} - \mathbf{M}_Z)$  into orthogonal projections of ranks  $s - r$  and  $n - s$  comes from [Theorem 6.5.1](#), and [Theorem 4.4.4](#) gives (c) apart from the monotonicity. Parts (a) and (b) are the cases  $\gamma_1 = \gamma_2 = 0$  and  $\gamma_2 = 0$ . For the monotonicity, write  $\Pr(F > c) = \mathbb{E}[\Pr(U > c'V \mid V)]$  with  $c' = c(s - r)/(n - s)$ . For each value  $V > 0$  the inner probability is strictly increasing in  $\gamma_1$  by [Proposition 4.1.3](#). Similarly  $\Pr(F > c) = \mathbb{E}[\Pr(V < U/c' \mid U)]$ , and  $\Pr\{\chi^2(n - s, \gamma_2) < u\}$  is strictly decreasing in  $\gamma_2$  for  $u > 0$ . ■

The size is exact however  $\mathbf{Z}$  is chosen without using  $\mathbf{Y}$ ; with random regressors,  $\mathbf{Z}$  may be any function of  $\mathbf{X}$  ([Theorem 14.1.2](#)). The pure-error test is the case  $\mathbf{Z} = \text{cell means}$ . Part (c) describes what happens under the alternatives that matter, where the true mean lies in neither model. Departures captured by the larger model raise  $\gamma_1$  and help; departures outside it raise  $\gamma_2$ , inflate the denominator and hurt. A departure orthogonal to  $\mathcal{C}(\mathbf{Z})$  gives power *below* the

level (Exercise (B3)).

## 14.7.2 Near replicates, partitions and the rainbow test

Without exact replicates, the natural substitute is to group rows that are *nearly* identical. A cluster analysis of the standardized rows of  $\mathbf{X}$  forms  $g$  clusters of near replicates, with indicator matrix  $\mathbf{D}$ . Testing  $\mathbf{X}$  against  $\mathbf{Z} = [\mathbf{X}, \mathbf{D}]$  allows each cluster its own shift from the fitted plane. This is the near-replicate test studied by Christensen (1989); Shillington (1979) proposed a variant based on the cluster averages of the regressors, and Christensen (1991) characterized the kinds of lack of fit each detects. With exact replicates all reduce to the pure-error test.

A second family partitions the cases instead of the regressor space. If the rows are split into two groups, with model matrices  $\mathbf{X}_1$  and  $\mathbf{X}_2$ , the model that fits a separate regression to each group has  $\mathbf{Z} = \begin{pmatrix} \mathbf{X}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_2 \end{pmatrix}$ , whose column space contains  $\mathcal{C}(\mathbf{X})$ , and  $\text{SSE}(\mathbf{Z})$  is the sum of the two residual sums of squares (Exercise (B1)). A straight line fitted to a curved mean is beaten by two lines, one for each half of the range. Utts (1982) proposed an extreme partition, the **rainbow test**: fit the model to a central subset of  $m$  cases, those with the smallest leverages, and give each of the remaining  $n - m$  cases a free mean of its own. Then  $\text{SSE}(\mathbf{Z})$  is the residual sum of squares of the central fit, and the test asks whether the model fitted to the middle of the design extrapolates to its edges.

All of these are exact by Proposition 14.7.1, and all require a decision made from  $\mathbf{X}$  alone: how many clusters, where to split, how large a central subset. Choosing the clusters or the split after looking at the residuals destroys the exactness, because  $\mathbf{Z}$  then depends on  $\mathbf{Y}$ .

## 14.7.3 Richer mean functions

The most direct alternative is a larger parametric model: add squares and cross-products of the regressors, or a regression spline. For a single regressor, a cubic spline with interior knots  $\kappa_1, \dots, \kappa_m$  adds the columns  $x^2$ ,  $x^3$  and  $(x - \kappa_l)_+^3$ ,  $l = 1, \dots, m$ , and can follow any smooth curve if the knots are dense enough. These are reduced-model  $F$  tests (Theorem 11.1.1), exact by Proposition 14.7.1(a) provided the larger model is fixed before seeing the data. Their power is concentrated on departures that the added columns can represent. A quadratic term is excellent against smooth curvature and nearly useless against a departure confined to a small part of the range.

## 14.7.4 Regressors built from the fitted values

A classical alternative looks for departures along the fitted values: if the mean is a nonlinear function of  $\mathbf{x}^\top \boldsymbol{\beta}$ , then powers of  $\hat{y}$  should help predict  $y$ . But  $\hat{\mathbf{Y}} = \mathbf{M}\mathbf{Y}$  depends on the response, so Proposition 14.7.1 does not apply. Remarkably, the  $F$  test is exact anyway.

**Theorem 14.7.2** — Regressors that are functions of the fitted values

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$  with  $\text{rank}(\mathbf{X}) = r$ . Let  $\mathbf{W} = w(\mathbf{M}\mathbf{Y})$  be an  $n \times q$  matrix computed from the fitted values alone, with  $\text{rank}[\mathbf{X}, \mathbf{W}] = r + q < n$  with probability one. Then

$$F = \frac{[\text{SSE}(\mathbf{X}) - \text{SSE}([\mathbf{X}, \mathbf{W}])]/q}{\text{SSE}([\mathbf{X}, \mathbf{W}])/(n - r - q)} \sim F(q, n - r - q).$$

*Proof.* Let  $\tilde{\mathbf{W}} = (\mathbf{I} - \mathbf{M})\mathbf{W}$ , which has rank  $q$ , and let  $\mathbf{P}$  project onto  $\mathcal{C}(\tilde{\mathbf{W}})$ . As in Lemma 6.6.1,  $\mathcal{C}([\mathbf{X}, \mathbf{W}]) = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\tilde{\mathbf{W}})$  with orthogonal summands, so the projection onto  $\mathcal{C}([\mathbf{X}, \mathbf{W}])$  is  $\mathbf{M} + \mathbf{P}$ . With  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y}$ , and since  $\mathbf{P}\mathbf{M} = \mathbf{0}$ ,

$$\begin{aligned} \text{SSE}(\mathbf{X}) - \text{SSE}([\mathbf{X}, \mathbf{W}]) &= \|\mathbf{P}\mathbf{Y}\|^2 = \|\mathbf{P}\hat{\boldsymbol{\varepsilon}}\|^2, \\ \text{SSE}([\mathbf{X}, \mathbf{W}]) &= \|\hat{\boldsymbol{\varepsilon}}\|^2 - \|\mathbf{P}\hat{\boldsymbol{\varepsilon}}\|^2. \end{aligned}$$

By Theorem 4.4.4,  $\mathbf{M}\mathbf{Y}$  and  $\hat{\boldsymbol{\varepsilon}}$  are independent, and  $\hat{\boldsymbol{\varepsilon}} \sim N_n(\mathbf{0}, \sigma^2(\mathbf{I} - \mathbf{M}))$ . So, given  $\mathbf{M}\mathbf{Y}$ , the matrix  $\mathbf{P}$  is fixed and  $\hat{\boldsymbol{\varepsilon}}$  keeps its distribution, which is that of  $(\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}^*$  for  $\boldsymbol{\varepsilon}^* \sim N_n(\mathbf{0}, \sigma^2\mathbf{I})$ . Because  $\mathcal{C}(\mathbf{P}) \subseteq \mathcal{C}(\mathbf{X})^\perp$ ,  $\mathbf{P}(\mathbf{I} - \mathbf{M}) = \mathbf{P}$ , and the two sums of squares become  $\|\mathbf{P}\boldsymbol{\varepsilon}^*\|^2$  and  $\|(\mathbf{I} - \mathbf{M} - \mathbf{P})\boldsymbol{\varepsilon}^*\|^2$ . These are quadratic forms in orthogonal projections of ranks  $q$  and  $n - r - q$ , independent  $\sigma^2\chi^2(q)$  and  $\sigma^2\chi^2(n - r - q)$  variables by Theorem 4.4.4. The conditional law of  $F$  is therefore  $F(q, n - r - q)$  whatever  $\mathbf{M}\mathbf{Y}$  is, and by Lemma 14.1.1 it is also the unconditional law. ■

The theorem is due to Milliken and Graybill (1970). Its best-known case is Ramsey's (1969) **RESET**, which adds the squares and cubes of the fitted values, entry by entry. Tukey's (1949) one degree of freedom for nonadditivity, which tests a two-way additive layout against a multiplicative interaction, can be written in the same form with the squared fitted values. With a single regressor the construction brings nothing new: when  $\hat{\beta}_1 \neq 0$ , the columns  $\hat{\mathbf{y}}^2$  and  $\hat{\mathbf{y}}^3$  span, together with  $\mathbf{1}$  and  $\mathbf{x}$ , the same space as  $x^2$  and  $x^3$ , so RESET is the cubic polynomial test (Exercise (B4)). Its value is in many dimensions, where it spends two degrees of freedom on the one direction along which the fit varies.

### 14.7.5 Smoothing-based tests

Instead of committing to a parametric alternative, one can ask whether the residuals contain any smooth structure. Let  $\mathbf{S}$  be a fixed smoother matrix, for instance kernel weights that average each residual with those of its neighbours in the regressor space, and consider

$$T = \frac{\|\mathbf{S}\hat{\boldsymbol{\varepsilon}}\|^2}{\|\hat{\boldsymbol{\varepsilon}}\|^2}. \quad (14.7.1)$$

If the model is right, the residuals are noise and smoothing them leaves little; if the mean has structure the model misses, the smoothed residuals keep it and  $T$  is large.

#### Proposition 14.7.3 — The smoother statistic is pivotal

If  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ , the distribution of  $T$  in Eq. 14.7.1 depends only on  $\mathbf{X}$  and  $\mathbf{S}$ , not on  $\boldsymbol{\beta}$  or  $\sigma^2$ . It equals the distribution of  $\|\mathbf{S}(\mathbf{I} - \mathbf{M})\mathbf{u}\|^2 / \|(\mathbf{I} - \mathbf{M})\mathbf{u}\|^2$  with  $\mathbf{u} \sim N_n(\mathbf{0}, \mathbf{I})$ .

*Proof.*  $\hat{\varepsilon} = (\mathbf{I} - \mathbf{M})\mathbf{Y} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$ , and  $\boldsymbol{\varepsilon} = \sigma\mathbf{u}$  with  $\mathbf{u} \sim N_n(\mathbf{0}, \mathbf{I})$ . The factor  $\sigma^2$  cancels in the ratio. ■

So the null distribution of  $T$  can be simulated as precisely as desired, from the design alone, and the test has exact size apart from Monte Carlo error. Statistics of this kind were proposed by Azzalini and Bowman (1993), who compare the residual sums of squares of the parametric fit and of a nonparametric smoother, and they are developed at length by Hart (1997). The power depends on the amount of smoothing, and a bandwidth chosen after inspecting the data breaks the exact size. With several regressors, neighbourhoods become sparse, so in practice one smooths along a single direction, such as the fitted values. And exactness depends, as for every test in this section, on normal homoscedastic errors.

### 14.7.6 Stack loss with all three regressors

#### Example. (Is the three-variable model adequate?)

Return to the stack loss data with air flow, water temperature and acid concentration,  $n = 21$  and  $r = 4$ . Ward clustering of the standardized rows into 8 clusters and testing  $\mathbf{X}$  against  $[\mathbf{X}, \mathbf{D}]$  gives  $F = 1.43$  on 7 and 10 degrees of freedom ( $p$ -value 0.292). Adding the squares of the three regressors gives  $F = 1.92$  on 3 and 14 ( $p$ -value 0.172). The rainbow test with the 11 cases of smallest leverage as the central subset gives  $F = 1.04$  on 10 and 7 ( $p$ -value 0.497), and RESET gives  $F = 1.17$  on 2 and 15 ( $p$ -value 0.336). None of the tests finds lack of fit. The strong lack of fit of the line in air flow alone (Example (Stack loss and air flow)) was the omitted water temperature, which varies within the groups of equal air flow. With 21 cases none of these tests is powerful, and a few unusual days deserve the scrutiny of Chapter 20.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
from scipy.cluster.hierarchy import fcluster, linkage
from statsmodels.stats.diagnostic import linear_reset

def proj(A):
    """Orthogonal projection onto  $C(A)$ , via an orthonormal basis of the column space
    ."""
    U, s, _ = np.linalg.svd(A, full_matrices=False)
    U = U[:, s > s[0] * 1e-10]
    return U @ U.T

def f_test(y, X, Zbig):
    """F test of  $C(X)$  against a larger  $C(Zbig)$   $C(X)$ ; returns  $F$ ,  $df1$ ,  $df2$ ,  $p$ ."""
    n = len(y)
    M, MZ = proj(X), proj(Zbig)
    r, s = round(np.trace(M)), round(np.trace(MZ))
    sse, sse_z = y @ (y - M @ y), y @ (y - MZ @ y)
    F = ((sse - sse_z) / (s - r)) / (sse_z / (n - s))
    return F, s - r, n - s, stats.f.sf(F, s - r, n - s)

data = sm.datasets.stackloss.load_pandas().data
```

```

y = data["STACKLOSS"].to_numpy()
R = data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]].to_numpy()
n = len(y)
X = np.column_stack([np.ones(n), R])
fit = sm.OLS(y, X).fit()

# exact replicates: rows of X that occur more than once
_, rows = np.unique(R, axis=0, return_inverse=True)
print("distinct rows:", rows.max() + 1, "of", n)

# near replicates: cluster the standardized regressor rows (Ward linkage), c
clusters
Rs = (R - R.mean(axis=0)) / R.std(axis=0)
clusters = fcluster(linkage(Rs, method="ward"), t=8, criterion="maxclust")
D = np.eye(clusters.max())[clusters - 1] # cluster indicators
tests = {"near replicates": f_test(y, X, np.column_stack([X, D])),
        "added squares": f_test(y, X, np.column_stack([X, R ** 2]))}

# rainbow: refit on the half of the cases with the smallest leverages
h = np.diag(proj(X))
central = np.argsort(h)[: n // 2 + 1]
E = np.eye(n)[:, np.setdiff1d(np.arange(n), central)] # one free mean per outer
case
tests["rainbow"] = f_test(y, X, np.column_stack([X, E]))

# RESET: add squares and cubes of the fitted values
reset = linear_reset(fit, power=3, test_type="fitted", use_f=True)
tests["RESET"] = (reset.fvalue, reset.df_num, reset.df_denom, reset.pvalue)
for name, (F, d1, d2, p) in tests.items():
    print(f"{name:16s} F = {F:6.2f} on ({d1:.0f}, {d2:.0f}) df, p = {p:.3f}")

```

### 14.7.7 Which test?

No lack-of-fit test is best against every alternative: by [Proposition 14.7.1\(c\)](#), a test gains power from departures its larger model can represent and loses power from the others.

#### Example. (Two departures, five tests)

Take  $n = 50$  equally spaced values of  $x$  in  $(0, 1)$ , a straight-line working model, standard normal errors, and five tests at level 0.05: adding  $x^2$ ; RESET with squared and cubed fitted values; a cubic spline with knots at 0.25, 0.5 and 0.75; near replicates formed by 10 clusters of 5 neighbouring values; and the smoother statistic [Eq. 14.7.1](#) with Gaussian kernel weights of bandwidth 0.06, its critical value simulated from 50000 null residual vectors. Over 4000 samples from the straight line, the rejection rates are 0.050, 0.054, 0.054, 0.052 and 0.054: all five tests are exact.

Against smooth curvature, a mean of  $1 + 2x + 6(x - 0.5)^2$ , the powers are 0.874 for the quadratic term, 0.786 for RESET (which here is the cubic polynomial test), 0.606 for the spline, 0.422 for near replicates and 0.723 for the smoother. Against a narrow bump of height 2.5 near  $x = 0.8$ , the order is reversed: the smoother has power 0.622, the spline 0.501, near replicates 0.459, RESET 0.227, and the quadratic term 0.029, *below* the level: the bump is nearly orthogonal to  $x^2$  after the line is removed, so it inflates the denominator of that test without feeding its numerator. [Figure 14.7.1](#) shows the power



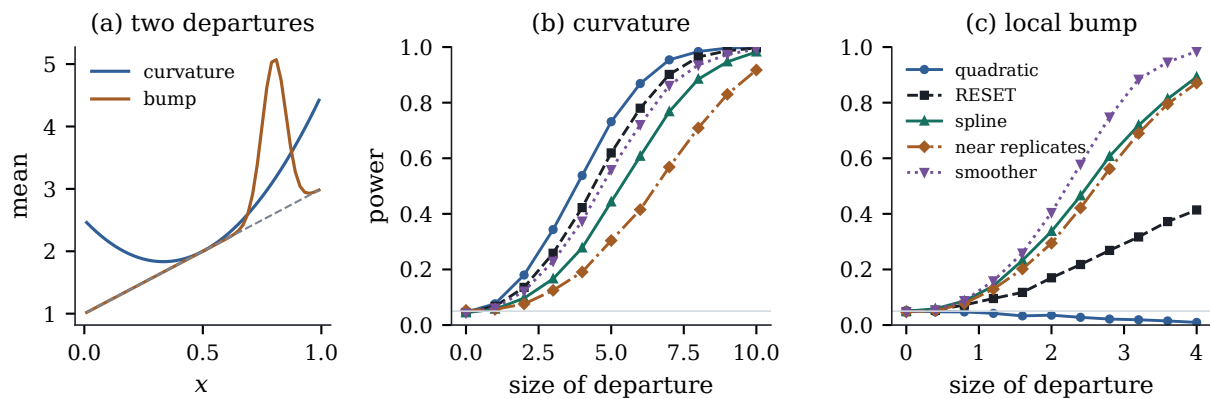


Figure 14.7.1: Power of five exact lack-of-fit tests of a straight line ( $n = 50$ , level 0.05). (a) The two departures. (b) Smooth curvature. (c) A local bump, against which the quadratic test has power below its level.

curves.

```

rng = np.random.default_rng(1471)
n2, reps = 50, 4000
x = (np.arange(n2) + 0.5) / n2                                # fixed design on (0, 1)
X2 = np.column_stack([np.ones(n2), x])
M2 = proj(X2)
knots = (0.25, 0.5, 0.75)
spline = np.column_stack([x ** 2, x ** 3] + [np.clip(x - k, 0, None) ** 3 for k in
    knots])
cells = np.eye(10)[np.arange(n2) // 5]                        # 10 clusters of 5
    neighbouring x
bigger = {"quadratic": np.column_stack([X2, x ** 2]),
    "spline": np.column_stack([X2, spline]),
    "near replicates": np.column_stack([X2, cells])}
proj_big = {k: proj(v) for k, v in bigger.items()}

# kernel smoother of the residuals;  $T = |S e|^2 / |e|^2$ , null law free of beta and
    sigma
bw = 0.06
K = np.exp(-0.5 * ((x[:, None] - x[None, :]) / bw) ** 2)
S = K / K.sum(axis=1, keepdims=True)
null_e = (np.eye(n2) - M2) @ rng.normal(size=(n2, 50_000))
T_null = np.sum((S @ null_e) ** 2, axis=0) / np.sum(null_e ** 2, axis=0)
T_crit = np.quantile(T_null, 0.95)

def powers(mean):
    """Rejection rates at level 0.05 of each test when  $E(Y) = \text{mean}$ ."""
    Y = mean[:, None] + rng.normal(size=(n2, reps))
    e = Y - M2 @ Y
    sse = np.sum(e ** 2, axis=0)
    out = {}
    for name, P in proj_big.items():
        s = round(np.trace(P))
        sse_z = np.sum((Y - P @ Y) ** 2, axis=0)
        F = ((sse - sse_z) / (s - 2)) / (sse_z / (n2 - s))
        out[name] = np.mean(F > stats.f.ppf(0.95, s - 2, n2 - s))
    Fr = np.empty(reps)
    for j in range(reps):                                # RESET: regressors depend on
        y

```



```

    yh = M2 @ Y[:, j]
    W = np.column_stack([yh ** 2, yh ** 3])
    Wt = W - M2 @ W
    g, *_ = np.linalg.lstsq(Wt, e[:, j], rcond=None)
    ss_add = e[:, j] @ (Wt @ g)
    Fr[j] = (ss_add / 2) / ((sse[j] - ss_add) / (n2 - 4))
    out["RESET"] = np.mean(Fr > stats.f.ppf(0.95, 2, n2 - 4))
    T = np.sum((S @ e) ** 2, axis=0) / sse
    out["smoother"] = np.mean(T > T_crit)
    return out

curve = (x - 0.5) ** 2                                # smooth curvature
bump = np.exp(-0.5 * ((x - 0.8) / 0.05) ** 2)         # a local bump
sizes = powers(1 + 2 * x)
print("sizes:", {k: round(float(v), 3) for k, v in sizes.items()})
reported = {}
for label, shape, a in [("curvature", curve, 6.0), ("bump", bump, 2.5)]:
    reported[label] = powers(1 + 2 * x + a * shape)
print(f"{label:9s}:", {k: round(float(v), 3) for k, v in reported[label].items()})
})

```

The lesson is to decide which inadequacy matters before testing: a low-degree polynomial or RESET for smooth curvature, a flexible alternative or a smoother for local departures. Only the pure-error test of [Theorem 14.6.2](#) has a larger model containing every mean function of the regressors. Residual plots, which look in all directions informally, remain indispensable ([Chapter 20](#)).

## 14.7.8 Exercises

### 14.7.8.1 A. Check your understanding

#### Exercise A1.

A regression with an intercept and two regressors has  $n = 30$  cases, whose rows are grouped into 8 clusters of near replicates. Assuming no combination of cluster indicators lies in  $\mathcal{C}(\mathbf{X})$  except the constant, give the degrees of freedom of the near-replicate test of  $\mathbf{X}$  against  $[\mathbf{X}, \mathbf{D}]$ .

### 14.7.8.2 B. Practice

#### Exercise B1.

Let  $\mathbf{X} = \begin{pmatrix} \mathbf{X}_1 \\ \mathbf{X}_2 \end{pmatrix}$  and  $\mathbf{Z} = \begin{pmatrix} \mathbf{X}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{X}_2 \end{pmatrix}$ . Show that  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ , that the projection onto  $\mathcal{C}(\mathbf{Z})$  is  $\begin{pmatrix} \mathbf{M}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{M}_2 \end{pmatrix}$ , and that  $\text{SSE}(\mathbf{Z})$  is the sum of the residual sums of squares of the two separate regressions.

*Solution.*  $\mathbf{X} = \mathbf{Z} \begin{pmatrix} \mathbf{I} \\ \mathbf{0} \end{pmatrix}$ , so  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ . The block-diagonal matrix of the two projections is symmetric and idempotent, and its column space is  $\mathcal{C}(\mathbf{X}_1) \times \mathcal{C}(\mathbf{X}_2) = \mathcal{C}(\mathbf{Z})$ , so it is the projection onto  $\mathcal{C}(\mathbf{Z})$  ([Theorem 6.3.3](#)). Applying  $\mathbf{I}$  minus it to  $\mathbf{y} = (\mathbf{y}_1^\top, \mathbf{y}_2^\top)^\top$  gives the two residual vectors  $(\mathbf{I} - \mathbf{M}_1)\mathbf{y}_1$  and  $(\mathbf{I} - \mathbf{M}_2)\mathbf{y}_2$ , whose squared lengths add.

**Exercise B2.**

In the rainbow test, order the cases so that the central subset of size  $m$  comes first, and let its model matrix  $\mathbf{X}_c$  have rank  $r$ . Show that  $\mathbf{Z} = \begin{pmatrix} \mathbf{X}_c & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_{n-m} \end{pmatrix}$  has  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ , that  $\text{SSE}(\mathbf{Z})$  is the residual sum of squares of the central fit, and that the test has  $n - m$  and  $m - r$  degrees of freedom.

**Exercise B3.**

In [Proposition 14.7.1](#), suppose  $\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}$  with  $\boldsymbol{\delta} \neq \mathbf{0}$  orthogonal to  $\mathcal{C}(\mathbf{Z})$ . Show that the power of the test is strictly less than its size.

*Solution.* Here  $(\mathbf{M}_Z - \mathbf{M})\boldsymbol{\theta} = \mathbf{M}_Z\boldsymbol{\delta} - \mathbf{M}\boldsymbol{\delta} = \mathbf{0}$ , since  $\boldsymbol{\delta} \perp \mathcal{C}(\mathbf{Z}) \supseteq \mathcal{C}(\mathbf{X})$ , so  $\gamma_1 = 0$ , while  $\gamma_2 = \|\boldsymbol{\delta}\|^2/\sigma^2 > 0$ . By part (c),  $\Pr(F > c)$  is strictly smaller than its value at  $\gamma_2 = 0$ , which is the size by part (a).

**Exercise B4.**

With a single regressor and an intercept, show that if  $\hat{\beta}_1 \neq 0$ , then  $\mathcal{C}([\mathbf{1}, \mathbf{x}, \hat{\mathbf{y}}^2, \hat{\mathbf{y}}^3]) = \mathcal{C}([\mathbf{1}, \mathbf{x}, \mathbf{x}^2, \mathbf{x}^3])$ , powers taken entry by entry. Conclude that RESET is then the  $F$  test of the cubic polynomial.

**14.7.8.3 C. Going deeper****Exercise C1.**

In a two-way layout with one observation per cell and the additive model  $E(y_{ij}) = \mu + \alpha_i + \beta_j$ , Tukey's statistic tests the added regressor  $(\hat{\alpha}_i \hat{\beta}_j)$ . Show that, after the additive model is projected out, this regressor spans the same space as the entrywise square of the fitted values, and deduce from [Theorem 14.7.2](#) that Tukey's test has an exact  $F(1, (a-1)(b-1)-1)$  null distribution.

## 14.8 Summary and notes

### 14.8.1 Summary

*Idea.*

1. If the errors are normal, homoscedastic and independent of the regressors, every  $t$  and  $F$  statistic has the same law given every design, so tests and intervals stay exact with random regressors of any distribution. Power depends on the design drawn and is averaged over it ([Lemma 14.1.1](#), [Theorem 14.1.2](#)).
2. The estimates themselves change law. With normal regressors the slope of a simple regression is a scaled  $t(n-1)$  variable with variance  $\sigma^2/(\sigma_X^2(n-3))$ , and with  $k$  regressors the slopes have covariance  $\sigma^2 \boldsymbol{\Sigma}_{XX}^{-1}/(n-k-2)$  ([Proposition 14.1.4](#),

[Lemma 14.2.4](#), [Corollary 14.2.5](#)).

3. The conditional mean is the only predictor whose errors are uncorrelated with every function of the regressors, and the correlation ratio bounds the correlation any predictor can reach ([Proposition 14.2.1](#)).
4. A fitted linear predictor pays for estimation. With normal regressors its expected error on a new case is  $\sigma^2(1 + 1/n)(n - 2)/(n - k - 2)$ , while its in-sample error has mean  $\sigma^2(n - k - 1)/n$ ; with a fixed design the gap is  $2r\sigma^2/n$  ([Proposition 14.2.2](#), [Theorem 14.2.6](#), [Proposition 14.2.7](#)).
5. When the new error is correlated with the data, the best linear unbiased predictor adds to the estimated mean the best linear prediction of that error from the generalized least squares residuals ([Theorem 14.2.8](#)).
6.  $R^2$  is the maximum likelihood estimate of  $\rho_{Y \cdot X}^2$ . Given the design it is a transformed noncentral  $F$  variable; with normal regressors its noncentrality is  $\lambda\chi^2(n-1)$ ,  $\lambda = \rho^2/(1 - \rho^2)$ , which gives its exact law, its bias, the power of the overall  $F$  test and an exact interval for  $\rho^2$  ([Proposition 14.3.1](#), [Theorem 14.3.2](#)).
7. The sample correlation has an exact law expressed through noncentral  $t$  probabilities, and it is stochastically increasing in  $\rho$ . Fisher's  $z = \tanh^{-1} r$  is asymptotically  $N(\zeta, 1/n)$  whatever  $\rho$ , and in practice  $N(\zeta, 1/(n - 3))$  ([Proposition 14.4.1](#), [Theorem 14.4.3](#)).
8. A sample partial correlation given  $q$  variables behaves exactly like an ordinary correlation from  $n - q$  cases, and its test is the  $t$  test of the corresponding regression coefficient ([Theorem 14.5.1](#), [Proposition 14.5.2](#), [Theorem 14.5.3](#)).
9. With replicated rows, the residual sum of squares splits into pure error and lack of fit, and their ratio is an exact  $F$  test of the model against every mean function of the regressors ([Lemma 14.6.1](#), [Theorem 14.6.2](#)).
10. Without replicates, any larger model chosen from  $\mathbf{X}$  alone, any regressors built from the fitted values, and any smoother of the residuals give exact tests. Each has power only against departures its alternative can represent, and some departures reduce its power below the level ([Proposition 14.7.1](#), [Theorem 14.7.2](#), [Proposition 14.7.3](#)).

## 14.8.2 Notes and sources

**Source texts.** The chapter covers Christensen (2020, sections 6.3 to 6.7) on prediction theory, multiple and partial correlation, best linear unbiased prediction and lack of fit; Rencher and Schaalje (2008, chapter 10) on regression with random regressors; and Agresti (2015, sections 3.3 and 3.4), whose worked example includes partial correlations and model checking. Christensen's near-replicate tests and his view of prediction as the goal of regression shaped [Section 14.2](#) and [Section 14.7](#). Rencher and Schaalje give the normal-theory formulas for  $R^2$ ,  $z$  and partial correlations. Agresti's intervals of section 3.3 belong to [Chapter 12](#).

**Random regressors and prediction.** Reporting precision conditionally on an ancillary design is Fisher's conditionality principle; Cox (1958) gave the standard discussion. The Wishart distribution is due to Wishart (1928), and Anderson (2003) treats it and the mean of its inverse through densities; the proof of [Lemma 14.2.4](#) here uses only conditioning and symmetry. Breiman and Freedman (1983) use the prediction error of regression with normal regressors to study how many variables to enter, a question [Chapter 29](#) takes up. Best linear unbiased prediction is due to Goldberger (1962); [Chapter 31](#) and [Chapter 32](#) develop it for general covariance structures

and mixed models ([Theorem 32.3.1](#)).

**Correlation coefficients.** Fisher found the exact distribution of  $r$  (1915), introduced  $z$  with its approximate mean and variance (1921), showed that partial correlations follow the same law with the sample size reduced by the number of conditioning variables (1924), and derived the distribution of the multiple correlation (1928). Hotelling (1953) refined the moments of  $r$  and  $z$ . [Lemma 14.4.2](#) is a case of Isserlis's (1918) formula, and the mean of  $R^2$  in [Example \(How biased is R-squared?\)](#) follows from Fisher's (1928) distribution; see Olkin and Pratt (1958), who also give the unbiased estimator of  $\rho^2$ . For the limit theorems of [Section 14.4](#), see van der Vaart (1998).

**Lack of fit.** The pure-error test is Fisher's (1922). Near-replicate tests were proposed by Shillington (1979) and analysed by Christensen (1989, 1991); the rainbow test is due to Utts (1982). Ramsey (1969) introduced RESET, and Milliken and Graybill (1970) proved [Theorem 14.7.2](#), which also covers Tukey's (1949) one degree of freedom for nonadditivity. Smoothing-based tests were proposed by Azzalini and Bowman (1993) and are the subject of Hart (1997).

**Data.** The stack loss data are from Brownlee (1965). The RAND Health Insurance Experiment is described by Manning et al. (1987); the extract and the 2009 state data are public-domain data distributed with statsmodels.

**What is new here.** The conditional derivations of the laws of  $R^2$ ,  $r$  and partial correlations ([Theorem 14.3.2](#), [Proposition 14.4.1](#), [Theorem 14.5.1](#)), the proof of [Lemma 14.2.4](#), the organization of lack-of-fit tests around [Proposition 14.7.1](#), and the examples and simulations are this book's.

### 14.8.3 References

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# Chapter 15

## Analysis of Variance as Subspace Comparison

An experimenter assigns treatments to units at random and measures a response. The simplest such experiment has one factor, and its analysis is the prototype for Part IV. The mean vector lies in the space of vectors constant within groups, that space splits orthogonally into the constants and a treatment space, and every question about the treatments concerns the component of the mean in the treatment space. This chapter turns that geometry into a working analysis. It states the one-way model and what its parameters mean, shows that single contrasts, families of contrasts and the overall  $F$  test are projections of the data onto pieces of one subspace, and derives the analysis of variance table and its expected mean squares. It splits the treatment sum of squares into single degrees of freedom along directions chosen for their meaning, including polynomial trends in a quantitative factor. Finally it asks what balance buys: plain orthogonality, the largest guaranteed power, and a test that tolerates unequal variances. When the design is unbalanced and the variances differ, the classical test can go badly wrong, and the chapter describes the heteroscedasticity-robust tests of Welch and of Brown and Forsythe.

**What you need.** The one-way projection and its generalized inverses from [Chapter 6](#) ([Example \(A rank-deficient layout\)](#), [Example \(The one-way layout\)](#), [Theorem 6.5.1](#)); estimability, side conditions and contrasts from [Chapter 8](#) ([Theorem 8.7.1](#), [Proposition 8.7.2](#), [Theorem 8.4.2](#)); orthogonal decompositions and expected mean squares from [Chapter 9](#) ([Theorem 9.6.1](#), [Theorem 9.4.4](#)); the  $F$  test and its power from [Chapter 11](#) ([Theorem 11.1.1](#), [Theorem 11.5.1](#)); the  $t$  interval from [Chapter 12](#) ([Theorem 12.1.1](#)); and the simultaneous methods of [Chapter 13](#) ([Theorem 13.2.2](#), [Theorem 13.3.2](#), [Theorem 13.4.1](#)). The pure-error lack-of-fit test of [Chapter 14](#) ([Theorem 14.6.2](#)) reappears as trend analysis.

### 15.0.1 Roadmap

- [The one-way layout as a linear model](#)
- [Estimable contrasts](#)
- [The analysis of variance table from projections](#)
- [Orthogonal contrasts](#)
- [Balanced and unbalanced one-way layouts](#)
- [Summary and notes](#)

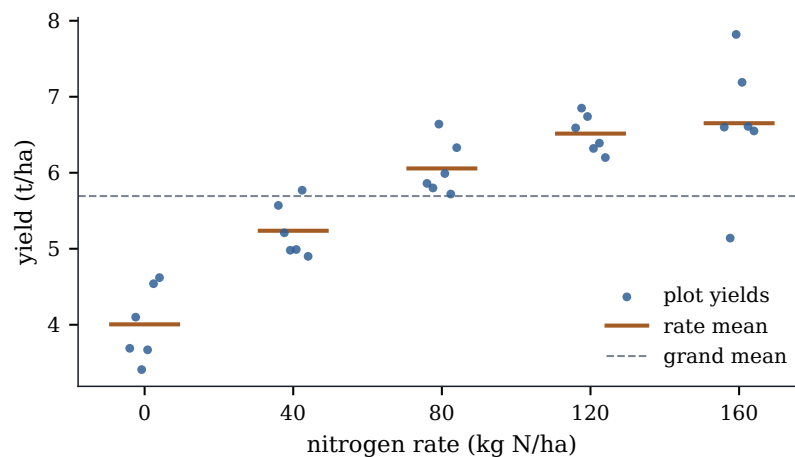


Figure 15.1.1: The nitrogen trial: yields of thirty plots at five nitrogen rates, six plots per rate (slightly spread horizontally), with the rate means and the grand mean.

## 15.1 The one-way layout as a linear model

Earlier chapters used the one-way layout as an example of rank deficiency ([Chapter 6](#)), estimability ([Chapter 8](#)), orthogonal decomposition ([Chapter 9](#)) and simultaneous inference ([Chapter 13](#)). Here it is the subject. This section fixes the model, recalls its projection, and settles what its parameters mean and where they come from.

### 15.1.1 A designed experiment

An agronomist wants to know how the yield of a wheat variety responds to nitrogen fertilizer. Thirty field plots of equal size are available. Five application rates are chosen, 0, 40, 80, 120 and 160 kilograms of nitrogen per hectare, and each rate is assigned to six plots chosen at random from the thirty. At harvest the grain yield of each plot is recorded in tonnes per hectare. This is a **completely randomized design** with one **factor** (nitrogen) at five **levels**, also called **treatments**, with six **replicates** of each. The data used throughout this chapter were simulated from a smooth response curve with plot-to-plot standard deviation 0.5 t/ha, which the analysis does not know about.

| Rate (kg N/ha) | Yields (t/ha)                      | Mean  | Standard deviation |
|----------------|------------------------------------|-------|--------------------|
| 0              | 3.69, 4.10, 3.41, 3.67, 4.54, 4.62 | 4.005 | 0.498              |
| 40             | 5.57, 5.21, 4.98, 4.99, 5.77, 4.90 | 5.237 | 0.357              |
| 80             | 5.86, 5.80, 6.64, 5.99, 5.72, 6.33 | 6.057 | 0.357              |
| 120            | 6.59, 6.85, 6.74, 6.32, 6.39, 6.20 | 6.515 | 0.254              |
| 160            | 6.60, 5.14, 7.82, 7.19, 6.61, 6.55 | 6.652 | 0.889              |

[Figure 15.1.1](#) shows the thirty yields. Mean yield rises with the rate, steeply at first and then more slowly, and the plots within a rate scatter by roughly half a tonne. The later sections compare particular treatments ([Section 15.2](#)), ask whether the rates differ at all ([Section 15.3](#)), and describe the shape of the response ([Section 15.4](#)).

## 15.1.2 The model in two forms

Index the groups by  $k = 1, \dots, g$  and the observations within group  $k$  by  $j = 1, \dots, n_k$ . The observations may be listed in any order in the vector  $\mathbf{Y}$ ; what matters is which group each belongs to.

### Definition 15.1.1 — The one-way layout

A **one-way layout** has  $g \geq 2$  groups with  $n_k \geq 1$  observations in group  $k$  and  $n = \sum_k n_k > g$  observations in all. Its **cell-means form** is

$$Y_{kj} = \mu_k + \varepsilon_{kj}, \quad \text{that is,} \quad \mathbf{Y} = \mathbf{Z}\boldsymbol{\mu} + \boldsymbol{\varepsilon},$$

where  $\mathbf{Z}$  is the  $n \times g$  matrix whose column  $k$  is the indicator  $\mathbf{z}_k$  of group  $k$  and  $\boldsymbol{\mu} = (\mu_1, \dots, \mu_g)^\top$ . Its **effects form** is

$$Y_{kj} = \mu + \alpha_k + \varepsilon_{kj}, \quad \text{that is,} \quad \mathbf{Y} = [\mathbf{1}, \mathbf{Z}] \begin{pmatrix} \mu \\ \boldsymbol{\alpha} \end{pmatrix} + \boldsymbol{\varepsilon}.$$

In both, the errors have  $E(\boldsymbol{\varepsilon}) = \mathbf{0}$  and  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ . The **normal one-way model** adds  $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ . The layout is **balanced** if  $n_1 = \dots = n_g = m$ , say. The treatment parameters  $\mu_k$  (or  $\alpha_k$ ) are **fixed effects**: unknown constants attached to the  $g$  treatments actually studied.

The two forms describe the same mean vectors: since  $\mathbf{z}_1 + \dots + \mathbf{z}_g = \mathbf{1}$ , the column space of  $[\mathbf{1}, \mathbf{Z}]$  is  $\mathcal{C}(\mathbf{Z})$ , the vectors constant within each group (Example (A rank-deficient layout)). The effects matrix has  $g + 1$  columns but rank  $g$ , so its parameters are not identified. Fitted values, residuals, sums of squares and tests depend only on  $\mathcal{C}(\mathbf{Z})$ , so the choice of form decides which parameters one talks about, never the results of the analysis.

A word on notation. Group means are written  $\bar{y}_k$  (the literature also writes  $\bar{y}_{k\cdot}$ ), the grand mean is  $\bar{y} = n^{-1} \sum_{k,j} y_{kj}$ , and  $\bar{\mu} = n^{-1} \sum_k n_k \mu_k$  is the *weighted* average of the group means, which is  $E(\bar{Y})$ . The unweighted average  $g^{-1} \sum_k \mu_k$  is a different quantity unless the layout is balanced.

## 15.1.3 The model space and its projection

The orthogonal projection onto  $\mathcal{C}(\mathbf{Z})$  was computed in Example (The one-way layout):

$$\mathbf{M} = \sum_{k=1}^g \frac{\mathbf{z}_k \mathbf{z}_k^\top}{n_k}, \quad (\mathbf{M}\mathbf{y})_{kj} = \bar{y}_k, \quad (15.1.1)$$

so the fitted value of every observation is its group mean and the residual is its deviation from that mean. Whatever the parameterization and whatever generalized inverse is used to solve the normal equations, this is the fit (Theorem 6.3.7). In the cell-means form the least squares estimate is unique,  $\hat{\boldsymbol{\mu}} = (\bar{y}_1, \dots, \bar{y}_g)^\top$ , because  $\mathbf{Z}^\top \mathbf{Z} = \text{diag}(n_1, \dots, n_g)$  is invertible.

The residual sum of squares pools the variation inside the groups:



$$\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \sum_{k=1}^g \sum_{j=1}^{n_k} (y_{kj} - \bar{y}_k)^2 = \sum_{k=1}^g (n_k - 1)s_k^2, \quad (15.1.2)$$

where  $s_k^2$  is the sample variance of group  $k$  (taken as zero when  $n_k = 1$ ). The error space  $\mathcal{C}(\mathbf{Z})^\perp$  has dimension  $n - g$ , so the unbiased estimator of  $\sigma^2$  ([Theorem 5.6.1](#)) is

$$s^2 = \frac{\text{SSE}}{n - g} = \sum_{k=1}^g \frac{n_k - 1}{n - g} s_k^2,$$

a weighted average of the group variances with weights proportional to their degrees of freedom. In the nitrogen trial  $s = 0.5211$  t/ha on 25 degrees of freedom, and since the design is balanced  $s^2$  is the plain average of the five group variances.

```
import numpy as np

rates = np.array([0, 40, 80, 120, 160])          # kg N per hectare
g, m = len(rates), 6                             # 5 rates, 6 plots per rate
true_means = 6.8 - 2.7 * np.exp(-0.018 * rates)  # unknown to the analyst
rng = np.random.default_rng(1515)
group = np.repeat(np.arange(g), m)               # plot i received rate group[i]
y = np.round(true_means[group] + rng.normal(0, 0.5, g * m), 2) # yield, t/ha
n = len(y)

Z = (group[:, None] == np.arange(g)).astype(float) # cell-means model matrix
n_k = Z.sum(axis=0)
means = Z.T @ y / n_k                             # least squares: the group means
fitted = Z @ means                                 # M y
resid = y - fitted                                 # (I - M) y
s2 = resid @ resid / (n - g)                       # pooled within-group variance
print("group means:", means.round(3))
print(f"s = {np.sqrt(s2):.4f} on {n - g} degrees of freedom")
```

### 15.1.4 What the parameters mean

The group means  $\mu_k$  are estimable, and so is every linear function of them. The effects form adds one parameter that the data cannot determine. By [Theorem 8.7.1](#), a function  $\lambda_0\mu + \sum_k \lambda_k\alpha_k$  is estimable iff  $\lambda_0 = \sum_k \lambda_k$ ; in particular no single  $\alpha_k$  and not  $\mu$  itself is estimable, while every difference  $\alpha_k - \alpha_l$  is.

Software nevertheless reports values of  $\mu$  and  $\alpha_k$ , by adding a side condition, one nonestimable linear restriction that picks a single solution of the normal equations ([Theorem 8.4.2](#)). Under the restriction, each parameter becomes a particular estimable function of the group means ([Proposition 8.4.3](#)), and it is worth knowing which:

| Side condition            | $\mu$ becomes                                                | $\alpha_k$ becomes            |
|---------------------------|--------------------------------------------------------------|-------------------------------|
| $\sum_k \alpha_k = 0$     | $g^{-1} \sum_l \mu_l$ , the unweighted average               | $\mu_k - g^{-1} \sum_l \mu_l$ |
| $\sum_k n_k \alpha_k = 0$ | $\bar{\mu} = n^{-1} \sum_l n_l \mu_l$ , the weighted average | $\mu_k - \bar{\mu}$           |
| $\alpha_1 = 0$            | $\mu_1$ , the reference group                                | $\mu_k - \mu_1$               |

Each row solves the  $g$  equations  $\mu + \alpha_k = \mu_k$  together with the side condition, and the estimates put  $\bar{y}_k$  for  $\mu_k$  ([Exercise \(B1\)](#)). The second gives the classical  $\hat{\mu} = \bar{y}$ ,  $\hat{\alpha}_k = \bar{y}_k - \bar{y}$ . The first two agree in a balanced layout; in an unbalanced one they define different “overall means”, and a reported  $\hat{\alpha}_k$  means different things under each ([Example \(Five side conditions, one fit\)](#)). Contrasts are unaffected by the choice.

*Warning.* A side condition is a convention for printing, not a statement about the treatments. “The effect of treatment 3 is 0.4” means nothing until the side condition is named. Report group means and contrasts.

### 15.1.5 Where the model comes from

In a randomized experiment much of the one-way model can be derived rather than assumed. Suppose each plot  $i$  has a yield  $u_i$  it would give without nitrogen, reflecting its soil and position, and suppose rate  $k$  adds a fixed amount  $\tau_k$ , the same on every plot: this is **unit–treatment additivity**. Random assignment then makes the set of plots receiving rate  $k$  a simple random sample of the thirty. The observed yields in group  $k$  are  $\tau_k$  plus a sample of the  $u_i$ , so each has expectation  $\tau_k + \bar{u}$ , with  $\bar{u}$  the average of all thirty plot values. The expectations therefore have exactly the cell-means structure, with  $\mu_k = \tau_k + \bar{u}$ , whatever the pattern of fertility in the field. Had the six plots given the highest rate been chosen for convenience, say all at one end of the field, a fertility gradient would be confounded with the rate, and no analysis could separate the two.

The derived errors are not exactly those of [Definition 15.1.1](#): sampling without replacement makes them slightly negatively correlated ([Exercise \(B3\)](#)), and nothing makes them normal. But under the hypothesis that no treatment has any effect, every rearrangement of the yields among the treatments was equally likely, so the **randomization distribution** of the  $F$  statistic of [Section 15.3](#) can be computed exactly by permuting the data, and for moderate designs the normal-theory  $F$  distribution approximates it closely. [Chapter 23](#) develops permutation tests ([Theorem 23.4.2](#)). Here the normal model is the working model, and randomization is the reason for trusting it.

### 15.1.6 Fixed and random effects

The five nitrogen rates were chosen deliberately, and the conclusions are about them: the treatment parameters are fixed constants, the setting of this chapter. In other studies the levels are themselves a sample. A laboratory measuring an enzyme activity in batches of reagent from a production line wants to know how much batches vary, not how batch 3 compares with batch 5. The natural model then takes the effects  $\alpha_k$  to be random, uncorrelated with mean zero and variance  $\sigma_a^2$ , and independent of the errors. The mean vector is  $\mu\mathbf{1}$  and the covariance matrix is  $\sigma^2\mathbf{I} + \sigma_a^2\mathbf{Z}\mathbf{Z}^\top$ , so the model is no longer a linear model with covariance  $\sigma^2\mathbf{I}$ .

The projections and sums of squares are the same in both models, but their expectations differ ([Section 15.3](#)), and so does the target: contrasts among the treatments studied, or the variance components  $\sigma_a^2$  and  $\sigma^2$  ([Exercise \(B1\)](#)). [Chapter 32](#) develops random and mixed effects ([Proposition 32.1.1](#)).

## 15.1.7 Exercises

### 15.1.7.1 A. Check your understanding

#### Exercise A1.

For a one-way layout with group sizes 2, 1, 3, write down  $\mathbf{Z}$ ,  $\mathbf{M}$  and  $\mathbf{I} - \mathbf{M}$  with the observations listed group by group. What is the residual of the observation in group 2, and why? How many degrees of freedom does the error space have, and which groups contribute to them?

#### Exercise A2.

Show that  $\mu_1 = \cdots = \mu_g$  iff  $\sum_k c_k \mu_k = 0$  for every contrast  $\mathbf{c}$  ( $\sum_k c_k = 0$ ), and iff  $\mu_k - \mu_g = 0$  for  $k = 1, \dots, g-1$ .

*Solution.* If all means are equal to  $\mu$ , then  $\sum_k c_k \mu_k = \mu \sum_k c_k = 0$ . Conversely, the differences  $\mu_k - \mu_g$  are contrasts, and if they vanish the means are equal. So the three conditions are equivalent. The differences  $\mu_k - \mu_g$  span the space of contrasts, which has dimension  $g-1$ .

### 15.1.7.2 B. Practice

#### Exercise B1.

Derive the three rows of the table in “What the parameters mean”. Then show that under  $\sum_k n_k \alpha_k = 0$  the least squares estimates are  $\hat{\mu} = \bar{y}$  and  $\hat{\alpha}_k = \bar{y}_k - \bar{y}$ , and that  $\sum_k n_k \hat{\alpha}_k^2 = \sum_k n_k (\bar{y}_k - \bar{y})^2$ .

*Solution.* A solution of the normal equations ([Theorem 6.4.2](#)) reproduces the fitted values, so  $\hat{\mu} + \hat{\alpha}_k = \bar{y}_k$  for every  $k$ . With  $\sum_k \alpha_k = 0$ , summing over  $k$  gives  $g\hat{\mu} = \sum_k \bar{y}_k$ . With  $\sum_k n_k \alpha_k = 0$ , multiplying by  $n_k$  and summing gives  $n\hat{\mu} = \sum_k n_k \bar{y}_k = n\bar{y}$ . With  $\alpha_1 = 0$ , the first equation gives  $\hat{\mu} = \bar{y}_1$ . In each case  $\hat{\alpha}_k = \bar{y}_k - \hat{\mu}$ . The same algebra with  $\mu_k$  in place of  $\bar{y}_k$  gives the parameter column. The last identity is immediate from  $\hat{\alpha}_k = \bar{y}_k - \bar{y}$ .

#### Exercise B2.

For  $g = 2$ , show that the pooled variance of the one-way model is the pooled variance of the two-sample  $t$  test, and that the  $t$  statistic for  $\mu_1 - \mu_2$  is  $(\bar{y}_1 - \bar{y}_2)/(s\sqrt{1/n_1 + 1/n_2})$ . ([Section 15.3](#) shows that its square is the one-way  $F$  statistic.)

#### Exercise B3.

In the setting of “Where the model comes from”, suppose  $N$  units with values  $u_1, \dots, u_N$  are divided at random into groups of fixed sizes  $n_1, \dots, n_g$ , and unit  $i$  in group  $k$  yields  $u_i + \tau_k$ . Show that  $E(\bar{Y}_k) = \tau_k + \bar{u}$  and that  $\text{Cov}(Y_{kj}, Y_{k'j'}) = -S_u^2/N$  for two different units, where  $S_u^2 = \sum_i (u_i - \bar{u})^2/(N-1)$ . *Hint:* the sum of the  $N$  deviations  $u_i - \bar{u}$  is

zero.

*Solution.* Under random assignment the unit placed in position  $(k, j)$  is equally likely to be any of the  $N$ , so  $E(Y_{kj}) = \tau_k + \bar{u}$ , and so is the average  $E(\bar{Y}_k)$ . For two different positions, the pair of units is equally likely to be any ordered pair of distinct units, so  $\text{Cov}(Y_{kj}, Y_{k'j'}) = E\{(u_I - \bar{u})(u_J - \bar{u})\}$  with  $(I, J)$  a random ordered pair of distinct indices. Since  $\sum_i (u_i - \bar{u}) = 0$ ,

$$\sum_{i \neq l} (u_i - \bar{u})(u_l - \bar{u}) = \left( \sum_i (u_i - \bar{u}) \right)^2 - \sum_i (u_i - \bar{u})^2 = -(N-1)S_u^2,$$

and dividing by the  $N(N-1)$  ordered pairs gives  $-S_u^2/N$ . The variance of a single observation is  $(N-1)S_u^2/N$ , so the correlation between two observations is  $-1/(N-1)$ , small when  $N$  is large.

### 15.1.7.3 C. Going deeper

#### Exercise C1.

Let  $\mathcal{G}$  be the set of  $n \times n$  permutation matrices that permute observations only within groups. Show that

$$\mathbf{M} = \frac{1}{|\mathcal{G}|} \sum_{\mathbf{\Pi} \in \mathcal{G}} \mathbf{\Pi},$$

so the one-way projection is the average over the symmetries of the design. Deduce that  $\mathcal{C}(\mathbf{Z})$  is exactly the set of vectors fixed by every  $\mathbf{\Pi} \in \mathcal{G}$ .

*Solution.* Take observations  $a$  and  $b$ . If they are in different groups, no  $\mathbf{\Pi} \in \mathcal{G}$  sends  $b$  to  $a$ , so entry  $(a, b)$  of the average is  $0 = m_{ab}$ . If both are in group  $k$ , the permutations of group  $k$  that send  $b$  to  $a$  are a fraction  $1/n_k$  of all permutations of group  $k$ , and the other groups are permuted freely, so entry  $(a, b)$  is  $1/n_k = m_{ab}$ . A vector fixed by every  $\mathbf{\Pi}$  is constant within groups, because a transposition of two members of a group exchanges their entries. Conversely, a vector constant within groups is fixed by each  $\mathbf{\Pi}$ , hence by their average  $\mathbf{M}$ , so it lies in  $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{Z})$ .

#### Exercise C2.

In the random-effects model of “Fixed and random effects”, show that  $\text{Var}(\bar{Y}) = \sigma^2/n + \sigma_a^2 \sum_k n_k/n^2$ . For fixed  $n$  and  $g$ , which group sizes make the grand mean most precise? Contrast this with the fixed-effects model, where  $\text{Var}(\bar{Y}) = \sigma^2/n$  whatever the group sizes.

## 15.2 Estimable contrasts

Most questions about the treatments of a one-way experiment are contrasts. Does fertilizer help at all? Is the highest rate worth its cost? Which pairs of treatments differ? Earlier chapters supplied the theory: estimability (Chapter 8), the  $t$  test and interval (Chapter 11, Chapter 12) and simultaneous methods (Chapter 13). This section collects them for the one-way analysis,

adds a geometric fact about power, and applies them to the nitrogen trial.

## 15.2.1 Contrasts are the treatment space

Recall that a contrast is a linear function  $\psi = \sum_k c_k \mu_k$  with  $\sum_k c_k = 0$  and  $\mathbf{c} \neq \mathbf{0}$  (Definition 8.7.1). In the effects form it equals  $\sum_k c_k \alpha_k$ , and the contrasts are exactly the estimable functions of the effects alone (Theorem 8.7.1). Their estimates are  $\hat{\psi} = \sum_k c_k \bar{y}_k$ , whatever side condition or coding was used to fit the model.

Behind this lies a simple geometry. Let

$$\mathcal{T} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$$

be the **treatment space**: the vectors that are constant within groups and sum to zero. Its dimension is  $g - 1$  (Theorem 6.5.1). A vector  $\mathbf{u} \in \mathcal{T}$  takes some value  $v_k$  on every observation of group  $k$ , with  $\sum_k n_k v_k = 0$ . Put  $c_k = n_k v_k$ . Then  $\mathbf{c}$  is a contrast and  $\mathbf{u}$  is the vector  $\mathbf{u}_{\mathbf{c}} = \mathbf{ZD}^{-1}\mathbf{c}$  of Eq. 8.7.2, with  $\mathbf{D} = \text{diag}(n_1, \dots, n_g)$ . Conversely every  $\mathbf{u}_{\mathbf{c}}$  lies in  $\mathcal{T}$  (Proposition 8.7.2). So contrasts and nonzero vectors of the treatment space correspond one to one, and  $\hat{\psi} = \mathbf{u}_{\mathbf{c}}^\top \mathbf{y}$ . A contrast is a direction in the treatment space, and its estimate is the component of the data along that direction.

Equal treatment means say that the mean vector  $\boldsymbol{\theta} = \mathbf{E}(\mathbf{Y})$  has no component in  $\mathcal{T}$ , that is, that every contrast vanishes (Exercise (A2)). The overall test of Section 15.3 looks at all of  $\mathcal{T}$ ; a single contrast looks along one line of it.

## 15.2.2 The test and interval for one contrast

For a contrast  $\mathbf{c}$  write

$$v_{\mathbf{c}} = \sum_{k=1}^g \frac{c_k^2}{n_k},$$

so that  $\text{Var}(\hat{\psi}) = \sigma^2 v_{\mathbf{c}}$  (Eq. 8.7.1).

### Proposition 15.2.1 — Sum of squares and test for a contrast

In the normal one-way model, let  $\psi = \sum_k c_k \mu_k$  be a contrast,  $\hat{\psi} = \sum_k c_k \bar{Y}_k$ , and  $\psi_0 \in \mathbb{R}$ . Then:

- (a)  $\hat{\psi} \sim N(\psi, \sigma^2 v_{\mathbf{c}})$ , and  $\hat{\psi}$  is independent of  $s^2$ ;
- (b) the statistic

$$T = \frac{\hat{\psi} - \psi_0}{s\sqrt{v_{\mathbf{c}}}} \sim t(n - g, \delta), \quad \delta = \frac{\psi - \psi_0}{\sigma\sqrt{v_{\mathbf{c}}}},$$

so the test of  $\psi = \psi_0$  rejects when  $|T| > t_{n-g, \alpha/2}$ , and  $\hat{\psi} \pm t_{n-g, \alpha/2} s\sqrt{v_{\mathbf{c}}}$  is an exact  $1 - \alpha$  confidence interval for  $\psi$ ;

- (c) the **sum of squares of the contrast**

$$\text{SS}(\mathbf{c}) = \frac{\hat{\psi}^2}{v_{\mathbf{c}}} = \|\mathbf{P}_{\mathbf{c}} \mathbf{Y}\|^2,$$

where  $\mathbf{P}_{\mathbf{c}}$  projects onto the line spanned by  $\mathbf{u}_{\mathbf{c}}$ , has one degree of freedom, and  $SS(\mathbf{c})/s^2 = T^2$  for  $\psi_0 = 0$  has the  $F(1, n - g, \delta^2)$  distribution. Replacing  $\mathbf{c}$  by  $a\mathbf{c}$  with  $a \neq 0$  changes neither  $SS(\mathbf{c})$  nor the test;

(d)  $SS(\mathbf{c}) \leq \sum_k n_k (\bar{Y}_k - \bar{Y})^2$ , with equality when  $c_k = n_k(\bar{Y}_k - \bar{Y})$  for every  $k$ .

*Proof.* Use the cell-means form, with model matrix  $\mathbf{Z}$  of full rank  $g$ , so that  $(\mathbf{Z}^\top \mathbf{Z})^{-1} = \mathbf{D}^{-1}$  and  $\mathbf{c}^\top \mathbf{D}^{-1} \mathbf{c} = v_{\mathbf{c}}$ . The function  $\mathbf{c}^\top \boldsymbol{\mu}$  is estimable with estimate  $\mathbf{c}^\top \hat{\boldsymbol{\mu}} = \hat{\psi}$ . (a) and (b) are [Theorem 11.6.1\(a\)](#) and [Corollary 7.5.2](#) with  $\boldsymbol{\lambda} = \mathbf{c}$ ,  $r = g$ , and the interval is [Theorem 12.1.1](#). The independence of  $\hat{\psi}$  and  $s^2$  is part of [Theorem 7.5.1](#). (c) The identity  $\hat{\psi}^2/v_{\mathbf{c}} = \|\mathbf{P}_{\mathbf{c}} \mathbf{Y}\|^2$  is [Proposition 8.7.2\(b\)](#), and  $T^2$  is the  $F$  statistic of the hypothesis  $\psi = 0$  by [Theorem 11.6.1\(b\)](#), with noncentrality  $\delta^2$ . Scaling  $\mathbf{c}$  by  $a$  multiplies  $\hat{\psi}^2$  and  $v_{\mathbf{c}}$  by  $a^2$ , and leaves the line spanned by  $\mathbf{u}_{\mathbf{c}}$  unchanged. (d) is [Exercise \(B1\)](#). ■

By part (c) a contrast's sum of squares is a projection onto a line of  $\mathcal{T}$ , and by part (d) no line captures more than the whole space. The line that captures all of it is chosen by the data, which is why its  $t$  statistic needs Scheffé's multiplier ([Theorem 13.2.2](#)).

### 15.2.3 Planned contrasts in the nitrogen trial

Before the trial the agronomist wrote down two questions. First, does nitrogen raise yield at all? The natural contrast compares the average of the four fertilized rates with the control,  $\psi_1 = \frac{1}{4}(\mu_2 + \mu_3 + \mu_4 + \mu_5) - \mu_1$ . Second, is the highest rate better than the moderate one? That is  $\psi_2 = \mu_5 - \mu_3$ , the difference between 160 and 80 kg N/ha.

#### Example. (Two planned contrasts)

The pooled standard deviation is  $s = 0.5211$  on 25 degrees of freedom. For  $\psi_1$ ,  $v_{\mathbf{c}} = \frac{1}{6}(1 + 4 \cdot \frac{1}{16})$ , the estimate is 2.110 t/ha with standard error 0.238, so  $t = 8.87$  and the 95% interval is [1.620, 2.600]. Fertilizer raises yield by roughly two tonnes per hectare. Its sum of squares is 21.370.

For  $\psi_2$  the estimate is 0.595 t/ha with standard error 0.301,  $t = 1.98$ , two-sided  $p = 0.059$ , and the interval [−0.025, 1.215]. The data are consistent with anything from no gain to a gain of more than a tonne: with six plots per rate, half a tonne is only about 1.7 standard errors of a pairwise difference.

```
import numpy as np
from scipy import stats

rates = np.array([0, 40, 80, 120, 160])           # kg N per hectare
g, m = len(rates), 6                             # 5 rates, 6 plots per rate
true_means = 6.8 - 2.7 * np.exp(-0.018 * rates)  # unknown to the analyst
rng = np.random.default_rng(1515)
group = np.repeat(np.arange(g), m)                # plot i received rate group[i]
y = np.round(true_means[group] + rng.normal(0, 0.5, g * m), 2) # yield, t/ha
n = len(y)

n_k = np.bincount(group).astype(float)
```

```

means = np.bincount(group, weights=y) / n_k
nu = n - g
s = np.sqrt(np.sum((y - means[group]) ** 2) / nu)

def contrast(c, psi0=0.0, mult=None):
    """Estimate, standard error, t statistic, SS and interval for sum_k c_k mu_k."""
    est = c @ means
    v = np.sum(c ** 2 / n_k) # Var(estimate) / sigma^2
    se = s * np.sqrt(v)
    mult = stats.t.ppf(0.975, nu) if mult is None else mult
    return est, se, (est - psi0) / se, est ** 2 / v, (est - mult * se, est + mult * se)

c_fert = np.array([-1, 0.25, 0.25, 0.25, 0.25]) # fertilized (average) minus control
c_top = np.array([0, 0, -1, 0, 1]) # 160 against 80 kg N/ha
for name, c in (("fertilized - control", c_fert), ("160 - 80", c_top)):
    est, se, t, ss, (lo, hi) = contrast(c)
    print(f"{name:21s} {est:6.3f} se {se:.3f} t = {t:5.2f} SS = {ss:6.3f}"
          f" 95% interval [{lo:.3f}, {hi:.3f}]")

```

Each was a single planned question, so each gets an unadjusted  $t$  interval; joint coverage for the pair would use Bonferroni's  $t_{25,0.0125}$ . Writing the questions down in advance fixes a small family (Section 13.6).

## 15.2.4 Families of comparisons

Three families come up in almost every one-way analysis, and Chapter 13 supplies a method for each. The multipliers below multiply the standard error of a difference of two means,  $s\sqrt{2/m} = 0.3009$  here.

**Comparisons with a control.** The four differences  $\mu_k - \mu_1$ ,  $k = 2, \dots, 5$ , form a planned family of size four. Bonferroni's method (Theorem 13.4.1) uses the multiplier  $t_{25,0.05/8} = 2.692$  in place of the unadjusted 2.060:

| Rate against control | Difference | Bonferroni 95% interval |
|----------------------|------------|-------------------------|
| 40                   | 1.232      | [0.422, 2.041]          |
| 80                   | 2.052      | [1.242, 2.861]          |
| 120                  | 2.510      | [1.700, 3.320]          |
| 160                  | 2.647      | [1.837, 3.456]          |

Every fertilized rate beats the control. The four estimates share  $\bar{y}_1$ , so they are positively correlated, with correlation  $\frac{1}{2}$  in a balanced design (Exercise (B4)). Dunnett (1955) computed the exact joint distribution of the four  $t$  statistics under this correlation; his multiplier here is 2.606 (Exercise (B2) characterizes it), a few percent shorter than Bonferroni's. We do not derive its tables.

**All pairs.** Tukey's method (Theorem 13.3.2) covers all ten differences  $\mu_k - \mu_l$  with exact joint coverage in a balanced design. With  $q_{0.05}(5, 25) = 4.153$  the half-width is  $q s / \sqrt{m} = 0.884$ , and six of the ten pairs are declared different. The four that are not are 40 and 80, 80 and 120, 80 and 160, and 120 and 160. The conclusions are not transitive: 40 is not distinguished from

80, nor 80 from 160, yet 40 and 160 are declared different, since  $6.652 - 5.237 = 1.415$  exceeds 0.884. “Not declared different” means “not enough evidence of a difference”, which is not an equivalence relation, so displays that sort treatments into groups of “equal” means should be read with care, especially in unbalanced designs.

**Contrasts suggested by the data.** Looking at Figure 15.1.1, one might ask whether the response has levelled off, by comparing the average of the two highest rates with 80 kg N/ha,  $\frac{1}{2}(\mu_4 + \mu_5) - \mu_3$ . The estimate is 0.527 with standard error 0.261. The unadjusted interval  $[-0.010, 1.063]$  is not legitimate, because the contrast was chosen after seeing the means. Scheffé’s interval (Eq. 13.2.4), with multiplier  $\sqrt{4F_{0.05}(4, 25)} = 3.322$ , is  $[-0.339, 1.392]$ . It is valid however the contrast was found, and it says the question is open.

```
import itertools

k_ctrl = g - 1                                # four comparisons with the
control
bonf = stats.t.ppf(1 - 0.05 / (2 * k_ctrl), nu) # Bonferroni multiplier
se_pair = s * np.sqrt(2 / m)
for k in range(1, g):
    d = means[k] - means[0]
    print(f"{rates[k]:3d} - 0: {d:6.3f} Bonferroni interval"
          f" [{d - bonf * se_pair:.3f}, {d + bonf * se_pair:.3f}]")

q = stats.studentized_range.ppf(0.95, g, nu)    # q_{0.05}(g, nu)
hsd = q * s / np.sqrt(m)                       # Tukey half-width, all pairs
pairs = list(itertools.combinations(range(g), 2))
different = [(int(rates[k]), int(rates[l])) for k, l in pairs if abs(means[k] -
    means[l]) > hsd]
print(f"Tukey half-width {hsd:.3f}; pairs declared different: {len(different)} of {
    len(pairs)}")
print("not declared different:",
      [(int(rates[k]), int(rates[l])) for k, l in pairs if abs(means[k] - means[l])
    <= hsd])
```

## 15.2.5 Why planned contrasts are powerful

A single contrast is tested with one numerator degree of freedom, the overall test with  $g - 1$ . When the scientific question is sharp, the contrast is the better test, and the reason is geometric. Write  $\mathbf{P}_{\mathcal{T}}$  for the projection onto the treatment space and

$$\gamma_{\mathcal{T}} = \frac{\|\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}\|^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{k=1}^g n_k (\mu_k - \bar{\mu})^2,$$

the squared length of the treatment component of the mean in units of  $\sigma^2$ . (Section 15.3 shows that it is the noncentrality of the overall  $F$  test.)

### Proposition 15.2.2 — The noncentrality of a contrast

Let  $\mathbf{c}$  be a contrast with vector  $\mathbf{u}_{\mathbf{c}} \in \mathcal{T}$ , and suppose  $\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta} \neq \mathbf{0}$ . The noncentrality of the test of  $\psi = 0$  in Proposition 15.2.1 is

$$\delta^2 = \frac{\psi^2}{\sigma^2 v_{\mathbf{c}}} = \gamma_{\mathcal{T}} \cos^2 \varphi,$$



where  $\varphi$  is the angle between  $\mathbf{u}_c$  and  $\mathbf{P}_T\boldsymbol{\theta}$ . In particular  $\delta^2 \leq \gamma_T$ , with equality iff  $\mathbf{u}_c$  is parallel to  $\mathbf{P}_T\boldsymbol{\theta}$ , that is, iff  $c_k \propto n_k(\mu_k - \bar{\mu})$ .

*Proof.* By Proposition 8.7.2(a) applied to the mean vector,  $\psi = \mathbf{u}_c^\top \boldsymbol{\theta}$ , and  $v_c = \|\mathbf{u}_c\|^2$  by the computation in the proof of that result. Since  $\mathbf{u}_c \in \mathcal{T}$ ,  $\mathbf{u}_c^\top \boldsymbol{\theta} = \mathbf{u}_c^\top \mathbf{P}_T \boldsymbol{\theta}$ . So

$$\delta^2 = \frac{(\mathbf{u}_c^\top \mathbf{P}_T \boldsymbol{\theta})^2}{\sigma^2 \|\mathbf{u}_c\|^2} = \frac{\|\mathbf{P}_T \boldsymbol{\theta}\|^2}{\sigma^2} \cdot \frac{(\mathbf{u}_c^\top \mathbf{P}_T \boldsymbol{\theta})^2}{\|\mathbf{u}_c\|^2 \|\mathbf{P}_T \boldsymbol{\theta}\|^2} = \gamma_T \cos^2 \varphi$$

by Eq. 6.2.2. Equality in  $\cos^2 \varphi \leq 1$  holds iff the two vectors are parallel. The vector  $\mathbf{P}_T \boldsymbol{\theta}$  has entry  $\mu_k - \bar{\mu}$  in group  $k$  (Theorem 6.5.1), so it is  $\mathbf{u}_c$  for  $c_k = n_k(\mu_k - \bar{\mu})$ . ■

If the departure from equal means lies along the planned contrast, the contrast test gets all of the noncentrality on one degree of freedom, while the overall test spreads the same noncentrality over  $g - 1$ . For the nitrogen design, with 25 error degrees of freedom, the powers at level 0.05 are:

| $\gamma_T$ , all along the contrast | 4     | 8     | 12    |
|-------------------------------------|-------|-------|-------|
| contrast test,<br>$F(1, 25)$        | 0.485 | 0.776 | 0.914 |
| overall test,<br>$F(4, 25)$         | 0.269 | 0.516 | 0.714 |

The price is paid when the question is wrong: a contrast orthogonal to  $\mathbf{P}_T \boldsymbol{\theta}$  has power  $\alpha$  however large the treatment differences, and a direction chosen at random captures on average only  $1/(g - 1)$  of  $\gamma_T$  (Exercise (C1)).

```
def power_f(gamma, q, nu, alpha=0.05):
    return stats.ncf.sf(stats.f.ppf(1 - alpha, q, nu), q, nu, gamma)

for gamma in (4.0, 8.0, 12.0):
    # all of it along the planned contrast
    print(f"gamma = {gamma:4.1f}: contrast test {power_f(gamma, 1, nu):.3f},"
          f" omnibus F test {power_f(gamma, g - 1, nu):.3f}")
```

## 15.2.6 Exercises

### 15.2.6.1 A. Check your understanding

#### Exercise A1.

In the nitrogen trial, compute by hand the estimate and standard error of  $\frac{1}{2}(\mu_4 + \mu_5) - \frac{1}{2}(\mu_1 + \mu_2)$ , the average of the two highest rates minus the average of the two lowest, from the rate means of Section 15.1 and  $s = 0.5211$ .

*Solution.* The estimate is  $\frac{1}{2}(6.515 + 6.652) - \frac{1}{2}(4.005 + 5.237) = 6.5835 - 4.621 = 1.9625$ . With  $m = 6$ ,  $v_{\mathbf{c}} = 4 \cdot \frac{1}{4}/6 = 1/6$ , so the standard error is  $0.5211/\sqrt{6} \approx 0.213$ .

### Exercise A2.

Show directly that the  $t$  statistic of Proposition 15.2.1(b) with  $\psi_0 = 0$  is unchanged when  $\mathbf{c}$  is multiplied by  $a > 0$  and changes sign when  $a < 0$ . What happens to the confidence interval?

## 15.2.6.2 B. Practice

### Exercise B1.

In a balanced layout with  $m$  observations per group, show that the noncentrality of the test of a contrast is  $\delta^2 = m\psi^2/(\sigma^2 \sum_k c_k^2)$ . How many observations per group are needed for the test of  $\mu_5 - \mu_3$  in the nitrogen trial to have power 0.8 at level 0.05 when the true difference is 0.5 t/ha and  $\sigma = 0.5$ ? (Use software for the noncentral  $F$ , or the normal approximation.)

*Solution.* Here  $v_{\mathbf{c}} = \sum_k c_k^2/m$ , so  $\delta^2 = \psi^2/(\sigma^2 v_{\mathbf{c}}) = m\psi^2/(\sigma^2 \sum_k c_k^2)$ . For  $\mu_5 - \mu_3$ ,  $\sum_k c_k^2 = 2$ ,  $\psi/\sigma = 1$ , and  $\delta^2 = m/2$ . The normal approximation asks for  $\delta \approx z_{0.025} + z_{0.2} = 1.96 + 0.84 = 2.80$ , so  $m \approx 2 \cdot 2.80^2 \approx 15.7$ . The exact noncentral  $F$  with  $5(m-1)$  error degrees of freedom gives  $m = 17$ , since the  $t$  quantile exceeds the normal one. The trial's  $m = 6$  is far short.

### Exercise B2.

A single contrast  $\mathbf{c}$  matters above all. For a fixed total  $n$ , show that  $v_{\mathbf{c}} = \sum_k c_k^2/n_k$  is minimized, over positive real  $n_k$  with  $\sum_k n_k = n$ , by  $n_k \propto |c_k|$ , with minimum  $(\sum_k |c_k|)^2/n$ . What does this say about how to allocate plots between two treatments, and between a control and the average of four treatments?

*Solution.* By the Cauchy–Schwarz inequality,  $(\sum_k |c_k|)^2 = (\sum_k (|c_k|/\sqrt{n_k})\sqrt{n_k})^2 \leq \sum_k (c_k^2/n_k) \sum_k n_k = nv_{\mathbf{c}}$ , with equality iff  $|c_k|/\sqrt{n_k} \propto \sqrt{n_k}$ , that is,  $n_k \propto |c_k|$ . Groups with  $c_k = 0$  get no observations. For  $\mu_1 - \mu_2$ , equal allocation is best. For  $\mu_1 - \frac{1}{4}(\mu_2 + \dots + \mu_5)$ , the control should get as many plots as the four treatments together,  $n_1 = n/2$  and  $n_k = n/8$  for the others.

### Exercise B3.

In a comparison of  $g - 1$  treatments with a control, suppose the control gets  $n_0$  observations and each treatment gets  $n_t$ , with  $n_0 + (g - 1)n_t = n$  fixed. Show that  $\sum_k \text{Var}(\bar{Y}_k - \bar{Y}_0)$  is minimized when  $n_0 = \sqrt{g-1} n_t$ . For  $g = 5$  and  $n = 30$ , what allocation does this suggest?

*Solution.* The sum is  $\sigma^2(g-1)(1/n_0 + 1/n_t)$ . Minimizing  $1/n_0 + 1/n_t$  subject to  $n_0 + (g-1)n_t = n$  with a Lagrange multiplier gives  $1/n_0^2 = \lambda$  and  $1/n_t^2 = (g-1)\lambda$ , so  $n_0/n_t = \sqrt{g-1}$ . For  $g = 5$  the control should get twice as many observations as each

treatment:  $n_0 = 10$  and  $n_t = 5$ . Contrast [Exercise \(B2\)](#): for the average of the four comparisons the optimal control share is  $n/2$ , that is  $n_0 = 4n_t$ . The best allocation depends on whether the four comparisons or their average is the target.

#### Exercise B4.

With  $n_0$  observations on the control and  $n_t$  on each treatment, show that the estimates  $\bar{Y}_k - \bar{Y}_0$  and  $\bar{Y}_l - \bar{Y}_0$  of two treatment-versus-control differences have correlation  $n_t/(n_t + n_0)$ . In a balanced design it is  $\frac{1}{2}$ .

*Solution.* The covariance is  $\text{Var}(\bar{Y}_0) = \sigma^2/n_0$ , since the treatment means are independent of each other and of  $\bar{Y}_0$ . Each variance is  $\sigma^2(1/n_t + 1/n_0)$ . The ratio is  $(1/n_0)/(1/n_t + 1/n_0) = n_t/(n_t + n_0)$ .

### 15.2.6.3 C. Going deeper

#### Exercise C1.

Let  $q = g - 1 \geq 2$  and let  $\mathbf{u}$  be uniformly distributed on the unit sphere of the  $q$ -dimensional space  $\mathcal{T}$  (for example  $\mathbf{u} = \mathbf{w}/\|\mathbf{w}\|$  with  $\mathbf{w} \sim N$  on  $\mathcal{T}$  with covariance  $\mathbf{P}_{\mathcal{T}}$ ). Show that  $E \cos^2 \varphi = 1/q$ , where  $\varphi$  is the angle between  $\mathbf{u}$  and a fixed nonzero  $\mathbf{v} \in \mathcal{T}$ . Interpret this with [Proposition 15.2.2](#).

*Solution.* Choose an orthonormal basis of  $\mathcal{T}$  whose first vector is  $\mathbf{v}/\|\mathbf{v}\|$ . In these coordinates  $\cos^2 \varphi = u_1^2$ . By symmetry the  $u_i^2$  have a common mean, and they sum to 1, so  $E u_1^2 = 1/q$ . By [Proposition 15.2.2](#), a contrast whose direction is unrelated to the truth captures on average a fraction  $1/q$  of  $\gamma_{\mathcal{T}}$  on its one degree of freedom, the same average share per degree of freedom as the overall test. Knowledge, not the count of degrees of freedom, is what makes a planned contrast more powerful.

#### Exercise C2.

The Scheffé multiplier for a family of contrasts depends on the dimension of the space the family spans ([Theorem 13.2.2](#)). Show that the  $g - 1$  comparisons with a control span  $\mathcal{T}$ , and that the family of the four “successive differences”  $\mu_{k+1} - \mu_k$  does too. Why, then, are Bonferroni or Dunnett intervals shorter than Scheffé’s for these families?

## 15.3 The analysis of variance table from projections

Whether the treatments differ at all is answered by the  $F$  test, with the analysis of variance table as its bookkeeping. Both come from one orthogonal decomposition of the observation space into the constants, the treatment space and the error space, and every entry of the table is a property of one piece.

### 15.3.1 Three subspaces

Write  $\mathbf{P}_0 = n^{-1}\mathbf{1}\mathbf{1}^\top$  for the projection onto the constants and  $\mathbf{M}$  for the projection onto  $\mathcal{C}(\mathbf{Z})$  (Eq. 15.1.1). Since  $\mathbf{1} \in \mathcal{C}(\mathbf{Z})$ , Theorem 6.5.1 applies to the pair  $\text{span}(\mathbf{1}) \subseteq \mathcal{C}(\mathbf{Z})$  and gives the orthogonal decomposition

$$\mathbb{R}^n = \underbrace{\text{span}(\mathbf{1})}_{\dim 1} \oplus \underbrace{\mathcal{T}}_{\dim g-1} \oplus \underbrace{\mathcal{C}(\mathbf{Z})^\perp}_{\dim n-g}, \quad \mathbf{I} = \mathbf{P}_0 + (\mathbf{M} - \mathbf{P}_0) + (\mathbf{I} - \mathbf{M}), \quad (15.3.1)$$

with the treatment space  $\mathcal{T} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$  of Section 15.2 and  $\mathbf{P}_{\mathcal{T}} = \mathbf{M} - \mathbf{P}_0$ . The three projections act on the data in the familiar way:

$$(\mathbf{P}_0 \mathbf{y})_{kj} = \bar{y}, \quad (\mathbf{P}_{\mathcal{T}} \mathbf{y})_{kj} = \bar{y}_k - \bar{y}, \quad ((\mathbf{I} - \mathbf{M}) \mathbf{y})_{kj} = y_{kj} - \bar{y}_k.$$

So every observation is split as  $y_{kj} = \bar{y} + (\bar{y}_k - \bar{y}) + (y_{kj} - \bar{y}_k)$ , grand mean plus treatment deviation plus residual, and the three vectors so formed are mutually orthogonal. Pythagoras gives

$$\sum_{k,j} y_{kj}^2 = n\bar{y}^2 + \sum_k n_k(\bar{y}_k - \bar{y})^2 + \sum_{k,j} (y_{kj} - \bar{y}_k)^2. \quad (15.3.2)$$

The middle term is the **between-groups sum of squares**  $\text{SSB} = \|\mathbf{P}_{\mathcal{T}} \mathbf{y}\|^2$ , and the last is the **within-groups** or residual sum of squares  $\text{SSE} = \|(\mathbf{I} - \mathbf{M}) \mathbf{y}\|^2$  of Eq. 15.1.2. Their sum  $\sum_{k,j} (y_{kj} - \bar{y})^2$  is the corrected total sum of squares. The same decomposition, with the distributions attached, was the first application of Cochran's theorem (Example (The one-way layout)).

### 15.3.2 The $F$ test

#### Theorem 15.3.1 — The one-way $F$ test

In the normal one-way model of Definition 15.1.1, with mean vector  $\boldsymbol{\theta} = \mathbf{Z}\boldsymbol{\mu}$ , put

$$\text{MSB} = \frac{\text{SSB}}{g-1}, \quad s^2 = \text{MSE} = \frac{\text{SSE}}{n-g}, \quad F = \frac{\text{MSB}}{\text{MSE}}.$$

- (a)  $\bar{Y}$ ,  $\text{SSB} = \sum_k n_k(\bar{Y}_k - \bar{Y})^2$  and  $\text{SSE} = \sum_{k,j} (Y_{kj} - \bar{Y}_k)^2$  are mutually independent.
- (b)  $\text{SSE}/\sigma^2 \sim \chi^2(n-g)$ , and  $\text{SSB}/\sigma^2 \sim \chi^2(g-1, \gamma)$  with

$$\gamma = \frac{\|\mathbf{P}_{\mathcal{T}} \boldsymbol{\theta}\|^2}{\sigma^2} = \frac{1}{\sigma^2} \sum_{k=1}^g n_k(\mu_k - \bar{\mu})^2, \quad \bar{\mu} = \frac{1}{n} \sum_{k=1}^g n_k \mu_k. \quad (15.3.3)$$

- (c)  $F \sim F(g-1, n-g, \gamma)$ , and  $\gamma = 0$  iff  $\mu_1 = \dots = \mu_g$ . In particular, when the means are equal,  $F \sim F(g-1, n-g)$  whatever their common value and  $\sigma^2$ .
- (d)  $F$  is the statistic of the  $F$  test of the reduced model  $\mathbf{E}(\mathbf{Y}) \in \text{span}(\mathbf{1})$  against  $\mathcal{C}(\mathbf{Z})$ , and of the testable hypothesis  $\alpha_1 = \dots = \alpha_g$  in the effects form. The test that rejects when  $F > F_\alpha(g-1, n-g)$  is the likelihood ratio test of equal means. Its

power is a continuous, strictly increasing function of  $\gamma$ , equal to  $\alpha$  at  $\gamma = 0$ .

*Proof.* By Eq. 15.3.1 the matrices  $\mathbf{P}_0$ ,  $\mathbf{P}_T$  and  $\mathbf{I} - \mathbf{M}$  are symmetric, idempotent and mutually orthogonal, with ranks 1,  $g - 1$  and  $n - g$ . Theorem 4.4.4 gives (a), since  $\bar{Y}$  is a function of  $\mathbf{P}_0\mathbf{Y}$ , and it gives the chi-squared laws of (b) with noncentralities  $\|\mathbf{P}_T\boldsymbol{\theta}\|^2/\sigma^2$  and  $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2/\sigma^2$ . The second is zero because  $\boldsymbol{\theta} \in \mathcal{C}(\mathbf{Z})$ . The vector  $\boldsymbol{\theta}$  has entry  $\mu_k$  in group  $k$ , so  $\mathbf{M}\boldsymbol{\theta} = \boldsymbol{\theta}$  and  $\mathbf{P}_0\boldsymbol{\theta} = \bar{\mu}\mathbf{1}$ , and  $\mathbf{P}_T\boldsymbol{\theta}$  has entry  $\mu_k - \bar{\mu}$  in group  $k$ . Its squared length is  $\sum_k n_k(\mu_k - \bar{\mu})^2$ , which proves Eq. 15.3.3.

For (c): by (a) and (b),  $F$  is a ratio of independent chi-squared variables divided by their degrees of freedom, the numerator noncentral, which is the noncentral  $F$  law (Definition 4.1.2). Since every  $n_k \geq 1$ ,  $\gamma = 0$  iff  $\mu_k = \bar{\mu}$  for every  $k$ .

For (d), take  $\mathbf{X}_0 = \mathbf{1}$  inside  $\mathbf{X} = \mathbf{Z}$ . Then  $\text{SSE}_0 - \text{SSE} = \|(\mathbf{M} - \mathbf{P}_0)\mathbf{Y}\|^2 = \text{SSB}$  by Eq. 6.5.2, with  $r - r_0 = g - 1$ , so  $F$  is the statistic of Theorem 11.1.1. In the effects form the hypothesis is  $\alpha_k - \alpha_g = 0$ ,  $k < g$ :  $g - 1$  estimable, linearly independent contrasts, so it is testable (Definition 11.3.1). The mean vectors it allows are the constant vectors, so by Theorem 11.3.2(c) its  $F$  statistic is the same one. The likelihood ratio property is Theorem 11.4.1, and the power statement is Theorem 11.5.1(a). ■

The noncentrality Eq. 15.3.3 is a weighted sum of squared deviations of the group means from their *weighted* mean. The weights are the group sizes, because a group of  $n_k$  observations carries  $n_k$  copies of its mean in  $\boldsymbol{\theta}$ . In a balanced layout  $\gamma = m \sum_k (\mu_k - \bar{\mu})^2 / \sigma^2$ , the case studied for planning in Proposition 11.5.2 and Example (How many observations per group?). The dependence on the group sizes in unbalanced layouts is taken up in Section 15.5.

### 15.3.3 Expected mean squares

The distributions in Theorem 15.3.1 need normality. The expectations of the mean squares do not, and they explain why the ratio of two mean squares is the right statistic.

#### Theorem 15.3.2 — Expected mean squares in the one-way layout

In the one-way model of Definition 15.1.1 (uncorrelated errors with common variance  $\sigma^2$ , not necessarily normal):

- (a)  $E(\text{MSE}) = \sigma^2$ ;
- (b)  $E(\text{MSB}) = \sigma^2 + \frac{1}{g-1} \sum_{k=1}^g n_k(\mu_k - \bar{\mu})^2$ ;
- (c) in the effects form,  $\sum_k n_k(\mu_k - \bar{\mu})^2 = \sum_k n_k(\alpha_k - \bar{\alpha})^2$  with  $\bar{\alpha} = n^{-1} \sum_k n_k \alpha_k$ , which is  $\sum_k n_k \alpha_k^2$  under the side condition  $\sum_k n_k \alpha_k = 0$ . In a balanced layout it is  $m \sum_k (\mu_k - \bar{\mu})^2$  with  $\bar{\mu} = g^{-1} \sum_k \mu_k$ , which is  $m \sum_k \alpha_k^2$  under  $\sum_k \alpha_k = 0$ ;
- (d)  $E(\text{MSB}) \geq E(\text{MSE})$ , with equality iff  $\mu_1 = \dots = \mu_g$ .

*Proof.* The spaces of Eq. 15.3.1 form an analysis of variance decomposition of  $\mathcal{C}(\mathbf{Z})$  in the sense of Theorem 9.6.1, which gives  $E(\text{SSE}) = \sigma^2(n - g) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2 = \sigma^2(n - g)$

and  $E(\text{SSB}) = \sigma^2(g-1) + \|\mathbf{P}_{\mathcal{T}}\boldsymbol{\theta}\|^2$ . The squared length was computed in the proof of [Theorem 15.3.1](#). Dividing by the degrees of freedom gives (a) and (b). For (c),  $\mu_k = \mu + \alpha_k$  and  $\bar{\mu} = \mu + \bar{\alpha}$ , so  $\mu_k - \bar{\mu} = \alpha_k - \bar{\alpha}$ . In a balanced layout  $\bar{\mu} = \tilde{\mu}$  and  $\bar{\alpha} = g^{-1} \sum_k \alpha_k$ . (d) follows from (a) and (b). Part (b) is also [Exercise \(A2\)](#), which computes it directly from the group means. ■

Both mean squares estimate  $\sigma^2$  when the treatments do not differ; otherwise MSB estimates  $\sigma^2$  plus the treatment variation per degree of freedom. Every  $F$  ratio in an analysis of variance follows this logic: the denominator's expectation equals the numerator's exactly when the hypothesis holds ([Section 9.6](#)).

The same projections serve the random-effects model of [Section 15.1](#), but the expectations change. With  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$  and a constant mean,  $E(\text{MSE}) = \sigma^2$  still, while  $E(\text{MSB}) = \sigma^2 + n_0 \sigma_a^2$  with  $n_0 = (n - \sum_k n_k^2/n)/(g-1)$ , which is  $m$  in a balanced layout ([Exercise \(C1\)](#)). The ratio  $F$  then tests  $\sigma_a^2 = 0$ , and the difference of the two mean squares divided by  $n_0$  estimates  $\sigma_a^2$ . [Chapter 32](#) develops this ([Proposition 32.1.1](#)).

## 15.3.4 The table, derived

Collecting the pieces gives the one-way analysis of variance table. Each row is a subspace, its degrees of freedom are the dimension, its sum of squares is the squared length of the projection of  $\mathbf{y}$ , and its expected mean square is  $\sigma^2$  plus the squared length of the projection of  $\boldsymbol{\theta}$  per dimension.

| Source          | Subspace                        | df    | Sum of squares                       | Mean square | $F$     | Expected mean square                                      |
|-----------------|---------------------------------|-------|--------------------------------------|-------------|---------|-----------------------------------------------------------|
| treatments      | $\mathcal{T}$                   | $g-1$ | $\sum_k n_k (\bar{y}_k - \bar{y})^2$ | MSB         | MSB/MSE | $\sigma^2 + \frac{\sum_k n_k (\mu_k - \bar{\mu})^2}{g-1}$ |
| error           | $\mathcal{C}(\mathbf{Z})^\perp$ | $n-g$ | $\sum_{k,j} (y_{kj} - \bar{y}_k)^2$  | MSE         |         | $\sigma^2$                                                |
| corrected total | $\mathbf{1}^\perp$              | $n-1$ | $\sum_{k,j} (y_{kj} - \bar{y})^2$    |             |         |                                                           |

Some tables add a row for the mean, with 1 degree of freedom and sum of squares  $n\bar{y}^2 = \|\mathbf{P}_0 \mathbf{y}\|^2$ , and an uncorrected total  $\sum y_{kj}^2$ ; its  $F$  ratio would test  $\bar{\mu} = 0$ , rarely of interest.

Hand computation used the group totals  $T_k$  and grand total  $T$ :  $\|\mathbf{M}\mathbf{y}\|^2 = \sum_k T_k^2/n_k$  and  $\|\mathbf{P}_0 \mathbf{y}\|^2 = T^2/n$ , the “correction factor”, so  $\text{SSB} = \sum_k T_k^2/n_k - T^2/n$ . Such differences of nearly equal numbers lose accuracy in floating point ([Chapter 10](#)); computing deviations first, as the projections do, avoids this.

### Example. (The analysis of variance for the nitrogen trial)

For the thirty yields of [Section 15.1](#), with grand mean  $\bar{y} = 5.6930$  t/ha:

| Source       | df | Sum of squares | Mean square | $F$   |
|--------------|----|----------------|-------------|-------|
| nitrogen     | 4  | 28.7074        | 7.1768      | 26.43 |
| rates        |    |                |             |       |
| plots within | 25 | 6.7887         | 0.2715      |       |
| rates        |    |                |             |       |
| corrected    | 29 | 35.4960        |             |       |
| total        |    |                |             |       |

The mean row would add  $n\bar{y}^2 = 972.307$  on one degree of freedom. The critical value is  $F_{0.05}(4, 25) = 2.759$ , and the  $p$ -value is  $1.2 \times 10^{-8}$ . The table says only that the treatment component of the mean is not zero; the contrasts of [Section 15.2](#) and [Section 15.4](#) say what it is. Since the data were simulated, the true noncentrality is known:  $\gamma = 103.4$ , far more than needed for power near one.

```
import numpy as np
from scipy import stats

rates = np.array([0, 40, 80, 120, 160])          # kg N per hectare
g, m = len(rates), 6                             # 5 rates, 6 plots per rate
true_means = 6.8 - 2.7 * np.exp(-0.018 * rates)  # unknown to the analyst
rng = np.random.default_rng(1515)
group = np.repeat(np.arange(g), m)               # plot i received rate group[i]
y = np.round(true_means[group] + rng.normal(0, 0.5, g * m), 2) # yield, t/ha
n = len(y)
Z = (group[:, None] == np.arange(g)).astype(float) # cell-means model matrix
n_k = Z.sum(axis=0)

M = Z @ np.diag(1 / n_k) @ Z.T
P0 = np.full((n, n), 1 / n)                     # projection onto span(1)
ss_mean = y @ P0 @ y                            # n * ybar^2
ss_between = y @ (M - P0) @ y                    # sum_k n_k (ybar_k - ybar)^2
ss_within = y @ (np.eye(n) - M) @ y              # sum_kj (y_kj - ybar_k)^2
df_between, df_within = g - 1, n - g
F = (ss_between / df_between) / (ss_within / df_within)
p_value = stats.f.sf(F, df_between, df_within)
print(f"{'source':10s} {'df':>3s} {'SS':>9s} {'MS':>8s}")
print(f"{'rates':10s} {'df_between':3d} {'ss_between':9.4f} {'ss_between / df_between':8.4f}")
print(f"{'plots':10s} {'df_within':3d} {'ss_within':9.4f} {'ss_within / df_within':8.4f}")
print(f"{'corrected':10s} {'n - 1':3d} {'ss_between + ss_within':9.4f}")
print(f"F = {F:.2f} on ({df_between}, {df_within}) df, p = {p_value:.2g}")
```

### 15.3.5 Three ways to read $F$

**As a proportion of variation.** Let  $\eta^2 = \text{SSB}/(\text{SSB} + \text{SSE})$ , the fraction of the corrected total sum of squares that lies in the treatment space. It is the  $R^2$  of the regression of  $\mathbf{y}$  on the group indicators, and the sample version of the correlation ratio of [Eq. 14.2.1](#). By [Theorem 9.5.2](#),

$$F = \frac{\eta^2/(g-1)}{(1-\eta^2)/(n-g)},$$

and in the normal model, under equal means,  $\eta^2$  has the Beta law of that theorem, with  $E(\eta^2) = (g-1)/(n-1)$ , not small with many groups and few observations. In the nitrogen trial  $\eta^2 = 0.809$ : four fifths of the variation among plots is variation among rates.

**As the largest contrast.** By Proposition 15.2.1(d), every contrast has  $SS(c) \leq SSB$ , with equality for  $c_k \propto n_k(\bar{y}_k - \bar{y})$ . Dividing by  $s^2$ , the largest squared  $t$  statistic over all contrasts is  $(g-1)F$ . Hence the  $F$  test rejects iff some contrast, perhaps not an interesting one, is declared nonzero by Scheffé's method (Corollary 13.2.3).

**As a two-sample test, for  $g = 2$ .** Then  $\mathcal{T}$  is a line, the only contrast is  $\mu_1 - \mu_2$  up to scale, and  $F = T^2$  for the pooled two-sample  $t$  statistic of Exercise (B2), whose sensitivity to unequal variances the one-way test inherits (Section 15.5).

## 15.3.6 Exercises

### 15.3.6.1 A. Check your understanding

#### Exercise A1.

A one-way experiment with four treatments and six observations per treatment gives a treatment mean square of 12.0 and a corrected total sum of squares of 96.0. Complete the analysis of variance table, compute  $F$  and  $\eta^2$ , and state the reference distribution of  $F$  under equal means.

*Solution.* The degrees of freedom are 3, 20 and 23. The treatment sum of squares is  $3 \times 12 = 36$ , so the error sum of squares is  $96 - 36 = 60$  and  $MSE = 3$ . Then  $F = 12/3 = 4$ ,  $\eta^2 = 36/96 = 0.375$ , and under equal means  $F \sim F(3, 20)$ .

#### Exercise A2.

Show from Eq. 15.3.2 that  $SSB = \sum_k T_k^2/n_k - T^2/n$  and  $SSE = \sum_{k,j} y_{kj}^2 - \sum_k T_k^2/n_k$ .

### 15.3.6.2 B. Practice

#### Exercise B1.

For  $g = 2$  with  $n_1 + n_2 = n$  fixed, show that  $\gamma = (\mu_1 - \mu_2)^2 n_1 n_2 / (n\sigma^2)$ , and that it is largest for  $n_1 = n_2$ . By how much does the noncentrality fall for the split  $n_1 = n/4$ ,  $n_2 = 3n/4$ ?

*Solution.* With  $\bar{\mu} = (n_1\mu_1 + n_2\mu_2)/n$ ,  $\mu_1 - \bar{\mu} = n_2(\mu_1 - \mu_2)/n$  and  $\mu_2 - \bar{\mu} = -n_1(\mu_1 - \mu_2)/n$ . So  $\sigma^2\gamma = (\mu_1 - \mu_2)^2(n_1n_2^2 + n_2n_1^2)/n^2 = (\mu_1 - \mu_2)^2 n_1 n_2 / n$ . The product  $n_1 n_2$  with fixed sum is largest at  $n_1 = n_2 = n/2$ , where it is  $n^2/4$ . For the split  $(n/4, 3n/4)$  it is  $3n^2/16$ , so the noncentrality falls by a quarter.

#### Exercise B2.

In the normal model with equal means  $E(\eta^2) = (g-1)/(n-1)$ , which is not zero, so  $\eta^2$



overstates the treatment variation in small experiments. Show that the numerator of

$$\hat{\omega}^2 = \frac{\text{SSB} - (g-1)\text{MSE}}{\text{SSB} + \text{SSE}}$$

has expectation  $\sum_k n_k(\mu_k - \bar{\mu})^2$ , whatever the means, so that it removes the bias from the numerator. When is  $\hat{\omega}^2$  negative?

*Solution.* By [Theorem 15.3.2](#),  $E(\text{SSB}) = (g-1)\sigma^2 + \sum_k n_k(\mu_k - \bar{\mu})^2$  and  $E\{(g-1)\text{MSE}\} = (g-1)\sigma^2$ , so the difference has the stated expectation. The numerator is negative iff  $\text{MSB} < \text{MSE}$ , that is, iff  $F < 1$ , which often has null probability above one half: 0.57 for the nitrogen design's  $F(4, 25)$  in the normal model. The ratio itself is not unbiased, since its denominator is random.

### Exercise B3.

In the nitrogen trial, compute the Scheffé-maximal contrast  $c_k = n_k(\bar{y}_k - \bar{y})$  from the rate means, and verify that its squared  $t$  statistic is  $4F$ . Describe in words what it compares.

### 15.3.6.3 C. Going deeper

#### Exercise C1.

In the random-effects model  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I} + \sigma_a^2\mathbf{Z}\mathbf{Z}^\top$ ,  $E(\mathbf{Y}) = \mu\mathbf{1}$ , use [Theorem 2.3.1](#) to show that  $E(\text{SSB}) = (g-1)\sigma^2 + (n - \sum_k n_k^2/n)\sigma_a^2$  and  $E(\text{SSE}) = (n-g)\sigma^2$ . Show that  $n_0 = (n - \sum_k n_k^2/n)/(g-1) \leq n/g$ , with equality iff the layout is balanced.

*Solution.* By [Theorem 2.3.1](#),  $E(\mathbf{Y}^\top\mathbf{A}\mathbf{Y}) = \text{tr}(\mathbf{A}\Sigma) + \boldsymbol{\theta}^\top\mathbf{A}\boldsymbol{\theta}$ , and  $\boldsymbol{\theta} = \mu\mathbf{1}$  is annihilated by both  $\mathbf{P}_\mathcal{T}$  and  $\mathbf{I} - \mathbf{M}$ . Now  $\text{tr}(\mathbf{P}_\mathcal{T}) = g-1$  and  $\text{tr}(\mathbf{P}_\mathcal{T}\mathbf{Z}\mathbf{Z}^\top) = \text{tr}(\mathbf{Z}^\top\mathbf{M}\mathbf{Z}) - \text{tr}(\mathbf{Z}^\top\mathbf{P}_0\mathbf{Z})$ . Since  $\mathbf{M}\mathbf{Z} = \mathbf{Z}$ , the first term is  $\text{tr}(\mathbf{Z}^\top\mathbf{Z}) = n$ . The second is  $n^{-1}\|\mathbf{Z}^\top\mathbf{1}\|^2 = \sum_k n_k^2/n$ . For the error,  $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$ , so only  $\sigma^2\text{tr}(\mathbf{I} - \mathbf{M}) = (n-g)\sigma^2$  remains. Finally  $\sum_k n_k^2 \geq n^2/g$  by the Cauchy-Schwarz inequality, with equality iff all  $n_k = n/g$ , so  $(g-1)n_0 \leq n - n/g = (g-1)n/g$ .

#### Exercise C2.

Show that  $F$  is unchanged when  $\mathbf{y}$  is replaced by  $a\mathbf{y} + b\mathbf{1}$  with  $a \neq 0$ , and when the observations are permuted within groups. Show that it changes when a constant is added to the observations of one group only, and explain why it should.

## 15.4 Orthogonal contrasts

Once the  $F$  test of [Section 15.3](#) says that the treatment component of the mean is not zero, the next step is to split that component into interpretable pieces. A complete set of orthogonal contrasts chooses an orthogonal basis of the treatment space, one direction per question, and splits the between-groups sum of squares along it. [Chapter 8](#) proved the arithmetic ([Propo-](#)

sition 8.7.2) and Chapter 9 placed it in the theory of single-degree-of-freedom decompositions (Theorem 9.4.4). This section adds the distribution theory, builds orthogonal sets, and analyses the shape of the nitrogen response.

### 15.4.1 The decomposition

Contrasts  $\mathbf{c}$  and  $\mathbf{d}$  are orthogonal for the design if  $\sum_k c_k d_k / n_k = 0$  (Definition 8.7.1), which says exactly that  $\mathbf{u}_c \perp \mathbf{u}_d$  (Proposition 8.7.2(c)). Since  $\dim \mathcal{T} = g - 1$ , a set of  $g - 1$  pairwise orthogonal contrasts is as large as possible; we call it **complete**.

#### Theorem 15.4.1 — Orthogonal contrasts decompose the treatment sum of squares

In the one-way model, let  $\mathbf{c}_1, \dots, \mathbf{c}_{g-1}$  be contrasts that are pairwise orthogonal for the design,

$$\sum_{k=1}^g \frac{c_{ik} c_{lk}}{n_k} = 0 \quad (i \neq l), \quad (15.4.1)$$

and write  $\psi_i = \sum_k c_{ik} \mu_k$  and  $F_i = \text{SS}(\mathbf{c}_i) / s^2$ .

- (a) The vectors  $\mathbf{u}_{\mathbf{c}_1}, \dots, \mathbf{u}_{\mathbf{c}_{g-1}}$  form an orthogonal basis of the treatment space  $\mathcal{T}$ , and

$$\text{SSB} = \sum_{i=1}^{g-1} \text{SS}(\mathbf{c}_i).$$

- (b) In the normal one-way model,  $\text{SS}(\mathbf{c}_1), \dots, \text{SS}(\mathbf{c}_{g-1})$  and SSE are mutually independent, and  $\text{SS}(\mathbf{c}_i) / \sigma^2 \sim \chi^2(1, \gamma_i)$  with  $\gamma_i = \psi_i^2 / (\sigma^2 v_{\mathbf{c}_i})$ . The noncentralities add up to that of the  $F$  test:  $\sum_i \gamma_i = \gamma$  of Eq. 15.3.3.
- (c) The  $F$  statistic is the average of the single-degree-of-freedom statistics:  $F = (g - 1)^{-1} \sum_i F_i$ .
- (d) In a balanced layout, Eq. 15.4.1 reduces to  $\sum_k c_{ik} c_{lk} = 0$ .
- (e) Conversely, every orthogonal basis  $\mathbf{u}_1, \dots, \mathbf{u}_{g-1}$  of  $\mathcal{T}$  arises in this way: if  $\mathbf{u}_i$  takes the value  $v_{ik}$  on group  $k$ , then  $c_{ik} = n_k v_{ik}$  defines a complete set of orthogonal contrasts with  $\mathbf{u}_{\mathbf{c}_i} = \mathbf{u}_i$ .

*Proof.* Part (a) is Proposition 8.7.2(d), whose proof shows that the  $\mathbf{u}_{\mathbf{c}_i}$  are nonzero, pairwise orthogonal vectors of  $\mathcal{T}$ , hence a basis, and that the projections  $\mathbf{P}_i$  onto the lines they span sum to  $\mathbf{P}_{\mathcal{T}}$ . (b) The  $\mathbf{P}_i$  are mutually orthogonal projections of rank one, and each is orthogonal to  $\mathbf{I} - \mathbf{M}$  because its range lies in  $\mathcal{C}(\mathbf{Z})$ . By Theorem 4.4.4 the sums of squares  $\|\mathbf{P}_i \mathbf{Y}\|^2 = \text{SS}(\mathbf{c}_i)$  (Proposition 15.2.1(c)) and SSE are independent, with noncentralities  $\|\mathbf{P}_i \boldsymbol{\theta}\|^2 / \sigma^2$ . By the computation in the proof of Proposition 15.2.2,  $\|\mathbf{P}_i \boldsymbol{\theta}\|^2 = (\mathbf{u}_{\mathbf{c}_i}^\top \boldsymbol{\theta})^2 / \|\mathbf{u}_{\mathbf{c}_i}\|^2 = \psi_i^2 / v_{\mathbf{c}_i}$ . Since  $\sum_i \mathbf{P}_i = \mathbf{P}_{\mathcal{T}}$  and the  $\mathbf{P}_i \boldsymbol{\theta}$  are orthogonal,  $\sum_i \|\mathbf{P}_i \boldsymbol{\theta}\|^2 = \|\mathbf{P}_{\mathcal{T}} \boldsymbol{\theta}\|^2$ , which is  $\sigma^2 \gamma$  by Theorem 15.3.1. (c) Divide (a) by  $(g - 1)s^2$ . (d) Put  $n_k = m$ . (e) In Section 15.2 we saw that each  $\mathbf{u} \in \mathcal{T}$  equals  $\mathbf{u}_c$  for the contrast

$c_k = n_k v_k$ , and by Proposition 8.7.2(c),  $\sum_k c_{ik} c_{lk} / n_k = \mathbf{u}_{\mathbf{c}_i}^\top \mathbf{u}_{\mathbf{c}_l}$ , which is zero for  $i \neq l$ . ■

By part (e) there are infinitely many complete orthogonal sets, one for each orthogonal basis of  $\mathcal{T}$ , and by part (a) all of them split the same total. The individual pieces depend on the basis (Exercise (C1)): choosing the basis is choosing the questions, and the design of the study, not the mathematics, should make that choice.

*Warning.* Orthogonality is a convenience, not a requirement. Planned contrasts that are not orthogonal are valid, each with the test and interval of Proposition 15.2.1. What is lost is additivity: their sums of squares need not total SSB, and their estimates are correlated. No contrast should be replaced by a less relevant one merely to achieve orthogonality.

## 15.4.2 Building orthogonal sets

Three constructions cover most designed experiments.

**Splitting the treatments in stages.** Many questions form a hierarchy: control against treatments, then old treatments against new, then one new formulation against another. Each splits a set of levels into two parts, and the comparisons are orthogonal for any group sizes provided each part is summarized by the mean of all its observations.

### Proposition 15.4.2 — Contrasts from successive splits

For a set  $S$  of levels split into disjoint nonempty parts  $A$  and  $B$ , let  $N_A = \sum_{k \in A} n_k$  and  $N_B = \sum_{k \in B} n_k$ , and define the contrast

$$c_k^{(S)} = \begin{cases} n_k/N_A, & k \in A, \\ -n_k/N_B, & k \in B, \\ 0, & k \notin S. \end{cases}$$

Its estimate is  $\bar{y}_A - \bar{y}_B$ , the difference between the means of all observations in  $A$  and in  $B$ , and

$$\text{SS}(\mathbf{c}^{(S)}) = \frac{N_A N_B}{N_A + N_B} (\bar{y}_A - \bar{y}_B)^2.$$

- (a) Every contrast  $\mathbf{d}$  that is zero outside  $A$  (or outside  $B$ ) is orthogonal to  $\mathbf{c}^{(S)}$  for the design.
- (b) Split the set of all  $g$  levels into two parts, split each part with more than one level into two, and continue until every part is a single level. The  $g - 1$  contrasts of the splits form a complete set of orthogonal contrasts, whatever the group sizes.

*Proof.* The estimate is  $\sum_{k \in A} n_k \bar{y}_k / N_A - \sum_{k \in B} n_k \bar{y}_k / N_B = \bar{y}_A - \bar{y}_B$ , and  $v_{\mathbf{c}} = \sum_{k \in A} n_k / N_A^2 + \sum_{k \in B} n_k / N_B^2 = 1/N_A + 1/N_B$ , which gives the sum of squares. (a) If  $\mathbf{d}$  is zero outside  $A$ , then  $\sum_k c_k^{(S)} d_k / n_k = \sum_{k \in A} (n_k / N_A) d_k / n_k = N_A^{-1} \sum_{k \in A} d_k = 0$ , since  $\mathbf{d}$  is a contrast; the case of  $B$  is the same. (b) A sequence of splits ending in single levels is a binary tree with  $g$  leaves, so it has  $g - 1$  internal nodes, one split each. Take two different splits,

of sets  $S$  and  $S'$ . By construction either  $S$  and  $S'$  are disjoint, in which case the two contrasts have disjoint supports and are orthogonal, or one of them, say  $S'$ , lies inside one part of the split of  $S$ , in which case (a) applies. So the  $g - 1$  contrasts are pairwise orthogonal, and they form a complete set. ■

The weighting matters: comparing plain averages of the levels in  $A$  and  $B$  gives orthogonal contrasts, in general, only in a balanced layout, as [Example \(Splitting the education sum of squares\)](#) showed numerically. The Helmert contrasts of [Exercise \(B3\)](#), which compare each level with the average of those before it, are the special case of the proposition that splits off one level at a time; with the weights  $n_k/N_A$  they stay orthogonal for unequal group sizes.

**Factorial treatments.** Four treatments may be the combinations of two fertilizers, each present or absent: none,  $P$  only,  $K$  only, and both. The contrasts

$$\begin{aligned} \text{effect of } P &: (-1, 1, -1, 1), & \text{effect of } K &: (-1, -1, 1, 1), \\ \text{interaction} &: (1, -1, -1, 1) \end{aligned}$$

are orthogonal in a balanced layout. The third asks whether the effect of  $P$  depends on  $K$ . A one-way analysis with these contrasts is a two-factor analysis in disguise; [Chapter 16](#) treats such layouts directly and defines interaction in general ([Definition 16.2.1](#)).

**Polynomial trends.** When the levels are values of a quantitative variable, such as a dose or a fertilizer rate, the orthogonal polynomial contrasts describe the shape of the response one degree at a time. For equally spaced levels and equal group sizes they are the integer vectors obtained from Gram–Schmidt applied to  $1, s, s^2, \dots$  in the level scores ([Section 8.7](#), [Section 9.4](#)). For five levels they are

$$\begin{aligned} \text{linear} & (-2, -1, 0, 1, 2), & \text{quadratic} & (2, -1, -2, -1, 2), \\ \text{cubic} & (-1, 2, 0, -2, 1), & \text{quartic} & (1, -4, 6, -4, 1). \end{aligned}$$

The degree- $d$  contrast is orthogonal to every polynomial of lower degree in the score, so it vanishes whenever the means lie on a polynomial of degree less than  $d$ . With unequal spacing the Gram–Schmidt step uses the actual level values ([Exercise \(B2\)](#)), and with unequal group sizes it uses the weighted inner product  $\sum_k n_k p_k q_k$  and the contrasts  $c_k = n_k p_k$  ([Example \(Trend in ideology across party identification\)](#)).

### 15.4.3 Trend analysis of the nitrogen trial

**Example. (The shape of the nitrogen response)**

The five rates are equally spaced and the design is balanced, so the integer contrasts above apply. The four single-degree-of-freedom sums of squares, each tested against  $s^2 = 0.2715$  on 25 degrees of freedom, are:

| Component     | Sum of squares       | $F$                  | $p$                   | Share of SSB |
|---------------|----------------------|----------------------|-----------------------|--------------|
| linear        | 25.9121              | 95.42                | $5.1 \times 10^{-10}$ | 90.3%        |
| quadratic     | 2.7904               | 10.28                | 0.0037                | 9.7%         |
| cubic         | 0.0049               | 0.02                 | 0.89                  | 0.0%         |
| quartic       | $8.6 \times 10^{-6}$ | $3.2 \times 10^{-5}$ | 0.996                 | 0.0%         |
| rates (total) | 28.7074              | 26.43                |                       |              |

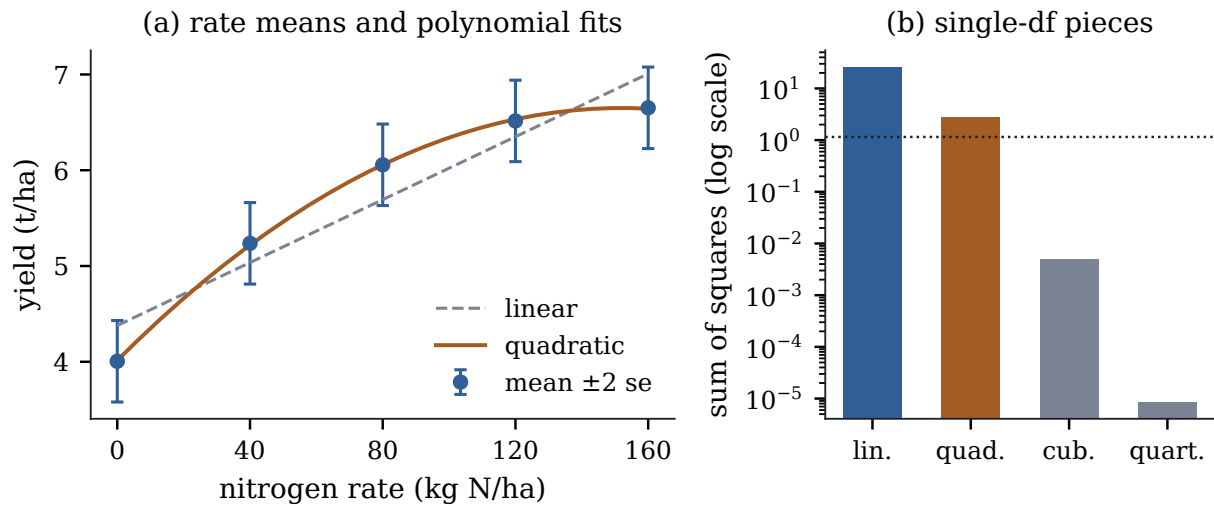


Figure 15.4.1: Trend analysis of the nitrogen trial. (a) Rate means with bars of two standard errors, and the least squares linear and quadratic curves in the rate fitted to all thirty plots. (b) The four orthogonal polynomial sums of squares (logarithmic scale); the dotted line is the value a single degree of freedom needs to be significant at level 0.05,  $s^2 F_{0.05}(1, 25)$ .

The pieces add to the between-rates sum of squares of [Example \(The analysis of variance for the nitrogen trial\)](#), and  $F$  is the average of the four single-degree-of-freedom  $F$  values. Yield rises with the rate and the rise flattens: the quadratic component is clearly significant. The cubic and quartic components together test whether a quadratic in the rate describes the five means; their sum of squares 0.0049 on two degrees of freedom gives  $F = 0.009$  and  $p = 0.99$  ([Figure 15.4.1](#)).

```
import numpy as np
from scipy import stats

rates = np.array([0, 40, 80, 120, 160])          # kg N per hectare
g, m = len(rates), 6                             # 5 rates, 6 plots per rate
true_means = 6.8 - 2.7 * np.exp(-0.018 * rates)  # unknown to the analyst
rng = np.random.default_rng(1515)
group = np.repeat(np.arange(g), m)               # plot i received rate group[i]
y = np.round(true_means[group] + rng.normal(0, 0.5, g * m), 2) # yield, t/ha
n = len(y)

means = np.array([y[group == k].mean() for k in range(g)])
nu = n - g
s2 = np.sum((y - means[group]) ** 2) / nu
ss_between = m * np.sum((means - means.mean()) ** 2)

C_poly = np.array([[-2, -1, 0, 1, 2],           # linear
                   [2, -1, -2, -1, 2],         # quadratic
                   [-1, 2, 0, -2, 1],          # cubic
                   [1, -4, 6, -4, 1]], float)    # quartic

def contrast_ss(c):
    return (c @ means) ** 2 / (c @ c / m)        # balanced: sum c_k^2 / n_k = c'
    c / m

ss_poly = np.array([contrast_ss(c) for c in C_poly])
F_poly = ss_poly / s2
```

```

p_poly = stats.f.sf(F_poly, 1, nu)
for name, ss, F, p in zip(["linear", "quadratic", "cubic", "quartic"], ss_poly,
    F_poly, p_poly):
    print(f"{name:9s} SS = {ss:8.4f} F = {F:7.2f} p = {p:.2g}")
print(f"sum = {ss_poly.sum():.4f}    between rates = {ss_between:.4f}")

```

The test of the cubic and quartic components is not a new test. Every model in which the mean depends on the rate alone has its mean space inside  $\mathcal{C}(\mathbf{Z})$  (Lemma 14.6.1), so the one-way residual sum of squares is the pure-error sum of squares of Section 14.6. The quadratic regression on the rate misses exactly the cubic and quartic directions of  $\mathcal{C}(\mathbf{Z})$ , so their sum of squares is its lack-of-fit sum of squares, and the trend test is the pure-error test of Theorem 14.6.2 (Exercise (C2)). The listing checks the identity: the quadratic regression has residual sum of squares 6.7935, the lack of fit plus the within-rates sum of squares.

```

ss_lof = ss_poly[2] + ss_poly[3]                                # cubic + quartic, 2 df
F_lof = (ss_lof / 2) / s2
X2 = np.column_stack([np.ones(n), rates[group], rates[group] ** 2])
coef, *_ = np.linalg.lstsq(X2, y, rcond=None)                  # quadratic regression on the
    rate
sse_quad = np.sum((y - X2 @ coef) ** 2)
print(f"lack of fit of the quadratic: SS = {ss_lof:.4f}, F = {F_lof:.3f},"
    f" p = {stats.f.sf(F_lof, 2, nu):.2f}")
print(f"quadratic regression SSE = {sse_quad:.4f} = lack of fit + within = {ss_lof +
    s2 * nu:.4f}")

```

The fitted quadratic is  $4.014 + 0.03466 N - 0.0001139 N^2$  t/ha at  $N$  kg N/ha. It has a maximum at  $N = 0.03466 / (2 \times 0.0001139) \approx 152$ , inside the range of rates, and it declines after that. That decline is a property of the parabola, not of the crop: the data were generated from a response that rises at every rate, and five noisy means cannot distinguish a parabola that turns over from a curve that levels off. A trend analysis describes the means at the rates used and should not be extrapolated.

### 15.4.4 One data set, two decompositions

A different set of questions gives a different complete set. Suppose the agronomist had planned to ask first whether fertilizer helps at all, and then how the response behaves among the fertilized rates. The contrast of the control against the four fertilized rates is a split of the kind in Proposition 15.4.2, with coefficients proportional to  $(-4, 1, 1, 1, 1)$ . The linear, quadratic and cubic polynomial contrasts among the four fertilized rates,  $(0, -3, -1, 1, 3)$ ,  $(0, 1, -1, -1, 1)$  and  $(0, -1, 3, -3, 1)$ , are zero at the control, so they are orthogonal to it by part (a) of that proposition and to each other by balance. Their sums of squares are 21.3701 ( $F = 78.70$ ), 6.6364 ( $F = 24.44$ ), 0.7004 ( $F = 2.58$ ,  $p = 0.12$ ) and 0.0005, and they add to the same 28.7074.

```

C_ctrl = np.array([[-4, 1, 1, 1, 1],                          # control against the four
    fertilized
                    [0, -3, -1, 1, 3],                          # linear among 40, ..., 160
                    [0, 1, -1, -1, 1],                          # quadratic among them
                    [0, -1, 3, -3, 1]], float)                  # cubic among them
ss_ctrl = np.array([contrast_ss(c) for c in C_ctrl])
print("second set:", ss_ctrl.round(4), " sum =", round(ss_ctrl.sum(), 4))

```

The two decompositions tell different but compatible stories. In the first, curvature is significant;

in the second, the curvature among the fertilized rates is not, because most of the bend is the jump from no nitrogen to 40 kg, which the second set assigns to the control contrast. Computing several decompositions and reporting the one whose pieces look best is the data snooping of [Section 13.6](#).

## 15.4.5 Exercises

### 15.4.5.1 A. Check your understanding

#### Exercise A1.

Check that the four polynomial contrasts for five levels are pairwise orthogonal. Show that the quadratic contrast vanishes when  $\mu_k = a + bk$ , and the cubic contrast vanishes when  $\mu_k = a + bk + ck^2$ .

#### Exercise A2.

For the four fertilizer combinations of “Factorial treatments”, verify that the three contrasts are orthogonal when every group has the same size, and that the  $P$  and  $K$  contrasts are not orthogonal for group sizes 3, 3, 3, 5 (in the order none,  $P$ ,  $K$ , both). What does the lack of orthogonality mean for the estimates?

*Solution.* With equal sizes, orthogonality is the plain inner product:  $(-1, 1, -1, 1) \cdot (-1, -1, 1, 1) = 1 - 1 - 1 + 1 = 0$ , and similarly for the other two pairs. With sizes 3, 3, 3, 5,  $\sum_k c_k d_k / n_k = \frac{1}{3} - \frac{1}{3} - \frac{1}{3} + \frac{1}{5} = -\frac{2}{15} \neq 0$ . By [Proposition 8.7.2\(c\)](#) the two estimates are then correlated, with covariance  $-\frac{2}{15}\sigma^2$ , and their sums of squares do not add to the part of SSB that the two contrasts span.

### 15.4.5.2 B. Practice

#### Exercise B1.

Show that the two planned contrasts of [Example \(Two planned contrasts\)](#),  $(-1, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4})$  and  $(0, 0, -1, 0, 1)$ , are orthogonal. Find two further contrasts that complete them to a complete orthogonal set, and say what they compare.

*Solution.* The design is balanced, and  $-1 \cdot 0 + \frac{1}{4}(0 - 1 + 0 + 1) = 0$ . A contrast  $\mathbf{e}$  orthogonal to the first satisfies  $-e_1 + \frac{1}{4}(e_2 + e_3 + e_4 + e_5) = 0$ , and since  $e_2 + e_3 + e_4 + e_5 = -e_1$  this forces  $e_1 = 0$ . Orthogonality to the second forces  $e_3 = e_5$ . So  $\mathbf{e} = (0, a, b, c, b)$  with  $a + 2b + c = 0$ . Two orthogonal choices are  $(0, 1, 0, -1, 0)$ , comparing 40 with 120, and  $(0, 1, -1, 1, -1)$ , comparing 40 and 120 together with 80 and 160 together. Their inner product is  $1 - 1 = 0$ . The completion is valid but not very natural: completing a planned set is optional.

#### Exercise B2.

Five plots at each of four rates 0, 1, 2 and 4 (in units of 50 kg N/ha) are to be analysed for trend. Show that the linear and quadratic orthogonal polynomial contrasts are

proportional to  $(-7, -3, 1, 9)$  and  $(7, -4, -8, 5)$ . Check that the quadratic contrast vanishes when the means are linear in the rate.

*Solution.* The rates have mean  $7/4$ , so the linear contrast is proportional to  $4(s_k - 7/4) = (-7, -3, 1, 9)$ . For the quadratic, centre  $s_k^2 = (0, 1, 4, 16)$  at its mean  $21/4$  and multiply by four:  $(-21, -17, -5, 43)$ . Its inner product with  $(-7, -3, 1, 9)$  is 580, and the linear vector has squared length 140, so subtract  $\frac{580}{140}(-7, -3, 1, 9)$  to get  $(8, -\frac{32}{7}, -\frac{64}{7}, \frac{40}{7})$ , which is proportional to  $(7, -4, -8, 5)$ . Its entries sum to zero and its inner product with the linear contrast is  $-49 + 12 - 8 + 45 = 0$ . For means  $a + bs_k$  the quadratic contrast gives  $b(0 - 4 - 16 + 20) = 0$ .

### Exercise B3.

Four equally spaced levels have group sizes 2, 4, 4, 2. Show that the classical linear contrast  $(-3, -1, 1, 3)$  is orthogonal for this design to the quadratic  $(1, -1, -1, 1)$  but not to the cubic  $(-1, 3, -3, 1)$ . Explain the first fact by symmetry.

*Solution.* For linear and quadratic,  $\sum_k c_k d_k / n_k = -\frac{3}{2} + \frac{1}{4} - \frac{1}{4} + \frac{3}{2} = 0$ . For linear and cubic, it is  $\frac{3}{2} - \frac{3}{4} - \frac{3}{4} + \frac{3}{2} = \frac{3}{2} \neq 0$ . The sizes are symmetric about the middle of the scale. The linear contrast is odd under the reflection  $k \mapsto 5 - k$ , the quadratic is even, and the weights  $1/n_k$  are even, so the weighted inner product of an odd and an even vector vanishes. Linear and cubic are both odd, and nothing forces their weighted inner product to vanish.

### Exercise B4.

Use the formula of [Proposition 15.4.2](#) to recompute the sum of squares of the control against the fertilized rates in the nitrogen trial from the rate means, and compare it with the value 21.3701 in the text.

*Solution.* Here  $N_A = 24$ ,  $N_B = 6$ ,  $\bar{y}_A = \frac{1}{4}(5.237 + 6.057 + 6.515 + 6.652) = 6.11525$  and  $\bar{y}_B = 4.005$ . The sum of squares is  $\frac{24 \cdot 6}{30}(6.11525 - 4.005)^2 = 4.8 \times 2.11025^2 \approx 21.375$ . The small difference from 21.3701 comes from using the rounded rate means.

## 15.4.5.3 C. Going deeper

### Exercise C1.

Let  $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_{g-1}]$  and  $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_{g-1}]$  be two orthonormal bases of the treatment space. Show that  $\mathbf{V} = \mathbf{U}\mathbf{Q}$  for an orthogonal  $(g-1) \times (g-1)$  matrix  $\mathbf{Q}$ , that the vectors of squared coordinates of  $\mathbf{y}$  in the two bases have the same sum SSB, and that the individual squared coordinates generally differ. More generally, show that for any symmetric idempotent  $\mathbf{A}$  of rank  $q$  and  $\mathbf{Y} \sim N_n(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ ,  $\mathbf{Y}^\top \mathbf{A} \mathbf{Y}$  splits into  $q$  independent scaled noncentral  $\chi^2(1)$  variables, one for each vector of an orthonormal basis of  $\mathcal{C}(\mathbf{A})$ .

*Solution.* Both  $\mathbf{U}\mathbf{U}^\top$  and  $\mathbf{V}\mathbf{V}^\top$  equal  $\mathbf{P}_{\mathcal{T}}$ , the unique projection onto  $\mathcal{T}$ . Put  $\mathbf{Q} = \mathbf{U}^\top \mathbf{V}$ . Then  $\mathbf{U}\mathbf{Q} = \mathbf{P}_{\mathcal{T}} \mathbf{V} = \mathbf{V}$  and  $\mathbf{Q}^\top \mathbf{Q} = \mathbf{V}^\top \mathbf{P}_{\mathcal{T}} \mathbf{V} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}$ . The coordinate vectors satisfy  $\mathbf{V}^\top \mathbf{y} = \mathbf{Q}^\top \mathbf{U}^\top \mathbf{y}$ , and an orthogonal matrix preserves length, so both



have squared length  $\mathbf{y}^\top \mathbf{P}_\tau \mathbf{y} = \text{SSB}$ . The individual coordinates are mixed by  $\mathbf{Q}$  and generally change. For the general statement, write  $\mathbf{A} = \sum_{i=1}^q \mathbf{w}_i \mathbf{w}_i^\top$  with  $\mathbf{w}_1, \dots, \mathbf{w}_q$  an orthonormal basis of  $\mathcal{C}(\mathbf{A})$  (Proposition 6.3.2). The rank-one projections  $\mathbf{w}_i \mathbf{w}_i^\top$  are mutually orthogonal, so by Theorem 4.4.4 the terms  $(\mathbf{w}_i^\top \mathbf{Y})^2$  are independent, and  $(\mathbf{w}_i^\top \mathbf{Y})^2 / \sigma^2 \sim \chi^2(1, (\mathbf{w}_i^\top \boldsymbol{\theta})^2 / \sigma^2)$ . Eigenvectors of  $\mathbf{A}$  for the eigenvalue one are one such basis.

### Exercise C2.

For  $g$  equally spaced levels with equal group sizes, show that the sum of the sums of squares of the polynomial contrasts of degrees  $d+1, \dots, g-1$  is the lack-of-fit sum of squares of the polynomial regression of degree  $d < g-1$  on the level score, and that the sum for degrees  $1, \dots, d$  is that regression's sum of squares after the mean.

*Solution.* Let  $\mathcal{P}_d \subseteq \mathcal{C}(\mathbf{Z})$  be the mean space of the degree- $d$  regression, spanned by the vectors whose entry in group  $k$  is  $s_k^i$ ,  $i = 0, \dots, d$  (Lemma 14.6.1). The contrast vector of the degree- $i$  orthogonal polynomial  $p_i$  has entry  $p_i(s_k)/m$  in group  $k$ , and in a balanced layout orthogonality of these vectors is orthogonality of the  $p_i$  in the plain inner product. By construction  $p_0, \dots, p_d$  span the polynomials of degree at most  $d$  on the scores, so the vectors of degrees  $1, \dots, d$  span  $\mathcal{P}_d \cap \mathbf{1}^\perp$ , and those of degrees  $d+1, \dots, g-1$  span  $\mathcal{C}(\mathbf{Z}) \cap \mathcal{P}_d^\perp$ . The projection onto the first space gives the regression sum of squares after the mean (Theorem 6.5.1), and the projection onto the second gives  $\|(\mathbf{M} - \mathbf{P}_{\mathcal{P}_d})\mathbf{y}\|^2$ , the lack-of-fit sum of squares of Theorem 14.6.2. Each projection is the sum of the projections onto the lines, by Theorem 6.3.8.

## 15.5 Balanced and unbalanced one-way layouts

Every result so far holds for any group sizes, yet experimenters aim for balance. This section says what it buys: plain orthogonality, the largest guaranteed power for a fixed number of observations, and a test of equal means that tolerates unequal variances. The third matters most in practice, because unbalanced data with unequal variances are common and the classical  $F$  test can then go badly wrong.

### 15.5.1 Orthogonality and simplicity

In a balanced layout, orthogonality of contrasts is plain orthogonality  $\sum_k c_k d_k = 0$  (Theorem 15.4.1(d)), so tabulated sets such as the integer polynomial contrasts can be used as they stand. The side conditions  $\sum_k \alpha_k = 0$  and  $\sum_k n_k \alpha_k = 0$  define the same parameters, the pooled variance is the plain average of the group variances, and every pairwise difference has the standard error  $s\sqrt{2/m}$ . Tukey's intervals have exact simultaneous coverage (Theorem 13.3.2), whereas with unequal sizes the Tukey–Kramer intervals (Eq. 13.3.3) are conservative by an unknown amount.

## 15.5.2 Balance maximizes guaranteed power

The noncentrality [Eq. 15.3.3](#) depends on the group sizes as well as the means. No allocation of a fixed total is best for every configuration of means, and the planner does not know the means, so the relevant criterion is the power *guaranteed* against every configuration whose means span at least a range  $\Delta$ , as in [Proposition 11.5.2](#). For unequal group sizes, [Exercise \(B1\)](#) already identifies the least favourable configuration; what is new here is the comparison of allocations.

### Proposition 15.5.1 — Balance maximizes the guaranteed noncentrality

In a one-way layout with group sizes  $n_1, \dots, n_g$ ,  $g \geq 2$ , let

$$h = \min_{k \neq l} \frac{n_k n_l}{n_k + n_l} = \min_{k \neq l} \left( \frac{1}{n_k} + \frac{1}{n_l} \right)^{-1}.$$

- (a) For all means with  $\max_k \mu_k - \min_k \mu_k \geq \Delta > 0$ , the noncentrality of the  $F$  test satisfies  $\gamma \geq h\Delta^2/\sigma^2$ , and there is a configuration with range exactly  $\Delta$  at which equality holds.
- (b) For fixed  $n = \sum_k n_k$ ,  $h \leq n/(2g)$ , with equality iff the layout is balanced.

*Proof.* Part (a) is [Exercise \(B1\)](#), rewritten with  $h$ : its bound is  $\Delta^2/(\sigma^2(1/n_{(1)} + 1/n_{(2)})) = h\Delta^2/\sigma^2$ , since  $1/n_k + 1/n_l$  is largest for the two smallest groups. Equality holds when those two groups carry means  $\Delta$  apart and every other mean equals their weighted average.

For (b): as just noted,  $1/n_k + 1/n_l$  is largest for the two smallest groups. Call their sizes  $a \leq b$ . The other  $g-2$  groups have at least  $b$  observations each, so  $n \geq a + (g-1)b$ , and

$$\left( \frac{1}{a} + \frac{1}{b} \right) n \geq \left( \frac{1}{a} + \frac{1}{b} \right) (a + (g-1)b) = g + (g-1)t + \frac{1}{t},$$

where  $t = b/a \geq 1$ . Now  $(g-1)t + 1/t - g = ((g-1)t - 1)(t-1)/t \geq 0$ , because both factors are nonnegative when  $t \geq 1$  and  $g \geq 2$ . Hence  $1/a + 1/b \geq 2g/n$ , that is,  $h \leq n/(2g)$ . Equality requires  $t = 1$  and  $n = a + (g-1)b$ , so all groups have the same size. Conversely, a balanced layout has  $h = n/2 = n/(2g)$ . ■

A departure is hardest to detect when it sits in the least informative pair of groups. The balanced value  $h = n/2$  is the bound of [Proposition 11.5.2](#).

#### Example. (Forty observations in four groups)

With  $n = 40$  and  $g = 4$ , the guaranteed noncentrality per unit of  $(\Delta/\sigma)^2$  is 5.000 for the balanced allocation 10, 10, 10, 10, 3.938 for the mildly unbalanced 7, 9, 11, 13, and 2.400 for 4, 6, 10, 20. At  $\Delta = \sigma$  the guaranteed powers of the level-0.05 test are 0.399, 0.320 and 0.205. To guarantee power 0.8, the range of the means must be at least  $1.56\sigma$ ,  $1.76\sigma$  and  $2.25\sigma$  respectively ([Figure 15.5.1](#)). The mild imbalance gives the same

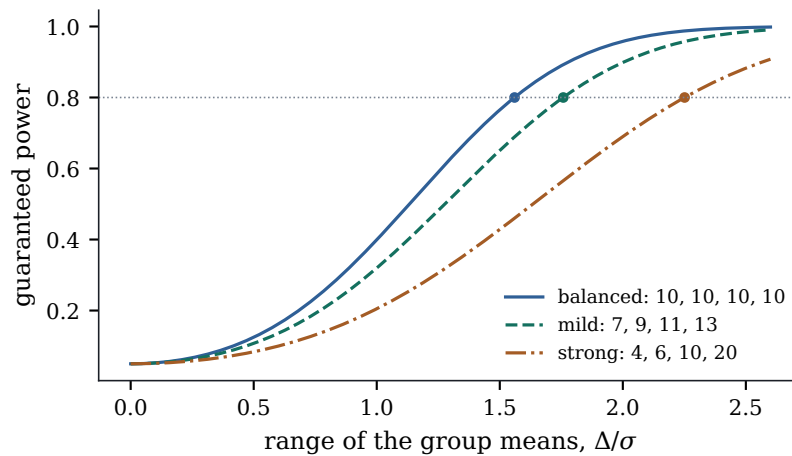


Figure 15.5.1: Guaranteed power of the level-0.05  $F$  test with 40 observations in four groups, against the range of the group means in units of  $\sigma$ , for three allocations. The curves are the power at the least favourable configuration of Proposition 15.5.1; dots mark where each reaches 0.8.

guarantee as a balanced design with  $8 \times 3.938 \approx 31.5$  observations, so it wastes a fifth of the experiment.

```
import itertools

import numpy as np
from scipy import stats

def guaranteed_gamma(sizes, delta_over_sigma):
    """Smallest noncentrality over all mean configurations with range at least Delta
    """
    h = min(a * b / (a + b) for a, b in itertools.combinations(sizes, 2))
    return delta_over_sigma ** 2 * h

def power_f(gamma, q, nu, alpha=0.05):
    return stats.ncf.sf(stats.f.ppf(1 - alpha, q, nu), q, nu, gamma)

allocations = {"balanced": [10, 10, 10, 10], "mild": [7, 9, 11, 13], "strong": [4,
    6, 10, 20]}
for name, sizes in allocations.items():
    gam = guaranteed_gamma(sizes, 1.0)
    print(f"{name:9s} {sizes}: guaranteed gamma = {gam:.3f},"
        f" guaranteed power at Delta = sigma: {power_f(gam, 3, 36):.3f}")
```

Balance is optimal for the overall test and for the family of all pairwise comparisons. It is not optimal for every purpose. A single planned contrast is estimated most precisely with  $n_k \propto |c_k|$  (Exercise (B2)), and comparisons with a control favour a larger control group (Exercise (B3)). The allocation should follow the questions.

### 15.5.3 Unequal variances

The one-way model assumes a common variance. Suppose instead that the observations in group  $k$  have variance  $\sigma_k^2$ , with everything else as before. What do the two mean squares then

estimate?

### Theorem 15.5.2 — Expected mean squares under unequal variances

Let the  $Y_{kj}$  be uncorrelated with  $E(Y_{kj}) = \mu_k$  and  $\text{Var}(Y_{kj}) = \sigma_k^2$ , and write  $\bar{n} = n/g$ . Then:

- (a)  $E(\text{SSE}) = \sum_k (n_k - 1)\sigma_k^2$ , so  $E(\text{MSE})$  is the average of the  $\sigma_k^2$  weighted by  $n_k - 1$ ;
- (b)  $E(\text{SSB}) = \sum_k (1 - n_k/n)\sigma_k^2 + \sum_k n_k(\mu_k - \bar{\mu})^2$ ;
- (c) when the means are equal,

$$E(\text{MSB}) - E(\text{MSE}) = \frac{g(n-1)}{n(g-1)(n-g)} \sum_{k=1}^g (\bar{n} - n_k) \sigma_k^2. \quad (15.5.1)$$

In a balanced layout this is zero, and both mean squares estimate  $g^{-1} \sum_k \sigma_k^2$ . In an unbalanced layout it is positive when the larger variances belong to the smaller groups and negative when they belong to the larger groups.

*Proof.* The argument of [Theorem 9.6.1](#) applies with a diagonal covariance. Now  $\text{Cov}(\mathbf{Y}) = \mathbf{\Sigma}$ , diagonal with  $\sigma_k^2$  on the observations of group  $k$ , and  $E(\mathbf{Y}) = \boldsymbol{\theta}$ . By [Theorem 2.3.1](#),  $E(\mathbf{Y}^\top \mathbf{A} \mathbf{Y}) = \text{tr}(\mathbf{A} \mathbf{\Sigma}) + \boldsymbol{\theta}^\top \mathbf{A} \boldsymbol{\theta}$ . (a) With  $\mathbf{A} = \mathbf{I} - \mathbf{M}$ , whose diagonal entries in group  $k$  are  $1 - 1/n_k$ ,  $\text{tr}(\mathbf{A} \mathbf{\Sigma}) = \sum_k n_k(1 - 1/n_k)\sigma_k^2$ , and  $\mathbf{A} \boldsymbol{\theta} = \mathbf{0}$ . (b) With  $\mathbf{A} = \mathbf{M} - \mathbf{P}_0$ , the diagonal entries in group  $k$  are  $1/n_k - 1/n$ , so  $\text{tr}(\mathbf{A} \mathbf{\Sigma}) = \sum_k (1 - n_k/n)\sigma_k^2$ , and  $\boldsymbol{\theta}^\top \mathbf{A} \boldsymbol{\theta} = \sum_k n_k(\mu_k - \bar{\mu})^2$  as in [Theorem 15.3.1](#). (c) With equal means, multiply the difference of the two expectations by  $(g-1)(n-g)$ :

$$\begin{aligned} & (n-g) \sum_k (1 - n_k/n) \sigma_k^2 - (g-1) \sum_k (n_k - 1) \sigma_k^2 \\ &= \sum_k (n-1 - g(n-1)n_k/n) \sigma_k^2 = \frac{g(n-1)}{n} \sum_k (\bar{n} - n_k) \sigma_k^2, \end{aligned}$$

where the middle step collects the coefficient of  $\sigma_k^2$ :  $(n-g) - (n-g)n_k/n - (g-1)n_k + (g-1) = n-1 - gn_k + gn_k/n$ . Since  $\sum_k (\bar{n} - n_k) = 0$ , the sum equals  $-\sum_k (n_k - \bar{n})(\sigma_k^2 - \bar{\sigma}^2)$  for any constant  $\bar{\sigma}^2$ , which is negative when  $n_k$  and  $\sigma_k^2$  move together and positive when they move in opposite directions. In a balanced layout every  $\bar{n} - n_k$  is zero, and then  $E(\text{MSE}) = \sum_k (m-1)\sigma_k^2/(g(m-1)) = g^{-1} \sum_k \sigma_k^2$ . ■

The theorem predicts the behaviour of the  $F$  test. In a balanced layout both mean squares estimate the same average variance under the hypothesis, and although the ratio is no longer exactly  $F$ -distributed, Box (1954) showed that the effect on the level is modest. In an unbalanced layout MSE leans towards the variances of the large groups and MSB towards those of the small groups. When the small groups are the variable ones the test rejects too often; when the large groups are, it rejects too rarely and loses power.

#### Example. (How far the level moves)

Four normal groups have equal means and standard deviations increasing geometrically

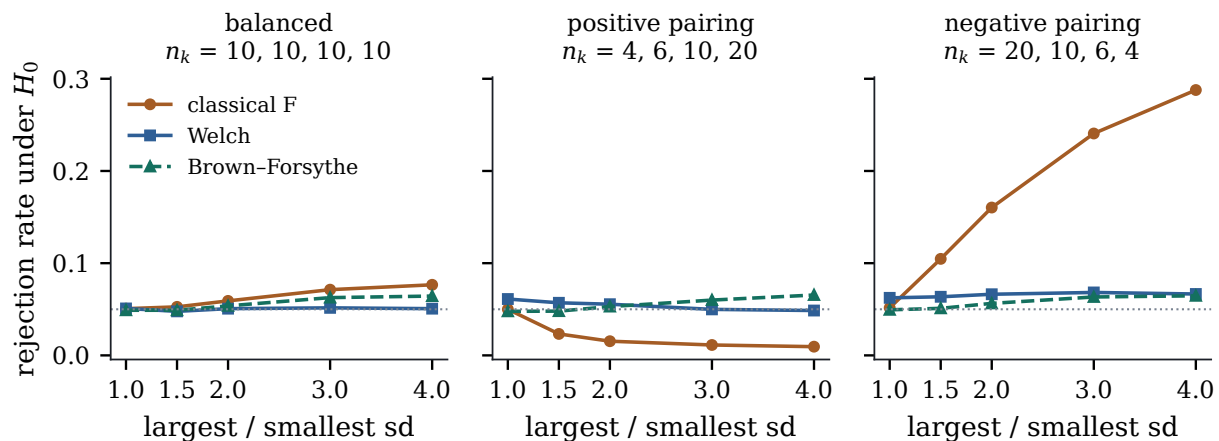


Figure 15.5.2: Rejection rates under equal means of three tests at nominal level 0.05 (dotted), for four normal groups whose standard deviations increase from 1 to the ratio on the horizontal axis. Each point is based on 40000 simulated data sets.

from 1 to  $R$ . Three designs of 40 observations are compared: balanced (10 per group), *positive pairing* (sizes 4, 6, 10, 20, the largest group most variable) and *negative pairing* (sizes 20, 10, 6, 4, the smallest group most variable). For each design and each  $R$ , 40000 data sets were simulated and three level-0.05 tests applied: the classical  $F$  test and the tests of Welch and of Brown and Forsythe described below (Figure 15.5.2).

At  $R = 3$ , Eq. 15.5.1 gives expected mean square ratios  $E(\text{MSB})/E(\text{MSE})$  of 1.000, 0.559 and 1.923 for the three designs. The simulated levels of the classical test follow: 0.071 when balanced, 0.011 with positive pairing, and 0.241 with negative pairing, almost five times the nominal level. At  $R = 4$  the negative pairing gives 0.288. Welch's test stays between 0.048 and 0.068 throughout, and the Brown-Forsythe test between 0.048 and 0.065.

The simulation supports the common rule of thumb that a ratio of about two between the largest and smallest standard deviations does little harm, but only with balance: at  $R = 2$  the balanced level is 0.059, while the two unbalanced designs give 0.015 and 0.160.

## 15.5.4 Tests that allow unequal variances

If the variances  $\sigma_k^2$  were known, weighted least squares (Section 6.9) would compare the group means with their precision-weighted average  $\tilde{\mu} = \sum_k w_k \bar{Y}_k / \sum_k w_k$ ,  $w_k = n_k / \sigma_k^2$ , and  $\sum_k w_k (\bar{Y}_k - \tilde{\mu})^2$  would have an exact  $\chi^2(g-1)$  law under equal means. In practice the  $\sigma_k^2$  are replaced by the  $s_k^2$  and the reference distribution is approximated. We describe the two approximations in common use without deriving them; Chapter 21 treats heteroscedasticity in general.

**Welch's test.** Welch (1951) replaced  $\sigma_k^2$  by  $s_k^2$  in the weights,  $w_k = n_k / s_k^2$ , and divided the weighted mean square by a correction factor:

$$F_W = \frac{\sum_k w_k (\bar{y}_k - \tilde{\mu})^2 / (g-1)}{1 + \frac{2(g-2)}{g^2-1} \sum_k h_k}, \quad h_k = \frac{(1 - w_k / \sum_l w_l)^2}{n_k - 1},$$

referred to  $F(g-1, \nu_W)$  with  $\nu_W = (g^2 - 1) / (3 \sum_k h_k)$ . The correction and the degrees of freedom come from an approximation to the statistic's null distribution that is accurate to first

order in the quantities  $1/(n_k - 1)$ . For two groups  $F_W$  is the square of Welch's two-sample  $t$  statistic with Satterthwaite's degrees of freedom ([Exercise \(B1\)](#)).

**The Brown–Forsythe test.** Brown and Forsythe (1974a) kept the unweighted numerator and changed the denominator. By [Theorem 15.5.2\(b\)](#), under equal means SSB has expectation  $\sum_k (1 - n_k/n) \sigma_k^2$ , whatever the variances. Replacing  $\sigma_k^2$  by  $s_k^2$  gives an unbiased estimate of that expectation, and the statistic

$$F_{BF} = \frac{\text{SSB}}{\sum_k (1 - n_k/n) s_k^2}$$

compares SSB with an estimate of what it should be under the hypothesis. It is referred to  $F(g - 1, f)$ , where  $1/f = \sum_k c_k^2 / (n_k - 1)$  with  $c_k = (1 - n_k/n) s_k^2 / \sum_l (1 - n_l/n) s_l^2$ , Satterthwaite's approximation to the degrees of freedom of a linear combination of independent variance estimates. In a balanced layout the denominator is  $(g - 1) s^2$ , so  $F_{BF} = F$  and only the degrees of freedom change.

Individual contrasts need the same care, since  $s\sqrt{v_c}$  estimates the wrong variance when the  $\sigma_k^2$  differ, even in a balanced layout ([Exercise \(C2\)](#)). The **Welch–Satterthwaite** interval uses

$$\widehat{\text{Var}}(\hat{\psi}) = \sum_k \frac{c_k^2 s_k^2}{n_k}, \quad \nu = \frac{(\sum_k c_k^2 s_k^2 / n_k)^2}{\sum_k (c_k^2 s_k^2 / n_k)^2 / (n_k - 1)},$$

and the multiplier  $t_{\nu, \alpha/2}$  ([Exercise \(B3\)](#)), adjusted as in [Chapter 13](#) for simultaneous inference.

**Example. (An unbalanced trial with unequal variances)**

A repair mortar was made with four admixture formulations, and its setting time was measured in minutes. The standard formulation A was tested most often, and the newer formulations least (simulated data, rounded to 0.1 minute):

| Formulation | $n_k$ | Setting times (min)                                                    | Mean   | $s_k$ |
|-------------|-------|------------------------------------------------------------------------|--------|-------|
| A           | 12    | 18.9, 20.2, 20.6, 21.2, 20.3, 19.0, 20.2, 19.0, 20.5, 20.6, 20.1, 20.3 | 20.075 | 0.73  |
| B           | 9     | 22.6, 21.3, 22.2, 22.8, 19.3, 20.7, 21.5, 20.8, 18.8                   | 21.111 | 1.38  |
| C           | 6     | 20.7, 18.2, 18.6, 23.6, 21.3, 22.6                                     | 20.833 | 2.14  |
| D           | 4     | 20.1, 22.6, 28.3, 25.3                                                 | 24.075 | 3.53  |

The standard deviations grow as the group sizes shrink, the negative pairing of [Example \(How far the level moves\)](#). The classical analysis gives  $F = 5.34$  on  $(3, 27)$  degrees of freedom and  $p = 0.0051$ , apparently strong evidence, driven by the high mean of D. But  $\text{MSE} = 3.015$  gives the variance of D weight  $3/27$ , while the null expectation of  $\text{MSB}$  in [Theorem 15.5.2\(b\)](#) gives it weight  $(1 - 4/31)/3 \approx 0.29$ ; estimated with the  $s_k^2$ , that expectation is 5.407, far above MSE. Welch's test gives  $F_W = 2.64$  on  $(3, 8.3)$  degrees of freedom and  $p = 0.119$ , and the Brown–Forsythe test gives  $F_{BF} = 2.98$  on  $(3, 6.2)$  degrees of freedom and  $p = 0.115$ . The difference between D and A is 4.00 minutes. With the pooled standard error 1.003 its 95% interval is  $[1.94, 6.06]$ ; with the Welch–Satterthwaite standard error 1.776 on 3.1 degrees of freedom it is  $[-1.56, 9.56]$ . Four observations with a standard deviation of 3.5 minutes cannot establish a difference of four minutes.

The weighted and unweighted averages of the group means are 21.039 and 21.524, so the side conditions  $\sum_k n_k \alpha_k = 0$  and  $\sum_k \alpha_k = 0$  of [Section 15.1 \(Theorem 8.4.2\)](#) define overall means that differ by half a minute here.

```
import numpy as np
from scipy import stats
from statsmodels.stats.oneway import anova_oneway

labels = ["A", "B", "C", "D"]
n_k = np.array([12, 9, 6, 4])
rng = np.random.default_rng(1520)
samples = [np.round(rng.normal(mu, sd, k), 1)
            for mu, sd, k in zip([20.0, 20.6, 21.0, 22.2], [0.8, 1.2, 2.0, 3.0], n_k)
          ]

g, n = len(samples), n_k.sum()
means = np.array([x.mean() for x in samples])
s2_k = np.array([x.var(ddof=1) for x in samples])      # group variances
s2 = np.sum((n_k - 1) * s2_k) / (n - g)              # pooled within-group
              variance
grand = means @ n_k / n
ss_b = np.sum(n_k * (means - grand) ** 2)
F = ss_b / (g - 1) / s2
print("means:", means.round(3), " sds:", np.sqrt(s2_k).round(2))
print(f"classical F = {F:.2f}, p = {stats.f.sf(F, g - 1, n - g):.4f}")
# null expectation of MS_between if the variances really are s2_k (the theorem of
  this section)
print(f"MS_within = {s2:.3f}; estimated null E(MS_between) = "
      f"{np.sum((1 - n_k / n) * s2_k) / (g - 1):.3f}")

welch = anova_oneway(samples, use_var="unequal")
bf = anova_oneway(samples, use_var="bf")
print(f"Welch: F = {welch.statistic:.2f} on ({g - 1}, {welch.df_denom:.1f}) df, p = "
      f"{welch.pvalue:.3f}")
print(f"Brown-Forsythe: F = {bf.statistic:.2f} on ({g - 1}, {bf.df2[1]:.1f}) df, p = "
      f"{bf.pvalue2:.3f}")
lev = stats.levene(*samples, center="median")          # dispersion test on |y -
              median|
print(f"variance test on absolute deviations from medians: p = {lev.pvalue:.4f}")
```

(By default `statsmodels` refines the numerator degrees of freedom of the Brown–Forsythe test; `pvalue2` is the original version.)



The simulation of [Example \(How far the level moves\)](#) used the same formulas, applied to group means and variances drawn directly from their sampling distributions:

```
import numpy as np
from scipy import stats

def three_tests(ybar, s2, n_k):
    """p-values of the classical F, Welch and Brown-Forsythe tests; arrays of shape
    (reps, g)."""
    g, n = n_k.size, n_k.sum()
    grand = ybar @ n_k / n
    ss_b = ((ybar - grand[:, None]) ** 2) @ n_k
    ss_w = s2 @ (n_k - 1)
    p_f = stats.f.sf(ss_b / (g - 1) / (ss_w / (n - g)), g - 1, n - g)
    # Welch (1951): weights n_k / s_k^2 and a Satterthwaite-type denominator df
    w = n_k / s2
    mu_w = np.sum(w * ybar, axis=1) / w.sum(axis=1)
    h = (1 - w / w.sum(axis=1)[:, None]) ** 2 / (n_k - 1)
    a = np.sum(w * (ybar - mu_w[:, None]) ** 2, axis=1) / (g - 1)
    b = 1 + 2 * (g - 2) / (g ** 2 - 1) * h.sum(axis=1)
    p_w = stats.f.sf(a / b, g - 1, (g ** 2 - 1) / (3 * h.sum(axis=1)))
    # Brown-Forsythe (1974): SS_between over its null expectation under unequal
    variances
    d = (1 - n_k / n) * s2
    c = d / d.sum(axis=1)[:, None]
    p_bf = stats.f.sf(ss_b / d.sum(axis=1), g - 1, 1 / np.sum(c ** 2 / (n_k - 1),
axis=1))
    return p_f, p_w, p_bf

def size(n_k, sds, reps=40000, seed=0, alpha=0.05):
    """Rejection rates under equal means, from group means and variances drawn
    exactly."""
    rng = np.random.default_rng(seed)
    n_k, sds = np.asarray(n_k, float), np.asarray(sds, float)
    ybar = rng.normal(0, sds / np.sqrt(n_k), (reps, n_k.size))
    s2 = sds ** 2 * rng.chisquare(n_k - 1, (reps, n_k.size)) / (n_k - 1)
    return [np.mean(p < alpha) for p in three_tests(ybar, s2, n_k)]

designs = {"balanced": [10, 10, 10, 10],
          "positive pairing": [4, 6, 10, 20],          # big groups get the big
          variances
          "negative pairing": [20, 10, 6, 4]}          # small groups get the big
          variances
ratios = [1, 1.5, 2, 3, 4]
res = {}
for name, sizes in designs.items():                  # sds grow geometrically from 1
    to R
    res[name] = np.array([size(sizes, R ** (np.arange(4) / 3), seed=int(10 * R)) for
    R in ratios])
for R, row in zip(ratios, res[name]):
    print(f"{name:17s} R = {R:3.1f}: F, Welch, BF sizes = {np.round(row, 3)}")
```

Robustness has a small price. With equal variances and a group of only four observations, Welch's test rejects slightly too often, 0.061 and 0.062 in the two unbalanced designs. The practical advice is simple: with unequal group sizes, use a heteroscedasticity-robust test unless there is good reason to believe the variances are equal; with equal sizes the classical test is acceptable, although individual contrasts still need care.



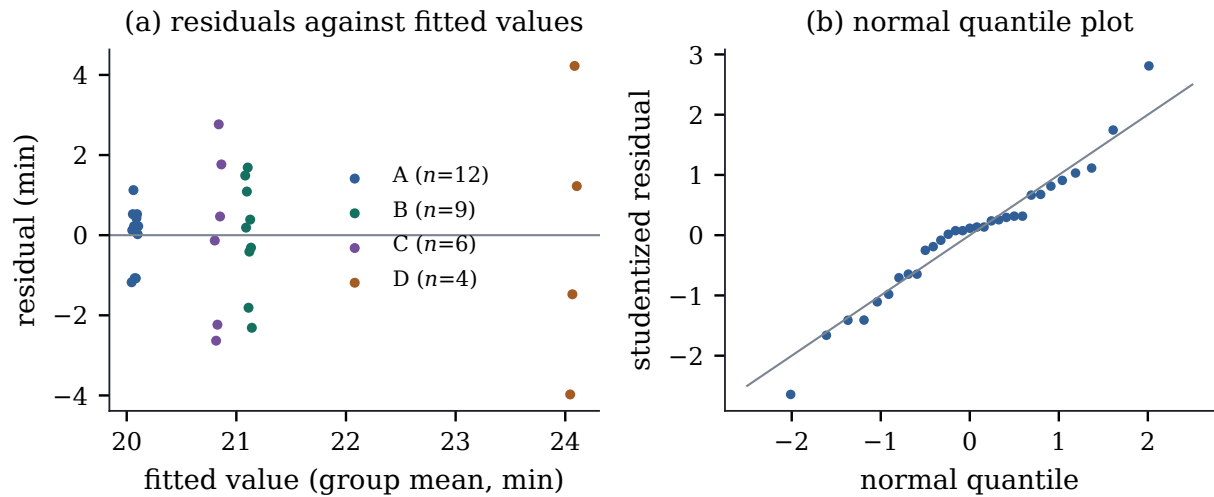


Figure 15.5.3: Residual plots for the mortar trial. (a) Residuals against fitted values: the spread grows from formulation A to formulation D. (b) Normal quantile plot of the studentized residuals: the curvature at both ends reflects the mixture of groups with different variances, not a heavy-tailed error distribution.

### 15.5.5 Checking the assumptions

Independence is secured by the design, above all by randomization, and cannot be checked well from the data. Correlation within groups, which arises when the observations in a group share a batch, an operator or a day, inflates the  $F$  statistic badly ([Exercise \(C1\)](#)). A common variance and normality are checked with the residuals  $e_{kj} = y_{kj} - \bar{y}_k$ .

The leverage of an observation in group  $k$  is  $1/n_k$  ([Proposition 6.8.1](#)), so  $\text{Var}(e_{kj}) = \sigma^2(1 - 1/n_k)$ . Residuals from small groups are shrunk towards zero, so they are compared on the common scale  $r_{kj} = e_{kj}/(s\sqrt{1 - 1/n_k})$ . A plot of residuals against the group means shows whether the spread differs between groups, and a spread growing with the mean suggests a transformation ([Chapter 22](#)). A normal quantile plot of the studentized residuals shows skewness and heavy tails. [Chapter 20](#) develops residual diagnostics in general.

[Figure 15.5.3](#) shows the two plots for the mortar trial. The fan shape in panel (a) is unmistakable. The classical tests of equal variances, which compare the logarithms of the  $s_k^2$ , are so sensitive to nonnormality that a rejection says as much about the tails as about the variances. Levene (1960) proposed instead to apply the one-way  $F$  test to the absolute deviations  $|y_{kj} - \bar{y}_k|$ , which are larger on average in a more variable group; Brown and Forsythe (1974b) used deviations from the group *medians*, which behave better for skewed data. For the mortar trial that version gives  $p = 0.0019$ . Using such a test as a gate for the classical  $F$  test is not recommended, since it has little power in small groups, exactly where unequal variances do the most harm. The analysis should be chosen in advance, from the design and from what is known about the response.

Nonnormality is usually the least serious problem for tests about means: the level of the  $F$  test is fairly robust to it, more so in balanced designs ([Proposition 19.3.5](#)), though heavy tails cost power by inflating MSE. Transformations ([Chapter 22](#)) and resampling ([Chapter 23](#)), for example the Kruskal–Wallis permutation test on ranks, are the usual remedies.

## 15.5.6 Exercises

### 15.5.6.1 A. Check your understanding

#### Exercise A1.

For the balanced design with  $m = 10$  and variances 1, 2, 4, 8, compute  $E(\text{MSB})$  under equal means and  $E(\text{MSE})$  from [Theorem 15.5.2](#). Repeat for the sizes 4, 6, 10, 20 and for 20, 10, 6, 4.

*Solution.* Balanced: both equal  $(1 + 2 + 4 + 8)/4 = 3.75$ . Sizes 4, 6, 10, 20 ( $n = 40$ ):  $E(\text{SSB}) = 0.9 + 0.85 \cdot 2 + 0.75 \cdot 4 + 0.5 \cdot 8 = 9.6$ , so  $E(\text{MSB}) = 3.2$ ;  $E(\text{SSE}) = 3 + 10 + 36 + 152 = 201$ , so  $E(\text{MSE}) = 201/36 \approx 5.58$ . The test is conservative. Sizes 20, 10, 6, 4:  $E(\text{SSB}) = 0.5 + 0.75 \cdot 2 + 0.85 \cdot 4 + 0.9 \cdot 8 = 12.6$ , so  $E(\text{MSB}) = 4.2$ ;  $E(\text{SSE}) = 19 + 18 + 20 + 24 = 81$ , so  $E(\text{MSE}) = 2.25$ . The test is liberal.

#### Exercise A2.

Show that the hat matrix of the one-way model has diagonal entries  $1/n_k$  and that  $\text{Var}(e_{kj}) = \sigma^2(1 - 1/n_k)$ . What is the residual of an observation that is alone in its group, and why is it useless for checking the model?

*Solution.* The hat matrix is  $\mathbf{M}$  of [Eq. 15.1.1](#), whose diagonal entry for an observation of group  $k$  is  $1/n_k$ . By [Proposition 5.5.3](#),  $\text{Cov}(\hat{\boldsymbol{\epsilon}}) = \sigma^2(\mathbf{I} - \mathbf{M})$ , so  $\text{Var}(e_{kj}) = \sigma^2(1 - 1/n_k)$ . An observation alone in its group has leverage one, its fitted value is itself, and its residual is identically zero: it carries no information about the variance or about the model.

### 15.5.6.2 B. Practice

#### Exercise B1.

Show that for  $g = 2$  the Welch statistic  $F_W$  equals  $t_W^2$ , where  $t_W = (\bar{y}_1 - \bar{y}_2)/\sqrt{s_1^2/n_1 + s_2^2/n_2}$ , and that  $\nu_W$  is Satterthwaite's  $(s_1^2/n_1 + s_2^2/n_2)^2 / [(s_1^2/n_1)^2/(n_1 - 1) + (s_2^2/n_2)^2/(n_2 - 1)]$ .

*Solution.* With  $g = 2$  the correction factor is 1. Write  $v_k = s_k^2/n_k = 1/w_k$  and  $W = w_1 + w_2$ . The weighted sum of squares of two numbers about their weighted mean is  $w_1 w_2 (\bar{y}_1 - \bar{y}_2)^2 / W = (\bar{y}_1 - \bar{y}_2)^2 / (v_1 + v_2)$ , which is  $t_W^2$ . Next  $1 - w_1/W = w_2/W = v_1/(v_1 + v_2)$ , so  $h_1 = v_1^2 / \{(v_1 + v_2)^2(n_1 - 1)\}$  and similarly for  $h_2$ . Then  $\nu_W = 3 / (3 \sum_k h_k) = (v_1 + v_2)^2 / \{v_1^2/(n_1 - 1) + v_2^2/(n_2 - 1)\}$ .

#### Exercise B2.

Show that for  $g = 2$  the Brown–Forsythe statistic also equals  $t_W^2$ , and that its degrees of freedom  $f$  equal  $\nu_W$ . So the two robust tests coincide for two groups and differ only for three or more.

*Solution.* With two groups,  $SSB = n_1 n_2 (\bar{y}_1 - \bar{y}_2)^2 / n$ , and the denominator is  $(n_2 s_1^2 + n_1 s_2^2) / n$ . Their ratio is  $(\bar{y}_1 - \bar{y}_2)^2 / (s_1^2 / n_1 + s_2^2 / n_2) = t_W^2$ . The weights are  $c_1 = (n_2 s_1^2 / n) / \{(n_2 s_1^2 + n_1 s_2^2) / n\} = v_1 / (v_1 + v_2)$  and  $c_2 = v_2 / (v_1 + v_2)$ , so  $1/f = \sum_k c_k^2 / (n_k - 1) = 1/\nu_W$  by the previous exercise.

### Exercise B3.

Let  $\hat{V} = \sum_k c_k^2 s_k^2 / n_k$  with independent  $s_k^2 \sim \sigma_k^2 \chi^2(n_k - 1) / (n_k - 1)$  under normality. Show that  $E \hat{V} = V = \text{Var}(\hat{\psi})$  and  $\text{Var} \hat{V} = 2 \sum_k (c_k^2 \sigma_k^2 / n_k)^2 / (n_k - 1)$ . Choose  $\nu$  so that  $\nu \hat{V} / V$  has the mean and variance of a  $\chi^2(\nu)$  variable, and show that replacing  $\sigma_k^2$  by  $s_k^2$  in the answer gives the degrees of freedom of the Welch–Satterthwaite interval.

*Solution.* Since  $E s_k^2 = \sigma_k^2$ ,  $E \hat{V} = \sum_k c_k^2 \sigma_k^2 / n_k = V$ , which is  $\text{Var}(\hat{\psi})$  because the group means are independent with variances  $\sigma_k^2 / n_k$ .  $\text{Var}(s_k^2) = 2\sigma_k^4 / (n_k - 1)$ , and the terms are independent, which gives  $\text{Var} \hat{V}$ . A  $\chi^2(\nu)$  variable has mean  $\nu$  and variance  $2\nu$ . The mean of  $\nu \hat{V} / V$  is  $\nu$  for every  $\nu$ , and its variance is  $\nu^2 \text{Var}(\hat{V}) / V^2$ . Setting this equal to  $2\nu$  gives  $\nu = 2V^2 / \text{Var} \hat{V} = V^2 / \sum_k (c_k^2 \sigma_k^2 / n_k)^2 / (n_k - 1)$ . With  $s_k^2$  for  $\sigma_k^2$  this is the formula in the text.

### Exercise B4.

For  $n = 30$  observations in five groups, compare the quantity  $h$  of [Proposition 15.5.1](#) for the balanced allocation and for the allocation 10, 5, 5, 5, 5 suggested by [Exercise \(B3\)](#) for comparisons with a control. What does each allocation optimize?

*Solution.* Balanced:  $h = 6/2 = 3$ . For 10, 5, 5, 5, 5, the least informative pair is two treatments of size 5, with  $h = 25/10 = 2.5$ . The guaranteed noncentrality of the overall test is  $3(\Delta/\sigma)^2$  against  $2.5(\Delta/\sigma)^2$ . But each comparison with the control has variance  $\sigma^2(1/10 + 1/5) = 0.3\sigma^2$  instead of  $\sigma^2/3$ . Balance protects the overall test and the comparisons among treatments; the unbalanced allocation sharpens the comparisons with the control.

## 15.5.6.3 C. Going deeper

### Exercise C1.

In a balanced layout, suppose that under equal means the errors within a group are equicorrelated,  $\text{Cov}(\varepsilon_{kj}, \varepsilon_{kj'}) = \rho\sigma^2$  for  $j \neq j'$  with  $\rho \geq 0$ , and independent between groups. Show that  $E(\text{MSE}) = \sigma^2(1 - \rho)$  and  $E(\text{MSB}) = \sigma^2(1 - \rho + m\rho)$ . What happens to the level of the  $F$  test with  $m = 10$  and  $\rho = 0.1$ ?

*Solution.* The covariance matrix is  $\sigma^2\{(1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top\}$ , the random-effects form with  $\sigma_a^2 = \rho\sigma^2$  and error variance  $(1 - \rho)\sigma^2$ . By [Exercise \(C1\)](#) with  $n_0 = m$ ,  $E(\text{MSE}) = (1 - \rho)\sigma^2$  and  $E(\text{MSB}) = (1 - \rho)\sigma^2 + m\rho\sigma^2$ . Under normality  $\text{MSB}/E(\text{MSB})$  and  $\text{MSE}/E(\text{MSE})$  are independent scaled chi-squared variables, so  $F$  is  $1 + m\rho/(1 - \rho)$  times an  $F(g - 1, n - g)$  variable. With  $m = 10$  and  $\rho = 0.1$  the factor is  $1 + 1/0.9 \approx 2.1$ , and a nominal 5% test rejects far more often: with  $g = 4$  groups its level is 0.271. A small intraclass correlation does great damage when groups are large.

**Exercise C2.**

In a balanced layout with variances  $\sigma_k^2$ , show that the pooled variance estimate of the pairwise difference  $\bar{Y}_k - \bar{Y}_l$ ,  $2s^2/m$ , has expectation  $2\bar{\sigma}^2/m$  with  $\bar{\sigma}^2 = g^{-1} \sum_i \sigma_i^2$ , while the true variance is  $(\sigma_k^2 + \sigma_l^2)/m$ . Conclude that balance protects the overall test but not individual comparisons, and find the ratio of true to estimated variance for the two most variable groups when the variances are 1, 2, 4, 8.

*Solution.* By [Theorem 15.5.2\(c\)](#), in a balanced layout  $E(s^2) = \bar{\sigma}^2$ , so  $E(2s^2/m) = 2\bar{\sigma}^2/m$ , and the true variance is  $\sigma_k^2/m + \sigma_l^2/m$  because the means are independent. For the groups with variances 4 and 8, the true variance is  $12/m$  and the pooled estimate targets  $2 \cdot 3.75/m = 7.5/m$ , a ratio of 1.6. Intervals for that difference built from  $s^2$  are too short by a factor of about  $\sqrt{1.6} \approx 1.26$ , and intervals for the difference of the two least variable groups are too long.

## 15.6 Summary and Notes

### 15.6.1 Summary

*Idea.*

1. The one-way layout has mean space  $\mathcal{C}(\mathbf{Z})$ , the vectors constant within groups, in both its forms ([Definition 15.1.1](#)). The fit is the group means, and  $s^2$  pools the group variances. Different side conditions define different “overall means” when the layout is unbalanced ([Section 15.1](#)).
2. Contrasts are the directions of the treatment space  $\mathcal{T} = \mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$ , and a contrast’s test, interval and sum of squares come from the projection onto its line ([Proposition 15.2.1](#)). Families use Bonferroni or Dunnett for comparisons with a control, Tukey for all pairs and Scheffé for contrasts suggested by the data ([Section 15.2](#)).
3. A contrast test captures the fraction  $\cos^2 \varphi$  of the overall noncentrality,  $\varphi$  being its angle with the true treatment component ([Proposition 15.2.2](#)): a well-chosen planned contrast beats the overall test, a badly chosen one has no power.
4. The decomposition  $\mathbf{I} = \mathbf{P}_0 + (\mathbf{M} - \mathbf{P}_0) + (\mathbf{I} - \mathbf{M})$  gives the analysis of variance table. The  $F$  statistic has the  $F(g-1, n-g, \gamma)$  law with  $\gamma = \sum_k n_k (\mu_k - \bar{\mu})^2 / \sigma^2$ , and it is the likelihood ratio test of equal means ([Theorem 15.3.1](#)). Without normality,  $E(\text{MSE}) = \sigma^2$  and  $E(\text{MSB}) = \sigma^2 + \sum_k n_k (\mu_k - \bar{\mu})^2 / (g-1)$  ([Theorem 15.3.2](#)). The quantity  $(g-1)F$  is the largest squared contrast  $t$  statistic.
5. A complete set of orthogonal contrasts, orthogonal in the sense  $\sum_k c_k d_k / n_k = 0$ , splits SSB into  $g-1$  independent single-degree-of-freedom pieces whose noncentralities add up to  $\gamma$ , and  $F$  is the average of their  $F$  statistics ([Theorem 15.4.1](#)). Successive splits of the treatments give such sets for any group sizes ([Proposition 15.4.2](#)), and polynomial contrasts give a trend analysis, whose higher-degree pieces are the pure-error lack-of-fit test of a low-degree curve ([Example \(The shape of the nitrogen response\)](#)).
6. Balance makes design orthogonality plain orthogonality and maximizes the noncentrality guaranteed against every configuration of means with a given range

([Proposition 15.5.1](#)).

7. With unequal variances, MSB and MSE estimate the same average variance under equal means if the layout is balanced, but not otherwise. The classical test is liberal when the small groups are the variable ones and conservative when the large groups are ([Theorem 15.5.2](#)). The tests of Welch and of Brown and Forsythe hold their level approximately, and Welch–Satterthwaite intervals do the same for contrasts ([Section 15.5](#)).
8. Residuals have variance  $\sigma^2(1 - 1/n_k)$ ; plotted against the group means and on a normal quantile plot after studentizing, they check the variances and normality. Within-group correlation, which the data cannot reveal, is the most damaging failure.

## 15.6.2 Notes and sources

**Origins.** The analysis of variance is Fisher’s. It appears in *Statistical Methods for Research Workers* (Fisher 1925), developed for the field experiments at Rothamsted, and *The Design of Experiments* (Fisher 1935) made randomization the logical basis of the analysis. The derivation of the one-way model from unit–treatment additivity and random assignment, sketched in [Section 15.1](#), is developed in detail by Kempthorne (1952). Scheffé (1959) remains the classic treatment of the fixed-effects theory, including the effects of departures from the assumptions.

**The textbook treatments.** Christensen (2020, chapter 4) derives the one-way analysis entirely from the projections  $\mathbf{M}$  and  $\mathbf{M} - \mathbf{P}_0$ , including the characterization of contrasts through the space onto which  $\mathbf{M} - \mathbf{P}_0$  projects; this chapter follows his geometric approach, with the treatment space as the organizing object. Rencher and Schaalje (2008, chapter 13) give the balanced case through full-and-reduced models and through the general linear hypothesis, derive the expected mean squares both ways, and treat orthogonal and orthogonal polynomial contrasts, including the result that the overall  $F$  statistic averages the single-degree-of-freedom statistics of a complete orthogonal set. Their section 12.7 covers the testable hypotheses of non-full-rank models used in [Theorem 15.3.1](#). Seber and Lee (2003, section 8.2) discuss simultaneous intervals and the assumptions, including unequal variances. Note that Rencher and Schaalje and Christensen define noncentrality as half of our  $\gamma$ .

**Contrasts and multiple comparisons.** Tables of orthogonal polynomials for equally spaced levels go back at least to Fisher and Yates (1938). The exact method for comparisons with a control is due to Dunnett (1955), and the conservativeness of the Tukey–Kramer intervals for unequal group sizes was proved by Hayter (1984); [Chapter 13 \(Exercise \(B2\)\)](#) characterizes the many-to-one multiplier and gives further references. [Proposition 15.2.2](#) and [Proposition 15.4.2](#) are elementary consequences of the projection picture. Part (a) of [Proposition 15.5.1](#) is [Exercise \(B1\)](#) of Chapter 11; part (b), the optimality of balance, is a short calculation.

**Unequal variances.** Box (1954) computed the effect of unequal variances on the one-way  $F$  test, showing that it is small when the groups are of equal size and can be large otherwise, in the directions described by [Theorem 15.5.2](#). The test of Welch (1951) weights the groups by their estimated precisions; the test of Brown and Forsythe (1974a) keeps the unweighted numerator and estimates its null expectation. Welch obtained his correction factor and degrees of freedom from a series expansion of the statistic’s distribution to first order in  $1/(n_k - 1)$ ; the Brown–Forsythe degrees of freedom, like those of the contrast intervals, are Satterthwaite’s (1946) moment match. The robust test of equal variances by analysis of variance of absolute deviations is due to Levene (1960), and the version based on deviations from medians to Brown

and Forsythe (1974b). The simulation of [Example \(How far the level moves\)](#) is ours. [Chapter 21](#) returns to heteroscedasticity with weighted least squares and heteroscedasticity-consistent standard errors, and [Chapter 32](#) to random effects.

### 15.6.3 References

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# Chapter 16

## Multi-Way Layouts and Interaction

An experiment with several factors asks more than one question. It asks whether each factor matters, and also whether the effect of one factor depends on the level of another. In a balanced layout, where every combination of levels is observed equally often, these questions correspond to mutually orthogonal subspaces of observation space. Each subspace has a projection that is a Kronecker product of averaging and centring matrices, and the analysis of variance is Pythagoras applied to that decomposition. This chapter builds the decomposition for two factors, with and without replication, for any number of crossed factors, and for factors nested inside other factors. The algebra is the easy part. The harder part is interpretation: what an interaction is, when a transformation removes it, what a main effect means when interaction is present, and when it is safe to pool one term into another.

**What you need.** Kronecker products ([Section 1.10](#)); independence of projections of a normal vector and the noncentral  $F$  distribution ([Chapter 4](#)); nested projections and the two-factor projection identity of [Example \(Additive two-factor model, one observation per cell\)](#) ([Section 6.5](#)); estimability with two factors ([Section 8.6](#)); analysis of variance decompositions, orthogonal designs and expected mean squares ([Chapter 9](#));  $F$  tests of reduced models ([Chapter 11](#)); simultaneous inference ([Chapter 13](#)); the lack-of-fit test and regressors built from fitted values ([Sections 14.6 and 14.7](#)); and the one-way analysis of variance ([Chapter 15](#)).

### 16.0.1 Roadmap

- [Two-way additive models](#)
- [Interaction and its interpretation](#)
- [Balanced two-way layouts via orthogonal projections](#)
- [Higher-way layouts](#)
- [Nested factors](#)
- [Summary and notes](#)

## 16.1 Two-way additive models

Two factors are observed together. Factor  $A$  has levels  $i = 1, \dots, a$  and factor  $B$  has levels  $j = 1, \dots, b$ , with  $a, b \geq 2$ . Every one of the  $ab$  combinations, or **cells**, is observed the same



number  $m \geq 1$  of times. We write  $y_{ijk}$  for the  $k$ th observation in cell  $(i, j)$ , so that  $n = abm$ . [Example \(Additive two-factor model, one observation per cell\)](#) found the projections of the additive model for  $m = 1$  by a direct argument. This section redoes it with the tools used throughout the chapter, and adds replication, estimable functions, tests, and a test of additivity with one observation per cell.

### 16.1.1 Averaging and centring matrices

Stack the observations with the replicate index  $k$  running fastest, then  $j$ , then  $i$ :

$$\mathbf{y} = (y_{111}, \dots, y_{11m}, y_{121}, \dots, y_{12m}, \dots, y_{ab1}, \dots, y_{abm})^\top.$$

With this order a matrix of the form  $\mathbf{E}_a \otimes \mathbf{E}_b \otimes \mathbf{E}_m$  acts on each index separately, as in [Example \(The balanced two-way layout\)](#). For  $k \geq 1$  write

$$\bar{\mathbf{J}}_k = \frac{1}{k} \mathbf{1}_k \mathbf{1}_k^\top, \quad \mathbf{C}_k = \mathbf{I}_k - \bar{\mathbf{J}}_k.$$

The **averaging matrix**  $\bar{\mathbf{J}}_k$  replaces each entry of a  $k$ -vector by the mean of the entries. The **centring matrix**  $\mathbf{C}_k$  subtracts that mean. With the usual dot notation for means, the following products replace  $y_{ijk}$  by

$$\begin{aligned} (\mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{ij\cdot}, & (\mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{i\cdot\cdot}, \\ (\bar{\mathbf{J}}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{\cdot j\cdot}, & (\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m) \mathbf{y} &: \bar{y}_{\cdot\cdot\cdot}, \end{aligned}$$

the cell, row, column and grand means. Each operator averages over the indices where it has a  $\bar{\mathbf{J}}$  and leaves the others alone. Every projection in this chapter is a Kronecker product of factors  $\mathbf{I}$ ,  $\bar{\mathbf{J}}$  and  $\mathbf{C}$ , and the following lemma is all the algebra they need.

#### Lemma 16.1.1 — Kronecker products of averaging and centring matrices

Let  $l_1, \dots, l_t \geq 1$ .

- (a)  $\bar{\mathbf{J}}_k$  and  $\mathbf{C}_k$  are symmetric and idempotent, with ranks 1 and  $k - 1$ , and  $\bar{\mathbf{J}}_k \mathbf{C}_k = \mathbf{0}$ ,  $\bar{\mathbf{J}}_k + \mathbf{C}_k = \mathbf{I}_k$ .
- (b) If each  $\mathbf{E}_s$  is one of  $\mathbf{I}_{l_s}$ ,  $\bar{\mathbf{J}}_{l_s}$ ,  $\mathbf{C}_{l_s}$ , then  $\mathbf{E}_1 \otimes \dots \otimes \mathbf{E}_t$  is an orthogonal projection whose rank is the product of the ranks of the  $\mathbf{E}_s$ .
- (c) If each  $\mathbf{E}_s$  and each  $\mathbf{F}_s$  is one of  $\bar{\mathbf{J}}_{l_s}$ ,  $\mathbf{C}_{l_s}$ , and  $\mathbf{E}_s \neq \mathbf{F}_s$  for at least one  $s$ , then  $(\mathbf{E}_1 \otimes \dots \otimes \mathbf{E}_t)(\mathbf{F}_1 \otimes \dots \otimes \mathbf{F}_t) = \mathbf{0}$ .
- (d) The  $2^t$  products  $\mathbf{E}_1 \otimes \dots \otimes \mathbf{E}_t$  with each  $\mathbf{E}_s \in \{\bar{\mathbf{J}}_{l_s}, \mathbf{C}_{l_s}\}$  sum to  $\mathbf{I}$ .

*Proof.*

- (a)  $\bar{\mathbf{J}}_k$  is symmetric and  $\bar{\mathbf{J}}_k^2 = k^{-2} \mathbf{1}(\mathbf{1}^\top \mathbf{1}) \mathbf{1}^\top = \bar{\mathbf{J}}_k$ . Its trace is 1. Then  $\mathbf{C}_k = \mathbf{I} - \bar{\mathbf{J}}_k$  is the complementary projection, with trace  $k - 1$ , and  $\bar{\mathbf{J}}_k \mathbf{C}_k = \bar{\mathbf{J}}_k - \bar{\mathbf{J}}_k^2 = \mathbf{0}$ . Ranks equal traces by [Proposition 6.3.4](#). (b) By [Proposition 1.10.1\(e\)](#) a Kronecker product of symmetric idempotent matrices is symmetric and idempotent, and by [Proposition 1.10.1\(c\)](#) its rank is the product of the ranks. (c) By the mixed-product rule the product equals  $\mathbf{E}_1 \mathbf{F}_1 \otimes \dots \otimes \mathbf{E}_t \mathbf{F}_t$ , and the factor with  $\mathbf{E}_s \neq \mathbf{F}_s$  is  $\bar{\mathbf{J}} \mathbf{C} = \mathbf{0}$  or  $\mathbf{C} \bar{\mathbf{J}} = \mathbf{0}$ . (d) Expand  $\mathbf{I} = (\bar{\mathbf{J}}_{l_1} + \mathbf{C}_{l_1}) \otimes \dots \otimes (\bar{\mathbf{J}}_{l_t} + \mathbf{C}_{l_t})$  by bilinearity of the Kronecker product.



The rank  $k - 1$  of  $\mathbf{C}_k$  is the source of every “levels minus one” in the analysis of variance, and part (d) is the source of every analysis of variance table in this chapter. Part (b) allows  $k = 1$ : then  $\bar{\mathbf{J}}_1 = \mathbf{1}$  and  $\mathbf{C}_1 = \mathbf{0}$ .

```
import itertools

import numpy as np

def Jbar(k):
    return np.full((k, k), 1.0 / k)          # averaging: replaces a k-vector by its
    mean

def Cen(k):
    return np.eye(k) - Jbar(k)              # centring: subtracts the mean

def kron_all(mats):
    out = np.ones((1, 1))
    for A in mats:
        out = np.kron(out, A)
    return out

def factorial_projections(levels, m):
    """P_S for every subset S of the factors (a 0/1 tuple), and the error projection
    ."""
    P = {}
    for S in itertools.product([0, 1], repeat=len(levels)):
        P[S] = kron_all([Cen(l) if s else Jbar(l) for l, s in zip(levels, S)] + [
            Jbar(m)])
    P["error"] = kron_all([np.eye(l) for l in levels] + [Cen(m)])
    return P

def rank(A):
    return int(round(np.trace(A)))           # rank = trace for a projection
```

## 16.1.2 The additive model

The **additive two-way model** is

$$E(y_{ijk}) = \mu + \alpha_i + \beta_j, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}, \quad (16.1.1)$$

with model matrix  $\mathbf{X} = [\mathbf{1}_n, \mathbf{Z}_A, \mathbf{Z}_B]$ , where  $\mathbf{Z}_A = \mathbf{I}_a \otimes \mathbf{1}_b \otimes \mathbf{1}_m$  and  $\mathbf{Z}_B = \mathbf{1}_a \otimes \mathbf{I}_b \otimes \mathbf{1}_m$  are the indicator matrices of the two factors. It says that the difference between two levels of  $A$  is the same in every column and, symmetrically, that the difference between two levels of  $B$  is the same in every row. The model has  $1 + a + b$  parameters and, by [Theorem 8.6.1\(a\)](#), rank  $a + b - 1$ . Define

$$\begin{aligned}
\mathbf{P}_0 &= \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, \\
\mathbf{P}_A &= \mathbf{C}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, \\
\mathbf{P}_B &= \bar{\mathbf{J}}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m,
\end{aligned} \tag{16.1.2}$$

and  $\mathbf{R} = \mathbf{I} - \mathbf{P}_0 - \mathbf{P}_A - \mathbf{P}_B$ . By the table of means above,  $\mathbf{P}_A \mathbf{y}$  has entry  $\bar{y}_{i..} - \bar{y}_{...}$  in every position of row  $i$ , and  $\mathbf{P}_B \mathbf{y}$  has entry  $\bar{y}_{.j.} - \bar{y}_{...}$  in every position of column  $j$ .

### 16.1.3 Additivity with one observation per cell

Many two-factor data sets have one observation per cell, such as methods each run once on several test problems. By [Example \(Additive two-factor model, one observation per cell\)](#) the residual space of the additive model then has dimension  $(a-1)(b-1)$ , and the residuals are the interaction terms  $y_{ij} - \bar{y}_{i.} - \bar{y}_{.j} + \bar{y}_{...}$ . The residual sum of squares then serves two purposes. It estimates  $\sigma^2$  if the model is additive, and it is the only place where nonadditivity could show. There is no model-free estimate of  $\sigma^2$ , and no test of additivity against a general alternative.

A test is still possible against a *specific* kind of departure. Suppose the cell means are additive on some other scale,  $\mu_{ij} = f(\nu + a_i + b_j)$ , for a smooth increasing function  $f$  and small effects  $a_i, b_j$  with  $\sum a_i = \sum b_j = 0$ . A second-order Taylor expansion about  $\nu$  gives

$$\mu_{ij} \approx f(\nu) + f'(\nu)(a_i + b_j) + \frac{1}{2}f''(\nu)(a_i^2 + b_j^2) + f''(\nu)a_ib_j.$$

The terms in  $a_i^2$  and  $b_j^2$  are additive. Only the product term is not. To first order the row and column effects of  $\mu_{ij}$  are  $\alpha_i \approx f'(\nu)a_i$  and  $\beta_j \approx f'(\nu)b_j$ , so the interaction is approximately

$$\mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{...} \approx \theta \alpha_i \beta_j, \quad \theta = \frac{f''(\nu)}{f'(\nu)^2}. \tag{16.1.3}$$

Tukey (1949) proposed to look for interaction of exactly this shape. The row and column effects are estimated by  $r_i = \bar{y}_{i.} - \bar{y}_{...}$  and  $c_j = \bar{y}_{.j} - \bar{y}_{...}$  (with  $m = 1$  we drop the third subscript). The residuals are then regressed on the products  $r_i c_j$ . The table  $(r_i c_j)$  has zero row and column sums, so it lies in the residual space. The regression through the origin of the residual table on it has slope and sum of squares

$$\hat{\theta} = \frac{\sum_{i,j} r_i c_j y_{ij}}{\sum_i r_i^2 \sum_j c_j^2}, \quad \text{SS}_N = \frac{(\sum_{i,j} r_i c_j y_{ij})^2}{\sum_i r_i^2 \sum_j c_j^2}. \tag{16.1.4}$$

The regressor is computed from the same data, so [Proposition 14.7.1](#) does not apply. But a regressor that is a function of the fitted values alone leaves the  $F$  test exact ([Theorem 14.7.2](#)), and part (f) of the next theorem shows that Tukey's regressor is of this kind.

### 16.1.4 Estimation and tests

The following theorem collects what the additive model delivers in a balanced layout, with and without replication.

**Theorem 16.1.2 — The additive two-way model**

Assume Eq. 16.1.1 with  $m \geq 1$  observations in every cell.

- (a)  $\mathbf{P}_0, \mathbf{P}_A, \mathbf{P}_B$  and  $\mathbf{R}$  are mutually orthogonal projections summing to  $\mathbf{I}$ , with ranks 1,  $a-1$ ,  $b-1$  and  $n-a-b+1$ . The projection onto  $\mathcal{C}(\mathbf{X})$  is  $\mathbf{M} = \mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$ , and the fitted values are  $\hat{y}_{ijk} = \bar{y}_{i..} + \bar{y}_{.j.} - \bar{y}_{...}$ .
- (b)  $\lambda_0\mu + \sum_i \lambda_i \alpha_i + \sum_j \kappa_j \beta_j$  is estimable iff  $\lambda_0 = \sum_i \lambda_i = \sum_j \kappa_j$ . The BLUE of a contrast  $\sum_i c_i \alpha_i$  is  $\sum_i c_i \bar{y}_{i..}$ , with variance  $\sigma^2 \sum_i c_i^2 / (bm)$ . These are the estimate and variance of the one-way layout in  $A$  that ignores  $B$ . The estimate of a contrast in the  $\alpha_i$  is uncorrelated with the estimate of any contrast in the  $\beta_j$ .
- (c) Under normal errors,  $SS_A = \|\mathbf{P}_A \mathbf{y}\|^2 = bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2$ ,  $SS_B = \|\mathbf{P}_B \mathbf{y}\|^2 = am \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2$  and  $SSE = \|\mathbf{R} \mathbf{y}\|^2$  are independent,  $SSE/\sigma^2 \sim \chi^2(n-a-b+1)$ , and

$$F_A = \frac{SS_A/(a-1)}{SSE/(n-a-b+1)} \sim F(a-1, n-a-b+1, \gamma_A), \quad \gamma_A = \frac{bm}{\sigma^2} \sum_i (\alpha_i - \bar{\alpha})^2,$$

with  $\bar{\alpha} = a^{-1} \sum_i \alpha_i$ . So  $F_A$  is an exact test of  $\alpha_1 = \dots = \alpha_a$ . Its numerator is that of the one-way  $F$  test in  $A$  that ignores  $B$  (Theorem 15.3.1); only the error term differs. The same holds for  $B$ , with  $\gamma_B = (am/\sigma^2) \sum_j (\beta_j - \bar{\beta})^2$ .

- (d) If  $m \geq 2$ , then  $\mathbf{R} = \mathbf{P}_{AB} + \mathbf{P}_E$  with  $\mathbf{P}_{AB} = \mathbf{C}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m$  and  $\mathbf{P}_E = \mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{C}_m$ , of ranks  $(a-1)(b-1)$  and  $ab(m-1)$ . The squared lengths  $\|\mathbf{P}_{AB} \mathbf{y}\|^2$  and  $\|\mathbf{P}_E \mathbf{y}\|^2$  are the lack-of-fit and pure-error sums of squares of the additive model (Definition 14.6.1). If the cell means  $\mu_{ij}$  are arbitrary, the lack-of-fit  $F$  test of additivity has noncentrality

$$\gamma_{AB} = \frac{m}{\sigma^2} \sum_{i,j} (\mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{..})^2.$$

- (e) If  $m = 1$ , then  $\mathbf{R} = \mathbf{C}_a \otimes \mathbf{C}_b$ , and the residual space consists of the  $a \times b$  tables whose rows and columns all sum to zero. There is no pure error.
- (f) (*Tukey's test.*) If  $m = 1$ , the errors are normal and  $(a-1)(b-1) \geq 2$ , then with  $SS_N$  as in Eq. 16.1.4,

$$F_N = \frac{SS_N}{(SSE - SS_N)/((a-1)(b-1)-1)} \sim F(1, (a-1)(b-1)-1).$$

*Proof.*

- (a) By Lemma 16.1.1(c) the three projections  $\mathbf{P}_0, \mathbf{P}_A, \mathbf{P}_B$  are mutually orthogonal, and their ranks are 1,  $a-1$ ,  $b-1$  by Lemma 16.1.1(b). Moreover  $\mathbf{P}_0 + \mathbf{P}_A = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m$  replaces each observation by its row mean. It is the projection onto  $\mathcal{C}(\mathbf{Z}_A)$ , the vectors that are constant on rows (Example (The one-way layout)). Likewise  $\mathbf{P}_0 + \mathbf{P}_B$  projects onto  $\mathcal{C}(\mathbf{Z}_B)$ . Hence

$$\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{Z}_A) + \mathcal{C}(\mathbf{Z}_B) = (\mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_A)) + (\mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_B)) = \mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_A) \oplus \mathcal{C}(\mathbf{P}_B),$$

a sum of mutually orthogonal subspaces. By Theorem 6.3.8 its projection is  $\mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$ . Then  $\mathbf{R} = \mathbf{I} - \mathbf{M}$  is the complementary projection, orthogonal to the other three, with rank  $n - (a+b-1)$ . The fitted values are  $\mathbf{M} \mathbf{y} = (\mathbf{P}_0 + \mathbf{P}_A) \mathbf{y} +$

$$(\mathbf{P}_0 + \mathbf{P}_B)\mathbf{y} - \mathbf{P}_0\mathbf{y}.$$

- (b) The design is complete, hence connected, and the estimability condition is [Exercise \(C1\)](#) with a single component. For the estimator,  $E(\bar{y}_{i..}) = \mu + \alpha_i + \bar{\beta}$ , so  $\sum_i c_i \bar{y}_{i..}$  is unbiased for  $\sum_i c_i \alpha_i$  when  $\sum_i c_i = 0$ . It equals  $\boldsymbol{\rho}^\top \mathbf{y}$  with  $\boldsymbol{\rho} = (bm)^{-1} \mathbf{c} \otimes \mathbf{1}_b \otimes \mathbf{1}_m$ , and  $\boldsymbol{\rho} = \mathbf{P}_A \boldsymbol{\rho} \in \mathcal{C}(\mathbf{X})$  because  $\mathbf{c} \perp \mathbf{1}_a$ . So  $(\mathbf{I} - \mathbf{M})\boldsymbol{\rho} = \mathbf{0}$ , and by [Eq. 7.2.1](#) its variance equals that of the BLUE. By [Theorem 7.2.1\(c\)](#) it is the BLUE. Its variance is  $\sigma^2 \|\boldsymbol{\rho}\|^2 = \sigma^2 \sum_i c_i^2 / (bm)$ . The one-way layout in  $A$  has  $bm$  observations per level, and [Theorem 8.7.1](#) gives the same estimator and variance. A contrast in the  $\beta_j$  has coefficient vector  $\boldsymbol{\rho}_B \in \mathcal{C}(\mathbf{P}_B)$ , and  $\text{Cov}(\boldsymbol{\rho}^\top \mathbf{Y}, \boldsymbol{\rho}_B^\top \mathbf{Y}) = \sigma^2 \boldsymbol{\rho}^\top \boldsymbol{\rho}_B = 0$ .
- (c) The spaces  $\mathcal{C}(\mathbf{P}_0), \mathcal{C}(\mathbf{P}_A), \mathcal{C}(\mathbf{P}_B)$  form an analysis of variance decomposition of  $\mathcal{C}(\mathbf{X})$  ([Definition 9.1.1](#)), so [Corollary 9.6.2](#) applies. The mean vector has entries  $\mu + \alpha_i + \beta_j$ . Its row means minus its grand mean are  $\alpha_i - \bar{\alpha}$ , so  $\|\mathbf{P}_A \boldsymbol{\mu}\|^2 = bm \sum_i (\alpha_i - \bar{\alpha})^2$ . This is zero iff all  $\alpha_i$  are equal. The formula for  $\text{SS}_A$  is the same computation applied to  $\mathbf{y}$ .
- (d) By [Lemma 16.1.1\(d\)](#) with  $t = 3$ ,  $\mathbf{I}$  is the sum of the eight products  $\mathbf{E}_a \otimes \mathbf{E}_b \otimes \mathbf{E}_m$ . The four with  $\mathbf{E}_m = \bar{\mathbf{J}}_m$  are  $\mathbf{P}_0, \mathbf{P}_A, \mathbf{P}_B, \mathbf{P}_{AB}$ . The four with  $\mathbf{E}_m = \mathbf{C}_m$  add up to  $\mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{C}_m = \mathbf{P}_E$ . So  $\mathbf{R} = \mathbf{P}_{AB} + \mathbf{P}_E$ , with ranks from [Lemma 16.1.1\(b\)](#). The matrix  $\mathbf{I} - \mathbf{P}_E = \mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m$  replaces each observation by its cell mean. It is the projection  $\mathbf{M}_Z$  onto the cell-indicator space, the  $\mathbf{Z}$  of [Definition 14.6.1](#). Hence  $\mathbf{P}_E = \mathbf{I} - \mathbf{M}_Z$  and  $\mathbf{P}_{AB} = \mathbf{M}_Z - \mathbf{M}$ . By [Theorem 14.6.2](#) the noncentrality is  $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\theta}\|^2 / \sigma^2$ , where  $\boldsymbol{\theta}$  has entries  $\mu_{ij}$ . Here  $(\mathbf{I} - \mathbf{M})\boldsymbol{\theta} = \mathbf{P}_{AB}\boldsymbol{\theta}$ , because  $\mathbf{P}_E\boldsymbol{\theta} = \mathbf{0}$ . Its entries are the displayed interaction terms, each repeated  $m$  times.
- (e) With  $m = 1$ ,  $\bar{\mathbf{J}}_1 = \mathbf{1}$  and  $\mathbf{C}_1 = \mathbf{0}$ , so  $\mathbf{P}_E = \mathbf{0}$  and  $\mathbf{R} = \mathbf{P}_{AB} = \mathbf{C}_a \otimes \mathbf{C}_b$ . A vector  $\mathbf{v}$ , read as an  $a \times b$  table  $\mathbf{V}$ , satisfies  $(\mathbf{C}_a \otimes \mathbf{C}_b)\mathbf{v} = \mathbf{v}$  iff  $\mathbf{C}_a \mathbf{V} \mathbf{C}_b = \mathbf{V}$ , which holds iff  $\mathbf{V} \mathbf{1}_b = \mathbf{0}$  and  $\mathbf{1}_a^\top \mathbf{V} = \mathbf{0}^\top$ .
- (f) Write  $\hat{y}_{ij} = \bar{y}_{..} + r_i + c_j$ , let  $\mathbf{r} = (r_i)$  and  $\mathbf{c} = (c_j)$ , and put  $\mathbf{U} = \mathbf{r} \otimes \mathbf{c}$ , the table  $(r_i c_j)$ . Expanding,

$$\hat{y}_{ij}^2 = (\bar{y}_{..}^2 + r_i^2 + 2\bar{y}_{..}r_i) + (c_j^2 + 2\bar{y}_{..}c_j) + 2r_i c_j.$$

The first bracket depends on  $i$  alone and the second on  $j$  alone, so both lie in  $\mathcal{C}(\mathbf{X})$ . Since  $\mathbf{1}_a^\top \mathbf{r} = \mathbf{1}_b^\top \mathbf{c} = 0$ ,  $\mathbf{R}\mathbf{U} = \mathbf{C}_a \mathbf{r} \otimes \mathbf{C}_b \mathbf{c} = \mathbf{U}$  by (e). Hence the column  $\mathbf{W} = (\hat{y}_{ij}^2)$ , a function of  $\mathbf{M}\mathbf{Y}$ , satisfies  $(\mathbf{I} - \mathbf{M})\mathbf{W} = 2\mathbf{U}$ . By [Theorem 9.2.2](#) the extra sum of squares from adding  $\mathbf{W}$  to the additive model is  $(\mathbf{U}^\top \mathbf{y})^2 / \mathbf{U}^\top \mathbf{U}$ , and  $\mathbf{U}^\top \mathbf{U} = \sum_i r_i^2 \sum_j c_j^2$ , so it is  $\text{SS}_N$ , and  $\text{SSE} - \text{SS}_N$  is the residual sum of squares of the augmented model. The rank condition of [Theorem 14.7.2](#) holds because  $\mathbf{U} \neq \mathbf{0}$  with probability one:  $\mathbf{U} = \mathbf{0}$  only if  $\mathbf{P}_A \mathbf{Y} = \mathbf{0}$  or  $\mathbf{P}_B \mathbf{Y} = \mathbf{0}$ , and each event has probability zero, because  $\|\mathbf{P}_A \mathbf{Y}\|^2 / \sigma^2$  has a continuous (noncentral  $\chi^2$ ) distribution on  $a - 1 \geq 1$  degrees of freedom, and likewise for  $B$ . So  $\text{rank}[\mathbf{X}, \mathbf{W}] = a + b$ , which is less than  $n = ab$  because  $(a - 1)(b - 1) \geq 2$ . [Theorem 14.7.2](#) with  $q = 1$  and  $n - r - q = (a - 1)(b - 1) - 1$  gives the distribution of  $F_N$ .

■

Part (b) is the practical content of balance: rows are compared exactly as if the columns did not exist. The columns still matter, because they remove their own variation from the error.

The listing builds the five projections of part (d) for  $a = 3$ ,  $b = 4$ ,  $m = 2$  and checks their ranks, orthogonality and sum. The script also checks each identification in the proof against projections computed from indicator columns.

```

a, b, m = 3, 4, 2
P = factorial_projections([a, b], m)
names = {(0, 0): "mean", (1, 0): "A", (0, 1): "B", (1, 1): "AB", "error": "error"}
for key, name in names.items():
    print(f"{name:6s} rank {rank(P[key])}")
keys = list(names)
worst = max(np.abs(P[s] @ P[t]).max() for s, t in itertools.combinations(keys, 2))
print("largest entry of any product P_s P_t, s != t:", f"{worst:.1e}")
print("sum is the identity:", np.allclose(sum(P[k] for k in keys), np.eye(a * b * m)
))

```

## 16.1.5 Tukey's test in practice

Tukey's test ([Theorem 16.1.2\(f\)](#)) is exact against every additive model, whatever the values of  $\mu$ ,  $\alpha_i$ ,  $\beta_j$ . It has power against interactions roughly proportional to  $\alpha_i\beta_j$ , the kind produced by measuring on the wrong scale, and little against anything else. The estimated slope says which scale. If  $f$  is the inverse of a power transformation, so that  $y^p$  is additive, then  $f(t) = t^{1/p}$ , and [Eq. 16.1.3](#) gives  $\theta = (1 - p)/f(\nu)$  ([Exercise \(B3\)](#)). Replacing  $f(\nu)$  by the grand mean suggests the power

$$\hat{p} = 1 - \hat{\theta} \bar{y}_{..}, \quad (16.1.5)$$

with  $\hat{p} = 0$  read as the logarithm. The rule is rough, but it turns a vague complaint into a specific suggestion.

### Example. (Solvers on benchmark problems)

Six solvers were each run once on eight benchmark problems, and the running time was recorded in seconds. The data are synthetic, generated by the script `tukey_runtime.py` so that the logarithm of time is additive in solver and problem, with a little noise:

|    | P1  | P2  | P3   | P4   | P5   | P6    | P7    | P8    |
|----|-----|-----|------|------|------|-------|-------|-------|
| S1 | 1.7 | 3.4 | 5.6  | 14.1 | 28.1 | 50.2  | 82.9  | 249.4 |
| S2 | 2.1 | 5.7 | 9.0  | 14.7 | 36.9 | 76.0  | 116.7 | 292.0 |
| S3 | 1.3 | 3.1 | 5.1  | 10.5 | 18.8 | 37.6  | 82.2  | 148.3 |
| S4 | 3.3 | 9.1 | 12.7 | 19.1 | 52.8 | 110.0 | 183.1 | 371.8 |
| S5 | 2.0 | 4.7 | 7.2  | 13.5 | 31.4 | 65.2  | 111.1 | 289.3 |
| S6 | 1.5 | 3.4 | 5.4  | 10.4 | 26.0 | 43.2  | 106.3 | 172.0 |

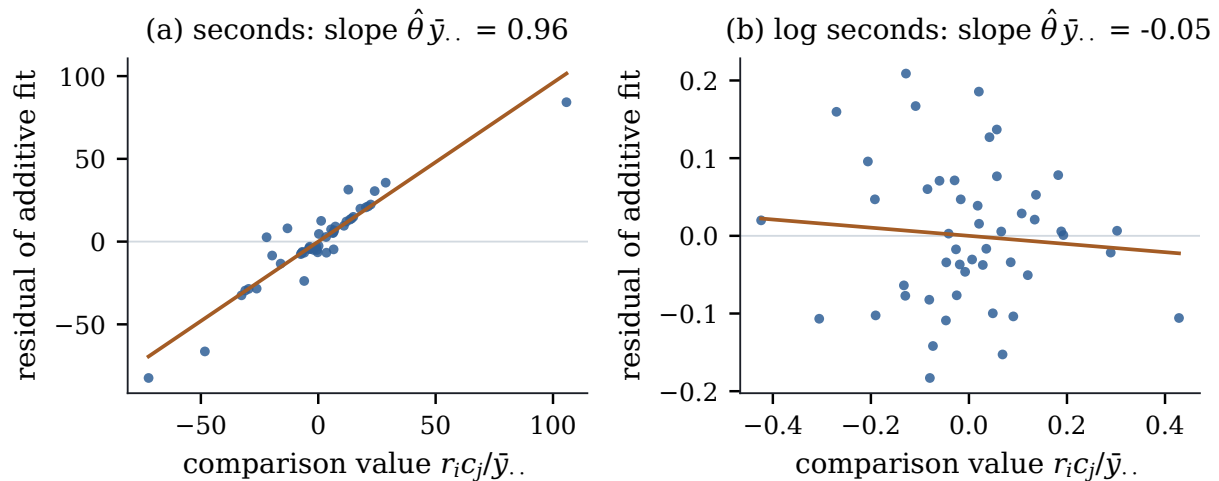


Figure 16.1.1: Tukey's diagnostic plot for the solver data: residuals of the additive fit against the comparison values  $r_i c_j / \bar{y}_{..}$ , with the fitted line of slope  $\hat{\theta}\bar{y}_{..} = 1 - \hat{p}$ . (a) In seconds the residuals lie along a line of slope near one, which points to the logarithm. (b) In log seconds there is no trend.

The additive model in seconds leaves a residual sum of squares of 29396 on 35 degrees of freedom. Tukey's single direction takes  $SS_N = 26126$  of it, a fraction 0.889, and  $F_N = 271.7$  on 1 and 34 degrees of freedom ( $p = 9 \times 10^{-18}$ ). The interaction has exactly the shape of Eq. 16.1.3: fast problems separate the solvers by fractions of a second, slow problems by minutes. The slope is  $\hat{\theta} = 0.01563$  and the grand mean is 61.46, so Eq. 16.1.5 suggests the power 0.04, essentially the logarithm. Figure 16.1.1 shows the diagnostic plot. On the log scale,  $F_N = 0.27$  ( $p = 0.60$ ), and the residuals show no trend against  $r_i c_j / \bar{y}_{..}$ .

The scale changes the conclusion about the solvers. In seconds,  $F_A = 3.88$  ( $p = 0.007$ ), because the nonadditivity inflates the error term. In log seconds,  $F_A = 77.7$ . The residual standard deviation is  $s = 0.105$ , so differences between solvers are measured in proportions of about ten per cent. By Exercise (B1) the solver means of log time are independent with common variance  $\sigma^2/8$  and independent of  $s^2$ . Tukey's method therefore gives simultaneous intervals for all 15 ratios of solver speeds, with multiplier  $q_{0.05}(6, 35) = 4.261$ . Solver 4, the slowest, takes 2.50 times as long as solver 3, the fastest, with simultaneous 95% interval [2.14, 2.94].

```
import numpy as np
from scipy import stats

gen_rng = np.random.default_rng(16_01)
a, b = 6, 8
solver_log = np.array([0.0, 0.25, -0.35, 0.55, 0.1, -0.2]) # log speed
factors
problem_log = np.log([2.0, 4.5, 7.0, 12.0, 25.0, 60.0, 110.0, 240.0]) # typical
seconds
noise = gen_rng.normal(scale=0.12, size=(a, b))
times = np.round(np.exp(solver_log[:, None] + problem_log[None, :] + noise), 1)

def additive_fit(Y):
    """Row effects, column effects and residuals of the additive fit to an a x b
    table."""
    g = Y.mean()
```

```

r = Y.mean(axis=1) - g                                # ybar_i. - ybar..
c = Y.mean(axis=0) - g                                # ybar_.j - ybar..
resid = Y - g - r[:, None] - c[None, :]
return g, r, c, resid

def tukey_test(Y):
    """Tukey's one-degree-of-freedom test for nonadditivity in an a x b table."""
    a, b = Y.shape
    g, r, c, resid = additive_fit(Y)
    u = np.outer(r, c)                                # lies in the interaction space
    ss_n = (u * Y).sum() ** 2 / (u * u).sum()
    sse = (resid ** 2).sum()
    df = (a - 1) * (b - 1) - 1
    F = ss_n / ((sse - ss_n) / df)
    theta = (u * Y).sum() / (u * u).sum()             # coefficient of a_i b_j
    return F, stats.f.sf(F, 1, df), theta, g, sse, ss_n

for label, Y in [("seconds", times), ("log seconds", np.log(times))]:
    F, p, theta, g, sse, ss_n = tukey_test(Y)
    print(f"{label:12s} F = {F:9.2f} p = {p:.2g} theta = {theta:.4f}"
          f" suggested power 1 - theta*ybar = {1 - theta * g:.2f}")

```

The listing below checks [Theorem 16.1.2\(f\)](#) by simulation. It draws 20000 tables from an additive normal model with  $a = 6$  and  $b = 8$ . The 5% test rejects in a fraction 0.0508 of them, and a Kolmogorov–Smirnov comparison of the simulated statistics with  $F(1, 34)$  gives  $p = 0.98$ .

```

rng = np.random.default_rng(20260919)
reps = 20_000
Fs = np.empty(reps)
for t in range(reps):
    Y = 3.0 + np.arange(a)[:, None] * 0.5 + np.arange(b)[None, :] + rng.normal(size=(a, b))
    Fs[t] = tukey_test(Y)[0]
size = np.mean(Fs > stats.f.ppf(0.95, 1, (a - 1) * (b - 1) - 1))
ks = stats.kstest(Fs, stats.f(1, (a - 1) * (b - 1) - 1).cdf)
print(f"simulated size of the 5% test: {size:.4f} Kolmogorov-Smirnov p-value: {ks.pvalue:.2f}")

```

*Warning.* A nonsignificant Tukey test does not establish additivity. It rules out only interaction of the form  $\theta\alpha_i\beta_j$ . A single aberrant cell or row can leave  $F_N$  small and still distort every conclusion, so look at the residual table and an interaction plot ([Section 16.2](#)) as well.

## 16.1.6 Exercises

### 16.1.6.1 A. Check your understanding

#### Exercise A1.

For  $a = b = 2$  and  $m = 1$ , write out the  $4 \times 4$  matrices  $\mathbf{P}_0$ ,  $\mathbf{P}_A$ ,  $\mathbf{P}_B$  and  $\mathbf{R}$  of [Eq. 16.1.2](#). Show that  $\mathbf{R}\mathbf{y} = \frac{1}{4}(y_{11} - y_{12} - y_{21} + y_{22})(1, -1, -1, 1)^\top$ , and explain why Tukey's test

is not available here.

### Exercise A2.

In the additive model with  $m$  observations per cell, show that the sum of squares for testing a single contrast  $\sum_i c_i \alpha_i = 0$  is  $(\sum_i c_i \bar{y}_{i..})^2 / (\sum_i c_i^2 / (bm))$ , the same as in the one-way layout in  $A$  ([Proposition 15.2.1](#)). What differs between the two  $F$  tests of this contrast, the one here and the one-way test that ignores  $B$ ?

### 16.1.6.2 B. Practice

#### Exercise B1.

In the additive model, suppose  $\alpha_1 = \delta$  and  $\alpha_2 = \dots = \alpha_a = 0$ . Find  $\gamma_A$ . With  $a = 4$ ,  $b = 5$  and  $\delta = \sigma$ , compare the power of the 5% test for  $m = 1$  and  $m = 2$  using [Theorem 11.5.1](#), and explain why doubling  $m$  more than doubles the error degrees of freedom.

*Solution.* Here  $\bar{\alpha} = \delta/a$ , and  $\sum_i (\alpha_i - \bar{\alpha})^2 = \delta^2(1 - 1/a)^2 + (a - 1)\delta^2/a^2 = \delta^2(a - 1)/a$ . So  $\gamma_A = bm\delta^2(a - 1)/(a\sigma^2)$ . With  $a = 4$ ,  $b = 5$ ,  $\delta = \sigma$  this is  $3.75m$ . The error degrees of freedom are  $n - a - b + 1 = 20m - 8$ , which is 12 for  $m = 1$  and 32 for  $m = 2$ . The power is  $\Pr\{F(3, 20m - 8, 3.75m) > F_{0.05}(3, 20m - 8)\}$ , about 0.25 for  $m = 1$  and 0.56 for  $m = 2$  (computed by the script `exercise_checks.py`). The noncentrality doubles, and the error degrees of freedom more than double because the  $a + b - 1$  degrees of freedom spent on the mean structure are a fixed cost. Both effects raise the power.

#### Exercise B2.

Show that Tukey's statistic  $F_N$  is unchanged if  $\mathbf{y}$  is replaced by  $\lambda\mathbf{y} + \mathbf{t}$ , where  $\lambda \neq 0$  and  $\mathbf{t}$  is any additive table  $t_{ij} = u_i + v_j$ . Conclude that  $F_N$  has the same distribution under every additive model with the same  $\sigma^2$ .

#### Exercise B3.

Suppose  $y^p$  is additive for some  $p \neq 0$ , so that  $\mu_{ij} = f(\nu + a_i + b_j)$  with  $f(t) = t^{1/p}$ . Verify that  $\theta = f''(\nu)/f'(\nu)^2 = (1 - p)/f(\nu)$ , and derive [Eq. 16.1.5](#). What does  $\hat{\theta}\bar{y}_{..} \approx 1$  suggest, and why is  $p = 0$  the right reading?

*Solution.*  $f'(t) = p^{-1}t^{1/p-1}$  and  $f''(t) = p^{-1}(p^{-1} - 1)t^{1/p-2}$ . So

$$\frac{f''(\nu)}{f'(\nu)^2} = \frac{p^{-1}(p^{-1} - 1)\nu^{1/p-2}}{p^{-2}\nu^{2/p-2}} = p\left(\frac{1}{p} - 1\right)\nu^{-1/p} = \frac{1 - p}{f(\nu)}.$$

Solving  $\theta = (1 - p)/f(\nu)$  for  $p$ , and estimating  $\theta$  by  $\hat{\theta}$  and  $f(\nu)$ , the typical size of a cell mean, by  $\bar{y}_{..}$ , gives  $\hat{p} = 1 - \hat{\theta}\bar{y}_{..}$ . A value  $\hat{\theta}\bar{y}_{..} \approx 1$  gives  $\hat{p} \approx 0$ . For  $f(t) = e^t$ , which makes  $\log y$  additive,  $f''/f'^2 = e^{-\nu} = 1/f(\nu)$ , which is the limit  $p \rightarrow 0$  of the power formula. This matches the convention that  $(y^p - 1)/p \rightarrow \log y$ .



**Exercise B4.**

With  $m = 1$  and normal errors, suppose the cell means are  $\mu_{ij} = \mu + \alpha_i + \beta_j + \theta\alpha_i\beta_j$  with  $\sum_i \alpha_i = \sum_j \beta_j = 0$ , exactly the shape of [Eq. 16.1.3](#). Show that  $r_i = \alpha_i + \bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}$ , and that, given  $\mathbf{P}_A \mathbf{Y}$  and  $\mathbf{P}_B \mathbf{Y}$ ,

$$E(\hat{\theta} \mid \mathbf{P}_A \mathbf{Y}, \mathbf{P}_B \mathbf{Y}) = \theta \frac{\sum_i r_i \alpha_i}{\sum_i r_i^2} \cdot \frac{\sum_j c_j \beta_j}{\sum_j c_j^2}.$$

Explain why  $\hat{\theta}$  is nearly unbiased when the row and column effects are large compared with  $\sigma$ , and why it tends to be pulled towards zero when they are not. What does this imply for the power rule [Eq. 16.1.5](#)?

*Solution.* The interaction terms  $\theta\alpha_i\beta_j$  have zero row and column means, so  $\bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot} = \alpha_i$  and  $r_i = \alpha_i + \bar{\varepsilon}_{i\cdot} - \bar{\varepsilon}_{\cdot\cdot}$ ; likewise for  $c_j$ . With  $\mathbf{U} = \mathbf{r} \otimes \mathbf{c}$  as in the proof of [Theorem 16.1.2\(f\)](#),  $\hat{\theta} = \mathbf{U}^\top \mathbf{R} \mathbf{Y} / \mathbf{U}^\top \mathbf{U}$ . By [Theorem 4.4.4](#),  $\mathbf{R} \mathbf{Y}$  is independent of  $(\mathbf{P}_A \mathbf{Y}, \mathbf{P}_B \mathbf{Y})$ , whatever the mean, so given those two vectors  $\mathbf{U}$  is fixed and  $E(\mathbf{R} \mathbf{Y}) = \mathbf{R} \boldsymbol{\mu}$ , the table  $(\theta\alpha_i\beta_j)$ . Hence the conditional mean is  $\theta \sum_{i,j} r_i c_j \alpha_i \beta_j / (\sum_i r_i^2 \sum_j c_j^2)$ , which factorizes as displayed. If the effects dominate the noise,  $\mathbf{r} \approx \boldsymbol{\alpha}$  and both factors are near one. If not,  $\sum_i r_i^2$  carries the extra noise  $\sigma^2(a-1)/b$  on average while  $\sum_i r_i \alpha_i$  does not: the attenuation of a regression on a noisy regressor. The power rule then suggests a  $\hat{p}$  too close to one, too weak a transformation.

**16.1.6.3 C. Going deeper****Exercise C1.**

Extend Tukey's test to  $m \geq 2$  observations per cell. Replace  $y_{ij}$  by the cell mean  $\bar{y}_{ij}$  in [Eq. 16.1.4](#) and multiply the result by  $m$ , and compare it with the pure-error mean square. Show that under the additive model the resulting statistic has the  $F(1, ab(m-1))$  distribution. Show also that  $SS_{AB} - SS_N$  gives a test of the remaining nonadditivity.

## 16.2 Interaction and its interpretation

The additive model says that the difference between two levels of  $A$  is the same at every level of  $B$ . *Interaction* is the failure of that statement. This section defines interaction through the cell means, so that its meaning does not depend on a parameterization. It then shows how to see interaction in a plot and when a change of scale removes it, and explains why a “main effect” in the presence of interaction is a weighted average whose weights someone has to choose.

### 16.2.1 Cell means and the definition of interaction

Write  $\mu_{ij}$  for the mean response in cell  $(i, j)$ . The **cell-means model**  $E(y_{ijk}) = \mu_{ij}$  places no restriction on the  $ab$  means. By [Proposition 8.6.2\(a\)](#) it has the same mean space as the overparameterized model  $\mu + \alpha_i + \beta_j + \gamma_{ij}$ , and every cell mean of an occupied cell is estimable. Averages with equal weights split the table of means into four parts:

$$\mu_{ij} = \bar{\mu}_{..} + (\bar{\mu}_{i.} - \bar{\mu}_{..}) + (\bar{\mu}_{.j} - \bar{\mu}_{..}) + (\mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{..}). \quad (16.2.1)$$

This is an algebraic identity. The four parts are the images of the vector of cell means under  $\bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$ ,  $\mathbf{C}_a \otimes \bar{\mathbf{J}}_b$ ,  $\bar{\mathbf{J}}_a \otimes \mathbf{C}_b$  and  $\mathbf{C}_a \otimes \mathbf{C}_b$ , which are mutually orthogonal projections (Lemma 16.1.1).

### Definition 16.2.1 — Interaction

Let  $\mu_{ij}$  be the cell means of a two-way layout.

- (a) An **interaction contrast** is a linear function  $\sum_{i,j} d_{ij} \mu_{ij}$  whose coefficient table  $(d_{ij})$  is not zero and has every row sum and every column sum equal to zero. The simplest is the **tetrad difference**  $\mu_{ij} - \mu_{il} - \mu_{kj} + \mu_{kl}$  of rows  $i \neq k$  and columns  $j \neq l$ . It is the difference between columns  $j$  and  $l$  in row  $i$ , minus the same difference in row  $k$ .
- (b) The factors **interact** if some interaction contrast is nonzero. Otherwise the layout is **additive**.
- (c) In the *effects form*  $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$  of Eq. 16.2.1, the **main effects** are  $\alpha_i = \bar{\mu}_{i.} - \bar{\mu}_{..}$  and  $\beta_j = \bar{\mu}_{.j} - \bar{\mu}_{..}$ , and the **interaction effects** are  $\gamma_{ij} = \mu_{ij} - \bar{\mu}_{i.} - \bar{\mu}_{.j} + \bar{\mu}_{..}$ . They satisfy  $\sum_i \alpha_i = \sum_j \beta_j = 0$ , and every row and column of  $(\gamma_{ij})$  sums to zero.

The cell means are what the data estimate. The effects in (c) are one coordinate system for them, fixed by the choice of equal weights. The next result shows that whether interaction is present does not depend on that choice.

### Proposition 16.2.1 — Equivalent forms of additivity

For a table of cell means  $(\mu_{ij})$ , the following are equivalent:

- (i) every tetrad difference is zero;
- (ii) for all rows  $i, k$ , the difference  $\mu_{ij} - \mu_{kj}$  does not depend on  $j$ ;
- (iii)  $\mu_{ij} = u_i + v_j$  for some numbers  $u_1, \dots, u_a$  and  $v_1, \dots, v_b$ ;
- (iv) every interaction contrast is zero;
- (v) for some pair of weight vectors  $\mathbf{s} \in \mathbb{R}^a$  and  $\mathbf{w} \in \mathbb{R}^b$  with  $\sum_k s_k = \sum_l w_l = 1$ ,

$$\mu_{ij} - \sum_l w_l \mu_{il} - \sum_k s_k \mu_{kj} + \sum_{k,l} s_k w_l \mu_{kl} = 0 \quad \text{for all } i, j;$$

- (vi) the condition in (v) holds for every such pair of weight vectors, in particular  $\gamma_{ij} = 0$  for all  $i, j$ .

Moreover, the coefficient tables of interaction contrasts, together with the zero table, form a subspace of dimension  $(a-1)(b-1)$ , and  $\sum_{i,j} d_{ij} \mu_{ij} = \sum_{i,j} d_{ij} \gamma_{ij}$  for each of them.

*Proof.* (i) $\Rightarrow$ (ii): a zero tetrad difference says  $\mu_{ij} - \mu_{kj} = \mu_{il} - \mu_{kl}$ . (ii) $\Rightarrow$ (iii): put  $u_i = \mu_{i1} - \mu_{11}$  and  $v_j = \mu_{1j}$ . By (ii),  $\mu_{ij} - \mu_{1j} = \mu_{i1} - \mu_{11}$ , that is,  $\mu_{ij} = u_i + v_j$ . (iii) $\Rightarrow$ (iv):  $\sum_{i,j} d_{ij}(u_i + v_j) = \sum_i u_i \sum_j d_{ij} + \sum_j v_j \sum_i d_{ij} = 0$ . (iv) $\Rightarrow$ (i): a tetrad difference is an interaction contrast. (iii) $\Rightarrow$ (vi): substituting  $u_i + v_j$ , the four terms are

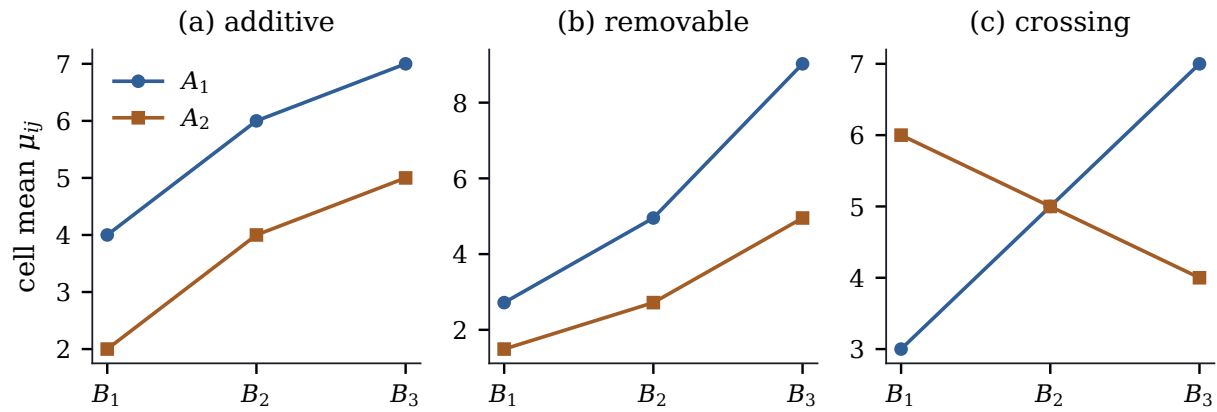


Figure 16.2.1: Three tables of cell means for a  $2 \times 3$  layout. (a) Additive: parallel profiles. (b) Interaction that the logarithm removes: the means are exp of an additive table. (c) Crossing interaction: the order of the rows reverses between  $B_1$  and  $B_3$ , and no increasing transformation can make the profiles parallel.

$u_i + v_j$ ,  $u_i + \sum_l w_l v_l$ ,  $\sum_k s_k u_k + v_j$  and  $\sum_k s_k u_k + \sum_l w_l v_l$ , and the signed sum is zero. (vi) $\Rightarrow$ (v) is trivial. (v) $\Rightarrow$ (iii): (v) writes  $\mu_{ij}$  as  $\sum_l w_l \mu_{il}$ , which depends only on  $i$ , plus  $\sum_k s_k \mu_{kj} - \sum_{k,l} s_k w_l \mu_{kl}$ , which depends only on  $j$ .

For the last statement, read tables as vectors in  $\mathbb{R}^{ab}$  with  $j$  running fastest. A table has zero row and column sums iff it is fixed by  $\mathbf{C}_a \otimes \mathbf{C}_b$ , as in the proof of Theorem 16.1.2(e). So the coefficient tables form  $\mathcal{C}(\mathbf{C}_a \otimes \mathbf{C}_b)$ , of dimension  $(a-1)(b-1)$  by Lemma 16.1.1(b). Such a table is orthogonal to the first three parts of Eq. 16.2.1, which leaves  $\sum d_{ij} \gamma_{ij}$ .

■

So interaction is a property of the table of cell means, with  $(a-1)(b-1)$  dimensions. In the overparameterized model  $\mu + \alpha_i + \beta_j + \gamma_{ij}$  without side conditions,  $\alpha_i - \alpha_k$  is not estimable (Proposition 8.6.2). The function  $\alpha_i - \alpha_k + \bar{\gamma}_i - \bar{\gamma}_k$  is estimable, because it equals  $\bar{\mu}_{i\cdot} - \bar{\mu}_{k\cdot}$ . Software that prints a “main effect” in such a model has chosen side conditions (Theorem 8.4.2) and hence a set of weights, and the printed number depends on that choice. The last part of this section explains why.

## 16.2.2 Interaction plots

An **interaction plot** draws the estimated cell means  $\bar{y}_{ij}$  against the levels of one factor, with one line (a *profile*) for each level of the other. By Proposition 16.2.1(ii), the layout is additive iff the profiles of the true means are parallel, that is, vertical translates of each other. Figure 16.2.1 shows three patterns of true means. In (a) the profiles are parallel. In (b) they fan out, but the row that is higher in one column is higher in every column. In (c) they cross.

Estimated profiles are never exactly parallel. Whether the departures exceed the noise is a question for the  $F$  test of Section 16.3. The plot shows *where* the interaction is, which is usually more useful.

### Example. (Flour and proofing temperature)

A bakery trial measured the volume (ml) of loaves made from three flours (white, whole-

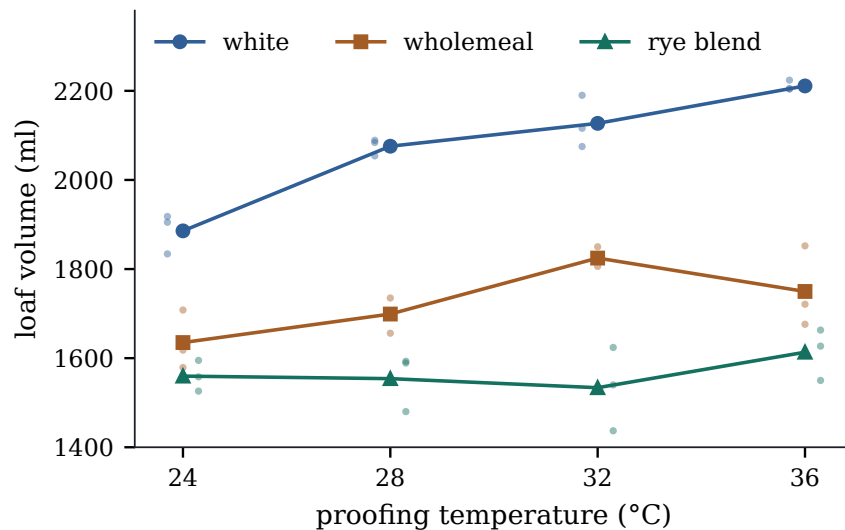


Figure 16.2.2: Interaction plot for the bread trial: cell means of loaf volume against proofing temperature, one profile per flour, with the individual loaves as faint points.

meal and a rye blend), with the dough proofed at four temperatures (24, 28, 32 and 36 °C). Three loaves were baked for each of the 12 combinations, 36 in all, in random order. The data are synthetic, generated by the script `bread_factorial.py`. The cell means are

|           | 24 °C  | 28 °C  | 32 °C  | 36 °C  | mean   |
|-----------|--------|--------|--------|--------|--------|
| white     | 1885.7 | 2075.7 | 2127.0 | 2211.0 | 2074.8 |
| wholemeal | 1635.0 | 1699.0 | 1824.7 | 1749.7 | 1727.1 |
| rye blend | 1559.7 | 1554.0 | 1533.7 | 1613.3 | 1565.2 |
| mean      | 1693.4 | 1776.2 | 1828.4 | 1858.0 | 1789.0 |

Figure 16.2.2 is the interaction plot. White flour gains volume steadily with temperature. Wholemeal gains up to 32 °C and then loses. The rye blend hardly responds. The profiles are far from parallel. The within-cell standard deviation, estimated in Section 16.3, is  $s = 56.4$  ml, so a cell mean of three loaves has standard error  $56.4/\sqrt{3} \approx 32.6$  ml, much smaller than the departures from parallelism. Yet the flours never change order: at every temperature white beats wholemeal, which beats the rye blend. The interaction changes the *size* of the flour differences but not their *direction*.

### 16.2.3 Removable and nonremovable interaction

Panel (b) of Figure 16.2.1 shows the pattern of the solver data of Example (Solvers on benchmark problems): the effect of  $A$  is a constant *ratio*, and on the logarithmic scale the table is additive. Such interaction is **removable**: some strictly increasing  $g$  makes the table  $g(\mu_{ij})$  additive. When can no transformation remove an interaction?

#### Proposition 16.2.2 — An order condition for removable interaction

If  $g$  is strictly increasing and the table  $g(\mu_{ij})$  is additive, then the rows are ordered in

the same way in every column, and the columns in the same way in every row. That is, for all  $i, k$  the sign of  $\mu_{ij} - \mu_{kj}$  does not depend on  $j$ , and for all  $j, l$  the sign of  $\mu_{ij} - \mu_{il}$  does not depend on  $i$ . In particular, if  $\mu_{ij} > \mu_{kj}$  and  $\mu_{il} < \mu_{kl}$  for some  $i, k, j, l$ , no strictly increasing transformation removes the interaction.

*Proof.* Write  $g(\mu_{ij}) = u_i + v_j$  (Proposition 16.2.1(iii)). Since  $g$  is strictly increasing,  $\mu_{ij} > \mu_{kj}$  iff  $g(\mu_{ij}) > g(\mu_{kj})$  iff  $u_i > u_k$ , a condition that does not involve  $j$ . The same holds with  $>$  replaced by  $=$  or  $<$ . The statement for columns is symmetric. ■

This justifies a distinction that is often drawn informally. An interaction is **quantitative** (or *ordinal*) if the orders are consistent in the sense of the proposition, and **qualitative** (or *crossing, disordinal*) if the order of some pair of levels reverses. Qualitative interaction is a fact about the phenomenon on every monotone scale. Quantitative interaction may be a fact about the scale of measurement. Whether an interaction is qualitative can depend on which factor one looks along. In the bread trial the flours keep their order at every temperature, but the temperatures do not keep their order for every flour. White flour gives more volume at 36 °C than at 32 °C, and wholemeal gives less. If the true means follow the same pattern as the estimates, the proposition says that no rescaling of volume makes the table additive. The condition is necessary but not sufficient. With three or more levels of each factor, a table can have consistent orders and still admit no additive rescaling (Exercise (C1)).

The listing checks the three tables of Figure 16.2.1. It also shows that the logarithm *creates* interaction in the additive table (a). Additivity is a property of a table on a given scale, and so is its absence.

```
import itertools

import numpy as np

additive = np.array([[4.0, 6.0, 7.0], [2.0, 4.0, 5.0]])           # rows: levels of
A
removable = np.exp(np.array([[1.0, 1.6, 2.2], [0.4, 1.0, 1.6]])) # exp of an
additive table
crossing = np.array([[3.0, 5.0, 7.0], [6.0, 5.0, 4.0]])

def interaction(mu):
    """mu_ij - mu_i. - mu_.j + mu_.. (equal weights)."""
    return mu - mu.mean(1, keepdims=True) - mu.mean(0, keepdims=True) + mu.mean()

def same_orders(mu):
    """Necessary condition for removability: every column orders the rows alike, and
    every row the columns."""
    a, b = mu.shape
    rows_ok = all(len(set(np.sign(mu[i] - mu[k]))) == 1 for i, k in itertools.
combinations(range(a), 2))
    cols_ok = all(len(set(np.sign(mu[:, j] - mu[:, l]))) == 1 for j, l in itertools.
combinations(range(b), 2))
    return rows_ok and cols_ok

for name, mu in [("additive", additive), ("removable", removable), ("crossing",
```

```
crossing)]:
print(f"{name:9s} max |interaction| raw {np.abs(interaction(mu)).max():.3f}"
      f"   log {np.abs(interaction(np.log(mu))).max():.3f}   same orders: {
same_orders(mu)}")
```

Two cautions apply to transforming data rather than means. Since  $E g(Y) \neq g(E Y)$  for nonlinear  $g$ , additivity of  $g(y)$  and of  $g(\mu_{ij})$  agree only approximately, for small errors. And a transformation also changes the shape and variance of the errors. In [Example \(Solvers on benchmark problems\)](#) the logarithm happened to fix both, because the noise was multiplicative. The choice of scale is the subject of [Chapter 22](#).

## 16.2.4 Main effects in the presence of interaction

Without interaction, “the effect of  $A$ ” is the common difference  $\mu_{ij} - \mu_{kj}$ . With interaction, the difference between rows  $i$  and  $k$  is a list of **simple effects**  $\delta_j = \mu_{ij} - \mu_{kj}$ , and a single summary must average them with weights. For a weight vector  $\mathbf{w}$  with  $w_j \geq 0$  and  $\sum_j w_j = 1$ , define the **w-weighted main effect** difference of rows  $i$  and  $k$  as  $\sum_j w_j \delta_j = \sum_j w_j \mu_{ij} - \sum_j w_j \mu_{kj}$ .

### Proposition 16.2.3 — Main effects depend on the weights

Fix rows  $i \neq k$  and let  $\delta_j = \mu_{ij} - \mu_{kj}$ .

- (a) If the  $\delta_j$  are all equal, every weighted main effect difference equals their common value.
- (b) If they are not all equal, the values of  $\sum_j w_j \delta_j$  over weight vectors with every  $w_j > 0$  fill the open interval  $(\min_j \delta_j, \max_j \delta_j)$ . In particular, if the simple effects change sign, some weighting with all weights positive makes the main effect of  $A$  for these two rows exactly zero, and other weightings give it either sign.
- (c) If all  $\delta_j > 0$ , every weighted main effect difference is positive.

*Proof.*

- (a) and (c) are immediate, and a weighted average with positive weights of numbers that are not all equal lies strictly between their minimum and maximum. Conversely, let  $\min_j \delta_j < t < \max_j \delta_j$ , attained at columns  $j_0$  and  $j_1$ , and let  $\bar{\delta} = b^{-1} \sum_j \delta_j$ . For  $0 < \varepsilon < 1$  put  $t_\varepsilon = (t - \varepsilon \bar{\delta}) / (1 - \varepsilon)$ . Since  $t_\varepsilon \rightarrow t$  as  $\varepsilon \rightarrow 0$ , there is an  $\varepsilon$  with  $\delta_{j_0} < t_\varepsilon < \delta_{j_1}$ . Then  $t_\varepsilon = \lambda \delta_{j_0} + (1 - \lambda) \delta_{j_1}$  for some  $\lambda \in (0, 1)$ . The weights  $\varepsilon/b$  on every column, plus  $(1 - \varepsilon)\lambda$  on  $j_0$  and  $(1 - \varepsilon)(1 - \lambda)$  on  $j_1$ , are all positive and give  $\varepsilon \bar{\delta} + (1 - \varepsilon) t_\varepsilon = t$ .

■

In table (c) of [Figure 16.2.1](#) the simple effects of  $A_1$  against  $A_2$  are  $-3, 0$  and  $3$ . With equal weights the main effect of  $A$  is exactly zero, although  $A$  matters in two of the three columns. With weights  $(\frac{1}{2}, \frac{1}{2}, 0)$  it is  $-1.5$ , and with  $(0, \frac{1}{2}, \frac{1}{2})$  it is  $+1.5$ .

The balanced analysis of [Section 16.3](#) uses equal weights: its sum of squares for  $A$  tests  $\bar{\mu}_{1.} = \dots = \bar{\mu}_{a.}$ . In unbalanced data the various sums of squares use different weights, some determined by the cell counts ([Theorem 9.3.2](#)), and [Chapter 17](#) sorts them out ([Theorem 17.2.2](#)). Equal

weights are a convention, natural when the levels of  $B$  were chosen as a representative spread.

For the bread trial the simple effects of white against wholemeal flour are 250.7, 376.7, 302.3 and 461.3 ml at the four temperatures. The equal-weight main effect is their average, 347.7 ml. A bakery that proofs mostly warm might weight the temperatures in the proportions 1 : 1 : 2 : 4, and then the relevant difference is 384.7 ml. Both figures answer different questions correctly. Because all four simple effects are positive, part (c) guarantees that white flour gives larger loaves than wholemeal under every weighting, a statement that needs no choice of weights.

*Idea.* Report main effects when the interaction is small or quantitative, and say what they average over. When the interaction is qualitative, report simple effects. A main effect that averages a positive and a negative effect answers a question nobody asked.

## 16.2.5 Exercises

### 16.2.5.1 A. Check your understanding

#### Exercise A1.

For the crossing table  $\mu = \begin{pmatrix} 3 & 5 & 7 \\ 6 & 5 & 4 \end{pmatrix}$ , compute  $\bar{\mu}_{..}$ ,  $\alpha_i$ ,  $\beta_j$  and  $\gamma_{ij}$  of Definition 16.2.1(c). Check that the tetrad difference of columns 1 and 3 equals  $\gamma_{11} - \gamma_{13} - \gamma_{21} + \gamma_{23}$ .

*Solution.*  $\bar{\mu}_{..} = 5$ , row means 5, 5, so  $\alpha_1 = \alpha_2 = 0$ . Column means 4.5, 5, 5.5, so  $\beta = (-0.5, 0, 0.5)$ . Then  $\gamma_{1j} = \mu_{1j} - 5 - \beta_j = (-1.5, 0, 1.5)$  and  $\gamma_{2j} = (1.5, 0, -1.5)$ . The tetrad difference is  $3 - 7 - 6 + 4 = -6$ , and  $\gamma_{11} - \gamma_{13} - \gamma_{21} + \gamma_{23} = -1.5 - 1.5 - 1.5 - 1.5 = -6$ .

#### Exercise A2.

From the table of cell means in Example (Flour and proofing temperature), compute the simple effects of wholemeal against the rye blend at the four temperatures. Is their interaction quantitative or qualitative? Which pair of temperatures, if any, reverses its order for some flour?

### 16.2.5.2 B. Practice

#### Exercise B1.

Show that the  $(a-1)(b-1)$  tetrad differences  $\mu_{ij} - \mu_{ib} - \mu_{aj} + \mu_{ab}$ ,  $i < a$ ,  $j < b$ , have linearly independent coefficient tables. Conclude that they form a basis of the interaction contrasts, so that additivity is equivalent to  $(a-1)(b-1)$  linear restrictions on the cell means.

*Solution.* The coefficient table of the  $(i, j)$  tetrad has entry 1 at  $(i, j)$  and its other nonzero entries in row  $a$  or column  $b$ . In a linear combination  $\sum_{i < a, j < b} c_{ij}(\text{table}_{ij})$ , the entry at position  $(i, j)$  with  $i < a$ ,  $j < b$  is  $c_{ij}$ , because no other table in the list has a nonzero entry there. So the combination vanishes only if all  $c_{ij} = 0$ . There are  $(a-1)(b-1)$  independent tables in a space of that dimension (Proposition 16.2.1), so

they form a basis. Additivity means all interaction contrasts vanish, which is equivalent to the basis contrasts vanishing.

### Exercise B2.

Let  $\mu_{ij} = u_i + v_j > 0$  be an additive table. Show that  $\log \mu_{ij}$  is additive only if all  $u_i$  are equal or all  $v_j$  are equal. So on a positive scale a transformation can create interaction as well as remove it.

*Solution.* By Proposition 16.2.1(i),  $\log \mu_{ij}$  is additive iff  $\mu_{ij}\mu_{kl} = \mu_{il}\mu_{kj}$  for all  $i, k, j, l$ . Expanding  $(u_i + v_j)(u_k + v_l) - (u_i + v_l)(u_k + v_j) = u_i v_l + u_k v_j - u_i v_j - u_k v_l = -(u_i - u_k)(v_j - v_l)$ . This vanishes for all index choices iff  $u_i = u_k$  for all  $i, k$  or  $v_j = v_l$  for all  $j, l$ , since if some  $u_i \neq u_k$  and some  $v_j \neq v_l$ , that choice of indices gives a nonzero product.

### Exercise B3.

Show that the converse of Proposition 16.2.2 holds for  $2 \times 2$  tables. If the rows are ordered alike in both columns and the columns alike in both rows, then some strictly increasing function  $g$  makes  $g(\mu_{ij})$  additive.

#### 16.2.5.3 C. Going deeper

### Exercise C1.

Consider the  $3 \times 3$  table

$$\mu = \begin{pmatrix} 1 & 3 & 6 \\ 4 & 7 & 9 \\ 5 & 10 & 11 \end{pmatrix}.$$

Check that every row and every column is increasing, so that the order condition of Proposition 16.2.2 holds. Show that nevertheless no strictly increasing  $g$  makes  $g(\mu_{ij})$  additive. (*Hint:* compare  $\mu_{21}$  with  $\mu_{12}$ ,  $\mu_{32}$  with  $\mu_{23}$ , and  $\mu_{31}$  with  $\mu_{13}$ .) The condition violated here is the *double cancellation* axiom of conjoint measurement.

*Solution.* The rows  $(1, 3, 6)$ ,  $(4, 7, 9)$ ,  $(5, 10, 11)$  and the columns  $(1, 4, 5)$ ,  $(3, 7, 10)$ ,  $(6, 9, 11)$  are increasing. Suppose  $g(\mu_{ij}) = u_i + v_j$  with  $g$  strictly increasing. From  $\mu_{21} = 4 > 3 = \mu_{12}$  we get  $u_2 + v_1 > u_1 + v_2$ . From  $\mu_{32} = 10 > 9 = \mu_{23}$  we get  $u_3 + v_2 > u_2 + v_3$ . Adding,  $u_3 + v_1 > u_1 + v_3$ , so  $\mu_{31} > \mu_{13}$ . But  $\mu_{31} = 5 < 6 = \mu_{13}$ , a contradiction.

### Exercise C2.

For a weight vector  $\mathbf{w}$ , let  $H_A^{\mathbf{w}}$  be the hypothesis that  $\sum_j w_j \mu_{ij}$  is the same for all  $i$ . Show that  $H_A^{\mathbf{w}}$  and  $H_A^{\mathbf{w}'}$  define the same set of cell-mean tables only if  $\mathbf{w} = \mathbf{w}'$ . Show also that on the set of additive tables all of them coincide with  $\alpha_1 = \cdots = \alpha_a$ .



## 16.3 Balanced two-way layouts via orthogonal projections

With  $m \geq 2$  observations in every cell, the replicates supply an estimate of  $\sigma^2$  that is valid whatever the cell means are, and interaction can be tested without any assumption about its form. This section gives the complete analysis of the balanced two-way layout: the orthogonal decomposition, exact tests, contrasts, and simultaneous inference. [Exercise \(C1\)](#) asked for the five projections, and [Theorem 16.1.2\(d\)](#) has already met four of them.

### 16.3.1 The decomposition

The model is the cell-means model

$$E(y_{ijk}) = \mu_{ij}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}, \quad i \leq a, j \leq b, k \leq m, \quad (16.3.1)$$

written in the effects form of [Definition 16.2.1](#) as  $\mu_{ij} = \mu + \alpha_i + \beta_j + \gamma_{ij}$ , with equal-weight effects. Its mean space  $\mathcal{C}(\mathbf{X})$  is the set of vectors that are constant within cells, of dimension  $ab$  ([Proposition 8.6.2](#)). With the ordering and notation of [Section 16.1](#), define

$$\begin{aligned} \mathbf{P}_0 &= \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_A &= \mathbf{C}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_B &= \bar{\mathbf{J}}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m, \\ \mathbf{P}_{AB} &= \mathbf{C}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_E &= \mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{C}_m. \end{aligned} \quad (16.3.2)$$

#### Theorem 16.3.1 — The balanced two-way layout

Assume [Eq. 16.3.1](#) with  $a, b \geq 2$  and  $m \geq 2$ , and let  $\alpha_i, \beta_j, \gamma_{ij}$  be the equal-weight effects of [Definition 16.2.1](#).

- (a) The five matrices [Eq. 16.3.2](#) are mutually orthogonal projections that sum to  $\mathbf{I}$ , with ranks 1,  $a - 1$ ,  $b - 1$ ,  $(a - 1)(b - 1)$  and  $ab(m - 1)$ . The projection onto  $\mathcal{C}(\mathbf{X})$  is  $\mathbf{M} = \mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B + \mathbf{P}_{AB}$ , and that onto the additive model space is  $\mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$ . So  $\mathcal{C}(\mathbf{P}_{AB})$  is the orthogonal complement of the additive model within the cell-means model.
- (b) The projections act by

$$\begin{aligned} (\mathbf{P}_A \mathbf{y})_{ijk} &= \bar{y}_{i..} - \bar{y}_{...}, \\ (\mathbf{P}_B \mathbf{y})_{ijk} &= \bar{y}_{.j.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...}, \\ (\mathbf{P}_E \mathbf{y})_{ijk} &= y_{ijk} - \bar{y}_{ij.}, \end{aligned}$$

and symmetrically for  $\mathbf{P}_B$ . The sums of squares are

$$\begin{aligned} \text{SS}_A &= bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2, \\ \text{SS}_B &= am \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2, \\ \text{SS}_{AB} &= m \sum_{i,j} (\bar{y}_{ij.} - \bar{y}_{i..} - \bar{y}_{.j.} + \bar{y}_{...})^2, \\ \text{SSE} &= \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij.})^2. \end{aligned}$$

(c) The mean squares have expectations

$$\begin{aligned} \text{E MS}_A &= \sigma^2 + \frac{bm}{a-1} \sum_i \alpha_i^2, \\ \text{E MS}_B &= \sigma^2 + \frac{am}{b-1} \sum_j \beta_j^2, \\ \text{E MS}_{AB} &= \sigma^2 + \frac{m}{(a-1)(b-1)} \sum_{i,j} \gamma_{ij}^2, \\ \text{E MSE} &= \sigma^2. \end{aligned}$$

(d) Under normal errors the four sums of squares are independent,  $\text{SSE}/\sigma^2 \sim \chi^2(ab(m-1))$ , and

$$\begin{aligned} F_A &= \frac{\text{MS}_A}{\text{MSE}} \sim F\left(a-1, ab(m-1), \frac{bm}{\sigma^2} \sum_i \alpha_i^2\right), \\ F_{AB} &= \frac{\text{MS}_{AB}}{\text{MSE}} \sim F\left((a-1)(b-1), ab(m-1), \frac{m}{\sigma^2} \sum_{i,j} \gamma_{ij}^2\right), \end{aligned}$$

and likewise for  $F_B$ . So  $F_{AB}$  is an exact test of additivity,  $F_A$  of  $\bar{\mu}_{1.} = \cdots = \bar{\mu}_{a.}$ , and  $F_B$  of  $\bar{\mu}_{.1} = \cdots = \bar{\mu}_{.b}$ .

(e)  $\text{SS}_A$  is the sum of squares of the hypothesis  $\alpha_1 = \cdots = \alpha_a = 0$  in the cell-means model. It is also the sequential sum of squares of  $A$  whether  $A$  is entered before or after  $B$ , and it equals the Type II and Type III sums of squares of  $A$ . The same holds for  $B$ , and  $\text{SS}_{AB}$  is the sequential, Type II and Type III sum of squares of the interaction.

*Proof.*

- (a) Orthogonality, ranks and the sum  $\mathbf{I}$  follow from [Lemma 16.1.1](#), as in the proof of [Theorem 16.1.2\(d\)](#). The sum of the first four is  $(\bar{\mathbf{J}}_a + \mathbf{C}_a) \otimes (\bar{\mathbf{J}}_b + \mathbf{C}_b) \otimes \bar{\mathbf{J}}_m = \mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m$ . This matrix replaces each observation by its cell mean, so it is the projection onto the vectors that are constant within cells, which is  $\mathcal{C}(\mathbf{X})$ . The additive projection is [Theorem 16.1.2\(a\)](#).
- (b) Expand each Kronecker product with  $\mathbf{C} = \mathbf{I} - \bar{\mathbf{J}}$  and read off the means from the table in [Section 16.1](#). For example,

$$\begin{aligned} \mathbf{P}_{AB} &= \mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m - \mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m \\ &\quad - \bar{\mathbf{J}}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m + \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m. \end{aligned}$$

Each sum of squares is a sum over the  $n$  positions of a quantity that is constant over  $m$ ,  $bm$  or  $am$  of them.

- (c) Apply (b) to the mean vector  $\boldsymbol{\mu}$ , whose entries are  $\mu_{ij}$ . Then  $\mathbf{P}_A \boldsymbol{\mu}$  has entries  $\bar{\mu}_{i.} - \bar{\mu}_{..} = \alpha_i$ , so  $\|\mathbf{P}_A \boldsymbol{\mu}\|^2 = bm \sum_i \alpha_i^2$ . Likewise  $\|\mathbf{P}_{AB} \boldsymbol{\mu}\|^2 = m \sum_{i,j} \gamma_{ij}^2$ , and  $\mathbf{P}_E \boldsymbol{\mu} = \mathbf{0}$ . The four subspaces  $\mathcal{C}(\mathbf{P}_0), \dots, \mathcal{C}(\mathbf{P}_{AB})$  form an analysis of variance decomposition of  $\mathcal{C}(\mathbf{X})$ , so [Theorem 9.6.1](#) gives the expectations.
- (d) This is [Corollary 9.6.2](#) with the noncentralities of (c). The noncentrality of  $F_A$  vanishes iff every  $\alpha_i = 0$ , that is, iff the row means  $\bar{\mu}_{i.}$  are equal. The noncentrality of  $F_{AB}$  vanishes iff every  $\gamma_{ij} = 0$ , which is additivity by [Proposition 16.2.1\(vi\)](#).
- (e) Each  $\alpha_i$  is a linear function of the cell means, so the hypothesis is testable. The mean vectors it allows are the  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$  with  $\mathbf{P}_A \boldsymbol{\mu} = \mathbf{0}$ . Since  $\boldsymbol{\mu} = \mathbf{P}_0 \boldsymbol{\mu} + \mathbf{P}_A \boldsymbol{\mu} + \mathbf{P}_B \boldsymbol{\mu} + \mathbf{P}_{AB} \boldsymbol{\mu}$ , these form  $\mathcal{S}_0 = \mathcal{C}(\mathbf{P}_0) \oplus \mathcal{C}(\mathbf{P}_B) \oplus \mathcal{C}(\mathbf{P}_{AB})$ . By [Theorem 11.3.2\(b\)](#) the sum of squares of the hypothesis is  $\|(\mathbf{M} - \mathbf{P}_{\mathcal{S}_0}) \mathbf{y}\|^2 = \|\mathbf{P}_A \mathbf{y}\|^2$ . The rest is Chapter 9: equal counts are proportional ([Theorem 9.4.3](#)), so [Theorem 9.4.1](#) applies, and [Exercise \(B1\)](#) shows that the sequential sums of squares of  $A$  in either order and its Type II and Type III sums of squares all equal  $\|\mathbf{P}_A \mathbf{y}\|^2$ . The interaction enters last in every order, so all its sums of squares are  $\|(\mathbf{M} - \mathbf{P}_0 - \mathbf{P}_A - \mathbf{P}_B) \mathbf{y}\|^2 = \|\mathbf{P}_{AB} \mathbf{y}\|^2$ .

■

The theorem is summarized in the analysis of variance table:

| Source          | df               | Sum of squares                                 | E(MS)                                                      |
|-----------------|------------------|------------------------------------------------|------------------------------------------------------------|
| $A$             | $a - 1$          | $\ \mathbf{P}_A \mathbf{y}\ ^2$                | $\sigma^2 + bm \sum_i \alpha_i^2 / (a - 1)$                |
| $B$             | $b - 1$          | $\ \mathbf{P}_B \mathbf{y}\ ^2$                | $\sigma^2 + am \sum_j \beta_j^2 / (b - 1)$                 |
| $AB$            | $(a - 1)(b - 1)$ | $\ \mathbf{P}_{AB} \mathbf{y}\ ^2$             | $\sigma^2 + m \sum_{i,j} \gamma_{ij}^2 / ((a - 1)(b - 1))$ |
| error           | $ab(m - 1)$      | $\ \mathbf{P}_E \mathbf{y}\ ^2$                | $\sigma^2$                                                 |
| corrected total | $abm - 1$        | $\ (\mathbf{I} - \mathbf{P}_0) \mathbf{y}\ ^2$ |                                                            |

The test for  $A$  is valid with or without interaction, because  $\mathcal{C}(\mathbf{P}_{AB}) \perp \mathcal{C}(\mathbf{P}_A)$ . What the interaction changes is the *meaning* of  $H_A$ : equality of equal-weight row means is one hypothesis among many when the simple effects differ ([Section 16.2](#)). Part (e) is special to balance. With unequal cell counts the sequential, Type II and Type III sums of squares of  $A$  test hypotheses with different weights ([Theorem 9.3.2](#)), the subject of [Chapter 17](#).

## 16.3.2 Contrasts and their sums of squares

The  $F$  tests say whether a term matters. Contrasts say how. A **main-effect contrast** in  $A$  is  $\psi = \sum_i c_i \bar{\mu}_{i.} = \sum_i c_i \alpha_i$  with  $\sum_i c_i = 0$ . An **interaction contrast** is  $\sum_{i,j} d_{ij} \mu_{ij} = \sum_{i,j} d_{ij} \gamma_{ij}$ , with  $(d_{ij})$  having zero row and column sums ([Proposition 16.2.1](#)). Their least squares estimators replace means by sample means. Since  $\bar{y}_{i.}$  averages  $bm$  independent observations and  $\bar{y}_{ij.}$  averages  $m$ ,

$$\text{Var}\left(\sum_i c_i \bar{y}_{i.}\right) = \frac{\sigma^2}{bm} \sum_i c_i^2, \quad \text{Var}\left(\sum_{i,j} d_{ij} \bar{y}_{ij.}\right) = \frac{\sigma^2}{m} \sum_{i,j} d_{ij}^2.$$

The sums of squares of the corresponding single-degree-of-freedom hypotheses are the squared estimates divided by these variances without the  $\sigma^2$  ([Theorem 9.4.4](#)), as for a one-way contrast

(Proposition 15.2.1):

$$SS(\mathbf{c}) = \frac{(\sum_i c_i \bar{y}_{i\cdot})^2}{\sum_i c_i^2 / (bm)}, \quad SS(\mathbf{d}) = \frac{(\sum_{i,j} d_{ij} \bar{y}_{ij\cdot})^2}{\sum_{i,j} d_{ij}^2 / m}. \quad (16.3.3)$$

For main effects, a complete set of  $a - 1$  mutually orthogonal contrasts ( $\sum_i c_i c'_i = 0$ ) splits  $SS_A$  into  $a - 1$  independent pieces. This is the one-way result [Theorem 15.4.1](#) applied to the row means, which are one-way means with  $bm$  observations each. For interaction, orthogonal contrasts are most easily built as products.

### Proposition 16.3.2 — Product contrasts

Let  $\mathbf{c}^{(1)}, \dots, \mathbf{c}^{(a-1)}$  be mutually orthogonal contrast vectors in  $\mathbb{R}^a$ , and  $\mathbf{e}^{(1)}, \dots, \mathbf{e}^{(b-1)}$  mutually orthogonal contrast vectors in  $\mathbb{R}^b$ . Then the  $(a - 1)(b - 1)$  tables  $\mathbf{d}^{(rs)} = \mathbf{c}^{(r)} \mathbf{e}^{(s)\top}$ , with entries  $c_i^{(r)} e_j^{(s)}$ , are coefficient tables of mutually orthogonal interaction contrasts, and

$$\sum_{r=1}^{a-1} \sum_{s=1}^{b-1} SS(\mathbf{d}^{(rs)}) = SS_{AB}.$$

Under normality the  $(a - 1)(b - 1)$  sums of squares are independent of each other and of SSE.

*Proof.* The row sums of  $\mathbf{d}^{(rs)}$  are  $c_i^{(r)} \sum_j e_j^{(s)} = 0$  and the column sums vanish likewise. Let  $\mathbf{v}_{rs} = \mathbf{c}^{(r)} \otimes \mathbf{e}^{(s)} \otimes \mathbf{1}_m \in \mathbb{R}^n$ . Since  $\mathbf{C}_a \mathbf{c}^{(r)} = \mathbf{c}^{(r)}$  and  $\mathbf{C}_b \mathbf{e}^{(s)} = \mathbf{e}^{(s)}$ , the mixed-product rule gives  $\mathbf{P}_{AB} \mathbf{v}_{rs} = \mathbf{v}_{rs}$ , so  $\mathbf{v}_{rs} \in \mathcal{C}(\mathbf{P}_{AB})$ . Also  $\mathbf{v}_{rs}^\top \mathbf{v}_{r's'} = m (\mathbf{c}^{(r)\top} \mathbf{c}^{(r')})(\mathbf{e}^{(s)\top} \mathbf{e}^{(s')})$ , which is zero unless  $r = r'$  and  $s = s'$ . So the  $\mathbf{v}_{rs}$  are  $(a - 1)(b - 1)$  nonzero orthogonal vectors in a space of that dimension, an orthogonal basis of  $\mathcal{C}(\mathbf{P}_{AB})$ . Hence  $\|\mathbf{P}_{AB} \mathbf{y}\|^2 = \sum_{r,s} (\mathbf{v}_{rs}^\top \mathbf{y})^2 / \|\mathbf{v}_{rs}\|^2$ . Finally  $\mathbf{v}_{rs}^\top \mathbf{y} = m \sum_{i,j} d_{ij}^{(rs)} \bar{y}_{ij\cdot}$  and  $\|\mathbf{v}_{rs}\|^2 = m \sum_{i,j} (d_{ij}^{(rs)})^2$ , so each term is  $SS(\mathbf{d}^{(rs)})$  of [Eq. 16.3.3](#). Independence follows from [Theorem 4.4.4](#) applied to the projections onto the lines  $\text{span}(\mathbf{v}_{rs})$  and  $\mathbf{P}_E$ . ■

Product contrasts suit factors with structure. If  $\mathbf{c}$  compares a control with the treatments and  $\mathbf{e}$  is a linear trend in a quantitative factor, then  $\mathbf{c}\mathbf{e}^\top$  asks whether the trend differs between them. The cell-means model of a balanced layout with quantitative levels  $w_i$  and  $z_j$  is the same as the polynomial model with all terms  $w^r z^s$ ,  $r < a$ ,  $s < b$  ([Exercise \(B3\)](#)), and the product contrasts of orthogonal polynomials give its sequential sums of squares.

## 16.3.3 Simultaneous inference

The tests and contrasts above come in families, and the methods of [Chapter 13](#) apply directly.

- **Tukey for the levels of one factor.** The row means  $\bar{y}_{i\cdot}$  are independent, each with variance  $\sigma^2 / (bm)$ . They are functions of  $(\mathbf{I} - \mathbf{P}_E) \mathbf{Y}$ , hence independent of MSE, with or without interaction ([Exercise \(B2\)](#)). So [Theorem 13.3.2](#) applies with  $g = a$ , multiplier  $q_\alpha(a, ab(m - 1))$  and standard error  $s / \sqrt{bm}$ .

- **Scheffé for interaction contrasts.** Taking  $\mathcal{V} = \mathcal{C}(\mathbf{P}_{AB})$  in [Theorem 13.2.2](#) gives intervals for all interaction contrasts at once, with multiplier  $\sqrt{(a-1)(b-1)F_\alpha((a-1)(b-1), ab(m-1))}$ . By [Corollary 13.2.3](#), some interaction contrast is Scheffé-significant iff the  $F_{AB}$  test rejects. This is the honest way to report a contrast suggested by the interaction plot. The most extreme contrast always accounts for the whole interaction sum of squares ([Exercise \(C1\)](#)); the question is whether it is interpretable.
- **Bonferroni or Holm for planned families** ([Theorem 13.4.1](#), [Theorem 13.4.5](#)), such as the comparison of two levels of  $A$  separately at each level of  $B$ .

#### Example. (Analysis of the bread trial)

For the data of [Example \(Flour and proofing temperature\)](#), the decomposition of [Theorem 16.3.1](#) gives

| Source      | df | Sum of squares | Mean square | $F$    | $p$                   |
|-------------|----|----------------|-------------|--------|-----------------------|
| flour       | 2  | 1627629        | 813814      | 255.74 | $6.6 \times 10^{-17}$ |
| temperature | 3  | 140499         | 46833       | 14.72  | $1.2 \times 10^{-5}$  |
| interaction | 6  | 98947          | 16491       | 5.18   | $1.5 \times 10^{-3}$  |
| error       | 24 | 76373          | 3182        |        |                       |

so  $s = 56.4$  ml, and the interaction is clearly present. The script confirms that statsmodels reports the same sums of squares for Types I, II and III, as part (e) says.

*Where is the interaction?* Take the flour contrasts  $(2, -1, -1)$  (white against the other two) and  $(0, 1, -1)$  (wholemeal against rye), and the orthogonal polynomial contrasts  $(-3, -1, 1, 3)$ ,  $(1, -1, -1, 1)$ ,  $(-1, 3, -3, 1)$  for the equally spaced temperatures. The six products split  $SS_{AB}$  as in [Proposition 16.3.2](#):

|                  | linear                | quadratic            | cubic                |
|------------------|-----------------------|----------------------|----------------------|
| white vs others  | 52152 ( $F = 16.39$ ) | 3134 ( $F = 0.98$ )  | 6011 ( $F = 1.89$ )  |
| wholemeal vs rye | 8118 ( $F = 2.55$ )   | 18872 ( $F = 5.93$ ) | 10660 ( $F = 3.35$ ) |

The six add up to 98947. The largest piece, a fraction 0.527 of the interaction, says that white flour responds to temperature more steeply than the others. The second is the wholemeal peak at 32 °C. Does the white-by-linear component survive the most generous correction? Its estimate is 1444.3. The temperature contrast has  $\sum_j e_j t_j = -72 - 28 + 32 + 108 = 40$ , and the flour contrast is twice “white minus the average of the others”, so the estimate divided by 80 is the difference between white flour’s slope and the average slope of the other two: 18.05 ml per °C. The Scheffé multiplier for interaction contrasts is 3.879, and the simultaneous interval for this slope difference is  $[0.75, 35.35]$  ml per °C. It excludes zero even though Scheffé’s correction allows for every contrast the plot might have suggested. The temperature main effect is almost entirely linear, with sequential sums of squares 134098, 6373 and 28 for the linear, quadratic and cubic components.

*Comparing the flours.* The equal-weight flour means are 2074.8, 1727.1 and 1565.2 ml. Tukey’s multiplier  $q_{0.05}(3, 24) = 3.532$  with standard error  $s/\sqrt{12}$  gives half-width 57.5 ml. White minus wholemeal is 347.7, with interval  $[290.2, 405.3]$ . White minus rye blend is 509.7  $[452.2, 567.2]$ , and wholemeal minus rye blend is 161.9  $[104.4, 219.4]$ . Because of the interaction, these average over the four temperatures the trial happened to use. The simple effects of white against wholemeal, with Bonferroni intervals for the four temperatures (multiplier  $t_{24, 0.05/8} = 2.700$ , half-width 124.4), are 250.7  $[126.3, 375.0]$ , 376.7  $[252.3, 501.0]$ , 302.3  $[178.0, 426.7]$  and 461.3  $[337.0, 585.7]$ . All four lie above zero, so with simultaneous confidence 95% white flour is better at every temperature, and hence ([Proposition 16.2.3\(c\)](#)) under any weighting of the temperatures.

```
import itertools

import numpy as np
from scipy import stats

flours = ["white", "wholemeal", "rye blend"]
temps = [24, 28, 32, 36]
true_means = np.array([[1880, 2010, 2130, 2190],
                        [1660, 1760, 1810, 1750],
                        [1500, 1540, 1580, 1590]], dtype=float)

a, b, m = 3, 4, 3
data_rng = np.random.default_rng(16_03)
volume = np.round(true_means[:, :, None] + data_rng.normal(scale=55, size=(a, b, m))
                  )

def Jbar(k):
    return np.full((k, k), 1.0 / k)
```

```

def Cen(k):
    return np.eye(k) - Jbar(k)

y = volume.ravel()                                # replicate fastest, then
                                                temperature, then flour
P = {"flour": np.kron(np.kron(Cen(a), Jbar(b)), Jbar(m)),
     "temperature": np.kron(np.kron(Jbar(a), Cen(b)), Jbar(m)),
     "interaction": np.kron(np.kron(Cen(a), Cen(b)), Jbar(m)),
     "error": np.kron(np.kron(np.eye(a), np.eye(b)), Cen(m))}
ss = {k: y @ Pk @ y for k, Pk in P.items()}
df = {k: int(round(np.trace(Pk))) for k, Pk in P.items()}
mse = ss["error"] / df["error"]
for k in ["flour", "temperature", "interaction", "error"]:
    F = ss[k] / df[k] / mse
    tail = f"F = {F:7.2f}  p = {stats.f.sf(F, df[k], df['error']):.2g}" if k != "
    error" else ""
    print(f"{k:12s} df {df[k]:2d}  SS {ss[k]:10.0f}  MS {ss[k] / df[k]:9.0f}  {tail}"
          ")

```

```

cell = volume.mean(axis=2)
flour_c = {"white vs others": np.array([2.0, -1, -1]), "wholemeal vs rye": np.array
          ([0.0, 1, -1])}
temp_c = {"linear": np.array([-3.0, -1, 1, 3]), "quadratic": np.array([1.0, -1, -1,
1]),
          "cubic": np.array([-1.0, 3, -3, 1])}
rows = []
for (fn, c), (tn, d) in itertools.product(flour_c.items(), temp_c.items()):
    D = np.outer(c, d)                                # an interaction contrast in
    the cell means
    est = (D * cell).sum()
    ss_c = est ** 2 / ((D ** 2).sum() / m)                # its one-degree-of-freedom
    sum of squares
    rows.append((fn, tn, est, ss_c, ss_c / mse))
    print(f"{fn:17s} x {tn:9s}  estimate {est:8.1f}  SS {ss_c:8.0f}  F {ss_c / mse
:6.2f}")
print(f"sum of the six: {sum(r[3] for r in rows):.0f}  interaction SS: {ss['
interaction']:.0f}")

```

```

fmean, tmean, grand = cell.mean(1), cell.mean(0), cell.mean()
s = np.sqrt(mse)
nu = df["error"]
q = stats.studentized_range.ppf(0.95, a, nu)
half_tukey = q * s / np.sqrt(b * m)                    # flour means average b*m
loaves
for i, k in itertools.combinations(range(a), 2):
    d = fmean[i] - fmean[k]
    print(f"{flours[i]:9s} - {flours[k]:9s}: {d:7.1f}  Tukey [{d - half_tukey:7.1f},
{d + half_tukey:7.1f}]")
t_bonf = stats.t.ppf(1 - 0.05 / (2 * b), nu)            # white - wholemeal at each
temperature
half_simple = t_bonf * s * np.sqrt(2 / m)
for j in range(b):
    d = cell[0, j] - cell[1, j]
    print(f"white - wholemeal at {temps[j]} C: {d:6.1f}  Bonferroni [{d -

```

```
half_simple:6.1f}, {d + half_simple:6.1f}])")
c_scheffe = np.sqrt(df["interaction"] * stats.f.ppf(0.95, df["interaction"], nu))
```

*Warning.* It is tempting to drop a nonsignificant interaction and “pool” its sum of squares into the error. With  $ab(m-1)$  error degrees of freedom already available this gains little. If the interaction is real, the pooled mean square overestimates  $\sigma^2$ , and if the decision to pool depends on the data, the nominal levels no longer hold (Section 16.4).

## 16.3.4 Exercises

### 16.3.4.1 A. Check your understanding

#### Exercise A1.

Let  $SS_{\text{cells}} = m \sum_{i,j} (\bar{y}_{ij} - \bar{y}_{...})^2$ , the between-cells sum of squares of the one-way layout with  $ab$  groups. Show that  $SS_{\text{cells}} = SS_A + SS_B + SS_{AB}$ , and identify each term with a projection.

### 16.3.4.2 B. Practice

#### Exercise B1.

Suppose all cell means are equal except  $\mu_{11}$ , which is larger by  $\delta$ . Find  $\alpha_i$ ,  $\beta_j$  and  $\gamma_{ij}$ , and show that the noncentralities of  $F_A$ ,  $F_B$  and  $F_{AB}$  are

$$\frac{m\delta^2}{\sigma^2} \cdot \frac{a-1}{ab}, \quad \frac{m\delta^2}{\sigma^2} \cdot \frac{b-1}{ab}, \quad \frac{m\delta^2}{\sigma^2} \cdot \frac{(a-1)(b-1)}{ab}.$$

Which test is most likely to detect a single deviant cell when  $a$  and  $b$  are large?

*Solution.* The row means are  $\bar{\mu}_{1.} = \mu + \delta/b$  and  $\bar{\mu}_{i.} = \mu$  otherwise, and the grand mean is  $\mu + \delta/(ab)$ . So  $\alpha_1 = \delta(a-1)/(ab)$  and  $\alpha_i = -\delta/(ab)$  for  $i \geq 2$ , and

$$\sum_i \alpha_i^2 = \frac{\delta^2}{a^2 b^2} ((a-1)^2 + (a-1)) = \frac{\delta^2(a-1)}{ab^2},$$

so  $bm \sum_i \alpha_i^2 / \sigma^2 = m\delta^2(a-1)/(ab\sigma^2)$ . By symmetry the noncentrality for  $B$  is  $m\delta^2(b-1)/(ab\sigma^2)$ . For the interaction,  $\gamma_{ij} = \delta(g_i - 1/a)(h_j - 1/b)$ , where  $g_1 = h_1 = 1$  and  $g_i = h_j = 0$  for  $i, j \geq 2$ , so  $\sum \gamma_{ij}^2 = \delta^2 \cdot \frac{a-1}{a} \cdot \frac{b-1}{b}$ , and the noncentrality is  $m\delta^2(a-1)(b-1)/(ab\sigma^2)$ . For large  $a, b$  the interaction noncentrality approaches  $m\delta^2/\sigma^2$ , while the main-effect noncentralities approach  $m\delta^2/(b\sigma^2)$  and  $m\delta^2/(a\sigma^2)$ . Almost all of a single deviant cell shows up as interaction. It is, however, spread over  $(a-1)(b-1)$  degrees of freedom, and the Scheffé-type contrast aimed at that cell is the one to look at.

#### Exercise B2.

In the cell-means model Eq. 16.3.1 with normal errors, show that  $\bar{y}_{1..}, \dots, \bar{y}_{a..}$  are in-



dependent, each with variance  $\sigma^2/(bm)$ , and jointly independent of SSE, whatever the  $\mu_{ij}$ . Deduce Tukey intervals for all  $\bar{\mu}_{i\cdot} - \bar{\mu}_{k\cdot}$  with exact simultaneous coverage.

*Solution.* Different rows use disjoint sets of the independent observations, so the row means are independent, and each averages  $bm$  observations. The vector of row means is a linear function of  $(\mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m)\mathbf{Y}$ , whose range lies in  $\mathcal{C}(\mathbf{P}_0 + \mathbf{P}_A) \perp \mathcal{C}(\mathbf{P}_E)$ . By Theorem 4.4.4,  $(\mathbf{P}_0 + \mathbf{P}_A)\mathbf{Y}$  and  $\mathbf{P}_E\mathbf{Y}$  are independent, and  $\text{SSE} = \|\mathbf{P}_E\mathbf{Y}\|^2$ . The row means have expectations  $\bar{\mu}_{i\cdot}$ . These are exactly the conditions used in the proof of Theorem 13.3.2, with  $g = a$ , common variance  $\sigma^2/(bm)$  and  $\nu = ab(m-1)$ . So the intervals  $\bar{y}_{i\cdot} - \bar{y}_{k\cdot} \pm q_\alpha(a, ab(m-1)) s/\sqrt{bm}$  have simultaneous coverage exactly  $1 - \alpha$ .

### Exercise B3.

Let the levels of  $A$  and  $B$  be distinct numbers  $w_1, \dots, w_a$  and  $z_1, \dots, z_b$ . Show that the vectors with entries  $w_i^r z_j^s$  (repeated over replicates),  $0 \leq r < a$ ,  $0 \leq s < b$ , span the cell-means space  $\mathcal{C}(\mathbf{X})$ . Show that the additive model corresponds to the terms with  $r = 0$  or  $s = 0$ . What does the coefficient of  $wz$  measure?

#### 16.3.4.3 C. Going deeper

### Exercise C1.

Let  $\hat{\gamma}_{ij} = \bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot} - \bar{y}_{\cdot j\cdot} + \bar{y}_{\cdot\cdot\cdot}$ . Show that among all interaction contrasts,  $\text{SS}(\mathbf{d})$  of Eq. 16.3.3 is largest for  $d_{ij} \propto \hat{\gamma}_{ij}$ , and that the maximum is  $\text{SS}_{AB}$ . Relate this to Corollary 13.2.3(a). Why does it mean that the  $F$  statistic of a contrast chosen after inspecting the interaction plot cannot be compared with  $F(1, ab(m-1))$ ?

*Solution.* Since  $(d_{ij})$  has zero margins,  $\sum d_{ij}\bar{y}_{ij\cdot} = \sum d_{ij}\hat{\gamma}_{ij}$  (Proposition 16.2.1 applied to the table of cell means). By the Cauchy–Schwarz inequality,  $(\sum d_{ij}\hat{\gamma}_{ij})^2 \leq \sum d_{ij}^2 \sum \hat{\gamma}_{ij}^2$ , with equality iff  $\mathbf{d} \propto \hat{\boldsymbol{\gamma}}$ . Hence  $\text{SS}(\mathbf{d}) \leq m \sum \hat{\gamma}_{ij}^2 = \text{SS}_{AB}$ , with equality at  $\mathbf{d} = \hat{\boldsymbol{\gamma}}$ , which is itself a zero-margin table. In the notation of Corollary 13.2.3(a) this says that the largest single-contrast  $F$  ratio is  $(a-1)(b-1)F_{AB}$ . A contrast chosen for its large value is, in the extreme, this maximizing contrast. Its null distribution is that of  $(a-1)(b-1)$  times an  $F((a-1)(b-1), ab(m-1))$  variable, not  $F(1, ab(m-1))$ .

### Exercise C2.

For each column  $j$ , let  $\text{SS}_{A|j} = m \sum_i (\bar{y}_{ij\cdot} - \bar{y}_{\cdot j\cdot})^2$  be the sum of squares for  $A$  computed within column  $j$  alone. Show that  $\sum_j \text{SS}_{A|j} = \text{SS}_A + \text{SS}_{AB}$ , and find the projection whose squared length is the left side. What hypothesis does the corresponding  $F$  test with  $b(a-1)$  numerator degrees of freedom address?

## 16.4 Higher-way layouts

Nothing in the two-way analysis depended on there being two factors. With  $t$  crossed factors, all combinations observed equally often, the observation space splits into  $2^t$  mutually orthogonal

pieces plus error, one for each set of factors. This section proves that decomposition, interprets the interaction of three or more factors, specializes to two-level factorial designs, and ends with a warning about pooling “negligible” terms into error.

### 16.4.1 The general balanced factorial

Let factors  $F_1, \dots, F_t$  have  $l_1, \dots, l_t \geq 2$  levels. Suppose all  $N = l_1 \cdots l_t$  combinations are observed  $m \geq 1$  times each, so  $n = Nm$ . Stack the observations with the replicate index fastest, then the level of  $F_t$ , and so on, with  $F_1$  slowest. For a subset  $S \subseteq \{1, \dots, t\}$ , a *term*, define

$$\begin{aligned} \mathbf{P}_S &= \mathbf{E}_1 \otimes \cdots \otimes \mathbf{E}_t \otimes \bar{\mathbf{J}}_m, & \mathbf{E}_s &= \begin{cases} \mathbf{C}_{l_s}, & s \in S, \\ \bar{\mathbf{J}}_{l_s}, & s \notin S, \end{cases} \\ \mathbf{P}_E &= \mathbf{I}_{l_1} \otimes \cdots \otimes \mathbf{I}_{l_t} \otimes \mathbf{C}_m, \end{aligned} \quad (16.4.1)$$

and the **marginal averaging operator**

$$\mathbf{A}_S = \mathbf{G}_1 \otimes \cdots \otimes \mathbf{G}_t \otimes \bar{\mathbf{J}}_m, \quad \mathbf{G}_s = \begin{cases} \mathbf{I}_{l_s}, & s \in S, \\ \bar{\mathbf{J}}_{l_s}, & s \notin S. \end{cases}$$

The operator  $\mathbf{A}_S$  replaces each observation by the mean of all observations that share its levels of the factors in  $S$ . For example, with three factors  $A, B, C$ ,  $\mathbf{A}_{\{A,C\}}\mathbf{y}$  has entries  $\bar{y}_{i.k.}$ . Let  $\mathcal{V}_S$  be the set of vectors whose entries depend only on the levels of the factors in  $S$ . It is the column space of the indicator matrix of the combinations of those factors. So  $\mathcal{V}_\emptyset = \text{span}(\mathbf{1})$ , and  $\mathcal{V}_{\{1, \dots, t\}}$  is the space of the full cell-means model.

#### Theorem 16.4.1 — The balanced factorial decomposition

In the setting above:

- (a) The  $2^t$  matrices  $\mathbf{P}_S$  and  $\mathbf{P}_E$  are mutually orthogonal projections that sum to  $\mathbf{I}$ . Their ranks are  $r_S = \prod_{s \in S} (l_s - 1)$  (with  $r_\emptyset = 1$ ) and  $N(m - 1)$ .
- (b)  $\mathbf{A}_S$  is the orthogonal projection onto  $\mathcal{V}_S$ , and  $\mathbf{A}_S = \sum_{T \subseteq S} \mathbf{P}_T$ . So  $\mathcal{V}_S = \bigoplus_{T \subseteq S} \mathcal{C}(\mathbf{P}_T)$ .
- (c)  $\mathbf{P}_S = \sum_{T \subseteq S} (-1)^{|S|-|T|} \mathbf{A}_T$ . Thus  $\mathbf{P}_S \mathbf{y}$  is the alternating sum of the marginal means over the subsets of  $S$ .
- (d) Let  $\mathcal{H}$  be a *hierarchical* family of terms:  $\emptyset \in \mathcal{H}$ , and  $T \subseteq S \in \mathcal{H}$  implies  $T \in \mathcal{H}$ . The model  $\mathbf{E}(\mathbf{Y}) \in \sum_{S \in \mathcal{H}} \mathcal{V}_S$ , which contains the terms in  $\mathcal{H}$  and nothing else, has projection  $\sum_{S \in \mathcal{H}} \mathbf{P}_S$ . If terms are entered one at a time so that each term follows all of its subsets, the sequential sum of squares of term  $S$  is  $\|\mathbf{P}_S \mathbf{y}\|^2$ , whatever the order.
- (e) If  $\mathbf{E}(\mathbf{Y}) = \boldsymbol{\mu} \in \mathcal{V}_{\{1, \dots, t\}}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , then  $\mathbf{E}\|\mathbf{P}_S \mathbf{Y}\|^2 = \sigma^2 r_S + \|\mathbf{P}_S \boldsymbol{\mu}\|^2$ . Under normality the  $\|\mathbf{P}_S \mathbf{Y}\|^2$  and  $\text{SSE} = \|\mathbf{P}_E \mathbf{Y}\|^2$  are independent. When  $m \geq 2$ ,

$$F_S = \frac{\|\mathbf{P}_S \mathbf{Y}\|^2 / r_S}{\text{SSE} / (N(m - 1))} \sim F(r_S, N(m - 1), \|\mathbf{P}_S \boldsymbol{\mu}\|^2 / \sigma^2).$$

*Proof.*

- (a) By Lemma 16.1.1(d) with  $t + 1$  factors,  $\mathbf{I}$  is the sum of the  $2^{t+1}$  products with  $\bar{\mathbf{J}}$  or

- $\mathbf{C}$  in each position. Those with  $\bar{\mathbf{J}}_m$  last are the  $\mathbf{P}_S$ . Those with  $\mathbf{C}_m$  last add up to  $\mathbf{I} \otimes \cdots \otimes \mathbf{I} \otimes \mathbf{C}_m = \mathbf{P}_E$ . Orthogonality and ranks come from [Lemma 16.1.1](#)(b) and (c).
- (b) In each position  $s \in S$  write  $\mathbf{I}_{l_s} = \bar{\mathbf{J}}_{l_s} + \mathbf{C}_{l_s}$  and expand. This gives  $\mathbf{A}_S = \sum_{T \subseteq S} \mathbf{P}_T$ , a sum of orthogonal projections, hence a projection ([Theorem 6.3.8](#)) onto  $\bigoplus_{T \subseteq S} \mathcal{C}(\mathbf{P}_T)$ . The entries of  $\mathbf{A}_S \mathbf{y}$  are means over sets of observations with the same levels of the factors in  $S$ , so  $\mathcal{C}(\mathbf{A}_S) \subseteq \mathcal{V}_S$ . Conversely, averaging a vector in  $\mathcal{V}_S$  over such a set leaves it unchanged, so  $\mathcal{V}_S \subseteq \mathcal{C}(\mathbf{A}_S)$ .
- (c) In each position  $s \in S$  write  $\mathbf{C}_{l_s} = \mathbf{I}_{l_s} - \bar{\mathbf{J}}_{l_s}$  and expand. Choosing  $\mathbf{I}$  in the positions of  $T \subseteq S$  and  $-\bar{\mathbf{J}}$  in the rest of  $S$  gives  $(-1)^{|S|-|T|} \mathbf{A}_T$ .
- (d) By (b),  $\sum_{S \in \mathcal{H}} \mathcal{V}_S = \sum_{S \in \mathcal{H}} \bigoplus_{T \subseteq S} \mathcal{C}(\mathbf{P}_T)$ . Every  $T$  appearing belongs to  $\mathcal{H}$ , and every  $T \in \mathcal{H}$  appears (take  $S = T$ ). So the model space is  $\bigoplus_{T \in \mathcal{H}} \mathcal{C}(\mathbf{P}_T)$ , with projection  $\sum_{T \in \mathcal{H}} \mathbf{P}_T$  ([Theorem 6.3.8](#)). If the terms enter in an order in which each follows its subsets, every intermediate model is hierarchical. Two consecutive models differ by one term  $S$ , so their projections differ by  $\mathbf{P}_S$  ([Definition 9.3.1](#)).
- (e) The subspaces  $\mathcal{C}(\mathbf{P}_S)$  form an analysis of variance decomposition of  $\mathcal{V}_{\{1, \dots, t\}}$  by (a) and (b). Apply [Theorem 9.6.1](#) and [Corollary 9.6.2](#).

■

With  $t = 2$  the theorem is [Theorem 16.3.1](#), since  $\mathbf{P}_{\{1\}} = \mathbf{P}_A$ ,  $\mathbf{P}_{\{2\}} = \mathbf{P}_B$  and  $\mathbf{P}_{\{1,2\}} = \mathbf{P}_{AB}$ . Part (c) gives the familiar formulas. For three factors  $A, B, C$  with levels indexed by  $i, j, k$  and replicates by  $r$ , the three-factor term has entries

$$(\mathbf{P}_{ABC} \mathbf{y})_{ijk r} = \bar{y}_{ijk \cdot} - \bar{y}_{ij \cdot \cdot} - \bar{y}_{i \cdot k \cdot} - \bar{y}_{\cdot j k \cdot} + \bar{y}_{i \cdot \cdot} + \bar{y}_{\cdot j \cdot \cdot} + \bar{y}_{\cdot \cdot k \cdot} - \bar{y}_{\cdot \cdot \cdot \cdot}.$$

Part (d) explains why balanced factorials have one analysis of variance table and not several. The qualification “each term follows its subsets” matters: entering  $AB$  before  $A$  gives  $A$  nothing, because  $\mathcal{V}_A \subseteq \mathcal{V}_{AB}$  ([Exercise \(B1\)](#)). The listing builds the decomposition for a  $2 \times 3 \times 4$  layout with  $m = 2$  and checks the ranks of part (a). The script also checks parts (b), (c) and (d).

```
import itertools

import numpy as np

def Jbar(k):
    return np.full((k, k), 1.0 / k)          # averaging: replaces a k-vector by its
    mean

def Cen(k):
    return np.eye(k) - Jbar(k)              # centring: subtracts the mean

def kron_all(mats):
    out = np.ones((1, 1))
    for A in mats:
        out = np.kron(out, A)
    return out

def factorial_projections(levels, m):
    """P_S for every subset S of the factors (a 0/1 tuple), and the error projection
    ."""
```

```

P = {}
for S in itertools.product([0, 1], repeat=len(levels)):
    P[S] = kron_all([Cen(1) if s else Jbar(1) for l, s in zip(levels, S)] + [
Jbar(m)])
P["error"] = kron_all([np.eye(1) for l in levels] + [Cen(m)])
return P

def rank(A):
    return int(round(np.trace(A)))          # rank = trace for a projection

levels, m3 = [2, 3, 4], 2
P3 = factorial_projections(levels, m3)
for S in P3:
    if S == "error":
        continue
    label = "".join(f for f, s in zip("ABC", S) if s) or "mean"
    print(f"{label:5s} rank {rank(P3[S]):2d}    predicted {int(np.prod([1 - 1 for l,
s in zip(levels, S) if s]))}")
print("error rank", rank(P3["error"]), "    n =", int(np.prod(levels)) * m3)

```

## 16.4.2 Three-factor interaction

In a three-factor layout with cell means  $\mu_{ijk}$ , the two-factor interaction of  $A$  and  $B$  can be computed separately at each level  $k$  of  $C$ :

$$\gamma_{ij}^{(k)} = \mu_{ijk} - \bar{\mu}_{i..k} - \bar{\mu}_{.jk} + \bar{\mu}_{...k}.$$

The  $AB$  interaction of the analysis of variance, the entries of  $\mathbf{P}_{AB}\boldsymbol{\mu}$ , is the average of these over  $k$ . A three-factor interaction is present when they are not all the same.

### Proposition 16.4.2 — Three-factor interaction

For the cell means  $\mu_{ijk}$  of a three-factor layout, the following are equivalent:

- (i)  $\mathbf{P}_{ABC}\boldsymbol{\mu} = \mathbf{0}$ ;
- (ii) the  $AB$  interaction effects  $\gamma_{ij}^{(k)}$  at level  $k$  of  $C$  do not depend on  $k$ ;
- (iii) every  $AB$  interaction contrast  $\sum_{i,j} d_{ij}\mu_{ijk}$  has the same value at every level  $k$  of  $C$ ;
- (iv) statements (ii) and (iii) with the roles of the three factors permuted in any way;
- (v)  $\mu_{ijk} = f_{ij} + g_{ik} + h_{jk}$  for some numbers  $f_{ij}, g_{ik}, h_{jk}$ .

*Proof.* It suffices to consider  $m = 1$ , since the replicate factor  $\bar{\mathbf{J}}_m$  only repeats entries. Then  $\mathbf{P}_{ABC} = \mathbf{C}_a \otimes \mathbf{C}_b \otimes \mathbf{C}_c = \mathbf{C}_a \otimes \mathbf{C}_b \otimes \mathbf{I}_c - \mathbf{C}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_c$ . Applied to  $\boldsymbol{\mu}$ , the first term produces  $\gamma_{ij}^{(k)}$  in position  $(i, j, k)$ , because it centres over  $i$  and over  $j$  within each  $k$ . The second produces the average of  $\gamma_{ij}^{(k)}$  over  $k$ . So (i) holds iff  $\gamma_{ij}^{(k)}$  equals its average over  $k$  for all  $i, j, k$ , which is (ii). By Proposition 16.2.1,  $\sum_{i,j} d_{ij}\mu_{ijk} = \sum_{i,j} d_{ij}\gamma_{ij}^{(k)}$  for every zero-margin table  $(d_{ij})$ . Since the tables  $\gamma^{(k)}$  themselves have zero margins,  $\gamma^{(k)} = \gamma^{(k')}$  iff their inner products with every zero-margin  $\mathbf{d}$  agree (take  $\mathbf{d} = \gamma^{(k)} - \gamma^{(k')}$ ). This proves (ii)  $\Leftrightarrow$  (iii). Statement (i) is symmetric in the three factors, which gives (iv).

Finally, the vectors of the form in (v) make up  $\mathcal{V}_{AB} + \mathcal{V}_{AC} + \mathcal{V}_{BC}$ , the model space of the hierarchical family of all terms except  $ABC$ . By [Theorem 16.4.1\(d\)](#) this is the orthogonal complement of  $\mathcal{C}(\mathbf{P}_{ABC})$  in the cell-means space, so  $\boldsymbol{\mu}$  lies in it iff  $\mathbf{P}_{ABC}\boldsymbol{\mu} = \mathbf{0}$ .

■

So the three-factor interaction is an “interaction of interactions”. It measures how much the pattern of  $AB$  interaction changes from one level of  $C$  to another, and by (iv) it measures the same thing whichever factor plays the role of  $C$ . When it is present, the  $AB$  interaction in the table is an equal-weight average over the levels of  $C$ , with all the caveats of [Section 16.2](#). Interpretation proceeds from the top down: examine the highest-order interactions clearly present, then read lower-order terms as averages or look at simple effects. Models are usually required to be hierarchical, because a model with  $AB$  but without  $A$  constrains the equal-weight average  $\bar{\mu}_{i..}$ , and so depends on an arbitrary choice of weights.

### 16.4.3 Two-level factorial designs

When every factor has two levels, each  $\mathbf{C}_2$  has rank one, and every term of [Theorem 16.4.1](#) has a single degree of freedom. Code the levels of each factor as  $-1$  (“low”) and  $+1$  (“high”). For a nonempty term  $S$ , let  $\mathbf{x}_S$  be the column whose entry for an observation is the product of the codes of the factors in  $S$ . For example,  $\mathbf{x}_{AC}$  is  $+1$  when  $A$  and  $C$  are at the same level and  $-1$  otherwise. The **effect** of  $S$  is

$$\hat{E}_S = \bar{y}(\mathbf{x}_S = +1) - \bar{y}(\mathbf{x}_S = -1),$$

the mean response where  $\mathbf{x}_S$  is  $+1$  minus the mean where it is  $-1$ . For a main effect this is the average change in response when the factor moves from low to high. For  $AC$  it is half the difference between the effect of  $C$  at high  $A$  and at low  $A$  ([Exercise \(B2\)](#)).

#### Proposition 16.4.3 — Effects in a $2^t$ factorial

In a  $2^t$  factorial with  $m$  replicates,  $n = 2^t m$ , and a nonempty term  $S$ :

- (a)  $\mathbf{P}_S = \mathbf{x}_S \mathbf{x}_S^\top / n$ . The columns  $\mathbf{x}_S$  are mutually orthogonal and orthogonal to  $\mathbf{1}$ .
- (b)  $\hat{E}_S = 2\mathbf{x}_S^\top \mathbf{y} / n$ , and  $\text{Var}(\hat{E}_S) = 4\sigma^2 / n$ . Estimated effects of different terms are uncorrelated.
- (c) The sum of squares of  $S$  is  $\|\mathbf{P}_S \mathbf{y}\|^2 = n\hat{E}_S^2 / 4$ .
- (d) In the regression of  $\mathbf{y}$  on  $\mathbf{1}$  and all  $2^t - 1$  columns  $\mathbf{x}_S$ , the coefficient of  $\mathbf{x}_S$  is  $\hat{E}_S / 2$ , and it is the same in every submodel that contains  $\mathbf{x}_S$ .

*Proof.*

- (a) With  $\mathbf{u} = (-1, 1)^\top$ ,  $\mathbf{C}_2 = \frac{1}{2}\mathbf{u}\mathbf{u}^\top$  and  $\bar{\mathbf{J}}_2 = \frac{1}{2}\mathbf{1}_2\mathbf{1}_2^\top$ . By the mixed-product rule  $\mathbf{P}_S = (2^t m)^{-1} \mathbf{x} \mathbf{x}^\top$ , where  $\mathbf{x}$  is the Kronecker product with  $\mathbf{u}$  in the positions of  $S$ ,  $\mathbf{1}_2$  elsewhere, and  $\mathbf{1}_m$  last. Its entries are the products of the codes of the factors in  $S$ , so  $\mathbf{x} = \mathbf{x}_S$ . Orthogonality follows from [Theorem 16.4.1\(a\)](#), since  $\mathbf{x}_S \in \mathcal{C}(\mathbf{P}_S)$  and  $\mathbf{1} \in \mathcal{C}(\mathbf{P}_\emptyset)$ .
- (b) Since  $\mathbf{x}_S \perp \mathbf{1}$ , exactly  $n/2$  entries are  $+1$ , so  $\mathbf{x}_S^\top \mathbf{y} = \frac{n}{2}(\bar{y}(+) - \bar{y}(-))$ . Then  $\text{Var}(\hat{E}_S) = (4/n^2)\sigma^2 \|\mathbf{x}_S\|^2 = 4\sigma^2 / n$ . Covariances are  $(4/n^2)\sigma^2 \mathbf{x}_S^\top \mathbf{x}_T = 0$ .

- (c)  $\|\mathbf{P}_S \mathbf{y}\|^2 = (\mathbf{x}_S^\top \mathbf{y})^2 / n = n \hat{E}_S^2 / 4$ .
- (d) With orthogonal columns each coefficient is  $\mathbf{x}_S^\top \mathbf{y} / \|\mathbf{x}_S\|^2 = \mathbf{x}_S^\top \mathbf{y} / n$  and does not depend on the other columns ([Proposition 7.2.4](#)).

■

**Example. (Heat seals: an unreplicated  $2^4$  design)**

A packaging line makes heat seals on plastic film. Four factors were each set at two levels:  $A$ , jaw temperature;  $B$ , jaw pressure;  $C$ , dwell time;  $D$ , film gauge. One seal was made at each of the 16 combinations, in random order, and its peel strength was measured in newtons. The data are synthetic, generated by the script `two_level.py`. In standard order ( $A$  changing fastest), the strengths are

| run | 1    | 2    | 3    | 4    | 5    | 6    | 7    | 8    | 9    | 10   | 11   | 12   | 13   | 14   | 15   | 16   |
|-----|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|------|
| $A$ |      | +    |      | +    |      | +    |      | +    |      | +    |      | +    |      | +    |      | +    |
| $B$ |      |      | +    | +    |      |      | +    | +    |      |      | +    | +    |      |      | +    | +    |
| $C$ |      |      |      |      | +    | +    | +    | +    |      |      |      |      | +    | +    | +    | +    |
| $D$ |      |      |      |      |      |      |      |      | +    | +    | +    | +    | +    | +    | +    | +    |
| $y$ | 28.5 | 32.1 | 26.7 | 31.1 | 29.0 | 37.4 | 27.2 | 37.4 | 25.8 | 26.6 | 24.2 | 28.8 | 24.9 | 34.3 | 25.9 | 36.7 |

The mean is 29.79. The largest estimated effects are  $A$  (6.52),  $C$  (3.62),  $AC$  (3.17) and  $D$  (−2.78). They are followed by  $BD$  (1.08) and  $AB$  (0.98), and none of the other nine exceeds 0.53 in absolute value. There is no pure error. The experimenters had decided *before the trial* to treat the five interactions of three or four factors as negligible and to use their sums of squares as error. That gives  $s^2 = 0.514$  on 5 degrees of freedom. Against the critical value  $F_{0.05}(1, 5) = 6.61$ , the tests reject for  $A$ ,  $C$  and  $AC$  ( $F = 331.0, 102.2, 78.4$ , each  $p < 0.001$ ), for  $D$  ( $F = 59.9, p = 0.001$ ), and, nominally, for  $BD$  ( $p = 0.030$ ) and  $AB$  ( $p = 0.042$ ). The last two are among ten tests, however. Holm's procedure at familywise level 0.05 ([Theorem 13.4.5](#)) accepts  $A$ ,  $C$ ,  $AC$  and  $D$  and stops at  $BD$ , whose  $p$ -value exceeds 0.05/6. The data were in fact generated with exactly those four effects.

The interaction  $AC$  matters in practice. The effect of dwell time is  $\hat{E}_C + \hat{E}_{AC} = 3.62 + 3.17 = 6.79$  N at high temperature and  $\hat{E}_C - \hat{E}_{AC} = 0.45$  N at low temperature ([Exercise B2](#)). Longer dwell helps only when the jaws are hot. [Figure 16.4.1](#) shows the half-normal plot of Daniel (1959): absolute effects against half-normal quantiles. Noise falls near a line through the origin and real effects stand out above it, with no estimate of  $\sigma$  needed.

```
import itertools

import numpy as np
from scipy import stats

k = 4
runs = np.array(list(itertools.product([-1, 1], repeat=k))[:, :-1]) # standard
                        order: A changes fastest
names = []
columns = []
```

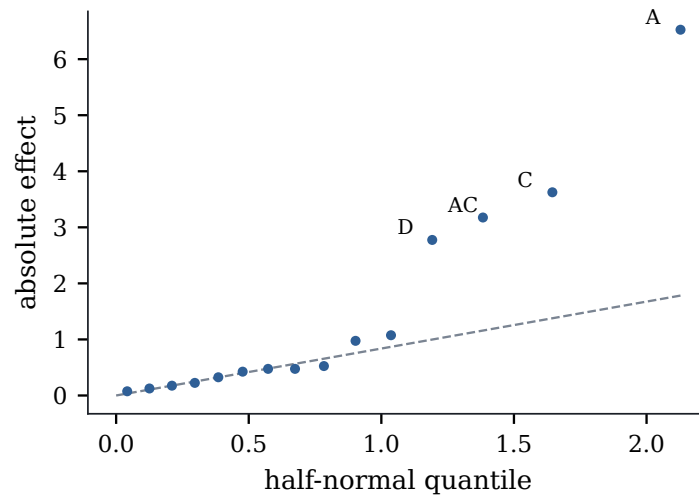


Figure 16.4.1: Half-normal plot of the 15 absolute effects of the heat-seal experiment. The dashed line through the origin is fitted to the ten smallest. *A*, *C*, *AC* and *D* stand clear of it.

```
for size in range(1, k + 1):
    for S in itertools.combinations(range(k), size):
        names.append("".join("ABCD"[i] for i in S))
        columns.append(np.prod(runs[:, list(S)], axis=1))
Xc = np.column_stack(columns)           # 16 x 15 matrix of +-1 contrast
columns
print("columns orthogonal:", np.allclose(Xc.T @ Xc, 16 * np.eye(15)))
```

```
true = 30 + 3.0 * runs[:, 0] + 2.0 * runs[:, 2] + 1.5 * runs[:, 0] * runs[:, 2] -
    1.25 * runs[:, 3]
rng = np.random.default_rng(16_04)
y = np.round(true + rng.normal(scale=1.0, size=16), 1)
n = len(y)
effects = Xc.T @ y / (n / 2)           # mean at + minus mean at -
ss = n * effects ** 2 / 4              # one degree of freedom each
order = np.argsort(-np.abs(effects))
for i in order[:6]:
    print(f"{names[i]:4s} effect {effects[i]:6.2f}    SS {ss[i]:7.2f}")
high = [i for i, nm in enumerate(names) if len(nm) >= 3]  # ABC, ABD, ACD, BCD,
ABCD
ss_pooled = ss[high].sum()
s2 = ss_pooled / len(high)
F = ss / s2
p = stats.f.sf(F, 1, len(high))
for i in order[:6]:
    print(f"{names[i]:4s} F = {F[i]:7.2f}    p = {p[i]:.3f}")
```

Two-level designs are economical: every main effect is estimated with the precision  $4\sigma^2/n$  of the whole experiment, as if each factor had been studied alone with all  $n$  runs. With many factors, *fractional* factorials use a chosen subset of the  $2^t$  runs and deliberately confound some effects with others. Box, Hunter and Hunter (2005) and Wu and Hamada (2009) develop that theory.

## 16.4.4 Pooling and its dangers

In [Example \(Heat seals: an unreplicated  \$2^4\$  design\)](#) the error was built from the sums of squares of terms assumed to be zero. This is **pooling**: replacing SSE by  $\text{SSE} + \sum_{T \in \mathcal{Q}} \|\mathbf{P}_T \mathbf{y}\|^2$ , with  $\nu_E + \sum_{T \in \mathcal{Q}} r_T$  degrees of freedom, for a set  $\mathcal{Q}$  of terms. Without replication it is the only source of an error term. The theory is simple when  $\mathcal{Q}$  is fixed in advance.

### Proposition 16.4.4 — Pooling a fixed set of terms

In the setting of [Theorem 16.4.1\(e\)](#) with normal errors, let  $\mathcal{Q}$  be a set of terms fixed before the data are seen. Let  $\nu$  be the pooled degrees of freedom and  $\text{MS}_{\text{pool}}$  the pooled mean square. Let  $S \notin \mathcal{Q}$  and  $F_S^{\text{pool}} = (\|\mathbf{P}_S \mathbf{Y}\|^2 / r_S) / \text{MS}_{\text{pool}}$ .

- (a)  $\text{E MS}_{\text{pool}} = \sigma^2 + \nu^{-1} \sum_{T \in \mathcal{Q}} \|\mathbf{P}_T \boldsymbol{\mu}\|^2$ .
- (b) If  $\mathbf{P}_T \boldsymbol{\mu} = \mathbf{0}$  for every  $T \in \mathcal{Q}$ , then  $F_S^{\text{pool}} \sim F(r_S, \nu, \|\mathbf{P}_S \boldsymbol{\mu}\|^2 / \sigma^2)$ . The pooled test is exact and has more denominator degrees of freedom.
- (c) If  $\mathbf{P}_S \boldsymbol{\mu} = \mathbf{0}$  but some pooled term is not null, then  $\Pr\{F_S^{\text{pool}} > F_\alpha(r_S, \nu)\} < \alpha$ . The test is conservative, and its power is reduced for the same reason.

*Proof.* The matrix  $\mathbf{P}_{\text{pool}} = \mathbf{P}_E + \sum_{T \in \mathcal{Q}} \mathbf{P}_T$  is a projection of rank  $\nu$ , orthogonal to  $\mathbf{P}_S$  ([Theorem 16.4.1\(a\)](#)). By [Theorem 4.4.4](#),  $U = \|\mathbf{P}_S \mathbf{Y}\|^2 / \sigma^2$  and  $V = \|\mathbf{P}_{\text{pool}} \mathbf{Y}\|^2 / \sigma^2$  are independent, with  $V \sim \chi^2(\nu, \delta)$  and  $\delta = \sum_{T \in \mathcal{Q}} \|\mathbf{P}_T \boldsymbol{\mu}\|^2 / \sigma^2$ , because  $\mathbf{P}_E \boldsymbol{\mu} = \mathbf{0}$ . Part (a) is the mean  $\nu + \delta$  of  $V$ , times  $\sigma^2 / \nu$ . If  $\delta = 0$ , (b) is the definition of the  $F$  distribution. For (c),  $U \sim \chi^2(r_S)$ , and with  $c = F_\alpha(r_S, \nu)$ ,

$$\Pr\{F_S^{\text{pool}} > c\} = \text{E} \left[ \Pr \left\{ V < \frac{\nu U}{r_S c} \mid U \right\} \right].$$

By [Proposition 4.1.3](#),  $\Pr\{\chi^2(\nu, \delta) < x\}$  is strictly smaller for  $\delta > 0$  than for  $\delta = 0$ , for every  $x > 0$ . Taking expectations gives a probability smaller than its value at  $\delta = 0$ , which is  $\alpha$ . ■

The dangers come when the choice of  $\mathcal{Q}$  depends on the data. There are two common ways that happens.

*Pooling after a preliminary test.* A common rule pools an interaction into error if its own  $F$  test is not significant at a lenient level such as 0.25. Consider a  $2 \times 3 \times 4$  layout with  $m = 2$ , so  $\nu_E = 24$  and  $r_{ABC} = 6$ , and test a null main effect  $A$  at level 0.05 after this preliminary test of  $ABC$ . By [Theorem 16.4.1\(e\)](#),  $\text{SS}_A$ ,  $\text{SS}_{ABC}$  and  $\text{SSE}$  are independent, with distributions  $\sigma^2 \chi^2(1)$ ,  $\sigma^2 \chi^2(6, \gamma_{ABC})$  and  $\sigma^2 \chi^2(24)$ . So the rule can be simulated exactly by drawing these three variables ([Figure 16.4.2](#)). When  $ABC$  is absent, the rule pools in 75% of samples, precisely those with small  $\text{MS}_{ABC}$ . The pooled denominator is biased downwards, and the test of  $A$  rejects a true null with probability 0.0551 instead of 0.05. When  $ABC$  is large the rule rarely pools, and the level returns to 0.05 (0.0504 at  $\gamma_{ABC} = 30$ ). Always pooling is very conservative there (0.0064), as [Proposition 16.4.4\(c\)](#) predicts. Here the damage is modest, but a preliminary test buys degrees of freedom with a size that is no longer what it claims to be (Bancroft 1944).

```
import numpy as np
from scipy import stats
```



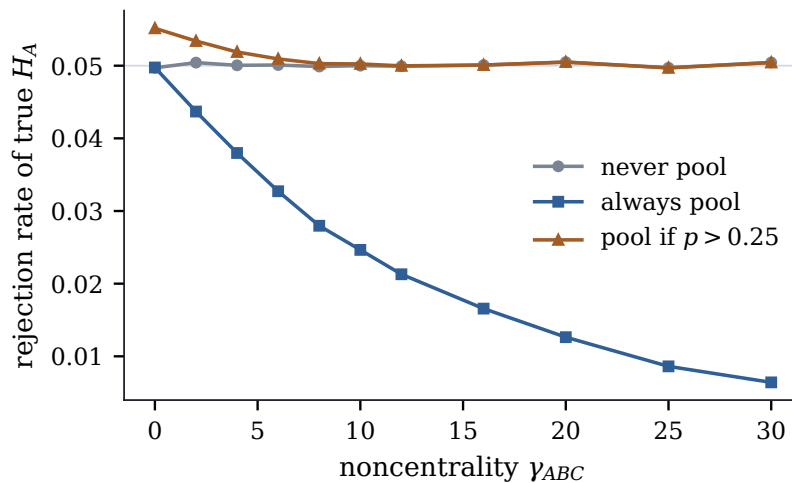


Figure 16.4.2: Rejection rate of the level-0.05 test of a true null main effect in a  $2 \times 3 \times 4$  layout with two replicates, as the three-factor noncentrality  $\gamma_{ABC}$  grows. Never pooling is exact. Always pooling is conservative once  $ABC$  is present. Pooling after a preliminary test at level 0.25 is anticonservative near  $\gamma_{ABC} = 0$ . Each point is based on 400000 simulated samples.

```

alpha, alpha_pre = 0.05, 0.25
df_A, df_ABC, df_E = 1, 6, 24

def rejection_rates(gamma, reps, rng):
    """Size of the level-0.05 test of A under three pooling rules, when A has no
    effect."""
    ssA = rng.chisquare(df_A, reps)
    ssABC = rng.noncentral_chisquare(df_ABC, gamma, reps) if gamma > 0 else rng.
    chisquare(df_ABC, reps)
    ssE = rng.chisquare(df_E, reps)
    never = ssA / (ssE / df_E) > stats.f.ppf(1 - alpha, df_A, df_E)
    pooled_F = ssA / ((ssE + ssABC) / (df_E + df_ABC))
    always = pooled_F > stats.f.ppf(1 - alpha, df_A, df_E + df_ABC)
    pool = ssABC / df_ABC / (ssE / df_E) <= stats.f.ppf(1 - alpha_pre, df_ABC, df_E)
    pretest = np.where(pool, always, never)
    return never.mean(), always.mean(), pretest.mean(), pool.mean()

rng = np.random.default_rng(20260919)
gammas = np.array([0, 2, 4, 6, 8, 10, 12, 16, 20, 25, 30])
rates = np.array([rejection_rates(g, 400_000, rng) for g in gammas])
for g, (nv, al, pt, pl) in zip(gammas, rates):
    print(f"gamma {g:3d}: never {nv:.4f}  always {al:.4f}  pre-test {pt:.4f} (
    pooled in {pl:.2f})")

```

*Pooling the smallest effects.* The second temptation is much worse. In an unreplicated  $2^t$  design one may look at the estimated effects, declare the smallest few to be “error”, and test the rest against them. If no factor does anything, the 15 estimated effects of a  $2^4$  design are independent normal variables with a common variance (Proposition 16.4.3(b)). Pool the five smallest in absolute value and test the largest against them with the  $F(1, 5)$  critical value. It is declared significant in a fraction 0.999 of null data sets (and so, of course, at least one effect is declared significant). The five smallest of fifteen normal variables are the bottom third of the noise, and a mean square built from them underestimates the variance by a factor of 11.5 on average. This

is why the error terms in [Example \(Heat seals: an unreplicated  \$2^4\$  design\)](#) were chosen before the data were seen, and why the half-normal plot and Lenth's (1989) robust standard error ([Exercise \(C1\)](#)) are the standard tools for unreplicated designs.

```
rng2 = np.random.default_rng(16_0404)
reps = 200_000
effects = rng2.normal(size=(reps, 15))           # 15 null effect
           estimates, common variance
sq = np.sort(effects ** 2, axis=1)
F_largest = sq[:, -1] / sq[:, :5].mean(axis=1)    # largest against the
           five smallest
size_largest = np.mean(F_largest > stats.f.ppf(1 - alpha, 1, 5))
pooled_ms = sq[:, :5].mean(axis=1)               # the "error" mean
           square; true variance 1
underestimate = 1 / pooled_ms.mean()             # factor by which it
           falls short on average
print(f"largest effect declared significant: {size_largest:.3f};"
      f" pooled mean square averages {pooled_ms.mean():.3f}, an underestimate by a
      factor {underestimate:.1f}")
```

*Idea.* Pooling is safe when the pooled terms are chosen before the data are seen and are genuinely negligible. It is conservative when they are chosen in advance but are not negligible. It is invalid when they are chosen because they look small.

## 16.4.5 Exercises

### 16.4.5.1 A. Check your understanding

#### Exercise A1.

List the terms and their degrees of freedom for a balanced  $3 \times 2 \times 4$  layout with  $m = 2$ , and check that they add up to  $n - 1$ .

#### Exercise A2.

For  $t$  factors each with  $l$  levels, show that the number of terms involving exactly  $q$  factors is  $\binom{t}{q}$ , each with  $(l - 1)^q$  degrees of freedom, and that these add up to  $l^t - 1$ .

### 16.4.5.2 B. Practice

#### Exercise B1.

In a balanced two-way layout, show that if  $AB$  (the cell indicators) is entered before  $A$ , the sequential sum of squares of  $A$  is zero. What does that say about software output for models written in a non-hierarchical order?

#### Exercise B2.

In a  $2^t$  factorial, show that the effect of  $C$  computed only from the runs with  $A$  at its

high level is  $\hat{E}_C + \hat{E}_{AC}$ , and at the low level it is  $\hat{E}_C - \hat{E}_{AC}$ . Conclude that  $\hat{E}_{AC}$  is half the difference between the two simple effects of  $C$ .

*Solution.* Let  $\bar{y}_{ac}$  be the mean of the runs with codes  $a, c \in \{-1, 1\}$  for  $A$  and  $C$ , each averaging  $n/4$  runs. Then  $\hat{E}_C = \frac{1}{2}(\bar{y}_{1,1} + \bar{y}_{-1,1}) - \frac{1}{2}(\bar{y}_{1,-1} + \bar{y}_{-1,-1})$ , and, since  $\mathbf{x}_{AC} = +1$  when the codes agree,  $\hat{E}_{AC} = \frac{1}{2}(\bar{y}_{1,1} + \bar{y}_{-1,-1}) - \frac{1}{2}(\bar{y}_{1,-1} + \bar{y}_{-1,1})$ . Adding,  $\hat{E}_C + \hat{E}_{AC} = \bar{y}_{1,1} - \bar{y}_{-1,-1}$ , the effect of  $C$  at high  $A$ . Subtracting gives  $\bar{y}_{-1,1} - \bar{y}_{1,-1}$ . Half the difference of the two is  $\hat{E}_{AC}$ .

### Exercise B3.

Use [Theorem 16.4.1\(c\)](#) to derive the displayed formula for  $(\mathbf{P}_{ABC}\mathbf{y})_{ijkl}$ . Write the analogous formula for  $(\mathbf{P}_{AB}\mathbf{y})_{ijkl}$  in a three-factor layout, and check that it does not involve  $k$ .

### 16.4.5.3 C. Going deeper

#### Exercise C1.

In an unreplicated  $2^t$  design, let  $\hat{E}_1, \dots, \hat{E}_K$  be the  $K = 2^t - 1$  estimated effects. Lenth (1989) proposed  $s_0 = 1.5 \text{median}|\hat{E}_S|$  and the *pseudo standard error*  $\text{PSE} = 1.5 \text{median}\{|\hat{E}_S| : |\hat{E}_S| < 2.5s_0\}$ . Show that if all effects are null and  $\tau^2 = 4\sigma^2/n$ , then  $\text{median}|\hat{E}_S| \rightarrow 0.6745\tau$  as  $K \rightarrow \infty$ , so that  $s_0$  estimates  $1.01\tau$ . Explain why the trimming at  $2.5s_0$  makes PSE insensitive to a few large active effects, and why that is the property pooling lacks.

*Solution.* Under the null hypothesis the  $\hat{E}_S$  are independent  $N(0, \tau^2)$  variables ([Proposition 16.4.3\(b\)](#)), so  $|\hat{E}_S|$  has the half-normal distribution with median  $\tau \Phi^{-1}(0.75) = 0.6745\tau$ . The sample median converges to it, and  $1.5 \times 0.6745 \approx 1.01$ , so  $s_0 \approx \tau$ . The median is unaffected by a few large values. The trimmed median in PSE additionally removes effects that are large compared with the first estimate before re-estimating, so active effects enter neither estimate as long as they are few (*effect sparsity*). Pooling the smallest effects uses the bottom of the distribution only and so is biased downwards. The median of all effects is instead a consistent estimate of the scale when most effects are null.

## 16.5 Nested factors

So far every level of one factor has appeared with every level of the other. Many studies are built differently. Pupils are taught in classrooms, and each classroom follows one curriculum. Specimens are cut from batches, and each batch comes from one supplier. Classroom 3 under one curriculum has nothing to do with classroom 3 under another: the factors are *nested*, not crossed. This section gives the decomposition for balanced nested layouts, explains why a nested factor has no main effect, and shows why the right denominator for the outer factor depends on what the inner levels represent.

## 16.5.1 Nested and crossed factors

### Definition 16.5.1 — Nested factors

Factor  $B$  is **nested** in factor  $A$  if every level of  $B$  occurs together with exactly one level of  $A$ . Factors  $A$  and  $B$  are **crossed** if every level of  $B$  occurs with every level of  $A$ . A **balanced nested layout** has  $a$  levels of  $A$ ,  $b$  levels of  $B$  within each level of  $A$ , and  $m$  observations at each level of  $B$ . We write  $B(A)$  for the nested factor, index its levels within level  $i$  of  $A$  by  $j = 1, \dots, b$ , and write  $y_{ijk}$ ,  $k = 1, \dots, m$ .

The labels  $j$  are local:  $(i, j)$  names a level of  $B$ , but  $j$  alone names nothing. The model is

$$E(y_{ijk}) = \mu + \alpha_i + \beta_{j(i)}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}, \quad (16.5.1)$$

with  $1 + a + ab$  parameters. Every level of  $B$  has its own mean  $\mu_{ij} = \mu + \alpha_i + \beta_{j(i)}$ , so the mean space is the set of vectors constant on each level of  $B$ , of dimension  $ab$ . This is the mean space of the crossed cell-means model [Eq. 16.3.1](#) on an array of the same shape. Nesting changes not the space of possible means but which questions about them make sense. In a crossed layout, averaging over  $i$  at fixed  $j$  is meaningful. In a nested layout it is not, so there is no main effect of  $B$  and no  $AB$  interaction, only variation among the levels of  $B$  within each level of  $A$ .

## 16.5.2 The nested decomposition

With the observations stacked as in [Section 16.1](#) ( $k$  fastest, then  $j$ , then  $i$ ), define

$$\begin{aligned} \mathbf{P}_0 &= \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_A &= \mathbf{C}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m, \\ \mathbf{P}_{B(A)} &= \mathbf{I}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m, & \mathbf{P}_E &= \mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{C}_m. \end{aligned} \quad (16.5.2)$$

### Theorem 16.5.1 — The balanced nested layout

Assume [Eq. 16.5.1](#) for a balanced nested layout with  $a \geq 2$ ,  $b \geq 2$ ,  $m \geq 2$ . Write  $\mu_{ij}$  for the mean of level  $(i, j)$  of  $B$  and  $\bar{\mu}_i = b^{-1} \sum_j \mu_{ij}$ .

- (a) The four matrices [Eq. 16.5.2](#) are mutually orthogonal projections summing to  $\mathbf{I}$ , with ranks 1,  $a - 1$ ,  $a(b - 1)$  and  $ab(m - 1)$ . The first three sum to the projection onto the mean space. Moreover  $\mathbf{P}_{B(A)} = \mathbf{P}_B + \mathbf{P}_{AB}$ , where  $\mathbf{P}_B$  and  $\mathbf{P}_{AB}$  are the projections [Eq. 16.3.2](#) of the crossed layout of the same shape.
- (b) The sums of squares are

$$SS_A = bm \sum_i (\bar{y}_{i..} - \bar{y}_{...})^2, \quad SS_{B(A)} = m \sum_i \sum_j (\bar{y}_{ij.} - \bar{y}_{i..})^2, \quad SSE = \sum_{i,j,k} (y_{ijk} - \bar{y}_{ij.})^2.$$

So  $SS_{B(A)}$  is the sum over  $i$  of the between-levels sums of squares of  $B$  computed separately within each level of  $A$ .

- (c) The function  $\lambda_0 \mu + \sum_i \lambda_i \alpha_i + \sum_{i,j} \kappa_{ij} \beta_{j(i)}$  is estimable iff  $\lambda_i = \sum_j \kappa_{ij}$  for every  $i$  and  $\lambda_0 = \sum_i \lambda_i$ . In particular  $\beta_{j(i)} - \beta_{l(i)}$  is estimable, but  $\beta_{j(i)} - \beta_{l(k)}$  with  $i \neq k$

is not, and  $\alpha_i - \alpha_k$  is not, while  $\alpha_i - \alpha_k + \bar{\beta}_{\cdot(i)} - \bar{\beta}_{\cdot(k)} = \bar{\mu}_{i\cdot} - \bar{\mu}_{k\cdot}$  is.

(d) The expected mean squares are

$$E \text{ MS}_A = \sigma^2 + \frac{bm}{a-1} \sum_i (\bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot})^2, \quad E \text{ MS}_{B(A)} = \sigma^2 + \frac{m}{a(b-1)} \sum_{i,j} (\mu_{ij} - \bar{\mu}_{i\cdot})^2, \quad E \text{ MSE} = \sigma^2.$$

Under normal errors the three sums of squares are independent, and  $F_A = \text{MS}_A/\text{MSE}$  and  $F_{B(A)} = \text{MS}_{B(A)}/\text{MSE}$  have noncentral  $F$  distributions with  $(a-1, ab(m-1))$  and  $(a(b-1), ab(m-1))$  degrees of freedom and noncentralities equal to  $\sigma^{-2}$  times the sums in the numerators above. So  $F_A$  is an exact test of  $\bar{\mu}_{1\cdot} = \cdots = \bar{\mu}_{a\cdot}$ , and  $F_{B(A)}$  of  $\mu_{ij} = \bar{\mu}_{i\cdot}$  for all  $i, j$ .

*Proof.*

- (a) Ranks and symmetric idempotence follow from [Lemma 16.1.1\(b\)](#). The mixed-product rule gives  $\mathbf{P}_0 \mathbf{P}_A = \mathbf{0}$  (from  $\bar{\mathbf{J}}_a \mathbf{C}_a$ ),  $\mathbf{P}_0 \mathbf{P}_{B(A)} = \mathbf{P}_A \mathbf{P}_{B(A)} = \mathbf{0}$  (from  $\bar{\mathbf{J}}_b \mathbf{C}_b$ ), and each of the first three is orthogonal to  $\mathbf{P}_E$  (from  $\bar{\mathbf{J}}_m \mathbf{C}_m$ ). Next,  $\mathbf{P}_0 + \mathbf{P}_A = \mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m$ , and adding  $\mathbf{P}_{B(A)}$  gives  $\mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m$ , the projection onto the vectors constant on each level of  $B$ , which is the mean space. Adding  $\mathbf{P}_E$  gives  $\mathbf{I}$ . Finally,  $\mathbf{I}_a \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m = (\bar{\mathbf{J}}_a + \mathbf{C}_a) \otimes \mathbf{C}_b \otimes \bar{\mathbf{J}}_m = \mathbf{P}_B + \mathbf{P}_{AB}$ .
- (b)  $\mathbf{P}_{B(A)} = \mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m - \mathbf{I}_a \otimes \bar{\mathbf{J}}_b \otimes \bar{\mathbf{J}}_m$  maps  $y_{ijk}$  to  $\bar{y}_{ij\cdot} - \bar{y}_{i\cdot\cdot}$ . The other two are as in [Theorem 16.3.1\(b\)](#).
- (c) By [Theorem 8.2.2](#) the estimable functions are the linear combinations  $\sum_{i,j} w_{ij}(\mu + \alpha_i + \beta_{j(i)})$  of the expected responses. Such a combination has  $\kappa_{ij} = w_{ij}$ ,  $\lambda_i = \sum_j w_{ij}$  and  $\lambda_0 = \sum_{i,j} w_{ij}$ , which gives the conditions. Conversely, if they hold,  $w_{ij} = \kappa_{ij}$  reproduces the function. The three examples follow by checking the conditions. For  $\alpha_i - \alpha_k$  all  $\kappa$  are zero but  $\lambda_i = 1 \neq 0$ .
- (d) Apply [Theorem 9.6.1](#) and [Corollary 9.6.2](#) to the decomposition in (a). The vector  $\mathbf{P}_A \boldsymbol{\mu}$  has entries  $\bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot\cdot}$ , repeated  $bm$  times, and  $\mathbf{P}_{B(A)} \boldsymbol{\mu}$  has entries  $\mu_{ij} - \bar{\mu}_{i\cdot}$ , repeated  $m$  times.

■

Part (a) is the precise sense in which a nested factor “absorbs” the interaction. If nested data are analysed as though crossed, with the local labels  $j$  treated as real levels, the sums of squares for  $B$  and  $AB$  add up to  $\text{SS}_{B(A)}$ , but each depends on the arbitrary labelling ([Exercise \(B1\)](#)). Part (c) is the nested analogue of [Proposition 8.6.2](#). The contrasts among the outer levels that can be estimated are contrasts among averages over their own inner levels.

More stages work the same way. With  $A, B$  nested in  $A$ , and  $C$  nested in  $B$ , the projections are  $\mathbf{C} \otimes \bar{\mathbf{J}} \otimes \bar{\mathbf{J}} \otimes \bar{\mathbf{J}}, \mathbf{I} \otimes \mathbf{C} \otimes \bar{\mathbf{J}} \otimes \bar{\mathbf{J}}, \mathbf{I} \otimes \mathbf{I} \otimes \mathbf{C} \otimes \bar{\mathbf{J}}$  and the error  $\mathbf{I} \otimes \mathbf{I} \otimes \mathbf{I} \otimes \mathbf{C}$  ([Exercise \(B3\)](#)). Each nested term has an identity matrix in the positions of the factors it is nested in and a centring matrix in its own position. The listing checks the identity of part (a) and the ranks for  $a = 3, b = 4, m = 2$ .

```
import itertools

import numpy as np

def Jbar(k):
    return np.full((k, k), 1.0 / k)           # averaging: replaces a k-vector by its
```

```

mean

def Cen(k):
    return np.eye(k) - Jbar(k)          # centring: subtracts the mean

def kron_all(mats):
    out = np.ones((1, 1))
    for A in mats:
        out = np.kron(out, A)
    return out

def factorial_projections(levels, m):
    """P_S for every subset S of the factors (a 0/1 tuple), and the error projection
    ."""
    P = {}
    for S in itertools.product([0, 1], repeat=len(levels)):
        P[S] = kron_all([Cen(l) if s else Jbar(l) for l, s in zip(levels, S)] + [
            Jbar(m)])
    P["error"] = kron_all([np.eye(l) for l in levels] + [Cen(m)])
    return P

def rank(A):
    return int(round(np.trace(A)))        # rank = trace for a projection

a, b, m = 3, 4, 2
PB_in_A = kron_all([np.eye(a), Cen(b), Jbar(m)])          # B within A
P2 = factorial_projections([a, b], m)
print("B(A) = B + AB of the crossed layout:", np.allclose(PB_in_A, P2[(0, 1)] + P2
    [(1, 1)]))
print("rank of B(A):", rank(PB_in_A), " = a(b-1) =", a * (b - 1))

```

### 16.5.3 What the inner levels represent

The tests of [Theorem 16.5.1\(d\)](#) treat the levels of  $B$  as fixed. Then  $H_A$  says that the averages over *these particular* inner levels are equal, and the error is the variation within inner levels. That is right when the inner levels matter in themselves, such as three specific machines in each of two factories.

Often, though, the inner levels are a sample. The classrooms in a curriculum study stand for all classrooms that might follow that curriculum. Then the question about  $A$  concerns populations of inner levels, and the variation *between* inner levels is part of the noise against which  $A$  must be judged. A preview of [Chapters 32](#) and [33](#) makes this precise. If the  $\beta_{j(i)}$  are independent  $N(0, \sigma_B^2)$  variables, independent of normal errors, then  $E MS_{B(A)} = \sigma^2 + m\sigma_B^2$ , and  $E MS_A$  has the same expression plus a term that vanishes under  $H_A$  ([Exercise \(C1\)](#)). The ratio  $MS_A/MS_{B(A)}$  then has an exact  $F(a-1, a(b-1))$  null distribution. The ratio  $MS_A/MSE$  does not, because its numerator is inflated by  $m\sigma_B^2$  even when  $H_A$  is true. With  $a(b-1)$  denominator degrees of freedom instead of  $ab(m-1)$ , the correct test is based, in effect, on the number of inner levels rather than the number of observations ([Exercise \(B2\)](#)).

**Example. (Curricula and classrooms)**

Seventy-two pupils took the same reading test after a year in one of three reading curricula. Each curriculum was followed in four classrooms, with six pupils tested per classroom. The data are synthetic, generated by the script `nested_classrooms.py` with random classroom effects (standard deviation 4 points), pupil-level noise (standard deviation 8), and curriculum means differing by a few points. The classroom means are

| curriculum     | classroom 1 | classroom 2 | classroom 3 | classroom 4 | mean |
|----------------|-------------|-------------|-------------|-------------|------|
| phonics        | 77.0        | 67.8        | 68.7        | 60.8        | 68.6 |
| whole language | 52.7        | 64.8        | 64.3        | 60.8        | 60.7 |
| mixed          | 61.8        | 61.8        | 63.8        | 58.5        | 61.5 |

and the nested analysis of variance is

| Source                | df | Sum of squares | Mean square |
|-----------------------|----|----------------|-------------|
| curriculum            | 2  | 908.3          | 454.2       |
| classroom(curriculum) | 9  | 1445.8         | 160.6       |
| pupils (error)        | 60 | 3063.3         | 51.1        |

Classrooms within curricula differ:  $F_{B(A)} = 3.15$  on 9 and 60 degrees of freedom,  $p = 0.004$ . For curricula, the fixed-effects test of [Theorem 16.5.1\(d\)](#) gives  $F = 8.90$  on 2 and 60 degrees of freedom ( $p = 0.0004$ ). It says, correctly, that these twelve classrooms differ between curricula on average. Against the classroom mean square the ratio is  $F = 2.83$  on 2 and 9 degrees of freedom ( $p = 0.111$ ). For classrooms in general, no difference has been shown. [Figure 16.5.1](#) shows why: with only four classrooms per curriculum, the gap rests on a handful of classroom means. One phonics classroom (77.0) contributes much of it, and the lowest phonics classroom (60.8) sits among those of the other curricula.

The listing checks by simulation which test to trust. It generates 20000 data sets from the same design with random classroom effects and *no* curriculum effect. The test against pupils rejects at the nominal 5% level in a fraction 0.289 of them, and the test against classrooms in a fraction 0.051. Under the null hypothesis the ratio of the two expected mean squares is  $(8^2 + 6 \cdot 4^2)/8^2 = 2.5$ .

```
import numpy as np
from scipy import stats

a, b, m = 3, 4, 6
data_rng = np.random.default_rng(16_05)
curriculum_effect = np.array([3.0, -2.0, 0.0])
classroom_effect = data_rng.normal(scale=4.0, size=(a, b))
scores = np.round(62 + curriculum_effect[:, None, None] + classroom_effect[:, :,
    None]
    + data_rng.normal(scale=8.0, size=(a, b, m)))

def Jbar(k):
    return np.full((k, k), 1.0 / k)
```

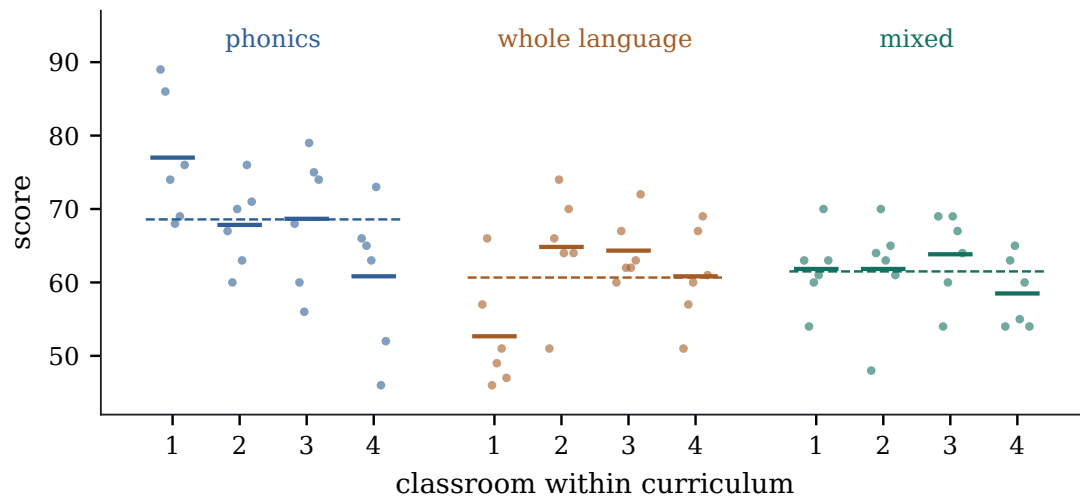


Figure 16.5.1: Reading scores of the 72 pupils by classroom, grouped by curriculum. Solid bars mark classroom means and dashed lines curriculum means. Classrooms vary visibly within each curriculum.

```
def Cen(k):
    return np.eye(k) - Jbar(k)

y = scores.ravel()                                # pupils fastest, then
                                                # classroom, then curriculum
P = {"curriculum": np.kron(np.kron(Cen(a), Jbar(b)), Jbar(m)),
     "classroom(curriculum)": np.kron(np.kron(np.eye(a), Cen(b)), Jbar(m)),
     "error": np.kron(np.kron(np.eye(a), np.eye(b)), Cen(m))}
ss = {k: y @ Pk @ y for k, Pk in P.items()}
df = {k: int(round(np.trace(Pk))) for k, Pk in P.items()}
ms = {k: ss[k] / df[k] for k in P}
for k in P:
    print(f"{k:22s} df {df[k]:2d} SS {ss[k]:8.1f} MS {ms[k]:7.1f}")
F_cls = ms["classroom(curriculum)"] / ms["error"]
F_cur_fixed = ms["curriculum"] / ms["error"]
F_cur_random = ms["curriculum"] / ms["classroom(curriculum)"]
print(f"classrooms within curricula: F = {F_cls:.2f}, p = {stats.f.sf(F_cls, df['classroom(curriculum)'], df['error']):.3f}")
print(f"curricula against pupils: F = {F_cur_fixed:.2f}, p = {stats.f.sf(F_cur_fixed, 2, df['error']):.3f}")
print(f"curricula against classrooms: F = {F_cur_random:.2f}, p = {stats.f.sf(F_cur_random, 2, df['classroom(curriculum)']):.3f}")
```

```
rng = np.random.default_rng(20260919)
reps = 20_000
Pc, Pb, Pe = P["curriculum"], P["classroom(curriculum)"], P["error"]
rej_fixed = rej_random = 0
for _ in range(reps):
    sim = (rng.normal(scale=4.0, size=(a, b))[:, :, None] + rng.normal(scale=8.0,
    size=(a, b, m))).ravel()
    msc, msb, mse = sim @ Pc @ sim / 2, sim @ Pb @ sim / 9, sim @ Pe @ sim / 60
    rej_fixed += msc / mse > stats.f.ppf(0.95, 2, 60)
    rej_random += msc / msb > stats.f.ppf(0.95, 2, 9)
```



```
print(f"no curriculum effect, random classrooms: rejection rate {rej_fixed / reps:.3f} "
      f"against pupils, {rej_random / reps:.3f} against classrooms")
```

*Warning.* The mistake of testing an outer factor against the within-level error, when the inner levels are a sample, is common enough to have a name, *pseudoreplication* (Hurlbert 1984). The pupils are not independent replicates of a curriculum. The classrooms are. With a fixed number of pupils, adding classrooms improves the curriculum comparison much more than adding pupils per classroom.

Nested and crossed structure are often combined, for example when every classroom is tested both before and after the year. The balanced decomposition is again a sum of Kronecker products, one for each meaningful term ([Exercise \(C2\)](#)). Unbalanced nested data lose the Kronecker structure, and random inner levels lead to the mixed models of Chapters 32 and 33 ([Definition 32.2.1](#)).

## 16.5.4 Exercises

### 16.5.4.1 A. Check your understanding

#### Exercise A1.

Say whether each pair of factors is crossed or nested, and if nested, which is inside which. (a) Four fertilizers applied to plots of each of three wheat varieties. (b) Two suppliers, each delivering five batches, with several specimens tested from each batch. (c) Hospitals within regions, with patients within hospitals. (d) Three laboratories, each measuring portions of the same six reference samples.

### 16.5.4.2 B. Practice

#### Exercise B1.

In a balanced nested layout, permute the local labels  $j$  of  $B$  within one level  $i$  of  $A$ , leaving the data unchanged. Show that  $SS_{B(A)}$  does not change, but that  $SS_B$  and  $SS_{AB}$  of the crossed analysis generally do. Conclude that only their sum has a meaning for nested data.

*Solution.*  $SS_{B(A)} = m \sum_i \sum_j (\bar{y}_{ij.} - \bar{y}_{i..})^2$  sums over  $j$  within each  $i$ , so the order of the  $j$  within a level of  $A$  does not matter.  $SS_B = am \sum_j (\bar{y}_{.j.} - \bar{y}_{...})^2$  depends on which inner levels share a label  $j$  across different  $i$ . For example, with  $a = b = 2$ ,  $m = 1$  and inner means  $(1, 3)$  and  $(1, 3)$ , the column means are 1 and 3 and  $SS_B = 4$ . Swapping the labels in the second row gives inner means  $(1, 3)$  and  $(3, 1)$ , column means 2 and 2, and  $SS_B = 0$ . By [Theorem 16.5.1\(a\)](#),  $SS_B + SS_{AB} = SS_{B(A)}$  is unchanged, so the interaction absorbs the difference.

#### Exercise B2.

Show that  $MS_A/MS_{B(A)}$  equals the one-way  $F$  statistic for  $A$  computed from the  $ab$

inner-level means  $\bar{y}_{ij\cdot}$ , treated as  $b$  observations at each of the  $a$  levels. Interpret this as an analysis in which the inner level is the unit.

*Solution.* Let  $z_{ij} = \bar{y}_{ij\cdot}$ . The one-way analysis of the  $z_{ij}$  has between sum of squares  $b \sum_i (\bar{z}_i - \bar{z}_{..})^2$  on  $a - 1$  degrees of freedom and within sum of squares  $\sum_{i,j} (z_{ij} - \bar{z}_i)^2$  on  $a(b - 1)$ . Since  $\bar{z}_i = \bar{y}_{i..}$ , these are  $SS_A/m$  and  $SS_{B(A)}/m$  by [Theorem 16.5.1\(b\)](#). The factor  $m$  cancels in the ratio of mean squares. So the test of  $A$  against  $B(A)$  is exactly the one-way analysis of variance of the inner-level means. The individual observations enter only through those means.

### Exercise B3.

For a three-stage nested layout ( $a$  levels of  $A$ ,  $b$  levels of  $B$  in each,  $c$  levels of  $C$  in each level of  $B$ ,  $m$  observations at each level of  $C$ ), verify that the four projections given in the text and  $\mathbf{P}_0$  are mutually orthogonal and sum to  $\mathbf{I}$ , and find their ranks. Give the sum of squares for  $C(B)$  in terms of means.

#### 16.5.4.3 C. Going deeper

### Exercise C1.

In the balanced nested layout of [Theorem 16.5.1](#), let  $y_{ijk} = \mu + \alpha_i + b_{ij} + \varepsilon_{ijk}$ , where the  $b_{ij}$  and  $\varepsilon_{ijk}$  are uncorrelated random variables with means zero and variances  $\sigma_B^2$  and  $\sigma^2$ . Show that  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} + m\sigma_B^2(\mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m)$ , and use [Theorem 2.3.1](#) to show that

$$E \text{MS}_A = \sigma^2 + m\sigma_B^2 + \frac{bm}{a-1} \sum_i (\alpha_i - \bar{\alpha})^2, \quad E \text{MS}_{B(A)} = \sigma^2 + m\sigma_B^2.$$

Compare [Exercise \(B1\)](#).

*Solution.* The vector of  $b_{ij}$ , each repeated  $m$  times, is  $(\mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{1}_m)\mathbf{b}$ , with covariance  $\sigma_B^2(\mathbf{I}_a \otimes \mathbf{I}_b \otimes \mathbf{1}_m \mathbf{1}_m^\top) = m\sigma_B^2(\mathbf{I}_a \otimes \mathbf{I}_b \otimes \bar{\mathbf{J}}_m)$ . Adding the error covariance gives  $\Sigma = \text{Cov}(\mathbf{Y})$ . By [Theorem 2.3.1](#),  $E(\mathbf{Y}^\top \mathbf{P} \mathbf{Y}) = \text{tr}(\mathbf{P} \Sigma) + \boldsymbol{\mu}^\top \mathbf{P} \boldsymbol{\mu}$ . For  $\mathbf{P} = \mathbf{P}_A$ :  $\mathbf{P}_A(\mathbf{I} \otimes \mathbf{I} \otimes \bar{\mathbf{J}}_m) = \mathbf{P}_A$ , so  $\text{tr}(\mathbf{P}_A \Sigma) = (\sigma^2 + m\sigma_B^2)(a - 1)$ , and  $\boldsymbol{\mu}^\top \mathbf{P}_A \boldsymbol{\mu} = bm \sum_i (\alpha_i - \bar{\alpha})^2$ . For  $\mathbf{P}_{B(A)}$  likewise  $\text{tr} = (\sigma^2 + m\sigma_B^2)a(b - 1)$ , and  $\mathbf{P}_{B(A)} \boldsymbol{\mu} = \mathbf{0}$  because the mean  $\mu + \alpha_i$  does not vary within levels of  $A$ . Dividing by the degrees of freedom gives the result. Under  $H_A$  both mean squares have expectation  $\sigma^2 + m\sigma_B^2$ , which is why the ratio  $\text{MS}_A/\text{MS}_{B(A)}$  is the natural test statistic.

### Exercise C2.

Suppose each classroom of [Example \(Curricula and classrooms\)](#) is tested on two occasions (factor  $C$ , crossed with classroom), with  $m$  pupils per classroom and occasion. Order the observations by curriculum, classroom, occasion and replicate. Write the Kronecker projections for the terms  $A$ ,  $B(A)$ ,  $C$ ,  $AC$ ,  $C \times B(A)$  and error, check that they are mutually orthogonal and sum to  $\mathbf{I}$ , and find their ranks.

## 16.6 Summary and Notes

### 16.6.1 Summary

*Idea.*

1. In a balanced layout every projection of interest is a Kronecker product of averaging matrices  $\bar{\mathbf{J}}$ , centring matrices  $\mathbf{C}$  and identities. Products that differ by  $\bar{\mathbf{J}}$  against  $\mathbf{C}$  in some position are orthogonal, and ranks multiply ([Lemma 16.1.1](#)).
2. The additive two-way model has projection  $\mathbf{P}_0 + \mathbf{P}_A + \mathbf{P}_B$ . Row contrasts are estimated and judged as in a one-way layout that ignores the columns. With replication, the residual splits into interaction (lack of fit) and pure error. With one observation per cell, Tukey's one-degree-of-freedom test is an exact test against the kind of nonadditivity a wrong scale produces, and its slope suggests a power transformation ([Theorem 16.1.2](#)).
3. Interaction is a property of the table of cell means. It is present iff some interaction contrast (a zero-margin combination of cell means) is nonzero, and it has  $(a - 1)(b - 1)$  dimensions whatever parameterization is used ([Definition 16.2.1](#), [Proposition 16.2.1](#)).
4. An increasing transformation can remove interaction only if the orders of the levels are the same everywhere. Qualitative (crossing) interaction cannot be removed. Main effects in the presence of interaction are weighted averages of simple effects, and their values, even their signs, depend on the weights ([Proposition 16.2.2](#), [Proposition 16.2.3](#)).
5. The balanced two-way layout with replication splits into grand mean,  $A$ ,  $B$ ,  $AB$  and error, with ranks 1,  $a - 1$ ,  $b - 1$ ,  $(a - 1)(b - 1)$ ,  $ab(m - 1)$ . The  $F$  tests are exact, and the sequential, Type II and Type III sums of squares all coincide ([Theorem 16.3.1](#)).
6. Products of orthogonal contrasts split the interaction sum of squares into single degrees of freedom. Tukey's method covers the levels of one factor, Scheffé's all interaction contrasts, and Bonferroni or Holm planned families ([Proposition 16.3.2](#)).
7. A balanced factorial with  $t$  factors has one term for each subset of factors, with projection  $\mathbf{P}_S$  built from centring matrices in the positions of  $S$ . Hierarchical models have order-free sums of squares, and  $\mathbf{P}_S \mathbf{y}$  is an alternating sum of marginal means ([Theorem 16.4.1](#)). The three-factor interaction measures how a two-factor interaction changes across the third factor ([Proposition 16.4.2](#)).
8. In a  $2^t$  factorial every term has one degree of freedom, and its effect is a difference of two means with variance  $4\sigma^2/n$  ([Proposition 16.4.3](#)). Pooling terms into error is exact if they are null and chosen in advance, and conservative if they are chosen in advance but not null. It is invalid if they are chosen because they look small ([Proposition 16.4.4](#)).
9. A nested factor has no main effect. Its sum of squares is the crossed  $B$  plus  $AB$ , the variation among inner levels within each outer level ([Definition 16.5.1](#), [Theorem 16.5.1](#)). When the inner levels are a sample, the outer factor must be tested against the inner mean square, not against the within-level error.

## 16.6.2 Notes and sources

**Sources.** The coverage of this chapter was checked against Rencher and Schaalje (2008, chapter 14), Christensen (2020, chapter 7) and Seber and Lee (2003, sections 8.4–8.6). The projection approach follows the vector-space treatment of Christensen (2020), which develops interaction contrasts and their product construction ([Proposition 16.3.2](#)) with particular care. Rencher and Schaalje compute the expected mean squares directly, and projections replace that computation here. Scheffé (1959) remains the classical account of fixed-effects multiway layouts, including main effects and interactions defined with general weights (compare [Proposition 16.2.1\(v\)](#)).

**Factorial experiments.** Fisher (1935) argued for factorial experiments over one-factor-at-a-time trials: every observation contributes to every main effect, and interactions can be studied at all. Yates (1937) developed the analysis of  $2^t$  and  $3^t$  designs, including the algorithm for  $2^t$  effects that carries his name. Box, Hunter and Hunter (2005) and Wu and Hamada (2009) are standard modern texts.

**Tukey's test.** The one-degree-of-freedom test is due to Tukey (1949). Its exact distribution follows from the theorem of Milliken and Graybill (1970) on regressors computed from the fitted values ([Theorem 14.7.2](#)), because Tukey's regressor is, after the additive model is projected out, the table of squared fitted values; [Exercise \(C1\)](#) made the same connection. The power rule [Eq. 16.1.5](#) is a heuristic, and [Chapter 22](#) treats transformations systematically.

**Interaction and scale.** The dependence of interaction on the scale of measurement, and the order condition of [Proposition 16.2.2](#), belong to the theory of conjoint measurement of Luce and Tukey (1964). The counterexample of [Exercise \(C1\)](#) is an instance of their double cancellation condition. In clinical trials a treatment effect that changes sign across patient subgroups is called a *qualitative interaction*, and Gail and Simon (1985) gave a likelihood ratio test for it.

**Unreplicated factorials and pooling.** The half-normal plot is due to Daniel (1959), and the pseudo standard error of [Exercise \(C1\)](#) to Lenth (1989). The effect of preliminary tests on subsequent inference was first studied by Bancroft (1944), and the simulation of [Figure 16.4.2](#) illustrates his conclusions in a simple case.

**Nested designs.** Hurlbert (1984) named *pseudoreplication*, treating observations within an experimental unit as independent replicates of the unit's treatment, and documented how widespread it was in ecology. Random inner levels are taken up with variance components in [Chapters 32](#) and [33](#). Unbalanced multiway data are the subject of [Chapter 17](#), and randomized blocks and Latin squares, additive layouts whose structure comes from randomization, of [Chapter 18](#).

## 16.6.3 References

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# Chapter 17

## Unbalanced Data, Cell Means and Empty Cells

Balanced layouts give orthogonal decompositions, and then every sum of squares answers one clear question. Real experiments lose plots and patients, and observational data are unbalanced from the start. With unequal cell counts the same data produce several different “sums of squares for  $A$ ”, each testing its own hypothesis. This chapter starts from the parameterization in which nothing is ambiguous, a mean for every cell, and writes every hypothesis as linear constraints on the cell means. That shows exactly what the Type I, II and III sums of squares test, why Type III needs contrast codings, how the additive model is a constrained cell-means model, and which questions survive when some cells are empty.

**What you need.** Projections and nested models from [Chapter 6](#) ([Theorem 6.5.1](#), [Theorem 6.5.2](#)). Estimability, factor codings and connectedness from [Chapter 8](#) ([Theorem 8.3.1](#), [Definition 8.5.1](#), [Theorem 8.6.1](#), [Proposition 8.6.2](#)). The definitions of the Type I, II and III sums of squares and the first cell-mean results in [Chapter 9](#) ([Definition 9.3.1](#), [Definition 9.3.2](#), [Theorem 9.3.2](#), [Theorem 9.2.4](#)). Testable hypotheses and restricted least squares from [Chapter 11](#) ([Theorem 11.3.2](#), [Proposition 11.3.3](#)). The one-way layout of [Chapter 15](#) and the balanced two-way layout of [Chapter 16](#).

### 17.0.1 Roadmap

- [The cell-means model](#)
- [Types of sums of squares and what they test](#)
- [Constrained models](#)
- [Empty cells and estimability](#)
- [Summary and notes](#)

## 17.1 The cell-means model

With several factors the observations fall into *cells*, one for each combination of levels. The effects model of [Chapter 8](#) writes the mean of a cell as  $\mu + \alpha_i + \beta_j + \gamma_{ij}$  and pays for it with side conditions and codings. The cell-means model gives each cell its own parameter and nothing

else. It fits the same values as the effects model with all interactions, but every question about the factors becomes a statement about a table of means, and the counts enter only through the precision with which that table is estimated.

### 17.1.1 The model

#### Definition 17.1.1 — Cell-means model

Let factors  $F_1, \dots, F_k$  have  $a_1, \dots, a_k$  levels. A **cell** is a combination  $\omega = (i_1, \dots, i_k)$  of levels, and there are  $p = a_1 a_2 \cdots a_k$  cells. Cell  $\omega$  contains  $n_\omega \geq 0$  observations  $y_{\omega 1}, \dots, y_{\omega n_\omega}$ . It is **occupied** if  $n_\omega \geq 1$  and **empty** otherwise. Write  $O$  for the set of occupied cells,  $m = |O|$  and  $n = \sum_\omega n_\omega$ . The **cell-means model** is

$$y_{\omega l} = \mu_\omega + \varepsilon_{\omega l}, \quad l = 1, \dots, n_\omega,$$

with errors that are uncorrelated with mean zero and variance  $\sigma^2$ , and independent  $N(0, \sigma^2)$  when we test. In matrix form  $E(\mathbf{Y}) = \mathbf{W}\boldsymbol{\mu}$ , where  $\boldsymbol{\mu} \in \mathbb{R}^p$  collects the cell means and  $\mathbf{W}$  is the  $n \times p$  matrix whose column  $\omega$  indicates the observations in cell  $\omega$ . We write  $\mathbf{D} = \mathbf{W}^\top \mathbf{W} = \text{diag}(n_\omega)$ , and  $\bar{y}_\omega$  for the average of an occupied cell.

Empty cells keep their entries in  $\boldsymbol{\mu}$ : their means do not enter the distribution of the data, but they are properties of the population (Section 17.4). In two-factor notation, cells are pairs  $(i, j)$ , counts are  $n_{ij}$ , and we order the cells row by row:  $\boldsymbol{\mu} = (\mu_{11}, \dots, \mu_{1b}, \mu_{21}, \dots, \mu_{ab})^\top$ . With this ordering, a Kronecker product  $\mathbf{P} \otimes \mathbf{Q}$  acts on  $\boldsymbol{\mu}$  by transforming the  $a \times b$  table  $(\mu_{ij})$  into  $\mathbf{P}(\mu_{ij})\mathbf{Q}^\top$  (Proposition 1.10.1(f) applied to the transposed table).

By Proposition 8.6.2(a), the overparameterized two-way model with interaction has column space  $\mathcal{C}(\mathbf{W})$ , so the cell-means model is its identifiable parameterization. It is also a one-way layout (Definition 15.1.1) whose groups are the  $m$  occupied cells. The factorial structure plays no part in fitting.

#### Proposition 17.1.1 — Inference with cell means

In the model of Definition 17.1.1, let  $\mathbf{D}^+$  be the diagonal matrix with entries  $1/n_\omega$  for occupied cells and 0 for empty ones, and let  $\bar{\mathbf{y}} \in \mathbb{R}^p$  hold  $\bar{y}_\omega$  for occupied cells and 0 for empty ones. Then:

- (a)  $\text{rank}(\mathbf{W}) = m$ , and  $\mathbf{M} = \mathbf{W}\mathbf{D}^+\mathbf{W}^\top$  replaces each observation by the average of its cell;
- (b)  $\boldsymbol{\lambda}^\top \boldsymbol{\mu}$  is estimable iff  $\lambda_\omega = 0$  for every empty cell. Its best linear unbiased estimator is  $\boldsymbol{\lambda}^\top \bar{\mathbf{y}} = \sum_{\omega \in O} \lambda_\omega \bar{y}_\omega$ , with variance  $\sigma^2 \sum_{\omega \in O} \lambda_\omega^2 / n_\omega$ ;
- (c)  $\text{SSE} = \sum_{\omega \in O} \sum_l (y_{\omega l} - \bar{y}_\omega)^2$  has  $n - m$  degrees of freedom, and  $s^2 = \text{SSE} / (n - m)$ , the pooled within-cell variance, is unbiased for  $\sigma^2$ ;
- (d) if  $\mathbf{L}$  is a  $q \times p$  matrix of rank  $q$  whose rows vanish on the empty cells,  $\mathbf{d} \in \mathbb{R}^q$ , and  $n > m$ , the  $F$  statistic for  $H : \mathbf{L}\boldsymbol{\mu} = \mathbf{d}$  is

$$F = \frac{(\mathbf{L}\bar{\mathbf{y}} - \mathbf{d})^\top (\mathbf{L}\mathbf{D}^+\mathbf{L}^\top)^{-1} (\mathbf{L}\bar{\mathbf{y}} - \mathbf{d})}{q s^2} \sim F(q, n - m, \gamma_H), \quad (17.1.1)$$

where  $\gamma_H = (\mathbf{L}\boldsymbol{\mu} - \mathbf{d})^\top (\mathbf{L}\mathbf{D}^+ \mathbf{L}^\top)^{-1} (\mathbf{L}\boldsymbol{\mu} - \mathbf{d}) / \sigma^2$ .

*Proof.*

- (a) The columns of  $\mathbf{W}$  for occupied cells are nonzero indicators with disjoint supports, and those for empty cells are zero. So  $\text{rank}(\mathbf{W}) = m$ . Since  $\mathbf{D}\mathbf{D}^+\mathbf{D} = \mathbf{D}$ ,  $\mathbf{D}^+$  is a generalized inverse of  $\mathbf{W}^\top \mathbf{W}$ , and  $\mathbf{M} = \mathbf{W}\mathbf{D}^+ \mathbf{W}^\top$  by Theorem 6.3.7. The entry of  $\mathbf{W}\mathbf{D}^+ \mathbf{W}^\top \mathbf{y}$  for an observation in cell  $\omega$  is  $n_\omega^{-1} \sum_l y_{\omega l} = \bar{y}_\omega$ .
- (b) The rows of  $\mathbf{W}$  are the coordinate vectors  $\mathbf{e}_\omega^\top$  of the occupied cells, so  $\mathcal{C}(\mathbf{W}^\top) = \text{span}\{\mathbf{e}_\omega : \omega \in O\}$ , and Theorem 8.3.1 gives the criterion. The vector  $\mathbf{D}^+ \mathbf{W}^\top \mathbf{y} = \bar{\mathbf{y}}$  solves the normal equations (Theorem 6.4.2), so  $\boldsymbol{\lambda}^\top \bar{\mathbf{y}}$  is the least squares estimator of an estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\mu}$ , and it is best linear unbiased by the Gauss–Markov theorem (Theorem 7.2.1). The cell averages are uncorrelated with variances  $\sigma^2/n_\omega$ , which gives the variance.
- (c)  $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2$  is the displayed sum by (a), and  $E(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2(n - m)$  by Theorem 2.3.1, since  $(\mathbf{I} - \mathbf{M})\mathbf{W}\boldsymbol{\mu} = \mathbf{0}$ .
- (d) Apply Theorem 11.3.2 with  $\mathbf{G} = \mathbf{D}^+$ ,  $\hat{\boldsymbol{\beta}} = \bar{\mathbf{y}}$  and  $\boldsymbol{\Lambda}^\top = \mathbf{L}$ , which is testable by (b). The matrix  $\mathbf{L}\mathbf{D}^+ \mathbf{L}^\top$  is nonsingular: for  $\mathbf{x} \neq \mathbf{0}$ ,  $\mathbf{x}^\top \mathbf{L}\mathbf{D}^+ \mathbf{L}^\top \mathbf{x} = \sum_{\omega \in O} (\mathbf{L}^\top \mathbf{x})_\omega^2 / n_\omega$ . If this were zero,  $\mathbf{L}^\top \mathbf{x}$  would vanish on  $O$ , and hence everywhere, because the rows of  $\mathbf{L}$  vanish off  $O$ . That contradicts  $\text{rank}(\mathbf{L}) = q$ .

■

The counts enter only through  $\mathbf{D}^+$ , the variances of the cell averages. An unbalanced cell-means analysis has one fitting step, averaging within cells, and one inferential tool, Eq. 17.1.1. The work lies in choosing  $\mathbf{L}$ .

## 17.1.2 Standard hypotheses as Kronecker products

Let every cell of a two-way layout be occupied. Let  $\mathbf{C}_a$  be an  $a \times (a - 1)$  matrix whose columns form a basis of the contrasts  $\{\mathbf{u} \in \mathbb{R}^a : \mathbf{1}^\top \mathbf{u} = 0\}$ , and define  $\mathbf{C}_b$  in the same way. These are coding matrices in the sense of Definition 8.5.1. The  $a \times a$  centring matrix of Chapter 16 has the same column space, but its columns are linearly dependent, so we keep the thinner matrix here and count restrictions by its columns. The three classical hypotheses are

$$H_A : (\mathbf{C}_a^\top \otimes b^{-1} \mathbf{1}_b^\top) \boldsymbol{\mu} = \mathbf{0}, \quad H_B : (a^{-1} \mathbf{1}_a^\top \otimes \mathbf{C}_b^\top) \boldsymbol{\mu} = \mathbf{0}, \quad H_{AB} : (\mathbf{C}_a^\top \otimes \mathbf{C}_b^\top) \boldsymbol{\mu} = \mathbf{0}. \quad (17.1.2)$$

By the remark on Kronecker products above,  $H_A$  says that  $\mathbf{C}_a^\top$  annihilates the vector of **unweighted row means**  $\bar{\mu}_{i\cdot} = b^{-1} \sum_j \mu_{ij}$ , that is, that these means are all equal.  $H_B$  says the same of the unweighted column means.  $H_{AB}$  says that  $\mathbf{C}_a^\top (\mu_{ij}) \mathbf{C}_b = \mathbf{0}$ . Because  $\mathcal{C}(\mathbf{C}_a) = \mathbf{1}_a^\perp$  and  $\mathcal{C}(\mathbf{C}_b) = \mathbf{1}_b^\perp$ , this holds iff the doubly centred table  $\mu_{ij} - \bar{\mu}_{i\cdot} - \bar{\mu}_{\cdot j} + \bar{\mu}_{\cdot\cdot}$  vanishes, that is, iff the cell means are additive (Definition 16.2.1). The ranks are  $a - 1$ ,  $b - 1$  and  $(a - 1)(b - 1)$  (Proposition 1.10.1(c)), and the hypotheses do not depend on the basis of contrasts (Exercise (B2)). They are the hypotheses tested in the balanced layout by Theorem 16.3.1, where every type of sum of squares tests them.



### 17.1.3 Weighted and unweighted marginal means

A row of the table can be summarized by any average of its cells. Two choices dominate practice. The **weighted row mean**  $\bar{\mu}_i^w = n_{i.}^{-1} \sum_j n_{ij} \mu_{ij}$  weights each cell by its count. Its natural estimator is the raw average  $\bar{y}_{i.}$  of all observations in row  $i$ . The **unweighted row mean**  $\bar{\mu}_i$  gives each column the same weight. Its estimator is the average of the cell averages,  $\bar{y}_i = b^{-1} \sum_j \bar{y}_{ij}$ . More generally, one may fix weights  $w_1, \dots, w_b$  summing to one, such as the shares of the levels of  $B$  in a target population, and estimate  $\sum_j w_j \mu_{ij}$ .

#### Proposition 17.1.2 — Precision of marginal means

Fix a row  $i$  with all  $n_{ij} \geq 1$ , and weights  $w_1, \dots, w_b$  with  $\sum_j w_j = 1$ . The estimator  $\sum_j w_j \bar{y}_{ij}$  of  $\sum_j w_j \mu_{ij}$  is unbiased with variance  $\sigma^2 \sum_j w_j^2 / n_{ij}$ . Among all such weights the variance is smallest, and equal to  $\sigma^2 / n_{i.}$ , exactly for the count weights  $w_j = n_{ij} / n_{i.}$ . In particular

$$\text{Var}(\bar{y}_i) = \frac{\sigma^2}{b^2} \sum_j \frac{1}{n_{ij}} \geq \frac{\sigma^2}{n_{i.}} = \text{Var}(\bar{y}_{i.}),$$

with equality iff the counts  $n_{i1}, \dots, n_{ib}$  are equal.

*Proof.* Unbiasedness and the variance come from Proposition 17.1.1(b). By the Cauchy–Schwarz inequality,

$$1 = \left( \sum_j \frac{w_j}{\sqrt{n_{ij}}} \sqrt{n_{ij}} \right)^2 \leq \left( \sum_j \frac{w_j^2}{n_{ij}} \right) n_{i.},$$

with equality iff  $w_j / \sqrt{n_{ij}}$  is proportional to  $\sqrt{n_{ij}}$ , that is,  $w_j \propto n_{ij}$ . The unweighted mean is the case  $w_j = 1/b$ , which is proportional to  $n_{ij}$  iff the counts are equal. ■

The count weights are the most *precise*, but precision is the wrong criterion for choosing a target. The weighted mean  $\bar{\mu}_i^w$  is the mean of row  $i$  with the levels of  $B$  mixed as they happened to be mixed in this study. If the counts are an accident of the design, such as lost plots, that mixture describes nothing. If they are proportional to a real population, as in a random sample,  $\bar{\mu}_i^w$  may be exactly the target. When the rows have different mixes, comparing weighted means also compares mixes of  $B$ , and that is what the Type I sum of squares of a factor entered first tests (Section 17.2). Choose the weights to describe a population, and accept the variance that follows.

### 17.1.4 An unbalanced field trial

#### Example. (Three varieties at three sites)

A (synthetic) trial compared three wheat varieties V1, V2 and V3 at three sites S1, S2 and S3, with yield in tonnes per hectare. Each site grew mostly the variety it already knew. There are 37 plots in the  $m = 9$  cells:

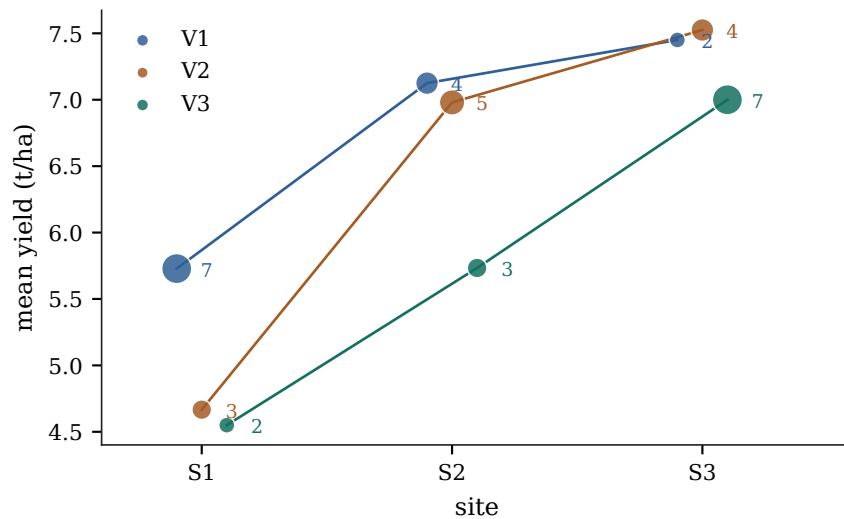


Figure 17.1.1: Cell means of the variety trial; point areas and labels give the cell counts.

|    | S1            | S2      | S3      | all sites |
|----|---------------|---------|---------|-----------|
| V1 | 7 plots, 5.73 | 4, 7.12 | 2, 7.45 | 13, 6.423 |
| V2 | 3, 4.67       | 5, 6.98 | 4, 7.53 | 12, 6.583 |
| V3 | 2, 4.55       | 3, 5.73 | 7, 7.00 | 12, 6.275 |

The last column gives each variety's raw average, the estimate of its weighted mean  $\bar{\mu}_i^w$ . The pooled within-cell variance is  $s^2 = 0.3136$  on  $n - m = 28$  degrees of freedom, so the standard errors of the cell averages range from 0.212 (seven plots) to 0.396 (two plots).

Figure 17.1.1 shows the table. V1 is the best variety at S1 and S2 and nearly ties with V2 at S3, yet its raw average, 6.423, is below that of V2, 6.583, because V1 was grown mostly at the poorest site. The unweighted means 6.768, 6.391 and 5.761 compare the varieties over the same three sites, and V1 comes first. Their standard errors, 0.176, 0.165 and 0.184, exceed those of the raw averages (0.155, 0.162 and 0.162), as Proposition 17.1.2 requires. For V1 the ratio depends only on the counts:  $\sqrt{13(1/7 + 1/4 + 1/2)}/3 = \sqrt{11.607}/3 \approx 1.136$ .

The tests Eq. 17.1.2, computed with Eq. 17.1.1, give  $F = 7.90$  on 2 and 28 degrees of freedom for equal unweighted variety means ( $p = 0.0019$ ),  $F = 44.06$  for equal unweighted site means, and  $F = 1.71$  on 4 and 28 degrees of freedom for additivity ( $p = 0.1755$ ). The difference between V1 and V2, averaged over sites without weights, is estimated as 0.377 with standard error 0.242. With  $t_{28,0.025} = 2.048$ , the 95% interval is from  $-0.12$  to  $0.87$ . The raw averages give a difference of  $-0.160$ , with the opposite sign.

```
import numpy as np
import pandas as pd
from scipy import stats
counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                   [3, 5, 4],
                   [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                       [5.0, 6.5, 7.5],
                       [4.4, 5.9, 7.2]])
```

```
rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
         for i in range(3) for j in range(3) for _ in range(counts[i, j])]
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])
```

```
cells = trial.groupby(["variety", "site"])["y"]
ybar = cells.mean().unstack().to_numpy()           # 3 x 3 table of cell means
n_ij = cells.size().unstack().to_numpy()
n, m = n_ij.sum(), (n_ij > 0).sum()
sse = ((trial["y"] - cells.transform("mean")) ** 2).sum()
s2 = sse / (n - m)                                # pooled within-cell variance
se_cell = np.sqrt(s2 / n_ij)
weighted_rows = (n_ij * ybar).sum(axis=1) / n_ij.sum(axis=1) # = raw variety
                        averages
unweighted_rows = ybar.mean(axis=1)
se_weighted = np.sqrt(s2 / n_ij.sum(axis=1))
se_unweighted = np.sqrt(s2 * (1 / n_ij).sum(axis=1)) / 3
print(np.round(ybar, 3))
print("weighted :", weighted_rows.round(3), " se", se_weighted.round(3))
print("unweighted:", unweighted_rows.round(3), " se", se_unweighted.round(3))
```

```
def hypothesis_test(L, ybar, n_ij, s2, dfe):
    """F test of  $L\mu = 0$  in the cell-means model, all cells occupied (cells in row-
    major order)."""
    u = L @ ybar.ravel()
    ss = u @ np.linalg.solve(L @ np.diag(1 / n_ij.ravel()) @ L.T, u)
    q = np.linalg.matrix_rank(L)
    F = ss / q / s2
    return ss, q, F, stats.f.sf(F, q, dfe)

C3 = np.array([[1.0, 0.0], [0.0, 1.0], [-1.0, -1.0]]) # any 3 x 2 matrix of
                        contrasts will do
avg3 = np.ones((1, 3)) / 3
L_A = np.kron(C3.T, avg3)           # unweighted variety means equal
L_B = np.kron(avg3, C3.T)           # unweighted site means equal
L_AB = np.kron(C3.T, C3.T)          # all interaction contrasts zero
for name, L in [("variety", L_A), ("site", L_B), ("interaction", L_AB)]:
    ss, q, F, p = hypothesis_test(L, ybar, n_ij, s2, n - m)
    print(f"{name:12s} SS = {ss:6.3f} df = {q} F = {F:6.2f} p = {p:.4f}")
```

The script also checks that the variety sum of squares 4.958 is the increase in residual sum of squares when the cell means are fitted subject to  $H_A$  (Theorem 11.3.2(b)), and that a Helmert-type contrast matrix gives the same test. With one factor, counts only decide which contrasts split the between-groups sum of squares orthogonally (Theorem 15.4.1). With several they decide which hypotheses the printed sums of squares address.

## 17.1.5 Exercises

### 17.1.5.1 A. Check your understanding

#### Exercise A1.

A  $2 \times 3$  layout has counts  $n_{11} = 4$ ,  $n_{12} = 0$ ,  $n_{13} = 2$ ,  $n_{21} = 3$ ,  $n_{22} = 5$ ,  $n_{23} = 1$ . Give  $m$ ,

$n$ , the rank of  $\mathbf{W}$  and the error degrees of freedom of the cell-means model. Which of  $\mu_{11} - \mu_{21}$ ,  $\bar{\mu}_{1\cdot} - \bar{\mu}_{2\cdot}$  and  $(\mu_{11} + \mu_{13})/2 - (\mu_{21} + \mu_{23})/2$  are estimable?

### Exercise A2.

In [Example \(Three varieties at three sites\)](#), compute from the counts alone the ratio of the standard errors of the unweighted and weighted means of V3, and check it against the quoted standard errors.

## 17.1.5.2 B. Practice

### Exercise B1.

Show that the  $F$  statistic of [Eq. 17.1.1](#) for a single contrast  $\boldsymbol{\lambda}^\top \boldsymbol{\mu}$  with all cells occupied is the one-way contrast statistic of [Proposition 15.2.1](#), applied to the one-way layout whose groups are the cells. Which  $t$  statistic is it the square of, and what is its distribution under the hypothesis?

*Solution.* With  $q = 1$ ,  $\mathbf{L} = \boldsymbol{\lambda}^\top$  and  $\mathbf{d} = 0$ , [Eq. 17.1.1](#) is  $F = (\sum_{\omega} \lambda_{\omega} \bar{y}_{\omega})^2 / (s^2 \sum_{\omega} \lambda_{\omega}^2 / n_{\omega})$ . This is the contrast sum of squares  $(\sum \lambda_{\omega} \bar{y}_{\omega})^2 / \sum (\lambda_{\omega}^2 / n_{\omega})$  of the one-way layout with groups  $\omega$ , divided by the one-way error mean square, which is the pooled within-cell variance. Its signed root is  $t = \boldsymbol{\lambda}^\top \bar{\mathbf{y}} / (s \sqrt{\sum \lambda_{\omega}^2 / n_{\omega}})$ , which has a  $t(n - m)$  distribution under the hypothesis. The factorial labels of the cells play no role.

### Exercise B2.

Let  $\mathbf{C}_a$  and  $\mathbf{C}'_a$  be two  $a \times (a - 1)$  matrices whose columns are bases of  $\mathbf{1}_a^\perp$ . Show that  $\mathbf{C}'_a = \mathbf{C}_a \mathbf{T}$  for a nonsingular  $\mathbf{T}$ , and deduce that the hypotheses  $H_A$  and  $H_{AB}$  of [Eq. 17.1.2](#), and their  $F$  statistics, are the same for both.

*Solution.* Both column spaces equal  $\mathbf{1}_a^\perp$ , so each column of  $\mathbf{C}'_a$  is a combination of the columns of  $\mathbf{C}_a$ :  $\mathbf{C}'_a = \mathbf{C}_a \mathbf{T}$ . The rank of  $\mathbf{C}'_a$  is  $a - 1$ , so  $\mathbf{T}$  is nonsingular. By [Proposition 1.10.1\(a\)](#),  $(\mathbf{C}'_a)^\top \otimes \mathbf{v}^\top = (\mathbf{T}^\top \otimes \mathbf{1})(\mathbf{C}_a^\top \otimes \mathbf{v}^\top)$  for any row vector  $\mathbf{v}^\top$ , and similarly for  $(\mathbf{C}'_a)^\top \otimes \mathbf{C}_b^\top$ . So the new hypothesis matrix is  $\mathbf{S}\mathbf{L}$  with  $\mathbf{S}$  nonsingular. Then  $\mathbf{S}\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$  iff  $\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$ . In [Eq. 17.1.1](#),

$$(\mathbf{S}\mathbf{L}\bar{\mathbf{y}})^\top (\mathbf{S}\mathbf{L}\mathbf{D}^+ \mathbf{L}^\top \mathbf{S}^\top)^{-1} \mathbf{S}\mathbf{L}\bar{\mathbf{y}} = (\mathbf{L}\bar{\mathbf{y}})^\top (\mathbf{L}\mathbf{D}^+ \mathbf{L}^\top)^{-1} \mathbf{L}\bar{\mathbf{y}}.$$

### Exercise B3.

In [Example \(Three varieties at three sites\)](#), suppose that sites like S1, S2 and S3 make up 50%, 30% and 20% of the region where the varieties would be grown. Write down the estimate of the region-weighted mean yield of V1 and its standard error, and do the same for the region-weighted difference between V1 and V3. Compute them with the listing. Which of the three kinds of marginal mean in this section would you report to a grower in the region?

## 17.1.5.3 C. Going deeper

**Exercise C1.**

A row of a two-way layout will receive  $N$  observations, to be allocated as  $n_{i1}, \dots, n_{ib}$ , with the aim of estimating  $\sum_j w_j \mu_{ij}$  for fixed weights  $w_j > 0$  summing to one. Treating the  $n_{ij}$  as real numbers, show that the variance of  $\sum_j w_j \bar{y}_{ij}$  is minimized by  $n_{ij} = N w_j$ , with minimum  $\sigma^2/N$ . What does this say about how the counts should relate to the target of inference?

*Solution.* Minimize  $\sum_j w_j^2/n_{ij}$  subject to  $\sum_j n_{ij} = N$ . By Cauchy–Schwarz,  $1 = (\sum_j w_j)^2 = (\sum_j (w_j/\sqrt{n_{ij}})\sqrt{n_{ij}})^2 \leq N \sum_j w_j^2/n_{ij}$ , with equality iff  $n_{ij} \propto w_j$ , that is,  $n_{ij} = N w_j$ . So the variance is at least  $\sigma^2/N$ , with equality exactly there. When the counts are proportional to the weights of the target, the target is estimated by the raw row average, the weighted and target means coincide, and no precision is lost. A design should allocate observations in proportion to the target weights, which for an unweighted target means equal replication.

**Exercise C2.**

In an  $a \times b \times d$  layout, where  $C$  has  $d$  levels and all cells are occupied, order the cells lexicographically. Write, as Kronecker products, hypothesis matrices for: (a) equality of the unweighted  $A$  means  $\bar{\mu}_{i..}$ ; (b) absence of the  $AB$  interaction in the table  $\bar{\mu}_{ij.}$  of means averaged over  $C$ ; (c) absence of the three-factor interaction ([Proposition 16.4.2](#)). Give their ranks.

## 17.2 Types of sums of squares and what they test

[Section 9.3](#) defined the Type I, II and III sums of squares by the subspaces they project onto ([Definition 9.3.1](#), [Definition 9.3.2](#)). It also found the cell-mean hypotheses of three of them in the two-way layout: equal weighted row means for a factor entered first, equal unweighted row means for Type III, and a single column of the table for last-in sums of squares under reference coding ([Theorem 9.3.2](#)). This section completes the list with one lemma, and then shows that the hypothesis of a main effect's last-in sum of squares is decided by the coding of the *other* factor.

### 17.2.1 One lemma for every sum of squares

Every term of a factorial model has columns that are constant within cells, so every model matrix we meet has the form  $\mathbf{W}\mathbf{A}$  for a matrix  $\mathbf{A}$  with  $p$  rows, one for each cell. On the cell scale, projections become weighted by the counts: write

$$\mathbf{H}_\mathbf{A} = \mathbf{A}(\mathbf{A}^\top \mathbf{D} \mathbf{A})^{-1} \mathbf{A}^\top \mathbf{D},$$

the operator that fits a table of cell means by the columns of  $\mathbf{A}$  with weights  $n_\omega$ .

**Lemma 17.2.1 — The hypothesis of an extra sum of squares**

Let  $\mathbf{X}_1 = \mathbf{W}\mathbf{A}_1$  and  $\mathbf{X}_2 = \mathbf{W}\mathbf{A}_2$ , let  $\mathbf{P}_1$  project onto  $\mathcal{C}(\mathbf{X}_1)$ , and let  $\mathbf{Q}$  be the projection onto  $\mathcal{C}([\mathbf{X}_1, \mathbf{X}_2])$  minus  $\mathbf{P}_1$ , so that  $\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1) = \mathbf{y}^\top \mathbf{Q}\mathbf{y}$ . For every table of cell means  $\boldsymbol{\mu}$ :

- (a)  $\mathbf{P}_1 \mathbf{W}\boldsymbol{\mu} = \mathbf{W}\mathbf{H}_{\mathbf{A}_1} \boldsymbol{\mu}$ ;
- (b)  $\mathbf{Q}\mathbf{W}\boldsymbol{\mu} = \mathbf{0}$  iff  $\mathbf{A}_2^\top \mathbf{D}(\mathbf{I} - \mathbf{H}_{\mathbf{A}_1})\boldsymbol{\mu} = \mathbf{0}$ ;
- (c) in the cell-means model,  $\text{SS}(\mathbf{X}_2 \mid \mathbf{X}_1)$  is the sum of squares of the hypothesis in (b), and its  $F$  ratio with the cell-means error mean square is the  $F$  statistic Eq. 17.1.1 of that hypothesis.

*Proof.*

- (a)  $\mathbf{X}_1^\top \mathbf{X}_1 = \mathbf{A}_1^\top \mathbf{D}\mathbf{A}_1$ , so by Theorem 6.3.7  $\mathbf{P}_1 \mathbf{W}\boldsymbol{\mu} = \mathbf{W}\mathbf{A}_1(\mathbf{A}_1^\top \mathbf{D}\mathbf{A}_1)^{-1} \mathbf{A}_1^\top \mathbf{W}^\top \mathbf{W}\boldsymbol{\mu} = \mathbf{W}\mathbf{H}_{\mathbf{A}_1} \boldsymbol{\mu}$ .
- (b) By Lemma 9.2.1,  $\mathbf{Q}$  projects onto  $\mathcal{C}(\mathbf{T})$  with  $\mathbf{T} = (\mathbf{I} - \mathbf{P}_1)\mathbf{X}_2$ . A projection annihilates a vector iff the vector is orthogonal to its range, so  $\mathbf{Q}\mathbf{W}\boldsymbol{\mu} = \mathbf{0}$  iff  $\mathbf{T}^\top \mathbf{W}\boldsymbol{\mu} = \mathbf{0}$ . By (a),

$$\mathbf{T}^\top \mathbf{W}\boldsymbol{\mu} = \mathbf{X}_2^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{W}\boldsymbol{\mu} = \mathbf{A}_2^\top \mathbf{W}^\top \mathbf{W}(\mathbf{I} - \mathbf{H}_{\mathbf{A}_1})\boldsymbol{\mu} = \mathbf{A}_2^\top \mathbf{D}(\mathbf{I} - \mathbf{H}_{\mathbf{A}_1})\boldsymbol{\mu}.$$

- (c) Let  $\mathcal{S}_0 = \{\mathbf{W}\boldsymbol{\mu} : \mathbf{Q}\mathbf{W}\boldsymbol{\mu} = \mathbf{0}\}$ , the mean vectors allowed by the hypothesis. Since  $\mathcal{C}(\mathbf{Q}) \subseteq \mathcal{C}(\mathbf{W})$ , the orthogonal complement of  $\mathcal{S}_0$  within  $\mathcal{C}(\mathbf{W})$  is  $\mathcal{C}(\mathbf{Q})$ , and  $\mathbf{y}^\top \mathbf{Q}\mathbf{y}$  is the drop in residual sum of squares from  $\mathcal{S}_0$  to  $\mathcal{C}(\mathbf{W})$  (Eq. 6.5.2). That is the sum of squares of the hypothesis (Theorem 11.3.2(b)), whose  $F$  statistic is Eq. 17.1.1 by Proposition 17.1.1(d). ■

Part (b) says that the extra sum of squares of  $\mathbf{X}_2$  tests whether the residual of the true cell means, after the terms already fitted and weighted by the counts, is orthogonal to the new term. Unless  $\mathbf{D}$  is a multiple of the identity, the counts enter the hypothesis. That is the root of every difference between the types.

**17.2.2 The complete list for two factors**

Let the  $a \times b$  layout have all  $n_{ij} \geq 1$ , with row totals  $n_{i.}$ , column totals  $n_{.j}$  and  $n = \sum_{ij} n_{ij}$ . Besides the weighted row means  $\bar{\mu}_i^w$  of Section 17.1, write

$$\bar{\mu}_{.j}^w = \frac{1}{n_{.j}} \sum_i n_{ij} \mu_{ij}, \quad \bar{\mu}^w = \frac{1}{n} \sum_{i,j} n_{ij} \mu_{ij}$$

for the weighted column means and the weighted grand mean. As in Chapter 9,  $\text{SS}(B \mid \mu, A)$  denotes the extra sum of squares of the columns of  $B$  given the intercept and  $A$ .

**Theorem 17.2.2 — The cell-mean hypotheses of the types of sums of squares**

In the two-way layout with all cells occupied:

- (a)  $SS(A \mid \mu)$ , the Type I sum of squares of  $A$  entered first, addresses  $\bar{\mu}_1^w = \cdots = \bar{\mu}_a^w$ ;
- (b)  $SS(B \mid \mu, A)$ , which is both the Type I sum of squares of  $B$  entered second and the Type II sum of squares of  $B$ , addresses

$$\sum_i n_{ij} (\mu_{ij} - \bar{\mu}_i^w) = 0 \quad \text{for } j = 1, \dots, b; \quad (17.2.1)$$

- (c)  $SS(A \mid \mu, B)$ , the Type II sum of squares of  $A$ , addresses

$$\sum_j n_{ij} (\mu_{ij} - \bar{\mu}_{.j}^w) = 0 \quad \text{for } i = 1, \dots, a; \quad (17.2.2)$$

- (d) the Type III sum of squares of  $A$  addresses  $\bar{\mu}_{1.} = \cdots = \bar{\mu}_{a.}$ ;
- (e)  $SS(AB \mid \mu, A, B)$ , the interaction entry of every type, addresses additivity:  $\mu_{ij} - \mu_{il} - \mu_{kj} + \mu_{kl} = 0$  for all  $i, k, j, l$ .

The hypotheses have  $a - 1$ ,  $b - 1$ ,  $a - 1$ ,  $a - 1$  and  $(a - 1)(b - 1)$  degrees of freedom, and each sum of squares is the sum of squares Eq. 17.1.1 of its hypothesis. The statements for  $B$  follow by exchanging the factors.

*Proof.* Parts (a) and (d) are Theorem 9.3.2(a) and (b). By Lemma 17.2.1(c), each sum of squares is the sum of squares of the hypothesis given by Lemma 17.2.1(b), and it remains to identify that hypothesis and count its degrees of freedom.

- (b) Take  $\mathbf{X}_1 = [\mathbf{1}, \mathbf{Z}_A]$  and  $\mathbf{X}_2 = \mathbf{Z}_B$ . The columns of  $\mathbf{X}_1$  span the vectors that are constant within rows, so  $\mathbf{P}_1$  replaces each entry of a vector by the average of its row. Applied to  $\mathbf{W}\boldsymbol{\mu}$ , this gives  $\bar{\mu}_i^w$  in row  $i$ . The entry  $j$  of  $\mathbf{Z}_B^\top (\mathbf{I} - \mathbf{P}_1) \mathbf{W}\boldsymbol{\mu}$  sums  $\mu_{ij} - \bar{\mu}_i^w$  over the observations in column  $j$ , which is the left side of Eq. 17.2.1. Part (c) is the same argument with the factors exchanged.

For part (e), the projection is  $\mathbf{Q} = \mathbf{M} - \mathbf{P}_{A+B}$ , where  $\mathbf{P}_{A+B}$  projects onto the additive model space, so  $\mathbf{QW}\boldsymbol{\mu} = \mathbf{0}$  iff  $\mathbf{W}\boldsymbol{\mu}$  lies in that space, that is, iff  $\mu_{ij} = \alpha_i + \beta_j$  for some  $\alpha, \beta$ . Additive tables have vanishing tetrads. Conversely, if all tetrads vanish, then  $\mu_{ij} = \mu_{i1} + \mu_{1j} - \mu_{11}$ , which is additive.

Degrees of freedom. A complete layout is connected, so the additive model matrix has rank  $a + b - 1$  (Theorem 8.6.1(a)). The degrees of freedom of the extra sums of squares are therefore  $a - 1$  for (a),  $(a + b - 1) - a = b - 1$  for (b),  $a - 1$  for (c) and  $ab - (a + b - 1)$  for (e). In each case this is  $\text{rank}(\mathbf{Q})$ , the number of independent linear restrictions that the hypothesis places on  $\boldsymbol{\mu}$ . ■

In the Type II hypothesis Eq. 17.2.2, row  $i$  is compared cell by cell with the weighted means  $\bar{\mu}_{.j}^w$  of its columns, with the row's own counts as weights. The  $a$  equations sum to zero (Exercise (B1)). With two rows the hypothesis is  $\sum_j h_j (\mu_{1j} - \mu_{2j}) = 0$  with  $h_j = n_{1j}n_{2j}/n_{.j}$  (Exercise (B2)), which favours columns where *both* rows are well replicated. That is the most precise comparison when the row difference is the same in every column. When it is not, the weights are set by the design rather than by anyone's question.

Under additivity,  $\mu_{ij} = \alpha_i + \beta_j$ , the hypotheses separate cleanly. The Type II and Type III

| Type I, variety first |       |       |       | Type II |       |       |       | Type III |       |       |       |
|-----------------------|-------|-------|-------|---------|-------|-------|-------|----------|-------|-------|-------|
| V1                    | +0.35 | +0.20 | +0.10 | V1      | +0.22 | +0.21 | +0.13 | V1       | +0.22 | +0.22 | +0.22 |
| V2                    | -0.08 | -0.14 | -0.11 | V2      | -0.13 | -0.13 | -0.05 | V2       | -0.11 | -0.11 | -0.11 |
| V3                    | -0.05 | -0.08 | -0.19 | V3      | -0.09 | -0.08 | -0.08 | V3       | -0.11 | -0.11 | -0.11 |
|                       | S1    | S2    | S3    |         | S1    | S2    | S3    |          | S1    | S2    | S3    |

Figure 17.2.1: Coefficients on the cell means of the first equation of each variety hypothesis in [Example \(Three varieties at three sites\)](#) (Type II divided by  $n_1$ ).

hypotheses both reduce to  $\alpha_1 = \dots = \alpha_a$ : Type III because unweighted row means are  $\alpha_i + \bar{\beta}$ , and Type II by [Theorem 9.3.2\(d\)](#). The Type I hypothesis becomes

$$\alpha_i + \sum_j \frac{n_{ij}}{n_{i.}} \beta_j \text{ the same for all } i,$$

which involves the column effects unless the counts are proportional ([Theorem 9.4.3](#)). A factor entered first is credited with the effects of the factors it is confounded with.

In [Figure 17.2.1](#) the Type I coefficients (V1's weighted mean minus the weighted grand mean) do not sum to zero within columns, so that hypothesis compares sites as well as varieties. The Type II and Type III tables have zero column sums and compare varieties within sites. Type II gives site  $j$  the weight  $n_{1j}(n_{.j} - n_{1j})/n_{.j}$ , which is large only where V1 and the other varieties are both well replicated. Type III weights the sites equally.

### 17.2.3 Why Type III needs contrast codings

[Definition 9.3.2](#) defines Type III through sum-to-zero coding, and [Theorem 9.3.2\(c\)](#) showed that reference coding gives a different hypothesis. The next result settles which codings give Type III and why.

#### Proposition 17.2.3 — Which coding a last-in sum of squares depends on

In the two-way layout with all cells occupied, code  $A$  by an  $a \times (a - 1)$  coding matrix  $\mathbf{C}_A$  and  $B$  by a  $b \times (b - 1)$  coding matrix  $\mathbf{C}_B$ , with  $[\mathbf{1}_a, \mathbf{C}_A]$  and  $[\mathbf{1}_b, \mathbf{C}_B]$  nonsingular ([Definition 8.5.1](#)). Let the interaction columns be the products of the coded columns of  $A$  and  $B$ , and let  $\mathbf{w} \neq \mathbf{0}$  span the line  $\mathcal{C}(\mathbf{C}_B)^\perp \subseteq \mathbb{R}^b$ . Then the partial sum of squares of the columns of  $A$ , with the intercept,  $B$  and the interaction in the model, addresses

$$\sum_j w_j \mu_{ij} \text{ is the same for all } i. \quad (17.2.3)$$

The hypothesis depends neither on  $\mathbf{C}_A$  nor on the counts. It is the Type III hypothesis



$\bar{\mu}_{1\cdot} = \cdots = \bar{\mu}_{a\cdot}$  iff every column of  $\mathbf{C}_B$  sums to zero.

*Proof.* Put  $\mathbf{K}_A = [\mathbf{1}_a, \mathbf{C}_A]$  and  $\mathbf{K}_B = [\mathbf{1}_b, \mathbf{C}_B]$ . For an observation in cell  $(i, j)$ , the product of the  $k$ th coded column of  $A$  and the  $l$ th coded column of  $B$  takes the value  $(\mathbf{C}_A)_{ik}(\mathbf{C}_B)_{jl}$ , the entry in row  $(i, j)$  of the corresponding column of  $\mathbf{C}_A \otimes \mathbf{C}_B$ . So the full model matrix is  $\mathbf{W}$  times the columns of  $\mathbf{K}_A \otimes \mathbf{K}_B$ , namely  $\mathbf{1}_a \otimes \mathbf{1}_b$ ,  $\mathbf{1}_a \otimes \mathbf{C}_B$ ,  $\mathbf{C}_A \otimes \mathbf{1}_b$  and  $\mathbf{C}_A \otimes \mathbf{C}_B$ . The matrix  $\mathbf{K}_A \otimes \mathbf{K}_B$  is nonsingular (Proposition 1.10.1(b)), so the full model space is  $\mathcal{C}(\mathbf{W})$  and its  $ab$  columns are linearly independent. Deleting the  $a - 1$  columns  $\mathbf{C}_A \otimes \mathbf{1}_b$  leaves the reduced model  $\mathbf{WF}$  with

$$\mathbf{F} = [\mathbf{1}_a \otimes \mathbf{1}_b, \mathbf{1}_a \otimes \mathbf{C}_B, \mathbf{C}_A \otimes \mathbf{C}_B], \quad \text{rank}(\mathbf{F}) = ab - (a - 1).$$

Let  $\mathcal{V} = \{\boldsymbol{\mu} : (\mathbf{u} \otimes \mathbf{w})^\top \boldsymbol{\mu} = 0 \text{ for all } \mathbf{u} \in \mathbf{1}_a^\perp\}$ . Each column of  $\mathbf{F}$  lies in  $\mathcal{V}$ : by the mixed-product rule,  $(\mathbf{u}^\top \otimes \mathbf{w}^\top)(\mathbf{1}_a \otimes \mathbf{v}) = (\mathbf{u}^\top \mathbf{1}_a)(\mathbf{w}^\top \mathbf{v}) = 0$  for any  $\mathbf{v}$ , and  $(\mathbf{u}^\top \otimes \mathbf{w}^\top)(\mathbf{C}_A \otimes \mathbf{C}_B) = (\mathbf{u}^\top \mathbf{C}_A) \otimes (\mathbf{w}^\top \mathbf{C}_B) = \mathbf{0}$  because  $\mathbf{w} \perp \mathcal{C}(\mathbf{C}_B)$ . The functionals  $\mathbf{u} \otimes \mathbf{w}$ ,  $\mathbf{u} \in \mathbf{1}_a^\perp$ , form a space of dimension  $a - 1$ , since  $\mathbf{u} \mapsto \mathbf{u} \otimes \mathbf{w}$  is injective. So  $\dim \mathcal{V} = ab - (a - 1) = \text{rank}(\mathbf{F})$ , and  $\mathcal{C}(\mathbf{F}) = \mathcal{V}$ . The partial sum of squares is the drop in residual sum of squares from  $\mathcal{C}(\mathbf{WF})$  to  $\mathcal{C}(\mathbf{W})$ , so it addresses the hypothesis  $\boldsymbol{\mu} \in \mathcal{V}$ . Now  $(\mathbf{u} \otimes \mathbf{w})^\top \boldsymbol{\mu} = \sum_i u_i \sum_j w_j \mu_{ij}$  vanishes for every contrast  $\mathbf{u}$  iff  $\sum_j w_j \mu_{ij}$  does not depend on  $i$ , which is Eq. 17.2.3. Finally,  $\mathbf{w} \propto \mathbf{1}_b$  iff  $\mathbf{1}_b \perp \mathcal{C}(\mathbf{C}_B)$ , that is, iff  $\mathbf{1}_b^\top \mathbf{C}_B = \mathbf{0}$ . ■

Sum-to-zero, Helmert and orthogonal polynomial codings of  $B$  all give the Type III sum of squares for  $A$ , whatever coding  $A$  receives. Reference coding of  $B$  at level  $r$  gives  $\mathbf{w} = \mathbf{e}_r$  and the simple effect of Theorem 9.3.2(c). With more factors (Exercise (C2)), the last-in sum of squares of a main effect is the Type III one when every *other* factor is coded by contrasts. In statsmodels, write `C(x, Sum)` for every factor.

## 17.2.4 The variety trial, five ways

**Example. (Five tables for one data set)**

The table has the layout of Example (Party, education and political self-placement) in Chapter 9, and the listing computes its entries the same way, as squared lengths of projections. For the trial of Example (Three varieties at three sites):

| Term        | df | Type I,<br>variety<br>first | Type I,<br>site first | Type II | Type III | reference<br>coding |
|-------------|----|-----------------------------|-----------------------|---------|----------|---------------------|
| variety     | 2  | 0.571                       | 6.184                 | 6.184   | 4.958    | 3.601               |
| site        | 2  | 31.175                      | 25.562                | 31.175  | 27.634   | 7.456               |
| interaction | 4  | 2.146                       | 2.146                 | 2.146   | 2.146    | 2.146               |
| residual    | 28 | 8.781                       | 8.781                 | 8.781   | 8.781    | 8.781               |

Only the Type I columns add up to the model sum of squares 33.892; the Type II and Type III entries add up to 39.506 and 34.738. The verdicts on varieties differ. Entered first,  $F = 0.91$  and  $p = 0.414$ : the raw averages 6.423, 6.583 and 6.275 mix each variety with the sites it was grown at. After sites (Type II),  $F = 9.86$  and  $p = 0.001$ . Type III gives  $F = 7.90$  and  $p = 0.002$ , the test of [Example \(Three varieties at three sites\)](#). The last column compares the varieties at S1 only ( $F = 5.74$ ,  $p = 0.008$ ).

The script checks every entry against statsmodels' `anova_lm` (types 1, 2 and 3, with sum-to-zero and Helmert coding for type 3), and recovers each entry from the matrix **L** of its part of [Theorem 17.2.2](#). It also confirms [Proposition 17.2.3](#): reference coding of the varieties with contrast coding of the sites reproduces 4.958, and a random coding of the sites gives the sum of squares of [Eq. 17.2.3](#).

```
import numpy as np
import pandas as pd
counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                  [3, 5, 4],
                  [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                       [5.0, 6.5, 7.5],
                       [4.4, 5.9, 7.2]])
rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
        for i in range(3) for j in range(3) for _ in range(counts[i, j])]
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])

def proj(*blocks):
    """Orthogonal projection onto the span of the given column blocks (any rank)."""
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]]
    return U @ U.T

y = trial["y"].to_numpy()
n = len(y)
one = np.ones((n, 1))
ZA = pd.get_dummies(trial["variety"]).to_numpy(float)      # variety indicators
ZB = pd.get_dummies(trial["site"]).to_numpy(float)          # site indicators
W = np.column_stack([ZA[:, i] * ZB for i in range(3)])      # the nine cell indicators

def ss(P):
    return y @ P @ y

P0, PA, PB, PAB, M = proj(one), proj(one, ZA), proj(one, ZB), proj(one, ZA, ZB),
proj(W)
typeI_AB = {"A": ss(PA - P0), "B": ss(PAB - PA), "AB": ss(M - PAB)} # variety
first
typeI_BA = {"B": ss(PB - P0), "A": ss(PAB - PB), "AB": ss(M - PAB)} # site first
typeII = {"A": ss(PAB - PB), "B": ss(PAB - PA), "AB": ss(M - PAB)}

def coding(labels, kind):
    """Columns coding a factor: sum-to-zero, or reference level (treatment)."""
    D = pd.get_dummies(labels).to_numpy(float)
    g = D.shape[1]
    C = np.vstack([np.eye(g - 1), -np.ones(g - 1)]) if kind == "sum" else np.eye(g)
   [:, 1:]
```

```

return D @ C

def last_in(CA, CB):
    """Partial (last-in) sums of squares of A and B with interaction columns =
    products."""
    CAB = np.column_stack([CA[:, [i]] * CB for i in range(CA.shape[1])])
    full = proj(one, CA, CB, CAB)
    assert np.allclose(full, M) # every coding spans the cell-means
    space
    return {"A": ss(full - proj(one, CB, CAB)), "B": ss(full - proj(one, CA, CAB))}

typeIII = last_in(coding(trial["variety"], "sum"), coding(trial["site"], "sum"))
reference = last_in(coding(trial["variety"], "treatment"), coding(trial["site"], "
    treatment"))
sse = ss(np.eye(n) - M)
for name, t in [("Type I, variety first", typeI_AB), ("Type I, site first", typeI_BA
    ),
                ("Type II", typeII), ("Type III (sum)", typeIII), ("reference coding
    ", reference)]:
    print(f"{name:22s}", {k: round(float(v), 3) for k, v in t.items()})

```

The next cell builds the Type II hypothesis [Eq. 17.2.2](#) and the Type I hypothesis as matrices on the nine cell means, and [Eq. 17.1.1](#) recovers their entries in the table.

```

cells = trial.groupby(["variety", "site"])["y"]
ybar = cells.mean().to_numpy() # nine cell means, row-major
N = cells.size().to_numpy().reshape(3, 3) # counts n_ij
npi, npj = N.sum(axis=1), N.sum(axis=0)

def hypothesis_ss(L):
    """Sum of squares of L mu = 0 in the cell-means model (redundant rows allowed).
    """
    u = L @ ybar
    return u @ np.linalg.pinv(L @ np.diag(1 / N.ravel()) @ L.T) @ u

# Type II for variety: row i has coefficient n_ij (1[i = k] - n_kj / n_.j) on cell (
    k, j)
L2A = np.array([[N[i, j] * ((i == k) - N[k, j] / npj[j]) for k in range(3) for j in
    range(3)]
                for i in range(3)])
# Type I, variety first: the count-weighted variety means are equal
wrow = np.array([[ (i == k) * N[k, j] / npi[k] for k in range(3) for j in range(3)]
                for i in range(3)])
L1 = wrow[[0]] - wrow[1:]
print(f"Type II hypothesis: SS = {hypothesis_ss(L2A):.3f}; Type I hypothesis: SS =
    {hypothesis_ss(L1):.3f}")

```

## 17.2.5 Choosing a sum of squares

The advice of [Section 9.3](#) stands: decide the hypothesis first, use Type I only when the order of entry is part of the question, and without interaction prefer Type II. The cell-mean hypotheses sharpen it in three ways.

1. *Test the interaction first.* It is the same in every type and tests additivity ([Theorem 17.2.2\(e\)](#)).

Under additivity Type II and Type III test the same hypothesis, and Type II is never less powerful ([Exercise \(C1\)](#)).

2. *If the factors interact*, the weights of a main-effect hypothesis decide the answer. Choose weights that describe a population (equal weights give Type III), or report simple effects within each column with multiplicity-adjusted intervals ([Chapter 13](#)).
3. *Report the hypothesis, not the type*. Anyone can read  $\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$ , and [Eq. 17.1.1](#) tests it. With empty cells this is the only safe course ([Section 17.4](#)).

## 17.2.6 Exercises

### 17.2.6.1 A. Check your understanding

#### Exercise A1.

In a  $2 \times 2$  layout with counts  $n_{11} = 6$ ,  $n_{12} = 2$ ,  $n_{21} = 1$ ,  $n_{22} = 3$ , write the hypothesis of the Type I sum of squares of  $A$  entered first as one linear equation in the four cell means. Show that for the additive table  $\mu_{ij} = \beta_j$ , in which  $A$  has no effect, the hypothesis holds only if  $\beta_1 = \beta_2$ . In particular it fails for  $\beta_1 = 0$ ,  $\beta_2 = 1$ .

### 17.2.6.2 B. Practice

#### Exercise B1.

Show that the left sides of the  $a$  equations [Eq. 17.2.2](#) add up to zero for every table  $\boldsymbol{\mu}$ . Show also that, for a connected design, the equations have rank exactly  $a - 1$ . *Hint:* count degrees of freedom.

*Solution.* Summing over  $i$  gives

$$\sum_{ij} n_{ij} \mu_{ij} - \sum_j n_{.j} \bar{\mu}_{.j}^w = \sum_{ij} n_{ij} \mu_{ij} - \sum_j \sum_i n_{ij} \mu_{ij} = 0.$$

So at most  $a - 1$  of the equations are independent. By [Lemma 17.2.1\(b\)](#) the equations are  $\mathbf{Z}_A^\top (\mathbf{I} - \mathbf{P}_B) \mathbf{W} \boldsymbol{\mu} = \mathbf{0}$ , where  $\mathbf{P}_B$  projects onto  $\mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$ . The number of independent restrictions they impose on  $\boldsymbol{\mu}$  is the rank of the projection onto  $\mathcal{C}((\mathbf{I} - \mathbf{P}_B) \mathbf{Z}_A)$ , which is  $\text{rank}[\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B] - \text{rank}[\mathbf{1}, \mathbf{Z}_B] = (a + b - 1) - b = a - 1$  for a connected design ([Theorem 8.6.1\(a\)](#)).

#### Exercise B2.

Show that when  $a = 2$ , the Type II hypothesis [Eq. 17.2.2](#) for  $A$  is the single equation  $\sum_j h_j (\mu_{1j} - \mu_{2j}) = 0$  with  $h_j = n_{1j} n_{2j} / (n_{1j} + n_{2j})$ . Show that  $\sigma^2 / h_j$  is the variance of  $\bar{y}_{1j} - \bar{y}_{2j}$ , and conclude that under additivity the Type II hypothesis compares the rows with inverse-variance weights.

*Solution.* The first equation is

$$\sum_j n_{1j} (\mu_{1j} - (n_{1j} \mu_{1j} + n_{2j} \mu_{2j}) / n_{.j}) = \sum_j (n_{1j} n_{2j} / n_{.j}) (\mu_{1j} - \mu_{2j}),$$

and the second is its negative (Exercise B1). The variance of  $\bar{y}_{1j} - \bar{y}_{2j}$  is  $\sigma^2(1/n_{1j} + 1/n_{2j}) = \sigma^2 n_{\cdot j} / (n_{1j} n_{2j}) = \sigma^2 / h_j$ . If the difference  $\delta = \mu_{1j} - \mu_{2j}$  is the same in every column, the most precise combination of the column differences weights them in proportion to  $h_j$ , and the hypothesis says that this combination is zero.

### Exercise B3.

Suppose the counts are proportional,  $n_{ij} = n_{i\cdot} n_{\cdot j} / n$ . Show that the hypotheses of Theorem 17.2.2(a) and (c) coincide, and that both say that the row means weighted by the column shares  $n_{\cdot j} / n$  are equal. Why does this agree with Theorem 9.4.3?

*Solution.* With proportional counts,  $n_{ij} / n_{i\cdot} = n_{\cdot j} / n =: \pi_j$  for every  $i$ , so  $\bar{\mu}_i^w = \sum_j \pi_j \mu_{ij}$ . The left side of Eq. 17.2.2 is  $n_{i\cdot} \sum_j \pi_j (\mu_{ij} - \bar{\mu}_{\cdot j}^w) = n_{i\cdot} (\bar{\mu}_i^w - \sum_j \pi_j \bar{\mu}_{\cdot j}^w)$ , and  $\sum_j \pi_j \bar{\mu}_{\cdot j}^w = \bar{\mu}^w$ . So (c) says  $\bar{\mu}_i^w = \bar{\mu}^w$  for all  $i$ , which is (a). In Theorem 9.4.3 the centred factor spaces are orthogonal, so  $\text{SS}(A \mid \mu, B) = \text{SS}(A \mid \mu)$  for every data set. Two sums of squares that agree for every  $\mathbf{y}$  must address the same hypothesis.

### Exercise B4.

Use Proposition 17.2.3 to find the hypothesis of the last-in sum of squares of  $A$  when  $B$  (with  $b = 3$ ) is coded by (a) the Helmert columns  $(-1, 1, 0)^\top, (-1, -1, 2)^\top$ ; (b) the orthogonal polynomial columns  $(-1, 0, 1)^\top, (1, -2, 1)^\top$ ; (c) the columns  $(1, 0, 0)^\top, (0, 0, 1)^\top$ , reference coding with level 2 as reference; (d) the columns  $(1, 0, 0)^\top, (0, 1, 0)^\top$  plus  $\frac{1}{2}(1, 1, 1)^\top$  added to both.

## 17.2.6.3 C. Going deeper

### Exercise C1.

Suppose the cell means are additive. Let  $\mathbf{P}_{\text{II}} = \mathbf{P}_{A+B} - \mathbf{P}_B$  and let  $\mathbf{P}_{\text{III}}$  be the projection of the Type III sum of squares of  $A$ . Both have rank  $a-1$ , and both are used with the cell-means error mean square. Show that the noncentrality parameters satisfy  $\|\mathbf{P}_{\text{II}} \mathbf{W}\boldsymbol{\mu}\|^2 \geq \|\mathbf{P}_{\text{III}} \mathbf{W}\boldsymbol{\mu}\|^2$ , so that the Type II test is at least as powerful. *Hint:* both are squared distances from  $\mathbf{W}\boldsymbol{\mu}$  to a subspace.

*Solution.* Write  $\mathbf{v} = \mathbf{W}\boldsymbol{\mu}$ , which lies in the additive space  $\mathcal{C}(\mathbf{P}_{A+B})$ . Then  $\mathbf{P}_{\text{II}} \mathbf{v} = \mathbf{v} - \mathbf{P}_B \mathbf{v}$ , so  $\|\mathbf{P}_{\text{II}} \mathbf{v}\|^2$  is the squared distance from  $\mathbf{v}$  to  $\mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$ . Also  $\mathbf{P}_{\text{III}} = \mathbf{M} - \mathbf{P}_R$ , where  $\mathbf{P}_R$  projects onto the reduced space  $\mathcal{C}([\mathbf{1}, \mathbf{C}_B\text{-columns, interaction columns}])$ , and  $\mathbf{M}\mathbf{v} = \mathbf{v}$ . So  $\|\mathbf{P}_{\text{III}} \mathbf{v}\|^2$  is the squared distance from  $\mathbf{v}$  to  $\mathcal{C}(\mathbf{P}_R)$ . This space contains  $\mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$ , and the distance to a larger subspace is smaller. The two  $F$  statistics have the same degrees of freedom, and the power of an  $F$  test increases with its noncentrality (Theorem 11.5.1).

### Exercise C2.

In an  $a \times b \times d$  layout, where  $C$  has  $d$  levels and all cells are occupied, code the factors by  $\mathbf{C}_A, \mathbf{C}_B, \mathbf{C}_C$ , and let the model contain all main effects and all two- and three-factor interactions, formed as products of coded columns. Show that the last-in sum of squares

of the columns of  $A$  addresses equality over  $i$  of  $\sum_{j,k} v_j w_k \mu_{ijk}$ , where  $\mathbf{v}$  spans  $\mathcal{C}(\mathbf{C}_B)^\perp$  and  $\mathbf{w}$  spans  $\mathcal{C}(\mathbf{C}_C)^\perp$ . When is this the unweighted hypothesis  $\bar{\mu}_{1..} = \cdots = \bar{\mu}_{a..}$ ?

## 17.3 Constrained models

The cell-means model assumes nothing about the table of means. Smaller models, such as the additive model or a three-way model without the three-factor interaction, are the cell-means model with linear constraints  $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$ . This section fits and tests them with the restricted least squares of Section 11.3, and shows that main-effect hypotheses can merge once the constraints hold.

### 17.3.1 Estimation and tests under constraints

#### Theorem 17.3.1 — Constrained cell-means models

In the cell-means model of Definition 17.1.1, let  $\mathbf{R}$  be a  $g \times p$  matrix of rank  $g$ , and let the columns of the  $p \times (p - g)$  matrix  $\mathbf{K}$  form a basis of  $\mathcal{N}(\mathbf{R})$ . Assume that  $\mathbf{WK}$  has full column rank  $p - g$ . The **constrained model** is  $E(\mathbf{Y}) = \mathbf{W}\boldsymbol{\mu}$  with  $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$ .

- (a) The constrained model is the full-rank linear model  $E(\mathbf{Y}) = \mathbf{WK}\boldsymbol{\theta}$ , with  $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\theta}$ . Every cell mean, including the means of empty cells, is estimable in it. The least squares estimate is

$$\hat{\boldsymbol{\mu}}_{\mathbf{R}} = \mathbf{V}_{\mathbf{R}}\mathbf{D}\bar{\mathbf{y}}, \quad \mathbf{V}_{\mathbf{R}} = \mathbf{K}(\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1}\mathbf{K}^\top, \quad \text{Cov}(\hat{\boldsymbol{\mu}}_{\mathbf{R}}) = \sigma^2 \mathbf{V}_{\mathbf{R}}, \quad (17.3.1)$$

and its residual sum of squares  $\text{SSE}_{\mathbf{R}}$  has  $n - p + g$  degrees of freedom.

- (b) If every cell is occupied, then  $\mathbf{V}_{\mathbf{R}} = \mathbf{D}^{-1} - \mathbf{D}^{-1}\mathbf{R}^\top(\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top)^{-1}\mathbf{R}\mathbf{D}^{-1}$ , so  $\hat{\boldsymbol{\mu}}_{\mathbf{R}} = \bar{\mathbf{y}} - \mathbf{D}^{-1}\mathbf{R}^\top(\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top)^{-1}\mathbf{R}\bar{\mathbf{y}}$ , and  $\text{SSE}_{\mathbf{R}} - \text{SSE} = (\mathbf{R}\bar{\mathbf{y}})^\top(\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top)^{-1}\mathbf{R}\bar{\mathbf{y}}$  is the sum of squares, on  $g$  degrees of freedom, of the hypothesis  $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$  in the unconstrained model.
- (c) Under normality, for an  $\ell \times p$  matrix  $\mathbf{L}$  with  $q = \text{rank}(\mathbf{L}\mathbf{K}) \geq 1$ , the  $F$  statistic for  $H : \mathbf{L}\boldsymbol{\mu} = \mathbf{0}$  within the constrained model is

$$F = \frac{(\mathbf{L}\hat{\boldsymbol{\mu}}_{\mathbf{R}})^\top(\mathbf{L}\mathbf{V}_{\mathbf{R}}\mathbf{L}^\top)^{-1}\mathbf{L}\hat{\boldsymbol{\mu}}_{\mathbf{R}}}{q s_{\mathbf{R}}^2}, \quad s_{\mathbf{R}}^2 = \frac{\text{SSE}_{\mathbf{R}}}{n - p + g}, \quad (17.3.2)$$

for any generalized inverse. It has the  $F(q, n - p + g, \gamma)$  distribution with  $\gamma = (\mathbf{L}\boldsymbol{\mu})^\top(\mathbf{L}\mathbf{V}_{\mathbf{R}}\mathbf{L}^\top)^{-1}\mathbf{L}\boldsymbol{\mu}/\sigma^2$ .

- (d) Within the constrained model,  $\mathbf{L}_1\boldsymbol{\mu} = \mathbf{0}$  and  $\mathbf{L}_2\boldsymbol{\mu} = \mathbf{0}$  are the same hypothesis iff  $\mathbf{L}_1\mathbf{K}$  and  $\mathbf{L}_2\mathbf{K}$  have the same row space, and then their  $F$  statistics coincide.

*Proof.*

- (a) A table satisfies  $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$  iff  $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\theta}$  for a unique  $\boldsymbol{\theta}$ , so the constrained model is the linear model with model matrix  $\mathbf{WK}$ , of full column rank by assumption. Every coordinate of  $\boldsymbol{\theta}$  is then estimable, and so is every linear function of  $\boldsymbol{\mu} = \mathbf{K}\boldsymbol{\theta}$ .

Since  $(\mathbf{W}\mathbf{K})^\top \mathbf{W}\mathbf{K} = \mathbf{K}^\top \mathbf{D}\mathbf{K}$  and  $\mathbf{W}^\top \mathbf{y} = \mathbf{D}\bar{\mathbf{y}}$ , the least squares estimate is  $\hat{\boldsymbol{\theta}} = (\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1} \mathbf{K}^\top \mathbf{D}\bar{\mathbf{y}}$ , with covariance  $\sigma^2(\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1}$  (Theorem 5.5.1). Multiplying by  $\mathbf{K}$  gives Eq. 17.3.1. The model matrix has rank  $p - g$ , which gives the degrees of freedom.

- (b) With every cell occupied,  $\mathbf{W}$  has full column rank,  $\mathbf{D}^{-1}$  is the inverse of  $\mathbf{W}^\top \mathbf{W}$ , and  $\bar{\mathbf{y}}$  is the unrestricted estimate. The constrained model is the reduced model of the testable hypothesis  $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$  in the unconstrained one. So Proposition 11.3.3(a) and (b), with  $\boldsymbol{\Lambda}^\top = \mathbf{R}$  and  $\mathbf{G} = \mathbf{D}^{-1}$ , give the displayed formula for  $\hat{\boldsymbol{\mu}}_{\mathbf{R}}$  and  $\text{SSE}_{\mathbf{R}} = \text{SSE} + (\mathbf{R}\bar{\mathbf{y}})^\top (\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top)^{-1} \mathbf{R}\bar{\mathbf{y}}$ , the sum of squares Eq. 17.1.1 of the hypothesis. For the covariance, let  $\mathbf{V}'$  be the right side of the claimed identity. On vectors  $\mathbf{D}\mathbf{K}\mathbf{a}$  both matrices give  $\mathbf{K}\mathbf{a}$ , because  $\mathbf{R}\mathbf{K} = \mathbf{0}$ . On vectors  $\mathbf{R}^\top \mathbf{b}$  both give  $\mathbf{0}$ , because  $\mathbf{K}^\top \mathbf{R}^\top = \mathbf{0}$  and  $\mathbf{V}'\mathbf{R}^\top \mathbf{b} = \mathbf{D}^{-1}\mathbf{R}^\top \mathbf{b} - \mathbf{D}^{-1}\mathbf{R}^\top \mathbf{b}$ . These vectors span  $\mathbb{R}^p$ : the spaces  $\mathcal{C}(\mathbf{D}\mathbf{K})$  and  $\mathcal{C}(\mathbf{R}^\top)$  have dimensions  $p - g$  and  $g$ , and if  $\mathbf{D}\mathbf{K}\mathbf{a} = \mathbf{R}^\top \mathbf{b}$ , then  $\mathbf{a}^\top \mathbf{K}^\top \mathbf{D}\mathbf{K}\mathbf{a} = (\mathbf{R}\mathbf{K}\mathbf{a})^\top \mathbf{b} = 0$ , so  $\mathbf{K}\mathbf{a} = \mathbf{0}$  because  $\mathbf{D}$  is positive definite. Two matrices that agree on a spanning set are equal.
- (c) In the full-rank model  $\mathbf{W}\mathbf{K}\boldsymbol{\theta}$  the hypothesis is  $\mathbf{L}\mathbf{K}\boldsymbol{\theta} = \mathbf{0}$ , which is testable. Theorem 11.3.2 with  $\boldsymbol{\Lambda}^\top = \mathbf{L}\mathbf{K}$  and  $\mathbf{G} = (\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1}$  gives the statistic, since  $\mathbf{L}\mathbf{K}\hat{\boldsymbol{\theta}} = \mathbf{L}\hat{\boldsymbol{\mu}}_{\mathbf{R}}$  and  $\mathbf{L}\mathbf{K}(\mathbf{K}^\top \mathbf{D}\mathbf{K})^{-1} \mathbf{K}^\top \mathbf{L}^\top = \mathbf{L}\mathbf{V}_{\mathbf{R}}\mathbf{L}^\top$ .
- (d) Within the model,  $\mathbf{L}\boldsymbol{\mu} = \mathbf{0}$  iff  $\boldsymbol{\theta} \in \mathcal{N}(\mathbf{L}\mathbf{K})$ , and two matrices have the same null space iff they have the same row space. The hypotheses then have the same reduced model, and by Theorem 11.3.2(b) the same  $F$  statistic.

■

Part (d) is the new phenomenon: hypotheses that differ in the cell-means model can coincide once the constraints hold. Part (b) pulls the cell averages onto the constraint set along the directions  $\mathbf{D}^{-1}\mathbf{R}^\top$ . The rank assumption on  $\mathbf{W}\mathbf{K}$  holds whenever every cell is occupied; with empty cells it is a real condition (Exercise (C2)).

## 17.3.2 The additive model

Take  $\mathbf{R} = \mathbf{C}_a^\top \otimes \mathbf{C}_b^\top$ , the interaction hypothesis matrix of Eq. 17.1.2, with  $g = (a - 1)(b - 1)$ . Its null space is the space of additive tables  $\mu_{ij} = \alpha_i + \beta_j$  (Section 17.1), so the constrained model is the additive model Eq. 8.6.1 written in cell means.

### Corollary 17.3.2 — Main effects in the additive model

In the two-way layout with all cells occupied, let  $\mathbf{R} = \mathbf{C}_a^\top \otimes \mathbf{C}_b^\top$ . Then:

- (a)  $\hat{\boldsymbol{\mu}}_{\mathbf{R}}$  is the table of fitted cell means of the additive model, and  $\text{SSE}_{\mathbf{R}} - \text{SSE} = \text{SS}(AB \mid \mu, A, B)$ ;
- (b) for any weights  $\mathbf{w} \in \mathbb{R}^b$  with  $\sum_j w_j \neq 0$ , the hypothesis that  $\sum_j w_j \mu_{ij}$  is the same for all  $i$  is, within the additive model, the hypothesis  $\alpha_1 = \cdots = \alpha_a$ . Its sum of squares is  $\text{SS}(A \mid \mu, B)$ , whatever  $\mathbf{w}$ ;
- (c) weights that differ from row to row do not have this property: the hypothesis that  $\sum_j w_{ij} \mu_{ij}$  is the same for all  $i$  depends on the column effects unless  $\sum_j w_{ij} \beta_j$  does not depend on  $i$ .



*Proof.*

- (a) The constrained mean vectors  $\mathbf{W}\boldsymbol{\mu}$  with  $\boldsymbol{\mu}$  additive form  $\mathcal{C}([\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B])$ , so the constrained fit is the additive fit. The difference of residual sums of squares is then  $SS(AB \mid \mu, A, B)$  by definition. (b) For an additive table,  $\sum_j w_j \mu_{ij} = (\sum_j w_j) \alpha_i + \sum_j w_j \beta_j$ , which is the same for all  $i$  iff the  $\alpha_i$  are equal. So all these hypotheses define the same subspace of the constrained model, and by [Theorem 17.3.1\(d\)](#) the same test. Taking  $\mathbf{w} = \mathbf{e}_1$  shows that the subspace is  $\{\mu_{ij} = \beta_j\}$ , the model with  $B$  only, and the drop in residual sum of squares from it to the additive model is  $SS(A \mid \mu, B)$ . (c) Now  $\sum_j w_{ij} \mu_{ij} = (\sum_j w_{ij}) \alpha_i + \sum_j w_{ij} \beta_j$ , and the second term varies with  $i$  in general.

■

So once additivity is accepted, the argument about types disappears for the main effects: any common weighting, and even a single column, gives the comparison  $\alpha_i - \alpha_k$ , tested by the Type II sum of squares (and by Type III, which coincides with it there). Only the count-weighted means of Type I stay contaminated, by part (c). The usual summaries after an additive fit are the **least squares means** (estimated marginal means), the unweighted row means of  $\hat{\boldsymbol{\mu}}_{\mathbf{R}}$ , with standard errors from  $\mathbf{V}_{\mathbf{R}}$ .

#### Example. (The variety trial under additivity)

In [Example \(Five tables for one data set\)](#) the interaction sum of squares 2.146 gave  $F = 1.71$  on 4 and 28 degrees of freedom,  $p = 0.176$ . Suppose we accept additivity. The constrained fit [Eq. 17.3.1](#) gives the table of fitted means

|    | S1   | S2   | S3   |
|----|------|------|------|
| V1 | 5.56 | 7.17 | 7.95 |
| V2 | 5.11 | 6.73 | 7.51 |
| V3 | 4.48 | 6.09 | 6.87 |

with  $SSE_{\mathbf{R}} = 10.927$  on  $37 - 9 + 4 = 32$  degrees of freedom and  $s_{\mathbf{R}}^2 = 0.3415$ . The fitted mean of V1 at S3 is 7.95, although its two plots averaged 7.45: additivity carries V1's advantage at S1 and S2 over to S3. The test of equal unweighted variety means, [Eq. 17.3.2](#), gives 6.184, the Type II sum of squares, with  $F = 9.06$  on 2 and 32 degrees of freedom,  $p = 0.00076$ . The script checks that site weights 0.5, 0.3, 0.2, and the column S1 alone, give the same 6.184 ([Corollary 17.3.2\(b\)](#)), while each variety's own count weights give 0.571, the Type I value. The least squares means are 6.894, 6.449 and 5.811 (standard errors 0.169, 0.170, 0.178), and V1 exceeds V2 by 0.445 (standard error 0.243).

The generating means had a small interaction: V1's advantage over V2 is 0.8 at S1, 0.7 at S2 and 0.1 at S3, and the test on 4 and 28 degrees of freedom has little power to detect it. A non-significant interaction test is not evidence of additivity.

```
import numpy as np
import pandas as pd
from scipy import stats
from scipy.linalg import null_space
```



```

counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                  [3, 5, 4],
                  [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                      [5.0, 6.5, 7.5],
                      [4.4, 5.9, 7.2]])
rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
        for i in range(3) for j in range(3) for _ in range(counts[i, j])]
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])

cells = trial.groupby(["variety", "site"])["y"]
ybar = cells.mean().to_numpy()          # 9 cell means, row-major
n_c = cells.size().to_numpy()
Dinv = np.diag(1 / n_c)
C3 = np.array([[1.0, 0.0], [0.0, 1.0], [-1.0, -1.0]])
G = np.kron(C3.T, C3.T)                # 4 interaction contrasts: G
mu = 0
K = null_space(G)                      # 9 x 5 basis of additive
                                     tables

V_G = K @ np.linalg.inv(K.T @ np.diag(n_c) @ K) @ K.T          # Cov(mu_hat_G) /
sigma^2
mu_G = V_G @ np.diag(n_c) @ ybar          # constrained
estimate
mu_G_direct = ybar - Dinv @ G.T @ np.linalg.solve(G @ Dinv @ G.T, G @ ybar)
print("additive fitted cell means:\n", mu_G.reshape(3, 3).round(3))

n, m, g = n_c.sum(), 9, G.shape[0]
sse = ((trial["y"] - cells.transform("mean")) ** 2).sum()
Gy = G @ ybar
ss_constraint = Gy @ np.linalg.solve(G @ Dinv @ G.T, Gy)        # test of additivity
sse_G = sse + ss_constraint
s2_G = sse_G / (n - m + g)          # residual mean square of the
additive model

def constrained_test(L):
    """F test of L mu = 0 within the constrained model G mu = 0."""
    u = L @ mu_G
    V = L @ V_G @ L.T
    ss = u @ np.linalg.pinv(V) @ u
    q = np.linalg.matrix_rank(L @ K)
    F = ss / q / s2_G
    return ss, q, F, stats.f.sf(F, q, n - m + g)

avg3 = np.ones((1, 3)) / 3
print("variety, unweighted:", np.round(constrained_test(np.kron(C3.T, avg3)), 4))
print("site, unweighted:", np.round(constrained_test(np.kron(avg3, C3.T)), 4))

lsmeans = (mu_G.reshape(3, 3)).mean(axis=1)          # estimated unweighted variety
means
L_rows = np.kron(np.eye(3), avg3)
se_ls = np.sqrt(np.diag(L_rows @ V_G @ L_rows.T) * s2_G)
d = np.array([1.0, -1.0, 0.0]) @ L_rows
diff12, se_diff12 = d @ mu_G, np.sqrt(d @ V_G @ d * s2_G)
print("least squares means:", lsmeans.round(3), " se", se_ls.round(3))
print(f"V1 - V2: {diff12:.3f} (se {se_diff12:.3f})")

```

In the listing the constraint matrix is called **G**. The script checks the fit, the identity of [Theorem 17.3.1\(b\)](#) and the tests against statsmodels.

### 17.3.3 Several factors: choosing among constrained models

With three or more factors the useful constrained models are the hierarchical ones, named by their highest interactions. In an  $a \times b \times d$  layout, where  $C$  has  $d$  levels,  $[AB][C]$  contains the  $AB$  interaction, the main effect of  $C$ , and everything below them. The model  $[AB][AC][BC]$  sets the  $(a-1)(b-1)(d-1)$  contrasts  $\mathbf{C}_a^\top \otimes \mathbf{C}_b^\top \otimes \mathbf{C}_d^\top$  to zero. Unbalanced data make these models non-orthogonal, so we remove one term at a time, judging each step by its drop in residual sum of squares over the error mean square of the full model.

#### Example. (Simplifying a three-factor curing study)

A (synthetic) study of a resin coating measured hardness after curing at three temperatures ( $T$ ), with two hardeners ( $H$ ) and resin from three suppliers ( $S$ ). The 18 cells received one to three panels each, 37 in all, so the full model  $[THS]$  leaves 19 error degrees of freedom, with mean square 0.408. The backward path:

| Step | From           | To             | Term removed | df | SS     | $F$   | $p$    |
|------|----------------|----------------|--------------|----|--------|-------|--------|
| 1    | $[THS]$        | $[TH][TS][HS]$ | $THS$        | 4  | 1.137  | 0.70  | 0.6036 |
| 2    | $[TH][TS][HS]$ | $[TH][HS]$     | $TS$         | 4  | 1.845  | 1.13  | 0.3719 |
| 3    | $[TH][HS]$     | $[TH][S]$      | $HS$         | 2  | 4.132  | 5.06  | 0.0173 |
| 4    | $[TH][HS]$     | $[T][HS]$      | $TH$         | 2  | 13.716 | 16.81 | 0.0001 |
| 5    | $[TH][HS]$     | $[HS]$         | $T$ and $TH$ | 4  | 19.100 | 11.70 | 0.0001 |

Steps 1 and 2 cost little. Step 1 is the test of the constraints  $\mathbf{G}_3\boldsymbol{\mu} = \mathbf{0}$  with  $\mathbf{G}_3 = \mathbf{C}_3^\top \otimes \mathbf{C}_2^\top \otimes \mathbf{C}_3^\top$ , and the script recovers its sum of squares from the cell means through [Theorem 17.3.1\(b\)](#). Steps 3 and 4 show that neither interaction with the hardener can go, so the path stops at  $[TH][HS]$ . In that model the effect of temperature under a given hardener is the same for all three suppliers, so whether temperature matters at all is a single question inside the model, answered by step 5.

```
import statsmodels.formula.api as smf
rng3 = np.random.default_rng(31)
levels = [(i, j, k) for i in range(3) for j in range(2) for k in range(3)]
n3 = rng3.integers(1, 4, size=len(levels)) # 1 to 3 runs per cell

def true_hardness(i, j, k):
    """Temperature i, hardener j, supplier k: hardener interacts with both, nothing else does."""
    return (40 + [0, 1.0, 2.0][i] + 1.5 * j + [0, 0.5, -0.5][k]
            + j * [0, -1.0, -2.0][i] + j * [0, 1.2, -0.8][k])

cure = pd.DataFrame([(f"t{i + 1}", f"h{j + 1}", f"s{k + 1}", true_hardness(i, j, k)
                      + rng3.normal(0, 0.7))
```

```

        for (i, j, k), r in zip(levels, n3) for _ in range(r)],
        columns=["temp", "hardener", "supplier", "y"])

def fit(formula):
    f = smf.ols(formula, cure).fit()
    return f.ssr, int(f.df_resid)

sse_full, df_full = fit("y ~ temp * hardener * supplier")
mse_full = sse_full / df_full
path = [("THS", "[TH] [TS] [HS]", "y ~ temp * hardener * supplier", "y ~ (temp +
    hardener + supplier) ** 2"),
        ("TH [TS] [HS]", "[TH] [HS]", "y ~ (temp + hardener + supplier) ** 2", "y ~
    temp * hardener + hardener * supplier"),
        ("TH [HS]", "[TH] [S]", "y ~ temp * hardener + hardener * supplier", "y ~
    temp * hardener + supplier"),
        ("TH [HS]", "[T] [HS]", "y ~ temp * hardener + hardener * supplier", "y ~
    temp + hardener * supplier"),
        ("TH [HS]", "[HS]", "y ~ temp * hardener + hardener * supplier", "y ~
    hardener * supplier")]
steps = []
for big, small, f_big, f_small in path:
    (s1, d1), (s0, d0) = fit(f_big), fit(f_small)
    F = (s0 - s1) / (d0 - d1) / mse_full
    steps.append((big, small, d0 - d1, s0 - s1, F, stats.f.sf(F, d0 - d1, df_full)))
print(f"{big:13s} -> {small:13s} df = {d0 - d1}  SS = {s0 - s1:6.3f}  F = {F:5.2
f}  p = {steps[-1][5]:.4f}")

```

Such comparisons can look contradictory: a factor may matter alone but not after another. That is the unbalanced analogue of collinearity, and it should be reported as such, not resolved by choosing a type of sum of squares.

## 17.3.4 Exercises

### 17.3.4.1 A. Check your understanding

#### Exercise A1.

In an  $a \times b \times d$  layout with every cell occupied and  $n$  observations, how many constraints define the model  $[AB][AC][BC]$ , and how many error degrees of freedom does it have? Check your answer against step 1 of [Example \(Simplifying a three-factor curing study\)](#).

### 17.3.4.2 B. Practice

#### Exercise B1.

For a  $2 \times 2$  layout with all cells occupied, take  $\mathbf{R} = (1, -1, -1, 1)$ . Use [Theorem 17.3.1\(b\)](#) to show that the additive fit is  $\hat{\mu}_{ij} = \bar{y}_{ij} - (-1)^{i+j} \hat{\delta} / (n_{ij}H)$ , where  $\hat{\delta} = \bar{y}_{11} - \bar{y}_{12} - \bar{y}_{21} + \bar{y}_{22}$  and  $H = \sum_{ij} 1/n_{ij}$ . Which cells move most?

*Solution.* Here  $\mathbf{R}\bar{\mathbf{y}} = \hat{\delta}$  and  $\mathbf{R}\mathbf{D}^{-1}\mathbf{R}^\top = H$ , so  $\hat{\boldsymbol{\mu}}_{\mathbf{R}} = \bar{\mathbf{y}} - \mathbf{D}^{-1}\mathbf{R}^\top \hat{\delta} / H$ . The  $(i, j)$  entry of  $\mathbf{D}^{-1}\mathbf{R}^\top$  is  $(-1)^{i+j} / n_{ij}$ . Each cell average moves by  $\hat{\delta} / (n_{ij}H)$  in the direction that removes the interaction, so the least replicated cells move most. The four moves remove

$\hat{\delta}$  exactly:  $\sum_{ij} (-1)^{i+j} \cdot (-1)^{i+j} \hat{\delta} / (n_{ij} H) = \hat{\delta}$ .

### Exercise B2.

In a  $2 \times 3$  layout with every cell occupied, impose only  $\mu_{11} - \mu_{12} - \mu_{21} + \mu_{22} = 0$ . Describe the constrained model in words. How many degrees of freedom does its residual have, and what does the  $F$  test of the remaining interaction contrast within it test?

### Exercise B3.

In the additive model with all cells occupied, show that the least squares means satisfy  $\hat{\mu}_{i\cdot} - \hat{\mu}_{k\cdot} = \hat{\alpha}_i - \hat{\alpha}_k$  for any solution of the normal equations, and that the same difference is obtained if the fitted table is averaged over  $j$  with any fixed weights summing to one. Why does the choice of weights still matter for the least squares means themselves?

*Solution.* The fitted table is  $\hat{\mu}_{ij} = \hat{m} + \hat{\alpha}_i + \hat{\beta}_j$ , unique as a table (Theorem 6.4.2). Averaging over  $j$  with weights  $w_j$  summing to one gives  $\hat{m} + \hat{\alpha}_i + \sum_j w_j \hat{\beta}_j$ , and the last term does not depend on  $i$ . So the differences are  $\hat{\alpha}_i - \hat{\alpha}_k$ , an estimable function whose value does not depend on the solution chosen. The means themselves contain  $\sum_j w_j \hat{\beta}_j$ , which does depend on the weights. They are levels, not comparisons, and a level is a mean over some population of columns.

#### 17.3.4.3 C. Going deeper

### Exercise C1.

The *method of unweighted means* estimates  $\alpha_1 - \alpha_2$  in a  $2 \times b$  additive layout by  $b^{-1} \sum_j (\bar{y}_{1j} - \bar{y}_{2j})$ , as if every cell had the same count. Show that the least squares estimate is  $\sum_j h_j (\bar{y}_{1j} - \bar{y}_{2j}) / \sum_j h_j$  with  $h_j = n_{1j} n_{2j} / n_{\cdot j}$ . Deduce that both are unbiased under additivity, and that the least squares estimate has the smaller variance unless all  $h_j$  are equal.

*Solution.* Under additivity each  $\bar{y}_{1j} - \bar{y}_{2j}$  is unbiased for  $\delta = \alpha_1 - \alpha_2$ , with variance  $\sigma^2/h_j$  (Exercise (B2)), and the  $b$  differences are independent. Any weighted average with weights summing to one is unbiased. Its variance  $\sigma^2 \sum_j v_j^2 / h_j$  is minimized, by the Cauchy-Schwarz argument of Proposition 17.1.2, at  $v_j \propto h_j$ . The minimum variance is  $\sigma^2 / \sum_j h_j$ . The least squares estimate is the best linear unbiased estimator (Theorem 7.2.1), and it is a linear function of the cell averages, which are sufficient here. So it is the  $h$ -weighted average. The unweighted average has  $v_j = 1/b$ , and it is optimal only if all  $h_j$  are equal.

### Exercise C2.

Allow empty cells. Show that **WK** in Theorem 17.3.1 has full column rank iff no nonzero table satisfying  $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$  vanishes on every occupied cell. Deduce that for the additive constraints this holds iff the design graph of Theorem 8.6.1 is connected.

*Solution.*  $\mathbf{WK}\boldsymbol{\theta} = \mathbf{0}$  iff the table  $\mathbf{K}\boldsymbol{\theta}$  vanishes on every occupied cell, since the rows of  $\mathbf{W}$  pick out the occupied cells. The columns of  $\mathbf{K}$  are independent, so  $\mathbf{WK}$  has a nonzero null vector iff some nonzero table satisfying the constraints vanishes on  $O$ . For additive tables  $\alpha_i + \beta_j$ , vanishing on  $O$  means  $\alpha_i = -\beta_j$  for every occupied  $(i, j)$ . As in the proof of [Theorem 8.6.1](#), this forces  $\alpha_i = t_C$  and  $\beta_j = -t_C$  on each component  $C$  of the design graph. The table is nonzero iff the  $t_C$  are not all equal, which is possible iff there are at least two components.

## 17.4 Empty cells and estimability

Some combinations of levels are never observed, or are impossible. An empty cell's mean does not appear in the distribution of the data, so no unconstrained analysis can estimate it ([Proposition 17.1.1\(b\)](#)). This section works out what survives, building on connectedness ([Theorem 8.6.1](#)) and the estimability of interaction contrasts ([Proposition 8.6.2](#)).

Throughout, the two-way layout has occupied cells  $O$ , with  $m = |O|$ , every row and every column contains at least one occupied cell, and  $c$  is the number of connected components of the design graph of [Theorem 8.6.1](#). A row is **complete** if none of its cells is empty.

### 17.4.1 What remains estimable

#### **Theorem 17.4.1 — Estimability and tests with empty cells**

In the two-way layout described above:

- (a) In the cell-means model,  $\sum_i u_i \sum_j w_j \mu_{ij}$  is estimable iff  $u_i w_j = 0$  for every empty cell  $(i, j)$ . In particular the unweighted mean  $\bar{\mu}_i$  is estimable iff row  $i$  is complete.
- (b) The estimable interaction contrasts, the tables  $(d_{ij})$  with zero row and column sums and  $d_{ij} = 0$  on every empty cell ([Proposition 8.6.2\(c\)](#)), form a space of dimension  $m - a - b + c$ . The interaction sum of squares  $\text{SS}(AB \mid \mu, A, B)$  has  $m - a - b + c$  degrees of freedom. It addresses the hypothesis that all these contrasts vanish, which holds iff the occupied cell means are additive:  $\mu_{ij} = \alpha_i + \beta_j$  for all  $(i, j) \in O$  and some  $\alpha, \beta$ .
- (c) If the design is connected, the additive model estimates every cell mean, including those of empty cells, and under additivity  $\text{SS}(A \mid \mu, B)$  addresses  $\alpha_1 = \cdots = \alpha_a$ , with  $a - 1$  degrees of freedom.
- (d) Code both factors by contrasts and form the interaction columns as products. If at least one row is incomplete and  $k$  rows are complete, the last-in sum of squares of  $A$  addresses equality of the unweighted means of the complete rows, and it has  $\max(k - 1, 0)$  degrees of freedom.

*Proof.*

- (a) The coefficient of  $\mu_{ij}$  is  $u_i w_j$ , and [Proposition 17.1.1\(b\)](#) applies. For  $\bar{\mu}_i$  the coefficients are  $1/b$  on row  $i$  and zero elsewhere.
- (b) Let  $T : \mathbb{R}^O \rightarrow \mathbb{R}^a \times \mathbb{R}^b$  send a table  $(d_{ij})_{(i,j) \in O}$  to its row sums and column sums.

The estimable interaction contrasts form  $\mathcal{N}(T)$ . The transpose sends  $(\mathbf{u}, \mathbf{v})$  to the table  $(u_i + v_j)_{(i,j) \in O}$ . As in the proof of [Theorem 8.6.1](#),  $u_i + v_j = 0$  on  $O$  forces  $u_i = t_C$  and  $v_j = -t_C$  on each component  $C$ , with arbitrary  $t_C$ . So  $\mathcal{N}(T^\top)$  has dimension  $c$ ,  $\text{rank}(T) = a + b - c$ , and  $\dim \mathcal{N}(T) = m - a - b + c$ . The tables  $(u_i + v_j)_{(i,j) \in O}$  are exactly the occupied parts of additive tables. They form  $\mathcal{C}(T^\top)$ , whose orthogonal complement in  $\mathbb{R}^O$  is  $\mathcal{N}(T)$ . So the occupied cell means are additive iff they are orthogonal to every estimable interaction contrast, that is, iff all those contrasts vanish. The interaction sum of squares is  $\mathbf{y}^\top (\mathbf{M} - \mathbf{P}_{A+B}) \mathbf{y}$ , which addresses the hypothesis that  $\mathbf{W}\boldsymbol{\mu}$  lies in the additive model space, that is, that the occupied cell means are additive. Its degrees of freedom are  $m - (a + b - c)$  by [Proposition 17.1.1\(a\)](#) and [Theorem 8.6.1\(a\)](#).

- (c) The first claim is [Theorem 8.6.1\(c\)](#). For the second, let  $\mu_{ij} = \alpha_i + \beta_j$ . Then  $(\mathbf{P}_{A+B} - \mathbf{P}_B)\mathbf{W}\boldsymbol{\mu} = \mathbf{0}$  iff  $\mathbf{W}\boldsymbol{\mu} \in \mathcal{C}([\mathbf{1}, \mathbf{Z}_B])$ , that is, iff  $\alpha_i + \beta_j = \beta'_j$  on  $O$  for some  $\beta'$ . Then  $\alpha_i + (\beta_j - \beta'_j) = 0$  on  $O$ , and in a connected design this forces all  $\alpha_i$  to be equal, as in (b). Conversely, equal  $\alpha_i$  give such a  $\beta'$ . The degrees of freedom are  $\text{rank}[\mathbf{1}, \mathbf{Z}_A, \mathbf{Z}_B] - \text{rank}[\mathbf{1}, \mathbf{Z}_B] = (a + b - 1) - b$ .
- (d) As in the proof of [Proposition 17.2.3](#), the full model matrix is  $\mathbf{W}$  times the  $ab$  independent columns of  $\mathbf{K}_A \otimes \mathbf{K}_B$ , so its column space is  $\mathcal{C}(\mathbf{W})$ . The reduced model is  $\mathbf{W}\mathbf{F}$ . The argument in that proof does not use the counts, and with  $\mathbf{w} = \mathbf{1}$  it shows that  $\mathcal{C}(\mathbf{F})$  is the space  $\mathcal{V}$  of tables with equal unweighted row means. Hence the reduced mean vectors are  $\{\mathbf{W}\boldsymbol{\mu} : \boldsymbol{\mu} \in \mathcal{V}\}$ : the tables of occupied cell means that can be completed, by choosing values in the empty cells, to a table with equal unweighted row means. Suppose the complete rows share a common unweighted mean  $t$ , with  $t$  arbitrary if there are none. Each incomplete row can then be given the mean  $t$  by choosing the value in one of its empty cells, so the occupied means can be completed. Conversely, a completion cannot change the means of complete rows. So the hypothesis is equality of the unweighted means of the complete rows. It consists of the  $k - 1$  restrictions  $\bar{\mu}_i = \bar{\mu}_{i_0}$ , for the complete rows  $i \neq i_0$ , which are independent because they involve disjoint sets of cells, and of none if  $k \leq 1$ . The last-in sum of squares is the drop in residual sum of squares from the reduced model to  $\mathcal{C}(\mathbf{W})$ , so it addresses this hypothesis ([Eq. 6.5.2](#)), and its degrees of freedom are the number of independent restrictions. ■

In part (b), a table with zero row and column sums on the edges of the design graph is a *circulation*. The simplest ones run around cycles with alternating signs. A four-cycle is the familiar tetrad  $\mu_{ij} - \mu_{il} - \mu_{kj} + \mu_{kl}$ ; longer cycles give contrasts on six or more cells that survive when every tetrad through them is broken. The number  $m - a - b + c$  counts the independent cycles ([Exercise \(C1\)](#)).

Part (d) is the result to remember about Type III. With empty cells it ignores every incomplete row, and it may test nothing at all: an empty cell can hold whatever value makes its row's unweighted mean agree, and such means are not estimable, by (a).

## 17.4.2 An example with two combinations never grown

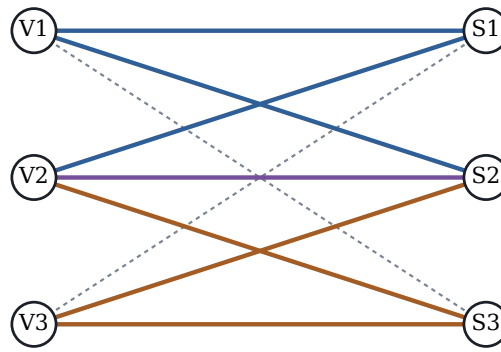


Figure 17.4.1: Design graph of [Example \(The variety trial with two empty cells\)](#) (dashed: empty cells). The blue and orange tetrads share the purple edge V2–S2.

### Example. (The variety trial with two empty cells)

Suppose that in the trial of [Example \(Three varieties at three sites\)](#) V1 had never been grown at S3, nor V3 at S1. That leaves  $n = 33$  plots in  $m = 7$  cells, with 26 error degrees of freedom and error mean square 0.3096. The design graph ([Figure 17.4.1](#)) is connected, so the estimable interaction contrasts have dimension  $7 - 3 - 3 + 1 = 2$ , spanned by two tetrads. The interaction sum of squares is 1.218,  $F = 1.97$ ,  $p = 0.160$ , and the script checks that the tetrads give the same value through [Eq. 17.1.1](#). The six-cell contrast around V1–S1–V2–S3–V3–S2–V1 is the difference of the tetrads ([Exercise \(B3\)](#)).

The unweighted means of V1 and V3 are no longer estimable. V1 and V2 share S1 and S2, and the difference of their averages there is 0.603 (standard error 0.268). The additive model is still identified. Its Type II sum of squares for varieties is 5.766 on 2 degrees of freedom, and it fills the empty cells with 8.32 for V1 at S3 and 4.20 for V3 at S1. The first exceeds every observed cell mean: V1's advantage at S1 and S2 is carried to a site where V1 was never grown, although in the generating means it had nearly vanished there (the true mean is 7.6).

The “Type III” sum of squares for varieties is 0 on 0 degrees of freedom: V2 is the only complete row ([Theorem 17.4.1\(d\)](#)). If only V1 at S3 is removed, it is 2.027 on 1 degree of freedom, the sum of squares of  $\bar{\mu}_{2.} = \bar{\mu}_{3.}$ , for every ordering of the levels.

```
import numpy as np
import pandas as pd
counts = np.array([[7, 4, 2],          # rows: varieties V1-V3; columns: sites S1-S3
                  [3, 5, 4],
                  [2, 3, 7]])
true_means = np.array([[5.8, 7.2, 7.6],
                      [5.0, 6.5, 7.5],
                      [4.4, 5.9, 7.2]])
rng = np.random.default_rng(20172)
rows = [(f"V{i + 1}", f"S{j + 1}", round(true_means[i, j] + rng.normal(0, 0.5), 1))
        for i in range(3) for j in range(3) for _ in range(counts[i, j])]
trial = pd.DataFrame(rows, columns=["variety", "site", "y"])
```



```

def proj(*blocks):
    """Orthogonal projection onto the span of the given column blocks (any rank)."""
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
    U = U[:, s > 1e-10 * s[0]]
    return U @ U.T

def rank(*blocks):
    return np.linalg.matrix_rank(np.column_stack(blocks))

missing = ((trial["variety"] == "V1") & (trial["site"] == "S3")) | \
           ((trial["variety"] == "V3") & (trial["site"] == "S1"))
sub = trial[~missing].reset_index(drop=True)          # V1 never grown at S3, V3
                                                    # never at S1
y = sub["y"].to_numpy()
n = len(y)
one = np.ones((n, 1))
ZA = pd.get_dummies(sub["variety"]).to_numpy(float)
ZB = pd.get_dummies(sub["site"]).to_numpy(float)
W = np.column_stack([ZA[:, [i]] * ZB for i in range(3)])
m = rank(W)                                           # occupied cells
M, PAB, PA, PB = proj(W), proj(one, ZA, ZB), proj(one, ZA), proj(one, ZB)
sse = y @ (np.eye(n) - M) @ y
df_inter = m - rank(one, ZA, ZB)                     # m - a - b + 1 for a
                                                    # connected design
ss_inter = y @ (M - PAB) @ y
F_inter = ss_inter / df_inter / (sse / (n - m))
print(f"occupied cells {m}, interaction df {df_inter}, SS {ss_inter:.3f}, F {F_inter
:.2f}")
ss_typeII_A = y @ (PAB - PB) @ y                    # well defined, a - 1 = 2 df

def sum_coding(labels):
    D = pd.get_dummies(labels).to_numpy(float)
    return D[:, :-1] - D[:, [-1]]

CA, CB = sum_coding(sub["variety"]), sum_coding(sub["site"])
CAB = np.column_stack([CA[:, [i]] * CB for i in range(2)])
df_III_A = m - rank(one, CB, CAB)
ss_III_A = y @ (M - proj(one, CB, CAB)) @ y
print(f"last-in SS for variety with sum coding: {abs(ss_III_A):.3f} on {df_III_A} df
")

```

### 17.4.3 How software behaves

A program fitting the rank-deficient effects model picks one coefficient vector and computes its sums of squares from it. The script fits the data of [Example \(The variety trial with two empty cells\)](#) with statsmodels 0.15 under three namings of the levels. It warns that the design is rank-deficient and prints tables anyway. The Type I interaction appears on 4 degrees of freedom instead of 2, with sums of squares 3.96, 2.15 and 2.05 instead of 1.218. The Type II interaction entries are 23.46, 53.21 and 3.17, and the Type III variety entries 9.43, 41.10 and 34.25 on 2 degrees of freedom, where there is nothing to test. The projections give the same answers under every naming, as they must. SAS's "Type IV" sums of squares for empty cells build hypotheses from the occupied cells in a way that depends on the order of the levels (Goodnight 1980). Whatever the program, check that the printed degrees of freedom equal the rank of the



projection, and write down the hypothesis.

## 17.4.4 Recommended practice

1. *Tabulate the counts first*, draw the design graph and count its components. Rows in different components cannot be compared, even under additivity ([Theorem 8.6.1\(b\)](#)).
2. *Ask why the cells are empty*. Cells lost at random fit the theory above. Impossible combinations have no mean, and a main effect averaged over them is fiction. Combinations avoided *because* they were expected to do badly bias every additive fill-in, and no analysis of the observed cells can detect it.
3. *Test the interaction on its  $m-a-b+c$  degrees of freedom*. If there are few cycles, additivity is barely checked.
4. *If additivity is credible*, use the additive model and  $SS(A \mid \mu, B)$ , and flag least squares means that involve empty cells as extrapolations.
5. *If not*, ask estimable questions of the cell means: simple effects, comparisons over shared columns, or a complete sub-table, and state which cells each comparison uses. Do not report a Type III or Type IV table without identifying its hypotheses.

One can also impose just enough interaction constraints to identify the empty cells ([Section 17.3](#)). The resulting estimates are then set by constraints the data cannot test ([Exercise \(B4\)](#)), and should be presented as assumptions.

## 17.4.5 Exercises

### 17.4.5.1 A. Check your understanding

#### Exercise A1.

A  $3 \times 4$  layout has every cell occupied except  $(1, 4)$ ,  $(2, 1)$  and  $(3, 1)$ . Find  $m$ , the number of components of the design graph, the degrees of freedom of the interaction, and the tetrads that are estimable.

#### Exercise A2.

In [Example \(The variety trial with two empty cells\)](#), which comparisons between V1 and V3 are estimable in the cell-means model? Why is the V1–V2 comparison over S1 and S2 estimable when neither unweighted mean is?

### 17.4.5.2 B. Practice

#### Exercise B1.

In a  $2 \times 2$  layout with cell  $(2, 2)$  empty, show that the additive model estimates  $\mu_{22}$  by  $\bar{y}_{21} + \bar{y}_{12} - \bar{y}_{11}$ , and find its variance. Show that the last-in sum of squares of  $A$  with sum-to-zero coding is identically zero. Compare with [Exercise \(B4\)](#).

*Solution.* With three occupied cells and a connected graph, the additive model has  $1 + 2 + 2 - 1 = 3 = m$  free parameters. Its fitted values are the three cell averages, since  $\text{SS}(AB \mid \mu, A, B)$  has  $3 - 2 - 2 + 1 = 0$  degrees of freedom. Additivity gives  $\mu_{22} = \mu_{21} + \mu_{12} - \mu_{11}$ , estimated by  $\bar{y}_{21} + \bar{y}_{12} - \bar{y}_{11}$  with variance  $\sigma^2(1/n_{21} + 1/n_{12} + 1/n_{11})$ . Row 1 is the only complete row, so [Theorem 17.4.1\(d\)](#) with  $k = 1$  gives zero degrees of freedom, and the sum of squares vanishes for every data set. Directly, the reduced model with columns **1**, the  $B$  column and the product column already has rank 3 on the three occupied cells.

### Exercise B2.

Take the  $4 \times 5$  design of [Example \(Two islands\)](#), with occupied cells  $(1, 1), (1, 2), (2, 2), (2, 3), (3, 4), (4, 4), (4, 5)$ . Find the degrees of freedom of the interaction and the Type II sum of squares of  $A$ . Which comparisons of rows can the additive model make?

### Exercise B3.

In [Example \(The variety trial with two empty cells\)](#), show that the six-cell contrast  $\mu_{11} - \mu_{12} - \mu_{21} + \mu_{23} + \mu_{32} - \mu_{33}$  has zero row and column sums and vanishes on the empty cells. Write it as a combination of the two tetrads, and trace the cycle it corresponds to in [Figure 17.4.1](#).

*Solution.* Row sums:  $1 - 1 = 0$ ,  $-1 + 1 = 0$ ,  $1 - 1 = 0$ . Column sums:  $1 - 1 = 0$  for S1,  $-1 + 1 = 0$  for S2,  $1 - 1 = 0$  for S3. It has no coefficient on  $(1, 3)$  or  $(3, 1)$ . With  $T_1 = \mu_{11} - \mu_{12} - \mu_{21} + \mu_{22}$  and  $T_2 = \mu_{22} - \mu_{23} - \mu_{32} + \mu_{33}$ , the contrast is  $T_1 - T_2$ , because  $\mu_{22}$  cancels. The cycle is V1–S1–V2–S3–V3–S2–V1. It uses every occupied cell except V2–S2, the edge the two tetrads share.

### Exercise B4.

In a  $2 \times 3$  layout with only cell  $(2, 3)$  empty, impose the single constraint  $\mu_{12} - \mu_{13} - \mu_{22} + \mu_{23} = 0$ . Show that every cell mean becomes estimable and find the estimate of  $\mu_{23}$ . Can the constraint be tested?

*Solution.* The constraint gives  $\mu_{23} = \mu_{22} + \mu_{13} - \mu_{12}$ , so any table satisfying it that vanishes on the occupied cells vanishes everywhere, and [Exercise \(C2\)](#) gives identifiability. The five occupied cell means are unrestricted, so their estimates remain the cell averages, and  $\hat{\mu}_{23} = \bar{y}_{22} + \bar{y}_{13} - \bar{y}_{12}$ . The constraint involves the empty cell with a nonzero coefficient, so it is not estimable in the cell-means model and cannot be tested ([Proposition 11.3.1](#)). The fit is the same whether it holds or not. It only decides what we say about  $\mu_{23}$ .

## 17.4.5.3 C. Going deeper

### Exercise C1.

Let  $F$  be a spanning forest of the design graph, with  $a + b - c$  edges. Show that each occupied cell  $e$  not in  $F$  closes a unique cycle in  $F \cup \{e\}$ , and that the alternating-

sign contrast  $d^{(e)}$  of that cycle is an estimable interaction contrast. Show that the  $m - a - b + c$  contrasts  $d^{(e)}$  are linearly independent and hence form a basis of the space in [Theorem 17.4.1\(b\)](#).

*Solution.* Adding an edge  $e = (i, j)$  to a forest joins two vertices of the same tree (the graph and the forest have the same components), so it closes exactly one cycle, the tree path from  $i$  to  $j$  followed by  $e$ . A cycle in a bipartite graph has even length and alternates between row and column vertices. Giving its edges alternating signs  $\pm 1$  makes the two edges at each vertex cancel, so the row and column sums vanish, and the table is supported on occupied cells. The contrast  $d^{(e)}$  is the only one of the family with a nonzero entry at  $e$ , because the other cycles use only forest edges and their own extra edge. So in any vanishing linear combination the coefficient of  $d^{(e)}$  is zero, for each  $e$ . There are  $m - (a + b - c)$  of them, the dimension of the space, so they form a basis.

### Exercise C2.

In the trial with only V1 at S3 removed, what does the “Type III” sum of squares for sites test, and on how many degrees of freedom? Verify your answer by modifying the listing.

## 17.5 Summary and Notes

### 17.5.1 Summary

*Idea.*

1. The cell-means model is the one-way layout on the occupied cells. Any hypothesis  $\mathbf{L}\boldsymbol{\mu} = \mathbf{d}$  on occupied cells is tested by [Eq. 17.1.1](#) ([Definition 17.1.1](#), [Proposition 17.1.1](#)).
2. Count-weighted marginal means are the most precise, but they estimate the mixture the design happened to produce. Choose the target before the weights ([Proposition 17.1.2](#)).
3. An extra sum of squares tests whether the count-weighted residual of the true cell means, after the terms already fitted, is orthogonal to the new term ([Lemma 17.2.1](#)). This gives the hypotheses of all the types ([Theorem 17.2.2](#)).
4. The last-in sum of squares of a main effect depends on the coding of the *other* factors. It is the Type III sum of squares exactly when they are coded by contrasts ([Proposition 17.2.3](#)).
5. Constrained models are cell-means models with  $\mathbf{R}\boldsymbol{\mu} = \mathbf{0}$ , fitted and tested by restricted least squares ([Theorem 17.3.1](#)). Under additivity every common weighting of the row means gives the Type II test ([Corollary 17.3.2](#)).
6. With empty cells, the interaction has  $m - a - b + c$  degrees of freedom, one per independent cycle of the design graph, and the “Type III” sum of squares compares only the complete rows ([Theorem 17.4.1](#)).

## 17.5.2 Notes and sources

**Cell means.** Analysing unbalanced data through cell means goes back to Yates (1934), whose weighted squares of means test hypotheses about unweighted means. Speed, Hocking and Hackney (1978) identified the cell-mean hypotheses of the sums of squares computed by the programs of their day, and Hocking (1985) and Searle (1987) give book-length treatments. Rencher and Schaalje (2008, chapter 15) presents unconstrained and constrained cell-means models and empty cells, and was the main source for this chapter’s coverage. Christensen (2020, sections 7.5–7.6) treats unbalanced layouts as a problem of comparing models, an emphasis reflected in [Section 17.3](#).

**Types of sums of squares.** The Type I, II and III terminology comes from statistical software. Nelder (1977) argued from the marginality of model terms that main effects should not be tested in the presence of their interactions, and Nelder and Lane (1995) restated the case. Langsrud (2003) argued for Type II on grounds of power, which [Exercise \(C1\)](#) makes precise. The hypothesis formulas of [Theorem 17.2.2\(b\)](#) and (c) are classical. Christensen (2020, section 7.5) states them for the two-way layout with interaction, and [Lemma 17.2.1](#) is our route to them. The characterization of the coding dependence in [Proposition 17.2.3](#), stated through the orthogonal complement of the other factor’s coding columns, is this book’s formulation of the familiar advice to use sum-to-zero contrasts.

**Empty cells.** Goodnight (1980) described the construction of the hypotheses behind SAS’s Type IV sums of squares for designs with empty cells, and noted that they are not unique. Milliken and Johnson (2009) discuss the analysis of two-way layouts with missing cells at length, with an emphasis on estimable questions and on identifying the hypotheses that software tests. The reading of estimable interaction contrasts as the cycle space of the design graph ([Theorem 17.4.1\(b\)](#)) uses the standard fact that a graph with  $m$  edges,  $v$  vertices and  $c$  components has  $m - v + c$  independent cycles. The description of the Type III hypothesis with empty cells in [Theorem 17.4.1\(d\)](#) and the software comparison of [Section 17.4](#) are this book’s.

**Data.** The variety trial and the curing study are synthetic, generated by the scripts in `code/ch17/` with fixed seeds, so that the true cell means are known.

## 17.5.3 References

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# Chapter 18

## Analysis of Covariance and Designed Experiments

An experiment is designed before any response is measured. The experimenter decides which units receive which treatment, which nuisance variation to remove by grouping units into blocks, and which extra measurements to take on each unit before treatment begins. Each decision shows up in the linear model: blocks as additional factors, extra measurements as covariates, and the randomization as the reason the error assumptions are believable. This chapter works through the standard designed structures with the projection tools of Parts II and III. Analysis of covariance turns out to be the Frisch–Waugh–Lovell theorem applied to a factor model. Adjusted means are estimable functions with a correction term. The test of parallel slopes compares nested subspaces. Randomized blocks, Latin squares and balanced incomplete blocks are layouts whose balance makes the projections simple. A lost observation is handled exactly by least squares on the observations that remain, and the classical ways of filling in the gap are shortcuts to the same answer.

**What you need.** [Chapter 6](#) (nested projections, [Theorem 6.5.1](#), and the Frisch–Waugh–Lovell theorem, [Theorem 6.6.2](#)), [Chapter 8](#) (estimability, and the parallel and separate lines models of [Section 8.6](#)), [Chapter 9](#) (extra sums of squares and orthogonal designs), [Chapter 11](#) and [Chapter 13](#) ( $F$  tests and simultaneous inference), [Chapter 12](#) ( $t$  intervals, prediction intervals and the Working–Hotelling band), and, from this part, [Chapter 15](#) and [Chapter 16](#) (the one-way and additive two-way layouts) and [Chapter 17](#) (cell means and least squares means in unbalanced layouts).

### 18.0.1 Roadmap

- [Analysis of covariance as a projection problem](#)
- [Adjusted treatment means](#)
- [Heterogeneous slopes](#)
- [Randomized complete blocks](#)
- [Latin squares and incomplete blocks](#)
- [Missing observations in designed experiments](#)
- [Summary and notes](#)

# 18.1 Analysis of covariance as a projection problem

In many experiments something about each unit is known before the treatments are applied: a pretest score, the weight of an animal at the start of a feeding trial, the yield of a plot in the previous season, a patient's blood pressure at enrolment. Such a measurement is a **covariate**. If it predicts the response, much of the variation that the one-way analysis (Chapter 15) calls error is not error at all. Part of it is the covariate at work, and it can be removed.

The **analysis of covariance** (ANCOVA) adds covariates to a factor model. Section 8.6 treated one factor and one covariate (Proposition 8.6.3). This section states the result for any design part, rank deficient or not, and any number of covariates, and then asks what a covariate does in a randomized experiment.

## 18.1.1 The model

Write the mean of the response as the sum of a *design part* and a *covariate part*,

$$E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}_n. \quad (18.1.1)$$

Here  $\mathbf{X}$  is  $n \times p$  with rank  $r$ , built from the indicator columns of the treatment factor and of any blocking factors. It is usually overparameterized. The  $n \times q$  matrix  $\mathbf{Z}$  holds the covariate values, and  $\boldsymbol{\gamma} \in \mathbb{R}^q$  their coefficients. Let  $\mathbf{M}$  be the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$ . Everything below is written with the three matrices

$$\mathbf{E}_{zz} = \mathbf{Z}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Z}, \quad \mathbf{E}_{zy} = \mathbf{Z}^\top (\mathbf{I} - \mathbf{M}) \mathbf{y}, \quad E_{yy} = \mathbf{y}^\top (\mathbf{I} - \mathbf{M}) \mathbf{y}. \quad (18.1.2)$$

The letter  $E$  stands for *error* or *within*: in the one-way layout  $(\mathbf{I} - \mathbf{M})\mathbf{y}$  holds the deviations from the group means (Example (The one-way layout)), so these are within-group sums of squares and products. We assume throughout that

$$\text{rank}[\mathbf{X}, \mathbf{Z}] = \text{rank}(\mathbf{X}) + q, \quad \text{equivalently} \quad (\mathbf{I} - \mathbf{M})\mathbf{Z} \text{ has full column rank } q. \quad (18.1.3)$$

The two forms are equivalent by Lemma 9.2.1, since  $\mathcal{C}([\mathbf{X}, \mathbf{Z}]) = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}((\mathbf{I} - \mathbf{M})\mathbf{Z})$ . The condition fails exactly when some combination of the covariates lies in the design space. With one factor and one covariate, it fails when the covariate is constant within every group. Then the covariate cannot be separated from the group effects (Exercise (B1)).

## 18.1.2 The main theorem

### Theorem 18.1.1 — Analysis of covariance

Assume Eq. 18.1.1 and Eq. 18.1.3. Let  $\mathbf{M}_W$  be the orthogonal projection onto  $\mathcal{C}([\mathbf{X}, \mathbf{Z}])$ .

- (a)  $\mathbf{M}_W = \mathbf{M} + (\mathbf{I} - \mathbf{M})\mathbf{Z}\mathbf{E}_{zz}^{-1}\mathbf{Z}^\top(\mathbf{I} - \mathbf{M})$ , and the two terms are orthogonal projections

with orthogonal ranges.

- (b)  $\boldsymbol{\gamma}$  is estimable, with the unique least squares estimate  $\hat{\boldsymbol{\gamma}} = \mathbf{E}_{zz}^{-1}\mathbf{E}_{zy}$ .  
 (c) A function  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is estimable in Eq. 18.1.1 iff it is estimable in the model  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  without covariates. If  $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$ , its least squares estimate is

$$\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}), \quad (18.1.4)$$

so the fitted design part is  $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}})$ : the covariance-adjusted response  $\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}$  analysed by the formulas of the model without covariates.

- (d) The residual sum of squares is

$$\text{SSE} = E_{yy} - \mathbf{E}_{zy}^\top \mathbf{E}_{zz}^{-1} \mathbf{E}_{zy}, \quad (18.1.5)$$

on  $n - r - q$  degrees of freedom.

- (e)  $\text{Cov}(\hat{\boldsymbol{\gamma}}) = \sigma^2 \mathbf{E}_{zz}^{-1}$ ,  $\text{Cov}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}}) = -\sigma^2 \boldsymbol{\rho}^\top \mathbf{M} \mathbf{Z} \mathbf{E}_{zz}^{-1}$ , and

$$\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 (\boldsymbol{\rho}^\top \mathbf{M} \boldsymbol{\rho} + \boldsymbol{\rho}^\top \mathbf{M} \mathbf{Z} \mathbf{E}_{zz}^{-1} \mathbf{Z}^\top \mathbf{M} \boldsymbol{\rho}). \quad (18.1.6)$$

- (f) Let  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  with rank  $r_0 < r$  and projection  $\mathbf{M}_0$ , and define  $\mathbf{T}_{zz}, \mathbf{T}_{zy}, T_{yy}$  as in Eq. 18.1.2 with  $\mathbf{M}_0$  in place of  $\mathbf{M}$ . Then Eq. 18.1.3 holds for  $\mathbf{X}_0$  as well, and the hypothesis that the design part of the mean lies in  $\mathcal{C}(\mathbf{X}_0)$ , with  $\boldsymbol{\gamma}$  unrestricted, has the **adjusted sum of squares**

$$\text{SS}_H = (T_{yy} - \mathbf{T}_{zy}^\top \mathbf{T}_{zz}^{-1} \mathbf{T}_{zy}) - (E_{yy} - \mathbf{E}_{zy}^\top \mathbf{E}_{zz}^{-1} \mathbf{E}_{zy}) \quad (18.1.7)$$

on  $r - r_0$  degrees of freedom. If  $\mathbf{Y}$  is normal,

$$F = \frac{\text{SS}_H / (r - r_0)}{\text{SSE} / (n - r - q)} \sim F(r - r_0, n - r - q, \gamma),$$

where the noncentrality  $\gamma$ , a scalar not to be confused with the coefficient vector  $\boldsymbol{\gamma}$ , is given by  $\sigma^2 \gamma$  equal to the squared distance from  $\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma}$  to  $\mathcal{C}([\mathbf{X}_0, \mathbf{Z}])$ .

*Proof.*

- (a) By Lemma 9.2.1,  $\mathcal{C}([\mathbf{X}, \mathbf{Z}]) = \mathcal{C}(\mathbf{X}) \oplus \mathcal{C}(\tilde{\mathbf{Z}})$  with  $\tilde{\mathbf{Z}} = (\mathbf{I} - \mathbf{M})\mathbf{Z}$ , and the summands are orthogonal.  $\tilde{\mathbf{Z}}$  has full column rank by Eq. 18.1.3, so its projection is  $\tilde{\mathbf{Z}}(\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}})^{-1} \tilde{\mathbf{Z}}^\top$  (Theorem 6.3.7), and  $\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} = \mathbf{E}_{zz}$  because  $\mathbf{I} - \mathbf{M}$  is symmetric and idempotent. Theorem 6.3.8 adds the two projections.  
 (b) and (c). Put  $\hat{\boldsymbol{\gamma}} = \mathbf{E}_{zz}^{-1} \mathbf{E}_{zy}$ . By (a),

$$\mathbf{M}_W \mathbf{y} = \mathbf{M} \mathbf{y} + (\mathbf{I} - \mathbf{M}) \mathbf{Z} \hat{\boldsymbol{\gamma}} = \mathbf{M}(\mathbf{y} - \mathbf{Z} \hat{\boldsymbol{\gamma}}) + \mathbf{Z} \hat{\boldsymbol{\gamma}}.$$

The first term lies in  $\mathcal{C}(\mathbf{X})$ , so it equals  $\mathbf{X}\mathbf{b}$  for some  $\mathbf{b}$ , and  $(\mathbf{b}, \hat{\boldsymbol{\gamma}})$  is a least squares estimate (Theorem 6.4.1). If  $(\mathbf{b}', \boldsymbol{\gamma}')$  is another one, then  $\mathbf{X}(\mathbf{b} - \mathbf{b}') = \mathbf{Z}(\boldsymbol{\gamma}' - \hat{\boldsymbol{\gamma}})$ . Applying  $\mathbf{I} - \mathbf{M}$  gives  $\tilde{\mathbf{Z}}(\boldsymbol{\gamma}' - \hat{\boldsymbol{\gamma}}) = \mathbf{0}$ , and so  $\boldsymbol{\gamma}' = \hat{\boldsymbol{\gamma}}$ . The coefficient vector of the covariates is therefore the same in every least squares estimate, and so  $\boldsymbol{\gamma}$  is estimable (Theorem 8.3.1(g)). The same argument shows that  $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}})$  for every least squares estimate.

For estimability of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ : in the covariance model this function has coefficient vector  $(\boldsymbol{\lambda}^\top, \mathbf{0}^\top)$ , which is estimable iff  $\mathbf{a}^\top [\mathbf{X}, \mathbf{Z}] = (\boldsymbol{\lambda}^\top, \mathbf{0}^\top)$  for some  $\mathbf{a}$  (Theorem 8.2.2). Such



an  $\mathbf{a}$  gives  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ . Conversely, if  $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$ , put

$$\mathbf{a} = \mathbf{M}\boldsymbol{\rho} - \tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{M}\boldsymbol{\rho}.$$

Then  $\mathbf{a}^\top\mathbf{X} = \boldsymbol{\rho}^\top\mathbf{M}\mathbf{X} - \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{X} = \boldsymbol{\lambda}^\top$ , because  $\tilde{\mathbf{Z}}^\top\mathbf{X} = \mathbf{Z}^\top(\mathbf{I} - \mathbf{M})\mathbf{X} = \mathbf{0}$ , and  $\mathbf{a}^\top\mathbf{Z} = \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z} - \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{Z} = \mathbf{0}^\top$ , because  $\tilde{\mathbf{Z}}^\top\mathbf{Z} = \mathbf{E}_{zz}$ . So  $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$  is estimable. Since  $\mathbf{a}$  lies in  $\mathcal{C}([\mathbf{X}, \mathbf{Z}])$ , its least squares estimate is  $\mathbf{a}^\top\mathbf{M}_W\mathbf{y} = \mathbf{a}^\top\mathbf{y}$  (Proposition 8.3.2(a)):

$$\mathbf{a}^\top\mathbf{y} = \boldsymbol{\rho}^\top\mathbf{M}\mathbf{y} - \boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}\mathbf{E}_{zy} = \boldsymbol{\rho}^\top\mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}).$$

- (d)  $\text{SSE} = \|(\mathbf{I} - \mathbf{M}_W)\mathbf{y}\|^2 = \mathbf{y}^\top(\mathbf{I} - \mathbf{M})\mathbf{y} - \mathbf{y}^\top\tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{y}$  by (a), and  $\tilde{\mathbf{Z}}^\top\mathbf{y} = \mathbf{E}_{zy}$ . The rank of  $\mathbf{M}_W$  is  $r + q$ .
- (e)  $\hat{\boldsymbol{\gamma}} = \mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\mathbf{Y}$  has covariance  $\sigma^2\mathbf{E}_{zz}^{-1}\tilde{\mathbf{Z}}^\top\tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1} = \sigma^2\mathbf{E}_{zz}^{-1}$ . The estimate of  $\boldsymbol{\lambda}^\top\boldsymbol{\beta}$  is  $\mathbf{a}^\top\mathbf{Y}$ , and  $\mathbf{M}\boldsymbol{\rho}$  is orthogonal to  $\tilde{\mathbf{Z}}$ . So  $\text{Var}(\mathbf{a}^\top\mathbf{Y}) = \sigma^2\|\mathbf{a}\|^2$  is the sum of the two squared lengths in Eq. 18.1.6, and  $\text{Cov}(\mathbf{a}^\top\mathbf{Y}, \hat{\boldsymbol{\gamma}}) = \sigma^2\mathbf{a}^\top\tilde{\mathbf{Z}}\mathbf{E}_{zz}^{-1} = -\sigma^2\boldsymbol{\rho}^\top\mathbf{M}\mathbf{Z}\mathbf{E}_{zz}^{-1}$ .
- (f) If  $(\mathbf{I} - \mathbf{M}_0)\mathbf{Z}\mathbf{c} = \mathbf{0}$ , then  $\mathbf{Z}\mathbf{c} \in \mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$ , so  $(\mathbf{I} - \mathbf{M})\mathbf{Z}\mathbf{c} = \mathbf{0}$  and  $\mathbf{c} = \mathbf{0}$ . So Eq. 18.1.3 holds for  $\mathbf{X}_0$ , and (d) applied to  $[\mathbf{X}_0, \mathbf{Z}]$  gives the first bracket in Eq. 18.1.7 as the residual sum of squares of the reduced model. The rank of the reduced model is  $r_0 + q$ , and  $\mathcal{C}([\mathbf{X}_0, \mathbf{Z}]) \subseteq \mathcal{C}([\mathbf{X}, \mathbf{Z}])$ . The distribution is Theorem 11.1.1. ■

The proof is the Frisch–Waugh–Lovell theorem (Theorem 6.6.2) with the covariates as the residualized columns, stated through  $\mathbf{M}$  so that the design part may be rank deficient.

*Idea.* Analysis of covariance is two ordinary analyses joined by one projection. The within-group regression of  $\mathbf{y}$  on the covariates gives  $\hat{\boldsymbol{\gamma}}$ , and the analysis of variance of the adjusted response  $\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}$  gives the treatment estimates. Their variances Eq. 18.1.6 carry an extra term for the uncertainty in  $\hat{\boldsymbol{\gamma}}$ . Tests must compare models, as in Eq. 18.1.7: the between-groups sum of squares of the adjusted response is not the numerator of any  $F$  statistic, because the reduced model must be allowed to re-estimate  $\boldsymbol{\gamma}$ .

### 18.1.3 The one-way layout with one covariate

For a single factor with  $g$  levels and a single covariate  $x$ , the matrices of Eq. 18.1.2 are scalars:

$$E_{xx} = \sum_k \sum_{i \in k} (x_i - \bar{x}_k)^2, \quad E_{xy} = \sum_k \sum_{i \in k} (x_i - \bar{x}_k)(y_i - \bar{y}_k),$$

and  $E_{yy}$  is the within-groups sum of squares of  $y$ . With  $\mathbf{X}_0 = \mathbf{1}$ , the  $T$  quantities are the same sums around the grand means. Theorem 18.1.1 gives:

- (i) slope  $\hat{\beta} = E_{xy}/E_{xx}$ ;
- (ii)  $\text{SSE} = E_{yy} - E_{xy}^2/E_{xx}$  on  $n - g - 1$  degrees of freedom;
- (iii) adjusted treatment sum of squares  $\text{SS}_{\text{trt}} = (T_{yy} - T_{xy}^2/T_{xx}) - (E_{yy} - E_{xy}^2/E_{xx})$  on  $g - 1$  degrees of freedom;
- (iv) level means adjusted to a common covariate value,  $\bar{y}_k - \hat{\beta}(\bar{x}_k - x_0)$ , which differ by  $(\bar{y}_k - \bar{y}_l) - \hat{\beta}(\bar{x}_k - \bar{x}_l)$ , as in Proposition 8.6.3;
- (v) the test of  $\beta = 0$ , with sum of squares  $E_{xy}^2/E_{xx}$  on one degree of freedom (Corollary 9.2.3).

**Example. (Four study formats with a pretest)**

In a synthetic randomized experiment,  $n = 40$  students are assigned at random, ten to each of four study formats for an introductory statistics unit. A pretest score  $x$  is recorded before the assignment, and the response  $y$  is the score on the final examination. The data were generated from a parallel lines model. Randomization balanced the pretest only on average: the four groups have pretest means 62.1, 57.2, 59.9 and 60.5, and final means 72.86, 75.15, 69.09 and 78.09. Figure 18.1.1 shows the data, and the sums of squares and products are:

|                 | $xx$   | $xy$   | $yy$    | df |
|-----------------|--------|--------|---------|----|
| between formats | 124.9  | -31.4  | 432.94  | 3  |
| within formats  | 3643.9 | 3159.2 | 4146.41 | 36 |
| total           | 3768.8 | 3127.8 | 4579.35 | 39 |

Without the covariate, the one-way analysis has error mean square  $4146.41/36 = 115.2$  and treatment  $F = 1.25$  on 3 and 36 degrees of freedom ( $p = 0.305$ ). There is no evidence that the formats differ.

With the covariate, the within-groups slope is  $\hat{\beta} = 3159.2/3643.9 = 0.8670$  points on the final per point on the pretest. The residual sum of squares falls to

$$\text{SSE} = 4146.41 - 2738.94 = 1407.47$$

on 35 degrees of freedom, so the error mean square falls from 115.2 to 40.2. The residual standard deviation falls from 10.73 to 6.34. The reduced model, one line for all forty students, has residual sum of squares  $4579.35 - 2595.83 = 1983.52$ . The adjusted treatment sum of squares is therefore  $1983.52 - 1407.47 = 576.05$ , and

$$F = \frac{576.05/3}{1407.47/35} = 4.77$$

on 3 and 35 degrees of freedom, with  $p = 0.0068$ : clear differences between the formats. The test of  $\beta = 0$  has  $F = 68.1$ . The adjusted treatment sum of squares, 576.05, is even *larger* than the unadjusted 432.94: format 2 had the weakest students on entry and still finished second, and adjustment gives it credit for that.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(84)
g, m = 4, 10                                # four formats, ten students
    each
n = g * m
group = rng.permutation(np.repeat(np.arange(g), m)) # the randomization
x = np.round(rng.normal(60, 10, n))             # pretest, measured before
    assignment
mu_true = np.array([70.0, 74.0, 71.0, 77.0])
y = np.round(mu_true[group] + 0.8 * (x - 60) + rng.normal(0, 6, n), 1)

Z = np.eye(g)[group]                         # indicator matrix of the
    formats
n_k = Z.sum(axis=0)
```

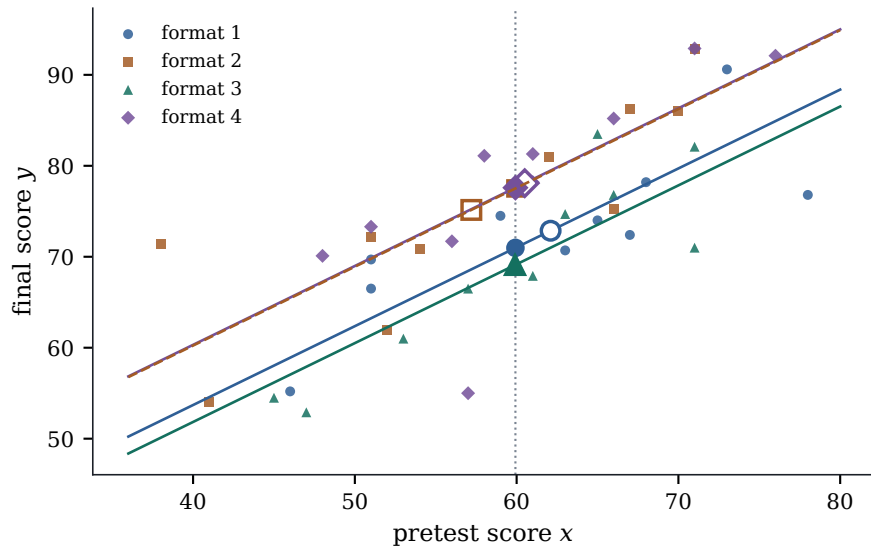


Figure 18.1.1: The tutoring experiment: final score against pretest score for the four formats, with the fitted parallel lines of the covariance model. Open symbols mark the raw group means  $(\bar{x}_k, \bar{y}_k)$ , and filled symbols on the dotted vertical line mark the adjusted means at the overall pretest mean  $\bar{x}$ . The lines of formats 2 (dashed, drawn on top) and 4 almost coincide.

```
x_bar, y_bar = Z.T @ x / n_k, Z.T @ y / n_k           # group means

def within(v):
    """(I - M) v: deviations from the group means (M projects onto C(Z))."""
    return v - (Z.T @ v / n_k)[group]

E_xx, E_xy, E_yy = within(x) @ within(x), within(x) @ within(y), within(y) @ within(
    y)
beta = E_xy / E_xx                                   # slope from within-group
            variation only
sse = E_yy - E_xy ** 2 / E_xx                        # ANCOVA residual sum of
            squares
nu = n - g - 1

T_xx = np.sum((x - x.mean()) ** 2)                   # the same with I - MO, MO =
            mean only
T_xy = np.sum((x - x.mean()) * (y - y.mean()))
T_yy = np.sum((y - y.mean()) ** 2)
sse0 = T_yy - T_xy ** 2 / T_xx                       # reduced model: one line for
            everyone
ss_trt = sse0 - sse                                  # adjusted treatment sum of
            squares
F_adj = (ss_trt / (g - 1)) / (sse / nu)
F_unadj = ((T_yy - E_yy) / (g - 1)) / (E_yy / (n - g))
print(f"slope {beta:.4f}; SSE: ANOVA {E_yy:.2f} on {n - g} df, ANCOVA {sse:.2f} on
      {nu} df")
print(f"treatments: unadjusted F = {F_unadj:.2f} (p = {stats.f.sf(F_unadj, g - 1, n
      - g):.3f}),"
      f" adjusted F = {F_adj:.2f} (p = {stats.f.sf(F_adj, g - 1, nu):.4f})")
```

## 18.1.4 What a covariate does in a randomized experiment

In a randomized experiment the groups differ on the covariate only by chance. Adjustment still helps, for two reasons.

**Precision.** The error variance of the covariance model is the variance of  $y$  given  $x$ , which is smaller whenever the two are correlated. The price is the estimated slope, which adds the second term in Eq. 18.1.6. The next result balances gain against price when the covariate is random, as it is in practice.

### Proposition 18.1.2 — The precision of adjustment

Consider a one-way layout with  $g$  groups of  $m$  units,  $N = gm$ . Suppose the covariate is measured before randomization, so that the  $x_i$  are independent  $N(\xi, \tau^2)$  whatever their group, and that given all the  $x$ 's the  $y_i$  are independent and normal, with  $E(y_i | x_i) = \mu_k + \beta x_i$  for unit  $i$  in group  $k$  and  $\text{Var}(y_i | x_i) = \sigma^2$ . Let  $\rho$  be the within-group correlation of  $x$  and  $y$ . For two groups  $k \neq l$ , the unadjusted estimate  $\bar{y}_k - \bar{y}_l$  and the adjusted estimate  $(\bar{y}_k - \bar{y}_l) - \hat{\beta}(\bar{x}_k - \bar{x}_l)$  are both unbiased for  $\mu_k - \mu_l$ . If  $N - g > 2$ ,

$$\frac{\text{Var}(\text{adjusted})}{\text{Var}(\text{unadjusted})} = (1 - \rho^2) \left( 1 + \frac{1}{N - g - 2} \right). \quad (18.1.8)$$

*Proof.* Write  $\bar{y}_k = \mu_k + \beta \bar{x}_k + \bar{\varepsilon}_k$ , where  $\varepsilon_i = y_i - \mu_k - \beta x_i$  has mean zero and variance  $\sigma^2$  and is independent of all the  $x$ 's. Unconditionally  $\bar{y}_k - \bar{y}_l$  has mean  $\mu_k - \mu_l$ , because  $E \bar{x}_k = E \bar{x}_l$ , and variance  $(2/m)(\beta^2 \tau^2 + \sigma^2)$ . The within-group variance of  $y$  is  $\beta^2 \tau^2 + \sigma^2$ , and  $\sigma^2 = (1 - \rho^2)(\beta^2 \tau^2 + \sigma^2)$ .

Given all the  $x$ 's, the covariance model holds with fixed covariates. By Theorem 18.1.1(e), with  $\mathbf{p}$  the difference of the two scaled group indicators (so  $\mathbf{M}\mathbf{p} = \mathbf{p}$  and  $\mathbf{p}^\top \mathbf{Z} = \bar{x}_k - \bar{x}_l$ ), the adjusted estimate is conditionally unbiased with conditional variance  $\sigma^2 \{2/m + (\bar{x}_k - \bar{x}_l)^2 / E_{xx}\}$ . Its unconditional mean is therefore  $\mu_k - \mu_l$ , and its unconditional variance is the mean of the conditional variance. The group means of  $x$  are independent of the within-group sum of squares  $E_{xx}$  (Corollary 4.4.5, applied group by group), with  $\bar{x}_k - \bar{x}_l \sim N(0, 2\tau^2/m)$  and  $E_{xx}/\tau^2 \sim \chi^2(N - g)$ . For  $V \sim \chi^2(\nu)$  with  $\nu > 2$ ,  $E(1/V) = 1/(\nu - 2)$  (Exercise (B1)). Hence

$$E \frac{(\bar{x}_k - \bar{x}_l)^2}{E_{xx}} = \frac{2\tau^2}{m} \cdot \frac{1}{\tau^2(N - g - 2)}, \quad \text{Var}(\text{adjusted}) = \frac{2\sigma^2}{m} \left( 1 + \frac{1}{N - g - 2} \right).$$

Dividing by the unadjusted variance gives Eq. 18.1.8. ■

The factor  $1 - \rho^2$  is what adjustment would achieve if the slope were known. The factor  $1 + 1/(N - g - 2)$  is the cost of estimating it. For the design of Example (Four study formats with a pretest) the cost is 1.0294. Adjustment pays whenever  $\rho^2 > 1/(N - g - 1)$ , that is, when  $|\rho|$  exceeds 0.169. At  $\rho = 0.8$ , the correlation used to generate the tutoring data, the adjusted comparison has 0.371 of the variance of the unadjusted one (simulation: 0.372; Figure 18.1.2). Without the covariate the experiment would need almost three times as many students.

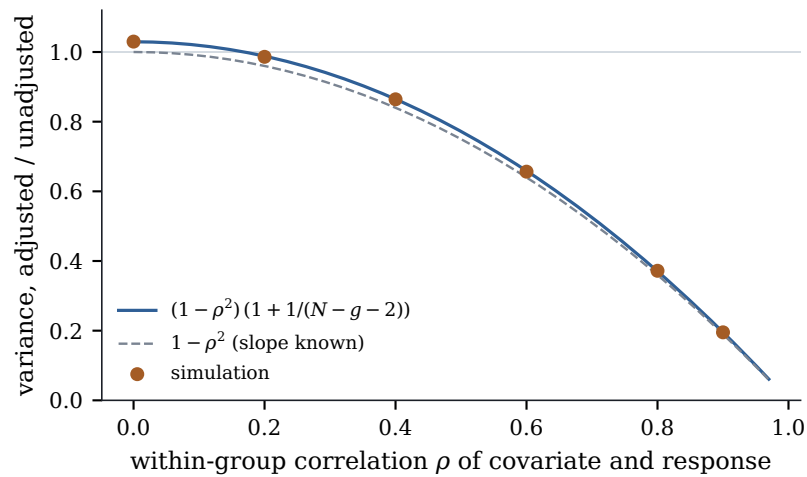


Figure 18.1.2: Variance of the adjusted difference of two group means relative to the unadjusted difference, for four groups of ten, as a function of the within-group correlation between covariate and response. The curve is Eq. 18.1.8, and the points are simulated from 100000 experiments each. The dashed curve is the ratio when the slope is known.

**Chance imbalance.** Randomization makes  $\bar{y}_k - \bar{y}_l$  unbiased *over repetitions of the experiment*. In the experiment actually run, the proof above shows that, given the covariates,

$$E(\bar{y}_k - \bar{y}_l \mid x\text{'s}) = \mu_k - \mu_l + \beta(\bar{x}_k - \bar{x}_l),$$

which is *not*  $\mu_k - \mu_l$  unless the covariate means agree. Once the pretests are known, the raw difference has an error of predictable sign. In [Example \(Four study formats with a pretest\)](#), format 1 started 4.9 points ahead of format 2 on the pretest, which predicts  $0.8670 \times 4.9 \approx 4.25$  points of the final difference. The raw difference between formats 1 and 2 is  $-2.29$ , and the conditionally unbiased adjusted difference is  $-2.29 - 4.25 = -6.54$ .

All this assumes the covariate was chosen in advance. Choosing, after seeing the data, the covariates that make the effect look best is a form of multiple testing, which is why trial protocols name their covariates beforehand.

*Remark* (Random covariates). The theorem ([Theorem 18.1.1](#)) treats  $\mathbf{Z}$  as fixed. If, given random covariates, the model holds with normal errors, every  $F$  test and  $t$  interval of the theorem keeps its level unconditionally ([Theorem 14.1.2](#), applied after reparameterizing the design part to full rank, which changes neither the tests nor the estimable functions). Only the power depends on the covariate values that occur.

## 18.1.5 Adjusting for a covariate that the treatment affects

*Warning.* A covariate must not be affected by the treatment. If it is measured after assignment and the treatment changes it, adjusting for it removes the part of the treatment effect that acts through the covariate, and the adjusted comparison no longer estimates the effect of assigning the treatment.

If hours of study were used as a covariate, a format that works *by* getting students to study more would be compared with the others at equal hours of study, which removes the very thing

it does. In a synthetic randomized experiment with two arms of 100, let the treatment raise a post-assignment variable  $w$  by 2 units and let  $E(y) = \theta T + 0.75w$  with direct effect  $\theta = 1$ . The effect of assigning the treatment is  $1 + 0.75 \times 2 = 2.50$ . Over 20000 replications, the unadjusted difference of means averages 2.500, and the coefficient of the treatment after adjustment for  $w$  averages 1.000, the direct effect, which is not what a decision about the treatment needs.

```
import numpy as np

rng = np.random.default_rng(2025)
n, reps = 200, 20_000
theta, delta, gamma = 1.0, 2.0, 0.75           # direct effect; T -> w; w ->
y
total = theta + gamma * delta                  # effect of assigning
treatment
est_unadj, est_adj = np.empty(reps), np.empty(reps)
for r in range(reps):
    T = rng.permutation(np.repeat([0.0, 1.0], n // 2)) # randomized assignment
    w = rng.normal(size=n) + delta * T                # measured after treatment
    y = theta * T + gamma * w + rng.normal(size=n)
    X = np.column_stack([np.ones(n), T, w])
    est_unadj[r] = y[T == 1].mean() - y[T == 0].mean()
    est_adj[r] = np.linalg.lstsq(X, y, rcond=None)[0][1]
print(f"total effect {total:.2f}: unadjusted mean {est_unadj.mean():.3f}, "
      f" adjusted for w mean {est_adj.mean():.3f}")
```

Chapter 25 develops the language for this in observational data: mediators, on the causal path from treatment to response, must not be adjusted for when the total effect is wanted. In a randomized experiment the safe rule is simple: *adjust only for variables fixed before randomization*. By Proposition 18.1.2 this costs little even when the covariate turns out to be useless. A test of whether the groups are “balanced” on such a covariate is pointless, since its null hypothesis is true by construction. What matters is whether the covariate predicts the response.

## 18.1.6 Exercises

### 18.1.6.1 A. Check your understanding

#### Exercise A1.

In the one-way layout with one covariate, suppose  $x$  is constant within each group. Show that Eq. 18.1.3 fails, and describe the set of least squares estimates of  $\beta$ . Is the adjusted difference of two group means estimable?

*Solution.* If  $x_i = c_k$  for every unit in group  $k$ , then  $\mathbf{x}$  is a combination of the group indicators, so  $\mathbf{x} \in \mathcal{C}(\mathbf{X})$ ,  $(\mathbf{I} - \mathbf{M})\mathbf{x} = \mathbf{0}$ , and Eq. 18.1.3 fails. The mean of group  $k$  is  $\theta_k = \mu_k + \beta c_k$ , and only the  $\theta_k$  are estimable. Every value of  $\beta$  belongs to some least squares estimate: for any  $b$ , the group parameters  $\bar{y}_k - bc_k$  together with  $b$  reproduce the fitted group means. The adjusted difference  $\mu_k - \mu_l = (\theta_k - \theta_l) - \beta(c_k - c_l)$  is therefore estimable iff  $c_k = c_l$ . The data cannot separate a covariate that varies only between groups from the group effects.

**Exercise A2.**

From the numbers in [Example \(Four study formats with a pretest\)](#), compute the  $F$  statistic for  $\beta = 0$  and check that its square root is the  $t$  statistic  $\hat{\beta}/(s/\sqrt{E_{xx}})$ .

**18.1.6.2 B. Practice****Exercise B1.**

Let  $V \sim \chi^2(\nu)$  with  $\nu > 2$ . Show that  $E(1/V) = 1/(\nu - 2)$ , and that  $E(1/V) = \infty$  for  $\nu \leq 2$ .

*Solution.* The density of  $V$  is  $f_\nu(v) = v^{\nu/2-1}e^{-v/2}/\{2^{\nu/2}\Gamma(\nu/2)\}$ . Then

$$E(1/V) = \int_0^\infty \frac{v^{\nu/2-2}e^{-v/2}}{2^{\nu/2}\Gamma(\nu/2)} dv = \frac{2^{\nu/2-1}\Gamma(\nu/2-1)}{2^{\nu/2}\Gamma(\nu/2)} = \frac{1}{2(\nu/2-1)} = \frac{1}{\nu-2},$$

using  $\Gamma(s+1) = s\Gamma(s)$ . The integral converges at 0 iff  $\nu/2 - 2 > -1$ , that is, iff  $\nu > 2$ .

**Exercise B2.**

In the one-way layout with one covariate, write  $B_{xx} = T_{xx} - E_{xx}$ , and similarly  $B_{xy}$  and  $B_{yy}$ , for the between-groups sums of squares and products. Show that the adjusted treatment sum of squares equals

$$B_{yy} - \left( \frac{T_{xy}^2}{T_{xx}} - \frac{E_{xy}^2}{E_{xx}} \right),$$

and deduce that it exceeds  $B_{yy}$  iff  $E_{xy}^2/E_{xx} > T_{xy}^2/T_{xx}$ . Show that this happens in [Example \(Four study formats with a pretest\)](#), and explain it in terms of  $B_{xy}$ .

*Solution.* The formula is [Eq. 18.1.7](#) with  $T_{yy} - E_{yy} = B_{yy}$ . In the example  $E_{xy}^2/E_{xx} = 2738.94$  and  $T_{xy}^2/T_{xx} = 2595.83$ , so the adjusted sum of squares is larger, by 143.11. The total regression uses both the within-group and the between-group variation of  $x$ . Here the between-group cross-product  $B_{xy} = -31.4$  is negative: across the four formats, higher pretest means go with *lower* final means, the opposite of the within-group relationship. The single line for all students therefore explains less than the within-group lines do, and the difference is attributed to the formats.

**Exercise B3.**

In the one-way layout with  $q$  covariates, use [Eq. 18.1.6](#) to show that the adjusted difference  $(\bar{y}_k - \bar{y}_l) - (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)^\top \hat{\boldsymbol{\gamma}}$  has variance  $\sigma^2\{1/n_k + 1/n_l + (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)\}$ .

**18.1.6.3 C. Going deeper****Exercise C1.**

In the one-way covariance model, show that replacing  $x$  by  $a + cx$  with  $c \neq 0$  leaves the

fitted values, SSE, the adjusted treatment sum of squares and the adjusted differences unchanged. Then replace  $x$  by  $x + d_k$  on group  $k$ , with group-specific constants  $d_k$ . Show that  $\hat{\beta}$  and SSE are unchanged but the adjusted differences and the adjusted treatment sum of squares change. Explain why a covariate recorded with a group-specific calibration error is dangerous.

## 18.2 Adjusted treatment means

A covariance analysis is usually reported as a table of treatment means “adjusted” for the covariate. The raw mean  $\bar{y}_k$  estimates the mean response of treatment  $k$  at whatever covariate values its units happened to have; the adjusted mean estimates it at a covariate value common to all treatments. This section defines adjusted means, gives their standard errors, and shows how to compare them.

### 18.2.1 Definition

Take the one-way covariance model with  $g$  groups of sizes  $n_1, \dots, n_g$  and  $q$  covariates,

$$E(y_i) = \mu_k + \mathbf{z}_{(i)}^\top \boldsymbol{\gamma} \quad \text{for unit } i \text{ in group } k, \quad (18.2.1)$$

where  $\mathbf{z}_{(i)} \in \mathbb{R}^q$  is the covariate vector of unit  $i$ , and write  $\bar{\mathbf{z}}_k$  for the mean covariate vector of group  $k$  and  $\bar{\mathbf{z}}$  for the overall mean. We assume [Eq. 18.1.3](#), and write  $\mathbf{E}_{zz}$  and  $\hat{\boldsymbol{\gamma}}$  as in [Theorem 18.1.1](#).

#### Definition 18.2.1 — Adjusted treatment means

For a fixed covariate vector  $\mathbf{z}_0 \in \mathbb{R}^q$ , the **adjusted mean** of treatment  $k$  at  $\mathbf{z}_0$  is  $\mu_k + \mathbf{z}_0^\top \boldsymbol{\gamma}$ , the mean response of a unit with covariate  $\mathbf{z}_0$  under treatment  $k$ . Its estimate is

$$\hat{\mu}_k(\mathbf{z}_0) = \bar{y}_k - (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \hat{\boldsymbol{\gamma}}. \quad (18.2.2)$$

When no value is named,  $\mathbf{z}_0 = \bar{\mathbf{z}}$ , the overall mean of the covariates, and  $\hat{\mu}_k(\bar{\mathbf{z}})$  is called simply the adjusted mean of treatment  $k$ .

The fitted line (or plane) of treatment  $k$  passes through  $(\bar{\mathbf{z}}_k, \bar{y}_k)$  with slope  $\hat{\boldsymbol{\gamma}}$ ; the adjusted mean is its height at  $\mathbf{z}_0$ . In [Figure 18.1.1](#) the filled symbols on the dashed line are the adjusted means.

The adjusted mean is estimable:  $\mu_k + \mathbf{z}_0^\top \boldsymbol{\gamma}$  is the sum of  $\mu_k + \bar{\mathbf{z}}_k^\top \boldsymbol{\gamma}$ , the mean of the fitted values of group  $k$ , and  $(\mathbf{z}_0 - \bar{\mathbf{z}}_k)^\top \boldsymbol{\gamma}$ , a multiple of the estimable  $\boldsymbol{\gamma}$  ([Theorem 18.1.1\(b\)](#)). By [Theorem 18.1.1\(c\)](#) with  $\boldsymbol{\rho}$  the scaled indicator of group  $k$ , the least squares estimate of  $\mu_k$  is  $\bar{y}_k - \bar{\mathbf{z}}_k^\top \hat{\boldsymbol{\gamma}}$ , which gives [Eq. 18.2.2](#).

### 18.2.2 Standard errors



### Proposition 18.2.1 — Moments of adjusted means

In the model Eq. 18.2.1 with  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , and for a fixed  $\mathbf{z}_0$ :

- (a)  $\bar{y}_1, \dots, \bar{y}_g$  are uncorrelated with  $\hat{\gamma}$ , and

$$\text{Var}(\hat{\mu}_k(\mathbf{z}_0)) = \sigma^2 \left( \frac{1}{n_k} + (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}}_k - \mathbf{z}_0) \right); \quad (18.2.3)$$

- (b) for  $k \neq l$ ,  $\text{Cov}(\hat{\mu}_k(\mathbf{z}_0), \hat{\mu}_l(\mathbf{z}_0)) = \sigma^2 (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}}_l - \mathbf{z}_0)$ ;

- (c) for a contrast  $\psi = \sum_k c_k \mu_k$ , the estimate  $\hat{\psi} = \sum_k c_k \hat{\mu}_k(\mathbf{z}_0) = \sum_k c_k \bar{y}_k - \bar{\mathbf{z}}_c^\top \hat{\gamma}$ , with  $\bar{\mathbf{z}}_c = \sum_k c_k \bar{\mathbf{z}}_k$ , does not depend on  $\mathbf{z}_0$ , and

$$\text{Var}(\hat{\psi}) = \sigma^2 \left( \sum_k \frac{c_k^2}{n_k} + \bar{\mathbf{z}}_c^\top \mathbf{E}_{zz}^{-1} \bar{\mathbf{z}}_c \right); \quad (18.2.4)$$

- (d) with  $\mathbf{z}_0 = \bar{\mathbf{z}}$ , the weighted average of the adjusted means is the grand mean:  $\sum_k n_k \hat{\mu}_k(\bar{\mathbf{z}}) = \sum_i y_i$ .

*Proof.* Let  $\mathbf{M}$  project onto the span of the group indicators. The group mean  $\bar{y}_k$  is  $\mathbf{u}_k^\top \mathbf{Y}$  with  $\mathbf{u}_k$  the indicator of group  $k$  divided by  $n_k$ , a vector in  $\mathcal{C}(\mathbf{M})$ . The estimate  $\hat{\gamma} = \mathbf{E}_{zz}^{-1} \mathbf{Z}^\top (\mathbf{I} - \mathbf{M}) \mathbf{Y}$  is a linear function of  $(\mathbf{I} - \mathbf{M}) \mathbf{Y}$ . Hence  $\text{Cov}(\bar{y}_k, \hat{\gamma}) = \sigma^2 \mathbf{u}_k^\top (\mathbf{I} - \mathbf{M}) \mathbf{Z} \mathbf{E}_{zz}^{-1} = \mathbf{0}$ , since  $(\mathbf{I} - \mathbf{M}) \mathbf{u}_k = \mathbf{0}$ . Then (a) and (b) follow from  $\text{Var}(\bar{y}_k) = \sigma^2/n_k$ ,  $\text{Cov}(\bar{y}_k, \bar{y}_l) = 0$  and  $\text{Cov}(\hat{\gamma}) = \sigma^2 \mathbf{E}_{zz}^{-1}$  (Theorem 18.1.1(e)). For (c),  $\sum_k c_k = 0$  removes  $\mathbf{z}_0$ , and the variance follows in the same way. For (d),  $\sum_k n_k \hat{\mu}_k(\bar{\mathbf{z}}) = \sum_k n_k \bar{y}_k - (\sum_k n_k \bar{\mathbf{z}}_k - n \bar{\mathbf{z}})^\top \hat{\gamma}$ , and the bracket is zero. ■

Adjusted means are correlated, because their errors share  $\hat{\gamma}$ . Their variances grow with the distance of  $\bar{\mathbf{z}}_k$  from  $\mathbf{z}_0$ : an adjusted mean is a prediction from the treatment's line, and for a group whose covariate values barely overlap the others' it rests on extrapolation, as in Section 12.3. Differences do not depend on  $\mathbf{z}_0$ ; only the common level and the individual standard errors do. The overall mean  $\bar{\mathbf{z}}$  keeps the adjusted means on the scale of the data (d) and minimizes  $\sum_k n_k \text{Var}(\hat{\mu}_k(\mathbf{z}_0))$  (Exercise (B1)), but any stated value, such as a clinically meaningful baseline, is legitimate.

## 18.2.3 Comparing adjusted means

Pairwise differences are contrasts, so Eq. 18.2.4 gives

$$\text{se}(\hat{\mu}_k - \hat{\mu}_l) = s \sqrt{\frac{1}{n_k} + \frac{1}{n_l} + (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)^\top \mathbf{E}_{zz}^{-1} (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}_l)}, \quad s^2 = \frac{\text{SSE}}{n - g - q}, \quad (18.2.5)$$

and a single comparison uses the  $t$  interval of Theorem 12.1.1 with  $n - g - q$  degrees of freedom. For simultaneous comparisons the correlations matter.

- (i) *Scheffé*. The adjusted treatment  $F$  test of Theorem 18.1.1(f) is a test of  $\mu_1 = \dots = \mu_g$ , a  $(g - 1)$ -dimensional space of estimable contrasts. By Theorem 13.2.2, the intervals

- $\hat{\psi} \pm \sqrt{(g-1)F_{\alpha}(g-1, n-g-q)} \text{se}(\hat{\psi})$  have simultaneous coverage exactly  $1 - \alpha$  over *all* contrasts  $\psi$ . Some interval excludes zero iff the adjusted  $F$  test rejects ([Corollary 13.2.3](#)).
- (ii) *Bonferroni*. For a planned list of  $m$  comparisons,  $t$  intervals at level  $\alpha/m$  have coverage at least  $1 - \alpha$  whatever the correlations ([Theorem 13.4.1](#)).
  - (iii) *Tukey-type intervals*. The half-width  $(q_{\alpha}(g, n-g-q)/\sqrt{2}) \text{se}(\hat{\mu}_k - \hat{\mu}_l)$  copies the Tukey–Kramer rule [Eq. 13.3.3](#). It is exact when the covariate means are equal and the groups balanced ([Exercise \(C1\)](#)). Otherwise its coverage is only approximately  $1 - \alpha$ : Hayter’s theorem covers unequal group sizes in the one-way layout, but not the correlations that the covariate introduces. When the covariate means are close, as they usually are after randomization, the departure is small ([Exercise \(C2\)](#) asks you to measure it). When they are far apart, use Scheffé’s intervals, or compute the exact multiplier for the given correlation matrix by simulation, as [Section 13.3](#) did for Tukey–Kramer.

### Example. (Adjusted means for the four formats)

In [Example \(Four study formats with a pretest\)](#), the overall pretest mean is  $\bar{x} = 59.92$ . Format 2 had pretest mean 57.2, so its adjusted mean is

$$75.15 - 0.8670(57.2 - 59.92) = 75.15 + 2.36 = 77.51.$$

The four adjusted means are 70.97, 77.51, 69.11 and 77.59, averaging to the grand mean 73.80, with standard errors 2.02, 2.03, 2.01 and 2.01. These barely differ, because the pretest means are close to  $\bar{x}$  compared with the within-group standard deviation of the pretest, 10.06.

The six pairwise comparisons, with  $s = 6.34$  on 35 degrees of freedom, are as follows. The Tukey-type multiplier is  $q_{0.05}(4, 35)/\sqrt{2} = 2.697$ , and Scheffé’s is  $\sqrt{3F_{0.05}(3, 35)} = 2.936$ .

| formats | raw difference | adjusted difference | se   | Tukey-type half-width | Scheffé half-width |
|---------|----------------|---------------------|------|-----------------------|--------------------|
| 1 2     | −2.29          | −6.54               | 2.88 | 7.77                  | 8.46               |
| 1 3     | 3.77           | 1.86                | 2.85 | 7.67                  | 8.36               |
| 1 4     | −5.23          | −6.62               | 2.84 | 7.66                  | 8.34               |
| 2 3     | 6.06           | 8.40                | 2.85 | 7.69                  | 8.37               |
| 2 4     | −2.94          | −0.08               | 2.86 | 7.71                  | 8.39               |
| 3 4     | −9.00          | −8.48               | 2.84 | 7.65                  | 8.33               |

Both methods declare formats 2 and 4 better than format 3 and resolve nothing else, although the 2 3 comparison clears Scheffé’s bound by only 0.03. Without the covariate the standard error of a difference would be  $10.73\sqrt{2/10} = 4.80$ . The raw difference between formats 3 and 4, −9.00, would then have had a Tukey half-width near 13, and nothing would have been resolved.

Adjustment moved comparisons in both directions: format 2’s advantage over format 1 grew from 2.29 to 6.54, while format 4’s lead over format 3 shrank from 9.00 to 8.48.

```
import itertools
import numpy as np
from scipy import stats
```

```
rng = np.random.default_rng(84)
```

```

g, m = 4, 10                                # four formats, ten students
    each
n = g * m
group = rng.permutation(np.repeat(np.arange(g), m)) # the randomization
x = np.round(rng.normal(60, 10, n))            # pretest, measured before
    assignment
mu_true = np.array([70.0, 74.0, 71.0, 77.0])
y = np.round(mu_true[group] + 0.8 * (x - 60) + rng.normal(0, 6, n), 1)
Z = np.eye(g)[group]                          # indicator matrix of the
    formats
n_k = Z.sum(axis=0)
x_bar, y_bar = Z.T @ x / n_k, Z.T @ y / n_k    # group means

def within(v):
    """(I - M) v: deviations from the group means (M projects onto C(Z))."""
    return v - (Z.T @ v / n_k)[group]

E_xx, E_xy, E_yy = within(x) @ within(x), within(x) @ within(y), within(y) @ within(
    y)
beta = E_xy / E_xx                            # slope from within-group
    variation only
sse = E_yy - E_xy ** 2 / E_xx                  # ANCOVA residual sum of
    squares
nu = n - g - 1

s2 = sse / nu
x0 = x.mean()                                # the covariate value for
    adjustment
adj = y_bar - beta * (x_bar - x0)              # adjusted means
se_adj = np.sqrt(s2 * (1 / n_k + (x_bar - x0) ** 2 / E_xx))
pairs = list(itertools.combinations(range(g), 2))
q = stats.studentized_range.ppf(0.95, g, nu) / np.sqrt(2) # Tukey-Kramer
    multiplier
c_s = np.sqrt((g - 1) * stats.f.ppf(0.95, g - 1, nu))    # Scheffe multiplier
for k, l in pairs:
    d = adj[k] - adj[l]
    se_d = np.sqrt(s2 * (1 / n_k[k] + 1 / n_k[l] + (x_bar[k] - x_bar[l]) ** 2 / E_xx))
    print(f"{k + 1}-{l + 1}: raw {y_bar[k] - y_bar[l]:6.2f} adjusted {d:6.2f} se {
        se_d:.2f}")
        f" Tukey-type +-{q * se_d:.2f} Scheffe +-{c_s * se_d:.2f}")
print("adjusted means", adj.round(2), " standard errors", se_adj.round(2))

```

## 18.2.4 Adjusted means in other designs

For any design part, the adjusted version at  $\mathbf{z}_0$  of an estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ , such as a treatment mean averaged over blocks, is  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} + \mathbf{z}_0^\top \boldsymbol{\gamma}$ , estimated by  $\boldsymbol{\rho}^\top \mathbf{M}(\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}) + \mathbf{z}_0^\top \hat{\boldsymbol{\gamma}}$  with variance from [Theorem 18.1.1\(e\)](#). Software calls these **least squares means**. In unbalanced multi-way data the choice of  $\boldsymbol{\lambda}$ , equal or count-proportional weights over cells, matters as much as  $\mathbf{z}_0$  ([Chapter 17](#), [Theorem 17.2.2](#)); it is safest to write the intended function of the cell means ([Definition 17.1.1](#)) explicitly.

*Remark* (The adjusted mean is not a prediction interval). The standard error [Eq. 18.2.3](#) describes the uncertainty about the *mean* response of treatment  $k$  at  $\mathbf{z}_0$ . A new unit

with covariate  $\mathbf{z}_0$  given treatment  $k$  will vary around that mean with variance  $\sigma^2$  as well. Its prediction interval uses  $s\sqrt{1 + 1/n_k + (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}}_k - \mathbf{z}_0)}$ , as in [Theorem 12.3.1](#).

## 18.2.5 Exercises

### 18.2.5.1 A. Check your understanding

#### Exercise A1.

From the numbers in [Example \(Four study formats with a pretest\)](#), compute the adjusted means of formats 1 and 3 by hand and check them against the values in [Example \(Adjusted means for the four formats\)](#).

#### Exercise A2.

Software that fits the model  $E(y) = \mu_k + \beta x$  reports  $\hat{\mu}_k$ , the adjusted mean at  $x_0 = 0$ . For a pretest scored from 0 to 100, what does this number describe? Show that its standard error can be far larger than that of the adjusted mean at  $\bar{x}$ , although both give the same differences.

### 18.2.5.2 B. Practice

#### Exercise B1.

Show that  $\sum_k n_k \text{Var}(\hat{\mu}_k(\mathbf{z}_0))$  is minimized over  $\mathbf{z}_0$  by  $\mathbf{z}_0 = \bar{\mathbf{z}}$ .

*Solution.* By [Eq. 18.2.3](#), the sum is  $\sigma^2(g + \sum_k n_k (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}}_k - \mathbf{z}_0))$ . Writing  $\bar{\mathbf{z}}_k - \mathbf{z}_0 = (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}) + (\bar{\mathbf{z}} - \mathbf{z}_0)$  and using  $\sum_k n_k (\bar{\mathbf{z}}_k - \bar{\mathbf{z}}) = \mathbf{0}$ , the cross terms vanish and the sum becomes

$$\sigma^2 \left( g + \sum_k n_k (\bar{\mathbf{z}}_k - \bar{\mathbf{z}})^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}}_k - \bar{\mathbf{z}}) + n(\bar{\mathbf{z}} - \mathbf{z}_0)^\top \mathbf{E}_{zz}^{-1}(\bar{\mathbf{z}} - \mathbf{z}_0) \right).$$

The last term is nonnegative, and zero iff  $\mathbf{z}_0 = \bar{\mathbf{z}}$ , because  $\mathbf{E}_{zz}$  is positive definite.

#### Exercise B2.

A common alternative to covariance adjustment analyses the *change*  $y - x$ . Show that this is the covariance model with  $\beta$  fixed at 1. Using [Proposition 18.1.2](#)'s setting with  $\tau^2$  the variance of  $x$ , find the variance of the difference of mean changes between two groups, and show that it exceeds the variance of the adjusted difference (ignoring the  $1/(N - g - 2)$  term) unless  $\beta = 1$ .

*Solution.* The mean change in group  $k$  is  $\bar{y}_k - \bar{x}_k = \mu_k + (\beta - 1)\bar{x}_k + \bar{\varepsilon}_k$ . The difference of two groups has variance  $(2/m)\{(\beta - 1)^2\tau^2 + \sigma^2\}$ . The adjusted difference, with the slope known, has variance  $2\sigma^2/m$ . The change-score analysis is therefore worse by  $(2/m)(\beta - 1)^2\tau^2$ , which is zero only when  $\beta = 1$ . It beats the unadjusted analysis, which

has  $(\beta - 1)^2$  replaced by  $\beta^2$ , only if  $\beta > 1/2$ . Covariance adjustment estimates the slope rather than assuming one.

### 18.2.5.3 C. Going deeper

#### Exercise C1.

Suppose all groups have size  $m$  and the covariate means  $\bar{\mathbf{z}}_1, \dots, \bar{\mathbf{z}}_g$  are equal. Show that the adjusted differences  $\hat{\mu}_k - \hat{\mu}_l$  are the raw differences  $\bar{y}_k - \bar{y}_l$ , that these are independent of  $s^2$  from the covariance model, and that the Tukey intervals built from  $s$  and  $q_\alpha(g, n - g - q)$  have simultaneous coverage exactly  $1 - \alpha$ .

*Solution.* With equal covariate means,  $\bar{\mathbf{z}}_c = \sum_k c_k \bar{\mathbf{z}}_k = \mathbf{0}$  for every contrast, so the adjusted contrast equals the raw one and has variance  $\sigma^2 \sum_k c_k^2 / m$  by Eq. 18.2.4. The group means are functions of  $\mathbf{M}\mathbf{Y}$ . The residual vector  $(\mathbf{I} - \mathbf{M}_W)\mathbf{Y}$  is orthogonal to  $\mathcal{C}(\mathbf{M}_W) \supseteq \mathcal{C}(\mathbf{M})$ , so under normality  $s^2$  is independent of  $\bar{y}_1, \dots, \bar{y}_g$  (Theorem 4.4.4), and  $(n - g - q)s^2/\sigma^2 \sim \chi^2(n - g - q)$ . The means are independent normal with common variance  $\sigma^2/m$ . The maximum of  $|(\bar{y}_k - \bar{y}_l) - (\mu_k - \mu_l)|/(s/\sqrt{m})$  therefore has the studentized range distribution with parameters  $g$  and  $n - g - q$ , and the proof of Theorem 13.3.2 applies unchanged.

#### Exercise C2.

For four groups of ten and one covariate, simulate the simultaneous coverage of the Tukey-type intervals of Example (Adjusted means for the four formats) when the covariate means are held at (62, 57, 60, 61) and when they are held at (70, 50, 60, 60), with the within-group covariate variation as in the example. How far from 0.95 is the coverage in each case?

## 18.3 Heterogeneous slopes

The covariance model of Section 18.1 assumes a common slope, so each treatment comparison is a vertical shift between parallel lines, one number. If the slopes differ, the treatment effect depends on the covariate, and “the” adjusted difference no longer exists. This section tests parallelism and says what to report when it fails.

### 18.3.1 The separate lines model and the test of parallelism

Keep one factor with  $g$  levels and one covariate, and allow each level its own line:

$$E(y_i) = \mu_k + \beta_k x_i \quad \text{for unit } i \text{ in group } k.$$

This is the separate lines model Eq. 8.6.4 of Section 8.6. The parallel lines model is the special case  $\beta_1 = \dots = \beta_g$ . For group  $k$  write

$$E_{xx,k} = \sum_{i \in k} (x_i - \bar{x}_k)^2, \quad E_{xy,k} = \sum_{i \in k} (x_i - \bar{x}_k)(y_i - \bar{y}_k),$$

so that the pooled within-group quantities of Section 18.1 are  $E_{xx} = \sum_k E_{xx,k}$  and  $E_{xy} = \sum_k E_{xy,k}$ .

### Theorem 18.3.1 — The test of parallel slopes

Assume  $x$  varies within every group, so that every  $E_{xx,k} > 0$ , and  $n > 2g$ .

- (a) In the separate lines model,  $\hat{\beta}_k = E_{xy,k}/E_{xx,k}$ , these estimates are uncorrelated with variances  $\sigma^2/E_{xx,k}$ , and the residual sum of squares is

$$\text{SSE}_{\text{sep}} = \sum_k \left( E_{yy,k} - \frac{E_{xy,k}^2}{E_{xx,k}} \right)$$

on  $n - 2g$  degrees of freedom.

- (b) The pooled slope of the parallel lines model is the weighted mean  $\hat{\beta} = \sum_k E_{xx,k} \hat{\beta}_k / \sum_k E_{xx,k}$ , and the extra sum of squares for the separate slopes is

$$\text{SS}_{\text{slopes}} = \sum_k E_{xx,k} (\hat{\beta}_k - \hat{\beta})^2 = \sum_k \frac{E_{xy,k}^2}{E_{xx,k}} - \frac{E_{xy}^2}{E_{xx}} \quad (18.3.1)$$

on  $g - 1$  degrees of freedom.

- (c) If  $\mathbf{Y}$  is normal,

$$F = \frac{\text{SS}_{\text{slopes}}/(g-1)}{\text{SSE}_{\text{sep}}/(n-2g)} \sim F(g-1, n-2g, \gamma),$$

with

$$\gamma = \frac{1}{\sigma^2} \sum_k E_{xx,k} (\beta_k - \bar{\beta})^2,$$

where  $\bar{\beta} = \sum_k E_{xx,k} \beta_k / \sum_k E_{xx,k}$ . The test that rejects for large  $F$  is the  $F$  test of parallelism.

*Proof.* Let  $\mathbf{M}$  project onto the span of the group indicators, and let  $\mathbf{w}_k$  be the vector that equals  $x_i - \bar{x}_k$  for units of group  $k$  and zero elsewhere. The  $k$ th slope column of the separate lines model is  $\mathbf{x}$  restricted to group  $k$ , and its residual after projection onto  $\mathcal{C}(\mathbf{M})$  is  $\mathbf{w}_k$ . By Lemma 9.2.1 the separate lines model space is  $\mathcal{C}(\mathbf{M}) \oplus \text{span}\{\mathbf{w}_1, \dots, \mathbf{w}_g\}$ , and the parallel lines space is  $\mathcal{C}(\mathbf{M}) \oplus \text{span}\{\mathbf{w}\}$  with  $\mathbf{w} = \sum_k \mathbf{w}_k = (\mathbf{I} - \mathbf{M})\mathbf{x}$ . The vectors  $\mathbf{w}_k$  have disjoint supports, so they are mutually orthogonal, with  $\|\mathbf{w}_k\|^2 = E_{xx,k}$  and  $\mathbf{w}_k^\top \mathbf{y} = E_{xy,k}$ .

- (a) The projection of  $\mathbf{y}$  onto  $\text{span}\{\mathbf{w}_k\}$  is  $\sum_k (E_{xy,k}/E_{xx,k})\mathbf{w}_k$ . Reading off the coefficient of the  $k$ th slope column gives  $\hat{\beta}_k$ , which is  $\mathbf{w}_k^\top \mathbf{y} / \|\mathbf{w}_k\|^2$ . Hence  $\text{Cov}(\hat{\beta}_k, \hat{\beta}_l) = \sigma^2 \mathbf{w}_k^\top \mathbf{w}_l / (E_{xx,k} E_{xx,l})$ , which is  $\sigma^2/E_{xx,k}$  for  $k = l$  and zero otherwise. The residual sum of squares is  $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 - \sum_k E_{xy,k}^2/E_{xx,k}$ , and  $\|(\mathbf{I} - \mathbf{M})\mathbf{y}\|^2 = \sum_k E_{yy,k}$ . The model has rank  $2g$ .

- (b) The pooled slope is  $\mathbf{w}^\top \mathbf{y} / \|\mathbf{w}\|^2 = E_{xy} / E_{xx}$ , and  $E_{xy} = \sum_k E_{xx,k} \hat{\beta}_k$ . By [Theorem 6.5.1](#) the extra sum of squares is the squared length of the projection onto the orthogonal complement of  $\mathbf{w}$  in  $\text{span}\{\mathbf{w}_k\}$ , which equals  $\sum_k E_{xy,k}^2 / E_{xx,k} - E_{xy}^2 / E_{xx}$ . Writing  $E_{xy,k} = E_{xx,k} \hat{\beta}_k$  and expanding  $\sum_k E_{xx,k} (\hat{\beta}_k - \hat{\beta})^2 = \sum_k E_{xx,k} \hat{\beta}_k^2 - \hat{\beta}^2 E_{xx}$  gives the first form.
- (c) By [Theorem 11.1.1](#), with  $\sigma^2 \gamma$  equal to the extra sum of squares evaluated at the mean vector. The mean vector has  $\mathbf{w}_k^\top \boldsymbol{\mu} = \beta_k E_{xx,k}$ , because  $\mathbf{w}_k$  is orthogonal to the group indicators. Substituting into (b) with  $\hat{\beta}_k$  replaced by  $\beta_k$  gives  $\gamma$ .

■

The test is a weighted test of homogeneity of  $g$  independent slope estimates, with weights equal to their precisions  $E_{xx,k} / \sigma^2$ .

### Example. (Parallelism in the tutoring experiment)

For the four formats of [Example \(Four study formats with a pretest\)](#), the separate slopes are 0.703, 0.823, 1.035 and 0.992, and their weighted mean is the pooled slope 0.8670. The separate lines model has residual sum of squares 1347.47 on 32 degrees of freedom, against 1407.47 on 35 for parallel lines. The test gives  $F = 0.47$  on 3 and 32 degrees of freedom,  $p = 0.70$ , and the covariance analysis stands.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(84)
g, m = 4, 10                                # four formats, ten students
    each
n = g * m
group = rng.permutation(np.repeat(np.arange(g), m)) # the randomization
x = np.round(rng.normal(60, 10, n))           # pretest, measured before
    assignment
mu_true = np.array([70.0, 74.0, 71.0, 77.0])
y = np.round(mu_true[group] + 0.8 * (x - 60) + rng.normal(0, 6, n), 1)
Z = np.eye(g)[group]                         # indicator matrix of the
    formats
n_k = Z.sum(axis=0)

def within(v):
    """(I - M) v: deviations from the group means (M projects onto C(Z))."""
    return v - (Z.T @ v / n_k)[group]

E_xx, E_xy, E_yy = within(x) @ within(x), within(x) @ within(y), within(y) @ within(
    y)
sse = E_yy - E_xy ** 2 / E_xx                 # ANCOVA residual sum of
    squares
X_sep = np.column_stack([Z, Z * x[:, None]]) # a separate line for each
    format
E_xx_k = np.array([within(x)[group == k] @ within(x)[group == k] for k in range(g)])
E_xy_k = np.array([within(x)[group == k] @ within(y)[group == k] for k in range(g)])
slopes = E_xy_k / E_xx_k
sse_sep = sse - (np.sum(E_xy_k ** 2 / E_xx_k) - E_xy ** 2 / E_xx)
F_par = ((sse - sse_sep) / (g - 1)) / (sse_sep / (n - 2 * g))
print("separate slopes", slopes.round(3),
```

```
f" F = {F_par:.2f}, p = {stats.f.sf(F_par, g - 1, n - 2 * g):.3f}"
```

With ten students per group the test has little power, so the case for parallel lines rests mostly on the science. When a treatment is meant to help some units more than others, such as a remedial programme aimed at weak students, the separate lines model should be the starting point rather than a check.

## 18.3.2 When the slopes differ

If the slopes differ, the difference between treatments  $k$  and  $l$  at covariate value  $x$  is the line

$$\delta(x) = (\mu_k - \mu_l) + (\beta_k - \beta_l)x,$$

estimated by  $\hat{\delta}(x) = (\hat{\mu}_k - \hat{\mu}_l) + (\hat{\beta}_k - \hat{\beta}_l)x$ . The two lines are independent simple regressions, so by [Corollary 5.5.4](#)

$$\text{Var}(\hat{\delta}(x)) = \sigma^2 v(x), \quad v(x) = \frac{1}{n_k} + \frac{(x - \bar{x}_k)^2}{E_{xx,k}} + \frac{1}{n_l} + \frac{(x - \bar{x}_l)^2}{E_{xx,l}}. \quad (18.3.2)$$

Honest reports include:

- (i) Report  $\hat{\delta}(x)$  with its standard error at a few covariate values chosen for their meaning, for instance the quartiles of  $x$ , or a clinical threshold. Each is a  $t$  interval ([Theorem 12.1.1](#)) with  $n - 2g$  degrees of freedom.
- (ii) Draw  $\hat{\delta}(x)$  as a function with a confidence band. The pointwise band uses  $t_{n-2g, \alpha/2}$ . A band that covers the whole line  $\delta(\cdot)$  with probability exactly  $1 - \alpha$  replaces this multiplier by  $\sqrt{2F_\alpha(2, n - 2g)}$ , because  $\{\delta(x) : x \in \mathbb{R}\}$  lies in a two-dimensional space of estimable functions. This is the argument of [Theorem 13.2.2](#), as used for the Working–Hotelling band ([Theorem 12.4.1](#)).
- (iii) Give the set of covariate values at which the treatments differ significantly. This is the **Johnson–Neyman region** ([Proposition 18.3.2](#) below).

Reporting  $\hat{\delta}(\bar{x})$  alone as *the* treatment effect hides the slope, which is the whole point.

*Remark* (The average effect in a randomized experiment). There is one exception. In a randomized experiment,  $\delta$  at the population mean of the covariate is the *average* treatment effect, because  $\delta$  is linear. Its estimate  $\hat{\delta}(\bar{x})$  is the treatment coefficient in the regression with a treatment-by-covariate interaction and the covariate centred at  $\bar{x}$  ([Exercise \(C1\)](#)). This “interacted” adjustment is asymptotically never less precise than the unadjusted difference, even when the model is wrong. [Chapter 25](#) returns to it ([Theorem 25.6.2](#)).

## 18.3.3 The Johnson–Neyman region

### Proposition 18.3.2 — Johnson–Neyman region

In the separate lines model with normal errors, let  $s^2 = \text{SSE}_{\text{sep}}/(n - 2g)$  and fix a multiplier  $c > 0$ . Write  $\Delta_\mu = \hat{\mu}_k - \hat{\mu}_l$  and  $\Delta_\beta = \hat{\beta}_k - \hat{\beta}_l$ . The set of  $x$  with  $|\hat{\delta}(x)| >$



$cs\sqrt{v(x)}$  is  $\{x : Ax^2 + Bx + C > 0\}$ , where

$$A = \Delta_\beta^2 - c^2 s^2 \left( \frac{1}{E_{xx,k}} + \frac{1}{E_{xx,l}} \right), \quad B = 2\Delta_\mu \Delta_\beta + 2c^2 s^2 \left( \frac{\bar{x}_k}{E_{xx,k}} + \frac{\bar{x}_l}{E_{xx,l}} \right),$$

$$C = \Delta_\mu^2 - c^2 s^2 \left( \frac{1}{n_k} + \frac{1}{n_l} + \frac{\bar{x}_k^2}{E_{xx,k}} + \frac{\bar{x}_l^2}{E_{xx,l}} \right).$$

If  $A > 0$ , which happens iff the  $t$  test of  $\beta_k = \beta_l$  with multiplier  $c$  rejects, the region is the whole line with at most an interval removed:  $\delta$  is significantly positive on one side and significantly negative on the other. If  $A < 0$ , the region is a bounded interval or empty. With  $c = t_{n-2g, \alpha/2}$  each point of the region is significant at level  $\alpha$ . With  $c = \sqrt{2F_\alpha(2, n-2g)}$  the statement “ $\delta(x) \neq 0$  for every  $x$  in the region” holds with probability at least  $1 - \alpha$ .

*Proof.* Square both sides of  $|\hat{\delta}(x)| > cs\sqrt{v(x)}$  and expand  $\hat{\delta}(x)^2 = \Delta_\mu^2 + 2\Delta_\mu \Delta_\beta x + \Delta_\beta^2 x^2$  and  $v(x)$  from Eq. 18.3.2 as polynomials in  $x$ . Collecting terms gives  $Ax^2 + Bx + C$ . The coefficient  $A$  is positive iff  $\Delta_\beta^2 / \{s^2(1/E_{xx,k} + 1/E_{xx,l})\} > c^2$ , which is the squared  $t$  statistic for  $\beta_k - \beta_l$  exceeding  $c^2$ . A quadratic with  $A > 0$  is positive outside the interval between its real roots (everywhere if it has none), and with  $A < 0$  it is positive between its real roots (nowhere if it has none). The last statement follows from the simultaneous band of item (ii) above: on the event that the band covers  $\delta(\cdot)$  everywhere,  $\delta(x)$  cannot be zero at any point where the band excludes zero. ■

### Example. (Two versions of a reading programme)

In a second synthetic randomized experiment, 48 children with baseline reading fluency between 42 and 118 words per minute are assigned, 24 each, to the current and a new version of a reading programme. The response is the gain in fluency. The new version was designed for weak readers, and the fitted lines of Figure 18.3.1(a) show why that matters. The slope is 0.0156 for the current version and  $-0.1615$  for the new one. The test of parallelism has  $F = 11.2$  on 1 and 44 degrees of freedom,  $p = 0.0017$ . The pooled slope of the parallel lines model would be  $-0.0638$ , a compromise that describes neither group.

The estimated lines cross at  $x = 107.7$ . At a baseline of 60 words per minute the new version adds 8.46 words per minute (standard error 1.56). At 100 it adds 1.37 (standard error 1.46), which is not significant. With  $c = t_{44, 0.025}$ , the Johnson–Neyman region (Proposition 18.3.2) is  $x < 93.5$  together with  $x > 152.0$ . The second part lies far beyond the largest baseline in the data, and it is an artefact of extending two fitted lines. The useful conclusion is that the new version is better for children who start below about 93 words per minute, which covers 33 of the 48 children in the study. With the simultaneous multiplier  $\sqrt{2F_{0.05}(2, 44)}$ , the region shrinks slightly, to  $x < 90.8$  (and  $x > 195.6$ ).

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(31)
n_per = 24
```

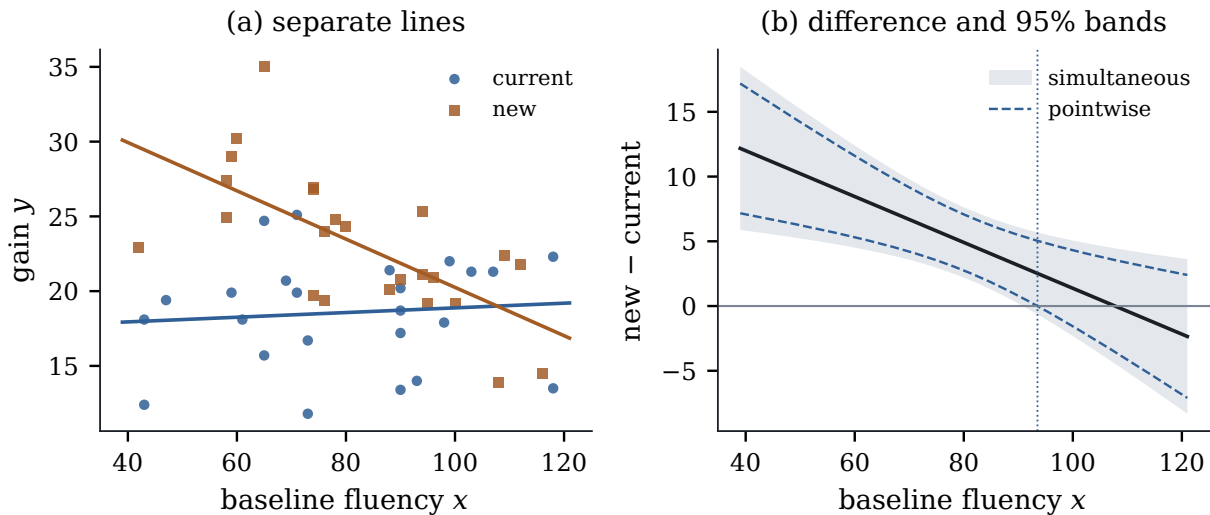


Figure 18.3.1: The reading programme. (a) Gain against baseline fluency with the separate least squares lines. (b) The estimated difference  $\hat{\delta}(x)$  (new minus current) with a pointwise 95% band (dashed) and a simultaneous 95% band (shaded). The dotted line marks the lower end of the Johnson–Neyman region,  $x = 93.5$ ; to its left the difference is significantly positive.

```

arm = rng.permutation(np.repeat([0, 1], n_per))      # 0 = current version, 1 = new
version
x = np.round(rng.uniform(40, 120, 2 * n_per))      # baseline fluency
y = np.where(arm == 0, 14 + 0.06 * x, 36 - 0.16 * x) + rng.normal(0, 4, 2 * n_per)
y = np.round(y, 1)
n = len(y)

def line(k):
    """Least squares line within arm k, and the arm's summaries."""
    xs, ys = x[arm == k], y[arm == k]
    Sxx = np.sum((xs - xs.mean()) ** 2)
    b = np.sum((xs - xs.mean()) * (ys - ys.mean())) / Sxx
    a = ys.mean() - b * xs.mean()
    sse = np.sum((ys - a - b * xs) ** 2)
    return a, b, xs.mean(), Sxx, len(xs), sse

(a0, b0, xb0, S0, n0, e0), (a1, b1, xb1, S1, n1, e1) = line(0), line(1)
nu = n - 4
s2 = (e0 + e1) / nu                                # separate-lines model

def significant_region(crit2):
    """Roots of  $d(x)^2 - \text{crit2} * s2 * v(x) = 0$ , a quadratic in  $x$ ."""
    A = (b1 - b0) ** 2 - crit2 * s2 * (1 / S0 + 1 / S1)
    B = 2 * (a1 - a0) * (b1 - b0) + 2 * crit2 * s2 * (xb0 / S0 + xb1 / S1)
    C = (a1 - a0) ** 2 - crit2 * s2 * (1 / n0 + 1 / n1 + xb0 ** 2 / S0 + xb1 ** 2 / S1)
    return np.sort(np.roots([A, B, C]).real)

t2 = stats.t.ppf(0.975, nu) ** 2                    # pointwise, 5% level
w2 = 2 * stats.f.ppf(0.95, 2, nu)                  # simultaneous over all x (
    Scheffe,  $q = 2$ )
x_cross = -(a1 - a0) / (b1 - b0)
print(f"lines cross at x = {x_cross:.1f}")
print("pointwise 5% region: difference significant outside", significant_region(t2).
    round(1))

```

```
print("simultaneous region: difference significant outside", significant_region(w2).
      round(1))
```

The region is only as good as the linearity of each line, it extrapolates outside the data, and it marks significance, not practical importance: with enough data every value except the crossing point becomes significant.

### 18.3.4 Several covariates, several factors

With  $q$  covariates, the test of a common  $\gamma$  against  $\gamma_1, \dots, \gamma_g$  has  $(g-1)q$  and  $n - g(q+1)$  degrees of freedom. With two factors, slope terms can be built term by term like the means model of [Chapter 16](#). Each such test compares nested column spaces ([Theorem 11.1.1](#)).

### 18.3.5 Exercises

#### 18.3.5.1 A. Check your understanding

##### Exercise A1.

For  $g = 2$ , show that [Eq. 18.3.1](#) reduces to  $SS_{\text{slopes}} = \frac{E_{xx,1}E_{xx,2}}{E_{xx,1} + E_{xx,2}} (\hat{\beta}_1 - \hat{\beta}_2)^2$ , and that the  $F$  statistic is the square of the two-sample  $t$  statistic for the difference of slopes.

##### Exercise A2.

In [Example \(Two versions of a reading programme\)](#), the estimated lines cross at  $x = 107.7$ . Explain why the Johnson–Neyman region never contains the crossing point, whatever the multiplier.

#### 18.3.5.2 B. Practice

##### Exercise B1.

Give a numerical example of two groups for which  $A < 0$  and the Johnson–Neyman region of [Proposition 18.3.2](#) is a bounded interval. Interpret the region.

##### Exercise B2.

In the tutoring experiment, suppose the true slopes were 0.8, 0.8, 0.8, 1.2 and  $\sigma = 6$ . Using the  $E_{xx,k}$  of the example (their sum is 3643.9; take them equal for the calculation), compute the noncentrality of the test of parallelism and its power at level 0.05 ([Theorem 11.5.1](#)).

##### Exercise B3.

Prove the claim in item (ii): with  $c = \sqrt{2F_\alpha(2, n-2g)}$ , the band  $\hat{\delta}(x) \pm cs\sqrt{v(x)}$

contains  $\delta(x)$  for all  $x$  simultaneously with probability at least  $1 - \alpha$ . Is the probability exactly  $1 - \alpha$ ?

*Solution.* The functions  $\delta(x) = \lambda_x^\top \beta$  for  $x \in \mathbb{R}$  have  $\lambda_x$  in a two-dimensional space  $\mathcal{L}$  spanned by the coefficient vectors of  $\mu_k - \mu_l$  and  $\beta_k - \beta_l$ , both estimable. By [Theorem 13.2.2\(b\)](#) with  $q = 2$ , with probability exactly  $1 - \alpha$  every  $\lambda \in \mathcal{L}$  satisfies  $|\lambda^\top \hat{\beta} - \lambda^\top \beta| \leq c \text{se}(\lambda^\top \hat{\beta})$ . The functions  $\lambda_x$  form a subset of  $\mathcal{L}$ , so the probability for them is at least  $1 - \alpha$ . It is in fact exactly  $1 - \alpha$ . In the coordinates  $(\mu_k - \mu_l, \beta_k - \beta_l)$  we have  $\lambda_x = (1, x)$ , and the studentized ratio  $(\lambda^\top \hat{\beta} - \lambda^\top \beta)^2 / \text{se}(\lambda^\top \hat{\beta})^2$  is unchanged when  $\lambda$  is rescaled and is continuous on  $\mathcal{L} \setminus \{0\}$ . Every direction of  $\mathcal{L}$  except that of  $(0, 1)$  is the direction of some  $\lambda_x$ , and  $\lambda_x / \|\lambda_x\| \rightarrow \pm(0, 1)$  as  $x \rightarrow \pm\infty$ . So the supremum of the ratio over  $x \in \mathbb{R}$  equals its supremum over  $\mathcal{L} \setminus \{0\}$ , and the two events coincide. This is the argument of [Theorem 12.4.1\(b\)](#) for a single line.

### 18.3.5.3 C. Going deeper

#### Exercise C1.

In a two-group randomized experiment, show that the treatment coefficient in the least squares fit of  $y$  on  $1, T, x - \bar{x}$  and  $T(x - \bar{x})$ , where  $\bar{x}$  is the overall mean, equals  $(\bar{y}_1 - \hat{\beta}_1(\bar{x}_1 - \bar{x})) - (\bar{y}_0 - \hat{\beta}_0(\bar{x}_0 - \bar{x}))$ , with  $\hat{\beta}_k$  the within-group slopes. Compare with the adjusted difference of the parallel lines model.

*Solution.* The model with an interaction is the separate lines model, so its fitted lines are the two within-group least squares lines,  $\bar{y}_k + \hat{\beta}_k(x - \bar{x}_k)$ . In the given parameterization the line for group  $k$  is  $(\beta_0 + \beta_T T) + (\beta_x + \beta_{Tx} T)(x - \bar{x})$ , so the coefficient of  $T$  is the difference of the two lines at  $x = \bar{x}$ , which is the stated expression. The parallel lines model replaces both  $\hat{\beta}_k$  by the pooled slope. The two estimates agree when the slopes are equal or when  $\bar{x}_0 = \bar{x}_1 = \bar{x}$ .

## 18.4 Randomized complete blocks

A covariate removes nuisance variation by regression. **Blocking** removes it by design: similar units (plots in one part of a field, animals from one litter, strips cut from one board) are grouped into blocks, and treatments are compared *within* blocks. This section analyses the randomized complete block design, explains how randomization justifies its model, and measures what blocking gains.

### 18.4.1 The design and its model

#### Definition 18.4.1 — Randomized complete block design

There are  $t$  treatments and  $b$  blocks of  $t$  units each. In every block, the  $t$  treatments are assigned to the  $t$  units by a random permutation, chosen uniformly and independently for different blocks. The response of the unit in block  $j$  that received treatment  $i$  is

written  $y_{ij}$ .

The usual model is the additive two-way model with one observation per cell,

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2), \quad (18.4.1)$$

with treatment effects  $\tau_i$  and block effects  $\beta_j$ . Its geometry is [Example \(Additive two-factor model, one observation per cell\)](#) and its analysis of variance is that of [Theorem 16.1.2](#); we restate it in the form the design needs.

### Theorem 18.4.1 — Analysis of the randomized complete block design

Assume [Eq. 18.4.1](#) with  $t, b \geq 2$ . Let  $\bar{y}_{i\cdot}$ ,  $\bar{y}_{\cdot j}$  and  $\bar{y}_{\cdot\cdot}$  be the treatment, block and grand means.

- (a) The sums of squares

$$\begin{aligned} SS_T &= b \sum_i (\bar{y}_{i\cdot} - \bar{y}_{\cdot\cdot})^2, & SS_B &= t \sum_j (\bar{y}_{\cdot j} - \bar{y}_{\cdot\cdot})^2, \\ SSE &= \sum_{i,j} (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y}_{\cdot\cdot})^2 \end{aligned}$$

are squared lengths of projections onto mutually orthogonal subspaces of dimensions  $t-1$ ,  $b-1$  and  $(t-1)(b-1)$ , and with the correction for the mean they add up to  $\sum_{i,j} y_{ij}^2$ .

- (b) A function  $\sum_i c_i \tau_i$  is estimable iff  $\sum_i c_i = 0$ . Its least squares estimate is  $\sum_i c_i \bar{y}_{i\cdot}$ , and

$$\sum_i c_i \bar{y}_{i\cdot} = \sum_i c_i \tau_i + \sum_i c_i \bar{\varepsilon}_i.$$

identically, whatever the block effects. Its variance is  $\sigma^2 \sum_i c_i^2 / b$ .

- (c) The treatment  $F$  statistic satisfies

$$F_T = \frac{SS_T / (t-1)}{SSE / ((t-1)(b-1))} \sim F(t-1, (t-1)(b-1), \gamma),$$

with

$$\gamma = \frac{b}{\sigma^2} \sum_i (\tau_i - \bar{\tau})^2.$$

- (d)  $E(SS_B / (b-1)) = \sigma^2 + t \sum_j (\beta_j - \bar{\beta})^2 / (b-1)$ , and  $E(SSE / ((t-1)(b-1))) = \sigma^2$ .

*Proof.* Model [Eq. 18.4.1](#) is the additive two-way model of [Theorem 16.1.2](#) with factor  $A$  the treatments, factor  $B$  the blocks,  $a = t$  and  $m = 1$ , so that  $n - a - b + 1 = (t-1)(b-1)$ . Parts (a), (b) and (c) are [Theorem 16.1.2](#)(a), (b) and (c), with (e) for the form of the residual space. The only new statement is the identity in (b): averaging [Eq. 18.4.1](#) over  $j$  gives  $\bar{y}_{i\cdot} = \mu + \tau_i + \bar{\beta} + \bar{\varepsilon}_i$ , and  $\sum_i c_i (\mu + \bar{\beta}) = 0$  for a contrast. Part (d) is [Theorem 9.6.1](#) applied to the decomposition in (a); the block projection sends the mean vector to the

vector with entries  $\beta_j - \bar{\beta}$ , whose squared length is  $t \sum_j (\beta_j - \bar{\beta})^2$ .

■

Part (b) is the reason to block. However large the block effects, they cancel exactly from every treatment comparison and from the error mean square, which estimates the variation *within* blocks. The between-block variation moves into its own line,  $SS_B$ . A test of blocks is rarely of interest, since blocks were chosen because they differ; a large  $MS_B$  measures what blocking achieved ([Proposition 18.4.4](#) below).

Any block-by-treatment interaction is absorbed into the error, and without replication within cells it cannot be separated from it. Tukey's one-degree-of-freedom test ([Theorem 16.1.2\(f\)](#)) checks for its most common form. If treatment differences genuinely vary between blocks, the comparisons refer to the average over the blocks used.

## 18.4.2 Randomization as the justification of the model

The units of an experiment are not a sample from a normal population; they are the boards or patients at hand. What the experimenter controls is the randomization, and it alone supports much of the analysis. Give each unit  $u$  a fixed value  $\xi_u$ , and assume **unit–treatment additivity**: unit  $u$  responds to treatment  $i$  with  $\xi_u + \tau_i$ . The only probability below is that of the randomization.

### Proposition 18.4.2 — Randomization moments of the block analysis

Let the units of block  $j$  have values  $\xi_{j1}, \dots, \xi_{jt}$ , with mean  $\bar{\xi}_j$  and  $W_j = \sum_s (\xi_{js} - \bar{\xi}_j)^2$ , and put

$$S_W^2 = \frac{1}{b(t-1)} \sum_j W_j,$$

the average within-block variance of the unit values. Under unit–treatment additivity and the randomization of [Definition 18.4.1](#):

- (a) every contrast estimate  $\sum_i c_i \bar{y}_i$  has randomization mean  $\sum_i c_i \tau_i$  and randomization variance  $S_W^2 \sum_i c_i^2 / b$ ;
- (b)  $E(\text{SSE} / ((t-1)(b-1))) = S_W^2$ ;
- (c)  $E(\text{SS}_T / (t-1)) = S_W^2 + b \sum_i (\tau_i - \bar{\tau})^2 / (t-1)$ .

So the error mean square is unbiased for the variance that the model attributes to  $\sigma^2$ , and when the treatments have no effect the treatment and error mean squares have the same expectation.

*Proof.* Let  $\pi_j(i)$  be the unit of block  $j$  that receives treatment  $i$ , and  $w_{ij} = \xi_{j\pi_j(i)} - \bar{\xi}_j$ . Then  $y_{ij} = \bar{\xi}_j + \tau_i + w_{ij}$ . For fixed  $j$ , the vector  $(w_{1j}, \dots, w_{tj})$  is a uniformly random permutation of the deviations  $\xi_{js} - \bar{\xi}_j$ , which sum to zero. Hence  $E w_{ij} = 0$ ,  $E w_{ij}^2 = W_j / t$ , and, since  $\sum_k w_{kj} = 0$ ,  $E w_{ij} w_{kj} = -W_j / \{t(t-1)\}$  for  $i \neq k$ . Different blocks are independent.

- (a)  $\sum_i c_i \bar{y}_i = \sum_i c_i \tau_i + b^{-1} \sum_j \sum_i c_i w_{ij}$ , because  $\sum_i c_i = 0$  removes the  $\bar{\xi}_j$ . The mean follows. For the variance,

$$\text{Var}\left(\sum_i c_i w_{ij}\right) = \frac{W_j}{t} \sum_i c_i^2 - \frac{W_j}{t(t-1)} \sum_{i \neq k} c_i c_k = \frac{W_j}{t-1} \sum_i c_i^2,$$

since  $\sum_{i \neq k} c_i c_k = (\sum_i c_i)^2 - \sum_i c_i^2 = -\sum_i c_i^2$ . Summing over the independent blocks and dividing by  $b^2$  gives  $S_W^2 \sum_i c_i^2 / b$ .

- (b) The block means are  $\bar{y}_{.j} = \bar{\xi}_j + \bar{\tau}$ , and the treatment means are  $\bar{y}_i = \bar{\xi} + \tau_i + \bar{w}_{i.}$ , with  $\bar{\xi} = b^{-1} \sum_j \bar{\xi}_j$ . So the residual is  $w_{ij} - \bar{w}_{i.}$ , and

$$\text{SSE} = \sum_{i,j} w_{ij}^2 - b \sum_i \bar{w}_{i.}^2.$$

The first sum is  $\sum_j W_j$  for every randomization. By independence across blocks,  $E \bar{w}_{i.}^2 = b^{-2} \sum_j E w_{ij}^2 = b^{-2} \sum_j W_j / t$ , so  $E(b \sum_i \bar{w}_{i.}^2) = b^{-1} \sum_j W_j$ . Hence  $E(\text{SSE}) = (1 - 1/b) \sum_j W_j = (t-1)(b-1)S_W^2$ .

- (c)  $\text{SS}_T = b \sum_i (\tau_i - \bar{\tau} + \bar{w}_{i.})^2$ , using  $\sum_i \bar{w}_{i.} = 0$ . The cross terms have mean zero, and  $E(b \sum_i \bar{w}_{i.}^2) = (t-1)S_W^2$  by the computation in (b).

■

Independent measurement errors of variance  $\sigma_e^2$  add  $\sigma_e^2$  to every expectation. The proposition does not make  $F_T$  exactly  $F$ -distributed, but it shows that the estimates and their estimated variances are unbiased because of the randomization, whatever made the units differ. A systematic assignment could line treatments up with the  $\xi$ 's, and then neither statement would hold.

**Randomization tests.** Under the *sharp* null hypothesis that every unit would respond the same under every treatment, the responses are fixed and only the labels are random. The proportion of the  $(t!)^b$  equally likely randomizations giving an  $F_T$  at least as large as the observed one is an exact  $p$ -value, estimated by sampling randomizations when there are too many. [Chapter 23](#) develops permutation tests for linear models.

### 18.4.3 An example

#### Example. (Five wood adhesives on six boards)

In a synthetic experiment, five adhesives are compared for the shear strength of glued joints. Timber varies from board to board, so each of six boards is cut into five strips, assigned to the adhesives by a random permutation. The strengths (MPa) are:

| adhesive | board 1 | board 2 | board 3 | board 4 | board 5 | board 6 |
|----------|---------|---------|---------|---------|---------|---------|
| 1        | 7.89    | 8.42    | 8.55    | 8.30    | 9.26    | 9.15    |
| 2        | 8.68    | 8.69    | 8.73    | 8.62    | 9.72    | 9.51    |
| 3        | 9.13    | 9.28    | 9.81    | 9.43    | 10.30   | 9.41    |
| 4        | 8.48    | 8.60    | 8.46    | 8.95    | 9.42    | 9.02    |
| 5        | 8.90    | 9.17    | 9.79    | 9.57    | 10.24   | 9.09    |

The analysis of variance is:

| source          | df | sum of squares | mean square | $F$   |
|-----------------|----|----------------|-------------|-------|
| adhesives       | 4  | 4.106          | 1.0266      | 14.88 |
| boards          | 5  | 4.068          | 0.8136      | 11.79 |
| error           | 20 | 1.380          | 0.0690      |       |
| corrected total | 29 | 9.555          |             |       |

The adhesives differ clearly ( $p = 8.6 \times 10^{-6}$ ). The adhesive means are 8.595, 8.992, 9.560, 8.822 and 9.460, and every difference has standard error  $\sqrt{2 \times 0.0690/6} = 0.152$ . Tukey's honestly significant difference at level 0.05,  $q_{0.05}(5, 20)\sqrt{0.0690/6}$ , is 0.454 ([Theorem 13.3.2](#) applies, with the error mean square of the block analysis, as in [Exercise \(B1\)](#)). It separates adhesives 3 and 5 from adhesives 1, 2 and 4: the significant pairs are 1–3, 1–5, 2–3, 2–5, 3–4 and 4–5. Tukey's test for nonadditivity gives  $F = 0.07$  ( $p = 0.80$ ), so there is no sign that the adhesive differences depend on the strength of the board.

Of 20000 randomizations sampled from the  $2.99 \times 10^{12}$  possible plans, 1 gave an adhesive  $F$  at least as large as 14.88: the randomization  $p$ -value is  $(1 + 1)/(20000 + 1) = 0.00010$ , in line with the normal theory.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(606)
t, b = 5, 6                                # adhesives, boards
plan = np.array([rng.permutation(t) for _ in range(b)]) # which strip of each
board gets which adhesive
tau = np.array([0.0, 0.40, 0.65, 0.15, 0.80]) # adhesive effects (unknown in
practice)
board = rng.normal(0, 0.75, b)               # board effects
y = np.round(9 + tau[:, None] + board[None, :] + rng.normal(0, 0.3, (t, b)), 2)

grand = y.mean()
trt_mean, blk_mean = y.mean(axis=1), y.mean(axis=0)
resid = y - trt_mean[:, None] - blk_mean[None, :] + grand
ss_trt = b * np.sum((trt_mean - grand) ** 2)
ss_blk = t * np.sum((blk_mean - grand) ** 2)
ss_err = np.sum(resid ** 2)
df_err = (t - 1) * (b - 1)
ms_trt, ms_blk, ms_err = ss_trt / (t - 1), ss_blk / (b - 1), ss_err / df_err
F_trt = ms_trt / ms_err
print(f"treatments SS {ss_trt:.2f}, blocks SS {ss_blk:.2f}, error SS {ss_err:.2f} on
{df_err} df")
print(f"F for adhesives = {F_trt:.2f}, p = {stats.f.sf(F_trt, t - 1, df_err):.2g}")
print("adhesive means", trt_mean.round(2), "se of a difference", np.sqrt(2 * ms_err
/ b).round(3))
```

```
def f_stat(yy):
    tm, bm, gm = yy.mean(axis=1), yy.mean(axis=0), yy.mean()
    sst = b * np.sum((tm - gm) ** 2)
    sse = np.sum((yy - tm[:, None] - bm[None, :] + gm) ** 2)
    return (sst / (t - 1)) / (sse / df_err)
```



```

rng_perm = np.random.default_rng(7)
draws = 20_000
F_perm = np.empty(draws)
for r in range(draws):
    # re-randomize: permute the adhesive labels independently within every board
    perm = np.array([rng_perm.permutation(t) for _ in range(b)]).T
    F_perm[r] = f_stat(np.take_along_axis(y, perm, axis=0))
p_rand = (1 + np.sum(F_perm >= F_trt)) / (draws + 1)
print(f"randomization p-value {p_rand:.4f} (F-test p-value {stats.f.sf(F_trt, t -
    1, df_err):.2g})")

```

### 18.4.4 How much did blocking gain?

Had the adhesives been assigned to the thirty strips completely at random, the completely randomized design (CRD) would have had the board-to-board variation in its error. The block experiment itself can estimate how large that error would have been, using a fact about sampling from a finite population.

#### Lemma 18.4.3 — Sample variance without replacement

Let  $u_1, \dots, u_N$  be fixed numbers with mean  $\bar{u}$  and  $S^2 = \sum_a (u_a - \bar{u})^2 / (N - 1)$ . If  $\mathcal{S}$  is a simple random sample of  $m \geq 2$  of the indices, drawn without replacement, then

$$E \sum_{a \in \mathcal{S}} (u_a - \bar{u}_{\mathcal{S}})^2 = (m - 1)S^2,$$

where  $\bar{u}_{\mathcal{S}}$  is the sample mean.

*Proof.* For any  $m$  numbers,  $\sum_a (u_a - \bar{u}_{\mathcal{S}})^2 = \frac{1}{2m} \sum_{a,c} (u_a - u_c)^2$ , the sum being over ordered pairs of sample members. The  $m(m - 1)$  ordered pairs with  $a \neq c$  are each a uniformly random ordered pair of distinct population members. For such a pair,  $E(u_a - u_c)^2 = \frac{1}{N(N-1)} \sum_{a \neq c} (u_a - u_c)^2 = \frac{2N \sum_a (u_a - \bar{u})^2}{N(N-1)} = 2S^2$ . So the expectation is  $\frac{1}{2m} m(m - 1) 2S^2 = (m - 1)S^2$ . ■

#### Proposition 18.4.4 — What a completely randomized design would have given

Let the  $N = bt$  units follow Eq. 18.4.1, with fixed block effects  $\beta_j$ , and write  $S_{\beta}^2 = t \sum_j (\beta_j - \bar{\beta})^2 / (N - 1)$  for the variance of the block effects over the  $N$  units.

- (a) If instead the treatments were assigned to the same units completely at random,  $b$  units each, the within-treatment mean square of the resulting one-way analysis would have expectation  $\sigma^2 + S_{\beta}^2$ , over the errors and the randomization.
- (b) In the block design,

$$\hat{\sigma}_{\text{CR}}^2 = \frac{(b - 1)\text{MS}_B + b(t - 1)\text{MSE}}{bt - 1} \quad (18.4.2)$$

is unbiased for  $\sigma^2 + S_\beta^2$ .

*Proof.*

- (a) The  $b$  units given treatment  $i$  form a simple random sample of the  $N$  units. Their responses are  $\mu + \tau_i + \beta_{j(u)} + \varepsilon_u$ , so the within-treatment sum of squares of treatment  $i$  is that of the numbers  $\beta_{j(u)} + \varepsilon_u$  over the sample. Given the assignment, the errors contribute  $(b-1)\sigma^2$  in expectation, and the cross terms have mean zero. By [Lemma 18.4.3](#) the block effects contribute  $(b-1)S_\beta^2$  in expectation over the assignment. Summing over the  $t$  treatments and dividing by  $t(b-1)$  gives  $\sigma^2 + S_\beta^2$ .
- (b) By [Theorem 18.4.1](#)(d),  $E[(b-1)MS_B] = (b-1)\sigma^2 + t \sum_j (\beta_j - \bar{\beta})^2$  and  $E[b(t-1)MSE] = b(t-1)\sigma^2$ . The sum has expectation  $(bt-1)\sigma^2 + (N-1)S_\beta^2$ . ■

The **relative efficiency**  $\hat{\sigma}_{CR}^2 / \text{MSE}$  estimates how many units a CRD would need per unit of the block design for the same precision. Some authors multiply it by Fisher's factor  $(\nu_B + 1)(\nu_C + 3) / \{(\nu_B + 3)(\nu_C + 1)\}$ , with  $\nu_B = (t-1)(b-1)$  and  $\nu_C = t(b-1)$  the two error degrees of freedom, to allow for estimating  $\sigma^2$  with fewer degrees of freedom; this is a convention, not a theorem.

For the adhesives,

$$\hat{\sigma}_{CR}^2 = \frac{5 \times 0.8136 + 24 \times 0.0690}{29} = 0.1974,$$

and the relative efficiency is  $0.1974/0.0690 = 2.86$  (2.81 with Fisher's factor). Blocking was worth almost three times as many strips; without it the adhesive  $F$  would have been about  $1.0266/0.1974 = 5.20$ .

```
s2_crd = ((b - 1) * ms_blk + b * (t - 1) * ms_err) / (b * t - 1)  # estimated CRD
# error variance
re = s2_crd / ms_err
nu_b, nu_c = df_err, b * t - t                                     # error df: blocked,
# completely randomized
re_fisher = re * (nu_b + 1) * (nu_c + 3) / ((nu_b + 3) * (nu_c + 1))
F_crd = ms_trt / s2_crd                                           # what the treatment F would
# roughly have been
print(f"estimated CRD error variance {s2_crd:.2f} against {ms_err:.2f}:"
      f" relative efficiency {re:.2f} ({re_fisher:.2f} with Fisher's df factor)")
```

*Remark* (Fixed or random blocks). Treating the boards as a random sample of boards changes nothing about treatment comparisons in a complete block design, since block effects cancel from every contrast ([Theorem 18.4.1](#)(b)). It matters for incomplete blocks ([Section 18.5](#)) and for predicting a new block; [Chapter 32](#) treats such mixed models.

## 18.4.5 Exercises

### 18.4.5.1 A. Check your understanding

**Exercise A1.**

An experiment has  $t = 4$  treatments in  $b = 8$  blocks. Give the degrees of freedom of the block design's error, and of the error of a completely randomized design with the same 32 units. How large must the relative efficiency (without Fisher's factor) be for blocking to be worthwhile?

**Exercise A2.**

Verify from the table in [Example \(Five wood adhesives on six boards\)](#) that the treatment mean of adhesive 3 is 9.560, and compute the difference between adhesives 3 and 5 with its standard error. Is it significant by Tukey's criterion?

**18.4.5.2 B. Practice****Exercise B1.**

The strips cut from a board differ systematically: strips from the outer edge are stronger. How would you modify the design and analysis to remove this source of variation as well? Which design of the next section does your answer lead to?

**Exercise B2.**

In [Proposition 18.4.2](#), show that  $\text{Cov}(\sum_i c_i \bar{y}_i, \sum_i d_i \bar{y}_i) = S_W^2 \sum_i c_i d_i / b$  for two contrasts. Conclude that orthogonal contrasts are uncorrelated over the randomization, as they are under the normal model.

*Solution.* As in the proof,  $\text{Cov}(\sum_i c_i w_{ij}, \sum_k d_k w_{kj}) = \frac{W_j}{t} \sum_i c_i d_i - \frac{W_j}{t(t-1)} \sum_{i \neq k} c_i d_k$ , and  $\sum_{i \neq k} c_i d_k = (\sum_i c_i)(\sum_k d_k) - \sum_i c_i d_i = -\sum_i c_i d_i$ . So the covariance is  $W_j \sum_i c_i d_i / (t-1)$ . Summing over blocks and dividing by  $b^2$  gives  $S_W^2 \sum_i c_i d_i / b$ , which is zero for orthogonal contrasts.

**Exercise B3.**

Suppose the block effects are random,  $\beta_j \stackrel{\text{iid}}{\sim} N(0, \sigma_\beta^2)$ , independent of the errors. Show that  $E(\hat{\sigma}_{\text{CR}}^2) = \sigma^2 + \sigma_\beta^2 t(b-1)/(bt-1)$ , and interpret the factor  $t(b-1)/(bt-1)$ .

**18.4.5.3 C. Going deeper****Exercise C1.**

In [Eq. 18.4.1](#) let the block effects be random,  $\beta_j \stackrel{\text{iid}}{\sim} N(0, \sigma_\beta^2)$ , independent of the errors. Show that the treatment means  $\bar{y}_i$  are equicorrelated, with variance  $(\sigma^2 + \sigma_\beta^2)/b$  and covariance  $\sigma_\beta^2/b$ , while every contrast estimate keeps the distribution given by [Theorem 18.4.1\(b\)](#). Show that  $E(\text{MS}_B) = \sigma^2 + t\sigma_\beta^2$ , and that  $(\text{MS}_B + (t-1)\text{MSE})/(tb)$

is unbiased for  $\text{Var}(\bar{y}_{i.})$ . Why is  $s/\sqrt{b}$  the right scale for comparing treatments but the wrong standard error for a single treatment mean?

*Solution.* Averaging over blocks,  $\bar{y}_{i.} = \mu + \tau_i + \bar{\beta} + \bar{\varepsilon}_{i.}$ , where  $\bar{\beta} \sim N(0, \sigma_\beta^2/b)$  is common to all treatments and the  $\bar{\varepsilon}_{i.}$  are independent  $N(0, \sigma^2/b)$  and independent of  $\bar{\beta}$ . Hence  $\text{Var}(\bar{y}_{i.}) = (\sigma^2 + \sigma_\beta^2)/b$  and  $\text{Cov}(\bar{y}_{i.}, \bar{y}_{k.}) = \text{Var}(\bar{\beta}) = \sigma_\beta^2/b$  for  $i \neq k$ . In a contrast  $\bar{\beta}$  is multiplied by  $\sum_i c_i = 0$ , so  $\sum_i c_i \bar{y}_{i.} = \sum_i c_i \tau_i + \sum_i c_i \bar{\varepsilon}_{i.}$  exactly as with fixed blocks. The block means are  $\bar{y}_{.j} = \mu + \bar{\tau} + \beta_j + \bar{\varepsilon}_{.j}$ , and the  $\beta_j + \bar{\varepsilon}_{.j}$  are independent with variance  $\sigma_\beta^2 + \sigma^2/t$ , so  $E(\text{SS}_B) = t(b-1)(\sigma_\beta^2 + \sigma^2/t)$  and  $E(\text{MS}_B) = \sigma^2 + t\sigma_\beta^2$ . The error sum of squares is unchanged, because the residual projection annihilates every vector that is constant within blocks, so  $E(\text{MSE}) = \sigma^2$ . Then

$$E\left(\frac{\text{MS}_B + (t-1)\text{MSE}}{tb}\right) = \frac{t\sigma^2 + t\sigma_\beta^2}{tb} = \frac{\sigma^2 + \sigma_\beta^2}{b}.$$

A comparison of treatments uses only within-block information, in which  $\bar{\beta}$  cancels, so  $s/\sqrt{b}$  is its scale and the Tukey intervals of [Exercise \(B1\)](#) remain exact. A single treatment mean is a statement about the population of blocks, and its uncertainty includes the sampling of the  $b$  blocks:  $s^2/b$  misses the term  $\sigma_\beta^2/b$ . The estimate above mixes two mean squares with different degrees of freedom, so an interval built on it is only approximately  $t$ ; the proper treatment is the subject of [Chapter 32](#) ([Proposition 32.5.1](#)).

## 18.5 Latin squares and incomplete blocks

A **Latin square** removes two crossed sources of nuisance variation at once. A **balanced incomplete block design** allows blocks smaller than the number of treatments, at a computable price in precision. The Latin square is orthogonal; the incomplete block design is not, and its treatment comparisons are adjusted for blocks exactly as they were adjusted for a covariate in [Section 18.1](#).

### 18.5.1 Latin squares

#### Definition 18.5.1 — Latin square design

An  $a \times a$  **Latin square** is an arrangement of  $a$  symbols in an  $a \times a$  array such that each symbol occurs exactly once in each row and exactly once in each column. A **Latin square design** compares  $a$  treatments on  $a^2$  units classified by two crossed blocking factors with  $a$  levels each (rows and columns). The treatment applied to the unit in row  $i$  and column  $j$  is the symbol in cell  $(i, j)$  of a Latin square chosen at random.

The cyclic square  $k(i, j) = i + j$  modulo  $a$  exists for every  $a$ ; the usual randomization permutes its rows, columns and symbols at random, which makes every unit equally likely to receive every treatment. With  $k(i, j)$  the treatment in cell  $(i, j)$ , the model is additive in all three classifications:

$$y_{ij} = \mu + \rho_i + \kappa_j + \tau_{k(i,j)} + \varepsilon_{ij}, \quad \varepsilon_{ij} \stackrel{\text{iid}}{\sim} N(0, \sigma^2). \quad (18.5.1)$$

### Theorem 18.5.1 — Analysis of the Latin square

Assume Eq. 18.5.1 with  $a \geq 3$ . Write  $\bar{y}_{i\cdot}$ ,  $\bar{y}_{\cdot j}$  and  $\bar{y}^{(k)}$  for the means of row  $i$ , column  $j$  and treatment  $k$ , each an average of  $a$  observations, and  $\bar{y}$  for the grand mean.

- (a) The centred row, column and treatment spaces are mutually orthogonal. So  $\mathbf{I}$  is the sum of five mutually orthogonal projections: onto the mean, rows, columns, treatments (ranks 1,  $a - 1$ ,  $a - 1$ ,  $a - 1$ ) and error (rank  $(a - 1)(a - 2)$ ). The corresponding sums of squares are

$$\begin{aligned} SS_R &= a \sum_i (\bar{y}_{i\cdot} - \bar{y})^2, & SS_C &= a \sum_j (\bar{y}_{\cdot j} - \bar{y})^2, \\ SS_T &= a \sum_k (\bar{y}^{(k)} - \bar{y})^2, \end{aligned}$$

$$\text{and } SSE = \sum_{i,j} (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} - \bar{y}^{(k(i,j))} + 2\bar{y})^2.$$

- (b) Every contrast  $\sum_k c_k \tau_k$  is estimable, with least squares estimate  $\sum_k c_k \bar{y}^{(k)}$ , free of the row and column effects, and variance  $\sigma^2 \sum_k c_k^2 / a$ .
- (c) The treatment  $F$  statistic satisfies

$$F_T = \frac{SS_T / (a - 1)}{SSE / ((a - 1)(a - 2))} \sim F(a - 1, (a - 1)(a - 2), \gamma),$$

with

$$\gamma = \frac{a}{\sigma^2} \sum_k (\tau_k - \bar{\tau})^2.$$

*Proof.*

- (a) Take any two of the three factors, say rows and treatments. Each treatment occurs exactly once in each row, so the two-way table of counts has every entry 1, with all row and column totals  $a$ , and it satisfies the proportional frequencies condition Eq. 9.4.1. By Theorem 9.4.3 the two centred spaces are orthogonal. The same argument applies to rows and columns (each cell once) and to columns and treatments. By Theorem 6.3.8 the projection onto the model space is  $\mathbf{P}_0 + (\mathbf{P}_R - \mathbf{P}_0) + (\mathbf{P}_C - \mathbf{P}_0) + (\mathbf{P}_T - \mathbf{P}_0)$ , of rank  $1 + 3(a - 1)$ . The error projection is  $\mathbf{I}$  minus this, with rank  $a^2 - 3a + 2 = (a - 1)(a - 2)$ . Applying the projections to  $\mathbf{y}$  gives the stated sums of squares and the residual.
- (b) Treatment  $k$  occurs once in each row and once in each column, so averaging Eq. 18.5.1 over its  $a$  cells gives  $\bar{y}^{(k)} = \mu + \bar{\rho} + \bar{\kappa} + \tau_k + \bar{\varepsilon}^{(k)}$ . For a contrast the constant cancels. Estimability follows as in Theorem 18.4.1(b), and by (a) and Theorem 9.4.1 the least squares estimate is the one computed from the treatment means alone. (c) is Corollary 9.6.2. ■

The price is the error: two degrees of freedom for  $a = 3$  and six for  $a = 4$ , so small squares

are often replicated. The error consists of pieces of the row-by-column, row-by-treatment and column-by-treatment interactions, which cannot be told apart ([Exercise \(B2\)](#)). An interaction, such as a treatment working better in some rows, distorts the treatment comparisons: the design is only as good as its additivity assumption.

**Example. (Keyboard layouts, typists and sessions)**

In a synthetic experiment, five typists each type a standard passage in five sessions, comparing five keyboard layouts. Typists differ greatly, and speed rises with practice. A randomized  $5 \times 5$  Latin square gives each layout once to each typist and once in each session:

|          | S1 | S2 | S3 | S4 | S5 |
|----------|----|----|----|----|----|
| typist 1 | 2  | 3  | 1  | 4  | 5  |
| typist 2 | 3  | 5  | 4  | 2  | 1  |
| typist 3 | 5  | 1  | 2  | 3  | 4  |
| typist 4 | 4  | 2  | 5  | 1  | 3  |
| typist 5 | 1  | 4  | 3  | 5  | 2  |

The analysis of variance of the speeds (words per minute) is:

| source          | df | sum of squares | mean square | $F$  |
|-----------------|----|----------------|-------------|------|
| typists         | 4  | 689.2          | 172.3       | 82.7 |
| sessions        | 4  | 89.8           | 22.44       | 10.8 |
| layouts         | 4  | 73.6           | 18.40       | 8.83 |
| error           | 12 | 24.99          | 2.083       |      |
| corrected total | 24 | 877.6          |             |      |

The layouts differ ( $p = 0.0015$ ). Their means are 47.06, 50.10, 45.92, 50.30 and 48.94 words per minute, and a difference has standard error  $\sqrt{2 \times 2.083/5} = 0.913$ .

Had the sessions been ignored, with layouts randomized within typists, the error variance would have been about  $(MS_C + (a - 1)MSE)/a$  ([Exercise \(B1\)](#)). Relative to the Latin square this is

$$\frac{22.44 + 4 \times 2.083}{5 \times 2.083} = 2.95.$$

Ignoring the typists instead would have cost a factor of 17.35, and no blocking at all a factor of 16.25.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(62)
a = 5
cyclic = (np.arange(a)[: , None] + np.arange(a)[None, :]) % a      # a standard square
square = cyclic[rng.permutation(a)][: , rng.permutation(a)]      # permute rows and
    columns
square = rng.permutation(a)[square]                                # and relabel the
    layouts
typist = rng.normal(0, 8, a)                                       # typists differ a lot
session = np.array([-3.0, -1.0, 0.0, 1.5, 2.5])                  # practice effect over
    sessions
layout = np.array([0.0, 2.0, -1.0, 3.0, 1.0])
y = np.round(55 + typist[: , None] + session[None, :] + layout[square])
```

```

        + rng.normal(0, 1.5, (a, a)), 1)

grand = y.mean()
row_m, col_m = y.mean(axis=1), y.mean(axis=0)
trt_m = np.array([y[square == k].mean() for k in range(a)])
ss_row = a * np.sum((row_m - grand) ** 2)
ss_col = a * np.sum((col_m - grand) ** 2)
ss_trt = a * np.sum((trt_m - grand) ** 2)
ss_tot = np.sum((y - grand) ** 2)
ss_err = ss_tot - ss_row - ss_col - ss_trt
df_err = (a - 1) * (a - 2)
ms_err = ss_err / df_err
F_trt = ss_trt / (a - 1) / ms_err
print(f"SS rows {ss_row:.1f}, columns {ss_col:.1f}, layouts {ss_trt:.1f}, error {
    ss_err:.2f} on {df_err} df")
print(f"F for layouts = {F_trt:.2f}, p = {stats.f.sf(F_trt, a - 1, df_err):.2g}")
print("layout means", trt_m.round(2), " se of a difference", np.sqrt(2 * ms_err / a)
    .round(3))

```

Because the sessions are ordered in time, a layout may affect speed on the next one. Such **carryover** effects are not in [Eq. 18.5.1](#); *crossover designs*, such as Latin squares balanced for the preceding treatment, are built to estimate them.

## 18.5.2 Balanced incomplete block designs

Often a block cannot hold every treatment: a taster can judge only a few samples, a batch suffices for only a few runs. With blocks of size  $k < t$ , treatment comparisons are no longer free of block effects, and the fairest arrangement treats every pair alike.

### Definition 18.5.2 — Balanced incomplete block design

A **balanced incomplete block design** (BIBD) with parameters  $(t, b, r, k, \lambda)$  arranges  $t$  treatments in  $b$  blocks of size  $k < t$ , so that each block contains  $k$  distinct treatments, each treatment appears in  $r$  blocks, and each pair of distinct treatments appears together in exactly  $\lambda$  blocks. Its **incidence matrix** is the  $t \times b$  matrix  $\mathbf{N}$  with  $n_{ij} = 1$  if treatment  $i$  is in block  $j$  and 0 otherwise. Treatments are assigned to the units within each block at random, and the labels of treatments and blocks are randomized as well.

### Lemma 18.5.2 — Identities for balanced incomplete block designs

For a BIBD:

- (a)  $bk = tr$  and  $\lambda(t - 1) = r(k - 1)$ ;
- (b)  $\mathbf{N}\mathbf{N}^\top = (r - \lambda)\mathbf{I}_t + \lambda\mathbf{J}_t$ , where  $\mathbf{J}_t = \mathbf{1}\mathbf{1}^\top$ ;
- (c)  $r > \lambda$ , and  $b \geq t$  (**Fisher's inequality**).

*Proof.*

- (a) Both sides of the first identity count the units. For the second, fix a treatment  $i$ .

The  $r$  blocks containing it hold  $r(k-1)$  other units, and each of the other  $t-1$  treatments occurs in exactly  $\lambda$  of these blocks.

- (b) The  $(i, i')$  entry of  $\mathbf{NN}^\top$  counts the blocks containing both  $i$  and  $i'$ . It is  $r$  on the diagonal and  $\lambda$  off it.
- (c)  $k < t$  gives  $k-1 < t-1$ , so  $\lambda = r(k-1)/(t-1) < r$ . The matrix in (b) has eigenvalue  $r - \lambda + \lambda t$  on  $\mathbf{1}$  and  $r - \lambda$  on  $\mathbf{1}^\perp$ , all positive. So  $\mathbf{NN}^\top$  is nonsingular, and  $t = \text{rank}(\mathbf{NN}^\top) \leq \text{rank}(\mathbf{N}) \leq b$ .

■

The model is the additive block model, observed only on the occupied cells:

$$y_{ij} = \mu + \tau_i + \beta_j + \varepsilon_{ij} \quad \text{for the } bk \text{ pairs with } n_{ij} = 1. \quad (18.5.2)$$

Let  $T_i$  be the total of the observations on treatment  $i$ ,  $B_j$  the total of block  $j$ , and define the **adjusted treatment totals**

$$Q_i = T_i - \frac{1}{k} \sum_j n_{ij} B_j, \quad (18.5.3)$$

treatment  $i$ 's total minus the average level of the blocks it happened to be in. Then  $\sum_i Q_i = 0$ .

### Theorem 18.5.3 — Intra-block analysis of a balanced incomplete block design

Assume Eq. 18.5.2 for a BIBD, with  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ .

- (a)  $\sum_i c_i \tau_i$  is estimable iff  $\sum_i c_i = 0$ . Its least squares estimate and variance are

$$\sum_i c_i \hat{\tau}_i = \frac{k}{\lambda t} \sum_i c_i Q_i, \quad \text{Var}\left(\sum_i c_i \hat{\tau}_i\right) = \frac{k}{\lambda t} \sigma^2 \sum_i c_i^2. \quad (18.5.4)$$

- (b) The sum of squares for treatments adjusted for blocks is  $\text{SS}_{T|B} = \frac{k}{\lambda t} \sum_i Q_i^2$  on  $t-1$  degrees of freedom. The residual sum of squares is  $\text{SSE} = \text{SS}_{\text{tot}} - \text{SS}_B - \text{SS}_{T|B}$ , where  $\text{SS}_B = \sum_j B_j^2/k - (\sum_j B_j)^2/(bk)$  is the unadjusted block sum of squares, on  $bk - b - t + 1$  degrees of freedom.
- (c) Under normality,

$$F = \frac{\text{SS}_{T|B}/(t-1)}{\text{SSE}/(bk - b - t + 1)} \sim F(t-1, bk - b - t + 1, \gamma),$$

with

$$\gamma = \frac{\lambda t}{k \sigma^2} \sum_i (\tau_i - \bar{\tau})^2.$$

- (d) A randomized complete block design with  $r$  blocks estimates each contrast with variance  $\sigma^2 \sum_i c_i^2/r$ . The ratio of that variance to Eq. 18.5.4 is the **efficiency**



**factor**

$$E = \frac{\lambda t}{rk} = \frac{t(k-1)}{(t-1)k} < 1.$$

*Proof.* Let  $\mathbf{X}_B$  be the  $n \times b$  indicator matrix of blocks (with the intercept in its span) and  $\mathbf{Z}$  the  $n \times t$  indicator matrix of treatments, with  $n = bk$ . Let  $\mathbf{M}_B$  project onto  $\mathcal{C}(\mathbf{X}_B)$ , which replaces every observation by its block mean, and put  $\tilde{\mathbf{Z}} = (\mathbf{I} - \mathbf{M}_B)\mathbf{Z}$ . By Lemma 9.2.1,  $\mathcal{C}([\mathbf{X}_B, \mathbf{Z}]) = \mathcal{C}(\mathbf{X}_B) \oplus \mathcal{C}(\tilde{\mathbf{Z}})$ , orthogonally. This is the structure of Theorem 18.1.1 with the treatment indicators in the role of covariates, except that  $\tilde{\mathbf{Z}}$  is not of full rank:  $\mathbf{Z}\mathbf{1} = \mathbf{1}$  lies in  $\mathcal{C}(\mathbf{X}_B)$ , so  $\tilde{\mathbf{Z}}\mathbf{1} = \mathbf{0}$ .

The matrix  $\mathbf{C} = \tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} = \mathbf{Z}^\top \mathbf{Z} - \mathbf{Z}^\top \mathbf{M}_B \mathbf{Z}$  is the *information matrix* of the design. Here  $\mathbf{Z}^\top \mathbf{Z} = r\mathbf{I}$ , and  $\mathbf{Z}^\top \mathbf{M}_B \mathbf{Z} = \mathbf{N}\mathbf{N}^\top/k$ , because  $\mathbf{Z}^\top \mathbf{X}_B = \mathbf{N}$  and  $\mathbf{X}_B^\top \mathbf{X}_B = k\mathbf{I}$ . By Lemma 18.5.2(b) and (a),

$$\mathbf{C} = r\mathbf{I} - \frac{(r-\lambda)\mathbf{I} + \lambda\mathbf{J}}{k} = \frac{r(k-1) + \lambda}{k}\mathbf{I} - \frac{\lambda}{k}\mathbf{J} = \frac{\lambda t}{k}\left(\mathbf{I} - \frac{1}{t}\mathbf{J}\right).$$

Also  $\tilde{\mathbf{Z}}^\top \mathbf{y} = \mathbf{Z}^\top \mathbf{y} - \mathbf{Z}^\top \mathbf{M}_B \mathbf{y} = \mathbf{Q}$ , since  $\mathbf{Z}^\top \mathbf{y}$  is the vector of treatment totals and  $\mathbf{Z}^\top \mathbf{M}_B \mathbf{y}$  has entries  $\sum_j n_{ij} B_j/k$ .

- (a) A vector  $(\mathbf{0}, \mathbf{c})$ , zero on the block and intercept parameters, is estimable iff  $\mathbf{c}^\top = \mathbf{a}^\top \mathbf{Z}$  for some  $\mathbf{a}$  with  $\mathbf{a}^\top \mathbf{X}_B = \mathbf{0}$  (Theorem 8.2.2). Such an  $\mathbf{a}$  satisfies  $\mathbf{a} = (\mathbf{I} - \mathbf{M}_B)\mathbf{a}$ , so  $\mathbf{c} \in \mathcal{C}(\tilde{\mathbf{Z}}^\top) = \mathcal{C}(\mathbf{C})$ , which is the space of contrasts because  $\mathbf{C}$  is a positive multiple of the centring projection. Conversely, for a contrast  $\mathbf{c} = \mathbf{C}\mathbf{h}$ , the vector  $\mathbf{a} = \tilde{\mathbf{Z}}\mathbf{h}$  works. Its least squares estimate is  $\mathbf{a}^\top \mathbf{y} = \mathbf{h}^\top \mathbf{Q}$ . With the generalized inverse  $\mathbf{C}^- = \frac{k}{\lambda t}\mathbf{I}$ , which satisfies  $\mathbf{C}\mathbf{C}^-\mathbf{C} = \mathbf{C}$  because  $\mathbf{I} - \mathbf{J}/t$  is idempotent, we have  $\mathbf{C}\mathbf{C}^-\mathbf{c} = \mathbf{c}$  for a contrast, so we can take  $\mathbf{h} = \mathbf{C}^-\mathbf{c}$ , and the estimate is  $\frac{k}{\lambda t}\mathbf{c}^\top \mathbf{Q}$ . Its variance is  $\sigma^2 \|\tilde{\mathbf{Z}}\mathbf{h}\|^2 = \sigma^2 \mathbf{h}^\top \mathbf{C}\mathbf{h} = \sigma^2 \mathbf{c}^\top \mathbf{C}^-\mathbf{c} = \frac{k}{\lambda t}\sigma^2 \|\mathbf{c}\|^2$ .
- (b) The extra sum of squares for treatments after blocks is the squared length of the projection onto  $\mathcal{C}(\tilde{\mathbf{Z}})$ ,  $\mathbf{y}^\top \tilde{\mathbf{Z}}\mathbf{C}^-\tilde{\mathbf{Z}}^\top \mathbf{y} = \mathbf{Q}^\top \mathbf{C}^-\mathbf{Q} = \frac{k}{\lambda t} \sum_i Q_i^2$  (Lemma 9.2.1). The block sum of squares is  $\|(\mathbf{M}_B - \mathbf{P}_0)\mathbf{y}\|^2$ , and the block, adjusted treatment and residual projections are mutually orthogonal and sum to  $\mathbf{I} - \mathbf{P}_0$  (Theorem 6.3.8). The model has rank  $b + t - 1$ .
- (c) is Theorem 11.1.1. The noncentrality is the same quadratic form evaluated at the mean, where  $\tilde{\mathbf{Z}}^\top \boldsymbol{\mu} = \mathbf{C}\boldsymbol{\tau}$ , giving  $\boldsymbol{\tau}^\top \mathbf{C}\mathbf{C}^-\mathbf{C}\boldsymbol{\tau}/\sigma^2 = \frac{\lambda t}{k} \sum_i (\tau_i - \bar{\tau})^2/\sigma^2$ .
- (d) The ratio is  $(1/r)/(k/(\lambda t))$ , and  $\lambda = r(k-1)/(t-1)$  by Lemma 18.5.2(a). Since  $k < t$ ,  $(k-1)/k < (t-1)/t$ .

■

The adjusted total  $Q_i$  compares treatment  $i$  with the blocks it appeared in, and  $E Q_i = \frac{\lambda t}{k}(\tau_i - \bar{\tau})$  with no  $\beta$ 's left (Exercise B3): this is the **intra-block analysis**. The efficiency factor is the fraction of information that survives the elimination of blocks: about one half for  $k = 2$ , close to one for  $k = t - 1$ . Smaller blocks are often more homogeneous, and the smaller  $\sigma^2$  can more than make up for  $E$ . Every pair sharing a block makes the design connected (Theorem 8.6.1), and balance gives every  $\tau_i - \tau_{i'}$  the same variance  $2k\sigma^2/(\lambda t)$ . Unbalanced connected designs are analysed the same way with a different information matrix  $\mathbf{C}$ .

**Example. (Seven recipes, fourteen tasters)**

In a synthetic tasting experiment, seven recipes of a sports drink are rated on a 0–10 scale by 14 tasters, each of whom can judge only three. The blocks are the lines of two Fano planes, the cyclic sets  $\{0, 1, 3\} + s$  and  $\{0, 1, 5\} + s$  modulo 7 for  $s = 0, \dots, 6$  (recipes numbered  $0, \dots, 6$  in this description and  $1, \dots, 7$  below). This is a BIBD with  $t = 7$ ,  $b = 14$ ,  $k = 3$ ,  $r = 6$  and  $\lambda = 2$ . There are 42 ratings and 22 error degrees of freedom.

Tasters use the scale very differently: the unadjusted block sum of squares is 69.72 out of a corrected total of 80.71. The raw recipe means, 6.15, 6.10, 5.40, 6.60, 6.30, 6.85 and 6.02, therefore mix recipe quality with the generosity of the tasters each recipe met. The adjusted totals are  $Q = (-1.30, 0.90, -3.43, 3.13, 0.43, -1.50, 1.77)$ , and the adjusted means  $\bar{y} + \frac{k}{\lambda t} Q_i$  are 5.92, 6.40, 5.47, 6.87, 6.30, 5.88 and 6.58. Recipe 6 had the highest raw mean. Its adjusted mean is

$$6.202 + \frac{3}{14} \times (-1.50) = 5.88,$$

among the lowest: recipe 6 happened to meet generous tasters, and the adjustment removes 0.97 from its mean.

The adjusted recipe sum of squares is 6.36, and the residual sum of squares is 4.64, so

$$F = \frac{6.36/6}{4.64/22} = 5.03$$

on 6 and 22 degrees of freedom ( $p = 0.0022$ ). A difference of two recipes has standard error  $\sqrt{2 \times 3 \times 0.2108/14} = 0.301$ . A complete block design with six tasters tasting all seven recipes, with the same  $\sigma^2$ , would give 0.265; the variance ratio is the efficiency factor  $7 \times 2/(6 \times 3) = 0.7778$ . A taster judging seven drinks would probably not rate as consistently, though.

```
import numpy as np
from scipy import stats

t, k = 7, 3
blocks = [sorted((d + s) % t for d in base) for base in ([0, 1, 3], [0, 1, 5]) for s
           in range(t)]
b = len(blocks)                                # 14 tasters
N = np.zeros((t, b), dtype=int)                # incidence: recipe i tasted
                                           by taster j
for j, blk in enumerate(blocks):
    N[blk, j] = 1
r = N.sum(axis=1)[0]
lam = (N @ N.T)[0, 1]

rng = np.random.default_rng(77)
tau = np.array([0.0, 0.4, -0.3, 0.8, 0.1, -0.6, 0.5]) # recipe effects
taster = rng.normal(0, 1.2, b)                    # tasters use the scale
                                           differently
rows = [(i, j) for j, blk in enumerate(blocks) for i in blk]
recipe = np.array([i for i, j in rows])
block = np.array([j for i, j in rows])
y = np.round(6 + tau[recipe] + taster[block] + rng.normal(0, 0.5, len(rows)), 1)
n = len(y)
```

```

T_tot = np.bincount(recipe, weights=y, minlength=t)           # recipe totals
B_tot = np.bincount(block, weights=y, minlength=b)           # taster totals
Q = T_tot - N @ B_tot / k                                     # adjusted recipe
    totals
tau_hat = k * Q / (lam * t)                                    # intra-block estimates
    (sum to zero)
ss_trt_adj = k / (lam * t) * np.sum(Q ** 2)
ss_blk = np.sum(B_tot ** 2) / k - y.sum() ** 2 / n           # blocks, ignoring
    recipes
ss_tot = np.sum((y - y.mean()) ** 2)
ss_err = ss_tot - ss_blk - ss_trt_adj
df_err = n - b - t + 1
ms_err = ss_err / df_err
F = ss_trt_adj / (t - 1) / ms_err
se_diff = np.sqrt(2 * k * ms_err / (lam * t))
eff = lam * t / (r * k)                                       # efficiency factor
print("raw recipe means      ", (T_tot / r).round(2))
print("adjusted recipe means", (y.mean() + tau_hat).round(2))
print(f"adjusted recipe SS {ss_trt_adj:.3f}, error SS {ss_err:.3f} on {df_err} df,"
      f" F = {F:.2f}, p = {stats.f.sf(F, t - 1, df_err):.2g}")
print(f"se of a difference {se_diff:.3f}; efficiency factor {eff:.4f}")

```

*Remark* (Recovering inter-block information). The block totals, with  $E B_j = k\mu + \sum_i n_{ij}\tau_i + k\beta_j$ , also carry treatment information. With random block effects they give an independent *inter-block* estimate of every contrast, whose variance grows with  $\sigma_\beta^2$ . Combining the two estimates by inverse-variance weights, as Yates proposed, is generalized least squares in a mixed model (Chapter 32). With blocks as different as these tasters, the inter-block estimate adds little.

## 18.5.3 Exercises

### 18.5.3.1 A. Check your understanding

#### Exercise A1.

Show that the cyclic array  $k(i, j) = i + j \bmod a$  is a Latin square, and that permuting its rows, its columns, or its symbols gives a Latin square. For  $a = 3$ , how many distinct Latin squares are there, and does the randomization described above reach all of them?

#### Exercise A2.

Check the identities of Lemma 18.5.2 for the design of Example (Seven recipes, fourteen tasters). Is there a BIBD with  $t = 7$ ,  $k = 3$  and  $b = 7$ ? With  $t = 6$ ,  $k = 3$ ,  $b = 6$ ?

#### Exercise A3.

From the numbers in Example (Seven recipes, fourteen tasters), compute the estimate

of  $\tau_4 - \tau_3$  and its standard error, and decide whether recipe 4 is better than recipe 3 at level 0.05, considered alone.

### 18.5.3.2 B. Practice

#### Exercise B1.

In a Latin square, suppose the columns had not been used as a blocking factor, so that the treatments were randomized within rows only. Following [Proposition 18.4.4](#), show that the expected error mean square of that design is  $\sigma^2 + \sum_j (\kappa_j - \bar{\kappa})^2 / (a - 1)$ , and that  $(\text{MS}_C + (a - 1)\text{MSE})/a$  is unbiased for it in the Latin square.

*Solution.* Randomizing within rows, the units of row  $i$  are assigned to the treatments by a random permutation. Within a row, the column effects  $\kappa_1, \dots, \kappa_a$  play the role of the unit values, with within-row variance  $S_\kappa^2 = \sum_j (\kappa_j - \bar{\kappa})^2 / (a - 1)$ , the same for every row. By [Proposition 18.4.2\(b\)](#), applied to the unit values  $\kappa_j + \varepsilon_{ij}$  and averaged over the errors, the expected error mean square of the row-blocked design is  $\sigma^2 + S_\kappa^2$ . In the Latin square,  $\text{E MS}_C = \sigma^2 + a S_\kappa^2$  and  $\text{E MSE} = \sigma^2$  ([Theorem 9.6.1](#)), so  $\text{E}\{\text{MS}_C + (a - 1)\text{MSE}\}/a = \sigma^2 + S_\kappa^2$ .

#### Exercise B2.

In a  $4 \times 4$  Latin square, the row-by-column interaction space of the 16 cells has dimension 9. Show that the treatment space (centred) lies inside it, and that the error space is its orthogonal complement within it, of dimension 6. What does this imply about the effect of a row-by-column interaction on the treatment comparisons?

*Solution.* The row-by-column interaction space is the orthogonal complement, within  $\mathbb{R}^{16}$ , of the span of the mean, row and column indicators; it has dimension  $16 - 1 - 3 - 3 = 9$ . By [Theorem 18.5.1\(a\)](#) the centred treatment space is orthogonal to the mean, row and column spaces, so it lies in this complement, with dimension 3. The error space is the orthogonal complement of the mean, row, column and treatment spaces, so it is the rest of the interaction space, of dimension 6. A row-by-column interaction in the mean vector is a vector of the 9-dimensional space, and its component in the treatment space is indistinguishable from a treatment effect. Unless the interaction happens to be orthogonal to the treatment space, it biases the treatment comparisons.

#### Exercise B3.

Show that  $\text{E } Q_i = \frac{\lambda t}{k}(\tau_i - \bar{\tau})$  in a BIBD, directly from [Eq. 18.5.2](#) and [Lemma 18.5.2](#).

*Solution.*  $\text{E } T_i = r\mu + r\tau_i + \sum_j n_{ij}\beta_j$  and  $\text{E } B_j = k\mu + \sum_{i'} n_{i'j}\tau_{i'} + k\beta_j$ . So

$$\text{E } Q_i = r\tau_i - \frac{1}{k} \sum_j \sum_{i'} n_{ij} n_{i'j} \tau_{i'} = r\tau_i - \frac{1}{k} (r\tau_i + \lambda \sum_{i' \neq i} \tau_{i'}),$$

the  $\mu$ 's and  $\beta$ 's cancelling. Writing  $\sum_{i' \neq i} \tau_{i'} = t\bar{\tau} - \tau_i$  gives  $\text{E } Q_i = \tau_i(r - (r - \lambda)/k) - \lambda t\bar{\tau}/k = \frac{\lambda t}{k}(\tau_i - \bar{\tau})$ , using  $r(k - 1) + \lambda = \lambda t$ .

**Exercise B4.**

A **Graeco-Latin square** superimposes two Latin squares of the same order so that every ordered pair of symbols occurs exactly once. Write down one of order 3, state the model with rows, columns and two treatment factors, and give the analysis of variance table with its degrees of freedom. How many error degrees of freedom remain for  $a = 3, 4$  and  $5$ ?

**18.5.3.3 C. Going deeper****Exercise C1.**

For a general incomplete block design with incidence matrix  $\mathbf{N}$ , replication  $r_i$  and block sizes  $k_j$ , show that the information matrix is  $\mathbf{C} = \text{diag}(r_i) - \mathbf{N} \text{diag}(1/k_j) \mathbf{N}^\top$ , that  $\mathbf{C}\mathbf{1} = \mathbf{0}$ , and that every treatment contrast is estimable iff  $\text{rank}(\mathbf{C}) = t - 1$ . Relate this to the connectedness of [Theorem 8.6.1](#).

## 18.6 Missing observations in designed experiments

Designed experiments lose observations: a plot floods, a sample is dropped, a strip of timber splits. The balanced formulas of the last two sections then no longer apply. The correct response is simple: the model still holds for the remaining observations, so fit it to them and test by comparing models ([Chapter 11](#)). Before computers, experimenters instead *filled in the missing value so that the balanced formulas give the exact least squares answer*. The trick shows exactly what the balanced formulas get right and wrong with a filled-in value, and it is the simplest instance of the EM algorithm.

### 18.6.1 Filling in, exactly

Let  $\mathbf{X}$  be the  $n \times p$  model matrix of the complete design, of rank  $r$ , and suppose the observations in a set of  $m$  rows are missing. Reorder so that  $\mathbf{X} = \begin{pmatrix} \mathbf{X}_o \\ \mathbf{X}_m \end{pmatrix}$  and  $\mathbf{y} = \begin{pmatrix} \mathbf{y}_o \\ \cdot \end{pmatrix}$ , with  $\mathbf{y}_o$  the  $n - m$  observed values. Let  $\mathbf{M}$  be the projection onto  $\mathcal{C}(\mathbf{X})$  for the *complete* design, partitioned conformably into blocks  $\mathbf{M}_{oo}, \mathbf{M}_{om}, \mathbf{M}_{mo}, \mathbf{M}_{mm}$ , and let  $\mathbf{D} = \begin{pmatrix} \mathbf{0} \\ \mathbf{I}_m \end{pmatrix}$  be the  $n \times m$  matrix of indicators of the missing rows. The key assumption is that the loss does not destroy estimability:

$$\text{rank}(\mathbf{X}_o) = \text{rank}(\mathbf{X}) = r. \quad (18.6.1)$$

Since  $\mathcal{N}(\mathbf{X}) \subseteq \mathcal{N}(\mathbf{X}_o)$  always, [Eq. 18.6.1](#) says that  $\mathcal{N}(\mathbf{X}_o) = \mathcal{N}(\mathbf{X})$ : every function estimable in the complete design remains estimable. A single missing observation in a randomized block design with  $t, b \geq 2$ , or in a Latin square with  $a \geq 3$ , never violates it. Losing every observation on some treatment does, and so does losing the only observation of a cell in an unreplicated factorial fitted with all its interactions.

**Theorem 18.6.1 — Missing observations**

Assume Eq. 18.6.1, and let  $\hat{\beta}_o$  be any least squares estimate from the observed data alone, that is, for the model  $E(\mathbf{Y}_o) = \mathbf{X}_o\beta$ . Put  $\mathbf{z}^* = \mathbf{X}_m\hat{\beta}_o$ , the fitted values at the missing positions, and for  $\mathbf{z} \in \mathbb{R}^m$  let  $\mathbf{y}(\mathbf{z}) = \begin{pmatrix} \mathbf{y}_o \\ \mathbf{z} \end{pmatrix}$  be the data completed with  $\mathbf{z}$ .

- (a) (*Filling in.*)  $\hat{\beta}_o$  is a least squares estimate for the complete data  $\mathbf{y}(\mathbf{z}^*)$ , and the residual sum of squares of the complete-data analysis of  $\mathbf{y}(\mathbf{z}^*)$  equals  $\text{SSE}_o$ , the residual sum of squares of the observed data.
- (b) (*Characterization.*) The eigenvalues of  $\mathbf{M}_{mm}$  lie in  $[0, 1)$ , and  $\mathbf{z}^*$  is the unique solution of

$$\mathbf{z} = \mathbf{M}_{mo}\mathbf{y}_o + \mathbf{M}_{mm}\mathbf{z}, \quad (18.6.2)$$

that is, the unique  $\mathbf{z}$  that equals its own complete-data fitted value.

- (c) (*Iteration.*) For any starting value  $\mathbf{z}_0$ , the sequence  $\mathbf{z}_{s+1} = \mathbf{M}_{mo}\mathbf{y}_o + \mathbf{M}_{mm}\mathbf{z}_s$ , which fills in the current values, fits the complete-data model and replaces the missing values by their fitted values, converges to  $\mathbf{z}^*$ . The error  $\mathbf{z}_s - \mathbf{z}^*$  equals  $\mathbf{M}_{mm}^s(\mathbf{z}_0 - \mathbf{z}^*)$ .
- (d) (*Covariate method.*) For any  $\mathbf{w} \in \mathbb{R}^m$ , the model  $E \mathbf{y}(\mathbf{w}) = \mathbf{X}\beta + \mathbf{D}\gamma$  has rank  $r + m$ . Its least squares fit has  $\mathbf{X}_o\hat{\beta} = \mathbf{X}_o\hat{\beta}_o$ ,  $\hat{\gamma} = \mathbf{w} - \mathbf{z}^*$ , and residual sum of squares  $\text{SSE}_o$ , on  $n - m - r$  degrees of freedom.
- (e) (*Tests.*) Let  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  with  $\text{rank}(\mathbf{X}_{0,o}) = \text{rank}(\mathbf{X}_0)$ . The correct sum of squares for the reduced model is  $\text{SSE}_{0,o} - \text{SSE}_o$ , the difference of the residual sums of squares of the two models fitted to the observed data, on  $r - r_0$  degrees of freedom, with error degrees of freedom  $n - m - r$ . The sum of squares computed from the complete-data formulas applied to  $\mathbf{y}(\mathbf{z}^*)$  is at least as large.

*Proof.*

- (a) For any  $\mathbf{b}$ , the complete-data criterion splits as

$$\begin{aligned} \|\mathbf{y}(\mathbf{z}^*) - \mathbf{X}\mathbf{b}\|^2 &= \|\mathbf{y}_o - \mathbf{X}_o\mathbf{b}\|^2 + \|\mathbf{z}^* - \mathbf{X}_m\mathbf{b}\|^2 \\ &\geq \|\mathbf{y}_o - \mathbf{X}_o\mathbf{b}\|^2 \geq \text{SSE}_o. \end{aligned}$$

At  $\mathbf{b} = \hat{\beta}_o$  both inequalities are equalities, because  $\mathbf{z}^* = \mathbf{X}_m\hat{\beta}_o$  and  $\hat{\beta}_o$  minimizes the observed-data criterion. So  $\hat{\beta}_o$  minimizes the complete-data criterion, and the minimum is  $\text{SSE}_o$ .

- (b) By (a), the complete-data fitted vector of  $\mathbf{y}(\mathbf{z}^*)$  is  $\mathbf{X}\hat{\beta}_o$ , whose missing part is  $\mathbf{z}^*$ . The fitted vector is  $\mathbf{M}\mathbf{y}(\mathbf{z}^*)$ , whose missing part is  $\mathbf{M}_{mo}\mathbf{y}_o + \mathbf{M}_{mm}\mathbf{z}^*$ . So  $\mathbf{z}^*$  solves Eq. 18.6.2. For uniqueness, note that  $\mathbf{M}_{mm} = \mathbf{D}^\top\mathbf{M}\mathbf{D}$  is symmetric and nonnegative definite, and for  $\mathbf{u} \in \mathbb{R}^m$ ,

$$\mathbf{u}^\top\mathbf{M}_{mm}\mathbf{u} = \|\mathbf{M}\mathbf{D}\mathbf{u}\|^2 \leq \|\mathbf{D}\mathbf{u}\|^2 = \|\mathbf{u}\|^2,$$

with equality iff  $\mathbf{D}\mathbf{u} \in \mathcal{C}(\mathbf{X})$ , that is, iff  $\mathbf{D}\mathbf{u} = \mathbf{X}\mathbf{b}$  for some  $\mathbf{b}$ . Such a  $\mathbf{b}$  has  $\mathbf{X}_o\mathbf{b} = \mathbf{0}$  and  $\mathbf{X}_m\mathbf{b} = \mathbf{u}$ . Under Eq. 18.6.1,  $\mathbf{X}_o\mathbf{b} = \mathbf{0}$  implies  $\mathbf{X}\mathbf{b} = \mathbf{0}$ , so  $\mathbf{u} = \mathbf{0}$ . Hence  $\mathbf{u}^\top\mathbf{M}_{mm}\mathbf{u} < \|\mathbf{u}\|^2$  for  $\mathbf{u} \neq \mathbf{0}$ , every eigenvalue of  $\mathbf{M}_{mm}$  is less than 1, and

$\mathbf{I} - \mathbf{M}_{mm}$  is nonsingular. Equation Eq. 18.6.2 is  $(\mathbf{I} - \mathbf{M}_{mm})\mathbf{z} = \mathbf{M}_{mo}\mathbf{y}_o$ , which has exactly one solution.

- (c) Subtracting Eq. 18.6.2 for  $\mathbf{z}^*$  from the recursion gives  $\mathbf{z}_{s+1} - \mathbf{z}^* = \mathbf{M}_{mm}(\mathbf{z}_s - \mathbf{z}^*)$ . The spectral radius of the symmetric matrix  $\mathbf{M}_{mm}$  is its largest eigenvalue, which is less than 1 by (b), so  $\mathbf{M}_{mm}^s \rightarrow \mathbf{0}$ .
- (d) The argument of (b) shows that  $\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{D}) = \{\mathbf{0}\}$ , so  $\text{rank}[\mathbf{X}, \mathbf{D}] = r + m$ . For any  $\mathbf{b}$  and  $\mathbf{g}$ ,

$$\|\mathbf{y}(\mathbf{w}) - \mathbf{X}\mathbf{b} - \mathbf{D}\mathbf{g}\|^2 = \|\mathbf{y}_o - \mathbf{X}_o\mathbf{b}\|^2 + \|\mathbf{w} - \mathbf{X}_m\mathbf{b} - \mathbf{g}\|^2.$$

For each  $\mathbf{b}$  the second term is minimized, at zero, by  $\mathbf{g} = \mathbf{w} - \mathbf{X}_m\mathbf{b}$ . What remains is the observed-data criterion, so the minimizing  $\mathbf{b}$  are the observed-data least squares estimates,  $\mathbf{X}_o\hat{\boldsymbol{\beta}} = \mathbf{X}_o\hat{\boldsymbol{\beta}}_o$ , and the minimum is  $\text{SSE}_o$ . Since  $\mathbf{X}_m\mathbf{b}$  is the same for every least squares estimate (by Eq. 18.6.1, every row of  $\mathbf{X}$  is an estimable function in the observed-data model),  $\hat{\boldsymbol{\gamma}} = \mathbf{w} - \mathbf{X}_m\hat{\boldsymbol{\beta}}_o = \mathbf{w} - \mathbf{z}^*$ .

- (e) The first statement is Theorem 11.1.1 for the observed-data model, of rank  $r$  with  $n - m$  observations. For the second, the complete-data reduced-model fit of  $\mathbf{y}(\mathbf{z}^*)$  has residual sum of squares  $\min_{\mathbf{b}} (\|\mathbf{y}_o - \mathbf{X}_{0,o}\mathbf{b}\|^2 + \|\mathbf{z}^* - \mathbf{X}_{0,m}\mathbf{b}\|^2) \geq \text{SSE}_{0,o}$ , while by (a) the complete-data full-model fit has residual sum of squares exactly  $\text{SSE}_o$ . Their difference is therefore at least  $\text{SSE}_{0,o} - \text{SSE}_o$ .

■

Part (a) is the classical claim. Part (e) gives the two classical corrections. The residual sum of squares is right, but it has  $n - m - r$  degrees of freedom, not the  $n - r$  that the complete-data formulas assign, since the filled-in values fit perfectly. And the treatment sum of squares is biased upward, because the filled-in value suits the full model and the reduced model cannot choose its own. Part (d) is Theorem 18.1.1 with indicator covariates: each gap gets its own parameter, the model fits it exactly and learns nothing from it, and the indicator's coefficient measures how far the number in the gap was from the least squares value.

*Idea.* A missing observation can be filled in with the value that the model, fitted to the remaining data, predicts for it. Then the balanced analysis of the completed data gives the exact least squares estimates and residual sum of squares. Subtract one error degree of freedom for each filled-in value, and compute tests by fitting the reduced model to the observed data.

## 18.6.2 The randomized block design with one missing value

For the randomized complete block design, Eq. 18.6.2 can be solved by hand. The projection of the complete design is  $\mathbf{M} = \mathbf{P}_T + \mathbf{P}_B - \mathbf{P}_0$  (Example (Additive two-factor model, one observation per cell)), so the entry of  $\mathbf{M}$  for two units is

$$m_{uv} = \frac{[\text{same treatment}]}{b} + \frac{[\text{same block}]}{t} - \frac{1}{bt},$$

where  $[\cdot]$  is 1 if the condition holds and 0 otherwise.

**Corollary 18.6.2 — Yates's missing value formula**

In a randomized complete block design with  $t$  treatments and  $b$  blocks, suppose the observation on treatment  $i$  in block  $j$  is missing. Let  $T'$  and  $B'$  be the totals of the observed values of treatment  $i$  and of block  $j$ , and  $G'$  the total of all observed values. The least squares estimate of the missing value is

$$z^* = \frac{tT' + bB' - G'}{(t-1)(b-1)}. \quad (18.6.3)$$

The iteration of [Theorem 18.6.1\(c\)](#) reduces the error by the factor  $1/b + 1/t - 1/(bt)$  at every step.

*Proof.* Here  $m = 1$ , and  $\mathbf{M}_{mm} = 1/b + 1/t - 1/(bt)$ , so  $1 - \mathbf{M}_{mm} = (b-1)(t-1)/(bt)$ . The row of  $\mathbf{M}$  belonging to the missing unit, restricted to the observed units, sums the observed values of treatment  $i$  with weight  $1/b - 1/(bt)$ , those of block  $j$  with weight  $1/t - 1/(bt)$ , and all others with weight  $-1/(bt)$ . No observed unit shares both the treatment and the block. So  $\mathbf{M}_{mo}\mathbf{y}_o = T'/b + B'/t - G'/(bt)$ . Solving [Eq. 18.6.2](#),

$$z^* = \frac{bt}{(b-1)(t-1)} \left( \frac{T'}{b} + \frac{B'}{t} - \frac{G'}{bt} \right) = \frac{tT' + bB' - G'}{(t-1)(b-1)}.$$

The rate is  $\mathbf{M}_{mm}$  by [Theorem 18.6.1\(c\)](#). ■

**Example. (A strip that split)**

Suppose that in [Example \(Five wood adhesives on six boards\)](#) the strip from board 4 for adhesive 2 had split while being cut, so its strength (in fact 8.62) was never measured. The observed totals are  $T' = 45.33$  for adhesive 2,  $B' = 36.25$  for board 4, and  $G' = 263.95$  overall. Yates's formula gives

$$z^* = \frac{5 \times 45.33 + 6 \times 36.25 - 263.95}{4 \times 5} = 9.010.$$

The iteration started from the mean of the observed values, 9.102, gives 9.041, 9.020, 9.013, .... The distance to 9.010 shrinks by the factor  $1/6 + 1/5 - 1/30 = 1/3$  at each step. Filling the gap with 0 and adding an indicator column gives  $\hat{\gamma} = -9.010$ , so the covariate method recovers the same value.

The completed table gives residual sum of squares 1.2788, exactly that of the 29 observed values, but on 19 degrees of freedom: the error mean square is  $1.2788/19 = 0.0673$ , not 0.0639. Its adhesive sum of squares, 4.0534, exceeds the correct 4.0512 by 0.0022. The correct  $F$  is 15.05 on 4 and 19 degrees of freedom; the naive analysis would report 15.85, mostly because of the degrees of freedom.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(606)
```



```

t, b = 5, 6                                # adhesives, boards
plan = np.array([rng.permutation(t) for _ in range(b)]) # which strip of each
board gets which adhesive
tau = np.array([0.0, 0.40, 0.65, 0.15, 0.80])    # adhesive effects (unknown in
practice)
board = rng.normal(0, 0.75, b)                  # board effects
y = np.round(9 + tau[:, None] + board[None, :] + rng.normal(0, 0.3, (t, b)), 2)

n = t * b
trt = np.repeat(np.arange(t), b)                # y.ravel() lists adhesive 1
first
blk = np.tile(np.arange(b), t)
X = np.column_stack([np.ones(n), np.eye(t)[trt], np.eye(b)[blk]]) # complete-data
design
miss = np.array([1 * b + 3])                    # adhesive 2, board 4
obs = np.setdiff1d(np.arange(n), miss)
yv = y.ravel().copy()
yv[miss] = np.nan                               # the value is lost

def fit(Xm, yy):
    coef = np.linalg.lstsq(Xm, yy, rcond=None)[0]
    return coef, np.sum((yy - Xm @ coef) ** 2)

coef_obs, sse_obs = fit(X[obs], yv[obs])          # exact least squares, 29
observations
df_obs = len(obs) - np.linalg.matrix_rank(X[obs]) # 29 - 10 = 19
X_red = np.column_stack([np.ones(n), np.eye(b)[blk]]) # blocks only
_, sse_red = fit(X_red[obs], yv[obs])
F_exact = ((sse_red - sse_obs) / (t - 1)) / (sse_obs / df_obs)
print(f"exact: SSE {sse_obs:.3f} on {df_obs} df, adhesive SS {sse_red - sse_obs:.3f}
",
      f" F = {F_exact:.2f}; fitted value for the lost cell {X[miss] @ coef_obs}")

```

```

Y_obs = np.where(np.isnan(yv), 0.0, yv).reshape(t, b)
T_i = Y_obs[1].sum()                          # observed total of adhesive 2
B_j = Y_obs[:, 3].sum()                       # observed total of board 4
G = Y_obs.sum()
z_yates = (t * T_i + b * B_j - G) / ((t - 1) * (b - 1))
print(f"Yates's value {z_yates:.3f}")

d = np.zeros(n)
d[miss] = 1.0                                # one indicator column per
missing cell
y_zero = np.where(np.isnan(yv), 0.0, yv)       # any number will do in the
gap
coef_cov, sse_cov = fit(np.column_stack([X, d]), y_zero)
print(f"covariate method: gamma-hat = {coef_cov[-1]:.3f}, so the filled value is {-
coef_cov[-1]:.3f};"
      f" SSE {sse_cov:.3f}")

z = np.array([np.nanmean(yv)])                # start from the grand mean
history = [z[0]]
for step in range(30):
    y_fill = yv.copy()
    y_fill[miss] = z
    coef_fill, _ = fit(X, y_fill)              # complete-data analysis
    z = X[miss] @ coef_fill                    # replace the gap by its
fitted value

```

```
history.append(z[0])
print("iterates", np.round(history[:6], 3))
```

With several missing values, [Eq. 18.6.2](#) is an  $m \times m$  system, and the iteration converges at the rate of the largest eigenvalue of  $\mathbf{M}_{mm}$ . In the adhesive experiment with a second value also lost (adhesive 5 on board 1), that rate is 0.367, a little slower than  $1/3$ , because the two gaps are coupled through the grand mean ([Exercise \(B2\)](#)).

### 18.6.3 Why the method works, and when it does not

The iteration of [Theorem 18.6.1\(c\)](#) is the EM algorithm for the normal linear model with missing responses: the E-step replaces each missing value by its current fitted value, and the M-step refits. Part (b) proves convergence directly.

Everything assumes that the loss is unrelated to what the value would have been. A strip that split before gluing qualifies; a joint that failed in handling because its glue was weak does not, and then both filling in and dropping make the adhesive look stronger than it is. The observed data cannot reveal this. [Chapter 41](#) treats missing data in general, including the kinds of missingness ([Definition 41.1.1](#)) and multiple imputation ([Definition 41.5.1](#)).

*Warning.* Never analyse filled-in values as if they were data without the corrections of [Theorem 18.6.1\(e\)](#). The residual degrees of freedom must be reduced by one for each filled-in value, and tests must be computed by comparing models fitted to the observed data. Standard errors computed from the completed data with the balanced formulas are also too small, because they treat  $\mathbf{z}^*$  as observed.

### 18.6.4 Exercises

#### 18.6.4.1 A. Check your understanding

##### Exercise A1.

Verify the arithmetic of Yates's formula in [Example \(A strip that split\)](#), and check that the filled-in value 9.010 makes the residual of that cell in the completed table exactly zero.

##### Exercise A2.

In a randomized block design with  $t = 3$  and  $b = 2$ , suppose both observations on treatment 1 are lost. Show that [Eq. 18.6.1](#) fails, and that  $\mathbf{M}_{mm}$  has an eigenvalue equal to 1. What happens to the iteration?

##### Exercise A3.

Derive the analogue of Yates's formula for one missing value in an  $a \times a$  Latin square:  $z^* = \{a(R' + C' + T') - 2G'\} / \{(a-1)(a-2)\}$ , where  $R'$ ,  $C'$  and  $T'$  are the observed

totals of the row, column and treatment of the missing cell.

#### 18.6.4.2 B. Practice

##### Exercise B1.

In the randomized block design with one missing value, show that the treatment sum of squares computed from the completed table exceeds the correct one by

$$\frac{\{B' - (t-1)z^*\}^2}{t(t-1)}.$$

Check the formula against the numbers in [Example \(A strip that split\)](#).

*Solution.* The correct treatment sum of squares is  $\text{SSE}_{0,o} - \text{SSE}_o$ , where  $\text{SSE}_{0,o}$  is the residual sum of squares of the blocks-only model on the observed data. The complete-data sum of squares is  $\text{SSE}_0(\mathbf{y}(z^*)) - \text{SSE}_o$  by [Theorem 18.6.1\(a\)](#). So the bias is  $\text{SSE}_0(\mathbf{y}(z^*)) - \text{SSE}_{0,o}$ . In the blocks-only model, the missing cell's fitted value from the observed data is the mean of the observed values of its block,  $B'/(t-1)$ , and by [Theorem 18.6.1\(a\)](#) applied to that model, filling the gap with  $B'/(t-1)$  would give  $\text{SSE}_{0,o}$  exactly. Only the within-block sum of squares of block  $j$  depends on the filled value  $z$ . It equals a constant plus  $z^2 - (B' + z)^2/t$ , a quadratic in  $z$  with leading coefficient  $1 - 1/t$  and minimum at  $z = B'/(t-1)$ . Hence  $\text{SSE}_0(\mathbf{y}(z)) - \text{SSE}_{0,o} = (1 - 1/t)\{z - B'/(t-1)\}^2 = \{(t-1)z - B'\}^2/(t(t-1))$ . With  $B' = 36.25$ ,  $t = 5$  and  $z^* = 9.010$ :  $(36.25 - 36.04)^2/20 = 0.0022$ , as in the example.

##### Exercise B2.

Suppose two values are missing in a randomized block design, in different treatments and different blocks. Write  $\mathbf{M}_{mm}$  explicitly and find its eigenvalues. Check the rate 0.367 quoted for the adhesive experiment.

*Solution.* With the two missing cells in different treatments and different blocks, they share neither, so  $\mathbf{M}_{mm}$  has diagonal entries  $d = 1/b + 1/t - 1/(bt)$  and off-diagonal entries  $-1/(bt)$ . Its eigenvalues are  $d \pm 1/(bt)$ , and the larger is  $1/b + 1/t = 1/6 + 1/5 = 0.367$  for  $t = 5$ ,  $b = 6$ .

#### 18.6.4.3 C. Going deeper

##### Exercise C1.

Show that the covariance method of [Theorem 18.6.1\(d\)](#) gives correct standard errors for estimable functions of  $\boldsymbol{\beta}$ , without any correction, when the software's estimated covariance matrix of  $(\hat{\boldsymbol{\beta}}, \hat{\boldsymbol{\gamma}})$  is used. Why does this not contradict the warning at the end of the section?

## 18.7 Summary and Notes

### 18.7.1 Summary

*Idea.*

1. Analysis of covariance is a projection problem. With design part  $\mathbf{X}$  (any rank) and covariates  $\mathbf{Z}$ , the model space splits orthogonally as  $\mathcal{C}(\mathbf{X}) \oplus \mathcal{C}((\mathbf{I} - \mathbf{M})\mathbf{Z})$ . The covariate coefficients are estimated from within-group variation,  $\hat{\boldsymbol{\gamma}} = \mathbf{E}_{zz}^{-1} \mathbf{E}_{zy}$ , and the design part is fitted to the adjusted response  $\mathbf{y} - \mathbf{Z}\hat{\boldsymbol{\gamma}}$ . Tests compare residual sums of squares of nested models ([Theorem 18.1.1](#)).
2. In a randomized experiment a covariate measured before randomization reduces the variance of treatment comparisons by the factor  $(1 - \rho^2)\{1 + 1/(N - g - 2)\}$  ([Proposition 18.1.2](#)), and it removes the conditional bias caused by chance imbalance ([Section 18.1](#)). A covariate affected by the treatment must not be adjusted for (same section).
3. Adjusted means are estimable functions  $\mu_k + \mathbf{z}_0^\top \boldsymbol{\gamma}$ , estimated by  $\bar{y}_k - (\bar{\mathbf{z}}_k - \mathbf{z}_0)^\top \hat{\boldsymbol{\gamma}}$ . They are correlated, their variances grow with the distance of  $\bar{\mathbf{z}}_k$  from  $\mathbf{z}_0$ , and their differences do not depend on  $\mathbf{z}_0$  ([Definition 18.2.1](#) and [Proposition 18.2.1](#)). Scheffé's intervals are exact for all their contrasts; Tukey-type intervals are approximate.
4. The test of parallel slopes is a weighted test of homogeneity of the within-group slopes ([Theorem 18.3.1](#)). When slopes differ, the treatment difference is a line in the covariate, to be reported with a band or a Johnson–Neyman region ([Proposition 18.3.2](#)).
5. In a randomized complete block design, block effects cancel from every treatment contrast. The analysis is the additive two-way analysis ([Theorem 18.4.1](#)). Randomization alone makes the contrast estimates unbiased and the error mean square unbiased for their variance ([Proposition 18.4.2](#)). The block experiment estimates the error variance a completely randomized design would have had ([Proposition 18.4.4](#)).
6. A Latin square blocks in two crossed directions with mutually orthogonal row, column and treatment spaces. Its error consists of confounded interaction pieces ([Theorem 18.5.1](#)).
7. A balanced incomplete block design has information matrix  $(\lambda t/k)(\mathbf{I} - \mathbf{J}/t)$ . Contrasts are estimated from the adjusted totals  $Q_i$  with variance  $(k/\lambda t)\sigma^2 \sum c_i^2$ , and the efficiency factor relative to complete blocks is  $\lambda t/(rk)$  ([Theorem 18.5.3](#)).
8. A lost observation is handled exactly by least squares on the remaining data. Filling it with its fitted value, found by Yates's formula, by iteration or by the covariate method, reproduces the exact estimates and residual sum of squares. The error degrees of freedom must be reduced, and tests must be computed by comparing models ([Theorem 18.6.1](#) and [Corollary 18.6.2](#)).

### 18.7.2 Notes and sources

**Analysis of covariance.** Fisher introduced the analysis of covariance in the 1930s, in later editions of *Statistical Methods for Research Workers* (Fisher 1925 and later editions), as a

way of using a concomitant measurement to increase the precision of agricultural experiments. Cochran's review (Cochran 1957) remains an excellent account of its uses, including the dangers of adjusting for a covariate affected by treatment and of adjusting non-randomized groups. Rencher and Schaalje (2008, chapter 16) treat one-way, two-way and multi-covariate layouts and the test of homogeneous slopes. Christensen (2020, chapter 9) derives the analysis through the projection  $(\mathbf{I} - \mathbf{M})\mathbf{Z}$ , which is the route taken in [Theorem 18.1.1](#), and uses the same device for missing data and for incomplete block designs. The case against testing baseline balance in randomized trials was made forcefully by Senn (1994). The modern debate on regression adjustment in randomized experiments, which asks whether adjustment can hurt when the model is wrong, was opened by Freedman (2008) and answered, for the interacted estimator of [Exercise \(C1\)](#), by Lin (2013). [Chapter 25](#) returns to it ([Theorem 25.6.2](#)).

**Heterogeneous slopes.** The region of significance for non-parallel lines is due to Johnson and Neyman (1936), who developed it for comparisons of teaching methods with an ability covariate. Simultaneous versions use the Scheffé argument of [Chapter 13](#).

**Block designs and randomization.** Randomization, blocking and the Latin square as a design are Fisher's, set out in *The Design of Experiments* (Fisher 1935). The randomization moments of [Proposition 18.4.2](#) and the view of the linear model as an approximation justified by randomization are developed in Kempthorne (1952) and Hinkelmann and Kempthorne (2008). Cox (1958) remains the best short account of the ideas behind designed experiments, and Bailey (2008) gives a modern treatment of block designs built on the same orthogonal decompositions as this chapter. Relative efficiencies such as [Eq. 18.4.2](#) are standard in the design literature, for instance Cochran and Cox (1957). Christensen (2020, chapter 8) presents the randomized complete block and Latin square models as additive multi-way layouts and discusses their assumptions.

**Incomplete blocks.** Balanced incomplete block designs were introduced by Yates (1936), who also showed how to recover inter-block information (Yates 1940). Fisher (1940) proved the inequality  $b \geq t$  of [Lemma 18.5.2](#). Christensen (2020, section 9.4) derives the intra-block analysis as an analysis of covariance with the treatment indicators as covariates, as in the proof of [Theorem 18.5.3](#).

**Missing values.** Yates (1933) derived the missing value formula [Eq. 18.6.3](#) and the rule of subtracting a degree of freedom for each estimated value. Bartlett (1937) proposed the covariance method with indicator covariates, and Healy and Westmacott (1956) the iterative method of [Theorem 18.6.1\(c\)](#), convenient for early computers. Dempster, Laird and Rubin (1977) placed the iteration within the EM algorithm. Christensen (2020, section 9.3) treats missing observations through the covariance model.

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# Chapter 19

## What Goes Wrong: A Theory of Departures

Parts II to IV derived everything from a short list of assumptions: the mean is  $\mathbf{X}\boldsymbol{\beta}$  for the regressors actually used, the errors are uncorrelated with a common variance, they are normal, and the regressors are fixed and measured without error. Real data meet each of these only approximately. This chapter asks, one assumption at a time, what a specific departure does to the least squares estimator, to  $s^2$  and to the tests and intervals built from them. Omitting a regressor biases the coefficients, and yet the biased estimator can still have the smaller mean squared error. A wrong covariance matrix leaves the estimator nearly efficient but can make its standard error wrong by a factor of two or more. With non-normal errors the  $t$  and  $F$  tests are asymptotically valid when no observation has large leverage, while tests and intervals for  $\sigma^2$  are not valid in any sample size. A single bad observation can move the estimate arbitrarily far. With random regressors and a curved mean, the usual standard errors describe the wrong target. Collinearity breaks no assumption at all, but it inflates variances. Chapters 20–22 turn to diagnosis and remedies, and Chapter 23 to inference that relies less on the assumptions.

**What you need.** Chapter 5 (the moments of  $\hat{\boldsymbol{\beta}}$  under departures, Proposition 5.5.7, and omitted regressors, Proposition 5.7.2), Chapter 6 (the Frisch–Waugh–Lovell theorem, leverage, Kruskal’s theorem and the population projection), Chapter 7 (Gauss–Markov and Aitken), Chapter 10 (the deletion formulas, Proposition 10.6.2), and the tests and intervals of Chapter 11 and Chapter 12. Section 14.1 treats random regressors under the normal model, and Section 19.5 builds on it.

### 19.0.1 Roadmap

- [Omitting and adding regressors](#)
- [An incorrect covariance matrix](#)
- [Non-normal errors](#)
- [A single bad observation](#)
- [Random regressors and a curved mean](#)
- [Collinearity, previewed](#)
- [Summary and notes](#)

## 19.1 Omitting and adding regressors

The assumption  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  claims that the columns of  $\mathbf{X}$  are the right ones. This section compares two ways of getting them wrong. *Underfitting* leaves out regressors that matter. *Overfitting* includes regressors that do not. This is a first look at the bias–variance trade-off of [Part VI](#).

### 19.1.1 The setting

Throughout the section the truth is

$$E(\mathbf{Y}) = \mathbf{X}_1\boldsymbol{\beta}_1 + \mathbf{X}_2\boldsymbol{\beta}_2, \quad \text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}_n,$$

where  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$  has full column rank  $p = p_1 + p_2 < n$ . The *long* regression fits both blocks and gives  $\hat{\boldsymbol{\beta}} = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\boldsymbol{\beta}}_2^\top)^\top$  and  $s^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2 / (n - p)$ . The *short* regression fits  $\mathbf{X}_1$  alone and gives

$$\hat{\boldsymbol{\beta}}_1^S = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{Y}, \quad s_1^2 = \frac{\|(\mathbf{I} - \mathbf{M}_1)\mathbf{Y}\|^2}{n - p_1},$$

where the superscript S marks the short fit and  $\mathbf{M}_1$  projects onto  $\mathcal{C}(\mathbf{X}_1)$ . Two quantities recur. The first is  $\boldsymbol{\Pi} = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{X}_2$ , the coefficients from regressing each column of  $\mathbf{X}_2$  on  $\mathbf{X}_1$ , as in [Proposition 5.7.2](#). The second is

$$\gamma = \frac{\|(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\boldsymbol{\beta}_2\|^2}{\sigma^2} = \frac{\boldsymbol{\beta}_2^\top \mathbf{X}_2^\top (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2 \boldsymbol{\beta}_2}{\sigma^2}. \quad (19.1.1)$$

Since  $(\mathbf{M} - \mathbf{M}_1)\mathbf{X}\boldsymbol{\beta} = (\mathbf{I} - \mathbf{M}_1)\mathbf{X}\boldsymbol{\beta} = (\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\boldsymbol{\beta}_2$ , this is the noncentrality parameter [Eq. 11.1.2](#) of the  $F$  test of  $\boldsymbol{\beta}_2 = \mathbf{0}$  in the long model. It measures how far the true mean lies from the short model's column space, in units of  $\sigma^2$ . It is zero iff  $\boldsymbol{\beta}_2 = \mathbf{0}$ , because  $\mathbf{X}_2\boldsymbol{\beta}_2 \in \mathcal{C}(\mathbf{X}_1)$  forces  $\boldsymbol{\beta}_2 = \mathbf{0}$  when  $\mathbf{X}$  has full rank.

Both regressions are linked by an exact identity and by one covariance.

#### Lemma 19.1.1 — Short and long regressions

In this setting:

- (a)  $\hat{\boldsymbol{\beta}}_2 = (\mathbf{X}_2^\top (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2)^{-1} \mathbf{X}_2^\top (\mathbf{I} - \mathbf{M}_1) \mathbf{Y}$  and  $\hat{\boldsymbol{\beta}}_1 = \hat{\boldsymbol{\beta}}_1^S - \boldsymbol{\Pi} \hat{\boldsymbol{\beta}}_2$ ;
- (b)  $\text{Cov}(\hat{\boldsymbol{\beta}}_1^S, \hat{\boldsymbol{\beta}}_2) = \mathbf{0}$  and  $\text{Cov}(\hat{\boldsymbol{\beta}}_2) = \sigma^2 \mathbf{C}$ , where  $\mathbf{C} = (\mathbf{X}_2^\top (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2)^{-1}$ .

*Proof.*

- (a) The first formula is [Theorem 6.6.2\(a\)](#) with the roles of the two blocks exchanged, and the second is [Proposition 5.7.2\(a\)](#). (b) Both estimators are linear in  $\mathbf{Y}$ . By [Theorem 2.2.1](#),

$$\text{Cov}(\hat{\boldsymbol{\beta}}_1^S, \hat{\boldsymbol{\beta}}_2) = \sigma^2 (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top (\mathbf{I} - \mathbf{M}_1) \mathbf{X}_2 \mathbf{C} = \mathbf{0},$$

because  $\mathbf{X}_1^\top (\mathbf{I} - \mathbf{M}_1) = \mathbf{0}$ . Similarly  $\text{Cov}(\hat{\boldsymbol{\beta}}_2) = \sigma^2 \mathbf{C} \mathbf{X}_2^\top (\mathbf{I} - \mathbf{M}_1)^2 \mathbf{X}_2 \mathbf{C} = \sigma^2 \mathbf{C}$ .



## 19.1.2 Underfitting

### Theorem 19.1.2 — Omitted regressors

Fit the short model when the long model is true.

- (a)  $E(\hat{\beta}_1^S) = \beta_1 + \Pi\beta_2$  and  $\text{Cov}(\hat{\beta}_1^S) = \sigma^2(\mathbf{X}_1^\top \mathbf{X}_1)^{-1}$ . The bias  $\Pi\beta_2$  vanishes iff  $\mathbf{X}_1^\top \mathbf{X}_2\beta_2 = \mathbf{0}$ , and it vanishes for every  $\beta_2$  iff  $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$ .
- (b) The fitted values have mean  $\mathbf{X}\beta - (\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2$ . The residuals have mean  $(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2$  and covariance  $\sigma^2(\mathbf{I} - \mathbf{M}_1)$ , as if the short model were right.
- (c)  $E(s_1^2) = \sigma^2(1 + \gamma/(n - p_1))$ , so  $s_1^2$  overestimates  $\sigma^2$  unless  $\beta_2 = \mathbf{0}$ .
- (d) At a new point with regressors  $(\mathbf{x}_{10}, \mathbf{x}_{20})$ , the short prediction  $\mathbf{x}_{10}^\top \hat{\beta}_1^S$  has bias  $(\Pi^\top \mathbf{x}_{10} - \mathbf{x}_{20})^\top \beta_2$  for the mean  $\mathbf{x}_{10}^\top \beta_1 + \mathbf{x}_{20}^\top \beta_2$ .

*Proof.*

- (a) The mean is Proposition 5.7.2(b). The covariance is Theorem 5.5.1(b) for the model matrix  $\mathbf{X}_1$ , which needs only  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ . Since  $(\mathbf{X}_1^\top \mathbf{X}_1)^{-1}$  is nonsingular,  $\Pi\beta_2 = \mathbf{0}$  iff  $\mathbf{X}_1^\top \mathbf{X}_2\beta_2 = \mathbf{0}$ , and this holds for all  $\beta_2$  iff  $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$ .
- (b)  $E(\mathbf{M}_1\mathbf{Y}) = \mathbf{X}_1\beta_1 + \mathbf{M}_1\mathbf{X}_2\beta_2 = \mathbf{X}\beta - (\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2$ , and  $E((\mathbf{I} - \mathbf{M}_1)\mathbf{Y}) = (\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2$  because  $(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_1 = \mathbf{0}$ . The covariance is  $(\mathbf{I} - \mathbf{M}_1)\sigma^2\mathbf{I}(\mathbf{I} - \mathbf{M}_1) = \sigma^2(\mathbf{I} - \mathbf{M}_1)$ .
- (c) By Eq. 2.3.1 with  $\mathbf{M}_1$  of rank  $p_1$  and mean  $\mathbf{X}\beta$ ,  $E\|(\mathbf{I} - \mathbf{M}_1)\mathbf{Y}\|^2 = \sigma^2(n - p_1) + \|(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2\|^2 = \sigma^2(n - p_1 + \gamma)$ .
- (d) Take expectations in  $\mathbf{x}_{10}^\top \hat{\beta}_1^S$  using (a), and subtract the true mean.

Nothing in the second moments signals the omission: the covariances of  $\hat{\beta}_1^S$  and of the residuals are exactly what the short model claims. The damage is in the first moments. The estimator is centred at the wrong place, and the residuals carry the systematic component  $(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\beta_2$  that added-variable plots (Theorem 6.6.2) and lack-of-fit tests (Theorem 14.6.2) look for. Because  $s_1^2$  is inflated, the reported variances are on average too large, yet as  $n$  grows they shrink while the bias does not, so the coverage of intervals for  $\beta_1$  tends to zero. Nothing requires a coefficient to be omitted: for any true mean  $\mathbf{X}_1\beta_1 + \delta$ , such as a curved mean fitted by a line, (a)–(c) hold with  $\mathbf{X}_2\beta_2$  replaced by  $\delta$  (Proposition 5.5.7(a)).

## 19.1.3 Overfitting

### Theorem 19.1.3 — Irrelevant regressors

Suppose  $\beta_2 = \mathbf{0}$ , so the short model is correct, but the long model is fitted.

- (a)  $E(\hat{\beta}_1) = \beta_1$ ,  $E(\hat{\beta}_2) = \mathbf{0}$  and  $E(s^2) = \sigma^2$ .
- (b)  $\text{Cov}(\hat{\beta}_1) = \text{Cov}(\hat{\beta}_1^S) + \sigma^2 \Pi C \Pi^\top$ . Hence  $\text{Var}(\mathbf{a}^\top \hat{\beta}_1) \geq \text{Var}(\mathbf{a}^\top \hat{\beta}_1^S)$  for every  $\mathbf{a} \in \mathbb{R}^{p_1}$ , with equality iff  $\Pi^\top \mathbf{a} = \mathbf{0}$ , and  $\text{Cov}(\hat{\beta}_1) = \text{Cov}(\hat{\beta}_1^S)$  iff  $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$ .
- (c) At a new point  $\mathbf{x}_0 = (\mathbf{x}_{10}^\top, \mathbf{x}_{20}^\top)^\top$ ,

$$\begin{aligned} \text{Var}(\mathbf{x}_0^\top \hat{\beta}) &= \text{Var}(\mathbf{x}_{10}^\top \hat{\beta}_1^S) \\ &\quad + \sigma^2 (\mathbf{x}_{20} - \Pi^\top \mathbf{x}_{10})^\top \mathbf{C} (\mathbf{x}_{20} - \Pi^\top \mathbf{x}_{10}), \end{aligned}$$

and summed over the design points  $E\|\mathbf{X}\hat{\beta} - \mathbf{X}\beta\|^2 = \sigma^2 p$ , against  $\sigma^2 p_1$  for the short fit.

*Proof.*

- (a) The long model is correct (with  $\beta_2 = \mathbf{0}$ ), so [Theorem 5.5.1\(a\)](#) and [Theorem 5.6.1](#) apply.
- (b) By [Lemma 19.1.1\(a\)](#),  $\hat{\beta}_1 = \hat{\beta}_1^S - \Pi \hat{\beta}_2$  with the two terms uncorrelated, so  $\text{Cov}(\hat{\beta}_1) = \text{Cov}(\hat{\beta}_1^S) + \Pi \text{Cov}(\hat{\beta}_2) \Pi^\top$ , and [Lemma 19.1.1\(b\)](#) gives  $\text{Cov}(\hat{\beta}_2)$ . Because  $\mathbf{C}$  is positive definite,  $\mathbf{a}^\top \Pi C \Pi^\top \mathbf{a} = 0$  iff  $\Pi^\top \mathbf{a} = \mathbf{0}$ , and  $\Pi C \Pi^\top = \mathbf{0}$  iff  $\Pi = \mathbf{0}$ , iff  $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$ .
- (c) Write  $\mathbf{x}_0^\top \hat{\beta} = \mathbf{x}_{10}^\top \hat{\beta}_1^S + (\mathbf{x}_{20} - \Pi^\top \mathbf{x}_{10})^\top \hat{\beta}_2$  using [Lemma 19.1.1\(a\)](#), and use the zero covariance of (b) of that lemma. For the sum,  $\mathbf{X}\hat{\beta} = \mathbf{M}\mathbf{Y}$  is unbiased for  $\mathbf{X}\beta$  with covariance  $\sigma^2 \mathbf{M}$ , so  $E\|\mathbf{M}\mathbf{Y} - \mathbf{X}\beta\|^2 = \sigma^2 \text{tr } \mathbf{M} = \sigma^2 p$ . The short fit  $\mathbf{M}_1 \mathbf{Y}$  is also unbiased, and its trace is  $p_1$ .

■

An irrelevant regressor costs no bias, only variance, and the price grows with its correlation with the relevant columns: for one extra column the variance of a coefficient is multiplied by  $1/(1 - r^2)$ , with  $r$  the partial correlation of the two regressors ([Eq. 6.6.2](#)).

## 19.1.4 Bias against variance

When  $\beta_2$  is small but not zero, the short regression is biased but less variable. Which is better? Compare the two estimators of  $\beta$  by their mean squared error matrices, writing  $\hat{\beta}^S = ((\hat{\beta}_1^S)^\top, \mathbf{0}^\top)^\top$  for the short estimator, which sets  $\beta_2$  to zero.

### Lemma 19.1.4 — A rank-one comparison

Let  $\mathbf{C}$  be a positive definite  $q \times q$  matrix,  $\mathbf{b} \in \mathbb{R}^q$  and  $c > 0$ . Then  $c\mathbf{C} - \mathbf{b}\mathbf{b}^\top$  is nonnegative definite iff  $\mathbf{b}^\top \mathbf{C}^{-1} \mathbf{b} \leq c$ .

*Proof.* By [Lemma 12.2.3](#) with  $\mathbf{W} = \mathbf{C}$  and  $\mathbf{u} = \mathbf{b}$ , the inequality  $\mathbf{b}^\top \mathbf{C}^{-1} \mathbf{b} \leq c$  holds iff  $(\mathbf{a}^\top \mathbf{b})^2 \leq c \mathbf{a}^\top \mathbf{C} \mathbf{a}$  for every  $\mathbf{a}$ , that is, iff  $\mathbf{a}^\top (c\mathbf{C} - \mathbf{b}\mathbf{b}^\top) \mathbf{a} \geq 0$  for every  $\mathbf{a}$ .

■

**Theorem 19.1.5 — When the short regression wins**

In the setting of this section, let  $\mathbf{G} = \begin{pmatrix} -\mathbf{\Pi} \\ \mathbf{I}_{p_2} \end{pmatrix}$ , a  $p \times p_2$  matrix.

- (a) The mean squared error matrix of  $\hat{\boldsymbol{\beta}}^S$  is

$$E[(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})^\top] = \text{Cov}(\hat{\boldsymbol{\beta}}) - \mathbf{G}(\sigma^2 \mathbf{C} - \boldsymbol{\beta}_2 \boldsymbol{\beta}_2^\top) \mathbf{G}^\top.$$

- (b)  $E(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}^S - \boldsymbol{\lambda}^\top \boldsymbol{\beta})^2 \leq \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})$  for every  $\boldsymbol{\lambda} \in \mathbb{R}^p$  iff  $\gamma \leq 1$ . If  $\gamma > 1$ , the reverse strict inequality holds for some  $\boldsymbol{\lambda}$ .
- (c) Over the design points,  $E\|\mathbf{X}\hat{\boldsymbol{\beta}}^S - \mathbf{X}\boldsymbol{\beta}\|^2 = \sigma^2(p_1 + \gamma)$ , against  $\sigma^2 p$  for the long fit. The short fit is better there iff  $\gamma < p_2$ .

*Proof.*

- (a) By Lemma 19.1.1(a),  $\hat{\boldsymbol{\beta}} = \hat{\boldsymbol{\beta}}^S + \mathbf{G}\hat{\boldsymbol{\beta}}_2$ , and  $\hat{\boldsymbol{\beta}}^S$  is uncorrelated with  $\hat{\boldsymbol{\beta}}_2$  by part (b) of the lemma. Since  $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$  and  $E(\hat{\boldsymbol{\beta}}_2) = \boldsymbol{\beta}_2$ ,  $E(\hat{\boldsymbol{\beta}}^S) - \boldsymbol{\beta} = -\mathbf{G}\boldsymbol{\beta}_2$ . Expand  $\text{Cov}(\hat{\boldsymbol{\beta}}) = E[(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta} + \mathbf{G}\hat{\boldsymbol{\beta}}_2)(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta} + \mathbf{G}\hat{\boldsymbol{\beta}}_2)^\top]$ . The cross term is  $E[(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})\hat{\boldsymbol{\beta}}_2^\top] = \text{Cov}(\hat{\boldsymbol{\beta}}^S, \hat{\boldsymbol{\beta}}_2) + (E\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})\boldsymbol{\beta}_2^\top = -\mathbf{G}\boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top$ , and  $E(\hat{\boldsymbol{\beta}}_2\hat{\boldsymbol{\beta}}_2^\top) = \sigma^2 \mathbf{C} + \boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top$ . So

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = E[(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})(\hat{\boldsymbol{\beta}}^S - \boldsymbol{\beta})^\top] + \mathbf{G}(\sigma^2 \mathbf{C} + \boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top - 2\boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top) \mathbf{G}^\top.$$

- (b) Put  $\mathbf{A} = \sigma^2 \mathbf{C} - \boldsymbol{\beta}_2\boldsymbol{\beta}_2^\top$ . The claimed inequality for all  $\boldsymbol{\lambda}$  says that  $\mathbf{GAG}^\top$  is nonnegative definite. The matrix  $\mathbf{L} = [\mathbf{0}, \mathbf{I}_{p_2}]$  satisfies  $\mathbf{LG} = \mathbf{I}$ , so  $\mathbf{A} = \mathbf{L}(\mathbf{GAG}^\top)\mathbf{L}^\top$ . Hence  $\mathbf{GAG}^\top$  is nonnegative definite iff  $\mathbf{A}$  is, and by Lemma 19.1.4 iff  $\boldsymbol{\beta}_2^\top \mathbf{C}^{-1} \boldsymbol{\beta}_2 \leq \sigma^2$ , which is  $\gamma \leq 1$ . If  $\gamma > 1$ , the same lemma gives  $\mathbf{a}$  with  $\mathbf{a}^\top \mathbf{A} \mathbf{a} < 0$ , and  $\boldsymbol{\lambda} = \mathbf{L}^\top \mathbf{a}$  satisfies  $\mathbf{G}^\top \boldsymbol{\lambda} = \mathbf{a}$ , so  $\boldsymbol{\lambda}^\top \mathbf{GAG}^\top \boldsymbol{\lambda} < 0$ .
- (c)  $\mathbf{X}\hat{\boldsymbol{\beta}}^S = \mathbf{M}_1 \mathbf{Y}$ , whose mean is  $\mathbf{M}_1 \mathbf{X}\boldsymbol{\beta}$  and covariance  $\sigma^2 \mathbf{M}_1$ . So  $E\|\mathbf{M}_1 \mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 = \sigma^2 p_1 + \|(\mathbf{I} - \mathbf{M}_1)\mathbf{X}\boldsymbol{\beta}\|^2 = \sigma^2(p_1 + \gamma)$ . The long fit is Theorem 19.1.3(c), whose proof did not use  $\boldsymbol{\beta}_2 = \mathbf{0}$  for this part. ■

The threshold in (b) depends only on the noncentrality, not on the design or on  $n$ . It covers all linear functions at once; for a particular function the short estimator can win for larger  $\gamma$ , as (c) shows. Since  $\gamma$  is unknown, neither criterion applies directly. But the  $F$  statistic for  $\boldsymbol{\beta}_2 = \mathbf{0}$  depends on the parameters only through  $\gamma$  (Theorem 11.1.1), so a small  $F$  is evidence that  $\gamma$  is small. Turning this into a procedure, and paying for having chosen the model from the data, is the subject of Chapter 29.

**Example. (A biased estimator that wins)**

Take  $n = 30$  observations of two centred regressors with unit sample variances and sample correlation  $r = 0.9$ , an intercept,  $\sigma = 1$  and  $\beta_2 = 0.301$ , chosen so that  $\gamma = 0.5$ . Here  $\mathbf{\Pi}$  has slope entry  $r$ , so the short slope on  $x_1$  has bias  $r\beta_2 = 0.271$ . Its standard deviation is 0.186, against 0.426 for the long slope, and the ratio of mean squared error to long variance is 0.595. A simulation with 200000 data sets gives 0.594.

For two standardized regressors the ratio has a closed form (Exercise (B2)):

$$\frac{E(\hat{\beta}_1^S - \beta_1)^2}{\text{Var}(\hat{\beta}_1)} = 1 - r^2 + r^2\gamma = 1 + r^2(\gamma - 1), \quad (19.1.2)$$

which is  $1 - 0.81 + 0.81 \times 0.5 = 0.595$ . The short slope wins iff  $\gamma < 1$ , as Theorem 19.1.5(b) requires, and the stakes, gain or loss, are proportional to  $r^2$ . This is panel (a) of Figure 19.1.1. The short model's  $s_1^2$  has mean  $1 + \gamma/(n-2) = 1.0179$ , a bias too small to notice.

```
import numpy as np
rng = np.random.default_rng(1901)
n, r, sigma = 30, 0.9, 1.0

# two centred, unit-variance regressors with sample correlation exactly r:
# orthonormalize (1, z1, z2) and keep the last two columns, which are centred
Z = np.linalg.qr(np.column_stack([np.ones(n), rng.normal(size=(n, 2))]))[0][:, 1:]
Z *= np.sqrt(n - 1)
x1 = Z[:, 0]
x2 = r * Z[:, 0] + np.sqrt(1 - r**2) * Z[:, 1]
X1 = np.column_stack([np.ones(n), x1]) # the short model
X = np.column_stack([X1, x2])         # the long model

M1 = X1 @ np.linalg.solve(X1.T @ X1, X1.T)
S22_1 = x2 @ (np.eye(n) - M1) @ x2      # ||(I - M1) x2||^2
beta2 = np.sqrt(0.5 * sigma**2 / S22_1)  # chosen so that gamma = 0.5
gamma = beta2**2 * S22_1 / sigma**2

var_long = sigma**2 * np.linalg.inv(X.T @ X)[1, 1] # Var of the long slope on x1
Pi = np.linalg.solve(X1.T @ X1, X1.T @ x2)[1]     # slope of x2 on x1
var_short = sigma**2 * np.linalg.inv(X1.T @ X1)[1, 1]
mse_short = var_short + (Pi * beta2) ** 2           # variance + squared bias
print(f"gamma = {gamma:.3f}, MSE(short)/Var(long) = {mse_short / var_long:.4f}")
print(f"formula 1 - r^2 + r^2 gamma = {1 - r**2 + r**2 * gamma:.4f}")

reps = 200_000
beta = np.array([1.0, 1.0, beta2])
Y = X @ beta + sigma * rng.normal(size=(reps, n))
b_long = np.linalg.solve(X.T @ X, X.T @ Y.T)[1]
b_short = np.linalg.solve(X1.T @ X1, X1.T @ Y.T)[1]
print("simulated MSE ratio:", np.mean((b_short - 1) ** 2) / np.mean((b_long - 1) ** 2))
```

### Example. (Choosing a polynomial degree)

Part (c) of Theorem 19.1.5 applies to any nested sequence of models. Take  $n = 40$  equally spaced points on  $[-1, 1]$ , the mean  $\mu(x) = e^x \sin 3x$ , which is not a polynomial, and  $\sigma = 0.25$ . A polynomial of degree  $d$  has  $p = d + 1$  columns, and its risk at the design points is  $\sigma^2(d + 1) + \|(\mathbf{I} - \mathbf{M}_d)\boldsymbol{\mu}\|^2$ . Panel (b) of Figure 19.1.1 shows the two terms. In units of  $\sigma^2$  the risk is 161.41 for a straight line, falls to 5.52 at degree 4, and climbs back to 11.00 at degree 10, where the fit is almost unbiased and pays only for its variance. Exercise (B3) shows that adding one term lowers the risk iff that term's own noncentrality exceeds 1.

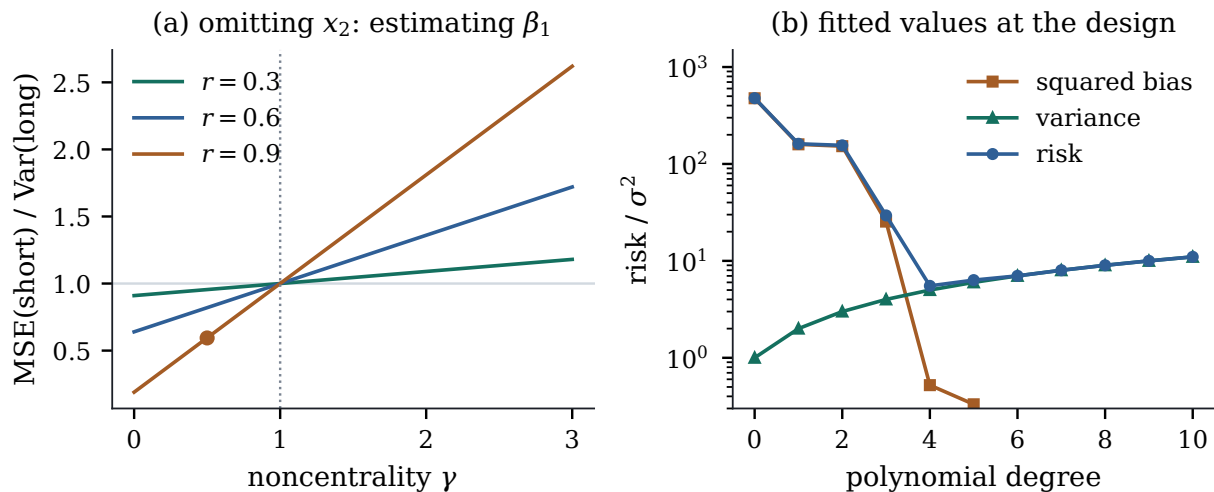


Figure 19.1.1: (a) Mean squared error of the short slope relative to the variance of the long slope, Eq. 19.1.2, for three correlations. All lines cross at  $\gamma = 1$ ; the point is the simulation of [Example \(A biased estimator that wins\)](#). (b) Squared bias, variance and their sum for polynomial fits of increasing degree ([Example \(Choosing a polynomial degree\)](#)), in units of  $\sigma^2$  and on a logarithmic scale.

*Idea.* Omitting a regressor that matters costs bias; keeping one that does not costs variance, and for  $\gamma < 1$  the bias is the cheaper. This does not make the biased coefficient *interpretable*: it still estimates  $\beta_1 + \Pi\beta_2$ , and whether that answers the question is a matter for [Chapter 25](#).

## 19.1.5 Exercises

### 19.1.5.1 A. Check your understanding

#### Exercise A1.

A straight line is fitted at  $x = 0, 1, 2, 3$  when the true mean is  $\beta_0 + \beta_1 x + \beta_2 x^2$ . Find the biases of the fitted intercept and slope.

#### Exercise A2.

True or false: if the omitted regressors are orthogonal to the included ones ( $\mathbf{X}_1^\top \mathbf{X}_2 = \mathbf{0}$ ), the short regression is harmless, including its estimate of  $\sigma^2$ .

### 19.1.5.2 B. Practice

#### Exercise B1.

In [Exercise \(A1\)](#), show that  $\|(\mathbf{I} - \mathbf{M}_1)\mathbf{X}_2\|^2 = 4$ . For which values of  $\beta_2/\sigma$  does the straight line have a smaller mean squared error matrix than the quadratic fit, in the sense of [Theorem 19.1.5\(b\)](#)? For which values is it better at the four design points?

**Exercise B2.**

Derive Eq. 19.1.2. Take an intercept and two centred regressors with  $\|\mathbf{x}_1\|^2 = \|\mathbf{x}_2\|^2 = S$  and sample correlation  $r$ . Show that  $\text{Var}(\hat{\beta}_1) = \sigma^2/(S(1-r^2))$ , that the short slope has variance  $\sigma^2/S$  and bias  $r\beta_2$ , and that  $\gamma = \beta_2^2 S(1-r^2)/\sigma^2$ .

*Solution.* The long variance is Eq. 6.6.2 with  $R_1^2 = r^2$ . The short slope is  $\mathbf{x}_1^\top \mathbf{Y}/S$ , with variance  $\sigma^2/S$  and mean  $\beta_1 + \beta_2 \mathbf{x}_1^\top \mathbf{x}_2/S = \beta_1 + r\beta_2$ . The residual of  $\mathbf{x}_2$  after regression on  $(\mathbf{1}, \mathbf{x}_1)$  has squared length  $S(1-r^2)$ , which gives  $\gamma$ . Then  $(\sigma^2/S + r^2\beta_2^2)S(1-r^2)/\sigma^2 = 1 - r^2 + r^2\gamma$ .

**Exercise B3.**

Let  $\mathcal{C}(\mathbf{X}_1) \subset \mathcal{C}(\mathbf{X}_2) \subset \dots$  be nested model spaces with dimensions  $p_1 < p_2 < \dots$ , and let  $\boldsymbol{\mu} = \mathbf{E}(\mathbf{Y})$  be arbitrary. Show that the risk  $\mathbf{E}\|\mathbf{M}_k \mathbf{Y} - \boldsymbol{\mu}\|^2$  of model  $k$  exceeds that of model  $k+1$  iff  $\gamma_k = \|(\mathbf{M}_{k+1} - \mathbf{M}_k)\boldsymbol{\mu}\|^2/\sigma^2 > p_{k+1} - p_k$ .

*Solution.* As in Theorem 19.1.5(c), the risk of model  $k$  is  $\sigma^2 p_k + \|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2$ . By Theorem 6.5.1,  $\mathbf{I} - \mathbf{M}_k = (\mathbf{I} - \mathbf{M}_{k+1}) + (\mathbf{M}_{k+1} - \mathbf{M}_k)$  with orthogonal ranges, so  $\|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2 = \|(\mathbf{I} - \mathbf{M}_{k+1})\boldsymbol{\mu}\|^2 + \sigma^2 \gamma_k$ . The difference of the two risks is  $\sigma^2(\gamma_k - (p_{k+1} - p_k))$ .

**19.1.5.3 C. Going deeper****Exercise C1.**

Assume normal errors. Compare  $s_1^2$  (short) and  $s^2$  (long) as estimators of  $\sigma^2$  by mean squared error, when  $\boldsymbol{\beta}_2 \neq \mathbf{0}$ . Show that  $s_1^2$  is better iff  $2(n - p_1) + 4\gamma + \gamma^2 < 2(n - p_1)^2/(n - p)$ , and interpret the condition when  $n$  is large.

## 19.2 An incorrect covariance matrix

Suppose the mean is right but the errors are correlated or have unequal variances. Write

$$\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}, \quad (19.2.1)$$

with  $\mathbf{V}$  positive definite and  $\mathbf{X}$  of full rank  $p < n$ . The analyst fits ordinary least squares and reports  $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$  as if  $\mathbf{V} = \mathbf{I}$ . Two things go wrong:  $\hat{\boldsymbol{\beta}}$  is no longer the best linear unbiased estimator (Corollary 7.2.3), and  $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$  no longer estimates its covariance. The second failure is usually much more serious.

## 19.2.1 What happens to the estimator and its standard error

The bounds below need a fact about traces of projections, a special case of an inequality of Ky Fan.

### Lemma 19.2.1 — Trace of a compressed matrix

Let  $\mathbf{A}$  be symmetric with eigenvalues  $\lambda_1 \geq \cdots \geq \lambda_n$ , and let  $\mathbf{P}$  be a symmetric idempotent matrix of rank  $k$ . Then

$$\sum_{i=n-k+1}^n \lambda_i \leq \operatorname{tr}(\mathbf{PA}) \leq \sum_{i=1}^k \lambda_i.$$

*Proof.* By Theorem 1.7.2,  $\mathbf{A} = \sum_i \lambda_i \mathbf{q}_i \mathbf{q}_i^\top$  with orthonormal  $\mathbf{q}_i$ . So  $\operatorname{tr}(\mathbf{PA}) = \sum_i \lambda_i w_i$  with  $w_i = \mathbf{q}_i^\top \mathbf{P} \mathbf{q}_i = \|\mathbf{P} \mathbf{q}_i\|^2$ . Each  $w_i$  lies in  $[0, 1]$  because the eigenvalues of  $\mathbf{P}$  are 0 or 1 (Proposition 6.3.4), so  $0 \leq \mathbf{q}_i^\top \mathbf{P} \mathbf{q}_i \leq \mathbf{q}_i^\top \mathbf{q}_i = 1$ , and  $\sum_i w_i = \operatorname{tr}(\mathbf{P} \sum_i \mathbf{q}_i \mathbf{q}_i^\top) = \operatorname{tr} \mathbf{P} = k$ . Then

$$\begin{aligned} \sum_i \lambda_i w_i - \sum_{i \leq k} \lambda_i &= \sum_{i \leq k} \lambda_i (w_i - 1) + \sum_{i > k} \lambda_i w_i \\ &\leq \lambda_k \sum_{i \leq k} (w_i - 1) + \lambda_k \sum_{i > k} w_i = \lambda_k \left( \sum_i w_i - k \right) = 0, \end{aligned}$$

since  $w_i - 1 \leq 0$  and  $\lambda_i \geq \lambda_k$  for  $i \leq k$ , while  $w_i \geq 0$  and  $\lambda_i \leq \lambda_k$  for  $i > k$ . The lower bound follows by applying this to  $-\mathbf{A}$ . ■

### Theorem 19.2.2 — Least squares under a wrong covariance

Under Eq. 19.2.1, let  $\mathbf{a} \in \mathbb{R}^p$  be nonzero and let  $v = s^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$  be the usual estimate of  $\operatorname{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})$ .

- (a)  $\mathbb{E}(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$  and  $\operatorname{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$ .
- (b)  $\mathbb{E}(s^2) = \sigma^2 \operatorname{tr}((\mathbf{I} - \mathbf{M})\mathbf{V}) / (n - p)$ .
- (c) With  $\mathbf{u} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a} / \|\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}\|$ , a unit vector in  $\mathcal{C}(\mathbf{X})$ ,

$$\frac{\mathbb{E}(v)}{\operatorname{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})} = \frac{\operatorname{tr}((\mathbf{I} - \mathbf{M})\mathbf{V}) / (n - p)}{\mathbf{u}^\top \mathbf{V} \mathbf{u}}. \quad (19.2.2)$$

- (d) Let  $\lambda_1 \geq \cdots \geq \lambda_n > 0$  be the eigenvalues of  $\mathbf{V}$ , let  $\bar{\lambda}_-$  be the mean of the  $n - p$  smallest and  $\bar{\lambda}_+$  the mean of the  $n - p$  largest. Then

$$\frac{\bar{\lambda}_-}{\lambda_1} \leq \frac{\mathbb{E}(v)}{\operatorname{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})} \leq \frac{\bar{\lambda}_+}{\lambda_n}.$$

*Proof.*

- (a) is [Proposition 5.5.7\(b\)](#) with  $\Sigma = \sigma^2 \mathbf{V}$ . (b) is [Theorem 2.3.1](#) with the mean term vanishing, since  $(\mathbf{I} - \mathbf{M})\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ . (c) Put  $\mathbf{c} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{a}$ . By (a),  $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{c}^\top \mathbf{V} \mathbf{c}$ , and  $\mathbf{c}^\top \mathbf{c} = \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$ . By (b),  $E(v) = \sigma^2 \mathbf{c}^\top \mathbf{c} \text{tr}((\mathbf{I} - \mathbf{M})\mathbf{V})/(n - p)$ . Divide, and write  $\mathbf{u} = \mathbf{c}/\|\mathbf{c}\|$ .
- (b)  $\mathbf{I} - \mathbf{M}$  is symmetric idempotent of rank  $n - p$ , so by [Lemma 19.2.1](#) the numerator of [Eq. 19.2.2](#) lies in  $[\bar{\lambda}_-, \bar{\lambda}_+]$ . By [Theorem 1.7.7](#) the denominator lies in  $[\lambda_n, \lambda_1]$ . ■

Part (c) shows what the usual standard error gets wrong. The numerator of [Eq. 19.2.2](#) averages  $\mathbf{V}$  over the residual space, which is what  $s^2$  measures. The denominator evaluates  $\mathbf{V}$  in the one direction  $\mathbf{u}$  that  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  uses. When the errors vary more along  $\mathbf{u}$  than on average, the usual standard error is too small. The bounds in (d), due to Swindel (1968), hold for every design and linear function and can be attained ([Exercise \(C1\)](#)). They are typically wide.

## 19.2.2 When ordinary least squares loses little

Generalized least squares has variance  $\sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{a}$  for  $\mathbf{a}^\top \boldsymbol{\beta}$  ([Corollary 7.2.3](#)). How much larger can the variance of  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  be? The answer again depends only on the extreme eigenvalues of  $\mathbf{V}$ .

### Theorem 19.2.3 — Efficiency of least squares

Under [Eq. 19.2.1](#), with  $\lambda_1$  and  $\lambda_n$  the largest and smallest eigenvalues of  $\mathbf{V}$ ,

$$\frac{\mathbf{a}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{a}}{\mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}} \geq \frac{4\lambda_1 \lambda_n}{(\lambda_1 + \lambda_n)^2}$$

for every  $\mathbf{a} \neq \mathbf{0}$ . The bound is attained for  $p = 1$  and  $\mathbf{X} = \mathbf{x} = (\mathbf{q}_1 + \mathbf{q}_n)/\sqrt{2}$ , where  $\mathbf{q}_1, \mathbf{q}_n$  are unit eigenvectors for  $\lambda_1, \lambda_n$ .

*Proof.* Write  $\mathbf{X} = \mathbf{Q}\mathbf{R}$  with  $\mathbf{Q}$  having orthonormal columns and  $\mathbf{R}$  nonsingular ([Theorem 1.5.2](#)), and put  $\mathbf{b} = \mathbf{R}^{-\top} \mathbf{a}$ ,  $\mathbf{A} = \mathbf{Q}^\top \mathbf{V} \mathbf{Q}$  and  $\mathbf{B} = \mathbf{Q}^\top \mathbf{V}^{-1} \mathbf{Q}$ , both positive definite. Then  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{A} \mathbf{R}^{-\top}$  and  $(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} = \mathbf{R}^{-1} \mathbf{B}^{-1} \mathbf{R}^{-\top}$ , so the ratio is  $\mathbf{b}^\top \mathbf{B}^{-1} \mathbf{b} / \mathbf{b}^\top \mathbf{A} \mathbf{b}$ . It suffices to show  $\mathbf{A} \leq \kappa \mathbf{B}^{-1}$  in the nonnegative definite order, with  $\kappa = (\lambda_1 + \lambda_n)^2 / (4\lambda_1 \lambda_n)$ .

Every eigenvalue  $\lambda$  of  $\mathbf{V}$  satisfies  $(\lambda - \lambda_1)(\lambda - \lambda_n) \leq 0$ , that is,  $\lambda + \lambda_1 \lambda_n / \lambda \leq \lambda_1 + \lambda_n$ . By the spectral decomposition this says  $\mathbf{V} + \lambda_1 \lambda_n \mathbf{V}^{-1} \leq (\lambda_1 + \lambda_n) \mathbf{I}$ . Multiplying by  $\mathbf{Q}^\top$  on the left and  $\mathbf{Q}$  on the right preserves the order, so

$$\mathbf{A} \leq (\lambda_1 + \lambda_n) \mathbf{I} - \lambda_1 \lambda_n \mathbf{B}.$$

Now let  $\mu > 0$  be any eigenvalue of  $\mathbf{B}$ . By the inequality between arithmetic and geometric means,  $\kappa/\mu + \lambda_1 \lambda_n \mu \geq 2\sqrt{\kappa \lambda_1 \lambda_n} = \lambda_1 + \lambda_n$ . Applied along the eigenvectors of  $\mathbf{B}$ , this gives  $(\lambda_1 + \lambda_n) \mathbf{I} - \lambda_1 \lambda_n \mathbf{B} \leq \kappa \mathbf{B}^{-1}$ , and hence  $\mathbf{A} \leq \kappa \mathbf{B}^{-1}$ .

For the attainment, with  $\mathbf{x} = (\mathbf{q}_1 + \mathbf{q}_n)/\sqrt{2}$  we have  $\mathbf{x}^\top \mathbf{x} = 1$ ,  $\mathbf{x}^\top \mathbf{V} \mathbf{x} = (\lambda_1 + \lambda_n)/2$  and



$\mathbf{x}^\top \mathbf{V}^{-1} \mathbf{x} = (1/\lambda_1 + 1/\lambda_n)/2$ . The ratio is  $1/((\mathbf{x}^\top \mathbf{V} \mathbf{x})(\mathbf{x}^\top \mathbf{V}^{-1} \mathbf{x})) = 4\lambda_1 \lambda_n / (\lambda_1 + \lambda_n)^2$ . ■

The scalar inequality behind the proof,  $(\mathbf{x}^\top \mathbf{V} \mathbf{x})(\mathbf{x}^\top \mathbf{V}^{-1} \mathbf{x}) \leq \kappa(\mathbf{x}^\top \mathbf{x})^2$ , is Kantorovich's inequality, and the proof above is a matrix form of it. Bloomfield and Watson (1975) and Knott (1975) found the corresponding sharp bound when efficiency is measured by the determinant of the whole covariance matrix.

The bound is a worst case over all designs. For particular designs ordinary least squares can be fully efficient. Kruskal's theorem (Theorem 6.9.3) says that  $\hat{\boldsymbol{\beta}}$  equals the generalized least squares estimate for every  $\mathbf{Y}$  iff  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , and then the efficiency is one. Even then the standard errors may be wrong.

### Example. (Equicorrelated errors)

Let  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ , with  $-1/(n-1) < \rho < 1$  so that  $\mathbf{V}$  is positive definite, and let the model contain an intercept,  $\mathbf{X} = [\mathbf{1}, \mathbf{X}_1]$ . It arises when all observations share one random shock. Since  $\mathbf{V}\mathbf{X} = (1 - \rho)\mathbf{X} + \rho\mathbf{1}(\mathbf{1}^\top \mathbf{X})$  has columns in  $\mathcal{C}(\mathbf{X})$ , Kruskal's condition holds and  $\hat{\boldsymbol{\beta}}$  is the best linear unbiased estimator. Its covariance is not  $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ , however. Because  $\mathbf{X}\mathbf{e}_1 = \mathbf{1}$  for the first standard basis vector  $\mathbf{e}_1$ , we have  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{1} = \mathbf{e}_1$ , and Theorem 19.2.2(a) gives

$$\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(1 - \rho)(\mathbf{X}^\top \mathbf{X})^{-1} + \sigma^2 \rho \mathbf{e}_1 \mathbf{e}_1^\top.$$

By Exercise (B2),  $E(s^2) = \sigma^2(1 - \rho)$ . So  $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$  is exactly unbiased for the covariance of the slopes, whose information comes from comparisons in which the common shock cancels. For the intercept it misses the term  $\sigma^2 \rho$ . When  $\rho > 0$  this term does not shrink as  $n$  grows, so the intercept is not even consistent, and its reported standard error goes to zero anyway (Exercise (B2)). The same holds for any fitted mean  $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ , since  $\mathbf{x}_0^\top \mathbf{e}_1 = 1$  whenever  $\mathbf{x}_0$  has a leading 1.

## 19.2.3 Autocorrelated errors

### Example. (A trend with autocorrelated errors)

Take  $n = 20$  equally spaced times and errors with  $v_{st} = \rho^{|s-t|}$ , the covariance of a stationary first-order autoregression, with  $\rho = 0.6$ . Fit a straight line in two designs, a linear trend  $x_t = t$  and an alternating regressor  $x_t = (-1)^t$ . For the trend, the expected usual variance of the slope is 0.256 times its true variance, so the root of the expected usual variance is 0.506 of the true standard error. Yet ordinary least squares is nearly efficient: generalized least squares has 0.905 of its variance. For the alternating regressor the error goes the other way. The ratio is 3.271, so the standard error is too large by a factor 1.809, and the efficiency is 0.950. Swindel's bounds for this  $\mathbf{V}$  are 0.193 and 4.31, so both designs lie well towards the extremes. The Kantorovich bound, 0.2351 from  $\lambda_1 = 3.757$  and  $\lambda_n = 0.2514$ , is far below either efficiency.

In 100000 simulated series with normal errors, the nominal 95%  $t$  interval for the slope

covered the true value in 0.693 of them for the trend and 0.9976 for the alternating regressor. [Figure 19.2.1](#) shows the whole range of  $\rho$ .

```
import numpy as np
n = 20
t = np.arange(1, n + 1)
designs = {"trend": t.astype(float), "alternating": (-1.0) ** t}

def ar1(rho):
    return rho ** np.abs(t[:, None] - t[None, :])

def slope_summary(x, V):
    """True variance, mean of the usual estimate, GLS variance of the slope (sigma = 1)."""
    X = np.column_stack([np.ones(n), x])
    G = np.linalg.inv(X.T @ X)
    A = G @ X.T                                # beta_hat = A y
    true_var = (A @ V @ A.T)[1, 1]              # the sandwich
    M = X @ A
    Es2 = np.trace((np.eye(n) - M) @ V) / (n - 2) # E(s^2)
    usual = Es2 * G[1, 1]                        # E of s^2 [(X'X)^{-1}]_{11}
    gls = np.linalg.inv(X.T @ np.linalg.solve(V, X))[1, 1]
    return true_var, usual, gls

rho = 0.6
for name, x in designs.items():
    true_var, usual, gls = slope_summary(x, ar1(rho))
    print(f"{name:12s} usual/true = {usual / true_var:.3f}    GLS/OLS variance = {gls / true_var:.3f}")

from scipy import stats
rng = np.random.default_rng(1902)
reps = 100_000
L = np.linalg.cholesky(ar1(rho))
E = rng.normal(size=(reps, n)) @ L.T          # rows are AR(1) error vectors
tq = stats.t.ppf(0.975, n - 2)
for name, x in designs.items():
    X = np.column_stack([np.ones(n), x])
    G = np.linalg.inv(X.T @ X)
    B = E @ (G @ X.T).T                        # beta_hat - beta, one row per
    data set
    s2 = np.sum((E - B @ X.T) ** 2, axis=1) / (n - 2)
    cover = np.mean(np.abs(B[:, 1]) <= tq * np.sqrt(s2 * G[1, 1]))
    print(f"{name:12s} coverage of the nominal 95% interval: {cover:.4f}")
```

[Eq. 19.2.2](#) explains the pattern. A smooth regressor puts  $\mathbf{u}$  along slowly varying directions, where positively correlated errors have their largest variance; an alternating one puts it where positive correlation cancels. Most time-series regressors are smooth, which is why positive autocorrelation usually makes ordinary standard errors too small.

*Idea.* A wrong covariance matrix is mainly a problem for *standard errors*, not *estimates*:  $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$  can be wrong by a large factor in either direction, and nothing in the fitted model reveals this. [Chapter 21](#) shows how to estimate the correct covariance [Eq. 5.5.3](#) without knowing  $\mathbf{V}$ , and how to detect serial correlation. [Chapter 31](#) treats generalized least squares when  $\mathbf{V}$  is known up to a few parameters ([Definition 31.4.1](#)).

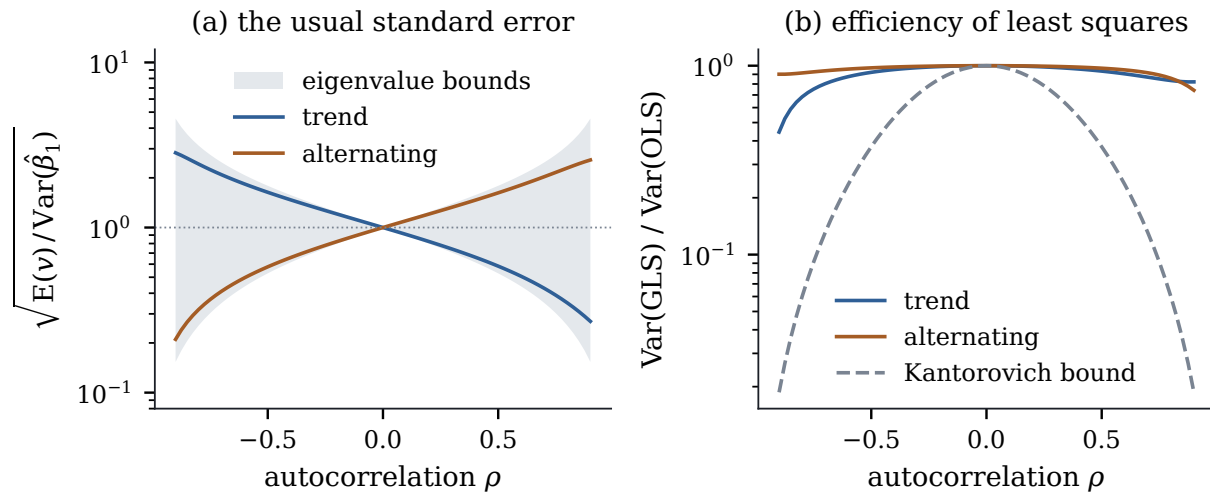


Figure 19.2.1: Straight-line regression with  $n = 20$  and first-order autoregressive errors. (a) The square root of the ratio of the expected usual variance of the slope to its true variance, for a trend and an alternating regressor, with Swindel's bounds (shaded). (b) The efficiency of ordinary least squares relative to generalized least squares, with the Kantorovich bound of Theorem 19.2.3.

## 19.2.4 Exercises

### 19.2.4.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{V} = \text{diag}(v_1, \dots, v_n)$ . Show that  $E(s^2) = \sigma^2 \sum_i (1 - h_{ii})v_i / (n - p)$ , a weighted average of the variances  $\sigma^2 v_i$ .

#### Exercise A2.

Show that the bounds of Theorem 19.2.2(d) do not change when  $\mathbf{V}$  is replaced by  $c\mathbf{V}$  for  $c > 0$ , and that both equal 1 iff  $\mathbf{V}$  is a multiple of  $\mathbf{I}$ .

### 19.2.4.2 B. Practice

#### Exercise B1.

For the regression through the origin  $E(Y_i) = \beta x_i$  with  $\text{Var}(Y_i) = \sigma^2 v_i$ , show that  $\text{Var}(\hat{\beta}) = \sigma^2 \sum_i h_i v_i / \sum_i x_i^2$  and  $E(v) = \sigma^2 \sum_i (1 - h_i)v_i / ((n - 1) \sum_i x_i^2)$ , where  $h_i = x_i^2 / \sum_j x_j^2$  are the leverages. Conclude that the usual standard error is too small on average iff the leverage-weighted average of the  $v_i$  exceeds the  $(1 - h_i)$ -weighted average.

*Solution.*  $\hat{\beta} = \sum_i x_i Y_i / S$  with  $S = \sum_j x_j^2$ , so  $\text{Var}(\hat{\beta}) = \sigma^2 \sum_i x_i^2 v_i / S^2 = \sigma^2 \sum_i h_i v_i / S$ . Exercise (A1) with  $p = 1$  gives  $E(s^2)$ , and  $v = s^2 / S$ . The weights  $h_i$  sum to 1, and so do  $(1 - h_i) / (n - 1)$ . The usual variance is too small on average iff  $\sum_i h_i v_i > \sum_i (1 - h_i)v_i / (n - 1)$ . In particular, when the error variance grows with  $|x_i|$ , the points that dominate  $\hat{\beta}$  are the noisiest, and the usual standard error understates the truth.

**Exercise B2.**

In [Example \(Equicorrelated errors\)](#) with  $\rho > 0$ , show that  $\text{Var}(\hat{\beta}_0) \geq \sigma^2 \rho$  for every  $n$ , while  $E(v) \rightarrow 0$  for the intercept when  $[(\mathbf{X}^\top \mathbf{X})^{-1}]_{11} \rightarrow 0$ . Explain the result by writing  $\varepsilon_i = \sqrt{\rho} \sigma Z + \sqrt{1-\rho} \sigma U_i$  with independent standardized  $Z, U_1, \dots, U_n$ .

*Solution.* From the covariance in the example,  $\text{Var}(\hat{\beta}_0) = \sigma^2(1-\rho)[(\mathbf{X}^\top \mathbf{X})^{-1}]_{11} + \sigma^2 \rho \geq \sigma^2 \rho$ , while  $E(v) = \sigma^2(1-\rho)[(\mathbf{X}^\top \mathbf{X})^{-1}]_{11}$ . The representation has covariance  $\sigma^2 \rho \mathbf{1}\mathbf{1}^\top + \sigma^2(1-\rho)\mathbf{I} = \sigma^2 \mathbf{V}$ . The shared term  $\sqrt{\rho} \sigma Z \mathbf{1}$  lies in  $\mathcal{C}(\mathbf{X})$  and is absorbed entirely by the intercept, since  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{1} = \mathbf{e}_1$ . It never averages out, and it leaves no trace in the residuals, so  $s^2$  cannot see it.

**19.2.4.3 C. Going deeper****Exercise C1.**

Show that both bounds of [Theorem 19.2.2\(d\)](#) are attained. *Hint:* take  $\mathcal{C}(\mathbf{X})$  spanned by eigenvectors of  $\mathbf{V}$ , and choose  $\mathbf{a}$  so that  $\mathbf{u}$  is one of them.

*Solution.* Let  $\mathbf{q}_1, \dots, \mathbf{q}_n$  be orthonormal eigenvectors for  $\lambda_1 \geq \dots \geq \lambda_n$ . If  $\mathbf{X} = [\mathbf{q}_1, \dots, \mathbf{q}_p]$ , then  $\mathbf{I} - \mathbf{M} = \sum_{i>p} \mathbf{q}_i \mathbf{q}_i^\top$  and  $\text{tr}((\mathbf{I} - \mathbf{M})\mathbf{V}) = \sum_{i>p} \lambda_i = (n-p)\bar{\lambda}_-$ . With  $\mathbf{a} = \mathbf{e}_1$ ,  $\mathbf{u} = \mathbf{q}_1$  and  $\mathbf{u}^\top \mathbf{V} \mathbf{u} = \lambda_1$ , so the ratio [Eq. 19.2.2](#) equals  $\bar{\lambda}_-/\lambda_1$ . Symmetrically,  $\mathbf{X} = [\mathbf{q}_{n-p+1}, \dots, \mathbf{q}_n]$  and  $\mathbf{u} = \mathbf{q}_n$  give  $\bar{\lambda}_+/\lambda_n$ .

## 19.3 Non-normal errors

Normality entered Part III at one point only: it turned the moments of  $\hat{\boldsymbol{\beta}}$  and  $s^2$  into exact distributions ([Theorem 7.5.1](#)). Everything that needs only two moments holds for any error distribution with finite variance. Inference about the mean structure survives in large samples, provided no single observation dominates the fit. Inference about  $\sigma^2$  does not survive at any sample size, and neither does prediction of a single new response.

Throughout, the errors  $\varepsilon_1, \dots, \varepsilon_n$  are independent and identically distributed with mean 0 and variance  $\sigma^2 \in (0, \infty)$ , from a distribution that does not change with  $n$ . Where fourth moments are needed, the *excess kurtosis* is  $\kappa = \mu_4/\sigma^4 - 3$ , where  $\mu_4 = E(\varepsilon_i^4)$ . It is zero for normal errors, 6 for centred exponential errors,  $-1.2$  for uniform errors, and at least  $-2$  for every distribution, since  $\mu_4 \geq \sigma^4$  by Jensen's inequality.

### 19.3.1 Asymptotic normality of least squares

Large-sample statements concern a sequence of models  $\mathbf{Y}_n = \mathbf{X}_n \boldsymbol{\beta} + \boldsymbol{\varepsilon}_n$ ,  $n = 1, 2, \dots$ , with  $p$  fixed and  $\mathbf{X}_n$  of rank  $p$ . We write  $h_n^* = \max_i h_{ii}$  for the largest leverage of  $\mathbf{X}_n$ . Besides the three limit theorems recalled in [Section 14.4](#) (the central limit theorem, Slutsky's lemma and the delta method), we need two more standard results of probability theory.

*Remark* (Two more limit theorems).

- (i) *Lindeberg–Feller*. For each  $n$  let  $Z_{n1}, \dots, Z_{nn}$  be independent with mean zero and  $\sum_i \text{Var}(Z_{ni}) = 1$ . If  $\sum_i E[Z_{ni}^2 \mathbf{1}\{|Z_{ni}| > \eta\}] \rightarrow 0$  for every  $\eta > 0$ , then  $\sum_i Z_{ni} \rightarrow N(0, 1)$  in distribution (Billingsley 1995, section 27).
- (ii) *Cramér–Wold and continuous mapping*. Random vectors  $\mathbf{T}_n$  in  $\mathbb{R}^q$  converge in distribution to  $\mathbf{T}$  iff  $\mathbf{c}^\top \mathbf{T}_n \rightarrow \mathbf{c}^\top \mathbf{T}$  for every  $\mathbf{c} \in \mathbb{R}^q$ , and then  $g(\mathbf{T}_n) \rightarrow g(\mathbf{T})$  for every continuous  $g$  (van der Vaart 1998, chapter 2).

### Lemma 19.3.1 — Weighted sums of errors

Let  $\mathbf{w}_n = (w_{n1}, \dots, w_{nn})^\top$  be constant vectors with  $\|\mathbf{w}_n\| = 1$  and  $m_n = \max_i |w_{ni}| \rightarrow 0$ . Then  $\sigma^{-1} \sum_i w_{ni} \varepsilon_i \rightarrow N(0, 1)$  in distribution.

*Proof.* Put  $Z_{ni} = w_{ni} \varepsilon_i / \sigma$ , so that  $\sum_i \text{Var}(Z_{ni}) = \sum_i w_{ni}^2 = 1$ . With  $U = \varepsilon_1 / \sigma$ , for every  $\eta > 0$ ,

$$\begin{aligned} \sum_i E[Z_{ni}^2 \mathbf{1}\{|Z_{ni}| > \eta\}] &= \sum_i w_{ni}^2 E[U^2 \mathbf{1}\{|w_{ni} U| > \eta\}] \\ &\leq E[U^2 \mathbf{1}\{|U| > \eta/m_n\}], \end{aligned}$$

because  $|w_{ni}| \leq m_n$  and the weights  $w_{ni}^2$  sum to one. Since  $E U^2 = 1$  and  $\eta/m_n \rightarrow \infty$ , the right side tends to zero by dominated convergence. The Lindeberg–Feller theorem applies. ■

### Theorem 19.3.2 — Least squares with non-normal errors

In the sequence of models above, suppose  $h_n^* \rightarrow 0$ . Let  $s^2 = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}_n\|^2 / (n - p)$ .

- (a) For every sequence of nonzero vectors  $\mathbf{a}_n \in \mathbb{R}^p$ ,

$$\frac{\mathbf{a}_n^\top \hat{\boldsymbol{\beta}} - \mathbf{a}_n^\top \boldsymbol{\beta}}{\sigma \sqrt{\mathbf{a}_n^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{a}_n}} \rightarrow N(0, 1) \quad \text{in distribution.}$$

- (b)  $s^2 \rightarrow \sigma^2$  in probability. This part does not need  $h_n^* \rightarrow 0$ .
- (c) The  $t$  statistic, with  $s$  in place of  $\sigma$  in (a), tends to  $N(0, 1)$ , and the nominal  $1 - \alpha$  interval  $\mathbf{a}_n^\top \hat{\boldsymbol{\beta}} \pm t_{n-p, \alpha/2} s \sqrt{\mathbf{a}_n^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{a}_n}$  has coverage tending to  $1 - \alpha$ .
- (d) Let  $\boldsymbol{\Lambda}$  be a fixed  $p \times q$  matrix of rank  $q$ , and suppose  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$ . The statistic

$$F_n = \frac{(\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})^\top [\boldsymbol{\Lambda}^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \boldsymbol{\Lambda}]^{-1} (\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \mathbf{d})}{q s^2}$$

satisfies  $q F_n \rightarrow \chi^2(q)$  in distribution, and  $\Pr\{F_n > F_\alpha(q, n - p)\} \rightarrow \alpha$ .

*Proof.*

- (a) Put  $\mathbf{G}_n = (\mathbf{X}_n^\top \mathbf{X}_n)^{-1}$  and  $\tau_n^2 = \mathbf{a}_n^\top \mathbf{G}_n \mathbf{a}_n$ . Since  $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = \mathbf{G}_n \mathbf{X}_n^\top \boldsymbol{\varepsilon}_n$ , the nu-

merator is  $\sum_i c_{ni}\varepsilon_i$  with  $c_{ni} = \mathbf{a}_n^\top \mathbf{G}_n \mathbf{x}_{(i)}$ , and  $\sum_i c_{ni}^2 = \mathbf{a}_n^\top \mathbf{G}_n \mathbf{X}_n^\top \mathbf{X}_n \mathbf{G}_n \mathbf{a}_n = \tau_n^2$ . The Cauchy–Schwarz inequality in the inner product  $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}^\top \mathbf{G}_n \mathbf{v}$  (Proposition 1.1.1 applied to  $\mathbf{G}_n^{1/2} \mathbf{a}_n$  and  $\mathbf{G}_n^{1/2} \mathbf{x}_{(i)}$ ) gives

$$c_{ni}^2 \leq (\mathbf{a}_n^\top \mathbf{G}_n \mathbf{a}_n)(\mathbf{x}_{(i)}^\top \mathbf{G}_n \mathbf{x}_{(i)}) = \tau_n^2 h_{ii}.$$

So  $w_{ni} = c_{ni}/\tau_n$  has unit length and  $\max_i |w_{ni}| \leq \sqrt{h_n^*} \rightarrow 0$ . Apply Lemma 19.3.1.

- (b)  $(n-p)s^2 = \boldsymbol{\varepsilon}_n^\top \boldsymbol{\varepsilon}_n - \boldsymbol{\varepsilon}_n^\top \mathbf{M} \boldsymbol{\varepsilon}_n$ , since  $(\mathbf{I} - \mathbf{M})\mathbf{X}_n \boldsymbol{\beta} = \mathbf{0}$ . By the weak law of large numbers,  $\boldsymbol{\varepsilon}_n^\top \boldsymbol{\varepsilon}_n/n \rightarrow \sigma^2$  in probability. The second term is nonnegative with  $E(\boldsymbol{\varepsilon}_n^\top \mathbf{M} \boldsymbol{\varepsilon}_n) = \sigma^2 \text{tr} \mathbf{M} = \sigma^2 p$  (Theorem 2.3.1), so by Markov's inequality  $\Pr(\boldsymbol{\varepsilon}_n^\top \mathbf{M} \boldsymbol{\varepsilon}_n/n > \delta) \leq \sigma^2 p/(n\delta) \rightarrow 0$ . Since  $n/(n-p) \rightarrow 1$ ,  $s^2 \rightarrow \sigma^2$ . Under a finite fourth moment this is Corollary 5.6.4; the argument here needs only a finite variance.
- (c) The  $t$  statistic is the ratio in (a) multiplied by  $\sigma/s \rightarrow 1$ , so Slutsky's lemma gives the limit. The quantile  $t_{n-p, \alpha/2}$  tends to  $z_{\alpha/2}$ , because the  $t(\nu)$  distribution tends to  $N(0, 1)$  and the normal distribution function is continuous and strictly increasing. Hence  $|T_n| - t_{n-p, \alpha/2} \rightarrow |Z| - z_{\alpha/2}$ , and the coverage tends to  $\Pr(|Z| \leq z_{\alpha/2}) = 1 - \alpha$ .
- (d) Let  $\mathbf{W}_n = \boldsymbol{\Lambda}^\top \mathbf{G}_n \boldsymbol{\Lambda}$ , which is positive definite, and  $\mathbf{Z}_n = \mathbf{W}_n^{-1/2} \boldsymbol{\Lambda}^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/\sigma$ . Under the hypothesis the numerator of  $F_n$  is  $\sigma^2 \|\mathbf{Z}_n\|^2$ . For a unit vector  $\mathbf{c} \in \mathbb{R}^q$ ,  $\mathbf{c}^\top \mathbf{Z}_n = \mathbf{a}_n^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/\sigma$  with  $\mathbf{a}_n = \boldsymbol{\Lambda} \mathbf{W}_n^{-1/2} \mathbf{c}$ , and  $\mathbf{a}_n^\top \mathbf{G}_n \mathbf{a}_n = \mathbf{c}^\top \mathbf{W}_n^{-1/2} \mathbf{W}_n \mathbf{W}_n^{-1/2} \mathbf{c} = 1$ . By (a),  $\mathbf{c}^\top \mathbf{Z}_n \rightarrow N(0, 1)$ , which is the law of  $\mathbf{c}^\top \mathbf{Z}$  for  $\mathbf{Z} \sim N_q(\mathbf{0}, \mathbf{I})$ . By Cramér–Wold,  $\mathbf{Z}_n \rightarrow \mathbf{Z}$ , and by continuous mapping  $\|\mathbf{Z}_n\|^2 \rightarrow \chi^2(q)$ . Then  $qF_n = \|\mathbf{Z}_n\|^2 \sigma^2/s^2 \rightarrow \chi^2(q)$  by (b) and Slutsky's lemma. Finally,  $qF(q, \nu)$  is  $\chi^2(q)$  divided by an independent  $\chi^2(\nu)/\nu$ , which tends to one, so  $qF_\alpha(q, n-p)$  tends to the upper  $\alpha$  point of  $\chi^2(q)$ , and the rejection probability tends to  $\alpha$  as in (c).

■

Part (a) allows  $\mathbf{a}_n$  to depend on the design. The condition  $h_n^* \rightarrow 0$  says that no observation keeps a non-negligible share of the fit. For a straight line at  $x = 1, \dots, n$  the largest leverage is below  $4/n$  (Exercise (A1)). The condition is not a technicality, as the next example shows.

### Example. (An observation with its own parameter)

Fit an intercept together with an indicator of the first observation,  $\mathbf{X}_n = [\mathbf{1}, \mathbf{e}_1]$ , so that  $h_{11} = 1$  for every  $n$ . The least squares estimates are  $\hat{\mu} = \bar{Y}_{(1)}$ , the mean of observations  $2, \dots, n$ , and  $\hat{\delta} = Y_1 - \bar{Y}_{(1)}$ . Hence

$$\hat{\delta} - \delta = \varepsilon_1 - \bar{\varepsilon}_{(1)}, \quad \text{Var}(\hat{\delta}) = \sigma^2 \left( 1 + \frac{1}{n-1} \right),$$

and since  $\bar{\varepsilon}_{(1)} \rightarrow 0$  in probability,  $\hat{\delta} - \delta \rightarrow \varepsilon_1$  in distribution. The limit is the error distribution itself. No amount of other data can average away the error of one observation.

## 19.3.2 How far from normal in a finite sample

[Theorem 19.3.2](#) is a limit statement. A finite-sample measure of the approach comes from the third and fourth cumulants of the numerator, which leverage controls directly.

### Proposition 19.3.3 — Skewness and kurtosis of a linear estimate

Let the errors have finite fourth moment, skewness  $\gamma_1 = \mu_3/\sigma^3$  with  $\mu_3 = E(\varepsilon_i^3)$ , and excess kurtosis  $\kappa$ . For  $\mathbf{a} \neq \mathbf{0}$  let  $Z = (\mathbf{a}^\top \hat{\boldsymbol{\beta}} - \mathbf{a}^\top \boldsymbol{\beta})/\sqrt{\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})}$ , and let  $w_i = c_i/\|\mathbf{c}\|$  with  $\mathbf{c} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{a}$ . Then

$$E(Z^3) = \gamma_1 \sum_i w_i^3, \quad E(Z^4) - 3 = \kappa \sum_i w_i^4,$$

and  $|\sum_i w_i^3| \leq \sqrt{h^*}$ ,  $\sum_i w_i^4 \leq h^*$ , where  $h^* = \max_i h_{ii}$ .

*Proof.*  $Z = \sigma^{-1} \sum_i w_i \varepsilon_i$  with  $\sum_i w_i^2 = 1$ . Expanding the cube, every term with some index appearing exactly once has mean zero, by independence and  $E \varepsilon_i = 0$ . Only the terms  $w_i^3 \varepsilon_i^3$  remain, so  $E(Z^3) = \sigma^{-3} \mu_3 \sum_i w_i^3$ . In the fourth power the surviving terms are  $w_i^4 \varepsilon_i^4$  and, for  $i \neq j$ , the 3 pairings of  $w_i^2 \varepsilon_i^2 w_j^2 \varepsilon_j^2$ . So

$$\begin{aligned} E(Z^4) &= \frac{\mu_4}{\sigma^4} \sum_i w_i^4 + 3 \sum_{i \neq j} w_i^2 w_j^2 \\ &= (\kappa + 3) \sum_i w_i^4 + 3 \left(1 - \sum_i w_i^4\right), \end{aligned}$$

using  $\sum_{i \neq j} w_i^2 w_j^2 = (\sum_i w_i^2)^2 - \sum_i w_i^4$ . As in the proof of [Theorem 19.3.2\(a\)](#),  $w_i^2 \leq h_{ii} \leq h^*$ . Hence  $|\sum_i w_i^3| \leq \max_i |w_i| \sum_i w_i^2 \leq \sqrt{h^*}$  and  $\sum_i w_i^4 \leq \max_i w_i^2 \leq h^*$ . ■

So skewness is averaged away more slowly than heavy tails, although with symmetric weights it can cancel exactly ([Exercise \(B1\)](#)). The denominator  $s$  of the  $t$  statistic adds terms of the same order, so the proposition describes the leading effect only.

#### Example. (Coverage of the $t$ interval)

Fit a straight line with normal, centred exponential and scaled  $t(3)$  errors, each with variance one, in two designs. In the *spread* design  $x = 1, \dots, n$ , and  $h_n^*$  is 0.345, 0.096 and 0.025 at  $n = 10, 40$  and 160. In the *far-point* design the last value is moved to  $x = 4n$ , and  $h_n^*$  is 0.954, 0.800 and 0.485. With 100000 data sets per case, the coverages of the nominal 95% interval for the slope are:

| Design    | Errors      | $n = 10$ | $n = 40$ | $n = 160$ |
|-----------|-------------|----------|----------|-----------|
| spread    | normal      | 0.950    | 0.951    | 0.950     |
| spread    | exponential | 0.955    | 0.953    | 0.950     |
| spread    | $t(3)$      | 0.955    | 0.952    | 0.951     |
| far point | normal      | 0.951    | 0.950    | 0.951     |

|           |             |       |       |       |
|-----------|-------------|-------|-------|-------|
| far point | exponential | 0.933 | 0.948 | 0.950 |
| far point | $t(3)$      | 0.931 | 0.944 | 0.953 |

The two-sided coverages hide unbalanced tails. With exponential errors and the far point, the interval misses above the slope in 0.067 of the data sets at  $n = 10$  and below it in only 0.001. At  $n = 160$  the two proportions are 0.038 and 0.012, still far from 0.025 each, as the slowly falling leverage predicts. A one-sided test in this design has nearly three times its nominal size at  $n = 10$ . Panel (a) of [Figure 19.3.1](#) shows the skewed  $t$  statistic at  $n = 40$ . The  $t(3)$  errors have no fourth moment, but [Theorem 19.3.2](#) still applies.

```
import numpy as np
from scipy import stats

def errors(law, size, rng):
    """Errors with mean 0 and variance 1."""
    if law == "normal":
        return rng.normal(size=size)
    if law == "exponential":
        # skewed: skewness 2, excess
        # kurtosis 6
        return rng.exponential(size=size) - 1.0
    if law == "t3":
        # heavy tails: variance 3 before
        # scaling
        return rng.standard_t(3, size=size) / np.sqrt(3.0)
    if law == "uniform":
        # short tails: excess kurtosis
        # -1.2
        return rng.uniform(-np.sqrt(3), np.sqrt(3), size=size)

def design(kind, n):
    x = np.arange(1.0, n + 1)
    if kind == "far point":
        x[-1] = 4.0 * n
    return np.column_stack([np.ones(n), x])
```

```
def slope_coverage(kind, law, n, reps, rng):
    """Proportion of nominal 95% t intervals for the slope that cover it."""
    X = design(kind, n)
    G = np.linalg.inv(X.T @ X)
    A = G @ X.T
    E = errors(law, (reps, n), rng)
    B = E @ A.T
    # beta_hat - beta
    s2 = np.sum((E - B @ X.T) ** 2, axis=1) / (n - 2)
    T = B[:, 1] / np.sqrt(s2 * G[1, 1])
    tq = stats.t.ppf(0.975, n - 2)
    return np.mean(np.abs(T) <= tq), np.mean(T > tq), np.mean(T < -tq)
```

```
rng = np.random.default_rng(1903)
for kind in ["spread", "far point"]:
    for law in ["normal", "exponential", "t3"]:
        cov, up, lo = slope_coverage(kind, law, 10, 20_000, rng)
        print(f"{kind:9s} {law:11s} n=10: coverage {cov:.3f} (misses above {up:.3f},
              below {lo:.3f})")
```



### 19.3.3 Inference about the error variance

The interval for  $\sigma^2$  (Proposition 12.1.4) rests on SSE, a sum of *squared* errors, whose variance depends on the kurtosis (Proposition 5.6.3, Proposition 7.5.4). A central limit theorem applies, but with the wrong variance.

#### Proposition 19.3.4 — The interval for sigma squared

In the sequence of models above, let the errors have finite fourth moment with  $\kappa > -2$ . Then  $\sqrt{n}(s^2/\sigma^2 - 1) \rightarrow N(0, 2 + \kappa)$ , and the coverage of the nominal  $1 - \alpha$  interval of Proposition 12.1.4 tends to

$$2\Phi\left(z_{\alpha/2}\sqrt{2/(2+\kappa)}\right) - 1, \quad (19.3.1)$$

where  $\Phi$  is the standard normal distribution function. The limit equals  $1 - \alpha$  iff  $\kappa = 0$ .

*Proof.* As in the proof of Theorem 19.3.2(b),

$$\begin{aligned} \sqrt{n}(s^2 - \sigma^2) &= \frac{n}{n-p}\sqrt{n}\left(\frac{\boldsymbol{\varepsilon}_n^\top \boldsymbol{\varepsilon}_n}{n} - \sigma^2\right) \\ &\quad + \frac{\sqrt{n}p\sigma^2}{n-p} - \frac{\sqrt{n}}{n-p}\boldsymbol{\varepsilon}_n^\top \mathbf{M}\boldsymbol{\varepsilon}_n. \end{aligned}$$

The  $\varepsilon_i^2$  are independent with mean  $\sigma^2$  and variance  $\mu_4 - \sigma^4 = \sigma^4(2 + \kappa)$ , so by the central limit theorem the first term tends to  $N(0, \sigma^4(2 + \kappa))$ . The second term tends to zero, and so does the third in probability, because  $E(\boldsymbol{\varepsilon}_n^\top \mathbf{M}\boldsymbol{\varepsilon}_n) = \sigma^2 p$  and Markov's inequality applies to  $\boldsymbol{\varepsilon}_n^\top \mathbf{M}\boldsymbol{\varepsilon}_n/\sqrt{n}$ . Slutsky's lemma gives the first claim.

Write  $\nu = n - p$  and let  $c_\nu^-$  and  $c_\nu^+$  be the lower and upper  $\alpha/2$  points of  $\chi^2(\nu)$ . The interval covers  $\sigma^2$  iff  $c_\nu^- \leq \nu s^2/\sigma^2 \leq c_\nu^+$ , that is, iff

$$\frac{c_\nu^- - \nu}{\sqrt{\nu}} \leq \sqrt{\nu}\left(\frac{s^2}{\sigma^2} - 1\right) \leq \frac{c_\nu^+ - \nu}{\sqrt{\nu}}.$$

A  $\chi^2(\nu)$  variable is a sum of  $\nu$  independent squared standard normals, each with variance 2, so by the central limit theorem  $(c_\nu^\pm - \nu)/\sqrt{2\nu} \rightarrow \pm z_{\alpha/2}$ , and the bounds tend to  $\pm z_{\alpha/2}\sqrt{2}$ . The middle term tends to  $N(0, 2 + \kappa)$  because  $\nu/n \rightarrow 1$ . The coverage therefore tends to  $\Pr(|Z|\sqrt{2 + \kappa} \leq z_{\alpha/2}\sqrt{2})$ , which is Eq. 19.3.1. ■

For centred exponential errors,  $\kappa = 6$ , the nominal 95% interval has limiting coverage  $2\Phi(1.96/2) - 1 = 0.673$ . For uniform errors,  $\kappa = -1.2$ , it tends to 0.998. Panel (b) of Figure 19.3.1 shows simulated coverages with the spread design. For exponential errors they are 0.786 at  $n = 10$ , 0.717 at  $n = 40$  and 0.672 at  $n = 2560$ : more data make it *worse*. This answers Exercise (B3) and explains the failure reported in Section 12.1.

```
def sigma2_coverage(law, n, reps, rng):
    X = design("spread", n)
    E = errors(law, (reps, n), rng)
```

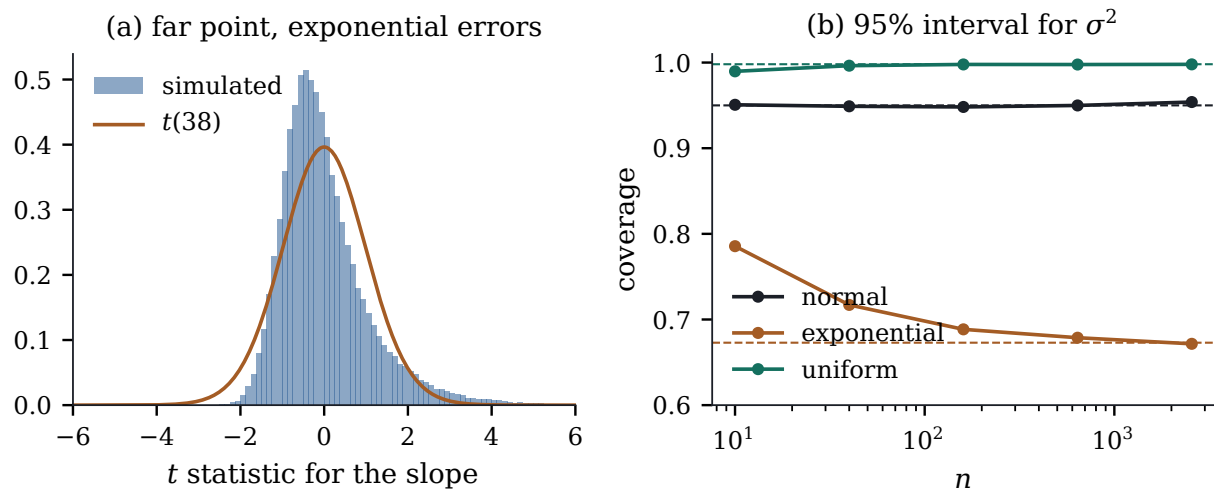


Figure 19.3.1: (a) The  $t$  statistic for the slope in the far-point design with  $n = 40$  and centred exponential errors (100000 data sets), with the  $t(38)$  density. (b) Coverage of the nominal 95% interval for  $\sigma^2$  in the spread design, against  $n$ , with the limits Eq. 19.3.1 dashed.

```
B = np.linalg.solve(X.T @ X, X.T @ E.T).T
sse = np.sum((E - B @ X.T) ** 2, axis=1) # sigma^2 = 1
lo, hi = stats.chi2.ppf([0.025, 0.975], n - 2)
return np.mean((sse >= lo) & (sse <= hi))

def limit(kappa, z=stats.norm.ppf(0.975)):
    """Limiting coverage of the nominal 95% chi-squared interval."""
    return 2 * stats.norm.cdf(z * np.sqrt(2 / (2 + kappa))) - 1

print("limits:", {k: round(limit(k), 3) for k in [0, 6, -1.2]})
```

Prediction intervals fail for the same reason: they must cover a single new error, which no averaging acts on (Proposition 12.3.3).

### 19.3.4 Why the F test is protected and the variance-ratio test is not

The  $F$  statistic is a ratio of two quadratic forms, each with a kurtosis-dependent variance. The effects can cancel, as Box and Watson (1962) and Atiqullah (1962) found. Since  $\log F$  is approximately the difference of the two mean squares divided by  $\sigma^2$ , it suffices to study that difference.

#### Proposition 19.3.5 — Kurtosis and a ratio of mean squares

Let  $\mathbf{P}_1$  and  $\mathbf{P}_2$  be symmetric idempotent with  $\mathbf{P}_1\mathbf{P}_2 = \mathbf{0}$ , ranks  $f_1$  and  $f_2$ , and  $\mathbf{P}_1\mathbf{X}\boldsymbol{\beta} = \mathbf{P}_2\mathbf{X}\boldsymbol{\beta} = \mathbf{0}$ . Let the errors have finite fourth moment and excess kurtosis  $\kappa$ , put  $S_j^2 = \mathbf{Y}^\top \mathbf{P}_j \mathbf{Y} / f_j$ , and let  $\mathbf{p}_j$  be the vector of diagonal entries of  $\mathbf{P}_j$ . Then  $L = (S_1^2 - S_2^2) / \sigma^2$  has mean zero and

$$\text{Var}(L) = \frac{2}{f_1} + \frac{2}{f_2} + \kappa \|\mathbf{d}\|^2, \quad \mathbf{d} = \frac{\mathbf{p}_1}{f_1} - \frac{\mathbf{p}_2}{f_2}.$$

In particular, if both  $\mathbf{P}_1$  and  $\mathbf{P}_2$  have constant diagonals, then  $\mathbf{d} = \mathbf{0}$  and  $\text{Var}(L)$  is the same as under normality.

*Proof.* Since  $\mathbf{P}_j \mathbf{Y} = \mathbf{P}_j \boldsymbol{\varepsilon}$ ,  $L = \boldsymbol{\varepsilon}^\top \mathbf{A} \boldsymbol{\varepsilon} / \sigma^2$  with  $\mathbf{A} = \mathbf{P}_1 / f_1 - \mathbf{P}_2 / f_2$ , whose diagonal is  $\mathbf{d}$ . By Theorem 2.3.1,  $E(L) = \text{tr} \mathbf{A} = 1 - 1 = 0$ . By Theorem 2.3.3 with zero means,  $\text{Var}(\boldsymbol{\varepsilon}^\top \mathbf{A} \boldsymbol{\varepsilon}) = (\mu_4 - 3\sigma^4) \mathbf{d}^\top \mathbf{d} + 2\sigma^4 \text{tr}(\mathbf{A}^2)$ , and  $\text{tr}(\mathbf{A}^2) = \text{tr} \mathbf{P}_1 / f_1^2 + \text{tr} \mathbf{P}_2 / f_2^2 - 2 \text{tr}(\mathbf{P}_1 \mathbf{P}_2) / (f_1 f_2) = 1/f_1 + 1/f_2$ . Divide by  $\sigma^4$ . If the diagonals are constant,  $\mathbf{p}_j = (f_j/n) \mathbf{1}$  because  $\text{tr} \mathbf{P}_j = f_j$ , and  $\mathbf{d} = \mathbf{0}$ . ■

For the  $F$  test of a linear hypothesis,  $\mathbf{P}_1 = \mathbf{M} - \mathbf{M}_0$  and  $\mathbf{P}_2 = \mathbf{I} - \mathbf{M}$ . Designs in which both have constant diagonals are called *quadratically balanced*. Every balanced one-way or complete factorial layout is an example. In such designs the kurtosis of the errors does not affect the  $F$  statistic to first order. The variance-ratio test for comparing the error variances of two independent samples is the opposite case. With  $n_1$  and  $n_2$  observations, each sample centred at its own mean,  $\mathbf{d}$  has entries  $1/n_1$  on the first sample and  $-1/n_2$  on the second (Exercise (B2)), so

$$\text{Var}(L) = \frac{2}{f_1} + \frac{2}{f_2} + \kappa \left( \frac{1}{n_1} + \frac{1}{n_2} \right),$$

and the kurtosis term is as large as the normal-theory term. The test is not robust at any sample size. For exponential errors and  $n_1 = n_2 = 20$ , the nominal 5% test rejected equal variances in 0.268 of 100000 simulated pairs of samples. Its large-sample limit is 0.327 (Exercise (B2)). The  $F$  test for equal means in a balanced one-way layout with three groups of eight had size 0.042, and with unbalanced groups of 2, 4 and 18 it had size 0.058. In the unbalanced layout  $\text{Var}(L)$  is 1.870 instead of 1.095, which overstates the effect on the size: in the tail, skewness and the curvature of the logarithm also matter.

*Idea.* Inference about  $\boldsymbol{\beta}$  is robust to the *shape* of the errors when no observation has large leverage. Inference about  $\sigma^2$ , variance ratios or single future responses is not, at any sample size.

## 19.3.5 Exercises

### 19.3.5.1 A. Check your understanding

#### Exercise A1.

For a straight line at  $x = 1, \dots, n$ , show that the largest leverage is  $1/n + 3(n-1)/(n(n+1)) < 4/n$ .

#### Exercise A2.

In the far-point design of Example (Coverage of the  $t$  interval), show that the leverage of the far point is approximately  $147/(n+147)$  for large  $n$ . Why does the table still show nearly nominal two-sided coverage at  $n = 160$ , when the leverage is almost one half?

## 19.3.5.2 B. Practice

**Exercise B1.**

In a one-way layout with  $m$  observations per group and skewed errors, compute  $\sum_i w_i^3$  of Proposition 19.3.3 for (i) a single group mean and (ii) the difference of two group means. What does this say about one-sample and two-sample  $t$  tests with skewed data?

*Solution.*

- (i) The mean of group 1 has  $c_i = 1/m$  on its  $m$  observations, so  $w_i = m^{-1/2}$  there and  $\sum_i w_i^3 = m \cdot m^{-3/2} = m^{-1/2}$ . (ii) The difference has  $c_i = \pm 1/m$  on the two groups, so  $w_i = \pm(2m)^{-1/2}$ , and the cubes cancel:  $\sum_i w_i^3 = 0$ . The numerator of a one-sample  $t$  statistic has skewness  $\gamma_1/\sqrt{m}$ . That of a two-sample comparison with equal group sizes has none, and its non-normality starts with the kurtosis term, of order  $1/m$ . Equal-sized two-sample comparisons are therefore much better protected against skewness.

**Exercise B2.**

Two independent samples of size  $m$  are each centred at their own mean. Let  $S_1^2$  and  $S_2^2$  be the sample variances. (i) Show that the vector  $\mathbf{d}$  of Proposition 19.3.5 has entries  $1/m$  and  $-1/m$ . (ii) Using the first part of Proposition 19.3.4 for each sample, show that when the two variances are equal the two-sided variance-ratio test of nominal size  $\alpha$  has size tending to  $2(1 - \Phi(z_{\alpha/2}\sqrt{2/(2+\kappa)}))$  as  $m \rightarrow \infty$ .

*Solution.*

- (i)  $\mathbf{P}_1$  is the centring matrix on the first sample and zero elsewhere, so its diagonal is  $1 - 1/m = f_1/m$  there, and  $\mathbf{p}_1/f_1$  has entries  $1/m$  on the first sample. Likewise  $\mathbf{p}_2/f_2$  has entries  $1/m$  on the second, which enter  $\mathbf{d}$  with a minus sign. (ii) By Proposition 19.3.4 (with  $p = 1$ ),  $\sqrt{m}(S_j^2/\sigma^2 - 1) \rightarrow N(0, 2 + \kappa)$  for  $j = 1, 2$ , independently. By the delta method,  $\sqrt{m} \log(S_1^2/S_2^2) \rightarrow N(0, 2(2 + \kappa))$ . Under normality the same statistic tends to  $N(0, 4)$ , and the critical values of the  $F(m-1, m-1)$  distribution satisfy  $\sqrt{m} \log F_{\alpha/2} \rightarrow 2z_{\alpha/2}$ . The rejection probability therefore tends to  $\Pr(|Z|\sqrt{2/(2+\kappa)} > 2z_{\alpha/2}) = 2(1 - \Phi(z_{\alpha/2}\sqrt{2/(2+\kappa)}))$ . For  $\kappa = 6$  and  $\alpha = 0.05$  this is 0.327.

## 19.3.5.3 C. Going deeper

**Exercise C1.**

Let the errors be independent with mean zero, variances  $\sigma_i^2 \in [m_0, M_0]$  with  $m_0 > 0$ , and  $\sup_i E|\varepsilon_i|^3 < \infty$ . Suppose  $h_n^* \rightarrow 0$ . Show that for every nonzero  $\mathbf{a}_n$ ,  $(\mathbf{a}_n^\top \hat{\boldsymbol{\beta}} - \mathbf{a}_n^\top \boldsymbol{\beta})/\sqrt{\text{Var}(\mathbf{a}_n^\top \hat{\boldsymbol{\beta}})} \rightarrow N(0, 1)$ , where the variance is the sandwich Eq. 5.5.3. Explain why this does not make the usual  $t$  statistic asymptotically standard normal.

*Solution.* With  $c_{ni}$  as in the proof of Theorem 19.3.2,  $\text{Var}(\mathbf{a}_n^\top \hat{\boldsymbol{\beta}}) = \sum_i c_{ni}^2 \sigma_i^2 \geq m_0 \tau_n^2$ . Put  $Z_{ni} = c_{ni} \varepsilon_i / \sqrt{\text{Var}(\mathbf{a}_n^\top \hat{\boldsymbol{\beta}})}$ . Lyapunov's bound  $E[Z^2 \mathbf{1}\{|Z| > \eta\}] \leq E|Z|^3/\eta$  gives the

Lindeberg sum at most

$$\begin{aligned} & \frac{\sup_i E|\varepsilon_i|^3 \sum_i |c_{ni}|^3}{\eta (m_0 \tau_n^2)^{3/2}} \\ & \leq \frac{\sup_i E|\varepsilon_i|^3}{\eta m_0^{3/2}} \max_i \frac{|c_{ni}|}{\tau_n} \\ & \leq \frac{\sup_i E|\varepsilon_i|^3}{\eta m_0^{3/2}} \sqrt{h_n^*} \rightarrow 0, \end{aligned}$$

using  $\sum_i |c_{ni}|^3 \leq \max_i |c_{ni}| \tau_n^2$ . The usual  $t$  statistic divides by  $s \tau_n$ , and  $s^2 \tau_n^2$  estimates  $\bar{\sigma}^2 \tau_n^2$  with  $\bar{\sigma}^2$  an average variance, not  $\sum_i c_{ni}^2 \sigma_i^2$ . The ratio of the two is the quantity of [Exercise \(B1\)](#), and it need not tend to one. [Chapter 21](#) replaces  $s^2 \tau_n^2$  by a consistent estimate of the sandwich.

## 19.4 A single bad observation

Some failures affect a single case, such as a misplaced decimal point. How much one bad case matters depends on its leverage and on how far its response is from the rest of the data. This section makes that exact; [Chapter 20](#) turns it into diagnostics.

### 19.4.1 The effect of contaminating one response

#### Theorem 19.4.1 — One contaminated response

Let  $\mathbf{X}$  have full column rank. For a data vector  $\mathbf{y}$ , a case  $k$  and a shift  $\Delta$ , let  $\mathbf{y}_\Delta = \mathbf{y} + \Delta \mathbf{e}_k$ , where  $\mathbf{e}_k$  is the  $k$ th standard basis vector.

- (a) The estimate moves by  $\hat{\boldsymbol{\beta}}(\mathbf{y}_\Delta) - \hat{\boldsymbol{\beta}}(\mathbf{y}) = \Delta(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(k)}$ . The fitted vector moves by  $\Delta \mathbf{M} \mathbf{e}_k$ , a vector of length  $|\Delta| \sqrt{h_{kk}}$ . The fitted value of case  $k$  moves by  $\Delta h_{kk}$ , and its residual by only  $\Delta(1 - h_{kk})$ .
- (b) If  $E(\mathbf{Y}) = \mathbf{X} \boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , then  $\hat{\boldsymbol{\beta}}(\mathbf{Y}_\Delta)$  has mean  $\boldsymbol{\beta} + \Delta(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(k)}$ , and the residual sum of squares has mean  $\sigma^2(n - p) + \Delta^2(1 - h_{kk})$ .
- (c) Suppose  $h_{kk} < 1$ , and let  $\hat{\boldsymbol{\beta}}_{(k)}$  be the estimate without case  $k$ . Then for every  $\mathbf{y}$ ,

$$\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(k)})\|^2 = \frac{h_{kk}}{1 - h_{kk}} \cdot \frac{\hat{\varepsilon}_k^2}{1 - h_{kk}}. \quad (19.4.1)$$

- (d) Under the conditions of (c), adding  $\mathbf{e}_k$  to the model matrix as an extra column gives the coefficients  $\hat{\boldsymbol{\beta}}_{(k)}$  for  $\mathbf{X}$  and  $\hat{\delta} = y_k - \mathbf{x}_{(k)}^\top \hat{\boldsymbol{\beta}}_{(k)} = \hat{\varepsilon}_k / (1 - h_{kk})$  for the new column.

*Proof.*

- (a)  $\hat{\boldsymbol{\beta}}(\mathbf{y}_\Delta) - \hat{\boldsymbol{\beta}}(\mathbf{y}) = \Delta(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{e}_k = \Delta(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(k)}$ . Multiplying by  $\mathbf{X}$  gives  $\Delta \mathbf{M} \mathbf{e}_k$ , with squared length  $\Delta^2 \mathbf{e}_k^\top \mathbf{M} \mathbf{e}_k = \Delta^2 h_{kk}$ , since  $\mathbf{M}$  is symmetric and idempotent. Its  $k$ th entry is  $\Delta h_{kk}$ , and the residual vector moves by  $\Delta(\mathbf{I} - \mathbf{M}) \mathbf{e}_k$ , whose  $k$ th entry is  $\Delta(1 - h_{kk})$ .
- (b) The mean follows from (a) and [Theorem 5.5.1\(a\)](#). The residual vector  $(\mathbf{I} - \mathbf{M}) \mathbf{Y}_\Delta$

has mean  $\Delta(\mathbf{I} - \mathbf{M})\mathbf{e}_k$ , so [Eq. 2.3.1](#) gives  $\sigma^2(n - p) + \Delta^2\|(\mathbf{I} - \mathbf{M})\mathbf{e}_k\|^2 = \sigma^2(n - p) + \Delta^2(1 - h_{kk})$ .

- (c) By [Proposition 10.6.2\(a\)](#),  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(k)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(k)} \hat{\varepsilon}_k / (1 - h_{kk})$ , and  $\|\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(k)}\|^2 = \mathbf{x}_{(k)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(k)} = h_{kk}$ .
- (d) With the extra column, the criterion is  $\sum_{i \neq k} (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2 + (y_k - \mathbf{x}_{(k)}^\top \mathbf{b} - \delta)^2$ . For any  $\mathbf{b}$  the choice  $\delta = y_k - \mathbf{x}_{(k)}^\top \mathbf{b}$  makes the last term zero, so the minimizing  $\mathbf{b}$  minimizes the first sum alone, which is  $\hat{\boldsymbol{\beta}}_{(k)}$ . The augmented matrix has full rank because  $h_{kk} < 1$  means  $\mathbf{e}_k \notin \mathcal{C}(\mathbf{X})$  ([Proposition 6.8.1](#)). Then  $\hat{\delta} = y_k - \mathbf{x}_{(k)}^\top \hat{\boldsymbol{\beta}}_{(k)}$ , which is  $\hat{\varepsilon}_k / (1 - h_{kk})$  by [Proposition 10.6.2\(b\)](#). ■

Part (a), which needs no model, says that a high-leverage case *hides* its own contamination: of a shift  $\Delta$ , the fraction  $h_{kk}$  goes into its fitted value and only  $1 - h_{kk}$  shows in its residual. Raw residuals therefore miss the cases that matter most, and [Chapter 20](#) rescales them.

Part (c) makes “influence is leverage times residual” exact: a leverage factor  $h_{kk}/(1 - h_{kk})$  times the squared *deleted* residual  $\hat{\varepsilon}_k^2/(1 - h_{kk})$ . Either factor alone is harmless. Divided by  $p s^2$ , [Eq. 19.4.1](#) is Cook’s distance ([Definition 20.4.1](#)). Part (d) is the *mean-shift outlier model*: giving case  $k$  its own parameter is the same as deleting it. Part (b) adds a second cost:  $s^2$  is inflated by about  $\Delta^2(1 - h_{kk})/(n - p)$ , which weakens every test, even for coefficients the bad case hardly moves.

### Example. (The last day of the stack loss data)

In the stack loss data of [Example \(Stack loss\)](#),  $n = 21$  days of a plant oxidizing ammonia, regress the stack loss on air flow, water temperature and acid concentration with an intercept. The average leverage is  $p/n = 0.190$ . By [Eq. 19.4.1](#), the case whose deletion moves the fitted vector most is day 21, with leverage 0.285 and residual  $-7.238$ . Deleting it moves the fitted vector by a distance of 5.396. The next largest move, for day 1, is 2.543, and the median over all days is 1.228. The largest leverage, 0.412 for day 17, comes with a residual of only  $-1.520$ , and that day has little influence.

Without day 21, the air-flow coefficient rises from 0.716 to 0.889, the water temperature coefficient falls from 1.295 to 0.817, and  $s$  drops from 3.243 to 2.569. The mean-shift estimate for day 21 is  $\hat{\delta} = -10.116$ . The listing computes every deletion effect from the single fit ([Theorem 19.4.1\(c\)](#)).

Adding  $\Delta = 10$  to the response of the lowest-leverage day, day 5 with  $h_{55} = 0.052$ , moves the fitted vector by 2.285 and its own residual by 9.478. The same shift on day 17 moves the fitted vector by 6.420 and its residual by only 5.879.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.stackloss.load_pandas().data
y = data["STACKLOSS"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["AIRFLOW", "WATERTEMP", "ACIDCONC"]].
                    to_numpy()])
n, p = X.shape

G = np.linalg.inv(X.T @ X)
```

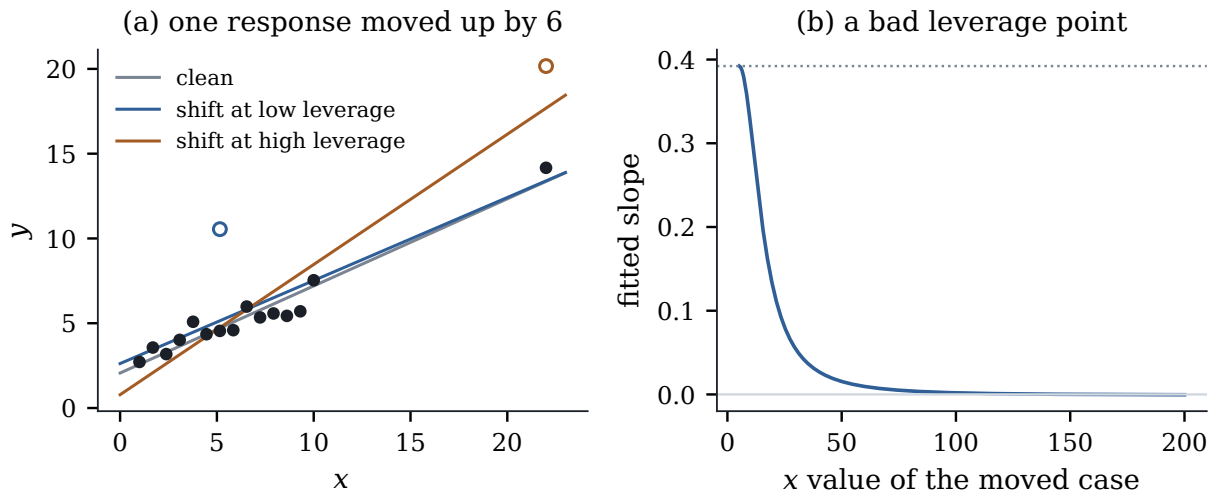


Figure 19.4.1: A straight line through 15 points. (a) The same upward shift of 6 applied to a point in the middle of the design barely moves the line; applied to the high-leverage point on the right, it tilts the line, and that point's residual stays small. (b) A *bad leverage point*: one case keeps its response while its  $x$  value is moved to the right, and the fitted slope (dotted: its original value) is dragged towards zero.

```

beta_hat = G @ X.T @ y
e_hat = y - X @ beta_hat
h = np.einsum("ij,jk,ik->i", X, G, X)                    # leverages

# the change in beta_hat from deleting case k, without refitting
change = (G @ X.T) * (e_hat / (1 - h))                    # column k is beta_hat - beta_hat_
(k)
shift = h * e_hat**2 / (1 - h) ** 2                        # ||X (beta_hat - beta_hat_(k))
//^2
k = int(np.argmax(shift))
print("most influential case:", k + 1, " leverage", round(h[k], 3), " residual",
      round(e_hat[k], 3))
print("beta_hat", np.round(beta_hat, 3))
print("beta_hat without it", np.round(beta_hat - change[:, k], 3))
print("mean-shift estimate e_k/(1-h_kk):", round(e_hat[k] / (1 - h[k]), 3))

```

```

Delta = 10.0
for j in [int(np.argmin(h)), int(np.argmax(h))]:
    y_c = y.copy()
    y_c[j] += Delta                                     # contaminate one
    response
    b_c = G @ X.T @ y_c
    print(f"case {j + 1:2d}: h = {h[j]:.3f}, ||X(b_c - b)|| = {np.linalg.norm(X @ (
    b_c - beta_hat)):.3f},"
          f" |Delta| sqrt(h) = {Delta * np.sqrt(h[j]):.3f}")

```

## 19.4.2 Breakdown

By [Theorem 19.4.1\(a\)](#) the effect of one case grows without bound with  $\Delta$ . Contaminated regressors can be as bad: in panel (b) of [Figure 19.4.1](#) a single case moved far out in  $x$  takes control of the slope. The following definition, from Donoho and Huber (1983), makes this precise.

**Definition 19.4.1 — Breakdown point**

Let  $T$  be an estimator that maps a data set  $Z$  of  $n$  cases  $(\mathbf{x}_{(i)}, y_i)$  to a vector in  $\mathbb{R}^p$ . For  $m \leq n$ , let  $b(m; T, Z)$  be the supremum of  $\|T(Z') - T(Z)\|$  over all data sets  $Z'$  obtained from  $Z$  by replacing any  $m$  of its cases with arbitrary values, for which  $T(Z')$  is defined. The **finite-sample breakdown point** of  $T$  at  $Z$  is

$$\varepsilon^*(T, Z) = \min \left\{ \frac{m}{n} : b(m; T, Z) = \infty \right\}.$$

**Proposition 19.4.2 — Least squares breaks down at one case**

For least squares and every data set with a full-rank model matrix,  $\varepsilon^* = 1/n$ .

*Proof.* Since  $\mathbf{X}$  has full rank, some row  $\mathbf{x}_{(k)}$  is nonzero, and then  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(k)} \neq \mathbf{0}$ . Replace case  $k$  by  $(\mathbf{x}_{(k)}, y_k + \Delta)$ . The model matrix is unchanged, so the estimate is defined, and by [Theorem 19.4.1\(a\)](#) it moves by  $|\Delta| \|(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(k)}\|$ , which is unbounded in  $\Delta$ . So  $b(1) = \infty$ . No smaller fraction than  $1/n$  is possible. ■

So least squares offers no protection against gross errors. The sample median, by contrast, has breakdown point close to one half ([Exercise A1](#)). High-breakdown regression estimators exist too, at a cost in efficiency discussed with robust fitting in [Section 20.6](#). Hampel (1974) introduced the related *influence function*, the effect of an infinitesimal contamination at a point  $(\mathbf{x}_0, y_0)$ . For least squares it is unbounded in both  $\mathbf{x}_0$  and  $y_0$  ([Exercise C1](#)).

*Idea.* High leverage makes a case both more influential and better hidden, and one case can move the estimate arbitrarily far.

**19.4.3 Exercises****19.4.3.1 A. Check your understanding****Exercise A1.**

Show that the sample median of  $n$  numbers has breakdown point  $\lfloor (n+1)/2 \rfloor / n$ , in the sense of [Definition 19.4.1](#) with  $p = 1$  and no regressors.

**Exercise A2.**

In [Theorem 19.4.1\(a\)](#), show that the residual of every other case  $i$  moves by  $-\Delta m_{ik}$ , where  $m_{ik}$  is an entry of  $\mathbf{M}$ . Which cases' residuals are most affected by a contaminated high-leverage case?

**19.4.3.2 B. Practice**



**Exercise B1.**

In a straight-line regression, move the  $x$  value of case  $k$  to  $t$  while keeping  $y_k$  fixed. Show that the fitted slope tends to 0 as  $t \rightarrow \infty$ . If instead  $y_k = ct + d$  moves with  $t$ , show that the slope tends to  $c$ .

*Solution.* Let  $\bar{x}_t$  and  $\bar{y}$  be the means with case  $k$  at  $t$ . The slope is  $\sum_i (x_i - \bar{x}_t)(y_i - \bar{y}) / \sum_i (x_i - \bar{x}_t)^2$ . As  $t \rightarrow \infty$ ,  $t - \bar{x}_t = (1 - 1/n)t + O(1)$  while the other deviations  $x_i - \bar{x}_t$  are  $-t/n + O(1)$ . The denominator is  $(1 - 1/n)t^2 + O(t)$ . With  $y_k$  fixed, the numerator is  $O(t)$ , since each term is a deviation of order  $t$  times a bounded  $y_i - \bar{y}$ , so the slope is  $O(1/t)$ . With  $y_k = ct + d$ , put  $r_i = y_i - cx_i$ , so that  $r_k = d$  and all the  $r_i$  stay bounded. Since  $y_i = cx_i + r_i$  and least squares is linear in the response, the slope is  $c$  plus the slope of  $r$  on  $x$ , which is  $O(1/t)$  as before. The far case dictates the slope.

**Exercise B2.**

Suppose case  $k$  is contaminated by  $\Delta$ , and consider the  $t$  statistic for a coefficient  $\beta_j$  whose estimate is unaffected,  $[(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(k)}]_j = 0$ . Using [Theorem 19.4.1\(b\)](#), explain how the contamination still changes the test, and show that its effect on  $E(s^2)$  is largest for low-leverage cases.

**19.4.3.3 C. Going deeper****Exercise C1.**

Let  $(\mathbf{x}, Y)$  have a joint distribution  $F$  with  $\Sigma = E(\mathbf{x}\mathbf{x}^\top)$  positive definite, and let  $\beta^*(F) = \Sigma^{-1} E(\mathbf{x}Y)$  be the population least squares coefficient. For the mixture  $F_\epsilon = (1 - \epsilon)F + \epsilon \delta_{(\mathbf{x}_0, y_0)}$ , which puts mass  $\epsilon$  at a single point, show that

$$\left. \frac{d}{d\epsilon} \beta^*(F_\epsilon) \right|_{\epsilon=0} = \Sigma^{-1} \mathbf{x}_0 (y_0 - \mathbf{x}_0^\top \beta^*(F)),$$

and compare with [Theorem 19.4.1\(c\)](#) for a sample of size  $n$ .

*Solution.* Under  $F_\epsilon$ ,  $\Sigma_\epsilon = (1 - \epsilon)\Sigma + \epsilon \mathbf{x}_0 \mathbf{x}_0^\top$  and  $\mathbf{m}_\epsilon = (1 - \epsilon)E(\mathbf{x}Y) + \epsilon \mathbf{x}_0 y_0$ , and  $\Sigma_\epsilon \beta^*(F_\epsilon) = \mathbf{m}_\epsilon$ . Differentiate at  $\epsilon = 0$ , writing  $\mathbf{d}$  for the derivative:  $(\mathbf{x}_0 \mathbf{x}_0^\top - \Sigma) \beta^* + \Sigma \mathbf{d} = \mathbf{x}_0 y_0 - E(\mathbf{x}Y)$ . Since  $\Sigma \beta^* = E(\mathbf{x}Y)$ , this gives  $\Sigma \mathbf{d} = \mathbf{x}_0 (y_0 - \mathbf{x}_0^\top \beta^*)$ . The influence is the product of a leverage-like factor  $\Sigma^{-1} \mathbf{x}_0$  and a residual, unbounded in both. In a sample, replacing  $F$  by the empirical distribution and  $\epsilon$  by  $1/n$  gives  $n^{-1}(\mathbf{X}^\top \mathbf{X}/n)^{-1} \mathbf{x}_{(k)} \hat{\epsilon}_k$ , which is [Proposition 10.6.2\(a\)](#) without the factor  $1/(1 - h_{kk})$ .

## 19.5 Random regressors and a curved mean

When the regressors are random, as in almost every observational study, the linear model is read conditionally on  $\mathbf{X}$ . [Section 14.1](#) settled the case in which the conditional model is right. If  $\mathbf{Y} | \mathbf{X} \sim N_n(\mathbf{X}\beta, \sigma^2 \mathbf{I})$ , every  $t$  and  $F$  procedure keeps its exact level unconditionally, whatever the distribution of the regressors ([Theorem 14.1.2](#)), and  $\hat{\beta}$  is unbiased with covariance

$\sigma^2 \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1}]$  (Corollary 5.5.5). Nothing further needs to be proved about that case.

Random regressors create new problems only when the conditional model fails, and it can fail in two essentially different ways. The regressors can be correlated with the errors, as when they are measured with error or when a common cause of regressor and response is omitted. Then least squares does not even estimate a coefficient of interest, a problem taken up in Chapter 24 and Chapter 25. Or the conditional mean  $\mathbb{E}(Y | \mathbf{x})$  is not linear in  $\mathbf{x}$ . This section treats the second case. Its message is that a curved mean, fitted by a linear model with random regressors, behaves like heteroscedastic error, and the usual standard errors are wrong even when the true errors are homoscedastic.

## 19.5.1 What least squares estimates, conditionally

### Proposition 19.5.1 — The design-dependent target

Let  $(\mathbf{X}, \mathbf{Y})$  be random with  $\text{rank}(\mathbf{X}) = p < n$  almost surely,  $\mathbb{E}(\mathbf{Y} | \mathbf{X}) = \boldsymbol{\mu}(\mathbf{X})$  for some function  $\boldsymbol{\mu}$  with values in  $\mathbb{R}^n$ , not necessarily in  $\mathcal{C}(\mathbf{X})$ , and  $\text{Cov}(\mathbf{Y} | \mathbf{X}) = \sigma^2 \mathbf{I}_n$ . Put  $\boldsymbol{\beta}_{\mathbf{X}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\mu}(\mathbf{X})$ .

- (a)  $\mathbb{E}(\hat{\boldsymbol{\beta}} | \mathbf{X}) = \boldsymbol{\beta}_{\mathbf{X}}$  and  $\text{Cov}(\hat{\boldsymbol{\beta}} | \mathbf{X}) = \sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ .
- (b) If the second moments below are finite,  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \mathbb{E}[(\mathbf{X}^\top \mathbf{X})^{-1}] + \text{Cov}(\boldsymbol{\beta}_{\mathbf{X}})$ .
- (c)  $\mathbb{E}(s^2 | \mathbf{X}) = \sigma^2 + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}(\mathbf{X})\|^2 / (n - p)$ .

*Proof.*

- (a) Given  $\mathbf{X}$ , apply Proposition 5.5.7 with  $\boldsymbol{\delta} = \boldsymbol{\mu}(\mathbf{X}) - \mathbf{X}\boldsymbol{\beta}_{\mathbf{X}}$ , which satisfies  $\mathbf{X}^\top \boldsymbol{\delta} = \mathbf{0}$ , and Theorem 5.5.1(b). (b) is the law of total covariance (Proposition 2.5.4) applied to (a). (c) is Eq. 2.3.1 given  $\mathbf{X}$ .

■

Given the design,  $\hat{\boldsymbol{\beta}}$  scatters around  $\boldsymbol{\beta}_{\mathbf{X}}$  exactly as the usual formula says. But  $\boldsymbol{\beta}_{\mathbf{X}}$ , the least squares coefficient of the true mean over the design drawn, changes from design to design. When  $\boldsymbol{\mu}(\mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$  it is constant (Corollary 5.5.5). Otherwise (b) adds a *between-design* component that no conditional calculation can see, while (c) inflates  $s^2$  by the misfit, which compensates only partly.

## 19.5.2 The large-sample picture

The natural fixed target is the population projection of Section 6.11,  $\boldsymbol{\beta}^* = \boldsymbol{\Sigma}^{-1} \mathbb{E}(\mathbf{x}Y)$ , which least squares estimates consistently whatever the form of the mean (Proposition 6.11.3). Its sampling distribution is the following.

### Theorem 19.5.2 — Least squares for the population projection

Let  $(\mathbf{x}_i, Y_i)$ ,  $i = 1, \dots, n$ , be independent copies of  $(\mathbf{x}, Y)$ , where  $\mathbf{x} \in \mathbb{R}^p$  (whose first component may be the constant 1) and  $Y$  have finite fourth moments and  $\boldsymbol{\Sigma} = \mathbb{E}(\mathbf{x}\mathbf{x}^\top)$

is positive definite. Let  $\beta^* = \Sigma^{-1} E(\mathbf{x}Y)$ ,  $U = Y - \mathbf{x}^\top \beta^*$  and  $\Omega = E(U^2 \mathbf{x}\mathbf{x}^\top)$ . Then  $E(\mathbf{x}U) = \mathbf{0}$ , and as  $n \rightarrow \infty$ :

- (a)  $\sqrt{n}(\hat{\beta} - \beta^*) \rightarrow N_p(\mathbf{0}, \Sigma^{-1}\Omega\Sigma^{-1})$  in distribution;
- (b)  $s^2 \rightarrow E(U^2)$  and  $n s^2(\mathbf{X}^\top \mathbf{X})^{-1} \rightarrow E(U^2) \Sigma^{-1}$  in probability;
- (c) if  $E(U^2 | \mathbf{x}) = E(U^2)$  almost surely, the two limiting covariances agree. Otherwise the usual interval for  $\mathbf{a}^\top \beta^*$  of nominal level  $1 - \alpha$  has coverage tending to  $2\Phi(z_{\alpha/2}\sqrt{\rho_{\mathbf{a}}}) - 1$ , with  $\rho_{\mathbf{a}} = E(U^2) \mathbf{a}^\top \Sigma^{-1} \mathbf{a} / \mathbf{a}^\top \Sigma^{-1} \Omega \Sigma^{-1} \mathbf{a}$ .

*Proof.*  $E(\mathbf{x}U) = E(\mathbf{x}Y) - \Sigma\beta^* = \mathbf{0}$ . Write  $\mathbf{U} = \mathbf{Y} - \mathbf{X}\beta^*$ , so that

$$\sqrt{n}(\hat{\beta} - \beta^*) = \left( \frac{\mathbf{X}^\top \mathbf{X}}{n} \right)^{-1} \frac{\mathbf{X}^\top \mathbf{U}}{\sqrt{n}}.$$

By the weak law of large numbers  $\mathbf{X}^\top \mathbf{X}/n = n^{-1} \sum_i \mathbf{x}_i \mathbf{x}_i^\top \rightarrow \Sigma$  in probability, so it is invertible with probability tending to one and its inverse tends to  $\Sigma^{-1}$ , inversion being continuous at  $\Sigma$ . The vectors  $\mathbf{x}_i U_i$  are independent with mean  $\mathbf{0}$  and covariance  $\Omega$ , which is finite because  $|E(U^2 x_j x_k)| \leq (E U^4 E x_j^2 x_k^2)^{1/2}$  and  $U$  has a finite fourth moment. By the central limit theorem  $\mathbf{X}^\top \mathbf{U}/\sqrt{n} = \sqrt{n} \cdot n^{-1} \sum_i \mathbf{x}_i U_i \rightarrow N_p(\mathbf{0}, \Omega)$ . If a random matrix  $\mathbf{A}_n$  tends in probability to a constant  $\mathbf{A}$  and a random vector  $\mathbf{T}_n$  tends in distribution to  $\mathbf{T}$ , then  $\mathbf{A}_n \mathbf{T}_n \rightarrow \mathbf{A}\mathbf{T}$ : the difference  $(\mathbf{A}_n - \mathbf{A})\mathbf{T}_n$  tends to zero in probability because  $\mathbf{T}_n$  is bounded in probability,  $\mathbf{A}\mathbf{T}_n \rightarrow \mathbf{A}\mathbf{T}$  by continuous mapping, and Slutsky's lemma combines the two. This proves (a).

- (b)  $(n-p)s^2 = \mathbf{U}^\top (\mathbf{I} - \mathbf{M}) \mathbf{U} = \mathbf{U}^\top \mathbf{U} - Q_n$ , where  $Q_n = (\mathbf{X}^\top \mathbf{U}/\sqrt{n})^\top (\mathbf{X}^\top \mathbf{X}/n)^{-1} (\mathbf{X}^\top \mathbf{U}/\sqrt{n})$  converges in distribution by the argument for (a) and continuous mapping. Hence  $Q_n/n \rightarrow 0$  in probability, while  $\mathbf{U}^\top \mathbf{U}/n \rightarrow E(U^2)$  by the law of large numbers. Then  $n s^2(\mathbf{X}^\top \mathbf{X})^{-1} = s^2(\mathbf{X}^\top \mathbf{X}/n)^{-1} \rightarrow E(U^2) \Sigma^{-1}$ .
- (c) If  $E(U^2 | \mathbf{x}) = c$ , then  $\Omega = E[E(U^2 | \mathbf{x}) \mathbf{x}\mathbf{x}^\top] = c \Sigma$ , with  $c = E(U^2)$ , and  $\Sigma^{-1} \Omega \Sigma^{-1} = c \Sigma^{-1}$ . In general, the  $t$  statistic for  $\mathbf{a}^\top \beta^*$  is  $\sqrt{n} \mathbf{a}^\top (\hat{\beta} - \beta^*)$  divided by  $(n s^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a})^{1/2}$ . By (a), (b) and Slutsky's lemma it tends to  $N(0, 1/\rho_{\mathbf{a}})$ , and the coverage follows as in [Theorem 19.3.2\(c\)](#).

■

The condition in (c) holds when the mean is linear and the errors are homoscedastic. It typically fails when either assumption fails ([Exercise B2](#)) gives an exception), and a curved mean alone is enough. If  $E(Y | \mathbf{x}) = m(\mathbf{x})$  and  $\text{Var}(Y | \mathbf{x}) = \sigma^2$ , then  $U = (m(\mathbf{x}) - \mathbf{x}^\top \beta^*) + \varepsilon$  and

$$E(U^2 | \mathbf{x}) = (m(\mathbf{x}) - \mathbf{x}^\top \beta^*)^2 + \sigma^2,$$

which varies with  $\mathbf{x}$  wherever the linear approximation is poor, exactly like a heteroscedastic error. The correct covariance  $\Sigma^{-1} \Omega \Sigma^{-1}$  is a population version of the sandwich [Eq. 5.5.3](#), and [Chapter 21](#) shows how to estimate it.

#### Example. (A line through a parabola)

Let  $X$  be uniform on  $(0, 2)$ ,  $E(Y | X) = X^2$  and  $\text{Var}(Y | X) = 0.04$ , with normal errors. A straight line is fitted. The population line has slope  $\beta^* = \text{Cov}(X, X^2)/\text{Var}(X) = 2$  and intercept  $-2/3$ , and the approximation error is  $U_0^2 - 1/3$  with  $U_0 = X - 1$

uniform on  $(-1, 1)$ . The slope's component of  $\Sigma^{-1}\Omega\Sigma^{-1}$  is  $E[(X - 1)^2 U^2] / \text{Var}(X)^2$  (Exercise A1), which equals  $9(44/945 + 0.04/3) = 0.5390$ . The usual formula estimates  $E(U^2) / \text{Var}(X) = 3(4/45 + 0.04) = 0.3867$  instead. The true standard error is 1.181 times what the usual one estimates, and by Theorem 19.5.2(c) the nominal 95% interval for  $\beta^*$  has limiting coverage 0.903.

Simulation agrees. With 40000 samples the interval covered  $\beta^* = 2$  in 0.896 of samples of size 50 and 0.900 of size 400; with 10000 samples of size 1600, in 0.908. Over the same samples,  $n$  times the variance of the slope was 0.584, 0.546 and 0.530, approaching the sandwich value 0.5390, not the usual 0.3867. As a statement about the design-dependent target  $\beta_{\mathbf{X}}$ , the same intervals are *conservative*. They covered it in 0.999 of the samples of size 50, because  $s^2$  estimates  $E(U^2) = 0.1289$  rather than  $\sigma^2 = 0.04$ . In a larger simulation of 200000 samples of size 50, the variance of the slope, 0.01163, splits as in Proposition 19.5.1(b) into 0.00249 from the errors and 0.00913 from the variation of  $\beta_{\mathbf{X}}$  between designs.

```
import numpy as np
from scipy import stats
from fractions import Fraction as Fr
sigma2 = Fr(1, 25)                                # sigma = 0.2
EU2, EU4, EU6 = Fr(1, 3), Fr(1, 5), Fr(1, 7)      # moments of U = X - 1 ~
                                                    Uniform(-1, 1)
Eu2 = EU4 - Fr(2, 3) * EU2 + Fr(1, 9) + sigma2    # u = U^2 - 1/3 + error
sandwich = (EU6 - Fr(2, 3) * EU4 + Fr(1, 9) * EU2 + sigma2 * EU2) / EU2**2
usual = Eu2 / EU2
print("n Var(slope): sandwich", float(sandwich), " usual", float(usual))
z = stats.norm.ppf(0.975)
print("limiting coverage of the usual interval:",
      2 * stats.norm.cdf(z * np.sqrt(float(usual / sandwich))) - 1)
```

```
def one_run(n, reps, rng):
    x = rng.uniform(0, 2, size=(reps, n))
    y = x**2 + 0.2 * rng.normal(size=(reps, n))
    xc = x - x.mean(axis=1, keepdims=True)
    Sxx = np.sum(xc**2, axis=1)
    b = np.sum(xc * y, axis=1) / Sxx                # least squares slope
    b_X = np.sum(xc * x**2, axis=1) / Sxx           # its conditional mean
                                                    given X
    a = y.mean(axis=1) - b * x.mean(axis=1)
    s2 = np.sum((y - a[:, None] - b[:, None] * x) ** 2, axis=1) / (n - 2)
    half = stats.t.ppf(0.975, n - 2) * np.sqrt(s2 / Sxx)
    return np.mean(np.abs(b - 2.0) <= half), np.mean(np.abs(b - b_X) <= half), n * b
    .var()

rng = np.random.default_rng(1905)
for n in [50, 400]:
    c_star, c_X, nvar = one_run(n, 4000, rng)
    print(f"n = {n}: covers beta* {c_star:.3f}, covers beta_X {c_X:.3f}, n Var(slope) {nvar:.3f}")
```

Which target is right depends on the question:  $\beta^*$  summarizes the population,  $\beta_{\mathbf{X}}$  this design. With a curved mean, neither describes  $E(Y | \mathbf{x})$  itself.

### 19.5.3 Controlled regressors

One situation with random regressors is more benign than it looks. In a controlled experiment the analyst sets a nominal value  $x_i$ , but the value actually delivered is  $x_i + \delta_i$ , with an error  $\delta_i$  independent of  $x_i$ . If the response follows  $\beta_0 + \beta_1(x_i + \delta_i) + \varepsilon_i$ , then regressing on the nominal values gives

$$Y_i = \beta_0 + \beta_1 x_i + (\beta_1 \delta_i + \varepsilon_i),$$

a correct linear model with a larger error variance ([Exercise \(B1\)](#)). Least squares on the nominal settings is unbiased, as Berkson (1950) pointed out. Contrast the *classical* measurement error model, in which the true value is fixed and the *recorded* value carries the error. There the error is correlated with the recorded regressor, and least squares is biased towards zero ([Theorem 24.1.1](#)).

Rounding can fall on either side. If the grid value is what was set, as with a dial, the model is Berkson's. If a continuous true value is *recorded* rounded to a grid of width  $w$ , the rounding error is roughly uniform with variance  $w^2/12$  and roughly independent of the true value, so the error is classical. In simple regression, classical error of variance  $\tau^2$  multiplies the slope by  $\text{Var}(x)/(\text{Var}(x) + \tau^2)$  ([Chapter 24](#)). So measurement error, rounding included, can be ignored when its variance is a small fraction of  $\text{Var}(x)$ .

*Idea.* With random regressors, what matters is whether the conditional model holds. If it does, nothing changes; if the mean is curved,  $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$  is not the covariance.

### 19.5.4 Exercises

#### 19.5.4.1 A. Check your understanding

##### Exercise A1.

With  $\mathbf{x} = (1, X)^\top$ , show that the slope entries of  $\Sigma^{-1}\Omega\Sigma^{-1}$  and  $E(U^2)\Sigma^{-1}$  are  $E[(X - \mu_X)^2 U^2] / \text{Var}(X)^2$  and  $E(U^2) / \text{Var}(X)$ .

*Solution.*  $\Sigma = \begin{pmatrix} 1 & \mu_X \\ \mu_X & E X^2 \end{pmatrix}$  has determinant  $\text{Var}(X)$ , and the second row of  $\Sigma^{-1}$  is  $(-\mu_X, 1) / \text{Var}(X)$ . So  $\mathbf{e}_2^\top \Sigma^{-1} \mathbf{x} = (X - \mu_X) / \text{Var}(X)$ , and  $\mathbf{e}_2^\top \Sigma^{-1} \Omega \Sigma^{-1} \mathbf{e}_2 = E[U^2 (X - \mu_X)^2] / \text{Var}(X)^2$ . Similarly  $[\Sigma^{-1}]_{22} = 1 / \text{Var}(X)$ .

##### Exercise A2.

In [Proposition 19.5.1](#), show that  $\text{Cov}(\boldsymbol{\beta}_\mathbf{X}) = \mathbf{0}$  when  $\boldsymbol{\mu}(\mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$ , and give an example of a nonlinear  $\boldsymbol{\mu}$  for which  $\text{Cov}(\boldsymbol{\beta}_\mathbf{X}) = \mathbf{0}$  anyway.

#### 19.5.4.2 B. Practice

##### Exercise B1.

In the controlled-regressor model, let  $\delta_i$  and  $\varepsilon_i$  be independent, with variances  $\tau^2$  and  $\sigma^2$ , independent of the settings. Show that least squares on the nominal  $x_i$  is unbiased

for  $(\beta_0, \beta_1)$ , that the usual standard errors are valid with error variance  $\sigma^2 + \beta_1^2 \tau^2$ , and that  $s^2$  estimates this variance.

*Solution.* The composite errors  $\eta_i = \beta_1 \delta_i + \varepsilon_i$  are independent of the fixed settings, with mean 0 and variance  $\sigma^2 + \beta_1^2 \tau^2$ , and uncorrelated across  $i$ . The model  $Y_i = \beta_0 + \beta_1 x_i + \eta_i$  therefore satisfies the second-moment assumptions of [Chapter 5](#), so [Theorem 5.5.1](#) and [Theorem 5.6.1](#) apply with  $\sigma^2$  replaced by  $\sigma^2 + \beta_1^2 \tau^2$ . If  $\delta_i$  and  $\varepsilon_i$  are normal, so is  $\eta_i$ , and the exact theory of Part III applies too.

### Exercise B2.

In [Theorem 19.5.2](#), show that  $\Sigma^{-1} \Omega \Sigma^{-1} = E(U^2) \Sigma^{-1}$  iff  $E[(E(U^2 | \mathbf{x}) - E(U^2)) \mathbf{x} \mathbf{x}^\top] = \mathbf{0}$ . Give an example of a nonconstant  $E(U^2 | \mathbf{x})$  for which the usual standard errors are nevertheless consistent.

### 19.5.4.3 C. Going deeper

#### Exercise C1.

In [Theorem 19.5.2](#), let  $E(Y | \mathbf{x}) = m(\mathbf{x})$ ,  $\text{Var}(Y | \mathbf{x}) = \sigma^2$ , and put  $d(\mathbf{x}) = m(\mathbf{x}) - \mathbf{x}^\top \beta^*$ . Show that  $\sqrt{n}(\beta_{\mathbf{X}} - \beta^*) \rightarrow N_p(\mathbf{0}, \Sigma^{-1} E(d^2 \mathbf{x} \mathbf{x}^\top) \Sigma^{-1})$ , and that  $\Sigma^{-1} \Omega \Sigma^{-1} = \sigma^2 \Sigma^{-1} + \Sigma^{-1} E(d^2 \mathbf{x} \mathbf{x}^\top) \Sigma^{-1}$ . Interpret the two terms in the light of [Proposition 19.5.1\(b\)](#).

*Solution.*  $\beta_{\mathbf{X}} - \beta^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{d}$  with  $\mathbf{d} = (d(\mathbf{x}_1), \dots, d(\mathbf{x}_n))^\top$ , and  $E(\mathbf{x} d(\mathbf{x})) = E(\mathbf{x} U) = \mathbf{0}$  because  $E(U | \mathbf{x}) = d(\mathbf{x})$ . The argument of [Theorem 19.5.2\(a\)](#) with  $U$  replaced by  $d(\mathbf{x})$  gives the first limit. Since  $U = d(\mathbf{x}) + \varepsilon$  with  $E(\varepsilon | \mathbf{x}) = 0$ ,  $E(U^2 | \mathbf{x}) = d(\mathbf{x})^2 + \sigma^2$ , and  $\Omega = \sigma^2 \Sigma + E(d^2 \mathbf{x} \mathbf{x}^\top)$ . The first term,  $\sigma^2 \Sigma^{-1}$ , is the asymptotic version of  $n \sigma^2 E[(\mathbf{X}^\top \mathbf{X})^{-1}]$ , the within-design part; the second is the asymptotic version of  $n \text{Cov}(\beta_{\mathbf{X}})$ , the between-design part. (They are limits of those moments only under uniform integrability, and for some designs the finite-sample moments do not even exist.)

## 19.6 Collinearity, previewed

Collinearity, near-linear dependence among the columns of  $\mathbf{X}$ , violates none of the assumptions of Part III. As long as  $\mathbf{X}$  has full rank, every result of Parts II and III holds. What collinearity does is make some of them unhelpful: intervals that are correct but very wide. [Chapter 26](#) develops diagnostics, and [Chapter 27](#) remedies that trade bias for variance.

### 19.6.1 Variance inflation

For a single coefficient, the Frisch–Waugh–Lovell theorem gave [Eq. 6.6.2](#):

$$\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{S_{jj}} \cdot \frac{1}{1 - R_j^2},$$

where  $S_{jj}$  is the centred sum of squares of  $\mathbf{x}_j$  and  $R_j^2$  is the coefficient of determination from regressing  $\mathbf{x}_j$  on the other regressors. The first factor is the variance  $\mathbf{x}_j$  would give if it were

orthogonal to the other regressors. The second, the **variance inflation factor**  $\text{VIF}_j = 1/(1 - R_j^2)$ , is the price of the part of  $\mathbf{x}_j$  that the others can already explain. A variance inflation factor of 100 means the standard error is ten times what an orthogonal design with the same spread would give.

The factors have a compact matrix form. Let  $\mathbf{R}$  be the correlation matrix of the  $k$  regressors other than the intercept. Then  $\text{VIF}_j = [\mathbf{R}^{-1}]_{jj}$  ([Exercise \(A1\)](#)), and the eigenvalues of  $\mathbf{R}$  show where the problem lies.

### Proposition 19.6.1 — Where collinearity costs variance

Let  $\mathbf{X}$  have full rank and let  $\mathbf{X}^\top \mathbf{X} = \sum_{\ell=1}^p \lambda_\ell \mathbf{v}_\ell \mathbf{v}_\ell^\top$  with orthonormal eigenvectors  $\mathbf{v}_\ell$  and eigenvalues  $\lambda_\ell > 0$ .

- (a) For every  $\mathbf{a} \in \mathbb{R}^p$ ,  $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \sum_{\ell} (\mathbf{a}^\top \mathbf{v}_\ell)^2 / \lambda_\ell$ .
- (b) The fitted values at the design points have average variance  $\sigma^2 p/n$ , whatever the eigenvalues.
- (c) With  $\mathbf{R}$  as above,  $\sum_j \text{VIF}_j = \text{tr } \mathbf{R}^{-1} = \sum_{\ell} 1/\lambda_\ell(\mathbf{R})$ , and every  $\text{VIF}_j \leq 1/\lambda_{\min}(\mathbf{R})$ .

*Proof.*

- (a) By [Theorem 1.7.2](#),  $(\mathbf{X}^\top \mathbf{X})^{-1} = \sum_{\ell} \lambda_\ell^{-1} \mathbf{v}_\ell \mathbf{v}_\ell^\top$ , and  $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{a}$  ([Corollary 5.5.2](#)). (b)  $\text{Cov}(\hat{\mathbf{Y}}) = \sigma^2 \mathbf{M}$ , and  $\text{tr } \mathbf{M} = p$  ([Proposition 6.3.4](#)). (c) The first equality is  $\text{VIF}_j = [\mathbf{R}^{-1}]_{jj}$  summed over  $j$ . The trace is the sum of the eigenvalues of  $\mathbf{R}^{-1}$ , which are  $1/\lambda_\ell(\mathbf{R})$ . Each diagonal entry of  $\mathbf{R}^{-1}$  is at most its largest eigenvalue, by [Theorem 1.7.7](#) applied to a standard basis vector.

■

Part (a) locates the damage. A small eigenvalue inflates the variance of  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  only through the component of  $\mathbf{a}$  along its eigenvector. Individual coefficients usually have such components, and so do predictions at points that break the pattern of the regressors; predictions that follow the pattern do not, and by (b) fitted values at the design points are as precise on average as in any design. Collinearity is a problem about *attributing* an effect to one of several regressors that moved together, not about fitting.

#### Example. (Longley's regression)

Longley's (1967) data, which [Example \(Longley's data, exactly\)](#) used to test the accuracy of least squares algorithms, regress total employment on six economic series over  $n = 16$  years. The variance inflation factors range from 3.59 for ARMED (the size of the armed forces) to 1788.5 for GNP. The eigenvalues of the correlation matrix run from 4.603 down to 0.00038, and the factors sum to 3119.4, which is the sum of the reciprocal eigenvalues, as [Proposition 19.6.1\(c\)](#) requires. Four of the series (the price deflator, GNP, population and the year itself) rose almost in step over the sixteen years, each with correlation above 0.99 with the year. The data can say much about their joint movement and almost nothing about any one of them with the others held fixed.

```
import numpy as np
import statsmodels.api as sm
```



```

data = sm.datasets.longley.load_pandas().data
Z = data.drop(columns="TOTEEMP").to_numpy()           # six regressors
n, k = Z.shape

def r_squared(v, W):
    W1 = np.column_stack([np.ones(len(v)), W])
    fitted = W1 @ np.linalg.lstsq(W1, v, rcond=None)[0]
    return 1 - np.sum((v - fitted) ** 2) / np.sum((v - v.mean()) ** 2)

vif = np.array([1 / (1 - r_squared(Z[:, j], np.delete(Z, j, axis=1))) for j in range
(k)])
R = np.corrcoef(Z, rowvar=False)
print(dict(zip(data.columns[1:], np.round(vif, 1))))
print("diagonal of R^{-1}:", np.round(np.diag(np.linalg.inv(R)), 1))
print("eigenvalues of R: ", np.round(np.linalg.eigvalsh(R), 5))

```

### Example. (Predicting along and across the pattern)

Draw  $m = 40$  pairs of regressors with sample correlation 0.9868, so that each has a variance inflation factor of 38.1, and fit  $E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ . In units of  $\sigma$ , the two slopes have standard deviations 0.904 and 0.907. At the new point  $(x_1, x_2) = (1, 1)$ , which follows the pattern of the data, the fitted mean has standard deviation 0.228, close to the root mean square over the design points,  $\sqrt{3/40} = 0.274$ . At  $(1, -1)$ , which is just as far from the origin but against the pattern, it is 1.809. [Figure 19.6.1](#) shows the contours of this standard deviation. They are long ellipses aligned with the data cloud.

```

rng = np.random.default_rng(1906)
m, rho = 40, 0.98
x1 = rng.normal(size=m)
x2 = rho * x1 + np.sqrt(1 - rho**2) * rng.normal(size=m)
X = np.column_stack([np.ones(m), x1, x2])
G = np.linalg.inv(X.T @ X)           # Cov(beta_hat) / sigma^2
along = np.array([1.0, 1.0, 1.0])    # a new point that follows the
    pattern
across = np.array([1.0, 1.0, -1.0])   # same distance, against the
    pattern
print("sd of beta_1, beta_2 (units of sigma):", np.round(np.sqrt(np.diag(G)[1:]), 3)
)
print("sd of the fitted mean, along :", round(np.sqrt(along @ G @ along), 3))
print("sd of the fitted mean, across:", round(np.sqrt(across @ G @ across), 3))
print("average over the design points:", round(np.trace(X @ G @ X.T) / m, 3), "= p/n
")

```

## 19.6.2 How collinearity interacts with the other departures

Collinearity causes no bias, but it amplifies the departures of the earlier sections. By [Eq. 19.1.2](#) the stakes of omitting  $x_2$  are  $r^2|\gamma - 1|$ , so the choice between short and long models matters most exactly when the data are least able to make it. Perturbations of the regressors, from rounding or measurement error, are governed by the condition number of  $\mathbf{X}$  ([Theorem 10.5.1](#)), which is large for the same reason. And a single case can have high leverage without being extreme in



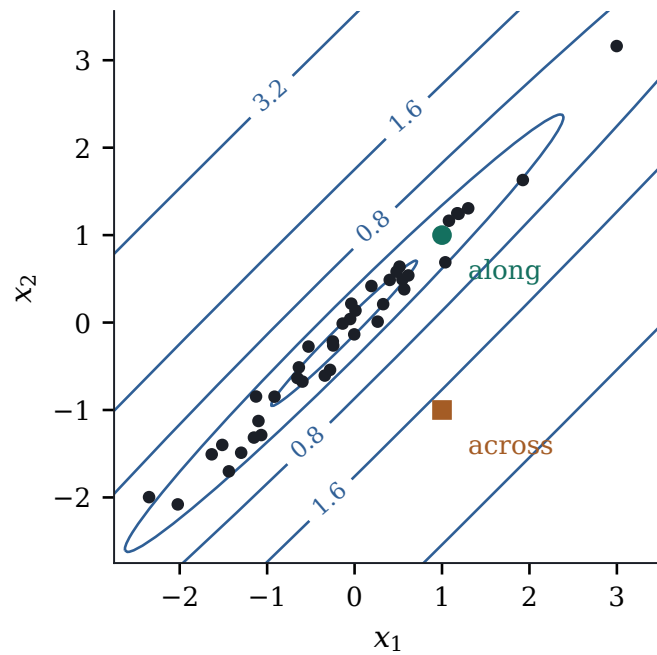


Figure 19.6.1: Standard deviation of the fitted mean, in units of  $\sigma$ , over the regressor plane for [Example \(Predicting along and across the pattern\)](#), with the 40 design points. The contours are elongated along the data: prediction at the circle, which follows the pattern, is precise; prediction at the square, which breaks it, is not.

any one regressor, because leverage measures distance in the metric of the regressors' covariance ([Proposition 6.8.2](#)).

*Idea.* Collinearity is a shortage of information, not a failure of the model, and it magnifies every other departure.

## 19.6.3 Exercises

### 19.6.3.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{Z}$  be the  $n \times k$  matrix of centred regressors, each scaled to unit length, so that  $\mathbf{Z}^\top \mathbf{Z} = \mathbf{R}$ . Use [Eq. 6.6.1](#) to show that  $[\mathbf{R}^{-1}]_{jj} = 1/(1 - R_j^2)$ .

*Solution.* By [Eq. 6.6.1](#) with  $\mathbf{X}_1 = \mathbf{z}_j$  and  $\mathbf{X}_2$  the other columns,  $[\mathbf{R}^{-1}]_{jj} = 1/\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{z}_j\|^2$ , where  $\mathbf{M}_{(j)}$  projects onto the other centred columns. Since  $\mathbf{z}_j$  is centred with unit length, the squared residual length is  $1 - R_j^2$ , as in [Exercise \(B1\)](#) (the intercept only removes means, and the columns are already centred).

#### Exercise A2.

Show that  $\text{VIF}_j \geq 1$ , with equality iff the centred  $\mathbf{x}_j$  is orthogonal to every other centred regressor.

## 19.6.3.2 B. Practice

**Exercise B1.**

For two centred regressors with  $\|\mathbf{x}_1\|^2 = \|\mathbf{x}_2\|^2 = S$  and correlation  $r$ , show that  $\text{Var}(\hat{\beta}_1 + \hat{\beta}_2) = 2\sigma^2/(S(1+r))$  and  $\text{Var}(\hat{\beta}_1 - \hat{\beta}_2) = 2\sigma^2/(S(1-r))$ . Relate the result to [Proposition 19.6.1\(a\)](#) and to [Figure 19.6.1](#).

*Solution.* The slope block of  $\mathbf{X}^\top \mathbf{X}$  is  $S \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$ , with eigenvalues  $S(1+r)$  and  $S(1-r)$  and eigenvectors  $(1, 1)^\top/\sqrt{2}$  and  $(1, -1)^\top/\sqrt{2}$ . The intercept is orthogonal to the centred slopes and plays no role. With  $\mathbf{a} = (1, \pm 1)^\top$ ,  $\mathbf{a}^\top \mathbf{v}$  is  $\sqrt{2}$  on one eigenvector and 0 on the other, so part (a) gives  $2\sigma^2/(S(1 \pm r))$ . When  $r$  is near one, the sum, the combined effect of two regressors that move together, is well determined, and the difference is not. In the figure, the new point  $(1, 1)$  asks for the sum and  $(1, -1)$  for the difference.

**Exercise B2.**

Show that  $\sum_j \text{VIF}_j \geq k$ , and that  $\max_j \text{VIF}_j \geq (\sum_j \text{VIF}_j)/k$ . Deduce that a correlation matrix with a small eigenvalue must have at least one large variance inflation factor, and conversely.

## 19.6.3.3 C. Going deeper

**Exercise C1.**

Show that if  $\mathbf{x}_0$  is a convex combination of the rows of  $\mathbf{X}$ , then  $\text{Var}(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}) \leq \sigma^2 \max_i h_{ii}$ . Is the converse true: does a small prediction variance mean  $\mathbf{x}_0$  lies inside the data? Compare with [Example \(Hidden extrapolation\)](#).

## 19.7 Summary and Notes

## 19.7.1 Summary

*Idea.*

1. Omitting regressors biases  $\hat{\boldsymbol{\beta}}_1^S$  by  $\boldsymbol{\Pi}\boldsymbol{\beta}_2$  and inflates  $s^2$ , while the covariances look exactly as the short model claims ([Theorem 19.1.2](#)). Irrelevant regressors cost only variance ([Theorem 19.1.3](#)).
2. The short regression has the smaller mean squared error matrix iff  $\gamma \leq 1$ ; at the design points the threshold is  $p_2$  ([Theorem 19.1.5](#)).
3. Under  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$  the usual variance estimate is off by a factor bounded by eigenvalue ratios of  $\mathbf{V}$  ([Theorem 19.2.2](#)), while the efficiency loss is bounded by Kantorovich's constant ([Theorem 19.2.3](#)).
4. With non-normal errors and  $\max_i h_{ii} \rightarrow 0$ , linear estimates are asymptotically normal and  $t$  and  $F$  tests asymptotically valid ([Theorem 19.3.2](#)); leverage bounds

the finite-sample skewness and kurtosis ([Proposition 19.3.3](#)).

5. Intervals for  $\sigma^2$ , variance-ratio tests and prediction intervals stay wrong in large samples ([Proposition 19.3.4](#), [Proposition 12.3.3](#)); quadratic balance protects  $F$  tests ([Proposition 19.3.5](#)).
6. One contaminated case moves  $\hat{\beta}$  by a leverage factor times its residual, hides itself in the fit ([Theorem 19.4.1](#)), and can move the estimate arbitrarily far ([Proposition 19.4.2](#)).
7. With random regressors and a curved mean, least squares estimates the population projection with a sandwich covariance that the usual formula misses ([Proposition 19.5.1](#), [Theorem 19.5.2](#)).
8. Collinearity breaks no assumption but inflates the variance of coefficients and of predictions against the pattern of the data ([Proposition 19.6.1](#)).

## 19.7.2 Notes and sources

**Coverage.** The chapter covers the topics of Seber and Lee (2003, chapter 9) with its own arguments, and adds [Theorem 19.1.5](#), [Proposition 19.3.3](#), the limiting coverages and the breakdown point. Measurement error is in [Chapter 24](#) (rounding in [Section 19.5](#)), perturbation theory in [Chapter 10](#).

**Underfitting and overfitting.** The omitted-variable formula is classical; it is [Proposition 5.7.2](#) in [Chapter 5](#). The criterion  $\gamma \leq 1$  for the restricted estimator to have the smaller mean squared error matrix is due to Toro-Vizcarrondo and Wallace (1968) (in their notation, with the factor  $1/2$  in the noncentrality, the condition reads  $\lambda \leq 1/2$ ), who also proposed testing it with the noncentral  $F$  distribution. Pretest estimators, which choose between the two models from the data, are taken up with model selection in [Section 29.2](#) and [Section 29.5](#).

**Wrong covariance.** The bounds on the bias of the usual variance estimate are Swindel's (1968). The efficiency bound of [Theorem 19.2.3](#) rests on Kantorovich's inequality. The sharp bound for the determinant of the whole covariance matrix is due to Bloomfield and Watson (1975) and Knott (1975). Grenander (1954) showed that least squares is asymptotically efficient for polynomial trends under stationary errors, the phenomenon of [Example \(A trend with autocorrelated errors\)](#). Kruskal's (1968) condition for exact equality is [Theorem 6.9.3](#).

**Non-normality.** The Lindeberg–Feller theorem is in Billingsley (1995), and Cramér–Wold, Slutsky and continuous mapping in van der Vaart (1998). The leverage condition  $h_n^* \rightarrow 0$  is the one used by Huber (1973) in his asymptotic theory of robust regression. That tests on variances are sensitive to non-normality while tests on means are not was emphasized by Box (1953). Box and Watson (1962) and Atiqullah (1962) studied the regression  $F$  test through its first two moments, and Atiqullah introduced quadratic balance. [Proposition 19.3.5](#) is a compact version of their calculation.

**Outliers and breakdown.** The breakdown point in its finite-sample form is from Donoho and Huber (1983), and the influence function from Hampel (1974). Regression estimators with breakdown point near one half exist, least median of squares (Rousseeuw 1984) being the first. The stack loss data are from Brownlee (1965), and their last observation is a much-analysed outlier. [Chapter 20](#) develops diagnostics for such cases.

**Random regressors.** The view of least squares as estimating a best linear approximation, with a sandwich covariance that the usual formula gets wrong, is due to White (1980). Berkson

(1950) distinguished controlled regressors from regressors measured with error. Longley's (1967) data are the standard example of severe collinearity.

### 19.7.3 References

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# Chapter 20

## Residuals, Leverage and Influence

Every result of Parts II–IV was proved for a model that holds for all  $n$  cases. A record is mistyped, one unit is unlike the others, or the model is right for most of the range and wrong at its edge. Least squares treats every case as equally credible, so a single bad case can move the fit a long way, and it can do so without leaving a large residual behind. This chapter builds the tools for finding such cases. Residuals, rescaled so that they can be compared, show cases that the model fits badly. Leverage shows cases whose position among the regressors gives them a strong pull on the fit. Deleting one case at a time, cheap by the updating formulas of [Chapter 10](#), turns these two ingredients into exact tests and into measures of influence. Two warnings run through the chapter. Diagnostics that look at one case at a time can be defeated by a few bad cases acting together. And a flagged case is a question to be investigated, not a verdict. The last section uses a robust fit as a second opinion: a fit that the bad cases cannot capture shows them by their residuals.

**What you need.** [Chapter 4](#) (independent projections), [Chapter 5](#) (moments of fitted values and residuals, [Proposition 5.5.3](#)), [Chapter 6](#) (the hat matrix and its properties, [Proposition 6.8.1](#) and [Proposition 6.8.2](#); the Frisch–Waugh–Lovell theorem, [Theorem 6.6.2](#)), [Chapter 10](#) (the deletion formulas, [Proposition 10.6.2](#)), [Chapter 11](#) ( $t$  and  $F$  tests), and [Chapter 13](#) (the Bonferroni inequality). [Chapter 19](#) describes in general terms what a contaminated case does to least squares ([Theorem 19.4.1](#)); this chapter is about finding such cases in a particular data set.

### 20.0.1 Roadmap

- [Kinds of residuals](#)
- [Leverage and the hat matrix](#)
- [Deletion diagnostics](#)
- [Cook's distance, DFFITS and DFBETAS](#)
- [Masking and swamping](#)
- [Robust fitting as a diagnostic](#)
- [Summary and notes](#)

## 20.1 Kinds of residuals

The model makes its assumptions about the errors, which are never observed. The residuals are their visible stand-ins, imperfect in a precise way: they are the projection of the errors onto a subspace of dimension  $n - p$ , so they are correlated, have unequal variances, and are shrunk most at the cases that matter most. This section defines rescaled residuals, finds their laws, and asks what residual plots reveal.

### 20.1.1 The setting

Throughout the chapter,

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I},$$

with  $\mathbf{X}$  of size  $n \times p$  and full column rank  $p < n$ . Results that need normal errors say so and then assume [Eq. 7.3.1](#). The hat matrix is  $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ , and  $h_{ij}$  denotes its  $(i, j)$  entry, so that  $h_{ii}$  is the leverage of case  $i$  ([Definition 6.8.1](#)). The  $i$ th row of  $\mathbf{X}$ , written as a column, is  $\mathbf{x}_{(i)}$ , and  $\mathbf{e}_i$  is the  $i$ th coordinate vector of  $\mathbb{R}^n$ . The residual vector is  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y}$ , with entries  $\hat{\varepsilon}_i$ , and  $s^2 = \|\hat{\boldsymbol{\varepsilon}}\|^2 / (n - p)$ .

Everything in the chapter depends on  $\mathbf{X}$  only through  $\mathcal{C}(\mathbf{X})$ . If  $\mathbf{X}$  has rank  $r < p$ , replace  $p$  by  $r$  in every degree of freedom and every formula that involves  $\mathbf{M}$  alone. Formulas that involve  $(\mathbf{X}^\top \mathbf{X})^{-1}$  hold for any full-rank reparameterization ([Theorem 6.7.1](#)).

Two facts from earlier chapters drive everything that follows. First, by [Proposition 6.4.3\(e\)](#),

$$\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}, \tag{20.1.1}$$

so the residuals are a fixed linear image of the errors. Second, by [Proposition 5.5.3\(b\)](#),  $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{M})$ . So  $\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii})$  and, for  $i \neq j$ ,

$$\text{Corr}(\hat{\varepsilon}_i, \hat{\varepsilon}_j) = \frac{-h_{ij}}{\sqrt{(1 - h_{ii})(1 - h_{jj})}}. \tag{20.1.2}$$

A case with leverage near one has a residual with variance near zero: the fit passes close to it whatever its response. Raw residuals are therefore not comparable across cases.

### 20.1.2 Four residuals

#### Definition 20.1.1 — Standardized and studentized residuals

Let  $h_{ii} < 1$ . For case  $i$ :

- (a) the **raw residual** is  $\hat{\varepsilon}_i = y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}$ ;
- (b) the **standardized residual** is  $z_i = \hat{\varepsilon}_i / (\sigma \sqrt{1 - h_{ii}})$ , which needs the true  $\sigma$ ;

(c) the **internally studentized residual** is

$$r_i = \frac{\hat{\varepsilon}_i}{s\sqrt{1-h_{ii}}};$$

(d) the **externally studentized residual** is

$$t_i = \frac{\hat{\varepsilon}_i}{s_{(i)}\sqrt{1-h_{ii}}},$$

where  $s_{(i)}^2 = \text{SSE}_{(i)}/(n-p-1)$  is the residual mean square of the least squares fit to the  $n-1$  cases other than  $i$ .

Under normal errors  $z_i$  is exactly  $N(0, 1)$ , but it cannot be computed. The studentized residuals replace  $\sigma$  by an estimate:  $s$ , which includes case  $i$ , or  $s_{(i)}$ , which does not, so that a gross error in  $y_i$  cannot inflate the yardstick it is measured against. [Section 20.3](#) shows that  $s_{(i)}$  costs nothing extra and that  $t_i$  has an exact  $t$  law.

*Remark* (Names). Terminology varies. Some authors call  $r_i$ , or even  $\hat{\varepsilon}_i/s$ , the standardized residual, and  $t_i$  is also called the studentized deleted residual or RSTUDENT. R's `rstandard` and `rstudent` return  $r_i$  and  $t_i$ , as do `resid.studentized.internal` and `resid.studentized.external` in `statsmodels`.

### 20.1.3 The law of the internally studentized residual

The residual  $r_i$  is not  $t$ -distributed, because  $\hat{\varepsilon}_i$  is part of  $s$ . Its law is a rescaled beta.

#### Proposition 20.1.1 — Law of the internally studentized residual

Assume [Eq. 7.3.1](#),  $n-p \geq 2$  and  $h_{ii} < 1$ . Then:

- (a)  $r_i^2/(n-p) \sim \text{Beta}(\frac{1}{2}, \frac{n-p-1}{2})$ ;
- (b)  $|r_i| \leq \sqrt{n-p}$ ,  $E(r_i) = 0$  and  $\text{Var}(r_i) = 1$ ;
- (c) the law of  $r_i$  is the same for every case and does not depend on  $\mathbf{X}$ ,  $\boldsymbol{\beta}$  or  $\sigma^2$ .

*Proof.* Put  $\mathbf{u} = (\mathbf{I} - \mathbf{M})\mathbf{e}_i/\sqrt{1-h_{ii}}$ . Since  $\|(\mathbf{I} - \mathbf{M})\mathbf{e}_i\|^2 = \mathbf{e}_i^\top (\mathbf{I} - \mathbf{M})\mathbf{e}_i = 1 - h_{ii}$ ,  $\mathbf{u}$  is a unit vector in  $\mathcal{C}(\mathbf{X})^\perp$ , and by [Eq. 20.1.1](#)  $\hat{\varepsilon}_i = \mathbf{e}_i^\top (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon} = \sqrt{1-h_{ii}} \mathbf{u}^\top \boldsymbol{\varepsilon}$ . Let  $\mathbf{P}_1 = \mathbf{u}\mathbf{u}^\top$  and  $\mathbf{P}_2 = \mathbf{I} - \mathbf{M} - \mathbf{u}\mathbf{u}^\top$ . Because  $(\mathbf{I} - \mathbf{M})\mathbf{u} = \mathbf{u}$ , both are symmetric and idempotent,  $\mathbf{P}_1\mathbf{P}_2 = \mathbf{0}$ , and their ranks are 1 and  $n-p-1$ . They split the residual sum of squares:  $\text{SSE} = \boldsymbol{\varepsilon}^\top (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon} = \|\mathbf{P}_1\boldsymbol{\varepsilon}\|^2 + \|\mathbf{P}_2\boldsymbol{\varepsilon}\|^2$ . By [Theorem 4.4.4](#), applied to  $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2\mathbf{I})$ , the variables  $U = (\mathbf{u}^\top \boldsymbol{\varepsilon})^2/\sigma^2 \sim \chi^2(1)$  and  $V = \|\mathbf{P}_2\boldsymbol{\varepsilon}\|^2/\sigma^2 \sim \chi^2(n-p-1)$  are independent. Now

$$\frac{r_i^2}{n-p} = \frac{\hat{\varepsilon}_i^2}{(1-h_{ii})\text{SSE}} = \frac{U}{U+V},$$

which is  $\text{Beta}(\frac{1}{2}, \frac{n-p-1}{2})$  by [Exercise \(B4\)](#). This is (a). A beta variable lies in  $[0, 1]$ , which gives the bound in (b). Replacing  $\boldsymbol{\varepsilon}$  by  $-\boldsymbol{\varepsilon}$  changes the sign of  $r_i$  and not its law, so  $E(r_i) = 0$ , and  $E(r_i^2) = (n-p) \cdot \frac{1}{2}/\frac{n-p}{2} = 1$ . Part (c) holds because the law in (a),

together with the symmetry, determines the law of  $r_i$ .

■

The bound in (b) bites when  $n - p$  is small: with  $n - p = 5$ , no  $|r_i|$  can exceed  $\sqrt{5} \approx 2.236$ , however wild the case, because a gross error inflates  $s$  along with  $\hat{\varepsilon}_i$ . The  $r_i$  are not independent either; their correlations are those of Eq. 20.1.2 (Exercise (C1)).

## 20.1.4 What a residual plot can show

A residual plot shows residuals against the fitted values, a regressor, an omitted variable or the order of collection, to reveal structure the model has missed. Least squares removes some structure automatically, and it is worth knowing exactly which.

### Proposition 20.1.2 — What least squares leaves in the residuals

Suppose  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ .

- (a) The residual vector is orthogonal to  $\mathbf{1}$ , to every column of  $\mathbf{X}$  and to  $\hat{\mathbf{Y}}$ . Hence the least squares line through a plot of the residuals against the fitted values, or against any regressor in the model, is the horizontal axis.
- (b) The least squares line through a plot of the residuals against the responses  $y_i$  has slope  $1 - R^2$ .
- (c) If in fact  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}$  for some  $\boldsymbol{\delta} \in \mathbb{R}^n$ , and  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ , then  $E(\hat{\boldsymbol{\varepsilon}}) = (\mathbf{I} - \mathbf{M})\boldsymbol{\delta}$  and  $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{M})$ .

*Proof.*

- (a) is Proposition 6.4.3(a) and (b). A line fitted with an intercept to points  $(a_i, \hat{\varepsilon}_i)$  has slope  $\sum_i (a_i - \bar{a})\hat{\varepsilon}_i / \sum_i (a_i - \bar{a})^2$ , and the numerator is  $\mathbf{a}^\top \hat{\boldsymbol{\varepsilon}} - \bar{a} \mathbf{1}^\top \hat{\boldsymbol{\varepsilon}} = 0$  when  $\mathbf{a}$  is  $\hat{\mathbf{Y}}$  or a column of  $\mathbf{X}$ .
- (b) With  $\mathbf{a} = \mathbf{y}$  the numerator is  $\mathbf{y}^\top \hat{\boldsymbol{\varepsilon}} = (\hat{\mathbf{Y}} + \hat{\boldsymbol{\varepsilon}})^\top \hat{\boldsymbol{\varepsilon}} = \|\hat{\boldsymbol{\varepsilon}}\|^2 = \text{SSE}$ , and the denominator is  $\sum_i (y_i - \bar{y})^2 = \text{SST}$ . The slope is  $\text{SSE}/\text{SST} = 1 - R^2$ .
- (c)  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{Y}$ , and Theorem 2.2.1 gives mean  $(\mathbf{I} - \mathbf{M})(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\delta}) = (\mathbf{I} - \mathbf{M})\boldsymbol{\delta}$  and covariance  $\sigma^2(\mathbf{I} - \mathbf{M})^2 = \sigma^2(\mathbf{I} - \mathbf{M})$ .

■

By (a), a plot against fitted values or regressors has had its linear trend removed, so any remaining trend is curved, a sign of a misspecified mean. By (b), a plot against the responses always slopes upward, even when the model is right, so it is not used.

Part (c) is the sobering one. The residuals are centred at  $(\mathbf{I} - \mathbf{M})\boldsymbol{\delta}$ , the part of the error in the mean orthogonal to the model space. The part of  $\boldsymbol{\delta}$  inside  $\mathcal{C}(\mathbf{X})$  leaves no trace, yet it is exactly this part,  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\delta}$ , that biases the coefficients (Proposition 5.5.7(a)). An omitted regressor highly correlated with the included ones biases their coefficients (Theorem 19.1.2) while leaving the residuals almost clean. Residual plots can show that the *fitted values* are wrong, never that the *coefficients* are right.



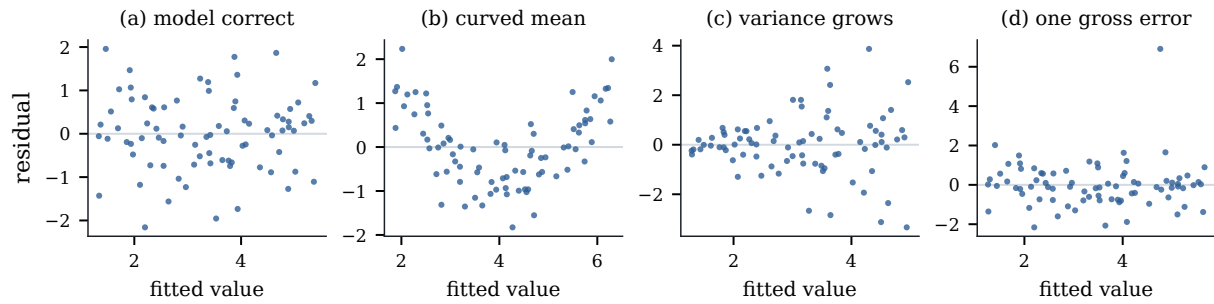


Figure 20.1.1: Residuals against fitted values for four straight-line fits: (a) correct model, (b) curved mean, (c) variance growing with the mean, (d) one gross error.

The standard plots are these. Against **fitted values**: curvature suggests a misspecified mean, a fan shape variance that grows with the mean, isolated points outliers. Against **each regressor**: curvature points to that regressor. Against a variable **not in the model**: better replaced by the added-variable plot of [Section 6.6](#), whose slope is the coefficient the variable would receive ([Exercise \(B1\)](#)). Against **collection order**: runs of equal sign suggest serial correlation, tested in [Chapter 21](#) ([Theorem 21.5.3](#)). A **normal quantile plot**: curvature suggests skewed or heavy-tailed errors ([Proposition 21.1.1](#)). Studentized residuals are preferable to raw ones throughout, since raw residuals make high-leverage cases look better than they are.

#### Example. (Four residual plots)

Each panel of [Figure 20.1.1](#) comes from 80 simulated cases with  $x$  uniform on  $(0, 10)$ , fitted by a straight line. In (a) the model is right and the residuals form a structureless band. In (b) the mean has a quadratic term, and the residuals show the parabola the line could not absorb. In (c) the error standard deviation grows with  $x$ , and the band opens into a fan. In (d) one response has an error of  $8\sigma$ ; it stands out, and the line has been pulled slightly towards it. In every panel the residuals are orthogonal to the fitted values, as [Proposition 20.1.2\(a\)](#) requires; the listing prints the inner product.

```
import numpy as np

def ls_fit(X, y):
    beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    return beta, X @ beta, y - X @ beta

rng = np.random.default_rng(7)
n = 80
x = rng.uniform(0, 10, n)
eps = rng.normal(size=n)
X = np.column_stack([np.ones(n), x])
responses = {
    "(a) model correct": 1 + 0.5 * x + eps,
    "(b) curved mean": 1 + 0.5 * x + 0.08 * (x - 5) ** 2 + 0.6 * eps,
    "(c) variance grows": 1 + 0.5 * x + (0.1 + 0.25 * x) * eps,
    "(d) one gross error": 1 + 0.5 * x + eps + 8.0 * (np.arange(n) == 17),
}
for name, y in responses.items():
    beta, fitted, resid = ls_fit(X, y)
    print(f"{name:22s} slope {beta[1]:.3f}    residuals . fitted = {resid @ fitted:.1e}")
```

## 20.1.5 Partial residuals

A plot of residuals against a regressor  $\mathbf{x}_j$  in the model has a flat least squares line ([Proposition 20.1.2\(a\)](#)). The **component-plus-residual plot**, or partial residual plot, adds the fitted linear component back. It plots

$$\hat{\varepsilon}_i + \hat{\beta}_j x_{ij} \quad \text{against} \quad x_{ij}, \quad i = 1, \dots, n.$$

Compare it with the added-variable plot of [Section 6.6](#), which plots  $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{y}$  against  $\tilde{\mathbf{x}}_j = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$ , where  $\mathbf{M}_{(j)}$  projects onto the span of the columns other than  $\mathbf{x}_j$ .

### Proposition 20.1.3 — Two plots with the same line

Let  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , and let  $\mathbf{x}_j$  be a column of  $\mathbf{X}$  that is not a multiple of  $\mathbf{1}$ .

- (a) The least squares line of  $\hat{\varepsilon} + \hat{\beta}_j \mathbf{x}_j$  on  $[\mathbf{1}, \mathbf{x}_j]$  has intercept 0, slope  $\hat{\beta}_j$ , and residual vector  $\hat{\varepsilon}$ .
- (b) The least squares line through the origin of  $\tilde{\mathbf{y}}$  on  $\tilde{\mathbf{x}}_j$  has slope  $\hat{\beta}_j$  and residual vector  $\hat{\varepsilon}$ .

*Proof.*

- (a) The vector  $\hat{\varepsilon}$  is orthogonal to  $\mathbf{1}$  and to  $\mathbf{x}_j$  by [Proposition 20.1.2\(a\)](#). So  $\hat{\varepsilon} + \hat{\beta}_j \mathbf{x}_j = \hat{\beta}_j \mathbf{x}_j + \hat{\varepsilon}$  is the sum of a vector in  $\mathcal{C}([\mathbf{1}, \mathbf{x}_j])$  and a vector orthogonal to that space. By [Theorem 6.4.1](#) the first is the fitted vector and the second the residual.
- (b) is [Theorem 6.6.2\(a\)](#) and (b) with  $\mathbf{X}_1 = \mathbf{x}_j$ .

■

So the two plots share their line and their scatter about it, and differ only in the horizontal axis. The added-variable plot uses the part of  $\mathbf{x}_j$  not explained by the other regressors: its spread is the information for  $\hat{\beta}_j$ , and cases far out along it pull on  $\hat{\beta}_j$  (partial leverage, [Section 20.2](#)). The component-plus-residual plot uses  $\mathbf{x}_j$  on its own scale, the scale on which a transformation would be chosen, so it shows the *form* of the dependence better. It can mislead when  $\mathbf{x}_j$  is related nonlinearly to the other regressors.

### Example. (Curvature seen two ways)

In 120 simulated cases,  $x_1$  is uniform on  $(0, 4)$ ,  $x_2$  is correlated with it (sample correlation 0.87), and  $E(Y) = 1 + x_2 + 0.7(x_1 - 2)^2$ . The fitted model is linear in both, with coefficient  $-0.144$  for  $x_1$ , the slope of both panels of [Figure 20.1.2](#). A quadratic fitted to the component-plus-residual plot (b) has leading coefficient 0.671, close to the true 0.7. In the added-variable plot (a) the curve is smeared, because its horizontal axis mixes  $x_1$  with  $x_2$ , and the corresponding coefficient is 0.309.

```
rng = np.random.default_rng(11)
n2 = 120
x1 = rng.uniform(0, 4, n2)
x2 = 0.8 * x1 + rng.normal(0, 0.6, n2)          # correlated with x1
y2 = 1 + x2 + 0.7 * (x1 - 2) ** 2 + rng.normal(0, 0.4, n2)
```

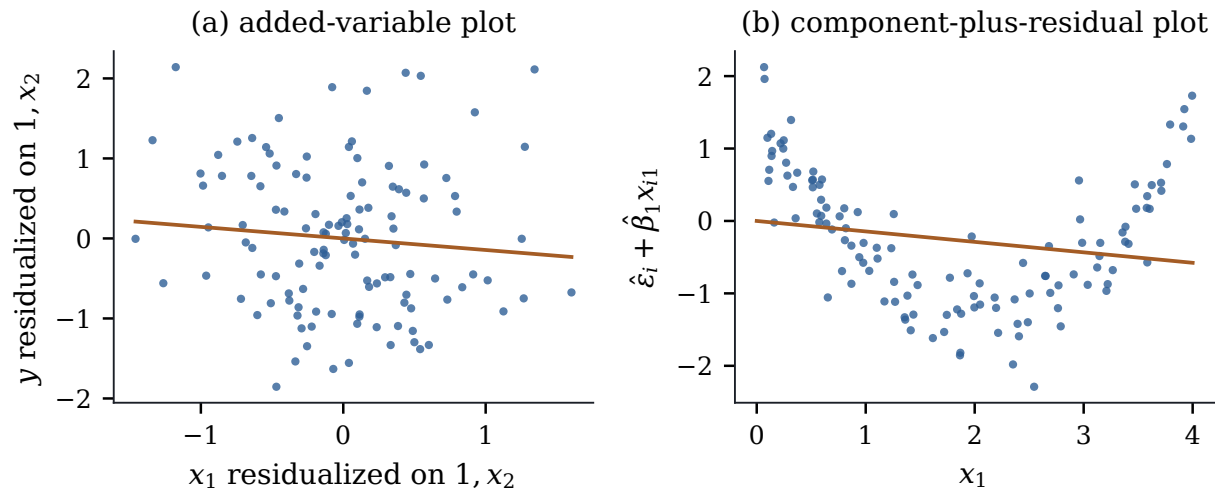


Figure 20.1.2: A curved effect of  $x_1$ , fitted linearly: (a) added-variable plot, (b) component-plus-residual plot. Same slope, same residuals (Proposition 20.1.3).

```
X2 = np.column_stack([np.ones(n2), x1, x2])          # the fitted model is linear in
x1                                                    # x1
beta2, fitted2, resid2 = ls_fit(X2, y2)

component = resid2 + beta2[1] * x1                    # component-plus-residual for x1
Z = np.column_stack([np.ones(n2), x2])              # everything except x1
x1_tilde = x1 - Z @ np.linalg.lstsq(Z, x1, rcond=None)[0]
y_tilde = y2 - Z @ np.linalg.lstsq(Z, y2, rcond=None)[0] # added-variable
coordinates

print("coefficient of x1:", beta2[1])
print("slope in component-plus-residual plot:", ls_fit(np.column_stack([np.ones(n2),
x1]), component)[0][1])
print("slope in added-variable plot:", (x1_tilde @ y_tilde) / (x1_tilde @ x1_tilde))
```

## 20.1.6 Exercises

### 20.1.6.1 A. Check your understanding

#### Exercise A1.

A straight line is fitted to  $n = 3$  points. Show that every internally studentized residual satisfies  $|r_i| = 1$  (unless all three residuals are zero), whatever the data. What does this say about using  $r_i$  to judge a fit with one residual degree of freedom?

*Solution.* Here  $n - p = 1$ , so  $\mathcal{C}(\mathbf{X})^\perp$  is spanned by one unit vector  $\mathbf{v}$ , and  $\hat{\boldsymbol{\varepsilon}} = c\mathbf{v}$  for a scalar  $c$ . Then  $\hat{\varepsilon}_i = cv_i$ ,  $s^2 = c^2$  and  $1 - h_{ii} = v_i^2$  (the  $i$ th diagonal entry of  $\mathbf{I} - \mathbf{M} = \mathbf{v}\mathbf{v}^\top$ ). So  $r_i^2 = c^2 v_i^2 / (c^2 v_i^2) = 1$  whenever  $v_i \neq 0$ . With one degree of freedom for error the studentized residuals carry no information at all about which case fits worst.

#### Exercise A2.

Show that  $\sum_i (1 - h_{ii}) r_i^2 = n - p$  for any data. Deduce that if every leverage is at most

$h_{\max} < 1$ , then at most  $(n - p)/((1 - h_{\max})c^2)$  cases can have  $|r_i| > c$ .

### 20.1.6.2 B. Practice

#### Exercise B1.

Let  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , let  $\mathbf{z} \notin \mathcal{C}(\mathbf{X})$  be a candidate regressor, let  $\hat{\gamma}$  be its coefficient when it is added to the model, and let  $R_z^2$  be the  $R^2$  of the regression of  $\mathbf{z}$  on  $\mathbf{X}$ . Show that the least squares slope of the plot of  $\hat{\mathbf{e}}$  against  $\mathbf{z}$  is  $\hat{\gamma}(1 - R_z^2)$ . Why is the added-variable plot the better tool for judging whether  $\mathbf{z}$  belongs in the model?

*Solution.* The slope is  $\sum_i (z_i - \bar{z})\hat{e}_i / S_{zz} = \mathbf{z}^\top \hat{\mathbf{e}} / S_{zz}$ , since  $\mathbf{1}^\top \hat{\mathbf{e}} = 0$ . Put  $\tilde{\mathbf{z}} = (\mathbf{I} - \mathbf{M})\mathbf{z}$ . Then  $\mathbf{z}^\top \hat{\mathbf{e}} = \mathbf{z}^\top (\mathbf{I} - \mathbf{M})\mathbf{y} = \tilde{\mathbf{z}}^\top \mathbf{y}$ , and by [Theorem 6.6.2\(a\)](#),  $\hat{\gamma} = \tilde{\mathbf{z}}^\top \mathbf{y} / \|\tilde{\mathbf{z}}\|^2$ . So  $\mathbf{z}^\top \hat{\mathbf{e}} = \hat{\gamma} \|\tilde{\mathbf{z}}\|^2$ , and  $\|\tilde{\mathbf{z}}\|^2 = S_{zz}(1 - R_z^2)$  is the residual sum of squares of  $\mathbf{z}$  on  $\mathbf{X}$  ([Exercise \(B1\)](#)). The slope is  $\hat{\gamma}(1 - R_z^2)$ . When  $\mathbf{z}$  is highly correlated with the columns of  $\mathbf{X}$ , the plot against  $\mathbf{z}$  shrinks the effect towards zero, while the added-variable plot shows  $\hat{\gamma}$  itself, against the horizontal spread  $\|\tilde{\mathbf{z}}\|$  that actually carries information about it.

#### Exercise B2.

Simple linear regression is fitted to  $x_i = i$ ,  $i = 1, \dots, n$ , but  $E(Y_i) = \beta_0 + \beta_1 x_i + \gamma x_i^2$ . Use [Proposition 20.1.2\(c\)](#) to show that the expected residuals lie on a parabola in  $i$ , and find its vertex.

### 20.1.6.3 C. Going deeper

#### Exercise C1.

Assume [Eq. 7.3.1](#) with  $n - p \geq 2$  and leverages below one. Show that  $E(r_i r_j) = -h_{ij} / \sqrt{(1 - h_{ii})(1 - h_{jj})}$  for  $i \neq j$ . *Hint:* write  $r_i = \sqrt{n - p} \mathbf{u}_i^\top \mathbf{W}$  with  $\mathbf{W} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon} / \|(\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}\|$ , and show that  $E(\mathbf{W}\mathbf{W}^\top) = (\mathbf{I} - \mathbf{M})/(n - p)$ .

*Solution.* With  $\mathbf{u}_i$  as in the proof of [Proposition 20.1.1](#),  $r_i = \hat{e}_i / (s\sqrt{1 - h_{ii}}) = \sqrt{n - p} \mathbf{u}_i^\top \mathbf{W}$ , because  $\mathbf{u}_i^\top (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon} = \mathbf{u}_i^\top \boldsymbol{\varepsilon}$ . For an orthogonal  $\mathbf{Q}$  with  $\mathbf{Q}(\mathbf{I} - \mathbf{M}) = (\mathbf{I} - \mathbf{M})\mathbf{Q}$  that maps  $\mathcal{C}(\mathbf{X})^\perp$  to itself,  $\mathbf{Q}\boldsymbol{\varepsilon}$  has the same law as  $\boldsymbol{\varepsilon}$ , so  $\mathbf{Q}\mathbf{W}$  has the same law as  $\mathbf{W}$  and  $\mathbf{S} = E(\mathbf{W}\mathbf{W}^\top)$  satisfies  $\mathbf{Q}\mathbf{S}\mathbf{Q}^\top = \mathbf{S}$ . Since  $\mathbf{W} \in \mathcal{C}(\mathbf{X})^\perp$ ,  $\mathbf{S}$  vanishes on  $\mathcal{C}(\mathbf{X})$ ; restricted to  $\mathcal{C}(\mathbf{X})^\perp$  it commutes with every rotation of that space and is therefore a multiple of the identity there. Its trace is  $E\|\mathbf{W}\|^2 = 1$ , so  $\mathbf{S} = (\mathbf{I} - \mathbf{M})/(n - p)$ . Hence  $E(r_i r_j) = (n - p) \mathbf{u}_i^\top \mathbf{S} \mathbf{u}_j = \mathbf{u}_i^\top \mathbf{u}_j = \mathbf{e}_i^\top (\mathbf{I} - \mathbf{M}) \mathbf{e}_j / \sqrt{(1 - h_{ii})(1 - h_{jj})}$ , which is the stated value. Since  $E(r_i) = 0$  and  $\text{Var}(r_i) = 1$ , this is also the correlation.

## 20.2 Leverage and the hat matrix

[Section 6.8](#) proved the basic properties of the leverage  $h_{ii}$ : it lies in  $[0, 1]$ , the leverages sum to  $p$ ,  $h_{ii} \geq 1/n$  with an intercept, and  $h_{ii}$  is an affine function of the Mahalanobis distance of  $\mathbf{x}_{(i)}$  from the centroid of the regressors ([Proposition 6.8.1](#) and [Proposition 6.8.2](#)). This section

asks when a leverage deserves attention, how designed and observational data differ, and how to measure the pull of a case on one coefficient.

The fitted value of case  $i$  is

$$\hat{y}_i = h_{ii}y_i + \sum_{j \neq i} h_{ij}y_j,$$

so  $h_{ii}$  is the weight of a case in its own fitted value. Since  $\sum_j h_{ij}^2 = h_{ii}$  (Proposition 6.8.1(a)), a case with  $h_{ii}$  near one leaves little weight for the others: its fitted value is essentially its own response, and the fit carries no information about whether  $y_i$  is right. Leverage is *potential* influence, a property of the design alone; Section 20.4 combines it with the residual.

## 20.2.1 How large is large?

The average leverage is  $p/n$ . The most widely used rule (Hoaglin and Welsch 1978) calls a case of high leverage if  $h_{ii} > 2p/n$ ; a stricter variant uses  $3p/n$ . To see what these rules mean, compute how often they fire under a reference design with multivariate normal regressors.

### Proposition 20.2.1 — Leverage under a Gaussian design

Let  $\mathbf{X} = [\mathbf{1}, \mathbf{Z}]$ , where the rows of the  $n \times k$  matrix  $\mathbf{Z}$  are independent  $N_k(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  vectors with  $\boldsymbol{\Sigma}$  positive definite, and let  $n \geq k + 2$ . Then  $\mathbf{X}$  has full column rank  $p = k + 1$  with probability one, and for every  $i$ ,

$$\frac{h_{ii} - 1/n}{1 - 1/n} \sim \text{Beta}\left(\frac{k}{2}, \frac{n - 1 - k}{2}\right).$$

*Proof. Reduction.* Let  $\mathbf{C} = \mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top$ . By Lemma 6.6.1 with  $\mathbf{X}_2 = \mathbf{1}$ ,  $\mathbf{M} = n^{-1}\mathbf{1}\mathbf{1}^\top + \mathbf{P}$ , where  $\mathbf{P}$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{CZ})$ , a subspace of  $\mathbf{1}^\perp$ . So  $h_{ii} - 1/n = \mathbf{e}_i^\top \mathbf{P} \mathbf{e}_i$ . Replacing  $\mathbf{Z}$  by  $(\mathbf{Z} - \mathbf{1}\boldsymbol{\mu}^\top)\boldsymbol{\Sigma}^{-1/2}$  does not change  $\mathcal{C}(\mathbf{CZ})$ , because  $\mathbf{C}\mathbf{1} = \mathbf{0}$  and  $\boldsymbol{\Sigma}^{-1/2}$  is nonsingular. So we may assume that the entries of  $\mathbf{Z}$  are independent  $N(0, 1)$ . Then  $\mathbf{CZ}$  has rank  $k$  with probability one (Exercise (C1)), and so does  $\mathbf{X}$ .

*Invariance.* Let  $\mathbf{v} = \mathbf{C}\mathbf{e}_i / \|\mathbf{C}\mathbf{e}_i\|$ , a unit vector in  $\mathbf{1}^\perp$ , with  $\|\mathbf{C}\mathbf{e}_i\|^2 = 1 - 1/n$ . Since  $\mathbf{P} = \mathbf{P}\mathbf{C} = \mathbf{C}\mathbf{P}$ ,

$$h_{ii} - \frac{1}{n} = (\mathbf{C}\mathbf{e}_i)^\top \mathbf{P}(\mathbf{C}\mathbf{e}_i) = \left(1 - \frac{1}{n}\right) \mathbf{v}^\top \mathbf{P} \mathbf{v}.$$

Write  $\mathbf{P}(\mathbf{Z})$  to show the dependence on  $\mathbf{Z}$ . Let  $\mathbf{w}$  be any other unit vector in  $\mathbf{1}^\perp$ , and let  $\mathbf{H} = \mathbf{I} - 2\mathbf{a}\mathbf{a}^\top / \mathbf{a}^\top \mathbf{a}$  with  $\mathbf{a} = \mathbf{v} - \mathbf{w}$  (or  $\mathbf{H} = \mathbf{I}$  if  $\mathbf{w} = \mathbf{v}$ ). This is a symmetric orthogonal matrix, and  $\mathbf{H}\mathbf{v} = \mathbf{w}$  because  $\|\mathbf{v}\| = \|\mathbf{w}\|$ . It fixes  $\mathbf{1}$ , because  $\mathbf{1} \perp \mathbf{a}$ , so  $\mathbf{H}$  commutes with  $\mathbf{C}$  and maps  $\mathcal{C}(\mathbf{CZ})$  onto  $\mathcal{C}(\mathbf{CHZ})$ . Hence  $\mathbf{P}(\mathbf{HZ}) = \mathbf{H}\mathbf{P}(\mathbf{Z})\mathbf{H}$ , and  $\mathbf{w}^\top \mathbf{P}(\mathbf{Z})\mathbf{w} = \mathbf{v}^\top \mathbf{H}\mathbf{P}(\mathbf{Z})\mathbf{H}\mathbf{v} = \mathbf{v}^\top \mathbf{P}(\mathbf{HZ})\mathbf{v}$ . Each column of  $\mathbf{HZ}$  is  $N_n(\mathbf{0}, \mathbf{H}\mathbf{H}^\top) = N_n(\mathbf{0}, \mathbf{I})$ , and the columns are independent, so  $\mathbf{HZ}$  has the law of  $\mathbf{Z}$ . Therefore the law of  $\mathbf{v}^\top \mathbf{P}(\mathbf{Z})\mathbf{v}$  is the same for every unit vector  $\mathbf{v} \in \mathbf{1}^\perp$ .

*Averaging.* Let  $\mathbf{g} \sim N_n(\mathbf{0}, \mathbf{C})$  be independent of  $\mathbf{Z}$ , and  $\mathbf{U} = \mathbf{g} / \|\mathbf{g}\|$ , a random unit vector in  $\mathbf{1}^\perp$ . By the previous paragraph, applied conditionally on  $\mathbf{U}$ , the variable  $\mathbf{U}^\top \mathbf{P}(\mathbf{Z})\mathbf{U}$  has the same law as  $\mathbf{v}^\top \mathbf{P}(\mathbf{Z})\mathbf{v}$ . Now condition on  $\mathbf{Z}$  instead. Then  $\mathbf{P}$  is a fixed projection of rank  $k$  onto a subspace of  $\mathbf{1}^\perp$ , and  $\mathbf{C} - \mathbf{P}$  is a projection of rank  $n - 1 - k$  orthogonal to it, so by Theorem 4.4.4  $\|\mathbf{P}\mathbf{g}\|^2 \sim \chi^2(k)$  and  $\|(\mathbf{C} - \mathbf{P})\mathbf{g}\|^2 \sim \chi^2(n - 1 - k)$  are independent.

Since  $\mathbf{g} = \mathbf{C}\mathbf{g}$ ,

$$\mathbf{U}^\top \mathbf{P}\mathbf{U} = \frac{\|\mathbf{P}\mathbf{g}\|^2}{\|\mathbf{P}\mathbf{g}\|^2 + \|(\mathbf{C} - \mathbf{P})\mathbf{g}\|^2} \sim \text{Beta}\left(\frac{k}{2}, \frac{n-1-k}{2}\right)$$

by [Exercise \(B4\)](#). This conditional law does not depend on  $\mathbf{Z}$ , so it is also the unconditional law. ■

The mean of the beta law is  $k/(n-1)$ , which gives  $E(h_{ii}) = 1/n + (1 - 1/n)k/(n-1) = p/n$ , as it must. As  $n \rightarrow \infty$  with  $k$  fixed,  $n(h_{ii} - 1/n)$  converges in law to  $\chi^2(k)$  ([Exercise \(B1\)](#)), so

$$\Pr\{h_{ii} > cp/n\} \approx \Pr\{\chi^2(k) > c(k+1) - 1\}.$$

With  $k = 3$ , the rule  $2p/n$  fires for 0.072 of the cases in large samples (0.067 when  $n = 50$ ), and the rule  $3p/n$  for 0.012. With  $k = 10$  and  $n = 500$ , the rule  $2p/n$  fires for only 0.020. “Twice the average” flags a different share of well-behaved cases for every  $p$ : it is a convention, not a test. [Figure 20.2.1\(a\)](#) checks the law against 4000 simulated designs.

```
import numpy as np
from scipy import stats

def leverages(X):
    Q, _ = np.linalg.qr(X)
    return np.sum(Q ** 2, axis=1)

rng = np.random.default_rng(2020)
n, k = 30, 3                                # intercept plus k Gaussian
regressors
p = k + 1
reps = 4000
h1 = np.empty(reps)
for r in range(reps):
    Z = rng.normal(size=(n, k)) @ np.array([[1.0, 0.0, 0.0], [0.8, 0.6, 0.0], [0.0,
    2.0, 1.0]])
    X = np.column_stack([np.ones(n), Z + 5.0])    # correlated, shifted regressors
    h1[r] = leverages(X)[0]                        # leverage of case 1
B = (h1 - 1 / n) / (1 - 1 / n)                  # should be Beta(k/2, (n-1-k)/2)
law = stats.beta(k / 2, (n - 1 - k) / 2)
print("mean of h_11:", h1.mean(), " theory p/n =", p / n)
print("Kolmogorov-Smirnov p-value against the Beta law:", stats.kstest(B, law.cdf).
pvalue)
```

## 20.2.2 Designed and observational data

In a designed experiment the analyst chooses the rows of  $\mathbf{X}$ . The largest leverage is at least the average  $p/n$ , with equality iff all are equal ([Exercise \(B2\)](#)), and many designs achieve it. If the columns are orthogonal with entries  $\pm 1$ , as in two-level factorials, then  $\mathbf{X}^\top \mathbf{X} = n\mathbf{I}$  and  $h_{ii} = \|\mathbf{x}_{(i)}\|^2/n = p/n$ : a  $2^3$  factorial run twice ( $n = 16$ ) with intercept, main effects and one interaction ( $p = 5$ ) gives every case leverage  $5/16 = 0.3125$ . A balanced one-way layout gives every case  $1/n_k = p/n$  ([Exercise \(B2\)](#)). Kiefer and Wolfowitz (1960) showed that, over a region of possible design points, the designs maximizing  $\det(\mathbf{X}^\top \mathbf{X})$  are exactly those minimizing the largest leverage  $\mathbf{x}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}$  over the region, with minimum  $p/n$  (for designs viewed as

probability measures). For D-optimal designs, good design and small maximum leverage are the same thing.

In observational data the rows are drawn, not chosen, and with light enough tails no case dominates in large samples.

### Proposition 20.2.2 — Leverage vanishes in large samples

Let the rows  $\mathbf{x}_{(1)}, \mathbf{x}_{(2)}, \dots$  be independent copies of a random vector  $\mathbf{x}$  with  $E\|\mathbf{x}\|^2 < \infty$  and  $E(\mathbf{x}\mathbf{x}^\top)$  positive definite, and let  $h_{ii}^{(n)}$  be the leverages of the first  $n$  rows. Then  $\max_{i \leq n} h_{ii}^{(n)} \rightarrow 0$  in probability.

*Proof.* Write  $\mathbf{S}_n = n^{-1}\mathbf{X}^\top\mathbf{X} = n^{-1}\sum_{i \leq n} \mathbf{x}_{(i)}\mathbf{x}_{(i)}^\top$  and let  $\lambda_n$  be its smallest eigenvalue. By the strong law of large numbers  $\mathbf{S}_n \rightarrow E(\mathbf{x}\mathbf{x}^\top)$  with probability one, and eigenvalues are continuous functions of a symmetric matrix, so  $\lambda_n \rightarrow \lambda > 0$ , the smallest eigenvalue of  $E(\mathbf{x}\mathbf{x}^\top)$ . Hence with probability one  $\mathbf{X}^\top\mathbf{X}$  is nonsingular for all large  $n$ , and the leverages are defined from then on. By Theorem 1.7.7 applied to  $(\mathbf{X}^\top\mathbf{X})^{-1}$ , whose largest eigenvalue is  $1/(n\lambda_n)$ ,

$$\max_{i \leq n} h_{ii}^{(n)} = \max_{i \leq n} \mathbf{x}_{(i)}^\top (\mathbf{X}^\top\mathbf{X})^{-1} \mathbf{x}_{(i)} \leq \frac{\max_{i \leq n} \|\mathbf{x}_{(i)}\|^2}{n\lambda_n}.$$

It remains to show that  $\max_{i \leq n} \|\mathbf{x}_{(i)}\|^2/n \rightarrow 0$  in probability. For  $\epsilon > 0$ , by the union bound and Markov's inequality,

$$\Pr\left\{\max_{i \leq n} \|\mathbf{x}_{(i)}\|^2 > \epsilon n\right\} \leq n \Pr\{\|\mathbf{x}\|^2 > \epsilon n\} \leq \frac{1}{\epsilon} E[\|\mathbf{x}\|^2 \mathbf{1}\{\|\mathbf{x}\|^2 > \epsilon n\}] \rightarrow 0,$$

by dominated convergence, since  $E\|\mathbf{x}\|^2 < \infty$ .

■

The condition matters for inference with non-normal errors. Huber (1973) showed that every fitted value  $\mathbf{x}^\top \hat{\boldsymbol{\beta}}$  is asymptotically normal, for every error law with finite variance, iff the largest leverage tends to zero, and Theorem 19.3.2 of Chapter 19 rests on a condition of this Lindeberg type. A case that keeps a fixed share of the leverage keeps a fixed share of its own error in its fitted value, and no averaging makes that error normal.

Figure 20.2.1(b) shows the median, over 300 replications, of the largest leverage with two independent regressors. For normal regressors it falls from 0.322 at  $n = 25$  to 0.0030 at  $n = 6400$ . For  $t_3$  regressors, heavy-tailed but with finite variance, it falls much more slowly, to 0.066. For Cauchy regressors, which have no variance, it does not fall at all (0.857 at  $n = 25$ , 0.831 at  $n = 6400$ ): one case keeps most of the pull however many are added.

#### Example. (Leverage in Engel's household budgets)

Engel's 1857 survey records the annual income and food expenditure of 235 Belgian working-class households. Income is strongly skewed (skewness 2.78): the median is 884 and the largest, row 138 of the file, is 4958. In the straight-line regression of food expenditure on income the average leverage is  $2/235 = 0.0085$ , and the richest household has leverage 0.255, about 30 times the average; the next largest is 0.058.

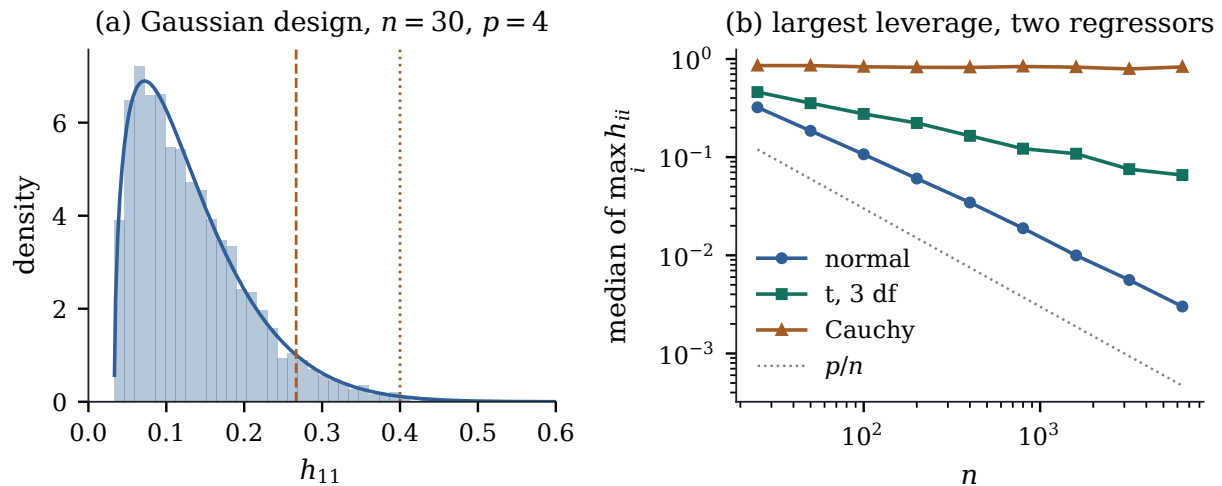


Figure 20.2.1: (a) Leverage of one case in 4000 Gaussian designs ( $n = 30$ ,  $p = 4$ ), the beta density of Proposition 20.2.1, and the cut-offs  $2p/n$  (dashed) and  $3p/n$  (dotted). (b) Median largest leverage for normal,  $t_3$  and Cauchy regressors.

Since  $h_{ii} - 1/n = (x_i - \bar{x})^2/S_{xx}$ , this one household supplies a quarter (0.251) of the spread of income about its mean, and the fitted line will pass near it whatever it spends on food. On the logarithmic scale its leverage is 0.070, still the largest but no longer in a class of its own. Choosing the scale (Chapter 22) decides which cases the fit depends on.

```
import numpy as np
import statsmodels.api as sm
engel = sm.datasets.engel.load_pandas().data

income = engel["income"].to_numpy()
food = engel["foodexp"].to_numpy()
n = len(income)
for label, x in [("income", income), ("log income", np.log(income))]:
    X = np.column_stack([np.ones(n), x])
    Q, _ = np.linalg.qr(X)
    h = np.sum(Q ** 2, axis=1)
    top = np.argsort(h)[::-1][:3]
    print(f"{label:11s} average p/n = {2 / n:.4f}; largest:",
          ", ".join(f"row {i + 1}: h = {h[i]:.3f}" for i in top))
```

Leverage also depends on the model: adding a column never decreases a leverage (Exercise (B2)), and a case ordinary for a line can be extreme for a quadratic (Exercise (B3)). A new point inside the range of every regressor can have high leverage because its *combination* of values is unusual (Example (Hidden extrapolation)).

### 20.2.3 Partial leverage

The leverage  $h_{ii}$  measures the pull of case  $i$  on the whole fitted surface. The Frisch–Waugh–Lovell theorem isolates one coefficient. Let  $\mathbf{M}_{(j)}$  project onto the span of the columns other than  $\mathbf{x}_j$ , and let  $\tilde{\mathbf{x}}_j = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$ .



**Definition 20.2.1 — Partial leverage**

The **partial leverage** of case  $i$  for the coefficient of  $\mathbf{x}_j$  is

$$\ell_{ij} = \frac{\tilde{x}_{ij}^2}{\|\tilde{\mathbf{x}}_j\|^2}.$$

**Proposition 20.2.3 — Properties of partial leverage**

Let  $h_{ii}^{(j)}$  be the leverages of the model without  $\mathbf{x}_j$ . Then:

- (a)  $h_{ii} = h_{ii}^{(j)} + \ell_{ij}$  for every  $i$ ;
- (b)  $0 \leq \ell_{ij} \leq 1$  and  $\sum_i \ell_{ij} = 1$ ;
- (c)  $\ell_{ij}$  is the leverage of case  $i$  in the regression through the origin of  $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{y}$  on  $\tilde{\mathbf{x}}_j$ , the regression drawn in the added-variable plot for  $\mathbf{x}_j$ ;
- (d)  $\hat{\beta}_j = \sum_i (\tilde{x}_{ij}/\|\tilde{\mathbf{x}}_j\|^2)y_i$ , so the weight of  $y_i$  in  $\hat{\beta}_j$  has square  $\ell_{ij}/\|\tilde{\mathbf{x}}_j\|^2 = \ell_{ij} \text{Var}(\hat{\beta}_j)/\sigma^2$ .

*Proof.* By Lemma 6.6.1(c) with  $\mathbf{X}_1 = \mathbf{x}_j$ ,  $\mathbf{M} = \mathbf{M}_{(j)} + \tilde{\mathbf{x}}_j\tilde{\mathbf{x}}_j^\top/\|\tilde{\mathbf{x}}_j\|^2$ . The diagonal entries give (a), and the second term is the hat matrix of the regression in (c). It is a rank-one projection, which gives (b). Part (d) is Theorem 6.6.2(a), together with  $\text{Var}(\hat{\beta}_j) = \sigma^2/\|\tilde{\mathbf{x}}_j\|^2$  (Eq. 6.6.2). ■

A case can have high leverage and negligible partial leverage for the coefficient of interest, when its unusual position lies along the other regressors. Partial leverages average  $1/n$ , so a few times  $1/n$  is already large.

**Example. (Where the District of Columbia's leverage comes from)**

In the murder-rate regression for the 51 jurisdictions, with poverty, single-parent households and urbanization as regressors, the District of Columbia has leverage 0.506 (Example (High-leverage states)) but partial leverage for poverty only 0.032: its leverage comes from its extreme single-parent and urban percentages, not from its poverty rate. The largest partial leverage for poverty is Alaska's, 0.141. Section 20.4 shows that the District nevertheless moves the poverty coefficient more than any other case, because influence also depends on the residual.

**20.2.4 Exercises****20.2.4.1 A. Check your understanding****Exercise A1.**

For regression through the origin on one regressor, show that  $h_{ii} = x_i^2/\sum_k x_k^2$ . Give

data with  $n = 10$  in which one leverage exceeds 0.99.

*Solution.*  $\mathbf{M} = \mathbf{xx}^\top / \mathbf{x}^\top \mathbf{x}$ , so  $h_{ii} = x_i^2 / \mathbf{x}^\top \mathbf{x}$ . With  $x_1 = 100$  and  $x_2 = \cdots = x_{10} = 1$ ,  $h_{11} = 10^4 / (10^4 + 9) > 0.99$ .

### Exercise A2.

Show that with an intercept the leverages are unchanged when the regressors are replaced by  $\mathbf{a} + \mathbf{Bx}$  with  $\mathbf{B}$  nonsingular, but not under a nonlinear transformation (Example (Leverage in Engel's household budgets)). Use Proposition 6.8.2.

## 20.2.4.2 B. Practice

### Exercise B1.

In the setting of Proposition 20.2.1, show that  $n(h_{ii} - 1/n)$  converges in law to  $\chi^2(k)$  as  $n \rightarrow \infty$ .

*Solution.* By the proof of Proposition 20.2.1,  $h_{ii} - 1/n$  has the law of  $(1 - 1/n)U/(U + V)$  with  $U \sim \chi^2(k)$  and  $V \sim \chi^2(n - 1 - k)$  independent. So  $n(h_{ii} - 1/n)$  has the law of  $(n - 1)U/(U + V) = U/[(U + V)/(n - 1)]$ . By the law of large numbers  $V/(n - 1) \rightarrow 1$  in probability, and  $U/(n - 1) \rightarrow 0$ , so the denominator tends to 1 in probability. Slutsky's theorem gives convergence in law to  $U \sim \chi^2(k)$ .

### Exercise B2.

Show that  $\max_i h_{ii} \geq p/n$ , with equality iff all leverages equal  $p/n$ . Show that if  $\mathbf{X}$  has orthogonal columns and every entry is  $\pm 1$ , then all leverages equal  $p/n$ .

### Exercise B3.

Take  $x = -3, -2, -1, 0, 1, 2, 3$ . Compute the leverage of the case at  $x = 0$  for the straight-line model and for the quadratic model  $E(Y) = \beta_0 + \beta_1 x + \beta_2 x^2$ . Explain the difference.

*Solution.* For the line,  $h = 1/7 + 0^2/28 = 1/7$ . For the quadratic, apply Proposition 20.2.3(a) with  $\mathbf{x}_j$  the column of squares:  $h = 1/7 + \ell$ , where  $\ell$  is the partial leverage of the quadratic term. The residual of  $x^2$  on  $(1, x)$  is  $x^2 - 4$ , because  $x^2$  has mean 4 and  $\sum_x x(x^2 - 4) = 0$  by symmetry. Its values are 5, 0, -3, -4, -3, 0, 5, with sum of squares 84. So  $\ell = 16/84 = 4/21$  and  $h = 1/7 + 4/21 = 1/3$ . The case at  $x = 0$  is the only one near the centre, and in a quadratic model the centre is where the curvature is pinned down.

## 20.2.4.3 C. Going deeper

### Exercise C1.

Complete the proof of Proposition 20.2.1 by showing that if  $\mathbf{Z}$  is  $n \times k$  with indepen-

dent  $N(0,1)$  entries and  $n \geq k + 2$ , then  $\mathbf{CZ}$  has rank  $k$  with probability one. *Hint:* add columns one at a time, and use that a nonzero normal vector with a nonsingular covariance restricted to a subspace falls in a fixed proper subspace with probability zero.

## 20.3 Deletion diagnostics

A case cannot fairly judge itself, since its own response pulled the fit towards it. So set the case aside, fit the other  $n - 1$  cases, and ask how well that fit predicts it and how far the fit moved. Chapter 10 showed that one regression gives all  $n$  answers (Proposition 10.6.2). This section collects the formulas and proves that the externally studentized residual has an exact  $t$  law, which turns it into a test for an outlier.

### 20.3.1 Leave-one-out formulas

Write  $\hat{\boldsymbol{\beta}}_{(i)}$  for the least squares estimate from the cases other than  $i$ ,  $\hat{\mathbf{Y}}_{(i)} = \mathbf{X}\hat{\boldsymbol{\beta}}_{(i)}$  for the fitted values it gives at *all*  $n$  cases (including case  $i$ ),  $\text{SSE}_{(i)}$  for its residual sum of squares and  $s_{(i)}^2 = \text{SSE}_{(i)}/(n - p - 1)$ .

#### Theorem 20.3.1 — Leave-one-out formulas

Let  $\mathbf{X}$  have full column rank, and let  $\mathbf{X}_{(i)}$  be  $\mathbf{X}$  without row  $i$ .

- (a)  $\mathbf{X}_{(i)}$  has full column rank iff  $h_{ii} < 1$ . Assume this from now on.
- (b)  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} \hat{\varepsilon}_i / (1 - h_{ii})$ .
- (c)  $\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} \hat{\varepsilon}_i / (1 - h_{ii})$ ; that is, the fitted value of case  $j$  moves by  $h_{ji} \hat{\varepsilon}_i / (1 - h_{ii})$ . In particular case  $i$ 's own fitted value moves by  $h_{ii} \hat{\varepsilon}_i / (1 - h_{ii})$ .
- (d) The **prediction residual** of case  $i$  is  $y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)} = \hat{\varepsilon}_i / (1 - h_{ii})$ .
- (e)  $(n - p - 1) s_{(i)}^2 = (n - p) s^2 - \hat{\varepsilon}_i^2 / (1 - h_{ii})$ .

*Proof.*

- (a)  $\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)} = \mathbf{X}^\top \mathbf{X} - \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ , whose determinant is  $(1 - h_{ii}) \det(\mathbf{X}^\top \mathbf{X})$  by Example (Deleting an observation). So it is nonsingular iff  $h_{ii} \neq 1$ , and  $h_{ii} \leq 1$  always. Parts (b), (d) and (e) are Proposition 10.6.2(a), (b) and (c), the last divided through by the degrees of freedom. Part (c) is (b) multiplied by  $\mathbf{X}$ ; its  $j$ th entry is  $\mathbf{x}_{(j)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} \hat{\varepsilon}_i / (1 - h_{ii})$ , and  $\mathbf{x}_{(j)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} = h_{ji}$ .

■

Every quantity is a residual times a function of the design: a case with zero residual changes nothing when deleted, and high leverage magnifies a residual by  $1/(1 - h_{ii})$ . When  $h_{ii} = 1$  the case determines one direction of  $\mathcal{C}(\mathbf{X})$  by itself (Proposition 6.8.1(d)), and without it a parameter is not estimable; no diagnostic can check it.

## 20.3.2 Prediction residuals and PRESS

The prediction residual  $\hat{\varepsilon}_i/(1 - h_{ii})$  is the error in predicting  $y_i$  by a fit that never saw it. Its variance is  $\sigma^2/(1 - h_{ii})$ , and dividing it by an estimate of  $\sigma/\sqrt{1 - \bar{h}_{ii}}$  gives back  $r_i$  (with  $s$ ) or  $t_i$  (with  $s_{(i)}$ ). The sum of squares of the prediction residuals is the PRESS statistic of Allen (1974),

$$\text{PRESS} = \sum_{i=1}^n \left( \frac{\hat{\varepsilon}_i}{1 - h_{ii}} \right)^2, \quad (20.3.1)$$

a leave-one-out estimate of prediction error that [Chapter 29](#) uses for choosing between models ([Theorem 29.3.2](#)).

### Proposition 20.3.2 — Expected PRESS

Under the second-moment model of [Section 20.1](#),

$$E(\text{PRESS}) = \sigma^2 \sum_{i=1}^n \frac{1}{1 - h_{ii}} \geq \sigma^2 \frac{n^2}{n - p},$$

with equality iff all leverages equal  $p/n$ . By comparison, the expected sum of squared errors in predicting new responses at the same  $n$  design points is  $\sigma^2(n + p)$  ([Proposition 14.2.7\(a\)](#)), which is smaller than  $\sigma^2 n^2/(n - p)$  by  $\sigma^2 p^2/(n - p)$ .

*Proof.*  $E(\hat{\varepsilon}_i^2) = \text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii})$ , and dividing by  $(1 - h_{ii})^2$  and summing gives the first identity. The function  $u \mapsto 1/(1 - u)$  is strictly convex on  $[0, 1)$ , so by Jensen's inequality  $n^{-1} \sum_i 1/(1 - h_{ii}) \geq 1/(1 - \bar{h})$  with  $\bar{h} = p/n$ , with equality iff all  $h_{ii}$  are equal. This gives the bound. Finally  $n^2 - (n + p)(n - p) = p^2$ . ■

So PRESS overcorrects the optimism of the in-sample error ([Proposition 14.2.7](#)) slightly, because each case is predicted from a design that lacks it. It also weights cases by  $(1 - h_{ii})^{-2}$ : at  $h_{ii} = 0.5$ , four times the squared raw residual. For the state regression with all 51 jurisdictions,  $\text{SSE} = 142.53$  and  $\text{PRESS} = 230.52$  ([Example \(Leaving out the District of Columbia\)](#)), and the District of Columbia alone contributes  $9.066^2$ , a share of 0.36. For this design  $E(\text{PRESS})/(n\sigma^2) = 1.096$ , against the bound  $n/(n - p) = 1.085$  and  $(n + p)/n = 1.078$ : the District's residual, not the unequal leverages, inflates PRESS.

## 20.3.3 The externally studentized residual

As a test statistic,  $r_i$  is bounded by  $\sqrt{n - p}$  and has a nonstandard law ([Proposition 20.1.1](#)). The externally studentized residual repairs both, and its law is clearest from the alternative it is meant to detect.

The **mean-shift outlier model** for case  $i$  is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \Delta \mathbf{e}_i + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I}), \quad (20.3.2)$$

in which case  $i$  alone has its mean displaced by an unknown  $\Delta$ , as by a misrecorded digit. Testing  $\Delta = 0$  tests whether case  $i$  is an outlier.

### Theorem 20.3.3 — Law of the externally studentized residual

Assume [Eq. 20.3.2](#) with  $h_{ii} < 1$  and  $n - p \geq 2$ .

- (a) In the model with columns  $[\mathbf{X}, \mathbf{e}_i]$ , the least squares estimate of the shift is  $\hat{\Delta} = \hat{\varepsilon}_i / (1 - h_{ii}) = y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)}$ , the estimate of  $\boldsymbol{\beta}$  is  $\hat{\boldsymbol{\beta}}_{(i)}$ , the residual mean square is  $s_{(i)}^2$ , and the  $t$  statistic for  $\Delta = 0$  is  $t_i$ .
- (b)  $t_i \sim t(n - p - 1, \Delta\sqrt{1 - h_{ii}}/\sigma)$ . In particular, if  $\Delta = 0$ , then  $t_i \sim t(n - p - 1)$ .
- (c)  $t_i = r_i \sqrt{(n - p - 1)/(n - p - r_i^2)}$ , an increasing function of  $r_i$  that maps  $(-\sqrt{n - p}, \sqrt{n - p})$  onto the whole real line.
- (d) (*Bonferroni outlier test.*) If no case is shifted, that is under [Eq. 7.3.1](#), and  $h_{jj} < 1$  for every  $j$ , then

$$\Pr\left\{\max_j |t_j| > t_{n-p-1, \alpha/(2n)}\right\} \leq \alpha.$$

Equivalently, the adjusted  $p$ -value  $\min(1, np_{(1)})$ , where  $p_{(1)}$  is the smallest of the single-case two-sided  $p$ -values from (b), is valid.

*Proof.*

- (a) Since  $h_{ii} < 1$ ,  $\mathbf{e}_i \notin \mathcal{C}(\mathbf{X})$  ([Proposition 6.8.1\(d\)](#)), so  $[\mathbf{X}, \mathbf{e}_i]$  has full column rank  $p + 1$ . By [Exercise \(B1\)](#),  $\hat{\Delta} = \hat{\varepsilon}_i / (1 - h_{ii})$ , the enlarged fit reproduces  $y_i$  exactly, and its estimate of  $\boldsymbol{\beta}$  is  $\hat{\boldsymbol{\beta}}_{(i)}$ . Its residuals are therefore zero at case  $i$  and equal to the residuals of the deletion fit elsewhere, so its residual sum of squares is  $\text{SSE}_{(i)}$  on  $n - p - 1$  degrees of freedom, and its residual mean square is  $s_{(i)}^2$ . By [Eq. 6.6.1](#), the diagonal entry of  $([\mathbf{X}, \mathbf{e}_i]^\top [\mathbf{X}, \mathbf{e}_i])^{-1}$  belonging to  $\mathbf{e}_i$  is  $1/\|(\mathbf{I} - \mathbf{M})\mathbf{e}_i\|^2 = 1/(1 - h_{ii})$ . So the standard error of  $\hat{\Delta}$  is  $s_{(i)}/\sqrt{1 - h_{ii}}$ , and

$$\frac{\hat{\Delta}}{s_{(i)}/\sqrt{1 - h_{ii}}} = \frac{\hat{\varepsilon}_i}{(1 - h_{ii})} \cdot \frac{\sqrt{1 - h_{ii}}}{s_{(i)}} = t_i.$$

- (b) The enlarged model is a normal linear model of full rank  $p + 1 < n$ , and  $\Delta$  is one of its coefficients. By [Theorem 11.6.1\(a\)](#) the  $t$  statistic for  $\Delta = 0$  has the  $t(n - p - 1, \delta)$  law with  $\delta = \Delta/(\sigma\sqrt{1/(1 - h_{ii})}) = \Delta\sqrt{1 - h_{ii}}/\sigma$ .
- (c) By [Theorem 20.3.1\(e\)](#) and  $\hat{\varepsilon}_i^2/(1 - h_{ii}) = r_i^2 s^2$ ,  $(n - p - 1)s_{(i)}^2 = s^2(n - p - r_i^2)$ . Hence  $t_i^2 = r_i^2 s^2/s_{(i)}^2 = r_i^2(n - p - 1)/(n - p - r_i^2)$ , and  $t_i$  has the sign of  $r_i$ . The map  $r \mapsto r\sqrt{(n - p - 1)/(n - p - r^2)}$  is increasing on  $(-\sqrt{n - p}, \sqrt{n - p})$  and tends to  $\pm\infty$  at the ends.
- (d) By (b) with  $\Delta = 0$ , each event  $A_j = \{|t_j| > t_{n-p-1, \alpha/(2n)}\}$  has probability exactly  $\alpha/n$ . Apply [Theorem 13.4.1\(a\)](#) to their union. The  $p$ -value form is [Theorem 13.4.1\(c\)](#) with every  $\alpha_j = \alpha/n$ .

■

By (c),  $r_i$  and  $t_i$  carry the same information, but  $t_i$  is unbounded and has a standard law, so it is the one to report. [Exercise \(B1\)](#) gives a second proof of (b). Part (b) also shows how leverage hides outliers: the noncentrality is  $\Delta\sqrt{1 - h_{ii}}/\sigma$ , because a shift at a high-leverage case is partly absorbed by the fit. At the state design, a shift of  $4\sigma$  at the District of Columbia ( $h_{ii} = 0.506$ ) gives noncentrality 2.81, and the Bonferroni test of (d) detects it with probability 0.258; at Missouri, the case of lowest leverage ( $h_{ii} = 0.021$ ), the noncentrality is 3.96 and the probability 0.666.

### 20.3.4 Testing for an outlier

A case suspected in advance can be tested with [Theorem 20.3.3\(b\)](#). Usually the suspect is chosen *because* its residual is largest, and the question is whether  $\max_i |t_i|$  is surprising under a model with no outliers. The  $t_i$  are dependent, but the Bonferroni test of [Theorem 20.3.3\(d\)](#) needs no joint law, and it is nearly exact because two cases rarely exceed the high cut-off together. At the state design ( $n = 51$ ,  $p = 4$ ), the simulated familywise level of the 5% test is 0.0497 (standard error 0.0005, from 200000 normal samples).

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statelcrime.load_pandas().data
X = np.column_stack([np.ones(len(data)), data[["poverty", "single", "urban"]]])

def max_abs_t(X, Y):
    """max_i |t_i| for each column of Y (one simulated response vector per column).
    """
    n, p = X.shape
    Q, _ = np.linalg.qr(X)
    h = np.sum(Q ** 2, axis=1)[:, None]
    E = Y - Q @ (Q.T @ Y)
    sse = np.sum(E ** 2, axis=0)
    s2_del = (sse - E ** 2 / (1 - h)) / (n - p - 1)
    return np.max(np.abs(E) / np.sqrt(s2_del * (1 - h)), axis=0)

n, p = X.shape
alpha = 0.05
crit = stats.t.isf(alpha / (2 * n), n - p - 1)
rng = np.random.default_rng(51)
reps = 4000
T = max_abs_t(X, rng.normal(size=(n, reps))) # no outliers: errors N(0, 1)
print(f"Bonferroni bound {alpha}, simulated familywise level {np.mean(T > crit):.4f}")
    "
```

#### Example. (Is the District of Columbia an outlier?)

For the murder-rate regression with all 51 jurisdictions, the District of Columbia has raw residual 4.475, leverage 0.506,  $r_i = 3.658$  and  $t_i = 4.278$  (as in [Example \(Leaving out the District of Columbia\)](#)); deleting it lowers  $s = 1.741$  to  $s_{(i)} = 1.489$ . Its single-case  $p$ -value is  $9.4 \times 10^{-5}$ , and the Bonferroni-adjusted value 0.0048 (critical value 3.522). The next case, Mississippi, has  $t_i = -2.877$  and adjusted  $p$ -value 0.310. Among the 50 states alone the most extreme is Louisiana,  $t_i = 3.299$ , adjusted  $p$ -value 0.095; such sequential testing is not covered by [Theorem 20.3.3\(d\)](#), and [Section 20.5](#) shows it can

fail badly.

The test says that the District's murder rate is incompatible with the linear relation that holds for the states. It does not say the number is wrong or that the case should be dropped. The District is a city, with regressors outside the range of the states; a defensible analysis restricts the population to states and says so.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statcrime.load_pandas().data           # District of Columbia
      kept
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
names = np.array(data.index)

n, p = X.shape
Q, R = np.linalg.qr(X)
beta = np.linalg.solve(R, Q.T @ y)
e = y - X @ beta                                         # raw residuals
h = np.sum(Q ** 2, axis=1)                               # leverages
s2 = e @ e / (n - p)
r = e / np.sqrt(s2 * (1 - h))                           # internally studentized
s2_del = (e @ e - e ** 2 / (1 - h)) / (n - p - 1)      # s^2 without case i
t = e / np.sqrt(s2_del * (1 - h))                       # externally studentized

p_single = 2 * stats.t.sf(np.abs(t), n - p - 1)          # one case, chosen in advance
p_bonf = np.minimum(1, n * p_single)                    # Bonferroni over all n cases
crit = stats.t.isf(0.05 / (2 * n), n - p - 1)           # 5% Bonferroni critical value
for i in np.argsort(-np.abs(t))[:3]:
    print(f"{names[i]:22s} h = {h[i]:.3f}  r = {r[i]:6.3f}  t = {t[i]:6.3f}"
          f"  Bonferroni p = {p_bonf[i]:.4f}")
print(f"critical value |t| > {crit:.3f}")
```

*Warning.* An outlier test tests *the model* at one case, including normality, constant variance and the mean function. A large  $|t_i|$  can come from a gross error, from heavy-tailed errors, from variance that grows where case  $i$  lies, or from a wrong mean function there, and the test cannot tell these apart (Chapter 21, Chapter 22).

## 20.3.5 Exercises

### 20.3.5.1 A. Check your understanding

#### Exercise A1.

Use Theorem 20.3.3(c) to compute the District of Columbia's  $t_i$  from its  $r_i = 3.658$ , with  $n = 51$  and  $p = 4$ .

*Solution.*  $t_i = 3.658\sqrt{46/(47 - 3.658^2)} = 3.658\sqrt{46/33.62} = 3.658 \times 1.1697 \approx 4.279$ , which agrees with the recorded 4.278 to the rounding of  $r_i$ .

**Exercise A2.**

A case has leverage 0.5 and residual 1. By how much does its own fitted value change when it is deleted, and what is its prediction residual? What is the prediction residual of a case with leverage 0.9 and the same raw residual?

**20.3.5.2 B. Practice****Exercise B1.**

Prove [Theorem 20.3.3](#)(b) with  $\Delta = 0$  directly. Show that  $y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)}$  is independent of  $s_{(i)}^2$ , that its variance is  $\sigma^2(1 + \mathbf{x}_{(i)}^\top (\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)})^{-1} \mathbf{x}_{(i)})$ , and that this equals  $\sigma^2/(1 - h_{ii})$ .

*Solution.* The fit without case  $i$  is a normal linear model for the other  $n - 1$  cases, and  $y_i$  is independent of those cases. So  $y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)}$  is the error of predicting a new observation at  $\mathbf{x}_{(i)}$ , and by [Theorem 12.3.1](#)(a) it is normal with mean zero and variance  $\sigma^2(1 + g)$ ,  $g = \mathbf{x}_{(i)}^\top (\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)})^{-1} \mathbf{x}_{(i)}$ , independent of  $s_{(i)}^2$ , with  $(y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)}) / (s_{(i)} \sqrt{1 + g}) \sim t(n - p - 1)$ . By [Example \(Deleting an observation\)](#),  $(\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)})^{-1} = (\mathbf{X}^\top \mathbf{X})^{-1} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} / (1 - h_{ii})$ , so  $g = h_{ii} + h_{ii}^2 / (1 - h_{ii}) = h_{ii} / (1 - h_{ii})$  and  $1 + g = 1 / (1 - h_{ii})$ . Using [Theorem 20.3.1](#)(d), the statistic is  $[\hat{\varepsilon}_i / (1 - h_{ii})] \sqrt{1 - h_{ii}} / s_{(i)} = t_i$ .

**Exercise B2.**

Show that  $s_{(i)} < s$  iff  $r_i^2 > 1$ . Interpret.

**Exercise B3.**

In the setting of [Theorem 20.3.3](#), the shift at case  $i$  is detected by the single-case test with probability  $\Pr\{|T| > c\}$ ,  $T \sim t(n - p - 1, \Delta \sqrt{1 - h_{ii}} / \sigma)$ . Show that for fixed  $\Delta$  and  $c$  this probability decreases as  $h_{ii}$  increases. *Hint:*  $T^2$  has a noncentral  $F$  law; use [Theorem 4.1.4](#).

**20.3.5.3 C. Going deeper****Exercise C1.**

Suppose the errors are independent with a  $t$  distribution on 3 degrees of freedom and the design is that of the state data. By simulation, estimate the probability that the 5% Bonferroni outlier test rejects. Explain the result in terms of the warning at the end of this section.



## 20.4 Cook's distance, DFFITS and DFBETAS

An outlier is a case the model fits badly; an **influential** case is one whose removal would change the conclusions. A remote case that follows the trend of the others exactly has no influence, and a large residual near the centroid moves the fit little. Influence measures, all from [Theorem 20.3.1](#), quantify different consequences of deleting a case.

### 20.4.1 Four measures

#### Definition 20.4.1 — Cook's distance, DFFITS, DFBETAS

Let  $h_{ii} < 1$ , and let  $c_{jj}$  be the  $j$ th diagonal entry of  $(\mathbf{X}^\top \mathbf{X})^{-1}$ , so that  $s^2 c_{jj}$  is the estimated variance of  $\hat{\beta}_j$ .

(a) **Cook's distance** is

$$D_i = \frac{(\hat{\beta}_{(i)} - \hat{\beta})^\top \mathbf{X}^\top \mathbf{X} (\hat{\beta}_{(i)} - \hat{\beta})}{p s^2}.$$

(b) **DFFITS** is the change in case  $i$ 's own fitted value in units of its standard error,

$$\text{DFFITS}_i = \frac{\hat{y}_i - \mathbf{x}_{(i)}^\top \hat{\beta}_{(i)}}{s_{(i)} \sqrt{h_{ii}}}.$$

(c) **DFBETAS** is the change in one coefficient in units of its standard error,

$$\text{DFBETAS}_{ij} = \frac{\hat{\beta}_j - \hat{\beta}_{j(i)}}{s_{(i)} \sqrt{c_{jj}}}.$$

(d) **COVRATIO** compares the generalized variances of the estimates with and without case  $i$ ,

$$\text{COVRATIO}_i = \frac{\det(s_{(i)}^2 (\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)})^{-1})}{\det(s^2 (\mathbf{X}^\top \mathbf{X})^{-1})}.$$

Cook (1977) proposed  $D_i$ ; DFFITS, DFBETAS and COVRATIO are from Belsley, Kuh and Welsch (1980). DFFITS and DFBETAS use  $s_{(i)}$ , for the same reason as  $t_i$ . None requires a refit.

#### Theorem 20.4.1 — Influence identities

Let  $h_{ii} < 1$  and  $n - p \geq 2$ , and let  $\tilde{\mathbf{x}}_j$  and  $\ell_{ij}$  be as in [Definition 20.2.1](#).

$$(a) \quad D_i = \frac{\|\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)}\|^2}{p s^2} = \frac{r_i^2}{p} \cdot \frac{h_{ii}}{1 - h_{ii}}.$$

$$(b) \quad \text{DFFITS}_i = t_i \sqrt{h_{ii}/(1 - h_{ii})}, \text{ and } D_i = \text{DFFITS}_i^2 s_{(i)}^2 / (p s^2).$$

(c)  $\text{DFBETAS}_{ij} = t_i \frac{\tilde{x}_{ij}}{\|\tilde{\mathbf{x}}_j\| \sqrt{1 - h_{ii}}}$ , so  $\text{DFBETAS}_{ij}^2 = t_i^2 \ell_{ij} / (1 - h_{ii})$ .

(d) For every  $\boldsymbol{\lambda} \neq \mathbf{0}$ ,

$$\frac{(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{(i)})^2}{s_{(i)}^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}} \leq \text{DFFITS}_i^2,$$

with equality at  $\boldsymbol{\lambda} = \mathbf{x}_{(i)}$ . In particular  $|\text{DFBETAS}_{ij}| \leq |\text{DFFITS}_i|$  for every  $j$ .

(e)  $\text{COVRATIO}_i = \frac{1}{1 - h_{ii}} \left( \frac{n - p}{n - p - 1 + t_i^2} \right)^p$ .

*Proof.* Put  $\mathbf{d} = \hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} g$  with  $g = \hat{\varepsilon}_i / (1 - h_{ii})$  (Theorem 20.3.1(b)).

(a)  $\hat{\mathbf{Y}} - \hat{\mathbf{Y}}_{(i)} = \mathbf{X} \mathbf{d}$ , and  $\|\mathbf{X} \mathbf{d}\|^2 = \mathbf{d}^\top \mathbf{X}^\top \mathbf{X} \mathbf{d} = g^2 \mathbf{x}_{(i)}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} = g^2 h_{ii}$ . So  $p D_i s^2 = \hat{\varepsilon}_i^2 h_{ii} / (1 - h_{ii})^2$ , and dividing by  $s^2$  and using  $r_i^2 = \hat{\varepsilon}_i^2 / \{s^2(1 - h_{ii})\}$  gives (a).

(b) The numerator of  $\text{DFFITS}_i$  is  $\mathbf{x}_{(i)}^\top \mathbf{d} = h_{ii} g$ , so  $\text{DFFITS}_i = \sqrt{h_{ii}} \hat{\varepsilon}_i / \{s_{(i)}(1 - h_{ii})\} = t_i \sqrt{h_{ii} / (1 - h_{ii})}$ . Then  $\text{DFFITS}_i^2 s_{(i)}^2 = \hat{\varepsilon}_i^2 h_{ii} / (1 - h_{ii})^2 = p D_i s^2$ .

(c) By Theorem 6.6.2(a) the  $j$ th row of  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  is  $\tilde{\mathbf{x}}_j^\top / \|\tilde{\mathbf{x}}_j\|^2$ , so the  $j$ th entry of  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)}$ , the  $i$ th column of that matrix, is  $\tilde{x}_{ij} / \|\tilde{\mathbf{x}}_j\|^2$ . By Eq. 6.6.1,  $c_{jj} = 1 / \|\tilde{\mathbf{x}}_j\|^2$ . Hence

$$\text{DFBETAS}_{ij} = \frac{\tilde{x}_{ij} g \|\tilde{\mathbf{x}}_j\|^{-2}}{s_{(i)} \|\tilde{\mathbf{x}}_j\|^{-1}} = \frac{\tilde{x}_{ij} \hat{\varepsilon}_i}{s_{(i)} \|\tilde{\mathbf{x}}_j\| (1 - h_{ii})},$$

which is the stated form.

(d)  $\boldsymbol{\lambda}^\top \mathbf{d} = g \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)}$ . Let  $\mathbf{K}^{1/2}$  be the symmetric square root of  $\mathbf{K} = (\mathbf{X}^\top \mathbf{X})^{-1}$  (Theorem 1.7.6). The Cauchy–Schwarz inequality (Proposition 1.1.1) for  $\mathbf{K}^{1/2} \boldsymbol{\lambda}$  and  $\mathbf{K}^{1/2} \mathbf{x}_{(i)}$  gives  $(\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)})^2 \leq \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda} \cdot h_{ii}$ , with equality when  $\boldsymbol{\lambda}$  is a multiple of  $\mathbf{x}_{(i)}$ . So the ratio is at most  $g^2 h_{ii} / s_{(i)}^2 = \text{DFFITS}_i^2$ . Taking  $\boldsymbol{\lambda} = \mathbf{e}_j$  gives  $\text{DFBETAS}_{ij}^2$ .

(e) By Example (Deleting an observation),  $\det(\mathbf{X}_{(i)}^\top \mathbf{X}_{(i)}) = (1 - h_{ii}) \det(\mathbf{X}^\top \mathbf{X})$ , so  $\text{COVRATIO}_i = (s_{(i)}^2 / s^2)^p / (1 - h_{ii})$ . From the proof of Theorem 20.3.3(c),  $s_{(i)}^2 / s^2 = (n - p - r_i^2) / (n - p - 1)$ , and solving  $t_i^2 = r_i^2 (n - p - 1) / (n - p - r_i^2)$  for  $r_i^2$  gives  $n - p - r_i^2 = (n - p)(n - p - 1) / (n - p - 1 + t_i^2)$ . Substitute.

■

## 20.4.2 What the measures mean

**Cook's distance** has two readings. By (a) it is the squared distance the fitted vector moves when case  $i$  is deleted, in units of  $p s^2$ . By Theorem 12.2.1 with  $\boldsymbol{\Lambda} = \mathbf{I}$ , the  $100(1 - \alpha)\%$  confidence ellipsoid for  $\boldsymbol{\beta}$  is  $\{\mathbf{b} : (\mathbf{b} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{X} (\mathbf{b} - \hat{\boldsymbol{\beta}}) \leq p s^2 F_\alpha(p, n - p)\}$ , so  $\hat{\boldsymbol{\beta}}_{(i)}$  lies on the boundary of the ellipsoid of level  $\Pr\{F(p, n - p) \leq D_i\}$ .

By the second equality in (a),  $D_i$  is misfit,  $r_i^2$ , times leverage,  $h_{ii} / (1 - h_{ii})$ : a case is influential only if it has both. DFFITS factors the same way with  $t_i$ . The **influence plot** of Figure

20.4.1 places each case by leverage and studentized residual, with the curves of constant  $D_i$ ,  $r^2 = pD(1 - h)/h$ .

**DFBETAS** isolates one coefficient: by (c) the partial leverage  $\ell_{ij}$  replaces the leverage in the product, but the factor  $1/(1 - h_{ii})$  still involves the overall leverage. By (d), the largest standardized change in *any* linear combination  $\lambda^\top \hat{\beta}$  is  $|\text{DFFITS}_i|$ , so a small DFFITS rules out a large change in every coefficient and contrast. **COVRATIO** measures precision rather than location: by (e) it exceeds one when  $h_{ii}$  is large and  $t_i$  small (the case adds precision) and falls below one when  $t_i$  is large (the case inflates  $s^2$ ).

### 20.4.3 The null law of Cook's distance and the choice of cut-offs

How large must an influence measure be before it matters? Since  $D_i$  is a multiple of  $r_i^2$ , its law under the model is known exactly.

#### Proposition 20.4.2 — Cook's distance under the model

Assume Eq. 7.3.1 with  $n - p \geq 2$  and all leverages below one. Then  $D_i = \frac{n - p}{p} \frac{h_{ii}}{1 - h_{ii}} B_i$  with  $B_i \sim \text{Beta}(\frac{1}{2}, \frac{n - p - 1}{2})$ , so

$$E(D_i) = \frac{h_{ii}}{p(1 - h_{ii})}, \quad \frac{1}{n} \sum_{i=1}^n E(D_i) \geq \frac{1}{n - p},$$

with equality in the last bound iff all leverages equal  $p/n$ .

*Proof.* Combine Theorem 20.4.1(a) with Proposition 20.1.1(a), and use  $E(B_i) = 1/(n - p)$ . For the bound,  $u \mapsto u/(1 - u)$  is strictly convex on  $[0, 1)$ , so by Jensen's inequality the average of  $h_{ii}/(1 - h_{ii})$  is at least  $\bar{h}/(1 - \bar{h}) = p/(n - p)$ , with equality iff the  $h_{ii}$  are equal. ■

So under the model a typical  $D_i$  is of order  $1/n$ , which explains the popular cut-off  $4/n$ . For the state design of Example (How much does the District of Columbia matter?) the average of  $E(D_i)$  is 0.0240, against the bound  $1/(n - p) = 0.0213$ . Comparing  $D_i$  with the median of  $F(p, n - p)$ , which is near one, is far stricter: it flags only cases whose deletion moves the estimate to the edge of a 50% confidence region. The usual cut-offs are these.

| Measure                 | Cut-off                                | Reasoning                                                     |
|-------------------------|----------------------------------------|---------------------------------------------------------------|
| $h_{ii}$                | $2p/n$                                 | twice the average leverage                                    |
| $D_i$                   | $4/n$ , or the median of $F(p, n - p)$ | four times the typical value; or the edge of the 50% region   |
| $ \text{DFFITS}_i $     | $2\sqrt{p/n}$                          | $ t_i  \approx 2$ at a case of average leverage $p/n$         |
| $ \text{DFBETAS}_{ij} $ | $2/\sqrt{n}$                           | $ t_i  \approx 2$ at a case of average partial leverage $1/n$ |

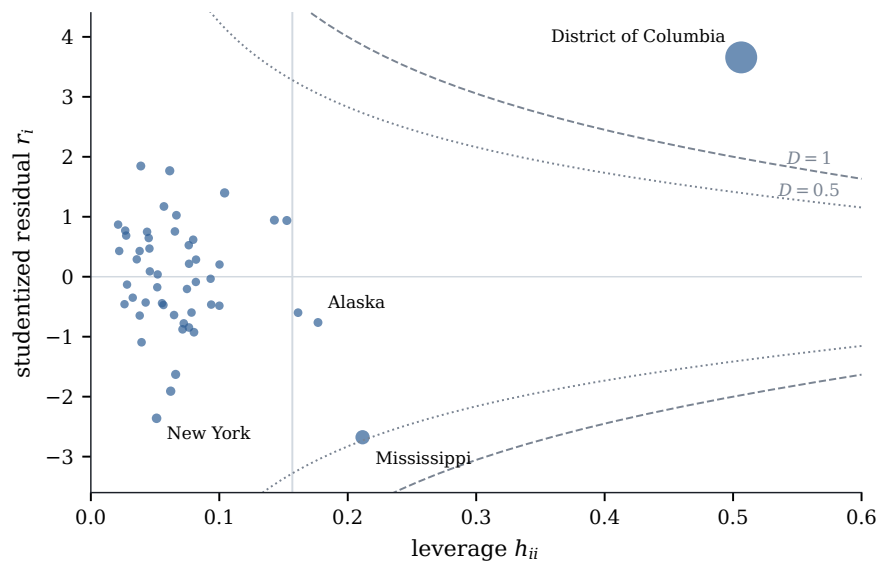


Figure 20.4.1: Influence plot for the state regression: leverage against  $r_i$ , area proportional to Cook's distance, curves  $D_i = 0.5$  (dotted) and  $D_i = 1$  (dashed), and the line  $h_{ii} = 2p/n$ .

| Measure                   | Cut-off | Reasoning                                                                  |
|---------------------------|---------|----------------------------------------------------------------------------|
| $ \text{COVRATIO}_i - 1 $ | $3p/n$  | first-order expansion of (e) at $ t_i  \approx 2$ or $h_{ii} \approx 2p/n$ |

All are arbitrary in the same way: each puts a case with studentized residual about two and typical leverage near the line. Calibrating  $D_i$  exactly by [Proposition 20.4.2](#) would not help: for a fixed design  $D_i$  is a multiple of  $r_i^2$ , so the resulting test is an outlier test with a case-dependent level, discarding the leverage factor that is the point of an influence measure. Influence is a consequence to be assessed, not a hypothesis: the right threshold is whether deleting the case changes a conclusion.

#### Example. (How much does the District of Columbia matter?)

In the murder-rate regression with all 51 jurisdictions, the District of Columbia has  $D_i = 3.430$ , seven times the next largest (Mississippi, 0.481); the median of  $F(4, 47)$  is 0.851, and deleting the District moves  $\hat{\beta}$  to the boundary of the 0.985 confidence ellipsoid. Its DFFITS is 4.333 (cut-off 0.560) and its DFBETAS for poverty  $-1.090$  (cut-off 0.280). Its partial leverage for poverty is only 0.032 ([Example \(Where the District of Columbia's leverage comes from\)](#)), but by [Theorem 20.4.1\(c\)](#) it is multiplied by  $t_i^2/(1 - h_{ii})$ :  $4.278^2 \times 0.032/(1 - 0.506) \approx 1.19$ , so  $|\text{DFBETAS}| \approx 1.09$ . Alaska, with the largest partial leverage (0.141), has  $t_i = -0.760$  and DFBETAS 0.315.

The consequences are substantive. With the District, the poverty coefficient is 0.1551 (standard error 0.1059,  $t = 1.47$ ); without it, 0.2538 (standard error 0.0934,  $t = 2.72$ ), the fit of [Example \(Poverty and murder rates\)](#). One jurisdiction decides whether poverty has a detectable association with murder rates given the other regressors, and the single-parent coefficient moves from 0.6256 to 0.3980. The cut-offs flag 2 cases by  $D_i > 4/n = 0.078$ , 3 by DFFITS, 5 by DFBETAS for poverty and 4 by leverage above  $2p/n = 0.157$ , but only the District changes a conclusion. Its COVRATIO is 0.578: it inflates the estimated generalized variance through its contribution to  $s^2$ .

```

import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statelcrime.load_pandas().data           # District of Columbia
    kept
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
names = np.array(data.index)
n, p = X.shape
Q, R = np.linalg.qr(X)
beta = np.linalg.solve(R, Q.T @ y)
e = y - X @ beta                                           # raw residuals
h = np.sum(Q ** 2, axis=1)                                  # leverages
s2 = e @ e / (n - p)
r = e / np.sqrt(s2 * (1 - h))                              # internally studentized
s2_del = (e @ e - e ** 2 / (1 - h)) / (n - p - 1)         # s^2 without case i
t = e / np.sqrt(s2_del * (1 - h))                          # externally studentized

cook = r ** 2 * h / (p * (1 - h))                          # Cook's distance
dffits = t * np.sqrt(h / (1 - h))
K = np.linalg.inv(X.T @ X)
C = K @ X.T                                                # row j: weights of y in beta_j
dfbetas = (C * e / (1 - h)).T / np.sqrt(np.outer(s2_del, np.diag(K)))
for i in np.argsort(-cook)[:3]:
    print(f"{names[i]:22s} D = {cook[i]:.3f}  DFFITS = {dffits[i]:6.3f}"
          f"  DFBETAS(poverty) = {dfbetas[i, 1]:6.3f}")
print("conventional cut-offs: D > 4/n =", round(4 / n, 3),
      " |DFFITS| > 2 sqrt(p/n) =", round(2 * np.sqrt(p / n), 3),
      " |DFBETAS| > 2/sqrt(n) =", round(2 / np.sqrt(n), 3))

```

## 20.4.4 Exercises

### 20.4.4.1 A. Check your understanding

#### Exercise A1.

In a regression with  $p = 3$ , case  $i$  has  $r_i = 2$ . Compute  $D_i$  when  $h_{ii} = 0.1$  and when  $h_{ii} = 0.6$ . Which case would you look at first, and why might neither be alarming?

*Solution.* By [Theorem 20.4.1\(a\)](#),  $D_i = (4/3) h_{ii} / (1 - h_{ii})$ :  $(4/3)(1/9) = 4/27 \approx 0.15$  and  $(4/3)(1.5) = 2$ . The second moves the coefficient vector to about the edge of a high-level confidence region. Neither is alarming by itself if the change it produces in the estimates of interest is small relative to their uncertainty, or if the case is known to be correct and the model to hold in its region.

#### Exercise A2.

Show that  $D_i = 0$  iff  $\hat{\varepsilon}_i = 0$ , whatever the leverage. Give an example of a case with leverage 0.9 and no influence.

## 20.4.4.2 B. Practice

**Exercise B1.**

Let  $\mathbf{C} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ . Show that the sum of squares of the  $j$ th row of  $\mathbf{C}$  is  $c_{jj}$ , and deduce that  $\sum_i \ell_{ij} = 1$  in a second way. Why does this suggest a cut-off of order  $1/\sqrt{n}$  for DFBETAS?

*Solution.*  $\mathbf{C}\mathbf{C}^\top = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} = (\mathbf{X}^\top \mathbf{X})^{-1}$ , whose  $j$ th diagonal entry is  $c_{jj}$ . The  $j$ th row of  $\mathbf{C}$  is  $\tilde{\mathbf{x}}_j^\top / \|\tilde{\mathbf{x}}_j\|^2$  (proof of [Theorem 20.4.1\(c\)](#)), so  $\sum_i \ell_{ij} = \sum_i \tilde{x}_{ij}^2 / \|\tilde{\mathbf{x}}_j\|^2 = c_{jj} \|\tilde{\mathbf{x}}_j\|^2 = 1$ . The partial leverages for a coefficient average  $1/n$ . With  $|t_i| \approx 2$  and  $h_{ii}$  small,  $|\text{DFBETAS}_{ij}| \approx 2\sqrt{\ell_{ij}} \approx 2/\sqrt{n}$  for a typical case.

**Exercise B2.**

For simple regression, show that the DFBETAS of case  $i$  for the slope is  $t_i(x_i - \bar{x}) / \sqrt{S_{xx}(1 - h_{ii})}$ . Which cases can change the slope most?

**Exercise B3.**

Use [Theorem 20.4.1\(e\)](#) to show that for large  $n$  and small  $p/n$ ,  $\text{COVRATIO}_i \approx 1 + h_{ii} - p(t_i^2 - 1)/n$ . Deduce the cut-off  $|\text{COVRATIO}_i - 1| > 3p/n$  from the two extreme cases  $h_{ii} \approx 0$ ,  $|t_i| = 2$  and  $h_{ii} = 2p/n$ ,  $t_i = 0$ .

*Solution.*  $1/(1 - h) \approx 1 + h$ , and  $((n - p)/(n - p - 1 + t^2))^p = (1 + (t^2 - 1)/(n - p))^{-p} \approx 1 - p(t^2 - 1)/n$ . Multiplying,  $\text{COVRATIO} \approx 1 + h - p(t^2 - 1)/n$  to first order. With  $h \approx 0$  and  $t^2 = 4$  this is  $1 - 3p/n$ : a case of no leverage and a residual of two standard errors lowers the precision by that much. With  $t = 0$  and  $h = 2p/n$  it is  $1 + 3p/n$ : a well-fitting case of high leverage raises it by the same amount. Deviations beyond  $3p/n$  in either direction are more extreme than both.

## 20.5 Masking and swamping

Every diagnostic so far removes *one* case. But bad cases often come in groups: a batch of records in the wrong units, or a sensor that failed for an afternoon. **Masking** is when a bad case escapes detection because others hide it: deleting it alone leaves its companions to hold the fit in place. **Swamping** is when good cases are flagged because the bad ones have dragged the fit away from them. This section deletes sets of cases and proves how a cluster masks itself and swamps the rest.

### 20.5.1 Deleting a set of cases

Let  $D \subset \{1, \dots, n\}$  be a set of  $d$  cases. Write  $\mathbf{X}_D$  for the  $d \times p$  matrix of their rows,  $\mathbf{X}_{(D)}$  for  $\mathbf{X}$  with those rows removed,  $\hat{\mathbf{e}}_D$  for their residuals in the full fit, and

$$\mathbf{H}_D = \mathbf{X}_D (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top$$

for the  $d \times d$  block of the hat matrix that belongs to them. Let  $\hat{\boldsymbol{\beta}}_{(D)}$  and  $\text{SSE}_{(D)}$  come from the least squares fit to the cases not in  $D$ , and let  $\mathbf{E}_D$  be the  $n \times d$  matrix whose columns are the coordinate vectors  $\mathbf{e}_i$ ,  $i \in D$ .

### Theorem 20.5.1 — Deleting a set of cases

Let  $\mathbf{X}$  and  $\mathbf{X}_{(D)}$  have full column rank. Then  $\mathbf{I} - \mathbf{H}_D$  is nonsingular, and

- (a)  $\mathbf{y}_D - \mathbf{X}_D \hat{\boldsymbol{\beta}}_{(D)} = (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\epsilon}}_D$ ;
- (b)  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(D)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\epsilon}}_D$ ;
- (c)  $\text{SSE}_{(D)} = \text{SSE} - \hat{\boldsymbol{\epsilon}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\epsilon}}_D$ ;
- (d) if  $n - p - d \geq 1$ , then under [Eq. 7.3.1](#)

$$F_D = \frac{\hat{\boldsymbol{\epsilon}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\epsilon}}_D / d}{\text{SSE}_{(D)} / (n - p - d)} \sim F(d, n - p - d),$$

and  $F_D$  is the  $F$  statistic for  $\boldsymbol{\gamma} = \mathbf{0}$  in the mean-shift model  $\mathbf{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{E}_D\boldsymbol{\gamma}$ .

*Proof.* Let  $\mathbf{Z} = [\mathbf{X}, \mathbf{E}_D]$ . If  $\mathbf{X}\mathbf{b} + \mathbf{E}_D\mathbf{c} = \mathbf{0}$ , the rows outside  $D$  give  $\mathbf{X}_{(D)}\mathbf{b} = \mathbf{0}$ , so  $\mathbf{b} = \mathbf{0}$ , and then  $\mathbf{c} = \mathbf{0}$ . So  $\mathbf{Z}$  has full column rank. By [Lemma 6.6.1\(b\)](#) with  $\mathbf{X}_1 = \mathbf{E}_D$ , the matrix  $(\mathbf{I} - \mathbf{M})\mathbf{E}_D$  has full column rank, so its Gram matrix  $\mathbf{E}_D^\top (\mathbf{I} - \mathbf{M})\mathbf{E}_D = \mathbf{I} - \mathbf{H}_D$  is nonsingular.

Fit the mean-shift model. By [Theorem 6.6.2\(a\)](#) with  $\mathbf{X}_1 = \mathbf{E}_D$  and  $\mathbf{X}_2 = \mathbf{X}$ , the estimated shifts are

$$\hat{\boldsymbol{\gamma}} = (\mathbf{E}_D^\top (\mathbf{I} - \mathbf{M})\mathbf{E}_D)^{-1} \mathbf{E}_D^\top (\mathbf{I} - \mathbf{M})\mathbf{y} = (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\epsilon}}_D.$$

On the other hand, for any  $\mathbf{b}$  the residual sum of squares  $\sum_{i \notin D} (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2 + \|\mathbf{y}_D - \mathbf{X}_D \mathbf{b} - \mathbf{c}\|^2$  is minimized over  $\mathbf{c}$  by  $\mathbf{c} = \mathbf{y}_D - \mathbf{X}_D \mathbf{b}$ , which removes the second term. What remains is minimized by  $\mathbf{b} = \hat{\boldsymbol{\beta}}_{(D)}$ , uniquely because  $\mathbf{X}_{(D)}$  has full column rank. So the mean-shift fit has coefficients  $\hat{\boldsymbol{\beta}}_{(D)}$  and  $\hat{\boldsymbol{\gamma}} = \mathbf{y}_D - \mathbf{X}_D \hat{\boldsymbol{\beta}}_{(D)}$ , and its residual sum of squares is  $\text{SSE}_{(D)}$ . Comparing the two expressions for  $\hat{\boldsymbol{\gamma}}$  gives (a). By [Theorem 6.6.2\(c\)](#), the coefficients of  $\mathbf{X}$  in the mean-shift fit are  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{y} - \mathbf{E}_D \hat{\boldsymbol{\gamma}}) = \hat{\boldsymbol{\beta}} - (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top \hat{\boldsymbol{\gamma}}$ , which gives (b). By [Lemma 6.6.1\(c\)](#) the projection onto  $\mathcal{C}(\mathbf{Z})$  is  $\mathbf{M}$  plus the projection onto  $\mathcal{C}((\mathbf{I} - \mathbf{M})\mathbf{E}_D)$ , so the fall in the residual sum of squares from the fit of  $\mathbf{X}$  to the fit of  $\mathbf{Z}$  is

$$\mathbf{y}^\top (\mathbf{I} - \mathbf{M})\mathbf{E}_D (\mathbf{I} - \mathbf{H}_D)^{-1} \mathbf{E}_D^\top (\mathbf{I} - \mathbf{M})\mathbf{y} = \hat{\boldsymbol{\epsilon}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\epsilon}}_D,$$

which gives (c). Finally  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$  with dimensions  $p$  and  $p + d$ , and  $F_D$  is the statistic [Eq. 11.1.1](#) for this pair of nested models, so (d) is [Theorem 11.1.1\(d\)](#). ■

With  $d = 1$  this is [Theorem 20.3.1](#), and  $F_D = t_i^2$  ([Exercise \(A2\)](#)). Part (d) is exact only for a set chosen in advance; a Bonferroni correction over all  $\binom{n}{d}$  subsets is valid but very conservative.

## 20.5.2 How a cluster masks itself

The simplest mechanism is a group of identical bad cases:  $n_0$  clean cases plus  $m$  copies of one bad case.

### Proposition 20.5.2 — Masking and swamping by a cluster

Let  $\mathbf{X}_0$  ( $n_0 \times p$ , full column rank) and  $\mathbf{y}_0$  be the clean cases, with least squares estimate  $\hat{\boldsymbol{\beta}}_0$ , residuals  $\tilde{\epsilon}_j = y_j - \mathbf{x}_{(j)}^\top \hat{\boldsymbol{\beta}}_0$  and residual sum of squares  $\text{SSE}_0$ . Add  $m \geq 1$  cases, each with row  $\mathbf{x}_\star^\top$  and response  $y_\star$ . Put  $\mathbf{A} = \mathbf{X}_0^\top \mathbf{X}_0$ ,

$$h_\star = \mathbf{x}_\star^\top \mathbf{A}^{-1} \mathbf{x}_\star, \quad \delta = y_\star - \mathbf{x}_\star^\top \hat{\boldsymbol{\beta}}_0,$$

the leverage of  $\mathbf{x}_\star$  relative to the clean design and the distance of the cluster from the clean fit. In the least squares fit to all  $n_0 + m$  cases:

- (a) each cluster case has leverage  $h = h_\star / (1 + mh_\star) < 1/m$ ;
- (b) each cluster case has residual  $\delta / (1 + mh_\star)$ ;
- (c) deleting one cluster case gives it prediction residual  $\delta / (1 + (m-1)h_\star)$  and changes the estimate by  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = \mathbf{A}^{-1} \mathbf{x}_\star \delta / [(1 + mh_\star)(1 + (m-1)h_\star)]$ ;
- (d) deleting the whole cluster gives each member prediction residual  $\delta$ , and  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(D)} = m(1 + (m-1)h_\star)(\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)})$ ;
- (e) clean case  $j$  has residual  $\tilde{\epsilon}_j - m\delta h_{j\star} / (1 + mh_\star)$ , where  $h_{j\star} = \mathbf{x}_{(j)}^\top \mathbf{A}^{-1} \mathbf{x}_\star$ , and  $\text{SSE} = \text{SSE}_0 + m\delta^2 / (1 + mh_\star)$ .

*Proof.* Let  $D$  be the cluster. Then  $\mathbf{X}^\top \mathbf{X} = \mathbf{A} + m\mathbf{x}_\star \mathbf{x}_\star^\top$ , and  $(\mathbf{A} + m\mathbf{x}_\star \mathbf{x}_\star^\top) \mathbf{A}^{-1} \mathbf{x}_\star = (1 + mh_\star) \mathbf{x}_\star$ , so

$$(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_\star = \frac{\mathbf{A}^{-1} \mathbf{x}_\star}{1 + mh_\star}. \quad (20.5.1)$$

- (a)  $h = \mathbf{x}_\star^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_\star = h_\star / (1 + mh_\star)$ , and  $mh = mh_\star / (1 + mh_\star) < 1$ .
- (b) and (d) Deleting  $D$  leaves the clean cases, so  $\hat{\boldsymbol{\beta}}_{(D)} = \hat{\boldsymbol{\beta}}_0$  and every cluster case has prediction residual  $\delta$ . All rows of  $\mathbf{X}_D$  equal  $\mathbf{x}_\star^\top$ , so  $\mathbf{H}_D = h \mathbf{1} \mathbf{1}^\top$ , and by Theorem 20.5.1(a)  $\hat{\boldsymbol{\epsilon}}_D = (\mathbf{I} - h \mathbf{1} \mathbf{1}^\top) \delta \mathbf{1} = (1 - mh) \delta \mathbf{1} = \delta \mathbf{1} / (1 + mh_\star)$ . By Theorem 20.5.1(b) and Eq. 20.5.1,  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_0 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}_D^\top \delta \mathbf{1} = m\delta (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_\star = m\delta \mathbf{A}^{-1} \mathbf{x}_\star / (1 + mh_\star)$ .
- (c) By Theorem 20.3.1(d) and (b), with  $1 - h = (1 + (m-1)h_\star) / (1 + mh_\star)$ , the prediction residual is  $\delta / (1 + (m-1)h_\star)$ . By Theorem 20.3.1(b) and Eq. 20.5.1,  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_{(i)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_\star \cdot \delta / (1 + (m-1)h_\star)$ , which is the stated vector. Comparing with the expression for  $\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_0$  in the previous step gives the ratio in (d).
- (d)  $y_j - \mathbf{x}_{(j)}^\top \hat{\boldsymbol{\beta}} = \tilde{\epsilon}_j - \mathbf{x}_{(j)}^\top (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_0)$ , and  $\mathbf{x}_{(j)}^\top (\hat{\boldsymbol{\beta}} - \hat{\boldsymbol{\beta}}_0) = m\delta h_{j\star} / (1 + mh_\star)$ . For the sum of squares, Theorem 20.5.1(c) gives  $\text{SSE} = \text{SSE}_0 + \hat{\boldsymbol{\epsilon}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\epsilon}}_D$ , and  $\hat{\boldsymbol{\epsilon}}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\boldsymbol{\epsilon}}_D = \delta^2 \mathbf{1}^\top (\mathbf{I} - h \mathbf{1} \mathbf{1}^\top)^{-1} \mathbf{1} = m\delta^2 (1 - mh)$ , using  $(\mathbf{I} - h \mathbf{1} \mathbf{1}^\top)^{-1} \hat{\boldsymbol{\epsilon}}_D = \delta \mathbf{1}$  from part (b).

■



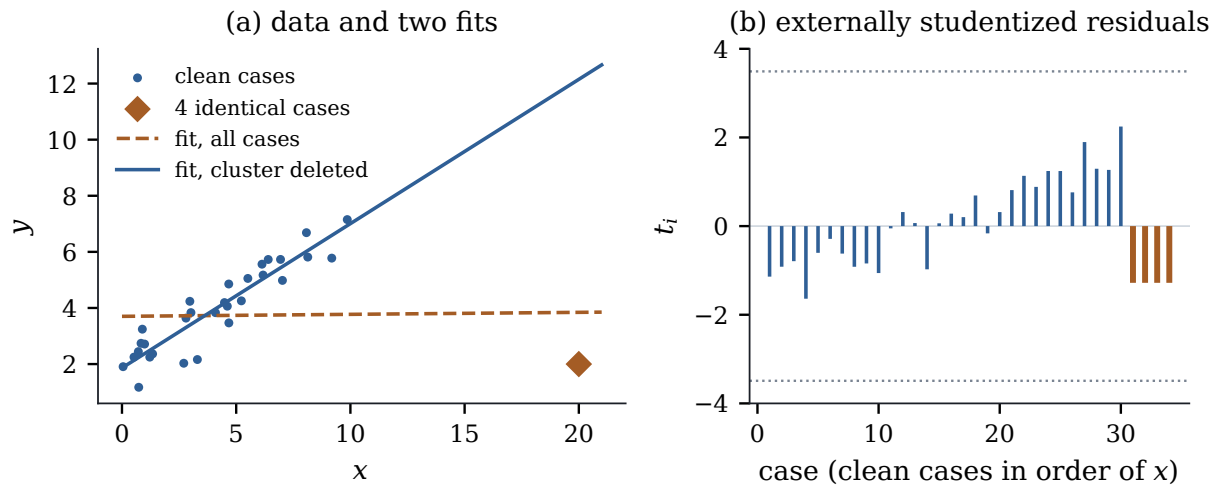


Figure 20.5.1: Masking and swamping. (a) Clean cases, four identical bad records (diamond), and least squares lines with and without them. (b) Externally studentized residuals (bad records last) and the 5% Bonferroni cut-offs.

Each part is a way of hiding. **Leverage is masked** (a): however remote the cluster, each copy has leverage below  $1/m$ ; a single copy would have  $h_*/(1 + h_*)$ , near one. **Residuals are masked** (b): each is only  $1/(1 + mh_*)$  of the cluster's distance from the clean fit. **Deletion is masked** (c, d): deleting one copy leaves  $m - 1$  to hold the fit, recovering only  $1/(1 + (m - 1)h_*)$  of  $\delta$ . **Good cases are swamped** (e): clean residuals shift in proportion to their cross-leverage with the cluster, and  $s^2$  is inflated.

#### Example. (Four identical bad records)

Thirty clean cases have  $x$  uniform on  $(0, 10)$  and  $y = 2 + 0.5x + \varepsilon$ ,  $\sigma = 0.5$ . Four identical records are added at  $x = 20$ ,  $y = 2$ , where the clean line predicts about 12 (Figure 20.5.1(a)). The slope drops from 0.514 (clean cases) to 0.007, and  $s$  rises from 0.608 to 1.632. Here  $h_* = 1.126$  (it can exceed one, since  $x = 20$  is outside the clean design, as in Proposition 12.3.2) and  $\delta = -10.150$ . Each bad case has leverage 0.205, below  $1/m = 0.25$ , where a single copy would have 0.530; residual  $-1.844$ ; prediction residual  $-2.318$ ; and  $t_i = -1.28$ . Deleting the group changes  $\hat{\beta}$  17.51 times as much as deleting one member.

The single-case diagnostics fail as predicted. The Bonferroni test (critical value 3.49) flags nothing, and the only case with  $|t_i| > 2$  is a clean one: case 30, the clean case with largest  $x$  (9.86), has  $t_i = 2.25$ , because the dragged line passes below it. Cook's distance does a little better: each bad case has  $D_i = 0.206$ , above  $4/n = 0.118$  (no clean case exceeds it; the largest is 0.100) but far below the median of  $F(2, 32)$ . A single bad record alone would have had  $t_i = -11.45$  and  $D_i = 13.46$ . Deleting the four together tells the truth: their prediction residuals are all  $\delta$ , and Theorem 20.5.1(d) gives  $F = 50.6$  on 4 and 28 degrees of freedom,  $p$ -value  $2.0 \times 10^{-12}$ . The difficulty is not in testing the cluster but in finding it.

```
import numpy as np
from scipy import stats

def single_case(X, y):
    """Leverages, residuals, s^2, externally studentized residuals and Cook's
```

```

distances."""
n, p = X.shape
Q, _ = np.linalg.qr(X)
h = np.sum(Q ** 2, axis=1)
e = y - Q @ (Q.T @ y)
s2 = e @ e / (n - p)
s2_del = (e @ e - e ** 2 / (1 - h)) / (n - p - 1)
t = e / np.sqrt(s2_del * (1 - h))
cook = e ** 2 * h / (p * s2 * (1 - h) ** 2)
return h, e, s2, t, cook

rng = np.random.default_rng(20)
n0, m = 30, 4                                # clean cases, size of the
cluster
x_clean = np.sort(rng.uniform(0, 10, n0))
y_clean = 2 + 0.5 * x_clean + rng.normal(0, 0.5, n0)
x0, y0 = 20.0, 2.0                            # four identical records
x = np.r_[x_clean, np.full(m, x0)]
y = np.r_[y_clean, np.full(m, y0)]
X = np.column_stack([np.ones(len(x)), x])
n, p = X.shape
cluster = np.arange(n0, n)

h, e, s2, t, cook = single_case(X, y)
crit = stats.t.isf(0.05 / (2 * n), n - p - 1)    # Bonferroni outlier test at 5%
print("slope with the cluster:", np.linalg.lstsq(X, y, rcond=None)[0][1])
print("cluster: h =", h[-1].round(3), " t =", t[-1].round(2), " Cook's D =", cook
      [-1].round(3))
i = int(np.argmax(np.abs(t)))
print(f"largest |t|: case {i + 1} at x = {x[i]:.2f}, t = {t[i]:.2f}; Bonferroni cut-
      off {crit:.2f}")

D = cluster                                # delete the whole cluster at
once
K = np.linalg.inv(X.T @ X)
H_D = X[D] @ K @ X[D].T
pred_D = np.linalg.solve(np.eye(m) - H_D, e[D])    # y_D minus its prediction from
the rest
sse = e @ e
sse_D = sse - e[D] @ pred_D
F = (e[D] @ pred_D / m) / (sse_D / (n - p - m))
print("group prediction residuals:", pred_D.round(3))
print(f"F = {F:.1f} on {m} and {n - p - m} degrees of freedom,"
      f" p-value {stats.f.sf(F, m, n - p - m):.1e}")

```

### 20.5.3 Finding the set

There are  $\binom{n}{d}$  candidate sets of each size  $d$ , far too many to search. The natural shortcut, deleting the case with the largest  $|t_i|$  and repeating, starts in the wrong place: in [Example \(Four identical bad records\)](#) it would first remove case 30, a clean case. Three ideas get around the combinatorics. First, **start from a fit the cluster cannot capture**, one determined by about half the data (the next section); the cases it flags form a candidate set to examine and test, remembering that it was chosen by looking. Second, **grow a clean subset**: the forward search of Atkinson and Riani (2000) adds cases one at a time in order of their residuals from a

robust start, and a cluster enters late and all at once, as a jump in the monitored residuals and estimates; Hadi and Simonoff (1993) proposed a related procedure. Third, **unmask leverage separately**: by [Proposition 20.5.2\(a\)](#) the hat matrix is fooled by clusters, since leverage is a Mahalanobis distance built from the sample covariance, which a cluster inflates in its own direction. Rousseeuw and van Zomeren (1990) replaced the mean and covariance by estimates that ignore up to half the cases.

*Idea.* Single-case diagnostics answer “what if this case alone were different?”, the wrong question when bad cases come in groups. Group deletion asks the right one, but only for a suspected group; finding the group needs a fit it cannot capture.

## 20.5.4 Exercises

### 20.5.4.1 A. Check your understanding

#### Exercise A1.

Check the leverage 0.205 and the residual  $-1.844$  in [Example \(Four identical bad records\)](#) from  $h_{\star} = 1.126$ ,  $\delta = -10.150$  and  $m = 4$ .

#### Exercise A2.

Show that [Theorem 20.5.1](#) with  $D = \{i\}$  gives parts (b), (d) and (e) of [Theorem 20.3.1](#), and that  $F_D = t_i^2$ .

*Solution.* With  $d = 1$ ,  $\mathbf{H}_D = h_{ii}$  and  $\hat{\mathbf{e}}_D = \hat{\varepsilon}_i$ , so (a) is the prediction residual  $\hat{\varepsilon}_i/(1 - h_{ii})$ , (b) is [Theorem 20.3.1\(b\)](#), and (c) is  $\text{SSE}_{(i)} = \text{SSE} - \hat{\varepsilon}_i^2/(1 - h_{ii})$ . Then  $F_D = \hat{\varepsilon}_i^2/\{(1 - h_{ii})s_{(i)}^2\} = t_i^2$ .

### 20.5.4.2 B. Practice

#### Exercise B1.

Show that for  $\mathbf{H}_D = h\mathbf{1}\mathbf{1}^\top$  ( $d \times d$ ) with  $dh \neq 1$ ,  $(\mathbf{I} - h\mathbf{1}\mathbf{1}^\top)^{-1} = \mathbf{I} + h\mathbf{1}\mathbf{1}^\top/(1 - dh)$ , and use it to recompute the group term  $\hat{\mathbf{e}}_D^\top(\mathbf{I} - \mathbf{H}_D)^{-1}\hat{\mathbf{e}}_D$  in the proof of [Proposition 20.5.2\(e\)](#) from  $\hat{\mathbf{e}}_D = \delta\mathbf{1}/(1 + mh_{\star})$ .

#### Exercise B2.

In simple regression with clean mean  $\bar{x}_0$  and  $S_0 = \sum(x_j - \bar{x}_0)^2$ , show that  $h_{j\star} = 1/n_0 + (x_j - \bar{x}_0)(x_{\star} - \bar{x}_0)/S_0$ . In [Example \(Four identical bad records\)](#), with  $x_{\star}$  far to the right and  $\delta < 0$ , which clean residuals does [Proposition 20.5.2\(e\)](#) push up, and which down? Explain why case 30 was flagged.

*Solution.* Use the centred rows  $(1, x - \bar{x}_0)$ , which span the same column space; then  $\mathbf{A}^{-1} = \text{diag}(1/n_0, 1/S_0)$ , so  $h_{j\star} = 1/n_0 + (x_j - \bar{x}_0)(x_{\star} - \bar{x}_0)/S_0$ . The shift in clean residual  $j$  is  $-\delta h_{j\star}/(1 + mh_{\star})$ , which has the sign of  $h_{j\star}$  because  $\delta < 0$ . Since  $x_{\star} - \bar{x}_0 > 0$ ,  $h_{j\star}$

grows with  $x_j$ : positive on the right of the clean data and negative on the far left (once  $(x_j - \bar{x}_0)(x_\star - \bar{x}_0)/S_0 < -1/n_0$ ). Residuals on the right are pushed up and those on the far left down. Case 30 is the clean case with the largest  $x$ , so it has the largest upward shift, and its studentized residual crosses two.

### Exercise B3.

In the setting of [Proposition 20.5.2](#), delete  $k < m$  of the cluster cases. Show that each deleted case has prediction residual  $\delta/(1 + (m - k)h_\star)$ . What does this say about searching for a cluster of unknown size by deleting sets of size  $k$ ?

#### 20.5.4.3 C. Going deeper

### Exercise C1.

Place two clusters of  $m$  identical bad cases at the two ends of a straight-line design, one above the clean line and one below. Describe, using [Theorem 20.5.1](#), what single-case deletion, deletion of one cluster, and deletion of both clusters report. Is it possible for each cluster to mask the other?

### Exercise C2.

For a set  $D$  of cases, define  $D_D = (\hat{\beta}_{(D)} - \hat{\beta})^\top \mathbf{X}^\top \mathbf{X} (\hat{\beta}_{(D)} - \hat{\beta}) / (p s^2)$ . Using [Theorem 20.5.1\(b\)](#), show that  $D_D = \hat{\epsilon}_D^\top (\mathbf{I} - \mathbf{H}_D)^{-1} \mathbf{H}_D (\mathbf{I} - \mathbf{H}_D)^{-1} \hat{\epsilon}_D / (p s^2)$ .

## 20.6 Robust fitting as a diagnostic

Least squares hides bad cases by accommodating them. A fit that refuses to accommodate them leaves them with large residuals, visible without any deletion. That is the diagnostic use of robust regression, and this section develops only as much theory as that use needs: does a fit the bad cases cannot capture tell a different story, and which cases account for the difference?

### 20.6.1 M-estimators

Least squares minimizes  $\sum_i \rho(y_i - \mathbf{x}_{(i)}^\top \mathbf{b})$  with  $\rho(u) = u^2/2$ , which lets one large residual outweigh many small ones. An M-estimator (Huber 1964) uses a  $\rho$  that grows more slowly.

#### Definition 20.6.1 — M-estimator of regression

Let  $\rho : \mathbb{R} \rightarrow [0, \infty)$  be even, nondecreasing on  $[0, \infty)$ , differentiable, with  $\rho(0) = 0$ , and put  $\psi = \rho'$  and  $w(u) = \psi(u)/u$  (with  $w(0) = \lim_{u \rightarrow 0} \psi(u)/u$ ). Given a scale  $\hat{\sigma} > 0$ , an

**M-estimate**  $\hat{\mathbf{b}}_M$  minimizes

$$Q(\mathbf{b}) = \sum_{i=1}^n \rho\left(\frac{y_i - \mathbf{x}_{(i)}^\top \mathbf{b}}{\hat{\sigma}}\right).$$

Two choices of  $\rho$  are standard.

- (a) **Huber:**  $\rho(u) = u^2/2$  for  $|u| \leq k$  and  $k|u| - k^2/2$  beyond, so  $\psi(u) = \max(-k, \min(k, u))$  and  $w(u) = \min(1, k/|u|)$ .
- (b) **Bisquare** (Tukey's biweight):  $\rho(u) = \frac{c^2}{6} [1 - (1 - u^2/c^2)^3]$  for  $|u| \leq c$  and  $c^2/6$  beyond, so  $\psi(u) = u(1 - u^2/c^2)^2$  and  $w(u) = (1 - u^2/c^2)^2$  for  $|u| < c$ , and  $\psi(u) = w(u) = 0$  beyond.

Setting the gradient of  $Q$  to zero gives the estimating equations

$$\sum_{i=1}^n \psi\left(\frac{y_i - \mathbf{x}_{(i)}^\top \mathbf{b}}{\hat{\sigma}}\right) \mathbf{x}_{(i)} = \mathbf{0} \iff \mathbf{X}^\top \mathbf{W}(\mathbf{b})(\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}, \quad (20.6.1)$$

where  $\mathbf{W}(\mathbf{b})$  is diagonal with entries  $w((y_i - \mathbf{x}_{(i)}^\top \mathbf{b})/\hat{\sigma})$ : the normal equations of weighted least squares (Section 6.9) with weights that depend on the fit. A residual beyond  $k\hat{\sigma}$  has its Huber influence capped; a residual beyond  $c\hat{\sigma}$  gets bisquare weight zero. The Huber  $\rho$  is convex; the bisquare  $\rho$  is not, and is called **redescending** because  $\psi$  falls back to zero.

A standard scale is  $\hat{\sigma} = \text{median}_i |y_i - \mathbf{x}_{(i)}^\top \mathbf{b}| / 0.6745$ , where  $0.6745 = \Phi^{-1}(0.75)$  makes it consistent for  $\sigma$  under centred normal errors; software re-estimates it at each step. The constants  $k = 1.345$  and  $c = 4.685$  make the location efficiency  $(E\psi'(Z))^2 / E\psi(Z)^2$ ,  $Z \sim N(0, 1)$ , equal to 0.950 for both (computed by numerical integration in the script). For convex  $\rho$  such as Huber's, with  $p$  fixed and under smoothness conditions on  $\psi$ , that this ratio governs the asymptotic variance of regression M-estimators when  $\max_i h_{ii} \rightarrow 0$  is a theorem of Huber (1973) that this book does not prove; redescending estimators such as the bisquare also need a consistent starting value, as in the MM-estimator of Yohai (1987).

## 20.6.2 Iteratively reweighted least squares

Eq. 20.6.1 suggests **iteratively reweighted least squares** (IRLS), studied for robust regression by Holland and Welsch (1977): from  $\mathbf{b}^{(t)}$ , compute  $w_i = w((y_i - \mathbf{x}_{(i)}^\top \mathbf{b}^{(t)})/\hat{\sigma})$  and let  $\mathbf{b}^{(t+1)}$  be the weighted least squares estimate; repeat. With the scale fixed, it never goes uphill.

### Proposition 20.6.1 — IRLS descends

Suppose  $\rho(u) = g(u^2)$ , where  $g : [0, \infty) \rightarrow [0, \infty)$  is concave, nondecreasing, and differentiable on  $(0, \infty)$  with a finite limit  $g'(0^+)$ . Then  $w(u) = 2g'(u^2) \geq 0$ . Hold  $\hat{\sigma}$  fixed, and suppose  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$  is nonsingular at every step. Then every IRLS step satisfies  $Q(\mathbf{b}^{(t+1)}) \leq Q(\mathbf{b}^{(t)})$ . The Huber and bisquare  $\rho$  satisfy the hypotheses.

*Proof.* Since  $\rho(u) = g(u^2)$ ,  $\psi(u) = 2ug'(u^2)$ , so  $w(u) = 2g'(u^2)$ , which is nonnegative because  $g$  is nondecreasing. A concave function lies below each of its tangent lines:  $g(v) \leq g(v_0) + g'(v_0)(v - v_0)$  for  $v \geq 0$ ,  $v_0 > 0$  (and for  $v_0 = 0$  with  $g'(0^+)$ , by continuity).

With  $v = u^2$  and  $v_0 = u_0^2$ ,

$$\rho(u) \leq \rho(u_0) + \frac{1}{2}w(u_0)(u^2 - u_0^2).$$

Write  $u_i(\mathbf{b}) = (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})/\hat{\sigma}$  and  $w_i = w(u_i(\mathbf{b}^{(t)}))$ . Summing over  $i$ ,

$$Q(\mathbf{b}) \leq G(\mathbf{b}) = Q(\mathbf{b}^{(t)}) + \frac{1}{2} \sum_i w_i(u_i(\mathbf{b})^2 - u_i(\mathbf{b}^{(t)})^2) \quad \text{for every } \mathbf{b},$$

with equality at  $\mathbf{b} = \mathbf{b}^{(t)}$ . Up to terms that do not involve  $\mathbf{b}$ ,  $G(\mathbf{b})$  is  $\frac{1}{2\hat{\sigma}^2} \sum_i w_i(y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$ , which the weighted least squares step minimizes. Hence  $Q(\mathbf{b}^{(t+1)}) \leq G(\mathbf{b}^{(t+1)}) \leq G(\mathbf{b}^{(t)}) = Q(\mathbf{b}^{(t)})$ .

For Huber,  $g(v) = v/2$  for  $v \leq k^2$  and  $k\sqrt{v} - k^2/2$  beyond, so  $g'(v) = \frac{1}{2}$  and then  $k/(2\sqrt{v})$ : continuous at  $v = k^2$ , nonincreasing, hence  $g$  is concave. For the bisquare,  $g(v) = \frac{c^2}{6}[1 - (1 - v/c^2)^3]$  for  $v \leq c^2$  and constant beyond, so  $g'(v) = \frac{1}{2}(1 - v/c^2)^2$ , falling to 0 at  $v = c^2$ , and 0 beyond: again nonincreasing and continuous.

■

This is majorization:  $Q$  is replaced by a quadratic that lies above it and touches it at the current point. For Huber's convex  $\rho$ , a fixed point of IRLS solves [Eq. 20.6.1](#) and is a global minimizer. For the bisquare,  $Q$  can have several local minima, and IRLS finds one near its start.

```
import numpy as np

def huber_weight(u, k=1.345):
    return np.minimum(1.0, k / np.maximum(np.abs(u), 1e-12))

def bisquare_weight(u, c=4.685):
    return np.where(np.abs(u) < c, (1 - (u / c) ** 2) ** 2, 0.0)

def mad_scale(r):
    return np.median(np.abs(r)) / 0.6745          # consistent for sigma at the
    normal

def irls(X, y, weight, beta=None, scale=None, tol=1e-10, max_iter=500):
    """M-estimate by iteratively reweighted least squares.

    Starts from least squares unless `beta` is given. If `scale` is None it is re-
    estimated
    from the current residuals at every step; otherwise it is held fixed."""
    if beta is None:
        beta, *_ = np.linalg.lstsq(X, y, rcond=None)
    for _ in range(max_iter):
        r = y - X @ beta
        s = mad_scale(r) if scale is None else scale
        w = weight(r / s)
        sw = np.sqrt(w)
        new, *_ = np.linalg.lstsq(X * sw[:, None], y * sw, rcond=None)
        if np.max(np.abs(new - beta)) < tol * (1 + np.max(np.abs(beta))):
            beta = new
            break
    beta = new
    r = y - X @ beta
    s = mad_scale(r) if scale is None else scale
```

```
return beta, weight(r / s), s
```

### 20.6.3 Breakdown

How many bad cases can an estimator tolerate? The finite-sample breakdown point of [Definition 19.4.1](#) counts the cases that must be replaced to make the estimate arbitrary, and least squares breaks down at a single case: changing  $y_i$  by  $\Delta$  moves  $\hat{\boldsymbol{\beta}}$  by  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_{(i)} \Delta$  ([Proposition 19.4.2](#)).

Provided no single row dominates the design, a Huber estimate cannot be broken by changing  $y_i$  alone, since  $\psi$  is bounded; how much a  $y$ -outlier can do depends on the leverages. But one case bad in both  $\mathbf{x}$  and  $y$ , a **bad leverage point**, breaks it: the fit turns towards the case, whose residual becomes moderate and whose large row then dominates the estimating equations ([Exercise \(C1\)](#)). This is masking again: M-estimates downweight by residual, and a high-leverage case controls its own residual. For regression M-estimates with monotone  $\psi$  the breakdown point is  $1/n$  (Maronna, Martin and Yohai 2006). Added to the thirty clean cases of [Example \(Four identical bad records\)](#), one case at  $x = 50$ ,  $y = 0$  turns the least squares slope from 0.514 to  $-0.025$ , and the Huber slope to  $-0.026$ .

### 20.6.4 High-breakdown fits

Rousseeuw (1984) proposed estimators that fit a majority of the data and ignore the rest.

#### Definition 20.6.2 — Least trimmed squares

Let  $q$  be an integer with  $n/2 \leq q \leq n$ . For  $\mathbf{b} \in \mathbb{R}^p$ , let  $r_{[1]}^2(\mathbf{b}) \leq \dots \leq r_{[n]}^2(\mathbf{b})$  be the ordered squared residuals  $(y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$ . The **least trimmed squares** (LTS) estimate minimizes

$$T(\mathbf{b}) = \sum_{k=1}^q r_{[k]}^2(\mathbf{b}),$$

the sum of the  $q$  smallest squared residuals. The **least median of squares** (LMS) estimate minimizes  $r_{[\lceil n/2 \rceil]}^2(\mathbf{b})$  instead.

With  $q = \lfloor (n + p + 1)/2 \rfloor$  the breakdown point of LTS is close to  $1/2$  (see Rousseeuw and Leroy 1987): the fit follows the best-fitting half of the data. Computing it is combinatorial, since  $\min T$  is the minimum over  $q$ -subsets of their least squares criteria ([Exercise \(B1\)](#)). The practical algorithm rests on the **concentration step** of Rousseeuw and Van Driessen (2006).

#### Proposition 20.6.2 — Concentration step

Let  $\mathbf{b} \in \mathbb{R}^p$ , let  $H$  be a set of  $q$  cases with the smallest squared residuals at  $\mathbf{b}$ , and suppose the rows of  $\mathbf{X}$  in  $H$  have full column rank. Let  $\mathbf{b}'$  be the least squares estimate computed from the cases in  $H$ . Then  $T(\mathbf{b}') \leq T(\mathbf{b})$ .

*Proof.* By the choice of  $H$ ,  $T(\mathbf{b}) = \sum_{i \in H} (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$ , and since  $\mathbf{b}'$  minimizes the sum of squares over  $H$ , this is at least  $\sum_{i \in H} (y_i - \mathbf{x}_{(i)}^\top \mathbf{b}')^2$ . Any  $q$  squared residuals at  $\mathbf{b}'$  sum to at least the  $q$  smallest,  $T(\mathbf{b}')$ . ■

Repeated concentration steps give a nonincreasing sequence of values of  $T$ , constant after finitely many steps since there are finitely many  $q$ -subsets. The algorithm starts from many random **elemental** fits through  $p$  cases and keeps the best result. It needs only one start whose  $p$  cases are clean: with a fraction  $\epsilon$  of bad cases,  $N$  starts all fail with probability about  $(1 - (1 - \epsilon)^p)^N$ , and with  $\epsilon = 0.2$ ,  $p = 4$ ,  $N = 9$  starts push this below 1% ([Exercise \(B2\)](#)). LTS with  $q \approx n/2$  is inefficient; the MM-estimator of Yohai (1987) uses a high-breakdown fit only to start a redescending M-estimate with a fixed robust scale, keeping the high breakdown point and recovering efficiency. A simplified version, bisquare IRLS from LTS with the scale fixed at the median absolute LTS residual over 0.6745, is used below.

```
def lts(X, y, q, n_starts=500, seed=0):
    """Least trimmed squares: minimize the sum of the q smallest squared residuals.

    Random elemental starts (p cases), each followed by concentration steps: fit
    least
    squares to the current q cases, then keep the q cases with the smallest
    residuals."""
    rng = np.random.default_rng(seed)
    n, p = X.shape
    best, best_obj = None, np.inf
    for _ in range(n_starts):
        idx = rng.choice(n, size=p, replace=False)
        if np.linalg.matrix_rank(X[idx]) < p:
            continue
        beta = np.linalg.solve(X[idx], y[idx])
        for _ in range(100): # concentration steps
            keep = np.argsort((y - X @ beta) ** 2)[:q]
            new, *_ = np.linalg.lstsq(X[keep], y[keep], rcond=None)
            if np.allclose(new, beta):
                break
            beta = new
        obj = np.sort((y - X @ beta) ** 2)[:q].sum()
        if obj < best_obj:
            best, best_obj = beta, obj
    return best, best_obj
```

### 20.6.5 Using a robust fit diagnostically

The procedure: (1) fit least squares and a high-breakdown or MM-type fit; (2) compare the coefficients, since a difference of more than a standard error means a minority of cases drives least squares; (3) standardize the robust residuals by the robust scale and list cases beyond about 2.5 or with small weights, plotting them against a robust leverage measure to separate outlying responses from bad leverage points (Rousseeuw and van Zomeren 1990); (4) examine the flagged set substantively, refit without it, and report both fits. [Theorem 20.5.1\(d\)](#) measures its incompatibility with the rest, with an optimistic  $p$ -value because the set was chosen from the data.



**Example. (Robust fits on the masking data)**

For the data of [Example \(Four identical bad records\)](#) ([Figure 20.6.1\(a\)](#)), least squares has slope 0.007, and the Huber fit 0.005, with Huber weight 1.000 for each bad record: they have captured the fit so completely that their residuals are not even downweighted. The bisquare fit started from least squares is captured too (slope 0.005). LTS with  $q = 18$  of the 34 cases has slope 0.546 and robust scale 0.587; in that scale each bad record has standardized residual  $-18.6$ , and no clean case exceeds 2.81. The bisquare fit started from LTS has slope 0.509 and intercept 1.930, close to the clean fit (0.514 and 1.862), and gives the four records weight zero.

```
rng = np.random.default_rng(20)
n0, m = 30, 4
x_clean = np.sort(rng.uniform(0, 10, n0))
y_clean = 2 + 0.5 * x_clean + rng.normal(0, 0.5, n0)
x = np.r_[x_clean, np.full(m, 20.0)]
y = np.r_[y_clean, np.full(m, 2.0)]
X = np.column_stack([np.ones(len(x)), x])
n, p = X.shape

b_ls, *_ = np.linalg.lstsq(X, y, rcond=None)
b_hub, w_hub, _ = irls(X, y, huber_weight)
b_lts, _ = lts(X, y, q=(n + p + 1) // 2)
s_lts = mad_scale(y - X @ b_lts)
b_bis, w_bis, _ = irls(X, y, bisquare_weight, beta=b_lts, scale=s_lts)
for name, b in [("least squares", b_ls), ("Huber", b_hub), ("LTS", b_lts), ("
    bisquare from LTS", b_bis)]:
    print(f"{name:18s} intercept {b[0]:6.3f} slope {b[1]:6.3f}")
print("scaled LTS residuals of the cluster:", ((y - X @ b_lts) / s_lts)[-m:].round
    (1))
```

**Example. (Engel's budgets, robustly)**

For food expenditure on income in Engel's data ([Example \(Leverage in Engel's household budgets\)](#)), least squares gives slope 0.485, Huber 0.537 and bisquare 0.553. The bisquare fit gives weight zero to 3 households, rows 59, 105, 138 ([Figure 20.6.1\(b\)](#)). Row 138, the case of highest leverage (income 4958, food 1827, far below the line through the others, Huber weight 0.117), pulls the least squares slope down; in the least squares fit its externally studentized residual is  $-8.39$  and its Cook's distance 9.28.

But the spread of food expenditure grows with income. On the logarithmic scale the three fits nearly agree (slopes 0.856, 0.868, 0.871) and the smallest bisquare weight is 0.16: nobody is rejected. The robust fit correctly reported that a few cases disagree with a constant-variance line; whether the data or the scale is wrong no fit can decide ([Chapter 21](#), [Chapter 22](#)).

```
import statsmodels.api as sm
engel = sm.datasets.engel.load_pandas().data
inc = engel["income"].to_numpy()
food = engel["foodexp"].to_numpy()
XE = np.column_stack([np.ones(len(inc)), inc])
fits = {
    "least squares": sm.OLS(food, XE).fit(),
```

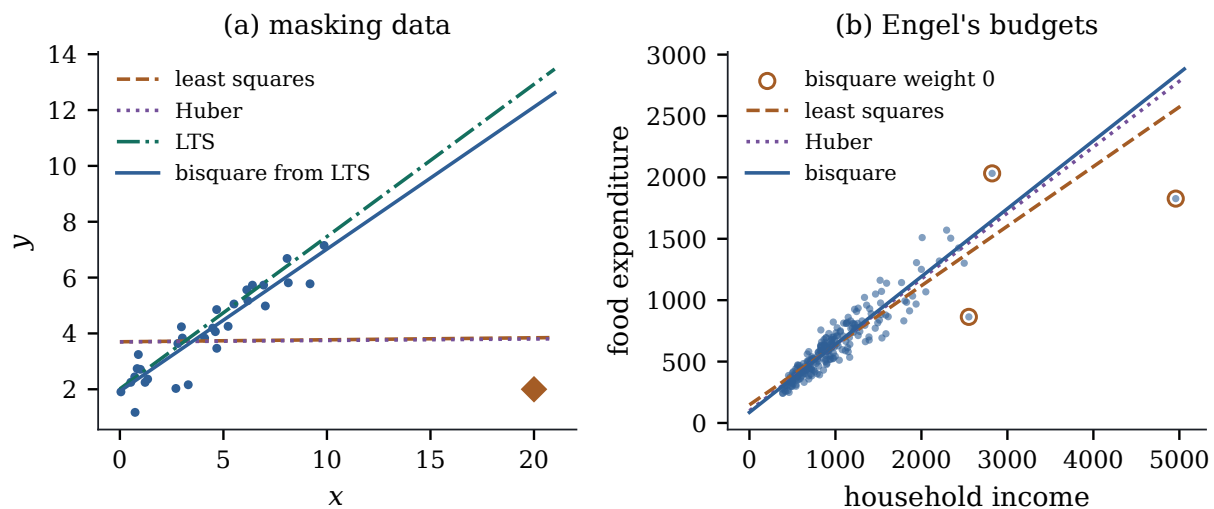


Figure 20.6.1: Robust fits. (a) The masking data: least squares and Huber are captured, LTS and bisquare from LTS are not. (b) Engel's budgets; circled points have bisquare weight zero.

```

"Huber": sm.RLM(food, XE, M=sm.robust.norms.HuberT()).fit(),
"bisquare": sm.RLM(food, XE, M=sm.robust.norms.TukeyBiweight()).fit(),
}
for name, f in fits.items():
    print(f"{name:14s} slope {f.params[1]:.4f}")
w = fits["bisquare"].weights
print("households with bisquare weight 0 (row numbers from 1):", np.where(w == 0)[0]
      + 1)

```

*Warning.* Deleting flagged cases and then reporting ordinary standard errors treats a data-driven choice as fixed in advance, and the intervals are too short. Report both analyses; [Chapter 23](#) discusses resampling that can include the selection step.

## 20.6.6 Exercises

### 20.6.6.1 A. Check your understanding

#### Exercise A1.

With  $k = 1.345$ , compute the Huber weights of standardized residuals 0.5, 2 and 10. With  $c = 4.685$ , compute the bisquare weights of the same residuals.

#### Exercise A2.

Show that least squares is the M-estimator with  $\rho(u) = u^2/2$ , that its weights are all one, and that IRLS started anywhere reaches it in one step.

### 20.6.6.2 B. Practice

**Exercise B1.**

Show that  $\min_{\mathbf{b}} T(\mathbf{b}) = \min_H \text{SSE}_H$ , where  $H$  ranges over the  $q$ -subsets of cases for which the rows of  $\mathbf{X}$  in  $H$  have full column rank and  $\text{SSE}_H$  is the residual sum of squares of least squares on  $H$  (assume at least one such subset exists). Deduce that an LTS estimate is the least squares estimate on some  $q$ -subset.

*Solution.* For any  $\mathbf{b}$  and any  $q$ -subset  $H$ ,  $T(\mathbf{b}) \leq \sum_{i \in H} (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$ , with equality when  $H$  consists of cases with the  $q$  smallest squared residuals. So  $T(\mathbf{b}) = \min_H \sum_{i \in H} (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$ , and  $\min_{\mathbf{b}} T(\mathbf{b}) = \min_H \min_{\mathbf{b}} \sum_{i \in H} (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2$ , the order of minimization being interchangeable. For a subset of full rank the inner minimum is  $\text{SSE}_H$ , attained at the least squares estimate on  $H$ . (Subsets without full rank can be handled by a limiting argument or excluded, as the statement assumes.) An LTS estimate is therefore the least squares estimate on a minimizing  $H$ .

**Exercise B2.**

Find the smallest number  $N$  of elemental starts for which  $(1 - (1 - \epsilon)^p)^N \leq 0.01$  when  $\epsilon = 0.2$  and  $p = 4$ , and when  $\epsilon = 0.4$  and  $p = 10$ . What does the second answer say about high-breakdown fitting with many regressors?

**Exercise B3.**

For the state murder-rate regression with all 51 jurisdictions, fit Huber and bisquare M-estimates with statsmodels and list the jurisdictions with the smallest weights. Compare the poverty coefficient with the least squares values with and without the District of Columbia ([Example \(How much does the District of Columbia matter?\)](#)).

**20.6.6.3 C. Going deeper****Exercise C1.**

Consider regression through the origin,  $E(Y) = \beta x$ , and an M-estimate with fixed scale  $\hat{\sigma} = 1$  and a continuous, nondecreasing, bounded, odd  $\psi$  with  $\psi(u) > 0$  for  $u > 0$ . Add to fixed data  $(x_i, y_i)$ ,  $i < n$ , one case  $(x_n, y_n) = (L, aL)$ . Show that as  $L \rightarrow \infty$  every solution  $\hat{\beta}_L$  of  $\sum_i \psi(y_i - \hat{\beta}_L x_i) x_i = 0$  tends to  $a$ . Conclude that the breakdown point is  $1/n$ .

*Solution.* Let  $B = \sup |\psi|$  and  $S = \sum_{i < n} |x_i|$ , so the first  $n - 1$  terms contribute at most  $BS$  in absolute value, whatever  $\hat{\beta}$ . Fix  $\eta > 0$ . If  $\hat{\beta} \leq a - \eta$ , the last term is  $\psi(L(a - \hat{\beta}))L \geq \psi(L\eta)L$ , which exceeds  $BS$  once  $L$  is large, because  $\psi(L\eta) \rightarrow \sup \psi > 0$  (as  $\psi$  is nondecreasing and positive on  $(0, \infty)$ ). Then the sum is positive and  $\hat{\beta}$  is not a solution. Symmetrically for  $\hat{\beta} \geq a + \eta$ . So for large  $L$  every solution lies in  $(a - \eta, a + \eta)$ . Since  $a$  is arbitrary, one case can place the estimate anywhere, and the breakdown point is  $1/n$ .

## 20.7 Summary and Notes

### 20.7.1 Summary

*Idea.*

1. Residuals are the projection of the errors onto  $\mathcal{C}(\mathbf{X})^\perp$ , with variances  $\sigma^2(1 - h_{ii})$ . The internally studentized residual  $r_i$  has a scaled beta law and is bounded by  $\sqrt{n-p}$ ; the externally studentized residual  $t_i$  uses a scale estimated without case  $i$  (Definition 20.1.1 and Proposition 20.1.1).
2. Residual plots show only the part of a misspecified mean that is orthogonal to  $\mathcal{C}(\mathbf{X})$ : they can reveal wrong fitted values, not biased coefficients. The added-variable and component-plus-residual plots share their line and their residuals and differ in the horizontal axis (Proposition 20.1.2 and Proposition 20.1.3).
3. Leverage is potential influence. Under a Gaussian design  $h_{ii} - 1/n$  is a scaled beta variable, so the  $2p/n$  rule fires at a rate that depends on  $p$ . With finite second moments the largest leverage vanishes as  $n$  grows; with heavy-tailed regressors it need not (Proposition 20.2.1 and Proposition 20.2.2).
4. Partial leverage isolates the pull of a case on one coefficient, and  $h_{ii}$  is the leverage without  $\mathbf{x}_j$  plus the partial leverage for  $\mathbf{x}_j$  (Definition 20.2.1 and Proposition 20.2.3).
5. Deleting one case changes the coefficients, fitted values and residual variance by explicit formulas in  $\hat{\varepsilon}_i$  and  $h_{ii}$ , and PRESS overestimates fixed-design prediction error slightly (Theorem 20.3.1 and Proposition 20.3.2).
6.  $t_i$  is the  $t$  statistic of the mean-shift outlier model and has the  $t(n-p-1, \Delta\sqrt{1-h_{ii}}/\sigma)$  law; the maximum over cases, with a Bonferroni correction, is an outlier test of level at most  $\alpha$  (Theorem 20.3.3).
7. Cook's distance, DFFITS and DFBETAS are products of a residual and a leverage factor. DFFITS is the largest standardized change in any linear combination of the coefficients. Their conventional cut-offs are scaled conventions, not tests (Definition 20.4.1, Theorem 20.4.1 and Proposition 20.4.2).
8. A set of cases can be deleted exactly, and the mean-shift  $F$  test checks a prespecified set (Theorem 20.5.1).
9. A cluster of  $m$  similar bad cases masks itself: each has leverage below  $1/m$ , its residual shrinks by  $1 + mh_*$ , single deletion recovers a fraction  $1/[m(1 + (m-1)h_*)]$  of the group effect, and clean cases are swamped (Proposition 20.5.2).
10. Huber and bisquare M-estimates are computed by IRLS, which never increases the objective with a fixed scale. They resist outlying responses at cases of moderate leverage, but not bad leverage points. High-breakdown fits such as LTS, computed by concentration steps, expose clusters that least squares and M-estimates absorb (Proposition 20.6.1, Proposition 19.4.2 and Proposition 20.6.2).

### 20.7.2 Notes and sources

**Sources.** Seber and Lee (2003, sections 10.2 and 10.6) derive the studentized residuals and their laws, the leave-one-out diagnostics including COVRATIO and the Andrews–Pregibon statistic,

the mean-shift test for a set of cases, and brief accounts of masking, swamping and robust residuals. Christensen (2020, sections 12.1 and 12.5–12.6) develops leverage from the projection  $\mathbf{M}$  and derives the updating formulae and predicted residuals. Sen and Srivastava (1990, chapter 8) and Rencher and Schaalje (2008, chapter 9) are briefer. The proofs here take the book's own route: the mean-shift model and the Frisch–Waugh–Lovell theorem for the law of  $t_i$  and for group deletion, orthogonal projections for the law of  $r_i$ , and partial leverage for DFBETAS.

**Residuals and influence.** Hoaglin and Welsch (1978) proposed the  $2p/n$  rule. The  $t$  law of the externally studentized residual is due to Beckman and Trussell (1974). Cook (1977) introduced his distance and its ellipsoid reading, Belsley, Kuh and Welsch (1980) DFFITS, DFBETAS and COVRATIO with their size-adjusted cut-offs, and Andrews and Pregibon (1978) a determinant-based measure for sets of cases. Book-length treatments are Cook and Weisberg (1982), Atkinson (1985) and Chatterjee and Hadi (1988); the review of Beckman and Cook (1983) surveys outlier testing, including the Bonferroni approach. PRESS is from Allen (1974), and partial residual plots were popularized by Larsen and McCleary (1972). [Proposition 20.2.1](#) is the regression form of the classical beta law for the Mahalanobis distance of a normal observation from the sample mean. Kiefer and Wolfowitz (1960) founded optimal design, which this book does not develop.

**Masking and robust fitting.** Beckman and Cook (1983) discuss masking and swamping at length; [Proposition 20.5.2](#) is a formal version for the simplest configuration. Hadi and Simonoff (1993), Atkinson and Riani (2000) and Rousseeuw and van Zomeren (1990) give the procedures mentioned in [Section 20.5](#). M-estimation began with Huber (1964); Huber (1973) treated regression, under the leverage condition of [Proposition 20.2.2](#). The bisquare is from Beaton and Tukey (1974), IRLS for robust regression was studied by Holland and Welsch (1977), and the finite-sample breakdown point is from Donoho and Huber (1983). LMS and LTS are from Rousseeuw (1984), with Rousseeuw and Leroy (1987) the standard account and Rousseeuw and Van Driessen (2006) the concentration step. The MM-estimator is Yohai's (1987). Maronna, Martin and Yohai (2006) give the theory; median and quantile regression are the subject of Chapter 45.

**Data.** Engel's budgets were made widely available by Koenker and Bassett (1982) and are distributed with statsmodels; the state data are those of [Chapter 6](#).

### 20.7.3 References

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# Chapter 21

## Non-Normality, Heteroscedasticity and Serial Correlation

The exact theory of Part III rests on three assumptions about the errors: they are normal, they have a common variance, and they are uncorrelated. [Chapter 19](#) worked out what least squares does when each of them fails. This chapter is about the data analyst's side of the same question: how to see a failure in the residuals, how large it must be before it matters, and what to do about it. The three assumptions do not matter equally. Coefficient inference survives non-normal errors in moderately large samples, but prediction intervals do not. Unequal variances leave least squares unbiased but make its usual standard errors wrong, sometimes by a factor of four or more, and they can be repaired either by weighting or by a variance estimator that does not trust the model. Serial correlation does the same damage in time-ordered data, and it often signals a mean function that is missing its dynamics. Every diagnostic in the chapter is computed from residuals, which are not the errors. How they differ is where the chapter starts.

**What you need.** Residuals, leverage and generalized least squares from [Chapter 6](#) ([Proposition 6.4.3](#), [Proposition 6.8.1](#), [Definition 6.9.1](#), [Theorem 6.9.1](#)), the Gauss–Markov and Aitken theorems and the normal sampling theory of [Chapter 7](#) ([Theorem 7.2.1](#), [Corollary 7.2.3](#), [Theorem 7.5.1](#)), the moments of quadratic forms from [Chapter 2](#) ([Theorem 2.3.1](#), [Theorem 2.3.3](#)) and their normal distribution theory from [Chapter 4](#) ([Theorem 4.2.1](#), [Corollary 4.3.3](#)), the deletion formulas of [Chapter 10](#) ([Proposition 10.6.2](#)), and prediction intervals from [Chapter 12](#) ([Theorem 12.3.1](#)) and their large-sample coverage under non-normal errors ([Proposition 12.3.3](#)). From [Chapter 19](#) we use the covariance of least squares under a general error covariance ([Theorem 19.2.2](#)) and its large-sample behaviour under non-normal errors ([Theorem 19.3.2](#)). From [Chapter 20](#) we use studentized residuals ([Definition 20.1.1](#), [Theorem 20.3.3](#)). The limit theorems of probability that the chapter needs are gathered, or pointed to, in [Section 21.2](#).

### 21.0.1 Roadmap

- [Checking normality](#)
- [Detecting heteroscedasticity](#)
- [Weighted least squares](#)
- [Heteroscedasticity-consistent covariance estimators](#)
- [Serial correlation and the Durbin–Watson test](#)
- [Autoregressive errors](#)



- [Summary and notes](#)

## 21.1 Checking normality

Normality turns the first two moments of least squares into exact distributions. How much of Part III needs it? [Chapter 19](#) answered the question for coefficients. When no observation carries much leverage,  $\hat{\beta}$  is approximately normal whatever the error law, and the  $t$  and  $F$  tests keep their nominal levels in large samples ([Theorem 19.3.2](#)). Two kinds of inference are not protected in this way. The interval for  $\sigma^2$  of [Section 12.1](#) depends on the fourth moment of the errors ([Proposition 7.5.4](#)). A prediction interval is a statement about a single new error, and no averaging makes a single error normal. So the error distribution is worth checking mainly when the analysis will predict individual responses, when the sample is small, or when a few observations may be of a different kind from the rest.

The errors are never observed, so the first task is to say how well the residuals stand in for them.

### 21.1.1 Residuals are not the errors

If  $\mathbf{Y} = \mathbf{X}\beta + \boldsymbol{\varepsilon}$  and  $\mathbf{M}$  is the hat matrix, the residual vector is  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$  ([Proposition 6.4.3](#)). Write  $m_{ij}$  for the entries of  $\mathbf{M}$ , so that  $h_{ii} = m_{ii}$  are the leverages, and  $c_{ij}$  for the entries of  $\mathbf{I} - \mathbf{M}$ . Each residual is a weighted sum of all  $n$  errors,

$$\hat{\varepsilon}_i = (1 - h_{ii})\varepsilon_i - \sum_{j \neq i} m_{ij}\varepsilon_j,$$

so even when the errors are independent and identically distributed the residuals are neither. They are correlated, and their variances  $\sigma^2(1 - h_{ii})$  differ.

#### Proposition 21.1.1 — What residuals reveal about the errors

Let  $\varepsilon_1, \dots, \varepsilon_n$  be independent and identically distributed with mean 0 and variance  $\sigma^2$ , and let  $\text{rank}(\mathbf{X}) = p$ .

- $\hat{\varepsilon}_i - \varepsilon_i = -(\mathbf{M}\boldsymbol{\varepsilon})_i$ , which has mean 0 and variance  $\sigma^2 h_{ii}$ . The total  $E\|\hat{\boldsymbol{\varepsilon}} - \boldsymbol{\varepsilon}\|^2$  equals  $\sigma^2 p$ .
- Let  $\hat{\varepsilon}_{(1)} \leq \dots \leq \hat{\varepsilon}_{(n)}$  and  $\varepsilon_{(1)} \leq \dots \leq \varepsilon_{(n)}$  be the ordered residuals and ordered errors. Then

$$\frac{1}{n} \sum_{i=1}^n (\hat{\varepsilon}_{(i)} - \varepsilon_{(i)})^2 \leq \frac{1}{n} \|\mathbf{M}\boldsymbol{\varepsilon}\|^2,$$

and the right-hand side has expectation  $\sigma^2 p/n$ .

- Suppose the errors have skewness  $\gamma_1 = E\varepsilon^3/\sigma^3$  and excess kurtosis  $\gamma_2 = E\varepsilon^4/\sigma^4 - 3$ . If  $h_{ii} < 1$ , the standardized residual  $\hat{\varepsilon}_i/(\sigma\sqrt{1 - h_{ii}})$  has skewness  $a_i\gamma_1$  and excess kurtosis  $b_i\gamma_2$ , where

$$a_i = \frac{\sum_j c_{ij}^3}{(1 - h_{ii})^{3/2}}, \quad b_i = \frac{\sum_j c_{ij}^4}{(1 - h_{ii})^2},$$



with  $|a_i| \leq 1$  and  $(1 - h_{ii})^2 \leq b_i \leq 1$ .

*Proof.*

- (a)  $\hat{\boldsymbol{\varepsilon}} - \boldsymbol{\varepsilon} = -\mathbf{M}\boldsymbol{\varepsilon}$ , with mean zero and covariance  $\sigma^2\mathbf{M}\mathbf{M}^\top = \sigma^2\mathbf{M}$  (Theorem 2.2.1), whose diagonal is  $\sigma^2 h_{ii}$  and whose trace is  $\sigma^2 p$ .
- (b) Let  $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ . Pairing the sorted values gives  $\sum_i u_{(i)} v_{(i)} \geq \sum_i u_i v_{\pi(i)}$  for every permutation  $\pi$ : if a pairing matches  $u_{(i)} < u_{(j)}$  with  $v$ -values  $b > b'$  respectively, exchanging the two partners changes the sum by  $(u_{(j)} - u_{(i)})(b - b') \geq 0$ , and finitely many such exchanges reach the sorted pairing. Since sorting does not change  $\|\mathbf{u}\|$  or  $\|\mathbf{v}\|$ , it follows that  $\sum_i (u_{(i)} - v_{(i)})^2 \leq \|\mathbf{u} - \mathbf{v}\|^2$ . Take  $\mathbf{u} = \hat{\boldsymbol{\varepsilon}}$  and  $\mathbf{v} = \boldsymbol{\varepsilon}$  and use (a).
- (c) Row  $i$  of  $\mathbf{I} - \mathbf{M}$  has  $\sum_j c_{ij}^2 = (\mathbf{I} - \mathbf{M})_{ii} = 1 - h_{ii}$  because  $\mathbf{I} - \mathbf{M}$  is symmetric and idempotent. Expand  $\hat{\varepsilon}_i = \sum_j c_{ij} \varepsilon_j$ . By independence and  $E \varepsilon_j = 0$ , every cross term with an index appearing once vanishes, so  $E \hat{\varepsilon}_i^3 = E \varepsilon^3 \sum_j c_{ij}^3$  and

$$\begin{aligned} E \hat{\varepsilon}_i^4 &= E \varepsilon^4 \sum_j c_{ij}^4 + 3\sigma^4 \sum_{j \neq k} c_{ij}^2 c_{ik}^2 \\ &= (E \varepsilon^4 - 3\sigma^4) \sum_j c_{ij}^4 + 3\sigma^4 (1 - h_{ii})^2. \end{aligned}$$

Dividing by  $\sigma^3(1 - h_{ii})^{3/2}$  and  $\sigma^4(1 - h_{ii})^2$  gives the skewness and the excess kurtosis. For the bounds,  $\max_j c_{ij}^2 \leq \sum_j c_{ij}^2 = 1 - h_{ii}$ , so  $|\sum_j c_{ij}^3| \leq (1 - h_{ii})^{1/2}(1 - h_{ii})$  and  $\sum_j c_{ij}^4 \leq (1 - h_{ii})^2$ , while  $\sum_j c_{ij}^4 \geq c_{ii}^4 = (1 - h_{ii})^4$ . ■

Parts (a) and (b) say that the residuals are good proxies for the errors when  $p/n$  is small. Part (b) is the statement that matters for plots, because a normal probability plot shows sorted values: on average the sorted residuals lie within  $\sigma\sqrt{p/n}$  of the sorted errors, in root mean square. Part (c) describes what happens when  $p/n$  is not small. Each residual is an average of many errors, and averaging pushes a distribution towards normality, so residuals look *more* normal than the errors that produced them. This is sometimes called **supernormality**. The factor  $b_i$  measures it exactly.

#### Example. (How much non-normality survives)

Take  $n = 30$  observations, an intercept, and  $p - 1$  columns of standard normal regressors. Averaged over the observations, the share of the errors' excess kurtosis that survives in the standardized residuals is 0.873 for  $p = 2$ , 0.467 for  $p = 10$  and 0.162 for  $p = 20$ . The share of skewness is 0.898, 0.532 and 0.162. With centred exponential errors (skewness 2, excess kurtosis 6) and  $p = 10$ , the residual with the largest leverage,  $h_{ii} = 0.57$ , has predicted excess kurtosis 1.23, against 1.26 in 200,000 simulated data sets. For the ordered values, the average squared distance between sorted residuals and sorted errors was 0.133, while the bound in (b) averaged 0.330, close to its expectation  $p/n = 1/3$ .

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(2101)
```

```

n = 30
designs = {}
for p in (2, 10, 20):
    X = np.column_stack([np.ones(n), rng.normal(size=(n, p - 1))])
    Q, _ = np.linalg.qr(X)
    M = Q @ Q.T                                # the hat matrix
    designs[p] = (M, np.diag(M).copy())

for p, (M, h) in designs.items():
    # residual i = (row i of C) @ errors
    C = np.eye(n) - M
    # excess kurtosis of residual i / that of an error
    kurt_kept = (C**4).sum(axis=1) / (1 - h) ** 2
    # skewness of residual i / that of an error
    skew_kept = (C**3).sum(axis=1) / (1 - h) ** 1.5
    print(f"p = {p:2d}: max leverage {h.max():.2f}, "
          f"average share of skewness kept {skew_kept.mean():.3f}, "
          f"of excess kurtosis kept {kurt_kept.mean():.3f}")

```

## 21.1.2 Normal probability plots

A **normal probability plot** graphs the sorted values of a sample against the corresponding quantiles of the standard normal distribution. If  $U_1, \dots, U_n$  are independent  $N(\mu, \tau^2)$ , the  $i$ th smallest is close to  $\mu + \tau q_i$ , where  $q_i$  approximates the expected  $i$ th order statistic of a standard normal sample. A common choice, due to Blom (1958), is

$$q_i = \Phi^{-1}\left(\frac{i - 3/8}{n + 1/4}\right).$$

Normal data give points near a straight line with intercept  $\mu$  and slope  $\tau$ . Heavy tails bend both ends away from the line, into an S shape. Skewness bends the plot into a curve that is convex for a long right tail, and isolated points at either end suggest outliers.

For residuals, the unequal variances  $\sigma^2(1 - h_{ii})$  should be removed first. The natural choice is the externally studentized residual  $t_i$ , which divides  $\hat{\varepsilon}_i$  by  $s_{(i)}\sqrt{1 - h_{ii}}$ , where  $s_{(i)}^2$  is the error variance estimate without observation  $i$  ([Definition 20.1.1](#)). Under normal errors every  $t_i$  has the  $t(n - p - 1)$  distribution ([Theorem 20.3.3](#)), so the  $t_i$  share a common marginal law and their sorted values can be compared with one reference. They are not independent, and that makes it hard to judge by eye how much wiggle is normal.

A simulated **envelope** answers that. Under normal errors the vector  $(t_1, \dots, t_n)$  is a function of  $\hat{\varepsilon}/\|\hat{\varepsilon}\| = (\mathbf{I} - \mathbf{M})\varepsilon/\|(\mathbf{I} - \mathbf{M})\varepsilon\|$  alone, so its joint distribution depends on the design but not on  $\beta$  or  $\sigma$  ([Exercise \(B1\)](#)). We can therefore simulate it exactly: generate normal errors, compute the sorted studentized residuals, repeat, and plot pointwise percentiles (Atkinson 1981). The envelope is pointwise, so a few points outside it are expected even for normal data. A run of consecutive points outside it, or an end of the plot leaving it, is the signal.

### Example. (Three error laws)

[Figure 21.1.1](#) shows plots for one simulated data set of  $n = 60$  observations with an intercept and two regressors ( $p = 3$ ), with three error laws of variance one. The normal case stays inside its envelope except for 3 scattered points, and the Shapiro–Wilk test described below gives  $p = 0.519$ . Errors from a  $t(3)$  distribution give the S shape of

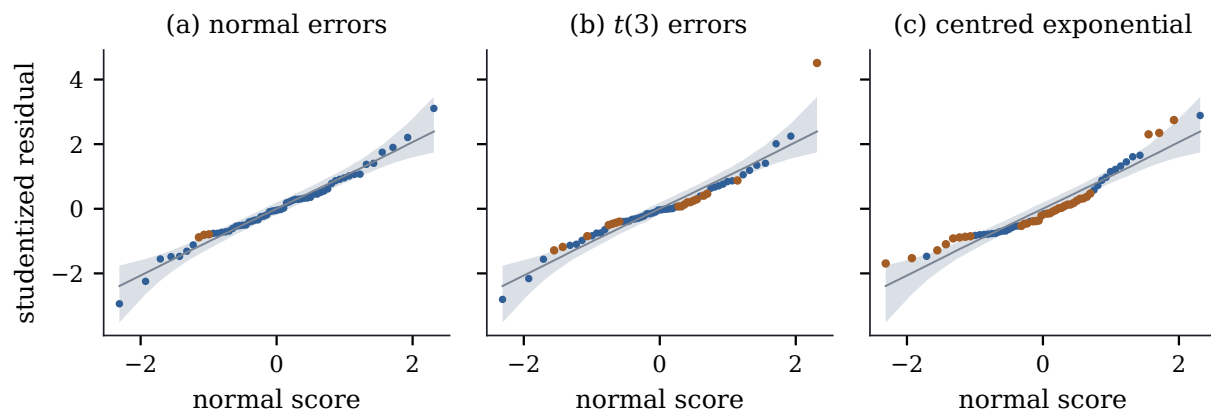


Figure 21.1.1: Normal probability plots of externally studentized residuals,  $n = 60$ ,  $p = 3$ , with pointwise 95% envelopes from 1999 data sets simulated with normal errors (shaded) and their pointwise median (line). Points outside the envelope are drawn in the second colour. (a) Normal errors. (b) Errors from  $t(3)$ . (c) Centred exponential errors.

heavy tails, with the largest residual far above the envelope (Shapiro–Wilk  $p < 0.001$ ). Centred exponential errors give a curved plot whose left end is too short and whose right end is too long ( $p = 0.001$ ).

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(2104)
n, p = 60, 3
X = np.column_stack([np.ones(n), rng.normal(size=(n, 2))])
Q, _ = np.linalg.qr(X)
h = np.sum(Q**2, axis=1) # leverages

def ext_studentized(y):
    """Externally studentized residuals  $t_i = e_i / (s_{-i} \sqrt{1 - h_{ii}})$ ."""
    e = y - Q @ (Q.T @ y)
    s2 = e @ e / (n - p)
    # squared internally studentized residuals
    r2 = e**2 / (s2 * (1 - h))
    return np.sign(e) * np.sqrt(r2 * (n - p - 1) / (n - p - r2))

# plotting positions
blom = stats.norm.ppf((np.arange(1, n + 1) - 0.375) / (n + 0.25))
sims = np.sort([ext_studentized(rng.normal(size=n)) for _ in range(1999)],
                axis=1)
# pointwise envelope
lower, upper = np.percentile(sims, [2.5, 97.5], axis=0)

laws = {
    "normal": rng.normal(size=n),
    "t(3)": rng.standard_t(3, size=n) / np.sqrt(3.0),
    "centred exponential": rng.exponential(size=n) - 1.0,
}
beta = np.array([1.0, 2.0, -1.0])
for name, eps in laws.items():
```

```
t = np.sort(ext_studentized(X @ beta + eps))
outside = np.sum((t < lower) | (t > upper))
print(f"{name:20s}: {outside} of {n} points outside the envelope, "
      f"Shapiro-Wilk p = {stats.shapiro(t).pvalue:.3f}")
```

### 21.1.3 Formal tests

Many tests of normality exist. Four kinds are in common use for residuals.

- The **Shapiro–Wilk** statistic (Shapiro and Wilk 1965) is  $W = (\sum_i a_i u_{(i)})^2 / \sum_i (u_i - \bar{u})^2$ . The weights  $a_i$  come from the means and the covariance matrix of standard normal order statistics, and they make the numerator the square of the best linear unbiased estimate of the standard deviation from the ordered sample, up to a constant. So  $W$  compares two estimates of scale, one that is efficient only under normality and one that is always valid. It lies in  $(0, 1]$ , and small values reject. Its null distribution has no closed form and is computed numerically.
- The **Shapiro–Francia** statistic (Shapiro and Francia 1972) replaces the weights by the plotting positions, which makes it the squared correlation between the sorted sample and the  $q_i$ . It is a one-number summary of how straight the normal probability plot is.
- **Moment tests** use the sample skewness  $\sqrt{b_1}$  and kurtosis  $b_2$ . The Jarque–Bera statistic  $n\{b_1/6 + (b_2 - 3)^2/24\}$  is approximately  $\chi^2(2)$  for large normal samples (Jarque and Bera 1980). These tests are directed at the two departures that matter most for inference.
- **Distance tests** compare the empirical and fitted normal distribution functions: Kolmogorov–Smirnov with estimated parameters (Lilliefors 1967) and Anderson–Darling (Anderson and Darling 1952). They are usually less powerful than Shapiro–Wilk.

Applied to residuals, none of these has its nominal null distribution, because the residuals are not a sample. The effect on the level is small. In the design of [Example \(How much non-normality survives\)](#) with  $p = 20$  and normal errors, the 5% Shapiro–Wilk test applied to the raw residuals rejected in 0.045 of 4000 simulated data sets, and applied to the internally studentized residuals in 0.035. The effect on power is not small. Supernormality hides the non-normality that the test is looking for.

#### Example. (Power lost to supernormality)

In the same designs ( $n = 30$ ), the 5% Shapiro–Wilk test applied to the unobservable centred exponential errors rejects in 0.96 to 0.97 of simulated data sets. Applied to the residuals it rejects in 0.92 for  $p = 2$ , 0.45 for  $p = 10$  and only 0.07 for  $p = 20$ . For errors from  $t(5)$ , rescaled to variance one, the rates are 0.24 to 0.25 for the errors and 0.23, 0.14 and 0.05 for the residuals.

```
def errors(kind, size, rng):
    """Mean-zero, variance-one errors."""
    if kind == "exp":
        return rng.exponential(size=size) - 1.0
    return rng.standard_t(5, size=size) / np.sqrt(5 / 3)

def power_table(reps, rng):
    """Rejection rates of the 5% Shapiro-Wilk test on errors and on residuals."""
    table = {}
```

```

for kind in ("exp", "t5"):
    for p, (M, h) in designs.items():
        E = errors(kind, (reps, n), rng)
        # residual vectors (M is symmetric)
        R = E - E @ M
        rej_err = np.mean([stats.shapiro(e).pvalue < 0.05 for e in E])
        rej_res = np.mean([stats.shapiro(r).pvalue < 0.05 for r in R])
        table[kind, p] = (rej_err, rej_res)
    print(f"{kind:3s} p = {p:2d}: Shapiro-Wilk rejects "
          f"errors {rej_err:.2f}, "
          f"residuals {rej_res:.2f}")
return table

# a quick version (the book uses 4000 replications)
quick = power_table(400, rng)

```

A test of normality also answers the wrong question. In small samples it passes error laws that would distort a prediction interval badly; in large ones it rejects departures too small to matter. The useful question is whether the departure is large enough to affect the inference at hand. For coefficient tests without high-leverage points it rarely is. For prediction intervals it can be, as the next result shows.

### 21.1.4 What non-normality does to prediction intervals

Under normal errors the interval  $\hat{Y}_0 \pm t_{n-p, \alpha/2} s \sqrt{1 + h_0}$  covers a new response  $Y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + \varepsilon_0$  with probability exactly  $1 - \alpha$ , where  $h_0 = \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0$  (Theorem 12.3.1). Without normality its coverage does not tend to  $1 - \alpha$  as the sample grows. By Proposition 12.3.3, when  $h_0 \rightarrow 0$  and the errors are independent and identically distributed with a continuous distribution function  $F$ , the coverage tends to  $\Pr(|\varepsilon_0| \leq z\sigma) = F(z\sigma) - F(-z\sigma)$ , where  $z = \Phi^{-1}(1 - \alpha/2)$ , and the probabilities of missing above and below tend to  $1 - F(z\sigma)$  and  $F(-z\sigma)$ .

The limit is the probability that a single error falls within  $z$  standard deviations of zero, and it depends on the whole shape of  $F$ . Section 12.3 evaluated it for centred exponential and uniform errors at the 80% and 95% levels. Table 21.1.1 extends that comparison to heavy-tailed laws and to the 99% level, and splits the misses between the two tails.

**Table 21.1.1.** Limiting coverage of the usual prediction interval (Proposition 12.3.3) when the errors are not normal, with the limiting probabilities of missing below and above in parentheses. All five laws have mean zero and variance one.

| error law           | nominal 80%          | nominal 95%          | nominal 99%          |
|---------------------|----------------------|----------------------|----------------------|
| normal              | 0.800 (0.100, 0.100) | 0.950 (0.025, 0.025) | 0.990 (0.005, 0.005) |
| $t(5)$              | 0.841 (0.079, 0.079) | 0.947 (0.026, 0.026) | 0.979 (0.010, 0.010) |
| $t(3)$              | 0.887 (0.057, 0.057) | 0.957 (0.021, 0.021) | 0.979 (0.010, 0.010) |
| uniform             | 0.740 (0.130, 0.130) | 1.000 (0.000, 0.000) | 1.000 (0.000, 0.000) |
| centred exponential | 0.898 (0.000, 0.102) | 0.948 (0.000, 0.052) | 0.972 (0.000, 0.028) |

Heavy tails make a 99% interval too short and an 80% interval too long. The 95% level lies near

the crossing point, so checking a 95% interval alone would miss the problem. For the skewed law every miss is above the interval, since a centred exponential error is never below  $-\sigma$ ; at the 99% level it misses above 0.028 of the time, more than five times the nominal 0.005 in that tail.

```
import numpy as np
from scipy import stats

laws = {
    "normal": stats.norm(),
    "t(5)": stats.t(5, scale=np.sqrt(3 / 5)),
    "t(3)": stats.t(3, scale=np.sqrt(1 / 3)),
    "uniform": stats.uniform(loc=-np.sqrt(3), scale=2 * np.sqrt(3)),
    "centred exponential": stats.expon(loc=-1.0),
}
for name, law in laws.items():
    assert abs(law.var() - 1) < 1e-9 and abs(law.mean()) < 1e-9
    row = []
    for level in (0.80, 0.95, 0.99):
        z = stats.norm.ppf(0.5 + level / 2)
        below, above = law.cdf(-z), law.sf(z)          # miss low, miss high
        row.append(f"{1 - below - above:.3f} ({below:.3f}, {above:.3f})")
    print(f"{name:20s}", " ".join(row))
```

The remedy is to use the residuals for what they are good at, estimating the error distribution: empirical quantiles of the residuals in place of  $\pm z\sigma$  give intervals whose coverage tends to the nominal level, and [Chapter 23](#) refines this by resampling. A transformation of the response ([Chapter 22](#)) can remove skewness at its source.

## 21.1.5 Exercises

### 21.1.5.1 A. Check your understanding

#### Exercise A1.

In simple regression with  $n = 5$  and  $x = (1, 2, 3, 4, 10)$ , compute the leverages and the variances of the five residuals when  $\sigma^2 = 1$ . Which residual is least like its error, in the sense of [Proposition 21.1.1\(a\)](#)?

### 21.1.5.2 B. Practice

#### Exercise B1.

Assume  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ . Show that the internally and externally studentized residuals are functions of  $\mathbf{g} = \hat{\boldsymbol{\varepsilon}}/\|\hat{\boldsymbol{\varepsilon}}\|$ , and that the law of  $\mathbf{g}$  does not depend on  $\boldsymbol{\beta}$  or  $\sigma$ . Conclude that the simulated envelope of this section is exact, whatever  $\boldsymbol{\beta}$  and  $\sigma$  are used in the simulation.

*Solution.*  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$ , and  $\boldsymbol{\varepsilon}/\sigma \sim N_n(\mathbf{0}, \mathbf{I})$  whatever  $\boldsymbol{\beta}$  and  $\sigma$  are, so  $\mathbf{g} = (\mathbf{I} - \mathbf{M})(\boldsymbol{\varepsilon}/\sigma)/\|(\mathbf{I} - \mathbf{M})(\boldsymbol{\varepsilon}/\sigma)\|$  has a fixed law. The internally studentized residual is  $r_i = \hat{\varepsilon}_i/(s\sqrt{1 - h_{ii}})$  with  $s = \|\hat{\boldsymbol{\varepsilon}}\|/\sqrt{n - p}$ , so  $r_i = \sqrt{n - p} g_i / \sqrt{1 - h_{ii}}$ . The externally studentized residual is the monotone function  $t_i = r_i \sqrt{(n - p - 1)/(n - p - r_i^2)}$  of  $r_i$  ([Def-](#)

inition 20.1.1). Hence the whole vector of sorted  $t_i$  has a law that depends only on  $\mathbf{M}$ , and it can be simulated with any  $\boldsymbol{\beta}$  and  $\sigma$ .

### Exercise B2.

For centred exponential errors,  $\varepsilon = E - 1$  with  $E$  standard exponential, show that the limit in Proposition 12.3.3 at the 95% level is  $1 - e^{-2.96}$ , and find the probabilities of missing below and above.

*Solution.* Here  $\sigma = 1$  and  $z = 1.96$ . The event  $|E - 1| \leq 1.96$  is  $E \leq 2.96$ , since  $E \geq 0 > -0.96$  always. Its probability is  $1 - e^{-2.96} = 1 - 0.0518 = 0.948$ . The interval never misses below, and it misses above with probability  $e^{-2.96} = 0.0518$ .

### Exercise B3.

In a one-way layout in which every group has  $m$  observations, show that the kurtosis factor of Proposition 21.1.1(c) is the same for every residual and equals

$$b = \frac{(1 - 1/m)^4 + (m - 1)/m^4}{(1 - 1/m)^2}.$$

Evaluate it for  $m = 2$  and  $m = 5$ , and explain why residual plots from small groups say little about the tails of the errors.

#### 21.1.5.3 C. Going deeper

### Exercise C1.

Under the assumptions of Proposition 21.1.1, suppose  $p/n \rightarrow 0$  along a sequence of designs. Let  $\hat{F}_n$  and  $F_n$  be the empirical distribution functions of the residuals and of the errors. Show that  $\hat{F}_n(x) - F_n(x) \rightarrow 0$  in probability at every continuity point  $x$  of  $F$ .

## 21.2 Detecting heteroscedasticity

The errors are **heteroscedastic** when their variances differ,  $\text{Var}(\varepsilon_i) = \sigma_i^2$ . Least squares remains unbiased, but its covariance becomes  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\Sigma} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$  with  $\boldsymbol{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$ , and  $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$  no longer estimates it (Theorem 19.2.2). Least squares also stops being the best linear unbiased estimator. How far the usual standard errors go wrong depends on how the variances line up with the regressors: the damage is greatest when the observations with the most influence on a coefficient also have the largest variances. Chapter 15 showed the same thing for the one-way layout, where unequal variances bias the  $F$  test only in unbalanced designs (Theorem 15.5.2).

Unequal variances have familiar sources. A response that is an average of  $m_i$  observations has variance proportional to  $1/m_i$ . Counts, sizes and amounts of money tend to vary more when they are larger. And if a coefficient itself varies from unit to unit,  $y_i = \mathbf{x}_{(i)}^\top (\boldsymbol{\beta} + \mathbf{u}_i)$  with random

$\mathbf{u}_i$ , the variance of  $y_i$  is a quadratic function of  $\mathbf{x}_{(i)}$  ([Exercise \(B1\)](#)).

## 21.2.1 Residual plots

Under heteroscedasticity the residuals  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$  have covariance  $(\mathbf{I} - \mathbf{M})\boldsymbol{\Sigma}(\mathbf{I} - \mathbf{M})$ , so

$$E \hat{\varepsilon}_i^2 = (1 - h_{ii})^2 \sigma_i^2 + \sum_{j \neq i} m_{ij}^2 \sigma_j^2. \quad (21.2.1)$$

When  $h_{ii}$  is small the first term dominates, and the squared residual tracks the variance of its own observation. At high leverage it does not: a point with  $h_{ii}$  near one has a small residual whatever its variance. Even with constant variance the residuals have unequal variances  $\sigma^2(1 - h_{ii})$ , so plots for heteroscedasticity should use the studentized residuals  $r_i = \hat{\varepsilon}_i / (s\sqrt{1 - h_{ii}})$ , which have constant variance under the model.

The standard displays are:

- $r_i$  against the fitted values. A fan that opens to the right says that variance grows with the mean.
- $r_i$  against each regressor, and against variables outside the model that might affect precision, such as time, the instrument or the interviewer.
- $|r_i|$ , or  $\log|r_i|$ , against the same variables. Absolute values fold the two halves of the fan onto each other and make a trend in spread easier to see.

The logarithmic plot also estimates a power law. If  $\sigma_i = \sigma v_i^\theta$  for a positive variable  $v_i$ , then  $\varepsilon_i = \sigma v_i^\theta \zeta_i$  with  $\zeta_i$  of constant law, and  $\log|\varepsilon_i| = \log \sigma + \theta \log v_i + \log|\zeta_i|$ . So the slope of  $\log|r_i|$  on  $\log v_i$  estimates  $\theta$ , with the residuals standing in for the errors.

### Example. (Engel's food expenditure data)

[Section 6.9](#) fitted Engel's data on the annual income and food expenditure of 235 Belgian households, and noted that the spread of expenditure grows with income. The least squares slope is 0.4852. [Figure 21.2.1](#) shows the residual plots. Panel (a) is a fan. Panel (b) plots  $|r_i|$  against income on logarithmic scales. The slope of the fitted line, 0.96, suggests a standard deviation roughly proportional to income, the weighting used in [Example \(Engel's food expenditure data\)](#). The most extreme point, at the far right of panel (a), is the household with the largest income. Its leverage is 0.255, about thirty times the average  $p/n$ , and its residual is large and negative.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)
X = np.column_stack([np.ones(n), income])
beta, *_ = np.linalg.lstsq(X, food, rcond=None)
fitted = X @ beta
e = food - fitted
Q, _ = np.linalg.qr(X)

# OLS residuals
```



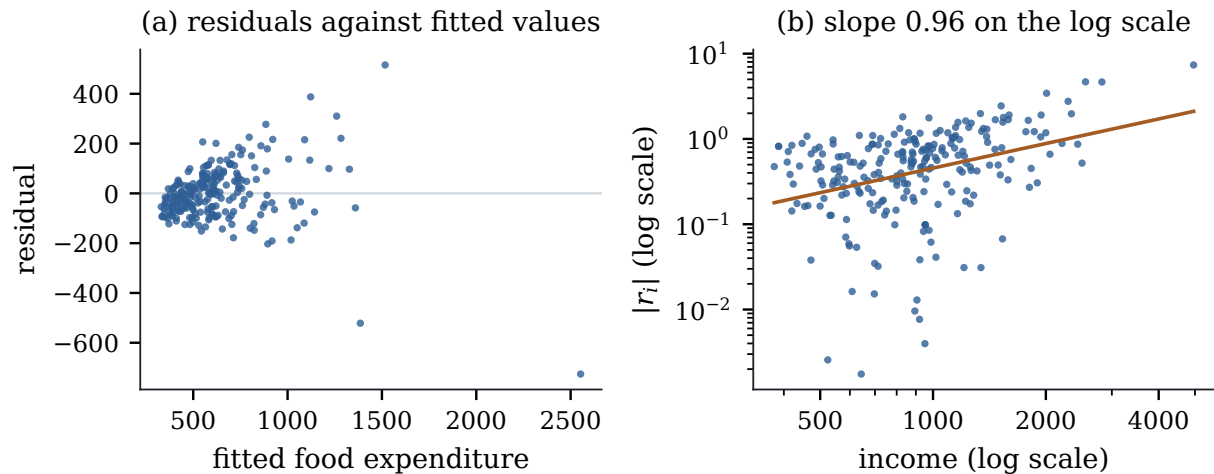


Figure 21.2.1: Residuals from the least squares fit to Engel's data. (a) Residuals against fitted values. (b) Absolute studentized residuals against income, both on logarithmic scales, with the least squares line.

```
h = np.sum(Q**2, axis=1)
s2 = e @ e / (n - 2)
# internally studentized residuals
r = e / np.sqrt(s2 * (1 - h))
```

Plots show the form of the variance function, which weighting will need. A formal test helps when the pattern is unclear or many models must be screened.

## 21.2.2 Limit theorems we need

Most results in the rest of the chapter are large-sample results. Three standard limit theorems, the central limit theorem, Slutsky's lemma and the delta method, were stated in [Section 14.4](#), and [Section 19.3](#) recalled three more: the Lindeberg–Feller central limit theorem, the Cramér–Wold device and the continuous mapping theorem. We add two pieces of notation and one law of large numbers.

*Remark* (Orders in probability and the weak law). A sequence  $T_n$  is **bounded in probability**, written  $T_n = O_p(1)$ , if for every  $\eta > 0$  there is a  $K$  with  $\Pr(|T_n| > K) < \eta$  for all  $n$ . If  $E|T_n|$  is bounded, then  $T_n = O_p(1)$  by Markov's inequality. We write  $T_n = o_p(1)$  if  $T_n \rightarrow 0$  in probability. Products and sums obey  $O_p(1)o_p(1) = o_p(1)$  and  $O_p(1) + o_p(1) = O_p(1)$ . The *weak law of large numbers* says that if  $W_1, W_2, \dots$  are independent copies of a random variable with finite mean, then  $\bar{W}_n \rightarrow EW_1$  in probability. Proofs are in van der Vaart (1998, chapter 2).

The regression version of the central limit theorem concerns a weighted sum  $\mathbf{c}^\top \boldsymbol{\xi}$  of independent variables, with weights determined by the design. It holds as long as no single weight dominates. [Lemma 19.3.1](#) proved it for identically distributed variables. The sandwich estimators of [Section 21.4](#) also need it for variables whose variances differ, and for that a moment slightly higher than the second must be bounded, in the form due to Lyapunov.

**Lemma 21.2.1 — A central limit theorem for weighted sums**

For each  $n$  let  $\xi_1, \dots, \xi_n$  be independent with mean zero, and let  $\mathbf{c} = \mathbf{c}_n \in \mathbb{R}^n$  be nonzero weights with

$$\max_i \frac{c_i^2}{\|\mathbf{c}\|^2} \rightarrow 0. \quad (21.2.2)$$

Suppose either (a) the  $\xi_i$  have a common distribution, not depending on  $n$ , with variance  $\tau^2 \in (0, \infty)$ , or (b) for some constants  $\delta, K, \tau_{\min} > 0$ ,  $\text{Var } \xi_i \geq \tau_{\min}^2$  and  $\text{E}|\xi_i|^{2+\delta} \leq K$  for all  $i$  and  $n$ . Then

$$\frac{\sum_i c_i \xi_i}{(\sum_i c_i^2 \text{Var } \xi_i)^{1/2}} \rightarrow N(0, 1) \quad \text{in distribution.}$$

*Proof.* Case (a) is the lemma on weighted sums of errors in [Section 19.3](#), applied with  $\xi_i$  in place of the errors and the unit weights  $\mathbf{c}/\|\mathbf{c}\|$ . For case (b), let  $v_n = \sum_i c_i^2 \text{Var } \xi_i$  and  $X_{ni} = c_i \xi_i / \sqrt{v_n}$ , so that  $\sum_i \text{Var } X_{ni} = 1$ , and write  $\mu_n = \max_i |c_i|/\|\mathbf{c}\| \rightarrow 0$ . We check Lindeberg's condition. Here  $v_n \geq \tau_{\min}^2 \|\mathbf{c}\|^2$ , and on  $\{|X_{ni}| > \eta\}$  we have  $X_{ni}^2 \leq |X_{ni}|^{2+\delta}/\eta^\delta$ . Hence the Lindeberg sum satisfies

$$\begin{aligned} \sum_i \text{E}[X_{ni}^2 \mathbf{1}\{|X_{ni}| > \eta\}] &\leq \frac{K \sum_i |c_i|^{2+\delta}}{\eta^\delta v_n^{1+\delta/2}} \\ &\leq \frac{K \mu_n^\delta \|\mathbf{c}\|^2 \|\mathbf{c}\|^\delta}{\eta^\delta \tau_{\min}^{2+\delta} \|\mathbf{c}\|^{2+\delta}} = \frac{K \mu_n^\delta}{\eta^\delta \tau_{\min}^{2+\delta}} \rightarrow 0, \end{aligned}$$

using  $\sum_i |c_i|^{2+\delta} \leq \max_i |c_i|^\delta \sum_i c_i^2$ . The Lindeberg–Feller theorem gives the conclusion. ■

For the coefficient vector  $\mathbf{c} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$  of a linear function  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$ , the condition [Eq. 21.2.2](#) is implied by  $\max_i h_{ii} \rightarrow 0$ , since  $c_i^2 \leq h_{ii} \|\mathbf{c}\|^2$ , as the proof of [Theorem 19.3.2](#) shows. This is the condition behind that theorem.

**21.2.3 The Breusch–Pagan score test**

To test for heteroscedasticity we need an alternative. A **variance function** model supposes that

$$\sigma_i^2 = h(\alpha_0 + \mathbf{z}_i^\top \boldsymbol{\alpha}), \quad (21.2.3)$$

where  $\mathbf{z}_i \in \mathbb{R}^q$  is a vector of observed variables, possibly some or all of the regressors, and  $h$  is a positive function. Constant variance is the hypothesis  $\boldsymbol{\alpha} = \mathbf{0}$ . Common choices are  $h(t) = e^t$ , which keeps variances positive, and  $h(t) = t$ . The score test of [Section 11.4](#) needs only the fit under the hypothesis, which is ordinary least squares, and it turns out not to depend on  $h$  at all.

Write  $\tilde{\sigma}^2 = \|\hat{\boldsymbol{\epsilon}}\|^2/n$  for the maximum likelihood estimate of  $\sigma^2$  under the hypothesis,  $u_i = \hat{\epsilon}_i^2/\tilde{\sigma}^2$

for the scaled squared residuals, and  $\tilde{\mathbf{Z}}$  for the  $n \times q$  matrix whose rows are  $(\mathbf{z}_i - \bar{\mathbf{z}})^\top$ . Let  $R^2$  be the coefficient of determination from regressing  $\hat{\varepsilon}_i^2$  on an intercept and  $\mathbf{z}_i$ , and let  $\hat{\kappa} = n \sum_i \hat{\varepsilon}_i^4 / (\sum_i \hat{\varepsilon}_i^2)^2$  be the sample kurtosis of the residuals. In this section  $\kappa$  denotes the raw kurtosis  $\mu_4/\sigma^4$ , which equals 3 for normal errors; [Section 19.3](#) used the same letter for the excess kurtosis  $\mu_4/\sigma^4 - 3$ .

### Theorem 21.2.2 — Breusch–Pagan and Koenker

Let  $\tilde{\mathbf{Z}}$  have full column rank  $q$ .

- (a) (*Score statistic.*) Suppose  $\varepsilon_i \sim N(0, \sigma_i^2)$  independently, with [Eq. 21.2.3](#),  $h$  continuously differentiable, and  $h' \neq 0$  at the null value of  $\alpha_0$ . The score statistic for  $\alpha = \mathbf{0}$  is

$$\text{BP} = \frac{1}{2}(\mathbf{u} - \mathbf{1})^\top \tilde{\mathbf{Z}}(\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}})^{-1} \tilde{\mathbf{Z}}^\top (\mathbf{u} - \mathbf{1}),$$

half the explained sum of squares from regressing  $u_i$  on an intercept and  $\mathbf{z}_i$ . It does not depend on  $h$ .

- (b) (*Studentized form.*) Let  $K = nR^2$ . Then  $\text{BP} = \frac{1}{2}(\hat{\kappa} - 1)K$ .
- (c) (*Null distributions.*) Suppose instead only that the errors are independent and identically distributed with mean 0, variance  $\sigma^2$ , finite fourth moment  $\mu_4$  and kurtosis  $\kappa = \mu_4/\sigma^4 > 1$ , that  $\text{rank}(\mathbf{X}) = p$  and  $q$  are fixed, that the  $\mathbf{z}_i$  are bounded, and that  $n^{-1}\tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}} \rightarrow \mathbf{Q}_z$ , positive definite. Then, as  $n \rightarrow \infty$ ,

$$K \rightarrow \chi^2(q), \quad \text{BP} \rightarrow \frac{1}{2}(\kappa - 1)\chi^2(q) \quad \text{in distribution.}$$

*Proof.*

- (a) The log-likelihood is  $\ell = -\frac{1}{2} \sum_i \log(2\pi\sigma_i^2) - \frac{1}{2} \sum_i (y_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta})^2 / \sigma_i^2$ . Put  $\boldsymbol{\theta} = (\alpha_0, \boldsymbol{\alpha}^\top)^\top$ ,  $\mathbf{z}_i^* = (1, \mathbf{z}_i^\top)^\top$  and  $\eta_i = \alpha_0 + \mathbf{z}_i^\top \boldsymbol{\alpha}$ . Since  $\partial\ell/\partial\sigma_i^2 = (\varepsilon_i^2/\sigma_i^2 - 1)/(2\sigma_i^2)$  and  $\partial\sigma_i^2/\partial\boldsymbol{\theta} = h'(\eta_i)\mathbf{z}_i^*$ ,

$$\begin{aligned} \frac{\partial\ell}{\partial\boldsymbol{\theta}} &= \sum_i \frac{h'(\eta_i)}{2\sigma_i^2} \left( \frac{\varepsilon_i^2}{\sigma_i^2} - 1 \right) \mathbf{z}_i^*, \\ \frac{\partial\ell}{\partial\boldsymbol{\beta}} &= \sum_i \frac{\varepsilon_i}{\sigma_i^2} \mathbf{x}_{(i)}. \end{aligned}$$

Under normality  $\varepsilon_i^2/\sigma_i^2$  has variance 2, and  $E\varepsilon_i(\varepsilon_i^2 - \sigma_i^2) = 0$ , so the information matrix is block diagonal between  $\boldsymbol{\beta}$  and  $\boldsymbol{\theta}$ , and its  $\boldsymbol{\theta}$  block is  $\sum_i h'(\eta_i)^2 \mathbf{z}_i^* \mathbf{z}_i^{*\top} / (2\sigma_i^4)$ . Under the hypothesis  $\eta_i = \alpha_0$  for all  $i$ , the restricted maximum likelihood estimates are  $\hat{\boldsymbol{\beta}}$  and  $\tilde{\sigma}^2$ , and  $h(\hat{\alpha}_0) = \tilde{\sigma}^2$ . Write  $g = h'(\hat{\alpha}_0)$ , let  $\mathbf{Z}$  be the  $n \times q$  matrix with rows  $\mathbf{z}_i^\top$ , and put  $\mathbf{Z}^* = [\mathbf{1}, \mathbf{Z}]$ . At the restricted estimate the  $\boldsymbol{\beta}$  score vanishes, and

$$\mathbf{S}_\theta = \frac{g}{2\tilde{\sigma}^2} \mathbf{Z}^{*\top} (\mathbf{u} - \mathbf{1}), \quad \mathcal{I}_{\theta\theta} = \frac{g^2}{2\tilde{\sigma}^4} \mathbf{Z}^{*\top} \mathbf{Z}^*.$$

By block diagonality the score statistic is  $\mathbf{S}_\theta^\top \mathcal{I}_{\theta\theta}^{-1} \mathbf{S}_\theta = \frac{1}{2}(\mathbf{u} - \mathbf{1})^\top \mathbf{Z}^* (\mathbf{Z}^{*\top} \mathbf{Z}^*)^{-1} \mathbf{Z}^{*\top} (\mathbf{u} - \mathbf{1})$ , in which  $g$  has cancelled. The matrix here is the projection onto  $\mathcal{C}(\mathbf{Z}^*) = \mathcal{C}(\mathbf{1}) \oplus \mathcal{C}(\tilde{\mathbf{Z}})$ , and  $\mathbf{1}^\top (\mathbf{u} - \mathbf{1}) = 0$  because  $\sum_i u_i = n$ , so only the projection onto  $\mathcal{C}(\tilde{\mathbf{Z}})$  contributes ([Lemma 6.6.1](#)).

- (b) Let  $\mathbf{w}$  have entries  $\hat{\varepsilon}_i^2$ . Its explained sum of squares on  $[\mathbf{1}, \mathbf{Z}]$  is  $\text{ESS}_w = (\mathbf{w} - \tilde{\sigma}^2 \mathbf{1})^\top \tilde{\mathbf{M}} (\mathbf{w} - \tilde{\sigma}^2 \mathbf{1})$ , where  $\tilde{\mathbf{M}}$  projects onto  $\mathcal{C}(\tilde{\mathbf{Z}})$ , and its total sum of squares is

$\text{TSS}_w = \sum_i (\hat{\varepsilon}_i^2 - \tilde{\sigma}^2)^2 = \sum_i \hat{\varepsilon}_i^4 - n\tilde{\sigma}^4$ . Since  $\mathbf{u} - \mathbf{1} = (\mathbf{w} - \tilde{\sigma}^2 \mathbf{1})/\tilde{\sigma}^2$ , (a) gives  $\text{BP} = \text{ESS}_w/(2\tilde{\sigma}^4)$ , while  $K = n \text{ESS}_w/\text{TSS}_w$ . Their ratio is  $\text{TSS}_w/(2n\tilde{\sigma}^4) = (\hat{\kappa} - 1)/2$ .

- (c) Put  $\mathbf{a} = \mathbf{M}\boldsymbol{\varepsilon}$ , so that  $\hat{\varepsilon}_i = \varepsilon_i - a_i$  and  $E\|\mathbf{a}\|^2 = \sigma^2 p$  (Proposition 21.1.1). Let  $|\tilde{z}_{ik}| \leq C$ . By (b),

$$K = \frac{\mathbf{T}_n^\top (n^{-1} \tilde{\mathbf{Z}}^\top \tilde{\mathbf{Z}})^{-1} \mathbf{T}_n}{\text{TSS}_w/n},$$

$$\mathbf{T}_n = n^{-1/2} \tilde{\mathbf{Z}}^\top \mathbf{w} = n^{-1/2} \tilde{\mathbf{Z}}^\top (\mathbf{w} - \sigma^2 \mathbf{1}),$$

using  $\tilde{\mathbf{Z}}^\top \mathbf{1} = \mathbf{0}$ . We show three things.

*The leading term.* Let  $\mathbf{T}_n^0 = n^{-1/2} \sum_i \tilde{\mathbf{z}}_i (\varepsilon_i^2 - \sigma^2)$ . The  $\xi_i = \varepsilon_i^2 - \sigma^2$  are independent and identically distributed with variance  $\mu_4 - \sigma^4 > 0$ . For fixed  $\mathbf{b} \in \mathbb{R}^q$  the weights  $c_i = \mathbf{b}^\top \tilde{\mathbf{z}}_i$  are bounded and  $\|\mathbf{c}\|^2/n \rightarrow \mathbf{b}^\top \mathbf{Q}_z \mathbf{b} > 0$ , so Eq. 21.2.2 holds, and Lemma 21.2.1(a) shows that  $\mathbf{b}^\top \mathbf{T}_n^0 \rightarrow N(0, (\mu_4 - \sigma^4) \mathbf{b}^\top \mathbf{Q}_z \mathbf{b})$ . By the Cramér–Wold device,  $\mathbf{T}_n^0 \rightarrow N_q(\mathbf{0}, (\mu_4 - \sigma^4) \mathbf{Q}_z)$ .

*The residuals do not matter.*  $\hat{\varepsilon}_i^2 - \varepsilon_i^2 = a_i^2 - 2a_i \varepsilon_i$ , so the  $k$ th coordinate of  $\mathbf{T}_n - \mathbf{T}_n^0$  is  $n^{-1/2} \sum_i \tilde{z}_{ik} a_i^2 - 2n^{-1/2} \boldsymbol{\varepsilon}^\top \mathbf{B} \boldsymbol{\varepsilon}$ , with  $\mathbf{B} = \frac{1}{2}(\mathbf{D}\mathbf{M} + \mathbf{M}\mathbf{D})$  and  $\mathbf{D} = \text{diag}(\tilde{z}_{1k}, \dots, \tilde{z}_{nk})$ . The first term is at most  $Cn^{-1/2} \|\mathbf{a}\|^2$ , whose mean  $C\sigma^2 p/\sqrt{n}$  tends to zero. For the second,  $|E \boldsymbol{\varepsilon}^\top \mathbf{B} \boldsymbol{\varepsilon}| = \sigma^2 |\text{tr}(\mathbf{D}\mathbf{M})| \leq C\sigma^2 p$  (Theorem 2.3.1), and by Theorem 2.3.3

$$\begin{aligned} \text{Var}(\boldsymbol{\varepsilon}^\top \mathbf{B} \boldsymbol{\varepsilon}) &\leq |\mu_4 - 3\sigma^4| \sum_i b_{ii}^2 + 2\sigma^4 \text{tr}(\mathbf{B}^2) \\ &\leq (|\mu_4 - 3\sigma^4| + 2\sigma^4) C^2 p, \end{aligned}$$

because  $b_{ii} = \tilde{z}_{ik} h_{ii}$ ,  $\sum_i h_{ii}^2 \leq \sum_i h_{ii} = p$ , and  $\text{tr}(\mathbf{B}^2) \leq \|\mathbf{D}\mathbf{M}\|_F^2 = \text{tr}(\mathbf{M}\mathbf{D}^2\mathbf{M}) \leq C^2 \text{tr} \mathbf{M}$ . So  $n^{-1/2} \boldsymbol{\varepsilon}^\top \mathbf{B} \boldsymbol{\varepsilon}$  has mean and variance tending to zero, and  $\mathbf{T}_n - \mathbf{T}_n^0 = o_p(1)$ .

*The denominator.*  $\tilde{\sigma}^2 = (\|\boldsymbol{\varepsilon}\|^2 - \|\mathbf{a}\|^2)/n \rightarrow \sigma^2$  by the weak law and  $\|\mathbf{a}\|^2 = O_p(1)$ . For the fourth powers, Minkowski's inequality in the norm  $\|\mathbf{v}\|_4 = (\sum_i v_i^4)^{1/4}$  gives  $|\|\hat{\boldsymbol{\varepsilon}}\|_4 - \|\boldsymbol{\varepsilon}\|_4| \leq \|\mathbf{a}\|_4 \leq \|\mathbf{a}\| = O_p(1)$ , while  $n^{-1} \|\boldsymbol{\varepsilon}\|_4^4 \rightarrow \mu_4$  by the weak law. Hence  $n^{-1} \sum_i \hat{\varepsilon}_i^4 \rightarrow \mu_4$ ,  $\text{TSS}_w/n \rightarrow \mu_4 - \sigma^4$  and  $\hat{\kappa} \rightarrow \kappa$ .

Combining these by Slutsky's lemma and continuous mapping,  $K$  converges in distribution to  $\mathbf{T}^\top \mathbf{Q}_z^{-1} \mathbf{T}/(\mu_4 - \sigma^4)$  with  $\mathbf{T} \sim N_q(\mathbf{0}, (\mu_4 - \sigma^4) \mathbf{Q}_z)$ , which is  $\chi^2(q)$  by Corollary 4.3.3. The statement about BP follows from (b). ■

Part (a) is the test of Breusch and Pagan (1979). Cook and Weisberg (1983) derived the same statistic, and used the regression of  $u_i$  on  $\mathbf{z}_i$  as a diagnostic plot. Part (c) exposes a weakness. BP is calibrated for normal errors, which have  $\kappa = 3$ . With heavier tails its null distribution is inflated by the factor  $(\kappa - 1)/2$ , and the test rejects too often, detecting kurtosis rather than heteroscedasticity. Koenker (1981) proposed the studentized form  $nR^2$ , whose null limit is the same for every error law with a fourth moment. It is the version to use.

#### Example. (Size of the two tests)

Take a straight-line regression with  $n = 100$  equally spaced values of  $x$ , constant variance, and  $\mathbf{z}_i = x_i$ . In 40,000 simulated data sets the 5% tests rejected as follows:

| error law | $\kappa$ | Breusch–Pagan | its limit | Koenker |
|-----------|----------|---------------|-----------|---------|
| normal    | 3        | 0.048         | 0.050     | 0.050   |
| Laplace   | 6        | 0.185         | 0.215     | 0.047   |
| uniform   | 1.8      | 0.003         | 0.002     | 0.049   |

The Laplace law makes the Breusch–Pagan test reject nearly four times too often; the uniform law makes it almost never reject. The studentized test holds its level in all three cases.

```
import numpy as np
from scipy import stats

rng = np.random.default_rng(2105)
n = 100
x = np.linspace(0, 10, n)
X = np.column_stack([np.ones(n), x])
Q, _ = np.linalg.qr(X)
# centred, unit-length variance regressor
zc = (x - x.mean()) / np.linalg.norm(x - x.mean())
crit = stats.chi2.ppf(0.95, 1)
laws = {
    "normal": (lambda size: rng.normal(size=size), 3.0),
    "Laplace": (lambda size: rng.laplace(scale=np.sqrt(0.5), size=size), 6.0),
    "uniform": (lambda size: rng.uniform(-np.sqrt(3), np.sqrt(3), size=size),
               1.8),
}

def size_table(reps):
    table = {}
    for name, (draw, kappa) in laws.items():
        E = draw((reps, n))
        R = E - (E @ Q) @ Q.T # residuals
        R2 = R**2
        sig2 = R2.mean(axis=1, keepdims=True)
        # explained SS of e^2 on [1, x]
        ess = (R2 @ zc) ** 2
        bp = ess / (2 * sig2[:, 0] ** 2)
        koenker = n * ess / np.sum((R2 - sig2) ** 2, axis=1)
        limit_bp = stats.chi2.sf(crit / ((kappa - 1) / 2), 1)
        table[name] = (np.mean(bp > crit), np.mean(koenker > crit), limit_bp)
    print(f"{name:8s}: Breusch-Pagan rejects {table[name][0]:.3f} "
          f"(limit {limit_bp:.3f}), "
          f"Koenker rejects {table[name][1]:.3f}")
    return table

# a quick version (the book uses 40000)
quick = size_table(2000)
```

**White's test.** For the least squares standard errors, what matters is whether  $\sigma_i^2$  is correlated with the entries of  $\mathbf{x}_{(i)}\mathbf{x}_{(i)}^\top$ , the terms of  $\mathbf{X}^\top\mathbf{\Sigma}\mathbf{X}$  (White 1980). White's test is Koenker's statistic with  $\mathbf{z}_i$  the regressors, their squares and their cross products; with many regressors it has many degrees of freedom and low power.

**Example. (Testing Engel's data)**

With  $z_i = \text{income}_i$ , the Breusch–Pagan statistic is 636.0 and Koenker's is 109.3, on one degree of freedom. Their ratio is  $5.820 = (\hat{\kappa} - 1)/2$ , since the residuals have sample kurtosis 12.64. The heavy tails of the residuals inflate BP about sixfold, but the evidence is overwhelming either way: Koenker's p-value is  $1.4 \times 10^{-25}$ . White's test, with income and its square, gives 181.1 on two degrees of freedom.

```
def explained_ss(v, Z):
    """Explained sum of squares of v on [1, Z], about the mean of v."""
    Z1 = np.column_stack([np.ones(len(v)), Z])
    coef, *_ = np.linalg.lstsq(Z1, v, rcond=None)
    return np.sum((Z1 @ coef - v.mean()) ** 2)

# maximum likelihood estimate under H0
sig2 = e @ e / n
u = e**2 / sig2
# Breusch-Pagan score statistic
bp = explained_ss(u, income) / 2
tss = np.sum((e**2 - sig2) ** 2)
# n R^2 from regressing e^2 on [1, income]
koenker = n * explained_ss(e**2, income) / tss
white = n * explained_ss(e**2, np.column_stack([income, income**2])) / tss
# sample kurtosis of the residuals
kappa = n * np.sum(e**4) / np.sum(e**2) ** 2
print(f"Breusch-Pagan {bp:.1f}, Koenker {koenker:.1f}, "
      f"White {white:.1f} (2 df)")
print(f"ratio BP/Koenker = {bp / koenker:.3f}, "
      f"(kurtosis - 1)/2 = {(kappa - 1) / 2:.3f}")

# Goldfeld-Quandt: sort, drop the middle fifth
order = np.argsort(income)
m = (n - 47) // 2
low, high = order[:m], order[-m:]
sse = []
for g in (low, high):
    b_g = np.linalg.lstsq(X[g], food[g], rcond=None)[0]
    sse.append(np.sum((food[g] - X[g] @ b_g) ** 2))
gq = (sse[1] / (m - 2)) / (sse[0] / (m - 2))
print(f"Goldfeld-Quandt F = {gq:.2f} on ({m - 2}, {m - 2}) df, "
      f"p = {stats.f.sf(gq, m - 2, m - 2):.1e}")
```

## 21.2.4 An exact test: Goldfeld–Quandt

When the variance is suspected to change monotonically with one variable  $v$  that does not depend on the responses, there is an exact test under normality. Sort the observations by  $v$ , set aside a block in the middle, and fit the model separately to the  $n_1$  observations with the smallest values of  $v$  and the  $n_2$  with the largest.

**Proposition 21.2.3 — Goldfeld–Quandt**

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ , let the two groups be chosen without reference to  $\mathbf{Y}$ , and let  $\mathbf{X}_1$  and  $\mathbf{X}_2$ , their rows of  $\mathbf{X}$ , have rank  $p < n_k$ . If  $\text{SSE}_1$  and  $\text{SSE}_2$  are the residual sums of squares of the separate fits, then

$$F_{\text{GQ}} = \frac{\text{SSE}_2/(n_2 - p)}{\text{SSE}_1/(n_1 - p)} \sim F(n_2 - p, n_1 - p).$$

*Proof.* Each group is a normal linear model with error variance  $\sigma^2$ , so  $\text{SSE}_k/\sigma^2 \sim \chi^2(n_k - p)$  by [Theorem 7.5.1\(b\)](#). The two sums of squares are functions of disjoint sets of independent errors, so they are independent, and the ratio has the stated law by [Definition 4.1.2](#) with  $\gamma = 0$ . ■

Dropping the middle block sharpens the contrast between the groups at the cost of degrees of freedom, and how much to drop is a matter of judgement; here we omit a fifth of the data. For Engel's data, with the middle 47 households omitted and 94 in each group,  $F_{\text{GQ}} = 10.22$  on  $(92, 92)$  degrees of freedom, with p-value  $1.7 \times 10^{-24}$ . The test is exact only under normality. Like every comparison of variances, it is sensitive to the tails of the error law. When the observations fall into groups, a more robust comparison is Levene's test (Levene 1960), the one-way analysis of variance of the absolute residuals  $|\hat{\varepsilon}_i|$ , or its version with deviations from group medians (Brown and Forsythe 1974). Box (1953) showed how badly the classical normal-theory test of equal group variances, Bartlett's, is affected by kurtosis.

**21.2.5 Exercises****21.2.5.1 A. Check your understanding****Exercise A1.**

Derive [Eq. 21.2.1](#). Show that under constant variance  $E \hat{\varepsilon}_i^2 = \sigma^2(1 - h_{ii})$ , and give an example in which  $\sigma_i^2$  is the largest variance but  $E \hat{\varepsilon}_i^2$  is the smallest.

**21.2.5.2 B. Practice****Exercise B1.**

Let  $y_i = \mathbf{x}_{(i)}^\top(\boldsymbol{\beta} + \mathbf{u}_i) + \varepsilon_i$ , where  $\mathbf{u}_i$  has mean  $\mathbf{0}$  and covariance  $\boldsymbol{\Omega}$ ,  $\varepsilon_i$  has variance  $\sigma^2$ , and all are independent. Show that  $\text{Var}(y_i)$  is a linear function of the squares and cross products of the regressors. Which test of this section is the score-type test for  $\boldsymbol{\Omega} = \mathbf{0}$ ?

*Solution.*  $y_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta} = \mathbf{x}_{(i)}^\top \mathbf{u}_i + \varepsilon_i$  has variance  $\sigma^2 + \mathbf{x}_{(i)}^\top \boldsymbol{\Omega} \mathbf{x}_{(i)} = \sigma^2 + \sum_{j,k} \omega_{jk} x_{ij} x_{ik}$ , linear in the products  $x_{ij} x_{ik}$  with coefficients  $\omega_{jk}$ . This is [Eq. 21.2.3](#) with  $h(t) = t$  and  $\mathbf{z}_i$  the distinct squares and cross products (the squares of the intercept column merge with the

intercept). The score test is the Breusch–Pagan test with these  $\mathbf{z}_i$ , whose studentized form is White’s test.

### Exercise B2.

Under the assumptions of [Proposition 21.2.3](#), suppose the observations are sorted by their fitted values  $\hat{Y}_i$  instead of by a fixed variable. Show that  $F_{\text{GQ}}$  still has the  $F(n_2 - p, n_1 - p)$  distribution. Show by an example that sorting by  $|\hat{\varepsilon}_i|$  destroys the result.

*Solution.* Fix a partition, with selection matrices  $\mathbf{S}_1, \mathbf{S}_2$ , and let  $\mathbf{M}_k$  project onto  $\mathcal{C}(\mathbf{X}_k)$ . The group residuals  $(\mathbf{I} - \mathbf{M}_k)\mathbf{S}_k\boldsymbol{\varepsilon}$  and the fitted values  $\mathbf{M}\boldsymbol{\varepsilon}$  are jointly normal with cross-covariance  $\sigma^2(\mathbf{I} - \mathbf{M}_k)\mathbf{S}_k\mathbf{M} = \sigma^2(\mathbf{I} - \mathbf{M}_k)\mathbf{X}_k(\mathbf{X}^\top\mathbf{X})^{-1}\mathbf{X}^\top = \mathbf{0}$ , since  $\mathbf{S}_k\mathbf{X} = \mathbf{X}_k$ . So, for every fixed partition,  $(\text{SSE}_1, \text{SSE}_2)$  is independent of  $\mathbf{MY}$ , and it has the law of [Proposition 21.2.3](#). Sorting by fitted values makes the partition a function of  $\mathbf{MY}$ , so given  $\mathbf{MY}$  the statistic has the  $F$  law, and therefore also unconditionally. Sorting by  $|\hat{\varepsilon}_i|$  puts the largest residuals in the second group: with  $p = 1$ ,  $\mathbf{X} = \mathbf{1}$  and  $n = 4$ , no middle block, the second group contains the two observations farthest from  $\bar{y}$ , and  $F_{\text{GQ}}$  exceeds one with probability one.

### 21.2.5.3 C. Going deeper

#### Exercise C1.

Suppose the observations fall into  $g$  groups and  $\mathbf{z}_i$  consists of  $g - 1$  group indicators. Show that Koenker’s statistic is  $n$  times the  $R^2$  of a one-way analysis of variance of the squared residuals, and compare it with Levene’s test.

## 21.3 Weighted least squares

Once heteroscedasticity has been found, one can transform the response ([Chapter 22](#)), keep least squares and correct its standard errors ([Section 21.4](#)), or give the noisier observations less weight. This section is about weighting. With known variances it is optimal and its theory is already in place. With estimated variances the gain is real but the theory is asymptotic, and a careless implementation can do worse than no weighting at all.

### 21.3.1 Known weights

Suppose  $\text{Var}(\varepsilon_i) = \sigma^2/w_i$ , with the  $w_i > 0$  known and  $\sigma^2$  unknown, and the errors uncorrelated. With  $\mathbf{W} = \text{diag}(w_1, \dots, w_n)$  this is  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{W}^{-1}$ . The generalized least squares estimate of [Definition 6.9.1](#) with  $\mathbf{V} = \mathbf{W}^{-1}$  minimizes

$$\sum_{i=1}^n w_i (y_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2, \quad \text{so} \quad \hat{\boldsymbol{\beta}}_W = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}. \quad (21.3.1)$$

This is the **weighted least squares** (WLS) estimate. [Section 6.9](#) described its geometry: it is



ordinary least squares after each row of  $\mathbf{X}$  and  $\mathbf{y}$  is multiplied by  $\sqrt{w_i}$ , which turns the errors into  $\sqrt{w_i}\varepsilon_i$  with common variance  $\sigma^2$ . Aitken's theorem ([Corollary 7.2.3](#)) makes it the best linear unbiased estimator. Everything in Part III then applies to the rescaled data. The error variance is estimated by

$$s_w^2 = \frac{1}{n-p} \sum_{i=1}^n w_i (y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_W)^2,$$

the covariance of  $\hat{\boldsymbol{\beta}}_W$  by  $s_w^2(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$ , and under normality the  $t$  and  $F$  procedures are exact. The residuals to plot are the rescaled ones,  $\sqrt{w_i}(y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_W)$ , studentized with the leverages of the weighted fit. The raw residuals  $y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_W$  should still fan out, since their variances still differ.

Known weights arise when  $y_i$  is the mean of  $m_i$  observations ( $w_i = m_i$ ), a total ( $w_i = 1/m_i$ ), or a measurement from an instrument of known precision. The weights are then fixed by the design, not by the data.

## 21.3.2 Estimating the weights

More often the variances are unknown, and the weights must come from the data. There are three common routes.

1. **Replicates.** If there are several observations at each distinct row of  $\mathbf{X}$ , or in each of a few groups known to differ in precision, the sample variance within each group estimates its  $\sigma_g^2$ .
2. **A variance function in other variables.** Model  $\sigma_i^2 = \sigma^2 \exp(\mathbf{z}_i^\top \boldsymbol{\gamma})$ , as in [Eq. 21.2.3](#), and estimate  $\boldsymbol{\gamma}$  by regressing  $\log \hat{\varepsilon}_i^2$  on an intercept and  $\mathbf{z}_i$ . The intercept of that regression is biased, because  $E \log \chi^2(1) \neq 0$ , but weights are needed only up to a common factor, so the intercept does not matter.
3. **A variance function of the mean.** Model  $\sigma_i = \sigma |\mu_i|^\theta$  with  $\mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , estimate  $\theta$  from the slope of  $\log |\hat{\varepsilon}_i|$  on  $\log |\hat{Y}_i|$ , and weight by  $|\hat{Y}_i|^{-2\theta}$ . Since the weights depend on the fit, the estimate is usually iterated.

The resulting estimate  $\hat{\boldsymbol{\beta}}_{\hat{W}}$  is called **feasible** (or estimated) weighted least squares.

### Example. (Engel's data, reweighted)

Regressing  $\log \hat{\varepsilon}_i^2$  on an intercept and  $\log \text{income}_i$  gives slope  $\hat{\gamma} = 1.90$ , so the fitted variance is proportional to  $\text{income}^{1.90}$ , close to the square law suggested by [Figure 21.2.1](#). Weighting by  $\text{income}^{-1.90}$  changes the slope from 0.4852 to 0.5711. The model-based standard error  $s_w\{(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}\}_{22}^{1/2}$  of the weighted slope is 0.0149. The sandwich standard error of [Section 21.4](#), which does not trust the variance model, is 0.0172 for the weighted slope and 0.0664 for the unweighted one. So weighting has cut the standard error by a factor of nearly four.

The variance function is not sharply determined by these data. Re-estimating  $\gamma$  from the residuals of the weighted fit, and repeating until it settles (10 iterations), moves  $\hat{\gamma}$  to 2.52 and the slope to 0.5875, about one standard error away. Reporting the weighted estimate with the sandwich standard error protects against this uncertainty in the weights.

```
import numpy as np
```

```

import statsmodels.api as sm
from scipy import stats

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)
X = np.column_stack([np.ones(n), income])

def wls(X, y, w):
    """Weighted least squares: coefficients, model-based and HC3 covariances."""
    sw = np.sqrt(w)
    Q, R = np.linalg.qr(X * sw[:, None])          # OLS on the whitened data
    b = np.linalg.solve(R, Q.T @ (y * sw))
    ew = (y - X @ b) * sw                          # whitened residuals
    # leverages in the weighted geometry
    hw = np.sum(Q**2, axis=1)
    Rinv = np.linalg.inv(R)
    bread = Rinv @ Rinv.T                          # (X'WX)^{-1}
    s2w = ew @ ew / (len(y) - X.shape[1])
    meat = (Q * (ew / (1 - hw))[:, None]).T @ (Q * (ew / (1 - hw))[:, None])
    return b, s2w * bread, Rinv @ meat @ Rinv.T

b_ols, V_ols, V_ols_hc3 = wls(X, food, np.ones(n))
e = food - X @ b_ols
Z = np.column_stack([np.ones(n), np.log(income)])
# variance ~ income^gamma
gamma = np.linalg.lstsq(Z, np.log(e**2), rcond=None)[0][1]
b_fw, V_fw, V_fw_hc3 = wls(X, food, income**-gamma)
print(f"estimated power gamma = {gamma:.2f}")
print(f"OLS slope {b_ols[1]:.4f} model se {np.sqrt(V_ols[1, 1]):.4f} "
      f"HC3 se {np.sqrt(V_ols_hc3[1, 1]):.4f}")
print(f"FWLS slope {b_fw[1]:.4f} model se {np.sqrt(V_fw[1, 1]):.4f} "
      f"HC3 se {np.sqrt(V_fw_hc3[1, 1]):.4f}")

```

### 21.3.3 What feasible weighting achieves

The weights in  $\hat{\beta}_W$  are random and depend on the same data as the fit, so none of the finite-sample results for known weights carries over directly. The next theorem collects what can be said in general. Part (b) shows that weighting with the wrong weights is harmless for bias. Part (c) is an exact finite-sample result, and part (d) is the large-sample justification in the setting of replicated groups.

#### Theorem 21.3.1 — Weighted least squares with known and estimated weights

Let  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $E\boldsymbol{\varepsilon} = \mathbf{0}$  and  $\text{rank}(\mathbf{X}) = p$ .

- (a) (*Known weights.*) If  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{W}^{-1}$  with  $\mathbf{W}$  known, then  $\hat{\beta}_W$  is the best linear unbiased estimator of  $\boldsymbol{\beta}$ ,  $\text{Cov}(\hat{\beta}_W) = \sigma^2 (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$  and  $E s_w^2 = \sigma^2$ . If  $\boldsymbol{\varepsilon}$  is normal,  $\hat{\beta}_W \sim N_p(\boldsymbol{\beta}, \sigma^2 (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1})$  independently of  $(n - p) s_w^2 / \sigma^2 \sim \chi^2(n - p)$ .

- (b) (*Wrong weights.*) If  $\text{Cov}(\boldsymbol{\varepsilon}) = \boldsymbol{\Sigma}$  is arbitrary and  $\mathbf{W}$  is any fixed positive diagonal matrix, then  $\hat{\boldsymbol{\beta}}_W$  is unbiased with

$$\text{Cov}(\hat{\boldsymbol{\beta}}_W) = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \boldsymbol{\Sigma} \mathbf{W} \mathbf{X} (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}.$$

- (c) (*Estimated weights, symmetric errors.*) Let  $\hat{\mathbf{W}} = \mathbf{W}(\hat{\boldsymbol{\varepsilon}})$  be a positive diagonal matrix that depends on the data only through the ordinary least squares residuals, with  $\mathbf{W}(-\mathbf{e}) = \mathbf{W}(\mathbf{e})$ . If  $-\boldsymbol{\varepsilon}$  has the same distribution as  $\boldsymbol{\varepsilon}$  and  $E\|\hat{\boldsymbol{\beta}}_{\hat{\mathbf{W}}}\| < \infty$ , then  $E\hat{\boldsymbol{\beta}}_{\hat{\mathbf{W}}} = \boldsymbol{\beta}$ .
- (d) (*Estimated weights from groups.*) Suppose the observations form  $G$  groups, fixed in advance, that the errors are independent, and that within group  $g$  they are identically distributed with variance  $\sigma_g^2 > 0$ . Let  $n_g \rightarrow \infty$  for each  $g$  with  $n^{-1} \mathbf{X}_g^\top \mathbf{X}_g \rightarrow \mathbf{A}_g$ , where  $\mathbf{X}_g$  holds the rows of group  $g$ , and let  $\mathbf{A}_w = \sum_g \mathbf{A}_g / \sigma_g^2$  be positive definite. Let  $\mathbf{W}$  have entries  $1/\sigma_g^2$ , and  $\hat{\mathbf{W}}$  have entries  $1/\hat{\sigma}_g^2$ , where  $\hat{\sigma}_g^2 = \|\hat{\boldsymbol{\varepsilon}}_g\|^2/n_g$  is the mean squared ordinary least squares residual in group  $g$ . Then

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\hat{\mathbf{W}}} - \hat{\boldsymbol{\beta}}_W) \rightarrow \mathbf{0} \quad \text{and} \quad n(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \rightarrow \mathbf{A}_w^{-1}$$

in probability, while  $n \text{Cov}(\hat{\boldsymbol{\beta}}_W) = n(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \rightarrow \mathbf{A}_w^{-1}$ .

*Proof.*

- (a) Multiply Eq. 21.3.1 through by  $\mathbf{W}^{1/2}$ : the model  $\mathbf{W}^{1/2} \mathbf{Y} = \mathbf{W}^{1/2} \mathbf{X} \boldsymbol{\beta} + \mathbf{W}^{1/2} \boldsymbol{\varepsilon}$  satisfies  $\text{Cov}(\mathbf{W}^{1/2} \boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ , and  $\hat{\boldsymbol{\beta}}_W$  is its least squares estimate. The claims are Corollary 7.2.3, Theorem 5.5.1, Theorem 5.6.1 and Theorem 7.5.1 applied to the rescaled model.
- (b)  $\hat{\boldsymbol{\beta}}_W - \boldsymbol{\beta} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \boldsymbol{\varepsilon}$  is a fixed linear map of  $\boldsymbol{\varepsilon}$ ; apply Theorem 2.2.1.
- (c) The residuals  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M}) \boldsymbol{\varepsilon}$  do not involve  $\boldsymbol{\beta}$ , and they change sign with  $\boldsymbol{\varepsilon}$ . So  $\hat{\boldsymbol{\beta}}_{\hat{\mathbf{W}}} - \boldsymbol{\beta} = \mathbf{g}(\boldsymbol{\varepsilon})$  with  $\mathbf{g}(\boldsymbol{\varepsilon}) = (\mathbf{X}^\top \mathbf{W}(\hat{\boldsymbol{\varepsilon}}) \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}(\hat{\boldsymbol{\varepsilon}}) \boldsymbol{\varepsilon}$ , and  $\mathbf{g}(-\boldsymbol{\varepsilon}) = -\mathbf{g}(\boldsymbol{\varepsilon})$  because  $\mathbf{W}(-\hat{\boldsymbol{\varepsilon}}) = \mathbf{W}(\hat{\boldsymbol{\varepsilon}})$ . If  $-\boldsymbol{\varepsilon}$  has the law of  $\boldsymbol{\varepsilon}$ , then  $\mathbf{g}(\boldsymbol{\varepsilon})$  has the law of  $\mathbf{g}(-\boldsymbol{\varepsilon}) = -\mathbf{g}(\boldsymbol{\varepsilon})$ , and its mean, which exists, equals its negative.
- (d) *Consistency of the group variances.* Write  $\mathbf{a} = \mathbf{M} \boldsymbol{\varepsilon}$ , so  $\hat{\boldsymbol{\varepsilon}}_g = \boldsymbol{\varepsilon}_g - \mathbf{a}_g$ . By the weak law,  $\|\boldsymbol{\varepsilon}_g\|^2/n_g \rightarrow \sigma_g^2$ . Also  $E\|\mathbf{a}\|^2 = \text{tr}(\mathbf{M} \boldsymbol{\Sigma}) \leq p \max_g \sigma_g^2$ , so  $\|\mathbf{a}_g\|^2/n_g \leq \|\mathbf{a}\|^2/n_g \rightarrow 0$  in probability. The triangle inequality  $|\|\hat{\boldsymbol{\varepsilon}}_g\| - \|\boldsymbol{\varepsilon}_g\|| \leq \|\mathbf{a}_g\|$  gives  $\hat{\sigma}_g^2 \rightarrow \sigma_g^2$ , hence  $\hat{w}_g = 1/\hat{\sigma}_g^2 \rightarrow w_g = 1/\sigma_g^2$ .

*Comparison.* Put  $\mathbf{A}_n(\mathbf{v}) = n^{-1} \sum_g v_g \mathbf{X}_g^\top \mathbf{X}_g$  for weights  $\mathbf{v} = (v_1, \dots, v_G)$ , and  $\mathbf{b}_g = n^{-1/2} \mathbf{X}_g^\top \boldsymbol{\varepsilon}_g$ . Then

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_{\hat{\mathbf{W}}} - \boldsymbol{\beta}) = \mathbf{A}_n(\hat{\mathbf{w}})^{-1} \sum_g \hat{w}_g \mathbf{b}_g,$$

$$\sqrt{n}(\hat{\boldsymbol{\beta}}_W - \boldsymbol{\beta}) = \mathbf{A}_n(\mathbf{w})^{-1} \sum_g w_g \mathbf{b}_g.$$

Each  $\mathbf{b}_g$  has mean zero and  $E\|\mathbf{b}_g\|^2 = \sigma_g^2 \text{tr}(n^{-1} \mathbf{X}_g^\top \mathbf{X}_g)$ , which is bounded, so  $\mathbf{b}_g = O_p(1)$ . Both  $\mathbf{A}_n(\hat{\mathbf{w}})$  and  $\mathbf{A}_n(\mathbf{w})$  converge in probability to  $\mathbf{A}_w$ , which is invertible, and matrix inversion is continuous there. The difference of the two displays is

$$\{\mathbf{A}_n(\hat{\mathbf{w}})^{-1} - \mathbf{A}_n(\mathbf{w})^{-1}\} \sum_g \hat{w}_g \mathbf{b}_g + \mathbf{A}_n(\mathbf{w})^{-1} \sum_g (\hat{w}_g - w_g) \mathbf{b}_g,$$

a sum of products of  $o_p(1)$  and  $O_p(1)$  terms, so it is  $o_p(1)$ . Finally  $n(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} =$

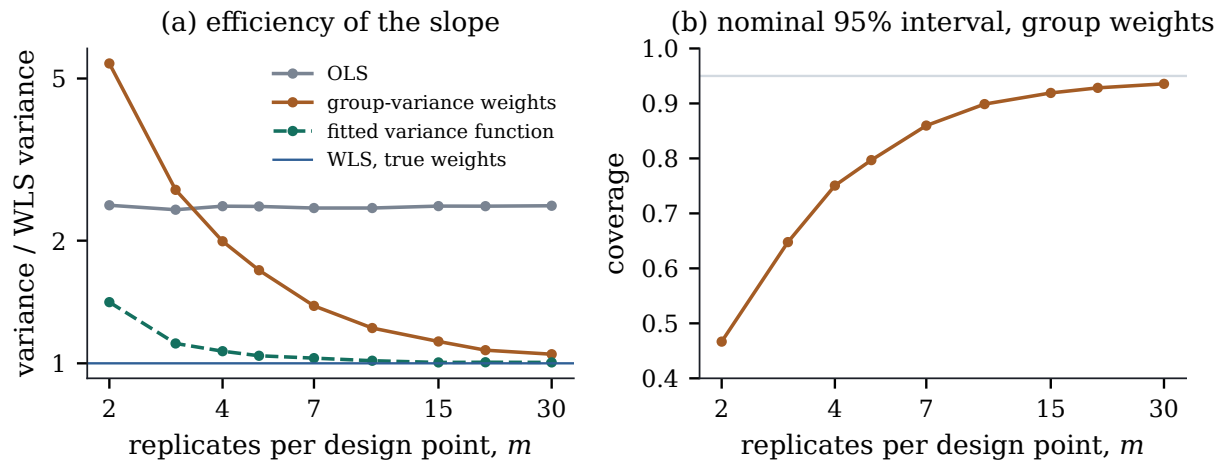


Figure 21.3.1: Feasible weighted least squares with  $m$  replicates at each of ten design points and standard deviation proportional to  $x$  (20,000 simulated data sets for each  $m$ ). (a) Variance of the slope estimate relative to weighted least squares with the true weights. (b) Coverage of the nominal 95% interval computed as if the within-group weights were known.

$$\mathbf{A}_n(\hat{\mathbf{w}})^{-1} \rightarrow \mathbf{A}_w^{-1}, \text{ and } n(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} = \mathbf{A}_n(\mathbf{w})^{-1} \rightarrow \mathbf{A}_w^{-1}.$$

Part (c) is exact but covers only weights computed from the least squares residuals, not those computed from fitted values ([Exercise \(B3\)](#)). Part (d) says that with enough replicates estimating the weights costs nothing in large samples. The scaled errors  $\sqrt{n}(\hat{\beta}_{\hat{W}} - \beta)$  and  $\sqrt{n}(\hat{\beta}_W - \beta)$  differ by  $o_p(1)$ , so whatever limiting distribution the optimal estimator has, the feasible one has the same; it is normal when, for instance, the rows of  $\mathbf{X}$  are bounded, by [Lemma 21.2.1\(a\)](#) applied within each group. The model-based covariance estimate is consistent as well. Carroll and Ruppert (1988) prove the same conclusion for smooth parametric variance functions, including functions of the mean, under conditions we do not reproduce. What (d) does not say is how many replicates are “enough”. A simulation shows that the answer can be surprisingly many.

#### Example. (Weights from a few replicates)

Take ten design points  $x = 1, \dots, 10$ ,  $m$  replicates at each, a straight-line mean, and normal errors with standard deviation proportional to  $x$ . [Figure 21.3.1](#) compares four slope estimates by their variance relative to weighted least squares with the true weights. Ordinary least squares has 2.44 times the optimal variance when  $m = 2$ , and about the same for every  $m$ . Weights  $1/s_g^2$  from the within-group sample variances are a disaster for small groups: with  $m = 2$  their variance is 5.44 times the optimum, twice that of the unweighted fit, and with  $m = 3$  it is still 2.67, worse than no weighting. They approach the optimum only slowly (1.22 for  $m = 10$ , 1.05 for  $m = 30$ ). Worse, the model-based interval, which treats the estimated weights as known, covers the true slope in only 0.47 of the data sets for  $m = 2$ , 0.65 for  $m = 3$  and 0.80 for  $m = 5$ . Weights from a fitted variance function, the regression of  $\log s_g^2$  on  $\log x_g$ , are far better: 1.41 times the optimum for  $m = 2$ , 1.12 for  $m = 3$  and 1.04 for  $m = 5$ .

```
rng = np.random.default_rng(2106)
# ten design points, sd proportional to x
xg = np.arange(1.0, 11.0)
```

```

def replicate_study(m, reps):
    """Variance of four slope estimators relative to WLS with true weights."""
    x = np.repeat(xg, m)
    Xr = np.column_stack([np.ones(len(x)), x])
    Y = 1 + 0.5 * x + rng.normal(size=(reps, len(x))) * x

    def slopes(W):
        A = np.einsum("ri,ij,ik->rjk", W, Xr, Xr)      # X'WX for each replicate
        c = np.einsum("ri,ij,ri->rj", W, Xr, Y)
        return np.linalg.solve(A, c[..., None])[..., 0], A

    b_o, _ = slopes(np.ones_like(Y))
    b_w, _ = slopes(np.tile(1 / x**2, (reps, 1)))
    # within-group variances
    s2g = Y.reshape(reps, len(xg), m).var(axis=2, ddof=1)
    Wg = np.repeat(1 / s2g, m, axis=1)
    b_g, A_g = slopes(Wg)
    # smooth model: log s2 on log x
    Zg = np.column_stack([np.ones(len(xg)), np.log(xg)])
    coef = np.linalg.lstsq(Zg, np.log(s2g).T, rcond=None)[0]
    b_m, _ = slopes(np.repeat(np.exp(-(Zg @ coef).T), m, axis=1))
    var = np.array([np.var(b[:, 1]) for b in (b_o, b_w, b_g, b_m)])
    # nominal interval for b_g
    res = Y - b_g @ Xr.T
    se = np.sqrt((Wg * res**2).sum(axis=1) / (len(x) - 2)
                  * np.linalg.inv(A_g)[:, 1, 1])
    q = stats.t.ppf(0.975, len(x) - 2)
    cover = np.mean(np.abs(b_g[:, 1] - 0.5) <= q * se)
    return var / var[1], cover

for m in (2, 3, 5, 10, 30):
    rel, cover = replicate_study(m, 2000)
    print(f"m = {m:2d}: variance relative to WLS  OLS {rel[0]:.2f}  "
          f"group weights {rel[2]:.2f}  fitted weights {rel[3]:.2f};  "
          f"coverage {cover:.2f}")

```

The failure of the group weights has a simple cause. With  $m = 2$  each  $s_g^2$  is  $\sigma_g^2 \chi^2(1)$ , which is often tiny, and  $1/s_g^2$  then gives one group an enormous weight on the strength of two nearly equal responses. In fact  $E(1/s_g^2) = \infty$  (Exercise (C1)). A fitted variance function pools the information from all groups into a few parameters and cannot be fooled by one lucky pair. The general lessons are to estimate a smooth variance function whenever possible, and to report feasible weighted estimates with standard errors that do not depend on the variance model being right. Part (b) of the theorem shows what those are: the sandwich form with  $\Sigma$  estimated from the residuals, which is the subject of the next section.

## 21.3.4 Exercises

### 21.3.4.1 A. Check your understanding

**Exercise A1.**

Under [Theorem 21.3.1\(a\)](#), show that  $E \sum_i w_i (Y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_W)^2 = \sigma^2(n - p)$ , and that the raw residual  $Y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_W$  has variance  $\sigma^2(1 - g_{ii})/w_i$ , where  $g_{ii}$  is the  $i$ th leverage of the weighted fit.

**Exercise A2.**

A laboratory reports only the mean  $\bar{y}_i$  of  $m_i$  replicate measurements at each of  $k$  settings  $\mathbf{x}_{(i)}$ , where the individual measurements have common variance  $\sigma^2$ . Which weights should be used? What does  $s_w^2$  then estimate, and why is it not the replicate variance  $\sigma^2$  unless the model for the means is correct?

**21.3.4.2 B. Practice****Exercise B1.**

In the model  $y_i = \beta x_i + \varepsilon_i$  with  $x_i > 0$  and  $\text{Var}(\varepsilon_i) = \sigma^2 x_i^2$ , show that the weighted estimate is the mean of the ratios  $y_i/x_i$ , and that the efficiency of ordinary least squares relative to it is  $(\sum_i x_i^2)^2 / (n \sum_i x_i^4) \leq 1$ . Evaluate the efficiency for  $x = 1, \dots, 10$ .

*Solution.* With  $w_i = x_i^{-2}$ ,  $\hat{\beta}_W = \sum_i w_i x_i y_i / \sum_i w_i x_i^2 = n^{-1} \sum_i y_i / x_i$ , with variance  $\sigma^2/n$ . Ordinary least squares gives  $\sum_i x_i y_i / \sum_i x_i^2$ , with variance  $\sigma^2 \sum_i x_i^4 / (\sum_i x_i^2)^2$ . The ratio is  $(\sum_i x_i^2)^2 / (n \sum_i x_i^4)$ , at most one by the Cauchy–Schwarz inequality applied to the vectors  $(x_i^2)$  and  $(1)$ . For  $x = 1, \dots, 10$ ,  $\sum x_i^2 = 385$  and  $\sum x_i^4 = 25,333$ , so the efficiency is  $148,225/253,330 = 0.585$ .

**Exercise B2.**

Some programs implement integer weights  $w_i$  by repeating row  $i$  of the data  $w_i$  times and running ordinary least squares. Show that this gives the correct  $\hat{\boldsymbol{\beta}}_W$  but divides the weighted residual sum of squares by  $\sum_i w_i - p$  instead of  $n - p$ . Which standard errors are then wrong, and in which direction?

**Exercise B3.**

Weights  $\hat{w}_i = |\hat{Y}_i|^{-2\theta}$  depend on the fitted values. Show that they are not functions of  $\hat{\boldsymbol{\varepsilon}}$  alone, and find where the symmetry argument of [Theorem 21.3.1\(c\)](#) fails. What happens for a single regressor through the origin,  $Y_i = \beta x_i + \varepsilon_i$  with every  $x_i \neq 0$ ?

*Solution.*  $\hat{\mathbf{Y}} = \mathbf{X}\boldsymbol{\beta} + \mathbf{M}\boldsymbol{\varepsilon}$  depends on  $\boldsymbol{\beta}$ , and replacing  $\boldsymbol{\varepsilon}$  by  $-\boldsymbol{\varepsilon}$  turns it into  $\mathbf{X}\boldsymbol{\beta} - \mathbf{M}\boldsymbol{\varepsilon}$ , which does not have the same absolute entries. So  $\hat{\mathbf{W}}$  is not an even function of  $\boldsymbol{\varepsilon}$ , and  $\mathbf{g}(-\boldsymbol{\varepsilon}) \neq -\mathbf{g}(\boldsymbol{\varepsilon})$  in general. The feasible estimate then need not be unbiased in finite samples, and the argument of [Theorem 21.3.1\(c\)](#) gives no guarantee.

The single regressor through the origin is an exception. There  $\hat{Y}_i = \hat{\beta}x_i$ , so  $\hat{w}_i = |\hat{\beta}|^{-2\theta} |x_i|^{-2\theta}$  whenever  $\hat{\beta} \neq 0$ , an event of probability one for continuously distributed errors. The random factor  $|\hat{\beta}|^{-2\theta}$  is common to all the weights and cancels from  $\hat{\beta}_{\hat{\mathbf{W}}} =$

$\sum_i \hat{w}_i x_i Y_i / \sum_i \hat{w}_i x_i^2$ . So the feasible estimate equals weighted least squares with the fixed weights  $|x_i|^{-2\theta}$ . It is unbiased by [Theorem 21.3.1\(b\)](#), and if  $\text{Var}(\varepsilon_i)$  is proportional to  $|x_i|^{2\theta}$  it is the best linear unbiased estimator by [Theorem 21.3.1\(a\)](#); for  $\theta = 1$  this is the ratio-type case of variance proportional to  $x_i^2$ . With two or more regressors the ratios  $\hat{w}_i/\hat{w}_j = |\hat{Y}_j/\hat{Y}_i|^{2\theta}$  depend on the data, and the argument fails again.

### 21.3.4.3 C. Going deeper

#### Exercise C1.

Show that  $E(1/U) = \infty$  if  $U \sim \chi^2(1)$ , and that  $E(1/U) = 1/(k-2)$  if  $U \sim \chi^2(k)$  with  $k \geq 3$ . Explain what this implies for weights  $1/s_g^2$  computed from  $m$  replicates.

*Solution.* The density of  $\chi^2(k)$  is proportional to  $u^{k/2-1}e^{-u/2}$ , so  $E(1/U)$  is proportional to  $\int_0^\infty u^{k/2-2}e^{-u/2} du$ , which diverges at zero when  $k/2 - 2 \leq -1$ , that is, for  $k \leq 2$ . For  $k \geq 3$  the integral is a gamma integral and gives  $E(1/U) = \Gamma(k/2 - 1)/\{2\Gamma(k/2)\} = 1/(k-2)$ . With  $m$  normal replicates,  $(m-1)s_g^2/\sigma_g^2 \sim \chi^2(m-1)$ , so  $1/s_g^2$  has infinite mean for  $m \leq 3$  and mean  $(m-1)/\{(m-3)\sigma_g^2\}$  for  $m \geq 4$ . The group weights are both inflated and wildly variable in small groups.

## 21.4 Heteroscedasticity-consistent covariance estimators

Weighting needs a model for the variances. Often there is none worth trusting, and the analyst would rather keep ordinary least squares, which is unbiased whatever the variances are, and fix its standard errors. The covariance of  $\hat{\beta}$  under  $\text{Cov}(\varepsilon) = \Sigma = \text{diag}(\sigma_1^2, \dots, \sigma_n^2)$  is

$$\text{Cov}(\hat{\beta}) = (\mathbf{X}^\top \mathbf{X})^{-1} \left( \sum_{i=1}^n \sigma_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right) (\mathbf{X}^\top \mathbf{X})^{-1} \quad (21.4.1)$$

This is the covariance given by [Theorem 19.2.2](#). The  $n$  variances cannot be estimated, one observation each. The idea of Eicker (1963), Huber (1967) and White (1980) is that they do not need to be. Only the  $p \times p$  matrix in the middle is needed, and a sum of  $n$  terms can be estimated consistently even when its terms cannot. The matrix is sandwiched between two copies of  $(\mathbf{X}^\top \mathbf{X})^{-1}$ , which gives the estimators their name.

### 21.4.1 The estimators

#### Definition 21.4.1 — Heteroscedasticity-consistent covariance estimators

Let  $\hat{\varepsilon}_i$  and  $h_{ii}$  be the least squares residuals and leverages, with  $h_{ii} < 1$ . For weights  $\omega_i$ ,



put

$$\hat{\mathbf{V}}_{\omega} = (\mathbf{X}^{\top} \mathbf{X})^{-1} \left( \sum_{i=1}^n \omega_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^{\top} \right) (\mathbf{X}^{\top} \mathbf{X})^{-1}.$$

The estimators **HC0** to **HC3** take

$$\omega_i = \hat{\varepsilon}_i^2, \quad \frac{n}{n-p} \hat{\varepsilon}_i^2, \quad \frac{\hat{\varepsilon}_i^2}{1-h_{ii}}, \quad \frac{\hat{\varepsilon}_i^2}{(1-h_{ii})^2}.$$

HC0 is White's estimator. HC1 applies the degrees-of-freedom correction of  $s^2$ . HC2 and HC3, proposed by MacKinnon and White (1985), correct each squared residual for its own leverage. Their finite-sample properties follow from [Eq. 21.2.1](#) and from the deletion formulas of [Chapter 10](#).

### Proposition 21.4.1 — Finite-sample properties

Let  $\boldsymbol{\lambda} \in \mathbb{R}^p$  and  $\mathbf{c} = \mathbf{X}(\mathbf{X}^{\top} \mathbf{X})^{-1} \boldsymbol{\lambda}$ , so that  $\boldsymbol{\lambda}^{\top} \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^{\top} \boldsymbol{\beta} = \mathbf{c}^{\top} \boldsymbol{\varepsilon}$  and  $\boldsymbol{\lambda}^{\top} \hat{\mathbf{V}}_{\omega} \boldsymbol{\lambda} = \sum_i c_i^2 \omega_i$ .

- (a) Under constant variance  $\sigma^2$ ,  $E(\boldsymbol{\lambda}^{\top} \hat{\mathbf{V}}_{\text{HC2}} \boldsymbol{\lambda}) = \text{Var}(\boldsymbol{\lambda}^{\top} \hat{\boldsymbol{\beta}})$ , while  $E(\boldsymbol{\lambda}^{\top} \hat{\mathbf{V}}_{\text{HC0}} \boldsymbol{\lambda}) = \text{Var}(\boldsymbol{\lambda}^{\top} \hat{\boldsymbol{\beta}}) - \sigma^2 \sum_i c_i^2 h_{ii} \leq \text{Var}(\boldsymbol{\lambda}^{\top} \hat{\boldsymbol{\beta}})$  and  $E(\boldsymbol{\lambda}^{\top} \hat{\mathbf{V}}_{\text{HC3}} \boldsymbol{\lambda}) \geq \text{Var}(\boldsymbol{\lambda}^{\top} \hat{\boldsymbol{\beta}})$ .
- (b) In general,  $E(\boldsymbol{\lambda}^{\top} \hat{\mathbf{V}}_{\text{HC0}} \boldsymbol{\lambda}) = \sum_i c_i^2 \sum_j (\mathbf{I} - \mathbf{M})_{ij}^2 \sigma_j^2$ .
- (c)  $\hat{\mathbf{V}}_{\text{HC0}} \preceq \hat{\mathbf{V}}_{\text{HC2}} \preceq \hat{\mathbf{V}}_{\text{HC3}}$  in the order of nonnegative definiteness.
- (d)  $\hat{\mathbf{V}}_{\text{HC3}} = \sum_{i=1}^n (\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}})(\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}})^{\top}$ , where  $\hat{\boldsymbol{\beta}}_{(i)}$  is the estimate with observation  $i$  deleted.

*Proof.*

- (a) and (b).  $E \hat{\varepsilon}_i^2 = \sum_j (\mathbf{I} - \mathbf{M})_{ij}^2 \sigma_j^2$  by [Eq. 21.2.1](#), which proves (b). Under constant variance it is  $\sigma^2(1-h_{ii})$ , so the HC2 weights have expectation  $\sigma^2$  and  $E \sum_i c_i^2 \omega_i = \sigma^2 \|\mathbf{c}\|^2 = \text{Var}(\mathbf{c}^{\top} \boldsymbol{\varepsilon})$ . The HC0 weights have expectation  $\sigma^2(1-h_{ii})$  and the HC3 weights  $\sigma^2/(1-h_{ii}) \geq \sigma^2$ .
- (b) The HC weights satisfy  $\hat{\varepsilon}_i^2 \leq \hat{\varepsilon}_i^2/(1-h_{ii}) \leq \hat{\varepsilon}_i^2/(1-h_{ii})^2$  term by term, and each difference  $\hat{\mathbf{V}}_{\omega'} - \hat{\mathbf{V}}_{\omega}$  with  $\omega'_i \geq \omega_i$  is  $(\mathbf{X}^{\top} \mathbf{X})^{-1} \{ \sum_i (\omega'_i - \omega_i) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^{\top} \} (\mathbf{X}^{\top} \mathbf{X})^{-1}$ , a nonnegative definite matrix.
- (c) By [Proposition 10.6.2\(a\)](#),  $\hat{\boldsymbol{\beta}}_{(i)} - \hat{\boldsymbol{\beta}} = -(\mathbf{X}^{\top} \mathbf{X})^{-1} \mathbf{x}_{(i)} \hat{\varepsilon}_i / (1-h_{ii})$ . The sum of the outer products of these vectors is  $\hat{\mathbf{V}}_{\text{HC3}}$ . ■

Part (d) says that HC3 measures how much the estimate moves when single observations are left out. This is the jackknife idea: the variability of an estimate is judged from its sensitivity to the data, not from a model for the errors. Part (a) makes HC0 biased downward when the model is right, and the bias is concentrated where  $c_i^2 h_{ii}$  is large, at high-leverage points that dominate the coefficient. Part (b) shows the same effect under heteroscedasticity: a high-leverage observation's squared residual understates its own variance, precisely when that variance matters most.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
```



```

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)
X = np.column_stack([np.ones(n), income])
Q, R = np.linalg.qr(X)
beta = np.linalg.solve(R, Q.T @ food)
e = food - X @ beta
h = np.sum(Q**2, axis=1) # leverages
Rinv = np.linalg.inv(R) # (X'X)^{-1} = Rinv Rinv'

def sandwich(omega):
    """(X'X)^{-1} X' diag(omega) X (X'X)^{-1}, computed through Q."""
    meat = (Q * omega[:, None]).T @ Q
    return Rinv @ meat @ Rinv.T

p = X.shape[1]
covs = {
    "classical": (e @ e / (n - p)) * (Rinv @ Rinv.T),
    "HC0": sandwich(e**2),
    "HC1": sandwich(e**2) * n / (n - p),
    "HC2": sandwich(e**2 / (1 - h)),
    "HC3": sandwich(e**2 / (1 - h) ** 2),
}
for name, V in covs.items():
    print(f"{name:9s} slope se {np.sqrt(V[1, 1]):.4f}")

```

```

loo = np.array([np.linalg.lstsq(np.delete(X, i, 0), np.delete(food, i),
                                rcond=None)[0]
                for i in range(n)])
# sum over i of (b_(i) - b)(b_(i) - b)'
jack = (loo - beta).T @ (loo - beta)
print("HC3 equals the sum of squared deletion changes:",
      np.allclose(jack, covs["HC3"]))

```

### Example. (Standard errors for Engel's slope)

For the least squares slope of food expenditure on income, the classical standard error  $s\{(\mathbf{X}^\top \mathbf{X})^{-1}\}_{22}^{1/2}$  is 0.0144. The heteroscedasticity-consistent standard errors are 0.0518 (HC0), 0.0520 (HC1), 0.0585 (HC2) and 0.0664 (HC3). The classical value is too small by a factor of 4.6 relative to HC3. Evidently the high-income households, which pin down the slope, are the noisy ones, so the classical formula, which averages the noise over all households, understates the uncertainty. The second listing above checks the jackknife identity of [Proposition 21.4.1\(d\)](#).

## 21.4.2 Consistency

The estimators are justified by a large-sample argument. We give it under simple conditions that make every step elementary: bounded regressors, errors with bounded fourth moments, and variances bounded away from zero. The theorem covers every  $\hat{\mathbf{V}}_\omega$  of [Definition 21.4.1](#).

### Theorem 21.4.2 — Consistency of the sandwich estimators

Let  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $p$  fixed. Assume that

- (H1) the errors are independent with mean zero and variances  $\sigma_i^2$ , with  $\sigma_i^2 \geq \sigma_{\min}^2 > 0$  and  $E\varepsilon_i^4 \leq K$  for all  $i$ ;
- (H2) the rows satisfy  $\|\mathbf{x}_{(i)}\| \leq C$  for all  $i$ ;
- (H3)  $n^{-1}\mathbf{X}^\top\mathbf{X} \rightarrow \mathbf{A}$ , a positive definite matrix.

Let  $\hat{\mathbf{V}}$  be any of HC0–HC3, and  $\mathbf{V} = \text{Cov}(\hat{\boldsymbol{\beta}})$  as in Eq. 21.4.1. Then:

- (a)  $n(\hat{\mathbf{V}} - \mathbf{V}) \rightarrow \mathbf{0}$  in probability;
- (b) for every fixed  $\boldsymbol{\lambda} \neq \mathbf{0}$ ,  $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/(\boldsymbol{\lambda}^\top \hat{\mathbf{V}} \boldsymbol{\lambda})^{1/2} \rightarrow N(0, 1)$  in distribution;
- (c) for every fixed  $p \times q$  matrix  $\boldsymbol{\Lambda}$  of rank  $q$ ,

$$W = (\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\Lambda}^\top \boldsymbol{\beta})^\top (\boldsymbol{\Lambda}^\top \hat{\mathbf{V}} \boldsymbol{\Lambda})^{-1} (\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\Lambda}^\top \boldsymbol{\beta})$$

tends to  $\chi^2(q)$  in distribution.

*Proof.* Write  $\mathbf{G}_n = n^{-1}\mathbf{X}^\top\mathbf{X}$ , and let  $\mathbf{B}_n = n^{-1} \sum_i \sigma_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$  and  $\hat{\mathbf{B}}_n = n^{-1} \sum_i \omega_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$  be the true and estimated middle matrices, so that  $n\mathbf{V} = \mathbf{G}_n^{-1} \mathbf{B}_n \mathbf{G}_n^{-1}$  and  $n\hat{\mathbf{V}} = \mathbf{G}_n^{-1} \hat{\mathbf{B}}_n \mathbf{G}_n^{-1}$ . Note  $\sigma_i^2 \leq (E\varepsilon_i^4)^{1/2} \leq K^{1/2}$ . By (H3),  $\mathbf{G}_n^{-1} \rightarrow \mathbf{A}^{-1}$ , and  $\lambda_{\min}(\mathbf{X}^\top\mathbf{X}) \geq cn$  for some  $c > 0$  and all large  $n$ .

*Step 1:  $\hat{\boldsymbol{\beta}}$  is consistent.*  $E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \text{tr } \mathbf{V} \leq K^{1/2} \text{tr}(\mathbf{X}^\top\mathbf{X})^{-1} \leq K^{1/2}p/(cn) \rightarrow 0$ .

*Step 2: the errors estimate the middle.* Each entry of  $n^{-1} \sum_i (\varepsilon_i^2 - \sigma_i^2) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$  is a sum of independent mean-zero terms with variance at most  $n^{-2} \sum_i KC^4 = KC^4/n$ , so it tends to zero in probability by Chebyshev's inequality.

*Step 3: the residuals estimate the middle.* Let  $d_i = \mathbf{x}_{(i)}^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ , so  $\hat{\varepsilon}_i = \varepsilon_i - d_i$  and  $|d_i| \leq C\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|$ . Then, in the spectral norm,

$$\begin{aligned} \left\| n^{-1} \sum_i (\hat{\varepsilon}_i^2 - \varepsilon_i^2) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right\| &\leq \frac{C^2}{n} \sum_i (d_i^2 + 2|d_i||\varepsilon_i|) \\ &\leq C^4 \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 + 2C^3 \|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\| \frac{1}{n} \sum_i |\varepsilon_i|, \end{aligned}$$

which tends to zero because  $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = o_p(1)$  and  $E n^{-1} \sum_i |\varepsilon_i| \leq K^{1/4}$ . With Step 2,  $\hat{\mathbf{B}}_n - \mathbf{B}_n \rightarrow \mathbf{0}$  for HC0.

*Step 4: the leverage corrections vanish.*  $h_{ii} \leq C^2/\lambda_{\min}(\mathbf{X}^\top\mathbf{X}) \leq C^2/(cn)$ , so  $\delta_n = (1 - \max_i h_{ii})^{-2} - 1 \rightarrow 0$ . For HC1–HC3,  $\hat{\varepsilon}_i^2 \leq \omega_i \leq (1 + \delta_n)\varepsilon_i^2$  for large  $n$  (with  $\delta_n$  replaced by  $p/(n-p)$  for HC1), so  $\mathbf{0} \leq \hat{\mathbf{B}}_n - \hat{\mathbf{B}}_n^{\text{HC0}} \leq \delta_n \hat{\mathbf{B}}_n^{\text{HC0}}$ , which tends to zero because  $\delta_n \rightarrow 0$  and  $\hat{\mathbf{B}}_n^{\text{HC0}} = \mathbf{B}_n + o_p(1)$  is bounded in probability, since  $\mathbf{0} \leq \mathbf{B}_n \leq K^{1/2}\mathbf{G}_n$  and  $\mathbf{G}_n$  converges. This proves (a), since  $n(\hat{\mathbf{V}} - \mathbf{V}) = \mathbf{G}_n^{-1}(\hat{\mathbf{B}}_n - \mathbf{B}_n)\mathbf{G}_n^{-1}$ .

- (c) Let  $\mathbf{V}_\Lambda = \boldsymbol{\Lambda}^\top \mathbf{V} \boldsymbol{\Lambda}$  and  $\mathbf{T}_n = \mathbf{V}_\Lambda^{-1/2} \boldsymbol{\Lambda}^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ , so that  $\text{Cov}(\mathbf{T}_n) = \mathbf{I}_q$ . For fixed  $\mathbf{a} \in \mathbb{R}^q$ ,  $\mathbf{a}^\top \mathbf{T}_n = \mathbf{c}^\top \boldsymbol{\varepsilon}$  with  $\mathbf{c} = \mathbf{X}(\mathbf{X}^\top\mathbf{X})^{-1}\boldsymbol{\lambda}$  and  $\boldsymbol{\lambda} = \boldsymbol{\Lambda} \mathbf{V}_\Lambda^{-1/2} \mathbf{a}$ . By the Cauchy–Schwarz inequality in the metric  $(\mathbf{X}^\top\mathbf{X})^{-1}$ ,  $c_i^2 \leq h_{ii}\|\mathbf{c}\|^2$ , so  $\max_i c_i^2/\|\mathbf{c}\|^2 \leq$

$\max_i h_{ii} \rightarrow 0$ . [Lemma 21.2.1](#)(b), with  $\delta = 2$ , shows that  $\mathbf{a}^\top \mathbf{T}_n / \|\mathbf{a}\| \rightarrow N(0, 1)$  for  $\mathbf{a} \neq \mathbf{0}$ . By the Cramér–Wold device,  $\mathbf{T}_n \rightarrow N_q(\mathbf{0}, \mathbf{I}_q)$ . Next,  $n\mathbf{V}_\Lambda = \mathbf{\Lambda}^\top \mathbf{G}_n^{-1} \mathbf{B}_n \mathbf{G}_n^{-1} \mathbf{\Lambda} \succeq \sigma_{\min}^2 \mathbf{\Lambda}^\top \mathbf{G}_n^{-1} \mathbf{\Lambda}$ , whose limit  $\sigma_{\min}^2 \mathbf{\Lambda}^\top \mathbf{A}^{-1} \mathbf{\Lambda}$  is positive definite; and  $n\mathbf{V}_\Lambda$  is bounded because  $\mathbf{B}_n \preceq K^{1/2} \mathbf{G}_n$ . So  $(n\mathbf{V}_\Lambda)^{-1/2}$  is bounded, and by (a)

$$\begin{aligned} \mathbf{V}_\Lambda^{-1/2} (\mathbf{\Lambda}^\top \hat{\mathbf{V}} \mathbf{\Lambda}) \mathbf{V}_\Lambda^{-1/2} &= \mathbf{I}_q + (n\mathbf{V}_\Lambda)^{-1/2} \mathbf{\Lambda}^\top \\ &\quad \times n(\hat{\mathbf{V}} - \mathbf{V}) \mathbf{\Lambda} (n\mathbf{V}_\Lambda)^{-1/2} \rightarrow \mathbf{I}_q. \end{aligned}$$

Since  $W = \mathbf{T}_n^\top \{ \mathbf{V}_\Lambda^{-1/2} (\mathbf{\Lambda}^\top \hat{\mathbf{V}} \mathbf{\Lambda}) \mathbf{V}_\Lambda^{-1/2} \}^{-1} \mathbf{T}_n$ , continuous mapping and Slutsky’s lemma give  $W \rightarrow \mathbf{T}^\top \mathbf{T}$  with  $\mathbf{T} \sim N_q(\mathbf{0}, \mathbf{I}_q)$ , which is  $\chi^2(q)$ . Part (b) is the case  $q = 1$ , with the sign kept. ■

The theorem does not assume that the errors are normal, that their variances follow any model, or that they are identically distributed. It does assume independence, which the next two sections relax. The bounded regressors of (H2) are stronger than needed. Eicker (1967) and White (1980) replace them by conditions that amount to  $\max_i h_{ii} \rightarrow 0$  together with moment bounds, and White’s version allows random regressors drawn independently from a common distribution, which covers most observational data. We do not prove these extensions.

In practice the statistic in (b) is compared with the  $t(n-p)$  distribution rather than the normal. That choice has no exact justification, but it costs nothing asymptotically and helps in small samples. A heteroscedasticity-robust test of  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{d}$  compares  $W/q$  with  $F(q, n-p)$ .

### 21.4.3 Finite-sample behaviour

Consistency says nothing about how large  $n$  must be. The answer depends on the leverages, as [Proposition 21.4.1](#) suggests, and it can be sobering.

#### Example. (Coverage in a skewed design)

Let the regressor values be the  $n$  quantiles  $x_i = \exp\{\Phi^{-1}((i-1/2)/n)\}$  of a standard lognormal distribution, a design with a long right tail like incomes. The largest leverage is 0.58 for  $n = 25$  and still 0.15 for  $n = 1000$ . [Figure 21.4.1](#) shows the coverage of nominal 95% intervals for the slope, with normal errors, over 20,000 simulated data sets for each  $n$ .

With constant variance (panel a) the classical interval is exact, and the robust intervals pay a price for their robustness: at  $n = 25$  they cover 0.852 (HC0), 0.866 (HC1), 0.895 (HC2) and 0.932 (HC3). With standard deviation proportional to  $x$  (panel b) the classical interval fails badly. Its coverage is 0.456 at  $n = 25$  and falls to 0.287 at  $n = 1000$ , because the growing right tail of the design gives ever more influence to the noisiest observations. The robust intervals improve with  $n$ , but slowly. At  $n = 25$  they cover 0.631 (HC0), 0.650 (HC1), 0.760 (HC2) and 0.853 (HC3); at  $n = 1000$ , 0.912, 0.912, 0.924 and 0.935. HC3 is the best of the four throughout, as Long and Ervin (2000) found in a wide range of designs.

```
rng = np.random.default_rng(2107)
```

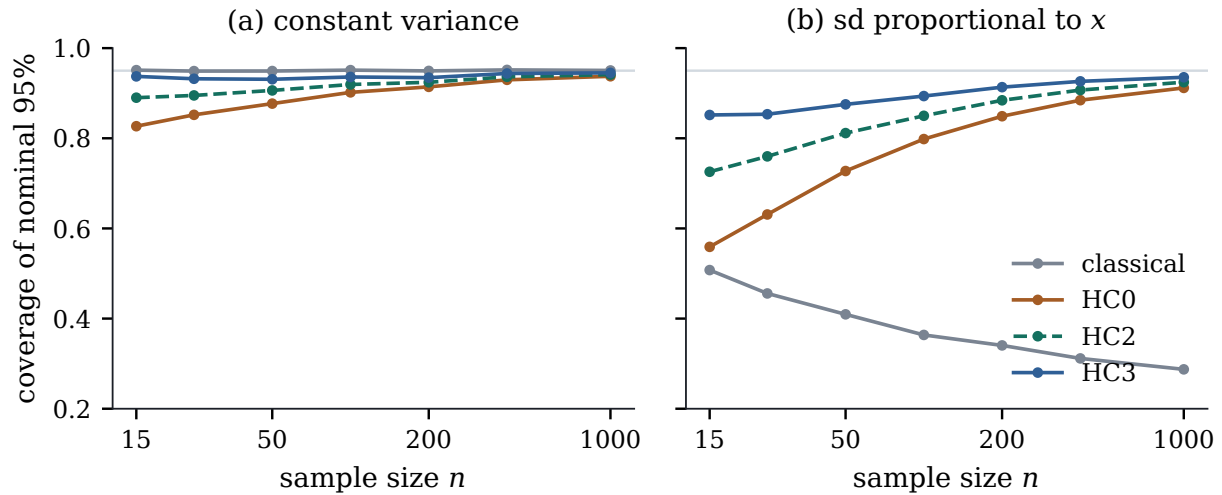


Figure 21.4.1: Coverage of nominal 95% intervals for the slope in a design with lognormal quantiles as regressor values. (a) Constant error variance. (b) Error standard deviation proportional to  $x$ . HC1 is omitted; it lies just above HC0.

```
def coverage(n, hetero, reps):
    """Coverage of nominal 95% slope intervals, classical and HC0-HC3."""
    # lognormal quantiles
    x = np.exp(stats.norm.ppf((np.arange(1, n + 1) - 0.5) / n))
    Xs = np.column_stack([np.ones(n), x])
    Qs, Rs = np.linalg.qr(Xs)
    hs = np.sum(Qs**2, axis=1)
    # slope estimate - slope = c @ errors
    c = Qs @ np.linalg.inv(Rs).T[:, 1]
    E = rng.normal(size=(reps, n)) * (x if hetero else 1.0)
    err = E @ c
    res = E - (E @ Qs) @ Qs.T
    var = {"classical": (res**2).sum(axis=1) / (n - 2) * (c @ c),
          "HC0": (res**2) @ c**2,
          "HC1": (res**2) @ c**2 * n / (n - 2),
          "HC2": (res**2 / (1 - hs)) @ c**2,
          "HC3": (res**2 / (1 - hs) ** 2) @ c**2}
    q = stats.t.ppf(0.975, n - 2)
    cover = {k: np.mean(np.abs(err) <= q * np.sqrt(v)) for k, v in var.items()}
    return cover, hs.max()

for n_sim in (25, 100, 400):
    cov, hmax = coverage(n_sim, hetero=True, reps=2000)
    print(f"n = {n_sim:3d} (max leverage {hmax:.2f}):",
          {k: round(v, 3) for k, v in cov.items()})
```

## 21.4.4 Leverage and the sandwich

The poor small-sample behaviour has a simple source. The variance estimate  $\sum_i c_i^2 \omega_i$  is a weighted sum of squared residuals, and when a few weights  $c_i^2$  dominate, it behaves like a variance estimate with very few degrees of freedom, however large  $n$  is. A  $t(n-p)$  reference distribution then has tails that are far too light.

**Example. (One household carries the standard error)**

In Engel's data the household with the largest income has leverage 0.255. Its term  $c_i^2 \hat{\varepsilon}_i^2 / (1 - h_{ii})^2$  makes up 0.85 of the HC3 variance of the slope. The robust standard error of [Example \(Standard errors for Engel's slope\)](#) is, in effect, an estimate of one household's error variance. That is not a reason to go back to the classical standard error, which is far worse. It is a reason to look at that household (the influence measures of [Chapter 20](#), such as [Definition 20.4.1](#), are the tools), to report how the conclusions change without it, and to prefer weighting, which here gives that household much less influence ([Example \(Engel's data, reweighted\)](#)).

Several refinements address the degrees-of-freedom problem. Bell and McCaffrey's (2002) adjustment, as recommended by Imbens and Kolesár (2016), combines HC2 with a Satterthwaite approximation to the degrees of freedom of  $\sum_i c_i^2 \omega_i$ . The wild bootstrap of [Chapter 23](#) ([Definition 23.2.1](#)) resamples the residuals with random signs, keeping each one attached to its own row, and often gives better calibrated intervals than any of the HC formulas.

## 21.4.5 Exercises

### 21.4.5.1 A. Check your understanding

**Exercise A1.**

Take the two-group model  $y_{gj} = \mu_g + \varepsilon_{gj}$ ,  $g = 1, 2$ ,  $j = 1, \dots, n_g$ . Show that HC2 for  $\hat{\mu}_1 - \hat{\mu}_2$  equals  $s_1^2/n_1 + s_2^2/n_2$ , the variance estimate of Welch's test, where  $s_g^2$  is the sample variance of group  $g$ . What does HC0 give?

*Solution.* Here  $h_{ii} = 1/n_g$ ,  $\hat{\varepsilon}_{gj} = y_{gj} - \bar{y}_g$ , and  $\mathbf{c}$  has entries  $1/n_1$  in group 1 and  $-1/n_2$  in group 2. HC2 is  $\sum_g \sum_j \hat{\varepsilon}_{gj}^2 / \{(1 - 1/n_g)n_g\} = \sum_g \sum_j (y_{gj} - \bar{y}_g)^2 / \{(n_g - 1)n_g\} = \sum_g s_g^2/n_g$ . HC0 gives  $\sum_g (n_g - 1)s_g^2/n_g^2$ , which is smaller.

**Exercise A2.**

What happens to HC0, HC2 and HC3 if some observation has  $h_{ii} = 1$ ? Explain why no estimator built from residuals can estimate that observation's contribution to  $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})$ .

### 21.4.5.2 B. Practice

**Exercise B1.**

In simple regression with an intercept and constant variance, show that the relative bias of HC0 for the slope variance is  $-\sum_i (x_i - \bar{x})^2 h_{ii} / S_{xx}$ . Evaluate it for  $x = (1, 2, 3, 4, 10)$  of [Exercise \(A1\)](#).

*Solution.* For the slope,  $c_i = (x_i - \bar{x}) / S_{xx}$  and  $\|\mathbf{c}\|^2 = 1/S_{xx}$ , so [Proposition 21.4.1\(a\)](#) gives relative bias  $-\sum_i c_i^2 h_{ii} / \|\mathbf{c}\|^2 = -\sum_i (x_i - \bar{x})^2 h_{ii} / S_{xx}$ . With  $S_{xx} = 50$  and the leverages  $(0.38, 0.28, 0.22, 0.20, 0.92)$ , the sum is  $9(0.38) + 4(0.28) + 1(0.22) + 0 + 36(0.92) =$

37.88, and the relative bias is  $-37.88/50 = -0.76$ . HC0 estimates the slope variance at a quarter of its true value, on average, because the point  $x = 10$  dominates the slope and has a tiny residual.

### Exercise B2.

Using [Theorem 21.3.1\(b\)](#), write down a sandwich estimator of the covariance of a weighted least squares estimate with fixed weights  $\mathbf{W}$ , and show that it reduces to HC0 applied to the rescaled data  $(\mathbf{W}^{1/2}\mathbf{X}, \mathbf{W}^{1/2}\mathbf{y})$ .

### Exercise B3.

White's test compares  $s^2\mathbf{X}^\top\mathbf{X}$  with  $\sum_i \hat{\varepsilon}_i^2 \mathbf{x}_{(i)}\mathbf{x}_{(i)}^\top$ . Show that the difference of these matrices, divided by  $n$ , has entries  $n^{-1} \sum_i (\hat{\varepsilon}_i^2 - s^2) x_{ij}x_{ik}$ , and explain why Koenker's statistic with  $\mathbf{z}_i$  the distinct products  $x_{ij}x_{ik}$  tests whether these are jointly zero.

### 21.4.5.3 C. Going deeper

#### Exercise C1.

Suppose the rows  $(\mathbf{x}_{(i)}, \varepsilon_i)$  are independent copies of a random pair with  $E(\varepsilon | \mathbf{x}) = 0$ ,  $E\|\mathbf{x}\|^4 < \infty$ ,  $E\varepsilon^4 < \infty$  and  $E\mathbf{x}\mathbf{x}^\top$  positive definite. Adapt the proof of [Theorem 21.4.2](#) to show that HC0 is consistent in the sense of part (a), replacing Chebyshev's inequality by the weak law of large numbers.

#### Exercise C2.

The usual jackknife variance is  $\frac{n-1}{n} \sum_i (\hat{\beta}_{(i)} - \mathbf{m})(\hat{\beta}_{(i)} - \mathbf{m})^\top$ , where  $\mathbf{m}$  is the mean of the  $\hat{\beta}_{(i)}$ . Express it in terms of  $\hat{\mathbf{V}}_{\text{HC3}}$  and the vector  $\sum_i \mathbf{x}_{(i)} \hat{\varepsilon}_i / (1 - h_{ii})$ , and show that the two agree to first order when  $\max_i h_{ii} \rightarrow 0$ .

## 21.5 Serial correlation and the Durbin–Watson test

When observations are taken in sequence, over time or along a transect, neighbouring errors are often alike. A shock to the economy lasts more than one quarter, a drifting instrument affects consecutive readings, and a variable missing from the model usually changes slowly, so its effect appears in the errors as a slow wave. Errors that are correlated with their neighbours are **serially correlated** or **autocorrelated**. As with heteroscedasticity, least squares stays unbiased and its usual standard errors become wrong ([Theorem 19.2.2](#)). The error is typically in the dangerous direction: with positive autocorrelation and a slowly varying regressor, the true variance of a slope can be many times its nominal value. [Section 21.6](#) measures this. This section is about detection.

## 21.5.1 Looking at residuals in time order

Three displays are standard. The residuals plotted against time show runs of the same sign when the autocorrelation is positive, and rapid alternation when it is negative. The **lag plot** of  $\hat{\varepsilon}_t$  against  $\hat{\varepsilon}_{t-1}$  shows the lag-one dependence as a tilt. And the **residual autocorrelations**

$$r_k = \frac{\sum_{t=k+1}^n \hat{\varepsilon}_t \hat{\varepsilon}_{t-k}}{\sum_{t=1}^n \hat{\varepsilon}_t^2}, \quad k = 1, 2, \dots,$$

summarize dependence at each lag. Under the model the residuals are themselves slightly correlated, with covariance  $\sigma^2(\mathbf{I} - \mathbf{M})$ , so the  $r_k$  are not exactly centred at zero, but for large  $n$  they are approximately independent with standard deviation  $n^{-1/2}$ .

### Example. (A quarterly Okun's-law regression)

Okun's law, named after the economist Arthur Okun, relates changes in unemployment to output growth. The statsmodels macroeconomic data give US quarterly series from 1959 to 2009. Regress the change in the unemployment rate (percentage points per quarter) on real GDP growth (percent, at an annual rate) for the  $n = 202$  quarters from the second quarter of 1959. The fitted line is

$$\widehat{\Delta u}_t = 0.227 - 0.0671 g_t,$$

so unemployment is stable when output grows at about  $0.227/0.0671 \approx 3.4$  percent a year, and each extra percentage point of growth lowers it by about 0.07 points per quarter. Figure 21.5.1(a,b) shows the residuals. They wander in long swings, and the lag plot tilts, with  $r_1 = 0.300$ .

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
from scipy.integrate import quad
from scipy.optimize import brentq

macro = sm.datasets.macrodatab.load_pandas().data
# change in unemployment rate
du = np.diff(macro["unemp"].to_numpy())
# GDP growth, % a year
growth = 400 * np.diff(np.log(macro["realgdp"].to_numpy()))
year = (macro["year"] + (macro["quarter"] - 1) / 4).to_numpy()[1:]
n = len(du)
X = np.column_stack([np.ones(n), growth])
beta, *_ = np.linalg.lstsq(X, du, rcond=None)
e = du - X @ beta
# Durbin-Watson statistic
d = np.sum(np.diff(e) ** 2) / np.sum(e ** 2)
# lag-one residual autocorrelation
r1 = np.sum(e[1:] * e[:-1]) / np.sum(e ** 2)
print(f"n = {n}, slope {beta[1]:.4f}, d = {d:.3f}, r1 = {r1:.3f}, "
      f"2(1 - r1) = {2 * (1 - r1):.3f}")
```

## 21.5.2 The Durbin–Watson statistic

**Definition 21.5.1 — Durbin–Watson statistic**

For residuals  $\hat{\varepsilon}_1, \dots, \hat{\varepsilon}_n$  in time order,

$$d = \frac{\sum_{t=2}^n (\hat{\varepsilon}_t - \hat{\varepsilon}_{t-1})^2}{\sum_{t=1}^n \hat{\varepsilon}_t^2} = \frac{\hat{\mathbf{\varepsilon}}^\top \mathbf{A} \hat{\mathbf{\varepsilon}}}{\hat{\mathbf{\varepsilon}}^\top \hat{\mathbf{\varepsilon}}}, \quad \mathbf{A} = \mathbf{D}^\top \mathbf{D},$$

where  $\mathbf{D}$  is the  $(n-1) \times n$  first-difference matrix, with rows  $(\dots, 0, -1, 1, 0, \dots)$ .

Expanding the numerator gives  $d = 2(1 - r_1) - (\hat{\varepsilon}_1^2 + \hat{\varepsilon}_n^2) / \sum_t \hat{\varepsilon}_t^2$ , so  $d \approx 2(1 - r_1)$ . Values near 2 indicate no lag-one correlation, values towards 0 positive correlation, and values towards 4 negative correlation. For the Okun regression  $d = 1.386$ , against  $2(1 - r_1) = 1.401$ .

The matrix  $\mathbf{A}$  is tridiagonal with diagonal  $(1, 2, \dots, 2, 1)$  and off-diagonal entries  $-1$ . Its spectrum is known exactly, and that is what makes the Durbin–Watson theory work.

**Lemma 21.5.1 — Eigenvalues of the difference matrix**

The eigenvalues of  $\mathbf{A}$  are

$$\lambda_j = 4 \sin^2 \left( \frac{\pi(j-1)}{2n} \right), \quad j = 1, \dots, n,$$

with eigenvectors  $\mathbf{v}_j = (\cos(\theta_j/2), \cos(3\theta_j/2), \dots, \cos((2n-1)\theta_j/2))^\top$ , where  $\theta_j = \pi(j-1)/n$ . In particular  $0 = \lambda_1 < \lambda_2 < \dots < \lambda_n < 4$ , and  $\mathbf{v}_1 = \mathbf{1}$ .

*Proof.* Let  $\theta = \theta_j$  and  $v_t = \cos((t-\frac{1}{2})\theta)$ , and extend the sequence by  $v_0 = \cos(-\theta/2) = v_1$  and  $v_{n+1} = \cos((n+\frac{1}{2})\theta)$ . Since  $n\theta = \pi(j-1)$ ,  $v_{n+1} = \cos(\pi(j-1) + \theta/2) = \cos(\pi(j-1) - \theta/2) = v_n$ . With these boundary values every row of  $\mathbf{A}\mathbf{v}$  has the interior form  $(\mathbf{A}\mathbf{v})_t = 2v_t - v_{t-1} - v_{t+1}$ , because the first row  $v_1 - v_2$  equals  $2v_1 - v_0 - v_2$  and similarly for the last. By the identity  $\cos(a-\theta) + \cos(a+\theta) = 2\cos a \cos \theta$ ,  $2v_t - v_{t-1} - v_{t+1} = 2(1 - \cos \theta)v_t = 4\sin^2(\theta/2)v_t$ . The  $n$  values  $\theta_j \in [0, \pi)$  give  $n$  distinct eigenvalues, so these are all of them. ■

The distribution theory also needs a classical fact about eigenvalues of compressed matrices.

**Lemma 21.5.2 — Interlacing**

Let  $\mathbf{B}$  be  $N \times N$  symmetric with eigenvalues  $\beta_1 \leq \dots \leq \beta_N$ , and let  $\mathbf{U}$  be  $N \times m$  with orthonormal columns. If  $\mu_1 \leq \dots \leq \mu_m$  are the eigenvalues of  $\mathbf{U}^\top \mathbf{B} \mathbf{U}$ , then  $\beta_k \leq \mu_k \leq \beta_{k+N-m}$  for  $k = 1, \dots, m$ .

*Proof.* Let  $\mathbf{f}_1, \dots, \mathbf{f}_m$  be orthonormal eigenvectors of  $\mathbf{U}^\top \mathbf{B} \mathbf{U}$  and  $\mathbf{q}_1, \dots, \mathbf{q}_N$  those of  $\mathbf{B}$ . The subspaces  $\mathcal{S} = \text{span}\{\mathbf{U}\mathbf{f}_1, \dots, \mathbf{U}\mathbf{f}_m\}$ , of dimension  $m$ , and  $\mathcal{T} = \text{span}\{\mathbf{q}_k, \dots, \mathbf{q}_N\}$ , of dimension  $N - k + 1$ , have dimensions adding to  $N + 1$ , so they share a nonzero vector  $\mathbf{x} = \mathbf{U}\mathbf{f}$ . Since  $\mathbf{U}$  preserves lengths,  $\mathbf{x}^\top \mathbf{B} \mathbf{x} / \mathbf{x}^\top \mathbf{x} = \mathbf{f}^\top \mathbf{U}^\top \mathbf{B} \mathbf{U} \mathbf{f} / \mathbf{f}^\top \mathbf{f} \leq \mu_k$ , because  $\mathbf{f} \in \text{span}\{\mathbf{f}_1, \dots, \mathbf{f}_m\}$ ; and  $\mathbf{x}^\top \mathbf{B} \mathbf{x} / \mathbf{x}^\top \mathbf{x} \geq \beta_k$ , because  $\mathbf{x} \in \mathcal{T}$ . Both bounds are [Theo-](#)



rem 1.7.7 applied to the restriction of the quadratic form to the span of eigenvectors. So  $\beta_k \leq \mu_k$ . The upper bound follows in the same way from  $\text{span}\{\mathbf{U}\mathbf{f}_k, \dots, \mathbf{U}\mathbf{f}_m\}$  and  $\text{span}\{\mathbf{q}_1, \dots, \mathbf{q}_{k+N-m}\}$ , whose dimensions add to  $N + 1$ .

■

## 21.5.3 The null distribution of $d$

### Theorem 21.5.3 — Durbin–Watson

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$  with  $\text{rank}(\mathbf{X}) = p < n$ , let the columns of the  $n \times (n-p)$  matrix  $\mathbf{Q}_\perp$  be an orthonormal basis of  $\mathcal{C}(\mathbf{X})^\perp$ , and let  $\nu_1 \leq \dots \leq \nu_{n-p}$  be the eigenvalues of  $\mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp$ . Put  $m = n - p$ .

- (a)  $d$  has the distribution of  $\sum_k \nu_k \xi_k^2 / \sum_k \xi_k^2$ , where  $\xi_1, \dots, \xi_m$  are independent  $N(0, 1)$ . It depends on  $\mathbf{X}$  but not on  $\boldsymbol{\beta}$  or  $\sigma^2$ , and  $d$  is independent of  $\text{SSE} = \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}}$ .
- (b)  $E d = \bar{\nu} = m^{-1} \sum_k \nu_k = \text{tr}\{(\mathbf{I} - \mathbf{M})\mathbf{A}\}/m$  and

$$\text{Var } d = \frac{2 \sum_k (\nu_k - \bar{\nu})^2}{m(m+2)}.$$

- (c)  $\Pr(d \leq c) = \Pr\{\sum_k (\nu_k - c) \xi_k^2 \leq 0\}$  for every  $c$ .
- (d) (*Bounds.*) If  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , then  $\lambda_{k+1} \leq \nu_k \leq \lambda_{k+p}$  for  $k = 1, \dots, m$ , with  $\lambda_j$  as in Lemma 21.5.1. Consequently  $d_L \leq d \leq d_U$  with probability one, where, for the same  $\xi_k$ ,

$$d_L = \frac{\sum_k \lambda_{k+1} \xi_k^2}{\sum_k \xi_k^2}, \quad d_U = \frac{\sum_k \lambda_{k+p} \xi_k^2}{\sum_k \xi_k^2},$$

whose distributions depend only on  $n$  and  $p$ .

*Proof.*

- (a)  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon} = \mathbf{Q}_\perp \mathbf{Q}_\perp^\top \boldsymbol{\varepsilon}$ , because  $\mathbf{Q}_\perp \mathbf{Q}_\perp^\top$  is the projection onto  $\mathcal{C}(\mathbf{X})^\perp$ . Let  $\boldsymbol{\zeta} = \mathbf{Q}_\perp^\top \boldsymbol{\varepsilon} / \sigma \sim N_m(\mathbf{0}, \mathbf{I}_m)$  (Theorem 3.3.1), and take a spectral decomposition  $\mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp = \mathbf{P} \text{diag}(\nu_k) \mathbf{P}^\top$  with  $\mathbf{P}$  orthogonal. Then  $\boldsymbol{\xi} = \mathbf{P}^\top \boldsymbol{\zeta} \sim N_m(\mathbf{0}, \mathbf{I}_m)$ ,  $\hat{\boldsymbol{\varepsilon}}^\top \mathbf{A} \hat{\boldsymbol{\varepsilon}} = \sigma^2 \boldsymbol{\zeta}^\top \mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp \boldsymbol{\zeta} = \sigma^2 \sum_k \nu_k \xi_k^2$  and  $\hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}} = \sigma^2 \|\boldsymbol{\zeta}\|^2 = \sigma^2 \sum_k \xi_k^2$ . For independence, put  $G_k = \xi_k^2/2$ , which are independent gamma variables with shape  $\frac{1}{2}$  and density proportional to  $g^{-1/2} e^{-g}$ . Change variables from  $(G_1, \dots, G_m)$  to  $S = \sum_k G_k$  and  $W_k = G_k/S$ ,  $k < m$ ; the Jacobian is  $S^{m-1}$ , and the joint density becomes proportional to  $S^{m/2-1} e^{-S} \prod_k w_k^{-1/2}$  (with  $w_m = 1 - \sum_{k < m} w_k$ ), a product of a function of  $S$  and a function of the  $W_k$ . So  $(W_1, \dots, W_m)$  is independent of  $S$ . Now  $d = \sum_k \nu_k W_k$  and  $\text{SSE} = 2\sigma^2 S$ .
- (b) Let  $N = \sum_k \nu_k \xi_k^2$  and  $S_2 = \sum_k \xi_k^2$ , so  $N = d S_2$  with  $d$  independent of  $S_2$ . Then  $E N = E d E S_2$  and  $E N^2 = E d^2 E S_2^2$ . Here  $E N = \sum_k \nu_k$ ,  $E S_2 = m$ ,  $E N^2 = \text{Var } N + (E N)^2 = 2 \sum_k \nu_k^2 + (\sum_k \nu_k)^2$  and  $E S_2^2 = 2m + m^2$ . Dividing gives  $E d = \bar{\nu}$  and  $E d^2 = \{2 \sum_k \nu_k^2 + (\sum_k \nu_k)^2\} / \{m(m+2)\}$ , and  $\text{Var } d = E d^2 - \bar{\nu}^2$  simplifies to the stated form. Finally  $\sum_k \nu_k = \text{tr}(\mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp) = \text{tr}(\mathbf{A} \mathbf{Q}_\perp \mathbf{Q}_\perp^\top) = \text{tr}\{\mathbf{A}(\mathbf{I} - \mathbf{M})\}$ .
- (c)  $d \leq c$  iff  $\sum_k \nu_k \xi_k^2 \leq c \sum_k \xi_k^2$ .
- (d) Let  $\mathbf{P}_1$  be the  $n \times (n-1)$  matrix whose columns are the normalized eigenvectors

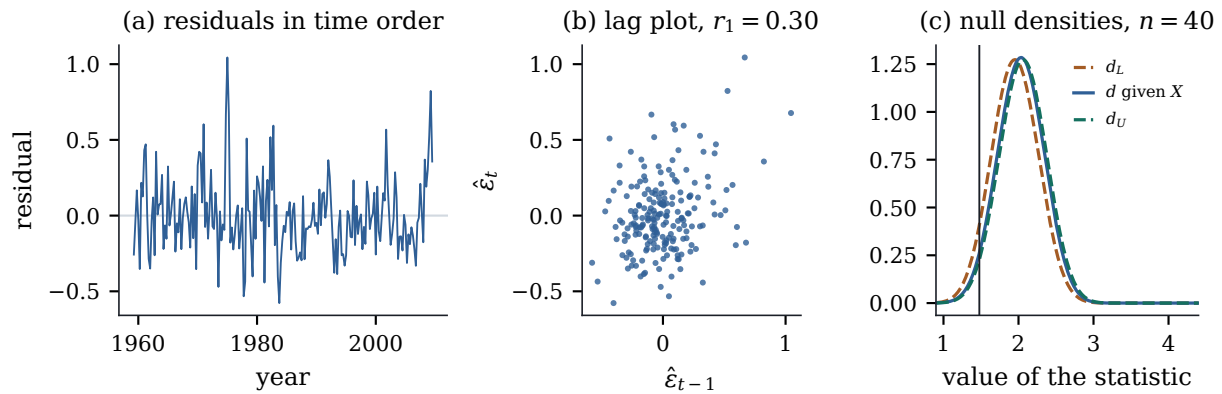


Figure 21.5.1: The Okun regression. (a) Residuals in time order. (b) Lag plot. (c) For the 40-quarter window 1999–2009: the exact null density of  $d$  given the regressors, between the densities of the bounds  $d_L$  and  $d_U$ ; the vertical line marks the observed  $d$ .

$\mathbf{v}_2, \dots, \mathbf{v}_n$  of  $\mathbf{A}$ ; they span  $\mathbf{1}^\perp$ . Since  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ ,  $\mathcal{C}(\mathbf{Q}_\perp) \subseteq \mathbf{1}^\perp$ , so  $\mathbf{Q}_\perp = \mathbf{P}_1 \mathbf{U}$  for an  $(n-1) \times m$  matrix  $\mathbf{U} = \mathbf{P}_1^\top \mathbf{Q}_\perp$  with orthonormal columns. Then  $\mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp = \mathbf{U}^\top \text{diag}(\lambda_2, \dots, \lambda_n) \mathbf{U}$ , and Lemma 21.5.2 with  $N = n-1$  gives  $\lambda_{k+1} \leq \nu_k \leq \lambda_{k+1+(n-1-m)} = \lambda_{k+p}$ . Multiplying by  $\xi_k^2$ , summing, and dividing by  $\sum_k \xi_k^2$  gives  $d_L \leq d \leq d_U$ . The  $\lambda_j$  depend only on  $n$ .

■

Part (a) reduces the exact null distribution to that of a ratio of quadratic forms in normal variables, of the kind studied in Chapter 4. Part (c) turns its distribution function into that of a single indefinite quadratic form, whose distribution function Imhof (1961) showed how to compute by numerical inversion of its characteristic function. So the exact p-value, given  $\mathbf{X}$ , is a routine computation, and the listing below does it. When Durbin and Watson (1950, 1951) proposed the test, that computation was impractical, and part (d) was their way around it. The bounds  $d_L$  and  $d_U$  do not depend on  $\mathbf{X}$ , so their percentage points can be tabulated once for each  $n$  and  $p$ . The **bounds test** of the hypothesis of no autocorrelation against positive autocorrelation, at level  $\alpha$ , rejects if  $d < d_{L,\alpha}$ , accepts if  $d > d_{U,\alpha}$ , and is inconclusive in between. To test against negative autocorrelation, use  $4 - d$ .

#### Example. (The exact test for the Okun regression)

Under the hypothesis of independent normal errors, and given the growth series,  $d$  has mean 2.0031 and standard deviation 0.1399, and its 5% point is 1.773. The bounds for  $n = 202$ ,  $p = 2$  are  $d_{L,0.05} = 1.760$  and  $d_{U,0.05} = 1.780$ , close together for a sample this large. The observed  $d = 1.386$  is far below both, and the exact one-sided p-value is  $3.3 \times 10^{-6}$ .

The bounds matter in small samples. For the 40 quarters from the second quarter of 1999 to the first quarter of 2009, the same regression gives  $d = 1.478$ , between the bounds 1.442 and 1.544: the bounds test is inconclusive. The exact p-value, 0.037, settles it. Figure 21.5.1(c) shows the three null densities for this window.

```
def dw_matrix(n):
    """A = D'D, where D takes first differences, so that d = e'Ae / e'e."""
    D = np.diff(np.eye(n), axis=0)
```

```

    return D.T @ D

def null_eigenvalues(X):
    """The  $n - p$  eigenvalues  $\nu_k$  of the DW numerator on  $C(X)$ -perp."""
    n, p = X.shape
    Qfull, _ = np.linalg.qr(X, mode="complete")
    # orthonormal basis of  $C(X)$ -perp
    Qp = Qfull[:, p:]
    return np.linalg.eigvalsh(Qp.T @ dw_matrix(n) @ Qp)

def prob_negative(c):
    """Pr( $\sum_k c_k x_{i_k}^2 < 0$ ),  $x_{i_k}$  iid standard normal (Imhof's formula)."""
    def integrand(u):
        theta = 0.5 * np.sum(np.arctan(c * u))
        rho = np.exp(0.25 * np.sum(np.log1p((c * u) ** 2)))
        return np.sin(theta) / (u * rho)
    return 0.5 - quad(integrand, 0, np.inf, limit=400)[0] / np.pi

nu = null_eigenvalues(X)
# Pr( $d \leq \text{observed}$ ) under  $H_0$ 
p_value = prob_negative(nu - d)
m = len(nu)
mean_d = nu.mean()
var_d = 2 * (m * np.sum(nu**2) - np.sum(nu) ** 2) / (m**2 * (m + 2))
print(f"exact one-sided p-value {p_value:.2e}; "
      f"null mean {mean_d:.4f}, sd {np.sqrt(var_d):.4f}")

w = slice(160, 200)
Xw, yw = X[w], du[w]
ew = yw - Xw @ np.linalg.lstsq(Xw, yw, rcond=None)[0]
d_w = np.sum(np.diff(ew) ** 2) / np.sum(ew**2)
nw_, pw_ = Xw.shape
lam_w = 4 * np.sin(np.pi * np.arange(nw_) / (2 * nw_)) ** 2
dL_w = brentq(lambda t: prob_negative(lam_w[1:nw_ - pw_ + 1] - t) - 0.05,
              0.5, 2.0)
dU_w = brentq(lambda t: prob_negative(lam_w[pw_:] - t) - 0.05, 0.5, 2.0)
p_w = prob_negative(null_eigenvalues(Xw) - d_w)
print(f"1999Q2-2009Q1: d = {d_w:.3f}, bounds ({dL_w:.3f}, {dU_w:.3f}), "
      f"exact p = {p_w:.3f}")

```

Three limitations should be kept in mind. The exact theory needs normal errors and regressors fixed in the sense of [Chapter 14](#): conditional on the regressors the argument goes through as long as they are not functions of past responses. It fails when a lagged response is among the regressors, and then  $d$  is biased towards 2, hiding autocorrelation; Durbin (1970) gave an alternative for that case. The bounds need an intercept. And the statistic is aimed at lag-one dependence. It is sensitive to other forms only insofar as they produce lag-one correlation, and it can miss, for example, quarterly seasonal correlation at lag four.

## 21.5.4 The Breusch–Godfrey test

A more flexible test comes from the score principle of [Section 21.2](#). Regress the residual  $\hat{\varepsilon}_t$  on the original regressors  $\mathbf{x}_{(t)}$  and on the lagged residuals  $\hat{\varepsilon}_{t-1}, \dots, \hat{\varepsilon}_{t-q}$ , with lagged values before the first observation set to zero, and let  $R^2$  be the coefficient of determination. The statistic  $nR^2$  is the score statistic for the hypothesis of no autocorrelation against autoregressive, or moving-average, errors of order  $q$ , and under the hypothesis it is approximately  $\chi^2(q)$  in large samples (Breusch 1978; Godfrey 1978). Unlike  $d$ , it remains valid when lagged responses are among the regressors, and it looks at several lags at once. We state its null limit without proof; the argument parallels that of [Theorem 21.2.2\(c\)](#).

For the Okun regression with  $q = 4$ ,  $nR^2 = 22.43$  on four degrees of freedom, with p-value  $1.6 \times 10^{-4}$ . Portmanteau statistics such as that of Ljung and Box (1978), based on  $\sum_{k \leq q} r_k^2$ , serve the same purpose in time-series modelling.

```
lags = np.column_stack([np.concatenate([np.zeros(k), e[:-k]])
                        for k in range(1, 5)])
Z = np.column_stack([X, lags])
coef, *_ = np.linalg.lstsq(Z, e, rcond=None)
# n R^2
bg = n * (1 - np.sum((e - Z @ coef) ** 2) / np.sum((e - e.mean()) ** 2))
print(f"Breusch-Godfrey (4 lags) = {bg:.2f}, p = {stats.chi2.sf(bg, 4):.1e}")
```

## 21.5.5 Exercises

### 21.5.5.1 A. Check your understanding

#### Exercise A1.

Show that  $0 \leq d < 4$  and that  $d = 2(1 - r_1) - (\hat{\varepsilon}_1^2 + \hat{\varepsilon}_n^2) / \sum_t \hat{\varepsilon}_t^2$ . Which residual patterns give values near 0 and near 4?

#### Exercise A2.

For the intercept-only model  $\mathbf{X} = \mathbf{1}$ , show that  $Ed = 2$  exactly under the hypothesis.

*Solution.* By [Theorem 21.5.3\(b\)](#),  $Ed = \text{tr}\{(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)\mathbf{A}\} / (n-1) = \{\text{tr} \mathbf{A} - n^{-1}\mathbf{1}^\top \mathbf{A} \mathbf{1}\} / (n-1)$ . The diagonal of  $\mathbf{A}$  sums to  $2(n-1)$ , and  $\mathbf{A} \mathbf{1} = \mathbf{0}$ . So  $Ed = 2$ .

### 21.5.5.2 B. Practice

#### Exercise B1.

If  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{V}$ , show that  $E(\hat{\boldsymbol{\varepsilon}}^\top \mathbf{A} \hat{\boldsymbol{\varepsilon}}) = \sigma^2 \text{tr}\{\mathbf{A}(\mathbf{I} - \mathbf{M})\mathbf{V}(\mathbf{I} - \mathbf{M})\}$ . For  $n = 3$ ,  $\mathbf{X} = \mathbf{1}$  and  $v_{st} = \rho^{|s-t|}$ , compute it and show that it decreases in  $\rho$ .

**Exercise B2.**

Suppose  $\mathcal{C}(\mathbf{X})$  is spanned by the eigenvectors  $\mathbf{v}_1, \dots, \mathbf{v}_p$  of  $\mathbf{A}$ . Show that  $\nu_k = \lambda_{k+p}$  for all  $k$ , so that  $d = d_U$ . Which columns of  $\mathbf{X}$  have this property, and what do they look like?

*Solution.* Then  $\mathcal{C}(\mathbf{X})^\perp$  is spanned by  $\mathbf{v}_{p+1}, \dots, \mathbf{v}_n$ , and  $\mathbf{Q}_\perp^\top \mathbf{A} \mathbf{Q}_\perp$  has eigenvalues  $\lambda_{p+1}, \dots, \lambda_n$ . The  $\mathbf{v}_j$  are cosines of increasing frequency, starting with the constant  $\mathbf{v}_1$ ;  $\mathbf{v}_2$  is a single slow half-wave, close to a centred linear trend. Regressors that are smooth functions of time are close to this case, so  $d$  is close to  $d_U$  for trend regressions, and the upper bound is the relevant one.

**Exercise B3.**

Show that  $d$  is unchanged if  $\mathbf{y}$  is replaced by  $c\mathbf{y} + \mathbf{X}\mathbf{a}$  for any  $c \neq 0$  and  $\mathbf{a}$ . Use this to explain why its null distribution cannot depend on  $\boldsymbol{\beta}$  or  $\sigma$ , without appeal to normality.

**21.5.5.3 C. Going deeper****Exercise C1.**

For the intercept-only model, show that under the hypothesis  $\text{Var } d = 4(n-2)/\{(n-1)(n+1)\}$  exactly, so that the standard deviation of  $d$  is close to  $2/\sqrt{n}$ . (Use [Theorem 21.5.3\(b\)](#),  $\sum_j \lambda_j^2 = \text{tr}(\mathbf{A}^2)$ , and the entries of  $\mathbf{A}$ .) Compare with the Okun regression of [Example \(The exact test for the Okun regression\)](#).

*Solution.* Here  $m = n - 1$  and the  $\nu_k$  are  $\lambda_2, \dots, \lambda_n$ , so  $\sum_k \nu_k = \text{tr } \mathbf{A} = 2(n-1)$  and  $\sum_k \nu_k^2 = \text{tr}(\mathbf{A}^2)$ , the sum of the squared entries of  $\mathbf{A}$ :  $1 + 4(n-2) + 1$  from the diagonal and  $2(n-1)$  from the off-diagonal, in total  $6n-8$ . Then  $m \sum_k \nu_k^2 - (\sum_k \nu_k)^2 = (n-1)(6n-8) - 4(n-1)^2 = 2(n-1)(n-2)$ , and  $\text{Var } d = 4(n-1)(n-2)/\{(n-1)^2(n+1)\} = 4(n-2)/\{(n-1)(n+1)\}$ . For  $n = 202$ ,  $2/\sqrt{202} = 0.141$ , close to the standard deviation 0.1399 computed for the Okun design, which has one more column.

**Exercise C2.**

In the model  $y_t = \beta y_{t-1} + \varepsilon_t$  with independent errors, fitted by least squares on  $t = 2, \dots, n$ , argue that the residuals are less autocorrelated than the errors would be under an autocorrelated alternative, so that  $d$  is biased towards 2. Relate this to the requirement in [Theorem 21.5.3](#) that  $\mathbf{X}$  be fixed.

## 21.6 Autoregressive errors

Serial correlation raises the same choice as heteroscedasticity: model it and use generalized least squares, or keep least squares and estimate its covariance without a model. Either way dependence must be described by few parameters, and the simplest description, the one behind the Durbin–Watson test, is the first-order autoregression.

## 21.6.1 The AR(1) model

### Definition 21.6.1 — First-order autoregressive errors

The errors follow a **first-order autoregressive process**, AR(1), if

$$\varepsilon_t = \rho \varepsilon_{t-1} + u_t, \quad t = 2, \dots, n,$$

with  $|\rho| < 1$ , where  $u_2, \dots, u_n$  are uncorrelated with mean zero and variance  $\sigma_u^2$ , and the starting value  $\varepsilon_1$  has mean zero, variance  $\sigma_u^2/(1 - \rho^2)$  and is uncorrelated with the  $u_t$ .

The variance of  $\varepsilon_1$  is the one that makes the process **stationary**: it is the variance an AR(1) process reaches after running for a long time, and with it every  $\varepsilon_t$  has the same variance.

### Theorem 21.6.1 — Regression with AR(1) errors

Let  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $\text{rank}(\mathbf{X}) = p$  and AR(1) errors.

- (a)  $\text{Cov}(\varepsilon_s, \varepsilon_t) = \sigma_u^2 \rho^{|s-t|}/(1 - \rho^2)$ . Write  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma_u^2 \mathbf{V}_\rho$ .
- (b) (*Prais–Winsten transformation.*) Let  $\mathbf{L}_\rho$  be the  $n \times n$  lower bidiagonal matrix with  $(\mathbf{L}_\rho)_{11} = \sqrt{1 - \rho^2}$ ,  $(\mathbf{L}_\rho)_{tt} = 1$  and  $(\mathbf{L}_\rho)_{t,t-1} = -\rho$  for  $t \geq 2$ . Then  $\text{Cov}(\mathbf{L}_\rho \boldsymbol{\varepsilon}) = \sigma_u^2 \mathbf{I}$ , so  $\mathbf{V}_\rho^{-1} = \mathbf{L}_\rho^\top \mathbf{L}_\rho$  and  $\det \mathbf{V}_\rho = 1/(1 - \rho^2)$ . For known  $\rho$ , the generalized least squares estimate is ordinary least squares on  $(\mathbf{L}_\rho \mathbf{y}, \mathbf{L}_\rho \mathbf{X})$ , and it is the best linear unbiased estimator.
- (c) (*Likelihood.*) If the errors are normal, the log-likelihood is

$$\begin{aligned} \ell(\boldsymbol{\beta}, \rho, \sigma_u^2) = & -\frac{n}{2} \log(2\pi\sigma_u^2) + \frac{1}{2} \log(1 - \rho^2) \\ & - \frac{\|\mathbf{L}_\rho(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\|^2}{2\sigma_u^2}. \end{aligned}$$

- (d) (*Cochrane–Orcutt.*) Let  $S(\boldsymbol{\beta}, \rho) = \sum_{t=2}^n \{(y_t - \rho y_{t-1}) - (\mathbf{x}_{(t)} - \rho \mathbf{x}_{(t-1)})^\top \boldsymbol{\beta}\}^2$ , the sum of squares of the last  $n - 1$  rows of  $\mathbf{L}_\rho(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$ . Alternate the steps

$$\begin{aligned} \rho_{k+1} &= \frac{\sum_{t \geq 2} e_t e_{t-1}}{\sum_{t \geq 2} e_{t-1}^2} \quad \text{with } \mathbf{e} = \mathbf{y} - \mathbf{X}\boldsymbol{\beta}_k, \\ \boldsymbol{\beta}_{k+1} &= \arg \min_{\mathbf{b}} S(\mathbf{b}, \rho_{k+1}). \end{aligned}$$

Then  $S(\boldsymbol{\beta}_{k+1}, \rho_{k+1}) \leq S(\boldsymbol{\beta}_k, \rho_{k+1}) \leq S(\boldsymbol{\beta}_k, \rho_k)$ , so  $S$  decreases along the iterations, and any limit point  $(\boldsymbol{\beta}^*, \rho^*)$  with  $|\rho^*| < 1$  at which the steps are continuous is a stationary point of  $S$ .

*Proof.*

- (a) By induction  $\text{Var } \varepsilon_t = \sigma_u^2/(1 - \rho^2)$  for all  $t$ , because  $\text{Var } \varepsilon_t = \rho^2 \text{Var } \varepsilon_{t-1} + \sigma_u^2$ ,  $\varepsilon_{t-1}$  being a linear combination of  $\varepsilon_1, u_2, \dots, u_{t-1}$  and so uncorrelated with  $u_t$ . For  $k \geq 1$ , iterating the recursion gives  $\varepsilon_t = \rho^k \varepsilon_{t-k} + \sum_{j=0}^{k-1} \rho^j u_{t-j}$ , and the sum is uncorrelated with  $\varepsilon_{t-k}$ , so  $\text{Cov}(\varepsilon_t, \varepsilon_{t-k}) = \rho^k \text{Var } \varepsilon_{t-k}$ .

- (b)  $\mathbf{L}_\rho \boldsymbol{\varepsilon} = (\sqrt{1-\rho^2} \varepsilon_1, u_2, \dots, u_n)^\top$ , whose entries are uncorrelated with variance  $\sigma_u^2$ . So  $\sigma_u^2 \mathbf{L}_\rho \mathbf{V}_\rho \mathbf{L}_\rho^\top = \sigma_u^2 \mathbf{I}$ , and since  $\mathbf{L}_\rho$  is nonsingular,  $\mathbf{V}_\rho^{-1} = \mathbf{L}_\rho^\top \mathbf{L}_\rho$ .  $\mathbf{L}_\rho$  is triangular with determinant  $\sqrt{1-\rho^2}$ , so  $\det \mathbf{V}_\rho = 1/(1-\rho^2)$ . The last statement is [Theorem 6.9.1](#) and [Corollary 7.2.3](#) with the whitening matrix  $\mathbf{L}_\rho$ , as in [Section 6.9](#).
- (c) The normal density of  $\boldsymbol{\varepsilon}$  with covariance  $\sigma_u^2 \mathbf{V}_\rho$  has  $-\frac{1}{2} \log \det(\sigma_u^2 \mathbf{V}_\rho) = -\frac{n}{2} \log \sigma_u^2 + \frac{1}{2} \log(1-\rho^2)$  and quadratic form  $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}_\rho^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) / \sigma_u^2 = \|\mathbf{L}_\rho(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\|^2 / \sigma_u^2$ .
- (d) For fixed  $\boldsymbol{\beta}$ ,  $S(\boldsymbol{\beta}, \rho) = \sum_{t \geq 2} (e_t - \rho e_{t-1})^2$  is a quadratic in  $\rho$  minimized by  $\rho_{k+1}$ , which gives the second inequality. For fixed  $\rho$ ,  $S$  is a least squares criterion in  $\mathbf{b}$ , minimized by  $\boldsymbol{\beta}_{k+1}$ , which gives the first. At a limit point both steps leave  $(\boldsymbol{\beta}^*, \rho^*)$  unchanged, so  $\partial S / \partial \rho = 0$  and  $\partial S / \partial \boldsymbol{\beta} = \mathbf{0}$  there. ■

The whitening in (b) is the transformation of Prais and Winsten (1954). Row  $t \geq 2$  of the transformed model is the **quasi-difference**  $y_t - \rho y_{t-1} = (\mathbf{x}_{(t)} - \rho \mathbf{x}_{(t-1)})^\top \boldsymbol{\beta} + u_t$ , and row one is the first observation rescaled by  $\sqrt{1-\rho^2}$ . The inverse covariance  $\mathbf{V}_\rho^{-1}$  is tridiagonal, with diagonal  $(1, 1+\rho^2, \dots, 1+\rho^2, 1)$  and off-diagonal entries  $-\rho$  ([Exercise \(A1\)](#)), which is why AR(1) models are cheap to fit however long the series. Cochrane and Orcutt (1949) dropped the first row, which leaves the conditional sum of squares  $S$  of (d). Dropping it loses little when the regressors are stationary, but it can lose a great deal when they trend, because the first observation is then the only one whose transformed regressor is not nearly constant. Maximizing the likelihood of (c) keeps the first row and the term  $\frac{1}{2} \log(1-\rho^2)$ , which keeps  $\hat{\rho}$  inside  $(-1, 1)$ .

## 21.6.2 Feasible generalized least squares

In practice  $\rho$  is unknown. The two-step **feasible** estimator estimates it from the ordinary least squares residuals,  $\hat{\rho} = \sum_{t \geq 2} \hat{e}_t \hat{e}_{t-1} / \sum_{t \geq 2} \hat{e}_{t-1}^2$ , and then applies the Prais–Winsten transformation with  $\hat{\rho}$ . Iterating the two steps gives either the Cochrane–Orcutt estimate or, keeping the first row, a close relative of the maximum likelihood estimate.

*Remark* (What is known about feasible GLS with AR(1) errors). Suppose the errors are a stationary AR(1) process with independent innovations of finite variance, and the regressors are stationary, or have polynomial trends, with suitable limits for their sample moments. Then  $\hat{\rho} \rightarrow \rho$  in probability, and  $\sqrt{n}(\hat{\boldsymbol{\beta}}_{\text{FGLS}} - \hat{\boldsymbol{\beta}}_{\text{GLS}}) \rightarrow \mathbf{0}$  after the appropriate normalization for trending regressors, so the feasible estimator has the limiting distribution of the optimal one and the usual standard errors from the transformed regression are valid in large samples. We do not prove this. A sketch:  $\hat{\rho}$  differs from the sample autocorrelation of the true errors by terms bounded by multiples of  $\|\mathbf{M}\boldsymbol{\varepsilon}\| \|\boldsymbol{\varepsilon}\|/n$  and  $\|\mathbf{M}\boldsymbol{\varepsilon}\|^2/n$ , which tend to zero because  $\|\mathbf{M}\boldsymbol{\varepsilon}\|$  is bounded in probability (compare the proof of [Theorem 21.2.2\(c\)](#)), and that autocorrelation is consistent by a law of large numbers for stationary sequences; the comparison of  $\hat{\boldsymbol{\beta}}_{\text{FGLS}}$  with  $\hat{\boldsymbol{\beta}}_{\text{GLS}}$  then follows the pattern of [Theorem 21.3.1\(d\)](#), because  $\mathbf{L}_\rho$  depends smoothly on  $\rho$ . Fuller (1996) gives complete statements and proofs. The finite-sample behaviour must be checked by simulation.

### Example. (Okun's law with AR(1) errors)

For the regression of [Example \(A quarterly Okun's-law regression\)](#),  $\hat{\rho} = 0.303$ . The two-step Prais–Winsten estimate of the slope is  $-0.0506$  with standard error  $0.0049$ , against

the least squares  $-0.0671$  with nominal standard error  $0.0050$ . Iterating Cochrane–Orcutt to convergence takes 17 iterations and ends at  $\hat{\rho} = 0.548$  and slope  $-0.0380$ . The estimate of the effect of growth on unemployment thus changes by almost half depending on the treatment of the errors, and the residuals of the transformed regression still have lag-one autocorrelation  $0.148$ . Both facts suggest that the AR(1) error model does not describe these data. We return to this below.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

macro = sm.datasets.macrodatab.load_pandas().data
du = np.diff(macro["unemp"].to_numpy())
growth = 400 * np.diff(np.log(macro["realgdp"].to_numpy()))
n = len(du)
X = np.column_stack([np.ones(n), growth])

def prais_winsten(X, y, rho):
    """Whiten AR(1) errors: row 1 times sqrt(1 - rho^2), row t - rho row t-1."""
    Xs, ys = X.copy(), y.copy()
    Xs[1:], ys[1:] = X[1:] - rho * X[:-1], y[1:] - rho * y[:-1]
    Xs[0], ys[0] = np.sqrt(1 - rho**2) * X[0], np.sqrt(1 - rho**2) * y[0]
    return Xs, ys

def ols(X, y):
    b, *_ = np.linalg.lstsq(X, y, rcond=None)
    e = y - X @ b
    cov = (e @ e / (len(y) - X.shape[1])) * np.linalg.inv(X.T @ X)
    return b, e, cov

b_ols, e, V_ols = ols(X, du)
# regress e_t on e_{t-1}
rho = np.sum(e[1:] * e[:-1]) / np.sum(e[:-1]**2)
# two-step Prais-Winsten
b_pw, e_pw, V_pw = ols(*prais_winsten(X, du, rho))
print(f"rho-hat {rho:.3f}")
print(f"OLS slope {b_ols[1]:.4f} (se {np.sqrt(V_ols[1, 1]):.4f}); "
      f"Prais-Winsten {b_pw[1]:.4f} (se {np.sqrt(V_pw[1, 1]):.4f})")

def cond_ss(b, r):
    """Conditional sum of squares S(beta, rho) over t = 2, ..., n."""
    u = (du[1:] - r * du[:-1]) - (X[1:] - r * X[:-1]) @ b
    return u @ u

b_co, r_co, path = b_ols, 0.0, []
for it in range(100):
    res = du - X @ b_co
    r_new = np.sum(res[1:] * res[:-1]) / np.sum(res[:-1]**2) # rho step
    path.append(cond_ss(b_co, r_new))
    # beta step
    b_co, *_ = np.linalg.lstsq(X[1:] - r_new * X[:-1], du[1:] - r_new * du[:-1],
                              rcond=None)
    path.append(cond_ss(b_co, r_new))
```



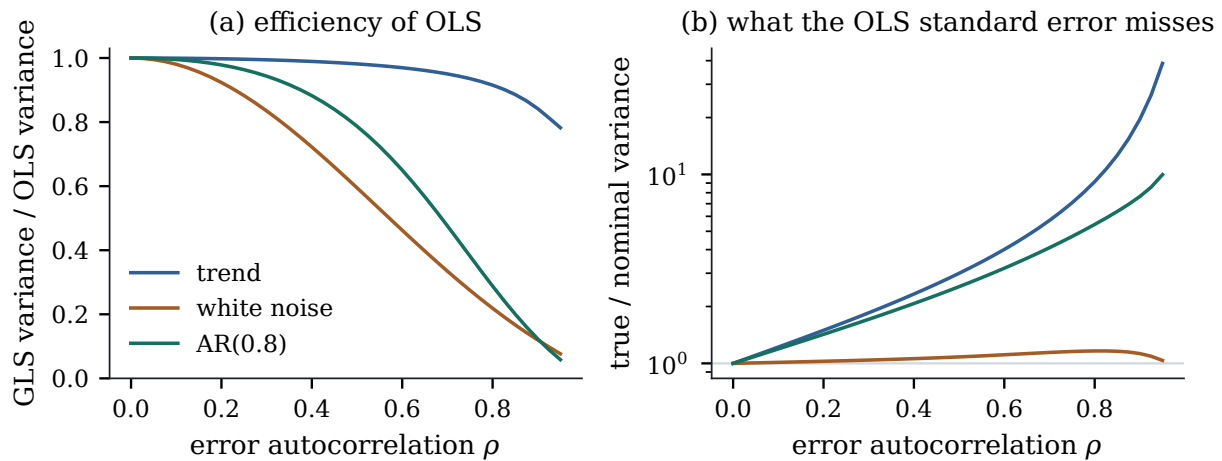


Figure 21.6.1: Least squares with AR(1) errors,  $n = 100$ , for three regressors. (a) Efficiency: variance of the generalized least squares slope divided by that of the ordinary one. (b) True variance of the least squares slope divided by its expected nominal variance (log scale).

```

if abs(r_new - r_co) < 1e-10:
    break
r_co = r_new
print(f"Cochrane-Orcutt after {it + 1} iterations: rho {r_co:.4f}, "
      f"slope {b_co[1]:.4f}")

```

### 21.6.3 Efficiency and validity of least squares

How much is lost by ignoring AR(1) errors? There are two separate questions: how much less efficient least squares is than generalized least squares, and how wrong its nominal standard error  $s\{(\mathbf{X}^\top \mathbf{X})^{-1}\}_{jj}^{1/2}$  is. The answers depend on the regressor, and they run in opposite directions.

#### Example. (Three regressors)

Take  $n = 100$ , an intercept, and one of three regressors: a linear trend, a fixed draw of white noise, and a fixed draw of an AR(1) series with coefficient 0.8 (its sample lag-one autocorrelation is 0.87). For AR(1) errors, Figure 21.6.1 shows (a) the ratio of the generalized to the ordinary least squares variance of the slope, and (b) the ratio of the true least squares variance to the expected nominal one,  $E(s^2)\{(\mathbf{X}^\top \mathbf{X})^{-1}\}_{22}$ , with  $E s^2$  from Theorem 2.3.1. All are exact computations.

At  $\rho = 0.6$ : for the trend, least squares is 0.970 efficient but its nominal variance is too small by a factor of 4.01. For white noise it is only 0.462 efficient, yet its nominal variance is nearly right (factor 1.11). The autocorrelated regressor suffers both ways: efficiency 0.650, variance understated by a factor of 3.19. At  $\rho = 0.9$  the factors for the trend become 0.842 and 19.59.

The trend repeats the lesson of the autocorrelated-errors example of Section 19.2; the other two regressors are new.

The pattern has a simple explanation. The variance of  $\hat{\beta}_1$  in a simple regression is  $\sum_{s,t} (x_s - \bar{x})(x_t - \bar{x}) \text{Cov}(\varepsilon_s, \varepsilon_t) / S_{xx}^2$ . When the regressor is itself autocorrelated, neighbouring terms add

up, and for a stationary regressor with lag-one autocorrelation  $\phi$  the ratio of true to nominal variance is approximately  $(1 + \rho\phi)/(1 - \rho\phi)$  (Exercise (B2)). For the AR(0.8) regressor this gives 3.18 at  $\rho = 0.6$ , against the exact 3.19. A white-noise regressor has  $\phi \approx 0$ , so the nominal variance is nearly right; but generalized least squares, which quasi-differences  $x_t$ , turns a rough regressor into an even rougher one with more information, and least squares' efficiency tends to  $(1 - \rho^2)/(1 + \rho^2)$ , which is 0.471 at  $\rho = 0.6$  (Exercise (B1)). For polynomial trends, Grenander (1954) showed that least squares is asymptotically fully efficient, which the trend curve in panel (a) approaches; but its standard error is then at its most misleading.

## 21.6.4 Newey–West standard errors

The analogue of the sandwich of Section 21.4 for serially correlated errors keeps least squares and estimates the middle matrix  $\mathbf{X}^\top \text{Cov}(\boldsymbol{\varepsilon})\mathbf{X} = \sum_{s,t} \text{Cov}(\varepsilon_s, \varepsilon_t) \mathbf{x}_{(s)} \mathbf{x}_{(t)}^\top$  without a parametric model. The products  $\hat{\varepsilon}_s \hat{\varepsilon}_t$  can stand in for the covariances only when  $s$  and  $t$  are close, since distant covariances are small but their estimates are not. With  $\mathbf{v}_t = \mathbf{x}_{(t)} \hat{\varepsilon}_t$  and a truncation lag  $L$ , Newey and West (1987) proposed

$$\begin{aligned} \hat{\boldsymbol{\Omega}}_L &= \sum_{k=-L}^L \left(1 - \frac{|k|}{L+1}\right) \hat{\mathbf{r}}_k, \\ \hat{\mathbf{r}}_k &= \sum_{t=k+1}^n \mathbf{v}_t \mathbf{v}_{t-k}^\top, \quad \hat{\mathbf{r}}_{-k} = \hat{\mathbf{r}}_k^\top, \end{aligned} \tag{21.6.1}$$

and the covariance estimate  $(\mathbf{X}^\top \mathbf{X})^{-1} \hat{\boldsymbol{\Omega}}_L (\mathbf{X}^\top \mathbf{X})^{-1}$ . The declining **Bartlett weights**  $1 - |k|/(L+1)$  are not a matter of taste.

### Proposition 21.6.2 — Newey–West estimates are nonnegative definite

For any vectors  $\mathbf{v}_1, \dots, \mathbf{v}_n$  and any  $L \geq 0$ ,  $\hat{\boldsymbol{\Omega}}_L$  in Eq. 21.6.1 is nonnegative definite. Without the weights,  $\sum_{|k| \leq L} \hat{\mathbf{r}}_k$  need not be.

*Proof.* Set  $\mathbf{v}_t = \mathbf{0}$  for  $t \leq 0$  and  $t > n$ , and let  $\mathbf{s}_j = \sum_{i=0}^L \mathbf{v}_{j-i}$  for every integer  $j$ . Then

$$\sum_j \mathbf{s}_j \mathbf{s}_j^\top = \sum_j \sum_{i,l=0}^L \mathbf{v}_{j-i} \mathbf{v}_{j-l}^\top = \sum_{k=-L}^L (L+1-|k|) \sum_t \mathbf{v}_t \mathbf{v}_{t-k}^\top,$$

because for fixed  $t = j - i$  and lag  $k = l - i$  there are exactly  $L+1-|k|$  pairs  $(i, l)$  in  $\{0, \dots, L\}^2$  with  $l - i = k$ . Dividing by  $L+1$  shows that  $\hat{\boldsymbol{\Omega}}_L = (L+1)^{-1} \sum_j \mathbf{s}_j \mathbf{s}_j^\top$ , a sum of nonnegative definite matrices. Exercise (C1) gives an example for the unweighted sum. ■

Newey and West (1987) proved that the estimator is consistent when  $L$  grows with  $n$  slowly enough, for errors and regressors that are stationary with dependence dying out fast enough. We do not prove this. Its practical weakness is the choice of  $L$ , which trades bias from truncation

against variance from estimating many autocovariances; Andrews (1991) studied data-based choices. A common rule of thumb is  $L = \lfloor 4(n/100)^{2/9} \rfloor$ . For the Okun regression, with  $L = 4$ , the Newey–West standard error of the slope is 0.0086, against the nominal 0.0050. With  $L = 0$  the estimator is HC0 ([Exercise \(A2\)](#)).

```
def newey_west(X, e, L):
    """Bartlett-weighted HAC covariance of the OLS estimate, truncation lag L."""
    v = X * e[:, None]
    S = v.T @ v
    for k in range(1, L + 1):
        G = v[k:].T @ v[:-k]
        S += (1 - k / (L + 1)) * (G + G.T)
    B = np.linalg.inv(X.T @ X)
    return B @ S @ B

V_nw = newey_west(X, e, 4)
print(f"Newey-West (L = 4) se of slope {np.sqrt(V_nw[1, 1]):.4f}")
```

### Example. (Coverage with AR(1) errors)

Let the regressor be a fixed draw of an AR(1) series with coefficient 0.8, and the errors AR(1) with  $\rho = 0.6$  and normal innovations. Over 10,000 simulated data sets the nominal 95% intervals for the slope covered as follows. For  $n = 50$ : least squares with the usual standard error 0.747, least squares with Newey–West 0.784, two-step Prais–Winsten 0.913, and generalized least squares with the true  $\rho$  0.946. For  $n = 200$ : 0.746, 0.871, 0.944 and 0.950. The Newey–West interval is valid only asymptotically, and converges slowly; when the AR(1) model is right, using it pays.

```
def ar1_draws(n, rho, reps, rng):
    u = rng.normal(size=(reps, n))
    x = np.empty((reps, n))
    x[:, 0] = u[:, 0] / np.sqrt(1 - rho**2)
    for t in range(1, n):
        x[:, t] = rho * x[:, t - 1] + u[:, t]
    return x

def coverage(n, reps, rng):
    """Coverage of 95% slope intervals: OLS, Newey-West, Prais-Winsten, GLS."""
    x = ar1_draws(n, 0.8, 1, rng)[0]
    Xc = np.column_stack([np.ones(n), x])
    L = int(4 * (n / 100) ** (2 / 9))
    q = stats.t.ppf(0.975, n - 2)
    hits = np.zeros(4)
    for y in ar1_draws(n, 0.6, reps, rng):          # true slope 0
        b, res, V = ols(Xc, y)
        r = np.sum(res[1:] * res[:-1]) / np.sum(res[:-1] ** 2)
        b_f, _, V_f = ols(*prais_winsten(Xc, y, r))
        b_g, _, V_g = ols(*prais_winsten(Xc, y, 0.6))
        ses = [V[1, 1], newey_west(Xc, res, L)[1, 1], V_f[1, 1], V_g[1, 1]]
        hits += [abs(bb[1]) <= q * np.sqrt(v)
                  for bb, v in zip((b, b_f, b_g), ses)]
    return hits / reps
```

```

rng = np.random.default_rng(2109)
for n_sim in (50, 200):
    print(n_sim, "OLS, Newey-West, Prais-Winsten, GLS:",
          np.round(coverage(n_sim, 500, rng), 3))

```

### 21.6.5 Serial correlation as a symptom

An AR(1) error model is a restriction on a dynamic regression. Multiply the model at time  $t-1$  by  $\rho$  and subtract:

$$y_t = \rho y_{t-1} + \mathbf{x}_{(t)}^\top \boldsymbol{\beta} - \rho \mathbf{x}_{(t-1)}^\top \boldsymbol{\beta} + u_t.$$

This is a regression of  $y_t$  on its own lag and on current and lagged regressors, with the coefficients of the lagged regressors constrained to equal  $-\rho$  times those of the current ones, the **common factor restriction**. Residual autocorrelation is often the trace of a missing lag of  $y$  or of a regressor, and then the dynamic regression without the restriction is the better model. Hendry and Mizon (1978) argued that the restriction should be tested before the error model is adopted.

#### Example. (Okun's law, dynamically)

Regressing  $\Delta u_t$  on an intercept,  $\Delta u_{t-1}$ ,  $g_t$  and  $g_{t-1}$  gives coefficients 0.2160, 0.3571 (standard error 0.0574),  $-0.0487$  (0.0043) and  $-0.0162$  (0.0055). The AR(1) model would require the last coefficient to be  $-0.3571 \times (-0.0487) = 0.0174$ . The estimate has the opposite sign, and the delta-method test of the restriction gives  $z = -8.1$ . The unrestricted dynamic regression has no remaining residual autocorrelation: its Breusch–Godfrey statistic with four lags is 4.42, with p-value 0.35. Unemployment responds to growth with a lag and with persistence; the AR(1) error model mistook this for correlated noise, and the slope estimates of [Example \(Okun's law with AR\(1\) errors\)](#), which move with  $\hat{\rho}$ , were artefacts of that mistake.

```

# the AR(1) error model is the dynamic regression
#   du_t = a + rho du_(t-1) + b g_t + c g_(t-1) + u_t
#   with the restriction c = -rho b
Xd = np.column_stack([np.ones(n - 1), du[:-1], growth[1:], growth[:-1]])
bd, ed, Vd = ols(Xd, du[1:])
# zero under the restriction
restr = bd[3] + bd[1] * bd[2]
grad = np.array([0.0, bd[2], bd[1], 1.0]) # delta method
z_cf = restr / np.sqrt(grad @ Vd @ grad)
print("dynamic regression: const, du(t-1), g(t), g(t-1) =", np.round(bd, 4))
print(f"common-factor restriction c + rho b = {restr:.4f}, z = {z_cf:.1f}")

```

An extreme case is the regression of one random walk on another, independent one, which gives large  $t$  statistics and a small  $d$  far more often than the nominal level suggests (Granger and Newbold 1974). There the residual autocorrelation signals a meaningless regression, and no correction of the standard errors repairs it.

## 21.6.6 Exercises

### 21.6.6.1 A. Check your understanding

#### Exercise A1.

Compute  $\mathbf{L}_\rho^\top \mathbf{L}_\rho$  and show that it is tridiagonal with diagonal  $(1, 1 + \rho^2, \dots, 1 + \rho^2, 1)$  and off-diagonal entries  $-\rho$ . Check  $\mathbf{V}_\rho \mathbf{L}_\rho^\top \mathbf{L}_\rho = \mathbf{I}$  for  $n = 2$ .

*Solution.* Column  $t$  of  $\mathbf{L}_\rho$  has entry 1 (or  $\sqrt{1 - \rho^2}$  for  $t = 1$ ) in row  $t$  and  $-\rho$  in row  $t + 1$  (for  $t < n$ ). The inner product of columns  $t$  and  $t$  is  $1 + \rho^2$  for  $1 < t < n$ ,  $(1 - \rho^2) + \rho^2 = 1$  for  $t = 1$  and 1 for  $t = n$ ; columns  $t$  and  $t + 1$  share only row  $t + 1$ , giving  $-\rho$ ; other pairs share nothing. For  $n = 2$ ,  $\mathbf{V}_\rho = (1 - \rho^2)^{-1} \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}$  and  $\mathbf{L}_\rho^\top \mathbf{L}_\rho = \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$ , whose product is  $(1 - \rho^2)^{-1}(1 - \rho^2)\mathbf{I} = \mathbf{I}$ .

#### Exercise A2.

Show that the Newey–West estimator with  $L = 0$  is HC0, and that with  $\mathbf{X} = \mathbf{1}$  and  $L \geq 1$  it estimates the long-run variance  $\sum_k \gamma_k$  of a stationary series, scaled by  $n$ .

### 21.6.6.2 B. Practice

#### Exercise B1.

Let the regressor  $x_t$  be white noise with variance  $\tau^2$ , independent of AR(1) errors, and consider the slope in a regression through the origin. Show heuristically that  $n \text{Var}(\hat{\beta}_{\text{OLS}}) \rightarrow \sigma_u^2 / \{(1 - \rho^2)\tau^2\}$  and  $n \text{Var}(\hat{\beta}_{\text{GLS}}) \rightarrow \sigma_u^2 / \{(1 + \rho^2)\tau^2\}$ , so that the efficiency of least squares, the ratio bounded in [Theorem 19.2.3](#), tends to  $(1 - \rho^2)/(1 + \rho^2)$ .

#### Exercise B2.

Let the regressor be a stationary AR(1) series with coefficient  $\phi$ , independent of AR(1) errors with coefficient  $\rho$ . Show heuristically that the ratio of the true variance of the least squares slope to its nominal variance tends to  $(1 + \rho\phi)/(1 - \rho\phi)$ .

*Solution.* Conditionally on the regressor,  $\text{Var}(\hat{\beta}) = \sum_{s,t} \tilde{x}_s \tilde{x}_t \gamma_0 \rho^{|s-t|} / S_{xx}^2$ , where  $\tilde{x}_t = x_t - \bar{x}$ . For large  $n$ ,  $\sum_t \tilde{x}_t \tilde{x}_{t-k} \approx S_{xx} \phi^{|k|}$ , so the variance is about  $(\gamma_0 / S_{xx}) \sum_k (\rho\phi)^{|k|} = (\gamma_0 / S_{xx})(1 + \rho\phi)/(1 - \rho\phi)$ . The nominal variance  $E(s^2)/S_{xx}$  is about  $\gamma_0 / S_{xx}$ , since  $s^2$  estimates the marginal error variance  $\gamma_0$  when  $p/n$  is small.

#### Exercise B3.

In the model  $y_t = \alpha + \beta t + \varepsilon_t$ ,  $t = 1, \dots, n$ , with AR(1) errors, show that the Cochrane–Orcutt regression has columns  $(1 - \rho)\mathbf{1}$  and  $(1 - \rho)t + \rho$  on  $t = 2, \dots, n$ , and that the information it carries about  $\beta$  is  $(1 - \rho)^2 \sum_{t \geq 2} (t - \bar{t})^2 / \sigma_u^2$ , where  $\bar{t}$  is the mean of  $2, \dots, n$ . What happens as  $\rho \rightarrow 1$  with  $n$  fixed, and why does the Prais–Winsten first row change

the picture?

### 21.6.6.3 C. Going deeper

#### Exercise C1.

Give vectors  $v_1, v_2, v_3$  (with  $p = 1$ ) for which the unweighted sum  $\sum_{|k| \leq 1} \hat{\Gamma}_k$  is negative.

*Solution.* Take  $(v_1, v_2, v_3) = (1, -1, 1)$ . Then  $\hat{\Gamma}_0 = 3$  and  $\hat{\Gamma}_1 = \hat{\Gamma}_{-1} = v_2 v_1 + v_3 v_2 = -2$ , so the unweighted sum is  $3 - 4 = -1$ . With Bartlett weights it is  $3 - 2 = 1$ .

## 21.7 Summary and Notes

### 21.7.1 Summary

*Idea.*

1. Residuals mix all the errors. Each differs from its error with variance  $\sigma^2 h_{ii}$ , and residuals keep only part of the errors' skewness and kurtosis, so they look more normal than the errors when  $p/n$  is not small ([Proposition 21.1.1](#)).
2. Normal plots of studentized residuals with simulated envelopes are the main check of normality. Coefficient inference is protected by the central limit theorem; prediction intervals are not, and their coverage tends to  $\Pr(|\varepsilon_0| \leq z\sigma)$  ([Proposition 12.3.3](#)), which [Table 21.1.1](#) evaluates for five error laws.
3. The Breusch–Pagan statistic is half the explained sum of squares of  $\hat{\varepsilon}_i^2/\tilde{\sigma}^2$  on the variance variables, whatever the variance function. Non-normal errors inflate its null law by  $(\kappa - 1)/2$ , with  $\kappa$  the raw kurtosis, and Koenker's  $nR^2$  corrects this ([Theorem 21.2.2](#), using [Lemma 21.2.1](#)). The Goldfeld–Quandt test is exact under normality ([Proposition 21.2.3](#)).
4. Weighted least squares is optimal with known weights, unbiased with wrong ones, unbiased under symmetric errors with weights from residuals, and asymptotically optimal with weights from well-replicated groups ([Theorem 21.3.1](#)). Weights from a few replicates can be worse than none.
5. HC0–HC3 estimate the covariance of least squares without a variance model ([Definition 21.4.1](#)). HC2 is unbiased under homoscedasticity and HC3 is the jackknife ([Proposition 21.4.1](#)). All are consistent ([Theorem 21.4.2](#)), but high leverage makes them unreliable in small samples.
6. Given  $\mathbf{X}$  and normal errors, the Durbin–Watson  $d \approx 2(1 - r_1)$  is a ratio of quadratic forms with a computable law, bracketed by design-free bounds when there is an intercept ([Theorem 21.5.3](#), [Lemma 21.5.1](#), [Lemma 21.5.2](#)).
7. AR(1) errors are whitened by the Prais–Winsten transformation, and Cochrane–Orcutt iteration decreases the conditional sum of squares ([Definition 21.6.1](#), [Theorem 21.6.1](#)). Smooth regressors make least squares efficient but its standard errors wrong. Newey–West estimates are nonnegative definite ([Proposition 21.6.2](#)). Residual autocorrelation often signals missing dynamics.

## 21.7.2 Notes and sources

**Sources.** The chapter covers Sen and Srivastava (1990, chapters 5–6 and sections 7.1–7.3 and 7.6), Seber and Lee (2003, sections 10.4–10.5) and Christensen (2020, sections 12.2–12.4). It adds proofs of the facts about residuals behind the diagnostics, the null limits of both forms of the Breusch–Pagan test, the properties of feasible weighting in the grouped case, the consistency of the sandwich estimators, and the Durbin–Watson bounds through interlacing. The limit theorems are from van der Vaart (1998).

**Normality.** Blom (1958) gave the plotting positions, Shapiro and Wilk (1965) and Shapiro and Francia (1972) the two correlation-type tests, and Jarque and Bera (1980) the skewness–kurtosis score test. Lilliefors (1967) tabulated the Kolmogorov–Smirnov test with estimated parameters, and Anderson and Darling (1952) introduced their tail-weighted distance. Simulated envelopes are due to Atkinson (1981). That residuals look more normal than errors is an old observation; [Proposition 21.1.1\(c\)](#) quantifies it.

**Heteroscedasticity.** The score test is due to Breusch and Pagan (1979) and to Cook and Weisberg (1983); the score principle is Rao’s (1948). Koenker (1981) exposed the sensitivity to kurtosis and gave the studentized form. White (1980) proposed the direct test and the covariance estimator, and Goldfeld and Quandt (1965) the exact  $F$  test. Levene (1960) and Brown and Forsythe (1974) gave robust tests for groups, and Box (1953) showed the non-robustness of normal-theory variance tests. Carroll and Ruppert (1988) is the standard monograph on weighting; Carroll (1982) showed that weights estimated by smoothing can be fully efficient asymptotically. The symmetry argument of [Theorem 21.3.1\(c\)](#) goes back to Kakwani (1967).

**Sandwich estimators.** Eicker (1963) gave conditions for asymptotic normality of least squares with unequal variances and Eicker (1967) the consistency of the covariance estimator; Huber (1967) introduced the sandwich form for general estimators, and White (1980) made it standard in econometrics. MacKinnon and White (1985) proposed HC2 and HC3. Long and Ervin (2000) recommended HC3, and Imbens and Kolesár (2016) recommend HC2 with the degrees-of-freedom adjustment of Bell and McCaffrey (2002).

**Serial correlation.** Durbin and Watson (1950, 1951, 1971) introduced the statistic and the bounds and studied its exact distribution; Imhof (1961) gave the inversion formula. Durbin (1970) treated lagged responses, Breusch (1978) and Godfrey (1978) the score test for higher-order errors, and Ljung and Box (1978) the portmanteau statistic. The transformations are those of Cochran and Orcutt (1949) and Prais and Winsten (1954). Grenander (1954) proved the asymptotic efficiency of least squares for polynomial trends. Newey and West (1987) proposed the Bartlett-weighted estimator and Andrews (1991) studied its bandwidth. Hendry and Mizon (1978) argued for testing the common factor restriction, and Granger and Newbold (1974) exposed spurious regressions. Fuller (1996) gives the asymptotic theory of regression with autocorrelated errors in full.

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# Chapter 22

## Transformations

A regression is fitted on a scale, and nothing in least squares says which one. Food expenditure can be modelled in currency units or in logarithms, a count as it stands or through its square root. The choice affects all three claims of the normal linear model: a linear mean, a constant variance and normal errors. When the three failures share a cause, one transformation repairs them together; when they do not, the chapter says which to fix first and at what price. It treats the choice of scale as estimation: the Box–Cox profile likelihood for the response, the Box–Tidwell procedure and component-plus-residual plots for regressors, and the delta method for transformations that stabilize a variance tied to the mean. It ends with what a transformed model says on the scale the reader cares about: medians, means, prediction intervals and elasticities.

**What you need.** Least squares as projection and the Frisch–Waugh–Lovell theorem ([Chapter 6](#), [Theorem 6.6.2](#)), maximum likelihood and profile likelihood in the normal linear model ([Chapter 7](#), [Theorem 7.3.1](#) and [Definition 7.3.1](#)), the interpretation of log-scale coefficients ([Section 5.7](#)), prediction intervals ([Theorem 12.3.1](#)), the delta method as used in [Section 14.4](#), and the diagnostics of [Chapter 20](#) and [Chapter 21](#).

### 22.0.1 Roadmap

- [Transforming the response](#)
- [The Box–Cox family](#)
- [Transforming regressors](#)
- [Variance-stabilizing transformations](#)
- [Interpreting a transformed model](#)
- [Summary and notes](#)

## 22.1 Transforming the response

The normal linear model claims that the mean of the response is linear in the regressors, that its variance is constant, and that its distribution is normal. [Chapter 19](#) worked out what each failure costs, and [Chapter 20](#) and [Chapter 21](#) showed how to see them. Replacing  $y$  by  $g(y)$ , for a strictly increasing  $g$ , acts on all three claims at once. If the failures have a common cause, one

$g$  removes them together; if not, the  $g$  that straightens the mean may worsen the variance.

## 22.1.1 Engel's data on two scales

### Example. (Food expenditure on the raw and the log scale)

Engel's survey of 235 Belgian households ([Example \(Engel's food expenditure data\)](#)) records annual income, from 377 to 4958 in the survey's currency, and expenditure on food. The least squares line on the raw scale has slope 0.4852, and panel (b) of [Figure 22.1.1](#) shows a fan: the residual standard deviation among the richest third of households is 3.55 times that among the poorest third. The residuals have excess kurtosis 9.64 (zero for a normal distribution), largely a by-product of the fan, since a mixture of normals with different spreads has heavy tails.

On the log-log scale both defects nearly disappear. The spread ratio falls to 1.32, the excess kurtosis to  $-0.02$ , and the residual plot, panel (d), shows a horizontal band. The slope is 0.8559 with standard error 0.0203 and 95% interval [0.816, 0.896]. It is an elasticity ([Section 5.7](#)): households with 1 percent more income spend about 0.86 percent more on food. That the interval lies below one is Engel's law, that the share of the budget spent on food falls as income rises.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)

raw = sm.OLS(food, sm.add_constant(income)).fit() # food on income
loglog = sm.OLS(np.log(food), sm.add_constant(np.log(income))).fit() # log food on
    log income

def spread_ratio(fit):
    """Residual SD among the richest third over that among the poorest third."""
    order = np.argsort(income)
    low, high = order[: n // 3], order[-(n // 3):]
    return fit.resid[high].std(ddof=1) / fit.resid[low].std(ddof=1)

for name, fit in [("raw", raw), ("log-log", loglog)]:
    print(f"{name:8s} slope {fit.params[1]:.4f} spread ratio {spread_ratio(fit):.2f}
    {name:8s} excess kurtosis of residuals {stats.kurtosis(fit.resid):.2f}")
```

## 22.1.2 Multiplicative models

The logarithm worked because the variation in food expenditure behaves like a *percentage* of its level. A model in which the regressors and the error act multiplicatively makes this precise.

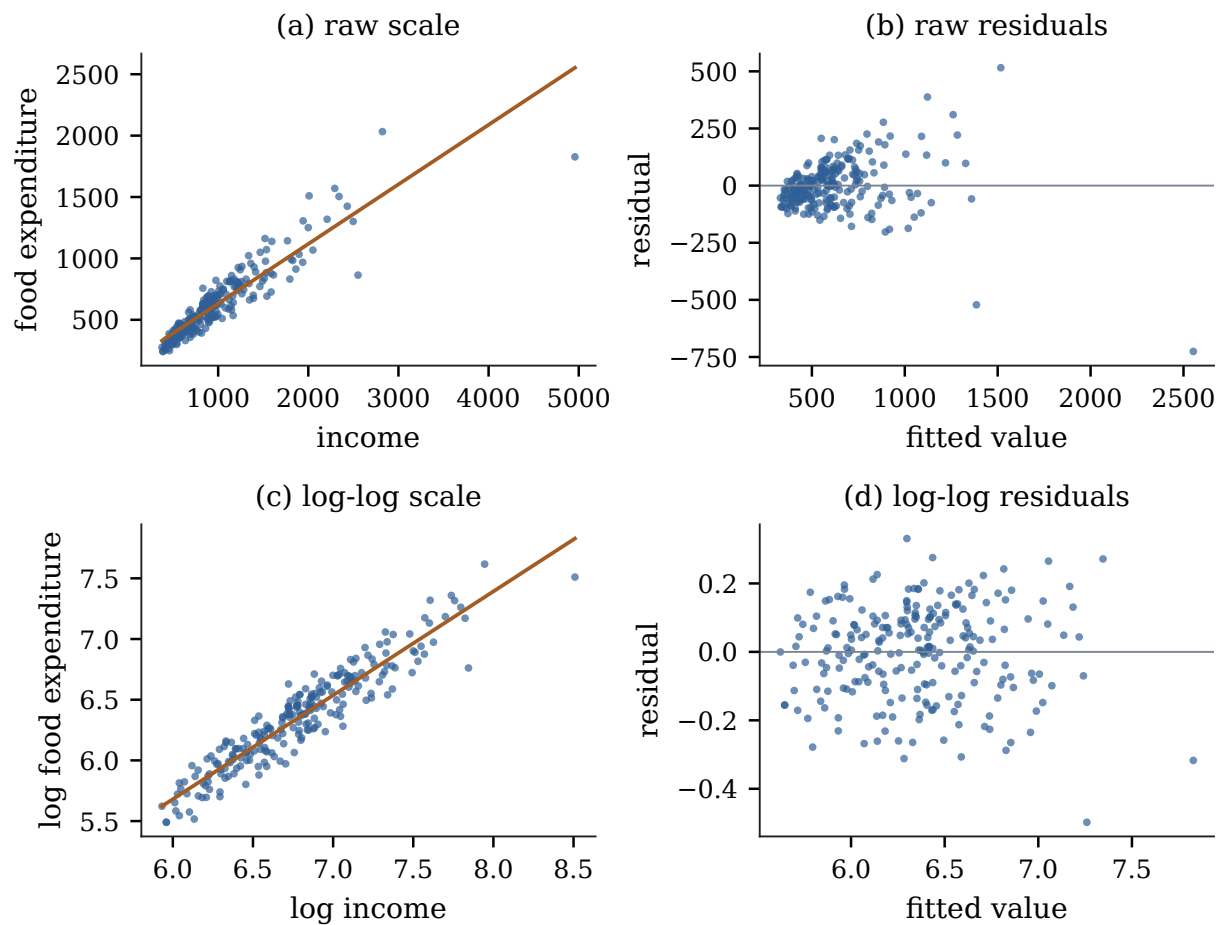


Figure 22.1.1: Engel's data. (a) Food expenditure against income with the least squares line. (b) Its residuals against fitted values: the spread grows with the level. (c) The same data on the log-log scale. (d) The log-log residuals form a band of nearly constant width.

**Proposition 22.1.1 — Multiplicative errors**

Let the model matrix  $\mathbf{X}$  have first column  $\mathbf{1}$ , with  $\beta_0$  the corresponding coefficient, and let

$$Y_i = \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}) U_i, \quad i = 1, \dots, n, \quad (22.1.1)$$

where  $U_1, \dots, U_n$  are independent and identically distributed positive random variables with  $\nu = E(\log U_i)$  and  $\tau^2 = \text{Var}(\log U_i)$  finite.

- (a) The logarithms follow the linear model with constant variance:  $\log Y_i = (\beta_0 + \nu) + \sum_{j \geq 1} x_{ij} \beta_j + \varepsilon_i$ , where  $\varepsilon_i = \log U_i - \nu$  are independent with mean 0 and variance  $\tau^2$ . The slopes are  $\beta_1, \dots, \beta_{p-1}$ , unchanged; only the intercept absorbs  $\nu$ .
- (b) If  $E(U_i^2) < \infty$ , then  $E(Y_i) = \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}) E(U_1)$  and the standard deviation of  $Y_i$  is  $\exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$  times that of  $U_1$ . The standard deviation is proportional to the mean, so the coefficient of variation is the same for every case.
- (c) For two regressor profiles  $\mathbf{x}$  and  $\mathbf{x}'$ , every quantile of  $Y$  at  $\mathbf{x}'$  is  $\exp\{(\mathbf{x}' - \mathbf{x})^\top \boldsymbol{\beta}\}$  times the same quantile at  $\mathbf{x}$ . The same ratio holds for the geometric means  $\exp E(\log Y)$ , and for the means when they are finite.

*Proof.*

- (a) Take logarithms in [Eq. 22.1.1](#) and add and subtract  $\nu$ . The  $\varepsilon_i$  are functions of the independent  $U_i$ , hence independent, and have the stated moments.
- (b)  $\exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$  is a constant, so it factors out of the mean and the standard deviation.
- (c) If  $u_q$  is a  $q$ -quantile of  $U_1$ , then  $\exp(\mathbf{x}^\top \boldsymbol{\beta}) u_q$  is a  $q$ -quantile of  $Y$  at  $\mathbf{x}$ , because multiplication by a positive constant preserves order. The geometric mean at  $\mathbf{x}$  is  $\exp(\mathbf{x}^\top \boldsymbol{\beta} + \nu)$ . In each case the ratio of the value at  $\mathbf{x}'$  to the value at  $\mathbf{x}$  is  $\exp\{(\mathbf{x}' - \mathbf{x})^\top \boldsymbol{\beta}\}$ .

■

Part (c) makes log-scale coefficients easy to report:  $(\mathbf{x}' - \mathbf{x})^\top \boldsymbol{\beta}$  is the log of the *same* ratio for medians, upper quartiles, geometric means and means. Part (b) is the diagnostic signature, a raw-scale spread proportional to the fitted value, as in panel (b) of [Figure 22.1.1](#).

*Warning.* The proposition needs the error to *multiply* the mean. If instead  $Y_i = \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}) + \varepsilon_i$  with errors of constant variance, then  $\log Y_i$  is not a linear function of the regressors plus an error of mean zero. For large means  $\log Y_i \approx \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i \exp(-\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ , so the logarithm *creates* a variance that falls with the level ([Exercise \(B1\)](#)). A mean that is exponential in the regressors combined with errors of constant variance calls for fitting the mean directly, by nonlinear least squares or by a generalized linear model with a logarithmic link ([Chapter 34](#)), not for transforming the data.

### 22.1.3 The ladder of powers

The logarithm belongs to Tukey's (1977) *ladder* of powers  $y^p$ ,  $p = \dots, -1, -\frac{1}{2}, 0, \frac{1}{2}, 1, 2, \dots$ , with  $\log y$  in the place of  $y^0$  and  $-y^p$  for  $p < 0$  so that every rung is increasing. Rungs below  $p = 1$  are concave: they pull in a long right tail, reduce a spread that grows with the level, and straighten a relationship that curves upwards. Rungs above  $p = 1$  do the opposite. [Section 22.2](#) makes the ordering precise ([Proposition 22.2.1](#)) and estimates the rung, and [Section 22.4](#) proves the rule linking rung and variance: a standard deviation proportional to  $\mu^a$  calls for the power  $1 - a$ , the logarithm when  $a = 1$  as in [Proposition 22.1.1\(b\)](#).

### 22.1.4 Three goals and one function

The three claims are not equally important. With non-normal errors, least squares and the usual tests remain asymptotically valid under mild conditions ([Theorem 19.3.2](#)). A non-constant variance leaves the coefficients unbiased but the standard errors wrong ([Theorem 19.2.2](#)). A wrong mean function biases everything. So the mean comes first, then the variance, then the shape of the errors.

Counts show the goals pulling apart. Suppose  $Y_i$  is Poisson with mean  $\mu_i = \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ . The logarithm linearizes the mean exactly, but the variance of  $\log Y_i$  is, to first order in the delta method, about  $1/\mu_i$  ([Exercise \(B2\)](#)), which changes with the regressors. The square root stabilizes the variance at about  $\frac{1}{4}$  ([Section 22.4](#)), but its mean is about  $\exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}/2)$ , which is not linear in the regressors. (Strictly, zeros have probability  $e^{-\mu_i} > 0$ , so the statement concerns  $\log Y_i$  given  $Y_i > 0$ , or  $\log(Y_i + c)$ .) No power does both. The remedy is a model that states the mean and the variance separately, the Poisson generalized linear model of [Chapter 34](#). A transformation can also remove an interaction, under the order condition of [Proposition 16.2.2](#), as the logarithm did for the running times of [Section 16.1](#).

### 22.1.5 Zeros and negative values

The logarithm and the negative powers need  $y > 0$ . The common fix  $\log(y + c)$  for a “start”  $c > 0$  makes the coefficients depend on  $c$  when zeros are frequent. The start can be estimated with the power ([Exercise \(B3\)](#)), but a spike at zero usually means two processes, whether a case is positive and how large it is, which no transformation separates; two-part models and generalized linear models ([Chapter 34](#)) are more natural. For responses of both signs, Yeo and Johnson (2000) give a family defined on the whole line.

### 22.1.6 Exercises

#### 22.1.6.1 A. Check your understanding

##### Exercise A1.

In the log–log fit of [Example \(Food expenditure on the raw and the log scale\)](#) the slope is 0.8559. By what factor does the median food expenditure differ between two households whose incomes differ by 10 percent? By what factor between households whose incomes differ by a factor of two? Why is the answer the same for the median, the geometric

mean and, under Eq. 22.1.1, the mean?

*Solution.* The log of the ratio of incomes is  $\log 1.1$ , so the ratio of medians is  $1.1^{0.8559} = \exp(0.8559 \log 1.1)$ , about 1.085: a difference of about 8.5 percent. For a factor of two the ratio is  $2^{0.8559}$ , about 1.81. By Proposition 22.1.1(c) every quantile, the geometric mean and the mean scale by the same factor  $\exp\{(\mathbf{x}' - \mathbf{x})^\top \boldsymbol{\beta}\}$ , because the error multiplies the whole distribution.

### 22.1.6.2 B. Practice

#### Exercise B1.

Let  $Y = \mu + \varepsilon$  with  $\mu > 0$ ,  $E\varepsilon = 0$ ,  $\text{Var}\varepsilon = \sigma^2$  and  $\sigma/\mu$  small. Use  $\log(1+u) = u - u^2/2 + O(u^3)$  to show that, to the leading order,  $E \log Y \approx \log \mu - \sigma^2/(2\mu^2)$  and  $\text{Var} \log Y \approx \sigma^2/\mu^2$ . Conclude that when errors are additive with constant variance, the log scale is heteroscedastic, with the variance falling as the mean rises, and its mean is not exactly  $\log \mu$ .

*Solution.* Write  $\log Y = \log \mu + \log(1+u)$  with  $u = \varepsilon/\mu$ . Then  $\log Y = \log \mu + u - u^2/2 + O(u^3)$ . Taking expectations,  $E u = 0$  and  $E u^2 = \sigma^2/\mu^2$ , which gives the mean. For the variance, the leading term of  $\log Y - E \log Y$  is  $u$ , with variance  $\sigma^2/\mu^2$ . Both depend on  $\mu$ , hence on the regressors, unless  $\sigma$  is proportional to  $\mu$ , which is the multiplicative case.

#### Exercise B2.

Let  $Y$  be Poisson with a large mean  $\mu$ . Use the first-order expansion of the delta method to show that  $\text{Var}(\log Y) \approx 1/\mu$  and  $\text{Var}(\sqrt{Y}) \approx \frac{1}{4}$ , and that  $E \sqrt{Y} \approx \sqrt{\mu}$ . (Read  $\log Y$  given  $Y > 0$ , since  $\Pr(Y = 0) = e^{-\mu}$ .) If  $\mu = \exp(\mathbf{x}^\top \boldsymbol{\beta})$ , which scale has a linear mean and which a constant variance?

#### Exercise B3.

A colleague fits the same regressors to  $y$  and to  $\log y$  and prefers the log model because its  $R^2$  is larger. Explain why the two values of  $R^2$  are not comparable. Then take  $x = (0, 1, 2)$  and  $y = (1, 8, 7)$ , fit a straight line to  $y$  and to  $\log y$ , and compare the two values of  $R^2$  with the residual sums of squares on the original scale, using  $\exp$  of the fitted values for the log model.

*Solution.*  $R^2$  is the fraction of the total sum of squares of the response on its own scale that the model explains (Theorem 6.7.2). The two fits explain different totals, so their fractions answer different questions. A likelihood on a common scale, such as the Box–Cox profile of Section 22.2, which includes the Jacobian, does compare them. In the example the raw line has  $R^2 = 0.628$  and the log line  $R^2 = 0.699$ . On the original scale the raw line leaves a residual sum of squares of  $32/3$ , about 10.67, while the exponentiated log fit leaves 27.37, more than twice as much. The larger  $R^2$  belongs to the fit that predicts  $y$  worse, because it is measured on a scale where the large values count for less.

## 22.1.6.3 C. Going deeper

**Exercise C1.**

In a  $2 \times 2$  layout let  $\mu_{ij} = (a_i + b_j)^2$  with positive  $a_i, b_j$ . Show that the interaction contrast  $\mu_{11} - \mu_{12} - \mu_{21} + \mu_{22}$  is  $2(a_1 - a_2)(b_1 - b_2)$ , and that it is zero on the square-root scale. Now let  $Y_{ij} = \mu_{ij} + \varepsilon_{ij}$  with independent errors of variance  $\sigma^2$ . Is the table of  $E \sqrt{Y_{ij}}$  exactly additive? Relate your answer to the caution in [Section 16.2](#).

## 22.2 The Box–Cox family

Choosing a rung of the ladder by eye works when the evidence is strong, but it gives no measure of uncertainty and no way to compare rungs that fit almost equally well. Box and Cox (1964) turned the choice into a parameter: embed the powers in a family indexed by a real number  $\lambda$ , assume that for some  $\lambda$  the normal linear model holds on the transformed scale, and estimate  $\lambda$  by maximum likelihood along with everything else.

### 22.2.1 The family

**Definition 22.2.1 — The Box–Cox family**

For  $y > 0$  and  $\lambda \in \mathbb{R}$ , the **Box–Cox transformation** of  $y$  is

$$y^{(\lambda)} = \begin{cases} \frac{y^\lambda - 1}{\lambda}, & \lambda \neq 0, \\ \log y, & \lambda = 0. \end{cases} \quad (22.2.1)$$

For a sample  $y_1, \dots, y_n > 0$  with geometric mean  $\dot{y} = \exp(n^{-1} \sum_i \log y_i)$ , the **normalized** transformation is  $z_i^{(\lambda)} = y_i^{(\lambda)} / \dot{y}^{\lambda-1}$ , and  $\mathbf{z}^{(\lambda)} = (z_1^{(\lambda)}, \dots, z_n^{(\lambda)})^\top$ .

In a model with an intercept  $y^{(\lambda)}$  and  $y^\lambda$  give the same fit. Subtracting 1 and dividing by  $\lambda$  make the family *continuous* in  $\lambda$ , with the logarithm in its place, and increasing for every  $\lambda$ .

**Proposition 22.2.1 — Properties of the family**

Let  $y > 0$ .

- (a)  $y^{(\lambda)} \rightarrow \log y$  as  $\lambda \rightarrow 0$ , so  $\lambda \mapsto y^{(\lambda)}$  is continuous.
- (b) For each  $\lambda$ ,  $y \mapsto y^{(\lambda)}$  is strictly increasing on  $(0, \infty)$ , with derivative  $y^{\lambda-1}$ . The normalized transformation has derivative  $(y/\dot{y})^{\lambda-1}$ , which equals 1 at  $y = \dot{y}$  for every  $\lambda$ .
- (c) If  $p < q$ , then  $y^{(q)} = \varphi(y^{(p)})$  for a strictly increasing and strictly convex function  $\varphi$ . Equivalently, moving down the ladder from  $q$  to  $p$  applies a strictly concave increasing function.



- (d) As  $y$  ranges over  $(0, \infty)$ ,  $y^{(\lambda)}$  ranges over  $(-1/\lambda, \infty)$  if  $\lambda > 0$ , over  $(-\infty, -1/\lambda)$  if  $\lambda < 0$ , and over  $\mathbb{R}$  if  $\lambda = 0$ .

*Proof.*

- (a)  $y^\lambda = \exp(\lambda \log y) = 1 + \lambda \log y + O(\lambda^2)$ , so  $(y^\lambda - 1)/\lambda = \log y + O(\lambda)$ .
- (b) For  $\lambda \neq 0$  the derivative of  $(y^\lambda - 1)/\lambda$  is  $y^{\lambda-1} > 0$ , and for  $\lambda = 0$  the derivative of  $\log y$  is  $y^{-1}$ . Dividing by the constant  $y^{\lambda-1}$  gives the second statement.
- (c) Let  $u = y^{(p)}$ . By (b) the map  $y \mapsto u$  has an increasing inverse  $y = \psi(u)$  with  $\psi'(u) = y^{1-p}$ . Then  $\varphi = (\cdot)^{(q)} \circ \psi$  has derivative  $\varphi'(u) = y^{q-1} y^{1-p} = \psi(u)^{q-p}$ . Since  $q - p > 0$  and  $\psi$  is strictly increasing,  $\varphi'$  is positive and strictly increasing, so  $\varphi$  is strictly increasing and strictly convex. The inverse of a strictly increasing convex function is strictly increasing and concave.
- (d)  $y^\lambda$  takes every value in  $(0, \infty)$ . For  $\lambda > 0$  subtracting 1 and dividing by  $\lambda$  gives  $(-1/\lambda, \infty)$ , and for  $\lambda < 0$  the division reverses the interval to  $(-\infty, -1/\lambda)$ .

■

Part (c) is the ladder's advice made precise: a plot of  $y^{(q)}$  that bends upwards is a convex function of  $y^{(p)}$ , and a lower power removes some of the bend. Part (d) is a warning. For  $\lambda \neq 0$  the transformed response is bounded on one side, so it *cannot* be exactly normal. The Box-Cox model is always an approximation, harmless when the bound is many standard deviations from the data.

## 22.2.2 The likelihood

The working model is that for some  $\lambda$  the  $Y_i^{(\lambda)}$  are independent with  $Y_i^{(\lambda)} \sim N(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \sigma^2)$ , ignoring the small normal probability beyond the bound of Proposition 22.2.1(d). The likelihood must be written for the *observed*  $y_i$ , since otherwise changing  $\lambda$  changes the data being explained. That is where the Jacobian enters, and the geometric mean with it.

### Theorem 22.2.2 — Profile likelihood of the Box-Cox parameter

Let  $\mathbf{X}$  have rank  $r < n$ , let  $y_1, \dots, y_n > 0$ , and for each  $\lambda$  let  $\text{SSE}_z(\lambda) = \|(\mathbf{I} - \mathbf{M})\mathbf{z}^{(\lambda)}\|^2$ , assumed positive. Under the model above:

- (a) the log-likelihood of  $(\boldsymbol{\beta}, \sigma^2, \lambda)$  is

$$\begin{aligned} \ell(\boldsymbol{\beta}, \sigma^2, \lambda) = & -\frac{n}{2} \log(2\pi\sigma^2) - \frac{\|\mathbf{y}^{(\lambda)} - \mathbf{X}\boldsymbol{\beta}\|^2}{2\sigma^2} \\ & + (\lambda - 1) \sum_{i=1}^n \log y_i; \end{aligned} \quad (22.2.2)$$

- (b) for fixed  $\lambda$  it is maximized by the least squares fit of  $\mathbf{y}^{(\lambda)}$  on  $\mathbf{X}$ , and the profile log-likelihood of  $\lambda$  is

$$\ell_p(\lambda) = -\frac{n}{2} \left\{ \log \left( \frac{2\pi \text{SSE}_z(\lambda)}{n} \right) + 1 \right\}, \quad (22.2.3)$$

so the maximum likelihood estimate  $\hat{\lambda}$  is the value that minimizes  $\text{SSE}_z(\lambda)$ ;

- (c) the likelihood ratio statistic for  $H : \lambda = \lambda_0$  is  $2\{\ell_p(\hat{\lambda}) - \ell_p(\lambda_0)\} = n \log\{\text{SSE}_z(\lambda_0)/\text{SSE}_z(\hat{\lambda})\}$ , and the likelihood ratio confidence set of level  $1 - \alpha$  is

$$\left\{ \lambda_0 : n \log \frac{\text{SSE}_z(\lambda_0)}{\text{SSE}_z(\hat{\lambda})} \leq \chi_{1,\alpha}^2 \right\}, \quad (22.2.4)$$

where  $\chi_{1,\alpha}^2$  is the upper  $\alpha$  point of  $\chi^2(1)$ .

*Proof.*

- (a) The map  $y \mapsto y^{(\lambda)}$  is strictly increasing and differentiable with derivative  $y^{\lambda-1}$  (Proposition 22.2.1(b)). By the change-of-variables formula the density of  $Y_i$  at  $y_i$  is the  $N(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \sigma^2)$  density at  $y_i^{(\lambda)}$  times  $y_i^{\lambda-1}$ . The  $Y_i$  are independent, so the log-likelihood is the sum of the logs, which is Eq. 22.2.2.
- (b) The Jacobian term does not involve  $(\boldsymbol{\beta}, \sigma^2)$ , and the other two terms are the log-likelihood of the normal linear model with response  $\mathbf{y}^{(\lambda)}$ . By Theorem 7.3.1 its maximum over  $(\boldsymbol{\beta}, \sigma^2)$  is attained at any least squares fit and equals  $-\frac{n}{2}\{\log(2\pi \text{SSE}(\mathbf{y}^{(\lambda)})/n) + 1\}$ . Now  $\mathbf{y}^{(\lambda)} = \dot{y}^{\lambda-1} \mathbf{z}^{(\lambda)}$ , so  $\text{SSE}(\mathbf{y}^{(\lambda)}) = \dot{y}^{2(\lambda-1)} \text{SSE}_z(\lambda)$ , and

$$-\frac{n}{2} \log \dot{y}^{2(\lambda-1)} = -n(\lambda-1) \log \dot{y} = -(\lambda-1) \sum_{i=1}^n \log y_i.$$

This cancels the Jacobian term exactly and leaves Eq. 22.2.3, which is a decreasing function of  $\text{SSE}_z(\lambda)$ .

- (c) Subtract two values of Eq. 22.2.3. ■

The theorem is algebraic. The *coverage* of Eq. 22.2.4 is a separate, asymptotic matter: if the model holds at  $\lambda_0$  and the regularity conditions of Wilks's theorem are satisfied, the probability that the set contains  $\lambda_0$  tends to  $1 - \alpha$  (van der Vaart 1998, chapter 16). That result is not proved here, and by Proposition 22.2.1(d) its hypothesis can hold exactly only at  $\lambda_0 = 0$ . Otherwise, as Hernandez and Johnson (1980) showed,  $\hat{\lambda}$  converges to the value whose normal model is closest in Kullback–Leibler divergence to the true distribution, and the interval describes the precision of that best approximation.

*Remark* (Why normalize). Residual sums of squares of  $\mathbf{y}^{(\lambda)}$  are in the units of  $y^\lambda$  and cannot be compared across  $\lambda$ . The geometric mean builds the Jacobian into the response: every  $\mathbf{z}^{(\lambda)}$  is in the units of  $y$ , with unit slope at  $\dot{y}$  (Proposition 22.2.1(b)), so  $s_z(\lambda) = \{\text{SSE}_z(\lambda)/(n-r)\}^{1/2}$  can be compared across  $\lambda$ , and minimizing it is maximizing the likelihood.

The normalization also makes the analysis independent of the units in which  $y$  is recorded.

### Proposition 22.2.3 — Units of the response

Suppose  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ . If every  $y_i$  is multiplied by the same  $c > 0$ , then  $\text{SSE}_z(\lambda)$  is multiplied

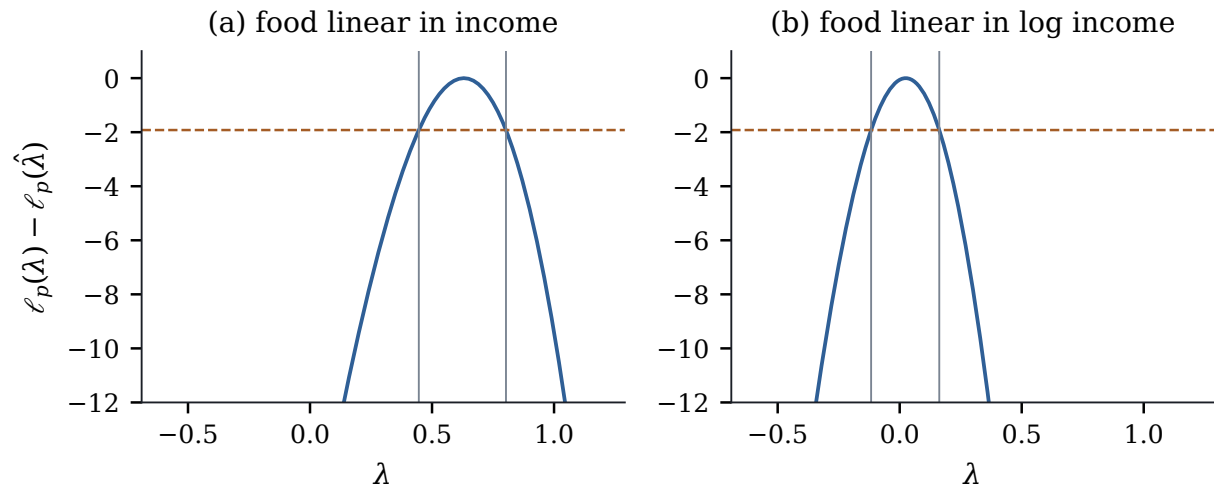


Figure 22.2.1: Box–Cox profile log-likelihoods for Engel’s data, relative to their maxima. (a) Food expenditure linear in income. (b) Linear in log income. The dashed line is at  $-\chi^2_{1,0.05}/2$ , and the vertical lines mark the 95% likelihood ratio intervals.

by  $c^2$  for every  $\lambda$ . Consequently  $\ell_p$  changes by a constant, and  $\hat{\lambda}$ , the likelihood ratio statistics and the set [Eq. 22.2.4](#) are unchanged.

*Proof.* The geometric mean of  $cy_1, \dots, cy_n$  is  $c\bar{y}$ . For  $\lambda \neq 0$ ,

$$\frac{(cy_i)^\lambda - 1}{\lambda(c\bar{y})^{\lambda-1}} = c \frac{y_i^\lambda - 1}{\lambda\bar{y}^{\lambda-1}} + c \frac{1 - c^{-\lambda}}{\lambda\bar{y}^{\lambda-1}},$$

and for  $\lambda = 0$ ,  $c\bar{y} \log(cy_i) = c\bar{y} \log y_i + c\bar{y} \log c$ . In both cases the new normalized vector is  $c\mathbf{z}^{(\lambda)} + k\mathbf{1}$  for a constant  $k$ . Since  $(\mathbf{I} - \mathbf{M})\mathbf{1} = \mathbf{0}$ , its residual vector is  $c(\mathbf{I} - \mathbf{M})\mathbf{z}^{(\lambda)}$ , and  $\text{SSE}_z$  is multiplied by  $c^2$ . By [Eq. 22.2.3](#),  $\ell_p$  changes by  $-n \log c$ , a constant. ■

### Example. (Box–Cox for Engel’s data)

First let transformed food expenditure be linear in income. The profile log-likelihood, panel (a) of [Figure 22.2.1](#), peaks at  $\hat{\lambda} = 0.63$ , and the 95% interval given by [Eq. 22.2.4](#) is  $[0.45, 0.80]$ . Both the raw scale and the logarithm are rejected: the likelihood ratio statistics are 18.8 for  $\lambda = 1$  and 37.2 for  $\lambda = 0$ , against the critical value 3.84.

Now let food expenditure be linear in *log* income. The profile, panel (b), peaks at  $\hat{\lambda} = 0.025$ , with interval  $[-0.117, 0.162]$ . The logarithm is well inside, with statistic 0.12, while the square root (46.6) and the raw scale (181.4) are firmly rejected. With log income as the regressor, the normalized residual standard deviation is 78.6 on the log scale and 115.6 on the raw scale, both in the currency of the survey (the geometric mean of food expenditure is 574.4).

The answers differ because a response transformation is chosen to make a *particular* mean function linear. [Section 22.3](#) estimates the two powers together.

```
import numpy as np
import statsmodels.api as sm
```

```

from scipy import optimize, stats

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)

def box_cox_z(y, lam):
    """Box-Cox transform normalized by the geometric mean, so every lambda is in the
    units of y."""
    gm = np.exp(np.mean(np.log(y)))
    if abs(lam) < 1e-12:
        return gm * np.log(y)
    return np.expm1(lam * np.log(y)) / (lam * gm ** (lam - 1))  # (y^lam - 1)/(lam
    gm^(lam-1))

def sse(v, X):
    coef, *_ = np.linalg.lstsq(X, v, rcond=None)
    r = v - X @ coef
    return r @ r

def profile(lam, y, X):
    """Profile log-likelihood of lambda, up to an additive constant."""
    return -0.5 * len(y) * np.log(sse(box_cox_z(y, lam), X))

def box_cox_fit(y, X, level=0.95):
    """MLE of lambda and the likelihood-ratio interval {lambda: 2(l_max - l_p) <=
    chi2_1}."""
    lam_hat = optimize.minimize_scalar(lambda l: -profile(l, y, X), bounds=(-3, 3),
    method="bounded",
                                options={"xatol": 1e-10}).x

    l_max = profile(lam_hat, y, X)
    cut = stats.chi2.ppf(level, 1) / 2
    g = lambda l: l_max - profile(l, y, X) - cut
    lo = optimize.brentq(g, lam_hat - 3, lam_hat)
    hi = optimize.brentq(g, lam_hat, lam_hat + 3)
    return lam_hat, (lo, hi), l_max

X_lin = np.column_stack([np.ones(n), income])  # food linear in income
X_log = np.column_stack([np.ones(n), np.log(income)])  # food linear in log income
for name, X in [("income", X_lin), ("log income", X_log)]:
    lam_hat, (lo, hi), l_max = box_cox_fit(food, X)
    lr = {lam: 2 * (l_max - profile(lam, food, X)) for lam in (0, 0.5, 1)}
    print(f"{name:10s} lambda_hat {lam_hat:.3f} 95% interval [{lo:.3f}, {hi:.3f}]
    LR at 0, 1/2, 1:",
        " ".join(f"{v:.1f}" for v in lr.values()))

```

### 22.2.3 Choosing a convenient value

Nobody thinks in units of food expenditure raised to the power 0.025. The usual practice is to pick a simple value inside the interval, here  $\lambda = 0$ , and to analyse on that scale as if it had

been chosen in advance. Simple values have interpretations: the logarithm gives percentages and elasticities, the reciprocal turns a time into a rate.

Treating the chosen scale as known ignores that the data chose it. Bickel and Doksum (1981) showed that when  $\lambda$  is estimated, the unconditional variance of  $\hat{\beta}$  can be many times its fixed- $\lambda$  value. Box and Cox (1982) replied that the unconditional variance of a coefficient whose *units* change with  $\lambda$  has no scientific meaning, and Hinkley and Runger (1984) argued for inference conditional on the selected scale. Quantities whose meaning does not depend on  $\lambda$ , such as a prediction of  $y$  or the hypothesis that a regressor has no effect at all ([Exercise \(C1\)](#)), avoid the difficulty, and resampling ([Chapter 23](#)) can include the choice of scale in their uncertainty.

## 22.2.4 The constructed variable

A single extra regression gives a quick check of a proposed  $\lambda_0$  and shows which observations drive the evidence. Let

$$\mathbf{w}^{(\lambda_0)} = \left. \frac{\partial \mathbf{z}^{(\lambda)}}{\partial \lambda} \right|_{\lambda=\lambda_0}, \quad (22.2.5)$$

computed entry by entry. Differentiating  $(y^\lambda - 1)y^{1-\lambda}/\lambda$  at  $\lambda_0 = 1$  gives  $w_i = y_i\{\log(y_i/\bar{y}) - 1\} + 1 + \log \bar{y}$ , and at  $\lambda_0 = 0$  it gives  $w_i = \frac{1}{2}\dot{y}\{\log(y_i/\bar{y})\}^2 - \frac{1}{2}\dot{y}(\log \bar{y})^2$  ([Exercise \(B1\)](#)). In a model with an intercept the additive constants do not matter, so the **constructed variables** are  $y_i\{\log(y_i/\bar{y}) - 1\}$  for checking the raw scale and  $\frac{1}{2}\dot{y}\{\log(y_i/\bar{y})\}^2$  for checking the logarithm.

### Proposition 22.2.4 — A constructed variable for the response power

Let  $\hat{\mathbf{e}}_0 = (\mathbf{I} - \mathbf{M})\mathbf{z}^{(\lambda_0)}$  and  $\mathbf{w} = \mathbf{w}^{(\lambda_0)}$ , with  $\mathbf{w} \notin \mathcal{C}(\mathbf{X})$ .

- (a)  $\left. \frac{d}{d\lambda} \text{SSE}_z(\lambda) \right|_{\lambda_0} = 2\hat{\mathbf{e}}_0^\top \mathbf{w}$ , and so  $\ell'_p(\lambda_0) = -n \hat{\mathbf{e}}_0^\top \mathbf{w} / \text{SSE}_z(\lambda_0)$ .
- (b) In the least squares fit of  $\mathbf{z}^{(\lambda_0)}$  on  $[\mathbf{X}, \mathbf{w}]$ , the coefficient of  $\mathbf{w}$  is  $\hat{\gamma} = \hat{\mathbf{e}}_0^\top \mathbf{w} / \|(\mathbf{I} - \mathbf{M})\mathbf{w}\|^2$ . Hence  $\hat{\gamma}$  and  $\ell'_p(\lambda_0)$  have opposite signs, and  $\hat{\gamma} = 0$  iff  $\lambda_0$  is a stationary point of the profile log-likelihood. In particular  $\hat{\gamma} = 0$  at an interior maximum  $\lambda_0 = \hat{\lambda}$ .

*Proof.*

- (a)  $\text{SSE}_z(\lambda) = \mathbf{z}^{(\lambda)\top}(\mathbf{I} - \mathbf{M})\mathbf{z}^{(\lambda)}$ . Its derivative is  $2\mathbf{z}^{(\lambda)\top}(\mathbf{I} - \mathbf{M})\partial\mathbf{z}^{(\lambda)}/\partial\lambda$ , because  $\mathbf{I} - \mathbf{M}$  is symmetric. At  $\lambda_0$  this is  $2\hat{\mathbf{e}}_0^\top \mathbf{w}$ . The chain rule applied to [Eq. 22.2.3](#) gives the derivative of  $\ell_p$ .
- (b) By [Theorem 6.6.2](#) the coefficient of  $\mathbf{w}$  is  $\tilde{\mathbf{w}}^\top \mathbf{z}^{(\lambda_0)} / \|\tilde{\mathbf{w}}\|^2$  with  $\tilde{\mathbf{w}} = (\mathbf{I} - \mathbf{M})\mathbf{w} \neq \mathbf{0}$ , and  $\tilde{\mathbf{w}}^\top \mathbf{z}^{(\lambda_0)} = \mathbf{w}^\top (\mathbf{I} - \mathbf{M})\mathbf{z}^{(\lambda_0)} = \mathbf{w}^\top \hat{\mathbf{e}}_0$ . Comparing with (a) gives the rest. ■

To first order  $\mathbf{z}^{(\lambda)} \approx \mathbf{z}^{(\lambda_0)} + (\lambda - \lambda_0)\mathbf{w}$ , so if the model holds at  $\lambda$ , then  $\mathbf{z}^{(\lambda_0)} \approx \mathbf{X}\beta - (\lambda - \lambda_0)\mathbf{w} + \boldsymbol{\varepsilon}$  and  $\lambda_0 - \hat{\gamma}$  is a rough one-step estimate of  $\lambda$ . The added-variable plot of  $\mathbf{w}$  ([Theorem 6.6.2](#)) shows which observations carry the evidence, the use Atkinson (1985) made of it. The  $t$  statistic

of  $\hat{\gamma}$  tests  $\lambda = \lambda_0$ , but not exactly:  $\mathbf{w}$  is a function of  $\mathbf{y}$  itself, not only of the fitted values, so [Theorem 14.7.2](#) does not apply.

**Example. (The constructed variable for Engel's data)**

In the model with log income, the constructed variable for  $\lambda_0 = 1$  has coefficient  $\hat{\gamma} = 0.831$  and  $t = 17.96$ : overwhelming evidence against the raw scale. The one-step estimate  $1 - 0.831 = 0.169$  is far from  $\hat{\lambda} = 0.025$ , as expected from so far away. For  $\lambda_0 = 0$  the coefficient is  $-0.031$  with  $t = -0.38$ .

In 4000 data sets simulated from the fitted log-log model, with normal errors and Engel's incomes (so  $\lambda = 0$  is true), the 95% likelihood ratio interval covers 0 in a fraction 0.945, close to nominal, while the nominal 5% constructed-variable test rejects in a fraction 0.075. The constructed variable is a diagnostic; the likelihood interval is for inference.

Deleting one household at a time, the largest change moves  $\hat{\lambda}$  from 0.025 to 0.081, when the household with income 2823 and food expenditure 2033, the largest in the survey, is removed. It lies well above the line in panels (a) and (c) of [Figure 22.1.1](#), at a point of high leverage (the leverage of Engel's households is examined in [Section 20.2](#)). The logarithm stays inside the interval whichever household is removed. For a fixed scale [Chapter 20](#) gives deletion formulas ([Theorem 20.3.1](#), and [Proposition 10.6.2](#) for their computation); for  $\hat{\lambda}$  the refits are cheap enough to do directly.

```
def constructed(y, lam0, h=1e-5):
    """w = d z(lambda)/d lambda at lam0 (numerical derivative of the normalized
    transform)."""
    return (box_cox_z(y, lam0 + h) - box_cox_z(y, lam0 - h)) / (2 * h)

def constructed_test(y, X, lam0):
    """Regress z(lam0) on [X, w]: the coefficient of w and its t statistic."""
    w = constructed(y, lam0)
    fit = sm.OLS(box_cox_z(y, lam0), np.column_stack([X, w])).fit()
    return fit.params[-1], fit.tvalues[-1]

for lam0 in (1, 0):
    g, t = constructed_test(food, X_log, lam0)
    print(f"lambda0 = {lam0}: coefficient of w {g:.3f}, t = {t:.2f}, one-step
    estimate {lam0 - g:.3f}")
```

In the browser the simulation uses 200 replications, so its fractions are rough.

```
def simulate(reps, seed):
    rng = np.random.default_rng(seed)
    fit = sm.OLS(np.log(food), X_log).fit()           # the log-log fit: data
    generated with lambda = 0
    mu, sigma = fit.fittedvalues, np.sqrt(fit.scale)
    cover = reject = 0
    for _ in range(reps):
        y_sim = np.exp(mu + sigma * rng.standard_normal(n))
        lam_s, (lo_s, hi_s), _ = box_cox_fit(y_sim, X_log)
        cover += lo_s <= 0 <= hi_s
        reject += abs(constructed_test(y_sim, X_log, 0)[1]) > stats.t.ppf(0.975, n -
        3)
```

```

return cover / reps, reject / reps

coverage, size = simulate(reps=200, seed=2201)
print(f"coverage of the 95% LR interval {coverage:.3f}, size of the 5% constructed-
      variable test {size:.3f}")

```

*Warning.*  $\hat{\lambda}$  responds to anything that spoils the normal linear model: curvature, a variance that changes with the level, skewness, single outlying cases. A value that makes the residuals look better has not necessarily corrected the defect that mattered. After transforming, look at the residual plots of [Chapter 20](#) and [Chapter 21](#) again, on the new scale.

## 22.2.5 Exercises

### 22.2.5.1 A. Check your understanding

#### Exercise A1.

Compute  $4^{(\lambda)}$  for  $\lambda = -1, 0, \frac{1}{2}, 1, 2$ . Check that the values increase with  $\lambda$ , and explain, using the definition, why  $y^{(\lambda)}$  is increasing in  $\lambda$  for every fixed  $y > 0$ .

#### Exercise A2.

Show that if every  $y_i$  is multiplied by  $c > 0$ , the residual sum of squares of the *unnormalized*  $\mathbf{y}^{(\lambda)}$  in a model with an intercept is multiplied by  $c^{2\lambda}$ . Conclude that choosing  $\lambda$  by minimizing the unnormalized residual sum of squares would give an answer that depends on the units of  $y$ .

*Solution.* For  $\lambda \neq 0$ ,  $((cy)^\lambda - 1)/\lambda = c^\lambda(y^\lambda - 1)/\lambda + (c^\lambda - 1)/\lambda$ , a multiple  $c^\lambda$  of  $y^{(\lambda)}$  plus a constant, which the intercept absorbs. So the residual sum of squares is multiplied by  $c^{2\lambda}$ , a factor that depends on  $\lambda$ . Changing the units reweights the comparison between different values of  $\lambda$ : as  $c$  grows the comparison is pushed towards ever more negative  $\lambda$ , and as  $c$  shrinks towards ever larger  $\lambda$ . By [Proposition 22.2.3](#) the normalized version has the common factor  $c^2$  and no such problem.

### 22.2.5.2 B. Practice

#### Exercise B1.

Derive the constructed variable at  $\lambda_0 = 0$ . Write  $z^{(\lambda)} = f(\lambda) \dot{y}^{1-\lambda}$  with  $f(\lambda) = (y^\lambda - 1)/\lambda$ , expand  $f(\lambda) = \log y + \frac{1}{2}\lambda(\log y)^2 + O(\lambda^2)$ , and show that  $\partial z^{(\lambda)}/\partial \lambda$  at 0 is  $\frac{1}{2}\dot{y}\{\log(y/\dot{y})\}^2 - \frac{1}{2}\dot{y}(\log \dot{y})^2$ .

*Solution.* The expansion follows from  $y^\lambda = 1 + \lambda \log y + \frac{1}{2}\lambda^2(\log y)^2 + O(\lambda^3)$ , so  $f(0) = \log y$  and  $f'(0) = \frac{1}{2}(\log y)^2$ . Also  $\dot{y}^{1-\lambda} = \dot{y} \exp(-\lambda \log \dot{y})$  has value  $\dot{y}$  and derivative

$-\dot{y} \log \dot{y}$  at 0. By the product rule the derivative of  $z^{(\lambda)}$  at 0 is

$$\dot{y} \left\{ \frac{1}{2} (\log y)^2 - \log y \log \dot{y} \right\} = \frac{1}{2} \dot{y} \{ (\log y - \log \dot{y})^2 - (\log \dot{y})^2 \}.$$

### Exercise B2.

In [Example \(Box-Cox for Engel's data\)](#), with log income as regressor, the normalized residual standard deviations are 78.6 at  $\lambda = 0$  and 115.6 at  $\lambda = 1$ . Show from [Eq. 22.2.3](#) that the difference between the likelihood ratio statistics at  $\lambda = 1$  and  $\lambda = 0$  is  $2n \log\{s_z(1)/s_z(0)\}$ , and check it against the values 181.4 and 0.12 of the example.

### Exercise B3.

For data that may include zeros, consider the shifted family  $(y+c)^{(\lambda)}$  with an unknown  $c$  such that every  $y_i + c > 0$ . Write the log-likelihood of  $(\boldsymbol{\beta}, \sigma^2, \lambda, c)$ , show that the profile log-likelihood of  $(\lambda, c)$  is [Eq. 22.2.3](#) with  $\mathbf{z}^{(\lambda)}$  computed from  $y_i + c$  and its geometric mean, and state the likelihood ratio confidence region for  $(\lambda, c)$ .

*Solution.* The Jacobian of  $y \mapsto (y+c)^{(\lambda)}$  is  $(y+c)^{\lambda-1}$ , so the log-likelihood is [Eq. 22.2.2](#) with  $y_i$  replaced by  $y_i + c$  throughout. The proof of [Theorem 22.2.2](#) goes through word for word for each fixed  $c$ , with  $\dot{y}$  the geometric mean of the  $y_i + c$ . The region is  $\{(\lambda, c) : n \log\{\text{SSE}_z(\lambda, c)/\text{SSE}_z(\hat{\lambda}, \hat{c})\} \leq \chi_{2,\alpha}^2\}$ , two degrees of freedom because two parameters are profiled. The maximization needs care near the boundary  $c \rightarrow -\min_i y_i$ : there  $\log(\min_i y_i + c) \rightarrow -\infty$ , so for  $\lambda < 1$  the Jacobian term  $(\lambda-1) \sum_i \log(y_i + c)$  grows without bound, and the likelihood can be unbounded, as it is for the three-parameter lognormal distribution.

## 22.2.5.3 C. Going deeper

### Exercise C1.

Suppose that the conditional distribution of  $Y$  given  $(x_1, \dots, x_k)$  does not depend on  $x_1$ . Show that the same holds for  $g(Y)$  for every strictly increasing  $g$ , so the hypothesis that  $x_1$  has no effect is the same hypothesis on every scale. Why does the converse direction, from a zero coefficient on one scale to a zero coefficient on another, fail when the conditional distribution does depend on  $x_1$ ? Give an example with two groups whose medians are equal but whose means are not.

### Exercise C2.

Let  $\hat{\lambda}$  be an interior maximizer of  $\ell_p$ , and write  $s(\lambda) = \hat{\mathbf{e}}(\lambda)^\top \mathbf{w}^{(\lambda)}$ , which is zero at  $\hat{\lambda}$  by [Proposition 22.2.4](#). Suppose case  $i$  is deleted. Show, by one Newton step for the equation  $s = 0$ , that the change in  $\hat{\lambda}$  is approximately proportional to the contribution  $\hat{\varepsilon}_{(i)} \tilde{w}_{(i)}$  of that case, where both factors are computed with the case deleted. Which cases does this predict to be influential for  $\hat{\lambda}$ , and how does leverage in the regression of  $\mathbf{w}$  on  $\mathbf{X}$  enter?



## 22.3 Transforming regressors

Transforming a regressor changes only the mean function. The errors are untouched, and there is no Jacobian, because the data being explained stay the same. What is at stake is bias. If  $E(Y) = \mathbf{X}_2\boldsymbol{\beta}_2 + \beta_1 g(\mathbf{x}_1)$  and the analyst fits  $\mathbf{x}_1$  itself, the missing part  $\beta_1\{g(\mathbf{x}_1) - \mathbf{x}_1\}$  acts as an omitted regressor (Proposition 5.5.7(a) and Theorem 19.1.2). It shows up as curvature in the residuals, and tests against a larger model detect it (Proposition 14.7.1 and Theorem 14.7.2). This section finds the right  $g$ , first by looking, then by estimating a power.

### 22.3.1 Component-plus-residual plots

The component-plus-residual plot of Section 20.1 plots the partial residuals  $\hat{\epsilon} + \hat{\beta}_1\mathbf{x}_1$  against  $\mathbf{x}_1$ . Its least squares line has slope  $\hat{\beta}_1$  and residuals  $\hat{\epsilon}$ , as in the added-variable plot, and that section warns that it can mislead when  $\mathbf{x}_1$  is related nonlinearly to the other regressors. The next result measures how far.

#### Proposition 22.3.1 — What the plot shows

Let  $\mathbf{X} = [\mathbf{x}_1, \mathbf{X}_2]$  have full column rank, with  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_2)$ , and let  $\hat{\boldsymbol{\beta}}$  and  $\hat{\epsilon}$  be the coefficients and residuals of the least squares fit of  $\mathbf{y}$  on  $\mathbf{X}$ . Let  $\mathbf{M}$  and  $\mathbf{M}_2$  project onto  $\mathcal{C}(\mathbf{X})$  and  $\mathcal{C}(\mathbf{X}_2)$ , and  $\tilde{\mathbf{x}}_1 = (\mathbf{I} - \mathbf{M}_2)\mathbf{x}_1$ .

- (a) If  $E(\mathbf{Y}) = \mathbf{g} + \mathbf{X}_2\boldsymbol{\beta}_2$ , where  $\mathbf{g}$  has entries  $g(x_{i1})$  for some function  $g$ , then

$$E(\hat{\epsilon} + \hat{\beta}_1\mathbf{x}_1) = \mathbf{g} - \mathbf{M}_2(\mathbf{g} - b\mathbf{x}_1), \quad b = E(\hat{\beta}_1) = \frac{\tilde{\mathbf{x}}_1^\top \mathbf{g}}{\|\tilde{\mathbf{x}}_1\|^2}. \quad (22.3.1)$$

- (b) In particular, if  $\mathcal{C}(\mathbf{X}_2)$  is spanned by  $\mathbf{1}$  alone, the expected partial residuals are  $\mathbf{g}$  shifted by a constant, so the plot shows  $g$  exactly apart from its level.

*Proof.*

- (a) Since  $\mathbf{X}_2\boldsymbol{\beta}_2 \in \mathcal{C}(\mathbf{X})$ ,  $E(\hat{\epsilon}) = (\mathbf{I} - \mathbf{M})(\mathbf{g} + \mathbf{X}_2\boldsymbol{\beta}_2) = (\mathbf{I} - \mathbf{M})\mathbf{g}$ . By Theorem 6.6.2(a),  $\hat{\beta}_1 = \tilde{\mathbf{x}}_1^\top \mathbf{Y} / \|\tilde{\mathbf{x}}_1\|^2$ , whose mean is  $\tilde{\mathbf{x}}_1^\top \mathbf{g} / \|\tilde{\mathbf{x}}_1\|^2 = b$  because  $\tilde{\mathbf{x}}_1 \perp \mathcal{C}(\mathbf{X}_2)$ . By Lemma 6.6.1(c),  $\mathbf{M} = \mathbf{M}_2 + \tilde{\mathbf{x}}_1\tilde{\mathbf{x}}_1^\top / \|\tilde{\mathbf{x}}_1\|^2$ , so  $(\mathbf{I} - \mathbf{M})\mathbf{g} = \mathbf{g} - \mathbf{M}_2\mathbf{g} - b\tilde{\mathbf{x}}_1$ . Adding  $b\mathbf{x}_1 = b\tilde{\mathbf{x}}_1 + b\mathbf{M}_2\mathbf{x}_1$  gives  $\mathbf{g} - \mathbf{M}_2\mathbf{g} + b\mathbf{M}_2\mathbf{x}_1$ , which is Eq. 22.3.1.
- (b) If  $\mathcal{C}(\mathbf{X}_2) = \text{span}\{\mathbf{1}\}$ , then  $\mathbf{M}_2\mathbf{v} = \bar{v}\mathbf{1}$  for every  $\mathbf{v}$ .

■

The distortion  $\mathbf{M}_2(\mathbf{g} - b\mathbf{x}_1)$  is the part of the *nonlinearity*  $\mathbf{g} - b\mathbf{x}_1$  that the other regressors reproduce linearly. Correlation with  $\mathbf{x}_1$  is not in itself harmful; a regressor that resembles the curvature is. Cook (1993) showed for random regressors that the plot recovers  $g$ , up to a linear term, when the conditional mean of the other regressors given  $x_1$  is linear in  $x_1$ , and proposed CERES plots for other cases. The **augmented partial residual plot** of Mallows (1986) adds  $\mathbf{x}_1^2$  to the fit and plots  $\hat{\epsilon} + \hat{\beta}_1\mathbf{x}_1 + \hat{\beta}_{11}\mathbf{x}_1^2$ . Its expectation is  $\mathbf{g}$  minus the part of  $\mathbf{g} - b_1\mathbf{x}_1 - b_{11}\mathbf{x}_1^2$  that the other regressors reproduce, small when  $g$  is nearly quadratic (Exercise (B3)).

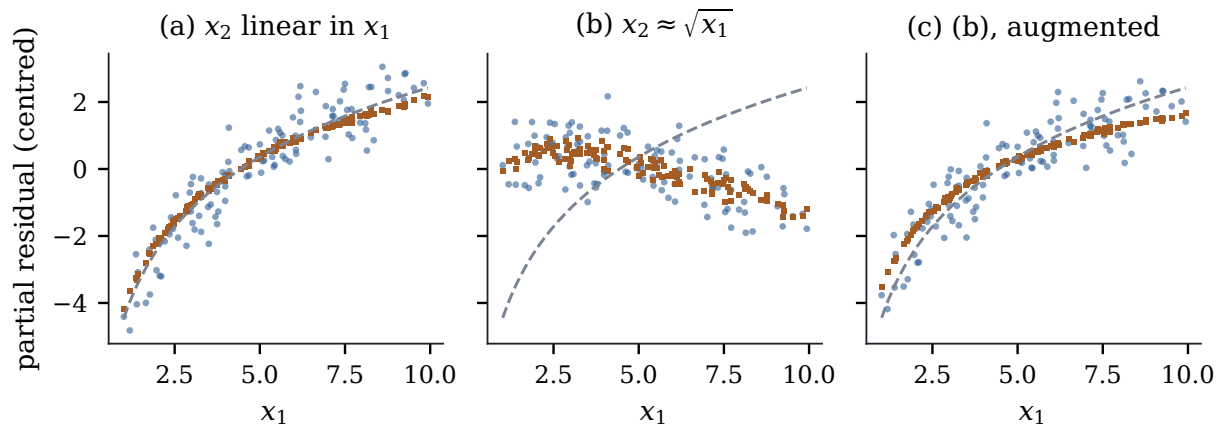


Figure 22.3.1: Component-plus-residual plots for simulated data with  $E(Y) = 3 \log x_1 + x_2$ . Points: partial residuals. Dashed: the true  $g$ , centred. Small squares: the expected partial residuals Eq. 22.3.1, which scatter because they depend on the simulated  $x_2$ . (a)  $x_2$  linearly related to  $x_1$ . (b)  $x_2$  close to  $\sqrt{x_1}$ : the plot hides the curvature. (c) The augmented plot for the data of (b).

#### Example. (When the plot misleads)

The script simulates 120 cases with  $E(Y) = 3 \log x_1 + x_2$ ,  $x_1$  uniform on  $[1, 10]$ , and measures how much of the curvature of  $g$  the expected plot keeps (the coefficient of the nonlinear part of  $g$  in that of Eq. 22.3.1; 1 is faithful). In panel (a) of Figure 22.3.1,  $x_2$  is linear in  $x_1$  plus noise, with correlation 0.95, and the plot keeps 0.99. In panel (b),  $x_2$  is a noisy square root of  $x_1$ , with correlation 0.99; it resembles the logarithm and takes over much of the curvature, and the plot keeps only 0.45, its slope even changing sign. The augmented plot, panel (c), restores 0.88.

```
import numpy as np

rng = np.random.default_rng(2204)
n = 120
x1 = rng.uniform(1, 10, n)
g = 3 * np.log(x1)                                     # the true shape in x1
standardize = lambda v: (v - v.mean()) / v.std()
x2_lin = standardize(x1) + rng.normal(0, 0.3, n)         # linear in x1, plus noise
x2_sqrt = standardize(np.sqrt(x1)) + rng.normal(0, 0.1, n) # nonlinear in x1, plus
noise

def lstsq(X, v):
    coef, *_ = np.linalg.lstsq(X, v, rcond=None)
    return coef

def partial_residuals(y, x2, augmented=False):
    """e + b1 x1 (+ b11 x1^2 if augmented), from the fit of y on 1, x1 (, x1^2), x2.
    """
    X = np.column_stack([np.ones(len(y)), x1] + ([x1**2] if augmented else []) + [x2])
    coef = lstsq(X, y)
    e = y - X @ coef
```

```

return e + coef[1] * x1 + (coef[2] * x1**2 if augmented else 0)

def curvature_retained(v):
    """Coefficient of the nonlinear part of g in the nonlinear part of v (1 = shape
    kept)."""
    Z = np.column_stack([np.ones(n), x1])
    nonlin = lambda u: u - Z @ lstsq(Z, u)
    return nonlin(v) @ nonlin(g) / (nonlin(g) @ nonlin(g))

noise = rng.normal(0, 0.6, n)
results = {}
for name, x2, aug in [("linear", x2_lin, False), ("square root", x2_sqrt, False),
                     ("square root, augmented", x2_sqrt, True)]:
    mean = g + x2 # E(Y); partial residuals
                  # are linear in y
    results[name] = (partial_residuals(mean + noise, x2, aug), partial_residuals(
    mean, x2, aug))
    print(f"{name:23s} corr(x1, x2) {np.corrcoef(x1, x2)[0, 1]:.2f}    curvature
    retained "
          f"{curvature_retained(results[name][1]):.2f}")

```

## 22.3.2 Power transformations: Box and Tidwell

When the plot suggests a power, it can be estimated. Box and Tidwell (1962) considered the mean function

$$E(\mathbf{Y}) = \mathbf{X}_2 \boldsymbol{\beta}_2 + \beta_1 \mathbf{x}^{(\alpha)}, \quad \mathbf{x}^{(\alpha)} = (x_1^{(\alpha)}, \dots, x_n^{(\alpha)})^\top, \quad (22.3.2)$$

for a positive regressor  $\mathbf{x}$ , with  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_2)$  and Eq. 22.2.1 applied to the regressor. For fixed  $\alpha$  this is a linear model, and  $\hat{\alpha}$  minimizes the residual sum of squares  $\text{SSE}(\alpha)$  of the fit of  $\mathbf{y}$  on  $[\mathbf{X}_2, \mathbf{x}^{(\alpha)}]$ . Under normal errors it is the maximum likelihood estimate, and  $\{\alpha : n \log\{\text{SSE}(\alpha)/\text{SSE}(\hat{\alpha})\} \leq \chi_{1,0.05}^2\}$  is an asymptotic 95% interval under the conditions of Wilks's theorem, which include  $\beta_1 \neq 0$  (Exercise (C1)). The derivative of  $x^{(\alpha)}$  with respect to  $\alpha$  at  $\alpha_0 \neq 0$  is  $x^{\alpha_0} \log x / \alpha_0 - x^{(\alpha_0)} / \alpha_0$ , and at  $\alpha_0 = 0$  it is  $\frac{1}{2}(\log x)^2$  (Exercise (B1)). Since  $x^{(\alpha_0)}$  and  $\mathbf{1}$  are already in the model, the part that matters is the **constructed variable**

$$\mathbf{w} = \begin{cases} (x_i^{\alpha_0} \log x_i / \alpha_0)_{i=1}^n, & \alpha_0 \neq 0, \\ (\frac{1}{2}(\log x_i)^2)_{i=1}^n, & \alpha_0 = 0. \end{cases} \quad (22.3.3)$$

For  $\alpha_0 = 1$  it is  $x \log x$ .

### Theorem 22.3.2 — Box–Tidwell

Let  $\mathbf{X}_0 = [\mathbf{X}_2, \mathbf{x}^{(\alpha_0)}]$  have full column rank  $p$ , with  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_2)$ , let  $\mathbf{w}$  be the constructed variable Eq. 22.3.3, and suppose  $[\mathbf{X}_0, \mathbf{w}]$  has full column rank  $p+1 < n$ . Let  $\hat{\beta}_1$  be the coefficient of  $\mathbf{x}^{(\alpha_0)}$  and  $\hat{\boldsymbol{\varepsilon}}_0$  the residual vector in the least squares fit of  $\mathbf{y}$  on  $\mathbf{X}_0$ , and let  $\hat{\gamma}$  be the coefficient of  $\mathbf{w}$  in the fit of  $\mathbf{y}$  on  $[\mathbf{X}_0, \mathbf{w}]$ .

- (a) **Exact test.** If  $\mathbf{Y} \sim N_n(\mathbf{X}_0\boldsymbol{\beta}, \sigma^2\mathbf{I})$ , which is Eq. 22.3.2 with  $\alpha = \alpha_0$ , then the  $t$  statistic of  $\hat{\gamma}$  has the  $t(n - p - 1)$  distribution.
- (b) **The step.** If  $\hat{\beta}_1 \neq 0$ , one Gauss–Newton step for minimizing  $\|\mathbf{y} - \mathbf{X}_2\boldsymbol{\beta}_2 - \beta_1\mathbf{x}^{(\alpha)}\|^2$ , started from the least squares coefficients at  $\alpha_0$ , moves  $\alpha$  to

$$\alpha_1 = \alpha_0 + \hat{\gamma}/\hat{\beta}_1. \quad (22.3.4)$$

- (c) **Stationarity.**  $\text{SSE}(\alpha)$  is differentiable at  $\alpha_0$  with  $\text{SSE}'(\alpha_0) = -2\hat{\beta}_1\hat{\boldsymbol{\epsilon}}_0^\top \mathbf{w} = -2\hat{\beta}_1\hat{\gamma}\|(\mathbf{I} - \mathbf{M}_0)\mathbf{w}\|^2$ , where  $\mathbf{M}_0$  projects onto  $\mathcal{C}(\mathbf{X}_0)$ . So if  $\hat{\beta}_1 \neq 0$ , then  $\hat{\gamma} = 0$  iff  $\alpha_0$  is a stationary point of SSE, and every fixed point of the iteration Eq. 22.3.4 is a stationary point.

*Proof.* Write  $\mathbf{d} = \partial\mathbf{x}^{(\alpha)}/\partial\alpha$  at  $\alpha_0$ . By the derivative computed above,  $\mathbf{d} - \mathbf{w}$  is a multiple of  $\mathbf{x}^{(\alpha_0)}$ , so  $\mathbf{d} - \mathbf{w} \in \mathcal{C}(\mathbf{X}_0)$  and  $(\mathbf{I} - \mathbf{M}_0)\mathbf{d} = (\mathbf{I} - \mathbf{M}_0)\mathbf{w} =: \tilde{\mathbf{w}}$ , which is nonzero because  $[\mathbf{X}_0, \mathbf{w}]$  has full rank. By Theorem 6.6.2(a),  $\hat{\gamma} = \tilde{\mathbf{w}}^\top \mathbf{y}/\|\tilde{\mathbf{w}}\|^2$ , and  $\tilde{\mathbf{w}}^\top \mathbf{y} = \mathbf{w}^\top (\mathbf{I} - \mathbf{M}_0)\mathbf{y} = \mathbf{w}^\top \hat{\boldsymbol{\epsilon}}_0$ .

- (a) The column  $\mathbf{w}$  depends on  $\mathbf{x}$  alone, not on  $\mathbf{Y}$ . The  $t$  test for  $\hat{\gamma}$  is therefore the  $F$  test of  $\mathcal{C}(\mathbf{X}_0)$  against the fixed larger space  $\mathcal{C}([\mathbf{X}_0, \mathbf{w}])$ , with  $t^2 = F$  and a symmetric null distribution. By Proposition 14.7.1(a),  $F \sim F(1, n - p - 1)$  when  $\mathbf{E}(\mathbf{Y}) \in \mathcal{C}(\mathbf{X}_0)$ , so  $t \sim t(n - p - 1)$ .
- (b) The mean function  $\boldsymbol{\mu}(\boldsymbol{\beta}_2, \beta_1, \alpha) = \mathbf{X}_2\boldsymbol{\beta}_2 + \beta_1\mathbf{x}^{(\alpha)}$  has derivative matrix  $\mathbf{J} = [\mathbf{X}_2, \mathbf{x}^{(\alpha_0)}, \hat{\beta}_1\mathbf{d}]$  at the starting point, where the residual is  $\hat{\boldsymbol{\epsilon}}_0$ . The Gauss–Newton step is the least squares coefficient vector of  $\hat{\boldsymbol{\epsilon}}_0$  on  $\mathbf{J}$ , and its last entry is the change in  $\alpha$ . By Theorem 6.6.2(a) that entry is

$$\frac{\{(\mathbf{I} - \mathbf{M}_0)\hat{\beta}_1\mathbf{d}\}^\top \hat{\boldsymbol{\epsilon}}_0}{\|(\mathbf{I} - \mathbf{M}_0)\hat{\beta}_1\mathbf{d}\|^2} = \frac{1}{\hat{\beta}_1} \frac{\tilde{\mathbf{w}}^\top \hat{\boldsymbol{\epsilon}}_0}{\|\tilde{\mathbf{w}}\|^2} = \frac{\hat{\gamma}}{\hat{\beta}_1},$$

using  $\tilde{\mathbf{w}}^\top \hat{\boldsymbol{\epsilon}}_0 = \tilde{\mathbf{w}}^\top \mathbf{y}$ , which holds because  $\tilde{\mathbf{w}} \perp \mathcal{C}(\mathbf{X}_0)$ .

- (c) Let  $\mathbf{X}(\alpha) = [\mathbf{X}_2, \mathbf{x}^{(\alpha)}]$ , which has full column rank near  $\alpha_0$ , and let  $\hat{\boldsymbol{\beta}}(\alpha)$  be its least squares coefficients, a differentiable function of  $\alpha$ . Then  $\text{SSE}(\alpha) = \|\mathbf{y} - \mathbf{X}(\alpha)\hat{\boldsymbol{\beta}}(\alpha)\|^2$ , and

$$\text{SSE}'(\alpha_0) = -2\hat{\boldsymbol{\epsilon}}_0^\top \{\hat{\beta}_1\mathbf{d} + \mathbf{X}_0\hat{\boldsymbol{\beta}}'(\alpha_0)\} = -2\hat{\beta}_1\hat{\boldsymbol{\epsilon}}_0^\top \mathbf{d} = -2\hat{\beta}_1\hat{\boldsymbol{\epsilon}}_0^\top \mathbf{w},$$

because  $\hat{\boldsymbol{\epsilon}}_0 \perp \mathcal{C}(\mathbf{X}_0)$  and  $\mathbf{d} - \mathbf{w} \in \mathcal{C}(\mathbf{X}_0)$ . Finally  $\hat{\boldsymbol{\epsilon}}_0^\top \mathbf{w} = \hat{\gamma}\|\tilde{\mathbf{w}}\|^2$  from the first paragraph. ■

Part (a) is exact because this constructed variable is built from the fixed regressor, not from the response as in Proposition 22.2.4. Part (b) identifies the Box–Tidwell iteration with Gauss–Newton, fast near a well-determined minimum and liable to oscillate otherwise, so in one dimension it is safer to minimize  $\text{SSE}(\alpha)$  directly; by part (c) the two agree at convergence. Several powers can be stepped at once (Exercise (C2)).

### Example. (Transforming income)

With  $\log(\text{food})$  as the response and  $\text{income}^{(\alpha)}$  as the regressor, starting from  $\alpha_0 = 1$ ,

the Box–Tidwell iteration gives  $-0.053$  after one step and  $-0.182$  after two, where it stays. The direct minimization agrees:  $\hat{\alpha} = -0.182$ , with 95% likelihood ratio interval  $[-0.334, -0.036]$ . The constructed-variable  $t$  statistic is  $-16.58$  at  $\alpha_0 = 1$ , so untransformed income is decisively rejected, and  $-2.47$  at  $\alpha_0 = 0$ , beyond the critical value  $1.97$ , so the logarithm is rejected at the 5% level as well. As a check on part (a), 4000 data sets simulated from the log–log fit give a rejection rate of  $0.047$  for this test.

The rejection rests largely on one household. The richest, with income 4958 and food expenditure 1827, sits far to the right in panel (c) of [Figure 22.1.1](#), below the line. Without it,  $\hat{\alpha} = -0.119$ , and the likelihood ratio statistic for  $\alpha = 0$  falls from  $5.98$  to  $1.77$ . A negative power says food expenditure flattens among the richest households, where the data are few; Cook’s distance ([Definition 20.4.1](#)) pursues this.

```
import numpy as np
import statsmodels.api as sm

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)
y = np.log(food)

def bc(x, a):
    """Box-Cox transform (x^a - 1)/a, with log x at a = 0."""
    return np.log(x) if abs(a) < 1e-12 else np.expml(a * np.log(x)) / a

def fit(y, x, a):
    return sm.OLS(y, np.column_stack([np.ones(len(y)), bc(x, a)]).fit()

def box_tidwell_step(y, x, a0):
    """One Box-Tidwell (Gauss-Newton) step: a1 = a0 + gamma/beta1."""
    beta1 = fit(y, x, a0).params[1] # slope of x^(a0) in the null
    fit
    w = x**a0 * np.log(x) / a0 if a0 != 0 else np.log(x) ** 2 / 2 # d bc(x, a)/da,
    up to C(X0)
    aug = sm.OLS(y, np.column_stack([np.ones(len(y)), bc(x, a0), w])).fit()
    return a0 + aug.params[2] / beta1, aug.tvalues[2]

a, path = 1.0, [1.0]
for _ in range(8):
    a, t = box_tidwell_step(y, income, a)
    path.append(a)
print("Box-Tidwell iterates from alpha = 1:", " ".join(f"{v:.4f}" for v in path))
```

### 22.3.3 Choosing both scales together

The best power for the response depends on the scale of the regressor, and conversely, so the two can be estimated together. With response  $\mathbf{z}^{(\lambda)}$  and model matrix  $[\mathbf{1}, \mathbf{x}^{(\alpha)}]$  the proof of [Theorem 22.2.2](#) goes through unchanged, since the Jacobian involves only  $\mathbf{y}$ : the joint profile log-likelihood is  $-\frac{n}{2} \log \text{SSE}_z(\lambda, \alpha)$  plus a constant.

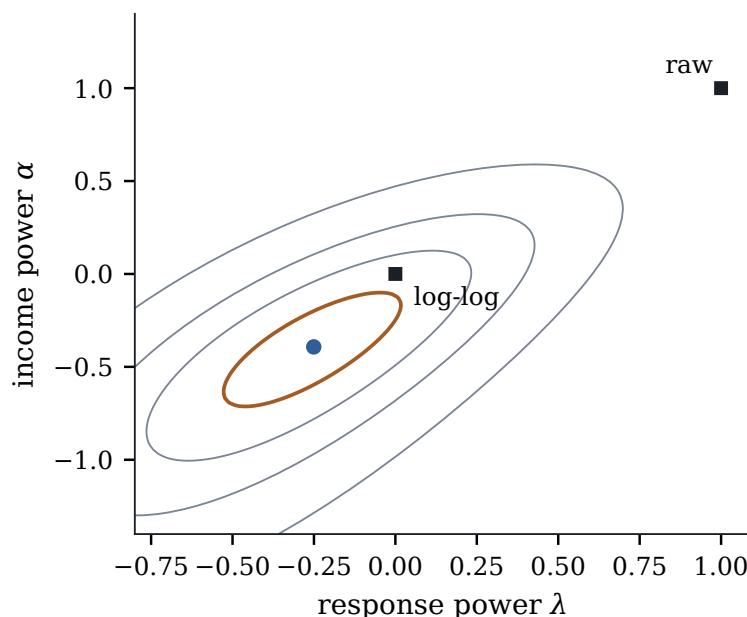


Figure 22.3.2: Joint profile log-likelihood of the response power  $\lambda$  and the income power  $\alpha$  for Engel's data. The thick contour bounds the 95% likelihood ratio region, the thin contours are 10, 20 and 40 units of log-likelihood below the maximum (the dot). Squares mark the log-log and raw models.

#### Example. (Two powers for Engel's data)

Figure 22.3.2 shows the joint profile. The maximum is at  $(\hat{\lambda}, \hat{\alpha}) = (-0.25, -0.39)$ . The raw scale  $(1, 1)$  has likelihood ratio statistic 186.5, and the best pair with untransformed income,  $(0.63, 1)$  from [Example \(Box-Cox for Engel's data\)](#), has 167.8: whatever is done to the response, income needs transforming. The log-log pair  $(0, 0)$  has statistic 11.2, outside the 95% region, whose cut-off is 5.99 for two parameters. Without the richest household the statistic falls to 6.55, still outside.

The region is a long tilted ellipse, since a lower response power can be partly compensated by a lower income power ([Exercise \(B2\)](#)). The log-log model lies outside the region (11.2 against 5.99), largely because of the richest household. It is still the natural working choice for interpretation, since its slope is an elasticity, with the caveat that the data reject a constant elasticity and that it may fall among the richest households ([Proposition 22.5.3](#)).

### 22.3.4 Polynomials and other alternatives

Powers suit positive regressors with monotone, smoothly bending effects. Otherwise the alternative is to add columns rather than change one: a polynomial or a spline. These buy flexibility with interpretability, since no single coefficient measures the effect of  $x$  ([Definition 5.7.1](#), third qualification), and high-degree polynomials behave badly at the edges of the data. Royston and Altman's (1994) *fractional polynomials* use one or two terms from a small set of powers including the logarithm. Chapter 42 treats polynomial and spline regression.

## 22.3.5 Exercises

### 22.3.5.1 A. Check your understanding

#### Exercise A1.

At  $\alpha_0 = 1$  a fit gives  $\hat{\beta}_1 = 2.5$ , and adding  $x \log x$  gives it the coefficient  $\hat{\gamma} = -1.5$ . What is the next Box–Tidwell value of  $\alpha$ ? What does the sign of  $\hat{\gamma}$  say about the curvature of the relation?

### 22.3.5.2 B. Practice

#### Exercise B1.

Show that  $\partial x^{(\alpha)} / \partial \alpha = x^\alpha \log x / \alpha - x^{(\alpha)} / \alpha$  for  $\alpha \neq 0$ , and that its limit as  $\alpha \rightarrow 0$  is  $\frac{1}{2}(\log x)^2$ . Conclude that in a model containing **1** and  $\mathbf{x}^{(\alpha_0)}$  the constructed variable may be taken to be [Eq. 22.3.3](#).

*Solution.* Differentiate  $(x^\alpha - 1)/\alpha$ : the quotient rule gives  $\{\alpha x^\alpha \log x - (x^\alpha - 1)\}/\alpha^2 = x^\alpha \log x / \alpha - x^{(\alpha)} / \alpha$ . As  $\alpha \rightarrow 0$ , expand  $x^\alpha = 1 + \alpha \log x + \frac{1}{2}\alpha^2(\log x)^2 + O(\alpha^3)$ : the numerator is  $\alpha \log x + \alpha^2(\log x)^2 - \alpha \log x - \frac{1}{2}\alpha^2(\log x)^2 + O(\alpha^3) = \frac{1}{2}\alpha^2(\log x)^2 + O(\alpha^3)$ , and dividing by  $\alpha^2$  gives the limit. The term  $-\mathbf{x}^{(\alpha_0)}/\alpha_0$  lies in the model space, and removing it changes neither the residualized column  $\tilde{\mathbf{w}}$  nor  $\hat{\gamma}$ .

#### Exercise B2.

Suppose  $y = c x^\beta$  exactly, with  $c, \beta > 0$ . Show that for  $\lambda \neq 0$ ,  $y^{(\lambda)}$  is an affine function of  $x^{(\alpha)}$  iff  $\alpha = \beta\lambda$ , and that for  $\lambda = 0$  it is affine in  $x^{(\alpha)}$  iff  $\alpha = 0$ . Explain why data close to a power law give a joint likelihood for  $(\lambda, \alpha)$  with a ridge, and why the slope of the ridge need not be exactly  $\beta$  when the variance also depends on  $\lambda$ .

*Solution.* For  $\lambda \neq 0$ ,  $y^\lambda = c^\lambda x^{\beta\lambda}$ , so  $y^{(\lambda)} = (c^\lambda x^{\beta\lambda} - 1)/\lambda$ , an affine function of  $x^{\beta\lambda}$ . An affine function of  $x^{\beta\lambda}$  is affine in  $x^\alpha$  (equivalently in  $x^{(\alpha)}$ ) for all  $x > 0$  iff  $\alpha = \beta\lambda$ , since distinct powers, and a power and the logarithm, are affinely independent functions on  $(0, \infty)$ . For  $\lambda = 0$ ,  $\log y = \log c + \beta \log x$  is affine in  $x^{(\alpha)}$  iff  $\alpha = 0$ , which is again  $\alpha = \beta\lambda$ . So every pair on the line  $\alpha = \beta\lambda$  makes the mean exactly linear, and nearby data make the residual sum of squares small along it. The likelihood also rewards a constant variance and normal errors, which depend on  $\lambda$  alone, so the maximum lies somewhere along the tilted valley, and its exact direction mixes the two requirements.

#### Exercise B3.

Let  $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_1^2, \mathbf{X}_2]$ , with  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_2)$  and full column rank, and let  $E(\mathbf{Y}) = \mathbf{g} + \mathbf{X}_2\boldsymbol{\beta}_2$ . Show that the augmented partial residuals  $\hat{\mathbf{e}} + \hat{\beta}_1\mathbf{x}_1 + \hat{\beta}_{11}\mathbf{x}_1^2$  have expectation  $\mathbf{g} - \mathbf{M}_2(\mathbf{g} - b_1\mathbf{x}_1 - b_{11}\mathbf{x}_1^2)$ , where  $(b_1, b_{11})$  are the expectations of  $(\hat{\beta}_1, \hat{\beta}_{11})$ . Conclude that the plot is exact, up to a vertical shift, when  $g$  is a quadratic.

*Solution.* Let  $\mathbf{Q} = [\mathbf{x}_1, \mathbf{x}_1^2]$  and  $\tilde{\mathbf{Q}} = (\mathbf{I} - \mathbf{M}_2)\mathbf{Q}$ . As in the proof of [Proposition 22.3.1](#),  $\mathbf{M} = \mathbf{M}_2 + \tilde{\mathbf{Q}}(\tilde{\mathbf{Q}}^\top \tilde{\mathbf{Q}})^{-1} \tilde{\mathbf{Q}}^\top$  and the coefficient vector  $\mathbf{b}$  of  $\mathbf{Q}$  has expectation  $(\tilde{\mathbf{Q}}^\top \tilde{\mathbf{Q}})^{-1} \tilde{\mathbf{Q}}^\top \mathbf{g}$ . Then  $E(\hat{\mathbf{e}}) = (\mathbf{I} - \mathbf{M})\mathbf{g} = \mathbf{g} - \mathbf{M}_2\mathbf{g} - \mathbf{Q}\mathbf{b}$ , and adding  $\mathbf{Q}\mathbf{b} = \tilde{\mathbf{Q}}\mathbf{b} + \mathbf{M}_2\mathbf{Q}\mathbf{b}$  gives  $\mathbf{g} - \mathbf{M}_2(\mathbf{g} - \mathbf{Q}\mathbf{b})$ . If  $\mathbf{g} = c_0\mathbf{1} + c_1\mathbf{x}_1 + c_2\mathbf{x}_1^2$ , then  $\mathbf{g}$  lies in  $\mathcal{C}(\mathbf{X})$ ,  $\mathbf{b} = (c_1, c_2)^\top$ , and  $\mathbf{g} - \mathbf{Q}\mathbf{b} = c_0\mathbf{1}$ , whose projection is  $c_0\mathbf{1}$ .

#### Exercise B4.

In the setting of [Proposition 22.3.1](#), let  $\mathbf{X}_2 = [\mathbf{1}, \mathbf{q}]$  with  $q_i = x_{i1}^2$ , so the second regressor is an exact quadratic in the first, and write  $\boldsymbol{\beta}_2 = (\beta_{20}, \delta)^\top$ .

- If  $g(x) = cx^2$ , show that  $b = 0$  and that the expected partial residuals are  $\mathbf{0}$ : the plot is flat, although  $E(Y)$  curves in  $x_1$  when  $\delta \neq -c$ .
- If  $g(x) = cx$ , show that the plot is exact.
- Explain why no plot against  $x_1$  can tell which of the two regressors carries the curvature in (a).

*Solution.*

- Here  $\mathbf{g} = c\mathbf{q} \in \mathcal{C}(\mathbf{X}_2)$ , so  $\tilde{\mathbf{x}}_1^\top \mathbf{g} = 0$  and  $b = 0$ ; then  $\mathbf{M}_2(\mathbf{g} - b\mathbf{x}_1) = \mathbf{M}_2\mathbf{g} = \mathbf{g}$  and [Eq. 22.3.1](#) is  $\mathbf{0}$ . The whole of  $g$  is the nonlinearity  $\mathbf{g} - b\mathbf{x}_1$ , and  $\mathbf{q}$  reproduces all of it. The mean  $E(Y) = \beta_{20} + (c + \delta)x_1^2$  curves unless  $c + \delta = 0$ .
- Now  $b = c\tilde{\mathbf{x}}_1^\top \mathbf{x}_1 / \|\tilde{\mathbf{x}}_1\|^2 = c$ , because  $\tilde{\mathbf{x}}_1^\top \mathbf{x}_1 = \|\tilde{\mathbf{x}}_1\|^2$ , so  $\mathbf{g} - b\mathbf{x}_1 = \mathbf{0}$  and the expected partial residuals are  $\mathbf{g}$ .
- The models  $E(Y) = \beta_{20} + cx_1^2 + \delta x_1^2$  with the same  $c + \delta$  have the same mean for every case: the split between  $g$  and  $\mathbf{X}_2\boldsymbol{\beta}_2$  is not identified from the data, so no display of them can recover it.

#### 22.3.5.3 C. Going deeper

##### Exercise C1.

Suppose  $\beta_1 = 0$  in [Eq. 22.3.2](#). Show that the model does not depend on  $\alpha$ , so  $\alpha$  is not identified. Show that the test of [Theorem 22.3.2\(a\)](#) still has exact size, and argue that its power against changes in  $\alpha$  tends to zero as  $\beta_1 \rightarrow 0$ . What happens to the Box–Tidwell step, and to the likelihood ratio interval for  $\alpha$ ?

##### Exercise C2.

Let  $E(\mathbf{Y}) = \mathbf{X}_3\boldsymbol{\beta}_3 + \beta_1\mathbf{x}_1^{(\alpha_1)} + \beta_2\mathbf{x}_2^{(\alpha_2)}$ . Show that one Gauss–Newton step from  $(\alpha_1^0, \alpha_2^0)$  and the least squares coefficients there changes  $\alpha_j$  by  $\hat{\gamma}_j/\hat{\beta}_j$ , where  $\hat{\gamma}_1, \hat{\gamma}_2$  are the coefficients of the two constructed variables in a single regression of  $\mathbf{y}$  on the model matrix augmented by both.



## 22.4 Variance-stabilizing transformations

For many responses the variance is tied to the mean by the nature of the measurement: counts, proportions, amounts measured with a fixed *relative* precision. Then  $\text{Var}(Y) = \phi v(\mu)$  for a known or plausible variance function  $v$ , where  $\mu = E(Y)$  and  $\phi$  is a constant. The transformation that makes the variance constant turns out to be determined by  $v$  up to an affine change.

### 22.4.1 The idea

If  $g$  is smooth and  $Y$  is concentrated near  $\mu$ , then  $g(Y) \approx g(\mu) + g'(\mu)(Y - \mu)$ , so

$$\text{Var } g(Y) \approx g'(\mu)^2 \phi v(\mu).$$

This is constant in  $\mu$  exactly when  $g'(\mu)$  is proportional to  $v(\mu)^{-1/2}$ , that is, when

$$g(\mu) = \int^\mu \frac{dt}{\sqrt{v(t)}}. \quad (22.4.1)$$

Part (a) of the theorem below makes this precise, with  $m$  measuring the information; it is [Exercise \(B1\)](#), where Fisher's  $z$  arose this way. Part (b) adds uniqueness.

#### Theorem 22.4.1 — Variance-stabilizing transformations

Let  $I$  be an open interval and  $v : I \rightarrow (0, \infty)$  continuous. Suppose that for every  $\theta \in I$  the random variables  $T_m$  satisfy  $\sqrt{m}(T_m - \theta) \rightarrow N(0, v(\theta))$  in distribution as  $m \rightarrow \infty$ . Fix  $t_0 \in I$  and let  $g(t) = \int_{t_0}^t v(s)^{-1/2} ds$  for  $t \in I$ , extended in any way outside  $I$ .

- (a) For every  $\theta \in I$ ,  $\sqrt{m}\{g(T_m) - g(\theta)\} \rightarrow N(0, 1)$  in distribution.
- (b) If  $h : I \rightarrow \mathbb{R}$  is continuously differentiable and  $\sqrt{m}\{h(T_m) - h(\theta)\} \rightarrow N(0, 1)$  for every  $\theta \in I$ , then  $h = g + c$  or  $h = -g + c$  on  $I$  for a constant  $c$ .

*Proof.*

- (a) By the fundamental theorem of calculus  $g' = v^{-1/2}$  on  $I$ , so this is [Exercise \(B1\)](#) with  $s = v^{1/2}$ ; the values of  $g$  outside  $I$  do not matter because  $T_m \rightarrow \theta$  in probability.
- (b) By the delta method (stated in [Section 14.4](#)),  $\sqrt{m}\{h(T_m) - h(\theta)\} \rightarrow N(0, h'(\theta)^2 v(\theta))$ . Limits in distribution are unique, so  $h'(\theta)^2 v(\theta) = 1$  and  $|h'(\theta)| = v(\theta)^{-1/2}$  for every  $\theta \in I$ . The continuous function  $h'$  never vanishes on the interval  $I$ , so by the intermediate value theorem it has one sign throughout. Hence  $h' = g'$  or  $h' = -g'$  on  $I$ , and integrating gives the result. ■

Convergence in distribution does not by itself imply that the *variance* of  $g(T_m)$  is close to  $1/m$ ; that needs uniform integrability, which the theorem does not assert ([Exercise \(C2\)](#) supplies it for Poisson counts).

**Corollary 22.4.2 — The classical transformations**

- (a) **Counts.** If  $Y \sim \text{Poisson}(\mu)$ , then  $2(\sqrt{Y} - \sqrt{\mu}) \rightarrow N(0, 1)$  as  $\mu \rightarrow \infty$ .
- (b) **Proportions.** If  $S \sim \text{Binomial}(m, p)$  with  $0 < p < 1$  fixed and  $\hat{p} = S/m$ , then  $2\sqrt{m}\{\arcsin \sqrt{\hat{p}} - \arcsin \sqrt{p}\} \rightarrow N(0, 1)$  as  $m \rightarrow \infty$ .
- (c) **Constant coefficient of variation.** If  $Y = \mu U$  for a positive random variable  $U$  whose distribution does not depend on  $\mu$ , then  $\text{Var}(\log Y) = \text{Var}(\log U)$  exactly, for every  $\mu$ , whenever it is finite.
- (d) **Power variance functions.** If  $v(\mu) = \mu^{2a}$  on  $(0, \infty)$ , then [Eq. 22.4.1](#) gives  $\mu^{1-a}/(1-a)$  for  $a \neq 1$  and  $\log \mu$  for  $a = 1$ . Up to an affine change this is the Box–Cox transformation with  $\lambda = 1 - a$ .

*Proof.*

- (a) The characteristic function of  $(Y - \mu)/\sqrt{\mu}$  is  $\exp\{\mu(e^{it/\sqrt{\mu}} - 1 - it/\sqrt{\mu})\}$ , which tends to  $e^{-t^2/2}$  because  $e^{iu} - 1 - iu = -u^2/2 + O(u^3)$ . By the continuity theorem of Lévy (van der Vaart 1998, chapter 2),  $(Y - \mu)/\sqrt{\mu} \rightarrow N(0, 1)$ . Now write

$$2(\sqrt{Y} - \sqrt{\mu}) = \frac{Y - \mu}{\sqrt{\mu}} \cdot \frac{2\sqrt{\mu}}{\sqrt{Y} + \sqrt{\mu}}.$$

Since  $\text{Var}(Y/\mu) = 1/\mu$ , Chebyshev's inequality gives  $Y/\mu \rightarrow 1$  in probability, so the second factor tends to 1 in probability, and Slutsky's lemma gives the limit.

- (b) By the central limit theorem  $\sqrt{m}(\hat{p} - p) \rightarrow N(0, p(1-p))$ . With  $v(p) = p(1-p)$  on  $(0, 1)$ , the derivative of  $\arcsin \sqrt{p}$  is  $\frac{1}{2}\{p(1-p)\}^{-1/2}$ , so [Eq. 22.4.1](#) gives  $g(p) = 2 \arcsin \sqrt{p}$ , and [Theorem 22.4.1\(a\)](#) applies.
- (c)  $\log Y = \log \mu + \log U$ , and adding a constant does not change a variance.
- (d) Integrate  $t^{-a}$ .

■

Part (c) is exact: for gamma data with shape  $k$ ,  $\text{Var}(\log Y)$  is the trigamma function at  $k$  whatever the mean, 0.2213 for  $k = 5$  against the first-order value  $1/k = 0.2$ . Part (d) is the rule quoted in [Section 22.1](#). A known variance function determines the power; otherwise the Box–Cox likelihood estimates one, chosen partly to stabilize the variance.

## 22.4.2 How well they work

The limits say nothing about moderate means, but the variances can be computed exactly by summing probability mass functions. [Figure 22.4.1](#) and the table show them, scaled so that the target is 1.

| Poisson mean                    |       |       |       |       |       |
|---------------------------------|-------|-------|-------|-------|-------|
| $\mu$                           | 1     | 2     | 5     | 10    | 20    |
| $4 \text{ Var } \sqrt{Y}$       | 1.609 | 1.560 | 1.144 | 1.045 | 1.020 |
| $4 \text{ Var } \sqrt{Y + 3/8}$ | 0.717 | 0.924 | 1.002 | 1.001 | 1.000 |

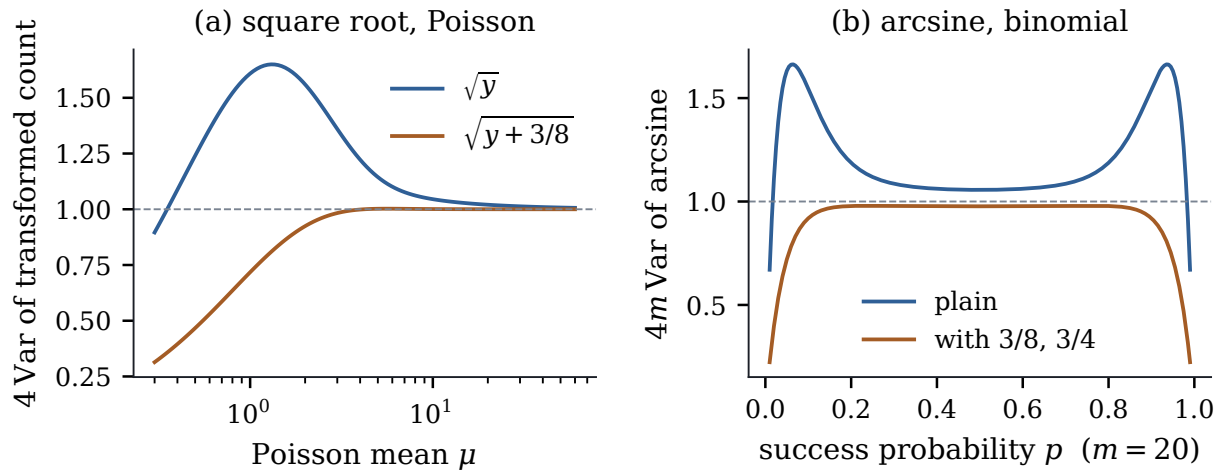


Figure 22.4.1: Exact variances of transformed counts and proportions, scaled so that the target is 1 (dashed). (a) Poisson counts:  $\sqrt{y}$  and Anscombe's  $\sqrt{y + 3/8}$ , against the mean on a log scale. (b) Binomial proportions with  $m = 20$  trials: the arcsine square root, plain and with Anscombe's constants.

Without transformation the variance of a Poisson count grows in proportion to the mean, a factor of 20 between  $\mu = 1$  and  $\mu = 20$ . After the square root the scaled variance stays between 1 and 1.7 over the same range, and from  $\mu = 5$  on it is within 15 percent of the target. Anscombe's (1948) modification  $\sqrt{Y + c} = \frac{1}{4} + (3 - 8c)/(32\mu) + O(\mu^{-2})$  (Exercise (C1)), and  $c = 3/8$  removes the  $1/\mu$  term. For binomial proportions from  $m = 20$  trials,  $4m \text{Var}(\arcsin \sqrt{\hat{p}})$  is 1.056 at  $p = 0.5$ , 1.118 at  $p = 0.25$ , 1.546 at  $p = 0.1$  and 1.632 at  $p = 0.05$ . Anscombe's version  $\arcsin \sqrt{(S + 3/8)/(m + 3/4)}$  gives 0.977, 0.979, 0.919 and 0.720. Both transformations fail near the boundary, where most of the probability sits on a few values and no smooth function can equalize anything.

```
import numpy as np
from scipy import stats

def poisson_var(h, mu):
    """Var h(Y) for Y ~ Poisson(mu), by summing the probability mass function."""
    k = np.arange(0, int(mu + 40 * np.sqrt(mu) + 60))
    p = stats.poisson.pmf(k, mu)
    m1 = p @ h(k)
    return p @ (h(k) - m1) ** 2

def binomial_var(h, m, prob):
    """Var h(S) for S ~ Binomial(m, prob)."""
    s = np.arange(m + 1)
    p = stats.binom.pmf(s, m, prob)
    m1 = p @ h(s)
    return p @ (h(s) - m1) ** 2

for mu in (1, 2, 5, 10, 20):
    print(f"Poisson mean {mu:2d}: 4 Var sqrt(Y) = {4 * poisson_var(np.sqrt, mu):.3f}")
    print(f"4 Var sqrt(Y + 3/8) = {4 * poisson_var(lambda k: np.sqrt(k + 3 / 8),
```

```

    mu):.3f}")
m = 20
for prob in (0.05, 0.1, 0.25, 0.5):
    plain = 4 * m * binomial_var(lambda s: np.arcsin(np.sqrt(s / m)), m, prob)
    anscombe = 4 * m * binomial_var(lambda s: np.arcsin(np.sqrt((s + 3 / 8) / (m + 3
        / 4))), m, prob)
    print(f"binomial m = {m}, p = {prob:.2f}: 4m Var = {plain:.3f}    with 3/8: {
        anscombe:.3f}")

```

A stabilized scale also gives a check: for Poisson counts the residual variance on the square-root scale should be near  $\frac{1}{4}$ . A much larger value says the counts are *overdispersed*, and then the square root no longer stabilizes ([Exercise \(B1\)](#)).

### 22.4.3 Estimating the variance function

With replicates at several settings, plot the log standard deviation of each group against the log of its mean. If  $s \propto \mu^a$  the points lie near a line of slope  $a$ , and the power  $1 - a$  stabilizes the variance ([Exercise \(B3\)](#)), an idea that goes back to Bartlett (1947). Without replicates, residuals can be grouped by fitted value, but misfit of the mean then inflates some spreads, so the plot should come *after* the mean function is roughly right. Whether the variance depends on the regressors at all is tested formally by [Theorem 21.2.2](#).

### 22.4.4 Transform, weight, or model the variance

Transformation is one of three remedies for a variance that depends on the mean.

1. **Transform.** Fit the linear model to  $g(y)$ . The variance becomes nearly constant, but the *mean* is now modelled on the new scale, and statements about  $y$  need the retransformation of [Section 22.5](#).
2. **Weight.** Keep the scale and fit by weighted least squares with weights  $1/v(\hat{\mu}_i)$  ([Theorem 21.3.1](#) and [Definition 6.9.1](#)), or keep ordinary least squares with heteroscedasticity-consistent standard errors ([Theorem 21.4.2](#)).
3. **Model.** Specify the mean through a link function and the variance through  $v$ : a generalized linear model, such as Poisson or logistic regression, whose coefficients refer to the mean of  $y$  itself. [Chapter 34](#) develops this modern default.

The first two can be combined. The *logit* of a binomial proportion is not variance-stabilizing, but its variance is about  $1/\{m p(1 - p)\}$ , so weighted least squares on empirical logits with weights  $m\hat{p}(1 - \hat{p})$  gives a linear model for the log odds ([Exercise \(B2\)](#)), the approach of Grizzle, Starmer and Koch (1969) for categorical data.

### 22.4.5 Exercises

#### 22.4.5.1 A. Check your understanding

##### Exercise A1.

Find the variance-stabilizing transformation for  $v(\mu) = \mu^3$ , the variance function of the

inverse Gaussian distribution. Which Box–Cox power is it?

### Exercise A2.

Let  $T_m$  be the mean of  $m$  independent exponential waiting times with mean  $\theta > 0$ . Find the variance-stabilizing transformation. Use [Theorem 22.4.1\(b\)](#) to show that the square root, although it stabilizes Poisson counts, does not stabilize  $T_m$ , and find the limiting variance of  $\sqrt{m}(\sqrt{T_m} - \sqrt{\theta})$ .

## 22.4.5.2 B. Practice

### Exercise B1.

Negative binomial counts have  $v(\mu) = \mu + \mu^2/\kappa$  for a shape parameter  $\kappa > 0$ . Show that [Eq. 22.4.1](#) gives  $g(\mu) = 2\sqrt{\kappa} \sinh^{-1} \sqrt{\mu/\kappa}$ . Show that  $g$  behaves like  $2\sqrt{\mu}$  for  $\mu \ll \kappa$  and like  $\sqrt{\kappa} \log \mu$  plus a constant for  $\mu \gg \kappa$ , and interpret.

*Solution.* Substitute  $t = \kappa u^2$ :  $dt = 2\kappa u du$  and  $t + t^2/\kappa = \kappa u^2(1 + u^2)$ , so the integrand becomes  $2\kappa u du / \{\sqrt{\kappa} u \sqrt{1 + u^2}\} = 2\sqrt{\kappa} du / \sqrt{1 + u^2}$ , whose integral is  $2\sqrt{\kappa} \sinh^{-1} u$  with  $u = \sqrt{\mu/\kappa}$ . For small  $u$ ,  $\sinh^{-1} u \approx u$ , giving  $2\sqrt{\mu}$ . For large  $u$ ,  $\sinh^{-1} u = \log(2u) + o(1)$ , giving  $\sqrt{\kappa} \log \mu$  plus a constant. Small counts behave like Poisson counts and need a square root; large counts have a nearly constant coefficient of variation and need a logarithm.

### Exercise B2.

Let  $S \sim \text{Binomial}(m, p)$  with  $0 < p < 1$ . Show by the delta method that  $\sqrt{m} \{\text{logit}(S/m) - \text{logit } p\} \rightarrow N(0, 1/\{p(1-p)\})$ , where  $\text{logit } p = \log\{p/(1-p)\}$ . For independent groups  $i = 1, \dots, k$  with  $S_i \sim \text{Binomial}(m_i, p_i)$  and  $\text{logit } p_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , write down the weighted least squares estimate of  $\boldsymbol{\beta}$  from the empirical logits, and its approximate covariance matrix.

*Solution.* The derivative of  $\text{logit } p$  is  $1/\{p(1-p)\}$ , and  $\sqrt{m}(S/m - p) \rightarrow N(0, p(1-p))$ , so the limiting variance is  $p(1-p)/\{p(1-p)\}^2$ . With  $\ell_i = \text{logit}(S_i/m_i)$  and  $\mathbf{W} = \text{diag}\{m_i \hat{p}_i(1 - \hat{p}_i)\}$ , the estimate is  $\boldsymbol{\beta} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \boldsymbol{\ell}$  ([Definition 6.9.1](#)), with approximate covariance  $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$ , since the weights are the approximate inverse variances and no  $\sigma^2$  needs estimating. The empirical logit is infinite when  $S_i = 0$  or  $m_i$ , which is why  $\frac{1}{2}$  is often added to  $S_i$  and to  $m_i - S_i$ .

### Exercise B3.

Replicate observations at four settings have means 10, 40, 160, 640 and standard deviations 2, 5.7, 16, 45. Fit the spread–level line  $\log s = c + a \log \mu$  described above by eye or by least squares, and use [Corollary 22.4.2\(d\)](#) to choose the power.

## 22.4.5.3 C. Going deeper

**Exercise C1.**

Let  $Y \sim \text{Poisson}(\mu)$  and  $c \geq 0$ . Expand  $\sqrt{Y+c} = \sqrt{\mu}(1+u)^{1/2}$  with  $u = (Y-\mu+c)/\mu$  to third order in  $u$ , use the Poisson central moments  $E(Y-\mu)^2 = \mu$ ,  $E(Y-\mu)^3 = \mu$ ,  $E(Y-\mu)^4 = 3\mu^2 + \mu$ , and show formally that  $\text{Var} \sqrt{Y+c} = \frac{1}{4} + (3-8c)/(32\mu) + O(\mu^{-2})$ . (The expansion is formal: making the remainder rigorous needs a bound on the tail of  $Y$ .)

**Exercise C2.**

[Corollary 22.4.2\(a\)](#) is a statement about distributions. Show that the variance converges too:  $4 \text{Var} \sqrt{Y} \rightarrow 1$  as  $\mu \rightarrow \infty$  for  $Y \sim \text{Poisson}(\mu)$ . Use the bound  $(\sqrt{Y} - \sqrt{\mu})^2 \leq (Y-\mu)^2/\mu$ , and the fact that a family of nonnegative random variables that converges in distribution, and whose means converge to the mean of the limit, is uniformly integrable.

*Solution.* The bound holds because  $\sqrt{Y} - \sqrt{\mu} = (Y-\mu)/(\sqrt{Y} + \sqrt{\mu})$  and  $\sqrt{Y} + \sqrt{\mu} \geq \sqrt{\mu}$ . The variables  $V_\mu = (Y-\mu)^2/\mu$  converge in distribution to  $\chi^2(1)$  (by the proof of [Corollary 22.4.2\(a\)](#)) and have mean exactly 1, the mean of the limit, so they are uniformly integrable, and so is the dominated family  $W_\mu = 4(\sqrt{Y} - \sqrt{\mu})^2$ . Since  $W_\mu \rightarrow \chi^2(1)$  in distribution, uniform integrability gives  $E W_\mu \rightarrow 1$ . The same argument applied to  $2(\sqrt{Y} - \sqrt{\mu})$ , which is dominated by  $1 + W_\mu$ , gives  $E\{2(\sqrt{Y} - \sqrt{\mu})\} \rightarrow 0$ . Hence  $4 \text{Var} \sqrt{Y} = E W_\mu - [E\{2(\sqrt{Y} - \sqrt{\mu})\}]^2 \rightarrow 1$ .

## 22.5 Interpreting a transformed model

A model fitted on a transformed scale answers questions on that scale, but readers ask about the original one: how much will a household with income 2000 spend on food, within what range, and how much more if its income rises? The dividing line is simple. Quantiles pass through a monotone transformation unchanged; means do not.

Throughout this section  $g$  is strictly increasing with inverse  $h$ , and the model on the transformed scale is

$$g(Y_i) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i, \quad i = 1, \dots, n, \quad (22.5.1)$$

where  $\varepsilon_1, \dots, \varepsilon_n$  are independent with distribution  $F$ , mean 0 and variance  $\sigma^2$ .

### 22.5.1 Medians and means

#### **Proposition 22.5.1 — Retransformation bias and the smearing estimate**

Assume [Eq. 22.5.1](#).

- (a) **Quantiles.** If  $\xi_q$  is a  $q$ -quantile of  $F$ , then  $h(\mathbf{x}^\top \boldsymbol{\beta} + \xi_q)$  is a  $q$ -quantile of  $Y$  at  $\mathbf{x}$ .

In particular, if  $F$  has median 0 (for instance, if it is symmetric), the median of  $Y$  at  $\mathbf{x}$  is  $h(\mathbf{x}^\top \boldsymbol{\beta})$ .

- (b) **Means.**  $E(Y | \mathbf{x}) = \int h(\mathbf{x}^\top \boldsymbol{\beta} + e) dF(e)$  whenever this is finite. If  $h$  is strictly convex and  $\sigma^2 > 0$ , then  $E(Y | \mathbf{x}) > h(\mathbf{x}^\top \boldsymbol{\beta})$ . For  $h = \exp$ ,  $E(Y | \mathbf{x}) = \exp(\mathbf{x}^\top \boldsymbol{\beta}) E e^\varepsilon$ , and for normal errors  $E e^\varepsilon = \exp(\sigma^2/2)$ .
- (c) **Smearing.** Let  $h = \exp$ , let  $\mathbf{X}$  have full column rank  $p$  for every  $n$ , with leverages  $h_{ii}$ , and let the  $\varepsilon_i$  be the first  $n$  terms of an independent and identically distributed sequence with  $E e^\varepsilon < \infty$ . If  $\max_i h_{ii} \rightarrow 0$  as  $n \rightarrow \infty$ , then the **smearing factor** satisfies

$$\frac{1}{n} \sum_{i=1}^n \exp(\hat{\varepsilon}_i) \rightarrow E e^\varepsilon \quad \text{in probability.} \quad (22.5.2)$$

If moreover  $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \rightarrow \mathbf{x}_0^\top \boldsymbol{\beta}$  in probability, the **smearing estimate**  $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}) n^{-1} \sum_i \exp(\hat{\varepsilon}_i)$  converges in probability to  $E(Y | \mathbf{x}_0)$ .

*Proof.*

- (a)  $Y \leq h(\mathbf{x}^\top \boldsymbol{\beta} + \xi_q)$  iff  $g(Y) \leq \mathbf{x}^\top \boldsymbol{\beta} + \xi_q$  iff  $\varepsilon \leq \xi_q$ , because  $g$  is strictly increasing, and likewise with strict inequalities. So the two events have the probabilities that define a  $q$ -quantile.
- (b)  $Y = h(\mathbf{x}^\top \boldsymbol{\beta} + \varepsilon)$ . If  $h$  is strictly convex and  $\varepsilon$  is not degenerate, Jensen's inequality is strict:  $E h(\mathbf{x}^\top \boldsymbol{\beta} + \varepsilon) > h(\mathbf{x}^\top \boldsymbol{\beta} + E \varepsilon) = h(\mathbf{x}^\top \boldsymbol{\beta})$ . For  $h = \exp$  the factor  $\exp(\mathbf{x}^\top \boldsymbol{\beta})$  comes out of the expectation, and the normal moment generating function (Theorem 3.2.1) at 1 is  $\exp(\sigma^2/2)$ .
- (c) Put  $\mathbf{d} = \mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \mathbf{M}\boldsymbol{\varepsilon}$ , so that  $\hat{\varepsilon}_i = \varepsilon_i - d_i$ . With  $\mathbf{e}_i$  the  $i$ th standard basis vector,  $d_i = \mathbf{e}_i^\top \mathbf{M}\boldsymbol{\varepsilon} = (\mathbf{M}\mathbf{e}_i)^\top (\mathbf{M}\boldsymbol{\varepsilon})$ , so by the Cauchy-Schwarz inequality and Proposition 6.8.1(a),

$$|d_i| \leq \|\mathbf{M}\mathbf{e}_i\| \|\mathbf{M}\boldsymbol{\varepsilon}\| = \sqrt{h_{ii}} \|\mathbf{M}\boldsymbol{\varepsilon}\|.$$

Since  $E\|\mathbf{M}\boldsymbol{\varepsilon}\|^2 = \sigma^2 \text{tr}(\mathbf{M}) = p\sigma^2$ , Markov's inequality gives  $\Pr(\|\mathbf{M}\boldsymbol{\varepsilon}\| > K) \leq p\sigma^2/K^2$  for every  $K$ , and therefore  $\delta_n = \max_i |d_i| \leq (\max_i h_{ii})^{1/2} \|\mathbf{M}\boldsymbol{\varepsilon}\| \rightarrow 0$  in probability. Let  $A_n = n^{-1} \sum_i \exp(\varepsilon_i)$ , which tends to  $E e^\varepsilon$  in probability by the weak law of large numbers. Because  $\hat{\varepsilon}_i = \varepsilon_i - d_i$ , we have

$$e^{-\delta_n} A_n \leq \frac{1}{n} \sum_{i=1}^n \exp(\hat{\varepsilon}_i) \leq e^{\delta_n} A_n,$$

and both bounds tend to  $E e^\varepsilon$  in probability. This proves Eq. 22.5.2. The last statement follows because  $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}) \rightarrow \exp(\mathbf{x}_0^\top \boldsymbol{\beta})$  in probability, and products of convergent sequences converge. ■

By part (a), a transformed model is naturally read through medians:  $h(\mathbf{x}^\top \hat{\boldsymbol{\beta}})$  estimates the conditional median when the errors are symmetric. Part (b) is the **retransformation bias**: for the exponential,  $h(\mathbf{x}^\top \hat{\boldsymbol{\beta}})$  underestimates the mean by the factor  $E e^\varepsilon$ . Under normal errors this suggests  $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + \sigma^2/2)$ , but the factor depends on the whole error distribution, and with non-normal errors that estimate is in general inconsistent: it converges to  $\exp(\sigma^2/2)$ , not to  $E e^\varepsilon$ .

Part (c) is Duan's (1983) remedy, the empirical distribution of the residuals standing in for  $F$ , under the familiar condition that no observation keep a fixed share of the fit (Section 6.8). For general  $h$  the smearing estimate is  $n^{-1} \sum_i h(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + \hat{\varepsilon}_i)$  (Exercise (C1)).

### Example. (Skewed errors)

The script simulates the log-linear model  $\log Y = 1 + 0.5x + \varepsilon$  with  $n = 100$  equally spaced values of  $x$  in  $[0, 2]$  and right-skewed errors  $\varepsilon = 0.6(E - 1)$ ,  $E$  exponential with mean 1. The errors have mean 0 and variance 0.36. The true factor is  $E e^\varepsilon = e^{-0.6}/0.4$ , about 1.372, while the normal-theory factor  $\exp(0.36/2)$  is about 1.197. At  $x_0 = 1$ , averaged over 4000 samples, the ratio of the estimate to the true mean is 0.730 for the naive  $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}})$ , 0.877 for the normal-theory correction and 0.986 for the smearing estimate. Its remaining bias of about one percent reflects that residuals vary slightly less than errors.

```
import numpy as np

def simulate(reps, n_sim, seed):
    """Log-linear model with right-skewed errors  $e = 0.6 (E - 1)$ ,  $E \sim \text{Exp}(1)$ :
    estimate / true mean."""
    rng = np.random.default_rng(seed)
    xs = np.linspace(0, 2, n_sim)
    Xs = np.column_stack([np.ones(n_sim), xs])
    beta, sig = np.array([1.0, 0.5]), 0.6
    x0 = np.array([1.0, 1.0])
    true_mean = np.exp(x0 @ beta) * np.exp(-sig) / (1 - sig)  #  $E \exp(e) = e^{(-sig)/(1-sig)}$ 
    out = np.empty((reps, 3))
    for r in range(reps):
        e = sig * (rng.exponential(size=n_sim) - 1)
        z = Xs @ beta + e
        coef, *_ = np.linalg.lstsq(Xs, z, rcond=None)
        res = z - Xs @ coef
        s2_sim = res @ res / (n_sim - 2)
        naive = np.exp(x0 @ coef)
        out[r] = [naive, naive * np.exp(s2_sim / 2), naive * np.mean(np.exp(res))]
    return out.mean(axis=0) / true_mean

ratios = simulate(reps=300, n_sim=100, seed=2205)
print("mean of estimate / true mean:  naive %.3f   lognormal %.3f   smearing %.3f" %
      tuple(ratios))
```

The browser cell uses 300 samples, so its ratios differ slightly from those quoted.

*Warning.* Smearing needs identically distributed errors on the transformed scale. If their variance depends on the regressors, so does  $E e^\varepsilon$ , and a single average corrects too much where the variance is small and too little where it is large (Exercise (C2)); Manning (1998) showed how large the errors can be. Check the residual spread on the transformed scale (Chapter 21) before retransforming a mean.



## 22.5.2 Intervals on the original scale

### Proposition 22.5.2 — Intervals after a transformation

Assume [Eq. 22.5.1](#), and let  $Y_0$  be a new response at  $\mathbf{x}_0$ , independent of  $\mathbf{Y}$  and following the same model.

- (a) If  $L \leq U$  are statistics with values in the range of  $g$  such that  $\Pr\{L \leq g(Y_0) \leq U\} = 1 - \alpha$ , then  $\Pr\{h(L) \leq Y_0 \leq h(U)\} = 1 - \alpha$ . Under normal errors the prediction interval of [Theorem 12.3.1](#) therefore transforms into an exact prediction interval for  $Y_0$ .
- (b) If  $[L, U]$  is a confidence interval for  $\mathbf{x}_0^\top \boldsymbol{\beta}$  with coverage  $1 - \alpha$  and  $F$  has median 0, then  $[h(L), h(U)]$  is a confidence interval for the median of  $Y_0$  with coverage  $1 - \alpha$ .
- (c) Let  $g = \log$ , let the errors be normal, and let  $\mathbf{X}$  have rank  $r < n$  with  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$  and  $h_0 = \mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^- \mathbf{x}_0$ . The logarithm of the mean,  $\theta = \mathbf{x}_0^\top \boldsymbol{\beta} + \sigma^2/2$ , is estimated by  $\hat{\theta} = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + s^2/2$ , which is unbiased with

$$\text{Var}(\hat{\theta}) = \sigma^2 h_0 + \frac{\sigma^4}{2(n-r)}. \quad (22.5.3)$$

*Proof.*

- (a) and (b): since  $g$  is strictly increasing with inverse  $h$ , the events  $\{L \leq g(Y_0) \leq U\}$  and  $\{h(L) \leq Y_0 \leq h(U)\}$  are equal, and so are  $\{L \leq \mathbf{x}_0^\top \boldsymbol{\beta} \leq U\}$  and  $\{h(L) \leq h(\mathbf{x}_0^\top \boldsymbol{\beta}) \leq h(U)\}$ ; by [Proposition 22.5.1\(a\)](#),  $h(\mathbf{x}_0^\top \boldsymbol{\beta})$  is the median of  $Y_0$ . (c) By [Theorem 7.5.1](#),  $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$  is normal with mean  $\mathbf{x}_0^\top \boldsymbol{\beta}$  and variance  $\sigma^2 h_0$ ,  $(n-r)s^2/\sigma^2 \sim \chi^2(n-r)$ , and the two are independent. So  $E\hat{\theta} = \theta$ ,  $\text{Var}(s^2) = 2\sigma^4/(n-r)$ , and the variances add. ■

Part (c) gives the approximate interval  $\exp\{\hat{\theta} \pm z_{\alpha/2}(s^2 h_0 + s^4/\{2(n-r)\})^{1/2}\}$  for the mean, approximate because the variance is estimated and  $\hat{\theta}$  is only approximately normal, and dependent on the lognormal assumption. The intervals of parts (a) and (b) are exact under the normal model on the log scale.

### Example. (Engel's data on the original scale)

With the log-log fit of [Example \(Food expenditure on the raw and the log scale\)](#), [Figure 22.5.1](#) shows the data on the original scale with the fitted median curve, the smearing estimate of the mean and a 95% prediction band. For a household with income 500 the estimated median food expenditure is 352.2, with 95% confidence interval [342.2, 362.5], and a 95% prediction interval for a single household ([Theorem 12.3.1](#) and [Proposition 22.5.2\(a\)](#)) is [269, 462]. At income 2000 the median is 1153.7, with interval [1111.7, 1197.3], and the prediction interval is [879, 1514]. A fraction 0.455 of the households lie below the fitted median curve.

The log-scale residual variance  $s^2 = 0.01871$  is small, and so is the retransformation bias: the smearing factor is 1.0091 and  $\exp(s^2/2)$  is 1.0094. The estimated mean at income 2000 is 1164.2 by smearing and 1164.5 under normality, about one percent above the

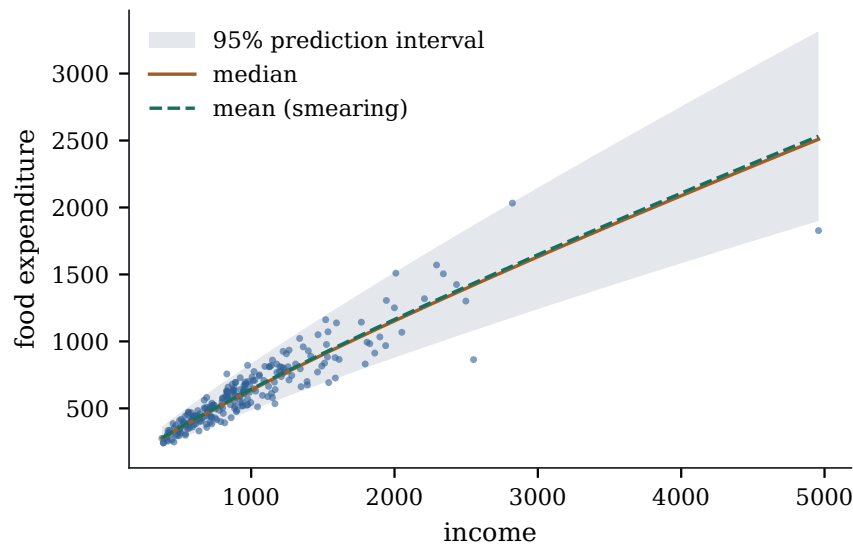


Figure 22.5.1: Engel's data on the original scale with the log-log fit transformed back: the median curve  $\exp(\hat{\beta}_0 + \hat{\beta}_1 \log x)$ , the smearing estimate of the mean (dashed), nearly indistinguishable here because  $s^2$  is small, and the 95% prediction band, which widens and becomes asymmetric as income grows.

median. The approximate 95% interval for the mean from Eq. 22.5.3 is [1122.4, 1208.3] at income 2000 and [345.4, 365.9] at income 500.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

df = sm.datasets.engel.load_pandas().data
income, food = df["income"].to_numpy(), df["foodexp"].to_numpy()
n = len(food)
X = np.column_stack([np.ones(n), np.log(income)])
fit = sm.OLS(np.log(food), X).fit()
b, s2, resid = fit.params, fit.scale, fit.resid
XtX_inv = np.linalg.inv(X.T @ X)
t = stats.t.ppf(0.975, n - 2)

smear = np.mean(np.exp(resid))          # Duan's smearing factor
lognormal = np.exp(s2 / 2)             # the normal-theory factor
print(f"smearing factor {smear:.4f}, lognormal factor {lognormal:.4f}")

for x0 in (500, 2000):
    v0 = np.array([1, np.log(x0)])
    eta, h0 = v0 @ b, v0 @ XtX_inv @ v0
    median = np.exp(eta)
    ci_med = np.exp(eta + np.array([-1, 1]) * t * np.sqrt(s2 * h0))    # for
    the median, exact
    pi = np.exp(eta + np.array([-1, 1]) * t * np.sqrt(s2 * (1 + h0)))    # for a
    new household, exact
    print(f"income {x0}: median {median:.1f}, CI [{ci_med[0]:.1f}, {ci_med[1]:.1f}];
    "
          f" mean {median * smear:.1f} (smearing), {median * lognormal:.1f} (
    lognormal);"
          f" 95% prediction interval [{pi[0]:.0f}, {pi[1]:.0f}]"
    )
```

### 22.5.3 Elasticities and marginal effects

In a transformed model a coefficient is not a rate of change of  $y$  per unit of  $x$ ; rates and elasticities follow by differentiating through both transformations.

#### Proposition 22.5.3 — Elasticities in a transformed model

Let  $x_1 > 0$ , and suppose that the median  $m(\mathbf{x})$  of  $Y$  satisfies  $m(\mathbf{x})^{(\lambda)} = \beta_0 + \beta_1 x_1^{(\alpha)} + c(\mathbf{x}_{-1})$ , where  $c$  does not involve  $x_1$  and the Box–Cox transformation Eq. 22.2.1 is applied to both  $m$  and  $x_1$ . Then

$$\frac{\partial m}{\partial x_1} = \beta_1 x_1^{\alpha-1} m^{1-\lambda}, \quad \frac{x_1}{m} \frac{\partial m}{\partial x_1} = \beta_1 x_1^\alpha m^{-\lambda}. \quad (22.5.4)$$

In particular the elasticity is the constant  $\beta_1$  in the log–log model ( $\lambda = \alpha = 0$ ),  $\beta_1 x_1$  in the log–level model ( $\lambda = 0, \alpha = 1$ ),  $\beta_1/m$  in the level–log model ( $\lambda = 1, \alpha = 0$ ) and  $\beta_1 x_1/m$  in the untransformed model. Under Eq. 22.1.1 the mean is a constant multiple of the median, so for  $\lambda = 0$  the elasticity of the mean is the same.

*Proof.* Differentiate the identity with respect to  $x_1$ . By Proposition 22.2.1(b) the derivative of  $m^{(\lambda)}$  with respect to  $m$  is  $m^{\lambda-1}$  and that of  $x_1^{(\alpha)}$  is  $x_1^{\alpha-1}$ , so  $m^{\lambda-1} \partial m / \partial x_1 = \beta_1 x_1^{\alpha-1}$ . Multiply by  $m^{1-\lambda}$ , and then by  $x_1/m$ . If the mean is  $k m(\mathbf{x})$  for a constant  $k$ , the factor  $k$  cancels in the elasticity. ■

The model fixes the *shape* of the elasticity profile before the data have a say. For Engel’s data the log–log slope 0.8559 (interval [0.816, 0.896]) is one elasticity for all households. The untransformed linear fit implies an elasticity  $\hat{\beta}_1 x / \hat{m}$  rising from 0.62 at income 500 to 0.87 at income 2000, only because a line with a positive intercept forces that shape. The Box–Tidwell estimate  $\hat{\alpha} = -0.182$  of Example (Transforming income), with  $\lambda = 0$ , implies an elasticity proportional to  $x^{-0.182}$ , which *falls* between the same incomes by the factor  $4^{-0.182} = \exp(-0.182 \log 4)$ , about 0.78. Which shape the data support is a question about the mean function (Section 22.3).

### 22.5.4 Which scale to report

A transformation chosen for statistical reasons need not be the reporting scale. The logarithm is often worth keeping, since its coefficients are ratios and elasticities (Proposition 22.1.1). For other powers, report on the original scale: medians and prediction intervals, which transform exactly, and means, which need a correction factor and a constant variance. Compare models fitted on different scales on a common scale, through the Box–Cox likelihood (Section 22.2) or the accuracy of predictions of  $y$  (Chapter 29), not through  $R^2$  (Exercise (B3)).

### 22.5.5 Exercises

#### 22.5.5.1 A. Check your understanding

**Exercise A1.**

For Engel's log-log fit  $s^2 = 0.01871$ . Check that  $\exp(s^2/2)$  is 1.0094. The smearing factor is 1.0091. Why are the two so close here, and in what kind of data would they differ substantially?

**Exercise A2.**

Explain why the prediction interval of [Example \(Engel's data on the original scale\)](#), obtained from [Theorem 12.3.1](#), is longer above the median than below it, and why its coverage is nevertheless exactly 95% under the normal model on the log scale.

**22.5.5.2 B. Practice****Exercise B1.**

Under normal errors on the log scale, show that  $E \exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}) = \exp(\mathbf{x}_0^\top \boldsymbol{\beta} + \sigma^2 h_0/2)$ , so the naive median estimate is biased upwards by a factor that tends to one as  $h_0 \rightarrow 0$ . Suggest an approximately mean-unbiased alternative, and explain why  $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}})$  is exactly median-unbiased.

*Solution.*  $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} \sim N(\mathbf{x}_0^\top \boldsymbol{\beta}, \sigma^2 h_0)$  ([Theorem 7.5.1](#)), and the normal moment generating function ([Theorem 3.2.1](#)) at 1 gives the mean. The median of  $\mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$  is  $\mathbf{x}_0^\top \boldsymbol{\beta}$ , and  $\exp$  preserves medians, so  $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}})$  is exactly median-unbiased for  $\exp(\mathbf{x}_0^\top \boldsymbol{\beta})$ . For mean-unbiasedness one can use  $\exp(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} - s^2 h_0/2)$ , which is approximately unbiased. The factor  $\exp(\sigma^2 h_0/2)$  is tiny when  $h_0$  is small, so the choice rarely matters.

**Exercise B2.**

Let  $\sqrt{Y_i} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i$  with  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ . Show that  $E(Y \mid \mathbf{x}_0) = (\mathbf{x}_0^\top \boldsymbol{\beta})^2 + \sigma^2$  and that the smearing estimate  $n^{-1} \sum_i (\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + \hat{\varepsilon}_i)^2$  equals  $(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}})^2 + \text{SSE}/n$ . How does this compare with the plug-in estimate  $(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}})^2 + s^2$ ?

**Exercise B3.**

For Engel's data at income 500,  $s^2 = 0.01871$ ,  $h_0 = 0.01147$  and  $n - r = 233$ . Evaluate the two terms of [Eq. 22.5.3](#) with  $\sigma^2$  replaced by  $s^2$ , and check the standard error 0.0147 used for the interval [345.4, 365.9]. Which term dominates, and why?

**22.5.5.3 C. Going deeper****Exercise C1.**

Extend [Proposition 22.5.1\(c\)](#) to a general strictly increasing  $h$  that is continuously differentiable with  $|h'(t)| \leq K(1 + e^{\kappa|t|})$  for some constants, assuming  $E e^{\kappa'|\varepsilon|} < \infty$  for some

$\kappa' > \kappa$ . Show that  $n^{-1} \sum_i h(\mathbf{x}_0^\top \hat{\boldsymbol{\beta}} + \hat{\varepsilon}_i)$  converges in probability to  $E(Y \mid \mathbf{x}_0)$ . (Use the mean value theorem and the bound on  $\max_i |\hat{\varepsilon}_i - \varepsilon_i|$  from the proof.)

### Exercise C2.

Suppose that on the log scale  $\varepsilon_i \sim N(0, \sigma^2(\mathbf{x}_{(i)}))$ , independent, with the variance depending on the regressors. Show that  $E(Y \mid \mathbf{x}_0) = \exp\{\mathbf{x}_0^\top \boldsymbol{\beta} + \sigma^2(\mathbf{x}_0)/2\}$ , and that, when the leverages tend to zero and the average  $n^{-1} \sum_i \exp\{\sigma^2(\mathbf{x}_{(i)})/2\}$  converges to a limit  $\bar{c}$ , the smearing factor tends to  $\bar{c}$ . For which  $\mathbf{x}_0$  does the smearing estimate overestimate the mean, and for which does it underestimate it?

*Solution.* Given  $\mathbf{x}_0$ ,  $\log Y$  is normal with mean  $\mathbf{x}_0^\top \boldsymbol{\beta}$  and variance  $\sigma^2(\mathbf{x}_0)$ , and the moment generating function (Theorem 3.2.1) gives the mean. The argument of Proposition 22.5.1(c) goes through with  $A_n = n^{-1} \sum_i e^{\varepsilon_i}$ , whose mean is  $n^{-1} \sum_i \exp\{\sigma^2(\mathbf{x}_{(i)})/2\}$ ; if the variances are bounded, its variance tends to zero, so  $A_n \rightarrow \bar{c}$ . The smearing estimate therefore targets  $\exp(\mathbf{x}_0^\top \boldsymbol{\beta}) \bar{c}$ . It overestimates the mean where  $\exp\{\sigma^2(\mathbf{x}_0)/2\} < \bar{c}$ , that is, where the log-scale variance is below its average in this sense, and underestimates it where the variance is above.

## 22.6 Summary and Notes

### 22.6.1 Summary

*Idea.*

1. A strictly increasing transformation of the response acts on linearity, constant variance and normality at once. Multiplicative errors make the logarithm exact for the first two, and log-scale coefficients are then ratios of quantiles and means (Proposition 22.1.1). The mean comes first.
2. The Box–Cox family is continuous in  $\lambda$ , increasing, ordered by convexity, and bounded on one side except at the logarithm (Definition 22.2.1 and Proposition 22.2.1).
3. With the geometric-mean normalization the profile log-likelihood of  $\lambda$  is  $-\frac{n}{2} \log \text{SSE}_z(\lambda)$  plus a constant, which gives the likelihood ratio interval (Theorem 22.2.2), free of the units of  $y$  (Proposition 22.2.3). The constructed variable locates its stationary points (Proposition 22.2.4).
4. Component-plus-residual plots show the shape of a regressor's effect, distorted by the part of the nonlinearity the other regressors reproduce (Proposition 22.3.1). The Box–Tidwell constructed variable gives an exact test of a regressor's power, and its iteration is Gauss–Newton (Theorem 22.3.2).
5.  $\int v^{-1/2}$  stabilizes a variance function  $v$ , uniquely up to sign and a constant (Theorem 22.4.1): square roots for counts, arcsine square roots for proportions, logarithms for a constant coefficient of variation (Corollary 22.4.2).
6. Quantiles pass through a monotone transformation, means do not. The back-transformed fit estimates medians; the mean needs a factor, which smearing sup-

plies without normality ([Proposition 22.5.1](#)). Prediction intervals and intervals for medians transform exactly ([Proposition 22.5.2](#)), and elasticities follow by differentiating through both scales ([Proposition 22.5.3](#)).

## 22.6.2 Notes and sources

**Source books.** Coverage was checked against Sen and Srivastava (1990, chapter 9), Seber and Lee (2003, section 10.3 and parts of sections 10.4–10.5) and Christensen (2020, section 12.7). Proofs, examples and exercises here are this book’s own.

**Power transformations.** Tukey (1957) studied the powers systematically, and Tukey (1977) popularized the ladder. Box and Cox (1964) introduced the likelihood approach and the geometric-mean normalization, and Hernandez and Johnson (1980) gave its large-sample theory when no power is exactly right. On whether inference should allow for estimating  $\lambda$ , see Bickel and Doksum (1981), Box and Cox (1982) and Hinkley and Runger (1984). Carroll and Ruppert (1988) is the standard monograph on transformation and weighting, including the *transform-both-sides* model, which transforms the response and a known nonlinear mean function alike. Yeo and Johnson (2000) give a family defined on the whole line.

**Diagnostics.** Constructed variables and their added-variable plots are developed by Atkinson (1985) and Cook and Weisberg (1982); Atkinson and Cook and Wang (1983) study the influence of single cases on  $\hat{\lambda}$ . Larsen and McCleary (1972) analysed partial residual plots, Mallows (1986) augmented them, and Cook (1993) explained when they work. Box and Tidwell (1962) proposed the regressor transformation, and Royston and Altman (1994) fractional polynomials.

**Variance stabilization and retransformation.** Bartlett (1947) reviewed variance-stabilizing transformations and the spread-versus-level idea, Anscombe (1948) derived the constants  $3/8$ , and Grizzle, Starmer and Koch (1969) developed weighted least squares on transformed proportions. The limit theorems are in van der Vaart (1998). The smearing estimate is due to Duan (1983). Manning (1998) showed how heteroscedasticity on the log scale biases retransformed means, one reason why generalized linear models with a log link ([Chapter 34](#)) are now the usual alternative for skewed positive responses.

## 22.6.3 References

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# Chapter 23

## Bootstrap and Permutation Inference

Every test and interval in Part III took its reference distribution from the normal model. This chapter asks whether the reference distribution can be taken from the data instead. The bootstrap imitates the process that produced the data by resampling residuals, whole cases, or residuals with random signs, and each choice assumes a different model. A permutation test rearranges the data in ways that leave their distribution unchanged under the null hypothesis, and is exact whenever such rearrangements exist. In regression they exist only for simple hypotheses, and the Freedman–Lane scheme shows how to get close in other cases.

**What you need.** Chapter 6 (projections, leverage, Proposition 6.8.1, and the Frisch–Waugh–Lovell theorem, Theorem 6.6.2), Chapter 7 (sampling distributions in the normal model, Theorem 7.5.1 and Corollary 7.5.2), Chapter 10 (the deletion formulas, Proposition 10.6.2), Chapter 11 (the  $F$  test of a linear hypothesis, Theorem 11.1.1), Chapter 14 (random regressors and conditional inference) and Chapter 18 (randomized blocks). From this part: Chapter 19 (asymptotic normality without normal errors), Chapter 20 (studentized residuals) and Chapter 21 (sandwich covariance estimators and serial correlation).

### 23.0.1 Roadmap

- Resampling cases versus resampling residuals
- The wild bootstrap
- Bootstrap confidence intervals
- Permutation tests for linear hypotheses
- Freedman–Lane and related schemes
- When resampling fails
- Summary and notes

## 23.1 Resampling cases versus resampling residuals

Let  $\theta = \mathbf{c}^\top \boldsymbol{\beta}$  be estimated by  $\hat{\theta} = \mathbf{c}^\top \hat{\boldsymbol{\beta}}$ . An interval or a test for  $\theta$  needs the distribution of a **root** such as  $\hat{\theta} - \theta$ , or a studentized version of it. That distribution depends on the mechanism



$P$  that produced the data. In Part III,  $P$  lay in the normal linear model, where the studentized root is  $t(n-r)$  whatever the parameters. Without that assumption the distribution is unknown.

The **bootstrap** (Efron 1979) replaces  $P$  by an estimate  $\hat{P}$ . Draw a data set  $\mathbf{Y}^*$  from  $\hat{P}$ , compute  $\hat{\theta}^*$  from it, and use the distribution of  $\hat{\theta}^* - \hat{\theta}$ , given the data, as the approximation to that of  $\hat{\theta} - \theta$ . Here  $\hat{\theta}$  plays the true value, because it is the value of  $\theta$  under  $\hat{P}$ . Probabilities, expectations and covariances under  $\hat{P}$  with the data held fixed are written  $\Pr_*$ ,  $E_*$  and  $\text{Cov}_*$ . In practice  $B$  resamples approximate the bootstrap distribution. Their Monte Carlo error shrinks as  $B$  grows; the statistical error, from the gap between  $\hat{P}$  and  $P$ , does not, and it is the subject of this chapter.

### 23.1.1 Two ways to estimate the data-generating mechanism

In the **fixed-design model** the rows  $\mathbf{x}_{(i)}^\top$  of  $\mathbf{X}$  are constants and  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with independent, identically distributed errors from an unknown law  $F$ . Estimate  $\boldsymbol{\beta}$  by  $\hat{\boldsymbol{\beta}}$  and  $F$  by the empirical distribution of the residuals. In the **random-design model** the cases  $(\mathbf{x}_{(i)}, Y_i)$  are independent draws from an unknown joint distribution. Estimate it by the empirical distribution of the  $n$  cases.

#### Definition 23.1.1 — Residual and case bootstraps

Let  $\mathbf{X}$  be  $n \times p$  of rank  $p$ , with least squares estimate  $\hat{\boldsymbol{\beta}}$  and residuals  $\hat{\varepsilon}_1, \dots, \hat{\varepsilon}_n$  with mean  $\bar{\hat{\varepsilon}}$ .

- (a) The **residual bootstrap** draws  $\varepsilon_1^*, \dots, \varepsilon_n^*$  independently and uniformly from the centred residuals  $\tilde{\varepsilon}_i = \hat{\varepsilon}_i - \bar{\hat{\varepsilon}}$ , sets  $\mathbf{Y}^* = \mathbf{X}\hat{\boldsymbol{\beta}} + \boldsymbol{\varepsilon}^*$ , and computes  $\hat{\boldsymbol{\beta}}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}^*$ . The design is the same in every resample.
- (b) The **case bootstrap** (or pairs bootstrap) draws  $n$  indices  $I_1, \dots, I_n$  independently and uniformly from  $\{1, \dots, n\}$  and computes the least squares estimate  $\hat{\boldsymbol{\beta}}^*$  from the cases  $(\mathbf{x}_{(I_k)}, Y_{I_k})$ ,  $k = 1, \dots, n$ . The design changes from resample to resample.

Variants of (a) resample the **leverage-adjusted residuals**  $r_i = \hat{\varepsilon}_i / \sqrt{1 - h_{ii}}$ , centred, or the residuals rescaled by  $\sqrt{n/(n-p)}$ .

With an intercept the residuals already sum to zero (Proposition 6.4.3(b)). Without one, centring prevents a bias in the bootstrap world (Exercise (A1)).

### 23.1.2 The moments of the residual bootstrap

#### Proposition 23.1.1 — Moments of the residual bootstrap

In the residual bootstrap, put  $\hat{\sigma}_*^2 = n^{-1} \sum_i \tilde{\varepsilon}_i^2$ , the variance of the resampled law. Then:

- (a)  $E_*(\hat{\boldsymbol{\beta}}^*) = \hat{\boldsymbol{\beta}}$  and  $\text{Cov}_*(\hat{\boldsymbol{\beta}}^*) = \hat{\sigma}_*^2 (\mathbf{X}^\top \mathbf{X})^{-1}$ ;
- (b) if  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , then  $\hat{\sigma}_*^2 = \text{SSE}/n = (n-p)s^2/n$ ;

- (c) if the data follow the linear model with  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ , then  $E(r_i^2) = \sigma^2$  for each  $i$  with  $h_{ii} < 1$ .

*Proof.*

- (a) Given the data,  $\hat{\boldsymbol{\beta}}^* = \hat{\boldsymbol{\beta}} + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}^*$ , where the  $\varepsilon_i^*$  are independent with  $E_*(\varepsilon_i^*) = n^{-1} \sum_j \tilde{\varepsilon}_j = 0$  and  $\text{Var}_*(\varepsilon_i^*) = \hat{\sigma}_*^2$ . Apply [Theorem 5.5.1](#) in the bootstrap world. (b) The residuals sum to zero, so  $\tilde{\varepsilon}_i = \hat{\varepsilon}_i$ . (c)  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$  by [Proposition 6.4.3\(e\)](#), so  $E(\hat{\varepsilon}_i) = 0$  and  $\text{Var}(\hat{\varepsilon}_i) = \sigma^2(1 - h_{ii})$ . ■

By (b) the plain residual bootstrap shrinks the usual variance by the factor  $(n - p)/n$ : residuals are shorter than errors because least squares fits part of the noise ([Theorem 5.6.1](#)). The factor matters only when  $p/n$  is not small ([Section 23.6](#)). Rescaling by  $\sqrt{n/(n - p)}$  removes it exactly when the model has an intercept. The leverage-adjusted residuals remove it observation by observation. They are  $s$  times the internally studentized residuals of [Chapter 20](#) ([Definition 20.1.1](#)).

### 23.1.3 Consistency for a linear contrast

Consider designs  $\mathbf{X}_n$  of rank  $p$ , with  $p$  fixed, and errors  $\varepsilon_{n1}, \dots, \varepsilon_{nn}$  drawn independently from a fixed law  $F$  with mean zero and variance  $\sigma^2 \in (0, \infty)$ . For  $\mathbf{c}_n \neq \mathbf{0}$  put

$$\begin{aligned} \mathbf{a}_n &= \mathbf{X}_n (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{c}_n, & \mathbf{c}_n^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) &= \mathbf{a}_n^\top \boldsymbol{\varepsilon}, \\ \|\mathbf{a}_n\|^2 &= \mathbf{c}_n^\top (\mathbf{X}_n^\top \mathbf{X}_n)^{-1} \mathbf{c}_n, \end{aligned}$$

so the contrast estimate is a weighted sum of the errors with standard deviation  $\sigma \|\mathbf{a}_n\|$ . The condition we need is that no single weight dominates:

$$m_n = \max_i \frac{|a_{ni}|}{\|\mathbf{a}_n\|} \rightarrow 0. \quad (23.1.1)$$

The Cauchy–Schwarz bound  $a_{ni}^2 \leq h_{ii} \|\mathbf{a}_n\|^2$  in the proof of [Theorem 19.3.2](#) shows that [Eq. 23.1.1](#) holds for every contrast when the largest leverage tends to zero. The limit theorem used there is the Lindeberg–Feller theorem of [Chapter 19](#); for a triangular array  $U_{ni}$  its condition holds in particular when  $\sum_i E U_{ni}^4 \rightarrow 0$  (Lyapunov), since the Lindeberg sum is at most  $\eta^{-2} \sum_i E U_{ni}^4$ . We add one more result from probability, used without proof (van der Vaart 1998, chapter 2).

*Remark* (Pólya’s theorem). If distribution functions  $G_n$  converge pointwise to a continuous distribution function  $G$ , then  $\sup_x |G_n(x) - G(x)| \rightarrow 0$ .

A bootstrap distribution is a conditional distribution that changes with the data, so we need a conditional form of the Lindeberg–Feller theorem.

#### Lemma 23.1.2 — A conditional central limit theorem

For each  $n$ , given the data  $\mathcal{D}_n$ , let  $U_{n1}^*, \dots, U_{nn}^*$  be independent with  $E_*(U_{ni}^*) = 0$  and

$\sum_i E_*(U_{ni}^{*2}) = 1$ . If for every  $\eta > 0$

$$L_n^*(\eta) = \sum_i E_*[U_{ni}^{*2} \mathbf{1}\{|U_{ni}^*| > \eta\}] \rightarrow 0 \quad \text{in probability,}$$

then  $\Delta_n = \sup_x |\Pr_*(\sum_i U_{ni}^* \leq x) - \Phi(x)| \rightarrow 0$  in probability.

*Proof.* Random variables converge to zero in probability if and only if every subsequence has a further subsequence along which they converge almost surely. Given a subsequence, a diagonal argument gives a further subsequence along which  $L_n^*(1/k) \rightarrow 0$  almost surely for every  $k = 1, 2, \dots$  at once. Since  $L_n^*(\eta)$  decreases in  $\eta$ , on this event  $L_n^*(\eta) \rightarrow 0$  for every  $\eta > 0$ . For each outcome in the event the conditional laws satisfy the Lindeberg condition, so  $\Delta_n \rightarrow 0$  by the Lindeberg–Feller and Pólya theorems. ■

### Lemma 23.1.3 — Residuals have uniformly small tails

In the setting above,  $\hat{\sigma}_*^2 \rightarrow \sigma^2$  in probability, and for every sequence of constants  $K_n \rightarrow \infty$ ,  $n^{-1} \sum_i \tilde{\varepsilon}_i^2 \mathbf{1}\{|\tilde{\varepsilon}_i| > K_n\} \rightarrow 0$  in probability.

*Proof.* Write  $\tilde{\varepsilon}_i = \varepsilon_i - \delta_i$  with  $\boldsymbol{\delta} = \mathbf{M}\boldsymbol{\varepsilon} + \bar{\varepsilon}\mathbf{1}$ . Here  $E\|\mathbf{M}\boldsymbol{\varepsilon}\|^2 = \sigma^2 p$ , and  $\bar{\varepsilon} = n^{-1}\mathbf{1}^\top(\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$  has  $E(\bar{\varepsilon}^2) = \sigma^2 n^{-2}\mathbf{1}^\top(\mathbf{I} - \mathbf{M})\mathbf{1} \leq \sigma^2/n$ . So  $E\|\boldsymbol{\delta}\|^2 \leq 2\sigma^2(p+1)$ , and  $n^{-1}\|\boldsymbol{\delta}\|^2 \rightarrow 0$  in probability.

First claim:  $\hat{\sigma}_*^2 = n^{-1}\|\boldsymbol{\varepsilon}\|^2 - n^{-1}\|\mathbf{M}\boldsymbol{\varepsilon}\|^2 - \bar{\varepsilon}^2$ . The first term tends to  $\sigma^2$  by the weak law of large numbers (the errors of the  $n$ th model are an independent sample from  $F$ ), and the others tend to zero.

Second claim: for  $K > 0$  and reals  $u, d$ , with  $v = u - d$ ,

$$v^2 \mathbf{1}\{|v| > K\} \leq 2u^2 \mathbf{1}\{|u| > K/2\} + 4d^2.$$

If  $|u| > K/2$ , use  $v^2 \leq 2u^2 + 2d^2$ . If  $|u| \leq K/2$  and  $|v| > K$ , then  $|d| \geq |v| - |u| > K/2 \geq |u|$ , so  $|v| \leq 2|d|$ . Otherwise the left side is zero. With  $u = \varepsilon_i$ ,  $d = \delta_i$  and  $K = K_n$ , averaging gives

$$\begin{aligned} \frac{1}{n} \sum_i \tilde{\varepsilon}_i^2 \mathbf{1}\{|\tilde{\varepsilon}_i| > K_n\} &\leq \frac{2}{n} \sum_i \varepsilon_i^2 \mathbf{1}\{|\varepsilon_i| > K_n/2\} \\ &\quad + \frac{4}{n} \|\boldsymbol{\delta}\|^2. \end{aligned}$$

The first term has expectation  $2E[\varepsilon^2 \mathbf{1}\{|\varepsilon| > K_n/2\}] \rightarrow 0$  by dominated convergence, and the second tends to zero in probability. ■

### Theorem 23.1.4 — Consistency of the residual bootstrap

In the fixed-design setting above, assume [Eq. 23.1.1](#). Let

$$G_n(x) = \Pr(\mathbf{c}_n^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \leq x \|\mathbf{a}_n\|),$$

$$\hat{G}_n(x) = \Pr_*(\mathbf{c}_n^\top (\hat{\boldsymbol{\beta}}^* - \hat{\boldsymbol{\beta}}) \leq x \|\mathbf{a}_n\|),$$

where  $\hat{\boldsymbol{\beta}}^*$  comes from the residual bootstrap. Then:

- (a)  $\sup_x |G_n(x) - \Phi(x/\sigma)| \rightarrow 0$ ;
- (b)  $\sup_x |\hat{G}_n(x) - \Phi(x/\sigma)| \rightarrow 0$  in probability;
- (c)  $\sup_x |\hat{G}_n(x) - G_n(x)| \rightarrow 0$  in probability.

*Proof.* Put  $w_i = a_{ni}/\|\mathbf{a}_n\|$ , so  $\sum_i w_i^2 = 1$  and  $\max_i |w_i| = m_n \rightarrow 0$ .

- (a)  $\mathbf{c}_n^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})/\|\mathbf{a}_n\| = \sum_i w_i \varepsilon_i$ . When  $\max_i h_{ii} \rightarrow 0$  the claim is [Theorem 19.3.2\(a\)](#) with Pólya's theorem. The same Lindeberg argument needs only [Eq. 23.1.1](#): with  $U_{ni} = w_i \varepsilon_i/\sigma$ , the Lindeberg sum is at most  $E[\sigma^{-2} \varepsilon_i^2 \mathbf{1}\{|\varepsilon_i| > \eta\sigma/m_n\}] \rightarrow 0$ , by dominated convergence.
- (b) Similarly  $\mathbf{c}_n^\top (\hat{\boldsymbol{\beta}}^* - \hat{\boldsymbol{\beta}})/\|\mathbf{a}_n\| = \sum_i w_i \varepsilon_i^*$ . Take  $U_{ni}^* = w_i \varepsilon_i^*/\hat{\sigma}_*$ . As in (a),

$$L_n^*(\eta) \leq \frac{1}{n\hat{\sigma}_*^2} \sum_i \tilde{\varepsilon}_i^2 \mathbf{1}\{|\tilde{\varepsilon}_i| > \eta\hat{\sigma}_*/m_n\}.$$

By [Lemma 23.1.3](#),  $\hat{\sigma}_* \rightarrow \sigma$  in probability. On the event  $\hat{\sigma}_* > \sigma/2$ , whose probability tends to one, the right side is at most  $4\sigma^{-2}n^{-1} \sum_i \tilde{\varepsilon}_i^2 \mathbf{1}\{|\tilde{\varepsilon}_i| > K_n\}$  with  $K_n = \eta\sigma/(2m_n) \rightarrow \infty$ , which tends to zero in probability by the lemma. So [Lemma 23.1.2](#) gives  $\sup_x |\hat{G}_n(x) - \Phi(x/\hat{\sigma}_*)| \rightarrow 0$  in probability. Finally  $\sup_x |\Phi(x/\hat{\sigma}_*) - \Phi(x/\sigma)|$  is a continuous function of  $\hat{\sigma}_*/\sigma$  that vanishes at 1, so it tends to zero in probability.

- (c) follows from (a) and (b) by the triangle inequality. ■

The theorem holds for the rescaled residuals too, since they differ by the factor  $\sqrt{n/(n-p)} \rightarrow 1$ ; for the leverage-adjusted ones the proof adapts when  $\max_i h_{ii} \rightarrow 0$ , with small changes we omit. It needs neither normal errors nor moments beyond the second. Freedman (1981) proved it in a stronger metric, together with a statement for the case bootstrap. It does not say that the bootstrap beats the normal approximation  $\Phi(x/s)$ , which is consistent under the same conditions. The bootstrap earns its keep through studentization ([Section 23.3](#)) and through its flexibility for statistics with no convenient standard error.

*Warning.* The residual bootstrap treats the residuals as exchangeable. It gives  $\text{Cov}_*(\hat{\boldsymbol{\beta}}^*) = \hat{\sigma}_*^2(\mathbf{X}^\top \mathbf{X})^{-1}$  whatever the data look like, which is the classical formula, so it inherits that formula's failure under heteroscedasticity ([Theorem 19.2.2](#), [Exercise \(B1\)](#)). Wu (1986) made this point forcefully.

### 23.1.4 The case bootstrap

The case bootstrap assumes only that the cases are independent draws from a population. By [Proposition 6.11.3](#), least squares then estimates the population projection coefficient whether

or not the regression function is linear, and the case bootstrap estimates the variability of  $\hat{\beta}$  around that target.

Let  $W_i$  be the number of times case  $i$  is drawn, so that  $(W_1, \dots, W_n)$  is multinomial with  $n$  trials and equal probabilities, and put  $\mathbf{W} = \text{diag}(W_i)$ . When  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$  is invertible,  $\hat{\beta}^* = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{y}$ , and since  $\mathbf{y} = \mathbf{X} \hat{\beta} + \hat{\epsilon}$ ,

$$\hat{\beta}^* - \hat{\beta} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \hat{\epsilon}. \quad (23.1.2)$$

The matrix  $n^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X}$  has expectation  $n^{-1} \mathbf{X}^\top \mathbf{X}$  and is close to it for large  $n$ . Replacing it by its expectation gives a linear approximation whose covariance is exact.

### Proposition 23.1.5 — The case bootstrap, linearized, is the sandwich

Let  $\mathbf{L}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \hat{\epsilon}$ . Then  $E_*(\mathbf{L}^*) = \mathbf{0}$  and

$$\text{Cov}_*(\mathbf{L}^*) = (\mathbf{X}^\top \mathbf{X})^{-1} \left( \sum_i \hat{\epsilon}_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right) (\mathbf{X}^\top \mathbf{X})^{-1},$$

the sandwich covariance estimator HC0 of Chapter 21 (Theorem 21.4.2).

*Proof.* For multinomial counts with  $n$  trials and probabilities  $1/n$ ,  $E(W_i) = 1$ ,  $\text{Var}(W_i) = 1 - 1/n$  and  $\text{Cov}(W_i, W_j) = -1/n$  for  $i \neq j$ . So  $E_*(\mathbf{W} \hat{\epsilon}) = \hat{\epsilon}$  and  $\text{Cov}_*(\mathbf{W} \hat{\epsilon}) = \text{diag}(\hat{\epsilon})(\mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top) \text{diag}(\hat{\epsilon})$ . Hence  $E_*(\mathbf{L}^*) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \hat{\epsilon} = \mathbf{0}$  by Proposition 6.4.3(a), and  $\text{Cov}_*(\mathbf{X}^\top \mathbf{W} \hat{\epsilon}) = \sum_i \hat{\epsilon}_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top - n^{-1} (\mathbf{X}^\top \hat{\epsilon})(\mathbf{X}^\top \hat{\epsilon})^\top = \sum_i \hat{\epsilon}_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ . ■

To first order, then, the case bootstrap estimates the variance of  $\hat{\beta}$  by the sandwich, not the classical formula. The price is a varying design: a case of high leverage appears in some resamples several times and in others not at all, and with indicator columns some resamples are singular (Exercise (B2)).

#### Example. (Engel's food expenditure)

Return to Engel's data on the income and food expenditure of 235 households (Example (Engel's food expenditure data)). The least squares slope is 0.4852, and the scatter fans out (Figure 23.1.1(a)): the residual standard deviation is 2.36 times larger for the richer half of the households.

The classical standard error of the slope is 0.01437. The residual bootstrap with  $B = 4000$  resamples gives 0.01433, near its exact value 0.01431, the classical value times  $\sqrt{(n-2)/n} = 0.9957$  (Proposition 23.1.1); with leverage-adjusted residuals the exact value is 0.01484. The case bootstrap gives 0.05032, more than three times as large and close to the HC0 value 0.05177 of Proposition 23.1.5; Chapter 21 computed this and the other sandwich standard errors for the same slope. The residual bootstrap's answer rests on a constant variance that the data contradict.

```
import numpy as np
import statsmodels.api as sm
```

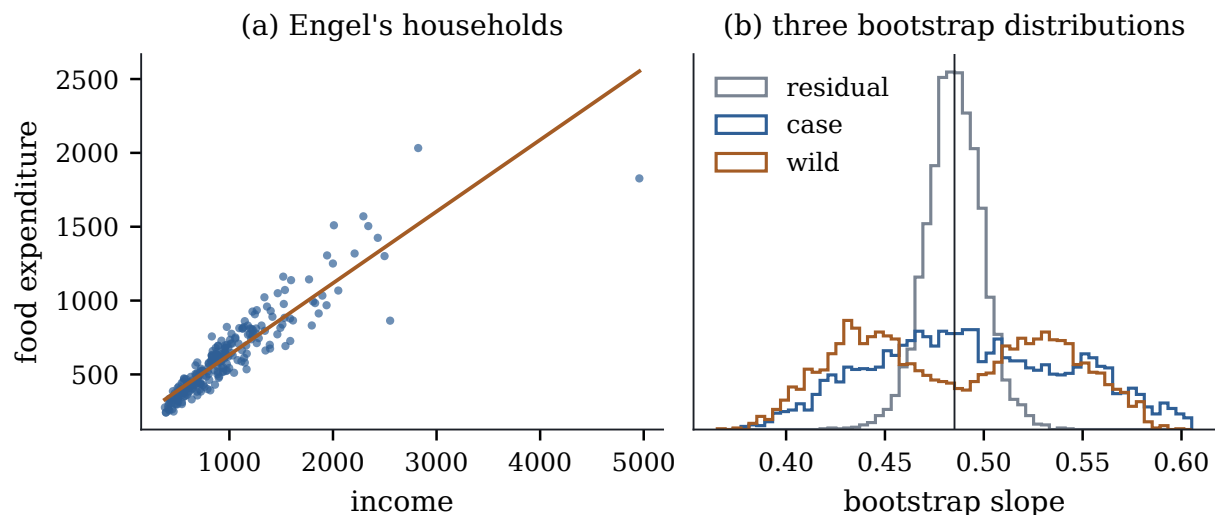


Figure 23.1.1: (a) Engel's households, with the least squares line. (b) Bootstrap distributions of the slope: residual (grey), case (blue) and wild (orange) resampling.

```
data = sm.datasets.engel.load_pandas().data

y = data["foodexp"].to_numpy()
x = data["income"].to_numpy()
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape
XtX_inv = np.linalg.inv(X.T @ X)
beta_hat = XtX_inv @ X.T @ y
e_hat = y - X @ beta_hat
h = np.sum((X @ XtX_inv) * X, axis=1)          # leverages
s2 = e_hat @ e_hat / (n - p)
se_classical = np.sqrt(s2 * XtX_inv[1, 1])
print(f"slope {beta_hat[1]:.4f}, classical se {se_classical:.5f}, max leverage {h.
      max():.3f}")

rng = np.random.default_rng(2301)
B = 4000
A = XtX_inv @ X.T                               # beta_hat = A y
fit = X @ beta_hat
e_c = e_hat - e_hat.mean()                       # centred residuals
r = e_hat / np.sqrt(1 - h)                       # leverage-adjusted residuals
r_c = r - r.mean()

def residual_bootstrap(resid, B):
    """Slopes from  $y^* = X \beta_{\text{hat}} + e^*$ , with  $e^*$  drawn with replacement from resid.
    """
    idx = rng.integers(0, n, size=(B, n))
    y_star = fit + resid[idx]
    return (y_star @ A.T)[: , 1]

slopes_res = residual_bootstrap(e_c, B)
slopes_mod = residual_bootstrap(r_c, B)
print(f"residual bootstrap se: raw {slopes_res.std():.5f}, leverage-adjusted {
      slopes_mod.std():.5f}")

def pairs_bootstrap(B):
    """Slopes refitted to  $n$  cases drawn with replacement from the  $(x_i, y_i)$ ."""
```

```

idx = rng.integers(0, n, size=(B, n))
xs, ys = x[idx], y[idx]
xc = xs - xs.mean(axis=1, keepdims=True)
return np.sum(xc * ys, axis=1) / np.sum(xc ** 2, axis=1)

slopes_pairs = pairs_bootstrap(B)
print(f"case bootstrap se {slopes_pairs.std():.5f}")

```

The choice between the two is a choice of model. The residual bootstrap suits a linear mean with exchangeable errors, even for sampled cases: when the error law does not depend on  $\mathbf{X}$ , conditioning on the design is justified as in [Section 14.1](#), where [Theorem 14.1.2](#) makes it exact under normality. It is then the more accurate, since it adds no noise from a varying design. Under heteroscedasticity or a nonlinear mean the case bootstrap is safer. The wild bootstrap of the next section keeps the design fixed and still respects heteroscedasticity.

## 23.1.5 Exercises

### 23.1.5.1 A. Check your understanding

#### Exercise A1.

In a regression through the origin,  $E(Y_i) = \beta x_i$ , the residuals need not sum to zero. Show that if the residual bootstrap resampled the uncentred residuals, then  $E_*(\hat{\beta}^*) - \hat{\beta} = \bar{\tilde{\varepsilon}} \sum_i x_i / \sum_i x_i^2$ .

*Solution.* Here  $\hat{\beta}^* - \hat{\beta} = \sum_i x_i \varepsilon_i^* / \sum_i x_i^2$ , and each uncentred draw has  $E_*(\varepsilon_i^*) = n^{-1} \sum_j \hat{\varepsilon}_j = \bar{\tilde{\varepsilon}}$ . Taking expectations gives the stated bias, which is not zero unless  $\bar{\tilde{\varepsilon}} = 0$  or  $\sum_i x_i = 0$ .

#### Exercise A2.

A case-bootstrap resample of size  $n$  contains each original case with probability  $1 - (1 - 1/n)^n$ . Show that the expected number of distinct cases in a resample is  $n(1 - (1 - 1/n)^n)$ , about  $0.632n$  for large  $n$ .

### 23.1.5.2 B. Practice

#### Exercise B1.

Suppose the errors are independent with variances  $\sigma_i^2$ , and the model has an intercept. Show that the residual bootstrap variance of  $\mathbf{c}^\top \hat{\boldsymbol{\beta}}^*$  has expectation  $\|\mathbf{a}\|^2 n^{-1} \sum_i (1 - h_{ii}) \sigma_i^2$ , whereas  $\text{Var}(\mathbf{c}^\top \hat{\boldsymbol{\beta}}) = \sum_i a_i^2 \sigma_i^2$ . Construct a two-group example in which the ratio of the two is as small as you like.

*Solution.* By [Proposition 23.1.1](#) the bootstrap variance is  $\|\mathbf{a}\|^2 \text{SSE}/n$ , and  $E(\text{SSE}) = \text{tr}((\mathbf{I} - \mathbf{M})\boldsymbol{\Sigma}) = \sum_i (1 - h_{ii}) \sigma_i^2$  for diagonal  $\boldsymbol{\Sigma}$ . The true variance is  $\mathbf{a}^\top \boldsymbol{\Sigma} \mathbf{a}$ . Take two groups of sizes  $n_1$  and  $n_2$  and the contrast of the group means, so  $a_i = 1/n_1$  in group 1 and  $-1/n_2$  in group 2. With  $\sigma_i^2 = \sigma_1^2$  in the small group and 0 in the large one, the true

variance is  $\sigma_1^2/n_1$  and the expected bootstrap variance is  $(1/n_1 + 1/n_2)(n_1 - 1)\sigma_1^2/n$ . With  $n_1$  fixed the ratio of the second to the first tends to 0 as  $n_2 \rightarrow \infty$ .

### Exercise B2.

In a one-way layout with  $g$  groups of  $m$  observations each, find the probability that a case-bootstrap resample omits some group entirely, so that the cell-means model cannot be fitted. Evaluate it for  $g = 5$ ,  $m = 4$ , and describe a resampling scheme that avoids the problem.

## 23.2 The wild bootstrap

The residual bootstrap keeps the design but moves every residual to a random position, destroying any link between the size of an error and where it occurred. The case bootstrap keeps that link but changes the design. The **wild bootstrap** keeps both: each residual stays with its own row of  $\mathbf{X}$ , and only a random multiplier changes (Wu 1986; Liu 1988; Mammen 1993).

### Definition 23.2.1 — Wild bootstrap

Let  $\mathbf{X}$  have rank  $p$ , with least squares estimate  $\hat{\boldsymbol{\beta}}$ , residuals  $\hat{\varepsilon}_i$  and leverages  $h_{ii} < 1$ . Choose transformed residuals  $u_i$ , one of

$$u_i = \hat{\varepsilon}_i, \quad u_i = \frac{\hat{\varepsilon}_i}{\sqrt{1 - h_{ii}}}, \quad u_i = \frac{\hat{\varepsilon}_i}{1 - h_{ii}},$$

and a law for the **multipliers**, with mean 0 and variance 1. The wild bootstrap draws  $v_1, \dots, v_n$  independently from that law, independently of the data, sets

$$Y_i^* = \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}} + u_i v_i, \quad i = 1, \dots, n,$$

and computes  $\hat{\boldsymbol{\beta}}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}^*$ . Two standard multiplier laws are the **Rademacher** law,  $v = \pm 1$  with probability 1/2 each, and **Mammen's** two-point law

$$\Pr\left(v = \frac{1 - \sqrt{5}}{2}\right) = \frac{\sqrt{5} + 1}{2\sqrt{5}}, \quad \Pr\left(v = \frac{1 + \sqrt{5}}{2}\right) = \frac{\sqrt{5} - 1}{2\sqrt{5}}.$$

In the bootstrap world the error of case  $i$  is  $u_i v_i$ , with mean zero and its own variance  $u_i^2$ . One residual is a poor estimate of a variance, but  $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$  needs only a weighted average of the variances, as for the sandwich.

### Proposition 23.2.1 — Moments of the wild bootstrap

In the wild bootstrap, with  $\mathbf{U} = \text{diag}(u_1, \dots, u_n)$  and  $\mathbf{a} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}$ :

(a)  $E_*(\hat{\boldsymbol{\beta}}^*) = \hat{\boldsymbol{\beta}}$  and



$$\text{Cov}_*(\hat{\boldsymbol{\beta}}^*) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{U}^2 \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}, \quad (23.2.1)$$

which is the sandwich estimator HC0, HC2 or HC3 (Theorem 21.4.2) for the three choices of  $u_i$ ;

$$(b) \ E_*[(\mathbf{c}^\top \hat{\boldsymbol{\beta}}^* - \mathbf{c}^\top \hat{\boldsymbol{\beta}})^3] = E(v^3) \sum_i a_i^3 u_i^3.$$

*Proof.*  $\hat{\boldsymbol{\beta}}^* - \hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{U} \mathbf{v}$ , where  $\mathbf{v} = (v_1, \dots, v_n)^\top$  has mean  $\mathbf{0}$  and covariance  $\mathbf{I}$ . So  $E_*(\hat{\boldsymbol{\beta}}^*) = \hat{\boldsymbol{\beta}}$  and  $\text{Cov}_*(\hat{\boldsymbol{\beta}}^*) = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{U} \mathbf{U} \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$ . For (b),  $\mathbf{c}^\top (\hat{\boldsymbol{\beta}}^* - \hat{\boldsymbol{\beta}}) = \sum_i a_i u_i v_i$  is a sum of independent terms with mean zero, and the third moment of such a sum is the sum of the third moments, because every cross term contains some  $v_i$  to the first power. ■

So the wild bootstrap reproduces the sandwich exactly, not only to first order as the case bootstrap does (Proposition 23.1.5). Part (b) explains the choice of multipliers. The true third moment is  $\sum_i a_i^3 E(\varepsilon_i^3)$ , not zero for skewed errors. Rademacher multipliers have  $E(v^3) = 0$  and give a symmetric bootstrap distribution; Mammen's law is the two-point law with third moment 1 (Exercise (B1)) and carries the skewness of the residuals. Davidson and Flachaire (2008) nonetheless favour Rademacher weights for tests, because matching the third moment costs accuracy in the fourth.

## 23.2.1 Consistency under heteroscedasticity

Take the fixed designs of Section 23.1, with contrasts  $\mathbf{c}_n$ , weights  $\mathbf{a}_n$ ,  $w_i = a_{ni}/\|\mathbf{a}_n\|$  and Eq. 23.1.1. The errors  $\varepsilon_{n1}, \dots, \varepsilon_{nn}$  are independent with mean zero and variances  $\sigma_{ni}^2$ , and for some constants  $0 < \underline{\sigma}^2$  and  $K < \infty$ ,

$$\sigma_{ni}^2 \geq \underline{\sigma}^2, \quad E \varepsilon_{ni}^4 \leq K \quad \text{for all } n, i. \quad (23.2.2)$$

The variance of the standardized contrast is  $\tau_n^2 = \sum_i w_i^2 \sigma_{ni}^2$ , which lies between  $\underline{\sigma}^2$  and  $\sqrt{K}$ , since  $\sigma_{ni}^2 \leq (E \varepsilon_{ni}^4)^{1/2}$ .

### Theorem 23.2.2 — Consistency of the wild bootstrap

Under Eq. 23.1.1 and Eq. 23.2.2, let  $\hat{\boldsymbol{\beta}}^*$  come from the wild bootstrap with  $u_i = \varepsilon_i$  and multipliers from any fixed law with mean 0 and variance 1, and let  $G_n$  and  $\hat{G}_n$  be defined as in Theorem 23.1.4, with this  $\hat{\boldsymbol{\beta}}^*$ . Then  $\sup_x |\hat{G}_n(x) - G_n(x)| \rightarrow 0$  in probability, and both are uniformly close to  $\Phi(x/\tau_n)$ .

*Proof.* Write  $\mathbf{d} = \mathbf{M}\boldsymbol{\varepsilon}$ , so that  $\hat{\varepsilon}_i = \varepsilon_i - d_i$ , and put  $m_n = \max_i |w_i|$ . Four steps.

*Step 1: the bootstrap variance.* Let  $S_n^2 = \sum_i w_i^2 \hat{\varepsilon}_i^2$ , the bootstrap variance of  $\mathbf{c}_n^\top (\hat{\boldsymbol{\beta}}^* - \hat{\boldsymbol{\beta}})/\|\mathbf{a}_n\|$  by Proposition 23.2.1. We show  $S_n^2 - \tau_n^2 \rightarrow 0$  in probability. Expand  $S_n^2 = \sum_i w_i^2 \varepsilon_i^2 - 2 \sum_i w_i^2 \varepsilon_i d_i + \sum_i w_i^2 d_i^2$ . The first sum has mean  $\tau_n^2$  and variance  $\sum_i w_i^4 \text{Var}(\varepsilon_i^2) \leq K m_n^2 \sum_i w_i^2 = K m_n^2 \rightarrow 0$ . The third is at most  $m_n^2 \|\mathbf{d}\|^2$ , and  $E\|\mathbf{d}\|^2 = \text{tr}(\mathbf{M} \text{Cov}(\boldsymbol{\varepsilon})) \leq$

$p\sqrt{K}$ , so it tends to zero in probability. The middle sum is at most  $2(\sum_i w_i^2 \varepsilon_i^2)^{1/2}(\sum_i w_i^2 d_i^2)^{1/2}$  by Cauchy–Schwarz, a bounded factor times one that tends to zero.

*Step 2: no dominant term.*  $\max_i |w_i \hat{\varepsilon}_i| \leq \max_i |w_i \varepsilon_i| + m_n \|\mathbf{d}\|$ . The second term tends to zero in probability, and for  $\eta > 0$

$$\Pr(\max_i |w_i \varepsilon_i| > \eta) \leq \sum_i \frac{w_i^4 \mathbb{E} \varepsilon_i^4}{\eta^4} \leq \frac{K m_n^2}{\eta^4},$$

which tends to zero. So  $\max_i |w_i \hat{\varepsilon}_i| \rightarrow 0$  in probability.

*Step 3: the bootstrap limit.* Given the data,  $U_{ni}^* = w_i \hat{\varepsilon}_i v_i / S_n$  are independent with mean zero and variances summing to one. With  $g(t) = \mathbb{E}[v^2 \mathbf{1}\{|v| > t\}]$ , which decreases to zero as  $t \rightarrow \infty$ ,

$$\begin{aligned} L_n^*(\eta) &= \sum_i \frac{w_i^2 \hat{\varepsilon}_i^2}{S_n^2} g\left(\frac{\eta S_n}{|w_i \hat{\varepsilon}_i|}\right) \\ &\leq g\left(\frac{\eta S_n}{\max_i |w_i \hat{\varepsilon}_i|}\right). \end{aligned}$$

By Steps 1 and 2 the argument of  $g$  tends to infinity in probability, because  $S_n^2 \geq \sigma^2/2$  with probability tending to one. So  $L_n^*(\eta) \rightarrow 0$  in probability, and [Lemma 23.1.2](#) gives  $\sup_x |\Pr_*(\sum_i w_i \hat{\varepsilon}_i v_i \leq x) - \Phi(x/S_n)| \rightarrow 0$  in probability. Since  $S_n/\tau_n \rightarrow 1$  in probability, the same holds with  $\Phi(x/\tau_n)$ , as at the end of the proof of [Theorem 23.1.4](#).

*Step 4: the true distribution.*  $\mathbf{c}_n^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) / \|\mathbf{a}_n\| = \sum_i w_i \varepsilon_i$ . With  $U_{ni} = w_i \varepsilon_i / \tau_n$ , Lyapunov's condition holds:  $\sum_i \mathbb{E} U_{ni}^4 \leq K m_n^2 / \sigma^4 \rightarrow 0$ . So  $\sum_i w_i \varepsilon_i / \tau_n \rightarrow N(0, 1)$ , and by Pólya's theorem  $\sup_x |\Pr(\sum_i w_i \varepsilon_i \leq x) - \Phi(x/\tau_n)| \rightarrow 0$ . The triangle inequality completes the proof. ■

An argument like Step 1 shows why the residual bootstrap fails here. Its variance for the standardized contrast is  $\hat{\sigma}_*^2$ , whose expectation  $n^{-1} \sum_i (1 - h_{ii}) \sigma_{ni}^2$  is an unweighted average of the error variances ([Exercise \(B1\)](#)), not the weighted average  $\tau_n^2$ . So it settles on a normal law with the variance the classical formula gives. The wild bootstrap succeeds because each resampled error keeps the variance of its own case.

#### Example. (The wild bootstrap for Engel's data)

For the line of [Example \(Engel's food expenditure\)](#), 4000 wild resamples with  $u_i = \hat{\varepsilon}_i$  give standard errors 0.05119 (Rademacher) and 0.05217 (Mammen), Monte Carlo estimates of the HC0 value 0.05177. With  $u_i = \hat{\varepsilon}_i / (1 - h_{ii})$  they give 0.06653, estimating HC3, 0.06639.

The shape is more interesting. The wild distribution in [Figure 23.1.1\(b\)](#) has two modes, at 0.439 and 0.531. The cause is the richest household, whose leverage 0.255 was examined in [Chapter 20](#) and whose share of the HC3 variance was measured in [Chapter 21](#). Its term  $a_i^2 \hat{\varepsilon}_i^2$  accounts for a fraction 0.78 of the HC0 variance  $\sum_j a_j^2 \hat{\varepsilon}_j^2$ , so the condition behind Step 2 of the proof is badly violated. The bootstrap slope is dominated by the contribution  $a_i \hat{\varepsilon}_i v_i$  of this one household, which with Rademacher multipliers is  $\pm 0.046$ , and the modes sit at  $\hat{\beta}_1 \mp 0.046$ . The sampling distribution of the slope is dominated

by that household's error too, and one observation cannot reveal the law it came from; the remedy is the one given in those chapters.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.engel.load_pandas().data
y = data["foodexp"].to_numpy()
x = data["income"].to_numpy()
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape
XtX_inv = np.linalg.inv(X.T @ X)
beta_hat = XtX_inv @ X.T @ y
e_hat = y - X @ beta_hat
h = np.sum((X @ XtX_inv) * X, axis=1) # leverages
rng = np.random.default_rng(2301)
B = 4000
A = XtX_inv @ X.T # beta_hat = A y
fit = X @ beta_hat

golden = (1 + np.sqrt(5)) / 2
mammen_values = np.array([1 - golden, golden]) # -0.618 and 1.618
mammen_probs = np.array([golden / np.sqrt(5), 1 - golden / np.sqrt(5)])

def wild_bootstrap(resid, B, weights="rademacher"):
    """Slopes from  $y_i = x_i' \beta + u_i$ , with independent multipliers  $u_i$ ."""
    if weights == "rademacher":
        v = rng.choice([-1.0, 1.0], size=(B, n))
    else:
        v = rng.choice(mammen_values, p=mammen_probs, size=(B, n))
    y_star = fit + resid * v
    return (y_star @ A.T)[:, 1]

slopes_wild = wild_bootstrap(e_hat, B)
slopes_mammen = wild_bootstrap(e_hat, B, weights="mammen")
slopes_wild3 = wild_bootstrap(e_hat / (1 - h), B)
se_hc0 = np.sqrt(A[1] ** 2 @ e_hat ** 2)
se_hc3 = np.sqrt(A[1] ** 2 @ (e_hat / (1 - h)) ** 2)
print(f"wild se: Rademacher {slopes_wild.std():.5f}, Mammen {slopes_mammen.std():.5f}
    }, "
      f"HC3-type {slopes_wild3.std():.5f}")
print(f"sandwich se: HC0 {se_hc0:.5f}, HC3 {se_hc3:.5f}")
```

## 23.2.2 Bootstrap or sandwich?

For a standard error the wild bootstrap adds nothing to the sandwich it reproduces. It earns its place through the *studentized* statistic  $(\mathbf{c}^\top \hat{\boldsymbol{\beta}} - \mathbf{c}^\top \boldsymbol{\beta}) / \widehat{\text{se}}_{\text{HC}}$ , whose noisy denominator makes its small-sample distribution far from  $t$  when some leverages are large; the simulations of [Section 23.3](#) show the wild bootstrap- $t$  beating the HC2  $t$  interval. For a test of  $\mathbf{c}^\top \boldsymbol{\beta} = d$ , the resamples should come from the restricted fit, so that the bootstrap approximates the null distribution even when the null hypothesis is false (Davidson and Flachaire 2008).

## 23.2.3 Exercises

### 23.2.3.1 A. Check your understanding

#### Exercise A1.

With Rademacher multipliers and  $u_i = \hat{\varepsilon}_i$ , how many distinct wild resamples are there? Show that the bootstrap variance of  $\mathbf{c}^\top \hat{\boldsymbol{\beta}}^*$  computed by averaging over all of them equals the HC0 variance exactly.

*Solution.* There are  $2^n$  sign vectors, equally likely. Averaging  $(\sum_i a_i \hat{\varepsilon}_i v_i)^2$  over them, every cross term  $a_i a_j \hat{\varepsilon}_i \hat{\varepsilon}_j v_i v_j$  with  $i \neq j$  averages to zero, because half of the sign vectors give  $v_i v_j = 1$  and half give  $-1$ . What remains is  $\sum_i a_i^2 \hat{\varepsilon}_i^2$ , the  $(\mathbf{c}, \mathbf{c})$  entry of HC0. The bootstrap mean is  $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$ , so this is the variance.

#### Exercise A2.

Two groups of sizes  $n_1$  and  $n_2$  are compared through the difference of their means, fitted by least squares with one indicator column per group. Show that the wild bootstrap with  $u_i = \hat{\varepsilon}_i$  gives the variance  $\hat{\sigma}_1^2/n_1 + \hat{\sigma}_2^2/n_2$ , where  $\hat{\sigma}_k^2$  is the within-group variance with divisor  $n_k$ , and that  $u_i = \hat{\varepsilon}_i/\sqrt{1 - h_{ii}}$  gives the Welch variance  $s_1^2/n_1 + s_2^2/n_2$ .

### 23.2.3.2 B. Practice

#### Exercise B1.

Let  $v$  take two values  $a < 0 < b$ . Show that  $E(v) = 0$  and  $E(v^2) = 1$  force  $ab = -1$  and  $\Pr(v = a) = b/(b - a)$ , and that then  $E(v^3) = a + b$ . Deduce that Mammen's law of [Definition 23.2.1](#) is the only two-point law with mean 0, variance 1 and third moment 1, and compute its fourth moment.

*Solution.* With  $q = \Pr(v = a)$ ,  $qa + (1 - q)b = 0$  gives  $q = b/(b - a)$ . Then  $E(v^2) = (ba^2 - ab^2)/(b - a) = -ab$ , so  $ab = -1$ . Similarly  $E(v^3) = (ba^3 - ab^3)/(b - a) = -ab(a + b) = a + b$ . Third moment one means  $a + b = 1$  and  $ab = -1$ , so  $a$  and  $b$  are the roots of  $t^2 - t - 1 = 0$ , namely  $(1 \mp \sqrt{5})/2$ , and  $q = b/(b - a) = (1 + \sqrt{5})/(2\sqrt{5})$ . The fourth moment is  $(ba^4 - ab^4)/(b - a) = -ab(a^2 + ab + b^2) = (a + b)^2 - ab = 2$ , compared with 1 for Rademacher weights and 3 for the normal.

### 23.2.3.3 C. Going deeper

#### Exercise C1.

Suppose the cases fall into clusters and errors are correlated within clusters but independent between them. Show that the wild bootstrap of [Definition 23.2.1](#) gives the HC0 covariance, which ignores the correlation. Propose a version in which all cases in a cluster share one multiplier, and show that its bootstrap covariance is the cluster-robust sandwich  $(\mathbf{X}^\top \mathbf{X})^{-1} \sum_g \mathbf{X}_g^\top \hat{\boldsymbol{\varepsilon}}_g \hat{\boldsymbol{\varepsilon}}_g^\top \mathbf{X}_g (\mathbf{X}^\top \mathbf{X})^{-1}$ .

## 23.3 Bootstrap confidence intervals

A bootstrap distribution gives a confidence interval through its quantiles, in several ways. They agree to first order, which is all the consistency theorems guarantee, and differ in how fast their coverage approaches the nominal level.

Let  $se$  be a standard error for  $\hat{\theta} = \mathbf{c}^\top \hat{\boldsymbol{\beta}}$ , classical or sandwich, and let  $\hat{\theta}^*$  and  $se^*$  be computed from a resample in the same way. Write  $\hat{q}(\gamma)$  and  $\hat{t}(\gamma)$  for the  $\gamma$  quantiles of the bootstrap distributions of  $\hat{\theta}^*$  and of the **bootstrap  $t$  statistic**  $T^* = (\hat{\theta}^* - \hat{\theta})/se^*$ .

### Definition 23.3.1 — Percentile, basic and studentized intervals

The nominal  $1 - \alpha$  bootstrap intervals for  $\theta$  are:

- (a) the **percentile** interval  $[\hat{q}(\alpha/2), \hat{q}(1 - \alpha/2)]$ ;
- (b) the **basic** interval  $[2\hat{\theta} - \hat{q}(1 - \alpha/2), 2\hat{\theta} - \hat{q}(\alpha/2)]$ ;
- (c) the **studentized** (or **bootstrap- $t$** ) interval  $[\hat{\theta} - \hat{t}(1 - \alpha/2)se, \hat{\theta} - \hat{t}(\alpha/2)se]$ .

The basic interval translates the bootstrap idea directly: if  $\hat{\theta} - \theta$  had the distribution of  $\hat{\theta}^* - \hat{\theta}$ , then  $\hat{q}(\alpha/2) - \hat{\theta} \leq \hat{\theta} - \theta \leq \hat{q}(1 - \alpha/2) - \hat{\theta}$  would have probability  $1 - \alpha$ , and solving for  $\theta$  gives (b). The studentized interval does the same with the root  $(\hat{\theta} - \theta)/se$ , as the  $t$  interval of [Corollary 7.5.2](#) does. The percentile interval looks naive, but if some increasing  $\phi$  makes  $\phi(\hat{\theta}) - \phi(\theta)$  a pivot symmetric about zero, it is exact without knowing  $\phi$  ([Exercise \(B1\)](#)). It is also **transformation respecting**, because quantiles commute with increasing functions; the basic and studentized intervals are not.

### 23.3.1 First-order correctness

The consistency theorems translate into coverage through a lemma about quantiles.

#### Lemma 23.3.1 — Consistent bootstrap quantiles

Let  $R_n$  be random variables with distribution functions  $H_n$ , and let  $\hat{H}_n$  be random distribution functions with  $\sup_x |\hat{H}_n(x) - H_n(x)| \rightarrow 0$  in probability. Suppose  $H_n(x) \rightarrow H(x)$  for every  $x$ , where  $H$  is continuous and strictly increasing on  $\mathbb{R}$ . For  $0 < \gamma < 1$ , let  $\hat{\xi}_n(\gamma) = \inf\{x : \hat{H}_n(x) \geq \gamma\}$  and  $\xi(\gamma) = H^{-1}(\gamma)$ . Then  $\hat{\xi}_n(\gamma) \rightarrow \xi(\gamma)$  in probability, and  $\Pr(R_n \leq \hat{\xi}_n(\gamma)) \rightarrow \gamma$ .

*Proof.* By Pólya's theorem  $\sup_x |H_n(x) - H(x)| \rightarrow 0$ , so  $\delta_n = \sup_x |\hat{H}_n(x) - H(x)| \rightarrow 0$  in probability. Fix  $\epsilon > 0$ , write  $\xi = \xi(\gamma)$ , and let  $\kappa = \min\{\gamma - H(\xi - \epsilon), H(\xi + \epsilon) - \gamma\} > 0$ . On the event  $\delta_n < \kappa$ ,  $\hat{H}_n(\xi - \epsilon) < \gamma$  and  $\hat{H}_n(\xi + \epsilon) > \gamma$ , so  $\xi - \epsilon \leq \hat{\xi}_n(\gamma) \leq \xi + \epsilon$ . This proves the first claim. For the second,

$$\begin{aligned} \Pr(R_n \leq \hat{\xi}_n) &\geq H_n(\xi - \epsilon) - \Pr(|\hat{\xi}_n - \xi| > \epsilon), \\ \Pr(R_n \leq \hat{\xi}_n) &\leq H_n(\xi + \epsilon) + \Pr(|\hat{\xi}_n - \xi| > \epsilon). \end{aligned}$$

Let  $n \rightarrow \infty$  and then  $\epsilon \rightarrow 0$ , using the continuity of  $H$  at  $\xi$ .

■

### Corollary 23.3.2 — Bootstrap intervals are first-order correct

Under the conditions of [Theorem 23.1.4](#), the percentile and basic intervals built from the residual bootstrap have coverage probability tending to  $1 - \alpha$ .

*Proof.* Take  $R_n = (\hat{\theta} - \theta)/\|\mathbf{a}_n\|$ , with  $\hat{H}_n$  the bootstrap distribution of  $(\hat{\theta}^* - \hat{\theta})/\|\mathbf{a}_n\|$  and  $H(x) = \Phi(x/\sigma)$ . [Theorem 23.1.4](#) gives the hypotheses of [Lemma 23.3.1](#). The basic interval covers  $\theta$  if and only if  $\hat{q}(\alpha/2) - \hat{\theta} \leq \hat{\theta} - \theta \leq \hat{q}(1 - \alpha/2) - \hat{\theta}$ , that is,  $\hat{\xi}_n(\alpha/2) \leq R_n \leq \hat{\xi}_n(1 - \alpha/2)$  with  $\hat{\xi}_n$  the quantiles of  $\hat{H}_n$ . By the lemma the probability tends to  $(1 - \alpha/2) - \alpha/2$ , the lower endpoint contributing no mass in the limit because  $H$  is continuous. The percentile interval covers  $\theta$  if and only if  $-\hat{\xi}_n(1 - \alpha/2) \leq R_n \leq -\hat{\xi}_n(\alpha/2)$ . Since  $H$  is symmetric,  $-\xi(1 - \alpha/2) = \xi(\alpha/2)$ , and the same argument applies.

■

The same argument applies to the wild bootstrap under the conditions of [Theorem 23.2.2](#). Since  $\tau_n$  need not converge, take  $R_n = (\hat{\theta} - \theta)/(\|\mathbf{a}_n\|\tau_n)$  and  $H = \Phi$ ; dividing by the constant  $\tau_n$  leaves the intervals unchanged. It applies also to studentized intervals whenever the standard error is consistent. None of this separates the methods from one another or from the normal interval  $\hat{\theta} \pm z_{\alpha/2} \text{se}$ .

## 23.3.2 Pivots and higher-order accuracy

The case for studentizing is clearest when the root is an exact pivot.

### Proposition 23.3.3 — The parametric bootstrap in the normal model

In the normal linear model with  $\text{rank}(\mathbf{X}) = p < n$ , let the **parametric bootstrap** draw  $\mathbf{Y}^* \sim N_n(\mathbf{X}\hat{\boldsymbol{\beta}}, s^2\mathbf{I})$ . Then for every data set:

- (a)  $T^* = (\mathbf{c}^\top \hat{\boldsymbol{\beta}}^* - \mathbf{c}^\top \hat{\boldsymbol{\beta}})/(s^*\|\mathbf{a}\|) \sim t(n - p)$  under  $\text{Pr}_*$ , so the studentized interval is the  $t$  interval of [Corollary 7.5.2](#), apart from Monte Carlo error;
- (b)  $\mathbf{c}^\top \hat{\boldsymbol{\beta}}^* \sim N(\mathbf{c}^\top \hat{\boldsymbol{\beta}}, s^2\|\mathbf{a}\|^2)$  under  $\text{Pr}_*$ , so the percentile and basic intervals are  $\hat{\theta} \pm z_{\alpha/2} s\|\mathbf{a}\|$ , whose coverage is  $\Pr(|T| \leq z_{\alpha/2}) < 1 - \alpha$  with  $T \sim t(n - p)$ .

*Proof.* Given the data,  $\mathbf{Y}^*$  follows the normal linear model with coefficient vector  $\hat{\boldsymbol{\beta}}$  and variance  $s^2$ . (a) is [Corollary 7.5.2](#) applied to that model, with true value  $\mathbf{c}^\top \hat{\boldsymbol{\beta}}$ . (b) is [Theorem 7.5.1\(a\)](#) for the same model. The interval  $\hat{\theta} \pm z_{\alpha/2} s\|\mathbf{a}\|$  covers  $\theta$  if and only if  $|T| \leq z_{\alpha/2}$ , where  $T = (\hat{\theta} - \theta)/(s\|\mathbf{a}\|) \sim t(n - p)$  by [Corollary 7.5.2](#), and  $t(n - p)$  has heavier tails than  $N(0, 1)$ .

■

The studentized root has the same distribution in both worlds, so the bootstrap reproduces it exactly. The unstudentized root depends on  $\sigma$ , and the bootstrap substitutes  $s$ , losing the spread due to estimating  $\sigma$ . The nonparametric bootstrap is never exact, but the same mechanism operates: an **asymptotic pivot** is reproduced more accurately than a root whose distribution depends strongly on the unknown law. Hall (1988) made this precise with Edgeworth expansions. Under smoothness conditions the one-sided studentized interval has coverage error of order  $n^{-1}$ , the one-sided percentile and basic intervals of order  $n^{-1/2}$ . We do not prove these rates (see Hall 1992). The price of studentizing is a standard error recomputed in each resample, and instability when it is poorly estimated.

**BCa.** Efron (1987) corrected the percentile interval to the same order of accuracy without a standard error. The **bias-corrected and accelerated** (BCa) interval uses the percentile interval at adjusted levels

$$\gamma_k = \Phi\left(\hat{z}_0 + \frac{\hat{z}_0 + z_k}{1 - \hat{a}(\hat{z}_0 + z_k)}\right), \quad z_1 = z_{(\alpha/2)}, \quad z_2 = z_{(1-\alpha/2)},$$

where  $z_{(\gamma)} = \Phi^{-1}(\gamma)$ , the **bias correction**  $\hat{z}_0 = \Phi^{-1}(\Pr_*(\hat{\theta}^* < \hat{\theta}))$  measures median bias, and the **acceleration**  $\hat{a}$ , the rate at which the standard error changes with  $\theta$  on a normalizing scale, is usually estimated from the jackknife values  $\hat{\theta}_{(i)}$ , computed without case  $i$ :

$$\hat{a} = \frac{\sum_i d_i^3}{6(\sum_i d_i^2)^{3/2}}, \quad d_i = \bar{\theta}_{(\cdot)} - \hat{\theta}_{(i)}.$$

In regression no refitting is needed:  $\hat{\theta}_{(i)} = \hat{\theta} - a_i \hat{\varepsilon}_i / (1 - h_{ii})$  by [Proposition 10.6.2\(a\)](#) ([Exercise \(B2\)](#)). With  $\hat{z}_0 = \hat{a} = 0$  BCa is the percentile interval. Its derivation and second-order accuracy are in Efron (1987) and DiCiccio and Efron (1996), for resampling cases, to which the jackknife formula above belongs. Pairing it with the residual bootstrap, as below, is a common heuristic that Efron's proof does not cover.

### 23.3.3 An example and a simulation

#### Example. (Intervals for the poverty coefficient)

Return to the state murder rates of [Example \(Poverty and murder rates\)](#), with the District of Columbia omitted, and the coefficient 0.2538 of poverty in the regression on poverty, single parenthood and urbanization. Seven 95% intervals from  $B = 9999$  resamples:

| method                          | interval       | length |
|---------------------------------|----------------|--------|
| $t$ , classical standard error  | (0.066, 0.442) | 0.376  |
| residual bootstrap, percentile  | (0.064, 0.437) | 0.374  |
| residual bootstrap, basic       | (0.070, 0.444) | 0.374  |
| residual bootstrap, studentized | (0.071, 0.449) | 0.378  |
| residual bootstrap, BCa         | (0.040, 0.418) | 0.378  |
| case bootstrap, percentile      | (0.080, 0.428) | 0.348  |
| wild bootstrap- $t$ , HC2       | (0.079, 0.428) | 0.348  |



The residual bootstrap used leverage-adjusted residuals. The first four intervals agree closely: with fifty states and a well-behaved fit, the classical interval is hard to improve on. BCa is shifted down; its bias correction  $\hat{z}_0 = -0.016$  is negligible, but the jackknife acceleration  $\hat{a} = -0.0458$  moves both endpoints. The shift comes from that heuristic pairing, not from BCa as such, and opposes the small upward shift of the studentized interval. The case and wild bootstraps give shorter intervals, because the HC2 standard error 0.0869 is smaller than the classical 0.0934 for these data. All seven exclude zero.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.statelcrime.load_pandas().data
data = data.drop(index="District of Columbia")

rng = np.random.default_rng(2303)
B = 9999
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
XtX_inv = np.linalg.inv(X.T @ X)
A = XtX_inv @ X.T
a, c = A[1], np.sqrt(XtX_inv[1, 1])          # the poverty coefficient is a @ y
h = np.sum((X @ XtX_inv) * X, axis=1)
fit = X @ (A @ y)
e = y - fit
b = a @ y
s = np.sqrt(e @ e / (n - p))
q = [0.025, 0.975]
out = {"t": b + np.array([-1, 1]) * stats.t.ppf(0.975, n - p) * s * c}

r = e / np.sqrt(1 - h)                      # residual bootstrap, modified
residuals
r -= r.mean()
Ystar = fit + r[rng.integers(0, n, size=(B, n))]
bstar = Ystar @ a
sstar = np.sqrt(np.sum((Ystar - (Ystar @ A.T) @ X.T) ** 2, axis=1) / (n - p))
tstar = (bstar - b) / (sstar * c)
lo, hi = np.quantile(bstar, q)
out["percentile"] = np.array([lo, hi])
out["basic"] = np.array([2 * b - hi, 2 * b - lo])
out["studentized"] = b - np.quantile(tstar, q[::-1]) * s * c

z0 = stats.norm.ppf(np.mean(bstar < b))      # BCa
loo = b - a * e / (1 - h)                   # case-jackknife values (heuristic
with residual resampling)
dev = loo.mean() - loo
acc = np.sum(dev ** 3) / (6 * np.sum(dev ** 2) ** 1.5)
zs = z0 + stats.norm.ppf(q)
out["BCa"] = np.quantile(bstar, stats.norm.cdf(z0 + zs / (1 - acc * zs)))

idx = rng.integers(0, n, size=(B, n))       # case bootstrap
bcase = np.array([np.linalg.lstsq(X[i], y[i], rcond=None)[0][1] for i in idx])
out["case"] = np.quantile(bcase, q)

se2 = np.sqrt(np.sum(a ** 2 * e ** 2 / (1 - h))) # wild bootstrap-t with HC2
Ystar = fit + (e / np.sqrt(1 - h)) * rng.choice([-1.0, 1.0], size=(B, n))
Estar = Ystar - (Ystar @ A.T) @ X.T
```



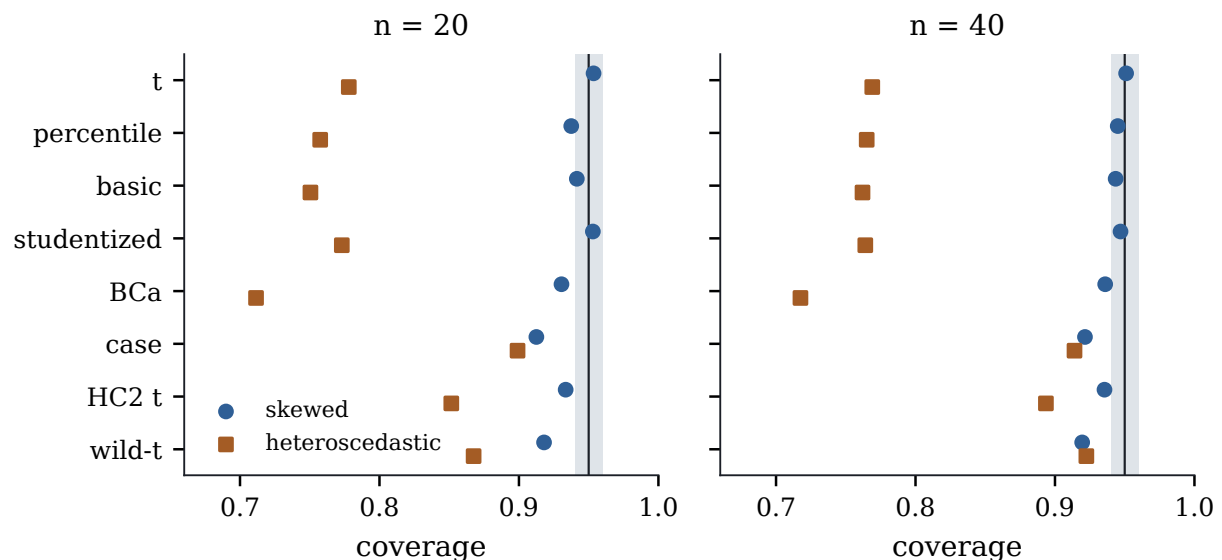


Figure 23.3.1: Coverage of eight nominal 95% intervals for a slope. Circles: skewed errors, constant variance. Squares: normal errors with variance increasing in  $x$ . The band is two Monte Carlo standard errors around 0.95.

```
wstar = (Ystar @ a - b) / np.sqrt((Estar ** 2 / (1 - h)) @ a ** 2)
out["wild-t"] = b - np.quantile(wstar, q[::-1]) * se2

for m, (l, u) in out.items():
    print(f"{m:12s} ({l:.3f}, {u:.3f})    length {u - l:.3f}")
```

One data set cannot show which interval is right. Figure 23.3.1 reports coverages for the slope of a line, from 2000 simulated data sets with  $B = 999$  resamples each, and Monte Carlo standard error about 0.005. The fixed design is skewed, with largest leverage 0.433 at  $n = 20$  and 0.309 at  $n = 40$ . The errors are either centred exponential with constant variance, or normal with standard deviation proportional to  $x$ , so that the largest variances sit at the high-leverage end.

Three lessons stand out.

1. *With constant variance, studentization is what matters.* At  $n = 20$  the classical  $t$  interval (0.954) and the studentized residual bootstrap (0.953) are on target despite skewed errors. The percentile (0.938) and BCa (0.930) intervals are a little short, as Proposition 23.3.3 predicts for intervals that ignore the variability of  $s$ . The robust methods (case 0.912, wild- $t$  0.918) pay for estimating a variance that did not need estimating.
2. *With heteroscedastic errors the classical  $t$  interval and the four residual-bootstrap intervals all fail, and more data do not help:* all five cover between 0.71 and 0.78 at both sample sizes (Exercise (B1)).
3. *The robust methods improve with  $n$  but start below nominal.* The case bootstrap, the HC2  $t$  interval and the wild bootstrap- $t$  cover 0.899, 0.852 and 0.868 at  $n = 20$ , and 0.914, 0.893 and 0.922 at  $n = 40$ . A high-leverage point with a large error variance is hard for all of them, since one residual carries much of the information about the variance that matters (Example (The wild bootstrap for Engel's data)).

```
import numpy as np
from scipy import stats
level = 0.95
lo_q, hi_q = (1 - level) / 2, (1 + level) / 2
```

```

beta = np.array([1.0, 1.0])
methods = ["t", "percentile", "basic", "studentized", "BCa", "case", "HC2 t", "wild-
t"]

def make_design(n):
    """A skewed fixed design and the quantities every data set reuses."""
    x = np.exp(0.5 * stats.norm.ppf((np.arange(1, n + 1) - 0.5) / n))
    X = np.column_stack([np.ones(n), x])
    XtX_inv = np.linalg.inv(X.T @ X)
    A = XtX_inv @ X.T                                # beta_hat = A y
    h = np.sum((X @ XtX_inv) * X, axis=1)            # leverages
    sd = x / np.sqrt(np.mean(x ** 2))                # heteroscedastic sd, mean square
    one
    return dict(n=n, x=x, X=X, A=A, h=h, a=A[1], c=np.sqrt(XtX_inv[1, 1]), sd=sd)

def errors(rng, law, d):
    if law == "skewed":                               # centred exponential, constant
        variance
        return rng.exponential(size=d["n"]) - 1.0
    return d["sd"] * rng.normal(size=d["n"])          # normal, sd proportional to x

def one_data_set(rng, law, d, B):
    """Return, for each method, whether its 95% interval covers the true slope."""
    n, x, X, A, h, a, c = (d[k] for k in ("n", "x", "X", "A", "h", "a", "c"))
    y = X @ beta + errors(rng, law, d)
    b = a @ y
    e = y - X @ (A @ y)
    s = np.sqrt(e @ e / (n - 2))
    cover = {"t": abs(b - beta[1]) <= stats.t.ppf(hi_q, n - 2) * s * c}
    # residual bootstrap from leverage-adjusted, centred residuals
    r = e / np.sqrt(1 - h)
    r -= r.mean()
    Ystar = X @ (A @ y) + r[rng.integers(0, n, size=(B, n))]
    bstar = Ystar @ a
    Estar = Ystar - (Ystar @ A.T) @ X.T
    tstar = (bstar - b) / (np.sqrt(np.sum(Estar ** 2, axis=1) / (n - 2)) * c)
    q_lo, q_hi = np.quantile(bstar, [lo_q, hi_q])
    t_lo, t_hi = np.quantile(tstar, [lo_q, hi_q])
    cover["percentile"] = q_lo <= beta[1] <= q_hi
    cover["basic"] = 2 * b - q_hi <= beta[1] <= 2 * b - q_lo
    cover["studentized"] = b - t_hi * s * c <= beta[1] <= b - t_lo * s * c
    # BCa: residual-bootstrap bias correction, case-jackknife acceleration (a
    heuristic pairing)
    z0 = stats.norm.ppf(np.clip(np.mean(bstar < b), 1 / B, 1 - 1 / B))
    loo = b - a * e / (1 - h)                          # slopes with one case deleted
    dev = loo.mean() - loo
    acc = np.sum(dev ** 3) / (6 * np.sum(dev ** 2) ** 1.5)
    zs = z0 + stats.norm.ppf([lo_q, hi_q])
    bca_lo, bca_hi = np.quantile(bstar, stats.norm.cdf(z0 + zs / (1 - acc * zs)))
    cover["BCa"] = bca_lo <= beta[1] <= bca_hi
    # case bootstrap percentile
    idx = rng.integers(0, n, size=(B, n))
    xs, ys = x[idx], y[idx]
    xc = xs - xs.mean(axis=1, keepdims=True)
    sxx = np.sum(xc ** 2, axis=1)
    ok = sxx > 1e-12                                    # drop the (rare) resamples with
    one distinct x
    bcase = np.sum(xc[ok] * ys[ok], axis=1) / sxx[ok]

```

```

c1, ch = np.quantile(bcase, [lo_q, hi_q])
cover["case"] = c1 <= beta[1] <= ch
# HC2 t interval, and the wild bootstrap-t studentized by HC2
se2 = np.sqrt(np.sum(a ** 2 * e ** 2 / (1 - h)))
cover["HC2 t"] = abs(b - beta[1]) <= stats.t.ppf(hi_q, n - 2) * se2
v = rng.choice([-1.0, 1.0], size=(B, n))
Ystar = X @ (A @ y) + (e / np.sqrt(1 - h)) * v
Estar = Ystar - (Ystar @ A.T) @ X.T
wstar = (Ystar @ a - b) / np.sqrt((Estar ** 2 / (1 - h)) @ a ** 2)
w_lo, w_hi = np.quantile(wstar, [lo_q, hi_q])
cover["wild-t"] = b - w_hi * se2 <= beta[1] <= b - w_lo * se2
return cover

def coverage(law, n, reps, B, seed):
    rng = np.random.default_rng(seed)
    d = make_design(n)
    hits = {m: 0 for m in methods}
    for _ in range(reps):
        for m, ok in one_data_set(rng, law, d, B).items():
            hits[m] += ok
    return {m: hits[m] / reps for m in methods}

for law in ["skewed", "heteroscedastic"]:
    cov = coverage(law, n=20, reps=200, B=499, seed=2331)
    print(f"[law:16s]" + " ".join(f"{m} {v:.2f}" for m, v in cov.items()))

```

With 200 data sets the cell's coverages have Monte Carlo standard errors of about  $\sqrt{0.95 \times 0.05/200} = 0.015$ .

## 23.3.4 Bootstrap tests of a linear hypothesis

A test of  $H: \Lambda^\top \beta = \mathbf{d}$  can take its critical value from the bootstrap instead of the  $F$  table. In each residual-bootstrap resample from the full model, compute  $F_H^*$  of Eq. 11.3.1 for the hypothesis  $\Lambda^\top \beta = \Lambda^\top \hat{\beta}$ , which is true in the bootstrap world, and report the proportion of resamples with  $F_H^* \geq F_H$ . This is the testing counterpart of the studentized interval (Proposition 12.1.2). Alternatively, resample the residuals of the restricted fit, so that the null hypothesis holds in the bootstrap world; that is the bootstrap analogue of the Freedman–Lane scheme of Section 23.5.

## 23.3.5 Exercises

### 23.3.5.1 A. Check your understanding

#### Exercise A1.

Show that the percentile and basic intervals coincide when the bootstrap distribution of  $\hat{\theta}^*$  is symmetric about  $\hat{\theta}$ . With  $\hat{\theta} = 2$ ,  $\hat{q}(0.025) = 1.5$  and  $\hat{q}(0.975) = 3.5$ , compute both intervals and explain why they lean in opposite directions.

*Solution.* Symmetry about  $\hat{\theta}$  means  $\hat{q}(\gamma) - \hat{\theta} = \hat{\theta} - \hat{q}(1 - \gamma)$ , so  $2\hat{\theta} - \hat{q}(1 - \alpha/2) = \hat{q}(\alpha/2)$ , and similarly for the other endpoint. In the example the percentile interval is  $[1.5, 3.5]$

and the basic interval is  $[4 - 3.5, 4 - 1.5] = [0.5, 2.5]$ . The bootstrap distribution has a long right tail. The percentile interval copies it. The basic interval reflects it, reasoning that if  $\hat{\theta}$  tends to fall far above  $\theta$  with some probability, then  $\theta$  may lie far below  $\hat{\theta}$ .

### 23.3.5.2 B. Practice

#### Exercise B1.

Suppose there is a strictly increasing  $\phi$  such that  $\phi(\hat{\theta}) - \phi(\theta)$  has a continuous distribution  $K$ , symmetric about zero and the same for every  $\theta$ , and suppose the bootstrap reproduces it exactly:  $\phi(\hat{\theta}^*) - \phi(\hat{\theta})$  has distribution  $K$  under  $\text{Pr}_*$ . Show that the percentile interval has coverage exactly  $1 - \alpha$ .

*Solution.* Let  $k_\gamma$  be the  $\gamma$  quantile of  $K$ , so  $k_{\alpha/2} = -k_{1-\alpha/2}$ . Under  $\text{Pr}_*$ ,  $\phi(\hat{\theta}^*) = \phi(\hat{\theta}) +$  a  $K$ -variable, so the  $\gamma$  quantile of  $\hat{\theta}^*$  is  $\hat{q}(\gamma) = \phi^{-1}(\phi(\hat{\theta}) + k_\gamma)$ . Hence  $\hat{q}(\alpha/2) \leq \theta \leq \hat{q}(1 - \alpha/2)$  if and only if  $k_{\alpha/2} \leq \phi(\theta) - \phi(\hat{\theta}) \leq k_{1-\alpha/2}$ , that is,  $-k_{1-\alpha/2} \leq \phi(\hat{\theta}) - \phi(\theta) \leq -k_{\alpha/2}$ . By symmetry this is  $k_{\alpha/2} \leq \phi(\hat{\theta}) - \phi(\theta) \leq k_{1-\alpha/2}$ , which has probability  $1 - \alpha$ .

#### Exercise B2.

Show that the jackknife values used in the BCa acceleration satisfy  $\hat{\theta}_{(i)} = \hat{\theta} - a_i \hat{\varepsilon}_i / (1 - h_{ii})$ , where  $\mathbf{a} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{c}$ . Deduce that the jackknife variance estimate  $\frac{n-1}{n} \sum_i (\hat{\theta}_{(i)} - \bar{\theta}_{(\cdot)})^2$  is close to the HC3 variance  $\sum_i a_i^2 \hat{\varepsilon}_i^2 / (1 - h_{ii})^2$ . (Chapter 21 shows that the uncentred sum  $\sum_i (\hat{\theta}_{(i)} - \hat{\theta})^2$  equals it exactly.)

### 23.3.5.3 C. Going deeper

#### Exercise C1.

The coverage of a bootstrap interval can itself be estimated by a second level of bootstrapping: treat each first-level resample as data, and ask how often its interval covers  $\hat{\theta}$ . Describe how this can be used to recalibrate the nominal level, and count the least squares fits needed with  $B_1 = B_2 = 999$ . How does the design-fixed residual bootstrap reduce the cost?

## 23.4 Permutation tests for linear hypotheses

The bootstrap replaces an unknown distribution by an estimate. A permutation test instead looks for transformations of the data that leave their distribution unchanged under the null hypothesis, and compares the observed statistic with its values on the transformed data. When such transformations exist the test is exact at every sample size, whatever the error law. The idea is Fisher's (1935), developed by Pitman (1937); in designed experiments the transformations are the rearrangements the randomization could have produced (Section 18.4).

## 23.4.1 The general construction

Let  $\mathcal{G}$  be a finite group of  $n \times n$  permutation matrices. A random vector  $\mathbf{Y}$  is  **$\mathcal{G}$ -invariant in distribution** if  $\mathbf{PY}$  has the distribution of  $\mathbf{Y}$  for every  $\mathbf{P} \in \mathcal{G}$ . For the group of all permutations this is exchangeability.

### Definition 23.4.1 — Permutation test

Let  $T : \mathbb{R}^n \rightarrow \mathbb{R}$  be a statistic, large values of which count against the null hypothesis.

- (a) The **permutation p-value** is  $p(\mathbf{y}) = |\mathcal{G}|^{-1} \#\{\mathbf{P} \in \mathcal{G} : T(\mathbf{PY}) \geq T(\mathbf{y})\}$ .
- (b) The **Monte Carlo permutation p-value** draws  $\mathbf{P}_1, \dots, \mathbf{P}_B$  independently and uniformly from  $\mathcal{G}$ , independently of  $\mathbf{Y}$ , and is

$$p_B(\mathbf{y}) = \frac{1 + \#\{b : T(\mathbf{P}_b \mathbf{y}) \geq T(\mathbf{y})\}}{B + 1}. \quad (23.4.1)$$

The test of level  $\alpha$  rejects when the p-value is at most  $\alpha$ .

The 1 in Eq. 23.4.1 counts the observed arrangement among those compared; without it the p-value can be zero and the test is not exact (Phipson and Smyth 2010). Exactness rests on a counting fact.

### Lemma 23.4.1 — Ranks of exchangeable values

For real numbers  $t_0, \dots, t_m$ , let  $R_j = \#\{i : t_i \geq t_j\}$ . Then  $\#\{j : R_j \leq k\} \leq k$  for every  $k$ . Consequently, if  $(T_0, \dots, T_m)$  is an exchangeable random vector and  $R_0 = \#\{i : T_i \geq T_0\}$ , then  $\Pr(R_0 \leq k) \leq k/(m+1)$ .

*Proof.* Let  $S = \{j : R_j \leq k\}$ , and suppose it is not empty. Choose  $j^* \in S$  with the smallest  $t_{j^*}$ . Every  $j \in S$  has  $t_j \geq t_{j^*}$ , so  $R_{j^*} \geq |S|$ , and  $|S| \leq R_{j^*} \leq k$ . For the second claim, exchangeability gives  $\Pr(R_j \leq k) = \Pr(R_0 \leq k)$  for every  $j$ , so  $(m+1)\Pr(R_0 \leq k) = \mathbb{E} \#\{j : R_j \leq k\} \leq k$ . ■

### Theorem 23.4.2 — Exactness of permutation tests

If  $\mathbf{Y}$  is  $\mathcal{G}$ -invariant in distribution, then for every statistic  $T$ , every  $\alpha \in [0, 1]$  and every  $B \geq 1$ ,

$$\Pr(p(\mathbf{Y}) \leq \alpha) \leq \alpha, \quad \Pr(p_B(\mathbf{Y}) \leq \alpha) \leq \alpha.$$

*Proof.* Let  $\mathbf{\Gamma}$  be uniform on  $\mathcal{G}$ , independent of everything else. Invariance gives  $\mathbf{\Gamma Y} \stackrel{d}{=} \mathbf{Y}$ , since this holds conditionally on each value of  $\mathbf{\Gamma}$ .

*Full group.*  $p(\mathbf{Y})$  has the distribution of  $p(\mathbf{\Gamma Y})$ . As  $\mathbf{P}$  runs over  $\mathcal{G}$  so does  $\mathbf{P}\mathbf{\Gamma}$ , so  $p(\mathbf{\Gamma Y}) = |\mathcal{G}|^{-1} \#\{\mathbf{P} \in \mathcal{G} : T(\mathbf{PY}) \geq T(\mathbf{\Gamma Y})\}$ . Given  $\mathbf{Y}$ , list the values  $t_{\mathbf{P}} = T(\mathbf{PY})$ ,  $\mathbf{P} \in \mathcal{G}$ . Then  $|\mathcal{G}| p(\mathbf{\Gamma Y}) = R_{\mathbf{\Gamma}}$  in the notation of Lemma 23.4.1, with  $\mathbf{\Gamma}$  uniform over the

$|\mathcal{G}|$  indices. By the first part of the lemma,  $\Pr(R_{\mathbf{r}} \leq k \mid \mathbf{Y}) \leq k/|\mathcal{G}|$ . With  $k = \lfloor \alpha|\mathcal{G}| \rfloor$  this gives  $\Pr(p \leq \alpha \mid \mathbf{Y}) \leq \alpha$ , and we take expectations.

*Monte Carlo.* Put  $\mathbf{P}_0 = \mathbf{I}$  and  $T_b = T(\mathbf{P}_b \mathbf{Y})$ , so that  $(B+1)p_B = R_0$ . We show  $(T_0, \dots, T_B)$  is exchangeable. Since  $\mathbf{Y}$  is independent of  $(\mathbf{P}_1, \dots, \mathbf{P}_B)$  and  $\mathbf{\Gamma Y} \stackrel{d}{=} \mathbf{Y}$  with  $\mathbf{\Gamma Y}$  also independent of them,  $(T_0, \dots, T_B)$  has the distribution of  $(T(\mathbf{\Gamma Y}), T(\mathbf{P}_1 \mathbf{\Gamma Y}), \dots, T(\mathbf{P}_B \mathbf{\Gamma Y}))$ . The map  $(\mathbf{G}_0, \mathbf{G}_1, \dots, \mathbf{G}_B) \mapsto (\mathbf{G}_0, \mathbf{G}_1 \mathbf{G}_0, \dots, \mathbf{G}_B \mathbf{G}_0)$  is a bijection of  $\mathcal{G}^{B+1}$ , so  $\mathbf{\Gamma}, \mathbf{P}_1 \mathbf{\Gamma}, \dots, \mathbf{P}_B \mathbf{\Gamma}$  are independent and uniform on  $\mathcal{G}$ , and independent of  $\mathbf{Y}$ . Given  $\mathbf{Y}$ , the  $B+1$  values are therefore independent and identically distributed, hence exchangeable, and so they are unconditionally. The second part of Lemma 23.4.1 with  $k = \lfloor \alpha(B+1) \rfloor$  completes the proof. ■

Nothing about  $T$  was used: the statistic affects power, never validity. The full-group test has size exactly  $\alpha$  when  $\alpha|\mathcal{G}|$  is an integer and the values  $T(\mathbf{P Y})$  are almost surely distinct. In the Monte Carlo test a draw can repeat the identity or another draw, and the resulting ties make  $R_0$  not exactly uniform, so its size is at most  $\alpha$  and approaches  $\alpha$  as  $|\mathcal{G}|$  grows, when  $\alpha(B+1)$  is an integer. And the theorem says nothing about  $\beta$ . The hypothesis it tests is the invariance, and translating a regression hypothesis into an invariance is the whole difficulty.

## 23.4.2 Simple regression

In the straight-line model with fixed  $x_i$  and independent, identically distributed errors,  $\beta_1 = 0$  says  $\mathbf{Y} = \beta_0 \mathbf{1} + \boldsymbol{\varepsilon}$ , which is exchangeable, so a permutation test of  $\beta_1 = 0$  is exact for any error law. The usual statistics give the same test.

### Proposition 23.4.3 — Equivalent statistics for the slope

With  $\mathcal{G}$  the group of all permutations, the statistics  $|\hat{\beta}_1|$ ,  $|r|$ ,  $|t|$  and  $F = t^2$  for the simple regression of  $\mathbf{y}$  on  $\mathbf{x}$  give the same permutation p-value, for every data set and, in the Monte Carlo version, for every draw of permutations.

*Proof.* Permuting  $\mathbf{y}$  leaves  $\bar{y}$  and  $S_{yy} = \sum_i (y_i - \bar{y})^2$  unchanged, and  $S_{xx}$  does not involve  $\mathbf{y}$ . Now  $\hat{\beta}_1 = S_{xy}/S_{xx}$  and  $r = S_{xy}/\sqrt{S_{xx}S_{yy}}$ , so  $|\hat{\beta}_1|$  and  $|r|$  are the same increasing function of  $|S_{xy}|$  for every rearrangement. And  $t^2 = (n-2)r^2/(1-r^2)$ , which increases with  $r^2$  on  $[0, 1)$ . So all four order the arrangements  $\mathbf{P y}$  in the same way, and the counts in Definition 23.4.1 agree. ■

### Example. (Acid concentration and stack loss)

Brownlee's stack loss data (Example (Stack loss)) record 21 days of operation of a plant oxidizing ammonia. Regress stack loss on the acid concentration of the absorbing liquid alone. The slope is 0.7590, with  $t = 1.901$  and a two-sided  $t$ -test p-value of 0.0725. The  $5.11 \times 10^{19}$  arrangements are too many to enumerate; Monte Carlo with  $B = 19999$  permutations gives the p-value 0.0698, with Monte Carlo standard error 0.0018. Figure

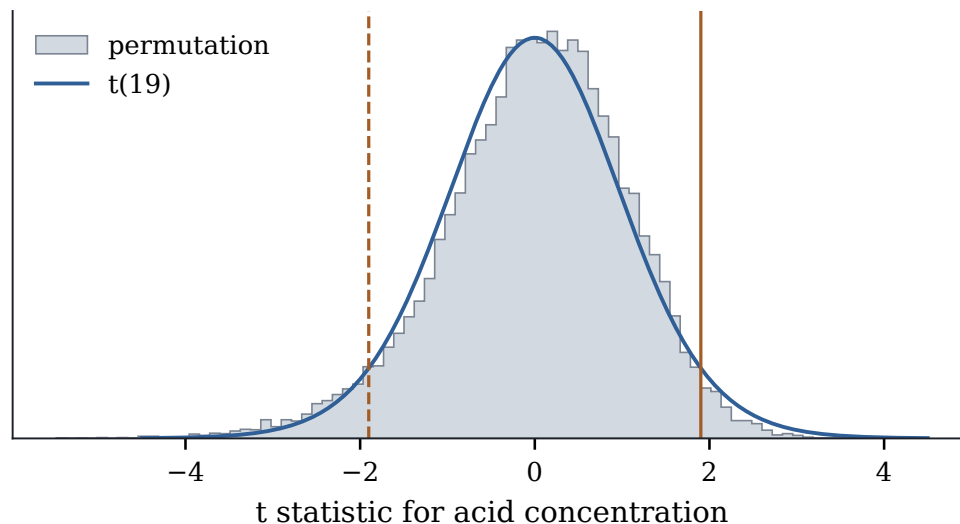


Figure 23.4.1: Permutation distribution of  $t$  for stack loss on acid concentration, with the  $t(19)$  density and the observed  $\pm t$ .

23.4.1 shows the permutation distribution of  $t$  with the  $t(19)$  density. The agreement is close, but the permutation p-value assumes nothing about the error law.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.stackloss.load_pandas().data

rng = np.random.default_rng(2304)
y = data["STACKLOSS"].to_numpy()
x = data["ACIDCONC"].to_numpy()
n = len(y)
xc = x - x.mean()

def slope_stats(Y):
    """Slope, correlation and t statistic for each row of Y, regressed on the fixed
    x."""
    Yc = Y - Y.mean(axis=-1, keepdims=True)
    b = Yc @ xc / (xc @ xc)
    r = Yc @ xc / np.sqrt((xc @ xc) * np.sum(Yc ** 2, axis=-1))
    t = r * np.sqrt(n - 2) / np.sqrt(1 - r ** 2)
    return b, r, t

b_obs, r_obs, t_obs = slope_stats(y)
p_t = 2 * stats.t.sf(abs(t_obs), n - 2)

B = 19_999
Yperm = y[np.argsort(rng.random((B, n)), axis=1)] # B random permutations of y
b_perm, r_perm, t_perm = slope_stats(Yperm)
p_perm = (1 + np.sum(np.abs(b_perm) >= abs(b_obs))) / (B + 1)
print(f"slope {b_obs:.4f}, t = {t_obs:.3f}, t-test p = {p_t:.4f}, permutation p = {
    p_perm:.4f}")
```

### 23.4.3 What the permutation test tests

Without identically distributed errors, invariance and  $\beta_1 = 0$  come apart. If independent cases have  $E(Y | X) = 0$  but  $\text{Var}(Y | X)$  depending on  $X$ , the slope is zero but the responses are not exchangeable given the  $x$ 's. The permutation test tests independence, and rejects too often as a test of zero slope. By the central limit theorem for the sample covariance, when  $\text{Cov}(X, Y) = 0$ ,  $\sqrt{n}r \rightarrow N(0, \kappa^2)$  with

$$\kappa^2 = \frac{E[(X - \mu_X)^2(Y - \mu_Y)^2]}{\text{Var}(X) \text{Var}(Y)},$$

which equals 1 under independence but not in general. The permutation distribution of  $\sqrt{n}r$  tends to  $N(0, 1)$  under mild conditions whatever the dependence, because permuting creates independence; we do not prove this permutation central limit theorem (see Lehmann and Romano 2005, chapter 15). Let  $Y = X\varepsilon$ ,  $X$  and  $\varepsilon$  independent standard normals. Then  $\text{Var}(Y) = 1$  and  $E(X^2Y^2) = E X^4 E \varepsilon^2 = 3$ , so  $\kappa^2 = 3$ . A nominal 5% permutation test based on  $|r|$ , or by [Proposition 23.4.3](#) on  $|\hat{\beta}_1|$ , has limiting size

$$\Pr(|N(0, 3)| > 1.96) = 2(1 - \Phi(1.96/\sqrt{3})) = 2(1 - \Phi(1.1316)) = 0.258.$$

A simulation with 4000 data sets and 399 permutations each gives 0.236 at  $n = 20$  and 0.248 at  $n = 200$ .

The remedy (DiCiccio and Romano 2017) is to studentize:  $\hat{\beta}_1/\widehat{\text{se}}_{\text{HC0}}$  has a standard normal limit under zero slope whether or not  $X$  and  $Y$  are independent ([Exercise \(C1\)](#)), and so does its permutation distribution. The test is exact under independence and asymptotically valid for zero slope. In the simulation its size is 0.117 at  $n = 20$  and 0.056 at  $n = 200$ : much better, though at  $n = 20$  still far from exact.

```
import numpy as np

def perm_sizes(n, reps, B, rng, alpha=0.05):
    """Rejection rates of permutation tests of zero slope when y = x * eps (x, eps
    iid normal)."""
    reject = {"raw": 0, "studentized": 0}
    for _ in range(reps):
        x = rng.normal(size=n)
        y = x * rng.normal(size=n) # E(y | x) = 0, Var(y | x) = x^2
        Y = np.vstack([y, y[np.argsort(rng.random((B, n))), axis=1]]]) # row 0: the
        data
        xc = x - x.mean()
        Yc = Y - Y.mean(axis=1, keepdims=True)
        b = Yc @ xc / (xc @ xc)
        e = Yc - b[:, None] * xc
        se_hc0 = np.sqrt((e ** 2) @ xc ** 2) / (xc @ xc)
        for name, T in [("raw", np.abs(b)), ("studentized", np.abs(b / se_hc0))]:
            p = np.mean(T >= T[0]) # (1 + #{T_b >= T_0}) / (B + 1)
            reject[name] += p <= alpha
    return {k: v / reps for k, v in reject.items()}

rng = np.random.default_rng(2305)
for n_sim in (20, 200):
    print(n_sim, perm_sizes(n_sim, reps=400, B=199, rng=rng))
```



## 23.4.4 Nuisance regressors

Most regression hypotheses concern some coefficients in the presence of others:

$$\mathbf{Y} = \mathbf{Z}\boldsymbol{\gamma} + \mathbf{X}_1\boldsymbol{\beta}_1 + \boldsymbol{\varepsilon}, \quad H_0 : \boldsymbol{\beta}_1 = \mathbf{0}, \quad (23.4.2)$$

with  $\mathbf{Z}$  of size  $n \times p_0$  containing the nuisance regressors (including the intercept) and  $\mathbf{X}_1$  of size  $n \times q$ . Under  $H_0$ ,  $\mathbf{Y} = \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon}$ , which is not exchangeable unless  $\mathbf{Z}\boldsymbol{\gamma}$  is a multiple of  $\mathbf{1}$ . There is an exact test only when some group of permutations leaves the nuisance part alone.

### Proposition 23.4.4 — Permutations that fix the nuisance regressors

Suppose  $\boldsymbol{\varepsilon}$  is  $\mathcal{G}$ -invariant in distribution, and  $\mathbf{P}\mathbf{Z} = \mathbf{Z}$  for every  $\mathbf{P} \in \mathcal{G}$ . Then under  $H_0$  of Eq. 23.4.2 the response  $\mathbf{Y}$  is  $\mathcal{G}$ -invariant in distribution, and the permutation test over  $\mathcal{G}$  with any statistic is exact, whatever the value of  $\boldsymbol{\gamma}$ .

*Proof.*  $\mathbf{P}\mathbf{Y} = \mathbf{P}\mathbf{Z}\boldsymbol{\gamma} + \mathbf{P}\boldsymbol{\varepsilon} = \mathbf{Z}\boldsymbol{\gamma} + \mathbf{P}\boldsymbol{\varepsilon} \stackrel{d}{=} \mathbf{Z}\boldsymbol{\gamma} + \boldsymbol{\varepsilon} = \mathbf{Y}$ . Apply Theorem 23.4.2. ■

The condition  $\mathbf{P}\mathbf{Z} = \mathbf{Z}$  says that  $\mathbf{P}$  permutes only cases with identical rows of  $\mathbf{Z}$ . For block indicators,  $\mathcal{G}$  is the group of permutations within blocks, and the proposition gives, under errors exchangeable within blocks, the same test as the design-based randomization test of a randomized complete block design (Section 18.4, Definition 18.4.1), which needs no error model at all. When  $\mathbf{Z}$  contains a continuous covariate with distinct values only the identity qualifies, and no exact permutation test exists. The schemes of the next section give up exactness for generality.

## 23.4.5 Exercises

### 23.4.5.1 A. Check your understanding

#### Exercise A1.

With  $B = 99$  random permutations, what is the smallest possible value of Eq. 23.4.1? For which  $\alpha$  between 0.01 and 0.10 is the size of the Monte Carlo test exactly  $\alpha$  when there are no ties (ignoring the chance of drawing the same arrangement twice)?

*Solution.* The smallest value is  $1/100$ , reached when no permuted statistic reaches the observed one. When there are no ties (ignoring the chance of drawing the same arrangement twice),  $R_0$  is uniform on  $\{1, \dots, 100\}$ , so  $\Pr(p_B \leq \alpha) = \lfloor 100\alpha \rfloor / 100$ , which equals  $\alpha$  exactly when  $100\alpha$  is an integer:  $\alpha = 0.01, 0.02, \dots, 0.10$ .

### 23.4.5.2 B. Practice

**Exercise B1.**

For  $x = (1, 2, 3, 4)$  and  $y = (1.2, 0.8, 2.9, 3.1)$ , enumerate all 24 arrangements of  $y$  and compute the exact two-sided permutation p-value for the slope. Compare it with the  $t$ -test p-value, and explain why the permutation p-value cannot be smaller than  $2/24$  for any data with  $n = 4$  and distinct values.

*Solution.* By [Proposition 23.4.3](#) it suffices to rank the arrangements by  $|S_{xy}|$ , where  $S_{xy} = \sum_i (x_i - \bar{x})y_{\pi(i)}$  with centred  $x$  equal to  $(-1.5, -0.5, 0.5, 1.5)$ . For the data,  $S_{xy} = -1.8 - 0.4 + 1.45 + 4.65 = 3.9$ . The largest values of  $S_{xy}$  come from putting the two largest responses at  $x = 3, 4$  and the two smallest at  $x = 1, 2$ ; the four such arrangements give 3.7, 3.9, 4.1 and 4.3, and every other arrangement gives at most 2.6. Reversing an arrangement negates  $S_{xy}$ . So 6 of the 24 arrangements have  $|S_{xy}| \geq 3.9$ , and the p-value is  $6/24 = 0.25$ . The  $t$ -test gives 0.139: with four points the permutation distribution is coarse. The observed arrangement and its reversal always count, so  $p \geq 2/24$ .

**Exercise B2.**

Compute  $\kappa^2$  when  $Y = \varepsilon\sqrt{1 + \lambda X^2}$  with  $X$  and  $\varepsilon$  independent standard normals, and the limiting size of the nominal 5% permutation test of zero slope as a function of  $\lambda$ . What happens as  $\lambda \rightarrow \infty$ ?

**23.4.5.3 C. Going deeper****Exercise C1.**

For independent pairs with  $\text{Cov}(X, Y) = 0$  and finite fourth moments, show that  $\hat{\beta}_1/\widehat{\text{se}}_{\text{HC0}} \rightarrow N(0, 1)$  in distribution. (Use the central limit theorem for  $n^{-1/2} \sum_i (X_i - \mu_X)(Y_i - \mu_Y)$  and Slutsky's lemma.)

## 23.5 Freedman–Lane and related schemes

With a continuous nuisance regressor in [Eq. 23.4.2](#), no permutation of the responses leaves the null distribution unchanged. Several schemes permute something else, or permute the responses and hope. None is exact in general, but their differences can be understood with the projection algebra of [Chapter 6](#).

### 23.5.1 Five schemes

Keep the notation of [Eq. 23.4.2](#):  $\mathbf{Z}$  is  $n \times p_0$  with  $\mathbf{1} \in \mathcal{C}(\mathbf{Z})$ ,  $\mathbf{X}_1$  is  $n \times q$ , the full model matrix  $[\mathbf{Z}, \mathbf{X}_1]$  has rank  $p = p_0 + q < n$ ,  $\mathbf{M}_0$  projects onto  $\mathcal{C}(\mathbf{Z})$  and  $\mathbf{M}$  onto  $\mathcal{C}([\mathbf{Z}, \mathbf{X}_1])$ . The statistic for  $H_0 : \beta_1 = \mathbf{0}$  is the  $F$  statistic of [Theorem 11.1.1](#), written as a function of the response vector:

$$F(\mathbf{u}) = \frac{\|(\mathbf{M} - \mathbf{M}_0)\mathbf{u}\|^2/q}{\|(\mathbf{I} - \mathbf{M})\mathbf{u}\|^2/(n-p)}. \quad (23.5.1)$$

Since  $(\mathbf{M} - \mathbf{M}_0)\mathbf{Z} = \mathbf{0}$  and  $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$ ,  $F(\mathbf{u} + \mathbf{Zg}) = F(\mathbf{u})$  for every  $\mathbf{g}$ . Let

$$\hat{\boldsymbol{\varepsilon}}_0 = (\mathbf{I} - \mathbf{M}_0)\mathbf{y}, \quad \hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$$

be the residuals of the reduced and the full model. By [Theorem 6.6.2](#),  $\mathbf{M} - \mathbf{M}_0$  projects onto  $\mathcal{C}(\tilde{\mathbf{X}}_1)$  with  $\tilde{\mathbf{X}}_1 = (\mathbf{I} - \mathbf{M}_0)\mathbf{X}_1$ .

### Definition 23.5.1 — Permutation schemes for a partial hypothesis

For a permutation matrix  $\mathbf{P}$ , each scheme computes a permuted statistic; its p-value is [Eq. 23.4.1](#) with  $T(\mathbf{y})$  the observed  $F(\mathbf{y})$  and  $T(\mathbf{P}_b\mathbf{y})$  replaced by the scheme's permuted statistic.

- (a) **Raw permutation** (Manly 1997):  $F(\mathbf{P}\mathbf{y})$ , refitting the full model to permuted responses.
- (b) **Permuting the regressors of interest** (Draper and Stoneman 1966): the  $F$  statistic computed from  $\mathbf{y}$  with model matrix  $[\mathbf{Z}, \mathbf{P}\mathbf{X}_1]$ .
- (c) **Kennedy's scheme** (Kennedy 1995): regress  $\mathbf{P}\hat{\boldsymbol{\varepsilon}}_0$  on  $\tilde{\mathbf{X}}_1$  alone and compute  $F_K(\mathbf{P}\hat{\boldsymbol{\varepsilon}}_0)$ , where  $F_K(\mathbf{u}) = \frac{\|(\mathbf{M} - \mathbf{M}_0)\mathbf{u}\|^2/q}{(\|\mathbf{u}\|^2 - \|(\mathbf{M} - \mathbf{M}_0)\mathbf{u}\|^2)/(n-p)}$ .
- (d) **Freedman–Lane** (Freedman and Lane 1983): form  $\mathbf{y}^* = \mathbf{M}_0\mathbf{y} + \mathbf{P}\hat{\boldsymbol{\varepsilon}}_0$ , the reduced-model fit plus permuted reduced-model residuals, and compute  $F(\mathbf{y}^*)$ .
- (e) **ter Braak's scheme** (ter Braak 1992): form  $\mathbf{y}^* = \mathbf{M}\mathbf{y} + \mathbf{P}\hat{\boldsymbol{\varepsilon}}$ , the full-model fit plus permuted full-model residuals, and compute the  $F$  statistic for the hypothesis  $\beta_1 = \hat{\beta}_1$ , which is true in the permuted world.

## 23.5.2 What can be proved

### Proposition 23.5.1 — The Freedman–Lane scheme

In the setting above:

- (a)  $F(\mathbf{M}_0\mathbf{y} + \mathbf{P}\hat{\boldsymbol{\varepsilon}}_0) = F(\mathbf{P}\hat{\boldsymbol{\varepsilon}}_0)$  for every  $\mathbf{P}$ , and  $F(\mathbf{y}) = F(\hat{\boldsymbol{\varepsilon}}_0)$ . ter Braak's permuted statistic equals  $F(\mathbf{P}\hat{\boldsymbol{\varepsilon}})$ .
- (b) Under  $H_0$ ,  $\hat{\boldsymbol{\varepsilon}}_0 = (\mathbf{I} - \mathbf{M}_0)\boldsymbol{\varepsilon}$ . The p-values of the Freedman–Lane, ter Braak, Kennedy and Draper–Stoneman schemes depend on  $\mathbf{y}$  only through  $(\mathbf{I} - \mathbf{M}_0)\mathbf{y}$ , so under  $H_0$  their distributions do not depend on  $\boldsymbol{\gamma}$ . The raw permutation statistic involves  $\boldsymbol{\gamma}$ .
- (c) Under  $H_0$ , if  $\mathbf{P}\mathbf{Z} = \mathbf{Z}$  for every  $\mathbf{P}$  in a group  $\mathcal{G}$  and  $\boldsymbol{\varepsilon}$  is  $\mathcal{G}$ -invariant in distribution, then the Freedman–Lane test over  $\mathcal{G}$  is exact, and it coincides with the raw permutation test over  $\mathcal{G}$ .
- (d) Suppose the errors are independent draws from one law with mean zero and variance  $\sigma^2 > 0$ ,  $\mathbf{P}$  is uniform over all permutations and independent of  $\boldsymbol{\varepsilon}$ , and  $p$  is fixed as  $n \rightarrow \infty$ . Then under  $H_0$ ,  $F(\mathbf{P}\hat{\boldsymbol{\varepsilon}}_0) - F(\mathbf{P}\boldsymbol{\varepsilon}) \rightarrow 0$  in probability, and so does  $F(\mathbf{P}\hat{\boldsymbol{\varepsilon}}) - F(\mathbf{P}\boldsymbol{\varepsilon})$ .

*Proof.*

- (a)  $\mathbf{M}_0\mathbf{y} \in \mathcal{C}(\mathbf{Z})$ , so the invariance of Eq. 23.5.1 gives the first claim, and  $\mathbf{y} = \mathbf{M}_0\mathbf{y} + \hat{\boldsymbol{\varepsilon}}_0$  gives the second. In ter Braak's scheme, the  $F$  statistic for  $\boldsymbol{\beta}_1 = \hat{\boldsymbol{\beta}}_1$  computed from  $\mathbf{y}^*$  is the  $F$  statistic for  $\boldsymbol{\beta}_1 = \mathbf{0}$  computed from  $\mathbf{y}^* - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1$ , whose reduced model is the set of means  $\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 + \mathcal{C}(\mathbf{Z})$ : the numerators agree by Theorem 11.3.2(b), and the denominators agree because subtracting  $\mathbf{X}_1\hat{\boldsymbol{\beta}}_1 \in \mathcal{C}(\mathbf{X})$  leaves the residuals unchanged. Since  $\mathbf{y}^* - \mathbf{X}_1\hat{\boldsymbol{\beta}}_1 = \mathbf{Z}\boldsymbol{\gamma} + \mathbf{P}\hat{\boldsymbol{\varepsilon}}$  with  $\mathbf{Z}\boldsymbol{\gamma} \in \mathcal{C}(\mathbf{Z})$ , the statistic is  $F(\mathbf{P}\hat{\boldsymbol{\varepsilon}})$ .
- (b) Under  $H_0$ ,  $(\mathbf{I} - \mathbf{M}_0)\mathbf{y} = (\mathbf{I} - \mathbf{M}_0)\boldsymbol{\varepsilon}$ . Freedman–Lane and Kennedy use  $\mathbf{y}$  only through  $\hat{\boldsymbol{\varepsilon}}_0$ , and ter Braak through  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\hat{\boldsymbol{\varepsilon}}_0$ . The Draper–Stoneman statistic is unchanged when a vector in  $\mathcal{C}(\mathbf{Z})$  is added to  $\mathbf{y}$ . In each case the observed statistic is  $F(\mathbf{y}) = F(\hat{\boldsymbol{\varepsilon}}_0)$  by (a). The raw permutation statistic  $F(\mathbf{P}\mathbf{y}) = F(\mathbf{P}\mathbf{Z}\boldsymbol{\gamma} + \mathbf{P}\boldsymbol{\varepsilon})$  involves  $\boldsymbol{\gamma}$  whenever  $\mathbf{P}\mathbf{Z}\boldsymbol{\gamma} \notin \mathcal{C}(\mathbf{Z})$ , and its null distribution in general depends on  $\boldsymbol{\gamma}$ , as Figure 23.5.1 shows; we do not prove this.
- (c)  $\mathbf{P}\mathbf{Z} = \mathbf{Z}$  implies  $\mathbf{P}\mathbf{M}_0\mathbf{P}^\top = \mathbf{M}_0$ , since  $\mathbf{P}\mathbf{M}_0\mathbf{P}^\top$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{P}\mathbf{Z}) = \mathcal{C}(\mathbf{Z})$ . So  $\mathbf{P}\hat{\boldsymbol{\varepsilon}}_0 = \mathbf{P}(\mathbf{I} - \mathbf{M}_0)\boldsymbol{\varepsilon} = (\mathbf{I} - \mathbf{M}_0)\mathbf{P}\boldsymbol{\varepsilon}$  under  $H_0$ , and  $F(\mathbf{P}\hat{\boldsymbol{\varepsilon}}_0) = F(\mathbf{P}\boldsymbol{\varepsilon})$  by invariance. Also  $F(\mathbf{y}) = F(\boldsymbol{\varepsilon})$ . So the Freedman–Lane p-value is the permutation p-value of the statistic  $F$  applied to  $\boldsymbol{\varepsilon}$ , which is exact by Theorem 23.4.2. The raw permutation statistic is  $F(\mathbf{Z}\boldsymbol{\gamma} + \mathbf{P}\boldsymbol{\varepsilon}) = F(\mathbf{P}\boldsymbol{\varepsilon})$  as well.
- (d) Put  $\mathbf{v} = \mathbf{M}_0\boldsymbol{\varepsilon}$ , so  $\mathbf{P}\hat{\boldsymbol{\varepsilon}}_0 = \mathbf{P}\boldsymbol{\varepsilon} - \mathbf{P}\mathbf{v}$ . For a fixed vector  $\mathbf{v}$  and a uniform random permutation,  $E(\mathbf{P}\mathbf{v}) = \bar{v}\mathbf{1}$  and  $\text{Cov}(\mathbf{P}\mathbf{v}) = \frac{S_v}{n-1}(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)$ , where  $S_v = \sum_i (v_i - \bar{v})^2 \leq \|\mathbf{v}\|^2$  (Exercise (B1)). Since  $(\mathbf{M} - \mathbf{M}_0)\mathbf{1} = \mathbf{0}$ ,

$$\begin{aligned} E[\|(\mathbf{M} - \mathbf{M}_0)\mathbf{P}\mathbf{v}\|^2 \mid \boldsymbol{\varepsilon}] &= \frac{S_v}{n-1} \text{tr}(\mathbf{M} - \mathbf{M}_0) \\ &\leq \frac{q \|\mathbf{M}_0\boldsymbol{\varepsilon}\|^2}{n-1}, \end{aligned}$$

and  $E\|\mathbf{M}_0\boldsymbol{\varepsilon}\|^2 = \sigma^2 p_0$ , so  $\|(\mathbf{M} - \mathbf{M}_0)\mathbf{P}\mathbf{v}\| \rightarrow 0$  in probability. The numerators of the two statistics are  $\|\mathbf{N}\|^2/q$  and  $\|\mathbf{N} - \boldsymbol{\delta}\|^2/q$  with  $\mathbf{N} = (\mathbf{M} - \mathbf{M}_0)\mathbf{P}\boldsymbol{\varepsilon}$  and  $\boldsymbol{\delta} = (\mathbf{M} - \mathbf{M}_0)\mathbf{P}\mathbf{v}$ . Here  $\mathbf{P}\boldsymbol{\varepsilon}$  has the distribution of  $\boldsymbol{\varepsilon}$ , so  $E\|\mathbf{N}\|^2 = q\sigma^2$ , and  $|\|\mathbf{N} - \boldsymbol{\delta}\|^2 - \|\mathbf{N}\|^2| \leq 2\|\mathbf{N}\|\|\boldsymbol{\delta}\| + \|\boldsymbol{\delta}\|^2 \rightarrow 0$  in probability. For the denominators,  $\|(\mathbf{I} - \mathbf{M})\mathbf{P}\hat{\boldsymbol{\varepsilon}}_0\| - \|(\mathbf{I} - \mathbf{M})\mathbf{P}\boldsymbol{\varepsilon}\| \leq \|\mathbf{v}\| = O_p(1)$ , while  $\|(\mathbf{I} - \mathbf{M})\mathbf{P}\boldsymbol{\varepsilon}\|^2/(n-p)$  has the distribution of  $s^2$  and tends to  $\sigma^2$  in probability (Corollary 5.6.4 needs fourth moments; the argument of Lemma 23.1.3 needs only the second). Both norms are  $O_p(\sqrt{n})$ , so the difference of their squares is at most  $\|\mathbf{v}\|$  times  $O_p(\sqrt{n})$ . Dividing by  $n-p$ , the two denominators differ by a quantity that tends to zero, and both tend to  $\sigma^2$ . The claim for  $F(\mathbf{P}\hat{\boldsymbol{\varepsilon}})$  is the same argument with  $\mathbf{v} = \mathbf{M}\boldsymbol{\varepsilon}$ ,  $E\|\mathbf{M}\boldsymbol{\varepsilon}\|^2 = \sigma^2 p$ . ■

Part (d) compares Freedman–Lane with an **oracle** that permutes the true errors. The oracle is exact:  $F(\mathbf{y}) = F(\boldsymbol{\varepsilon})$  under  $H_0$ , and it compares  $F(\boldsymbol{\varepsilon})$  with  $F(\mathbf{P}_b\boldsymbol{\varepsilon})$ , a permutation test for the exchangeable  $\boldsymbol{\varepsilon}$ . Freedman–Lane replaces  $\mathbf{P}_b\boldsymbol{\varepsilon}$  by  $\mathbf{P}_b\hat{\boldsymbol{\varepsilon}}_0$ , which changes the statistics by vanishing amounts. To conclude that its size tends to  $\alpha$ , one needs also that the oracle statistics have a continuous joint limit, so that vanishing perturbations rarely change their ranks. That holds under conditions like Eq. 23.1.1 by a permutation central limit theorem that we do not prove; Anderson and Robinson (2001) give the argument.

Part (b) is the practical point: four of the schemes see the data only after  $\mathbf{Z}$  has been projected

out, while for large  $\mathbf{Z}\boldsymbol{\gamma}$  the raw permutation mostly permutes  $\mathbf{Z}\boldsymbol{\gamma}$ , not the errors. Kennedy's scheme can be pinned down exactly.

### Proposition 23.5.2 — Kennedy's scheme is anticonservative relative to Freedman–Lane

For every permutation  $\mathbf{P}$ ,  $F_K(\mathbf{P}\hat{\boldsymbol{\epsilon}}_0) \leq F(\mathbf{P}\hat{\boldsymbol{\epsilon}}_0)$ , with equality when  $\mathbf{P} = \mathbf{I}$ . Consequently, for any set of permutations  $\mathbf{P}_1, \dots, \mathbf{P}_B$ , the Kennedy p-value is at most the Freedman–Lane p-value.

*Proof.* The two statistics have the same numerator. Write  $\mathbf{u} = \mathbf{P}\hat{\boldsymbol{\epsilon}}_0$ . Since  $\mathbf{I} = \mathbf{M}_0 + (\mathbf{M} - \mathbf{M}_0) + (\mathbf{I} - \mathbf{M})$  with orthogonal ranges,

$$\begin{aligned}\|(\mathbf{I} - \mathbf{M})\mathbf{u}\|^2 &= \|\mathbf{u}\|^2 - \|(\mathbf{M} - \mathbf{M}_0)\mathbf{u}\|^2 - \|\mathbf{M}_0\mathbf{u}\|^2 \\ &\leq \|\mathbf{u}\|^2 - \|(\mathbf{M} - \mathbf{M}_0)\mathbf{u}\|^2,\end{aligned}$$

so Kennedy's denominator is at least as large. For  $\mathbf{P} = \mathbf{I}$ ,  $\mathbf{M}_0\hat{\boldsymbol{\epsilon}}_0 = \mathbf{0}$ , and the two agree. Each permuted Kennedy statistic is therefore at most the corresponding Freedman–Lane one, while the observed statistics coincide, so fewer permuted values reach the observed one. ■

The discrepancy  $\|\mathbf{M}_0\mathbf{P}\hat{\boldsymbol{\epsilon}}_0\|^2$ , the part of the permuted residuals that  $\mathbf{Z}$  could have explained, has expectation about  $(p_0 - 1)\sigma^2$ , small beside  $(n - p)\sigma^2$  for large  $n$ ; so Kennedy's scheme is liberal only in small samples.

## 23.5.3 Stack loss again, and a simulation

### Example. (Acid concentration given air flow and water temperature)

In the full stack loss regression, test whether acid concentration matters once air flow and water temperature are in the model. The  $F$  statistic is 0.947 on 1 and 17 degrees of freedom, with p-value 0.344. With 19999 permutations each, the schemes give: raw permutation 0.346, Draper–Stoneman 0.343, Kennedy 0.315, Freedman–Lane 0.345 and ter Braak 0.345. Only Kennedy's differs noticeably, in the direction [Proposition 23.5.2](#) predicts. The marginal association of [Example \(Acid concentration and stack loss\)](#) disappears once the correlated regressors are in the model.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats
data = sm.datasets.stackloss.load_pandas().data

def proj(Z):
    """Orthogonal projection onto C(Z), from a QR factorization."""
    Q, _ = np.linalg.qr(Z)
    return Q @ Q.T

def perm_rows(v, B, rng):
```

```

"""B random permutations of the vector v, one per row."""
return v[np.argsort(rng.random((B, len(v))), axis=1)]

def perm_pvalues(y, x, Z, B, rng):
    """p-values for H0: beta = 0 in E(y) = Z gamma + x beta, by six methods."""
    n, p = len(y), Z.shape[1] + 1
    MZ = proj(Z)
    xt = x - MZ @ x # x residualized on Z
    yt = y - MZ @ y # reduced-model residuals
    e_full = yt - xt * (xt @ yt) / (xt @ xt) # full-model residuals (FWL)

    def F(Y):
        """F statistic for beta = 0, for each row of Y (model matrix [Z, x] fixed)."""
        num = (Y @ xt) ** 2 / (xt @ xt)
        rss_Z = np.sum(Y ** 2, axis=-1) - np.sum((Y @ MZ) * Y, axis=-1)
        return num / ((rss_Z - num) / (n - p))

    F_obs = F(y)
    out = {"F test": stats.f.sf(F_obs, 1, n - p)}

    def pval(F_star, F0=F_obs):
        return (1 + np.sum(F_star >= F0)) / (len(F_star) + 1)

    out["Manly"] = pval(F(perm_rows(y, B, rng))) # permute y
    Xs = perm_rows(x, B, rng) # permute x (
    Draper-Stoneman)
    xts = Xs - Xs @ MZ
    num = (xts @ y) ** 2 / np.sum(xts ** 2, axis=1)
    out["Draper-Stoneman"] = pval(num / ((yt @ yt - num) / (n - p)))
    Yk = perm_rows(yt, B, rng) # Kennedy: no Z
    in the refit
    num = (Yk @ xt) ** 2 / (xt @ xt)
    out["Kennedy"] = pval(num / ((np.sum(Yk ** 2, axis=1) - num) / (n - p)))
    out["Freedman-Lane"] = pval(F(perm_rows(yt, B, rng))) # permute
    reduced residuals
    out["ter Braak"] = pval(F(perm_rows(e_full, B, rng))) # permute full
    residuals
    return out, F_obs

rng = np.random.default_rng(2307)
y = data["STACKLOSS"].to_numpy()
x = data["ACIDCONC"].to_numpy()
Z = np.column_stack([np.ones(len(y)), data[["AIRFLOW", "WATERTEMP"]]])
pvals, F_obs = perm_pvalues(y, x, Z, B=19_999, rng=rng)
print(f"F = {F_obs:.3f}")
for m, pv in pvals.items():
    print(f"{m:16s} p = {pv:.4f}")

```

Real data with a p-value near 0.34 cannot separate the schemes. The simulation in [Figure 23.5.1](#) is designed to. It has  $n = 12$  cases, normal errors, a nuisance regressor  $z$  correlated with the regressor of interest  $x$ , and one high-leverage value of  $x$ . The null hypothesis holds, and the nuisance coefficient  $\gamma$  ranges from 0 to 30. Each point is the rejection rate of a nominal 5% test in 5000 data sets, with 199 permutations each, so the Monte Carlo standard error is about 0.0031.

The same random numbers are used for every  $\gamma$ , so by [Proposition 23.5.1\(b\)](#) five of the six tests

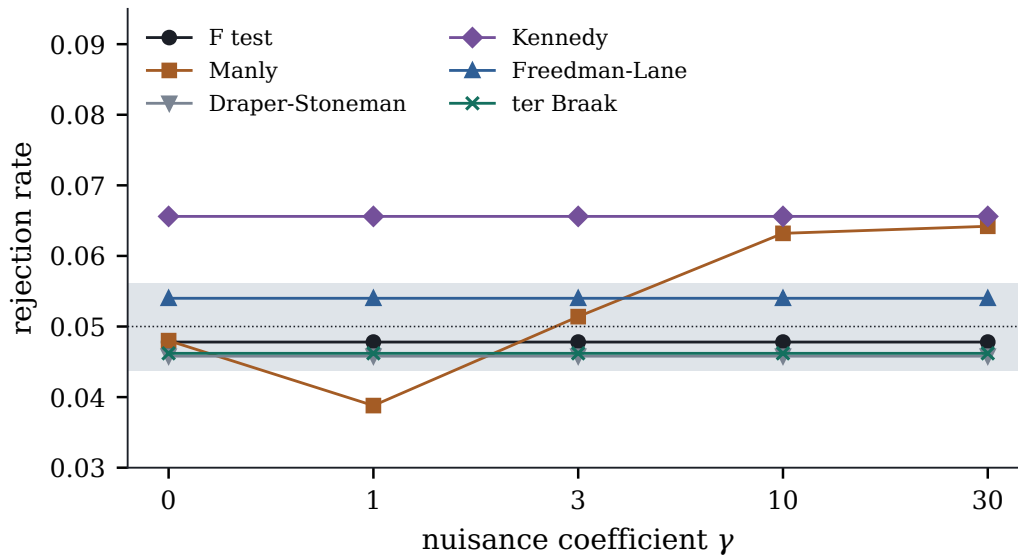


Figure 23.5.1: Null rejection rates of nominal 5% tests of one coefficient,  $n = 12$ , against the nuisance coefficient  $\gamma$ . The band is two Monte Carlo standard errors around 0.05.

make identical decisions at every  $\gamma$ , and their lines are flat. The  $F$  test rejects in 0.048 of the data sets, Freedman–Lane in 0.054, ter Braak and Draper–Stoneman in 0.046, and Kennedy in 0.066, the liberal bias of [Proposition 23.5.2](#) made visible by the small sample. The raw permutation test is exact at  $\gamma = 0$  (0.048), but its size moves with  $\gamma$ , reaching 0.064 at  $\gamma = 30$ .

```
def size_study(gamma, reps, B, seed, n=12, alpha=0.05):
    """Rejection rates at level alpha when beta = 0 and the nuisance coefficient is
    gamma."""
    rng = np.random.default_rng(seed)
    z = np.linspace(-1, 1, n)
    x = 0.8 * z + 0.6 * np.random.default_rng(1).normal(size=n) # fixed,
    # correlated with z
    x[0] = 6.0 # one high-leverage value of x
    Z = np.column_stack([np.ones(n), z])
    rejections = {}
    for _ in range(reps):
        y = 1.0 + gamma * z + rng.normal(size=n) # the null hypothesis beta = 0
        # holds
        for m, pv in perm_pvalues(y, x, Z, B, rng)[0].items():
            rejections[m] = rejections.get(m, 0) + (pv <= alpha)
    return {m: r / reps for m, r in rejections.items()}

for gamma in (0.0, 30.0):
    print(gamma, size_study(gamma, reps=300, B=99, seed=2308))
```

As Anderson and Legendre (1999) concluded from a much larger simulation, Freedman–Lane is the sensible default: exact whenever an exact test exists, blind to the nuisance coefficients, and close to the oracle in large samples. ter Braak’s scheme behaves similarly, and since its reference distribution does not need the null hypothesis, it suits the inversion of tests into intervals. Winkler and coauthors (2014) review the schemes for the general linear model.

## 23.5.4 Exercises

### 23.5.4.1 A. Check your understanding

#### Exercise A1.

Show that when  $\mathbf{Z} = \mathbf{1}$  the Freedman–Lane, Kennedy and raw permutation tests all coincide with the permutation test of [Section 23.4](#).

*Solution.* With  $\mathbf{Z} = \mathbf{1}$ , every permutation fixes  $\mathbf{Z}$ , so by [Proposition 23.5.1\(c\)](#) Freedman–Lane and the raw permutation coincide and are exact. Kennedy’s scheme differs only through  $\|\mathbf{M}_0 \mathbf{P} \hat{\mathbf{e}}_0\|^2 = n \overline{(\mathbf{P} \hat{\mathbf{e}}_0)^2}$ , which is zero because  $\hat{\mathbf{e}}_0 = \mathbf{y} - \bar{y} \mathbf{1}$  sums to zero and permuting does not change the sum.

#### Exercise A2.

In ter Braak’s scheme, what is the permuted statistic for  $\mathbf{P} = \mathbf{I}$ ? Explain why the observed statistic must still be counted in the numerator and denominator of [Eq. 23.4.1](#).

### 23.5.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{v} \in \mathbb{R}^n$  be fixed and  $\mathbf{P}$  a uniformly random permutation matrix. Show that  $E(\mathbf{P}\mathbf{v}) = \bar{v} \mathbf{1}$  and  $\text{Cov}(\mathbf{P}\mathbf{v}) = \frac{S_v}{n-1} (\mathbf{I} - n^{-1} \mathbf{1} \mathbf{1}^\top)$  with  $S_v = \sum_i (v_i - \bar{v})^2$ . (The computation is the one in the proof of [Proposition 18.4.2](#), there for permutations within a block.)

### 23.5.4.3 C. Going deeper

#### Exercise C1.

Suppose the errors are independent with variances that depend on  $\mathbf{X}_1$ . Explain why the schemes of [Definition 23.5.1](#) need not be asymptotically valid for  $H_0 : \boldsymbol{\beta}_1 = \mathbf{0}$ , and where the argument of [Proposition 23.5.1\(d\)](#) breaks down. Propose a studentized version in the spirit of [Section 23.4](#).

## 23.6 When resampling fails

The consistency theorems of this chapter assume exchangeable resampled quantities, a smooth statistic, a fixed number of parameters, a finite error variance, no dominant case, independence, and a sample large enough for limits to be relevant. When any of these fails, the bootstrap can fail silently, producing a distribution that looks reasonable and is wrong.



## 23.6.1 A statistic that is not smooth: the maximum

The cleanest failure is not specific to regression. Let  $Y_1, \dots, Y_n$  be independent and uniform on  $[0, \theta]$ , and estimate  $\theta$  by the sample maximum  $Y_{(n)}$ .

### Proposition 23.6.1 — The bootstrap of a maximum

- (a)  $n(\theta - Y_{(n)})/\theta$  converges in distribution to the standard exponential law.
- (b) For every sample with distinct values, the case bootstrap gives  $\Pr_*(Y_{(n)}^* = Y_{(n)}) = 1 - (1 - 1/n)^n \rightarrow 1 - e^{-1}$ .

So the bootstrap distribution of  $n(Y_{(n)} - Y_{(n)}^*)/Y_{(n)}$  has an atom at zero of mass at least  $1 - e^{-1}$  for every sample, and it cannot approach the continuous exponential law.

*Proof.*

- (a) For  $0 \leq t \leq n$ ,  $\Pr(n(\theta - Y_{(n)})/\theta > t) = \Pr(\text{all } Y_i < \theta(1 - t/n)) = (1 - t/n)^n \rightarrow e^{-t}$ .
- (b)  $Y_{(n)}^* = Y_{(n)}$  unless none of the  $n$  draws picks the largest observation, which happens with probability  $(1 - 1/n)^n$ . And  $(1 - 1/n)^n = \exp(n \log(1 - 1/n)) \rightarrow e^{-1}$ , from below, since  $n \log(1 - 1/n) < -1$ .

■

With  $n = 50$  the atom is  $1 - 0.98^{50} = 0.636$ , and a simulation with 20000 resamples finds 0.639. The bootstrap needs the statistic to depend smoothly on the empirical distribution, and extremes do not. The  $m$ -out-of- $n$  bootstrap, with resamples of size  $m$  and  $m/n \rightarrow 0$ , repairs many such failures at a price in efficiency (Bickel, Götze and van Zwet 1997). In regression the analogue is any statistic driven by the most extreme residual, such as the largest studentized residual of the outlier test (Theorem 20.3.3).

```
import numpy as np
rng = np.random.default_rng(2310)
n = 50
u = rng.uniform(size=n)
B = 20_000
boot_max = u[rng.integers(0, n, size=(B, n))].max(axis=1)
atom = np.mean(boot_max == u.max())
print(f"bootstrap P*(max* = max) = {atom:.3f}; 1 - (1 - 1/n)^n = {1 - (1 - 1 / n) ** n:.3f}")
```

## 23.6.2 Heavy tails

Without a finite error variance the sample mean, and any least squares coefficient, is not asymptotically normal. Athreya (1987) showed that the bootstrap of the mean then converges not to the correct stable limit but to a *random* limit that differs from sample to sample: resampling cannot learn a tail that shows up only through a few enormous values.

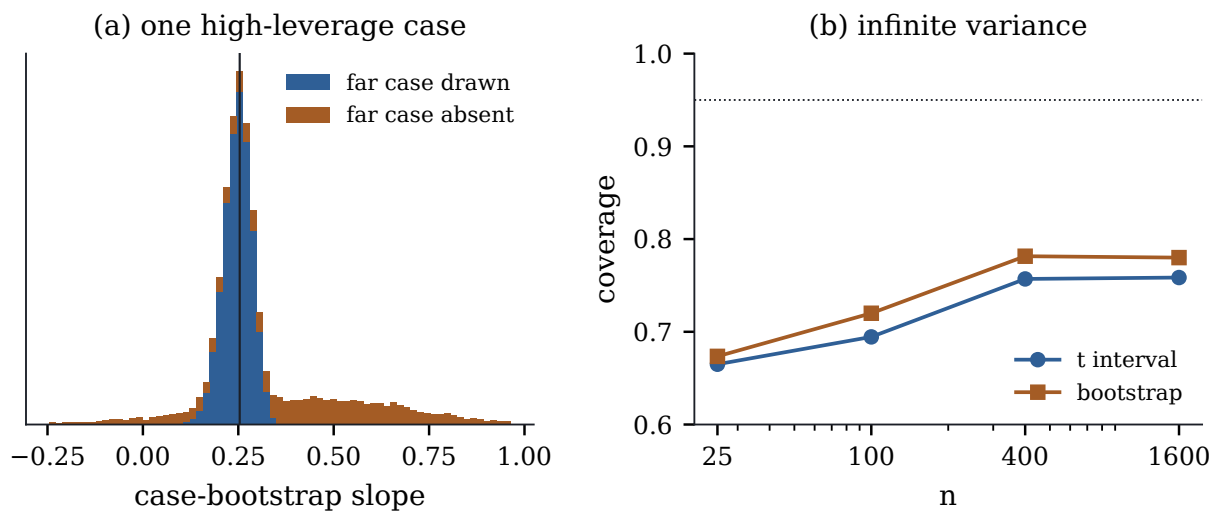


Figure 23.6.1: (a) Case-bootstrap slopes when one of 15 cases has leverage 0.902, split by whether the resample contains it. (b) Coverage of nominal 95% intervals for a Pareto mean with infinite variance.

Figure 23.6.1(b) shows the consequence for coverage. The data are samples from a Pareto law with tail index 1.5, which has a finite mean and an infinite variance, and the target is the mean. The nominal 95%  $t$  interval covers 0.665 of the time at  $n = 25$  and 0.758 at  $n = 1600$ ; the bootstrap percentile interval covers 0.673 and 0.780. Neither improves between  $n = 400$  and  $n = 1600$ . The same holds for regression coefficients, which is one reason to prefer estimators less sensitive to extreme errors, such as the least absolute deviations fit of Chapter 45, to resampling least squares.

### 23.6.3 One dominant case

When one case has leverage near one, Eq. 23.1.1 fails and the case bootstrap splits into resamples that contain the case and resamples, about a third of them, that do not.

#### Example. (A slope determined by one case)

Take 15 cases, fourteen with  $x$  spread over  $[0, 1]$  and one at  $x = 4$ , with leverage 0.902, and a response that puts the far case a little below the line of the others. The least squares slope is 0.254. Without the far case it would be 0.437. In 20000 case-bootstrap resamples, a fraction 0.356 omit the far case, close to  $(14/15)^{15} = 0.355$ . Figure 23.6.1(a) shows the result. The resamples that contain the far case give slopes tightly concentrated near the full-data value, with standard deviation 0.037. Those that omit it give slopes spread around the fourteen-case value, with standard deviation 0.274. The bootstrap distribution mixes two answers to two questions, and neither its spread nor its percentiles describe the uncertainty of  $\hat{\beta}_1$ .

```
import numpy as np
rng = np.random.default_rng(2313)
n_1 = 15
x_1 = np.append(np.linspace(0, 1, n_1 - 1), 4.0)    # one case far from the rest
X_1 = np.column_stack([np.ones(n_1), x_1])
h_1 = np.sum((X_1 @ np.linalg.inv(X_1.T @ X_1)) * X_1, axis=1)
```

```

y_1 = 1 + 0.5 * x_1 + 0.3 * rng.normal(size=n_1)
y_1[-1] -= 1.0                                # and not quite on the line of
                                                the others
idx = rng.integers(0, n_1, size=(20_000, n_1))
xs, ys = x_1[idx], y_1[idx]
xc = xs - xs.mean(axis=1, keepdims=True)
b_case = np.sum(xc * ys, axis=1) / np.sum(xc ** 2, axis=1)
absent = ~np.any(idx == n_1 - 1, axis=1)
print(f"leverage of the far case {h_1[-1]:.3f}; resamples without it: {absent.mean():.3f}")

```

The residual and wild bootstraps keep the far case, but inherit the problem ([Example \(The wild bootstrap for Engel's data\)](#)): its residual has variance  $\sigma^2(1 - h_{ii})$ , nearly zero, so the data say almost nothing about its error.

## 23.6.4 Many parameters

With  $p$  proportional to  $n$ , the plain residual bootstrap underestimates variances by the non-vanishing factor  $(n - p)/n$  of [Proposition 23.1.1\(b\)](#). The case bootstrap errs the other way: a resample has only about  $0.632n$  distinct cases ([Exercise \(A2\)](#)), and when  $p/n$  is not small its design is much worse conditioned than the original.

A simulation with  $n = 60$ ,  $p = 30$ , a Gaussian design and normal errors shows both effects. The median ratio of bootstrap variance to true variance for one coefficient, over 200 data sets, is 0.47 for the plain residual bootstrap, 0.97 after rescaling the residuals by  $\sqrt{n/(n - p)}$ , and 3.38 for the case bootstrap. El Karoui and Purdom (2018) study these failures, and Mammen (1993) gives conditions under which the residual and wild bootstraps remain valid as  $p$  grows. [Chapter 28](#) takes up regression with more parameters than cases.

```

import numpy as np

def variance_ratios(n, p, reps, B, rng):
    """Bootstrap variance of the first coefficient divided by its true variance (
    errors  $N(0, 1)$ )."""
    X = rng.normal(size=(n, p))                    # fixed design
    XtX_inv = np.linalg.inv(X.T @ X)
    true_var = XtX_inv[0, 0]
    H = X @ XtX_inv @ X.T
    out = {"residual": [], "rescaled": [], "case": []}
    for _ in range(reps):
        y = rng.normal(size=n)                    # beta = 0
        e = y - H @ y
        fit = H @ y
        for name, r in [("residual", e), ("rescaled", e * np.sqrt(n / (n - p)))]:
            Ys = fit + r[rng.integers(0, n, size=(B, n))]
            out[name].append(np.var(Ys @ (X @ XtX_inv)[: , 0]) / true_var)
        bs = []
        for _ in range(B):
            i = rng.integers(0, n, size=n)
            bs.append(np.linalg.lstsq(X[i], y[i], rcond=None)[0][0])
        out["case"].append(np.var(bs) / true_var)
    return {k: float(np.median(v)) for k, v in out.items()}

rng = np.random.default_rng(2314)
print(variance_ratios(60, 30, reps=20, B=100, rng=rng))

```

### 23.6.5 Dependence

Resampling independently destroys serial correlation. With positively autocorrelated errors and a smooth regressor, the true variance of a slope far exceeds the independent-errors formula ([Theorem 19.2.2](#)), which every bootstrap of this chapter reproduces. Take a straight line in time with  $n = 100$  equally spaced points and AR(1) errors with parameter 0.7. Over 2000 series the slope has standard deviation 1.130, while the residual bootstrap's average standard error is 0.449, a ratio of 0.40.

The **moving-block bootstrap** (Künsch 1989) resamples blocks of consecutive residuals and so keeps the dependence within blocks. With blocks of length 10 its average standard error is 0.818, a ratio of 0.72: better, still short, because dependence between blocks is lost (Lahiri 2003). When the dependence has a known form it is better to model it, as in [Theorem 21.6.1](#), and resample the estimated innovations.

```
import numpy as np

def dependence_study(n, phi, reps, B, block, rng):
    """True sd of the slope, and the average iid and moving-block residual-bootstrap
    standard errors."""
    x = np.linspace(0, 1, n)
    X = np.column_stack([np.ones(n), x])
    A = np.linalg.solve(X.T @ X, X.T)
    slopes, se_iid, se_block = [], [], []
    starts_max = n - block + 1
    for _ in range(reps):
        eps = np.empty(n)
        eps[0] = rng.normal() / np.sqrt(1 - phi ** 2)
        for i in range(1, n):
            eps[i] = phi * eps[i - 1] + rng.normal()    # AR(1) errors
        y = 1 + 2 * x + eps
        b = A @ y
        e = y - X @ b
        slopes.append(b[1])
        Ys = X @ b + e[rng.integers(0, n, size=(B, n))]
        se_iid.append(np.std(Ys @ A[1]))
        starts = rng.integers(0, starts_max, size=(B, n // block))
        idx = (starts[:, :, None] + np.arange(block)).reshape(B, -1)
        Ys = X @ b + e[idx]
        se_block.append(np.std(Ys @ A[1]))
    return np.std(slopes), np.mean(se_iid), np.mean(se_block)

rng = np.random.default_rng(2316)
print(dependence_study(100, 0.7, reps=100, B=200, block=10, rng=rng))
```

### 23.6.6 Small samples

A bootstrap distribution computed from eight residuals is an estimate from eight numbers. Take a straight line with  $n = 8$  equally spaced points and normal errors, the most favourable case. In 4000 simulated data sets, the classical  $t$  interval covers 0.953, as it must. The residual

bootstrap percentile interval covers only 0.857. The reason is [Proposition 23.1.1\(b\)](#) combined with [Proposition 23.3.3\(b\)](#). The percentile interval is roughly  $\hat{\beta}_1 \pm 1.96\sqrt{(n-2)/n} s\|\mathbf{a}\|$ , so its coverage is about

$$\Pr(|t_6| \leq 1.96\sqrt{6/8}) = \Pr(|t_6| \leq 1.697) = 0.859.$$

The studentized residual bootstrap covers 0.952. In small samples, studentize, and compare with the classical interval.

```
import numpy as np
from scipy import stats

def small_n_coverage(n, reps, B, rng):
    """Coverage of 95% intervals for a slope with normal errors: t, residual
    percentile, bootstrap-t."""
    x = np.linspace(0, 1, n)
    X = np.column_stack([np.ones(n), x])
    XtX_inv = np.linalg.inv(X.T @ X)
    A = XtX_inv @ X.T
    c = np.sqrt(XtX_inv[1, 1])
    hits = {"t": 0, "percentile": 0, "studentized": 0}
    for _ in range(reps):
        y = rng.normal(size=n)                # true slope 0
        b = A[1] @ y
        e = y - X @ (A @ y)
        s = np.sqrt(e @ e / (n - 2))
        hits["t"] += abs(b) <= stats.t.ppf(0.975, n - 2) * s * c
        Ys = X @ (A @ y) + e[rng.integers(0, n, size=(B, n))]
        bs = Ys @ A[1]
        ss = np.sqrt(np.sum((Ys - (Ys @ A.T) @ X.T) ** 2, axis=1) / (n - 2))
        lo, hi = np.quantile(bs, [0.025, 0.975])
        hits["percentile"] += lo <= 0 <= hi
        tl, th = np.quantile((bs - b) / (ss * c), [0.025, 0.975])
        hits["studentized"] += b - th * s * c <= 0 <= b - tl * s * c
    return {k: v / reps for k, v in hits.items()}

rng = np.random.default_rng(2318)
print(small_n_coverage(8, reps=300, B=499, rng=rng))
```

*Idea.* Before trusting a bootstrap, check each assumption listed at the start of this section, and studentize the root. A permutation test needs only exchangeability under the null hypothesis, and is exact or not at all.

## 23.6.7 Exercises

### 23.6.7.1 A. Check your understanding

#### Exercise A1.

Compute  $1 - (1 - 1/n)^n$  for  $n = 10$  and  $n = 1000$ . Does the failure of [Proposition 23.6.1](#) become less severe with more data?

*Solution.* The values are 0.651 and 0.632. The atom decreases slightly toward its limit  $1 - e^{-1}$  but never vanishes, so the failure does not go away.

**Exercise A2.**

In the moving-block bootstrap with blocks of length  $\ell$ , explain why the bootstrap variance of a sample mean tends to be too small when  $\ell$  is much shorter than the range of dependence, and why a very long block makes the estimate noisy.

**23.6.7.2 B. Practice****Exercise B1.**

In [Example \(A slope determined by one case\)](#), show that the probability that the far case appears at least twice in a resample is  $1 - (1 - 1/n)^n - (1 - 1/n)^{n-1}$ , and evaluate it for  $n = 15$ . What does a resample with the far case drawn three times estimate?

*Solution.* The number of times the case is drawn is binomial with  $n$  trials and probability  $1/n$ . It is zero with probability  $(1 - 1/n)^n$  and one with probability  $n(1/n)(1 - 1/n)^{n-1} = (1 - 1/n)^{n-1}$ , which gives the formula. For  $n = 15$  this is  $1 - 0.3553 - 0.3806 = 0.2641$ . A resample with the case drawn three times fits a line in which that case has triple weight, a weighted least squares estimate that pulls even harder toward the far case, and not a draw from anything like the sampling distribution of  $\hat{\beta}_1$ .

**23.6.7.3 C. Going deeper****Exercise C1.**

For the uniform maximum of [Proposition 23.6.1](#), show that the  $m$ -out-of- $n$  bootstrap with  $m \rightarrow \infty$  and  $m/n \rightarrow 0$  gives  $\Pr_*(Y_{(m)}^* = Y_{(n)}) = 1 - (1 - 1/n)^m \rightarrow 0$ , and that the bootstrap distribution of  $m(Y_{(n)} - Y_{(m)}^*)/Y_{(n)}$  approaches the standard exponential law.

## 23.7 Summary and Notes

### 23.7.1 Summary

*Idea.*

1. The bootstrap estimates the distribution of  $\hat{\theta} - \theta$  by that of  $\hat{\theta}^* - \hat{\theta}$  under an estimated mechanism: resampled residuals with the design fixed, or resampled cases ([Definition 23.1.1](#)).
2. The residual bootstrap has covariance  $\hat{\sigma}_*^2(\mathbf{X}^\top \mathbf{X})^{-1}$ , biased by  $(n - p)/n$  ([Proposition 23.1.1](#)), and is consistent for contrasts under exchangeable errors when no case dominates ([Theorem 23.1.4](#)); it fails under heteroscedasticity.
3. The case bootstrap, linearized, has the HC0 sandwich covariance ([Proposition 23.1.5](#)). The wild bootstrap reproduces the sandwich exactly ([Proposition 23.2.1](#)) and is consistent under heteroscedasticity ([Theorem 23.2.2](#)).
4. Percentile, basic and studentized intervals are first-order correct ([Corollary 23.3.2](#));

studentizing reproduces exact pivots exactly ([Proposition 23.3.3](#)) and, by Hall's Edgeworth analysis (not proved here), gives higher-order accuracy.

5. Permutation tests are exact for any statistic under invariance ([Theorem 23.4.2](#)): in simple regression with exchangeable errors, and with nuisance regressors only for permutations that fix them ([Proposition 23.4.4](#)). Heteroscedasticity turns them into tests of independence unless the statistic is studentized.
6. The Freedman–Lane scheme is exact when an exact test exists, ignores the nuisance coefficients, and its permuted statistics are asymptotically those of an exact oracle ([Proposition 23.5.1\(d\)](#)); its asymptotic size is due to Anderson and Robinson (2001). Kennedy's scheme is more liberal ([Proposition 23.5.2](#)).
7. Resampling fails for extremes ([Proposition 23.6.1](#)), heavy tails, dominant cases, large  $p/n$ , dependence and, unless studentized, small samples ([Section 23.6](#)).

## 23.7.2 Notes and sources

**The bootstrap.** Efron (1979) introduced the bootstrap. Freedman (1981) set out residual and case resampling for regression and proved both consistent; [Theorem 23.1.4](#) is a version for linear contrasts. Bickel and Freedman (1981) developed the general theory, and Wu (1986) showed that the residual bootstrap fails under heteroscedasticity. Efron and Tibshirani (1993) and Davison and Hinkley (1997) are the standard books; the latter uses the leverage-adjusted residuals adopted here. Sen and Srivastava (1990, section 5.4) describe the residual bootstrap test of a linear hypothesis.

**The wild bootstrap.** The idea is in Wu (1986); Liu (1988) and Mammen (1993) developed its theory, and Mammen introduced the two-point law. Davidson and Flachaire (2008) recommend Rademacher weights with restricted residuals for tests.

**Intervals.** Hall (1988) compared bootstrap intervals by Edgeworth expansions, and Hall (1992) is the full account, including the iterated bootstrap of [Exercise \(C1\)](#). Efron (1987) introduced BCa; DiCiccio and Efron (1996) review the methods. Prediction intervals for a new response, which add a resampled error to the resampled estimate in place of the normal-theory interval of [Theorem 12.3.1](#), are treated in Davison and Hinkley (1997, chapter 6).

**Permutation tests.** Fisher (1935) and Pitman (1937) introduced permutation tests. The proof of [Theorem 23.4.2](#) follows Lehmann and Romano (2005, chapter 15); on the Monte Carlo count see Phipson and Smyth (2010). DiCiccio and Romano (2017) proved the validity of studentized permutation tests under dependence without correlation.

**Partial tests.** The schemes are due to Freedman and Lane (1983), ter Braak (1992), Kennedy (1995), Draper and Stoneman (1966) and Manly (1997). Anderson and Legendre (1999) compared them by simulation, Anderson and Robinson (2001) analytically, and Winkler and coauthors (2014) review them for the general linear model.

**Failures.** Bickel, Götze and van Zwet (1997) analyse the  $m$ -out-of- $n$  bootstrap, Athreya (1987) the infinite-variance mean, and El Karoui and Purdom (2018) large  $p/n$ . Künsch (1989) introduced the moving-block bootstrap; Lahiri (2003) treats dependent data at length.

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# Chapter 24

## Errors in Variables

Every earlier chapter takes the regressors as recorded without error. Often they are not: one clinic reading stands in for long-run blood pressure, a questionnaire for usual diet. The scientific model concerns the true value  $X$ , but the data contain a noisy  $W$ , and least squares on  $W$  estimates the best linear predictor from what was recorded. This chapter measures the gap and closes it. Classical error flattens a slope by a factor no sample size removes, makes correctly measured covariates look important, and in the normal model leaves the slope unidentified. Information from outside the regression rescues the analysis: a known error variance, repeated measurements, a known variance ratio, an instrument, or validation data. The moment correction, instrumental variables, regression calibration and SIMEX turn that information into estimates, each with its own conditions and costs.

**What you need.** [Chapter 6](#) (the population projection, [Theorem 6.11.1](#) and [Proposition 6.11.3](#), and [Theorem 6.3.7](#)). [Chapter 5](#) (moments of least squares under departures, [Proposition 5.5.7](#)). [Chapter 14](#) (random regressors, [Theorem 14.1.2](#), and the limit theorems of [Section 14.4](#)). From this part, [Chapter 19](#) (omitted-variable bias, [Theorem 19.1.2](#)), [Chapter 21](#) (sandwich covariance estimators, [Theorem 21.4.2](#)) and [Chapter 23](#) (the pairs bootstrap, [Definition 23.1.1](#)).

### 24.0.1 Roadmap

- [Attenuation bias](#)
- [The classical measurement error model](#)
- [Reliability and moment correction](#)
- [Instrumental variables](#)
- [Regression calibration and SIMEX](#)
- [Summary and notes](#)

## 24.1 Attenuation bias

When a regressor is recorded with error, conditioning on the recorded value is still legitimate, but it describes how the response varies with the *reading*, not with the underlying *quantity*. This section measures the difference.

## 24.1.1 A running example

One simulated study is used throughout the chapter. For  $n = 300$  patients, let  $x_i$  be long-run systolic blood pressure, the average that many visits would give, with mean 130 mmHg and standard deviation 12. Age has mean 55 years, standard deviation 10, and correlation 0.5 with  $x_i$ . Each patient has two clinic readings,  $w_{ij} = x_i + u_{ij}$ , with independent errors  $u_{ij} \sim N(0, 9^2)$ . The outcome is

$$y_i = -4 + 0.08 x_i + 0 \cdot \text{age}_i + \varepsilon_i, \quad \varepsilon_i \sim N(0, 1),$$

so age has no effect given long-run pressure. In a real study  $x_i$  is unseen; here it lets every method be checked.

## 24.1.2 Classical error

Throughout the chapter, a regression model is stated in terms of *true* regressors  $\mathbf{x}$  (the ones measured with error) and  $\mathbf{z}$  (the ones measured exactly):

$$Y = \alpha + \boldsymbol{\beta}_x^\top \mathbf{x} + \boldsymbol{\beta}_z^\top \mathbf{z} + \varepsilon, \quad (24.1.1)$$

where  $\varepsilon$  has mean zero and variance  $\sigma^2$  and is uncorrelated with  $(\mathbf{x}, \mathbf{z})$ . We write a generic observation without an index. The data are  $n$  independent copies of it unless stated otherwise.

### Definition 24.1.1 — Classical measurement error

Suppose  $\mathbf{x}$  is observed only through  $\mathbf{w} = \mathbf{x} + \mathbf{u}$ . The error  $\mathbf{u}$  is **classical** if  $E(\mathbf{u}) = \mathbf{0}$ ,  $\text{Cov}(\mathbf{u}) = \boldsymbol{\Sigma}_{uu}$ , and  $\mathbf{u}$  is independent of  $(\mathbf{x}, \mathbf{z}, \varepsilon)$ .

The error is *additive and unrelated to the true value*, and *nondifferential*: independent of  $\varepsilon$ , so the reading says nothing about the response beyond what  $\mathbf{x}$  says. (People with a disease may recall past exposures differently, a differential error.) This section uses only that  $\mathbf{u}$  has mean zero and is *uncorrelated* with  $\mathbf{x}$ ,  $\mathbf{z}$  and  $\varepsilon$ .

### Definition 24.1.2 — Reliability ratio

The **reliability ratio** of  $W = X + U$  is

$$\lambda = \frac{\sigma_x^2}{\sigma_x^2 + \sigma_u^2} = \frac{\text{Var}(X)}{\text{Var}(W)}.$$

If  $\mathbf{z}$  is measured exactly, let  $\sigma_{x \cdot z}^2 = \sigma_x^2 - \boldsymbol{\sigma}_{zx}^\top \boldsymbol{\Sigma}_{zz}^{-1} \boldsymbol{\sigma}_{zx}$  be the residual variance of  $X$  after its best linear prediction from  $\mathbf{z}$  (Theorem 6.11.1), where  $\boldsymbol{\sigma}_{zx} = \text{Cov}(\mathbf{z}, X)$ . The **reliability given  $\mathbf{z}$**  is

$$\lambda_{x|z} = \frac{\sigma_{x \cdot z}^2}{\sigma_{x \cdot z}^2 + \sigma_u^2}.$$

In the running example a single reading has  $\lambda = 144/(144 + 81) = 0.64$ . Age explains a quarter of the variance of long-run pressure, leaving  $\sigma_{x \cdot z}^2 = 108$ , so  $\lambda_{x|z} = 108/189 = 0.571$ . Covariates remove part of the signal in  $W$  and none of the noise, so  $\lambda_{x|z} \leq \lambda$ .

### 24.1.3 Least squares on the reading

Substituting  $X = W - U$  in  $Y = \alpha + \beta X + \varepsilon$  gives

$$Y = \alpha + \beta W + (\varepsilon - \beta U). \quad (24.1.2)$$

Its error is correlated with the regressor,  $\text{Cov}(W, \varepsilon - \beta U) = -\beta\sigma_u^2$ , so least squares does not estimate  $\beta$ . It estimates the population projection of  $Y$  on the recorded regressors ([Proposition 6.11.3](#)), which the next theorem computes.

#### Theorem 24.1.1 — Attenuation

Let  $(\mathbf{x}_{(i)}, \mathbf{z}_{(i)}, \mathbf{u}_{(i)}, \varepsilon_i)$ ,  $i = 1, 2, \dots$ , be independent copies of  $(\mathbf{x}, \mathbf{z}, \mathbf{u}, \varepsilon)$  with finite second moments, satisfying [Eq. 24.1.1](#) and [Definition 24.1.1](#). Write  $\mathbf{t} = (\mathbf{x}^\top, \mathbf{z}^\top)^\top$  for the true regressors,  $\mathbf{s} = (\mathbf{w}^\top, \mathbf{z}^\top)^\top$  for the observed ones,  $\boldsymbol{\theta} = (\boldsymbol{\beta}_x^\top, \boldsymbol{\beta}_z^\top)^\top$ ,  $\boldsymbol{\Sigma}_{tt} = \text{Cov}(\mathbf{t})$ , assumed positive definite, and  $\tilde{\boldsymbol{\Sigma}} = \text{diag}(\boldsymbol{\Sigma}_{uu}, \mathbf{0})$  for the covariance matrix of the error in  $\mathbf{s}$ . Let  $\hat{\alpha}_n$  and  $\hat{\boldsymbol{\theta}}_n$  be the least squares intercept and slopes from regressing  $Y$  on  $\mathbf{s}$  in the first  $n$  observations. Then, with probability one:

- (a)  $\hat{\boldsymbol{\theta}}_n \rightarrow \boldsymbol{\theta}_w = (\boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}})^{-1} \boldsymbol{\Sigma}_{tt} \boldsymbol{\theta} = \boldsymbol{\theta} - (\boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}})^{-1} \tilde{\boldsymbol{\Sigma}} \boldsymbol{\theta}$ , and  $\hat{\alpha}_n \rightarrow \alpha + \boldsymbol{\mu}_t^\top (\boldsymbol{\theta} - \boldsymbol{\theta}_w)$ , where  $\boldsymbol{\mu}_t = \text{E}(\mathbf{t})$ ;
- (b) with a single regressor and no  $\mathbf{z}$ , the slope converges to  $\lambda\beta$ ;
- (c) with a single mismeasured scalar  $X$  and exactly measured  $\mathbf{z}$ , the coefficient of  $W$  converges to  $\lambda_{x|z}\beta_x$  and the coefficients of  $\mathbf{z}$  converge to  $\boldsymbol{\beta}_z + (1 - \lambda_{x|z})\beta_x\boldsymbol{\pi}$ , where  $\boldsymbol{\pi} = \boldsymbol{\Sigma}_{zz}^{-1}\boldsymbol{\sigma}_{zx}$  is the vector of coefficients from the best linear prediction of  $X$  from  $\mathbf{z}$  ([Theorem 2.5.2](#));
- (d) the residual sum of squares divided by  $n$  (or by  $n$  minus the number of coefficients) converges to  $\sigma^2 + \boldsymbol{\theta}^\top \boldsymbol{\Sigma}_{tt} (\boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}})^{-1} \tilde{\boldsymbol{\Sigma}} \boldsymbol{\theta}$ , which in case (b) equals  $\sigma^2 + \lambda\beta^2\sigma_u^2$ .

*Proof.* Because  $\mathbf{u}$  is uncorrelated with  $\mathbf{t}$ ,  $\text{Cov}(\mathbf{s}) = \boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}}$ , which is positive definite as the sum of a positive definite and a nonnegative definite matrix. By [Proposition 6.11.3](#), the least squares coefficients converge with probability one to those of the best linear predictor of  $Y$  from  $\mathbf{s}$ , which by [Theorem 6.11.1](#) has slopes  $\text{Cov}(\mathbf{s})^{-1} \text{Cov}(\mathbf{s}, Y)$  and intercept  $\text{E}Y - \boldsymbol{\theta}_w^\top \text{E}\mathbf{s}$ . Now

$$\text{Cov}(\mathbf{s}, Y) = \text{Cov}(\mathbf{t} + (\mathbf{u}^\top, \mathbf{0}^\top)^\top, \mathbf{t}^\top \boldsymbol{\theta} + \varepsilon) = \boldsymbol{\Sigma}_{tt} \boldsymbol{\theta},$$

since  $\mathbf{u}$  is uncorrelated with  $\mathbf{t}$  and  $\varepsilon$ , and  $\varepsilon$  with  $\mathbf{t}$ . This is the first form in (a). The second follows from  $\boldsymbol{\Sigma}_{tt} = (\boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}}) - \tilde{\boldsymbol{\Sigma}}$ . For the intercept,  $\text{E}\mathbf{s} = \boldsymbol{\mu}_t$  and  $\text{E}Y = \alpha + \boldsymbol{\theta}^\top \boldsymbol{\mu}_t$ .

(b) With scalars,  $\theta_w = \sigma_x^2\beta/(\sigma_x^2 + \sigma_u^2) = \lambda\beta$ .

(c) Here  $\tilde{\boldsymbol{\Sigma}}\boldsymbol{\theta} = \sigma_u^2\beta_x\mathbf{e}_1$ , so by the second form in (a) the bias  $\boldsymbol{\theta} - \boldsymbol{\theta}_w$  is  $\sigma_u^2\beta_x$  times the first column of  $\text{Cov}(\mathbf{s})^{-1}$ . Order the observed regressors as  $(\mathbf{z}, W)$  for the moment, so that

$$\text{Cov}(\mathbf{s}) = \begin{pmatrix} \boldsymbol{\Sigma}_{zz} & \boldsymbol{\sigma}_{zx} \\ \boldsymbol{\sigma}_{zx}^\top & \sigma_x^2 + \sigma_u^2 \end{pmatrix}.$$

The Schur complement of  $\boldsymbol{\Sigma}_{zz}$  is  $\sigma_x^2 + \sigma_u^2 - \boldsymbol{\sigma}_{zx}^\top \boldsymbol{\Sigma}_{zz}^{-1} \boldsymbol{\sigma}_{zx} = \sigma_{x \cdot z}^2 + \sigma_u^2$ . By [Theorem 1.6.4](#) the last column of the inverse is  $(-\boldsymbol{\pi}^\top, 1)^\top / (\sigma_{x \cdot z}^2 + \sigma_u^2)$ . So the bias in the coefficient of  $W$  is

$\sigma_u^2 \beta_x / (\sigma_{x|z}^2 + \sigma_u^2) = (1 - \lambda_{x|z}) \beta_x$ , and the bias in the coefficients of  $\mathbf{z}$  is  $-(1 - \lambda_{x|z}) \beta_x \boldsymbol{\pi}$ .

(d) The residual sum of squares over  $n$  is  $n^{-1} \sum_i (Y_i - \hat{\alpha}_n - \hat{\boldsymbol{\theta}}_n^\top \mathbf{s}_i)^2$ . Expanded, it is a continuous function of the coefficients and of second sample moments, which converge by (a) and the strong law, so the limit is  $E(Y - \alpha_w - \boldsymbol{\theta}_w^\top \mathbf{s})^2$ , the mean squared error of the best linear predictor. By [Eq. 6.11.1](#) it equals  $\text{Var}(Y) - \boldsymbol{\theta}_w^\top \text{Cov}(\mathbf{s}) \boldsymbol{\theta}_w$ , which is

$$\sigma^2 + \boldsymbol{\theta}^\top \boldsymbol{\Sigma}_{tt} \boldsymbol{\theta} - \boldsymbol{\theta}^\top \boldsymbol{\Sigma}_{tt} (\boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}})^{-1} \boldsymbol{\Sigma}_{tt} \boldsymbol{\theta} = \sigma^2 + \boldsymbol{\theta}^\top \boldsymbol{\Sigma}_{tt} (\boldsymbol{\Sigma}_{tt} + \tilde{\boldsymbol{\Sigma}})^{-1} \tilde{\boldsymbol{\Sigma}} \boldsymbol{\theta}.$$

In case (b) the last term is  $\beta^2 \sigma_x^2 \sigma_u^2 / (\sigma_x^2 + \sigma_u^2) = \lambda \beta^2 \sigma_u^2$ .

■

Part (b) is **attenuation**: the slope is shrunk towards zero by the reliability ratio, whatever the sample size. More data give a more precise estimate of the wrong number  $\lambda \beta$ .

#### Example. (One reading instead of long-run pressure)

In the running example, regressing  $y$  on the true  $x$  gives slope 0.0755 (standard error 0.0050). Regressing on the first clinic reading gives 0.0499 (standard error 0.0046), a ratio of 0.661, close to  $\lambda = 0.64$ ; the limit is  $\lambda \beta = 0.0512$ . The usual 95% interval, [0.0409, 0.0589], misses 0.08, and in 4000 repeated samples the observed coverage was 0.000. The standard error measures variability around  $\lambda \beta$ , not the distance from  $\lambda \beta$  to  $\beta$ . The residual variance tends to  $1 + 0.64 \cdot 0.08^2 \cdot 81 = 1.3318$  by [Theorem 24.1.1](#)(d).

## 24.1.4 Several regressors: bias spreads

Part (c) is more troubling. Correctly measured covariates lower the relevant reliability to  $\lambda_{x|z}$ , and they inherit part of the effect of  $X$ . The part of  $X$  that the reading misses acts like an omitted regressor ([Theorem 19.1.2](#)), and covariates correlated with  $X$  pick up its effect in proportion to  $\boldsymbol{\pi}$ . This is **residual confounding**: adjusting for a noisy measurement of a confounder only partly adjusts for it.

#### Example. (Age looks important)

Add age to the regression on the first reading. Theory predicts a coefficient of  $\lambda_{x|z} \beta = 0.0457$  for the reading and  $(1 - 0.571) \cdot 0.08 \cdot 0.6 = 0.0206$  per year for age, which truly has no effect. The sample gives 0.0455 for the reading and 0.0182 for age, with standard error 0.0070 and  $p$ -value 0.0098. In 4000 repeated samples the  $t$  test of “no age effect” rejected at the 5% level in a proportion 0.812 of them. [Figure 24.1.1](#)(b) shows both limits as functions of the reliability.

```
import numpy as np
import statsmodels.api as sm

rng = np.random.default_rng(2401)
n = 300
age = rng.normal(55, 10, n)
x = 130 + 0.6 * (age - 55) + rng.normal(0, np.sqrt(108), n)  # long-run blood
                        pressure, sd 12
```

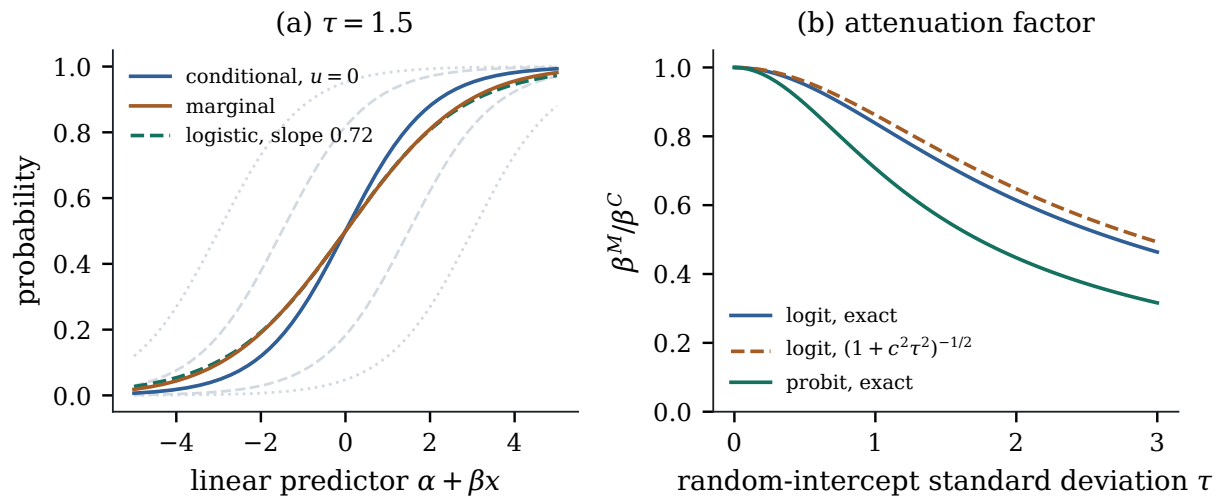


Figure 24.1.1: (a) The outcomes against true pressure (grey) and one reading (blue), with their least squares lines. (b) Limits of the coefficients of the reading and of age, in units of  $\beta_x$ , against the reliability; dashed,  $\lambda$  itself.

```
w = x[:, None] + rng.normal(0, 9, (n, 2)) # two clinic readings
y = -4 + 0.08 * x + 0.0 * age + rng.normal(0, 1.0, n) # age has no effect
given x

oracle = sm.OLS(y, sm.add_constant(x)).fit() # uses the true x
naive = sm.OLS(y, sm.add_constant(w[:, 0])).fit() # uses one reading
lam = 144 / (144 + 81) # reliability of one
reading

print(f"slope on x: {oracle.params[1]:.4f}")
print(f"slope on reading: {naive.params[1]:.4f} limit lambda*beta = {lam *
0.08:.4f}")
print("naive 95% interval:", naive.conf_int()[1].round(4))

both = sm.OLS(y, sm.add_constant(np.column_stack([w[:, 0], age]))).fit()
print(f"with age: reading {both.params[1]:.4f}, age {both.params[2]:.4f} (p = {both.
pvalues[2]:.4f})")
```

The repeated-sample figures come from this loop (with fewer samples in the cell).

```
def repeat(reps, seed):
    """Coverage of the naive 95% interval for the slope, and the rate at which age
    is 'significant'."""
    r = np.random.default_rng(seed)
    cover = reject = 0
    for _ in range(reps):
        a = r.normal(55, 10, n)
        xx = 130 + 0.6 * (a - 55) + r.normal(0, np.sqrt(108), n)
        ww = xx + r.normal(0, 9, n)
        yy = -4 + 0.08 * xx + r.normal(0, 1.0, n)
        lo, hi = sm.OLS(yy, sm.add_constant(ww)).fit().conf_int()[1]
        cover += lo <= 0.08 <= hi
        reject += sm.OLS(yy, sm.add_constant(np.column_stack([ww, a]))).fit().
pvalues[2] < 0.05
    return cover / reps, reject / reps

print("coverage of naive interval, rate of rejecting 'no age effect':", repeat(200,
```

2404))

With several mismeasured regressors,  $(\Sigma_{tt} + \tilde{\Sigma})^{-1}\Sigma_{tt}$  need not shrink each coefficient. A coefficient can grow, change sign, or appear from nothing ([Exercise \(B3\)](#)).

## 24.1.5 What survives

Two things survive classical error intact.

**Error in the response.** Suppose instead that the response is measured with error,  $Y^* = Y + V$ , where  $V$  has mean zero and variance  $\sigma_v^2$  and is independent of the regressors and of  $\varepsilon$ .

### Proposition 24.1.2 — Error in the response

Under these assumptions  $Y^* = \alpha + \beta^\top \mathbf{x} + (\varepsilon + V)$ , a linear model with the same coefficients and error variance  $\sigma^2 + \sigma_v^2$ . If  $\varepsilon$  and  $V$  are normal, so is their sum, and all the exact theory of Parts II and III applies to  $Y^*$  with  $\sigma^2$  replaced by  $\sigma^2 + \sigma_v^2$ .

The proof is the display. Error in the response costs precision, not bias, unless it depends on the regressors (heavier people under-reporting weight) or is correlated with a regressor's error (both from the same self-report).

**Tests of no effect.** Since  $\lambda\beta = 0$  iff  $\beta = 0$ , attenuation can hide an effect of the mismeasured regressor but cannot create one. The protection does not extend to the covariates.

### Proposition 24.1.3 — Which tests survive classical error

Assume [Eq. 24.1.1](#) and [Definition 24.1.1](#) with a single mismeasured scalar  $X$ ,  $\varepsilon \sim N(0, \sigma^2)$  independent of  $(X, \mathbf{z})$ , and the covariance matrix of  $(X, \mathbf{z}^\top)^\top$  positive definite. Consider the least squares regression of  $Y$  on  $(1, W, \mathbf{z})$ .

- (a) If  $\beta_x = 0$ , the usual  $t$  test of “the coefficient of  $W$  is zero” has exactly its nominal level.
- (b) If  $\beta_x \neq 0$  and  $\boldsymbol{\pi} \neq \mathbf{0}$ , the usual  $F$  test of “the coefficients of  $\mathbf{z}$  equal  $\boldsymbol{\beta}_z$ ” rejects with probability tending to one as  $n \rightarrow \infty$ .

*Proof.* (a) If  $\beta_x = 0$ , then  $Y = \alpha + \beta_z^\top \mathbf{z} + \varepsilon$ , and  $\varepsilon$  is independent of  $(W, \mathbf{z})$  because it is independent of  $(X, \mathbf{z}, U)$ : it is independent of  $(X, \mathbf{z})$  by hypothesis, and  $U$  is independent of  $(X, \mathbf{z}, \varepsilon)$  by [Definition 24.1.1](#). So, conditionally on the observed regressors,  $Y$  follows the normal linear model of [Definition 14.1.1](#) with regressors  $(1, W, \mathbf{z})$ , in which the coefficient of  $W$  is zero. [Theorem 14.1.2](#) gives the exact  $t$  distribution.

(b) By [Theorem 24.1.1\(c\)](#),  $\hat{\boldsymbol{\beta}}_z - \boldsymbol{\beta}_z \rightarrow \mathbf{c} = (1 - \lambda_{x|z})\beta_x\boldsymbol{\pi} \neq \mathbf{0}$ . The  $F$  statistic is  $n(\hat{\boldsymbol{\beta}}_z - \boldsymbol{\beta}_z)^\top \mathbf{K}_n(\hat{\boldsymbol{\beta}}_z - \boldsymbol{\beta}_z)$ , where  $\mathbf{K}_n = [n(\mathbf{S}^\top \mathbf{S})^{-1}]_{zz}^{-1}/(qs^2)$ ,  $\mathbf{S}$  is the observed model matrix and  $q$  the dimension of  $\mathbf{z}$ . By the strong law and [Theorem 24.1.1\(d\)](#),  $\mathbf{K}_n$  converges to a positive definite limit, so the statistic grows like  $n$ , while the critical value tends to a finite limit.

So “is there any association with the underlying quantity?” can be tested with the reading, with reduced power. “Does this covariate matter once the underlying quantity is accounted for?” cannot.

## 24.1.6 Exercises

### 24.1.6.1 A. Check your understanding

#### Exercise A1.

Show that  $\lambda = \text{Corr}(W, X)^2$ . If  $W_1 = X + U_1$  and  $W_2 = X + U_2$  are two readings with independent classical errors of the same variance, show that  $\text{Corr}(W_1, W_2) = \lambda$ . This is why test–retest correlations are used as reliability estimates.

*Solution.*  $\text{Cov}(W, X) = \sigma_x^2$ , so  $\text{Corr}(W, X)^2 = \sigma_x^4 / \{\sigma_x^2(\sigma_x^2 + \sigma_u^2)\} = \lambda$ . For two readings,  $\text{Cov}(W_1, W_2) = \text{Var}(X) = \sigma_x^2$  because the errors are independent of each other and of  $X$ , and  $\text{Var}(W_1) = \text{Var}(W_2) = \sigma_x^2 + \sigma_u^2$ . The correlation is  $\sigma_x^2 / (\sigma_x^2 + \sigma_u^2) = \lambda$ .

### 24.1.6.2 B. Practice

#### Exercise B1.

Let  $\bar{W}$  be the mean of  $m$  readings with independent classical errors of variance  $\sigma_u^2$ , and let  $\lambda$  be the reliability of one reading. Show that the reliability of  $\bar{W}$  is

$$\lambda_m = \frac{m\lambda}{1 + (m-1)\lambda}.$$

How many readings are needed in the running example for the attenuation factor to reach 0.9?

*Solution.*  $\lambda_m = \sigma_x^2 / (\sigma_x^2 + \sigma_u^2/m)$ . Dividing through by  $\sigma_x^2 + \sigma_u^2$  and using  $\sigma_u^2 / (\sigma_x^2 + \sigma_u^2) = 1 - \lambda$  gives  $\lambda / \{\lambda + (1 - \lambda)/m\}$ , the formula. With  $\lambda = 0.64$ ,  $\lambda_m \geq 0.9$  iff  $0.64m \geq 0.9 + 0.576(m - 1)$ , so  $m \geq 5.0625$ : six readings (without age in the model).

#### Exercise B2.

In the setting of [Theorem 24.1.1\(b\)](#), show that the population  $R^2$  of  $Y$  on  $W$  equals  $\lambda$  times the population  $R^2$  of  $Y$  on  $X$ . Measurement error lowers the apparent strength of a relation twice: once through the slope and once through the residual variance.

#### Exercise B3.

Let  $\mathbf{t} = (X_1, X_2)$  have unit variances and correlation  $\rho$ , let  $X_1$  be measured exactly and  $X_2$  with classical error of variance  $\sigma_u^2$ , and let  $\boldsymbol{\theta} = (\beta_1, \beta_2)$ . Show that the limit of the coefficient of  $X_1$  is  $\beta_1 + (1 - \lambda_{2|1})\rho\beta_2$ , where  $\lambda_{2|1} = (1 - \rho^2)/(1 - \rho^2 + \sigma_u^2)$ . With  $\rho = 0.5$ ,



$\sigma_u^2 = 1$ ,  $\beta_2 = 1$  and  $\beta_1 = -0.2$ , show that the sign of the first coefficient is reversed.

*Solution.* This is [Theorem 24.1.1\(c\)](#) with  $\mathbf{z} = X_1$ ,  $X = X_2$  and  $\boldsymbol{\pi} = \rho$ , since the coefficient of the best linear prediction of  $X_2$  from  $X_1$  is  $\rho$  and  $\sigma_{x.z}^2 = 1 - \rho^2$ . With the numbers given,  $\lambda_{2|1} = 0.75/1.75 = 3/7$ , so the limit is  $-0.2 + (4/7)(0.5) = -0.2 + 2/7 \approx 0.086 > 0$ . A regressor with a negative effect acquires a positive coefficient.

### 24.1.6.3 C. Going deeper

#### Exercise C1.

Regressors are often recorded after rounding. Model this as  $\mathbf{S} = \mathbf{T} + \boldsymbol{\Delta}$  with  $\mathbf{T}$  the true model matrix and  $\boldsymbol{\Delta}$  a *fixed* matrix of rounding errors, and suppose  $E(\mathbf{Y}) = \mathbf{T}\boldsymbol{\theta}$ ,  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ . Use [Proposition 5.5.7](#) to show that least squares on  $\mathbf{S}$  has bias  $-(\mathbf{S}^\top\mathbf{S})^{-1}\mathbf{S}^\top\boldsymbol{\Delta}\boldsymbol{\theta}$ , that its covariance matrix is exactly  $\sigma^2(\mathbf{S}^\top\mathbf{S})^{-1}$ , and that the expected residual mean square is at least  $\sigma^2$ .

*Solution.* Here  $E(\mathbf{Y}) = \mathbf{S}\boldsymbol{\theta} + \boldsymbol{\delta}$  with  $\boldsymbol{\delta} = -\boldsymbol{\Delta}\boldsymbol{\theta}$ . By [Proposition 5.5.7\(a\)](#) with model matrix  $\mathbf{S}$ , the expectation is  $\boldsymbol{\theta} + (\mathbf{S}^\top\mathbf{S})^{-1}\mathbf{S}^\top\boldsymbol{\delta}$ , which gives the bias. The covariance is  $\sigma^2(\mathbf{S}^\top\mathbf{S})^{-1}$  by [Proposition 5.5.7\(b\)](#) with  $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$ : nothing random has been added. For the residual mean square, [Theorem 2.3.1](#) gives  $E\|(\mathbf{I} - \mathbf{M}_S)\mathbf{Y}\|^2 = \sigma^2(n-p) + \boldsymbol{\delta}^\top(\mathbf{I} - \mathbf{M}_S)\boldsymbol{\delta}$ , where  $\mathbf{M}_S$  projects onto  $\mathcal{C}(\mathbf{S})$  and  $(\mathbf{I} - \mathbf{M}_S)\mathbf{S}\boldsymbol{\theta} = \mathbf{0}$ . The last term is nonnegative. Unlike random error, fixed rounding error does not inflate the variance of the estimator. It only biases it, and the bias is small when rounding is fine compared with the spread of the regressors.

## 24.2 The classical measurement error model

This section sets up the two versions of the classical model, shows that its normal version cannot identify the slope from the observed data alone, and separates classical error from Berkson error, which does no harm to a linear regression.

### 24.2.1 Structural and functional models

In a **structural** model the true values are random draws, as in [Theorem 24.1.1](#). In a **functional** model they are fixed unknown constants, one nuisance parameter per observation, which cannot be conditioned on because they are not observed. Attenuation is the same in both.

#### Proposition 24.2.1 — Attenuation with fixed true values

Let  $\mathbf{t}_1, \mathbf{t}_2, \dots \in \mathbb{R}^p$  be fixed vectors with first coordinate 1, such that  $n^{-1} \sum_{i=1}^n \mathbf{t}_i \mathbf{t}_i^\top \rightarrow \mathbf{A}$ , a positive definite matrix. Let  $y_i = \mathbf{t}_i^\top \boldsymbol{\theta} + \varepsilon_i$  and  $\mathbf{s}_i = \mathbf{t}_i + \tilde{\mathbf{u}}_i$ , where the pairs  $(\tilde{\mathbf{u}}_i, \varepsilon_i)$  are independent and identically distributed,  $\tilde{\mathbf{u}}_i$  and  $\varepsilon_i$  are independent with mean zero and finite variances, the first coordinate of  $\tilde{\mathbf{u}}_i$  is zero (so that  $\mathbf{s}_i$  keeps the constant and  $\tilde{\boldsymbol{\Sigma}}$  has zero first row and column), and  $\text{Cov}(\tilde{\mathbf{u}}_i) = \tilde{\boldsymbol{\Sigma}}$ . Then the least squares coefficient

vector from regressing  $y$  on  $\mathbf{s}$  converges in probability to  $(\mathbf{A} + \tilde{\Sigma})^{-1} \mathbf{A}\theta$ .

*Proof.* Stack the vectors as rows of  $\mathbf{T}$ ,  $\mathbf{S}$  and  $\tilde{\mathbf{U}}$ . Then

$$\frac{1}{n} \mathbf{S}^\top \mathbf{S} = \frac{1}{n} \mathbf{T}^\top \mathbf{T} + \frac{1}{n} (\mathbf{T}^\top \tilde{\mathbf{U}} + \tilde{\mathbf{U}}^\top \mathbf{T}) + \frac{1}{n} \tilde{\mathbf{U}}^\top \tilde{\mathbf{U}}.$$

The last term tends to  $\tilde{\Sigma}$  in probability by the weak law of large numbers. An entry  $n^{-1} \sum_i t_{ij} \tilde{u}_{ik}$  of  $n^{-1} \mathbf{T}^\top \tilde{\mathbf{U}}$  has mean zero and variance  $\tilde{\Sigma}_{kk} n^{-2} \sum_i t_{ij}^2 \rightarrow 0$ , so it tends to zero by Chebyshev's inequality. Hence  $n^{-1} \mathbf{S}^\top \mathbf{S} \rightarrow \mathbf{A} + \tilde{\Sigma}$ , which is positive definite. In the same way,  $n^{-1} \mathbf{S}^\top \mathbf{y} = n^{-1} \mathbf{S}^\top \mathbf{T}\theta + n^{-1} \mathbf{T}^\top \boldsymbol{\varepsilon} + n^{-1} \tilde{\mathbf{U}}^\top \boldsymbol{\varepsilon}$ . Here  $n^{-1} \mathbf{S}^\top \mathbf{T} \rightarrow \mathbf{A}$  by the argument just given,  $n^{-1} \mathbf{T}^\top \boldsymbol{\varepsilon} \rightarrow \mathbf{0}$  by Chebyshev's inequality, and  $n^{-1} \tilde{\mathbf{U}}^\top \boldsymbol{\varepsilon} \rightarrow \mathbf{0}$  by the weak law, since  $E(\tilde{u}_i \varepsilon_i) = 0$  by independence. Matrix inversion is continuous at a nonsingular matrix, which finishes the proof. ■

For a straight line the slope tends to  $\beta s_x^2 / (s_x^2 + \sigma_u^2)$ , where  $s_x^2$  is the limiting variance of the fixed values: what matters is the *spread* of the true values relative to the error. Because the number of parameters grows with  $n$ , maximum likelihood can fail in the functional model ([Example \(Many means, one variance\)](#), [Exercise \(C1\)](#)).

## 24.2.2 What the data cannot tell you

In the **normal structural model**,  $X \sim N(\mu_x, \sigma_x^2)$ ,  $U \sim N(0, \sigma_u^2)$  and  $\varepsilon \sim N(0, \sigma^2)$  are independent,  $W = X + U$  and  $Y = \alpha + \beta X + \varepsilon$ . Then  $(W, Y)$  is bivariate normal ([Theorem 3.3.1](#)) with mean  $(\mu_x, \alpha + \beta \mu_x)$  and covariance matrix

$$\begin{pmatrix} \sigma_{ww} & \sigma_{wy} \\ \sigma_{wy} & \sigma_{yy} \end{pmatrix} = \begin{pmatrix} \sigma_x^2 + \sigma_u^2 & \beta \sigma_x^2 \\ \beta \sigma_x^2 & \beta^2 \sigma_x^2 + \sigma^2 \end{pmatrix}. \quad (24.2.1)$$

A bivariate normal distribution is fixed by five numbers, but the model has six parameters  $(\mu_x, \alpha, \beta, \sigma_x^2, \sigma_u^2, \sigma^2)$ , so data on  $(W, Y)$  cannot determine all six.

### Proposition 24.2.2 — The slope is identified only up to an interval

Let  $(W, Y)$  be bivariate normal with covariance matrix entries  $\sigma_{ww}, \sigma_{wy}, \sigma_{yy}$ , where  $\sigma_{ww} \sigma_{yy} > \sigma_{wy}^2$ . If  $\sigma_{wy} \neq 0$ , a number  $b$  is the slope of some normal structural model with  $\sigma_x^2 > 0$ ,  $\sigma_u^2 \geq 0$  and  $\sigma^2 \geq 0$  that produces this distribution iff  $b$  lies in the closed interval with end points

$$b_1 = \frac{\sigma_{wy}}{\sigma_{ww}}, \quad b_2 = \frac{\sigma_{yy}}{\sigma_{wy}}.$$

If  $\sigma_{wy} = 0$ , the only such slope is  $b = 0$ .

*Proof.* A normal distribution is determined by its mean and covariance matrix, so a model reproduces the distribution iff it reproduces both. The means can always be matched by  $\mu_x = EW$  and  $\alpha = EY - b\mu_x$ . By [Eq. 24.2.1](#) the covariances are matched iff

$$\sigma_x^2 + \sigma_u^2 = \sigma_{ww}, \quad b \sigma_x^2 = \sigma_{wy}, \quad b^2 \sigma_x^2 + \sigma^2 = \sigma_{yy}.$$

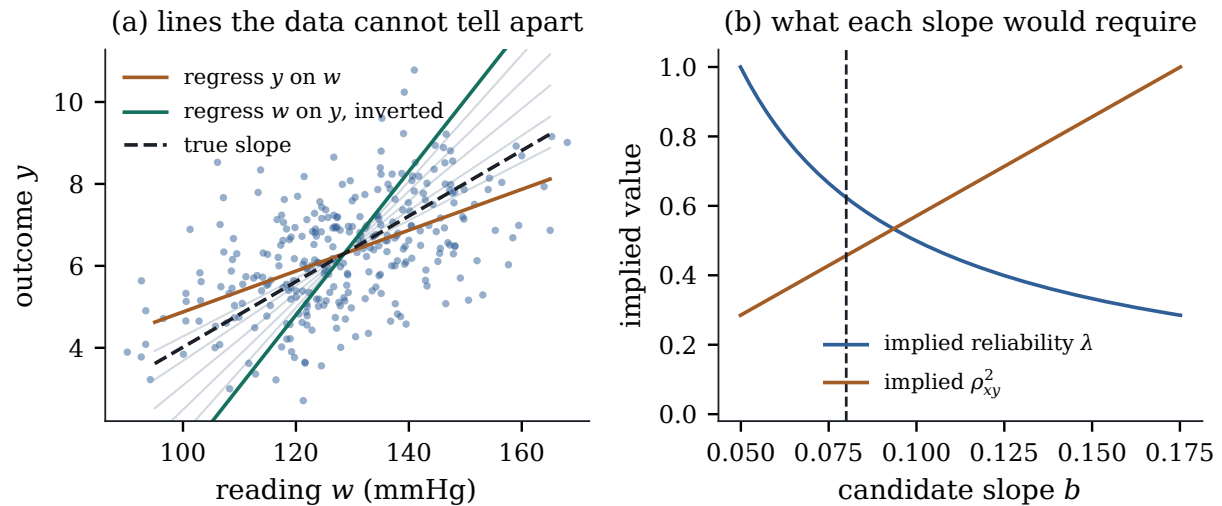


Figure 24.2.1: (a) Direct and reverse regression lines and slopes in between (grey), each fitting a normal structural model exactly; dashed, the true slope. (b) The reliability  $b_1/b$  and squared correlation  $b/b_2$  that slope  $b$  would require.

If  $\sigma_{wy} = 0$ , the middle equation with  $\sigma_x^2 > 0$  forces  $b = 0$ , and then  $\sigma_x^2 \in (0, \sigma_{ww}]$  and  $\sigma^2 = \sigma_{yy}$  complete a model. If  $\sigma_{wy} \neq 0$ , then  $b \neq 0$  and the equations force  $\sigma_x^2 = \sigma_{wy}/b$ ,  $\sigma_u^2 = \sigma_{ww} - \sigma_{wy}/b$  and  $\sigma^2 = \sigma_{yy} - b\sigma_{wy}$ . These are admissible iff  $b$  has the sign of  $\sigma_{wy}$  (so  $\sigma_x^2 > 0$ ),  $|b| \geq |\sigma_{wy}|/\sigma_{ww} = |b_1|$  (so  $\sigma_u^2 \geq 0$ ) and  $|b| \leq \sigma_{yy}/|\sigma_{wy}| = |b_2|$  (so  $\sigma^2 \geq 0$ ). Since  $b_1$  and  $b_2$  share the sign of  $\sigma_{wy}$ , these conditions say that  $b$  lies between  $b_1$  and  $b_2$ . The interval is not empty, because  $|b_1| \leq |b_2|$  is equivalent to  $\sigma_{wy}^2 \leq \sigma_{ww}\sigma_{yy}$ .

■

The end point  $b_1$  is the least squares slope of  $Y$  on  $W$ , correct if there is no measurement error. The end point  $b_2$  is the inverted slope of  $W$  on  $Y$ , correct if there is no equation error. Their ratio is the squared correlation, so the interval is short only when the correlation is high.

### Example. (The interval for the blood-pressure study)

For the first reading in the running example, the sample covariances are  $s_{ww} = 216.21$ ,  $s_{wy} = 10.786$  and  $s_{yy} = 1.889$ . The compatible slopes form the interval  $[0.0499, 0.1752]$ , and the squared correlation is 0.285. It contains the true 0.08, whose implied variances  $\sigma_x^2 = 134.8$ ,  $\sigma_u^2 = 81.4$  and  $\sigma^2 = 1.026$  are close to the generating 144, 81 and 1, but the data cannot prefer them to those of any other slope in the interval (Figure 24.2.1).

```
import numpy as np

rng = np.random.default_rng(2401)
n = 300
age = rng.normal(55, 10, n)
x = 130 + 0.6 * (age - 55) + rng.normal(0, np.sqrt(108), n)  # long-run blood
                    pressure, sd 12
w = x[:, None] + rng.normal(0, 9, (n, 2))                  # two clinic readings
y = -4 + 0.08 * x + 0.0 * age + rng.normal(0, 1.0, n)      # age has no effect
                    given x

S = np.cov(w[:, 0], y)                                     # sample covariance matrix of (reading,
```

```

outcome)
s_ww, s_wy, s_yy = S[0, 0], S[0, 1], S[1, 1]
b_direct = s_wy / s_ww          # assumes no measurement error (sigma_u
                                ^2 = 0)
b_reverse = s_yy / s_wy        # assumes no equation error (sigma^2 =
                                0)
print(f"slopes compatible with the data: [{b_direct:.4f}, {b_reverse:.4f}]")

for b in [b_direct, 0.08, b_reverse]:          # the variances each slope would imply
    sx2 = s_wy / b
    print(f"b = {b:.4f}: sigma_x^2 = {sx2:7.2f}, sigma_u^2 = {s_ww - sx2:7.2f}, "
          f"sigma^2 = {s_yy - b * s_wy:.4f}")

```

The sign of  $\beta$ , whether  $\beta = 0$ , and the lower bound  $|b_1|$  on  $|\beta|$  are identified. Any information that pins down one variance identifies everything:  $\sigma_u^2$ ,  $\lambda$ , the ratio  $\sigma^2/\sigma_u^2$ , or a bound  $\lambda \geq \lambda_0$  that narrows the interval ([Exercise \(A2\)](#)). The rest of the chapter is about where such information comes from.

*Remark* (Non-normal true values). Reiersol (1950) showed that when  $\beta \neq 0$  and the errors are normal and independent of  $X$ , the slope is identified exactly when  $X$  is *not* normal; higher moments then carry the missing information. We do not prove this. Estimators based on higher moments are highly variable unless  $X$  is far from normal, and rarely replace external information.

### 24.2.3 Berkson error

Sometimes the truth scatters around the recorded value instead: a furnace set to  $w$  degrees, every resident of a district assigned the district monitor's pollution level, a solution prepared at a nominal concentration. Then

$$X = W + U, \quad U \text{ independent of } W,$$

which is **Berkson error** (Berkson, 1950). The difference from classical error is which variable the error is independent of. It cannot be settled from data on  $(W, Y)$ , only from how the recorded value was produced. The scalar case with fixed settings appeared in [Section 19.5](#); the proposition below allows several regressors and random recorded values.

#### Proposition 24.2.3 — Berkson error in linear regression

Let  $Y = \alpha + \beta^\top \mathbf{x} + \varepsilon$  and  $\mathbf{x} = \mathbf{w} + \mathbf{u}$ , where  $\mathbf{w}$  is fixed or random and, given  $\mathbf{w}$ , the errors  $\mathbf{u}$  and  $\varepsilon$  are independent, with means zero, covariance matrix  $\Sigma_{uu}$  and variance  $\sigma^2$ , not depending on  $\mathbf{w}$ . Then, given  $\mathbf{w}$ ,

$$E(Y | \mathbf{w}) = \alpha + \beta^\top \mathbf{w}, \quad \text{Var}(Y | \mathbf{w}) = \sigma^2 + \beta^\top \Sigma_{uu} \beta,$$

and if  $\mathbf{u}$  and  $\varepsilon$  are normal, so is  $Y$  given  $\mathbf{w}$ . Independent observations of this kind therefore satisfy the linear model in the recorded  $\mathbf{w}$ , with the same coefficients and error variance  $\sigma^2 + \beta^\top \Sigma_{uu} \beta$ .

*Proof.* Given  $\mathbf{w}$ ,  $Y = \alpha + \beta^\top \mathbf{w} + (\beta^\top \mathbf{u} + \varepsilon)$ , and the composite error is a linear combination of the independent  $\mathbf{u}$  and  $\varepsilon$  ([Theorem 2.2.1](#)).

So least squares on the settings is unbiased, and under normality the  $t$  and  $F$  procedures are exact, with  $s^2$  estimating the enlarged error variance. Nothing needs correcting.

### Example. (Concentrations set on a dial)

Five solutions are prepared at each nominal concentration 10, 20, ..., 60 mg/L ( $n = 30$ ); the achieved concentration adds  $N(0, 4^2)$  error, and the response is  $2 + 0.5x + \varepsilon$ ,  $\varepsilon \sim N(0, 3^2)$ . Regressing on the nominal concentrations in 20000 simulated experiments gives a mean slope of 0.4994 and a mean  $s^2$  of 13.01, matching  $\sigma^2 + \beta^2 \sigma_u^2 = 9 + 0.25 \cdot 16 = 13$ . The nominal 95%  $t$  intervals for the slope covered 0.5 in a proportion 0.952 of the experiments.

```
from scipy import stats

rng_b = np.random.default_rng(2410)
settings = np.repeat([10.0, 20.0, 30.0, 40.0, 50.0, 60.0], 5)      # dial settings,
    n = 30
Xd = np.column_stack([np.ones(30), settings])
reps = 20_000
achieved = settings + rng_b.normal(0, 4, (reps, 30))              # x = w + u, u
    independent of w
Y = 2 + 0.5 * achieved + rng_b.normal(0, 3, (reps, 30))
B = np.linalg.solve(Xd.T @ Xd, Xd.T @ Y.T)                        # least squares
    on the settings
s2 = np.sum((Y.T - Xd @ B) ** 2, axis=0) / 28
se = np.sqrt(s2 * np.linalg.inv(Xd.T @ Xd)[1, 1])
covered = np.abs(B[1] - 0.5) <= stats.t.ppf(0.975, 28) * se
print(f"mean slope {B[1].mean():.4f}, mean s^2 {s2.mean():.2f} (sigma^2 + beta^2
    sigma_u^2 = 13), "
      f"coverage {covered.mean():.3f}")
```

*Warning.* Berkson error is harmless only because the model is linear in  $\mathbf{x}$ . For nonlinear  $g$ ,  $E\{g(\mathbf{x}) \mid \mathbf{w}\} \neq g(\mathbf{w})$  in general ([Exercise B1](#)). Real measurements often mix Berkson and classical components, which must then be handled separately.

## 24.2.4 Exercises

### 24.2.4.1 A. Check your understanding

#### Exercise A1.

Classify each regressor error as classical, Berkson or neither, and say whether it biases the slope: (a) weight from a noisy scale; (b) a prescribed dose, when patients take a random fraction of it with mean one; (c) income rounded down to the nearest 10,000.

**Exercise A2.**

In [Proposition 24.2.2](#), suppose it is known only that the reliability is at least  $\lambda_0 \in (0, 1]$ . Show that the compatible slopes are those between  $b_1$  and  $\min(|b_2|, |b_1|/\lambda_0)$  in magnitude, with the sign of  $b_1$ . Evaluate the interval for [Example \(The interval for the blood-pressure study\)](#) with  $\lambda_0 = 0.5$ .

*Solution.* The implied reliability of slope  $b$  is  $\sigma_x^2/\sigma_{ww} = b_1/b$ , so  $\lambda \geq \lambda_0$  iff  $|b| \leq |b_1|/\lambda_0$ . Intersect with the interval of the proposition. For the example,  $b_1/\lambda_0 = 0.0499/0.5 = 0.0998 < 0.1752$ , so the interval shrinks to  $[0.0499, 0.0998]$ , which still contains 0.08.

**24.2.4.2 B. Practice****Exercise B1.**

Let  $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \varepsilon$  and  $X = W + U$  with Berkson error  $U \sim N(0, \sigma_u^2)$  independent of  $W$  and  $\varepsilon$ . Compute  $E(Y | W)$ . Which coefficients of a cubic regression on  $W$  are biased, and by how much? What happens for a quadratic?

*Solution.* With  $EU = EU^3 = 0$  and  $EU^2 = \sigma_u^2$ ,  $E(X^2 | W) = W^2 + \sigma_u^2$  and  $E(X^3 | W) = W^3 + 3W\sigma_u^2$ . So

$$E(Y | W) = (\beta_0 + \beta_2 \sigma_u^2) + (\beta_1 + 3\beta_3 \sigma_u^2)W + \beta_2 W^2 + \beta_3 W^3.$$

The cubic regression on  $W$  estimates the intercept with bias  $\beta_2 \sigma_u^2$  and the linear coefficient with bias  $3\beta_3 \sigma_u^2$ . The top two coefficients are unbiased. For a quadratic ( $\beta_3 = 0$ ) only the intercept is biased. The conditional variance also changes with  $W$ , so the errors are heteroscedastic.

**Exercise B2.**

In the normal structural model, show that the reciprocal of the least squares slope of  $W$  on  $Y$  converges to  $b_2 = \beta + \sigma^2/(\beta\sigma_x^2)$ , so that it overstates  $|\beta|$  exactly when there is equation error. Evaluate  $b_2$  for the population of the running example and compare it with [Example \(The interval for the blood-pressure study\)](#).

**24.2.4.3 C. Going deeper****Exercise C1.**

In the functional model  $w_i = x_i + u_i$ ,  $y_i = \alpha + \beta x_i + \varepsilon_i$ , with  $u_i, \varepsilon_i$  independent  $N(0, \sigma^2)$  (equal, unknown variances) and  $x_1, \dots, x_n$  unknown constants:

- show that, for fixed  $(\alpha, \beta)$ , maximizing the likelihood over each  $x_i$  leaves  $d_i^2 = (y_i - \alpha - \beta w_i)^2/(1 + \beta^2)$ , the squared perpendicular distance from  $(w_i, y_i)$  to the line;
- deduce that the maximum likelihood line is the orthogonal regression line of [Section 24.3](#), and that  $\hat{\sigma}_{ML}^2 = (2n)^{-1} \sum_i \hat{d}_i^2$ ;
- show that  $\hat{\sigma}_{ML}^2 \rightarrow \sigma^2/2$  in probability, assuming the line is estimated consistently.

Compare [Example \(Many means, one variance\)](#).

*Solution.* (a) The log likelihood is  $-n \log(2\pi\sigma^2) - \sum_i \{(w_i - x_i)^2 + (y_i - \alpha - \beta x_i)^2\} / (2\sigma^2)$ . For each  $i$  the bracket is the squared distance from  $(w_i, y_i)$  to the point  $(x_i, \alpha + \beta x_i)$  of the line, and its minimum over  $x_i$  is the squared perpendicular distance  $d_i^2$ . (b) The profile log likelihood is  $-n \log(2\pi\sigma^2) - \sum_i d_i^2 / (2\sigma^2)$ . It is maximized over the line by minimizing  $\sum_i d_i^2$ , which is orthogonal regression, and over  $\sigma^2$  by  $\sum_i d_i^2 / (2n)$ . (c) At the true line,  $y_i - \alpha - \beta w_i = \varepsilon_i - \beta u_i$  has variance  $\sigma^2(1 + \beta^2)$ , so  $n^{-1} \sum_i d_i^2 \rightarrow \sigma^2$  by the law of large numbers. A consistent estimate of the line gives the same limit, so  $\hat{\sigma}_{\text{ML}}^2 \rightarrow \sigma^2/2$ : each nuisance  $x_i$  uses up half of its observation's two degrees of freedom.

## 24.3 Reliability and moment correction

This section supplies the missing information of [Section 24.2](#) directly, as a reliability ratio, an error covariance or a variance ratio, and solves [Theorem 24.1.1](#) for  $\theta$ .

### 24.3.1 Correcting with a known reliability

If  $\lambda$  is known, for instance from a test–retest correlation ([Exercise \(A1\)](#) in [Section 24.1](#)), then  $\hat{\beta}/\lambda$  is consistent in simple regression. For correlations this is Spearman's (1904) **correction for attenuation**,  $\text{Corr}(X_1, X_2) = \text{Corr}(W_1, W_2) / \sqrt{\lambda_1 \lambda_2}$  ([Exercise \(A1\)](#)). With covariates the right divisor is  $\lambda_{x|z}$  ([Exercise \(B1\)](#)). Dividing by  $\lambda$  undercorrects and leaves the covariates' bias untouched.

### 24.3.2 A known error covariance: the corrected estimator

Work with raw second moments, so the intercept needs no special treatment. Collect the true regressors, *including* the constant, in  $\mathbf{t} = (1, \mathbf{x}^\top, \mathbf{z}^\top)^\top \in \mathbb{R}^p$ , and the observed ones in  $\mathbf{s} = \mathbf{t} + \tilde{\mathbf{u}}$ , where  $\tilde{\mathbf{u}} = (0, \mathbf{u}^\top, \mathbf{0}^\top)^\top$  has zeros for the constant and for the exactly measured  $\mathbf{z}$ . Write the model as  $Y = \mathbf{t}^\top \theta + \varepsilon$ . Under classical error,

$$E(\mathbf{s}\mathbf{s}^\top) = E(\mathbf{t}\mathbf{t}^\top) + \tilde{\Sigma}, \quad E(\mathbf{s}Y) = E(\mathbf{t}\mathbf{t}^\top)\theta, \quad \tilde{\Sigma} = E(\tilde{\mathbf{u}}\tilde{\mathbf{u}}^\top),$$

since the cross terms vanish. So  $\theta = \{E(\mathbf{s}\mathbf{s}^\top) - \tilde{\Sigma}\}^{-1} E(\mathbf{s}Y)$ : subtract the error's share of the moment matrix before inverting. To cover a known  $\tilde{\Sigma}$  and one estimated from replicates at once, let observation  $i$  carry a matrix  $\mathbf{D}_i$ , computed from its own data, with expectation  $\tilde{\Sigma}$ . If  $\tilde{\Sigma}$  is known,  $\mathbf{D}_i = \tilde{\Sigma}$ .

#### Theorem 24.3.1 — Method-of-moments correction

Let  $(\mathbf{t}_i, \tilde{\mathbf{u}}_i, \varepsilon_i, \mathbf{D}_i)$ ,  $i = 1, 2, \dots$ , be independent copies of  $(\mathbf{t}, \tilde{\mathbf{u}}, \varepsilon, \mathbf{D})$ , where  $\mathbf{s}_i = \mathbf{t}_i + \tilde{\mathbf{u}}_i$  and  $y_i = \mathbf{t}_i^\top \theta + \varepsilon_i$  are observed, and  $\mathbf{D}_i$  is a symmetric  $p \times p$  matrix computed from the data of observation  $i$ . Assume that  $E(\varepsilon) = 0$ ,  $E(\mathbf{t}\varepsilon) = \mathbf{0}$ ,  $E(\tilde{\mathbf{u}}\varepsilon) = \mathbf{0}$ ,  $E(\mathbf{t}\tilde{\mathbf{u}}^\top) = \mathbf{0}$ ,  $E(\mathbf{D}) = \tilde{\Sigma} = E(\tilde{\mathbf{u}}\tilde{\mathbf{u}}^\top)$ , and that  $\mathbf{A} = E(\mathbf{t}\mathbf{t}^\top)$  is positive definite. Let

$$\hat{\mathbf{A}}_n = \frac{1}{n} \sum_{i=1}^n (\mathbf{s}_i \mathbf{s}_i^\top - \mathbf{D}_i), \quad \hat{\boldsymbol{\theta}}_{\text{mm}} = \hat{\mathbf{A}}_n^{-1} \frac{1}{n} \sum_{i=1}^n \mathbf{s}_i y_i. \quad (24.3.1)$$

- (a) With probability one,  $\hat{\mathbf{A}}_n$  is nonsingular for all large  $n$  and  $\hat{\boldsymbol{\theta}}_{\text{mm}} \rightarrow \boldsymbol{\theta}$ .  
 (b) If  $\mathbf{t}$ ,  $\tilde{\mathbf{u}}$  and  $\varepsilon$  have finite fourth moments and  $\mathbf{D}$  has finite second moments, then

$$\sqrt{n}(\hat{\boldsymbol{\theta}}_{\text{mm}} - \boldsymbol{\theta}) \rightarrow N_p(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}) \quad \text{in distribution,}$$

where  $\mathbf{B} = E(\boldsymbol{\psi} \boldsymbol{\psi}^\top)$  and  $\boldsymbol{\psi} = \mathbf{s}(y - \mathbf{s}^\top \boldsymbol{\theta}) + \mathbf{D} \boldsymbol{\theta}$ .

- (c) Under the conditions of (b),  $\hat{\mathbf{A}}_n^{-1} \hat{\mathbf{B}}_n \hat{\mathbf{A}}_n^{-1} \rightarrow \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1}$  in probability, where  $\hat{\mathbf{B}}_n = n^{-1} \sum_i \hat{\boldsymbol{\psi}}_i \hat{\boldsymbol{\psi}}_i^\top$  and  $\hat{\boldsymbol{\psi}}_i$  is  $\boldsymbol{\psi}_i$  with  $\hat{\boldsymbol{\theta}}_{\text{mm}}$  in place of  $\boldsymbol{\theta}$ .

*Proof.* (a) By the strong law of large numbers,  $n^{-1} \sum_i \mathbf{s}_i \mathbf{s}_i^\top \rightarrow E(\mathbf{s} \mathbf{s}^\top) = \mathbf{A} + \tilde{\boldsymbol{\Sigma}}$  (the cross terms vanish since  $E(\mathbf{t} \tilde{\mathbf{u}}^\top) = \mathbf{0}$ ) and  $n^{-1} \sum_i \mathbf{D}_i \rightarrow \tilde{\boldsymbol{\Sigma}}$ . So  $\hat{\mathbf{A}}_n \rightarrow \mathbf{A}$ , which is nonsingular, and  $\hat{\mathbf{A}}_n$  is nonsingular for all large  $n$  because the determinant is continuous. Also  $n^{-1} \sum_i \mathbf{s}_i y_i \rightarrow E\{(\mathbf{t} + \tilde{\mathbf{u}})(\mathbf{t}^\top \boldsymbol{\theta} + \varepsilon)\} = \mathbf{A} \boldsymbol{\theta}$ . By continuity of the inverse,  $\hat{\boldsymbol{\theta}}_{\text{mm}} \rightarrow \mathbf{A}^{-1} \mathbf{A} \boldsymbol{\theta} = \boldsymbol{\theta}$ .

(b) Directly from the definitions,

$$\frac{1}{n} \sum_i \mathbf{s}_i y_i - \hat{\mathbf{A}}_n \boldsymbol{\theta} = \frac{1}{n} \sum_i \{\mathbf{s}_i y_i - \mathbf{s}_i \mathbf{s}_i^\top \boldsymbol{\theta} + \mathbf{D}_i \boldsymbol{\theta}\} = \frac{1}{n} \sum_i \boldsymbol{\psi}_i,$$

so  $\sqrt{n}(\hat{\boldsymbol{\theta}}_{\text{mm}} - \boldsymbol{\theta}) = \hat{\mathbf{A}}_n^{-1} n^{-1/2} \sum_i \boldsymbol{\psi}_i$ . The  $\boldsymbol{\psi}_i$  are independent and identically distributed with mean

$$E(\boldsymbol{\psi}) = E(\mathbf{s}Y) - E(\mathbf{s} \mathbf{s}^\top) \boldsymbol{\theta} + E(\mathbf{D}) \boldsymbol{\theta} = \mathbf{A} \boldsymbol{\theta} - (\mathbf{A} + \tilde{\boldsymbol{\Sigma}}) \boldsymbol{\theta} + \tilde{\boldsymbol{\Sigma}} \boldsymbol{\theta} = \mathbf{0}.$$

Each entry of  $\boldsymbol{\psi}$  is a sum of products of at most two entries of  $(\mathbf{t}, \tilde{\mathbf{u}}, \varepsilon)$ , plus a linear function of  $\mathbf{D}$ , so the moment conditions give  $\mathbf{B}$  finite entries. By the central limit theorem (the remark in [Section 14.4](#)),  $\boldsymbol{\xi}_n = n^{-1/2} \sum_i \boldsymbol{\psi}_i \rightarrow N_p(\mathbf{0}, \mathbf{B})$ . Split

$$\hat{\mathbf{A}}_n^{-1} \boldsymbol{\xi}_n = \mathbf{A}^{-1} \boldsymbol{\xi}_n + (\hat{\mathbf{A}}_n^{-1} - \mathbf{A}^{-1}) \boldsymbol{\xi}_n.$$

By continuous mapping ([Section 19.3](#)) the first term converges in distribution to  $\mathbf{A}^{-1} N_p(\mathbf{0}, \mathbf{B}) = N_p(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1})$ . Each entry of the second is a sum of products of an entry of  $\hat{\mathbf{A}}_n^{-1} - \mathbf{A}^{-1}$ , tending to 0, and an entry of  $\boldsymbol{\xi}_n$ ; by the scalar Slutsky lemma it tends to 0 in probability. The sum form of Slutsky's lemma finishes the proof.

(c) Write  $\mathbf{G}_i = \mathbf{s}_i \mathbf{s}_i^\top - \mathbf{D}_i$  and  $\boldsymbol{\delta} = \hat{\boldsymbol{\theta}}_{\text{mm}} - \boldsymbol{\theta}$ . Then  $\hat{\boldsymbol{\psi}}_i = \boldsymbol{\psi}_i - \mathbf{G}_i \boldsymbol{\delta}$  and

$$\hat{\mathbf{B}}_n = \frac{1}{n} \sum_i \boldsymbol{\psi}_i \boldsymbol{\psi}_i^\top - \frac{1}{n} \sum_i (\boldsymbol{\psi}_i \boldsymbol{\delta}^\top \mathbf{G}_i^\top + \mathbf{G}_i \boldsymbol{\delta} \boldsymbol{\psi}_i^\top) + \frac{1}{n} \sum_i \mathbf{G}_i \boldsymbol{\delta} \boldsymbol{\delta}^\top \mathbf{G}_i^\top.$$

The first term tends to  $\mathbf{B}$  by the strong law. Every entry of the second is at most  $2 \|\boldsymbol{\delta}\| n^{-1} \sum_i \|\boldsymbol{\psi}_i\| \|\mathbf{G}_i\|$  in absolute value (with the Frobenius norm for  $\mathbf{G}_i$ ). The average converges to  $E(\|\boldsymbol{\psi}\| \|\mathbf{G}\|)$ , which is finite by the Cauchy–Schwarz inequality, and  $\|\boldsymbol{\delta}\| \rightarrow 0$



by (a). Similarly the third term is at most  $\|\delta\|^2 n^{-1} \sum_i \|\mathbf{G}_i\|^2 \rightarrow 0$ . Combine with  $\hat{\mathbf{A}}_n \rightarrow \mathbf{A}$ .

■

In simple regression with known error variance, Eq. 24.3.1 is  $s_{wy}/(s_{ww} - \sigma_u^2)$ , with sample moments taken with divisor  $n$ , the naive slope divided by the estimated reliability. The covariance  $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$  has the sandwich form of Theorem 21.4.2, to which it reduces when  $\tilde{\mathbf{u}} = \mathbf{0}$  and  $\mathbf{D} = \mathbf{0}$ . No assumption about the distribution of  $X$  or homoscedasticity is needed.

**Replicates.** Suppose each unit has  $m \geq 2$  readings  $\mathbf{w}_{ij} = \mathbf{x}_i + \mathbf{u}_{ij}$ ,  $j = 1, \dots, m$ , with errors independent across  $j$ , independent of  $(\mathbf{x}_i, \mathbf{z}_i, \varepsilon_i)$ , and with covariance  $\Sigma_{uu}$ . Use the mean  $\bar{\mathbf{w}}_i$  as the observed regressor. Its error  $\tilde{\mathbf{u}}_i$  has covariance  $\Sigma_{uu}/m$ , and an unbiased estimate of that covariance from unit  $i$  alone is

$$\frac{1}{m(m-1)} \sum_{j=1}^m (\mathbf{w}_{ij} - \bar{\mathbf{w}}_i)(\mathbf{w}_{ij} - \bar{\mathbf{w}}_i)^\top,$$

padded with zeros to make  $\mathbf{D}_i$  (Exercise A2). For  $m = 2$  and a scalar regressor this is  $(w_{i1} - w_{i2})^2/4$ . Because  $\mathbf{D}_i$  enters  $\boldsymbol{\psi}_i$ , the sandwich of (c) includes the uncertainty from estimating the error variance. An external estimate treated as known would need that uncertainty added separately.

#### Example. (Correcting with two readings)

In the running example, regress  $y$  on  $(1, \bar{w}, \text{age})$ , where  $\bar{w}$  is the mean of the two readings. The reliability of the mean is  $144/(144 + 40.5) = 0.780$ , and given age it is  $108/(108 + 40.5) = 0.727$ . The naive coefficients are 0.0562 (standard error 0.0054) for blood pressure and 0.0120 (standard error 0.0070) for age. The replicates estimate  $\sigma_u^2$  as 74.1 (true 81), and Eq. 24.3.1 gives

$$\tilde{\beta}_x = 0.0757 \text{ (0.0076)}, \quad \tilde{\beta}_{\text{age}} = 0.0018 \text{ (0.0076)},$$

with sandwich standard errors. The 95% interval for  $\beta_x$  is [0.0608, 0.0905], and the spurious age effect is gone. The standard error is 1.40 times the naive one: the bias is paid for in variance. With  $\sigma_u^2 = 81$  known the estimate would be 0.0782. In 10000 repeated samples the corrected intervals covered 0.08 in a proportion 0.949, the naive ones in 0.013.

```
import numpy as np

rng = np.random.default_rng(2401)
n = 300
age = rng.normal(55, 10, n)
x = 130 + 0.6 * (age - 55) + rng.normal(0, np.sqrt(108), n) # long-run blood
                    pressure, sd 12
w = x[:, None] + rng.normal(0, 9, (n, 2)) # two clinic readings
y = -4 + 0.08 * x + 0.0 * age + rng.normal(0, 1.0, n) # age has no effect
              given x

def corrected(w, age, y):
    """Moment-corrected coefficients of (1, x, age) and their sandwich standard
    errors."""
    n = len(y)
```

```

W = np.column_stack([np.ones(n), w.mean(axis=1), age]) # observed regressors
D = np.zeros((n, 3, 3))
D[:, 1, 1] = (w[:, 0] - w[:, 1]) ** 2 / 4 # E D_i = Var(error
of the mean)
A = (W.T @ W - D.sum(axis=0)) / n
theta = np.linalg.solve(A, W.T @ y / n)
psi = W * (y - W @ theta)[:, None] + D @ theta # estimating
function, one row per patient
B = psi.T @ psi / n
Ainv = np.linalg.inv(A)
cov = Ainv @ B @ Ainv / n
return theta, np.sqrt(np.diag(cov))

theta, se = corrected(w, age, y)
W = np.column_stack([np.ones(n), w.mean(axis=1), age])
naive = np.linalg.lstsq(W, y, rcond=None)[0]
s2_u = np.mean((w[:, 0] - w[:, 1]) ** 2) / 2 # estimate of sigma_u
^2 (true 81)
print(f"sigma_u^2 estimate {s2_u:.1f}")
print("naive      (1, x, age):", naive.round(4))
print("corrected (1, x, age):", theta.round(4), " se:", se.round(4))

```

```

def coverage(reps, seed):
    """Coverage of nominal 95% intervals for beta_x = 0.08: corrected (sandwich) and
    naive."""
    r = np.random.default_rng(seed)
    hits_c = hits_n = 0
    for _ in range(reps):
        a = r.normal(55, 10, n)
        xx = 130 + 0.6 * (a - 55) + r.normal(0, np.sqrt(108), n)
        ww = xx[:, None] + r.normal(0, 9, (n, 2))
        yy = -4 + 0.08 * xx + r.normal(0, 1.0, n)
        th, s = corrected(ww, a, yy)
        hits_c += abs(th[1] - 0.08) <= 1.96 * s[1]
        Wr = np.column_stack([np.ones(n), ww.mean(axis=1), a])
        b = np.linalg.lstsq(Wr, yy, rcond=None)[0]
        e = yy - Wr @ b
        sb = np.sqrt(e @ e / (n - 3) * np.linalg.inv(Wr.T @ Wr)[1, 1])
        hits_n += abs(b[1] - 0.08) <= 1.96 * sb
    return hits_c / reps, hits_n / reps

print("coverage (corrected, naive):", coverage(500, 2421))

```

*Warning.* The theorem is asymptotic, and the correction is fragile when the reliability is low.  $\hat{\mathbf{A}}_n$  can fail to be positive definite, and under normality the corrected slope has no finite mean ([Exercise \(C1\)](#)), as with [Proposition 12.5.2](#); Fuller (1987) gives modified estimators. Gleser and Hwang (1987) showed that, as in [Proposition 12.5.4](#), any confidence set with guaranteed coverage has infinite expected length. Wald intervals are reliable only when the estimated reliability is well away from zero.

### 24.3.3 Deming and orthogonal regression

Sometimes only a *ratio* of error variances is known. In a method-comparison study two instruments measure the same specimens,  $W = X + U$  and  $Y = \alpha + \beta X + \varepsilon$ , with  $\varepsilon$  the second

instrument's error, and repeatability experiments give

$$\delta = \frac{\text{Var}(\varepsilon)}{\text{Var}(U)}.$$

The question is whether  $\alpha = 0$  and  $\beta = 1$ .

### Proposition 24.3.2 — Deming regression

Let  $(W, Y)$  have covariances  $\sigma_{ww}, \sigma_{wy}, \sigma_{yy}$  with  $\sigma_{wy} \neq 0$  and  $\sigma_{ww}\sigma_{yy} \geq \sigma_{wy}^2$ , and let  $\delta > 0$  be known.

- (a) The equations  $\sigma_{ww} = \sigma_x^2 + \sigma_u^2$ ,  $\sigma_{wy} = \beta\sigma_x^2$ ,  $\sigma_{yy} = \beta^2\sigma_x^2 + \delta\sigma_u^2$  have exactly one solution with  $\sigma_x^2 > 0$ , namely

$$\beta_\delta = \frac{\sigma_{yy} - \delta\sigma_{ww} + \sqrt{(\sigma_{yy} - \delta\sigma_{ww})^2 + 4\delta\sigma_{wy}^2}}{2\sigma_{wy}}, \quad (24.3.2)$$

and for it  $\sigma_u^2 \geq 0$ . The slope  $\beta_\delta$  lies in the interval of Proposition 24.2.2. It tends to  $b_2$  as  $\delta \rightarrow 0$  and to  $b_1$  as  $\delta \rightarrow \infty$ .

- (b) For  $\delta = 1$ ,  $\beta_1$  is the slope of the line that minimizes the mean squared *perpendicular* distance from  $(W, Y)$ , whose direction is an eigenvector of the covariance matrix of  $(W, Y)$  for its largest eigenvalue. For general  $\delta$ ,  $\beta_\delta$  is  $\sqrt{\delta}$  times the  $\delta = 1$  slope computed for  $(W, Y/\sqrt{\delta})$ .

*Proof.* (a) For  $\beta \neq 0$  the first two equations give  $\sigma_x^2 = \sigma_{wy}/\beta$  and  $\sigma_u^2 = \sigma_{ww} - \sigma_{wy}/\beta$ . Substituting in the third and multiplying by  $\beta$ ,

$$q(\beta) = \sigma_{wy}\beta^2 + (\delta\sigma_{ww} - \sigma_{yy})\beta - \delta\sigma_{wy} = 0.$$

The product of the roots is  $-\delta < 0$ , so there is one root of each sign, and  $\sigma_x^2 > 0$  selects the root with the sign of  $\sigma_{wy}$ . That root is Eq. 24.3.2. Take  $\sigma_{wy} > 0$  (otherwise replace  $Y$  by  $-Y$ ). Then

$$q(b_1) = \frac{\sigma_{wy}}{\sigma_{ww}^2}(\sigma_{wy}^2 - \sigma_{ww}\sigma_{yy}) \leq 0, \quad q(b_2) = \frac{\delta}{\sigma_{wy}}(\sigma_{ww}\sigma_{yy} - \sigma_{wy}^2) \geq 0,$$

and  $q$  is an upward parabola with  $q(0) < 0$ , so its positive root lies in  $[b_1, b_2]$ . Then  $\sigma_u^2 = \sigma_{ww}(1 - b_1/\beta_\delta) \geq 0$ . As  $\delta \rightarrow 0$  the formula tends to  $(\sigma_{yy} + \sigma_{yy})/(2\sigma_{wy}) = b_2$ . Multiplying the numerator and denominator of Eq. 24.3.2 by the conjugate of the numerator gives  $\beta_\delta = 2\delta\sigma_{wy}/\{\sqrt{(\sigma_{yy} - \delta\sigma_{ww})^2 + 4\delta\sigma_{wy}^2} + \delta\sigma_{ww} - \sigma_{yy}\}$ , and the denominator is  $2\delta\sigma_{ww} + O(1)$ , so  $\beta_\delta \rightarrow b_1$  as  $\delta \rightarrow \infty$ .

(b) Let  $\Sigma$  be the covariance matrix of  $\mathbf{v} = (W, Y)^\top$ . A line through the point  $\mathbf{E}\mathbf{v}$  with unit normal vector  $\mathbf{n}$  has mean squared perpendicular distance  $\mathbf{E}\{\mathbf{n}^\top(\mathbf{v} - \mathbf{E}\mathbf{v})\}^2 = \mathbf{n}^\top \Sigma \mathbf{n}$ , and a line through any other point  $\mathbf{c}$  adds  $\{\mathbf{n}^\top(\mathbf{E}\mathbf{v} - \mathbf{c})\}^2 \geq 0$ . By Theorem 1.7.7,  $\mathbf{n}^\top \Sigma \mathbf{n}$  is minimized by an eigenvector for the smallest eigenvalue, so the line's direction is an eigenvector  $(v_1, v_2)$  for the largest eigenvalue  $\mu$ . From  $(\sigma_{ww} - \mu)v_1 + \sigma_{wy}v_2 = 0$ , the slope is  $v_2/v_1 = (\mu - \sigma_{ww})/\sigma_{wy}$ . Put  $b = v_2/v_1$ . Then  $\mu - \sigma_{ww} = b\sigma_{wy}$ , and the second eigenvector equation  $\sigma_{wy}v_1 + (\sigma_{yy} - \mu)v_2 = 0$  gives  $\mu - \sigma_{yy} = \sigma_{wy}/b$ . Subtracting,  $\sigma_{yy} - \sigma_{ww} = b\sigma_{wy} - \sigma_{wy}/b$ , which is  $q(b) = 0$  with  $\delta = 1$ . Since  $\mu \geq \sigma_{ww}$ ,  $b$  has the sign of  $\sigma_{wy}$ , so  $b = \beta_1$ . For the last claim, substitute  $\sigma_{yy}/\delta$  and  $\sigma_{wy}/\sqrt{\delta}$  for  $\sigma_{yy}$  and  $\sigma_{wy}$  in Eq. 24.3.2 with  $\delta = 1$  and multiply by  $\sqrt{\delta}$ .

With sample moments, Eq. 24.3.2 is **Deming regression**, consistent because it is continuous where  $\sigma_{wy} \neq 0$ . With  $\delta = 1$  it is **orthogonal regression** (total least squares), whose line is the first principal axis of the scatter.

*Warning.* Orthogonal regression makes sense only when the errors have equal variances in the units used. Changing the units of  $Y$  changes the fitted line by more than a change of scale. Deming regression, with  $\delta$  rescaled along with the units, does not have this problem.

### Example. (Comparing two assays)

Sixty specimens with true concentrations uniform on  $[2, 20]$  are measured by method A with error standard deviation 0.8 and by method B, which reads  $0.3 + 1.05x$  with error standard deviation 1.0, so  $\delta = 1.5625$ . The four slopes are

| Method                                   | Slope  |
|------------------------------------------|--------|
| least squares of B on A                  | 1.0042 |
| Deming, $\delta = 1.5625$                | 1.0308 |
| orthogonal ( $\delta = 1$ , unjustified) | 1.0382 |
| least squares of A on B, inverted        | 1.0709 |

with Deming intercept 0.482. Over 20000 simulated studies the mean slopes were 1.0262 for least squares, attenuated by  $27/(27 + 0.64) = 0.9768$ , and 1.0505 for Deming. With B in units ten times smaller, orthogonal regression gives 1.0703 (converted back to the original units) instead of 1.0382.

```
import numpy as np

def deming(w, y, delta):
    """Slope and intercept of Deming regression; delta = Var(error in y) / Var(error
    in w)."""
    S = np.cov(w, y)
    s_ww, s_wy, s_yy = S[0, 0], S[0, 1], S[1, 1]
    d = s_yy - delta * s_ww
    b = (d + np.sqrt(d ** 2 + 4 * delta * s_wy ** 2)) / (2 * s_wy)
    return b, y.mean() - b * w.mean()

rng = np.random.default_rng(2430)
x = rng.uniform(2, 20, 60)          # true concentrations
w = x + rng.normal(0, 0.8, 60)      # method A
y = 0.3 + 1.05 * x + rng.normal(0, 1.0, 60)  # method B
delta = 1.0 ** 2 / 0.8 ** 2

b_ls = np.polyfit(w, y, 1)[0]        # regress y on w
b_rev = 1 / np.polyfit(y, w, 1)[0]   # regress w on y, inverted
b_dem, a_dem = deming(w, y, delta)
b_orth, _ = deming(w, y, 1.0)        # orthogonal regression
print(f"least squares {b_ls:.4f}, Deming {b_dem:.4f} (intercept {a_dem:.3f}), "
```

```
f"orthogonal {b_orth:.4f}, reverse {b_rev:.4f}")
```

## 24.3.4 Exercises

### 24.3.4.1 A. Check your understanding

#### Exercise A1.

Prove Spearman's correction: if  $W_k = X_k + U_k$  for  $k = 1, 2$ , with classical errors independent of each other and of  $(X_1, X_2)$ , then  $\text{Corr}(W_1, W_2) = \sqrt{\lambda_1 \lambda_2} \text{Corr}(X_1, X_2)$ .

#### Exercise A2.

Show that the matrix  $\mathbf{D}_i$  defined from  $m$  replicates has expectation  $\Sigma_{uu}/m$ , the covariance matrix of  $\bar{\mathbf{u}}_i$ , padded with zeros.

*Solution.*  $\mathbf{w}_{ij} - \bar{\mathbf{w}}_i = \mathbf{u}_{ij} - \bar{\mathbf{u}}_i$ , so  $\mathbf{x}_i$  drops out. For independent vectors with common covariance  $\Sigma_{uu}$ , the sample covariance matrix  $(m-1)^{-1} \sum_j (\mathbf{u}_{ij} - \bar{\mathbf{u}}_i)(\mathbf{u}_{ij} - \bar{\mathbf{u}}_i)^\top$  is unbiased for  $\Sigma_{uu}$  (Proposition 2.7.1). Dividing by  $m$  gives expectation  $\Sigma_{uu}/m = \text{Cov}(\bar{\mathbf{u}}_i)$ .

### 24.3.4.2 B. Practice

#### Exercise B1.

For a scalar  $X$  and a scalar exactly measured  $Z$ , show that

$$\lambda_{x|z} = \frac{\lambda - \rho_{wz}^2}{1 - \rho_{wz}^2},$$

where  $\rho_{wz}$  is the correlation between  $W$  and  $Z$ . Check the formula on the running example, and deduce that  $\lambda_{x|z} < \lambda$  whenever  $\rho_{wz} \neq 0$  and  $\lambda < 1$ .

*Solution.*  $\sigma_{x \cdot z}^2 = \sigma_x^2 - \sigma_{xz}^2/\sigma_z^2$ , and since  $\sigma_{wz} = \sigma_{xz}$ ,  $\sigma_{xz}^2/\sigma_z^2 = \rho_{wz}^2 \sigma_w^2$ . Divide the numerator and the denominator of  $\lambda_{x|z} = \sigma_{x \cdot z}^2/(\sigma_{x \cdot z}^2 + \sigma_u^2)$  by  $\sigma_w^2$ . The numerator becomes  $\lambda - \rho_{wz}^2$  and the denominator  $\lambda - \rho_{wz}^2 + (1 - \lambda) = 1 - \rho_{wz}^2$ . In the example  $\rho_{wz}^2 = 60^2/(225 \cdot 100) = 0.16$ , so  $\lambda_{x|z} = (0.64 - 0.16)/0.84 = 0.571$ , as in Section 24.1. Finally  $(\lambda - \rho^2)/(1 - \rho^2) < \lambda$  iff  $\lambda - \rho^2 < \lambda - \lambda \rho^2$  iff  $\rho^2(1 - \lambda) > 0$ .

#### Exercise B2.

In simple regression with centred variables and known  $\sigma_u^2$ , take  $\mathbf{D} = \text{diag}(0, \sigma_u^2)$  in Theorem 24.3.1 and suppose  $(X, U, \varepsilon)$  are jointly normal and independent. Show that the asymptotic variance of  $\sqrt{n}(\hat{\beta} - \beta)$  is

$$\frac{\sigma_w^2 \sigma_v^2 + \beta^2 \sigma_u^4}{\sigma_x^4}, \quad \sigma_v^2 = \sigma^2 + \beta^2 \sigma_u^2.$$

Compare it with  $\sigma^2/\sigma_x^2$ , the asymptotic variance when  $X$  is observed. What does this

give for the running example with the mean of two readings?

*Solution.* The slope's estimating function (the intercept decouples for centred variables) is  $\psi = W(Y - W\beta) + \sigma_u^2\beta = Wv + \sigma_u^2\beta$  with  $v = \varepsilon - \beta U$ , and the slope entry of  $\mathbf{A}$  is  $\sigma_x^2$ . Now  $(W, v)$  is bivariate normal with  $\text{Cov}(W, v) = -\beta\sigma_u^2$ , and for such a pair  $E(W^2v^2) = \sigma_w^2\sigma_v^2 + 2\text{Cov}(W, v)^2$ . So  $\text{Var}(\psi) = E(W^2v^2) - \{E(Wv)\}^2 = \sigma_w^2\sigma_v^2 + \beta^2\sigma_u^4$ , and the sandwich is  $\text{Var}(\psi)/\sigma_x^4$ . It exceeds  $\sigma^2/\sigma_x^2$  because  $\sigma_w^2\sigma_v^2 > \sigma_x^2\sigma^2$ . For the example, use  $\sigma_u^2 = 40.5$  for the mean of two readings:  $\sigma_w^2 = 184.5$ ,  $\sigma_v^2 = 1 + 0.0064 \cdot 40.5 = 1.2592$ , and the variance is  $(184.5 \cdot 1.2592 + 0.0064 \cdot 1640.25)/144^2 \approx 0.01171$ , a standard error of about  $\sqrt{0.01171/300} \approx 0.0062$ , below the 0.0076 of [Example \(Correcting with two readings\)](#), where  $\sigma_u^2$  is estimated and age lowers the reliability.

### 24.3.4.3 C. Going deeper

#### Exercise C1.

In the normal structural model with known  $\sigma_u^2$  and  $n \geq 3$ , show that  $\tilde{\beta} = s_{wy}/(s_{ww} - \sigma_u^2)$  has  $E|\tilde{\beta}| = \infty$ . (Condition on  $W_1, \dots, W_n$ , and use the fact that  $s_{ww}$  has a continuous density that is positive at  $\sigma_u^2$ .)

*Solution.* Given the  $W_i$ ,  $(W_i, Y_i)$  is jointly normal and  $E(Y | W) = a + \lambda\beta W$  with constant conditional variance  $\tau^2 > 0$  ([Theorem 3.5.2](#)). So, given the  $W_i$ ,  $s_{wy} \sim N(\lambda\beta s_{ww}, \tau^2 s_{ww}/(n-1))$ , and  $E(|s_{wy}| | W) \geq \tau\sqrt{2s_{ww}/\{\pi(n-1)\}}$ , because a normal variable's absolute mean is at least that of its centred version. Hence  $E|\tilde{\beta}| \geq c E\{\sqrt{s_{ww}}/|s_{ww} - \sigma_u^2|\}$ . Here  $(n-1)s_{ww}/\sigma_w^2 \sim \chi^2(n-1)$ , whose density is continuous and positive on  $(0, \infty)$ . So the density of  $s_{ww}$  is bounded below near  $\sigma_u^2$ , and the integral of  $1/|s - \sigma_u^2|$  diverges there.

## 24.4 Instrumental variables

The corrections so far used a *number* from outside the regression. Instrumental variables use a *variable*, related to the mismeasured regressor and unrelated to everything that makes least squares fail. The idea goes back to Philip Wright (1928), who used it to estimate supply and demand curves.

### 24.4.1 Endogeneity

By [Eq. 24.1.2](#),  $Y = \alpha + \beta W + e$  with  $e = \varepsilon - \beta U$  correlated with  $W$ . An omitted common cause ([Theorem 19.1.2](#)) or a regressor partly determined by the response has the same effect. Such regressors are called **endogenous**. In every case the equation

$$y_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + e_i, \quad E(\mathbf{x}_{(i)} e_i) \neq \mathbf{0}, \quad (24.4.1)$$

describes a relation of scientific interest, not the best linear predictor that least squares estimates. Here  $\mathbf{x}_{(i)}$  holds all the regressors, including the constant. In this section  $\mathbf{x}$  collects all regressors and  $\mathbf{z}$  all instruments; the exactly measured covariates of earlier sections appear in both.

**Definition 24.4.1 — Instruments**

A random  $q$ -vector  $\mathbf{z}$  is a vector of **instruments** for Eq. 24.4.1, with random regressor  $p$ -vector  $\mathbf{x}$ , if

- (i) *exogeneity*:  $E(\mathbf{z}e) = \mathbf{0}$ ;
- (ii) *relevance*:  $\Sigma_{zx} = E(\mathbf{z}\mathbf{x}^\top)$  has rank  $p$ , which requires  $q \geq p$ ;

and  $\Sigma_{zz} = E(\mathbf{z}\mathbf{z}^\top)$  is positive definite.

Regressors uncorrelated with  $e$ , such as the constant, are their own instruments; the rank condition asks the *excluded* instruments to carry extra information about the endogenous regressors. With  $q = p$  the equation is **just identified**, and with  $q > p$  **overidentified**.

In the blood-pressure study the second reading instruments the first:  $\mathbf{x} = (1, W_1, \text{age})$ ,  $\mathbf{z} = (1, W_2, \text{age})$  and  $e = \varepsilon - \beta U_1$ , which is uncorrelated with  $W_2 = X + U_2$ . Nothing about  $\sigma_u^2$  needs to be known, but the two errors must be *independent*, which rules out two readings that share a transient cause (Exercise (A2)).

**24.4.2 The estimators**

With the  $n \times p$  matrix  $\mathbf{X}$  of regressors and the  $n \times q$  matrix  $\mathbf{Z}$  of instruments, the sample version of  $E\{\mathbf{z}(y - \mathbf{x}^\top \boldsymbol{\beta})\} = \mathbf{0}$  in the just-identified case is  $\mathbf{Z}^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0}$ , with solution

$$\hat{\boldsymbol{\beta}}_{\text{IV}} = (\mathbf{Z}^\top \mathbf{X})^{-1} \mathbf{Z}^\top \mathbf{y}.$$

When  $q > p$  these are more equations than unknowns. **Two-stage least squares** (2SLS) combines them through the projection  $\mathbf{M}_Z = \mathbf{Z}(\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top$  onto  $\mathcal{C}(\mathbf{Z})$ :

$$\hat{\boldsymbol{\beta}}_{2\text{SLS}} = (\mathbf{X}^\top \mathbf{M}_Z \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{M}_Z \mathbf{y}. \quad (24.4.2)$$

The name comes from a two-step computation.

**Lemma 24.4.1 — Two-stage least squares as projection**

Let  $\mathbf{Z}$  have full column rank and  $\mathbf{X}^\top \mathbf{M}_Z \mathbf{X}$  be nonsingular, and let  $\hat{\mathbf{X}} = \mathbf{M}_Z \mathbf{X}$  be the matrix of first-stage fitted values, obtained by regressing each column of  $\mathbf{X}$  on  $\mathbf{Z}$ .

- (a)  $\hat{\boldsymbol{\beta}}_{2\text{SLS}}$  is the least squares coefficient of  $\mathbf{y}$  on  $\hat{\mathbf{X}}$ , and also the IV estimator with instruments  $\hat{\mathbf{X}}$ :  $\hat{\boldsymbol{\beta}}_{2\text{SLS}} = (\hat{\mathbf{X}}^\top \hat{\mathbf{X}})^{-1} \hat{\mathbf{X}}^\top \mathbf{y} = (\hat{\mathbf{X}}^\top \mathbf{X})^{-1} \hat{\mathbf{X}}^\top \mathbf{y}$ .
- (b) Columns of  $\mathbf{X}$  that are also columns of  $\mathbf{Z}$  are unchanged by the first stage.
- (c) If  $q = p$  and  $\mathbf{Z}^\top \mathbf{X}$  is nonsingular,  $\hat{\boldsymbol{\beta}}_{2\text{SLS}} = \hat{\boldsymbol{\beta}}_{\text{IV}}$ .

*Proof.* (a)  $\mathbf{M}_Z$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{Z})$  (Theorem 6.3.7), so it is symmetric and idempotent (Theorem 6.3.3). Hence  $\hat{\mathbf{X}}^\top \hat{\mathbf{X}} = \mathbf{X}^\top \mathbf{M}_Z \mathbf{M}_Z \mathbf{X} = \mathbf{X}^\top \mathbf{M}_Z \mathbf{X}$ , and  $\hat{\mathbf{X}}^\top \mathbf{X} = \mathbf{X}^\top \mathbf{M}_Z \mathbf{X}$  and  $\hat{\mathbf{X}}^\top \mathbf{y} = \mathbf{X}^\top \mathbf{M}_Z \mathbf{y}$  as well. Substitute these in Eq. 24.4.2. (b) A column in  $\mathcal{C}(\mathbf{Z})$  is fixed by  $\mathbf{M}_Z$ . (c) With  $\mathbf{Z}^\top \mathbf{X}$  square and invertible,  $(\mathbf{X}^\top \mathbf{Z}(\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{X})^{-1} = (\mathbf{Z}^\top \mathbf{X})^{-1} (\mathbf{Z}^\top \mathbf{Z})(\mathbf{X}^\top \mathbf{Z})^{-1}$ , and multiplying by  $\mathbf{X}^\top \mathbf{Z}(\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{Z}^\top \mathbf{y}$  leaves  $(\mathbf{Z}^\top \mathbf{X})^{-1} \mathbf{Z}^\top \mathbf{y}$ .

The first stage replaces each endogenous regressor by the part of it that the instruments explain, which is uncorrelated with  $e$ . As with the residualized regressions of [Section 6.6](#), the second-stage *coefficients* are right but its *residuals*  $\mathbf{y} - \hat{\mathbf{X}}\hat{\boldsymbol{\beta}}$  are not; the equation's residuals are  $\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$ .

### Theorem 24.4.2 — Consistency and asymptotic normality of IV

Let  $(\mathbf{x}_{(i)}, \mathbf{z}_{(i)}, e_i)$  be independent copies of  $(\mathbf{x}, \mathbf{z}, e)$  with finite second moments, satisfying [Eq. 24.4.1](#) and [Definition 24.4.1](#). Let  $\mathbf{H} = \boldsymbol{\Sigma}_{xz}\boldsymbol{\Sigma}_{zz}^{-1}\boldsymbol{\Sigma}_{zx}$ , where  $\boldsymbol{\Sigma}_{xz} = \boldsymbol{\Sigma}_{zx}^\top$ .

- (a)  $\hat{\boldsymbol{\beta}}_{2\text{SLS}} \rightarrow \boldsymbol{\beta}$  with probability one.
- (b) If  $E(e^2|\mathbf{z}) < \infty$ , then  $\sqrt{n}(\hat{\boldsymbol{\beta}}_{2\text{SLS}} - \boldsymbol{\beta}) \rightarrow N_p(\mathbf{0}, \mathbf{V})$  in distribution, where

$$\mathbf{V} = \mathbf{H}^{-1}\boldsymbol{\Sigma}_{xz}\boldsymbol{\Sigma}_{zz}^{-1}\boldsymbol{\Omega}\boldsymbol{\Sigma}_{zz}^{-1}\boldsymbol{\Sigma}_{zx}\mathbf{H}^{-1}, \quad \boldsymbol{\Omega} = E(e^2\mathbf{z}\mathbf{z}^\top).$$

If  $E(e^2 | \mathbf{z}) = \sigma_e^2$  is constant, then  $\mathbf{V} = \sigma_e^2\mathbf{H}^{-1}$ .

- (c)  $\hat{\sigma}_e^2 = n^{-1}\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{2\text{SLS}}\|^2 \rightarrow \sigma_e^2 = E(e^2)$  with probability one.

*Proof.* (a) Write  $\hat{\mathbf{C}} = n^{-1}\mathbf{Z}^\top\mathbf{X}$  and  $\hat{\mathbf{S}} = n^{-1}\mathbf{Z}^\top\mathbf{Z}$ . Substituting  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  in [Eq. 24.4.2](#),

$$\hat{\boldsymbol{\beta}}_{2\text{SLS}} - \boldsymbol{\beta} = (\hat{\mathbf{C}}^\top\hat{\mathbf{S}}^{-1}\hat{\mathbf{C}})^{-1}\hat{\mathbf{C}}^\top\hat{\mathbf{S}}^{-1}\frac{1}{n}\mathbf{Z}^\top\boldsymbol{\varepsilon}.$$

By the strong law,  $\hat{\mathbf{C}} \rightarrow \boldsymbol{\Sigma}_{zx}$ ,  $\hat{\mathbf{S}} \rightarrow \boldsymbol{\Sigma}_{zz}$  and  $n^{-1}\mathbf{Z}^\top\boldsymbol{\varepsilon} \rightarrow E(\mathbf{z}e) = \mathbf{0}$ . The matrix  $\mathbf{H}$  is positive definite: for  $\mathbf{a} \neq \mathbf{0}$ ,  $\boldsymbol{\Sigma}_{zx}\mathbf{a} \neq \mathbf{0}$  by the rank condition, so  $\mathbf{a}^\top\mathbf{H}\mathbf{a} = (\boldsymbol{\Sigma}_{zx}\mathbf{a})^\top\boldsymbol{\Sigma}_{zz}^{-1}(\boldsymbol{\Sigma}_{zx}\mathbf{a}) > 0$ . By continuity the right side tends to  $\mathbf{H}^{-1}\boldsymbol{\Sigma}_{xz}\boldsymbol{\Sigma}_{zz}^{-1}\mathbf{0} = \mathbf{0}$ .

(b) Multiply the display by  $\sqrt{n}$ . The vectors  $\mathbf{z}_{(i)}e_i$  are independent and identically distributed with mean  $\mathbf{0}$  and covariance  $\boldsymbol{\Omega}$ , finite by assumption, so  $n^{-1/2}\mathbf{Z}^\top\boldsymbol{\varepsilon} \rightarrow N_q(\mathbf{0}, \boldsymbol{\Omega})$  by the central limit theorem. The matrix in front converges to  $\mathbf{L} = \mathbf{H}^{-1}\boldsymbol{\Sigma}_{xz}\boldsymbol{\Sigma}_{zz}^{-1}$ , and the splitting used for [Theorem 24.3.1\(b\)](#) gives the limit  $\mathbf{L}N_q(\mathbf{0}, \boldsymbol{\Omega}) = N_p(\mathbf{0}, \mathbf{V})$ . If  $E(e^2 | \mathbf{z}) = \sigma_e^2$ , then  $\boldsymbol{\Omega} = E\{E(e^2 | \mathbf{z})\mathbf{z}\mathbf{z}^\top\} = \sigma_e^2\boldsymbol{\Sigma}_{zz}$ , and  $\mathbf{V}$  collapses to  $\sigma_e^2\mathbf{H}^{-1}$ .

(c) With  $\boldsymbol{\delta} = \hat{\boldsymbol{\beta}}_{2\text{SLS}} - \boldsymbol{\beta}$ ,  $\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{2\text{SLS}} = \boldsymbol{\varepsilon} - \mathbf{X}\boldsymbol{\delta}$ , so

$$\hat{\sigma}_e^2 = \frac{1}{n}\boldsymbol{\varepsilon}^\top\boldsymbol{\varepsilon} - 2\boldsymbol{\delta}^\top\frac{1}{n}\mathbf{X}^\top\boldsymbol{\varepsilon} + \boldsymbol{\delta}^\top\left(\frac{1}{n}\mathbf{X}^\top\mathbf{X}\right)\boldsymbol{\delta}.$$

The averages converge to  $E(e^2)$ ,  $E(\mathbf{x}e)$  and  $E(\mathbf{x}\mathbf{x}^\top)$ , all finite, and  $\boldsymbol{\delta} \rightarrow \mathbf{0}$  by (a).

In practice  $\mathbf{V}$  is estimated with sample moments and  $\hat{\boldsymbol{\Omega}} = n^{-1}\sum_i \hat{e}_i^2\mathbf{z}_{(i)}\mathbf{z}_{(i)}^\top$ , as in [Theorem 21.4.2](#), or under homoscedasticity by  $\hat{\sigma}_e^2(\mathbf{X}^\top\mathbf{M}_\mathbf{Z}\mathbf{X})^{-1}$ . Among estimators using linear combinations of the instruments, 2SLS has the smallest asymptotic covariance under homoscedasticity ([Exercise \(C2\)](#)).

#### Example. (The second reading as an instrument)

Instrument the first reading with the second, and let age and the constant instrument themselves. The IV estimates are 0.0708 for blood pressure, with standard error 0.0085



(0.0081 from the sandwich), and 0.0054 for age (standard error 0.0081). The spurious age effect has gone. The first-stage  $F$  statistic for the second reading is 160.1, a strong instrument. The correct residuals give  $\hat{\sigma}_e^2 = 1.438$ , estimating  $\sigma^2 + \beta^2 \sigma_u^2 = 1.5184$ ; the second-stage residuals give 1.376. The moment correction of [Example \(Correcting with two readings\)](#) gave 0.0757 with standard error 0.0076. It uses both readings symmetrically, while IV uses one only as an instrument and pays in precision. But IV needs no estimate of the error variance, and the instrument need only be correlated with  $X$ , not a second unbiased reading of it.

```
import numpy as np

rng = np.random.default_rng(2401)
n = 300
age = rng.normal(55, 10, n)
x = 130 + 0.6 * (age - 55) + rng.normal(0, np.sqrt(108), n)  # long-run blood
                        pressure, sd 12
w = x[:, None] + rng.normal(0, 9, (n, 2))                  # two clinic readings
y = -4 + 0.08 * x + 0.0 * age + rng.normal(0, 1.0, n)      # age has no effect
                        given x

X = np.column_stack([np.ones(n), w[:, 0], age])            # regressors: first reading, age
Z = np.column_stack([np.ones(n), w[:, 1], age])            # instruments: second reading,
                        age
b_iv = np.linalg.solve(Z.T @ X, Z.T @ y)                  # residuals use X, not the first
e = y - X @ b_iv                                           -stage fit
ZXinv = np.linalg.inv(Z.T @ X)
cov_h = e @ e / n * ZXinv @ (Z.T @ Z) @ ZXinv.T           # homoscedastic
cov_s = ZXinv @ ((Z * e[:, None] ** 2).T @ Z) @ ZXinv.T    # heteroscedasticity-
                        robust
print("IV (1, x, age):", b_iv.round(4))
print("se, homoscedastic:", np.sqrt(np.diag(cov_h)).round(4))
print("se, sandwich:      ", np.sqrt(np.diag(cov_s)).round(4))

# first stage: regress the first reading on the instruments; F for the excluded
                        instrument
g, rss1 = np.linalg.lstsq(Z, w[:, 0], rcond=None)[:2]
rss0 = np.sum((w[:, 0] - np.column_stack([np.ones(n), age]) @
                        np.linalg.lstsq(np.column_stack([np.ones(n), age]), w[:, 0], rcond=
                        None)[0]) ** 2)
F1 = (rss0 - rss1[0]) / (rss1[0] / (n - 3))
print(f"first-stage F = {F1:.1f}")
```

Wald's (1940) grouping estimator ([Section 7.1](#)) is an IV estimator whose instrument is a group indicator ([Exercise \(B1\)](#)).

### 24.4.3 Weak instruments

If the instruments explain little of the endogenous regressor, the limit theory of [Theorem 24.4.2](#) is a poor guide at realistic  $n$ . The simplest case can be solved exactly.

**Proposition 24.4.3 — The exact distribution with one instrument**

Let  $y_i = \beta x_i + e_i$  and  $x_i = \pi z_i + v_i$ ,  $i = 1, \dots, n$ , with fixed  $z_i$ ,  $S_{zz} = \sum_i z_i^2 > 0$ , and  $(e_i, v_i)$  independent bivariate normal with means zero, standard deviations  $\sigma_e, \sigma_v$  and correlation  $\rho$ ,  $|\rho| < 1$ . Let  $\hat{\beta} = \sum_i z_i y_i / \sum_i z_i x_i$  and  $\mu = \pi \sqrt{S_{zz}} / \sigma_v$ .

(a)  $\hat{\beta} - \beta$  has the same distribution as

$$\frac{\sigma_e}{\sigma_v} \cdot \frac{\rho\eta + \sqrt{1-\rho^2}\xi}{\mu + \eta}, \quad \eta, \xi \stackrel{\text{iid}}{\sim} N(0, 1). \quad (24.4.3)$$

(b)  $E|\hat{\beta}| = \infty$  for every  $\mu$ .

(c) As  $\mu \rightarrow \infty$ ,  $(\mu\sigma_v/\sigma_e)(\hat{\beta} - \beta) \rightarrow N(0, 1)$  in distribution. If  $\mu = 0$ ,  $\hat{\beta} - \beta$  has median  $\rho\sigma_e/\sigma_v$ , which is the large-sample bias of least squares when  $\pi = 0$ .

(d)  $(\sum_i z_i x_i)^2 / (\sigma_v^2 S_{zz}) \sim \chi^2(1, \mu^2)$ , with mean  $1 + \mu^2$ .

*Proof.* (a) Put  $\eta = \sum_i z_i v_i / (\sigma_v \sqrt{S_{zz}})$  and  $\zeta = \sum_i z_i e_i / (\sigma_e \sqrt{S_{zz}})$ . They are jointly normal (Theorem 3.3.1) with unit variances and covariance  $\sum_i z_i^2 \rho \sigma_e \sigma_v / (\sigma_e \sigma_v S_{zz}) = \rho$ . By Theorem 3.5.2,  $\xi = (\zeta - \rho\eta) / \sqrt{1-\rho^2}$  is standard normal and independent of  $\eta$ . Now  $\sum_i z_i x_i = \pi S_{zz} + \sigma_v \sqrt{S_{zz}} \eta = \sigma_v \sqrt{S_{zz}}(\mu + \eta)$ , and  $\hat{\beta} - \beta = \sum_i z_i e_i / \sum_i z_i x_i = \sigma_e \zeta / \{\sigma_v(\mu + \eta)\}$ .

(b) Given  $\eta$ , the numerator  $\rho\eta + \sqrt{1-\rho^2}\xi$  is normal with standard deviation  $\sqrt{1-\rho^2} > 0$ , so its conditional absolute mean is at least  $c = \sqrt{2(1-\rho^2)}/\pi$ . Hence  $E|\hat{\beta} - \beta| \geq (\sigma_e/\sigma_v) c E|\mu + \eta|^{-1} = \infty$ , because the density of  $\mu + \eta$  is continuous and positive at zero, as in the proof of Proposition 12.5.2. Finally  $|\hat{\beta}| \geq |\hat{\beta} - \beta| - |\beta|$ .

(c)  $(\mu\sigma_v/\sigma_e)(\hat{\beta} - \beta)$  has the law of  $(\rho\eta + \sqrt{1-\rho^2}\xi)/(1 + \eta/\mu)$ . The numerator is standard normal and  $\eta/\mu \rightarrow 0$  in probability, so Slutsky's lemma applies. For  $\mu = 0$  the representation is  $(\sigma_e/\sigma_v)(\rho + \sqrt{1-\rho^2}\xi/\eta)$ , and  $\xi/\eta$  is symmetric about zero, so the median is  $\rho\sigma_e/\sigma_v$ . When  $\pi = 0$ ,  $x = v$  and least squares converges to  $\beta + \text{Cov}(v, e)/\sigma_v^2 = \beta + \rho\sigma_e/\sigma_v$ .

(d)  $\sum_i z_i x_i / (\sigma_v \sqrt{S_{zz}}) = \mu + \eta \sim N(\mu, 1)$ , and its square is  $\chi^2(1, \mu^2)$ , whose mean is  $1 + \mu^2$  (the convention of Chapter 4, without a factor  $\frac{1}{2}$ ).

■

The design and  $\pi$  enter only through the **concentration parameter**  $\mu^2$ , which acts as a sample size, and by (d) the first-stage statistic estimates  $1 + \mu^2$ . Hence the rule of thumb of Staiger and Stock (1997): an instrument is weak if the first-stage  $F$  is below about 10. A useless instrument ( $\mu = 0$ ) gives an estimator centred on the least squares bias, with Cauchy tails.

**Example. (How weak is too weak?)**

With  $n = 200$ ,  $\beta = 1$ ,  $\sigma_e = \sigma_v = 1$  and  $\rho = 0.9$ , least squares tends to about 1.9 for a weak instrument. In 20000 samples for each concentration:

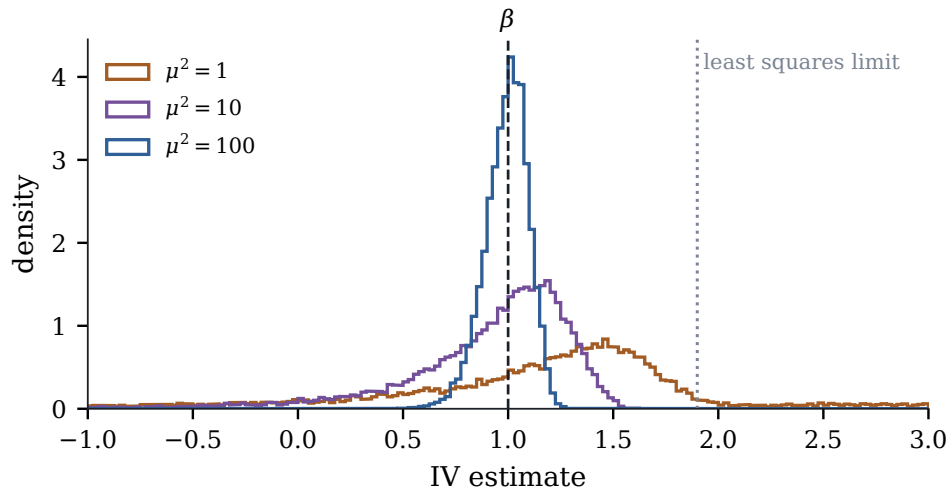


Figure 24.4.1: Sampling distributions of the IV estimate of  $\beta = 1$  for  $\mu^2 = 1, 10, 100$  on  $[-1, 3]$ ; dotted, the least squares limit.

| $\mu^2$ | median | quartiles      | coverage of nominal<br>95% Wald interval |
|---------|--------|----------------|------------------------------------------|
| 1       | 1.298  | 0.711 to 1.646 | 0.843                                    |
| 10      | 1.008  | 0.740 to 1.181 | 0.921                                    |
| 100     | 1.002  | 0.929 to 1.064 | 0.947                                    |

With  $\mu^2 = 1$  the median moves a third of the way to the least squares limit, a proportion 0.200 of estimates fall outside  $[-1, 3]$ , and the Wald interval undercovers, as it still does at  $\mu^2 = 10$  (Figure 24.4.1).

```
def iv_draws(mu2, reps, seed, n=200, rho=0.9):
    """IV estimates of beta = 1 and Wald-interval coverage when the concentration is
    mu2."""
    r = np.random.default_rng(seed)
    z = r.normal(size=n)
    z = z * np.sqrt(n / (z @ z))          # fixed instrument with sum z^2 =
    n
    pi = np.sqrt(mu2 / n)
    ev = r.multivariate_normal([0, 0], [[1, rho], [rho, 1]], size=(reps, n))
    xx = pi * z + ev[:, :, 1]
    yy = 1.0 * xx + ev[:, :, 0]
    b = (yy @ z) / (xx @ z)
    res = yy - b[:, None] * xx
    se = np.sqrt((res ** 2).mean(axis=1) * n / (xx @ z) ** 2)
    return b, np.abs(b - 1) <= 1.96 * se

for mu2 in [1, 10, 100]:
    b, hit = iv_draws(mu2, 2000, 2440)
    print(f"mu^2 = {mu2:3d}: median {np.median(b):.3f}, "
          f"quartiles {np.percentile(b, 25):.3f} to {np.percentile(b, 75):.3f}, "
          f"coverage {hit.mean():.3f}")
```

With many weak instruments, 2SLS is biased towards least squares, and Bound, Jaeger and

Baker (1995) showed that it can nearly reproduce least squares; we do not prove this. For inference, invert a test whose null distribution does not depend on instrument strength, such as the Anderson–Rubin test ([Exercise \(C1\)](#)). Its confidence sets are of Fieller type ([Proposition 12.5.1](#)) and can be unbounded, as they must sometimes be ([Proposition 12.5.4](#)).

*Warning.* Exogeneity cannot be checked in a just-identified model: all the equations  $E(\mathbf{z}e) = \mathbf{0}$  are used to estimate  $\beta$ . With more instruments, tests of overidentifying restrictions (Sargan, 1958) check only that the instruments *agree*. An instrument is justified by knowledge of how the data arose; for replicates, by the independence of their errors. Instruments for causal effects are taken up in [Chapter 25](#).

## 24.4.4 Exercises

### 24.4.4.1 A. Check your understanding

#### Exercise A1.

For  $\mathbf{x} = (1, W)$  and  $\mathbf{z} = (1, Z)$ , show that the IV slope is  $s_{zy}/s_{zw}$ , the ratio of two sample covariances. Show that the population ratio equals  $\beta$  when  $Z$  is correlated with  $X$  and uncorrelated with  $U$  and  $\varepsilon$ .

#### Exercise A2.

If the errors of two readings are correlated,  $\text{Cov}(U_1, U_2) = \kappa$ , find the limit of the IV slope in simple regression that instruments  $W_1$  with  $W_2$ . In which direction is it biased when  $\kappa > 0$ ?

*Solution.*  $\text{Cov}(W_2, Y) = \beta\sigma_x^2$  and  $\text{Cov}(W_2, W_1) = \sigma_x^2 + \kappa$ , so the limit is  $\beta\sigma_x^2/(\sigma_x^2 + \kappa)$ . With  $\kappa > 0$  it is attenuated, though less than least squares unless  $\kappa = \sigma_u^2$ . The instrument is not exogenous:  $\text{Cov}(W_2, \varepsilon - \beta U_1) = -\beta\kappa$ .

### 24.4.4.2 B. Practice

#### Exercise B1.

Let  $g_i \in \{0, 1\}$  be a group indicator, and let  $\mathbf{z}_{(i)} = (1, g_i)$  instrument  $\mathbf{x}_{(i)} = (1, w_i)$ . Show that the IV slope is  $(\bar{y}_1 - \bar{y}_0)/(\bar{w}_1 - \bar{w}_0)$ , the difference of group means of  $y$  divided by that of  $w$ . This is Wald's (1940) estimator. Show that it is consistent if  $g$  is independent of  $(U, \varepsilon)$  and  $E(X | g)$  differs between the groups. Explain why forming the groups by splitting at the median of  $w$  itself, as Wald proposed, does not satisfy these conditions under classical error.

*Solution.* By [Exercise \(A1\)](#) the slope is  $s_{gy}/s_{gw}$ . For a binary  $g$  with  $n_1$  ones,  $\sum_i (g_i - \bar{g})y_i = n_1(\bar{y}_1 - \bar{y})$  and  $\bar{y}_1 - \bar{y} = (n_0/n)(\bar{y}_1 - \bar{y}_0)$ . The same holds for  $w$ , and the factors cancel. Exogeneity needs  $\text{Cov}(g, \varepsilon - \beta U) = 0$ , which holds when  $g$  is independent of  $(U, \varepsilon)$ . Relevance needs  $\text{Cov}(g, W) = \text{Cov}(g, X) \neq 0$ , that is, different group means of  $X$ . If the groups are formed from  $w$ , membership depends on  $U$ : readings with large positive errors tend to land in the upper group, so  $\text{Cov}(g, U) > 0$  and exogeneity fails. The grouping must come from information independent of the measurement error, such

as a second reading or a design variable.

#### 24.4.4.3 C. Going deeper

##### Exercise C1.

In [Eq. 24.4.1](#) with  $\mathbf{x} = (1, W)$ , let  $\mathbf{z} = (1, \mathbf{z}_1)$  with  $\mathbf{z}_1$  a  $k$ -vector of excluded instruments, and suppose  $e$  is independent of  $\mathbf{z}$  and  $N(0, \sigma_e^2)$ . To test  $H_0 : \beta = \beta_0$ , regress  $y - \beta_0 w$  on  $\mathbf{z}$  and compute the  $F$  statistic for  $\mathbf{z}_1$ . Show that under  $H_0$  it has an exact  $F(k, n - k - 1)$  distribution, whatever the strength of the instruments. This is the test of Anderson and Rubin (1949). Inverting it gives a confidence set for  $\beta$ . Show that for  $k = 1$  the set is of the form in [Proposition 12.5.1](#).

*Solution.* Under  $H_0$ ,  $y - \beta_0 w = \alpha + e$  with  $e$  normal and independent of  $\mathbf{z}$ : given  $\mathbf{Z}$ , a normal linear model in which the coefficients of  $\mathbf{z}_1$  are zero. The  $F$  statistic of [Theorem 11.1.1](#) is exactly  $F(k, n - k - 1)$  given  $\mathbf{Z}$ , hence unconditionally, and  $w$  plays no role. For  $k = 1$ , let  $\hat{a}$  and  $\hat{b}$  be the slopes of  $y$  and of  $w$  on  $(1, z)$ , let  $v = 1 / \sum_i (z_i - \bar{z})^2$ , and let  $s_{11}$ ,  $s_{12}$ ,  $s_{22}$  be the residual mean squares and mean cross-product of  $y$  and  $w$  from these two regressions. The regression of  $y - \beta_0 w$  on  $(1, z)$  has slope  $\hat{a} - \beta_0 \hat{b}$  and residual mean square  $s_{11} - 2\beta_0 s_{12} + \beta_0^2 s_{22}$ , so  $F = t^2$  and the acceptance region is

$$(\hat{a} - \beta_0 \hat{b})^2 \leq t_{n-2, \alpha/2}^2 v (s_{11} - 2\beta_0 s_{12} + \beta_0^2 s_{22}).$$

This is [Eq. 12.5.1](#) with  $s^2 v_{ij}$  replaced by  $v s_{ij}$ : a quadratic inequality in  $\beta_0$ . Like Fieller's set, it can be an interval, the complement of an interval, or the whole line.

##### Exercise C2.

Let  $\mathbf{F}$  be a fixed  $q \times p$  matrix with  $\Sigma_{zx}^\top \mathbf{F}$  nonsingular, and consider the IV estimator with instruments  $\mathbf{ZF}$ . Under homoscedasticity its asymptotic covariance is  $\sigma_e^2 (\mathbf{F}^\top \Sigma_{zx})^{-1} \mathbf{F}^\top \Sigma_{zz} \mathbf{F} (\Sigma_{zx}^\top \mathbf{F})^{-1}$ . Show that this is never smaller, in the nonnegative definite ordering, than  $\sigma_e^2 \mathbf{H}^{-1}$ , with equality when  $\mathbf{F} = \Sigma_{zz}^{-1} \Sigma_{zx}$ , which is the population version of 2SLS.

## 24.5 Regression calibration and SIMEX

The moment correction and IV are tied to linear moment equations. Two more portable ideas were designed for logistic, Cox and other nonlinear regressions, where they are approximations. **Regression calibration** replaces the unobserved regressor by its best prediction from what was observed. **SIMEX** watches the naive estimate deteriorate as error is added, and extrapolates back to no error. In the linear model both can be exact.

### 24.5.1 Regression calibration

Error is **nondifferential** if the distribution of  $Y$  given  $(\mathbf{x}, \mathbf{z}, \mathbf{w})$  does not depend on  $\mathbf{w}$ , as for classical error.

**Proposition 24.5.1 — Regression calibration**

Suppose  $E(Y | \mathbf{x}, \mathbf{z}) = \alpha + \beta_x^\top \mathbf{x} + \beta_z^\top \mathbf{z}$  and the error is nondifferential, and let  $\mathbf{m}(\mathbf{w}, \mathbf{z}) = E(\mathbf{x} | \mathbf{w}, \mathbf{z})$ .

- (a)  $E(Y | \mathbf{w}, \mathbf{z}) = \alpha + \beta_x^\top \mathbf{m}(\mathbf{w}, \mathbf{z}) + \beta_z^\top \mathbf{z}$ .
- (b) If moreover  $\text{Var}(Y | \mathbf{x}, \mathbf{z}) = \sigma^2$  and  $\text{Cov}(\mathbf{x} | \mathbf{w}, \mathbf{z}) = \mathbf{C}$  does not depend on  $(\mathbf{w}, \mathbf{z})$ , as when  $(\mathbf{x}, \mathbf{z}, \mathbf{u})$  is jointly normal, then  $\text{Var}(Y | \mathbf{w}, \mathbf{z}) = \sigma^2 + \beta_x^\top \mathbf{C} \beta_x$ . The regression of  $Y$  on  $(1, \mathbf{m}(\mathbf{w}, \mathbf{z}), \mathbf{z})$  is then a homoscedastic linear model with the original coefficients.
- (c) Without distributional assumptions, let the error be classical with finite second moments. With  $\mathbf{t} = (1, \mathbf{x}^\top, \mathbf{z}^\top)^\top$ ,  $\mathbf{s} = (1, \mathbf{w}^\top, \mathbf{z}^\top)^\top$ ,  $\mathbf{A} = E(\mathbf{t}\mathbf{t}^\top)$  positive definite and  $\mathbf{S} = E(\mathbf{s}\mathbf{s}^\top)$ , the best linear predictor of  $\mathbf{t}$  from  $\mathbf{s}$  is  $\hat{\mathbf{t}} = \mathbf{A}\mathbf{S}^{-1}\mathbf{s}$ , and the best linear predictor of  $Y$  from  $\hat{\mathbf{t}}$  is  $\boldsymbol{\theta}^\top \hat{\mathbf{t}}$ , where  $\boldsymbol{\theta} = (\alpha, \beta_x^\top, \beta_z^\top)^\top$ .
- (d) The sample version of (c) reproduces the method-of-moments estimator exactly. If  $\hat{\mathbf{S}} = n^{-1} \sum_i \mathbf{s}_i \mathbf{s}_i^\top$ ,  $\hat{\mathbf{A}} = \hat{\mathbf{S}} - \mathbf{D}$  for a symmetric matrix  $\mathbf{D}$  with  $\hat{\mathbf{A}}$  nonsingular, and  $\hat{\mathbf{t}}_i = \hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}\mathbf{s}_i$ , then the least squares coefficients of  $y$  on the  $\hat{\mathbf{t}}_i$  are  $\hat{\mathbf{A}}^{-1}n^{-1} \sum_i \mathbf{s}_i y_i$ , which is [Eq. 24.3.1](#) with  $\mathbf{D}_i = \mathbf{D}$ .

*Proof.* (a) By the tower property and nondifferential error,  $E(Y | \mathbf{w}, \mathbf{z}) = E\{E(Y | \mathbf{x}, \mathbf{z}, \mathbf{w}) | \mathbf{w}, \mathbf{z}\} = E\{\alpha + \beta_x^\top \mathbf{x} + \beta_z^\top \mathbf{z} | \mathbf{w}, \mathbf{z}\}$ , which is the claim.

(b) The law of total variance, conditioning on  $\mathbf{x}$  inside  $(\mathbf{w}, \mathbf{z})$ , gives  $\text{Var}(Y | \mathbf{w}, \mathbf{z}) = E\{\text{Var}(Y | \mathbf{x}, \mathbf{z}, \mathbf{w}) | \mathbf{w}, \mathbf{z}\} + \text{Var}\{E(Y | \mathbf{x}, \mathbf{z}, \mathbf{w}) | \mathbf{w}, \mathbf{z}\} = \sigma^2 + \beta_x^\top \mathbf{C} \beta_x$ . Under joint normality the conditional covariance of  $\mathbf{x}$  given  $(\mathbf{w}, \mathbf{z})$  does not depend on the conditioning values ([Theorem 3.5.2](#)).

(c) The best linear predictor ([Definition 6.11.1](#), [Theorem 2.5.2](#)), with the constant among the predictors, is the projection in  $L^2$  onto the span of the entries of  $\mathbf{s}$ , so it is  $E(\mathbf{t}\mathbf{s}^\top)\mathbf{S}^{-1}\mathbf{s}$ . Under classical error  $E(\mathbf{t}\mathbf{s}^\top) = E\{\mathbf{t}(\mathbf{t} + \tilde{\mathbf{u}})^\top\} = \mathbf{A}$ , and  $\mathbf{A}\mathbf{S}^{-1}\mathbf{s}$  follows. The entries of  $\hat{\mathbf{t}}$  span the same space as those of  $\mathbf{s}$ , since  $\mathbf{A}\mathbf{S}^{-1}$  is nonsingular. Their second-moment matrix is  $\mathbf{A}\mathbf{S}^{-1}\mathbf{S}\mathbf{S}^{-1}\mathbf{A} = \mathbf{A}\mathbf{S}^{-1}\mathbf{A}$ , and  $E(\hat{\mathbf{t}}Y) = \mathbf{A}\mathbf{S}^{-1}E(\mathbf{s}Y) = \mathbf{A}\mathbf{S}^{-1}\mathbf{A}\boldsymbol{\theta}$ , because  $E(\mathbf{s}Y) = \mathbf{A}\boldsymbol{\theta}$  as in [Section 24.3](#). The projection coefficients are  $(\mathbf{A}\mathbf{S}^{-1}\mathbf{A})^{-1}\mathbf{A}\mathbf{S}^{-1}\mathbf{A}\boldsymbol{\theta} = \boldsymbol{\theta}$ .

(d) With the calibrated regressors  $\hat{\mathbf{t}}_i = \hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}\mathbf{s}_i$ , the normal equations ([Theorem 6.4.2](#)) have matrix  $n^{-1} \sum_i \hat{\mathbf{t}}_i \hat{\mathbf{t}}_i^\top = \hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}\hat{\mathbf{A}}$  and right side  $\hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}n^{-1} \sum_i \mathbf{s}_i y_i$ . Solving,  $(\hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}\hat{\mathbf{A}})^{-1}\hat{\mathbf{A}}\hat{\mathbf{S}}^{-1}n^{-1} \sum_i \mathbf{s}_i y_i = \hat{\mathbf{A}}^{-1}n^{-1} \sum_i \mathbf{s}_i y_i$ . ■

The coordinates of  $\hat{\mathbf{t}}$  for the constant and  $\mathbf{z}$  reproduce them, so only the mismeasured regressors are replaced. By (c) and (d), linear regression calibration needs no distributional assumption and is the moment correction, with the standard errors of [Theorem 24.3.1](#). Second-stage standard errors, which treat the calibrated values as data, are too small. A **validation subsample** in which  $\mathbf{x}$  is measured identifies the calibration directly, by regressing  $\mathbf{x}$  on  $(\mathbf{w}, \mathbf{z})$  there, and needs only (a): nondifferential error and a correct model for  $E(\mathbf{x} | \mathbf{w}, \mathbf{z})$ . In a logistic regression, (a) gives  $E\{G(\alpha + \beta X) | w\}$ , not  $G(\alpha + \beta m(w))$ , so calibration is an approximation, good when  $\text{Var}(X | W)$  is small (Carroll, Ruppert, Stefanski and Crainiceanu, 2006).

*Remark* (Two meanings of “calibration”). In [Section 12.5](#) *calibration* meant inverse prediction of  $x_0$  from a new response. Regression calibration instead predicts the regressor from its measurement.

### Example. (Calibrating the mean reading)

The replicates estimate the error variance of the mean reading as 37.04. The calibrated pressure  $\hat{x}_i$  has standard deviation 10.36, against 13.11 for the mean reading: each reading is shrunk towards what age predicts. Regressing  $y$  on  $(1, \hat{x}, \text{age})$  gives 0.0757 and 0.0018, exactly the moment correction of [Example \(Correcting with two readings\)](#), by [Proposition 24.5.1\(d\)](#).

```
import numpy as np

rng = np.random.default_rng(2401)
n = 300
age = rng.normal(55, 10, n)
x = 130 + 0.6 * (age - 55) + rng.normal(0, np.sqrt(108), n)  # long-run blood
                        pressure, sd 12
w = x[:, None] + rng.normal(0, 9, (n, 2))                  # two clinic readings
y = -4 + 0.08 * x + 0.0 * age + rng.normal(0, 1.0, n)      # age has no effect
                        given x

wbar = w.mean(axis=1)
s2_uvar = np.mean((w[:, 0] - w[:, 1]) ** 2) / 4              # error variance of
                        the mean reading
W = np.column_stack([np.ones(n), wbar, age])
S = W.T @ W / n                                             # raw second moments
                        of (1, wbar, age)
Suu = np.diag([0.0, s2_uvar, 0.0])

# calibration: best linear predictor of (1, x, age) from (1, wbar, age), using E(w x
# ^T) = S - Suu
Gamma = np.linalg.solve(S, S - Suu)
x_hat = (W @ Gamma)[:, 1]                                  # predicted long-run
                        pressure
b_rc = np.linalg.lstsq(np.column_stack([np.ones(n), x_hat, age]), y, rcond=None)[0]
b_mom = np.linalg.solve(S - Suu, W.T @ y / n)              # method of moments,
                        Section 24.3
print("regression calibration:", b_rc.round(4))
print("method of moments:      ", b_mom.round(4))
```

## 24.5.2 SIMEX

Cook and Stefanski (1994) treat the naive estimator as a black box. Data cannot be made more accurate, but they can be made *less* accurate by adding simulated error; watching the naive estimate move as the error variance grows, we extrapolate back to zero. Let  $\sigma_u^2$  be the known or estimated error variance of  $W$ .

1. **Simulation.** For each  $\zeta$  in a grid such as  $\{0.5, 1, 1.5, 2\}$  and  $b = 1, \dots, B$ , form remeasured data  $W_{b,i}(\zeta) = W_i + \sqrt{\zeta} \sigma_u V_{b,i}$ , with  $V_{b,i} \stackrel{\text{iid}}{\sim} N(0, 1)$  independent of the data, and compute



the naive estimate  $\hat{\theta}_b(\zeta)$  from  $(W_b(\zeta), \mathbf{z}, Y)$ . Average over  $b$  to get  $\hat{\theta}(\zeta)$ , and put  $\hat{\theta}(0)$  equal to the naive estimate.

2. **Extrapolation.** Fit a function  $g(\zeta; \gamma)$  of  $\zeta$  to the points  $(\zeta, \hat{\theta}(\zeta))$ , one coefficient at a time, and report  $g(-1; \hat{\gamma})$ .

The remeasured error  $U + \sqrt{\zeta}\sigma_u V$  is still classical, with variance  $(1 + \zeta)\sigma_u^2$ , so  $\zeta = -1$  means no error. Averaging over  $b$  removes simulation noise only.

### Proposition 24.5.2 — SIMEX in the linear model

Assume the conditions of [Theorem 24.1.1\(c\)](#), with one mismeasured scalar regressor of error variance  $\sigma_u^2$ , and fix  $\zeta \geq 0$ . For each  $b$ , the naive least squares coefficients from the remeasured data converge with probability one to  $\mathbf{g}(\zeta) = (g_x(\zeta), \mathbf{g}_z(\zeta))$ , where

$$g_x(\zeta) = \frac{\beta_x \sigma_{x \cdot z}^2}{\sigma_{x \cdot z}^2 + (1 + \zeta)\sigma_u^2}, \quad \mathbf{g}_z(\zeta) = \boldsymbol{\beta}_z + \frac{(1 + \zeta)\sigma_u^2}{\sigma_{x \cdot z}^2 + (1 + \zeta)\sigma_u^2} \beta_x \boldsymbol{\pi}, \quad (24.5.1)$$

and so does their average over  $b$ . Moreover:

- (a)  $g_x(-1) = \beta_x$  and  $\mathbf{g}_z(-1) = \boldsymbol{\beta}_z$ ;
- (b) every coefficient has the form  $a + b/(c + \zeta)$  with  $c = 1 + \sigma_{x \cdot z}^2/\sigma_u^2$ ; in particular  $1/g_x(\zeta)$  is linear in  $\zeta$  when  $\beta_x \neq 0$ ;
- (c) if a polynomial of fixed degree is fitted by least squares to the points  $(\zeta_k, \hat{\theta}(\zeta_k))$  on a fixed grid, the SIMEX estimate converges to the value at  $-1$  of the polynomial fitted in the same way to  $(\zeta_k, \mathbf{g}(\zeta_k))$ , which in general differs from  $\mathbf{g}(-1)$ .

*Proof.* The remeasured data satisfy [Definition 24.1.1](#) with error  $U + \sqrt{\zeta}\sigma_u V$ , whose variance is  $(1 + \zeta)\sigma_u^2$ , and the observations  $(X_i, \mathbf{z}_i, U_i, V_{b,i}, \varepsilon_i)$  are independent over  $i$ . By [Theorem 24.1.1\(c\)](#), with  $\sigma_u^2$  replaced by  $(1 + \zeta)\sigma_u^2$ , the coefficients converge to  $\lambda_\zeta \beta_x$  and  $\boldsymbol{\beta}_z + (1 - \lambda_\zeta)\beta_x \boldsymbol{\pi}$ , where  $\lambda_\zeta = \sigma_{x \cdot z}^2 / \{\sigma_{x \cdot z}^2 + (1 + \zeta)\sigma_u^2\}$ . This is [Eq. 24.5.1](#). A finite average of sequences converging with probability one converges to the average of the limits. (a) At  $\zeta = -1$ ,  $\lambda_\zeta = 1$ . (b) The denominator is  $\sigma_u^2(c + \zeta)$ . So  $g_x(\zeta) = (\beta_x \sigma_{x \cdot z}^2 / \sigma_u^2) / (c + \zeta)$ , and  $(1 + \zeta)/(c + \zeta) = 1 - (c - 1)/(c + \zeta)$  gives  $\mathbf{g}_z(\zeta) = \boldsymbol{\beta}_z + \beta_x \boldsymbol{\pi} - (c - 1)\beta_x \boldsymbol{\pi} / (c + \zeta)$ . (c) On a fixed grid the fitted polynomial's coefficients are a fixed linear function of the fitted points (the least squares solution with a fixed model matrix), so its value at  $-1$  is a continuous function of the points, and the points converge. ■

So the rational form matches the limit exactly, and the rational SIMEX estimate is consistent provided the nonlinear fit is unique and continuous in the points, which we do not verify. The quadratic, a common default because it is linear in its parameters, recovers only part of the attenuation ([Exercise \(B1\)](#)). With several mismeasured regressors, or in nonlinear models, no simple extrapolant is exact and SIMEX is only approximately consistent. Its appeal is that it needs nothing but the naive estimator.

#### Example. (SIMEX for the blood-pressure study)

Apply SIMEX to the regression on the mean reading and age, with error variance 37.04,



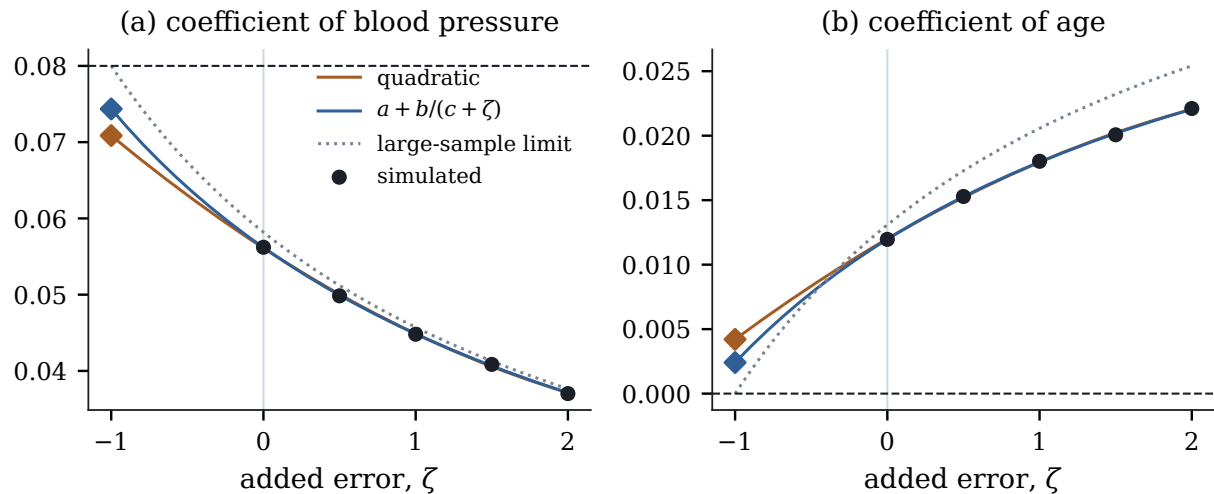


Figure 24.5.1: SIMEX: averaged naive coefficients (dots), quadratic and rational extrapolants with their values at  $\zeta = -1$  (diamonds), the limits Eq. 24.5.1 (dotted) and the truth (dashed).

grid  $\{0, 0.5, 1, 1.5, 2\}$  and  $B = 200$  (Figure 24.5.1). Extrapolating to  $\zeta = -1$ :

| Coefficient       | naive  | SIMEX,<br>quadratic | SIMEX,<br>$a + b/(c + \zeta)$ | moment<br>correction | truth |
|-------------------|--------|---------------------|-------------------------------|----------------------|-------|
| blood<br>pressure | 0.0562 | 0.0709              | 0.0744                        | 0.0757               | 0.08  |
| age               | 0.0120 | 0.0042              | 0.0024                        | 0.0018               | 0     |

The rational extrapolant nearly reproduces the moment correction; the quadratic stops short, and by Proposition 24.5.2(c) its large-sample limit is 0.0749, only 0.77 of the way from the naive limit 0.0582 to 0.08, and 0.0030 for age. A pairs bootstrap (Definition 23.1.1) over patients, re-estimating the error variance and rerunning SIMEX ( $B = 50, 200$  resamples), gives standard error 0.0076 for the rational SIMEX slope, as for the moment correction.

```
from scipy.optimize import curve_fit

def naive_fits(wcol, age, y):
    """Least squares coefficients of y on (1, wcol, age) for a stack of remeasured
    columns."""
    m, k = wcol.shape # m remeasured data
    sets of size k
    Xs = np.stack([np.ones((m, k)), wcol, np.broadcast_to(age, (m, k))], axis=2)
    XtX = np.einsum("mki,mkj->mij", Xs, Xs)
    Xty = np.einsum("mki,k->mi", Xs, y)
    return np.linalg.solve(XtX, Xty[:, :, None])[ :, :, 0]

def simex(wbar, age, y, s2_ubar, zetas, B, rng):
    """Average naive coefficients after adding error of variance zeta * s2_ubar."""
    out = [naive_fits(wbar[None, :], age, y)[0]]
    for z in zetas[1:]:
        extra = rng.normal(0, np.sqrt(z * s2_ubar), (B, len(y)))
        out.append(naive_fits(wbar + extra, age, y).mean(axis=0))
    return np.array(out) # one row per zeta
```

```

def rational(z, a, b, c):
    return a + b / (c + z)

zetas = np.array([0.0, 0.5, 1.0, 1.5, 2.0])
G = simex(wbar, age, y, s2_ubar, zetas, 200, np.random.default_rng(2450))
est_quad, est_rat = [], []
for j in [1, 2]:                                # coefficient of x, of
    age
    est_quad.append(np.polyval(np.polyfit(zetas, G[:, j], 2), -1.0))
    p, _ = curve_fit(rational, zetas, G[:, j], p0=[0.0, G[0, j], 1.5], maxfev=20000)
    est_rat.append(rational(-1.0, *p))
print("naive (zeta = 0):   ", G[0, 1:].round(4))
print("SIMEX, quadratic:  ", np.round(est_quad, 4))
print("SIMEX, rational:   ", np.round(est_rat, 4))

```

### 24.5.3 Which correction?

In the linear model the moment correction and regression calibration coincide, rational SIMEX agrees up to simulation noise, and IV trades precision for not needing an error variance. Beyond it, only regression calibration and SIMEX carry over. All rest on information from outside the regression of  $Y$  on  $W$ . Without it, [Proposition 24.2.2](#) is the honest summary: a range of slopes, with the naive estimate at one end.

### 24.5.4 Exercises

#### 24.5.4.1 A. Check your understanding

##### Exercise A1.

In simple regression without  $\mathbf{z}$ , show directly that regression calibration ([Proposition 24.5.1](#)) with the linear calibration  $\hat{x} = \bar{w} + \hat{\lambda}(w - \bar{w})$ ,  $\hat{\lambda} = (s_{ww} - \sigma_u^2)/s_{ww}$ , gives the slope  $\hat{\beta}/\hat{\lambda}$ , where  $\hat{\beta}$  is the naive slope.

#### 24.5.4.2 B. Practice

##### Exercise B1.

In simple regression with reliability  $\lambda$ , write  $g(\zeta) = \lambda\beta/\{1 + \zeta(1 - \lambda)\}$ . Show that the quadratic through  $(\zeta, g(\zeta))$  for  $\zeta = 0, 1, 2$  takes the value  $3g(0) - 3g(1) + g(2)$  at  $\zeta = -1$ . Evaluate it for  $\lambda = 0.5$  and  $\lambda = 0.64$ , as fractions of  $\beta$ .

*Solution.* The quadratic through three equally spaced points is determined by the forward differences. Written as  $p(\zeta) = g_0 + \zeta\Delta g_0 + \frac{1}{2}\zeta(\zeta - 1)\Delta^2 g_0$ , it gives  $p(-1) = g_0 - \Delta g_0 + \Delta^2 g_0 = g_0 - (g_1 - g_0) + (g_2 - 2g_1 + g_0) = 3g_0 - 3g_1 + g_2$ . With  $\lambda = 0.5$ :  $g_0 = \beta/2$ ,  $g_1 = \beta/3$ ,  $g_2 = \beta/4$ , so  $p(-1) = \beta(3/2 - 1 + 1/4) = 0.75\beta$ . With  $\lambda = 0.64$ :  $g_0 = 0.64\beta$ ,  $g_1 = 0.64\beta/1.36$ ,  $g_2 = 0.64\beta/1.72$ , so  $p(-1) = \beta(1.92 - 1.4118 + 0.3721) \approx 0.880\beta$ . The

quadratic removes most of the attenuation but not all of it, and it does worse as the reliability falls.

### Exercise B2.

A validation subsample of  $n_v$  patients has the true  $x$  measured along with  $(w, z)$ , and the main sample has only  $(w, z, y)$ . Describe regression calibration using the validation data. Which of the assumptions of [Proposition 24.5.1](#) does it need, and which does it avoid? Why does using the validation patients'  $x$  directly in the outcome regression not help if their  $y$  is not recorded?

#### 24.5.4.3 C. Going deeper

### Exercise C1.

Simulate a logistic regression  $\Pr(Y = 1 \mid X) = 1/(1 + e^{-(\alpha + \beta X)})$  with  $X \sim N(0, 1)$ ,  $\beta = 1$ ,  $n = 2000$ , and a reading  $W = X + U$  with  $\sigma_u^2 = 0.5$ . Compare the naive estimate, regression calibration ([Proposition 24.5.1](#): replace  $W$  by  $E(X \mid W)$ ), and SIMEX with quadratic and rational extrapolants, over repeated samples. Which is closest to  $\beta$ ?

## 24.6 Summary and Notes

### 24.6.1 Summary

*Idea.*

1. Under classical, nondifferential error, least squares estimates the best linear predictor from the *reading*. A simple-regression slope is attenuated by the reliability  $\lambda$ , whatever  $n$  ([Theorem 24.1.1](#)).
2. Exactly measured covariates absorb part of the effect and can look important when they are not ([Example \(Age looks important\)](#)). Error in the response only adds variance ([Proposition 24.1.2](#)). The test of no effect of the mismeasured regressor keeps its level; tests on correlated, correctly measured covariates do not ([Proposition 24.1.3](#)).
3. Attenuation is the same with fixed true values ([Proposition 24.2.1](#)). The normal structural model identifies the slope only up to the interval between the direct and reverse regressions ([Proposition 24.2.2](#)). Berkson error causes no bias in a linear model ([Proposition 24.2.3](#)).
4. A known or replicate-estimated error covariance gives the moment correction, with a sandwich covariance ([Theorem 24.3.1](#)); a known variance ratio gives Deming regression ([Proposition 24.3.2](#)).
5. IV and two-stage least squares are consistent under exogeneity and relevance ([Lemma 24.4.1](#), [Theorem 24.4.2](#)); with weak instruments they have no mean, drift towards least squares, and Wald intervals undercover ([Proposition 24.4.3](#)).
6. Linear regression calibration is exact and equals the moment correction ([Proposi-](#)

tion 24.5.1). SIMEX extrapolates the naive estimate to zero error; in the linear model the rational form matches the limit exactly, so the rational extrapolant is consistent under regularity conditions for its fit ([Proposition 24.5.2](#)).

## 24.6.2 Notes and sources

**Sources.** None of the six source books treats errors in variables at length. Seber and Lee (2003, section 9.6) surveys random regressors, fixed regressors measured with error, the structural identification problem, rounding errors and Berkson’s model, covered here in [Theorem 24.1.1](#), [Proposition 24.2.1](#), [Proposition 24.2.2](#), [Exercise \(C1\)](#) and [Proposition 24.2.3](#). Fuller (1987) is the classical treatment of linear measurement error models, including moment estimators, their asymptotics and small-sample modifications. Carroll, Ruppert, Stefanski and Crainiceanu (2006) is the standard reference for nonlinear models, regression calibration and SIMEX, and discusses the choice of SIMEX extrapolant. All data in the chapter are simulated.

**Attenuation and identification.** Spearman (1904) introduced the correction for attenuation in mental testing. Reiersol (1950) proved that non-normality of the true regressor restores identification. The functional model’s incidental parameters are an instance of Neyman and Scott (1948). Gleser and Hwang (1987) proved that errors-in-variables confidence sets with guaranteed coverage have infinite expected length. Berkson (1950) distinguished controlled regressors from classical error.

**Orthogonal and Deming regression.** Orthogonal regression goes back to Adcock in the 1870s; the version with a known variance ratio carries Deming’s name from Deming (1943). In numerical analysis it is total least squares (Golub and Van Loan, 1980). Wald (1940) proposed the grouping estimator of [Exercise \(B1\)](#).

**Instrumental variables.** Instrumental variables go back to Wright (1928). Two-stage least squares was developed in the 1950s by Theil and by Basmann (1957), among others, and Sargan (1958) gave the general theory, including the test of overidentifying restrictions. Anderson and Rubin (1949) introduced the test of [Exercise \(C1\)](#). Bound, Jaeger and Baker (1995) showed the practical danger of weak instruments, and Staiger and Stock (1997) gave the asymptotics behind the first-stage  $F$  rule of thumb. Angrist and Pischke (2009) give a modern treatment, including the causal use of IV taken up in [Chapter 25](#).

**Regression calibration and SIMEX.** Regression calibration developed in epidemiology in the 1980s to correct relative risks for exposure measurement error. SIMEX is due to Cook and Stefanski (1994).

## 24.6.3 References

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# Chapter 25

## Causal Interpretation of Coefficients

Earlier chapters treated a regression coefficient as a feature of a joint distribution. People who fit regressions usually want more: what would happen to the response if a regressor were *changed*, if a patient took the drug or a school hired more teachers. No amount of data on the undisturbed system answers that without further assumptions. This chapter states the assumptions and proves what they buy: potential outcomes and randomization, graphs and the back-door criterion for observational data, the bias from adjusting for mediators and colliders, and covariate adjustment in experiments, which can hurt unless the regression is interacted. This makes precise the warnings of [Section 5.7](#), of the remark after the Frisch–Waugh–Lovell theorem ([Section 6.6](#)) and of [Section 18.1](#).

**What you need.** [Chapter 5](#) (what a coefficient measures, [Definition 5.7.1](#), and the omitted-variable formula, [Proposition 5.7.2](#)), [Chapter 6](#) (the Frisch–Waugh–Lovell theorem, [Theorem 6.6.2](#), and projection in the population, [Theorem 6.11.1](#) and [Proposition 6.11.3](#)), [Chapter 3](#) and [Chapter 14](#) (partial correlation, [Proposition 3.6.1](#) and [Proposition 14.5.2](#)), [Chapter 18](#) (analysis of covariance and randomization, [Theorem 18.1.1](#), [Proposition 18.1.2](#) and [Lemma 18.4.3](#)), and, from this part, [Chapter 19](#) (omitted-variable bias of least squares, [Theorem 19.1.2](#)) and [Chapter 21](#) (sandwich standard errors, [Theorem 21.4.2](#)).

### 25.0.1 Roadmap

- [Prediction versus intervention](#)
- [Potential outcomes and the regression coefficient](#)
- [Confounding and causal graphs](#)
- [Adjustment sets and the back-door criterion](#)
- [Bad controls: mediators and colliders](#)
- [Regression adjustment in randomized experiments](#)
- [Summary and notes](#)

## 25.1 Prediction versus intervention

A regression of  $Y$  on  $X$  answers a question about *seeing*: among units in which  $X = x$  was observed, what is  $Y$  typically? The answer is  $E(Y \mid X = x)$ , or its best linear approximation

$L(Y | X)$  (Theorem 6.11.1), which least squares estimates whatever produced the data (Proposition 6.11.3). A decision asks about *doing*: if  $X$  were set to  $x$  by an outside agent, what would  $Y$  typically be? This section makes the second question precise and shows, in the simplest model, that the answers differ, that more data do not close the gap, and that the better predictor can be the worse guide to action.

## 25.1.1 Structural causal models

A joint distribution says how variables co-vary, not which would move if another were changed. For that we need a model of the mechanism, a list of the processes that generate each variable. The following definition goes back to Wright (1921) and Haavelmo (1943).

### Definition 25.1.1 — Structural causal model

Let  $V_1, \dots, V_m$  be random variables. A **structural causal model** (SCM) for them consists of

- (a) for each  $j$ , a set  $\text{pa}(j) \subseteq \{1, \dots, j-1\}$  of indices, the **parents** of  $V_j$ ;
- (b) for each  $j$ , a function  $f_j$ , and mutually independent random variables  $U_1, \dots, U_m$ , the **disturbances**, such that

$$V_j = f_j(V_{\text{pa}(j)}, U_j), \quad j = 1, \dots, m, \quad (25.1.1)$$

where  $V_{\text{pa}(j)}$  is the vector of parents of  $V_j$ .

Each line of Eq. 25.1.1 is an **assignment**: it says how nature sets  $V_j$ , not merely that an equation holds. Solving the assignments in the order  $j = 1, \dots, m$  writes every  $V_j$  as a function of  $U_1, \dots, U_j$ . For a variable  $X = V_k$  and a value  $x$ , the **intervention**  $\text{do}(X = x)$  replaces the  $k$ th assignment by  $V_k = x$  and keeps every other assignment and every disturbance unchanged. The distribution of the variables in the modified model is the **interventional distribution**, and expectations under it are written  $E(Y | \text{do}(X = x))$ . The model is **linear** if each  $f_j(\mathbf{v}, u) = c_j + \sum_{l \in \text{pa}(j)} b_{jl}v_l + u$  with constants  $c_j, b_{jl}$ , and disturbances with mean zero and finite variance.

Section 25.3 draws the parent sets as a graph. The substantive assumption is that an intervention changes one assignment and leaves the other mechanisms intact. Conditioning and intervening are then different operations:  $E(Y | X = x)$  looks at the part of the *original* distribution where  $X = x$  happened, while  $E(Y | \text{do}(X = x))$  describes a *new* distribution in which  $X = x$  was imposed.

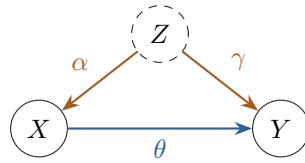
## 25.1.2 A common cause

The simplest model in which the two differ has a background variable  $Z$  that influences both  $X$  and  $Y$ :

$$Z = c_Z + U_Z, \quad X = c_X + \alpha Z + U_X, \quad Y = c_Y + \theta X + \gamma Z + U_Y, \quad (25.1.2)$$

with independent disturbances of variances  $\sigma_Z^2 > 0$ ,  $\sigma_X^2 > 0$  and  $\sigma_Y^2$  (Figure 25.1.1). The

coefficient  $\theta$  is what the assignment for  $Y$  does with its argument  $X$ .



**Figure 25.1.1.** The model [Eq. 25.1.2](#). An arrow points from each parent to its child and is labelled with its coefficient. The dashed circle marks a variable that is not recorded in [Example \(Clinic visits and health\)](#).

### Proposition 25.1.1 — Seeing and doing with a common cause

In the model [Eq. 25.1.2](#):

- (a)  $E(Y \mid \text{do}(X = x)) = c_Y + \gamma c_Z + \theta x$ , so the interventional slope is  $\theta$ ;
- (b) the best linear predictor  $L(Y \mid X)$  has slope

$$\beta^* = \theta + \frac{\gamma \alpha \sigma_Z^2}{\alpha^2 \sigma_Z^2 + \sigma_X^2}; \quad (25.1.3)$$

- (c) the coefficient of  $X$  in  $L(Y \mid X, Z)$  is  $\theta$ ;
- (d) if the disturbances are normal,  $E(Y \mid X = x) = L(Y \mid X)$  evaluated at  $x$ , so the slopes of seeing and doing differ whenever  $\gamma \alpha \neq 0$ .

*Proof.*

- (a) Under  $\text{do}(X = x)$  the assignment for  $Z$  is unchanged and does not involve  $X$ , so  $Z = c_Z + U_Z$  still, and  $Y = c_Y + \theta x + \gamma Z + U_Y$ . Take expectations.
- (b) By [Theorem 6.11.1](#) the slope is  $\text{Cov}(X, Y) / \text{Var}(X)$ . Here  $\text{Var}(X) = \alpha^2 \sigma_Z^2 + \sigma_X^2$  and  $\text{Cov}(X, Y) = \theta \text{Var}(X) + \gamma \text{Cov}(X, Z) = \theta \text{Var}(X) + \gamma \alpha \sigma_Z^2$ .
- (c) The error  $Y - c_Y - \theta X - \gamma Z = U_Y$  has mean zero and is uncorrelated with  $X$  and  $Z$ , which are functions of  $U_Z$  and  $U_X$ . So  $c_Y + \theta X + \gamma Z$  satisfies the population normal equations of [Theorem 6.11.1](#) and is the projection.
- (d)  $(X, Y)$  is a linear function of the normal vector  $(U_Z, U_X, U_Y)$ , hence bivariate normal, and its conditional mean is the best linear predictor ([Theorem 3.5.2](#)).

■

The extra term in [Eq. 25.1.3](#) is association that  $X$  borrows from  $Z$ . Part (c) is the familiar remedy, including the common cause as a regressor, and it needs  $Z$  to be recorded. [Section 25.4](#) says in general which variables must be included.

#### Example. (Clinic visits and health)

In a synthetic population,  $Z$  is the severity of a chronic illness, which nobody records;  $X$  is the number of clinic visits in a year; and  $Y$  is a health score measured at the end of the year. Sicker people visit more ( $\alpha = 1.5$ ) and end the year less healthy ( $\gamma = -6$ ),



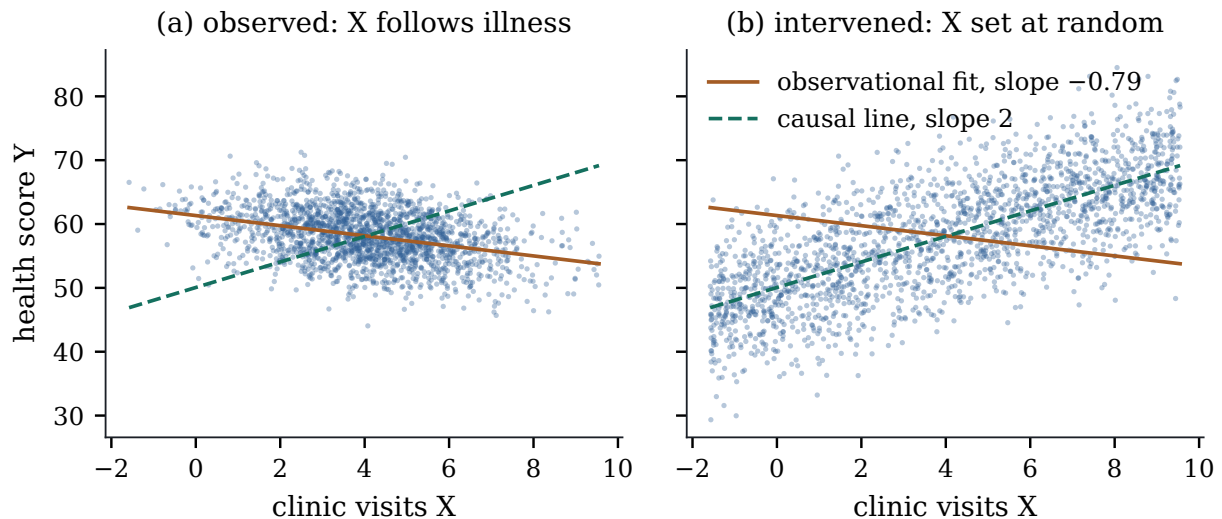


Figure 25.1.1: The model of [Example \(Clinic visits and health\)](#). (a) Observational data: the fitted line slopes down. (b) Data in which the number of visits is set at random by the analyst: the same mechanism for  $Y$ , but now  $X$  carries no information about illness. The solid line is the observational fit in both panels; the dashed line has the causal slope  $\theta = 2$ .

and each visit helps ( $\theta = 2$ ). With  $\sigma_Z = \sigma_X = 1$  and  $\sigma_Y = 2$ , [Eq. 25.1.3](#) gives

$$\beta^* = 2 + \frac{(-6)(1.5)}{1.5^2 + 1} = 2 - \frac{9}{3.25} = -0.769.$$

In a sample of  $n = 2000$  people the least squares slope is  $-0.790$ : each extra visit goes with a *lower* health score. A second data set in which visits are assigned at random, which is an intervention on  $X$ , has least squares slope  $1.999$  ([Figure 25.1.2](#)). A health authority that capped visits on the strength of the first regression would harm its patients.

```
import numpy as np

rng = np.random.default_rng(2501)
n = 2000
alpha, theta, gamma = 1.5, 2.0, -6.0          # Z -> X, X -> Y, Z -> Y

def draw(n, do_x=None):
    """Draw from the model; with do_x, X is set by the analyst instead of by Z."""
    Z = rng.normal(size=n)                    # illness severity, never recorded
    X = 4 + alpha * Z + rng.normal(size=n) if do_x is None else do_x
    Y = 50 + theta * X + gamma * Z + rng.normal(scale=2.0, size=n)
    return X, Y

x_obs, y_obs = draw(n)                       # observational data
x_do = rng.uniform(x_obs.min(), x_obs.max(), size=n) # X assigned at random
x_int, y_int = draw(n, do_x=x_do)           # interventional data

slope_obs, icpt_obs = np.polyfit(x_obs, y_obs, 1)
slope_int, icpt_int = np.polyfit(x_int, y_int, 1)
print(f"slope from observational data: {slope_obs:.3f}")
print(f"slope from interventional data: {slope_int:.3f} (theta = {theta})")
```

### 25.1.3 A good predictor can be a bad causal model

The observational line in [Example \(Clinic visits and health\)](#) is the best linear predictor of health from visits in the population that generated the data, and the causal line, with slope  $\theta$  through the same means, predicts *worse* there. On the observational data the fitted line has mean squared error 15.0 and the causal line 40.7, which is worse than predicting every person by the sample mean (the variance of  $Y$  is 17.1). On the interventional data the positions reverse: 123.3 for the fitted line and 39.2 for the causal line.

```
def mse(x, y, slope, icpt):
    return np.mean((y - icpt - slope * x) ** 2)

causal_icpt = np.mean(y_obs) - theta * np.mean(x_obs)  # causal slope, fitted level
for label, x, y in [("observational", x_obs, y_obs), ("interventional", x_int, y_int)]:
    print(f"{label:15s} MSE: regression line {mse(x, y, slope_obs, icpt_obs):6.2f}, "
          f" causal line {mse(x, y, theta, causal_icpt):6.2f}")
```

Observationally, any line through the means other than the projection loses exactly  $(\text{slope} - \beta^*)^2 \text{Var}(X)$  ([Exercise \(B1\)](#)). Under intervention a large  $X$  no longer signals a large  $Z$ , so the part of the slope borrowed from  $Z$  becomes pure error: the causal line describes the mechanism both regimes share, the regression line a distribution that the intervention destroys.

Confounding by a common cause is one of three standard ways in which a predictor misleads about action. In **reverse causation**  $Y$  causes  $X$ , and  $X$  can predict  $Y$  perfectly while setting it does nothing ([Exercise \(B2\)](#)): a thermometer predicts the temperature, but moving its needle does not warm the room. In **selection** the data contain only units selected by a common effect of  $X$  and  $Y$ , which creates association without effect ([Section 25.5](#)).

More data do not help: the least squares slope converges to  $\beta^*$ , not  $\theta$  ([Proposition 6.11.3](#)), and no procedure using only the distribution of  $(X, Y)$  can do better, because models with different  $\theta$  can produce the same distribution ([Exercise \(B3\)](#)). Causal conclusions need either an intervention, as in panel (b) of [Figure 25.1.2](#), or assumptions about the mechanism that link the interventional distribution to the observational one.

### 25.1.4 When seeing is doing

Conditioning and intervening agree for every mechanism when nothing in the model influences  $X$ .

#### Proposition 25.1.2 — An exogenous cause

In a structural causal model, suppose  $X = V_k$  has no parents, so that  $X = f_k(U_k)$ . Then a version of the conditional distribution of the other variables given  $X = x$  is their distribution under  $\text{do}(X = x)$ ; in particular, when  $E|Y| < \infty$ ,  $E(Y \mid X = x) = E(Y \mid \text{do}(X = x))$  for almost every  $x$  (with respect to the distribution of  $X$ ), and for every  $x$  with  $\Pr(X = x) > 0$ .

*Proof.* Solve the assignments in order. Every variable is a function  $h(X, \mathbf{U}_{-k})$  of  $X$  and the disturbances  $\mathbf{U}_{-k}$  other than  $U_k$ , because  $U_k$  enters the other assignments only

through  $X$ . The same function describes the model under  $\text{do}(X = x)$ , with  $x$  in place of  $X$ , since only the  $k$ th assignment changed. Now  $X = f_k(U_k)$  is independent of  $\mathbf{U}_{-k}$ , and for independent  $X$  and  $\mathbf{U}_{-k}$  a version of the conditional distribution of  $h(X, \mathbf{U}_{-k})$  given  $X = x$  is the distribution of  $h(x, \mathbf{U}_{-k})$  (Billingsley 1995), which is the interventional distribution.

■

A randomized experiment makes the treatment exogenous by construction: the random device is the only parent of  $X$  and influences nothing else, so what is seen is what would be done. [Section 25.2](#) turns this into exact finite-sample statements that need no structural model.

## 25.1.5 Exercises

### 25.1.5.1 A. Check your understanding

#### Exercise A1.

In the model [Eq. 25.1.2](#) with  $\theta = 2$ ,  $\gamma = -6$  and  $\sigma_Z = \sigma_X = 1$ , for which values of  $\alpha$  is the observational slope  $\beta^*$  exactly zero? What would an analyst who saw  $\beta^* = 0$  be tempted to conclude?

### 25.1.5.2 B. Practice

#### Exercise B1.

Let  $(x_i, y_i)$ ,  $i = 1, \dots, n$ , be any data with least squares slope  $\hat{\beta}$ . For a number  $b$ , let  $\ell_b(x) = \bar{y} + b(x - \bar{x})$  be the line with slope  $b$  through the means. Show that

$$\frac{1}{n} \sum_i (y_i - \ell_b(x_i))^2 = \frac{1}{n} \sum_i (y_i - \ell_{\hat{\beta}}(x_i))^2 + (b - \hat{\beta})^2 \frac{1}{n} \sum_i (x_i - \bar{x})^2,$$

and deduce the ordering of the two observational mean squared errors in [Example \(Clinic visits and health\)](#).

*Solution.* Write  $y_i - \ell_b(x_i) = (y_i - \ell_{\hat{\beta}}(x_i)) + (\hat{\beta} - b)(x_i - \bar{x})$ . The cross term is  $(\hat{\beta} - b) \sum_i \hat{e}_i (x_i - \bar{x})$ , where  $\hat{e}_i$  are the least squares residuals. The residuals are orthogonal to  $\mathbf{1}$  and to  $\mathbf{x}$  ([Theorem 6.4.1](#)), so the cross term is zero, and squaring gives the identity. With  $b = \theta = 2$  the causal line through the means loses  $(2 - \hat{\beta})^2$  times the variance of the visits, which is why its mean squared error exceeds that of the fitted line. The script checks the identity to rounding error.

#### Exercise B2.

Let  $Y = c + U_Y$  and  $X = \delta Y + U_X$ , with independent disturbances of variances  $\sigma_Y^2$  and  $\sigma_X^2$ . Find the slope of  $L(Y | X)$ , the squared correlation of  $X$  and  $Y$ , and  $E(Y | \text{do}(X = x))$ . Show that the squared correlation can be arbitrarily close to one while setting  $X$  has no effect on  $Y$ .

**Exercise B3.**

Take  $c_Z = c_X = c_Y = 0$  in Eq. 25.1.2 with normal disturbances, and suppose only  $(X, Y)$  is observed. Show that the parameter values

$$(\alpha, \theta, \gamma, \sigma_Z^2, \sigma_X^2, \sigma_Y^2) = (1, 1, 0, 1, 1, 1) \quad \text{and} \quad (1, 0, \frac{4}{3}, \frac{3}{2}, \frac{1}{2}, \frac{1}{3})$$

give the same distribution of  $(X, Y)$ , although  $X$  has an effect on  $Y$  in the first model and none in the second.

*Solution.* Both distributions are bivariate normal with mean zero, so it suffices to compare covariance matrices, using  $\text{Var}(X) = \alpha^2 \sigma_Z^2 + \sigma_X^2$ ,  $\text{Cov}(X, Y) = \theta \text{Var}(X) + \gamma \alpha \sigma_Z^2$  and  $\text{Var}(Y) = \theta^2 \text{Var}(X) + \gamma^2 \sigma_Z^2 + 2\theta \gamma \alpha \sigma_Z^2 + \sigma_Y^2$ . In the first model these are 2, 2 and  $2 + 1 = 3$ . In the second they are  $\frac{3}{2} + \frac{1}{2} = 2$ ,  $\frac{4}{3} \cdot \frac{3}{2} = 2$  and  $\frac{16}{9} \cdot \frac{3}{2} + \frac{1}{3} = 3$ . The covariance matrices agree, so no function of the  $(X, Y)$  data can distinguish  $\theta = 1$  from  $\theta = 0$ .

## 25.2 Potential outcomes and the regression coefficient

Whether a treatment works needs much less than a model of the whole mechanism. Each unit has two possible responses, with and without the treatment, and the effect on that unit is their difference. This language of potential outcomes, due to Neyman (1923) for field trials and extended to observational studies by Rubin (1974), needs no model for the responses, and under randomization it gives exact results. The regression coefficient of a treatment indicator is the natural estimator; its usual standard error is wrong in a specific way.

### 25.2.1 Potential outcomes

#### Definition 25.2.1 — Potential outcomes and average effects

Consider  $n$  units and a binary treatment. Unit  $i$  has two **potential outcomes**:  $y_i(1)$ , the response it would show if treated, and  $y_i(0)$ , the response it would show if not. With  $T_i \in \{0, 1\}$  the treatment it actually receives, the observed response is

$$Y_i = T_i y_i(1) + (1 - T_i) y_i(0). \quad (25.2.1)$$

The **unit effect** is  $\tau_i = y_i(1) - y_i(0)$ , and the **average treatment effect** of the  $n$  units is

$$\tau = \frac{1}{n} \sum_{i=1}^n \tau_i = \bar{y}(1) - \bar{y}(0).$$

When the units are instead drawn at random from a population, so that  $(Y(0), Y(1), T)$  is a random vector, the population **average treatment effect** is  $E(Y(1) - Y(0))$  and the **average effect on the treated** is  $E(Y(1) - Y(0) \mid T = 1)$ .

Equation [Eq. 25.2.1](#) assumes that unit  $i$ 's response depends only on its own treatment (**no interference**, which fails for vaccines) and that there is one version of each treatment: Rubin's *stable unit treatment value assumption*. In a structural causal model,  $Y(t)$  is the value of  $Y$ , as a function of the disturbances, in the model modified by  $\text{do}(T = t)$ , so  $E Y(t) = E(Y \mid \text{do}(T = t))$ . On the event  $T = t$ , solving the original and the modified assignments in order gives every variable the same value, so  $Y = Y(t)$  there, which is [Eq. 25.2.1](#).

For each unit one potential outcome is observed and the other is **counterfactual**, so no unit effect is ever observed (Holland's (1986) fundamental problem of causal inference). Averages can still be estimated from *different* units, if treated and untreated units are comparable.

### Proposition 25.2.1 — Selection bias

Let  $(Y(0), Y(1), T)$  be random with finite means and  $0 < \Pr(T = 1) < 1$ . Then

$$E(Y \mid T = 1) - E(Y \mid T = 0) = E(Y(1) - Y(0) \mid T = 1) + [E(Y(0) \mid T = 1) - E(Y(0) \mid T = 0)]. \quad (25.2.2)$$

*Proof.* By [Eq. 25.2.1](#),  $E(Y \mid T = 1) = E(Y(1) \mid T = 1)$  and  $E(Y \mid T = 0) = E(Y(0) \mid T = 0)$ . Add and subtract  $E(Y(0) \mid T = 1)$ . ■

The observed difference is the effect on the treated plus a **selection bias**, the difference between treated and untreated units in the response they would have shown untreated. In [Example \(Clinic visits and health\)](#) heavy users of the clinic would have been less healthy than light users even without their visits. The data on  $(T, Y)$  cannot reveal the bias, because  $Y(0)$  is never observed for the treated.

## 25.2.2 Randomization

A **completely randomized experiment** treats  $n_1$  of the  $n$  units, every subset of size  $n_1$  being equally likely, and leaves the other  $n_0 = n - n_1$  as controls. In Neyman's framework the potential outcomes are fixed numbers and the only randomness is the assignment: no linearity, normality or common variance is assumed, and the units need not be a sample from anything.

Write  $\bar{Y}_1$  and  $\bar{Y}_0$  for the mean observed responses of the treated and control units,  $s_1^2$  and  $s_0^2$  for their sample variances, and

$$S_t^2 = \frac{1}{n-1} \sum_{i=1}^n (y_i(t) - \bar{y}(t))^2, \quad S_\tau^2 = \frac{1}{n-1} \sum_{i=1}^n (\tau_i - \tau)^2$$

for the variances of the potential outcomes and of the unit effects over the  $n$  units. The treated units are a simple random sample from the  $n$  units, and the moments of such samples are as follows (compare [Lemma 18.4.3](#)).

**Lemma 25.2.2 — Means of a simple random sample**

Let  $u_1, \dots, u_N$  and  $v_1, \dots, v_N$  be fixed numbers with means  $\bar{u}$ ,  $\bar{v}$  and  $S_{uv} = \sum_a (u_a - \bar{u})(v_a - \bar{v})/(N-1)$ . If  $\mathcal{S}$  is a simple random sample of  $m$  of the indices,  $1 \leq m \leq N$ , with sample means  $\bar{u}_{\mathcal{S}}$  and  $\bar{v}_{\mathcal{S}}$ , then

$$E \bar{u}_{\mathcal{S}} = \bar{u}, \quad \text{Cov}(\bar{u}_{\mathcal{S}}, \bar{v}_{\mathcal{S}}) = \left( \frac{1}{m} - \frac{1}{N} \right) S_{uv}.$$

*Proof.* Let  $W_a$  indicate that  $a \in \mathcal{S}$ . Then  $E W_a = m/N$ , and for  $a \neq c$ ,  $E W_a W_c = m(m-1)/\{N(N-1)\}$ . Hence  $\text{Var } W_a = q$  and  $\text{Cov}(W_a, W_c) = m(m-1)/\{N(N-1)\} - m^2/N^2 = -q/(N-1)$ , where  $q = (m/N)(1 - m/N)$ . Put  $\tilde{u}_a = u_a - \bar{u}$  and  $\tilde{v}_a = v_a - \bar{v}$ . Since  $\sum_a W_a = m$ ,  $\bar{u}_{\mathcal{S}} - \bar{u} = m^{-1} \sum_a W_a \tilde{u}_a$ , whose mean is  $N^{-1} \sum_a \tilde{u}_a = 0$ . For the covariance,

$$m^2 \text{Cov}(\bar{u}_{\mathcal{S}}, \bar{v}_{\mathcal{S}}) = \sum_{a,c} \text{Cov}(W_a, W_c) \tilde{u}_a \tilde{v}_c = q \left( \sum_a \tilde{u}_a \tilde{v}_a - \frac{1}{N-1} \sum_{a \neq c} \tilde{u}_a \tilde{v}_c \right).$$

Because  $\sum_a \tilde{u}_a = 0$ ,  $\sum_{a \neq c} \tilde{u}_a \tilde{v}_c = -\sum_a \tilde{u}_a \tilde{v}_a$ , and the bracket is  $\sum_a \tilde{u}_a \tilde{v}_a N/(N-1) = N S_{uv}$ . So the covariance is  $q N S_{uv}/m^2 = (1/m - 1/N) S_{uv}$ . ■

**Theorem 25.2.3 — Randomization and the difference in means**

In a completely randomized experiment with  $n_1 \geq 1$  and  $n_0 \geq 1$ , let  $\hat{\tau} = \bar{Y}_1 - \bar{Y}_0$ .

- (a)  $\hat{\tau}$  is unbiased for the average treatment effect:  $E \hat{\tau} = \tau$ .
- (b) Its variance over the randomization is

$$\text{Var}(\hat{\tau}) = \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{S_{\tau}^2}{n}. \quad (25.2.3)$$

- (c) If  $n_1, n_0 \geq 2$ , the estimator  $\hat{V} = s_1^2/n_1 + s_0^2/n_0$  has  $E \hat{V} = \text{Var}(\hat{\tau}) + S_{\tau}^2/n \geq \text{Var}(\hat{\tau})$ , with equality iff the unit effects are all equal.
- (d)  $\hat{\tau}$  is the least squares coefficient of  $T$  in the regression of the observed responses on an intercept and the treatment indicator.
- (e) If instead the units are independent copies of  $(Y(0), Y(1), T)$  with  $T$  independent of  $(Y(0), Y(1))$ ,  $0 < \Pr(T=1) < 1$  and finite means, then  $E(Y | T=1) - E(Y | T=0) = E(Y(1) - Y(0))$ , and the average effect on the treated equals the average treatment effect.

*Proof.* Let  $\mathcal{S}$  be the set of treated units, a simple random sample of size  $n_1$ . Then  $\bar{Y}_1$  is the sample mean of the  $y_i(1)$  over  $\mathcal{S}$ , and  $\bar{Y}_0$  is the mean of the  $y_i(0)$  over the complement.

- (a) The complement of  $\mathcal{S}$  is a simple random sample of size  $n_0$ . By [Lemma 25.2.2](#),  $E \bar{Y}_1 = \bar{y}(1)$  and  $E \bar{Y}_0 = \bar{y}(0)$ .
- (b) Let  $\bar{y}_{\mathcal{S}}(0)$  be the mean of the  $y_i(0)$  over  $\mathcal{S}$ . The control mean is  $\bar{Y}_0 = (n \bar{y}(0) - n_1 \bar{y}_{\mathcal{S}}(0))/n_0$ , so

$$\hat{\tau} = \bar{u}_S - \frac{n}{n_0} \bar{y}(0), \quad u_i = y_i(1) + \frac{n_1}{n_0} y_i(0). \quad (25.2.4)$$

By [Lemma 25.2.2](#),  $\text{Var}(\hat{\tau}) = (1/n_1 - 1/n)S_u^2 = \{n_0/(nn_1)\}S_u^2$ . Expanding,  $S_u^2 = S_1^2 + (n_1/n_0)^2 S_0^2 + 2(n_1/n_0)S_{10}$ , where  $S_{10}$  is the covariance of the two potential outcomes over the units, and  $S_\tau^2 = S_1^2 + S_0^2 - 2S_{10}$ . Eliminating  $S_{10}$ ,

$$S_u^2 = \frac{n}{n_0} S_1^2 + \frac{nn_1}{n_0^2} S_0^2 - \frac{n_1}{n_0} S_\tau^2,$$

and multiplying by  $n_0/(nn_1)$  gives [Eq. 25.2.3](#).

- (c) The treated responses are a simple random sample of  $n_1 \geq 2$  of the numbers  $y_i(1)$ , so  $E s_1^2 = S_1^2$  by [Lemma 18.4.3](#); likewise  $E s_0^2 = S_0^2$ . Hence  $E \hat{V} = S_1^2/n_1 + S_0^2/n_0$ , which exceeds [Eq. 25.2.3](#) by  $S_\tau^2/n$ . This is zero iff  $\tau_i = \tau$  for every  $i$ .
- (d) This is [Proposition 5.4.4](#) with two groups and control as the reference.
- (e) By [Eq. 25.2.1](#) and independence,  $E(Y \mid T = 1) = E(Y(1) \mid T = 1) = E Y(1)$ , and likewise for  $T = 0$ . The same independence gives  $E(Y(1) - Y(0) \mid T = 1) = E(Y(1) - Y(0))$ .

■

Part (e) shows randomization removing the selection bias of [Eq. 25.2.2](#); part (a) needs no sampling model at all. The last term of (b) cannot be estimated, because it depends on how  $y_i(1)$  is paired with  $y_i(0)$  within units, which no unit reveals. Part (c) accepts a conservative answer, exact when the treatment adds the same amount to every unit.

### Example. (All randomizations of twenty units)

A synthetic population of  $n = 20$  units has  $\tau = 2.100$ ,  $S_1^2 = 12.129$ ,  $S_0^2 = 5.678$  and  $S_\tau^2 = 1.698$ ; the effect grows with  $y_i(0)$ . With  $n_1 = 8$  there are 125970 assignments, few enough to enumerate. Over all of them  $\hat{\tau}$  has mean 2.100 and variance 1.9043, which is [Eq. 25.2.3](#) to every digit. Neyman's  $\hat{V}$  has mean 1.9892, too large by  $S_\tau^2/n = 0.0849$ . The pooled least squares variance estimate averages 1.7055, below the true variance, because it gives the more variable treated arm the weight of the smaller arm ([Exercise \(B1\)](#)). [Figure 25.2.1](#) shows the distribution of  $\hat{\tau}$ .

```
import itertools
import numpy as np

rng = np.random.default_rng(2502)
n, n1 = 20, 8
n0 = n - n1
y0 = np.round(rng.normal(10, 3, size=n), 1) # potential outcome
      , control
y1 = np.round(y0 + 2 + 0.5 * (y0 - 10) + rng.normal(0, 1, n), 1) # potential
      outcome, treated
tau = np.mean(y1 - y0) # the sample
      average effect
S1, S0, St = np.var(y1, ddof=1), np.var(y0, ddof=1), np.var(y1 - y0, ddof=1)
print(f"tau = {tau:.3f}; S1^2 = {S1:.3f}, S0^2 = {S0:.3f}, S_tau^2 = {St:.3f}")
```

```
idx = np.array(list(itertools.combinations(range(n), n1))) # every possible
```

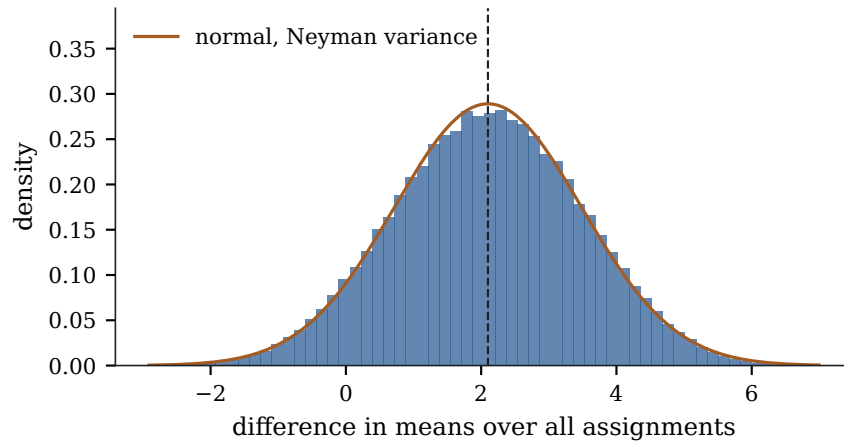


Figure 25.2.1: The randomization distribution of the difference in means in [Example \(All randomizations of twenty units\)](#), over all 125970 assignments of 8 of the 20 units to treatment. The dashed line is  $\tau$ ; the curve is the normal density with Neyman’s variance [Eq. 25.2.3](#).

```

treated set
treated = np.zeros((len(idx), n), dtype=bool)
treated[np.arange(len(idx))[:, None], idx] = True
assignment
m1, m0 = (treated @ y1) / n1, (~treated @ y0) / n0
row per assignment
diff = m1 - m0
means
s1 = (treated @ y1 ** 2 - n1 * m1 ** 2) / (n1 - 1)
variances
s0 = (~treated @ y0 ** 2 - n0 * m0 ** 2) / (n0 - 1)
V_hat = s1 / n1 + s0 / n0
variance estimate
V_neyman = S1 / n1 + S0 / n0 - St / n
print(f"{len(diff)} assignments")
print(f"mean of difference in means {diff.mean():.3f}    (tau = {tau:.3f})")
print(f"variance over assignments {diff.var():.4f}    (Neyman's formula {V_neyman:.4f})")
print(f"mean of variance estimate {V_hat.mean():.4f}    (excess S_tau^2/n = {St / n:.4f})")

```

Finite-population central limit theorems (Hájek 1960; Li and Ding 2017), whose proofs are beyond this chapter, make  $(\hat{\tau} - \tau) / \sqrt{\widehat{\text{Var}} \hat{\tau}}$  asymptotically standard normal under conditions on the potential outcomes, so  $\hat{\tau} \pm 1.96 \sqrt{\widehat{V}}$  has coverage at least 95% in the limit. In the example, with twenty units, it covers  $\tau$  for a fraction 0.931 of the assignments.

### 25.2.3 What the regression coefficient estimates

By part (d), the least squares coefficient of  $T$  is unbiased for the average effect because of the design, not because the model  $E(Y_i) = \beta_0 + \beta_1 T_i$  is true: its “errors” are not random draws and have different variances in the two arms. What the model gets wrong is the standard error. The pooled estimate of  $\sigma^2(1/n_1 + 1/n_0)$  ([Theorem 5.5.1](#)) can be too small or too large when  $n_1 \neq n_0$  ([Exercise \(B1\)](#)), while Neyman’s  $\widehat{V}$  is exactly the HC2 sandwich estimate of [Chapter 21](#) ([Exercise \(B2\)](#), [Theorem 21.4.2](#)). Covariates measured before randomization can be added



to this regression to gain precision; [Section 25.6](#) shows how, and finds a surprise.

*Remark* (Fisher's sharp null). Fisher (1935) tested the *sharp* null  $y_i(1) = y_i(0)$  for every unit, under which the randomization distribution of any statistic is known exactly. [Chapter 23](#) develops these permutation tests ([Theorem 23.4.2](#)).

## 25.2.4 Exercises

### 25.2.4.1 A. Check your understanding

#### Exercise A1.

Four units have potential outcomes  $(y_i(0), y_i(1)) = (1, 4), (3, 3), (5, 9), (7, 8)$ . Two are treated by complete randomization. List the six possible values of  $\hat{\tau}$ , and verify [Theorem 25.2.3](#)(a) and (b) directly.

*Solution.* The unit effects are 3, 0, 4, 1, so  $\tau = 2$ . For the treated pairs  $\{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 3\}, \{2, 4\}, \{3, 4\}$  the treated means are 3.5, 6.5, 6, 6, 5.5, 8.5 and the control means 6, 5, 4, 4, 3, 2, giving  $\hat{\tau} = -2.5, 1.5, 2, 2, 2.5, 6.5$ . Their mean is  $12/6 = 2 = \tau$ . Their squared deviations from 2 sum to  $20.25 + 0.25 + 0 + 0 + 0.25 + 20.25 = 41$ , so  $\text{Var } \hat{\tau} = 41/6$ . The formula:  $S_1^2 = 26/3$ ,  $S_0^2 = 20/3$ ,  $S_\tau^2 = 10/3$ , and  $(26/3)/2 + (20/3)/2 - (10/3)/4 = 46/6 - 5/6 = 41/6$ .

### 25.2.4.2 B. Practice

#### Exercise B1.

Let  $s_p^2 = \{(n_1 - 1)s_1^2 + (n_0 - 1)s_0^2\}/(n - 2)$  and  $\hat{V}_{\text{OLS}} = s_p^2(1/n_1 + 1/n_0)$ , the least squares variance estimate for the coefficient of  $T$ . Show that  $\hat{V}_{\text{OLS}} = \hat{V}$  when  $n_1 = n_0$ , and that in general

$$\text{E } \hat{V}_{\text{OLS}} - \text{E } \hat{V} = \frac{(n - 1)(n_1 - n_0)(S_1^2 - S_0^2)}{n_1 n_0 (n - 2)}.$$

Conclude that when the smaller arm has the more variable potential outcomes, the least squares variance estimate is smaller on average than Neyman's, and it can be smaller than the true variance.

*Solution.* If  $n_1 = n_0 = m$ , then  $s_p^2 = (s_1^2 + s_0^2)/2$  and  $\hat{V}_{\text{OLS}} = s_p^2 \cdot 2/m = \hat{V}$ . In general  $\text{E } s_t^2 = S_t^2$  ([Lemma 18.4.3](#)), so  $\text{E } \hat{V}_{\text{OLS}} = n\{(n_1 - 1)S_1^2 + (n_0 - 1)S_0^2\}/\{n_1 n_0 (n - 2)\}$ . The coefficient of  $S_1^2$  in  $\text{E } \hat{V}_{\text{OLS}} - \text{E } \hat{V}$  is

$$\frac{n(n_1 - 1) - n_0(n - 2)}{n_1 n_0 (n - 2)} = \frac{n(n_1 - n_0) - (n_1 - n_0)}{n_1 n_0 (n - 2)} = \frac{(n - 1)(n_1 - n_0)}{n_1 n_0 (n - 2)},$$

using  $n = n_1 + n_0$ ; by symmetry the coefficient of  $S_0^2$  is its negative. If  $n_1 < n_0$  and  $S_1^2 > S_0^2$ , the difference is negative. Since  $\text{E } \hat{V}$  exceeds  $\text{Var } \hat{\tau}$  only by  $S_\tau^2/n$ , the least squares estimate underestimates the variance on average whenever the negative difference is larger than that. In [Example \(All randomizations of twenty units\)](#),  $\text{E } \hat{V}_{\text{OLS}} = 1.7055$  while  $\text{Var } \hat{\tau} = 1.9043$ .

**Exercise B2.**

In the regression of the observed  $Y_i$  on  $\mathbf{1}$  and the treatment indicator, the leverages are  $h_{ii} = 1/n_1$  for treated units and  $1/n_0$  for controls (Proposition 6.8.1). The HC2 estimate of  $\text{Cov}(\hat{\beta})$  is  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \text{diag}(\hat{e}_i^2 / (1 - h_{ii})) \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1}$ . Show that its entry for the coefficient of  $T$  is exactly Neyman's  $\hat{V}$ .

*Solution.* Reparameterize with the two arm indicators as columns, which changes neither the fit nor the estimate of the difference of the arm means (Proposition 5.7.1). Then  $\mathbf{X}^\top \mathbf{X} = \text{diag}(n_1, n_0)$ , the residuals are  $Y_i - \bar{Y}_1$  and  $Y_i - \bar{Y}_0$ , and the sandwich is diagonal with entries  $n_1^{-2} \sum_{\text{treated}} \hat{e}_i^2 / (1 - 1/n_1) = s_1^2 / n_1$  and similarly  $s_0^2 / n_0$ . The variance of the difference of the two coefficients is their sum,  $\hat{V}$ .

**25.2.4.3 C. Going deeper****Exercise C1.**

Show that  $S_\tau^2 \geq (S_1 - S_0)^2$ , and hence that

$$\text{Var}(\hat{\tau}) \leq \frac{S_1^2}{n_1} + \frac{S_0^2}{n_0} - \frac{(S_1 - S_0)^2}{n},$$

a bound that is attained when  $y_i(1)$  is an increasing affine function of  $y_i(0)$ . Propose a conservative variance estimator that is never larger than  $\hat{V}$ .

## 25.3 Confounding and causal graphs

The functions and disturbances of a structural causal model are usually unknown; its parent sets, saying which variables can influence which, are qualitative and best displayed as a graph. This section shows how the graph constrains the joint distribution and identifies confounding as a property of paths. For linear models the connection is exact, and the omitted-variable bias of Proposition 5.7.2 becomes a sum over paths.

### 25.3.1 Graphs

#### Definition 25.3.1 — Directed acyclic graph

A **directed graph** on nodes  $1, \dots, m$  is a set of ordered pairs  $(l, j)$ ,  $l \neq j$ , drawn as arrows  $l \rightarrow j$ . It is **acyclic** (a DAG) if no sequence of arrows leads from a node back to itself. In a DAG:

- (a)  $l$  is a **parent** of  $j$ , and  $j$  a **child** of  $l$ , if  $l \rightarrow j$ ;
- (b) a **path** between  $l$  and  $j$  is a sequence of distinct nodes, starting at  $l$  and ending at  $j$ , in which consecutive nodes are joined by an arrow in either direction;
- (c) a path is **directed** (from  $l$  to  $j$ ) if every arrow on it points towards  $j$ ; then  $l$  is an **ancestor** of  $j$ , and  $j$  a **descendant** of  $l$ ;

- (d) an interior node  $c$  of a path is a **collider** on the path if both arrows of the path that meet at  $c$  point into it,  $\rightarrow c \leftarrow$ ; every other interior node is a **non-collider** on the path.

The **graph of a structural causal model** has an arrow  $l \rightarrow j$  iff  $l \in \text{pa}(j)$ . It is acyclic because parents have smaller indices.

Being a collider is a property of a node *on a given path*: in Figure 25.3.1(c),  $C$  is a collider on  $X \rightarrow C \leftarrow Y$  but not on a path that continues along an arrow  $C \rightarrow D$ .

## 25.3.2 What the graph says about the distribution

Assume that for each  $j$  and each value  $\mathbf{v}$  of the parents, the variable  $f_j(\mathbf{v}, U_j)$  has a density  $p_j(\cdot \mid \mathbf{v})$  with respect to Lebesgue or counting measure.

### Proposition 25.3.1 — Markov factorization and truncated factorization

Under this assumption:

- (a)  $(V_1, \dots, V_m)$  has density

$$p(v_1, \dots, v_m) = \prod_{j=1}^m p_j(v_j \mid v_{\text{pa}(j)}), \quad (25.3.1)$$

and  $p_j(\cdot \mid v_{\text{pa}(j)})$  is the conditional density of  $V_j$  given  $V_1, \dots, V_{j-1}$ , which depends on them only through the parents;

- (b) under  $\text{do}(V_k = x)$ , the other variables have density  $\prod_{j \neq k} p_j(v_j \mid v_{\text{pa}(j)})$ , evaluated with  $v_k = x$ .

*Proof.*

- (a) The variables  $V_1, \dots, V_{j-1}$  are functions of  $U_1, \dots, U_{j-1}$ , which are independent of  $U_j$ . For independent inputs, the conditional distribution of  $f_j(V_{\text{pa}(j)}, U_j)$  given  $(V_1, \dots, V_{j-1}) = (v_1, \dots, v_{j-1})$  is the distribution of  $f_j(v_{\text{pa}(j)}, U_j)$  (Billingsley 1995), whose density is  $p_j(\cdot \mid v_{\text{pa}(j)})$ . The factorization follows by multiplying these conditional densities in the order  $j = 1, \dots, m$ .
- (b) The modified model is a structural causal model for the variables other than  $V_k$ , in which  $V_k$  is the constant  $x$  and every other assignment and disturbance is unchanged. Apply (a) to it.

■

So an intervention deletes one factor and keeps the others, which are observational conditional densities; with every variable observed, the interventional distribution is computable from the observational one.

The linear case has an explicit form. Write a linear model as  $\mathbf{V} = \mathbf{c} + \mathbf{B}\mathbf{V} + \mathbf{U}$ , where  $\mathbf{B} = (b_{jl})$  has  $b_{jl} \neq 0$  only if  $l \in \text{pa}(j)$ , so  $\mathbf{B}$  is strictly lower triangular, and  $\mathbf{\Omega} = \text{Cov}(\mathbf{U}) =$

$\text{diag}(\omega_1, \dots, \omega_m)$ . The coefficient  $b_{jl}$  labels the arrow  $l \rightarrow j$ .

### Proposition 25.3.2 — Path tracing

In a linear structural causal model:

- (a)  $\mathbf{I} - \mathbf{B}$  is invertible,  $\mathbf{V} = \mathbf{A}(\mathbf{c} + \mathbf{U})$  with  $\mathbf{A} = (\mathbf{I} - \mathbf{B})^{-1}$ , and  $\text{Cov}(\mathbf{V}) = \mathbf{A}\mathbf{\Omega}\mathbf{A}^\top$ ;
- (b) for  $j \neq l$ , the entry  $a_{jl}$  of  $\mathbf{A}$  is the sum, over all directed paths from  $l$  to  $j$ , of the products of the coefficients on the arrows of the path (zero if there is none), and  $a_{jj} = 1$ ;
- (c)  $E(V_j \mid \text{do}(V_l = x)) = E(V_j \mid \text{do}(V_l = 0)) + a_{jl}x$ .

The number  $a_{jl}$  is the **total effect** of  $V_l$  on  $V_j$ : the change in the interventional mean of  $V_j$  per unit change in the value imposed on  $V_l$ .

*Proof.*

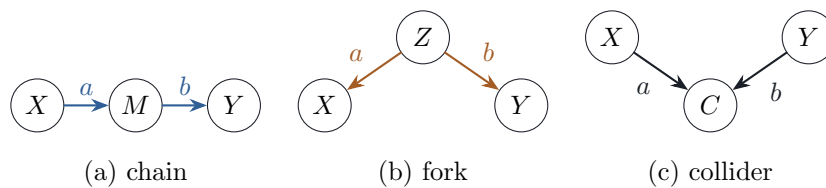
- (a)  $\mathbf{I} - \mathbf{B}$  is unit lower triangular, hence invertible, and  $(\mathbf{I} - \mathbf{B})\mathbf{V} = \mathbf{c} + \mathbf{U}$ . The covariance follows from [Theorem 2.2.1](#).
- (b)  $\mathbf{B}$  is strictly lower triangular, so  $\mathbf{B}^m = \mathbf{0}$  and  $(\mathbf{I} - \mathbf{B})(\mathbf{I} + \mathbf{B} + \dots + \mathbf{B}^{m-1}) = \mathbf{I} - \mathbf{B}^m = \mathbf{I}$ . The  $(j, l)$  entry of  $\mathbf{B}^r$  is  $\sum b_{jk_1}b_{k_1k_2} \dots b_{k_{r-1}l}$  over all  $k_1, \dots, k_{r-1}$ ; a term is nonzero only if  $l \rightarrow k_{r-1} \rightarrow \dots \rightarrow k_1 \rightarrow j$  are arrows, which is a directed path of length  $r$  (its nodes are distinct because the graph is acyclic). Summing over  $r$  gives (b).
- (c) The intervened model is linear with  $\mathbf{B}$  replaced by  $\tilde{\mathbf{B}}$ , in which row  $l$  is zero, and with  $c_l + U_l$  replaced by  $x$ . A directed path out of  $l$  never uses an arrow into  $l$ , so the directed paths from  $l$  in the graph of  $\tilde{\mathbf{B}}$  are those of  $\mathbf{B}$ , and by (b) the  $(j, l)$  entry of  $(\mathbf{I} - \tilde{\mathbf{B}})^{-1}$  is  $a_{jl}$ . By (a) applied to the intervened model,  $V_j$  is  $a_{jl}x$  plus terms not involving  $x$ .

■

This is Wright's method of path coefficients (Wright 1921, 1934); the covariance formula in (a) gives his rule for correlations ([Exercise \(C1\)](#)).

### 25.3.3 Chains, forks and colliders

Three configurations of three variables are the building blocks of everything the graph says about conditional independence ([Figure 25.3.1](#)).



**Figure 25.3.1.** The three structures of [Proposition 25.3.3](#). In (a) and (b) the middle node

transmits association between  $X$  and  $Y$ , and conditioning on it removes the association. In (c) the middle node transmits nothing until it is conditioned on.

### Proposition 25.3.3 — Chains, forks and colliders

Consider linear models with independent disturbances,  $\text{Var } U_X = \sigma_X^2 > 0$  and so on, and write  $\sigma_{XY \cdot W}$  for the partial covariance of  $X$  and  $Y$  given  $W$  (Definition 3.6.1).

- (a) **Chain**  $X = U_X$ ,  $M = aX + U_M$ ,  $Y = bM + U_Y$ :  $\text{Cov}(X, Y) = ab\sigma_X^2$  and  $\sigma_{XY \cdot M} = 0$ .
- (b) **Fork**  $Z = U_Z$ ,  $X = aZ + U_X$ ,  $Y = bZ + U_Y$ :  $\text{Cov}(X, Y) = ab\sigma_Z^2$  and  $\sigma_{XY \cdot Z} = 0$ .
- (c) **Collider**  $X = U_X$ ,  $Y = U_Y$ ,  $C = aX + bY + U_C$ :  $\text{Cov}(X, Y) = 0$  and

$$\sigma_{XY \cdot C} = -\frac{ab\sigma_X^2\sigma_Y^2}{a^2\sigma_X^2 + b^2\sigma_Y^2 + \sigma_C^2}. \quad (25.3.2)$$

With normal disturbances, a zero partial covariance means conditional independence (Proposition 3.6.1), so in (a) and (b)  $X$  and  $Y$  are dependent but conditionally independent given the middle node, and in (c) they are independent but, if  $ab \neq 0$ , conditionally dependent given  $C$ .

*Proof.* The covariances follow from Proposition 25.3.2(a). (a)  $Y - bM = U_Y$  is uncorrelated with  $X$  and  $M$ , so  $L(Y | M, X) = bM$  has no  $X$  term, and by Proposition 14.5.2  $\sigma_{XY \cdot M} = 0$ . (b) The same argument with  $Y - bZ = U_Y$ . (c) By definition,

$$\sigma_{XY \cdot C} = \text{Cov}(X, Y) - \frac{\text{Cov}(X, C)\text{Cov}(Y, C)}{\text{Var}(C)} = 0 - \frac{(a\sigma_X^2)(b\sigma_Y^2)}{\text{Var}(C)}.$$

■

In the collider, independent causes become dependent once their common effect is held fixed. In the listing, with  $a = b = 1$  and unit variances, the correlations of  $X$  and  $Y$  in the chain and the fork are 0.577 and 0.500 (simulated: 0.582 and 0.505), and both partial correlations given the middle node are zero. In the collider the correlation is zero and the partial correlation given  $C$  is  $-0.500$  (simulated  $-0.501$ ).

```
import numpy as np

rng = np.random.default_rng(2503)
n = 100_000
a, b = 1.0, 1.0                                     # the two edge coefficients

def corr_and_partial(x, y, w):
    """Correlation of x and y, and their partial correlation given w (residuals on
    1, w)."""
    W = np.column_stack([np.ones(len(w)), w])
    rx = x - W @ np.linalg.lstsq(W, x, rcond=None)[0]
    ry = y - W @ np.linalg.lstsq(W, y, rcond=None)[0]
    return np.corrcoef(x, y)[0, 1], np.corrcoef(rx, ry)[0, 1]

U = rng.normal(size=(3, n))
x = U[0]; m = a * x + U[1]; y = b * m + U[2]         # chain    X -> M -> Y
```

```

chain = corr_and_partial(x, y, m)
z = U[0]; x = a * z + U[1]; y = b * z + U[2]          # fork      X <- Z -> Y
fork = corr_and_partial(x, y, z)
x = U[0]; y = U[1]; c = a * x + b * y + U[2]          # collider  X -> C <- Y
collider = corr_and_partial(x, y, c)
for name, (r, rp) in [("chain", chain), ("fork", fork), ("collider", collider)]:
    print(f"{name:9s} corr {r:6.3f}    partial corr given middle {rp:6.3f}")

```

### 25.3.4 d-separation

In a larger graph association can travel along many paths at once; d-separation combines the three structures.

#### Definition 25.3.2 — d-separation

Let  $\mathcal{Z}$  be a set of nodes. A path is **blocked** by  $\mathcal{Z}$  if

- (i) some non-collider on the path is in  $\mathcal{Z}$ , or
- (ii) some collider on the path is not in  $\mathcal{Z}$  and has no descendant in  $\mathcal{Z}$ .

Otherwise the path is **open** given  $\mathcal{Z}$ . Disjoint sets of nodes  $\mathcal{A}$  and  $\mathcal{B}$  are **d-separated** by  $\mathcal{Z}$  (disjoint from both) if every path between a node of  $\mathcal{A}$  and a node of  $\mathcal{B}$  is blocked by  $\mathcal{Z}$ .

A path is open when each non-collider is left free and each collider is conditioned on, directly or through a descendant, a noisy measurement of it ([Exercise \(B1\)](#)).

#### Theorem 25.3.4 — d-separation and conditional independence

Let  $\mathbf{V}$  follow a structural causal model with graph  $G$ , and let  $\mathcal{A}, \mathcal{B}, \mathcal{Z}$  be disjoint sets of nodes.

- (a) If  $\mathcal{Z}$  d-separates  $\mathcal{A}$  and  $\mathcal{B}$  in  $G$ , then  $\mathbf{V}_{\mathcal{A}}$  and  $\mathbf{V}_{\mathcal{B}}$  are conditionally independent given  $\mathbf{V}_{\mathcal{Z}}$ .
- (b) If  $\mathcal{Z}$  does not d-separate them, there is a linear structural causal model on  $G$  with normal disturbances in which they are conditionally dependent; in fact this holds for all edge coefficients and disturbance variances outside a set of Lebesgue measure zero.

*Proof. Sketch; full proofs are in the references.* For (a) with densities (Lauritzen, Dawid, Larsen and Leimer 1990): the nodes of  $\mathcal{A} \cup \mathcal{B} \cup \mathcal{Z}$  and their ancestors form a set  $\mathcal{N}$  whose density, by [Proposition 25.3.1](#), is a product of functions of the sets  $\{j\} \cup \text{pa}(j)$ ,  $j \in \mathcal{N}$ . These sets are complete in the *moral graph* on  $\mathcal{N}$ , which joins each node to its parents and any two parents of a common child. If  $\mathcal{Z}$  separates  $\mathcal{A}$  from  $\mathcal{B}$  there, the factors split into two groups, giving the conditional independence; a combinatorial lemma shows that d-separation in  $G$  is equivalent to this separation. Geiger, Verma and Pearl (1990) give another proof. For (b), partial correlations of a linear normal model are rational functions of the parameters, not identically zero along an open path, and so vanish only

on a set of measure zero (Spirtes, Glymour and Scheines 2000). ■

Part (b) says the converse of (a) fails only by accident, as in [Exercise \(A1\)](#), where the effect of visits and the association through illness cancel exactly. Distributions with no such cancellation are called **faithful** to the graph.

## 25.3.5 Confounding as an open path

### Definition 25.3.3 — Back-door paths and confounding

A **back-door path** from  $X$  to  $Y$  is a path between them whose arrow at  $X$  points into  $X$ ,  $X \leftarrow \cdots Y$ . The effect of  $X$  on  $Y$  is **confounded** if some back-door path from  $X$  to  $Y$  is open given the empty set, that is, contains no collider.

Directed paths from  $X$  to  $Y$  carry the effect of  $X$ ; back-door paths carry association that reflects whatever caused  $X$ . In a linear model the split is exact.

### Proposition 25.3.5 — Confounding bias in a linear model

In a linear structural causal model with disturbances of positive variance, let  $X = V_k$ ,  $Y = V_j$ , and let  $\tau = a_{jk}$  be the total effect of  $X$  on  $Y$ . For each  $l$ , let  $d_l$  be the sum over the directed paths from  $V_l$  to  $Y$  that do not pass through  $X$  of the products of coefficients (so  $d_k = 0$ , and  $d_j = 1$  from the trivial path of  $Y$  to itself).

- (a)  $Y = \tau X + R$ , where  $R = \sum_l d_l(c_l + U_l)$  is the value of  $Y$  under  $\text{do}(X = 0)$ . Under  $\text{do}(X = x)$ ,  $Y = \tau x + R$ .
- (b) The slope of the best linear predictor  $L(Y | X)$  is

$$\tau + \frac{\text{Cov}(R, X)}{\text{Var}(X)}, \quad \text{Cov}(R, X) = \sum_l \omega_l d_l a_{kl}. \quad (25.3.3)$$

- (c) If the effect of  $X$  on  $Y$  is not confounded, the slope of  $L(Y | X)$  is  $\tau$ .

*Proof.*

- (a) A directed path from  $V_l$  to  $Y$  either avoids  $X$  or passes through it exactly once, and a path of the second kind splits at  $X$  into a directed path from  $V_l$  to  $X$  and one from  $X$  to  $Y$ . Summing products over paths with [Proposition 25.3.2\(b\)](#),  $a_{jl} = d_l + a_{kl}\tau$  for every  $l$  (for  $l = k$ ,  $a_{kk} = 1$  and  $d_k = 0$ ; for  $l = j$ ,  $a_{jj} = 1 = d_j$  and  $a_{kj} = 0$ ). Hence  $Y = \sum_l a_{jl}(c_l + U_l) = \tau \sum_l a_{kl}(c_l + U_l) + R = \tau X + R$ . Under  $\text{do}(X = x)$ , as in the proof of [Proposition 25.3.2\(c\)](#), the arrows into  $X$  are removed, so the directed paths from  $V_l$ ,  $l \neq k$ , to  $Y$  in the intervened graph are exactly those that avoid  $X$ . Thus  $Y = \tau x + \sum_{l \neq k} d_l(c_l + U_l) = \tau x + R$ .
- (b) The slope is  $\text{Cov}(X, Y) / \text{Var}(X) = \tau + \text{Cov}(R, X) / \text{Var}(X)$  by [Theorem 6.11.1](#), and  $\text{Cov}(R, X) = \sum_{l, l'} d_l a_{kl'} \text{Cov}(U_l, U_{l'}) = \sum_l \omega_l d_l a_{kl}$ , since the disturbances are uncorrelated.

- (c) Suppose  $\omega_l d_l a_{kl} \neq 0$  for some  $l$ . Then  $l \neq k$ , and there are a directed path  $\pi_1$  from  $V_l$  to  $X$  and a directed path  $\pi_2$  from  $V_l$  to  $Y$  that avoids  $X$ . Let  $w$  be the node of  $\pi_1$  closest to  $X$  that also lies on  $\pi_2$ ; it exists because  $V_l$  lies on both, and  $w \neq X$  because  $\pi_2$  avoids  $X$ . Follow  $\pi_1$  backwards from  $X$  to  $w$ , then  $\pi_2$  forwards from  $w$  to  $Y$ . By the choice of  $w$  the nodes are distinct, so this is a path. Its arrow at  $X$  is the last arrow of  $\pi_1$ , which points into  $X$ , and every arrow on it points away from  $w$ , so it has no collider. The effect is therefore confounded. Contrapositively, if it is not confounded, every term of  $\text{Cov}(R, X)$  vanishes.

■

Part (a) is the potential outcome  $Y(x) = \tau x + Y(0)$ : linear structural models have constant effects. Part (b) writes the regression slope as the effect plus one contribution for each common ancestor of  $X$  and  $Y$ . In the model of [Figure 25.1.1](#),  $R = c_Y + \gamma Z + U_Y$  and the only term comes from  $Z$ :  $\omega da = \sigma_Z^2 \cdot \gamma \cdot \alpha$ , which is [Eq. 25.1.3](#).

This is the omitted-variable bias of least squares: by [Proposition 5.7.2](#) the short regression estimates  $\theta + \pi\gamma$ , with  $\pi$  the sample slope of  $\mathbf{z}$  on  $\mathbf{x}$ , whose limit is  $\alpha\sigma_Z^2/\text{Var}(X)$  (see also [Theorem 19.1.2](#) in [Chapter 19](#)). The graph adds that an omitted variable causes bias only if it lies on an open back-door path; an omitted mediator should stay omitted ([Section 25.5](#)).

## 25.3.6 Exercises

### 25.3.6.1 A. Check your understanding

#### Exercise A1.

A DAG has arrows  $A \rightarrow X$ ,  $A \rightarrow B$ ,  $B \rightarrow Y$ ,  $X \rightarrow Y$ ,  $X \rightarrow C$ ,  $Y \rightarrow C$  and  $D \rightarrow C$ . List every path between  $X$  and  $Y$ , mark the back-door paths, and say which paths are open given  $\emptyset$ , given  $\{B\}$  and given  $\{C\}$ .

*Solution.* There are three paths:  $X \rightarrow Y$ ;  $X \leftarrow A \rightarrow B \rightarrow Y$ , a back-door path; and  $X \rightarrow C \leftarrow Y$ , with collider  $C$ . (The arrow  $D \rightarrow C$  gives no path to  $Y$  except through  $C$ , where  $D \rightarrow C \leftarrow Y$  would require revisiting  $C$ .) Given  $\emptyset$ : the first two are open and the third is blocked by the collider. Given  $\{B\}$ : only  $X \rightarrow Y$  is open. Given  $\{C\}$ : all three are open, since conditioning on the collider opens the third.

### 25.3.6.2 B. Practice

#### Exercise B1.

Extend the collider by  $D = C + U_D$ , with  $\text{Var } U_D = \delta$  and all other variances one. Compute  $\sigma_{XY \cdot D}$  and show that it lies between 0 and  $\sigma_{XY \cdot C}$ , approaching them as  $\delta \rightarrow \infty$  and  $\delta \rightarrow 0$ .

*Solution.*  $\text{Cov}(X, D) = a$ ,  $\text{Cov}(Y, D) = b$  and  $\text{Var } D = a^2 + b^2 + 1 + \delta$ , so  $\sigma_{XY \cdot D} = -ab/(a^2 + b^2 + 1 + \delta)$ . At  $\delta = 0$  this is [Eq. 25.3.2](#), and it decreases in absolute value to zero as  $\delta$  grows. A proxy for a collider opens the path partially, the more so the better the proxy.



## 25.3.6.3 C. Going deeper

**Exercise C1.**

Show from [Proposition 25.3.2\(a\)](#) that in a linear model  $\text{Cov}(V_j, V_k) = \sum_l \omega_l a_{jl} a_{kl}$ . Interpret each term as a sum over pairs of directed paths from  $V_l$  to  $V_j$  and to  $V_k$ , and use this to compute  $\text{Cov}(X, Y)$  in the chain of [Proposition 25.3.3](#).

## 25.4 Adjustment sets and the back-door criterion

If confounding is association along back-door paths, the remedy is to block them by conditioning on variables on them, that is, by including those variables as regressors. This section says which sets do the job, proves that they do, and compares their precision.

### 25.4.1 The criterion

#### Definition 25.4.1 — Back-door criterion

Let  $X$  and  $Y$  be nodes of a DAG. A set  $\mathcal{Z}$  of nodes, containing neither  $X$  nor  $Y$ , satisfies the **back-door criterion** relative to  $(X, Y)$  if

- (i) no node in  $\mathcal{Z}$  is a descendant of  $X$ , and
- (ii)  $\mathcal{Z}$  blocks every back-door path from  $X$  to  $Y$  ([Definition 25.3.3](#)).

Such a set is called a (back-door) **adjustment set**.

Condition (ii) closes the non-causal routes. Condition (i) protects the causal ones: a descendant of  $X$  may be a mediator, whose conditioning removes part of the effect, or a collider, whose conditioning opens a path ([Section 25.5](#)). The empty set qualifies exactly when the effect is not confounded, and the parents of  $X$  always qualify, provided  $Y$  is not one of them, since every back-door path begins  $X \leftarrow P$  with  $P$  a parent and a non-collider. The graph itself is an assumption that the data cannot check ([Exercise \(B3\)](#)); the criterion turns it into a precise statement about which regression to run.

Recall from [Section 25.2](#) that  $Y(x)$  is the value of  $Y$  in the model modified by  $\text{do}(X = x)$ , and that  $Y = Y(x)$  on the event  $X = x$ . Conditional independence of  $Y(x)$  and  $X$  given  $\mathbf{Z}$  is the potential-outcome form of “no confounding given  $\mathbf{Z}$ ”, which Rosenbaum and Rubin (1983), adding positivity, called *strong ignorability*.

#### Theorem 25.4.1 — Back-door adjustment

Let  $\mathbf{V}$  follow a structural causal model with graph  $G$ , let  $X$  and  $Y$  be two of its variables, and let  $\mathbf{Z}$  be the vector of the variables in a set  $\mathcal{Z}$  of other nodes.

- (a) **Adjustment formula.** Suppose  $X$  takes finitely many values,  $\Pr(X = x \mid \mathbf{Z}) > 0$  with probability one,  $E|Y(x)| < \infty$ , and  $Y(x)$  is conditionally independent of  $X$  given  $\mathbf{Z}$ . Then

$$E(Y \mid \text{do}(X = x)) = E[E(Y \mid X = x, \mathbf{Z})], \quad (25.4.1)$$

where the outer expectation is over the distribution of  $\mathbf{Z}$ .

- (b) **The criterion gives ignorability.** If  $\mathcal{Z}$  satisfies the back-door criterion relative to  $(X, Y)$ , then  $Y(x)$  is conditionally independent of  $X$  given  $\mathbf{Z}$ , for every  $x$ .
- (c) **Linear models.** In a linear structural causal model whose disturbances have finite positive variances, if  $\mathcal{Z}$  satisfies the back-door criterion, the coefficient of  $X$  in  $L(Y \mid X, \mathbf{Z})$  is the total effect  $\tau$  of  $X$  on  $Y$ . Consequently, for independent copies of  $\mathbf{V}$ , the least squares coefficient of  $X$  in the regression of  $Y$  on 1,  $X$  and  $\mathbf{Z}$  converges to  $\tau$  with probability one.

The proofs of (b) and (c) below are complete when  $\mathcal{Z}$  contains every parent of  $X$ ; for general adjustment sets they are sketches, since the proof of (c) uses (b).

*Proof.*

- (a) Using the tower rule, then the conditional independence together with positivity, then  $Y = Y(x)$  on  $\{X = x\}$ ,

$$\begin{aligned} E Y(x) &= E[E(Y(x) \mid \mathbf{Z})] \\ &= E[E(Y(x) \mid X = x, \mathbf{Z})] = E[E(Y \mid X = x, \mathbf{Z})], \end{aligned}$$

and  $E Y(x) = E(Y \mid \text{do}(X = x))$  by definition.

- (b) *When  $\mathcal{Z}$  contains the parents of  $X$ .* Let  $X = V_k$ , and write  $\mathbf{U}_{-k}$  for the disturbances other than  $U_k$ . A node that is not a descendant of  $X$  has no directed path from  $X$ , so solving the assignments shows that it is a function of  $\mathbf{U}_{-k}$  alone; by (i), so is  $\mathbf{Z}$ . In the model modified by  $\text{do}(X = x)$ , the disturbance  $U_k$  appears nowhere, so  $Y(x)$  is also a function of  $\mathbf{U}_{-k}$ . Hence  $U_k$  is independent of  $(Y(x), \mathbf{Z})$ , and given  $\mathbf{Z}$  it remains independent of  $Y(x)$ , with its unconditional distribution. Finally  $X = f_k(\mathbf{Z}_{\text{pa}}, U_k)$ , where  $\mathbf{Z}_{\text{pa}}$  is the part of  $\mathbf{Z}$  formed by the parents of  $X$ . Given  $\mathbf{Z}$ ,  $X$  is a function of  $U_k$  alone, and so is conditionally independent of  $Y(x)$ . *General adjustment sets (sketch).* Form a graph containing both the original variables and their counterparts in the model modified by  $\text{do}(X = x)$ , joined through shared disturbance nodes (a *twin network*);  $Y(x)$  is a node of it, and  $X$  has no arrows into the counterpart variables. A path from  $X$  to  $Y(x)$  that leaves  $X$  through a child of  $X$  must turn back somewhere, at the latest at a disturbance node shared with a counterpart variable, so it meets a collider that is a descendant of  $X$ ; by (i) neither that collider nor its descendants are in  $\mathcal{Z}$ , and the path is blocked. A path that leaves  $X$  through a parent starts like a back-door path, and one checks, using (ii) and again (i), that  $\mathcal{Z}$  blocks it. So  $\mathcal{Z}$  d-separates  $X$  from  $Y(x)$ , and [Theorem 25.3.4\(a\)](#) gives the conditional independence. Pearl (1995, 2009) gives the criterion and its proof in the language of interventions, and Shpitser, VanderWeele and Robins (2010) characterize exactly which sets satisfy [Eq. 25.4.1](#) for every model on  $G$ .
- (c) By [Proposition 25.3.5\(a\)](#),  $Y = \tau X + R$ , where  $R$  is the value of  $Y$  under  $\text{do}(X = 0)$ , that is,  $R = Y(0)$ . The variables  $R$ ,  $X$  and  $\mathbf{Z}$  are linear functions of the disturbances with coefficients determined by  $\mathbf{B}$ , so their covariance matrix depends

only on  $\mathbf{B}$  and  $\mathbf{\Omega}$ . The covariance matrix of  $(X, \mathbf{Z})$  is positive definite, because  $\text{Cov}(\mathbf{V}) = \mathbf{A}\mathbf{\Omega}\mathbf{A}^\top$  with  $\mathbf{A}$  invertible and  $\mathbf{\Omega}$  positive definite. By [Proposition 14.5.2](#) the coefficient of  $X$  in  $L(Y \mid X, \mathbf{Z})$  is

$$\frac{\sigma_{YX \cdot Z}}{\sigma_{XX \cdot Z}} = \tau + \frac{\sigma_{RX \cdot Z}}{\sigma_{XX \cdot Z}},$$

with  $\sigma_{XX \cdot Z} > 0$ . Consider the linear model with the same  $\mathbf{B}$  and  $\mathbf{\Omega}$  but normal disturbances. It has the same covariance matrix of  $(R, X, \mathbf{Z})$ , hence the same partial covariance  $\sigma_{RX \cdot Z}$ . In it  $(R, X, \mathbf{Z})$  is jointly normal and, by (b),  $R = Y(0)$  is conditionally independent of  $X$  given  $\mathbf{Z}$ , so  $\sigma_{RX \cdot Z} = 0$  by [Proposition 3.6.1\(b\)](#). Therefore the coefficient is  $\tau$ . (This step is complete when  $\mathcal{Z}$  contains the parents of  $X$ , and otherwise inherits the sketch of (b).) The last statement is [Proposition 6.11.3](#). ■

Part (a) averages the conditional mean at  $X = x$  over the population distribution of  $\mathbf{Z}$ , not over its distribution among units with  $X = x$  (direct standardization), and needs positivity: where  $X = x$  never occurs, the data say nothing about  $E(Y \mid X = x, \mathbf{Z})$ . Part (c) answers the question left open after the Frisch–Waugh–Lovell theorem: the multiple-regression coefficient of  $X$  in a linear model is the effect of changing  $X$  whenever the other regressors form an adjustment set.

*Warning.* Only the coefficient of  $X$  acquires a causal meaning. The coefficients of the adjustment variables are generally not their effects, because  $X$  may mediate them; reporting them as effects is the “Table 2 fallacy” (Westreich and Greenland 2013), computed in [Exercise \(B2\)](#).

## 25.4.2 An example with eight variables

**Example. (Choosing an adjustment set)**

[Figure 25.4.1](#) shows a linear model with unit-variance disturbances and

$$\begin{aligned} W_2 &= 0.8W_1 + U, & X &= 0.7W_1 + 0.6W_3 + 0.9I + U, & M &= 0.8X + U, \\ Y &= 0.5X + 0.6M + 0.7W_2 + 0.8W_3 + P + U, \end{aligned}$$

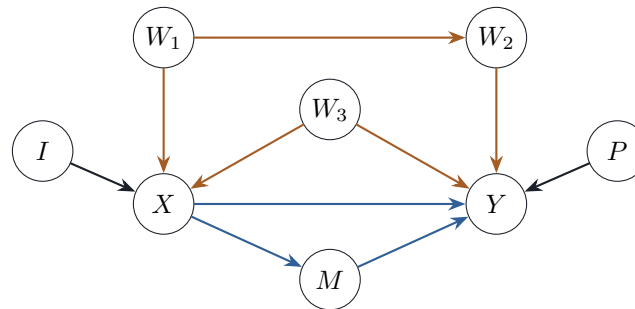
with a separate disturbance  $U$  in each equation and  $W_1, W_3, I, P$  parentless. Summing over the two directed paths,  $\tau = 0.5 + 0.8 \times 0.6 = 0.98$ . The back-door paths  $X \leftarrow W_1 \rightarrow W_2 \rightarrow Y$  and  $X \leftarrow W_3 \rightarrow Y$  are blocked by  $W_1$  or  $W_2$ , and by  $W_3$ , and  $M$  is a descendant of  $X$ . So a set is valid iff it contains  $W_3$  and one of  $W_1, W_2$ , and not  $M$ ;  $I$  and  $P$  are optional.

The listing computes the coefficient of  $X$  in  $L(Y \mid X, \mathbf{Z})$  exactly from  $\text{Cov}(\mathbf{V}) = \mathbf{A}\mathbf{A}^\top$ , with the factor  $\sigma_{Y \cdot XZ}/\sigma_{X \cdot Z}$  that governs precision (see below), and the standard deviation over 4000 simulated samples of size  $n = 500$ :

| adjustment set | valid | coefficient of $X$ | $\sigma_{Y \cdot XZ}/\sigma_{X \cdot Z}$ | simulated sd |
|----------------|-------|--------------------|------------------------------------------|--------------|
| none           | no    | 1.308              | 1.15                                     | 0.052        |
| $W_3$          | no    | 1.150              | 1.16                                     | 0.052        |

|                       |     |       |      |       |
|-----------------------|-----|-------|------|-------|
| $W_1, W_3$            | yes | 0.980 | 1.25 | 0.056 |
| $W_2, W_3$            | yes | 0.980 | 1.06 | 0.047 |
| $W_2, W_3, P$         | yes | 0.980 | 0.80 | 0.036 |
| $W_1, W_3, I$         | yes | 0.980 | 1.69 | 0.076 |
| $W_1, W_3, M$         | no  | 0.500 | 1.72 | 0.077 |
| $W_1, W_2, W_3, I, P$ | yes | 0.980 | 1.17 | 0.052 |

Every valid set gives  $\tau$ ; the invalid sets leave a back-door path open or, with  $M$ , keep only the direct effect 0.5. Among valid sets the standard deviation varies by a factor of two (Figure 25.4.2).



**Figure 25.4.1.** The graph of Example (Choosing an adjustment set). Blue arrows form the directed paths from  $X$  to  $Y$ ; brown arrows form the two back-door paths.  $I$  influences only  $X$ , and  $P$  only  $Y$ .

```
import numpy as np

names = ["W1", "W3", "I", "P", "W2", "X", "M", "Y"]      # a topological order
k = {v: j for j, v in enumerate(names)}
B = np.zeros((8, 8))      # B[j, i]: coefficient of
    i in j's equation
for child, parent, coef in [
    ("W2", "W1", 0.8), ("X", "W1", 0.7), ("X", "W3", 0.6), ("X", "I", 0.9),
    ("M", "X", 0.8), ("Y", "X", 0.5), ("Y", "M", 0.6), ("Y", "W2", 0.7),
    ("Y", "W3", 0.8), ("Y", "P", 1.0)]:
    B[k[child], k[parent]] = coef
A = np.linalg.inv(np.eye(8) - B)      # V = A U
Sigma = A @ A.T      # unit disturbance
    variances
tau = A[k["Y"], k["X"]]      # total effect: sum over
    directed paths

def population_fit(Z):
    """Coefficient of X in L(Y | X, Z) and the large-sample sd factor sigma_{Y.XZ}/
    sigma_{X.Z}."""
    R = [k["X"]] + [k[z] for z in Z]
    coef = np.linalg.solve(Sigma[np.ix_(R, R)], Sigma[R, k["Y"]])
    res_y = Sigma[k["Y"], k["Y"]] - Sigma[k["Y"], R] @ coef
    Zi = [k[z] for z in Z]
    res_x = Sigma[k["X"], k["X"]] - (Sigma[k["X"], Zi] @ np.linalg.solve(Sigma[np.
    ix_(Zi, Zi)], Sigma[Zi, k["X"]]))
    if Z else 0.0)
```

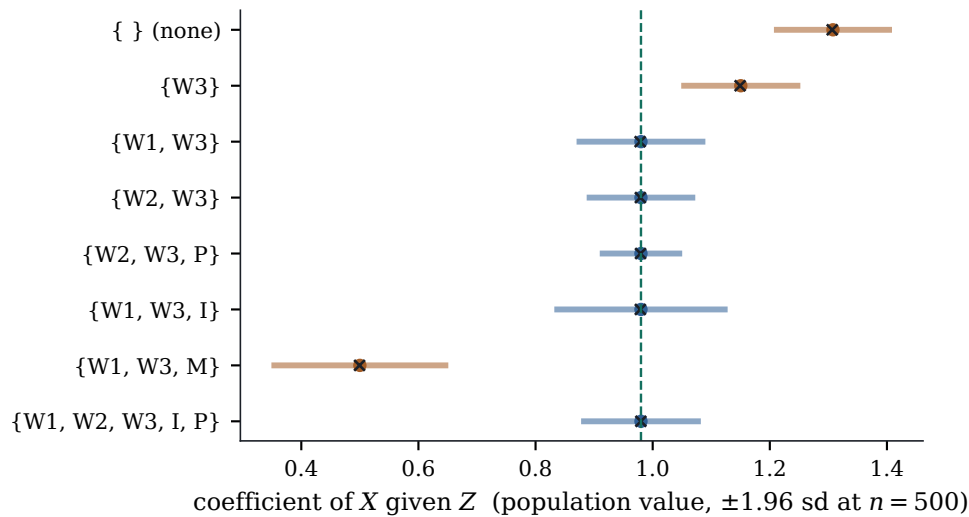


Figure 25.4.1: The coefficient of  $X$  given each candidate set in [Example \(Choosing an adjustment set\)](#): population value (dot), a band of  $\pm 1.96$  large-sample standard deviations at  $n = 500$ , and the mean over the simulated samples (cross). Blue sets satisfy the back-door criterion; the dashed line is  $\tau = 0.98$ .

```

return coef[0], np.sqrt(res_y / res_x)

sets = [[], ["W3"], ["W1", "W3"], ["W2", "W3"], ["W2", "W3", "P"], ["W1", "W3", "I"],
        ["W1", "W3", "M"], ["W1", "W2", "W3", "I", "P"]]
print(f"total effect of X on Y: {tau:.3f}")
for Z in sets:
    c, f = population_fit(Z)
    print(f"Z = {{{', '.join(Z)}}} ".ljust(26) + f"coefficient {c:6.3f}    sd factor {f:5.3f}")

```

### 25.4.3 Which adjustment set is most precise

All valid sets estimate  $\tau$ , so the choice among them is a question of variance. With normal disturbances,  $Y$  given  $(X, \mathbf{Z})$  follows a linear model with variance  $\sigma_{Y \cdot XZ}^2$ , and given the regressors the coefficient of  $X$  has variance  $\sigma_{Y \cdot XZ}^2 / \|(\mathbf{I} - \mathbf{M}_Z)\mathbf{x}\|^2$ , where  $\mathbf{M}_Z$  projects onto the span of  $\mathbf{1}$  and the columns of  $\mathbf{Z}$  ([Theorem 6.6.2](#) and [Eq. 6.6.2](#)). Divided by  $n$ , the squared length in the denominator is the residual mean square of the regression of  $X$  on  $\mathbf{Z}$ , which converges to  $\sigma_{X \cdot Z}^2$  with probability one ([Proposition 6.11.3](#)). So  $n$  times the variance tends to

$$v(\mathcal{Z}) = \frac{\sigma_{Y \cdot XZ}^2}{\sigma_{X \cdot Z}^2}, \quad (25.4.2)$$

the square of the factor in the table. A good adjustment set explains much of  $Y$  and little of  $X$ .

#### Proposition 25.4.2 — Adding a variable to an adjustment set

Let  $\mathcal{Z}$  and  $\mathcal{Z} \cup \{W\}$  both satisfy the back-door criterion, with all covariance matrices positive definite.

- (a) If  $\sigma_{YW \cdot XZ} = 0$ , then  $v(\mathcal{Z} \cup \{W\}) \geq v(\mathcal{Z})$ .
- (b) If  $\sigma_{XW \cdot Z} = 0$ , then  $v(\mathcal{Z} \cup \{W\}) \leq v(\mathcal{Z})$ .

*Proof.* Partial variances are the mean squared errors of best linear predictors, and projecting onto a larger space can only reduce them ([Theorem 2.5.2\(c\)](#)). (a) By [Proposition 14.5.2](#) applied with  $(X, \mathbf{Z})$  as conditioning variables,  $\sigma_{YW \cdot XZ} = 0$  means that  $W$  has coefficient zero in  $L(Y \mid X, \mathbf{Z}, W)$ , so this predictor equals  $L(Y \mid X, \mathbf{Z})$  and the numerator of [Eq. 25.4.2](#) is unchanged; the denominator can only decrease. (b) Likewise  $\sigma_{XW \cdot Z} = 0$  means  $L(X \mid \mathbf{Z}, W) = L(X \mid \mathbf{Z})$ , so the denominator is unchanged, while the numerator can only decrease. ■

In [Example \(Choosing an adjustment set\)](#), a direct computation from  $\text{Cov}(\mathbf{V}) = \mathbf{A}\mathbf{A}^\top$  gives  $\sigma_{YI \cdot XW_1W_3} = 0$  and  $\sigma_{XP \cdot W_2W_3} = 0$ : the pure cause  $I$  of  $X$  satisfies (a) and the pure cause  $P$  of  $Y$  satisfies (b). Accordingly, adding  $I$  to  $\{W_1, W_3\}$  raises the factor from 1.25 to 1.69; adding  $P$  to  $\{W_2, W_3\}$  lowers it from 1.06 to 0.80. Replacing  $W_1$  by  $W_2$ , closer to  $Y$  on the back-door path, also helps. Henckel, Perković and Maathuis (2022) turn these observations into a graphical rule for the valid set of smallest asymptotic variance in linear models: roughly, the parents of  $Y$  and of the mediators, other than the mediators and  $X$ .

If every adjustment set contains an unrecorded variable, as in [Example \(Clinic visits and health\)](#), other structure must be used: an instrumental variable ([Theorem 24.4.2](#) in [Chapter 24](#)), or the front-door criterion of Pearl (1995), which uses an observed mediator; or a sensitivity analysis asks how strong an unobserved confounder would have to be, using formulas such as [Eq. 25.3.3](#).

## 25.4.4 Exercises

### 25.4.4.1 A. Check your understanding

#### Exercise A1.

In [Example \(Choosing an adjustment set\)](#), decide which of  $\{W_1, W_2\}$ ,  $\{W_3, P\}$  and  $\{W_2, W_3, M\}$  satisfy the back-door criterion, and explain the coefficients of  $X$  that the listing gives for them: 1.201, 1.150 and 0.500.

*Solution.* None does.  $\{W_1, W_2\}$  leaves  $X \leftarrow W_3 \rightarrow Y$  open, and  $\{W_3, P\}$  leaves  $X \leftarrow W_1 \rightarrow W_2 \rightarrow Y$  open, so both coefficients exceed  $\tau = 0.98$ . The set  $\{W_2, W_3, M\}$  blocks both back-door paths but contains the mediator  $M$ , violating (i); holding  $M$  fixed leaves only the direct arrow, with coefficient 0.5.

### 25.4.4.2 B. Practice

#### Exercise B1.

Let  $X \in \{0, 1\}$  and let  $Z$  take values  $1, \dots, K$ . In a sample, stratum  $z$  has  $n_z$  units, a fraction  $\hat{p}_z \in (0, 1)$  of them with  $X = 1$ , and difference of mean responses  $\hat{\delta}_z = \bar{y}_{1z} - \bar{y}_{0z}$ . Show that the sample version of [Eq. 25.4.1](#) for the effect  $E(Y \mid \text{do}(X = 1)) - E(Y \mid \text{do}(X = 0))$  is  $\sum_z (n_z/n) \hat{\delta}_z$ , while the coefficient of  $X$  in the least squares regression of

$Y$  on  $X$  and the  $K$  stratum indicators is

$$\hat{\beta}_X = \frac{\sum_z n_z \hat{p}_z (1 - \hat{p}_z) \hat{\delta}_z}{\sum_z n_z \hat{p}_z (1 - \hat{p}_z)}.$$

When do they agree?

*Solution.* Replacing  $E(Y \mid X = x, Z = z)$  by  $\bar{y}_{xz}$  and the distribution of  $Z$  by the sample fractions  $n_z/n$  gives the first expression. For the second, apply [Theorem 6.6.2](#) with  $\mathbf{X}_2$  the stratum indicators. Projecting onto their span replaces each value by its stratum mean ([Example \(The one-way layout\)](#)), so  $\tilde{x}_i = x_i - \hat{p}_{z(i)}$ . Then  $\hat{\beta}_X = \sum_i \tilde{x}_i y_i / \sum_i \tilde{x}_i^2$ . Within stratum  $z$ ,  $\sum_i \tilde{x}_i^2 = n_z \hat{p}_z (1 - \hat{p}_z)$  and  $\sum_i \tilde{x}_i y_i = n_z \hat{p}_z (1 - \hat{p}_z) (\bar{y}_{1z} - \bar{y}_{0z})$ , since the treated contribute  $(1 - \hat{p}_z) n_z \hat{p}_z \bar{y}_{1z}$  and the untreated  $-\hat{p}_z n_z (1 - \hat{p}_z) \bar{y}_{0z}$ . The two agree when the  $\hat{\delta}_z$  are all equal or when  $\hat{p}_z$  is the same in every stratum. Otherwise the regression weights strata by the variance of  $X$  within them, not by their size, and estimates a weighted average effect that is not the average effect (Angrist 1998).

### Exercise B2.

In [Example \(Choosing an adjustment set\)](#), show that the coefficient of  $W_3$  in  $L(Y \mid X, W_1, W_3)$  is 0.80, the coefficient of the arrow  $W_3 \rightarrow Y$ , while the total effect of  $W_3$  on  $Y$  is 1.388. Which set of variables would one adjust for to estimate the effect of  $W_3$ ?

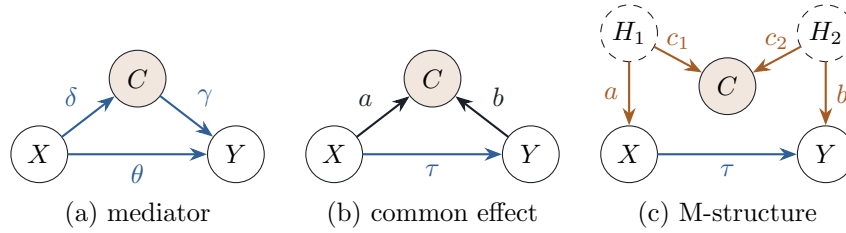
*Solution.* Write  $Y = 0.5X + 0.6(0.8X + U_M) + 0.7(0.8W_1 + U_{W_2}) + 0.8W_3 + P + U_Y = 0.98X + 0.56W_1 + 0.8W_3 + E$ , where  $E = 0.6U_M + 0.7U_{W_2} + P + U_Y$  is uncorrelated with  $X$ ,  $W_1$  and  $W_3$  (none of these depends on the disturbances in  $E$ ). So  $L(Y \mid X, W_1, W_3) = 0.98X + 0.56W_1 + 0.8W_3$ . The total effect of  $W_3$  is the sum over its directed paths to  $Y$ : 0.8 directly plus  $0.6 \times 0.98$  through  $X$ , which is 1.388. The coefficient in the regression misses the part that goes through  $X$ , because  $X$  is held fixed.  $W_3$  has no parents and no back-door paths, so the empty set is an adjustment set for its effect: the simple regression of  $Y$  on  $W_3$  estimates 1.388.

### Exercise B3.

Suppose  $X$  is randomized, so it has no parents, and  $P$  is a pre-treatment cause of  $Y$ . Show that both  $\emptyset$  and  $\{P\}$  are adjustment sets and that  $v(\{P\}) = (1 - \rho_{Y|P,X}^2) v(\emptyset)$ , where  $\rho_{Y|P,X}$  is the partial correlation of  $Y$  and  $P$  given  $X$ . Compare with [Proposition 18.1.2](#).

## 25.5 Bad controls: mediators and colliders

A regression that “controls for” many variables is not better for it. A control helps when it blocks a back-door path, costs at most precision when it lies on no path, and does harm when it lies on the causal path or when conditioning on it opens a closed path. Angrist and Pischke (2009) call the last two kinds **bad controls**. This section computes the damage in the linear models of [Figure 25.5.1](#) and describes the related bias from selecting the sample on a collider.



**Figure 25.5.1.** Three bad controls  $C$  (shaded). (a)  $C$  is a mediator. (b)  $C$  is a common effect of  $X$  and  $Y$ . (c)  $C$  is measured before  $X$  but is a collider on the back-door path through the unrecorded  $H_1$  and  $H_2$ .

## 25.5.1 Three computations

### Theorem 25.5.1 — Bad controls in linear models

In each model below the disturbances  $U_X, U_Y, U_C$  (and, in (c), the unrecorded variables  $H_1, H_2$ ) are independent with mean zero and variances  $\sigma_X^2, \sigma_Y^2, \sigma_C^2 > 0$  (and 1). Let  $\tau$  be the total effect of  $X$  on  $Y$ . In every case the slope of  $L(Y | X)$  equals  $\tau$ , so the simple regression is consistent, while the coefficient  $\beta_{X|C}$  of  $X$  in  $L(Y | X, C)$  is as follows.

- (a) **Mediator:**  $X = U_X$ ,  $C = \delta X + U_C$ ,  $Y = \theta X + \gamma C + U_Y$ . Here  $\tau = \theta + \gamma\delta$  and  $\beta_{X|C} = \theta$ : adjustment removes the indirect effect  $\gamma\delta$ .
- (b) **Common effect:**  $X = U_X$ ,  $Y = \tau X + U_Y$ ,  $C = aX + bY + U_C$ . Then

$$\beta_{X|C} = \tau - \frac{(a + b\tau) b \sigma_Y^2}{b^2 \sigma_Y^2 + \sigma_C^2}. \quad (25.5.1)$$

- (c) **M-structure:**  $X = aH_1 + U_X$ ,  $C = c_1H_1 + c_2H_2 + U_C$ ,  $Y = \tau X + bH_2 + U_Y$ . Then

$$\beta_{X|C} = \tau - \frac{a b c_1 c_2}{D}, \quad (25.5.2)$$

$$D = (a^2 + \sigma_X^2)(c_1^2 + c_2^2 + \sigma_C^2) - a^2 c_1^2 > 0.$$

*Proof.*

- (a)  $Y - \theta X - \gamma C = U_Y$  is uncorrelated with  $X$  and  $C$ , so  $L(Y | X, C) = \theta X + \gamma C$ . Without  $C$ ,  $Y = (\theta + \gamma\delta)X + (\gamma U_C + U_Y)$ , and the bracket is uncorrelated with  $X$ , so the slope is  $\tau$ .
- (b) Without  $C$ ,  $U_Y$  is uncorrelated with  $X$  and the slope is  $\tau$ . With  $C$ , write  $E = C - (a + b\tau)X = bU_Y + U_C$ , which is uncorrelated with  $X$ . The span of  $1, X, C$  is the span of  $1, X, E$ , with  $E$  orthogonal to  $1$  and  $X$ , so  $L(U_Y | X, C) = L(U_Y | X) + L(U_Y | E) = 0 + \lambda E$ , where  $\lambda = \text{Cov}(U_Y, E) / \text{Var}(E) = b\sigma_Y^2 / (b^2\sigma_Y^2 + \sigma_C^2)$ . Hence  $L(Y | X, C) = \tau X + \lambda C - \lambda(a + b\tau)X$ , whose coefficient of  $X$  is Eq. 25.5.1.
- (c) Without  $C$ ,  $bH_2 + U_Y$  is uncorrelated with  $X$  and the slope is  $\tau$ . With  $C$ ,  $L(Y | X, C) = \tau X + bL(H_2 | X, C)$ , since  $U_Y$  is uncorrelated with  $X$  and  $C$ . The coefficients of  $L(H_2 | X, C)$  solve the population equations of Theorem 6.11.1,



with covariance matrix of  $(X, C)$  equal to

$$\begin{pmatrix} a^2 + \sigma_X^2 & ac_1 \\ ac_1 & c_1^2 + c_2^2 + \sigma_C^2 \end{pmatrix}$$

and right-hand side  $(\text{Cov}(X, H_2), \text{Cov}(C, H_2)) = (0, c_2)$ . The determinant is  $D$ , positive since the matrix is a positive definite covariance matrix, and the first coefficient is  $-ac_1c_2/D$ . Multiply by  $b$ .

■

In (a) the adjusted coefficient is the **direct effect**, with the mediator held fixed by intervention. That may be of interest, but only if nothing confounds the mediator and the outcome ([Exercise \(B2\)](#)); a policy decision needs  $\tau$  ([Section 18.1](#) gave the experimental version).

In (b) the bias exists even when  $\tau = 0$ : adjusting for a consequence of both variables produces an “effect”  $-ab\sigma_Y^2/(b^2\sigma_Y^2 + \sigma_C^2)$  where there is none, negative when  $a$  and  $b$  have the same sign. This is the collider of [Proposition 25.3.3\(c\)](#).

In (c) the control is measured before  $X$ , so the advice to adjust only for pre-treatment variables does not exclude it, yet adjustment opens the back-door path  $X \leftarrow H_1 \rightarrow C \leftarrow H_2 \rightarrow Y$ . This **M-bias**, named after the graph’s shape, was pointed out by Greenland, Pearl and Robins (1999). It is a product of four path coefficients over  $D$  and tends to be small unless all four are large (Ding and Miratrix 2015). When the collider is also a confounder, with arrows  $C \rightarrow X$  and  $C \rightarrow Y$ , neither choice removes the bias; Ding and Miratrix call this butterfly bias.

### Example. (Three bad controls by simulation)

Take, with unit variances throughout:

- a mediator with  $\theta = 0.5$ ,  $\delta = 1.2$ ,  $\gamma = 1$ , so that  $\tau = 1.7$  and  $\beta_{X|C} = 0.5$ ;
- a common effect with  $\tau = 1$ ,  $a = 0.5$ ,  $b = 1$ , for which [Eq. 25.5.1](#) gives  $\beta_{X|C} = 1 - (1.5)(1)/2 = 0.25$ ;
- an M-structure with  $\tau = 1$  and  $a = b = c_1 = c_2 = 1.5$ , for which  $D = \frac{13}{4} \cdot \frac{11}{2} - \frac{81}{16} = \frac{205}{16} = 12.8125$  and  $\beta_{X|C} = 1 - \frac{81}{205} = 0.605$ .

Over 4000 samples of  $n = 400$ , the least squares coefficients of  $X$  without and with  $C$  average 1.700 and 0.499 for the mediator, 1.000 and 0.251 for the common effect, and 0.999 and 0.605 for the M-structure ([Figure 25.5.2](#)). In the M-structure the adjusted estimate is also slightly *less* variable (standard deviation 0.046 against 0.050), a reminder that a smaller standard error says nothing about bias.

```
import numpy as np

rng = np.random.default_rng(2505)

def mediator(n):
    # X -> C -> Y and X -> Y; total effect 0.5 + 1.2 * 1.0 = 1.7
    x = rng.normal(size=n); c = 1.2 * x + rng.normal(size=n)
    return x, c, 0.5 * x + 1.0 * c + rng.normal(size=n)

def collider(n):
    # X -> Y, and X -> C <- Y; total effect 1
    x = rng.normal(size=n); y = 1.0 * x + rng.normal(size=n)
```

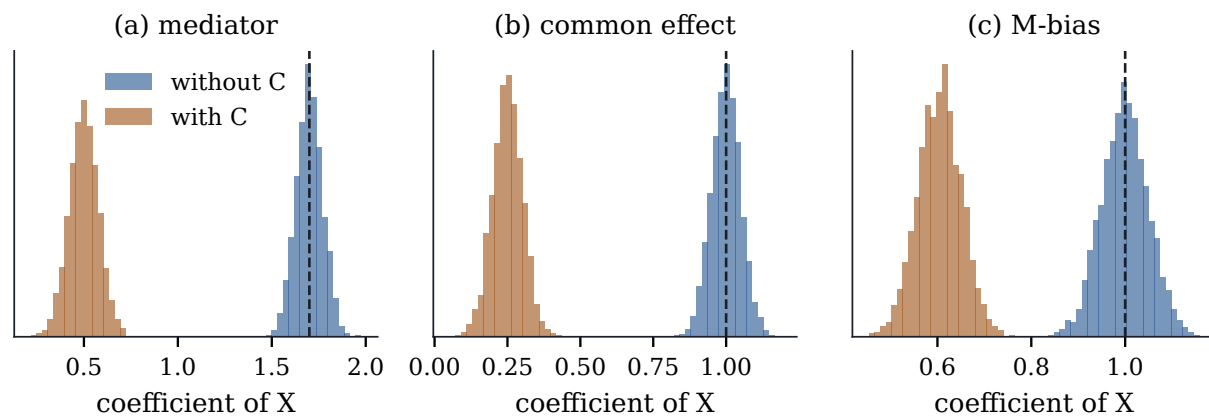


Figure 25.5.1: Sampling distributions of the least squares coefficient of  $X$  without (blue) and with (brown) adjustment for  $C$ , in the three models of [Example \(Three bad controls by simulation\)](#); 4000 samples of size 400. The dashed line is the total effect  $\tau$ .

```

return x, 0.5 * x + 1.0 * y + rng.normal(size=n), y

def m_bias(n):
    # X <- H1 -> C <- H2 -> Y, and X -> Y; total effect 1
    h1, h2 = rng.normal(size=(2, n))
    x = 1.5 * h1 + rng.normal(size=n)
    c = 1.5 * h1 + 1.5 * h2 + rng.normal(size=n)
    return x, c, 1.0 * x + 1.5 * h2 + rng.normal(size=n)

def slopes(x, c, y):
    """Coefficient of x without and with adjustment for c."""
    one = np.ones(len(x))
    b_short = np.linalg.lstsq(np.column_stack([one, x]), y, rcond=None)[0][1]
    b_long = np.linalg.lstsq(np.column_stack([one, x, c]), y, rcond=None)[0][1]
    return b_short, b_long

```

```

n, reps = 400, 500
for name, model in [("mediator", mediator), ("collider", collider), ("M-bias",
    m_bias)]:
    est = np.array([slopes(*model(n)) for _ in range(reps)])
    print(f"{name:9s} mean without C {est[:, 0].mean():.3f}    mean with C {est[:,
    1].mean():.3f}")

```

## 25.5.2 Post-treatment variables in general

A variable affected by  $X$  removes part of the effect if it lies on a directed path to  $Y$ , and opens a path if  $Y$  or a cause of  $Y$  also affects it; its descendants act partially ([Exercise \(B1\)](#)). A descendant of  $X$  with no other connection to  $Y$  costs only precision ([Exercise \(B1\)](#)). “Adjust only for pre-treatment variables” is sound in randomized experiments ([Section 25.6](#)); in observational data it is neither sufficient, by the M-structure, nor necessary, since a later measurement can proxy an unrecorded confounder. The graph, not the timing, decides. Adjusting for a pure cause of  $X$  can even *increase* the bias from an unrecorded confounder ([Exercise \(B3\)](#)); Pearl (2010) calls such variables bias amplifiers.

### 25.5.3 Selection on a collider

A sample selected on a common effect of  $X$  and  $Y$  is conditioned on a collider without any adjustment. Berkson (1946) saw two diseases, independent in the population, associated among hospital patients, since either brings a patient to hospital; Elwert and Winship (2014) survey this *endogenous selection bias*. In the common-effect model of [Example \(Three bad controls by simulation\)](#), observing only the units with  $C > 1$  (about 0.31 of them) moves the slope of  $Y$  on  $X$  from 1.002 to 0.584.

```
x, c, y = collider(200_000)
keep = c > 1                                # only units with a large C
                                         are observed
slope_all = np.polyfit(x, y, 1)[0]
slope_sel = np.polyfit(x[keep], y[keep], 1)[0]
print(f"slope of Y on X: all units {slope_all:.3f}, units with C > 1 {slope_sel:.3f}")
```

*Warning.* A variable belongs in a causal regression because of its position in the graph, not because it improves  $R^2$  or is “significant”; predictive model selection ([Chapter 29](#)) readily selects mediators and colliders.

### 25.5.4 Exercises

#### 25.5.4.1 A. Check your understanding

##### Exercise A1.

A study of the effect of a training programme ( $X$ ) on earnings two years later ( $Y$ ) considers adjusting for: (i) education before the programme; (ii) the job obtained six months after the programme; (iii) whether the person was still living in the region at the end of the study, which depends on both training and earnings; (iv) the day of the week on which the person applied. Classify each as a confounder, mediator, collider or irrelevant, under a plausible graph, and say which should be adjusted for.

#### 25.5.4.2 B. Practice

##### Exercise B1.

Let  $X = U_X$ ,  $Y = \tau X + U_Y$  and  $D = dX + U_D$ , where  $D$  is affected by  $X$  but has no connection with  $Y$ . Show that the coefficient of  $X$  in  $L(Y | X, D)$  is  $\tau$ , and that the asymptotic variance of the least squares coefficient is larger with  $D$  than without it by the factor  $(d^2\sigma_X^2 + \sigma_D^2)/\sigma_D^2$ .

##### Exercise B2.

In the mediator model add an unrecorded  $U$ , with variance  $\sigma_U^2$ , that affects both the

mediator and the outcome:  $C = \delta X + gU + U_C$ ,  $Y = \theta X + \gamma C + hU + U_Y$ . Show that

$$\beta_{X|C} = \theta - \frac{\delta g h \sigma_U^2}{g^2 \sigma_U^2 + \sigma_C^2},$$

so that adjusting for the mediator does not even recover the direct effect  $\theta$ .

*Solution.*  $Y = \theta X + \gamma C + hU + U_Y$  and  $U_Y$  is uncorrelated with  $(X, C)$ , so  $\beta_{X|C} = \theta + h\kappa$ , where  $\kappa$  is the coefficient of  $X$  in  $L(U | X, C)$ . Put  $E = C - \delta X = gU + U_C$ , uncorrelated with  $X$ . As in the proof of [Theorem 25.5.1\(b\)](#),  $L(U | X, C) = L(U | E) = \lambda(C - \delta X)$  with  $\lambda = g\sigma_U^2/(g^2\sigma_U^2 + \sigma_C^2)$ , so  $\kappa = -\lambda\delta$ . The mediator is a collider on the path  $X \rightarrow C \leftarrow U \rightarrow Y$ , which adjustment opens.

### Exercise B3.

Let  $U$  be unrecorded,  $X = \pi I + gU + U_X$  and  $Y = \tau X + hU + U_Y$ , with  $I, U, U_X, U_Y$  independent,  $\text{Var } I = \text{Var } U = 1$  and  $\text{Var } U_X = \sigma^2$ . Show that the slope of  $L(Y | X)$  is  $\tau + gh/(\pi^2 + g^2 + \sigma^2)$ , and that the coefficient of  $X$  in  $L(Y | X, I)$  is  $\tau + gh/(g^2 + \sigma^2)$ . Adjusting for the instrument  $I$  enlarges the bias.

*Solution.*  $\text{Cov}(X, Y) = \tau \text{Var } X + gh$  and  $\text{Var } X = \pi^2 + g^2 + \sigma^2$ , which gives the first slope. For the second, by [Proposition 14.5.2](#) the coefficient is  $\sigma_{YX \cdot I}/\sigma_{XX \cdot I}$ . Removing the linear effect of  $I$  leaves  $X - \pi I = gU + U_X$ , with variance  $g^2 + \sigma^2$ , and  $Y - L(Y | I) = \tau(gU + U_X) + hU + U_Y$ , so  $\sigma_{YX \cdot I} = \tau(g^2 + \sigma^2) + gh$ . Divide. The confounded covariance  $gh$  is the same in both, but adjustment divides it by a smaller variance.

## 25.6 Regression adjustment in randomized experiments

In an experiment covariates are not needed to remove bias ([Theorem 25.2.3](#)); they are used to reduce variance, which analysis of covariance does under a normal linear model with a common slope ([Proposition 18.1.2](#)). Freedman (2008a) analysed adjustment in Neyman's framework of [Section 25.2](#), with no model, and found that the adjusted estimator is biased in finite samples, that its usual standard error can be wrong, and that it can be *less* precise than the difference in means. Lin (2013) showed that interacting the covariates with the treatment repairs the precision problem. This section proves both halves.

### 25.6.1 The setting

A completely randomized experiment assigns  $n_1$  of  $n$  units to treatment and  $n_0 = n - n_1$  to control. Each unit has potential outcomes  $y_i(1), y_i(0)$  and a vector  $\mathbf{x}_i \in \mathbb{R}^q$  of covariates measured before randomization, hence fixed numbers unaffected by treatment, which we centre:

$\sum_i \mathbf{x}_i = \mathbf{0}$ . Write

$$\mathbf{S}_{xx} = \frac{1}{n-1} \sum_i \mathbf{x}_i \mathbf{x}_i^\top, \quad \mathbf{s}_t = \frac{1}{n-1} \sum_i \mathbf{x}_i (y_i(t) - \bar{y}(t)),$$

$$\boldsymbol{\beta}_t = \mathbf{S}_{xx}^{-1} \mathbf{s}_t \quad (t = 0, 1),$$

assuming  $\mathbf{S}_{xx}$  positive definite. The vector  $\boldsymbol{\beta}_t$  is the least squares slope of the potential outcomes  $y_i(t)$  on  $\mathbf{x}_i$  over *all*  $n$  units, a finite-population projection coefficient in the sense of [Section 6.11](#). No linear model is assumed.

With  $\bar{Y}_t$  and  $\bar{\mathbf{x}}_t$  the arm means, three estimators are in use: the **difference in means**  $\hat{\tau}_D = \bar{Y}_1 - \bar{Y}_0$ ; **Fisher's**  $\hat{\tau}_F$ , the coefficient of  $T$  in the least squares regression of  $Y$  on  $1, T, \mathbf{x}$  (the analysis of covariance of [Section 18.1](#)); and **Lin's**  $\hat{\tau}_L$ , the coefficient of  $T$  in the regression on  $1, T, \mathbf{x}, T\mathbf{x}$ . All three belong to one family. For fixed vectors  $\mathbf{b}_1, \mathbf{b}_0 \in \mathbb{R}^q$  put

$$\hat{\tau}(\mathbf{b}_1, \mathbf{b}_0) = (\bar{Y}_1 - \mathbf{b}_1^\top \bar{\mathbf{x}}_1) - (\bar{Y}_0 - \mathbf{b}_0^\top \bar{\mathbf{x}}_0). \quad (25.6.1)$$

Each arm mean is corrected for the chance imbalance of its covariate mean. Then  $\hat{\tau}_D = \hat{\tau}(\mathbf{0}, \mathbf{0})$ . By [Theorem 18.1.1](#)(b,c), the analysis of covariance estimates the slope by the pooled within-arm slope  $\hat{\boldsymbol{\beta}}_p$  and the treatment effect by the difference of the arm means adjusted with it, so  $\hat{\tau}_F = \hat{\tau}(\hat{\boldsymbol{\beta}}_p, \hat{\boldsymbol{\beta}}_p)$ . The regression with the interaction fits a separate line in each arm, so by [Exercise \(C1\)](#) (whose argument is unchanged for a vector covariate)  $\hat{\tau}_L = \hat{\tau}(\hat{\boldsymbol{\beta}}_1, \hat{\boldsymbol{\beta}}_0)$ , with  $\hat{\boldsymbol{\beta}}_t$  the least squares slope within arm  $t$ .

## 25.6.2 Fixed coefficients: an exact result

If the coefficients in [Eq. 25.6.1](#) are fixed in advance, everything is exact.

### Lemma 25.6.1 — Adjustment with fixed coefficients

For fixed  $\mathbf{b}_1, \mathbf{b}_0$ , put  $\mathbf{c} = (n_0 \mathbf{b}_1 + n_1 \mathbf{b}_0)/n$  and  $\mathbf{c}^* = (n_0 \boldsymbol{\beta}_1 + n_1 \boldsymbol{\beta}_0)/n$ . Then:

- (a)  $E \hat{\tau}(\mathbf{b}_1, \mathbf{b}_0) = \tau$ ;
- (b) the variance is

$$\text{Var } \hat{\tau}(\mathbf{b}_1, \mathbf{b}_0) = \text{Var } \hat{\tau}_D + \frac{n}{n_0 n_1} \left[ (\mathbf{c} - \mathbf{c}^*)^\top \mathbf{S}_{xx} (\mathbf{c} - \mathbf{c}^*) - \mathbf{c}^{*\top} \mathbf{S}_{xx} \mathbf{c}^* \right].$$

In particular the variance depends on  $(\mathbf{b}_1, \mathbf{b}_0)$  only through  $\mathbf{c}$ , and it is smallest, over all fixed coefficients, exactly when  $\mathbf{c} = \mathbf{c}^*$ , for example when  $(\mathbf{b}_1, \mathbf{b}_0) = (\boldsymbol{\beta}_1, \boldsymbol{\beta}_0)$ .

*Proof.* Define adjusted potential outcomes  $\tilde{y}_i(t) = y_i(t) - \mathbf{b}_t^\top \mathbf{x}_i$ . Because the treated units are exactly those whose  $\tilde{y}_i(1)$  is observed,  $\hat{\tau}(\mathbf{b}_1, \mathbf{b}_0)$  is the difference in means for the  $\tilde{y}$ 's, and their average effect is  $\tau - (\mathbf{b}_1 - \mathbf{b}_0)^\top \bar{\mathbf{x}} = \tau$  since the covariates are centred. Part (a) is [Theorem 25.2.3](#)(a). By [Eq. 25.2.4](#) and [Lemma 25.2.2](#), the variance is  $\{n_0/(nn_1)\} S_{\tilde{u}}^2$ , where

$$\tilde{u}_i = \tilde{y}_i(1) + \frac{n_1}{n_0} \tilde{y}_i(0) = u_i - \frac{n}{n_0} \mathbf{c}^\top \mathbf{x}_i, \quad u_i = y_i(1) + \frac{n_1}{n_0} y_i(0).$$

Expanding,

$$S_u^2 = S_u^2 - 2(n/n_0)\mathbf{c}^\top \mathbf{s}_{xu} + (n/n_0)^2 \mathbf{c}^\top \mathbf{S}_{xx} \mathbf{c},$$

with

$$\mathbf{s}_{xu} = \mathbf{s}_1 + (n_1/n_0)\mathbf{s}_0 = \mathbf{S}_{xx}(\boldsymbol{\beta}_1 + (n_1/n_0)\boldsymbol{\beta}_0) = (n/n_0)\mathbf{S}_{xx}\mathbf{c}^*.$$

Completing the square,

$$S_u^2 = S_u^2 + \frac{n^2}{n_0^2} \left[ (\mathbf{c} - \mathbf{c}^*)^\top \mathbf{S}_{xx} (\mathbf{c} - \mathbf{c}^*) - \mathbf{c}^{*\top} \mathbf{S}_{xx} \mathbf{c}^* \right].$$

Multiplying by  $n_0/(nn_1)$ , and noting that  $\{n_0/(nn_1)\}S_u^2 = \text{Var } \hat{\tau}_D$ , gives (b). The bracket is smallest iff  $\mathbf{c} = \mathbf{c}^*$ , because  $\mathbf{S}_{xx}$  is positive definite. ■

The optimal  $\mathbf{c}^*$  weights the treated slope by the *control* fraction and the control slope by the *treated* fraction: by Eq. 25.2.4,  $\mathbf{c}^*$  is, up to the factor  $n_0/n$ , the slope of  $u_i = y_i(1) + (n_1/n_0)y_i(0)$  on  $\mathbf{x}_i$ .

### 25.6.3 Estimated coefficients: Freedman's critique and Lin's repair

Estimated coefficients cost exact unbiasedness but, in large experiments, nothing else. Consider a sequence of finite populations and experiments indexed by  $n$ , with covariates centred in each, and assume:

- (A1)  $n_1/n \rightarrow p \in (0, 1)$ ;
- (A2) the mean and the covariance matrix, over the  $n$  units, of  $\mathbf{w}_i = (\mathbf{x}_i^\top, y_i(1), y_i(0))^\top$  converge to finite limits, and the limit of  $\mathbf{S}_{xx}$  is positive definite;
- (A3)  $n^{-1} \sum_i \|\mathbf{w}_i - \bar{\mathbf{w}}\|^4$  is bounded.

A sequence of random vectors  $\mathbf{g}_n$  is **bounded in probability** if for every  $\epsilon > 0$  there is an  $M$  with  $\Pr(\|\mathbf{g}_n\| > M) < \epsilon$  for all  $n$ . If  $\mathbf{g}_n$  is bounded in probability and  $\mathbf{h}_n \rightarrow \mathbf{0}$  in probability, then  $\mathbf{h}_n^\top \mathbf{g}_n \rightarrow 0$  in probability.

#### Theorem 25.6.2 — Regression adjustment in a completely randomized experiment

Under (A1)–(A3), let  $\mathbf{b}_n = (n_1\boldsymbol{\beta}_1 + n_0\boldsymbol{\beta}_0)/n$ .

- (a)  $\hat{\boldsymbol{\beta}}_t - \boldsymbol{\beta}_t \rightarrow \mathbf{0}$  for  $t = 0, 1$ , and  $\hat{\boldsymbol{\beta}}_p - \mathbf{b}_n \rightarrow \mathbf{0}$ , in probability.
- (b)  $\sqrt{n} \{\hat{\tau}_L - \hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0)\} \rightarrow 0$  and  $\sqrt{n} \{\hat{\tau}_F - \hat{\tau}(\mathbf{b}_n, \mathbf{b}_n)\} \rightarrow 0$  in probability. So, to order  $n^{-1/2}$ , Lin's estimator behaves like the optimal fixed-coefficient estimator and Fisher's like the fixed-coefficient estimator with the common coefficient  $\mathbf{b}_n$ .
- (c) The variances of these approximating estimators satisfy

$$\text{Var } \hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0) \leq \text{Var } \hat{\tau}(\mathbf{b}, \mathbf{b}) \quad \text{for every } \mathbf{b},$$

in particular  $\text{Var } \hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0) \leq \text{Var } \hat{\tau}_D$ ; and

$$\text{Var } \hat{\tau}(\mathbf{b}_n, \mathbf{b}_n) - \text{Var } \hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0) = \frac{(n_1 - n_0)^2}{nn_0n_1} (\boldsymbol{\beta}_1 - \boldsymbol{\beta}_0)^\top \mathbf{S}_{xx} (\boldsymbol{\beta}_1 - \boldsymbol{\beta}_0), \quad (25.6.2)$$

which is zero iff  $n_1 = n_0$  or  $\boldsymbol{\beta}_1 = \boldsymbol{\beta}_0$ . Fisher's approximating estimator has *larger* variance than the difference in means iff  $\mathbf{b}_n^\top \mathbf{S}_{xx} \mathbf{b}_n > 2 \mathbf{b}_n^\top \mathbf{S}_{xx} \mathbf{c}^*$ .

*Proof.*

- (a) Fix components  $j, l$  of  $\mathbf{w}$ , and let  $z_i$  be  $w_{ij} - \bar{w}_j$  or  $(w_{ij} - \bar{w}_j)(w_{il} - \bar{w}_l)$ . The treated units are a simple random sample of size  $n_1$ , so by Lemma 25.2.2 the mean of the  $z_i$  over the treated arm has mean equal to their mean over all units and variance at most  $S_z^2/n_1$ , where  $S_z^2 \leq (n-1)^{-1} \sum_i z_i^2 \leq (n-1)^{-1} \sum_i (1 + \|\mathbf{w}_i - \bar{\mathbf{w}}\|^4)$ , which is bounded by (A3). By (A1) and Chebyshev's inequality, every such arm mean differs from the corresponding mean over all units by a quantity tending to zero in probability. The within-arm covariance matrix of  $\mathbf{w}$  is  $n_1/(n_1 - 1)$  times the arm mean of the products minus the product of the arm means, so it differs from the covariance matrix over all units by a quantity tending to zero in probability. In particular the arm's  $\mathbf{x}$ -covariance matrix and its covariance vector of  $\mathbf{x}$  with  $y(1)$  are close to  $\mathbf{S}_{xx}$  and  $\mathbf{s}_1$ . Since the limit of  $\mathbf{S}_{xx}$  is positive definite and matrix inversion is continuous there, and  $\boldsymbol{\beta}_1$  converges by (A2),  $\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1 \rightarrow \mathbf{0}$  in probability. The control arm is the same. The pooled slope is

$$\begin{aligned} \hat{\boldsymbol{\beta}}_p &= [(n_1 - 1)\hat{\mathbf{S}}_{xx,1} + (n_0 - 1)\hat{\mathbf{S}}_{xx,0}]^{-1} \\ &\quad \times [(n_1 - 1)\hat{\mathbf{s}}_1 + (n_0 - 1)\hat{\mathbf{s}}_0] \end{aligned}$$

in terms of the arm moments, which by the same argument is close to  $[(n - 2)\mathbf{S}_{xx}]^{-1} \mathbf{S}_{xx} [(n_1 - 1)\boldsymbol{\beta}_1 + (n_0 - 1)\boldsymbol{\beta}_0]$ , and this differs from  $\mathbf{b}_n$  by  $O(1/n)$ .

- (b) By Eq. 25.6.1,

$$\hat{\tau}_L - \hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0) = -(\hat{\boldsymbol{\beta}}_1 - \boldsymbol{\beta}_1)^\top \bar{\mathbf{x}}_1 + (\hat{\boldsymbol{\beta}}_0 - \boldsymbol{\beta}_0)^\top \bar{\mathbf{x}}_0.$$

By Lemma 25.2.2,  $E \bar{\mathbf{x}}_1 = \mathbf{0}$  and  $E \|\bar{\mathbf{x}}_1\|^2 = (1/n_1 - 1/n) \text{tr } \mathbf{S}_{xx}$ , so  $n E \|\bar{\mathbf{x}}_1\|^2$  is bounded, and  $\sqrt{n} \bar{\mathbf{x}}_1$  is bounded in probability by Markov's inequality; likewise  $\sqrt{n} \bar{\mathbf{x}}_0$ . With (a), both products tend to zero in probability. The argument for  $\hat{\tau}_F$  is identical, with  $\hat{\boldsymbol{\beta}}_p - \mathbf{b}_n$  in both places.

- (c) Apply Lemma 25.6.1. The pair  $(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0)$  has  $\mathbf{c} = \mathbf{c}^*$ , the minimum, and  $(\mathbf{b}, \mathbf{b})$  has  $\mathbf{c} = \mathbf{b}$ . For  $\mathbf{b} = \mathbf{b}_n$ ,  $\mathbf{b}_n - \mathbf{c}^* = \{(n_1 - n_0)/n\}(\boldsymbol{\beta}_1 - \boldsymbol{\beta}_0)$ , which gives Eq. 25.6.2. Comparing with  $\mathbf{b} = \mathbf{0}$ , the variance is larger iff

$$(\mathbf{b}_n - \mathbf{c}^*)^\top \mathbf{S}_{xx} (\mathbf{b}_n - \mathbf{c}^*) > \mathbf{c}^{*\top} \mathbf{S}_{xx} \mathbf{c}^*,$$

which simplifies to the stated condition. ■

Part (b) concerns the leading term only. Because the slopes and  $\bar{\mathbf{x}}_t$  come from the same assignment,  $\hat{\tau}_L$  and  $\hat{\tau}_F$  have a bias of order  $1/n$  under moment conditions, negligible beside a standard error of order  $n^{-1/2}$  (Freedman 2008a; Lin 2013). With a finite-population central limit theorem (Li and Ding 2017), part (b) makes both estimators asymptotically normal with the variances

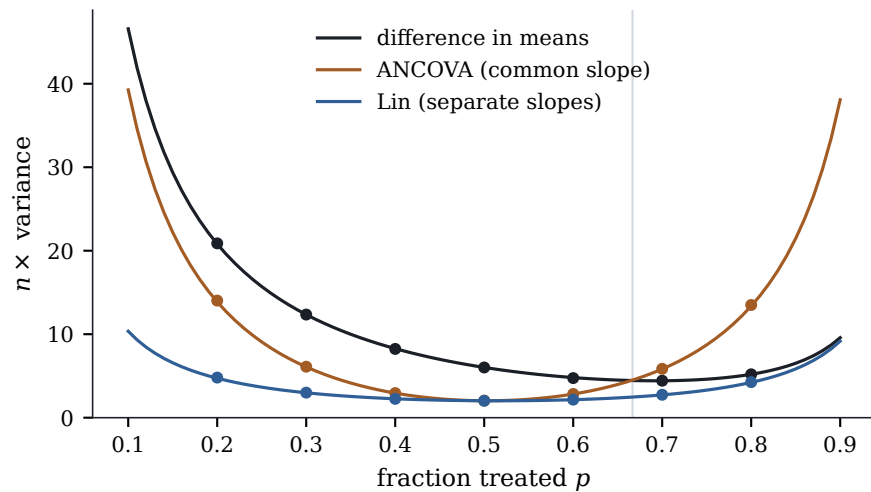


Figure 25.6.1:  $n$  times the variance of the three estimators in [Example \(When adjustment hurts\)](#) as a function of the fraction treated. Curves: exact variances of the fixed-coefficient estimators that the estimators approach ([Theorem 25.6.2](#)). Points: simulation over 20000 randomizations. The vertical line is  $p = 2/3$ , beyond which the analysis of covariance loses to the difference in means.

of part (c).

**Freedman’s critique** is part (c) for Fisher’s estimator: its common slope converges to  $\mathbf{b}_n$ , which weights each arm’s slope by the arm’s *own* size, while the optimal  $\mathbf{c}^*$  uses the *other* arm’s size. With an unbalanced design and an effect that varies with the covariate, the analysis of covariance can lose to no adjustment at all. **Lin’s repair** is part (c) for the interacted regression, which converges to the optimal coefficients and is asymptotically at least as precise as both alternatives *whether or not the linear model is true*. When  $n_1 = n_0$  the two adjusted estimators agree to first order.

#### Example. (When adjustment hurts)

A synthetic population of  $n = 400$  units has one standardized covariate, on which the response depends under treatment ( $\beta_1 = 2.007$ ) but not under control ( $\beta_0 = -0.009$ );  $\tau = 1.003$ . With  $\beta_0 \approx 0$  and  $S_{xx} = 1$ , the condition of part (c) reads  $p\beta_1(p\beta_1 - 2(1-p)\beta_1) > 0$ , that is  $p > 2/3$ . [Figure 25.6.1](#) compares  $n$  times the variance of each estimator over 20000 randomizations with the exact variances of the approximating fixed-coefficient estimators:

| fraction treated | difference in means | Fisher (ANCOVA) | Lin (interacted) |
|------------------|---------------------|-----------------|------------------|
| 0.2              | 20.88 (20.81)       | 14.03 (13.87)   | 4.80 (4.73)      |
| 0.5              | 6.01 (6.02)         | 2.04 (2.03)     | 2.03 (2.03)      |
| 0.8              | 5.20 (5.20)         | 13.50 (13.37)   | 4.26 (4.23)      |

(simulated; values from [Lemma 25.6.1](#) in brackets). At  $p = 0.8$  the analysis of covariance has more than twice the variance of the unadjusted estimate while Lin’s estimator still gains; at  $p = 0.5$  both cut it by two thirds.

```
import numpy as np
```



```

rng = np.random.default_rng(2506)
n = 400
x = rng.normal(size=n)
x = (x - x.mean()) / x.std(ddof=1)           # centred,  $S_{xx} = 1$ 
y0 = rng.normal(size=n)                     # control: no dependence on  $x$ 
y1 = 1 + 2 * x + rng.normal(size=n)         # treated: slope 2
tau = np.mean(y1 - y0)
beta1, beta0 = np.polyfit(x, y1, 1)[0], np.polyfit(x, y0, 1)[0]

def estimators(T):
    """Difference in means, ANCOVA and Lin's estimator for assignment rows  $T$  (reps  $x$ 
     $n$ )."""
    n1 = T.sum(axis=1, keepdims=True); n0 = n - n1
    yobs = np.where(T, y1, y0)
    xb1, xb0 = (T * x).sum(1, keepdims=True) / n1, (~T * x).sum(1, keepdims=True) /
    n0
    yb1, yb0 = (T * yobs).sum(1, keepdims=True) / n1, (~T * yobs).sum(1, keepdims=
    True) / n0
    dx = np.where(T, x - xb1, x - xb0)       # deviations from the arm
    means
    dy = np.where(T, yobs - yb1, yobs - yb0)
    b1 = (T * dx * dy).sum(1) / (T * dx ** 2).sum(1) # within-arm slopes
    b0 = (~T * dx * dy).sum(1) / (~T * dx ** 2).sum(1)
    bp = (dx * dy).sum(1) / (dx ** 2).sum(1)       # pooled within-arm slope
    xb1, xb0, yb1, yb0 = xb1[:, 0], xb0[:, 0], yb1[:, 0], yb0[:, 0]
    unadj = yb1 - yb0
    ancova = unadj - bp * (xb1 - xb0)
    lin = (yb1 - b1 * xb1) - (yb0 - b0 * xb0)       #  $x$  is centred, so  $\bar{x} = 0$ 
    return unadj, ancova, lin

p, reps = 0.8, 2000
T = np.array([rng.permutation(n) < p * n for _ in range(reps)])
for name, est in zip(["difference in means", "ANCOVA", "Lin (interacted)"],
    estimators(T)):
    print(f"{name:20s} mean {est.mean():6.3f} (tau = {tau:.3f})    n * variance {n *
    est.var():6.2f}")

```

## 25.6.4 Standard errors

Freedman (2008a) showed that the classical standard error of the analysis of covariance can be inconsistent in either direction. Lin (2013) showed that the sandwich standard errors of [Chapter 21 \(Theorem 21.4.2\)](#) for the interacted regression are consistent or asymptotically conservative, for the reason seen in [Exercise \(B2\)](#): the residuals estimate the variances of [Exercise \(B1\)](#) arm by arm, and the one term that cannot be estimated enters with a negative sign.

In [Example \(When adjustment hurts\)](#) at  $p = 0.8$ , over 10000 randomizations, nominal 95% intervals covered  $\tau$  with frequency 0.997 for the difference in means with Neyman's standard error, 0.921 for the analysis of covariance with its classical standard error, and 0.984 for Lin's estimator with the HC2 standard error. The first and last are conservative because the effect varies across units.

*Idea.* In a randomized experiment, adjust only for covariates measured before randomization; include their interactions with the treatment (with the covariates centred at their overall mean); and use sandwich standard errors. Then adjustment cannot hurt in

large samples, whatever the true relation between covariates and responses, and it helps as much as a linear adjustment can.

Freedman (2008b) extends the critique to several treatments; the interacted regression, with one set of slopes per arm, extends Lin's repair. With many covariates relative to  $n$  the  $O(1/n)$  terms are no longer negligible.

## 25.6.5 Exercises

### 25.6.5.1 A. Check your understanding

#### Exercise A1.

With one covariate and a fixed common coefficient  $b$ , show from [Lemma 25.6.1](#) that  $\hat{\tau}(b, b)$  has smaller variance than the difference in means iff  $b$  lies strictly between 0 and  $2c^*$ . What does this say about guessing a slope from a previous study?

*Solution.* With  $\mathbf{c} = b$  and  $S_{xx} > 0$ , the variance difference is  $\{n/(n_0n_1)\}S_{xx}[(b - c^*)^2 - c^{*2}] = \{n/(n_0n_1)\}S_{xx}b(b - 2c^*)$ , which is negative iff  $b$  is strictly between 0 and  $2c^*$ . A slope guessed from earlier data, fixed before randomization, keeps the estimator exactly unbiased, and it reduces the variance as long as it has the right sign and is less than twice the optimal value: a crude guess is enough.

### 25.6.5.2 B. Practice

#### Exercise B1.

Let  $e_i(t) = y_i(t) - \bar{y}(t) - \boldsymbol{\beta}_t^\top \mathbf{x}_i$  be the residuals of the finite-population regressions. Show that

$$\text{Var } \hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0) = \frac{S_{e(1)}^2}{n_1} + \frac{S_{e(0)}^2}{n_0} - \frac{S_{e(1)-e(0)}^2}{n},$$

Neyman's formula with the potential outcomes replaced by their residuals. Deduce that the proportional gain over the difference in means is large when the covariates explain much of both potential outcomes.

*Solution.* As in the proof of [Lemma 25.6.1](#),  $\hat{\tau}(\boldsymbol{\beta}_1, \boldsymbol{\beta}_0)$  is the difference in means for the adjusted outcomes  $y_i(t) - \boldsymbol{\beta}_t^\top \mathbf{x}_i = \bar{y}(t) + e_i(t)$ . Adding constants changes nothing in [Eq. 25.2.3](#), so its variance is Neyman's formula for the  $e_i(t)$ , whose unit effects are  $e_i(1) - e_i(0)$  up to a constant. The first two terms are  $(1 - R_t^2)$  times those of the unadjusted formula, where  $R_t^2$  is the finite-population  $R^2$  of  $y(t)$  on  $\mathbf{x}$ .

#### Exercise B2.

In the setting of [Example \(When adjustment hurts\)](#), let  $\beta_0 = 0$ ,  $\beta_1 = \beta \neq 0$  and  $S_{xx} = 1$ , and write  $p = n_1/n$ . Show that the excess [Eq. 25.6.2](#), multiplied by  $n$ , is  $(2p - 1)^2\beta^2/\{p(1 - p)\}$ , and that the analysis of covariance loses to the difference in means iff  $p > 2/3$ . What happens if instead  $\beta_1 = 0$  and  $\beta_0 = \beta$ ?

## 25.6.5.3 C. Going deeper

**Exercise C1.**

Show that Lin's estimator is the average over *all*  $n$  units of the difference between the two arms' fitted regression lines: if  $\hat{\alpha}_t + \hat{\beta}_t^\top \mathbf{x}$  is the least squares fit in arm  $t$ , then

$$\hat{\tau}_L = n^{-1} \sum_i \{(\hat{\alpha}_1 + \hat{\beta}_1^\top \mathbf{x}_i) - (\hat{\alpha}_0 + \hat{\beta}_0^\top \mathbf{x}_i)\}.$$

Relate each arm's term to the regression estimator of a population mean in survey sampling.

## 25.7 Summary and Notes

### 25.7.1 Summary

*Idea.*

1.  $E(Y \mid \text{do}(X = x))$ , defined by a structural causal model (Definition 25.1.1), differs from  $E(Y \mid X = x)$  under confounding, at every sample size (Proposition 25.1.1).
2. Potential outcomes (Definition 25.2.1): the naive comparison is the effect on the treated plus a selection bias (Proposition 25.2.1).
3. Under complete randomization the difference in means is unbiased, with Neyman's variance; the estimate  $s_1^2/n_1 + s_0^2/n_0$  is conservative and equals the HC2 sandwich (Theorem 25.2.3).
4. An intervention deletes one factor of the Markov factorization (Proposition 25.3.1); linear effects are sums over paths (Proposition 25.3.2); d-separation reads conditional independence off the graph (Proposition 25.3.3, Theorem 25.3.4).
5. Confounding is an open back-door path, and in linear models its bias is a sum over common ancestors (Proposition 25.3.5).
6. An adjustment set identifies the effect; in linear models the coefficient of  $X$  is then the total effect (Theorem 25.4.1). Causes of  $Y$  improve precision, pure causes of  $X$  worsen it (Proposition 25.4.2).
7. Mediators, common effects and pre-treatment colliders are bad controls (Theorem 25.5.1).
8. In experiments, analysis of covariance can lose to no adjustment, and the interacted regression is asymptotically optimal (Lemma 25.6.1, Theorem 25.6.2).

### 25.7.2 Notes and sources

**Structural models and graphs.** Wright (1921, 1934) introduced path analysis (Proposition 25.3.2), and Haavelmo (1943) read structural equations as autonomous mechanisms that interventions change one at a time. The intervention as an explicit operation and the truncated factorization (Proposition 25.3.1) are from Pearl (1995), and in parallel the manipulation theo-

rem of Spirtes, Glymour and Scheines (2000; first edition 1993); Pearl (2009) is the systematic account. d-separation is due to Pearl (1988), with proofs by Geiger, Verma and Pearl (1990) and Lauritzen, Dawid, Larsen and Leimer (1990); Spirtes, Glymour and Scheines (2000) develop faithfulness.

**Potential outcomes.** [Definition 25.2.1](#) and [Eq. 25.2.3](#) are Neyman’s (1923), in a paper on field trials translated in 1990. Rubin (1974) extended the framework to observational studies, Holland (1986) named the fundamental problem of causal inference, and Imbens and Rubin (2015) set out the finite-population view. Finite-population central limit theorems go back to Hájek (1960); Li and Ding (2017) give forms suited to causal inference. Samii and Aronow (2012) show that Neyman’s estimate is the HC2 sandwich ([Exercise \(B2\)](#)).

**Adjustment and bad controls.** The back-door and front-door criteria are from Pearl (1995); Rosenbaum and Rubin (1983) introduced strong ignorability and the propensity score; Shpitser, VanderWeele and Robins (2010) give a necessary and sufficient adjustment criterion; and Henckel, Perković and Maathuis (2022) find the most efficient adjustment set in linear models. [Exercise \(B1\)](#) is Angrist (1998)’s observation, and the “Table 2 fallacy” is named by Westreich and Greenland (2013). Angrist and Pischke (2009) popularized the term bad control. Berkson (1946) described selection on a common effect, Greenland, Pearl and Robins (1999) showed that pre-treatment colliders can bias adjustment, Ding and Miratrix (2015) quantify M-bias and butterfly bias, Elwert and Winship (2014) survey collider bias, and Pearl (2010) analyses bias amplification.

**Adjustment in experiments.** Covariance adjustment of experiments is Fisher’s (1935). The design-based critique is Freedman (2008a, 2008b), and the interacted estimator and its sandwich standard errors are Lin (2013). The route through the exact variance of fixed-coefficient estimators ([Lemma 25.6.1](#)), which reduces the comparison to completing a square, is this book’s.

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# Chapter 26

## Collinearity: Diagnosis and Consequences

Part VI gives up unbiasedness in exchange for smaller variance, and collinearity is the situation that first made that trade attractive. When the columns of the model matrix are nearly linearly dependent, the data answer some questions well and others barely at all, and the badly answered ones are often those the analyst asked: the separate effects of regressors that moved together. This chapter says exactly which linear functions lose precision, builds the standard diagnostics (variance inflation factors and their generalization to groups of columns, condition indices and the Belsley–Kuh–Welsch variance decomposition), shows why predictions that follow the pattern of the data survive intact, and ends with the remedies, the last of which, biased estimation, is where [Chapter 27](#) begins.

**What you need.** [Chapter 1](#) (the spectral theorem and Rayleigh quotients, [Theorem 1.7.2](#) and [Theorem 1.7.7](#); the singular value decomposition and the condition number, [Theorem 1.8.1](#) and [Definition 1.8.1](#)), [Chapter 6](#) (the Frisch–Waugh–Lovell theorem and the variance identity [Eq. 6.6.2](#); leverage, [Proposition 6.8.1](#)), [Chapter 8](#) (estimability), [Chapter 10](#) (conditioning and column scaling, [Theorem 10.5.1](#) and [Proposition 10.5.2](#)) and [Chapter 12](#) (confidence ellipsoids and prediction at a new point). [Section 19.6](#) previewed the subject ([Proposition 19.6.1](#)); this chapter builds on that preview rather than repeating it.

### 26.0.1 Roadmap

- [What collinearity does to variance](#)
- [Variance inflation factors](#)
- [Condition indices and variance decomposition](#)
- [Collinearity and prediction](#)
- [What to do about collinearity](#)
- [Summary and notes](#)

## 26.1 What collinearity does to variance

The columns of a model matrix are **collinear** when a nontrivial linear combination of them is zero, and **nearly collinear** when a combination with coefficients of reasonable size is a short vector. The exact case is rank deficiency, settled in [Chapter 8](#). Here  $\mathbf{X}$  has full column rank and every linear function of  $\boldsymbol{\beta}$  is estimable, yet some are estimated so poorly that in practice they have not been estimated at all; “collinearity” below means this near case.

[Section 19.6](#) wrote the variance of  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  as a sum over the eigenvectors of  $\mathbf{X}^\top \mathbf{X}$  ([Proposition 19.6.1](#)). This section says which functions collinearity damages, links them to estimability, and separates the statistical problem from the numerical one of [Chapter 10](#).

### 26.1.1 Canonical coordinates

Throughout the section  $\mathbf{X}$  is  $n \times p$  of rank  $p$ ,  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ . Write the spectral decomposition ([Theorem 1.7.2](#)) as

$$\mathbf{X}^\top \mathbf{X} = \sum_{\ell=1}^p \lambda_\ell \mathbf{v}_\ell \mathbf{v}_\ell^\top, \quad \lambda_1 \geq \lambda_2 \geq \cdots \geq \lambda_p > 0,$$

with orthonormal eigenvectors  $\mathbf{v}_1, \dots, \mathbf{v}_p$ , and put  $\gamma_\ell = \mathbf{v}_\ell^\top \boldsymbol{\beta}$ . The numbers  $\gamma_\ell$  are the coordinates of  $\boldsymbol{\beta}$  in the eigenvector basis, the **canonical coordinates** of the regression. Since  $\mathbf{X}\mathbf{v}_\ell$  has squared length  $\lambda_\ell$ , the mean vector is

$$\mathbf{X}\boldsymbol{\beta} = \sum_{\ell=1}^p \gamma_\ell \mathbf{X}\mathbf{v}_\ell = \sum_{\ell=1}^p \sqrt{\lambda_\ell} \gamma_\ell \mathbf{u}_\ell, \quad \mathbf{u}_\ell = \mathbf{X}\mathbf{v}_\ell / \sqrt{\lambda_\ell}, \quad (26.1.1)$$

and the vectors  $\mathbf{u}_\ell$  are the left singular vectors of  $\mathbf{X}$  ([Theorem 1.8.1](#)). A small  $\lambda_\ell$  means that a unit change in  $\gamma_\ell$  moves the mean vector only by  $\sqrt{\lambda_\ell}$ . The data then carry little information about  $\gamma_\ell$ , and the following theorem makes this precise.

#### Theorem 26.1.1 — Variances in canonical coordinates

Under the assumptions above, let  $\hat{\gamma}_\ell = \mathbf{v}_\ell^\top \hat{\boldsymbol{\beta}}$ .

- (a)  $\hat{\gamma}_\ell = \mathbf{u}_\ell^\top \mathbf{Y} / \sqrt{\lambda_\ell}$ , and  $\hat{\boldsymbol{\beta}} = \sum_{\ell} \hat{\gamma}_\ell \mathbf{v}_\ell$ . The  $\hat{\gamma}_\ell$  are uncorrelated, with  $\text{Var}(\hat{\gamma}_\ell) = \sigma^2 / \lambda_\ell$ .
- (b) For every nonzero  $\mathbf{a} \in \mathbb{R}^p$ ,

$$\frac{\sigma^2 \|\mathbf{a}\|^2}{\lambda_1} \leq \text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) \leq \frac{\sigma^2 \|\mathbf{a}\|^2}{\lambda_p},$$

with equality on the left at  $\mathbf{a} = \mathbf{v}_1$  and on the right at  $\mathbf{a} = \mathbf{v}_p$ . The ratio of the largest to the smallest variance of  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  over unit vectors  $\mathbf{a}$  is  $\kappa(\mathbf{X})^2$ .

- (c) If  $\mathbf{a} \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ , then  $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) \leq \sigma^2 \|\mathbf{a}\|^2 / \lambda_k$ : the eigenvalues  $\lambda_{k+1}, \dots, \lambda_p$  do not enter. For every  $\mathbf{a}$ ,  $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) \geq \sigma^2 (\mathbf{a}^\top \mathbf{v}_p)^2 / \lambda_p$ .
- (d)  $E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \sigma^2 \sum_{\ell} 1/\lambda_\ell \geq \sigma^2 / \lambda_p$ , while, by [Proposition 19.6.1\(b\)](#),  $E\|\mathbf{X}\hat{\boldsymbol{\beta}} - \mathbf{X}\boldsymbol{\beta}\|^2 = \sigma^2 p$  whatever the eigenvalues.

- (e) For each column,  $\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2 \geq \lambda_p$ , where  $\mathbf{M}_{(j)}$  projects onto the span of the other columns. Hence  $\text{Var}(\hat{\beta}_j) \leq \sigma^2/\lambda_p$ .

*Proof.*

- (a) By [Theorem 5.5.1](#),  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$  with covariance  $\sigma^2 (\mathbf{X}^\top \mathbf{X})^{-1} = \sigma^2 \sum_{\ell} \lambda_{\ell}^{-1} \mathbf{v}_{\ell} \mathbf{v}_{\ell}^\top$ . Hence  $\mathbf{v}_{\ell}^\top \hat{\boldsymbol{\beta}} = \lambda_{\ell}^{-1} (\mathbf{X} \mathbf{v}_{\ell})^\top \mathbf{Y} = \mathbf{u}_{\ell}^\top \mathbf{Y} / \sqrt{\lambda_{\ell}}$ , and  $\text{Cov}(\mathbf{v}_{\ell}^\top \hat{\boldsymbol{\beta}}, \mathbf{v}_m^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \mathbf{v}_{\ell}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{v}_m$ , which is  $\sigma^2/\lambda_{\ell}$  if  $\ell = m$  and zero otherwise. The expansion of  $\hat{\boldsymbol{\beta}}$  holds because the  $\mathbf{v}_{\ell}$  form an orthonormal basis.
- (b) Write  $c_{\ell} = \mathbf{a}^\top \mathbf{v}_{\ell}$ , so that  $\sum_{\ell} c_{\ell}^2 = \|\mathbf{a}\|^2$ . By [Proposition 19.6.1\(a\)](#),  $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}) = \sigma^2 \sum_{\ell} c_{\ell}^2 / \lambda_{\ell}$ , which lies between  $\sigma^2 \|\mathbf{a}\|^2 / \lambda_1$  and  $\sigma^2 \|\mathbf{a}\|^2 / \lambda_p$ , with the stated equality cases. The singular values of  $\mathbf{X}$  are  $\sqrt{\lambda_{\ell}}$  ([Theorem 1.8.1](#)), so the ratio of the extremes is  $\lambda_1/\lambda_p = \kappa(\mathbf{X})^2$  by [Definition 1.8.1](#).
- (c) If  $\mathbf{a} \in \text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\}$ , then  $c_{\ell} = 0$  for  $\ell > k$  and  $\sum_{\ell \leq k} c_{\ell}^2 / \lambda_{\ell} \leq \|\mathbf{a}\|^2 / \lambda_k$ . The lower bound keeps only the term  $\ell = p$ .
- (d)  $E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \text{tr Cov}(\hat{\boldsymbol{\beta}}) = \sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$ , and the eigenvalues of  $(\mathbf{X}^\top \mathbf{X})^{-1}$  are the  $1/\lambda_{\ell}$ . The statement about fitted values is [Proposition 19.6.1\(b\)](#) multiplied by  $n$ .
- (e)  $(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j = \mathbf{x}_j - \mathbf{X}_{(j)}\mathbf{c}$  for some  $\mathbf{c}$ , which is  $\mathbf{X}\mathbf{b}$  for a vector  $\mathbf{b}$  with  $b_j = 1$ . By [Theorem 1.7.7](#),  $\|\mathbf{X}\mathbf{b}\|^2 \geq \lambda_p \|\mathbf{b}\|^2 \geq \lambda_p$ . The variance bound follows from [Proposition 7.2.4](#). ■

Part (b) is the precise sense in which  $\kappa(\mathbf{X})^2$  measures collinearity: it is the ratio of the worst to the best precision among linear functions of the same length. Part (e) is the Frisch–Waugh–Lovell picture of [Theorem 6.6.2](#): a coefficient is imprecise when its column's part orthogonal to the others is short, and no such part is shorter than  $\sqrt{\lambda_p}$ .

*Idea.* Collinearity does not make least squares imprecise in general. It makes it imprecise in a few directions of coefficient space, those of the eigenvectors with small eigenvalues, and precise in the others. Whether it matters depends on whether the question asked points along a weak direction.

### Example. (Two regressors that move together)

Draw  $n = 30$  values  $x_{i1}$  from a standard normal distribution and set  $x_{i2} = x_{i1} + 0.15z_i$  with independent standard normal  $z_i$ . The sample correlation is 0.9951. Centre both regressors and fit  $E(Y) = \beta_0 + \beta_1 x_1 + \beta_2 x_2$ , so that the intercept is orthogonal to the slopes and the slope block of  $\mathbf{X}^\top \mathbf{X}$  carries all the collinearity. Its eigenvalues are 70.05 and 0.1718. The weak eigenvector is close to  $(1, -1)^\top / \sqrt{2}$  (its first coordinate is 0.718 in absolute value), so the weak canonical coordinate is essentially the difference  $\beta_1 - \beta_2$ . In units of  $\sigma$ , the standard deviations are

| Function                        | Standard deviation |
|---------------------------------|--------------------|
| $\hat{\beta}_1$                 | 1.734              |
| $\hat{\beta}_2$                 | 1.682              |
| $\hat{\beta}_1 + \hat{\beta}_2$ | 0.177              |
| $\hat{\beta}_1 - \hat{\beta}_2$ | 3.412              |



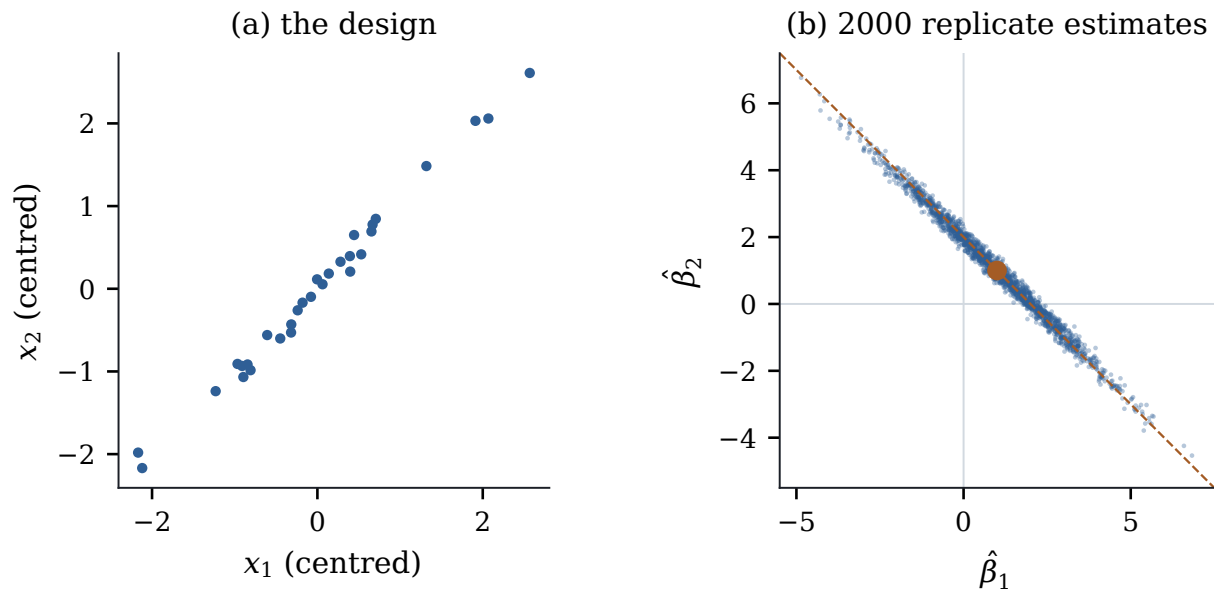


Figure 26.1.1: (a) The design of [Example \(Two regressors that move together\)](#). (b) Estimates  $(\hat{\beta}_1, \hat{\beta}_2)$  from 2000 replicate responses, true value marked: long in the weak direction  $(1, -1)$ , thin along  $(1, 1)$ ; dashed,  $\beta_1 + \beta_2 = 2$ .

Had  $x_1$  been orthogonal to  $x_2$  with the same spread, the first entry would have been 0.171, smaller by the factor  $10.1 = \sqrt{102.4}$ , the root of the variance inflation factor of Section 26.2. The sum of the slopes is estimated about as well as a slope in an orthogonal design, the difference worst of all. The ratio of the largest to the smallest variance over unit vectors is 407.8, the squared condition number of the centred slope columns ([Theorem 26.1.1\(b\)](#)).

[Figure 26.1.1](#) shows what this means for a single data set. With the design fixed and true slopes  $\beta_1 = \beta_2 = 1$ , the estimates from 2000 replicate responses lie along the line  $\beta_1 + \beta_2 = 2$ . The estimate of  $\beta_2$  ranges from  $-4.535$  to  $7.359$  and is negative in 26.1 per cent of the replicates, while  $\hat{\beta}_1 + \hat{\beta}_2$  stays between 1.440 and 2.639. Every data set says clearly that the two regressors together matter, yet a quarter of them give  $\beta_2$  the wrong sign, and 54.1 per cent give at least one of the two slopes the wrong sign.

```
import numpy as np
rng = np.random.default_rng(2601)
n, sigma = 30, 1.0
x1 = rng.normal(size=n)
x2 = x1 + 0.15 * rng.normal(size=n)           # x2 nearly repeats x1
x1c, x2c = x1 - x1.mean(), x2 - x2.mean()      # centring makes the intercept
                                                # orthogonal
X = np.column_stack([np.ones(n), x1c, x2c])
G = np.linalg.inv(X.T @ X)                     # Cov(beta_hat) / sigma^2

lam, V = np.linalg.eigh(X[:, 1:].T @ X[:, 1:])  # canonical coordinates of the
                                                # slopes
print("correlation of x1 and x2:", round(np.corrcoef(x1, x2)[0, 1], 4))
print("eigenvalues:", np.round(lam, 3), " weak direction:", np.round(V[:, 0], 3))

def sd(a):
    """Standard deviation of a' beta_hat, in units of sigma."""
    return np.sqrt(a @ G @ a)
```

```
print("sd of b1, b2   :", round(sd(np.array([0, 1, 0])), 3), round(sd(np.array([0,
    0, 1])), 3))
print("sd of b1 + b2   :", round(sd(np.array([0, 1, 1])), 3))
print("sd of b1 - b2   :", round(sd(np.array([0, 1, -1])), 3))
```

```
beta = np.array([0.0, 1.0, 1.0])           # equal true slopes
reps = 2000
Y = X @ beta + sigma * rng.normal(size=(reps, n))  # 2000 replicate responses,
    same design
B = np.linalg.solve(X.T @ X, X.T @ Y.T).T         # one row of estimates per
    replicate
print("share of replicates with b2 < 0:", np.mean(B[:, 2] < 0))
print("share with b1 < 0 or b2 < 0:", np.mean((B[:, 1:] < 0).any(axis=1)))
print("range of b1 + b2:", np.round([B[:, 1:].sum(1).min(), B[:, 1:].sum(1).max()],
    3))
```

## 26.1.2 The symptoms

The classical symptoms all follow from [Theorem 26.1.1](#): large standard errors for single coefficients alongside small ones for some combinations; coefficients of implausible size or sign; instability under deletion of a few cases; and a significant  $F$  test for a group in which no single  $t$  test is significant, because the group statistic is governed by the best combination of its members ([Proposition 11.6.2](#)). None of them means the model is wrong.

## 26.1.3 Near-estimability

When the near-dependence is a small perturbation of an exact one, the functions that stay precise are exactly those estimable in the limiting, rank-deficient design.

### Proposition 26.1.2 — Near-estimability

Let  $\mathbf{X}_1$  be  $n \times (p-1)$  of full column rank,  $\mathbf{c} \in \mathbb{R}^{p-1}$ , and  $\mathbf{z}$  a unit vector orthogonal to  $\mathcal{C}(\mathbf{X}_1)$ . For  $\delta > 0$  let  $\mathbf{X}_\delta = [\mathbf{X}_1, \mathbf{X}_1\mathbf{c} + \delta\mathbf{z}]$ , with least squares estimate  $\hat{\boldsymbol{\beta}}_\delta = (\hat{\boldsymbol{\beta}}_1^\top, \hat{\beta}_2)^\top$ , and let  $\mathbf{X}_0 = [\mathbf{X}_1, \mathbf{X}_1\mathbf{c}]$ , which has rank  $p-1$ . Write  $\mathbf{a} = (\mathbf{a}_1^\top, a_2)^\top$ .

- (a)  $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}_\delta) = \sigma^2 \mathbf{a}_1^\top (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{a}_1 + \sigma^2 (a_2 - \mathbf{c}^\top \mathbf{a}_1)^2 / \delta^2$ .
- (b)  $\mathbf{a}^\top \boldsymbol{\beta}$  is estimable in the model with matrix  $\mathbf{X}_0$  iff  $a_2 = \mathbf{c}^\top \mathbf{a}_1$ . For such  $\mathbf{a}$ ,  $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}_\delta)$  does not depend on  $\delta$  and equals the variance of the least squares estimator of  $\mathbf{a}^\top \boldsymbol{\beta}$  in the limiting model. For every other  $\mathbf{a}$  it grows like  $\delta^{-2}$  as  $\delta \rightarrow 0$ .

*Proof.*

- (a) Apply [Theorem 6.6.2](#) with  $\mathbf{X}_2 = \mathbf{X}_1$  and the last column in the role of  $\mathbf{X}_1$  there. The last column residualized on  $\mathcal{C}(\mathbf{X}_1)$  is  $(\mathbf{I} - \mathbf{M}_1)(\mathbf{X}_1\mathbf{c} + \delta\mathbf{z}) = \delta\mathbf{z}$ , so  $\hat{\beta}_2 = \mathbf{z}^\top \mathbf{Y} / \delta$ .

By part (c) of the same theorem and  $\mathbf{X}_1^\top \mathbf{z} = \mathbf{0}$ ,

$$\hat{\boldsymbol{\beta}}_1 = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top (\mathbf{Y} - (\mathbf{X}_1 \mathbf{c} + \delta \mathbf{z}) \hat{\beta}_2) = \hat{\boldsymbol{\beta}}_1^S - \mathbf{c} \hat{\beta}_2, \quad \hat{\boldsymbol{\beta}}_1^S = (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{Y}.$$

Hence  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}_\delta = \mathbf{a}_1^\top \hat{\boldsymbol{\beta}}_1^S + (a_2 - \mathbf{c}^\top \mathbf{a}_1) \hat{\beta}_2$ . The two terms are uncorrelated, because  $\text{Cov}(\hat{\boldsymbol{\beta}}_1^S, \hat{\beta}_2) = \sigma^2 (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{X}_1^\top \mathbf{z} / \delta = \mathbf{0}$ , and  $\text{Var}(\hat{\beta}_2) = \sigma^2 / \delta^2$ .

- (b) By [Theorem 8.3.1](#),  $\mathbf{a}^\top \boldsymbol{\beta}$  is estimable under  $\mathbf{X}_0$  iff  $\mathbf{a} \in \mathcal{C}(\mathbf{X}_0^\top)$ . Since  $\mathbf{X}_1$  has full column rank,  $\mathcal{C}(\mathbf{X}_1^\top) = \mathbb{R}^{p-1}$ , so  $\mathcal{C}(\mathbf{X}_0^\top) = \{(\mathbf{a}_1^\top, \mathbf{c}^\top \mathbf{a}_1)^\top : \mathbf{a}_1 \in \mathbb{R}^{p-1}\}$ . For such  $\mathbf{a}$  the second term in (a) vanishes. In the limiting model  $\mathbf{X}_0 \boldsymbol{\beta} = \mathbf{X}_1(\boldsymbol{\beta}_1 + \mathbf{c} \beta_2)$ , so  $\mathbf{a}^\top \boldsymbol{\beta} = \mathbf{a}_1^\top (\boldsymbol{\beta}_1 + \mathbf{c} \beta_2)$  is estimated by  $\mathbf{a}_1^\top \hat{\boldsymbol{\beta}}_1^S$ , with variance  $\sigma^2 \mathbf{a}_1^\top (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{a}_1$ . If  $a_2 \neq \mathbf{c}^\top \mathbf{a}_1$ , the second term is a positive multiple of  $\delta^{-2}$ .

■

The orthogonality of  $\mathbf{z}$  to  $\mathcal{C}(\mathbf{X}_1)$  matters only for the size of the bounded term. A general perturbation  $\mathbf{w}$  splits as  $\mathbf{M}_1 \mathbf{w} + (\mathbf{I} - \mathbf{M}_1) \mathbf{w}$ , and the first part, being  $\mathbf{X}_1 \mathbf{d}$  for some  $\mathbf{d}$ , can be absorbed into  $\mathbf{c}$ . The dichotomy survives: a function estimable in the limit keeps a variance that does not depend on  $\delta$ , and every other function's variance grows like  $\delta^{-2}$ . But the bounded value can exceed the limiting-model variance ([Exercise \(B2\)](#)). [Figure 8.3.1](#) showed the  $\delta^{-2}$  growth numerically; the proposition identifies the rate and the exact set of functions that escape it.

## 26.1.4 Statistical, not numerical

[Chapter 10](#) met ill-conditioning as a numerical problem: a large  $\kappa(\mathbf{X})$  amplifies rounding errors ([Theorem 10.5.1](#)). The statistical cost is different. It is  $\sigma^2 / \lambda_\ell$  along each weak direction, so it involves the error variance and the absolute size of the eigenvalues, not only their ratio. Rescaling a column changes  $\kappa(\mathbf{X})$  ([Exercise \(B2\)](#)) but no estimand's variance, so diagnostics must be computed for a stated scaling ([Section 26.3](#)) or be scale-free ([Section 26.2](#)). And the eigenvalues grow with  $n$ : if the rows are drawn independently from a distribution with nonsingular second-moment matrix  $\boldsymbol{\Sigma}$ , then  $\lambda_p / n \rightarrow \lambda_{\min}(\boldsymbol{\Sigma})$  by the strong law of large numbers and [Proposition 10.4.2](#), so collinearity in the population is paid for by a larger  $n$ . Longley's regression shows how far apart the two problems can be: QR computes its coefficients to about eleven digits ([Example \(Longley's data, exactly\)](#)), yet its variance inflation factors reach 1788.5 ([Example \(Longley's regression\)](#)).

Measurement error in the regressors sits between the two. It perturbs  $\mathbf{X}$  as rounding does, by an amount that more digits cannot reduce, and a perturbation comparable to  $\sqrt{\lambda_p}$  can turn the weak eigenvector and change which combinations appear well estimated ([Theorem 10.5.1](#), with  $\kappa(\mathbf{X})$  as the amplifier). The directions that collinearity leaves ill-determined are thus the least robust to errors in  $\mathbf{X}$ ; [Chapter 24](#) treats the resulting bias ([Theorem 24.1.1](#)).

## 26.1.5 Exercises

### 26.1.5.1 A. Check your understanding

**Exercise A1.**

In [Example \(Two regressors that move together\)](#), the difference  $\beta_1 - \beta_2$  corresponds to  $\mathbf{a} = (0, 1, -1)^\top$ , of squared length 2. Check that the variance quoted for  $\hat{\beta}_1 - \hat{\beta}_2$  lies between the two bounds of [Theorem 26.1.1\(b\)](#), applied to the slope block, and explain why it is close to the upper one.

*Solution.* Computed without rounding, the variance is  $11.640\sigma^2$ , and the bounds are  $2\sigma^2/70.05 = 0.0286\sigma^2$  and  $2\sigma^2/\lambda_2 = 11.643\sigma^2$ . (The check needs more digits than the table gives:  $3.412^2$  and  $2/0.1718$  agree to four figures.) The variance is  $\sigma^2 \sum_\ell (\mathbf{a}^\top \mathbf{v}_\ell)^2 / \lambda_\ell$ , and  $(\mathbf{a}^\top \mathbf{v}_2)^2 = 1.9995$  of the total  $\|\mathbf{a}\|^2 = 2$ , because  $\mathbf{a}$  points almost exactly along the weak eigenvector. Nearly all of its weight sits on the term with the small eigenvalue.

**Exercise A2.**

Under normal errors,  $\hat{\beta}_2 \sim N(\beta_2, \text{Var}(\hat{\beta}_2))$ . With  $\beta_2 = 1$  and the standard deviation of [Example \(Two regressors that move together\)](#), compute  $\Pr(\hat{\beta}_2 < 0)$  and compare it with the share of replicates in which  $\hat{\beta}_2$  was negative.

**Exercise A3.**

Let  $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2]$  have orthogonal columns with  $\|\mathbf{x}_1\| = 1$  and  $\|\mathbf{x}_2\| = 10$ . Find the canonical coordinates and the weak direction. Now measure the second regressor in units ten times larger, so that the column becomes  $\mathbf{x}_2/10$ . What happens to the weak direction and to  $\kappa(\mathbf{X})$ ? Which variances of estimable quantities have changed?

**26.1.5.2 B. Practice****Exercise B1.**

Show that  $\max_j \text{Var}(\hat{\beta}_j) \geq \sigma^2 / (p\lambda_p)$ , so that a small  $\lambda_p$  always hurts at least one coefficient. Give an example with  $p = 3$  in which  $\lambda_p$  is tiny but only one coefficient has a large variance.

*Solution.* By [Theorem 26.1.1\(d\)](#),  $\sum_j \text{Var}(\hat{\beta}_j) = \sigma^2 \sum_\ell 1/\lambda_\ell \geq \sigma^2/\lambda_p$ , and the largest of  $p$  numbers is at least their average. For the example take  $\mathbf{X}$  with orthonormal columns and multiply the third column by  $\epsilon$ . Then  $\mathbf{X}^\top \mathbf{X} = \text{diag}(1, 1, \epsilon^2)$ ,  $\lambda_p = \epsilon^2$ , and only  $\text{Var}(\hat{\beta}_3) = \sigma^2/\epsilon^2$  is large: a short column, not a near-dependence, which is why [Section 26.3](#) scales the columns.

**Exercise B2.**

In [Proposition 26.1.2](#), replace  $\mathbf{z}$  by an arbitrary vector  $\mathbf{w} \notin \mathcal{C}(\mathbf{X}_1)$ . Write  $\mathbf{w} = \mathbf{X}_1 \mathbf{d} + \|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\| \mathbf{z}$  with  $\mathbf{z}$  a unit vector orthogonal to  $\mathcal{C}(\mathbf{X}_1)$ , and show that

$$\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}_\delta) = \sigma^2 \mathbf{a}_1^\top (\mathbf{X}_1^\top \mathbf{X}_1)^{-1} \mathbf{a}_1 + \sigma^2 \frac{(a_2 - (\mathbf{c} + \delta \mathbf{d})^\top \mathbf{a}_1)^2}{\delta^2 \|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\|^2}.$$

Deduce that if  $a_2 = \mathbf{c}^\top \mathbf{a}_1$  the variance does not depend on  $\delta$  and equals the limiting-model variance plus  $\sigma^2(\mathbf{d}^\top \mathbf{a}_1)^2 / \|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\|^2$ , which vanishes iff  $\mathbf{d}^\top \mathbf{a}_1 = 0$ , and that for every other  $\mathbf{a}$  it still grows like  $\delta^{-2}$ .

*Solution.*  $\mathbf{X}_1 \mathbf{c} + \delta \mathbf{w} = \mathbf{X}_1(\mathbf{c} + \delta \mathbf{d}) + \delta \|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\| \mathbf{z}$ , which is the setting of the proposition with  $\mathbf{c}$  replaced by  $\mathbf{c} + \delta \mathbf{d}$  and  $\delta$  by  $\delta \|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\|$ . Part (a) gives the formula. If  $a_2 = \mathbf{c}^\top \mathbf{a}_1$ , the numerator of the second term is  $\delta^2(\mathbf{d}^\top \mathbf{a}_1)^2$ , the  $\delta^2$  cancels, and the second term is the constant  $\sigma^2(\mathbf{d}^\top \mathbf{a}_1)^2 / \|(\mathbf{I} - \mathbf{M}_1)\mathbf{w}\|^2$ ; the first is the limiting-model variance, by part (b) of the proposition. Otherwise the numerator tends to  $(a_2 - \mathbf{c}^\top \mathbf{a}_1)^2 > 0$  and the term grows like  $\delta^{-2}$ .

### Exercise B3.

Section 8.3 used twelve equally spaced  $t_i$  on  $[-1, 1]$ ,  $\mathbf{X}_1 = [\mathbf{1}, \mathbf{t}]$ ,  $\mathbf{c} = (0, 1)^\top$  and  $\mathbf{w} = (\cos \pi t_i)$ , and reported  $\text{Var}(\hat{\beta}_1 + \hat{\beta}_2) = 0.212 \sigma^2$  whatever  $\delta$ . Use Exercise (B2) to explain the number: show that  $\mathbf{d} = (d_1, 0)^\top$  because  $\mathbf{w}$  is even and  $\mathbf{t}$  is odd, and that the variance equals  $\sigma^2 / \sum_i t_i^2$ .

*Solution.* Here  $\mathbf{a} = (0, 1, 1)^\top$ , so  $\mathbf{a}_1 = (0, 1)^\top$  and  $a_2 = \mathbf{c}^\top \mathbf{a}_1 = 1$ : the function is estimable in the limit. The coefficients  $\mathbf{d}$  of the projection of  $\mathbf{w}$  on  $\mathcal{C}(\mathbf{1}, \mathbf{t})$  are  $d_1 = \bar{w}$  and  $d_2 = \mathbf{t}^\top \mathbf{w} / \mathbf{t}^\top \mathbf{t} = 0$ , because the  $t_i$  are symmetric about zero and  $\cos$  is even. So  $\mathbf{d}^\top \mathbf{a}_1 = d_2 = 0$  and the second term vanishes. The first is  $\sigma^2[(\mathbf{X}_1^\top \mathbf{X}_1)^{-1}]_{22} = \sigma^2 / \sum_i t_i^2$ , since  $\mathbf{1} \perp \mathbf{t}$ . With  $t_i = -1 + 2i/11$ ,  $i = 0, \dots, 11$ ,  $\sum_i t_i^2 = (4/121) \sum_{i=0}^{11} (i - 5.5)^2 = (4/121) \cdot 143 = 4.727$ , and  $1/4.727 = 0.2115$ , which rounds to 0.212.

## 26.2 Variance inflation factors

Analysts usually want something coarser than the eigenvalues: one number per coefficient saying how much its precision has suffered. Chapter 6 supplied that number as a by-product of the Frisch–Waugh–Lovell theorem. This section states what it measures and what it does not, and extends it to a group of columns such as the indicators of a factor.

### 26.2.1 Definition and meaning

Throughout, the model contains an intercept,  $\mathbf{x}_j$  is one of the other columns, and  $\mathbf{X}_{(j)}$  is  $\mathbf{X}$  with  $\mathbf{x}_j$  removed, with projection  $\mathbf{M}_{(j)}$ . Write  $S_{jj} = \|\mathbf{x}_j - \bar{x}_j \mathbf{1}\|^2$  for the centred sum of squares of  $\mathbf{x}_j$ .

#### Definition 26.2.1 — Variance inflation factor

Let  $R_j^2$  be the coefficient of determination of the least squares regression of  $\mathbf{x}_j$  on the columns of  $\mathbf{X}_{(j)}$ , which include  $\mathbf{1}$ , and suppose  $R_j^2 < 1$ . The **variance inflation factor** of  $\mathbf{x}_j$  is

$$\text{VIF}_j = \frac{1}{1 - R_j^2},$$

and  $\text{TOL}_j = 1 - R_j^2 = 1/\text{VIF}_j$  is its **tolerance**.

The identity Eq. 6.6.2, proved with the Frisch–Waugh–Lovell theorem, reads  $\text{Var}(\hat{\beta}_j) = \sigma^2 \text{VIF}_j / S_{jj}$ . The next proposition collects what follows from it.

### Proposition 26.2.1 — What a variance inflation factor measures

- (a)  $\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2 = S_{jj}/\text{VIF}_j$ , and  $\text{VIF}_j$  is the ratio of  $\text{Var}(\hat{\beta}_j)$  to  $\sigma^2/S_{jj}$ , the variance  $\hat{\beta}_j$  would have if the centred  $\mathbf{x}_j$  were orthogonal to every other centred column. By Exercise (A2),  $\text{VIF}_j \geq 1$ .
- (b)  $\text{VIF}_j$  depends on  $\mathbf{X}$  only through  $\mathbf{x}_j$  and  $\mathcal{C}(\mathbf{X}_{(j)})$ . It is unchanged when  $\mathbf{x}_j$  is replaced by  $a\mathbf{x}_j + b\mathbf{1}$  with  $a \neq 0$ , and when  $\mathbf{X}_{(j)}$  is replaced by any matrix with the same column space.
- (c) If  $\mathbf{R}$  is the correlation matrix of the non-intercept columns, then  $\text{VIF}_j = [\mathbf{R}^{-1}]_{jj}$ .
- (d) The  $t$  interval for  $\beta_j$  at any level is  $\sqrt{\text{VIF}_j}$  times as long, on average, as in the orthogonal design of (a), and the noncentrality of the  $F$  test of  $\beta_j = 0$  (the square of the  $t$  statistic's) is divided by  $\text{VIF}_j$ :

$$\frac{\beta_j^2}{\text{Var}(\hat{\beta}_j)} = \frac{\beta_j^2 S_{jj}}{\sigma^2} \cdot \frac{1}{\text{VIF}_j}.$$

*Proof.*

- (a) Since  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_{(j)})$ , the vector  $(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$  is the residual of the regression of  $\mathbf{x}_j$  on  $\mathbf{X}_{(j)}$ , and its squared length is  $S_{jj}(1 - R_j^2)$  by the definition of  $R_j^2$  (Exercise (B1)). The variance ratio is Eq. 6.6.2 divided by  $\sigma^2/S_{jj}$ , which is the variance in the orthogonal case by Proposition 7.2.4 applied to the model with centred columns.
- (b) By (a),  $1/\text{VIF}_j = \|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2/S_{jj}$ , and  $\mathbf{M}_{(j)}$  depends only on  $\mathcal{C}(\mathbf{X}_{(j)})$ . Replacing  $\mathbf{x}_j$  by  $a\mathbf{x}_j + b\mathbf{1}$  multiplies the numerator by  $a^2$ , because  $(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{1} = \mathbf{0}$ , and multiplies  $S_{jj}$  by  $a^2$ .
- (c) This is Exercise (A1).
- (d) The  $t$  interval has half-length  $t_{\alpha/2}(n-p) s \sqrt{\text{VIF}_j/S_{jj}}$ , and  $s$  has the same distribution in both designs. The noncentrality is  $\beta_j^2/\text{Var}(\hat{\beta}_j)$  with the variance from (a) (Theorem 11.6.1).

■

By (b) the factor is scale-free: a property of the angle between the centred  $\mathbf{x}_j$  and the space of the other centred columns, whose squared cosine is  $R_j^2$  (Theorem 6.7.2). By (a), with the spread per observation fixed,  $\hat{\beta}_j$  is as precise as in an orthogonal design with  $n/\text{VIF}_j$  observations.

*Idea.* A variance inflation factor of 10 costs  $\hat{\beta}_j$  as much precision as discarding nine observations in ten from an orthogonal design with the same spread.

## 26.2.2 Reading variance inflation factors

Rules of thumb flag a regressor when  $VIF_j$  exceeds 10, or 5, or 4. The thresholds are arbitrary, and they measure the wrong thing. The factor compares the design with a hypothetical orthogonal design of the same size; it says nothing about whether the actual standard error is small enough for the purpose, which depends also on  $\sigma^2$  and on  $S_{jj}$ , hence on  $n$ . As  $n$  grows with the regressors drawn from a fixed population,  $VIF_j$  settles to a population value while  $S_{jj}$  grows like  $n$ , so a factor that is ruinous at  $n = 20$  can be irrelevant at  $n = 20,000$ . O'Brien (2007) makes this case in detail. A large factor is a reason to look at the interval, never a verdict by itself.

### Example. (Large factors, precise estimate)

The quarterly US macroeconomic series of [Example \(A macroeconomic design\)](#) run from 1959 to 2009,  $n = 203$ . Regress real personal consumption on real disposable income, population, the consumer price index, the money stock M1, the three-month Treasury bill rate and the unemployment rate, with an intercept. The variance inflation factors are

| Regressor | income | population | CPI   | M1   | T-bill | unemployment |
|-----------|--------|------------|-------|------|--------|--------------|
| $VIF_j$   | 154.5  | 183.7      | 140.4 | 97.0 | 2.9    | 1.3          |

By every rule of thumb, four regressors are badly collinear. Yet the coefficient of income is 1.0659 with standard error 0.0210, a  $t$  statistic of 50.9: the standard error is  $\sqrt{154.5} \approx 12.4$  times what an orthogonal design would give, and still small, because income varies widely and consumption tracks it closely. (The errors are autocorrelated, so all the standard errors are too small, [Chapter 21](#); the factors depend on  $\mathbf{X}$  alone.)

```
import numpy as np
import statsmodels.api as sm

def vifs(Z):
    """VIFs of the columns of Z (an intercept is always included): diagonal of R
    ~{-1}."""
    return np.diag(np.linalg.inv(np.corrcoef(Z, rowvar=False)))

macro = sm.datasets.macrodta.load_pandas().data
names = ["realdpi", "pop", "cpi", "m1", "tbilrate", "unemp"]
Z = macro[names].to_numpy()
fit = sm.OLS(macro["realcons"].to_numpy(), sm.add_constant(Z)).fit()
for name, v, b, se in zip(names, vifs(Z), fit.params[1:], fit.bse[1:]):
    print(f"{name:9s} VIF {v:7.1f} estimate {b:9.4f} se {se:7.4f} t {b / se:6.1f}")
```

## 26.2.3 Collinearity that a reparameterization removes

Some large factors are produced by the parameterization, not by the data. Polynomial and interaction terms built from regressors whose origin lies far from the data are the standard case.

Such collinearity is sometimes called *nonessential*.

**Example. (A quadratic far from the origin)**

Take  $n = 21$  equally spaced values  $x_i = 10, 10.5, \dots, 20$  and the quadratic model  $E(Y) = \beta_0 + \beta_1 x + \beta_2 x^2$ . Over this range  $x$  and  $x^2$  are almost perfectly correlated:  $R^2 = 0.9920$ , and both have  $\text{VIF} = 124.6$ . Rewrite the model in the centred regressors  $x - 15$  and  $(x - 15)^2$ . The fitted values are identical, and both variance inflation factors are now exactly 1, because the design is symmetric about 15 and an odd function of  $x - 15$  is orthogonal to an even one once both are centred.

What changed is the question. In the raw coding  $\beta_1$  is the slope of the mean function at  $x = 0$ , ten units below the data, with standard deviation  $0.804\sigma$ . In the centred coding the coefficient of  $x - 15$  is the slope at  $x = 15$ , with standard deviation  $0.0721\sigma$ , and the raw coding estimates the same slope,  $\beta_1 + 30\beta_2$ , with exactly the same standard deviation. The large factors described the difficulty of a question, the slope at the origin, that the analyst probably did not mean to ask ([Exercise \(B2\)](#) derives 124.6 by hand).

```
x = np.linspace(10, 20, 21)                                # a regressor far from its
                                                           origin

raw = np.column_stack([x, x**2])
cen = np.column_stack([x - x.mean(), (x - x.mean()) ** 2])
print("VIFs, raw      :", np.round(vifs(raw), 1))
print("VIFs, centred :", np.round(vifs(cen), 3))

X_raw = np.column_stack([np.ones_like(x), raw])
X_cen = np.column_stack([np.ones_like(x), cen])
G_raw, G_cen = np.linalg.inv(X_raw.T @ X_raw), np.linalg.inv(X_cen.T @ X_cen)
a_mean = np.array([0, 1, 2 * x.mean()])                    # slope at x = 15 in the raw
                                                           coefficients
print("sd of the slope at 0 (raw b1)      :", round(np.sqrt(G_raw[1, 1]), 3))
print("sd of the slope at 15, raw coding  :", round(np.sqrt(a_mean @ G_raw @ a_mean
), 4))
print("sd of the slope at 15, centred (b1) :", round(np.sqrt(G_cen[1, 1]), 4))
```

## 26.2.4 Groups of columns

A factor enters the model through several indicator columns, a polynomial through several powers. The individual coefficients of such a block depend on the coding ([Definition 8.5.1](#)), and so do their variance inflation factors. What is intrinsic is the joint precision of the block, which the next definition, due to Fox and Monette (1992), measures.

For the rest of the section, split the non-intercept columns of  $\mathbf{X}$  into  $\mathbf{Z}_1$  ( $n \times q$ ) and  $\mathbf{Z}_2$  ( $n \times (k - q)$ ), let  $\tilde{\mathbf{Z}}_1$  and  $\tilde{\mathbf{Z}}_2$  be the centred versions, and let  $\tilde{\mathbf{M}}_2$  project onto  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ . Assume  $\mathbf{X} = [\mathbf{1}, \mathbf{Z}_1, \mathbf{Z}_2]$  has full column rank.

**Definition 26.2.2 — Generalized variance inflation factor**



The **generalized variance inflation factor** of the block  $\mathbf{Z}_1$  is

$$\text{GVIF}_1 = \frac{\det(\tilde{\mathbf{Z}}_1^\top \tilde{\mathbf{Z}}_1)}{\det(\tilde{\mathbf{Z}}_1^\top (\mathbf{I} - \tilde{\mathbf{M}}_2) \tilde{\mathbf{Z}}_1)}.$$

The **canonical correlations** between the two blocks are defined through orthonormal bases  $\mathbf{Q}_1$  of  $\mathcal{C}(\tilde{\mathbf{Z}}_1)$  and  $\mathbf{Q}_2$  of  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ : they are the singular values  $\rho_1 \geq \rho_2 \geq \dots \geq \rho_m$ ,  $m = \min(q, k - q)$ , of the  $q \times (k - q)$  matrix  $\mathbf{Q}_1^\top \mathbf{Q}_2$ . The largest,  $\rho_1$ , is the largest correlation between a linear combination of the centred columns of  $\mathbf{Z}_1$  and a linear combination of the centred columns of  $\mathbf{Z}_2$  ([Exercise \(C1\)](#)).

### Proposition 26.2.2 — Properties of the generalized factor

Let  $\boldsymbol{\beta}_1$  be the coefficients of  $\mathbf{Z}_1$ .

- (a)  $\det \text{Cov}(\hat{\boldsymbol{\beta}}_1) = \text{GVIF}_1 \cdot \det \text{Cov}_\perp$ , where  $\text{Cov}_\perp = \sigma^2(\tilde{\mathbf{Z}}_1^\top \tilde{\mathbf{Z}}_1)^{-1}$  is the covariance  $\hat{\boldsymbol{\beta}}_1$  would have if  $\mathcal{C}(\tilde{\mathbf{Z}}_1)$  were orthogonal to  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ . So  $\text{GVIF}_1$  is the squared ratio of the volumes of the joint confidence ellipsoids for  $\boldsymbol{\beta}_1$  in the two designs.
- (b)  $\text{GVIF}_1 = \prod_{i=1}^m 1/(1 - \rho_i^2)$ .
- (c)  $\text{GVIF}_1$  is unchanged when  $\mathbf{Z}_1$  is replaced by  $\mathbf{Z}_1 \mathbf{A} + \mathbf{1} \mathbf{f}^\top$  and  $\mathbf{Z}_2$  by  $\mathbf{Z}_2 \mathbf{B} + \mathbf{1} \mathbf{g}^\top$ , for nonsingular  $\mathbf{A}$  and  $\mathbf{B}$  and any vectors  $\mathbf{f}$  and  $\mathbf{g}$ . In particular it does not depend on the coding of a factor. It is symmetric:  $\text{GVIF}_1 = \text{GVIF}_2$ .
- (d) If  $q = 1$ ,  $\text{GVIF}_1$  is the variance inflation factor of [Definition 26.2.1](#). In general, with  $\mathbf{R}$  the correlation matrix of  $[\mathbf{Z}_1, \mathbf{Z}_2]$  partitioned conformably,  $\text{GVIF}_1 = \det \mathbf{R}_{11} \det \mathbf{R}_{22} / \det \mathbf{R}$ .

*Proof.*

- (a) Since  $\mathcal{C}(\mathbf{1}, \mathbf{Z}_2) = \mathcal{C}(\mathbf{1}) \oplus \mathcal{C}(\tilde{\mathbf{Z}}_2)$  with orthogonal summands, the projection onto  $\mathcal{C}(\mathbf{1}, \mathbf{Z}_2)$  is  $n^{-1} \mathbf{1} \mathbf{1}^\top + \tilde{\mathbf{M}}_2$  ([Theorem 6.3.8](#)), and the residual of  $\mathbf{Z}_1$  after it is  $(\mathbf{I} - \tilde{\mathbf{M}}_2) \tilde{\mathbf{Z}}_1$ . By [Theorem 6.6.2](#) and [Eq. 6.6.1](#),  $\text{Cov}(\hat{\boldsymbol{\beta}}_1) = \sigma^2(\tilde{\mathbf{Z}}_1^\top (\mathbf{I} - \tilde{\mathbf{M}}_2) \tilde{\mathbf{Z}}_1)^{-1}$ . If  $\mathcal{C}(\tilde{\mathbf{Z}}_1) \perp \mathcal{C}(\tilde{\mathbf{Z}}_2)$ , then  $\tilde{\mathbf{M}}_2 \tilde{\mathbf{Z}}_1 = \mathbf{0}$  and the covariance is  $\text{Cov}_\perp$ . Take the ratio of determinants. By [Theorem 12.2.1\(b\)](#), the volume of the confidence ellipsoid is proportional to  $(\det \boldsymbol{\Sigma})^{1/2}$ , where  $\boldsymbol{\Sigma}$  is the covariance matrix, with the same constant in both designs.
- (b) Centring annihilates  $\mathbf{1} \mathbf{f}^\top$  and commutes with multiplication on the right, so replacing  $\mathbf{Z}_1$  by  $\mathbf{Z}_1 \mathbf{A} + \mathbf{1} \mathbf{f}^\top$  replaces  $\tilde{\mathbf{Z}}_1$  by  $\tilde{\mathbf{Z}}_1 \mathbf{A}$ , and both determinants in [Definition 26.2.2](#) are multiplied by  $\det(\mathbf{A})^2$ . For the same reason  $\mathbf{Z}_2 \mapsto \mathbf{Z}_2 \mathbf{B} + \mathbf{1} \mathbf{g}^\top$  replaces  $\tilde{\mathbf{Z}}_2$  by  $\tilde{\mathbf{Z}}_2 \mathbf{B}$ , which leaves  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ , and so  $\tilde{\mathbf{M}}_2$ , unchanged. We may therefore take  $\tilde{\mathbf{Z}}_1 = \mathbf{Q}_1$ . Then  $\tilde{\mathbf{M}}_2 = \mathbf{Q}_2 \mathbf{Q}_2^\top$ , and with  $\mathbf{C} = \mathbf{Q}_1^\top \mathbf{Q}_2$ ,

$$\text{GVIF}_1 = \frac{\det(\mathbf{Q}_1^\top \mathbf{Q}_1)}{\det(\mathbf{Q}_1^\top \mathbf{Q}_1 - \mathbf{C} \mathbf{C}^\top)} = \frac{1}{\det(\mathbf{I}_q - \mathbf{C} \mathbf{C}^\top)}.$$

The eigenvalues of  $\mathbf{C} \mathbf{C}^\top$  are  $\rho_1^2, \dots, \rho_m^2$  and  $q - m$  zeros, so the determinant is  $\prod_i (1 - \rho_i^2)$ .

- (c) The invariance was shown in the proof of (b). Symmetry follows from (b), because  $\mathbf{Q}_1^\top \mathbf{Q}_2$  and  $\mathbf{Q}_2^\top \mathbf{Q}_1$  have the same singular values.
- (d) For  $q = 1$  the ratio is  $S_{jj} / \|(\mathbf{I} - \mathbf{M}_{(j)}) \mathbf{x}_j\|^2 = \text{VIF}_j$  by [Proposition 26.2.1\(a\)](#). For the

determinant form, let  $\mathbf{D}$  be the diagonal matrix with  $\mathbf{R} = \mathbf{D}[\tilde{\mathbf{Z}}_1, \tilde{\mathbf{Z}}_2]^\top [\tilde{\mathbf{Z}}_1, \tilde{\mathbf{Z}}_2] \mathbf{D}$ , with diagonal blocks  $\mathbf{D}_1, \mathbf{D}_2$ . By [Theorem 1.6.3](#),  $\det \mathbf{R} = \det \mathbf{R}_{22} \det(\mathbf{R}_{11} - \mathbf{R}_{12} \mathbf{R}_{22}^{-1} \mathbf{R}_{21})$ , and the Schur complement is  $\mathbf{D}_1 \tilde{\mathbf{Z}}_1^\top (\mathbf{I} - \tilde{\mathbf{M}}_2) \tilde{\mathbf{Z}}_1 \mathbf{D}_1$ . Since  $\mathbf{R}_{11} = \mathbf{D}_1 \tilde{\mathbf{Z}}_1^\top \tilde{\mathbf{Z}}_1 \mathbf{D}_1$ , the factors  $\det(\mathbf{D}_1)^2$  cancel in  $\det \mathbf{R}_{11} \det \mathbf{R}_{22} / \det \mathbf{R}$ , which leaves  $\text{GVIF}_1$ . ■

A generalized factor inflates a  $q$ -dimensional volume, so for comparison with single columns Fox and Monette suggest  $\text{GVIF}_1^{1/(2q)}$ , the geometric mean of the factors by which the axes of the confidence ellipsoid lengthen; for  $q = 1$  it is  $\sqrt{\text{VIF}_j}$ .

### Example. (A factor that explains a regressor)

Grunfeld's panel ([Example \(Grunfeld's investment data\)](#)) has 220 firm-years of gross investment, market value and capital stock for 11 firms. Regress investment on value, capital and the firm, entered as 10 indicator columns with the first firm as reference. The generalized factors are

| Term    | columns $q$ | GVIF  | $\text{GVIF}^{1/(2q)}$ |
|---------|-------------|-------|------------------------|
| value   | 1           | 18.32 | 4.280                  |
| capital | 1           | 2.04  |                        |
| firm    | 10          | 23.61 | 1.171                  |

The canonical correlations between the firm block and the pair (value, capital) are 0.9694 and 0.5459, and  $1/((1 - 0.9694^2)(1 - 0.5459^2)) \approx 23.6$  as [Proposition 26.2.2\(b\)](#) requires. Market value is largely a characteristic of the firm. With the firm indicators in the model, the coefficient of value is identified by movements within firms (the within estimator of [Section 6.6](#)), and the part of value that neither firm nor capital explains is  $1/18.32$ , about 5.5 per cent, of its variation.

The ordinary factors of the ten indicator columns range from 1.8 to 16.8; with the last firm as reference the largest is 12.7. The generalized factors are the same under every coding, including sum-to-zero coding ([Proposition 26.2.2\(c\)](#)).

```
import pandas as pd
grun = sm.datasets.grunfeld.load_pandas().data
D = pd.get_dummies(grun["firm"], drop_first=True).to_numpy(dtype=float)  # 10
    indicator columns
Zg = np.column_stack([grun["value"], grun["capital"], D])

def gvif(Z, block):
    """det(R_11) det(R_22) / det(R) for the columns in block against the rest."""
    R = np.corrcoef(Z, rowvar=False)
    rest = [k for k in range(Z.shape[1]) if k not in block]
    return (np.linalg.det(R[np.ix_(block, block)]) * np.linalg.det(R[np.ix_(rest,
    rest)]))
        / np.linalg.det(R))

firm = list(range(2, Zg.shape[1]))
print("GVIF value  :", round(gvif(Zg, [0]), 2))
```

```
print("GVIF capital:", round(gvif(Zg, [1]), 2))
print("GVIF firm   :", round(gvif(Zg, firm), 2), " per dimension:", round(gvif(Zg,
    firm) ** (1 / 20), 3))
print("VIFs of the firm indicators:", np.round(vifs(Zg)[2:], 1))
```

## 26.2.5 Exercises

### 26.2.5.1 A. Check your understanding

#### Exercise A1.

Which values of  $R_j^2$  correspond to variance inflation factors of 4, 5 and 10? By what factor does each multiply the length of a confidence interval for  $\beta_j$ , relative to the orthogonal design of [Proposition 26.2.1\(a\)](#)?

#### Exercise A2.

For a model with an intercept and two regressors with sample correlation  $r$ , show that  $VIF_1 = VIF_2 = 1/(1 - r^2)$ . What does [Proposition 26.2.1\(a\)](#) say about the design of [Example \(Two regressors that move together\)](#)?

### 26.2.5.2 B. Practice

#### Exercise B1.

Show that appending a column  $\mathbf{w} \notin \mathcal{C}(\mathbf{X})$  to  $\mathbf{X}$  cannot decrease the variance inflation factor of any column already in the model, and that it leaves  $VIF_j$  unchanged iff the residual of  $\mathbf{w}$  on  $\mathbf{X}_{(j)}$  is orthogonal to the residual of  $\mathbf{x}_j$  on  $\mathbf{X}_{(j)}$ . Is  $\text{Var}(\hat{\beta}_j)$  also nondecreasing?

*Solution.* Appending a column enlarges  $\mathcal{C}(\mathbf{X}_{(j)})$ , so  $\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|^2$  cannot increase, and by [Proposition 26.2.1\(a\)](#)  $VIF_j$  cannot decrease. Let  $\tilde{\mathbf{w}} \neq \mathbf{0}$  be the residual of  $\mathbf{w}$  on  $\mathbf{X}_{(j)}$ . The enlarged space is  $\mathcal{C}(\mathbf{X}_{(j)}) \oplus \mathcal{C}(\tilde{\mathbf{w}})$  with orthogonal summands, so the squared residual of  $\mathbf{x}_j$  drops by  $(\tilde{\mathbf{w}}^\top \mathbf{x}_j)^2 / \|\tilde{\mathbf{w}}\|^2$ , and  $\tilde{\mathbf{w}}^\top \mathbf{x}_j = \tilde{\mathbf{w}}^\top (\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j$ . So  $VIF_j$  is unchanged iff the two residuals are orthogonal. Since  $S_{jj}$  does not change,  $\text{Var}(\hat{\beta}_j) = \sigma^2 VIF_j / S_{jj}$  is nondecreasing too. This is the variance half of [Theorem 19.1.3](#).

#### Exercise B2.

Let  $x_i = m + u_i$ , where the  $u_i$  are symmetric about zero. Write  $v$  for the mean of the  $u_i^2$  and  $w$  for the variance of the  $u_i^2$  (both with divisor  $n$ ). Show that the variance inflation factor of  $x$  and of  $x^2$  in the quadratic model is

$$VIF = 1 + \frac{4m^2v}{w}.$$

Check the value in [Example \(A quadratic far from the origin\)](#).

*Solution.* With one other regressor,  $\text{VIF} = 1/(1 - r^2)$ , where  $r$  is the correlation of  $x$  and  $x^2$  (Exercise A2). Since  $x^2 = m^2 + 2mu + u^2$ , the covariance of  $x$  and  $x^2$  is that of  $u$  and  $2mu + u^2$ , which is  $2mv$  because the third moment of  $u$  vanishes. The variance of  $x^2$  is  $4m^2v + w$ , since  $u$  and  $u^2$  are uncorrelated. So  $r^2 = 4m^2v^2/(v(4m^2v + w)) = 4m^2v/(4m^2v + w)$  and  $1/(1 - r^2) = 1 + 4m^2v/w$ . In the example,  $m = 15$  and  $u$  runs over  $-5, -4.5, \dots, 5$ :  $v = 192.5/21 \approx 9.167$ , the mean of  $u^4$  is  $3166.6/21 \approx 150.79$ , so  $w \approx 150.79 - 84.03 = 66.76$  and  $\text{VIF} \approx 1 + 900 \cdot 9.167/66.76 \approx 124.6$ .

### Exercise B3.

Let  $\mathbf{x}_j$  be a column of the block  $\mathbf{Z}_1$ . Show that  $\text{GVIF}_1 \geq 1/(1 - R_{j|2}^2)$ , where  $R_{j|2}^2$  is the coefficient of determination of  $\mathbf{x}_j$  regressed on  $\mathbf{1}$  and  $\mathbf{Z}_2$  alone. Why is there no such inequality between  $\text{GVIF}_1$  and the ordinary  $\text{VIF}_j$  computed in the full model?

*Solution.*  $R_{j|2}$  is the correlation between the centred  $\mathbf{x}_j$  and its projection on  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ , which is a correlation between a vector  $\mathbf{Q}_1\mathbf{s}$  of  $\mathcal{C}(\tilde{\mathbf{Z}}_1)$  and a vector  $\mathbf{Q}_2\mathbf{t}$  of  $\mathcal{C}(\tilde{\mathbf{Z}}_2)$ , with  $\mathbf{s}$  and  $\mathbf{t}$  unit vectors. That correlation is  $\mathbf{s}^\top \mathbf{Q}_1^\top \mathbf{Q}_2 \mathbf{t} \leq \rho_1$  (Proposition 1.8.2; this is the first part of Exercise C1). Hence  $\text{GVIF}_1 \geq 1/(1 - \rho_1^2) \geq 1/(1 - R_{j|2}^2)$  by Proposition 26.2.2(b). The ordinary  $\text{VIF}_j$  regresses  $\mathbf{x}_j$  also on the other columns of its own block, which can explain it well even when  $\mathbf{Z}_2$  cannot. The indicators of a factor with a small reference level are an example: each is nearly determined by the others.

### 26.2.5.3 C. Going deeper

#### Exercise C1.

Show that  $\rho_1 = \max\{\cos \angle(\mathbf{u}, \mathbf{v}) : \mathbf{u} \in \mathcal{C}(\tilde{\mathbf{Z}}_1), \mathbf{v} \in \mathcal{C}(\tilde{\mathbf{Z}}_2), \mathbf{u}, \mathbf{v} \neq \mathbf{0}\}$ . More generally, show that the generalized eigenvalues of  $\text{Cov}(\hat{\boldsymbol{\beta}}_1)$  relative to  $\text{Cov}_\perp$ , the roots  $\theta$  of  $\det(\text{Cov}(\hat{\boldsymbol{\beta}}_1) - \theta \text{Cov}_\perp) = 0$ , are  $1/(1 - \rho_i^2)$ ,  $i \leq m$ , together with  $q - m$  ones. Deduce that for every  $\mathbf{a} \neq \mathbf{0}$  the ratio  $\text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}}_1)/\mathbf{a}^\top \text{Cov}_\perp \mathbf{a}$  lies between 1 and  $1/(1 - \rho_1^2)$ .

*Solution.* Write  $\mathbf{u} = \mathbf{Q}_1\mathbf{s}$  and  $\mathbf{v} = \mathbf{Q}_2\mathbf{t}$  with unit  $\mathbf{s}, \mathbf{t}$ . Then  $\cos \angle(\mathbf{u}, \mathbf{v}) = \mathbf{s}^\top \mathbf{Q}_1^\top \mathbf{Q}_2 \mathbf{t}$ , whose maximum over unit vectors is the largest singular value of  $\mathbf{Q}_1^\top \mathbf{Q}_2$  (Proposition 1.8.2). For the second part, take  $\tilde{\mathbf{Z}}_1 = \mathbf{Q}_1$  by Proposition 26.2.2(c), which changes both matrices by the same congruence and so leaves the generalized eigenvalues unchanged. Then  $\text{Cov}_\perp = \sigma^2 \mathbf{I}$  and  $\text{Cov}(\hat{\boldsymbol{\beta}}_1) = \sigma^2(\mathbf{I} - \mathbf{C}\mathbf{C}^\top)^{-1}$ , whose eigenvalues are the stated ones. The ratio in the last part is a ratio of quadratic forms, bounded by Corollary 1.7.8, so it lies between the smallest and the largest generalized eigenvalue.

## 26.3 Condition indices and variance decomposition

A variance inflation factor says how much one coefficient has suffered, but not which other regressors are responsible, how many near-dependencies there are, or whether the intercept is involved. The eigenvalue analysis of Section 26.1 answers those questions but depends on

the units of the columns. This section fixes a scaling and derives the variance-decomposition proportions of Belsley, Kuh and Welsch (1980).

### 26.3.1 Scaling the columns

The canonical directions of  $\mathbf{X}^\top \mathbf{X}$  change when a column is rescaled ([Exercise \(A3\)](#)), and a merely short column produces a small eigenvalue without any near-dependence ([Exercise \(B1\)](#)). The diagnostics are therefore computed for the design with every column, including  $\mathbf{1}$ , scaled to unit length:

$$\tilde{\mathbf{X}} = \mathbf{X}\mathbf{D}_0, \quad \mathbf{D}_0 = \text{diag}(1/\|\mathbf{x}_1\|, \dots, 1/\|\mathbf{x}_p\|).$$

By [Proposition 10.5.2](#) this is within a factor  $\sqrt{p}$  of the best diagonal scaling for the condition number. The coefficients of  $\tilde{\mathbf{X}}$  are  $\tilde{\beta}_j = \|\mathbf{x}_j\|\beta_j$ .

#### Definition 26.3.1 — Condition indices

Let  $\tilde{\mathbf{X}}$  have singular values  $\mu_1 \geq \mu_2 \geq \dots \geq \mu_p > 0$ . The **condition indices** of  $\mathbf{X}$  are

$$\eta_k = \frac{\mu_1}{\mu_k}, \quad k = 1, \dots, p.$$

The largest,  $\eta_p = \kappa(\tilde{\mathbf{X}})$ , is the **scaled condition number**.

Whether to centre before scaling is a real choice. Let  $\mathbf{R}$  be the correlation matrix of the non-intercept columns, the cross-product matrix of the centred and scaled design, and write  $s_j^2 = S_{jj}/n$  and  $\text{CV}_j = s_j/|\bar{x}_j|$  for the **coefficient of variation** of a column with nonzero mean.

#### Proposition 26.3.1 — Scaled condition numbers

Let  $\mathbf{X}$  contain the column  $\mathbf{1}$  and have full column rank, and let  $\mathbf{R}$  be nonsingular.

- (a)  $\sum_k \mu_k^2 = p$ , so  $1 \leq \mu_1^2 \leq p$ . Every diagonal entry of  $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$  is at most  $1/\mu_p^2 \leq \eta_p^2$ .
- (b)  $\lambda_{\min}(\mathbf{R}) \geq \mu_p^2$ . Hence  $\eta_p^2 \geq 1/\lambda_{\min}(\mathbf{R}) \geq \max_j \text{VIF}_j$ : every near-dependency among the centred regressors is at least as severe in the uncentred scaled design.
- (c) If a non-intercept column  $\mathbf{x}_j$  has  $\bar{x}_j \neq 0$ , then  $\mu_p^2 \leq \text{CV}_j^2/(\text{CV}_j^2 + 2)$ , so that  $\eta_p \geq \sqrt{(\text{CV}_j^2 + 2)/\text{CV}_j^2} > \sqrt{2}/\text{CV}_j$ .

*Proof.*

- (a)  $\sum_k \mu_k^2 = \|\tilde{\mathbf{X}}\|_F^2 = p$  ([Proposition 1.8.2](#)), since each column has unit length. The largest of  $p$  numbers with sum  $p$  is between 1 and  $p$ . The  $j$ th diagonal entry of  $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$  is a Rayleigh quotient of a matrix whose largest eigenvalue is  $1/\mu_p^2$  ([Theorem 1.7.7](#)), and  $1/\mu_p^2 = \eta_p^2/\mu_1^2 \leq \eta_p^2$ .
- (b) Number the columns so that  $\mathbf{x}_0 = \mathbf{1}$  and  $\mathbf{x}_1, \dots, \mathbf{x}_k$  are the others. The centred and scaled columns are  $\mathbf{z}_j = (\mathbf{x}_j - \bar{x}_j \mathbf{1})/(\sqrt{n} s_j)$ , so that  $\mathbf{R} = \mathbf{Z}^\top \mathbf{Z}$  with  $\mathbf{Z} = [\mathbf{z}_1, \dots, \mathbf{z}_k]$ .

Since  $\mathbf{1} = \sqrt{n} \tilde{\mathbf{x}}_0$  and  $\mathbf{x}_j = \|\mathbf{x}_j\| \tilde{\mathbf{x}}_j$ , where  $\tilde{\mathbf{x}}_j$  are the columns of  $\tilde{\mathbf{X}}$ ,

$$\mathbf{Zc} = \sum_{j=1}^k c_j \frac{\|\mathbf{x}_j\| \tilde{\mathbf{x}}_j - \sqrt{n} \bar{x}_j \tilde{\mathbf{x}}_0}{\sqrt{n} s_j} = \tilde{\mathbf{X}} \mathbf{b},$$

where

$$b_j = \frac{c_j \|\mathbf{x}_j\|}{\sqrt{n} s_j} \quad (j \geq 1), \quad b_0 = -\sum_{j=1}^k \frac{c_j \bar{x}_j}{s_j}.$$

Because  $\|\mathbf{x}_j\|^2 = n(s_j^2 + \bar{x}_j^2) \geq ns_j^2$ ,  $|b_j| \geq |c_j|$ , so  $\|\mathbf{b}\| \geq \|\mathbf{c}\|$ . For a unit vector  $\mathbf{c}$ ,  $\mathbf{c}^\top \mathbf{R} \mathbf{c} = \|\tilde{\mathbf{X}} \mathbf{b}\|^2 \geq \mu_p^2 \|\mathbf{b}\|^2 \geq \mu_p^2$  by [Theorem 1.7.7](#); minimizing over  $\mathbf{c}$  gives  $\lambda_{\min}(\mathbf{R}) \geq \mu_p^2$ . Then  $\eta_p^2 \geq 1/\mu_p^2 \geq 1/\lambda_{\min}(\mathbf{R})$  by (a), and  $1/\lambda_{\min}(\mathbf{R}) \geq \text{VIF}_j$  by [Proposition 19.6.1\(c\)](#).

- (c) Take  $\mathbf{b}$  with  $b_j = \|\mathbf{x}_j\|$ ,  $b_0 = -\sqrt{n} \bar{x}_j$  and all other entries zero. Then  $\tilde{\mathbf{X}} \mathbf{b} = \mathbf{x}_j - \bar{x}_j \mathbf{1}$ , with squared length  $ns_j^2$ , while  $\|\mathbf{b}\|^2 = n(s_j^2 + \bar{x}_j^2) + n\bar{x}_j^2$ . So  $\mu_p^2 \leq ns_j^2/(n(s_j^2 + 2\bar{x}_j^2)) = \text{CV}_j^2/(\text{CV}_j^2 + 2)$ , and  $\eta_p \geq 1/\mu_p$  by (a). ■

Part (b) is the formal content of the argument of Belsley (1984) for not centring: centring can only hide near-dependencies, never create them. Part (c) shows what it hides. A regressor that varies little relative to its mean is nearly a multiple of the column of ones, and that near-dependency with the intercept is invisible in  $\mathbf{R}$  and in every variance inflation factor. The design of [Example \(A macroeconomic design\)](#) shows the size of the effect: scaled to unit length, its condition number is 383; centred and scaled, 124.

Whether the hidden dependency matters depends on the question, as in [Example \(A quadratic far from the origin\)](#). It inflates the variance of the intercept, the mean response at the origin, and of the coefficients whose columns take part in it. If the origin is far from the data, it concerns a question nobody asked; if the origin is meaningful, the uncentred diagnostic is the honest one. This chapter uses the uncentred scaled design, as Belsley, Kuh and Welsch do, and reports the centred condition number alongside.

## 26.3.2 The variance-decomposition proportions

Let the singular value decomposition of the scaled design be

$$\tilde{\mathbf{X}} = \sum_{k=1}^p \mu_k \mathbf{u}_k \mathbf{t}_k^\top,$$

with orthonormal right singular vectors  $\mathbf{t}_k = (t_{1k}, \dots, t_{pk})^\top$ . The  $p \times p$  matrix with columns  $\mathbf{t}_k$  is orthogonal, so its rows also have unit length:  $\sum_k t_{jk}^2 = 1$  for every  $j$ .

### Theorem 26.3.2 — Variance-decomposition proportions

Let  $\mathbf{X}$  have rank  $p$ ,  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , and let  $\phi_{jk} = t_{jk}^2 / \mu_k^2$ .

(a)  $\text{Var}(\hat{\beta}_j) = \frac{\sigma^2}{\|\mathbf{x}_j\|^2} \sum_{k=1}^p \phi_{jk}$ . The **variance-decomposition proportions**

$$\pi_{kj} = \frac{\phi_{jk}}{\sum_{l=1}^p \phi_{jl}}$$

satisfy  $\sum_k \pi_{kj} = 1$ , do not depend on  $\sigma^2$ , and vanish iff  $t_{jk} = 0$ . The term  $\pi_{kj}$  is the share of  $\text{Var}(\hat{\beta}_j)$  contributed by the canonical coordinate with condition index  $\eta_k$ .

- (b) For each  $k$ ,  $\|\sum_j t_{jk} \tilde{\mathbf{x}}_j\| = \mu_k$  while  $\|\mathbf{t}_k\| = 1$ : the vector  $\mathbf{t}_k$  holds the coefficients of a near-dependency among the scaled columns, of strength  $\mu_k$ .
- (c) Let  $S$  be a nonempty proper subset of  $\{1, \dots, p\}$  such that  $\mu_k \leq \mu_S$  for  $k \in S$  and  $\mu_k \geq \mu_L$  for  $k \notin S$ , with  $\mu_S < \mu_L$ . Let  $w_j = \sum_{k \in S} t_{jk}^2$  measure the participation of column  $j$  in the near-dependencies indexed by  $S$ . Then

$$\sum_{k \in S} \pi_{kj} \geq \frac{w_j}{w_j + (1 - w_j) \mu_S^2 / \mu_L^2}, \quad \text{Var}(\hat{\beta}_j) \geq \frac{\sigma^2 w_j}{\|\mathbf{x}_j\|^2 \mu_S^2}.$$

*Proof.*

- (a) Since  $\mathbf{X}\boldsymbol{\beta} = \tilde{\mathbf{X}}\mathbf{D}_0^{-1}\boldsymbol{\beta}$ , the least squares coefficients of  $\tilde{\mathbf{X}}$  are  $\hat{\tilde{\beta}}_j = \|\mathbf{x}_j\| \hat{\beta}_j$ , and by [Theorem 26.1.1\(a\)](#) applied to  $\tilde{\mathbf{X}}$ , whose cross-product matrix has eigenvalues  $\mu_k^2$  and eigenvectors  $\mathbf{t}_k$ ,  $\text{Cov}(\hat{\tilde{\beta}}) = \sigma^2 \sum_k \mathbf{t}_k \mathbf{t}_k^\top / \mu_k^2$ . Its  $j$ th diagonal entry is  $\sigma^2 \sum_k \phi_{jk}$ . The statements about the proportions follow from the definition, since  $\mu_k > 0$ .
- (b)  $\sum_j t_{jk} \tilde{\mathbf{x}}_j = \tilde{\mathbf{X}} \mathbf{t}_k = \mu_k \mathbf{u}_k$ .
- (c) Put  $A = \sum_{k \in S} \phi_{jk}$  and  $B = \sum_{k \notin S} \phi_{jk}$ . Then  $A \geq w_j / \mu_S^2$  and, since  $\sum_{k \notin S} t_{jk}^2 = 1 - w_j$ ,  $B \leq (1 - w_j) / \mu_L^2$ . The sum of the proportions over  $S$  is  $A / (A + B)$ , which increases with  $A$  and decreases with  $B$ , so it is at least  $(w_j / \mu_S^2) / (w_j / \mu_S^2 + (1 - w_j) / \mu_L^2)$ , which is the stated bound. The variance bound is  $\text{Var}(\hat{\beta}_j) \geq \sigma^2 A / \|\mathbf{x}_j\|^2$  together with  $A \geq w_j / \mu_S^2$ . ■

Part (c) is why the proportions locate near-dependencies. If the condition indices in  $S$  are at least ten times those outside it,  $\mu_S^2 / \mu_L^2 \leq 0.01$ , then any column with even a tenth of its unit row inside  $S$ ,  $w_j \geq 0.1$ , has at least  $0.1 / (0.1 + 0.9 \times 0.01) \approx 0.92$  of its variance attributed to  $S$ , and its variance is inflated accordingly. A column that does not take part has proportions near zero there. When several small singular values are close together, the individual  $\mathbf{t}_k$  are poorly determined, and a column's proportions can be split between their rows in unpredictable ratios. Part (c) bounds the *sum* over the group, which is stable, and that is what should be read ([Exercise \(C1\)](#)).

*Idea.* Belsley, Kuh and Welsch's reading of the table: scale the columns to unit length; flag condition indices of about 30 or more (indices near 10 signal weak dependencies); in each flagged row, or group of rows with similar indices, look for two or more coefficients with large proportions, conventionally above 0.5. Those columns form a near-dependency. Confirm it by regressing one of them on the others.

**Example. (A consumption function)**

Return to the regression of [Example \(Large factors, precise estimate\)](#), with an intercept and six regressors. The scaled design has condition indices and proportions

| Index | const. | income | pop.  | CPI   | M1    | T-bill | unemp. |
|-------|--------|--------|-------|-------|-------|--------|--------|
| 1.0   | 0.000  | 0.000  | 0.000 | 0.000 | 0.000 | 0.001  | 0.001  |
| 3.4   | 0.000  | 0.000  | 0.000 | 0.000 | 0.001 | 0.053  | 0.007  |
| 7.7   | 0.000  | 0.000  | 0.000 | 0.001 | 0.001 | 0.347  | 0.090  |
| 13.4  | 0.002  | 0.001  | 0.000 | 0.001 | 0.002 | 0.001  | 0.688  |
| 38.9  | 0.005  | 0.149  | 0.000 | 0.001 | 0.290 | 0.049  | 0.036  |
| 74.3  | 0.007  | 0.130  | 0.000 | 0.995 | 0.569 | 0.241  | 0.116  |
| 270.9 | 0.985  | 0.720  | 0.999 | 0.001 | 0.138 | 0.307  | 0.060  |

[Figure 26.3.1](#) shows the table graphically. Three indices exceed 30, and the table separates them.

- *Index 270.9: intercept, income and population* (proportions 0.985, 0.720, 0.999). The singular vector has entries  $-0.522$ ,  $-0.236$  and  $0.817$  on these columns and small ones elsewhere: population, which grew steadily with income, is nearly an affine function of it ( $R^2 = 0.9885$ ). Its small coefficient of variation, 0.156, alone forces an index of at least 9.1 by [Proposition 26.3.1\(c\)](#); the shared trend does the rest.
- *Index 74.3: the price level and money*. CPI has proportion 0.995 and M1 0.569. Regressing CPI on income and M1 gives  $R^2 = 0.9868$ .
- *Index 38.9: a weaker dependency involving M1 and income*. Taken with the previous row, M1 has  $0.290 + 0.569 = 0.859$  of its variance in the two rows, and income has  $0.149 + 0.130 + 0.720 = 0.999$  in the three rows above 30. Income takes part in all three near-dependencies.

A proportion is a share, not a size: the 0.307 of the T-bill rate at index 270.9 is a share of a variance that collinearity has barely inflated (its variance inflation factor is 2.9). Centred and scaled, the design has condition number 37.9 against 270.9 uncentred: centring removes the intercept from the largest near-dependency, and what is left is the relation between income and population that the variance inflation factors 154.5 and 183.7 already showed.

```
import numpy as np
import statsmodels.api as sm
macro = sm.datasets.macrodatab.load_pandas().data
names = ["const", "realdpi", "pop", "cpi", "m1", "tbilrate", "unemp"]
X = sm.add_constant(macro[names[1:]].to_numpy())
X_s = X / np.linalg.norm(X, axis=0)                                # unit-length columns, not
    centred

U, mu, Vt = np.linalg.svd(X_s, full_matrices=False)              # X_s = U diag(mu) V'
eta = mu[0] / mu                                                  # condition indices
phi = Vt.T ** 2 / mu**2                                           # phi[j, k] = v_jk^2 / mu_k^2
prop = (phi / phi.sum(axis=1, keepdims=True)).T                  # row k: proportions for index
    eta_k
```



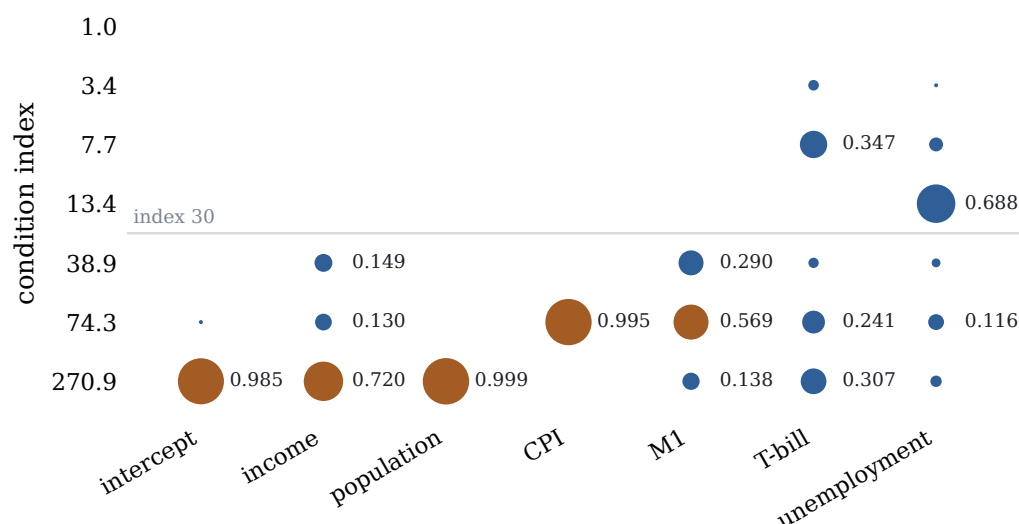


Figure 26.3.1: Variance-decomposition proportions of [Example \(A consumption function\)](#): rows are condition indices, columns coefficients, dot area proportional to  $\pi_{kj}$ . Highlighted:  $\pi_{kj} \geq 0.5$  with index at least 30.

```
print("index   " + " ".join(f"{nm[:7]:>7s}" for nm in names))
for k in range(len(mu)):
    print(f"{eta[k]:7.1f} " + " ".join(f"{p:7.3f}" for p in prop[k]))
```

```
Z = X[:, 1:] - X[:, 1:].mean(axis=0)
Z_s = Z / np.linalg.norm(Z, axis=0) # centred, then unit length
mu_c = np.linalg.svd(Z_s, compute_uv=False)
print("scaled condition number, not centred:", round(eta[-1], 1))
print("scaled condition number, centred   :", round(mu_c[0] / mu_c[-1], 1))
cv = X[:, 1:].std(axis=0) / X[:, 1:].mean(axis=0)
print("coefficients of variation:", dict(zip(names[1:], np.round(cv, 3))))
```

```
def r_squared(target, others):
    fit = sm.OLS(macro[target], sm.add_constant(macro[others])).fit()
    return fit.rsquared

print("pop on realdpi      : R^2 =", round(r_squared("pop", ["realdpi"]), 4))
print("cpi on realdpi, m1 : R^2 =", round(r_squared("cpi", ["realdpi", "m1"]), 4))
```

The script behind the example checks the bound of [Theorem 26.3.2\(c\)](#) for the groups formed by the last one, two and three indices, and the inequalities of [Proposition 26.3.1](#).

### 26.3.3 Single cases

Collinearity diagnostics can be dominated by a few cases: one far out along a near-dependency can create it, one lying across it can hide it. The deletion formulas of [Chapter 20](#) show that only cases of high leverage can have much effect.

### Proposition 26.3.3 — Deleting a case

Let  $\mathbf{X}$  have rank  $p$ , let  $\mathbf{x}_{(i)}$  be its  $i$ th row with leverage  $h_{ii} < 1$ , let  $\mathbf{X}_{[i]}$  be  $\mathbf{X}$  without that row ([Theorem 20.3.1](#) writes  $\mathbf{X}_{(i)}$ ; square brackets here avoid a clash with  $\mathbf{X}_{(j)}$ ), the matrix without column  $j$ ), and write  $\mathbf{G} = (\mathbf{X}^\top \mathbf{X})^{-1}$ ,  $\mathbf{G}_{[i]} = (\mathbf{X}_{[i]}^\top \mathbf{X}_{[i]})^{-1}$ .

- (a)  $\mathbf{a}^\top \mathbf{G}_{[i]} \mathbf{a} = \mathbf{a}^\top \mathbf{G} \mathbf{a} + (\mathbf{a}^\top \mathbf{G} \mathbf{x}_{(i)})^2 / (1 - h_{ii})$  for every  $\mathbf{a}$ .
- (b)  $\mathbf{a}^\top \mathbf{G} \mathbf{a} \leq \mathbf{a}^\top \mathbf{G}_{[i]} \mathbf{a} \leq \mathbf{a}^\top \mathbf{G} \mathbf{a} / (1 - h_{ii})$ : deleting case  $i$  multiplies the variance of every linear function of  $\hat{\boldsymbol{\beta}}$  by a factor between 1 and  $1/(1 - h_{ii})$ .
- (c)  $(1 - h_{ii}) \lambda_{\min}(\mathbf{X}^\top \mathbf{X}) \leq \lambda_{\min}(\mathbf{X}_{[i]}^\top \mathbf{X}_{[i]}) \leq \lambda_{\min}(\mathbf{X}^\top \mathbf{X})$ .

*Proof.*

- (a) This is the rank-one deletion formula  $\mathbf{G}_{[i]} = \mathbf{G} + \mathbf{G} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \mathbf{G} / (1 - h_{ii})$  of [Example \(Deleting an observation\)](#), on which [Theorem 20.3.1](#) rests, sandwiched between  $\mathbf{a}^\top$  and  $\mathbf{a}$ .
- (b) The lower bound is clear from (a). For the upper, apply the Cauchy–Schwarz inequality to the inner product  $\langle \mathbf{a}, \mathbf{b} \rangle = \mathbf{a}^\top \mathbf{G} \mathbf{b}$ , which is legitimate because  $\mathbf{G}$  is positive definite:  $(\mathbf{a}^\top \mathbf{G} \mathbf{x}_{(i)})^2 \leq (\mathbf{a}^\top \mathbf{G} \mathbf{a}) h_{ii}$ . Then  $\mathbf{a}^\top \mathbf{G}_{[i]} \mathbf{a} \leq \mathbf{a}^\top \mathbf{G} \mathbf{a} (1 + h_{ii} / (1 - h_{ii}))$ .
- (c) For a positive definite matrix,  $1/\lambda_{\min}$  is the largest eigenvalue of the inverse, which is  $\max_{\|\mathbf{a}\|=1} \mathbf{a}^\top (\cdot)^{-1} \mathbf{a}$  ([Theorem 1.7.7](#)). Taking the maximum in (b) over unit vectors gives the two inequalities. ■

So removing a case of small leverage changes the smallest eigenvalue by at most the factor  $1 - h_{ii}$ , and no variance by more than  $1/(1 - h_{ii})$ . In the consumption regression the largest leverage is 0.161, at 2009Q3, so by (c) no single quarter can reduce the smallest eigenvalue of the scaled cross-product matrix by more than the factor 1.192, with the column scaling of the full data held fixed. The largest reduction actually achieved by deleting one of the 203 quarters is by the factor 1.055. The near-dependencies of this design are carried by the whole sample, not by a few quarters.

Adding a case never increases a variance, by (b) read the other way round, but it can increase every variance inflation factor and the condition number ([Exercise \(B2\)](#)).

## 26.3.4 Exercises

### 26.3.4.1 A. Check your understanding

#### Exercise A1.

In the table of [Example \(A consumption function\)](#), why is the intercept's variance almost entirely attributed to the largest index, although the intercept is not an economic variable? Which coefficients would you expect to change most if a few more years of data were added in which population grew faster than income?

## 26.3.4.2 B. Practice

**Exercise B1.**

Let  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$  with  $\bar{x} \neq 0$ . Show that the scaled design has  $\mu_{1,2}^2 = 1 \pm \cos \theta$ , where  $\cos \theta = 1/\sqrt{1 + \text{CV}^2}$  is the cosine of the angle between  $\mathbf{1}$  and  $\mathbf{x}$ , and hence  $\eta_2^2 = (1 + \cos \theta)/(1 - \cos \theta)$ . Compare with the bound of Proposition 26.3.1(c). What is the variance inflation factor of  $\mathbf{x}$ ?

*Solution.* The scaled columns are unit vectors at angle  $\theta$ , so  $\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} = \begin{pmatrix} 1 & c \\ c & 1 \end{pmatrix}$  with  $c = \cos \theta$ , whose eigenvalues are  $1 \pm c$  (Exercise (B1)). Now  $c = \mathbf{1}^\top \mathbf{x} / (\sqrt{n} \|\mathbf{x}\|) = n\bar{x} / \sqrt{n \cdot n(s^2 + \bar{x}^2)} = 1/\sqrt{1 + \text{CV}^2}$  (taking  $\bar{x} > 0$ ). So  $\eta_2^2 = (1 + c)/(1 - c)$ . The bound of the proposition gives  $\mu_2^2 \leq \text{CV}^2/(\text{CV}^2 + 2)$ . For small CV,  $1 - c \approx \text{CV}^2/2$ , so  $\mu_2^2 = 1 - c \approx \text{CV}^2/2$ , and the bound is attained to first order. With a single regressor  $R_j^2 = 0$  and  $\text{VIF} = 1$ : the variance inflation factor cannot see this near-dependency at all.

**Exercise B2.**

Take  $n = 20$  cases with  $(x_1, x_2)$  at the points of a  $4 \times 5$  grid in  $[-1, 1]^2$ , so that the regressors are uncorrelated, and add a twenty-first case at  $(t, t)$ . Show that the correlation of the two regressors tends to one as  $t \rightarrow \infty$ , so that both variance inflation factors grow without bound, while no variance of any linear function of  $\hat{\beta}$  increases. Reconcile the two statements.

*Solution.* By Proposition 26.3.3(b) read backwards, adding a case can only decrease  $\mathbf{a}^\top \mathbf{G} \mathbf{a}$  for every  $\mathbf{a}$ . The sample covariance of  $x_1, x_2$  in the enlarged data is  $t^2(1 - 1/21)/21$  (the grid contributes nothing, being centred and uncorrelated), and the sample variance of  $x_j$  is  $(S_j + t^2(1 - 1/21))/21$ , where  $S_j$  is the grid's sum of squares for  $x_j$ . As  $t \rightarrow \infty$  the correlation tends to one and  $\text{VIF}_j = 1/(1 - r^2) \rightarrow \infty$ . But  $\text{Var}(\hat{\beta}_j) = \sigma^2 \text{VIF}_j / S_{jj}$ , and  $S_{jj}$  grows like  $t^2$  as fast as  $\text{VIF}_j$  does, so the variances converge to finite limits no larger than before. The variance inflation factor compares with an orthogonal design with the *new* spread, which the added point has made enormous; the point added information along  $(1, 1)$  and none across it.

## 26.3.4.3 C. Going deeper

**Exercise C1.**

Suppose  $\mu_k = \mu$  for all  $k \in S$ . Show that the singular vectors  $\mathbf{t}_k$ ,  $k \in S$ , are determined only up to an orthogonal transformation of their span, that the individual proportions  $\pi_{kj}$ ,  $k \in S$ , depend on the choice, and that  $\sum_{k \in S} \pi_{kj}$  does not.

*Solution.* If  $\mathbf{T}_S$  has columns  $\mathbf{t}_k$ ,  $k \in S$ , and  $\mathbf{O}$  is orthogonal, then  $\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}} \mathbf{T}_S \mathbf{O} = \mu^2 \mathbf{T}_S \mathbf{O}$ , so the columns of  $\mathbf{T}_S \mathbf{O}$  are equally valid singular vectors, and the individual  $t_{jk}^2$  change with  $\mathbf{O}$ . But  $\sum_{k \in S} \phi_{jk} = \mu^{-2} \sum_{k \in S} t_{jk}^2 = \mu^{-2} \mathbf{e}_j^\top \mathbf{T}_S \mathbf{T}_S^\top \mathbf{e}_j$ , and  $\mathbf{T}_S \mathbf{T}_S^\top$  is the projection onto the span, which does not depend on  $\mathbf{O}$ . The denominator  $\sum_l \phi_{jl}$  is a diagonal entry of  $(\tilde{\mathbf{X}}^\top \tilde{\mathbf{X}})^{-1}$  and does not depend on the choice either.

## 26.4 Collinearity and prediction

By [Theorem 26.1.1\(d\)](#), collinearity leaves the total variance of the fitted values, the predictions at the design points, untouched. At a new point  $\mathbf{x}_0$ , collinearity damages a prediction exactly to the extent that the point asks for information along a weak direction. [Example \(Predicting along and across the pattern\)](#) showed this contrast on simulated data; this section proves it in general and measures it on real data.

### 26.4.1 The variance of a prediction

Keep the notation of [Section 26.1](#):  $\mathbf{X}^\top \mathbf{X} = \sum_{\ell} \lambda_{\ell} \mathbf{v}_{\ell} \mathbf{v}_{\ell}^\top$ , and let

$$m_{\ell} = \frac{\lambda_{\ell}}{n} = \frac{1}{n} \sum_{i=1}^n (\mathbf{x}_{(i)}^\top \mathbf{v}_{\ell})^2$$

be the mean square of the design rows along the direction  $\mathbf{v}_{\ell}$ . A small  $m_{\ell}$  means that the rows hardly vary along  $\mathbf{v}_{\ell}$ : the data cloud is thin in that direction. The estimate of the mean response at  $\mathbf{x}_0$  is  $\hat{Y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}$ .

#### Proposition 26.4.1 — Prediction under collinearity

Let  $\mathbf{X}$  have rank  $p$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ .

- (a)  $\text{Var}(\hat{Y}_0) = \frac{\sigma^2}{n} \sum_{\ell=1}^p \frac{(\mathbf{x}_0^\top \mathbf{v}_{\ell})^2}{m_{\ell}}$ .
- (b) If  $\mathbf{x}_0 = \mathbf{X}^\top \mathbf{w}$  for some  $\mathbf{w} \in \mathbb{R}^n$ , then  $\text{Var}(\hat{Y}_0) = \sigma^2 \|\mathbf{M}\mathbf{w}\|^2 \leq \sigma^2 \|\mathbf{w}\|^2$ . If  $\mathbf{x}_0$  is a convex combination of the rows of  $\mathbf{X}$ , then  $\text{Var}(\hat{Y}_0) \leq \sigma^2 \max_i h_{ii} \leq \sigma^2$ . Neither bound involves the eigenvalues.
- (c) Let  $\mathbf{x}_0$  be random, independent of  $\mathbf{Y}$ , with  $\text{E}(\mathbf{x}_0 \mathbf{x}_0^\top) = \boldsymbol{\Sigma}_0$ . Then

$$\text{E}\{\text{Var}(\hat{Y}_0 \mid \mathbf{x}_0)\} = \frac{\sigma^2}{n} \sum_{\ell=1}^p \frac{\mathbf{v}_{\ell}^\top \boldsymbol{\Sigma}_0 \mathbf{v}_{\ell}}{m_{\ell}}.$$

If  $\mathbf{x}_0$  is drawn uniformly from the rows of  $\mathbf{X}$ , this equals  $\sigma^2 p/n$ , whatever the eigenvalues. For every  $\boldsymbol{\Sigma}_0$  it is at least  $(\sigma^2/n) \mathbf{v}_p^\top \boldsymbol{\Sigma}_0 \mathbf{v}_p / m_p$ .

*Proof.*

- (a) This is [Proposition 19.6.1\(a\)](#) with  $\mathbf{a} = \mathbf{x}_0$  and  $\lambda_{\ell} = nm_{\ell}$ . The second expression for  $m_{\ell}$  is  $\|\mathbf{X}\mathbf{v}_{\ell}\|^2/n$ .
- (b) By [Theorem 6.3.7](#),

$$\text{Var}(\hat{Y}_0) = \sigma^2 \mathbf{w}^\top \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{w} = \sigma^2 \mathbf{w}^\top \mathbf{M}\mathbf{w} = \sigma^2 \|\mathbf{M}\mathbf{w}\|^2,$$

and  $\|\mathbf{M}\mathbf{w}\| \leq \|\mathbf{w}\|$  ([Proposition 6.3.4](#)). If  $\mathbf{x}_0 = \sum_i w_i \mathbf{x}_{(i)}$  with  $w_i \geq 0$  and  $\sum_i w_i =$

1, then  $\mathbf{x}_0 = \mathbf{X}^\top \mathbf{w}$ , and by the triangle inequality and [Proposition 6.8.1\(a\)](#),

$$\|\mathbf{M}\mathbf{w}\| \leq \sum_i w_i \|\mathbf{M}\mathbf{e}_i\| = \sum_i w_i \sqrt{h_{ii}} \leq \max_i \sqrt{h_{ii}},$$

and  $h_{ii} \leq 1$ .

- (c) Conditionally on  $\mathbf{x}_0$ , (a) holds. Taking expectations,  $E(\mathbf{x}_0^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{x}_0) = \text{tr}((\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\Sigma}_0) = \sum_\ell \mathbf{v}_\ell^\top \boldsymbol{\Sigma}_0 \mathbf{v}_\ell / \lambda_\ell$ . If  $\mathbf{x}_0$  is uniform on the rows,  $\boldsymbol{\Sigma}_0 = \mathbf{X}^\top \mathbf{X} / n$ , and the trace is  $\text{tr}(\mathbf{I}_p) / n = p / n$ . The lower bound keeps the term  $\ell = p$ , since all terms are non-negative. ■

Part (b) answers the first question of [Exercise \(C1\)](#): inside the convex hull of the data, a prediction is never worse than the worst fitted value, however collinear the regressors. Part (a) says what “the pattern of the data” means: a point outside the hull is still predicted well if its coordinates along the weak directions are small compared with  $\sqrt{m_\ell}$ , and a point inside the range of every regressor separately can be far off the pattern ([Example \(Hidden extrapolation\)](#)). By part (c), if future regressors are generated the way the design was, the average prediction variance is  $\sigma^2 p / n$ ; if a relation between the regressors breaks down, the term  $\mathbf{v}_p^\top \boldsymbol{\Sigma}_0 \mathbf{v}_p / m_p$  can be enormous.

*Warning* (Variance is not the whole risk). [Proposition 26.4.1](#) is about variance under a correct model. Off the pattern of the data the model itself has not been checked, so its bias there is unknown ([Section 12.3](#)). A small prediction variance is not a guarantee.

#### Example. (Predicting consumption along and across the pattern)

In the consumption regression of [Example \(Large factors, precise estimate\)](#),  $s = 58.1$  (billions of 2005 dollars). At 1985Q1, a design point with population 237.9 million, the estimated mean has standard deviation 8.5, below the design average  $s\sqrt{p/n} = 10.8$ . Midway between the first quarters of 1969 and 1999 the standard deviation is 7.8, and by [Proposition 26.4.1\(b\)](#) no point in the convex hull of the data can exceed  $s\sqrt{\max_i h_{ii}} = 23.3$ .

Now keep every regressor at its 1985 value except population, which is lowered by ten per cent, to 214.1 million. Population did take that value, in 1974, and income took its 1985 value, but never together. The standard deviation of the estimated mean becomes 36.8, 4.3 times as large. Decomposing the variance over the canonical directions of the scaled design, as in the proof of [Theorem 26.3.2](#), the direction with condition index 270.9, the income–population near-dependency of [Example \(A consumption function\)](#), contributes 93.9 per cent of the variance at the new point and 0.2 per cent at the 1985 quarter itself.

[Figure 26.4.1](#) shows 2000 parametric replicates: responses generated from the fitted model with normal errors of standard deviation  $s$ , each refitted. The income and population coefficients are strongly negatively correlated: their exact correlation is  $-0.842$  ( $-0.832$  in the replicates). The prediction at 1985Q1 barely moves while the one off the pattern spreads widely.

```
import numpy as np
import statsmodels.api as sm
```

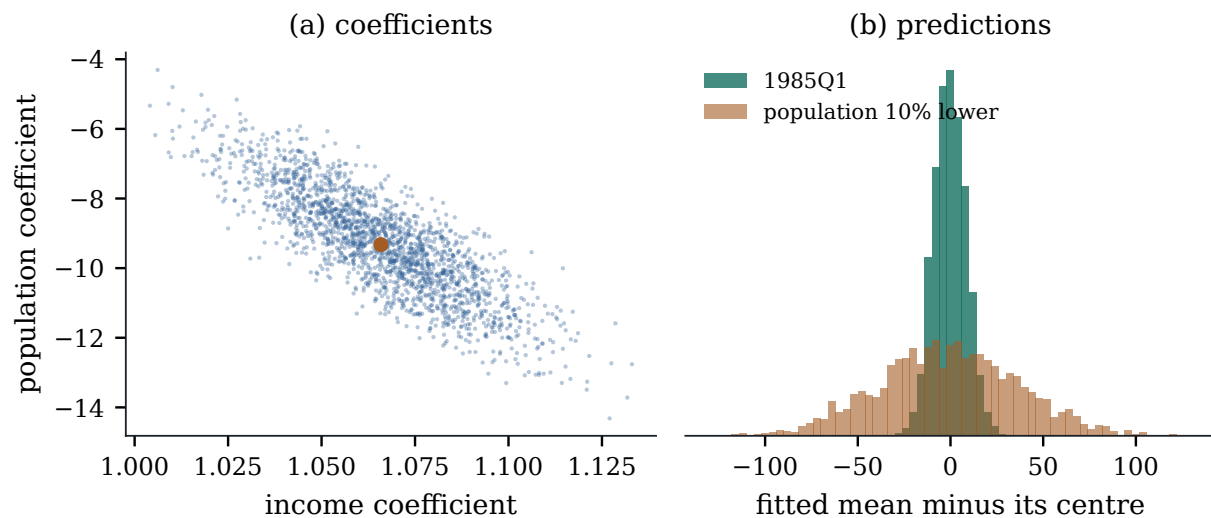


Figure 26.4.1: Parametric replicates for [Example \(Predicting consumption along and across the pattern\)](#). (a) Income and population coefficients. (b) Estimated mean, centred, at 1985Q1 and at the same point with population ten per cent lower.

```
macro = sm.datasets.macrodatab.load_pandas().data
names = ["realdpi", "pop", "cpi", "m1", "tbilrate", "unemp"]
X = sm.add_constant(macro[names].to_numpy())
y = macro["realcons"].to_numpy()
n, p = X.shape
fit = sm.OLS(y, X).fit()
s = np.sqrt(fit.scale)                                # estimate of sigma
G = np.linalg.inv(X.T @ X)

x_on = X[104]                                         # 1985Q1, a design point
x_mix = 0.5 * (X[40] + X[160])                       # midway between 1969Q1 and
1999Q1
x_off = x_on.copy()
x_off[2] *= 0.90                                     # 1985Q1 with population 10%
lower
for label, x0 in [("1985Q1", x_on), ("midway", x_mix), ("off pattern", x_off)]:
    print(f"[label:13s] fitted {x0 @ fit.params:8.1f}    sd {s * np.sqrt(x0 @ G @ x0)
:6.1f}")
```

```
d = np.linalg.norm(X, axis=0)
U, mu, Vt = np.linalg.svd(X / d, full_matrices=False) # scaled design, as in Section
26.3

def shares(x0):
    """Contributions of the canonical directions to Var(x0' beta_hat), as fractions.
    """
    c = (Vt @ (x0 / d)) ** 2 / mu**2                # (x0_scaled' v_k)^2 / mu_k^2
    return c / c.sum()

np.set_printoptions(precision=3, suppress=True)
print("condition indices:", mu[0] / mu)
print("shares, 1985Q1      :", shares(x_on))
print("shares, off pattern :", shares(x_off))
```

```
rng = np.random.default_rng(2604)
```

```

reps = 2000
Ystar = X @ fit.params + s * rng.normal(size=(reps, n))      # parametric replicates
                        , same design
Bstar = np.linalg.solve(X.T @ X, X.T @ Ystar.T).T
print("sd of income, population coefficients:", Bstar[:, 1:3].std(axis=0))
print("correlation of the two:", np.corrcoef(Bstar[:, 1], Bstar[:, 2])[0, 1])
print("sd of predictions, 1985Q1 / off:", (Bstar @ x_on).std(), (Bstar @ x_off).std()
      ())

```

## 26.4.2 Coefficients against predictions

A coefficient  $\beta_j$  is the difference between the mean responses at two points that differ by one unit in  $x_j$  and agree in every other regressor; with collinear data at least one of them lies off the pattern. A regression built for forecasting, where the relations among the regressors are expected to persist, can ignore collinearity; one built to attribute effects to individual regressors cannot.

## 26.4.3 Exercises

### 26.4.3.1 A. Check your understanding

#### Exercise A1.

For the design of [Example \(Two regressors that move together\)](#), rank the following new points by the variance of the fitted mean, without computing: the centroid;  $(x_1, x_2) = (3, 3)$ ;  $(x_1, x_2) = (0.5, -0.5)$ ; the average of the two design points with the largest  $x_1$ . Which bound of [Proposition 26.4.1](#) applies to each? Compare [Figure 19.6.1](#).

### 26.4.3.2 B. Practice

#### Exercise B1.

Show that the bound  $\text{Var}(\hat{Y}_0) \leq \sigma^2 \|\mathbf{w}\|^2$  of [Proposition 26.4.1\(b\)](#) is attained iff  $\mathbf{w} \in \mathcal{C}(\mathbf{X})$ , and that  $\min\{\|\mathbf{w}\|^2 : \mathbf{X}^\top \mathbf{w} = \mathbf{x}_0\}$  equals  $\text{Var}(\hat{Y}_0)/\sigma^2$ . Compare with [Proposition 12.3.2\(a\)](#).

*Solution.*  $\|\mathbf{M}\mathbf{w}\|^2 = \|\mathbf{w}\|^2 - \|(\mathbf{I} - \mathbf{M})\mathbf{w}\|^2$ , so equality holds iff  $(\mathbf{I} - \mathbf{M})\mathbf{w} = \mathbf{0}$ , that is,  $\mathbf{w} \in \mathcal{C}(\mathbf{X})$ . Every  $\mathbf{w}$  with  $\mathbf{X}^\top \mathbf{w} = \mathbf{x}_0$  has the same  $\mathbf{M}\mathbf{w}$ , because two such vectors differ by an element of  $\mathcal{N}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})^\perp$ . The minimum norm is therefore attained at  $\mathbf{w} = \mathbf{M}\mathbf{w}$  and equals  $\|\mathbf{M}\mathbf{w}\|^2 = \text{Var}(\hat{Y}_0)/\sigma^2$ . This is [Proposition 12.3.2\(a\)](#) with  $h_0 = \text{Var}(\hat{Y}_0)/\sigma^2$ : the leverage of a new point is the smallest squared norm of a set of weights that builds  $\mathbf{x}_0$  from the rows of the design.

#### Exercise B2.

Let the model have an intercept and centred regressors, so that the rows of  $\mathbf{X}$  are  $(1, \mathbf{z}_{(i)}^\top)$  with  $\sum_i \mathbf{z}_{(i)} = \mathbf{0}$ . For a design row  $i$  and  $t > 1$ , let  $\mathbf{x}_0 = (1, t\mathbf{z}_{(i)}^\top)^\top$ , an extrapolation

from the centroid along the ray through a design point. Show that  $\text{Var}(\hat{Y}_0) = \sigma^2(1/n + t^2(h_{ii} - 1/n))$ . Why does collinearity not enter, although  $\mathbf{x}_0$  is outside the convex hull of the data when  $t$  is large?

*Solution.* With centred regressors  $\mathbf{X}^\top \mathbf{X} = \text{diag}(n, \mathbf{Z}^\top \mathbf{Z})$ , where  $\mathbf{Z}$  has rows  $\mathbf{z}_{(i)}^\top$ , so  $\text{Var}(\hat{Y}_0) = \sigma^2(1/n + t^2 \mathbf{z}_{(i)}^\top (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{z}_{(i)})$ , and  $h_{ii} = 1/n + \mathbf{z}_{(i)}^\top (\mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{z}_{(i)}$  by [Proposition 6.8.2](#). The slope part of  $\mathbf{x}_0$  is  $t$  times that of a row, so its coordinate along every eigenvector is  $t$  times the row's, and in each term of [Proposition 26.4.1\(a\)](#) it is penalized relative to the data exactly as much as the row is, times  $t^2$ . Collinearity could enter only through a component along a weak direction that is large compared with the spread of the data there, and the row itself has no such component.

### 26.4.3.3 C. Going deeper

#### Exercise C1.

In the design of [Example \(Two regressors that move together\)](#), suppose future regressor values satisfy  $x_2 - x_1 = \tau\zeta$  with  $\zeta$  independent of  $x_1$ ,  $E\zeta = 0$ ,  $\text{Var}\zeta = 1$ , and  $x_1$  distributed as in the sample. Using [Proposition 26.4.1\(c\)](#), show that the average prediction variance is approximately  $(\sigma^2/n)(2 + \tau^2/(2m_2))$ , where  $m_2 = \lambda_2/n$  is the mean square of the design along the weak slope direction. Check that this is  $\sigma^2 p/n$  when  $\tau^2$  equals the within-sample mean square of  $x_2 - x_1$ , and discuss what happens when  $\tau$  is several times larger. What would a sensible forecaster do?

## 26.5 What to do about collinearity

Refitting the same data in another way cannot supply information the design lacks, and every genuine remedy does one of four things: it gets more information of the right kind, changes the question to one the data can answer, adds information from outside the data, or accepts bias in exchange for variance. Before choosing among them, decide whether anything needs to be done at all.

### 26.5.1 Does it matter?

By [Theorem 26.1.1\(c\)](#) and [Proposition 26.4.1](#), collinearity is harmless for quantities with no component along the weak directions: fitted values, predictions that follow the pattern of the data, and combinations such as the sum in [Example \(Two regressors that move together\)](#). If those are the targets, report them and stop.

### 26.5.2 Collect better data

The only remedy that keeps the question is more information along the weak directions. The next proposition shows how much a single well-placed observation can supply.



**Proposition 26.5.1 — One new observation**

Let  $\mathbf{A} = \mathbf{X}^\top \mathbf{X}$  be positive definite with eigenvalues  $\lambda_1 \geq \dots \geq \lambda_p$  and orthonormal eigenvectors  $\mathbf{v}_1, \dots, \mathbf{v}_p$ ,  $p \geq 2$ , and let a case with regressor vector  $\mathbf{x}$  be added, so that  $\mathbf{A}$  becomes  $\mathbf{A} + \mathbf{x}\mathbf{x}^\top$ . Write  $\mathbf{G} = \mathbf{A}^{-1}$ .

- (a) For every  $\mathbf{a}$ , the variance of  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  falls by  $\sigma^2(\mathbf{a}^\top \mathbf{G}\mathbf{x})^2/(1 + \mathbf{x}^\top \mathbf{G}\mathbf{x})$ ; in particular it never increases.
- (b) If  $\|\mathbf{x}\| \leq c$ , then  $\lambda_{\min}(\mathbf{A} + \mathbf{x}\mathbf{x}^\top) \leq \min(\lambda_p + c^2, \lambda_{p-1})$ , with equality when  $\mathbf{x} = c\mathbf{v}_p$ .
- (c) Among all  $\mathbf{x}$  with  $\|\mathbf{x}\| = c$ , the choice  $\mathbf{x} = c\mathbf{v}_p$  minimizes the largest variance of  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  over unit vectors  $\mathbf{a}$ .

*Proof.*

- (a) By Eq. 1.6.7,  $(\mathbf{A} + \mathbf{x}\mathbf{x}^\top)^{-1} = \mathbf{G} - \mathbf{G}\mathbf{x}\mathbf{x}^\top \mathbf{G}/(1 + \mathbf{x}^\top \mathbf{G}\mathbf{x})$ .
- (b) By Theorem 1.7.7,  $\lambda_{\min}(\mathbf{A} + \mathbf{x}\mathbf{x}^\top) \leq \mathbf{v}_p^\top (\mathbf{A} + \mathbf{x}\mathbf{x}^\top) \mathbf{v}_p = \lambda_p + (\mathbf{x}^\top \mathbf{v}_p)^2 \leq \lambda_p + c^2$ . For the second bound, the plane  $\text{span}\{\mathbf{v}_{p-1}, \mathbf{v}_p\}$  contains a unit vector  $\mathbf{b}$  orthogonal to  $\mathbf{x}$ , since the linear map  $\mathbf{b} \mapsto \mathbf{x}^\top \mathbf{b}$  from the plane to  $\mathbb{R}$  has a nontrivial null space. Then  $\lambda_{\min}(\mathbf{A} + \mathbf{x}\mathbf{x}^\top) \leq \mathbf{b}^\top \mathbf{A} \mathbf{b} \leq \lambda_{p-1}$ . If  $\mathbf{x} = c\mathbf{v}_p$ , then  $\mathbf{A} + \mathbf{x}\mathbf{x}^\top$  has the same eigenvectors as  $\mathbf{A}$  and eigenvalues  $\lambda_1, \dots, \lambda_{p-1}, \lambda_p + c^2$ , whose minimum is  $\min(\lambda_p + c^2, \lambda_{p-1})$ .
- (c) By Theorem 26.1.1(b), the largest variance over unit vectors is  $\sigma^2/\lambda_{\min}$ , and (b) says that  $\mathbf{x} = c\mathbf{v}_p$  maximizes  $\lambda_{\min}$ .

■

The new case should lie along the weakest direction, where the existing data are thin. Such a point breaks the pattern of the regressors, and in observational data it may not exist; in an experiment, or when the sample can be chosen, the proposition says what to look for.

**Example. (The value of one observation)**

In the design of Example (Two regressors that move together), add one case at the centred values  $(x_1, x_2) = (1.077, -1.044)$ , the point at distance 1.5 from the centroid along the weak slope direction. Each coordinate is well inside the range of its regressor; only the combination is new. The slope estimates are governed by the slope block with the intercept partialled out (Theorem 6.6.2), and the new case, which has an intercept entry, changes that block from  $\mathbf{S}$  to  $\mathbf{S} + \frac{n}{n+1} \mathbf{x}\mathbf{x}^\top$ , where  $\mathbf{x}$  holds its two slope entries: the columns must be centred again about the new mean, and that costs the factor  $n/(n+1)$ . Applied to this block, Proposition 26.5.1(b) predicts that the smaller eigenvalue rises from 0.172 to  $0.172 + 1.5^2 \cdot 30/31 = 2.349$ , which is what the script finds. The standard deviation of  $\hat{\beta}_1$  falls from  $1.734\sigma$  to  $0.476\sigma$ . To achieve the same reduction with more data of the old pattern, the whole sample would have to be repeated 13.3 times, about 370 further cases in place of one. The script also checks, on the centred block, that no other direction does better.

```
import numpy as np
rng = np.random.default_rng(2601)
n = 30
```

*# the design of Section 26.1*

```

x1 = rng.normal(size=n)
x2 = x1 + 0.15 * rng.normal(size=n)
X = np.column_stack([np.ones(n), x1 - x1.mean(), x2 - x2.mean()])
G = np.linalg.inv(X.T @ X)
lam, V = np.linalg.eigh(X.T @ X)           # ascending eigenvalues
v_weak = V[:, 0]                          # the weak direction (it has
    no intercept part)

```

```

c = 1.5
x_new = np.r_[1.0, c * v_weak[1:] / np.linalg.norm(v_weak[1:])] # breaks the
    pattern
print("new point (centred x1, x2):", np.round(x_new[1:], 3))
G_new = np.linalg.inv(X.T @ X + np.outer(x_new, x_new))
print("sd of b1 before / after:", round(np.sqrt(G[1, 1]), 3), round(np.sqrt(G_new[1,
    1]), 3))
S_old = X[:, 1:].T @ X[:, 1:]              # slope block: the intercept
    is orthogonal
xs = x_new[1:]
S_new = S_old + n / (n + 1) * np.outer(xs, xs) # centred slope block after
    recentring
print("smallest slope eigenvalue before / after:", round(np.linalg.eigvalsh(S_old)
    [0], 3),
    round(np.linalg.eigvalsh(S_new)[0], 3))
# repeating the old pattern k times multiplies X'X by k: Var(b1) falls by the factor
    1/k
k_equiv = G[1, 1] / G_new[1, 1]
print("equivalent number of copies of the whole data set:", round(k_equiv, 1))

```

### 26.5.3 Change the question

A reparameterization  $\mathbf{X}\boldsymbol{\beta} = (\mathbf{X}\mathbf{T})(\mathbf{T}^{-1}\boldsymbol{\beta})$  leaves the least squares estimate of every linear function of  $\boldsymbol{\beta}$  unchanged ([Theorem 6.7.1](#)), so it cannot make  $\beta_1 - \beta_2$  better determined. It can make the reported parameters the well-determined ones. Replacing  $x_1$  and  $x_2$  by their average and their difference gives a coefficient for the average that estimates  $\beta_1 + \beta_2$  precisely, and leaves all the imprecision in the coefficient of the difference. Centring polynomial terms ([Example \(A quadratic far from the origin\)](#)) is the same move.

Two common changes of model are really linear restrictions on  $\boldsymbol{\beta}$ , and should be judged as such. *Dropping a regressor* imposes  $\beta_2 = 0$ . By [Theorem 19.1.2](#) the short slope estimates  $\beta_1 + \pi\beta_2$ , where  $\pi$  is the slope of  $x_2$  on  $x_1$ ; in [Example \(Two regressors that move together\)](#)  $\pi = 1.0261$ , and the short slope has standard deviation  $0.171\sigma$  against  $1.734\sigma$ . It is precise because it answers a different question, giving  $x_1$  the credit for both regressors: with true slopes  $(1, 1)$  it centres at about 2.03. [Theorem 19.1.5](#) gives the condition under which it has the smaller mean squared error. *Combining regressors into an index* imposes a restriction such as  $\beta_1 = \beta_2$ . It concerns the weak direction  $\beta_1 - \beta_2$ , whose estimate has standard deviation  $3.412\sigma$  in the example, so an  $F$  test of it has almost no power: the restriction is cheap in variance and nearly untestable, for the same reason.

## 26.5.4 Add information from outside the data

An exact restriction known from theory is fitted by restricted least squares ([Proposition 11.3.3](#)). An approximate one can be written as extra observations  $\mathbf{r} = \mathbf{L}\boldsymbol{\beta} + \mathbf{u}$  and combined with the data by generalized least squares, the mixed estimation of Theil and Goldberger (1961) ([Exercise \(B1\)](#)). The Bayesian version is a prior: with the conjugate prior of [Theorem 7.6.2](#), under which  $\boldsymbol{\beta}$  given  $\sigma^2$  has covariance  $\sigma^2\mathbf{V}_0$ , the posterior mean solves equations with matrix  $\mathbf{V}_0^{-1} + \mathbf{X}^\top\mathbf{X}$ . A prior informative along the weak eigenvector raises the smallest eigenvalue as a new observation does, and the answer along that direction then comes mostly from the prior.

## 26.5.5 Accept bias

The last option gives up unbiasedness. Principal component regression drops the canonical coordinates with small eigenvalues, the truncated SVD estimator of [Proposition 10.4.3](#); ridge regression shrinks every canonical coordinate towards zero, the weak ones most. [Chapter 27](#) develops both ([Theorem 27.2.2](#) and [Definition 27.3.1](#)). Neither creates information: they trade variance along the weak directions for bias along the same directions.

## 26.5.6 What not to do

- Do not drop regressors because their variance inflation factors exceed a threshold; the fit may hardly change while the meaning of every remaining coefficient does.
- Do not expect centring or rescaling to cure collinearity that is not created by the parameterization.
- Do not let an automatic search choose among collinear regressors: noise decides which enters ([Chapter 29](#)).

## 26.5.7 Exercises

### 26.5.7.1 A. Check your understanding

#### Exercise A1.

Explain, using [Theorem 26.1.1](#), why no reparameterization of [Example \(Two regressors that move together\)](#) can estimate  $\beta_1 - \beta_2$  with a standard deviation smaller than  $3.412\sigma$ , although the coefficient of the new difference regressor may have any standard deviation you like.

#### Exercise A2.

In [Example \(Two regressors that move together\)](#), what does the short regression of  $y$  on  $x_1$  alone estimate if the true slopes are  $(2, 0)$ ? If they are  $(0, 2)$ ? What does this say about using the short regression to decide which regressor matters?

### 26.5.7.2 B. Practice

**Exercise B1.**

Suppose that, besides  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$ , there is independent information  $\mathbf{r} = \mathbf{L}\boldsymbol{\beta} + \mathbf{u}$  with  $E(\mathbf{u}) = \mathbf{0}$  and  $\text{Cov}(\mathbf{u}) = \sigma^2 \boldsymbol{\Omega}$ ,  $\boldsymbol{\Omega}$  known and positive definite. Show that the generalized least squares estimator from the stacked data is

$$\tilde{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X} + \mathbf{L}^\top \boldsymbol{\Omega}^{-1} \mathbf{L})^{-1} (\mathbf{X}^\top \mathbf{Y} + \mathbf{L}^\top \boldsymbol{\Omega}^{-1} \mathbf{r}),$$

that it is unbiased with covariance  $\sigma^2 (\mathbf{X}^\top \mathbf{X} + \mathbf{L}^\top \boldsymbol{\Omega}^{-1} \mathbf{L})^{-1}$ , and that it equals the posterior mean of [Theorem 7.6.2](#) for a suitable prior. With  $\mathbf{L} = \mathbf{v}_p^\top$ , how does the smallest eigenvalue change?

*Solution.* Stack  $(\mathbf{Y}^\top, \mathbf{r}^\top)^\top = (\mathbf{X}^\top, \mathbf{L}^\top)^\top \boldsymbol{\beta} + (\boldsymbol{\varepsilon}^\top, \mathbf{u}^\top)^\top$ , with error covariance  $\sigma^2 \text{diag}(\mathbf{I}, \boldsymbol{\Omega})$ . Generalized least squares ([Corollary 7.2.3](#)) gives the stated estimator, and it is unbiased with covariance  $\sigma^2$  times the inverse of the weighted cross-product matrix. If  $\mathbf{L}$  is square and nonsingular, the prior  $\boldsymbol{\beta} \sim N(\mathbf{L}^{-1} \mathbf{r}, \sigma^2 \mathbf{L}^{-1} \boldsymbol{\Omega} (\mathbf{L}^{-1})^\top)$  has precision  $\mathbf{L}^\top \boldsymbol{\Omega}^{-1} \mathbf{L} / \sigma^2$  and mean  $\mathbf{L}^{-1} \mathbf{r}$ , and the posterior mean of [Theorem 7.6.2](#) is the same expression. With  $\mathbf{L} = \mathbf{v}_p^\top$  and  $\boldsymbol{\Omega} = \omega^2$ , the matrix becomes  $\mathbf{X}^\top \mathbf{X} + \mathbf{v}_p \mathbf{v}_p^\top / \omega^2$ , whose smallest eigenvalue is  $\min(\lambda_p + 1/\omega^2, \lambda_{p-1})$ : exactly the effect of a new observation of length  $1/\omega$  along  $\mathbf{v}_p$  ([Proposition 26.5.1\(b\)](#)).

**Exercise B2.**

Let  $x_1, x_2$  be centred with  $\|\mathbf{x}_1\|^2 = S_1$ ,  $\|\mathbf{x}_2\|^2 = S_2$  and  $\mathbf{x}_1^\top \mathbf{x}_2 = S_{12}$ , and fit the intercept and the single composite  $\mathbf{s} = \mathbf{x}_1 + \mathbf{x}_2$ . Show that its coefficient has expectation  $\beta_1 + (\beta_2 - \beta_1)(S_2 + S_{12}) / (S_1 + S_2 + 2S_{12})$ . When is this  $(\beta_1 + \beta_2)/2$ ?

*Solution.* The coefficient is  $\mathbf{s}^\top \mathbf{Y} / \mathbf{s}^\top \mathbf{s}$ , because  $\mathbf{s}$  is centred and so orthogonal to  $\mathbf{1}$ . Its expectation is  $\mathbf{s}^\top (\mathbf{x}_1 \beta_1 + \mathbf{x}_2 \beta_2) / \mathbf{s}^\top \mathbf{s}$ , with  $\mathbf{s}^\top \mathbf{x}_1 = S_1 + S_{12}$ ,  $\mathbf{s}^\top \mathbf{x}_2 = S_2 + S_{12}$  and  $\mathbf{s}^\top \mathbf{s} = S_1 + S_2 + 2S_{12}$ . Writing  $\beta_1(S_1 + S_{12}) = \beta_1(\mathbf{s}^\top \mathbf{s}) - \beta_1(S_2 + S_{12})$  gives the formula. It is  $(\beta_1 + \beta_2)/2$  for all  $\boldsymbol{\beta}$  iff  $S_2 + S_{12} = (S_1 + S_2 + 2S_{12})/2$ , that is, iff  $S_1 = S_2$ : the composite estimates the average slope when the two regressors have equal spread, whatever their correlation.

**Exercise B3.**

For  $p = 2$  and  $\mathbf{A} = \text{diag}(\lambda_1, \lambda_2)$  with  $\lambda_1 > \lambda_2$ , let  $\mathbf{x} = c(\cos \theta, \sin \theta)^\top$ . Compute  $\lambda_{\min}(\mathbf{A} + \mathbf{x}\mathbf{x}^\top)$  and show directly that it is maximized at  $\theta = \pi/2$ .

*Solution.* The trace is  $T = \lambda_1 + \lambda_2 + c^2$ , independent of  $\theta$ , and the determinant is  $D = (\lambda_1 + c^2 \cos^2 \theta)(\lambda_2 + c^2 \sin^2 \theta) - c^4 \cos^2 \theta \sin^2 \theta = \lambda_1 \lambda_2 + c^2(\lambda_1 \sin^2 \theta + \lambda_2 \cos^2 \theta)$ . The smaller eigenvalue is  $(T - \sqrt{T^2 - 4D})/2$ , which increases with  $D$  for fixed  $T$ . Since  $\lambda_1 > \lambda_2$ ,  $D$  is largest when  $\sin^2 \theta = 1$ .

## 26.6 Summary and Notes

### 26.6.1 Summary

*Idea.*

1. Collinearity breaks no assumption. It inflates the variance of  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  when  $\mathbf{a}$  points along an eigenvector of  $\mathbf{X}^\top \mathbf{X}$  with a small eigenvalue; the worst-to-best ratio is  $\kappa(\mathbf{X})^2$ , and fitted values keep total variance  $\sigma^2 p$  (Theorem 26.1.1).
2. Near a rank-deficient design, the functions with bounded variance are those estimable in the limit (Proposition 26.1.2).
3. The variance inflation factor is the variance ratio to an orthogonal design of the same spread: scale-free, but a ratio, not a verdict (Definition 26.2.1, Proposition 26.2.1). Large factors that vanish under centring belong to questions about the origin (Example (A quadratic far from the origin)).
4. For a block of columns, the generalized factor  $\prod_i 1/(1 - \rho_i^2)$  measures the inflation of the joint confidence ellipsoid and does not depend on coding (Definition 26.2.2, Proposition 26.2.2).
5. Condition indices of the unit-length scaled design and the variance-decomposition proportions locate near-dependencies (Definition 26.3.1, Theorem 26.3.2). Centring can only hide them, namely those involving the intercept (Proposition 26.3.1), and only high-leverage cases can mask them (Proposition 26.3.3).
6. Predictions inside the convex hull, or along the pattern of the regressors, are unharmed (Proposition 26.4.1).
7. Remedies: data along the weak direction (Proposition 26.5.1), a question the data can answer, outside information, or biased estimation (Chapter 27).

### 26.6.2 Notes and sources

**The problem.** Frisch introduced the word “multicollinearity” in the 1930s. The statistical treatment rests on the observation that the eigenvectors of  $\mathbf{X}^\top \mathbf{X}$  sort linear functions of  $\boldsymbol{\beta}$  into well and badly estimated ones; Silvey (1969) set this out, including the advice to take new observations along the weak eigenvectors behind Proposition 26.5.1. Goldberger (1991) insisted that collinearity is a shortage of data, not a disease of the model. Stewart (1987) gave a numerical analyst’s account.

**Diagnostics.** The name “variance inflation factor” is usually traced to Marquardt (1970). The generalized factor and its  $1/(2q)$  power are due to Fox and Monette (1992); the canonical-correlation form of Proposition 26.2.2(b) is implicit there. O’Brien (2007) examined the rules of thumb. Condition indices and variance-decomposition proportions, with the working rule of indices above about 30 and two or more proportions above one half, are due to Belsley, Kuh and Welsch (1980, chapter 3); Belsley (1991) is a book-length development, and Belsley (1984), with its discussion, argues for scaling without centring. The bounds Theorem 26.3.2(c) and Proposition 26.3.3 are stated in the form this book needs and follow from standard eigenvalue inequalities. Among the textbooks used for coverage, Seber and Lee (2003, sections 9.7 and 10.7) treat variance inflation factors, eigenvalues, centring and influential cases; Christensen (2020, sections 13.1–13.2) develops collinearity through ill-defined directions and tolerances; and Sen

and Srivastava (1990, chapter 10) give detailed variance-decomposition examples.

**Data and remedies.** The US macroeconomic series and Grunfeld's panel are public-domain data distributed with statsmodels; Longley (1967) built his data to test regression programs. Mixed estimation is due to Theil and Goldberger (1961). Shrinkage is the subject of [Chapter 27](#), and selection among collinear regressors of [Chapter 29](#).

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# Chapter 27

## Shrinkage: Ridge, Principal Components and the Lasso

Least squares is the best linear unbiased estimator, and under normal errors the best unbiased estimator of any kind. With nearly dependent columns it can still be very bad, because its variance along poorly determined directions swamps everything else. This chapter gives up unbiasedness on purpose, measures estimators by mean squared error, and studies three ways of buying a large cut in variance with a little bias. Ridge regression shrinks every direction of the coefficient space, the weakly determined ones most; principal component regression discards the weakly determined directions; the lasso shrinks coefficients with an absolute-value penalty and sets some exactly to zero. The chapter ends with Stein's paradox: in three or more dimensions a shrinkage estimator beats least squares for every value of the parameter, for an empirical Bayes reason that also explains the other methods.

**What you need.** [Chapter 1](#) (the singular value decomposition, [Theorem 1.8.1](#)), [Chapter 6](#) (projections and the variance inflation identity [Eq. 6.6.2](#)), [Chapter 7](#) (the Gauss–Markov theorem, [Theorem 7.2.1](#), and the conjugate normal model, [Theorem 7.6.2](#) and [Proposition 7.6.3](#)), [Chapter 10](#) (least squares through the SVD, [Theorem 10.4.1](#) and [Proposition 10.4.3](#)), [Chapter 19](#) (the short regression as a biased estimator, [Theorem 19.1.5](#)), [Chapter 20](#) (the leave-one-out formulas, [Theorem 20.3.1](#)) and [Chapter 26](#) (what collinearity does to variances, [Theorem 26.1.1](#)). The James–Stein section uses the noncentral chi-squared distribution of [Chapter 4](#) ([Theorem 4.1.2](#)).

### 27.0.1 Roadmap

- [The bias–variance trade-off](#)
- [Ridge regression](#)
- [Principal component regression](#)
- [The lasso](#)
- [Stein's paradox and James–Stein estimation](#)
- [Summary and notes](#)



## 27.1 The bias–variance trade-off

Chapter 26 showed where collinearity does its damage (Theorem 26.1.1). The least squares estimate remains unbiased, but along a direction  $\mathbf{v}$  in which the regressors barely vary, the variance of  $\mathbf{v}^\top \hat{\boldsymbol{\beta}}$  is  $\sigma^2$  divided by a tiny eigenvalue; for a single coefficient the same damage appears as the factor  $1/(1 - R_j^2)$  of Eq. 6.6.2, the variance inflation factor of Definition 26.2.1. No assumption of Parts II and III is violated. The Gauss–Markov theorem (Theorem 7.2.1) concerns the class of *linear unbiased* estimators, within which least squares cannot be improved, so improving on it means leaving the class. This section sets up the criterion that makes biased and unbiased estimators comparable.

### 27.1.1 The mean squared error matrix

#### Definition 27.1.1 — Mean squared error

Let  $\tilde{\boldsymbol{\theta}}$  be an estimator of  $\boldsymbol{\theta} \in \mathbb{R}^q$  with finite second moments. Its **mean squared error matrix** is

$$\text{MSE}(\tilde{\boldsymbol{\theta}}) = \mathbb{E}[(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta})(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta})^\top],$$

its **bias** is  $\boldsymbol{\delta} = \mathbb{E}(\tilde{\boldsymbol{\theta}}) - \boldsymbol{\theta}$ , and its **total mean squared error** is  $\text{tr MSE}(\tilde{\boldsymbol{\theta}}) = \mathbb{E}\|\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2$ . For estimators  $\tilde{\boldsymbol{\theta}}_1, \tilde{\boldsymbol{\theta}}_2$  we say that  $\tilde{\boldsymbol{\theta}}_1$  is **at least as good in the matrix sense** if  $\text{MSE}(\tilde{\boldsymbol{\theta}}_2) - \text{MSE}(\tilde{\boldsymbol{\theta}}_1)$  is nonnegative definite.

All three quantities depend on the unknown parameter, so one estimator may be better for some parameter values and worse for others. For unbiased estimators the mean squared error matrix is the covariance matrix, and the matrix sense is the ordering in which Gauss–Markov declares least squares the winner (Theorem 7.2.1(d)).

#### Theorem 27.1.1 — Bias–variance decomposition for linear estimators

- (a) For any estimator with finite second moments,  $\text{MSE}(\tilde{\boldsymbol{\theta}}) = \text{Cov}(\tilde{\boldsymbol{\theta}}) + \boldsymbol{\delta}\boldsymbol{\delta}^\top$ . For every  $\mathbf{a} \in \mathbb{R}^q$ ,  $\mathbb{E}(\mathbf{a}^\top \tilde{\boldsymbol{\theta}} - \mathbf{a}^\top \boldsymbol{\theta})^2 = \mathbf{a}^\top \text{MSE}(\tilde{\boldsymbol{\theta}}) \mathbf{a}$ , and for every nonnegative definite  $\mathbf{W}$ ,  $\mathbb{E}[(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta})^\top \mathbf{W}(\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta})] = \text{tr}(\mathbf{W} \text{MSE}(\tilde{\boldsymbol{\theta}}))$ .
- (b) Suppose  $\mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , and let  $\tilde{\boldsymbol{\beta}} = \mathbf{A}\mathbf{Y}$  be a linear estimator of  $\boldsymbol{\beta}$ . Then  $\boldsymbol{\delta} = (\mathbf{A}\mathbf{X} - \mathbf{I})\boldsymbol{\beta}$  and  $\text{MSE}(\tilde{\boldsymbol{\beta}}) = \sigma^2 \mathbf{A}\mathbf{A}^\top + (\mathbf{A}\mathbf{X} - \mathbf{I})\boldsymbol{\beta}\boldsymbol{\beta}^\top (\mathbf{A}\mathbf{X} - \mathbf{I})^\top$ .
- (c) In the setting of (b), let  $\mathbf{X}$  have full column rank, let  $\hat{\boldsymbol{\beta}}$  be least squares, and put  $\boldsymbol{\Delta}_{\mathbf{A}} = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1} - \sigma^2 \mathbf{A}\mathbf{A}^\top$ , the covariance that  $\tilde{\boldsymbol{\beta}}$  saves. The following are equivalent:
  - (i)  $\tilde{\boldsymbol{\beta}}$  is at least as good as  $\hat{\boldsymbol{\beta}}$  in the matrix sense;
  - (ii)  $\mathbb{E}(\mathbf{a}^\top \tilde{\boldsymbol{\beta}} - \mathbf{a}^\top \boldsymbol{\beta})^2 \leq \text{Var}(\mathbf{a}^\top \hat{\boldsymbol{\beta}})$  for every  $\mathbf{a}$ ;
  - (iii)  $\mathbb{E}[(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{W}(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})] \leq \mathbb{E}[(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{W}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})]$  for every nonnegative definite  $\mathbf{W}$ ;
  - (iv)  $\boldsymbol{\Delta}_{\mathbf{A}}$  is nonnegative definite,  $\boldsymbol{\delta} \in \mathcal{C}(\boldsymbol{\Delta}_{\mathbf{A}})$  and  $\boldsymbol{\delta}^\top \boldsymbol{\Delta}_{\mathbf{A}}^+ \boldsymbol{\delta} \leq 1$ .
 When  $\boldsymbol{\Delta}_{\mathbf{A}}$  is positive definite, (iv) reads  $\boldsymbol{\delta}^\top \boldsymbol{\Delta}_{\mathbf{A}}^{-1} \boldsymbol{\delta} \leq 1$ .



*Proof.*

- (a) Write  $\tilde{\boldsymbol{\theta}} - \boldsymbol{\theta} = (\tilde{\boldsymbol{\theta}} - \mathbb{E}\tilde{\boldsymbol{\theta}}) + \boldsymbol{\delta}$ . The cross terms have expectation zero, which gives the decomposition. The scalar identities follow from  $\mathbb{E}(\mathbf{a}^\top \mathbf{w})^2 = \mathbf{a}^\top \mathbb{E}(\mathbf{w}\mathbf{w}^\top) \mathbf{a}$  and  $\mathbf{w}^\top \mathbf{W} \mathbf{w} = \text{tr}(\mathbf{W} \mathbf{w} \mathbf{w}^\top)$  with  $\mathbf{w} = \tilde{\boldsymbol{\theta}} - \boldsymbol{\theta}$ .
- (b)  $\mathbb{E}(\mathbf{A}\mathbf{Y}) = \mathbf{A}\mathbf{X}\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{A}\mathbf{Y}) = \sigma^2 \mathbf{A}\mathbf{A}^\top$  (Theorem 2.2.1); apply (a).
- (c) By (a),  $\text{MSE}(\hat{\boldsymbol{\beta}}) - \text{MSE}(\tilde{\boldsymbol{\beta}}) = \boldsymbol{\Delta}_\mathbf{A} - \boldsymbol{\delta}\boldsymbol{\delta}^\top$ , and (i) says this matrix is nonnegative definite. By (a), (ii) says  $\mathbf{a}^\top (\boldsymbol{\Delta}_\mathbf{A} - \boldsymbol{\delta}\boldsymbol{\delta}^\top) \mathbf{a} \geq 0$  for all  $\mathbf{a}$ , which is (i). Next, (i) implies (iii) because  $\text{tr}(\mathbf{W}\mathbf{N}) \geq 0$  when  $\mathbf{W}$  and  $\mathbf{N}$  are both nonnegative definite: with  $\mathbf{W} = \sum_k \omega_k \mathbf{q}_k \mathbf{q}_k^\top$  (Theorem 1.7.2),  $\text{tr}(\mathbf{W}\mathbf{N}) = \sum_k \omega_k \mathbf{q}_k^\top \mathbf{N} \mathbf{q}_k \geq 0$ . And (iii) with  $\mathbf{W} = \mathbf{a}\mathbf{a}^\top$  is (ii). For (i)  $\Leftrightarrow$  (iv), put  $\mathbf{N} = \boldsymbol{\Delta}_\mathbf{A} - \boldsymbol{\delta}\boldsymbol{\delta}^\top$ . If  $\mathbf{N}$  is nonnegative definite, so is  $\boldsymbol{\Delta}_\mathbf{A} = \mathbf{N} + \boldsymbol{\delta}\boldsymbol{\delta}^\top$ . If  $\boldsymbol{\delta} \notin \mathcal{C}(\boldsymbol{\Delta}_\mathbf{A})$ , there is  $\mathbf{w} \in \mathcal{N}(\boldsymbol{\Delta}_\mathbf{A})$  with  $\mathbf{w}^\top \boldsymbol{\delta} \neq 0$  (Proposition 1.5.1), and then  $\mathbf{w}^\top \mathbf{N} \mathbf{w} = -(\mathbf{w}^\top \boldsymbol{\delta})^2 < 0$ . So (i) forces  $\boldsymbol{\delta} = \boldsymbol{\Delta}_\mathbf{A} \mathbf{c}$  for some  $\mathbf{c}$ , and  $\mathbf{c}^\top \mathbf{N} \mathbf{c} = \mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c} (1 - \mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c}) \geq 0$  gives  $\boldsymbol{\delta}^\top \boldsymbol{\Delta}_\mathbf{A}^+ \boldsymbol{\delta} = \mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c} \leq 1$  (when  $\mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c} = 0$  there is nothing to prove). Here  $\boldsymbol{\delta}^\top \boldsymbol{\Delta}_\mathbf{A}^+ \boldsymbol{\delta} = \mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \boldsymbol{\Delta}_\mathbf{A}^+ \boldsymbol{\Delta}_\mathbf{A} \mathbf{c} = \mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c}$  by the first Penrose condition. Conversely, under (iv) the Cauchy–Schwarz inequality for the semi-inner product  $(\mathbf{w}, \mathbf{z}) \mapsto \mathbf{w}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{z}$  gives, for every  $\mathbf{w}$ ,

$$(\mathbf{w}^\top \boldsymbol{\delta})^2 = (\mathbf{w}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c})^2 \leq (\mathbf{w}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{w})(\mathbf{c}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{c}) \leq \mathbf{w}^\top \boldsymbol{\Delta}_\mathbf{A} \mathbf{w},$$

so  $\mathbf{w}^\top \mathbf{N} \mathbf{w} \geq 0$ . When  $\boldsymbol{\Delta}_\mathbf{A}$  is invertible,  $\boldsymbol{\Delta}_\mathbf{A}^+ = \boldsymbol{\Delta}_\mathbf{A}^{-1}$ ; this case is also Lemma 19.1.4. ■

Part (c) makes the trade-off precise: a biased linear estimator is better in every direction and under every quadratic loss exactly when its squared bias, measured in the metric of the covariance it saves, is at most one. Since  $\boldsymbol{\delta} = (\mathbf{A}\mathbf{X} - \mathbf{I})\boldsymbol{\beta}$ , condition (iv) holds when  $\boldsymbol{\beta}$  is close to what the estimator shrinks towards, and fails far from it. No linear estimator with  $\mathbf{A}\mathbf{X} \neq \mathbf{I}$  wins everywhere (Exercise (B1)). Theorem 19.1.5 was the first instance: the short regression is a biased linear estimator, and its winning region  $\gamma \leq 1$  is condition (iv) for that choice of  $\mathbf{A}$ .

## 27.1.2 What unbiasedness buys, and what it costs

Unbiasedness concerns an estimator's centre, not its accuracy, and it does not survive transformation:  $\|\hat{\boldsymbol{\beta}}\|^2$  overestimates  $\|\boldsymbol{\beta}\|^2$  by  $\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$  on average (Exercise (B3)), which under collinearity is enormous. Least squares estimates are systematically *too long*, and shrinking them is a correction, not a distortion. When many parameters are estimated at once, small biases spread over many coordinates can buy a large cut in total variance; Section 27.5 turns this into a theorem. The cost is in inference: the bias depends on the unknown  $\boldsymbol{\beta}$ , so an interval “estimate  $\pm$  multiple of standard error” centred on a shrinkage estimate has no guaranteed coverage. Chapter 28 and Chapter 29 return to inference after shrinkage and selection.

## 27.1.3 Shrinkage in canonical coordinates

The collinearity problem has a natural coordinate system. Let  $\mathbf{X}$  have full column rank and let  $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$  be its thin singular value decomposition (Theorem 1.8.1), with  $\mathbf{U} = [\mathbf{u}_1, \dots, \mathbf{u}_p]$  and  $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_p]$  having orthonormal columns and  $\mathbf{D} = \text{diag}(d_1, \dots, d_p)$ ,  $d_1 \geq \dots \geq d_p > 0$ . We write  $d_j$  instead of the  $\sigma_j$  of Chapter 1 because  $\sigma$  is the error standard deviation. Put

$$\boldsymbol{\alpha} = \mathbf{V}^\top \boldsymbol{\beta}, \quad \hat{\boldsymbol{\alpha}} = \mathbf{V}^\top \hat{\boldsymbol{\beta}} = \mathbf{D}^{-1} \mathbf{U}^\top \mathbf{Y}, \quad c_j = \mathbf{u}_j^\top \mathbf{Y} = d_j \hat{\alpha}_j. \quad (27.1.1)$$

The  $\alpha_j$  are the **canonical coefficients**: the coordinates of  $\boldsymbol{\beta}$  along the eigenvectors  $\mathbf{v}_j$  of  $\mathbf{X}^\top \mathbf{X}$ , whose eigenvalues are  $d_j^2$ . By [Theorem 10.4.1\(d\)](#), the  $\hat{\alpha}_j$  are uncorrelated with  $E(\hat{\alpha}_j) = \alpha_j$  and  $\text{Var}(\hat{\alpha}_j) = \sigma^2/d_j^2$ . A small  $d_j$  is a direction  $\mathbf{v}_j$  along which the regressors hardly move, and along which the data therefore say little ([Proposition 19.6.1](#)).

Many shrinkage estimators act on the canonical coordinates one at a time. For numbers  $f_1, \dots, f_p$ , called **shrinkage factors** or filter factors, define

$$\tilde{\boldsymbol{\beta}}_{\mathbf{f}} = \sum_{j=1}^p f_j \hat{\alpha}_j \mathbf{v}_j = \mathbf{V} \mathbf{F} \hat{\boldsymbol{\alpha}}, \quad \mathbf{F} = \text{diag}(f_1, \dots, f_p).$$

Least squares has every  $f_j = 1$ . Principal component regression ([Section 27.3](#)) has  $f_j \in \{0, 1\}$ , and ridge regression ([Section 27.2](#)) has  $f_j = d_j^2/(d_j^2 + \lambda)$ . The signal-to-noise ratio of direction  $j$  is

$$\tau_j^2 = \frac{\alpha_j^2}{\text{Var}(\hat{\alpha}_j)} = \frac{d_j^2 \alpha_j^2}{\sigma^2},$$

the squared mean of the standardized coordinate  $\hat{\alpha}_j/\sqrt{\text{Var}(\hat{\alpha}_j)}$ . Under normal errors it is the noncentrality of  $c_j^2/\sigma^2 \sim \chi^2(1, \tau_j^2)$ .

### Proposition 27.1.2 — Shrinking along the principal directions

Let  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ ,  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$  and  $\text{rank}(\mathbf{X}) = p$ . For fixed factors  $f_j$ :

- (a)  $E\|\tilde{\boldsymbol{\beta}}_{\mathbf{f}} - \boldsymbol{\beta}\|^2 = \sum_j \{f_j^2 \sigma^2/d_j^2 + (1-f_j)^2 \alpha_j^2\}$ ;
- (b)  $E\|\mathbf{X}\tilde{\boldsymbol{\beta}}_{\mathbf{f}} - \mathbf{X}\boldsymbol{\beta}\|^2 = \sum_j \{f_j^2 \sigma^2 + (1-f_j)^2 d_j^2 \alpha_j^2\}$ ;
- (c) both are minimized by the same factors  $f_j^* = \tau_j^2/(1+\tau_j^2)$ , and the minima are  $\sum_j (\sigma^2/d_j^2) \tau_j^2/(1+\tau_j^2)$  and  $\sigma^2 \sum_j \tau_j^2/(1+\tau_j^2)$ ;
- (d) replacing  $f_j = 1$  by some  $f_j \in [0, 1]$ , with the other factors fixed, lowers both quantities iff  $\tau_j^2 < (1+f_j)/(1-f_j)$ . In particular dropping direction  $j$  ( $f_j = 0$ ) helps iff  $\tau_j^2 < 1$ .

*Proof.*  $\tilde{\boldsymbol{\beta}}_{\mathbf{f}} - \boldsymbol{\beta} = \mathbf{V}(\mathbf{F}\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha})$  and  $\mathbf{V}$  is orthogonal, so  $\|\tilde{\boldsymbol{\beta}}_{\mathbf{f}} - \boldsymbol{\beta}\|^2 = \sum_j (f_j \hat{\alpha}_j - \alpha_j)^2$ . Each term has expectation  $\text{Var}(f_j \hat{\alpha}_j) + (f_j \alpha_j - \alpha_j)^2 = f_j^2 \sigma^2/d_j^2 + (1-f_j)^2 \alpha_j^2$ , which is (a). For (b),  $\mathbf{X}(\tilde{\boldsymbol{\beta}}_{\mathbf{f}} - \boldsymbol{\beta}) = \mathbf{U}\mathbf{D}(\mathbf{F}\hat{\boldsymbol{\alpha}} - \boldsymbol{\alpha})$  and  $\mathbf{U}$  has orthonormal columns, so each term of (a) is multiplied by  $d_j^2$ . (c) Each term of (a) is a convex quadratic in  $f_j$  with derivative  $2f_j \sigma^2/d_j^2 - 2(1-f_j)\alpha_j^2$ , which vanishes at  $f_j = \alpha_j^2/(\alpha_j^2 + \sigma^2/d_j^2) = \tau_j^2/(1+\tau_j^2)$ . Terms of (b) are those of (a) times  $d_j^2$ , so they have the same minimizer; substituting gives the minima. (d) The  $j$ th term of (a) falls below its value  $\sigma^2/d_j^2$  at  $f_j = 1$  iff  $(1-f_j)^2 \alpha_j^2 < (1-f_j^2) \sigma^2/d_j^2$ . Dividing by  $(1-f_j)\sigma^2/d_j^2 > 0$  gives  $(1-f_j)\tau_j^2 < 1+f_j$ . The same holds for (b). ■

The ideal shrinkage depends on  $\tau_j^2$ , which combines the spread of the regressors in direction  $j$  and the size of the coefficient there, so a direction with tiny  $d_j$  but large  $\alpha_j$  should be kept. The

ideal factors are the same for estimating  $\beta$  and  $\mathbf{X}\beta$ . The oracle factors depend on  $\beta$  and  $\sigma^2$ ; each method of this chapter chooses factors from the data instead, ridge by one tuning constant, principal components by a cut-off, and James–Stein by an unbiased estimate of the risk. Part (d) with  $f_j = 0$  is the criterion of [Proposition 10.4.3](#).

**Example. (Two nearly collinear regressors)**

Take  $n = 50$  observations of two centred regressors, scaled so that  $\mathbf{X}^\top \mathbf{X} = n\mathbf{R}$  where  $\mathbf{R}$  is a correlation matrix with off-diagonal entry 0.95. Let  $\sigma = 1$  and  $\beta = (1, 0.5)^\top$ . The eigenvalues of  $\mathbf{X}^\top \mathbf{X}$  are  $d_1^2 = n(1 + 0.95) = 97.5$  and  $d_2^2 = n(1 - 0.95) = 2.5$ , with eigenvectors  $(1, 1)^\top/\sqrt{2}$  and  $(1, -1)^\top/\sqrt{2}$ . The canonical coefficients have sizes  $|\alpha_1| = 1.0607$  and  $|\alpha_2| = 0.3536$ , and the signal-to-noise ratios are  $\tau_1^2 = 109.7$  and  $\tau_2^2 = 0.3125$ . The first direction, the sum of the two regressors, is estimated precisely. The second, their difference, is not.

| Estimator                  | Factors $(f_1, f_2)$ | Total MSE |
|----------------------------|----------------------|-----------|
| least squares              | (1, 1)               | 0.4103    |
| drop the second direction  | (1, 0)               | 0.1353    |
| halve the second direction | (1, 0.5)             | 0.1415    |
| oracle $f_j^*$             | (0.991, 0.238)       | 0.1054    |

Dropping the difference direction cuts the total mean squared error by two thirds, because  $\tau_2^2 < 1$ , and wins in the matrix sense too. Here  $\Delta_{\mathbf{A}} = \sigma^2 \mathbf{v}_2 \mathbf{v}_2^\top / d_2^2$  is singular, the bias is  $\delta = -\alpha_2 \mathbf{v}_2 \in \mathcal{C}(\Delta_{\mathbf{A}})$ , and  $\delta^\top \Delta_{\mathbf{A}}^+ \delta = \alpha_2^2 d_2^2 / \sigma^2 = \tau_2^2 = 0.3125 \leq 1$ . So by [Theorem 27.1.1\(c\)](#), every linear combination  $\mathbf{a}^\top \beta$  is estimated at least as accurately. Had the truth been  $\beta = (1, -0.5)^\top$ , with the same length but pointing along the difference, then  $\tau_2^2 = 2.5 \times 1.125 = 2.81$  and dropping would have increased the total error.

```
import numpy as np
n, rho, sigma = 50, 0.95, 1.0
XtX = n * np.array([[1.0, rho], [rho, 1.0]])
beta = np.array([1.0, 0.5])

evals, V = np.linalg.eigh(XtX)
order = np.argsort(evals)[::-1]
d2, V = evals[order], V[:, order]          # d_1^2 >= d_2^2, eigenvectors v_1, v_2
alpha = V.T @ beta                         # canonical coefficients
tau2 = d2 * alpha ** 2 / sigma ** 2         # squared signal-to-noise ratio of each
direction

def mse(f):
    """Total MSE of the estimator that multiplies alpha_hat_j by f_j."""
    f = np.asarray(f, dtype=float)
    return np.sum(f ** 2 * sigma ** 2 / d2 + (1 - f) ** 2 * alpha ** 2)

f_oracle = tau2 / (1 + tau2)
for label, f in [("least squares", [1, 1]), ("drop v_2", [1, 0]),
                ("halve v_2", [1, 0.5]), ("oracle", f_oracle)]:
    print(f"{label:14s} f = {np.round(f, 3)}    MSE = {mse(f):.4f}")
print("tau^2 =", np.round(tau2, 3))
```

## 27.1.4 Exercises

### 27.1.4.1 A. Check your understanding

#### Exercise A1.

Let  $\bar{Y}$  be the mean of  $n$  independent observations with mean  $\mu$  and variance  $\sigma^2$ . For a constant  $c$ , find the mean squared error of  $c\bar{Y}$  as an estimator of  $\mu$ . Show that it is minimized at  $c^* = \mu^2/(\mu^2 + \sigma^2/n)$  and that  $c\bar{Y}$  beats  $\bar{Y}$  iff  $(1 - c)\mu^2 < (1 + c)\sigma^2/n$ . Why is  $c^*$  not an estimator?

#### Exercise A2.

In [Example \(Two nearly collinear regressors\)](#), verify the least squares total mean squared error  $\sigma^2(1/d_1^2 + 1/d_2^2) = 0.4103$  by hand, and check the entry for “drop the second direction” using [Proposition 27.1.2\(a\)](#).

### 27.1.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{X}$  have full column rank and let  $\tilde{\boldsymbol{\beta}} = \mathbf{A}\mathbf{Y}$  with  $\mathbf{A}\mathbf{X} \neq \mathbf{I}$ . Show that there is a  $\boldsymbol{\beta}$  for which  $E\|\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 > E\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2$ . So no biased linear estimator dominates least squares.

*Solution.* By [Theorem 27.1.1\(b\)](#),  $E\|\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}\|^2 = \sigma^2 \text{tr}(\mathbf{A}\mathbf{A}^\top) + \|(\mathbf{A}\mathbf{X} - \mathbf{I})\boldsymbol{\beta}\|^2$ . The first term does not depend on  $\boldsymbol{\beta}$ . Since  $\mathbf{A}\mathbf{X} - \mathbf{I} \neq \mathbf{0}$  there is a unit vector  $\mathbf{v}$  with  $(\mathbf{A}\mathbf{X} - \mathbf{I})\mathbf{v} \neq \mathbf{0}$ . Along  $\boldsymbol{\beta} = t\mathbf{v}$  the second term is  $t^2\|(\mathbf{A}\mathbf{X} - \mathbf{I})\mathbf{v}\|^2$ , which exceeds the constant  $\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$  for large  $t$ .

#### Exercise B2.

A **linear smoother** estimates the mean vector  $\boldsymbol{\mu} = E(\mathbf{Y})$  by  $\mathbf{S}\mathbf{Y}$  for a fixed  $n \times n$  matrix  $\mathbf{S}$ . Show that  $E\|\mathbf{S}\mathbf{Y} - \boldsymbol{\mu}\|^2 = \sigma^2 \text{tr}(\mathbf{S}\mathbf{S}^\top) + \|(\mathbf{I} - \mathbf{S})\boldsymbol{\mu}\|^2$  when  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ . For the projection  $\mathbf{S} = \mathbf{M}$  and  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ , recover  $\sigma^2 r$  with  $r = \text{rank}(\mathbf{X})$ .

*Solution.*  $\mathbf{S}\mathbf{Y} - \boldsymbol{\mu} = \mathbf{S}(\mathbf{Y} - \boldsymbol{\mu}) - (\mathbf{I} - \mathbf{S})\boldsymbol{\mu}$ . The first term has mean zero and covariance  $\sigma^2\mathbf{S}\mathbf{S}^\top$ , so its expected squared length is  $\sigma^2 \text{tr}(\mathbf{S}\mathbf{S}^\top)$  ([Theorem 2.3.1](#)). The second term is constant, and the cross term has mean zero. For  $\mathbf{S} = \mathbf{M}$ ,  $\text{tr}(\mathbf{M}\mathbf{M}^\top) = \text{tr} \mathbf{M} = r$  and  $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \mathbf{0}$ .

#### Exercise B3.

Show that  $E\|\hat{\boldsymbol{\beta}}\|^2 = \|\boldsymbol{\beta}\|^2 + \sigma^2 \sum_j d_j^{-2}$ . For which  $c$  does  $c\hat{\boldsymbol{\beta}}$  have smaller total mean squared error than  $\hat{\boldsymbol{\beta}}$ ? Express the answer through  $\|\boldsymbol{\beta}\|^2$  and  $\sigma^2 \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$ .

### 27.1.4.3 C. Going deeper

**Exercise C1.**

Least squares is equivariant under rescaling: replacing  $\mathbf{X}$  by  $\mathbf{X}\mathbf{G}^{-1}$ , with  $\mathbf{G}$  diagonal and positive, turns the estimate into  $\mathbf{G}\hat{\boldsymbol{\beta}}$ . By rescaling one column in [Example \(Two nearly collinear regressors\)](#) by 10, show that  $\hat{\boldsymbol{\beta}}_{\mathbf{f}}$  with  $f_1 = 1$ ,  $f_2 = 0$  is not equivariant. Why are shrinkage estimators usually applied to standardized regressors?

## 27.2 Ridge regression

The oldest remedy for collinearity is also the simplest: when  $\mathbf{X}^\top \mathbf{X}$  has eigenvalues close to zero, add a positive constant to its diagonal before inverting. Hoerl and Kennard (1970) proposed this in statistics; numerical analysts knew it as Tikhonov regularization. The constant bounds the variance at the price of bias towards the origin. [Exercise \(B1\)](#) and [Exercise \(C1\)](#) met the estimator briefly.

### 27.2.1 Centring, scaling and the intercept

A penalty should not touch the intercept, or shifting the response by a constant would change the fitted slopes. Leaving the intercept unpenalized is equivalent to centring, for every penalty.

#### Proposition 27.2.1 — An unpenalized intercept

Let  $\mathbf{X}_0$  be an  $n \times p$  matrix of regressors with column means  $\bar{\mathbf{x}}_0$ , let  $\mathbf{X} = \mathbf{X}_0 - \mathbf{1}\bar{\mathbf{x}}_0^\top$  and  $\tilde{\mathbf{y}} = \mathbf{y} - \bar{y}\mathbf{1}$  be the centred regressors and response, and let  $P : \mathbb{R}^p \rightarrow [0, \infty]$  be any function. Then  $(\beta_0, \mathbf{b})$  minimizes

$$\|\mathbf{y} - \beta_0\mathbf{1} - \mathbf{X}_0\mathbf{b}\|^2 + P(\mathbf{b})$$

iff  $\mathbf{b}$  minimizes  $\|\tilde{\mathbf{y}} - \mathbf{X}\mathbf{b}\|^2 + P(\mathbf{b})$  and  $\beta_0 = \bar{y} - \bar{\mathbf{x}}_0^\top \mathbf{b}$ .

*Proof.* For fixed  $\mathbf{b}$ , the first term is the residual sum of squares of regressing  $\mathbf{y} - \mathbf{X}_0\mathbf{b}$  on  $\mathbf{1}$ , minimized uniquely by the mean  $\beta_0 = \bar{y} - \bar{\mathbf{x}}_0^\top \mathbf{b}$ , with minimum  $\|(\mathbf{I} - n^{-1}\mathbf{1}\mathbf{1}^\top)(\mathbf{y} - \mathbf{X}_0\mathbf{b})\|^2 = \|\tilde{\mathbf{y}} - \mathbf{X}\mathbf{b}\|^2$ . So the joint minimum is the minimum over  $\mathbf{b}$  of  $\|\tilde{\mathbf{y}} - \mathbf{X}\mathbf{b}\|^2 + P(\mathbf{b})$ , attained exactly at the pairs described. ■

From now on the regressors are centred and the response is centred, so  $\mathbf{X}$  has no intercept column and  $\mathbf{1}^\top \mathbf{X} = \mathbf{0}^\top$ . A penalty also treats all coefficients as commensurable, so the regressors should be on a common scale ([Exercise \(C1\)](#)). In this chapter's examples every column has unit standard deviation, so  $\mathbf{X}^\top \mathbf{X} = (n-1)\mathbf{R}$  with  $\mathbf{R}$  the correlation matrix, and a coefficient is the change in mean response per standard deviation of its regressor.

### 27.2.2 The estimator

**Definition 27.2.1 — Ridge regression**

For  $\lambda \geq 0$ , a **ridge estimate**  $\hat{\beta}_\lambda^R$  minimizes

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda\|\mathbf{b}\|^2. \quad (27.2.1)$$

The constant  $\lambda$  is the **ridge parameter** (or tuning parameter). The **ridge trace** is the curve  $\lambda \mapsto \hat{\beta}_\lambda^R$ .

For  $\lambda = 0$  this is least squares. For  $\lambda > 0$  it is uniquely defined whatever the rank of  $\mathbf{X}$ , and its behaviour is transparent in the coordinates of Eq. 27.1.1. Let  $\mathbf{X} = \sum_{j \leq r} d_j \mathbf{u}_j \mathbf{v}_j^\top$  be the SVD of  $\mathbf{X}$ , of rank  $r$ , and  $c_j = \mathbf{u}_j^\top \mathbf{y}$ .

**Theorem 27.2.2 — Ridge regression**

Let  $\lambda > 0$ .

- (a) (*Closed form.*) The criterion Eq. 27.2.1 has the unique minimizer

$$\hat{\beta}_\lambda^R = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y} = \sum_{j \leq r} \frac{d_j}{d_j^2 + \lambda} c_j \mathbf{v}_j.$$

If  $r = p$ , then  $\hat{\beta}_\lambda^R = \sum_j f_j(\lambda) \hat{\alpha}_j \mathbf{v}_j$  with **shrinkage factors**  $f_j(\lambda) = d_j^2 / (d_j^2 + \lambda) \in (0, 1)$ , decreasing in  $\lambda$  and smallest for the smallest  $d_j$ .

- (b) (*Fitted values.*)  $\mathbf{X} \hat{\beta}_\lambda^R = \mathbf{H}_\lambda \mathbf{y}$  with  $\mathbf{H}_\lambda = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top = \sum_{j \leq r} f_j(\lambda) \mathbf{u}_j \mathbf{u}_j^\top$ . The **effective degrees of freedom**  $\text{df}(\lambda) = \text{tr} \mathbf{H}_\lambda = \sum_{j \leq r} f_j(\lambda)$  decrease strictly from  $r$  (as  $\lambda \rightarrow 0$ ) to 0 (as  $\lambda \rightarrow \infty$ ).

- (c) (*Moments.*) If  $\mathbf{E}(\mathbf{Y}) = \mathbf{X}\beta$ ,  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$  and  $r = p$ , then

$$\mathbf{E} \hat{\beta}_\lambda^R - \beta = -\lambda (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \beta,$$

$$\text{Cov}(\hat{\beta}_\lambda^R) = \sigma^2 \sum_j \frac{d_j^2}{(d_j^2 + \lambda)^2} \mathbf{v}_j \mathbf{v}_j^\top,$$

and, with  $\alpha = \mathbf{V}^\top \beta$ , the total mean squared error is

$$\mathbf{E} \|\hat{\beta}_\lambda^R - \beta\|^2 = \sigma^2 \sum_j \frac{d_j^2}{(d_j^2 + \lambda)^2} + \lambda^2 \sum_j \frac{\alpha_j^2}{(d_j^2 + \lambda)^2}. \quad (27.2.2)$$

- (d) (*The path.*) Unless  $\mathbf{X}^\top \mathbf{y} = \mathbf{0}$ ,  $\|\hat{\beta}_\lambda^R\|$  decreases strictly in  $\lambda$ , with limit 0 as  $\lambda \rightarrow \infty$ ; and  $\hat{\beta}_\lambda^R \rightarrow \mathbf{X}^+ \mathbf{y}$ , the least squares estimate of smallest norm, as  $\lambda \rightarrow 0$ .
- (e) (*Bayesian reading.*) If  $\mathbf{Y} \mid \beta \sim N_n(\mathbf{X}\beta, \sigma^2 \mathbf{I})$  with  $\sigma^2$  known and  $\beta \sim N_p(\mathbf{0}, (\sigma^2/\lambda) \mathbf{I})$ , then  $\beta \mid \mathbf{Y} \sim N_p(\hat{\beta}_\lambda^R, \sigma^2 (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1})$ . With  $\sigma^2$  unknown and the normal-inverse-gamma prior of Theorem 7.6.2 with  $\mathbf{m}_0 = \mathbf{0}$  and  $\mathbf{V}_0 = \lambda^{-1} \mathbf{I}$ , the posterior mean of  $\beta$  is again  $\hat{\beta}_\lambda^R$ .

*Proof.*

- (a) Uniqueness and the first formula are [Exercise \(A2\)](#):  $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$  is positive definite, and [Proposition 1.11.2](#) applies. Complete  $\mathbf{v}_1, \dots, \mathbf{v}_r$  to an orthonormal basis  $\mathbf{v}_1, \dots, \mathbf{v}_p$  of  $\mathbb{R}^p$ . Then  $\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I} = \sum_{j \leq r} (d_j^2 + \lambda) \mathbf{v}_j \mathbf{v}_j^\top + \sum_{j > r} \lambda \mathbf{v}_j \mathbf{v}_j^\top$ , whose inverse has the same eigenvectors and reciprocal eigenvalues ([Theorem 1.7.2](#)). Since  $\mathbf{X}^\top \mathbf{y} = \sum_{j \leq r} d_j c_j \mathbf{v}_j$ , the second formula follows. When  $r = p$ ,  $d_j c_j / (d_j^2 + \lambda) = f_j(\lambda) c_j / d_j = f_j(\lambda) \hat{\alpha}_j$ .
- (b)  $\mathbf{X} \mathbf{v}_j = d_j \mathbf{u}_j$ , so  $\mathbf{X} \hat{\boldsymbol{\beta}}_\lambda^R = \sum_{j \leq r} f_j(\lambda) c_j \mathbf{u}_j = \sum_j f_j(\lambda) \mathbf{u}_j \mathbf{u}_j^\top \mathbf{y}$ . The trace of  $\mathbf{u}_j \mathbf{u}_j^\top$  is one. Each  $f_j$  decreases strictly from 1 to 0.
- (c) These are the formulas of [Exercise \(B1\)](#) in the present notation (its  $d_j$  and  $\boldsymbol{\gamma}$  are our  $d_j^2$  and  $\boldsymbol{\alpha}$ ); they are also [Proposition 27.1.2\(a\)](#) with the factors  $f_j(\lambda)$ .
- (d)  $\|\hat{\boldsymbol{\beta}}_\lambda^R\|^2 = \sum_{j \leq r} d_j^2 c_j^2 / (d_j^2 + \lambda)^2$ . Every term is nonincreasing in  $\lambda$  and tends to zero, and a term with  $c_j \neq 0$  decreases strictly. Some  $c_j \neq 0$  iff  $\mathbf{X}^\top \mathbf{y} \neq \mathbf{0}$ . As  $\lambda \rightarrow 0$ ,  $d_j / (d_j^2 + \lambda) \rightarrow 1/d_j$ , and  $\sum_{j \leq r} (c_j / d_j) \mathbf{v}_j = \mathbf{X}^+ \mathbf{y}$  by [Theorem 10.4.1\(b\)](#).
- (e) With  $\sigma^2$  known, the posterior density is proportional to  $\exp\{-(\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|^2) / (2\sigma^2)\}$ . Completing the square as in the proof of [Theorem 7.6.2](#), with  $\mathbf{P}_0 = \lambda \mathbf{I}$  and  $\mathbf{m}_0 = \mathbf{0}$ ,

$$\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda \|\boldsymbol{\beta}\|^2 = (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_\lambda^R)^\top (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}) (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_\lambda^R) + C,$$

with  $C$  free of  $\boldsymbol{\beta}$ . This is the stated normal density. The last statement is [Proposition 7.6.3\(d\)](#). ■

Part (a) says what ridge does. It keeps the directions  $\mathbf{v}_j$  and shrinks the coordinate along each by  $f_j(\lambda)$ : directions with  $d_j^2 \gg \lambda$  are hardly touched, and those with  $d_j^2 \ll \lambda$  are shrunk almost to zero, whatever the size of the coefficients. Comparing with the oracle factors  $f_j^* = d_j^2 / (d_j^2 + \sigma^2 / \alpha_j^2)$  of [Proposition 27.1.2\(c\)](#), ridge is the oracle when all canonical coefficients have the same size,  $\alpha_j^2 = \sigma^2 / \lambda$ . Part (e) says the same in Bayesian terms: the prior  $N(\mathbf{0}, (\sigma^2 / \lambda) \mathbf{I})$  treats every direction alike.

The effective degrees of freedom in (b) count each dimension by the fraction  $f_j(\lambda)$  in which it is used; [Chapter 29](#) uses them in model-selection criteria, and [Exercise \(C1\)](#) gives them a second meaning.

*Remark* (Penalized and constrained forms). For any penalty  $P \geq 0$ , a minimizer  $\hat{\mathbf{b}}$  of  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda P(\mathbf{b})$ , with  $\lambda > 0$ , also minimizes  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  subject to  $P(\mathbf{b}) \leq t$ , where  $t = P(\hat{\mathbf{b}})$ . Indeed, if  $P(\mathbf{b}) \leq t$ , then

$$\begin{aligned} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 &\geq \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda\{P(\mathbf{b}) - t\} \\ &\geq \|\mathbf{y} - \mathbf{X}\hat{\mathbf{b}}\|^2 + \lambda\{P(\hat{\mathbf{b}}) - t\} = \|\mathbf{y} - \mathbf{X}\hat{\mathbf{b}}\|^2. \end{aligned}$$

So ridge is least squares restricted to a ball,  $\|\mathbf{b}\|^2 \leq t$ , whose radius is fixed by  $\lambda$  ([Exercise \(B3\)](#)). The lasso of [Section 27.4](#) replaces the ball by a cross-polytope.

### 27.2.3 Ridge beats least squares

[Exercise \(B1\)](#) showed that the total mean squared error of ridge falls as  $\lambda$  leaves 0. Hoerl and Kennard made the range explicit, and Theobald (1974) added the matrix sense.

#### Theorem 27.2.3 — Hoerl–Kennard

Let  $\mathbf{X}$  have full column rank,  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ ,  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$  with  $\sigma^2 > 0$ , and let  $\alpha_{\max}^2 = \max_j \alpha_j^2$  with  $\boldsymbol{\alpha} = \mathbf{V}^\top \boldsymbol{\beta}$ . Let  $m(\lambda)$  be the total mean squared error [Eq. 27.2.2](#), so that  $m(0) = \sigma^2 \sum_j d_j^{-2}$  is that of least squares.

- (a)  $m(\lambda) < m(0)$  for every  $0 < \lambda < 2\sigma^2/\alpha_{\max}^2$ , and in particular for every  $0 < \lambda < 2\sigma^2/\|\boldsymbol{\beta}\|^2$ . If  $\boldsymbol{\beta} = \mathbf{0}$ , for every  $\lambda > 0$ .
- (b)  $m$  is strictly decreasing on  $[0, \sigma^2/\alpha_{\max}^2]$ .
- (c)  $\hat{\boldsymbol{\beta}}_\lambda^R$  is at least as good as  $\hat{\boldsymbol{\beta}}$  in the matrix sense iff

$$\frac{\lambda}{\sigma^2} \sum_j \frac{d_j^2 \alpha_j^2}{2d_j^2 + \lambda} \leq 1,$$

which holds whenever  $0 < \lambda \leq 2\sigma^2/\|\boldsymbol{\beta}\|^2$ .

*Proof.*

- (a) Compare term by term. The  $j$ th term of [Eq. 27.2.2](#) is below its value  $\sigma^2/d_j^2$  at  $\lambda = 0$  iff  $d_j^2(\sigma^2 d_j^2 + \lambda^2 \alpha_j^2) < \sigma^2(d_j^2 + \lambda)^2$ , that is, iff  $\lambda^2 d_j^2 \alpha_j^2 < 2\lambda \sigma^2 d_j^2 + \lambda^2 \sigma^2$ , or, dividing by  $\lambda > 0$ ,

$$\lambda(d_j^2 \alpha_j^2 - \sigma^2) < 2\sigma^2 d_j^2.$$

If  $d_j^2 \alpha_j^2 \leq \sigma^2$ , this holds for every  $\lambda > 0$ . Otherwise it holds for  $\lambda < 2\sigma^2 d_j^2/(d_j^2 \alpha_j^2 - \sigma^2)$ , and the right side exceeds  $2\sigma^2 d_j^2/(d_j^2 \alpha_j^2) = 2\sigma^2/\alpha_j^2 \geq 2\sigma^2/\alpha_{\max}^2$ . So for  $0 < \lambda < 2\sigma^2/\alpha_{\max}^2$  every term decreases, and so does the sum. Finally  $\alpha_{\max}^2 \leq \|\boldsymbol{\alpha}\|^2 = \|\boldsymbol{\beta}\|^2$ .

- (b) By [Exercise \(B1\)](#), in the present notation,  $m'(\lambda) = 2 \sum_j d_j^2 (\lambda \alpha_j^2 - \sigma^2)/(d_j^2 + \lambda)^3$ . For  $0 \leq \lambda < \sigma^2/\alpha_{\max}^2$  every numerator is negative, so  $m' < 0$  there, and  $m$  is strictly decreasing on the closed interval.
- (c) Write  $\mathbf{S} = \mathbf{X}^\top \mathbf{X}$  and  $\mathbf{A} = (\mathbf{S} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top$ . In the notation of [Theorem 27.1.1\(c\)](#), all matrices involved have the eigenvectors  $\mathbf{v}_j$ , with eigenvalues

$$\Delta_{\mathbf{A}} : \sigma^2 \left( \frac{1}{d_j^2} - \frac{d_j^2}{(d_j^2 + \lambda)^2} \right) = \sigma^2 \frac{\lambda(2d_j^2 + \lambda)}{d_j^2(d_j^2 + \lambda)^2} > 0,$$

and the bias is  $\boldsymbol{\delta} = -\sum_j \lambda \alpha_j (d_j^2 + \lambda)^{-1} \mathbf{v}_j$ . So  $\Delta_{\mathbf{A}}$  is positive definite and  $\boldsymbol{\delta}^\top \Delta_{\mathbf{A}}^{-1} \boldsymbol{\delta} = \sum_j \lambda^2 \alpha_j^2 (d_j^2 + \lambda)^{-2} \cdot d_j^2 (d_j^2 + \lambda)^2 / \{\sigma^2 \lambda (2d_j^2 + \lambda)\}$ , which is the stated sum. Each factor  $d_j^2/(2d_j^2 + \lambda)$  is below  $1/2$ , so the sum is at most  $\lambda \|\boldsymbol{\alpha}\|^2/(2\sigma^2) = \lambda \|\boldsymbol{\beta}\|^2/(2\sigma^2)$ . Apply [Theorem 27.1.1\(c\)](#). ■

The theorem is an existence result. The good range of  $\lambda$  depends on the unknown  $\boldsymbol{\beta}$  and  $\sigma^2$ , and



it says nothing about a  $\lambda$  chosen from the data. Replacing  $\boldsymbol{\beta}$  by  $\hat{\boldsymbol{\beta}}$  in the bound is misleading, because  $\|\hat{\boldsymbol{\beta}}\|^2$  overestimates  $\|\boldsymbol{\beta}\|^2$  by  $\sigma^2 \sum_j d_j^{-2}$  on average ([Exercise \(B3\)](#)), which is large exactly when ridge is needed. What the theorem does establish is that least squares is never the best member of the ridge family.

## 27.2.4 Choosing the ridge parameter

Hoerl and Kennard suggested choosing the smallest  $\lambda$  at which the ridge trace “stabilizes”, a subjective judgement. Hoerl, Kennard and Baldwin (1975) proposed the plug-in  $\lambda = p\hat{\sigma}^2/\|\hat{\boldsymbol{\beta}}\|^2$ , the optimum of [Exercise \(A2\)](#) with estimates substituted; since  $\|\hat{\boldsymbol{\beta}}\|^2$  is too large on average, it tends to shrink too little under collinearity. A more defensible choice minimizes an estimate of prediction error. Because ridge is a linear smoother,  $\hat{\mathbf{y}} = \mathbf{H}_\lambda \mathbf{y}$ , the leave-one-out prediction errors need only one fit: the residual of case  $i$  divided by  $1 - h_{ii}(\lambda)$ , with  $h_{ii}(\lambda)$  the  $i$ th diagonal entry of  $\mathbf{H}_\lambda$  ([Exercise \(B1\)](#); add  $1/n$  for an unpenalized intercept), the ridge version of [Theorem 20.3.1](#). **Generalized cross-validation** replaces each  $h_{ii}(\lambda)$  by the average  $\text{tr } \mathbf{H}_\lambda/n$ :

$$\text{GCV}(\lambda) = \frac{n^{-1} \|\mathbf{y} - \mathbf{H}_\lambda \mathbf{y}\|^2}{(1 - \text{tr } \mathbf{H}_\lambda/n)^2}$$

(with an unpenalized intercept,  $\text{tr } \mathbf{H}_\lambda$  becomes  $1 + \text{df}(\lambda)$ ). [Chapter 29](#) develops both criteria for general linear smoothers ([Theorem 29.3.2](#) and [Definition 29.3.2](#)). A Bayesian alternative estimates the prior variance  $\sigma^2/\lambda$  from the marginal likelihood, an empirical Bayes idea that returns in [Section 27.5](#).

### Example. (Ridge regression for Longley’s data)

Longley’s regression ([Example \(Longley’s regression\)](#)) relates total employment, in thousands, to six economic series over  $n = 16$  years. With the regressors standardized, the eigenvalues  $d_j^2$  of  $\mathbf{X}^\top \mathbf{X}$  run from 69.05 down to 0.0057. The least squares coefficients, in thousands of persons per standard deviation, are 163 for the GNP deflator,  $-3560$  for GNP,  $-1888$  for unemployment,  $-719$  for the armed forces,  $-355$  for population and 8709 for the year. The GNP and YEAR coefficients are large and of opposite sign: the two series rose together, and the data can hardly separate them.

[Figure 27.2.1\(a\)](#) shows the ridge trace against the effective degrees of freedom. As  $\lambda$  grows, the GNP and YEAR coefficients approach each other, and the GNP coefficient changes sign at  $\lambda = 0.0110$ , where  $\text{df} = 5.1$ . Leave-one-out cross-validation chooses  $\lambda = 0.0026$ , with  $\text{df} = 5.61$ , and GCV chooses  $\lambda = 0.0030$ , with  $\text{df} = 5.56$ . Both prefer little shrinkage. At the leave-one-out choice the coefficients are  $-6$ ,  $-1809$ ,  $-1645$ ,  $-669$ ,  $-825$  and  $7415$ . The largest changes are in the coefficients of GNP, YEAR and population, the three series that rose together, and the deflator coefficient is essentially zero. The minimized root mean squared leave-one-out error is 402, against 425 for least squares, but this minimum is optimistic, since the same errors chose  $\lambda$ . Choosing  $\lambda$  afresh within each leave-one-out fold (*nested* cross-validation) gives an honest 441: on these sixteen years tuned ridge does not predict better than least squares.

Panel (b) asks what the theorem guarantees in a world where the truth is the least squares fit,  $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}$ , with  $\sigma = s = 304.9$ . There, least squares has total mean squared error 19.33 million, and the best ridge estimate, at  $\lambda = 0.0019$  with  $\text{df} = 5.69$ , has 15.01 million, a ratio of 0.777. [Theorem 27.2.3\(a\)](#) guarantees an improvement only for  $\lambda < 0.00383$ , and part (c) gives dominance in the matrix sense for  $\lambda \leq 0.00200$ ; in fact the improvement continues up to  $\lambda = 0.0058$ . The ranges are narrow because this

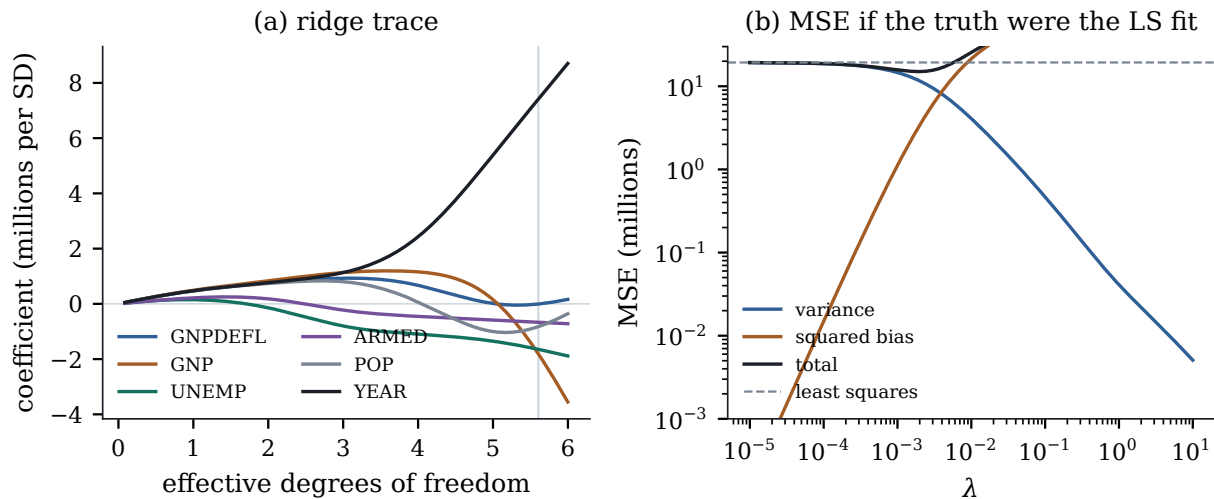


Figure 27.2.1: Ridge regression for Longley's data. (a) The ridge trace: standardized coefficients against the effective degrees of freedom  $df(\lambda)$ ; the vertical line marks the leave-one-out choice. (b) Variance, squared bias and total mean squared error of ridge, computed exactly from Eq. 27.2.2 when the true coefficients are the least squares estimates and  $\sigma = s$ ; the dashed line is least squares.

“truth” inherits the least squares estimate's large components along the weak directions.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
names = ["GNPDEFL", "GNP", "UNEMP", "ARMED", "POP", "YEAR"]
Z = data[names].to_numpy()
y = data["TOTEMP"].to_numpy()
n, p = Z.shape

X = (Z - Z.mean(axis=0)) / Z.std(axis=0, ddof=1)    # centred, unit standard
                                                deviation
yc = y - y.mean()                                  # centred response (the
                                                intercept is ybar)

U, d, Vt = np.linalg.svd(X, full_matrices=False)    # X = U diag(d) V^T
V = Vt.T
c = U.T @ yc                                        # coordinates of y along u_1,
                                                ..., u_p

def ridge(lam):
    """Ridge coefficients for the centred problem, through the SVD."""
    return V @ (d / (d ** 2 + lam) * c)

def edf(lam):
    """Effective degrees of freedom of the slopes: sum of the shrinkage factors."""
    return np.sum(d ** 2 / (d ** 2 + lam))

beta_ls = V @ (c / d)
print("singular values squared:", np.round(d ** 2, 4))
print("least squares:", np.round(beta_ls, 1))
for lam in [0.01, 0.1, 1.0]:
    print(f"lambda = {lam:5.2f}  edf = {edf(lam):.2f}  ridge =", np.round(ridge(lam), 1))
```

The second cell computes the leverages of the ridge smoother, the leave-one-out and GCV criteria, and the two choices of  $\lambda$ .

```
def hat_diag(lam):
    """Leverages of the ridge smoother with an unpenalized intercept."""
    f = d ** 2 / (d ** 2 + lam)
    return 1 / n + np.sum(U ** 2 * f, axis=1)

def loo_and_gcv(lam):
    resid = yc - X @ ridge(lam)
    h = hat_diag(lam)
    loo = np.mean((resid / (1 - h)) ** 2) # leave-one-out shortcut
    gcv = np.mean(resid ** 2) / (1 - (1 + edf(lam)) / n) ** 2
    return loo, gcv

grid = np.logspace(-4, 2, 601)
scores = np.array([loo_and_gcv(l) for l in grid])
lam_loo = grid[np.argmin(scores[:, 0])]
lam_gcv = grid[np.argmin(scores[:, 1])]
print(f"L00 chooses lambda = {lam_loo:.4f} (edf {edf(lam_loo):.2f})")
print(f"GCV chooses lambda = {lam_gcv:.4f} (edf {edf(lam_gcv):.2f})")
print("ridge at the L00 choice:", np.round(ridge(lam_loo), 1))
```

The script also checks the closed form against a direct solve, augmented least squares ([Exercise \(A1\)](#)) and the posterior mean of [Theorem 7.6.2](#), and the leave-one-out shortcut against sixteen refits.

## 27.2.5 Exercises

### 27.2.5.1 A. Check your understanding

#### Exercise A1.

Show that  $\hat{\beta}_\lambda^R$  is the least squares estimate for the augmented data  $\tilde{\mathbf{y}} = (\mathbf{y}^\top, \mathbf{0}^\top)^\top$  and  $\tilde{\mathbf{X}} = (\mathbf{X}^\top, \sqrt{\lambda}\mathbf{I})^\top$ . Show that  $\kappa(\tilde{\mathbf{X}})^2 = (d_1^2 + \lambda)/(d_p^2 + \lambda)$ , so ridge can be computed stably by QR ([Chapter 10](#)), and that the augmented problem has “residual sum of squares”  $\|\mathbf{y} - \mathbf{X}\hat{\beta}_\lambda^R\|^2 + \lambda\|\hat{\beta}_\lambda^R\|^2$ .

*Solution.*  $\|\tilde{\mathbf{y}} - \tilde{\mathbf{X}}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \|\mathbf{0} - \sqrt{\lambda}\mathbf{b}\|^2$ , which is [Eq. 27.2.1](#), so the minimizers coincide and the minimum is the stated sum.  $\tilde{\mathbf{X}}^\top\tilde{\mathbf{X}} = \mathbf{X}^\top\mathbf{X} + \lambda\mathbf{I}$  has eigenvalues  $d_j^2 + \lambda$ , and the singular values of  $\tilde{\mathbf{X}}$  are their square roots ([Theorem 1.8.1](#)). The condition number is the ratio of the largest to the smallest ([Definition 1.8.1](#)).

#### Exercise A2.

Suppose  $\mathbf{X}^\top\mathbf{X} = \mathbf{I}$ . Show that  $\hat{\beta}_\lambda^R = \hat{\beta}/(1 + \lambda)$ ,  $\text{df}(\lambda) = p/(1 + \lambda)$ , and that the value of  $\lambda$  minimizing the total mean squared error is  $p\sigma^2/\|\hat{\beta}\|^2$ .

### 27.2.5.2 B. Practice

**Exercise B1.**

Let  $\hat{\boldsymbol{\beta}}_{\lambda,(i)}^R$  be the ridge estimate computed without case  $i$  (no intercept), and  $h_{ii}$  the  $i$ th diagonal entry of  $\mathbf{H}_\lambda$ . Show that

$$y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{\lambda,(i)}^R = \frac{y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_\lambda^R}{1 - h_{ii}}.$$

*Solution.* Put  $\mathbf{K} = \mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}$  and  $\mathbf{x} = \mathbf{x}_{(i)}$ , so  $h_{ii} = \mathbf{x}^\top \mathbf{K}^{-1} \mathbf{x} < 1$ . Without case  $i$  the matrix is  $\mathbf{K} - \mathbf{x}\mathbf{x}^\top$  and the right side is  $\mathbf{X}^\top \mathbf{y} - \mathbf{x}y_i$ . By [Eq. 1.6.7](#),  $(\mathbf{K} - \mathbf{x}\mathbf{x}^\top)^{-1} = \mathbf{K}^{-1} + \mathbf{K}^{-1} \mathbf{x}\mathbf{x}^\top \mathbf{K}^{-1} / (1 - h_{ii})$ . Hence

$$\mathbf{x}^\top \hat{\boldsymbol{\beta}}_{\lambda,(i)}^R = \left( \mathbf{x}^\top \mathbf{K}^{-1} + \frac{h_{ii} \mathbf{x}^\top \mathbf{K}^{-1}}{1 - h_{ii}} \right) (\mathbf{X}^\top \mathbf{y} - \mathbf{x}y_i) = \frac{\mathbf{x}^\top \hat{\boldsymbol{\beta}}_\lambda^R - h_{ii} y_i}{1 - h_{ii}},$$

and  $y_i$  minus this is  $(y_i - \mathbf{x}^\top \hat{\boldsymbol{\beta}}_\lambda^R) / (1 - h_{ii})$ . The argument is that of [Theorem 20.3.1](#) with  $\mathbf{X}^\top \mathbf{X}$  replaced by  $\mathbf{K}$ ; only the rank-one update is used.

**Exercise B2.**

**Generalized ridge regression** uses a separate constant for each principal direction:  $\hat{\boldsymbol{\beta}}_{\mathbf{K}} = (\mathbf{X}^\top \mathbf{X} + \mathbf{V}\mathbf{K}\mathbf{V}^\top)^{-1} \mathbf{X}^\top \mathbf{y}$  with  $\mathbf{K} = \text{diag}(k_1, \dots, k_p)$ ,  $k_j \geq 0$ . Show that its shrinkage factors are  $d_j^2 / (d_j^2 + k_j)$ , and that the choice  $k_j = \sigma^2 / \alpha_j^2$  gives the oracle factors of [Proposition 27.1.2\(c\)](#).

**Exercise B3.**

Show that for every  $t$  with  $0 < t < \|\hat{\boldsymbol{\beta}}\|^2$  (full column rank) there is exactly one  $\lambda > 0$  with  $\|\hat{\boldsymbol{\beta}}_\lambda^R\|^2 = t$ , and that  $\hat{\boldsymbol{\beta}}_\lambda^R$  is then the unique minimizer of  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  subject to  $\|\mathbf{b}\|^2 \leq t$ .

**27.2.5.3 C. Going deeper****Exercise C1.**

For any estimator  $\hat{\boldsymbol{\mu}} = \mathbf{S}\mathbf{Y}$  of  $\boldsymbol{\mu} = \mathbf{E}(\mathbf{Y})$  with  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , show that  $\sigma^{-2} \sum_i \text{Cov}(\hat{\mu}_i, Y_i) = \text{tr } \mathbf{S}$ . Deduce that  $\text{df}(\lambda)$  measures how strongly the ridge fit follows its own data, and that  $\mathbf{E}\|\mathbf{Y} - \mathbf{H}_\lambda \mathbf{Y}\|^2 = \mathbf{E}\|\mathbf{H}_\lambda \mathbf{Y} - \boldsymbol{\mu}\|^2 + n\sigma^2 - 2\sigma^2 \text{df}(\lambda)$ .

**Exercise C2.**

Let  $p > n$ . Show that  $\hat{\boldsymbol{\beta}}_\lambda^R = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1} \mathbf{y}$ , which needs only an  $n \times n$  inverse, and that  $\hat{\boldsymbol{\beta}}_\lambda^R \in \mathcal{C}(\mathbf{X}^\top)$ . What is the limit as  $\lambda \rightarrow 0$ ? [Chapter 28](#) studies this limit.

*Solution.* The identity  $\mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top + \lambda \mathbf{I}_n) = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p) \mathbf{X}^\top$  gives  $(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I}_p)^{-1} \mathbf{X}^\top = \mathbf{X}^\top (\mathbf{X}\mathbf{X}^\top + \lambda \mathbf{I}_n)^{-1}$  after multiplying by the two inverses. The right side is in  $\mathcal{C}(\mathbf{X}^\top)$ . By [Theorem 27.2.2\(d\)](#) the limit is  $\mathbf{X}^+ \mathbf{y}$ , the least squares solution of smallest norm, which

interpolates the data when  $\text{rank}(\mathbf{X}) = n$ .

## 27.3 Principal component regression

Ridge regression keeps every principal direction and shrinks each by a factor between zero and one. Principal component regression makes the factors zero or one: it keeps the directions in which the regressors vary most and discards the rest. Principal components (Hotelling 1933) are an old remedy for collinearity (Massy 1965), and [Proposition 10.4.3](#) computed the bias and variance of regression on the leading ones in uncentred form. This section proves when it beats least squares and makes precise the warning that goes with it.

### 27.3.1 Definition

Keep the conventions of [Section 27.2](#):  $\mathbf{X}$  is  $n \times p$ , centred, of full column rank, with its columns on a common scale, and  $\mathbf{X} = \mathbf{U}\mathbf{D}\mathbf{V}^\top$  with  $d_1 \geq \dots \geq d_p > 0$ .

#### Definition 27.3.1 — Principal component regression

The **principal components** of the regressors are the vectors  $\mathbf{z}_j = \mathbf{X}\mathbf{v}_j = d_j\mathbf{u}_j$ ,  $j = 1, \dots, p$ . Their sample variances are  $d_j^2/(n-1)$ , they are mutually orthogonal, and  $\mathbf{v}_j$  is the  $j$ th **loading vector**. For  $0 \leq k \leq p$ , the **principal component regression** estimate with  $k$  components is

$$\hat{\boldsymbol{\beta}}_k^{\text{PC}} = \sum_{j \leq k} \hat{\alpha}_j \mathbf{v}_j = \mathbf{V}_k \mathbf{V}_k^\top \hat{\boldsymbol{\beta}}, \quad \mathbf{V}_k = [\mathbf{v}_1, \dots, \mathbf{v}_k],$$

obtained by regressing  $\mathbf{y}$  on  $\mathbf{z}_1, \dots, \mathbf{z}_k$  and transforming the coefficients back.

Because the  $\mathbf{z}_j$  are orthogonal, the coefficient of  $\mathbf{z}_j$  in the regression on  $\mathbf{z}_1, \dots, \mathbf{z}_k$  is  $\mathbf{z}_j^\top \mathbf{y} / \|\mathbf{z}_j\|^2 = c_j/d_j = \hat{\alpha}_j$ , the same for every  $k \geq j$ ; the regression on the leading components is least squares with the trailing canonical coordinates set to zero. The shrinkage factors are  $f_j = 1$  for  $j \leq k$  and  $f_j = 0$  for  $j > k$ .

#### Proposition 27.3.1 — Bias and variance of principal component regression

Let  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ ,  $\text{Cov}(\mathbf{Y}) = \sigma^2\mathbf{I}$ ,  $\boldsymbol{\alpha} = \mathbf{V}^\top \boldsymbol{\beta}$  and  $\tau_j^2 = d_j^2 \alpha_j^2 / \sigma^2$ .

- $\hat{\boldsymbol{\beta}}_k^{\text{PC}}$  is the least squares estimate subject to the  $p-k$  restrictions  $\mathbf{v}_j^\top \mathbf{b} = 0$ ,  $j > k$ . Its fitted vector is  $\mathbf{M}_k \mathbf{y}$ , where  $\mathbf{M}_k = \sum_{j \leq k} \mathbf{u}_j \mathbf{u}_j^\top$  is the orthogonal projection onto  $\text{span}(\mathbf{z}_1, \dots, \mathbf{z}_k)$ .
- $E \hat{\boldsymbol{\beta}}_k^{\text{PC}} - \boldsymbol{\beta} = -\sum_{j > k} \alpha_j \mathbf{v}_j$  and  $\text{Cov}(\hat{\boldsymbol{\beta}}_k^{\text{PC}}) = \sigma^2 \sum_{j \leq k} d_j^{-2} \mathbf{v}_j \mathbf{v}_j^\top$ . The total mean squared error is  $\sum_{j > k} \alpha_j^2 + \sigma^2 \sum_{j \leq k} d_j^{-2}$ , and the mean squared error of the fitted values is  $E \|\mathbf{X} \hat{\boldsymbol{\beta}}_k^{\text{PC}} - \mathbf{X} \boldsymbol{\beta}\|^2 = k\sigma^2 + \sum_{j > k} d_j^2 \alpha_j^2$ .

- (c)  $\hat{\beta}_k^{\text{PC}}$  is at least as good as  $\hat{\beta}$  in the matrix sense iff  $\sum_{j>k} \tau_j^2 \leq 1$ .
- (d) Among all estimators  $\tilde{\beta}_f$  with every  $f_j \in \{0, 1\}$ , both mean squared errors in
  - (a) are minimized by keeping exactly the directions with  $\tau_j^2 > 1$  (ties may go either way). Principal component regression with  $k$  components is this best selection iff  $\tau_j^2 \geq 1$  for  $j \leq k$  and  $\tau_j^2 \leq 1$  for  $j > k$ .

*Proof.*

- (a) The restrictions say that  $\mathbf{b} \in \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_k)$ , that is,  $\mathbf{b} = \mathbf{V}_k \mathbf{g}$ . Then  $\mathbf{X}\mathbf{b} = \mathbf{U}_k \mathbf{D}_k \mathbf{g}$  with  $\mathbf{U}_k, \mathbf{D}_k$  the leading blocks, and  $\|\mathbf{y} - \mathbf{U}_k \mathbf{D}_k \mathbf{g}\|^2$  is minimized at  $\mathbf{D}_k \mathbf{g} = \mathbf{U}_k^\top \mathbf{y}$ , so  $\mathbf{g} = (\hat{\alpha}_1, \dots, \hat{\alpha}_k)^\top$  and the fit is  $\mathbf{U}_k \mathbf{U}_k^\top \mathbf{y} = \mathbf{M}_k \mathbf{y}$ .
- (b) The moments are [Proposition 10.4.3](#), and the mean squared errors are [Proposition 27.1.2\(a\)](#) and (b) with the factors 0 and 1.
- (c) By [Theorem 27.1.1\(c\)](#) with the saving  $\Delta_{\mathbf{A}} = \sigma^2 \sum_{j>k} d_j^{-2} \mathbf{v}_j \mathbf{v}_j^\top$ , which is nonnegative definite with column space  $\text{span}(\mathbf{v}_{k+1}, \dots, \mathbf{v}_p)$ , and bias  $\boldsymbol{\delta} = -\sum_{j>k} \alpha_j \mathbf{v}_j$ , which lies in that column space. The Moore–Penrose inverse is  $\Delta_{\mathbf{A}}^+ = \sigma^{-2} \sum_{j>k} d_j^2 \mathbf{v}_j \mathbf{v}_j^\top$ , so  $\boldsymbol{\delta}^\top \Delta_{\mathbf{A}}^+ \boldsymbol{\delta} = \sum_{j>k} d_j^2 \alpha_j^2 / \sigma^2$ .
- (d) The losses are sums over directions, and by [Proposition 27.1.2\(d\)](#) the  $j$ th term is smaller with  $f_j = 0$  than with  $f_j = 1$  iff  $\tau_j^2 < 1$ .

■

By part (c), discarding directions improves the estimate of *every* linear combination exactly when the total signal they carry, in noise units, is at most one. Part (b) shows that the two losses charge a dropped direction differently: for  $\beta$  it costs  $\alpha_j^2$  and saves  $\sigma^2/d_j^2$ , which is large when  $d_j$  is small; for the mean it costs  $d_j^2 \alpha_j^2$  and saves only  $\sigma^2$ . Collinearity is a problem for coefficients, not for fitted values ([Proposition 26.4.1](#)).

### 27.3.2 Low-variance components need not be irrelevant

Part (d) is the crux. The right directions to drop are those with small  $\tau_j^2 = d_j^2 \alpha_j^2 / \sigma^2$ ; principal component regression drops those with small  $d_j^2$ . The two orderings agree only if the  $\alpha_j$  do not grow too fast as  $d_j$  shrinks, and nothing in the model says they should: the components are computed from  $\mathbf{X}$  alone. Jolliffe (1982) gave published examples in which low-variance components were important predictors, and Hadi and Ling (1998) showed how a component with a tiny variance can carry most of a regression ([Exercise \(B2\)](#)).

The signal-to-noise ratios can be estimated: under normal errors  $c_j^2/\sigma^2 - 1$  is unbiased for  $\tau_j^2$ , and the component  $t$  statistics  $t_j = c_j/s$  estimate  $\tau_j$  ([Exercise \(B1\)](#)). Part (d) then suggests keeping the components with  $t_j^2 > 2$ . Such a supervised selection is not a fixed projection, and the leave-one-out and standard-error formulas do not apply to it without adjustment ([Chapter 29](#)).

**Example. (Principal components of Longley’s regressors)**

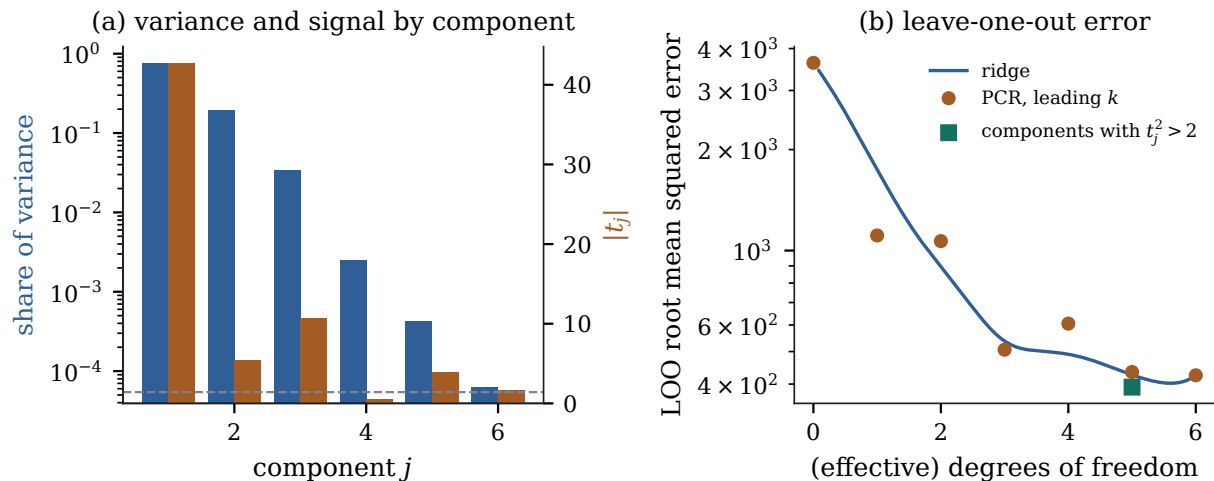


Figure 27.3.1: Principal components of Longley’s regressors. (a) Share of the regressors’ variance (log scale, left) and the absolute component  $t$  statistic (right; dashed line at  $\sqrt{2}$ ). (b) Leave-one-out root mean squared prediction error of principal component regression with the leading  $k$  components, of ridge regression against its effective degrees of freedom, and of the components with  $t_j^2 > 2$ , each for a fixed choice of components or  $\lambda$ .

The six standardized Longley regressors have principal components whose shares of the total variance are 76.723%, 19.589%, 3.390%, 0.249%, 0.043% and 0.006%, so the usual rules of thumb keep two or three. The component  $t$  statistics are 42.66, 5.40, 10.66,  $-0.55$ , 3.96 and 1.72 (Figure 27.3.1(a)). The fifth component, with less than a two-thousandth of the variance, is nearly as strongly related to employment as the second, while the fourth, with more variance than the fifth and sixth together, is irrelevant.

Because principal component regression fits a projection ([Proposition 27.3.1\(a\)](#)), the leave-one-out prediction errors follow from [Theorem 20.3.1](#) with the leverages  $1/n + \sum_{j \leq k} u_{ij}^2$ . The root mean squared leave-one-out errors for  $k = 0, 1, \dots, 6$  are 3627, 1109, 1067, 506, 606, 435 and 425, in thousands. Adding the fourth component makes predictions *worse*, and adding the fifth makes them much better; the best number of leading components is 6, which is least squares itself. The rule  $t_j^2 > 2$  keeps components 1, 2, 3, 5, 6. Applied once to all sixteen years and then treated as fixed, it gives 392, but that figure ignores the selection. Redoing the selection within each fold (*nested* leave-one-out) gives 444, worse than least squares. Searching all 64 subsets of components picks 1, 2, 3, 5 with 389, optimistic for the same reason. With three leading components the coefficients are 1023, 1260,  $-1085$ ,  $-417$ , 1070 and 966: all six regressors are retained, with the GNP and YEAR coefficients now positive and comparable.

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
names = ["GNPDEFL", "GNP", "UNEMP", "ARMED", "POP", "YEAR"]
Z = data[names].to_numpy()
y = data["TOTEMP"].to_numpy()
n, p = Z.shape
X = (Z - Z.mean(axis=0)) / Z.std(axis=0, ddof=1)
yc = y - y.mean()
U, d, Vt = np.linalg.svd(X, full_matrices=False)
c = U.T @ yc
```

*#  $c_j = u_j^T y$ , the fitted*

*coordinate of component  $j$*



```

def fit_components(keep):
    """Least squares on the chosen component scores: coefficients and leave-one-out
    error."""
    keep = list(keep)
    beta = Vt[keep].T @ (c[keep] / d[keep])
    resid = yc - X @ beta
    h = 1 / n + np.sum(U[:, keep] ** 2, axis=1)           # leverages, intercept
    included
    return beta, np.mean((resid / (1 - h)) ** 2)

resid_ls = yc - U @ c
s = np.sqrt(resid_ls @ resid_ls / (n - p - 1))
t = c / s                                               # t statistic of each component
share = d ** 2 / np.sum(d ** 2)                       # share of the regressors' variance
for k in range(p + 1):
    beta_k, loo_k = fit_components(range(k))
    print(f"k = {k}  L00 = {loo_k:9.0f}  coefficients", np.round(beta_k, 0))
print("variance shares:", np.round(share, 5))
print("component t statistics:", np.round(t, 2))

```

### 27.3.3 Ridge and principal components compared

Both methods order the directions by  $d_j$  alone, ridge softly and principal components hard, and neither dominates. If the trailing canonical coefficients are zero, principal components with the right  $k$  is close to the oracle, while ridge must leave variance in the trailing directions or bias the leading ones; if all the  $\alpha_j^2$  are equal, ridge with  $\lambda = \sigma^2/\alpha_j^2$  is the oracle and no zero-one choice matches it ([Exercise \(B3\)](#)). In the simulations of Frank and Friedman (1993) ridge was generally at least as good as principal components. For Longley's data neither method gains over least squares (leading components never do, and tuned ridge has nested leave-one-out error 441 against 425), because the fifth direction matters and both treat it as close to noise. Partial least squares, which builds its directions from  $\mathbf{y}$  as well as  $\mathbf{X}$ , is one response ([Exercise \(C1\)](#)).

### 27.3.4 Exercises

#### 27.3.4.1 A. Check your understanding

##### Exercise A1.

Show that for  $k < p$  the estimate  $\hat{\beta}_k^{\text{PC}}$  generally has no zero coordinates, so that principal component regression reduces dimension without selecting variables. When does it set the coefficient of regressor  $j$  to zero?

##### Exercise A2.

Show that the fitted values of principal component regression with  $k$  components, with an unpenalized intercept, are  $(n^{-1}\mathbf{1}\mathbf{1}^\top + \mathbf{M}_k)\mathbf{y}$ , that this matrix is an orthogonal projection of rank  $k + 1$ , and that its leverages are  $1/n + \sum_{j \leq k} u_{ij}^2$ . Why does [Theorem 20.3.1](#) apply to principal component regression but not to ridge regression?



## 27.3.4.2 B. Practice

**Exercise B1.**

Under the normal linear model with  $\text{rank}(\mathbf{X}) = p < n - 1$  (centred data), show that the  $c_j = \mathbf{u}_j^\top \mathbf{Y}$  are independent  $N(d_j \alpha_j, \sigma^2)$ , independent of the residual sum of squares, and that  $t_j = c_j/s$  has a noncentral  $t$  distribution with  $n - p - 1$  degrees of freedom and noncentrality  $\tau_j$ . Show that  $c_j^2/\sigma^2 - 1$  is unbiased for  $\tau_j^2$ .

*Solution.*  $\mathbf{U}^\top \mathbf{Y}$  is normal with mean  $\mathbf{U}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{D} \mathbf{V}^\top \boldsymbol{\beta} = \mathbf{D} \boldsymbol{\alpha}$  and covariance  $\sigma^2 \mathbf{U}^\top \mathbf{U} = \sigma^2 \mathbf{I}$ , so the  $c_j$  are independent with the stated laws. They are functions of the fitted values  $\mathbf{M} \mathbf{Y} = \mathbf{U} \mathbf{U}^\top \mathbf{Y}$ , which are independent of the residual sum of squares ([Theorem 7.5.1\(c\)](#)); with the intercept removed by centring the residual degrees of freedom are  $n - p - 1$ . Hence  $t_j = (c_j/\sigma)/(s/\sigma)$  is a normal with mean  $\tau_j$  (up to sign) divided by an independent  $\sqrt{\chi^2(n - p - 1)/(n - p - 1)}$ , which is the noncentral  $t$  ([Definition 4.1.3](#)). Finally  $E(c_j^2/\sigma^2) = \text{Var}(c_j/\sigma) + \{E(c_j/\sigma)\}^2 = 1 + \tau_j^2$ .

**Exercise B2.**

Let  $x_1, x_2$  be standardized regressors with correlation  $\rho = 0.99$  and let  $E(Y) = x_1 - x_2$ . Find the canonical coefficients and show that  $\alpha_1 = 0$ . Compute  $\tau_1^2$  and  $\tau_2^2$  with  $n = 100$ , unit-variance columns and  $\sigma = 1$ . Which principal component regression is best, and what does a one-component principal component regression estimate?

*Solution.* With  $\mathbf{X}^\top \mathbf{X} = (n - 1)\mathbf{R}$ , the eigenvectors are  $\mathbf{v}_1 = (1, 1)^\top/\sqrt{2}$  and  $\mathbf{v}_2 = (1, -1)^\top/\sqrt{2}$ , with  $d_1^2 = (n - 1)(1 + \rho)$  and  $d_2^2 = (n - 1)(1 - \rho)$ . For  $\boldsymbol{\beta} = (1, -1)^\top$ ,  $\alpha_1 = \mathbf{v}_1^\top \boldsymbol{\beta} = 0$  and  $\alpha_2 = \sqrt{2}$ . So  $\tau_1^2 = 0$  and  $\tau_2^2 = (n - 1)(1 - \rho) \cdot 2/\sigma^2 = 99 \times 0.01 \times 2 = 1.98$ . By [Proposition 27.3.1\(d\)](#), the best zero-one selection keeps only the *second* component. Principal component regression with one component keeps the first, estimates  $\alpha_1 = 0$  and has expected value  $\mathbf{0}$ : it discards the whole regression.

**Exercise B3.**

Consider a design with two principal directions. (i) If  $\alpha_2 = 0 \neq \alpha_1$ , show that the oracle factors of [Proposition 27.1.2\(c\)](#) are  $(\tau_1^2/(1 + \tau_1^2), 0)$ , that no ridge estimate attains them, and that principal component regression with  $k = 1$  exceeds the oracle's total mean squared error by only  $(\sigma^2/d_1^2)/(1 + \tau_1^2)$ . (ii) If  $\alpha_1^2 = \alpha_2^2 > 0$ , show that ridge with  $\lambda = \sigma^2/\alpha_1^2$  attains the oracle, and that every principal component regression is strictly worse.

*Solution.* Part (i):  $\tau_2^2 = 0$  gives  $f_2^* = 0$ . Ridge has  $f_2(\lambda) = d_2^2/(d_2^2 + \lambda) > 0$  for every finite  $\lambda$ , so it never attains  $f_2^* = 0$ . Principal components with  $k = 1$  has factors  $(1, 0)$ . By [Proposition 27.1.2\(a\)](#) its first term is  $\sigma^2/d_1^2$  against the oracle's  $(\sigma^2/d_1^2)\tau_1^2/(1 + \tau_1^2)$ , and the second terms are both zero; the difference is  $(\sigma^2/d_1^2)/(1 + \tau_1^2)$ . (ii) With  $\alpha_1^2 = \alpha_2^2 = a$  and  $\lambda = \sigma^2/a$ , ridge has factors  $d_j^2/(d_j^2 + \sigma^2/a) = d_j^2 a/(d_j^2 a + \sigma^2) = \tau_j^2/(1 + \tau_j^2)$ , the oracle. Each term of [Proposition 27.1.2\(a\)](#) is a strictly convex quadratic in  $f_j$ , so the oracle is its unique minimizer, and since  $0 < \tau_j^2/(1 + \tau_j^2) < 1$ , every choice of factors in  $\{0, 1\}$  is strictly worse.

## 27.3.4.3 C. Going deeper

**Exercise C1.**

The first direction of **partial least squares** is  $\mathbf{w}_1 \propto \mathbf{X}^\top \mathbf{y}$ , the direction maximizing the sample covariance of  $\mathbf{X}\mathbf{w}$  with  $\mathbf{y}$  over unit vectors  $\mathbf{w}$ . Show that  $\mathbf{w}_1 \propto \sum_j d_j c_j \mathbf{v}_j$ , and compare with the first principal direction  $\mathbf{v}_1$ . Show that the one-dimensional fit along  $\mathbf{w}_1$  never ignores a direction  $j$  with  $c_j \neq 0$ .

## 27.4 The lasso

Ridge regression and principal components shrink, but neither selects ([Exercise \(A1\)](#)). Subset selection does, but it is a combinatorial search whose output jumps when the data change slightly. Tibshirani (1996) replaced the squared penalty of ridge by the sum of the absolute coefficients. The resulting **lasso** (“least absolute shrinkage and selection operator”) is convex, like ridge, and sets some coefficients exactly to zero, like subset selection; Chen, Donoho and Saunders (1998) introduced the same criterion as basis pursuit denoising. The theory for  $p > n$  is in [Chapter 28](#), and generalizations in [Chapter 30](#).

### 27.4.1 Definition and optimality conditions

The conventions of [Section 27.2](#) remain:  $\mathbf{X}$  is centred with columns on a common scale,  $\mathbf{y}$  is centred, and the intercept is  $\bar{y}$  ([Proposition 27.2.1](#)). Here  $\mathbf{X}$  may have any rank.

#### Definition 27.4.1 — The lasso

For  $\lambda \geq 0$ , a **lasso estimate**  $\hat{\boldsymbol{\beta}}_\lambda^L$  minimizes

$$L_\lambda(\mathbf{b}) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda \|\mathbf{b}\|_1, \quad \|\mathbf{b}\|_1 = \sum_j |b_j|. \quad (27.4.1)$$

This is the **penalized form**; the **constrained form** minimizes  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  subject to  $\|\mathbf{b}\|_1 \leq t$ . A vector  $\mathbf{s}$  is a **subgradient** of  $\|\cdot\|_1$  at  $\mathbf{b}$ , written  $\mathbf{s} \in \partial \|\mathbf{b}\|_1$ , if  $s_j = \text{sign}(b_j)$  when  $b_j \neq 0$  and  $s_j \in [-1, 1]$  when  $b_j = 0$ . The **Karush–Kuhn–Tucker (KKT) conditions** for [Eq. 27.4.1](#) are

$$\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \lambda \mathbf{s} \quad \text{for some } \mathbf{s} \in \partial \|\mathbf{b}\|_1; \quad (27.4.2)$$

that is,  $|\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\mathbf{b})| \leq \lambda$  for every  $j$ , with  $\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \lambda \text{sign}(b_j)$  whenever  $b_j \neq 0$ .

The factor  $\frac{1}{2}$  removes a 2 from the KKT conditions; [Chapter 28](#) writes the criterion as  $(2n)^{-1} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda \|\mathbf{b}\|_1$ , whose  $\lambda$  is the present one divided by  $n$ . The KKT conditions say that no regressor’s inner product with the residual exceeds  $\lambda$  in absolute value, and that only regressors at the bound are active.

**Theorem 27.4.1 — Lasso solutions**

Let  $\lambda > 0$ .

- (a)  $L_\lambda$  has at least one minimizer, and the set of minimizers is convex.
- (b)  $\mathbf{b}$  minimizes  $L_\lambda$  iff it satisfies the KKT conditions [Eq. 27.4.2](#).
- (c) All minimizers have the same fitted vector  $\mathbf{X}\mathbf{b}$  and the same norm  $\|\mathbf{b}\|_1$ . If  $\mathbf{X}$  has full column rank, the minimizer is unique.
- (d)  $\mathbf{0}$  is a minimizer iff  $\lambda \geq \lambda_{\max} = \|\mathbf{X}^\top \mathbf{y}\|_\infty = \max_j |\mathbf{x}_j^\top \mathbf{y}|$ .
- (e) Every minimizer  $\hat{\mathbf{b}}$  also minimizes  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  subject to  $\|\mathbf{b}\|_1 \leq \|\hat{\mathbf{b}}\|_1$ .

*Proof.*

- (a)  $L_\lambda$  is continuous and  $L_\lambda(\mathbf{b}) \geq \lambda \|\mathbf{b}\|_1$ , so the set  $\{\mathbf{b} : L_\lambda(\mathbf{b}) \leq L_\lambda(\mathbf{0})\}$  is closed and bounded, and  $L_\lambda$  attains its minimum there. Both terms of  $L_\lambda$  are convex, so if  $\mathbf{b}, \mathbf{b}'$  are minimizers, every point between them has value at most the minimum, and is a minimizer too.
- (b) *Sufficiency.* Let  $\mathbf{b}$  satisfy [Eq. 27.4.2](#) with subgradient  $\mathbf{s}$ , and let  $\mathbf{c}$  be arbitrary. Expanding the square,

$$\begin{aligned} \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{c}\|^2 &= \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 - (\mathbf{c} - \mathbf{b})^\top \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) \\ &\quad + \frac{1}{2} \|\mathbf{X}(\mathbf{c} - \mathbf{b})\|^2 \\ &\geq \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 - \lambda \mathbf{s}^\top (\mathbf{c} - \mathbf{b}). \end{aligned}$$

Since  $|s_j| \leq 1$  and  $s_j b_j = |b_j|$ , we have  $\|\mathbf{c}\|_1 \geq \mathbf{s}^\top \mathbf{c} = \|\mathbf{b}\|_1 + \mathbf{s}^\top (\mathbf{c} - \mathbf{b})$ . Adding  $\lambda$  times this to the previous inequality gives  $L_\lambda(\mathbf{c}) \geq L_\lambda(\mathbf{b})$ . *Necessity.* Let  $\mathbf{b}$  be a minimizer,  $\mathbf{r} = \mathbf{y} - \mathbf{X}\mathbf{b}$  and  $g_j = \mathbf{x}_j^\top \mathbf{r}$ . For fixed  $j$  consider  $\phi(t) = L_\lambda(\mathbf{b} + t\mathbf{e}_j)$ , which is minimized at  $t = 0$ . Its one-sided derivatives at zero are

$$\phi'(0^+) = -g_j + \lambda \sigma_+, \quad \phi'(0^-) = -g_j + \lambda \sigma_-,$$

with  $\sigma_\pm = \text{sign}(b_j)$  if  $b_j \neq 0$ , and  $\sigma_+ = 1, \sigma_- = -1$  if  $b_j = 0$ . A minimum at zero requires  $\phi'(0^+) \geq 0 \geq \phi'(0^-)$ . If  $b_j \neq 0$  this gives  $g_j = \lambda \text{sign}(b_j)$ ; if  $b_j = 0$  it gives  $-\lambda \leq g_j \leq \lambda$ . These are the KKT conditions with  $s_j = g_j/\lambda$ .

- (c) Let  $\mathbf{b}, \mathbf{b}'$  be minimizers with  $\mathbf{X}\mathbf{b} \neq \mathbf{X}\mathbf{b}'$ , and let  $\mathbf{m} = (\mathbf{b} + \mathbf{b}')/2$ . The function  $\mathbf{u} \mapsto \|\mathbf{y} - \mathbf{u}\|^2$  is strictly convex, so  $\|\mathbf{y} - \mathbf{X}\mathbf{m}\|^2 < \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}'\|^2$ , while  $\|\mathbf{m}\|_1 \leq \frac{1}{2} \|\mathbf{b}\|_1 + \frac{1}{2} \|\mathbf{b}'\|_1$ . Then  $L_\lambda(\mathbf{m}) < L_\lambda(\mathbf{b})$ , a contradiction. Equal fitted vectors give equal residual sums of squares, and equal values of  $L_\lambda$  then force equal  $\ell_1$  norms. If  $\mathbf{X}$  has full column rank,  $\mathbf{X}\mathbf{b} = \mathbf{X}\mathbf{b}'$  implies  $\mathbf{b} = \mathbf{b}'$ .
- (d) At  $\mathbf{b} = \mathbf{0}$  every  $s_j$  may be chosen in  $[-1, 1]$ , so [Eq. 27.4.2](#) holds iff  $|\mathbf{x}_j^\top \mathbf{y}| \leq \lambda$  for all  $j$ . Apply (b).
- (e) This is the remark on penalized and constrained forms in [Section 27.2](#) with  $P = \|\cdot\|_1$ .

■

By (b) and (c), all solutions share the inner products  $\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\mathbf{b})$  and vanish outside the set of  $j$  at which these equal  $\pm\lambda$ , a set determined by the data even when the solution is not.

The converse of (e) is Lagrangian duality for convex problems (Boyd and Vandenberghe 2004, chapter 5), which we shall not need.

The constrained forms show why the lasso produces exact zeros and ridge does not. The constrained estimate is where the smallest contour of  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$ , an ellipsoid, touches the constraint region. The ridge ball has a single normal direction at each boundary point, so the contact lies on a coordinate subspace only by accident. The lasso's cross-polytope has vertices and edges in coordinate subspaces, each with a whole cone of normal directions, so for an open set of data the contact is at such a face and some coordinates are exactly zero. Algebraically, at  $b_j = 0$  the subgradient may be anything in  $[-1, 1]$ .

## 27.4.2 The orthonormal case: soft thresholding

Under an orthonormal design the lasso, ridge regression and subset selection are all explicit. Define the **soft** and **hard thresholding** functions

$$S_\lambda(z) = \text{sign}(z) (|z| - \lambda)_+, \quad H_\lambda(z) = z \mathbf{1}\{|z| > \lambda\}.$$

### Theorem 27.4.2 — Three kinds of shrinkage under an orthonormal design

Suppose  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ , so that  $\hat{\boldsymbol{\beta}} = \mathbf{X}^\top \mathbf{y}$ , and let  $\lambda > 0$ .

- (a) (*Lasso.*) The unique minimizer of Eq. 27.4.1 is given by  $\hat{\beta}_{\lambda,j}^L = S_\lambda(\hat{\beta}_j)$ .
- (b) (*Ridge.*) The minimizer of Eq. 27.2.1 is  $\hat{\boldsymbol{\beta}}/(1 + \lambda)$ .
- (c) (*Subset selection.*) The minimizers of  $\frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \frac{1}{2}\lambda^2 \#\{j : b_j \neq 0\}$  are given by  $b_j = H_\lambda(\hat{\beta}_j)$  (with either choice when  $|\hat{\beta}_j| = \lambda$ ), and for each  $k$  the best subset of size  $k$  consists of the  $k$  largest  $|\hat{\beta}_j|$ .

*Proof.* Since  $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{0}$ ,  $\mathbf{y} - \mathbf{X}\mathbf{b} = (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) + \mathbf{X}(\hat{\boldsymbol{\beta}} - \mathbf{b})$  is an orthogonal sum, and  $\|\mathbf{X}\mathbf{w}\| = \|\mathbf{w}\|$ , so

$$\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + \sum_j (\hat{\beta}_j - b_j)^2.$$

Each criterion therefore separates into one-dimensional problems. (a) Minimize  $\frac{1}{2}(\hat{\beta}_j - b)^2 + \lambda|b|$  over  $b$ . For  $b > 0$  the derivative is  $b - \hat{\beta}_j + \lambda$ , which vanishes at  $b = \hat{\beta}_j - \lambda$ , admissible iff  $\hat{\beta}_j > \lambda$ ; symmetrically for  $b < 0$ ; otherwise the convex function is minimized at  $b = 0$ . This is  $S_\lambda(\hat{\beta}_j)$ , and it is unique because the problem is strictly convex. (b) Minimize  $(\hat{\beta}_j - b)^2 + \lambda b^2$ , a quadratic with minimum at  $\hat{\beta}_j/(1 + \lambda)$ . (c) The  $j$ th problem costs  $\frac{1}{2}\hat{\beta}_j^2$  with  $b_j = 0$ , and at best  $\frac{1}{2}\lambda^2$ , at  $b_j = \hat{\beta}_j$ , with  $b_j \neq 0$ . So  $b_j = \hat{\beta}_j$  iff  $\hat{\beta}_j^2 > \lambda^2$ . For a fixed subset  $A$  of size  $k$  the residual sum of squares is  $\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\|^2 + \sum_{j \notin A} \hat{\beta}_j^2$ , smallest when  $A$  holds the  $k$  largest  $|\hat{\beta}_j|$ . ■

Figure 27.4.1(a) draws the three rules. Ridge shrinks every estimate by the same factor; hard thresholding keeps large estimates and kills small ones, with a jump at  $\pm\lambda$ ; soft thresholding kills

small estimates and moves the others towards zero by  $\lambda$ , continuous like ridge and sparse like subset selection, at the price of a bias  $\lambda$  on large coefficients. Soft thresholding of independent normal estimates was analysed by Donoho and Johnstone (1994), whose threshold  $\sigma\sqrt{2\log p}$  reappears in [Chapter 28](#). The lasso is also a posterior *mode* under independent Laplace priors (Tibshirani 1996; [Exercise \(B2\)](#)).

### 27.4.3 The lasso path

As  $\lambda$  decreases from  $\lambda_{\max}$  the lasso moves from  $\mathbf{0}$  towards a least squares fit, and the KKT conditions show that it does so along straight lines.

#### Proposition 27.4.3 — Piecewise linearity

Suppose that for all  $\lambda$  in an interval  $[\lambda_1, \lambda_2]$  with  $\lambda_1 > 0$  the lasso solution is unique, has the same active set  $A = \{j : \hat{\beta}_{\lambda,j}^L \neq 0\}$  and the same signs  $\mathbf{s}_A$ , and that  $\mathbf{X}_A$ , the columns of  $\mathbf{X}$  in  $A$ , has full column rank. Then on that interval

$$\hat{\beta}_{\lambda,A}^L = (\mathbf{X}_A^\top \mathbf{X}_A)^{-1}(\mathbf{X}_A^\top \mathbf{y} - \lambda \mathbf{s}_A),$$

an affine function of  $\lambda$ , and the other coordinates are zero.

*Proof.* The KKT equations for  $j \in A$  read  $\mathbf{X}_A^\top(\mathbf{y} - \mathbf{X}_A \hat{\beta}_{\lambda,A}^L) = \lambda \mathbf{s}_A$ , and  $\mathbf{X}_A^\top \mathbf{X}_A$  is invertible. ■

So the path is made of line segments joined where a variable enters or leaves the active set, and the whole path costs about as much as one least squares fit (Osborne, Presnell and Turlach 2000; Efron, Hastie, Johnstone and Tibshirani 2004; [Section 30.5](#) returns to this). A variable leaves when its affine path reaches zero, which happens with correlated regressors but never under an orthonormal design ([Exercise \(B3\)](#)).

### 27.4.4 Computing the lasso by coordinate descent

As a function of one coordinate  $b_j$ , with the others fixed, [Eq. 27.4.1](#) is a one-dimensional lasso, solved by soft thresholding. Cycling through the coordinates gives a simple, fast algorithm, the “shooting” method of Fu (1998), popularized by Friedman, Hastie, Höfling and Tibshirani (2007).

#### Lemma 27.4.4 — Coordinate updates

Fix  $\mathbf{b}$  and  $j$  with  $\mathbf{x}_j \neq \mathbf{0}$ , and let  $\mathbf{r}_{(j)} = \mathbf{y} - \sum_{k \neq j} \mathbf{x}_k b_k$  be the partial residual.

- (a) The minimizer of  $L_\lambda$  over  $b_j$ , with the other coordinates fixed, is  $S_\lambda(\mathbf{x}_j^\top \mathbf{r}_{(j)})/\|\mathbf{x}_j\|^2$ .
- (b) If  $\mathbf{b}$  is unchanged by the update of every coordinate, then  $\mathbf{b}$  satisfies the KKT conditions [Eq. 27.4.2](#), and is therefore a lasso solution.

*Proof.*

- (a) With  $z = \mathbf{x}_j^\top \mathbf{r}_{(j)}$  and  $q = \|\mathbf{x}_j\|^2$ ,  $\frac{1}{2}\|\mathbf{r}_{(j)} - \mathbf{x}_j b_j\|^2 = \frac{1}{2}q(b_j - z/q)^2 + \text{const.}$  Dividing the one-dimensional criterion by  $q$  gives  $\frac{1}{2}(b_j - z/q)^2 + (\lambda/q)|b_j|$ , minimized by  $S_{\lambda/q}(z/q) = S_\lambda(z)/q$  as in [Theorem 27.4.2\(a\)](#).
- (b) A fixed point has  $b_j = S_\lambda(z_j)/q_j$  for every  $j$ , which by the one-dimensional case of [Theorem 27.4.1\(b\)](#) means  $z_j - q_j b_j \in \lambda \partial|b_j|$ . But  $z_j - q_j b_j = \mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\mathbf{b})$ . So [Eq. 27.4.2](#) holds, and [Theorem 27.4.1\(b\)](#) applies.

■

**Cyclic coordinate descent for the lasso.** Start from  $\mathbf{b} = \mathbf{0}$  (or a warm start) and  $\mathbf{r} = \mathbf{y} - \mathbf{X}\mathbf{b}$ . Repeat until no coordinate changes by more than a tolerance: for  $j = 1, \dots, p$ , compute  $z = \mathbf{x}_j^\top \mathbf{r} + \|\mathbf{x}_j\|^2 b_j$ , set  $b_j^{\text{new}} = S_\lambda(z)/\|\mathbf{x}_j\|^2$ , update  $\mathbf{r} \leftarrow \mathbf{r} - \mathbf{x}_j(b_j^{\text{new}} - b_j)$  and  $b_j \leftarrow b_j^{\text{new}}$ .

Each update lowers  $L_\lambda$  or leaves it unchanged. Tseng (2001) proved, for criteria that are smooth and convex plus a sum of convex functions of single coordinates, that every limit point of cyclic coordinate descent is a minimizer; we do not reproduce the proof. The lemma gives what a computation needs: at a fixed point the KKT conditions hold, and their violation can be measured. In practice the lasso is computed on a decreasing grid of  $\lambda$  from  $\lambda_{\max}$ , each solution the warm start for the next (Friedman, Hastie and Tibshirani 2010). Highly correlated regressors slow convergence, because single-coordinate moves zigzag along a narrow valley.

#### Example. (The lasso for Longley's data)

For the standardized Longley data,  $\lambda_{\max} = 51813$ , and the variables enter in the order GNP, UNEMP, ARMED, YEAR, POP; the GNP deflator has not entered by  $\lambda_{\max}/1000$ . [Figure 27.4.1\(b\)](#) shows the path against the ratio of  $\|\hat{\boldsymbol{\beta}}_\lambda^L\|_1$  to the  $\ell_1$  norm of least squares. GNP, the regressor most correlated with employment, grows to over 3000; when YEAR enters, the two compete for the same direction, and GNP falls back to exactly zero and leaves the active set. Coordinate descent needs 42 sweeps at  $\lambda_{\max}/2$ , 157 at  $\lambda_{\max}/50$  and 3479 at  $\lambda_{\max}/1000$  for a tolerance of  $10^{-10}$ : the zigzag effect of collinearity.

Leave-one-out cross-validation, refitting the path without each year in turn, chooses  $\lambda = 130$ , about  $0.0025\lambda_{\max}$ , with 3 nonzero coefficients:  $-1303$  for unemployment,  $-484$  for the armed forces and  $4475$  for the year. Its minimized root mean squared error is 373, but as with ridge the minimum is optimistic; nested leave-one-out gives 428, level with least squares (425). What the lasso offers here is a simpler fit of equal accuracy, not a better one, and its choice of YEAR over GNP is exactly the kind that collinearity makes fragile.

The script checks the solutions at five values of  $\lambda$  three ways: the KKT conditions hold to within  $10^{-6}\lambda$ ; a brute-force search over all  $3^6 = 729$  active sets and signs, solving the equations of [Proposition 27.4.3](#), finds the same solution; and a bound-constrained optimizer on the split  $\mathbf{b} = \mathbf{u} - \mathbf{v}$ ,  $\mathbf{u}, \mathbf{v} \geq \mathbf{0}$ , finds no smaller criterion value.

```
import numpy as np
import statsmodels.api as sm
def soft(z, t):
    """Soft thresholding: sign(z) * max(|z| - t, 0)."""
    return np.sign(z) * np.maximum(np.abs(z) - t, 0.0)
```

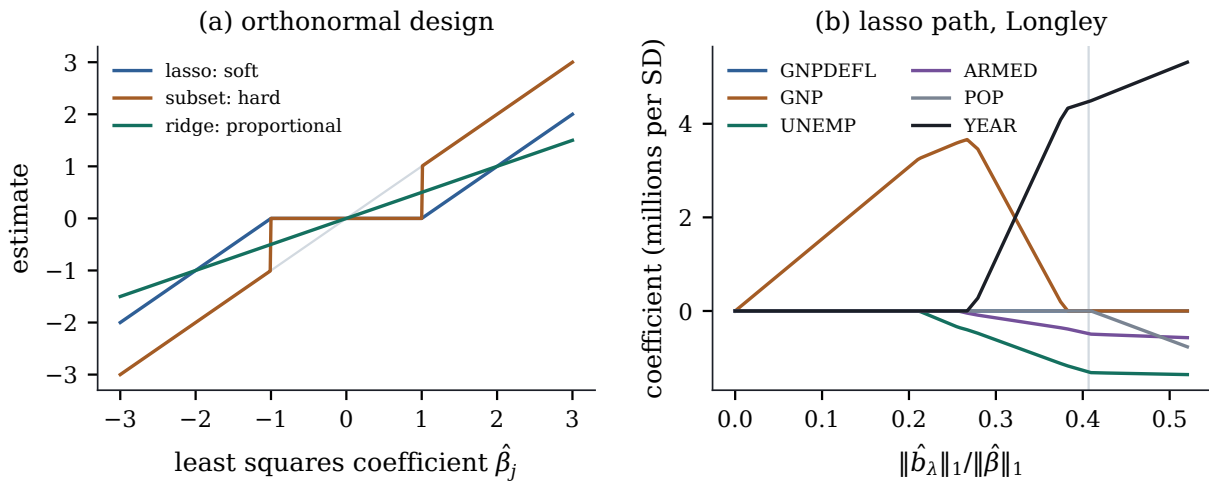


Figure 27.4.1: (a) Lasso (soft thresholding), subset selection (hard thresholding) and ridge regression (proportional shrinkage) as functions of the least squares coefficient under an orthonormal design, with  $\lambda = 1$  (Theorem 27.4.2). (b) The lasso path for the standardized Longley data; the vertical line marks the leave-one-out choice of  $\lambda$ .

```
def lasso_cd(X, y, lam, b=None, tol=1e-10, max_sweeps=100_000):
    """Cyclic coordinate descent for  $(1/2)\|y - Xb\|^2 + \lambda \|b\|_1$ ."""
    p = X.shape[1]
    b = np.zeros(p) if b is None else b.copy()
    r = y - X @ b
    sq = np.sum(X ** 2, axis=0)
    for sweep in range(1, max_sweeps + 1):
        biggest = 0.0
        for j in range(p):
            z = X[:, j] @ r + sq[j] * b[j]
            new = soft(z, lam) / sq[j]
            if new != b[j]:
                r -= X[:, j] * (new - b[j])
                biggest = max(biggest, abs(new - b[j]) * np.sqrt(sq[j]))
                b[j] = new
        if biggest < tol:
            break
    return b, sweep

def kkt_violation(X, y, b, lam):
    """Largest violation of the KKT conditions  $X^T(y - Xb) = \lambda * s$ ,  $s$  in the subdifferential."""
    g = X.T @ (y - X @ b)
    active = b != 0
    return max(np.max(np.abs(g[active] - lam * np.sign(b[active]))), initial=0.0),
           np.max(np.abs(g[~active]) - lam, initial=0.0))
```

The second cell computes the path on a grid of 61 values of  $\lambda$  with warm starts.

```
data = sm.datasets.longley.load_pandas().data
names = ["GNPDEFL", "GNP", "UNEMP", "ARMED", "POP", "YEAR"]
Z = data[names].to_numpy()
y = data["TOTEMP"].to_numpy()
n, p = Z.shape
X = (Z - Z.mean(axis=0)) / Z.std(axis=0, ddof=1)
```

```

yc = y - y.mean()

lam_max = np.max(np.abs(X.T @ yc))           # smallest lam with b = 0
lams = lam_max * np.logspace(0, -3, 61)
path, b = [], np.zeros(p)
for lam in lams:                             # warm starts along a
    decreasing grid
    b, _ = lasso_cd(X, yc, lam, b, tol=1e-6)
    path.append(b.copy())
path = np.array(path)
first = {names[j]: np.argmax(path[:, j] != 0) for j in range(p) if np.any(path[:, j]
    != 0)}
entry = sorted(first, key=first.get)          # variables in order of entry
print(f"lambda_max = {lam_max:.0f}; order of entry:", entry)
print("GNP coefficient along the path:", np.round(path[:6, 1], 0))

```

## 27.4.5 Lasso, ridge and subset selection compared

[Theorem 27.4.2](#) suggests that hard thresholding suits a few large coefficients among zeros, proportional shrinkage many moderate ones, and soft thresholding lies between. A simulation with correlated regressors tests this.

### Example. (Three coefficient patterns)

Fix a design with  $n = 50$  and  $p = 10$ , its rows drawn once from a normal distribution with correlations  $0.5^{|j-k|}$ , and columns scaled to  $\|\mathbf{x}_j\|^2 = n$ ; the errors are normal with  $\sigma = 2$ . The coefficients are *sparse*,  $(3, -2, 1.5, 0, \dots, 0)$ ; *dense*, ten coefficients  $\pm 0.5$  with alternating signs; or *intermediate*, five coefficients  $\pm 1$  and five zeros. Each method's tuning constant ( $\lambda$ , or the subset size, with the best subsets found among all 1024) minimizes the error on an independent copy of the responses, an idealized validation set. The table gives the average of  $\|\mathbf{X}\tilde{\boldsymbol{\beta}} - \mathbf{X}\boldsymbol{\beta}\|^2 / (p\sigma^2)$  over 300 data sets; least squares has expectation one ([Exercise \(B2\)](#)), and the Monte Carlo standard error of each entry is about 0.02.

| Pattern      | least squares | ridge | lasso | best subset |
|--------------|---------------|-------|-------|-------------|
| sparse       | 1.02          | 0.95  | 0.74  | 0.46        |
| dense        | 0.99          | 0.73  | 0.83  | 1.03        |
| intermediate | 0.99          | 0.84  | 0.86  | 0.90        |

Best subset selection is far ahead in the sparse case and worst in the dense case, where it cannot shrink and finds no zeros; ridge is the reverse; the lasso is never best and never worst. In the intermediate case the paired difference between the lasso and ridge is 0.02 with a standard error of 0.01. Hastie, Tibshirani and Tibshirani (2020) reached similar conclusions in a much larger study, adding that the ranking depends strongly on the signal-to-noise ratio ([Figure 27.4.2](#) displays the table).

The cell below reruns the sparse scenario with 10 data sets; each one searches all 1024 subsets, so in the browser it may take half a minute. The script uses 300 per scenario.

```
import itertools
```



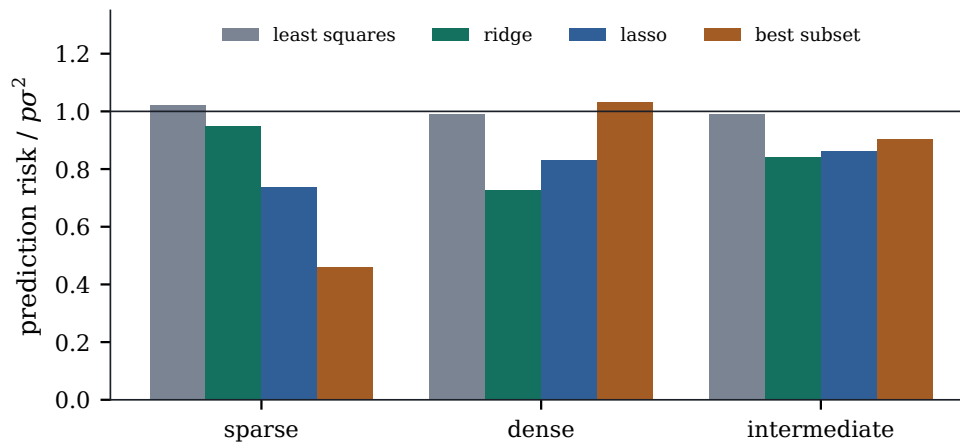


Figure 27.4.2: Prediction risk at the design points, relative to  $p\sigma^2$ , for least squares, ridge, the lasso and best subset selection, each tuned on a validation copy of the responses, under three patterns of true coefficients (300 data sets each).

```
import numpy as np
def soft(z, t):
    return np.sign(z) * np.maximum(np.abs(z) - t, 0.0)

def lasso_cd(X, y, lam, b=None, tol=1e-8, max_sweeps=10_000):
    """Cyclic coordinate descent for  $(1/2)||y - Xb||^2 + \lambda ||b||_1$ ."""
    p = X.shape[1]
    b = np.zeros(p) if b is None else b.copy()
    r = y - X @ b
    sq = np.sum(X ** 2, axis=0)
    for _ in range(max_sweeps):
        biggest = 0.0
        for j in range(p):
            new = soft(X[:, j] @ r + sq[j] * b[j], lam) / sq[j]
            if new != b[j]:
                r -= X[:, j] * (new - b[j])
                biggest = max(biggest, abs(new - b[j]))
                b[j] = new
        if biggest < tol:
            break
    return b

rng = np.random.default_rng(2027)
n, p, sigma = 50, 10, 2.0
Sigma = 0.5 ** np.abs(np.subtract.outer(np.arange(p), np.arange(p)))
X = rng.multivariate_normal(np.zeros(p), Sigma, size=n)
X = (X - X.mean(axis=0)) / np.sqrt(np.mean((X - X.mean(axis=0)) ** 2, axis=0)) #
     $||x_j||^2 = n$ 
U, d, Vt = np.linalg.svd(X, full_matrices=False)
subsets = [list(S) for k in range(p + 1) for S in itertools.combinations(range(p), k)]
ridge_grid = np.logspace(-2, 4, 40)
scenarios = {
    "sparse": np.r_[3.0, -2.0, 1.5, np.zeros(p - 3)],
    "dense": 0.5 * (-1.0) ** np.arange(p),
    "intermediate": np.r_[1.0, -1.0, 1.0, -1.0, 1.0, np.zeros(p - 5)],
}
```

```

def fits(y):
    """Candidate fitted vectors of each method along its tuning path."""
    c = U.T @ y
    out = {"least squares": [U @ c]}
    out["ridge"] = [U @ (d ** 2 / (d ** 2 + lam) * c) for lam in ridge_grid]
    lam_max = np.max(np.abs(X.T @ y))
    b, lasso = np.zeros(p), []
    for lam in lam_max * np.logspace(0, -3, 30):
        b = lasso_cd(X, y, lam, b)
        lasso.append(X @ b)
    out["lasso"] = lasso
    best = {} # best subset of each size, by
    residual sum of squares
    for S in subsets:
        fit = X[:, S] @ np.linalg.lstsq(X[:, S], y, rcond=None)[0] if S else np.
        zeros(n)
        rss = np.sum((y - fit) ** 2)
        if len(S) not in best or rss < best[len(S)][0]:
            best[len(S)] = (rss, fit)
    out["best subset"] = [best[k][1] for k in range(p + 1)]
    return out

def run(beta, reps, seed=1):
    """Mean of ||X b - X beta||^2 / (p sigma^2) for each method, tuned on a
    validation copy."""
    gen = np.random.default_rng(seed)
    mu = X @ beta
    loss = {m: [] for m in ["least squares", "ridge", "lasso", "best subset"]}
    for _ in range(reps):
        y = mu + sigma * gen.normal(size=n)
        y_val = mu + sigma * gen.normal(size=n)
        for m, cand in fits(y).items():
            chosen = min(cand, key=lambda f: np.sum((y_val - f) ** 2))
            loss[m].append(np.sum((chosen - mu) ** 2) / (p * sigma ** 2))
    return {m: np.array(v) for m, v in loss.items()}

quick = run(scenarios["sparse"], reps=10)
print({m: round(float(v.mean()), 2) for m, v in quick.items()})

```

## 27.4.6 Exercises

### 27.4.6.1 A. Check your understanding

#### Exercise A1.

Suppose  $\max_j |\mathbf{x}_j^\top \mathbf{y}|$  is attained at a single  $j^*$ . Show that for  $\lambda$  slightly below  $\lambda_{\max}$  the lasso solution has one nonzero coordinate,  $j^*$ , with the sign of  $\mathbf{x}_{j^*}^\top \mathbf{y}$  and absolute value  $(|\mathbf{x}_{j^*}^\top \mathbf{y}| - \lambda) / \|\mathbf{x}_{j^*}\|^2$ .

*Solution.* Try  $\mathbf{b} = b\mathbf{e}_{j^*}$  with  $b = \text{sign}(g)(|g| - \lambda)/q$ ,  $g = \mathbf{x}_{j^*}^\top \mathbf{y}$ ,  $q = \|\mathbf{x}_{j^*}\|^2$ . Then  $\mathbf{x}_{j^*}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = g - qb = \lambda \text{sign}(g)$ , as required. For  $k \neq j^*$ ,  $|\mathbf{x}_k^\top (\mathbf{y} - \mathbf{X}\mathbf{b})| \leq |\mathbf{x}_k^\top \mathbf{y}| + |b| |\mathbf{x}_k^\top \mathbf{x}_{j^*}|$ . The first term is strictly less than  $\lambda_{\max}$ , and  $b \rightarrow 0$  as  $\lambda \rightarrow \lambda_{\max}$ , so the bound is below  $\lambda$  for  $\lambda$  close enough to  $\lambda_{\max}$ . By Theorem 27.4.1(b)  $\mathbf{b}$  is a solution; it is unique among

solutions supported on  $\{j^*\}$ , and every solution is supported on the set where the inner product with the residual equals  $\pm\lambda$ , which is  $\{j^*\}$ .

### Exercise A2.

With an orthonormal design and  $\hat{\boldsymbol{\beta}} = (3, -0.5, 1.2)^\top$ , compute the lasso, ridge and hard-thresholding estimates of [Theorem 27.4.2](#) with  $\lambda = 1$ .

## 27.4.6.2 B. Practice

### Exercise B1.

Suppose  $\mathbf{x}_1 = \mathbf{x}_2$  and let  $\hat{\mathbf{b}}$  be a lasso solution with  $\hat{b}_1 + \hat{b}_2 = \gamma \neq 0$ . Show that every  $\mathbf{b}$  with  $b_1 + b_2 = \gamma$ ,  $b_1, b_2$  of the sign of  $\gamma$ , and the other coordinates as in  $\hat{\mathbf{b}}$ , is also a solution, whereas ridge regression gives  $b_1 = b_2$ .

*Solution.* Such a  $\mathbf{b}$  has  $\mathbf{X}\mathbf{b} = \mathbf{X}\hat{\mathbf{b}}$  and  $|b_1| + |b_2| = |\gamma| \leq |\hat{b}_1| + |\hat{b}_2|$ , so  $L_\lambda(\mathbf{b}) \leq L_\lambda(\hat{\mathbf{b}})$ , with equality by optimality. For ridge, replacing  $(b_1, b_2)$  by their average leaves  $\mathbf{X}\mathbf{b}$  unchanged and does not increase  $b_1^2 + b_2^2$ , strictly decreasing it unless  $b_1 = b_2$ ; since the ridge solution is unique, it has  $b_1 = b_2$ . The elastic net of [Chapter 30](#) combines the two penalties to get the ridge behaviour with lasso sparsity.

### Exercise B2.

Let  $\mathbf{Y} \mid \boldsymbol{\beta} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I})$  with  $\sigma^2$  known, and let the  $\beta_j$  be independent with density  $(\lambda/2\sigma^2) \exp(-\lambda|\beta_j|/\sigma^2)$ . Show that the posterior mode is the lasso estimate. Is the posterior mean sparse?

*Solution.* The log posterior is  $-\{\frac{1}{2}\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|_1\}/\sigma^2 + \text{const}$ , maximized at the minimizer of [Eq. 27.4.1](#). The posterior mean is an average over a continuous distribution with positive density everywhere, so every coordinate is nonzero with probability one: the sparsity of the lasso is a property of the mode, not of the Bayesian model.

### Exercise B3.

Under an orthonormal design, describe the lasso path: show that it is piecewise linear with kinks at  $\lambda = |\hat{\beta}_j|$ , that variables enter in decreasing order of  $|\hat{\beta}_j|$ , and that none ever leaves.

## 27.4.6.3 C. Going deeper

### Exercise C1.

In the orthonormal case with normal errors, show that  $\sigma^{-2} \sum_j \text{Cov}(S_\lambda(\hat{\beta}_j), \hat{\beta}_j) = \mathbb{E} \#\{j : |\hat{\beta}_j| > \lambda\}$ : the expected number of nonzero coefficients plays the role of degrees of freedom (compare [Exercise \(C1\)](#)). Hint: integrate by parts, one coordinate at a time.

**Exercise C2.**

Show that for every  $\lambda > 0$  there is a lasso solution with at most  $\min(n, p)$  nonzero coordinates. Hint: if the active columns of a solution are linearly dependent, move along a direction in their null space that keeps the signs and the fitted vector until a coordinate hits zero.

## 27.5 Stein's paradox and James–Stein estimation

The shrinkage estimators so far beat least squares only for *some* parameter values, which is unavoidable for linear estimators ([Exercise \(B1\)](#)). Stein (1956) discovered that a *nonlinear* shrinkage can do better everywhere: when three or more means are estimated together under total squared error, the estimator that is at once least squares, maximum likelihood and best unbiased is inadmissible. James and Stein (1961) gave an explicit estimator that dominates it. This section proves their theorem and explains it as an empirical Bayes phenomenon.

### 27.5.1 The normal means problem

Let  $\mathbf{Z} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$  with  $\sigma^2$  known, and measure an estimator  $\hat{\boldsymbol{\theta}}(\mathbf{Z})$  by its **risk**  $R(\boldsymbol{\theta}, \hat{\boldsymbol{\theta}}) = E\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2$ , its total mean squared error. The estimator  $\mathbf{Z}$  has risk  $p\sigma^2$  for every  $\boldsymbol{\theta}$ . An estimator **dominates** another if its risk is never larger and is smaller for some  $\boldsymbol{\theta}$ ; an estimator dominated by none is **admissible**.

This is regression in canonical form. If  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ , then  $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2 \mathbf{I})$  ([Theorem 7.5.1\(a\)](#)). For a general full-rank design,  $\mathbf{Z} = (\mathbf{X}^\top \mathbf{X})^{1/2} \hat{\boldsymbol{\beta}}$  is  $N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$  with  $\boldsymbol{\theta} = (\mathbf{X}^\top \mathbf{X})^{1/2} \boldsymbol{\beta}$ , and for any estimator  $\tilde{\boldsymbol{\beta}}$  with  $\hat{\boldsymbol{\theta}} = (\mathbf{X}^\top \mathbf{X})^{1/2} \tilde{\boldsymbol{\beta}}$ ,

$$\|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}\|^2 = (\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{X}^\top \mathbf{X} (\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}) = \|\mathbf{X}\tilde{\boldsymbol{\beta}} - \mathbf{X}\boldsymbol{\beta}\|^2,$$

the prediction loss at the design points, with the square root of [Theorem 1.7.6](#) as the change of coordinates.

### 27.5.2 Stein's lemma

The proof rests on an integration-by-parts identity for the normal distribution.

#### Lemma 27.5.1 — Stein's lemma

- (a) Let  $Z \sim N(\mu, \sigma^2)$ , and let  $g : \mathbb{R} \rightarrow \mathbb{R}$  be absolutely continuous with  $E|g'(Z)| < \infty$ . Then  $E[(Z - \mu)g(Z)] < \infty$  and

$$E[(Z - \mu)g(Z)] = \sigma^2 E g'(Z).$$

- (b) Let  $\mathbf{Z} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ , and let  $\mathbf{g} = (g_1, \dots, g_p) : \mathbb{R}^p \rightarrow \mathbb{R}^p$  be such that, for each  $i$  and almost every value of the other coordinates,  $g_i$  is absolutely continuous in  $z_i$ , with  $E|\partial g_i(\mathbf{Z})/\partial z_i| < \infty$  and  $E|(Z_i - \theta_i)g_i(\mathbf{Z})| < \infty$ . Then

$$E[(\mathbf{Z} - \boldsymbol{\theta})^\top \mathbf{g}(\mathbf{Z})] = \sigma^2 E \sum_{i=1}^p \frac{\partial g_i}{\partial z_i}(\mathbf{Z}). \quad (27.5.1)$$

*Proof.*

- (a) Write  $Z = \mu + \sigma W$  with  $W \sim N(0, 1)$  and  $h(w) = g(\mu + \sigma w)$ , so that  $h'(w) = \sigma g'(\mu + \sigma w)$ . The claim becomes  $E[Wh(W)] = E h'(W)$ , so we may take  $\mu = 0$ ,  $\sigma = 1$ ,  $g = h$ . Let  $\varphi$  be the standard normal density. Since  $\varphi'(t) = -t\varphi(t)$ ,

$$\begin{aligned} \varphi(w) &= \int_w^\infty t\varphi(t) dt \quad (w \geq 0), \\ \varphi(w) &= -\int_{-\infty}^w t\varphi(t) dt \quad (w \leq 0). \end{aligned}$$

Hence

$$\begin{aligned} E h'(W) &= \int_0^\infty h'(w) \int_w^\infty t\varphi(t) dt dw \\ &\quad - \int_{-\infty}^0 h'(w) \int_{-\infty}^w t\varphi(t) dt dw. \end{aligned}$$

The double integrals converge absolutely: replacing  $h'$  by  $|h'|$  and  $t$  by  $|t|$  gives  $\int |h'(w)|\varphi(w) dw = E|h'(W)| < \infty$ . By Fubini's theorem we may reverse the order of integration. The first term becomes  $\int_0^\infty t\varphi(t) \int_0^t h'(w) dw dt = \int_0^\infty t\varphi(t)\{h(t) - h(0)\} dt$ , and the second  $-\int_{-\infty}^0 t\varphi(t) \int_t^0 h'(w) dw dt = \int_{-\infty}^0 t\varphi(t)\{h(t) - h(0)\} dt$ , using absolute continuity of  $h$ . Their sum is  $E[W\{h(W) - h(0)\}] = E[Wh(W)]$ , since  $EW = 0$ . The same bound shows  $E|W\{h(W) - h(0)\}| \leq E|h'(W)|$ , so  $E|Wh(W)| < \infty$ .

- (b) The coordinates of  $\mathbf{Z}$  are independent. Fix  $i$  and condition on the other coordinates  $\mathbf{Z}_{-i}$ . By Fubini,  $E[|\partial g_i/\partial z_i| \mid \mathbf{Z}_{-i}] < \infty$  for almost every value of  $\mathbf{Z}_{-i}$ , and for those values part (a), applied to  $z_i \mapsto g_i(\mathbf{Z})$  with  $Z_i \sim N(\theta_i, \sigma^2)$ , gives  $E[(Z_i - \theta_i)g_i(\mathbf{Z}) \mid \mathbf{Z}_{-i}] = \sigma^2 E[\partial g_i(\mathbf{Z})/\partial z_i \mid \mathbf{Z}_{-i}]$ . Take expectations and sum over  $i$ .

■

The lemma turns an expectation involving the unknown  $\boldsymbol{\theta}$  into the expectation of a function of  $\mathbf{Z}$  alone. Stein (1981) built a theory of risk estimation on it ([Exercise \(C2\)](#)), and it computes covariance degrees of freedom ([Exercise \(C1\)](#)) for nonlinear fits such as the lasso ([Exercise \(C1\)](#)).

### 27.5.3 The James–Stein theorem

#### Theorem 27.5.2 — James–Stein

Let  $\mathbf{Z} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$  with  $p \geq 3$  and  $\sigma^2$  known, and for a constant  $c$  let  $\hat{\boldsymbol{\theta}}_c = (1 - c\sigma^2/\|\mathbf{Z}\|^2)\mathbf{Z}$ . Then  $E\|\mathbf{Z}\|^{-2} < \infty$  and:

- (a)  $E\|\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}\|^2 = p\sigma^2 - c\{2(p-2) - c\}\sigma^4 E\|\mathbf{Z}\|^{-2}$ . This is below  $p\sigma^2$  for every  $\boldsymbol{\theta}$  iff  $0 < c < 2(p-2)$ , and smallest at  $c = p-2$ . The **James–Stein estimator** is

$$\hat{\boldsymbol{\theta}}^{\text{JS}} = \left(1 - \frac{(p-2)\sigma^2}{\|\mathbf{Z}\|^2}\right)\mathbf{Z},$$

with risk

$$E\|\hat{\boldsymbol{\theta}}^{\text{JS}} - \boldsymbol{\theta}\|^2 = p\sigma^2 - (p-2)^2\sigma^4 E\|\mathbf{Z}\|^{-2} < p\sigma^2. \quad (27.5.2)$$

- (b) With  $\gamma = \|\boldsymbol{\theta}\|^2/\sigma^2$  and  $K \sim \text{Poisson}(\gamma/2)$ ,  $\sigma^2 E\|\mathbf{Z}\|^{-2} = E\{1/(p-2+2K)\}$ . The risk of  $\hat{\boldsymbol{\theta}}^{\text{JS}}$  depends on  $\boldsymbol{\theta}$  only through  $\gamma$ , equals  $2\sigma^2$  at  $\boldsymbol{\theta} = \mathbf{0}$ , increases strictly in  $\gamma$ , and tends to  $p\sigma^2$  as  $\gamma \rightarrow \infty$ .

*Proof. Finiteness.* The density of  $\mathbf{Z}$  is at most  $(2\pi\sigma^2)^{-p/2}$ , so

$$\begin{aligned} E\|\mathbf{Z}\|^{-2} &\leq (2\pi\sigma^2)^{-p/2} \int_{\|\mathbf{z}\| \leq 1} \|\mathbf{z}\|^{-2} d\mathbf{z} + 1 \\ &= (2\pi\sigma^2)^{-p/2} A_p \int_0^1 r^{p-3} dr + 1 < \infty, \end{aligned}$$

where  $A_p$  is the surface area of the unit sphere in  $\mathbb{R}^p$  and the integral is finite because  $p \geq 3$ . (a) Apply Eq. 27.5.1 to  $\mathbf{g}(\mathbf{z}) = \mathbf{z}/\|\mathbf{z}\|^2$ . For  $p \geq 3$  the other coordinates  $\mathbf{Z}_{-i}$  are nonzero with probability one, and then  $g_i$  is a smooth function of  $z_i$  with

$$\frac{\partial g_i}{\partial z_i} = \frac{1}{\|\mathbf{z}\|^2} - \frac{2z_i^2}{\|\mathbf{z}\|^4}, \quad \left| \frac{\partial g_i}{\partial z_i} \right| \leq \frac{3}{\|\mathbf{z}\|^2}, \quad \sum_i \frac{\partial g_i}{\partial z_i} = \frac{p-2}{\|\mathbf{z}\|^2}.$$

The bound makes  $E|\partial g_i/\partial z_i|$  finite, and  $|(Z_i - \theta_i)g_i(\mathbf{Z})| \leq |Z_i - \theta_i|/\|\mathbf{Z}\|$  has finite expectation by the Cauchy–Schwarz inequality. So  $E[(\mathbf{Z} - \boldsymbol{\theta})^\top \mathbf{Z}/\|\mathbf{Z}\|^2] = (p-2)\sigma^2 E\|\mathbf{Z}\|^{-2}$ . Now expand

$$\|\hat{\boldsymbol{\theta}}_c - \boldsymbol{\theta}\|^2 = \|\mathbf{Z} - \boldsymbol{\theta}\|^2 - 2c\sigma^2 \frac{(\mathbf{Z} - \boldsymbol{\theta})^\top \mathbf{Z}}{\|\mathbf{Z}\|^2} + \frac{c^2\sigma^4}{\|\mathbf{Z}\|^2}$$

and take expectations:  $p\sigma^2 - 2c(p-2)\sigma^4 E\|\mathbf{Z}\|^{-2} + c^2\sigma^4 E\|\mathbf{Z}\|^{-2}$ . The factor  $c\{2(p-2) - c\}$  is positive iff  $0 < c < 2(p-2)$  and largest at  $c = p-2$ , where it equals  $(p-2)^2$ . (b)  $\mathbf{Z}/\sigma \sim N_p(\boldsymbol{\theta}/\sigma, \mathbf{I})$ , so  $U = \|\mathbf{Z}\|^2/\sigma^2 \sim \chi^2(p, \gamma)$  (Definition 4.1.1), and by Theorem 4.1.2(d), given  $K = k$ ,  $U \sim \chi^2(p+2k)$ . For  $m \geq 3$ ,  $E\{1/\chi^2(m)\} = 1/(m-2)$  (Exercise (A1)), which gives the formula. At  $\gamma = 0$ ,  $K = 0$  and the risk is  $p\sigma^2 - (p-2)\sigma^2 = 2\sigma^2$ . Since  $E(1/U) = \int_0^\infty \Pr(U < 1/s) ds$  and each  $\Pr(U < 1/s)$  decreases strictly in  $\gamma$  (Proposition 4.1.3),  $E(1/U)$  decreases strictly and the risk increases. As  $\gamma \rightarrow \infty$ ,  $K \rightarrow \infty$  in probability and  $1/(p-2+2K) \leq 1/(p-2)$ , so  $E\{1/(p-2+2K)\} \rightarrow 0$  by dominated convergence. ■

The shrinkage factor is close to one when  $\|\mathbf{Z}\|^2$  is large compared with its null expectation  $p\sigma^2$ , and small when the data are consistent with  $\boldsymbol{\theta} \approx \mathbf{0}$ . Nothing is special about the origin: shrinking towards any point fixed in advance also dominates  $\mathbf{Z}$ , with the largest gains near that point (Exercise (B1)).

The paradox is that the coordinates need not be related. If  $\theta_1$  is a physical constant,  $\theta_2$  a crop yield and  $\theta_3$  a batting average, the James–Stein estimate of the crop yield depends on the

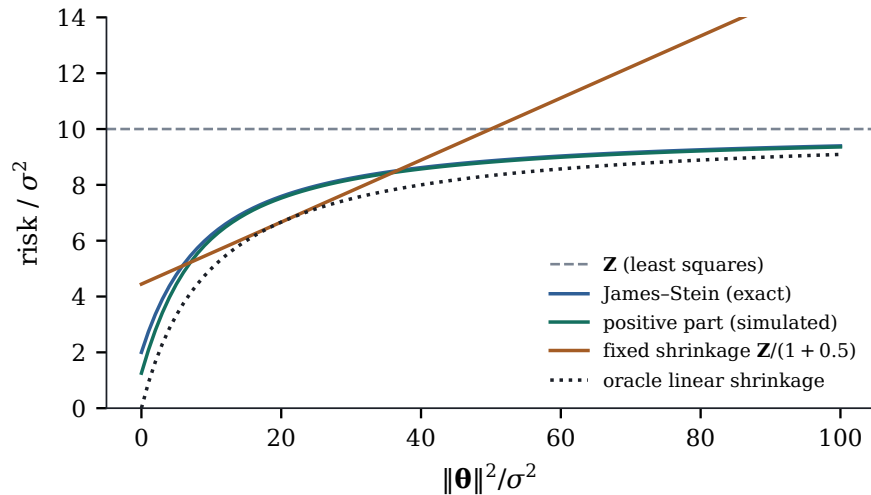


Figure 27.5.1: Risk of estimators of ten normal means with  $\sigma = 1$ , against  $\|\theta\|^2$ : the unbiased estimator  $\mathbf{Z}$ , James–Stein (exact, from the Poisson mixture), its positive part (simulated, 4000 draws per point), a fixed proportional shrinkage, and the best linear shrinkage for each  $\theta$ .

other two measurements. The resolution is that the theorem concerns the *total* risk; a single coordinate's risk can exceed  $\sigma^2$  (Exercise (B2)).

#### Example. (The risk function for ten means)

For  $p = 10$  and  $\sigma = 1$ , the exact risk from Theorem 27.5.2(b) is 2 at  $\theta = \mathbf{0}$ , 4.784 at  $\|\theta\|^2 = 5$ , 7.587 at 20 and 9.257 at 80, always below the constant risk 10 of  $\mathbf{Z}$  (Figure 27.5.1). The fixed shrinkage  $\mathbf{Z}/(1 + 0.5)$ , which is ridge regression with  $\lambda = 0.5$  in an orthonormal design, has risk  $(p + 0.25\|\theta\|^2)/2.25$  (Exercise (A2)). It beats James–Stein only on a middle range,  $5.9 < \|\theta\|^2 < 36.6$ . At the origin it is worse ( $10/2.25 = 4.44$  against 2), it exceeds  $p$  once  $\|\theta\|^2 > 5p = 50$ , and it grows without bound. The dotted curve is the risk  $p\|\theta\|^2/(p + \|\theta\|^2)$  of the best linear shrinkage  $f\mathbf{Z}$  for each  $\theta$ , the oracle of Proposition 27.1.2(c) with a common factor; James–Stein stays within 2 of it without knowing  $\theta$ .

```
import numpy as np
from scipy import stats
p = 10

def js_risk(gamma, p=p):
    """Exact risk of  $(1 - (p-2)/\|Z\|^2) Z$  when  $Z \sim N_p(\theta, I)$  and  $\gamma = \|\theta\|^2$ ."""
    k = np.arange(0, 400)
    w = stats.poisson.pmf(k, gamma / 2)          #  $\|Z\|^2 \sim \chi^2(p, \gamma)$ :
    Poisson(gamma/2) mixture
    return p - (p - 2) * 2 * np.sum(w / (p - 2 + 2 * k))

def js(z, positive=False):
    factor = 1 - (len(z) - 2) / np.sum(z ** 2)
    return (max(factor, 0.0) if positive else factor) * z

rng = np.random.default_rng(27)
for gamma in [0.0, 5.0, 20.0, 80.0]:
    theta = np.sqrt(gamma / p) * np.ones(p)
```

```

Zs = theta + rng.normal(size=(2_000, p))
loss_js = np.mean([np.sum((js(z) - theta) ** 2) for z in Zs])
loss_pp = np.mean([np.sum((js(z, True) - theta) ** 2) for z in Zs])
print(f"||theta||^2 = {gamma:5.1f}: exact JS risk {js_risk(gamma):.3f}, "
      f"simulated {loss_js:.3f}, positive part {loss_pp:.3f} (MLE: {p})")

```

## 27.5.4 The positive part

When  $\|\mathbf{Z}\|^2 < (p-2)\sigma^2$  the James–Stein factor is negative, and the estimator reverses the sign of every coordinate, which is absurd. Truncating the factor at zero helps.

### Proposition 27.5.3 — Positive-part James–Stein

In the setting of [Theorem 27.5.2](#), the estimator

$$\hat{\boldsymbol{\theta}}^{\text{JS}+} = \left(1 - \frac{(p-2)\sigma^2}{\|\mathbf{Z}\|^2}\right)_+ \mathbf{Z}$$

has strictly smaller risk than  $\hat{\boldsymbol{\theta}}^{\text{JS}}$  for every  $\boldsymbol{\theta}$ .

*Proof.* Write  $\hat{\boldsymbol{\theta}}^{\text{JS}} = h\mathbf{Z}$  with  $h = 1 - (p-2)\sigma^2/\|\mathbf{Z}\|^2$ , and let  $\psi = \max(-h, 0) \geq 0$ , a function of  $\|\mathbf{Z}\|$ . Where  $h \geq 0$  the estimators agree. Where  $h < 0$ , the positive part is  $\mathbf{0}$ , and

$$\|h\mathbf{Z} - \boldsymbol{\theta}\|^2 - \|\boldsymbol{\theta}\|^2 = h^2\|\mathbf{Z}\|^2 - 2h\boldsymbol{\theta}^\top\mathbf{Z} = \psi^2\|\mathbf{Z}\|^2 + 2\psi\boldsymbol{\theta}^\top\mathbf{Z}.$$

So the difference of risks is  $\mathbb{E}[\psi^2\|\mathbf{Z}\|^2] + 2\mathbb{E}[\psi\boldsymbol{\theta}^\top\mathbf{Z}]$ . The first term is positive, because  $\Pr(h < 0) = \Pr\{\|\mathbf{Z}\|^2 < (p-2)\sigma^2\} > 0$ . For the second, choose an orthogonal  $\mathbf{Q}$  with  $\mathbf{Q}\boldsymbol{\theta} = m\mathbf{e}_1$ ,  $m = \|\boldsymbol{\theta}\|$ . Then  $\mathbf{w} = \mathbf{Q}\mathbf{Z} \sim N_p(m\mathbf{e}_1, \sigma^2\mathbf{I})$ ,  $\|\mathbf{w}\| = \|\mathbf{Z}\|$  and  $\boldsymbol{\theta}^\top\mathbf{Z} = mw_1$ . Given  $R = \sum_{i \geq 2} w_i^2$ , which is independent of  $w_1 \sim N(m, \sigma^2)$ ,

$$\begin{aligned} \mathbb{E}[\psi w_1 \mid R] &= \int_0^\infty \psi(\sqrt{t^2 + R}) t \{\varphi_\sigma(t - m) - \varphi_\sigma(t + m)\} dt \\ &\geq 0, \end{aligned}$$

where  $\varphi_\sigma$  is the  $N(0, \sigma^2)$  density: the integrand pairs  $t$  with  $-t$ , and  $(t-m)^2 \leq (t+m)^2$  for  $t, m \geq 0$ . (The expectations are finite since  $\psi \leq (p-2)\sigma^2/\|\mathbf{Z}\|^2$ .) Hence  $\mathbb{E}[\psi\boldsymbol{\theta}^\top\mathbf{Z}] \geq 0$  and the risk difference is positive. ■

The positive-part estimator is itself inadmissible, but improvements on it are tiny (Lehmann and Casella 1998, chapter 5). For  $p = 10$  its risk at the origin is  $\mathbb{E}[\{1 - (p-2)/U\}_+^2 U] = 1.258$  with  $U \sim \chi^2(10)$ , against 2 for James–Stein, and the two curves in [Figure 27.5.1](#) merge as negative factors become rare.



## 27.5.5 The empirical Bayes explanation

Efron and Morris (1973) explained the paradox by drawing the means themselves from a normal distribution.

### Proposition 27.5.4 — James–Stein as empirical Bayes

Let  $\boldsymbol{\theta} \sim N_p(\mathbf{0}, \omega^2 \mathbf{I})$  and  $\mathbf{Z} \mid \boldsymbol{\theta} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$ ,  $p \geq 3$ , and put  $B = \sigma^2 / (\sigma^2 + \omega^2)$ .

- (a) The posterior mean is  $E(\boldsymbol{\theta} \mid \mathbf{Z}) = (1-B)\mathbf{Z}$ , and its Bayes risk is  $E\|(1-B)\mathbf{Z} - \boldsymbol{\theta}\|^2 = p\sigma^2(1-B)$ .
- (b) Marginally  $\mathbf{Z} \sim N_p(\mathbf{0}, (\sigma^2 + \omega^2)\mathbf{I})$ , and  $(p-2)\sigma^2/\|\mathbf{Z}\|^2$  is an unbiased estimator of  $B$ .
- (c) The Bayes risk of  $\hat{\boldsymbol{\theta}}^{\text{JS}}$  is  $p\sigma^2(1-B) + 2\sigma^2 B$ .

*Proof.*

- (a)  $(\boldsymbol{\theta}, \mathbf{Z})$  is jointly normal with  $\text{Cov}(\boldsymbol{\theta}, \mathbf{Z}) = \omega^2 \mathbf{I}$  and  $\text{Cov}(\mathbf{Z}) = (\sigma^2 + \omega^2)\mathbf{I}$ . By [Theorem 3.5.2](#),  $\boldsymbol{\theta} \mid \mathbf{Z} \sim N_p(\omega^2(\sigma^2 + \omega^2)^{-1}\mathbf{Z}, \sigma^2\omega^2(\sigma^2 + \omega^2)^{-1}\mathbf{I})$ ; the mean is  $(1-B)\mathbf{Z}$ , and the Bayes risk is the expected trace of the posterior covariance,  $p\sigma^2\omega^2/(\sigma^2 + \omega^2) = p\sigma^2(1-B)$ .
- (b)  $\mathbf{Z} = \boldsymbol{\theta} + (\mathbf{Z} - \boldsymbol{\theta})$  is a sum of independent normal vectors. So  $\|\mathbf{Z}\|^2/(\sigma^2 + \omega^2) \sim \chi^2(p)$ , and  $E\{(p-2)\sigma^2/\|\mathbf{Z}\|^2\} = (p-2)\sigma^2/\{(\sigma^2 + \omega^2)(p-2)\} = B$  by [Exercise \(A1\)](#).
- (c) Average [Eq. 27.5.2](#) over the prior: the Bayes risk is  $p\sigma^2 - (p-2)^2\sigma^4 E\|\mathbf{Z}\|^{-2}$  with the marginal expectation  $E\|\mathbf{Z}\|^{-2} = 1/\{(\sigma^2 + \omega^2)(p-2)\}$ , which gives  $p\sigma^2 - (p-2)\sigma^2 B = p\sigma^2(1-B) + 2\sigma^2 B$ . ■

James–Stein plugs an unbiased estimate of  $B$ , based on the spread of the  $p$  observations, into the Bayes rule, and part (c) shows that not knowing  $\omega^2$  costs at most  $2\sigma^2$  in total. This is the **empirical Bayes** idea: the ensemble of parameters supplies its own prior. [Theorem 27.5.2](#) says more, that the gain is guaranteed even if no such prior exists.

In an orthonormal design ridge regression is the Bayes rule  $\mathbf{Z}/(1+\lambda)$  with  $\omega^2 = \sigma^2/\lambda$  ([Theorem 27.2.2\(e\)](#)). A fixed  $\lambda$  is a guess at the prior variance, and a bad guess is punished without limit, as the fixed-shrinkage curve in [Figure 27.5.1](#) shows; choosing  $\lambda$  from the data is the regression version of James–Stein. With a general design the  $g$ -prior of [Proposition 7.6.3\(c\)](#) shrinks  $\hat{\boldsymbol{\beta}}$  by the common factor  $g/(1+g)$ , and estimating that factor gives the next corollary.

## 27.5.6 Shrinking a regression with unknown variance

In regression  $\sigma^2$  is unknown but estimated independently of  $\hat{\boldsymbol{\beta}}$ , which is enough.

**Corollary 27.5.5 — James–Stein for regression**

Assume the normal linear model with  $\mathbf{X}$  of full column rank  $p \geq 3$ , let SSE have  $m$  degrees of freedom (so  $\text{SSE}/\sigma^2 \sim \chi^2(m)$ , independent of  $\hat{\boldsymbol{\beta}}$ ), and for  $c > 0$  let

$$\tilde{\boldsymbol{\beta}}_c = \left(1 - \frac{c \text{SSE}}{\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}}\right) \hat{\boldsymbol{\beta}}.$$

Then, with  $\mathbf{Z} = (\mathbf{X}^\top \mathbf{X})^{1/2} \hat{\boldsymbol{\beta}}$ ,

$$\mathbb{E}[(\tilde{\boldsymbol{\beta}}_c - \boldsymbol{\beta})^\top \mathbf{X}^\top \mathbf{X} (\tilde{\boldsymbol{\beta}}_c - \boldsymbol{\beta})] = p\sigma^2 - cm \{2(p-2) - c(m+2)\} \sigma^4 \mathbb{E}\|\mathbf{Z}\|^{-2}.$$

So  $\tilde{\boldsymbol{\beta}}_c$  dominates least squares in this loss iff  $0 < c < 2(p-2)/(m+2)$ , and the best constant is  $c = (p-2)/(m+2)$ . In terms of the  $F$  statistic  $F = (\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}/p)/(\text{SSE}/m)$  for the hypothesis  $\boldsymbol{\beta} = \mathbf{0}$ , the shrinkage factor is  $1 - cm/(pF)$ .

*Proof.* By Theorem 7.5.1,  $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta}, \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})$  and SSE is independent of it. So  $\mathbf{Z} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$  with  $\boldsymbol{\theta} = (\mathbf{X}^\top \mathbf{X})^{1/2} \boldsymbol{\beta}$ , and the loss is  $\|(1 - a/\|\mathbf{Z}\|^2)\mathbf{Z} - \boldsymbol{\theta}\|^2$  with  $a = c \text{SSE}$ . Conditionally on SSE,  $a$  is a constant and the computation in the proof of Theorem 27.5.2(a), with  $c\sigma^2$  replaced by  $a$ , gives the conditional risk  $p\sigma^2 - \{2a(p-2)\sigma^2 - a^2\} \mathbb{E}\|\mathbf{Z}\|^{-2}$ . Now  $\mathbb{E}(a) = cm\sigma^2$  and  $\mathbb{E}(a^2) = c^2\sigma^4 \mathbb{E}\{\chi^2(m)^2\} = c^2m(m+2)\sigma^4$  (Theorem 4.1.2(b)). Taking expectations gives the formula, and  $c\{2(p-2) - c(m+2)\}$  is positive iff  $0 < c < 2(p-2)/(m+2)$  and largest at the midpoint. The last statement is  $\text{SSE}/(\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{X} \hat{\boldsymbol{\beta}}) = m/(pF)$ . ■

The factor  $1 - (p-2)m/\{(m+2)pF\}$  is close to  $1 - 1/F$  when  $p$  and  $m$  are large: a regression that clearly fits is barely shrunk, and one indistinguishable from noise is shrunk hard. The loss concerns fitted values, not individual coefficients. Shrinking towards the fit of a reduced model, using its  $F$  statistic, works in the same way when the full model has at least three more parameters (Exercise (C1)).

**Example. (James–Stein for Longley's regression)**

For Longley's six slopes there are  $m = 9$  residual degrees of freedom and the overall  $F$  statistic is 330.3. The James–Stein factor with  $c = (p-2)/(m+2)$  is 0.99835: the regression fits far too well for shrinkage towards zero to help. The script therefore also simulates responses at Longley's design with the least squares estimates divided by 60 as the truth, a weak signal whose noncentrality  $\boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{X} \boldsymbol{\beta}/\sigma^2$  is about 0.6, with  $\sigma$  equal to Longley's  $s$ . Over 20000 simulated data sets the ratio of the James–Stein prediction risk to that of least squares is 0.501.

```
import statsmodels.api as sm
data = sm.datasets.longley.load_pandas().data
Zr = data.drop(columns="TOTEMP").to_numpy()
y = data["TOTEMP"].to_numpy()
n, k = Zr.shape
Xc = Zr - Zr.mean(axis=0)                                # centred slopes; the
intercept is ybar
```

```

yc = y - y.mean()
b = np.linalg.lstsq(Xc, yc, rcond=None)[0]
m = n - k - 1                                # residual degrees of freedom
sse = np.sum((yc - Xc @ b) ** 2)
fit_ss = b @ Xc.T @ Xc @ b                  # b^T X^T X b, the regression
                                             sum of squares
F = (fit_ss / k) / (sse / m)
factor = 1 - (k - 2) / (m + 2) * sse / fit_ss
print(f"F = {F:.1f}; James-Stein factor {factor:.5f} = 1 - (k-2) m / ((m+2) k F)")

```

## 27.5.7 Exercises

### 27.5.7.1 A. Check your understanding

#### Exercise A1.

Show that  $E\{1/\chi^2(m)\} = 1/(m-2)$  for  $m \geq 3$  and that the expectation is infinite for  $m \leq 2$ . Deduce that the risk of the James-Stein estimator at  $\boldsymbol{\theta} = \mathbf{0}$  is  $2\sigma^2$  for every  $p \geq 3$ , however large  $p$  is.

*Solution.* With the  $\chi^2(m)$  density  $x^{m/2-1}e^{-x/2}/\{2^{m/2}\Gamma(m/2)\}$ ,

$$E \frac{1}{\chi^2(m)} = \frac{\int_0^\infty x^{m/2-2} e^{-x/2} dx}{2^{m/2}\Gamma(m/2)} = \frac{2^{m/2-1}\Gamma(m/2-1)}{2^{m/2}\Gamma(m/2)} = \frac{1}{2(m/2-1)} = \frac{1}{m-2},$$

where the integral converges at zero iff  $m/2 - 2 > -1$ , that is,  $m > 2$ . At  $\boldsymbol{\theta} = \mathbf{0}$ ,  $\|\mathbf{Z}\|^2/\sigma^2 \sim \chi^2(p)$ , so Eq. 27.5.2 gives  $p\sigma^2 - (p-2)^2\sigma^2/(p-2) = 2\sigma^2$ .

### 27.5.7.2 B. Practice

#### Exercise B1.

- (a) Show that for a fixed  $\boldsymbol{\theta}_0$ , the estimator  $\boldsymbol{\theta}_0 + (1 - (p-2)\sigma^2/\|\mathbf{Z} - \boldsymbol{\theta}_0\|^2)(\mathbf{Z} - \boldsymbol{\theta}_0)$  has risk  $p\sigma^2 - (p-2)^2\sigma^4 E\|\mathbf{Z} - \boldsymbol{\theta}_0\|^{-2}$ . (b) Lindley's estimator shrinks towards the grand mean:  $\hat{\theta}_i = \bar{Z} + (1 - (p-3)\sigma^2/S)(Z_i - \bar{Z})$  with  $S = \sum_i (Z_i - \bar{Z})^2$ . Show that it dominates  $\mathbf{Z}$  when  $p \geq 4$ .

*Solution.*

- (a)  $\mathbf{Z} - \boldsymbol{\theta}_0 \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I})$ , and the estimation error is that of the James-Stein estimator of  $\boldsymbol{\theta} - \boldsymbol{\theta}_0$  from  $\mathbf{Z} - \boldsymbol{\theta}_0$ ; apply Eq. 27.5.2.
- (b) Choose an orthogonal  $\mathbf{Q}$  whose first row is  $\mathbf{1}^\top/\sqrt{p}$ . In the coordinates  $\mathbf{w} = \mathbf{QZ}$ , the first coordinate is  $\sqrt{p}\bar{Z}$  and the remaining  $p-1$  are a  $N_{p-1}$  vector with squared length  $S$ . The estimator leaves the first coordinate alone and applies James-Stein in dimension  $p-1 \geq 3$  to the others, so its risk is  $\sigma^2$  plus less than  $(p-1)\sigma^2$  (Theorem 27.5.2).

**Exercise B2.**

Let  $p$  be large and  $\boldsymbol{\theta} = (\sqrt{p}\sigma, 0, \dots, 0)^\top$ . Using the approximation  $\|\mathbf{Z}\|^2 \approx \|\boldsymbol{\theta}\|^2 + p\sigma^2$ , show that the James–Stein estimate of  $\theta_1$  has mean squared error of roughly  $p\sigma^2/4$ , far above the  $\sigma^2$  of  $Z_1$ . Why does this not contradict [Theorem 27.5.2](#)?

*Solution.* With  $\|\mathbf{Z}\|^2 \approx 2p\sigma^2$ , the factor is about  $1 - (p-2)/(2p) \approx \frac{1}{2}$ , so  $\hat{\theta}_1 \approx Z_1/2$ , with bias  $-\theta_1/2$  and squared bias  $p\sigma^2/4$ . Halving the other  $p-1$  coordinates, whose means are zero, cuts each of their risks from  $\sigma^2$  to about  $\sigma^2/4$ . The total is roughly  $p\sigma^2/4 + (p-1)\sigma^2/4 \approx p\sigma^2/2 < p\sigma^2$ . One coordinate pays heavily for the others' gains; the theorem only controls the sum.

**Exercise B3.**

In [Proposition 27.5.4](#), show that the ratio of the Bayes risk of James–Stein to that of the Bayes rule is  $1 + 2\sigma^2/(p\omega^2)$ . When is the cost of estimating  $\omega^2$  negligible?

**27.5.7.3 C. Going deeper****Exercise C1.**

Let  $\mathbf{P}$  be the orthogonal projection onto a fixed  $q$ -dimensional subspace of  $\mathbb{R}^p$ , with  $p-q \geq 3$ . Show that  $\mathbf{P}\mathbf{Z} + (1 - (p-q-2)\sigma^2/\|(\mathbf{I}-\mathbf{P})\mathbf{Z}\|^2)(\mathbf{I}-\mathbf{P})\mathbf{Z}$  dominates  $\mathbf{Z}$ , and translate this into shrinking a regression fit towards a nested reduced model using its  $F$  statistic.

**Exercise C2.**

Let  $\hat{\boldsymbol{\theta}} = \mathbf{Z} + \mathbf{g}(\mathbf{Z})$  with  $\mathbf{g}$  satisfying the conditions of [Lemma 27.5.1](#). Show that

$$\hat{R} = p\sigma^2 + \|\mathbf{g}(\mathbf{Z})\|^2 + 2\sigma^2 \sum_i \frac{\partial g_i}{\partial z_i}(\mathbf{Z})$$

is unbiased for the risk of  $\hat{\boldsymbol{\theta}}$  (Stein's unbiased risk estimate). Compute it for James–Stein and for soft thresholding  $\hat{\theta}_i = S_\lambda(Z_i)$ ; how could it choose  $\lambda$ ?

## 27.6 Summary and Notes

### 27.6.1 Summary

*Idea.*

1. Mean squared error is variance plus squared bias. A linear estimator beats least squares in the matrix sense iff its squared bias, measured in the metric of the covariance it saves, is at most one ([Theorem 27.1.1](#)); none wins for every  $\boldsymbol{\beta}$  ([Exercise](#)

- (B1)).
2. Shrinking the  $j$ th canonical coordinate is governed by its signal-to-noise ratio  $\tau_j^2 = d_j^2 \alpha_j^2 / \sigma^2$ : the oracle factor is  $\tau_j^2 / (1 + \tau_j^2)$ , and dropping the direction helps iff  $\tau_j^2 < 1$  ([Proposition 27.1.2](#)).
  3. Ridge regression shrinks the  $j$ th direction by  $d_j^2 / (d_j^2 + \lambda)$ , has effective degrees of freedom  $\sum_j d_j^2 / (d_j^2 + \lambda)$ , and is the posterior mean under an exchangeable normal prior ([Theorem 27.2.2](#)). Every  $\lambda < 2\sigma^2 / \max_j \alpha_j^2$  beats least squares, and every  $\lambda \leq 2\sigma^2 / \|\boldsymbol{\beta}\|^2$  does so in the matrix sense ([Theorem 27.2.3](#)).
  4. Principal component regression keeps the leading  $k$  directions, and beats least squares in the matrix sense iff the dropped directions carry total signal  $\sum_{j>k} \tau_j^2 \leq 1$ . The directions worth dropping are those with small  $\tau_j^2$ , not small  $d_j^2$  ([Proposition 27.3.1](#), [Example \(Principal components of Longley's regressors\)](#)). Tuning or selecting from the data makes the minimized leave-one-out error optimistic; nested cross-validation is honest.
  5. The lasso minimizes  $\frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda \|\mathbf{b}\|_1$ . Its solutions are the solutions of the KKT conditions and share one fitted vector ([Definition 27.4.1](#), [Theorem 27.4.1](#)). Under an orthonormal design it is soft thresholding, against proportional shrinkage for ridge and hard thresholding for subset selection ([Theorem 27.4.2](#)). Coordinate descent computes it by repeated soft thresholding ([Lemma 27.4.4](#)).
  6. By simulation, best subset selection wins when a few coefficients are large, ridge wins when many are moderate, and the lasso sits between ([Example \(Three coefficient patterns\)](#)).
  7. With Stein's lemma ([Lemma 27.5.1](#)), the James–Stein estimator  $(1 - (p-2)\sigma^2 / \|\mathbf{Z}\|^2) \mathbf{Z}$  has risk below  $p\sigma^2$  for every  $\boldsymbol{\theta}$  when  $p \geq 3$  ([Theorem 27.5.2](#)), its positive part does better still ([Proposition 27.5.3](#)), and in regression, with an estimated variance, the factor is about  $1 - 1/F$  ([Corollary 27.5.5](#)).
  8. James–Stein is an empirical Bayes rule: it estimates the shrinkage factor  $\sigma^2 / (\sigma^2 + \omega^2)$  of a normal prior without bias, and pays at most  $2\sigma^2$  for not knowing  $\omega^2$  ([Proposition 27.5.4](#)). Ridge with a fixed  $\lambda$  is the same rule with a guessed prior variance.

## 27.6.2 Notes and sources

**Coverage.** Christensen (2020, sections 13.3–13.5) treats canonical form, principal component and ridge regression; Sen and Srivastava (1990, chapter 12) add a Stein-type shrinkage of least squares; Fahrmeir, Kneib, Lang and Marx (2021, section 4.2) present ridge and the lasso as penalized least squares. This chapter organizes them around the shrinkage factors of [Proposition 27.1.2](#). Examples, simulations and exercises are the book's own.

**Mean squared error.** The matrix comparison of estimators and its equivalence with every quadratic loss are due to Theobald (1974), who used them for the matrix-sense part of [Theorem 27.2.3](#).

**Ridge regression.** Hoerl and Kennard (1970) named the method, introduced the ridge trace and proved the existence theorem. The same regularization of ill-posed problems is due to Tikhonov (Tikhonov and Arsenin 1977). Generalized cross-validation for ridge is due to Golub, Heath and Wahba (1979). Hoerl, Kennard and Baldwin (1975) proposed the plug-in choice of  $\lambda$ , and Draper and Van Nostrand (1979) review the debate over data-chosen  $\lambda$ .

**Principal components.** Hotelling (1933) introduced principal components and Massy (1965) regression on them. The warnings about low-variance components are from Jolliffe (1982) and Hadi and Ling (1998); Jolliffe (2002) is the standard monograph. Frank and Friedman (1993) compare principal components, partial least squares and ridge regression.

**The lasso.** Tibshirani (1996) introduced the lasso, Chen, Donoho and Saunders (1998) the same criterion as basis pursuit, and Park and Casella (2008) a Bayesian version. Tibshirani (2013) treats uniqueness and sparsity of solutions ([Exercise \(C2\)](#)). Path algorithms are due to Osborne, Presnell and Turlach (2000) and Efron, Hastie, Johnstone and Tibshirani (2004); coordinate descent to Fu (1998), Friedman, Hastie, Höfling and Tibshirani (2007) and Friedman, Hastie and Tibshirani (2010), with convergence theory by Tseng (2001). Donoho and Johnstone (1994) analysed soft thresholding, and Hastie, Tibshirani and Tibshirani (2020) compare subset selection and the lasso at scale. Hastie, Tibshirani and Wainwright (2015) and Hastie, Tibshirani and Friedman (2009, chapter 3) cover these methods for prediction.

**Stein's paradox.** The inadmissibility of  $\mathbf{Z}$  for  $p \geq 3$  is due to Stein (1956), and the explicit estimator, including an estimated variance, to James and Stein (1961). The proof through [Lemma 27.5.1](#) follows Stein (1981); the empirical Bayes reading is from Efron and Morris (1973). Lehmann and Casella (1998, chapter 5) discuss admissibility and the positive-part estimator; Efron and Hastie (2016, chapter 7) link James–Stein and ridge regression.

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# Chapter 28

## High-Dimensional Regression: $p > n$

Every chapter so far has assumed, openly or quietly, that there are more observations than coefficients. Once  $p > n$  the columns of  $\mathbf{X}$  cannot be linearly independent, least squares fits the data exactly whatever they are, and no individual coefficient is estimable. This chapter asks what can still be done. The first answer is to keep least squares and pick the shortest of its many solutions. Its risk can be computed exactly, and it produces the “double descent” curve. The second answer is to assume that only a few coefficients matter and to use the lasso. Here a restricted form of the eigenvalue conditions of Part II replaces full rank, and short non-asymptotic proofs show that the lasso predicts and estimates almost as well as an oracle that knew which coefficients were zero. Selecting exactly the right variables needs a stronger condition, which can fail even when prediction is good. The chapter ends with inference. Intervals computed after selection are wrong, and a one-step correction of the lasso restores approximately normal estimates of single coefficients.

**What you need.** [Chapter 1](#) (the singular value decomposition, [Theorem 1.8.1](#), and the Moore–Penrose inverse, [Definition 1.3.2](#)), [Chapter 6](#) (projections, minimum-norm least squares, [Proposition 6.10.1](#), and the Frisch–Waugh–Lovell theorem, [Theorem 6.6.2](#)), [Chapter 8](#) (estimable functions, [Theorem 8.2.2](#)), [Chapter 10](#) (least squares through the SVD, [Theorem 10.4.1](#)), [Chapter 13](#) (the union bound, [Theorem 13.4.1](#)) and [Chapter 14](#) (random regressors and the inverse Wishart mean, [Lemma 14.2.4](#)). From this part: the lasso, its optimality conditions and soft thresholding ([Chapter 27](#), [Definition 27.4.1](#) and [Theorem 27.4.2](#)) and ridge regression ([Theorem 27.2.2](#)). Selection bias and cross-validation ([Chapter 29](#)) are referred to but not needed.

### 28.0.1 Roadmap

- [Minimum-norm interpolation](#)
- [Restricted eigenvalue conditions](#)
- [Prediction and estimation bounds for the lasso](#)
- [Support recovery](#)
- [Post-selection and debiased inference](#)
- [Summary and notes](#)



## 28.1 Minimum-norm interpolation

Gene expression studies measure tens of thousands of genes on a few hundred patients, and a model with many splines, interactions or random features easily has more columns than rows. This section asks what least squares does when  $p > n$ . It does not fail, but it no longer picks out one answer, and we study the answer that the SVD and the common algorithms return.

### 28.1.1 Least squares when $p$ exceeds $n$

Suppose  $\text{rank}(\mathbf{X}) = n < p$ , the usual case (a matrix with continuously distributed entries has full rank with probability one). Three consequences follow from Parts I and II.

First,  $\mathcal{C}(\mathbf{X}) = \mathbb{R}^n$ , so  $\mathbf{M} = \mathbf{I}$ : every least squares fit reproduces the data, the residuals are zero,  $R^2 = 1$ , and the estimator  $s^2$  of [Theorem 5.6.1](#) has  $n - r = 0$  in its denominator. The data contain no internal information about  $\sigma^2$ .

Second,  $\mathcal{N}(\mathbf{X})$  has dimension  $p - n \geq 1$ , and the least squares estimates form the affine set  $\mathbf{X}^+\mathbf{y} + \mathcal{N}(\mathbf{X})$  ([Theorem 10.4.1](#)). Coefficient vectors that differ by an element of  $\mathcal{N}(\mathbf{X})$  give the same mean, so the data cannot tell them apart.

Third,  $\lambda^\top \beta$  is estimable iff  $\lambda \in \mathcal{C}(\mathbf{X}^\top)$  ([Theorem 8.2.2](#)), and a coordinate vector  $\mathbf{e}_j$  lies in the  $n$ -dimensional row space only if it is orthogonal to  $\mathcal{N}(\mathbf{X})$ , which fails for generic designs. For a generic  $\mathbf{X}$  with  $p > n$ , *no single coefficient is estimable*. Any method that answers questions about individual coefficients is adding information, explicitly or through its algorithm.

#### Example. (Two interpolators)

Take  $n = 4$  observations on  $p = 6$  regressors, with the design of the listing below and  $\mathbf{y} = (1, 2, 0, -1)^\top$ . The minimum-norm solution  $\hat{\beta}^+ = \mathbf{X}^+\mathbf{y}$  ([Proposition 1.4.3](#), [Proposition 6.10.1](#)) has length 1.322. Adding twice a unit vector of  $\mathcal{N}(\mathbf{X})$  gives a second solution of length 2.398. Both reproduce  $\mathbf{y}$  exactly. They disagree about the coefficients: the first, for instance, is  $-0.002$  in one and  $-0.059$  in the other. They also disagree about predictions at a point that was not observed: at  $\mathbf{x}_0 = \mathbf{1}$  they predict 2.131 and 2.588. The data give no reason to prefer either. The minimum-norm solution is singled out by a choice, not by the data.

```
import numpy as np

rng = np.random.default_rng(2801)
n, p = 4, 6
X = np.round(rng.normal(size=(n, p)), 1)          # a 4 x 6 design, rank 4
y = np.array([1.0, 2.0, 0.0, -1.0])

b_plus = X.T @ np.linalg.solve(X @ X.T, y)        # X^T (X X^T)^{-1} y
print("min-norm solution ", np.round(b_plus, 3))
print("residual norm      ", np.linalg.norm(y - X @ b_plus))

# another interpolator: add a null-space vector of X
_, _, Vt = np.linalg.svd(X)
b_other = b_plus + 2.0 * Vt[-1]                    # Vt[-1] spans part of N(X)
print("other solution     ", np.round(b_other, 3))
```

```

print("residual norm      ", np.linalg.norm(y - X @ b_other))
print("norms", np.linalg.norm(b_plus), np.linalg.norm(b_other))

x0 = np.ones(p) # a new point
print("predictions at x0:", x0 @ b_plus, x0 @ b_other)

```

## 28.1.2 The minimum-norm solution

In this chapter the singular values of  $\mathbf{X}$  are written  $d_1 \geq \dots \geq d_r > 0$ , so that  $\sigma$  stays free for the error standard deviation. By [Theorem 1.8.1](#),

$$\mathbf{X} = \sum_{i=1}^r d_i \mathbf{u}_i \mathbf{v}_i^\top, \quad \mathbf{X}^+ = \sum_{i=1}^r d_i^{-1} \mathbf{v}_i \mathbf{u}_i^\top,$$

with orthonormal  $\mathbf{u}_1, \dots, \mathbf{u}_r \in \mathbb{R}^n$  and  $\mathbf{v}_1, \dots, \mathbf{v}_r \in \mathbb{R}^p$ . The matrix

$$\mathbf{P} = \mathbf{X}^+ \mathbf{X} = \sum_{i=1}^r \mathbf{v}_i \mathbf{v}_i^\top \quad (28.1.1)$$

is the orthogonal projection onto the row space  $\mathcal{C}(\mathbf{X}^\top) = \text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r)$ . The next theorem collects what is known about  $\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^+ \mathbf{y}$  when  $p > n$ . Part (a) restates earlier results in the form needed here. Parts (b) and (c) show that  $\hat{\boldsymbol{\beta}}^+$  is what two common procedures produce when they are pushed to the limit. Part (d) gives its bias and variance.

### Theorem 28.1.1 — The minimum-norm interpolator

Let  $\mathbf{X}$  be  $n \times p$  with rank  $r$  and singular values  $d_1 \geq \dots \geq d_r > 0$ , and let  $\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^+ \mathbf{y}$ .

- (a) (*Characterization.*) The least squares estimates are the vectors  $\hat{\boldsymbol{\beta}}^+ + \mathbf{v}$  with  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ . Among them  $\hat{\boldsymbol{\beta}}^+$  is the only one in  $\mathcal{C}(\mathbf{X}^\top)$ , and it has the smallest Euclidean norm. If  $r = n$ , then  $\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^\top (\mathbf{X} \mathbf{X}^\top)^{-1} \mathbf{y}$  and every least squares estimate interpolates:  $\mathbf{X} \hat{\boldsymbol{\beta}}^+ = \mathbf{y}$ .
- (b) (*Ridgeless limit.*) For  $\lambda > 0$  let  $\hat{\boldsymbol{\beta}}_\lambda = (\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top \mathbf{y}$  be the ridge estimate. Then  $\|\hat{\boldsymbol{\beta}}_\lambda - \hat{\boldsymbol{\beta}}^+\| \leq (\lambda/d_r^2) \|\hat{\boldsymbol{\beta}}^+\|$ , so  $\hat{\boldsymbol{\beta}}_\lambda \rightarrow \hat{\boldsymbol{\beta}}^+$  as  $\lambda \downarrow 0$ .
- (c) (*Gradient descent.*) Let  $\mathbf{b}_0 \in \mathbb{R}^p$ ,  $0 < \eta < 2/d_1^2$  and  $\mathbf{b}_{t+1} = \mathbf{b}_t + \eta \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \mathbf{b}_t)$ . Then  $\mathbf{b}_t \rightarrow \hat{\boldsymbol{\beta}}^+ + (\mathbf{I} - \mathbf{P}) \mathbf{b}_0$ . In particular, gradient descent started at  $\mathbf{b}_0 = \mathbf{0}$  converges to  $\hat{\boldsymbol{\beta}}^+$ .
- (d) (*Bias and variance.*) If  $\mathbf{E}(\mathbf{Y}) = \mathbf{X} \boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , then  $\mathbf{E}(\mathbf{X}^+ \mathbf{Y}) = \mathbf{P} \boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{X}^+ \mathbf{Y}) = \sigma^2 \sum_{i \leq r} d_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$ . For every  $\mathbf{a} \in \mathbb{R}^p$ ,

$$\mathbf{E}(\mathbf{a}^\top \mathbf{X}^+ \mathbf{Y} - \mathbf{a}^\top \boldsymbol{\beta})^2 = (\mathbf{a}^\top (\mathbf{I} - \mathbf{P}) \boldsymbol{\beta})^2 + \sigma^2 \sum_{i=1}^r \frac{(\mathbf{a}^\top \mathbf{v}_i)^2}{d_i^2}, \quad (28.1.2)$$

and  $\mathbf{E} \|\mathbf{X}^+ \mathbf{Y} - \boldsymbol{\beta}\|^2 = \|(\mathbf{I} - \mathbf{P}) \boldsymbol{\beta}\|^2 + \sigma^2 \sum_{i \leq r} d_i^{-2}$ .

*Proof.*

- (a) The first two statements are [Theorem 10.4.1](#), and [Proposition 1.4.3](#) applied to the normal equations of [Theorem 6.4.2](#);  $\hat{\beta}^+ = \sum_i (\mathbf{u}_i^\top \mathbf{y} / d_i) \mathbf{v}_i$  lies in  $\text{span}(\mathbf{v}_1, \dots, \mathbf{v}_r) = \mathcal{C}(\mathbf{X}^\top)$ . If  $r = n$ , then  $\mathbf{X}\mathbf{X}^\top$  is a nonsingular  $n \times n$  matrix, and  $\mathbf{X}^\top(\mathbf{X}\mathbf{X}^\top)^{-1}$  satisfies the four conditions of [Definition 1.3.2](#), so it equals  $\mathbf{X}^+$ . Moreover  $\mathcal{C}(\mathbf{X}) = \mathbb{R}^n$ , so the projection onto  $\mathcal{C}(\mathbf{X})$  is  $\mathbf{I}$  and every least squares fit equals  $\mathbf{y}$ .
- (b) Write  $c_i = \mathbf{u}_i^\top \mathbf{y}$ . By [Theorem 27.2.2\(a\)](#),  $\hat{\beta}_\lambda = \sum_{i \leq r} d_i c_i \mathbf{v}_i / (d_i^2 + \lambda)$ , and [Theorem 27.2.2\(d\)](#) gives the limit  $\hat{\beta}^+ = \sum_{i \leq r} (c_i / d_i) \mathbf{v}_i$ ; only the rate is new. Subtracting,  $\hat{\beta}^+ - \hat{\beta}_\lambda = \sum_{i \leq r} \{\lambda / (d_i^2 + \lambda)\} (c_i / d_i) \mathbf{v}_i$ , so by orthonormality  $\|\hat{\beta}^+ - \hat{\beta}_\lambda\|^2 \leq (\lambda / d_r^2)^2 \sum_i (c_i / d_i)^2 = (\lambda / d_r^2)^2 \|\hat{\beta}^+\|^2$ .
- (c) Complete  $\mathbf{v}_1, \dots, \mathbf{v}_r$  to an orthonormal basis of  $\mathbb{R}^p$  with vectors  $\mathbf{v}_{r+1}, \dots, \mathbf{v}_p$  spanning  $\mathcal{N}(\mathbf{X})$ . Since  $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b}) = \sum_{i \leq r} d_i (c_i - d_i \mathbf{v}_i^\top \mathbf{b}) \mathbf{v}_i$ , the coordinates of  $\mathbf{b}_t$  evolve separately:

$$\mathbf{v}_i^\top \mathbf{b}_{t+1} = (1 - \eta d_i^2) \mathbf{v}_i^\top \mathbf{b}_t + \eta d_i c_i \quad (i \leq r), \quad \mathbf{v}_i^\top \mathbf{b}_{t+1} = \mathbf{v}_i^\top \mathbf{b}_t \quad (i > r).$$

For  $i \leq r$ , subtracting the fixed point  $c_i / d_i$  gives  $\mathbf{v}_i^\top \mathbf{b}_t - c_i / d_i = (1 - \eta d_i^2)^t (\mathbf{v}_i^\top \mathbf{b}_0 - c_i / d_i) \rightarrow 0$ , because  $0 < \eta d_i^2 < 2$ . The coordinates with  $i > r$  never move. So  $\mathbf{b}_t \rightarrow \sum_{i \leq r} (c_i / d_i) \mathbf{v}_i + \sum_{i > r} (\mathbf{v}_i^\top \mathbf{b}_0) \mathbf{v}_i = \hat{\beta}^+ + (\mathbf{I} - \mathbf{P}) \mathbf{b}_0$ .

- (d)  $\mathbf{E}(\mathbf{X}^+ \mathbf{Y}) = \mathbf{X}^+ \mathbf{X} \boldsymbol{\beta} = \mathbf{P} \boldsymbol{\beta}$ , and  $\text{Cov}(\mathbf{X}^+ \mathbf{Y}) = \sigma^2 \mathbf{X}^+ (\mathbf{X}^+)^T = \sigma^2 \sum_i d_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$  ([Theorem 2.2.1](#)). The mean squared error of  $\mathbf{a}^\top \mathbf{X}^+ \mathbf{Y}$  is its squared bias  $(\mathbf{a}^\top (\mathbf{P} - \mathbf{I}) \boldsymbol{\beta})^2$  plus its variance, which is [Eq. 28.1.2](#). Summing [Eq. 28.1.2](#) over  $\mathbf{a} = \mathbf{e}_1, \dots, \mathbf{e}_p$  gives the last formula. ■

Parts (b) and (c) explain why  $\hat{\beta}^+$  matters although nobody chooses it on purpose. Gradient iterates started at zero never leave  $\mathcal{C}(\mathbf{X}^\top)$ , because every gradient  $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b})$  lies in it. Algorithms for large models are typically started near zero and run until the training error vanishes, and in the linear case they return the minimum-norm interpolator. This is called the *implicit regularization* of the algorithm.

```
for lam in [1.0, 0.1, 0.01, 0.001]:
    b_ridge = np.linalg.solve(X.T @ X + lam * np.eye(p), X.T @ y)
    print(f"lambda = {lam:6.3f}    distance to min-norm {np.linalg.norm(b_ridge -
    b_plus):.2e}")

# gradient descent on ||y - X b||^2 / 2 started at zero
step = 1.0 / np.linalg.norm(X, 2) ** 2
b = np.zeros(p)
for t in range(5000):
    b = b + step * X.T @ (y - X @ b)
print("gradient descent: distance to min-norm", np.linalg.norm(b - b_plus))
```

For the design of [Example \(Two interpolators\)](#) the ridge estimate at  $\lambda = 0.01$  is within 0.0132 of  $\hat{\beta}^+$ , and the distance shrinks in proportion to  $\lambda$ , as part (b) predicts.

In part (d), the bias  $(\mathbf{I} - \mathbf{P}) \boldsymbol{\beta}$  is the component of  $\boldsymbol{\beta}$  in the null space, which the data cannot see and the minimum-norm solution sets to zero. The variance is the familiar  $\sigma^2 / d_i^2$  of [Theo-](#)

rem 10.4.1, summed over the directions the data do see. So  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}^+$  is unbiased for all  $\boldsymbol{\beta}$  exactly when  $\mathbf{a}^\top \boldsymbol{\beta}$  is estimable (Exercise (A2)). For any other  $\mathbf{a}$ , its bias depends on the invisible part of  $\boldsymbol{\beta}$  and can be anything.

### 28.1.3 The risk with random regressors

To say whether interpolation predicts well, the design must be random, since prediction is at new regressor values. The cleanest case has independent standard normal regressors, where both terms of part (d) can be averaged exactly.

#### Proposition 28.1.2 — Risk under isotropic regressors

Let the rows of  $\mathbf{X}$  be independent  $N_p(\mathbf{0}, \mathbf{I}_p)$  vectors, let  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$  independent of  $\mathbf{X}$ , and let  $\mathbf{x}_0 \sim N_p(\mathbf{0}, \mathbf{I}_p)$  be independent of both. The prediction risk  $\mathcal{R} = E(\mathbf{x}_0^\top \mathbf{X}^+ \mathbf{Y} - \mathbf{x}_0^\top \boldsymbol{\beta})^2$  is

$$\mathcal{R} = \begin{cases} \sigma^2 \frac{p}{n-p-1}, & p \leq n-2, \\ \|\boldsymbol{\beta}\|^2 \left(1 - \frac{n}{p}\right) + \sigma^2 \frac{n}{p-n-1}, & p \geq n+2. \end{cases} \quad (28.1.3)$$

*Proof.* Given  $\mathbf{X}$  and  $\mathbf{Y}$ , the vector  $\hat{\boldsymbol{\beta}}^+ = \mathbf{X}^+ \mathbf{Y}$  is fixed and  $E\{(\mathbf{x}_0^\top (\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta}))^2 \mid \mathbf{X}, \mathbf{Y}\} = \|\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta}\|^2$ , because  $\text{Cov}(\mathbf{x}_0) = \mathbf{I}$ . So  $\mathcal{R} = E\|\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta}\|^2$ , and by Theorem 28.1.1(d) applied conditionally on  $\mathbf{X}$ ,

$$\mathcal{R} = E\|(\mathbf{I} - \mathbf{P})\boldsymbol{\beta}\|^2 + \sigma^2 E \sum_{i=1}^r d_i^{-2}.$$

*Case  $p \leq n-2$ .*  $\mathbf{X}^\top \mathbf{X} = \sum_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$  has the Wishart law  $W_p(n, \mathbf{I})$  (Definition 14.2.1), so by Lemma 14.2.4 it is nonsingular with probability one and  $E(\mathbf{X}^\top \mathbf{X})^{-1} = \mathbf{I}/(n-p-1)$ . Then  $r = p$ ,  $\mathbf{P} = \mathbf{I}$ , the bias term vanishes, and  $\sum_i d_i^{-2} = \text{tr}(\mathbf{X}^\top \mathbf{X})^{-1}$  has mean  $p/(n-p-1)$ .

*Case  $p \geq n+2$ .* The  $p$  columns of  $\mathbf{X}$  are independent  $N_n(\mathbf{0}, \mathbf{I}_n)$  vectors, so  $\mathbf{X}\mathbf{X}^\top \sim W_n(p, \mathbf{I}_n)$ . By Lemma 14.2.4 again,  $\mathbf{X}\mathbf{X}^\top$  is nonsingular with probability one, so  $r = n$ . The numbers  $d_i^2$  are also the eigenvalues of  $\mathbf{X}\mathbf{X}^\top$ , so  $\sum_i d_i^{-2} = \text{tr}(\mathbf{X}\mathbf{X}^\top)^{-1}$ , with mean  $n/(p-n-1)$ . For the bias, let  $\mathbf{Q}$  be any orthogonal  $p \times p$  matrix. The rows of  $\mathbf{X}\mathbf{Q}$  are independent  $N_p(\mathbf{0}, \mathbf{Q}^\top \mathbf{Q}) = N_p(\mathbf{0}, \mathbf{I})$  vectors (Theorem 3.3.1), so  $\mathbf{X}\mathbf{Q}$  has the same law as  $\mathbf{X}$ . One checks from Definition 1.3.2 that  $(\mathbf{X}\mathbf{Q})^+ = \mathbf{Q}^\top \mathbf{X}^+$ , so the row-space projection of  $\mathbf{X}\mathbf{Q}$  is  $\mathbf{Q}^\top \mathbf{P} \mathbf{Q}$ . Hence  $\bar{\mathbf{P}} = \mathbf{E} \mathbf{P}$  satisfies  $\bar{\mathbf{P}} = \mathbf{Q}^\top \bar{\mathbf{P}} \mathbf{Q}$  for every orthogonal  $\mathbf{Q}$ . Then  $\mathbf{u}^\top \bar{\mathbf{P}} \mathbf{u}$  takes the same value  $c$  at every unit vector  $\mathbf{u}$ , since any unit vector is  $\mathbf{Q} \mathbf{e}_1$  for some  $\mathbf{Q}$ . A symmetric matrix whose quadratic form is constant on the unit sphere is  $c\mathbf{I}$ , by Theorem 1.7.7. Taking traces,  $pc = E \text{tr} \mathbf{P} = E r = n$ . Therefore  $E\|(\mathbf{I} - \mathbf{P})\boldsymbol{\beta}\|^2 = \boldsymbol{\beta}^\top (\mathbf{I} - E \mathbf{P}) \boldsymbol{\beta} = \|\boldsymbol{\beta}\|^2 (1 - n/p)$ . ■

Below  $p = n$  the fit is unbiased and its variance grows like  $p/(n-p-1)$ , which explodes as  $p$

approaches  $n$ . Above  $p = n$  the variance *falls* again, like  $n/(p - n - 1)$ . The fit has to interpolate  $n$  noisy values, and with many more directions available the interpolator can spread the noise over them thinly. The price is the bias: the fit recovers only the fraction  $n/p$  of  $\|\boldsymbol{\beta}\|^2$ . The formulas do not cover  $|p - n| \leq 1$ .

## 28.1.4 Double descent

In applications  $p$  is usually a choice, such as the number of basis functions. To compare choices of  $p$  on one problem, let the response depend on a pool of  $D$  features and let the fit use only the first  $p$ . The features left out act as extra noise.

### Corollary 28.1.3 — Fitting the first $p$ of $D$ features

Let  $\mathbf{x} \sim N_D(\mathbf{0}, \mathbf{I}_D)$ ,  $Y = \mathbf{x}^\top \boldsymbol{\beta} + \varepsilon$  with  $\varepsilon \sim N(0, \sigma^2)$  independent of  $\mathbf{x}$ , and let  $n$  independent copies be observed. For  $p \leq D$ , split  $\boldsymbol{\beta} = (\boldsymbol{\beta}_{(1)}^\top, \boldsymbol{\beta}_{(2)}^\top)^\top$  into its first  $p$  and last  $D - p$  entries, write  $\omega^2 = \|\boldsymbol{\beta}_{(2)}\|^2$ , fit the minimum-norm least squares estimate  $\hat{\boldsymbol{\beta}}^+$  (Proposition 6.10.1) to the first  $p$  features, and predict with  $(\hat{\boldsymbol{\beta}}^+, \mathbf{0}^\top)^\top$ . With  $\sigma_p^2 = \sigma^2 + \omega^2$ , the excess prediction risk at an independent new  $\mathbf{x}_0$  is  $\omega^2$  plus the right-hand side of Eq. 28.1.3 with  $\boldsymbol{\beta}_{(1)}$  in place of  $\boldsymbol{\beta}$  and  $\sigma_p^2$  in place of  $\sigma^2$ .

*Proof.* Split  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$  in the same way. Then  $\mathbf{Y} = \mathbf{X}_1 \boldsymbol{\beta}_{(1)} + (\mathbf{X}_2 \boldsymbol{\beta}_{(2)} + \boldsymbol{\varepsilon})$ . The entries of  $\mathbf{X}_2$  are independent of  $\mathbf{X}_1$ , so the bracket is a  $N_n(\mathbf{0}, \sigma_p^2 \mathbf{I})$  vector independent of  $\mathbf{X}_1$ . Proposition 28.1.2 applies with  $\mathbf{X}_1$ ,  $\boldsymbol{\beta}_{(1)}$  and  $\sigma_p^2$ , and gives  $E\|\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta}_{(1)}\|^2$ . At the new point, the prediction error  $\mathbf{x}_{0,1}^\top (\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta}_{(1)}) - \mathbf{x}_{0,2}^\top \boldsymbol{\beta}_{(2)}$  is a sum of two uncorrelated terms with mean zero, so its mean square is  $E\|\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta}_{(1)}\|^2 + \omega^2$ . ■

### Example. (Two shapes of signal)

Take  $n = 50$  observations, a pool of  $D = 200$  features,  $\sigma^2 = 0.25$  and  $\|\boldsymbol{\beta}\|^2 = 1$ . Figure 28.1.1 plots the exact risk of Corollary 28.1.3 against  $p$  for two coefficient vectors, with simulated values as a check. It also shows the risk of ridge regression (Theorem 27.2.2) with the best penalty for each  $p$ .

In panel (a) all 200 coefficients are equal, so each feature left out adds to the noise. No model with  $p < n$  beats predicting zero, which has risk 1.000. The risk climbs to 12.31 at  $p = 45$ , is 12.94 at  $p = 55$ , and then falls. The overall minimum is at  $p = 200$ : 0.834. Here the best prediction comes from interpolating with every feature.

In panel (b) the coefficients decay geometrically, and the classical picture holds: a small model with  $p = 9$  has risk 0.078, and nothing to the right of the peak comes close. At  $p = D$  the risk is 0.834 again. When all features are used, Eq. 28.1.3 depends on  $\boldsymbol{\beta}$  only through  $\|\boldsymbol{\beta}\|$ .

A risk curve that falls, peaks at  $p = n$  and falls again is called **double descent**. Three cautions keep it in proportion.

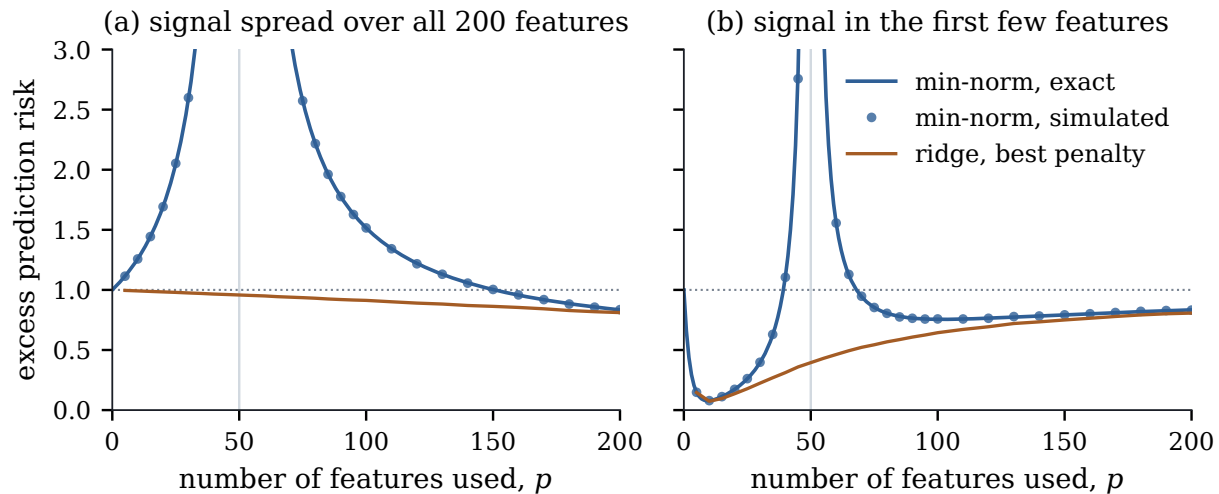


Figure 28.1.1: Excess prediction risk of the minimum-norm least squares fit on the first  $p$  of  $D = 200$  isotropic features, with  $n = 50$  (vertical line) and  $\sigma^2 = 0.25$ : the exact formula of [Corollary 28.1.3](#) (line) and simulation (dots), and ridge regression with the best penalty for each  $p$ . The dotted line is the risk of predicting zero. (a) Equal coefficients. (b) Geometrically decaying coefficients.

- *The peak belongs to the estimator, not to the problem.* It comes from the variance term  $\sigma^2 \sum_i d_i^{-2}$ , large when the smallest nonzero singular value is near zero, as for  $p \approx n$ . Ridge with the best penalty has no peak, and its best values, 0.811 and 0.073, are slightly below the best minimum-norm values, as [Theorem 28.1.1\(b\)](#) requires.
- *The second descent need not win.* That depends on how the signal is spread, as the panels show.
- *Isotropic features are a special case.* Hastie, Montanari, Rosset and Tibshirani (2022) treat general covariances as  $n, p \rightarrow \infty$  with  $p/n$  fixed, and Bartlett, Long, Lugosi and Tsigler (2020) find when interpolation is *benign*, with risk still tending to that of the best predictor.

```
import numpy as np

def exact_risk(p, n, beta, sigma2):
    """Excess prediction risk of min-norm least squares on the first p of D
    isotropic features."""
    kept, left = np.sum(beta[:p] ** 2), np.sum(beta[p:] ** 2)
    noise = sigma2 + left # omitted features act as extra noise
    if p == 0:
        return left
    if p <= n - 2:
        return left + noise * p / (n - p - 1)
    if p >= n + 2:
        return left + kept * (1 - n / p) + noise * n / (p - n - 1)
    return np.inf # |p - n| <= 1: not covered by the formula

def simulated_risk(p, n, beta, sigma2, reps, rng, lams=()):
    """Average over designs of the conditional risk given X; also ridge for each
    penalty."""
    bS, left = beta[:p], np.sum(beta[p:] ** 2)
    noise = sigma2 + left
    out, ridge = [], np.zeros(len(lams))
```

```

for _ in range(reps):
    X = rng.normal(size=(n, p))
    _, d, Vt = np.linalg.svd(X, full_matrices=False)
    c = Vt @ bS                                     # coordinates of beta_S in the
row space
    bias2 = bS @ bS - c @ c                         # ||(I - P) beta_S||^2
    out.append(left + bias2 + noise * np.sum(1 / d ** 2))
    if len(lams):
        shrink = d ** 2 / (d ** 2 + np.asarray(lams)[: , None]) # one row per
penalty
        ridge += left + bias2 + np.sum(((1 - shrink) * c) ** 2, axis=1) \
            + noise * np.sum(shrink ** 2 / d ** 2, axis=1)
return np.mean(out), np.std(out) / np.sqrt(reps), ridge / reps

n, D, sigma2 = 20, 100, 0.25
beta = np.full(D, 1 / np.sqrt(D))                 # signal spread evenly, ||beta
||^2 = 1
rng = np.random.default_rng(2802)
for p in [5, 10, 15, 25, 40, 100]:
    sim, se, _ = simulated_risk(p, n, beta, sigma2, reps=200, rng=rng)
    print(f"p = {p:3d}: formula {exact_risk(p, n, beta, sigma2):.3f} simulation {
sim:.3f}")

```

For the rest of the chapter we take a different route. Instead of letting the algorithm decide which of the many solutions to report, we *assume* that the truth is simple: only a few coefficients are nonzero. That assumption is strong, but it is explicit, and it can be exploited through the lasso.

## 28.1.5 Exercises

### 28.1.5.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{X}$  be  $n \times p$  with an intercept column and  $\text{rank}(\mathbf{X}) = n < p$ . Show that every least squares fit has  $R^2 = 1$ , and explain why neither the  $F$  test of [Theorem 11.1.1](#) nor the residual-based diagnostics of [Chapter 20](#) can be used. What are the leverages  $h_{ii}$ ?

#### Exercise A2.

In the setting of [Theorem 28.1.1\(d\)](#), show that  $\mathbf{a}^\top \mathbf{X}^+ \mathbf{Y}$  is unbiased for  $\mathbf{a}^\top \boldsymbol{\beta}$  for every  $\boldsymbol{\beta}$  iff  $\mathbf{a} \in \mathcal{C}(\mathbf{X}^\top)$ . Show also that for such  $\mathbf{a}$  its variance is  $\sigma^2 \mathbf{a}^\top (\mathbf{X}^\top \mathbf{X})^+ \mathbf{a}$ , and that it equals the least squares estimator  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}$  of [Theorem 6.4.4](#) for any least squares estimate  $\hat{\boldsymbol{\beta}}$ .

*Solution.* The bias is  $\mathbf{a}^\top (\mathbf{P} - \mathbf{I}) \boldsymbol{\beta}$ , which vanishes for all  $\boldsymbol{\beta}$  iff  $(\mathbf{I} - \mathbf{P}) \mathbf{a} = \mathbf{0}$ , that is, iff  $\mathbf{a} \in \mathcal{C}(\mathbf{P}) = \mathcal{C}(\mathbf{X}^\top)$ . The variance is  $\sigma^2 \sum_i (\mathbf{a}^\top \mathbf{v}_i)^2 / d_i^2$ , and  $\sum_i d_i^{-2} \mathbf{v}_i \mathbf{v}_i^\top$  is the Moore–Penrose inverse of  $\mathbf{X}^\top \mathbf{X} = \sum_i d_i^2 \mathbf{v}_i \mathbf{v}_i^\top$  ([Theorem 1.8.1](#)). Finally, any least squares estimate is  $\hat{\boldsymbol{\beta}}^+ + \mathbf{v}$  with  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$ , and  $\mathbf{a}^\top \mathbf{v} = 0$  because  $\mathbf{a} \in \mathcal{C}(\mathbf{X}^\top) = \mathcal{N}(\mathbf{X})^\perp$ , so  $\mathbf{a}^\top \hat{\boldsymbol{\beta}} = \mathbf{a}^\top \hat{\boldsymbol{\beta}}^+$ .

## 28.1.5.2 B. Practice

**Exercise B1.**

Show directly from the update in [Theorem 28.1.1\(c\)](#) that  $(\mathbf{I} - \mathbf{P})\mathbf{b}_t = (\mathbf{I} - \mathbf{P})\mathbf{b}_0$  for all  $t$ . Deduce that gradient descent started at  $\mathbf{b}_0$  returns the least squares estimate closest to  $\mathbf{b}_0$ .

**Exercise B2.**

Fix  $\mathbf{X}$  of rank  $r$  and let  $m(\lambda) = \mathbb{E}\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|^2$  for the ridge estimate, with  $m(0)$  the risk of  $\hat{\boldsymbol{\beta}}^+$  in [Theorem 28.1.1\(d\)](#). Show that

$$m(\lambda) = \|(\mathbf{I} - \mathbf{P})\boldsymbol{\beta}\|^2 + \sum_{i=1}^r \left[ \left( \frac{\lambda}{d_i^2 + \lambda} \right)^2 (\mathbf{v}_i^\top \boldsymbol{\beta})^2 + \sigma^2 \frac{d_i^2}{(d_i^2 + \lambda)^2} \right],$$

and that  $m'(0) < 0$  whenever  $\sigma > 0$ . So some ridge estimate beats the minimum-norm interpolator for every  $\boldsymbol{\beta}$ . Compare [Theorem 27.2.3](#).

*Solution.* The linear representation in [Theorem 27.2.2\(a\)](#) holds for any rank  $r$ . Applied to  $\mathbb{E}(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ , whose coordinates are  $\mathbf{u}_i^\top \mathbf{X}\boldsymbol{\beta} = d_i \mathbf{v}_i^\top \boldsymbol{\beta}$ , it gives  $\mathbb{E} \hat{\boldsymbol{\beta}}_\lambda = \sum_{i \leq r} \{d_i^2 / (d_i^2 + \lambda)\} (\mathbf{v}_i^\top \boldsymbol{\beta}) \mathbf{v}_i$ , and the covariance is  $\sigma^2 \sum_{i \leq r} d_i^2 (d_i^2 + \lambda)^{-2} \mathbf{v}_i \mathbf{v}_i^\top$ , as in [Theorem 27.2.2\(c\)](#). What  $r < p$  adds is the bias component  $-(\mathbf{I} - \mathbf{P})\boldsymbol{\beta}$  in the null space, orthogonal to the rest of the bias  $-\sum_i \{\lambda / (d_i^2 + \lambda)\} (\mathbf{v}_i^\top \boldsymbol{\beta}) \mathbf{v}_i$ ; it contributes  $\|(\mathbf{I} - \mathbf{P})\boldsymbol{\beta}\|^2$ . Adding the squared bias and the trace of the covariance gives  $m(\lambda)$ . Differentiating term by term, the first bracket has derivative  $2\lambda d_i^2 (\mathbf{v}_i^\top \boldsymbol{\beta})^2 / (d_i^2 + \lambda)^3$ , which is zero at  $\lambda = 0$ . The second has derivative  $-2\sigma^2 d_i^2 / (d_i^2 + \lambda)^3$ , which equals  $-2\sigma^2 / d_i^4 < 0$  at  $\lambda = 0$ . Hence  $m'(0) = -2\sigma^2 \sum_i d_i^{-4} < 0$ , and  $m(\lambda) < m(0)$  for small  $\lambda > 0$ .

## 28.1.5.3 C. Going deeper

**Exercise C1.**

Check [Eq. 28.1.3](#) for  $n = 1$  directly. With one observation  $(\mathbf{x}, y)$ , show that  $\hat{\boldsymbol{\beta}}^+ = \mathbf{x}y / \|\mathbf{x}\|^2$ , and use  $\mathbb{E}(1/\chi^2(p)) = 1/(p-2)$  for  $p \geq 3$  to show that  $\mathcal{R} = \|\boldsymbol{\beta}\|^2(1 - 1/p) + \sigma^2/(p-2)$ .

*Solution.* Here  $\mathbf{X} = \mathbf{x}^\top$  and  $\mathbf{X}\mathbf{X}^\top = \|\mathbf{x}\|^2$ , so  $\hat{\boldsymbol{\beta}}^+ = \mathbf{x}y / \|\mathbf{x}\|^2$  and  $\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta} = -(\mathbf{I} - \mathbf{x}\mathbf{x}^\top / \|\mathbf{x}\|^2)\boldsymbol{\beta} + \mathbf{x}\varepsilon / \|\mathbf{x}\|^2$ , a sum of orthogonal vectors. So

$$\|\hat{\boldsymbol{\beta}}^+ - \boldsymbol{\beta}\|^2 = \|\boldsymbol{\beta}\|^2 - \frac{(\mathbf{x}^\top \boldsymbol{\beta})^2}{\|\mathbf{x}\|^2} + \frac{\varepsilon^2}{\|\mathbf{x}\|^2}.$$

The matrix  $\mathbf{P} = \mathbf{x}\mathbf{x}^\top / \|\mathbf{x}\|^2$  is the row-space projection, and the rotation argument in the proof of [Proposition 28.1.2](#) gives  $\mathbb{E} \mathbf{P} = \mathbf{I}/p$ , so  $\mathbb{E}(\mathbf{x}^\top \boldsymbol{\beta})^2 / \|\mathbf{x}\|^2 = \boldsymbol{\beta}^\top (\mathbb{E} \mathbf{P}) \boldsymbol{\beta} = \|\boldsymbol{\beta}\|^2 / p$ . And  $\|\mathbf{x}\|^2 \sim \chi^2(p)$  is independent of  $\varepsilon$ , so  $\mathbb{E} \varepsilon^2 / \|\mathbf{x}\|^2 = \sigma^2 / (p-2)$ . This agrees with [Eq. 28.1.3](#) at  $n = 1$ .



**Exercise C2.**

In [Proposition 28.1.2](#) let  $n, p \rightarrow \infty$  with  $n/p \rightarrow \gamma \in (0, 1)$  and  $\|\beta\|^2$  fixed. Show that  $\mathcal{R} \rightarrow \|\beta\|^2(1 - \gamma) + \sigma^2\gamma/(1 - \gamma)$ , and that this is below the risk  $\|\beta\|^2$  of predicting zero iff  $\|\beta\|^2/\sigma^2 > 1/(1 - \gamma)$ . What happens to the limit as  $\gamma \rightarrow 0$ ?

## 28.2 Restricted eigenvalue conditions

In Parts II and III, the precision of least squares was governed by the smallest eigenvalue of  $\mathbf{X}^\top \mathbf{X}$ : the worst variance of  $\mathbf{a}^\top \hat{\beta}$  over unit vectors  $\mathbf{a}$  is  $\sigma^2/\lambda_{\min}(\mathbf{X}^\top \mathbf{X})$  ([Proposition 19.6.1](#)). When  $p > n$  that eigenvalue is zero. This section explains which weaker condition takes its place once  $\beta$  is assumed sparse, and shows with numbers how demanding it is.

### 28.2.1 Notation

Throughout Sections 28.2–28.5 the columns of  $\mathbf{X}$  are scaled so that  $\|\mathbf{x}_j\|^2 = n$  for every  $j$ , and  $\hat{\Sigma} = \mathbf{X}^\top \mathbf{X}/n$  is the sample Gram matrix, with unit diagonal. For a vector  $\mathbf{v} \in \mathbb{R}^p$  and a set  $S \subseteq \{1, \dots, p\}$ ,  $\mathbf{v}_S$  is the vector that agrees with  $\mathbf{v}$  on  $S$  and is zero off  $S$ , so that  $\mathbf{v} = \mathbf{v}_S + \mathbf{v}_{S^c}$ . The  $\ell_1$  and  $\ell_\infty$  norms are  $\|\mathbf{v}\|_1 = \sum_j |v_j|$  and  $\|\mathbf{v}\|_\infty = \max_j |v_j|$ , and  $\|\mathbf{v}\|$  is still the Euclidean norm. The **support** of  $\beta$  is  $S = \{j : \beta_j \neq 0\}$ , and  $s = |S|$ . A vector with at most  $s$  nonzero entries is  **$s$ -sparse**. By Cauchy–Schwarz ([Proposition 1.1.1](#)), for every  $\mathbf{v}$ ,

$$\|\mathbf{v}_S\|_1 \leq \sqrt{s} \|\mathbf{v}_S\| \leq \sqrt{s} \|\mathbf{v}\|. \quad (28.2.1)$$

### 28.2.2 Sparsity restores identifiability

If  $\beta$  is only known to lie in  $\mathbb{R}^p$ , then  $\beta$  and  $\beta + \mathbf{v}$  with  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$  give the same mean, and nothing can separate them. If  $\beta$  is known to be sparse, most of those alternatives are excluded.

#### **Proposition 28.2.1 — Identifiability of sparse vectors**

Distinct  $s$ -sparse vectors  $\beta \neq \beta'$  always have  $\mathbf{X}\beta \neq \mathbf{X}\beta'$  iff every set of  $2s$  columns of  $\mathbf{X}$  is linearly independent. In particular this requires  $2s \leq n$ .

*Proof.* If  $\beta, \beta'$  are  $s$ -sparse and  $\mathbf{X}\beta = \mathbf{X}\beta'$ , then  $\mathbf{v} = \beta - \beta'$  is a vector in  $\mathcal{N}(\mathbf{X})$  with at most  $2s$  nonzero entries. It is zero if every  $2s$  columns are independent. Conversely, if some  $2s$  columns are dependent, there is a nonzero  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$  supported on them. Split its support into two sets of at most  $s$  indices,  $T_1$  and  $T_2$ , and put  $\beta = \mathbf{v}_{T_1}$ ,  $\beta' = -\mathbf{v}_{T_2}$ . These are distinct  $s$ -sparse vectors with  $\mathbf{X}\beta - \mathbf{X}\beta' = \mathbf{X}\mathbf{v} = \mathbf{0}$ . Finally,  $2s$  vectors in  $\mathbb{R}^n$  can be independent only if  $2s \leq n$ . ■

Identifiability is a statement about exactly sparse vectors and noiseless data. For estimation we need a quantitative version that allows for noise and for estimates that are only approximately sparse. The key observation, proved in [Section 28.3](#), is that the error of the lasso,  $\Delta = \hat{\beta}_\lambda - \beta$ , lies with high probability in a particular cone. Its  $\ell_1$  mass off the support is at most three times its mass on the support. Curvature of the least squares criterion is needed only in the directions of that cone.

### 28.2.3 Two conditions on a cone

#### Definition 28.2.1 — Compatibility and restricted eigenvalue conditions

Let  $S \subseteq \{1, \dots, p\}$  have  $s \geq 1$  elements, and let

$$\mathcal{K}(S) = \{\mathbf{v} \in \mathbb{R}^p : \|\mathbf{v}_{S^c}\|_1 \leq 3\|\mathbf{v}_S\|_1\}. \quad (28.2.2)$$

The **compatibility constant** and the **restricted eigenvalue** of the design for  $S$  are

$$\phi^2(S) = \min_{\mathbf{v} \in \mathcal{K}(S), \mathbf{v} \neq \mathbf{0}} \frac{\mathbf{v}^\top \hat{\Sigma} \mathbf{v}}{\|\mathbf{v}_S\|_1^2}, \quad \kappa^2(S) = \min_{\mathbf{v} \in \mathcal{K}(S), \mathbf{v} \neq \mathbf{0}} \frac{\mathbf{v}^\top \hat{\Sigma} \mathbf{v}}{\|\mathbf{v}\|^2}.$$

The design satisfies the **compatibility condition** for  $S$  if  $\phi^2(S) > 0$ , and the **restricted eigenvalue condition** for  $S$  if  $\kappa^2(S) > 0$ . The same definitions with a nonnegative definite matrix  $\Sigma$  in place of  $\hat{\Sigma}$  give  $\phi_\Sigma^2(S)$  and  $\kappa_\Sigma^2(S)$ .

A nonzero  $\mathbf{v} \in \mathcal{K}(S)$  has  $\mathbf{v}_S \neq \mathbf{0}$ , since otherwise  $\|\mathbf{v}_{S^c}\|_1 \leq 0$ . So both ratios are defined. Both are unchanged when  $\mathbf{v}$  is rescaled, and the sets  $\{\mathbf{v} \in \mathcal{K}(S) : \|\mathbf{v}\| = 1\}$  and  $\{\mathbf{v} \in \mathcal{K}(S) : \|\mathbf{v}_S\|_1 = 1\}$  are closed and bounded. So the minima are attained, which justifies writing min. The constant 3 comes from the choice of penalty in [Section 28.3](#). Other choices give other constants and the same theory.

The cone contains the  $s$ -dimensional subspace of vectors supported on  $S$  but no larger subspace ([Exercise \(C1\)](#)). So the null space of  $\mathbf{X}$ , of dimension at least  $p - n$ , can still avoid it.

#### Proposition 28.2.2 — Basic facts

- (a)  $\lambda_{\min}(\hat{\Sigma}) \leq \kappa^2(S) \leq \phi^2(S)$ . If  $p > n$ , then  $\lambda_{\min}(\hat{\Sigma}) = 0$ .
- (b)  $\kappa^2(S) > 0$  iff  $\phi^2(S) > 0$  iff  $\mathcal{N}(\mathbf{X}) \cap \mathcal{K}(S) = \{\mathbf{0}\}$ .
- (c) (*Transfer.*) If  $\Sigma$  is symmetric and  $\max_{j,k} |\hat{\Sigma}_{jk} - \Sigma_{jk}| \leq \delta$ , then  $\kappa^2(S) \geq \kappa_\Sigma^2(S) - 16s\delta$  and  $\phi^2(S) \geq \phi_\Sigma^2(S) - 16s\delta$ .
- (d) (*Coherence.*) If  $\mu = \max_{j \neq k} |\hat{\Sigma}_{jk}|$  is the **mutual coherence** of the scaled design, then  $\kappa^2(S) \geq 1 - 16s\mu$  and  $\phi^2(S) \geq 1 - 16s\mu$ .

*Proof.*

- (a) The first inequality holds because  $\kappa^2(S)$  minimizes the Rayleigh quotient of  $\hat{\Sigma}$  over a subset of nonzero vectors, while  $\lambda_{\min}(\hat{\Sigma})$  minimizes it over all of them ([Theo-](#)

rem 1.7.7). For the second, Eq. 28.2.1 gives  $\|\mathbf{v}_S\|_1^2 \leq s\|\mathbf{v}\|^2$ , so  $s\mathbf{v}^\top \hat{\Sigma}\mathbf{v}/\|\mathbf{v}_S\|_1^2 \geq \mathbf{v}^\top \hat{\Sigma}\mathbf{v}/\|\mathbf{v}\|^2 \geq \kappa^2(S)$  for every  $\mathbf{v} \in \mathcal{K}(S)$ . If  $p > n$ , then  $\text{rank}(\hat{\Sigma}) \leq n < p$ , so  $\hat{\Sigma}$  is singular.

- (b)  $\mathbf{v}^\top \hat{\Sigma}\mathbf{v} = \|\mathbf{X}\mathbf{v}\|^2/n$  vanishes iff  $\mathbf{X}\mathbf{v} = \mathbf{0}$ . Since the minima are attained, each constant is zero iff some nonzero  $\mathbf{v} \in \mathcal{K}(S)$  has  $\mathbf{X}\mathbf{v} = \mathbf{0}$ .
- (c) For  $\mathbf{v} \in \mathcal{K}(S)$ ,  $\|\mathbf{v}\|_1 = \|\mathbf{v}_S\|_1 + \|\mathbf{v}_{S^c}\|_1 \leq 4\|\mathbf{v}_S\|_1$ . Hence

$$|\mathbf{v}^\top (\hat{\Sigma} - \Sigma)\mathbf{v}| \leq \sum_{j,k} |v_j| |v_k| \delta = \delta \|\mathbf{v}\|_1^2 \leq 16\delta \|\mathbf{v}_S\|_1^2 \leq 16s\delta \|\mathbf{v}\|^2,$$

the last step by Eq. 28.2.1. Dividing by  $\|\mathbf{v}\|^2$  gives  $\mathbf{v}^\top \hat{\Sigma}\mathbf{v}/\|\mathbf{v}\|^2 \geq \kappa_{\Sigma}^2(S) - 16s\delta$ . Dividing the middle bound by  $\|\mathbf{v}_S\|_1^2/s$  gives  $s\mathbf{v}^\top \hat{\Sigma}\mathbf{v}/\|\mathbf{v}_S\|_1^2 \geq \phi_{\Sigma}^2(S) - 16s\delta$ .

- (d) Take  $\Sigma = \mathbf{I}$  in (c), with  $\delta = \mu$  (the diagonals agree). Then  $\kappa_{\mathbf{I}}^2(S) = 1$ , and  $\phi_{\mathbf{I}}^2(S) \geq \kappa_{\mathbf{I}}^2(S) = 1$  by (a) applied to  $\mathbf{I}$ .

■

By (b) the two conditions agree about *whether* the constants vanish, but by (a) their sizes, which enter the bounds, can differ a great deal: a lower bound on  $\kappa^2(S)$  is the stronger assumption. The fast prediction rate of Section 28.3 uses  $\phi^2(S)$ , the  $\ell_2$  bound  $\kappa^2(S)$ . Part (c) carries a population condition to the sample when  $\hat{\Sigma}$  is uniformly close to  $\Sigma$ , and (d) is the case  $\Sigma = \mathbf{I}$ .

## 28.2.4 When the conditions hold

**Populations.** If  $\lambda_{\min}(\Sigma) > 0$ , part (a) applied to  $\Sigma$  gives  $\phi_{\Sigma}^2(S) \geq \kappa_{\Sigma}^2(S) \geq \lambda_{\min}(\Sigma)$ . For example, the equicorrelation matrix  $\Sigma = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$ ,  $0 \leq \rho < 1$ , has  $\mathbf{v}^\top \Sigma\mathbf{v} = (1 - \rho)\|\mathbf{v}\|^2 + \rho(\mathbf{1}^\top \mathbf{v})^2 \geq (1 - \rho)\|\mathbf{v}\|^2$ , so both constants are at least  $1 - \rho$  for every  $p$  and  $S$ . The difficulty lies in going from  $\Sigma$  to  $\hat{\Sigma}$  when  $p > n$ .

**Random designs.** Suppose the rows of  $\mathbf{X}$  are independent  $N_p(\mathbf{0}, \Sigma)$  vectors with unit variances. A union bound over the  $p^2$  entries of  $\hat{\Sigma} - \Sigma$ , each an average of  $n$  products of normal variables, gives  $\max_{j,k} |\hat{\Sigma}_{jk} - \Sigma_{jk}| \leq C\sqrt{\log p/n}$  with high probability once  $n \geq C'\log p$ . The tail bound for each entry is not the normal bound of the next section, since products of normal variables have heavier tails; it is a Bernstein-type inequality, not proved here (see Wainwright 2019, chapter 2). Part (c) then gives a positive restricted eigenvalue once  $n$  exceeds a multiple of  $s^2 \log p$ . This is wasteful, because it bounds  $\mathbf{v}^\top (\hat{\Sigma} - \Sigma)\mathbf{v}$  by the worst entry. Raskutti, Wainwright and Yu (2010) proved, by an argument we do not reproduce, that with probability tending to one

$$\frac{\|\mathbf{X}\mathbf{v}\|}{\sqrt{n}} \geq c_1 \|\Sigma^{1/2}\mathbf{v}\| - c_2 \sqrt{\frac{\log p}{n}} \|\mathbf{v}\|_1 \quad \text{for all } \mathbf{v} \in \mathbb{R}^p,$$

for constants  $c_1, c_2 > 0$ . On the cone,  $\|\mathbf{v}\|_1 \leq 4\sqrt{s}\|\mathbf{v}\|$ , so this gives  $\kappa(S) \geq c_1 \lambda_{\min}(\Sigma)^{1/2} - 4c_2 \sqrt{s \log p/n}$ , which is positive once  $n$  exceeds a multiple of  $s \log p$ , possibly far smaller than  $p$ .

## 28.2.5 When they fail

The conditions fail when a vector of the cone is (nearly) invisible to  $\mathbf{X}$ .

*Duplicated columns.* If  $\mathbf{x}_k = \mathbf{x}_j$  with  $j \in S$  and  $k \notin S$ , then  $\mathbf{v} = \mathbf{e}_j - \mathbf{e}_k$  has  $\|\mathbf{v}_{S^c}\|_1 = \|\mathbf{v}_S\|_1 = 1$ ,

so it lies in the cone, and  $\mathbf{X}\mathbf{v} = \mathbf{0}$ . Both constants are zero. Nearly equal columns give nearly zero constants ([Exercise \(B1\)](#)). No estimator can say whether the effect belongs to  $\mathbf{x}_j$  or to its copy.

*Short combinations of inactive columns.* More generally, suppose an active column is a combination of inactive ones,  $\mathbf{x}_j = \mathbf{X}_{S^c}\mathbf{a}$  with  $\|\mathbf{a}\|_1 \leq 3$ . Then  $\mathbf{v} = \mathbf{e}_j - \mathbf{a}$  (with  $\mathbf{a}$  placed in the coordinates of  $S^c$ ) lies in the cone and  $\mathbf{X}\mathbf{v} = \mathbf{0}$ . When  $p > n$  the inactive columns usually span  $\mathbb{R}^n$ , so the conditions require every such combination to have a large  $\ell_1$  norm.

*Too many active variables for the sample size.* Even with independent regressors and a population compatibility constant of one, the sample constant can be close to zero if  $s$  is large relative to  $n/\log p$ . A combination of hundreds of inactive columns with total  $\ell_1$  weight 3 can then imitate  $\mathbf{X}_S\mathbf{v}_S$  closely. The next example measures this.

### Example. (Compatibility constants of Gaussian designs)

Take  $p = 1000$  regressors,  $S = \{1, \dots, s\}$ , and three random designs, with columns scaled so that  $\|\mathbf{x}_j\|^2 = n$ : independent standard normal entries with  $n = 200$  (the design used in [Section 28.3](#)) and with  $n = 400$ , and equicorrelated normal regressors with  $\rho = 0.5$  and  $n = 400$ . The compatibility constant is a minimum of a convex quadratic over the cone. For each sign pattern of  $\mathbf{v}_S$  the problem is convex, and a duality gap certifies the computed value. The table gives  $\phi^2(S)$  and the mutual coherence  $\mu$ ; its rows are labelled by the common correlation  $\rho$ .

| Design                      | $\mu$ | $s = 1$ | $s = 2$ | $s = 5$ | Population |
|-----------------------------|-------|---------|---------|---------|------------|
| $\rho = 0$ ,<br>$n = 200$   | 0.350 | 0.328   | 0.146   | 0.0045  | 1          |
| $\rho = 0$ ,<br>$n = 400$   | 0.238 | 0.479   | 0.384   | 0.177   | 1          |
| $\rho = 0.5$ ,<br>$n = 400$ | 0.641 | 0.266   | 0.192   | 0.089   | $\geq 0.5$ |

The sample constants lie far below the population values, even for  $s = 1$ , and fall quickly with  $s$ . At  $n = 200$  and  $s = 5$  the constant 0.0045 makes the fast-rate bound of the next section numerically useless for that design; doubling  $n$  helps a great deal. The coherence bound of [Proposition 28.2.2\(d\)](#) is useless throughout: with  $\mu = 0.350$ ,  $1 - 16s\mu$  is  $-4.60$  already at  $s = 1$ .

```
import numpy as np

def proj_simplex(u):
    """Euclidean projection onto {u >= 0, sum(u) = 1}."""
    v = np.sort(u)[::-1]
    css = np.cumsum(v) - 1
    k = np.flatnonzero(v - css / np.arange(1, len(u) + 1) > 0)[-1]
    return np.maximum(u - css[k] / (k + 1), 0)

def proj_l1(w, radius):
    """Euclidean projection onto the l1 ball of the given radius."""
    if np.abs(w).sum() <= radius:
        return w
    return np.sign(w) * proj_simplex(np.abs(w) / radius) * radius
```

```

def compatibility(X, S, radius=3.0, iters=4000):
    """Return  $\phi^2(S)$  and a certified lower bound (value minus duality gap)."""
    n, p = X.shape
    s = len(S)
    Sc = np.setdiff1d(np.arange(p), S)
    A, C = X[:, S], X[:, Sc]
    L = 2 * s / n * np.linalg.norm(X, 2) ** 2          # Lipschitz constant of the
    gradient
    best, lower = np.inf, np.inf
    for code in range(2 ** (s - 1)):                  # sign patterns of  $v_S$  up to a
    global sign
        sg = np.array([1.0] + [-1.0 if (code >> k) & 1 else 1.0 for k in range(s -
    1)])
        u, w = np.full(s, 1 / s), np.zeros(len(Sc))
        yu, yw, t = u.copy(), w.copy(), 1.0
        for _ in range(iters):                        # accelerated projected
        gradient
            r = A @ (sg * yu) + C @ yw
            gu, gw = 2 * s / n * sg * (A.T @ r), 2 * s / n * (C.T @ r)
            u_new, w_new = proj_simplex(yu - gu / L), proj_l1(yw - gw / L, radius)
            t_new = (1 + np.sqrt(1 + 4 * t * t)) / 2
            yu = u_new + (t - 1) / t_new * (u_new - u)
            yw = w_new + (t - 1) / t_new * (w_new - w)
            u, w, t = u_new, w_new, t_new
            r = A @ (sg * u) + C @ w
            value = s / n * r @ r
            gu, gw = 2 * s / n * sg * (A.T @ r), 2 * s / n * (C.T @ r)
            gap = gu @ u - gu.min() + gw @ w + radius * np.abs(gw).max()
            best, lower = min(best, value), min(lower, value - gap)
    return best, lower

rng = np.random.default_rng(2820)
n, p = 100, 300
X = rng.normal(size=(n, p))
X /= np.sqrt(np.sum(X ** 2, axis=0) / n)              #  $\|x_j\|^2 = n$ 
print("smallest eigenvalue of  $X^T X / n$ :", np.linalg.eigvalsh(X.T @ X / n)[0])
for s in [1, 2, 3]:
    value, lower = compatibility(X, np.arange(s), iters=1500)
    print(f"s = {s}:  $\phi^2 = \{value:.3f\}$ ")

```

*Remark* (Checking the conditions in practice). The constants depend on the unknown  $S$  and cannot be checked from data. They describe the designs for which the theory is informative: those in which no short combination of columns imitates another. With groups of nearly equivalent regressors, common in genomics, the lasso may still predict well (by the slow rate of [Section 28.3](#)), but its choice among them is arbitrary.

## 28.2.6 Exercises

### 28.2.6.1 A. Check your understanding

#### Exercise A1.

Let  $p = 2$  and  $S = \{1\}$ . Sketch  $\mathcal{K}(S)$  and show that it is not convex. Show that for general  $S$  it is the union of  $2^s$  convex cones, one for each sign pattern of  $\mathbf{v}_S$ . This is why the example above solves one convex problem per sign pattern.

### 28.2.6.2 B. Practice

#### Exercise B1.

Let  $j \in S$  and  $k \notin S$ . Show that  $\phi^2(S) \leq s\|\mathbf{x}_j - \mathbf{x}_k\|^2/n$  and  $\kappa^2(S) \leq \|\mathbf{x}_j - \mathbf{x}_k\|^2/(2n)$ . If  $\mathbf{x}_j^\top \mathbf{x}_k/n = r$  (the cosine of the angle between the two columns, which is their sample correlation when the columns are centred), conclude that  $\phi^2(S) \leq 2s(1 - r)$ .

*Solution.* Take  $\mathbf{v} = \mathbf{e}_j - \mathbf{e}_k \in \mathcal{K}(S)$ . Then  $\|\mathbf{v}_S\|_1 = 1$ ,  $\|\mathbf{v}\|^2 = 2$  and  $\mathbf{v}^\top \hat{\Sigma} \mathbf{v} = \|\mathbf{x}_j - \mathbf{x}_k\|^2/n$ , which gives both bounds. With  $\|\mathbf{x}_j\|^2 = \|\mathbf{x}_k\|^2 = n$ ,  $\|\mathbf{x}_j - \mathbf{x}_k\|^2/n = 2 - 2\mathbf{x}_j^\top \mathbf{x}_k/n = 2(1 - r)$ .

#### Exercise B2.

For  $\Sigma = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$  with  $0 \leq \rho < 1$  and  $p \geq 2$ , show that  $\kappa_\Sigma^2(S) = 1 - \rho$  for every  $S$ . Is the equicorrelation structure a hard case for the restricted eigenvalue condition?

*Solution.* The lower bound  $1 - \rho$  was shown in the text. For equality, it is enough to find a nonzero  $\mathbf{v} \in \mathcal{K}(S)$  with  $\mathbf{1}^\top \mathbf{v} = 0$ , since then  $\mathbf{v}^\top \Sigma \mathbf{v} = (1 - \rho)\|\mathbf{v}\|^2$ . If  $s \geq 2$ , take  $\mathbf{e}_j - \mathbf{e}_k$  with  $j, k \in S$ . If  $s = 1$ , take  $\mathbf{e}_j - \mathbf{e}_k$  with  $S = \{j\}$  and any  $k \neq j$ ; it lies in the cone because  $1 \leq 3$ . The population condition is therefore as good as the smallest eigenvalue, and equicorrelation is not a hard case in the population. It is harder in the sample (the table above), and harder still for exact support recovery, as [Section 28.4](#) shows for a related design.

### 28.2.6.3 C. Going deeper

#### Exercise C1.

Show that every linear subspace contained in  $\mathcal{K}(S)$  has dimension at most  $s$ , and that the bound is attained. Deduce that if  $\dim \mathcal{N}(\mathbf{X}) > p - s$  then the conditions fail for every  $S$  of size  $s$ , whereas  $\dim \mathcal{N}(\mathbf{X}) = p - n$  with  $n \geq s$  is not excluded by this dimension count.

#### Exercise C2.

(Exact recovery by  $\ell_1$  minimization.) Say  $\mathbf{X}$  has the **null space property** for  $S$  if every nonzero  $\mathbf{v} \in \mathcal{N}(\mathbf{X})$  has  $\|\mathbf{v}_{S^c}\|_1 > \|\mathbf{v}_S\|_1$ . Show that this holds iff, for every  $\beta$  supported on  $S$ ,  $\beta$  is the unique solution of: minimize  $\|\mathbf{b}\|_1$  subject to  $\mathbf{X}\mathbf{b} = \mathbf{X}\beta$ . Relate the property to [Proposition 28.2.2\(b\)](#) with the constant 3 replaced by 1.

*Solution.* Suppose the property holds,  $\beta$  is supported on  $S$ , and  $\mathbf{X}\mathbf{b} = \mathbf{X}\beta$  with  $\mathbf{b} \neq \beta$ . Then  $\mathbf{v} = \mathbf{b} - \beta \in \mathcal{N}(\mathbf{X})$  is nonzero, and  $\mathbf{b}_S = \beta + \mathbf{v}_S$ ,  $\mathbf{b}_{S^c} = \mathbf{v}_{S^c}$ , so  $\|\mathbf{b}\|_1 \geq \|\beta\|_1 - \|\mathbf{v}_S\|_1 + \|\mathbf{v}_{S^c}\|_1 > \|\beta\|_1$ . So  $\beta$  is the unique minimizer. Conversely, if some nonzero

$\mathbf{v} \in \mathcal{N}(\mathbf{X})$  has  $\|\mathbf{v}_{S^c}\|_1 \leq \|\mathbf{v}_S\|_1$ , put  $\boldsymbol{\beta} = \mathbf{v}_S$  and  $\mathbf{b} = -\mathbf{v}_{S^c}$ . Then  $\mathbf{X}\mathbf{b} = \mathbf{X}\boldsymbol{\beta} - \mathbf{X}\mathbf{v} = \mathbf{X}\boldsymbol{\beta}$ ,  $\mathbf{b} \neq \boldsymbol{\beta}$  and  $\|\mathbf{b}\|_1 \leq \|\boldsymbol{\beta}\|_1$ , so  $\boldsymbol{\beta}$  is not the unique minimizer. The property says exactly that  $\mathcal{N}(\mathbf{X})$  meets the cone  $\{\|\mathbf{v}_{S^c}\|_1 \leq \|\mathbf{v}_S\|_1\}$  only at zero. This is part (b) with the constant 1 in place of 3. The noisy problem needs the larger cone because the penalty must also absorb the noise.

## 28.3 Prediction and estimation bounds for the lasso

This section proves that the lasso works when  $p > n$ , in the form of explicit inequalities valid for every  $n$  and  $p$ . The proofs use three ingredients: the lasso minimizes its criterion, the largest of  $p$  normal variables is small, and the conditions of [Section 28.2](#) hold.

### 28.3.1 Setting

Throughout this section and the next two,  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $\mathbf{X}$  a fixed  $n \times p$  matrix whose columns satisfy  $\|\mathbf{x}_j\|^2 = n$ ,  $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ , and  $p$  possibly much larger than  $n$ . The support of  $\boldsymbol{\beta}$  is  $S$ , with  $s = |S|$ . A **lasso estimate** is any

$$\hat{\boldsymbol{\beta}}_\lambda \in \arg \min_{\mathbf{b} \in \mathbb{R}^p} \left\{ \frac{1}{2n} \|\mathbf{Y} - \mathbf{X}\mathbf{b}\|^2 + \lambda \|\mathbf{b}\|_1 \right\}, \quad \lambda > 0. \quad (28.3.1)$$

This is the lasso of [Definition 27.4.1](#) with the penalty measured per observation. Other scalings of the criterion only rename  $\lambda$ . By [Theorem 27.4.1](#), a minimizer exists, all minimizers share the fitted values  $\mathbf{X}\hat{\boldsymbol{\beta}}_\lambda$  and the norm  $\|\hat{\boldsymbol{\beta}}_\lambda\|_1$ , and the optimality conditions of [Definition 27.4.1](#) are necessary and sufficient. When  $p > n$  the minimizer need not be unique, and everything below holds for every minimizer. In the present scaling they say that  $\mathbf{b}$  is a lasso estimate iff

$$\begin{aligned} \frac{1}{n} \mathbf{X}^\top (\mathbf{Y} - \mathbf{X}\mathbf{b}) &= \lambda \mathbf{z} \quad \text{with} \\ z_j &= \text{sign}(b_j) \text{ if } b_j \neq 0, \quad |z_j| \leq 1 \text{ if } b_j = 0. \end{aligned} \quad (28.3.2)$$

In words, every column's correlation with the residual is at most  $\lambda$  in absolute value, and exactly  $\lambda$ , with the sign of the coefficient, for every column in the fit. The error of the estimate is written  $\boldsymbol{\Delta} = \hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}$ .

The benchmark is the **oracle** estimator, least squares on the columns in  $S$ , which could be computed if  $S$  were known. If  $\mathbf{X}_S$  has full column rank and  $\mathbf{M}_S$  is the projection onto  $\mathcal{C}(\mathbf{X}_S)$ , its fitted values are  $\mathbf{M}_S \mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{M}_S \boldsymbol{\varepsilon}$ , so its in-sample prediction error is  $E\|\mathbf{M}_S \boldsymbol{\varepsilon}\|^2/n = \sigma^2 \text{tr}(\mathbf{M}_S)/n = \sigma^2 s/n$  ([Proposition 6.3.4](#)). No estimator that does not know  $S$  can be expected to do this well. The question is how much is lost.

### 28.3.2 The basic inequality

### Lemma 28.3.1 — Basic inequality

Every lasso estimate satisfies

$$\frac{1}{2n} \|\mathbf{X}\Delta\|^2 \leq \frac{1}{n} \boldsymbol{\varepsilon}^\top \mathbf{X}\Delta + \lambda(\|\boldsymbol{\beta}\|_1 - \|\hat{\boldsymbol{\beta}}_\lambda\|_1). \quad (28.3.3)$$

*Proof.* The criterion at  $\hat{\boldsymbol{\beta}}_\lambda$  is at most its value at  $\boldsymbol{\beta}$ :

$$\frac{1}{2n} \|\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_\lambda\|^2 + \lambda\|\hat{\boldsymbol{\beta}}_\lambda\|_1 \leq \frac{1}{2n} \|\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}\|^2 + \lambda\|\boldsymbol{\beta}\|_1.$$

Since  $\mathbf{Y} - \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\varepsilon}$  and  $\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_\lambda = \boldsymbol{\varepsilon} - \mathbf{X}\Delta$ , the left side contains  $\|\boldsymbol{\varepsilon} - \mathbf{X}\Delta\|^2 = \|\boldsymbol{\varepsilon}\|^2 - 2\boldsymbol{\varepsilon}^\top \mathbf{X}\Delta + \|\mathbf{X}\Delta\|^2$ . Cancelling  $\|\boldsymbol{\varepsilon}\|^2/(2n)$  and rearranging gives Eq. 28.3.3. ■

The inequality compares a quadratic in  $\Delta$  on the left with a linear noise term and a difference of  $\ell_1$  norms on the right. By Hölder's inequality,  $|\boldsymbol{\varepsilon}^\top \mathbf{X}\Delta|/n \leq \|\mathbf{X}^\top \boldsymbol{\varepsilon}/n\|_\infty \|\Delta\|_1$ . So the noise enters only through the largest correlation between the errors and a column of  $\mathbf{X}$ ,

$$\|\mathbf{X}^\top \boldsymbol{\varepsilon}/n\|_\infty = \max_{j \leq p} \frac{|\mathbf{x}_j^\top \boldsymbol{\varepsilon}|}{n}.$$

The strategy is to choose  $\lambda$  large enough to dominate this quantity with high probability.

## 28.3.3 The noise event

### Lemma 28.3.2 — Maximum of normal variables

Let  $W_1, \dots, W_m$  be normal random variables, not necessarily independent, with mean zero and variances at most  $v$ . For every  $t > 0$ ,

$$\Pr\left(\max_{j \leq m} |W_j| > t\right) \leq 2m \exp\left(-\frac{t^2}{2v}\right). \quad (28.3.4)$$

In particular  $\max_j |W_j| \leq \sqrt{2v \log(2m/\delta)}$  with probability at least  $1 - \delta$ .

*Proof.* Let  $W \sim N(0, v_j)$  with  $0 < v_j \leq v$ . For  $u > 0$ , the function  $e^{uw}$  is increasing, so Markov's inequality ( $\Pr(U \geq a) \leq \mathbb{E}U/a$  for  $U \geq 0$ ) and the normal moment generating function (Theorem 3.2.1) give

$$\Pr(W > t) = \Pr(e^{uW} \geq e^{ut}) \leq e^{-ut} \mathbb{E} e^{uW} = \exp\left(\frac{1}{2}u^2 v_j - ut\right).$$

The exponent is smallest at  $u = t/v_j$ , where it equals  $-t^2/(2v_j) \leq -t^2/(2v)$ . By symmetry  $\Pr(|W| > t) \leq 2e^{-t^2/(2v)}$ , and this holds trivially when  $v_j = 0$ . The union bound (Theorem 13.4.1(a)) over  $j = 1, \dots, m$  gives Eq. 28.3.4. Setting the right side equal to



$\delta$  and solving for  $t$  gives the last statement. ■

Each  $\mathbf{x}_j^\top \boldsymbol{\varepsilon}/n$  is normal with mean zero and variance  $\sigma^2 \|\mathbf{x}_j\|^2/n^2 = \sigma^2/n$ . For  $0 < \delta < 1$  put

$$\lambda_0 = \sigma \sqrt{\frac{2 \log(2p/\delta)}{n}}, \quad \mathcal{T} = \left\{ \|\mathbf{X}^\top \boldsymbol{\varepsilon}/n\|_\infty \leq \lambda_0 \right\}. \quad (28.3.5)$$

By Lemma 28.3.2,  $\Pr(\mathcal{T}) \geq 1 - \delta$ , whatever the correlations among the columns. The threshold grows only like  $\sqrt{\log p}$ , which is why  $p \gg n$  is not hopeless.

## 28.3.4 Prediction bounds

### Theorem 28.3.3 — Prediction bounds for the lasso

Let  $\lambda \geq 2\lambda_0$ , and suppose the event  $\mathcal{T}$  of Eq. 28.3.5 occurs. Then every lasso estimate satisfies:

- (a) (*Cone.*)  $\frac{1}{n} \|\mathbf{X}\boldsymbol{\Delta}\|^2 + \lambda \|\boldsymbol{\Delta}_{S^c}\|_1 \leq 3\lambda \|\boldsymbol{\Delta}_S\|_1$ . In particular  $\boldsymbol{\Delta} \in \mathcal{K}(S)$ .
- (b) (*Slow rate.*)  $\frac{1}{n} \|\mathbf{X}\boldsymbol{\Delta}\|^2 \leq 3\lambda \|\boldsymbol{\beta}\|_1$ , with no condition on  $\mathbf{X}$ .
- (c) (*Fast rate.*) If  $\phi^2(S) > 0$ , then  $\frac{1}{n} \|\mathbf{X}\boldsymbol{\Delta}\|^2 \leq \frac{9\lambda^2 s}{\phi^2(S)}$ .

Consequently, with  $\lambda = 2\lambda_0$ , with probability at least  $1 - \delta$ ,

$$\frac{1}{n} \|\mathbf{X}(\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta})\|^2 \leq \min \left\{ 6\sigma \|\boldsymbol{\beta}\|_1 \sqrt{\frac{2 \log(2p/\delta)}{n}}, \frac{72 \sigma^2 s \log(2p/\delta)}{\phi^2(S) n} \right\}. \quad (28.3.6)$$

*Proof.* On  $\mathcal{T}$ , Hölder's inequality gives  $|\boldsymbol{\varepsilon}^\top \mathbf{X}\boldsymbol{\Delta}|/n \leq \lambda_0 \|\boldsymbol{\Delta}\|_1 \leq (\lambda/2) \|\boldsymbol{\Delta}\|_1$ .

- (a) Because  $\boldsymbol{\beta}$  vanishes off  $S$ ,  $\|\hat{\boldsymbol{\beta}}_\lambda\|_1 = \|\boldsymbol{\beta}_S + \boldsymbol{\Delta}_S\|_1 + \|\boldsymbol{\Delta}_{S^c}\|_1 \geq \|\boldsymbol{\beta}\|_1 - \|\boldsymbol{\Delta}_S\|_1 + \|\boldsymbol{\Delta}_{S^c}\|_1$ . Substituting both bounds in Eq. 28.3.3, and writing  $\|\boldsymbol{\Delta}\|_1 = \|\boldsymbol{\Delta}_S\|_1 + \|\boldsymbol{\Delta}_{S^c}\|_1$ ,

$$\begin{aligned} \frac{1}{2n} \|\mathbf{X}\boldsymbol{\Delta}\|^2 &\leq \frac{\lambda}{2} (\|\boldsymbol{\Delta}_S\|_1 + \|\boldsymbol{\Delta}_{S^c}\|_1) + \lambda (\|\boldsymbol{\Delta}_S\|_1 - \|\boldsymbol{\Delta}_{S^c}\|_1) \\ &= \frac{3\lambda}{2} \|\boldsymbol{\Delta}_S\|_1 - \frac{\lambda}{2} \|\boldsymbol{\Delta}_{S^c}\|_1. \end{aligned}$$

Multiplying by 2 gives the inequality. Its left side is at least  $\lambda \|\boldsymbol{\Delta}_{S^c}\|_1$ , so  $\|\boldsymbol{\Delta}_{S^c}\|_1 \leq 3\|\boldsymbol{\Delta}_S\|_1$ .

(b) Use instead  $\|\Delta\|_1 \leq \|\hat{\beta}_\lambda\|_1 + \|\beta\|_1$  in Eq. 28.3.3:

$$\begin{aligned} \frac{1}{2n} \|\mathbf{X}\Delta\|^2 &\leq \frac{\lambda}{2} (\|\hat{\beta}_\lambda\|_1 + \|\beta\|_1) + \lambda \|\beta\|_1 - \lambda \|\hat{\beta}_\lambda\|_1 \\ &\leq \frac{3\lambda}{2} \|\beta\|_1. \end{aligned}$$

(c) Write  $a = \|\mathbf{X}\Delta\|/\sqrt{n}$ . By (a),  $\Delta \in \mathcal{K}(S)$ , so Definition 28.2.1 gives  $\phi^2(S) \leq s a^2 / \|\Delta_S\|_1^2$ , that is,  $\|\Delta_S\|_1 \leq \sqrt{s} a / \phi(S)$ . (If  $\Delta = \mathbf{0}$  there is nothing to prove.) Then (a) gives  $a^2 \leq 3\lambda \|\Delta_S\|_1 \leq 3\lambda \sqrt{s} a / \phi(S)$ , so  $a \leq 3\lambda \sqrt{s} / \phi(S)$ .

For Eq. 28.3.6, substitute  $\lambda = 2\lambda_0$  in (b) and (c), and use  $\Pr(\mathcal{T}) \geq 1 - \delta$ .

■

The **slow rate** needs nothing of  $\mathbf{X}$ : the lasso predicts consistently whenever  $\|\beta\|_1 \sqrt{\log p/n} \rightarrow 0$ , which allows  $p$  to grow exponentially in  $n$ . Greenshtein and Ritov (2004) called this *persistence*. The **fast rate**, of order  $\sigma^2 s \log p/n$ , loses against the oracle's  $\sigma^2 s/n$  a factor  $\log p$  and the constant  $72/\phi^2(S)$ . The logarithm cannot be removed: no estimator that does not know  $S$  has worst-case error below a multiple of  $\sigma^2 s \log(p/s)/n$  under mild conditions on the design (Raskutti, Wainwright and Yu 2011; not proved here). The constants are not sharp.

*Idea.* The proof has two separate parts. A *probabilistic* part shows that the noise is small in the  $\ell_\infty$  norm dual to the penalty, at the level  $\lambda_0 \asymp \sigma \sqrt{\log p/n}$ . A *deterministic* part says that on this event the estimate's error lies in a cone and is controlled by the geometry of  $\mathbf{X}$  on that cone. Changing the noise distribution changes only the first part. Changing the penalty changes the norm and the cone.

## 28.3.5 Estimation bounds

Prediction error measures  $\Delta$  through  $\mathbf{X}$ . Bounds on the coefficients themselves need the design to separate the directions of the cone, which is what the restricted eigenvalue measures.

### Theorem 28.3.4 — Estimation bounds for the lasso

Let  $\lambda \geq 2\lambda_0$ , and suppose  $\mathcal{T}$  occurs. Then every lasso estimate satisfies:

- (a) if  $\phi^2(S) > 0$ , then  $\|\hat{\beta}_\lambda - \beta\|_1 \leq \frac{4\lambda s}{\phi^2(S)}$ ;
- (b) if  $\kappa^2(S) > 0$ , then  $\|\hat{\beta}_\lambda - \beta\| \leq \frac{3\lambda\sqrt{s}}{\kappa^2(S)}$ .

With  $\lambda = 2\lambda_0$ , both hold with probability at least  $1 - \delta$ , and then  $\|\hat{\beta}_\lambda - \beta\|_1 \leq 8\lambda_0 s / \phi^2(S)$  and  $\|\hat{\beta}_\lambda - \beta\| \leq 6\lambda_0 \sqrt{s} / \kappa^2(S)$ .

*Proof.* Keep the notation  $a = \|\mathbf{X}\Delta\|/\sqrt{n}$  of the previous proof.

- (a) Add  $\lambda \|\Delta_S\|_1$  to both sides of Theorem 28.3.3(a), and use  $\|\Delta_S\|_1 \leq \sqrt{s} a / \phi(S)$  and

$$2xy \leq x^2 + y^2:$$

$$a^2 + \lambda \|\Delta\|_1 \leq 4\lambda \|\Delta_S\|_1 \leq 2 \cdot a \cdot \frac{2\lambda\sqrt{s}}{\phi(S)} \leq a^2 + \frac{4\lambda^2 s}{\phi^2(S)}.$$

Cancel  $a^2$  and divide by  $\lambda$ .

(b) By [Theorem 28.3.3\(a\)](#) and [Eq. 28.2.1](#), and since  $\Delta \in \mathcal{K}(S)$ , by [Definition 28.2.1](#),

$$\kappa^2(S) \|\Delta\|^2 \leq a^2 \leq 3\lambda \|\Delta_S\|_1 \leq 3\lambda\sqrt{s} \|\Delta\|.$$

Dividing by  $\|\Delta\|$  (when it is not zero) gives the bound.

■

With  $\lambda \asymp \sigma\sqrt{\log p/n}$ , the  $\ell_2$  error is of order  $\sigma\sqrt{s \log p/n}$ . The oracle has  $\ell_2$  error of order  $\sigma\sqrt{s/n}$ . Again only a logarithm is lost. Bickel, Ritov and Tsybakov (2009) proved bounds of this form for the lasso and the Dantzig selector together. The compatibility condition was introduced by van de Geer (see van de Geer and Bühlmann 2009), and Bühlmann and van de Geer (2011, chapter 6) present the prediction bound in the form given here.

*Remark* (What the theorems assume, and what they do not). The theorems are *deterministic* on  $\mathcal{T}$ ; normality entered only through  $\Pr(\mathcal{T}) \geq 1 - \delta$ , so errors with light (sub-Gaussian) tails need the same  $\lambda_0$  up to a constant. For a random design they apply conditionally on  $\mathbf{X}$ . The choice  $\lambda_0$  needs  $\sigma$ ; the *square-root lasso* of Belloni, Chernozhukov and Wang (2011), which minimizes  $\|\mathbf{Y} - \mathbf{X}\mathbf{b}\|/\sqrt{n} + \lambda\|\mathbf{b}\|_1$ , and the *scaled lasso* of Sun and Zhang (2012) avoid this. In practice  $\lambda$  is chosen by cross-validation ([Chapter 29](#)), without these guarantees.

## 28.3.6 The bounds in numbers

### Example. (How tight are the bounds?)

Take the independent Gaussian design of [Example \(Compatibility constants of Gaussian designs\)](#) with  $n = 200$  and  $p = 1000$ , five nonzero coefficients equal to  $\pm 1$ ,  $\sigma = 1$  and  $\delta = 0.05$ . Then  $\lambda_0 = 0.3255$ . In 200 simulated data sets the event  $\mathcal{T}$  occurred in a fraction 0.990, more than the guaranteed 0.95, because the Chernoff bound and the union bound are both conservative. The cone property of [Theorem 28.3.3\(a\)](#) held in a fraction 1.000 of the data sets.

At the theoretical choice  $\lambda = 2\lambda_0 = 0.651$  the average prediction error  $\|\mathbf{X}\Delta\|^2/n$  is 2.3643, against the oracle's  $\sigma^2 s/n = 0.025$ . The slow-rate bound  $3\lambda\|\beta\|_1$  is 9.77. With the compatibility constant 0.00447 of this design, the fast-rate bound is 4269.8 (computed from the unrounded values), which is useless here. The same is true of the  $\ell_1$  bound 2914.8, against an actual average  $\ell_1$  error of 3.594.

Almost all of the error at  $\lambda = 2\lambda_0$  is shrinkage. The lasso pulls each selected coefficient towards zero by about  $\lambda$ , and  $s\lambda^2 = 2.12$  is close to the observed error. Panel (a) of [Figure 28.3.1](#) shows the whole path. The smallest average error on the path is 0.1893, at  $\lambda \approx 0.39\lambda_0$ , well below the theoretical choice. Below that the lasso admits many false variables and the error rises again, to 0.587 at  $\lambda = 0.1\lambda_0$ .

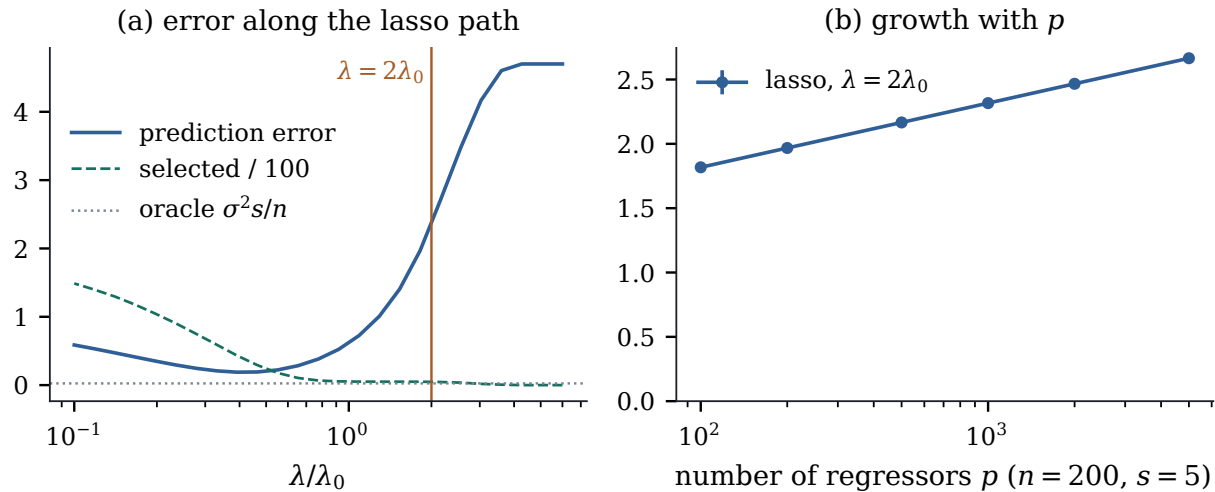


Figure 28.3.1: The lasso with  $n = 200$ ,  $s = 5$ ,  $\sigma = 1$ . (a) Average prediction error  $\|\mathbf{X}(\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta})\|^2/n$  and average number of selected variables (divided by 100) along the path for  $p = 1000$ , against  $\lambda/\lambda_0$ . The vertical line is the theoretical choice  $\lambda = 2\lambda_0$ , the dotted line the oracle error. (b) Prediction error at  $\lambda = 2\lambda_0$  as  $p$  grows (logarithmic axis), with two standard errors.

Panel (b) holds  $n = 200$  and  $s = 5$  fixed and lets  $p$  grow from 100 to 5000 (the first  $p$  columns of one larger design), with  $\lambda = 2\lambda_0$  for each  $p$ . The error grows from 1.818 to 2.665, a factor of 1.466, close to the ratio 1.472 of the values of  $\log(2p/\delta)$ . The error tracks the  $\lambda_0^2$  in the bounds, which is logarithmic in  $p$ . The theorem gives only upper bounds, so this agreement is an observation about the example, not a consequence of the theorem.

The bounds are worst-case statements over designs and noise vectors, and for a particular design they can be very loose. What they get right is the *form* of the dependence on  $s$ ,  $p$  and  $n$ , with one tuning parameter of order  $\sigma\sqrt{\log p/n}$  that works for all sparse  $\boldsymbol{\beta}$ . They also expose the cost of shrinkage, an error of order  $s\lambda^2$  even when the right variables are selected, which motivates the next two sections.

```
import numpy as np

def soft(z, t):
    """Soft thresholding  $S(z, t) = \text{sign}(z) \max(|z| - t, 0)$ ."""
    return np.sign(z) * np.maximum(np.abs(z) - t, 0.0)

def lasso_cd(X, Y, lam, B=None, tol=1e-9, max_sweeps=2000):
    """Minimize  $\|y - Xb\|_2^2 / (2n) + \text{lam} \|b\|_1$  for every column  $y$  of  $Y$ , by
    coordinate descent."""
    n, p = X.shape
    Y = Y.reshape(n, -1)
    B = np.zeros((p, Y.shape[1])) if B is None else B.copy()
    R = Y - X @ B  # residuals, one column per response
    scale = np.sum(X ** 2, axis=0) / n
    active = np.arange(p)
    for sweep in range(max_sweeps):
        change = 0.0
        for j in active:
            z = X[:, j] @ R / n + scale[j] * B[j]
            new = soft(z, lam) / scale[j]
```

```

        d = new - B[j]
        if np.any(d != 0):
            R -= np.outer(X[:, j], d)
            B[j] = new
            change = max(change, np.max(np.abs(d)))
        if change < tol:
            if len(active) == p:
                # converged over all
                coordinates
                break
            active = np.arange(p)
            # check every coordinate once
            more
        else:
            active = np.flatnonzero(np.any(B != 0, axis=1)) if sweep % 5 else np.
            arange(p)
        return B

rng = np.random.default_rng(2832)
n, p, s, sigma, delta = 100, 300, 5, 1.0, 0.05
X = rng.normal(size=(n, p))
X /= np.sqrt(np.sum(X ** 2, axis=0) / n)
# columns scaled to ||x_j||^2 = n
beta = np.zeros(p)
beta[s] = [1.0, -1.0, 1.0, -1.0, 1.0]
lam0 = sigma * np.sqrt(2 * np.log(2 * p / delta) / n)
E = sigma * rng.normal(size=(n, 50))
Y = (X @ beta)[:, None] + E
print("lambda_0 =", round(lam0, 3))
print("frequency of the event:", np.mean(np.max(np.abs(X.T @ E) / n, axis=0) <= lam0
    ))
for ratio in [2.0, 1.0, 0.5]:
    B = lasso_cd(X, Y, ratio * lam0)
    err = np.sum((X @ (B - beta[:, None])) ** 2, axis=0) / n
    print(f"lambda = {ratio} lambda_0: mean prediction error {err.mean():.3f},"
        f" mean number selected {np.mean(np.sum(B != 0, axis=0)):.1f}")
print("oracle sigma^2 s / n =", sigma ** 2 * s / n, " slow-rate bound at 2 lambda_0
    :", 6 * lam0 * s)

```

## 28.3.7 Exercises

### 28.3.7.1 A. Check your understanding

#### Exercise A1.

By Theorem 27.4.1(d),  $\mathbf{0}$  is a lasso estimate iff  $\lambda \geq \lambda_{\max} = \|\mathbf{X}^\top \mathbf{Y}\|_\infty / n$  (check this from Eq. 28.3.2). Show that it is then the only lasso estimate. Why does the theory above not rule out  $2\lambda_0 > \lambda_{\max}$ , and what happens then?

*Solution.* All lasso estimates have the same  $\ell_1$  norm (Theorem 27.4.1(c)), and that of  $\mathbf{0}$  is zero; so every lasso estimate is  $\mathbf{0}$ . If  $2\lambda_0 > \lambda_{\max}$  and  $\lambda \geq \lambda_{\max}$ , the lasso returns  $\mathbf{0}$ , and the theorems still hold: the error is  $-\beta$ , and the bounds then say that  $\beta$  is small compared with the noise level. They are not violated, only uninformative.

**Exercise A2.**

Show that for every  $\mathbf{b} \in \mathbb{R}^p$  a lasso estimate satisfies

$$\frac{1}{2n} \|\mathbf{X}(\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta})\|^2 \leq \frac{1}{2n} \|\mathbf{X}(\mathbf{b} - \boldsymbol{\beta})\|^2 + \frac{1}{n} \boldsymbol{\varepsilon}^\top \mathbf{X}(\hat{\boldsymbol{\beta}}_\lambda - \mathbf{b}) + \lambda(\|\mathbf{b}\|_1 - \|\hat{\boldsymbol{\beta}}_\lambda\|_1),$$

and that  $\mathbf{b} = \boldsymbol{\beta}$  gives [Lemma 28.3.1](#). Deduce that on  $\mathcal{T}$ , with  $\lambda \geq 2\lambda_0$ ,

$$\frac{1}{n} \|\mathbf{X}(\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta})\|^2 \leq \min_{\mathbf{b} \in \mathbb{R}^p} \left\{ \frac{1}{n} \|\mathbf{X}(\mathbf{b} - \boldsymbol{\beta})\|^2 + 3\lambda \|\mathbf{b}\|_1 \right\}.$$

This oracle inequality suits an approximately sparse  $\boldsymbol{\beta}$ :  $\mathbf{b}$  can keep its large entries and drop the small ones.

*Solution.* Compare the criterion [Eq. 28.3.1](#) at  $\hat{\boldsymbol{\beta}}_\lambda$  and at  $\mathbf{b}$ , expanding  $\|\mathbf{X}(\boldsymbol{\beta} - \mathbf{b}) + \boldsymbol{\varepsilon}\|^2 = \|\mathbf{X}(\mathbf{b} - \boldsymbol{\beta})\|^2 - 2\boldsymbol{\varepsilon}^\top \mathbf{X}(\mathbf{b} - \boldsymbol{\beta}) + \|\boldsymbol{\varepsilon}\|^2$  and likewise at  $\hat{\boldsymbol{\beta}}_\lambda$ ; the  $\|\boldsymbol{\varepsilon}\|^2$  terms cancel. On  $\mathcal{T}$ , Hölder's inequality gives  $|\boldsymbol{\varepsilon}^\top \mathbf{X}(\hat{\boldsymbol{\beta}}_\lambda - \mathbf{b})|/n \leq \lambda_0 \|\hat{\boldsymbol{\beta}}_\lambda - \mathbf{b}\|_1 \leq (\lambda/2)(\|\hat{\boldsymbol{\beta}}_\lambda\|_1 + \|\mathbf{b}\|_1)$ . So the right side is at most  $\|\mathbf{X}(\mathbf{b} - \boldsymbol{\beta})\|^2/(2n) + \frac{3}{2}\lambda \|\mathbf{b}\|_1 - \frac{1}{2}\lambda \|\hat{\boldsymbol{\beta}}_\lambda\|_1$ . Drop the last term, multiply by 2, and minimize over  $\mathbf{b}$ . At  $\mathbf{b} = \boldsymbol{\beta}$  this is the slow rate of [Theorem 28.3.3\(b\)](#). An analogue with  $\lambda^2 s/\phi^2$  in place of  $3\lambda \|\mathbf{b}\|_1$ , under a compatibility condition, is in Bühlmann and van de Geer (2011, chapter 6).

**28.3.7.2 B. Practice****Exercise B1.**

Let  $p \leq n$  and  $\hat{\boldsymbol{\Sigma}} = \mathbf{I}$ . Using [Theorem 27.4.2](#), the lasso estimate is  $\hat{\beta}_{\lambda,j} = \text{sign}(z_j)(|z_j| - \lambda)_+$  with  $\mathbf{z} = \mathbf{X}^\top \mathbf{Y}/n = \boldsymbol{\beta} + \mathbf{X}^\top \boldsymbol{\varepsilon}/n$ . On  $\mathcal{T}$ , with  $\lambda \geq 2\lambda_0$ , show that  $\hat{\beta}_{\lambda,j} = 0$  for  $j \notin S$ , that  $|\hat{\beta}_{\lambda,j} - \beta_j| \leq \lambda + \lambda_0$  for  $j \in S$ , and hence  $\|\mathbf{X}\boldsymbol{\Delta}\|^2/n \leq \frac{9}{4}\lambda^2 s$ . Show also that if every  $|\beta_j|$ ,  $j \in S$ , exceeds  $\lambda + \lambda_0$ , then  $\|\mathbf{X}\boldsymbol{\Delta}\|^2/n \geq \frac{1}{4}\lambda^2 s$ . So the order  $\lambda^2 s$  of [Theorem 28.3.3\(c\)](#) is attained.

*Solution.* With  $\hat{\boldsymbol{\Sigma}} = \mathbf{I}$ ,  $\|\mathbf{X}\boldsymbol{\Delta}\|^2/n = \|\boldsymbol{\Delta}\|^2$ . For  $j \notin S$ ,  $|z_j| = |\mathbf{x}_j^\top \boldsymbol{\varepsilon}|/n \leq \lambda_0 < \lambda$ , so the estimate is zero. For  $j \in S$ , soft thresholding moves  $z_j$  by at most  $\lambda$ , and  $|z_j - \beta_j| \leq \lambda_0$ , so  $|\hat{\beta}_{\lambda,j} - \beta_j| \leq \lambda + \lambda_0 \leq \frac{3}{2}\lambda$ . Summing squares over  $S$  gives the upper bound. If  $|\beta_j| > \lambda + \lambda_0$ , then  $|z_j| > \lambda$ , so  $\hat{\beta}_{\lambda,j} = z_j - \lambda \text{sign}(z_j)$  with the sign of  $\beta_j$ , and  $|\hat{\beta}_{\lambda,j} - \beta_j| \geq \lambda - \lambda_0 \geq \lambda/2$ . Summing gives the lower bound. The error of order  $\lambda^2 s$  is pure shrinkage.

**28.3.7.3 C. Going deeper****Exercise C1.**

(*The Dantzig selector.*) Candès and Tao (2007) proposed to estimate  $\boldsymbol{\beta}$  by a solution  $\tilde{\boldsymbol{\beta}}$  of: minimize  $\|\mathbf{b}\|_1$  subject to  $\|\mathbf{X}^\top (\mathbf{Y} - \mathbf{X}\mathbf{b})\|_\infty/n \leq \lambda_0$ . On  $\mathcal{T}$ , show that  $\boldsymbol{\beta}$  is feasible, that  $\boldsymbol{\Delta} = \tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}$  satisfies  $\|\boldsymbol{\Delta}_{S^c}\|_1 \leq \|\boldsymbol{\Delta}_S\|_1$ , and that  $\|\mathbf{X}\boldsymbol{\Delta}\|^2/n \leq 16\lambda_0^2 s/\phi^2(S)$ .

*Solution.* On  $\mathcal{T}$ ,  $\|\mathbf{X}^\top (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})\|_\infty/n = \|\mathbf{X}^\top \boldsymbol{\varepsilon}\|_\infty/n \leq \lambda_0$ , so  $\boldsymbol{\beta}$  is feasible and  $\|\tilde{\boldsymbol{\beta}}\|_1 \leq \|\boldsymbol{\beta}\|_1$ . The argument of [Theorem 28.3.3\(a\)](#) gives  $\|\boldsymbol{\beta}\|_1 \geq \|\tilde{\boldsymbol{\beta}}\|_1 \geq \|\boldsymbol{\beta}\|_1 - \|\boldsymbol{\Delta}_S\|_1 +$

$\|\Delta_{S^c}\|_1$ , which is the cone property with constant 1. Since both  $\tilde{\beta}$  and  $\beta$  are feasible,  $\|\mathbf{X}^\top \mathbf{X} \Delta\|_\infty / n \leq 2\lambda_0$ , and by Hölder  $\|\mathbf{X} \Delta\|^2 / n = \Delta^\top \mathbf{X}^\top \mathbf{X} \Delta / n \leq 2\lambda_0 \|\Delta\|_1 \leq 4\lambda_0 \|\Delta_S\|_1$ . The cone with constant 1 lies inside  $\mathcal{K}(S)$ , so  $\|\Delta_S\|_1 \leq \sqrt{s} \|\mathbf{X} \Delta\| / (\sqrt{n} \phi(S))$ . Combining,  $\|\mathbf{X} \Delta\| / \sqrt{n} \leq 4\lambda_0 \sqrt{s} / \phi(S)$ , and squaring gives the bound.

## 28.4 Support recovery

Good prediction does not require the right variables: the bounds of [Section 28.3](#) allow false variables with small coefficients and missed true ones. Often, though, the variables are the object of interest. This section asks when the lasso selects *exactly* the right variables with the right signs. The answer needs a condition on the design much stronger than those of [Section 28.2](#), and it fails in simple, realistic examples.

### 28.4.1 Uniqueness

The question “which variables does the lasso select?” needs an answer that does not depend on which minimizer is reported.

#### Lemma 28.4.1 — Uniqueness of the lasso fit

- (a) All lasso estimates at a given  $\lambda$  have the same fitted values  $\mathbf{X} \hat{\beta}_\lambda$  and the same  $\ell_1$  norm. Hence they share the vector  $\hat{\mathbf{z}} = \mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \hat{\beta}_\lambda) / (n\lambda)$  of [Eq. 28.3.2](#).
- (b) Let  $\hat{\beta}_\lambda$  be a lasso estimate with support  $\hat{S}$ . If  $|\hat{z}_j| < 1$  for every  $j \notin \hat{S}$  and  $\mathbf{X}_{\hat{S}}$  has full column rank, then  $\hat{\beta}_\lambda$  is the only lasso estimate.

*Proof.*

- (a) The first statement is [Theorem 27.4.1\(c\)](#). The vector  $\hat{\mathbf{z}}$  depends on  $\hat{\beta}_\lambda$  only through  $\mathbf{X} \hat{\beta}_\lambda$ .
- (b) Let  $\mathbf{b}$  be any lasso estimate. By (a) its optimality conditions [Eq. 28.3.2](#) hold with the same  $\hat{\mathbf{z}}$ . If  $b_j \neq 0$  then  $\hat{z}_j = \text{sign}(b_j)$ , so  $|\hat{z}_j| = 1$  and  $j \in \hat{S}$ . Thus  $\mathbf{b}$  is supported on  $\hat{S}$ , and  $\mathbf{X}_{\hat{S}} \mathbf{b}_{\hat{S}} = \mathbf{X} \mathbf{b} = \mathbf{X} \hat{\beta}_\lambda = \mathbf{X}_{\hat{S}} \hat{\beta}_{\lambda, \hat{S}}$ . Full column rank of  $\mathbf{X}_{\hat{S}}$  gives  $\mathbf{b} = \hat{\beta}_\lambda$ . ■

### 28.4.2 The irrepresentable condition

When will the lasso choose exactly  $S$ ? Suppose for a moment that it does, with the correct signs  $\mathbf{z}_S = \text{sign}(\beta_S)$ . Then [Eq. 28.3.2](#) on  $S$  determines the estimate:  $\mathbf{X}_S^\top (\mathbf{Y} - \mathbf{X}_S \hat{\beta}_S) / n = \lambda \mathbf{z}_S$ , so

$$\begin{aligned} \hat{\beta}_S &= (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} (\mathbf{X}_S^\top \mathbf{Y} - n\lambda \mathbf{z}_S) \\ &= \beta_S + (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{X}_S^\top \boldsymbol{\varepsilon} - \lambda \hat{\Sigma}_{SS}^{-1} \mathbf{z}_S, \end{aligned} \tag{28.4.1}$$

where  $\hat{\Sigma}_{SS} = \mathbf{X}_S^\top \mathbf{X}_S / n$ . The estimate is least squares on the true support, shifted towards zero by  $\lambda \hat{\Sigma}_{SS}^{-1} \mathbf{z}_S$ . For each  $k \notin S$ , the optimality conditions also require  $|\mathbf{x}_k^\top (\mathbf{Y} - \mathbf{X}_S \hat{\boldsymbol{\beta}}_S)| \leq n\lambda$ . Since  $\mathbf{Y} - \mathbf{X}_S \hat{\boldsymbol{\beta}}_S = (\mathbf{I} - \mathbf{M}_S) \boldsymbol{\varepsilon} + n\lambda \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{z}_S$ , where  $\mathbf{M}_S$  projects onto  $\mathcal{C}(\mathbf{X}_S)$ , this reads

$$\left| \underbrace{\mathbf{x}_k^\top \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{z}_S}_{a_k} + \frac{\mathbf{x}_k^\top (\mathbf{I} - \mathbf{M}_S) \boldsymbol{\varepsilon}}{n\lambda} \right| \leq 1. \quad (28.4.2)$$

The first term does not involve the noise at all. The vector  $(\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{X}_S^\top \mathbf{x}_k$  holds the least squares coefficients of  $\mathbf{x}_k$  on the active columns, and  $a_k$  is their sum weighted by the true signs. If an inactive column is well explained by the active ones, with coefficients that line up with the signs of  $\boldsymbol{\beta}_S$ , then  $|a_k|$  is large and the condition is hard to meet.

### Definition 28.4.1 — Irrepresentable and beta-min conditions

Let  $\mathbf{X}_S$  have full column rank. The design satisfies the **irrepresentable condition** for  $\boldsymbol{\beta}$  with margin  $\eta \in (0, 1]$  if

$$\max_{k \notin S} |\mathbf{x}_k^\top \mathbf{X}_S (\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \text{sign}(\boldsymbol{\beta}_S)| \leq 1 - \eta.$$

A **beta-min condition** is a lower bound on  $\beta_{\min} = \min_{j \in S} |\beta_j|$ .

The condition depends on the design and on the signs of the true coefficients, but not on their sizes. It holds with  $\eta = 1$  when the inactive columns are orthogonal to the active ones. It is much stronger than the compatibility condition. van de Geer and Bühlmann (2009) compare the two and show that a uniform version of the irrepresentable condition implies compatibility; the example below shows that the converse fails.

## 28.4.3 Sufficient conditions

### Theorem 28.4.2 — Sign recovery by the lasso

Suppose  $\lambda_{\min}(\hat{\Sigma}_{SS}) \geq C_{\min} > 0$  and the irrepresentable condition holds with margin  $\eta$ . Let  $\lambda > 0$ , put  $g = \lambda \|\hat{\Sigma}_{SS}^{-1} \text{sign}(\boldsymbol{\beta}_S)\|_\infty$ , and suppose  $\beta_{\min} > g$ . Consider the events

$$\mathcal{A} = \left\{ \max_{k \notin S} \frac{|\mathbf{x}_k^\top (\mathbf{I} - \mathbf{M}_S) \boldsymbol{\varepsilon}|}{n} < \eta\lambda \right\},$$

$$\mathcal{B} = \left\{ \|(\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{X}_S^\top \boldsymbol{\varepsilon}\|_\infty < \beta_{\min} - g \right\}.$$

- (a) On  $\mathcal{A} \cap \mathcal{B}$  the lasso estimate is unique and  $\text{sign}(\hat{\boldsymbol{\beta}}_\lambda) = \text{sign}(\boldsymbol{\beta})$ . In particular its support is exactly  $S$ .
- (b) The probabilities of the complements satisfy

$$\Pr(\mathcal{A}^c) \leq 2(p - s) \exp\left\{-\frac{n\eta^2\lambda^2}{2\sigma^2}\right\}, \quad \Pr(\mathcal{B}^c) \leq 2s \exp\left\{-\frac{nC_{\min}(\beta_{\min} - g)^2}{2\sigma^2}\right\}.$$



(c) Consequently, if

$$\lambda = \frac{\sigma}{\eta} \sqrt{\frac{2 \log\{4(p-s)/\delta\}}{n}} \quad \text{and} \quad \beta_{\min} \geq g + \sigma \sqrt{\frac{2 \log(4s/\delta)}{nC_{\min}}},$$

the lasso recovers the signs of  $\boldsymbol{\beta}$  with probability at least  $1 - \delta$ .

*Proof.*

(a) Construct a candidate and check that it is optimal. This is the *primal-dual witness* argument. Let  $\mathbf{z}_S = \text{sign}(\boldsymbol{\beta}_S)$ , let  $\check{\boldsymbol{\beta}}$  be zero off  $S$  with  $\check{\boldsymbol{\beta}}_S$  given by the right side of Eq. 28.4.1, and let  $\check{\mathbf{z}} = \mathbf{X}^\top (\mathbf{Y} - \mathbf{X}\check{\boldsymbol{\beta}})/(n\lambda)$ .

On  $S$ . By construction  $\mathbf{X}_S^\top (\mathbf{Y} - \mathbf{X}_S\check{\boldsymbol{\beta}}_S) = n\lambda \mathbf{z}_S$ , so  $\check{\mathbf{z}}_S = \mathbf{z}_S$ . By Eq. 28.4.1, for  $j \in S$ ,  $|\check{\beta}_j - \beta_j| \leq \|(\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{X}_S^\top \boldsymbol{\varepsilon}\|_\infty + g < \beta_{\min} \leq |\beta_j|$  on  $\mathcal{B}$ . So  $\check{\beta}_j$  is nonzero with the sign of  $\beta_j$ , and  $\check{z}_j = \text{sign}(\check{\beta}_j)$ .

Off  $S$ . Since  $\mathbf{X}\boldsymbol{\beta} = \mathbf{X}_S\boldsymbol{\beta}_S \in \mathcal{C}(\mathbf{X}_S)$ ,  $(\mathbf{I} - \mathbf{M}_S)\mathbf{Y} = (\mathbf{I} - \mathbf{M}_S)\boldsymbol{\varepsilon}$ , and

$$\mathbf{Y} - \mathbf{X}_S\check{\boldsymbol{\beta}}_S = \mathbf{Y} - \mathbf{M}_S\mathbf{Y} + n\lambda \mathbf{X}_S(\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{z}_S = (\mathbf{I} - \mathbf{M}_S)\boldsymbol{\varepsilon} + n\lambda \mathbf{X}_S(\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{z}_S.$$

So for  $k \notin S$ ,  $\check{z}_k = a_k + \mathbf{x}_k^\top (\mathbf{I} - \mathbf{M}_S)\boldsymbol{\varepsilon}/(n\lambda)$ , with  $|a_k| \leq 1 - \eta$  by the irrepresentable condition. On  $\mathcal{A}$ ,  $|\check{z}_k| < 1 - \eta + \eta = 1$ .

So  $\check{\boldsymbol{\beta}}$  and  $\check{\mathbf{z}}$  satisfy Eq. 28.3.2, and  $\check{\boldsymbol{\beta}}$  is a lasso estimate, because the conditions are sufficient (Theorem 27.4.1(b)). Its support is  $S$ ,  $\mathbf{X}_S$  has full column rank because  $C_{\min} > 0$ , and  $|\check{z}_k| < 1$  off  $S$ . By Lemma 28.4.1(b) it is the only lasso estimate.

(b)  $\mathbf{x}_k^\top (\mathbf{I} - \mathbf{M}_S)\boldsymbol{\varepsilon}/n$  is normal with mean zero and variance  $\sigma^2 \|(\mathbf{I} - \mathbf{M}_S)\mathbf{x}_k\|^2/n^2 \leq \sigma^2 \|\mathbf{x}_k\|^2/n^2 = \sigma^2/n$ . The  $j$ th entry of  $(\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \mathbf{X}_S^\top \boldsymbol{\varepsilon}$  is normal with mean zero and variance  $\sigma^2 [(\mathbf{X}_S^\top \mathbf{X}_S)^{-1}]_{jj} \leq \sigma^2/(nC_{\min})$ , by Theorem 1.7.7 applied to  $(\mathbf{X}_S^\top \mathbf{X}_S)^{-1}$ , whose largest eigenvalue is  $1/(nC_{\min})$  at most. Apply Lemma 28.3.2 with  $t = \eta\lambda$  and  $t = \beta_{\min} - g$ . The events in the lemma use  $>$ , and the events here use  $\geq$ , which have the same probability for normal variables with positive variance (and probability zero for those with zero variance).

(c) With the stated  $\lambda$ , the first bound in (b) equals  $\delta/2$ . With the stated  $\beta_{\min}$ , the second is at most  $\delta/2$ .

■

The three hypotheses ask that the active columns be well conditioned ( $C_{\min}$ ), that the inactive ones not be representable by them ( $\eta$ ), and that the coefficients stand out from noise and shrinkage ( $\beta_{\min}$ ). Since  $g \leq \lambda\sqrt{s/C_{\min}}$ , the last asks in the worst case for every nonzero coefficient to exceed a multiple of  $\sigma\sqrt{s \log p/n}$ . The theorem in this form is due to Wainwright (2009).

## 28.4.4 The condition is also necessary

### Proposition 28.4.3 — Failure of the irrepresentable condition

Let  $\mathbf{X}_S$  have full column rank, and suppose that for some  $k \notin S$

$$|a_k| = |\mathbf{x}_k^\top \mathbf{X}_S(\mathbf{X}_S^\top \mathbf{X}_S)^{-1} \text{sign}(\boldsymbol{\beta}_S)| > 1.$$

Then the probability that for some  $\lambda > 0$  some lasso estimate has  $\text{sign}(\hat{\beta}_\lambda) = \text{sign}(\beta)$  is at most  $\frac{1}{2}$ .

*Proof.* Suppose a lasso estimate at some  $\lambda > 0$  has the signs of  $\beta$ . Its support is  $S$ , so by the calculation leading to Eq. 28.4.2,  $|a_k + W_k/(n\lambda)| \leq 1$ , where  $W_k = \mathbf{x}_k^\top (\mathbf{I} - \mathbf{M}_S) \varepsilon$ . Since  $|a_k| > 1$ , this forces  $\text{sign}(a_k) W_k/(n\lambda) \leq 1 - |a_k| < 0$ , so  $\text{sign}(a_k) W_k < 0$ . The event in question is therefore contained in  $\{\text{sign}(a_k) W_k < 0\}$ , which does not depend on  $\lambda$ .  $W_k$  is normal with mean zero, so this event has probability  $\frac{1}{2}$  if  $(\mathbf{I} - \mathbf{M}_S) \mathbf{x}_k \neq \mathbf{0}$ , and probability zero otherwise. ■

The argument is due to Zhao and Yu (2006), who showed in addition that the condition with  $\leq 1$  is necessary for sign consistency. The bound  $\frac{1}{2}$  does not improve with more data. Sampling more observations cannot repair a design in which an inactive variable imitates the active ones with coefficients aligned with their signs. In the noiseless limit the lasso *never* recovers the signs (Exercise (C1)).

#### Example. (An inactive variable that looks active)

Let  $x_1, x_2$  be independent standard normal regressors, let  $x_3$  have correlation  $r$  with each of them, and let  $x_4, \dots, x_{500}$  be independent noise variables, with  $Y = 0.5x_1 + 0.5x_2 + \varepsilon$  and  $\sigma = 1$ . The population coefficients of  $x_3$  on  $(x_1, x_2)$  are  $(r, r)$ , so  $a_3 = 2r$  (Exercise (A1)). With  $r = 0.3$  the condition holds with margin 0.4; with the moderate correlation  $r = 0.6$  it fails. Both population designs have  $\lambda_{\min}(\Sigma) > 0$ , so both are benign for prediction.

Figure 28.4.1(a) shows, for each sample size, the fraction of 200 data sets (fixed design, new errors) for which *some* penalty on a fine grid gives exactly the right signs. When the condition holds, this rises from 0.12 at  $n = 50$  to 0.66 at  $n = 100$  and 1.000 at  $n = 800$ . When it fails, it rises only slowly, from 0.03 at  $n = 100$  to 0.155 at  $n = 800$ , and it can never exceed the bound  $\frac{1}{2}$  of Proposition 28.4.3, however large  $n$  is. The sample irrepresentable quantities at  $n = 800$  are 0.66 and 1.21.

Panel (b) shows why. Along the lasso path for one data set with  $r = 0.6$  and  $n = 800$ , the inactive  $x_3$  is the first variable to enter, because its inner product  $\mathbf{x}_3^\top \mathbf{X} \beta / n$  with the signal (the mean of the  $\mathbf{x}_3^\top \mathbf{Y} / n$  that decides which variable enters first) is 0.62 in this sample (0.6 in the population) and exceeds that of  $x_1$  and  $x_2$  (0.5 each). It never leaves. The lasso finds  $x_1$  and  $x_2$  later, but always together with  $x_3$ .

```
import numpy as np

def soft(z, t):
    """Soft thresholding  $S(z, t) = \text{sign}(z) \max(|z| - t, 0)$ ."""
    return np.sign(z) * np.maximum(np.abs(z) - t, 0.0)

def lasso_cd(X, Y, lam, B=None, tol=1e-9, max_sweeps=2000):
    """Minimize  $\|y - Xb\|_2^2 / (2n) + \text{lam} \|b\|_1$  for every column  $y$  of  $Y$ , by
    coordinate descent."""
    n, p = X.shape
    Y = Y.reshape(n, -1)
    B = np.zeros((p, Y.shape[1])) if B is None else B.copy()
```

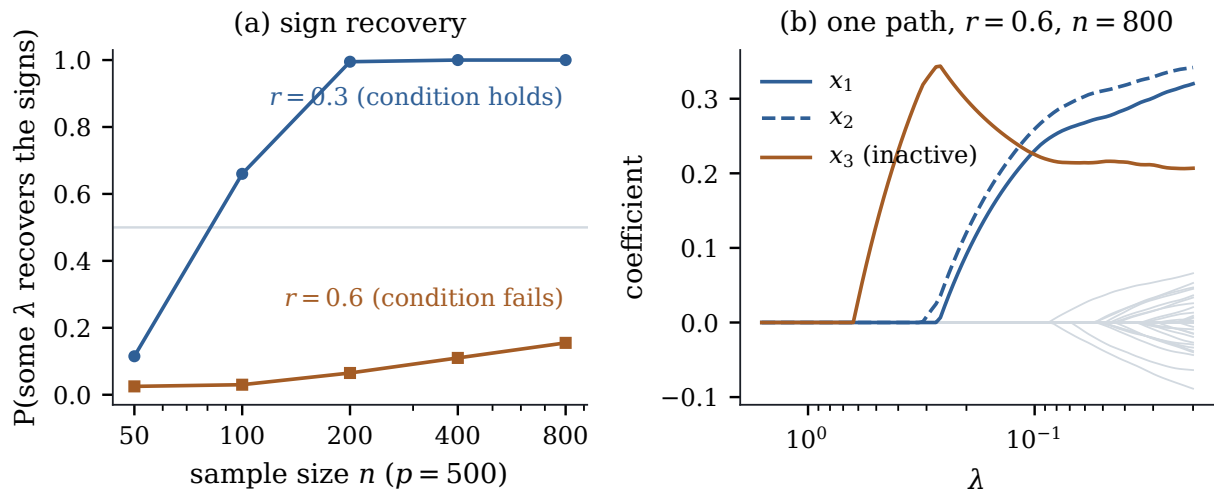


Figure 28.4.1: Support recovery with  $p = 500$ , two active regressors and an inactive regressor correlated  $r$  with both. (a) Fraction of data sets in which some penalty gives exactly the right signs, against  $n$ . (b) The lasso path of one data set with  $r = 0.6$ ,  $n = 800$ : the inactive  $x_3$  enters first and stays; grey lines are the noise variables.

```

R = Y - X @ B                                     # residuals, one column per
response
scale = np.sum(X ** 2, axis=0) / n
active = np.arange(p)
for sweep in range(max_sweeps):
    change = 0.0
    for j in active:
        z = X[:, j] @ R / n + scale[j] * B[j]
        new = soft(z, lam) / scale[j]
        d = new - B[j]
        if np.any(d != 0):
            R -= np.outer(X[:, j], d)
            B[j] = new
            change = max(change, np.max(np.abs(d)))
    if change < tol:
        if len(active) == p:                         # converged over all
coordinates
            break
        active = np.arange(p)                         # check every coordinate once
more
    else:
        active = np.flatnonzero(np.any(B != 0, axis=1)) if sweep % 5 else np.
arange(p)
return B

def make_design(rng, n, p, r):
    """x_1, x_2 independent; x_3 correlated r with each; x_4, ..., x_p independent
noise."""
    Z = rng.normal(size=(n, p))
    Z[:, 2] = r * Z[:, 0] + r * Z[:, 1] + np.sqrt(1 - 2 * r ** 2) * Z[:, 2]
    return Z / np.sqrt(np.sum(Z ** 2, axis=0) / n) # ||x_j||^2 = n

def irrepresentable(X, S, signs):
    """max over j outside S of |x_j^T X_S (X_S^T X_S)^{-1} sign(beta_S)|."""
    A = X[:, S]
    w = A @ np.linalg.solve(A.T @ A, signs)

```

```

out = np.delete(X.T @ w, S)
return np.max(np.abs(out))

def recovers_signs(X, Y, beta, lams):
    """For each column of Y: does some penalty in lams give exactly sign(beta)?"""
    ok = np.zeros(Y.shape[1], dtype=bool)
    B = None
    for lam in lams:                                     # from large to small, warm
        starts                                         starts
        B = lasso_cd(X, Y, lam, B)
        ok |= np.all(np.sign(B) == np.sign(beta)[: , None], axis=0)
    return ok

rng = np.random.default_rng(2840)
n, p, sigma = 100, 200, 1.0
beta = np.zeros(p)
beta[:2] = 0.5
S = np.array([0, 1])
lams = np.geomspace(1.0, 0.1, 30)
for r in [0.3, 0.6]:
    X = make_design(rng, n, p, r)
    Y = (X @ beta)[: , None] + sigma * rng.normal(size=(n, 100))
    print(f"r = {r}: irrepresentable quantity {irrepresentable(X, S, np.ones(2)):.2f}
        , "
          f" sign recovery in {recovers_signs(X, Y, beta, lams).mean():.2f} of 100
        data sets")

```

## 28.4.5 Screening instead of selection

A weaker goal, often enough in practice, is **screening**: the selected set should *contain*  $S$ , with false variables left for a second stage to remove. It needs only the restricted eigenvalue and a beta-min condition.

### Proposition 28.4.4 — Variable screening

Suppose  $\lambda \geq 2\lambda_0$ , the event  $\mathcal{T}$  of Eq. 28.3.5 occurs,  $\kappa^2(S) > 0$ , and  $\beta_{\min} > 3\lambda\sqrt{s}/\kappa^2(S)$ . Then every lasso estimate has  $\hat{S} \supseteq S$ .

*Proof.* If  $j \in S$  and  $\hat{\beta}_{\lambda,j} = 0$ , then  $|\hat{\beta}_{\lambda,j} - \beta_j| = |\beta_j| \geq \beta_{\min}$ . By Theorem 28.3.4(b),  $|\hat{\beta}_{\lambda,j} - \beta_j| \leq \|\hat{\beta}_\lambda - \beta\| \leq 3\lambda\sqrt{s}/\kappa^2(S) < \beta_{\min}$ , a contradiction. ■

In Example (An inactive variable that looks active) with  $r = 0.6$ , screening succeeds where selection fails:  $x_1$  and  $x_2$  are found, together with  $x_3$ . Second stages build on this. The **thresholded lasso** discards small coefficients and needs no irrepresentable condition (Exercise (B2)). The **adaptive lasso** of Zou (2006) refits with penalty weights  $1/|\hat{\beta}_j|$  from a first stage, and **stability selection** (Meinshausen and Bühlmann 2010) keeps the variables selected in most subsamples. All need beta-min conditions; none can find a coefficient below the noise level.

*Warning.* The lasso's selected set is a random set that depends on small features of the data, especially among correlated regressors. Reporting it as “the important variables” without an assessment of its stability invites overinterpretation. [Section 28.5](#) shows what goes wrong if the selected set is then treated as fixed.

## 28.4.6 Exercises

### 28.4.6.1 A. Check your understanding

#### Exercise A1.

In the population version of [Example \(An inactive variable that looks active\)](#), with  $\Sigma_{SS} = \mathbf{I}_2$  and  $\Sigma_{3S} = (r, r)$ , compute  $\Sigma_{3S}\Sigma_{SS}^{-1}\text{sign}(\boldsymbol{\beta}_S)$  for  $\boldsymbol{\beta}_S = (0.5, 0.5)$  and for  $\boldsymbol{\beta}_S = (0.5, -0.5)$ . For which signs does the irrepresentable condition fail?

*Solution.*  $\Sigma_{3S}\Sigma_{SS}^{-1} = (r, r)$ . With signs  $(1, 1)$  the quantity is  $2r$ , which exceeds 1 for  $r > \frac{1}{2}$ . With signs  $(1, -1)$  it is 0, so the condition holds with  $\eta = 1$ . The trouble arises only when  $x_3$  imitates the combination  $x_1 + x_2$  that appears in the signal.

### 28.4.6.2 B. Practice

#### Exercise B1.

Let  $\Sigma = (1-\rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$  with  $0 \leq \rho < 1$ . Show that  $\Sigma_{kS}\Sigma_{SS}^{-1}\mathbf{z}_S = \rho(\mathbf{1}^\top\mathbf{z}_S)/\{1+(s-1)\rho\}$  for  $k \notin S$ , and conclude that the population irrepresentable condition holds for every sign pattern, with margin at least  $(1-\rho)/\{1+(s-1)\rho\}$ .

#### Exercise B2.

(*Thresholded lasso.*) Under the assumptions of [Theorem 28.3.4\(b\)](#), let  $B = 3\lambda\sqrt{s}/\kappa^2(S)$  and suppose  $\beta_{\min} > 2B$ . Show that  $\{j : |\hat{\beta}_{\lambda,j}| > B\} = S$ , and that the signs of the retained coefficients are correct. No irrepresentable condition is needed.

*Solution.* Every coordinate satisfies  $|\hat{\beta}_{\lambda,j} - \beta_j| \leq \|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\| \leq B$ . For  $j \notin S$ ,  $|\hat{\beta}_{\lambda,j}| \leq B$ , so  $j$  is discarded. For  $j \in S$ ,  $|\hat{\beta}_{\lambda,j}| \geq |\beta_j| - B > 2B - B = B$ , so  $j$  is kept, and  $\hat{\beta}_{\lambda,j}$  lies within  $B < |\beta_j|$  of  $\beta_j$ , so it has the same sign. The price is the stronger beta-min condition  $\beta_{\min} > 6\lambda\sqrt{s}/\kappa^2(S)$  and a threshold that depends on the unknown  $\kappa$  and  $s$ .

### 28.4.6.3 C. Going deeper

#### Exercise C1.

Let  $\sigma = 0$ , so  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta}$ , and let  $\mathbf{X}_S$  have full column rank. Show that if  $\max_{k \notin S} |a_k| < 1$ , then every  $\lambda$  with  $0 < \lambda < \beta_{\min}/\|\hat{\Sigma}_{SS}^{-1}\text{sign}(\boldsymbol{\beta}_S)\|_\infty$  gives a unique lasso estimate with the correct signs, and that if  $|a_k| > 1$  for some  $k \notin S$ , then no  $\lambda > 0$  does.

*Solution.* With  $\varepsilon = \mathbf{0}$ , the events  $\mathcal{A}$  and  $\mathcal{B}$  of [Theorem 28.4.2](#) hold whenever  $\eta > 0$  and  $\beta_{\min} > g$ , which is the stated range of  $\lambda$ , with  $\eta = 1 - \max_k |a_k|$ . Part (a) of the theorem applies, and its proof uses no probability. Conversely, if some lasso estimate had the correct signs, [Eq. 28.4.2](#) with  $\varepsilon = \mathbf{0}$  would require  $|a_k| \leq 1$  for all  $k \notin S$ . So with perfect data, the irrepresentable condition (in its weak form) is necessary and, with strict inequality, sufficient.

## 28.5 Post-selection and debiased inference

This section turns to confidence intervals and tests for a single coefficient  $\beta_j$  when  $p > n$ . Two tempting approaches fail: least squares intervals for the model the lasso selected, as if it had been fixed in advance, and intervals around the lasso estimate itself, whose shrinkage bias is at least as large as its standard error. The fix is a one-step correction of the lasso, which turns out to be the Frisch–Waugh–Lovell theorem with a lasso in place of a projection.

### 28.5.1 Why intervals after selection fail

The least squares interval of [Theorem 12.1.1](#) has coverage  $1 - \alpha$  for a model fixed before the data are seen. After selection, the interval for  $\beta_j$  is computed only in the samples where  $x_j$  was selected, which are those in which its estimated effect happened to be large. Conditional on selection the estimate is biased away from zero, and the interval is centred in the wrong place. This is the winner’s curse of [Chapter 29](#) ([Proposition 29.5.1](#)), and the multiplicity problem of [Chapter 13](#) in another form. With the lasso and  $p > n$  it is severe, because the selection is among hundreds or thousands of candidates.

Leeb and Pötscher (2006) proved that this cannot be repaired by estimating the distribution of the post-selection estimator, which depends on the unknown coefficients in a way that no estimator can track uniformly. One can change the target (the selected model’s coefficients, conditional on the selection) or the estimator. This section changes the estimator.

The lasso estimate itself is no better as a centre for an interval. On the event of [Theorem 28.4.2](#), [Eq. 28.4.1](#) shows that it equals least squares on  $S$  minus  $\lambda \hat{\Sigma}_{SS}^{-1} \text{sign}(\beta_S)$ . The shift is of order  $\lambda \asymp \sigma \sqrt{\log p/n}$ , which is larger than the standard error  $\sigma/\sqrt{n}$  of the oracle estimate by the factor  $\sqrt{\log p}$ . An interval of the usual width centred at the lasso estimate misses the truth more and more often as  $p$  grows.

### 28.5.2 The debiased lasso

Write  $\hat{\Sigma} = \mathbf{X}^\top \mathbf{X}/n$ , and let  $\hat{\Theta}$  be a  $p \times p$  matrix computed from  $\mathbf{X}$  alone, meant to act like an inverse of  $\hat{\Sigma}$  (which does not exist when  $p > n$ ). Its rows are  $\hat{\theta}_1^\top, \dots, \hat{\theta}_p^\top$ .

#### Definition 28.5.1 — Debiased lasso

Given a lasso estimate  $\hat{\beta}_\lambda$  and  $\hat{\Theta}$ , the **debiased** (or **desparsified**) lasso is

$$\hat{\mathbf{b}} = \hat{\boldsymbol{\beta}}_\lambda + \frac{1}{n} \hat{\boldsymbol{\Theta}} \mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}}_\lambda). \quad (28.5.1)$$

Its  $j$ th entry uses only the row  $\hat{\boldsymbol{\theta}}_j$ . The quantity  $\mu_j = \|\hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\theta}}_j - \mathbf{e}_j\|_\infty$  measures how far  $\hat{\boldsymbol{\theta}}_j$  is from being the  $j$ th row of an inverse of  $\hat{\boldsymbol{\Sigma}}$ .

The correction adds to the lasso a multiple of the residual correlations  $\mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \hat{\boldsymbol{\beta}}_\lambda)/n$ . If  $\hat{\boldsymbol{\Sigma}}$  were invertible and  $\hat{\boldsymbol{\Theta}} = \hat{\boldsymbol{\Sigma}}^{-1}$ ,  $\hat{\mathbf{b}}$  would be the least squares estimate (Exercise (A1)).

### Theorem 28.5.1 — Debiased lasso

In the setting of Section 28.3, let  $\hat{\boldsymbol{\Theta}}$  depend on  $\mathbf{X}$  only, and let  $\hat{\mathbf{b}}$  be the debiased lasso.

- (a) (*Decomposition.*)  $\sqrt{n}(\hat{\mathbf{b}} - \boldsymbol{\beta}) = \mathbf{w} + \mathbf{r}$ , where  $\mathbf{w} = \hat{\boldsymbol{\Theta}} \mathbf{X}^\top \boldsymbol{\varepsilon} / \sqrt{n} \sim N_p(\mathbf{0}, \sigma^2 \hat{\boldsymbol{\Theta}} \hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\Theta}}^\top)$  exactly, and  $\mathbf{r} = -\sqrt{n}(\hat{\boldsymbol{\Theta}} \hat{\boldsymbol{\Sigma}} - \mathbf{I})(\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta})$ .
- (b) (*Remainder.*)  $|r_j| \leq \sqrt{n} \mu_j \|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|_1$ . If  $\lambda \geq 2\lambda_0$ ,  $\phi^2(S) > 0$  and  $\mathcal{T}$  occurs, then  $|r_j| \leq 4\sqrt{n} \mu_j \lambda s / \phi^2(S)$ .
- (c) (*Variance.*) If  $\mu_j < 1$ , then  $\omega_j^2 = \hat{\boldsymbol{\theta}}_j^\top \hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\theta}}_j \geq (1 - \mu_j)^2$ .
- (d) (*Asymptotic normality.*) Consider a sequence of such problems with  $n \rightarrow \infty$ , in which  $p, s, \mathbf{X}, \boldsymbol{\beta}$  and  $\hat{\boldsymbol{\Theta}}$  may change with  $n$  and  $\sigma$  is fixed. Suppose  $\lambda = 2\lambda_0$  with  $\delta = \delta_n \rightarrow 0$  and  $\log(1/\delta_n) = O(\log p)$ ;  $\phi^2(S) \geq \phi_0^2 > 0$ ;  $\mu_j \leq C\sqrt{\log p/n}$  for a constant  $C$ ;  $s \log p / \sqrt{n} \rightarrow 0$ ; and  $\hat{\sigma}/\sigma \rightarrow 1$  in probability. Then

$$T_j = \frac{\sqrt{n}(\hat{b}_j - \beta_j)}{\hat{\sigma} \omega_j} \rightarrow N(0, 1) \quad \text{in distribution,} \quad (28.5.2)$$

and the interval  $\hat{b}_j \pm z_{\alpha/2} \hat{\sigma} \omega_j / \sqrt{n}$  has coverage tending to  $1 - \alpha$ .

*Proof.*

- (a) Substitute  $\mathbf{Y} = \mathbf{X} \boldsymbol{\beta} + \boldsymbol{\varepsilon}$  in Eq. 28.5.1 and write  $\boldsymbol{\Delta} = \hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}$ :

$$\hat{\mathbf{b}} - \boldsymbol{\beta} = \boldsymbol{\Delta} + \frac{1}{n} \hat{\boldsymbol{\Theta}} \mathbf{X}^\top (\boldsymbol{\varepsilon} - \mathbf{X} \boldsymbol{\Delta}) = \frac{1}{n} \hat{\boldsymbol{\Theta}} \mathbf{X}^\top \boldsymbol{\varepsilon} - (\hat{\boldsymbol{\Theta}} \hat{\boldsymbol{\Sigma}} - \mathbf{I}) \boldsymbol{\Delta}.$$

Multiply by  $\sqrt{n}$ . Since  $\hat{\boldsymbol{\Theta}}$  is a fixed matrix,  $\mathbf{w}$  is a linear function of  $\boldsymbol{\varepsilon}$ , hence normal with mean zero and covariance  $\hat{\boldsymbol{\Theta}} \mathbf{X}^\top (\sigma^2 \mathbf{I}) \mathbf{X} \hat{\boldsymbol{\Theta}}^\top / n = \sigma^2 \hat{\boldsymbol{\Theta}} \hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\Theta}}^\top$  (Theorem 3.3.1).

- (b) The  $j$ th row of  $\hat{\boldsymbol{\Theta}} \hat{\boldsymbol{\Sigma}} - \mathbf{I}$  is  $(\hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\theta}}_j - \mathbf{e}_j)^\top$ , because  $\hat{\boldsymbol{\Sigma}}$  is symmetric. By Hölder's inequality  $|r_j| \leq \sqrt{n} \mu_j \|\boldsymbol{\Delta}\|_1$ . The second bound follows from Theorem 28.3.4(a).
- (c) The columns of  $\mathbf{X}$  have  $\|\mathbf{x}_j\|^2 = n$ , so  $\mathbf{e}_j^\top \hat{\boldsymbol{\Sigma}} \mathbf{e}_j = 1$ . The  $j$ th entry of  $\hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\theta}}_j$  differs from 1 by at most  $\mu_j$ , so  $\mathbf{e}_j^\top \hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\theta}}_j \geq 1 - \mu_j > 0$ . The Cauchy–Schwarz inequality for the nonnegative definite form  $\hat{\boldsymbol{\Sigma}}$  (Proposition 1.1.1 applied to  $\hat{\boldsymbol{\Sigma}}^{1/2} \mathbf{e}_j$  and  $\hat{\boldsymbol{\Sigma}}^{1/2} \hat{\boldsymbol{\theta}}_j$ ) gives  $(1 - \mu_j)^2 \leq (\mathbf{e}_j^\top \hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\theta}}_j)^2 \leq (\mathbf{e}_j^\top \hat{\boldsymbol{\Sigma}} \mathbf{e}_j) \omega_j^2 = \omega_j^2$ .
- (d) Write  $T_j = (\sigma/\hat{\sigma})\{w_j/(\sigma \omega_j) + r_j/(\sigma \omega_j)\}$ . By (a),  $w_j/(\sigma \omega_j)$  has exactly the  $N(0, 1)$  law. For the remainder,  $\log(2p/\delta_n) = \log 2 + \log p + \log(1/\delta_n) \leq C' \log p$  for a constant  $C'$  and large  $n$ , so  $\lambda = 2\lambda_0 \leq 2\sigma\sqrt{2C' \log p/n}$ . On  $\mathcal{T}$ , which has

probability at least  $1 - \delta_n \rightarrow 1$ , part (b) gives

$$\begin{aligned} |r_j| &\leq 4\sqrt{n} \cdot C \sqrt{\frac{\log p}{n}} \cdot 2\sigma \sqrt{\frac{2C' \log p}{n}} \cdot \frac{s}{\phi_0^2} \\ &= \frac{8\sqrt{2C'} C \sigma}{\phi_0^2} \cdot \frac{s \log p}{\sqrt{n}} \rightarrow 0. \end{aligned}$$

By (c) and  $\mu_j \rightarrow 0$ ,  $\omega_j \geq \frac{1}{2}$  for large  $n$ . Hence  $r_j/(\sigma\omega_j) \rightarrow 0$  in probability, and Slutsky's lemma (with  $\sigma/\hat{\sigma} \rightarrow 1$ ) gives Eq. 28.5.2. The coverage statement is  $\Pr(|T_j| \leq z_{\alpha/2}) \rightarrow 1 - \alpha$ . ■

The theorem separates what the design must deliver (a  $\hat{\Theta}$  with small  $\mu_j$ , and  $\phi^2(S)$  bounded below) from what the lasso delivers (an  $\ell_1$  error of order  $s\sqrt{\log p/n}$ ). The condition  $s \log p/\sqrt{n} \rightarrow 0$  is stronger than the  $s \log p/n \rightarrow 0$  that suffices for prediction: the remainder must be small compared with the standard error  $1/\sqrt{n}$ .

### 28.5.3 Constructing the approximate inverse

The remaining question is how to find a  $\hat{\Theta}$  with small  $\mu_j$  when  $\hat{\Sigma}$  is singular. The *node-wise lasso* regresses each column on all the others.

#### Lemma 28.5.2 — Node-wise lasso

For a fixed  $j$ , let  $\hat{\gamma}_j \in \mathbb{R}^{p-1}$  be a lasso estimate, with penalty  $\lambda_j$ , for the regression of  $\mathbf{x}_j$  on the other columns  $\mathbf{X}_{-j}$ . Put  $\hat{\mathbf{z}}_j = \mathbf{x}_j - \mathbf{X}_{-j}\hat{\gamma}_j$  and  $\hat{\tau}_j^2 = \mathbf{x}_j^\top \hat{\mathbf{z}}_j/n$ , and let  $\hat{\theta}_j$  have entry  $1/\hat{\tau}_j^2$  in position  $j$  and  $-\hat{\gamma}_j/\hat{\tau}_j^2$  elsewhere. Then:

- (a)  $\hat{\tau}_j^2 = \|\hat{\mathbf{z}}_j\|^2/n + \lambda_j \|\hat{\gamma}_j\|_1 \geq 0$ ;
- (b) if  $\hat{\tau}_j^2 > 0$ , then  $(\hat{\Sigma}\hat{\theta}_j)_j = 1$  and  $\mu_j = \|\hat{\Sigma}\hat{\theta}_j - \mathbf{e}_j\|_\infty \leq \lambda_j/\hat{\tau}_j^2$ ;
- (c) the  $j$ th debiased estimate is

$$\hat{b}_j = \frac{\hat{\mathbf{z}}_j^\top (\mathbf{Y} - \mathbf{X}_{-j}\hat{\beta}_{\lambda, -j})}{\hat{\mathbf{z}}_j^\top \mathbf{x}_j}. \quad (28.5.3)$$

*Proof.* The optimality conditions Eq. 28.3.2 for the node-wise regression say that

$$\mathbf{X}_{-j}^\top \hat{\mathbf{z}}_j/n = \lambda_j \hat{\mathbf{k}},$$

where  $\hat{\mathbf{k}}$  has entries of size at most 1 and equal to  $\text{sign}(\hat{\gamma}_{j,k})$  where  $\hat{\gamma}_{j,k} \neq 0$ . Hence  $\hat{\gamma}_j^\top \mathbf{X}_{-j}^\top \hat{\mathbf{z}}_j/n = \lambda_j \|\hat{\gamma}_j\|_1$ , and since  $\mathbf{x}_j = \hat{\mathbf{z}}_j + \mathbf{X}_{-j}\hat{\gamma}_j$ ,

$$\hat{\tau}_j^2 = \frac{\mathbf{x}_j^\top \hat{\mathbf{z}}_j}{n} = \frac{\|\hat{\mathbf{z}}_j\|^2}{n} + \lambda_j \|\hat{\gamma}_j\|_1,$$

which is (a). For (b),  $\mathbf{X}\hat{\theta}_j = (\mathbf{x}_j - \mathbf{X}_{-j}\hat{\gamma}_j)/\hat{\tau}_j^2 = \hat{\mathbf{z}}_j/\hat{\tau}_j^2$ , so the  $k$ th entry of  $\hat{\Sigma}\hat{\theta}_j$  is



$\mathbf{x}_k^\top \hat{\mathbf{z}}_j / (n\hat{\tau}_j^2)$ . For  $k = j$  this is 1 by the definition of  $\hat{\tau}_j^2$ . For  $k \neq j$  it is  $\lambda_j \hat{\kappa}_k / \hat{\tau}_j^2$ , of size at most  $\lambda_j / \hat{\tau}_j^2$ . For (c),  $\hat{b}_j$  is  $\hat{\beta}_{\lambda,j} + \hat{\mathbf{z}}_j^\top (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_\lambda) / (n\hat{\tau}_j^2)$ . Put  $n\hat{\tau}_j^2 = \hat{\mathbf{z}}_j^\top \mathbf{x}_j$  and  $\mathbf{X}\hat{\boldsymbol{\beta}}_\lambda = \mathbf{x}_j \hat{\beta}_{\lambda,j} + \mathbf{X}_{-j} \hat{\boldsymbol{\beta}}_{\lambda,-j}$ ; the terms in  $\hat{\beta}_{\lambda,j}$  cancel. ■

Part (c) shows what the debiased lasso is. By the Frisch–Waugh–Lovell theorem (Theorem 6.6.2), the least squares coefficient of  $\mathbf{x}_j$  comes from regressing  $\mathbf{Y}$  on the part of  $\mathbf{x}_j$  orthogonal to the other columns, which is zero when  $p > n$ . The node-wise lasso replaces it by a lasso residual  $\hat{\mathbf{z}}_j$ , nearly orthogonal to every other column by (b), and removes the other regressors' effect approximately through  $\mathbf{X}_{-j} \hat{\boldsymbol{\beta}}_{\lambda,-j}$ . Javanmard and Montanari (2014) instead choose  $\hat{\boldsymbol{\theta}}_j$  to minimize  $\hat{\boldsymbol{\theta}}^\top \hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\theta}}$  subject to  $\|\hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\theta}} - \mathbf{e}_j\|_\infty \leq \mu$ .

*Remark* (Sketch: when the assumptions of part (d) hold). This is a sketch; the full arguments are in van de Geer, Bühlmann, Ritov and Dezeure (2014) and Javanmard and Montanari (2014). Suppose the rows of  $\mathbf{X}$  are independent  $N_p(\mathbf{0}, \boldsymbol{\Sigma})$ , the eigenvalues of  $\boldsymbol{\Sigma}$  are bounded away from 0 and  $\infty$ , and row  $j$  of  $\boldsymbol{\Sigma}^{-1}$  has at most  $s_j$  nonzero entries. The population regression of  $x_j$  on the others then has  $s_j$  nonzero coefficients and residual variance  $\tau_j^2 = 1/(\boldsymbol{\Sigma}^{-1})_{jj}$ . By the lemma  $\mu_j \leq \lambda_j / \hat{\tau}_j^2$  *deterministically*, so with  $\lambda_j \asymp \sqrt{\log p/n}$  it remains to show  $\hat{\tau}_j^2 \approx \tau_j^2$ . This follows from Section 28.3 applied to the node-wise regression, with the random-design bound of Section 28.2, when  $s_j \log p/n \rightarrow 0$ ; the scaled lasso (Sun and Zhang 2012) supplies  $\hat{\sigma}$ . Then  $\omega_j^2 \rightarrow (\boldsymbol{\Sigma}^{-1})_{jj}$ , which van de Geer and coauthors show to be the smallest asymptotic variance of a regular estimator.

*Warning.* The theorem covers one prespecified coefficient, or a fixed finite set. Coefficients chosen *because* they look large bring back the selection problem, and simultaneous intervals need a multiplicity correction such as Bonferroni (Theorem 13.4.1). With moderate  $n$  and a truth that is not very sparse, the remainder  $r_j$  can be comparable to the noise, and coverage suffers.

## 28.5.4 A simulation

### Example. (Coverage of three intervals)

Take  $n = 200$ ,  $p = 400$ , a fixed Gaussian design with correlations  $0.5^{|j-k|}$ ,  $\sigma = 1$ , and  $\beta_1 = 1$ ,  $\beta_6 = 0.15$ ,  $\beta_{11} = -1$ ,  $\beta_{16} = 0.15$ , all other coefficients zero. The targets are a strong coefficient  $\beta_1$ , a weak one  $\beta_6$ , and  $\beta_2 = 0$ , whose regressor is correlated 0.5 with  $x_1$ . In each of 1000 data sets we fit the lasso with  $\lambda = \sigma \sqrt{2 \log p/n} = 0.245$ , below  $2\lambda_0$  for the reason seen in Example (How tight are the bounds?), and compute three 95% intervals:

- the **naïve** interval: least squares on the selected variables and its  $t$  interval, reported only when the target was selected;
- the **debiased** interval  $\hat{b}_j \pm 1.96 \hat{\sigma} \omega_j / \sqrt{n}$ , with node-wise lasso rows ( $\lambda_j = 0.024$ ) and  $\hat{\sigma}^2$  the lasso residual sum of squares divided by  $n$  minus the number of selected variables;
- the **oracle**  $t$  interval from least squares on the true support plus the target.

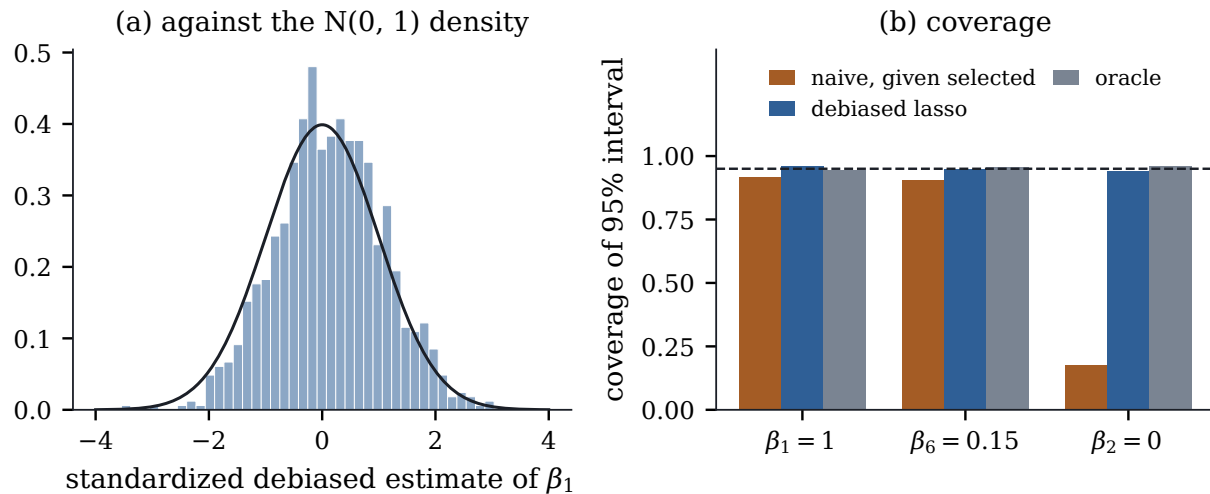


Figure 28.5.1: Inference for single coefficients with  $n = 200$ ,  $p = 400$ , 1000 data sets. (a) Histogram of the standardized debiased estimates of  $\beta_1$  and the  $N(0, 1)$  density. (b) Coverage of nominal 95% intervals: naive least squares after lasso selection (computed only when the target is selected), the debiased lasso, and the oracle.

The results are in Figure 28.5.1. For the null coefficient  $\beta_2$ , the lasso selected  $x_2$  in a fraction 0.069 of the data sets, and in those the naive interval covered zero only in a fraction 0.174. For the weak coefficient, selected in a fraction 0.329, the naive coverage was 0.903, and for the strong one 0.915. The debiased intervals covered  $\beta_1$ ,  $\beta_6$  and  $\beta_2$  in fractions 0.960, 0.946 and 0.940 respectively, close to the oracle's 0.943, 0.956 and 0.958 for the same three coefficients.

On the strong coefficient the lasso's average error is  $-0.210$ , close to  $-\lambda$ , and the debiased estimate's is 0.014. The standardized estimates  $T_1$  have mean 0.14 and standard deviation 0.95 (panel (a)). Two costs remain.

The debiased intervals are wider than the oracle's (0.393 against 0.288 for  $\beta_1$ ): part of this is the price of not knowing the support, and part is tuning. The node-wise penalty  $\lambda_j = 0.024$ , a tenth of  $\sqrt{2 \log p/n}$ , keeps the bound  $\mu_j \leq \lambda_j / \hat{\tau}_j^2$  of Lemma 28.5.2, and so the remainder, small. But it lets the node-wise fits explain much of each column:  $\hat{\tau}_j^2$  (0.28, 0.25 and 0.27 for  $\beta_1$ ,  $\beta_6$  and  $\beta_2$ ) is well below the population value  $1/(\Sigma^{-1})_{jj}$ , which is  $\frac{3}{4}$  for  $j = 1$  and  $\frac{3}{5}$  otherwise, and that allows a larger  $\omega_j$  (Exercise (B2)).

The second cost is a remainder bias: 0.049 for  $\beta_2$  and 0.046 for the weak  $\beta_6$ , about a third of its value. This is the term  $r_j/\sqrt{n}$  of Theorem 28.5.1, at most  $\mu_j \|\hat{\beta}_\lambda - \beta\|_1$ , with  $\mu_j$  between 0.088 and 0.099 and an average lasso  $\ell_1$  error of 0.74. At  $n = 200$  that product is not negligible.

```
import numpy as np
from scipy import stats

def soft(z, t):
    """Soft thresholding  $S(z, t) = \text{sign}(z) \max(|z| - t, 0)$ ."""
    return np.sign(z) * np.maximum(np.abs(z) - t, 0.0)

def lasso_cd(X, Y, lam, B=None, tol=1e-9, max_sweeps=2000):
    """Minimize  $\|y - Xb\|_2^2 / (2n) + \text{lam} \|b\|_1$  for every column  $y$  of  $Y$ , by
    coordinate descent."""
    n, p = X.shape
```

```

Y = Y.reshape(n, -1)
B = np.zeros((p, Y.shape[1])) if B is None else B.copy()
R = Y - X @ B                                     # residuals, one column per
response
scale = np.sum(X ** 2, axis=0) / n
active = np.arange(p)
for sweep in range(max_sweeps):
    change = 0.0
    for j in active:
        z = X[:, j] @ R / n + scale[j] * B[j]
        new = soft(z, lam) / scale[j]
        d = new - B[j]
        if np.any(d != 0):
            R -= np.outer(X[:, j], d)
            B[j] = new
            change = max(change, np.max(np.abs(d)))
    if change < tol:
        if len(active) == p:                         # converged over all
coordinates
            break
        active = np.arange(p)                         # check every coordinate once
more
    else:
        active = np.flatnonzero(np.any(B != 0, axis=1)) if sweep % 5 else np.
arange(p)
return B

def nodewise_row(X, j, lam_j):
    """Row theta_j of Theta from the lasso of x_j on the other columns, and tau_j^2.
    """
    n, p = X.shape
    others = np.delete(np.arange(p), j)
    gamma = lasso_cd(X[:, others], X[:, j], lam_j)[:, 0]
    z = X[:, j] - X[:, others] @ gamma                # lasso residual of x_j on the
rest
    tau2 = X[:, j] @ z / n
    theta = np.zeros(p)
    theta[j] = 1.0
    theta[others] = -gamma
    return theta / tau2, tau2

def debiased_interval(X, Y, B, j, theta, level=0.95):
    """Debiased estimate of beta_j and its normal-theory interval, one per column of
    Y."""
    n = X.shape[0]
    R = Y - X @ B
    b = B[j] + theta @ X.T @ R / n                    # one-step correction
    sigma_hat = np.sqrt(np.sum(R ** 2, axis=0) / (n - np.sum(B != 0, axis=0)))
    G = X.T @ X / n
    se = sigma_hat * np.sqrt(theta @ G @ theta / n)
    z = stats.norm.ppf((1 + level) / 2)
    return b, b - z * se, b + z * se

def toeplitz_design(rng, n, p, rho):
    idx = np.arange(p)
    L = np.linalg.cholesky(rho ** np.abs(idx[:, None] - idx[None, :]))
    X = rng.normal(size=(n, p)) @ L.T
    return X / np.sqrt(np.sum(X ** 2, axis=0) / n)

```

```

rng = np.random.default_rng(2850)
n, p, sigma = 200, 400, 1.0
X = toeplitz_design(rng, n, p, rho=0.5)
beta = np.zeros(p)
beta[[0, 5, 10, 15]] = [1.0, 0.15, -1.0, 0.15]
lam = sigma * np.sqrt(2 * np.log(p) / n)
targets = {"strong": 0, "weak": 5, "null": 1}

Ys = (X @ beta)[:, None] + sigma * rng.normal(size=(n, 200))
Bs = lasso_cd(X, Ys, lam)
for name, j in targets.items():
    theta, tau2 = nodewise_row(X, j, 0.1 * np.sqrt(2 * np.log(p) / n))
    b, lo, hi = debiased_interval(X, Ys, Bs, j, theta)
    print(f"{name:6s}: lasso bias {Bs[j].mean() - beta[j]:+.3f}, debiased bias {b.
mean() - beta[j]:+.3f},
        f" coverage {np.mean((lo <= beta[j]) & (beta[j] <= hi)):.3f}")

```

## 28.5.5 Other routes to valid inference

Other approaches answer different questions. **Sample splitting** (Wasserman and Roeder 2009) selects on one half of the data and fits on the other ([Exercise B1](#)). **Selective inference** (Lee, Sun, Sun and Taylor 2016) conditions on the lasso having chosen the reported model, a union of polyhedra in  $\mathbf{Y}$ . **Simultaneous inference over all models** (Berk, Brown, Buja, Zhang and Zhao 2013) widens the intervals, as Scheffé's method does. These target the coefficients of the *selected* model, which generally differ from  $\beta_j$ . **Double selection** (Belloni, Chernozhukov and Hansen 2014) targets  $\beta_j$ : it keeps the controls that predict the response or  $x_j$  and refits least squares, the idea of [Eq. 28.5.3](#).

## 28.5.6 Exercises

### 28.5.6.1 A. Check your understanding

#### Exercise A1.

Suppose  $p < n$ ,  $\mathbf{X}$  has full column rank and  $\hat{\Theta} = \hat{\Sigma}^{-1}$ . Show that the debiased lasso [Eq. 28.5.1](#) equals the least squares estimate, whatever  $\hat{\beta}_\lambda$  is, and that  $\mu_j = 0$  and  $\mathbf{r} = \mathbf{0}$  in [Theorem 28.5.1](#).

*Solution.*  $\hat{\mathbf{b}} = \hat{\beta}_\lambda + (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top (\mathbf{Y} - \mathbf{X} \hat{\beta}_\lambda) = \hat{\beta}_\lambda + \hat{\beta} - \hat{\beta}_\lambda = \hat{\beta}$ , where  $\hat{\beta} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$ . Then  $\hat{\Sigma} \hat{\Theta} = \mathbf{I}$ , so every  $\mu_j = 0$  and  $\mathbf{r} = \mathbf{0}$ , and part (a) of the theorem reduces to  $\sqrt{n}(\hat{\beta} - \beta) \sim N_p(\mathbf{0}, \sigma^2 \hat{\Sigma}^{-1})$  ([Theorem 7.5.1](#)).

#### Exercise A2.

On the event of [Theorem 28.4.2](#), with  $\hat{\Sigma}_{SS} = \mathbf{I}$ , show that the interval  $\hat{\beta}_{\lambda,j} \pm 1.96 \sigma / \sqrt{n}$  for  $j \in S$  covers  $\beta_j$  iff  $|\mathbf{x}_j^\top \boldsymbol{\varepsilon} / \sqrt{n} \sigma - \sqrt{n} \lambda \text{sign}(\beta_j) / \sigma| \leq 1.96$ . If  $\lambda = 2\lambda_0$ , what happens to its coverage as  $p \rightarrow \infty$ ?

## 28.5.6.2 B. Practice

**Exercise B1.**

(*Sample splitting.*) Split the observations into halves  $I_1$  and  $I_2$  chosen independently of the data. Select a set  $\hat{S}$  of columns using only the data in  $I_1$ , and suppose  $\mathbf{X}_{I_2, \hat{S}}$  has full column rank and  $|\hat{S}| < |I_2|$ . Show that if  $\hat{S} \supseteq S$ , then conditionally on the data in  $I_1$ , the usual  $t$  interval for  $\beta_j$  ( $j \in \hat{S}$ ) from the least squares fit of  $\mathbf{Y}_{I_2}$  on  $\mathbf{X}_{I_2, \hat{S}}$  has exact coverage  $1 - \alpha$ . What does the fit estimate when  $\hat{S}$  misses an active variable?

*Solution.* The errors in  $I_2$  are independent of those in  $I_1$ , so conditionally on the data in  $I_1$  the set  $\hat{S}$  is fixed and  $\mathbf{Y}_{I_2} \sim \mathcal{N}(\mathbf{X}_{I_2} \boldsymbol{\beta}, \sigma^2 \mathbf{I})$ . If  $\hat{S} \supseteq S$ , then  $\mathbf{X}_{I_2} \boldsymbol{\beta} = \mathbf{X}_{I_2, \hat{S}} \boldsymbol{\beta}_{\hat{S}}$ , so the fit on  $\mathbf{X}_{I_2, \hat{S}}$  is a correctly specified normal linear model with coefficients  $\boldsymbol{\beta}_{\hat{S}}$ , and the  $t$  interval of [Theorem 12.1.1](#) has exact conditional coverage. Averaging over the data in  $I_1$ , coverage is exact on the event  $\hat{S} \supseteq S$ . If  $\hat{S}$  misses an active variable, the fit estimates the projected coefficients  $(\mathbf{X}_{I_2, \hat{S}}^\top \mathbf{X}_{I_2, \hat{S}})^{-1} \mathbf{X}_{I_2, \hat{S}}^\top \mathbf{X}_{I_2} \boldsymbol{\beta}$  ([Proposition 5.5.7](#)), which differ from  $\boldsymbol{\beta}_{\hat{S}}$  when the missed variables are correlated with the selected ones, and the residual mean square overestimates  $\sigma^2$ .

**Exercise B2.**

In [Lemma 28.5.2](#), show that  $\omega_j^2 = \hat{\boldsymbol{\theta}}_j^\top \hat{\boldsymbol{\Sigma}} \hat{\boldsymbol{\theta}}_j = \|\hat{\mathbf{z}}_j\|^2 / (n \hat{\tau}_j^4)$ , and hence  $\omega_j^2 \leq 1 / \hat{\tau}_j^2$ . Interpret: a small  $\hat{\tau}_j^2$  (a column that the others explain well) means a large variance for  $\hat{b}_j$ . Compare with the variance inflation factor ([Definition 26.2.1](#)).

*Solution.*  $\mathbf{X} \hat{\boldsymbol{\theta}}_j = \hat{\mathbf{z}}_j / \hat{\tau}_j^2$ , so  $\omega_j^2 = \|\mathbf{X} \hat{\boldsymbol{\theta}}_j\|^2 / n = \|\hat{\mathbf{z}}_j\|^2 / (n \hat{\tau}_j^4)$ . By [Lemma 28.5.2\(a\)](#),  $\|\hat{\mathbf{z}}_j\|^2 / n \leq \hat{\tau}_j^2$ , so  $\omega_j^2 \leq 1 / \hat{\tau}_j^2$ . With least squares in place of the lasso ( $p < n$ ,  $\lambda_j = 0$ ),  $\hat{\tau}_j^2 = \|(\mathbf{I} - \mathbf{M}_{(j)}) \mathbf{x}_j\|^2 / n$ , and  $\omega_j^2 = 1 / \hat{\tau}_j^2$  is the variance factor  $n / \{S_{jj}(1 - R_j^2)\}$  of [Eq. 6.6.2](#) for a column with  $\|\mathbf{x}_j\|^2 = n$  (and an intercept among the other columns). The node-wise lasso residual plays the role of the FWL residual, and  $1 / \hat{\tau}_j^2$  the role of the variance inflation.

## 28.6 Summary and Notes

### 28.6.1 Summary

*Idea.*

1. When  $p > n$ , least squares interpolates and no single coefficient is estimable. The minimum-norm solution  $\mathbf{X}^+ \mathbf{y}$  is the ridgeless limit of ridge and the limit of gradient descent from zero; its error splits into the invisible null-space part of  $\boldsymbol{\beta}$  and the variance  $\sigma^2 \sum_i d_i^{-2}$  ([Theorem 28.1.1](#)).
2. For isotropic Gaussian regressors its risk is exact ([Proposition 28.1.2](#)). Fitting  $p$  of  $D$  features gives the double descent curve ([Corollary 28.1.3](#)), whose peak at  $p = n$  belongs to the estimator and disappears with a tuned ridge penalty.
3. Sparsity restores identifiability ([Proposition 28.2.1](#)). The compatibility and re-

stricted eigenvalue conditions ask for curvature only on a cone of approximately sparse vectors (Definition 28.2.1); they transfer from populations to samples with a loss of  $16s\delta$  (Proposition 28.2.2), and can be small in samples even when they are large in the population.

4. On the event  $\|\mathbf{X}^\top \boldsymbol{\varepsilon}/n\|_\infty \leq \lambda_0 = \sigma\sqrt{2\log(2p/\delta)/n}$ , which has probability at least  $1 - \delta$  (Lemma 28.3.2), every lasso estimate with  $\lambda \geq 2\lambda_0$  has its error in the cone and prediction error at most  $3\lambda\|\boldsymbol{\beta}\|_1$  without conditions and  $9\lambda^2s/\phi^2(S)$  under compatibility (Theorem 28.3.3). Its  $\ell_1$  and  $\ell_2$  errors are of order  $\lambda s$  and  $\lambda\sqrt{s}$  (Theorem 28.3.4).
5. Exact sign recovery holds under the irrepresentable and beta-min conditions (Theorem 28.4.2). Without the irrepresentable condition it has probability at most  $\frac{1}{2}$  for every sample size (Proposition 28.4.3); screening needs only the restricted eigenvalue and beta-min conditions (Proposition 28.4.4).
6. Intervals computed after selection under-cover. The debiased lasso adds a one-step correction; its error is exactly normal plus a remainder bounded by  $\sqrt{n}\mu_j\|\hat{\boldsymbol{\beta}}_\lambda - \boldsymbol{\beta}\|_1$ , which vanishes when  $s\log p/\sqrt{n} \rightarrow 0$  (Theorem 28.5.1). With node-wise lasso rows it is the Frisch–Waugh–Lovell estimator with a lasso residual (Lemma 28.5.2).

## 28.6.2 Notes and sources

**Scope.** None of the planning sources treats  $p > n$  in depth; Agresti (2015, section 11.2) notes it briefly, through a Bayesian prior. Bühlmann and van de Geer (2011), Hastie, Tibshirani and Wainwright (2015) and Wainwright (2019) develop all the topics of the chapter at length.

**Interpolation.** The term *double descent* is from Belkin, Hsu, Ma and Mandal (2019). Belkin, Hsu and Xu (2020) give exact risk formulas for Gaussian features, including omitted features as in Corollary 28.1.3. Hastie, Montanari, Rosset and Tibshirani (2022) treat general covariances and ridge asymptotically, and Bartlett, Long, Lugosi and Tsigler (2020) characterize benign overfitting.

**Design conditions and the lasso.** The lasso is due to Tibshirani (1996); Tibshirani (2013) studies uniqueness. Greenshtein and Ritov (2004) introduced persistence, behind the slow rate. The restricted eigenvalue condition and the fast rates are from Bickel, Ritov and Tsybakov (2009), with the Dantzig selector of Candès and Tao (2007); the basic-inequality presentation follows Bühlmann and van de Geer (2011, chapter 6), and van de Geer and Bühlmann (2009) compare the conditions. Donoho and Huo (2001) used mutual coherence, and Cohen, Dahmen and DeVore (2009) the null space property. Raskutti, Wainwright and Yu (2010, 2011) prove the random-design bound and the minimax lower bound. Belloni, Chernozhukov and Wang (2011) and Sun and Zhang (2012) free  $\lambda$  from  $\sigma$ .

**Selection and inference.** The irrepresentable condition is due to Zhao and Yu (2006) and Meinshausen and Bühlmann (2006), the primal–dual witness proof to Wainwright (2009); Zou (2006) and Meinshausen and Bühlmann (2010) propose the adaptive lasso and stability selection. Leeb and Pötscher (2005, 2006) analyse post-selection distributions and prove that they cannot be estimated uniformly. The debiased lasso is due to Zhang and Zhang (2014), van de Geer, Bühlmann, Ritov and Dezeure (2014) and Javanmard and Montanari (2014). The alternatives are due to Wasserman and Roeder (2009), Meinshausen, Meier and Bühlmann (2009), Lee, Sun, Sun and Taylor (2016), Berk, Brown, Buja, Zhang and Zhao (2013) and Belloni, Chernozhukov and Hansen (2014). Chapter 29 treats selection bias for classical rules, and Chapter 30 extends the lasso to other penalties.

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# Chapter 29

## Model Selection and Prediction

A fitted model is judged by how well it predicts responses it has not seen, and its error on the responses it was fitted to flatters it. For any fitting procedure the flattery is twice the covariance between the fit and the data,  $2\sigma^2 p/n$  for least squares whether the model is right or wrong. Correcting for it gives Mallows'  $C_p$  and AIC. BIC answers a different question, which model is most probable: it finds a true finite model with probability tending to one, while AIC predicts almost as well as the best candidate when no finite model is true. Cross-validation estimates prediction error with fewer assumptions, and for least squares and ridge it needs no refitting. The search over subsets is unstable, and every estimate and interval computed after it is biased by it, unless the inference is designed to survive the search.

**What you need.** Chapter 6 (projections, Theorem 6.5.1, and the Frisch–Waugh–Lovell theorem, Theorem 6.6.2), Chapter 7 (the likelihood of the normal model, Theorem 7.3.1), Chapter 9 (adjusted  $R^2$ , Proposition 9.5.3), Chapter 11 (the  $F$  test, Theorem 11.1.1), Chapter 13 (Scheffé and Bonferroni, Theorem 13.2.2 and Theorem 13.4.1), Chapter 14 (prediction error and its optimism, Proposition 14.2.7 and Theorem 14.2.6), Chapter 19 (omitted and irrelevant regressors, Theorem 19.1.5) and Chapter 20 (the deletion formulas and PRESS, Theorem 20.3.1). From this part: Chapter 26 (collinearity) and Chapter 27 (ridge regression, thresholding and Stein's lemma, Lemma 27.5.1).

### 29.0.1 Roadmap

- Prediction error and optimism
- Mallows'  $C_p$ , AIC and BIC
- Cross-validation
- All subsets and stepwise search
- Selection bias and honest inference
- Summary and notes

## 29.1 Prediction error and optimism

Chapter 19 compared a short regression with a long one and found that neither wins everywhere. Leaving out regressors whose coefficients are small buys a drop in variance that can outweigh

the bias it causes, and [Theorem 19.1.5](#) located the break-even point exactly: at the design points the short fit is better iff the noncentrality [Eq. 19.1.1](#) of the omitted block is less than the number of columns omitted. The noncentrality is unknown, so the theorem does not choose a model. It does say what a good choice should aim at, namely the error of the fitted values as predictions, and this chapter builds that into procedures.

The residual mean square describes how well a fit reproduces the data it was fitted to, and every fitting procedure bends towards those data. This section makes the size of that bend exact for any procedure, and turns it into a definition of the degrees of freedom a procedure spends.

### 29.1.1 In-sample prediction error

Throughout the chapter the data follow

$$\mathbf{Y} = \boldsymbol{\mu} + \boldsymbol{\varepsilon}, \quad \mathbf{E}(\boldsymbol{\varepsilon}) = \mathbf{0}, \quad \text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n, \quad (29.1.1)$$

where the mean vector  $\boldsymbol{\mu} \in \mathbb{R}^n$  is unrestricted. A candidate linear model  $\mathcal{C}(\mathbf{X})$  may or may not contain  $\boldsymbol{\mu}$ . That is the point of model selection: we want procedures whose properties do not depend on a candidate being right. A **fitting procedure** is any function  $\hat{\boldsymbol{\mu}} = \hat{\boldsymbol{\mu}}(\mathbf{Y})$  from  $\mathbb{R}^n$  to  $\mathbb{R}^n$  with  $\mathbf{E}\|\hat{\boldsymbol{\mu}}(\mathbf{Y})\|^2 < \infty$ : least squares on a fixed model, least squares on a model chosen from the data, ridge regression, thresholding.

#### Definition 29.1.1 — Training error, prediction error, optimism

Let  $\mathbf{Y}^* = \boldsymbol{\mu} + \boldsymbol{\varepsilon}^*$ , where  $\boldsymbol{\varepsilon}^*$  is independent of  $\boldsymbol{\varepsilon}$  and has the same distribution: new responses at the same design points. For a fitting procedure  $\hat{\boldsymbol{\mu}}$ :

- (a) the **training error** is  $\overline{\text{err}} = n^{-1}\|\mathbf{Y} - \hat{\boldsymbol{\mu}}\|^2$ ;
- (b) the **in-sample prediction error** is  $\text{Err}_{\text{in}} = n^{-1} \mathbf{E}(\|\mathbf{Y}^* - \hat{\boldsymbol{\mu}}\|^2 \mid \mathbf{Y})$ ;
- (c) the **optimism** is  $\text{Err}_{\text{in}} - \overline{\text{err}}$ , and its mean  $\omega = \mathbf{E}(\text{Err}_{\text{in}} - \overline{\text{err}})$  is the **expected optimism**.

Expanding  $\mathbf{Y}^* - \hat{\boldsymbol{\mu}} = \boldsymbol{\varepsilon}^* + (\boldsymbol{\mu} - \hat{\boldsymbol{\mu}})$  and using the independence of  $\boldsymbol{\varepsilon}^*$  and  $\mathbf{Y}$ ,

$$\text{Err}_{\text{in}} = \sigma^2 + \frac{1}{n}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2. \quad (29.1.2)$$

So the in-sample prediction error is the irreducible  $\sigma^2$  plus the average squared error of the fitted values as estimates of the mean. Ranking procedures by  $\mathbf{E}(\text{Err}_{\text{in}})$  is the same as ranking them by the risk  $\mathbf{E}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2$ , which is the criterion of [Theorem 19.1.5\(c\)](#). *In-sample* refers to the design points: the responses are new, the regressor values old. New regressor values are taken up at the end of the section.

### 29.1.2 The covariance penalty

**Theorem 29.1.1 — Optimism is a covariance**

Under Eq. 29.1.1, let  $\hat{\boldsymbol{\mu}}$  be a fitting procedure.

- (a) The expected optimism is

$$\omega = \frac{2}{n} \sum_{i=1}^n \text{Cov}(\hat{\mu}_i, Y_i).$$

- (b) If  $\hat{\boldsymbol{\mu}} = \mathbf{S}\mathbf{Y}$  for a fixed  $n \times n$  matrix  $\mathbf{S}$ , then  $\omega = 2\sigma^2 \text{tr}(\mathbf{S})/n$  and  $\mathbb{E}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 = \|(\mathbf{I} - \mathbf{S})\boldsymbol{\mu}\|^2 + \sigma^2 \text{tr}(\mathbf{S}\mathbf{S}^\top)$ .
- (c) For least squares on a model matrix  $\mathbf{X}$  of rank  $p$ , whether or not  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ ,

$$\begin{aligned} \mathbb{E}(\overline{\text{err}}) &= \sigma^2 \frac{n-p}{n} + \frac{\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2}{n}, \\ \mathbb{E}(\text{Err}_{\text{in}}) &= \sigma^2 \frac{n+p}{n} + \frac{\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2}{n}, \end{aligned}$$

and  $\omega = 2\sigma^2 p/n$ .

*Proof.*

- (a) Write  $\mathbf{Y} - \hat{\boldsymbol{\mu}} = (\mathbf{Y} - \boldsymbol{\mu}) - (\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})$ . Then

$$\|\mathbf{Y} - \hat{\boldsymbol{\mu}}\|^2 = \|\boldsymbol{\varepsilon}\|^2 + \|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2 - 2\boldsymbol{\varepsilon}^\top(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}).$$

Take expectations and compare with Eq. 29.1.2:  $n \mathbb{E}(\text{Err}_{\text{in}}) = n\sigma^2 + \mathbb{E}\|\hat{\boldsymbol{\mu}} - \boldsymbol{\mu}\|^2$ , so  $n\omega = 2\mathbb{E}[\boldsymbol{\varepsilon}^\top(\hat{\boldsymbol{\mu}} - \boldsymbol{\mu})] = 2\sum_i \mathbb{E}[(Y_i - \mu_i)\hat{\mu}_i]$ , since  $\boldsymbol{\mu}$  is constant and  $\mathbb{E}(\boldsymbol{\varepsilon}) = \mathbf{0}$ . Because  $\mathbb{E}(Y_i - \mu_i) = 0$ , the last expectation is  $\text{Cov}(\hat{\mu}_i, Y_i)$ . The moments exist by the Cauchy–Schwarz inequality.

- (b)  $\text{Cov}(\mathbf{S}\mathbf{Y}, \mathbf{Y}) = \sigma^2 \mathbf{S}$ , whose diagonal sums to  $\sigma^2 \text{tr}(\mathbf{S})$ . For the risk,  $\mathbf{S}\mathbf{Y} - \boldsymbol{\mu} = \mathbf{S}\boldsymbol{\varepsilon} - (\mathbf{I} - \mathbf{S})\boldsymbol{\mu}$  has mean  $-(\mathbf{I} - \mathbf{S})\boldsymbol{\mu}$  and covariance  $\sigma^2 \mathbf{S}\mathbf{S}^\top$ , and Theorem 2.3.1 with  $\mathbf{A} = \mathbf{I}$  gives the sum of the squared mean and the trace.
- (c) Take  $\mathbf{S} = \mathbf{M}$ , with  $\text{tr}(\mathbf{M}) = p$  and  $\mathbf{M}\mathbf{M}^\top = \mathbf{M}$  (Proposition 6.3.4). Then  $\mathbb{E}(\text{Err}_{\text{in}}) = \sigma^2 + n^{-1}\{\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2 + \sigma^2 p\}$ , and subtracting  $\omega$  gives  $\mathbb{E}(\overline{\text{err}})$ . ■

Part (c) with  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$  is Proposition 14.2.7(a). What is new is that the optimism of least squares does not depend on  $\boldsymbol{\mu}$ : a wrong model has larger training and prediction errors, but both grow by the same squared bias, so a penalty depending only on  $p$  is a sensible correction even for wrong models. Part (a) asks nothing of the procedure but finite second moments: a fit is optimistic to the extent that each fitted value follows its own response.

*Idea.* The training error of a procedure understates its prediction error by  $2/n$  times the total covariance between fitted values and responses. For least squares on  $p$  columns the covariance is  $\sigma^2 p$ , right model or wrong.

**Example. (Optimism of a wrong model)**

Take  $n = 30$  equally spaced points  $t_i$  on  $[-1, 1]$ , a curved mean  $\mu_i = 1 + t_i + 1.5t_i^2$ ,  $\sigma = 1$ , and fit by least squares a straight line in  $t$  together with two columns of pure noise, so  $p = 4$ . The model is wrong: its squared bias per observation is  $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/n = 0.2054$ . Over 20000 simulated data sets the average optimism is 0.2688, against  $2\sigma^2 p/n = 0.2667$  from [Theorem 29.1.1\(c\)](#). Ridge regression ([Theorem 27.2.2](#)) on the same columns with penalty  $\lambda = 5$  is a linear smoother with  $\text{tr}(\mathbf{S}) = 3.207$ ; its simulated optimism is 0.2163, against  $2 \times 3.207/30 = 0.2138$  from part (b).

```
import numpy as np
rng = np.random.default_rng(2901)
n, sigma, reps = 30, 1.0, 20_000

t = np.linspace(-1, 1, n)
X = np.column_stack([np.ones(n), t, rng.normal(size=(n, 2))])  # line + 2
                        irrelevant columns
mu = 1 + t + 1.5 * t**2                                         # the true mean is
                        curved
p = X.shape[1]
M = X @ np.linalg.solve(X.T @ X, X.T)                          # projection onto C(
                        X)

Y = mu + sigma * rng.normal(size=(reps, n))                    # one data set per
                        row
fit = Y @ M                                                      # M is symmetric
train = np.mean((Y - fit) ** 2, axis=1)                         # training error
err_in = sigma**2 + np.mean((fit - mu) ** 2, axis=1)            # in-sample
                        prediction error
print(f"average optimism {np.mean(err_in - train):.4f}, 2 sigma^2 p / n = {2 *
      sigma**2 * p / n:.4f}")

lam = 5.0
S = X @ np.linalg.solve(X.T @ X + lam * np.eye(p), X.T)        # ridge smoother
fit_r = Y @ S.T
train_r = np.mean((Y - fit_r) ** 2, axis=1)
err_in_r = sigma**2 + np.mean((fit_r - mu) ** 2, axis=1)
omega_r = np.mean(err_in_r - train_r)
print(f"ridge: optimism {omega_r:.4f}, 2 sigma^2 tr(S) / n = {2 * np.trace(S) / n
      :.4f}")
```

### 29.1.3 Degrees of freedom of a procedure

[Theorem 29.1.1](#) suggests measuring the complexity of any procedure by the covariance it creates.

#### Definition 29.1.2 — Degrees of freedom

The **degrees of freedom** of a fitting procedure  $\hat{\boldsymbol{\mu}}$  under [Eq. 29.1.1](#) is

$$\text{df}(\hat{\boldsymbol{\mu}}) = \frac{1}{\sigma^2} \sum_{i=1}^n \text{Cov}(\hat{\mu}_i, Y_i),$$

so that  $\omega = 2\sigma^2 \text{df}(\hat{\boldsymbol{\mu}})/n$ .

For least squares on a fixed model  $\text{df} = p$ , the number of free coefficients, and for a linear smoother  $\text{df} = \text{tr}(\mathbf{S})$ . For ridge regression with singular values  $d_j$  of  $\mathbf{X}$  this is  $\sum_j d_j^2/(d_j^2 + \lambda)$ , the effective degrees of freedom of [Theorem 27.2.2](#), which falls from  $\text{rank}(\mathbf{X})$  to 0 as  $\lambda$  grows. If  $\text{df}$  is known,

$$\widehat{\text{Err}}_{\text{in}} = \overline{\text{err}} + \frac{2\sigma^2}{n} \text{df}(\hat{\boldsymbol{\mu}}) \quad (29.1.3)$$

is an unbiased estimate of  $E(\text{Err}_{\text{in}})$  by [Theorem 29.1.1\(a\)](#). [Section 29.2](#) turns this into Mallows'  $C_p$ . For a procedure that searches, the covariance is not a count and has to be computed. Under normal errors it is computed by Stein's lemma, [Lemma 27.5.1\(b\)](#), proved in [Section 27.5](#) for the James–Stein theorem: if  $\mathbf{Z} \sim N_p(\boldsymbol{\theta}, \sigma^2 \mathbf{I})$  and each  $g_i$  is absolutely continuous in  $z_i$  with the integrability stated there, then  $E[(\mathbf{Z} - \boldsymbol{\theta})^\top \mathbf{g}(\mathbf{Z})] = \sigma^2 E \sum_i \partial g_i(\mathbf{Z})/\partial z_i$ .

### Proposition 29.1.2 — Degrees of freedom as a divergence

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$ . Suppose that for each  $i$ , for almost every value of the other coordinates,  $y_i \mapsto \hat{\mu}_i(\mathbf{y})$  is absolutely continuous, and that  $E|\partial \hat{\mu}_i/\partial y_i|$  and  $E\hat{\mu}_i^2$  are finite. Then

$$\text{df}(\hat{\boldsymbol{\mu}}) = E \sum_{i=1}^n \frac{\partial \hat{\mu}_i}{\partial y_i}(\mathbf{Y}),$$

and  $\overline{\text{err}} + 2\sigma^2 n^{-1} \sum_i \partial \hat{\mu}_i/\partial y_i$  is an unbiased estimate of  $E(\text{Err}_{\text{in}})$  that does not involve  $\boldsymbol{\mu}$ .

*Proof.* Since  $E(Y_i - \mu_i) = 0$ ,  $\sum_i \text{Cov}(\hat{\mu}_i, Y_i) = E[(\mathbf{Y} - \boldsymbol{\mu})^\top \hat{\boldsymbol{\mu}}(\mathbf{Y})]$ , and each term is integrable by the Cauchy–Schwarz inequality. [Lemma 27.5.1\(b\)](#) with  $\mathbf{g} = \hat{\boldsymbol{\mu}}$  turns this into  $\sigma^2 E \sum_i \partial \hat{\mu}_i/\partial y_i$ . The unbiasedness statement is then [Theorem 29.1.1\(a\)](#). ■

The estimate in the proposition is **Stein's unbiased risk estimate** (Stein 1981). For a linear smoother the divergence is  $\text{tr}(\mathbf{S})$ , and nothing is new. The interesting cases are adaptive.

**Soft thresholding.** Take an orthonormal design,  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}_p$ , and put  $\mathbf{Z} = \mathbf{X}^\top \mathbf{Y} \sim N_p(\mathbf{X}^\top \boldsymbol{\mu}, \sigma^2 \mathbf{I})$ . For a fit  $\hat{\boldsymbol{\mu}} = \mathbf{X} \hat{\boldsymbol{\beta}}(\mathbf{Z})$ ,  $\sum_i \text{Cov}(\hat{\mu}_i, Y_i) = \text{tr Cov}(\mathbf{X} \hat{\boldsymbol{\beta}}, \mathbf{Y}) = \text{tr Cov}(\hat{\boldsymbol{\beta}}, \mathbf{X}^\top \mathbf{Y}) = \sum_j \text{Cov}(\hat{\beta}_j, Z_j)$ , so the degrees of freedom can be computed coordinate by coordinate in the sequence model  $Z_j \sim N(\theta_j, \sigma^2)$ . The lasso in this design is soft thresholding,  $\hat{\beta}_j = \text{sign}(Z_j)(|Z_j| - \lambda)_+$  ([Theorem 27.4.2](#)). It is Lipschitz, with derivative  $1\{|Z_j| > \lambda\}$  except at  $\pm\lambda$ , so by [Proposition 29.1.2](#) its degrees of freedom are the expected number of nonzero coefficients, as an exercise of [Section 27.4](#) found by direct integration. The shrinkage exactly pays for the search.

**Hard thresholding.** Subset selection in the same design keeps or drops each coordinate whole,  $\hat{\beta}_j = Z_j 1\{|Z_j| > \lambda\}$  ([Theorem 27.4.2](#) again). This jumps by  $\lambda$  at  $\pm\lambda$ , so Stein's lemma does not apply, and the degrees of freedom exceed the expected count.

### Proposition 29.1.3 — Degrees of freedom of hard thresholding

Let  $Z \sim N(\theta, \sigma^2)$  and  $\lambda > 0$ . Then

$$\begin{aligned} \frac{\text{Cov}(Z1\{|Z| > \lambda\}, Z)}{\sigma^2} &= \Pr(|Z| > \lambda) \\ &\quad + \frac{\lambda}{\sigma} \left[ \phi\left(\frac{\lambda - \theta}{\sigma}\right) + \phi\left(\frac{\lambda + \theta}{\sigma}\right) \right], \end{aligned}$$

where  $\phi$  is the standard normal density.

*Proof.* Write  $Z = \theta + \sigma W$  with  $W \sim N(0, 1)$ , and put  $a = (\lambda - \theta)/\sigma$ ,  $b = (-\lambda - \theta)/\sigma$ , so that  $|Z| > \lambda$  iff  $W > a$  or  $W < b$ . The covariance is  $E[\sigma W(\theta + \sigma W)1\{W > a \text{ or } W < b\}]$ . The truncated moments of the standard normal are  $E[W1\{W > a\}] = \phi(a)$ ,  $E[W1\{W < b\}] = -\phi(b)$ ,  $E[W^21\{W > a\}] = 1 - \Phi(a) + a\phi(a)$  and  $E[W^21\{W < b\}] = \Phi(b) - b\phi(b)$ , each by one integration by parts. Collecting terms,

$$\sigma\theta(\phi(a) - \phi(b)) + \sigma^2(1 - \Phi(a) + \Phi(b) + a\phi(a) - b\phi(b)).$$

Since  $\sigma\theta + \sigma^2a = \sigma\lambda$  and  $-\sigma\theta - \sigma^2b = \sigma\lambda$ , this is  $\sigma^2 \Pr(|Z| > \lambda) + \sigma\lambda(\phi(a) + \phi(b))$ , and  $\phi(b) = \phi((\lambda + \theta)/\sigma)$ . ■

The first term is the expected number of coefficients kept. The second is the price of the search: the fit follows  $Z$  most closely just where  $Z$  decides whether to keep the coordinate. At the threshold  $\lambda = \sigma\sqrt{2}$  (which [Section 29.2](#) shows is where  $C_p$  and AIC put it) and  $\theta = 0$ , the expected count is 0.157 but the degrees of freedom are 0.572: the search term 0.415 is more than twice the count. At  $\theta = \sigma$  the degrees of freedom are 0.896, and at  $\theta = 3\sigma$  they are 1.104, more than one full parameter for a single coordinate.

### Example. (Best subset of a given size)

Keep the  $k$  largest of  $m = 20$  values  $|Z_j|$  and set the rest to zero, with  $Z_j \sim N(\theta_j, 1)$  independent. For a subset fixed in advance the degrees of freedom would be  $k$ . [Figure 29.1.1](#) shows them by simulation. When all  $\theta_j = 0$ , the best single coordinate already costs 4.91 degrees of freedom, the best five cost 13.94 and the best ten 18.27. When five means equal 6 and the rest are zero, choosing the best five is almost free (5.06), because the five are obvious; choosing a sixth means choosing among fifteen pure-noise coordinates, and the degrees of freedom jump to 9.47. Searching among many candidates of equal merit is expensive, and choosing between clear winners and clear losers is cheap.

```
m, reps_s = 20, 20_000
Zs = rng.normal(size=(reps_s, m))

def subset_df(theta, k):
    """Monte Carlo df of 'keep the k largest |Y_j|' when Y ~ N(theta, I)."""
    Yb = theta + Zs
    keep = np.argsort(-np.abs(Yb), axis=1)[:k]
    fit = np.zeros_like(Yb)
    np.put_along_axis(fit, keep, np.take_along_axis(Yb, keep, axis=1), axis=1)
    return np.mean(np.sum(fit * Zs, axis=1)) # sum_j Cov(fit_j, Y_j)
```

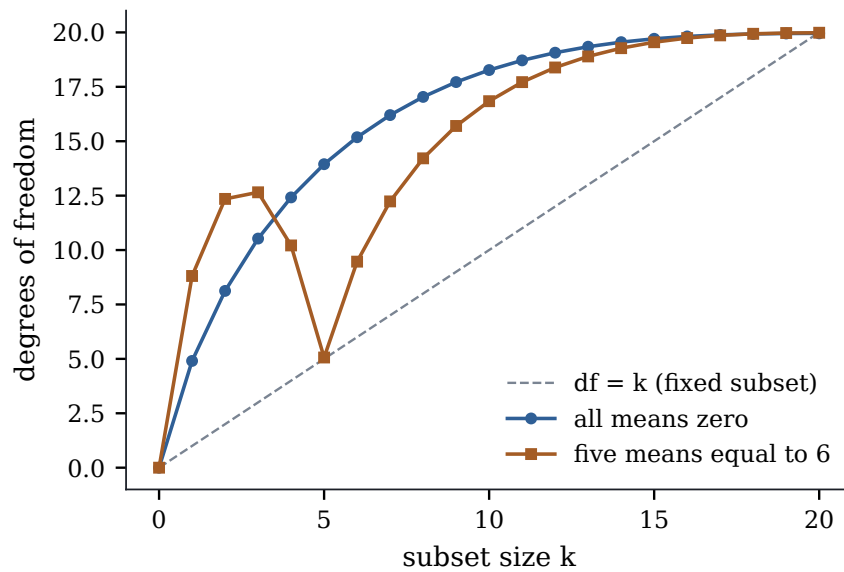


Figure 29.1.1: Degrees of freedom of the best subset of size  $k$  among 20 orthonormal coordinates, by simulation (20000 data sets). The dashed line is  $df = k$ , correct only when the subset is fixed in advance.

```

null = np.zeros(m)
sparse = np.r_[np.full(5, 6.0), np.zeros(m - 5)] # five large means
for k in (1, 5, 10):
    print(f"k = {k:2d}: df null {subset_df(null, k):5.2f}, df sparse {subset_df(
        sparse, k):5.2f}")

```

So a penalty of  $2\sigma^2 p/n$  is right for a model chosen in advance and too small for a model chosen by search, a point [Section 29.5](#) returns to.

## 29.1.4 New regressor values

The in-sample error keeps the design fixed. For a new case with its own regressors the error is larger, and for least squares with  $k$  normal regressors and a correct model [Theorem 14.2.6](#) gives it exactly; since  $E(s^2) = \sigma^2$ ,  $s^2(1 + 1/n)(n - 2)/(n - k - 2)$  estimates it without bias. Breiman and Spector (1992) found that fixed-design estimates understate the error when the design is random, and recommended five- or ten-fold cross-validation. Cross-validation ([Section 29.3](#)) estimates the random-design error directly.

## 29.1.5 Exercises

### 29.1.5.1 A. Check your understanding

#### Exercise A1.

The procedure  $\hat{\mu}(\mathbf{Y}) = \mathbf{Y}$  interpolates the data. Compute its training error, its in-sample prediction error and its degrees of freedom, and check [Theorem 29.1.1\(a\)](#).

## 29.1.5.2 B. Practice

**Exercise B1.**

Let  $\mathbf{S}_\lambda = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + \lambda \mathbf{I})^{-1} \mathbf{X}^\top$  be the ridge smoother, whose trace  $\sum_j d_j^2 / (d_j^2 + \lambda)$  was found in [Theorem 27.2.2](#). Show that  $\text{tr}(\mathbf{S}_\lambda \mathbf{S}_\lambda^\top) < \text{tr}(\mathbf{S}_\lambda)$  for every  $\lambda > 0$  with  $\mathbf{X} \neq \mathbf{0}$ , and that for a symmetric smoother  $\mathbf{S}$  with eigenvalues in  $[0, 1]$ ,  $\text{tr}(\mathbf{S}\mathbf{S}^\top) = \text{tr}(\mathbf{S})$  iff  $\mathbf{S}$  is an orthogonal projection.

*Solution.* A symmetric  $\mathbf{S}$  with eigenvalues  $s_j \in [0, 1]$  has  $\text{tr}(\mathbf{S}\mathbf{S}^\top) = \text{tr}(\mathbf{S}^2) = \sum_j s_j^2 \leq \sum_j s_j = \text{tr}(\mathbf{S})$ , with equality iff every  $s_j \in \{0, 1\}$ , that is, iff  $\mathbf{S}$  is idempotent, an orthogonal projection ([Theorem 6.3.3](#)). The ridge smoother is symmetric with eigenvalues  $d_j^2 / (d_j^2 + \lambda)$ , which lie in  $(0, 1)$  for  $d_j > 0$ , and at least one  $d_j$  is positive. So the variance term of [Theorem 29.1.1\(b\)](#),  $\sigma^2 \text{tr}(\mathbf{S}\mathbf{S}^\top)$ , is strictly smaller than the optimism term  $\sigma^2 \text{tr}(\mathbf{S})$ .

**Exercise B2.**

Arrange  $n$  observations on a circle, and let  $\hat{\mu}_i$  be the average of  $Y_{i-q}, \dots, Y_{i+q}$  (indices modulo  $n$ ,  $2q + 1 \leq n$ ). Find the degrees of freedom and the expected optimism, and explain why a wider window is “simpler”.

## 29.1.5.3 C. Going deeper

**Exercise C1.**

In the sequence model  $Z_j \sim N(\theta_j, \sigma^2)$ ,  $j = 1, \dots, p$ , independent, soft thresholding at  $\lambda$  has Stein’s unbiased risk estimate

$$\text{SURE}(\lambda) = \sum_{j=1}^p \min(Z_j^2, \lambda^2) - p\sigma^2 + 2\sigma^2 \#\{j : |Z_j| > \lambda\}$$

by [Proposition 29.1.2](#), because  $Z_j - \hat{\theta}_j$  is  $Z_j$  when  $|Z_j| \leq \lambda$  and  $\pm\lambda$  otherwise. Let  $0 = a_0 \leq a_1 \leq \dots \leq a_p$  be 0 and the ordered  $|Z_j|$ . Show that SURE is nondecreasing on each interval  $[a_k, a_{k+1})$ , so it is minimized over  $\lambda \geq 0$  at one of  $a_0, \dots, a_p$ . Compute  $\text{SURE}(0)$ , and explain why the minimized value is biased downwards. (Minimizing SURE is the SureShrink rule of Donoho and Johnstone 1995.)

*Solution.* On  $[a_k, a_{k+1})$  exactly  $p - k$  of the  $|Z_j|$  exceed  $\lambda$  (for distinct values), so  $\text{SURE}(\lambda) = \sum_{|Z_j| \leq a_k} Z_j^2 + (p - k)\lambda^2 - p\sigma^2 + 2\sigma^2(p - k)$ , which is nondecreasing in  $\lambda$  and smallest at  $a_k$ . At  $\lambda = 0$  nothing is shrunk and  $\text{SURE}(0) = p\sigma^2$ , the exact risk of  $\mathbf{Z}$ . Each  $\text{SURE}(a_k)$  is unbiased for its own risk, so their minimum has expectation at most the smallest risk: the optimism of a minimized estimate studied in [Section 29.5](#).

**Exercise C2.**

In [Example \(Best subset of a given size\)](#) with all  $\theta_j = 0$  and  $k = 1$ , show that the degrees of freedom equal  $E \max_j Z_j^2$  (with  $\sigma = 1$ ). Deduce that they are at least 1, and



use  $E \max_j Z_j^2 \geq (E \max_j |Z_j|)^2$  to explain why they grow without bound as  $m \rightarrow \infty$ .

## 29.2 Mallows' Cp, AIC and BIC

A **selection criterion** assigns a number to each candidate model and picks the smallest.  $C_p$  estimates the prediction error of [Section 29.1](#), AIC a likelihood-based discrepancy, and BIC the posterior probability of the model. The first two nearly always agree; the third answers a different question, and the difference shows up in large samples.

### 29.2.1 Candidates

The candidates are a finite collection of models  $\mathcal{C}(\mathbf{X}_k)$ ,  $k \in \mathcal{K}$ , with  $\text{rank}(\mathbf{X}_k) = p_k$ , each fitted by least squares:  $\hat{\boldsymbol{\mu}}_k = \mathbf{M}_k \mathbf{Y}$ , with residual sum of squares  $\text{SSE}_k = \|(\mathbf{I} - \mathbf{M}_k) \mathbf{Y}\|^2$ . Typically  $\mathbf{X}_k$  consists of the intercept and a subset of the columns of a full model matrix  $\mathbf{X}_F$  of rank  $P < n$ , and  $s^2 = \text{SSE}_F / (n - P)$  is the usual variance estimate from the full model. The data follow [Eq. 29.1.1](#). By [Theorem 29.1.1\(c\)](#) the scaled risk of candidate  $k$  is

$$J_k = \frac{E \|\hat{\boldsymbol{\mu}}_k - \boldsymbol{\mu}\|^2}{\sigma^2} = p_k + \gamma_k, \quad \gamma_k = \frac{\|(\mathbf{I} - \mathbf{M}_k) \boldsymbol{\mu}\|^2}{\sigma^2}, \quad (29.2.1)$$

variance plus squared bias. The bias term  $\gamma_k$  is the noncentrality [Eq. 19.1.1](#) of the test of candidate  $k$  against any correct larger model.

### 29.2.2 Mallows' Cp

#### Definition 29.2.1 — Mallows' $C_p$

The  $C_p$  statistic of candidate  $k$  is

$$C_k = \frac{\text{SSE}_k}{s^2} - n + 2p_k.$$

With  $\sigma^2$  in place of  $s^2$ ,  $\sigma^2(C_k + n)/n$  is the covariance-penalty estimate [Eq. 29.1.3](#) of the in-sample prediction error of candidate  $k$ . The scaling makes  $C_k$  comparable with  $p_k$ .

#### Proposition 29.2.1 — Properties of $C_p$

- (a) Under [Eq. 29.1.1](#),  $C_k^\sigma = \text{SSE}_k / \sigma^2 - n + 2p_k$  is unbiased for  $J_k$ .
- (b) Assume normal errors,  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}_F)$ ,  $\mathcal{C}(\mathbf{X}_k) \subseteq \mathcal{C}(\mathbf{X}_F)$  and  $n - P > 2$ . Then

$$E(C_k) = J_k + \frac{2(P - p_k + \gamma_k)}{n - P - 2}.$$

- (c) Let  $F_k = [(\text{SSE}_k - \text{SSE}_F) / (P - p_k)] / s^2$  be the  $F$  statistic for testing candidate  $k$

against the full model ( $p_k < P$ ). Then

$$C_k = (P - p_k)(F_k - 1) + p_k.$$

In particular  $C_F = P$ , and  $C_k \leq p_k$  iff  $F_k \leq 1$ .

*Proof.*

- (a) By [Theorem 29.1.1\(c\)](#),  $E(\text{SSE}_k) = \sigma^2(n - p_k) + \|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2 = \sigma^2(n - p_k + \gamma_k)$ .  
 (b) Write  $\text{SSE}_k = \text{SSE}_F + (\text{SSE}_k - \text{SSE}_F)$ . By [Theorem 11.1.1\(a\)](#), applied to  $\mathcal{C}(\mathbf{X}_k) \subseteq \mathcal{C}(\mathbf{X}_F)$ , the two terms are independent,  $\text{SSE}_F/\sigma^2 \sim \chi^2(n - P)$ , and  $(\text{SSE}_k - \text{SSE}_F)/\sigma^2$  is noncentral  $\chi^2$  with  $P - p_k$  degrees of freedom and noncentrality  $\|(\mathbf{M}_F - \mathbf{M}_k)\boldsymbol{\mu}\|^2/\sigma^2 = \gamma_k$ , so its mean is  $P - p_k + \gamma_k$ . By [Lemma 14.1.3](#),  $E(\sigma^2/s^2) = (n - P)/(n - P - 2)$ . Hence

$$\begin{aligned} E\left(\frac{\text{SSE}_k}{s^2}\right) &= (n - P) + (P - p_k + \gamma_k) \frac{n - P}{n - P - 2} \\ &= n - p_k + \gamma_k + \frac{2(P - p_k + \gamma_k)}{n - P - 2}, \end{aligned}$$

and subtracting  $n - 2p_k$  gives the result.

- (c) By the definition of  $F_k$ ,  $\text{SSE}_k/s^2 = (n - P) + (P - p_k)F_k$ . Subtract  $n - 2p_k$ . ■

By (b), estimating  $\sigma^2$  costs little unless the full model uses up most of the data. Mallows (1973) proposed plotting  $C_k$  against  $p_k$ : points near the line  $C_p = p$  have small bias, and the lowest point estimates the best predictor. By (c), adding  $q$  columns changes  $C_p$  by  $2q - (\text{SSE}_{\text{small}} - \text{SSE}_{\text{big}})/s^2$ , so  $C_p$  prefers the larger model iff the  $F$  statistic of the added columns, with the full-model  $s^2$ , exceeds 2.

### 29.2.3 AIC

If  $g$  is the true density and  $f$  a model density, the **Kullback–Leibler divergence**  $\int g \log(g/f) \geq 0$  is zero iff  $f = g$  (Kullback and Leibler 1951), and up to a term free of  $f$  it is  $E_g[-\log f(\mathbf{Y}^*)]$  for a new data set  $\mathbf{Y}^* \sim g$ . A fitted model  $f(\cdot; \hat{\boldsymbol{\theta}})$  is therefore judged by the **discrepancy**

$$\Delta = E_{\mathbf{Y}^*}[-2 \log f(\mathbf{Y}^*; \hat{\boldsymbol{\theta}}(\mathbf{Y}))],$$

with  $\mathbf{Y}^*$  independent of  $\mathbf{Y}$ . The maximized log-likelihood evaluates the same function at the data themselves, and is optimistic in the same way as the training error.

#### Definition 29.2.2 — AIC and BIC

For a model with  $d_k$  free parameters and maximized log-likelihood  $\ell_k$ ,

$$\text{AIC}_k = -2\ell_k + 2d_k, \quad \text{BIC}_k = -2\ell_k + d_k \log n.$$

In the normal linear model with unknown variance,  $d_k = p_k + 1$  and  $-2\ell_k = n \log(2\pi \text{SSE}_k/n) + n$  ([Theorem 7.3.1](#)), so up to a constant common to all candidates,

$$\text{AIC}_k = n \log \frac{\text{SSE}_k}{n} + 2(p_k + 1), \quad \text{BIC}_k = n \log \frac{\text{SSE}_k}{n} + (p_k + 1) \log n. \quad (29.2.2)$$

Akaike (1973, 1974) showed that  $2d$  is the leading term of the optimism of  $-2\ell$  for a correct model fitted by maximum likelihood, in large samples. In the normal linear model the optimism can be computed exactly, without asymptotics.

### Theorem 29.2.2 — The expected discrepancy of a normal linear model

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$  with  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}_k)$  and  $n > p_k + 2$ , and let  $\hat{\sigma}_k^2 = \text{SSE}_k/n$ . For the fitted density  $N_n(\hat{\boldsymbol{\mu}}_k, \hat{\sigma}_k^2 \mathbf{I})$ ,

$$E(\Delta_k) = E[n \log(2\pi \hat{\sigma}_k^2)] + \frac{n(n + p_k)}{n - p_k - 2}.$$

Consequently

$$\text{AICc}_k = -2\ell_k + 2(p_k + 1) + \frac{2(p_k + 1)(p_k + 2)}{n - p_k - 2} \quad (29.2.3)$$

is an unbiased estimate of  $E(\Delta_k)$ .

*Proof.* For a new data set  $\mathbf{Y}^* = \boldsymbol{\mu} + \boldsymbol{\varepsilon}^*$ ,  $-2 \log f(\mathbf{Y}^*) = n \log(2\pi \hat{\sigma}_k^2) + \|\mathbf{Y}^* - \hat{\boldsymbol{\mu}}_k\|^2 / \hat{\sigma}_k^2$ , and as in Eq. 29.1.2  $E(\|\mathbf{Y}^* - \hat{\boldsymbol{\mu}}_k\|^2 \mid \mathbf{Y}) = n\sigma^2 + \|\hat{\boldsymbol{\mu}}_k - \boldsymbol{\mu}\|^2$ . So

$$\Delta_k = n \log(2\pi \hat{\sigma}_k^2) + \frac{n\sigma^2 + \|\mathbf{M}_k \boldsymbol{\varepsilon}\|^2}{\hat{\sigma}_k^2},$$

using  $\hat{\boldsymbol{\mu}}_k - \boldsymbol{\mu} = \mathbf{M}_k \boldsymbol{\varepsilon}$  when  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X}_k)$ . Now  $\|\mathbf{M}_k \boldsymbol{\varepsilon}\|^2 / \sigma^2 \sim \chi^2(p_k)$  and  $n\hat{\sigma}_k^2 / \sigma^2 = \|(\mathbf{I} - \mathbf{M}_k) \boldsymbol{\varepsilon}\|^2 / \sigma^2 \sim \chi^2(n - p_k)$  are independent (Theorem 7.5.1), and  $E(\sigma^2 / \hat{\sigma}_k^2) = n / (n - p_k - 2)$  by Lemma 14.1.3. Hence the second term has mean  $n(n + p_k) / (n - p_k - 2)$ . For the second statement,  $-2\ell_k = n \log(2\pi \hat{\sigma}_k^2) + n$ , and the identity  $n(n + p) = (n - p - 2)(n + 2(p + 1)) + 2(p + 1)(p + 2)$  shows that  $n(n + p) / (n - p - 2) = n + 2(p + 1) + 2(p + 1)(p + 2) / (n - p - 2)$ . ■

The corrected criterion AICc is due to Sugiura (1978) and Hurvich and Tsai (1989). Its extra term vanishes as  $n \rightarrow \infty$  with  $p_k$  fixed, which recovers AIC. It matters when  $p_k$  is not small compared with  $n$ . With  $n = 20$  and  $p_k = 5$ , AIC charges  $2(p_k + 1) = 12$ , while the exact charge in the proof is  $2(p_k + 1)n / (n - p_k - 2) = 18.46$ . A simulation of 100000 data sets gives a mean discrepancy of 68.11, equal to the theorem's value to two decimals. AIC underestimates it by 6.46, and the underestimate favours the larger models.

```
import numpy as np
rng = np.random.default_rng(2902)
n_c, p_c, reps_c = 20, 5, 100_000
Xc = np.column_stack([np.ones(n_c), rng.normal(size=(n_c, p_c - 1))])
mu_c = Xc @ np.linspace(1, 2, p_c) # the model is correct
H = Xc @ np.linalg.solve(Xc.T @ Xc, Xc.T)
```

```

Yc = mu_c + rng.normal(size=(reps_c, n_c))
fit_c = Yc @ H
s2ml = np.mean((Yc - fit_c) ** 2, axis=1) # ML variance estimate
# discrepancy: expected -2 log-likelihood of an independent copy Y*, at the fitted
# values
disc = n_c * np.log(2 * np.pi * s2ml) + (n_c + np.sum((fit_c - mu_c) ** 2, axis=1))
      / s2ml
target = np.mean(n_c * np.log(2 * np.pi * s2ml)) + n_c * (n_c + p_c) / (n_c - p_c -
2)
print(f"E(discrepancy) {disc.mean():.2f}, formula {target:.2f}")
print(f"penalty in AIC {2 * (p_c + 1)}, in AICc {2 * (p_c + 1) * n_c / (n_c - p_c -
2):.2f}")

```

For a wrong candidate the penalty is no longer free of  $\mu$ , and  $2d_k$  is an approximation, justified when the candidate is close to the truth (Konishi and Kitagawa 2008 discuss corrections). In practice AIC and  $C_p$  rank candidates almost identically (Exercise (B1)).

## 29.2.4 BIC

Suppose the candidates carry prior probabilities  $\pi_k$ , and the parameters  $\theta_k$  of model  $k$  a prior density  $\pi(\theta_k | k)$ . The posterior probability of model  $k$  is proportional to  $\pi_k m_k(\mathbf{y})$ , where

$$m_k(\mathbf{y}) = \int f_k(\mathbf{y} | \theta_k) \pi(\theta_k | k) d\theta_k$$

is the **marginal likelihood** of the model. For the conjugate normal prior of Theorem 7.6.2 the integral has a closed form. BIC is an approximation that needs no particular prior.

*Remark* (Sketch: where  $\log n$  comes from). This is a heuristic derivation; Schwarz (1978) proved the result for exponential families, and Kass and Raftery (1995) give conditions and error terms. Let  $\hat{\theta}$  maximize  $\ell(\theta) = \log f_k(\mathbf{y} | \theta)$ , and let  $\mathbf{J}_n = -\nabla^2 \ell(\hat{\theta})$  be the observed information, which for independent observations grows like  $n$ :  $\mathbf{J}_n = n\tilde{\mathbf{J}}$  with  $\tilde{\mathbf{J}}$  of order one. Expanding  $\ell$  to second order about  $\hat{\theta}$  (Laplace's method, Tierney and Kadane 1986),

$$\begin{aligned}
m_k(\mathbf{y}) &\approx f_k(\mathbf{y} | \hat{\theta}) \pi(\hat{\theta} | k) \int e^{-(\theta - \hat{\theta})^\top \mathbf{J}_n (\theta - \hat{\theta})/2} d\theta \\
&= f_k(\mathbf{y} | \hat{\theta}) \pi(\hat{\theta} | k) (2\pi)^{d_k/2} \det(\mathbf{J}_n)^{-1/2}.
\end{aligned}$$

Since  $\log \det \mathbf{J}_n = d_k \log n + \log \det \tilde{\mathbf{J}}$ ,

$$-2 \log m_k(\mathbf{y}) = -2\ell_k + d_k \log n + O(1),$$

where the  $O(1)$  collects the prior density at  $\hat{\theta}$ ,  $\det \tilde{\mathbf{J}}$  and  $2\pi$ , none of which grows with  $n$ . So  $\text{BIC}_k$  approximates  $-2 \log m_k$ , and with equal prior probabilities the model with the smallest BIC is approximately the most probable.

The error is of order one, so  $\exp(-\text{BIC}_k/2)$  is a crude estimate of a posterior probability, but the rate is right: each parameter costs  $\log n$ . For a “unit information” prior the error is of order  $n^{-1/2}$  (Kass and Wasserman 1995).

## 29.2.5 Three criteria, one data set, one threshold

Adjusted  $R^2$  of [Chapter 9](#) is sometimes used as a criterion too.

### Proposition 29.2.3 — Adjusted $R^2$ as a criterion

Maximizing  $\bar{R}^2$  over candidates that contain the intercept is minimizing  $s_k^2 = \text{SSE}_k/(n - p_k)$ , and for two nested candidates the larger has the larger  $\bar{R}^2$  iff its added columns have  $F > 1$ .

*Proof.* These are [Proposition 9.5.3\(d\)](#) and (c). ■

The threshold 1 is lower than any of the others, so  $\bar{R}^2$  chooses the largest models. Since  $E(s_k^2) = \sigma^2(1 + \gamma_k/(n - p_k))$  ([Theorem 19.1.2\(c\)](#)), minimizing  $s_k^2$  aims at the smallest *bias*, not the smallest prediction error.

For two nested candidates, each criterion is a test with its own critical value. Let the larger model add  $q$  columns, have residual degrees of freedom  $\nu$ , and let  $F = [(\text{SSE}_{\text{small}} - \text{SSE}_{\text{big}})/q]/(\text{SSE}_{\text{big}}/\nu)$ , so that  $\text{SSE}_{\text{small}}/\text{SSE}_{\text{big}} = 1 + qF/\nu$ . The larger model is preferred by

| criterion                  | when                                           | threshold for $F$ ,<br>approximately |
|----------------------------|------------------------------------------------|--------------------------------------|
| adjusted $R^2$             | $F > 1$ ( <a href="#">Proposition 29.2.3</a> ) | 1                                    |
| $C_p$                      | $F > 2$ , with $s^2$ from the full<br>model    | 2                                    |
| AIC                        | $n \log(1 + qF/\nu) > 2q$                      | 2                                    |
| BIC                        | $n \log(1 + qF/\nu) > q \log n$                | $\log n$                             |
| $F$ test at level $\alpha$ | $F > F_\alpha(q, \nu)$                         | $F_\alpha(q, \nu)$                   |

The approximations use  $\log(1 + x) \approx x$  and  $\nu \approx n$ . With  $n = 50$ ,  $q = 1$  and  $\nu = 47$ , the exact thresholds are 1.92 for AIC and 3.82 for BIC ( $\log n = 3.91$ ), against  $F_{0.05}(1, 47) = 4.05$ . As tests of one column, AIC acts at level 0.17 and BIC at level 0.056, close to the conventional 5% at this sample size. BIC's level falls as  $n$  grows, and AIC's does not.

### Example. (Violent crime in the states)

Return to the 50 states of [Chapter 6](#) (the District of Columbia left out) and take as response the logarithm of the violent crime rate in 2009. The candidate regressors are the percentages of adults with a high-school diploma, of people below the poverty line, of single-parent households, of white residents and of urban residents, so there are  $2^5 = 32$  candidates, all with an intercept. The full model has  $P = 6$  and  $s^2 = 0.1035$ . The eight candidates with the smallest  $C_p$  are

| regressors       | $p$ | $C_p$ | $\Delta\text{AIC}$ | $\Delta\text{BIC}$ | $\bar{R}^2$ |
|------------------|-----|-------|--------------------|--------------------|-------------|
| single,<br>urban | 3   | 2.20  | 0.00               | 0.42               | 0.471       |

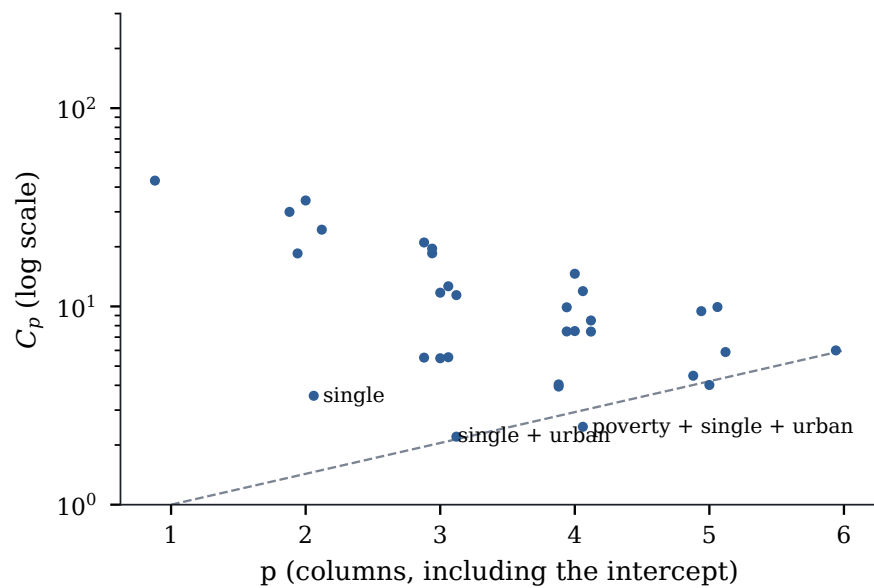


Figure 29.2.1:  $C_p$  against the number of columns for the 32 candidate models for log violent crime in the 50 states. The dashed line is  $C_p = p$ . The three labelled models are the choices of BIC, of  $C_p$  and AIC, and of adjusted  $R^2$ .

|                                              |   |      |      |      |       |
|----------------------------------------------|---|------|------|------|-------|
| poverty,<br>single,<br>urban                 | 4 | 2.47 | 0.09 | 2.42 | 0.480 |
| single                                       | 2 | 3.54 | 1.49 | 0.00 | 0.445 |
| high school,<br>single,<br>urban             | 4 | 3.94 | 1.72 | 4.05 | 0.463 |
| high school,<br>poverty,<br>single,<br>urban | 5 | 4.01 | 1.57 | 5.82 | 0.474 |
| single,<br>white,<br>urban                   | 4 | 4.04 | 1.82 | 4.16 | 0.462 |
| poverty,<br>single,<br>white,<br>urban       | 5 | 4.47 | 2.09 | 6.34 | 0.468 |
| high school,<br>single                       | 3 | 5.48 | 3.42 | 3.85 | 0.434 |

where  $\Delta$  is the difference from the smallest value over all 32 candidates.  $C_p$  and AIC choose single parenthood and urbanization; BIC drops urbanization;  $\bar{R}^2$  adds poverty. The first three rows are within two units of each other on every scale, and none of the criteria separates them convincingly. Figure 29.2.1 is Mallows' plot. Every candidate without single parenthood lies far above the line  $C_p = p$ : it is badly biased.

```
import itertools
import numpy as np
```

```

import statsmodels.api as sm
data = sm.datasets.statecrime.load_pandas().data.drop(index="District of Columbia")
names = ["hs_grad", "poverty", "single", "white", "urban"]
y = np.log(data["violent"].to_numpy())
n = len(y)

def sse(cols):
    X = np.column_stack([np.ones(n)] + [data[c].to_numpy() for c in cols])
    b, *_ = np.linalg.lstsq(X, y, rcond=None)
    return np.sum((y - X @ b) ** 2), X.shape[1]

sse_full, P = sse(names)
s2 = sse_full / (n - P)                                     # sigma^2 estimate from the
    largest model
sst = np.sum((y - y.mean()) ** 2)
rows = []
for k in range(len(names) + 1):
    for cols in itertools.combinations(names, k):
        e, p = sse(cols)
        rows.append(dict(model=cols, p=p, sse=e,
                           cp=e / s2 - n + 2 * p,
                           aic=n * np.log(e / n) + 2 * (p + 1),
                           bic=n * np.log(e / n) + np.log(n) * (p + 1),
                           adjr2=1 - (e / (n - p)) / (sst / (n - 1))))
for crit in ("cp", "aic", "bic"):
    best = min(rows, key=lambda r: r[crit])
    print(f"{crit:4s} chooses {best['model']}")
print("adj R2 chooses", max(rows, key=lambda r: r["adjr2"])["model"])

```

## 29.2.6 Consistency and efficiency

The thresholds explain the large-sample behaviour. The  $F$  statistic of an irrelevant column stays of order one, so a growing threshold such as  $\log n$  eventually excludes it and a fixed one does not; that of a relevant column grows like  $n$ .

### Proposition 29.2.4 — AIC and BIC in large samples

Let the candidate set  $\mathcal{K}$  be finite and fixed, and consider a sequence of designs and means with  $n \rightarrow \infty$ . Let the errors be independent and identically distributed with mean zero and variance  $\sigma^2$ , and let  $\sup_n \|\mu\|^2/n < \infty$ . Call  $k$  *correct* if  $\mu \in \mathcal{C}(\mathbf{X}_k)$ . Assume that there is a correct  $k_0$  such that every other correct candidate has  $p_k > p_{k_0}$ , and that  $\liminf_n \|(\mathbf{I} - \mathbf{M}_k)\mu\|^2/n > 0$  for every candidate that is not correct. Select  $\hat{k}$  by minimizing  $\text{IC}_k = n \log(\text{SSE}_k/n) + c_n p_k$ .

- (a) If  $c_n \rightarrow \infty$  and  $c_n/n \rightarrow 0$ , then  $\Pr(\hat{k} = k_0) \rightarrow 1$ . In particular BIC ( $c_n = \log n$ ) is consistent.
- (b) If  $c_n = c$  is constant, the errors are normal, and  $k$  is a correct candidate with  $\mathcal{C}(\mathbf{X}_{k_0}) \subset \mathcal{C}(\mathbf{X}_k)$  and  $q = p_k - p_{k_0}$ , then  $\Pr(\text{IC}_k < \text{IC}_{k_0}) \rightarrow \Pr\{\chi^2(q) > cq\} > 0$ . In particular AIC ( $c = 2$ ) is not consistent when such a candidate is available.
- (c) (*Efficiency, stated with a sketch.*) Suppose instead that no candidate is correct,

and that the candidate sets grow with  $n$  so that the smallest risk over candidates tends to infinity. Then, under moment conditions on the errors and a condition limiting the number of candidates, with  $\sigma^2$  known or consistently estimated in  $C_p$ , and for AIC with the largest candidate dimension  $o(n)$ , the loss  $\|\hat{\boldsymbol{\mu}}_k - \boldsymbol{\mu}\|^2$  of the  $C_p$  or AIC choice divided by  $\min_k \|\hat{\boldsymbol{\mu}}_k - \boldsymbol{\mu}\|^2$  tends to one in probability (Shibata 1981; Li 1987). A criterion with  $c_n \rightarrow \infty$  does not have this property in general.

*Proof.* Throughout,  $\text{SSE}_k = \|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2 + 2\boldsymbol{\mu}^\top (\mathbf{I} - \mathbf{M}_k)\boldsymbol{\varepsilon} + \|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\varepsilon}\|^2$ . The last term is  $\|\boldsymbol{\varepsilon}\|^2 - \|\mathbf{M}_k\boldsymbol{\varepsilon}\|^2$ , where  $\|\boldsymbol{\varepsilon}\|^2/n \rightarrow \sigma^2$  in probability by the weak law of large numbers and  $\mathbb{E}\|\mathbf{M}_k\boldsymbol{\varepsilon}\|^2 = \sigma^2 p_k$ , so  $\|\mathbf{M}_k\boldsymbol{\varepsilon}\|^2 = O_p(1)$  by Markov's inequality. The middle term has mean zero and variance  $4\sigma^2\|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2 = O(n)$ , so it is  $O_p(\sqrt{n})$  by Chebyshev's inequality.

- (a) It suffices to show  $\Pr(\text{IC}_k > \text{IC}_{k_0}) \rightarrow 1$  for each  $k \neq k_0$ , since  $\mathcal{K}$  is finite. For  $k_0$ ,  $\text{SSE}_{k_0} = \|\boldsymbol{\varepsilon}\|^2 - \|\mathbf{M}_{k_0}\boldsymbol{\varepsilon}\|^2$ , so  $\text{SSE}_{k_0}/n \rightarrow \sigma^2$ .

A correct  $k \neq k_0$ . Then  $|\text{SSE}_k - \text{SSE}_{k_0}| \leq \|\mathbf{M}_k\boldsymbol{\varepsilon}\|^2 + \|\mathbf{M}_{k_0}\boldsymbol{\varepsilon}\|^2 = O_p(1)$ , and  $n \log(\text{SSE}_k/\text{SSE}_{k_0}) = n \log(1 + (\text{SSE}_k - \text{SSE}_{k_0})/\text{SSE}_{k_0}) = O_p(1)$ , because the ratio inside the logarithm is  $O_p(1/n)$ . Hence  $\text{IC}_k - \text{IC}_{k_0} = O_p(1) + c_n(p_k - p_{k_0})$ , which tends to  $+\infty$  in probability since  $p_k > p_{k_0}$  and  $c_n \rightarrow \infty$ .

A candidate that is not correct. Let  $0 < 2b < \liminf_n \|(\mathbf{I} - \mathbf{M}_k)\boldsymbol{\mu}\|^2/n$ . By the expansion,  $\text{SSE}_k/n \geq \sigma^2 + b$  with probability tending to one, so  $\log(\text{SSE}_k/\text{SSE}_{k_0}) \geq \delta$  with probability tending to one, where  $\delta = \frac{1}{2} \log(1 + b/\sigma^2) > 0$ . Then  $\text{IC}_k - \text{IC}_{k_0} \geq n\delta - c_n|p_k - p_{k_0}|$ , which tends to  $+\infty$  because  $c_n/n \rightarrow 0$ .

- (b) Here  $D = \text{SSE}_{k_0} - \text{SSE}_k = \|(\mathbf{M}_k - \mathbf{M}_{k_0})\boldsymbol{\varepsilon}\|^2$ , and  $D/\sigma^2 \sim \chi^2(q)$  exactly for every  $n$  (Theorem 4.3.2). With  $\text{SSE}_k/n \rightarrow \sigma^2$ , Slutsky's theorem gives  $nD/\text{SSE}_k \rightarrow \chi^2(q)$  in distribution, and  $n \log(1 + D/\text{SSE}_k) - nD/\text{SSE}_k \rightarrow 0$  in probability because  $D/\text{SSE}_k = O_p(1/n)$  and  $|\log(1 + x) - x| \leq x^2$  for  $x \geq 0$ . So  $n \log(\text{SSE}_{k_0}/\text{SSE}_k) \rightarrow \chi^2(q)$  in distribution. Now  $\text{IC}_k < \text{IC}_{k_0}$  iff  $n \log(\text{SSE}_{k_0}/\text{SSE}_k) > cq$ , and  $cq$  is a continuity point of the limit.

- (c) *Sketch.* By Proposition 29.2.1(a),  $C_k^\sigma$  is unbiased for the scaled risk of each candidate, and  $C_k^\sigma - J_k$  and the loss minus the risk are sums of centred quadratic and linear forms in  $\boldsymbol{\varepsilon}$ . The conditions make these fluctuations small compared with the risks, uniformly over the candidates, so minimizing  $C_k$  is asymptotically the same as minimizing the loss. Li (1987) gives such conditions for  $C_p$ , cross-validation and generalized cross-validation: finite moments of order  $4m$  for the errors and  $\sum_k (nR_k)^{-m} \rightarrow 0$  for some  $m$ , where  $R_k$  is the risk of candidate  $k$  per observation. AIC differs from  $C_p$  by terms of order  $p_k^2/n$ , which are negligible when the candidate dimensions are  $o(n)$  (Shibata 1981). A growing penalty  $c_n$  biases the choice towards candidates whose risk is larger by a factor that does not tend to one, as Figure 29.2.2(b) shows. ■

For AIC the limits in (b) are 0.157, 0.135 and 0.075 for  $q = 1, 2, 5$ : one extra column is kept about one time in six however large the sample. Nishii (1984) proved (a) and (b) for the normal linear model.

### Example. (Consistency against efficiency)

Panel (a) of Figure 29.2.2 uses six independent normal regressors, three of them with



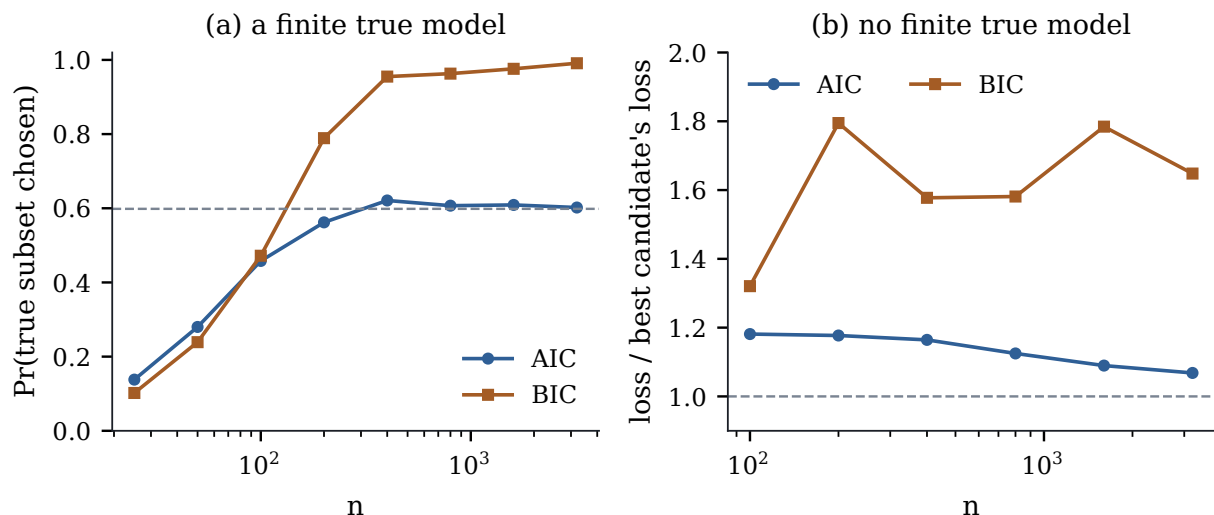


Figure 29.2.2: (a) Probability of selecting the true subset among 64, when a finite true model exists; dashed: AIC's limit 0.598. (b) Loss of the selected model relative to the best candidate, when no candidate is true. 1000 simulated data sets per point in (a), 400 in (b).

slopes 1, 0.5 and 0.25 and three with slope zero, and all 64 subsets as candidates. With  $n = 50$  the true subset is found by AIC with probability 0.28 and by BIC with probability 0.24, because the smallest slope is hard to detect. With  $n = 3200$  BIC finds it with probability 0.99, while AIC stays at 0.60. That is the limit  $\Pr\{\chi^2(1) \leq 2\}^3 = 0.598$ , because in large samples AIC keeps each irrelevant regressor independently when its squared  $t$  statistic exceeds about 2.

In panel (b) the truth has 1000 regressors with slopes  $2/j$ , all nonzero, and the candidates are the nested models containing the first  $k$  regressors,  $k \leq n/4$ . No candidate is correct. The ratio of the loss of the chosen model to the smallest loss among the candidates falls for AIC, from 1.18 at  $n = 100$  to 1.07 at  $n = 3200$ , while for BIC it stays between 1.3 and 1.8 at every sample size.

```
import itertools
import numpy as np
rng = np.random.default_rng(2903)
beta = np.array([1.0, 0.5, 0.25, 0.0, 0.0, 0.0]) # three relevant, three
            irrelevant
true_set = (0, 1, 2)
subsets = [s for k in range(7) for s in itertools.combinations(range(6), k)]

def choose(X, y, penalty):
    """All-subsets choice by  $n \log(\text{SSE}/n) + \text{penalty} * (\text{number of columns})$ ."""
    n = len(y)
    best, arg = np.inf, None
    for s in subsets:
        Xs = np.column_stack([np.ones(n), X[:, list(s)]])
        b, *_ = np.linalg.lstsq(Xs, y, rcond=None)
        crit = n * np.log(np.sum((y - Xs @ b) ** 2) / n) + penalty * Xs.shape[1]
        if crit < best:
            best, arg = crit, s
    return arg
```

```

def hit_rates(n, reps):
    X = rng.normal(size=(n, 6))                # the design, fixed for
    this n
    hits = np.zeros(2)
    for _ in range(reps):
        y = 1 + X @ beta + rng.normal(size=n)
        hits += [choose(X, y, 2.0) == true_set, choose(X, y, np.log(n)) == true_set]
    return hits / reps

for n in (50, 200, 800):
    aic, bic = hit_rates(n, 100)
    print(f"n = {n:4d}: AIC finds the true subset {aic:.2f} of the time, BIC {bic:.2f}")

```

The two properties cannot be combined: Yang (2005) showed that no rule can be consistent and also attain the best rate of prediction risk. The choice between AIC and BIC is a choice of goal: prediction (AIC,  $C_p$ , cross-validation) or identifying a small true set of regressors (BIC).

## 29.2.7 Selection in an orthonormal design

When the columns are orthonormal the search disappears, and the criteria can be compared exactly.

### Proposition 29.2.5 — Selection as hard thresholding

Let  $\mathbf{X}$  have orthonormal columns,  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}_m$ , put  $\mathbf{Z} = \mathbf{X}^\top \mathbf{Y}$ , and let  $\sigma^2$  be known. Among all subsets  $S$  of the columns,  $\text{SSE}_S/\sigma^2 + c^2|S|$  is minimized (uniquely, with probability one) by  $S = \{j : |Z_j| > c\sigma\}$ . The selected fit has coefficients  $Z_j 1\{|Z_j| > c\sigma\}$ .

*Proof.* The projection onto the columns in  $S$  is  $\sum_{j \in S} \mathbf{x}_j \mathbf{x}_j^\top$ , so  $\text{SSE}_S = \|\mathbf{Y}\|^2 - \sum_{j \in S} Z_j^2$  and the criterion is  $\|\mathbf{Y}\|^2/\sigma^2 + \sum_{j \in S} (c^2 - Z_j^2/\sigma^2)$ . A column lowers it iff  $Z_j^2 > c^2\sigma^2$ , and ties have probability zero. ■

With known  $\sigma^2$ ,  $C_p$  and AIC take  $c = \sqrt{2}$  and BIC takes  $c = \sqrt{\log n}$ . Testing each coefficient at level 5% and keeping the significant ones (the **pretest estimator** of Bancroft 1944) takes  $c = 1.96$ . This is the hard thresholding of [Theorem 27.4.2](#), and each coordinate can be studied alone. With  $\sigma = 1$  and  $Z \sim N(\theta, 1)$ , the risk of  $Z 1\{|Z| > c\}$  is

$$\begin{aligned}
 R(\theta; c) &= \theta^2 D + 1 - D + (c - \theta)\phi(c - \theta) + (c + \theta)\phi(c + \theta), \\
 D &= \Phi(c - \theta) - \Phi(-c - \theta),
 \end{aligned}
 \tag{29.2.4}$$

where  $D$  is the probability that the coordinate is dropped ([Exercise \(B2\)](#)). Least squares has risk 1, and dropping the coordinate always has risk  $\theta^2$ .

[Figure 29.2.3](#) shows the trade. At  $\theta = 0$  the risks are 0.572 for  $c = \sqrt{2}$ , 0.279 for the pretest and 0.075 for BIC with  $n = 1000$ . (At zero the risk equals the degrees of freedom of [Proposition 29.1.3](#): both are  $E[Z^2 1\{|Z| > c\}]$ .) The curves cross 1 at  $\theta = 0.78, 0.84$  and  $0.92$  and peak

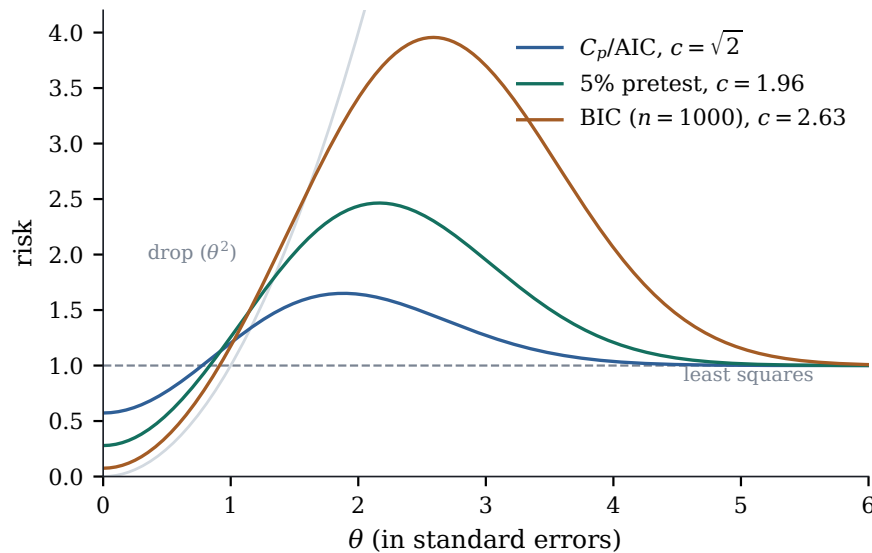


Figure 29.2.3: Risk of hard thresholding  $Z1\{|Z| > c\}$ ,  $Z \sim N(\theta, 1)$ , for the thresholds of  $C_p$  and AIC, a 5% pretest and BIC with  $n = 1000$ . Least squares has risk 1 (dashed); always dropping the coordinate has risk  $\theta^2$  (grey).

at 1.651, 2.464 and 3.956, at  $\theta = 1.88$ , 2.16 and 2.59. A larger threshold gains more at zero and loses more at moderate  $\theta$ , and no threshold, least squares included, is uniformly best. This is [Theorem 19.1.5](#) again: dropping a coordinate helps iff  $\theta^2 < 1$ , and the data cannot tell which side of the boundary they are on. Shrinkage avoids the bump: the James–Stein estimator of [Theorem 27.5.2](#) beats least squares everywhere when  $m \geq 3$ .

```
import numpy as np
from scipy.stats import norm

def hard_risk(theta, c):
    """E(Z 1{|Z| > c} - theta)^2 for Z ~ N(theta, 1)."""
    D = norm.cdf(c - theta) - norm.cdf(-c - theta) # Pr(|Z| <= c): coordinate
    dropped
    return theta**2 * D + 1 - D + (c - theta) * norm.pdf(c - theta) + (c + theta) *
    norm.pdf(c + theta)

thetas = np.linspace(0, 6, 601)
for label, c in [("C_p / AIC", np.sqrt(2)), ("5% pretest", 1.96), ("BIC, n = 1000",
    np.sqrt(np.log(1000)))]:
    r = hard_risk(thetas, c)
    print(f"{label:14s} c = {c:.3f}: risk at 0 = {r[0]:.3f}, "
        f"maximum {r.max():.3f} at theta = {thetas[r.argmax():.2f}")
```

## 29.2.8 Exercises

### 29.2.8.1 A. Check your understanding

**Exercise A1.**

In [Example \(Violent crime in the states\)](#) the model with single parenthood and urbanization has  $\text{SSE} = 4.782$ . Compute its  $C_p$  from  $s^2$ ,  $n$  and  $p$ , and the  $F$  statistic for testing it against the full model.

*Solution.*  $C_p = 4.782/0.1035 - 50 + 6 = 46.20 - 44 = 2.20$ . By [Proposition 29.2.1\(c\)](#),  $2.20 = 3(F - 1) + 3$ , so  $F = 0.73$ : the three regressors left out together have an  $F$  statistic below one.

**29.2.8.2 B. Practice****Exercise B1.**

Let  $\hat{\sigma}^2$  be any fixed positive number and write  $\text{SSE}_k/n = \hat{\sigma}^2(1 + \delta_k)$ . Show that

$$\text{AIC}_k - \text{AIC}_j = \frac{\text{SSE}_k - \text{SSE}_j}{\hat{\sigma}^2} + 2(p_k - p_j) + n O(\delta_k^2 + \delta_j^2),$$

and conclude that with  $\hat{\sigma}^2 = s^2$  AIC and  $C_p$  differences agree when every candidate's residual mean square is close to  $s^2$ .

**Exercise B2.**

Derive [Eq. 29.2.4](#). (Split the risk according to whether the coordinate is kept, and use  $\int_a^b w^2 \phi(w) dw = \Phi(b) - \Phi(a) - b\phi(b) + a\phi(a)$ .)

*Solution.* Let  $Z = \theta + W$ . When the coordinate is dropped ( $|Z| \leq c$ , probability  $D$ ) the loss is  $\theta^2$ ; when kept it is  $W^2$ . So  $R = \theta^2 D + E[W^2 1\{|\theta + W| > c\}]$ . The event  $|\theta + W| \leq c$  is  $-c - \theta \leq W \leq c - \theta$ , so  $E[W^2 1\{|\theta + W| > c\}] = 1 - \int_{-c-\theta}^{c-\theta} w^2 \phi(w) dw = 1 - D + (c - \theta)\phi(c - \theta) + (c + \theta)\phi(c + \theta)$ , using  $\phi(-x) = \phi(x)$ .

**Exercise B3.**

Show that AIC prefers the larger of two nested candidates iff  $F > (\nu/q)(e^{2q/n} - 1)$ , and BIC iff  $F > (\nu/q)(n^{q/n} - 1)$ . Find the limits of both thresholds as  $n \rightarrow \infty$  with  $q$  and  $n - \nu$  fixed.

**29.2.8.3 C. Going deeper****Exercise C1.**

In panel (a) of [Example \(Consistency against efficiency\)](#), suppose the three regressors with zero slopes are orthogonal to each other and to the relevant ones. Show that for large  $n$  AIC selects the true subset iff each of them has squared  $t$  statistic below about 2, and deduce the limit  $\Pr\{\chi^2(1) \leq 2\}^3$ .

*Solution.* Every candidate that omits a relevant regressor loses with probability tending to one (Proposition 29.2.4, proof of (a), second case, which holds for any  $c_n$  with  $c_n/n \rightarrow 0$ ). Among candidates containing the relevant block, orthogonality makes  $\text{SSE}_{k_0 \cup A} = \text{SSE}_{k_0} - \sum_{j \in A} \|\mathbf{M}_{\{j\}} \boldsymbol{\epsilon}\|^2$ , and  $n \log(\text{SSE}_{k_0}/\text{SSE}_{k_0 \cup A}) = \sum_{j \in A} T_j^2 + o_p(1)$  with  $T_j^2 \rightarrow \chi^2(1)$  independent. AIC adds  $A$  iff this exceeds  $2|A|$ , so it chooses  $A = \{j : T_j^2 > 2\}$  up to  $o_p(1)$ , and chooses  $A = \emptyset$  with probability tending to  $\prod_j \Pr(T_j^2 \leq 2) = \Pr\{\chi^2(1) \leq 2\}^3$ .

## 29.3 Cross-validation

The criteria of Section 29.2 correct the training error by a formula that rests on assumptions. Cross-validation instead predicts cases that the fit has not seen. It needs neither  $\sigma^2$  nor degrees of freedom, and applies to any procedure, including one that selects a model. Its price is computation, a bias that is easy to describe, and a variance that is not.

### 29.3.1 K-fold and leave-one-out

#### Definition 29.3.1 — Cross-validation

Let the cases  $(\mathbf{x}_{(i)}, y_i)$ ,  $i = 1, \dots, n$ , be divided into  $K$  disjoint **folds**  $I_1, \dots, I_K$ . For each  $k$ , let  $\hat{f}^{(-k)}$  be the prediction rule obtained by applying the procedure to the cases outside  $I_k$ . The  **$K$ -fold cross-validation** estimate of prediction error is

$$\text{CV}_K = \frac{1}{n} \sum_{k=1}^K \sum_{i \in I_k} (y_i - \hat{f}^{(-k)}(\mathbf{x}_{(i)}))^2.$$

With  $K = n$ , each fold is a single case, and  $\text{CV}_n$  is **leave-one-out** cross-validation. With  $K = 2$  and one fold used only for fitting, the estimate is a **holdout** or **validation-set** estimate.

The idea is old (Stone 1974; Geisser 1975; Allen 1974 for least squares). What cross-validation estimates is clearest when the cases are a random sample.

#### Proposition 29.3.1 — What K-fold cross-validation estimates

Let the cases be independent draws of  $(\mathbf{X}, Y)$ , let the folds be fixed in advance with equal sizes  $n/K$ , and let  $\text{Err}(m) = \mathbb{E}(Y_0 - \hat{f}_m(\mathbf{X}_0))^2$  be the expected squared error of the procedure fitted to  $m$  cases, for an independent new case  $(\mathbf{X}_0, Y_0)$ . Then

$$\mathbb{E}(\text{CV}_K) = \text{Err}(n - n/K).$$

*Proof.* For  $i \in I_k$ , the rule  $\hat{f}^{(-k)}$  is the procedure applied to  $n - n/K$  independent cases, and case  $i$  is independent of them and has the same distribution as  $(\mathbf{X}_0, Y_0)$ . So each term of the double sum has expectation  $\text{Err}(n - n/K)$ . ■

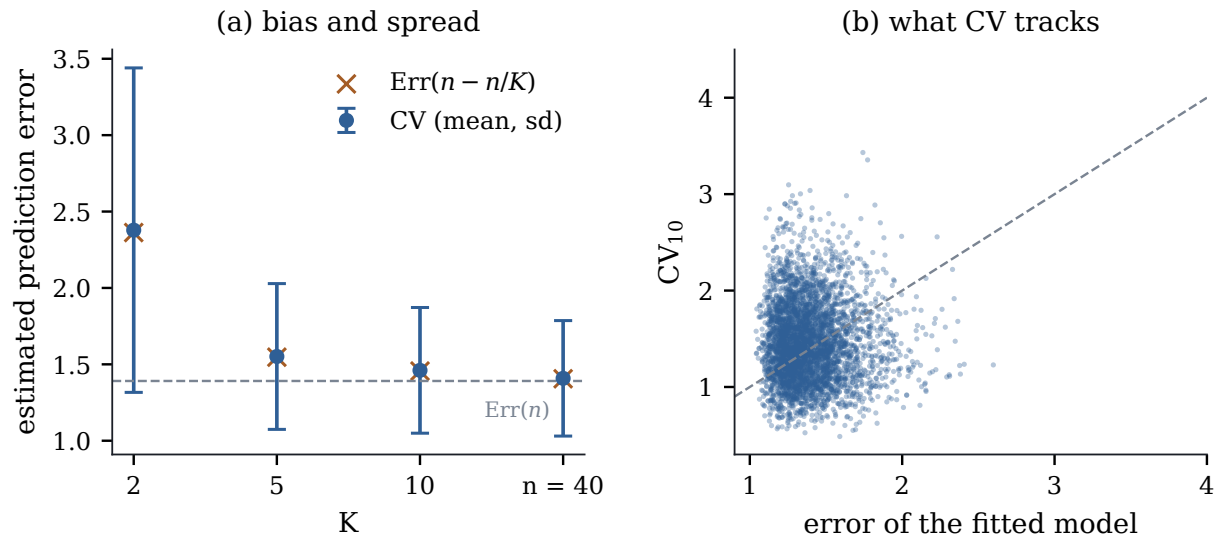


Figure 29.3.1: (a) Mean and standard deviation of  $CV_K$  over 4000 training sets ( $n = 40$ , ten regressors), with the expected error at sample size  $n - n/K$  (crosses) and at  $n$  (dashed). (b)  $CV_{10}$  against the prediction error of the model actually fitted.

The expectation is over training sets as well as new cases. Cross-validation therefore estimates the expected error of the *procedure* at a slightly smaller sample size, not the error of the particular fit computed from all  $n$  cases. For least squares with  $k$  normal regressors and a correct model, [Theorem 14.2.6](#) gives  $Err(m) = \sigma^2(1 + 1/m)(m - 2)/(m - k - 2)$ , so the bias of  $CV_K$  can be computed exactly.

#### Example. (Bias and spread of K-fold cross-validation)

Take  $n = 40$  cases with  $k = 10$  independent standard normal regressors, an intercept, slopes 0.5 and  $\sigma = 1$ , and fit by least squares. The target is  $Err(40) = 1.391$ . Over 4000 training sets, the averages of  $CV_K$  for  $K = 2, 5, 10, 40$  are 2.378, 1.551, 1.461 and 1.408, against  $Err(n - n/K)$  of 2.363, 1.547, 1.456 and 1.406 from [Proposition 29.3.1](#). Fitting to half the data inflates the error by 70% here, because ten slopes are estimated from twenty cases. Their standard deviations across training sets are 1.062, 0.477, 0.411 and 0.378. For this stable procedure leave-one-out has both the smallest bias and the smallest variance.

The error of the fitted model itself,  $\sigma^2 + \|\hat{\beta} - \beta\|^2$  for a new standard normal case, varies too: its mean is 1.395 and its standard deviation 0.203. Panel (b) of [Figure 29.3.1](#) plots  $CV_{10}$  against it. The correlation is  $-0.005$  ( $-0.014$  for leave-one-out). Cross-validation reports how well least squares does on average with forty such cases, not how well this fit will do.

```
import numpy as np
rng = np.random.default_rng(2905)
n_s, k_s, sigma = 40, 10, 1.0
beta = np.r_[1.0, np.full(k_s, 0.5)]  # intercept and ten
    slopes

def err_theory(m):
    """Expected error of a least squares fit to m cases (Chapter 14, normal
    regressors)."""
```

```

    return sigma**2 * (1 + 1 / m) * (m - 2) / (m - k_s - 2)

def cv_kfold(X, y, K):
    folds = np.arange(len(y)) % K                                # equal folds (cases
    are exchangeable)
    sq = np.empty(len(y))
    for f in range(K):
        tr, te = folds != f, folds == f
        b, *_ = np.linalg.lstsq(X[tr], y[tr], rcond=None)
        sq[te] = (y[te] - X[te] @ b) ** 2
    return sq.mean()

Ks = [2, 5, 10, n_s]

def simulate(reps):
    cv = np.empty((reps, len(Ks)))
    err_cond = np.empty(reps)                                     # error of the fit at
    hand
    for r in range(reps):
        Xr = np.column_stack([np.ones(n_s), rng.normal(size=(n_s, k_s))])
        yr = Xr @ beta + sigma * rng.normal(size=n_s)
        b, *_ = np.linalg.lstsq(Xr, yr, rcond=None)
        err_cond[r] = sigma**2 + np.sum((b - beta) ** 2)           # new case  $X_0 \sim N(0, I)$ 
        cv[r] = [cv_kfold(Xr, yr, K) for K in Ks]
    return cv, err_cond

cv, err_cond = simulate(300)                                     # the book's figures
    use 4000
for j, K in enumerate(Ks):
    print(f"K = {K:2d}: mean {cv[:, j].mean():.3f} (theory {err_theory(n_s - n_s //
    K):.3f}), sd {cv[:, j].std():.3f}")
print(f"error of the fit at hand: mean {err_cond.mean():.3f} (theory {err_theory(n_s
):.3f})")
print("correlation of CV_10 with it:", np.corrcoef(cv[:, 2], err_cond)[0, 1])

```

Bates, Hastie and Tibshirani (2024) study this phenomenon for linear models. The spread falling with  $K$  is typical of least squares, whose fits change little when a few cases are removed; for procedures that jump, such as subset selection, leave-one-out can be the more variable choice (Breiman and Spector 1992). Five or ten folds are the usual compromise.

## 29.3.2 The leverage shortcut

Leave-one-out looks expensive:  $n$  refits. For least squares and for ridge regression it costs one fit. The reason is a small fixed-point argument.

### Theorem 29.3.2 — Leave-one-out without refitting

Let  $\Omega$  be a  $p \times p$  nonnegative definite matrix, let  $\hat{\beta}$  minimize  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \mathbf{b}^\top \Omega \mathbf{b}$ , and let  $\hat{\beta}_{(i)}$  minimize the same criterion with case  $i$  removed. Assume that  $\mathbf{A}_{(i)} = \mathbf{X}_{(i)}^\top \mathbf{X}_{(i)} + \Omega$

is nonsingular for every  $i$ , where  $\mathbf{X}_{(i)}$  is  $\mathbf{X}$  without row  $i$ . Let  $\mathbf{S} = \mathbf{X}(\mathbf{X}^\top \mathbf{X} + \mathbf{\Omega})^{-1} \mathbf{X}^\top$  and  $\hat{\mathbf{y}} = \mathbf{S}\mathbf{y}$ . Then:

- (a)  $s_{ii} < 1$  for every  $i$ ;
- (b)  $y_i - \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)} = (y_i - \hat{y}_i)/(1 - s_{ii})$ ;
- (c) consequently

$$\text{CV}_n = \frac{1}{n} \sum_{i=1}^n \left( \frac{y_i - \hat{y}_i}{1 - s_{ii}} \right)^2. \quad (29.3.1)$$

For least squares ( $\mathbf{\Omega} = \mathbf{0}$ ) this is [Theorem 20.3.1\(d\)](#), and  $\text{CV}_n = \text{PRESS}/n$  with PRESS as in [Eq. 20.3.1](#).

*Proof.*

- (a) Write  $\mathbf{x} = \mathbf{x}_{(i)}$  and  $\mathbf{A} = \mathbf{A}_{(i)}$ , which is positive definite because it is nonnegative definite and nonsingular. Then  $\mathbf{X}^\top \mathbf{X} + \mathbf{\Omega} = \mathbf{A} + \mathbf{x}\mathbf{x}^\top$ , and  $(\mathbf{A} + \mathbf{x}\mathbf{x}^\top)^{-1} \mathbf{x} = \mathbf{x}(1 + a)^{-1}$  with  $a = \mathbf{x}^\top \mathbf{A}^{-1} \mathbf{x} \geq 0$ . Hence  $(\mathbf{A} + \mathbf{x}\mathbf{x}^\top)^{-1} \mathbf{x} = \mathbf{A}^{-1} \mathbf{x} / (1 + a)$  and  $s_{ii} = \mathbf{x}^\top (\mathbf{A} + \mathbf{x}\mathbf{x}^\top)^{-1} \mathbf{x} = a / (1 + a) < 1$ .
- (b) Let  $\tilde{y}_i = \mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}}_{(i)}$ , the leave-one-out prediction, and let  $\tilde{\mathbf{y}}$  be  $\mathbf{y}$  with  $y_i$  replaced by  $\tilde{y}_i$ . For any  $\mathbf{b}$ , the full criterion evaluated at  $\tilde{\mathbf{y}}$  is the leave-one-out criterion plus  $(\tilde{y}_i - \mathbf{x}_{(i)}^\top \mathbf{b})^2 \geq 0$ . At  $\mathbf{b} = \hat{\boldsymbol{\beta}}_{(i)}$  the extra term is zero and the leave-one-out criterion is at its minimum. So  $\hat{\boldsymbol{\beta}}_{(i)}$  minimizes the full criterion for the data  $\tilde{\mathbf{y}}$ , and since that minimizer is unique ( $\mathbf{X}^\top \mathbf{X} + \mathbf{\Omega} = \mathbf{A} + \mathbf{x}\mathbf{x}^\top$  is positive definite),  $\mathbf{X} \hat{\boldsymbol{\beta}}_{(i)} = \mathbf{S} \tilde{\mathbf{y}}$ . Its  $i$ th entry reads

$$\tilde{y}_i = (\mathbf{S} \tilde{\mathbf{y}})_i = (\mathbf{S} \mathbf{y})_i - s_{ii}(y_i - \tilde{y}_i) = \hat{y}_i - s_{ii}(y_i - \tilde{y}_i).$$

Rearranging,  $(1 - s_{ii})(y_i - \tilde{y}_i) = y_i - \hat{y}_i$ , and (a) allows division by  $1 - s_{ii}$ .

- (c) follows from (b), and for  $\mathbf{\Omega} = \mathbf{0}$ ,  $s_{ii} = h_{ii}$  and (b) is [Theorem 20.3.1\(d\)](#). ■

The argument uses only that deleting a case gives the same answer as replacing its response by its own prediction. It covers the ridge estimators of [Chapter 27](#) and the smoothing splines of [Chapter 43](#), but not the lasso or subset selection, where deleting a case can change which coefficients are nonzero.

[Section 27.2](#) already used the leave-one-out shortcut and GCV to choose the ridge  $\lambda$ ; the theorem proves the shortcut for every quadratic penalty. Replacing each  $s_{ii}$  by its average  $\text{tr}(\mathbf{S})/n$  gives a criterion that needs only the trace.

### Definition 29.3.2 — Generalized cross-validation

For a linear smoother  $\hat{\mathbf{y}} = \mathbf{S}\mathbf{y}$  with  $\text{tr}(\mathbf{S}) < n$ ,

$$\text{GCV} = \frac{n^{-1} \|\mathbf{y} - \hat{\mathbf{y}}\|^2}{(1 - \text{tr}(\mathbf{S})/n)^2}.$$



Craven and Wahba (1979) introduced GCV for smoothing splines and Golub, Heath and Wahba (1979) for ridge regression, where it is leave-one-out after a rotation that equalizes the diagonal of  $\mathbf{S}$ . Since  $(1 - x)^{-2} = 1 + 2x + O(x^2)$ ,  $\text{GCV} \approx \overline{\text{err}}(1 + 2\text{df}/n)$ , the covariance penalty Eq. 29.1.3 with  $\sigma^2$  replaced by  $\overline{\text{err}}$ . For least squares  $\text{GCV} = ns^2/(n - p)$ , and for a correct model ( $\mu \in \mathcal{C}(\mathbf{X})$ ), by Proposition 20.3.2,  $E(\text{PRESS})/n \geq \sigma^2 n/(n - p) = E(\text{GCV})$ : high-leverage cases inflate leave-one-out.

For the state regression of Example (Violent crime in the states) with all five regressors,  $\text{PRESS}/n = 0.1426$  and  $\text{GCV} = 0.1176$ . For ridge regression on the standardized regressors with  $\lambda = 5$  (intercept unpenalized),  $\text{tr}(\mathbf{S}) = 4.870$ , and the shortcut gives  $\text{CV}_n = 0.1309$ , equal to the brute-force value from fifty refits to rounding error, against  $\text{GCV} = 0.1140$ .

```
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
names = ["hs_grad", "poverty", "single", "white", "urban"]
y = np.log(data["violent"].to_numpy())
Zs = data[names].to_numpy()
Zs = (Zs - Zs.mean(axis=0)) / Zs.std(axis=0)           # standardized regressors
X = np.column_stack([np.ones(len(y)), Zs])
n, p = X.shape
Omega = np.diag([0.0] + [1.0] * (p - 1))              # ridge penalty,
                                                    intercept left free

def ridge_fit(X, y, lam):
    return np.linalg.solve(X.T @ X + lam * Omega, X.T @ y)

lam = 5.0
S = X @ np.linalg.solve(X.T @ X + lam * Omega, X.T)   # the smoother matrix
resid = y - S @ y
loo_short = resid / (1 - np.diag(S))                  # the shortcut
loo_brute = np.array([y[i] - X[i] @ ridge_fit(np.delete(X, i, 0), np.delete(y, i),
    lam)
                    for i in range(n)])                # n refits
print("largest discrepancy:", np.max(np.abs(loo_short - loo_brute)))
cv_loo = np.mean(loo_short**2)
gcv = np.mean(resid**2) / (1 - np.trace(S) / n) ** 2
print(f"CV_n = {cv_loo:.5f}, GCV = {gcv:.5f}, tr(S) = {np.trace(S):.3f}")
```

*Remark* (Sketch: leave-one-out behaves like AIC). If every leverage is close to  $p/n$ , then  $(1 - h_{ii})^{-2} \approx 1 + 2p/n$  and  $\text{CV}_n \approx (\text{SSE}/n)(1 + 2p/n)$ , so  $n \log \text{CV}_n \approx n \log(\text{SSE}/n) + 2p$ , the AIC of Eq. 29.2.2 up to a constant. Stone (1977) made this equivalence precise. Leave-one-out therefore shares AIC's large-sample behaviour: efficient when no candidate is correct (Li 1987), not consistent when one is. Shao (1993) showed that consistency requires leaving out a fraction of the cases that tends to one, the opposite of what prediction suggests;  $K$ -fold with fixed  $K$  is not consistent either.

### 29.3.3 Choosing a model by cross-validation

To choose among candidates, take the smallest  $\text{CV}_K$ . The fold errors  $\text{CV}^{(k)}$  give a rough standard error  $\text{sd}(\text{CV}^{(1)}, \dots, \text{CV}^{(K)})/\sqrt{K}$ , which treats the folds as independent although their

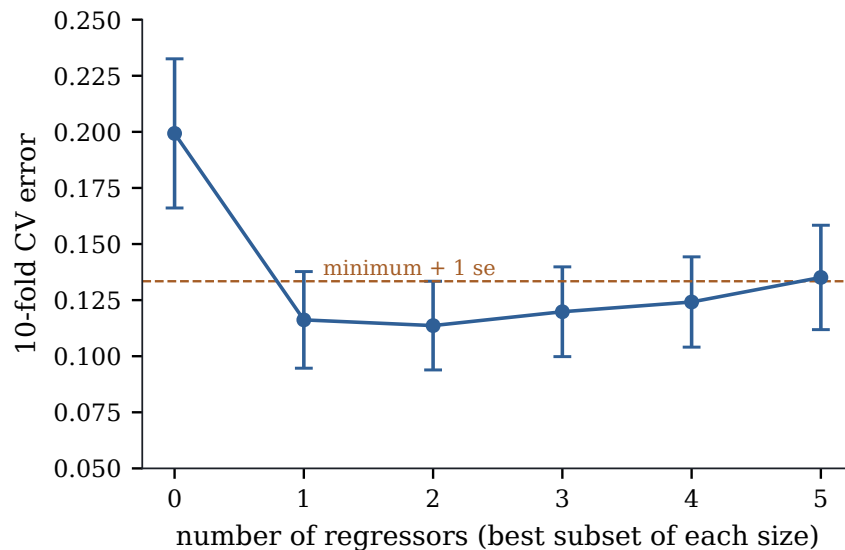


Figure 29.3.2: Ten-fold cross-validation error of the best subset of each size for log violent crime in the 50 states, with one-standard-error bars. Dashed: the minimum plus one standard error.

training sets overlap; no unbiased estimate of the variance of  $CV_K$  exists (Bengio and Grandvalet 2004). The **one-standard-error rule** (Breiman, Friedman, Olshen and Stone 1984) takes the simplest candidate within one standard error of the minimum, trading a little estimated error for a smaller model.

#### Example. (Cross-validating the state regressions)

For each size from 0 to 5, take the best subset of the state regressors by residual sum of squares and estimate its prediction error by ten-fold cross-validation, with one random split into folds of five states. Figure 29.3.2 shows the result. The minimum, 0.1136 with standard error 0.0198, is at two regressors (single parenthood and urbanization), the choice of  $C_p$  and AIC. Single parenthood alone has 0.1162, well within one standard error, so the one-standard-error rule chooses it, the choice of BIC. The full model has 0.1351 and the intercept alone 0.1993. Leave-one-out over all 32 candidates gives 0.1129 for single parenthood and urbanization and 0.1130 for single parenthood alone, a tie, and 0.1184 for poverty with single parenthood and urbanization.

```
import itertools
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
names = ["hs_grad", "poverty", "single", "white", "urban"]
y = np.log(data["violent"].to_numpy())
n = len(y)

def design(cols):
    return np.column_stack([np.ones(n)] + [data[c].to_numpy() for c in cols])

def sse(cols):
    X = design(cols)
    b, *_ = np.linalg.lstsq(X, y, rcond=None)
    return np.sum((y - X @ b) ** 2)
```

```

# best subset of each size, by residual sum of squares
best = [min(itertools.combinations(names, k), key=sse) for k in range(len(names) +
1)]

rng = np.random.default_rng(2906)
K = 10
folds = rng.permutation(np.arange(n) % K)           # a random split into 10
            folds of 5
cv, se = [], []
for cols in best:
    X = design(cols)
    fold_err = []
    for f in range(K):
        tr, te = folds != f, folds == f
        b, *_ = np.linalg.lstsq(X[tr], y[tr], rcond=None)
        fold_err.append(np.mean((y[te] - X[te] @ b) ** 2))
    cv.append(np.mean(fold_err))
    se.append(np.std(fold_err, ddof=1) / np.sqrt(K))    # the usual (optimistic)
            standard error
cv, se = np.array(cv), np.array(se)
k_min = int(np.argmin(cv))
k_1se = int(np.min(np.nonzero(cv <= cv[k_min] + se[k_min])[0]))
for k, cols in enumerate(best):
    print(f"{k} regressors {cols}: CV {cv[k]:.4f} (se {se[k]:.4f})")
print("minimum CV at size", k_min, "; one-standard-error rule chooses size", k_1se)

```

The example has a flaw: the best subset of each size was found from all fifty states, and only its fitting was cross-validated.

### 29.3.4 Cross-validating the whole procedure

Cross-validation estimates the error of whatever is repeated inside the folds. Every decision that looked at the responses, such as screening regressors, choosing a subset, choosing  $\lambda$  or deleting outliers, must be repeated inside each fold. Otherwise the held-out cases have already influenced the fit that predicts them.

#### Example. (Screening, then cross-validating)

Take  $n = 50$  cases, 200 candidate regressors and a response that is independent of all of them, with  $\sigma = 1$ . The procedure keeps the 5 regressors most correlated with the response and fits least squares on them. Every procedure has expected prediction error at least  $\sigma^2 = 1$  here. Over 200 data sets:

- ten-fold cross-validation of the least squares fit, after screening on all the data, averages 0.720, and is below 1 in a fraction 0.93 of the data sets;
- ten-fold cross-validation that repeats the screening inside every fold averages 1.427;
- the actual prediction error of the screened and fitted model averages 1.372.

The wrong version reports that noise predicts the response. The right one is slightly pessimistic, because it uses 45 cases instead of 50 ([Proposition 29.3.1](#)).

```

import numpy as np
rng = np.random.default_rng(2907)
n, m, keep, K = 50, 200, 5, 10

def screen(X, y):
    """Indices of the `keep` columns most correlated with y."""
    Xc, yc = X - X.mean(axis=0), y - y.mean()
    corr = np.abs(Xc.T @ yc) / np.sqrt(np.sum(Xc**2, axis=0) * (yc @ yc))
    return np.argsort(-corr)[:keep]

def fit(X, y):
    A = np.column_stack([np.ones(len(y)), X])
    return np.linalg.lstsq(A, y, rcond=None)[0]

def predict(b, X):
    return b[0] + X @ b[1:]

def one_data_set():
    X = rng.normal(size=(n, m))
    y = rng.normal(size=n)                                # unrelated to every
    column
    folds = rng.permutation(np.arange(n) % K)
    cols = screen(X, y)                                    # screening on ALL the
    data
    wrong = right = 0.0
    for f in range(K):
        tr, te = folds != f, folds == f
        b = fit(X[tr][:, cols], y[tr])                    # wrong: screening done
        outside
        wrong += np.sum((y[te] - predict(b, X[te][:, cols])) ** 2)
        cols_f = screen(X[tr], y[tr])                    # right: screen again
        in the fold
        b = fit(X[tr][:, cols_f], y[tr])
        right += np.sum((y[te] - predict(b, X[te][:, cols_f])) ** 2)
    b = fit(X[:, cols], y)
    truth = 1 + b[0] ** 2 + np.sum(b[1:] ** 2)            # error on a new case
    return wrong / n, right / n, truth

res = np.array([one_data_set() for _ in range(200)])
print("average CV, screening outside the folds:", res[:, 0].mean().round(3))
print("average CV, screening inside the folds: ", res[:, 1].mean().round(3))
print("average true error of the fitted model: ", res[:, 2].mean().round(3))

```

When cross-validation is used both to choose among candidates and to report the error of the choice, the same problem appears in a milder form. The smallest of many cross-validation estimates is biased downwards as an estimate of the error of the model it picks (Section 29.5). The remedy is **nested cross-validation**: an outer loop holds out folds for assessment, and inside each outer training set an inner cross-validation makes the choice (Varma and Simon 2006). The outer estimate then describes the whole procedure, choice included, at sample size  $n - n/K$ .

## 29.3.5 Exercises

### 29.3.5.1 A. Check your understanding

#### Exercise A1.

In [Example \(Bias and spread of K-fold cross-validation\)](#), compute  $\text{Err}(20)$  from [Theorem 14.2.6](#) and confirm the value quoted for  $K = 2$ . Why is a holdout estimate that fits on half the data especially biased when the number of regressors is large?

*Solution.*  $\text{Err}(20) = (1 + 1/20)(18/8) = 1.05 \times 2.25 = 2.3625$ . The factor  $(m - 2)/(m - k - 2)$  blows up as the training size  $m$  approaches  $k + 2$ . With ten regressors, halving  $m$  from 40 to 20 moves it from  $38/28$  to  $18/8$ .

### 29.3.5.2 B. Practice

#### Exercise B1.

For the intercept-only model, show that the leave-one-out residual is  $n(y_i - \bar{y})/(n - 1)$  and that  $\text{CV}_n = ns^2/(n - 1)$ , where  $s^2$  is the sample variance. Check this against [Theorem 29.3.2](#).

*Solution.* Without case  $i$  the mean is  $\bar{y}_{(i)} = (n\bar{y} - y_i)/(n - 1)$ , so  $y_i - \bar{y}_{(i)} = (ny_i - n\bar{y})/(n - 1)$ . Squaring and averaging,  $\text{CV}_n = n^{-1}\{n/(n - 1)\}^2 \sum_i (y_i - \bar{y})^2 = n \sum_i (y_i - \bar{y})^2 / (n - 1)^2 = ns^2/(n - 1)$ . In [Theorem 29.3.2](#),  $s_{ii} = h_{ii} = 1/n$  and  $1/(1 - 1/n) = n/(n - 1)$ .

#### Exercise B2.

In [Theorem 29.3.2](#), suppose  $\mathbf{X}$  has full column rank. Show that  $s_{ii} \leq h_{ii}$ , where  $h_{ii}$  is the least squares leverage, so that the shortcut inflates ridge residuals less than least squares residuals.

*Solution.*  $\mathbf{X}^\top \mathbf{X} + \mathbf{\Omega} - \mathbf{X}^\top \mathbf{X} = \mathbf{\Omega}$  is nonnegative definite, so  $(\mathbf{X}^\top \mathbf{X} + \mathbf{\Omega})^{-1} \leq (\mathbf{X}^\top \mathbf{X})^{-1}$  in the nonnegative definite ordering (for positive definite  $\mathbf{A} \geq \mathbf{B}$ ,  $\mathbf{B}^{-1} - \mathbf{A}^{-1} = \mathbf{B}^{-1/2}(\mathbf{I} - \mathbf{C}^{-1})\mathbf{B}^{-1/2}$  with  $\mathbf{C} = \mathbf{B}^{-1/2}\mathbf{A}\mathbf{B}^{-1/2} \geq \mathbf{I}$ ). Evaluating both quadratic forms at  $\mathbf{x}_{(i)}$  gives  $s_{ii} \leq h_{ii}$ .

### 29.3.5.3 C. Going deeper

#### Exercise C1.

Generalize [Proposition 29.3.1](#) to folds of unequal sizes  $n_1, \dots, n_K$ , and to leave-one-out for a design that is fixed rather than random. In the fixed-design case, what does  $E(\text{CV}_n)$  equal for least squares, and how does it compare with the in-sample prediction error of [Theorem 29.1.1](#)?

**Exercise C2.**

Suppose  $\max_i |h_{ii} - p/n| \leq \epsilon p/n$ . Show that

$$\frac{\text{SSE}}{n} \left(1 - \frac{p}{n}(1 + \epsilon)\right)^{-2} \geq \text{CV}_n \geq \frac{\text{SSE}}{n} \left(1 - \frac{p}{n}(1 - \epsilon)\right)^{-2},$$

and deduce that  $n \log \text{CV}_n = n \log(\text{SSE}/n) + 2p + O(p\epsilon + p^2/n)$ . Which designs satisfy the hypothesis with small  $\epsilon$ ?

## 29.4 All subsets and stepwise search

With  $m$  candidate regressors and the intercept always in, there are  $2^m$  subsets. This section describes exhaustive search, branch and bound and the stepwise shortcuts, and shows that the subset they find depends on small features of the data, far more than the fitted values do.

### 29.4.1 Searching all subsets

The search splits into two stages, because every criterion based on the residual sum of squares agrees within a size.

#### Proposition 29.4.1 — One size, one ranking

Among candidates with the same number of columns  $p$ , the orderings by  $C_p$ , AIC, AICc, BIC,  $R^2$ ,  $\bar{R}^2$  and SSE coincide.

*Proof.* For fixed  $p$ ,  $n$  and  $s^2$ , each of  $C_p$ ,  $n \log(\text{SSE}/n)$  plus a penalty depending on  $p$ ,  $1 - \text{SSE}/\text{SST}$  and  $1 - (n-1)\text{SSE}/\{(n-p)\text{SST}\}$  is a monotone function of SSE, increasing for the criteria to be minimized and decreasing for the two  $R^2$ .

■

So it suffices to find, for each size, the subset with the smallest residual sum of squares, and then compare the  $m + 1$  winners with the criterion. Cross-validation does not reduce in this way, since its value is not a function of SSE and  $p$ ; in practice it is applied to the size winners, as in [Example \(Cross-validating the state regressions\)](#).

Visiting all  $2^m$  subsets is cheap per subset. Sweeping on one pivot ([Definition 10.2.1](#)) adds or removes a column in  $O(m^2)$  operations, and a Gray-code order, in which consecutive subsets differ in one column, gives every residual sum of squares with one sweep (Schatzoff, Tsao and Fienberg 1968). That makes  $m$  up to about 20 routine. Beyond that the number of subsets is the obstacle: finding the best subset of a given size is NP-hard in general (Natarajan 1995).

## 29.4.2 Branch and bound

The key fact is that removing columns never reduces the residual sum of squares: if  $\mathcal{C}(\mathbf{X}_A) \subseteq \mathcal{C}(\mathbf{X}_B)$ , then  $\text{SSE}_A - \text{SSE}_B = \|(\mathbf{M}_B - \mathbf{M}_A)\mathbf{y}\|^2 \geq 0$  by [Theorem 6.5.1](#). A subset whose residual sum of squares is already too large therefore cannot have a good subset below it.

Organize the subsets as a tree. The root is the full set  $\{1, \dots, m\}$ . A node is a subset  $S$  together with the index  $\ell$  of the last column deleted to reach it ( $\ell = 0$  at the root), and its children delete one more column  $j \in S$  with  $j > \ell$ . Deleting in increasing order of index reaches every subset along exactly one path. The descendants of  $(S, \ell)$  are the subsets  $T \subseteq S$  that contain  $F = \{j \in S : j \leq \ell\}$ , the columns that can no longer be deleted.

### Proposition 29.4.2 — Branch and bound for $C_p$

Let  $c^*$  be the smallest  $C_p$  found so far. If a node  $(S, \ell)$  satisfies

$$\frac{\text{SSE}_S}{s^2} - n + 2(|F| + 1) \geq c^*,$$

then no descendant of the node has  $C_p < c^*$ , and the node need not be expanded. A depth-first search that evaluates every expanded node and skips such nodes' descendants returns a subset of minimal  $C_p$ .

*Proof.* A descendant  $T$  contains  $F$  and is contained in  $S$ , so  $\text{SSE}_T \geq \text{SSE}_S$  and  $|T| \geq |F|$ , with the intercept counted separately. Hence its  $C_p = \text{SSE}_T/s^2 - n + 2(|T| + 1)$  is at least the left side of the display. The search only skips subsets that cannot beat a value already attained, so the minimum it reports is the minimum over all subsets. ■

The same bound works for AIC, BIC or the best subset of each size. Furnival and Wilson (1974) combined bounds of this kind with an efficient order of sweeps in the “leaps and bounds” algorithm, which handles 30 to 40 regressors. Bertsimas, King and Mazumder (2016) solved much larger problems by mixed-integer optimization, with a certificate of optimality.

### Example. (How much branch and bound saves)

Take  $n = 100$  cases and  $m = 15$  independent normal regressors, five with nonzero slopes (1, -0.8, 0.6, 0.4, 0.3) and ten with zero slopes. Exhaustive search evaluates 32768 subsets. The branch and bound of [Proposition 29.4.2](#) finds the same  $C_p$ -best subset, with 6 regressors (the four largest slopes and two irrelevant columns), after 1110 evaluations, a fraction 0.034 of the work.

```
import itertools
import numpy as np
rng = np.random.default_rng(2908)
n, m = 100, 15
X = rng.normal(size=(n, m))
beta = np.r_[1.0, -0.8, 0.6, 0.4, 0.3, np.zeros(m - 5)]
y = 2 + X @ beta + rng.normal(size=n)
Xc, yc = X - X.mean(axis=0), y - y.mean()
```

# centring accounts for

```

    the intercept
    G, g = Xc.T @ Xc, Xc.T @ yc

def sse(S):
    S = list(S)
    if not S:
        return yc @ yc
    return yc @ yc - g[S] @ np.linalg.solve(G[np.ix_(S, S)], g[S])

s2 = sse(range(m)) / (n - m - 1)

def cp(S):
    return sse(S) / s2 - n + 2 * (len(S) + 1)

def branch_and_bound():
    """Children delete a column with a larger index than the last one deleted, so
    every
    subset is visited at most once; a node is not expanded when no descendant can
    win."""
    best = [np.inf, None]
    count = [0]

    def visit(S, last):
        e = sse(S)
        count[0] += 1
        value = e / s2 - n + 2 * (len(S) + 1)
        if value < best[0]:
            best[:] = [value, S]
            forced = [j for j in S if j <= last] # kept by every descendant
            if e / s2 - n + 2 * (len(forced) + 1) >= best[0]:
                return # SSE only grows as
        columns go
        for j in S:
            if j > last:
                visit(tuple(k for k in S if k != j), j)

    visit(tuple(range(m)), -1)
    return best[1], best[0], count[0]

S_bb, cp_bb, fits = branch_and_bound()
print("branch and bound:", S_bb, round(cp_bb, 3), "after", fits, "fits of", 2**m)

```

### 29.4.3 Stepwise search

Stepwise methods give up the guarantee of finding the best subset in exchange for examining only  $O(m^2)$  of them.

#### Definition 29.4.1 — Stepwise selection



Let the current subset be  $S$ , with  $p_S$  columns including the intercept.

- (a) **Forward selection** starts from the intercept alone. At each step it adds the column  $j \notin S$  that most reduces SSE, provided its  **$F$ -to-enter**  $F_j = (\text{SSE}_S - \text{SSE}_{S+j}) / \{\text{SSE}_{S+j} / (n - p_S - 1)\}$  exceeds a threshold  $F_{\text{in}}$ , and stops otherwise.
- (b) **Backward elimination** starts from the full model. At each step it removes the column  $j \in S$  with the smallest  **$F$ -to-remove**  $F_j = (\text{SSE}_{S-j} - \text{SSE}_S) / \{\text{SSE}_S / (n - p_S)\}$ , provided this is below a threshold  $F_{\text{out}}$ , and stops otherwise.
- (c) **Stepwise regression** (Efroymson 1960) alternates: after each forward step it removes any column whose  $F$ -to-remove has fallen below  $F_{\text{out}}$ , with  $F_{\text{out}} \leq F_{\text{in}}$ .

The thresholds may be replaced by a criterion: add or remove the column that lowers AIC (or  $C_p$ , or BIC) most, and stop when no move lowers it.

The single step of forward selection can be described in several equivalent ways, all of them versions of the Frisch–Waugh–Lovell theorem.

### Proposition 29.4.3 — What a stepwise step maximizes

Let  $\tilde{\mathbf{y}} = (\mathbf{I} - \mathbf{M}_S)\mathbf{y}$  and  $\tilde{\mathbf{x}}_j = (\mathbf{I} - \mathbf{M}_S)\mathbf{x}_j \neq \mathbf{0}$ , and let  $r_j = \tilde{\mathbf{y}}^\top \tilde{\mathbf{x}}_j / (\|\tilde{\mathbf{y}}\| \|\tilde{\mathbf{x}}_j\|)$  be the partial correlation of  $\mathbf{y}$  and  $\mathbf{x}_j$  given the columns of  $S$  (Definition 14.5.1).

- (a)  $\text{SSE}_S - \text{SSE}_{S+j} = r_j^2 \text{SSE}_S$  and  $F_j = (n - p_S - 1) r_j^2 / (1 - r_j^2)$ . So the column that most reduces SSE is the one with the largest  $r_j^2$ , the largest  $F$ -to-enter, and the smallest  $C_p$ , AIC or BIC of the enlarged model.
- (b) For  $j \in S$ ,  $\text{SSE}_{S-j} - \text{SSE}_S = t_j^2 s_S^2$ , where  $t_j$  is the  $t$  statistic of  $\mathbf{x}_j$  in the model  $S$  and  $s_S^2 = \text{SSE}_S / (n - p_S)$ . So backward elimination removes the column with the smallest  $|t_j|$ , and its  $F$ -to-remove is  $t_j^2$ .

*Proof.*

- (a) By Theorem 6.6.2(b), the residual vector of the model  $S + j$  is the residual of the regression of  $\tilde{\mathbf{y}}$  on  $\tilde{\mathbf{x}}_j$ , so  $\text{SSE}_{S+j} = \|\tilde{\mathbf{y}}\|^2 - (\tilde{\mathbf{x}}_j^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}_j\|^2 = (1 - r_j^2) \text{SSE}_S$ . Then  $F_j = (n - p_S - 1) r_j^2 / (1 - r_j^2)$  is increasing in  $r_j^2$ . At a fixed size the criteria rank by SSE (Proposition 29.4.1).
- (b) Apply (a) to the model  $S - j$  and the column  $\mathbf{x}_j$ : the increase is  $(\tilde{\mathbf{x}}_j^\top \tilde{\mathbf{y}})^2 / \|\tilde{\mathbf{x}}_j\|^2$  with residuals now taken on  $S - j$ . By Theorem 6.6.2(a) this is  $\hat{\beta}_j^2 \|\tilde{\mathbf{x}}_j\|^2$ , and by Eq. 6.6.2  $\text{Var}(\hat{\beta}_j) = \sigma^2 / \|\tilde{\mathbf{x}}_j\|^2$ , so the increase is  $\hat{\beta}_j^2 s_S^2 / \widehat{\text{Var}}(\hat{\beta}_j) = t_j^2 s_S^2$ . ■

Criterion-based stopping is therefore threshold-based stopping with a particular threshold: forward selection by AIC adds a column iff its  $F$ -to-enter exceeds about 2, and by BIC iff it exceeds about  $\log n$  (the table in Section 29.2). The traditional thresholds  $F_{\text{in}} = F_{\text{out}} = 4$  correspond to BIC at  $n \approx 55$ .

For the state data, forward selection by AIC adds single parenthood and then urbanization and stops. Backward elimination by AIC removes white, high school, poverty (in that order) and stops at the same two regressors, which are also the all-subsets AIC choice. Forward

and backward computed the criterion for 13 and 15 subsets, against 32 for exhaustive search. Agreement is common but not guaranteed.

**Example. (A pair that forward selection cannot find)**

Take  $n = 60$  cases of three regressors,  $\mathbf{x}_1$  and  $\mathbf{x}_2$  with sample correlation 0.96 and  $\mathbf{x}_3$  independent of both, and let  $\mathbf{y} = 3(\mathbf{x}_1 - \mathbf{x}_2) + 0.4\mathbf{x}_3 + \boldsymbol{\varepsilon}$ . The signal lies mostly in the difference  $\mathbf{x}_1 - \mathbf{x}_2$ , a direction in which the regressors vary little. The sample correlations of  $\mathbf{y}$  with  $\mathbf{x}_1$ ,  $\mathbf{x}_2$  and  $\mathbf{x}_3$  are  $-0.02$ ,  $-0.19$  and  $0.41$ . Forward selection enters  $\mathbf{x}_3$  first and then  $\mathbf{x}_2$ , reaching  $R^2 = 0.19$ . The best pair is  $\{\mathbf{x}_1, \mathbf{x}_2\}$ , with  $R^2 = 0.39$ . Neither of its members looks useful alone, so a procedure that adds one column at a time never sees it. Backward elimination starts from a model that contains the pair and can keep it, but it has blind spots of its own ([Exercise \(C1\)](#)) and cannot start when  $m \geq n$ .

```
import itertools
import numpy as np
rng = np.random.default_rng(2909)
n2 = 60
z = rng.normal(size=(n2, 3))
x1 = z[:, 0]
x2 = 0.95 * z[:, 0] + np.sqrt(1 - 0.95**2) * z[:, 1]      # corr(x1, x2) about 0.95
x3 = z[:, 2]
Xf = np.column_stack([x1, x2, x3])
yf = 3 * (x1 - x2) + 0.4 * x3 + rng.normal(size=n2)      # the signal is in x1 - x2

def rss(cols):
    A = np.column_stack([np.ones(n2)] + [Xf[:, j] for j in cols])
    b, *_ = np.linalg.lstsq(A, yf, rcond=None)
    return np.sum((yf - A @ b) ** 2)

first = min(range(3), key=lambda j: rss([j]))
second = min((j for j in range(3) if j != first), key=lambda j: rss([first, j]))
best_pair = min(itertools.combinations(range(3), 2), key=rss)
print("forward chooses", sorted([first, second]), "RSS", round(rss([first, second]),
1))
print("best pair", list(best_pair), "RSS", round(rss(best_pair), 1))
```

Collinear regressors are exactly where this happens: a contrast between them can be highly predictive while each member is not ([Chapter 26](#) describes such directions through the eigenvectors of  $\mathbf{X}^\top \mathbf{X}$ ).

## 29.4.4 Comparing the procedures

**Example. (Selection procedures in a simulation)**

Take  $n = 60$  cases of eight normal regressors with all pairwise correlations 0.5,  $\sigma = 1$ , an intercept, and three nonzero slopes, either weak (1, 0.5, 0.25) or strong (1, 0.8, 0.6). Each procedure is scored by its loss at the design points,  $\|\mathbf{X}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})\|^2 / \sigma^2$ , with the intercept's expected contribution 1 included. Over 2000 data sets:

*Weak slopes:*

| procedure                  | loss | Pr(true) | size |
|----------------------------|------|----------|------|
| least squares, all eight   | 8.98 | 0.00     | 8.00 |
| true subset (oracle)       | 3.97 | 1.00     | 3.00 |
| all subsets, AIC           | 7.81 | 0.18     | 3.42 |
| all subsets, BIC           | 7.33 | 0.18     | 2.52 |
| all subsets, leave-one-out | 7.76 | 0.19     | 3.34 |
| forward, AIC               | 7.70 | 0.19     | 3.34 |
| backward, AIC              | 7.89 | 0.17     | 3.50 |

*Strong slopes:*

| procedure                  | loss | Pr(true) | size |
|----------------------------|------|----------|------|
| least squares, all eight   | 9.02 | 0.00     | 8.00 |
| true subset (oracle)       | 4.06 | 1.00     | 3.00 |
| all subsets, AIC           | 7.27 | 0.37     | 3.93 |
| all subsets, BIC           | 6.17 | 0.70     | 3.20 |
| all subsets, leave-one-out | 7.17 | 0.39     | 3.85 |
| forward, AIC               | 7.16 | 0.39     | 3.86 |
| backward, AIC              | 7.35 | 0.35     | 3.98 |

Every selection rule beats least squares on all eight regressors, whose expected loss is 9 ([Theorem 29.1.1\(c\)](#)), but none comes close to the oracle's 4. The search method hardly matters: forward selection by AIC returned the all-subsets AIC choice in a fraction 0.95 of the weak data sets and 0.96 of the strong ones. The criterion matters more: with strong effects BIC finds the true subset in a fraction 0.70 of data sets and has the smallest loss, while with a weak third slope every rule usually misses it, BIC keeping 2.52 regressors on average.

The simulation illustrates [Proposition 29.2.4](#) in a finite sample: where a small true model exists and its effects are large, BIC wins; with many small effects the comparison tilts towards AIC and cross-validation.

## 29.4.5 Instability

A procedure is **unstable** if small changes in the data change its output a great deal. Subset selection is a discontinuous function of the data: a column is in or out. Breiman (1996b) showed that this instability inflates the variance of the predictions that follow, and that averaging over perturbed data sets can reduce it.

### Example. (Bootstrapping the state regressions)

Resample the 50 states with replacement (the case bootstrap of [Section 23.1](#)) 1000 times and repeat the all-subsets AIC and BIC choices on each resample. AIC chooses 27 different subsets of the 32 and BIC 25. The AIC choice for the original data, single parenthood and urbanization, is chosen in a fraction 0.15 of resamples, and the most

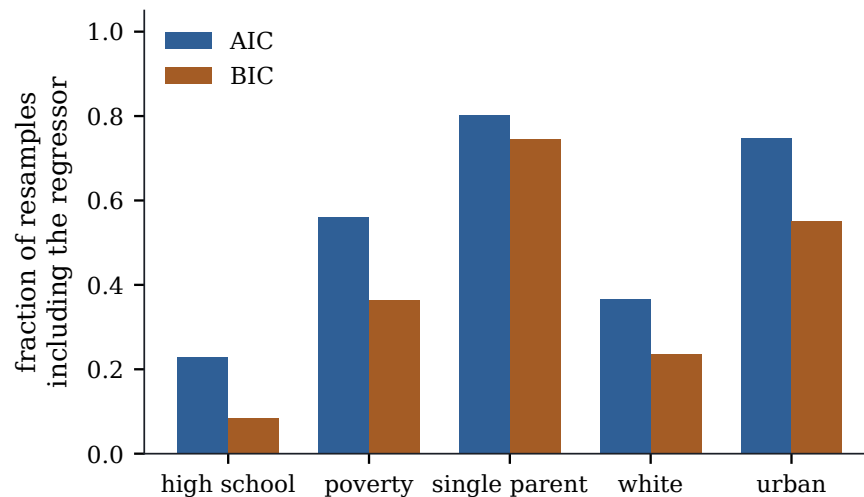


Figure 29.4.1: Fraction of 1000 bootstrap resamples of the 50 states in which each regressor is in the subset chosen by AIC or BIC (log violent crime, all 32 subsets).

frequent AIC choice, poverty with single parenthood and urbanization, in 0.20. The BIC choice, single parenthood alone, is its most frequent choice, at 0.31. Figure 29.4.1 shows how often each regressor is included. Even single parenthood, whose  $t$  statistic in the chosen model is about 5.5, is left out by AIC in a fifth of the resamples. It is correlated  $-0.70$  with the percentage of white residents, and the two can stand in for each other.

```
import itertools
from collections import Counter
import numpy as np
import statsmodels.api as sm
data = sm.datasets.statelcrime.load_pandas().data.drop(index="District of Columbia")
names = ["hs_grad", "poverty", "single", "white", "urban"]
y_all = np.log(data["violent"].to_numpy())
Z_all = data[names].to_numpy()
n = len(y_all)
subsets = [S for k in range(6) for S in itertools.combinations(range(5), k)]

def select(Z, y, penalty):
    def crit(S):
        X = np.column_stack([np.ones(len(y)), Z[:, list(S)]])
        b, *_ = np.linalg.lstsq(X, y, rcond=None)
        return n * np.log(np.sum((y - X @ b) ** 2) / n) + penalty * (len(S) + 2)
    return min(subsets, key=crit)

rng = np.random.default_rng(2911)
B = 200
uses 1000
chosen = {"AIC": Counter(), "BIC": Counter()}
for _ in range(B):
    idx = rng.integers(0, n, size=n)
    states
    chosen["AIC"][select(Z_all[idx], y_all[idx], 2.0)] += 1
    chosen["BIC"][select(Z_all[idx], y_all[idx], np.log(n))] += 1
for crit, counts in chosen.items():
    # the book's figure
    # resample whole
```

```
print(crit, "distinct models:", len(counts))
for S, c in counts.most_common(3):
    print("    ", [names[j] for j in S], c / B)
```

Instability is a property of the question as much as of the method. The three models favoured by the criteria and the bootstrap (single parenthood alone, with urbanization, and with poverty and urbanization) predict almost equally well (their leave-one-out errors in [Example \(Cross-validating the state regressions\)](#) lie between 0.1129 and 0.1184), so any rule that must name one will name different ones for similar data. The remedies are to report the uncertainty, as in the figure, or to stop requiring a single model: average over bootstrap resamples (bagging, Breiman 1996a) or over models weighted by posterior probability (Hoeting, Madigan, Raftery and Volinsky 1999), or shrink instead of selecting ([Chapter 27](#)).

## 29.4.6 Exercises

### 29.4.6.1 A. Check your understanding

#### Exercise A1.

How many subsets must exhaustive search examine for  $m = 20$  and  $m = 40$  candidate regressors? At a million subsets per second, how long does each take?

### 29.4.6.2 B. Practice

#### Exercise B1.

Show that in the tree of [Proposition 29.4.2](#) every subset of  $\{1, \dots, m\}$  is reached along exactly one path from the root, and that the descendants of  $(S, \ell)$  are exactly the subsets  $T$  with  $F \subseteq T \subseteq S$ .

*Solution.* A subset  $T$  with complement  $\{j_1 < \dots < j_r\}$  in  $\{1, \dots, m\}$  is reached by deleting  $j_1, \dots, j_r$  in increasing order, and every path deletes in increasing order, so this is the only path. A descendant of  $(S, \ell)$  deletes further columns of  $S$  with indices above  $\ell$ , so it contains  $F = \{j \in S : j \leq \ell\}$ ; conversely any  $T$  with  $F \subseteq T \subseteq S$  is reached from  $(S, \ell)$  by deleting the columns of  $S \setminus T$ , all of which exceed  $\ell$ , in increasing order.

#### Exercise B2.

In stepwise regression, show that the column just entered has  $F$ -to-remove equal to its  $F$ -to-enter. Deduce that it cannot be removed at the next step if  $F_{\text{out}} \leq F_{\text{in}}$ , and explain what can happen if  $F_{\text{out}} > F_{\text{in}}$ .

*Solution.* After  $j$  enters, the model is  $S + j$  with  $p_S + 1$  columns, and its  $F$ -to-remove is  $(\text{SSE}_S - \text{SSE}_{S+j}) / \{\text{SSE}_{S+j} / (n - p_S - 1)\}$ , which is the  $F$ -to-enter of  $j$  at the previous step. It exceeded  $F_{\text{in}} \geq F_{\text{out}}$ , so  $j$  stays. If  $F_{\text{out}} > F_{\text{in}}$ , a column with  $F_{\text{in}} < F_j < F_{\text{out}}$  is added and immediately removed, returning to  $S$ , where it is added again: the procedure cycles.

**Exercise B3.**

Show that a column's  $F$ -to-remove in backward elimination is the square of its  $t$  statistic, and that backward elimination with  $F_{\text{out}} = t_{\alpha/2}^2(\nu)$  is the rule “drop the least significant coefficient while any is not significant at level  $\alpha$ ”. Why do the  $p$ -values of the surviving coefficients not have their nominal meaning?

**29.4.6.3 C. Going deeper****Exercise C1.**

In [Example \(A pair that forward selection cannot find\)](#), show that the best subset of size one is  $\{\mathbf{x}_3\}$  and the best of size two is  $\{\mathbf{x}_1, \mathbf{x}_2\}$ , so the best subsets of consecutive sizes need not be nested. Why does this not contradict [Theorem 6.5.1](#)?

*Solution.* Among single columns the residual sum of squares is smallest for the largest squared correlation with  $\mathbf{y}$ , and  $0.41^2$  is far above  $0.19^2$  and  $0.02^2$ . The best pair was computed in the example. [Theorem 6.5.1](#) compares a subset with its own supersets, so  $\text{SSE}_{\{3\}} \geq \text{SSE}_{\{1,3\}}$  and  $\text{SSE}_{\{3\}} \geq \text{SSE}_{\{2,3\}}$ ; it says nothing about  $\{1, 2\}$ , which does not contain  $\{3\}$ . Because best subsets need not be nested, a search that only adds columns (forward) or only removes them (backward) can miss the best subset of some size.

## 29.5 Selection bias and honest inference

Everything in Part III assumed that the model was fixed before the data were seen. After a search, the reported model, its coefficients, its error estimate and its test statistics are the ones that looked best, and each is biased: effects look larger, errors smaller, and intervals cover less often than they claim. This section proves the direction of the bias, measures it, and describes three remedies.

### 29.5.1 The winner's curse

#### **Proposition 29.5.1** — What selection does to estimates and intervals

- (a) (*Selected maxima are biased upwards.*) Let  $Z_1, \dots, Z_m$  have finite means  $\theta_1, \dots, \theta_m$ , and let  $\hat{J}$  be an index at which  $Z_{\hat{J}} = \max_j Z_j$ . Then

$$E(Z_{\hat{J}}) \geq \max_j \theta_j \geq E(\theta_{\hat{J}}),$$

so  $E(Z_{\hat{J}} - \theta_{\hat{J}}) \geq 0$ . If the  $Z_j$  are independent  $N(\theta, 1)$  with a common mean and  $m \geq 2$ , the bias is  $E \max_j Z_j - \theta > 0$ .

- (b) (*Selected minima of risk estimates are optimistic.*) Let  $\hat{R}_k$  be an unbiased estimate

of  $R_k$  for each candidate  $k$  in a finite set, and let  $\hat{k}$  minimize  $\hat{R}_k$ . Then

$$E(\hat{R}_{\hat{k}}) \leq \min_k R_k \leq E(R_{\hat{k}}).$$

- (c) (*Selected intervals under-cover.*) Let  $Z \sim N(\theta, 1)$ , and report the interval  $Z \pm z_{\alpha/2}$  only when  $|Z| > c$ . If  $c \geq z_{\alpha/2}$ , the probability that the reported interval covers  $\theta$ , given that it is reported, is 0 at  $\theta = 0$ . It tends to  $1 - \alpha$  as  $|\theta| \rightarrow \infty$ .

*Proof.*

- (a)  $\max_j Z_j \geq Z_i$  for every  $i$ , so  $E \max_j Z_j \geq \theta_i$  for every  $i$ , which is the first inequality. The second holds pointwise:  $\theta_j \leq \max_j \theta_j$ . In the normal case,  $\max_j Z_j - \theta = \max_j (Z_j - \theta)$  has the distribution of the largest of  $m \geq 2$  independent standard normals, which exceeds  $Z_1 - \theta$  with positive probability and is never smaller, so its mean exceeds  $E(Z_1 - \theta) = 0$ .
- (b)  $\hat{R}_{\hat{k}} \leq \hat{R}_k$  for every  $k$ ; take expectations to get  $E \hat{R}_{\hat{k}} \leq R_k$  for every  $k$ . The second inequality holds pointwise.
- (c) At  $\theta = 0$  the interval covers iff  $|Z| \leq z_{\alpha/2}$ , which is incompatible with  $|Z| > c \geq z_{\alpha/2}$ . As  $|\theta| \rightarrow \infty$ ,  $\Pr(|Z| > c) \rightarrow 1$ , so the conditional coverage tends to the unconditional  $\Pr(|Z - \theta| \leq z_{\alpha/2}) = 1 - \alpha$ .

■

Part (a) is the **winner's curse**: the largest of ten unbiased estimates of equal means overstates its mean by 1.539 standard errors on average, and the coefficient and  $t$  statistic of a regressor that “won” a screening are biased in the same way. Part (b) applies to every criterion of this chapter, since  $C_p$  and cross-validation are unbiased for each fixed candidate ([Proposition 29.2.1\(a\)](#) and [Proposition 29.3.1](#)); the search degrees of freedom of [Section 29.1](#) measure the same effect.

### Example. (Freedman's paradox)

Freedman (1983) described the following procedure, which many analyses follow in some form. Fit  $n = 100$  cases on 50 regressors, none of which is related to the response. Keep those with  $p$ -value below 0.25, refit on the survivors, and report. In one simulated data set 15 regressors survive the first pass, 7 of them are significant at 5% in the refit, and the overall  $F$  test of the refit has  $p$ -value 0.0030. Over 2000 data sets, an average of 12.3 regressors survive and 5.6 are significant, and the overall  $F$  test rejects at 5% in a fraction 0.94 of them. Pure noise produces a model that every standard test endorses.

```
import numpy as np
from scipy import stats
rng = np.random.default_rng(2913)
n, m = 100, 50

def ols(X, y):
    """Coefficients, t statistics (slopes), overall F p-value; X without intercept.
    """
    A = np.column_stack([np.ones(len(y)), X])
    b, *_ = np.linalg.lstsq(A, y, rcond=None)
    e = y - A @ b
```

```

df = len(y) - A.shape[1]
s2 = e @ e / df
t = b / np.sqrt(s2 * np.diag(np.linalg.inv(A.T @ A)))
ssr = np.sum((A @ b - y.mean()) ** 2)
F = (ssr / X.shape[1]) / s2
return t[1:], df, stats.f.sf(F, X.shape[1], df)

def screen_and_refit():
    X = rng.normal(size=(n, m))
    y = rng.normal(size=n)
    t, df, _ = ols(X, y)
    keep = np.abs(t) > stats.t.ppf(1 - 0.25 / 2, df)
    t2, df2, p_F = ols(X[:, keep], y)
    survivors
    return keep.sum(), np.sum(np.abs(t2) > stats.t.ppf(0.975, df2)), p_F

kept, signif, p_F = screen_and_refit()
print(f"one data set: {kept} kept, {signif} significant at 5%, overall F p-value {
    p_F:.4f}")
runs = np.array([screen_and_refit() for _ in range(500)])
print("over 500 data sets: mean kept", runs[:, 0].mean(), " mean significant", runs
   [:, 1].mean(),
    " Pr(F significant)", np.mean(runs[:, 2] < 0.05))

```

## 29.5.2 Coverage after selection

The failure in [Proposition 29.5.1\(c\)](#) concerns intervals reported only after selection. A subtler failure affects intervals for a coefficient that is always reported, when the *model* around it was chosen from the data.

### Example. (Coverage after a pretest)

Take two standardized regressors with correlation  $r$ , known  $\sigma$ , and the question: what is  $\beta_1$ ? The analyst first tests  $\beta_2 = 0$  at level 5%. If the test rejects, she reports the usual 95% interval for  $\beta_1$  from the model with both regressors; otherwise, the interval from the model with  $\mathbf{x}_1$  alone. Each interval is correct for its own model. Panel (b) of [Figure 29.5.1](#) shows the coverage of the reported interval as a function of  $\delta = \beta_2 / \text{sd}(\hat{\beta}_2)$ , the standardized size of the coefficient being tested. When  $\delta = 0$  the short model is correct and coverage is close to nominal. When  $\delta$  is large the test always keeps  $\mathbf{x}_2$  and coverage is nominal. In between, the test often drops  $\mathbf{x}_2$  although the short slope is biased by  $r\beta_2$  ([Theorem 19.1.2](#)), and the coverage falls to a minimum of 0.873 (at  $\delta = 2.2$ ) for  $r = 0.5$ , 0.599 (at  $\delta = 1.8$ ) for  $r = 0.8$  and 0.234 (at  $\delta = 1.2$ ) for  $r = 0.95$ .

Panel (a) is the one-dimensional case of [Proposition 29.5.1\(c\)](#) with  $c = 1.96$ : the conditional coverage is 0 at  $\theta = 0$ , 0.844 at  $\theta = 1$ , 0.951 at  $\theta = 2$  and 0.971 at  $\theta = 3$ , where it slightly over-covers.

```

z = stats.norm.ppf(0.975)

def conditional_coverage(theta, c=z):

```



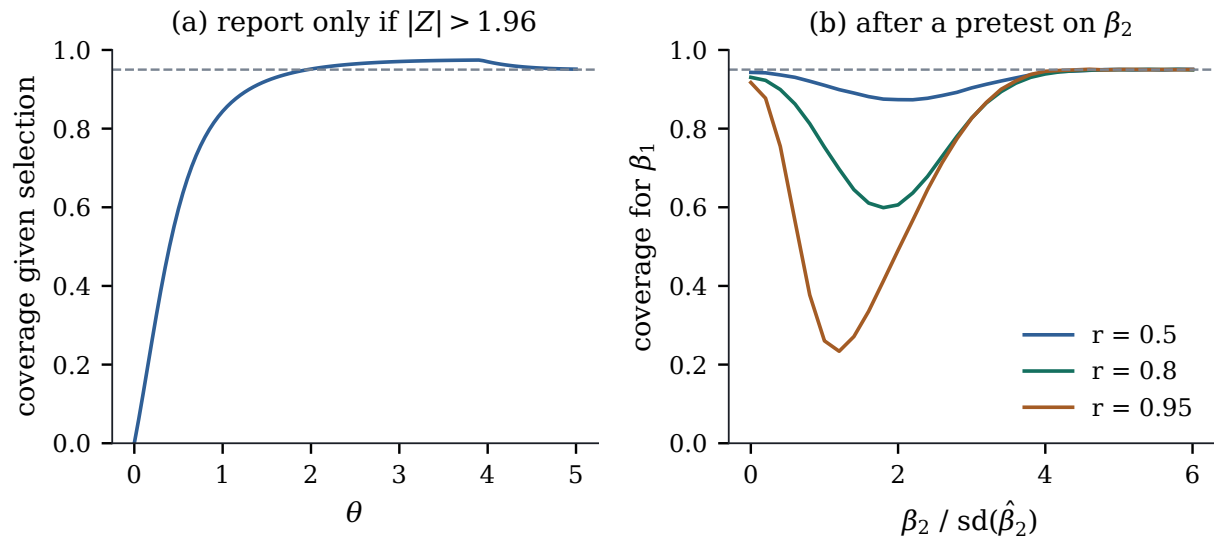


Figure 29.5.1: (a) Coverage of  $Z \pm 1.96$ ,  $Z \sim N(\theta, 1)$ , given that it is reported only when  $|Z| > 1.96$ . (b) Coverage for  $\beta_1$  of the usual 95% interval from the model chosen by a 5% test of  $\beta_2 = 0$ , for three correlations between the regressors (200000 simulations per point).

```

"""Pr(|Z - theta| <= z | |Z| > c) for Z ~ N(theta, 1): the reported interval."""
lo, hi = theta - z, theta + z
inside_sel = (stats.norm.cdf(max(hi, c) - theta) - stats.norm.cdf(max(lo, c) -
theta)) \
    + (stats.norm.cdf(min(hi, -c) - theta) - stats.norm.cdf(min(lo, -c) - theta)
)
return inside_sel / (stats.norm.sf(c - theta) + stats.norm.cdf(-c - theta))

def pretest_coverage(delta, r, reps=200_000):
    """Coverage of the usual 95% interval for beta_1 after a 5% pretest of beta_2 =
0.
Two standardized regressors with correlation r, sigma known; delta = beta_2 / sd
(beta_2 hat)."""
    sd2 = 1 / np.sqrt(1 - r**2) # sd of either long-
    model slope
    C = np.array([[1, -r], [-r, 1]]) / (1 - r**2) # covariance of (b1, b2)
    in the long model
    b = rng.normal(size=(reps, 2)) @ np.linalg.cholesky(C).T + [0.0, delta * sd2]
    keep = np.abs(b[:, 1] / sd2) > z # the pretest keeps x2
    b1_short = b[:, 0] + r * b[:, 1] # short slope (sd 1)
    cover_long = np.abs(b[:, 0]) <= z * sd2 # true beta_1 = 0
    cover_short = np.abs(b1_short) <= z
    return np.mean(np.where(keep, cover_long, cover_short))

for r, d_min in ((0.5, 2.2), (0.8, 1.8), (0.95, 1.2)): # the minimizing
    delta of the figure
    print(f"r = {r}: coverage at delta = 0, {d_min}, 5:",
        ", ".join(f"{pretest_coverage(d, r, 50_000):.3f}" for d in (0.0, d_min,
        5.0)))

```

The coverage depends on the unknown  $\delta$ , and it is worst where the data are least able to tell whether  $\mathbf{x}_2$  belongs. No simple correction repairs this: Leeb and Pötscher (2005, 2006) showed that the distribution of a post-selection estimator cannot be estimated consistently, uniformly in

the parameters, so bootstrapping the whole procedure does not rescue the naive interval either.

### 29.5.3 Honest inference, I: splitting the sample

The simplest remedy separates the two uses of the data. Choose the model on one part of the sample, and compute intervals from the other part, as if the model had been fixed in advance, which, for that part, it was (Cox 1975).

#### Proposition 29.5.2 — Sample splitting

Let  $\mathbf{Y} = (\mathbf{Y}_1^\top, \mathbf{Y}_2^\top)^\top$ , where  $\mathbf{Y}_2 \sim N_{n_2}(\boldsymbol{\mu}_2, \sigma^2 \mathbf{I})$  is independent of  $\mathbf{Y}_1$ , and let  $\mathbf{X}_2$  be the fixed design of the second part. Let  $\hat{M}$  be any rule that chooses a set of columns from  $\mathbf{Y}_1$  (and the design), with values in the set of  $M$  for which  $\mathbf{X}_{2,M}$  has full column rank  $p_M < n_2$ . For each such  $M$ , let  $\boldsymbol{\beta}_M = (\mathbf{X}_{2,M}^\top \mathbf{X}_{2,M})^{-1} \mathbf{X}_{2,M}^\top \boldsymbol{\mu}_2$ , and let  $I_{M,j}$  be the usual level  $1 - \alpha$   $t$  interval for the  $j$ th coefficient from the least squares fit of  $\mathbf{Y}_2$  on  $\mathbf{X}_{2,M}$ . Then, given  $\mathbf{Y}_1$ ,  $I_{\hat{M},j}$  covers  $\beta_{\hat{M},j}$  with probability at least  $1 - \alpha$ , with equality if  $\boldsymbol{\mu}_2 \in \mathcal{C}(\mathbf{X}_{2,\hat{M}})$ . The same holds unconditionally.

*Proof.* Given  $\mathbf{Y}_1$ , the model  $M = \hat{M}$  is fixed and  $\mathbf{Y}_2$  still has its distribution, by independence. For fixed  $M$ , write  $\mathbf{M}_M$  for the projection onto  $\mathcal{C}(\mathbf{X}_{2,M})$  and  $w = [(\mathbf{X}_{2,M}^\top \mathbf{X}_{2,M})^{-1}]_{jj}$ . The estimate  $\hat{\beta}_{M,j}$  is normal with mean  $\beta_{M,j}$  and variance  $\sigma^2 w$ , so  $Z = (\hat{\beta}_{M,j} - \beta_{M,j})/(\sigma\sqrt{w}) \sim N(0, 1)$ . The residual sum of squares  $\|(\mathbf{I} - \mathbf{M}_M)\mathbf{Y}_2\|^2$  is a function of  $(\mathbf{I} - \mathbf{M}_M)\mathbf{Y}_2$ , which is uncorrelated with  $\mathbf{M}_M\mathbf{Y}_2$  and hence independent of it (Theorem 3.4.1), and  $W = \|(\mathbf{I} - \mathbf{M}_M)\mathbf{Y}_2\|^2/\sigma^2 \sim \chi^2(\nu, \gamma)$  with  $\nu = n_2 - p_M$  and  $\gamma = \|(\mathbf{I} - \mathbf{M}_M)\boldsymbol{\mu}_2\|^2/\sigma^2$  (Theorem 4.3.2(a)). The interval covers iff  $|Z| \leq t_{\alpha/2}(\nu)\sqrt{W/\nu}$ , iff  $W > \nu Z^2/t_{\alpha/2}(\nu)^2$ . Conditioning on  $Z$  and using Proposition 4.1.3, this probability is nondecreasing in  $\gamma$ , and at  $\gamma = 0$  it is  $1 - \alpha$  by the definition of the  $t$  distribution. Averaging over  $\mathbf{Y}_1$  gives the unconditional statement. ■

The target  $\beta_{M,j}$  is the coefficient of  $\mathbf{x}_j$  in the best approximation to the mean by the chosen columns, which is what the fitted coefficient estimates, right model or wrong. The costs: selection sees only part of the data, the intervals are wider, and the result depends on the random split. Wasserman and Roeder (2009) and Rinaldo, Wasserman and G'Sell (2019) develop the idea for high-dimensional and assumption-lean settings.

### 29.5.4 Honest inference, II: covering every model

The second remedy keeps all the data and widens the intervals until they cover the coefficients of every model that could have been chosen, simultaneously. Then it does not matter which one was.

#### Proposition 29.5.3 — Simultaneous inference over submodels

Assume the normal model Eq. 7.3.1 with  $\text{rank}(\mathbf{X}) = P < n$ , let  $s^2$  be the full-model

estimate with  $\nu = n - P$ , and let  $\mathcal{M}$  be a collection of column subsets  $M$  with  $\mathbf{X}_M$  of full column rank. For  $M \in \mathcal{M}$  and  $j \in M$  let  $\mathbf{v}_{M,j} = \mathbf{X}_M(\mathbf{X}_M^\top \mathbf{X}_M)^{-1} \mathbf{e}_j$ , where  $\mathbf{e}_j$  picks out the coefficient of  $\mathbf{x}_j$ , and

$$\beta_{M,j} = \mathbf{v}_{M,j}^\top \mathbf{X} \boldsymbol{\beta}, \quad \hat{\beta}_{M,j} = \mathbf{v}_{M,j}^\top \mathbf{Y},$$

and for  $K > 0$  let

$$I_{M,j}(K) = [\hat{\beta}_{M,j} \pm K s \|\mathbf{v}_{M,j}\|].$$

Let  $N$  be the number of pairs  $(M, j)$  and  $q$  the dimension of the span of the  $\mathbf{v}_{M,j}$ .

- (a) With  $K = \sqrt{q F_\alpha(q, \nu)}$ ,  $\Pr(\beta_{M,j} \in I_{M,j}(K) \text{ for all } M \in \mathcal{M}, j \in M) \geq 1 - \alpha$ .
- (b) The same holds with  $K = t_{\alpha/(2N)}(\nu)$ .
- (c) If  $K$  has the property in (a), then for every selection rule  $\hat{M}$  with values in  $\mathcal{M}$ , a function of  $\mathbf{Y}$  that may also depend on the unknown  $\boldsymbol{\beta}$  and  $\sigma$ ,  $\Pr(\beta_{\hat{M},j} \in I_{\hat{M},j}(K) \text{ for all } j \in \hat{M}) \geq 1 - \alpha$ .
- (d) The smallest  $K$  with the property in (c) for every such rule, including rules that depend on the unknown parameters (the convention of Berk, Brown, Buja, Zhang and Zhao 2013), is  $K_{\text{PoSI}}$ , the  $1 - \alpha$  quantile of  $\max_{M,j} |\mathbf{v}_{M,j}^\top (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})| / (s \|\mathbf{v}_{M,j}\|)$ , whose distribution does not depend on  $\boldsymbol{\beta}$  or  $\sigma$ .

*Proof.* In each case  $\hat{\beta}_{M,j} - \beta_{M,j} = \mathbf{v}_{M,j}^\top (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})$  with  $\mathbf{v}_{M,j} \in \mathcal{C}(\mathbf{X})$ . (a) Let  $\mathcal{V}$  be the span of the  $\mathbf{v}_{M,j}$ , a subspace of  $\mathcal{C}(\mathbf{X})$  of dimension  $q$ . By Theorem 13.2.2(a), with probability exactly  $1 - \alpha$ ,  $|\mathbf{v}^\top (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})| \leq \sqrt{q F_\alpha(q, \nu)} s \|\mathbf{v}\|$  for every  $\mathbf{v} \in \mathcal{V}$ , in particular for every  $\mathbf{v}_{M,j}$ . (b) Each  $(\hat{\beta}_{M,j} - \beta_{M,j}) / (s \|\mathbf{v}_{M,j}\|)$  has the  $t(\nu)$  distribution, because  $\mathbf{v}_{M,j}^\top \mathbf{Y}$  is independent of  $s^2$  when  $\mathbf{v}_{M,j} \in \mathcal{C}(\mathbf{X})$  (Theorem 7.5.1). Apply Theorem 13.4.1(b) with  $\alpha_{M,j} = \alpha/N$ . (c) The event in (c) contains the simultaneous event of (a). (d) The simultaneous event of (a) is  $\{\max_{M,j} T_{M,j} \leq K\}$  with  $T_{M,j} = |\mathbf{v}_{M,j}^\top (\mathbf{Y} - \mathbf{X} \boldsymbol{\beta})| / (s \|\mathbf{v}_{M,j}\|)$ . Its distribution is free of  $\boldsymbol{\beta}$  and  $\sigma$  because  $\mathbf{Y} - \mathbf{X} \boldsymbol{\beta} = \sigma \mathbf{z}$  with  $\mathbf{z} \sim N_n(\mathbf{0}, \mathbf{I})$  and  $s = \sigma \|(\mathbf{I} - \mathbf{M})\mathbf{z}\| / \sqrt{\nu}$ . So  $K_{\text{PoSI}}$  has the property in (a) and hence in (c). If  $K < K_{\text{PoSI}}$ , take the rule that selects a model containing a pair at which  $T_{M,j}$  is largest, a rule that uses  $\boldsymbol{\beta}$  and so is allowed in (c); its coverage is  $\Pr(\max T_{M,j} \leq K) < 1 - \alpha$ . ■

This is the PoSI (post-selection inference) approach of Berk, Brown, Buja, Zhang and Zhao (2013): model selection treated as a multiple-comparison problem of Chapter 13. The Scheffé bound never exceeds  $\sqrt{P F_\alpha(P, \nu)}$ , however many models are searched, and the price is paid whether or not the search was aggressive.

#### Example. (Honest intervals for the state regressions)

For the state data, with the intercept always in, the 32 candidate models contain  $N = 80$  slope coefficients. Every  $\mathbf{v}_{M,j}$  is orthogonal to  $\mathbf{1}$ , so they span at most  $q = 5$  dimensions, and  $\nu = 44$ . The multipliers are 2.015 for the naive  $t$  interval, 3.12 for PoSI (by simulation), 3.484 for Scheffé and 3.684 for Bonferroni. In the model chosen by AIC, single parenthood and urbanization, the  $t$  statistics computed with the full-model  $s$  are 5.46 and 1.83. Single parenthood clears every multiplier. Urbanization clears none, not

even the naive one: AIC kept it because its squared  $t$  exceeds about 2 (the threshold of Section 29.2), which is a decision about prediction, not evidence of an effect.

```
import itertools
import statsmodels.api as sm
rng = np.random.default_rng(2914)
data = sm.datasets.statcrime.load_pandas().data.drop(index="District of Columbia")
names = ["hs_grad", "poverty", "single", "white", "urban"]
ys = np.log(data["violent"].to_numpy())
Xs = np.column_stack([np.ones(len(ys)), data[names].to_numpy()])
ns, P = Xs.shape
nu = ns - P

# unit vectors u with coefficient of x_j in submodel M = (u' y) / ||v||, for
# every M and j in M
U = []
for k in range(1, 6):
    for M in itertools.combinations(range(1, 6), k):
        XM = Xs[:, [0, *M]]
        V = XM @ np.linalg.inv(XM.T @ XM)
        # column i is v for
        # coefficient i
        for i in range(1, len(M) + 1):
            U.append(V[:, i] / np.linalg.norm(V[:, i]))
U = np.array(U)
# 80 unit vectors, all
# in C(X) minus 1
Qb, _ = np.linalg.qr(Xs)
Qb = Qb[:, 1:]
# orthonormal basis of
# C(X) orthogonal to 1
W = U @ Qb
# coordinates of the u'
# s in that basis
sims = 100_000
Zp = rng.normal(size=(sims, P - 1))
s = np.sqrt(rng.chisquare(nu, size=sims) / nu)
K_posi = np.quantile(np.max(np.abs(Zp @ W.T), axis=1) / s, 0.95)
K_t = stats.t.ppf(0.975, nu)
K_bonf = stats.t.ppf(1 - 0.025 / len(U), nu)
K_sch = np.sqrt((P - 1) * stats.f.ppf(0.95, P - 1, nu))
print(f"{len(U)} coefficients; multipliers: t {K_t:.3f}, PoSI {K_posi:.3f}, "
      f"Bonferroni {K_bonf:.3f}, Scheffe {K_sch:.3f}")
```

### Example. (Three ways to report after forward selection)

Take  $n = 100$  cases of ten normal regressors with pairwise correlations 0.5,  $\sigma = 1$ , choose a model by forward selection with AIC, and report 95% intervals for the chosen coefficients, with the projection coefficients  $\beta_{\hat{M},j}$  as targets. Over 3000 data sets, the fraction of reported intervals that cover their targets is:

|                                      | naive | split in half | Scheffé |
|--------------------------------------|-------|---------------|---------|
| no signal: coverage                  | 0.440 | 0.946         | 1.000   |
| no signal: mean half-width           | 0.228 | 0.351         | 0.516   |
| three slopes of 0.3: coverage        | 0.868 | 0.956         | 1.000   |
| three slopes of 0.3: mean half-width | 0.248 | 0.379         | 0.559   |

With no signal, fewer than half of the naive intervals cover: a coefficient is reported only because its estimate was large. Splitting restores the nominal level at the cost of intervals about one and a half times as wide. The Scheffé intervals, with multiplier  $\sqrt{10F_{0.05}(10, 89)} = 4.403$ , are valid for every model and every rule, and here they are very conservative, because forward selection explores only a small part of the 5120 coefficients they protect.

## 29.5.5 Honest inference, III: conditioning on the selection

The third remedy uses all the data and asks for coverage *given* that the model was selected. In the one-dimensional setting of [Proposition 29.5.1\(c\)](#) this can be done exactly.

### Example. (A selective interval)

Let  $Z \sim N(\theta, 1)$  be reported only when  $|Z| > 1.96$ , and suppose  $Z = 2.5$  is observed. Given selection,  $Z$  has the normal distribution truncated to  $|z| > 1.96$ . Let  $F_\theta$  be its distribution function. Then  $F_\theta(Z)$  is uniform on  $(0, 1)$  given selection, whatever  $\theta$ , and  $F_\theta(z)$  decreases in  $\theta$  ([Exercise C1](#)). So  $\{\theta : 0.025 \leq F_\theta(2.5) \leq 0.975\}$  is an interval with conditional coverage exactly 95% for every  $\theta$ . Numerically it is  $[-0.42, 4.33]$ , against the naive  $[0.54, 4.46]$ . An estimate just past the threshold is weak evidence once the threshold is taken into account, and the selective interval includes zero.

```
from scipy.optimize import brentq

def trunc_cdf(x, theta, c=z):
    """Distribution function at x of N(theta, 1) conditioned on |Z| > c."""
    den = stats.norm.sf(c - theta) + stats.norm.cdf(-c - theta)
    if x <= -c:
        num = stats.norm.cdf(x - theta)
    elif x < c:
        num = stats.norm.cdf(-c - theta)
    else:
        num = stats.norm.cdf(-c - theta) + stats.norm.cdf(x - theta) - stats.norm.
        cdf(c - theta)
    return num / den

z_obs = 2.5
lower = brentq(lambda th: trunc_cdf(z_obs, th) - 0.975, -10, z_obs)
upper = brentq(lambda th: trunc_cdf(z_obs, th) - 0.025, -10, 10)
print(f"naive interval [{z_obs - z:.2f}, {z_obs + z:.2f}], selective interval [{
    lower:.2f}, {upper:.2f}]"
```

Lee, Sun, Sun and Taylor (2016) extended the construction to selection events that are polyhedra  $\{\mathbf{AY} \leq \mathbf{b}\}$ , which covers the lasso at a fixed penalty, forward stepwise selection (Tibshirani, Taylor, Lockhart and Tibshirani 2016) and thresholding: a linear function of  $\mathbf{Y}$ , given the selection and the part of  $\mathbf{Y}$  orthogonal to it, is a truncated normal. The intervals can be very long near the selection boundary. Selecting on part of the data but using all of it for inference

(“data carving”, Fithian, Sun and Taylor 2014) shortens them. In high dimensions the debiased lasso of [Theorem 28.5.1](#) gives intervals for the full model’s coefficients instead.

None of the three routes is free. The cheapest honest inference is still to fix the model before seeing the data, and to use selection for prediction, where its errors can be measured.

## 29.5.6 Exercises

### 29.5.6.1 A. Check your understanding

#### Exercise A1.

In [Proposition 29.5.1\(c\)](#), take  $c = 1.5 < z_{0.025}$ . Compute the conditional coverage at  $\theta = 0$ .

### 29.5.6.2 B. Practice

#### Exercise B1.

In the orthonormal design of [Proposition 29.2.5](#) with  $\sigma = 1$  and a zero coefficient, show that  $E(|Z_j| \mid |Z_j| > c) = \phi(c)/(1 - \Phi(c))$ . Evaluate it for  $c = \sqrt{2}$  and  $c = 1.96$ , and interpret it as the average size of a selected null coefficient.

*Solution.* By symmetry  $E(|Z| \mid |Z| > c) = E(Z \mid Z > c) = \int_c^\infty z\phi(z) dz / (1 - \Phi(c)) = \phi(c)/(1 - \Phi(c))$ . For  $c = \sqrt{2}$  this is  $0.1468/0.0786 = 1.87$ ; for  $c = 1.96$  it is  $0.0584/0.0250 = 2.34$ . A coefficient that is truly zero, once selected, is reported with an estimate of about two standard errors on average: the winner’s curse of [Proposition 29.5.1\(a\)](#), in its purest form.

#### Exercise B2.

Verify that the 32 candidate models of [Example \(Honest intervals for the state regressions\)](#) contain 80 slope coefficients, and that every  $\mathbf{v}_{M,j}$  for a slope is orthogonal to  $\mathbf{1}$ . Why does this let Scheffé’s bound use  $q = 5$  instead of 6?

*Solution.* Each of the 5 regressors appears in  $2^4 = 16$  of the models, so there are  $5 \times 16 = 80$  pairs. The vector  $\mathbf{v}_{M,j} = \mathbf{X}_M(\mathbf{X}_M^\top \mathbf{X}_M)^{-1} \mathbf{e}_j$  satisfies  $\mathbf{X}_M^\top \mathbf{v}_{M,j} = \mathbf{e}_j$ , and the intercept column of  $\mathbf{X}_M$  is  $\mathbf{1}$ , with  $\mathbf{e}_j$  zero in that position, so  $\mathbf{1}^\top \mathbf{v}_{M,j} = 0$ . All the  $\mathbf{v}_{M,j}$  lie in  $\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp$ , of dimension 5, and [Theorem 13.2.2\(a\)](#) needs only a subspace containing them.

### 29.5.6.3 C. Going deeper

#### Exercise C1.

Let  $F_\theta$  be the distribution function of  $N(\theta, 1)$  truncated to a set  $A$  of positive probability. Show that for  $\theta_1 < \theta_2$  the density ratio  $f_{\theta_2}(z)/f_{\theta_1}(z)$  is increasing in  $z \in A$ , and deduce that  $F_{\theta_2}(z) \leq F_{\theta_1}(z)$  for every  $z$ . Why does this make the set in [Example \(A selective](#)

interval) an interval?

*Solution.* On  $A$ ,  $f_\theta(z) = \phi(z - \theta) / \Pr_\theta(A) = \phi(z)e^{\theta z - \theta^2/2} / \Pr_\theta(A)$ , so the ratio is a positive constant times  $e^{(\theta_2 - \theta_1)z}$ , increasing in  $z$ . Put  $g = f_{\theta_2}/f_{\theta_1}$  on  $A$ . If  $g(z) \leq 1$  then  $g \leq 1$  on  $A \cap (-\infty, z]$ , so  $F_{\theta_2}(z) \leq F_{\theta_1}(z)$ . If  $g(z) > 1$  then  $g > 1$  on  $A \cap [z, \infty)$ , so  $1 - F_{\theta_2}(z) \geq 1 - F_{\theta_1}(z)$ . Either way  $F_{\theta_2}(z) \leq F_{\theta_1}(z)$ . For fixed  $z$ ,  $\theta \mapsto F_\theta(z)$  is therefore nonincreasing (and continuous), so  $\{\theta : 0.025 \leq F_\theta(z) \leq 0.975\}$  is an interval.

### Exercise C2.

In [Example \(Coverage after a pretest\)](#), let  $\hat{\beta}_1^S$  be the short slope and  $\hat{\beta}_2$  the long-model estimate of  $\beta_2$ . Use [Lemma 19.1.1](#) to show that the two are independent, and write the coverage of the reported interval as

$$\Pr(|T_2| \leq z) \Pr(|\hat{\beta}_1^S - \beta_1| \leq z \text{sd}(\hat{\beta}_1^S)) + \Pr(|T_2| > z, |\hat{\beta}_1 - \beta_1| \leq z \text{sd}(\hat{\beta}_1)),$$

where  $T_2 = \hat{\beta}_2 / \text{sd}(\hat{\beta}_2)$ . Explain why only the first term depends on the bias  $r\beta_2$  of the short slope in this simple way.

## 29.6 Summary and Notes

### 29.6.1 Summary

*Idea.*

1. The training error of any fitting procedure understates its in-sample prediction error by  $2n^{-1} \sum_i \text{Cov}(\hat{\mu}_i, Y_i)$  on average. For least squares on  $p$  columns this is  $2\sigma^2 p/n$ , whether or not the model is correct, and for a linear smoother  $2\sigma^2 \text{tr}(\mathbf{S})/n$  ([Theorem 29.1.1](#)).
2. The degrees of freedom of a procedure are  $\sigma^{-2} \sum_i \text{Cov}(\hat{\mu}_i, Y_i)$  ([Definition 29.1.2](#)). Under normality they equal the expected divergence of the fit, by Stein's lemma ([Lemma 27.5.1](#), [Proposition 29.1.2](#)). Soft thresholding costs its expected number of nonzero coefficients, hard thresholding and subset search cost more ([Proposition 29.1.3](#) and [Example \(Best subset of a given size\)](#)).
3. Mallows'  $C_p$  is an unbiased estimate of the scaled risk of each candidate, and it prefers a larger model iff the added columns have  $F > 2$  ([Definition 29.2.1](#) and [Proposition 29.2.1](#)). AIC estimates a Kullback–Leibler discrepancy, and AICc estimates it without bias in the normal linear model ([Definition 29.2.2](#) and [Theorem 29.2.2](#)). BIC approximates a log marginal likelihood, with penalty  $\log n$  per parameter.
4. Criteria with a growing penalty such as BIC find a true finite model with probability tending to one. AIC keeps each irrelevant column with probability about  $\Pr\{\chi^2(1) > 2\}$  however large  $n$  is, but it is efficient for prediction when no candidate is true ([Proposition 29.2.4](#)).
5. In an orthonormal design selection is hard thresholding, and no threshold dominates another ([Proposition 29.2.5](#) and [Eq. 29.2.4](#)).



6.  $K$ -fold cross-validation estimates the expected error of the procedure fitted to  $n - n/K$  cases, not the error of the model at hand ([Proposition 29.3.1](#)). For quadratic penalties, leave-one-out needs no refitting, and GCV replaces the leverages by their average ([Theorem 29.3.2](#) and [Definition 29.3.2](#)). Every data-driven step must be repeated inside the folds ([Example \(Screening, then cross-validating\)](#)).
7. Criteria that depend on SSE agree within a size ([Proposition 29.4.1](#)). Branch and bound finds the best subset exactly with far fewer fits ([Proposition 29.4.2](#)). Stepwise steps maximize partial correlations ([Proposition 29.4.3](#)) and can miss the best subset, and the chosen subset is unstable ([Example \(Bootstrapping the state regressions\)](#)).
8. After selection, estimates are biased away from zero, minimized error estimates are optimistic, and naive intervals under-cover ([Proposition 29.5.1](#)). Sample splitting ([Proposition 29.5.2](#)), simultaneous intervals over all submodels ([Proposition 29.5.3](#)) and conditional intervals ([Example \(A selective interval\)](#)) restore validity at different prices.

## 29.6.2 Notes and sources

**Coverage.** The chapter covers Seber and Lee (2003, chapter 12), Christensen (2020, chapter 14), Sen and Srivastava (1990, chapter 11) and Fahrmeir, Kneib, Lang and Marx (2021, section 3.4) with its own arguments and examples. The comparison of short and long regressions behind selection is [Theorem 19.1.5](#). Shrinkage, the high-dimensional case, and penalized and boosted fits are in [Chapter 27](#), [Chapter 28](#) and [Chapter 30](#); bagging and random forests are outside the book's scope.

**Optimism and degrees of freedom.** The covariance form of the optimism is due to Efron (1986, 2004), and the covariance definition of degrees of freedom, with the observation that selection adds to them, to Ye (1998). Stein (1981) proved the identity behind [Proposition 29.1.2](#) and the unbiased risk estimate. Zou, Hastie and Tibshirani (2007) showed that the number of nonzero lasso coefficients is unbiased for its degrees of freedom, and Tibshirani (2015) studied the search degrees of freedom of best subset selection, of which [Proposition 29.1.3](#) is the orthonormal case. Breiman (1992) estimated the prediction error of a selected model by resampling residuals, and Breiman and Spector (1992) compared fixed and random designs and leave-one-out with five- and ten-fold cross-validation.

**Criteria.**  $C_p$  and its plot are due to Mallows (1973), AIC to Akaike (1973, 1974), and AICc to Sugiura (1978) and Hurvich and Tsai (1989), whose derivation [Theorem 29.2.2](#) follows. BIC is from Schwarz (1978); Kass and Raftery (1995) and Kass and Wasserman (1995) discuss its accuracy. Consistency for penalties with  $c_n \rightarrow \infty$ ,  $c_n/n \rightarrow 0$  is due to Nishii (1984), efficiency to Shibata (1981) and Li (1987), and their incompatibility to Yang (2005). Konishi and Kitagawa (2008) treat information criteria at book length. The pretest estimator goes back to Bancroft (1944).

**Cross-validation.** Cross-validation is due to Stone (1974) and Geisser (1975), PRESS to Allen (1974), the equivalence with AIC to Stone (1977), and the need for large validation sets for consistency to Shao (1993); Shao (1997) unifies these results. GCV is from Craven and Wahba (1979) and Golub, Heath and Wahba (1979), and the one-standard-error rule from Breiman, Friedman, Olshen and Stone (1984). Bengio and Grandvalet (2004) and Bates, Hastie and Tibshirani (2024) analyse its variance and its target. Varma and Simon (2006) and Hastie, Tibshirani and Friedman (2009, section 7.10) discuss cross-validation after screening.



**Search.** Enumeration by sweeps is due to Schatzoff, Tsao and Fienberg (1968), branch and bound to Furnival and Wilson (1974), and stepwise regression to Efroymson (1960). Miller (2002) is the standard monograph on subset selection. Natarajan (1995) proved NP-hardness, Bertsimas, King and Mazumder (2016) used mixed-integer optimization, and Hastie, Tibshirani and Tibshirani (2020) compare best subset, forward stepwise and the lasso. Breiman (1996b) analysed instability, Breiman (1996a) proposed bagging, Hoeting, Madigan, Raftery and Volinsky (1999) review model averaging, and Meinshausen and Bühlmann (2010) introduced stability selection.

**Inference after selection.** The screening paradox is Freedman’s (1983). Leeb and Pötscher (2005, 2006) showed that post-selection distributions cannot be estimated uniformly. Sample splitting goes back to Cox (1975), with modern versions by Wasserman and Roeder (2009) and Rinaldo, Wasserman and G’Sell (2019). [Proposition 29.5.3](#) is the PoSI approach of Berk, Brown, Buja, Zhang and Zhao (2013). Exact conditional inference is due to Lee, Sun, Sun and Taylor (2016) and Tibshirani, Taylor, Lockhart and Tibshirani (2016), and data carving to Fithian, Sun and Taylor (2014).

### 29.6.3 References

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# Chapter 30

## Regularization and Boosting

Ridge regression and the lasso are two members of one family: least squares plus a penalty on the coefficients. This chapter studies the family as a whole: when a minimizer exists and is unique, how to recognize one, what prior it corresponds to, and what is gained by giving up convexity. Two penalties then carry structure the lasso cannot see: the elastic net keeps correlated regressors together, and the group lasso selects whole factors. The second half turns to boosting, which fits a regression by many small steps along one coordinate at a time. Run to the end, it reproduces least squares; stopped early, it is a shrinkage estimator whose path, with small steps, is almost the lasso path. The number of steps plays the part of the penalty.

**What you need.** [Chapter 1](#) (the singular value decomposition, [Theorem 1.8.1](#), and the Cauchy–Schwarz inequality, [Proposition 1.1.1](#)), [Chapter 6](#) (least squares as projection, [Theorem 6.4.1](#), and the minimum-norm solution, [Proposition 6.10.1](#)), [Chapter 7](#) (the conjugate posterior, [Theorem 7.6.2](#) and [Proposition 7.6.3](#)), and indicator coding of factors ([Chapter 15](#)). From this part: [Chapter 27](#) (ridge regression, [Theorem 27.2.2](#), and the lasso, [Definition 27.4.1](#) and [Theorem 27.4.2](#)) and [Chapter 29](#) (covariance penalties, [Theorem 29.1.1](#), and cross-validation). [Chapter 28](#) is needed only for remarks.

### 30.0.1 Roadmap

- [Penalized least squares in general](#)
- [The elastic net](#)
- [Group penalties](#)
- [Componentwise boosting](#)
- [Boosting as regularization](#)
- [Summary and notes](#)

## 30.1 Penalized least squares in general

Ridge regression and the lasso of [Chapter 27](#) raise the same questions: is there a minimizer, is it unique, and how do we recognize one? This section answers them for a general, mainly convex, penalty, reads a penalty as a prior, and ends with penalties that give up convexity on purpose. Throughout,  $\mathbf{X}$  is  $n \times p$  of any rank, and  $p$  may exceed  $n$ .

### 30.1.1 The criterion

#### Definition 30.1.1 — Penalized least squares

A **penalty** is a function  $P : \mathbb{R}^p \rightarrow [0, \infty)$ . For  $\lambda \geq 0$ , the **penalized least squares criterion** is

$$Q_\lambda(\mathbf{b}) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda P(\mathbf{b}), \quad (30.1.1)$$

and a **penalized least squares estimate** is any minimizer  $\hat{\boldsymbol{\beta}}_\lambda$  of  $Q_\lambda$ . The penalty is **convex** if  $P(\alpha\mathbf{b} + (1 - \alpha)\mathbf{c}) \leq \alpha P(\mathbf{b}) + (1 - \alpha)P(\mathbf{c})$  for all  $\mathbf{b}, \mathbf{c}$  and  $\alpha \in (0, 1)$ , and **strictly convex** if the inequality is strict whenever  $\mathbf{b} \neq \mathbf{c}$ . The penalization is **partial** if  $\mathbf{b} = (\mathbf{b}_1^\top, \mathbf{b}_2^\top)^\top$  and  $P$  depends on  $\mathbf{b}_2$  alone; the coefficients in  $\mathbf{b}_1$  are then **unpenalized**.

Other scalings (no  $\frac{1}{2}$ , or a divisor  $n$ ) multiply  $\lambda$  by a constant; results are quoted here in the scaling of Eq. 30.1.1. Ridge ( $P = \frac{1}{2}\|\mathbf{b}\|^2$ , Theorem 27.2.2) and the lasso ( $P = \|\mathbf{b}\|_1$ , Definition 27.4.1) are convex; the best-subset penalty  $\#\{j : b_j \neq 0\}$  of Chapter 29 and the bridge penalties  $\sum_j |b_j|^q$ ,  $q < 1$ , are not. Sections 30.2 and 30.3 add penalties with structure.

**Intercept and scaling.** An intercept is not penalized. Minimizing over it first gives  $b_0 = \bar{y} - \bar{\mathbf{x}}^\top \mathbf{b}$  and leaves Eq. 30.1.1 with  $\mathbf{y}$  and the columns of  $\mathbf{X}$  centred (Exercise (B2)), the partialling-out of Theorem 6.6.2, so we centre and drop it. A penalty that treats coordinates alike is not equivariant under rescaling a column, so the columns are also scaled to a common length, a choice that is part of the model.

### 30.1.2 Existence and uniqueness

For least squares a minimizer always exists and its fit is unique (Theorem 6.4.1). For penalized criteria:

#### Theorem 30.1.1 — Existence and uniqueness of penalized estimates

Let  $P$  be continuous and  $\lambda \geq 0$ .

- (a) If  $Q_\lambda(\mathbf{b}) \rightarrow \infty$  as  $\|\mathbf{b}\| \rightarrow \infty$ , a minimizer exists. This holds in each of three cases:
  - (i)  $\mathbf{X}$  has full column rank; (ii)  $\lambda > 0$  and  $P(\mathbf{b}) \geq c\|\mathbf{b}\|$  for some  $c > 0$ ; (iii) the penalization is partial,  $\mathbf{X} = [\mathbf{X}_1, \mathbf{X}_2]$  with  $\mathbf{X}_1$  of full column rank,  $\lambda > 0$ , and  $P(\mathbf{b}) \geq c\|\mathbf{b}_2\|$  for some  $c > 0$ .
- (b) If  $P$  is convex, the set of minimizers is convex, and all minimizers have the same fitted vector  $\mathbf{X}\hat{\boldsymbol{\beta}}_\lambda$ . If in addition  $\lambda > 0$ , they all have the same penalty  $P(\hat{\boldsymbol{\beta}}_\lambda)$ .
- (c) If  $P$  is convex and either  $\mathbf{X}$  has full column rank, or  $\lambda > 0$  and  $P$  is strictly convex, the minimizer is unique.

*Proof.*

- (a) Let  $\mathcal{S} = \{\mathbf{b} : Q_\lambda(\mathbf{b}) \leq Q_\lambda(\mathbf{0})\}$ . It is closed because  $Q_\lambda$  is continuous, and bounded because  $Q_\lambda \rightarrow \infty$ , so  $Q_\lambda$  attains its minimum on  $\mathcal{S}$ . Any point outside  $\mathcal{S}$  has a larger value than  $\mathbf{0}$ , so this minimum is a global minimizer. In case (i), the smallest singular value  $s$  of  $\mathbf{X}$  is positive (Theorem 1.8.1), so  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\| \geq s\|\mathbf{b}\| - \|\mathbf{y}\|$ , which tends to infinity. In case (ii),  $Q_\lambda(\mathbf{b}) \geq \lambda c\|\mathbf{b}\|$ . In case (iii), suppose  $\|\mathbf{b}^{(k)}\| \rightarrow \infty$  along a sequence on which  $Q_\lambda$  stays bounded. Then  $\lambda c\|\mathbf{b}_2^{(k)}\|$  is bounded, so  $\|\mathbf{b}_1^{(k)}\| \rightarrow \infty$ . With  $s_1 > 0$  the smallest singular value of  $\mathbf{X}_1$ ,  $\|\mathbf{y} - \mathbf{X}\mathbf{b}^{(k)}\| \geq s_1\|\mathbf{b}_1^{(k)}\| - \|\mathbf{y}\| - \|\mathbf{X}_2\mathbf{b}_2^{(k)}\| \rightarrow \infty$ , a contradiction.
- (b) If  $P$  is convex, so is  $Q_\lambda$ , as a sum of convex functions, and the set of minimizers of a convex function is convex. Let  $\mathbf{b}$  and  $\mathbf{c}$  be minimizers with value  $q$ , and put  $\mathbf{m} = \frac{1}{2}(\mathbf{b} + \mathbf{c})$ . With  $\mathbf{u} = \mathbf{y} - \mathbf{X}\mathbf{b}$  and  $\mathbf{v} = \mathbf{y} - \mathbf{X}\mathbf{c}$ , the parallelogram identity gives

$$\|\mathbf{y} - \mathbf{X}\mathbf{m}\|^2 = \left\|\frac{1}{2}(\mathbf{u} + \mathbf{v})\right\|^2 = \frac{1}{2}\|\mathbf{u}\|^2 + \frac{1}{2}\|\mathbf{v}\|^2 - \frac{1}{4}\|\mathbf{X}(\mathbf{b} - \mathbf{c})\|^2.$$

Together with  $P(\mathbf{m}) \leq \frac{1}{2}P(\mathbf{b}) + \frac{1}{2}P(\mathbf{c})$  this yields  $q \leq Q_\lambda(\mathbf{m}) \leq q - \frac{1}{8}\|\mathbf{X}(\mathbf{b} - \mathbf{c})\|^2$ . Hence  $\mathbf{X}\mathbf{b} = \mathbf{X}\mathbf{c}$ . The two minimizers then have equal residual sums of squares, so when  $\lambda > 0$  their penalties are equal too.

- (c) If  $\mathbf{X}$  has full column rank,  $\mathbf{X}\mathbf{b} = \mathbf{X}\mathbf{c}$  forces  $\mathbf{b} = \mathbf{c}$ . If  $P$  is strictly convex and  $\mathbf{b} \neq \mathbf{c}$ , the inequality for  $P(\mathbf{m})$  in (b) is strict and  $Q_\lambda(\mathbf{m}) < q$ , which is impossible. ■

Norms satisfy (a)(ii), so the lasso, group lasso and elastic net always have minimizers. By (b), when  $p > n$  the lasso's minimizers, possibly many, share their fit and  $\ell_1$  norm (they are unique in general position; Tibshirani, 2013).

### 30.1.3 Optimality conditions

The corner of  $|b|$  at zero, which makes the lasso set coefficients to zero, calls for a substitute for the gradient.

#### Definition 30.1.2 — Subgradient

Let  $P$  be convex. A vector  $\mathbf{s}$  is a **subgradient** of  $P$  at  $\mathbf{b}$  if  $P(\mathbf{c}) \geq P(\mathbf{b}) + \mathbf{s}^\top(\mathbf{c} - \mathbf{b})$  for every  $\mathbf{c} \in \mathbb{R}^p$ . The set of all subgradients at  $\mathbf{b}$  is the **subdifferential**  $\partial P(\mathbf{b})$ .

Where  $P$  is differentiable,  $\partial P(\mathbf{b}) = \{\nabla P(\mathbf{b})\}$ . For  $P(b) = |b|$  on  $\mathbb{R}$ , the subdifferential is  $\{\text{sign}(b)\}$  if  $b \neq 0$  and  $[-1, 1]$  at 0.

#### Lemma 30.1.2 — Optimality condition

Let  $P$  be convex and  $\lambda > 0$ . A vector  $\hat{\boldsymbol{\beta}}$  minimizes  $Q_\lambda$  iff

$$\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) \in \lambda \partial P(\hat{\boldsymbol{\beta}}). \quad (30.1.2)$$

In that case  $Q_\lambda(\mathbf{b}) - Q_\lambda(\hat{\boldsymbol{\beta}}) \geq \frac{1}{2}\|\mathbf{X}(\mathbf{b} - \hat{\boldsymbol{\beta}})\|^2$  for every  $\mathbf{b}$ .

*Proof.* Write  $\mathbf{r} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$  and  $\mathbf{d} = \mathbf{b} - \hat{\boldsymbol{\beta}}$ . Expanding the square,

$$Q_\lambda(\mathbf{b}) - Q_\lambda(\hat{\boldsymbol{\beta}}) = -\mathbf{d}^\top \mathbf{X}^\top \mathbf{r} + \frac{1}{2} \|\mathbf{X}\mathbf{d}\|^2 + \lambda(P(\mathbf{b}) - P(\hat{\boldsymbol{\beta}})). \quad (30.1.3)$$

If  $\mathbf{X}^\top \mathbf{r} = \lambda \mathbf{s}$  with  $\mathbf{s} \in \partial P(\hat{\boldsymbol{\beta}})$ , then  $\lambda(P(\mathbf{b}) - P(\hat{\boldsymbol{\beta}})) \geq \lambda \mathbf{s}^\top \mathbf{d} = \mathbf{d}^\top \mathbf{X}^\top \mathbf{r}$ , and [Eq. 30.1.3](#) gives the stated lower bound, which is nonnegative. Conversely, let  $\hat{\boldsymbol{\beta}}$  be a minimizer and fix  $\mathbf{b}$ . For  $0 < \tau < 1$ , convexity gives  $P(\hat{\boldsymbol{\beta}} + \tau \mathbf{d}) - P(\hat{\boldsymbol{\beta}}) \leq \tau(P(\mathbf{b}) - P(\hat{\boldsymbol{\beta}}))$ . Apply [Eq. 30.1.3](#) to  $\hat{\boldsymbol{\beta}} + \tau \mathbf{d}$  in place of  $\mathbf{b}$ :

$$\begin{aligned} 0 &\leq Q_\lambda(\hat{\boldsymbol{\beta}} + \tau \mathbf{d}) - Q_\lambda(\hat{\boldsymbol{\beta}}) \\ &\leq -\tau \mathbf{d}^\top \mathbf{X}^\top \mathbf{r} + \frac{\tau^2}{2} \|\mathbf{X}\mathbf{d}\|^2 + \lambda \tau (P(\mathbf{b}) - P(\hat{\boldsymbol{\beta}})). \end{aligned}$$

Divide by  $\tau$  and let  $\tau \rightarrow 0$ . The result is  $\lambda(P(\mathbf{b}) - P(\hat{\boldsymbol{\beta}})) \geq \mathbf{d}^\top \mathbf{X}^\top \mathbf{r}$  for every  $\mathbf{b}$ , which says that  $\mathbf{X}^\top \mathbf{r} / \lambda \in \partial P(\hat{\boldsymbol{\beta}})$ . ■

Condition [Eq. 30.1.2](#) is the **Karush–Kuhn–Tucker** (KKT) condition; with  $\lambda = 0$  it is the normal equations. Most penalties below are sums of Euclidean norms of blocks.

### Lemma 30.1.3 — Subdifferential of a sum of block norms

Let  $G_1, \dots, G_K$  partition  $\{1, \dots, p\}$ , let  $w_1, \dots, w_K > 0$ , and let  $P(\mathbf{b}) = \sum_k w_k \|\mathbf{b}_{G_k}\|$ , where  $\mathbf{b}_{G_k}$  is the subvector of  $\mathbf{b}$  with coordinates in  $G_k$ . Then  $\mathbf{s} \in \partial P(\mathbf{b})$  iff, for every  $k$ ,

$$\mathbf{s}_{G_k} = w_k \frac{\mathbf{b}_{G_k}}{\|\mathbf{b}_{G_k}\|} \quad \text{if } \mathbf{b}_{G_k} \neq \mathbf{0}, \quad \|\mathbf{s}_{G_k}\| \leq w_k \quad \text{if } \mathbf{b}_{G_k} = \mathbf{0}.$$

With singleton blocks and unit weights,  $P = \|\cdot\|_1$ , and  $\mathbf{s} \in \partial \|\mathbf{b}\|_1$  iff  $s_j = \text{sign}(b_j)$  when  $b_j \neq 0$  and  $|s_j| \leq 1$  when  $b_j = 0$ .

*Proof.* First,  $\mathbf{s} \in \partial P(\mathbf{b})$  iff  $\mathbf{s}_{G_k} \in \partial(w_k \|\cdot\|)(\mathbf{b}_{G_k})$  for every  $k$ . Summing the blockwise inequalities gives one direction. For the other, apply the subgradient inequality to a  $\mathbf{c}$  that differs from  $\mathbf{b}$  only in block  $k$ . So it suffices to describe the subdifferential of  $f(\mathbf{u}) = w\|\mathbf{u}\|$  on  $\mathbb{R}^d$ . We show that  $\mathbf{s} \in \partial f(\mathbf{u})$  iff  $\|\mathbf{s}\| \leq w$  and  $\mathbf{s}^\top \mathbf{u} = w\|\mathbf{u}\|$ .

If these two conditions hold, then for every  $\mathbf{v}$ , by Cauchy–Schwarz ([Proposition 1.1.1](#)),  $f(\mathbf{v}) = w\|\mathbf{v}\| \geq \mathbf{s}^\top \mathbf{v} = f(\mathbf{u}) + \mathbf{s}^\top (\mathbf{v} - \mathbf{u})$ . Conversely, let  $\mathbf{s} \in \partial f(\mathbf{u})$ . Taking  $\mathbf{v} = \mathbf{0}$  and  $\mathbf{v} = 2\mathbf{u}$  gives  $\mathbf{s}^\top \mathbf{u} \geq w\|\mathbf{u}\|$  and  $\mathbf{s}^\top \mathbf{u} \leq w\|\mathbf{u}\|$ . Then  $w\|\mathbf{u} + \tau \mathbf{v}\| \geq w\|\mathbf{u}\| + \tau \mathbf{s}^\top \mathbf{v}$  for all  $\tau > 0$ . The triangle inequality bounds the left side by  $w\|\mathbf{u}\| + \tau w\|\mathbf{v}\|$ , so  $\mathbf{s}^\top \mathbf{v} \leq w\|\mathbf{v}\|$  for every  $\mathbf{v}$ . Taking  $\mathbf{v} = \mathbf{s}$  gives  $\|\mathbf{s}\| \leq w$ .

If  $\mathbf{u} = \mathbf{0}$ , the two conditions say  $\|\mathbf{s}\| \leq w$ . If  $\mathbf{u} \neq \mathbf{0}$ , then  $\mathbf{s}^\top \mathbf{u} = w\|\mathbf{u}\| \geq \|\mathbf{s}\|\|\mathbf{u}\|$  is equality in Cauchy–Schwarz, so  $\mathbf{s}$  is a nonnegative multiple of  $\mathbf{u}$  with norm  $w$ , that is,  $\mathbf{s} = w\mathbf{u}/\|\mathbf{u}\|$ . ■

Combining the two lemmas,  $\hat{\boldsymbol{\beta}}$  is a lasso estimate iff  $\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \lambda \text{sign}(\hat{\beta}_j)$  for every  $j$  with  $\hat{\beta}_j \neq 0$  and  $|\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})| \leq \lambda$  for every  $j$  with  $\hat{\beta}_j = 0$ , as in [Definition 27.4.1](#): active regressors



share the absolute correlation  $\lambda$  with the residual, and no inactive one exceeds it.

### 30.1.4 The Bayesian reading

Ridge regression is a posterior mean ([Proposition 7.6.3\(d\)](#)). For other penalties the matching statement is about the posterior *mode*; part (b) below restates, for completeness, the reading of the lasso found in [Exercise \(B2\)](#) ([Section 27.4](#)).

#### Proposition 30.1.4 — Penalized estimates as posterior modes

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$  with  $\sigma^2$  known, and give  $\boldsymbol{\beta}$  a prior density  $\pi(\mathbf{b}) \propto \exp\{-(\lambda/\sigma^2)P(\mathbf{b})\}$ , assumed integrable. Then the posterior density of  $\boldsymbol{\beta}$  is proportional to  $\exp\{-Q_\lambda(\mathbf{b})/\sigma^2\}$ . So the penalized least squares estimates are exactly the posterior modes. In particular:

- (a) a normal prior  $\boldsymbol{\beta} \sim N_p(\mathbf{0}, \tau^2\mathbf{I})$  gives ridge regression ([Theorem 27.2.2](#)) with  $\lambda = \sigma^2/\tau^2$ , and the posterior mode equals the posterior mean;
- (b) independent Laplace priors with density  $(a/2)e^{-a|b_j|}$  give the lasso with  $\lambda = a\sigma^2$ .

*Proof.* By Bayes' theorem the posterior density is proportional to the likelihood times the prior, that is, to  $\exp\{-\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2/(2\sigma^2)\} \exp\{-(\lambda/\sigma^2)P(\mathbf{b})\} = \exp\{-Q_\lambda(\mathbf{b})/\sigma^2\}$ , and  $t \mapsto e^{-t/\sigma^2}$  is decreasing. For (a),  $\tau^{-2}\|\mathbf{b}\|^2/2 = (\lambda/\sigma^2)\|\mathbf{b}\|^2/2$  with  $\lambda = \sigma^2/\tau^2$ . The posterior is normal, so its mode and mean coincide ([Theorem 7.6.2](#) with  $\sigma^2$  known, or [Proposition 7.6.3\(d\)](#)). For (b), as in [Exercise \(B2\)](#),  $a\|\mathbf{b}\|_1 = (\lambda/\sigma^2)\|\mathbf{b}\|_1$  with  $\lambda = a\sigma^2$ . ■

Under squared error loss, though, a Bayesian reports the posterior mean, which under a Laplace prior is never sparse.

#### Example. (Posterior mode and posterior mean under a Laplace prior)

Take one observation  $z \sim N(\theta, 1)$  and a Laplace prior on  $\theta$  with rate  $a = 1.5$ . By [Proposition 30.1.4\(b\)](#) the posterior mode is the lasso estimate with  $\lambda = 1.5$ , which is soft thresholding at 1.5 ([Theorem 27.4.2](#)). The posterior mean, computed by numerical integration in the listing below, behaves differently:

| $z$            | 0.5   | 1.0   | 2.0   | 4.0   |
|----------------|-------|-------|-------|-------|
| posterior mode | 0     | 0     | 0.500 | 2.500 |
| posterior mean | 0.174 | 0.364 | 0.859 | 2.509 |

The mode is zero for  $|z| \leq 1.5$ ; the mean is zero only at  $z = 0$ , and approaches  $z - 1.5$  for large  $z$  ([Figure 30.1.1\(c\)](#)). Exact zeros come from the mode, not from the prior; a prior that believes in exact zeros needs a point mass, as in the spike-and-slab priors of George and McCulloch (1993). [Chapter 39](#) returns to priors as regularizers ([Proposition 39.5.1](#)).

```
import numpy as np
from scipy import integrate
```

```

def soft(z, lam):
    return np.sign(z) * np.maximum(np.abs(z) - lam, 0.0)

def laplace_posterior_mean(z, rate):
    """E(theta | z) when z ~ N(theta, 1) and theta has density (rate/2) exp(-rate /
    theta/)."""
    w = lambda t: np.exp(-0.5 * (z - t) ** 2 - rate * abs(t))
    num = integrate.quad(lambda t: t * w(t), -30, 30, points=[0.0, z])[0]
    den = integrate.quad(w, -30, 30, points=[0.0, z])[0]
    return num / den

rate = 1.5
for z in [0.5, 1.0, 2.0, 4.0]:
    print(f"z = {z}: posterior mode {float(soft(z, rate)):.3f}, "
          f"posterior mean {laplace_posterior_mean(z, rate):.3f}")

```

### 30.1.5 Penalties that are not convex

Under an orthonormal design the lasso subtracts  $\lambda$  from every surviving coefficient, however large (Theorem 27.4.2). Fan and Li (2001) asked for a rule that is **sparse**, **continuous** in the data, and **nearly unbiased** (large estimates left alone). Soft thresholding fails the third, and hard thresholding, the best-subset rule (Theorem 27.4.2(c)), the second. No convex penalty has all three, since one that is flat far from zero on both sides is constant. Two penalties  $P(\mathbf{b}) = \sum_j \rho(b_j)$ , with  $\rho$  concave on  $[0, \infty)$ , are in common use.

**SCAD** (the smoothly clipped absolute deviation of Fan and Li, 2001), with a parameter  $a > 2$ , is defined through its derivative on  $t > 0$ :

$$\rho'(t) = \lambda \left\{ \mathbf{1}(t \leq \lambda) + \frac{(a\lambda - t)_+}{(a-1)\lambda} \mathbf{1}(t > \lambda) \right\}, \quad \rho(0) = 0, \quad \rho(-t) = \rho(t).$$

It is the lasso penalty up to  $\lambda$ ; its slope then falls linearly to zero at  $a\lambda$ . Here  $\lambda$  sits inside  $\rho$ , and the criterion is  $\frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \sum_j \rho(b_j)$ . Fan and Li suggest  $a = 3.7$ .

**MCP** (the minimax concave penalty of Zhang, 2010), with a parameter  $\gamma > 1$ , is

$$\rho(t) = \begin{cases} \lambda|t| - t^2/(2\gamma), & |t| \leq \gamma\lambda, \\ \gamma\lambda^2/2, & |t| > \gamma\lambda. \end{cases}$$

Its slope  $\lambda - |t|/\gamma$  starts to fall at once, rather than after  $\lambda$  as for SCAD.

#### Proposition 30.1.5 — MCP: convexity and firm thresholding

Let  $Q(\mathbf{b}) = \frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \sum_j \rho(b_j)$  with  $\rho$  the MCP penalty.

- If the smallest eigenvalue of  $\mathbf{X}^\top \mathbf{X}$  is at least  $1/\gamma$ , then  $Q$  is convex. If it exceeds  $1/\gamma$ , then  $Q$  is strictly convex and has a unique minimizer.
- If  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ , the unique minimizer is  $\hat{\beta}_j = \phi(z_j)$  with  $\mathbf{z} = \mathbf{X}^\top \mathbf{y}$ , where  $\phi$  is **firm**

**thresholding:**

$$\phi(z) = \begin{cases} 0, & |z| \leq \lambda, \\ \text{sign}(z) \frac{|z| - \lambda}{1 - 1/\gamma}, & \lambda < |z| \leq \gamma\lambda, \\ z, & |z| > \gamma\lambda. \end{cases}$$

*Proof.*

(a) Write

$$Q(\mathbf{b}) = \left\{ \frac{1}{2} \mathbf{b}^\top (\mathbf{X}^\top \mathbf{X} - \gamma^{-1} \mathbf{I}) \mathbf{b} - \mathbf{y}^\top \mathbf{X} \mathbf{b} + \frac{1}{2} \mathbf{y}^\top \mathbf{y} \right\} \\ + \sum_j \left\{ \rho(b_j) + \frac{b_j^2}{2\gamma} \right\}.$$

The first brace is a quadratic with nonnegative definite matrix, hence convex, and strictly convex if the matrix is positive definite. Each term of the second sum is  $\lambda|t|$  for  $|t| \leq \gamma\lambda$  and  $\gamma\lambda^2/2 + t^2/(2\gamma)$  beyond. On  $t > 0$  its derivative is  $\lambda$  up to  $\gamma\lambda$  and  $t/\gamma$  after. This derivative is continuous at  $\gamma\lambda$  and nondecreasing, so the function is convex (it is even, with a corner at 0). A sum of convex functions is convex, and adding a strictly convex function makes the sum strictly convex. A strictly convex function that tends to infinity has exactly one minimizer, by the argument of [Theorem 30.1.1\(a\)](#) and (c).

(b) With  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ , part (a) applies because  $\gamma > 1$ . Moreover  $Q(\mathbf{b}) = \frac{1}{2} \|\mathbf{z} - \mathbf{b}\|^2 + \sum_j \rho(b_j) + \text{const}$ , which separates into one problem per coordinate: minimize  $g(\theta) = \frac{1}{2}(z - \theta)^2 + \rho(\theta)$ . Take  $z \geq 0$ . For  $\theta > 0$ ,  $g(-\theta) - g(\theta) = 2z\theta \geq 0$ , so we may restrict to  $\theta \geq 0$ . There

$$g'(\theta) = \begin{cases} (1 - 1/\gamma)\theta - (z - \lambda), & 0 < \theta \leq \gamma\lambda, \\ \theta - z, & \theta > \gamma\lambda, \end{cases}$$

a continuous nondecreasing function. If  $z \leq \lambda$ , then  $g' \geq 0$  on  $(0, \infty)$ , and the minimizer is 0. If  $\lambda < z \leq \gamma\lambda$ , then  $g'$  vanishes at  $\theta = (z - \lambda)/(1 - 1/\gamma)$ , which is at most  $\gamma\lambda$ . If  $z > \gamma\lambda$ , then  $g'(\gamma\lambda) = \gamma\lambda - z < 0$ , and  $g'$  vanishes at  $\theta = z$ .

■

Firm thresholding has all three properties; it tends to soft thresholding as  $\gamma \rightarrow \infty$  and to hard thresholding as  $\gamma \downarrow 1$ . The SCAD rule under an orthonormal design is (Fan and Li, 2001)

$$\phi_{\text{SCAD}}(z) = \begin{cases} \text{sign}(z)(|z| - \lambda)_+, & |z| \leq 2\lambda, \\ \frac{(a-1)z - \text{sign}(z)a\lambda}{a-2}, & 2\lambda < |z| \leq a\lambda, \\ z, & |z| > a\lambda, \end{cases}$$

which [Exercise \(B1\)](#) derives. The script checks each rule of [Figure 30.1.1](#) against brute-force minimization.

```
import numpy as np

def soft(z, lam):
    return np.sign(z) * np.maximum(np.abs(z) - lam, 0.0)
```

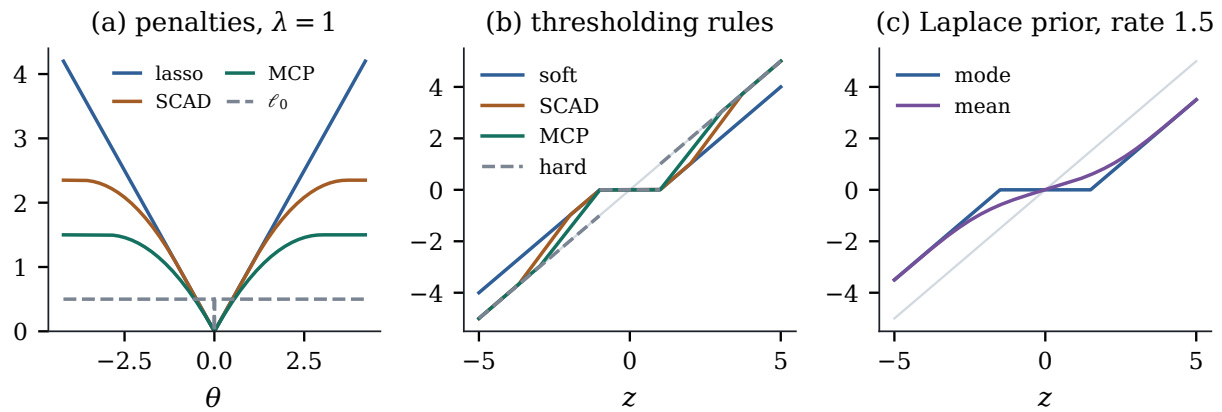


Figure 30.1.1: (a) The lasso, SCAD ( $a = 3.7$ ), MCP ( $\gamma = 3$ ) and  $\ell_0$  penalties with  $\lambda = 1$ . (b) The thresholding rules they induce under an orthonormal design (soft, SCAD, firm, and hard thresholding dashed); the grey line is the identity. (c) Posterior mode and posterior mean of  $\theta$  given  $z \sim N(\theta, 1)$  under a Laplace prior with rate 1.5 (Example (Posterior mode and posterior mean under a Laplace prior)).

```
def mcp_rule(z, lam, gam):
    # firm thresholding
    az = np.abs(z)
    mid = np.sign(z) * (az - lam) / (1 - 1 / gam)
    return np.where(az <= lam, 0.0, np.where(az <= gam * lam, mid, z))

def scad_rule(z, lam, a):
    az = np.abs(z)
    mid = ((a - 1) * z - np.sign(z) * a * lam) / (a - 2)
    return np.where(az <= 2 * lam, soft(z, lam), np.where(az <= a * lam, mid, z))

def hard_rule(z, lam):
    # penalty (lam^2 / 2) * 1{theta != 0}
    return np.where(np.abs(z) > lam, z, 0.0)
```

Without the eigenvalue condition there can be several local minima (Exercise (C1)). Replacing  $\rho$  by its tangent at a preliminary estimate gives a weighted lasso (Zou and Li, 2008), as does the adaptive lasso (Zou, 2006). Fan and Li (2001) proved an *oracle property*, which we state without proof: with  $p$  fixed,  $\lambda_n/n \rightarrow 0$ ,  $\lambda_n/\sqrt{n} \rightarrow \infty$  (in the scaling of Eq. 30.1.1; Fan and Li write the penalty as  $n \sum_j p_\lambda(|b_j|)$ , where the conditions read  $\lambda_n \rightarrow 0$ ,  $\sqrt{n}\lambda_n \rightarrow \infty$ ) and regularity conditions, some local SCAD minimizer finds the zero coefficients with probability tending to one and estimates the rest as well as least squares on the true support. The convergence is not uniform in  $\beta$  (Leeb and Pötscher, 2005), and it does not justify oracle standard errors (Chapter 29).

## 30.1.6 Exercises

### 30.1.6.1 A. Check your understanding

#### Exercise A1.

With  $\lambda = 1$ , compute the lasso, ridge (penalty  $\frac{1}{2}b^2$ ), hard, firm ( $\gamma = 3$ ) and SCAD ( $a = 3.7$ ) estimates of  $\theta$  for a single observation  $z = 2.5$  under an orthonormal design. Which rules shrink it, and by how much?

*Solution.* Lasso:  $2.5 - 1 = 1.5$ . Ridge:  $2.5/2 = 1.25$ . Hard: 2.5, since  $|z| > 1$ . Firm:  $\lambda < z \leq \gamma\lambda = 3$ , so  $(2.5 - 1)/(1 - 1/3) = 2.25$ . SCAD:  $2 < z \leq 3.7$ , so  $(2.7 \cdot 2.5 - 3.7)/1.7 = 3.05/1.7 = 1.79$ . Only hard thresholding leaves  $z$  alone. At  $z = 4$  the firm and SCAD rules would leave it alone as well, since 4 exceeds both  $\gamma\lambda = 3$  and  $a\lambda = 3.7$ .

### Exercise A2.

Show that the firm thresholding rule  $\phi$  of Proposition 30.1.5(b) is continuous at  $|z| = \lambda$  and at  $|z| = \gamma\lambda$ , whereas the hard thresholding rule  $z\mathbf{1}(|z| > \lambda)$  jumps by  $\lambda$  at  $|z| = \lambda$ . What is the largest slope of  $\phi$ , and what happens to it as  $\gamma \downarrow 1$ ?

*Solution.* Take  $z \geq 0$ . As  $z \downarrow \lambda$  the middle branch  $(z - \lambda)/(1 - 1/\gamma)$  tends to 0, the value below  $\lambda$ . At  $z = \gamma\lambda$  it equals  $(\gamma - 1)\lambda/\{(\gamma - 1)/\gamma\} = \gamma\lambda$ , the value of the upper branch. Hard thresholding is 0 at  $z = \lambda$  and tends to  $\lambda$  as  $z \downarrow \lambda$ . The slope of  $\phi$  is 0, then  $\gamma/(\gamma - 1)$ , then 1; the largest is  $\gamma/(\gamma - 1)$ , which tends to infinity as  $\gamma \downarrow 1$ , where  $\phi$  approaches the discontinuous hard rule.

### 30.1.6.2 B. Practice

#### Exercise B1.

Derive the SCAD thresholding rule displayed above. Minimize  $g(\theta) = \frac{1}{2}(z - \theta)^2 + \rho(\theta)$  for  $z \geq 0$  on the three pieces  $[0, \lambda]$ ,  $[\lambda, a\lambda]$  and  $[a\lambda, \infty)$ , and check that  $g$  is convex on  $[0, \infty)$  when  $a > 2$ .

*Solution.* On  $(0, \infty)$ ,  $g'(\theta) = \theta - z + \rho'(\theta)$ . This is  $\theta - z + \lambda$  on  $(0, \lambda]$ ,  $\theta - z + (a\lambda - \theta)/(a - 1)$  on  $(\lambda, a\lambda]$ , and  $\theta - z$  beyond. The middle piece has slope  $1 - 1/(a - 1) = (a - 2)/(a - 1) > 0$  when  $a > 2$ , and the pieces agree at  $\lambda$  and at  $a\lambda$ . So  $g'$  is continuous and increasing, and  $g$  is strictly convex on  $[0, \infty)$ . As in the proof of Proposition 30.1.5, a negative  $\theta$  is never better when  $z \geq 0$ . If  $z \leq \lambda$ , then  $g'(0+) = \lambda - z \geq 0$  and  $\hat{\theta} = 0$ . If  $\lambda < z \leq 2\lambda$ , the root  $z - \lambda$  lies in  $(0, \lambda]$ . If  $2\lambda < z \leq a\lambda$ , the middle piece vanishes at  $\theta = \{(a - 1)z - a\lambda\}/(a - 2)$ , which lies in  $(\lambda, a\lambda]$  exactly for these  $z$ . If  $z > a\lambda$ , then  $\hat{\theta} = z$ .

#### Exercise B2.

Let  $P$  be any penalty on the slopes, and let the model have an unpenalized intercept. By Proposition 27.2.1,  $(\hat{b}_0, \hat{\beta})$  minimizes  $\frac{1}{2} \sum_i (y_i - b_0 - \mathbf{x}_{(i)}^\top \mathbf{b})^2 + \lambda P(\mathbf{b})$  iff  $\hat{\beta}$  minimizes the centred criterion and  $\hat{b}_0 = \bar{y} - \bar{\mathbf{x}}^\top \hat{\beta}$ . Deduce that the fitted values of penalized regression with an intercept are unchanged when a constant is added to  $\mathbf{y}$ , and that the slopes are unchanged when a constant is added to a column of  $\mathbf{X}$ . Does the second statement survive standardizing the columns before fitting?

### 30.1.6.3 C. Going deeper

#### Exercise C1.

Let  $n = p = 2$ ,  $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$  with  $r \in (0, 1)$ , and use MCP with  $\gamma > 1$ . For which  $r$  does [Proposition 30.1.5\(a\)](#) guarantee convexity? Find a response  $\mathbf{y}$  for which the MCP criterion has two distinct local minima when  $r$  exceeds that bound.

## 30.2 The elastic net

When several regressors are near copies of one another (repeated measurements, genes in one pathway), the lasso tends to keep one and drop the rest, and which one can change with a small perturbation of the data. With two identical columns its solution is not even unique (Exercise B1 of [Section 27.4](#)). Ridge spreads the coefficient evenly over the copies but never sets anything to zero. Zou and Hastie (2005) proposed using both penalties.

### 30.2.1 Definition and basic properties

#### Definition 30.2.1 — Elastic net

For  $\lambda_1, \lambda_2 \geq 0$ , the **elastic net** criterion is

$$Q(\mathbf{b}) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda_1 \|\mathbf{b}\|_1 + \frac{1}{2} \lambda_2 \|\mathbf{b}\|^2, \quad (30.2.1)$$

and its minimizer  $\hat{\boldsymbol{\beta}}^{\text{en}}$  is the (naive) **elastic net estimate**. An equivalent parameterization is  $\lambda_1 = \alpha\lambda$ ,  $\lambda_2 = (1 - \alpha)\lambda$ , with a mixing parameter  $\alpha \in [0, 1]$  and an overall level  $\lambda \geq 0$ .

With  $\lambda_2 = 0$  this is the lasso, and with  $\lambda_1 = 0$  ridge. Here the columns of  $\mathbf{X}$  are centred with unit length and  $\mathbf{y}$  is centred, so  $\rho_{ij} = \mathbf{x}_i^\top \mathbf{x}_j$  is a sample correlation.

#### Proposition 30.2.1 — Elastic net: uniqueness, augmented data, grouping

Let  $\lambda_2 > 0$ .

- (a) The criterion [Eq. 30.2.1](#) has a unique minimizer, whatever the rank of  $\mathbf{X}$ . It is characterized by

$$\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) - \lambda_2 \hat{\beta}_j = \lambda_1 \text{sign}(\hat{\beta}_j) \quad (\hat{\beta}_j \neq 0), \quad |\mathbf{x}_j^\top (\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})| \leq \lambda_1 \quad (\hat{\beta}_j = 0). \quad (30.2.2)$$

- (b) (*Augmented data.*) Let  $\mathbf{X}^* = \begin{pmatrix} \mathbf{X} \\ \sqrt{\lambda_2} \mathbf{I}_p \end{pmatrix}$  and  $\mathbf{y}^* = \begin{pmatrix} \mathbf{y} \\ \mathbf{0} \end{pmatrix}$ . Then  $\hat{\boldsymbol{\beta}}^{\text{en}}$  is the lasso estimate, with parameter  $\lambda_1$ , for the data  $(\mathbf{X}^*, \mathbf{y}^*)$ . The matrix  $\mathbf{X}^*$  has full column rank  $p$ .
- (c) (*Orthonormal design.*) If  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ , then  $\hat{\beta}_j^{\text{en}} = \text{sign}(z_j)(|z_j| - \lambda_1)_+ / (1 + \lambda_2)$ , with  $\mathbf{z} = \mathbf{X}^\top \mathbf{y}$ .

(d) (*Grouping effect.*) If  $\hat{\beta}_i \hat{\beta}_j > 0$ , then

$$|\hat{\beta}_i - \hat{\beta}_j| \leq \frac{\|\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}\| \|\mathbf{x}_i - \mathbf{x}_j\|}{\lambda_2} \leq \frac{\|\mathbf{y}\| \sqrt{2(1 - \rho_{ij})}}{\lambda_2}. \quad (30.2.3)$$

(e) (*Identical columns.*) If  $\mathbf{x}_i = \mathbf{x}_j$ , then  $\hat{\beta}_i = \hat{\beta}_j$ .

*Proof.*

- (a) The penalty  $\lambda_1 \|\mathbf{b}\|_1 + \frac{1}{2} \lambda_2 \|\mathbf{b}\|^2$  is strictly convex, because  $\|\mathbf{b}\|^2$  is strictly convex and  $\|\mathbf{b}\|_1$  is convex. It is at least  $\frac{1}{2} \lambda_2 \|\mathbf{b}\|^2$ , so  $Q \rightarrow \infty$ . Existence and uniqueness follow as in [Theorem 30.1.1](#)(a) and (c). By [Lemma 30.1.2](#) applied to the whole penalty, with the gradient  $\lambda_2 \mathbf{b}$  of the smooth part moved to the left-hand side, and [Lemma 30.1.3](#), the minimizer is characterized by [Eq. 30.2.2](#).
- (b)  $\|\mathbf{y}^* - \mathbf{X}^* \mathbf{b}\|^2 = \|\mathbf{y} - \mathbf{X} \mathbf{b}\|^2 + \lambda_2 \|\mathbf{b}\|^2$ . So  $\frac{1}{2} \|\mathbf{y}^* - \mathbf{X}^* \mathbf{b}\|^2 + \lambda_1 \|\mathbf{b}\|_1$  is [Eq. 30.2.1](#). The last  $p$  rows of  $\mathbf{X}^*$  are  $\sqrt{\lambda_2} \mathbf{I}_p$ , which has rank  $p$ .
- (c) With  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ , [Eq. 30.2.1](#) equals  $\sum_j \{\frac{1}{2}(1 + \lambda_2)b_j^2 - z_j b_j + \lambda_1 |b_j|\}$  plus a constant. Dividing the  $j$ th term by  $1 + \lambda_2$  gives  $\frac{1}{2}\{b_j - z_j/(1 + \lambda_2)\}^2 + \{\lambda_1/(1 + \lambda_2)\}|b_j|$  plus a constant. By the univariate lasso solution ([Theorem 27.4.2](#)), this is minimized by soft thresholding  $z_j/(1 + \lambda_2)$  at  $\lambda_1/(1 + \lambda_2)$ , which is the stated formula.
- (d) Both coefficients are nonzero with a common sign  $s$ , so [Eq. 30.2.2](#) gives  $\mathbf{x}_i^\top \mathbf{r} - \lambda_2 \hat{\beta}_i = \lambda_1 s = \mathbf{x}_j^\top \mathbf{r} - \lambda_2 \hat{\beta}_j$ , where  $\mathbf{r} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$ . Subtracting,  $\lambda_2(\hat{\beta}_i - \hat{\beta}_j) = (\mathbf{x}_i - \mathbf{x}_j)^\top \mathbf{r}$ , and Cauchy–Schwarz gives the first inequality. For the second,  $\|\mathbf{x}_i - \mathbf{x}_j\|^2 = 2 - 2\rho_{ij}$  for unit-length columns, and  $\frac{1}{2}\|\mathbf{r}\|^2 \leq Q(\hat{\boldsymbol{\beta}}) \leq Q(\mathbf{0}) = \frac{1}{2}\|\mathbf{y}\|^2$ .
- (e) Swapping coordinates  $i$  and  $j$  of  $\hat{\boldsymbol{\beta}}$  changes neither  $\mathbf{X}\hat{\boldsymbol{\beta}}$  nor the penalty. So the swapped vector is also a minimizer, and by uniqueness it equals  $\hat{\boldsymbol{\beta}}$ .

■

By (b), the elastic net is a lasso on  $n + p$  observations with a full-rank design, so it can select all  $p$  regressors even when  $p > n$ ; a unique lasso solution has at most  $\min(n, p)$  nonzero coefficients ([Exercise \(C1\)](#)). Part (d) is the grouping effect of Zou and Hastie (2005): as  $\rho_{ij} \rightarrow 1$ , coefficients of the same sign are forced together at a rate set by  $\lambda_2$ . No such bound holds for the lasso.

### Example. (Three near-copies of one signal)

The script simulates  $n = 50$  observations of  $p = 10$  regressors. The first three are noisy copies of a common variable, with pairwise correlations between 0.93 and 0.95, and each has coefficient 2 (on the unit-length scale). The other seven are independent noise. With  $\lambda_1 = 0.8$ :

|                           | $\hat{\beta}_1$ | $\hat{\beta}_2$ | $\hat{\beta}_3$ | nonzero |
|---------------------------|-----------------|-----------------|-----------------|---------|
| lasso ( $\lambda_2 = 0$ ) | 2.335           | 0.000           | 2.522           | 2       |
| elastic net               | 1.248           | 1.103           | 1.236           | 4       |
| ( $\lambda_2 = 1$ )       |                 |                 |                 |         |
| rescaled elastic net      | 2.496           | 2.207           | 2.471           | 4       |

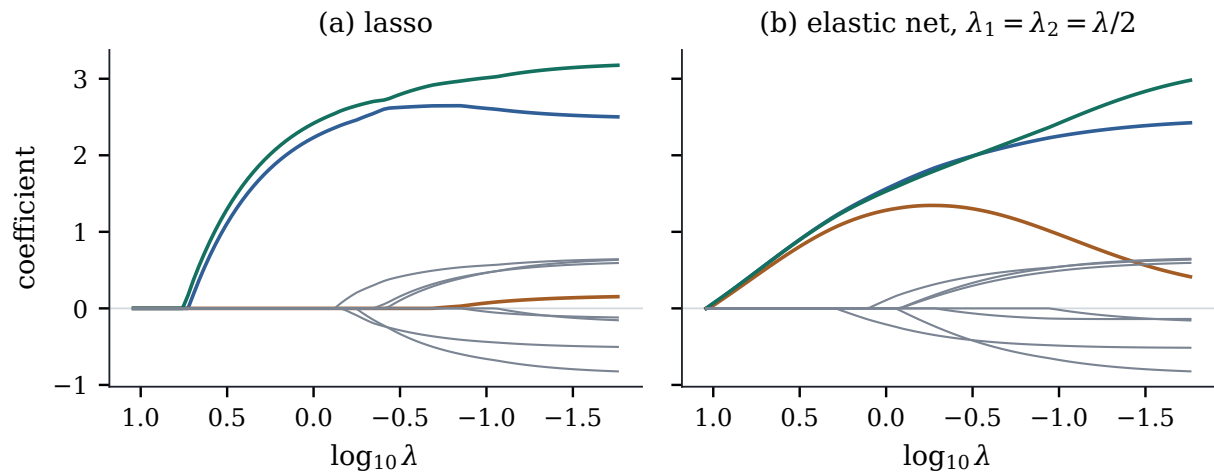


Figure 30.2.1: Coefficient paths for the data of [Example \(Three near-copies of one signal\)](#), plotted against  $\log_{10} \lambda$ , with  $\lambda$  decreasing to the right. The three correlated regressors are coloured and the seven noise regressors are grey. (a) The lasso. (b) The elastic net with  $\alpha = \frac{1}{2}$ , that is,  $\lambda_1 = \lambda_2 = \lambda/2$ .

The lasso drops the second copy. The elastic net keeps all three with similar coefficients (and one noise regressor, at 0.081). For the pair (1, 2), with correlation 0.927, the spread 0.144 is well inside [Eq. 30.2.3](#), whose middle term is 1.577 (residual norm 4.129) and right-hand side 2.532 ( $\|\mathbf{y}\| = 6.630$ ). Along the lasso path ([Figure 30.2.1](#)) the second copy enters only after  $\lambda$  has fallen by a factor of about 26; along the elastic net path all three enter at the same grid value.

## 30.2.2 Computing it: coordinate descent

**Cyclic coordinate descent**, used for the lasso in [Lemma 27.4.4](#), minimizes over one coordinate at a time, the others fixed, and sweeps until nothing changes.

### Proposition 30.2.2 — Coordinate update

Fix all coordinates of  $\mathbf{b}$  except  $b_j$ , and let  $\mathbf{r}_{(j)} = \mathbf{y} - \sum_{k \neq j} \mathbf{x}_k b_k$  be the partial residual. The value of  $b_j$  that minimizes [Eq. 30.2.1](#) is

$$b_j = \frac{\text{sign}(\mathbf{x}_j^\top \mathbf{r}_{(j)}) (|\mathbf{x}_j^\top \mathbf{r}_{(j)}| - \lambda_1)_+}{\|\mathbf{x}_j\|^2 + \lambda_2}. \quad (30.2.4)$$

*Proof.* As a function of  $b_j$  alone, [Eq. 30.2.1](#) equals  $\frac{1}{2}(\|\mathbf{x}_j\|^2 + \lambda_2)b_j^2 - b_j \mathbf{x}_j^\top \mathbf{r}_{(j)} + \lambda_1 |b_j|$  plus a constant. Divided by  $\|\mathbf{x}_j\|^2 + \lambda_2$ , this is a univariate lasso problem of the form in the proof of [Proposition 30.2.1\(c\)](#). Its solution is [Eq. 30.2.4](#). ■

Because the nonsmooth part is separable, every limit point of the iterates is a minimizer (Tseng, 2001; cited, not proved), so the iterates converge when the minimizer is unique, as for the elastic



net with  $\lambda_2 > 0$ . For a coupling penalty such as  $\sum_j |b_j - b_{j-1}|$ , coordinate descent can stall. Whole paths are computed on a decreasing grid of  $\lambda$ , each fit starting from the last (Friedman, Hastie, Höfling and Tibshirani, 2007; Friedman, Hastie and Tibshirani, 2010). The listing works through the Gram matrix, as the next proposition requires.

```
import numpy as np

def soft(z, lam):
    return np.sign(z) * np.maximum(np.abs(z) - lam, 0.0)

def cd_quadratic(G, c, lam1, b=None, tol=1e-12, max_sweeps=100000):
    """Minimize (1/2) b'Gb - c'b + lam1 ||b||_1 by cyclic coordinate descent."""
    p = len(c)
    b = np.zeros(p) if b is None else b.copy()
    for _ in range(max_sweeps):
        largest = 0.0
        for j in range(p):
            old = b[j]
            partial = c[j] - G[j] @ b + G[j, j] * old      # c_j - sum_{k != j} G_jk
            b_k
            b[j] = soft(partial, lam1) / G[j, j]
            largest = max(largest, abs(b[j] - old))
        if largest < tol:
            return b
    raise RuntimeError("coordinate descent did not converge")

def elastic_net(X, y, lam1, lam2, b=None):
    """The (naive) elastic net: (1/2)||y - Xb||^2 + lam1 ||b||_1 + (lam2/2)||b||^2.
    """
    G = X.T @ X + lam2 * np.eye(X.shape[1])
    return cd_quadratic(G, X.T @ y, lam1, b)

rng = np.random.default_rng(3002)
n, p = 50, 10
z = rng.normal(size=n)
X = rng.normal(size=(n, p))
X[:, :3] = z[:, None] + 0.3 * rng.normal(size=(n, 3))    # three near-copies of one
                  signal
X -= X.mean(axis=0)
X /= np.linalg.norm(X, axis=0)                          # centred, unit-length
                  columns
beta = np.r_[2.0, 2.0, 2.0, np.zeros(p - 3)]
y = X @ beta + 0.5 * rng.normal(size=n)
y -= y.mean()

lam1 = 0.8
b_lasso = elastic_net(X, y, lam1, 0.0)
b_naive = elastic_net(X, y, lam1, 1.0)
print("lasso          ", np.round(b_lasso[:5], 3))
print("naive EN       ", np.round(b_naive[:5], 3))
print("rescaled EN    ", np.round(2.0 * b_naive[:5], 3))    # (1 + lam2) * naive
```

### 30.2.3 Double shrinkage and the rescaled elastic net

By [Proposition 30.2.1\(c\)](#), under an orthonormal design the elastic net soft-thresholds and then divides by  $1 + \lambda_2$ : surviving coefficients are shrunk twice. Zou and Hastie (2005) proposed reporting  $(1 + \lambda_2)\hat{\boldsymbol{\beta}}^{\text{en}}$  instead, which in general is a lasso with a modified Gram matrix.

#### Proposition 30.2.3 — The rescaled elastic net

Let the columns of  $\mathbf{X}$  have unit length and let  $\lambda_2 > 0$ . Then  $\tilde{\boldsymbol{\beta}} = (1 + \lambda_2)\hat{\boldsymbol{\beta}}^{\text{en}}$  is the unique minimizer of

$$\frac{1}{2} \mathbf{c}^\top \boldsymbol{\Sigma}_{\lambda_2} \mathbf{c} - \mathbf{y}^\top \mathbf{X} \mathbf{c} + \lambda_1 \|\mathbf{c}\|_1, \quad \boldsymbol{\Sigma}_{\lambda_2} = \frac{\mathbf{X}^\top \mathbf{X} + \lambda_2 \mathbf{I}}{1 + \lambda_2}. \quad (30.2.5)$$

*Proof.* Up to the constant  $\frac{1}{2} \mathbf{y}^\top \mathbf{y}$ , [Eq. 30.2.1](#) is  $\frac{1}{2} \mathbf{b}^\top (\mathbf{X}^\top \mathbf{X} + \lambda_2 \mathbf{I}) \mathbf{b} - \mathbf{y}^\top \mathbf{X} \mathbf{b} + \lambda_1 \|\mathbf{b}\|_1$ . Substitute  $\mathbf{b} = \mathbf{c}/(1 + \lambda_2)$  and multiply by  $1 + \lambda_2 > 0$ . The result is [Eq. 30.2.5](#). The substitution is a bijection, so minimizers correspond, and the minimizer of [Eq. 30.2.1](#) is unique by [Proposition 30.2.1\(a\)](#). ■

The lasso has  $\mathbf{X}^\top \mathbf{X}$  in place of  $\boldsymbol{\Sigma}_{\lambda_2}$ , whose correlations are divided by  $1 + \lambda_2$ : the rescaled elastic net is a lasso computed as if the regressors were less correlated. The rescaling is a heuristic; in [Example \(Three near-copies of one signal\)](#), where the true coefficients are 2, the naive estimates are about half that and the rescaled ones somewhat too large. The glmnet software (Friedman, Hastie and Tibshirani, 2010) fits the naive criterion.

### 30.2.4 Exercises

#### 30.2.4.1 A. Check your understanding

##### Exercise A1.

Under an orthonormal design with  $\mathbf{z} = \mathbf{X}^\top \mathbf{y} = (3, 0.5, -1.2)^\top$ ,  $\lambda_1 = 1$  and  $\lambda_2 = 1$ , compute the lasso, ridge, naive elastic net and rescaled elastic net estimates.

*Solution.* Lasso ( $\lambda = 1$ ):  $(2, 0, -0.2)$ . Ridge ( $\lambda = 1$ ):  $\mathbf{z}/2 = (1.5, 0.25, -0.6)$ . Naive elastic net: the lasso divided by 2,  $(1, 0, -0.1)$ . Rescaled: multiply by  $1 + \lambda_2 = 2$ , which gives the lasso,  $(2, 0, -0.2)$ .

#### 30.2.4.2 B. Practice

##### Exercise B1.

Show that if  $\hat{\beta}_i \hat{\beta}_j < 0$ , then  $|\hat{\beta}_i + \hat{\beta}_j| \leq \|\mathbf{y}\| \sqrt{2(1 + \rho_{ij})}/\lambda_2$ . Interpret this for two

regressors with correlation close to  $-1$ .

*Solution.* With signs  $s$  and  $-s$ , Eq. 30.2.2 gives  $\mathbf{x}_i^\top \mathbf{r} - \lambda_2 \hat{\beta}_i = \lambda_1 s$  and  $\mathbf{x}_j^\top \mathbf{r} - \lambda_2 \hat{\beta}_j = -\lambda_1 s$ . Adding,  $\lambda_2(\hat{\beta}_i + \hat{\beta}_j) = (\mathbf{x}_i + \mathbf{x}_j)^\top \mathbf{r}$ , and  $\|\mathbf{x}_i + \mathbf{x}_j\|^2 = 2 + 2\rho_{ij}$ . Bound  $\|\mathbf{r}\| \leq \|\mathbf{y}\|$  as before. If  $\rho_{ij} \approx -1$ , then  $\mathbf{x}_j \approx -\mathbf{x}_i$ , and the bound forces  $\hat{\beta}_j \approx -\hat{\beta}_i$ . The two regressors are treated as one regressor with a sign flip.

### Exercise B2.

Let  $\mathbf{X} = [\mathbf{x}, \mathbf{x}]$  with  $\|\mathbf{x}\| = 1$ . Find the elastic net estimate in closed form and compare it with the ridge and lasso estimates.

### 30.2.4.3 C. Going deeper

### Exercise C1.

Suppose  $p > n$  and the lasso estimate  $\hat{\boldsymbol{\beta}}$  is unique with active set  $A$ . Use the KKT conditions to show that the columns  $\mathbf{X}_A$  are linearly independent, and deduce that  $|A| \leq n$ . (Hint: if  $\mathbf{X}_A \mathbf{d} = \mathbf{0}$  with  $\mathbf{d} \neq \mathbf{0}$ , move along  $\hat{\boldsymbol{\beta}} + \tau(\mathbf{d}, \mathbf{0})$  and use that the fit and the  $\ell_1$  norm are constant among minimizers.)

## 30.3 Group penalties

A factor with  $k$  levels enters a linear model as  $k - 1$  indicator columns (Chapter 15), and a smooth effect as several basis columns. The question is whether *the factor* matters. The lasso on the indicator columns can keep some levels and drop others, and its answer depends on the coding. Yuan and Lin (2006) proposed a penalty that treats a block of columns as a unit.

### 30.3.1 The group lasso

Partition the columns of  $\mathbf{X}$  into groups  $g = 1, \dots, G$ , with  $\mathbf{X}_g$  the  $n \times p_g$  block of group  $g$  and  $\mathbf{b}_g$  the matching block of coefficients, so that  $\mathbf{X}\mathbf{b} = \sum_g \mathbf{X}_g \mathbf{b}_g$ .

#### Definition 30.3.1 — Group lasso

For weights  $w_g > 0$ , the **group lasso** criterion is

$$Q(\mathbf{b}) = \frac{1}{2} \left\| \mathbf{y} - \sum_{g=1}^G \mathbf{X}_g \mathbf{b}_g \right\|^2 + \lambda \sum_{g=1}^G w_g \|\mathbf{b}_g\|, \quad (30.3.1)$$

with Euclidean norms  $\|\mathbf{b}_g\|$ , not squared. The usual weights are  $w_g = \sqrt{p_g}$ . The columns of each group are usually centred and **orthonormalized within the group** before fitting, so that  $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}_{p_g}$ . Different groups are not made orthogonal to each

other.

With singleton groups and unit weights this is the lasso. In general it acts like the lasso *across* groups and like ridge *within* them:  $\|\mathbf{b}_g\|$  has a corner only at  $\mathbf{b}_g = \mathbf{0}$ , so a group is either absent or present with, typically, all its coefficients nonzero.

### Proposition 30.3.1 — Group lasso and block soft thresholding

- (a) A minimizer of Eq. 30.3.1 exists, and all minimizers have the same fitted vector.  
 (b) (*KKT conditions*.)  $\hat{\boldsymbol{\beta}}$  is a minimizer iff, with  $\mathbf{r} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$ , for every group  $g$ ,

$$\mathbf{X}_g^\top \mathbf{r} = \lambda w_g \frac{\hat{\boldsymbol{\beta}}_g}{\|\hat{\boldsymbol{\beta}}_g\|} \quad \text{if } \hat{\boldsymbol{\beta}}_g \neq \mathbf{0}, \quad \|\mathbf{X}_g^\top \mathbf{r}\| \leq \lambda w_g \quad \text{if } \hat{\boldsymbol{\beta}}_g = \mathbf{0}. \quad (30.3.2)$$

- (c) Let  $\lambda > 0$ . Then  $\hat{\boldsymbol{\beta}} = \mathbf{0}$  iff  $\lambda \geq \lambda_{\max} = \max_g \|\mathbf{X}_g^\top \mathbf{y}\|/w_g$ .  
 (d) (*Orthonormal groups*.) If  $\mathbf{X}_g^\top \mathbf{X}_h = \mathbf{I}_{p_g}$  for  $g = h$  and  $\mathbf{0}$  for  $g \neq h$ , the unique minimizer is **block soft thresholding** of  $\mathbf{z}_g = \mathbf{X}_g^\top \mathbf{y}$ :

$$\hat{\boldsymbol{\beta}}_g = \left(1 - \frac{\lambda w_g}{\|\mathbf{z}_g\|}\right)_+ \mathbf{z}_g. \quad (30.3.3)$$

- (e) (*Block update; compare Proposition 30.2.2*.) If  $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}_{p_g}$ , then minimizing Eq. 30.3.1 over  $\mathbf{b}_g$  with the other blocks fixed gives Eq. 30.3.3 with  $\mathbf{z}_g$  replaced by  $\mathbf{s}_g = \mathbf{X}_g^\top \mathbf{r}_{(g)}$ , where  $\mathbf{r}_{(g)} = \mathbf{y} - \sum_{h \neq g} \mathbf{X}_h \mathbf{b}_h$  is the partial residual.

*Proof.*

- (a) The penalty is convex and continuous. Since  $\sum_g \|\mathbf{b}_g\| \geq \|\mathbf{b}\|$ , it is at least  $(\min_g w_g) \|\mathbf{b}\|$ . Apply Theorem 30.1.1(a)(ii) and (b).  
 (b) Combine Lemma 30.1.2 with Lemma 30.1.3, using the groups as the blocks.  
 (c) At  $\hat{\boldsymbol{\beta}} = \mathbf{0}$ , the residual is  $\mathbf{y}$ , and Eq. 30.3.2 says that  $\|\mathbf{X}_g^\top \mathbf{y}\| \leq \lambda w_g$  for every  $g$ . If this holds,  $\mathbf{0}$  is a minimizer. Every minimizer then has fitted vector  $\mathbf{0}$  and, by Theorem 30.1.1(b), penalty 0, so it is  $\mathbf{0}$ . If it fails,  $\mathbf{0}$  is not a minimizer.  
 (d) Under the assumption,  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 = \|\mathbf{y}\|^2 - 2 \sum_g \mathbf{z}_g^\top \mathbf{b}_g + \sum_g \|\mathbf{b}_g\|^2$ . So Eq. 30.3.1 separates into the problems  $\min_{\mathbf{b}_g} \{\frac{1}{2} \|\mathbf{z}_g - \mathbf{b}_g\|^2 + \lambda w_g \|\mathbf{b}_g\|\}$ , each strictly convex. Let  $c = \lambda w_g$  and let  $\mathbf{b}_g$  be given by Eq. 30.3.3. If  $\|\mathbf{z}_g\| \leq c$ , then  $\mathbf{b}_g = \mathbf{0}$  and  $\|\mathbf{z}_g - \mathbf{0}\| \leq c$ . If  $\|\mathbf{z}_g\| > c$ , then  $\mathbf{b}_g = (\|\mathbf{z}_g\| - c) \mathbf{z}_g / \|\mathbf{z}_g\| \neq \mathbf{0}$  and  $\mathbf{z}_g - \mathbf{b}_g = c \mathbf{z}_g / \|\mathbf{z}_g\| = c \mathbf{b}_g / \|\mathbf{b}_g\|$ . In both cases the one-group version of Eq. 30.3.2 holds.  
 (e) As a function of  $\mathbf{b}_g$ , the criterion is  $\frac{1}{2} \|\mathbf{r}_{(g)} - \mathbf{X}_g \mathbf{b}_g\|^2 + \lambda w_g \|\mathbf{b}_g\|$  plus a constant. With  $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}$ , this is the one-group problem of (d) with  $\mathbf{z}_g$  replaced by  $\mathbf{s}_g$ .

■

Block soft thresholding keeps the direction of the least squares vector  $\mathbf{z}_g$  and reduces its length by  $\lambda w_g$ : the lasso rule applied to a length. The weights come from (c): for pure noise and orthonormal groups,  $\|\mathbf{X}_g^\top \mathbf{y}\|^2 / \sigma^2 \sim \chi^2(p_g)$  has mean  $p_g$ , so without  $\sqrt{p_g}$  large groups would enter merely for being large (Exercise (A1)). Block coordinate descent with update (e) has every limit point a minimizer because the penalty is separable across blocks (Tseng, 2001), so it converges when the minimizer is unique; without orthonormality the block problem has no

closed form ([Exercise \(C1\)](#)).

## 30.3.2 Factors as groups

Orthonormalizing within groups also makes the group lasso independent of how a factor is coded.

### Proposition 30.3.2 — Invariance to the coding of a group

- (a) Let  $\mathbf{U}_g$  be  $p_g \times p_g$  orthogonal matrices, and replace every  $\mathbf{X}_g$  by  $\mathbf{X}_g \mathbf{U}_g$ . Then  $\hat{\boldsymbol{\beta}}$  minimizes the original criterion iff the vector with blocks  $\mathbf{U}_g^\top \hat{\boldsymbol{\beta}}_g$  minimizes the new one. The fitted contributions  $\mathbf{X}_g \hat{\boldsymbol{\beta}}_g$  are the same.
- (b) Suppose each group is orthonormalized within itself, so that  $\mathbf{X}_g$  is an orthonormal basis of a subspace  $\mathcal{V}_g$ . Then the minimizers, expressed through the fitted contributions  $\mathbf{X}_g \hat{\boldsymbol{\beta}}_g \in \mathcal{V}_g$ , depend on the groups only through the subspaces  $\mathcal{V}_g$ . In particular, for a factor they do not depend on the choice of reference level or of contrasts.

*Proof.*

- (a) With  $\mathbf{b}_g = \mathbf{U}_g \mathbf{c}_g$ , we have  $\mathbf{X}_g \mathbf{U}_g \mathbf{c}_g = \mathbf{X}_g \mathbf{b}_g$  and  $\|\mathbf{c}_g\| = \|\mathbf{b}_g\|$ . So the new criterion at  $\mathbf{c}$  equals the old criterion at  $\mathbf{b}$ , and  $\mathbf{b} \leftrightarrow \mathbf{c}$  is a bijection.
- (b) Two orthonormal bases  $\mathbf{X}_g$  and  $\mathbf{X}'_g$  of  $\mathcal{V}_g$  satisfy  $\mathbf{X}'_g = \mathbf{X}_g \mathbf{U}_g$  with  $\mathbf{U}_g = \mathbf{X}_g^\top \mathbf{X}'_g$  orthogonal. Apply (a). For a factor with  $k$  levels, the centred indicator columns of any reference coding, and the centred columns of any other full-rank contrast coding, span the same  $(k-1)$ -dimensional subspace: the vectors that are constant within levels and orthogonal to  $\mathbf{1}$ .

■

With orthonormalized groups  $\|\mathbf{b}_g\| = \|\mathbf{X}_g \mathbf{b}_g\|$ : the penalty is the total length of the groups' fitted contributions, a coding-free quantity. Simon and Tibshirani (2012) argue for this version.

### Example. (Selecting factors)

The script simulates  $n = 120$  observations of four factors  $A, B, C, D$  (with 4, 3, 5 and 3 levels) and three numeric regressors; the response depends on  $A$ , weakly on  $B$ , and on  $x_1$ . The group lasso path starts at  $\lambda_{\max} = 6.876$ , and groups enter in the order  $x_1, A, B, D$ . At  $\lambda = 2.0$  only  $A$  and  $x_1$  are selected, with  $\|\mathbf{X}_A \hat{\boldsymbol{\beta}}_A\| = 2.834$  and  $\|\mathbf{X}_{x_1} \hat{\boldsymbol{\beta}}_{x_1}\| = 4.861$ . Another reference level gives the same fitted contributions to machine precision. The lasso on the raw indicator columns does not: at  $\lambda = 6.0$  its fit moves by 8.6 per cent of its length when the reference levels change. Along its path the columns of  $A$  enter at values of  $\lambda$  spread over a factor of 2.8, and it admits its first column of a null factor ( $C$  or  $D$ ) at a residual sum of squares of 112.0, a worse fit than the 110.7 that the group lasso reaches with both null factors still excluded ([Figure 30.3.1](#)).

```
import numpy as np
```

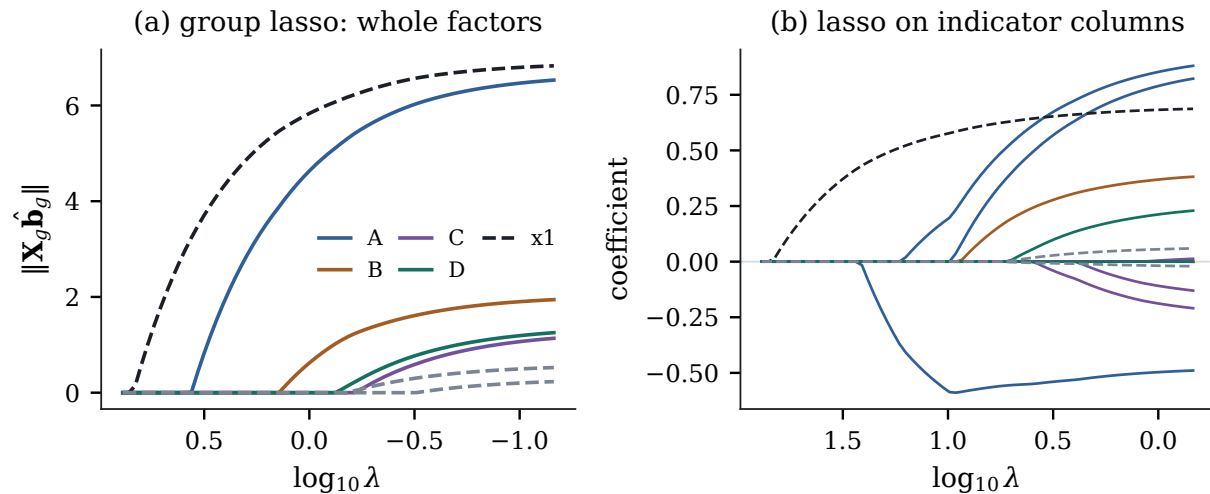


Figure 30.3.1: (a) Group lasso path for [Example \(Selecting factors\)](#): the length of each group's fitted contribution  $\|X_g \hat{b}_g\|$  against  $\log_{10} \lambda$ . Factors enter as wholes. (b) The lasso path for the same data, with each indicator column penalized separately, coloured by factor. The dashed curves are the numeric regressors.

```

rng = np.random.default_rng(3003)
n = 120
levels = {"A": 4, "B": 3, "C": 5, "D": 3}
factors = {f: rng.integers(0, k, size=n) for f, k in levels.items()}
numeric = rng.normal(size=(n, 3))
effect_A = np.array([0.0, 1.0, -0.5, 0.8])
effect_B = np.array([0.0, 0.3, -0.3])
y = effect_A[factors["A"]] + effect_B[factors["B"]] + 0.6 * numeric[:, 0] + rng.
    normal(size=n)
y -= y.mean()

def indicator_columns(codes, k, reference=0):
    """Indicator columns for the levels other than the reference level."""
    return np.column_stack([(codes == l).astype(float) for l in range(k) if l !=
        reference])

def orthonormal_group(Z):
    """Centre the columns, then an orthonormal basis of their span (Z = Q R)."""
    Z = Z - Z.mean(axis=0)
    Q, _ = np.linalg.qr(Z)
    return Q

def build(reference=0):
    blocks = [orthonormal_group(indicator_columns(factors[f], k, reference))
        for f, k in levels.items()]
    blocks += [orthonormal_group(numeric[:, [j]]) for j in range(3)]
    return blocks

def group_lasso(blocks, y, lam, b=None, tol=1e-12, max_sweeps=100000):
    """(1/2)||y - sum_g X_g b_g||^2 + lam sum_g sqrt(p_g) ||b_g||, with X_g'X_g = I.
    """
    b = [np.zeros(Xg.shape[1]) for Xg in blocks] if b is None else [v.copy() for v
        in b]
    r = y - sum(Xg @ bg for Xg, bg in zip(blocks, b))
    for _ in range(max_sweeps):
        largest = 0.0

```

```

    for g, Xg in enumerate(blocks):
        s = Xg.T @ r + b[g]                                # block least squares on the
        partial residual
        w = np.sqrt(Xg.shape[1])
        norm_s = np.linalg.norm(s)
        new = max(0.0, 1 - lam * w / norm_s) * s if norm_s > 0 else 0 * s
        r += Xg @ (b[g] - new)
        largest = max(largest, np.max(np.abs(new - b[g])))
        b[g] = new
    if largest < tol:
        return b
    raise RuntimeError("block coordinate descent did not converge")

blocks = build()
names = list(levels) + ["x1", "x2", "x3"]
lam = 2.0
b = group_lasso(blocks, y, lam)
for name, Xg, bg in zip(names, blocks, b):
    print(f"{name}: {Xg.shape[1]} columns, ||X_g b_g|| = {np.linalg.norm(Xg @ bg):.3f}")

```

The **sparse group lasso** adds an  $\ell_1$  term so that a selected group can have zero coefficients (Simon, Friedman, Hastie and Tibshirani, 2013).

### 30.3.3 Ordered coefficients: the fused lasso

When coefficients have a natural order (levels of an ordered factor, successive times), neighbours are expected to be similar. The **fused lasso** of Tibshirani, Saunders, Rosset, Zhu and Knight (2005) penalizes their differences:

$$Q(\mathbf{b}) = \frac{1}{2} \|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2 + \lambda_1 \sum_{j=1}^p |b_j| + \lambda_2 \sum_{j=2}^p |b_j - b_{j-1}|. \quad (30.3.4)$$

It produces exact zeros and exact ties  $\hat{\beta}_j = \hat{\beta}_{j-1}$ , so the coefficient sequence is piecewise constant. The penalty is convex but not separable, so coordinate descent can stall at a tie. For an ordered factor, fusion merges adjacent levels the data cannot distinguish (Gertheiss and Tutz, 2010). [Exercise \(B1\)](#) solves the smallest case.

### 30.3.4 Exercises

#### 30.3.4.1 A. Check your understanding

##### Exercise A1.

Under orthonormal groups and  $\mathbf{y} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ , show that  $P(\hat{\boldsymbol{\beta}}_g \neq \mathbf{0}) = P\{\chi^2(p_g) > \lambda^2 w_g^2 / \sigma^2\}$ . With  $w_g = 1$ , compare these probabilities for  $p_g = 1$  and  $p_g = 10$  when  $\lambda = 2\sigma$ . What do the weights  $\sqrt{p_g}$  achieve, and what do they not achieve?

*Solution.* By [Eq. 30.3.3](#),  $\hat{\beta}_g \neq \mathbf{0}$  iff  $\|\mathbf{z}_g\| > \lambda w_g$ . Here  $\mathbf{z}_g = \mathbf{X}_g^\top \mathbf{y} \sim N_{p_g}(\mathbf{0}, \sigma^2 \mathbf{I})$ , so  $\|\mathbf{z}_g\|^2 / \sigma^2 \sim \chi^2(p_g)$ . With  $w_g = 1$  and  $\lambda = 2\sigma$ , the probabilities are  $P\{\chi^2(1) > 4\} = 0.046$  and  $P\{\chi^2(10) > 4\} = 0.947$ , so a large null group almost always enters. With  $w_g = \sqrt{p_g}$  the threshold is  $4p_g$ , four times the mean of  $\chi^2(p_g)$ . This removes the gross imbalance, but it does not equalize the probabilities: they are 0.046 and  $P\{\chi^2(10) > 40\} = 1.7 \times 10^{-5}$ . Equal null entry probabilities would need thresholds at a common quantile of  $\chi^2(p_g)$ .

### 30.3.4.2 B. Practice

#### Exercise B1.

Solve the fused lasso [Eq. 30.3.4](#) with  $\mathbf{X} = \mathbf{I}_2$ ,  $\lambda_1 = 0$  and  $\lambda_2 = \lambda$ . Show that  $\hat{\beta}_1 = \hat{\beta}_2 = \frac{1}{2}(y_1 + y_2)$  if  $|y_1 - y_2| \leq 2\lambda$ . Otherwise, each estimate moves a distance  $\lambda$  from its observation towards the other.

*Solution.* The penalty is  $\lambda|b_1 - b_2| = \lambda|\mathbf{a}^\top \mathbf{b}|$  with  $\mathbf{a} = (1, -1)^\top$ . Its subgradients are  $\lambda s \mathbf{a}$  with  $s \in \partial|\cdot|(b_1 - b_2)$ . By [Lemma 30.1.2](#),  $y_1 - b_1 = \lambda s$  and  $y_2 - b_2 = -\lambda s$ . If  $b_1 = b_2 = b$ , adding the equations gives  $b = \frac{1}{2}(y_1 + y_2)$ , and then  $s = (y_1 - y_2)/(2\lambda)$  must lie in  $[-1, 1]$ . If  $|y_1 - y_2| > 2\lambda$ , then  $b_1 \neq b_2$ ,  $s = \text{sign}(b_1 - b_2)$ , and  $b_1 = y_1 - \lambda s$ ,  $b_2 = y_2 + \lambda s$ . This is consistent when  $s = \text{sign}(y_1 - y_2)$ , since then  $b_1 - b_2 = y_1 - y_2 - 2\lambda s$  has the sign of  $s$ .

#### Exercise B2.

Under an orthonormal design, compare the group lasso (one group of size 3,  $w = \sqrt{3}$ ) with the lasso on the same three columns, for  $\mathbf{z} = (2, 0.3, -0.2)^\top$  and  $\lambda = 1$ . Which coefficients are zero in each? Show that, under an orthonormal design, the group lasso never sets a single coordinate of a selected group to zero unless the corresponding  $z_j = 0$ .

### 30.3.4.3 C. Going deeper

#### Exercise C1.

Drop the assumption  $\mathbf{X}_g^\top \mathbf{X}_g = \mathbf{I}$ . Show that a nonzero solution of the block problem  $\min_{\mathbf{b}} \frac{1}{2} \|\mathbf{r} - \mathbf{X}_g \mathbf{b}\|^2 + c \|\mathbf{b}\|$  is  $\mathbf{b} = (\mathbf{X}_g^\top \mathbf{X}_g + (c/t)\mathbf{I})^{-1} \mathbf{X}_g^\top \mathbf{r}$ , where  $t = \|\mathbf{b}\|$  is the root of a one-dimensional equation. Show that this equation has a unique root when  $\|\mathbf{X}_g^\top \mathbf{r}\| > c$ , so the block update is a ridge regression ([Theorem 27.2.2](#)) with a data-dependent parameter.

## 30.4 Componentwise boosting

Boosting is an algorithm, not a criterion: it improves the fit by many small steps, each changing the single coefficient that helps most, and the number of steps plays the part of a penalty. Begun as a way to combine classifiers (Freund and Schapire, 1997) and recast as gradient descent on a loss (Friedman, Hastie and Tibshirani, 2000; Friedman, 2001), with squared error and single regressors it is the  $L_2$  boosting of Bühlmann and Yu (2003).



### 30.4.1 The algorithm

Let  $\mathbf{y}$  and the columns of  $\mathbf{X}$  be centred, and let no column be zero. For a vector  $\mathbf{r}$ , the least squares fit of  $\mathbf{r}$  on the single column  $\mathbf{x}_j$  has coefficient  $\mathbf{x}_j^\top \mathbf{r} / \|\mathbf{x}_j\|^2$ . It reduces the sum of squares by  $(\mathbf{x}_j^\top \mathbf{r})^2 / \|\mathbf{x}_j\|^2$ , by the one-column case of [Theorem 6.4.1](#).

#### Definition 30.4.1 — Componentwise $L_2$ boosting

Fix a **step size**  $\nu \in (0, 2)$ . Start from  $\mathbf{b}^{(0)} = \mathbf{0}$  and  $\mathbf{r}^{(0)} = \mathbf{y}$ . For  $m = 0, 1, 2, \dots$ :

1. choose  $j_m = \arg \max_j (\mathbf{x}_j^\top \mathbf{r}^{(m)})^2 / \|\mathbf{x}_j\|^2$ , the column whose simple regression fits the current residual best;
2. set  $b_{j_m}^{(m+1)} = b_{j_m}^{(m)} + \nu \mathbf{x}_{j_m}^\top \mathbf{r}^{(m)} / \|\mathbf{x}_{j_m}\|^2$ , leave the other coordinates unchanged, and put  $\mathbf{r}^{(m+1)} = \mathbf{y} - \mathbf{X}\mathbf{b}^{(m+1)}$ .

The estimate after  $m$  steps is  $\mathbf{b}^{(m)}$ , and the **stopping iteration**  $m$  is the tuning parameter.

With unit-length columns, step 1 picks the column most correlated with the residual. The residual is the negative gradient of  $\frac{1}{2}\|\mathbf{y} - \mathbf{f}\|^2$  in the fitted vector  $\mathbf{f}$ , so each step fits the negative gradient by the best single column and moves a fraction  $\nu$  of the way. The theory below allows  $\nu < 2$ ; with  $\nu > 1$  a step overshoots the simple regression fit, and in practice  $\nu \leq 1$ , often  $\nu = 0.1$ . With  $\nu = 1$  this is *matching pursuit* (Mallat and Zhang, 1993), which is not forward selection, since earlier coefficients are never refitted.

### 30.4.2 Convergence to least squares

Boosting converges to the least squares fit at a geometric rate, slow when the design is badly conditioned.

#### Theorem 30.4.1 — Convergence of componentwise $L_2$ boosting

Let  $\nu \in (0, 2)$ , let  $Q(\mathbf{b}) = \frac{1}{2}\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  with minimum  $Q^*$ , let  $s > 0$  be the smallest nonzero singular value of  $\mathbf{X}$ , and put

$$\kappa = \frac{\nu(2-\nu)s^2}{p \max_j \|\mathbf{x}_j\|^2}.$$

Then  $0 < \kappa \leq 1$ , and:

- (a) each step lowers the residual sum of squares by exactly

$$Q(\mathbf{b}^{(m)}) - Q(\mathbf{b}^{(m+1)}) = \nu \left(1 - \frac{\nu}{2}\right) \max_j \frac{(\mathbf{x}_j^\top \mathbf{r}^{(m)})^2}{\|\mathbf{x}_j\|^2}; \quad (30.4.1)$$

- (b)  $Q(\mathbf{b}^{(m)}) - Q^* \leq (1 - \kappa)^m (Q(\mathbf{0}) - Q^*)$ . Hence the fitted values converge geometrically to the least squares fit:  $\|\mathbf{X}\mathbf{b}^{(m)} - \mathbf{M}\mathbf{y}\|^2 = 2\{Q(\mathbf{b}^{(m)}) - Q^*\} \rightarrow 0$ ;

- (c) the coefficients  $\mathbf{b}^{(m)}$  converge to a least squares estimate. If  $\mathbf{X}$  has full column rank, the limit is  $\hat{\boldsymbol{\beta}}$ .

*Proof.*

- (a) Write  $\mathbf{r} = \mathbf{r}^{(m)}$ ,  $j = j_m$  and  $c = \mathbf{x}_j^\top \mathbf{r} / \|\mathbf{x}_j\|^2$ . Then  $\mathbf{r}^{(m+1)} = \mathbf{r} - \nu c \mathbf{x}_j$ , and

$$\begin{aligned} \|\mathbf{r} - \nu c \mathbf{x}_j\|^2 &= \|\mathbf{r}\|^2 - 2\nu c \mathbf{x}_j^\top \mathbf{r} + \nu^2 c^2 \|\mathbf{x}_j\|^2 \\ &= \|\mathbf{r}\|^2 - (2\nu - \nu^2) \frac{(\mathbf{x}_j^\top \mathbf{r})^2}{\|\mathbf{x}_j\|^2}. \end{aligned}$$

Halving gives [Eq. 30.4.1](#), since  $j$  maximizes the ratio.

- (b) Let  $\mathbf{u}_m = \mathbf{M}\mathbf{r}^{(m)} = \mathbf{M}\mathbf{y} - \mathbf{X}\mathbf{b}^{(m)}$ , which lies in  $\mathcal{C}(\mathbf{X})$ . The residual splits orthogonally as  $\mathbf{r}^{(m)} = (\mathbf{I} - \mathbf{M})\mathbf{y} + \mathbf{u}_m$ , so  $Q(\mathbf{b}^{(m)}) - Q^* = \frac{1}{2}\|\mathbf{u}_m\|^2$  and  $\mathbf{X}^\top \mathbf{r}^{(m)} = \mathbf{X}^\top \mathbf{u}_m$ . By the singular value decomposition ([Theorem 1.8.1](#)),  $\|\mathbf{X}^\top \mathbf{u}\| \geq s\|\mathbf{u}\|$  for every  $\mathbf{u} \in \mathcal{C}(\mathbf{X})$ . A maximum is at least an average, so

$$\begin{aligned} \max_j \frac{(\mathbf{x}_j^\top \mathbf{r}^{(m)})^2}{\|\mathbf{x}_j\|^2} &\geq \frac{\sum_j (\mathbf{x}_j^\top \mathbf{u}_m)^2}{p \max_j \|\mathbf{x}_j\|^2} \\ &= \frac{\|\mathbf{X}^\top \mathbf{u}_m\|^2}{p \max_j \|\mathbf{x}_j\|^2} \geq \frac{s^2 \|\mathbf{u}_m\|^2}{p \max_j \|\mathbf{x}_j\|^2}. \end{aligned}$$

With (a),  $Q(\mathbf{b}^{(m)}) - Q(\mathbf{b}^{(m+1)}) \geq \kappa\{Q(\mathbf{b}^{(m)}) - Q^*\}$ , that is,  $Q(\mathbf{b}^{(m+1)}) - Q^* \leq (1 - \kappa)\{Q(\mathbf{b}^{(m)}) - Q^*\}$ . Iterating gives (b). Finally  $\kappa > 0$ , and  $\kappa \leq 1$  because  $\nu(2 - \nu) \leq 1$  and  $s^2 \leq \text{tr}(\mathbf{X}^\top \mathbf{X}) = \sum_j \|\mathbf{x}_j\|^2 \leq p \max_j \|\mathbf{x}_j\|^2$ .

- (c) The  $m$ th step changes one coordinate, by  $\nu|\mathbf{x}_{j_m}^\top \mathbf{u}_m|/\|\mathbf{x}_{j_m}\|^2 \leq \nu\|\mathbf{x}_{j_m}\|\|\mathbf{u}_m\|/\|\mathbf{x}_{j_m}\|^2 \leq \nu\|\mathbf{u}_m\|/\min_j \|\mathbf{x}_j\|$ , using Cauchy–Schwarz. By (b),  $\|\mathbf{u}_m\| \leq (1 - \kappa)^{m/2}\|\mathbf{u}_0\|$ , which is summable. So  $\sum_m \|\mathbf{b}^{(m+1)} - \mathbf{b}^{(m)}\| < \infty$ , the sequence is Cauchy, and it converges to some  $\mathbf{b}^\infty$ . By continuity,  $\mathbf{X}\mathbf{b}^\infty = \lim \mathbf{X}\mathbf{b}^{(m)} = \mathbf{M}\mathbf{y}$ , so  $\mathbf{b}^\infty$  is a least squares estimate ([Theorem 6.4.1](#)). With full column rank there is only one. ■

The bound is crude, but it shows what controls the speed: the step size and the conditioning of  $\mathbf{X}$ .

*Remark* (When  $p > n$ ). If  $\text{rank } \mathbf{X} = n$ , as is typical when  $p > n$ , boosting run long enough *interpolates* the data, so stopping early is a necessity. Its limit depends on the order in which columns were chosen and is in general not the least squares estimate of smallest norm ([Proposition 6.10.1](#)), which full gradient descent from zero does reach ([Exercise \(B2\)](#); this is the ridgeless estimator of [Theorem 28.1.1](#)).

### 30.4.3 Early stopping as shrinkage

For a base learner linear in the data, the effect of stopping early is exact. Let an  $n \times n$  matrix  $\mathbf{S}$  map a residual vector to its fitted values; boosting updates  $\mathbf{f}^{(m+1)} = \mathbf{f}^{(m)} + \nu \mathbf{S}(\mathbf{y} - \mathbf{f}^{(m)})$  from  $\mathbf{f}^{(0)} = \mathbf{0}$ .

**Proposition 30.4.2 — The boosting operator**

- (a)  $\mathbf{f}^{(m)} = \mathbf{B}_m \mathbf{y}$  with  $\mathbf{B}_m = \mathbf{I} - (\mathbf{I} - \nu \mathbf{S})^m$ .  
 (b) Let  $\mathbf{S}$  be symmetric with eigenvalues  $d_k \geq 0$  and orthonormal eigenvectors  $\mathbf{u}_k$ , and let  $\nu d_k \leq 1$  for every  $k$ . Then

$$\mathbf{B}_m = \sum_k \{1 - (1 - \nu d_k)^m\} \mathbf{u}_k \mathbf{u}_k^\top.$$

Each factor  $1 - (1 - \nu d_k)^m$  lies in  $[0, 1]$ , is nondecreasing in  $m$ , and tends to 1 when  $d_k > 0$ . The trace  $\text{tr } \mathbf{B}_m$  increases with  $m$  towards  $\text{rank } \mathbf{S}$ .

- (c) (*Least squares learner.*) If  $\mathbf{S} = \mathbf{M}$ , then  $\mathbf{B}_m = \{1 - (1 - \nu)^m\} \mathbf{M}$ . Early stopping shrinks the least squares fit by a single factor.  
 (d) (*Gradient descent.*) The iteration  $\mathbf{b}^{(m+1)} = \mathbf{b}^{(m)} + \nu \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \mathbf{b}^{(m)})$  from  $\mathbf{b}^{(0)} = \mathbf{0}$  is the case  $\mathbf{S} = \mathbf{X} \mathbf{X}^\top$ . If  $\mathbf{X}$  has singular values  $\sigma_k$  and  $\nu \sigma_k^2 \leq 1$ , its fitted values shrink the  $k$ th singular component of the least squares fit by  $1 - (1 - \nu \sigma_k^2)^m$ . Ridge regression shrinks the same component by  $\sigma_k^2 / (\sigma_k^2 + \lambda)$ .

*Proof.*

- (a)  $\mathbf{y} - \mathbf{f}^{(m+1)} = (\mathbf{I} - \nu \mathbf{S})(\mathbf{y} - \mathbf{f}^{(m)})$ , so  $\mathbf{y} - \mathbf{f}^{(m)} = (\mathbf{I} - \nu \mathbf{S})^m \mathbf{y}$ . (b)  $(\mathbf{I} - \nu \mathbf{S})^m = \sum_k (1 - \nu d_k)^m \mathbf{u}_k \mathbf{u}_k^\top$  by the spectral theorem (Theorem 1.7.2), and  $0 \leq 1 - \nu d_k \leq 1$ .  
 (c)  $\mathbf{M}$  has eigenvalues 0 and 1, and it is idempotent.  
 (b)  $\mathbf{X} \mathbf{b}^{(m+1)} = \mathbf{X} \mathbf{b}^{(m)} + \nu \mathbf{X} \mathbf{X}^\top (\mathbf{y} - \mathbf{X} \mathbf{b}^{(m)})$ . The eigenvalues of  $\mathbf{X} \mathbf{X}^\top$  are the  $\sigma_k^2$  and zeros, and the eigenvectors for the  $\sigma_k^2$  are the left singular vectors (Theorem 1.8.1). For ridge, see Theorem 27.2.2.

■

For small  $\sigma_k^2$ , gradient descent keeps the fraction  $1 - (1 - \nu \sigma_k^2)^m \approx m \nu \sigma_k^2$  of a component and ridge keeps  $\approx \sigma_k^2 / \lambda$ ; for large  $\sigma_k^2$  both keep nearly everything. So  $m$  steps behave like ridge with  $\lambda \approx 1/(m \nu)$ . Part (c) is the proportional shrinkage of Proposition 7.6.3(c).

**Degrees of freedom.** For a linear smoother the complexity is  $\text{tr } \mathbf{B}_m$ , which equals the covariance penalty  $\sigma^{-2} \sum_i \text{Cov}(\hat{y}_i, y_i)$  (Theorem 29.1.1). Componentwise boosting is not linear, since the chosen columns depend on  $\mathbf{y}$ , but for a fixed sequence  $j_0, \dots, j_{m-1}$  its fitted values are  $\mathbf{B}_m \mathbf{y}$  with

$$\mathbf{B}_m = \mathbf{I} - (\mathbf{I} - \nu \mathbf{H}_{j_{m-1}}) \cdots (\mathbf{I} - \nu \mathbf{H}_{j_0}), \quad \mathbf{H}_j = \frac{\mathbf{x}_j \mathbf{x}_j^\top}{\|\mathbf{x}_j\|^2}, \quad (30.4.2)$$

by Proposition 30.4.2(a) applied step by step. Bühlmann (2006) used  $\text{tr } \mathbf{B}_m$  along the observed sequence as the degrees of freedom in a corrected Akaike criterion for choosing  $m$ . That treats the selection as fixed in advance, and the next example shows how much it can understate.

**Example. (Boosting a sparse regression)**

The script draws  $n = 60$  observations of  $p = 30$  normal regressors with correlations

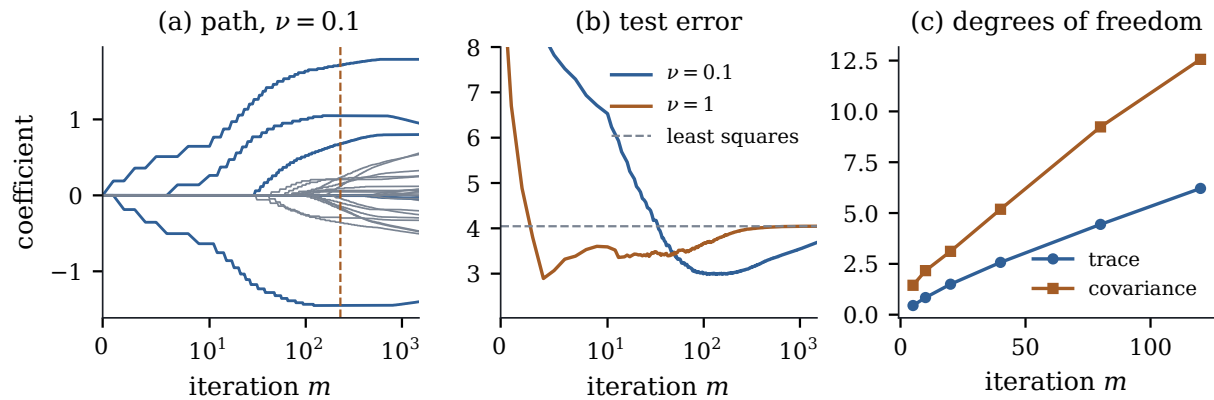


Figure 30.4.1: Componentwise boosting for [Example \(Boosting a sparse regression\)](#). (a) Coefficient paths with  $\nu = 0.1$  (true nonzero coefficients in blue); the dashed line is the cross-validated stopping iteration. (b) Exact risk along the path for  $\nu = 0.1$  and  $\nu = 1$ , with the least squares risk dashed. Both horizontal axes are linear up to 10 and logarithmic beyond. (c) Two degrees of freedom: the average trace of the boosting operator [Eq. 30.4.2](#), and the covariance degrees of freedom estimated from 2000 simulated responses.

$0.5^{|j-k|}$ , five nonzero coefficients and  $\sigma = 1.5$ , and boosts on the standardized regressors. The expected squared error on a new case (the *risk*) is computed exactly from the design law; its floor is  $\sigma^2 = 2.25$ .

*Convergence.* For  $\nu = 0.1$ ,  $\kappa = 3.18 \times 10^{-4}$ , and  $\|\mathbf{b}^{(m)} - \hat{\boldsymbol{\beta}}\|/\|\hat{\boldsymbol{\beta}}\|$  is 0.497, 0.161 and  $2.7 \times 10^{-8}$  after 100, 1000 and 50000 steps; the script checks [Eq. 30.4.1](#) and the bound (b) at every step.

*Early stopping.* Least squares has risk 4.046. With  $\nu = 0.1$  the risk is smallest, 2.981, after 108 steps (with 14 nonzero coefficients), and within 5 per cent of that from  $m = 72$  to  $m = 305$ ; five-fold cross-validation chooses  $m = 229$ , with risk 3.047. With  $\nu = 1$  the best risk, 2.893, occurs at  $m = 4$  and only there is the risk within 5 per cent of it: a needle that a data-based stopping rule is unlikely to hit ([Figure 30.4.1](#)).

*Degrees of freedom.* Over 2000 responses with the same design and mean, the covariance degrees of freedom after  $m = 20$  steps are 3.1 against an average trace of 1.5, and after  $m = 120$  steps 12.6 against 6.2. The trace does not see the adaptation of the chosen columns to the noise.

```
import numpy as np

def l2_boost(X, y, nu, steps):
    """Componentwise L2 boosting from b = 0; returns the coefficient path (steps +
    1) x p."""
    n, p = X.shape
    norms2 = np.sum(X ** 2, axis=0)
    b = np.zeros(p)
    r = y.astype(float).copy()
    path = [b.copy()]
    for _ in range(steps):
        corr = X.T @ r
        j = np.argmax(corr ** 2 / norms2)           # largest drop in the
        # residual sum of squares
        step = nu * corr[j] / norms2[j]
        b[j] += step
```

```

        r -= step * X[:, j]
        path.append(b.copy())
    return np.array(path)

rng = np.random.default_rng(3004)
n, p, sigma = 60, 30, 1.5
C = 0.5 ** np.abs(np.subtract.outer(np.arange(p), np.arange(p)))
L = np.linalg.cholesky(C)
beta = np.zeros(p)
beta[[0, 4, 9, 14, 19]] = [2.0, -1.5, 1.0, 1.0, -0.5]

def draw(m):
    Z = rng.normal(size=(m, p)) @ L.T
    return Z, Z @ beta + sigma * rng.normal(size=m)

X, y = draw(n)
mx, sx, my = X.mean(axis=0), X.std(axis=0), y.mean()
Xs, ys = (X - mx) / sx, y - my                                # centre and scale by the
                                                                training data

def risk(path):
    """Expected squared error on a new case, for each row b of path (exact, from the
    design law)."""
    d = beta[None, :] - path / sx                                # coefficient error on the
                                                                original scale
    offset = path @ (mx / sx) - my                                # error in the fitted
                                                                intercept
    return np.sum((d @ C) * d, axis=1) + offset ** 2 + sigma ** 2

b_ls = np.linalg.lstsq(Xs, ys, rcond=None)[0]
M = 1500
test_err = {}
for step in [0.1, 1.0]:
    test_err[step] = risk(l2_boost(Xs, ys, step, M))
err_ls = risk(b_ls[None, :])[0]
m_best = int(np.argmin(test_err[0.1]))
print(f"least squares: risk {err_ls:.3f}")
print(f"nu = 0.1: best m = {m_best}, risk {test_err[0.1][m_best]:.3f}")
print(f"nu = 1.0: best m = {np.argmin(test_err[1.0])}, risk {test_err[1.0].min():.3f}
    ")

```

The next cell repeats the comparison with 300 responses and  $m = 20$ ; the script's run of it gives 3.2 against 1.5.

```

mu = Xs @ (beta * sx)                                          # the true mean, in the
                                                                centred scale
def operator_trace(X, path, nu):
    """Trace of  $B_m = I - \text{prod}_k (I - \text{nu } H_{\{j_k\}})$ , with the selected columns held
    fixed."""
    n = X.shape[0]
    Rm = np.eye(n)
    for k in range(len(path) - 1):
        j = int(np.flatnonzero(path[k + 1] != path[k])[0])
        x = X[:, j]
        Rm -= nu * np.outer(x, x @ Rm) / (x @ x)
    return n - np.trace(Rm)

reps, m_df = 300, 20

```

```

rng_df = np.random.default_rng(30041)
Y = mu + sigma * rng_df.normal(size=(reps, n))           # repeated responses, same
                  design and mean
paths = [l2_boost(Xs, Y[r], 0.1, m_df) for r in range(reps)]
F = np.array([Xs @ P[-1] for P in paths])                 # fitted values after m_df
                  steps
cov_df = np.sum(np.mean((Y - Y.mean(axis=0)) * (F - F.mean(axis=0)), axis=0)) /
          sigma ** 2
trace_df = np.mean([operator_trace(Xs, P, 0.1) for P in paths])
print(f"m = {m_df}: mean trace {trace_df:.1f}, covariance df {cov_df:.1f}")

```

The gap is the optimism of selection ([Proposition 29.5.1](#)). For the lasso, by contrast, the covariance degrees of freedom equal the expected number of nonzero coefficients (Zou, Hastie and Tibshirani, 2007; we do not reproduce the proof): shrinkage offsets the cost of selection (compare [Exercise \(C1\)](#)). In practice  $m$  is chosen by cross-validation ([Chapter 29](#)), which needs no degrees of freedom.

## 30.4.4 Exercises

### 30.4.4.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ ,  $\mathbf{z} = \mathbf{X}^\top \mathbf{y} = (3, -1, 0.5)^\top$  and  $\nu = 0.1$ . Carry out the first three boosting steps by hand. Show that under an orthonormal design the current residual correlations are  $\mathbf{z} - \mathbf{b}^{(m)}$ , so boosting always moves the coordinate whose remaining correlation is largest in absolute value, and removes one tenth of that correlation.

*Solution.*  $\mathbf{X}^\top \mathbf{r}^{(m)} = \mathbf{X}^\top \mathbf{y} - \mathbf{X}^\top \mathbf{X} \mathbf{b}^{(m)} = \mathbf{z} - \mathbf{b}^{(m)}$ , with unit column lengths. Step 1: coordinate 1,  $b_1 = 0.3$ , residual correlation 2.7. Step 2: coordinate 1 again,  $b_1 = 0.3 + 0.27 = 0.57$ , correlation 2.43. Step 3: coordinate 1,  $b_1 = 0.57 + 0.243 = 0.813$ . Coordinate 1 keeps being chosen until its remaining correlation  $3(0.9)^m$  falls below 1, which first happens at  $m = 11$ . This is the approach to soft thresholding that [Section 30.5](#) makes precise.

### 30.4.4.2 B. Practice

#### Exercise B1.

Use [Eq. 30.4.1](#) to show that  $\sum_{m \geq 0} \max_j (\mathbf{x}_j^\top \mathbf{r}^{(m)})^2 / \|\mathbf{x}_j\|^2 \leq \|\mathbf{y}\|^2 / \{\nu(2 - \nu)\}$ . Deduce, without the singular value argument, that  $\mathbf{X}^\top \mathbf{r}^{(m)} \rightarrow \mathbf{0}$  for every  $\nu \in (0, 2)$ .

*Solution.* Summing [Eq. 30.4.1](#) over  $m = 0, \dots, M-1$  gives  $\nu(1 - \nu/2) \sum_{m < M} \max_j (\dots) = Q(\mathbf{0}) - Q(\mathbf{b}^{(M)}) \leq \frac{1}{2} \|\mathbf{y}\|^2$ . Multiply by  $2/\{\nu(2 - \nu)\}$  and let  $M \rightarrow \infty$ . The terms of a convergent series tend to zero. So  $\max_j |\mathbf{x}_j^\top \mathbf{r}^{(m)}| / \|\mathbf{x}_j\| \rightarrow 0$ , and hence every coordinate of  $\mathbf{X}^\top \mathbf{r}^{(m)}$  does.

**Exercise B2.**

For the gradient descent iteration of [Proposition 30.4.2\(d\)](#) with  $0 < \nu < 2/\sigma_1^2$ , show that every  $\mathbf{b}^{(m)}$  lies in  $\mathcal{C}(\mathbf{X}^\top)$ , and that  $\mathbf{b}^{(m)} \rightarrow \mathbf{X}^+\mathbf{y}$ , the minimum-norm least squares estimate ([Proposition 6.10.1](#)). Why does this argument fail for componentwise boosting?

*Solution.* Each increment  $\nu\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\mathbf{b}^{(m)})$  lies in  $\mathcal{C}(\mathbf{X}^\top)$ , and  $\mathbf{b}^{(0)} = \mathbf{0}$ . In the singular value basis,  $\mathbf{b}^{(m)} = \sum_k \{1 - (1 - \nu\sigma_k^2)^m\} \sigma_k^{-1}(\mathbf{u}_k^\top \mathbf{y}) \mathbf{v}_k$ , because the iteration acts on each component separately. With  $|1 - \nu\sigma_k^2| < 1$  this converges to  $\sum_k \sigma_k^{-1}(\mathbf{u}_k^\top \mathbf{y}) \mathbf{v}_k = \mathbf{X}^+\mathbf{y}$ . Componentwise boosting adds multiples of coordinate vectors  $\mathbf{e}_j$ , which need not lie in  $\mathcal{C}(\mathbf{X}^\top)$  when  $\text{rank } \mathbf{X} < p$ . So its limit can have a component in  $\mathcal{N}(\mathbf{X})$ .

**30.4.4.3 C. Going deeper****Exercise C1.**

Under an orthonormal design with  $\mathbf{y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ , the lasso fit with parameter  $\lambda$  has covariance degrees of freedom  $E \#\{j : |z_j| > \lambda\}$  (by Stein's lemma, [Lemma 27.5.1](#), applied coordinatewise). Show that hard thresholding at the same  $\lambda$  has covariance degrees of freedom  $E \#\{j : |z_j| > \lambda\} + \sum_j (\lambda/\sigma) \{\phi((\lambda - \beta_j)/\sigma) + \phi((\lambda + \beta_j)/\sigma)\}$ , where  $\phi$  is the standard normal density. The extra term is the price of selecting without shrinking.

## 30.5 Boosting as regularization

The boosting path of [Figure 30.4.1\(a\)](#) looks like a lasso path, and not by coincidence. A close relative of boosting with tiny steps traces the lasso path exactly under a condition on the design (Efron, Hastie, Johnstone and Tibshirani, 2004, under their positive cone condition; Hastie, Taylor, Tibshirani and Walther, 2007, in terms of monotone paths). This section proves the simplest case, sketches the general one, compares the procedures, and ends with boosting beyond linear least squares. Throughout,  $\mathbf{y}$  and the columns of  $\mathbf{X}$  are centred and the columns have unit length.

### 30.5.1 Forward stagewise regression

**Definition 30.5.1 — Incremental forward stagewise regression**

Fix  $\varepsilon > 0$ . Start from  $\mathbf{b}^{(0)} = \mathbf{0}$ . At step  $k$ , with residual  $\mathbf{r}^{(k)} = \mathbf{y} - \mathbf{X}\mathbf{b}^{(k)}$ , choose  $j = \arg \max_j |\mathbf{x}_j^\top \mathbf{r}^{(k)}|$  and set  $b_j^{(k+1)} = b_j^{(k)} + \varepsilon \text{sign}(\mathbf{x}_j^\top \mathbf{r}^{(k)})$ , leaving the other coordinates unchanged. Write  $\text{FS}_\varepsilon$  for this procedure, and  $\text{FS}_0$  for the limit of its paths as  $\varepsilon \rightarrow 0$ , when the limit exists.

Stagewise and componentwise boosting choose the same coordinate; boosting moves it by  $\nu$  times its correlation, stagewise by a fixed  $\varepsilon$ , so the  $\ell_1$  length of its path after  $k$  steps is exactly  $k\varepsilon$ .

## 30.5.2 The orthonormal case

Under an orthonormal design the lasso estimate at  $\lambda$  is  $\text{sign}(z_j)(|z_j| - \lambda)_+$  with  $\mathbf{z} = \mathbf{X}^\top \mathbf{y}$  (Theorem 27.4.2), and stagewise follows it to within one step.

### Proposition 30.5.1 — Stagewise under an orthonormal design

Let  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ ,  $\mathbf{z} = \mathbf{X}^\top \mathbf{y}$ , and let  $\tau_k = \max_j |z_j - b_j^{(k)}|$  be the largest absolute residual correlation after  $k$  steps of  $\text{FS}_\varepsilon$ . Let  $K$  be the last step before  $\tau$  first falls below  $\varepsilon$ , so that  $\tau_0, \dots, \tau_K \geq \varepsilon$ . For every  $k \leq K$ :

- (a)  $\tau_{k+1} \leq \tau_k$  (for  $k < K$ );
- (b) every coordinate satisfies  $|b_j^{(k)} - \text{sign}(z_j)(|z_j| - \tau_k)_+| \leq \varepsilon$ , that is,  $\text{FS}_\varepsilon$  is within  $\varepsilon$  of the lasso estimate at  $\lambda = \tau_k$ .

Consequently, the part of the stagewise path with  $\tau_k \geq \varepsilon$  and the lasso path are within  $3\varepsilon$  of each other, in the sense that every point of either is within  $3\varepsilon$  of some point of the other in every coordinate. As  $\varepsilon \rightarrow 0$  the stagewise path converges to the lasso path.

*Proof.* With  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ , the residual correlations are  $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{z} - \mathbf{b}$ . The coordinate chosen at step  $k$  has  $|z_j - b_j^{(k)}| = \tau_k \geq \varepsilon$ . Moving  $b_j$  by  $\varepsilon$  towards  $z_j$  reduces this distance to  $\tau_k - \varepsilon$  without overshooting, and the other coordinates are unchanged. This proves (a). It also shows that every move of  $b_j$  made while  $\tau \geq \varepsilon$  goes towards  $z_j$  without passing it. Starting from 0, the coordinate therefore stays on the segment from 0 to  $z_j$ , and  $b_j^{(k)} = \text{sign}(z_j)(|z_j| - d_j)$  with  $d_j = |z_j - b_j^{(k)}| \in [0, |z_j|]$ .

Fix  $j$  and  $k$ . If  $j$  has not been chosen before step  $k$ , then  $b_j^{(k)} = 0$  and  $|z_j| = d_j \leq \tau_k$ , so the lasso coordinate at  $\tau_k$  is also 0. Otherwise, let  $l < k$  be the last step at which  $j$  was chosen. Then  $d_j = \tau_l - \varepsilon \geq \tau_k - \varepsilon$  by (a), and  $d_j \leq \tau_k$  because  $\tau_k$  is a maximum. If  $|z_j| \geq \tau_k$ , the difference from the lasso coordinate is  $(|z_j| - d_j) - (|z_j| - \tau_k) = \tau_k - d_j \in [0, \varepsilon]$ . If  $|z_j| < \tau_k$ , the lasso coordinate is 0 and  $|b_j^{(k)}| = |z_j| - d_j \leq \tau_k - d_j \leq \varepsilon$ . This proves (b).

For the last statement, (b) puts every stagewise point with  $\tau_k \geq \varepsilon$  within  $\varepsilon$  of the lasso path. Conversely, with  $K$  as in the statement, each step lowers  $\tau$  by at most  $\varepsilon$ , because  $\tau_{k+1} \geq d_j = \tau_k - \varepsilon$  for the chosen  $j$ . Hence  $\tau_K < 2\varepsilon$ , and every  $\lambda \in [\tau_K, \tau_0]$  is within  $\varepsilon$  of some  $\tau_k$  with  $k \leq K$ . Soft thresholding is 1-Lipschitz in  $\lambda$ , so for such  $\lambda$  the lasso estimate is within  $2\varepsilon$  of  $\mathbf{b}^{(k)}$ . For  $\lambda \geq \tau_0 = \max_j |z_j|$  it is  $\mathbf{0} = \mathbf{b}^{(0)}$ . For  $\lambda < \tau_K$  it is within  $\tau_K - \lambda < 2\varepsilon$  of the lasso estimate at  $\tau_K$ , hence within  $3\varepsilon$  of  $\mathbf{b}^{(K)}$ . ■

For componentwise boosting the same argument gives tolerance  $\nu\tau_k$  (Exercise (B1)).

## 30.5.3 The general relation

In general the lasso path is piecewise linear with explicit directions. Part (a) below restates the segment formula of Proposition 27.4.3 in the notation of this chapter; what is new is the reading in terms of tied correlations.



### Proposition 30.5.2 — Stagewise and the lasso path

- (a) Suppose that for  $\lambda$  in an interval the lasso estimate is unique and has a fixed active set  $A$  and fixed signs  $\mathbf{s}_A$ , and  $\mathbf{X}_A$  has full column rank. Then on that interval

$$\hat{\boldsymbol{\beta}}_A(\lambda) = (\mathbf{X}_A^\top \mathbf{X}_A)^{-1} (\mathbf{X}_A^\top \mathbf{y} - \lambda \mathbf{s}_A), \quad (30.5.1)$$

so as  $\lambda$  decreases the active coefficients move along the direction  $\mathbf{d}_A = (\mathbf{X}_A^\top \mathbf{X}_A)^{-1} \mathbf{s}_A$ , and all active variables keep the common absolute correlation  $\lambda$  with the residual.

- (b) (*Sketch; Efron, Hastie, Johnstone and Tibshirani, 2004, who prove it under their positive cone condition on the design; Hastie et al., 2007, for monotone paths.*) If along the whole lasso path every direction  $\mathbf{d}_A$  has the signs  $\mathbf{s}_A$  coordinatewise, that is, if every lasso coefficient is monotone in  $\lambda$ , then  $\text{FS}_0$  exists and coincides with the lasso path.
- (c) (*Sketch; Hastie, Taylor, Tibshirani and Walther, 2007.*) In general  $\text{FS}_0$  is the path of the *monotone lasso*. Write  $\mathbf{b} = \mathbf{b}^+ - \mathbf{b}^-$  with  $\mathbf{b}^\pm \geq \mathbf{0}$ . Along the path, every coordinate of  $\mathbf{b}^+$  and of  $\mathbf{b}^-$  is nondecreasing, and the path lowers the residual sum of squares as fast as possible per unit of  $\ell_1$  arc length, subject to that constraint.

*Proof.*

- (a) The formula Eq. 30.5.1 is Proposition 27.4.3. By Lemma 30.1.2 and Lemma 30.1.3, the active coordinates satisfy  $\mathbf{X}_A^\top (\mathbf{y} - \mathbf{X}_A \hat{\boldsymbol{\beta}}_A) = \lambda \mathbf{s}_A$ , which says that the active correlations are  $\lambda \mathbf{s}_A$ .

*Sketch of (b) and (c).* Complete proofs need a careful limiting argument; see the papers cited. Choosing a variable of maximal absolute correlation lowers its correlation relative to the others, so in the limit the active correlations are tied, as for the lasso. The net movement is a direction  $\mathbf{d}$  with  $d_j s_j \geq 0$  on  $A$ , and the only direction keeping the correlations tied is  $\mathbf{d}_A$ . If  $\mathbf{d}_A$  has the signs  $\mathbf{s}_A$ , stagewise follows it; that is (b). Otherwise the lasso shrinks a coefficient against the sign of its correlation, which stagewise cannot do. Stagewise instead holds it and moves along a nonnegative least squares fit of the residual on the sign-adjusted active columns. Efron et al. (2004) build this into a modified least angle regression and prove it gives  $\text{FS}_0$ ; Hastie et al. (2007) identify the path with the monotone lasso. ■

The lasso at  $\ell_1$  norm  $t$  minimizes the residual sum of squares subject to  $\|\mathbf{b}\|_1 \leq t$  (the constrained form of Definition 27.4.1); stagewise budgets the  $\ell_1$  arc length and never undoes a move. For monotone paths the two budgets coincide (Exercise (A2)).

#### Example. (Stagewise and lasso paths)

*Orthonormal design.* With 5 orthonormal columns and  $\varepsilon = 0.01$ , the largest distance from the lasso estimate at  $\lambda = \tau_k$  over all steps is 0.0092.

*A non-monotone lasso path.* For  $n = 40$  observations of  $p = 6$  regressors sharing a common factor (correlations near 0.64), the lasso coefficient of regressor 6 grows to 0.308 at  $\|\mathbf{b}\|_1 = 4.12$  and then shrinks back.  $\text{FS}_\varepsilon$  with  $\varepsilon = 0.002$  agrees with the lasso to

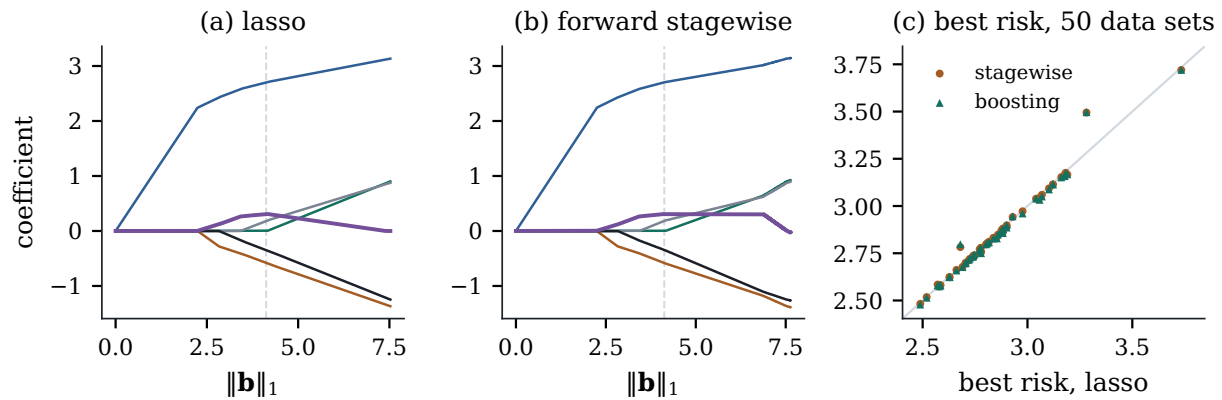


Figure 30.5.1: (a) The lasso path and (b) the forward stagewise path ( $\varepsilon = 0.002$ ) for the correlated design of [Example \(Stagewise and lasso paths\)](#), against  $\|\mathbf{b}\|_1$ . The thick purple curve is the coefficient that the lasso shrinks back after the dashed line; stagewise holds it. (c) Smallest risk along the path for 50 simulated data sets: forward stagewise and boosting ( $\nu = 0.05$ ) against the lasso.

within 0.039 until then, and afterwards holds the coefficient: halfway to the least squares norm 7.53, the two differ by 0.165 there ([Figure 30.5.1](#)). At every  $\ell_1$  norm checked, the lasso has the smaller residual sum of squares, as it must; the excess of stagewise is at most 0.011.

*Prediction.* Over 50 data sets from the design of [Example \(Boosting a sparse regression\)](#), the average smallest risk along the path is 2.857 for the lasso, 2.857 for stagewise and 2.856 for boosting with  $\nu = 0.05$ , against 4.607 for least squares. Data set by data set the points lie close to the diagonal (panel (c)): the best risks of stagewise and boosting never differ from the lasso's by more than 6.6 per cent.

```
import numpy as np

def soft(z, lam):
    return np.sign(z) * np.maximum(np.abs(z) - lam, 0.0)

def lasso_cd(X, y, lam, b=None, tol=1e-12):
    G, c = X.T @ X, X.T @ y
    b = np.zeros(X.shape[1]) if b is None else b.copy()
    while True:
        largest = 0.0
        for j in range(len(b)):
            old = b[j]
            b[j] = soft(c[j] - G[j] @ b + G[j, j] * old, lam) / G[j, j]
            largest = max(largest, abs(b[j] - old))
        if largest < tol:
            return b

def forward_stagewise(X, y, eps, steps):
    """Incremental forward stagewise: move the most correlated coefficient by eps.
    """
    b = np.zeros(X.shape[1])
    r = y.astype(float).copy()
    path = [b.copy()]
    for _ in range(steps):
        c = X.T @ r
```

```

        j = np.argmax(np.abs(c))
        delta = eps * np.sign(c[j])
        b[j] += delta
        r -= delta * X[:, j]
        path.append(b.copy())
    return np.array(path)

rng = np.random.default_rng(4)
n, p = 40, 6
z = rng.normal(size=(n, 1))
X_raw = 0.8 * z + 0.6 * rng.normal(size=(n, p))           # a shared factor:
                  correlations near 0.64
center = X_raw.mean(axis=0)
scale = np.linalg.norm(X_raw - center, axis=0)
X = (X_raw - center) / scale                             # centred, unit-length
                  columns
beta = np.array([3.0, -2.0, 2.0, 0.0, 0.0, -1.0])
y = X @ beta + 0.5 * rng.normal(size=n)
y_bar = y.mean()
y = y - y_bar

lam_max = np.max(np.abs(X.T @ y))
grid = lam_max * np.logspace(0, -3, 400)
lasso_path = []
b = np.zeros(p)
for lam in grid:
    b = lasso_cd(X, y, lam, b)
    lasso_path.append(b.copy())
lasso_path = np.array(lasso_path)

eps = 0.002
fs_path = forward_stagewise(X, y, eps, 6000)
l1_fs = np.abs(fs_path).sum(axis=1)
arc_fs = eps * np.arange(len(fs_path))                  # total variation of the
                  stagewise path
print("lasso at the end:      ", np.round(lasso_path[-1], 3))
print("stagewise at the end: ", np.round(fs_path[-1], 3))

```

Near least squares, stagewise dithers, stepping back and forth by  $\varepsilon$ : its arc length grows while its  $\ell_1$  norm does not.

### 30.5.4 Beyond linear least squares

For a differentiable loss  $\sum_i L(y_i, f_i)$ , boosting fits the negative gradient  $-\partial L(y_i, f_i)/\partial f_i$  in place of the residual (Friedman, 2001): absolute error gives residual signs, the Huber loss clipped residuals, and the exponential loss  $e^{-yf}$ ,  $y = \pm 1$ , recovers AdaBoost (Friedman, Hastie and Tibshirani, 2000). Small regression trees as base learners give gradient tree boosting; penalized splines in one regressor at a time give componentwise boosting of an additive model (Bühlmann and Yu, 2003; Bühlmann and Hothorn, 2007), a topic for Chapters 43 and 44.

Zhang and Yu (2005) prove consistency of early-stopped boosting for general convex losses over rich base classes; Bühlmann (2006) proves it for componentwise  $L_2$  boosting with  $p$  growing almost exponentially in  $n$  under sparsity. We state this without proof; it says that some sequence of stopping iterations works, not how to find it.

*Warning.* Early stopping is an implicit regularizer: the estimates minimize no stated criterion. For prediction, tune  $m$  like any smoothing parameter. For interpreting coefficients, boosting inherits every caution of [Chapter 29](#) about selection, and inference after boosting meets the obstacles of inference after the lasso ([Theorem 28.5.1](#)).

## 30.5.5 Exercises

### 30.5.5.1 A. Check your understanding

#### Exercise A1.

Let  $\mathbf{X}^\top \mathbf{X} = \mathbf{I}$ ,  $\mathbf{z} = (3, -1, 0.5)^\top$  and  $\varepsilon = 0.5$ . Run  $\text{FS}_\varepsilon$  by hand until the largest absolute residual correlation first falls below 1, and check [Proposition 30.5.1\(b\)](#) at each step.

#### Exercise A2.

Show that the  $\ell_1$  arc length of the  $\text{FS}_\varepsilon$  path after  $k$  steps is  $k\varepsilon$ , that it is at least  $\|\mathbf{b}^{(k)}\|_1$ , and that equality holds iff no coordinate has ever moved in both directions. Which of the two paths in [Figure 30.5.1](#) has equality up to the least squares region?

### 30.5.5.2 B. Practice

#### Exercise B1.

Adapt the proof of [Proposition 30.5.1](#) to componentwise boosting with step  $\nu \in (0, 1]$  under an orthonormal design. Show that  $\tau_k$  is nonincreasing and that  $|b_j^{(k)} - \text{sign}(z_j)(|z_j| - \tau_k)_+| \leq \nu\tau_k$  for all  $j$  and  $k$ .

*Solution.* The chosen coordinate's remaining correlation changes from  $\tau_k$  to  $(1 - \nu)\tau_k$ , with no overshoot since  $\nu \leq 1$ . So  $\tau$  is nonincreasing, and each coordinate stays between 0 and  $z_j$ . If  $j$  was last chosen at step  $l < k$ , then  $d_j = (1 - \nu)\tau_l \geq (1 - \nu)\tau_k$  and  $d_j \leq \tau_k$ . As in the proof, the distance from the lasso coordinate at  $\tau_k$  is at most  $\tau_k - d_j \leq \nu\tau_k$ . A coordinate that has never been chosen is 0, and so is the lasso coordinate.

#### Exercise B2.

On a segment of the lasso path with active set  $A$  and signs  $\mathbf{s}_A$ , show that the correlation of an inactive regressor is  $c_j(\lambda) = \mathbf{x}_j^\top (\mathbf{I} - \mathbf{M}_A) \mathbf{y} + \lambda \mathbf{x}_j^\top \mathbf{X}_A (\mathbf{X}_A^\top \mathbf{X}_A)^{-1} \mathbf{s}_A$ , where  $\mathbf{M}_A$  projects onto  $\mathcal{C}(\mathbf{X}_A)$ . Deduce the value of  $\lambda$  at which  $j$  joins the active set, and the value at which an active coefficient reaches zero. This is the lasso version of least angle regression.

*Solution.* By [Eq. 30.5.1](#),  $\mathbf{y} - \mathbf{X}_A \hat{\boldsymbol{\beta}}_A(\lambda) = (\mathbf{I} - \mathbf{M}_A) \mathbf{y} + \lambda \mathbf{X}_A (\mathbf{X}_A^\top \mathbf{X}_A)^{-1} \mathbf{s}_A$ . Take the inner product with  $\mathbf{x}_j$ . Regressor  $j$  joins when  $|c_j(\lambda)|$  reaches  $\lambda$ . Since  $c_j$  is affine in  $\lambda$ , this happens at the largest  $\lambda$  below the current one solving  $c_j(\lambda) = \pm\lambda$ . An active coefficient  $\hat{\beta}_k(\lambda)$  is affine in  $\lambda$  by [Eq. 30.5.1](#), and it leaves when it reaches 0. The next breakpoint is the largest of these candidate values. At a breakpoint,  $A$  and  $\mathbf{s}_A$  are updated, and the

path continues linearly.

### Exercise B3.

Show that for  $p = 2$  (unit-length columns, correlation  $\rho \in (-1, 1)$ ) the direction  $\mathbf{d}_A$  always has the signs  $\mathbf{s}_A$ . Conclude, using [Proposition 30.5.2\(b\)](#), that  $\text{FS}_0$  and the lasso path coincide for every data set with two regressors.

*Solution.* For  $A = \{j\}$ ,  $\mathbf{d}_A = s_j$ . For  $A = \{1, 2\}$ ,  $(\mathbf{X}_A^\top \mathbf{X}_A)^{-1} = (1 - \rho^2)^{-1} \begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$ . So  $\mathbf{d}_A = (1 - \rho^2)^{-1}(s_1 - \rho s_2, s_2 - \rho s_1)^\top$ . If  $s_1 = s_2 = s$ , this is  $s(1 - \rho)(1 - \rho^2)^{-1}(1, 1)^\top$ , with the signs of  $s$ . If  $s_2 = -s_1$ , it is  $(1 + \rho)(1 - \rho^2)^{-1}(s_1, s_2)^\top$ . In both cases the signs are right, because  $|\rho| < 1$ .

## 30.6 Summary and Notes

### 30.6.1 Summary

*Idea.*

1. A penalized criterion ([Definition 30.1.1](#)) has a minimizer when it is coercive; with a convex penalty all minimizers share their fit and penalty, and strict convexity or full rank gives uniqueness ([Theorem 30.1.1](#)).
2. For convex penalties,  $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}) \in \lambda \partial P(\hat{\boldsymbol{\beta}})$  characterizes minimizers ([Lemma 30.1.2](#), [Lemma 30.1.3](#)), and penalized estimates are posterior modes ([Proposition 30.1.4](#)).
3. SCAD and MCP are sparse, continuous and nearly unbiased; MCP is firm thresholding ([Proposition 30.1.5](#)).
4. The elastic net is unique, a lasso on augmented data, and groups correlated regressors ([Proposition 30.2.1](#), [Proposition 30.2.3](#)).
5. The group lasso selects whole groups by block soft thresholding ([Proposition 30.3.1](#)) and, orthonormalized, ignores the coding of factors ([Proposition 30.3.2](#)).
6. Componentwise boosting converges geometrically to least squares ([Theorem 30.4.1](#)); early stopping is a spectral filter like ridge ([Proposition 30.4.2](#)), and the operator trace understates its degrees of freedom.
7. Forward stagewise tracks the lasso path ([Proposition 30.5.1](#)); its limit is the lasso when paths are monotone and the monotone lasso otherwise ([Proposition 30.5.2](#)).

### 30.6.2 Notes and sources

**Coverage.** Fahrmeir, Kneib, Lang and Marx (2021, §4.2–4.3) treat penalties and componentwise boosting in applied terms; the theory and proofs are added here. Hastie, Tibshirani and Friedman (2009, chapters 3, 10 and 16) cover the same ground from the machine-learning side.

**Penalties.** Ridge regression is due to Hoerl and Kennard (1970), the lasso to Tibshirani (1996)

and bridge penalties to Frank and Friedman (1993); uniqueness of the lasso in general position is from Tibshirani (2013). Boyd and Vandenberghe (2004) cover subgradients, and Park and Casella (2008) develop the Bayesian lasso. SCAD and the oracle property are from Fan and Li (2001), MCP from Zhang (2010).

**Elastic net and groups.** The elastic net, its grouping effect and rescaling are from Zou and Hastie (2005); our grouping bound adds the residual-norm term. Pathwise coordinate descent is from Friedman, Hastie, Höfling and Tibshirani (2007) and Friedman, Hastie and Tibshirani (2010). The group lasso is due to Yuan and Lin (2006), its orthonormalized form is discussed by Simon and Tibshirani (2012), and the sparse group lasso is from Simon, Friedman, Hastie and Tibshirani (2013).

**Boosting.** AdaBoost is due to Freund and Schapire (1997); Friedman, Hastie and Tibshirani (2000) and Friedman (2001) gave the statistical and gradient view, and matching pursuit is from Mallat and Zhang (1993).  $L_2$  boosting and the boosting operator are from Bühlmann and Yu (2003), componentwise boosting in high dimensions from Bühlmann (2006), and a survey is Bühlmann and Hothorn (2007). Our proof of [Theorem 30.4.1](#) is the standard argument for greedy coordinate descent on a quadratic. Efron, Hastie, Johnstone and Tibshirani (2004) proved that  $FS_0$  is the lasso path under their positive cone condition, and Hastie, Taylor, Tibshirani and Walther (2007) described  $FS_0$  in general as the monotone lasso; [Proposition 30.5.1](#) is a special case proved directly.

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# Chapter 31

## The General Gauss–Markov Model

Every chapter so far has assumed that the errors are uncorrelated and equally variable. That assumption is a statement about the *geometry* of observation space: it says that Euclidean distance is the right way to measure how far the data are from the model. When observations share a common shock, follow one another in time, or are measured with different precision, Euclidean distance is the wrong ruler. This chapter replaces  $\sigma^2\mathbf{I}$  by a general  $\sigma^2\mathbf{V}$ : first with  $\mathbf{V}$  known and invertible, where the theory of Parts II and III survives a change of coordinates; then in the two places where it does not survive, when  $\mathbf{V}$  is singular and some functions of  $\boldsymbol{\beta}$  become known exactly, and when  $\mathbf{V}$  has to be estimated from the data that produced the fit.

**What you need.** Chapter 6, especially Section 6.9: projection in the  $\mathbf{V}^{-1}$  inner product (Definition 6.9.1, Theorem 6.9.1), oblique projections (Proposition 6.9.4) and Kruskal’s theorem (Theorem 6.9.3). Section 7.2 for the Gauss–Markov theorem (Theorem 7.2.1) and Aitken’s corollary (Corollary 7.2.3), which this chapter cites and does not reprove. From Chapter 1: generalized inverses, square roots (Theorem 1.7.6) and the Woodbury identity (Theorem 1.6.5). From Chapter 2: whitening (Proposition 2.2.4) and the support of a singular covariance (Theorem 2.2.3). Quadratic forms (Chapter 4), estimability (Chapter 8) and the  $F$  test (Chapter 11) are used throughout. Section 19.2 and Chapter 21 supply the diagnostic side: what a wrong covariance does (Theorem 19.2.2), how much efficiency it costs (Theorem 19.2.3), and the sandwich estimator (Theorem 21.4.2).

### 31.0.1 Roadmap

- Generalized least squares
- When ordinary least squares is best
- Singular covariance matrices
- Estimated covariance: feasible generalized least squares
- Summary and notes

## 31.1 Generalized least squares

The model of this chapter keeps the mean structure of every earlier chapter and loosens the second moments:



$$E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}, \quad (31.1.1)$$

where  $\mathbf{X}$  is  $n \times p$  of rank  $r$ , the matrix  $\mathbf{V}$  is known and nonnegative definite, and  $\sigma^2 > 0$  is unknown. This section and the next assume  $\mathbf{V}$  nonsingular; [Section 31.3](#) removes that assumption.

### 31.1.1 What is assumed and what is not

The split of the covariance into a known matrix  $\mathbf{V}$  and an unknown scalar  $\sigma^2$  is a convention, not extra knowledge. Only the product  $\sigma^2 \mathbf{V}$  enters [Eq. 31.1.1](#), so  $(\sigma^2, \mathbf{V})$  and  $(\sigma^2/c, c\mathbf{V})$  describe the same model for every  $c > 0$ ; it is usual to normalize  $\mathbf{V}$  so that  $\text{tr}(\mathbf{V})/n = 1$ , making  $\sigma^2$  the variance of a single observation on a reference scale. What the model really assumes is that the *shape* of the covariance — the ratios of variances and all the correlations — is known.

That is exactly true only when the design forces the shape, as when an observation is the mean of  $m_i$  replicates and has variance  $\sigma^2/m_i$  ([Section 21.3](#)), or when the construction of the data does, as under the linear transformations of [Section 31.3](#). Otherwise  $\mathbf{V}$  belongs to a family  $\mathbf{V}(\boldsymbol{\theta})$  estimated first and then treated as known, and this section's exactness is a fiction whose price is computed in [Section 31.4](#).

### 31.1.2 The estimator and the whitened model

Generalized least squares was defined in [Definition 6.9.1](#) as projection in the  $\mathbf{V}^{-1}$  inner product; we repeat it in the notation of this part.

#### Definition 31.1.1 — Generalized least squares

Under [Eq. 31.1.1](#) with  $\mathbf{V}$  positive definite, a **generalized least squares estimate** is any  $\hat{\boldsymbol{\beta}}_{\mathbf{V}}$  minimizing

$$Q_{\mathbf{V}}(\mathbf{b}) = (\mathbf{y} - \mathbf{X}\mathbf{b})^{\top} \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}),$$

equivalently any solution of the **generalized normal equations**  $\mathbf{X}^{\top} \mathbf{V}^{-1} \mathbf{X} \mathbf{b} = \mathbf{X}^{\top} \mathbf{V}^{-1} \mathbf{y}$ . The fitted vector is  $\mathbf{X} \hat{\boldsymbol{\beta}}_{\mathbf{V}} = \mathbf{A}_{\mathbf{V}} \mathbf{y}$ , where

$$\mathbf{A}_{\mathbf{V}} = \mathbf{X} (\mathbf{X}^{\top} \mathbf{V}^{-1} \mathbf{X})^{-} \mathbf{X}^{\top} \mathbf{V}^{-1} \quad (31.1.2)$$

for any generalized inverse, and the **generalized residual** is  $\hat{\boldsymbol{\varepsilon}}_{\mathbf{V}} = \mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}}_{\mathbf{V}}$ .

Everything algebraic about  $\mathbf{A}_{\mathbf{V}}$  is [Theorem 6.9.1](#) with  $\mathbf{A} = \mathbf{V}^{-1}$ : the choice of generalized inverse is immaterial, it is idempotent with column space  $\mathcal{C}(\mathbf{X})$ , and  $\mathbf{V}^{-1} \mathbf{A}_{\mathbf{V}}$  is symmetric although  $\mathbf{A}_{\mathbf{V}}$  is not — in the Euclidean geometry it is an oblique projection ([Proposition 6.9.4](#)). Everything statistical comes from one change of variables. Let  $\mathbf{V}^{-1/2}$  be the symmetric positive definite square root of  $\mathbf{V}^{-1}$  ([Theorem 1.7.6](#)) and set

$$\mathbf{Y}_* = \mathbf{V}^{-1/2} \mathbf{Y}, \quad \mathbf{X}_* = \mathbf{V}^{-1/2} \mathbf{X}. \quad (31.1.3)$$

By [Theorem 2.2.1](#),  $E(\mathbf{Y}_*) = \mathbf{X}_*\boldsymbol{\beta}$  and  $\text{Cov}(\mathbf{Y}_*) = \sigma^2\mathbf{I}$ : the **whitened model** satisfies the assumptions of Part II exactly. Since  $Q_{\mathbf{V}}(\mathbf{b}) = \|\mathbf{Y}_* - \mathbf{X}_*\mathbf{b}\|^2$ , generalized least squares for  $(\mathbf{Y}, \mathbf{X})$  is ordinary least squares for  $(\mathbf{Y}_*, \mathbf{X}_*)$ . Because  $\mathbf{V}^{-1/2}$  is nonsingular,  $\text{rank}(\mathbf{X}_*) = r$  and  $\mathcal{C}(\mathbf{X}_*) = \mathcal{C}(\mathbf{X}^\top)$ , so by [Theorem 8.3.1](#) the two models have the same estimable functions: nothing about identifiability changes. Any nonsingular  $\mathbf{L}$  with  $\mathbf{V} = \mathbf{L}\mathbf{L}^\top$  serves equally well, by [Proposition 2.2.4\(b\)](#); the Cholesky factor is the usual choice in computation.

Whitening is rarely carried out by forming  $\mathbf{V}^{-1/2}$ ; structured  $\mathbf{V}$  almost always has a shortcut. If the observations fall into  $G$  clusters of size  $m$  that share a random shock, so that  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top$  with  $\mathbf{Z}$  the  $n \times G$  matrix of cluster indicators, then [Theorem 1.6.5](#) gives

$$\mathbf{V}^{-1} = \frac{1}{1 - \rho} \left( \mathbf{I} - \frac{\rho}{1 + (m - 1)\rho} \mathbf{Z}\mathbf{Z}^\top \right), \quad (31.1.4)$$

and the whitening is the **quasi-demeaning**

$$y_{kj} \mapsto \frac{y_{kj} - c\bar{y}_k}{\sqrt{1 - \rho}}, \quad c = 1 - \sqrt{\frac{1 - \rho}{1 + (m - 1)\rho}}, \quad (31.1.5)$$

where  $\bar{y}_k$  is the mean of cluster  $k$  ([Exercise \(B1\)](#)). At  $\rho = 0$ ,  $c = 0$  and nothing happens; at  $\rho = 0.5$  with  $m = 12$ ,  $c = 0.723$ ; as  $\rho \rightarrow 1$ ,  $c \rightarrow 1$  and each cluster is fully centred, throwing away everything the shared shock contaminated. Generalized least squares interpolates, and [Chapter 32](#) will recognize [Eq. 31.1.5](#) as the transformation a random intercept induces.

### 31.1.3 Exact inference when the shape is known

#### Theorem 31.1.1 — Inference under a known covariance shape

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{V})$  with  $\mathbf{V}$  known and positive definite,  $r = \text{rank}(\mathbf{X}) < n$ . Put

$$\text{SSE}_{\mathbf{V}} = \hat{\boldsymbol{\epsilon}}_{\mathbf{V}}^\top \mathbf{V}^{-1} \hat{\boldsymbol{\epsilon}}_{\mathbf{V}} = \mathbf{Y}^\top \mathbf{V}^{-1} (\mathbf{I} - \mathbf{A}_{\mathbf{V}}) \mathbf{Y}, \quad s_{\mathbf{V}}^2 = \frac{\text{SSE}_{\mathbf{V}}}{n - r}. \quad (31.1.6)$$

- (a) For  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ , the statistic  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}$  is the same for every solution of the generalized normal equations, and

$$\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}} \sim N(\boldsymbol{\lambda}^\top \boldsymbol{\beta}, \sigma^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \boldsymbol{\lambda}).$$

It is the best linear unbiased estimator of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  ([Corollary 7.2.3](#)).

- (b)  $\text{SSE}_{\mathbf{V}}/\sigma^2 \sim \chi^2(n - r)$ , independently of  $\mathbf{X}\hat{\boldsymbol{\beta}}_{\mathbf{V}}$ .  
(c)  $E(s_{\mathbf{V}}^2) = \sigma^2$  and  $\text{Var}(s_{\mathbf{V}}^2) = 2\sigma^4/(n - r)$ .  
(d) For estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  with  $\boldsymbol{\lambda} \neq \mathbf{0}$ ,

$$\frac{\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta}}{s_{\mathbf{V}} \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \boldsymbol{\lambda}}} \sim t(n - r).$$

- (e) Let  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  with  $\text{rank}(\mathbf{X}_0) = r_0 < r$ , and let  $\text{SSE}_{\mathbf{V},0}$  be [Eq. 31.1.6](#) computed in the reduced model. Then

$$F = \frac{(\text{SSE}_{\mathbf{V},0} - \text{SSE}_{\mathbf{V}})/(r - r_0)}{s_{\mathbf{V}}^2} \sim F(r - r_0, n - r, \gamma), \quad (31.1.7)$$

with  $\gamma = \|(\mathbf{M}_* - \mathbf{M}_{*0})\mathbf{X}_*\boldsymbol{\beta}\|^2/\sigma^2$ , where  $\mathbf{M}_*$  and  $\mathbf{M}_{*0}$  are the orthogonal projections onto  $\mathcal{C}(\mathbf{V}^{-1/2}\mathbf{X})$  and  $\mathcal{C}(\mathbf{V}^{-1/2}\mathbf{X}_0)$ . The noncentrality is zero iff  $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$ .

*Proof.* Pass to the whitened model Eq. 31.1.3, which satisfies the normal linear model Eq. 7.3.1 with model matrix  $\mathbf{X}_*$  of rank  $r$ . As shown above,  $\hat{\boldsymbol{\beta}}_{\mathbf{V}}$  is exactly a least squares estimate for  $(\mathbf{Y}_*, \mathbf{X}_*)$ , and the two models have the same estimable functions. Moreover

$$\text{SSE}_{\mathbf{V}} = \|\mathbf{Y}_* - \mathbf{X}_*\hat{\boldsymbol{\beta}}_{\mathbf{V}}\|^2 = \|(\mathbf{I} - \mathbf{M}_*)\mathbf{Y}_*\|^2$$

is the residual sum of squares of the whitened problem, because  $\mathbf{V}^{-1/2}\hat{\boldsymbol{\varepsilon}}_{\mathbf{V}} = (\mathbf{I} - \mathbf{M}_*)\mathbf{Y}_*$  by Theorem 6.4.1, and  $\mathbf{X}^\top \mathbf{V}^{-1}\mathbf{X} = \mathbf{X}_*^\top \mathbf{X}_*$ .

Now apply Theorem 7.5.1 to  $(\mathbf{Y}_*, \mathbf{X}_*)$ . Part (a) of that theorem gives (a) here, since  $\boldsymbol{\lambda}^\top (\mathbf{X}_*^\top \mathbf{X}_*)^{-1}\boldsymbol{\lambda} = \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{V}^{-1}\mathbf{X})^{-1}\boldsymbol{\lambda}$  and invariance of the estimate follows from Theorem 6.4.4 applied in the whitened model. Parts (b) and (c) of Theorem 7.5.1 give (b) here, noting that  $\mathbf{X}\hat{\boldsymbol{\beta}}_{\mathbf{V}} = \mathbf{V}^{1/2}\mathbf{M}_*\mathbf{Y}_*$  is a function of  $\mathbf{M}_*\mathbf{Y}_*$ , and part (d) gives (c). Part (d) follows from (a) and (b) exactly as Theorem 12.1.1 does in the Euclidean case. For (e),  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  implies  $\mathcal{C}(\mathbf{V}^{-1/2}\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{V}^{-1/2}\mathbf{X})$  with ranks  $r_0 < r$ , so Definition 11.1.1 holds for the whitened pair and Theorem 11.1.1 applies;  $\text{SSE}_{\mathbf{V},0} = \|(\mathbf{I} - \mathbf{M}_{*0})\mathbf{Y}_*\|^2$  by the same computation. The noncentrality Eq. 11.1.2 is  $\|(\mathbf{M}_* - \mathbf{M}_{*0})\mathbf{X}_*\boldsymbol{\beta}\|^2/\sigma^2$ , and it vanishes iff  $\mathbf{X}_*\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_{*0})$ , that is, iff  $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)$  since  $\mathbf{V}^{-1/2}$  is nonsingular. ■

The theorem is not a new theory: it is Parts II and III read in another coordinate system, and the proof is nothing but the dictionary. Under normality  $\hat{\boldsymbol{\beta}}_{\mathbf{V}}$  also maximizes the likelihood (Exercise (B3)), and  $s_{\mathbf{V}}^2$  has smallest variance among unbiased estimates of  $\sigma^2$ , by Section 7.4 applied to the whitened model.

### 31.1.4 What changes, and what does not

The dictionary is exact, but several habits from Parts II and III do not survive translation.

*Warning.* Under Eq. 31.1.1 with  $\mathbf{V} \neq \mathbf{I}$ :

- The generalized residual is **not** orthogonal to  $\mathcal{C}(\mathbf{X})$ ; what holds is  $\mathbf{X}^\top \mathbf{V}^{-1}\hat{\boldsymbol{\varepsilon}}_{\mathbf{V}} = \mathbf{0}$ . The entries of  $\hat{\boldsymbol{\varepsilon}}_{\mathbf{V}}$  need not sum to zero even when the model has an intercept.
- $\text{SSE}_{\mathbf{V}}$  is **not**  $\sum_i (y_i - \hat{y}_i)^2$ , and the ordinary residual sum of squares of the generalized fit is necessarily *larger* than that of the ordinary fit, which minimizes it. A model comparison using  $\sum_i (y_i - \hat{y}_i)^2$  always prefers ordinary least squares and proves nothing.
- $R^2$  has no agreed meaning: the squared-cosine reading of Theorem 6.7.2 holds in the  $\mathbf{V}^{-1}$  geometry, not the Euclidean one.
- The diagonal entries of  $\mathbf{A}_{\mathbf{V}}$  still sum to  $r$ , but they are not leverages in the sense of Proposition 6.8.1 and can fall outside  $[0, 1]$  (Exercise (B2)).

What survives is everything inference needs: unbiasedness, optimality, the  $\chi^2$ ,  $t$  and  $F$  distribu-

tions, the degrees of freedom  $n - r$ , and  $\gamma$  as a squared distance from the reduced model space, measured now in the  $\mathbf{V}^{-1}$  metric.

**Example. (A trend in sea surface temperature)**

The El Niño data record the monthly averaged sea surface temperature of a region of the eastern Pacific for the 61 complete years 1950 to 2010, 732 observations in 61 blocks of 12. Fit a linear trend in time together with three harmonic pairs at one, two and three cycles per year, so  $p = 8$ . Warm and cold years are warm and cold in every month, so the natural working model is a random annual shock: equicorrelation  $\rho$  inside each year, independence across years. Take  $\rho = 0.5$  as known for the moment.

Ordinary least squares gives a trend of 0.1349 degrees per decade with the usual standard error 0.0223; generalized least squares gives 0.1347 with standard error 0.0512, larger by a factor of 2.30. The estimates agree to three decimals, the standard errors not at all. The annual cycle tells the same story in reverse: the coefficient of  $\sin(2\pi \text{ month}/12)$  is 2.7629 under both fits, with standard error 0.0554 from ordinary and 0.0354 from generalized least squares — this time the usual standard error is too *large*. [Section 31.2](#) explains both facts.

The  $F$  test of “no trend” is 36.7 if the annual shock is ignored and 6.92 once it is allowed for, on 1 and 724 degrees of freedom ( $p = 0.0087$ ): still clear evidence, but with a tenth of the apparent weight. The test that the second and third harmonics are absent moves the other way, from 10.4 to 25.6, because the annual shock inflates  $s^2$  and so deflates every  $F$  whose numerator lives inside years. [Figure 31.1.1](#)(a) shows the annual mean residuals, far too spread out for independent errors, and panel (b) the two confidence bands.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

frame = sm.datasets.elnino.load_pandas().data
years = frame["YEAR"].to_numpy()
temps = frame.iloc[:, 1:].to_numpy()           # 61 years by 12 months
G, m = temps.shape
n = G * m
y = temps.ravel()                             # the month varies fastest
year = np.repeat(years, m)
month = np.tile(np.arange(1, m + 1), G)
time = (year - 1980.0) + (month - 6.5) / 12.0   # decimal years, centred at
1980

cols = [np.ones(n), time / 10.0]               # the trend is per decade
for k in (1, 2, 3):
    cols += [np.cos(2 * np.pi * k * month / m), np.sin(2 * np.pi * k * month / m)]
X = np.column_stack(cols)
p = X.shape[1]

rho = 0.5                                     # taken as known for now

def gls_fit(X, y, rho, m):
    """GLS with equicorrelation rho inside blocks of m consecutive observations.
```

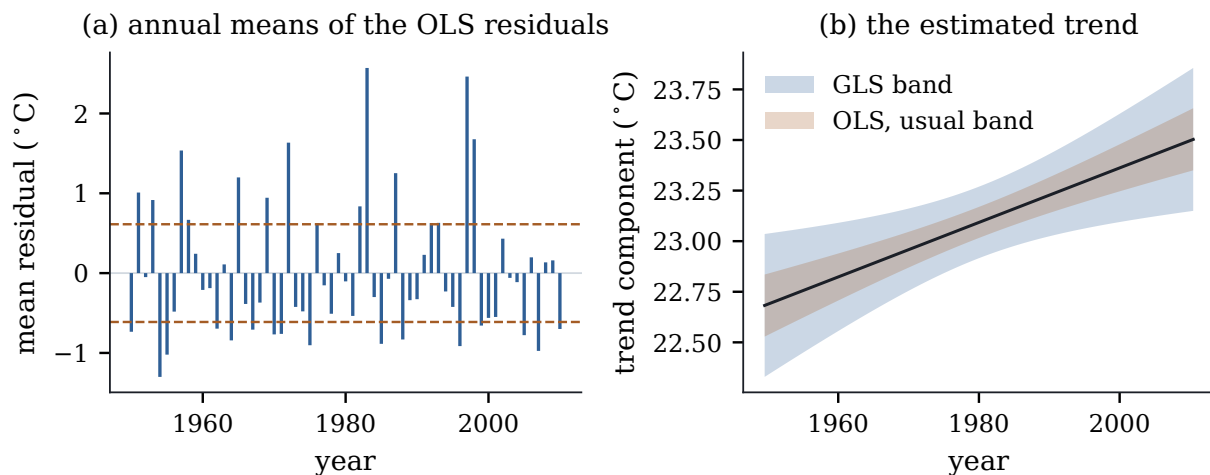


Figure 31.1.1: El Niño sea surface temperatures, 1950–2010. (a) The annual means of the ordinary least squares residuals, with the  $\pm 2$  standard deviation lines that independent errors would give (dashed). Whole years are warm or cold together. (b) The estimated trend component with nominal 95% bands, from ordinary least squares with the usual covariance and from generalized least squares with  $\rho = 0.5$ .

```

Whitening is the quasi-demeaning  $y \rightarrow (y - c \bar{y}) / \sqrt{1 - \rho}$  with
 $c = 1 - \sqrt{((1 - \rho) / (1 + (m - 1) \rho))}$ ; ordinary least squares on the
whitened
data is generalized least squares on the original data.
"""
c = 1.0 - np.sqrt((1 - rho) / (1 + (m - 1) * rho))

def whiten(a):
    b = a.reshape(len(a) // m, m, -1) if a.ndim > 1 else a.reshape(-1, m)
    return ((b - c * b.mean(axis=1, keepdims=True)) / np.sqrt(1 - rho)).reshape(
        a.shape)

Xw, yw = whiten(X), whiten(y)
b = np.linalg.solve(Xw.T @ Xw, Xw.T @ yw)
rss = float((yw - Xw @ b) @ (yw - Xw @ b))           # the  $V^{-1}$  residual sum of
squares
s2 = rss / (len(y) - X.shape[1])
return b, s2 * np.linalg.inv(Xw.T @ Xw), rss, c

b_ols = np.linalg.solve(X.T @ X, X.T @ y)
rss_ols = float((y - X @ b_ols) @ (y - X @ b_ols))
V_ols = rss_ols / (n - p) * np.linalg.inv(X.T @ X)   # the standard errors usually
reported
b_gls, V_gls, rss_gls, cq = gls_fit(X, y, rho, m)

print(f"trend per decade: OLS {b_ols[1]:+.4f} GLS {b_gls[1]:+.4f}")
print(f"standard error: OLS {np.sqrt(V_ols[1, 1]):.4f} GLS {np.sqrt(V_gls[1, 1]):.4f}")
print(f"quasi-demeaning constant c = {cq:.4f}, s_GLS = {np.sqrt(rss_gls / (n - p)):.4f}")

```

The  $F$  test of [Theorem 31.1.1\(e\)](#) is computed exactly as in Part III: fit the full and the reduced model in the same geometry and compare their  $\mathbf{V}^{-1}$  residual sums of squares. The listing does

this for both hypotheses of [Example \(A trend in sea surface temperature\)](#), using  $\rho = 0$  to recover the ordinary analysis.

```
def f_test(X, y, drop, rho, m):
    """F test that the coefficients listed in `drop` are zero, in the rho-geometry.
    """
    keep = [j for j in range(X.shape[1]) if j not in drop]
    full, null = gls_fit(X, y, rho, m), gls_fit(X[:, keep], y, rho, m)
    df1, df2 = len(drop), len(y) - X.shape[1]
    F = (null[2] - full[2]) / df1 / (full[2] / df2)
    return F, stats.f.sf(F, df1, df2)

F_trend_gls, p_trend_gls = f_test(X, y, [1], rho, m)
F_trend_ols, p_trend_ols = f_test(X, y, [1], 0.0, m)    # rho = 0 is ordinary least
               squares
F_harm_gls, p_harm_gls = f_test(X, y, [4, 5, 6, 7], rho, m)
F_harm_ols, p_harm_ols = f_test(X, y, [4, 5, 6, 7], 0.0, m)
print(f"trend:      F = {F_trend_ols:7.2f} (rho = 0)    F = {F_trend_gls:6.2f} (rho =
      0.5)")
print(f"harmonics:  F = {F_harm_ols:7.2f} (rho = 0)    F = {F_harm_gls:6.2f} (rho =
      0.5)")
```

*Idea.* Generalized least squares is one change of coordinates away from ordinary least squares, and everything in Parts II and III transfers — provided *all* of it does: the estimate, the residual sum of squares, the degrees of freedom and the sums of squares of every hypothesis. Mixing geometries, as in a generalized estimate with an ordinary standard error, is the one way to get this wrong.

## 31.1.5 Exercises

### 31.1.5.1 A. Check your understanding

#### Exercise A1.

Show that replacing  $(\sigma^2, \mathbf{V})$  by  $(\sigma^2/c, c\mathbf{V})$  for  $c > 0$  leaves  $\hat{\boldsymbol{\beta}}_{\mathbf{V}}$ ,  $\mathbf{A}_{\mathbf{V}}$  and every statistic in [Theorem 31.1.1](#)(d), (e) unchanged, and multiplies  $\text{SSE}_{\mathbf{V}}$  and  $s_{\mathbf{V}}^2$  by  $1/c$ . Conclude that no normalization of  $\mathbf{V}$  affects any inference about  $\boldsymbol{\beta}$ .

*Solution.* The criterion  $Q_{c\mathbf{V}}(\mathbf{b}) = Q_{\mathbf{V}}(\mathbf{b})/c$  is minimized at the same  $\mathbf{b}$ , and  $(c\mathbf{V})^{-1} = \mathbf{V}^{-1}/c$  cancels from [Eq. 31.1.2](#). Then  $\text{SSE}_{c\mathbf{V}} = \text{SSE}_{\mathbf{V}}/c$  and  $\boldsymbol{\lambda}^\top (\mathbf{X}^\top (c\mathbf{V})^{-1} \mathbf{X})^{-1} \boldsymbol{\lambda} = c \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \boldsymbol{\lambda}$ , so the factors  $c$  cancel in the  $t$  statistic; in  $F$  they cancel between numerator and denominator.

#### Exercise A2.

Show that  $\mathbf{1}^\top \hat{\boldsymbol{\epsilon}}_{\mathbf{V}} = 0$  for every  $\mathbf{y}$  iff  $\mathbf{1} \in \mathcal{C}(\mathbf{V}^{-1}\mathbf{X})$ . Give an example with an intercept in which this fails.

### 31.1.5.2 B. Practice

**Exercise B1.**

Let  $\mathbf{V}_0 = (1 - \rho)\mathbf{I}_m + \rho\mathbf{1}\mathbf{1}^\top$  with  $-1/(m-1) < \rho < 1$ . Verify [Eq. 31.1.4](#) by [Theorem 1.6.5](#), and show that  $\mathbf{L}_0 = (\mathbf{I}_m - c\mathbf{1}\mathbf{1}^\top/m)/\sqrt{1-\rho}$  satisfies  $\mathbf{L}_0\mathbf{V}_0\mathbf{L}_0^\top = \mathbf{I}_m$  for  $c = 1 - \sqrt{(1-\rho)/(1+(m-1)\rho)}$ . Deduce [Eq. 31.1.5](#).

*Solution.* Write  $\mathbf{P} = \mathbf{1}\mathbf{1}^\top/m$ , which is idempotent, so  $\mathbf{V}_0 = (1 - \rho)\mathbf{I} + \rho m\mathbf{P}$ . For any  $c$ ,  $(\mathbf{I} - c\mathbf{P})\mathbf{V}_0(\mathbf{I} - c\mathbf{P}) = (1 - \rho)\mathbf{I} + [(1 - \rho)(c^2 - 2c) + \rho m(1 - c)^2]\mathbf{P}$ , using  $\mathbf{P}^2 = \mathbf{P}$ . Setting  $a = 1 - \rho$  and  $b = \rho m$ , the bracket vanishes iff  $(a + b)c^2 - 2(a + b)c + b = 0$ , that is  $c^2 - 2c + b/(a + b) = 0$ , whose root in  $[0, 1)$  is  $c = 1 - \sqrt{a/(a + b)}$ . Dividing by  $1 - \rho$  gives  $\mathbf{L}_0\mathbf{V}_0\mathbf{L}_0^\top = \mathbf{I}$ . The Woodbury identity with  $\mathbf{U} = \mathbf{1}$  gives [Eq. 31.1.4](#) directly. Applying  $\mathbf{L}_0$  block by block is [Eq. 31.1.5](#), since  $\mathbf{1}\mathbf{1}^\top\mathbf{y}_k/m = \bar{y}_k\mathbf{1}$ .

**Exercise B2.**

Show that  $\text{tr } \mathbf{A}_\mathbf{V} = r$  for every positive definite  $\mathbf{V}$ . Then take  $n = 2$ ,  $p = 1$ ,

$$\mathbf{X} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}, \quad \mathbf{V} = \begin{pmatrix} 1 & 0.9 \\ 0.9 & 1 \end{pmatrix},$$

and show that the diagonal of  $\mathbf{A}_\mathbf{V}$  is  $(-4/7, 11/7)$ . Why does this not contradict [Proposition 6.8.1](#)?

**Exercise B3.**

Show that  $\hat{\boldsymbol{\varepsilon}}_\mathbf{V}^\top \hat{\boldsymbol{\varepsilon}}_\mathbf{V} \geq \hat{\boldsymbol{\varepsilon}}^\top \hat{\boldsymbol{\varepsilon}}$ , where  $\hat{\boldsymbol{\varepsilon}} = (\mathbf{I} - \mathbf{M})\mathbf{y}$  is the ordinary residual, with equality iff  $\mathbf{X}\hat{\boldsymbol{\beta}}_\mathbf{V} = \mathbf{M}\mathbf{y}$ . Show also that  $\hat{\boldsymbol{\varepsilon}}^\top \mathbf{V}^{-1} \hat{\boldsymbol{\varepsilon}} \geq \hat{\boldsymbol{\varepsilon}}_\mathbf{V}^\top \mathbf{V}^{-1} \hat{\boldsymbol{\varepsilon}}_\mathbf{V}$ . Explain why neither inequality says anything about which fit is better.

**Exercise B4.**

Observations  $Y_{kj}$ ,  $j = 1, \dots, m_k$ ,  $k = 1, \dots, G$ , have mean  $\mu$  and covariance  $\sigma^2\mathbf{V}$  with  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top$ ,  $\mathbf{Z}$  the cluster indicator matrix. Using [Eq. 31.1.4](#) cluster by cluster, show that

$$\hat{\mu}_\mathbf{V} = \frac{\sum_k w_k \bar{Y}_k}{\sum_k w_k}, \quad w_k = \frac{m_k}{1 - \rho + m_k \rho}, \quad (31.1.8)$$

with  $\text{Var}(\hat{\mu}_\mathbf{V}) = \sigma^2 / \sum_k w_k$ . Interpret the weights as  $\rho \rightarrow 0$  and  $\rho \rightarrow 1$ .

*Solution.* With  $\mathbf{V}$  block diagonal,  $\mathbf{V}^{-1}$  is block diagonal with blocks  $(\mathbf{I} - \frac{\rho}{1-\rho+m_k\rho}\mathbf{1}\mathbf{1}^\top)/(1-\rho)$ . Hence  $\mathbf{1}^\top \mathbf{V}^{-1} \mathbf{1} = \sum_k (m_k - \rho m_k^2 / (1 - \rho + m_k \rho)) / (1 - \rho) = \sum_k w_k$ , and  $\mathbf{1}^\top \mathbf{V}^{-1} \mathbf{y} = \sum_k w_k \bar{y}_k$  by the same computation. The generalized normal equation is  $(\mathbf{1}^\top \mathbf{V}^{-1} \mathbf{1})\mu = \mathbf{1}^\top \mathbf{V}^{-1} \mathbf{y}$ . As  $\rho \rightarrow 0$ ,  $w_k \rightarrow m_k$  and  $\hat{\mu}_\mathbf{V} \rightarrow \bar{y}$ ; as  $\rho \rightarrow 1$ ,  $w_k \rightarrow 1$  and every cluster counts once, however large, because the extra observations only repeat the same shock.

**31.1.5.3 C. Going deeper**

**Exercise C1.**

Suppose  $(\mathbf{Y}^\top, Y_0)^\top$  is jointly distributed with  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ ,  $E(Y_0) = \mathbf{x}_0^\top \boldsymbol{\beta}$ ,  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{V}$ ,  $\text{Cov}(\mathbf{Y}, Y_0) = \sigma^2 \mathbf{c}$  and  $\text{Var}(Y_0) = \sigma^2 v_0$ , with  $\mathbf{V}$  positive definite and  $\text{rank}(\mathbf{X}) = p$ . Among predictors  $\mathbf{a}^\top \mathbf{Y}$  with  $E(\mathbf{a}^\top \mathbf{Y} - Y_0) = 0$  for all  $\boldsymbol{\beta}$ , show that the one minimizing  $E(\mathbf{a}^\top \mathbf{Y} - Y_0)^2$  is

$$\hat{Y}_0 = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}} + \mathbf{c}^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\mathbf{V}}),$$

with prediction error variance  $\sigma^2[v_0 - \mathbf{c}^\top \mathbf{V}^{-1} \mathbf{c} + \mathbf{d}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{d}]$ ,  $\mathbf{d} = \mathbf{x}_0 - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{c}$ . This is Goldberger's formula; [Chapter 32](#) recognizes it as the best linear unbiased predictor of a random effect.

*Solution.* Unbiasedness requires  $\mathbf{X}^\top \mathbf{a} = \mathbf{x}_0$ , and  $E(\mathbf{a}^\top \mathbf{Y} - Y_0)^2 = \sigma^2(\mathbf{a}^\top \mathbf{V} \mathbf{a} - 2\mathbf{a}^\top \mathbf{c} + v_0)$ . By [Proposition 1.11.3](#) with the positive definite  $\mathbf{V}$ , the minimizer is  $\mathbf{a} = \mathbf{V}^{-1} \mathbf{c} + \mathbf{V}^{-1} \mathbf{X}(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{d}$ , and collecting terms gives  $\mathbf{a}^\top \mathbf{Y} = \mathbf{x}_0^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}} + \mathbf{c}^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}_{\mathbf{V}})$ ; the minimum value is the stated variance. Setting  $\mathbf{c} = \mathbf{0}$  recovers the ordinary prediction of [Chapter 12](#). A correlated future observation is predicted by correcting the fitted mean with the part of the current residual that predicts it.

**Exercise C2.**

Let  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{V}(\boldsymbol{\theta}))$  with  $\text{rank}(\mathbf{X}) = p$ . Show that for each fixed  $\boldsymbol{\theta}$  the log-likelihood is maximized over  $(\boldsymbol{\beta}, \sigma^2)$  at  $\hat{\boldsymbol{\beta}}_{\mathbf{V}(\boldsymbol{\theta})}$  and  $\hat{\sigma}^2(\boldsymbol{\theta}) = \text{SSE}_{\mathbf{V}(\boldsymbol{\theta})}/n$ , and that the profile log-likelihood is

$$\ell_p(\boldsymbol{\theta}) = -\frac{n}{2} \log \hat{\sigma}^2(\boldsymbol{\theta}) - \frac{1}{2} \log \det \mathbf{V}(\boldsymbol{\theta}) + \text{const.}$$

Why does the determinant term make the normalization of  $\mathbf{V}$  matter here, when [Exercise \(A1\)](#) says it does not matter for inference about  $\boldsymbol{\beta}$ ?

## 31.2 When ordinary least squares is best

The analysis in [Example \(A trend in sea surface temperature\)](#) left a puzzle: the generalized and the ordinary estimates of the trend agreed to three decimals and the seasonal coefficients to four, yet their standard errors disagreed by a factor of more than two in one direction and by half as much again in the other. Both halves have one explanation, the subject of this section: for a large class of designs ordinary least squares is *already* the best linear unbiased estimator, or very nearly so, while its reported standard error is wrong all the same.

Throughout,  $\mathbf{V}$  is nonnegative definite and may be singular,  $\mathbf{M}$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{X})$ , and “ $\mathbf{A}\mathbf{Y}$  is a BLUE of  $\mathbf{X}\boldsymbol{\beta}$ ” means that  $\boldsymbol{\rho}^\top \mathbf{A}\mathbf{Y}$  is best linear unbiased for  $\boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta}$  at every  $\boldsymbol{\rho} \in \mathbb{R}^n$ .

### 31.2.1 A criterion for a best estimator

Everything in this section, and most of the next, rests on one small lemma, the covariance version of [Exercise \(C3\)](#): a linear estimator is best exactly when it is uncorrelated with every



linear statistic that estimates zero. What is new is that  $\mathbf{V}$  may be singular, that the criterion is read off one vector  $\mathbf{c}$  at a time, and that uniqueness holds on  $\mathcal{C}(\mathbf{X} : \mathbf{V})$ , not merely with probability one.

### Lemma 31.2.1 — Criterion for a BLUE

Assume Eq. 31.1.1 with  $\mathbf{V}$  nonnegative definite. Let  $\mathbf{c} \in \mathbb{R}^n$  and  $\boldsymbol{\lambda} = \mathbf{X}^\top \mathbf{c}$ , so that  $\mathbf{c}^\top \mathbf{Y}$  is a linear unbiased estimator of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ .

- (a)  $\mathbf{c}^\top \mathbf{Y}$  is a BLUE of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  if and only if  $\mathbf{V}\mathbf{c} \in \mathcal{C}(\mathbf{X})$ .
- (b) Any two BLUEs of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  are equal with probability one. More precisely, if  $\mathbf{c}^\top \mathbf{Y}$  and  $\mathbf{b}^\top \mathbf{Y}$  are both BLUEs then  $\mathbf{c}^\top \mathbf{y} = \mathbf{b}^\top \mathbf{y}$  for every  $\mathbf{y} \in \mathcal{C}(\mathbf{X} : \mathbf{V})$ .
- (c) A matrix estimator  $\mathbf{A}\mathbf{Y}$  with  $\mathbf{A}\mathbf{X} = \mathbf{X}$  is a BLUE of  $\mathbf{X}\boldsymbol{\beta}$  if and only if  $\mathbf{A}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ , equivalently  $\mathcal{C}(\mathbf{V}\mathbf{A}^\top) \subseteq \mathcal{C}(\mathbf{X})$ .

*Proof.*

- (a) Every linear unbiased estimator of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is  $(\mathbf{c} + \mathbf{d})^\top \mathbf{Y}$  with  $\mathbf{X}^\top \mathbf{d} = \mathbf{0}$ , that is, with  $\mathbf{d} \in \mathcal{C}(\mathbf{X})^\perp$ . For real  $t$ ,

$$\text{Var}((\mathbf{c} + t\mathbf{d})^\top \mathbf{Y}) = \sigma^2(\mathbf{c}^\top \mathbf{V}\mathbf{c} + 2t\mathbf{c}^\top \mathbf{V}\mathbf{d} + t^2\mathbf{d}^\top \mathbf{V}\mathbf{d}),$$

a quadratic in  $t$  with nonnegative leading coefficient. If  $\mathbf{c}^\top \mathbf{Y}$  is best, this quadratic is minimized at  $t = 0$  for every such  $\mathbf{d}$ , which forces  $\mathbf{c}^\top \mathbf{V}\mathbf{d} = 0$ . Conversely, if  $\mathbf{c}^\top \mathbf{V}\mathbf{d} = 0$  for every  $\mathbf{d} \perp \mathcal{C}(\mathbf{X})$ , then  $\text{Var}((\mathbf{c} + \mathbf{d})^\top \mathbf{Y}) = \text{Var}(\mathbf{c}^\top \mathbf{Y}) + \sigma^2\mathbf{d}^\top \mathbf{V}\mathbf{d} \geq \text{Var}(\mathbf{c}^\top \mathbf{Y})$ . So  $\mathbf{c}^\top \mathbf{Y}$  is best iff  $\mathbf{V}\mathbf{c} \perp \mathcal{C}(\mathbf{X})^\perp$ , which is  $\mathbf{V}\mathbf{c} \in \mathcal{C}(\mathbf{X})$ .

- (b) Let  $\mathbf{c}^\top \mathbf{Y}$  and  $\mathbf{b}^\top \mathbf{Y}$  be BLUEs, with common variance  $v$  and covariance  $\kappa$ . The average  $\frac{1}{2}(\mathbf{c} + \mathbf{b})^\top \mathbf{Y}$  is unbiased, so  $v \leq \frac{1}{4}(v + v + 2\kappa)$ , that is  $v \leq \kappa$ . Therefore

$$\text{Var}((\mathbf{c} - \mathbf{b})^\top \mathbf{Y}) = 2v - 2\kappa \leq 0,$$

so it is zero, and  $(\mathbf{c} - \mathbf{b})^\top \mathbf{Y}$  has mean zero and zero variance: it vanishes with probability one. For the deterministic statement, zero variance means  $(\mathbf{c} - \mathbf{b})^\top \mathbf{V}(\mathbf{c} - \mathbf{b}) = 0$ , hence  $\mathbf{V}(\mathbf{c} - \mathbf{b}) = \mathbf{0}$  because  $\mathbf{V}$  is nonnegative definite, so  $\mathbf{c}$  and  $\mathbf{b}$  agree on  $\mathcal{C}(\mathbf{V})$ ; and  $\mathbf{X}^\top(\mathbf{c} - \mathbf{b}) = \mathbf{0}$  makes them agree on  $\mathcal{C}(\mathbf{X})$ . They therefore agree on  $\mathcal{C}(\mathbf{X}) + \mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{X} : \mathbf{V})$ .

- (c) Apply (a) to  $\mathbf{c} = \mathbf{A}^\top \boldsymbol{\rho}$ , which satisfies  $\mathbf{X}^\top \mathbf{A}^\top \boldsymbol{\rho} = (\mathbf{A}\mathbf{X})^\top \boldsymbol{\rho} = \mathbf{X}^\top \boldsymbol{\rho}$ , so  $\mathbf{c}^\top \mathbf{Y}$  is unbiased for  $\boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta}$ . The condition  $\mathbf{V}\mathbf{A}^\top \boldsymbol{\rho} \in \mathcal{C}(\mathbf{X})$  for every  $\boldsymbol{\rho}$  is  $\mathcal{C}(\mathbf{V}\mathbf{A}^\top) \subseteq \mathcal{C}(\mathbf{X})$ , which says  $(\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{A}^\top = \mathbf{0}$ ; transposing gives  $\mathbf{A}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ .

■

## 31.2.2 The characterization

### Theorem 31.2.2 — When ordinary least squares is best

Assume Eq. 31.1.1 with  $\mathbf{V}$  nonnegative definite, and let  $\mathbf{W}$  be any  $n \times (n - r)$  matrix with  $\mathcal{C}(\mathbf{W}) = \mathcal{C}(\mathbf{X})^\perp$ .

- (a) The following are equivalent.
- (i)  $\mathbf{MY}$  is a BLUE of  $\mathbf{X}\boldsymbol{\beta}$ .
  - (ii)  $\mathbf{MV}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ , equivalently  $\mathbf{MV} = \mathbf{VM}$ .
  - (iii)  $\mathcal{C}(\mathbf{VX}) \subseteq \mathcal{C}(\mathbf{X})$ .
  - (iv)  $\mathbf{V} = \mathbf{XAX}^\top + \mathbf{WBW}^\top$  for some nonnegative definite  $\mathbf{A}$  and  $\mathbf{B}$ .
  - (v)  $\mathcal{C}(\mathbf{X})$  has an orthonormal basis of eigenvectors of  $\mathbf{V}$ .
- (b) For  $\boldsymbol{\rho} \in \mathbb{R}^n$ , the estimator  $\boldsymbol{\rho}^\top \mathbf{MY}$  of  $\boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta}$  is a BLUE if and only if  $\mathbf{VM}\boldsymbol{\rho} \in \mathcal{C}(\mathbf{X})$ . When  $\text{rank}(\mathbf{X}) = p$  and  $\boldsymbol{\lambda} \in \mathbb{R}^p$ , this reads  $\mathbf{VX}(\mathbf{X}^\top \mathbf{X})^{-1}\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X})$ .
- (c) If  $\mathbf{V}$  is positive definite, the conditions of (a) are also equivalent to  $\mathbf{M}\mathbf{y} = \mathbf{A}_\mathbf{V}\mathbf{y}$  for every  $\mathbf{y}$ , and then ordinary and generalized least squares give the same fit, the same estimate of every estimable function, and the same  $\text{SSE}_\mathbf{V}$ .

*Proof.* For positive definite  $\mathbf{V}$  the chain (i)  $\Leftrightarrow$  (ii)  $\Leftrightarrow$  (iii) is [Exercise \(C3\)](#); the argument below needs no invertibility.

- (a) (i)  $\Leftrightarrow$  (ii). Take  $\mathbf{A} = \mathbf{M}$  in [Lemma 31.2.1\(c\)](#), which applies because  $\mathbf{MX} = \mathbf{X}$ . For the second form of (ii): if  $\mathbf{MV}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$  then  $\mathbf{MV} = \mathbf{MVM}$ , and  $\mathbf{MVM}$  is symmetric, so  $\mathbf{MV}$  is symmetric and equals  $(\mathbf{MV})^\top = \mathbf{VM}$ . Conversely  $\mathbf{MV} = \mathbf{VM}$  gives  $\mathbf{MV}(\mathbf{I} - \mathbf{M}) = \mathbf{VM} - \mathbf{VM} = \mathbf{0}$ .
- (ii)  $\Leftrightarrow$  (iii). If  $\mathbf{MV} = \mathbf{VM}$  then  $\mathbf{VX} = \mathbf{VMX} = \mathbf{MVX} \in \mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$ . Conversely, if  $\mathcal{C}(\mathbf{VX}) \subseteq \mathcal{C}(\mathbf{X})$  then  $\mathbf{MVX} = \mathbf{VX}$ . Since  $\mathcal{C}(\mathbf{M}) = \mathcal{C}(\mathbf{X})$ , every column of  $\mathbf{M}$  is a combination of columns of  $\mathbf{X}$ , so  $\mathbf{MVM} = \mathbf{VM}$ . The left side is symmetric, so  $\mathbf{VM} = (\mathbf{VM})^\top = \mathbf{MV}$ .
- (iii)  $\Rightarrow$  (iv). Split the identity as  $\mathbf{I} = \mathbf{M} + (\mathbf{I} - \mathbf{M})$ :

$$\begin{aligned} \mathbf{V} &= \mathbf{MVM} + \mathbf{MV}(\mathbf{I} - \mathbf{M}) \\ &\quad + (\mathbf{I} - \mathbf{M})\mathbf{VM} + (\mathbf{I} - \mathbf{M})\mathbf{V}(\mathbf{I} - \mathbf{M}) \\ &= \mathbf{MVM} + (\mathbf{I} - \mathbf{M})\mathbf{V}(\mathbf{I} - \mathbf{M}), \end{aligned}$$

the cross terms vanishing by (ii). Now  $\mathbf{M} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top$  by [Theorem 6.3.7](#), so  $\mathbf{MVM} = \mathbf{XAX}^\top$  with  $\mathbf{A} = (\mathbf{X}^\top \mathbf{X})^{-1}\mathbf{X}^\top \mathbf{VX}(\mathbf{X}^\top \mathbf{X})^{-1}$ , which is nonnegative definite because  $\mathbf{V}$  is. Likewise  $\mathbf{I} - \mathbf{M} = \mathbf{W}(\mathbf{W}^\top \mathbf{W})^{-1}\mathbf{W}^\top$ , since  $\mathbf{W}$  has full column rank and spans  $\mathcal{C}(\mathbf{X})^\perp$ , so  $(\mathbf{I} - \mathbf{M})\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{WBW}^\top$  with  $\mathbf{B} = (\mathbf{W}^\top \mathbf{W})^{-1}\mathbf{W}^\top \mathbf{VW}(\mathbf{W}^\top \mathbf{W})^{-1}$ , also nonnegative definite.

- (iv)  $\Rightarrow$  (iii).  $\mathbf{W}^\top \mathbf{X} = \mathbf{0}$ , so  $\mathbf{VX} = \mathbf{XAX}^\top \mathbf{X} \in \mathcal{C}(\mathbf{X})$ .
- (v)  $\Leftrightarrow$  (v). Condition (iii) says  $\mathcal{C}(\mathbf{X})$  is invariant under  $\mathbf{V}$ . If it is, the restriction of  $\mathbf{V}$  to  $\mathcal{C}(\mathbf{X})$  is a symmetric operator on that subspace, so by [Theorem 1.7.2](#) it has an orthonormal eigenbasis there, and these are eigenvectors of  $\mathbf{V}$ . Conversely, if  $\mathcal{C}(\mathbf{X})$  is spanned by eigenvectors of  $\mathbf{V}$  then  $\mathbf{V}$  maps each of them, and hence  $\mathcal{C}(\mathbf{X})$ , into  $\mathcal{C}(\mathbf{X})$ .
- (b) By [Lemma 31.2.1\(a\)](#) with  $\mathbf{c} = \mathbf{M}\boldsymbol{\rho}$ , which satisfies  $\mathbf{X}^\top \mathbf{M}\boldsymbol{\rho} = \mathbf{X}^\top \boldsymbol{\rho}$ , so that  $\boldsymbol{\rho}^\top \mathbf{MY}$  is unbiased for  $\boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta}$ . For the full-rank form,  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\rho}^\top \mathbf{MY}$  with  $\boldsymbol{\rho} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\boldsymbol{\lambda}$ , and then  $\mathbf{M}\boldsymbol{\rho} = \boldsymbol{\rho}$ .
- (c) With  $\mathbf{V}$  positive definite, (iii) is Kruskal's condition, and [Theorem 6.9.3](#) says it holds iff  $\mathbf{M}\mathbf{y} = \mathbf{A}_\mathbf{V}\mathbf{y}$  for every  $\mathbf{y}$ . Equal fits give equal estimates of every estimable function by [Theorem 6.4.4](#), and equal residuals, hence equal  $\text{SSE}_\mathbf{V}$ .

Form (iv) is Rao's **simple covariance structure**: the error splits as  $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_1 + \boldsymbol{\varepsilon}_2$  with  $\boldsymbol{\varepsilon}_1 \in \mathcal{C}(\mathbf{X})$ ,  $\boldsymbol{\varepsilon}_2 \perp \mathcal{C}(\mathbf{X})$  and the two parts uncorrelated ([Exercise \(B1\)](#)). Ordinary least squares is best exactly when the noise splits along the model space in this way, for then the residuals carry no usable information about  $\boldsymbol{\beta}$ . Form (v) is the one to check when the eigenstructure of  $\mathbf{V}$  is known.

### 31.2.3 Balanced layouts and a common shock

The most important case in practice is a balanced layout whose errors share a shock at some level of the design.

#### Example. (Equicorrelation in a balanced layout)

Let the  $n = Gm$  observations fall into  $G$  groups of equal size  $m$ , let  $\mathbf{Z}$  be the  $n \times G$  matrix of group indicators, and let  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top$ , the covariance produced by a group-level shock. Since  $\mathbf{Z}\mathbf{Z}^\top$  is  $m$  times the orthogonal projection onto the group-constant vectors, condition (iii) holds iff  $\mathcal{C}(\mathbf{X})$  is closed under replacing a vector by its group means — which happens whenever  $\mathcal{C}(\mathbf{X})$  is spanned by group-constant vectors together with vectors whose group means vanish, exactly what a balanced design gives through its between-group and within-group contrasts.

In the El Niño data of [Example \(A trend in sea surface temperature\)](#) the groups are years. If the seasonal part of the model is the full set of eleven month effects, every column of  $\mathbf{X}$  splits into a year-level part and a part whose year means vanish, both again in  $\mathcal{C}(\mathbf{X})$ , so the condition holds *exactly*: the two fits agree to nine decimals, whatever  $\rho$ . Their standard errors do not. At  $\rho = 0.5$  the root of the expected reported variance  $E(s^2)\{(\mathbf{X}^\top\mathbf{X})^{-1}\}_{22}$  is only 0.391 of the true standard error of the trend.

The same argument covers randomized complete block designs with random blocks, and any design whose every regressor is constant within a block or sums to zero in each block. The analysis of variance tables of [Part IV](#) therefore survive random block effects as *point estimates*; only their error terms need rethinking, and [Chapter 33](#) develops the split-plot case, where condition (iv) becomes a decomposition into error strata. Incomplete block designs break the condition: between-block comparisons then carry treatment information that a within-block analysis discards, and recovering it — Yates's recovery of interblock information — is generalized least squares.

### 31.2.4 When some contrasts are best and others are not

Part (b) of [Theorem 31.2.2](#) is finer than part (a): even when  $\mathbf{MY}$  is not best for  $\mathbf{X}\boldsymbol{\beta}$  as a whole, particular functions still can be. The mechanism is easiest to see when the balance of [Example \(Equicorrelation in a balanced layout\)](#) is broken.

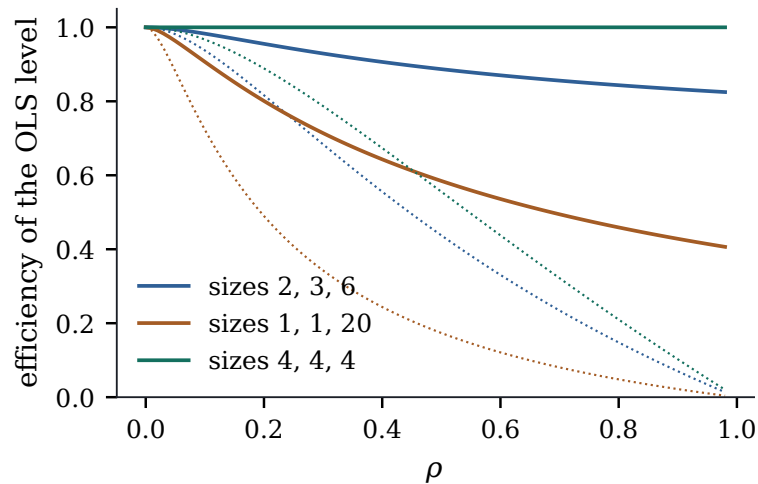


Figure 31.2.1: Efficiency of the ordinary least squares estimate of the overall level, for three cluster layouts with equicorrelation  $\rho$  inside clusters (solid). The balanced layout loses nothing at any  $\rho$ . The dotted curves are the Kantorovich bound of [Theorem 19.2.3](#) for the same  $\mathbf{V}$ , far below the efficiency the design actually attains.

### Example. (Unequal clusters)

Take  $G$  clusters of sizes  $m_1, \dots, m_G$ , a covariate  $\mathbf{w}$  centred inside every cluster, and the model  $\mathbf{X} = [\mathbf{1}, \mathbf{w}]$  with  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top$ . Because  $\mathbf{1}^\top \mathbf{w} = 0$ , the vector attached to the slope is  $\mathbf{c} = \mathbf{w}/\mathbf{w}^\top \mathbf{w}$ , and  $\mathbf{Z}\mathbf{Z}^\top \mathbf{w} = \mathbf{0}$  because  $\mathbf{w}$  has zero mean in each cluster; so  $\mathbf{V}\mathbf{c} = (1 - \rho)\mathbf{c} \in \mathcal{C}(\mathbf{X})$  and the within-cluster slope is estimated best by ordinary least squares, for every  $\rho$  and every set of sizes.

The vector attached to the level is  $\mathbf{c} = \mathbf{1}/n$ , and  $\mathbf{Z}\mathbf{Z}^\top \mathbf{1}$  is the vector whose entry for an observation in cluster  $k$  is  $m_k$ . That step function lies in  $\mathcal{C}(\mathbf{X}) = \text{span}\{\mathbf{1}, \mathbf{w}\}$  only if all the  $m_k$  are equal, since  $\mathbf{w}$  is not constant within clusters. So with unequal clusters the overall level is *not* estimated best by ordinary least squares; [Eq. 31.1.8](#) gives the estimator that is.

With sizes 2, 3, 6 and  $\rho = 0.5$  the weights of [Eq. 31.1.8](#) are 1.333, 1.500 and 1.714, summing to 4.5476, so the generalized estimate of the level has variance  $0.2199 \sigma^2$  against  $0.2479 \sigma^2$  for the sample mean: an efficiency of 0.887, well above the Kantorovich bound 0.437 that [Proposition 31.2.3\(c\)](#) allows for the eigenvalues 3.500 and 0.500 of this  $\mathbf{V}$ . The slope loses nothing. Meanwhile the usual standard errors are wrong for both: the root of the expected reported variance is 0.563 of the true standard error for the level and 1.314 of it for the slope.

```
import numpy as np

sizes = [2, 3, 6]
w = np.array([-1.0, 1.0, -1.0, 0.0, 1.0, -3.0, -2.0, -1.0, 1.0, 2.0, 3.0])
n = len(w)
Z = np.zeros((n, len(sizes))) # cluster indicators
Z[np.arange(n), np.repeat(np.arange(len(sizes)), sizes)] = 1.0
X = np.column_stack([np.ones(n), w])
rho = 0.5
V = (1 - rho) * np.eye(n) + rho * Z @ Z.T
```

```

Ginv = np.linalg.inv(X.T @ X)
Cov_ols = Ginv @ X.T @ V @ X @ Ginv # sigma^2 = 1 throughout
Cov_gls = np.linalg.inv(X.T @ np.linalg.solve(V, X))
print("efficiency of OLS: level %.4f slope %.4f"
      % (Cov_gls[0, 0] / Cov_ols[0, 0], Cov_gls[1, 1] / Cov_ols[1, 1]))

```

## 31.2.5 How much is lost, and what is actually broken

### Proposition 31.2.3 — The cost of ordinary least squares

Assume [Eq. 31.1.1](#) with  $\mathbf{V}$  positive definite and  $\text{rank}(\mathbf{X}) = p$ . Write  $\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{Y}$  and let  $\lambda_1 \geq \cdots \geq \lambda_n > 0$  be the eigenvalues of  $\mathbf{V}$ . For  $\boldsymbol{\lambda} \neq \mathbf{0}$  define the efficiency  $e(\boldsymbol{\lambda}) = \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}) / \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})$ .

- (a)  $\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\hat{\boldsymbol{\beta}}_{\mathbf{V}})$  is nonnegative definite, so  $0 < e(\boldsymbol{\lambda}) \leq 1$  for every  $\boldsymbol{\lambda}$ .
- (b)  $e(\boldsymbol{\lambda}) = 1$  if and only if  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}$  with probability one, which by [Theorem 31.2.2\(b\)](#) happens iff  $\mathbf{V}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X})$ .
- (c)  $e(\boldsymbol{\lambda}) \geq 4\lambda_1\lambda_n/(\lambda_1 + \lambda_n)^2$  for every  $\boldsymbol{\lambda}$  and every  $\mathbf{X}$ , and the bound is attained.

*Proof.*

- (a) Both  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  and  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}$  are linear unbiased estimators of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$ , and the latter is best by [Corollary 7.2.3](#), so  $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) \geq \text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}) > 0$ . Taking  $\boldsymbol{\lambda}$  over  $\mathbb{R}^p$  gives  $\boldsymbol{\lambda}^\top (\text{Cov}(\hat{\boldsymbol{\beta}}) - \text{Cov}(\hat{\boldsymbol{\beta}}_{\mathbf{V}})) \boldsymbol{\lambda} \geq 0$ .
- (b) If  $e(\boldsymbol{\lambda}) = 1$  then  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  attains the minimum variance and is therefore a BLUE, so it equals  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}}$  with probability one by [Lemma 31.2.1\(b\)](#); the converse is immediate. The condition is part (b) of [Theorem 31.2.2](#).
- (c) This is [Theorem 19.2.3](#), proved in [Section 19.2](#) by a matrix form of Kantorovich's inequality, together with the design given there that attains it.

■

Part (c) is the worst case over all designs and all  $\boldsymbol{\lambda}$ . A determinant version is also known: if  $p \leq n/2$ , then

$$\frac{\det \text{Cov}(\hat{\boldsymbol{\beta}}_{\mathbf{V}})}{\det \text{Cov}(\hat{\boldsymbol{\beta}})} \geq \prod_{j=1}^p \frac{4\lambda_j\lambda_{n-j+1}}{(\lambda_j + \lambda_{n-j+1})^2},$$

with equality for a design built from the corresponding pairs of eigenvectors. This is the Bloomfield–Watson–Knott inequality, proved independently by Bloomfield and Watson (1975) and by Knott (1975) in back-to-back papers. We state it without proof; the argument needs more matrix theory than this book develops.

In practice the loss is far smaller, because real designs are not adversarial. The El Niño model with three harmonics is typical: Kruskal's condition fails, with  $\max_{s,t} |\{(\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{X}\}_{st}| = 0.126$ , yet the efficiency of the trend coefficient is 0.9996 and that of every seasonal coefficient is 1.0000 to four decimals. The intercept is exactly best, because  $\mathbf{1}$  is orthogonal to every other column, so  $\mathbf{c} = \mathbf{1}/n$  and  $\mathbf{V}\mathbf{1} = \{1 + 11\rho\}\mathbf{1}$  is a multiple of  $\mathbf{1}$ . What is badly wrong is the standard

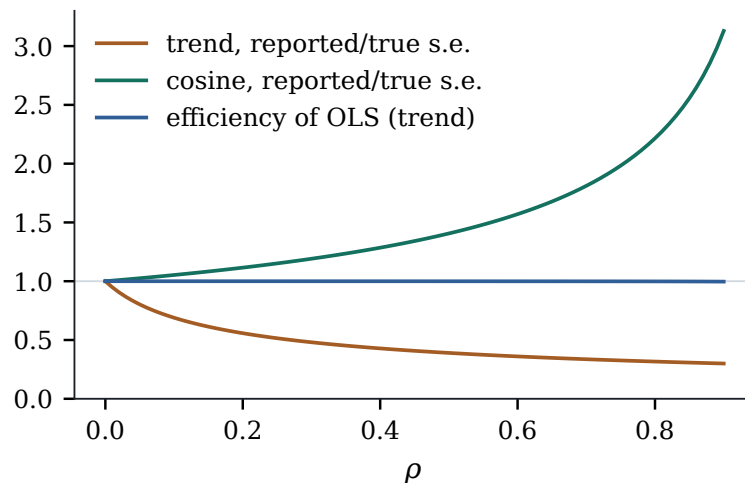


Figure 31.2.2: The El Niño model with three harmonics, as a function of the intraclass correlation  $\rho$ . The efficiency of the ordinary least squares trend estimate stays above 0.99, while the ratio of the reported to the true standard error falls to a third for the trend and rises past three for a seasonal coefficient.

error: the root of the expected value of  $s^2\{(\mathbf{X}^\top \mathbf{X})^{-1}\}_{jj}$  is 0.390 of the true standard error for the trend and 1.406 of it for the first cosine, and Figure 31.2.2 shows both ratios running away as  $\rho$  grows while the efficiency never falls below 0.99.

*Idea.* Correlated errors damage the *standard error* far more than the *estimate*. [Theorem 19.2.3](#) bounds the efficiency below by a quantity depending only on the extreme eigenvalues of  $\mathbf{V}$ , and for ordinary designs the loss is a few per cent. Nothing bounds the error in  $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$ , which can be too small or too large by any factor. If only one of the two problems can be fixed, fix the standard error.

That is what the sandwich estimator of [Section 21.4](#) does. It leaves  $\hat{\boldsymbol{\beta}}$  alone and replaces  $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$  by  $(\mathbf{X}^\top \mathbf{X})^{-1}(\sum_k \mathbf{X}_k^\top \hat{\boldsymbol{\varepsilon}}_k \hat{\boldsymbol{\varepsilon}}_k^\top \mathbf{X}_k)(\mathbf{X}^\top \mathbf{X})^{-1}$ , summed over units that can be treated as independent; for clustered data the unit is a cluster, and the argument of [Theorem 21.4.2](#) applies cluster by cluster rather than observation by observation. We do not prove that extension: [Theorem 21.4.2](#) assumes independent observations, consistency of the cluster form as the number of clusters grows is due to Liang and Zeger (1986), and [Exercise \(C1\)](#) derives the same matrix as a bootstrap covariance. It recovers no efficiency and needs many independent units, as [Section 31.4](#) shows; the two repairs are complementary.

## 31.2.6 Exercises

### 31.2.6.1 A. Check your understanding

#### Exercise A1.

Show directly from [Theorem 31.2.2\(a\)\(v\)](#) that ordinary least squares is best when the columns of  $\mathbf{X}$  are eigenvectors of  $\mathbf{V}$ , and use this to recover the equicorrelation result of [Exercise \(A1\)](#).

*Solution.* If each column of  $\mathbf{X}$  is an eigenvector of  $\mathbf{V}$ , then  $\mathcal{C}(\mathbf{X})$  is spanned by eigenvectors, which is (v). For  $\mathbf{V} = (1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top$  the vector  $\mathbf{1}$  is an eigenvector with eigenvalue  $1 + (n - 1)\rho$  and every vector orthogonal to  $\mathbf{1}$  is an eigenvector with eigenvalue  $1 - \rho$ . A model containing an intercept has  $\mathcal{C}(\mathbf{X}) = \text{span}\{\mathbf{1}\} \oplus (\mathcal{C}(\mathbf{X}) \cap \mathbf{1}^\perp)$ , a span of eigenvectors, so (v) holds.

### Exercise A2.

Show that  $\mathbf{M}\mathbf{V} = \mathbf{V}\mathbf{M}$  for every  $\mathbf{X}$  if and only if  $\mathbf{V}$  is a multiple of  $\mathbf{I}$ .

### 31.2.6.2 B. Practice

#### Exercise B1.

Assume the simple covariance structure  $\mathbf{V} = \mathbf{X}\mathbf{A}\mathbf{X}^\top + \mathbf{W}\mathbf{B}\mathbf{W}^\top$  of [Theorem 31.2.2\(a\)\(iv\)](#). Show that  $\text{Cov}(\mathbf{M}\mathbf{Y}, (\mathbf{I} - \mathbf{M})\mathbf{Y}) = \mathbf{0}$ , and that the error can be written  $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}_1 + \boldsymbol{\varepsilon}_2$  with  $\boldsymbol{\varepsilon}_1 \in \mathcal{C}(\mathbf{X})$ ,  $\boldsymbol{\varepsilon}_2 \in \mathcal{C}(\mathbf{X})^\perp$  and  $\text{Cov}(\boldsymbol{\varepsilon}_1, \boldsymbol{\varepsilon}_2) = \mathbf{0}$  with probability one.

*Solution.*  $\text{Cov}(\mathbf{M}\mathbf{Y}, (\mathbf{I} - \mathbf{M})\mathbf{Y}) = \sigma^2\mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$  by (ii). Put  $\boldsymbol{\varepsilon}_1 = \mathbf{M}\boldsymbol{\varepsilon}$  and  $\boldsymbol{\varepsilon}_2 = (\mathbf{I} - \mathbf{M})\boldsymbol{\varepsilon}$ ; these lie in the stated spaces, add to  $\boldsymbol{\varepsilon}$ , and have cross-covariance  $\sigma^2\mathbf{M}\mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ . Their covariance matrices are  $\sigma^2\mathbf{X}\mathbf{A}\mathbf{X}^\top$  and  $\sigma^2\mathbf{W}\mathbf{B}\mathbf{W}^\top$ .

#### Exercise B2.

Suppose  $\text{Cov}(\mathbf{Y}) = \boldsymbol{\Sigma}(\boldsymbol{\theta})$  with  $\boldsymbol{\theta}$  unknown, and that  $\mathcal{C}(\boldsymbol{\Sigma}(\boldsymbol{\theta})\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  for every  $\boldsymbol{\theta}$  in the parameter set. Show that  $\mathbf{M}\mathbf{Y}$  is a BLUE for every  $\boldsymbol{\theta}$ , so that the best estimator is known even though the covariance is not. Give an example with two unknown variance components.

#### Exercise B3.

In [Example \(Unequal clusters\)](#) with two clusters of sizes  $m_1$  and  $m_2$ , show that the efficiency of the sample mean as an estimate of the level is

$$e = \frac{(m_1 + m_2)^2}{\{(1 - \rho)(m_1 + m_2) + \rho(m_1^2 + m_2^2)\}(w_1 + w_2)}, \quad w_k = \frac{m_k}{1 - \rho + m_k\rho}.$$

Show that  $e = 1$  when  $m_1 = m_2$ , and find the limit as  $\rho \rightarrow 1$  with  $m_1 = 1$  and  $m_2 = m$ .

*Solution.*  $\text{Var}(\bar{Y}) = \sigma^2\mathbf{1}^\top\mathbf{V}\mathbf{1}/n^2$  with  $\mathbf{1}^\top\mathbf{V}\mathbf{1} = (1 - \rho)n + \rho\sum_k m_k^2$ , and  $\text{Var}(\hat{\mu}_{\mathbf{V}}) = \sigma^2/\sum_k w_k$  by [Eq. 31.1.8](#); divide. If  $m_1 = m_2 = m$  both variances are  $\sigma^2(1 - \rho + m\rho)/(2m)$ , so  $e = 1$ . With  $m_1 = 1$ ,  $m_2 = m$  and  $\rho \rightarrow 1$ ,  $w_k \rightarrow 1$  and  $e \rightarrow (1 + m)^2/\{2(1 + m^2)\}$ , which falls to  $1/2$  as  $m$  grows.

### 31.2.6.3 C. Going deeper

**Exercise C1.**

Let  $\text{rank}(\mathbf{X}) = p$  and put  $\mathcal{B} = \{\boldsymbol{\lambda} \in \mathbb{R}^p : \mathbf{V}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X})\}$ , the set of coefficient functions that ordinary least squares estimates best. Show that  $\mathcal{B}$  is a subspace, that  $\mathcal{B} = \mathbb{R}^p$  iff  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , and that  $\dim \mathcal{B} = p - \text{rank}((\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1})$ . Evaluate it for [Example \(Unequal clusters\)](#).

*Solution.* A vector  $\boldsymbol{\lambda}$  lies in  $\mathcal{B}$  iff  $(\mathbf{I} - \mathbf{M})\mathbf{V}\mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}\boldsymbol{\lambda} = \mathbf{0}$ , so  $\mathcal{B}$  is the null space of that matrix, hence a subspace, and [Theorem 1.2.2](#) gives its dimension. It is all of  $\mathbb{R}^p$  iff the matrix is zero, which is  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  because  $(\mathbf{X}^\top \mathbf{X})^{-1}$  is nonsingular. In [Example \(Unequal clusters\)](#),  $p = 2$  and the map kills the slope direction but not the level direction, so  $\dim \mathcal{B} = 1$ : the functions estimated best are exactly the multiples of the slope.

**Exercise C2.**

Suppose  $\mathbf{V}$  has only two distinct eigenvalues,  $\lambda_1$  with multiplicity  $k$  and  $\lambda_2$  with multiplicity  $n - k$ , and let  $\mathcal{E}_1$  be the eigenspace of  $\lambda_1$ . Show that  $\mathbf{M}\mathbf{V} = \mathbf{V}\mathbf{M}$  iff  $\mathcal{C}(\mathbf{X}) = (\mathcal{C}(\mathbf{X}) \cap \mathcal{E}_1) \oplus (\mathcal{C}(\mathbf{X}) \cap \mathcal{E}_1^\perp)$ . Deduce that when this splitting holds every  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  has efficiency 1, and that when it fails there is a design attaining the bound  $4\lambda_1\lambda_2/(\lambda_1 + \lambda_2)^2$ .

## 31.3 Singular covariance matrices

A covariance matrix is nonnegative definite, not necessarily positive definite. When it is singular, some linear combination of the observations has zero variance and is therefore a *constant*: the data satisfy an exact linear relation. This is not a pathology to be regularized away — it happens whenever a linear operation has destroyed information — and it changes the theory, because some functions of  $\boldsymbol{\beta}$  become known without error.

### 31.3.1 Where a singular covariance comes from

Three mechanisms cover most cases.

**Constraints built into the data.** Percentages adjusted to add to a hundred, effects reported under a sum-to-zero side condition ([Chapter 8](#)), residuals from an earlier fit: the reported vector satisfies an exact linear identity, and if  $\mathbf{a}^\top \mathbf{Y}$  is constant then  $\mathbf{V}\mathbf{a} = \mathbf{0}$  by [Theorem 2.2.3\(c\)](#).

**A deterministic component.** Some observations may be free of measurement error — a boundary value known exactly, a fitted value substituted for a missing one — and the corresponding rows and columns of  $\mathbf{V}$  vanish. So is  $\mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top$ , the covariance of a random-effects model  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}$  with no separate measurement error, as soon as  $\mathbf{Z}$  has fewer columns than rows; [Chapter 32](#) studies that family.

**Linear filtering with redundancy.** Replacing uncorrelated observations by more linear combinations than the rank of the transformation — differences together with their total, overlapping averages together with the underlying ones — produces a singular covariance. Redundancy is what matters, not filtering:  $n - 1$  successive differences of  $n$  uncorrelated observations have a



nonsingular tridiagonal covariance.

### 31.3.2 The consistency condition

*Idea.* A singular  $\mathbf{V}$  confines the *errors* to a subspace:  $\Pr(\boldsymbol{\varepsilon} \in \mathcal{C}(\mathbf{V})) = 1$ . With  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X})$ , this confines the data to  $\mathcal{C}(\mathbf{X} : \mathbf{V}) = \mathcal{C}(\mathbf{X}) + \mathcal{C}(\mathbf{V})$  and  $\mathbf{X}\boldsymbol{\beta}$  to the affine set  $\mathbf{y} + \mathcal{C}(\mathbf{V})$ . Estimation becomes a constrained problem: find the point of  $\mathcal{C}(\mathbf{X})$  nearest  $\mathbf{y}$  among those the data allow.

Write  $\mathcal{C}(\mathbf{X} : \mathbf{V})$  for  $\mathcal{C}(\mathbf{X}) + \mathcal{C}(\mathbf{V})$ , the column space of  $[\mathbf{X} \ \mathbf{V}]$ . The following small lemma is used repeatedly.

#### Lemma 31.3.1

If  $\mathbf{A}$  and  $\mathbf{B}$  are nonnegative definite of the same order, then  $\mathcal{C}(\mathbf{A} + \mathbf{B}) = \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})$ .

*Proof.*  $(\mathbf{A} + \mathbf{B})\mathbf{w} = \mathbf{A}\mathbf{w} + \mathbf{B}\mathbf{w}$  shows  $\mathcal{C}(\mathbf{A} + \mathbf{B}) \subseteq \mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B})$ . For the reverse, take  $\mathbf{w} \perp \mathcal{C}(\mathbf{A} + \mathbf{B})$ . Then  $\mathbf{w}^\top \mathbf{A}\mathbf{w} + \mathbf{w}^\top \mathbf{B}\mathbf{w} = 0$ , and both terms are nonnegative, so both vanish. By Theorem 1.7.6,  $\|\mathbf{A}^{1/2}\mathbf{w}\|^2 = \mathbf{w}^\top \mathbf{A}\mathbf{w} = 0$ , so  $\mathbf{A}\mathbf{w} = \mathbf{A}^{1/2}\mathbf{A}^{1/2}\mathbf{w} = \mathbf{0}$ , and likewise  $\mathbf{B}\mathbf{w} = \mathbf{0}$ . Hence  $\mathbf{w} \perp \mathcal{C}(\mathbf{A})$  and  $\mathbf{w} \perp \mathcal{C}(\mathbf{B})$ , so  $\mathcal{C}(\mathbf{A} + \mathbf{B})^\perp \subseteq (\mathcal{C}(\mathbf{A}) + \mathcal{C}(\mathbf{B}))^\perp$ , which gives the inclusion. ■

### 31.3.3 The general theorem

Fix an orthogonal splitting adapted to  $\mathbf{V}$ : let  $\mathbf{\Gamma}$  be  $n \times q$  and  $\mathbf{N}$  be  $n \times (n - q)$  with orthonormal columns,  $\mathcal{C}(\mathbf{\Gamma}) = \mathcal{C}(\mathbf{V})$  and  $\mathcal{C}(\mathbf{N}) = \mathcal{C}(\mathbf{V})^\perp$ , where  $q = \text{rank}(\mathbf{V})$ . Put  $\mathbf{\Lambda} = \mathbf{\Gamma}^\top \mathbf{V} \mathbf{\Gamma}$ , which is  $q \times q$  positive definite, and let  $\mathbf{V}^+$  be the Moore–Penrose inverse (Definition 1.3.2), so that  $\mathbf{V}^+ = \mathbf{\Gamma} \mathbf{\Lambda}^{-1} \mathbf{\Gamma}^\top$ . Write  $r = \text{rank}(\mathbf{X})$  and  $d = \dim(\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V}))$ , and let  $\mathbf{U}$  be any  $n \times d$  matrix of full column rank with  $\mathcal{C}(\mathbf{U}) = \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V})$ .

#### Theorem 31.3.2 — The general Gauss–Markov model

Assume Eq. 31.1.1 with  $\mathbf{V}$  nonnegative definite of rank  $q$ .

- (a) (*Consistency.*)  $\mathbf{N}^\top \mathbf{Y} = \mathbf{N}^\top \mathbf{X}\boldsymbol{\beta}$  with probability one; consequently  $\Pr(\mathbf{Y} \in \mathcal{C}(\mathbf{X} : \mathbf{V})) = 1$  and  $\Pr(\mathbf{X}\boldsymbol{\beta} \in \mathbf{Y} + \mathcal{C}(\mathbf{V})) = 1$ .
- (b) (*Estimability and perfect estimation.*)  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  has a linear unbiased estimator iff  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ , exactly as when  $\mathbf{V} = \mathbf{I}$ . It has a linear unbiased estimator of variance zero iff  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top \mathbf{N})$ , a subspace of dimension  $r - d$ , and then  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \mathbf{b}^\top \mathbf{N}^\top \mathbf{Y}$  with probability one for any  $\mathbf{b}$  with  $\mathbf{X}^\top \mathbf{N} \mathbf{b} = \boldsymbol{\lambda}$ .
- (c) (*Rao’s unified formula.*) Let  $\mathbf{T} = \mathbf{V} + \mathbf{X} \mathbf{X}^\top$ . Then  $\mathcal{C}(\mathbf{T}) = \mathcal{C}(\mathbf{X} : \mathbf{V})$ ,  $\mathbf{T} \mathbf{T}^+ \mathbf{X} = \mathbf{X}$ , and

$$\mathbf{A} = \mathbf{X}(\mathbf{X}^\top \mathbf{T}^+ \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{T}^+ \quad (31.3.1)$$

satisfies  $\mathbf{AX} = \mathbf{X}$  and  $\mathbf{AV}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ . Hence  $\mathbf{AY}$  is a BLUE of  $\mathbf{X}\boldsymbol{\beta}$ .

- (d) (*Uniqueness.*) Every estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  has a BLUE, and any two BLUEs of it agree for every  $\mathbf{y} \in \mathcal{C}(\mathbf{X} : \mathbf{V})$ , hence with probability one.
- (e) (*Projection form.*) Let  $\mathbf{y} \in \mathcal{C}(\mathbf{X} : \mathbf{V})$ . The set  $\mathcal{F}(\mathbf{y}) = \{\mathbf{b} : \mathbf{y} - \mathbf{Xb} \in \mathcal{C}(\mathbf{V})\}$  is nonempty, and

$$\tilde{\boldsymbol{\beta}} \in \arg \min_{\mathbf{b} \in \mathcal{F}(\mathbf{y})} (\mathbf{y} - \mathbf{Xb})^\top \mathbf{V}^+ (\mathbf{y} - \mathbf{Xb}) \quad (31.3.2)$$

determines  $\mathbf{X}\tilde{\boldsymbol{\beta}}$  uniquely; the map  $\mathbf{y} \mapsto \mathbf{X}\tilde{\boldsymbol{\beta}}$  is linear and  $\mathbf{X}\tilde{\boldsymbol{\beta}}$  is a BLUE of  $\mathbf{X}\boldsymbol{\beta}$ . Explicitly, with  $\mathbf{K} = \mathbf{N}^\top \mathbf{X}$  and any generalized inverse  $\mathbf{K}^-$ ,

$$\mathbf{X}\tilde{\boldsymbol{\beta}} = \mathbf{Xb}_* + \mathbf{U}(\mathbf{U}^\top \mathbf{V}^+ \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{V}^+ (\mathbf{y} - \mathbf{Xb}_*), \quad \mathbf{b}_* = \mathbf{K}^- \mathbf{N}^\top \mathbf{y}. \quad (31.3.3)$$

- (f) (*Error sum of squares.*) Put  $\hat{\boldsymbol{\varepsilon}} = \mathbf{Y} - \mathbf{X}\tilde{\boldsymbol{\beta}}$  and  $\text{SSE}_{\mathbf{V}} = \hat{\boldsymbol{\varepsilon}}^\top \mathbf{V}^+ \hat{\boldsymbol{\varepsilon}}$ . Then  $E(\text{SSE}_{\mathbf{V}}) = \sigma^2(q - d)$ , and  $q - d = \text{rank}(\mathbf{X} : \mathbf{V}) - \text{rank}(\mathbf{X})$ . If  $\mathbf{Y}$  is normal, then  $\text{SSE}_{\mathbf{V}}/\sigma^2 \sim \chi^2(q - d)$ , independently of  $\mathbf{X}\tilde{\boldsymbol{\beta}}$ .

*Proof.*

- (a) By Theorem 2.2.3(d),  $\Pr(\boldsymbol{\varepsilon} \in \mathcal{C}(\sigma^2 \mathbf{V})) = 1$ , and  $\mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{N})^\perp$ , so  $\mathbf{N}^\top \boldsymbol{\varepsilon} = \mathbf{0}$  almost surely, giving  $\mathbf{N}^\top \mathbf{Y} = \mathbf{N}^\top \mathbf{X}\boldsymbol{\beta}$ . Then  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon} \in \mathcal{C}(\mathbf{X}) + \mathcal{C}(\mathbf{V})$  and  $\mathbf{X}\boldsymbol{\beta} = \mathbf{Y} - \boldsymbol{\varepsilon} \in \mathbf{Y} + \mathcal{C}(\mathbf{V})$ .
- (b)  $\mathbf{c}^\top \mathbf{Y}$  is unbiased for  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  for every  $\boldsymbol{\beta}$  iff  $\mathbf{X}^\top \mathbf{c} = \boldsymbol{\lambda}$ , which is possible iff  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ . Its variance is  $\sigma^2 \mathbf{c}^\top \mathbf{V} \mathbf{c}$ , which vanishes iff  $\mathbf{V} \mathbf{c} = \mathbf{0}$  (as in Lemma 31.3.1), that is iff  $\mathbf{c} \in \mathcal{C}(\mathbf{V})^\perp = \mathcal{C}(\mathbf{N})$ , say  $\mathbf{c} = \mathbf{Nb}$ . Then  $\boldsymbol{\lambda} = \mathbf{X}^\top \mathbf{Nb}$  and  $\mathbf{c}^\top \mathbf{Y} = \mathbf{b}^\top \mathbf{N}^\top \mathbf{Y}$ , which equals  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  almost surely by (a). For the dimension,  $\text{rank}(\mathbf{X}^\top \mathbf{N}) = \text{rank}(\mathbf{N}) - \dim\{\mathbf{b} : \mathbf{Nb} \perp \mathcal{C}(\mathbf{X})\}$  by Theorem 1.2.2, and  $\{\mathbf{Nb} : \mathbf{Nb} \perp \mathcal{C}(\mathbf{X})\} = \mathcal{C}(\mathbf{V})^\perp \cap \mathcal{C}(\mathbf{X})^\perp = (\mathcal{C}(\mathbf{X} : \mathbf{V}))^\perp$ , of dimension  $n - \text{rank}(\mathbf{X} : \mathbf{V})$ . Since  $\text{rank}(\mathbf{X} : \mathbf{V}) = r + q - d$ , we get  $\text{rank}(\mathbf{X}^\top \mathbf{N}) = (n - q) - (n - r - q + d) = r - d$ .
- (c) By Lemma 31.3.1 and  $\mathcal{C}(\mathbf{X}\mathbf{X}^\top) = \mathcal{C}(\mathbf{X})$  (Corollary 6.2.4),  $\mathcal{C}(\mathbf{T}) = \mathcal{C}(\mathbf{V}) + \mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{X} : \mathbf{V})$ . Since  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{T})$ , the system  $\mathbf{TH} = \mathbf{X}$  is consistent (Theorem 1.4.1), so  $\mathbf{X} = \mathbf{TH}$  for some  $\mathbf{H}$  and  $\mathbf{TT}^+ \mathbf{X} = \mathbf{TT}^+ \mathbf{TH} = \mathbf{TH} = \mathbf{X}$ .

Write  $\mathbf{T}^+ = \mathbf{LL}^\top$ , possible because  $\mathbf{T}^+$  is symmetric nonnegative definite, and set  $\mathbf{X}_* = \mathbf{L}^\top \mathbf{X}$ , so that  $\mathbf{X}^\top \mathbf{T}^+ \mathbf{X} = \mathbf{X}_*^\top \mathbf{X}_*$ . If  $\mathbf{X}_* \mathbf{w} = \mathbf{0}$  then  $\mathbf{X} \mathbf{w} \in \mathcal{N}(\mathbf{L}^\top) = \mathcal{N}(\mathbf{T}^+) = \mathcal{C}(\mathbf{T})^\perp$ , while  $\mathbf{X} \mathbf{w} \in \mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{T})$ ; hence  $\mathbf{X} \mathbf{w} = \mathbf{0}$ . Every column of  $\mathbf{I} - (\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{X}_*$  lies in  $\mathcal{N}(\mathbf{X}_*^\top \mathbf{X}_*) = \mathcal{N}(\mathbf{X}_*)$ , so  $\mathbf{X}(\mathbf{I} - (\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{X}_*) = \mathbf{0}$ , which is  $\mathbf{AX} = \mathbf{X}(\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{L}^\top \mathbf{X} = \mathbf{X}(\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{X}_* = \mathbf{X}$ .

For the second condition, let  $\mathbf{d}$  satisfy  $\mathbf{X}^\top \mathbf{d} = \mathbf{0}$ . Using  $\mathbf{V} = \mathbf{T} - \mathbf{X}\mathbf{X}^\top$  and the symmetry of  $\mathbf{T}$  and  $\mathbf{T}^+$ ,

$$\mathbf{X}^\top \mathbf{T}^+ \mathbf{V} \mathbf{d} = \mathbf{X}^\top \mathbf{T}^+ \mathbf{T} \mathbf{d} - \mathbf{X}^\top \mathbf{T}^+ \mathbf{X} \mathbf{X}^\top \mathbf{d} = (\mathbf{TT}^+ \mathbf{X})^\top \mathbf{d} = \mathbf{X}^\top \mathbf{d} = \mathbf{0},$$

so  $\mathbf{AVd} = \mathbf{0}$ . As  $\mathbf{d}$  ranges over  $\mathcal{C}(\mathbf{X})^\perp$  this says  $\mathbf{AV}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ . Now Lemma 31.2.1(c) makes  $\mathbf{AY}$  a BLUE of  $\mathbf{X}\boldsymbol{\beta}$ .

- (d) Existence is (c): for estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \boldsymbol{\rho}^\top \mathbf{X}\boldsymbol{\beta}$ , the statistic  $\boldsymbol{\rho}^\top \mathbf{AY}$  is a BLUE. Uniqueness is Lemma 31.2.1(b).
- (e) If  $\mathbf{y} = \mathbf{Xb}_0 + \mathbf{v}$  with  $\mathbf{v} \in \mathcal{C}(\mathbf{V})$  then  $\mathbf{b}_0 \in \mathcal{F}(\mathbf{y})$ , so  $\mathcal{F}(\mathbf{y}) \neq \emptyset$ ; and  $\mathbf{b} \in \mathcal{F}(\mathbf{y})$  iff  $\mathbf{Kb} = \mathbf{N}^\top \mathbf{y}$ , a consistent system, whose solutions are  $\mathbf{K}^- \mathbf{N}^\top \mathbf{y}$  plus  $\mathcal{N}(\mathbf{K})$ . Since  $\mathbf{X}\mathcal{N}(\mathbf{K}) = \{\mathbf{Xh} : \mathbf{N}^\top \mathbf{Xh} = \mathbf{0}\} = \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{U})$ , the fitted vectors available

are  $\mathbf{Xb} = \mathbf{Xb}_* + \mathbf{Uc}$  with  $\mathbf{b}_* = \mathbf{K}^{-1}\mathbf{N}^\top \mathbf{y}$  and  $\mathbf{c} \in \mathbb{R}^d$  free. Also  $\mathbf{y} - \mathbf{Xb}_* \in \mathcal{C}(\mathbf{V})$ , and on  $\mathcal{C}(\mathbf{V})$  the form  $\mathbf{a}^\top \mathbf{V}^+ \mathbf{a}$  is positive definite; so minimizing over  $\mathbf{c}$  is an ordinary generalized least squares problem in the  $\mathbf{V}^+$  inner product with model matrix  $\mathbf{U}$  of full column rank, whose unique solution is  $\hat{\mathbf{c}} = (\mathbf{U}^\top \mathbf{V}^+ \mathbf{U})^{-1} \mathbf{U}^\top \mathbf{V}^+ (\mathbf{y} - \mathbf{Xb}_*)$ . This gives Eq. 31.3.3, shows that  $\mathbf{X}\tilde{\boldsymbol{\beta}}$  is unique and that the map is linear, say  $\mathbf{X}\tilde{\boldsymbol{\beta}} = \mathbf{A}_* \mathbf{y}$ .

It remains to check the two conditions of Lemma 31.2.1(c) for  $\mathbf{A}_*$ . If  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta}$  then  $\mathbf{Kb}_* = \mathbf{N}^\top \mathbf{X}\boldsymbol{\beta}$  and  $\mathbf{N}^\top (\mathbf{y} - \mathbf{Xb}_*) = \mathbf{0}$ , so  $\mathbf{y} - \mathbf{Xb}_* \in \mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{U})$ ; the  $\mathbf{V}^+$ -projection onto  $\mathcal{C}(\mathbf{U})$  leaves it fixed, and  $\mathbf{A}_* \mathbf{X}\boldsymbol{\beta} = \mathbf{Xb}_* + (\mathbf{y} - \mathbf{Xb}_*) = \mathbf{X}\boldsymbol{\beta}$ . Thus  $\mathbf{A}_* \mathbf{X} = \mathbf{X}$ . If  $\mathbf{y} = \mathbf{Vd}$  with  $\mathbf{X}^\top \mathbf{d} = \mathbf{0}$ , then  $\mathbf{N}^\top \mathbf{Vd} = \mathbf{0}$  so we may take  $\mathbf{b}_* = \mathbf{0}$ , and  $\mathbf{U}^\top \mathbf{V}^+ \mathbf{Vd} = \mathbf{U}^\top \mathbf{\Gamma} \mathbf{\Gamma}^\top \mathbf{d} = \mathbf{U}^\top \mathbf{d} = \mathbf{0}$ , using  $\mathbf{V}^+ \mathbf{V} = \mathbf{\Gamma} \mathbf{\Gamma}^\top$ ,  $\mathcal{C}(\mathbf{U}) \subseteq \mathcal{C}(\mathbf{V})$  and  $\mathcal{C}(\mathbf{U}) \subseteq \mathcal{C}(\mathbf{X})$ . So  $\mathbf{A}_* \mathbf{Vd} = \mathbf{0}$ , that is  $\mathbf{A}_* \mathbf{V}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ .

- (f) Put  $\mathbf{Z}_* = \mathbf{\Lambda}^{-1/2} \mathbf{\Gamma}^\top (\mathbf{I} - \mathbf{XK}^{-1} \mathbf{N}^\top) \mathbf{Y}$ , a random vector in  $\mathbb{R}^q$ . Because  $\mathbf{VN} = \mathbf{0}$ ,  $(\mathbf{I} - \mathbf{XK}^{-1} \mathbf{N}^\top) \mathbf{V}(\mathbf{I} - \mathbf{XK}^{-1} \mathbf{N}^\top)^\top = \mathbf{V}$ , so  $\text{Cov}(\mathbf{Z}_*) = \sigma^2 \mathbf{\Lambda}^{-1/2} \mathbf{\Gamma}^\top \mathbf{V} \mathbf{\Gamma} \mathbf{\Lambda}^{-1/2} = \sigma^2 \mathbf{I}_q$ . Its mean is  $\mathbf{\Lambda}^{-1/2} \mathbf{\Gamma}^\top \mathbf{X}(\mathbf{I} - \mathbf{K}^{-1} \mathbf{K}) \boldsymbol{\beta}$ , which lies in  $\mathbf{\Lambda}^{-1/2} \mathbf{\Gamma}^\top \mathcal{C}(\mathbf{U})$ , a subspace of dimension  $d$  (the map  $\mathbf{\Lambda}^{-1/2} \mathbf{\Gamma}^\top$  is injective on  $\mathcal{C}(\mathbf{V}) \supseteq \mathcal{C}(\mathbf{U})$ ). By the computation in (e),  $\text{SSE}_{\mathbf{V}} = \|(\mathbf{I} - \mathbf{M}_d) \mathbf{Z}_*\|^2$  where  $\mathbf{M}_d$  is the orthogonal projection of  $\mathbb{R}^q$  onto that  $d$ -dimensional subspace, and  $\mathbf{X}\tilde{\boldsymbol{\beta}}$  is a fixed vector plus a function of  $\mathbf{M}_d \mathbf{Z}_*$ , since  $\mathbf{N}^\top \mathbf{Y}$  is constant by (a). Now Theorem 2.3.1 gives  $E(\text{SSE}_{\mathbf{V}}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}_d) = \sigma^2(q - d)$  with no distributional assumption, and under normality Theorem 7.5.1 applied to  $\mathbf{Z}_*$  gives the  $\chi^2(q - d)$  law and the independence. Finally  $\text{rank}(\mathbf{X} : \mathbf{V}) = r + q - d$  by the dimension formula for a sum of subspaces, so  $q - d = \text{rank}(\mathbf{X} : \mathbf{V}) - r$ .

■

The singular model is the nonsingular model with a deterministic layer on top. Part (b) isolates that layer,  $r - d$  linear functions of  $\boldsymbol{\beta}$  known exactly once the data are seen; part (e) pins down a particular solution with them and then does generalized least squares inside  $\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V})$ , the part of the model space the noise can still reach; part (f) counts what is left. If  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{V})$  then  $d = r$ , nothing is known exactly, and Eq. 31.3.3 is generalized least squares with  $\mathbf{V}^+$  for  $\mathbf{V}^{-1}$  on  $q - r$  degrees of freedom; if  $\mathbf{V}$  is nonsingular then  $q = n$ ,  $d = r$ , and everything collapses to Theorem 31.1.1.

### 31.3.4 A worked example

#### Example. (Regression on estimated treatment effects)

A balanced one-way layout with  $a$  treatments and  $m$  observations each is analysed, and the estimated effects  $\hat{\alpha}_1, \dots, \hat{\alpha}_a$  are reported under the side condition  $\sum_k \hat{\alpha}_k = 0$  (Theorem 8.4.2). A second analyst, with access only to these numbers, regresses them on the dose  $x_k$  applied to treatment  $k$ , fitting  $E(\hat{\alpha}_k) = \beta_0 + \beta_1 x_k$  with an intercept.

The reported vector has  $\text{Cov}(\hat{\boldsymbol{\alpha}}) = \tau^2 \mathbf{V}$  with  $\mathbf{V} = \mathbf{I}_a - \mathbf{1}\mathbf{1}^\top/a$  and  $\tau^2 = \sigma^2/m$ , because  $\hat{\alpha}_k = \bar{Y}_k - \bar{Y}$  and the  $\bar{Y}_k$  are uncorrelated with variance  $\tau^2$ . Here  $q = a - 1$ ,  $\mathcal{C}(\mathbf{V}) = \mathbf{1}^\perp$ , and  $\mathbf{N} = \mathbf{1}/\sqrt{a}$ . With  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  we get  $d = 1$  and  $r = 2$ , so Theorem 31.3.2(b) gives exactly  $r - d = 1$  perfectly known function:  $\mathbf{X}^\top \mathbf{N} \propto (a, \sum_k x_k)^\top$ , so

$$a\beta_0 + \left(\sum_k x_k\right)\beta_1 = \mathbf{1}^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{1}^\top \hat{\boldsymbol{\alpha}} = 0$$

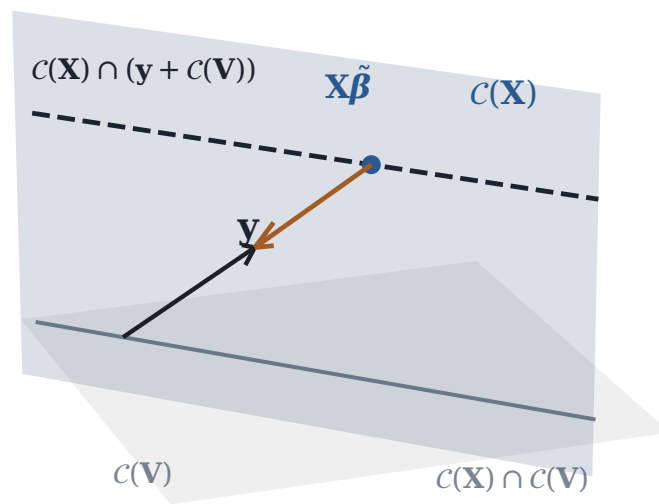


Figure 31.3.1: The geometry of a singular model with  $n = 3$ . The plane is  $\mathcal{C}(\mathbf{X})$ ;  $\mathcal{C}(\mathbf{V})$  is horizontal, so  $\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V})$  is the lower line. The data  $\mathbf{y}$  determine  $\mathbf{N}^\top \mathbf{X}\boldsymbol{\beta}$  exactly, which confines  $\mathbf{X}\boldsymbol{\beta}$  to the dashed line  $\mathcal{C}(\mathbf{X}) \cap (\mathbf{y} + \mathcal{C}(\mathbf{V}))$ ; the estimate is the point of that line nearest  $\mathbf{y}$  in the  $\mathbf{V}^+$  geometry, and the residual lies in  $\mathcal{C}(\mathbf{V})$ .

with probability one. In words: the intercept is determined by the slope,  $\beta_0 = -\bar{x}\beta_1$ , and the data say so without error. Fitting an intercept was a mistake, and the model itself reveals it.

Everything else is unchanged. Because  $\mathbf{V}\mathbf{X} = \mathbf{X} - \mathbf{1}(\mathbf{1}^\top \mathbf{X})/a$  has columns in  $\mathcal{C}(\mathbf{X})$ , [Theorem 31.2.2](#) applies and ordinary least squares is already best:  $\hat{\beta}_1$  is the usual centred slope and  $\hat{\beta}_0 = -\bar{x}\hat{\beta}_1$ . Since  $\mathbf{V}$  is idempotent,  $\mathbf{V}^+ = \mathbf{V}$  and  $\text{SSE}_{\mathbf{V}} = \hat{\mathbf{e}}^\top \hat{\mathbf{e}}$  is the usual residual sum of squares, on  $q - d = a - 2$  degrees of freedom. What is *not* unchanged is the covariance matrix. Because  $\mathbf{1}^\top \hat{\boldsymbol{\alpha}} = 0$  exactly,

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \frac{\tau^2}{S_{xx}}, & \text{Var}(\hat{\beta}_0) &= \bar{x}^2 \text{Var}(\hat{\beta}_1), \\ \text{Cov}(\hat{\beta}_0, \hat{\beta}_1) &= -\bar{x} \text{Var}(\hat{\beta}_1),\end{aligned}$$

whereas the ordinary formula  $s^2(\mathbf{X}^\top \mathbf{X})^{-1}$  reports  $s^2(1/a + \bar{x}^2/S_{xx})$  for the intercept. With  $a = 5$  doses 0, 1, 2, 4, 8,  $m = 6$  replicates and  $\sigma = 1.2$ , so  $\tau^2 = 0.24$ ,  $\bar{x} = 3.0$  and  $S_{xx} = 40.0$ : the true variances are 0.00600 for the slope and 0.0540 for the intercept, while the ordinary formula reports 0.1020 on average for the intercept, too large by a factor of 1.89. A simulation of 200,000 layouts reproduces 0.00600 and 0.0540.

```
import numpy as np

dose = np.array([0.0, 1.0, 2.0, 4.0, 8.0])           # a = 5 treatments
a, reps, sigma = len(dose), 6, 1.2
X = np.column_stack([np.ones(a), dose])
V = np.eye(a) - np.ones((a, a)) / a                # Cov(effects) = (sigma^2/reps)
V                                                    # V
xbar = dose.mean()
Sxx = float(((dose - xbar) ** 2).sum())

Vplus = V                                           # V is symmetric idempotent, so
V+ = V
N = np.ones((a, 1)) / np.sqrt(a)                  # an orthonormal basis of C(V)
perp
print("rank(V) =", np.linalg.matrix_rank(V), " rank([X V]) =",
      np.linalg.matrix_rank(np.column_stack([X, V])), " rank(X) =", np.linalg.
      matrix_rank(X))
print("the perfectly known function:", (X.T @ N).ravel())
```

```
rng = np.random.default_rng(3107)
trials = 200_000
cell = rng.normal(scale=sigma / np.sqrt(reps), size=(trials, a))   # treatment
means
cell += 0.35 * (dose - xbar)                                         # the true
effects
effects = cell - cell.mean(axis=1, keepdims=True)                  # sum to zero
exactly
B = effects @ np.linalg.solve(X.T @ X, X.T).T                     # ordinary
least squares

tau2 = sigma**2 / reps
print("Var(slope): %.5f simulated, %.5f from tau^2/Sxx" % (B[:, 1].var(), tau2 /
Sxx))
print("Var(intercept): %.5f simulated, %.5f from xbar^2 tau^2/Sxx"
      % (B[:, 0].var(), xbar**2 * tau2 / Sxx))
```

```
print("what OLS reports for the intercept: %.5f" % (tau2 * (1 / a + xbar**2 / Sxx)))
```

The script behind [Example \(Regression on estimated treatment effects\)](#) also computes the estimate four ways — by [Eq. 31.3.2](#), by the closed form [Eq. 31.3.3](#), by [Eq. 31.3.1](#) with  $\mathbf{T} = \mathbf{V} + \mathbf{X}\mathbf{X}^\top$ , and by ordinary least squares — and checks that all four agree, as [Theorem 31.3.2\(d\)](#) requires, and that [Eq. 31.3.3](#) gives the same answer for two different generalized inverses  $\mathbf{K}^-$ .

*Warning.* Software that inverts  $\mathbf{V}$  will fail here, and software that silently replaces  $\mathbf{V}^{-1}$  by a pseudo-inverse solves the wrong problem: it minimizes the  $\mathbf{V}^+$  criterion over all  $\mathbf{b}$  instead of over  $\mathcal{F}(\mathbf{y})$ , which is what [Theorem 31.3.2\(e\)](#) is for, so the point estimate can be wrong. In [Example \(Regression on estimated treatment effects\)](#) the formula  $(\mathbf{X}^\top \mathbf{V}^+ \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^+ \mathbf{y}$  always returns  $\hat{\beta}_0 = 0$ , contradicting  $\mathbf{1}^\top \hat{\boldsymbol{\alpha}} = 0$ . The smallest instance has two observations, the second measured without error:  $\mathbf{V} = \text{diag}(1, 0)$ ,  $\mathbf{X} = \mathbf{1}_2$ , so  $\beta = Y_2$  exactly, while the formula reports  $Y_1$ . The degrees of freedom are wrong too:  $n - r$  is not  $q - d$ . Here  $a - 2 = 3$  happens to agree with  $n - r$ , but with a  $\mathbf{V}$  of lower rank it would not; always compute  $\text{rank}(\mathbf{X} : \mathbf{V}) - \text{rank}(\mathbf{X})$ .

## 31.3.5 Testing

A reduced model  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  is tested as in [Theorem 31.1.1\(e\)](#), with two adjustments. The degrees of freedom are those of [Theorem 31.3.2\(f\)](#) for each model,  $\nu = \text{rank}(\mathbf{X} : \mathbf{V}) - \text{rank}(\mathbf{X})$  and  $\nu_0 = \text{rank}(\mathbf{X}_0 : \mathbf{V}) - \text{rank}(\mathbf{X}_0)$ , so the numerator carries  $\nu_0 - \nu$ . And the reduced model can be *refuted with certainty*: if  $\mathbf{Y} \notin \mathcal{C}(\mathbf{X}_0 : \mathbf{V})$  it is impossible and the test rejects at once. Otherwise [Eq. 31.1.7](#) has the  $F(\nu_0 - \nu, \nu, \gamma)$  distribution; [Exercise \(C1\)](#) works through the argument.

## 31.3.6 Exercises

### 31.3.6.1 A. Check your understanding

#### Exercise A1.

Show that if  $\mathbf{V}$  is singular with  $\mathbf{V}\mathbf{a} = \mathbf{0}$ , then  $\mathbf{a}^\top \mathbf{Y}$  is the same number for every possible data vector, and identify that number in terms of  $\boldsymbol{\beta}$ . What does this say about  $\mathbf{a}^\top \mathbf{X}\boldsymbol{\beta}$  when  $\mathbf{X}^\top \mathbf{a} = \mathbf{0}$ ?

#### Exercise A2.

Verify  $\text{rank}(\mathbf{X} : \mathbf{V}) = \text{rank}(\mathbf{X}) + \text{rank}(\mathbf{V}) - \dim(\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V}))$  and compute both sides for [Example \(Regression on estimated treatment effects\)](#).

*Solution.* This is the dimension formula for the sum of two subspaces,  $\dim(\mathcal{S} + \mathcal{T}) = \dim \mathcal{S} + \dim \mathcal{T} - \dim(\mathcal{S} \cap \mathcal{T})$ . In [Example \(Regression on estimated treatment effects\)](#),  $\mathcal{C}(\mathbf{X}) + \mathcal{C}(\mathbf{V}) = \mathbb{R}^a$  because  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  and  $\mathcal{C}(\mathbf{V}) = \mathbf{1}^\perp$ ; so the left side is  $a$ , and the right side is  $2 + (a - 1) - 1 = a$ .

### 31.3.6.2 B. Practice

**Exercise B1.**

Three fractions of a sample are measured and then rescaled so that they add to one. Model  $E(Y_k) = \mu + \delta_k$  with  $\sum_k \delta_k = 0$  known, and suppose  $\text{Cov}(\mathbf{Y}) = \sigma^2(\mathbf{I}_3 - \mathbf{1}\mathbf{1}^\top/3)$ . Which function of  $(\mu, \delta_1, \delta_2)$  is known exactly? Find the variance of  $Y_1 - Y_2$  as an estimate of  $\delta_1 - \delta_2$ , and explain why no other linear unbiased estimate exists.

*Solution.* Here  $\mathbf{N} = \mathbf{1}/\sqrt{3}$  and  $\mathbf{X}^\top \mathbf{N} \propto \mathbf{X}^\top \mathbf{1}$ ; with the parameterization  $\boldsymbol{\beta} = (\mu, \delta_1, \delta_2)^\top$  and  $\delta_3 = -\delta_1 - \delta_2$ , we get  $\mathbf{X}^\top \mathbf{1} = (3, 0, 0)^\top$ , so  $3\mu$  is known exactly:  $\mu = \mathbf{1}^\top \mathbf{Y}/3 = 1/3$ , the rescaling constant. Here  $\text{rank}(\mathbf{X}) = 3 = n$ , so  $\mathbf{X}^\top$  is nonsingular and the unbiasedness equation  $\mathbf{X}^\top \mathbf{c} = \boldsymbol{\lambda}$  has exactly one solution: every estimable function has a single linear unbiased estimate, which is therefore its BLUE. For  $\delta_1 - \delta_2$  that solution is  $\mathbf{c} = (1, -1, 0)^\top$ , and  $\text{Var}(Y_1 - Y_2) = \sigma^2 \mathbf{c}^\top \mathbf{V} \mathbf{c} = \sigma^2(2/3 + 2/3 + 2/3) = 2\sigma^2$ .

**Exercise B2.**

Suppose the first observation is measured without error, so  $\mathbf{V} = \begin{pmatrix} 0 & \mathbf{0}^\top \\ \mathbf{0} & \mathbf{V}_{22} \end{pmatrix}$  with  $\mathbf{V}_{22}$  positive definite, and  $\mathbf{X} = \begin{pmatrix} \mathbf{x}_{(1)}^\top \\ \mathbf{X}_2 \end{pmatrix}$ . Show that  $\mathbf{x}_{(1)}^\top \boldsymbol{\beta}$  is known exactly, and that the BLUE of any estimable  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is the generalized least squares estimate from the remaining  $n - 1$  observations *subject to* the constraint  $\mathbf{x}_{(1)}^\top \mathbf{b} = y_1$ .

*Solution.*  $\mathbf{N} = \mathbf{e}_1$  spans  $\mathcal{C}(\mathbf{V})^\perp$ , so  $\mathbf{N}^\top \mathbf{Y} = Y_1 = \mathbf{x}_{(1)}^\top \boldsymbol{\beta}$  with probability one. In Eq. 31.3.2 the feasible set is  $\{\mathbf{b} : \mathbf{y} - \mathbf{X}\mathbf{b} \in \mathcal{C}(\mathbf{V})\} = \{\mathbf{b} : \mathbf{x}_{(1)}^\top \mathbf{b} = y_1\}$ , and on  $\mathcal{C}(\mathbf{V})$  the form  $\mathbf{a}^\top \mathbf{V} \mathbf{a}$  is  $\mathbf{a}_2^\top \mathbf{V}_{22}^{-1} \mathbf{a}_2$ ; so the criterion is the generalized least squares criterion of the last  $n - 1$  observations, minimized subject to the constraint. This is restricted least squares (Theorem 9.2.5) in the  $\mathbf{V}_{22}^{-1}$  geometry.

**Exercise B3.**

Verify directly from Eq. 31.3.3 that  $\mathbf{X}\tilde{\boldsymbol{\beta}}$  does not depend on which generalized inverse  $\mathbf{K}^-$  is used, given  $\mathbf{y} \in \mathcal{C}(\mathbf{X} : \mathbf{V})$ .

**31.3.6.3 C. Going deeper****Exercise C1.**

In the setting of Theorem 31.3.2, let  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  with  $\nu_0 = \text{rank}(\mathbf{X}_0 : \mathbf{V}) - \text{rank}(\mathbf{X}_0)$  and  $\nu = \text{rank}(\mathbf{X} : \mathbf{V}) - \text{rank}(\mathbf{X})$ , and assume  $\mathbf{Y}$  is normal and  $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0 : \mathbf{V})$ . Working in the coordinates  $\mathbf{Z}_*$  of the proof of Theorem 31.3.2(f), show that  $\text{SSE}_{\mathbf{V},0} - \text{SSE}_{\mathbf{V}}$  and  $\text{SSE}_{\mathbf{V}}$  are independent with  $(\text{SSE}_{\mathbf{V},0} - \text{SSE}_{\mathbf{V}})/\sigma^2 \sim \chi^2(\nu_0 - \nu, \gamma)$ , and identify  $\gamma$ . Why is the assumption  $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0 : \mathbf{V})$  needed?

*Solution.* Because  $\mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0 : \mathbf{V})$  makes  $\mathbf{N}^\top \mathbf{Y} \in \mathcal{C}(\mathbf{N}^\top \mathbf{X}_0)$ , the same  $\mathbf{b}_*$  is feasible in both models. In the coordinates  $\mathbf{Z}_* \sim N_q(\boldsymbol{\mu}_*, \sigma^2 \mathbf{I})$  the two sums of squares are  $\|(\mathbf{I} - \mathbf{M}_d)\mathbf{Z}_*\|^2$  and  $\|(\mathbf{I} - \mathbf{M}_{d_0})\mathbf{Z}_*\|^2$  for nested subspaces of dimensions  $d_0 \leq d$ , with

$q - d = \nu$  and  $q - d_0 = \nu_0$ . Now [Theorem 11.1.1](#) applied to  $\mathbf{Z}_*$  gives independence, the  $\chi^2$  laws and  $\gamma = \|(\mathbf{M}_d - \mathbf{M}_{d_0})\boldsymbol{\mu}_*\|^2/\sigma^2$ . If  $\mathbf{X}\boldsymbol{\beta} \notin \mathcal{C}(\mathbf{X}_0 : \mathbf{V})$  the reduced model assigns probability zero to the observed  $\mathbf{Y}$ , so it is contradicted outright and no distribution theory is needed.

### Exercise C2.

Call  $\mathbf{X}\tilde{\boldsymbol{\beta}}$  a *least squares consistent estimate* when it minimizes the ordinary sum of squares  $\|\mathbf{y} - \mathbf{X}\mathbf{b}\|^2$  over  $\mathcal{F}(\mathbf{y})$ ; this is the criterion used by Christensen (2020). Show that it agrees with [Eq. 31.3.2](#) when  $\mathbf{V}$  is idempotent, and that in general the two differ. Show also that when  $\mathcal{C}(\mathbf{V}\mathbf{U}) \subseteq \mathcal{C}(\mathbf{U})$  the least squares consistent estimate is a BLUE.

## 31.4 Estimated covariance: feasible generalized least squares

Nothing in the previous three sections is usable until  $\mathbf{V}$  is known, and  $\mathbf{V}$  is almost never known. The practical procedure is to choose a small family  $\mathbf{V}(\boldsymbol{\theta})$ , estimate  $\boldsymbol{\theta}$ , and proceed as if the estimate were the truth. That costs asymptotically nothing and, in small samples, a great deal — the damage falling on the standard errors rather than on the estimate.

### 31.4.1 The two-step estimator

#### Definition 31.4.1 — Feasible generalized least squares

Let  $\{\mathbf{V}(\boldsymbol{\theta}) : \boldsymbol{\theta} \in \Theta\}$  be a family of positive definite  $n \times n$  matrices,  $\Theta \subseteq \mathbb{R}^k$ , and let  $\hat{\boldsymbol{\theta}}$  be an estimator of  $\boldsymbol{\theta}$  computed from the data. The **feasible generalized least squares** estimator is

$$\hat{\boldsymbol{\beta}}_F = (\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{Y}, \quad (31.4.1)$$

for  $\text{rank}(\mathbf{X}) = p$ , with reported covariance matrix  $s_F^2(\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{X})^{-1}$ , where  $s_F^2$  is [Eq. 31.1.6](#) evaluated at  $\mathbf{V}(\hat{\boldsymbol{\theta}})$ .

The families that matter are small. Equicorrelation inside clusters,  $\mathbf{V}(\rho) = (1 - \rho)\mathbf{I} + \rho\mathbf{Z}\mathbf{Z}^\top$ , and first-order autoregressive errors ([Definition 21.6.1](#)) both have  $k = 1$ ; a variance function  $\text{Var}(\varepsilon_i) = \sigma^2 \exp(\mathbf{z}_i^\top \boldsymbol{\gamma})$  has  $k = \dim \boldsymbol{\gamma}$ , the diagonal case treated in [Section 21.3](#); the linear mixed models of [Chapter 32](#) give  $\mathbf{V}(\boldsymbol{\theta}) = \mathbf{Z}\mathbf{G}(\boldsymbol{\theta})\mathbf{Z}^\top + \mathbf{R}(\boldsymbol{\theta})$  with a handful of variance components. In every case  $k$  stays fixed while  $n$  grows; a family with  $k$  of the order of  $n$  cannot be estimated at all, which is why “estimate  $\mathbf{V}$  freely” is not an option.

There are three routes to  $\hat{\boldsymbol{\theta}}$ . The **moment** route matches a few moments of the ordinary least squares residuals to the model, as in the analysis of variance estimate of an intraclass correlation used below. The **likelihood** route maximizes the profile log-likelihood of [Exercise \(C2\)](#), taken



up in [Chapter 32](#) with the restricted version that corrects its downward bias. The **iterative** route alternates between  $\beta$  and  $\theta$ , as in the Cochrane–Orcutt procedure of [Theorem 21.6.1\(d\)](#). Only consistency is needed below.

One case needs no asymptotics. If  $\mathcal{C}(\mathbf{V}(\theta)\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  for every  $\theta$ , the best estimator is known even though the covariance is not ([Theorem 31.2.2](#), [Exercise \(B2\)](#)): in a balanced layout with random block effects, only the standard errors need  $\hat{\theta}$ .

## 31.4.2 Large-sample equivalence

Write  $\mathbf{V} = \mathbf{V}(\theta_0)$  for the true covariance,  $\hat{\beta}_{\mathbf{V}}$  for the infeasible generalized least squares estimator that uses it, and  $\mathbf{D}(\theta) = \mathbf{V}(\theta)^{-1} - \mathbf{V}(\theta_0)^{-1}$ . The whole question is whether replacing  $\theta_0$  by  $\hat{\theta}$  disturbs the two quantities  $n^{-1}\mathbf{X}^\top \mathbf{V}^{-1}\mathbf{X}$  and  $n^{-1/2}\mathbf{X}^\top \mathbf{V}^{-1}\boldsymbol{\varepsilon}$  that determine  $\hat{\beta}_{\mathbf{V}}$ . The first is easy, because it does not involve the errors. The second is not, because  $\hat{\theta}$  is computed from the same  $\boldsymbol{\varepsilon}$ .

### Theorem 31.4.1 — Feasible generalized least squares

Assume [Eq. 31.1.1](#) with  $\text{rank}(\mathbf{X}) = p$  fixed, and consider a sequence of models indexed by  $n$ . Suppose:

- (G1) there are constants  $0 < c \leq C < \infty$  and a neighbourhood  $B$  of  $\theta_0$  such that every eigenvalue of  $\mathbf{V}(\theta)$  lies in  $[c, C]$  for all  $\theta \in B$  and all  $n$ ;
- (G2)  $\|\mathbf{D}(\theta)\|_2 \leq L\|\theta - \theta_0\|$  for  $\theta \in B$ , all  $n$ , with  $\|\cdot\|_2$  the spectral norm;
- (G3) the rows of  $\mathbf{X}$  satisfy  $\|\mathbf{x}_{(i)}\| \leq K$ , and  $n^{-1}\mathbf{X}^\top \mathbf{V}^{-1}\mathbf{X} \rightarrow \mathbf{A}$  for a positive definite  $\mathbf{A}$ ;
- (G4)  $\hat{\theta} \rightarrow \theta_0$  in probability.

Then:

- (a)  $n^{-1}\mathbf{X}^\top \mathbf{V}(\hat{\theta})^{-1}\mathbf{X} \rightarrow \mathbf{A}$  in probability, so the reported covariance factor is consistent up to the scale  $\sigma^2$ .
- (b) If in addition  $\sqrt{n}(\hat{\theta} - \theta_0) = O_p(1)$  and

- (G5)  $\sup_{\|\theta - \theta_0\| \leq \delta} \|n^{-1/2}\mathbf{X}^\top \mathbf{D}(\theta)\boldsymbol{\varepsilon}\| = O_p(\delta)$  as  $\delta \rightarrow 0$ , uniformly in  $n$ ,

then  $\sqrt{n}(\hat{\beta}_F - \hat{\beta}_{\mathbf{V}}) \rightarrow \mathbf{0}$  in probability. Consequently  $\hat{\beta}_F$  has whatever limiting distribution  $\hat{\beta}_{\mathbf{V}}$  has, and in particular  $\sqrt{n}(\hat{\beta}_F - \beta) \rightarrow N_p(\mathbf{0}, \sigma^2 \mathbf{A}^{-1})$  whenever  $\sqrt{n}(\hat{\beta}_{\mathbf{V}} - \beta)$  does.

*Proof.*

- (a) On the event  $\hat{\theta} \in B$ , whose probability tends to one by (G4),

$$\begin{aligned} \|n^{-1}\mathbf{X}^\top \mathbf{D}(\hat{\theta})\mathbf{X}\|_2 &\leq n^{-1}\|\mathbf{D}(\hat{\theta})\|_2\|\mathbf{X}\|_2^2 \\ &\leq n^{-1}L\|\hat{\theta} - \theta_0\| \cdot nK^2 = LK^2\|\hat{\theta} - \theta_0\|, \end{aligned}$$

using  $\|\mathbf{X}\|_2^2 \leq \|\mathbf{X}\|_F^2 = \sum_i \|\mathbf{x}_{(i)}\|^2 \leq nK^2$ . By (G4) this tends to zero in probability, and adding  $n^{-1}\mathbf{X}^\top \mathbf{V}^{-1}\mathbf{X} \rightarrow \mathbf{A}$  gives (a).

(b) Write  $\mathbf{A}_n(\boldsymbol{\theta}) = n^{-1} \mathbf{X}^\top \mathbf{V}(\boldsymbol{\theta})^{-1} \mathbf{X}$  and  $\mathbf{b}_n(\boldsymbol{\theta}) = n^{-1/2} \mathbf{X}^\top \mathbf{V}(\boldsymbol{\theta})^{-1} \boldsymbol{\varepsilon}$ , so that

$$\begin{aligned}\sqrt{n}(\hat{\boldsymbol{\beta}}_F - \boldsymbol{\beta}) &= \mathbf{A}_n(\hat{\boldsymbol{\theta}})^{-1} \mathbf{b}_n(\hat{\boldsymbol{\theta}}), \\ \sqrt{n}(\hat{\boldsymbol{\beta}}_V - \boldsymbol{\beta}) &= \mathbf{A}_n(\boldsymbol{\theta}_0)^{-1} \mathbf{b}_n(\boldsymbol{\theta}_0).\end{aligned}$$

By (G1) and (G3),  $E\|\mathbf{b}_n(\boldsymbol{\theta}_0)\|^2 = \sigma^2 \text{tr} \mathbf{A}_n(\boldsymbol{\theta}_0) \leq \sigma^2 p K^2 / c$  is bounded, so  $\mathbf{b}_n(\boldsymbol{\theta}_0) = O_p(1)$ . By (G5) with  $\delta_n = \|\hat{\boldsymbol{\theta}} - \boldsymbol{\theta}_0\| = O_p(n^{-1/2})$ ,

$$\mathbf{b}_n(\hat{\boldsymbol{\theta}}) - \mathbf{b}_n(\boldsymbol{\theta}_0) = n^{-1/2} \mathbf{X}^\top \mathbf{D}(\hat{\boldsymbol{\theta}}) \boldsymbol{\varepsilon} = O_p(n^{-1/2}) = o_p(1),$$

so  $\mathbf{b}_n(\hat{\boldsymbol{\theta}}) = \mathbf{b}_n(\boldsymbol{\theta}_0) + o_p(1)$  and both are  $O_p(1)$ . By (a),  $\mathbf{A}_n(\hat{\boldsymbol{\theta}})$  and  $\mathbf{A}_n(\boldsymbol{\theta}_0)$  both converge in probability to the invertible  $\mathbf{A}$ , and matrix inversion is continuous there, so  $\mathbf{A}_n(\hat{\boldsymbol{\theta}})^{-1} - \mathbf{A}_n(\boldsymbol{\theta}_0)^{-1} = o_p(1)$ . The difference of the two displays is therefore

$$\begin{aligned}&\{\mathbf{A}_n(\hat{\boldsymbol{\theta}})^{-1} - \mathbf{A}_n(\boldsymbol{\theta}_0)^{-1}\} \mathbf{b}_n(\hat{\boldsymbol{\theta}}) \\ &\quad + \mathbf{A}_n(\boldsymbol{\theta}_0)^{-1} \{\mathbf{b}_n(\hat{\boldsymbol{\theta}}) - \mathbf{b}_n(\boldsymbol{\theta}_0)\},\end{aligned}$$

a sum of products of  $o_p(1)$  and  $O_p(1)$  terms, hence  $o_p(1)$ . The last claim is Slutsky's theorem. ■

Condition (G5) is the substance of the theorem, and stating it as a hypothesis rather than proving it is the honest course: it separates “the weights are random” from “the weights might as well be fixed”, and it is not automatic. The naive bound  $\|n^{-1/2} \mathbf{X}^\top \mathbf{D} \boldsymbol{\varepsilon}\| \leq n^{-1/2} L K \sqrt{n} \delta \|\boldsymbol{\varepsilon}\|$  is of order  $\delta \sqrt{n}$ , far too crude, because it ignores the cancellation among the  $n$  terms. The next lemma supplies that cancellation for one fixed  $\boldsymbol{\theta}$  in the case that matters most, independent clusters of bounded size.

### Lemma 31.4.2 — The drift term for independent clusters

Suppose the observations fall into  $G$  independent clusters of size at most  $\bar{m}$ , that  $\mathbf{V}(\boldsymbol{\theta})$  is block diagonal with one block per cluster for every  $\boldsymbol{\theta}$ , and that (G1)–(G3) hold. Then for every fixed  $\boldsymbol{\theta} \in B$ ,

$$E\|n^{-1/2} \mathbf{X}^\top \mathbf{D}(\boldsymbol{\theta}) \boldsymbol{\varepsilon}\|^2 \leq \sigma^2 \bar{m} K^2 L^2 C \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|^2.$$

In particular  $n^{-1/2} \mathbf{X}^\top \mathbf{D}(\boldsymbol{\theta}) \boldsymbol{\varepsilon} = O_p(\|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|)$ .

*Proof.* Write  $\mathbf{X}_k$ ,  $\boldsymbol{\varepsilon}_k$  and  $\mathbf{D}_k$  for the blocks belonging to cluster  $k$ , so that  $\mathbf{X}^\top \mathbf{D} \boldsymbol{\varepsilon} = \sum_k \mathbf{X}_k^\top \mathbf{D}_k \boldsymbol{\varepsilon}_k$ . The summands are independent with mean zero, so their covariance matrices add and

$$E\|n^{-1/2} \mathbf{X}^\top \mathbf{D} \boldsymbol{\varepsilon}\|^2 = \frac{1}{n} \sum_k E\|\mathbf{X}_k^\top \mathbf{D}_k \boldsymbol{\varepsilon}_k\|^2 \leq \frac{1}{n} \sum_k \|\mathbf{X}_k\|_2^2 \|\mathbf{D}_k\|_2^2 E\|\boldsymbol{\varepsilon}_k\|^2.$$

Now  $\|\mathbf{X}_k\|_2^2 \leq m_k K^2$ ,  $\|\mathbf{D}_k\|_2 \leq \|\mathbf{D}\|_2 \leq L \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|$  by (G2), and  $E\|\boldsymbol{\varepsilon}_k\|^2 = \sigma^2 \text{tr} \mathbf{V}_{kk} \leq \sigma^2 m_k C$  by (G1). Since  $m_k \leq \bar{m}$  and  $\sum_k m_k = n$ , the sum is at most  $\sigma^2 \bar{m} K^2 L^2 C \|\boldsymbol{\theta} - \boldsymbol{\theta}_0\|^2$ . Chebyshev's inequality turns the moment bound into the stochastic order. ■

*Remark.* Lemma 31.4.2 gives the right order for each fixed  $\theta$ ; (G5) asks for it *uniformly* over a shrinking ball, which is what allows  $\theta$  to be replaced by the data-dependent  $\hat{\theta}$ . That passage is a stochastic equicontinuity argument we do not carry out: with  $\theta$  of fixed dimension, a chaining bound over the ball converts the pointwise bound into the uniform one under mild smoothness of  $\theta \mapsto \mathbf{V}(\theta)^{-1}$ . Amemiya (1985, ch. 6) proves the conclusion of Theorem 31.4.1(b) for several concrete families, and Carroll and Ruppert (1988, ch. 3) for smooth variance functions; (G5) is verified there, not assumed away.

### 31.4.3 What goes wrong in small samples

Theorem 31.4.1 says that  $\hat{\beta}_F$  and  $\hat{\beta}_V$  are indistinguishable in large samples. It says nothing about finite  $n$ , where three things go wrong.

**The estimator is no longer linear.**  $\hat{\beta}_F$  is a ratio of random quantities, so none of the exact theory of Theorem 31.1.1 applies. A symmetry argument rescues unbiasedness when  $\hat{\theta}$  is an even function of the residuals and the errors are symmetric, exactly as in Theorem 21.3.1(c), whose argument goes through verbatim for a non-diagonal  $\mathbf{V}(\hat{\theta})^{-1}$ .

**The variance exceeds the generalized least squares variance.** Estimating  $\theta$  adds noise, with an excess of order  $1/G$  in the cluster setting; Kackar and Harville (1984) give the standard expansion of it.

**The reported standard error ignores that noise.** The matrix  $s_F^2(\mathbf{X}^\top \mathbf{V}(\hat{\theta})^{-1} \mathbf{X})^{-1}$  is the covariance  $\hat{\beta}_V$  would have had if  $\hat{\theta}$  were the truth: it omits the extra variability, and is evaluated at a random  $\hat{\theta}$  drawn towards values that make the fit look good. Tests are liberal and intervals too short.

#### Example. (How many clusters are enough?)

Simulate clusters of sizes 2, 4, 8 repeated 1, 2, 4, 8, 16, 32 times, with a cluster-level regressor  $z$ , a within-cluster regressor  $w$ , and equicorrelation  $\rho = 0.6$  inside clusters. Estimate  $\rho$  by the analysis of variance moment estimator from the ordinary least squares residuals and form  $\hat{\beta}_F$ . Figure 31.4.1 shows 20,000 replications for each size.

The efficiency story is almost trivial. Ordinary least squares attains about 0.895 of the optimal precision for the coefficient of  $z$  — its variance is 1.12 times the optimal one — at every number of clusters, while feasible generalized least squares already reaches 0.987 with 3 clusters and 1.000 with 96. Estimating one smooth parameter costs almost nothing in efficiency, in sharp contrast with Example (Weights from a few replicates), where a separate variance was estimated in each of ten groups.

The inference story is the opposite. The nominal 95% interval based on  $s_F^2(\mathbf{X}^\top \mathbf{V}(\hat{\rho})^{-1} \mathbf{X})^{-1}$  covers the truth in 0.728 of samples with 3 clusters, 0.916 with 12, and 0.944 with 96, while the interval that uses the true  $\rho$  covers 0.950 at three clusters and stays within simulation error of 95% at every size, as Theorem 31.1.1 guarantees. The cluster-robust interval built on  $\hat{\beta}_F$  is uniformly a little better than the model-based one, but no cure when clusters are few — 0.764 at 3 — and the two become indistinguishable once there are several dozen, neither reaching the nominal level even at 96, where both stand at 0.944.

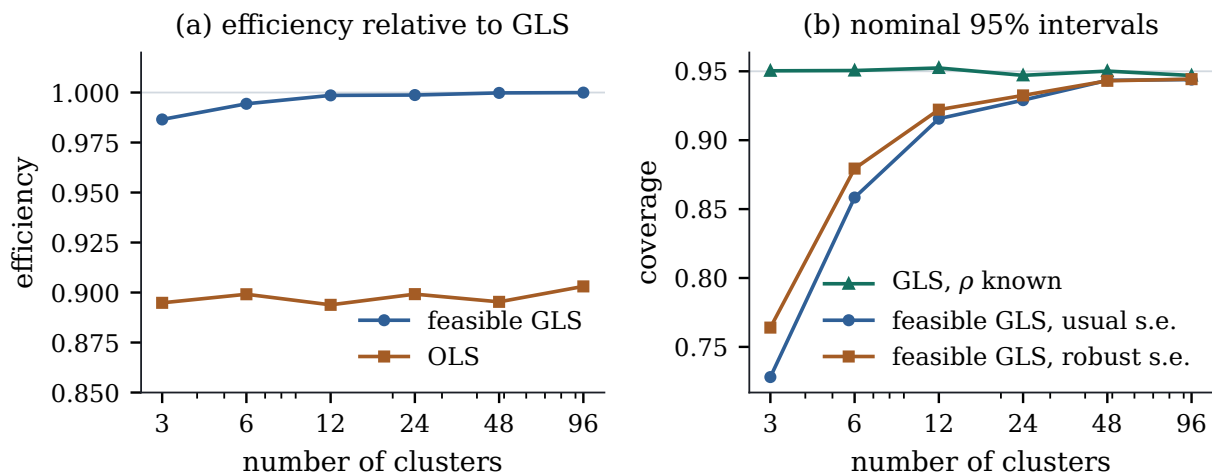


Figure 31.4.1: Feasible generalized least squares with equicorrelated clusters of sizes 2, 4, 8 and  $\rho = 0.6$ , for the coefficient of a cluster-level regressor (20,000 replications each). (a) Efficiency: the variance of generalized least squares with  $\rho$  known divided by the estimator's own variance. Feasible generalized least squares recovers essentially all of it at once. (b) Coverage of nominal 95% intervals: both intervals need dozens of clusters, and the cluster-robust one is no cure.

*Idea.* Estimating a covariance parameter buys back the efficiency almost at once and the correct standard error only slowly: efficiency depends on  $\hat{\theta}$  through a function whose derivative vanishes at the optimum, so a mediocre  $\hat{\theta}$  still gives nearly optimal weights, while the reported variance depends on  $\hat{\theta}$  to first order and inherits all its error.

### 31.4.4 Which repair to use

Ordinary least squares with robust standard errors assumes nothing about the shape of  $\mathbf{V}$ , loses the few per cent bounded by Proposition 31.2.3, and needs many clusters. Feasible generalized least squares recovers that efficiency, but only as far as  $\mathbf{V}(\theta)$  is right; a wrong shape leaves the estimate unbiased — by the argument of Theorem 21.3.1(b), applied with  $\mathbf{W} = \mathbf{V}(\theta_1)^{-1}$  for the limiting  $\theta_1$  — and the reported covariance wrong in an unknown direction. Both, a fitted  $\mathbf{V}(\hat{\theta})$  with a robust standard error, is the modern default for clustered data, and Chapter 33 develops it for longitudinal designs.

None of the three helps when the independent units are few: with three clusters  $\hat{\rho}$  is hopeless and a sandwich with three terms is not a covariance estimate. The remedy is a better small-sample approximation — the Satterthwaite and Kenward–Roger degrees of freedom of Chapter 32, or the bias-corrected sandwiches and cluster bootstrap of Chapter 23 — or more clusters.

#### Example. (The El Niño trend without assuming $\rho$ )

The analysis of Example (A trend in sea surface temperature) assumed  $\rho = 0.5$ . The moment estimator from the ordinary least squares residuals gives  $\hat{\rho} = 0.610$ , and the feasible generalized least squares fit then reports a trend of 0.1346 degrees per decade with standard error 0.0618. For comparison, the ordinary least squares trend, 0.1349, carries the usual standard error 0.0223 and a cluster-robust standard error of 0.0584 with years as clusters. The two honest routes agree closely and the naive one is out by a factor of nearly three. With 61 clusters both are in the range where Example (How

many clusters are enough?) found them reliable.

```
import numpy as np

def quasi_demean(A, Z, sizes, rho):
    """Whitening for equicorrelation rho: rows of A are vectors in  $R^n$ ."""
    c = 1.0 - np.sqrt((1 - rho) / (1 - rho + sizes * rho))
    return (A - (A @ Z / sizes * c) @ Z.T) / np.sqrt(1 - rho)

def fit(X, Y, Z, sizes, rho):
    """GLS coefficients and the model-based covariance factor, for many data sets at once."""
    Xw = quasi_demean(X.T, Z, sizes, rho).T
    Yw = quasi_demean(Y, Z, sizes, rho)
    A = np.linalg.inv(Xw.T @ Xw)
    B = Yw @ Xw @ A
    R = Yw - B @ Xw.T
    s2 = np.sum(R * R, axis=1) / (X.shape[0] - X.shape[1])
    return B, A, s2

def moment_rho(R, Z, sizes):
    """ANOVA estimate of the intraclass correlation from residual vectors R."""
    G, n = len(sizes), R.shape[1]
    means = R @ Z / sizes
    between = np.sum(means**2 * sizes, axis=1) / G
    within = (np.sum(R * R, axis=1) - np.sum(means**2 * sizes, axis=1)) / (n - G)
    mbar = (n - np.sum(sizes**2) / n) / (G - 1)
    est = (between - within) / (between + (mbar - 1) * within)
    return np.clip(est, 0.0, 0.95)
```

```
import numpy as np
import statsmodels.api as sm

frame = sm.datasets.elnino.load_pandas().data
temps = frame.iloc[:, 1:].to_numpy()
G, m = temps.shape
n = G * m
y = temps.ravel()
month = np.tile(np.arange(1, m + 1), G)
time = (np.repeat(frame["YEAR"].to_numpy(), m) - 1980.0) + (month - 6.5) / 12.0
cols = [np.ones(n), time / 10.0]
for k in (1, 2, 3):
    cols += [np.cos(2 * np.pi * k * month / m), np.sin(2 * np.pi * k * month / m)]
X = np.column_stack(cols)
Zn = np.zeros((n, G))
Zn[np.arange(n), np.repeat(np.arange(G), m)] = 1.0
sizes_n = np.full(G, m)

b_ols = np.linalg.solve(X.T @ X, X.T @ y)
rho_hat = float(moment_rho((y - X @ b_ols)[None, :], Zn, sizes_n)[0])
b_f, A_f, s2_f = fit(X, y[None, :], Zn, sizes_n, rho_hat)
se_f = np.sqrt(s2_f[0] * A_f[1, 1])
print(f"El Nino: rho_hat = {rho_hat:.3f}, trend {b_f[0, 1]:.4f} (s.e. {se_f:.4f})")
```

## 31.4.5 Exercises

### 31.4.5.1 A. Check your understanding

#### Exercise A1.

Explain why  $\hat{\beta}_F$  is not a linear function of  $\mathbf{Y}$ , and why this alone makes [Corollary 7.2.3](#) inapplicable to it. Which part of [Theorem 31.1.1](#) could still hold exactly if  $\hat{\theta}$  happened to be independent of  $\mathbf{Y}$ ?

#### Exercise A2.

Give a family  $\mathbf{V}(\theta)$  with  $k = 2$  for which no estimation of  $\theta$  is needed to obtain the best estimator of  $\mathbf{X}\beta$ , and say what  $\theta$  is still needed for.

### 31.4.5.2 B. Practice

#### Exercise B1.

For balanced clusters of size  $m$  with equicorrelation  $\rho$ , let  $B$  and  $W$  be the between- and within-cluster mean squares of the ordinary least squares residuals. Show that the moment estimator  $\hat{\rho} = (B - W)/\{B + (m - 1)W\}$  is the solution of matching  $EB$  and  $EW$  to their model values when the model is fitted without regressors, and explain why  $\hat{\rho}$  can be negative. What does truncating it at zero do to [Theorem 31.4.1](#)?

*Solution.* For a one-way layout with no regressors,  $EW = \sigma^2(1 - \rho)$  and  $EB = \sigma^2\{1 + (m - 1)\rho\}$  (this is [Exercise \(B1\)](#)). Solving  $B = \sigma^2\{1 + (m - 1)\rho\}$ ,  $W = \sigma^2(1 - \rho)$  for  $\rho$  gives the stated ratio. Since  $B$  and  $W$  are independent chi-squared multiples,  $B < W$  happens with positive probability whenever  $\rho$  is small, so  $\hat{\rho} < 0$  is possible; [Chapter 32](#) discusses negative variance-component estimates at length. Truncation leaves  $\hat{\rho}$  consistent whenever  $\rho > 0$ , which is all (G4) requires; at  $\rho = 0$  the truncated estimator is still consistent and still satisfies  $\sqrt{n}\hat{\rho} = O_p(1)$ , so [Theorem 31.4.1\(b\)](#) still applies to  $\hat{\beta}_F$ , but  $\hat{\rho}$  itself is no longer asymptotically normal and the usual Wald theory for it fails.

#### Exercise B2.

Suppose  $\mathbf{V}(\theta) = \text{diag}(1 + \theta z_1, \dots, 1 + \theta z_n)$  with known  $z_i \geq 0$  and  $\theta \geq 0$ . Verify (G1) and (G2) on a neighbourhood of an interior  $\theta_0$ , giving explicit constants, and show that (G3) holds when the  $z_i$  and the rows of  $\mathbf{X}$  are bounded.

#### Exercise B3.

In the balanced cluster model of [Exercise \(B4\)](#), all clusters of size  $m$ , consider estimating  $\mu$  with weights built from a wrong value  $\rho_1$  when the truth is  $\rho_0$ . Show that the resulting estimator is  $\bar{Y}$  for every  $\rho_1$ , and hence that the efficiency is exactly 1. Contrast with the unbalanced case and explain the flat curve in [Figure 31.4.1\(a\)](#).

*Solution.* With equal sizes, [Eq. 31.1.8](#) gives  $w_k = m/(1 - \rho_1 + m\rho_1)$ , the same for every  $k$ , so the weighted mean of the cluster means is  $\bar{Y}$  whatever  $\rho_1$  is: the estimator does not depend on the weights at all. With unequal sizes the weights do depend on  $\rho_1$ , but only through the smooth ratio  $m_k/(1 - \rho_1 + m_k\rho_1)$ , and the variance of the resulting estimator is stationary at  $\rho_1 = \rho_0$  because  $\rho_0$  minimizes it. A first-order error in  $\hat{\rho}$  therefore produces only a second-order loss of efficiency, which is why panel (a) is flat.

### 31.4.5.3 C. Going deeper

#### Exercise C1.

Show that the bound of [Lemma 31.4.2](#) cannot be improved in order: construct a family with  $k = 1$ , independent clusters of size 1, and a design for which  $E\|n^{-1/2}\mathbf{X}^\top \mathbf{D}(\theta)\boldsymbol{\varepsilon}\|^2$  is bounded below by a positive multiple of  $(\theta - \theta_0)^2$ .

#### Exercise C2.

Let  $\hat{\boldsymbol{\theta}}$  be independent of  $\mathbf{Y}$  with  $E\hat{\boldsymbol{\theta}} = \boldsymbol{\theta}_0$ , and write  $g(\boldsymbol{\theta}) = \mathbf{a}^\top (\mathbf{X}^\top \mathbf{V}(\boldsymbol{\theta})^{-1} \mathbf{X})^{-1} \mathbf{a}$  for the reported variance of  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}_F$  at scale  $\sigma^2 = 1$ , and  $h(\boldsymbol{\theta})$  for its true variance. Show that  $h(\boldsymbol{\theta}) \geq g(\boldsymbol{\theta}_0)$  for every  $\boldsymbol{\theta}$ , with equality at  $\boldsymbol{\theta}_0$ , and deduce that  $Eg(\hat{\boldsymbol{\theta}})$  understates  $Eh(\hat{\boldsymbol{\theta}})$  whenever  $g$  is concave near  $\boldsymbol{\theta}_0$ . Why does this not settle the question in general?

*Solution.* For fixed  $\boldsymbol{\theta}$  the statistic  $\mathbf{a}^\top \hat{\boldsymbol{\beta}}_{\mathbf{V}(\boldsymbol{\theta})}$  is a linear unbiased estimator of  $\mathbf{a}^\top \boldsymbol{\beta}$ , so [Corollary 7.2.3](#) gives  $h(\boldsymbol{\theta}) \geq h(\boldsymbol{\theta}_0) = g(\boldsymbol{\theta}_0)$ , with equality at  $\boldsymbol{\theta}_0$ . Conditionally on  $\hat{\boldsymbol{\theta}}$  the report is  $g(\hat{\boldsymbol{\theta}})$  and the truth is  $h(\hat{\boldsymbol{\theta}}) \geq g(\boldsymbol{\theta}_0)$ ; concavity of  $g$  and Jensen's inequality give  $Eg(\hat{\boldsymbol{\theta}}) \leq g(\boldsymbol{\theta}_0)$ . In general  $g$  need not be concave and  $\hat{\boldsymbol{\theta}}$  is not independent of  $\mathbf{Y}$ ; that dependence is what [Example \(How many clusters are enough?\)](#) measures.

## 31.5 Summary and Notes

### 31.5.1 Summary

*Idea.*

1. The general Gauss–Markov model [Eq. 31.1.1](#) keeps  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  and replaces  $\sigma^2\mathbf{I}$  by  $\sigma^2\mathbf{V}$ . Only the product is identified, so  $\mathbf{V}$  carries the *shape* of the covariance and  $\sigma^2$  its scale ([Exercise \(A1\)](#)).
2. With  $\mathbf{V}$  known and positive definite, generalized least squares ([Definition 31.1.1](#)) is ordinary least squares on the whitened data [Eq. 31.1.3](#), and everything in Parts II and III transfers: the  $\chi^2$  law of  $\text{SSE}_{\mathbf{V}}$ , the unbiased  $s_{\mathbf{V}}^2$ ,  $t$  intervals and the  $F$  test of a reduced model, all on  $n - r$  degrees of freedom ([Theorem 31.1.1](#)). What does not transfer is the Euclidean reading of the residual, which is  $\mathbf{V}^{-1}$ -orthogonal to  $\mathcal{C}(\mathbf{X})$ ;  $R^2$  and leverage lose their meaning.
3. For structured  $\mathbf{V}$  the whitening has a closed form. Equicorrelated clusters give the quasi-demeaning [Eq. 31.1.5](#), which interpolates between doing nothing and fully



centring each cluster.

4. A linear unbiased estimator  $\mathbf{c}^\top \mathbf{Y}$  of  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is best exactly when  $\mathbf{V}\mathbf{c} \in \mathcal{C}(\mathbf{X})$ , and any two best estimators agree on  $\mathcal{C}(\mathbf{X} : \mathbf{V})$  (Lemma 31.2.1). This one criterion drives the rest of the chapter.
5. Ordinary least squares is best for *everything* iff  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , iff  $\mathbf{M}\mathbf{V} = \mathbf{V}\mathbf{M}$ , iff  $\mathbf{V} = \mathbf{X}\mathbf{A}\mathbf{X}^\top + \mathbf{W}\mathbf{B}\mathbf{W}^\top$ , iff  $\mathcal{C}(\mathbf{X})$  is spanned by eigenvectors of  $\mathbf{V}$  (Theorem 31.2.2). Balanced layouts with a shared shock satisfy it; unbalanced ones do not, and then some contrasts are still estimated best and others are not.
6. The efficiency lost by ignoring the covariance is bounded below by the Kantorovich ratio and is usually small (Proposition 31.2.3). The error in the reported standard error is not bounded at all and is usually large. Fix the standard error first.
7. When  $\mathbf{V}$  is singular the errors live in  $\mathcal{C}(\mathbf{V})$ , so the data live in  $\mathcal{C}(\mathbf{X} : \mathbf{V})$  and  $r - d$  linear functions of  $\boldsymbol{\beta}$  are known exactly (Theorem 31.3.2). The BLUE uses those functions to pin down a particular solution and then does generalized least squares inside  $\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{V})$ , or comes in closed form from Rao's formula Eq. 31.3.1. The residual degrees of freedom are  $\text{rank}(\mathbf{X} : \mathbf{V}) - \text{rank}(\mathbf{X})$ , not  $n - r$ .
8. When  $\mathbf{V} = \mathbf{V}(\boldsymbol{\theta})$  must be estimated, the two-step estimator Eq. 31.4.1 is asymptotically equivalent to generalized least squares under stated regularity and a stochastic equicontinuity condition (Theorem 31.4.1, Lemma 31.4.2). In small samples the efficiency comes back almost at once while the reported standard error stays badly optimistic (Example (How many clusters are enough?)).

## 31.5.2 Notes and sources

**Generalized least squares.** The estimator and its optimality are due to Aitken (1935), who derived it by the same change of variables used here. Christensen (2020, §2.7) develops it coordinate-freely for arbitrary rank, the route Section 6.9 follows; Seber and Lee (2003, §3.10) and Rencher and Schaalje (2008, §7.8) give the full-rank version. That only  $\sigma^2\mathbf{V}$  is a parameter is standard but often left implicit, so “normalizing  $\mathbf{V}$ ” is sometimes mistaken for an assumption.

**When least squares is best.** The condition  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  has been found many times: Kruskal (1968) gave the coordinate-free statement that the two *fits* coincide, proved here as Theorem 6.9.3; Zyskind (1967) characterized through eigenvectors the covariance structures for which least squares is best, which is condition (v); and Rao (1967) gave the structural form  $\mathbf{V} = \mathbf{X}\mathbf{A}\mathbf{X}^\top + \mathbf{W}\mathbf{B}\mathbf{W}^\top$  now usually called simple covariance structure. Watson (1955) first asked how much efficiency is lost when the condition fails; the sharp scalar bound follows from Kantorovich's inequality, proved as Theorem 19.2.3 in Section 19.2, and the determinant bound quoted in Section 31.2 was proved independently by Bloomfield and Watson (1975) and by Knott (1975). Puntanen and Styan (1989) survey the whole equivalence. Recovery of interblock information in incomplete block designs, the applied face of the failure of the condition, goes back to Yates (1940).

**Singular covariance matrices.** The systematic theory is due to Rao (1971, 1973), whose unified theory of linear estimation produced the formula Eq. 31.3.1 with  $\mathbf{T} = \mathbf{V} + \mathbf{X}\mathbf{S}\mathbf{X}^\top$  for any nonnegative definite  $\mathbf{S}$  with  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{T})$ , and to Rao and Mitra (1971) for the generalized-inverse machinery. Christensen (2020, ch. 10) gives the vector-space treatment, including consistent estimation and the degrees-of-freedom count  $\text{rank}(\mathbf{X} : \mathbf{V}) - \text{rank}(\mathbf{X})$ . The proof given here is different: it splits the observation space orthogonally into  $\mathcal{C}(\mathbf{V})$  and its complement, turning the model into an ordinary generalized least squares problem subject to exact linear



constraints, and then verifies the two conditions of [Lemma 31.2.1](#). Our [Theorem 31.3.2\(c\)](#) uses the Moore–Penrose inverse throughout, which keeps it short; with an arbitrary generalized inverse of  $\mathbf{T}$  the same statements hold, but  $\mathbf{T}\mathbf{T}^-\mathbf{X} = \mathbf{X}$  and the invariance of [Eq. 31.3.1](#) then have to be established separately.

**Feasible generalized least squares.** Two-step estimation with an estimated covariance is old; the econometric literature traces it through Cochrane and Orcutt (1949) for autoregressive errors and Zellner (1962) for seemingly unrelated regressions. Amemiya (1985, ch. 6) states and proves the asymptotic equivalence for several concrete families, and Carroll and Ruppert (1988, ch. 3) for smooth variance functions; both need a uniformity condition of the kind labelled (G5) in [Theorem 31.4.1](#), and our [Lemma 31.4.2](#) isolates the cancellation that makes such a condition plausible. The finite-sample deficiency of the plug-in covariance is quantified by Kackar and Harville (1984), whose correction underlies the Kenward–Roger adjustment of [Chapter 32](#). The complementary strategy of leaving the estimator alone and correcting the standard error is White (1980) and, for clustered data, Liang and Zeger (1986); it appears in this book as [Theorem 21.4.2](#).

**What comes next.** [Chapter 32](#) makes  $\mathbf{V}$  the marginal covariance of a model with random effects, which supplies both a reason for a particular  $\mathbf{V}(\boldsymbol{\theta})$  and better estimators of  $\boldsymbol{\theta}$  than the moment estimators used here. [Chapter 33](#) applies the whole apparatus to split-plot, repeated-measures and longitudinal designs, where [Theorem 31.2.2](#) explains which classical analyses survive and which do not.

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# Chapter 32

## Random Effects and Linear Mixed Models

Part IV treated the levels of a factor as chosen: five nitrogen rates, three machines, the treatments the experimenter cared about. Many factors are not like that. The twelve laboratories in a proficiency study, the eleven firms in a panel, the patients in a trial and the classrooms in a school district are a *sample* of levels, and the question is not how laboratory 3 compares with laboratory 7 but how much laboratories vary. Declaring such a factor random changes three things at once. The parameters of interest become variances. The individual levels are *predicted* rather than estimated, and the prediction is shrunk towards the overall mean. And the covariance of the response stops being  $\sigma^2\mathbf{I}$ : observations sharing a level are correlated, which is what Part VII is about. This chapter develops the model that results, the best linear unbiased predictor and Henderson's equations that compute it, restricted maximum likelihood, and what inference survives when the covariance itself has been estimated from the same data.

**What you need.** Chapter 6 (projections and generalized least squares, Definition 6.9.1, Theorem 6.9.3), Chapter 7 (the Gauss–Markov and Aitken theorems, Theorem 7.2.1 and Corollary 7.2.3; maximum likelihood, Theorem 7.3.1; the conjugate posterior, Theorem 7.6.2), Chapter 4 (distributions of quadratic forms, Theorem 4.3.2 and Theorem 4.4.2), Chapter 9 (expected mean squares, Theorem 9.6.1 and Exercise (B1)), Chapter 14 (best linear unbiased prediction, Theorem 14.2.8, resting on the best linear predictor Theorem 2.5.2 of Chapter 2), Section 15.1 (the one-way layout and the first look at random effects, Definition 15.1.1) and Chapter 16 (nested and crossed layouts, Theorem 16.5.1). From Part V, Section 21.3 (weighted least squares with estimated weights) and Section 23.3 (the parametric bootstrap, Proposition 23.3.3); from Part VI, Chapter 27 (ridge and empirical Bayes, Theorem 27.2.2 and Proposition 27.5.4) and Chapter 29 (AIC and BIC, Definition 29.2.2). The generalized least squares theory of Chapter 31 is used throughout.

### 32.0.1 Roadmap

- Random effects
- The linear mixed model
- Prediction: BLUP and Henderson's equations
- Maximum likelihood and REML
- Inference when the covariance is estimated

- [Bayesian thread III: mixed models as hierarchical models](#)
- [Summary and notes](#)

## 32.1 Random effects

A one-way layout ([Definition 15.1.1](#)) has  $g$  groups and  $g$  group means. In Part IV those means were the object of the study: the five nitrogen rates were chosen, and every conclusion was about them. [Section 15.1](#) previewed the other possibility and left it here; this section carries the preview through. Consider a study of a different shape. A reference material with a known lead content is sent to twelve accredited laboratories, each of which analyses four aliquots of it. Nobody cares whether laboratory 3 reads higher than laboratory 7. The question is how much accredited laboratories disagree with one another, because that number, and not the twelve particular deviations, is what a client using an unnamed laboratory faces. The twelve are a sample from a population of laboratories, and the conclusion must reach the population.

### 32.1.1 When a factor is random

The test, met first in [Section 15.1](#), is about the *scope of the conclusion*. If the levels in the data exhaust the levels you want to talk about, the factor is fixed; if they are a sample and the conclusion should extend to levels you did not observe, it is random. Doses, treatments, varieties and the two arms of a trial are almost always fixed; laboratories, batches of reagent, subjects, litters, classrooms and interviewers are usually random. Three things change when a factor is declared random.

- (1) **The parameters change.** The  $g$  group effects are replaced by a single number, their variance. A study with  $g = 12$  laboratories estimates one variance, not twelve means, and its precision is governed by  $g$ , not by the total number of observations.
- (2) **Estimation becomes prediction.** The realized effect of laboratory 3 is a random variable, not a parameter. Asking for its value is a prediction problem, and [Section 32.3](#) shows that the best answer is not the laboratory's own average.
- (3) **The covariance changes.** Two aliquots analysed by the same laboratory share its effect, so they are correlated. The model is no longer a linear model with  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ , and this is the reason the chapter belongs to Part VII.

*Warning.* A variance is estimated from the number of *levels*, not the number of observations. With  $g = 3$  laboratories there are two degrees of freedom for the between-laboratory variance, and any interval for it is nearly useless however many aliquots each laboratory analyses. A treated and a control arm should never be a random factor: two levels carry no information about a population of levels, and there is no population. When a factor has few levels but its levels really are a sample, the honest options are to report the fixed-effect analysis alongside, or to supply prior information ([Section 32.6](#)).

### 32.1.2 The one-way random effects model

**Definition 32.1.1 — One-way random effects model**

Let  $g \geq 2$  groups contain  $n_k \geq 1$  observations each, with  $n = \sum_k n_k$ . The **one-way random effects model** is

$$Y_{kj} = \mu + a_k + \varepsilon_{kj}, \quad k = 1, \dots, g, \quad j = 1, \dots, n_k,$$

where  $\mu$  is an unknown constant, the **group effects**  $a_1, \dots, a_g$  are uncorrelated with mean 0 and variance  $\sigma_a^2 \geq 0$ , the errors  $\varepsilon_{kj}$  are uncorrelated with mean 0 and variance  $\sigma^2 > 0$ , and the two sets are uncorrelated with each other. In matrix form, with  $\mathbf{Z}$  the  $n \times g$  matrix of group indicators,

$$\mathbf{Y} = \mu \mathbf{1} + \mathbf{Z}\mathbf{a} + \boldsymbol{\varepsilon}.$$

The **normal** one-way random effects model adds  $\mathbf{a} \sim N_g(\mathbf{0}, \sigma_a^2 \mathbf{I})$  independent of  $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$ . The layout is **balanced** if  $n_1 = \dots = n_g = m$ . The pair  $(\sigma_a^2, \sigma^2)$  are the **variance components**.

The model has three parameters whatever  $g$  is, where Definition 15.1.1 had  $g + 1$ . That is the economy of random effects, and also their risk: it is bought by assuming that the  $g$  effects are an uncorrelated sample from one distribution.

**Proposition 32.1.1 — The one-way random effects model**

In Definition 32.1.1, write  $\gamma = \sigma_a^2/\sigma^2$  and let  $\mathbf{M}$  be the projection onto  $\mathcal{C}(\mathbf{Z})$ , which replaces each entry by its group mean.

- (a)  $E(\mathbf{Y}) = \mu \mathbf{1}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$ , a block diagonal matrix whose  $k$ th block is  $\sigma^2 \mathbf{I}_{n_k} + \sigma_a^2 \mathbf{1}\mathbf{1}^\top$ .
- (b) (*Intraclass correlation.*) Observations in different groups are uncorrelated, and two observations in the same group have correlation

$$\rho = \frac{\sigma_a^2}{\sigma_a^2 + \sigma^2} = \frac{\gamma}{1 + \gamma} \in [0, 1).$$

- (c) (*Moment estimates.*) Let MSB and MSE be the between and within mean squares of the one-way analysis of variance, on  $g - 1$  and  $n - g$  degrees of freedom. Then  $E(\text{MSE}) = \sigma^2$  and  $E(\text{MSB}) = \sigma^2 + m_0 \sigma_a^2$  with

$$m_0 = \frac{1}{g-1} \left( n - \frac{\sum_k n_k^2}{n} \right), \quad (32.1.1)$$

which equals  $m$  in a balanced layout. Hence  $\hat{\sigma}^2 = \text{MSE}$  and  $\hat{\sigma}_a^2 = (\text{MSB} - \text{MSE})/m_0$  are unbiased.

- (d) (*Distributions, balanced and normal.*) If the layout is balanced and the model is normal, then with  $\lambda_1 = \sigma^2 + m\sigma_a^2$ ,

$$\frac{\text{SSB}}{\lambda_1} \sim \chi^2(g-1) \quad \text{independently of} \quad \frac{\text{SSE}}{\sigma^2} \sim \chi^2(n-g),$$

and therefore  $\text{MSB}/\text{MSE} \sim (1 + m\gamma) F(g-1, n-g)$ .

*Proof.*

- (a) By [Theorem 2.2.1](#) applied to  $\mathbf{Y} = \mu\mathbf{1} + [\mathbf{Z}, \mathbf{I}](\mathbf{a}^\top, \boldsymbol{\varepsilon}^\top)^\top$ ,  $\text{Cov}(\mathbf{Y}) = \mathbf{Z}(\sigma_a^2\mathbf{I}_g)\mathbf{Z}^\top + \sigma^2\mathbf{I}$ . The matrix  $\mathbf{Z}\mathbf{Z}^\top$  has  $(i, i')$  entry 1 when observations  $i$  and  $i'$  are in the same group and 0 otherwise, which is the block form stated.
- (b) Read off (a): the covariance of two distinct observations in group  $k$  is  $\sigma_a^2$  and each variance is  $\sigma_a^2 + \sigma^2$ .
- (c) The two sums of squares are  $\text{SSB} = \mathbf{Y}^\top(\mathbf{M} - \mathbf{P}_1)\mathbf{Y}$  and  $\text{SSE} = \mathbf{Y}^\top(\mathbf{I} - \mathbf{M})\mathbf{Y}$ , where  $\mathbf{P}_1 = n^{-1}\mathbf{1}\mathbf{1}^\top$ . Both projections annihilate  $\mathbf{1}$ , so [Theorem 2.3.1](#) leaves only the trace term. Since the columns of  $\mathbf{Z}\mathbf{Z}^\top$  lie in  $\mathcal{C}(\mathbf{Z})$  we have  $\mathbf{M}\mathbf{Z}\mathbf{Z}^\top = \mathbf{Z}\mathbf{Z}^\top$ , and  $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$ , so

$$E(\text{SSE}) = \sigma^2 \text{tr}(\mathbf{I} - \mathbf{M}) = \sigma^2(n - g),$$

while  $\text{tr}(\mathbf{Z}\mathbf{Z}^\top) = n$  and  $\text{tr}(\mathbf{P}_1\mathbf{Z}\mathbf{Z}^\top) = n^{-1}\mathbf{1}^\top\mathbf{Z}\mathbf{Z}^\top\mathbf{1} = n^{-1}\sum_k n_k^2$  give

$$E(\text{SSB}) = \sigma^2(g - 1) + \sigma_a^2\left(n - \frac{\sum_k n_k^2}{n}\right).$$

Dividing by the degrees of freedom gives the two expectations, and unbiasedness of the two estimates follows by linearity. In a balanced layout  $n = gm$  and  $\sum_k n_k^2 = gm^2$ , so  $m_0 = \{gm - gm^2/(gm)\}/(g - 1) = m(g - 1)/(g - 1) = m$ .

- (d) In the balanced case  $\mathbf{Z}\mathbf{Z}^\top = m\mathbf{M}$ , so by (a)

$$\text{Cov}(\mathbf{Y}) = \lambda_1\mathbf{M} + \sigma^2(\mathbf{I} - \mathbf{M}), \quad (32.1.2)$$

a spectral decomposition into the two orthogonal pieces  $\mathcal{C}(\mathbf{Z})$  and  $\mathcal{C}(\mathbf{Z})^\perp$ . Write  $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$ . For the between projection  $\mathbf{A} = \mathbf{M} - \mathbf{P}_1$ , which satisfies  $\mathbf{A}\mathbf{M} = \mathbf{A}$  and  $\mathbf{A}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ , we get  $\mathbf{A}\boldsymbol{\Sigma} = \lambda_1\mathbf{A}$ , so  $\mathbf{A}\boldsymbol{\Sigma}/\lambda_1 = \mathbf{A}$  is idempotent of rank  $g - 1$  and  $\mathbf{A}\boldsymbol{\mu} = \mathbf{0}$ ; [Theorem 4.3.2](#) gives  $\text{SSB}/\lambda_1 \sim \chi^2(g - 1)$ . For  $\mathbf{B} = \mathbf{I} - \mathbf{M}$  we get  $\mathbf{B}\boldsymbol{\Sigma} = \sigma^2\mathbf{B}$  and  $\text{SSE}/\sigma^2 \sim \chi^2(n - g)$  in the same way. Independence follows from [Theorem 4.4.2](#), because  $\mathbf{A}\boldsymbol{\Sigma}\mathbf{B} = \lambda_1\mathbf{A}\mathbf{B} = \mathbf{0}$ . The ratio of the two mean squares is then  $(\lambda_1/\sigma^2)$  times an  $F(g - 1, n - g)$  variable, and  $\lambda_1/\sigma^2 = 1 + m\gamma$ . ■

Part (c) is [Exercise \(B1\)](#) carried to unbalanced layouts, and part (d) is the fact that makes the balanced case so pleasant: the same  $F$  ratio that tests equality of fixed group means tests  $\sigma_a^2 = 0$ , and it does more, because its distribution is known for every  $\gamma$ , not only at  $\gamma = 0$ . Inverting it gives an exact interval.

### Corollary 32.1.2 — Exact interval for the variance ratio

In the balanced normal one-way random effects model, let  $F = \text{MSB}/\text{MSE}$  and let  $F_q$  denote the  $q$  quantile of  $F(g - 1, n - g)$ . Then

$$\left[ \frac{1}{m} \left( \frac{F}{F_{1-\alpha/2}} - 1 \right), \frac{1}{m} \left( \frac{F}{F_{\alpha/2}} - 1 \right) \right] \quad (32.1.3)$$

is an exact  $1 - \alpha$  confidence interval for  $\gamma = \sigma_a^2/\sigma^2$ , and applying the increasing map  $\gamma \mapsto \gamma/(1 + \gamma)$  to its endpoints gives an exact  $1 - \alpha$  interval for the intraclass correlation

$\rho$ . Endpoints below zero are replaced by zero.

*Proof.* By [Proposition 32.1.1\(d\)](#),  $F/(1 + m\gamma) \sim F(g - 1, n - g)$ , so  $\Pr\{F_{\alpha/2} \leq F/(1 + m\gamma) \leq F_{1-\alpha/2}\} = 1 - \alpha$  whatever  $\gamma$  is. Solving the two inequalities for  $\gamma$  gives [Eq. 32.1.3](#), an event of the same probability. A strictly increasing transformation of both endpoints keeps the coverage, and  $\rho = \gamma/(1 + \gamma)$  is strictly increasing on  $[0, \infty)$ . Truncating at zero can only raise the coverage, since  $\gamma \geq 0$ . ■

Notice what [Eq. 32.1.3](#) is not: it is an interval for the *ratio*. No exact interval of this kind exists for  $\sigma_a^2$  alone, because  $\hat{\sigma}_a^2$  is a difference of two mean squares with different scale factors and its distribution depends on both components separately. Approximate intervals for  $\sigma_a^2$  come either from Satterthwaite's method ([Section 32.5](#)) or from the profile likelihood ([Section 32.4](#)).

### Example. (A laboratory proficiency study)

Twelve laboratories each analyse four aliquots of one homogeneous batch of soil for lead, in mg/kg; the data are simulated so that the truth is known, with  $\mu = 2.50$ ,  $\sigma_a = 0.20$  and  $\sigma = 0.25$ , so that the population intraclass correlation is 0.390. The analysis of variance gives  $MSB = 0.2166$  on 11 degrees of freedom and  $MSE = 0.0620$  on 36, so

$$\hat{\sigma}^2 = 0.0620, \quad \hat{\sigma}_a^2 = \frac{0.2166 - 0.0620}{4} = 0.0386,$$

that is, a within-laboratory (repeatability) standard deviation of 0.249 and a between-laboratory standard deviation of 0.197. The estimated intraclass correlation is 0.384: about two fifths of the variation in a single measurement is attributable to which laboratory made it.

The  $F$  ratio is 3.49, and [Eq. 32.1.3](#) gives  $\gamma \in (0.118, 2.439)$ , that is  $\rho \in (0.105, 0.709)$ . Twelve laboratories and 48 measurements pin the intraclass correlation down only to a factor of seven. That is not a defect of the method; it is what 11 degrees of freedom buy.

```
import numpy as np
from scipy import stats

g, m = 12, 4                                # laboratories, aliquots each
n = g * m
mu, sigma_a, sigma = 2.50, 0.20, 0.25        # the values used to simulate

rng = np.random.default_rng(20320048)
lab_effect = sigma_a * rng.standard_normal(g)
y = mu + np.repeat(lab_effect, m) + sigma * rng.standard_normal(n)

Y = y.reshape(g, m)                         # one row per laboratory
lab_mean, grand_mean = Y.mean(axis=1), Y.mean()

ss_between = m * np.sum((lab_mean - grand_mean) ** 2)
ss_within = np.sum((Y - lab_mean[:, None]) ** 2)
ms_between = ss_between / (g - 1)
ms_within = ss_within / (n - g)
```

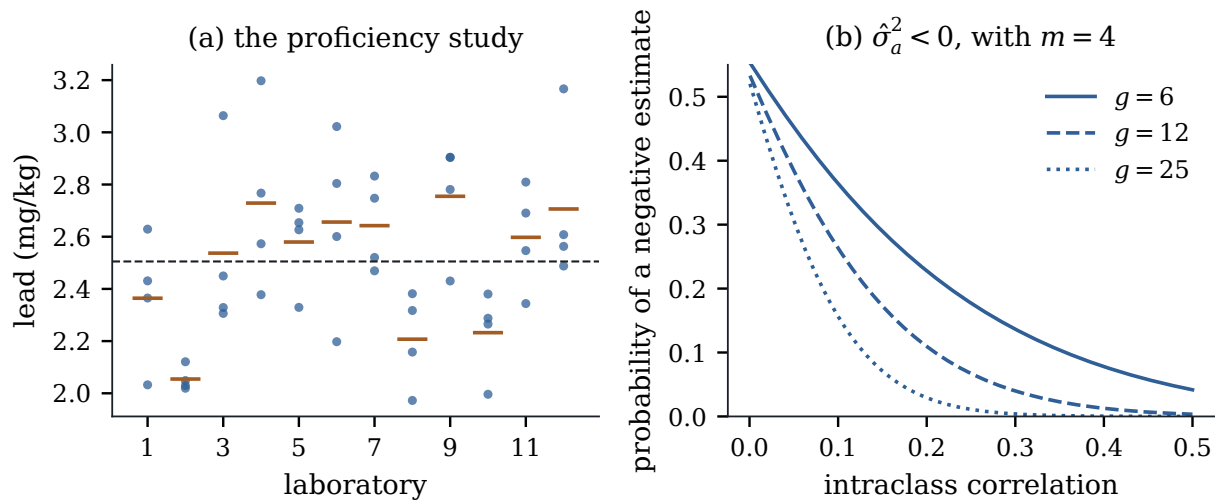


Figure 32.1.1: (a) The proficiency study: four aliquots in each of twelve laboratories, with the laboratory means (short bars) and the overall mean (dashed). (b) The probability that the moment estimate  $\hat{\sigma}_a^2$  is negative, as a function of the intraclass correlation, for  $g = 6, 12$  and  $25$  laboratories with  $m = 4$  aliquots each.

```
sigma2_hat = ms_within                                # E(MS within) = sigma^2
sigma2_a_hat = (ms_between - ms_within) / m           # E(MS between) = sigma^2 + m
               sigma_a^2
icc_hat = sigma2_a_hat / (sigma2_a_hat + sigma2_hat)

print(f"MS between {ms_between:.4f}    MS within {ms_within:.4f}")
print(f"sigma^2 {sigma2_hat:.4f}    sigma_a^2 {sigma2_a_hat:.4f}    ICC {icc_hat:.3f}"
      )
```

```
f_obs = ms_between / ms_within                        # ~ (1 + m gamma) F(g-1, n-g)
gamma_lo = (f_obs / stats.f.ppf(0.975, g - 1, n - g) - 1) / m
gamma_hi = (f_obs / stats.f.ppf(0.025, g - 1, n - g) - 1) / m
icc_lo, icc_hi = gamma_lo / (1 + gamma_lo), gamma_hi / (1 + gamma_hi)

print(f"F = {f_obs:.2f};  gamma in ({gamma_lo:.3f}, {gamma_hi:.3f})")
print(f"intraclass correlation in ({icc_lo:.3f}, {icc_hi:.3f})")
```

### 32.1.3 Negative estimates

Nothing in [Proposition 32.1.1\(c\)](#) prevents MSB from falling below MSE, and then the unbiased estimate of a variance is negative. In the balanced normal model the probability is exactly

$$\Pr(\hat{\sigma}_a^2 < 0) = \Pr\left(F(g-1, n-g) < \frac{1}{1+m\gamma}\right), \quad (32.1.4)$$

by [Proposition 32.1.1\(d\)](#). Panel (b) of [Figure 32.1.1](#) plots it. With  $g = 6$  laboratories,  $m = 4$  aliquots and an intraclass correlation of 0.05 the probability is 0.453; when  $\sigma_a^2$  is exactly zero it is 0.554, a little over one half, because  $\Pr\{F(5, 18) < 1\}$  exceeds  $1/2$ . In the study of [Example \(A laboratory proficiency study\)](#), with twelve laboratories and a true correlation of 0.390, it is only 0.0143.



A negative estimate is not a computational failure but evidence, and there are three honest responses.

- (1) **Report it.** A meta-analysis that averages several studies needs the unbiased value, negative or not; truncating every study at zero and then averaging biases the average upwards.
- (2) **Truncate.** The estimate  $\max(\hat{\sigma}_a^2, 0)$  is biased upwards but has smaller mean squared error for every parameter value ([Exercise \(C1\)](#)). Maximum likelihood and REML produce it automatically, since they maximize over  $\sigma_a^2 \geq 0$ .
- (3) **Question the model.** A persistently negative estimate can mean that observations within a group are genuinely *negatively* correlated, as happens when the group is a fixed resource the observations compete for: litter-mates sharing a food supply, or plots sharing water. The matrix  $\sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$  with  $\sigma_a^2 < 0$  is a perfectly good covariance as long as  $\sigma^2 + m\sigma_a^2 > 0$ , and fitting it as such is more honest than pretending a variance was estimated.

## 32.1.4 Unbalanced layouts, and beyond one way

Part (c) is the simplest instance of a general recipe: write down quadratic forms, equate them to their expectations, and solve. Henderson (1953) organized this into three methods for unbalanced layouts, of which the one used here is the first. The recipe always gives unbiased estimates and never needs normality, but it is not unique — different quadratic forms give different estimates, and in unbalanced layouts none of them is best. That is why [Section 32.4](#) turns to the likelihood.

The same calculation runs through any balanced layout. In the two-stage nested design of [Exercise \(C1\)](#), with  $a$  levels of  $A$ ,  $b$  of  $B$  within each and  $m$  observations per cell, that exercise gave  $E MS_{B(A)} = \sigma^2 + m\sigma_B^2$ , so the inner component is estimated by  $(MS_{B(A)} - MSE)/m$ . Taking  $A$  random too, with effects of variance  $\sigma_A^2$ ,  $MS_A$  is  $bm$  times the sample variance of  $a$  independent group means of variance  $\sigma_A^2 + \sigma_B^2/b + \sigma^2/(bm)$ , so  $E MS_A = \sigma^2 + m\sigma_B^2 + bm\sigma_A^2$  and the outer component is estimated by  $(MS_A - MS_{B(A)})/(bm)$ . Every mean square in a balanced layout turns out to estimate  $\sigma^2$  plus a positive combination of the components nested inside its factor, a fact [Chapter 33](#) establishes and uses to read a split-plot analysis off its strata.

Finally, a warning about the grand mean. In the fixed-effects model  $\text{Var}(\bar{Y}) = \sigma^2/n$ , so precision is bought with observations. Here it is different: [Exercise \(C2\)](#) showed that

$$\text{Var}(\bar{Y}) = \frac{\sigma^2}{n} + \sigma_a^2 \frac{\sum_k n_k^2}{n^2} \xrightarrow{\text{balanced}} \frac{\sigma^2 + m\sigma_a^2}{gm},$$

which for fixed  $g$  is bounded below by  $\sigma_a^2/g$  however large  $m$  grows. Measuring the same twelve laboratories a thousand times each does not improve the estimate of the population mean beyond a limit set by  $g = 12$ . This is the commonest way of overstating precision in clustered data, and [Chapter 33](#) quantifies it as the design effect.

## 32.1.5 Exercises

### 32.1.5.1 A. Check your understanding

#### Exercise A1.

For each study, say whether the named factor is better treated as fixed or random, and

what the corresponding target of inference is. (a) Four fertilizer formulations compared on a research farm. (b) Twenty operators sampled from a factory's workforce, each measuring the same set of parts. (c) The two arms of a randomized trial. (d) Thirty schools sampled from a district, in a study of a teaching method. (e) Three furnaces, the only three the plant owns, in a study of their temperature uniformity.

### 32.1.5.2 B. Practice

#### Exercise B1.

Verify [Eq. 32.1.1](#) directly from the group means: show that  $E\{\sum_k n_k(\bar{Y}_k - \bar{Y})^2\} = \sigma^2(g-1) + \sigma_a^2(n - \sum_k n_k^2/n)$  by computing  $\text{Var}(\bar{Y}_k)$ ,  $\text{Var}(\bar{Y})$  and  $\text{Cov}(\bar{Y}_k, \bar{Y})$ . Confirm that  $m_0 = m$  when the layout is balanced, and that  $m_0 \leq (n - \min_k n_k)/(g-1)$  in general.

*Solution.*  $\bar{Y}_k = \mu + a_k + \bar{\varepsilon}_k$  has variance  $\sigma_a^2 + \sigma^2/n_k$ , and  $\bar{Y} = \mu + \sum_k n_k a_k/n + \bar{\varepsilon}$  has variance  $\sigma_a^2 \sum_k n_k^2/n^2 + \sigma^2/n$ . Also  $\text{Cov}(\bar{Y}_k, \bar{Y}) = \sigma_a^2 n_k/n + \sigma^2/n$ . Hence  $E(\bar{Y}_k - \bar{Y})^2 = \sigma_a^2(1 - 2n_k/n + \sum_j n_j^2/n^2) + \sigma^2(1/n_k - 1/n)$ . Multiplying by  $n_k$  and summing, the  $\sigma^2$  terms give  $\sigma^2(g-1)$  and the  $\sigma_a^2$  terms give  $\sigma_a^2(n - 2\sum_k n_k^2/n + \sum_k n_k^2/n) = \sigma_a^2(n - \sum_k n_k^2/n)$ . For the balanced case,  $n = gm$  and  $\sum_k n_k^2 = gm^2$ , so  $m_0 = (gm - m)/(g-1) = m$ . In general  $\sum_k n_k^2 \geq n \min_k n_k$ , whence  $m_0 \leq (n - \min_k n_k)/(g-1)$ .

#### Exercise B2.

In the balanced normal model, use [Proposition 32.1.1\(d\)](#) to show that

$$\text{Var}(\hat{\sigma}_a^2) = \frac{2}{m^2} \left\{ \frac{(\sigma^2 + m\sigma_a^2)^2}{g-1} + \frac{\sigma^4}{n-g} \right\}.$$

With  $\sigma^2$  and  $\sigma_a^2$  fixed and  $n = gm$  held constant, is the variance smaller with many small groups or few large ones?

*Solution.* MSB and MSE are independent, with  $\text{Var}(\text{MSB}) = \lambda_1^2 \cdot 2/(g-1)$  and  $\text{Var}(\text{MSE}) = \sigma^4 \cdot 2/(n-g)$ , because a  $\chi^2(r)$  variable has variance  $2r$ . The formula follows from  $\hat{\sigma}_a^2 = (\text{MSB} - \text{MSE})/m$ . Neither extreme answers the second question. With  $g-1 \approx n/m$  and  $n-g \approx n$ ,

$$\text{Var}(\hat{\sigma}_a^2) \approx \frac{2}{n} \left( \frac{\sigma^4}{m} + 2\sigma^2\sigma_a^2 + m\sigma_a^4 + \frac{\sigma^4}{m^2} \right),$$

a decreasing part plus an increasing one, minimized near  $m = \sigma^2/\sigma_a^2$ , that is at  $m\gamma \approx 1$ : many small groups win only once each group is large enough for MSB to stand clear of MSE. In the exact formula with  $n = 100$  and  $\sigma^2 = 1$ , at  $\sigma_a^2 = 0.01$  the variance is 0.0206, 0.0056, 0.0029 and 0.0017 for  $m = 2, 5, 10, 25$  — four groups of twenty-five beat fifty pairs by a factor of twelve — while at  $\sigma_a^2 = 1$  it is 0.1018, 0.1526 and 0.2691 for  $m = 2, 5, 10$ , and the small groups win. The extreme  $m = 1$  always fails, because then the two components are not separately identified.

#### Exercise B3.

Using [Eq. 32.1.4](#), compute  $\Pr(\hat{\sigma}_a^2 < 0)$  when  $\sigma_a^2 = 0$ , for  $(g, m) = (5, 4)$  and  $(g, m) =$

(25, 4). Explain why the probability does not tend to 0 as  $g \rightarrow \infty$  with  $\sigma_a^2 = 0$  fixed, and why it does tend to 0 for any fixed  $\sigma_a^2 > 0$ .

### 32.1.5.3 C. Going deeper

#### Exercise C1.

Let  $T$  be any estimator of  $\sigma_a^2 \geq 0$  and  $T^+ = \max(T, 0)$ . Show that  $E(T^+ - \sigma_a^2)^2 \leq E(T - \sigma_a^2)^2$ , with strict inequality whenever  $\Pr(T < 0) > 0$ . Deduce that the truncated moment estimator dominates the unbiased one in mean squared error, and explain why this does not make the unbiased estimator useless.

*Solution.* On  $\{T \geq 0\}$  the two estimators agree. On  $\{T < 0\}$ ,  $|T^+ - \sigma_a^2| = \sigma_a^2 \leq \sigma_a^2 - T = |T - \sigma_a^2|$ , because  $T < 0$  and  $\sigma_a^2 \geq 0$ ; the inequality is strict unless  $T = 0$  almost surely on that event. Taking expectations gives the claim. The unbiased estimator remains useful because unbiasedness is preserved by averaging: combining  $K$  independent studies,  $K^{-1} \sum_k T_k$  is unbiased and consistent, whereas  $K^{-1} \sum_k T_k^+$  is biased upwards by an amount that does not vanish as  $K$  grows.

#### Exercise C2.

Consider a balanced two-way layout with  $a$  levels of  $A$ ,  $b$  levels of  $B$ ,  $m$  observations per cell, and *all* effects random:  $Y_{ijk} = \mu + \alpha_i + \beta_j + (\alpha\beta)_{ij} + \varepsilon_{ijk}$  with the four sets uncorrelated and variances  $\sigma_A^2, \sigma_B^2, \sigma_{AB}^2, \sigma^2$ . Using the Kronecker projections of [Section 16.3](#) and [Theorem 2.3.1](#), show that

$$E \text{MS}_A = \sigma^2 + m\sigma_{AB}^2 + bm\sigma_A^2, \quad E \text{MS}_{AB} = \sigma^2 + m\sigma_{AB}^2.$$

Which ratio tests  $\sigma_A^2 = 0$ ? Contrast this with the fixed-effects analysis, where  $\text{MS}_A/\text{MSE}$  is the right ratio.

## 32.2 The linear mixed model

The one-way random effects model of [Section 32.1](#) has one random factor and no regressors. Real studies have both: a trial has fixed treatment effects and random patient effects, a panel has fixed economic regressors and random firm effects, a field trial has fixed varieties and random blocks. The model that holds them all keeps the usual fixed part and adds a second design matrix whose coefficients are random.

### 32.2.1 Definition and the marginal model

#### Definition 32.2.1 — Linear mixed model

Let  $\mathbf{X}$  be a known  $n \times p$  matrix and  $\mathbf{Z}$  a known  $n \times q$  matrix. The **linear mixed model** is

$$\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon}, \quad (32.2.1)$$

where  $\boldsymbol{\beta} \in \mathbb{R}^p$  is a vector of unknown constants, the **fixed effects**;  $\mathbf{u}$  is a  $q \times 1$  random vector, the **random effects**, with  $E(\mathbf{u}) = \mathbf{0}$  and  $\text{Cov}(\mathbf{u}) = \mathbf{G}$ ;  $\boldsymbol{\varepsilon}$  is an  $n \times 1$  random vector with  $E(\boldsymbol{\varepsilon}) = \mathbf{0}$  and  $\text{Cov}(\boldsymbol{\varepsilon}) = \mathbf{R}$ ; and  $\text{Cov}(\mathbf{u}, \boldsymbol{\varepsilon}) = \mathbf{0}$ . The matrices  $\mathbf{G}$  and  $\mathbf{R}$  are nonnegative definite and are known functions  $\mathbf{G}(\boldsymbol{\theta})$ ,  $\mathbf{R}(\boldsymbol{\theta})$  of an unknown parameter  $\boldsymbol{\theta}$ , the **variance components**. The **normal** linear mixed model adds  $\mathbf{u} \sim N_q(\mathbf{0}, \mathbf{G})$  independent of  $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \mathbf{R})$ .

Taking expectations in Eq. 32.2.1 and using Theorem 2.2.1 gives the **marginal model**

$$E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}, \quad \text{Cov}(\mathbf{Y}) = \mathbf{V}(\boldsymbol{\theta}) = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R}, \quad (32.2.2)$$

and under normality  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \mathbf{V})$ . Everything Part VII knows about the general Gauss–Markov model applies to Eq. 32.2.2 at once. What distinguishes a mixed model from the general model of Chapter 31 is that  $\mathbf{V}$  is not an arbitrary unknown matrix but is built from a handful of parameters by a structure the experiment dictates. An  $n \times n$  covariance has  $n(n+1)/2$  free entries; a random intercept model has two.

*Remark* (Conditional and marginal). Two means live in Eq. 32.2.1. Conditionally on the random effects,  $E(\mathbf{Y} \mid \mathbf{u}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}$ : this is what the subjects in *this* sample did. Marginally,  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ : this is what the population does on average. In the linear model with normal errors the two lead to the same  $\boldsymbol{\beta}$ , and the distinction looks pedantic. It is not pedantic at all once the link is nonlinear. For a generalized linear mixed model (Chapter 40) the marginal mean of  $g^{-1}(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u})$  is not  $g^{-1}(\mathbf{X}\boldsymbol{\beta})$ , and the subject-specific and population-averaged coefficients genuinely differ (Proposition 40.1.1).

*Warning.* The pair  $(\mathbf{G}, \mathbf{R})$  is not identified by the marginal distribution; only  $\mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R}$  is. In the one-way layout with  $\mathbf{Z}$  the group indicators, the model with  $\mathbf{G} = \sigma_a^2 \mathbf{I}_g$ ,  $\mathbf{R} = \sigma^2 \mathbf{I}_n$  has exactly the same marginal distribution as the model with no random effects at all and  $\mathbf{R} = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$ , that is, with equicorrelated errors within groups. Every likelihood-based procedure in this chapter estimates  $\mathbf{V}$ , so it cannot tell the two apart. What the random-effects reading adds is a *target for prediction*: the vector  $\mathbf{u}$ , whose realized value Section 32.3 predicts. It also adds a constraint, since  $\sigma_a^2 \geq 0$  forces a nonnegative intraclass correlation, whereas the equicorrelated model allows  $\rho$  down to  $-1/(m-1)$ .

## 32.2.2 Inverting the covariance matrix

Formulas in this chapter are written with  $\mathbf{V}^{-1}$ , which is  $n \times n$ . When  $q$  is much smaller than  $n$ , or when the data fall into independent clusters,  $\mathbf{V}^{-1}$  never has to be formed.

### Proposition 32.2.1 — The inverse and determinant of $\mathbf{V}$

Let  $\mathbf{R}$  and  $\mathbf{G}$  be positive definite in Definition 32.2.1. Then  $\mathbf{V}$  is positive definite, and

$$\mathbf{V}^{-1} = \mathbf{R}^{-1} - \mathbf{R}^{-1}\mathbf{Z}(\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1}\mathbf{Z})^{-1}\mathbf{Z}^\top \mathbf{R}^{-1}, \quad (32.2.3)$$

$$\det \mathbf{V} = \det \mathbf{R} \det \mathbf{G} \det(\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1}\mathbf{Z}). \quad (32.2.4)$$

If in addition the observations fall into  $g$  clusters with  $\mathbf{R} = \bigoplus_k \mathbf{R}_k$ ,  $\mathbf{Z} = \bigoplus_k \mathbf{Z}_k$  and  $\mathbf{G} = \bigoplus_k \mathbf{G}_k$  block diagonal, the last saying that different clusters' random effects are independent, then  $\mathbf{V} = \bigoplus_k (\mathbf{Z}_k \mathbf{G}_k \mathbf{Z}_k^\top + \mathbf{R}_k)$  is block diagonal too, and everything decomposes cluster by cluster.

*Proof.*  $\mathbf{V} = \mathbf{R} + \mathbf{Z}\mathbf{G}\mathbf{Z}^\top$  is a sum of a positive definite and a nonnegative definite matrix, hence positive definite. Now use [Theorem 1.6.5](#) with the roles  $\mathbf{B} \mapsto \mathbf{R}$ ,  $\mathbf{U} \mapsto \mathbf{Z}$ ,  $\mathbf{C} \mapsto \mathbf{G}$  and (in the notation of that theorem) the second outer factor also  $\mathbf{Z}$ . The matrix  $\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1}\mathbf{Z}$  is positive definite, hence nonsingular, so the hypothesis holds and [Eq. 1.6.6](#) is [Eq. 32.2.3](#). For the determinant, [Theorem 1.6.3\(c\)](#) with  $\mathbf{B} = \mathbf{R}$ ,  $\mathbf{U} = \mathbf{Z}\mathbf{G}$  and the second factor  $\mathbf{Z}^\top$  gives

$$\begin{aligned} \det \mathbf{V} &= \det \mathbf{R} \det(\mathbf{I}_g + \mathbf{Z}^\top \mathbf{R}^{-1}\mathbf{Z}\mathbf{G}) \\ &= \det \mathbf{R} \det \mathbf{G} \det(\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1}\mathbf{Z}), \end{aligned}$$

the last step by  $\det(\mathbf{I}_g + \mathbf{A}\mathbf{G}) = \det(\mathbf{G}^{-1} + \mathbf{A}) \det \mathbf{G}$ . The block diagonal statement is immediate from the definitions. ■

A study of  $n = 10^5$  pupils in  $q = 500$  schools therefore needs a  $500 \times 500$  inverse, not a  $10^5 \times 10^5$  one, and if  $\mathbf{R}$  is diagonal the rest of [Eq. 32.2.3](#) costs nothing. [Section 32.3](#) improves on this again.

### 32.2.3 Four examples

#### Example. (Random intercepts)

Observations are grouped, group  $k$  contributing  $n_k$  of them. Take  $\mathbf{Z}$  to be the  $n \times g$  matrix of group indicators,  $\mathbf{G} = \sigma_a^2 \mathbf{I}_g$  and  $\mathbf{R} = \sigma^2 \mathbf{I}_n$ , so that  $\mathbf{V} = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$  as in [Proposition 32.1.1\(a\)](#). The fixed part  $\mathbf{X}\boldsymbol{\beta}$  may hold anything: an intercept alone, recovering [Definition 32.1.1](#); a treatment indicator constant within group, as in a cluster-randomized trial; or regressors that vary within a group, as in a longitudinal study. Each group's block of  $\mathbf{V}$  is  $\sigma^2 \mathbf{I}_{n_k} + \sigma_a^2 \mathbf{1}\mathbf{1}^\top$ , the **compound symmetric** or equicorrelated form.

#### Example. (Random intercepts and slopes)

Subject  $i$  is measured at times  $t_{i1}, \dots, t_{in_i}$ , and each subject has its own intercept and its own slope in time:

$$Y_{ij} = \mathbf{x}_{(ij)}^\top \boldsymbol{\beta} + u_{0i} + u_{1i} t_{ij} + \varepsilon_{ij}, \quad \begin{pmatrix} u_{0i} \\ u_{1i} \end{pmatrix} \sim N_2(\mathbf{0}, \mathbf{D}),$$

independently across subjects, with  $\boldsymbol{\varepsilon}_i \sim N_{n_i}(\mathbf{0}, \sigma^2 \mathbf{I})$ . Here  $\mathbf{Z}_i = [\mathbf{1}, \mathbf{t}_i]$ ,  $\mathbf{G} = \mathbf{I}_N \otimes \mathbf{D}$  for  $N$  subjects, and subject  $i$ 's covariance block is  $\mathbf{V}_i = \mathbf{Z}_i \mathbf{D} \mathbf{Z}_i^\top + \sigma^2 \mathbf{I}$ , with entries

$$\text{Cov}(Y_{ij}, Y_{ij'}) = d_{11} + d_{12}(t_{ij} + t_{ij'}) + d_{22}t_{ij}t_{ij'} + \sigma^2 \mathbf{1}\{j = j'\}.$$

Two features deserve notice. The variance of a response is quadratic in time, so a random slope model necessarily says that subjects spread apart, or, if  $d_{12} < 0$ , first converge and then spread. And the correlation between two occasions depends on both times, not only on their separation: it is not stationary. Whether this is the right shape is a modelling question that [Chapter 33](#) takes up.

### Example. (Nested and crossed random factors)

With pupils inside classrooms inside schools, write  $\mathbf{Z} = [\mathbf{Z}_{\text{sch}}, \mathbf{Z}_{\text{cls}}]$  and  $\mathbf{G} = \begin{pmatrix} \sigma_s^2 \mathbf{I} & \mathbf{0} \\ \mathbf{0} & \sigma_c^2 \mathbf{I} \end{pmatrix}$ . Because every classroom belongs to exactly one school,  $\mathbf{V}$  is block diagonal by school, and the analysis splits school by school; this is the covariance behind [Exercise \(C1\)](#). With subjects responding to items, each subject seeing every item, the two factors are **crossed**:  $\mathbf{Z}_{\text{sub}}$  and  $\mathbf{Z}_{\text{item}}$  generate overlapping patterns and  $\mathbf{V}$  has no block diagonal form at all. Nothing in the theory changes, but the computation does, and crossed random effects are the reason modern software stores  $\mathbf{Z}$  sparsely.

### Example. (Grunfeld's panel with a random firm effect)

[Section 6.6](#) fitted gross investment on market value and capital stock for 11 US firms over 20 years, once with a separate intercept per firm (the within estimator) and once with a single intercept. The mixed model puts the firm effect in between:  $\mathbf{X}$  holds the intercept, value and capital,  $\mathbf{Z}$  holds the firm indicators,  $\mathbf{G} = \sigma_f^2 \mathbf{I}_{11}$  and  $\mathbf{R} = \sigma^2 \mathbf{I}$ . Fitted by restricted maximum likelihood ([Section 32.4](#)), it gives  $\hat{\sigma}^2 = 2527.6$  and  $\hat{\sigma}_f^2 = 6740.9$ , a firm standard deviation of 82.1 and an intraclass correlation of 0.727: most of the variation in investment is between firms. The coefficients are 0.1094 for value and 0.3082 for capital, against 0.1101 and 0.3100 for the within estimator and 0.1145 and 0.2275 with a single intercept. The mixed estimate has landed essentially on the within estimate, and [Section 32.3](#) explains why: with 20 years per firm, the weight the model gives to between-firm comparisons is  $1 - 0.9816$ , less than two per cent.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

df = sm.datasets.grunfeld.load_pandas().data.sort_values(["firm", "year"])
firms = df["firm"].unique()
y = df["invest"].to_numpy()
X = np.column_stack([np.ones(len(y)), df["value"], df["capital"]])
blocks = [np.where(df["firm"].to_numpy() == f)[0] for f in firms]
n, p, g = len(y), X.shape[1], len(firms)
print(f"{n} observations, {g} firms, {n // g} years each")
```

## 32.2.4 When ordinary least squares is already best

The marginal model [Eq. 32.2.2](#) is a general Gauss–Markov model, so [Corollary 7.2.3](#) says the best linear unbiased estimator of an estimable  $\lambda^\top \beta$  is the generalized least squares estimate built from  $\mathbf{V}^{-1}$ , and [Theorem 31.2.2](#) says that ordinary least squares achieves this for every  $\mathbf{y}$  exactly when  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ . Mixed models supply the standard examples on both sides of that line.

### Example. (Cluster-constant regressors)

Take [Example \(Random intercepts\)](#) with a balanced layout,  $n_k = m$ , and suppose every column of  $\mathbf{X}$  is constant within groups, so that  $\mathcal{C}(\mathbf{X}) \subseteq \mathcal{C}(\mathbf{Z})$ . By [Eq. 32.1.2](#),  $\mathbf{V} = \lambda_1 \mathbf{M} + \sigma^2(\mathbf{I} - \mathbf{M})$  with  $\mathbf{M}$  the projection onto  $\mathcal{C}(\mathbf{Z})$ , so  $\mathbf{V}\mathbf{X} = \lambda_1 \mathbf{X}$  and  $\mathcal{C}(\mathbf{V}\mathbf{X}) = \mathcal{C}(\mathbf{X})$ . Ordinary least squares is therefore the best linear unbiased estimator, whatever the variance components are, and the analysis of a balanced cluster-randomized trial can be carried out on the cluster means without loss. The covariance of the estimate is *not*  $\sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ , however: it is  $\lambda_1(\mathbf{X}^\top \mathbf{X})^{-1}$ , larger by the factor  $1 + m\gamma$ . Getting the point estimate right and the standard error wrong is the classic error in clustered data.

### Example. (Regressors that vary within a cluster)

Keep the same  $\mathbf{V}$ , but let  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$  with  $\mathbf{x}$  varying within groups. Decomposing  $\mathbf{x}$  into its between and within parts,

$$\mathbf{V}\mathbf{x} = \lambda_1 \mathbf{M}\mathbf{x} + \sigma^2(\mathbf{I} - \mathbf{M})\mathbf{x},$$

which lies in  $\mathcal{C}(\mathbf{X})$  only if one of the two parts vanishes or  $\lambda_1 = \sigma^2$ . Ordinary least squares is then not efficient, and the reason is visible: generalized least squares downweights the between-group part of  $\mathbf{x}$  by  $\sigma^2/\lambda_1 = 1/(1 + m\gamma)$  relative to the within-group part, because between-group comparisons carry the extra variance  $\sigma_a^2$ . The mixed estimate is a compromise between the within estimator of [Section 6.6](#), which uses only  $(\mathbf{I} - \mathbf{M})\mathbf{x}$ , and the between estimator, which uses only  $\mathbf{M}\mathbf{x}$ ; as  $m\gamma \rightarrow \infty$  the weight on the between part goes to zero. [Example \(Grunfeld’s panel with a random firm effect\)](#) is that limit in practice.

*Idea.* A random effect is a statement about *where the information is*. Declaring the grouping factor random says that comparisons between groups carry an extra variance  $\sigma_a^2$  and should be downweighted accordingly, but not discarded. Fixed effects for the groups discard them entirely; ignoring the grouping keeps them at full weight. The mixed model interpolates, and the interpolation weight is estimated from the data.

The remaining difficulty is that  $\mathbf{V}$  depends on the unknown  $\theta$ . The estimator actually used is the feasible generalized least squares estimator of [Definition 31.4.1](#),

$$\hat{\beta}(\hat{\theta}) = (\mathbf{X}^\top \mathbf{V}(\hat{\theta})^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}(\hat{\theta})^{-1} \mathbf{Y}, \quad (32.2.5)$$

with  $\hat{\theta}$  from [Section 32.4](#). [Theorem 31.4.1](#) gives the large-sample justification; [Section 32.5](#) asks what it costs in small samples, and the answer is: more than people usually allow for.



## 32.2.5 Exercises

### 32.2.5.1 A. Check your understanding

#### Exercise A1.

A study measures 3 subjects on 2 occasions each, with a random subject effect of variance  $\sigma_s^2$  and independent errors of variance  $\sigma^2$ . Write  $\mathbf{Z}$ ,  $\mathbf{G}$ ,  $\mathbf{R}$  and  $\mathbf{V}$  explicitly as  $6 \times 6$  or  $6 \times 3$  matrices, and check Eq. 32.2.3 numerically for  $\sigma_s^2 = 2$ ,  $\sigma^2 = 1$ .

#### Exercise A2.

A colleague argues that in Example (Random intercepts) the two variance components cannot be separated, since raising  $\sigma_a^2$  and lowering  $\sigma^2$  both change the diagonal of  $\mathbf{V}$  in the same direction. Is the argument right? What happens if every group has  $n_k = 1$ ?

### 32.2.5.2 B. Practice

#### Exercise B1.

Specialize Eq. 32.2.3 to Example (Random intercepts) with group sizes  $n_k$ , and show that the  $k$ th block of  $\mathbf{V}^{-1}$  is

$$\frac{1}{\sigma^2} \left( \mathbf{I}_{n_k} - \frac{\sigma_a^2}{\sigma^2 + n_k \sigma_a^2} \mathbf{1}\mathbf{1}^\top \right).$$

Deduce that the generalized least squares criterion is  $\sigma^{-2} \sum_k \{ \sum_j r_{kj}^2 - w_k n_k \bar{r}_k^2 \}$  with  $w_k = n_k \sigma_a^2 / (\sigma^2 + n_k \sigma_a^2)$ , and interpret  $w_k$ .

*Solution.* Each block of  $\mathbf{V}$  is  $\sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{1}\mathbf{1}^\top$ ; apply Eq. 1.6.7 with  $\mathbf{B} = \sigma^2 \mathbf{I}$ ,  $\mathbf{u} = \mathbf{v} = \sigma_a \mathbf{1}$ : the correction is  $\sigma_a^2 \sigma^{-4} \mathbf{1}\mathbf{1}^\top / (1 + n_k \sigma_a^2 / \sigma^2)$ , which rearranges to the stated form. The criterion  $\mathbf{r}^\top \mathbf{V}^{-1} \mathbf{r}$  then equals  $\sigma^{-2} \sum_k \{ \sum_j r_{kj}^2 - n_k \bar{r}_k^2 w_k \}$  after writing  $\mathbf{1}^\top \mathbf{r}_k = n_k \bar{r}_k$ . The weight  $w_k \in (0, 1)$  increases with  $n_k$ : the model removes a fraction  $w_k$  of the group's mean residual from the criterion, because that part of the residual is explained by the group's random effect rather than counted against the fit.

#### Exercise B2.

Let  $\mathbf{V} = \lambda_1 \mathbf{M} + \sigma^2 (\mathbf{I} - \mathbf{M})$  as in Eq. 32.1.2. Show that  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  holds if and only if  $\mathcal{C}(\mathbf{X})$  decomposes as  $(\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{Z})) \oplus (\mathcal{C}(\mathbf{X}) \cap \mathcal{C}(\mathbf{Z})^\perp)$ , that is, iff  $\mathbf{M}$  maps  $\mathcal{C}(\mathbf{X})$  into itself. Give an  $\mathbf{X}$  with both between and within columns for which ordinary least squares is best.

*Solution.*  $\mathbf{V}\mathbf{X} = \sigma^2 \mathbf{X} + (\lambda_1 - \sigma^2) \mathbf{M}\mathbf{X}$ , so (assuming  $\lambda_1 \neq \sigma^2$ )  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  iff  $\mathcal{C}(\mathbf{M}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , that is,  $\mathbf{M}$  maps  $\mathcal{C}(\mathbf{X})$  into itself. A subspace invariant under an orthogonal projection splits into its parts inside  $\mathcal{C}(\mathbf{Z})$  and inside  $\mathcal{C}(\mathbf{Z})^\perp$ . Example:  $\mathbf{X} = [\mathbf{1}, \mathbf{x}]$  with  $\mathbf{1}$  constant within groups and  $\mathbf{x}$  having zero mean *within every group*, so



$\mathbf{M}\mathbf{x} = \mathbf{0}$ ; then  $\mathbf{MX} = [\mathbf{1}, \mathbf{0}]$  lies in  $\mathcal{C}(\mathbf{X})$ . A balanced split-plot design has exactly this structure, which is why [Chapter 33](#) can analyse it stratum by stratum.

### 32.2.5.3 C. Going deeper

#### Exercise C1.

Two crossed random factors with  $a$  and  $b$  levels and one observation per cell give  $\mathbf{Z} = [\mathbf{Z}_A, \mathbf{Z}_B]$  with  $n = ab$ ,  $q = a + b$ . Show that  $\text{rank}(\mathbf{Z}) = a + b - 1$ , so  $\mathbf{Z}^\top \mathbf{Z}$  is singular, and explain why this causes no difficulty for [Eq. 32.2.2](#) but would if one tried to treat  $\mathbf{u}$  as a parameter vector to be estimated. Compute  $\mathbf{V}$  and show it is *not* block diagonal under any ordering of the observations when  $a, b \geq 2$ .

## 32.3 Prediction: BLUP and Henderson's equations

The twelve laboratories of [Example \(A laboratory proficiency study\)](#) each have a realized effect  $a_k$ . It is not a parameter but the value taken by a random variable, so asking for it is a prediction problem. The answer is worth the trouble of setting the problem up properly: it is not the laboratory's own average, and the difference is shrinkage of exactly the kind [Chapter 27](#) studied, arriving here without anybody having decided to be biased.

### 32.3.1 The criterion

#### Definition 32.3.1 — Mixed functions and linear predictors

In [Definition 32.2.1](#), a **mixed function** is a random variable  $\tau = \boldsymbol{\lambda}^\top \boldsymbol{\beta} + \mathbf{c}^\top \mathbf{u}$  with known  $\boldsymbol{\lambda} \in \mathbb{R}^p$  and  $\mathbf{c} \in \mathbb{R}^q$ . A **linear predictor** of  $\tau$  is a statistic  $t = \mathbf{a}^\top \mathbf{Y} + b$  with known  $\mathbf{a}$  and  $b$ ; it is **unbiased** if  $E(t) = E(\tau)$  for every  $\boldsymbol{\beta} \in \mathbb{R}^p$ . The **best linear unbiased predictor** (BLUP) of  $\tau$  is the unbiased linear predictor minimizing the prediction mean squared error  $E(t - \tau)^2$ .

Since  $E(t) = \mathbf{a}^\top \mathbf{X}\boldsymbol{\beta} + b$  and  $E(\tau) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ , unbiasedness for every  $\boldsymbol{\beta}$  forces  $b = 0$  and  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ . So an unbiased linear predictor exists exactly when  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ , that is, when  $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$  is estimable in the sense of [Definition 8.2.1](#); the random part  $\mathbf{c}^\top \mathbf{u}$  places no restriction at all, because its mean is zero whatever  $\mathbf{c}$  is. Note also that the criterion averages over  $\mathbf{u}$  as well as over  $\boldsymbol{\epsilon}$ . This is the decision that makes the problem solvable, and the warning in the next subsection says what it costs.

The problem has already been solved. [Theorem 14.2.8](#) of [Section 14.2](#) predicted a new response  $y_0 = \mathbf{x}_0^\top \boldsymbol{\beta} + e_0$  whose error was correlated with the training errors, and  $\mathbf{c}^\top \mathbf{u}$  is exactly such an  $e_0$ . What follows is that theorem in mixed-model notation; its proof is the substitution, plus the uniqueness and the decomposition (c) that Chapter 14 did not need.

**Theorem 32.3.1 — Best linear unbiased prediction**

In [Definition 32.2.1](#) let  $\mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R}$  be positive definite and let  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ . Put

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}, \quad \hat{\mathbf{u}} = \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}). \quad (32.3.1)$$

- (a) The BLUP of  $\tau = \boldsymbol{\lambda}^\top \boldsymbol{\beta} + \mathbf{c}^\top \mathbf{u}$  is  $\hat{\tau} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{c}^\top \hat{\mathbf{u}}$ . It does not depend on which generalized inverse is used, and it is the unique minimizer.
- (b) Its prediction error variance is

$$\mathbb{E}(\hat{\tau} - \tau)^2 = \mathbf{c}^\top (\mathbf{G} - \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{Z}\mathbf{G}) \mathbf{c} + \mathbf{d}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{d}, \quad (32.3.2)$$

where  $\mathbf{d} = \boldsymbol{\lambda} - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Z}\mathbf{G}\mathbf{c}$ .

- (c) If  $\boldsymbol{\beta}$  were known, the best linear predictor of  $\tau$  would be  $\boldsymbol{\lambda}^\top \boldsymbol{\beta} + \mathbf{c}^\top \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$ , with prediction error variance the first term of [Eq. 32.3.2](#). The second term is the price of estimating  $\boldsymbol{\beta}$ .

*Proof.* Apply [Theorem 14.2.8](#) to  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + (\mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon})$ , whose error vector has covariance  $\mathbf{V}$ , with  $\sigma^2 = 1$ ,  $y_0 = \tau$ ,  $\mathbf{x}_0 = \boldsymbol{\lambda}$  and  $\mathbf{e}_0 = \mathbf{c}^\top \mathbf{u}$ . Since  $\text{Cov}(\mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon}, \mathbf{u}) = \mathbf{Z}\mathbf{G}$ , the two covariance inputs of that theorem are

$$\mathbf{v}_0 = \text{Cov}(\mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon}, \mathbf{c}^\top \mathbf{u}) = \mathbf{Z}\mathbf{G}\mathbf{c} =: \mathbf{s}, \quad v_{00} = \text{Var}(\mathbf{c}^\top \mathbf{u}) = \mathbf{c}^\top \mathbf{G}\mathbf{c},$$

and its hypothesis  $\mathbf{x}_0 \in \mathcal{C}(\mathbf{X}^\top)$  is our  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ . So [Eq. 14.2.3](#) reads  $\hat{\tau} = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{c}^\top \mathbf{G}\mathbf{Z}^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \mathbf{c}^\top \hat{\mathbf{u}}$ , and its mean squared error is [Eq. 32.3.2](#), because the vector called  $\mathbf{c}$  there is  $\mathbf{x}_0 - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{v}_0 = \boldsymbol{\lambda} - \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Z}\mathbf{G}\mathbf{c} = \mathbf{d}$  here. That is (a) and (b) apart from the two uniqueness claims.

For those, recall that an unbiased linear predictor is  $t = \mathbf{a}^\top \mathbf{Y}$  with  $\mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}$ , and that  $t - \tau$  then has mean zero, so

$$\mathbb{E}(t - \tau)^2 = \text{Var}(\mathbf{a}^\top \mathbf{Y} - \mathbf{c}^\top \mathbf{u}) = \mathbf{a}^\top \mathbf{V} \mathbf{a} - 2\mathbf{a}^\top \mathbf{s} + \mathbf{c}^\top \mathbf{G}\mathbf{c}.$$

The last term does not involve  $\mathbf{a}$  and  $\mathbf{V}$  is positive definite, so this is a strictly convex function of  $\mathbf{a}$  on the affine set  $\{\mathbf{a} : \mathbf{X}^\top \mathbf{a} = \boldsymbol{\lambda}\}$ , non-empty because  $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$ , and a strictly convex function has at most one minimizer on a convex set. So  $\mathbf{a}$ , and with it  $\hat{\tau}$ , is unique, by an argument that nowhere asks  $\mathbf{X}^\top$  to have full row rank. The generalized inverse then cancels, as one can also see directly:  $\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$  is the projection of  $\mathbf{Y}$  onto  $\mathcal{C}(\mathbf{X})$  in the  $\mathbf{V}^{-1}$  inner product ([Definition 6.9.1](#)), and both  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{r}^\top \mathbf{X}\hat{\boldsymbol{\beta}}$ , for  $\mathbf{r}$  with  $\boldsymbol{\lambda} = \mathbf{X}^\top \mathbf{r}$ , and  $\hat{\mathbf{u}}$  involve  $\hat{\boldsymbol{\beta}}$  only through  $\mathbf{X}\hat{\boldsymbol{\beta}}$ .

For (c), with  $\boldsymbol{\beta}$  known the problem is unconstrained prediction of  $\mathbf{c}^\top \mathbf{u}$  from  $\mathbf{Y} - \mathbf{X}\boldsymbol{\beta}$ , whose solution is the best linear predictor of [Theorem 2.5.2](#):  $\text{Cov}(\mathbf{c}^\top \mathbf{u}, \mathbf{Y}) \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$ , with error variance  $\mathbf{c}^\top \mathbf{G}\mathbf{c} - \mathbf{s}^\top \mathbf{V}^{-1} \mathbf{s}$ , which is the first term of [Eq. 32.3.2](#) because  $\mathbf{s} = \mathbf{Z}\mathbf{G}\mathbf{c}$ .

■

Two readings of [Eq. 32.3.1](#) are worth having. First,  $\hat{\mathbf{u}}$  is the best linear predictor of  $\mathbf{u}$  with  $\boldsymbol{\beta}$  replaced by its generalized least squares estimate; the whole content of “BLUP” is that this substitution is optimal, not merely convenient. Second, if the model is normal then  $\mathbb{E}(\mathbf{u} | \mathbf{Y}) =$

$\mathbf{GZ}^\top \mathbf{V}^{-1}(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$  by [Theorem 3.5.2](#), so the BLUP is the conditional mean with  $\boldsymbol{\beta}$  estimated. That is the door into [Section 32.6](#).

### 32.3.2 Shrinkage

Specialize to the balanced one-way model of [Definition 32.1.1](#), with  $\mathbf{X} = \mathbf{1}$ ,  $\mathbf{Z}$  the group indicators,  $\mathbf{G} = \sigma_a^2 \mathbf{I}_g$  and  $\mathbf{R} = \sigma^2 \mathbf{I}$ . By [Eq. 32.1.2](#),  $\mathbf{V}^{-1} = \lambda_1^{-1} \mathbf{M} + \sigma^{-2}(\mathbf{I} - \mathbf{M})$  with  $\lambda_1 = \sigma^2 + m\sigma_a^2$ , and since  $\mathbf{MZ} = \mathbf{Z}$  we have  $\mathbf{Z}^\top \mathbf{V}^{-1} = \lambda_1^{-1} \mathbf{Z}^\top$ . Also  $\mathbf{1} \in \mathcal{C}(\mathbf{Z})$ , so the generalized least squares estimate of  $\mu$  is the ordinary mean  $\bar{y}$  ([Example \(Cluster-constant regressors\)](#)). Hence

$$\hat{a}_k = \frac{m\sigma_a^2}{\sigma^2 + m\sigma_a^2} (\bar{y}_k - \bar{y}) = B (\bar{y}_k - \bar{y}), \quad B = \frac{m\gamma}{1 + m\gamma}, \quad (32.3.3)$$

with  $\gamma = \sigma_a^2/\sigma^2$ . The predicted group mean is  $\bar{y} + B(\bar{y}_k - \bar{y})$ : the group's own average pulled towards the overall average by  $B \in (0, 1)$ . The pull is strong when groups are small or when groups genuinely differ little, and it vanishes as either  $m$  or  $\gamma$  grows. Formula [Eq. 32.3.3](#) is the empirical Bayes shrinkage of [Proposition 27.5.4](#) with the prior variance supplied by the model, and it is a ridge estimate of the group effects with penalty  $\sigma^2/\sigma_a^2$  ([Theorem 27.2.2](#)); the next subsection makes that identification exact.

*Warning.* Unbiasedness in [Definition 32.3.1](#) averages over  $\mathbf{u}$ . Conditionally on the realized effects it fails, and deliberately so: from [Eq. 32.3.3](#),  $E(\hat{a}_k | a_1, \dots, a_g) = B(a_k - \bar{a})$ , which is nearer zero than  $a_k$ . Predicted effects are *shrunk*, so their spread understates the spread of the true effects, and ranking groups by  $\hat{a}_k$  is not the same as ranking them by  $a_k$ . A league table of hospitals or schools built from BLUPs will move small units towards the middle, which is right for predicting each unit's next observation and wrong if the question is "which unit is genuinely worst".

The shrinkage is paid for in mean squared error. Continuing the balanced one-way case, [Eq. 32.3.2](#) with  $\boldsymbol{\lambda} = \mathbf{0}$  and  $\mathbf{c} = \mathbf{e}_k$  gives, after substituting  $\mathbf{G} - \mathbf{GZ}^\top \mathbf{V}^{-1} \mathbf{ZG} = \sigma_a^2 \sigma^2 \lambda_1^{-1} \mathbf{I}$  and  $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} = n/\lambda_1$ ,

$$E(\hat{a}_k - a_k)^2 = \frac{\sigma_a^2}{\lambda_1} \left( \sigma^2 + \frac{m\sigma_a^2}{g} \right), \quad (32.3.4)$$

whereas the obvious predictor  $\bar{y}_k - \bar{y}$  has mean squared error  $\sigma^2(1 - 1/g)/m + \sigma_a^2/g$  ([Exercise \(B1\)](#)).

#### Example. (Shrinking twelve laboratories)

In [Example \(A laboratory proficiency study\)](#),  $\hat{\sigma}^2 = 0.0620$  and  $\hat{\sigma}_a^2 = 0.0386$  give a shrinkage factor  $B = 0.714$ . The laboratory deviations  $\bar{y}_k - \bar{y}$  run from  $-0.450$  to  $0.250$ ; the predicted effects run from  $-0.321$  to  $0.178$ . Each has been pulled about 29 per cent of the way towards zero.

The gain is real. By [Eq. 32.3.4](#) the prediction error variance is 0.01336, a standard error of 0.1156, against 0.01743 for the raw deviation, a ratio of 1.30 in mean squared error. Panel (b) of [Figure 32.3.1](#) shows the comparison over the whole range of intraclass

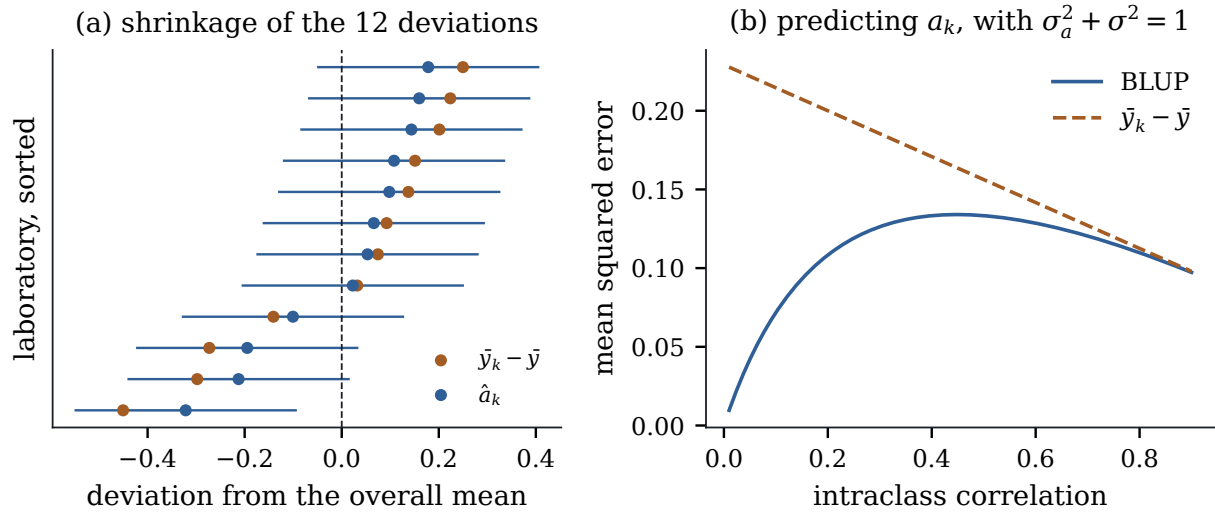


Figure 32.3.1: (a) The twelve laboratory deviations  $\bar{y}_k - \bar{y}$  and their predicted effects  $\hat{a}_k$ , sorted, with 95% prediction intervals from Eq. 32.3.4. (b) Mean squared error of the two predictors of  $a_k$ , as a function of the intraclass correlation, with  $\sigma_a^2 + \sigma^2 = 1$ ,  $g = 12$  and  $m = 4$ .

correlations: shrinkage helps most when the correlation is small, which is exactly when the raw deviations are mostly noise.

### 32.3.3 Henderson's mixed model equations

Formula Eq. 32.3.1 asks for  $\mathbf{V}^{-1}$ , an  $n \times n$  inverse. Henderson (1950) found a system of size  $p + q$  that delivers the generalized least squares estimate and the BLUP together, without ever forming  $\mathbf{V}$ .

#### Theorem 32.3.2 — Henderson's mixed model equations

Let  $\mathbf{G}$  and  $\mathbf{R}$  be positive definite and  $\text{rank}(\mathbf{X}) = p$ . Define

$$\mathbf{C} = \begin{pmatrix} \mathbf{X}^\top \mathbf{R}^{-1} \mathbf{X} & \mathbf{X}^\top \mathbf{R}^{-1} \mathbf{Z} \\ \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{X} & \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z} + \mathbf{G}^{-1} \end{pmatrix}, \quad \mathbf{h} = \begin{pmatrix} \mathbf{X}^\top \mathbf{R}^{-1} \mathbf{y} \\ \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{y} \end{pmatrix}. \quad (32.3.5)$$

- (a)  $\mathbf{C}$  is positive definite, and the unique solution of  $\mathbf{C}(\mathbf{b}^\top, \mathbf{v}^\top)^\top = \mathbf{h}$  is  $\mathbf{b} = \hat{\boldsymbol{\beta}}$ , the generalized least squares estimate, and  $\mathbf{v} = \hat{\mathbf{u}}$ , the BLUP of Eq. 32.3.1.
- (b) The same pair is the unique minimizer of the penalized least squares criterion

$$Q(\boldsymbol{\beta}, \mathbf{u}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u})^\top \mathbf{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}) + \mathbf{u}^\top \mathbf{G}^{-1} \mathbf{u}. \quad (32.3.6)$$

- (c) With  $\mathbf{W} = \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}$ , the blocks of  $\mathbf{C}^{-1}$  are

$$\begin{aligned} (\mathbf{C}^{-1})_{11} &= \mathbf{W}^{-1}, & (\mathbf{C}^{-1})_{12} &= -\mathbf{W}^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Z} \mathbf{G}, \\ (\mathbf{C}^{-1})_{22} &= \mathbf{G} - \mathbf{G} \mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{Z} \mathbf{G} + \mathbf{G} \mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{X} \mathbf{W}^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Z} \mathbf{G}. \end{aligned}$$

Its leading block is  $\text{Cov}(\hat{\boldsymbol{\beta}})$  and its trailing block is  $\text{Cov}(\hat{\mathbf{u}} - \mathbf{u})$ , the matrix whose quadratic forms are [Eq. 32.3.2](#).

*Proof.* We use repeatedly the *push-through identity*

$$(\mathbf{G}^{-1} + \mathbf{Z}^{\top} \mathbf{R}^{-1} \mathbf{Z})^{-1} \mathbf{Z}^{\top} \mathbf{R}^{-1} = \mathbf{G} \mathbf{Z}^{\top} \mathbf{V}^{-1}, \quad (32.3.7)$$

proved by multiplying on the right by  $\mathbf{V} = \mathbf{R} + \mathbf{Z} \mathbf{G} \mathbf{Z}^{\top}$ :

$$\begin{aligned} & (\mathbf{G}^{-1} + \mathbf{Z}^{\top} \mathbf{R}^{-1} \mathbf{Z})^{-1} (\mathbf{Z}^{\top} + \mathbf{Z}^{\top} \mathbf{R}^{-1} \mathbf{Z} \mathbf{G} \mathbf{Z}^{\top}) \\ &= (\mathbf{G}^{-1} + \mathbf{Z}^{\top} \mathbf{R}^{-1} \mathbf{Z})^{-1} (\mathbf{G}^{-1} + \mathbf{Z}^{\top} \mathbf{R}^{-1} \mathbf{Z}) \mathbf{G} \mathbf{Z}^{\top} \\ &= \mathbf{G} \mathbf{Z}^{\top}. \end{aligned}$$

(a) For  $(\mathbf{b}^{\top}, \mathbf{v}^{\top}) \neq \mathbf{0}$ ,

$$(\mathbf{b}^{\top}, \mathbf{v}^{\top}) \mathbf{C} (\mathbf{b}^{\top}, \mathbf{v}^{\top})^{\top} = \|\mathbf{R}^{-1/2}(\mathbf{X}\mathbf{b} + \mathbf{Z}\mathbf{v})\|^2 + \mathbf{v}^{\top} \mathbf{G}^{-1} \mathbf{v},$$

which vanishes only if  $\mathbf{v} = \mathbf{0}$  and  $\mathbf{X}\mathbf{b} = \mathbf{0}$ , hence only if  $\mathbf{b} = \mathbf{0}$  by full column rank. So  $\mathbf{C}$  is positive definite and the system has a unique solution. The second block row gives

$$\begin{aligned} \mathbf{v} &= (\mathbf{Z}^{\top} \mathbf{R}^{-1} \mathbf{Z} + \mathbf{G}^{-1})^{-1} \mathbf{Z}^{\top} \mathbf{R}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}) \\ &= \mathbf{G} \mathbf{Z}^{\top} \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}) \end{aligned}$$

by [Eq. 32.3.7](#). Substituting into the first block row, and using the residual identity

$$\begin{aligned} \mathbf{y} - \mathbf{X}\mathbf{b} - \mathbf{Z}\mathbf{v} &= (\mathbf{I} - \mathbf{Z} \mathbf{G} \mathbf{Z}^{\top} \mathbf{V}^{-1}) (\mathbf{y} - \mathbf{X}\mathbf{b}) \\ &= (\mathbf{V} - \mathbf{Z} \mathbf{G} \mathbf{Z}^{\top}) \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}) \\ &= \mathbf{R} \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}), \end{aligned}$$

the first block row  $\mathbf{X}^{\top} \mathbf{R}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b} - \mathbf{Z}\mathbf{v}) = \mathbf{0}$  becomes  $\mathbf{X}^{\top} \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{0}$ , the generalized normal equations of [Definition 6.9.1](#). Hence  $\mathbf{b} = \hat{\boldsymbol{\beta}}$  and then  $\mathbf{v} = \hat{\mathbf{u}}$ .

(b)  $Q$  is a quadratic in  $(\boldsymbol{\beta}, \mathbf{u})$  with matrix  $\mathbf{C}$  and linear term  $-2\mathbf{h}$ , and the display in (a) shows it is strictly convex. By [Proposition 1.11.2](#) its unique minimizer solves  $\mathbf{C}(\mathbf{b}^{\top}, \mathbf{v}^{\top})^{\top} = \mathbf{h}$ , which is [Eq. 32.3.5](#), whose solution is  $(\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}})$  by part (a).

(c) Write  $\mathbf{C}_{11} = \mathbf{X}^{\top} \mathbf{R}^{-1} \mathbf{X}$ ,  $\mathbf{C}_{12} = \mathbf{X}^{\top} \mathbf{R}^{-1} \mathbf{Z}$  and  $\mathbf{C}_{22} = \mathbf{Z}^{\top} \mathbf{R}^{-1} \mathbf{Z} + \mathbf{G}^{-1}$ , so that by [Eq. 32.2.3](#) the Schur complement of  $\mathbf{C}_{22}$  is

$$\begin{aligned} & \mathbf{C}_{11} - \mathbf{C}_{12} \mathbf{C}_{22}^{-1} \mathbf{C}_{12}^{\top} \\ &= \mathbf{X}^{\top} (\mathbf{R}^{-1} - \mathbf{R}^{-1} \mathbf{Z} (\mathbf{G}^{-1} + \mathbf{Z}^{\top} \mathbf{R}^{-1} \mathbf{Z})^{-1} \mathbf{Z}^{\top} \mathbf{R}^{-1}) \mathbf{X} \\ &= \mathbf{X}^{\top} \mathbf{V}^{-1} \mathbf{X} = \mathbf{W}. \end{aligned}$$

[Theorem 1.6.4](#) now gives  $(\mathbf{C}^{-1})_{11} = \mathbf{W}^{-1}$  and  $(\mathbf{C}^{-1})_{12} = -\mathbf{W}^{-1} \mathbf{C}_{12} \mathbf{C}_{22}^{-1} = -\mathbf{W}^{-1} \mathbf{X}^{\top} \mathbf{V}^{-1} \mathbf{Z} \mathbf{G}$ , the last step being the transpose of [Eq. 32.3.7](#). For the trailing block,

$$\begin{aligned} (\mathbf{C}^{-1})_{22} &= \mathbf{C}_{22}^{-1} + \mathbf{C}_{22}^{-1} \mathbf{C}_{12}^{\top} \mathbf{W}^{-1} \mathbf{C}_{12} \mathbf{C}_{22}^{-1} \\ &= \mathbf{C}_{22}^{-1} + \mathbf{G} \mathbf{Z}^{\top} \mathbf{V}^{-1} \mathbf{X} \mathbf{W}^{-1} \mathbf{X}^{\top} \mathbf{V}^{-1} \mathbf{Z} \mathbf{G}, \end{aligned}$$

and  $\mathbf{C}_{22}^{-1} = \mathbf{G} - \mathbf{GZ}^\top \mathbf{V}^{-1} \mathbf{ZG}$ , because

$$\begin{aligned} & (\mathbf{G} - \mathbf{GZ}^\top \mathbf{V}^{-1} \mathbf{ZG})(\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z}) \\ &= \mathbf{I} + \mathbf{GZ}^\top \mathbf{R}^{-1} \mathbf{Z} - \mathbf{GZ}^\top \mathbf{V}^{-1} (\mathbf{ZGZ}^\top + \mathbf{R}) \mathbf{R}^{-1} \mathbf{Z} = \mathbf{I}, \end{aligned}$$

using [Eq. 32.3.7](#) once more in the form  $\mathbf{GZ}^\top \mathbf{V}^{-1} \mathbf{V} = \mathbf{GZ}^\top$ . Finally,  $\text{Cov}(\hat{\boldsymbol{\beta}}) = \mathbf{W}^{-1}$  is [Corollary 7.2.3](#), and the trailing block is [Eq. 32.3.2](#) with  $\boldsymbol{\lambda} = \mathbf{0}$  read off entry by entry. ■

*Idea.* Henderson's system says that a mixed model is a penalized least squares problem ([Eq. 32.3.6](#)): fit the random effects as though they were free parameters, but charge  $\mathbf{u}^\top \mathbf{G}^{-1} \mathbf{u}$  for using them. With  $\mathbf{R} = \sigma^2 \mathbf{I}$  and  $\mathbf{G} = \sigma_u^2 \mathbf{I}$  the criterion is  $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}\|^2 + \lambda \|\mathbf{u}\|^2$  with  $\lambda = \sigma^2/\sigma_u^2$ : ridge regression on the random-effect columns only ([Theorem 27.2.2](#)). The difference from [Chapter 27](#) is that the penalty is not chosen by cross-validation; it is estimated, because the model says what  $\sigma^2/\sigma_u^2$  means. Chapter 43 exploits the same equivalence in the other direction, writing a smoothing spline as a mixed model so that its smoothing parameter can be estimated by REML.

```
import numpy as np

g, m = 12, 4
n = g * m
mu, sigma_a, sigma = 2.50, 0.20, 0.25          # the values used to simulate

rng = np.random.default_rng(20320048)
lab_effect = sigma_a * rng.standard_normal(g)
y = mu + np.repeat(lab_effect, m) + sigma * rng.standard_normal(n)

X = np.ones((n, 1))                            # one fixed effect, the grand mean
Z = np.repeat(np.eye(g), m, axis=0)             # laboratory indicators

Y = y.reshape(g, m)                            # moment estimates, Section 32.1
lab_mean = Y.mean(axis=1)
ms_between = m * np.sum((lab_mean - Y.mean()) ** 2) / (g - 1)
ms_within = np.sum((Y - lab_mean[:, None]) ** 2) / (n - g)
s2, s2a = ms_within, (ms_between - ms_within) / m

G = s2a * np.eye(g)
R = s2 * np.eye(n)
V = Z @ G @ Z.T + R
```

```
Vinv = np.linalg.inv(V)
beta_gls = np.linalg.solve(X.T @ Vinv @ X, X.T @ Vinv @ y)
u_blup = G @ Z.T @ Vinv @ (y - X @ beta_gls)

# Henderson's mixed model equations: one (p + q) x (p + q) system, no n x n inverse
Rinv = np.linalg.inv(R)
C = np.block([[X.T @ Rinv @ X, X.T @ Rinv @ Z],
               [Z.T @ Rinv @ X, Z.T @ Rinv @ Z + np.linalg.inv(G)]])
rhs = np.concatenate([X.T @ Rinv @ y, Z.T @ Rinv @ y])
solution = np.linalg.solve(C, rhs)
```

```
print("GLS estimate  ", beta_gls, solution[:1])
print("first two BLUPs", u_blup[:2], solution[1:3])
```

```
shrinkage = m * s2a / (s2 + m * s2a)           # the shrinkage factor B
print(f"B = {shrinkage:.4f}")
print("u_hat  ", np.round(u_blup[:4], 4))
print("B(ybar_k - ybar)", np.round(shrinkage * (lab_mean - y.mean())[:4], 4))
```

Two practical points close the section. First,  $\mathbf{C}$  is sparse whenever  $\mathbf{Z}$  is, which is always: each row of  $\mathbf{Z}$  has one non-zero entry per random factor, so Eq. 32.3.5 can be solved by sparse Cholesky factorization, and a model with hundreds of thousands of random effects is routine. Second,  $\mathbf{G}$  and  $\mathbf{R}$  are unknown in practice and estimates are substituted; the result is called the **empirical** BLUP, and Eq. 32.3.2 is then too small, since it ignores the variability of the estimated covariance. Kackar and Harville (1984) identified the extra term, which returns in Section 32.5 as the Kenward–Roger correction and in Section 32.6 as the gap between posterior and plug-in standard errors.

## 32.3.4 Exercises

### 32.3.4.1 A. Check your understanding

#### Exercise A1.

In Eq. 32.3.3, what happens to  $B$  as (a)  $m \rightarrow \infty$ , (b)  $\sigma_a^2 \rightarrow 0$ , (c)  $\sigma^2 \rightarrow 0$ ? Describe in words the predictor in each limit, and say which of the three limits makes the mixed model agree with a fixed-effects analysis.

### 32.3.4.2 B. Practice

#### Exercise B1.

Show that the predictor  $\bar{Y}_k - \bar{Y}$  of  $a_k$  in the balanced one-way model has  $E(\bar{Y}_k - \bar{Y} - a_k)^2 = \sigma^2(1 - 1/g)/m + \sigma_a^2/g$ , and verify that Eq. 32.3.4 is smaller for every  $\gamma > 0$ . Where does the difference come from when  $\gamma$  is very small?

*Solution.*  $\bar{Y}_k - \bar{Y} - a_k = \bar{\varepsilon}_k - \bar{\varepsilon} - \bar{a}$  with  $\bar{a} = g^{-1} \sum_j a_j$ . The three terms have variances  $\sigma^2/m$ ,  $\sigma^2/(gm)$  and  $\sigma_a^2/g$ , and  $\text{Cov}(\bar{\varepsilon}_k, \bar{\varepsilon}) = \sigma^2/(gm)$ , giving  $\sigma^2/m - \sigma^2/(gm) + \sigma_a^2/g$  as stated. In terms of  $\gamma = \sigma_a^2/\sigma^2$ , Eq. 32.3.4 equals  $\sigma^2\gamma(1 + m\gamma/g)/(1 + m\gamma)$  while the raw predictor's error is  $\sigma^2\{(1 - 1/g)/m + \gamma/g\}$ . Putting the two over the denominator  $1 + m\gamma$  and cancelling, their difference is exactly

$$\frac{\sigma^2}{m} \left(1 - \frac{1}{g}\right) \frac{1}{1 + m\gamma} = \frac{\sigma^2}{m} \left(1 - \frac{1}{g}\right) (1 - B),$$

which is strictly positive for every  $\gamma$ . As  $\gamma \rightarrow 0$  the BLUP's error tends to  $\sigma_a^2$  while the raw predictor's tends to  $\sigma^2/m$ , which is much larger: the raw predictor pays the full measurement error even when there is nothing to predict, whereas the BLUP predicts 0 and pays only  $\sigma_a^2$ .

**Exercise B2.**

Take  $\tau = \mu + a_k$ , the mean of group  $k$  in the balanced one-way model, so that  $\boldsymbol{\lambda} = 1$  and  $\mathbf{c} = \mathbf{e}_k$ . Use [Eq. 32.3.1](#) and [Eq. 32.3.2](#) to show that the BLUP is  $\bar{y} + B(\bar{y}_k - \bar{y})$  with prediction error variance  $\sigma^2 B/m + (1 - B)^2 \lambda_1/n$ . Show that this equals  $\lambda_1/n$  at  $B = 0$  and  $\sigma^2/m$  at  $B = 1$ ; that it is strictly below  $\sigma^2/m$ , the prediction error of the group's own mean; and that it is below  $\lambda_1/n$  as well exactly when  $m\sigma_a^2 > \sigma^2(g - 2)$ , so that it is no interpolation between the two — with  $g = 3$ ,  $m = 4$  and  $\sigma^2 = \sigma_a^2 = 1$  it beats both.

*Solution.* The predictor is immediate from [Eq. 32.3.1](#). For the variance,  $\mathbf{d} = 1 - \mathbf{1}^\top \mathbf{V}^{-1} \mathbf{Z} \mathbf{G} \mathbf{e}_k = 1 - m\sigma_a^2/\lambda_1 = 1 - B$ , and  $\mathbf{W}^- = \lambda_1/n$ , so the second term of [Eq. 32.3.2](#) is  $(1 - B)^2 \lambda_1/n$ ; the first term is  $\sigma_a^2 \sigma^2/\lambda_1 = \sigma^2 B/m$  since  $B = m\sigma_a^2/\lambda_1$ . Setting  $B = 0$  and  $B = 1$  gives the two stated endpoints. For the comparisons, use  $1 - B = \sigma^2/\lambda_1$  and  $n = gm$  to collapse the whole expression to

$$\frac{\sigma^2}{\lambda_1} \left( \sigma_a^2 + \frac{\sigma^2}{gm} \right),$$

and note that  $\sigma^2/m = (\sigma^2/\lambda_1)(\sigma_a^2 + \sigma^2/m)$ , so the first comparison holds whenever  $g > 1$ . The second asks  $\sigma^2 \sigma_a^2 gm + \sigma^4 < \lambda_1^2 = (\sigma^2 + m\sigma_a^2)^2$ , which after cancelling  $\sigma^4$  and dividing by  $m\sigma_a^2$  is  $\sigma^2 g < 2\sigma^2 + m\sigma_a^2$ . In the numerical case the expression is  $13/60$ , below both  $5/12$  and  $1/4$ : the quadratic in  $B$  is not an interpolation between two strategies but can beat both, and that gain is what the shrinkage buys.

**Exercise B3.**

Take  $\mathbf{X}$  empty (no fixed effects),  $\mathbf{R} = \sigma^2 \mathbf{I}$  and  $\mathbf{G} = \sigma_u^2 \mathbf{I}$  in [Theorem 32.3.2](#). Show that  $\hat{\mathbf{u}}$  is exactly the ridge estimate  $(\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I})^{-1} \mathbf{Z}^\top \mathbf{y}$  with  $\lambda = \sigma^2/\sigma_u^2$ , and that [Eq. 32.3.2](#) becomes  $\sigma_u^2 \mathbf{c}^\top (\mathbf{I} + \lambda^{-1} \mathbf{Z}^\top \mathbf{Z})^{-1} \mathbf{c}$ . Compare with the mean squared error of ridge in [Theorem 27.2.2](#): which is the same quantity and which is not?

*Solution.* With no  $\mathbf{X}$ , [Eq. 32.3.5](#) reduces to  $(\mathbf{Z}^\top \mathbf{Z}/\sigma^2 + \mathbf{I}/\sigma_u^2) \mathbf{v} = \mathbf{Z}^\top \mathbf{y}/\sigma^2$ , that is  $\mathbf{v} = (\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I})^{-1} \mathbf{Z}^\top \mathbf{y}$ . By [Theorem 32.3.2\(c\)](#) the prediction error covariance is  $\mathbf{C}^{-1} = \sigma^2 (\mathbf{Z}^\top \mathbf{Z} + \lambda \mathbf{I})^{-1} = \sigma_u^2 (\mathbf{I} + \lambda^{-1} \mathbf{Z}^\top \mathbf{Z})^{-1}$ . The quantity computed here averages over  $\mathbf{u}$ ; the mean squared error in [Theorem 27.2.2](#) is computed for a *fixed*  $\boldsymbol{\beta}$  and is therefore a function of that fixed value. Averaging the ridge mean squared error over  $\boldsymbol{\beta} \sim \mathbf{N}(\mathbf{0}, \sigma_u^2 \mathbf{I})$  reproduces the BLUP formula, which is the empirical Bayes reading of [Proposition 27.5.4](#).

**32.3.4.3 C. Going deeper****Exercise C1.**

Show that the BLUP is invariant to reparameterization of the random effects: if  $\mathbf{Z}^* = \mathbf{Z} \mathbf{T}^{-1}$  and  $\mathbf{G}^* = \mathbf{T} \mathbf{G} \mathbf{T}^\top$  for a nonsingular  $\mathbf{T}$ , with  $\mathbf{u}^* = \mathbf{T} \mathbf{u}$ , then  $\mathbf{V}$  is unchanged and  $\hat{\mathbf{u}}^* = \mathbf{T} \hat{\mathbf{u}}$ . What does this say about predicting a contrast  $\mathbf{c}^\top \mathbf{u}$  directly versus predicting  $\mathbf{u}$  and then forming the contrast?



**Exercise C2.**

Assume the normal mixed model and suppose  $\boldsymbol{\beta}$  is known. Show that among *all* predictors  $t(\mathbf{Y})$ , linear or not, the conditional mean  $E(\mathbf{c}^\top \mathbf{u} \mid \mathbf{Y})$  minimizes  $E(t - \mathbf{c}^\top \mathbf{u})^2$ , and that it coincides with the linear predictor of [Theorem 32.3.1\(c\)](#). Explain why the same argument does not show that  $\hat{\tau}$  is optimal among all predictors when  $\boldsymbol{\beta}$  is unknown.

## 32.4 Maximum likelihood and REML

Everything so far has treated the variance components as known. They never are. The moment estimates of [Section 32.1](#) work in balanced layouts and lose their footing in unbalanced ones, where no canonical set of quadratic forms presents itself. Under normality the likelihood has no such difficulty, but it has one defect, which this section diagnoses and repairs: maximum likelihood underestimates variance components, because it does not pay for the fixed effects it has fitted.

### 32.4.1 The likelihood and its profile

Under the normal linear mixed model,  $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \mathbf{V})$  with  $\mathbf{V} = \mathbf{V}(\boldsymbol{\theta})$ , so by [Theorem 3.2.2](#)

$$\ell(\boldsymbol{\beta}, \boldsymbol{\theta}) = -\frac{1}{2} \{n \log 2\pi + \log \det \mathbf{V} + (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})\}. \quad (32.4.1)$$

For fixed  $\boldsymbol{\theta}$  only the last term involves  $\boldsymbol{\beta}$ , and it is the generalized least squares criterion of [Definition 6.9.1](#), minimized by  $\hat{\boldsymbol{\beta}}(\boldsymbol{\theta})$ . Writing

$$\mathbf{P} = \mathbf{V}^{-1} - \mathbf{V}^{-1} \mathbf{X} (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1}, \quad (32.4.2)$$

the profile log-likelihood of [Definition 7.3.1](#) is

$$\ell_P(\boldsymbol{\theta}) = -\frac{1}{2} \{n \log 2\pi + \log \det \mathbf{V} + \mathbf{y}^\top \mathbf{P} \mathbf{y}\}. \quad (32.4.3)$$

#### Lemma 32.4.1 — The residual operator $\mathbf{P}$

Let  $\mathbf{V}$  be positive definite and  $r = \text{rank}(\mathbf{X})$ . The matrix  $\mathbf{P}$  of [Eq. 32.4.2](#) does not depend on the choice of generalized inverse, and

- (a)  $\mathbf{P}\mathbf{X} = \mathbf{0}$ ,  $\mathbf{PVP} = \mathbf{P}$ ,  $\mathbf{P}$  is symmetric and  $\text{rank}(\mathbf{P}) = n - r$ ;
- (b)  $\mathbf{P}\mathbf{y} = \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$  and  $\mathbf{y}^\top \mathbf{P} \mathbf{y} = \min_{\mathbf{b}} (\mathbf{y} - \mathbf{X}\mathbf{b})^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b})$ ;
- (c) if  $\mathbf{V}$  depends smoothly on  $\boldsymbol{\theta}$  and  $\mathbf{V}_j = \partial \mathbf{V} / \partial \theta_j$ , then  $\partial \mathbf{P} / \partial \theta_j = -\mathbf{P} \mathbf{V}_j \mathbf{P}$ .

*Proof.* Write  $\mathbf{X}_* = \mathbf{V}^{-1/2} \mathbf{X}$  ([Theorem 1.7.6](#)). Then  $\mathbf{P} = \mathbf{V}^{-1/2} (\mathbf{I} - \mathbf{M}_*) \mathbf{V}^{-1/2}$ , where  $\mathbf{M}_* = \mathbf{X}_* (\mathbf{X}_*^\top \mathbf{X}_*)^{-1} \mathbf{X}_*^\top$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{X}_*)$ , which by [Theorem 6.3.7](#) is the same for every generalized inverse and has rank  $r$ . Symmetry,  $\mathbf{P}\mathbf{X} = \mathbf{V}^{-1/2} (\mathbf{I} -$

$\mathbf{M}_*)\mathbf{X}_* = \mathbf{0}$ ,  $\mathbf{PVP} = \mathbf{V}^{-1/2}(\mathbf{I} - \mathbf{M}_*)^2\mathbf{V}^{-1/2} = \mathbf{P}$  and the rank statement follow at once, which is (a). For (b),  $\mathbf{Py} = \mathbf{V}^{-1}\mathbf{y} - \mathbf{V}^{-1}\mathbf{X}\hat{\boldsymbol{\beta}}$  by the definition of  $\hat{\boldsymbol{\beta}}$ , and  $\mathbf{y}^\top \mathbf{Py} = \|(\mathbf{I} - \mathbf{M}_*)\mathbf{V}^{-1/2}\mathbf{y}\|^2$ , the squared distance from  $\mathbf{V}^{-1/2}\mathbf{y}$  to  $\mathcal{C}(\mathbf{X}_*)$ , which is the stated minimum. Part (c) is proved after [Theorem 32.4.2](#), which supplies a shorter route. ■

## 32.4.2 Restricted maximum likelihood

The trouble with [Eq. 32.4.3](#) is visible in the simplest case. With  $\mathbf{V} = \sigma^2\mathbf{I}$  it reduces to the profile of [Theorem 7.3.1](#), whose maximizer is  $\hat{\sigma}^2 = \text{SSE}/n$ , biased downwards by the factor  $(n-p)/n$ . The likelihood has spent  $p$  dimensions fitting  $\boldsymbol{\beta}$  and then measured the spread of the data as if it had not. The repair is to compute the likelihood of a part of the data that never saw  $\boldsymbol{\beta}$ .

### Definition 32.4.1 — Error contrasts and restricted maximum likelihood

Let  $r = \text{rank}(\mathbf{X})$ . An **error contrast** is a random variable  $\mathbf{k}^\top \mathbf{Y}$  with  $\mathbf{X}^\top \mathbf{k} = \mathbf{0}$ ; its distribution does not involve  $\boldsymbol{\beta}$ . A **maximal set** of error contrasts is  $\mathbf{K}^\top \mathbf{Y}$  for an  $n \times (n-r)$  matrix  $\mathbf{K}$  of full column rank with  $\mathcal{C}(\mathbf{K}) = \mathcal{C}(\mathbf{X})^\perp$ . The **restricted** (or residual) **log-likelihood** is the log-likelihood of  $\mathbf{K}^\top \mathbf{Y}$ ,

$$\ell_R(\boldsymbol{\theta}) = -\frac{1}{2} \left\{ (n-r) \log 2\pi + \log \det(\mathbf{K}^\top \mathbf{V} \mathbf{K}) + \mathbf{y}^\top \mathbf{K}(\mathbf{K}^\top \mathbf{V} \mathbf{K})^{-1} \mathbf{K}^\top \mathbf{y} \right\}, \quad (32.4.4)$$

and the **REML estimate** of  $\boldsymbol{\theta}$  maximizes it over the parameter space.

### Theorem 32.4.2 — The restricted likelihood

In [Definition 32.4.1](#), with  $\mathbf{V}$  positive definite:

- (a)  $\mathbf{K}^\top \mathbf{Y} \sim N_{n-r}(\mathbf{0}, \mathbf{K}^\top \mathbf{V} \mathbf{K})$  with  $\mathbf{K}^\top \mathbf{V} \mathbf{K}$  positive definite, so [Eq. 32.4.4](#) is its log-likelihood.
- (b)  $\mathbf{K}(\mathbf{K}^\top \mathbf{V} \mathbf{K})^{-1} \mathbf{K}^\top = \mathbf{P}$  for every admissible  $\mathbf{K}$ . In particular the quadratic form in [Eq. 32.4.4](#) equals  $\mathbf{y}^\top \mathbf{Py}$  and does not depend on  $\mathbf{K}$ .
- (c) Two admissible  $\mathbf{K}_1, \mathbf{K}_2$  give restricted log-likelihoods differing by a constant free of  $\boldsymbol{\theta}$ . Hence the REML estimate does not depend on which maximal set of error contrasts is used.
- (d) If  $\text{rank}(\mathbf{X}) = p$ , then for any admissible  $\mathbf{K}$  with  $\mathbf{K}^\top \mathbf{K} = \mathbf{I}$ ,

$$\det(\mathbf{K}^\top \mathbf{V} \mathbf{K}) = \frac{\det \mathbf{V} \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})}{\det(\mathbf{X}^\top \mathbf{X})}, \quad (32.4.5)$$

so that

$$\begin{aligned}\ell_R(\boldsymbol{\theta}) &= \ell_P(\boldsymbol{\theta}) + \frac{p}{2} \log 2\pi - \frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) \\ &\quad + \frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{X}).\end{aligned}\tag{32.4.6}$$

*Proof.*

- (a) By [Theorem 3.3.1](#),  $\mathbf{K}^\top \mathbf{Y}$  is normal with mean  $\mathbf{K}^\top \mathbf{X}\boldsymbol{\beta} = \mathbf{0}$  and covariance  $\mathbf{K}^\top \mathbf{V}\mathbf{K}$ , which is positive definite because  $\mathbf{K}$  has full column rank and  $\mathbf{V}$  is positive definite. Its density is [Theorem 3.2.2](#).
- (b) Fix  $\mathbf{y}$  and consider

$$\min\{\mathbf{z}^\top \mathbf{V}^{-1} \mathbf{z} : \mathbf{K}^\top \mathbf{z} = \mathbf{K}^\top \mathbf{y}\}.$$

The feasible set is  $\mathbf{y} + \mathcal{C}(\mathbf{X})$ : indeed  $\mathbf{K}^\top(\mathbf{z} - \mathbf{y}) = \mathbf{0}$  says  $\mathbf{z} - \mathbf{y} \in \mathcal{C}(\mathbf{K})^\perp = \mathcal{C}(\mathbf{X})$ . So the minimum equals  $\min_{\mathbf{b}} (\mathbf{y} - \mathbf{X}\mathbf{b})^\top \mathbf{V}^{-1} (\mathbf{y} - \mathbf{X}\mathbf{b}) = \mathbf{y}^\top \mathbf{P}\mathbf{y}$  by [Lemma 32.4.1\(b\)](#). On the other hand the constrained problem can be solved directly. The criterion is strictly convex, so by [Proposition 1.11.3](#) the minimizer satisfies  $\mathbf{V}^{-1} \mathbf{z} = \mathbf{K}\boldsymbol{\theta}$ , that is  $\mathbf{z} = \mathbf{V}\mathbf{K}\boldsymbol{\theta}$ ; the constraint gives  $\mathbf{K}^\top \mathbf{V}\mathbf{K}\boldsymbol{\theta} = \mathbf{K}^\top \mathbf{y}$ , and the minimum value is  $\boldsymbol{\theta}^\top \mathbf{K}^\top \mathbf{V}\mathbf{K}\boldsymbol{\theta} = \mathbf{y}^\top \mathbf{K}(\mathbf{K}^\top \mathbf{V}\mathbf{K})^{-1} \mathbf{K}^\top \mathbf{y}$ . The two expressions agree for every  $\mathbf{y}$ , and both are quadratic forms with symmetric matrices, so the matrices coincide.

- (c) Since  $\mathcal{C}(\mathbf{K}_1) = \mathcal{C}(\mathbf{K}_2)$  and both have  $n - r$  columns of full rank,  $\mathbf{K}_2 = \mathbf{K}_1 \mathbf{A}$  for a nonsingular  $(n-r) \times (n-r)$  matrix  $\mathbf{A}$ . Then  $\det(\mathbf{K}_2^\top \mathbf{V}\mathbf{K}_2) = (\det \mathbf{A})^2 \det(\mathbf{K}_1^\top \mathbf{V}\mathbf{K}_1)$ , while the quadratic forms are equal by (b). So the two log-likelihoods differ by  $-\log|\det \mathbf{A}|$ , which does not involve  $\boldsymbol{\theta}$ .
- (d) Let  $\mathbf{T} = [\mathbf{X}, \mathbf{K}]$ , which is nonsingular. Because  $\mathbf{X}^\top \mathbf{K} = \mathbf{0}$  and  $\mathbf{K}^\top \mathbf{K} = \mathbf{I}$ ,

$$\det(\mathbf{T}^\top \mathbf{T}) = \det \begin{pmatrix} \mathbf{X}^\top \mathbf{X} & \mathbf{0} \\ \mathbf{0} & \mathbf{I} \end{pmatrix} = \det(\mathbf{X}^\top \mathbf{X}),$$

so  $(\det \mathbf{T})^2 = \det(\mathbf{X}^\top \mathbf{X})$  and  $\det(\mathbf{T}^\top \mathbf{V}\mathbf{T}) = \det(\mathbf{X}^\top \mathbf{X}) \det \mathbf{V}$ . Next,  $\mathbf{T}^{-1}$  has rows  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$  stacked on  $\mathbf{K}^\top$ , as one checks by multiplying. Hence the leading  $p \times p$  block of  $(\mathbf{T}^\top \mathbf{V}\mathbf{T})^{-1} = \mathbf{T}^{-1} \mathbf{V}^{-1} \mathbf{T}^{-\top}$  is  $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1}$ . By [Theorem 1.6.4](#) that block is the inverse of the Schur complement of the trailing block  $\mathbf{K}^\top \mathbf{V}\mathbf{K}$ , so by [Theorem 1.6.3\(b\)](#),

$$\det(\mathbf{T}^\top \mathbf{V}\mathbf{T}) = \det(\mathbf{K}^\top \mathbf{V}\mathbf{K}) \cdot \frac{\det(\mathbf{X}^\top \mathbf{X})^2}{\det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})}.$$

Equating the two expressions for  $\det(\mathbf{T}^\top \mathbf{V}\mathbf{T})$  gives [Eq. 32.4.5](#), and substituting it together with (b) into [Eq. 32.4.4](#) gives [Eq. 32.4.6](#). ■

The proof of [Lemma 32.4.1\(c\)](#) is now immediate: differentiating  $\mathbf{P} = \mathbf{K}(\mathbf{K}^\top \mathbf{V}\mathbf{K})^{-1} \mathbf{K}^\top$  with  $\mathbf{K}$  fixed gives  $\partial \mathbf{P} / \partial \theta_j = -\mathbf{K}(\mathbf{K}^\top \mathbf{V}\mathbf{K})^{-1} \mathbf{K}^\top \mathbf{V}_j \mathbf{K}(\mathbf{K}^\top \mathbf{V}\mathbf{K})^{-1} \mathbf{K}^\top = -\mathbf{P} \mathbf{V}_j \mathbf{P}$ .

*Idea.* REML fits  $\boldsymbol{\theta}$  to the residual directions only. It is the general form of the familiar divisor  $n-p$ : with  $\mathbf{V} = \sigma^2 \mathbf{I}$ ,  $\text{rank}(\mathbf{X}) = p$  and an orthonormal  $\mathbf{K}$  (that is  $\mathbf{K}^\top \mathbf{K} = \mathbf{I}$ , as in [Theorem 32.4.2\(d\)](#)), [Eq. 32.4.4](#) is the likelihood of  $\mathbf{K}^\top \mathbf{Y} \sim N_{n-p}(\mathbf{0}, \sigma^2 \mathbf{I})$ , maximized at  $\hat{\sigma}^2 = \|\mathbf{K}^\top \mathbf{y}\|^2 / (n-p) = \text{SSE} / (n-p) = s^2$ ; by [Theorem 32.4.2\(c\)](#) any other admissible  $\mathbf{K}$  gives the same answer. Ordinary maximum likelihood gives  $\text{SSE} / n$ . So  $s^2$ , the estimator every earlier chapter used, was the REML estimator all along. Its bias in a genuine

mixed model is computed exactly in [Proposition 32.4.3](#).

*Warning.* Since  $\mathbf{K}$  is determined by  $\mathcal{C}(\mathbf{X})$ , two models with *different fixed effects* have restricted likelihoods of different data, and comparing them is meaningless; [Section 32.5](#) makes the rule explicit. Worse, [Eq. 32.4.6](#) shows that the constant  $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{X})$  is part of the definition, and software often omits it; then the reported restricted log-likelihood changes if a column of  $\mathbf{X}$  is measured in thousands rather than units, by  $\log|\det \mathbf{S}|$  for the rescaling matrix  $\mathbf{S}$ . Comparisons of covariance structures with the *same*  $\mathbf{X}$  are unaffected, since the constant cancels.

### 32.4.3 Why maximum likelihood is biased, exactly

In the balanced one-way model both likelihoods can be maximized in closed form, and the bias can be computed rather than asserted.

#### Proposition 32.4.3 — Balanced one-way: ML and REML

In the balanced normal one-way random effects model, write  $\lambda_1 = \sigma^2 + m\sigma_a^2$  and  $\lambda_2 = \sigma^2$ , and treat  $(\lambda_1, \lambda_2)$  with  $\lambda_1 \geq \lambda_2 > 0$  as the parameters. Then, up to constants free of the parameters,

$$-2\ell_P = g \log \lambda_1 + (n - g) \log \lambda_2 + \frac{\text{SSB}}{\lambda_1} + \frac{\text{SSE}}{\lambda_2}, \quad (32.4.7)$$

$$-2\ell_R = (g - 1) \log \lambda_1 + (n - g) \log \lambda_2 + \frac{\text{SSB}}{\lambda_1} + \frac{\text{SSE}}{\lambda_2}. \quad (32.4.8)$$

The two criteria differ in one coefficient, but leave the interior of the parameter space at different moments.

- (a) When  $\text{MSB} \geq \text{MSE}$ , REML gives  $\hat{\lambda}_1 = \text{MSB}$ ,  $\hat{\lambda}_2 = \text{MSE}$ , so  $\hat{\sigma}^2 = \text{MSE}$  and  $\hat{\sigma}_a^2 = (\text{MSB} - \text{MSE})/m$ : the moment estimates of [Proposition 32.1.1\(c\)](#), which are unbiased.
- (b) When  $\text{MSB} < g \text{MSE}/(g - 1)$ , maximum likelihood gives  $\hat{\lambda}_1 = \text{SSB}/g$ ,  $\hat{\lambda}_2 = \text{MSE}$ , so  $\hat{\sigma}_{\text{ML}}^2 = \text{MSE}$  and

$$\hat{\sigma}_{a,\text{ML}}^2 = \frac{1}{m} \left( \frac{g-1}{g} \text{MSB} - \text{MSE} \right).$$

The right-hand side is the *unconstrained* maximizer whatever the data, it is non-negative exactly under the stated condition, and

$$\mathbb{E} \left\{ \frac{1}{m} \left( \frac{g-1}{g} \text{MSB} - \text{MSE} \right) \right\} - \sigma_a^2 = -\frac{\lambda_1}{gm}. \quad (32.4.9)$$

- (c) Otherwise the maximizer lies on the boundary  $\lambda_1 = \lambda_2$  and the method returns  $\hat{\sigma}_a^2 = 0$  with  $\hat{\sigma}^2$  the pooled mean square: for REML when  $\text{MSB} < \text{MSE}$ , for maximum likelihood on the wider range  $\text{MSB} < g \text{MSE}/(g - 1)$ . Between the two thresholds the REML estimate of  $\sigma_a^2$  is positive and the ML estimate is zero.

*Proof.* By Eq. 32.1.2,  $\mathbf{V} = \lambda_1 \mathbf{M} + \lambda_2 (\mathbf{I} - \mathbf{M})$ , so  $\det \mathbf{V} = \lambda_1^g \lambda_2^{n-g}$  and  $\mathbf{V}^{-1} = \lambda_1^{-1} \mathbf{M} + \lambda_2^{-1} (\mathbf{I} - \mathbf{M})$ . Since  $\mathbf{1} \in \mathcal{C}(\mathbf{Z})$ ,  $\mathbf{V}^{-1} \mathbf{1} = \lambda_1^{-1} \mathbf{1}$ , so the generalized least squares estimate of  $\mu$  is  $\bar{y}$  and the residual is  $\mathbf{y} - \bar{y} \mathbf{1}$ . Splitting it by Theorem 6.5.1 into its parts in  $\mathcal{C}(\mathbf{Z}) \cap \mathbf{1}^\perp$  and  $\mathcal{C}(\mathbf{Z})^\perp$ ,

$$\mathbf{y}^\top \mathbf{P} \mathbf{y} = (\mathbf{y} - \bar{y} \mathbf{1})^\top \mathbf{V}^{-1} (\mathbf{y} - \bar{y} \mathbf{1}) = \frac{\text{SSB}}{\lambda_1} + \frac{\text{SSE}}{\lambda_2}.$$

Substituting into Eq. 32.4.3 gives Eq. 32.4.7. For the restricted version,  $\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X} = n/\lambda_1$  and  $\mathbf{X}^\top \mathbf{X} = n$ , so the correction in Eq. 32.4.6 is  $-\frac{1}{2} \log(n/\lambda_1) + \frac{1}{2} \log n = \frac{1}{2} \log \lambda_1$ , which converts the coefficient  $g$  of  $\log \lambda_1$  into  $g - 1$  after multiplying by  $-2$ . That is Eq. 32.4.8.

Both right-hand sides have the form  $c_1 \log \lambda_1 + S_1/\lambda_1$  plus  $c_2 \log \lambda_2 + S_2/\lambda_2$ , a sum of two functions of separate arguments, each strictly convex in  $1/\lambda$  and minimized at  $\lambda = S/c$ . So without the constraint  $\lambda_1 \geq \lambda_2$ , REML gives  $\hat{\lambda}_1 = \text{SSB}/(g - 1) = \text{MSB}$  and  $\hat{\lambda}_2 = \text{SSE}/(n - g) = \text{MSE}$ , while maximum likelihood gives  $\hat{\lambda}_1 = \text{SSB}/g$  and the same  $\hat{\lambda}_2$ . These are feasible exactly when  $\hat{\lambda}_1 \geq \hat{\lambda}_2$ , that is when  $\text{MSB} \geq \text{MSE}$  for REML and when  $\text{SSB}/g = (g - 1)\text{MSB}/g \geq \text{MSE}$ , i.e.  $\text{MSB} \geq g \text{MSE}/(g - 1)$ , for maximum likelihood. Translating back through  $\sigma^2 = \lambda_2$  and  $\sigma_a^2 = (\lambda_1 - \lambda_2)/m$  gives (a) and (b), except for Eq. 32.4.9. For that, use  $\text{E MSB} = \lambda_1$  and  $\text{E MSE} = \sigma^2$ :

$$\text{E} \left\{ \frac{1}{m} \left( \frac{g-1}{g} \text{MSB} - \text{MSE} \right) \right\} = \frac{1}{m} \left( \frac{g-1}{g} \lambda_1 - \sigma^2 \right) = \sigma_a^2 - \frac{\lambda_1}{gm}.$$

For (c), suppose the unconstrained minimizer of one of the two objectives violates  $\lambda_1 \geq \lambda_2$ , which by the previous paragraph happens exactly on the stated range. Each objective is a sum of two terms, one in each argument, each strictly unimodal on  $(0, \infty)$  and tending to  $+\infty$  at both ends, so a constrained minimum exists; were it interior to  $\{\lambda_1 > \lambda_2\}$  the gradient would vanish there, making it the infeasible unconstrained minimizer. Hence it lies on the boundary  $\lambda_1 = \lambda_2 = \lambda$ , where the objective becomes  $c \log \lambda + (\text{SSB} + \text{SSE})/\lambda$  with  $c = n$  for ML and  $c = n - 1$  for REML, minimized at the corresponding pooled mean square. ■

The bias  $-\lambda_1/(gm)$  is  $\lambda_1$  divided by the total sample size: the price of the single degree of freedom used up by the grand mean, charged entirely to the between-group component. In the proficiency study,  $\lambda_1 = \sigma^2 + m\sigma_a^2 = 0.25^2 + 4(0.20)^2 = 0.2225$  and  $gm = 48$ , so the bias is  $-0.0046$ , which is 11.6 per cent of  $\sigma_a^2$ . A simulation of 40 000 studies reproduces it to the last recorded digit. Both the formula Eq. 32.4.9 and that simulation describe the *interior* expression of Proposition 32.4.3(b); truncating it at zero, as the estimator itself does, can only raise the estimate, so the estimator's own bias is slightly smaller in modulus. Here the expression is negative in 2.0 per cent of the simulated studies and the truncated bias is  $-0.0045$ . With  $p$  fixed effects instead of one the same calculation gives a bias of order  $p/(gm)$ , which is why REML matters most when the fixed-effects model is large relative to the data.

## 32.4.4 Score equations and algorithms

### Proposition 32.4.4 — Score equations

Let  $\mathbf{V}_j = \partial \mathbf{V} / \partial \theta_j$  and  $\mathbf{P}$  be as in Eq. 32.4.2. Then

$$\begin{aligned}\frac{\partial \ell_P}{\partial \theta_j} &= -\frac{1}{2} \{ \text{tr}(\mathbf{V}^{-1} \mathbf{V}_j) - \mathbf{y}^\top \mathbf{P} \mathbf{V}_j \mathbf{P} \mathbf{y} \}, \\ \frac{\partial \ell_R}{\partial \theta_j} &= -\frac{1}{2} \{ \text{tr}(\mathbf{P} \mathbf{V}_j) - \mathbf{y}^\top \mathbf{P} \mathbf{V}_j \mathbf{P} \mathbf{y} \},\end{aligned}\tag{32.4.10}$$

The restricted score exceeds the profile score by  $\frac{1}{2} \text{tr}\{(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{V}_j \mathbf{V}^{-1} \mathbf{X}\}$ , which is nonnegative whenever  $\mathbf{V}_j$  is nonnegative definite, as it is for a variance-component parameterization  $\mathbf{V} = \sum_j \theta_j \mathbf{Z}_j \mathbf{Z}_j^\top$ .

*Proof.* For the profile,  $\partial \log \det \mathbf{V} / \partial \theta_j = \text{tr}(\mathbf{V}^{-1} \mathbf{V}_j)$  and  $\partial(\mathbf{y}^\top \mathbf{P} \mathbf{y}) / \partial \theta_j = -\mathbf{y}^\top \mathbf{P} \mathbf{V}_j \mathbf{P} \mathbf{y}$  by Lemma 32.4.1(c). For the restricted version, write  $\ell_R$  through Eq. 32.4.4 with a fixed  $\mathbf{K}$ : then  $\partial \log \det(\mathbf{K}^\top \mathbf{V} \mathbf{K}) / \partial \theta_j = \text{tr}\{(\mathbf{K}^\top \mathbf{V} \mathbf{K})^{-1} \mathbf{K}^\top \mathbf{V}_j \mathbf{K}\} = \text{tr}(\mathbf{P} \mathbf{V}_j)$  by Theorem 32.4.2(b) and the cyclic property of the trace (Theorem 1.6.1), and the quadratic term differentiates as before. Subtracting, and using Eq. 32.4.2 for  $\mathbf{P}$ , gives the stated difference. Its value is the same for every generalized inverse, because  $\mathcal{C}(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{V}_j \mathbf{V}^{-1} \mathbf{X}) \subset \mathcal{C}(\mathbf{X}^\top) = \mathcal{C}(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})$ , so take the Moore–Penrose inverse, which is nonnegative definite; when  $\mathbf{V}_j$  is nonnegative definite too, the difference is a trace of a product of two nonnegative definite matrices, hence nonnegative. ■

Setting Eq. 32.4.10 to zero gives equations that are not solvable in closed form outside balanced designs, and three families of algorithms are used.

**Fisher scoring.** The expected information for the restricted likelihood has entries  $\frac{1}{2} \text{tr}(\mathbf{P} \mathbf{V}_j \mathbf{P} \mathbf{V}_k)$ , so a Newton-type step costs one trace per pair of components. Convergence is fast near the maximum and unreliable far from it, and nothing keeps the iterates inside the parameter space.

**Average information REML.** Gilmour, Thompson and Cullis (1995) noticed that the average of the observed and expected information has entries  $\frac{1}{2} \mathbf{y}^\top \mathbf{P} \mathbf{V}_j \mathbf{P} \mathbf{V}_k \mathbf{P} \mathbf{y}$ , which need no traces at all. Average information REML is the workhorse of large animal-breeding and genomic analyses.

**EM.** Treat  $\mathbf{u}$  as missing data (Dempster, Laird and Rubin, 1977). With  $\mathbf{R} = \sigma^2 \mathbf{I}$  and  $\mathbf{G} = \sigma_u^2 \mathbf{I}_q$ , the complete-data maximum likelihood estimates are averages of squares, so the E-step and M-step give the maximum likelihood update

$$\begin{aligned}\sigma_u^{2,\text{new}} &= \frac{1}{q} \{ \|\hat{\mathbf{u}}\|^2 + \text{tr} \mathbf{S} \}, \\ \sigma^{2,\text{new}} &= \frac{1}{n} \{ \|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{Z} \hat{\mathbf{u}}\|^2 + \text{tr}(\mathbf{Z} \mathbf{S} \mathbf{Z}^\top) \},\end{aligned}$$

because  $E(\mathbf{u}^\top \mathbf{u} \mid \mathbf{y}) = \|\hat{\mathbf{u}}\|^2 + \text{tr} \mathbf{S}$  and  $E(\boldsymbol{\varepsilon}^\top \boldsymbol{\varepsilon} \mid \mathbf{y}) = \|\mathbf{y} - \mathbf{X} \hat{\boldsymbol{\beta}} - \mathbf{Z} \hat{\mathbf{u}}\|^2 + \text{tr}(\mathbf{Z} \mathbf{S} \mathbf{Z}^\top)$ , where

$$\mathbf{S} = \text{Cov}(\mathbf{u} \mid \mathbf{y}) = (\mathbf{G}^{-1} + \mathbf{Z}^\top \mathbf{R}^{-1} \mathbf{Z})^{-1} = \mathbf{G} - \mathbf{G} \mathbf{Z}^\top \mathbf{V}^{-1} \mathbf{Z} \mathbf{G}$$

is the inverse of the trailing block of  $\mathbf{C}$  in Eq. 32.3.5, the matrix the push-through identity Eq. 32.3.7 was applied to. It is *not* the trailing block of  $\mathbf{C}^{-1}$  in Theorem 32.3.2(c), which carries a third term because  $\boldsymbol{\beta}$  has been estimated; that matrix gives an iteration maximizing neither likelihood. Each step is cheap, increases the likelihood and never leaves the parameter space; the price is slow convergence. A restricted version runs the same argument on the error contrasts.

Laird and Ware (1982) introduced this algorithm to longitudinal data analysis, and it remains the most reliable starting procedure.

*Warning.* One starting value is not enough. Fit the Grunfeld panel of [Example \(Grunfeld's panel with a random firm effect\)](#) with a random slope as well as a random intercept, on the logarithmic scale for the variances that keeps the iterates positive: a search begun with a negligible slope variance crawls along the boundary and stops at the random-intercept fit, with restricted log-likelihood  $-1176.53$  against the maximum  $-1159.01$ , a gap of 17.52 reported as successful convergence. Here thirty random starts all reach the same maximum, which is some evidence that it is the only one; but such likelihoods can also be genuinely multimodal, increasingly so as the covariance structure grows (Searle, Casella and McCulloch, 1992).

### 32.4.5 The proficiency study, computed

#### Example. (REML and ML for the twelve laboratories)

For the data of [Example \(A laboratory proficiency study\)](#), [Proposition 32.4.3](#) gives  $\hat{\sigma}^2 = 0.0620$  by both methods, and

$$\hat{\sigma}_{a,\text{REML}}^2 = 0.0386, \quad \hat{\sigma}_{a,\text{ML}}^2 = 0.0341,$$

the maximum likelihood estimate being 0.883 of the restricted one, because it replaces MSB by  $(g-1)\text{MSB}/g$  before subtracting MSE. The restricted log-likelihood at its maximum is  $-8.228$  and the ordinary log-likelihood is  $-8.360$ ; the two are not comparable, being likelihoods of 47 and 48 numbers respectively. Both agree with the general matrix formulas coded from [Eq. 32.4.3](#) and [Eq. 32.4.4](#), and with statsmodels' `MixedLM`.

[Figure 32.4.1](#) plots the two profiles against  $\sigma_a^2$ , each with  $\sigma^2$  maximized out: the same shape in different places, the restricted one flatter on the right, which is why REML intervals are wider. Cutting the restricted profile 1.92 units below its maximum gives  $\sigma_a^2 \in (0.0086, 0.1275)$ , an interval comparable to the exact one of [Corollary 32.1.2](#) and available in models where no exact interval exists.

```
import numpy as np

g, m = 12, 4
n = g * m
mu, sigma_a, sigma = 2.50, 0.20, 0.25          # the values used to simulate

rng = np.random.default_rng(20320048)
lab_effect = sigma_a * rng.standard_normal(g)
y = mu + np.repeat(lab_effect, m) + sigma * rng.standard_normal(n)

X = np.ones((n, 1))
Z = np.repeat(np.eye(g), m, axis=0)

def loglik(s2, s2a, reml=True):
    """Log-likelihood (reml=False) or restricted log-likelihood of the variances."""
    V = s2a * (Z @ Z.T) + s2 * np.eye(n)
    L = np.linalg.cholesky(V)
    Vi_X = np.linalg.solve(V, X)
    W = X.T @ Vi_X                                # X' V^{-1} X
```

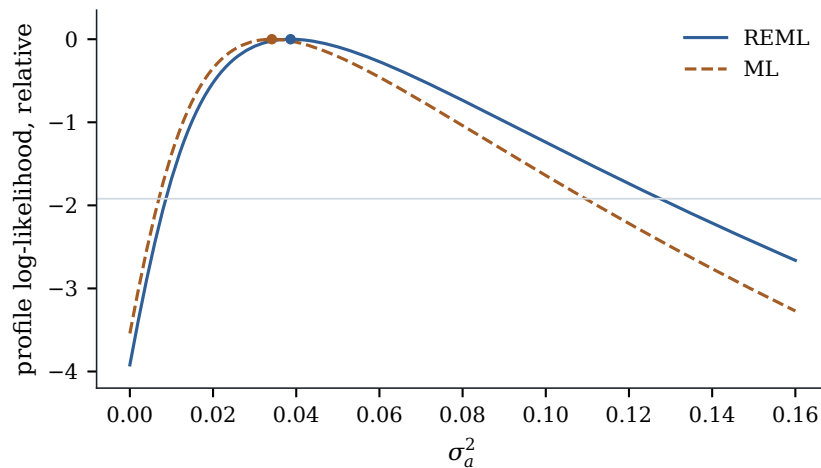


Figure 32.4.1: Profile log-likelihoods of  $\sigma_a^2$  for the proficiency study, with  $\sigma^2$  maximized out and each curve shifted to have maximum zero. The horizontal line is 1.92 below the maximum, the cut that gives an approximate 95% interval.

```

r = y - X @ np.linalg.solve(W, Vi_X.T @ y) # GLS residual
quad = r @ np.linalg.solve(V, r)           # = y' P y
logdetV = 2 * np.sum(np.log(np.diag(L)))
out = -0.5 * (n * np.log(2 * np.pi) + logdetV + quad)
if reml:
    out += -0.5 * (np.linalg.slogdet(W)[1] - np.linalg.slogdet(X.T @ X)[1]
                  - X.shape[1] * np.log(2 * np.pi))
return out

```

```

Y = y.reshape(g, m)
lab_mean = Y.mean(axis=1)
ms_between = m * np.sum((lab_mean - Y.mean()) ** 2) / (g - 1)
ms_within = np.sum((Y - lab_mean[:, None]) ** 2) / (n - g)

reml_s2, reml_s2a = ms_within, (ms_between - ms_within) / m
ml_s2 = ms_within
ml_s2a = ((g - 1) / g * ms_between - ms_within) / m

print(f"REML  sigma^2 {reml_s2:.4f}  sigma_a^2 {reml_s2a:.4f}")
print(f"ML    sigma^2 {ml_s2:.4f}   sigma_a^2 {ml_s2a:.4f}")
print("restricted log-likelihood", round(loglik(reml_s2, reml_s2a), 4))
print("log-likelihood", round(loglik(ml_s2, ml_s2a, reml=False), 4))

```

Two hundred EM steps from a deliberately poor start reach those maximum likelihood values to nine decimal places, which is what a correct **S** buys; the trailing block of  $\mathbf{C}^{-1}$  in its place converges instead to (0.0644, 0.0380), maximizing neither likelihood.

```

ZZt = Z @ Z.T
s2, s2a = 0.1, 0.1 # a deliberately poor starting value
for _ in range(200):
    V = s2a * ZZt + s2 * np.eye(n)
    Vi = np.linalg.inv(V)
    W = X.T @ Vi @ X
    beta_hat = np.linalg.solve(W, X.T @ Vi @ y)
    u_hat = s2a * Z.T @ Vi @ (y - X @ beta_hat)
    S = np.linalg.inv(np.eye(g) / s2a + Z.T @ Z / s2) # Cov(u | y), not (C^{-1})
    _22

```



```

resid = y - X @ beta_hat - Z @ u_hat
s2a = (u_hat @ u_hat + np.trace(S)) / g
s2 = (resid @ resid + np.trace(Z @ S @ Z.T)) / n

print(f"EM      sigma^2 {s2:.4f}  sigma_a^2 {s2a:.4f}")

```

## 32.4.6 Exercises

### 32.4.6.1 A. Check your understanding

#### Exercise A1.

Take  $\mathbf{V} = \sigma^2 \mathbf{I}$  and  $\text{rank}(\mathbf{X}) = p$ . Show from [Eq. 32.4.6](#) that  $-2\ell_R = (n - p) \log \sigma^2 + \text{SSE}/\sigma^2$  up to constants, and deduce that the REML estimate of  $\sigma^2$  is  $s^2$ . Then use [Proposition 7.3.2](#) to find which multiple of SSE has the smallest mean squared error, and show that the maximum likelihood estimator beats  $s^2$  for some  $(n, p)$  and loses to it for others.

*Solution.* With  $\mathbf{V} = \sigma^2 \mathbf{I}$ :  $\log \det \mathbf{V} = n \log \sigma^2$ ,  $\mathbf{y}^\top \mathbf{P} \mathbf{y} = \text{SSE}/\sigma^2$  and  $\det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) = \sigma^{-2p} \det(\mathbf{X}^\top \mathbf{X})$ , so  $\log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) - \log \det(\mathbf{X}^\top \mathbf{X}) = -p \log \sigma^2$ ; adding this to  $-2\ell_P = n \log \sigma^2 + \text{SSE}/\sigma^2$  leaves  $(n - p) \log \sigma^2 + \text{SSE}/\sigma^2$ . Its minimizer is  $\text{SSE}/(n - p) = s^2$ . By [Proposition 7.3.2](#) the best multiple of SSE is  $\text{SSE}/(n - p + 2)$ , which neither estimator in the question attains; writing  $k = n - p$ , the same proposition gives relative mean squared errors  $2/k$  for  $s^2$  and  $(2k + p^2)/n^2$  for  $\text{SSE}/n$ . Multiplying out  $n^2 = (k + p)^2$ , the inequality  $(2k + p^2)/n^2 \leq 2/k$  reduces to  $k(p - 4) \leq 2p$ . So maximum likelihood wins for every  $p \leq 4$ , whatever  $n$ , and loses once  $p$  is moderately large: with  $n = 100$  and  $p = 10$  it gives  $(180 + 100)/100^2 = 0.0280$  against  $2/90 = 0.0222$ . Fitting many fixed effects and then dividing by  $n$  is the bad case, which is the case REML exists for.

#### Exercise A2.

A colleague reports two restricted log-likelihoods for the same model fitted to the same data, differing by 6.91, the only difference between the runs being that one measured a regressor in units and the other in thousands. Explain the discrepancy using [Eq. 32.4.6](#), and confirm the value 6.91.

### 32.4.6.2 B. Practice

#### Exercise B1.

Verify [Theorem 32.4.2\(b\)](#) directly in the simplest case:  $n = 3$ ,  $\mathbf{X} = \mathbf{1}_3$ ,  $\mathbf{V} = \text{diag}(v_1, v_2, v_3)$ . Choose a basis  $\mathbf{K}$  of  $\mathbf{1}^\perp$ , compute  $\mathbf{K}(\mathbf{K}^\top \mathbf{V} \mathbf{K})^{-1} \mathbf{K}^\top$  and  $\mathbf{P}$ , and check they agree. Then change  $\mathbf{K}$  to a different basis and confirm that  $\det(\mathbf{K}^\top \mathbf{V} \mathbf{K})$  changes by a factor free of  $\mathbf{V}$ .

#### Exercise B2.

Two groups of sizes  $n_1, n_2$  have independent normal responses with a common mean and

variances  $\sigma^2$  and  $c\sigma^2$  with  $c$  known. Write down  $\ell_P$  and  $\ell_R$  for  $\sigma^2$ , find both maximizers, and confirm that REML divides by  $n_1 + n_2 - 1$  and ML by  $n_1 + n_2$ .

### Exercise B3.

Specialize the score equations [Eq. 32.4.10](#) of [Proposition 32.4.4](#) to the balanced one-way model with  $\boldsymbol{\theta} = (\lambda_1, \lambda_2)$ , for which  $\mathbf{V} = \lambda_1 \mathbf{M} + \lambda_2 (\mathbf{I} - \mathbf{M})$ ,  $\mathbf{V}_1 = \mathbf{M}$  and  $\mathbf{V}_2 = \mathbf{I} - \mathbf{M}$ . Show that the restricted score equations are  $\hat{\lambda}_1 = \text{MSB}$  and  $\hat{\lambda}_2 = \text{MSE}$ , recovering [Proposition 32.4.3\(a\)](#) without any optimization.

*Solution.* Here  $\mathbf{P} = \lambda_1^{-1}(\mathbf{M} - \mathbf{P}_1) + \lambda_2^{-1}(\mathbf{I} - \mathbf{M})$ , so  $\mathbf{P}\mathbf{V}_1 = \lambda_1^{-1}(\mathbf{M} - \mathbf{P}_1)$  with trace  $(g-1)/\lambda_1$ , and  $\mathbf{y}^\top \mathbf{P}\mathbf{V}_1 \mathbf{P}\mathbf{y} = \text{SSB}/\lambda_1^2$ . The first equation in [Eq. 32.4.10](#) therefore reads  $(g-1)/\lambda_1 = \text{SSB}/\lambda_1^2$ , that is  $\lambda_1 = \text{SSB}/(g-1)$ . Likewise  $\text{tr}(\mathbf{P}\mathbf{V}_2) = (n-g)/\lambda_2$  and  $\mathbf{y}^\top \mathbf{P}\mathbf{V}_2 \mathbf{P}\mathbf{y} = \text{SSE}/\lambda_2^2$ , giving  $\lambda_2 = \text{SSE}/(n-g)$ . Using  $\mathbf{V}^{-1}$  instead of  $\mathbf{P}$  replaces  $g-1$  by  $\text{tr}(\mathbf{V}^{-1}\mathbf{V}_1)\lambda_1 = g$ , which is the ML answer.

### 32.4.6.3 C. Going deeper

#### Exercise C1.

Show that

$$\int \exp\{\ell(\boldsymbol{\beta}, \boldsymbol{\theta})\} d\boldsymbol{\beta} = \exp\{\ell_R(\boldsymbol{\theta})\}$$

when  $\text{rank}(\mathbf{X}) = p$  and the constant  $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{X})$  is dropped from [Eq. 32.4.4](#). (Hint: complete the square in  $\boldsymbol{\beta}$  and use the normal integral.) This identity is the subject of [Section 32.6](#); state in words what it says about REML.

*Solution.* Write  $(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{P}\mathbf{y} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})$ , which is the decomposition behind [Lemma 32.4.1\(b\)](#). Integrating  $\exp\{-\frac{1}{2}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})\}$  over  $\mathbb{R}^p$  gives  $(2\pi)^{p/2} \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1/2}$ . Hence

$$\begin{aligned} \int e^\ell d\boldsymbol{\beta} &= e^{-\frac{1}{2}\{n \log 2\pi + \log \det \mathbf{V} + \mathbf{y}^\top \mathbf{P}\mathbf{y}\}} \\ &\quad \times (2\pi)^{p/2} \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1/2}, \end{aligned}$$

whose logarithm is [Eq. 32.4.6](#) without the  $\det(\mathbf{X}^\top \mathbf{X})$  term. In words: the restricted likelihood is the marginal likelihood of  $\boldsymbol{\theta}$  after integrating the fixed effects out with a flat prior.

## 32.5 Inference when the covariance is estimated

Every formula in the last three sections assumed  $\mathbf{V}$  known. In practice  $\mathbf{V}(\hat{\boldsymbol{\theta}})$  is substituted, and the substitution is not free: it perturbs the estimator, it perturbs the estimator's covariance, and it destroys the exactness of the  $F$  and  $t$  distributions Part III relied on. Testing a variance

component raises a separate difficulty, since the null value sits on the boundary of the parameter space and the usual chi-square calibration is simply wrong; [Theorem 32.5.2](#) computes the right one exactly in the balanced layout.

### 32.5.1 Feasible generalized least squares is still unbiased

#### Proposition 32.5.1 — Fixed-effect inference with an estimated covariance

Consider the normal linear mixed model with  $\text{rank}(\mathbf{X}) = p$ , and write

$$\hat{\boldsymbol{\beta}}(\boldsymbol{\theta}) = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$$

for the generalized least squares estimator.

- (a) (*Unbiasedness.*) Let  $\hat{\boldsymbol{\theta}} = \hat{\boldsymbol{\theta}}(\mathbf{Y})$  be **translation invariant**, meaning  $\hat{\boldsymbol{\theta}}(\mathbf{y} + \mathbf{X}\mathbf{b}) = \hat{\boldsymbol{\theta}}(\mathbf{y})$  for all  $\mathbf{b}$ , and **even**, meaning  $\hat{\boldsymbol{\theta}}(-\mathbf{y}) = \hat{\boldsymbol{\theta}}(\mathbf{y})$ . If  $E\|\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}})\| < \infty$ , then  $E\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}) = \boldsymbol{\beta}$ . Both REML and maximum likelihood estimators of  $\boldsymbol{\theta}$  have these two properties.
- (b) (*A design where everything is exact.*) Consider a balanced cluster-randomized trial:  $g$  clusters,  $g/2$  assigned to each arm,  $m$  units per cluster, with a random cluster effect of variance  $\sigma_a^2$  and independent errors of variance  $\sigma^2$ . Let  $\hat{\beta}$  be the difference of the two arms' means of cluster means, and let  $\text{MS}_C$  be the mean square between clusters within arms, on  $g - 2$  degrees of freedom, and  $\text{MSE}$  the mean square within clusters. Then, with  $\lambda_1 = \sigma^2 + m\sigma_a^2$ ,  $\text{Var}(\hat{\beta}) = 4\lambda_1/(gm)$  and

$$\frac{\hat{\beta} - \beta}{\sqrt{4\text{MS}_C/(gm)}} \sim t(g - 2) \quad (32.5.1)$$

exactly. If  $\text{MS}_C \geq \text{MSE}$  then  $\text{MS}_C$  is also the REML estimate of  $\lambda_1$ ; otherwise the REML estimate lies on the boundary  $\hat{\sigma}_a^2 = 0$ , as in [Proposition 32.4.3\(c\)](#).

- (c) (*Satterthwaite's approximation.*) Let  $U = \sum_i c_i \text{MS}_i > 0$ , where the  $\text{MS}_i$  are independent and  $\nu_i \text{MS}_i / E \text{MS}_i \sim \chi^2(\nu_i)$ . The two-moment match of [Section 4.3](#), fitting  $\nu U / E U \sim \chi^2(\nu)$  to the mean and variance of  $U$ , has

$$\nu = \frac{(E U)^2}{\sum_i c_i^2 (E \text{MS}_i)^2 / \nu_i}, \quad (32.5.2)$$

estimated in practice by replacing  $E \text{MS}_i$  by  $\text{MS}_i$ .

The companion statement for the variance components themselves, where the null value sits on the boundary of the parameter space, is [Theorem 32.5.2](#) below.

*Proof.*

- (a) Write  $\boldsymbol{\varepsilon} = \mathbf{Y} - \mathbf{X}\boldsymbol{\beta}$  and  $\mathbf{A}(\boldsymbol{\theta}) = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1}$ , so that  $\mathbf{A}(\boldsymbol{\theta})\mathbf{X} = \mathbf{I}$  and  $\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}) - \boldsymbol{\beta} = \mathbf{A}(\hat{\boldsymbol{\theta}}(\mathbf{Y}))\boldsymbol{\varepsilon}$ . Translation invariance gives  $\hat{\boldsymbol{\theta}}(\mathbf{Y}) = \hat{\boldsymbol{\theta}}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \hat{\boldsymbol{\theta}}(\boldsymbol{\varepsilon})$ , so  $\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}}) - \boldsymbol{\beta} = \mathbf{h}(\boldsymbol{\varepsilon})$  with  $\mathbf{h}(\mathbf{e}) = \mathbf{A}(\hat{\boldsymbol{\theta}}(\mathbf{e}))\mathbf{e}$ . Evenness gives  $\mathbf{h}(-\mathbf{e}) = -\mathbf{h}(\mathbf{e})$ , and  $\boldsymbol{\varepsilon}$  has the same distribution as  $-\boldsymbol{\varepsilon}$ , so  $E\mathbf{h}(\boldsymbol{\varepsilon}) = -E\mathbf{h}(\boldsymbol{\varepsilon})$  and the expectation is  $\mathbf{0}$ . This is the argument of [Theorem 21.3.1\(c\)](#) with the weights replaced by

a covariance. REML estimators are functions of  $\mathbf{K}^\top \mathbf{y}$  alone, hence translation invariant, and Eq. 32.4.4 is unchanged by  $\mathbf{y} \mapsto -\mathbf{y}$ , hence even; the same holds for maximum likelihood, whose profile Eq. 32.4.3 depends on  $\mathbf{y}$  only through  $\mathbf{y}^\top \mathbf{P} \mathbf{y}$  and is translation invariant because  $\mathbf{P} \mathbf{X} = \mathbf{0}$ .

- (b) Both columns of  $\mathbf{X} = [\mathbf{1}, \mathbf{T}]$  are constant within clusters, so by Example (Cluster-constant regressors) ordinary and generalized least squares agree and  $\hat{\beta}$  is the difference of arm means of cluster means. The  $g$  cluster means  $\bar{Y}_k$  are independent  $N(\beta_0 + \beta T_k, \lambda_1/m)$ , because  $\text{Var}(\bar{Y}_k) = \sigma_a^2 + \sigma^2/m = \lambda_1/m$ . Applying the exact normal theory of Chapter 11 to these  $g$  independent normal observations gives  $\hat{\beta} \sim N(\beta, 4\lambda_1/(gm))$ , since the two arm means are independent with variance  $\lambda_1/(m \cdot g/2)$  each. The quadratic form  $m \sum_k (\bar{Y}_k - \bar{Y}_{\text{arm}(k)})^2 / \lambda_1 = (g-2) \text{MS}_C / \lambda_1$  is the residual sum of squares of a two-group model fitted to the  $g$  cluster means, so it is  $\chi^2(g-2)$  and is independent of the fitted arm means by Theorem 4.4.1. Dividing gives Eq. 32.5.1.

For the REML claim, the error contrast space  $\mathcal{C}(\mathbf{X})^\perp$  decomposes orthogonally into  $\mathcal{C}(\mathbf{Z}) \cap \mathcal{C}(\mathbf{X})^\perp$ , of dimension  $g-2$ , on which  $\mathbf{V}$  acts as  $\lambda_1 \mathbf{I}$ , and  $\mathcal{C}(\mathbf{Z})^\perp$ , of dimension  $n-g$ , on which  $\mathbf{V}$  acts as  $\sigma^2 \mathbf{I}$ . Choosing  $\mathbf{K}$  adapted to this decomposition, Eq. 32.4.4 becomes  $-2\ell_R = (g-2) \log \lambda_1 + (n-g) \log \sigma^2 + \text{SS}_C / \lambda_1 + \text{SSE} / \sigma^2$  up to constants, to be minimized over  $\lambda_1 \geq \sigma^2 > 0$ , exactly as in Proposition 32.4.3. Its minimizers are  $\hat{\lambda}_1 = \text{MS}_C$  and  $\hat{\sigma}^2 = \text{MSE}$  when  $\text{MS}_C \geq \text{MSE}$ ; otherwise the constraint binds, by the argument of Proposition 32.4.3(c).

- (c) The two-moment match of Section 4.3 gives  $\nu = 2(\text{E} U)^2 / \text{Var}(U)$ , and here  $\text{Var}(U) = \sum_i c_i^2 \text{Var}(\text{MS}_i) = 2 \sum_i c_i^2 (\text{E} \text{MS}_i)^2 / \nu_i$ , which is Eq. 32.5.2 — the calculation already made for a linear combination of independent variance estimates in Section 15.5.

■

Part (b) isolates the problem. The standard error a mixed-model program reports for this design is *exactly right*: it is  $\sqrt{4 \text{MS}_C / (gm)}$  whenever the REML estimate of  $\lambda_1$  is interior, that is, unless  $\text{MS}_C < \text{MSE}$  — probability 0.157 in the design simulated below, and 0.558 when  $\sigma_a^2$  is in fact zero. What is wrong is the reference distribution. Treating  $\hat{\beta}/\text{se}$  as standard normal, which is what a Wald test does, ignores that  $\text{MS}_C$  came from  $g-2$  degrees of freedom. With  $g=8$  clusters that is six, and the  $t$  critical value 2.447 exceeds 1.96 by 25 per cent.

### Example. (Eight clusters)

Simulate the trial with  $g=8$  clusters of  $m=5$ , an intraclass correlation of 0.2, and no treatment effect. Over 200 000 trials, the nominal 95% normal Wald interval covers 0.901 of the time, while the  $t(g-2)$  interval covers 0.949. Figure 32.5.1 traces the coverage against the number of clusters: 0.812 with four clusters, still 0.943 with forty. The normal interval's coverage is exactly  $2F_{t(g-2)}(1.96) - 1$ , which is why it approaches but never reaches 0.95.

The coverage study is the null one; take now a single data set simulated from the same design but with a true effect of 0.8. On it the estimated effect is 1.154 with standard error 0.479. The normal interval is  $1.154 \pm 0.940$ , which excludes zero; the exact interval is  $1.154 \pm 1.173$ , which does not. The whole difference between “significant” and “not” is the reference distribution.

```
import numpy as np
```

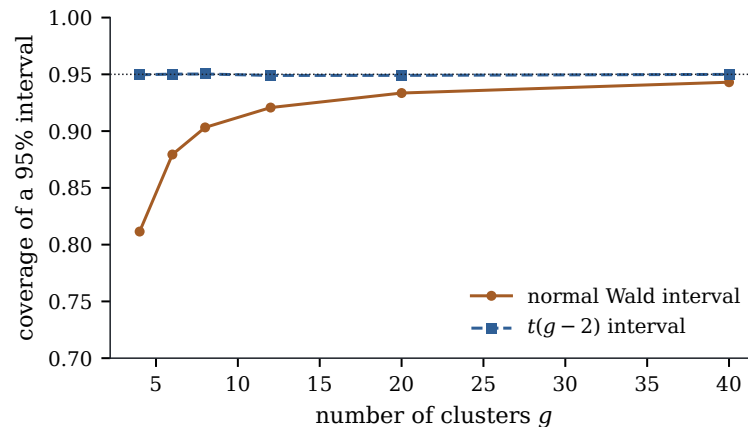


Figure 32.5.1: Coverage of a nominal 95% interval for the treatment effect in a balanced cluster-randomized trial with  $m = 5$  units per cluster and intraclass correlation 0.2, against the number of clusters. The normal Wald interval treats the estimated covariance as known; the  $t(g-2)$  interval is exact by Eq. 32.5.1.

```

from scipy import stats

def simulate(g, m, sigma_a, sigma, effect, B, seed):
    """B cluster-randomized trials; returns the estimate and its estimated variance.
    """
    rng = np.random.default_rng(seed)
    treat = np.tile([0.0, 1.0], g // 2)          # cluster k is treated if
    treat[k]=1
    Y = (effect * treat[:, None] + sigma_a * rng.standard_normal((B, g, 1))
         + sigma * rng.standard_normal((B, g, m)))
    cluster_mean = Y.mean(axis=2)
    diff = (cluster_mean[:, treat == 1].mean(axis=1)
           - cluster_mean[:, treat == 0].mean(axis=1))
    # mean square between clusters within arms, on g - 2 degrees of freedom
    centred = cluster_mean - np.where(treat == 1,
                                      cluster_mean[:, treat == 1].mean(axis=1,
                                      keepdims=True),
                                      cluster_mean[:, treat == 0].mean(axis=1,
                                      keepdims=True))
    lam1 = m * np.sum(centred**2, axis=1) / (g - 2)    # estimates sigma^2 + m
    sigma_a^2
    return diff, 4 * lam1 / (g * m)

g, m = 8, 5
# a shorter run than the 200 000 trials quoted in the text, so the cell finishes
quickly
diff, var_hat = simulate(g, m, 0.5, 1.0, 0.0, 20000, seed=5150)
t = diff / np.sqrt(var_hat)
print(f"normal interval covers {np.mean(np.abs(t) <= stats.norm.ppf(0.975)):.3f}")
print(f"t(g-2) interval covers {np.mean(np.abs(t) <= stats.t.ppf(0.975, g - 2)):.3f}")
")

```

## 32.5.2 Degrees of freedom in general

Outside designs as tidy as [Proposition 32.5.1\(b\)](#) there is no exact answer, and two approximations are in common use. Neither is proved here; both rest on simulation evidence rather than exact distribution theory.

**Satterthwaite.** The estimated variance  $\lambda^\top (\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{X})^{-1} \lambda$  is a smooth function of  $\hat{\boldsymbol{\theta}}$ . Approximating its distribution by a scaled chi-square and matching two moments, with the variance from the delta method and the asymptotic covariance of  $\hat{\boldsymbol{\theta}}$ , gives a denominator degrees of freedom that is in general not an integer. Giesbrecht and Burns (1985) developed this for mixed models; in the design of [Proposition 32.5.1\(b\)](#) it returns the exact  $g - 2$  ([Exercise \(B1\)](#)).

**Kenward and Roger.** Satterthwaite corrects the reference distribution but not the variance estimate, and the variance estimate is biased downwards: the plug-in covariance  $(\mathbf{X}^\top \mathbf{V}(\hat{\boldsymbol{\theta}})^{-1} \mathbf{X})^{-1}$  is what  $\hat{\boldsymbol{\beta}}$  would have if  $\hat{\boldsymbol{\theta}}$  were the truth, and the estimator actually used varies for a second reason, because  $\hat{\boldsymbol{\theta}}$  itself varies, adding a nonnegative definite term. Kackar and Harville (1984) approximated it; Kenward and Roger (1997) add an estimate of it, then scale the  $F$  statistic and match two moments for its degrees of freedom. The correction is the default in several packages and performs well in small samples; [Example \(The proficiency study, four ways\)](#) exhibits the inflation.

*Warning.* Unlike [Proposition 32.5.1\(a\)](#), these are approximations with no finite-sample guarantee. When the design is small, unbalanced and the covariance structure rich, the safest calibration is a parametric bootstrap: simulate from the fitted model, refit, and use the simulated distribution of the statistic ([Proposition 23.3.3](#) and [Section 23.3](#)). It costs a few hundred fits and needs no asymptotics.

## 32.5.3 Comparing models

Two rules govern likelihood comparisons in mixed models, and both follow from [Section 32.4](#).

**Different fixed effects: use maximum likelihood.** Since  $\mathbf{K}$  is determined by  $\mathcal{C}(\mathbf{X})$ , two fixed-effects structures give restricted likelihoods of different random vectors, and their difference means nothing. Likelihood ratio tests and information criteria across fixed-effects structures must use the ordinary likelihood [Eq. 32.4.1](#).

**Same fixed effects, different covariance: REML is preferred.** Here  $\mathbf{K}$  is the same, the restricted likelihoods are likelihoods of the same data, and the additive constants of [Theorem 32.4.2\(c\)](#) cancel. Restricted likelihood ratio tests, and the AIC and BIC of [Definition 29.2.2](#) computed from  $\ell_R$  with  $d_k$  the number of covariance parameters, are legitimate. For BIC the sample size to use is  $n - p$ , the dimension of the data actually modelled, not  $n$ , following Verbeke and Molenberghs (2000).

### Example. (Do the firms' slopes differ?)

Continue [Example \(Grunfeld's panel with a random firm effect\)](#). Adding a firm-specific slope on market value, with variance  $\sigma_s^2$  independent of the firm intercept, raises the restricted log-likelihood from  $-1176.53$  to  $-1159.01$ . The two models have the same fixed effects, so the comparison is legitimate: the statistic is 35.05 on one boundary parameter, far beyond any reasonable critical value, and AIC falls from 2357.1 to 2324.0, BIC from 2363.8 to 2334.2. The firms' responses to market value genuinely differ: the

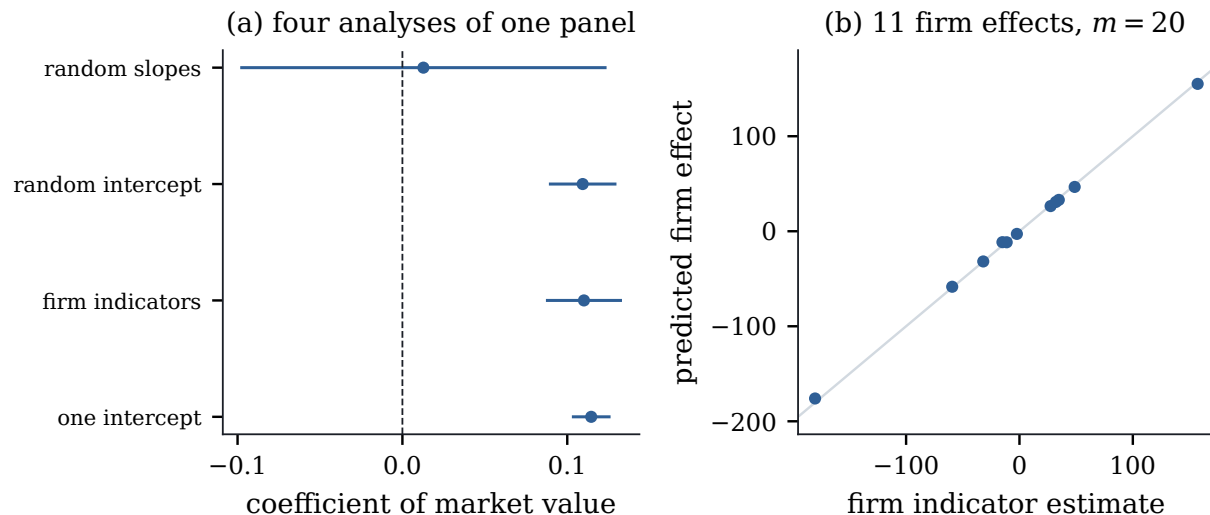


Figure 32.5.2: Grunfeld's panel. (a) The coefficient of market value with 95% intervals under four analyses: one common intercept, firm indicators, a random firm intercept, and random firm intercepts and slopes. (b) Firm effects from the firm-indicator fit against the predicted firm effects of the random-intercept model; with twenty years per firm there is nothing left to shrink.

estimated standard deviation of the firm-specific slopes is 0.1492.

The consequence for inference is dramatic. In the random-intercept model the coefficient of market value is 0.1094 with standard error 0.0100; in the random-slope model it is 0.0127 with standard error 0.0562, five and a half times larger. Nothing about the data changed. Two things did. The standard error grew because the effective sample size for a *population-average* slope is 11 firms, not 220 firm-years. The estimate moved because the two models estimate different functionals. Firm-specific intercepts and slopes with a common capital coefficient give eleven slopes of mean  $-0.0009$  and standard deviation 0.198: their spread dwarfs their average. The random-slope model estimates that average; the random-intercept model, forced to give every firm the same slope, reports a precision-weighted blend instead. Panel (a) of Figure 32.5.2 shows the four analyses side by side; panel (b) shows that with 20 years per firm the predicted firm effects are indistinguishable from the firm-indicator estimates.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from scipy import optimize

df = sm.datasets.grunfeld.load_pandas().data.sort_values(["firm", "year"])
firms = df["firm"].unique()
y = df["invest"].to_numpy()
X = np.column_stack([np.ones(len(y)), df["value"], df["capital"]])
blocks = [np.where(df["firm"].to_numpy() == f)[0] for f in firms]
n, p, g = len(y), X.shape[1], len(firms)

def restricted_loglik(theta, Zcols):
    """Restricted log-likelihood; theta holds log sigma^2 and the log variances of
    Zcols."""
    s2, dvar = np.exp(theta[0]), np.exp(theta[1:])
    logdet, W, XtVy, yVy = 0.0, np.zeros((p, p)), np.zeros(p), 0.0
    for ix in blocks:
```



```

Zi = Zcols[ix]
Vi = (Zi * dvar) @ Zi.T + s2 * np.eye(len(ix))
Vinv = np.linalg.inv(Vi)
logdet += np.linalg.slogdet(Vi)[1]
W += X[ix].T @ Vinv @ X[ix]
XtVy += X[ix].T @ Vinv @ y[ix]
yVy += y[ix] @ Vinv @ y[ix]
beta = np.linalg.solve(W, XtVy)
return -0.5 * (logdet + yVy - XtVy @ beta + np.linalg.slogdet(W)[1]
              - np.linalg.slogdet(X.T @ X)[1] + (n - p) * np.log(2 * np.pi)),
beta, W

Z_int = np.ones((n, 1)) # random intercept only
fit = optimize.minimize(lambda t: -restricted_loglik(t, Z_int)[0],
                        np.log([2000.0, 5000.0]), method="Nelder-Mead",
                        options={"xatol": 1e-9, "fatol": 1e-10, "maxiter": 5000})
ll_int, beta_int, W_int = restricted_loglik(fit.x, Z_int)
s2_hat, s2_firm = np.exp(fit.x)
se_int = np.sqrt(np.diag(np.linalg.inv(W_int)))

print(f"sigma^2 {s2_hat:.1f}    firm variance {s2_firm:.1f}")
print(f"value    {beta_int[1]:.4f} ({se_int[1]:.4f})")
print(f"capital {beta_int[2]:.4f} ({se_int[2]:.4f})")

```

### 32.5.4 Testing a variance component

The hypothesis  $\sigma_a^2 = 0$  sits on the edge of the parameter space, so the usual chi-square calibration of a likelihood ratio statistic does not apply: the standard argument expands the log-likelihood about an *interior* maximum, and here the unconstrained maximum falls outside the parameter space about half the time. In the balanced one-way model the correct answer can be computed exactly.

#### Theorem 32.5.2 — Exact null distribution of the restricted likelihood ratio

In the balanced normal one-way random effects model with  $n = gm$ , let

$$L = 2 \left\{ \sup_{\sigma_a^2 \geq 0} \ell_R - \sup_{\sigma_a^2 = 0} \ell_R \right\}$$

be the restricted likelihood ratio statistic for  $H_0 : \sigma_a^2 = 0$ , and write  $F = \text{MSB}/\text{MSE}$ ,  $a = (g - 1)/(n - 1)$ ,  $b = (n - g)/(n - 1)$ . Then

$$L = \begin{cases} 0, & F \leq 1, \\ (n - 1) \log(aF + b) - (g - 1) \log F, & F > 1. \end{cases} \quad (32.5.3)$$

Under  $H_0$ ,  $F \sim F(g - 1, n - g)$ , so the null distribution of  $L$  is free of  $\sigma^2$  and exactly computable. In particular  $L$  is a nondecreasing function of  $F$  and

$$\Pr(L = 0) = \Pr\{F(g - 1, n - g) \leq 1\},$$



a probability that tends to  $\frac{1}{2}$  as  $g$  and  $n - g$  grow.

*Proof.* By Eq. 32.4.8,  $-2\ell_R = (g - 1) \log \lambda_1 + (n - g) \log \lambda_2 + \text{SSB}/\lambda_1 + \text{SSE}/\lambda_2$  up to a constant, over the region  $\lambda_1 \geq \lambda_2 > 0$ . Under  $H_0$  the region collapses to  $\lambda_1 = \lambda_2 = \lambda$ , where the objective is  $(n - 1) \log \lambda + (\text{SSB} + \text{SSE})/\lambda$ , minimized at  $\hat{\lambda} = (\text{SSB} + \text{SSE})/(n - 1) = \text{MST}$ , with minimum  $(n - 1) \log \text{MST} + (n - 1)$ .

If  $\text{MSB} \geq \text{MSE}$  the unrestricted maximizers of Proposition 32.4.3(a) are feasible, and the minimum of  $-2\ell_R$  is  $(g - 1) \log \text{MSB} + (n - g) \log \text{MSE} + (n - 1)$ . Subtracting,

$$L = (n - 1) \log \text{MST} - (g - 1) \log \text{MSB} - (n - g) \log \text{MSE}.$$

Dividing inside every logarithm by MSE, which is legitimate because the coefficients  $(n - 1)$ ,  $-(g - 1)$  and  $-(n - g)$  sum to zero, and using  $\text{MST}/\text{MSE} = aF + b$ , gives Eq. 32.5.3. If  $\text{MSB} < \text{MSE}$  then by Proposition 32.4.3(c) the constrained maximizer is on the boundary, which is the null model, so  $L = 0$ .

Under  $H_0$  the distribution of  $F$  is  $F(g - 1, n - g)$  by Proposition 32.4.3 and Proposition 32.1.1(d) with  $\gamma = 0$ , and  $L$  is a function of  $F$  alone. It is nondecreasing in  $F$  because, using  $(n - 1)a = g - 1$  and then  $1 - a = b$ ,

$$\begin{aligned} \frac{dL}{dF} &= \frac{(n - 1)a}{aF + b} - \frac{g - 1}{F} = \frac{(g - 1)\{F - (aF + b)\}}{F(aF + b)} \\ &= \frac{(g - 1)b(F - 1)}{F(aF + b)} \geq 0 \end{aligned}$$

for  $F \geq 1$ . Hence  $\{L = 0\} = \{F \leq 1\}$ . Finally, as  $r = g - 1$  and  $s = n - g$  grow,  $\log F$  is asymptotically normal with variance  $2/r + 2/s \rightarrow 0$  and mean of order  $1/r + 1/s$ , which is small compared with the standard deviation; so  $\Pr(\log F \leq 0) \rightarrow 1/2$ . ■

*Remark* (Sketch of the chi-bar-squared limit). The exact result explains the shape of the general answer. Put  $\delta = F - 1$  and expand Eq. 32.5.3 for small  $\delta > 0$ :  $\log(1 + a\delta) = a\delta - a^2\delta^2/2 + O(\delta^3)$  and  $\log(1 + \delta) = \delta - \delta^2/2 + O(\delta^3)$ . Using  $(n - 1)a = g - 1$  and  $(n - 1)a^2 = (g - 1)a$ , the first-order terms cancel and

$$L = \frac{1}{2}(g - 1)(1 - a)\delta^2 + O((n - 1)\delta^3).$$

Under  $H_0$ ,  $\delta = F - 1$  has, to first order, the variance of  $\log F$ , namely  $\tau^2 = 2/(g - 1) + 2/(n - g) = 2(n - 1)/\{(g - 1)(n - g)\}$ , and  $\delta/\tau$  is asymptotically standard normal. Since  $(g - 1)(1 - a) = (g - 1)(n - g)/(n - 1)$ , the product  $(g - 1)(1 - a)\tau^2$  equals 2 exactly, so  $L \approx Z^2$  on  $\{\delta > 0\}$  for a standard normal  $Z$ , while  $L = 0$  on  $\{\delta \leq 0\}$ , an event of limiting probability  $1/2$ . So  $L$  converges in distribution to  $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$ , the **chi-bar-squared** mixture: a point mass at zero of weight one half, plus one half of a chi-square on one degree of freedom. This is a sketch, not a proof; the expansions above are heuristic. The general theorem, covering several variance components on the boundary simultaneously with weights given by the geometry of the constraint cone, is due to Self and Liang (1987), with the mixed-model case worked out by Stram and Lee (1994).

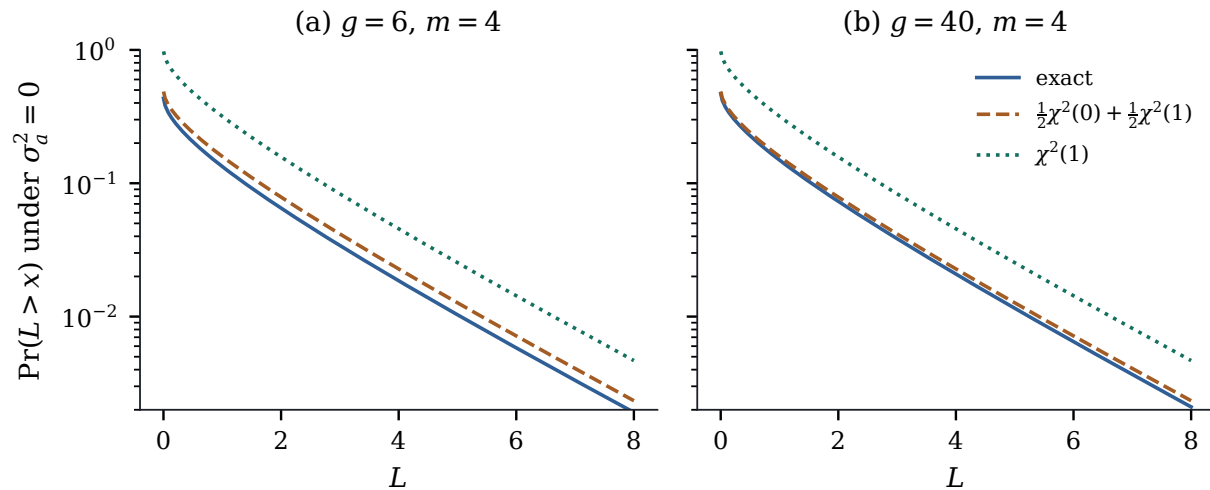


Figure 32.5.3: Null tail probabilities of the restricted likelihood ratio statistic for  $\sigma_a^2 = 0$  in a balanced one-way layout, on a logarithmic scale: exact (Eq. 32.5.3), the chi-bar-squared mixture, and  $\chi^2(1)$ . The mixture is conservative for small  $g$  and accurate for large  $g$ .

#### Example. (What the calibration costs)

Take  $g = 6$  groups of  $m = 4$ . Exactly 0.554 of the null distribution of  $L$  sits at zero, and a simulation of 200 000 data sets puts 0.554 there. Referring  $L$  to the chi-bar-squared mixture, whose 5 per cent point is the 90 per cent point 2.706 of  $\chi^2(1)$ , gives an actual size of 0.0411: slightly conservative. Referring it to  $\chi^2(1)$ , with critical value 3.841, gives 0.0204, badly conservative — the test throws away about half its power. With  $g = 40$  groups the mixture is nearly exact, at 0.0462. Figure 32.5.3 compares the three tail functions.

```
import numpy as np
from scipy import stats

def lrt_from_f(f, g, m):
    """Restricted likelihood ratio statistic for sigma_a^2 = 0, from the F ratio."""
    n = g * m
    a, b = (g - 1) / (n - 1), (n - g) / (n - 1)
    return np.where(f <= 1, 0.0,
                    (n - 1) * np.log(a * np.maximum(f, 1) + b) - (g - 1) * np.log(np
                    .maximum(f, 1)))

g, m = 6, 4
n = g * m
p_zero = stats.f.cdf(1.0, g - 1, n - g)
crit_mix = stats.chi2.ppf(0.90, 1)
print(f"g = {g}, m = {m}: P(L = 0) = {p_zero:.3f}, 5% point of the mixture {
    crit_mix:.3f}")
```

Crainiceanu and Ruppert (2004) showed that for models with a single variance component the exact null distribution can be computed in general, not only in the balanced case, and that the chi-bar-squared approximation can be poor when the design is unbalanced. Where their results do not apply, the parametric bootstrap is again the practical answer.

## 32.5.5 Intervals for variance components

Three routes, in decreasing order of reliability. In balanced designs, [Corollary 32.1.2](#) gives exact intervals for ratios of variance components and for the intraclass correlation. In general, the profile restricted likelihood of [Example \(REML and ML for the twelve laboratories\)](#) gives an interval by cutting 1.92 units below the maximum; it respects the boundary and is asymmetric, as the likelihood is. Last and least, the Wald interval from the inverse information matrix is symmetric, frequently includes negative values, and is badly calibrated whenever  $\hat{\sigma}_a^2$  is near zero. [Burdick and Graybill \(1992\)](#) is a book-length treatment of what can and cannot be done exactly.

## 32.5.6 Exercises

### 32.5.6.1 A. Check your understanding

#### Exercise A1.

In [Proposition 32.5.1\(b\)](#), the reported standard error is exactly right and the test is still wrong. Explain in one or two sentences what “exactly right” means here and what is wrong, and say why the problem gets worse as the intraclass correlation grows with  $n$  fixed.

#### Exercise A2.

A colleague tests whether a regressor belongs in the fixed part of a mixed model by comparing restricted log-likelihoods with and without it, referring twice the difference to  $\chi^2(1)$ . Give two separate reasons why this is wrong, and say what to do instead.

### 32.5.6.2 B. Practice

#### Exercise B1.

Apply [Eq. 32.5.2](#) to the estimated variance  $4\text{MS}_C/(gm)$  of [Proposition 32.5.1\(b\)](#), a single multiple of one mean square, and confirm that it returns  $\nu = g - 2$  exactly. Then apply it to  $\hat{\sigma}_a^2 = (\text{MSB} - \text{MSE})/m$  in the balanced one-way model and write down the resulting degrees of freedom.

*Solution.* With one term,  $U = c \text{MS}_1$ , [Eq. 32.5.2](#) gives  $\nu = c^2(\text{E MS}_1)^2 / \{c^2(\text{E MS}_1)^2 / \nu_1\} = \nu_1 = g - 2$ . For the variance component,  $c_1 = 1/m$  with  $\nu_1 = g - 1$  and  $c_2 = -1/m$  with  $\nu_2 = n - g$ , so

$$\hat{\nu} = \frac{(\hat{\sigma}_a^2)^2}{(\text{MSB}/m)^2/(g-1) + (\text{MSE}/m)^2/(n-g)}.$$

Note that this can be far below  $g - 1$  when MSB and MSE are close, which is the usual situation when  $\sigma_a^2$  is small.

#### Exercise B2.

Verify directly that the function  $F \mapsto (n-1)\log(aF+b) - (g-1)\log F$  in [Eq. 32.5.3](#)

vanishes at  $F = 1$ , is increasing for  $F > 1$  and tends to infinity as  $F \rightarrow \infty$ . What does the last fact say about the power of the test when  $\sigma_a^2$  is large?

### Exercise B3.

In the balanced one-way model, compute the Wald interval  $\hat{\sigma}_a^2 \pm 1.96 \text{se}$  for the proficiency study using [Exercise \(B2\)](#) for the standard error with the estimates plugged in, and compare it with the exact interval implied by [Corollary 32.1.2](#) and with the profile interval (0.0086, 0.1275) of [Example \(REML and ML for the twelve laboratories\)](#). Comment.

### 32.5.6.3 C. Going deeper

#### Exercise C1.

Consider testing  $\sigma_a^2 = 0$  when a second variance component  $\sigma_b^2 > 0$  is present and is *not* on the boundary. Argue informally that the limiting null distribution is still  $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$ , and explain why testing *both* components simultaneously gives a mixture over  $\chi^2(0), \chi^2(1), \chi^2(2)$  with weights determined by the probability that the unconstrained maximizer lands in each orthant. Which weights arise if the two estimates are asymptotically independent?

*Solution.* Locally the log-likelihood is approximately quadratic in the parameter, so the likelihood ratio behaves like the squared distance from the unconstrained maximizer to the constrained set, measured in the information metric. With one parameter on the boundary, the constraint acts only in that coordinate: half the time the unconstrained maximizer already satisfies it and the ratio is zero, half the time the constrained maximum is on the face and the ratio is a squared standard normal. With two parameters constrained, the ratio is zero when the maximizer lies in the nonnegative orthant, a  $\chi^2(1)$  when it lies in one of the two adjacent orthants, and a  $\chi^2(2)$  when it lies in the fully negative orthant. If the two coordinates are asymptotically independent, those four orthants have probabilities  $1/4, 1/4, 1/4, 1/4$ , giving weights  $1/4$  for  $\chi^2(0)$ ,  $1/2$  for  $\chi^2(1)$  and  $1/4$  for  $\chi^2(2)$ .

#### Exercise C2.

[Proposition 32.5.1\(a\)](#) needs the error distribution to be symmetric about zero. Give an example of a translation-invariant, even  $\hat{\boldsymbol{\theta}}$  and a skewed error distribution for which  $\hat{\boldsymbol{\beta}}(\hat{\boldsymbol{\theta}})$  is biased, and explain why symmetry cannot simply be dropped. (Compare the discussion after [Theorem 21.3.1](#).)

## 32.6 Bayesian thread III: mixed models as hierarchical models

The Bayesian thread began in [Section 7.6](#) with a conjugate prior on  $(\boldsymbol{\beta}, \sigma^2)$  and continued in [Section 12.6](#) with credible sets and predictive distributions. Mixed models are where the two traditions meet most closely, because a random effect *is* a prior: the statement  $\mathbf{u} \sim N_q(\mathbf{0}, \mathbf{G})$  is formally indistinguishable from a normal prior on  $q$  unknown constants. What separates the methods of [Section 32.4](#) from a fully Bayesian analysis is only that they estimate  $\mathbf{G}$  and then treat it as known, which is empirical Bayes ([Proposition 27.5.4](#)), while a Bayesian puts a prior on  $\mathbf{G}$  too and integrates.

### 32.6.1 The hierarchical reading

Write the model in three levels:

$$\mathbf{Y} \mid \boldsymbol{\beta}, \mathbf{u}, \boldsymbol{\theta} \sim N_n(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}, \mathbf{R}), \quad \mathbf{u} \mid \boldsymbol{\theta} \sim N_q(\mathbf{0}, \mathbf{G}), \quad (\boldsymbol{\beta}, \boldsymbol{\theta}) \sim \pi. \quad (32.6.1)$$

Level one is a linear model with known coefficients; level two is the prior on  $\mathbf{u}$  that the experiment's sampling of levels supplies; level three is the part that is genuinely a matter of choice.

#### **Proposition 32.6.1** — BLUP as a posterior mean, REML as a marginal posterior

Consider [Eq. 32.6.1](#) with  $\mathbf{G}$  and  $\mathbf{R}$  positive definite and  $\text{rank}(\mathbf{X}) = p$ .

- (a) (*BLUP is a posterior mean.*) Fix  $\boldsymbol{\theta}$  and take the improper flat prior  $\pi(\boldsymbol{\beta}) \propto 1$ . Then the joint posterior of  $(\boldsymbol{\beta}, \mathbf{u})$  is

$$\begin{pmatrix} \boldsymbol{\beta} \\ \mathbf{u} \end{pmatrix} \mid \mathbf{y} \sim N_{p+q} \left( \begin{pmatrix} \hat{\boldsymbol{\beta}} \\ \hat{\mathbf{u}} \end{pmatrix}, \mathbf{C}^{-1} \right),$$

with  $\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}}$  and  $\mathbf{C}$  as in [Theorem 32.3.2](#). In particular  $E(\mathbf{u} \mid \mathbf{y}) = \hat{\mathbf{u}}$  is the BLUP,  $E(\boldsymbol{\beta} \mid \mathbf{y}) = \hat{\boldsymbol{\beta}}$  is the generalized least squares estimate, and  $\text{Cov}(\mathbf{u} \mid \mathbf{y})$  is the prediction error covariance of [Eq. 32.3.2](#).

- (b) (*REML is a marginal posterior.*) With the flat prior on  $\boldsymbol{\beta}$  and any prior  $\pi(\boldsymbol{\theta})$ , the marginal posterior of the variance components is

$$\pi(\boldsymbol{\theta} \mid \mathbf{y}) \propto \pi(\boldsymbol{\theta}) \exp\{\ell_R(\boldsymbol{\theta})\}, \quad (32.6.2)$$

with  $\ell_R$  the restricted log-likelihood of [Definition 32.4.1](#). The REML estimate is therefore the mode of the marginal posterior under a flat prior on  $\boldsymbol{\theta}$ .

- (c) (*What the plug-in leaves out.*) Under (a) with  $\boldsymbol{\theta}$  replaced by an estimate, the reported posterior variance is  $\text{Cov}(\mathbf{u} \mid \mathbf{y}, \hat{\boldsymbol{\theta}})$ , whereas the correct marginal posterior variance is  $E\{\text{Cov}(\mathbf{u} \mid \mathbf{y}, \boldsymbol{\theta}) \mid \mathbf{y}\} + \text{Cov}\{E(\mathbf{u} \mid \mathbf{y}, \boldsymbol{\theta}) \mid \mathbf{y}\}$ . The plug-in omits the second term, so it is too small.

*Proof.*

- (a) With  $\pi(\boldsymbol{\beta}) \propto 1$  and  $\boldsymbol{\theta}$  fixed, the joint density of  $(\mathbf{y}, \mathbf{u})$  times the prior is proportional to  $\exp\{-\frac{1}{2}Q(\boldsymbol{\beta}, \mathbf{u})\}$  with  $Q$  the penalized criterion [Eq. 32.3.6](#), because  $-2 \log f(\mathbf{y} \mid \boldsymbol{\beta}, \mathbf{u}) = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u})^\top \mathbf{R}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u})$  and  $-2 \log f(\mathbf{u}) = \mathbf{u}^\top \mathbf{G}^{-1}\mathbf{u}$ , up to terms free of  $(\boldsymbol{\beta}, \mathbf{u})$ . By [Theorem 32.3.2\(b\)](#),  $Q$  is a quadratic in  $\mathbf{w} = (\boldsymbol{\beta}^\top, \mathbf{u}^\top)^\top$  with positive definite matrix  $\mathbf{C}$ , minimized at  $\hat{\mathbf{w}} = (\hat{\boldsymbol{\beta}}^\top, \hat{\mathbf{u}}^\top)^\top$ , so  $Q(\mathbf{w}) = (\mathbf{w} - \hat{\mathbf{w}})^\top \mathbf{C}(\mathbf{w} - \hat{\mathbf{w}}) + Q(\hat{\mathbf{w}})$ . The posterior is therefore  $N_{p+q}(\hat{\mathbf{w}}, \mathbf{C}^{-1})$ , and [Theorem 32.3.2\(c\)](#) identifies the blocks.
- (b) The joint posterior is  $\pi(\boldsymbol{\beta}, \boldsymbol{\theta} \mid \mathbf{y}) \propto \pi(\boldsymbol{\theta}) \exp\{\ell(\boldsymbol{\beta}, \boldsymbol{\theta})\}$  with  $\ell$  the marginal log-likelihood [Eq. 32.4.1](#), the random effects having already been integrated out in [Eq. 32.2.2](#). Completing the square as in [Lemma 32.4.1\(b\)](#),

$$(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})^\top \mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{y}^\top \mathbf{P}\mathbf{y} + (\boldsymbol{\beta} - \hat{\boldsymbol{\beta}})^\top \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}),$$

so integrating over  $\boldsymbol{\beta} \in \mathbb{R}^p$  contributes the factor  $(2\pi)^{p/2} \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1/2}$  and

$$\begin{aligned} \int e^{\ell(\boldsymbol{\beta}, \boldsymbol{\theta})} d\boldsymbol{\beta} &= \exp\left[-\frac{1}{2}\{(n-p) \log 2\pi + \log \det \mathbf{V} \right. \\ &\quad \left. + \log \det(\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X}) + \mathbf{y}^\top \mathbf{P}\mathbf{y}\}\right]. \end{aligned}$$

By [Eq. 32.4.5](#) the exponent is  $\ell_R(\boldsymbol{\theta})$  up to the additive constant  $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{X})$ , which does not involve  $\boldsymbol{\theta}$ . That is [Eq. 32.6.2](#).

- (c) This is the decomposition of a variance into the mean of the conditional variance plus the variance of the conditional mean ([Proposition 2.5.4](#)), applied to the posterior distribution with  $\boldsymbol{\theta}$  as the conditioning variable. The second term is nonnegative definite. ■

Part (b) is Harville's (1974) observation, and it settles a question that the frequentist derivation leaves slightly mysterious. REML looked like a technical device for recovering lost degrees of freedom; it is in fact the exact marginal likelihood of the variance components once the fixed effects have been integrated out with a flat prior. The " $n - p$ " is not a correction but a consequence.

*Idea.* Three readings of the same computation. Frequentist:  $\hat{\mathbf{u}}$  minimizes prediction mean squared error among unbiased linear predictors. Penalized: it minimizes a sum of squares with a ridge penalty on  $\mathbf{u}$ . Bayesian: it is the posterior mean of  $\mathbf{u}$  under a normal prior with covariance  $\mathbf{G}$ . Henderson's equations compute all three at once, and the three readings differ only in what they say about  $\mathbf{G}$ : a covariance to be estimated, a penalty to be tuned, or a prior to be chosen.

## 32.6.2 Priors for variance components

The conditional distributions in [Eq. 32.6.1](#) make the conjugate choices obvious. Given  $\mathbf{u}$  and  $\boldsymbol{\beta}$ , the sum of squares  $\|\mathbf{y} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}\|^2$  is an ordinary normal likelihood for  $\sigma^2$ , so an inverse gamma prior is conjugate; given  $\mathbf{u}$ , the sum  $\|\mathbf{u}\|^2$  plays the same role for  $\sigma_a^2$ . This is what makes the Gibbs sampler below so short. What is not obvious is which inverse gamma.

*Warning.* The prior  $\sigma_a^2 \sim \text{IG}(\epsilon, \epsilon)$  with  $\epsilon$  small is often described as non-informative. It is not, and the limit does not exist. The marginal (restricted) likelihood  $\exp\{\ell_R\}$  is continuous and *strictly positive* at  $\sigma_a^2 = 0$ , because  $\sigma_a^2 = 0$  is a perfectly good model. Any prior density whose integral diverges at the origin, such as  $\pi(\sigma_a^2) \propto 1/\sigma_a^2$  — equivalently a flat prior on  $\log \sigma_a$  — therefore gives an improper posterior:  $\int \pi(\sigma_a^2) e^{\ell_R} d\sigma_a^2 = \infty$ . For small  $\epsilon$  the  $\text{IG}(\epsilon, \epsilon)$  posterior is proper but arbitrarily close to improper, and with few groups the answer depends on  $\epsilon$ . Gelman (2006) made this point and recommended instead a prior that is flat or half- $t$  on the *standard deviation*  $\sigma_a$ , which has finite mass near zero.

A convenient weakly informative default is the half-Cauchy on  $\sigma_a$  with scale  $A$ , chosen large relative to the plausible size of a group effect. It keeps the Gibbs sampler conjugate through the scale-mixture representation

$$\sigma_a^2 \mid \xi \sim \text{IG}(\tfrac{1}{2}, 1/\xi), \quad \xi \sim \text{IG}(\tfrac{1}{2}, 1/A^2),$$

whose marginal for  $\sigma_a$  is half-Cauchy(0,  $A$ ). The auxiliary variable  $\xi$  is sampled alongside everything else.

### 32.6.3 A Gibbs sampler in a dozen lines

For the one-way model with a flat prior on  $\mu$ , the four full conditionals are all standard. Given  $\mu, \sigma^2, \sigma_a^2$ , the effects are independent with

$$a_k \mid \text{rest} \sim N\left(\frac{(m/\sigma^2)(\bar{y}_k - \mu)}{m/\sigma^2 + 1/\sigma_a^2}, \frac{1}{m/\sigma^2 + 1/\sigma_a^2}\right),$$

which is [Eq. 32.3.3](#) with the current parameter values; given the effects,  $\mu \sim N(\overline{y - \mathbf{Za}}, \sigma^2/n)$ ; and the two variances have inverse gamma conditionals with shapes  $n/2$  and  $g/2$  plus their prior shapes. Nothing needs tuning.

```
import numpy as np

g, m = 12, 4
n = g * m
mu, sigma_a, sigma = 2.50, 0.20, 0.25          # the values used to simulate

rng = np.random.default_rng(20320048)
lab_effect = sigma_a * rng.standard_normal(g)
y = mu + np.repeat(lab_effect, m) + sigma * rng.standard_normal(n)

Y = y.reshape(g, m)                          # the REML fit of Section 32.4
ms_between = m * np.sum((Y.mean(axis=1) - Y.mean()) ** 2) / (g - 1)
ms_within = np.sum((Y - Y.mean(axis=1)[:, None]) ** 2) / (n - g)
reml_s2, reml_s2a = ms_within, (ms_between - ms_within) / m

def gibbs(y, g, m, draws=40000, burn=2000, prior="inverse-gamma", A=1.0, seed=0):
    """Posterior draws of (mu, sigma^2, sigma_a^2, a) with a flat prior on mu."""
    rng = np.random.default_rng(seed)
    Y = y.reshape(g, m)
    n = g * m
    mu_, s2, s2a, xi = Y.mean(), Y.var(), 0.1, 1.0
    out = np.empty((draws, 3 + g))
    for t in range(draws + burn):
```

```

prec = m / s2 + 1 / s2a                                # a_k / rest
a = rng.normal((m / s2) * (Y.mean(axis=1) - mu_) / prec, np.sqrt(1 / prec))
mu_ = rng.normal((y - np.repeat(a, m)).mean(), np.sqrt(s2 / n)) # mu /
rest
resid = y - mu_ - np.repeat(a, m)
s2 = 1 / rng.gamma(0.001 + n / 2, 1 / (0.001 + resid @ resid / 2))
if prior == "inverse-gamma":                            # sigma_a^2 / rest
    s2a = 1 / rng.gamma(0.001 + g / 2, 1 / (0.001 + a @ a / 2))
else:                                                    # half-Cauchy(0, A)
    s2a = 1 / rng.gamma((g + 1) / 2, 1 / (1 / xi + a @ a / 2))
    xi = 1 / rng.gamma(1.0, 1 / (1 / A**2 + 1 / s2a))
if t >= burn:
    out[t - burn] = np.concatenate([[mu_, s2, s2a], a])
return out

```

```

# a shorter run than the 40 000 draws quoted in the text, so the cell finishes
quickly
short = gibbs(y, g, m, draws=4000, burn=500, prior="half-Cauchy", seed=12)
print(f"posterior median of sigma_a^2 {np.median(short[:, 2]):.4f}")
print(f"posterior mean of mu {short[:, 0].mean():.4f}")
print(f"REML estimate of sigma_a^2 {reml_s2a:.4f}")

```

### Example. (The proficiency study, four ways)

Run the sampler for 40 000 draws on the data of [Example \(A laboratory proficiency study\)](#). Under the  $\text{IG}(0.001, 0.001)$  prior the posterior median of  $\sigma_a^2$  is 0.0356; under the half-Cauchy prior with  $A = 1$  it is 0.0438. The REML estimate 0.0386 sits between them. The two priors therefore disagree by about a fifth of the estimate, with twelve laboratories — which is already a reason to report the prior.

The posterior means of the twelve effects track the BLUPs closely: their correlation exceeds 0.99 and the largest in absolute value is 0.294 against the BLUP's 0.321, slightly more shrunk — not because the posterior sits below the REML estimate, since its mean 0.0431 is above it, but because the shrinkage factor  $B = m\sigma_a^2 / (\sigma^2 + m\sigma_a^2)$  is concave in  $\sigma_a^2$ : averaging  $B$  over a posterior with mass near zero weights the data less than  $\hat{\sigma}_a^2$  does. Their posterior standard deviations average 0.1218, against the plug-in prediction standard error 0.1156 of [Eq. 32.3.4](#): 5 per cent wider, which is the uncertainty about  $\sigma_a^2$  that the plug-in throws away, exactly as [Proposition 32.6.1\(c\)](#) says. The same happens for the fixed effect: the posterior standard deviation of  $\mu$  is 0.0709, against the plug-in  $\sqrt{\hat{\lambda}_1/n} = 0.0672$ .

Panel (b) of [Figure 32.6.1](#) repeats the analysis on the first 5 laboratories. The posterior medians are now 0.0435 and 0.0693, a difference of sixty per cent, against a REML estimate of 0.0493. With five groups the prior is doing a large part of the work, and saying so is not optional.



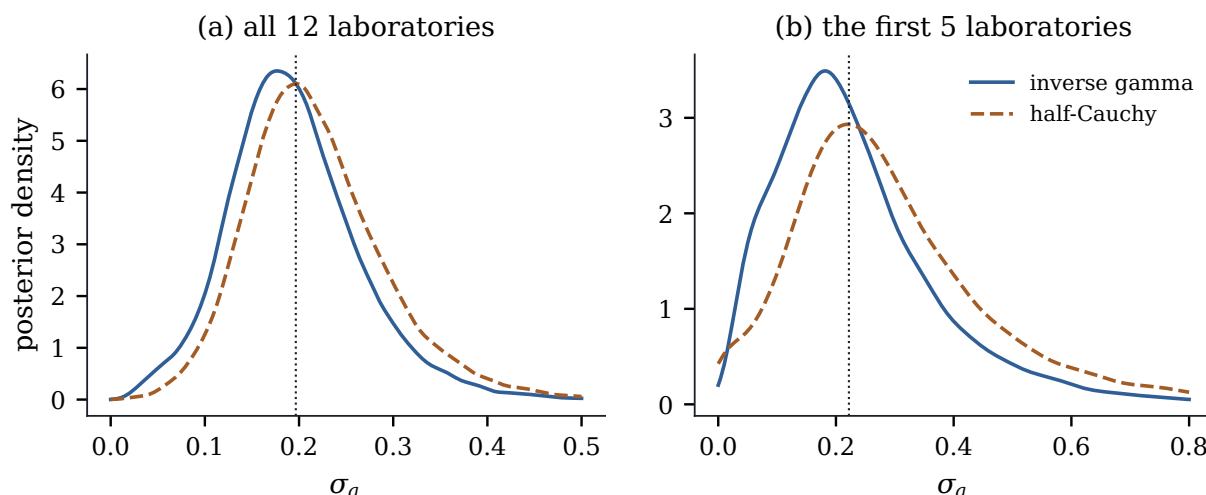


Figure 32.6.1: Posterior densities of the between-laboratory standard deviation  $\sigma_a$  under two priors, from a Gibbs sampler with 40 000 draws. The dotted line marks the REML estimate. (a) All twelve laboratories. (b) The first five, where the prior matters a great deal.

### 32.6.4 What the Bayesian version buys, and what it costs

Three gains. The uncertainty about  $\boldsymbol{\theta}$  is carried through automatically, so intervals for fixed effects and prediction intervals for random effects are honestly wider, with no need for the corrections of [Section 32.5](#). The boundary problem disappears: the posterior of  $\sigma_a^2$  is a proper distribution on  $[0, \infty)$ , with no point mass and no chi-bar-squared mixture to calibrate. And the same machinery handles models with no closed form, which is what [Chapter 33](#) needs for richer covariance structures and [Chapter 40](#) needs once the response is not normal.

Two costs. A prior for the variance components must be chosen, and with few groups it matters, as panel (b) shows; the only defensible practice is to state it and show the sensitivity. And the answers are not exactly the frequentist ones: the two frameworks agree on the point estimates, but they part company on the spread, and the frequentist coverage of a credible interval for a variance component is not guaranteed ([Proposition 12.6.4](#)).

### 32.6.5 Exercises

#### 32.6.5.1 A. Check your understanding

##### Exercise A1.

Explain in two sentences why a random effect and a prior are formally the same object, and name one thing that distinguishes them in practice.

#### 32.6.5.2 B. Practice

##### Exercise B1.

Show that in the one-way model the marginal posterior density of  $\sigma_a^2$  under the prior  $\pi(\sigma_a^2) \propto \sigma_a^{-2}$  is not integrable, by showing that  $\exp\{\ell_R(\sigma_a^2)\}$  tends to a strictly positive limit as  $\sigma_a^2 \downarrow 0$  with  $\sigma^2$  profiled out. Contrast with the behaviour of the prior  $\pi(\sigma_a) \propto 1$ .

*Solution.* By [Eq. 32.4.8](#),  $-2\ell_R$  is continuous in  $(\lambda_1, \lambda_2)$  on  $\lambda_1 \geq \lambda_2 > 0$ , and at  $\sigma_a^2 = 0$  it equals  $(n-1)\log \lambda_2 + (\text{SSB} + \text{SSE})/\lambda_2$ , finite and minimized at a positive  $\lambda_2$ . So  $\exp\{\ell_R\} \rightarrow c > 0$  as  $\sigma_a^2 \downarrow 0$ , and  $\int_0^\delta c \sigma_a^{-2} d\sigma_a^2 = \infty$ . With  $\pi(\sigma_a) \propto 1$  the Jacobian gives  $\pi(\sigma_a^2) \propto \sigma_a^{-1}$ , and  $\int_0^\delta \sigma_a^{-1} d(\sigma_a^2) = 2 \int_0^{\sqrt{\delta}} d\sigma_a < \infty$ : the posterior is proper near the origin.

### Exercise B2.

In [Proposition 32.6.1\(c\)](#), suppose  $\boldsymbol{\theta}$  takes only two values with posterior probabilities  $1/2$  each, giving conditional posterior means  $\hat{\mathbf{u}}_1, \hat{\mathbf{u}}_2$  and common conditional variance  $v$  for a coordinate. Compute the correct marginal posterior variance, and say how large the disagreement between  $\hat{u}_1$  and  $\hat{u}_2$  must be before the plug-in understates the standard deviation by ten per cent.

### 32.6.5.3 C. Going deeper

#### Exercise C1.

Formula [Eq. 32.6.2](#) shows REML is the posterior mode of  $\boldsymbol{\theta}$  under a flat prior on  $\boldsymbol{\theta}$ . A mode is not invariant to reparameterization: the mode of the posterior of  $\sigma_a^2$  is not the square of the mode of the posterior of  $\sigma_a$ . Reconcile this with [Theorem 32.4.2\(c\)](#), which says the REML estimate is well defined. What is invariant, and what is not?

#### Exercise C2.

Take the balanced one-way model with  $\sigma^2$  known,  $m = 1$  so that  $\bar{y}_k = y_k$ , and  $\mu$  known to be zero. Show that plugging the unbiased moment estimate of  $\sigma_a^2$  into [Eq. 32.3.3](#) gives a shrinkage factor of the form  $1 - c\sigma^2/\sum_k y_k^2$ , and compare with the James–Stein estimator of [Theorem 27.5.2](#). Which constant  $c$  does the plug-in give, and which does James–Stein use?

*Solution.* With  $m = 1$  and  $\mu = 0$  the data are  $y_k \sim N(a_k, \sigma^2)$  with  $a_k \sim N(0, \sigma_a^2)$ , so marginally  $y_k \sim N(0, \sigma^2 + \sigma_a^2)$ . The moment estimate from  $\sum_k y_k^2$  is  $\hat{\sigma}_a^2 = \sum_k y_k^2/g - \sigma^2$ , and

$$B = \frac{\hat{\sigma}_a^2}{\hat{\sigma}_a^2 + \sigma^2} = 1 - \frac{g\sigma^2}{\sum_k y_k^2},$$

so  $c = g$  in the stated form. The James–Stein estimator of [Theorem 27.5.2](#) uses  $1 - (g-2)\sigma^2/\sum_k y_k^2$ . The difference is the same degrees-of-freedom correction that separates ML from REML:  $g-2$  is the constant that makes  $(g-2)\sigma^2/\sum_k y_k^2$  unbiased for  $\sigma^2/(\sigma^2 + \sigma_a^2)$ , and it is what makes the estimator dominate the unshrunk one for every  $\boldsymbol{\alpha}$ , not merely on average.

## 32.7 Summary and Notes

### 32.7.1 Summary

*Idea.*

1. A factor is random when its levels are a sample and the conclusion is meant for the population of levels. The one-way random effects model has covariance  $\sigma^2\mathbf{I} + \sigma_a^2\mathbf{Z}\mathbf{Z}^\top$ , intraclass correlation  $\sigma_a^2/(\sigma_a^2 + \sigma^2)$ , and unbiased moment estimates from the analysis of variance ([Definition 32.1.1](#) and [Proposition 32.1.1](#)).
2. In a balanced normal layout the between and within sums of squares are independent scaled chi-squares, so MSB/MSE has an exact  $F$  distribution for every value of the variance ratio. Inverting it gives an exact interval for  $\sigma_a^2/\sigma^2$  and for the intraclass correlation ([Corollary 32.1.2](#)).
3. The moment estimate of a variance can be negative, with probability  $\Pr\{F(g-1, n-g) < 1/(1+m\gamma)\}$ , which is just over one half when  $\sigma_a^2 = 0$ . Reporting, truncating and re-thinking the model are all defensible responses ([Eq. 32.1.4](#)).
4. The linear mixed model  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u} + \boldsymbol{\varepsilon}$  has marginal mean  $\mathbf{X}\boldsymbol{\beta}$  and covariance  $\mathbf{V} = \mathbf{Z}\mathbf{G}\mathbf{Z}^\top + \mathbf{R}$ , a structured member of the general Gauss–Markov family. Woodbury’s identity inverts  $\mathbf{V}$  at the cost of a  $q \times q$  inverse ([Definition 32.2.1](#) and [Proposition 32.2.1](#)).
5. Ordinary least squares is efficient exactly when  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , which holds for cluster-constant regressors in a balanced design and fails as soon as a regressor varies within clusters; the mixed estimate then interpolates between the within and between estimators ([Example \(Cluster-constant regressors\)](#) and [Example \(Regressors that vary within a cluster\)](#)).
6. The best linear unbiased predictor of  $\boldsymbol{\lambda}^\top\boldsymbol{\beta} + \mathbf{c}^\top\mathbf{u}$  is  $\boldsymbol{\lambda}^\top\hat{\boldsymbol{\beta}} + \mathbf{c}^\top\mathbf{G}\mathbf{Z}^\top\mathbf{V}^{-1}(\mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}})$ : the best linear predictor with  $\boldsymbol{\beta}$  replaced by its generalized least squares estimate. Its prediction error variance has two terms, one for predicting  $\mathbf{u}$  and one for estimating  $\boldsymbol{\beta}$  ([Theorem 32.3.1](#), which is [Theorem 14.2.8](#) of [Section 14.2](#) written for mixed functions).
7. In the balanced one-way model the BLUP is  $B(\bar{y}_k - \bar{y})$  with  $B = m\gamma/(1+m\gamma)$ : shrinkage towards the overall mean, of the empirical Bayes kind, with a genuine gain in mean squared error and a conditional bias that matters for ranking ([Eq. 32.3.3](#)).
8. Henderson’s mixed model equations solve for  $(\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}})$  in one sparse system of size  $p+q$ ; they are the normal equations of a penalized least squares criterion with penalty  $\mathbf{u}^\top\mathbf{G}^{-1}\mathbf{u}$ , and the inverse of their coefficient matrix holds both  $\text{Cov}(\hat{\boldsymbol{\beta}})$  and the prediction error variance ([Theorem 32.3.2](#)).
9. Restricted maximum likelihood is the likelihood of a maximal set of error contrasts. It does not depend on which contrasts are used, it equals the profile likelihood plus  $-\frac{1}{2}\log\det(\mathbf{X}^\top\mathbf{V}^{-1}\mathbf{X})$  up to constants, and it reduces to  $s^2 = \text{SSE}/(n-p)$  when  $\mathbf{V} = \sigma^2\mathbf{I}$  ([Definition 32.4.1](#) and [Theorem 32.4.2](#)).
10. In the balanced one-way model, the interior maximum likelihood estimate underestimates  $\sigma_a^2$  by exactly  $\lambda_1/(gm)$ , while REML reproduces the unbiased moment estimates. The two score equations differ only in  $\text{tr}(\mathbf{V}^{-1}\mathbf{V}_j)$  against  $\text{tr}(\mathbf{P}\mathbf{V}_j)$  ([Proposition 32.4.3](#) and [Proposition 32.4.4](#)).

11. Feasible generalized least squares is still unbiased when the variance estimate is translation invariant and even. Its Wald statistic is not normal: in a balanced cluster-randomized trial it is exactly  $t(g-2)$ , and treating it as normal costs real coverage. Satterthwaite and Kenward–Roger approximate the right reference distribution in general ([Proposition 32.5.1](#)).
12. Testing  $\sigma_a^2 = 0$  puts the null on the boundary. In the balanced one-way model the restricted likelihood ratio is an explicit increasing function of the  $F$  ratio, zero with probability  $\Pr(F \leq 1)$ , and its exact null distribution follows; asymptotically it is the chi-bar-squared mixture  $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$  ([Theorem 32.5.2](#), the variance-component companion of [Proposition 32.5.1](#)).
13. Restricted likelihoods may be compared only across covariance structures with the same fixed effects; comparisons of fixed effects need the ordinary likelihood ([Section 32.5](#)).
14. In the hierarchical reading, the BLUP is a posterior mean and Henderson’s coefficient matrix is a posterior precision; the restricted likelihood is the marginal posterior of the variance components under a flat prior on  $\boldsymbol{\beta}$ . A plug-in analysis omits the variance of the conditional mean, so its intervals are too short ([Proposition 32.6.1](#)).

## 32.7.2 Notes and sources

**Fixed and random effects.** The separation of variability into components goes back to Fisher (1918), who decomposed the correlation between relatives. Eisenhart (1947) fixed the terminology “model I” and “model II” and made explicit that the choice changes the expected mean squares and therefore which ratio is a test. [Section 32.1](#) follows the expected-mean-squares route of [Chapter 9](#), and [Eq. 32.1.1](#) is the first of Henderson’s (1953) three methods for unbalanced data. Searle, Casella and McCulloch (1992) is the standard reference for variance component estimation, including methods this chapter omits such as Rao’s (1971) MINQUE; Burdick and Graybill (1992) treats intervals for variance components at book length.

**Prediction.** [Theorem 32.3.1](#) is [Section 14.2](#)’s theorem with a random effect in place of the new observation’s error, which is why its proof is a substitution. Best linear unbiased prediction is due to Henderson, in a sequence of papers beginning with the abstract Henderson (1950) and developed in Henderson, Kempthorne, Searle and von Krosigk (1959) and Henderson (1975); the motivating problem was ranking dairy sires from unbalanced records, where shrinkage is the difference between a usable and an unusable ranking. Harville (1976, 1977) gave the modern treatment, including the relation to Bayesian prediction. Robinson (1991) is the best single essay on what BLUP is and why several derivations converge on it, and the warning in [Section 32.3](#) about ranking shrunk predictions is developed there. The equivalence with penalized least squares, and hence with ridge regression ([Theorem 27.2.2](#)), is the bridge to smoothing: Ruppert, Wand and Carroll (2003) build semiparametric regression on it, and [Chapter 43](#) uses it to estimate a smoothing parameter by REML.

**REML.** Restricted maximum likelihood was introduced by Patterson and Thompson (1971) for recovering inter-block information in incomplete block designs, the problem Yates (1940) had posed; [Section 18.5](#) met it in its fixed-effects form. Harville (1974) proved the marginal-posterior characterization that [Proposition 32.6.1\(b\)](#) restates, and Harville (1977) surveyed the area; Hartley and Rao (1967) had treated maximum likelihood for the mixed model a few years earlier, and Corbeil and Searle (1976) established REML’s advantage for small designs by simulation. The determinant identity [Eq. 32.4.5](#) and the identity  $\mathbf{K}(\mathbf{K}^\top \mathbf{V} \mathbf{K})^{-1} \mathbf{K}^\top = \mathbf{P}$

are standard; the proof of the latter given here, through a constrained minimization, seems the shortest route and makes [Lemma 32.4.1\(c\)](#) immediate. On algorithms, Dempster, Laird and Rubin (1977) supplied EM, Laird and Ware (1982) brought it to longitudinal data, and Gilmour, Thompson and Cullis (1995) introduced average information REML. Bates, Mächler, Bolker and Walker (2015) describe a sparse-matrix implementation in which the profiled deviance is evaluated by a single sparse Cholesky update.

**Inference.** The unbiasedness result [Proposition 32.5.1\(a\)](#) is the covariance analogue of the weighted least squares argument of [Section 21.3](#); Kackar and Harville (1981) proved it in this generality, and Kackar and Harville (1984) gave the approximation to the extra variance of  $\hat{\beta}(\hat{\theta})$  that Kenward and Roger (1997) turned into a usable small-sample correction. The degrees-of-freedom approximation of [Eq. 32.5.2](#) is Satterthwaite (1946); Giesbrecht and Burns (1985) extended it to mixed models. For the boundary problem, Self and Liang (1987) gave the general asymptotic theory of likelihood ratios when the true parameter is on the boundary, and Stram and Lee (1994) worked out the mixed-model cases most often needed. Crainiceanu and Ruppert (2004) showed that for a single variance component the exact finite-sample null distribution of the restricted likelihood ratio can be computed, and documented how poor the chi-bar-squared approximation can be; the exact balanced calculation of [Theorem 32.5.2](#) is a special case that can be done by hand, and the proof sketch of the chi-bar-squared limit given after it is our own. One question left to the econometric literature is whether the random effects are uncorrelated with the regressors at all, as [Definition 32.2.1](#) assumes; if they are not, the mixed fit is inconsistent and the within fit of [Example \(Regressors that vary within a cluster\)](#) is not. Hausman (1978) tests this by comparing the two, using the fact that when one estimator is the BLUE the covariance of the difference is the difference of the covariances. Verbeke and Molenberghs (2000) and Pinheiro and Bates (2000) are the standard applied references, the first for the theory of inference in linear mixed models and the second for practice; McCulloch, Searle and Neuhaus (2008) covers both linear and generalized mixed models.

**Bayesian mixed models.** Gelman (2006) changed practice on priors for variance parameters, showing that the inverse gamma with small parameters is not a safe default and recommending half- $t$  priors on standard deviations; the scale-mixture representation used in our sampler is from that paper.

**Sources for the chapter.** The coverage checklist was built from Rencher and Schaalje (2008, chapter 17), Agresti (2015, sections 9.2–9.3), Fahrmeir, Kneib, Lang and Marx (2021, sections 7.1–7.4) and Sen and Srivastava (1990, sections 7.4–7.5). All the text, proofs, examples, data and exercises here are our own. The proficiency study is simulated from stated parameter values with a fixed seed; the investment panel is Grunfeld’s, in the public-domain reconstruction distributed with statsmodels, and continues the analysis begun in [Section 6.6](#).

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# Chapter 33

## Clustered, Longitudinal and Split-Plot Data

Most data arrive in groups. Plots lie in fields, pupils sit in classrooms, patients return for repeated visits, firms are followed for twenty years. Two observations from the same group are more alike than two from different groups, and the error covariance is no longer  $\sigma^2\mathbf{I}$ . This chapter is the applied half of Part VII: it takes the general Gauss–Markov theory of [Chapter 31](#) and the mixed models of [Chapter 32](#) and puts them to work on the three structures that account for most correlated data in practice. A split-plot experiment randomizes at two levels and therefore has two error terms, so some comparisons are precise and others are not. A repeated-measures analysis needs a condition on the covariance that is weaker than equicorrelation and stronger than nothing, and there are three standard ways of dealing with its failure. A longitudinal study needs a covariance model chosen from a small catalogue, checked against the data, and hedged with a standard error that survives its being wrong. The chapter closes with growth curves, where each subject has its own trajectory and the whole point is to borrow strength across subjects.

**What you need.** [Chapter 6](#) (projection in a general inner product, [Theorem 6.9.3](#)), [Chapter 9](#) (expected mean squares, [Theorem 9.6.1](#)), [Chapter 16](#) (balanced layouts as Kronecker products, [Lemma 16.1.1](#), [Theorem 16.3.1](#)), [Chapter 18](#) (randomized blocks, [Theorem 18.4.1](#)), [Chapter 19](#) (the effect of a wrong covariance, [Theorem 19.2.2](#)), [Chapter 21](#) (sandwich estimators, [Theorem 21.4.2](#), autoregressive errors, [Definition 21.6.1](#)), [Chapter 29](#) ([Definition 29.2.2](#)), and the two chapters before this one: generalized least squares and its inference ([Definition 31.1.1](#), [Theorem 31.1.1](#), [Theorem 31.2.2](#)) and the mixed model with its predictors and likelihoods ([Definition 32.2.1](#), [Theorem 32.3.1](#), [Theorem 32.3.2](#), [Definition 32.4.1](#)).

### 33.0.1 Roadmap

- [Split-plot designs](#)
- [Clustered data and compound symmetry](#)
- [Repeated measures](#)
- [Covariance models for longitudinal data](#)
- [Growth curves](#)
- [Summary and notes](#)



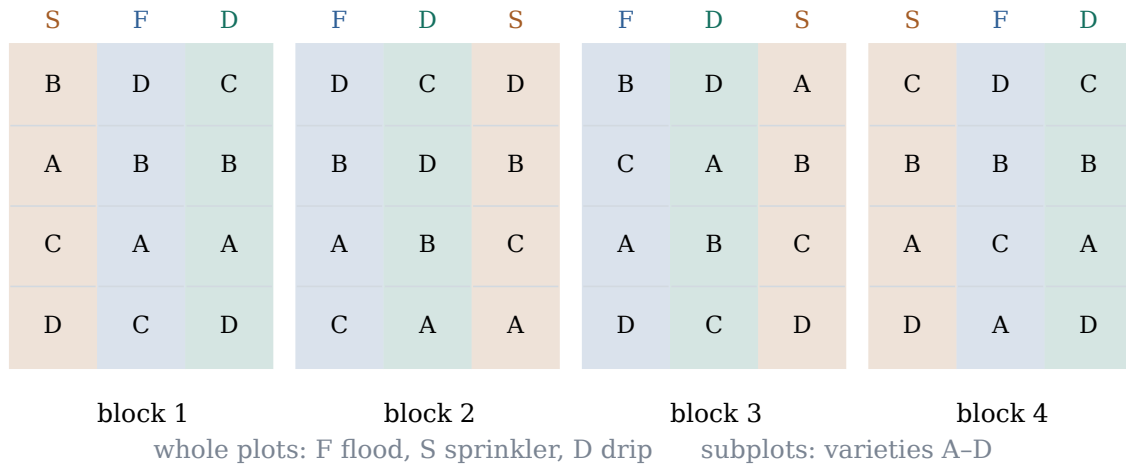


Figure 33.1.1: A split-plot layout with  $r = 4$  blocks,  $a = 3$  irrigation regimes assigned at random within each block, and  $b = 4$  varieties assigned at random within each whole plot.

## 33.1 Split-plot designs

Some treatments cannot be applied to small units: an irrigation rig waters a strip of field, not a square metre of it; a teaching method is delivered to a class, not to a pupil. An experiment combining such a treatment with one applied as finely as one likes gives the coarse treatment to large units and the fine treatment to subdivisions of them — a **split-plot** design, named after the field trials it came from. Two independent randomizations mean two error terms, so that an analysis ignoring the distinction gets both wrong, in opposite directions.

### 33.1.1 The design and its randomization

Take  $r$  blocks. Each is divided into  $a$  **whole plots**, carrying the  $a$  levels of a factor  $A$  in random order: a randomized complete block design (Definition 18.4.1). Each whole plot is then divided into  $b$  **subplots**, carrying the  $b$  levels of a factor  $C$  in random order. The two randomizations are independent, and the second is repeated separately inside every whole plot. Figure 33.1.1 shows one such layout.

Two things vary from subplot to subplot: whatever makes whole plots differ, shared by all  $b$  subplots of a whole plot, and whatever makes subplots differ within a whole plot, which is not. So the model carries two error terms.

#### Definition 33.1.1 — The balanced split-plot model

Index blocks by  $i = 1, \dots, r$ , whole-plot treatments by  $j = 1, \dots, a$  and subplot treatments by  $k = 1, \dots, b$ , and write  $y_{ijk}$  for the response on the subplot of block  $i$  receiving whole-plot treatment  $j$  and subplot treatment  $k$ . The **split-plot model** is

$$y_{ijk} = \mu + \xi_i + \omega_j + \eta_{ij} + \tau_k + \gamma_{jk} + e_{ijk}, \quad (33.1.1)$$

where  $\xi_i$ ,  $\omega_j$ ,  $\tau_k$  and  $\gamma_{jk}$  are fixed effects, the  $\eta_{ij}$  are uncorrelated **whole-plot errors** with mean zero and variance  $\sigma_w^2$ , the  $e_{ijk}$  are uncorrelated **subplot errors** with mean zero and variance  $\sigma_s^2$ , and the two sets are uncorrelated with each other. The **normal split-plot model** takes both sets to be normal.

The whole plot  $(i, j)$  is a cluster of  $b$  observations that share  $\eta_{ij}$ . Writing  $\mathbf{Z}_w$  for the  $n \times ra$  matrix of whole-plot indicators, with  $n = rab$ , the combined error  $\eta_{ij} + e_{ijk}$  has

$$\text{Cov}(\mathbf{Y}) = \mathbf{V} = \sigma_s^2 \mathbf{I} + \sigma_w^2 \mathbf{Z}_w \mathbf{Z}_w^\top = \sigma_s^2 \mathbf{I} + b \sigma_w^2 \mathbf{P}_W, \quad (33.1.2)$$

where  $\mathbf{P}_W = b^{-1} \mathbf{Z}_w \mathbf{Z}_w^\top$  replaces every observation by its whole-plot mean (Example (The one-way layout)). Every pair of subplots in a whole plot has correlation  $\rho = \sigma_w^2 / (\sigma_w^2 + \sigma_s^2)$ , pairs in different whole plots none. The next section studies this structure in general; here balance makes everything explicit.

Where does the equicorrelation come from? In a randomized experiment, random assignment plus unit-treatment additivity *produces* the model rather than assuming it (Section 15.1), and here the argument runs twice: the second randomization is performed anew in every whole plot, making the subplots of a whole plot exchangeable — exactly the covariance Eq. 33.1.2. When the subplot factor is not randomized, as it is not when that factor is time (Section 33.3), the justification disappears and with it the equicorrelation.

### 33.1.2 Two strata

Order the observations with  $k$  fastest, then  $j$ , then  $i$ , and use the averaging and centring matrices  $\bar{\mathbf{J}}_l$ ,  $\mathbf{C}_l$  of Section 16.1. Define

$$\begin{aligned} \mathbf{P}_0 &= \bar{\mathbf{J}}_r \otimes \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b, & \mathbf{P}_B &= \mathbf{C}_r \otimes \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b, & \mathbf{P}_A &= \bar{\mathbf{J}}_r \otimes \mathbf{C}_a \otimes \bar{\mathbf{J}}_b, \\ \mathbf{P}_{AB} &= \mathbf{C}_r \otimes \mathbf{C}_a \otimes \bar{\mathbf{J}}_b, & \mathbf{P}_C &= \bar{\mathbf{J}}_r \otimes \bar{\mathbf{J}}_a \otimes \mathbf{C}_b, & \mathbf{P}_{AC} &= \bar{\mathbf{J}}_r \otimes \mathbf{C}_a \otimes \mathbf{C}_b, \end{aligned} \quad (33.1.3)$$

and  $\mathbf{P}_E = \mathbf{C}_r \otimes \mathbf{I}_a \otimes \mathbf{C}_b$ , the terms of the balanced three-way layout with one observation per cell (Theorem 16.3.1):  $\mathbf{P}_{AB}$  is the block-by- $A$  interaction, and  $\mathbf{P}_E$  collects the block-by- $C$  and three-factor interactions. The split-plot analysis rests on the fact that these seven projections fall into two groups according to how  $\mathbf{V}$  acts on them.

#### Lemma 33.1.1 — The two error strata

Let  $\mathbf{V}$  be as in Eq. 33.1.2 and put  $\lambda_w = \sigma_s^2 + b \sigma_w^2$  and  $\lambda_s = \sigma_s^2$ . Then:

- $\mathbf{P}_W = \mathbf{I}_r \otimes \mathbf{I}_a \otimes \bar{\mathbf{J}}_b$ , and  $\mathbf{V} = \lambda_w \mathbf{P}_W + \lambda_s (\mathbf{I} - \mathbf{P}_W)$ , so  $\mathbf{V}$  has the two eigenvalues  $\lambda_w$  and  $\lambda_s$  with eigenspaces  $\mathcal{C}(\mathbf{P}_W)$  and  $\mathcal{C}(\mathbf{P}_W)^\perp$ , of dimensions  $ra$  and  $ra(b-1)$ .
- The six matrices of Eq. 33.1.3 together with  $\mathbf{P}_E$  are mutually orthogonal projections summing to  $\mathbf{I}$ , with ranks 1,  $r-1$ ,  $a-1$ ,  $(r-1)(a-1)$ ,  $b-1$ ,  $(a-1)(b-1)$  and  $(r-1)a(b-1)$ .
- $\mathbf{P}_0, \mathbf{P}_B, \mathbf{P}_A, \mathbf{P}_{AB}$  have their ranges inside  $\mathcal{C}(\mathbf{P}_W)$ , and  $\mathbf{P}_C, \mathbf{P}_{AC}, \mathbf{P}_E$  have their ranges inside  $\mathcal{C}(\mathbf{P}_W)^\perp$ . Hence  $\mathbf{V}\mathbf{P} = \lambda_w \mathbf{P}$  for the first four and  $\mathbf{V}\mathbf{P} = \lambda_s \mathbf{P}$  for the last three.

*Proof.*

- (a) The whole plot  $(i, j)$  consists of the  $b$  observations with those first two indices, so averaging within whole plots is  $\mathbf{I}_r \otimes \mathbf{I}_a \otimes \bar{\mathbf{J}}_b$ , which is symmetric and idempotent of rank  $ra$  by Lemma 16.1.1(b). Substituting  $\mathbf{P}_W$  in Eq. 33.1.2 gives  $\mathbf{V} = \sigma_s^2(\mathbf{I} - \mathbf{P}_W) + (\sigma_s^2 + b\sigma_w^2)\mathbf{P}_W$ .
- (b) Ranks, symmetry and idempotence are Lemma 16.1.1(b), noting  $\mathbf{P}_E = \mathbf{C}_r \otimes \bar{\mathbf{J}}_a \otimes \mathbf{C}_b + \mathbf{C}_r \otimes \mathbf{C}_a \otimes \mathbf{C}_b$ . Distinct terms differ in at least one slot, so they are orthogonal by Lemma 16.1.1(c), and they are the  $2^3$  products of Lemma 16.1.1(d) regrouped, so they sum to  $\mathbf{I}$ .
- (c) A product  $\mathbf{E}_1 \otimes \mathbf{E}_2 \otimes \bar{\mathbf{J}}_b$  satisfies  $\mathbf{P}_W \mathbf{P} = \mathbf{P}$  by the mixed-product rule and  $\bar{\mathbf{J}}_b \bar{\mathbf{J}}_b = \bar{\mathbf{J}}_b$ ; a product  $\mathbf{E}_1 \otimes \mathbf{E}_2 \otimes \mathbf{C}_b$  satisfies  $\mathbf{P}_W \mathbf{P} = \mathbf{0}$  because  $\bar{\mathbf{J}}_b \mathbf{C}_b = \mathbf{0}$ . The first four matrices are of the first kind and the last three of the second. Then  $\mathbf{V} \mathbf{P} = \lambda_w \mathbf{P}_W \mathbf{P} + \lambda_s (\mathbf{P} - \mathbf{P}_W \mathbf{P})$  gives the two cases. ■

Part (c) is the whole idea. The **whole-plot stratum**  $\mathcal{C}(\mathbf{P}_W)$  carries the whole-plot error intact, so its variance is  $\lambda_w$ ; in the **subplot stratum**  $\mathcal{C}(\mathbf{P}_W)^\perp$  that error is differenced away and only  $\sigma_s^2$  remains.

*Idea.* A split-plot design is two experiments sharing one field: a whole-plot experiment with  $ra$  units and error variance  $\sigma_s^2 + b\sigma_w^2$ , and a subplot experiment with  $rab$  units and error variance  $\sigma_s^2$ . Effects constant within whole plots belong to the first and effects averaging to zero within every whole plot to the second — the interaction among them, which is why such a design studies an interaction well and the coarse main effect badly.

### 33.1.3 The analysis

#### Theorem 33.1.2 — The balanced split-plot design

Assume Eq. 33.1.1 with  $r, a, b \geq 2$ , and write the cell means  $\mu_{ijk} = E(y_{ijk})$ . Let  $\mathbf{M}$  be the orthogonal projection onto the model space  $\mathcal{C}(\mathbf{X}) = \mathcal{C}(\mathbf{P}_0 + \mathbf{P}_B + \mathbf{P}_A + \mathbf{P}_C + \mathbf{P}_{AC})$ .

- (a)  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , so least squares is generalized least squares and every least squares estimate of an estimable function is its best linear unbiased estimator, whatever  $\sigma_w^2$  and  $\sigma_s^2$  are.
- (b) The sums of squares are those of the balanced three-way layout, with  $\text{SS}_A = rb \sum_j (\bar{y}_{\cdot j \cdot} - \bar{y}_{\dots})^2$ ,  $\text{SS}_C = ra \sum_k (\bar{y}_{\cdot \cdot k} - \bar{y}_{\dots})^2$ ,  $\text{SS}_{AC} = r \sum_{j,k} (\bar{y}_{\cdot jk} - \bar{y}_{\cdot j \cdot} - \bar{y}_{\cdot \cdot k} + \bar{y}_{\dots})^2$ , and the two error sums of squares  $\text{SS}_{AB} = \|\mathbf{P}_{AB}\mathbf{y}\|^2$  and  $\text{SSE} = \|\mathbf{P}_E\mathbf{y}\|^2$ .
- (c) The expected mean squares (Theorem 9.6.1) are

$$\begin{aligned}
E MS_B &= \lambda_w + \frac{ab}{r-1} \sum_i (\xi_i - \bar{\xi})^2, & E MS_{AB} &= \lambda_w = \sigma_s^2 + b\sigma_w^2, \\
E MS_A &= \lambda_w + \frac{rb}{a-1} \sum_j (\alpha_j - \bar{\alpha})^2, & E MSE &= \lambda_s = \sigma_s^2, \\
E MS_C &= \lambda_s + \frac{ra}{b-1} \sum_k (\kappa_k - \bar{\kappa})^2, & E MS_{AC} &= \lambda_s + \frac{r}{(a-1)(b-1)} \sum_{j,k} \delta_{jk}^2,
\end{aligned} \tag{33.1.4}$$

where  $\alpha_j = \omega_j + \bar{\gamma}_{j\cdot}$ ,  $\kappa_k = \tau_k + \bar{\gamma}_{\cdot k}$  and  $\delta_{jk} = \gamma_{jk} - \bar{\gamma}_{j\cdot} - \bar{\gamma}_{\cdot k} + \bar{\gamma}$  are the equal-weight effects of Definition 16.2.1 computed from  $\bar{\mu}_{\cdot jk}$ .

- (d) Under normality the six sums of squares are independent,  $SS_{AB}/\lambda_w \sim \chi^2((r-1)(a-1))$ ,  $SSE/\lambda_s \sim \chi^2((r-1)a(b-1))$ , and

$$\frac{MS_A}{MS_{AB}} \sim F\left(a-1, (r-1)(a-1), \frac{rb}{\lambda_w} \sum_j (\alpha_j - \bar{\alpha})^2\right),$$

while  $MS_C/MSE$  and  $MS_{AC}/MSE$  have noncentral  $F$  distributions on  $(b-1, (r-1)a(b-1))$  and  $((a-1)(b-1), (r-1)a(b-1))$  degrees of freedom with noncentralities  $ra \sum_k (\kappa_k - \bar{\kappa})^2/\lambda_s$  and  $r \sum_{j,k} \delta_{jk}^2/\lambda_s$ . So the whole-plot factor is tested against  $MS_{AB}$  and the subplot factor and the interaction against  $MSE$ .

- (e) A contrast  $\sum_j c_j \alpha_j$  is estimated by  $\sum_j c_j \bar{y}_{\cdot j}$  with variance  $\lambda_w \sum_j c_j^2/(rb)$ ; a contrast  $\sum_k d_k \kappa_k$  is estimated by  $\sum_k d_k \bar{y}_{\cdot k}$  with variance  $\lambda_s \sum_k d_k^2/(ra)$ . Both are best linear unbiased, and the two estimators are uncorrelated.
- (f) Fix a subplot treatment  $k$  and a contrast  $\sum_j c_j = 0$ . The simple effect  $\sum_j c_j \bar{\mu}_{\cdot jk}$  is estimated by  $\sum_j c_j \bar{y}_{\cdot jk}$ , best linear unbiased, with variance

$$\left(\sum_j c_j^2\right) \frac{\lambda_w + (b-1)\lambda_s}{rb} = \left(\sum_j c_j^2\right) \frac{\sigma_w^2 + \sigma_s^2}{r}, \tag{33.1.5}$$

which belongs to neither stratum: no single mean square estimates it, and the unbiased estimate is the combination  $(\sum_j c_j^2)\{MS_{AB} + (b-1)MSE\}/(rb)$ .

*Proof.*

- (a) Every  $\mathbf{v} \in \mathcal{C}(\mathbf{X})$  splits as  $\mathbf{v} = \mathbf{v}_1 + \mathbf{v}_2$  with  $\mathbf{v}_1 \in \mathcal{C}(\mathbf{P}_0 + \mathbf{P}_B + \mathbf{P}_A)$  and  $\mathbf{v}_2 \in \mathcal{C}(\mathbf{P}_C + \mathbf{P}_{AC})$ , because the five projections are mutually orthogonal and sum to  $\mathbf{M}$ . By Lemma 33.1.1(c),  $\mathbf{V}\mathbf{v} = \lambda_w \mathbf{v}_1 + \lambda_s \mathbf{v}_2$ , again in  $\mathcal{C}(\mathbf{X})$ . So  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , and Theorem 6.9.3 says the ordinary and the  $\mathbf{V}^{-1}$  projection onto  $\mathcal{C}(\mathbf{X})$  coincide. The estimator is therefore both the least squares and the generalized least squares estimator, which is the best linear unbiased estimator by Corollary 7.2.3. (This is the balanced case of Theorem 31.2.2.)
- (b) Each projection averages over the slots carrying  $\bar{\mathbf{J}}$  and centres in the slots carrying  $\mathbf{C}$ . For instance  $\mathbf{P}_A = \bar{\mathbf{J}}_r \otimes \mathbf{I}_a \otimes \bar{\mathbf{J}}_b - \bar{\mathbf{J}}_r \otimes \bar{\mathbf{J}}_a \otimes \bar{\mathbf{J}}_b$  puts  $\bar{y}_{\cdot j\cdot} - \bar{y}_{\cdot\cdot\cdot}$  in position  $(i, j, k)$ , a value repeated  $rb$  times, giving  $SS_A$ ; the same expansion gives  $SS_C$  and  $SS_{AC}$ , as in Theorem 16.3.1(b). The error sums of squares are the squared lengths of the remaining components.
- (c) Each  $\mathbf{P}$  in the list satisfies  $\mathbf{V}\mathbf{P} = \lambda\mathbf{P}$  with  $\lambda$  the eigenvalue of its stratum, so  $E(\mathbf{Y}^\top \mathbf{P}\mathbf{Y}) = \text{tr}(\mathbf{P}\mathbf{V}) + \boldsymbol{\mu}^\top \mathbf{P}\boldsymbol{\mu} = \lambda \text{rank}(\mathbf{P}) + \|\mathbf{P}\boldsymbol{\mu}\|^2$  by Theorem 2.3.1, where  $\boldsymbol{\mu}$

is the mean vector. Dividing by the rank gives  $E(\text{MS}) = \lambda + \|\mathbf{P}\boldsymbol{\mu}\|^2 / \text{rank}(\mathbf{P})$ . It remains to evaluate  $\mathbf{P}\boldsymbol{\mu}$ . Since  $\boldsymbol{\mu}$  has entries  $\mu + \xi_i + \omega_j + \tau_k + \gamma_{jk}$ , the projection  $\mathbf{P}_A$  returns  $\bar{\mu}_{\cdot j} - \bar{\mu}_{\dots} = \alpha_j - \bar{\alpha}$  on each of the  $rb$  observations with whole-plot treatment  $j$ ; likewise  $\mathbf{P}_C$  returns  $\kappa_k - \bar{\kappa}$  on each of  $ra$  observations,  $\mathbf{P}_{AC}$  returns  $\delta_{jk}$  on each of  $r$ , and  $\mathbf{P}_B$  returns  $\xi_i - \bar{\xi}$  on each of  $ab$ . Finally  $\mathbf{P}_{AB}\boldsymbol{\mu} = \mathbf{0}$  and  $\mathbf{P}_E\boldsymbol{\mu} = \mathbf{0}$ , because  $\boldsymbol{\mu}$  has no block-by-treatment terms.

- (d) Write  $\mathbf{V}^{1/2} = \lambda_w^{1/2}\mathbf{P}_W + \lambda_s^{1/2}(\mathbf{I} - \mathbf{P}_W)$ , a symmetric positive definite square root (Theorem 1.7.6), and put  $\mathbf{z} = \mathbf{V}^{-1/2}\mathbf{Y} \sim N_n(\mathbf{V}^{-1/2}\boldsymbol{\mu}, \mathbf{I})$  (Proposition 2.2.4). For any  $\mathbf{P}$  in the list, Lemma 33.1.1(c) gives  $\mathbf{V}^{1/2}\mathbf{P}\mathbf{V}^{1/2} = \lambda\mathbf{P}$ , so  $\mathbf{Y}^\top\mathbf{P}\mathbf{Y} = \lambda\mathbf{z}^\top\mathbf{P}\mathbf{z}$ . The projections are mutually orthogonal, so Theorem 4.4.4 applied to  $\mathbf{z}$  makes the quadratic forms  $\mathbf{z}^\top\mathbf{P}\mathbf{z}$  independent with noncentral chi-squared distributions on  $\text{rank}(\mathbf{P})$  degrees of freedom and noncentrality  $\|\mathbf{P}\mathbf{V}^{-1/2}\boldsymbol{\mu}\|^2 = \|\mathbf{P}\boldsymbol{\mu}\|^2/\lambda$ , the last equality because  $\mathbf{V}^{-1/2}$  acts on  $\mathcal{C}(\mathbf{P})$  as multiplication by  $\lambda^{-1/2}$ . Forming ratios and cancelling  $\lambda$  between numerator and denominator *within the same stratum* gives the stated  $F$  distributions (Definition 4.1.2). Ratios across strata would not cancel, which is precisely why  $\text{MS}_A/\text{MSE}$  is not an  $F$  statistic.
- (e) The coefficient vector of  $\sum_j c_j \bar{y}_{\cdot j}$  is  $\boldsymbol{\rho} = (rb)^{-1}\mathbf{1}_r \otimes \mathbf{c} \otimes \mathbf{1}_b \in \mathcal{C}(\mathbf{P}_A)$ , since  $\mathbf{c} \perp \mathbf{1}_a$ , so by Lemma 33.1.1(c)  $\text{Var}(\boldsymbol{\rho}^\top\mathbf{Y}) = \lambda_w\|\boldsymbol{\rho}\|^2 = \lambda_w \sum_j c_j^2/(rb)$ . The same computation with  $\boldsymbol{\rho}_C = (ra)^{-1}\mathbf{1}_r \otimes \mathbf{1}_a \otimes \mathbf{d} \in \mathcal{C}(\mathbf{P}_C)$  gives the second variance, and the two are uncorrelated because  $\boldsymbol{\rho}^\top\boldsymbol{\rho}_C = 0$ . Both are least squares estimates of estimable functions, hence best linear unbiased by (a).
- (f) Split  $\mathbf{e}_k = b^{-1}\mathbf{1}_b + (\mathbf{e}_k - b^{-1}\mathbf{1}_b)$ . The coefficient vector  $r^{-1}\mathbf{1}_r \otimes \mathbf{c} \otimes \mathbf{e}_k$  splits accordingly into  $\boldsymbol{\rho}_1 \in \mathcal{C}(\mathbf{P}_A)$  and  $\boldsymbol{\rho}_2 \in \mathcal{C}(\mathbf{P}_{AC})$ , of squared lengths  $\sum_j c_j^2/(rb)$  and  $(b-1)\sum_j c_j^2/(rb)$ . They lie in different strata and are orthogonal, so the variance is  $\lambda_w\|\boldsymbol{\rho}_1\|^2 + \lambda_s\|\boldsymbol{\rho}_2\|^2$ , which is Eq. 33.1.5; the estimate of it is unbiased by Eq. 33.1.4, and the estimator is best linear unbiased by (a).

■

Dropping  $(\xi\omega)_{ij}$  from a full three-way model is what turns the block-by- $A$  interaction into the error term for  $A$ , on only  $(r-1)(a-1)$  degrees of freedom: the formal content of “whole plots are the experimental units for  $A$ ”. And  $\text{MS}_A$  tests a contrast in  $\alpha_j = \omega_j + \bar{\gamma}_{j\cdot}$ , the whole-plot effect averaged over the subplot treatments used, not in  $\omega_j$  alone (Section 16.2).

Part (f) is the case most often got wrong. “The interaction is tested against MSE” tempts one to compare two whole-plot treatments *within a single subplot treatment* against MSE too; but that comparison carries the whole-plot error intact, and Eq. 33.1.5 prices it between the two strata. Its estimated variance combines two mean squares, so no exact  $t$  or  $F$  test exists, and Satterthwaite’s rule (Proposition 32.5.1) gives it

$$\nu = \frac{\{\text{MS}_{AB} + (b-1)\text{MSE}\}^2}{\text{MS}_{AB}^2/\{(r-1)(a-1)\} + (b-1)^2\text{MSE}^2/\{(r-1)a(b-1)\}} \quad (33.1.6)$$

approximate degrees of freedom, which lie between the two exact values.

#### Example. (An irrigation and variety trial)

Four fields are each divided into three whole plots, and the regimes flood, sprinkler and drip are assigned at random within each field; each whole plot is divided into four

subplots carrying four barley varieties in random order, giving  $n = 48$  yields in tonnes per hectare, simulated from Eq. 33.1.1 with  $\sigma_w = 0.35$  and  $\sigma_s = 0.25$  t/ha. The analysis of variance is

| Source                      | df | SS    | MS    | $F$    | $p$    |
|-----------------------------|----|-------|-------|--------|--------|
| blocks                      | 3  | 9.471 | 3.157 |        |        |
| irrigation $A$              | 2  | 9.215 | 4.608 | 10.637 | 0.0106 |
| whole-plot error            | 6  | 2.599 | 0.433 |        |        |
| variety $C$                 | 3  | 1.755 | 0.585 | 9.442  | 0.0002 |
| irrigation $\times$ variety | 6  | 0.958 | 0.160 | 2.577  | 0.0419 |
| subplot error               | 27 | 1.673 | 0.062 |        |        |

The two error mean squares differ by a factor of seven. Solving Eq. 33.1.4 gives  $\hat{\sigma}_s = \sqrt{\text{MSE}} = 0.2489$  and  $\hat{\sigma}_w = \sqrt{(\text{MS}_{AB} - \text{MSE})/b} = 0.3046$ , close to the generating values.

The irrigation means are 4.8075, 5.8000 and 5.6575 t/ha; a difference of two has standard error  $\sqrt{2 \text{MS}_{AB}/(rb)} = 0.2327$  on 6 degrees of freedom, so drip beats flood by 0.8500 t/ha with  $t = 3.653$ . The variety means are 5.134, 5.364, 5.574 and 5.614, and a difference of two has standard error  $\sqrt{2 \text{MSE}/(ra)} = 0.1016$  on 27 degrees of freedom — more than twice as precise, from the same 48 observations. Comparing drip with flood for one variety is neither: by Eq. 33.1.5 its standard error is  $\sqrt{\{\text{MS}_{AB} + 3 \text{MSE}\}/8} = 0.2782$ , on 11.77 Satterthwaite degrees of freedom.

```
import numpy as np

r, a, b = 4, 3, 4                                # blocks, irrigation regimes,
    varieties
n = r * a * b
block_eff = np.array([-0.30, 0.10, 0.05, 0.15])   # field-to-field fertility
irrig_eff = np.array([0.00, 0.45, 0.70])          # flood, sprinkler, drip
variety_eff = np.array([-0.30, -0.10, 0.15, 0.25])
inter = np.array([[0.10, -0.05, -0.10, 0.05],    # irrigation x variety
                  [-0.05, 0.10, 0.00, -0.05],
                  [-0.05, -0.05, 0.10, 0.00]])
sigma_w, sigma_s = 0.35, 0.25                    # whole-plot and subplot errors

rng = np.random.default_rng(3301)
eta = rng.normal(0, sigma_w, (r, a))             # one error per whole plot
e = rng.normal(0, sigma_s, (r, a, b))            # one error per subplot
mean = (5.0 + block_eff[:, None, None] + irrig_eff[None, :, None]
        + variety_eff[None, None, :] + inter[None, :, :])
Y = np.round(mean + eta[:, :, None] + e, 2)      # yields, t/ha
y = Y.reshape(-1)                                # stacked block, then regime,
    then variety
```

The analysis is seven inner products, each projection a Kronecker product as in Section 16.3, with two denominators in place of one.

```
from scipy import stats

def Jbar(k):
    return np.full((k, k), 1.0 / k)

def Cen(k):
```

```

    return np.eye(k) - Jbar(k)

def kron3(E1, E2, E3):
    return np.kron(np.kron(E1, E2), E3)

P = {"blocks": kron3(Cen(r), Jbar(a), Jbar(b)),
     "irrigation": kron3(Jbar(r), Cen(a), Jbar(b)),
     "whole-plot error": kron3(Cen(r), Cen(a), Jbar(b)),
     "varieties": kron3(Jbar(r), Jbar(a), Cen(b)),
     "irrigation x variety": kron3(Jbar(r), Cen(a), Cen(b)),
     "subplot error": kron3(Cen(r), np.eye(a), Cen(b))}
df = {k: int(round(np.trace(M))) for k, M in P.items()}
SS = {k: y @ M @ y for k, M in P.items()}
MS = {k: SS[k] / df[k] for k in P}

F_irrigation = MS["irrigation"] / MS["whole-plot error"]
F_variety = MS["varieties"] / MS["subplot error"]
F_inter = MS["irrigation x variety"] / MS["subplot error"]
for name, F, d in [("irrigation", F_irrigation, df["whole-plot error"]),
                  ("varieties", F_variety, df["subplot error"]),
                  ("irrigation x variety", F_inter, df["subplot error"])] :
    p = stats.f.sf(F, df[name], d)
    print(f"{name:22s} F({df[name]}, {d}) = {F:7.3f}    p = {p:.4f}")

```

### 33.1.4 What pooling costs

Pool the two error sums of squares into one residual mean square on 33 degrees of freedom — in [Example \(An irrigation and variety trial\)](#), 0.129 — and by [Eq. 33.1.4](#) the consequences are opposite in the two strata. The whole-plot denominator falls towards  $\lambda_s$ , inflating the irrigation  $F$  to 35.591 on (2, 33) degrees of freedom,  $p = 5.78 \times 10^{-9}$  against 0.0106, and shrinking the standard error of an irrigation difference from 0.2327 to 0.1272, a factor of 1.829 too small; the subplot denominator exceeds  $\lambda_s$ , and that of a variety difference rises from 0.1016 to 0.1469, a factor of 1.445 too large. Pooling manufactures significance for the treatment the design studies badly and destroys it for the one it studies well.

*Warning.* The two error terms are a property of the *randomization*, not of the data. A split-plot experiment analysed as a completely randomized three-factor experiment produces a plausible table with one residual line, and nothing in the numbers announces the mistake. Wherever one factor is applied to groups of units and another within them, look for two errors.

### 33.1.5 The same analysis as a mixed model

[Eq. 33.1.1](#) is a linear mixed model ([Definition 32.2.1](#)) with  $\mathbf{Z} = \mathbf{Z}_w$ ,  $\mathbf{G} = \sigma_w^2 \mathbf{I}_{ra}$  and  $\mathbf{R} = \sigma_s^2 \mathbf{I}_n$ , whose marginal covariance  $\mathbf{ZGZ}^\top + \mathbf{R}$  is [Eq. 33.1.2](#). In the balanced case, fitting it by restricted maximum likelihood ([Definition 32.4.1](#)) reproduces the analysis above exactly, provided  $\text{MS}_{AB} \geq \text{MSE}$ : the restricted likelihood depends on the data only through the two mean squares, and is maximized at the moment estimates of [Example \(An irrigation and variety trial\)](#) ([Exercise \(B2\)](#)). Satterthwaite's rule ([Proposition 32.5.1](#)) then returns exactly  $(r-1)(a-1)$  degrees of freedom for a whole-plot contrast, and [Eq. 33.1.6](#) for a simple effect.



The route earns its keep when the balance fails: a missing subplot, a whole plot lost to flooding, a covariate measured at the subplot level. The strata are then no longer orthogonal, no mean square estimates  $\lambda_w$  or  $\lambda_s$  on its own, and Henderson's equations ([Theorem 32.3.2](#)) do the computing. Two variations change nothing essential: the whole-plot design may be completely randomized or a Latin square ([Definition 18.5.1](#)), which changes only the whole-plot degrees of freedom, and splitting can be repeated, giving three strata ([Exercise \(C1\)](#)).

## 33.1.6 Exercises

### 33.1.6.1 A. Check your understanding

#### Exercise A1.

For the design of [Example \(An irrigation and variety trial\)](#), say which stratum each of the following belongs to, and on how many degrees of freedom its error is estimated: the difference between two block means; between two variety means; between two irrigation means; between two varieties within the drip regime; the difference in the effect of drip versus flood between two varieties; and the difference between drip and flood for the first variety alone.

*Solution.* The block and irrigation differences are constant within whole plots, so they lie in the whole-plot stratum and use  $MS_{AB}$  on 6 degrees of freedom. The next three are contrasts summing to zero within each whole plot, so they lie in the subplot stratum and use MSE on 27 degrees of freedom. The last lies in neither: it is [Theorem 33.1.2\(f\)](#), with estimated variance  $\{MS_{AB} + 3MSE\}/8$  on about 11.8 Satterthwaite degrees of freedom.

#### Exercise A2.

With  $r$ ,  $a$ ,  $\sigma_w^2$  and  $\sigma_s^2$  fixed, what happens to the variance of a whole-plot contrast, and to that of a subplot contrast, as  $b$  grows? Which kind of comparison does adding subplots help?

### 33.1.6.2 B. Practice

#### Exercise B1.

Verify [Theorem 33.1.2\(a\)](#) directly from [Theorem 6.9.3](#) by computing  $\mathbf{V}\mathbf{X}$  for the model matrix  $\mathbf{X} = [\mathbf{1}, \mathbf{B}, \mathbf{A}, \mathbf{C}, \mathbf{AC}]$  of indicator columns. Then show that adding the block-by- $A$  columns leaves the condition true but destroys the analysis, and say why.

#### Exercise B2.

Show that for the balanced split-plot model the restricted likelihood of  $(\sigma_w^2, \sigma_s^2)$  depends on the data only through  $SS_{AB}$  and  $SSE$ , and that it is maximized at  $\hat{\lambda}_w = MS_{AB}$ ,  $\hat{\lambda}_s = MSE$  provided  $MS_{AB} \geq MSE$ . *Hint:* the error contrasts span  $\mathcal{C}(\mathbf{P}_{AB}) \oplus \mathcal{C}(\mathbf{P}_E)$ , and  $\mathbf{V}$  acts as a scalar on each.



*Solution.* The error contrasts are the  $\mathbf{a}^\top \mathbf{y}$  with  $\mathbf{a} \in \mathcal{C}(\mathbf{X})^\perp = \mathcal{C}(\mathbf{P}_{AB}) \oplus \mathcal{C}(\mathbf{P}_E)$ . Take orthonormal bases of the two summands, of sizes  $f_w = (r-1)(a-1)$  and  $f_s = (r-1)a(b-1)$ . By Lemma 33.1.1(c) the transformed data are independent,  $N(0, \lambda_w)$  for the first basis and  $N(0, \lambda_s)$  for the second. By Definition 32.4.1 the restricted log likelihood is therefore  $-\frac{1}{2}\{f_w \log \lambda_w + \text{SS}_{AB}/\lambda_w + f_s \log \lambda_s + \text{SSE}/\lambda_s\}$  up to a constant, a sum of two ordinary normal-variance likelihoods maximized at the two mean squares. Since  $\sigma_s^2 = \lambda_s$  and  $\sigma_w^2 = (\lambda_w - \lambda_s)/b$ , the constraint  $\sigma_w^2 \geq 0$  is  $\text{MS}_{AB} \geq \text{MSE}$ .

### 33.1.6.3 C. Going deeper

#### Exercise C1.

A split-split-plot design divides each subplot into  $c$  sub-subplots and randomizes a third factor  $D$  among them, adding an error  $\zeta_{ijk}$  of variance  $\sigma_u^2$  shared by the sub-subplots of a subplot. Write down the covariance matrix in the style of Eq. 33.1.2, find its three eigenvalues and eigenspaces, and state which effects are tested against which of the three error mean squares.

## 33.2 Clustered data and compound symmetry

Whole plots are one instance of a pattern that is everywhere in applied work: observations arrive in groups — households, schools, hospitals, litters, survey clusters — and within a group resemble one another more than they resemble observations from elsewhere. The simplest description gives every pair within a group the same correlation and every pair across groups none. What that does to least squares is sharp and exactly computable: it inflates the variance of some coefficients and *deflates* that of others.

### 33.2.1 The equicorrelated cluster model

#### Definition 33.2.1 — Clustered errors with compound symmetry

The  $n$  observations fall into  $G$  **clusters**, cluster  $g$  containing  $m_g$  of them, with  $n = \sum_g m_g$ . The model is  $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $\text{E } \boldsymbol{\varepsilon} = \mathbf{0}$  and

$$\text{Cov}(\boldsymbol{\varepsilon}) = \mathbf{V} = \text{diag}(\boldsymbol{\Sigma}_1, \dots, \boldsymbol{\Sigma}_G), \quad \boldsymbol{\Sigma}_g = \sigma^2 \{(1 - \rho)\mathbf{I}_{m_g} + \rho \mathbf{1}\mathbf{1}^\top\}, \quad (33.2.1)$$

a block diagonal matrix with one block per cluster. This covariance is called **compound symmetry** or **equicorrelation**, and  $\rho$  is the **intraclass correlation**. The matrix is positive definite iff  $-1/(\max_g m_g - 1) < \rho < 1$ . The clusters are **balanced** if  $m_g = m$  for all  $g$ .

Compound symmetry is what a single shared random effect produces: if  $\varepsilon_i = u_{g(i)} + e_i$  with  $u_g$  and  $e_i$  uncorrelated, of variances  $\sigma_u^2$  and  $\sigma_e^2$ , then Eq. 33.2.1 holds with  $\sigma^2 = \sigma_u^2 + \sigma_e^2$

and  $\rho = \sigma_u^2/\sigma^2$  — the one-way random effects model of [Proposition 32.1.1](#) and [Exercise \(B1\)](#), whose power to inflate an  $F$  statistic was met at [Exercise \(C1\)](#). So read,  $\rho$  is the fraction of variance attributable to the cluster and is nonnegative; negative intraclass correlations occur — a fixed quota within a cluster, competition among littermates — but not from a random-effect representation.

The results below take the clusters balanced,  $m_g = m$ ; the clean statements fail otherwise ([Exercise \(B2\)](#)), balance being what makes the two strata of [Lemma 33.1.1](#) eigenspaces of  $\mathbf{V}$ .

### 33.2.2 Which coefficients suffer

Call  $\mathbf{v}$  **cluster-constant** if it is constant on every cluster and **cluster-centred** if its entries sum to zero within every cluster. With  $\mathbf{P}_W$  the orthogonal projection replacing each entry by its cluster mean, the two conditions are  $\mathbf{P}_W \mathbf{v} = \mathbf{v}$  and  $\mathbf{P}_W \mathbf{v} = \mathbf{0}$ , and every vector is the sum of one of each.

#### Proposition 33.2.1 — Compound symmetry and the design effect

Let [Eq. 33.2.1](#) hold with  $m_g = m$  for all  $g$ , let  $\lambda_w = \sigma^2\{1 + (m-1)\rho\}$  and  $\lambda_s = \sigma^2(1 - \rho)$ , and let  $\text{rank}(\mathbf{X}) = p$ .

- (a)  $\mathbf{V} = \lambda_w \mathbf{P}_W + \lambda_s (\mathbf{I} - \mathbf{P}_W)$ . Its eigenvalues are  $\lambda_w$  with multiplicity  $G$  and  $\lambda_s$  with multiplicity  $n - G$ , and

$$\mathbf{V}^{-1} = \frac{1}{\lambda_s} \left( \mathbf{I} - \frac{m\rho}{1 + (m-1)\rho} \mathbf{P}_W \right).$$

- (b)  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  iff  $\mathcal{C}(\mathbf{P}_W \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ . In particular, if  $\mathbf{X} = [\mathbf{X}_b, \mathbf{X}_w]$  with every column of  $\mathbf{X}_b$  cluster-constant and every column of  $\mathbf{X}_w$  cluster-centred, then least squares is generalized least squares and every least squares estimate of an estimable function is best linear unbiased.
- (c) Let  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  be estimable and let  $\mathbf{c} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-\boldsymbol{\lambda}}$ , so that  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \mathbf{c}^\top \mathbf{y}$ . If  $\mathbf{c}$  is cluster-constant then

$$\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = \{1 + (m-1)\rho\} \cdot \sigma^2 \|\mathbf{c}\|^2,$$

and if  $\mathbf{c}$  is cluster-centred then  $\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = (1 - \rho) \sigma^2 \|\mathbf{c}\|^2$ . In both cases  $\sigma^2 \|\mathbf{c}\|^2$  is the variance the estimate would have under  $\text{Cov}(\boldsymbol{\epsilon}) = \sigma^2 \mathbf{I}$ ; the multiplier is the **design effect**. In either case  $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}$  is best linear unbiased even if (b) fails.

- (d) Suppose  $\mathcal{C}(\mathbf{X}) = \mathcal{S}_b \oplus \mathcal{S}_w$  with  $\mathcal{S}_b \subseteq \mathcal{C}(\mathbf{P}_W)$  of dimension  $p_b$  and  $\mathcal{S}_w \subseteq \mathcal{C}(\mathbf{P}_W)^\perp$  of dimension  $p_w = p - p_b$ . Then the usual residual mean square has

$$E(s^2) = \frac{\lambda_w(G - p_b) + \lambda_s(n - G - p_w)}{n - p}, \quad (33.2.2)$$

and for any  $\mathbf{c}$  as in (c) the usual variance estimate  $v = s^2 \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-\boldsymbol{\lambda}}$  satisfies  $E(v)/\text{Var}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = E(s^2)/\lambda$ , with  $\lambda = \lambda_w$  or  $\lambda_s$  according to the stratum of  $\mathbf{c}$ .

*Proof.*

- (a) Within one cluster,  $\mathbf{1}\mathbf{1}^\top = m\bar{\mathbf{J}}_m$ , so the block is  $\sigma^2\{(1 - \rho)\mathbf{I}_m + m\rho\bar{\mathbf{J}}_m\} = \lambda_s(\mathbf{I}_m -$

$\bar{\mathbf{J}}_m) + \lambda_w \bar{\mathbf{J}}_m$ , using  $\lambda_s + m\rho\sigma^2 = \lambda_w$ . Assembling the blocks gives the stated form, since  $\mathbf{P}_W$  is block diagonal with blocks  $\bar{\mathbf{J}}_m$ . A matrix  $\lambda_w \mathbf{P} + \lambda_s(\mathbf{I} - \mathbf{P})$  with  $\mathbf{P}$  an orthogonal projection of rank  $G$  has the two eigenvalues with those multiplicities and inverse  $\lambda_w^{-1} \mathbf{P} + \lambda_s^{-1}(\mathbf{I} - \mathbf{P}) = \lambda_s^{-1} \{\mathbf{I} - (1 - \lambda_s/\lambda_w) \mathbf{P}\}$ , and  $1 - \lambda_s/\lambda_w = m\rho/\{1 + (m-1)\rho\}$ .

- (b) By (a),  $\mathbf{V}\mathbf{X} = \lambda_s \mathbf{X} + (\lambda_w - \lambda_s) \mathbf{P}_W \mathbf{X}$ , so  $\mathcal{C}(\mathbf{V}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  iff  $\mathcal{C}(\mathbf{P}_W \mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  (the case  $\lambda_w = \lambda_s$ , that is  $\rho = 0$ , being trivial). Under the stated partition,  $\mathbf{P}_W \mathbf{X}_b = \mathbf{X}_b$  and  $\mathbf{P}_W \mathbf{X}_w = \mathbf{0}$ , so  $\mathcal{C}(\mathbf{P}_W \mathbf{X}) = \mathcal{C}(\mathbf{X}_b) \subseteq \mathcal{C}(\mathbf{X})$ . The conclusion is [Theorem 6.9.3](#) together with [Corollary 7.2.3](#), as in [Theorem 33.1.2\(a\)](#); the general statement is [Theorem 31.2.2](#).
- (c)  $\text{Var}(\mathbf{c}^\top \mathbf{Y}) = \mathbf{c}^\top \mathbf{V} \mathbf{c}$ . If  $\mathbf{P}_W \mathbf{c} = \mathbf{c}$  then  $\mathbf{V} \mathbf{c} = \lambda_w \mathbf{c}$  and the variance is  $\lambda_w \|\mathbf{c}\|^2$ ; if  $\mathbf{P}_W \mathbf{c} = \mathbf{0}$  then it is  $\lambda_s \|\mathbf{c}\|^2$ . Under  $\text{Cov}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}$  it would be  $\sigma^2 \|\mathbf{c}\|^2$ , and dividing gives the two multipliers. For the last claim, write  $\lambda$  for  $\lambda_w$  or  $\lambda_s$  according to the stratum of  $\mathbf{c}$ , so that  $\mathbf{V} \mathbf{c} = \lambda \mathbf{c}$  in either case, and let  $\mathbf{a}^\top \mathbf{Y}$  be any linear unbiased estimate of  $\lambda^\top \boldsymbol{\beta}$ . Then  $\mathbf{X}^\top (\mathbf{a} - \mathbf{c}) = \mathbf{0}$ , so  $\mathbf{a} = \mathbf{c} + \mathbf{d}$  with  $\mathbf{d} \perp \mathcal{C}(\mathbf{X})$  and in particular  $\mathbf{c}^\top \mathbf{d} = 0$ . The cross term  $\mathbf{c}^\top \mathbf{V} \mathbf{d} = \lambda \mathbf{c}^\top \mathbf{d}$  vanishes and

$$\mathbf{a}^\top \mathbf{V} \mathbf{a} = \lambda \|\mathbf{c}\|^2 + \mathbf{d}^\top \mathbf{V} \mathbf{d} \geq \lambda \|\mathbf{c}\|^2,$$

with equality iff  $\mathbf{d}^\top \mathbf{V} \mathbf{d} = 0$ , that is  $\mathbf{d} = \mathbf{0}$ ,  $\mathbf{V}$  being positive definite. So  $\mathbf{c}^\top \mathbf{Y}$  is the unique best linear unbiased estimate. This is the criterion behind [Theorem 31.2.2](#): a linear estimate  $\mathbf{c}^\top \mathbf{Y}$  of an estimable function is best as soon as  $\mathbf{V} \mathbf{c} \in \mathcal{C}(\mathbf{X})$ .

- (d) By [Theorem 19.2.2\(b\)](#), whose part (b) uses  $\mathbf{X}$  only through  $\mathbf{M}$  and  $\text{rank}(\mathbf{X}) = p$  and so holds verbatim here,  $E(s^2) = \text{tr}\{(\mathbf{I} - \mathbf{M})\mathbf{V}\}/(n - p)$  with  $\mathbf{M}$  the projection onto  $\mathcal{C}(\mathbf{X})$ . Now  $\text{tr}\{(\mathbf{I} - \mathbf{M})\mathbf{V}\} = \lambda_w \text{tr}\{(\mathbf{I} - \mathbf{M})\mathbf{P}_W\} + \lambda_s \text{tr}\{(\mathbf{I} - \mathbf{M})(\mathbf{I} - \mathbf{P}_W)\}$ . Because  $\mathbf{M} = \mathbf{P}_{S_b} + \mathbf{P}_{S_w}$  with  $\mathbf{P}_{S_b} \mathbf{P}_W = \mathbf{P}_{S_b}$  and  $\mathbf{P}_{S_w} \mathbf{P}_W = \mathbf{0}$ ,

$$\text{tr}\{(\mathbf{I} - \mathbf{M})\mathbf{P}_W\} = \text{tr}(\mathbf{P}_W) - \text{tr}(\mathbf{P}_{S_b}) = G - p_b,$$

and similarly  $\text{tr}\{(\mathbf{I} - \mathbf{M})(\mathbf{I} - \mathbf{P}_W)\} = (n - G) - p_w$ . This gives [Eq. 33.2.2](#). Finally  $v = s^2 \|\mathbf{c}\|^2$  because  $\|\mathbf{c}\|^2 = \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{X} (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda} = \boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{X})^{-1} \boldsymbol{\lambda}$  for estimable  $\boldsymbol{\lambda}$ , so  $E(v)/\text{Var}(\mathbf{c}^\top \mathbf{Y}) = E(s^2) \|\mathbf{c}\|^2 / (\lambda \|\mathbf{c}\|^2)$ .

■

The design effect cuts both ways. A regressor measured at the cluster level — the school's curriculum, the village's water supply — is cluster-constant, and its variance is multiplied by  $1 + (m - 1)\rho$ : with  $m = 50$  and  $\rho = 0.05$  the factor is 3.45, so 5000 pupils in 100 schools carry the information of about 1450 independent pupils, the **effective sample size**  $n/\{1 + (m - 1)\rho\}$ . Since the factor grows linearly in  $m$ , extra clusters buy far more than extra members per cluster ([Exercise \(A2\)](#)). A cluster-centred regressor is in the opposite position: its design effect  $1 - \rho$  is less than one, the shared component cancelling out of within-cluster comparisons — the subplot stratum again.

*Idea.* Clustering does not simply lose information; it moves it from between-cluster to within-cluster comparisons. A design that can ask its question within clusters gains, one that must ask it between clusters loses, in proportion to the cluster size.

Part (d) says what the usual output does with all this:  $s^2$  averages  $\mathbf{V}$  over the residual space while the variance of an estimate reads  $\mathbf{V}$  along one direction, so reported and true variance stand in the ratio  $E(s^2)/\lambda$ , the same for every direction in a stratum.

**Example. (Forty clusters of five)**

Take  $G = 40$  clusters of  $m = 5$ , with  $\rho = 0.4$ ,  $\sigma^2 = 1$ , one cluster-level and one within-cluster regressor besides the intercept, so  $p_b = 2$  and  $p_w = 1$ . Then  $\lambda_w = 2.6$  and  $\lambda_s = 0.6$ , and the design effects are exactly 2.60 and 0.60, as the listing confirms from the full  $200 \times 200$  covariance matrix. By Eq. 33.2.2,  $E(s^2) = \{2.6(38) + 0.6(159)\}/197 = 0.9858$ , so the reported variance of the cluster-level coefficient is 0.379 times the truth and that of the within-cluster coefficient 1.643 times the truth. A nominal 95% interval for the first is about 1.62 times too short, and the simulation below finds its coverage to be 0.759.

```
import numpy as np

m, rho, sigma2 = 5, 0.4, 1.0                                # cluster size, intraclass
                                                         # correlation
S = sigma2 * ((1 - rho) * np.eye(m) + rho * np.ones((m, m)))
w = np.linalg.eigvalsh(S)                                    # one large, m-1 equal small
                                                         # eigenvalues
print("eigenvalues of the cluster covariance:", w.round(4))
print("large:", sigma2 * (1 + (m - 1) * rho), " small:", sigma2 * (1 - rho))

def design(G, m, seed):
    """Cluster indicator, a cluster-level regressor and a within-cluster regressor.
    """
    rng = np.random.default_rng(seed)
    g = np.repeat(np.arange(G), m)
    xb = np.repeat(rng.normal(0, 1, G), m)                  # constant inside each cluster
    xw = rng.normal(0, 1, G * m)
    xw = xw - np.repeat(xw.reshape(G, m).mean(axis=1), m)  # sums to zero inside
                                                         # each cluster
    X = np.column_stack([np.ones(G * m), xb, xw])
    return g, X

G = 40
g, X = design(G, m, 2024)
V = np.kron(np.eye(G), S)                                  # block diagonal, one block per
                                                         # cluster
XtXinv = np.linalg.inv(X.T @ X)
cov_true = XtXinv @ X.T @ V @ X @ XtXinv                  # the real covariance of the
                                                         # OLS estimate
cov_iid = sigma2 * XtXinv                                   # what the usual formula
                                                         # reports
deff = np.diag(cov_true) / np.diag(cov_iid)
print("design effects (intercept, between, within):", deff.round(4))
print("predicted:", [1 + (m - 1) * rho, 1 + (m - 1) * rho, 1 - rho])
```

### 33.2.3 Standard errors that survive the model

Generalized least squares with the right  $\mathbf{V}$  fixes everything, and by Proposition 33.2.1(b) is not even needed for the point estimate when the regressors split into cluster-constant and cluster-centred parts. But  $\rho$  is rarely known and compound symmetry is itself a guess. With many clusters the alternative is to keep ordinary least squares and estimate its covariance from the

clusters themselves: the sandwich idea of [Section 21.4](#) with the cluster in place of the observation. Writing  $\mathbf{X}_g$  and  $\hat{\boldsymbol{\varepsilon}}_g$  for the rows and least squares residuals of cluster  $g$ , the **cluster-robust** covariance estimator is, as in [Exercise \(C1\)](#),

$$\hat{\mathbf{U}}_{\text{CR}} = (\mathbf{X}^\top \mathbf{X})^{-1} \left( \sum_{g=1}^G \mathbf{X}_g^\top \hat{\boldsymbol{\varepsilon}}_g \hat{\boldsymbol{\varepsilon}}_g^\top \mathbf{X}_g \right) (\mathbf{X}^\top \mathbf{X})^{-1}. \quad (33.2.3)$$

Nothing about  $\mathbf{V}$  is assumed except that different clusters are uncorrelated: within a cluster the covariance may be arbitrary, and may differ from cluster to cluster — far weaker than [Eq. 33.2.1](#).

*Remark* (Consistency of the cluster-robust estimator). Let the clusters be independent, cluster  $g$  contributing rows  $\mathbf{X}_g$  and errors  $\boldsymbol{\varepsilon}_g$  of mean zero and covariance  $\boldsymbol{\Sigma}_g$ , with  $p$  fixed, and assume (K1)  $m_g \leq m_{\max}$  for all  $g$ , and  $\mathbb{E}\|\boldsymbol{\varepsilon}_g\|^4 \leq K$ ; (K2)  $\|\mathbf{x}_{(i)}\| \leq C$  for all  $i$ ; (K3)  $n^{-1} \mathbf{X}^\top \mathbf{X} \rightarrow \mathbf{A}$ , positive definite. Write  $\mathbf{U} = \text{Cov}(\hat{\boldsymbol{\beta}}) = (\mathbf{X}^\top \mathbf{X})^{-1} (\sum_g \mathbf{X}_g^\top \boldsymbol{\Sigma}_g \mathbf{X}_g) (\mathbf{X}^\top \mathbf{X})^{-1}$ . Then  $n(\hat{\mathbf{U}}_{\text{CR}} - \mathbf{U}) \rightarrow \mathbf{0}$  in probability as  $G \rightarrow \infty$ ; and if also  $n^{-1} \sum_g \mathbf{X}_g^\top \boldsymbol{\Sigma}_g \mathbf{X}_g \rightarrow \mathbf{B}$ , positive definite, then  $\sqrt{n}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \rightarrow N_p(\mathbf{0}, \mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1})$  and the Wald statistic of a hypothesis  $\mathbf{A}^\top \boldsymbol{\beta}$  of rank  $q$ , formed with  $\hat{\mathbf{U}}_{\text{CR}}$ , tends to  $\chi^2(q)$ .

The proof is that of [Theorem 21.4.2](#) with the cluster in place of the observation: the cluster score  $\mathbf{s}_g = \mathbf{X}_g^\top \boldsymbol{\varepsilon}_g$  replaces  $\mathbf{x}_{(i)} e_i$ , every sum has  $G$  independent terms instead of  $n$ , and each bound goes through with  $m_{\max} C$  in place of  $C$  ([Exercise \(C1\)](#)).

Note where the asymptotics live:  $G \rightarrow \infty$ , with cluster sizes bounded; the number of *observations* is irrelevant. A study of a million patients in eight hospitals has eight independent pieces of information about anything varying at the hospital level, and no covariance estimator manufactures more.

```
def cluster_sandwich(X, resid, g):
    """The cluster-robust covariance estimate: one outer product per cluster."""
    XtXinv = np.linalg.inv(X.T @ X)
    meat = np.zeros((X.shape[1], X.shape[1]))
    for k in np.unique(g):
        s = X[g == k].T @ resid[g == k]                # the cluster's score
        # contribution
        meat += np.outer(s, s)
    return XtXinv @ meat @ XtXinv

rng_one = np.random.default_rng(55)
n, p = X.shape
e_one = (np.linalg.cholesky(S) @ rng_one.standard_normal((m, G))).T.reshape(-1)
y_one = X[:, 1] + e_one                                # the between coefficient is 1
b_one = XtXinv @ X.T @ y_one
res_one = y_one - X @ b_one
s2_one = res_one @ res_one / (n - p)
se_naive_one = np.sqrt(s2_one * XtXinv[1, 1])
se_cr_one = np.sqrt(cluster_sandwich(X, res_one, g)[1, 1] * G / (G - 1))
se_true_one = np.sqrt(cov_true[1, 1])
print(f"between coefficient {b_one[1]:.4f}: naive se {se_naive_one:.4f}, "
      f"cluster-robust {se_cr_one:.4f}, true sd {se_true_one:.4f}")
```

On the data generated here the cluster-level coefficient is 1.0376 against a true value of 1, with a usual standard error of 0.0856, a cluster-robust one of 0.1600 (the listing scales the estimator by the simple factor  $G/(G-1)$ ) and an exact standard deviation of 0.1239: the naive interval

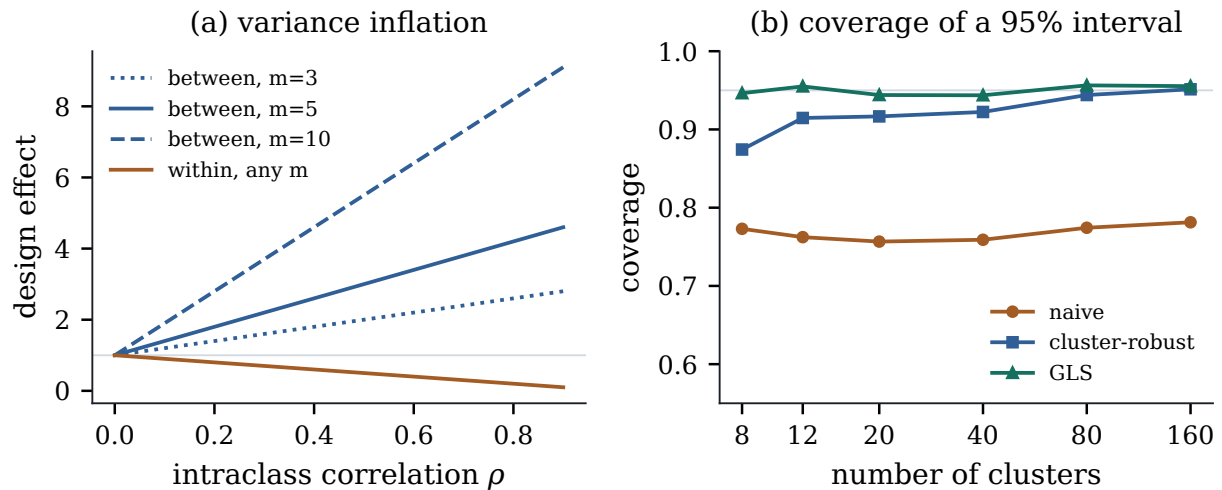


Figure 33.2.1: (a) The design effect of Proposition 33.2.1(c): inflation  $1 + (m - 1)\rho$  for a cluster-level regressor, deflation  $1 - \rho$  for a within-cluster one. (b) Simulated coverage of nominal 95% intervals for a cluster-level coefficient, with  $m = 5$  and  $\rho = 0.4$ .

is far too short, the robust one conservative on this sample.

Figure 33.2.1 shows the pattern over repeated samples, with the fuller correction  $G/(G - 1) \cdot (n - 1)/(n - p)$ . The naive interval covers about three quarters of the time — 0.757 to 0.781 across the six designs — whatever the number of clusters, its error being a bias rather than a variance. The cluster-robust interval is badly short with 8 clusters (coverage 0.874), still short with 20 (0.917) and acceptable by 160 (0.951); generalized least squares with the correct covariance is right throughout.

*Warning.* With few clusters the cluster-robust standard error is not a safe default: its bias is downward and substantial, and referring the statistic to  $t(G - 1)$ , the usual remedy and the one used in the figure, is not enough. Better are a bias correction of the residuals, a bootstrap over whole clusters (Exercise (C1)), or a covariance model fitted by restricted maximum likelihood, which does not need  $G$  large. Twenty clusters is a common rule of thumb; it is a floor, not a target.

## 33.2.4 Exercises

### 33.2.4.1 A. Check your understanding

#### Exercise A1.

Show from Proposition 33.2.1(a) that the compound symmetric matrix is positive definite iff  $-1/(m - 1) < \rho < 1$ , and that the random-effects representation  $\varepsilon_i = u_{g(i)} + e_i$  forces  $\rho \geq 0$ . Give a mechanism that produces  $\rho < 0$ .

#### Exercise A2.

A trial can afford  $n = 1200$  pupils and must randomize schools, not pupils, to the two

arms. The intraclass correlation of the outcome is  $\rho = 0.10$ . Compare the effective sample size for 12 schools of 100 pupils, 40 schools of 30, and 120 schools of 10.

*Solution.* The effective sample size is  $n/\{1+(m-1)\rho\}$ :  $1200/10.9 = 110$ ,  $1200/3.9 = 308$  and  $1200/1.9 = 632$  pupils. As  $m$  grows the effective size tends to  $n/(m\rho) = G/\rho$ , which depends only on the number of schools. For a school-level treatment the number of schools is nearly the whole story, and pupils within a school are cheap but nearly redundant.

### 33.2.4.2 B. Practice

#### Exercise B1.

Let  $\mathbf{X} = [\mathbf{X}_b, \mathbf{X}_w]$  as in Proposition 33.2.1(b), with  $\mathbf{1} \in \mathcal{C}(\mathbf{X}_b)$ . Show that  $\mathbf{M} = \mathbf{M}_b + \mathbf{M}_w$  with  $\mathbf{M}_b$  and  $\mathbf{M}_w$  the projections onto  $\mathcal{C}(\mathbf{X}_b)$  and  $\mathcal{C}(\mathbf{X}_w)$ , that the residual sum of squares splits into a between-cluster and a within-cluster part, and that the two corresponding mean squares are unbiased for  $\lambda_w$  and  $\lambda_s$ .

*Solution.* Any column of  $\mathbf{X}_b$  is fixed by  $\mathbf{P}_W$  and any column of  $\mathbf{X}_w$  is annihilated by it, so  $\mathcal{C}(\mathbf{X}_b) \perp \mathcal{C}(\mathbf{X}_w)$  and Theorem 6.3.8 gives  $\mathbf{M} = \mathbf{M}_b + \mathbf{M}_w$ . Then  $\mathbf{I} - \mathbf{M} = (\mathbf{P}_W - \mathbf{M}_b) + (\mathbf{I} - \mathbf{P}_W - \mathbf{M}_w)$ , a sum of two orthogonal projections of ranks  $G - p_b$  and  $n - G - p_w$ . By Proposition 33.2.1(a),  $\mathbf{V}$  acts as  $\lambda_w$  on the range of the first and  $\lambda_s$  on the range of the second, so Theorem 2.3.1 gives the expectations  $\lambda_w(G - p_b)$  and  $\lambda_s(n - G - p_w)$ . Dividing by the ranks gives the unbiased estimates, of which Eq. 33.2.2 is the pooled average.

#### Exercise B2.

Let the clusters have unequal sizes  $m_g$ . Show that  $\mathbf{V}$  of Eq. 33.2.1 has eigenvalues  $\sigma^2\{1 + (m_g - 1)\rho\}$ , one per cluster, together with  $\sigma^2(1 - \rho)$  with multiplicity  $n - G$ . Deduce that  $\mathbf{1}$  is an eigenvector only if all clusters have the same size, so that the intercept alone breaks Proposition 33.2.1(b).

### 33.2.4.3 C. Going deeper

#### Exercise C1.

Carry out in detail the three steps of the cluster-sandwich argument above, with the constants: show that (K1)–(K3) give  $\|\mathbf{X}_g\| \leq m_{\max}C$  and  $E\|\mathbf{X}_g^\top \boldsymbol{\varepsilon}_g\|^2 \leq m_{\max}^2 C^2 K^{1/2}$ ; that  $\text{Var}(\mathbf{a}^\top \mathbf{X}_g^\top (\boldsymbol{\varepsilon}_g \boldsymbol{\varepsilon}_g^\top - \boldsymbol{\Sigma}_g) \mathbf{X}_g \mathbf{a})$  is bounded uniformly in  $g$ ; and that  $n^{-1} \sum_g \|\mathbf{t}_g\|^2 \leq m_{\max}^2 C^2 (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})^\top \mathbf{A}_n (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ . Where exactly does the number of clusters, rather than of observations, enter?

#### Exercise C2.

Show that  $E(\mathbf{X}_g^\top \hat{\boldsymbol{\varepsilon}}_g \hat{\boldsymbol{\varepsilon}}_g^\top \mathbf{X}_g) \neq \mathbf{X}_g^\top \boldsymbol{\Sigma}_g \mathbf{X}_g$  in general, by writing  $\hat{\boldsymbol{\varepsilon}}_g = \boldsymbol{\varepsilon}_g - \mathbf{X}_g(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}$  and expanding, and show that the leading correction is of order  $p/G$ . Why is the bias downward, and what does this suggest about the factor  $G/(G-1) \cdot (n-1)/(n-p)$  used



in the coverage simulation behind [Figure 33.2.1](#)?

## 33.3 Repeated measures

A repeated-measures study observes the same subject on several occasions: a clustered design whose cluster is the subject and whose “treatments” are the occasions, laid out like the split-plot design of [Section 33.1](#), and for many years analysed that way.

The analogy has one weak joint. What makes the split-plot covariance compound symmetric is the *randomization* of subplot treatments within the whole plot, and occasions cannot be randomized: measurements close together in time are more alike than distant ones, so compound symmetry here is usually false. What is remarkable is that the univariate  $F$  test survives a much weaker condition, and that the condition can be written down exactly.

### 33.3.1 The univariate analysis

Let  $\mathbf{y}_1, \dots, \mathbf{y}_N$  be independent  $N_m(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  vectors, one per subject, with  $\boldsymbol{\Sigma}$  positive definite and unknown, and write  $y_{ij}$  for occasion  $j$  of subject  $i$ . The hypothesis of interest is

$$H : \mu_1 = \dots = \mu_m, \quad \text{that is,} \quad \boldsymbol{\mu} \in \text{span}\{\mathbf{1}\}.$$

Laid out with subjects as rows and occasions as columns the data have one observation per cell, and the classical analysis is the additive two-way analysis of variance of [Section 16.1](#), with the subject-by-occasion interaction as error. With  $\bar{y}_{i\cdot}$ ,  $\bar{y}_{\cdot j}$  and  $\bar{y}$  the subject, occasion and grand means,

$$SS_T = N \sum_{j=1}^m (\bar{y}_{\cdot j} - \bar{y})^2, \quad SS_E = \sum_{i=1}^N \sum_{j=1}^m (y_{ij} - \bar{y}_{i\cdot} - \bar{y}_{\cdot j} + \bar{y})^2, \quad (33.3.1)$$

and the statistic is  $F = \{SS_T/(m-1)\}/\{SS_E/((N-1)(m-1))\}$ . It is the subplot stratum of a split-plot design with one whole plot per subject, and under compound symmetry it is exact ([Theorem 33.3.2\(b\)](#) below, with [Exercise \(B1\)](#)). The question is how much of that exactness survives a general  $\boldsymbol{\Sigma}$ .

Everything reduces to the within-subject contrasts. Let  $\mathbf{U}$  be  $m \times (m-1)$  with orthonormal columns orthogonal to  $\mathbf{1}$ , so that  $\mathbf{U}\mathbf{U}^\top = \mathbf{I} - \bar{\mathbf{J}}_m$ , and set  $\mathbf{z}_i = \mathbf{U}^\top \mathbf{y}_i$ : independent  $N_{m-1}(\mathbf{U}^\top \boldsymbol{\mu}, \boldsymbol{\Psi})$  with  $\boldsymbol{\Psi} = \mathbf{U}^\top \boldsymbol{\Sigma} \mathbf{U}$ , with  $H$  saying exactly that  $\mathbf{U}^\top \boldsymbol{\mu} = \mathbf{0}$ . The subject means carry no information about  $H$ .

### 33.3.2 Sphericity

#### Definition 33.3.1 — Sphericity and Box’s measure

Let  $\boldsymbol{\Sigma}$  be an  $m \times m$  positive definite matrix and  $\mathbf{U}$  as above.  $\boldsymbol{\Sigma}$  satisfies **sphericity** if  $\mathbf{U}^\top \boldsymbol{\Sigma} \mathbf{U} = \lambda \mathbf{I}_{m-1}$  for some  $\lambda > 0$ . **Box’s measure of sphericity** is



$$\epsilon(\Sigma) = \frac{\{\text{tr } \Psi\}^2}{(m-1) \text{tr}(\Psi^2)}, \quad \Psi = \mathbf{U}^\top \Sigma \mathbf{U}. \quad (33.3.2)$$

Neither the condition nor  $\epsilon$  depends on the choice of  $\mathbf{U}$ : two such matrices satisfy  $\tilde{\mathbf{U}} = \mathbf{U}\mathbf{Q}$  with  $\mathbf{Q}$  orthogonal, so  $\tilde{\Psi} = \mathbf{Q}^\top \Psi \mathbf{Q}$  has the same trace, the same trace of square, and is a multiple of the identity exactly when  $\Psi$  is.

### Lemma 33.3.1 — Equivalent forms of sphericity

For a positive definite  $m \times m$  matrix  $\Sigma$  the following are equivalent:

- (i)  $\mathbf{U}^\top \Sigma \mathbf{U} = \lambda \mathbf{I}_{m-1}$ ;
- (ii)  $\text{Var}(Y_j - Y_k) = 2\lambda$  for every  $j \neq k$ ;
- (iii)  $\sigma_{jk} = a_j + a_k + \lambda \delta_{jk}$  for some  $a_1, \dots, a_m$ , where  $\delta_{jk}$  is one if  $j = k$  and zero otherwise;
- (iv)  $\Sigma = \lambda \mathbf{I} + \mathbf{1}\mathbf{a}^\top + \mathbf{a}\mathbf{1}^\top$  for some  $\mathbf{a} \in \mathbb{R}^m$ .

Moreover  $1/(m-1) \leq \epsilon(\Sigma) \leq 1$ , with  $\epsilon = 1$  iff sphericity holds, and compound symmetry implies sphericity but not conversely.

*Proof.* Write  $\mathbf{C} = \mathbf{I} - \bar{\mathbf{J}}_m = \mathbf{U}\mathbf{U}^\top$ . Since  $\mathbf{U}^\top \mathbf{U} = \mathbf{I}_{m-1}$ , condition (i) holds iff  $\mathbf{C}\Sigma\mathbf{C} = \lambda\mathbf{C}$ .

- (i)  $\Rightarrow$  (ii):  $\mathbf{e}_j - \mathbf{e}_k$  is orthogonal to  $\mathbf{1}$ , hence fixed by  $\mathbf{C}$ , so  $\text{Var}(Y_j - Y_k)$  equals  $(\mathbf{e}_j - \mathbf{e}_k)^\top \mathbf{C}\Sigma\mathbf{C}(\mathbf{e}_j - \mathbf{e}_k)$ , which is  $\lambda\|\mathbf{e}_j - \mathbf{e}_k\|^2 = 2\lambda$ .
- (ii)  $\Rightarrow$  (iii):  $\text{Var}(Y_j - Y_k) = \sigma_{jj} + \sigma_{kk} - 2\sigma_{jk} = 2\lambda$  gives  $\sigma_{jk} = (\sigma_{jj} - \lambda)/2 + (\sigma_{kk} - \lambda)/2$  for  $j \neq k$ . Put  $a_j = (\sigma_{jj} - \lambda)/2$ ; then  $\sigma_{jk} = a_j + a_k$  off the diagonal and  $\sigma_{jj} = 2a_j + \lambda$  on it, which is (iii).
- (iii)  $\Leftrightarrow$  (iv) is the same statement in matrix form.
- (iv)  $\Rightarrow$  (i):  $\mathbf{C}\mathbf{1} = \mathbf{0}$ , so  $\mathbf{C}\Sigma\mathbf{C} = \lambda\mathbf{C}$ .

For the bounds, let  $\psi_1, \dots, \psi_{m-1} > 0$  be the eigenvalues of  $\Psi$ . Then  $\epsilon = (\sum_l \psi_l)^2 / \{(m-1) \sum_l \psi_l^2\}$ . The Cauchy–Schwarz inequality gives  $(\sum_l \psi_l)^2 \leq (m-1) \sum_l \psi_l^2$  with equality iff all  $\psi_l$  are equal, so  $\epsilon \leq 1$  with equality iff  $\Psi = \lambda \mathbf{I}$ ; and  $\sum_l \psi_l^2 \leq (\sum_l \psi_l)^2$  for nonnegative numbers gives  $\epsilon \geq 1/(m-1)$ . Compound symmetry is (iv) with all  $a_j$  equal; taking the  $a_j$  unequal gives sphericity without compound symmetry, and such a  $\Sigma$  is positive definite as soon as  $\lambda$  is large enough. ■

Condition (ii) is the form to remember and to check: *all pairwise differences of the repeated measurements have the same variance*. Compound symmetry says more than the  $F$  test needs.

### Theorem 33.3.2 — Sphericity is necessary and sufficient for the exact F test

Let  $\mathbf{y}_1, \dots, \mathbf{y}_N$  be independent  $N_m(\boldsymbol{\mu}, \Sigma)$  with  $\Sigma$  positive definite,  $m \geq 2$ ,  $N \geq 2$ ,

and let  $F$  be the statistic built from Eq. 33.3.1. Put  $\Psi = \mathbf{U}^\top \Sigma \mathbf{U}$ ,  $\mathbf{q} = \sqrt{N} \bar{\mathbf{z}}$  and  $\mathbf{W} = \sum_i (\mathbf{z}_i - \bar{\mathbf{z}})(\mathbf{z}_i - \bar{\mathbf{z}})^\top$ .

- (a)  $SS_T = \|\mathbf{q}\|^2$  and  $SS_E = \text{tr}(\mathbf{W})$ , and under  $H$  the vector  $\mathbf{q} \sim N_{m-1}(\mathbf{0}, \Psi)$  is independent of  $\mathbf{W} \sim W_{m-1}(N-1, \Psi)$ . Hence  $F = N(N-1)\|\bar{\mathbf{z}}\|^2 / \text{tr}(\mathbf{W})$ .
- (b) If  $\Sigma$  satisfies sphericity, then under  $H$   $F \sim F(m-1, (N-1)(m-1))$  exactly, for every  $N \geq 2$ .
- (c) Conversely, if the null distribution of  $F$  is  $F(m-1, (N-1)(m-1))$  for every  $N \geq 2$ , then  $\Sigma$  satisfies sphericity.

*Proof.*

- (a) Since  $\mathbf{C} = \mathbf{U}\mathbf{U}^\top$  and  $\mathbf{U}$  has orthonormal columns,  $\sum_j (\bar{y}_{.j} - \bar{y})^2 = \|\mathbf{C}\bar{\mathbf{y}}\|^2 = \|\bar{\mathbf{z}}\|^2$ , so  $SS_T = N\|\bar{\mathbf{z}}\|^2 = \|\mathbf{q}\|^2$ ; likewise subject  $i$ 's contribution to  $SS_E$  is  $\|\mathbf{z}_i - \bar{\mathbf{z}}\|^2$ , summing to  $\text{tr}(\mathbf{W})$ . Under  $H$  the  $\mathbf{z}_i$  are independent  $N_{m-1}(\mathbf{0}, \Psi)$ , so Lemma 14.2.3 gives the Wishart law of  $\mathbf{W}$  and its independence of  $\bar{\mathbf{z}}$ . Finally  $F = \{\|\mathbf{q}\|^2 / (m-1)\} \{ (N-1)(m-1) / \text{tr} \mathbf{W} \}$ .
- (b) If  $\Psi = \lambda \mathbf{I}_{m-1}$ , then  $\mathbf{q}/\sqrt{\lambda}$  is standard normal, so  $\|\mathbf{q}\|^2/\lambda \sim \chi^2(m-1)$ . And  $\mathbf{W}/\lambda \sim W_{m-1}(N-1, \mathbf{I})$ , which is the matrix of sums of squares and cross products of  $(N-1)(m-1)$  independent standard normal variables, so  $\text{tr}(\mathbf{W})/\lambda \sim \chi^2((N-1)(m-1))$ . The two are independent, and  $\lambda$  cancels from the ratio, leaving the ratio of two independent chi-squared variables each divided by its degrees of freedom (Definition 4.1.2).
- (c) Write  $F_N$  for the statistic with  $N$  subjects. By (a),

$$F_N = \|\mathbf{q}\|^2 \cdot \frac{N-1}{\text{tr}(\mathbf{W})},$$

where  $\mathbf{q} \sim N_{m-1}(\mathbf{0}, \Psi)$  has the *same* distribution for every  $N$  and is independent of  $\mathbf{W}$ . Since  $\mathbf{W}/(N-1)$  is an average of  $N-1$  independent  $W_{m-1}(1, \Psi)$  matrices, the weak law of large numbers gives  $\text{tr}(\mathbf{W})/(N-1) \rightarrow \text{tr} \Psi$  in probability, so by Slutsky's lemma  $F_N \rightarrow \|\mathbf{q}\|^2 / \text{tr} \Psi$  in distribution. On the other hand  $F(m-1, \nu) \rightarrow \chi^2(m-1)/(m-1)$  in distribution as  $\nu \rightarrow \infty$ , because its denominator chi-squared divided by  $\nu$  tends to one in probability. If the exactness holds for every  $N$ , the two limits agree:

$$\|\mathbf{q}\|^2 \stackrel{d}{=} \frac{\text{tr} \Psi}{m-1} \chi^2(m-1).$$

Now match variances. By Theorem 4.2.3,  $\text{Var}(\|\mathbf{q}\|^2) = 2 \text{tr}(\Psi^2)$ , while the right-hand side has variance  $2\{\text{tr} \Psi\}^2/(m-1)$ . Equality says  $(m-1) \text{tr}(\Psi^2) = \{\text{tr} \Psi\}^2$ , that is  $\epsilon(\Sigma) = 1$ , and Lemma 33.3.1 makes that sphericity. ■

*Idea.* The repeated-measures  $F$  test does not require the occasions to be exchangeable, only that the  $m-1$  orthonormalized within-subject contrasts be uncorrelated with equal variances — that the contrast space look spherical, which series of measurements in time almost never do.

### 33.3.3 Two corrections and one alternative

When sphericity fails,  $\|\mathbf{q}\|^2$  and  $\text{tr}(\mathbf{W})$  are still weighted sums of chi-squared variables, and matching their first two moments to scaled chi-squared variables gives a usable approximation (Box 1954).

#### Proposition 33.3.3 — The moment-matched approximation

Under the assumptions of [Theorem 33.3.2](#), with  $H$  true, write  $\epsilon = \epsilon(\mathbf{\Sigma})$ ,  $d_1 = m - 1$  and  $d_2 = (N - 1)(m - 1)$ . Then

- (a)  $E \text{SS}_T = \text{tr } \mathbf{\Psi}$ ,  $\text{Var } \text{SS}_T = 2 \text{tr}(\mathbf{\Psi}^2)$ ,  $E \text{SS}_E = (N - 1) \text{tr } \mathbf{\Psi}$  and  $\text{Var } \text{SS}_E = 2(N - 1) \text{tr}(\mathbf{\Psi}^2)$ .
- (b) The scaled chi-squared variable  $g\chi^2(h)$  matching the first two moments of  $\text{SS}_T$  has  $h = \epsilon d_1$  and  $g = \text{tr}(\mathbf{\Psi}^2) / \text{tr } \mathbf{\Psi}$ ; the one matching  $\text{SS}_E$  has  $h' = \epsilon d_2$  and the *same*  $g$ . Since  $\text{SS}_T$  and  $\text{SS}_E$  are independent by [Theorem 33.3.2\(a\)](#), replacing each by its matched scaled chi-squared gives the approximation  $F \approx F(\epsilon d_1, \epsilon d_2)$ , which is exact when  $\epsilon = 1$ .

*Proof.*

- (a) The first two are [Theorem 4.2.3](#) with  $\mathbf{A} = \mathbf{I}$  and mean zero. Writing  $\mathbf{W} = \sum_{l=1}^{N-1} \mathbf{w}_l \mathbf{w}_l^\top$  with  $\mathbf{w}_l$  independent  $N_{m-1}(\mathbf{0}, \mathbf{\Psi})$  ([Lemma 14.2.3](#)),  $\text{tr } \mathbf{W} = \sum_l \|\mathbf{w}_l\|^2$  is a sum of  $N - 1$  independent copies of  $\|\mathbf{q}\|^2$ , so its mean and variance are  $N - 1$  times theirs.
- (b) Matching  $gh = \text{tr } \mathbf{\Psi}$  and  $2g^2h = 2 \text{tr}(\mathbf{\Psi}^2)$  gives  $g = \text{tr}(\mathbf{\Psi}^2) / \text{tr } \mathbf{\Psi}$  and  $h = \{\text{tr } \mathbf{\Psi}\}^2 / \text{tr}(\mathbf{\Psi}^2) = \epsilon d_1$ . For  $\text{SS}_E$  both moments are multiplied by  $N - 1$ , so  $g$  is unchanged and  $h' = (N - 1)h = \epsilon d_2$ . Since  $d_2/d_1 = h'/h$ , the ratio  $(\text{SS}_T/d_1)/(\text{SS}_E/d_2)$  becomes  $\{\chi^2(h)/h\}/\{\chi^2(h')/h'\}$ , an  $F(h, h')$  variable, the two chi-squared variables being independent by [Theorem 33.3.2\(a\)](#); at  $\epsilon = 1$  this is the exact result of [Theorem 33.3.2\(b\)](#). ■

By [Lemma 33.3.1](#),  $\epsilon \geq 1/(m - 1)$ , so the approximating distribution is never more dispersed than  $F(d_1/(m - 1), d_2/(m - 1)) = F(1, N - 1)$ , the largest that [Proposition 33.3.3\(b\)](#) can ask for. Referring  $F$  to  $F(1, N - 1)$  is therefore conservative within Box's approximation, and Geisser and Greenhouse (1958) proved it exactly conservative, for every  $\mathbf{\Sigma}$  and  $N$ . This is the **lower-bound** correction, turned into a recommendation by Greenhouse and Geisser (1959).

Part (b) becomes a procedure once  $\epsilon$  is estimated from the sample covariance matrix  $\mathbf{S}$  of the  $\mathbf{y}_i$ .

- The **Greenhouse–Geisser** correction refers  $F$  to  $F(\hat{\epsilon}d_1, \hat{\epsilon}d_2)$  with the plug-in  $\hat{\epsilon} = \epsilon(\mathbf{S})$ , which is biased downward, so the test is conservative.
- The **Huynh–Feldt** correction removes the leading bias. For a single group of  $N$  subjects it uses

$$\tilde{\epsilon} = \min \left\{ 1, \frac{N(m - 1)\hat{\epsilon} - 2}{(m - 1)\{(N - 1) - (m - 1)\hat{\epsilon}\}} \right\}, \quad (33.3.3)$$

which exceeds  $\hat{\epsilon}$  and gives a less conservative test; its own bias is upward when  $\epsilon$  is small.

- The **lower-bound** correction takes  $\epsilon = 1/(m-1)$ : safe, and usually too conservative to be useful.

A third option avoids the issue. Since the  $\mathbf{z}_i$  are independent  $N_{m-1}(\mathbf{U}^\top \boldsymbol{\mu}, \boldsymbol{\Psi})$  with  $\boldsymbol{\Psi}$  unrestricted,  $H$  says a multivariate normal mean is zero, and the classical test is Hotelling's.

### Proposition 33.3.4 — The multivariate test

Under the assumptions of Theorem 33.3.2 with  $N \geq m$ , let  $\mathbf{S}_z = \mathbf{W}/(N-1)$  and

$$T^2 = N \bar{\mathbf{z}}^\top \mathbf{S}_z^{-1} \bar{\mathbf{z}}.$$

Then  $T^2$  does not depend on the choice of  $\mathbf{U}$ , nor on replacing  $\mathbf{U}$  by  $\mathbf{U}\mathbf{K}$  for any nonsingular  $(m-1) \times (m-1)$  matrix  $\mathbf{K}$ ; and under  $H$ ,

$$\frac{N-m+1}{(N-1)(m-1)} T^2 \sim F(m-1, N-m+1),$$

whatever  $\boldsymbol{\Sigma}$  is. The distributional statement is Hotelling's and is not proved here; see Anderson (2003, §5.2).

*Proof.* Only the invariance is proved. Replacing  $\mathbf{U}$  by  $\mathbf{U}\mathbf{K}$  replaces  $\bar{\mathbf{z}}$  by  $\mathbf{K}^\top \bar{\mathbf{z}}$  and  $\mathbf{S}_z$  by  $\mathbf{K}^\top \mathbf{S}_z \mathbf{K}$ , leaving  $T^2$  unchanged; two orthonormal bases differ by an orthogonal  $\mathbf{K}$ . The requirement  $N \geq m$  makes  $\mathbf{S}_z$ , a Wishart matrix on  $N-1 \geq m-1$  degrees of freedom, invertible with probability one. ■

Exactness under any  $\boldsymbol{\Sigma}$  sounds decisive, but it costs power: the multivariate test estimates a whole covariance matrix and has  $N-m+1$  denominator degrees of freedom rather than  $(N-1)(m-1)$ . When sphericity nearly holds the univariate test is much more powerful, when it fails badly the two are comparable, and when  $N < m$  the multivariate test does not exist.

## 33.3.4 An example, four ways

### Example. (Grip strength over six visits)

Twenty-four patients have grip strength measured, in kilograms, at six weekly visits, simulated from  $N_6(\boldsymbol{\mu}, \boldsymbol{\Sigma})$  with  $\boldsymbol{\Sigma}$  first-order autoregressive,  $\sigma_{jk} = 4 \times 0.6^{|j-k|}$ , and a mean drifting slowly upward; the observed occasion means are 47.818, 48.099, 49.056, 49.177, 49.561 and 49.263 kg. The true Box measure is  $\epsilon = 0.700$ : adjacent visits correlate 0.6 and visits five weeks apart only  $0.6^5$ , so adjacent differences are much less variable than distant ones. The four analyses of  $H$  give

| Analysis                   | statistic | degrees of freedom | $p$    |
|----------------------------|-----------|--------------------|--------|
| univariate,<br>uncorrected | 4.770     | 5, 115             | 0.0005 |

|                                                  |       |            |        |
|--------------------------------------------------|-------|------------|--------|
| Greenhouse–<br>Geisser, $\hat{\epsilon} = 0.645$ | 4.770 | 3.23, 74.2 | 0.0035 |
| Huynh–Feldt,<br>$\tilde{\epsilon} = 0.763$       | 4.770 | 3.81, 87.7 | 0.0019 |
| multivariate                                     | 3.605 | 5, 19      | 0.0184 |

All four reject, but the evidence differs by a factor of nearly forty between first and last. Distrust the uncorrected test: its nominal degrees of freedom are too large by the factor  $\epsilon$ .

```
import numpy as np
from scipy import stats

N, m = 24, 6
mu_true = np.array([48.0, 48.5, 48.9, 49.2, 49.4, 49.5]) # kg, a slow improvement
sigma, rho_true = 2.0, 0.6
lags = np.abs(np.subtract.outer(np.arange(m), np.arange(m)))
Sigma_true = sigma**2 * rho_true**lags # AR(1) within a patient
rng = np.random.default_rng(3303)
Yobs = np.round(mu_true + rng.multivariate_normal(np.zeros(m), Sigma_true, N), 2)

def contrast_basis(m):
    """Orthonormal basis U of the space of contrasts among m repeated measures."""
    Q, _ = np.linalg.qr(np.column_stack([np.ones(m), np.eye(m)[:, 1:]]))
    return Q[:, 1:] # columns orthonormal and
                    # orthogonal to 1

def epsilon(Psi):
    """Box's measure of departure from sphericity, 1/(m-1) <= eps <= 1."""
    k = Psi.shape[0]
    return np.trace(Psi)**2 / (k * np.sum(Psi * Psi))

U = contrast_basis(m)
Z = Yobs @ U # N x (m-1) contrasts, one row
              # per patient
zbar = Z.mean(axis=0)
Sz = np.cov(Z, rowvar=False, ddof=1) # Wishart, (N-1) degrees of
                                     # freedom

F_uncorrected = N * (zbar @ zbar) / np.trace(Sz)
eps_hat = epsilon(Sz)
eps_hf = min(1.0, (N * (m - 1) * eps_hat - 2)
             / ((m - 1) * ((N - 1) - (m - 1) * eps_hat)))
d1, d2 = m - 1, (N - 1) * (m - 1)
p_un = stats.f.sf(F_uncorrected, d1, d2)
p_gg = stats.f.sf(F_uncorrected, d1 * eps_hat, d2 * eps_hat)
p_hf = stats.f.sf(F_uncorrected, d1 * eps_hf, d2 * eps_hf)
T2 = N * zbar @ np.linalg.solve(Sz, zbar) # Hotelling's statistic
F_mv = (N - m + 1) / ((N - 1) * (m - 1)) * T2
p_mv = stats.f.sf(F_mv, m - 1, N - m + 1)
print(f"F = {F_uncorrected:.3f} on ({d1}, {d2}) df, p = {p_un:.4f}")
print(f"Greenhouse-Geisser eps = {eps_hat:.3f}, p = {p_gg:.4f}")
print(f"Huynh-Feldt eps = {eps_hf:.3f}, p = {p_hf:.4f}")
print(f"multivariate F({m - 1}, {N - m + 1}) = {F_mv:.3f}, p = {p_mv:.4f}")
```

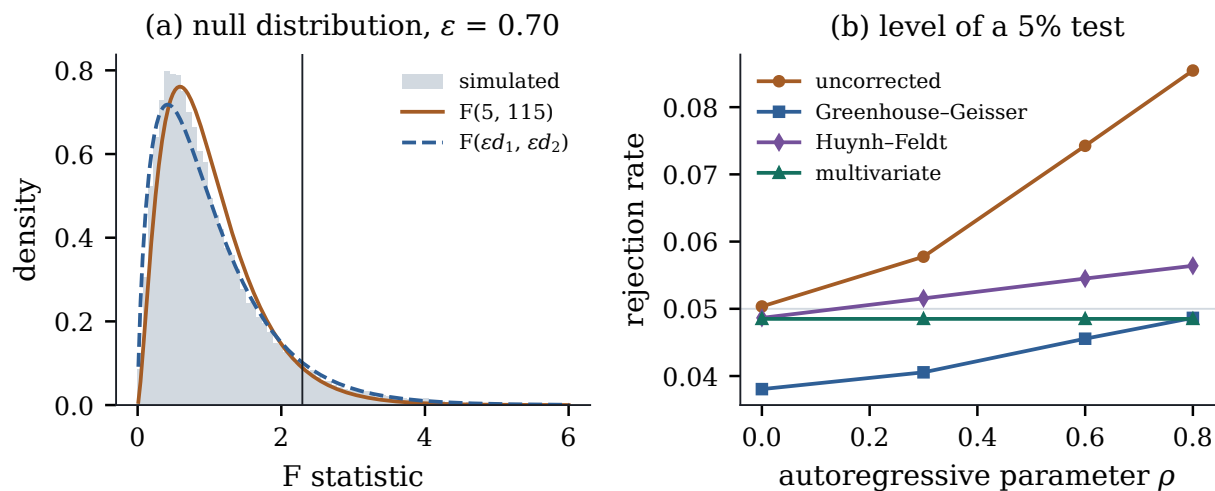


Figure 33.3.1: (a) The null distribution of the uncorrected  $F$  statistic with autoregressive errors, against the nominal  $F(5, 115)$  density and the moment-matched  $F(\epsilon d_1, \epsilon d_2)$  of Proposition 33.3.3, with the nominal 5% point marked. (b) Simulated level of four tests as  $\rho$  grows.

How much does the failure of sphericity cost? Figure 33.3.1 answers with a simulation under  $H$ . Panel (a) shows the null distribution of the uncorrected  $F$ : more dispersed than  $F(5, 115)$ , with a heavier right tail, and tracking Proposition 33.3.3 closely. Panel (b) traces the level of a nominal 5% test as the autoregressive parameter grows. At  $\rho = 0$  it is 0.050, as Theorem 33.3.2(b) requires; by  $\rho = 0.8$ , where  $\epsilon = 0.562$ , it is 0.085. The Greenhouse–Geisser test is conservative throughout (0.038 to 0.049), the Huynh–Feldt test close to nominal (0.049 to 0.056), and the multivariate test exact at 0.049 for every  $\rho$ . A spherical covariance of the form Lemma 33.3.1(iv) with unequal  $a_j$ , far from compound symmetric, gives the uncorrected test a level of 0.050: sphericity, not compound symmetry, is what matters.

### 33.3.5 The modern treatment

The corrections above patch a fixed analysis; the mixed-model approach fits the covariance instead. Write the data as a linear mixed model (Definition 32.2.1) with a fixed effect for each occasion and a within-subject covariance  $\mathbf{R}$  of chosen shape, estimate its parameters by restricted maximum likelihood (Definition 32.4.1), and test the fixed effects by a Wald statistic with estimated degrees of freedom (Proposition 32.5.1). Compound symmetric  $\mathbf{R}$  reproduces the uncorrected univariate  $F$  (Exercise (B2)), unstructured  $\mathbf{R}$  the multivariate test up to the degrees-of-freedom approximation, and anything between is a model the classical scheme has no room for. Unbalanced data and missing visits are handled too, whereas Eq. 33.3.1 and  $T^2$  need a complete rectangle. The price is that the covariance must be chosen, and the choice affects the standard errors: Section 33.4.

*Remark* (Between-subjects factors). With a between-subjects factor the within-subject analysis is the same, applied to the pooled within-group covariance, and the univariate tests are exact iff the pooled  $\Psi$  is spherical and the groups share a common  $\Sigma$ . Mauchly’s (1940) test of sphericity itself, which most software reports, is sensitive to non-normality and, like any preliminary test, distorts the level of what follows.

## 33.3.6 Exercises

### 33.3.6.1 A. Check your understanding

#### Exercise A1.

Show that every  $2 \times 2$  covariance matrix satisfies sphericity, so that with  $m = 2$  the repeated-measures  $F$  test is exact whatever  $\Sigma$  is. Identify the test in that case.

*Solution.* With  $m = 2$ ,  $\Psi = \mathbf{U}^\top \Sigma \mathbf{U}$  is  $1 \times 1$ , hence a multiple of  $\mathbf{I}_1$ . By [Theorem 33.3.2\(b\)](#) the test is exact. It is the paired  $t$  test: the single contrast is  $(y_{i1} - y_{i2})/\sqrt{2}$ , and  $F$  is the square of the paired  $t$  statistic on  $N - 1$  degrees of freedom.

#### Exercise A2.

For  $m = 3$  and an autoregressive covariance  $\sigma_{jk} = \rho^{|j-k|}$  with  $0 < \rho < 1$ , compute  $\Psi$  for the basis  $\mathbf{u}_1 = (1, -1, 0)^\top/\sqrt{2}$ ,  $\mathbf{u}_2 = (1, 1, -2)^\top/\sqrt{6}$  and show that sphericity fails. Which pairwise difference has the largest variance?

### 33.3.6.2 B. Practice

#### Exercise B1.

Show directly that compound symmetry  $\Sigma = \sigma^2\{(1 - \rho)\mathbf{I} + \rho\mathbf{1}\mathbf{1}^\top\}$  gives  $\Psi = \sigma^2(1 - \rho)\mathbf{I}_{m-1}$ , and hence that the split-plot analysis of [Theorem 33.1.2](#) is the repeated-measures  $F$  test with  $\epsilon = 1$ . Which  $\lambda$  of [Lemma 33.3.1](#) does it correspond to?

*Solution.*  $\mathbf{U}^\top \mathbf{1} = \mathbf{0}$ , so  $\Psi = \mathbf{U}^\top \Sigma \mathbf{U} = \sigma^2(1 - \rho)\mathbf{U}^\top \mathbf{U} = \sigma^2(1 - \rho)\mathbf{I}_{m-1}$ . Hence  $\epsilon = 1$  and  $\lambda = \sigma^2(1 - \rho)$ , which is the subplot stratum eigenvalue  $\lambda_s$  of [Lemma 33.1.1](#). In [Lemma 33.3.1\(iv\)](#) take  $\mathbf{a} = (\sigma^2\rho/2)\mathbf{1}$ .

#### Exercise B2.

Consider the mixed model  $y_{ij} = \mu_j + u_i + e_{ij}$  with  $u_i$  and  $e_{ij}$  independent normal, of variances  $\sigma_u^2$  and  $\sigma_e^2$ . Show that restricted maximum likelihood gives  $\hat{\sigma}_e^2 = \text{MS}_E$  and  $\hat{\sigma}_u^2 = (\text{MS}_S - \text{MS}_E)/m$  when this is nonnegative, where  $\text{MS}_S$  is the between-subject mean square, and that the Wald statistic for  $H: \mu_1 = \cdots = \mu_m$ , divided by  $m - 1$ , is the uncorrected  $F$  of [Eq. 33.3.1](#).

### 33.3.6.3 C. Going deeper

#### Exercise C1.

For  $m = 4$ ,  $N = 20$  and  $\Sigma$  compound symmetric, compare by simulation the power of the uncorrected univariate test, the Greenhouse–Geisser test and the multivariate test of [Proposition 33.3.4](#) against a linear trend in the occasion means. Repeat with an

autoregressive  $\Sigma$  of the same total variance. Which would you choose if you knew the alternative were a linear trend, and why is a single contrast better still?

## 33.4 Covariance models for longitudinal data

A longitudinal study follows subjects over time. Its data are clustered by subject, but with a structure the previous sections have not used: observations inside a cluster are *ordered*, and the distance between two of them matters. Compound symmetry ignores the ordering; an unstructured covariance uses it but pays  $m(m+1)/2$  parameters. Between the extremes lies a catalogue of models with two to four parameters, each a different story about where the correlation comes from.

### 33.4.1 Three sources of correlation

A longitudinal measurement is usefully read as the sum of three parts, a decomposition due to Diggle (1988):

$$y_{ij} = \mathbf{x}_{(ij)}^\top \boldsymbol{\beta} + \mathbf{z}_{(ij)}^\top \mathbf{u}_i + W_i(t_{ij}) + e_{ij}. \quad (33.4.1)$$

The vector  $\mathbf{u}_i$  holds **random effects**: a subject's own level or slope, anything persisting for the whole study. The process  $W_i(\cdot)$  is **serial correlation**, making nearby times alike and distant times nearly independent. And  $e_{ij}$  is **measurement error**. The three leave different fingerprints — correlation that does not decay with the gap, correlation that does, and a discontinuity at gap zero — and most models in use are [Eq. 33.4.1](#) with one or two parts dropped.

#### Definition 33.4.1 — Covariance models for a subject's observations

Let subject  $i$  be observed at times  $t_{i1} < \dots < t_{im_i}$ , and write  $\Sigma_i$  for the covariance matrix of its response vector, with entries  $\sigma_{jk}$  and time gap  $d_{jk} = |t_{ij} - t_{ik}|$ . The standard models are

| Model                                | $\sigma_{jk}$                                                      | Parameters |
|--------------------------------------|--------------------------------------------------------------------|------------|
| independence                         | $\sigma^2 \delta_{jk}$                                             | 1          |
| compound symmetry                    | $\sigma_u^2 + \sigma^2 \delta_{jk}$                                | 2          |
| first-order autoregressive           | $\sigma^2 \rho^{ j-k }$                                            | 2          |
| exponential (spatial power)          | $\sigma^2 \rho^{d_{jk}}$                                           | 2          |
| autoregressive plus random intercept | $\sigma_u^2 + \sigma^2 \rho^{ j-k }$                               | 3          |
| Toeplitz of order $q$                | $\sigma^2 c_{ j-k }$ , with $c_0 = 1$ and $c_u = 0$ for $u \geq q$ | $q$        |
| random intercept and slope           | $(1, t_{ij}) \mathbf{D} (1, t_{ik})^\top + \sigma^2 \delta_{jk}$   | 4          |
| unstructured                         | $\sigma_{jk}$ , unrestricted                                       | $m(m+1)/2$ |



Each model must also be nonnegative definite; the constraint is  $\sigma_u^2, \sigma^2 \geq 0$  and  $|\rho| < 1$  for the first five,  $\mathbf{D}$  nonnegative definite for the random-coefficient model, and a genuine restriction on  $(c_1, \dots, c_{q-1})$  for the Toeplitz model ([Exercise \(B2\)](#)).

Compound symmetry is [Definition 33.2.1](#), what a random intercept alone produces: it makes the correlation between the first and last visit equal to that between consecutive visits, which over five years it rarely is. The autoregressive model is [Definition 21.6.1](#) within a subject and needs equally spaced occasions; the exponential model is its continuous-time version. A random intercept added to either puts a floor under the decaying correlation, and that two-source version of [Eq. 33.4.1](#) fits much real longitudinal data. The Toeplitz model lets the decay take an arbitrary shape, truncated after  $q - 1$  lags. Unstructured assumes nothing, and is right when the occasions are few, common and complete and the subjects comfortably outnumber  $m(m + 1)/2$ : the covariance implicit in [Proposition 33.3.4](#).

*Warning.* The covariance model is a choice with consequences. It changes the standard errors of the fixed effects, and — unlike in the balanced cases of the previous sections — the estimates themselves whenever the design is unbalanced or the regressors vary within subjects.

## 33.4.2 Choosing, and what goes wrong if you choose badly

### **Proposition 33.4.1 — Working covariances, likelihood comparisons and empirical standard errors**

Let the  $N$  subjects be independent, with  $E(\mathbf{Y}_i) = \mathbf{X}_i\boldsymbol{\beta}$  and true covariance  $\boldsymbol{\Sigma}_i^0$ . Let  $\boldsymbol{\Sigma}_i(\boldsymbol{\theta})$  be a working covariance model whose eigenvalues are bounded above and away from zero uniformly in  $i$  and, for fixed  $\boldsymbol{\theta}$ , let

$$\tilde{\boldsymbol{\beta}}(\boldsymbol{\theta}) = \mathbf{A}(\boldsymbol{\theta})^{-1} \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i(\boldsymbol{\theta})^{-1} \mathbf{Y}_i, \quad \mathbf{A}(\boldsymbol{\theta}) = \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i(\boldsymbol{\theta})^{-1} \mathbf{X}_i,$$

assumed nonsingular.

- (a)  $\tilde{\boldsymbol{\beta}}(\boldsymbol{\theta})$  is unbiased for every  $\boldsymbol{\theta}$ , whether or not the working model is correct, with exact covariance  $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$  where  $\mathbf{B} = \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i^{-1} \boldsymbol{\Sigma}_i^0 \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i$ . If the working model is correct,  $\mathbf{B} = \mathbf{A}$  and the covariance is  $\mathbf{A}^{-1}$ .
- (b) The **empirical** covariance estimate  $\hat{\mathbf{U}}_E$  replaces  $\boldsymbol{\Sigma}_i^0$  by  $\mathbf{r}_i \mathbf{r}_i^\top$ , with  $\mathbf{r}_i = \mathbf{Y}_i - \mathbf{X}_i \tilde{\boldsymbol{\beta}}$ . Under the conditions (K1)–(K3) of [Section 33.2](#), applied with  $\boldsymbol{\Sigma}_i(\boldsymbol{\theta})^{-1} \mathbf{X}_i$  in place of  $\mathbf{X}_i$ , it satisfies  $n(\hat{\mathbf{U}}_E - \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}) \rightarrow \mathbf{0}$  in probability as  $N \rightarrow \infty$ , whatever  $\boldsymbol{\Sigma}_i^0$  is.
- (c) Let  $\mathbf{K}$  be any  $n \times (n - p)$  matrix whose columns form an orthonormal basis of  $\mathcal{C}(\mathbf{X})^\perp$ . The restricted log likelihood of a covariance model is the log likelihood of  $\mathbf{K}^\top \mathbf{Y}$  ([Definition 32.4.1](#)) and does not depend on the choice of  $\mathbf{K}$  ([Theorem 32.4.2](#)). Hence two covariance models fitted with the *same*  $\mathbf{X}$  have restricted likelihoods that are densities of one and the same random vector, and their difference is a genuine log likelihood ratio. If the models differ in  $\mathbf{X}$ , the two are densities of different random vectors and cannot be compared.

*Proof.*

- (a)  $\tilde{\boldsymbol{\beta}} = \mathbf{A}^{-1} \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i^{-1} \mathbf{Y}_i$  is a fixed linear function of  $\mathbf{Y}$ , so  $E \tilde{\boldsymbol{\beta}} = \mathbf{A}^{-1} \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i \boldsymbol{\beta} = \boldsymbol{\beta}$  and, by independence of the subjects,  $\text{Cov}(\tilde{\boldsymbol{\beta}}) = \mathbf{A}^{-1} \{ \sum_i \mathbf{X}_i^\top \boldsymbol{\Sigma}_i^{-1} \boldsymbol{\Sigma}_i^0 \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i \} \mathbf{A}^{-1}$  by [Theorem 2.2.1](#). If  $\boldsymbol{\Sigma}_i^0 = \boldsymbol{\Sigma}_i$ , the middle sum telescopes to  $\mathbf{A}$ .
- (b) Put  $\tilde{\mathbf{X}}_i = \boldsymbol{\Sigma}_i^{-1} \mathbf{X}_i$ , so that  $\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\sum_i \tilde{\mathbf{X}}_i^\top \mathbf{X}_i)^{-1} \sum_i \tilde{\mathbf{X}}_i^\top \boldsymbol{\varepsilon}_i$  and the empirical estimate is the same expression with  $\mathbf{r}_i \mathbf{r}_i^\top$  in the middle. The cluster-sandwich argument of [Section 33.2](#) — that is, the proof of [Theorem 21.4.2](#) with the cluster in place of the observation — goes through with  $\tilde{\mathbf{X}}_g$  in the meat and  $\mathbf{X}_g$  in the bread: the assumed bounds on the eigenvalues of  $\boldsymbol{\Sigma}_i$  make  $\|\tilde{\mathbf{X}}_i\|$  satisfy (K2) whenever  $\|\mathbf{X}_i\|$  does, and (K1) and (K3) are unchanged.
- (c) The first two sentences are [Definition 32.4.1](#) and [Theorem 32.4.2](#), and  $\mathbf{K}$  is determined up to an orthogonal rotation by  $\mathcal{C}(\mathbf{X})$ . Two models with the same  $\mathbf{X}$  therefore attach densities to the same  $(n - p)$ -dimensional random vector, so their ratio is a likelihood ratio; two models with different  $\mathbf{X}$  attach them to different random vectors, and the ratio is not even dimensionless when the two  $\mathbf{X}$  have different ranks.

■

Part (c) has a practical corollary: **fix the mean model first and compare covariance models by restricted likelihood; compare mean models by ordinary likelihood, or by a Wald or  $F$  test, holding the covariance fixed.** The criteria of [Chapter 29](#) then apply to the restricted likelihood as to any likelihood ([Definition 29.2.2](#)):

$$\text{AIC}_R = -2\ell_R + 2q, \quad \text{BIC}_R = -2\ell_R + q \log(n - p), \quad (33.4.2)$$

with  $q$  the number of covariance parameters and  $n - p$  the number of error contrasts, the dimension of the data whose likelihood  $\ell_R$  is.

Part (a) is why all this is less dangerous than it sounds: a wrong covariance model costs efficiency, not validity, since  $\boldsymbol{\beta}$  stays unbiased and part (b) supplies a standard error that is right anyway — the longitudinal version of the sandwich argument of [Section 21.4](#), and the basis of the generalized estimating equations ([Definition 40.4.1](#)) of [Chapter 40](#).

*Warning.* The empirical standard error protects against the covariance model and nothing else. If  $E(\mathbf{Y}_i) \neq \mathbf{X}_i \boldsymbol{\beta}$ , part (a) fails at the first line and there is nothing left to be robust about; it does not protect against missing data that depend on the unobserved responses; and it needs  $N$  large in the sense of (K1)–(K3), being worse than the model-based estimate with a dozen subjects.

### 33.4.3 Checking the fit of a covariance model

Two displays do most of the work, both built from residuals  $\hat{r}_{ij} = y_{ij} - \mathbf{x}_{(ij)}^\top \hat{\boldsymbol{\beta}}$  after a fit of the mean model. The **empirical variogram** plots, against the time gap  $u$ , the average of  $\frac{1}{2}(\hat{r}_{ij} - \hat{r}_{ik})^2$  over same-subject pairs at that gap. Its population version  $\gamma(u) = \frac{1}{2} \text{Var}(r_{ij} - r_{ik})$  is, for [Eq. 33.4.1](#),  $\sigma_e^2 + \sigma_s^2 \{1 - \rho(u)\}$ , rising from  $\sigma_e^2$  to  $\sigma_e^2 + \sigma_s^2$ ; the gap between that plateau and the total residual variance  $\sigma_u^2 + \sigma_s^2 + \sigma_e^2$  measures  $\sigma_u^2$ , the part of the correlation that

never decays. The **residual correlogram** plots the average correlation of residuals at gap  $u$ : compound symmetry predicts a horizontal line, the autoregressive model a geometric decay to zero, the combination a decay to a positive floor. Comparing empirical with fitted curves shows quickly which model does violence to the data.

**Example. (Investment in eleven firms)**

Grunfeld's panel records gross investment, market value and capital stock for 11 US firms in each of the 20 years from 1935 to 1954, a complete rectangle with  $n = 220$ . Take log investment as the response and the mean model  $E(y) = \beta_0 + \beta_1 t + \beta_2 \log(\text{value})$ , with  $t$  in decades from the middle of the record, so  $p = 3$ . An unstructured covariance would need 210 parameters from 11 subjects, which is hopeless. Fitting the alternatives by restricted maximum likelihood gives

| Covariance model                  | $q$ | $-2\ell_R$ | $AIC_R$ | $BIC_R$ |
|-----------------------------------|-----|------------|---------|---------|
| independence                      | 1   | 476.95     | 478.95  | 482.33  |
| compound symmetry                 | 2   | 114.83     | 118.83  | 125.59  |
| autoregressive                    | 2   | 49.89      | 53.89   | 60.65   |
| autoregressive + random intercept | 3   | 29.87      | 35.87   | 46.01   |
| random intercept and slope        | 4   | 98.66      | 106.66  | 120.18  |
| Toeplitz of order 4               | 4   | 147.97     | 155.97  | 169.49  |

Independence is hopeless: the firms differ enormously in size, and log investment is nearly constant within a firm relative to the spread between firms. Compound symmetry captures that and improves  $-2\ell_R$  by 362, the autoregressive model on the same two parameters does better still, and the combination is best on both criteria, by 82.96 of  $\text{AIC}_R$  over compound symmetry. The random-coefficient model, which gives each firm its own trend, describes these data worse than a persistent level plus decaying serial correlation.

Figure 33.4.1 shows why. The residual correlations are 0.953 at one year, 0.918 at two, 0.884 at five and 0.839 at ten: high everywhere, because the firm's level dominates, but visibly decaying, which compound symmetry cannot reproduce. The variogram rises to a plateau of about 0.099, far below the total residual variance of 0.4839; the gap is the between-firm variance. The selected model is not innocent either: its fitted correlogram flattens onto its own floor after three or four years while the data go on decaying. That is what the diagnostic is for, and a reason to read the empirical standard error of Proposition 33.4.1(b) alongside the model-based one.

With the selected model the coefficient of  $\log(\text{value})$  is 0.6295, with a model-based standard error of 0.0712 and an empirical one of 0.0608, and the coefficient of  $t$  is 0.4261 with standard errors 0.0600 and 0.0601. Ordinary least squares, treating the 220 observations as independent, reports 0.9833 and 0.3768 with standard errors 0.0351 and 0.0821. The two disagree substantially about the elasticity with respect to market value, because least squares reads it mostly from differences between firms while generalized least squares, having absorbed the firm levels into the covariance, reads it from movements within firms — the between-versus-within distinction of Section 6.6 again, and a question about the subject matter.

```
import numpy as np
import statsmodels.api as sm

df = sm.datasets.grunfeld.load_pandas().data
firms = sorted(df["firm"].unique())
years = np.sort(df["year"].unique())
N, m = len(firms), len(years)
t = (years - years.mean()) / 10

Ylist, Xlist = [], []
for f in firms:
    d = df[df["firm"] == f].sort_values("year")
    Ylist.append(np.log(d["invest"].to_numpy()))
    Xlist.append(np.column_stack([np.ones(m), t, np.log(d["value"].to_numpy())]))
p = Xlist[0].shape[1]
n = N * m
```

With a common  $\Sigma$ ,  $-2\ell_R$  is  $N \log|\Sigma| + \sum_i \mathbf{r}_i^\top \Sigma^{-1} \mathbf{r}_i + \log|\sum_i \mathbf{X}_i^\top \Sigma^{-1} \mathbf{X}_i|$  plus a constant, evaluated at the generalized least squares  $\hat{\beta}$ , the third term being what distinguishes it from the ordinary likelihood (Theorem 32.4.2).

```
from scipy.optimize import minimize

def reml(Sigma, Ylist, Xlist):
    """-2 x the restricted log likelihood, and the GLS estimate, for a common Sigma.
    """
    L = np.linalg.cholesky(Sigma)
    logdet = 2 * np.sum(np.log(np.diag(L)))
```

```

solve = lambda B: np.linalg.solve(Sigma, B)
XtWX = sum(X.T @ solve(X) for X in Xlist)
XtWy = sum(X.T @ solve(y) for X, y in zip(Xlist, Ylist))
beta = np.linalg.solve(XtWX, XtWy)
quad = sum((y - X @ beta) @ solve(y - X @ beta) for X, y in zip(Xlist, Ylist))
s, ld_info = np.linalg.slogdet(XtWX)
n, p = len(Ylist) * Sigma.shape[0], len(beta)
return len(Ylist) * logdet + quad + ld_info + (n - p) * np.log(2 * np.pi), beta

lag = np.abs(np.subtract.outer(np.arange(m), np.arange(m)))
Zrc = np.column_stack([np.ones(m), t]) # random intercept and slope
basis

def build(name, th):
    """Sigma for each covariance model, from an unconstrained parameter vector."""
    if name == "independence":
        return np.exp(th[0]) * np.eye(m)
    if name == "compound symmetry":
        return np.exp(th[0]) * np.eye(m) + np.exp(th[1]) * np.ones((m, m))
    if name == "AR(1)":
        return np.exp(th[0]) * np.tanh(th[1])**lag
    if name == "AR(1) + random intercept":
        return (np.exp(th[0]) * np.tanh(th[1])**lag + np.exp(th[2]) * np.ones((m, m)
    ))
    if name == "random intercept and slope":
        D = np.array([[np.exp(th[1]), th[3]], [th[3], np.exp(th[2])]])
        return np.exp(th[0]) * np.eye(m) + Zrc @ D @ Zrc.T
    if name == "Toeplitz(4)":
        a = np.concatenate([[1.0], th[1:4]]) # a moving average of order 3
        ...
        g = np.array([a[:4 - u] @ a[u:] for u in range(4)])
        c = np.concatenate([g / g[0], np.zeros(m - 4)]) # ... so the band is
        nonneg. definite
        return np.exp(th[0]) * c[lag]
    raise ValueError(name)

MODELS = {"independence": 1, "compound symmetry": 2, "AR(1)": 2,
          "AR(1) + random intercept": 3, "random intercept and slope": 4,
          "Toeplitz(4)": 4}

def fit(name, start):
    def obj(th):
        S = build(name, th)
        if np.min(np.linalg.eigvalsh(S)) <= 1e-9:
            return 1e6
        return reml(S, Ylist, Xlist)[0]
    best = min((minimize(obj, s, method="Nelder-Mead",
                        options={"maxiter": 20000, "xatol": 1e-9, "fatol": 1e-9})
                for s in start), key=lambda r: r.fun)
    S = build(name, best.x)
    d2l, beta = reml(S, Ylist, Xlist)
    q = MODELS[name]
    return {"m2ll": d2l, "aic": d2l + 2 * q, "bic": d2l + q * np.log(n - p),
            "q": q, "Sigma": S, "beta": beta}

starts = {"independence": [[-1.0]],
          "compound symmetry": [[-1.0, -1.0]],
          "AR(1)": [[-1.0, 1.0]],

```

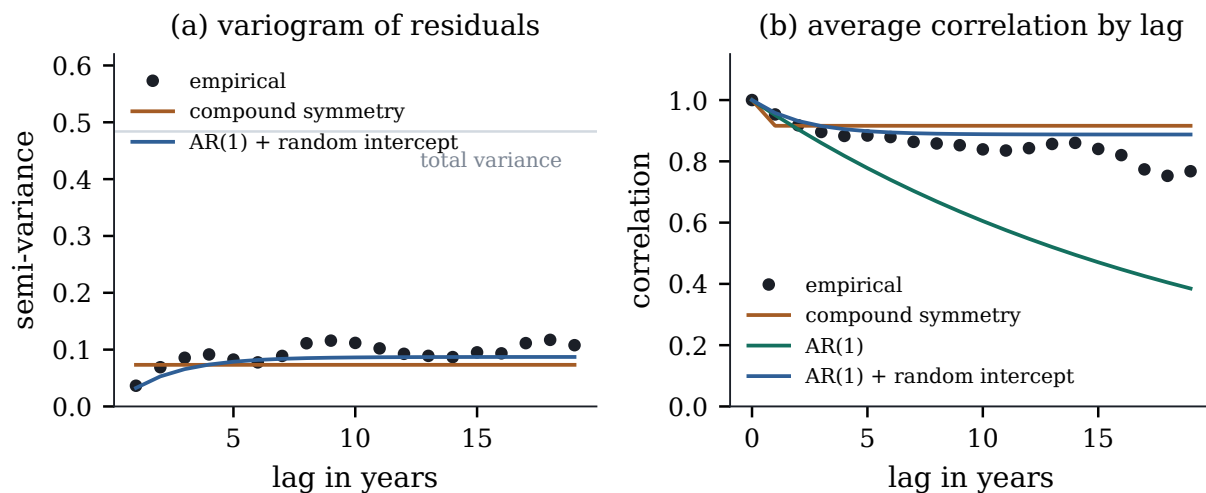


Figure 33.4.1: Checking the covariance model for the investment panel. (a) The empirical variogram of the residuals, with those implied by two fitted models; the gap between the grey total-variance line and the plateau is the between-firm variance. (b) The average residual correlation at each lag, with three fitted models.

```

"AR(1) + random intercept": [[-1.5, 1.0, -1.5]],
"random intercept and slope": [[-2.0, -1.0, -2.0, 0.0], [-3.0, -2.0, -3.0,
0.0]],
"Toeplitz(4)": [[-1.0, 1.5, 1.0, 0.5], [-1.0, 2.5, 2.0, 1.0]]
res = {name: fit(name, starts[name]) for name in MODELS}
for name in MODELS:
    r = res[name]
    print(f"{name:28s} q={r['q']} -2logL_R={r['m2ll']:9.3f} "
          f"AIC={r['aic']:9.3f} BIC={r['bic']:9.3f}")

```

```

best_name = min(MODELS, key=lambda k: res[k]["aic"])
Xall = np.vstack(Xlist)
yall = np.concatenate(Ylist)
Sbest = res[best_name]["Sigma"]
A = np.linalg.inv(sum(X.T @ np.linalg.solve(Sbest, X) for X in Xlist))
beta = res[best_name]["beta"]
resid = [y - X @ beta for X, y in zip(Xlist, Ylist)]
meat = sum(np.outer(X.T @ np.linalg.solve(Sbest, r), X.T @ np.linalg.solve(Sbest, r)
)
           for X, r in zip(Xlist, resid))
cov_robust = A @ meat @ A * N / (N - 1)                                # empirical ("sandwich")
covariance
XtXinv = np.linalg.inv(Xall.T @ Xall)
beta_ols = XtXinv @ Xall.T @ yall
s2_ols = np.sum((yall - Xall @ beta_ols)**2) / (n - p)
for j, lab in enumerate(["intercept", "time/10", "log value"]):
    print(f"{lab:10s} GLS {beta[j]: .4f} model se {np.sqrt(A[j, j]):.4f} "
          f"empirical se {np.sqrt(cov_robust[j, j]):.4f} "
          f"| OLS {beta_ols[j]: .4f} se {np.sqrt(s2_ols * XtXinv[j, j]):.4f}")

```

## 33.4.4 Missing data

Longitudinal data are rarely complete. The mixed-model machinery handles ragged data without complaint — [Proposition 33.4.1](#) never assumed a common  $m_i$  — an advantage over the classical analysis of [Section 33.3](#), which needs a complete rectangle. But “handles” means “computes”. If the probability of a missing value depends only on the covariates and the responses that *were* observed, the observed-data likelihood is the right one: the data are missing at random. If it depends on the value that would have been observed — dropout because a patient is not improving — the likelihood is wrong and no covariance model repairs it.

*Warning.* Software that fits a mixed model to incomplete longitudinal data makes a missing-at-random assumption on the analyst’s behalf, silently, and the empirical standard error of [Proposition 33.4.1\(b\)](#) does *not* protect against its failure. [Chapter 41](#) treats missing data properly.

## 33.4.5 Exercises

### 33.4.5.1 A. Check your understanding

#### Exercise A1.

A trial measures 60 patients at 8 visits. How many parameters does each model of [Definition 33.4.1](#) need? Which could not be fitted with only 10 patients, and why?

#### Exercise A2.

Sketch the population variogram  $\gamma(u)$  for compound symmetry, for the autoregressive model and for their combination, marking the total variance on each. Which feature identifies  $\sigma_u^2$ , and which identifies measurement error?

*Solution.* Compound symmetry gives  $\gamma(u) = \sigma^2$  for every  $u > 0$ , flat and well below the total variance  $\sigma^2 + \sigma_u^2$ . The autoregressive model gives  $\gamma(u) = \sigma^2(1 - \rho^u)$ , rising from 0 to its total variance  $\sigma^2$ . The combination rises from 0 to  $\sigma^2$  and stops, leaving the gap  $\sigma_u^2$  to the total. The gap between plateau and total variance is  $\sigma_u^2$ ; the intercept of the variogram as  $u \rightarrow 0$  is the measurement-error variance.

### 33.4.5.2 B. Practice

#### Exercise B1.

Two analysts fit the investment data of [Example \(Investment in eleven firms\)](#). The first uses the mean model of the example with an autoregressive covariance; the second adds a quadratic term in  $t$  and uses compound symmetry. They compare  $AIC_R$ . Explain what is wrong, and describe a correct sequence of comparisons.

*Solution.* By [Proposition 33.4.1\(c\)](#) the two restricted likelihoods are densities of different random vectors, of dimensions  $n - 3$  and  $n - 4$ , so their difference is not a log likelihood ratio and  $AIC_R$  cannot compare them. Correct sequence: fix a rich mean model and

compare covariance models by  $AIC_R$ ; then, with the chosen covariance fixed, compare mean models by a Wald or  $F$  test, or by AIC from the *full* likelihood; then refit the covariance.

### Exercise B2.

Show that the Toeplitz model of [Definition 33.4.1](#) with  $q = 3$  is nonnegative definite for every  $m$  iff  $(1, c_1, c_2, 0, \dots)$  is the autocorrelation sequence of a moving-average process of order two, and explain why the parameterization  $c_u \propto \sum_l a_l a_{l+u}$  used in the listing above satisfies this automatically.

### 33.4.5.3 C. Going deeper

### Exercise C1.

The autoregressive model is nested in the autoregressive-plus-random-intercept model at  $\sigma_u^2 = 0$ , a boundary of the parameter space. Explain why the likelihood ratio statistic comparing the two does not have a  $\chi^2(1)$  null distribution, state what it does have ([Proposition 32.5.1](#)), and say what this means for reading  $AIC_R$  differences as informal evidence.

## 33.5 Growth curves

A growth-curve model gives each subject a *curve of its own* — a straight line, a polynomial in time — the population curve being their average. How is the average curve estimated, and how is an individual's curve recovered from its own few observations? The second answer is the best linear unbiased predictor: a matrix-weighted average of the subject's own noisy curve and the precise population curve, weighted by signal against signal-plus-noise — the shrinkage of [Chapter 27](#) by a different road.

### 33.5.1 Two formulations

Let each of  $N$  subjects be observed at the same  $m$  times, and collect the data in an  $m \times N$  matrix  $\mathbf{Y}$  whose column  $i$  is subject  $i$ 's profile. Let  $\mathbf{X}$  be the  $m \times p$  matrix of a chosen basis in time, say  $[\mathbf{1}, \mathbf{t}]$  for a straight line.

The **Potthoff–Roy** formulation (Potthoff and Roy 1964) writes

$$\mathbf{E}(\mathbf{Y}) = \mathbf{X}\mathbf{\Xi}\mathbf{A}^\top, \quad \text{Cov}(\text{column } i) = \mathbf{\Sigma}, \text{ columns independent}, \quad (33.5.1)$$

where  $\mathbf{A}$  is an  $N \times k$  matrix of between-subject covariates (a column of ones if the subjects form one group, group indicators if not) and  $\mathbf{\Xi}$  is the  $p \times k$  matrix of curve coefficients, with  $\mathbf{\Sigma}$  unstructured. It is a multivariate regression with a design on *both* sides:  $\mathbf{X}$  acts on the times and  $\mathbf{A}$  on the subjects.

The **random-coefficient** formulation gives each subject its own coefficient vector,



$$\mathbf{y}_i = \mathbf{X}(\boldsymbol{\beta} + \mathbf{u}_i) + \boldsymbol{\varepsilon}_i, \quad \mathbf{u}_i \sim (\mathbf{0}, \mathbf{D}) \text{ independent of } \boldsymbol{\varepsilon}_i \sim (\mathbf{0}, \sigma^2 \mathbf{I}_m), \quad (33.5.2)$$

with  $\mathbf{u}_1, \dots, \mathbf{u}_N$  uncorrelated. This is the linear mixed model (Definition 32.2.1) with  $\mathbf{Z}_i = \mathbf{X}$ ,  $\mathbf{G} = \mathbf{D}$  and  $\mathbf{R} = \sigma^2 \mathbf{I}$ , and the implied within-subject covariance is

$$\boldsymbol{\Sigma} = \mathbf{XDX}^\top + \sigma^2 \mathbf{I}_m, \quad (33.5.3)$$

a covariance with  $p(p+1)/2 + 1$  parameters instead of the  $m(m+1)/2$  of Eq. 33.5.1 — four, for a straight line, whatever  $m$  is. The two describe the same *mean* structure: with one group of subjects, the random-coefficient model is Eq. 33.5.1 with  $\mathbf{A} = \mathbf{1}_N$  and  $\boldsymbol{\Sigma}$  restricted to Eq. 33.5.3. That restriction gives  $\mathcal{C}(\boldsymbol{\Sigma}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , so the population curve is ordinary least squares and needs no covariance parameter (Proposition 33.5.1(a) below); with  $\boldsymbol{\Sigma}$  unstructured the inclusion fails unless  $p = m$ , and the efficient estimate is a genuine generalized least squares fit.

## 33.5.2 Estimation

### Proposition 33.5.1 — The balanced random-coefficient growth model

Assume Eq. 33.5.2 with a common  $\mathbf{X}$  of full column rank  $p \leq m$ , and  $\mathbf{D}$  positive definite. Write  $\mathbf{K} = \mathbf{X}^\top \mathbf{X}$ ,  $\mathbf{C} = \sigma^2 \mathbf{K}^{-1}$ ,  $\mathbf{S} = \mathbf{D} + \mathbf{C}$ , and let  $\mathbf{b}_i = \mathbf{K}^{-1} \mathbf{X}^\top \mathbf{y}_i$  be subject  $i$ 's own least squares coefficient vector.

- (a)  $\mathcal{C}(\boldsymbol{\Sigma}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$ , so ordinary and generalized least squares coincide, and

$$\hat{\boldsymbol{\beta}} = \mathbf{K}^{-1} \mathbf{X}^\top \bar{\mathbf{y}} = \bar{\mathbf{b}}, \quad \text{Cov}(\hat{\boldsymbol{\beta}}) = \frac{\mathbf{D} + \sigma^2 \mathbf{K}^{-1}}{N} = \frac{\mathbf{S}}{N}, \quad (33.5.4)$$

exactly, for every  $N$ . No covariance parameter is needed to estimate the population curve.

- (b) The best linear unbiased predictor of  $\mathbf{u}_i$  (Theorem 32.3.1, which is Theorem 14.2.8 applied coordinatewise with  $\mathbf{x}_0 = \mathbf{0}$ ) is

$$\hat{\mathbf{u}}_i = \mathbf{DX}^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y}_i - \mathbf{X}\hat{\boldsymbol{\beta}}) = \mathbf{W}(\mathbf{b}_i - \hat{\boldsymbol{\beta}}), \quad \mathbf{W} = \mathbf{DS}^{-1}, \quad (33.5.5)$$

with  $\mathbf{W} = \mathbf{D}(\mathbf{D} + \mathbf{C})^{-1}$  and  $\text{Cov}(\hat{\mathbf{u}}_i - \mathbf{u}_i) = \mathbf{D} - (1 - 1/N)\mathbf{WD}$ . The predicted curve of subject  $i$  is  $\mathbf{X}(\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}_i)$ , with

$$\text{Cov}(\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}_i - \boldsymbol{\beta} - \mathbf{u}_i) = (\mathbf{D}^{-1} + \mathbf{C}^{-1})^{-1} + (\mathbf{I} - \mathbf{W})\frac{\mathbf{C}}{N}. \quad (33.5.6)$$

- (c) The restricted maximum likelihood estimates of  $\sigma^2$  and  $\mathbf{D}$  are

$$\hat{\sigma}^2 = \frac{1}{N(m-p)} \sum_{i=1}^N \|\mathbf{y}_i - \mathbf{Xb}_i\|^2, \quad \hat{\mathbf{D}} = \mathbf{S}_b - \hat{\sigma}^2 \mathbf{K}^{-1}, \quad (33.5.7)$$

where  $\mathbf{S}_b = (N-1)^{-1} \sum_i (\mathbf{b}_i - \bar{\mathbf{b}})(\mathbf{b}_i - \bar{\mathbf{b}})^\top$ , provided  $\hat{\mathbf{D}}$  is nonnegative definite.

*Proof.*

- (a)  $\Sigma \mathbf{X} = \mathbf{X} \mathbf{D} \mathbf{K} + \sigma^2 \mathbf{X} = \mathbf{X}(\mathbf{D} \mathbf{K} + \sigma^2 \mathbf{I})$ , whose columns lie in  $\mathcal{C}(\mathbf{X})$ . Stacking the subjects, the model matrix is  $\mathbf{1}_N \otimes \mathbf{X}$  and the covariance is  $\mathbf{I}_N \otimes \Sigma$ , so  $(\mathbf{I}_N \otimes \Sigma)(\mathbf{1}_N \otimes \mathbf{X}) = \mathbf{1}_N \otimes \Sigma \mathbf{X}$  has columns in  $\mathcal{C}(\mathbf{1}_N \otimes \mathbf{X})$ , and [Theorem 6.9.3](#) gives the equality of the two fits (this is [Theorem 31.2.2](#) in the balanced case). The ordinary least squares estimate from the stacked data is  $(N\mathbf{K})^{-1} \sum_i \mathbf{X}^\top \mathbf{y}_i = \mathbf{K}^{-1} \mathbf{X}^\top \bar{\mathbf{y}} = \bar{\mathbf{b}}$ . Its covariance is  $\mathbf{K}^{-1} \mathbf{X}^\top (\Sigma/N) \mathbf{X} \mathbf{K}^{-1}$ , and  $\mathbf{K}^{-1} \mathbf{X}^\top \Sigma \mathbf{X} \mathbf{K}^{-1} = \mathbf{K}^{-1}(\mathbf{K} \mathbf{D} \mathbf{K} + \sigma^2 \mathbf{K}) \mathbf{K}^{-1} = \mathbf{D} + \sigma^2 \mathbf{K}^{-1}$ .
- (b) The first expression is [Theorem 32.3.1](#). For the second, [Theorem 1.6.5](#) gives  $\Sigma^{-1} = \sigma^{-2} \{\mathbf{I} - \mathbf{X}(\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1} \mathbf{X}^\top\}$ , so  $\mathbf{D} \mathbf{X}^\top \Sigma^{-1} = \sigma^{-2} \mathbf{D} \{\mathbf{I} - \mathbf{K}(\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1}\} \mathbf{X}^\top$ . The braced matrix is  $\sigma^2 \mathbf{D}^{-1}(\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1}$ , so  $\mathbf{D} \mathbf{X}^\top \Sigma^{-1} = (\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1} \mathbf{X}^\top$ ; applying this to  $\mathbf{y}_i - \mathbf{X} \hat{\boldsymbol{\beta}}$  and inserting  $\mathbf{K} \mathbf{K}^{-1}$  gives  $\hat{\mathbf{u}}_i = (\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1} \mathbf{K}(\mathbf{b}_i - \hat{\boldsymbol{\beta}})$ , and  $(\sigma^2 \mathbf{D}^{-1} + \mathbf{K})^{-1} \mathbf{K} = (\sigma^2 \mathbf{I} + \mathbf{D} \mathbf{K})^{-1} \mathbf{D} \mathbf{K} = \mathbf{D}(\sigma^2 \mathbf{K}^{-1} + \mathbf{D})^{-1} = \mathbf{W}$ . For the prediction error covariance, [Theorem 32.3.1](#) gives  $\text{Cov}(\hat{\mathbf{u}} - \mathbf{u}) = \mathbf{G} - \mathbf{G} \mathbf{Z}^\top \mathbf{P} \mathbf{Z} \mathbf{G}$  with  $\mathbf{P} = \mathbf{V}^{-1} - \mathbf{V}^{-1} \mathbf{X}_* (\mathbf{X}_*^\top \mathbf{V}^{-1} \mathbf{X}_*)^{-1} \mathbf{X}_*^\top \mathbf{V}^{-1}$  for the stacked model. Here  $\mathbf{Z} = \mathbf{I}_N \otimes \mathbf{X}$ ,  $\mathbf{G} = \mathbf{I}_N \otimes \mathbf{D}$ ,  $\mathbf{X}_* = \mathbf{1}_N \otimes \mathbf{X}$  and  $\mathbf{V} = \mathbf{I}_N \otimes \Sigma$ , so  $\mathbf{Z}^\top \mathbf{P} \mathbf{Z} = (\mathbf{I}_N - N^{-1} \mathbf{1}_N \mathbf{1}_N^\top) \otimes (\mathbf{X}^\top \Sigma^{-1} \mathbf{X})$  and the  $i$ th diagonal block of  $\mathbf{G} - \mathbf{G} \mathbf{Z}^\top \mathbf{P} \mathbf{Z} \mathbf{G}$  is  $\mathbf{D} - (1 - 1/N) \mathbf{D} \mathbf{X}^\top \Sigma^{-1} \mathbf{X} \mathbf{D} = \mathbf{D} - (1 - 1/N) \mathbf{W} \mathbf{D}$ , using  $\mathbf{D} \mathbf{X}^\top \Sigma^{-1} \mathbf{X} = \mathbf{W}$  from the first part. For [Eq. 33.5.6](#), write  $\mathbf{b}_i = \boldsymbol{\beta} + \mathbf{u}_i + \mathbf{c}_i$  with  $\mathbf{c}_i = \mathbf{K}^{-1} \mathbf{X}^\top \boldsymbol{\varepsilon}_i \sim (\mathbf{0}, \mathbf{C})$  independent of  $\mathbf{u}_i$ , so  $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + \bar{\mathbf{u}} + \bar{\mathbf{c}}$  and  $\hat{\mathbf{u}}_i = \mathbf{W}(\mathbf{b}_i - \hat{\boldsymbol{\beta}})$ . Then  $\text{Cov}(\hat{\boldsymbol{\beta}}, \hat{\mathbf{u}}_i) = \{\text{Cov}(\bar{\mathbf{b}}, \mathbf{b}_i) - \text{Cov}(\bar{\mathbf{b}}, \bar{\mathbf{b}})\} \mathbf{W}^\top = \mathbf{0}$  and  $\text{Cov}(\hat{\boldsymbol{\beta}}, \mathbf{u}_i) = \mathbf{D}/N$ , so

$$\text{Cov}(\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}_i - \boldsymbol{\beta} - \mathbf{u}_i) = \frac{\mathbf{S}}{N} + \mathbf{D} - \left(1 - \frac{1}{N}\right) \mathbf{W} \mathbf{D} - \frac{2\mathbf{D}}{N}.$$

Now  $\mathbf{D} - \mathbf{W} \mathbf{D} = \mathbf{D} - \mathbf{D} \mathbf{S}^{-1} \mathbf{D} = \mathbf{D} \mathbf{S}^{-1} (\mathbf{S} - \mathbf{D}) = \mathbf{W} \mathbf{C}$  and  $\mathbf{I} - \mathbf{W} = (\mathbf{S} - \mathbf{D}) \mathbf{S}^{-1} = \mathbf{C} \mathbf{S}^{-1}$ , so the right-hand side is  $\mathbf{W} \mathbf{C} + (\mathbf{S} - 2\mathbf{D} + \mathbf{D} \mathbf{S}^{-1} \mathbf{D})/N = \mathbf{W} \mathbf{C} + \mathbf{C} \mathbf{S}^{-1} \mathbf{C}/N = \mathbf{W} \mathbf{C} + (\mathbf{I} - \mathbf{W}) \mathbf{C}/N$ . Finally  $\mathbf{W} \mathbf{C} = \mathbf{D} \mathbf{S}^{-1} \mathbf{C} = \{\mathbf{C}^{-1} \mathbf{S} \mathbf{D}^{-1}\}^{-1} = (\mathbf{C}^{-1} + \mathbf{D}^{-1})^{-1}$ , which is [Eq. 33.5.6](#).

- (c) An error contrast is a linear function of the data with mean zero for every  $\boldsymbol{\beta}$  ([Definition 32.4.1](#)). Let  $\mathbf{U}$  be  $m \times (m - p)$  with columns an orthonormal basis of  $\mathcal{C}(\mathbf{X})^\perp$ , and consider the  $Nm - p$  functions

$$\mathbf{w}_i = \mathbf{U}^\top \mathbf{y}_i \quad (i = 1, \dots, N), \quad \mathbf{b}_i - \bar{\mathbf{b}} \quad (i = 1, \dots, N).$$

They are error contrasts, and a maximal set: the map  $\mathbf{y}_i \mapsto (\mathbf{b}_i, \mathbf{w}_i)$  is a bijection, the  $\mathbf{w}_i$  contribute  $N(m - p)$  independent functions and the  $\mathbf{b}_i - \bar{\mathbf{b}}$  contribute  $(N - 1)p$ , totalling  $Nm - p$ . Now  $\mathbf{w}_i = \mathbf{U}^\top \boldsymbol{\varepsilon}_i \sim N_{m-p}(\mathbf{0}, \sigma^2 \mathbf{I})$  independently, while  $\mathbf{b}_i \sim N_p(\boldsymbol{\beta}, \mathbf{S})$  independently, and  $\text{Cov}(\mathbf{b}_i, \mathbf{w}_i) = \sigma^2 \mathbf{K}^{-1} \mathbf{X}^\top \mathbf{U} = \mathbf{0}$ ; the two blocks are independent, so the restricted likelihood factors as

$$L_R(\sigma^2, \mathbf{D}) = L_1(\sigma^2) \cdot L_2(\mathbf{S}), \quad \mathbf{S} = \mathbf{D} + \sigma^2 \mathbf{K}^{-1},$$

with  $L_1$  the likelihood of  $N(m - p)$  independent  $N(0, \sigma^2)$  variables and  $L_2$  that of  $\mathbf{S}$  from the contrasts  $\mathbf{b}_i - \bar{\mathbf{b}}$ . Since  $(\sigma^2, \mathbf{D}) \mapsto (\sigma^2, \mathbf{S})$  is a bijection onto  $\{\mathbf{S} - \sigma^2 \mathbf{K}^{-1} \succeq \mathbf{0}\}$ , the unconstrained maximizers are the restricted maximum likelihood estimates whenever they satisfy the constraint. Maximizing  $L_1$  gives  $\hat{\sigma}^2 = \sum_i \|\mathbf{y}_i - \mathbf{X} \mathbf{b}_i\|^2 / \{N(m - p)\}$ , and maximizing  $L_2$ , the restricted likelihood of a multivariate normal mean model with design  $\mathbf{1}_N$ , gives  $\hat{\mathbf{S}} = \mathbf{S}_b$  ([Theorem 32.4.2](#)). Subtracting yields  $\hat{\mathbf{D}}$ .

■

Part (a) is a statement about the restriction [Eq. 33.5.3](#), not about balance alone: the population curve is ordinary least squares on the *average profile*, which is also the average of the subjects' own curves, and nothing has to be estimated first. Potthoff and Roy's own proposal was not to estimate  $\Sigma$  at all: transforming each profile by a positive definite weight matrix reduces [Eq. 33.5.1](#) to an ordinary multivariate analysis of variance, whose tests are exact for weights chosen without looking at the data. The efficient estimate came later, with Rao (1965); estimating  $\Sigma$  from the residual profiles and weighting by it is feasible generalized least squares ([Definition 31.4.1](#)).

*Idea.*  $\mathbf{W} = \mathbf{D}(\mathbf{D} + \mathbf{C})^{-1}$  compares two covariances:  $\mathbf{D}$ , how much subjects really differ, and  $\mathbf{C} = \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1}$ , how noisily each subject's own curve is measured. When subjects differ a lot relative to the noise each keeps its own curve,  $\mathbf{W} \approx \mathbf{I}$ ; when they hardly differ every subject is given the population curve,  $\mathbf{W} \approx \mathbf{0}$ . Between those the predictor borrows strength across subjects, coordinate by coordinate: an intercept may be barely shrunk while a slope is shrunk by a quarter.

Part (c) says that restricted maximum likelihood here is the obvious moment estimator: pool the within-subject residual sums of squares for  $\sigma^2$ , and from the spread of the subjects' own coefficient vectors subtract the part that is estimation noise. The subtraction can produce a matrix that is not nonnegative definite — the growth-curve version of the negative variance component of [Proposition 32.1.1](#) — and the estimate then sits on the boundary, where the closed form no longer applies.

### 33.5.3 Whose curve

Two prediction problems must be kept apart. For a subject **in the data** the predicted curve is  $\mathbf{X}(\hat{\beta} + \hat{u}_i)$ , with error covariance [Eq. 33.5.6](#): the first term is what remains even with  $\beta$  known, the second the price of estimating it. For a subject **not in the data** nothing is known but its membership of the population, so the best predictor is  $\mathbf{X}\hat{\beta}$  with error covariance  $\mathbf{D} + \mathbf{S}/N$  — far larger, as it should be.

### 33.5.4 An example

#### Example. (The investment trajectories of eleven firms)

The panel of [Example \(Investment in eleven firms\)](#), seen a third time — after the random firm effect of [Chapter 32](#) and the covariance models of [Section 33.4](#) — now with a straight line in  $t$  (decades from the middle of the record) whose intercept *and slope* vary by firm. The within-firm residual standard deviation is  $\hat{\sigma} = 0.2797$ , and

$$\hat{\mathbf{D}} = \begin{pmatrix} 2.3635 & 0.0455 \\ 0.0455 & 0.0361 \end{pmatrix},$$

so firms' log-investment levels have standard deviation 1.5374 — enormous, the difference between General Motors and Diamond Match — and their slopes 0.1899 per decade, correlated 0.156 with the level. Maximizing the restricted likelihood numerically returns the same three values to four decimals, as [Proposition 33.5.1\(c\)](#) predicts.

The population curve has intercept 3.9109 and slope 0.5577 per decade, with standard

errors 0.4639 and 0.0659 from Eq. 33.5.4; the slope is 8.46 standard errors from zero. The shrinkage matrix is

$$\mathbf{W} = \begin{pmatrix} 0.9983 & 0.0016 \\ 0.0048 & 0.7496 \end{pmatrix},$$

so intercepts are left alone — with twenty observations per firm a firm's own level is known almost exactly — while slopes are pulled a quarter of the way towards the population slope. The firms' own slopes span 0.158 to 0.943, the predicted slopes 0.248 to 0.846, a standard deviation 0.755 times that of the raw slopes; the largest adjustment is IBM's, from 0.943 to 0.846. Figure 33.5.1 shows the eleven firms with all three lines.

For the last year of the record,  $t = 0.95$ , the predicted value on a firm's own curve has standard error 0.1102, against 1.6461 for a firm outside the data — fifteen times larger, and simulation from the fitted model confirms both, at 0.1103 and 1.6603. The interval for a new firm, 1.214 to 7.667 on the log scale, is nearly useless, and rightly so: without knowing a firm's size, its investment cannot be predicted within three orders of magnitude.

```
import numpy as np
import statsmodels.api as sm

df = sm.datasets.grunfeld.load_pandas().data
firms = sorted(df["firm"].unique())
years = np.sort(df["year"].unique())
N, m = len(firms), len(years)
t = (years - years.mean()) / 10                                # decades from the middle of
                                                                # the record
X = np.column_stack([np.ones(m), t])                          # the same design for every
                                                                # firm
p = X.shape[1]
Y = np.column_stack([np.log(df[df["firm"] == f].sort_values("year")["invest"]).
    to_numpy()
                                                                # m x N: one column per firm
                     for f in firms])

XtXinv = np.linalg.inv(X.T @ X)
B = XtXinv @ X.T @ Y                                          # p x N: each firm's own least
                                                                # squares line
beta_hat = B.mean(axis=1)                                      # the average of the firm lines
E = Y - X @ B                                                  # within-firm residuals
sigma2 = np.sum(E**2) / (N * (m - p))                          # pooled within-firm variance
Sb = np.cov(B, ddof=1)                                         # spread of the firm lines
D_hat = Sb - sigma2 * XtXinv                                    # ... minus what estimation
                                                                # noise adds
print("beta_hat", beta_hat.round(4), " sigma", round(np.sqrt(sigma2), 4))
print("D_hat\n", D_hat.round(4))
print("implied sd of firm intercepts and slopes:", np.sqrt(np.diag(D_hat)).round(4))

Sigma = X @ D_hat @ X.T + sigma2 * np.eye(m)
cov_beta = (D_hat + sigma2 * XtXinv) / N                      # exact covariance of beta_hat
W = D_hat @ np.linalg.inv(D_hat + sigma2 * XtXinv)            # the shrinkage matrix
B_blup = beta_hat[:, None] + W @ (B - beta_hat[:, None])
u_direct = D_hat @ X.T @ np.linalg.solve(Sigma, Y - (X @ beta_hat)[:, None])
print("shrinkage matrix\n", W.round(4))
print("largest difference from the direct formula:",
    np.abs(B_blup - beta_hat[:, None] - u_direct).max())
```

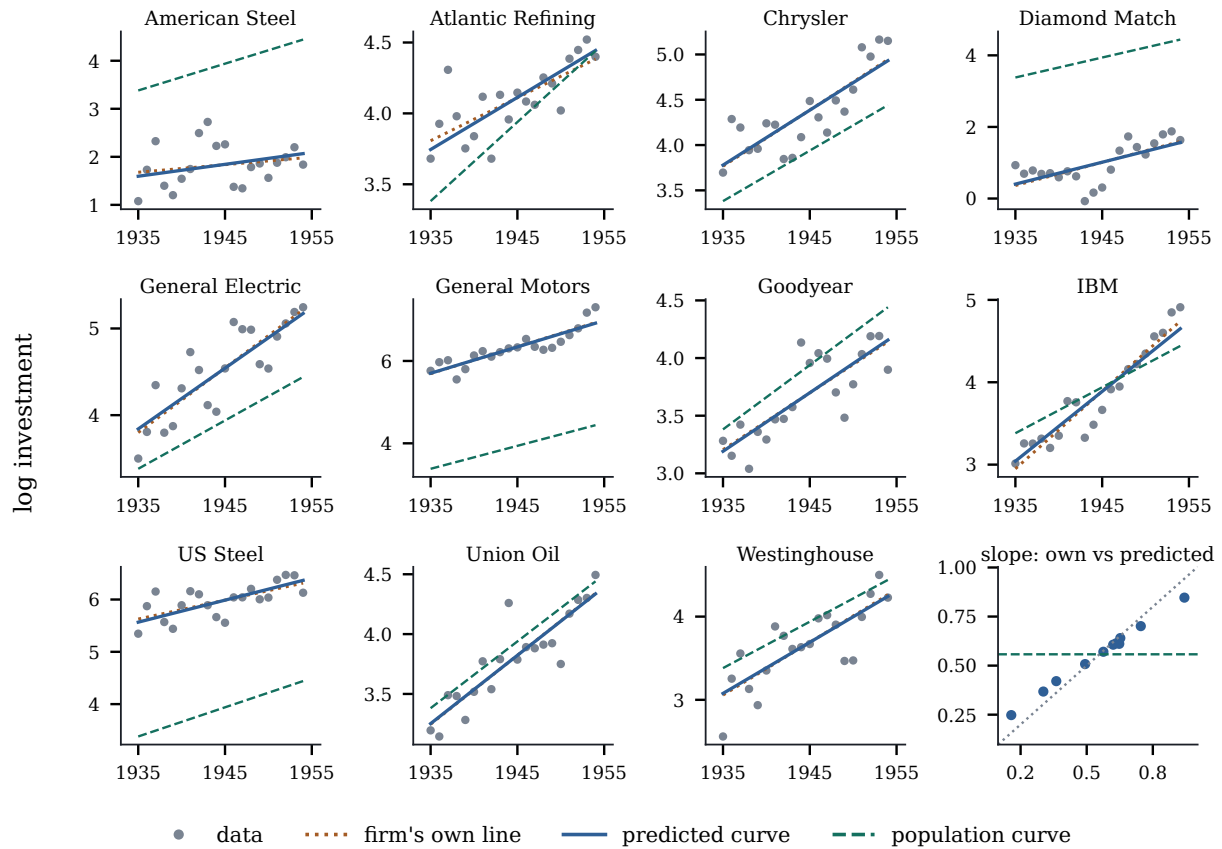


Figure 33.5.1: Log gross investment of eleven firms, 1935–1954, with each firm's own least squares line (dotted), its predicted curve  $\mathbf{X}(\hat{\boldsymbol{\beta}} + \hat{\mathbf{u}}_i)$  (solid) and the population curve  $\mathbf{X}\hat{\boldsymbol{\beta}}$  (dashed). The last panel plots own slope against predicted slope.

```

C = sigma2 * XtXinv                                     # noise in a firm's own line
V_known = np.linalg.inv(np.linalg.inv(D_hat) + np.linalg.inv(C)) \
    + (np.eye(p) - W) @ C / N                           # error of the predicted firm
    curve
V_new = D_hat + cov_beta                                # error for a firm not in the
    data
x_end = np.array([1.0, t[-1]])
se_curve_known = np.sqrt(x_end @ V_known @ x_end)
se_curve_new = np.sqrt(x_end @ V_new @ x_end)
pred_new = x_end @ beta_hat
print(f"1954 curve: se for a firm in the data {se_curve_known:.4f}, "
      f"for a new firm {se_curve_new:.4f}")

```

*Remark* (Beyond the balanced straight line). Three extensions matter in practice and none changes the ideas. Subjects observed at different times, or a different number of times: Eq. 33.5.4 fails, the fit becomes genuine generalized least squares with  $\boldsymbol{\Sigma}_i = \mathbf{X}_i \mathbf{D} \mathbf{X}_i^\top + \sigma^2 \mathbf{I}$ , and Henderson's equations (Theorem 32.3.2) do the computing. Between-subject covariates: replace  $\boldsymbol{\beta}$  by  $\boldsymbol{\Xi} \mathbf{a}_i$  as in the formulation Eq. 33.5.1. Serial correlation on top of the random coefficients: take  $\mathbf{R}$  autoregressive, as in Section 33.4. Beyond all of these lies the question whether a polynomial in time is the right shape at all: Chapter 42 and Chapter 43.

## 33.5.5 Exercises

### 33.5.5.1 A. Check your understanding

#### Exercise A1.

Take  $p = 1$ , so  $\mathbf{X} = \mathbf{1}_m$  and each subject has a single random level. Show that  $\mathbf{W}$  reduces to  $\sigma_u^2/(\sigma_u^2 + \sigma^2/m)$  and that the predictor is the shrunk mean  $\hat{\boldsymbol{\beta}} + \mathbf{W}(\bar{y}_i - \hat{\boldsymbol{\beta}})$ . What happens as  $m \rightarrow \infty$ , and as  $\sigma_u^2 \rightarrow 0$ ?

*Solution.* With  $\mathbf{X} = \mathbf{1}_m$ ,  $\mathbf{K} = m$ ,  $\mathbf{C} = \sigma^2/m$  and  $\mathbf{D} = \sigma_u^2$ , so  $\mathbf{W} = \sigma_u^2/(\sigma_u^2 + \sigma^2/m)$  and  $\mathbf{b}_i = \bar{y}_i$ . As  $m \rightarrow \infty$  the noise in  $\bar{y}_i$  vanishes,  $\mathbf{W} \rightarrow 1$  and no shrinkage occurs; as  $\sigma_u^2 \rightarrow 0$ ,  $\mathbf{W} \rightarrow 0$  and every subject is given the common mean, correctly, since the subjects are then identical.

#### Exercise A2.

The predicted curves are less spread out than the true ones. Is this a bias? Compute  $E(\hat{\mathbf{u}}_i)$  and  $\text{Cov}(\hat{\mathbf{u}}_i)$  and compare the latter with  $\mathbf{D}$ .

### 33.5.5.2 B. Practice

#### Exercise B1.

Suppose the fixed effects use a quadratic in time,  $\mathbf{X} = [\mathbf{1}, \mathbf{t}, \mathbf{t}^2]$ , but the random effects only a line,  $\mathbf{Z} = [\mathbf{1}, \mathbf{t}]$ . Show that  $\mathcal{C}(\boldsymbol{\Sigma}\mathbf{X}) \subseteq \mathcal{C}(\mathbf{X})$  still holds, and identify what changes in  $\text{Cov}(\hat{\boldsymbol{\beta}})$ . What if instead the random effects used the quadratic and the fixed effects only the line?

#### Exercise B2.

Let subject  $i$  be observed at its own times, giving  $\mathbf{X}_i$  of full column rank  $p$ , and let  $\mathbf{b}_i = (\mathbf{X}_i^\top \mathbf{X}_i)^{-1} \mathbf{X}_i^\top \mathbf{y}_i$ . Show that  $\mathbf{b}_i \sim (\boldsymbol{\beta}, \mathbf{D} + \sigma^2(\mathbf{X}_i^\top \mathbf{X}_i)^{-1})$  independently over  $i$ , and deduce that the generalized least squares estimate from the  $\mathbf{b}_i$  is  $\hat{\boldsymbol{\beta}} = (\sum_i \mathbf{W}_i)^{-1} \sum_i \mathbf{W}_i \mathbf{b}_i$  with  $\mathbf{W}_i = \{\mathbf{D} + \sigma^2(\mathbf{X}_i^\top \mathbf{X}_i)^{-1}\}^{-1}$ , a precision-weighted average of the subjects' own curves. Recover [Eq. 33.5.4](#) as the balanced case.

*Solution.*  $\mathbf{b}_i = \boldsymbol{\beta} + \mathbf{u}_i + (\mathbf{X}_i^\top \mathbf{X}_i)^{-1} \mathbf{X}_i^\top \boldsymbol{\varepsilon}_i$  is a sum of independent terms with covariances  $\mathbf{D}$  and  $\sigma^2(\mathbf{X}_i^\top \mathbf{X}_i)^{-1}$ , and different subjects are independent. The  $\mathbf{b}_i$  are therefore independent with common mean  $\boldsymbol{\beta}$  and covariances  $\mathbf{W}_i^{-1}$ , so [Corollary 7.2.3](#) applied to the stacked  $\mathbf{b}_i$  with design  $\mathbf{1}_N \otimes \mathbf{I}_p$  gives the weighted average. When every  $\mathbf{X}_i = \mathbf{X}$  the weights cancel, leaving  $\hat{\boldsymbol{\beta}} = \bar{\mathbf{b}}$  with covariance  $\mathbf{W}_1^{-1}/N$ , which is [Eq. 33.5.4](#).

### 33.5.5.3 C. Going deeper

#### Exercise C1.

Simulate data for which  $\hat{\mathbf{D}}$  of Eq. 33.5.7 is not nonnegative definite. Describe what the restricted maximum likelihood estimate does instead, relate it to the boundary problem of Proposition 32.5.1, and explain why the two-stage estimate is nevertheless unbiased for  $\mathbf{D}$ .

## 33.6 Summary and Notes

### 33.6.1 Summary

*Idea.*

1. A split-plot design randomizes twice and has two error terms: the response space splits into a whole-plot stratum with error variance  $\sigma_s^2 + b\sigma_w^2$  and a subplot stratum with error variance  $\sigma_s^2$  (Lemma 33.1.1).
2. In the balanced design ordinary least squares is still best, because  $\mathcal{C}(\mathbf{VX}) \subseteq \mathcal{C}(\mathbf{X})$ . The whole-plot factor is tested against the block-by-treatment mean square on  $(r-1)(a-1)$  degrees of freedom, the subplot factor and the interaction against the residual; a comparison of whole-plot treatments at one subplot level belongs to neither stratum and needs both mean squares (Theorem 33.1.2). Pooling inflates the whole-plot  $F$  and deflates the subplot one.
3. With compound symmetry inside clusters of size  $m$ , a cluster-level coefficient has its variance multiplied by the design effect  $1 + (m-1)\rho$  and a within-cluster coefficient by  $1 - \rho$ ; the usual standard error is wrong by  $\{E(s^2)/\lambda\}^{1/2}$ , with  $E(s^2)$  given exactly (Proposition 33.2.1). Clustering moves information from between-cluster to within-cluster comparisons.
4. The cluster-robust covariance estimator assumes only that clusters are independent, and is consistent as the number of clusters grows, not the number of observations. With few clusters it is badly biased downward.
5. The repeated-measures  $F$  test is exact if and only if the covariance satisfies sphericity: all pairwise differences of the repeated measurements have the same variance (Lemma 33.3.1, Theorem 33.3.2). Compound symmetry implies sphericity; the converse is false.
6. When sphericity fails, the moment-matched approximation multiplies both degrees of freedom by Box's  $\epsilon$  (Proposition 33.3.3); the Greenhouse–Geisser and Huynh–Feldt corrections estimate  $\epsilon$ , and Hotelling's  $T^2$  avoids the issue at a cost in power (Proposition 33.3.4).
7. Longitudinal covariance is built from random effects, serial correlation and measurement error, and the standard models keep one or two of the three (Definition 33.4.1). A wrong covariance model leaves the fixed effects unbiased and is repaired by the empirical standard error; restricted likelihoods compare only models with the same mean structure (Proposition 33.4.1).
8. In the balanced random-coefficient growth model the population curve is the average of the subjects' own curves, needing no covariance parameter; the predicted subject curve shrinks the subject's own curve towards it by  $\mathbf{D}(\mathbf{D} + \sigma^2(\mathbf{X}^\top \mathbf{X})^{-1})^{-1}$ ; and restricted maximum likelihood has the closed form of the two-stage moment estimator (Proposition 33.5.1).



## 33.6.2 Notes and sources

**Split plots.** The design and its two-stratum analysis are due to Yates (1935); Fisher (1935) had already made the choice of error term the central issue of experimental design. The route taken here is [Lemma 33.1.1](#), which reduces everything to [Theorem 16.3.1](#) plus Kruskal's condition ([Theorem 6.9.3](#)); Christensen (2020, chapter 11) develops split-plot models from a cluster sampling model in the same coordinate-free spirit. The cross-stratum simple effect of [Theorem 33.1.2\(f\)](#) is standard practical advice given a derivation here.

**Clusters and the design effect.** The design effect  $1 + (m - 1)\rho$  is Kish's (1965), from survey sampling; its damage to standard errors for aggregate regressors was pressed on econometricians by Moulton (1990). The cluster-robust estimator [Eq. 33.2.3](#) is the clustered form of the sandwich estimators of [Section 21.4](#), whose proof also gives its consistency, cluster by cluster in place of observation by observation; its small-sample failings and their remedies are surveyed by Cameron and Miller (2015).

**Sphericity.** Box (1954) computed the effect of an arbitrary covariance on the two-way analysis of variance and introduced  $\epsilon$  and the approximation of [Proposition 33.3.3](#). Huynh and Feldt (1970) isolated the necessary and sufficient condition of [Lemma 33.3.1](#), which is named after them, and the fixed- $N$  form of [Theorem 33.3.2\(c\)](#) is theirs; the proof here takes the easier route through the large-sample limit. Geisser and Greenhouse (1958) showed the lower-bound reference distribution to be exactly conservative, not merely so within Box's approximation; Greenhouse and Geisser (1959) made the plug-in and lower-bound corrections a recommendation, Huynh and Feldt (1976) gave the bias-corrected [Eq. 33.3.3](#), and Mauchly (1940) the likelihood ratio test of sphericity itself. The multivariate alternative is Hotelling's  $T^2$  (Hotelling 1931), whose distribution theory we quote from Anderson (2003).

**Longitudinal covariance.** The three-source decomposition [Eq. 33.4.1](#) and the empirical variogram are due to Diggle (1988); Diggle, Heagerty, Liang and Zeger (2002) is the standard account. That a working covariance can be wrong without invalidating the estimate, provided the standard error is computed empirically, is from Liang and Zeger (1986), whose generalized estimating equations are the subject of [Chapter 40](#). The rule of [Proposition 33.4.1\(c\)](#) follows from [Definition 32.4.1](#) and is a common source of error in practice. Degrees of freedom under an estimated covariance belong to Satterthwaite (1946) and Kenward and Roger (1997); missing data to [Chapter 41](#), the missing-at-random condition being Rubin's (1976).

**Growth curves.** Potthoff and Roy (1964) introduced [Eq. 33.5.1](#) and reduced it to an ordinary multivariate analysis of variance by a weight matrix fixed in advance; Rao (1965) gave the efficient estimate and recast growth curves in terms of random coefficients, [Eq. 33.5.2](#); Laird and Ware (1982) put the two together in the form that became standard software. The shrinkage [Eq. 33.5.5](#) is the multivariate signal-to-noise weighting of [Chapter 27](#); the closed form [Eq. 33.5.7](#) appears to be folklore, proved here by factorizing the error-contrast likelihood into within- and between-subject pieces.

**The data.** The investment panel of [Example \(Investment in eleven firms\)](#) and [Example \(The investment trajectories of eleven firms\)](#) is Grunfeld's (1958), in the eleven-firm version reconstructed by Kleibers and Zeileis (2010) and distributed with statsmodels — the panel already fitted with a random firm effect in [Chapter 32](#), seen here from two more angles. The split-plot and repeated-measures examples are simulated.



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# Chapter 34

## Exponential Families and the Generalized Linear Model

Every chapter so far has modelled a response whose scatter about a linear mean is, if not normal, at least additive and of constant shape. Counts, proportions, waiting times and insurance claims are none of these: their means are confined to an interval, their spread grows with their mean, their distributions are skewed by construction. This chapter builds the framework that handles them — a response from an **exponential dispersion family**, a **linear predictor**  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , and a **link** joining the two. Least squares gives way to maximum likelihood, the normal equations to the likelihood equations, and the closed-form estimate to an iteration that turns out to be weighted least squares, run until it stops changing. Exact distribution theory gives way to asymptotics, so every claim here comes with conditions attached.

**What you need.** Chapter 5 and Chapter 6 (the linear model, projection, leverage), Section 7.3 and Section 7.4 (maximum likelihood Theorem 7.3.1, the score and information Proposition 7.4.7, exponential families Lemma 7.4.2), Chapter 11 (the likelihood ratio test Theorem 11.4.1 and the trinity Proposition 11.4.2), Section 21.3 (weighted least squares Theorem 21.3.1). Chapter 20 (residuals and influence, and iteratively reweighted least squares Proposition 20.6.1) and Chapter 29 (AIC) are used in the last section.

### 34.0.1 Roadmap

- Exponential dispersion families
- Link functions
- The likelihood equations
- Iteratively reweighted least squares
- Deviance and asymptotic inference
- Summary and notes

## 34.1 Exponential dispersion families

The normal linear model makes two assumptions usually stated together but logically independent: that  $E(Y_i)$  is linear in the regressors, and that  $Y_i$  is normal with a variance not depending

on that mean. This section replaces the second by the least it can, a one-parameter family whose log-density is linear in the data with a single extra scale. [Section 7.4](#) already wrote the normal linear model in that form ([Eq. 7.4.1](#)); what follows names the ingredients, because in a generalized linear model each does a separate job.

### 34.1.1 The family

#### Definition 34.1.1 — Exponential dispersion family

Let  $\nu$  be a  $\sigma$ -finite measure on  $\mathbb{R}$ . A random variable  $Y$  belongs to the **exponential dispersion family** generated by  $\nu$  if, for a dispersion  $\phi > 0$  and a known **prior weight**  $w > 0$ , its density with respect to  $\nu$  is

$$f(y; \theta, \phi) = \exp \left\{ \frac{w \{y\theta - b(\theta)\}}{\phi} + c(y, \phi, w) \right\}, \quad (34.1.1)$$

where  $\theta$  lies in the **natural parameter space**

$$\Theta = \left\{ \theta \in \mathbb{R} : \kappa(\theta) = \int e^{wy\theta/\phi} e^{c(y, \phi, w)} d\nu(y) < \infty \right\}$$

and  $b(\theta) = (\phi/w) \log \kappa(\theta)$  is the **cumulant function**, required to be the *same* function of  $\theta$  for every admissible pair  $(\phi, w)$ . The parameter  $\theta$  is the **natural** (or canonical) parameter.

Four ingredients, four jobs. The measure  $\nu$  fixes what kind of thing the response is: Lebesgue measure on  $\mathbb{R}$  for the normal, on  $(0, \infty)$  for the gamma, counting measure on  $\{0, 1, \dots\}$  for the Poisson. The cumulant function  $b$  determines every moment, as this section shows. The dispersion  $\phi$  is a nuisance scale, unknown for the normal and the gamma, fixed at 1 for the Poisson and the binomial. The prior weight  $w$  is known and records how much data each observation summarizes: 1 for a single observation,  $m$  when  $y$  is an average of  $m$  of them. Only  $\phi/w$  appears, so  $w$  never needs estimating; and that one formalism covers  $w = 1$  and  $w = m$  at once is what the common- $b$  requirement buys, since it forces  $c$  to absorb the whole dependence on  $(\phi, w)$ . That is what makes the family a *dispersion* family (Jorgensen 1987).

The function  $c$  does no statistical work: it absorbs whatever is left once the term linear in  $y\theta$  has been extracted, and never depends on  $\theta$ . That is the point, because in the log-likelihood

$$\log f(y; \theta, \phi) = \frac{w \{y\theta - b(\theta)\}}{\phi} + c(y, \phi, w), \quad (34.1.2)$$

the data and the parameter of interest meet only in the product  $y\theta$ . By [Lemma 7.4.1](#),  $y$  itself is sufficient for  $\theta$ .

The normalization  $b = (\phi/w) \log \kappa$  is forced, because a density must integrate to one. The next lemma records the two properties of  $b$  that everything else rests on.

#### Lemma 34.1.1 — The natural parameter space and the cumulant function

$\Theta$  is a convex subset of  $\mathbb{R}$ , and  $b$  is convex on  $\Theta$ . If  $Y$  is not degenerate under some  $\theta_0 \in \Theta$ , then  $b$  is strictly convex on  $\Theta$ .

*Proof.* Write  $h(y) = e^{c(y, \phi, w)}$  and  $s = w/\phi > 0$ , so that  $\kappa(\theta) = \int e^{sy\theta} h(y) d\nu(y)$ . Let  $\theta_1, \theta_2 \in \Theta$  and  $\alpha \in (0, 1)$ . Hölder's inequality with exponents  $1/\alpha$  and  $1/(1-\alpha)$  applied to the factors  $(e^{sy\theta_1} h)^\alpha$  and  $(e^{sy\theta_2} h)^{1-\alpha}$  gives

$$\kappa(\alpha\theta_1 + (1-\alpha)\theta_2) \leq \kappa(\theta_1)^\alpha \kappa(\theta_2)^{1-\alpha} < \infty.$$

So  $\alpha\theta_1 + (1-\alpha)\theta_2 \in \Theta$ , which makes  $\Theta$  convex, and taking logarithms shows  $\log \kappa$ , hence  $b$ , is convex. Equality in Hölder's inequality requires  $e^{sy\theta_1} h$  and  $e^{sy\theta_2} h$  to be proportional  $\nu$ -almost everywhere, that is,  $e^{sy(\theta_1 - \theta_2)}$  constant on the set where  $h > 0$ . For  $\theta_1 \neq \theta_2$  this forces that set to be a single point, so  $Y$  is degenerate. ■

Throughout the chapter  $\Theta$  is assumed to have nonempty interior and  $\theta$  to lie in it — the same regularity [Lemma 7.4.2](#) needs, and what makes the differentiations below legal.

## 34.1.2 Moments from the cumulant function

### Proposition 34.1.2 — Mean, variance and cumulants

Let  $Y$  have density [Eq. 34.1.1](#) with  $\theta$  in the interior of  $\Theta$ . Then  $Y$  has moments of all orders, and

- (a)  $\mu := E(Y) = b'(\theta)$  and  $\text{Var}(Y) = \frac{\phi}{w} b''(\theta)$ ;
- (b) the map  $\theta \mapsto b'(\theta)$  is strictly increasing, hence a bijection from the interior of  $\Theta$  onto an open interval  $\mathcal{M}$ , the **mean space**;
- (c) the cumulant generating function of  $Y$  is  $K(t) = \log E e^{tY} = \frac{w}{\phi} \{b(\theta + t\phi/w) - b(\theta)\}$  for small  $|t|$ , so the  $r$ th cumulant of  $Y$  is  $(\phi/w)^{r-1} b^{(r)}(\theta)$ .

*Proof.* With  $h$  and  $s = w/\phi$  as in the proof of [Lemma 34.1.1](#), the density is  $\kappa(\theta)^{-1} e^{sy\theta} h(y)$ . The function  $\kappa$  is a two-sided Laplace transform of the measure  $h d\nu$  evaluated at  $-s\theta$ ; such a transform is analytic in the interior of its domain of convergence, and may be differentiated under the integral sign there, together with the finiteness of all moments (Widder 1941, chapter II; Brown 1986, chapter 2). Hence

$$\kappa'(\theta) = s \int y e^{sy\theta} h d\nu = s E(Y) \kappa(\theta), \quad \kappa''(\theta) = s^2 E(Y^2) \kappa(\theta).$$

Since  $b = s^{-1} \log \kappa$ , we have  $b' = \kappa'/(s\kappa) = E(Y)$ , which is (a) for the mean. Differentiating  $\kappa' = s b' \kappa$  once more gives  $\kappa'' = s b'' \kappa + s^2 (b')^2 \kappa$ , so  $E(Y^2) = b''/s + (b')^2$  and  $\text{Var}(Y) = b''(\theta)/s = \phi b''(\theta)/w$ .

- (b) By [Lemma 34.1.1](#),  $b$  is strictly convex, so  $b' = E(Y)$  is strictly increasing; it is continuous, so its range on an open interval is an open interval.
- (c) For  $t$  small enough that  $\theta + t/s \in \Theta$ ,  $E e^{tY} = \kappa(\theta)^{-1} \int e^{(s\theta+t)y} h d\nu = \kappa(\theta + t/s)/\kappa(\theta)$ ;

taking logarithms gives  $K(t) = s^{-1}\{b(\theta + t/s) - b(\theta)\}$  with  $s^{-1} = \phi/w$ , and the coefficient of  $t^r/r!$  in its expansion is  $s^{-(r-1)}b^{(r)}(\theta)$ .

■

Part (a) is the engine of the chapter: the variance is not free but determined by the mean, up to the single scalar  $\phi/w$ . Part (b) lets us reparameterize by the mean, which is the parameter a modeller wants.

### Definition 34.1.2 — Variance function

For  $\mu \in \mathcal{M}$  let  $\theta(\mu)$  be the inverse of  $b'$  from Proposition 34.1.2(b). The **variance function** of the family is  $V(\mu) = b''(\theta(\mu))$ , so that

$$\text{Var}(Y) = \frac{\phi}{w} V(\mu), \quad \frac{d\theta}{d\mu} = \frac{1}{V(\mu)}.$$

The last identity is the derivative of an inverse function,  $d\mu/d\theta = b'' = V$ , and is used constantly below. The pair  $(\mathcal{M}, V)$  determines the family: integrating  $V$  gives  $\theta(\mu)$ , hence  $b$ , hence the density.

*Idea.* A generalized linear model is specified by three separate choices: the **variance function**  $V$ , which fixes the family; the **link**; and the **linear predictor**  $\mathbf{X}\beta$ . Chapter 38 keeps the first two and throws the density away, which is possible because the estimating equations of Section 34.3 involve nothing else.

## 34.1.3 The standard families

The five classical families, plus the negative binomial with a known shape  $k$ , come from choosing  $\nu$  and reading off  $b$ . The gamma row is written for shape  $a$ , so that  $\phi = 1/a$ . The measure  $\nu$  is Lebesgue on  $\mathbb{R}$  for the normal and on  $(0, \infty)$  for the gamma and inverse Gaussian, counting measure on  $\{0, 1, 2, \dots\}$  for the Poisson and negative binomial, and counting measure on  $\{0, 1/m, \dots, 1\}$  for the binomial, whose response is a proportion out of  $m$ .

| Family           | $\theta(\mu)$            | $b(\theta)$             | $\phi$     | $V(\mu)$         | $\mathcal{M}$ |
|------------------|--------------------------|-------------------------|------------|------------------|---------------|
| normal           | $\mu$                    | $\theta^2/2$            | $\sigma^2$ | 1                | $\mathbb{R}$  |
| binomial         | $\log \frac{\mu}{1-\mu}$ | $\log(1 + e^\theta)$    | 1          | $\mu(1-\mu)$     | $(0, 1)$      |
| Poisson          | $\log \mu$               | $e^\theta$              | 1          | $\mu$            | $(0, \infty)$ |
| gamma            | $-1/\mu$                 | $-\log(-\theta)$        | $1/a$      | $\mu^2$          | $(0, \infty)$ |
| inverse Gaussian | $-1/(2\mu^2)$            | $-\sqrt{-2\theta}$      | $\sigma^2$ | $\mu^3$          | $(0, \infty)$ |
| neg. binomial    | $\log \frac{\mu}{\mu+k}$ | $-k \log(1 - e^\theta)$ | 1          | $\mu(1 + \mu/k)$ | $(0, \infty)$ |

Two entries deserve a full derivation, because they show the two ways the prior weight and the dispersion enter.

**Example. (The binomial as a family of proportions)**

Let  $S \sim \text{Binomial}(m, \mu)$  and let  $Y = S/m$  be the observed proportion. Its probability mass function at  $y = s/m$  is

$$\begin{aligned} \binom{m}{my} \mu^{my} (1 - \mu)^{m(1-y)} \\ = \exp \left\{ m \left( y \log \frac{\mu}{1 - \mu} + \log(1 - \mu) \right) + \log \binom{m}{my} \right\}. \end{aligned}$$

This is [Eq. 34.1.1](#) with  $w = m$ ,  $\phi = 1$ ,  $\theta = \log\{\mu/(1 - \mu)\}$  and  $b(\theta) = -\log(1 - \mu) = \log(1 + e^\theta)$ . Then  $b'(\theta) = e^\theta/(1 + e^\theta) = \mu$  and  $b''(\theta) = \mu(1 - \mu)$ , so [Proposition 34.1.2](#) gives  $E(Y) = \mu$  and  $\text{Var}(Y) = \mu(1 - \mu)/m$ : the number of trials acts entirely through the prior weight. The dispersion is fixed at 1, so a proportion more variable than  $\mu(1 - \mu)/m$  needs the different treatment of [Chapter 38](#).

**Example. (The gamma and its constant coefficient of variation)**

Let  $Y$  have the gamma density with shape  $a > 0$  and mean  $\mu$ ,

$$f(y) = \frac{(a/\mu)^a}{\Gamma(a)} y^{a-1} e^{-ay/\mu} \quad \text{on } (0, \infty).$$

Its logarithm is

$$a \left\{ y \left( -\frac{1}{\mu} \right) - \log \mu \right\} + (a - 1) \log y + a \log a - \log \Gamma(a),$$

which is [Eq. 34.1.1](#) with  $w = 1$ ,  $\phi = 1/a$ ,  $\theta = -1/\mu$  and  $b(\theta) = \log \mu = -\log(-\theta)$ . Then  $b'(\theta) = -1/\theta = \mu$  and  $b''(\theta) = 1/\theta^2 = \mu^2$ , so  $\text{Var}(Y) = \phi\mu^2$  and the coefficient of variation  $\sqrt{\text{Var}(Y)}/\mu = \sqrt{\phi}$  is free of the mean: the spread is *proportional* to the level, the defining feature of the gamma family as a regression model. With  $\mu = 4$  and  $\phi = 0.4$  the formula gives  $\text{Var}(Y) = 6.400$ , and by [Proposition 34.1.2\(c\)](#) the third cumulant is  $\phi^2 b'''(\theta) = 2\phi^2 \mu^3 = 20.480$ ; both agree with the exact gamma moments in the listing below.

```
import numpy as np
from scipy import stats

# Each family: theta(mu), b(theta) and its first three derivatives, and the exact
# distribution of Y, so that the exponential-family formulas can be checked.
FAMILIES = {
    "normal": dict(
        mu=2.0, phi=1.5, w=1.0,
        theta=lambda mu: mu,
        b=lambda t: t ** 2 / 2, b1=lambda t: t, b2=lambda t: 1.0 + 0 * t, b3=lambda
t: 0 * t,
        dist=lambda mu, phi, w: stats.norm(mu, np.sqrt(phi / w)),
    ),
    "binomial": dict(
        # Y = S/m, the sample
        proportion
        mu=0.3, phi=1.0, w=10.0,
        theta=lambda mu: np.log(mu / (1 - mu)),
        b=lambda t: np.log1p(np.exp(t)),
```

```

        b1=lambda t: 1 / (1 + np.exp(-t)),
        b2=lambda t: np.exp(-t) / (1 + np.exp(-t)) ** 2,
        b3=lambda t: np.exp(-t) * (np.exp(-t) - 1) / (1 + np.exp(-t)) ** 3,
        dist=None,                                # handled separately: Y is m
        ^-1 Binomial(m, mu)
    ),
    "Poisson": dict(
        mu=3.5, phi=1.0, w=1.0,
        theta=lambda mu: np.log(mu),
        b=lambda t: np.exp(t), b1=np.exp, b2=np.exp, b3=np.exp,
        dist=lambda mu, phi, w: stats.poisson(mu),
    ),
    "gamma": dict(                                # shape a = 1/phi
        mu=4.0, phi=0.4, w=1.0,
        theta=lambda mu: -1 / mu,
        b=lambda t: -np.log(-t), b1=lambda t: -1 / t, b2=lambda t: 1 / t ** 2,
        b3=lambda t: -2 / t ** 3,
        dist=lambda mu, phi, w: stats.gamma(a=w / phi, scale=mu * phi / w),
    ),
    "inverse Gaussian": dict(
        mu=2.0, phi=0.4, w=1.0,
        theta=lambda mu: -1 / (2 * mu ** 2),
        b=lambda t: -np.sqrt(-2 * t), b1=lambda t: (-2 * t) ** -0.5,
        b2=lambda t: (-2 * t) ** -1.5, b3=lambda t: 3 * (-2 * t) ** -2.5,
        dist=lambda mu, phi, w: stats.invgauss(mu=mu * phi / w, scale=w / phi),
    ),
}

def check(name, spec, k=None):
    """Compare b'(theta), phi b''(theta)/w and (phi/w)^2 b'''(theta) with the true
    moments."""
    mu, phi, w = spec["mu"], spec["phi"], spec["w"]
    t = spec["theta"](mu)
    mean_ef = spec["b1"](t)
    var_ef = phi * spec["b2"](t) / w
    cum3_ef = (phi / w) ** 2 * spec["b3"](t)
    if name == "binomial":                        # Y = S/m with S ~ Binomial(m
        , mu)
        m = int(w)
        s = np.arange(m + 1)
        p = stats.binom(m, mu).pmf(s)
        y = s / m
        mean, var = y @ p, (y - mu) ** 2 @ p
        cum3 = (y - mu) ** 3 @ p
    else:
        d = spec["dist"](mu, phi, w)
        mean, var = d.mean(), d.var()
        cum3 = d.stats("s") * var ** 1.5          # third cumulant = skewness *
        sd^3
    return mean_ef, mean, var_ef, var, cum3_ef, float(cum3)

for name, spec in FAMILIES.items():
    m_ef, m, v_ef, v, c_ef, c = check(name, spec)
    print(f"{name:17s} mean {m_ef:8.4f} {m:8.4f}    var {v_ef:8.4f} {v:8.4f}    cum3 {
c_ef:8.4f} {c:8.4f}")

```

The variance functions are drawn in [Figure 34.1.1](#). The normal, Poisson, gamma and inverse



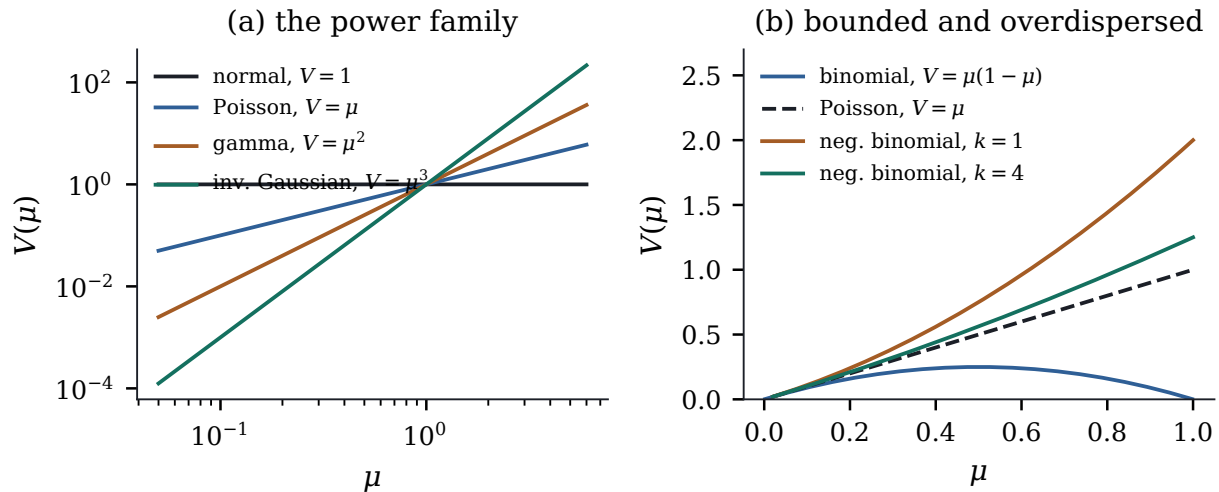


Figure 34.1.1: Variance functions. (a) The power family  $V(\mu) = \mu^\xi$  on logarithmic axes: normal, Poisson, gamma and inverse Gaussian are  $\xi = 0, 1, 2, 3$ . (b) Two variance functions that are not powers. The binomial variance vanishes at both ends; the negative binomial variance  $\mu(1 + \mu/k)$  lies above the Poisson line for every finite  $k$ .

Gaussian are powers,  $V(\mu) = \mu^\xi$  with  $\xi = 0, 1, 2, 3$ ; interpolating between them gives a continuum of families (Jorgensen 1987) that includes a useful model for a response with a point mass at zero and a continuous positive part. The remaining two are not powers: the binomial variance vanishes at both ends of  $(0, 1)$ , because a proportion pinned near 0 or 1 cannot vary, and the negative binomial variance exceeds the Poisson variance by an amount set by  $k$ .

**Example. (The negative binomial with known shape)**

Let  $\Pr(Y = y) = \frac{\Gamma(y+k)}{\Gamma(k)y!} \pi^y (1-\pi)^k$ ,  $y = 0, 1, 2, \dots$ , for a fixed  $k > 0$ . Writing  $\theta = \log \pi$  puts this in the form Eq. 34.1.1 with  $w = 1$ ,  $\phi = 1$  and  $b(\theta) = -k \log(1 - e^\theta)$ , so that

$$\mu = b'(\theta) = \frac{k\pi}{1-\pi}, \quad b''(\theta) = \frac{k\pi}{(1-\pi)^2} = \mu \left(1 + \frac{\mu}{k}\right)$$

after substituting  $\pi = \mu/(\mu + k)$ . With  $k = 2$  and  $\mu = 3$  this gives  $\text{Var}(Y) = 7.500$ , against 3 for a Poisson variable with the same mean.

The caveat is essential: this is an exponential dispersion family **only for fixed  $k$** . The shape enters the variance function, not the dispersion, so it cannot be divided out, and letting  $k$  vary leaves the form Eq. 34.1.1. Estimating  $k$  is outside this chapter's machinery; Chapter 37 takes it up.

## 34.1.4 Sufficiency, and what the natural parameter buys

Suppose  $Y_1, \dots, Y_n$  are independent, each with density Eq. 34.1.1, with a common  $b$  and  $\phi$ , known weights  $w_i$ , and natural parameters  $\theta_i$ . The joint log-likelihood is

$$\ell = \frac{1}{\phi} \sum_{i=1}^n w_i \{y_i \theta_i - b(\theta_i)\} + \sum_{i=1}^n c(y_i, \phi, w_i). \quad (34.1.3)$$

Nothing has been assumed yet about how the  $\theta_i$  depend on regressors — that is the next section's business — but Eq. 34.1.3 already shows what to expect. If the  $\theta_i$  are constrained to a  $p$ -dimensional linear family  $\theta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , then  $\sum_i w_i y_i \theta_i = \boldsymbol{\beta}^\top \mathbf{X}^\top \text{diag}(\mathbf{w}) \mathbf{y}$  and the data enter the likelihood only through the  $p$  numbers  $\mathbf{X}^\top \text{diag}(\mathbf{w}) \mathbf{y}$ . With  $\phi$  known the remaining term  $\sum_i c(y_i, \phi, w_i)$  is a function of the data alone, so Lemma 7.4.1 makes this vector sufficient for  $\boldsymbol{\beta}$ , and if the attainable  $\boldsymbol{\beta}$  contain an open subset of  $\mathbb{R}^p$ , Lemma 7.4.2 makes it complete. With  $\phi$  unknown that term depends on a parameter, both arguments fail, and a second statistic is needed — as in the normal linear model, the case  $b(\theta) = \theta^2/2$ ,  $w_i = 1$ , where  $\mathbf{X}^\top \mathbf{y}$  is accompanied by  $\mathbf{y}^\top \mathbf{y}$  for  $\sigma^2$  (Proposition 7.4.3).

That is a genuine reduction, and it is why the natural parameter deserves its name. What it does not give is any analogue of the Gauss–Markov theorem: the maximum likelihood estimate is a nonlinear function of  $\mathbf{y}$ , generally biased, with an exact distribution unknown outside the normal case. Everything finite-sample that Parts II and III delivered is gone, and Section 34.5 replaces it with asymptotics.

*Warning.* The reduction needs  $\theta_i$  — not  $\mu_i$ , and not some other function of it — to be linear in  $\boldsymbol{\beta}$ . That is the *canonical link*, one modelling choice among several. With any other link the reduction fails and the data enter the likelihood through the whole vector  $\mathbf{y}$ ; Section 34.2 explains why one might accept that.

## 34.1.5 Exercises

### 34.1.5.1 A. Check your understanding

#### Exercise A1.

Write the Poisson probability mass function in the form Eq. 34.1.1, identify  $\theta$ ,  $b$ ,  $\phi$ ,  $w$  and  $c$ , and verify  $E(Y) = \text{Var}(Y) = \mu$  using Proposition 34.1.2. What is  $\Theta$ ?

*Solution.*  $\Pr(Y = y) = e^{-\mu} \mu^y / y! = \exp\{y \log \mu - \mu - \log y!\}$ , so  $\theta = \log \mu$ ,  $b(\theta) = e^\theta$ ,  $\phi = w = 1$  and  $c(y) = -\log y!$ . Then  $b' = b'' = e^\theta = \mu$ , and Proposition 34.1.2 gives  $E(Y) = \text{Var}(Y) = \mu$ . The sum  $\sum_y e^{y\theta} / y! = e^{e^\theta}$  converges for every real  $\theta$ , so  $\Theta = \mathbb{R}$  and  $\mathcal{M} = (0, \infty)$ .

### 34.1.5.2 B. Practice

#### Exercise B1.

Verify the inverse Gaussian row of the table: starting from

$$f(y) = (2\pi\sigma^2 y^3)^{-1/2} \exp\left\{-\frac{(y - \mu)^2}{2\sigma^2 \mu^2 y}\right\} \quad \text{on } (0, \infty),$$

put it in the form [Eq. 34.1.1](#) and confirm  $V(\mu) = \mu^3$ . What is  $\Theta$ , and what happens at  $\theta = 0$ ?

*Solution.* Expanding the square,  $-(y-\mu)^2/(2\sigma^2\mu^2y) = -y/(2\sigma^2\mu^2) + 1/(\sigma^2\mu) - 1/(2\sigma^2y)$ . So the exponent is  $\sigma^{-2}\{y\theta - b(\theta)\}$  with  $\theta = -1/(2\mu^2)$  and  $b(\theta) = -1/\mu = -\sqrt{-2\theta}$ , while the remaining terms  $-\frac{1}{2}\log(2\pi\sigma^2y^3) - 1/(2\sigma^2y)$  make up  $c(y, \phi, w)$  with  $\phi = \sigma^2$ ,  $w = 1$ . Then  $b'(\theta) = (-2\theta)^{-1/2} = \mu$  and  $b''(\theta) = (-2\theta)^{-3/2} = \mu^3$ . Here  $\Theta = (-\infty, 0]$  and  $\theta = 0$  is a boundary point: the integral still converges there, but  $b'$  blows up, so  $\mathcal{M} = (0, \infty)$  is the image of the *interior* only.

### Exercise B2.

Show that the variance function determines the family: if  $V$  is given on an interval  $\mathcal{M}$ , then  $\theta(\mu) = \int^\mu dt/V(t)$  and  $b(\theta(\mu)) = \int^\mu t dt/V(t)$ , each up to constants that only relocate the density. Carry this out for  $V(\mu) = \mu(1-\mu)$  on  $(0, 1)$  and recover the binomial row.

*Solution.* By [Definition 34.1.2](#),  $d\theta/d\mu = 1/V(\mu)$ , which integrates to the stated  $\theta(\mu)$ . Also  $db/d\mu = (db/d\theta)(d\theta/d\mu) = \mu/V(\mu)$ , giving the second integral. For  $V(\mu) = \mu(1-\mu)$ , partial fractions give  $\int d\mu/\{\mu(1-\mu)\} = \log\{\mu/(1-\mu)\} = \theta$ , and  $\int \mu d\mu/\{\mu(1-\mu)\} = -\log(1-\mu) = \log(1+e^\theta) = b(\theta)$ , which is the binomial row.

### Exercise B3.

Use [Proposition 34.1.2\(c\)](#) to show that the skewness of  $Y$  is  $\gamma_1 = \sqrt{\phi/w} V'(\mu)/\sqrt{V(\mu)}$ . Evaluate it for the Poisson, the gamma and the binomial, and say in each case how the skewness behaves as the mean moves.

*Solution.* The third cumulant is  $(\phi/w)^2 b'''(\theta)$ , and  $b'''(\theta) = dV/d\theta = V'(\mu) d\mu/d\theta = V'(\mu)V(\mu)$ . Dividing by  $\text{Var}(Y)^{3/2} = \{(\phi/w)V\}^{3/2}$  gives  $\gamma_1 = \sqrt{\phi/w} V'(\mu)/\sqrt{V(\mu)}$ . For the Poisson ( $\phi = w = 1$ ,  $V = \mu$ ),  $\gamma_1 = \mu^{-1/2}$ : small counts are strongly skewed and large ones nearly symmetric. For the gamma,  $V = \mu^2$  gives  $\gamma_1 = 2\sqrt{\phi}$ , free of the mean. For the binomial with  $w = m$ ,  $\gamma_1 = (1-2\mu)/\sqrt{m\mu(1-\mu)}$ , which changes sign at  $\mu = \frac{1}{2}$ .

### 34.1.5.3 C. Going deeper

#### Exercise C1.

Show that the uniform family on  $(0, \vartheta)$  is not an exponential dispersion family for any choice of  $\nu$ , and that the Cauchy family with median  $\vartheta$  is not either. Which hypothesis of [Definition 34.1.1](#) fails in each case?

#### Exercise C2.

Suppose  $V(\mu) = \mu^\xi$  on  $(0, \infty)$  for some  $\xi \notin \{1, 2\}$ . Carry out [Exercise \(B2\)](#) to find  $\theta(\mu)$  and  $b(\theta)$ , and check that  $\xi = 0$  and  $\xi = 3$  reproduce the normal and inverse Gaussian rows. For  $1 < \xi < 2$  the family is the distribution of a Poisson number of independent gamma variables, with positive probability at 0 and a continuous part on

$(0, \infty)$  (Jorgensen 1987); confirm from the general formula that its mean space is  $(0, \infty)$ .

## 34.2 Link functions

The linear model says  $E(Y_i) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . For a count that must be nonnegative, or a probability that must lie in  $(0, 1)$ , that equation is a promise the right-hand side cannot keep: a linear function of unbounded regressors takes every real value. The resolution is to put a monotone function between the mean and the linear predictor.

### 34.2.1 The model

#### Definition 34.2.1 — Generalized linear model

Let  $\mathbf{X}$  be an  $n \times p$  model matrix with rows  $\mathbf{x}_{(i)}^\top$ , and let  $w_1, \dots, w_n > 0$  be known prior weights. A **generalized linear model** consists of

- (a) a **random component**:  $Y_1, \dots, Y_n$  are independent, and  $Y_i$  has the density [Eq. 34.1.1](#) with natural parameter  $\theta_i$ , weight  $w_i$  and a dispersion  $\phi$  that is the same for all  $i$ ;
- (b) a **linear predictor**  $\eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , with  $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta} \in \mathbb{R}^n$ ;
- (c) a **link function**  $g : \mathcal{M} \rightarrow \mathbb{R}$ , strictly monotone and twice continuously differentiable, with

$$g(\mu_i) = \eta_i, \quad \mu_i = E(Y_i) = b'(\theta_i).$$

The inverse  $h = g^{-1}$  is the **response function**, so  $\mu_i = h(\eta_i)$ . The **admissible set** of the model is  $\mathcal{B} = \{\boldsymbol{\beta} \in \mathbb{R}^p : \mathbf{x}_{(i)}^\top \boldsymbol{\beta} \in g(\mathcal{M}) \text{ for all } i\}$ .

The linear model of [Chapter 5](#) is the case: normal random component, identity link,  $w_i = 1$ ,  $\phi = \sigma^2$ ,  $\mathcal{B} = \mathbb{R}^p$ . Every classical result of Parts II and III is about that one cell of a large table.

Three remarks. The dispersion is common to all observations: the regressors move the mean, the variance follows through  $V$ , and nothing else may vary (modelling  $\phi$  too is the subject of [Chapter 45](#)). The weights  $w_i$  are known, not estimated, and usually record group sizes. And  $g$  transforms the mean, not the data: [Chapter 22](#) transformed  $Y$  itself, whereas here the response is left alone and the link acts inside the expectation.

#### Definition 34.2.2 — Canonical link

The **canonical link** of a family is  $g = (b')^{-1}$ , so that  $\theta_i = \eta_i$ : the natural parameter is the linear predictor. Equivalently,  $g'(\mu) = 1/V(\mu)$ .

| Family | canonical link | $g(\mu)$ | range of $\eta$ | other links in use |
|--------|----------------|----------|-----------------|--------------------|
| normal | identity       | $\mu$    | $\mathbb{R}$    | log, inverse       |

| Family               | canonical link        | $g(\mu)$              | range of $\eta$ | other links in use                            |
|----------------------|-----------------------|-----------------------|-----------------|-----------------------------------------------|
| binomial             | logit                 | $\log\{\mu/(1-\mu)\}$ | $\mathbb{R}$    | probit,<br>complementary<br>log-log, identity |
| Poisson              | log                   | $\log \mu$            | $\mathbb{R}$    | identity, square<br>root                      |
| gamma                | inverse               | $-1/\mu$              | $(-\infty, 0)$  | log, identity                                 |
| inverse Gaussian     | inverse square        | $-1/(2\mu^2)$         | $(-\infty, 0)$  | log                                           |
| negative<br>binomial | $\log\{\mu/(\mu+k)\}$ | as shown              | $(-\infty, 0)$  | log                                           |

The sign conventions for the gamma and inverse Gaussian are cosmetic:  $1/\mu$  and  $-1/\mu$  give the same fitted values with  $\boldsymbol{\beta}$  negated. The column headed “range of  $\eta$ ” is not. For the normal, binomial and Poisson the canonical link maps  $\mathcal{M}$  onto all of  $\mathbb{R}$ , so  $\mathcal{B} = \mathbb{R}^p$  and no constraint is ever active. For the gamma and inverse Gaussian it maps onto a half-line, and  $\mathcal{B}$  is the polyhedron  $\{\boldsymbol{\beta} : \mathbf{x}_{(i)}^\top \boldsymbol{\beta} < 0 \text{ for all } i\}$  — an open convex set with a boundary a fitting algorithm can walk into.

## 34.2.2 What a link choice costs and buys

*Idea.* The canonical link is a mathematical convenience, not a scientific claim. It makes the likelihood concave (Theorem 34.3.3), makes observed and expected information agree (Theorem 34.3.1), reduces the data to  $p$  sufficient statistics (Section 34.1), and forces the fitted values to reproduce the observed weighted column totals of  $\mathbf{X}$  exactly (Corollary 34.3.2). None of that is a reason to believe the mean depends on the regressors in that particular way.

Four considerations decide a link in practice.

**Range.** A link whose range is all of  $\mathbb{R}$  removes constraints from the fitting problem. The log link for a Poisson response guarantees  $\hat{\mu}_i > 0$  for every  $\boldsymbol{\beta} \in \mathbb{R}^p$ , and the logit link for a binomial response guarantees  $\hat{\mu}_i \in (0, 1)$ . An identity link for either does not.

### Example. (An identity link close to its boundary)

Eight counts were simulated from a Poisson model with mean  $12 - 1.2x$  at  $x = 1, \dots, 8$ . An identity-link Poisson fit recovers the declining mean, and every fitted value is positive — but the fitted line reaches zero at  $x = 9.89$ , barely outside the range of the data. Predicting at  $x = 10$  would return a negative mean, and a slightly different sample would push the boundary inside the observed range, where the likelihood is  $-\infty$ . The log link has no such boundary.

**Interpretation.** A log link makes the regressors act multiplicatively,  $\mu = e^{\beta_0} \prod_j e^{\beta_j x_j}$ , so  $\beta_j$  is a log rate ratio and  $e^{\beta_j}$  the factor multiplying the mean per unit of  $x_j$ . A logit link makes  $\beta_j$  a log odds ratio, the subject of Chapter 35 (Definition 35.1.1). An identity link keeps  $\beta_j$  a difference of means, often what a policy question asks about. These are different scientific statements, and the choice belongs on the scale where effects are plausibly additive, not on convenience.

**Fit and stability.** Different links give genuinely different mean curves, and a badly chosen one

shows up in the residuals like any other misspecification. Non-canonical links can also produce a log-likelihood with several stationary points, and links with a restricted range need the fitting algorithm to be protected; [Section 34.4](#) returns to this.

### Example. (Food expenditure: three models, two decisions)

Engel's classic budget survey records annual income and food expenditure for 235 working-class households. Food expenditure is positive, right-skewed, and visibly more variable at high incomes. Write  $u = \log(\text{income})$  and fit three models with the same linear predictor  $\beta_0 + \beta_1 u$ : a normal random component with the identity link (ordinary least squares), a gamma random component with the identity link, and a gamma random component with the log link. The first two differ only in the variance function, the last two only in the link.

The identity-link fits give  $\hat{\beta}_1 = 571.02$  (normal) and 474.54 (gamma) francs per unit of log income. The gamma/log fit gives  $\hat{\beta}_1 = 0.8629$ , an elasticity: a one per cent rise in income is associated with a 0.86 per cent rise in expected food expenditure. Its estimated dispersion is  $\hat{\phi} = 0.0177$ , so by [Example \(The gamma and its constant coefficient of variation\)](#) the fitted coefficient of variation is  $\sqrt{\hat{\phi}} = 0.133$ , the same at every income.

[Figure 34.2.1](#) separates the two decisions. In panel (a) the identity-link mean curve is concave in income and flattens, while the log-link curve is a power law that keeps climbing; at the largest income in the sample the two fitted means differ by a factor of 1.786. Panel (b) holds the link at the identity and changes the variance function, so its bands come from the identity-link fits, the gamma one using  $\hat{\phi} = 0.0240$  and a coefficient of variation of 0.155. The normal band has constant width and at low incomes dips below zero; the gamma band widens in proportion to the mean, as the data do.

```
import numpy as np
import statsmodels.api as sm

engel = sm.datasets.engel.load_pandas().data
y = engel["foodexp"].to_numpy()          # food expenditure
X = np.column_stack([np.ones(len(y)), np.log(engel["income"].to_numpy())])

gauss = sm.GLM(y, X, family=sm.families.Gaussian()).fit()          # identity link
gam_id = sm.GLM(y, X, family=sm.families.Gamma(sm.families.links.Identity())).fit()
gam_log = sm.GLM(y, X, family=sm.families.Gamma(sm.families.links.Log())).fit()

for name, fit in [("normal/identity", gauss), ("gamma/identity", gam_id), ("gamma/
log", gam_log)]:
    print(f"{name:16s} coefficients {fit.params[0]:9.3f} {fit.params[1]:8.3f}"
          f"    dispersion {fit.scale:8.4f}")
```

## 34.2.3 Links and transformations are not the same thing

Transforming the response and choosing a link both replace  $\mu$  by  $g(\mu)$ , but in different places. Modelling  $E\{g(Y)\} = \mathbf{x}^\top \boldsymbol{\beta}$  changes the quantity modelled: by Jensen's inequality  $E\{g(Y)\} \neq$

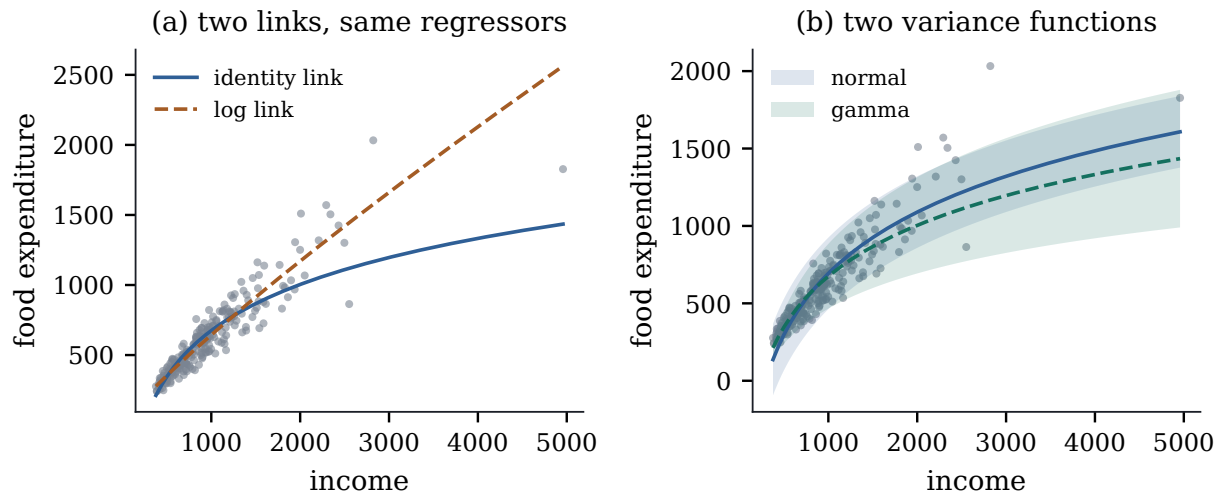


Figure 34.2.1: Engel's household budget data,  $n = 235$ . (a) Two links with the same gamma random component and the same regressors: the identity link (solid) and the log link (dashed). (b) Two variance functions with the same identity link: the normal band  $\hat{\mu} \pm 2\hat{\sigma}$  has constant width and dips below zero, while the gamma band  $\hat{\mu}(1 \pm 2\sqrt{\hat{\phi}})$ , with  $\hat{\phi}$  from the identity-link gamma fit, widens with the mean.

$g\{E(Y)\}$  unless  $g$  is affine, so the fitted curve estimates the mean of a transformed variable, and returning to the mean of  $Y$  needs the retransformation correction of [Proposition 22.5.1](#). Modelling  $g\{E(Y)\} = \mathbf{x}^\top \boldsymbol{\beta}$  leaves the response alone, and  $\hat{\mu}_i = h(\mathbf{x}_{(i)}^\top \hat{\boldsymbol{\beta}})$  estimates the mean of  $Y$  directly.

The transformation also had a second job the link does not do: stabilizing the variance ([Theorem 22.4.1](#)). A generalized linear model does not need that, because the variance function is already specified, and splitting one overloaded decision into two independent ones is the framework's main practical advantage. The weighted regression on empirical logits of [Exercise \(B2\)](#) is the historical compromise: the right link, but one weighted least squares step in place of the likelihood.

## 34.2.4 Exercises

### 34.2.4.1 A. Check your understanding

#### Exercise A1.

Describe the admissible set  $\mathcal{B}$  of [Definition 34.2.1](#) for (a) a Poisson model with the identity link, (b) a binomial model with the identity link, (c) a gamma model with the log link. Which of the three is all of  $\mathbb{R}^p$ ?

### 34.2.4.2 B. Practice

#### Exercise B1.

Let  $Y$  be positive with  $E(Y) = \mu$  and  $\text{Var}(Y) = \phi\mu^2$ , as in the gamma family. Using a second-order Taylor expansion of  $\log Y$  about  $\mu$ , show that  $E(\log Y) \approx \log \mu - \phi/2$ .

Deduce that a least squares regression of  $\log y_i$  on  $\mathbf{X}$  and a gamma/log generalized linear model estimate intercepts that differ by approximately  $\phi/2$ , while their slopes agree to this order. For the fit of [Example \(Food expenditure: three models, two decisions\)](#), how large is the discrepancy?

*Solution.* With  $g(y) = \log y$ ,  $g(Y) \approx g(\mu) + (Y - \mu)/\mu - (Y - \mu)^2/(2\mu^2)$ ; taking expectations gives  $E(\log Y) \approx \log \mu - \text{Var}(Y)/(2\mu^2) = \log \mu - \phi/2$ . Since the correction is the same constant for every observation, it is absorbed by the intercept when the model has one, and the slopes are unaffected to this order. For Engel's data  $\hat{\phi} = 0.0177$ , so the two intercepts differ by about  $\hat{\phi}/2 = 0.0177/2 = 0.009$ , less than one per cent on the expenditure scale. (For the gamma,  $E(\log Y)$  can be computed exactly with the digamma function; the approximation is good when  $\phi$  is small.)

### Exercise B2.

A Poisson model is sometimes fitted with the square-root link  $g(\mu) = \sqrt{\mu}$ , because [Theorem 22.4.1](#) makes  $\sqrt{Y}$  approximately variance-stabilized. Find  $\mathcal{B}$  for this link, find  $d\mu/d\eta$ , and explain why the link being variance-stabilizing for  $Y$  is irrelevant to whether it is a good link.

*Solution.*  $g(\mathcal{M}) = (0, \infty)$ , so  $\mathcal{B} = \{\boldsymbol{\beta} : \mathbf{x}_{(i)}^\top \boldsymbol{\beta} > 0 \ \forall i\}$ , and  $\mu = \eta^2$  gives  $d\mu/d\eta = 2\eta = 2\sqrt{\mu}$ . Variance stabilization matters when the model is fitted by *least squares on the transformed data*, because least squares wants constant variance. A generalized linear model already knows  $V(\mu) = \mu$  and weights each observation accordingly ([Theorem 34.3.1](#)), so the only question left about the link is whether  $\sqrt{EY}$  is plausibly linear in the regressors.

### Exercise B3.

The complementary log-log link for a binomial response is  $g(\mu) = \log\{-\log(1 - \mu)\}$ . Show that it is a bijection from  $(0, 1)$  onto  $\mathbb{R}$ , compute  $d\mu/d\eta$ , and show that it is *not* symmetric:  $g(\mu) \neq -g(1 - \mu)$ , unlike the logit and probit links. Which tail does it approach more slowly?

*Solution.*  $\mu \mapsto -\log(1 - \mu)$  maps  $(0, 1)$  onto  $(0, \infty)$  increasingly, and  $\log$  maps  $(0, \infty)$  onto  $\mathbb{R}$ , so  $g$  is an increasing bijection with inverse  $h(\eta) = 1 - \exp(-e^\eta)$  and  $d\mu/d\eta = e^\eta \exp(-e^\eta) = (1 - \mu)\{-\log(1 - \mu)\}$ . Symmetry would require  $h(-\eta) = 1 - h(\eta)$ , which fails, for example at  $\eta = 0$ , where  $h(0) = 1 - e^{-1} = 0.632 \neq \frac{1}{2}$ . As  $\eta \rightarrow -\infty$ ,  $\mu \approx e^\eta \rightarrow 0$  exponentially, while as  $\eta \rightarrow +\infty$ ,  $1 - \mu = \exp(-e^\eta) \rightarrow 0$  doubly exponentially: the approach to 1 is far faster than the approach to 0. Chapter 35 derives the link from a proportional-hazards argument ([Proposition 35.2.1](#)).

## 34.3 The likelihood equations

In the normal linear model the maximum likelihood estimate was available in closed form, because the log-likelihood was a quadratic in  $\boldsymbol{\beta}$  ([Theorem 7.3.1](#)). Outside that case there is no



closed form, and everything has to be read off the derivatives of the log-likelihood. This section computes them. The answer has the same shape as the normal equations  $\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) = \mathbf{0}$ , with two modifications: each observation is weighted according to its variance, and the residual  $y_i - \mu_i$  is converted from the mean scale to the linear-predictor scale.

Throughout, the model is [Definition 34.2.1](#),  $\text{rank}(\mathbf{X}) = p$ , and  $\boldsymbol{\beta}$  lies in the interior of the admissible set  $\mathcal{B}$ . Write

$$\mu_i = h(\eta_i), \quad \eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \quad \frac{d\mu_i}{d\eta_i} = h'(\eta_i),$$

and note that  $h' > 0$  everywhere, because  $g$  is strictly monotone (take  $g$  increasing; a decreasing link only flips signs).

### 34.3.1 The score

#### Theorem 34.3.1 — The likelihood equations and the information

Define the diagonal matrices

$$\mathbf{W} = \text{diag}\left\{\frac{w_i h'(\eta_i)^2}{V(\mu_i)}\right\}, \quad \mathbf{D} = \text{diag}\{h'(\eta_i)\}.$$

Then, with  $\ell(\boldsymbol{\beta})$  the log-likelihood of [Eq. 34.1.3](#):

(a) the score is

$$\mathbf{U}(\boldsymbol{\beta}) = \frac{\partial \ell}{\partial \boldsymbol{\beta}} = \frac{1}{\phi} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}), \quad (34.3.1)$$

so the likelihood equations are  $\sum_i x_{ij} w_i h'(\eta_i) \{y_i - \mu_i\} / V(\mu_i) = 0$  for  $j = 1, \dots, p$ , and they do not involve  $\phi$ ;

(b)  $\mathbf{E} \mathbf{U} = \mathbf{0}$  and the **expected (Fisher) information** is

$$\mathcal{I}(\boldsymbol{\beta}) = \text{Cov}\{\mathbf{U}(\boldsymbol{\beta})\} = \frac{1}{\phi} \mathbf{X}^\top \mathbf{W} \mathbf{X}; \quad (34.3.2)$$

(c) writing  $a(\eta) = h'(\eta) / V\{h(\eta)\}$ , the **observed information** is

$$-\frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}^\top} = \frac{1}{\phi} \mathbf{X}^\top (\mathbf{W} - \mathbf{D}_r) \mathbf{X},$$

where  $\mathbf{D}_r = \text{diag}\{w_i (y_i - \mu_i) a'(\eta_i)\}$ ;

(d) under the canonical link  $\mathbf{D}_r = \mathbf{0}$ , so the observed and expected information coincide,  $\mathbf{W} = \text{diag}\{w_i V(\mu_i)\}$ , and [Eq. 34.3.1](#) becomes

$$\mathbf{U}(\boldsymbol{\beta}) = \frac{1}{\phi} \mathbf{X}^\top \text{diag}(\mathbf{w}) (\mathbf{y} - \boldsymbol{\mu}). \quad (34.3.3)$$

*Proof.* By [Eq. 34.1.3](#),  $\ell = \phi^{-1} \sum_i w_i \{y_i \theta_i - b(\theta_i)\}$  plus a term free of  $\boldsymbol{\beta}$ . The chain rule gives

$$\frac{\partial \theta_i}{\partial \beta_j} = \frac{d\theta_i}{d\mu_i} \frac{d\mu_i}{d\eta_i} \frac{\partial \eta_i}{\partial \beta_j} = \frac{h'(\eta_i)}{V(\mu_i)} x_{ij},$$

using  $d\theta/d\mu = 1/V(\mu)$  from [Definition 34.1.2](#). Since  $b'(\theta_i) = \mu_i$ ,

$$\frac{\partial \ell}{\partial \beta_j} = \frac{1}{\phi} \sum_i w_i \{y_i - b'(\theta_i)\} \frac{\partial \theta_i}{\partial \beta_j} = \frac{1}{\phi} \sum_i x_{ij} \frac{w_i h'(\eta_i)}{V(\mu_i)} (y_i - \mu_i),$$

which is (a), since  $\mathbf{W}\mathbf{D}^{-1} = \text{diag}\{w_i h'(\eta_i)/V(\mu_i)\}$ . The dispersion appears only as the overall factor  $\phi^{-1}$ , so setting the score to zero gives equations free of  $\phi$ .

- (b)  $E(Y_i - \mu_i) = 0$  gives  $E\mathbf{U} = \mathbf{0}$ . The  $Y_i$  are independent with  $\text{Var}(Y_i) = \phi V(\mu_i)/w_i$  by [Proposition 34.1.2](#), so with the scalars  $a_i = w_i h'(\eta_i)/V(\mu_i)$ ,

$$\text{Cov}(\mathbf{U}) = \frac{1}{\phi^2} \sum_i a_i^2 \text{Var}(Y_i) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top = \frac{1}{\phi} \sum_i \frac{w_i h'(\eta_i)^2}{V(\mu_i)} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top,$$

which is [Eq. 34.3.2](#).

- (c) With  $a(\eta) = h'(\eta)/V\{h(\eta)\}$  we have  $\partial \ell / \partial \beta_j = \phi^{-1} \sum_i x_{ij} w_i (y_i - \mu_i) a(\eta_i)$ . Differentiating again, and using  $\partial \mu_i / \partial \beta_k = h'(\eta_i) x_{ik}$ ,

$$\frac{\partial^2 \ell}{\partial \beta_j \partial \beta_k} = \frac{1}{\phi} \sum_i x_{ij} x_{ik} w_i \{-h'(\eta_i) a(\eta_i) + (y_i - \mu_i) a'(\eta_i)\}.$$

Now  $w_i h'(\eta_i) a(\eta_i) = w_i h'(\eta_i)^2 / V(\mu_i)$  is the  $i$ th working weight, which gives the stated form. Taking expectations kills the second term and recovers (b), a second proof of [Eq. 34.3.2](#).

- (d) Under the canonical link  $\theta_i = \eta_i$ , so  $h = b'$  and  $h'(\eta) = b''(\theta) = V(\mu)$ . Hence  $a(\eta) \equiv 1$ ,  $a' \equiv 0$ ,  $\mathbf{D}_r = \mathbf{0}$ , and  $\mathbf{W} = \text{diag}\{w_i V(\mu_i)^2 / V(\mu_i)\} = \text{diag}\{w_i V(\mu_i)\}$ . Also  $\mathbf{W}\mathbf{D}^{-1} = \text{diag}\{w_i V(\mu_i) / V(\mu_i)\} = \text{diag}(\mathbf{w})$ , which is [Eq. 34.3.3](#). ■

For the normal family with the identity link and  $w_i = 1$ ,  $h' \equiv 1$  and  $V \equiv 1$ , so [Eq. 34.3.1](#) reads  $\mathbf{X}^\top (\mathbf{y} - \mathbf{X}\boldsymbol{\beta}) / \sigma^2$  and [Eq. 34.3.2](#) reads  $\mathbf{X}^\top \mathbf{X} / \sigma^2$ : the score is the normal equations divided by the variance, and [Proposition 7.4.7](#) is recovered.

Three features of [Eq. 34.3.1](#) are worth naming. The equations are **weighted** by  $w_i h'(\eta_i) / V(\mu_i)$ , so a more variable observation counts for less — the logic of weighted least squares ([Theorem 21.3.1\(a\)](#)), with weights dictated by the family rather than assumed. They are **free of**  $\phi$ , which multiplies the whole score, so  $\phi$  can be estimated afterwards without disturbing  $\hat{\boldsymbol{\beta}}$  ([Section 34.5](#)), just as least squares does not need  $\sigma^2$ . And they involve the family **only through**  $V$ : two densities with the same variance function give the same  $\hat{\boldsymbol{\beta}}$ , the seed of the quasi-likelihood of [Chapter 38](#) ([Definition 38.2.1](#)).

## 34.3.2 The fitted marginals

**Corollary 34.3.2** — Canonical links reproduce weighted column totals



```

logit = sm.GLM(y, X, family=sm.families.Binomial()).fit()
mu = logit.fittedvalues

print("level    n    observed Dole votes    fitted")
for k in range(7):
    inside = pid == k
    print(f"    {k}    {inside.sum():4d}    {y[inside].sum():8.0f}    {mu[inside].sum():12.6f}")
print(f"age column: X^T y = {age @ y:.6f},    X^T mu = {age @ mu:.6f}")

```

*Remark* (Where the marginal property is used). Chapter 37 turns [Corollary 34.3.2](#) into a tool: for a log-linear model of a contingency table the canonical link forces the fitted counts to match the observed margins of the terms in the model, which is what makes log-linear and multinomial fits agree ([Theorem 37.3.1](#)). In the binary case it means those totals carry no information about goodness of fit — a point [Proposition 35.4.1](#) takes up.

### 34.3.3 Concavity, existence and uniqueness

#### Theorem 34.3.3 — The canonical-link log-likelihood is concave

Under the canonical link,  $\ell(\boldsymbol{\beta})$  is concave on the convex set  $\mathcal{B}$ . If in addition  $\text{rank}(\mathbf{X}) = p$  and  $V(\mu_i) > 0$  for all  $i$ , then  $\ell$  is strictly concave, so it has at most one stationary point, and any stationary point is the unique global maximizer.

*Proof.* Under the canonical link  $\theta_i = \eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , so by [Eq. 34.1.3](#)

$$\ell(\boldsymbol{\beta}) = \frac{1}{\phi} \sum_{i=1}^n w_i \{y_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta} - b(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})\} + \text{const.}$$

The first term is linear in  $\boldsymbol{\beta}$ . In the second,  $b$  is convex by [Lemma 34.1.1](#) and  $\boldsymbol{\beta} \mapsto \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  is linear, so  $\boldsymbol{\beta} \mapsto b(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$  is convex; multiplying by  $w_i/\phi > 0$  and summing preserves convexity, and the minus sign makes  $\ell$  concave. The set  $\mathcal{B}$  is convex, being an intersection of the preimages of the interval  $g(\mathcal{M})$  under linear maps.

For strictness, [Theorem 34.3.1\(d\)](#) gives the Hessian  $-\phi^{-1} \mathbf{X}^\top \text{diag}\{w_i V(\mu_i)\} \mathbf{X}$ , which for  $\mathbf{v} \neq \mathbf{0}$  satisfies

$$\mathbf{v}^\top \left( \frac{1}{\phi} \mathbf{X}^\top \text{diag}\{w_i V(\mu_i)\} \mathbf{X} \right) \mathbf{v} = \frac{1}{\phi} \sum_i w_i V(\mu_i) (\mathbf{x}_{(i)}^\top \mathbf{v})^2 > 0,$$

because  $\text{rank}(\mathbf{X}) = p$  forces  $\mathbf{X}\mathbf{v} \neq \mathbf{0}$ . So the Hessian is negative definite everywhere on  $\mathcal{B}$  and  $\ell$  is strictly concave there. ■

Concavity settles uniqueness but not existence: a strictly concave function on an open set need not attain its supremum. Both failures happen in practice.

**Example. (Two ways the maximum fails to exist)**

*All counts zero.* For a Poisson log-linear model with an intercept, if every  $y_i = 0$  then  $\ell(\boldsymbol{\beta}) = -\sum_i e^{\mathbf{x}_{(i)}^\top \boldsymbol{\beta}}$ , which increases to its supremum 0 as  $\beta_0 \rightarrow -\infty$  and is never attained.

*Separated binary data.* If there is a vector  $\mathbf{v}$  with  $\mathbf{x}_{(i)}^\top \mathbf{v} > 0$  whenever  $y_i = 1$  and  $\mathbf{x}_{(i)}^\top \mathbf{v} < 0$  whenever  $y_i = 0$ , then along the ray  $t\mathbf{v}$  every fitted probability moves towards the observed response, so  $\ell$  increases without reaching a maximum. Chapter 35 proves that this is the only way the binary maximum likelihood estimate can fail to exist (Theorem 35.3.1).

The general statement belongs to exponential-family theory. Under a canonical link  $\mathbf{T} = \mathbf{X}^\top \text{diag}(\mathbf{w})\mathbf{y}$  is the sufficient statistic of Section 34.1, and by Corollary 34.3.2 the likelihood equations say  $\mathbf{X}^\top \text{diag}(\mathbf{w})\hat{\boldsymbol{\mu}} = \mathbf{T}$ . The attainable mean-value vectors  $\{\mathbf{X}^\top \text{diag}(\mathbf{w})\boldsymbol{\mu} : \boldsymbol{\mu} \in \mathcal{M}^n\}$  form an open convex set, and the maximum likelihood estimate exists precisely when  $\mathbf{T}$  lies inside it rather than on its boundary (Barndorff-Nielsen 1978, chapter 9; not reproduced here). Both examples above are boundary cases: all-zero counts put  $\mathbf{T}$  at the corner  $\mathbf{0}$ , separation puts it on a face. For non-canonical links even uniqueness can fail.

*Warning.* Concavity is a property of the canonical link, not of generalized linear models. Take the gamma family with the identity link, where  $\ell = \phi^{-1} \sum_i \{-y_i/\mu_i - \log \mu_i\}$  with  $\mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . The second derivative of one term with respect to  $\mu_i$  is  $(\mu_i - 2y_i)/\mu_i^3$ , positive whenever  $\mu_i > 2y_i$ , so the Hessian is a difference of nonnegative definite matrices and can be indefinite: an iteration can stop at a local maximum. Wedderburn (1976) settles existence and uniqueness link by link, cleanly for the binomial and Poisson with the standard links and not for the gamma with the identity link.

## 34.3.4 Exercises

### 34.3.4.1 A. Check your understanding

**Exercise A1.**

Write the likelihood equations for a Poisson log-linear model with an intercept. Show that they are  $\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$ , that  $\sum_i \hat{\mu}_i = \sum_i y_i$ , and that for the intercept-only model  $\hat{\mu}_i = \bar{y}$ . (This is Exercise (B2), now a special case of Corollary 34.3.2.)

### 34.3.4.2 B. Practice

**Exercise B1.**

For grouped binomial data with  $y_i = s_i/m_i$ , the logit link and  $w_i = m_i$ , write the likelihood equations in terms of the observed successes  $s_i$ . Show that the working weight is  $\mathbf{W}_{ii} = m_i\mu_i(1 - \mu_i)$  and that the information is  $\mathbf{X}^\top \text{diag}\{m_i\mu_i(1 - \mu_i)\}\mathbf{X}$ . Where have you seen that matrix before?

*Solution.* The canonical form Eq. 34.3.3 with  $w_i = m_i$  and  $\phi = 1$  gives  $\sum_i x_{ij} m_i (s_i/m_i - \mu_i) = \sum_i x_{ij} (s_i - m_i \mu_i) = 0$ : fitted and observed successes agree in every column of  $\mathbf{X}$ . By Theorem 34.3.1(d),  $\mathbf{W} = \text{diag}\{w_i V(\mu_i)\}$  with  $V(\mu) = \mu(1-\mu)$ , giving  $\mathbf{W}_{ii} = m_i \mu_i (1-\mu_i)$  and the stated information. It is the weight matrix of the empirical-logit weighted least squares fit in Exercise (B2), with  $\mu_i$  in place of  $\hat{p}_i$ : that procedure is one step of the iteration of Section 34.4 started from the data.

### Exercise B2.

Show that the observed information equals the expected information *identically in  $\mathbf{y}$*  if and only if the link is canonical (up to an affine change of  $\eta$ ). Assume that no row of  $\mathbf{X}$  is zero, an observation with  $\mathbf{x}_{(i)} = \mathbf{0}$  carrying no information about  $\boldsymbol{\beta}$ . Use Theorem 34.3.1(c).

*Solution.* By Theorem 34.3.1(c) the two agree for all  $\mathbf{y}$  iff  $\mathbf{X}^\top \mathbf{D}_r \mathbf{X} = \mathbf{0}$  for all  $\mathbf{y}$ , where  $(\mathbf{D}_r)_{ii} = w_i (y_i - \mu_i) a'(\eta_i)$ . Vary  $y_i$  over two admissible values with the other responses held fixed: subtracting the two identities leaves  $\delta_i a'(\eta_i) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top = \mathbf{0}$  with  $\delta_i = w_i (y_i - y'_i) \neq 0$ , and since  $\mathbf{x}_{(i)} \neq \mathbf{0}$  this forces  $a'(\eta_i) = 0$  for every  $i$ . (Two admissible values always exist: the  $y_i$  range over an interval for the normal and gamma families, and over  $\{0, 1/m, \dots, 1\}$  or  $\{0, 1, 2, \dots\}$  for the binomial and Poisson.) So  $a$  is constant, say  $a \equiv 1/\lambda$ , that is  $d\mu/d\eta = V(\mu)/\lambda$ . Comparing with the canonical link, for which  $d\mu/d\theta = V(\mu)$ , gives  $\eta = \lambda\theta + \text{const.}$

#### 34.3.4.3 C. Going deeper

### Exercise C1.

For a binary response,  $\ell(\boldsymbol{\beta}) = \sum_i \{y_i \log h(\eta_i) + (1 - y_i) \log(1 - h(\eta_i))\}$ . Show that  $\ell$  is concave for every design and every data set if and only if  $\log h$  and  $\log(1 - h)$  are both concave functions of  $\eta$ . Verify that the logit link satisfies this, and that  $h(\eta) = \Phi(\eta)$  does too, using the fact that the standard normal density is log-concave.

### Exercise C2.

An **offset** is a known column added to the linear predictor with coefficient fixed at 1:  $\eta_i = o_i + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . Show that Theorem 34.3.1 holds verbatim with  $\eta_i$  redefined this way, that Corollary 34.3.2 is unchanged, and that the offset does *not* appear among the columns whose totals are reproduced. Where would such a term come from in a Poisson model for rates? (Chapter 37 answers this with Proposition 37.2.1.)

## 34.4 Iteratively reweighted least squares

The likelihood equations Eq. 34.3.1 have no closed-form solution. Any general-purpose optimizer will solve them, but there is a better way, and it is why generalized linear models became practical: the natural Newton-type iteration is a *weighted least squares fit*, repeated. The apparatus of Chapter 6 and Section 21.3 is therefore reusable, including the stable routines

of [Chapter 10](#).

### 34.4.1 Fisher scoring is weighted least squares

Newton's method for maximizing  $\ell$  replaces the observed information by its inverse:  $\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \{-\nabla^2 \ell\}^{-1} \mathbf{U}$ . **Fisher scoring** replaces the observed information by the expected information  $\mathcal{I}$  instead. By [Theorem 34.3.1\(d\)](#) the two are the same under the canonical link; in general they differ, and the expected version is both simpler and guaranteed nonnegative definite.

#### Theorem 34.4.1 — Fisher scoring is iteratively reweighted least squares

Let  $\boldsymbol{\beta}^{(t)}$  be in the interior of  $\mathcal{B}$  with  $\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X}$  nonsingular, where  $\mathbf{W}^{(t)}$  is the working-weight matrix of [Theorem 34.3.1](#) evaluated at  $\boldsymbol{\beta}^{(t)}$ . Define the **working response**

$$z_i^{(t)} = \eta_i^{(t)} + \frac{y_i - \mu_i^{(t)}}{h'(\eta_i^{(t)})} = \eta_i^{(t)} + (y_i - \mu_i^{(t)}) g'(\mu_i^{(t)}). \quad (34.4.1)$$

Then the Fisher scoring step  $\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)} + \mathcal{I}(\boldsymbol{\beta}^{(t)})^{-1} \mathbf{U}(\boldsymbol{\beta}^{(t)})$  is exactly the weighted least squares estimate

$$\boldsymbol{\beta}^{(t+1)} = (\mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{(t)} \mathbf{z}^{(t)}, \quad (34.4.2)$$

that is, the regression of the working response  $\mathbf{z}^{(t)}$  on  $\mathbf{X}$  with weights  $\mathbf{W}^{(t)}$ . The dispersion  $\phi$  cancels and never enters the iteration.

*Proof.* By [Eq. 34.3.2](#) and [Eq. 34.3.1](#), with everything evaluated at  $\boldsymbol{\beta}^{(t)}$ ,

$$\begin{aligned} \mathcal{I}^{-1} \mathbf{U} &= \phi (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \cdot \frac{1}{\phi} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}) \\ &= (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}), \end{aligned}$$

so  $\phi$  has already cancelled. Adding  $\boldsymbol{\beta}^{(t)} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X} \boldsymbol{\beta}^{(t)}$  and using  $\mathbf{X} \boldsymbol{\beta}^{(t)} = \boldsymbol{\eta}^{(t)}$ ,

$$\boldsymbol{\beta}^{(t+1)} = (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \{ \boldsymbol{\eta}^{(t)} + \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}) \},$$

and the braced vector is  $\mathbf{z}^{(t)}$  by [Eq. 34.4.1](#), since  $\mathbf{D}^{-1} = \text{diag}\{1/h'(\eta_i)\}$  and  $g'(\mu) = 1/h'(\eta)$ . ■

*Idea.* The working response is a first-order Taylor expansion of the link applied to the data:  $g(y_i) \approx g(\mu_i) + (y_i - \mu_i)g'(\mu_i) = z_i$ . So each step regresses “the data, mapped to the linear-predictor scale” on  $\mathbf{X}$ . The working weight is the reciprocal of that quantity's approximate variance, because

$$\text{Var}(z_i) \approx g'(\mu_i)^2 \text{Var}(Y_i) = \frac{\phi}{w_i} \frac{V(\mu_i)}{h'(\eta_i)^2} = \frac{\phi}{\mathbf{W}_{ii}}.$$

Both the response and the weights depend on the current fit, which is why the regression has to be iterated.

Written out, the algorithm is short.

### Iteratively reweighted least squares.

1. Choose starting fitted means  $\mu_i^{(0)}$  and set  $\eta_i^{(0)} = g(\mu_i^{(0)})$ .
2. For  $t = 0, 1, 2, \dots$ : compute  $\mathbf{W}_{ii} = w_i h'(\eta_i^{(t)})^2 / V(\mu_i^{(t)})$  and  $z_i^{(t)}$  from [Eq. 34.4.1](#); solve the weighted least squares problem [Eq. 34.4.2](#) for  $\boldsymbol{\beta}^{(t+1)}$ ; set  $\boldsymbol{\eta}^{(t+1)} = \mathbf{X}\boldsymbol{\beta}^{(t+1)}$  and  $\mu_i^{(t+1)} = h(\eta_i^{(t+1)})$ .
3. Stop when  $\boldsymbol{\beta}$  (or the deviance of [Section 34.5](#)) stops changing; report  $\hat{\boldsymbol{\beta}} = \boldsymbol{\beta}^{(t+1)}$  and  $(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$ , with  $\hat{\mathbf{W}}$  the weights at the solution.

Because the map is a fixed-point iteration whose fixed points solve [Eq. 34.3.1](#), any limit is a solution of the likelihood equations; under the canonical link with  $\text{rank}(\mathbf{X}) = p$ , [Theorem 34.3.3](#) then makes it *the* maximum likelihood estimate.

**Starting values.** The usual device is to start from the data themselves: set  $\mu_i^{(0)} = y_i$ , adjusted if necessary to keep it inside  $\mathcal{M}$ . For a Poisson response take  $\mu_i^{(0)} = y_i + 0.1$ ; for a binomial proportion,  $\mu_i^{(0)} = (w_i y_i + \frac{1}{2}) / (w_i + 1)$ , the empirical-logit correction of [Exercise \(B2\)](#); for a gamma response,  $\mu_i^{(0)} = y_i$ . The first step is then the one-step weighted least squares fit that preceded generalized linear models historically, and the iteration refines it.

**Relation to the least squares chapters.** Each step of [Eq. 34.4.2](#) is the weighted least squares problem of [Theorem 21.3.1](#) with *known* weights, so its convergence theory is numerical, not statistical: the final  $(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$  is the inverse expected information of the likelihood, not the covariance of a fixed-weight estimate, and the two coincide asymptotically for a different reason ([Theorem 34.5.1](#)). Compute the step by the QR or Cholesky routines of [Section 6.10](#) applied to  $\mathbf{W}^{1/2} \mathbf{X}$  and  $\mathbf{W}^{1/2} \mathbf{z}$ , never by inverting  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ .

*Warning.* The name “iteratively reweighted least squares” is also used for the robust M-estimation algorithm of [Proposition 20.6.1](#), and the two are different. There the weights come from a  $\rho$  function chosen to resist outliers, and the iteration provably decreases the objective at every step. Here the weights come from the variance function, the iteration is Fisher scoring, and there is no monotonicity guarantee unless the link is canonical.

## 34.4.2 The algorithm in practice

The listing below implements [Eq. 34.4.2](#) in a dozen lines. A family is nothing but  $(V, g, h)$  together with  $h'$  and a starting rule; the solver never needs to know which distribution it is fitting, which is the computational content of the whole chapter.

```
import numpy as np
from scipy import stats

# A family is the variance function V, the link g, its inverse h, the derivative
# h' = d(mu)/d(eta), and a rule for the starting fitted means.
def poisson_log():
    return dict(name="Poisson/log", V=lambda mu: mu, g=np.log, h=np.exp,
                dmu=lambda eta: np.exp(eta), start=lambda y, w: y + 0.1)
```



```

def binomial_logit():
    return dict(name="binomial/logit", V=lambda mu: mu * (1 - mu),
               g=lambda mu: np.log(mu / (1 - mu)), h=lambda eta: 1 / (1 + np.exp(-
eta))),
               dmu=lambda eta: 1 / (2 + np.exp(eta) + np.exp(-eta)),
               start=lambda y, w: (w * y + 0.5) / (w + 1.0))

def binomial_probit():
    return dict(name="binomial/probit", V=lambda mu: mu * (1 - mu),
               g=stats.norm.ppf, h=stats.norm.cdf, dmu=stats.norm.pdf,
               start=lambda y, w: (w * y + 0.5) / (w + 1.0))

def gamma_log():
    return dict(name="gamma/log", V=lambda mu: mu ** 2, g=np.log, h=np.exp,
               dmu=lambda eta: np.exp(eta), start=lambda y, w: y)

def irls(X, y, family, w=None, tol=1e-12, max_iter=60, trace=False):
    """Fisher scoring for a generalized linear model, as weighted least squares.

    At each step the working response is  $z = \eta + (y - \mu)/h'(\eta)$  and the working
    weight is  $W = w h'(\eta)^2 / V(\mu)$ ;  $\beta$  is the weighted least squares
    coefficient
    of  $z$  on  $X$ . The dispersion  $\phi$  never enters, because it cancels.
    """
    w = np.ones(len(y)) if w is None else np.asarray(w, float)
    mu = family["start"](y, w)
    eta = family["g"](mu)
    beta, path = np.zeros(X.shape[1]), []
    for _ in range(max_iter):
        d = family["dmu"](eta) #  $h'(\eta)$ 
        W = w * d ** 2 / family["V"](mu) # working weights
        z = eta + (y - mu) / d # working response
        XtW = X.T * W
        beta_new = np.linalg.solve(XtW @ X, XtW @ z) # one weighted least
squares fit
        path.append(beta_new)
        eta = X @ beta_new
        mu = family["h"](eta)
        if np.max(np.abs(beta_new - beta)) < tol * (1 + np.max(np.abs(beta_new))):
            beta = beta_new
            break
        beta = beta_new
    W = w * family["dmu"](eta) ** 2 / family["V"](mu) # weights at the solution
    cov_unscaled = np.linalg.inv((X.T * W) @ X) #  $(X^T W X)^{-1}$ 
    return (beta, cov_unscaled, np.array(path)) if trace else (beta, cov_unscaled)

```

```

import statsmodels.api as sm

rng = np.random.default_rng(3401)
n = 300
Xs = np.column_stack([np.ones(n), rng.normal(size=n), rng.binomial(1, 0.4, n)])
beta_true = np.array([0.7, 0.5, -0.4])
counts = rng.poisson(np.exp(Xs @ beta_true))

beta_hat, cov = irls(Xs, counts.astype(float), poisson_log())
sm_fit = sm.GLM(counts, Xs, family=sm.families.Poisson()).fit()
print("from scratch:", np.round(beta_hat, 6))
print("statsmodels :", np.round(sm_fit.params, 6))

```

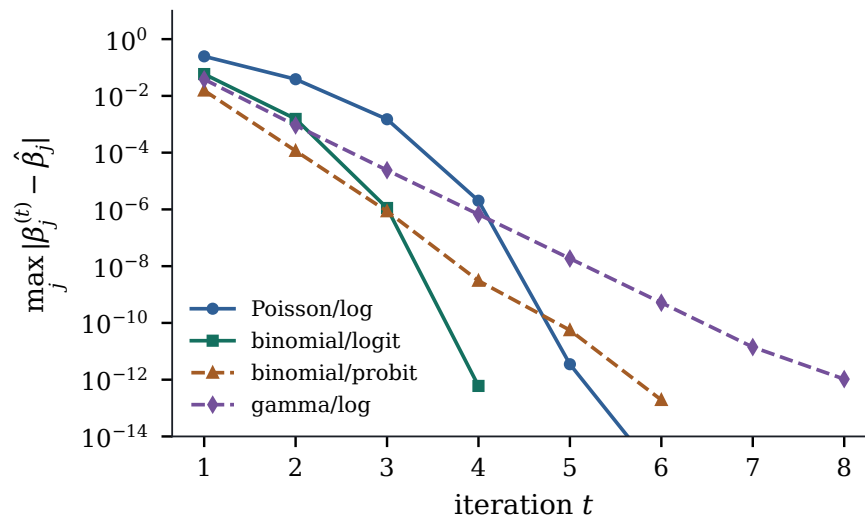


Figure 34.4.1: Convergence of iteratively reweighted least squares for four models. Solid curves are canonical links, dashed curves are not. On a logarithmic scale a quadratic rate bends downwards and a linear rate is straight.

```
print("standard errors:", np.round(np.sqrt(np.diag(cov)), 6), np.round(sm_fit.bse,
6))
```

The script behind this section runs the same solver on four models — Poisson/log, binomial/logit, binomial/probit and gamma/log — and checks the coefficients and standard errors against `statsmodels` to at least eight decimal places in every case. Two details matter. For the binomial fits the response passed to the solver is the *proportion*  $s_i/m_i$  with prior weight  $w_i = m_i$ , which is [Example \(The binomial as a family of proportions\)](#) in code. For the gamma fit the solver returns  $(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$ , so the standard errors need the extra factor  $\hat{\phi}$ , estimated separately; on Engel’s data that gives  $\hat{\boldsymbol{\beta}} = (0.5068, 0.8629)$  with  $\hat{\phi} = 0.0177$  and a standard error of 0.0198 for the elasticity.

### 34.4.3 Convergence and its failures

#### Example. (Two convergence rates)

[Figure 34.4.1](#) records  $\max_j |\beta_j^{(t)} - \hat{\beta}_j|$  against the iteration count for the four models. The canonical-link fits reach full double precision in 5 and 4 steps, with error curves that bend downwards on the logarithmic scale: the correct digits roughly double at each step. The non-canonical fits take 5 and 7 steps, along straight lines, so their correct digits grow by a fixed amount per step.

This is what the theory predicts. Under the canonical link Fisher scoring *is* Newton’s method ([Theorem 34.3.1\(d\)](#)), quadratically convergent near a nondegenerate maximum. Under any other link the two information matrices differ by  $\phi^{-1} \mathbf{X}^\top \mathbf{D}_r \mathbf{X}$ , which vanishes only in expectation, so the iteration is Newton’s method with a perturbed Hessian and converges linearly — in practice, one or two extra iterations.

Four things go wrong often enough to be worth recognizing.

**The estimate does not exist.** When the maximum of [Example \(Two ways the maximum fails to exist\)](#) is not attained the iteration cannot converge, and its behaviour is characteristic: the coefficients grow without limit, the deviance falls towards its infimum, the working weights collapse to zero. Software that stops on a small change in the deviance then reports a large but finite estimate with a large standard error and no warning.

**Example. (What separation looks like from inside the algorithm)**

Ten binary responses were generated with  $y_i = 1$  exactly when  $x_i > 0$ , so the data are completely separated. Running the solver on them gives

| step | $\hat{\beta}_1$ | deviance | largest working weight |
|------|-----------------|----------|------------------------|
| 1    | 1.36            | 4.589    | 0.245                  |
| 4    | 5.74            | 0.931    | 0.159                  |
| 7    | 16.95           | 0.0574   | 0.0142                 |
| 8    | —               | —        | —                      |

At the eighth step every fitted probability has reached 0 or 1 in double precision, so  $V(\hat{\mu}_i) = 0$ , the working weights are 0/0, and the iteration returns nothing. The slope was never going to settle: the supremum of the likelihood is approached only as  $\beta_1 \rightarrow \infty$ . [Chapter 35](#) characterizes exactly when this happens and what to do about it ([Theorem 35.3.1](#)).

```
x_sep = np.array([-2.4, -1.7, -1.1, -0.6, -0.2, 0.3, 0.8, 1.4, 1.9, 2.6])
y_sep = (x_sep > 0).astype(float) # the two groups do not
                                overlap
X_sep = np.column_stack([np.ones(10), x_sep])

for iters in range(1, 9):
    with np.errstate(divide="ignore", invalid="ignore"):
        b, _ = irls(X_sep, y_sep, binomial_logit(), tol=0.0, max_iter=iters)
        mu = 1 / (1 + np.exp(-(X_sep @ b)))
        dev = -2 * np.sum(np.where(y_sep == 1, np.log(mu), np.log(1 - mu)))
    print(f"{iters} steps: slope {b[1]:9.4f}    deviance {dev:.3e}"
          f"    largest working weight {np.max(mu * (1 - mu)):.3e}")
```

**The step leaves the admissible set.** With a link whose range is not all of  $\mathbb{R}$ , a full step can produce  $\eta_i^{(t+1)} \notin g(\mathcal{M})$  — a negative fitted mean for an identity-link Poisson model, where the variance function is then negative and the “weights” meaningless. The repair is **step halving**: if the proposed  $\beta^{(t+1)}$  is inadmissible, or the deviance increases, halve the step and retry. A few halvings almost always suffice, and software applies them automatically.

**The information matrix is nearly singular.** The matrix inverted at each step is  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ , not  $\mathbf{X}^\top \mathbf{X}$ , so full column rank is no longer enough: if the weights are tiny wherever a regressor varies, the weighted matrix can be numerically singular even for a well-conditioned  $\mathbf{X}$ . The diagnostics of [Chapter 26](#) apply to  $\mathbf{W}^{1/2} \mathbf{X}$ , not  $\mathbf{X}$ .

**The iteration converges to the wrong stationary point.** For non-canonical links the log-likelihood need not be concave, and Fisher scoring from a bad start can settle at a local maximum; the defence is several starting values compared by maximized log-likelihood.

## 34.4.4 Exercises

### 34.4.4.1 A. Check your understanding

#### Exercise A1.

Verify the approximation  $\text{Var}(z_i) \approx \phi/\mathbf{W}_{ii}$  stated in the key-idea box, and explain which two approximations are being made.

### 34.4.4.2 B. Practice

#### Exercise B1.

Take grouped binomial data with all  $0 < s_i < m_i$  and start the iteration at  $\mu_i^{(0)} = s_i/m_i$ . Show that the first working response is the empirical logit  $\log\{s_i/(m_i - s_i)\}$  and the first working weight is  $s_i(m_i - s_i)/m_i$ . Compare with [Exercise \(B2\)](#).

*Solution.* With  $\mu_i^{(0)} = s_i/m_i$  the starting linear predictor is already  $\eta_i^{(0)} = \log\{\mu_i^{(0)}/(1 - \mu_i^{(0)})\} = \log\{s_i/(m_i - s_i)\}$ , and  $y_i - \mu_i^{(0)} = 0$ , so  $z_i^{(0)} = \eta_i^{(0)}$  is the empirical logit. The working weight is  $\mathbf{W}_{ii} = m_i\mu_i^{(0)}(1 - \mu_i^{(0)}) = s_i(m_i - s_i)/m_i$ , which is the weight of [Exercise \(B2\)](#) with  $\hat{p}_i = s_i/m_i$ . The first iterate is therefore exactly the empirical-logit weighted least squares estimate; later iterates are not, which is why the two methods give different answers in small samples.

#### Exercise B2.

Modify [Eq. 34.4.2](#) for a model with an offset  $\eta_i = o_i + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , as in [Exercise \(C2\)](#). Show that the working response becomes  $z_i = \eta_i - o_i + (y_i - \mu_i)g'(\mu_i)$  and nothing else changes.

### 34.4.4.3 C. Going deeper

#### Exercise C1.

Show that under the canonical link, Fisher scoring is Newton's method applied to a strictly concave function, and deduce that when the maximum exists, the iteration converges to it from any starting point at which the step is well defined *provided the step is damped*: for each  $t$  there is  $s_t \in (0, 1]$  with  $\ell(\boldsymbol{\beta}^{(t)} + s_t \mathbf{d}^{(t)}) > \ell(\boldsymbol{\beta}^{(t)})$ , where  $\mathbf{d}^{(t)}$  is the Newton direction. Why does the undamped step not carry the same guarantee? (Compare [Proposition 20.6.1](#), where the undamped step *is* monotone, because the argument there is majorization rather than Newton.)

*Solution.* By [Theorem 34.3.3](#) the Hessian  $-\mathcal{I}$  is negative definite, so the Newton direction  $\mathbf{d} = \mathcal{I}^{-1}\mathbf{U}$  satisfies  $\mathbf{U}^\top \mathbf{d} = \mathbf{U}^\top \mathcal{I}^{-1}\mathbf{U} > 0$  whenever  $\mathbf{U} \neq \mathbf{0}$ : it is an ascent direction, so the directional derivative of  $\ell$  at  $\boldsymbol{\beta}^{(t)}$  along  $\mathbf{d}$  is positive and some small enough step increases  $\ell$ . The undamped step has length fixed by the local quadratic approximation, which can overshoot badly when  $\ell$  is far from quadratic, so it may decrease  $\ell$  or leave  $\mathcal{B}$  altogether.

## 34.5 Deviance and asymptotic inference

Parts II and III rested on exact distribution theory:  $\hat{\beta}$  was normal because  $\mathbf{Y}$  was, and sums of squares were chi-squared because they were quadratic forms in a normal vector ([Theorem 4.3.2](#)). None of that survives. This section states what replaces it in the limit, under which conditions, and — just as important — where the approximations fail.

### 34.5.1 Asymptotic normality

The conditions below are stated for the canonical link, where the argument can be given in full. Throughout  $\beta^0$  is the true parameter,  $\theta_i^0 = \mathbf{x}_{(i)}^\top \beta^0$ , and

$$\mathbf{A}_n = \mathcal{I}_n(\beta^0) = \frac{1}{\phi} \mathbf{X}^\top \text{diag}\{w_i b''(\theta_i^0)\} \mathbf{X}$$

is the expected information from [Eq. 34.3.2](#). Because the canonical-link Hessian does not involve  $\mathbf{y}$  ([Theorem 34.3.1\(d\)](#)),  $\mathbf{A}_n$  is a fixed matrix, not a random one. Define the scalars

$$q_{ni} = \frac{w_i b''(\theta_i^0)}{\phi} \mathbf{x}_{(i)}^\top \mathbf{A}_n^{-1} \mathbf{x}_{(i)}, \quad i = 1, \dots, n. \quad (34.5.1)$$

These are the exact analogue of leverages: nonnegative, summing to  $\text{tr}(\mathbf{A}_n^{-1} \mathbf{A}_n) = p$ , so they average  $p/n$  as in [Proposition 6.8.1\(b\)](#). The conditions are

- (G1) the responses are independent, the model is that of [Definition 34.2.1](#) with the canonical link,  $\text{rank}(\mathbf{X}) = p$  with  $p$  fixed, and  $\phi$  and the  $w_i$  are known;
- (G2)  $\beta^0$  is interior to  $\mathcal{B}$ , the natural parameters  $\theta_i^0$  lie in a fixed compact subset  $K$  of the interior of  $\Theta$ , and  $0 < w_{\min} \leq w_i \leq w_{\max} < \infty$ ;
- (G3)  $\max_{i \leq n} q_{ni} \rightarrow 0$ : no single observation carries a fixed share of the information;
- (G4)  $\lambda_{\min}(\mathbf{A}_n) \rightarrow \infty$ : information accumulates in every direction.

#### Theorem 34.5.1 — Consistency and asymptotic normality of the estimate

Under (G1)–(G4), with probability tending to one the likelihood equations have a solution  $\hat{\beta}_n$ ; it is the unique maximizer of  $\ell$ , it is consistent for  $\beta^0$ , and

$$\mathbf{A}_n^{1/2}(\hat{\beta}_n - \beta^0) \xrightarrow{d} N_p(\mathbf{0}, \mathbf{I}). \quad (34.5.2)$$

The same holds with  $\mathbf{A}_n$  replaced by  $\hat{\mathbf{A}}_n = \mathcal{I}_n(\hat{\beta}_n)$ . Consequently, for a fixed  $\lambda \neq \mathbf{0}$ ,

$$\frac{\lambda^\top \hat{\beta}_n - \lambda^\top \beta^0}{(\lambda^\top \hat{\mathbf{A}}_n^{-1} \lambda)^{1/2}} \xrightarrow{d} N(0, 1),$$

and  $\lambda^\top \hat{\beta}_n \pm z_{1-\alpha/2}(\lambda^\top \hat{\mathbf{A}}_n^{-1} \lambda)^{1/2}$  is an interval with asymptotic coverage  $1 - \alpha$ .

One result is imported, not proved: the standard corollary of **the convexity lemma** of Pollard (1991, section 6), that pointwise convergence of concave random functions is automatically uniform on compact sets. Let  $\Lambda_n$  be random concave functions on  $\mathbb{R}^p$ , allowed the value  $-\infty$  outside a convex set, of the form  $\Lambda_n(\mathbf{s}) = \mathbf{s}^\top \mathbf{z}_n - \frac{1}{2} \|\mathbf{s}\|^2 + r_n(\mathbf{s})$  with  $\mathbf{z}_n = O_p(1)$  and  $\sup_{\|\mathbf{s}\| \leq M} |r_n(\mathbf{s})| \rightarrow 0$  in probability for every  $M$ . Then a maximizer  $\hat{\mathbf{s}}_n$  exists with probability tending to one and  $\hat{\mathbf{s}}_n - \mathbf{z}_n \rightarrow \mathbf{0}$  in probability. Concavity does the work that compactness conditions do in the usual theory, and [Theorem 34.3.3](#) supplies it free.

*Proof. Step 1: the score is asymptotically normal.* By [Eq. 34.3.3](#),  $\mathbf{U}_n = \phi^{-1} \sum_i w_i (Y_i - \mu_i^0) \mathbf{x}_{(i)}$  is a sum of independent mean-zero vectors with  $\text{Cov}(\mathbf{U}_n) = \mathbf{A}_n$ . Put  $\mathbf{z}_n = \mathbf{A}_n^{-1/2} \mathbf{U}_n$ . For a fixed unit vector  $\mathbf{v}$  write  $\mathbf{v}^\top \mathbf{z}_n = \sum_i c_{ni} \xi_i$  with  $\xi_i = Y_i - \mu_i^0$  and the weights  $c_{ni} = \phi^{-1} w_i \mathbf{v}^\top \mathbf{A}_n^{-1/2} \mathbf{x}_{(i)}$ . This is the weighted sum of [Lemma 21.2.1](#), and its case (b) applies. Moments: by (G2) the cumulants of  $Y_i$  are continuous functions of  $\theta_i^0$  on a compact set ([Proposition 34.1.2\(c\)](#)) and the  $w_i$  are bounded away from 0 and  $\infty$ , so  $E \xi_i^4$  is bounded — case (b) with  $\delta = 2$  — and  $\tau_{\min}^2 \leq \text{Var} \xi_i = \phi b''(\theta_i^0)/w_i \leq \tau_{\max}^2$  for constants free of  $i$  and  $n$ . Weights: by construction  $\sum_i c_{ni}^2 \text{Var} \xi_i = \mathbf{v}^\top \mathbf{A}_n^{-1/2} \mathbf{A}_n \mathbf{A}_n^{-1/2} \mathbf{v} = 1$ , so the sum is already normalized and  $\|\mathbf{c}_n\|^2 \geq \tau_{\max}^{-2}$ ; while  $c_{ni}^2 \leq \phi^{-2} w_{\max}^2 \|\mathbf{A}_n^{-1/2} \mathbf{x}_{(i)}\|^2$  and  $\|\mathbf{A}_n^{-1/2} \mathbf{x}_{(i)}\|^2 = \mathbf{x}_{(i)}^\top \mathbf{A}_n^{-1} \mathbf{x}_{(i)} = \phi q_{ni} / \{w_i b''(\theta_i^0)\} \leq C' q_{ni}$  by [Eq. 34.5.1](#) and (G2) again. Hence  $\max_i c_{ni}^2 / \|\mathbf{c}_n\|^2 \leq C'' \max_i q_{ni} \rightarrow 0$  by (G3), which is [Eq. 21.2.2](#). So  $\mathbf{v}^\top \mathbf{z}_n \rightarrow N(0, 1)$  for every unit  $\mathbf{v}$ , and the Cramér–Wold device gives  $\mathbf{z}_n \rightarrow N_p(\mathbf{0}, \mathbf{I})$ .

*Step 2: the log-likelihood is asymptotically quadratic.* Put  $\Lambda_n(\mathbf{s}) = \ell(\boldsymbol{\beta}^0 + \mathbf{A}_n^{-1/2} \mathbf{s}) - \ell(\boldsymbol{\beta}^0)$ . Taylor’s theorem with integral remainder, applied to the smooth function  $\ell$ , gives

$$\Lambda_n(\mathbf{s}) = \mathbf{s}^\top \mathbf{z}_n - \int_0^1 (1-t) \mathbf{s}^\top \mathbf{A}_n^{-1/2} \mathcal{J}_n(\boldsymbol{\beta}_t) \mathbf{A}_n^{-1/2} \mathbf{s} dt,$$

where  $\boldsymbol{\beta}_t = \boldsymbol{\beta}^0 + t \mathbf{A}_n^{-1/2} \mathbf{s}$  and  $\mathcal{J}_n(\boldsymbol{\beta}) = \phi^{-1} \mathbf{X}^\top \text{diag}\{w_i b''(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})\} \mathbf{X}$  is the *nonrandom* observed information. Fix  $M > 0$  and let  $\|\mathbf{s}\| \leq M$ . Then  $|\mathbf{x}_{(i)}^\top \mathbf{A}_n^{-1/2} \mathbf{s}| \leq M \|\mathbf{A}_n^{-1/2} \mathbf{x}_{(i)}\| \leq M(C' q_{ni})^{1/2}$ , which by (G3) tends to 0 uniformly in  $i$ . So for large  $n$  every perturbed natural parameter  $\theta_i^0 + t \mathbf{x}_{(i)}^\top \mathbf{A}_n^{-1/2} \mathbf{s}$  lies in a fixed compact neighbourhood of  $K$  inside  $\Theta$ , on which  $b''$  is uniformly continuous and bounded away from 0 and  $\infty$ ; hence  $b''(\theta_i^0 + \delta_i) = b''(\theta_i^0) \{1 + \epsilon_{ni}\}$  with  $\sup_i |\epsilon_{ni}| \rightarrow 0$  uniformly in  $t \in [0, 1]$  and  $\|\mathbf{s}\| \leq M$ , and  $\mathbf{A}_n^{-1/2} \mathcal{J}_n(\cdot) \mathbf{A}_n^{-1/2} = \mathbf{I} + \mathbf{R}_n$  with  $\|\mathbf{R}_n\| \rightarrow 0$ , and

$$\Lambda_n(\mathbf{s}) = \mathbf{s}^\top \mathbf{z}_n - \frac{1}{2} \|\mathbf{s}\|^2 + r_n(\mathbf{s}), \quad \sup_{\|\mathbf{s}\| \leq M} |r_n(\mathbf{s})| \rightarrow 0. \quad (34.5.3)$$

*Step 3: the maximizer.* By [Theorem 34.3.3](#) each  $\Lambda_n$  is concave. It is defined only on the convex set  $\mathbf{A}_n^{1/2}(\mathcal{B} - \boldsymbol{\beta}^0)$ , a half-space for the canonical links of the gamma and inverse Gaussian; extend it by  $-\infty$  outside, which preserves concavity, and note that Step 2 put every ball  $\|\mathbf{s}\| \leq M$  inside that set for large  $n$ , so the expansion holds where the lemma needs it. Then [Eq. 34.5.3](#) is exactly the form the lemma requires, with  $\mathbf{z}_n = O_p(1)$  by Step 1 and a nonrandom remainder bound, so a maximizer  $\hat{\mathbf{s}}_n$  exists with probability tending to one and  $\hat{\mathbf{s}}_n - \mathbf{z}_n \rightarrow \mathbf{0}$  in probability. Since  $\hat{\mathbf{s}}_n = \mathbf{A}_n^{1/2}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}^0)$ , Slutsky’s theorem and Step 1 give [Eq. 34.5.2](#), and [Theorem 34.3.3](#) makes the maximizer unique.

*Consistency.*  $\|\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}^0\| \leq \lambda_{\min}(\mathbf{A}_n)^{-1/2} \|\hat{\mathbf{s}}_n\|$ , and  $\|\hat{\mathbf{s}}_n\| = O_p(1)$ , so (G4) gives  $\hat{\boldsymbol{\beta}}_n \rightarrow \boldsymbol{\beta}^0$  in probability.

*Plug-in.* Consistency and the uniform continuity of  $b''$  used in Step 2 give

$$\mathbf{A}_n^{-1/2} \hat{\mathbf{A}}_n \mathbf{A}_n^{-1/2} \rightarrow \mathbf{I}$$

in probability, whence  $\hat{\mathbf{A}}_n^{1/2}(\hat{\boldsymbol{\beta}}_n - \boldsymbol{\beta}^0) \rightarrow N_p(\mathbf{0}, \mathbf{I})$ . The last two displays follow by taking the  $\boldsymbol{\lambda}$  component. ■

*Remark* (What the conditions mean, and the non-canonical case). Condition (G3) is the honest content of “large sample”. It is the generalized-linear-model version of the requirement  $\max_i h_{ii} \rightarrow 0$  that [Chapter 21](#) needed, and Step 1 used it in exactly that role, to supply [Eq. 21.2.2](#). It fails when it should: a group with a single observation, a regressor that indicates one case, a binary design with a nearly empty cell. The canonical link is used twice, for the concavity and for the nonrandom Hessian, and the next proposition records what survives without it.

*Warning.* Nothing here is a finite-sample statement:  $\hat{\boldsymbol{\beta}}$  is biased, by  $O(n^{-1})$  against a standard error of  $O(n^{-1/2})$  — asymptotically negligible, not zero. Nor does  $p$  fixed cover many parameters: a model with a parameter per group and bounded group sizes violates (G3), and [Example \(Many means, one variance\)](#) showed what goes wrong.

**Intervals for a fitted mean.** For fixed  $\mathbf{x}_0$  the theorem with  $\boldsymbol{\lambda} = \mathbf{x}_0$  gives  $\hat{\eta}_0 \pm z_{1-\alpha/2}(\mathbf{x}_0^\top \hat{\mathbf{A}}_n^{-1} \mathbf{x}_0)^{1/2}$  for  $\eta_0 = \mathbf{x}_0^\top \boldsymbol{\beta}^0$ , the counterpart of [Theorem 12.1.1](#) with a normal quantile for a  $t$  one; applying the monotone  $h$  to both endpoints makes it an interval for  $\mu_0 = h(\eta_0)$  of the same asymptotic coverage. The *interval* transforms, not the endpoint formula, so it is not symmetric about  $\hat{\mu}_0$  — a feature, since a log link then keeps it positive and a logit link inside  $(0, 1)$ . The delta-method alternative  $\hat{\mu}_0 \pm z_{1-\alpha/2} h'(\hat{\eta}_0)(\mathbf{x}_0^\top \hat{\mathbf{A}}_n^{-1} \mathbf{x}_0)^{1/2}$  agrees to first order and is the worse wherever  $h$  curves sharply. Neither is a prediction interval: [Theorem 12.3.1](#) has no clean analogue here.

### Proposition 34.5.2 — Non-canonical links

Let the link be any twice continuously differentiable  $g$  with  $g' \neq 0$  on  $\mathcal{M}$ , and put  $\mathbf{A}_n = \phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X}$  with  $\mathbf{W}$  the working weights of [Theorem 34.3.1](#) evaluated at  $\boldsymbol{\beta}^0$ . Then every conclusion of [Theorem 34.5.1](#) — existence with probability tending to one, consistency, the limit [Eq. 34.5.2](#), the plug-in version and the Wald intervals — continues to hold, as do [Proposition 34.5.3](#) and [Theorem 34.5.5\(a\),\(b\)](#) below, which use the theorem only through the quadratic expansion [Eq. 34.5.3](#). The conditions are those of Fahrmeir and Kaufmann (1985): the divergence condition  $\lambda_{\min}(\mathbf{A}_n) \rightarrow \infty$ , and a continuity condition asking that  $\mathbf{A}_n(\boldsymbol{\beta}^0)^{-1/2} \mathbf{A}_n(\boldsymbol{\beta}) \mathbf{A}_n(\boldsymbol{\beta}^0)^{-1/2} \rightarrow \mathbf{I}$  uniformly over  $\boldsymbol{\beta}$  in neighbourhoods of  $\boldsymbol{\beta}^0$  that shrink in the information metric. The two together play the part of (G2)–(G4).

*Remark.* This proposition is quoted, not proved: without concavity and a nonrandom Hessian the argument above collapses, and the replacement is the content of Fahrmeir and Kaufmann (1985). It is recorded because most models in the chapters that follow use a non-canonical link somewhere — the complementary log–log and probit links of [Chapter 35](#), the gamma/log fit of [Example \(Food expenditure: three models, two decisions\)](#), the negative binomial with a log link in [Chapter 37](#).



## 34.5.2 Wald, score and likelihood ratio

Consider a reduced model  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  with  $\text{rank}(\mathbf{X}_0) = p - q$ , and let  $\tilde{\boldsymbol{\beta}}$  be the maximum likelihood estimate under it, written as a vector in  $\mathbb{R}^p$ . Since  $\mathbf{X}$  has full column rank,  $\{\boldsymbol{\beta} : \mathbf{X}\boldsymbol{\beta} \in \mathcal{C}(\mathbf{X}_0)\}$  is a subspace of dimension  $p - q$ , so it is  $\{\boldsymbol{\beta} : \boldsymbol{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}\}$  for some  $p \times q$  matrix  $\boldsymbol{\Lambda}$  of rank  $q$ , in the notation of [Definition 11.3.1](#).

### Proposition 34.5.3 — The three statistics

Define

$$\text{Wald} = (\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}})^\top (\boldsymbol{\Lambda}^\top \hat{\mathbf{A}}_n^{-1} \boldsymbol{\Lambda})^{-1} \boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}},$$

$$\text{Score} = \mathbf{U}(\tilde{\boldsymbol{\beta}})^\top \mathcal{I}_n(\tilde{\boldsymbol{\beta}})^{-1} \mathbf{U}(\tilde{\boldsymbol{\beta}}),$$

$$\text{LR} = 2\{\ell(\hat{\boldsymbol{\beta}}) - \ell(\tilde{\boldsymbol{\beta}})\}.$$

Under (G1)–(G4) and the reduced model, each of the three converges in distribution to  $\chi^2(q)$ , and any two of them differ by  $o_p(1)$ . For a single coefficient ( $q = 1$ ,  $\boldsymbol{\Lambda} = \mathbf{e}_j$ ) the Wald statistic is  $\hat{\beta}_j^2 / [\hat{\mathbf{A}}_n^{-1}]_{jj}$ , the square of the “estimate over standard error” ratio that every program prints.

*Proof.* Work in the coordinates  $\mathbf{s} = \mathbf{A}_n^{1/2}(\boldsymbol{\beta} - \boldsymbol{\beta}^0)$  of the proof of [Theorem 34.5.1](#), where  $\boldsymbol{\beta}^0$  is the true parameter, which lies in the reduced model, so  $\boldsymbol{\Lambda}^\top \boldsymbol{\beta}^0 = \mathbf{0}$ . The reduced model is the linear subspace  $\mathcal{S}_n = \mathbf{A}_n^{1/2} \mathcal{N}(\boldsymbol{\Lambda}^\top) = \mathcal{N}(\mathbf{B}_n)$  of  $\mathbb{R}^p$ , where  $\mathbf{B}_n = \boldsymbol{\Lambda}^\top \mathbf{A}_n^{-1/2}$  has rank  $q$ ; let  $\mathbf{P}_n$  be the orthogonal projection onto it, so that  $\mathbf{I} - \mathbf{P}_n = \mathbf{B}_n^\top (\mathbf{B}_n \mathbf{B}_n^\top)^{-1} \mathbf{B}_n$ .

For the likelihood ratio, [Eq. 34.5.3](#) gives  $\Lambda_n(\mathbf{s}) = \mathbf{s}^\top \mathbf{z}_n - \frac{1}{2} \|\mathbf{s}\|^2 + r_n(\mathbf{s})$  uniformly on compacts. Maximizing  $\mathbf{s}^\top \mathbf{z}_n - \frac{1}{2} \|\mathbf{s}\|^2$  over  $\mathbb{R}^p$  gives  $\frac{1}{2} \|\mathbf{z}_n\|^2$  at  $\mathbf{s} = \mathbf{z}_n$ ; over  $\mathcal{S}_n$  it gives  $\frac{1}{2} \|\mathbf{P}_n \mathbf{z}_n\|^2$  at  $\mathbf{s} = \mathbf{P}_n \mathbf{z}_n$ . Both maximizers are  $O_p(1)$ , so the remainder is negligible at both, and

$$\text{LR} = 2\left\{ \max_{\mathbb{R}^p} \Lambda_n - \max_{\mathcal{S}_n} \Lambda_n \right\} = \|(\mathbf{I} - \mathbf{P}_n) \mathbf{z}_n\|^2 + o_p(1).$$

If  $\mathbf{z} \sim N_p(\mathbf{0}, \mathbf{I})$  then  $\|(\mathbf{I} - \mathbf{P}) \mathbf{z}\|^2 \sim \chi^2(q)$  for *every* projection  $\mathbf{P}$  of rank  $p - q$  ([Theorem 4.3.2](#)). Projections of a fixed rank form a compact set, so along any subsequence  $\mathbf{P}_n \rightarrow \mathbf{P}$ , and the continuous mapping theorem gives  $\|(\mathbf{I} - \mathbf{P}_n) \mathbf{z}_n\|^2 \rightarrow \chi^2(q)$ ; the limit is the same for every subsequence, so the whole sequence converges.

For the Wald statistic, write  $\hat{\mathbf{s}}_n = \mathbf{A}_n^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0)$ . Then  $\boldsymbol{\Lambda}^\top \hat{\boldsymbol{\beta}} = \boldsymbol{\Lambda}^\top (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0) = \mathbf{B}_n \hat{\mathbf{s}}_n$  and  $\boldsymbol{\Lambda}^\top \hat{\mathbf{A}}_n^{-1} \boldsymbol{\Lambda} = \mathbf{B}_n \mathbf{B}_n^\top$ , so with  $\mathbf{A}_n$  in place of  $\hat{\mathbf{A}}_n$  the statistic is exactly  $\|(\mathbf{I} - \mathbf{P}_n) \hat{\mathbf{s}}_n\|^2$ . The proof of [Theorem 34.5.1](#) gives  $\hat{\mathbf{s}}_n = \mathbf{z}_n + o_p(1)$ , and  $\mathbf{A}_n^{-1/2} \hat{\mathbf{A}}_n \mathbf{A}_n^{-1/2} \rightarrow \mathbf{I}$  in probability makes the substitution of  $\hat{\mathbf{A}}_n$  cost  $o_p(1)$ ; hence  $\text{Wald} = \|(\mathbf{I} - \mathbf{P}_n) \mathbf{z}_n\|^2 + o_p(1)$ . The same argument inside the reduced model gives  $\mathbf{A}_n^{1/2}(\tilde{\boldsymbol{\beta}} - \boldsymbol{\beta}^0) = \mathbf{P}_n \mathbf{z}_n + o_p(1)$ , which is used next.

For the score statistic, differentiating [Eq. 34.5.3](#) — legitimate because pointwise convergence of differentiable concave functions forces convergence of their gradients (Rockafellar 1970, section 25) — gives  $\mathbf{A}_n^{-1/2} \mathbf{U}(\boldsymbol{\beta}^0 + \mathbf{A}_n^{-1/2} \mathbf{s}) = \mathbf{z}_n - \mathbf{s} + o_p(1)$  uniformly on compacts; at  $\mathbf{s} = \mathbf{P}_n \mathbf{z}_n + o_p(1)$  this is  $(\mathbf{I} - \mathbf{P}_n) \mathbf{z}_n + o_p(1)$ . ■

In the normal linear model the three are *exact* monotone functions of one another and of the



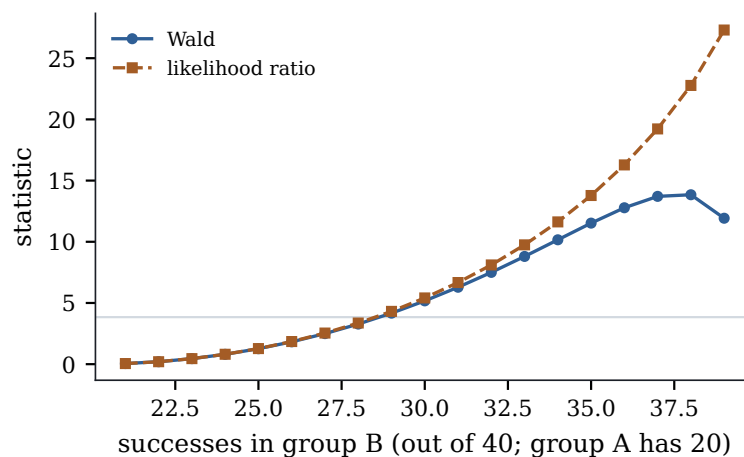


Figure 34.5.1: The Hauck–Donner effect. Two groups of 40 trials, with 20 successes in group A. As the successes in group B increase, the likelihood ratio statistic rises monotonically, while the Wald statistic turns over and falls. The horizontal line is the upper 5% point of  $\chi^2(1)$ .

$F$  statistic, with  $\text{Wald} > \text{LR} > \text{Score}$  in every sample (Proposition 11.4.2). Outside it that ordering is not guaranteed, and the three agree only to first order.

**The Wald statistic is not invariant.** Testing  $\beta_j = 0$  and testing  $e^{\beta_j} = 1$  are the same hypothesis with different Wald statistics, because the delta-method standard error of  $e^{\hat{\beta}_j}$  is not the transform of that of  $\hat{\beta}_j$ ; the likelihood ratio statistic, which compares maximized likelihoods, is invariant under every reparameterization.

**The Wald statistic can move the wrong way.** This failure has a name.

#### Example. (The Hauck–Donner effect)

Two groups of 40 trials, with 20 successes in group A and  $s_B$  in group B, fitted by a logistic model with an intercept and a group indicator. The coefficient is the log odds ratio  $\hat{\beta}_1 = \log\{s_B/(40 - s_B)\}$ , with standard error  $\{1/20 + 1/20 + 1/s_B + 1/(40 - s_B)\}^{1/2}$ . As  $s_B$  rises from 21 to 39 the evidence against equality strengthens monotonically and the likelihood ratio statistic records it, climbing to 27.301. The Wald statistic does not: it rises to 13.842 at  $s_B = 38$  and then *falls* to 11.924 at  $s_B = 39$ , though the fitted log odds ratio has grown to 3.664. The standard error grows faster than the coefficient.

At complete separation,  $s_B = 40$ , coefficient and standard error are both infinite and the Wald statistic is 0: the test accepts the null precisely where the data support it most strongly (Figure 34.5.1). The effect occurs whenever the log-likelihood is far from quadratic over the range where the estimate lies.

```
import numpy as np

m_hd, s_a = 40, 20                                     # group A: 20 successes out of
40                                                     # group B: from 21 to 39
s_b = np.arange(21, 40)
odds_ratio = (s_b / (m_hd - s_b)) / (s_a / (m_hd - s_a))
beta1 = np.log(odds_ratio)                             # the fitted log odds ratio
se1 = np.sqrt(1 / s_a + 1 / (m_hd - s_a) + 1 / s_b + 1 / (m_hd - s_b))
wald = (beta1 / se1) ** 2
```

```

def binom_ll(s, m, p):
    return s * np.log(p) + (m - s) * np.log(1 - p)

p_pool = (s_a + s_b) / (2 * m_hd)
lr = 2 * (binom_ll(s_a, m_hd, s_a / m_hd) + binom_ll(s_b, m_hd, s_b / m_hd)
          - binom_ll(s_a, m_hd, p_pool) - binom_ll(s_b, m_hd, p_pool))

for k in (0, 10, 15, 17, 18):
    print(f"successes in B = {s_b[k]:2d}:  log odds ratio {beta1[k]:6.3f}"
          f"    Wald {wald[k]:7.3f}    likelihood ratio {lr[k]:7.3f}")

```

When the log-likelihood *is* close to quadratic, the three agree closely. For the age coefficient of [Example \(Fitted votes by party identification\)](#) the script obtains the single-coefficient Wald = 2.046, Score = 2.054 and LR = 2.066, all far from the 5% point 3.84 of  $\chi^2(1)$ , so the three lead to the same conclusion.

The score statistic needs only the null fit, so it can test a term from the model without it; but it evaluates the information at  $\tilde{\beta}$ , the wrong place when the alternative is far away. The recommendation is simple: report likelihood ratio tests, and where an interval is wanted invert them,  $\{\beta_j : \text{LR}(\beta_j) \leq \chi^2_{1-\alpha}(1)\}$ , rather than using the Wald interval.

### 34.5.3 The deviance

#### Definition 34.5.1 — Deviance and scaled deviance

Let  $\hat{\mu}$  be the fitted means of a model and let  $\theta(\cdot)$  be as in [Definition 34.1.2](#). Assuming every  $y_i \in \mathcal{M}$ , the **saturated model** is the one that sets  $\mu_i = y_i$ . The **deviance** of the fit is

$$D(\mathbf{y}, \hat{\mu}) = \sum_{i=1}^n d_i, \quad (34.5.4)$$

$$d_i = 2w_i \left[ y_i \{ \theta(y_i) - \theta(\hat{\mu}_i) \} - b \{ \theta(y_i) \} + b \{ \theta(\hat{\mu}_i) \} \right],$$

and the **scaled deviance** is  $D^* = D/\phi = 2\{\ell_{\text{sat}} - \ell(\hat{\mu})\}$ .

The deviance does not involve  $\phi$ ; the scaled deviance does. For the normal family,  $\theta = \mu$  and  $b(\theta) = \theta^2/2$  give  $d_i = w_i(y_i - \hat{\mu}_i)^2$ : the deviance is the weighted residual sum of squares, the scaled deviance  $\text{SSE}/\sigma^2$ .

| Family                       | $d_i/w_i$                                                                                |
|------------------------------|------------------------------------------------------------------------------------------|
| normal                       | $(y - \hat{\mu})^2$                                                                      |
| binomial ( $y$ a proportion) | $2 \left[ y \log \frac{y}{\hat{\mu}} + (1 - y) \log \frac{1 - y}{1 - \hat{\mu}} \right]$ |
| Poisson                      | $2 \left[ y \log \frac{y}{\hat{\mu}} - (y - \hat{\mu}) \right]$                          |
| gamma                        | $2 \left[ -\log \frac{y}{\hat{\mu}} + \frac{y - \hat{\mu}}{\hat{\mu}} \right]$           |

| Family           | $d_i/w_i$                                 |
|------------------|-------------------------------------------|
| inverse Gaussian | $\frac{(y - \hat{\mu})^2}{y \hat{\mu}^2}$ |

with the convention  $0 \log 0 = 0$ .

### Proposition 34.5.4 — The deviance is nonnegative

Every  $d_i \geq 0$ , with  $d_i = 0$  if and only if  $\hat{\mu}_i = y_i$ . Consequently  $D \geq 0$ , with equality only for the saturated fit, and  $D$  decreases when a model is enlarged.

*Proof.* Write  $\tilde{\theta} = \theta(y_i)$ ,  $\hat{\theta} = \theta(\hat{\mu}_i)$ , so  $b'(\tilde{\theta}) = y_i$ . Then

$$\frac{d_i}{2w_i} = y_i(\tilde{\theta} - \hat{\theta}) - b(\tilde{\theta}) + b(\hat{\theta}) = b(\hat{\theta}) - b(\tilde{\theta}) - b'(\tilde{\theta})(\hat{\theta} - \tilde{\theta}) \geq 0$$

by convexity of  $b$  (Lemma 34.1.1): the right-hand side is the Bregman divergence of  $b$ , the amount by which  $b$  lies above its tangent at  $\tilde{\theta}$ . Strict convexity makes it strict unless  $\hat{\theta} = \tilde{\theta}$ , that is  $\hat{\mu}_i = y_i$ . Enlarging the model can only raise the maximized likelihood, and  $D = 2\phi\{\ell_{\text{sat}} - \ell(\hat{\mu})\}$ . ■

The other standard discrepancy is the **Pearson statistic**,

$$X^2 = \sum_{i=1}^n \frac{w_i(y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}, \quad (34.5.5)$$

which is  $\sum_i (y_i - \hat{\mu}_i)^2 / \widehat{\text{Var}}(Y_i)$  up to the factor  $\phi$ . It equals the deviance for the normal family and approximates it elsewhere, since a second-order expansion of  $d_i$  about  $\hat{\mu}_i = y_i$  gives  $w_i(y_i - \hat{\mu}_i)^2 / V(\hat{\mu}_i)$ .

## 34.5.4 Analysis of deviance

### Theorem 34.5.5 — Distribution theory for the deviance

Assume (G1)–(G4).

- (a) (*Nested models.*) Let  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  with  $\text{rank}(\mathbf{X}) - \text{rank}(\mathbf{X}_0) = q$ . Under the reduced model,

$$\frac{D_0 - D}{\phi} \xrightarrow{d} \chi^2(q),$$

where  $D_0$  and  $D$  are the deviances of the two fits. When  $\phi$  is unknown it is replaced by an estimate, and the ratio  $\{(D_0 - D)/q\}/\hat{\phi}$  is referred to  $F(q, n - p)$ , by analogy with Theorem 11.1.1.

- (b) (*Grouped data.*) Suppose the data consist of  $G$  groups,  $G$  and  $p$  fixed, with  $y_i$

an average of  $w_i = m_i$  observations and  $m_i \rightarrow \infty$  for every  $i$ . Then under the fitted model  $D/\phi \rightarrow \chi^2(G-p)$  and  $X^2/\phi \rightarrow \chi^2(G-p)$ , so both are goodness-of-fit statistics.

- (c) (*Ungrouped data.*) If the number of groups grows with the sample size, (b) does not apply and the limit is generally not  $\chi^2(n-p)$ . For binary responses with the canonical link the failure is total:  $D$  is a function of  $\hat{\beta}$  alone, so it carries no information about fit beyond what  $\hat{\beta}$  already contains.

*Proof.*

- (a) By [Definition 34.5.1](#) the saturated log-likelihood cancels in the difference, so  $(D_0 - D)/\phi = 2\{\ell(\hat{\beta}) - \ell(\tilde{\beta})\}$ , which is the likelihood ratio statistic of [Proposition 34.5.3](#).
- (b) Regard the data as the  $\sum_i m_i$  individual observations rather than the  $G$  averages, and take the saturated model to be the one with a separate parameter per group, so that its model matrix is the group indicators and it has  $G$  parameters, fixed as  $m_i \rightarrow \infty$ . For that model the quantities [Eq. 34.5.1](#) evaluated at the individual observations are  $q_i = 1/m_{g(i)}$ , with  $g(i)$  the group of observation  $i$ , so (G3) holds; and the information in group  $g$  is  $m_g b''(\theta_g)/\phi \rightarrow \infty$ , so (G4) holds. Part (a) applied to the fitted model against the saturated one, with  $q = G - p$ , gives  $D/\phi \rightarrow \chi^2(G-p)$ . For  $X^2$ , a third-order expansion of  $d_i$  about  $\hat{\mu}_i = y_i$  gives  $w_i(y_i - \hat{\mu}_i)^2/V(\hat{\mu}_i) + O(w_i|y_i - \hat{\mu}_i|^3)$ ; with  $w_i = m_i$  and  $y_i - \hat{\mu}_i = O_p(m_i^{-1/2})$  the remainder summed over the  $G$  groups is  $O_p(\max_i m_i^{-1/2})$ , so  $X^2 - D \rightarrow 0$  in probability.
- (c) For a binary response with the logit link,  $\mathcal{M} = (0, 1)$  and  $y_i \in \{0, 1\}$  is not in  $\mathcal{M}$ ; taking limits in [Eq. 34.5.4](#) with  $0 \log 0 = 0$  gives the saturated log-likelihood 0 and  $D = -2 \sum_i \{y_i \log \hat{\mu}_i + (1 - y_i) \log(1 - \hat{\mu}_i)\} = -2\ell(\hat{\beta})$ . Writing  $\hat{\eta}_i = \mathbf{x}_{(i)}^\top \hat{\beta}$  and using  $\log(1 - \hat{\mu}_i) = -\log(1 + e^{\hat{\eta}_i})$ ,

$$\ell(\hat{\beta}) = \sum_i \{y_i \hat{\eta}_i - \log(1 + e^{\hat{\eta}_i})\} = \hat{\beta}^\top \mathbf{X}^\top \mathbf{y} - \sum_i \log(1 + e^{\mathbf{x}_{(i)}^\top \hat{\beta}}).$$

By [Corollary 34.3.2](#),  $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \hat{\mu}$ , so  $\ell(\hat{\beta}) = \hat{\beta}^\top \mathbf{X}^\top \hat{\mu} - \sum_i \log(1 + e^{\mathbf{x}_{(i)}^\top \hat{\beta}})$  depends on the data only through  $\hat{\beta}$ , and two data sets with the same  $\hat{\beta}$  have the same deviance. ■

### Example. (Grouped and ungrouped, simulated)

A logistic model with two parameters was fitted to 2000 data sets simulated from it, in two formats with the same number of Bernoulli trials over the same range of  $x$ : as 40 groups of 20 trials, and as 800 single trials. (The listing below repeats the experiment with 200 replicates, so that it runs in a few seconds.)

For the grouped data  $G - p = 38$ , and the simulated deviances have mean 39.74 and standard deviation 9.43, against the  $\chi^2(38)$  values 38 and  $\sqrt{76} = 8.72$ : usable. For the ungrouped data  $n - p = 798$ , and the simulated deviances have mean 987.95 and standard deviation 20.62, against the  $\chi^2(798)$  values 798 and 39.95 — the mean off by nearly 200, the spread by a factor of two ([Figure 34.5.2](#)). Referring that deviance to

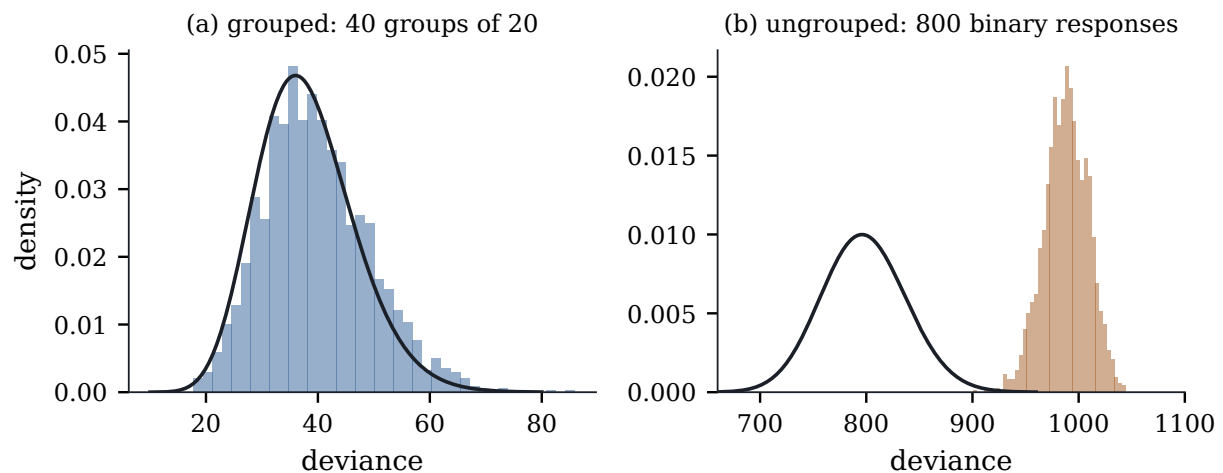


Figure 34.5.2: Simulated null distributions of the residual deviance, with the  $\chi^2$  density that the naive degrees-of-freedom count suggests. (a) Grouped data: the approximation works. (b) The same information as ungrouped binary responses: it does not.

$\chi^2(n - p)$  would reject a correct model with overwhelming confidence.

Nothing is wrong with the sample size. The two designs carry very nearly the same Fisher information — the script checks that the standard errors of  $\hat{\beta}$  agree to within 2% — so the estimates behave alike. What fails is the reference distribution, because the saturated model in the second format has 800 parameters and grows with the data.

```
def logit_irls(X, y, w, steps=25):
    """Maximum likelihood for a binomial model with logit link; y is a proportion.
    """
    mu = (w * y + 0.5) / (w + 1.0)
    eta = np.log(mu / (1 - mu))
    beta = np.zeros(X.shape[1])
    for _ in range(steps):
        d = mu * (1 - mu)
        W = w * d
        z = eta + (y - mu) / d
        beta = np.linalg.solve((X.T * W) @ X, (X.T * W) @ z)
        eta = X @ beta
        mu = 1 / (1 + np.exp(-eta))
    return beta, mu

def binomial_deviance(y, mu, w):
    t1 = np.where(y > 0, y * np.log(np.where(y > 0, y, 1.0) / mu), 0.0)
    t2 = np.where(y < 1, (1 - y) * np.log(np.where(y < 1, 1 - y, 1.0) / (1 - mu)),
    0.0)
    return 2 * np.sum(w * (t1 + t2))

def null_deviances(X, beta, m, reps, rng):
    """Residual deviance of the fitted model, on `reps` data sets simulated from it.
    """
    p = 1 / (1 + np.exp(-(X @ beta)))
    w = np.full(len(p), float(m))
    out = np.empty(reps)
    for r in range(reps):
```

```

    y = rng.binomial(m, p).astype(float) / m
    _, mu = logit_irls(X, y, w)
    out[r] = binomial_deviance(y, mu, w)
return out

rng = np.random.default_rng(3407)
beta0 = np.array([0.3, 0.9])
G, m = 40, 20                                # grouped: 40 groups of 20
    trials
Xg = np.column_stack([np.ones(G), np.linspace(-1.5, 1.5, G)])
N = 800                                        # ungrouped: 800 single trials
Xu = np.column_stack([np.ones(N), np.linspace(-1.5, 1.5, N)])

reps = 200
Dg = null_deviances(Xg, beta0, m, reps, rng)
Du = null_deviances(Xu, beta0, 1, reps, rng)
print(f"grouped   : mean {Dg.mean():8.2f} (df {G - 2}), sd {Dg.std():6.2f}"
      f" (sqrt(2 df) {np.sqrt(2 * (G - 2)):.2f})")
print(f"ungrouped : mean {Du.mean():8.2f} (df {N - 2}), sd {Du.std():6.2f}"
      f" (sqrt(2 df) {np.sqrt(2 * (N - 2)):.2f})")

```

*Idea.* Differences of deviances between nested models are reliable, and are what the analysis-of-deviance table uses. The deviance *itself* is a goodness-of-fit statistic only for grouped data with large groups. The two claims are constantly confused because software prints them in the same table.

The analysis-of-deviance table is built like the sequential analysis-of-variance table of [Definition 9.3.1](#): add terms in a chosen order, recording the drop in deviance at each step with its degrees of freedom. The reference distribution is chi-squared when  $\phi$  is known and  $F$  when it is estimated.

#### Example. (Analysis of deviance for Engel's data)

Continue the gamma/log model of [Example \(Food expenditure: three models, two decisions\)](#), with  $u$  the centred logarithm of income and  $n = 235$ . Fitting the three nested models gives

| model         | residual df | deviance | change | $F$    | $p$         |
|---------------|-------------|----------|--------|--------|-------------|
| 1             | 234         | 39.01    |        |        |             |
| 1 + $u$       | 233         | 4.27     | 34.73  | 1975.2 | $< 10^{-4}$ |
| 1 + $u + u^2$ | 232         | 4.18     | 0.0918 | 5.22   | 0.023       |

Here  $\phi$  is unknown, so each change is divided by  $\hat{\phi} = 0.01758$  and referred to  $F(1, 232)$ , following [Theorem 34.5.5\(a\)](#). One qualification: the log link is not the gamma's canonical link, so (G1) does not hold and the calibration rests not on the proof given but on [Proposition 34.5.2](#), which extends the limit to this case without proving it. Income matters overwhelmingly; the quadratic term is borderline. The residual deviance 4.18 is *not* to be compared with  $\chi^2(232)$ : the data are ungrouped, so [Theorem 34.5.5\(b\)](#) does not apply, and the small value reflects the small dispersion of the gamma fit, not a suspiciously good one.

```

import numpy as np
import statsmodels.api as sm
from scipy import stats

engel = sm.datasets.engel.load_pandas().data
y = engel["foodexp"].to_numpy()
u = np.log(engel["income"].to_numpy())
u = u - u.mean()                                # centred, so the terms are less
collinear
n = len(y)

def gamma_deviance(y, mu):
    """D = 2 sum { -log(y/mu) + (y - mu)/mu } for the gamma family."""
    return 2 * np.sum(-np.log(y / mu) + (y - mu) / mu)

models = {"1": np.ones((n, 1)),
          "1 + u": np.column_stack([np.ones(n), u]),
          "1 + u + u^2": np.column_stack([np.ones(n), u, u ** 2])}
fits, dev = {}, {}
for name, Xm in models.items():
    f = sm.GLM(y, Xm, family=sm.families.Gamma(sm.families.links.Log())).fit()
    fits[name], dev[name] = f, gamma_deviance(y, f.fittedvalues)

full = fits["1 + u + u^2"]
mu_full = full.fittedvalues
phi_pearson = np.sum((y - mu_full) ** 2 / mu_full ** 2) / (n - 3)
phi_deviance = dev["1 + u + u^2"] / (n - 3)

names = list(models)
print(f"{'term added':14s} {'df':>4s} {'deviance':>10s} {'change':>9s} {'F':>8s} {'p'
      ':>8s}")
prev = None
for name in names:
    df_res = n - models[name].shape[1]
    row = f"{'name':14s} {'df_res':4d} {'dev[name]:10.4f}"
    if prev is not None:
        change = dev[prev] - dev[name]
        F = change / phi_pearson
        row += f" {'change':9.4f} {'F':8.2f} {'stats.f.sf(F, 1, n - 3):8.4f}"
    print(row)
    prev = name
print(f"dispersion: Pearson {phi_pearson:.5f}    deviance {phi_deviance:.5f}")

```

### 34.5.5 Estimating the dispersion

For the binomial and Poisson families  $\phi = 1$  by construction; if the data contradict that, the remedy is a different model, which is [Chapter 38](#). For the normal, gamma and inverse Gaussian families  $\phi$  is free. Maximum likelihood would take it from the full likelihood, which for the gamma involves the digamma function and behaves badly under misspecification. The standard choice is instead the **Pearson estimate**

$$\hat{\phi} = \frac{X^2}{n - p} = \frac{1}{n - p} \sum_{i=1}^n \frac{w_i (y_i - \hat{\mu}_i)^2}{V(\hat{\mu}_i)}. \quad (34.5.6)$$

The motivation is the moment identity  $E\{w_i(Y_i - \mu_i)^2/V(\mu_i)\} = \phi$ , exact at the true mean; replacing  $\mu_i$  by  $\hat{\mu}_i$  costs  $p$  degrees of freedom, by analogy with  $s^2 = \text{SSE}/(n - p)$ , the exact normal-theory case ([Theorem 5.6.1](#)). The alternative  $D/(n - p)$  is also in use; for Engel's data the two give 0.01758 and 0.01803. They can differ sharply for skewed data, and the Pearson version is the more reliable, since it does not depend on the far tail of the assumed density.

Because the likelihood equations do not involve  $\phi$  ([Theorem 34.3.1\(a\)](#)), estimating it does not disturb  $\hat{\beta}$ . It does enter every standard error, through  $\hat{\mathbf{A}}_n^{-1} = \hat{\phi}(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$ , and its own variability is why the tests are then referred to an  $F$  distribution.

## 34.5.6 Residuals and diagnostics

The raw residual  $y_i - \hat{\mu}_i$  is uninformative on its own, because its variance depends on  $\hat{\mu}_i$ . Three standardizations are in use.

### Definition 34.5.2 — Pearson, deviance and Anscombe residuals

For a fitted generalized linear model, with  $d_i$  the deviance contributions of [Eq. 34.5.4](#) and  $h_{ii}$  the diagonal of the weighted hat matrix  $\mathbf{H} = \hat{\mathbf{W}}^{1/2} \mathbf{X}(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \mathbf{X}^\top \hat{\mathbf{W}}^{1/2}$ , define

$$r_i^P = \frac{\sqrt{w_i}(y_i - \hat{\mu}_i)}{\sqrt{\hat{\phi} V(\hat{\mu}_i)}}, \quad r_i^D = \text{sign}(y_i - \hat{\mu}_i) \sqrt{d_i / \hat{\phi}},$$

$$r_i^A = \frac{\sqrt{w_i} \{A(y_i) - A(\hat{\mu}_i)\}}{\sqrt{\hat{\phi} A'(\hat{\mu}_i) \sqrt{V(\hat{\mu}_i)}}},$$

where  $A(\mu) = \int^\mu V(t)^{-1/3} dt$ . The **standardized** versions divide by  $\sqrt{1 - h_{ii}}$ .

The Pearson residuals have squares adding to  $X^2/\hat{\phi}$ , the deviance residuals squares adding to  $D/\hat{\phi}$ ; each splits its own discrepancy measure into per-observation pieces, so a large deviance residual marks an observation the likelihood explains badly. The Anscombe residual first applies the transformation that brings the family closest to normal,  $A(\mu) = \frac{3}{2}\mu^{2/3}$  for the Poisson and  $3\mu^{1/3}$  for the gamma.

### Example. (Which residual is closest to normal)

Two thousand observations were simulated from a Poisson log-linear model with fitted means between about 1 and 6, and the model refitted. [Figure 34.5.3](#) shows normal quantile plots of the three residual types. The Pearson residuals are visibly right-skewed, with sample skewness 0.618 — the skewness  $\mu^{-1/2}$  of [Exercise \(B3\)](#), averaged over the fitted means. The deviance residuals have skewness  $-0.010$  and the Anscombe residuals  $-0.081$ ; both are far closer to normal, and neither is uniformly better. Discreteness remains visible in all three as short horizontal runs. The leverages sum to 2 ([Proposition 6.8.1\(b\)](#)) and the largest is 0.0029.

```
rng = np.random.default_rng(3405)
N = 2000
Xp = np.column_stack([np.ones(N), rng.uniform(-1, 1, N)])
mu_true = np.exp(Xp @ np.array([1.0, 0.8]))
```

*# means between about 1*



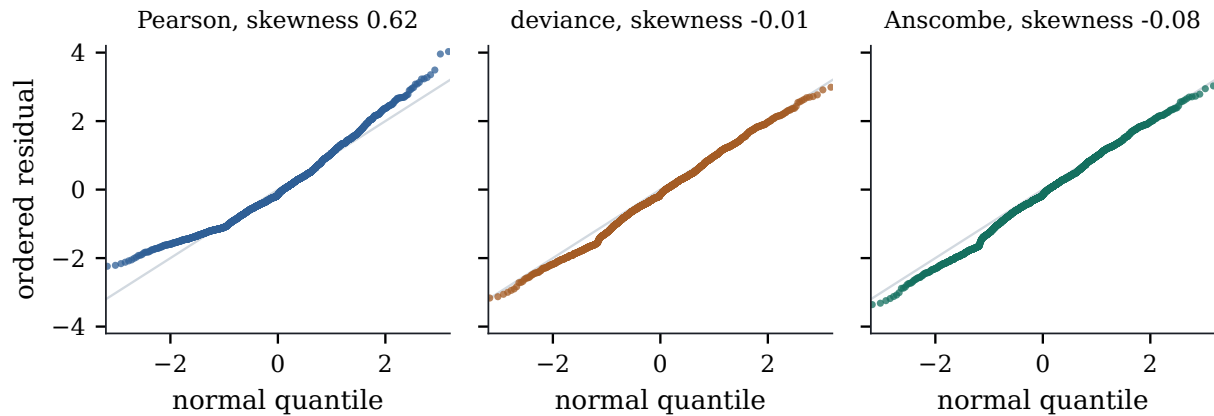


Figure 34.5.3: Normal quantile plots of the three residual types for a Poisson log-linear fit with small means. The reference line is the identity. Pearson residuals inherit the skewness of the family; deviance and Anscombe residuals largely remove it.

```

    and 6
yp = rng.poisson(mu_true).astype(float)
pois = sm.GLM(yp, Xp, family=sm.families.Poisson()).fit()
mu_p = pois.fittedvalues

pearson = (yp - mu_p) / np.sqrt(mu_p)                                # r = (y -
    mu)/sqrt(V)
d_i = 2 * (np.where(yp > 0, yp * np.log(np.maximum(yp, 1e-300) / mu_p), 0.0) - (yp -
    mu_p))
deviance_r = np.sign(yp - mu_p) * np.sqrt(d_i)                        # signed
    root of d_i
anscombe = 1.5 * (yp ** (2 / 3) - mu_p ** (2 / 3)) / mu_p ** (1 / 6)  # A(mu) =
    1.5 mu^{2/3}

W = mu_p                                                            # working weights for
    Poisson/log
root = np.sqrt(W)[: , None] * Xp
H = root @ np.linalg.inv(root.T @ root) @ root.T                    # the weighted hat matrix
h = np.diag(H)

for name, r in [("Pearson", pearson), ("deviance", deviance_r), ("Anscombe",
    anscombe)]:
    print(f"{name:9s} sd {r.std(ddof=2):6.3f} skewness {stats.skew(r):7.3f}")
print(f"leverages: sum {h.sum():.4f}, largest {h.max():.4f}")

```

The rest of [Chapter 20](#) transfers under the substitution  $\mathbf{X} \mapsto \hat{\mathbf{W}}^{1/2}\mathbf{X}$ , which is Pregibon's (1981) extension of the linear-model diagnostics: the leverages of [Proposition 6.8.1](#), a Cook-type measure  $C_i = (r_i^P)^2 h_{ii} / \{p(1-h_{ii})^2\}$  ([Definition 20.4.1](#)), and the one-step deletion approximations of [Theorem 20.3.1](#) with the working response in place of  $\mathbf{y}$ . The exact deletion identities do not, because dropping a case changes the weights too. For ungrouped binary data the residuals take only two values at each  $\hat{\mu}_i$ , so only smoothed plots carry a signal — a point [Proposition 35.4.1](#) develops.

### 34.5.7 Model selection

Because the fit is by maximum likelihood, [Definition 29.2.2](#) applies directly, with  $\text{AIC} = -2\ell(\hat{\boldsymbol{\beta}}) + 2d$  and  $d = p$  when  $\phi$  is known,  $p + 1$  when it is estimated; for a  $\phi = 1$  family comparing AIC is comparing  $D + 2p$ . The rest of [Chapter 29](#) transfers unchanged, [Proposition 29.5.1](#) included.

### 34.5.8 Exercises

#### 34.5.8.1 A. Check your understanding

##### Exercise A1.

Verify the normal and Poisson rows of the deviance table from [Eq. 34.5.4](#). For the Poisson, show that if the model has an intercept then  $\sum_i (y_i - \hat{\mu}_i) = 0$  by [Corollary 34.3.2](#), so the deviance reduces to  $2 \sum_i y_i \log(y_i / \hat{\mu}_i)$ .

##### Exercise A2.

A binomial model is fitted to 60 groups with 4 parameters, and the residual deviance is 71.2 on 56 degrees of freedom. When is comparing it with  $\chi^2(56)$  legitimate, and what would you need to know about the data first?

*Solution.* By [Theorem 34.5.5\(b\)](#) the comparison needs the number of groups fixed and the group sizes large, so one needs the  $m_i$ : if they are all in the hundreds the comparison is reasonable, and if many are 1 or 2 it is not, whatever the total sample size. A rough rule is that the expected counts  $m_i \hat{\mu}_i$  and  $m_i(1 - \hat{\mu}_i)$  should all exceed about five.

#### 34.5.8.2 B. Practice

##### Exercise B1.

Let  $\hat{\beta}$  have asymptotic standard error  $s$ . Compute the Wald statistic for  $H_0 : \beta = 0$  and, using the delta method, the Wald statistic for the equivalent hypothesis  $H_0 : e^\beta = 1$ . Show that their ratio is  $\{(e^{\hat{\beta}} - 1)/\hat{\beta}\}^2 e^{-2\hat{\beta}}$ , and evaluate it at  $\hat{\beta} = 1$  and  $\hat{\beta} = 3$ . Why is the likelihood ratio statistic unaffected?

##### Exercise B2.

Show that  $E\{w_i(Y_i - \mu_i)^2 / V(\mu_i)\} = \phi$  exactly, for every observation in an exponential dispersion family. Deduce that  $X^2/n$  is unbiased for  $\phi$  when the true means are used, and explain why  $X^2/(n - p)$  is only approximately unbiased when they are estimated.

##### Exercise B3.

Derive the Anscombe residual for the Poisson family from  $A(\mu) = \int^\mu t^{-1/3} dt$ , and check that the standardizing denominator is  $\hat{\mu}^{1/6}$ . Explain why  $A$  is *not* the variance-

stabilizing transformation  $2\sqrt{\mu}$  of [Corollary 22.4.2](#), and what each one is optimizing.

*Solution.*  $A(\mu) = \frac{3}{2}\mu^{2/3}$ , so  $A'(\mu) = \mu^{-1/3}$  and  $A'(\hat{\mu})\sqrt{V(\hat{\mu})} = \hat{\mu}^{-1/3}\hat{\mu}^{1/2} = \hat{\mu}^{1/6}$ , giving  $r^A = \frac{3}{2}(y^{2/3} - \hat{\mu}^{2/3})/\hat{\mu}^{1/6}$ . The variance-stabilizing transformation solves  $A' \propto V^{-1/2}$  and makes the *variance* constant; the Anscombe transformation solves  $A' \propto V^{-1/3}$  and makes the *third cumulant*, hence the skewness, vanish to the order considered. Different objectives, different exponents; the Anscombe residual is then standardized by hand, so it does not need the variance to be stable by itself.

#### Exercise B4.

Two summaries of predictive power for a generalized linear model are the sample correlation between  $\mathbf{y}$  and  $\hat{\boldsymbol{\mu}}$ , and the likelihood-based  $R_L^2 = (\ell_M - \ell_0)/(\ell_{\text{sat}} - \ell_0)$ , where  $\ell_0$ ,  $\ell_M$  and  $\ell_{\text{sat}}$  are the maximized log-likelihoods of the intercept-only model, the fitted model and the saturated model. Show that for a family with  $\phi = 1$  the second is  $(D_0 - D)/D_0$ , with  $D_0$  the null deviance, and deduce that it lies in  $[0, 1]$  and never decreases when a term is added. Why does that last property make it a poor basis for choosing a model?

*Solution.* With  $\phi = 1$ , [Definition 34.5.1](#) gives  $D = 2(\ell_{\text{sat}} - \ell_M)$  and  $D_0 = 2(\ell_{\text{sat}} - \ell_0)$ , so  $\ell_M - \ell_0 = (D_0 - D)/2$  and  $R_L^2 = (D_0 - D)/D_0$ . Both deviances are nonnegative by [Proposition 34.5.4](#) and  $D \leq D_0$ , since enlarging a model cannot lower the maximized likelihood; so  $R_L^2 \in [0, 1]$  and cannot decrease when a term is added. That is the defect of the unadjusted  $R^2$  of [Section 9.5](#), the point [Proposition 9.5.3](#) makes there: it rewards complexity unconditionally. AIC instead charges 2 per parameter against  $-2\ell$ , and so can rise.

### 34.5.8.3 C. Going deeper

#### Exercise C1.

Extend [Theorem 34.5.5\(c\)](#) to a Poisson log-linear model in which every  $y_i$  is 0 or 1: show that the deviance is again a function of  $\hat{\boldsymbol{\beta}}$  alone. Then show that the argument breaks down as soon as some  $y_i \geq 2$ , and explain in words what extra information the larger counts supply. (This is the reason the deviance is a serviceable goodness-of-fit statistic for Poisson data with large means and a useless one for sparse counts.)

*Solution.* For the Poisson deviance,  $D = 2 \sum_i \{y_i \log(y_i/\hat{\mu}_i) - (y_i - \hat{\mu}_i)\}$ . When every  $y_i \in \{0, 1\}$  the term  $y_i \log y_i$  is zero, so  $D = -2 \sum_i y_i \log \hat{\mu}_i + 2 \sum_i (y_i - \hat{\mu}_i)$ ; the second sum vanishes by [Corollary 34.3.2](#) if the model has an intercept, and the first is  $-2\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \mathbf{y} = -2\hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top \hat{\boldsymbol{\mu}}$ , again by [Corollary 34.3.2](#). So  $D$  is a function of  $\hat{\boldsymbol{\beta}}$ . If some  $y_i \geq 2$ , the term  $\sum_i y_i \log y_i$  is a nonzero function of the data that is *not* determined by  $\mathbf{X}^\top \mathbf{y}$ : two data sets with the same sufficient statistic, one with counts  $(2, 0)$  at two identical covariate values and one with  $(1, 1)$ , have the same  $\hat{\boldsymbol{\beta}}$  but different deviances. The larger counts carry information about the spread of the responses within a covariate pattern, which is precisely what a goodness-of-fit statistic needs and what binary data cannot supply.

## 34.6 Summary and Notes

### 34.6.1 Summary

*Idea.*

1. An exponential dispersion family has density  $\exp[w\{y\theta - b(\theta)\}/\phi + c(y, \phi, w)]$  with natural parameter  $\theta$ , cumulant function  $b$ , dispersion  $\phi$  and known weight  $w$  (Definition 34.1.1). The natural parameter space is convex and  $b$  is strictly convex on it (Lemma 34.1.1).
2. All moments come from  $b$ :  $\mu = b'(\theta)$ ,  $\text{Var}(Y) = \phi b''(\theta)/w$ , and the  $r$ th cumulant is  $(\phi/w)^{r-1} b^{(r)}(\theta)$  (Proposition 34.1.2). Reparameterizing by the mean gives the variance function  $V(\mu)$  (Definition 34.1.2), which determines the family:  $V = 1, \mu, \mu^2, \mu^3, \mu(1 - \mu), \mu(1 + \mu/k)$ .
3. A generalized linear model is a random component from such a family, a linear predictor  $\mathbf{X}\boldsymbol{\beta}$ , and a link  $g(\mu_i) = \eta_i$  (Definition 34.2.1); the canonical link  $g = (b')^{-1}$  makes  $\theta_i = \eta_i$  (Definition 34.2.2). The link and the variance function are separate decisions, and neither transforms the data.
4. The score is  $\phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu})$  with working weights  $\mathbf{W}_{ii} = w_i h'(\eta_i)^2 / V(\mu_i)$ ; the expected information is  $\phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X}$ , and under the canonical link the observed information equals it (Theorem 34.3.1). The likelihood equations do not involve  $\phi$ .
5. Under the canonical link the fit reproduces the weighted column totals of  $\mathbf{X}$  exactly (Corollary 34.3.2), and the log-likelihood is strictly concave, so the maximum is unique when it exists (Theorem 34.3.3). Existence can fail, as in separation.
6. Fisher scoring is weighted least squares on the working response  $z_i = \eta_i + (y_i - \mu_i)g'(\mu_i)$  with weights  $\mathbf{W}$ , iterated (Theorem 34.4.1). It is Newton's method exactly when the link is canonical, which is why canonical-link fits converge quadratically and others linearly.
7. Under stated regularity conditions — notably  $\max_i q_{ni} \rightarrow 0$ , the leverage-like condition Eq. 34.5.1 — the estimate exists with probability tending to one, is consistent, and  $\mathbf{A}_n^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0)$  is asymptotically standard normal (Theorem 34.5.1 for the canonical link, Proposition 34.5.2 for the rest). Wald, score and likelihood ratio statistics are asymptotically equivalent and  $\chi^2(q)$  (Proposition 34.5.3), but the Wald statistic is not invariant and can fall as the evidence grows (the Hauck–Donner effect).
8. The deviance is twice the gap to the saturated log-likelihood (Definition 34.5.1); every contribution is nonnegative (Proposition 34.5.4). Differences of deviances between nested models are  $\chi^2$  in the limit; the deviance itself is a goodness-of-fit statistic only for grouped data with large groups, and for ungrouped binary data it is a function of  $\hat{\boldsymbol{\beta}}$  alone (Theorem 34.5.5).
9. The dispersion is estimated by  $X^2/(n-p)$  (Eq. 34.5.6), which leaves  $\hat{\boldsymbol{\beta}}$  unchanged. Pearson, deviance and Anscombe residuals (Definition 34.5.2) and the weighted hat matrix carry the diagnostics of Chapter 20 across, and AIC applies as in Chapter 29.

## 34.6.2 Notes and sources

**The synthesis.** Nelder and Wedderburn (1972) created the subject. Every piece existed before — probit analysis, logistic regression, log-linear models for contingency tables, the gamma model for survival times, weighted least squares — and the paper’s contribution was to see that the exponential family structure made all of them one algorithm. That is what let a single program, GLIM, fit them all, and what makes the code of [Section 34.4](#) family-agnostic. McCullagh and Nelder (1989) remains the definitive treatment; its chapter 2 is the source for the deviance and residual definitions used here, chapter 12 for the diagnostics.

**Exponential dispersion families.** The class of [Definition 34.1.1](#), with  $\phi$  a genuine second parameter rather than a nuisance folded into  $\theta$ , is due to Jorgensen (1987), developed in Jorgensen (1997); that paper also classifies the power variance functions  $V(\mu) = \mu^\xi$ , the Tweedie families of [Exercise \(C2\)](#). The analytic facts behind [Proposition 34.1.2](#) are in Widder (1941); the statistical apparatus built on them, including the convexity of the natural parameter space and the mean-value mapping, is Barndorff-Nielsen (1978) and Brown (1986); the existence criterion quoted after [Example \(Two ways the maximum fails to exist\)](#) is Barndorff-Nielsen’s.

**Fitting.** The scoring method is older than the subject: it appears in Fisher’s appendix to Bliss (1935) on probit analysis, as a device for fitting a dose–response curve by hand. Nelder and Wedderburn (1972) identified the general form and recognized it as weighted least squares. Green (1984) is the standard reference for iteratively reweighted least squares as a general estimation device and for the convergence issues of [Section 34.4](#). Existence and uniqueness link by link are settled by Wedderburn (1976), the source of the gamma/identity warning in [Section 34.3](#); that only the variance function enters the estimating equations, so one may drop the density, is Wedderburn (1974), which [Chapter 38](#) follows.

**Asymptotics.** The complete theory — consistency, asymptotic normality, and existence with probability tending to one — is Fahrmeir and Kaufmann (1985), under conditions considerably weaker than (G1)–(G4) and covering non-canonical links and stochastic regressors; [Proposition 34.5.2](#) quotes what this book needs from it. The proof of [Theorem 34.5.1](#) takes the shortcut the canonical link allows: the log-likelihood is concave and its Hessian nonrandom, so the argument reduces to the central limit theorem of [Lemma 21.2.1](#) for the score, plus the convexity lemma of Pollard (1991), which makes concavity do the work usually done by uniform convergence arguments. The condition  $\max_i q_{ni} \rightarrow 0$  is the generalized-linear-model form of the bounded-leverage condition used throughout the book, and  $\sum_i q_{ni} = p$  is [Proposition 6.8.1\(b\)](#) in a new setting.

**The three tests.** Cox and Hinkley (1974, chapter 9) is the standard general account of their asymptotic equivalence. The failure of [Example \(The Hauck–Donner effect\)](#) was identified by Hauck and Donner (1977) and named after them; it is one case of a general phenomenon, since the Wald statistic uses a quadratic approximation to the log-likelihood centred at  $\hat{\beta}$ , worst exactly where the estimate is most extreme. None of this arises in the normal linear model, where the log-likelihood is exactly quadratic in  $\beta$  for fixed  $\sigma^2$ , and [Proposition 11.4.2](#) records the exact relationships that result.

**The deviance and its limits.** That the residual deviance of a binary logistic fit is a function of  $\hat{\beta}$  alone, hence useless as a goodness-of-fit statistic, is in McCullagh and Nelder (1989, section 4.4). The regime in which the deviance *is* a goodness-of-fit statistic — groups fixed, group sizes growing — is sometimes called grouped or small-dispersion asymptotics, against the  $n \rightarrow \infty$  asymptotics of [Theorem 34.5.1](#). The two give different answers for the same statistic, the source of most of the confusion in practice, and [Chapter 35](#) returns to it with [Proposition 35.4.1](#).

Anscombe residuals are from Anscombe (1953); the extension of the diagnostics of [Chapter 20](#), including the weighted hat matrix and one-step deletion approximations, is Pregibon (1981).

**Textbook treatments.** Agresti (2015, chapter 4) covers the same ground at a similar level and is the closest companion; its chapters 5–7 correspond to [Chapter 35](#), [Chapter 36](#) and [Chapter 37](#). Fahrmeir, Kneib, Lang and Marx (2021, section 5.4) handles the weight  $w$  and grouped data carefully, and is the source of the convention used here that  $y$  may be a group average. Dobson and Barnett (2018) is a gentler introduction; Rencher and Schaalje (2008, chapter 18) treats the subject briefly, as an extension of linear-model theory.

**What comes next.** [Chapter 38](#) keeps the score equations and drops the density, which handles data more variable than the family allows. [Chapter 39](#) takes up model building in practice and the Bayesian treatment, whose priors are the counterpart of the penalized likelihoods of [Chapter 27](#) and [Chapter 30](#). [Chapter 40](#) joins the subject to the mixed models of [Chapter 32](#), where the likelihood is no longer available in closed form.

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# Chapter 35

## Binary Responses

A response that can only be zero or one is the smallest departure from the normal linear model that still changes everything. Its mean is a probability, so it lives in an interval rather than on a line; its variance is determined by its mean, so there is no free dispersion parameter; and the exact distribution theory of Parts III and IV is gone. What remains is the likelihood, and the apparatus of [Chapter 34](#) applies word for word. This chapter is what that apparatus looks like pointed at the simplest possible response, and what that response does to it: odds ratios that survive sampling schemes no other model can tolerate, a maximum likelihood estimate that sometimes fails to exist, and a deviance that, for ungrouped data, carries no information about fit at all.

**What you need.** [Chapter 34](#) for the exponential dispersion family, the likelihood equations, iteratively reweighted least squares, the deviance and the asymptotics; [Chapter 7](#) for maximum likelihood and Fisher information ([Theorem 7.3.1](#), [Proposition 7.4.7](#)); [Chapter 11](#) for the likelihood ratio test in the normal model ([Theorem 11.4.1](#), [Proposition 11.4.2](#)); [Chapter 20](#) for residuals, leverage and influence; [Chapter 21](#) for weighted least squares; [Chapter 22](#) for empirical logits ([Exercise \(B2\)](#)); [Chapter 25](#) for the causal reading of a coefficient; [Chapter 27](#) and [Chapter 30](#) for penalized estimation; [Chapter 29](#) for AIC and cross-validation.

### 35.0.1 Roadmap

- [Logistic regression](#)
- [Probit and complementary log–log links](#)
- [Separation](#)
- [Goodness of fit](#)
- [Case–control sampling](#)
- [Summary and notes](#)

## 35.1 Logistic regression

A binary response records whether something happened: a patient recovered, a voter turned out, a loan defaulted. Then  $E(Y_i) = \Pr(Y_i = 1)$ , so modelling the mean *is* modelling a probability, and a linear model for it is in trouble before it starts:  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  ranges over the whole line while a



probability may not leave  $[0, 1]$ , and the variance  $\pi_i(1-\pi_i)$  moves with the mean. The generalized linear model of [Chapter 34](#) removes both problems by putting a link between the mean and the linear predictor.

### 35.1.1 The model

Let  $Y_1, \dots, Y_n$  be independent with  $Y_i \sim \text{Binomial}(m_i, \pi_i)$ ;  $m_i = 1$  throughout is **ungrouped** data, and larger  $m_i$  arise when several units share a covariate vector. Dropping the index,

$$\binom{m}{y} \pi^y (1-\pi)^{m-y} = \binom{m}{y} \exp\left\{y \log \frac{\pi}{1-\pi} + m \log(1-\pi)\right\},$$

an exponential dispersion family ([Definition 34.1.1](#)) for the mean  $\mu = m\pi$ , with natural parameter  $\theta = \log\{\pi/(1-\pi)\}$ , cumulant function  $b(\theta) = \log(1 + e^\theta)$ , dispersion  $\phi = 1$  and weight  $m$ . Differentiating  $b$  twice recovers the moments of [Proposition 34.1.2](#):  $b'(\theta) = \pi$  and  $b''(\theta) = \pi(1-\pi)$ , so the variance function is  $V(\pi) = \pi(1-\pi)$  and  $\text{Var}(Y) = m\pi(1-\pi)$ . The natural parameter is the **log odds**, and the canonical link is therefore the logit.

#### Definition 35.1.1 — Logistic regression

The **logistic regression model** for independent  $Y_i \sim \text{Binomial}(m_i, \pi_i)$  is

$$\text{logit } \pi_i = \log \frac{\pi_i}{1-\pi_i} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \quad \text{equivalently} \quad \pi_i = \frac{\exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})}. \quad (35.1.1)$$

The quantity  $\pi/(1-\pi)$  is the **odds** of the event, and the ratio of the odds at two covariate vectors is an **odds ratio**.

The logit is one choice among many ([Section 35.2](#)). Its claim to first place is that it is canonical, so [Theorem 34.3.1](#) takes its simplest form, and that its coefficients say something people want to hear.

### 35.1.2 Reading the coefficients

#### Proposition 35.1.1 — Three readings of a logistic coefficient

In [Eq. 35.1.1](#), write  $\eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  and let  $x_j$  be a covariate that appears in the linear predictor only through its own coefficient  $\beta_j$ . Then:

- (a) increasing  $x_j$  by  $c$  with the other covariates unchanged multiplies the odds by  $e^{c\beta_j}$ , whatever the values of the other covariates;
- (b)  $\partial\pi/\partial x_j = \beta_j \pi(1-\pi)$ , so  $|\partial\pi/\partial x_j| \leq |\beta_j|/4$ , with equality exactly at  $\pi = 1/2$ ;
- (c) the average of that derivative over the sample,  $n^{-1} \sum_i \beta_j \hat{\pi}_i(1-\hat{\pi}_i)$ , is the **average marginal effect**: the mean change in fitted probability per unit of  $x_j$ .

*Proof.*

- (a) The log odds at  $x_j + c$  minus the log odds at  $x_j$  is  $c\beta_j$ , and everything else in  $\eta$  cancels. (b) Differentiating  $\pi = e^\eta / (1 + e^\eta)$  gives  $d\pi/d\eta = \pi(1 - \pi)$ , and the chain rule supplies the factor  $\beta_j$ . The function  $\pi(1 - \pi)$  on  $[0, 1]$  is maximized at  $\pi = 1/2$  with value  $1/4$ , so the derivative has modulus at most  $|\beta_j|/4$ . (c) Immediate from (b).

■

Part (a) is why logistic regression is written as it is: one number summarizes the effect on the odds at every setting of the other covariates. Part (b) is Gelman and Hill’s “divide by four” rule, sharp near  $\pi = 1/2$ ; part (c) reports the same coefficient on the probability scale.

*Warning.* None of the three readings is causal: each compares two covariate vectors *within the fitted model*, and whether that estimates an intervention is the question of [Chapter 25](#). [Section 35.5](#) adds a wrinkle special to odds ratios.

### Example. (Voting intention in the 1996 American election)

The 1996 American National Election Studies survey recorded, for 944 respondents, the candidate they expected to vote for (393 said Dole), party identification from strong Democrat (0) to strong Republican (6), age, education on a seven-point scale and household income on a 24-point scale. Fitting [Eq. 35.1.1](#) with these four covariates and age in decades gives

| term                 | $\hat{\beta}_j$ | standard error |
|----------------------|-----------------|----------------|
| intercept            | −5.471          | 0.619          |
| party identification | 1.2188          | 0.0709         |
| age (decades)        | 0.1222          | 0.0714         |
| education            | 0.0147          | 0.0771         |
| income               | 0.0311          | 0.0210         |

One step along the party scale multiplies the odds of a Dole vote by  $e^{1.2188} = 3.383$ ; three steps multiply them by 38.72. Dividing by four bounds the effect on the probability by 0.305 per step, and the average marginal effect is 0.100: one step is worth about ten percentage points, averaged over the respondents ([Figure 35.1.1](#)).

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
X = pd.DataFrame({"const": 1.0, "PID": anes["PID"], "age": anes["age"] / 10,
                  "educ": anes["educ"], "income": anes["income"]})
y = anes["vote"] # 1 = Dole, 0 = Clinton

fit = sm.GLM(y, X, family=sm.families.Binomial()).fit()
pi_hat = fit.fittedvalues
```

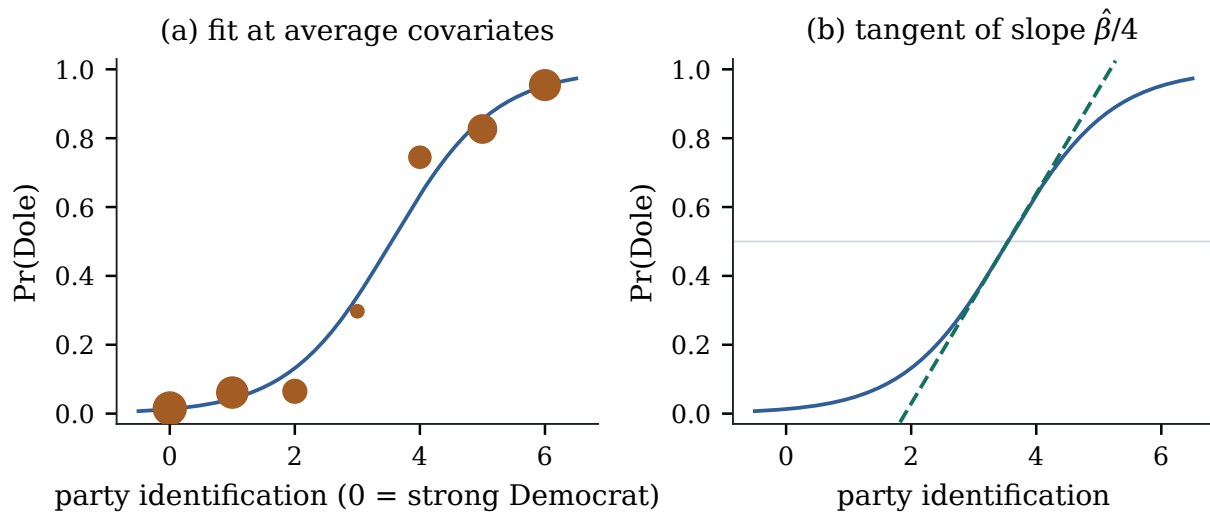


Figure 35.1.1: Expected vote for Dole in the 1996 ANES survey. (a) The fitted logistic curve against party identification, the other covariates held at their means; the discs are the observed proportions in the seven party groups, with area proportional to group size. (b) The tangent at the steepest point, of slope  $\hat{\beta}_j/4$ .

```
print(fit.params.round(4).to_string())
print("odds ratio for one step on the party scale:", round(np.exp(fit.params["PID"])
, 3))
print("divide by four:", round(fit.params["PID"] / 4, 3))
print("average marginal effect:", round((fit.params["PID"] * pi_hat * (1 - pi_hat)).
mean(), 3))
```

### 35.1.3 Grouped and ungrouped data

Two data sets recording the same information can look very different: one lists 944 individual zeros and ones, the other one binomial count per distinct covariate vector.

#### Proposition 35.1.2 — Grouping

Let  $\mathbf{y}$  be ungrouped binary data, and suppose the rows of  $\mathbf{X}$  take  $N$  distinct values  $\mathbf{x}_{(1)}^*, \dots, \mathbf{x}_{(N)}^*$ , the  $k$ th occurring  $m_k$  times with  $s_k$  successes. Then:

- (a) the two log-likelihoods differ by a constant free of  $\boldsymbol{\beta}$ , so they have the same maximizers, the same score, the same information and the same likelihood ratio statistics;
- (b) the two deviances of Definition 34.5.1 differ by  $-2\ell_{\text{sat}}$ , where  $\ell_{\text{sat}} = \sum_k \{s_k \log(s_k/m_k) + (m_k - s_k) \log(1 - s_k/m_k)\}$  is the maximized log-likelihood of the grouped saturated model;
- (c) if  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , then  $\sum_i m_i \hat{\pi}_i = \sum_i y_i$  in either form: the fit reproduces the total number of successes.

*Proof.*

(a) With  $\eta_k = \mathbf{x}_{(k)}^\top \boldsymbol{\beta}$ , the ungrouped log-likelihood is

$$\sum_{k=1}^N \{s_k \eta_k - m_k \log(1 + e^{\eta_k})\},$$

because the individual contributions within a group depend on the data only through  $s_k$ . The grouped log-likelihood is the same expression plus  $\sum_k \log \binom{m_k}{s_k}$ , a constant. (b) The deviance is twice the difference between the saturated and fitted log-likelihoods. The fitted log-likelihood is the same in both forms by (a); the ungrouped saturated model sets  $\hat{\pi}_i = y_i$  and attains log-likelihood 0, while the grouped saturated model sets  $\hat{\pi}_k = s_k/m_k$  and attains  $\ell_{\text{sat}} \leq 0$ . (c) Under the canonical link the likelihood equations are  $\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$  ([Theorem 34.3.1](#)); the column  $\mathbf{1}$  gives  $\sum_i (y_i - m_i \hat{\pi}_i) = 0$ .

■

Part (b) is worth pausing over: the deviance of a logistic fit depends on how the data were typed in. *Differences* of deviances between nested models do not, because the saturated term cancels, and those are what [Theorem 34.5.5](#) licenses.

#### Example. (The same fit, two deviances)

Fit logit  $\pi = \beta_0 + \beta_1(\text{party}) + \beta_2(\text{education})$  to the survey of [Example \(Voting intention in the 1996 American election\)](#). The 944 respondents occupy 47 distinct covariate cells. Grouped, the deviance is 45.99 on 44 degrees of freedom; ungrouped, it is 533.34 on 941. The coefficients and their standard errors agree to machine precision.

```
# the same model in grouped form: one binomial count per distinct covariate pattern
cols = ["PID", "educ"]
cells = anes.groupby(cols)["vote"].agg(["sum", "count"]).reset_index()
Xg = sm.add_constant(cells[cols])
counts = np.column_stack([cells["sum"], cells["count"] - cells["sum"]])
grouped = sm.GLM(counts, Xg, family=sm.families.Binomial()).fit()

Xu = sm.add_constant(anes[cols])
ungrouped = sm.GLM(anes["vote"], Xu, family=sm.families.Binomial()).fit()

print("grouped  ", grouped.params.round(4).to_numpy(), "deviance", round(grouped.
    deviance, 2))
print("ungrouped", ungrouped.params.round(4).to_numpy(), "deviance", round(ungrouped
    .deviance, 2))
```

### 35.1.4 Estimation

There is no closed form for  $\hat{\boldsymbol{\beta}}$ . Under the canonical link [Theorem 34.3.1](#) reduces to

$$\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}, \quad \mu_i = m_i \pi_i = \frac{m_i \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})}{1 + \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})}, \quad (35.1.2)$$

a system of  $p$  smooth equations in  $p$  unknowns. Specializing [Theorem 34.4.1](#) gives the algorithm every implementation uses.

### Proposition 35.1.3 — Fisher scoring for logistic regression

For [Eq. 35.1.1](#) the Fisher scoring step from  $\mathbf{b}$  is the weighted least squares fit of the working response  $\mathbf{z}$  on  $\mathbf{X}$  with weights  $\mathbf{W}$ , where

$$w_i = m_i \pi_i (1 - \pi_i), \quad z_i = \eta_i + \frac{y_i - m_i \pi_i}{m_i \pi_i (1 - \pi_i)},$$

both evaluated at  $\mathbf{b}$ . The observed and expected information coincide and equal  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ ; the log-likelihood is concave in  $\boldsymbol{\beta}$ , and strictly concave when  $\text{rank}(\mathbf{X}) = p$  and  $0 < \pi_i < 1$ .

*Proof.* The weights and working response are the general formulae of [Theorem 34.4.1](#) with  $V(\pi) = \pi(1 - \pi)$  and  $d\eta/d\mu = 1/\{m\pi(1 - \pi)\}$ . Differentiating [Eq. 35.1.2](#) once more gives the Hessian  $-\mathbf{X}^\top \mathbf{W} \mathbf{X}$ , so the observed information equals the expected information, as [Theorem 34.3.1](#) gives for every canonical link. The matrix  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$  is nonnegative definite, and positive definite when the  $w_i$  are positive and  $\mathbf{X}$  has full column rank, since  $\mathbf{b}^\top \mathbf{X}^\top \mathbf{W} \mathbf{X} \mathbf{b} = \sum_i w_i (\mathbf{x}_{(i)}^\top \mathbf{b})^2$  vanishes only if  $\mathbf{X} \mathbf{b} = \mathbf{0}$ . ■

Concavity means at most one local maximum, and the step is the weighted least squares of [Theorem 21.3.1](#) with response and weights refreshed each iteration. Convergence from  $\boldsymbol{\beta} = \mathbf{0}$  is usually fast, and took 6 iterations in [Example \(Voting intention in the 1996 American election\)](#); it is not guaranteed, and [Section 35.3](#) describes the configuration in which there is nothing to converge to.

## 35.1.5 Inference

With the estimate in hand, [Theorem 34.5.1](#) supplies its law: under the usual regularity conditions, with  $p$  fixed and the information growing without bound,  $\hat{\boldsymbol{\beta}}$  is treated as  $N_p(\boldsymbol{\beta}, (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1})$ . The growth condition is real: it fails if a covariate is almost constant, and it fails if the fitted probabilities are driven to 0 or 1, since then  $w_i = m_i \pi_i (1 - \pi_i) \rightarrow 0$ . Of the three statistics of [Proposition 34.5.3](#), the Wald statistic is what software prints, and it is the one the Hauck–Donner effect spoils; the likelihood ratio statistic is invariant and survives an infinite estimate, and the score statistic needs only the null fit.

### Example. (Tests and intervals in the voting model)

In [Example \(Voting intention in the 1996 American election\)](#), education has  $\hat{\beta} = 0.0147$  with standard error 0.0771; its Wald statistic is 0.036 and so is the likelihood ratio statistic from dropping it, while age has  $z = 1.71$  and  $p = 0.087$ . For the party coefficient the Wald interval is (1.080, 1.358) and the profile likelihood interval (1.086, 1.364): the log-likelihood is nearly quadratic here. [Example \(A separated subgroup\)](#) shows what happens when it is not.

The machinery of Part III survives in weakened form: simultaneous inference uses a quadratic

form in  $\mathbf{A}^\top \hat{\boldsymbol{\beta}} - \mathbf{d}$  with matrix  $\{\mathbf{A}^\top (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \mathbf{A}\}^{-1}$ , referred to  $\chi^2(q)$  in place of the  $F$  statistic of [Theorem 11.3.2](#). What does *not* survive is exactness. The right measure of how much there is to work with is the total weight  $\sum_i m_i \pi_i (1 - \pi_i)$ : a study of 500 subjects with twelve events has total weight about 12, the same as a balanced study of fifty, which is the arithmetic behind the events-per-variable rules of thumb in the applied literature.

## 35.1.6 Exercises

### 35.1.6.1 A. Check your understanding

#### Exercise A1.

A logistic model has  $\hat{\beta} = 0.7$  for a binary treatment indicator. Give the estimated odds ratio, the divide-by-four bound on the change in probability, and the actual change when the control probability is 0.5, 0.1 and 0.01.

*Solution.* The odds ratio is  $e^{0.7} = 2.014$  and the bound is  $0.7/4 = 0.175$ . Starting from odds 1, 1/9 and 1/99, the treated odds are 2.014, 0.2238 and 0.02034, so the probabilities become 0.668, 0.183 and 0.0199 and the changes are 0.168, 0.083 and 0.0099. The bound is nearly attained at  $\pi = 1/2$  and grossly overstates the change for rare events.

#### Exercise A2.

Show from [Eq. 35.1.2](#) that a logistic model containing an indicator column for a group reproduces that group's observed proportion exactly. What follows for a model whose covariates are a full set of indicators for a categorical variable?

### 35.1.6.2 B. Practice

#### Exercise B1.

In logit  $\pi = \beta_0 + \beta_1 x$  with  $\beta_1 \neq 0$ , the level at which  $\pi = 1/2$  is  $x_{1/2} = -\beta_0/\beta_1$  (in bioassay, the median effective dose). Derive the delta-method standard error of  $\hat{x}_{1/2}$ , and explain why it becomes useless as  $\hat{\beta}_1$  approaches zero.

#### Exercise B2.

For grouped data with all  $m_i$  large, [Exercise \(B2\)](#) fits  $\boldsymbol{\beta}$  by weighted least squares on the empirical logits  $\ell_i = \text{logit}(s_i/m_i)$  with weights  $m_i \hat{\pi}_i (1 - \hat{\pi}_i)$ . Show that one Fisher scoring step of [Proposition 35.1.3](#), started from the fit that reproduces the observed proportions, is exactly that estimate.

*Solution.* Start from a  $\boldsymbol{\beta}$  with  $\pi_i = s_i/m_i$ , so  $\eta_i = \ell_i$ . The working residual  $(y_i - m_i \pi_i)/\{m_i \pi_i (1 - \pi_i)\}$  vanishes, the working response is  $z_i = \ell_i$  and the weights are  $m_i \hat{\pi}_i (1 - \hat{\pi}_i)$ , so one step returns the weighted least squares fit. The empirical-logit fit is therefore a legitimate starting value; it is undefined when  $s_i \in \{0, m_i\}$ .

**Exercise B3.**

Two binary covariates enter with an interaction:  $\text{logit } \pi = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$ . Show that  $\beta_{12}$  is the log of a ratio of two odds ratios, and that  $\beta_{12} = 0$  does *not* make the effect of  $x_1$  on the probability scale the same at both levels of  $x_2$ .

**35.1.6.3 C. Going deeper****Exercise C1.**

Prove that  $\ell(\boldsymbol{\beta}) = -\sum_i \log\{1 + \exp(-s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta})\}$ , with  $s_i = 2y_i - 1$ , is concave in  $\boldsymbol{\beta}$ , without computing the Hessian.

*Solution.* The map  $u \mapsto \log(1 + e^u)$  is convex, since its second derivative  $e^u/(1 + e^u)^2$  is positive. A convex function composed with an affine map is convex, so each  $\boldsymbol{\beta} \mapsto \log\{1 + \exp(-s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta})\}$  is convex, and a sum of convex functions is convex. Hence  $\ell$  is concave.

## 35.2 Probit and complementary log-log links

Nothing forces a binary model to use the logit. Any function mapping the real line onto  $(0, 1)$  can serve as the response function  $\pi = h(\eta)$ , and its inverse is then the link. The candidates that survive in practice come with a story about how the binary outcome was produced, because such a story says what the coefficients mean. There are two such stories, and they give the probit and the complementary log-log link.

### 35.2.1 Thresholds

**Definition 35.2.1 — Threshold model**

Let  $\varepsilon_1, \dots, \varepsilon_n$  be independent with common continuous distribution function  $F$ , and let

$$Y_i = \begin{cases} 1, & \varepsilon_i \leq \mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \\ 0, & \text{otherwise.} \end{cases}$$

The variable  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta} - \varepsilon_i$  is the **latent response**, and  $Y_i$  records the sign of it.

Immediately  $\pi_i = \Pr(\varepsilon_i \leq \mathbf{x}_{(i)}^\top \boldsymbol{\beta}) = F(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ , so a threshold model *is* a binary regression with response function  $F$  and link  $F^{-1}$ , and the content lies in which  $F$  the situation suggests.

**Proposition 35.2.1 — Three links and where they come from**

In [Definition 35.2.1](#):

- (a) if  $F$  is the standard logistic distribution function  $F(u) = e^u/(1 + e^u)$ , the model is [Eq. 35.1.1](#); if  $F = \Phi$ , it is the **probit model**  $\Phi^{-1}(\pi_i) = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . Both  $F$  are symmetric about zero, so both links satisfy  $g(\pi) = -g(1 - \pi)$ ;
- (b) if  $F(u) = 1 - \exp(-e^u)$ , the distribution function of the smallest extreme value law, the model is the **complementary log-log** model  $\log\{-\log(1 - \pi_i)\} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . This  $F$  is not symmetric:  $\pi$  leaves 0 slowly and approaches 1 abruptly. Reflecting it, that is, applying the complementary log-log link to  $1 - \pi$  and changing the sign of  $\boldsymbol{\beta}$ , gives the **log-log** model  $-\log\{-\log \pi_i\} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , with the two tails exchanged;
- (c) suppose a positive lifetime  $T$  has hazard function  $\lambda(t | \mathbf{x}) = \lambda_0(t) \exp(\mathbf{x}^\top \boldsymbol{\gamma})$ , and that  $Y$  records whether the event has occurred by a fixed time  $t_0$ . Then  $Y$  follows a complementary log-log model with linear predictor  $\log \Lambda_0(t_0) + \mathbf{x}^\top \boldsymbol{\gamma}$ , where  $\Lambda_0$  is the integrated baseline hazard. Moreover, if  $g$  is a continuous strictly increasing link with the property that for every  $c > 0$  replacing the hazard  $\lambda$  by  $c\lambda$  changes  $g(\pi)$  by a quantity  $\kappa(c)$  depending on  $c$  alone, and not on  $\Lambda_0(t_0)$  or  $\mathbf{x}$ , then  $g$  is an increasing affine function of the complementary log-log link.

*Proof.*

- (a) and (b) are the identity  $\pi = F(\eta)$  read backwards, together with the computation  $F^{-1}(\pi) = \log\{\pi/(1 - \pi)\}$ ,  $\Phi^{-1}(\pi)$  and  $\log\{-\log(1 - \pi)\}$  for the three  $F$ . For the reflection in (b), if  $1 - \pi = 1 - \exp(-e^{-\eta})$  then  $\log\{-\log \pi\} = -\eta$ .
- (b) The survivor function of a hazard model is  $\Pr(T > t_0 | \mathbf{x}) = \exp\{-\int_0^{t_0} \lambda(u | \mathbf{x}) du\} = \exp\{-\Lambda_0(t_0)e^{\mathbf{x}^\top \boldsymbol{\gamma}}\}$ . Hence

$$1 - \pi = \exp(-e^{\log \Lambda_0(t_0) + \mathbf{x}^\top \boldsymbol{\gamma}}),$$

and taking  $\log(-\log \cdot)$  of both sides gives the claim. For uniqueness, write  $H = \Lambda_0(t_0)e^{\mathbf{x}^\top \boldsymbol{\gamma}}$  for the cumulative hazard and  $G(u) = g(1 - e^{-e^u})$  with  $u = \log H$ , so that  $u$  ranges over the whole line as  $c$  and the baseline vary. The hypothesis reads  $G(u + \log c) = G(u) + \kappa(c)$  for all real  $u$  and all  $c > 0$ , so that  $\kappa(v) = G(u + v) - G(u)$  does not depend on  $u$ , and hence  $\kappa(v + v') = \kappa(v) + \kappa(v')$  for all real  $v, v'$ . Continuity of  $g$  makes  $\kappa$  continuous, and the only continuous solutions of that additive Cauchy equation are the linear ones, so  $\kappa(v) = av$ ; monotonicity gives  $a > 0$ . Then  $G(u) = G(0) + au$ . Undoing the substitution,  $g(\pi) = a \log\{-\log(1 - \pi)\} + b$ .

■

Part (c) is why the complementary log-log link is standard for binary data derived from times: a “yes” by a fixed follow-up date, a machine failed within a warranty period. Its coefficients are log hazard ratios, whereas logistic coefficients for the same data would be log odds ratios depending on  $t_0$  ([Exercise \(B2\)](#)).

## 35.2.2 What the latent scale can and cannot tell you



### Proposition 35.2.2 — Scale is not identified

In [Definition 35.2.1](#), replace  $\varepsilon_i$  by  $\sigma\varepsilon_i + \tau$  with  $\sigma > 0$ , and  $\boldsymbol{\beta}$  by  $\sigma\boldsymbol{\beta} + \tau\mathbf{e}_1$ , where  $\mathbf{e}_1$  picks out an intercept column. The observed responses have exactly the same distribution. Consequently  $\boldsymbol{\beta}$  is identified only up to the location and scale of the latent errors, while ratios  $\beta_j/\beta_k$  of coefficients of non-intercept covariates are identified.

*Proof.* The event  $\{\sigma\varepsilon_i + \tau \leq \mathbf{x}_{(i)}^\top(\sigma\boldsymbol{\beta} + \tau\mathbf{e}_1)\}$  is  $\{\varepsilon_i \leq \mathbf{x}_{(i)}^\top\boldsymbol{\beta}\}$ , because the intercept column is identically 1 so  $\mathbf{x}_{(i)}^\top\mathbf{e}_1 = 1$  and the  $\tau$  cancels, after which dividing by  $\sigma$  removes it. The two models therefore give the same  $\pi_i$  for every  $i$ . Ratios of non-intercept coefficients are unchanged by the substitution. ■

This is why the probit fixes  $\text{Var}(\varepsilon) = 1$  and the logistic model fixes  $\text{Var}(\varepsilon) = \pi^2/3$ : a convention must be adopted. It is also why probit and logit coefficients are not comparable until one is rescaled. Matching the standard deviations multiplies a probit coefficient by  $\pi/\sqrt{3} = 1.8138$ ; matching the slope at  $\pi = 1/2$  instead, where the logistic density is  $1/4$  and the normal density is  $\phi(0)$ , multiplies it by  $4\phi(0) = 1.5958$ . The conventions differ by about fourteen per cent, a fair measure of how meaningless the comparison is in the tails.

*Warning.* The latent-variable story tempts one into calling the probit coefficients “the real effects, on the underlying continuous variable”. They are not: the latent variable is not observed and its scale is a convention that [Proposition 35.2.2](#) says the data cannot pin down. What the story buys is a rationale for a particular  $F$ .

## 35.2.3 How much does the choice matter?

Near  $\pi = 1/2$  all reasonable links agree, because smooth response curves look alike near their steepest point. They separate in the tails: the logistic tails are exponential, the normal tails decay like  $e^{-\eta^2/2}$ , and the complementary log-log is not symmetric at all.

### Example. (The survey under three links)

Fitting the model of [Example \(Voting intention in the 1996 American election\)](#) with the three links gives the following coefficients of party identification.

| link                     | $\hat{\beta}$ | rescaled by 1.8138 | maximized<br>log-likelihood |
|--------------------------|---------------|--------------------|-----------------------------|
| logit                    | 1.219         | —                  | −264.14                     |
| probit                   | 0.673         | 1.221              | −265.96                     |
| complementary<br>log-log | 0.801         | —                  | −273.48                     |

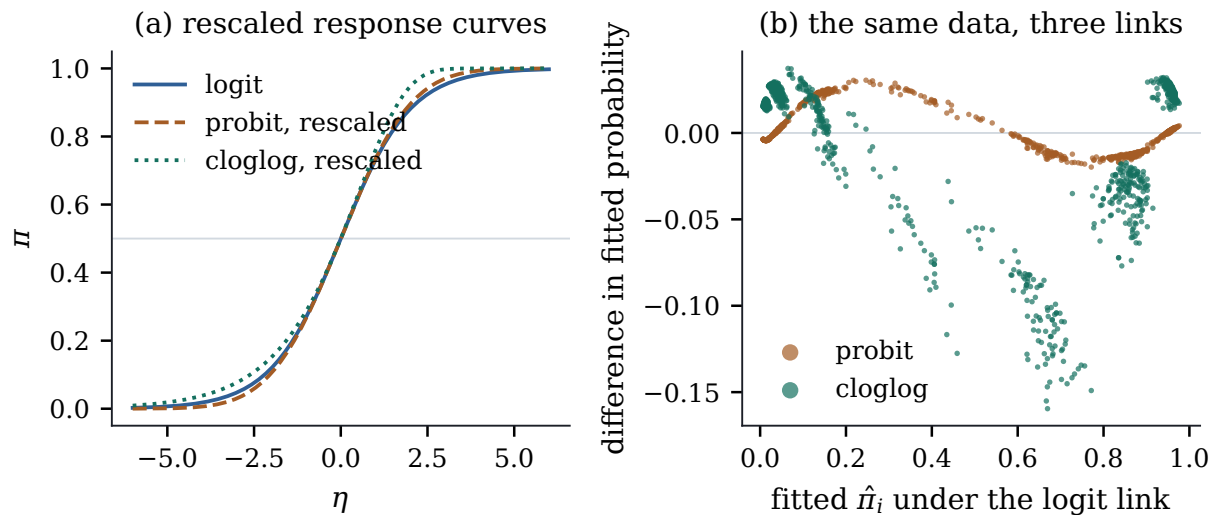


Figure 35.2.1: (a) The three response curves, each rescaled to pass through  $(0, 1/2)$  with slope  $1/4$ ; the symmetric ones are close everywhere. (b) Fitted probabilities for the survey of [Example \(Voting intention in the 1996 American election\)](#): the probit differs from the logit by at most 0.03, the complementary log-log by five times as much.

The rescaled probit coefficient reproduces the logistic one to two decimals, and the largest difference between the two sets of fitted probabilities is 0.0305. The complementary log-log fit is a different animal: its fitted probabilities differ from the logistic ones by as much as 0.1596, and with equal numbers of parameters AIC prefers the logit by 18.67. Here the choice between symmetric links is immaterial and the choice of an asymmetric one is not ([Figure 35.2.1](#)).

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
X = pd.DataFrame({"const": 1.0, "PID": anes["PID"], "age": anes["age"] / 10,
                  "educ": anes["educ"], "income": anes["income"]})
y = anes["vote"]

links = {"logit": sm.families.links.Logit(), "probit": sm.families.links.Probit(),
         "cloglog": sm.families.links.CLogLog()}
fits = {k: sm.GLM(y, X, family=sm.families.Binomial(link=v)).fit() for k, v in links.items()}

table = pd.DataFrame({k: f.params for k, f in fits.items()})
table["probit x 1.814"] = table["probit"] * np.pi / np.sqrt(3)
print(table.round(3).to_string())
print("maximized log-likelihoods:",
      {k: round(f.llf, 2) for k, f in fits.items()})
```

The choice therefore matters when the interesting probabilities are extreme, where the links disagree and few observations settle the disagreement; when a mechanism names the link, as in [Proposition 35.2.1\(c\)](#); and under the retrospective sampling of [Section 35.5](#), where the logit is the only link whose coefficients survive.

## 35.2.4 Exercises

### 35.2.4.1 A. Check your understanding

#### Exercise A1.

Show that the logit and probit links satisfy  $g(\pi) = -g(1 - \pi)$  while the complementary log-log link does not. Deduce that for the first two, exchanging the labels 0 and 1 merely reverses the sign of every coefficient, and say what the exchange does under the complementary log-log link.

*Solution.* For the logit,  $\log\{(1 - \pi)/\pi\} = -\log\{\pi/(1 - \pi)\}$ ; for the probit,  $\Phi^{-1}(1 - \pi) = -\Phi^{-1}(\pi)$ . Relabelling therefore turns  $\eta$  into  $-\eta$ , so  $\boldsymbol{\beta}$  becomes  $-\boldsymbol{\beta}$  and the fit is the same model. For the complementary log-log link, relabelling produces the *log-log* model of Proposition 35.2.1(b): a genuinely different fit.

### 35.2.4.2 B. Practice

#### Exercise B1.

Verify that  $F(u) = 1 - \exp(-e^u)$  is a distribution function, and that if  $\varepsilon$  has this law then  $E\varepsilon = -\gamma$  and  $\text{Var } \varepsilon = \pi^2/6$ , where  $\gamma = 0.5772$  is Euler's constant. Deduce the factor that makes a complementary log-log coefficient's latent scale comparable with a logistic one.

*Solution.*  $F$  is continuous and increasing with limits 0 and 1. Substituting  $v = e^u$  turns  $-\varepsilon$  into a standard Gumbel variable, whose mean is  $\gamma$  and whose variance is  $\pi^2/6$ ; reflecting gives the stated moments. The standard deviation is  $\pi/\sqrt{6} = 1.2825$  against  $\pi/\sqrt{3} = 1.8138$  for the logistic, so the multiplier is  $\sqrt{2}$ . The location differs too, which shifts the intercept, so only the slopes are comparable.

#### Exercise B2.

Follow-up is divided into intervals  $(t_{k-1}, t_k]$ . Let  $Y_{ik} = 1$  if subject  $i$  has the event in interval  $k$ , given that it has not had it before. Show that a proportional hazards model for the underlying continuous time implies  $\log\{-\log(1 - \Pr(Y_{ik} = 1))\} = \alpha_k + \mathbf{x}_{(i)}^\top \boldsymbol{\gamma}$  with  $\alpha_k = \log\{\Lambda_0(t_k) - \Lambda_0(t_{k-1})\}$ , and explain how to fit it with ordinary binary regression software.

### 35.2.4.3 C. Going deeper

#### Exercise C1.

Consider the asymmetric family of Aranda-Ordaz (1981),  $\pi = 1 - (1 + \alpha e^\eta)^{-1/\alpha}$  for  $\alpha > 0$ . Show that  $\alpha = 1$  gives the logistic response function and  $\alpha \rightarrow 0$  the complementary log-log one. Explain how to test  $\alpha = 1$  with grouped data, and why the test has little power with ungrouped data.

## 35.3 Separation

**Proposition 35.1.3** says the logistic log-likelihood is strictly concave when the model matrix has full column rank. That guarantees at most one maximum, not at least one, and for binary data the failure is common: it happens whenever some linear combination of the covariates predicts the response perfectly, as in small samples, with rare outcomes, and whenever a categorical covariate has a level with only one outcome in it.

### 35.3.1 Three configurations

Write  $s_i = 2y_i - 1 \in \{-1, 1\}$  for the signed response of an ungrouped binary observation. The logistic log-likelihood is then

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \{y_i \log \pi_i + (1 - y_i) \log(1 - \pi_i)\} = - \sum_{i=1}^n \log \{1 + \exp(-s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta})\}, \quad (35.3.1)$$

because  $\pi_i = 1/\{1 + \exp(-\mathbf{x}_{(i)}^\top \boldsymbol{\beta})\}$  and  $1 - \pi_i$  is the same expression with the sign of the linear predictor reversed. Every term is negative and rises towards zero as  $s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  increases, so the whole question is whether some direction raises all of them at once.

#### Definition 35.3.1 — Separation and overlap

Let  $\mathbf{X}$  be  $n \times p$  with rows  $\mathbf{x}_{(i)}^\top$ , and let  $s_i = 2y_i - 1$ . The data are

- (a) **completely separated** if there is a  $\mathbf{b} \neq \mathbf{0}$  with  $s_i \mathbf{x}_{(i)}^\top \mathbf{b} > 0$  for every  $i$ ;
- (b) **quasi-completely separated** if they are not completely separated but there is a  $\mathbf{b} \neq \mathbf{0}$  with  $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$  for every  $i$ ;
- (c) in **overlap** if  $\mathbf{b} = \mathbf{0}$  is the only  $\mathbf{b}$  with  $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$  at every observation.

A vector  $\mathbf{b}$  of the kind in (a) or (b) is a **separating direction**.

The three cases are mutually exclusive and exhaustive, and a linear programme decides which holds: maximize  $t$  over  $\mathbf{b}$  in the unit cube subject to  $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq t$ . A positive optimum is complete separation; an optimum of zero attained only at  $\mathbf{b} = \mathbf{0}$  is overlap, and otherwise quasi-complete separation.

### 35.3.2 Existence

#### Theorem 35.3.1 — Existence and uniqueness of the maximum likelihood estimate

Let  $\text{rank}(\mathbf{X}) = p$  and let  $\ell$  be the logistic log-likelihood of Eq. 35.3.1. Then:

- (a) if the data are completely or quasi-completely separated,  $\ell$  has no maximizer: for

- every  $\beta$  there is a  $\beta'$  with  $\ell(\beta') > \ell(\beta)$ ;
- (b) under complete separation  $\sup_{\beta} \ell(\beta) = 0$ , and the supremum is approached along any separating ray  $\beta = t\mathbf{b}$ ,  $t \rightarrow \infty$ , with every fitted probability tending to 0 or 1;
- (c) if the data are in overlap,  $\ell$  attains its supremum at exactly one  $\hat{\beta} \in \mathbb{R}^p$ , and  $\hat{\beta}$  solves Eq. 35.1.2.

*Proof.*

- (a) Let  $\mathbf{b} \neq \mathbf{0}$  satisfy  $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$  for all  $i$ . Not every one of these numbers can be zero: if  $\mathbf{x}_{(i)}^\top \mathbf{b} = 0$  for all  $i$  then  $\mathbf{X}\mathbf{b} = \mathbf{0}$ , and  $\text{rank}(\mathbf{X}) = p$  forces  $\mathbf{b} = \mathbf{0}$ . So  $s_k \mathbf{x}_{(k)}^\top \mathbf{b} > 0$  for at least one  $k$ . Each summand of Eq. 35.3.1 is a strictly decreasing function of  $-s_i \mathbf{x}_{(i)}^\top \beta$ , hence a nondecreasing function of  $t$  along  $\beta + t\mathbf{b}$ , and the  $k$ th is strictly increasing. Therefore  $\ell(\beta + \mathbf{b}) > \ell(\beta)$  for every  $\beta$ , and no maximizer exists.
- (b) Under complete separation put  $\delta = \min_i s_i \mathbf{x}_{(i)}^\top \mathbf{b} > 0$ . Then  $\ell(t\mathbf{b}) \geq -n \log(1 + e^{-t\delta}) \rightarrow 0$ , while  $\ell < 0$  everywhere, so the supremum is 0 and is not attained. Since  $s_i \mathbf{x}_{(i)}^\top(t\mathbf{b}) \rightarrow \infty$ , the fitted probability  $\pi_i$  tends to 1 when  $y_i = 1$  and to 0 when  $y_i = 0$ .
- (c) First,  $\ell$  is strictly concave. Its Hessian is  $-\mathbf{X}^\top \mathbf{W} \mathbf{X}$  with  $\mathbf{W} = \text{diag}\{\pi_i(1 - \pi_i)\}$  positive definite (Proposition 35.1.3), and  $\mathbf{b}^\top \mathbf{X}^\top \mathbf{W} \mathbf{X} \mathbf{b} = 0$  forces  $\mathbf{X}\mathbf{b} = \mathbf{0}$ , hence  $\mathbf{b} = \mathbf{0}$ . Second,  $\ell$  is coercive:  $\ell(\beta) \rightarrow -\infty$  as  $\|\beta\| \rightarrow \infty$ . Suppose not. Then there are  $\beta_k$  with  $\|\beta_k\| \rightarrow \infty$  and  $\ell(\beta_k) \geq -c$  for some  $c < \infty$ . Put  $\mathbf{b}_k = \beta_k / \|\beta_k\|$ ; since the unit sphere is compact, a subsequence converges to some  $\mathbf{b}$  with  $\|\mathbf{b}\| = 1$ . If  $s_i \mathbf{x}_{(i)}^\top \mathbf{b} < 0$  for some  $i$ , then along that subsequence  $s_i \mathbf{x}_{(i)}^\top \beta_k = \|\beta_k\| s_i \mathbf{x}_{(i)}^\top \mathbf{b}_k \rightarrow -\infty$ , so the  $i$ th term of Eq. 35.3.1 tends to  $-\infty$ ; as all the other terms are at most 0, this gives  $\ell(\beta_k) \rightarrow -\infty$ , a contradiction. Hence  $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$  for every  $i$  with  $\mathbf{b} \neq \mathbf{0}$ , contradicting overlap. So  $\ell$  is coercive, and the set  $\{\beta : \ell(\beta) \geq \ell(\mathbf{0})\}$  is closed and bounded. A continuous function attains its maximum on a compact set, and strict concavity makes the maximizer unique. It is an interior maximum of a differentiable function, so the gradient vanishes there, which is Eq. 35.1.2. ■

Parts (a) and (c) say that **the maximum likelihood estimate exists if and only if the data are in overlap**, a characterization due to Silvapulle (1981) and, in this form, to Albert and Anderson (1984).

*Remark* (What quasi-complete separation leaves behind). Under quasi-complete separation let  $\mathbf{b}$  be a separating direction and  $S = \{i : \mathbf{x}_{(i)}^\top \mathbf{b} = 0\}$ . Along  $\beta + t\mathbf{b}$  the terms outside  $S$  tend to zero and those inside do not move, so the supremum of  $\ell$  is that of the subsample  $S$  alone, approached with the coefficients along  $\mathbf{b}$  running off to infinity while the rest settle down: which is why software reports some coefficients that look sensible alongside others in the tens with standard errors in the millions.

### 35.3.3 What the computer does

Fisher scoring does not know any of this: given separated data it climbs the ray until it hits the iteration limit or the weights underflow, so the size of the reported coefficients is set by the iteration count, and the standard errors are larger still, because  $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$  explodes as the

weights vanish.

### Example. (A separated subgroup)

Of the 944 respondents of [Example \(Voting intention in the 1996 American election\)](#), 13 reported that their schooling stopped at or before the eighth grade. Among them party identification predicts the intended vote perfectly: the ten at 0, 1 or 2 on the party scale all expected to vote Clinton, the three at 4 or 6 all Dole. The direction  $\mathbf{b} = (-3, 1)^\top$ , the linear predictor party  $-3$ , satisfies  $s_i \mathbf{x}_{(i)}^\top \mathbf{b} > 0$  for all thirteen, so the data are completely separated and no maximum likelihood estimate exists.

Asked to fit logit  $\pi = \beta_0 + \beta_1 \text{ party}$ , software stopped at ten iterations reports  $\hat{\beta}_1 = 8.75$  with standard error 29.3; at twenty-five iterations it reports 23.75 with a standard error of order  $10^4$ . The deviance falls below  $10^{-3}$  after ten steps and below  $10^{-9}$  after twenty-five, and the Wald test is nowhere near significant at either stopping point.

Inference is nevertheless possible, from the likelihood ratio. The profile log-likelihood rises to its supremum of zero, so the set of  $\beta_1$  not rejected at the five per cent level is the half-line  $(0.902, \infty)$ : a one-sided interval, correctly reporting that the data bound the effect from below and not from above.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
sub = anes[anes["educ"] == 1]                # schooling stopped at or before
      grade eight
X = sm.add_constant(sub[["PID"]]).to_numpy()
y = sub["vote"].to_numpy()

print(pd.crosstab(sub["PID"], sub["vote"]))
for maxiter in (5, 10, 25, 50):
    f = sm.GLM(y, X, family=sm.families.Binomial()).fit(maxiter=maxiter, tol=1e-16)
    print(f"{maxiter:3d} steps: slope {f.params[1]:10.3f}    standard error {f.bse
          [1]:14.3f}"
          f"    deviance {f.deviance:.2e}")
```

### Example. (An empty cell)

Among the 127 respondents with a doctorate, model the vote by party identification as a seven-level factor with the independent-Republican category (PID = 4) as baseline, that being the only middle level with both outcomes present. Three levels have no variation at all, and exactly 3 of the six contrasts run off to infinity. The others behave normally: the contrast for weak Democrats is  $-3.140$  with standard error 0.916. The deviance, 53.808, is well defined, because the diverging coefficients drive their own cells' fitted probabilities to 0 or 1, contributing nothing to it.

### 35.3.4 Penalized likelihood

The estimate fails to exist because the likelihood has no interior maximum, so every remedy adds something that does; three are in general use. Firth (1993) proposed adding half the log determinant of the Fisher information to the log-likelihood, the log of the Jeffreys prior density, as a way of removing the leading  $O(n^{-1})$  bias of the maximum likelihood estimate; that it also cures separation was noticed later, by Heinze and Schemper (2002). The modified score equations take a memorable form.

#### Proposition 35.3.2 — Firth's modified score

For ungrouped logistic regression with  $\text{rank}(\mathbf{X}) = p$ , let  $\ell^*(\boldsymbol{\beta}) = \ell(\boldsymbol{\beta}) + \frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{W} \mathbf{X})$  with  $\mathbf{W} = \text{diag}\{\pi_i(1 - \pi_i)\}$ . Let  $h_i = \pi_i(1 - \pi_i) \mathbf{x}_{(i)}^\top (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{x}_{(i)}$  be the  $i$ th diagonal entry of the weighted hat matrix  $\mathbf{W}^{1/2} \mathbf{X} (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{1/2}$ . Then

$$\frac{\partial \ell^*}{\partial \boldsymbol{\beta}} = \mathbf{X}^\top \left\{ \mathbf{y} - \boldsymbol{\pi} + \mathbf{h} \circ \left( \frac{1}{2} \mathbf{1} - \boldsymbol{\pi} \right) \right\}, \quad (35.3.2)$$

where  $\circ$  is the entrywise product. Equivalently, the Firth estimate is a fixed point: it is the ordinary maximum likelihood estimate for pseudo-data in which observation  $i$  has  $y_i + h_i/2$  successes out of  $1 + h_i$  trials, with  $h_i$  evaluated at that same estimate. Along any ray  $t\mathbf{b}$  that completely separates the data,  $\ell^*(t\mathbf{b}) \rightarrow -\infty$ .

*Proof.* Write  $F(\boldsymbol{\beta}) = \mathbf{X}^\top \mathbf{W} \mathbf{X}$  and  $w_i = \pi_i(1 - \pi_i)$ . Since  $\partial \pi_i / \partial \beta_j = x_{ij} w_i$  and  $dw/d\pi = 1 - 2\pi$ ,

$$\frac{\partial w_i}{\partial \beta_j} = x_{ij} w_i (1 - 2\pi_i), \quad \frac{\partial F}{\partial \beta_j} = \mathbf{X}^\top \mathbf{D}_j \mathbf{X},$$

with  $\mathbf{D}_j = \text{diag}\{x_{ij} w_i (1 - 2\pi_i)\}$ . By the derivative of a log determinant,

$$\begin{aligned} \frac{\partial \log \det F}{\partial \beta_j} &= \text{tr}(F^{-1} \mathbf{X}^\top \mathbf{D}_j \mathbf{X}) = \sum_i x_{ij} w_i (1 - 2\pi_i) \mathbf{x}_{(i)}^\top F^{-1} \mathbf{x}_{(i)} \\ &= \sum_i x_{ij} (1 - 2\pi_i) h_i. \end{aligned}$$

Halving and adding the ordinary score  $\sum_i x_{ij} (y_i - \pi_i)$  gives Eq. 35.3.2, since  $\frac{1}{2}(1 - 2\pi_i) = \frac{1}{2} - \pi_i$ . For the pseudo-data statement, the ordinary score for  $y_i^* = y_i + h_i/2$  successes out of  $m_i^* = 1 + h_i$  trials is  $\sum_i x_{ij} (y_i^* - m_i^* \pi_i)$ , which is the same expression.

For the last claim, let  $\delta = \min_i s_i \mathbf{x}_{(i)}^\top \mathbf{b} > 0$ . At  $\boldsymbol{\beta} = t\mathbf{b}$  every fitted probability is within  $e^{-t\delta}$  of 0 or 1, so  $w_i \leq e^{-t\delta}$  and  $\mathbf{X}^\top \mathbf{W} \mathbf{X} \preceq e^{-t\delta} \mathbf{X}^\top \mathbf{X}$  in the nonnegative definite order. Hence  $\log \det(\mathbf{X}^\top \mathbf{W} \mathbf{X}) \leq -pt\delta + \log \det(\mathbf{X}^\top \mathbf{X})$ , while  $\ell \leq 0$ . Therefore  $\ell^*(t\mathbf{b}) \leq -\frac{1}{2}pt\delta + \text{constant} \rightarrow -\infty$ . ■

The penalty pulls the estimate towards zero exactly where the likelihood is flat: along the offending ray it falls linearly while the likelihood has nothing left to gain. That the Firth estimate is finite for *every* configuration with  $\text{rank}(\mathbf{X}) = p$  was proved by Kosmidis and Firth (2021), by a longer argument.

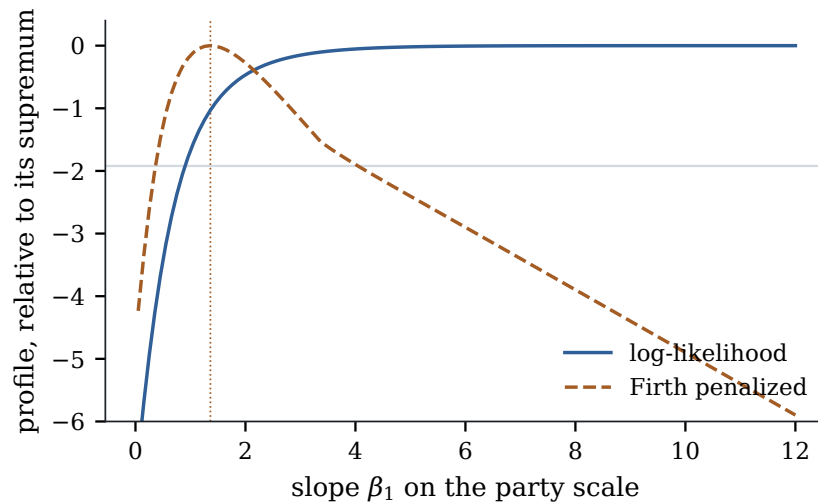


Figure 35.3.1: Profile log-likelihoods for the slope in [Example \(A separated subgroup\)](#), each relative to its own supremum. The ordinary profile (solid) increases to zero and is never attained, so the interval cut out by the horizontal line at  $-\chi_{0.95}^2(1)/2$  is a half-line. Firth's penalized profile (dashed) has an interior maximum, marked by the vertical line.

#### Example. (The separated subgroup, penalized)

Applied to [Example \(A separated subgroup\)](#), Firth's method gives  $\hat{\beta}_1 = 1.362$  with standard error 0.708 and intercept  $-4.049$ . The Wald statistic is 1.92, short of the conventional threshold, but the interval from inverting the *penalized* likelihood ratio is (0.364, 4.045), which excludes zero comfortably. With thirteen observations and a log-likelihood as asymmetric as [Figure 35.3.1](#) shows, the quadratic approximation behind the Wald statistic is simply not available.

```
def firth(X, y, steps=200):
    """Maximize the log-likelihood penalized by (1/2) log det(X'WX): Firth's
    estimate."""
    beta = np.zeros(X.shape[1])
    for _ in range(steps):
        mu = 1 / (1 + np.exp(-X @ beta))
        w = mu * (1 - mu)
        F = X.T @ (w[:, None] * X)
        # the expected
        information
        h = w * np.einsum("ij,jk,ik->i", X, np.linalg.inv(F), X)
        beta = beta + np.linalg.solve(F, X.T @ (y - mu + h * (0.5 - mu)))
    return beta

beta_firth = firth(X, y)
print("Firth estimate:", beta_firth.round(3))
```

### 35.3.5 Ridge and Cauchy priors

A quadratic penalty works too, for a simpler reason. Maximizing  $\ell(\boldsymbol{\beta}) - \frac{1}{2}\lambda\|\boldsymbol{\beta}\|^2$  is the logistic analogue of ridge regression ([Definition 27.2.1](#)), and the objective is strictly concave and coercive



whatever the data, so a unique maximizer always exists ([Exercise \(B1\)](#)). It is the posterior mode under independent  $N(0, \lambda^{-1})$  priors, which is how [Chapter 30](#) reads every penalty in [Proposition 30.1.4](#). In practice, and in the code below, the intercept is left out of the penalty, so that the prior reading is a flat one there; the maximizer still exists, provided both outcomes occur ([Exercise \(B1\)](#)). The cost is bias towards zero that depends on the scale of the covariates, and a  $\lambda$  that must come from somewhere, usually cross-validation ([Chapter 29](#)).

```
def ridge_logistic(X, y, lam, steps=200):
    """Maximize the log-likelihood minus (lam/2) times the squared norm of the
    slopes."""
    d = np.ones(X.shape[1])
    d[0] = 0.0                                     # leave the intercept
    unpenalized
    beta = np.zeros(X.shape[1])
    for _ in range(steps):
        mu = 1 / (1 + np.exp(-X @ beta))
        w = mu * (1 - mu)
        H = X.T @ (w[:, None] * X) + lam * np.diag(d)
        beta = beta + np.linalg.solve(H, X.T @ (y - mu) - lam * d * beta)
    return beta

for lam in (0.1, 1.0, 10.0):
    print(f"ridge, lambda = {lam:4.1f}: {ridge_logistic(X, y, lam).round(3)}")
```

On the separated subgroup the ridge estimate of the slope is 2.601 at  $\lambda = 0.1$ , 1.356 at  $\lambda = 1$  and 0.502 at  $\lambda = 10$ : the whole range is “the answer” for some penalty, which is the point. Gelman, Jakulin, Pittau and Su (2008) proposed instead independent Cauchy priors of scale 2.5 on standardized covariates, whose heavy tails leave a genuinely large coefficient nearly alone. Firth’s penalty alone is invariant to how the covariates are parameterized.

### 35.3.6 Exact conditional inference

A third route abandons the normal approximation. The sufficient statistic for  $\beta$  is  $\mathbf{X}^\top \mathbf{y}$ , by [Lemma 7.4.1](#) applied to [Eq. 35.3.1](#), so the conditional distribution of the component belonging to one coefficient, given the rest, depends on that coefficient alone. Enumerating it gives exact  $p$ -values and a median-unbiased estimate that is finite even under separation. The idea is Cox’s (1970) and the network algorithms are those of Hirji, Mehta and Patel (1987); the limit is computational.

*Idea.* Separation is not a defect of the data; it is the data saying that the effect is larger than the sample can measure, and every remedy decides how much larger. A one-sided likelihood ratio interval is the most honest of them, because it does not decide.

### 35.3.7 Exercises

#### 35.3.7.1 A. Check your understanding

**Exercise A1.**

A model contains an indicator for one level of a categorical covariate, and every observation at that level has  $y_i = 1$ . Exhibit a separating direction, say which of the three cases of [Definition 35.3.1](#) holds, and describe the fit.

*Solution.* Let  $\mathbf{b}$  have 1 in the coordinate of that indicator and zeros elsewhere. For observations at that level  $s_i \mathbf{x}_{(i)}^\top \mathbf{b} = +1 > 0$ ; for the others  $\mathbf{x}_{(i)}^\top \mathbf{b} = 0$ . So  $\mathbf{b}$  separates with some margins zero: quasi-complete separation, unless the remaining data are themselves completely separated. The coefficient of that indicator diverges, its fitted probabilities go to 1, and the other coefficients converge to the estimates from the remaining observations.

**35.3.7.2 B. Practice****Exercise B1.**

Show that for any data and any  $\lambda > 0$  the function  $\boldsymbol{\beta} \mapsto \ell(\boldsymbol{\beta}) - \frac{1}{2}\lambda\|\boldsymbol{\beta}\|^2$  has exactly one maximizer, without assuming  $\text{rank}(\mathbf{X}) = p$ . Show also that if  $\text{rank}(\mathbf{X}) = p$  and both outcomes occur the same is true when the intercept is left unpenalized.

*Solution.* By [Eq. 35.3.1](#),  $\ell \leq 0$ , so the objective is at most  $-\frac{1}{2}\lambda\|\boldsymbol{\beta}\|^2$ , which tends to  $-\infty$ : the objective is coercive and continuous, so it attains its maximum on a compact sublevel set. It is the sum of the concave  $\ell$  ([Exercise \(C1\)](#)) and a strictly concave quadratic, hence strictly concave, so the maximizer is unique. Neither step used the rank of  $\mathbf{X}$ . With the intercept  $\beta_0$  unpenalized, write  $\boldsymbol{\gamma}$  for the slopes. If  $\|\boldsymbol{\gamma}\| \rightarrow \infty$  the penalty alone sends the objective to  $-\infty$ , since  $\ell \leq 0$ ; if  $\boldsymbol{\gamma}$  stays bounded and  $|\beta_0| \rightarrow \infty$ , then some observation has  $s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta} \rightarrow -\infty$ , because an observation with  $y_i = 1$  supplies one as  $\beta_0 \rightarrow -\infty$  and one with  $y_i = 0$  as  $\beta_0 \rightarrow +\infty$ . So the objective is again coercive, and strict concavity now comes from  $\ell$  itself.

**Exercise B2.**

Take a  $2 \times 2$  table with a single binary covariate, coded as an intercept and an indicator. Compute the  $h_i$  of [Proposition 35.3.2](#), and show that Firth's pseudo-data amount to adding  $\frac{1}{2}$  to each of the four cell counts. What is the resulting estimate of the log odds ratio?

**Exercise B3.**

For a single binary covariate with fitted probabilities  $\hat{\pi}_1$  and  $\hat{\pi}_0$  in groups of sizes  $m_1$  and  $m_0$ , show that as  $\hat{\pi}_1 \rightarrow 1$ , with everything else fixed, the Wald statistic tends to zero while the likelihood ratio statistic increases to a positive limit.

**35.3.7.3 C. Going deeper****Exercise C1.**

Suppose the rows  $\mathbf{x}_{(i)}$  are independent draws from a distribution symmetric about the

origin in  $\mathbb{R}^p$  and the responses are independent coin flips. Cover's counting argument gives the probability of complete separation as  $2^{1-n} \sum_{k=0}^{p-1} \binom{n-1}{k}$ . Deduce that separation is almost certain when  $n < 2p$  and almost impossible when  $n \gg 2p$ , and relate this to [Chapter 28](#).

*Solution.* The sum is  $\Pr\{B \leq p-1\}$  with  $B \sim \text{Binomial}(n-1, \frac{1}{2})$ , so with  $E B = (n-1)/2$  it is near 1 when  $p-1$  exceeds  $(n-1)/2$ , roughly  $n < 2p$ , and near 0 when  $p$  is well below  $n/2$ , the transition having width of order  $\sqrt{n}$ . So with pure noise maximum likelihood fails to exist below about  $n = 2p$ : twice the sample size at which least squares begins to interpolate in [Chapter 28](#). Both thresholds mark the point past which an unpenalized fit reproduces the labels exactly, and both are why penalization is the default in high dimensions.

## 35.4 Goodness of fit

In the normal linear model the residual sum of squares both estimates  $\sigma^2$  and measures fit, and its known distribution makes a fit too poor to be chance announce itself ([Chapter 14](#)). Binary data have no free dispersion parameter, so the deviance has only the second role, and often cannot fill it either.

### 35.4.1 Two statistics for grouped data

#### Definition 35.4.1 — Pearson and deviance statistics

For grouped binomial data  $y_i \sim \text{Binomial}(m_i, \pi_i)$ ,  $i = 1, \dots, N$ , with fitted probabilities  $\hat{\pi}_i$  from a model with  $p$  parameters, the **Pearson statistic** and the **deviance** are

$$\begin{aligned} X^2 &= \sum_{i=1}^N \frac{(y_i - m_i \hat{\pi}_i)^2}{m_i \hat{\pi}_i (1 - \hat{\pi}_i)}, \\ D &= 2 \sum_{i=1}^N \left\{ y_i \log \frac{y_i}{m_i \hat{\pi}_i} + (m_i - y_i) \log \frac{m_i - y_i}{m_i - m_i \hat{\pi}_i} \right\}, \end{aligned} \tag{35.4.1}$$

with the convention  $0 \log 0 = 0$ . Both compare the fit with the **saturated** model that gives each group its own probability.

$X^2$  is the sum over all  $2N$  cells of successes and failures of  $(\text{observed} - \text{fitted})^2 / \text{fitted}$ , and  $D$  is twice the sum of  $\text{observed} \times \log(\text{observed}/\text{fitted})$ : the classical categorical-data statistics, specialized to the binomial.

#### Proposition 35.4.1 — What the two statistics measure

- (a) (*Grouped data.*) Fix  $N$  and the model, let  $m_i \rightarrow \infty$  for every  $i$  with  $\pi_i$  bounded away from 0 and 1, and suppose the model holds. Then  $X^2$  and  $D$  both converge

in distribution to  $\chi^2(N - p)$ , and  $X^2 - D \rightarrow 0$  in probability.

- (b) (*Ungrouped data, canonical link.*) If  $m_i = 1$  for every  $i$  and  $\hat{\beta}$  solves the logistic likelihood equations Eq. 35.1.2, then

$$D = -2 \sum_{i=1}^n \{ \hat{\pi}_i \log \hat{\pi}_i + (1 - \hat{\pi}_i) \log(1 - \hat{\pi}_i) \}. \quad (35.4.2)$$

The deviance is therefore a function of the fitted probabilities alone: it does not depend on which observations were successes, it is bounded above by  $2n \log 2$ , and it has no  $\chi^2(n - p)$  limit.

- (c) (*Differences.*) For nested models  $\omega \subset \Omega$  with  $q$  parameters between them,  $D_\omega - D_\Omega$  does not depend on the grouping and converges to  $\chi^2(q)$  under the smaller model, grouped or not.

*Proof.*

- (a) The statement about  $D$  is Theorem 34.5.5 specialized to the binomial family: with  $N$  fixed and every  $m_i \rightarrow \infty$ , the model is a smooth submodel of an  $N$ -parameter family whose maximum likelihood estimates are asymptotically normal, so the likelihood ratio statistic against the saturated model has  $N - p$  degrees of freedom. For  $X^2$ , put  $b = m_i \hat{\pi}_i$  and  $r_i = (y_i - b) / \{m_i \hat{\pi}_i (1 - \hat{\pi}_i)\}^{1/2}$ , so that  $r_i = O_p(1)$  and  $y_i - b = O_p(m_i^{1/2})$ . The  $i$ th deviance summand is  $g(y_i)$  for  $g(a) = 2\{a \log(a/b) + (m_i - a) \log((m_i - a)/(m_i - b))\}$ , and  $g(b) = g'(b) = 0$  and  $g''(b) = 2/\{m_i \hat{\pi}_i (1 - \hat{\pi}_i)\}$ , while the third derivative is  $O(m_i^{-2})$  near that point. Taylor's theorem therefore gives  $g(y_i) = r_i^2 + O_p(m_i^{-1/2})$ , using  $\hat{\pi}_i$  bounded away from 0 and 1. Summing over the  $N$  fixed groups,  $D - X^2 \rightarrow 0$  in probability, so  $X^2$  has the same limit as  $D$ .
- (b) Write  $\hat{\eta}_i = \mathbf{x}_{(i)}^\top \hat{\beta}$ , so that  $\hat{\eta}_i = \log \hat{\pi}_i - \log(1 - \hat{\pi}_i)$ . For ungrouped data the saturated model sets  $\hat{\pi}_i = y_i$  and has log-likelihood zero, so

$$\begin{aligned} D &= -2 \sum_i \{ y_i \log \hat{\pi}_i + (1 - y_i) \log(1 - \hat{\pi}_i) \} \\ &= -2 \sum_i \{ y_i \hat{\eta}_i + \log(1 - \hat{\pi}_i) \} \\ &= -2 \left\{ \hat{\beta}^\top \mathbf{X}^\top \mathbf{y} + \sum_i \log(1 - \hat{\pi}_i) \right\}. \end{aligned}$$

The likelihood equations say  $\mathbf{X}^\top \mathbf{y} = \mathbf{X}^\top \hat{\pi}$ , so  $\mathbf{X}^\top \mathbf{y}$  may be replaced by  $\mathbf{X}^\top \hat{\pi}$ , giving  $D = -2 \sum_i \{ \hat{\pi}_i \hat{\eta}_i + \log(1 - \hat{\pi}_i) \}$ , which is Eq. 35.4.2 after substituting for  $\hat{\eta}_i$ . Each summand of Eq. 35.4.2 is at most  $2 \log 2$ , attained at  $\hat{\pi}_i = 1/2$ , which gives the bound. For the last claim write  $H(\pi) = -\{\pi \log \pi + (1 - \pi) \log(1 - \pi)\}$  and  $f_n(\beta) = 2 \sum_i H(\pi_i)$  with  $\pi_i = \pi_i(\beta)$ , so that  $D = f_n(\hat{\beta})$  identically:  $D$  is a smooth deterministic function of  $\hat{\beta}$ , with no other dependence on the data. Now  $dH/d\eta = -\eta \pi(1 - \pi)$ , and  $\pi(1 - \pi) \leq e^{-|\eta|}$ , so  $|dH/d\eta| \leq |\eta| e^{-|\eta|} \leq e^{-1}$  and  $\|\nabla f_n\| \leq 2e^{-1} \sum_i \|\mathbf{x}_{(i)}\| = O(n)$  for bounded covariates. With  $\hat{\beta} - \beta = O_p(n^{-1/2})$  from Theorem 34.5.1, the mean value theorem gives  $D - f_n(\beta) = O_p(n^{1/2})$ , so

$$n^{-1} D - 2n^{-1} \sum_{i=1}^n H(\pi_i) \rightarrow 0 \text{ in probability,}$$

whereas  $n^{-1} \chi^2(n - p) \rightarrow 1$ . The two agree only by accident: the simulation of

Example (Three deviances that mean nothing) has  $n^{-1}D$  averaging  $206.8/200 = 1.03$  under one correct model and  $263.6/200 = 1.32$  under another. So  $D$  has no  $\chi^2(n-p)$  limit. The same bound leaves  $D$  fluctuating about  $f_n(\beta)$  by  $O_p(n^{1/2})$ , which is the order of the standard deviations in Example (Three deviances that mean nothing); what fails is not the spread of  $D$  but its location.

- (c) The saturated log-likelihood is common to both deviances and cancels, so the difference is twice a log-likelihood ratio between two nested smooth models with  $q$  free parameters between them, and Theorem 34.5.5 applies with  $q$  fixed.

■

### Example. (Is the party scale linear on the logit scale?)

Collapse the survey of Example (Voting intention in the 1996 American election) onto its seven party groups, of between 37 and 200 respondents each, and fit logit  $\pi = \beta_0 + \beta_1 \text{ party}$ . The deviance is 12.765 on 5 degrees of freedom ( $p = 0.0257$ ) and the Pearson statistic is 11.899 ( $p = 0.0362$ ). The Pearson residuals locate the misfit:  $-2.067$  at “independent-Democrat”, where the model predicts 0.132 against an observed 0.065, and 2.122 at “independent-Republican”, where it predicts 0.640 against 0.745. The transition across the middle of the scale is sharper than a straight line in the log odds allows, and a quadratic term does not help: the deviance falls only to 12.762, so the misfit is a step, not a curve.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
from scipy import stats

anes = sm.datasets.anes96.load_pandas().data
cells = anes.groupby("PID")["vote"].agg(["sum", "count"]).reset_index()
counts = np.column_stack([cells["sum"], cells["count"] - cells["sum"]])
Xg = sm.add_constant(cells[["PID"]])

g = sm.GLM(counts, Xg, family=sm.families.Binomial()).fit()
print(cells.to_string(index=False))
print(f"deviance {g.deviance:.3f} on {g.df_resid} df, p = {stats.chi2.sf(g.deviance,
    g.df_resid):.4f}")
print(f"Pearson {g.pearson_chi2:.3f} on {g.df_resid} df, "
    f"p = {stats.chi2.sf(g.pearson_chi2, g.df_resid):.4f}")
```

The conditions in part (a) are not decoration. If  $N$  grows with  $n$  instead, as it does whenever a continuous covariate is in the model, the approximation fails, worst of all in the extreme case  $m_i \equiv 1$  of part (b).

## 35.4.2 The ungrouped deviance measures nothing

Eq. 35.4.2 deserves to be read twice. The left-hand side looks like a discrepancy between data and fit; the right-hand side contains no data at all. It is twice the total Bernoulli entropy of the fitted probabilities, a measure of how *undecided* the model is: probabilities near  $1/2$  give a deviance near  $2n \log 2$  and extreme ones give a small deviance, whether or not the model is

correct.

### Example. (Three deviances that mean nothing)

Simulate  $n = 200$  observations with one covariate and an intercept in three ways: a correct model with a strong slope, a correct model with a weak slope, and a model that leaves a genuine quadratic effect out of the fitted predictor. All three fit two parameters, so  $n - p = 198$  and a chi-squared reference would have mean 198 and standard deviation 19.9. Over 2000 replications:

| data-generating mechanism | mean deviance | standard deviation |
|---------------------------|---------------|--------------------|
| correct, strong slope     | 206.8         | 14.3               |
| correct, weak slope       | 263.6         | 7.0                |
| quadratic omitted         | 252.2         | 8.5                |

Two *correct* models with the same  $n$  and  $p$  differ by nearly sixty in mean deviance, and the badly misspecified model has a *smaller* deviance than one of them: no reference distribution can serve all three (Figure 35.4.1).

The survey model of [Example \(Voting intention in the 1996 American election\)](#) has ungrouped deviance 528.29 on 939 degrees of freedom; dropping every covariate but education gives an obviously wrong model with deviance 1275.16 on 942. Neither number can be referred to a table.

```
# the ungrouped deviance is a function of the fitted probabilities alone
X = pd.DataFrame({"const": 1.0, "PID": anes["PID"], "age": anes["age"] / 10,
                  "educ": anes["educ"], "income": anes["income"]})
y = anes["vote"].to_numpy()
fit = sm.GLM(y, X, family=sm.families.Binomial()).fit()
p = fit.fittedvalues.to_numpy()

entropy = -2 * np.sum(p * np.log(p) + (1 - p) * np.log(1 - p))
print("deviance reported ", round(fit.deviance, 6))
print("entropy of the fit ", round(entropy, 6))
```

```
def deviance_null(n, B, seed=0):
    """Ungrouped deviances under three data-generating mechanisms: a correct model
    with a
    strong slope, a correct model with a weak slope, and a model that omits a real
    quadratic
    effect. All three fit an intercept and one slope to n observations."""
    rng = np.random.default_rng(seed)
    out = np.empty((B, 3))
    for b in range(B):
        x = rng.normal(size=n)
        Z = np.column_stack([np.ones(n), x])
        for j, eta in enumerate([0.4 + 1.5 * x, 0.4 + 0.3 * x, 0.4 + 1.5 * x - 1.8 *
                                x ** 2]):
            yb = (rng.random(n) < 1 / (1 + np.exp(-eta))).astype(float)
            out[b, j] = sm.GLM(yb, Z, family=sm.families.Binomial()).fit().deviance
    return out
```

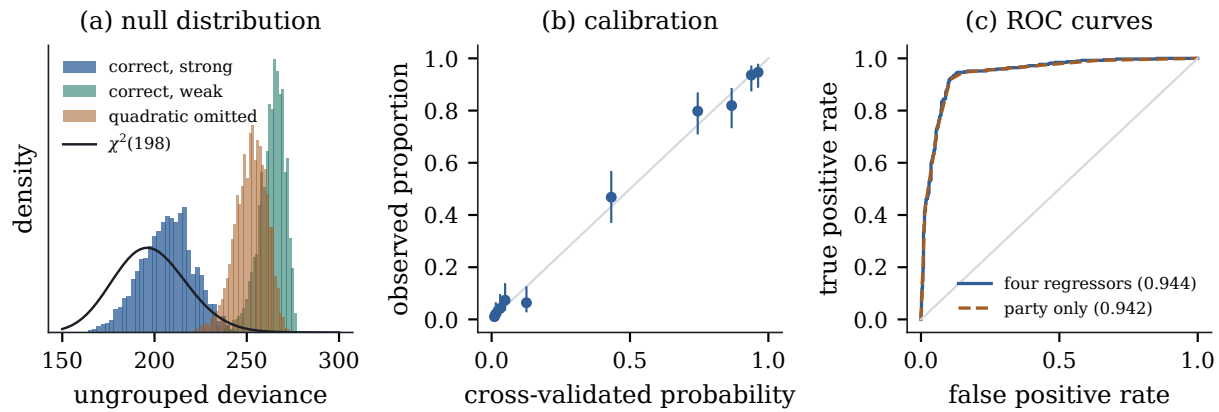


Figure 35.4.1: (a) Simulated null distributions of the ungrouped deviance with  $n = 200$  and two fitted parameters, under the three mechanisms of [Example \(Three deviances that mean nothing\)](#), against the  $\chi^2(198)$  density. (b) Cross-validated calibration of the survey model in ten bins, with pointwise intervals. (c) ROC curves for the four-regressor model and for party identification alone.

```
demo = deviance_null(200, 100, seed=1)
print("n = 200 and p = 2, so a chi-squared reference would have mean 198 and "
      f"standard deviation {np.sqrt(2 * 198):.1f}")
for j, label in enumerate(["correct, strong slope", "correct, weak slope ",
                           "quadratic omitted    "]):
    print(f"{label}: mean {demo[:, j].mean():6.1f}, "
          f"standard deviation {demo[:, j].std():4.1f}")
```

The Pearson statistic escapes [Eq. 35.4.2](#), since  $X^2 = \sum_i \{y_i/\hat{\pi}_i + (1 - y_i)/(1 - \hat{\pi}_i)\} - n$  does depend on  $\mathbf{y}$ , but gains nothing by it ([Exercise \(B1\)](#)). What survives is part (c): for ungrouped data, test the model against enlargements of it rather than against the saturated alternative, and use AIC or cross-validation ([Chapter 29](#)) when the comparisons are not nested.

### 35.4.3 Grouping by fitted values

Hosmer and Lemeshow (1980) proposed making grouped data out of ungrouped: sort by  $\hat{\pi}_i$ , cut into  $G$  groups of roughly equal size, and compare

$$\hat{C} = \sum_{k=1}^G \frac{(O_k - E_k)^2}{E_k(1 - E_k/n_k)}, \quad O_k = \sum_{i \in k} y_i, \quad E_k = \sum_{i \in k} \hat{\pi}_i,$$

referred to  $\chi^2(G - 2)$ . The reference distribution has no proof behind it: it was established by simulation, and the two lost degrees of freedom are an empirical finding. The statistic is a useful descriptive check and an unstable test.

#### Example. (The same model, three verdicts)

For the four-regressor model of [Example \(Voting intention in the 1996 American election\)](#):

| groups $G$ | $\hat{C}$ | $p$   |
|------------|-----------|-------|
| 8          | 8.670     | 0.193 |
| 10         | 12.232    | 0.141 |
| 12         | 16.119    | 0.096 |

The  $p$ -value halves as  $G$  goes from 8 to 12, and different software makes different choices about  $G$ , about how to break ties in  $\hat{\pi}_i$ , and about whether to use equal-sized groups or equally spaced cut-points. None of these choices is more principled than any other.

```
def hosmer_lemeshow(y, p, groups):
    """The Hosmer-Lemeshow statistic and its nominal p-value with the given number
    of groups."""
    order = np.argsort(p)
    chunks = np.array_split(order, groups)
    stat = 0.0
    for c in chunks:
        e = p[c].sum()
        stat += (y[c].sum() - e) ** 2 / (e * (1 - e / len(c)))
    return stat, stats.chi2.sf(stat, groups - 2)

for gr in (8, 10, 12):
    stat, pv = hosmer_lemeshow(y, p, gr)
    print(f"{gr:2d} groups: statistic {stat:6.3f}, p = {pv:.3f}")
```

### 35.4.4 Calibration and discrimination

Two questions hide inside “does the model fit”. **Calibration** asks whether the fitted probabilities are right: among observations with  $\hat{\pi}_i \approx 0.3$ , do about thirty per cent have  $y_i = 1$ ? **Discrimination** asks whether they order the observations correctly, and a model can have either without the other. Classification tables of sensitivity against specificity answer a third question, and depend on a threshold and on a loss they rarely state; pseudo- $R^2$  measures such as  $1 - D/D_{\text{null}}$  have no interpretation of their own, and by [Proposition 35.4.1](#)(b) the ungrouped version of that ratio is a ratio of entropies.

#### Proposition 35.4.2 — In-sample calibration is automatic

Let  $\hat{\beta}$  solve [Eq. 35.1.2](#) with  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ , let  $\hat{\eta}_i = \mathbf{x}_{(i)}^\top \hat{\beta}$ , and suppose  $\hat{\eta}$  is not constant. Fit the two-parameter logistic model  $\text{logit } \pi_i = a + b\hat{\eta}_i$  to the same responses. Then  $\hat{a} = 0$  and  $\hat{b} = 1$  exactly.

*Proof.* The likelihood equations of the two-parameter model at  $(a, b) = (0, 1)$  are  $\sum_i (y_i - \hat{\pi}_i) = 0$  and  $\sum_i \hat{\eta}_i (y_i - \hat{\pi}_i) = 0$ . The first holds because  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  and  $\mathbf{X}^\top (\mathbf{y} - \hat{\boldsymbol{\pi}}) = \mathbf{0}$ ; the second because  $\hat{\boldsymbol{\eta}} = \mathbf{X}\hat{\boldsymbol{\beta}} \in \mathcal{C}(\mathbf{X})$ , so  $\hat{\boldsymbol{\eta}}^\top (\mathbf{y} - \hat{\boldsymbol{\pi}}) = \hat{\boldsymbol{\beta}}^\top \mathbf{X}^\top (\mathbf{y} - \hat{\boldsymbol{\pi}}) = 0$ . So  $(0, 1)$  solves the two-parameter likelihood equations; it remains to see that a maximizer exists. The design  $[\mathbf{1}, \hat{\boldsymbol{\eta}}]$  has rank two, since  $\hat{\eta}$  is not constant. Suppose some  $(a, b) \neq (0, 1)$  had  $s_i(a + b\hat{\eta}_i) \geq 0$  for every  $i$ . As  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$  there is an  $\mathbf{e}$  with  $\mathbf{X}\mathbf{e} = \mathbf{1}$ , and  $\mathbf{b}^* = a\mathbf{e} + b\hat{\boldsymbol{\beta}}$  satisfies  $\mathbf{X}\mathbf{b}^* = a\mathbf{1} + b\hat{\boldsymbol{\eta}} \neq \mathbf{0}$ , so  $\mathbf{b}^* \neq \mathbf{0}$  separates  $\mathbf{X}$  — contradicting the overlap that the



finiteness of  $\hat{\beta}$  guarantees (Theorem 35.3.1). The two-parameter data are therefore in overlap, and Theorem 35.3.1(c) gives a unique maximizer, which must be the stationary point  $(0, 1)$ . ■

The **calibration slope**  $\hat{b}$  is therefore worthless on the data that produced  $\hat{\beta}$ . It becomes informative when  $\hat{\eta}$  comes from a held-out fold or a new sample, where  $\hat{b} < 1$  is the signature of overfitting. On the survey model of Example (Voting intention in the 1996 American election), ten-fold cross-validated predictions give a slope of 0.983 and an intercept of 0.005: four regressors and 944 observations is not a regime in which overfitting bites.

### Proposition 35.4.3 — What the area under the ROC curve is, and is not

Let  $\hat{s}_i$  be any scores, let  $I$  be a uniformly chosen index with  $y_I = 1$  and  $J$  an independent uniformly chosen index with  $y_J = 0$ . The area under the curve traced by (false positive rate, true positive rate) as a threshold moves through the scores equals

$$\text{AUC} = \Pr(\hat{s}_I > \hat{s}_J) + \frac{1}{2} \Pr(\hat{s}_I = \hat{s}_J),$$

the normalized Mann–Whitney statistic. Consequently the AUC depends on the scores only through their ranks, so it is unchanged by every strictly increasing transformation of  $\hat{s}$ , and in particular it says nothing whatever about calibration.

*Proof.* Order the distinct score values downwards. Moving the threshold past one of them advances the curve by the fraction  $\Delta\text{FPR}$  of controls and the fraction  $\Delta\text{TPR}$  of cases at that value, adding the trapezoid  $\Delta\text{FPR}\{\text{TPR} + \frac{1}{2}\Delta\text{TPR}\}$ . Summing, the first part counts pairs with  $\hat{s}_I > \hat{s}_J$  and the second counts ties with weight  $\frac{1}{2}$ , each divided by  $n_1 n_0$ . Invariance is immediate, since an increasing transformation preserves every comparison  $\hat{s}_I > \hat{s}_J$ . ■

So a model whose probabilities are all wrong by a factor of ten has the same AUC as the correctly calibrated model with the same ordering. The AUC is insensitive in the other direction too: for Example (Voting intention in the 1996 American election) party identification alone gives  $\text{AUC} = 0.942$ , and adding age, education and income raises it to 0.944. An increment of 0.002 in a rank-based summary is the wrong scale on which to judge the three covariates.

## 35.4.5 Residuals and influence

The diagnostics of Chapter 20 carry over, with the working weights as the metric.

### Definition 35.4.2 — Residuals for binomial data

With  $\hat{w}_i = m_i \hat{\pi}_i (1 - \hat{\pi}_i)$ ,  $\hat{\mathbf{W}} = \text{diag}(\hat{w}_i)$  and  $\hat{h}_i$  the  $i$ th diagonal entry of  $\hat{\mathbf{W}}^{1/2} \mathbf{X} (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \mathbf{X}^\top \hat{\mathbf{W}}^{1/2}$ ,

the **Pearson** and **deviance** residuals are

$$r_i = \frac{y_i - m_i \hat{\pi}_i}{\sqrt{\hat{w}_i}}, \quad d_i = \text{sign}(y_i - m_i \hat{\pi}_i) \sqrt{D_i},$$

where  $D_i$  is the  $i$ th summand of the deviance in [Eq. 35.4.1](#). Their **standardized** forms divide by  $(1 - \hat{h}_i)^{1/2}$ .

By construction  $\sum_i r_i^2 = X^2$  and  $\sum_i d_i^2 = D$ . The matrix in the definition is the hat matrix of the last weighted least squares step of [Proposition 35.1.3](#), so it is symmetric and idempotent with trace  $\text{rank}(\mathbf{X})$ , and [Section 6.8](#) applies to it unchanged. The analogue of Cook's distance ([Definition 20.4.1](#)) is  $r_i^2 \hat{h}_i / \{p(1 - \hat{h}_i)^2\}$ .

Two things change. For ungrouped data  $r_i$  takes only two values once  $\hat{\pi}_i$  is given, so plots against a covariate show two curves rather than a cloud and must be smoothed or binned. And  $\hat{h}_i$  carries the factor  $\hat{w}_i$ , so high leverage now means a covariate vector that is extreme *and* a fitted probability near 1/2; a point whose outcome the model regards as certain has low leverage wherever its covariates lie, the opposite of the linear-model intuition.

## 35.4.6 Exercises

### 35.4.6.1 A. Check your understanding

#### Exercise A1.

Use [Eq. 35.4.2](#) to find the largest and smallest possible values of the ungrouped deviance of a logistic fit, and describe the fits that attain them.

*Solution.* Each summand  $-2\{\hat{\pi} \log \hat{\pi} + (1 - \hat{\pi}) \log(1 - \hat{\pi})\}$  lies in  $[0, 2 \log 2]$ , so  $0 \leq D \leq 2n \log 2$ . The upper bound is attained when every  $\hat{\pi}_i = 1/2$ , as for the intercept-only model fitted to equal numbers of successes and failures. The lower bound is approached only as the fitted probabilities go to 0 and 1, which by [Theorem 35.3.1](#) needs complete separation, so 0 is an infimum and not a minimum.

### 35.4.6.2 B. Practice

#### Exercise B1.

Show that for ungrouped data  $X^2 = \sum_i \{y_i / \hat{\pi}_i + (1 - y_i) / (1 - \hat{\pi}_i)\} - n$ , and deduce that for the intercept-only model  $X^2 = n$  exactly, whatever the data. What does that say about  $X^2$  as a test of fit?

*Solution.* With  $m_i = 1$ ,  $(y_i - \hat{\pi}_i)^2 / \{\hat{\pi}_i(1 - \hat{\pi}_i)\}$  equals  $(1 - \hat{\pi}_i) / \hat{\pi}_i$  when  $y_i = 1$  and  $\hat{\pi}_i / (1 - \hat{\pi}_i)$  when  $y_i = 0$ , and both equal  $y_i / \hat{\pi}_i + (1 - y_i) / (1 - \hat{\pi}_i) - 1$ ; summing gives the identity. For the intercept-only model  $\hat{\pi}_i = \bar{y}$  for every  $i$ , so  $X^2 = n\bar{y}/\bar{y} + n(1 - \bar{y})/(1 - \bar{y}) - n = n$ . A statistic that equals the sample size for every data set cannot test anything.

#### Exercise B2.

Let  $\hat{\eta}$  be fitted on one sample and evaluated on an independent sample from the same population. Show that the calibration slope  $b$  has probability limit 1 if the model is correct and  $\hat{\beta}$  is consistent, and explain why a value below one indicates overfitting.

### Exercise B3.

Suppose the covariate is  $N(\mu_1, 1)$  among cases and  $N(\mu_0, 1)$  among controls. Compute the AUC of the correctly specified logistic model in terms of  $\Delta = \mu_1 - \mu_0$ , and find the  $\Delta$  needed for an AUC of 0.9.

*Solution.* The fitted score is monotone in the covariate, so the AUC is  $\Pr(X_1 > X_0) = \Phi(\Delta/\sqrt{2})$ , since  $X_1 - X_0 \sim N(\Delta, 2)$ . Setting this to 0.9 gives  $\Delta = \sqrt{2} \Phi^{-1}(0.9) = 1.812$ . An AUC of 0.65 needs only  $\Delta \approx 0.55$ , a difference highly significant in a large sample yet nearly useless for classifying an individual.

## 35.5 Case-control sampling

A rare disease affects one person in a thousand, so a prospective study hoping to see a hundred cases must follow a hundred thousand people, most of whom carry almost none of the weight  $\pi_i(1 - \pi_i)$ . The economy is to go looking for the cases. In a **retrospective**, or **case-control**, design the response decides who enters the sample and the covariates are observed afterwards. This inverts the conditioning, and one might expect nothing to be estimable. In fact, for the logit link and no other, almost everything is.

### 35.5.1 Only the intercept moves

#### Theorem 35.5.1 — Retrospective sampling under a logistic model

Let  $(\mathbf{X}, Y)$  follow  $\text{logit } \Pr(Y = 1 \mid \mathbf{X} = \mathbf{x}) = \alpha + \mathbf{x}^\top \boldsymbol{\beta}$  in the population, and let  $S \in \{0, 1\}$  indicate inclusion in the sample, with

$$\Pr(S = 1 \mid Y = y, \mathbf{X} = \mathbf{x}) = \rho_y \in (0, 1], \quad y = 0, 1,$$

depending on  $y$  but not on  $\mathbf{x}$ . Then:

- $\text{logit } \Pr(Y = 1 \mid \mathbf{X} = \mathbf{x}, S = 1) = \alpha + \log(\rho_1/\rho_0) + \mathbf{x}^\top \boldsymbol{\beta}$ . The slopes are unchanged; only the intercept moves, and it moves by a constant that does not involve  $\mathbf{x}$ ;
- let  $F$  be a continuous strictly increasing distribution function, suppose  $\Pr(Y = 1 \mid \mathbf{x}) = F(\alpha + \mathbf{x}^\top \boldsymbol{\beta})$ , and suppose that for every ratio  $r = \rho_1/\rho_0$  in some open interval containing 1 there is a constant  $c(r)$  with  $\Pr(Y = 1 \mid \mathbf{x}, S = 1) = F\{\alpha + c(r) + \mathbf{x}^\top \boldsymbol{\beta}\}$  for all  $\mathbf{x}$ . If the linear predictor  $\alpha + \mathbf{x}^\top \boldsymbol{\beta}$  ranges over an interval  $J$  with nonempty interior as  $\mathbf{x}$  varies, then  $F(u) = \{1 + e^{-(au+b)}\}^{-1}$  for every  $u \in J$  and some  $a > 0$ ,  $b$ : wherever the population identifies it, the link is logistic;
- if  $\rho_1$  and  $\rho_0$  are known, or if the population prevalence  $\bar{\pi} = \Pr(Y = 1)$  is known and the sample contains  $n_1$  cases and  $n_0$  controls, then  $\alpha$  itself is estimable: in the second case  $\hat{\alpha} = \hat{\alpha}_S - \log\{(n_1/n_0)(1 - \bar{\pi})/\bar{\pi}\}$ , where  $\hat{\alpha}_S$  is the intercept fitted

to the sample.

*Proof.*

- (a) By Bayes' theorem, writing  $\pi = \Pr(Y = 1 \mid \mathbf{x})$ ,

$$\Pr(Y = 1 \mid \mathbf{x}, S = 1) = \frac{\rho_1 \pi}{\rho_1 \pi + \rho_0 (1 - \pi)},$$

so that

$$\frac{\Pr(Y = 1 \mid \mathbf{x}, S = 1)}{\Pr(Y = 0 \mid \mathbf{x}, S = 1)} = \frac{\rho_1}{\rho_0} \cdot \frac{\pi}{1 - \pi}.$$

Taking logarithms and using  $\text{logit } \pi = \alpha + \mathbf{x}^\top \boldsymbol{\beta}$  gives the claim. The step needs  $\rho_y$  to be free of  $\mathbf{x}$ , and nothing else.

- (b) Put  $G = \text{logit} \circ F$ , a continuous strictly increasing function, and write  $\eta = \alpha + \mathbf{x}^\top \boldsymbol{\beta}$ . The computation in (a) used only Bayes' theorem, so it holds for any  $F$ :  $\text{logit } \Pr(Y = 1 \mid \mathbf{x}, S = 1) = \log r + G(\eta)$ . The hypothesis says this equals  $G\{\eta + c(r)\}$ . Hence

$$G\{\eta + c(r)\} - G(\eta) = \log r \quad \text{for all } \eta \text{ in an interval } J.$$

Since  $G$  is strictly increasing,  $c$  is strictly increasing in  $r$  with  $c(1) = 0$ , and as  $r$  runs over an open interval containing 1,  $v = c(r)$  runs over an open interval  $V$  containing 0. The displayed identity says that  $\kappa(v) = G(\eta + v) - G(\eta)$  is the same for every  $\eta \in J$  with  $\eta + v \in J$ . Taking  $v, v'$  and  $\eta$  with  $v, v', v + v' \in V$  and  $\eta, \eta + v, \eta + v + v'$  all in  $J$ , which the nonempty interior of  $J$  permits, gives  $\kappa(v + v') = \kappa(v) + \kappa(v')$ ;  $\kappa$  is continuous, so  $\kappa(v) = av$  near 0, with  $a > 0$  by monotonicity. Fix  $\eta_0$  in the interior of  $J$ . Then  $G(\eta) = G(\eta_0) + a(\eta - \eta_0)$  for  $\eta \in J$  within  $V$  of  $\eta_0$ , and chaining such steps along the interval  $J$  extends the identity to all of  $J$ . Inverting,  $F(u) = \text{logit}^{-1}(au + b)$  on  $J$ : nothing pins  $F$  down outside  $J$ , and nothing can, since the sampling scheme says nothing about linear predictors the population never produces.

- (c) By (a) the fitted intercept estimates  $\alpha + \log(\rho_1/\rho_0)$ , so knowing the ratio recovers  $\alpha$ . If instead  $\bar{\pi}$  is known and the population has  $N$  members, then  $\rho_1 = n_1/(N\bar{\pi})$  and  $\rho_0 = n_0/\{N(1 - \bar{\pi})\}$  in expectation, so  $\rho_1/\rho_0 = (n_1/n_0)(1 - \bar{\pi})/\bar{\pi}$ . ■

Part (a) is why the odds ratio dominates epidemiology. It is not that odds ratios are more natural than risk ratios — they are less natural — but that they are the only effect measure a retrospective study can estimate without knowing how the sample was drawn. Part (b) says this belongs to the logit alone.

*Remark* (Why the prospective likelihood is legitimate). Part (a) says the *model* transfers, not that maximizing the ordinary prospective likelihood on retrospective data is a correct procedure: the actual likelihood of such a sample is  $\prod_i f(\mathbf{x}_{(i)} \mid y_i)$ , which involves the unknown population distribution of  $\mathbf{X}$ . Prentice and Pyke (1979) settled this. Treating that distribution nonparametrically and profiling it out of the retrospective likelihood yields exactly the prospective logistic likelihood, so the ordinary fit is the semiparametric maximum likelihood estimate of  $\boldsymbol{\beta}$  and the usual inverse informa-

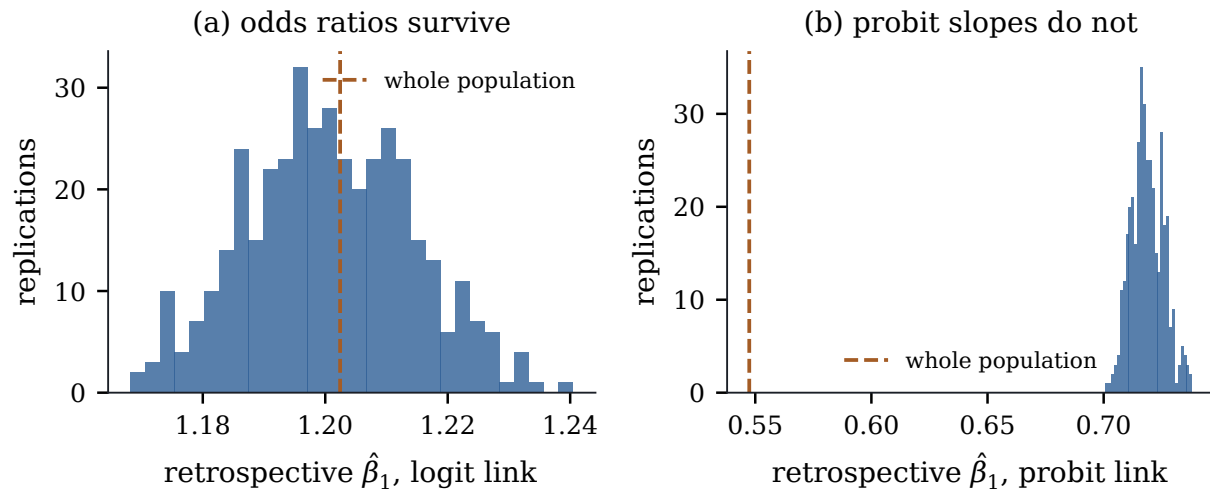


Figure 35.5.1: Retrospective estimates of a slope from 400 case-control samples, with the whole population estimate marked. (a) Logistic data under the logit link: centred on the population value. (b) Probit data under the probit link: systematically too large.

tion is its correct asymptotic covariance. Their argument profiles an infinite-dimensional nuisance parameter and is not reproduced here.

#### Example. (A simulated retrospective study)

Take a population of 200000 with a continuous exposure  $x_1 \sim N(0, 1)$ , an independent binary covariate  $x_2$ , and logit  $\pi = -5 + 1.2x_1 + 0.8x_2$ . The prevalence is 0.0178, so there are 3565 cases, and fitting the whole population gives  $\hat{\alpha} = -4.959$ ,  $\hat{\beta}_1 = 1.202$ ,  $\hat{\beta}_2 = 0.748$ .

Now take every case and four random controls per case, and fit the same model to those records as if they had been collected prospectively. Averaged over 400 control samples the slopes are recovered, at  $\hat{\beta}_1 = 1.201$  and  $\hat{\beta}_2 = 0.755$ , while the intercept averages  $-2.339$  against the predicted  $\hat{\alpha} + \log(\rho_1/\rho_0) = -4.959 + 2.623 = -2.337$ .

With data generated from a *probit* model and fitted with a probit link, the population slope is 0.547 while the retrospective samples average 0.719, too large by a factor of 1.31 (Figure 35.5.1).

```
import numpy as np
import statsmodels.api as sm

rng = np.random.default_rng(20)
N = 200_000
alpha, beta1, beta2 = -5.0, 1.2, 0.8
x1 = rng.normal(size=N) # a continuous exposure
x2 = (rng.random(N) < 0.3).astype(float) # a binary covariate
Xpop = np.column_stack([np.ones(N), x1, x2])
ypop = (rng.random(N) < 1 / (1 + np.exp(-(Xpop @ [alpha, beta1, beta2])))).astype(float)

prospective = sm.GLM(ypop, Xpop, family=sm.families.Binomial()).fit()
cases = np.flatnonzero(ypop == 1)
controls = np.flatnonzero(ypop == 0)
```

```

n1 = len(cases)

print("prevalence", ypop.mean().round(4), " cases", n1)
print("whole population:", prospective.params.round(3))

def case_control(rng, ratio=4, link=None):
    """Take every case and `ratio` controls per case; fit the same model
    prospectively."""
    keep = np.r_[cases, rng.choice(controls, ratio * n1, replace=False)]
    family = sm.families.Binomial(link=link) if link else sm.families.Binomial()
    return sm.GLM(ypop[keep], Xpop[keep], family=family).fit().params

rng2 = np.random.default_rng(7)
draws = np.array([case_control(rng2) for _ in range(400)])
shift = np.log(1.0 / (4 * n1 / len(controls))) # log(rho_1 / rho_0)

print("mean case-control estimate:", draws.mean(0).round(3))
print("predicted intercept:", round(prospective.params[0] + shift, 3))
print("slopes in the population:", prospective.params[1:].round(3))

```

## 35.5.2 Matched designs

Controls are often matched to cases on variables that matter but are not of interest: age, sex, neighbourhood. A matched set is one case and  $M$  controls sharing those variables, and the model gives each set its own intercept:

$$\text{logit } \Pr(Y_{kj} = 1 \mid \mathbf{x}_{kj}) = \alpha_k + \mathbf{x}_{kj}^\top \boldsymbol{\beta}, \quad k = 1, \dots, K, \quad j = 0, 1, \dots, M.$$

Estimating the  $K$  intercepts along with  $\boldsymbol{\beta}$  is the incidental-parameter situation of [Example \(Many means, one variance\)](#), in which maximum likelihood is not consistent. Conditioning removes them.

### Proposition 35.5.2 — Conditional logistic regression

In the model above, suppose exactly one member of each set is a case, and condition on that fact. Then:

- (a) the conditional probability that member  $j^*$  of set  $k$  is the case, given the covariates of the set and that exactly one member is, is

$$\frac{\exp(\mathbf{x}_{kj^*}^\top \boldsymbol{\beta})}{\sum_{j=0}^M \exp(\mathbf{x}_{kj}^\top \boldsymbol{\beta})}, \quad (35.5.1)$$

free of  $\alpha_k$ ;

- (b) the product of these over  $k$  is the **conditional likelihood**; maximizing it gives a consistent, asymptotically normal estimate of  $\boldsymbol{\beta}$  as  $K \rightarrow \infty$  with  $M$  fixed, and the ordinary inverse-information standard errors apply to it;
- (c) for  $M = 1$  (matched pairs), let  $\mathbf{d}_k = \mathbf{x}_{k1} - \mathbf{x}_{k0}$  be the case-minus-control difference in a discordant pair. The conditional log-likelihood is  $-\sum_k \log\{1 + \exp(-\boldsymbol{\beta}^\top \mathbf{d}_k)\}$ , while the profile log-likelihood obtained by maximizing the *unconditional* likelihood over the  $\alpha_k$  is  $-2 \sum_k \log\{1 + \exp(-\frac{1}{2} \boldsymbol{\beta}^\top \mathbf{d}_k)\}$ . Consequently the unconditional

estimate is exactly twice the conditional one, and converges to  $2\beta$ .

*Proof.*

- (a) With  $\pi_{kj} = \text{logit}^{-1}(\alpha_k + \mathbf{x}_{kj}^\top \boldsymbol{\beta})$ , the probability that  $j^*$  alone is a case is  $\pi_{kj^*} \prod_{j \neq j^*} (1 - \pi_{kj})$ . Dividing by the sum of the same expression over all choices, and cancelling  $\prod_j (1 - \pi_{kj})$  from numerator and denominator, leaves  $\{\pi_{kj^*} / (1 - \pi_{kj^*})\} / \sum_j \{\pi_{kj} / (1 - \pi_{kj})\}$ . Each odds is  $e^{\alpha_k} e^{\mathbf{x}_{kj}^\top \boldsymbol{\beta}}$ , and  $e^{\alpha_k}$  cancels, giving [Eq. 35.5.1](#).
- (b) The sets are independent and [Eq. 35.5.1](#) is a genuine probability model for a fixed parameter  $\boldsymbol{\beta}$  of fixed dimension, so the standard likelihood asymptotics apply as  $K \rightarrow \infty$ . (It is the likelihood of a multinomial logit model, which [Chapter 36](#) studies in its own right.)
- (c) Part (a) with  $M = 1$  gives  $\{1 + \exp(-\boldsymbol{\beta}^\top \mathbf{d}_k)\}^{-1}$ , whose logarithm summed over  $k$  is the stated conditional log-likelihood. For the unconditional calculation, write  $u = \alpha_k + \frac{1}{2} \boldsymbol{\beta}^\top (\mathbf{x}_{k1} + \mathbf{x}_{k0})$  and  $c = \frac{1}{2} \boldsymbol{\beta}^\top \mathbf{d}_k$ , so the two linear predictors are  $u + c$  and  $u - c$ . The pair's log-likelihood is  $(u + c) - \log(1 + e^{u+c}) - \log(1 + e^{u-c})$ ; differentiating in  $u$  gives  $1 - \text{logit}^{-1}(u + c) - \text{logit}^{-1}(u - c) = 0$ , that is,  $\text{logit}^{-1}(u + c) = \text{logit}^{-1}(c - u)$ , whose unique solution is  $u = 0$ . Substituting  $u = 0$ ,

$$c - \log(1 + e^c) - \log(1 + e^{-c}) = -2 \log(1 + e^{-c}),$$

which is the stated profile contribution with  $c = \frac{1}{2} \boldsymbol{\beta}^\top \mathbf{d}_k$ . A concordant pair contributes a supremum of 0, approached as  $\alpha_k \rightarrow \pm\infty$  and free of  $\boldsymbol{\beta}$ . The profile log-likelihood is therefore twice the conditional log-likelihood evaluated at  $\boldsymbol{\beta}/2$ , so its maximizer is twice the conditional maximizer; since the latter is consistent for  $\boldsymbol{\beta}$ , the former converges to  $2\boldsymbol{\beta}$ . ■

### Example. (Doubling the effect)

Simulate 3000 matched pairs with one covariate, stratum intercepts drawn from a normal distribution and  $\beta = 1$ , keeping only the discordant pairs. The conditional estimate is 0.942 with standard error 0.037. Profiling the unconditional likelihood numerically over the three thousand intercepts gives 1.884, twice that to three decimals as [Proposition 35.5.2\(c\)](#) requires, and about twenty-four standard errors from the truth.

```
from scipy.optimize import minimize

def matched_pairs(K=3000, beta=1.0, seed=11):
    """K strata, each with one case and one control, and a stratum-specific
    intercept."""
    rng = np.random.default_rng(seed)
    a = rng.normal(-1.0, 2.0, K)                # the nuisance intercepts
    xc, xk = np.empty(K), np.empty(K)
    for k in range(K):
        while True:                             # sample the stratum until
            it is discordant
            x = rng.normal(size=2)
            pr = 1 / (1 + np.exp(-(a[k] + beta * x)))
```

```

yy = rng.random(2) < pr
if yy[0] != yy[1]:
    xc[k], xk[k] = (x[0], x[1]) if yy[0] else (x[1], x[0])
    break
return xc, xk                                     # exposure of the case, of
the control

x_case, x_control = matched_pairs()
# the conditional likelihood for 1:1 matching depends on beta only through the
  difference
d = x_case - x_control
cond = minimize(lambda b: np.sum(np.log1p(np.exp(-b[0] * d))), [0.0]).x[0]
print("conditional estimate:", round(cond, 3))

```

Matching buys precision when the matching variables are strong confounders, and costs all information about them, since they are absorbed into the  $\alpha_k$ . Overmatching, matching on a variable on the causal path from exposure to outcome, removes part of the effect one is measuring: a bad control ([Theorem 25.5.1](#)).

### 35.5.3 Two warnings about odds ratios

#### Example. (The odds ratio does not collapse)

Generate a population of 200000 with two *independent* covariates,  $x_1$  standard normal and  $x_2$  Bernoulli with probability 0.3, and logit  $\pi = 1.0x_1 + 4.0x_2$ , giving a prevalence of 0.638. Fitting both covariates and an intercept recovers the coefficient of  $x_1$  as 1.006. Fitting an intercept and  $x_1$  alone, and thereby averaging over  $x_2$ , gives 0.745, a quarter smaller. There is no confounding here: the two covariates are independent by construction, and the marginal and conditional odds ratios simply differ.

The odds ratio is not *collapsible*: the best logistic approximation to an average of logistic curves is flatter than they are, by an amount growing with the variance of the omitted term ([Exercise \(C1\)](#)). Conditional and marginal odds ratios therefore answer different questions, and adding a covariate moves the other coefficients away from zero even when it is independent of them, so the reasoning “the coefficient changed, therefore the new variable is a confounder” is invalid here. Risk differences and risk ratios do collapse.

The second warning is causal. [Theorem 35.5.1](#) requires selection to depend on  $Y$  alone; if recruitment also depends on the exposure, say because the cases come from a hospital serving an exposed population, then  $S$  is a collider on a path between  $\mathbf{X}$  and  $Y$  and conditioning on  $S = 1$  opens it ([Proposition 25.2.1](#)). And the conditional odds ratio it estimates is the causal effect only if the covariates satisfy the backdoor criterion ([Definition 25.4.1](#) and [Theorem 25.4.1](#)), which no sampling scheme can supply.

*Idea.* Retrospective sampling is the one design in which the choice of link is forced: [Theorem 35.5.1](#) makes the odds ratio estimable under a scheme that destroys every other effect measure. The price is that odds ratios do not collapse, do not approximate risk ratios unless the outcome is rare, and are causal only under the usual assumptions.



## 35.5.4 Exercises

### 35.5.4.1 A. Check your understanding

#### Exercise A1.

A case-control study has 200 cases and 600 controls, and the disease affects 2% of the population. The fitted intercept is  $-0.8$ . Estimate the population intercept, and the probability of disease at the reference covariate values.

*Solution.* By Theorem 35.5.1(c),  $\log\{(200/600)(0.98/0.02)\} = \log(16.333) = 2.793$ , so  $\hat{\alpha} = -0.8 - 2.793 = -3.593$  and the probability is  $1/(1 + e^{3.593}) = 0.0268$ , close to the population prevalence as it should be.

### 35.5.4.2 B. Practice

#### Exercise B1.

Suppose  $\Pr(S = 1 \mid Y = y, \mathbf{X} = \mathbf{x}) = \rho_y g(\mathbf{x})$  for some  $g$  with values in  $(0, 1]$ . Show that Theorem 35.5.1(a) still holds. Then suppose instead that cases are sampled at rate  $\rho_1 e^{\mathbf{x}^\top \boldsymbol{\gamma}}$  while controls are sampled at a rate free of  $\mathbf{x}$ ; find the bias in the estimated slopes.

#### Exercise B2.

For matched pairs with a single binary exposure, let  $n_{10}$  be the number of pairs with an exposed case and unexposed control and  $n_{01}$  the reverse. Show that the conditional estimate of the log odds ratio is  $\log(n_{10}/n_{01})$ , and that the conditional test of  $\beta = 0$  is a binomial test of  $1/2$  in  $n_{10} + n_{01}$  trials.

*Solution.* With a binary exposure  $d_k \in \{-1, 0, 1\}$ , and pairs with  $d_k = 0$  contribute a constant. The conditional log-likelihood is  $-n_{10} \log(1 + e^{-\beta}) - n_{01} \log(1 + e^{\beta})$ ; setting the derivative to zero gives  $e^{\beta} = n_{10}/n_{01}$ . Writing  $p = e^{\beta}/(1 + e^{\beta})$ , the conditional likelihood is that of  $n_{10}$  successes in  $n_{10} + n_{01}$  Bernoulli trials, and  $\beta = 0$  is  $p = 1/2$ . The resulting test is McNemar's, its exact version the binomial test.

#### Exercise B3.

With  $n_1$  cases fixed and  $n_0 = Mn_1$  controls, show that for a single binary exposure and a rare disease the asymptotic variance of the log odds ratio is proportional to  $1 + 1/M$ , and deduce that there is little gain beyond  $M = 4$  or  $5$ .

### 35.5.4.3 C. Going deeper

#### Exercise C1.

Let  $\text{logit } \Pr(Y = 1 \mid X, Z) = \beta X + \gamma Z$  with  $Z$  independent of  $X$ . Show that  $\Pr(Y = 1 \mid X)$  is not of the form  $\text{logit}^{-1}(\beta^* X)$ , and explain why the best logistic approximation

has  $|\beta^*| < |\beta|$ . What happens as  $\gamma \rightarrow 0$ ?

*Solution.*  $\Pr(Y = 1 \mid X = x) = E_Z \text{logit}^{-1}(\beta x + \gamma Z)$  is a mixture of horizontally shifted logistic curves, and a nondegenerate such mixture is not itself logistic: its derivative at its median is strictly smaller than that of a single component with the same slope, by concavity of the response function above the median and convexity below. The marginal curve is flatter, so the best logistic fit has a smaller slope in modulus. As  $\gamma \rightarrow 0$ ,  $\beta^* \rightarrow \beta$ ; the attenuation grows with  $\gamma^2 \text{Var } Z$ , which is why the effect in [Example \(The odds ratio does not collapse\)](#) is large.

## 35.6 Summary and Notes

### 35.6.1 Summary

*Idea.*

1. A binary response is a binomial exponential dispersion family with variance function  $\pi(1 - \pi)$ , dispersion one and natural parameter the log odds; the canonical link is the logit ([Definition 35.1.1](#)).
2. A logistic coefficient is a log odds ratio, constant over the other covariates;  $|\beta_j|/4$  bounds its effect on the probability scale, and the average marginal effect converts it to a change in probability ([Proposition 35.1.1](#)).
3. Grouping observations with identical covariates changes the log-likelihood by a constant, so the estimate, its covariance and every likelihood ratio are unchanged; the deviance is not ([Proposition 35.1.2](#)).
4. Fisher scoring is iteratively reweighted least squares with weights  $m_i \pi_i(1 - \pi_i)$ ; observed and expected information coincide, and the log-likelihood is strictly concave when  $\text{rank}(\mathbf{X}) = p$  ([Proposition 35.1.3](#)).
5. Every binary regression is a threshold model: logistic, normal and smallest extreme-value latent errors give the logit, probit and complementary log-log links, and a proportional hazards process observed at a fixed time gives the last of them and no other ([Definition 35.2.1](#) and [Proposition 35.2.1](#)).
6. The latent scale is not identified, so probit and logit coefficients are comparable only after rescaling, by 1.8138 or 1.5958 depending on the convention ([Proposition 35.2.2](#)).
7. The maximum likelihood estimate exists if and only if the data are in overlap; under separation some coefficients diverge and the Wald test collapses while the likelihood ratio does not ([Definition 35.3.1](#) and [Theorem 35.3.1](#)).
8. Firth's penalty  $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{W} \mathbf{X})$  removes the leading bias and gives finite estimates; a ridge or Cauchy penalty does the same more crudely ([Proposition 35.3.2](#)).
9. Grouped Pearson and deviance statistics have  $\chi^2(N - p)$  limits as the group sizes grow. The ungrouped deviance is exactly  $-2 \sum_i \{\hat{\pi}_i \log \hat{\pi}_i + (1 - \hat{\pi}_i) \log(1 - \hat{\pi}_i)\}$ , a function of the fit alone with no  $\chi^2$  limit; differences of deviances remain valid ([Proposition 35.4.1](#)).
10. In-sample calibration is automatic, so calibration must be judged out of sample,

and the area under the ROC curve is a rank statistic, blind to calibration ([Proposition 35.4.2](#) and [Proposition 35.4.3](#)).

11. Retrospective sampling on  $Y$  alone shifts the intercept by  $\log(\rho_1/\rho_0)$  and leaves the slopes alone, and the logit is the only link with that property ([Theorem 35.5.1](#)).
12. Matched sets are handled by conditioning; fitting the stratum intercepts instead is inconsistent, and for matched pairs gives exactly twice the conditional estimate ([Proposition 35.5.2](#)).

## 35.6.2 Notes and sources

**Coverage.** The topics follow Agresti (2015, chapter 5) and Fahrmeir, Kneib, Lang and Marx (2021, sections 5.1–5.2), with proofs, examples and exercises of this book’s own. Collett (2003) and Hosmer, Lemeshow and Sturdivant (2013) are the standard applied accounts, and the source of the events-per-variable guidance alluded to in [Section 35.1](#); McCullagh and Nelder (1989, chapter 4) is the compact theoretical reference.

**Origins.** The logistic function entered statistics through bioassay: probit analysis came first, with Bliss (1934) and Finney (1971), and Berkson (1944) argued for the logistic alternative against considerable resistance. Cox (1958a) gave the modern regression treatment of binary sequences and Cox (1970) the first book on binary data. The complementary log–log link is the binary face of the proportional hazards model of Cox (1972), with the discrete-time version of [Exercise \(B2\)](#) due to Prentice and Gloeckler (1978); the link family of [Exercise \(C1\)](#) is Aranda-Ordaz’s (1981). The “divide by four” rule is from Gelman and Hill (2007).

**Separation.** Silvapulle (1981) proved existence of the estimate under a convexity condition equivalent to overlap; Albert and Anderson (1984) introduced the terminology of complete and quasi-complete separation and the linear-programming diagnostic, and Santner and Duffy (1986) corrected a gap in their argument. The proof of [Theorem 35.3.1](#) given here is the direct one: strict concavity leaves only coercivity to establish, and coercivity is exactly overlap. The counting formula of [Exercise \(C1\)](#) is Cover’s (1965). Firth (1993) introduced the Jeffreys-prior penalty as a bias-reduction device; Heinze and Schemper (2002) recognized it as a remedy for separation, and Kosmidis and Firth (2021) proved the resulting estimate always finite. The Cauchy prior is from Gelman, Jakulin, Pittau and Su (2008); exact conditional inference is due to Cox (1970) and Hirji, Mehta and Patel (1987); and the Wald pathology of [Exercise \(B3\)](#) is Hauck and Donner’s (1977).

**Goodness of fit.** That the ungrouped deviance depends on the data only through the fitted values is standard and appears in McCullagh and Nelder (1989, section 4.4); the exact identity [Eq. 35.4.2](#) and the simulation of [Example \(Three deviances that mean nothing\)](#) are this book’s way of making the point. The Hosmer–Lemeshow statistic is from Hosmer and Lemeshow (1980), its instability discussed at length in Hosmer, Lemeshow and Sturdivant (2013); a smoothed alternative is le Cessie and van Houwelingen (1991). Calibration in the sense of [Proposition 35.4.2](#) goes back to Cox (1958b). The ROC curve entered medical statistics through Hanley and McNeil (1982), and Pepe, Janes, Longton, Leisenring and Newcomb (2004) documented how insensitive its area is to genuinely informative covariates.

**Case–control studies.** The retrospective likelihood argument that justifies using the prospective fit is Prentice and Pyke (1979); Breslow and Day (1980) is the classical monograph and Breslow (1996) a readable retrospective on the design. The conditional likelihood of [Proposition 35.5.2](#) comes from the tradition of Andersen (1970), itself an instance of the incidental-parameter problem of Neyman and Scott (1948) met in [Example \(Many means, one variance\)](#).

Noncollapsibility is analysed by Greenland, Robins and Pearl (1999), who separate it carefully from confounding.

**What comes next.** Chapter 38 takes up quasi-likelihood and overdispersion, which matter for grouped binary data whose variance exceeds  $m\pi(1 - \pi)$ ; Chapter 39 treats fitting in practice and the Bayesian thread, including the priors touched on in Section 35.3; Chapter 40 joins generalized linear models to the random effects of Chapter 32; and Chapter 41 covers missing data. Responses with more than two categories are the subject of Chapter 36. The 1996 American National Election Studies survey used throughout is public domain; it illustrates method only, and nothing here should be read as a claim about the election.

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# Chapter 36

## Multinomial and Ordinal Responses

A response with two categories has one logit. A response with  $c$  categories has  $c - 1$  of them, and the modelling question is which  $c - 1$  comparisons to make. If the categories are names — a brand, a party, a mode of travel — the natural comparisons are with a fixed baseline, and the result is the *baseline-category logit model*, a generalized linear model whose response is a vector of counts. If the categories are ordered, that choice wastes information: a model ignoring the order needs  $(c - 1)p$  coefficients to say what a single slope may say. Ordinal models get back down to  $p$  by cutting the scale and giving every cut the same coefficients. Where the cut is made is the whole story. Cut *at* a level and you get the cumulative, or proportional odds, model; cut *conditionally on having got that far* and you get the sequential, or continuation-ratio, model. These answer different questions, and the data can tell them apart.

**What you need.** [Chapter 34](#) (exponential families, the likelihood equations, Fisher scoring, the deviance and the three test statistics) and [Chapter 35](#) (logistic regression, the alternative links, separation, goodness of fit). From earlier parts: [Chapter 11](#) for the likelihood ratio test [Theorem 11.4.1](#) and the three flavours of test statistic [Proposition 11.4.2](#), [Chapter 21](#) for weighted least squares and the sandwich estimator ([Theorem 21.4.2](#)), and [Chapter 29](#) for the information criteria ([Definition 29.2.2](#)). Nothing here needs normality of anything.

### 36.0.1 Roadmap

- [Baseline-category logits](#)
- [Cumulative logit models](#)
- [Proportional odds and its checks](#)
- [Sequential models](#)
- [Summary and notes](#)

## 36.1 Baseline-category logits

Let the response of observation  $i$  fall in one of  $c$  categories, recorded as a vector of counts  $\mathbf{Y}_i = (Y_{i1}, \dots, Y_{ic})^\top$  with  $\sum_r Y_{ir} = m_i$ . When each row is a single subject,  $m_i = 1$  and  $\mathbf{Y}_i$  is an indicator vector; when rows are covariate patterns,  $m_i$  counts the subjects sharing that pattern. As with grouped and ungrouped binary data in [Chapter 35](#), the two cases are the same model

with different bookkeeping, and the distinction matters only for the deviance as a goodness-of-fit statistic.

### 36.1.1 The multinomial distribution is an exponential family

Chapter 34 was built on a one-dimensional exponential dispersion family (Definition 34.1.1). A categorical response is a *vector* with  $c - 1$  free coordinates, and the structure survives the move, with the natural parameter and the cumulant function becoming vector-valued.

#### Lemma 36.1.1 — The multinomial as an exponential family

Let  $\mathbf{Y} \sim \text{Mult}(m, \boldsymbol{\pi})$  with  $\pi_r > 0$  for every  $r$  and  $\sum_{r=1}^c \pi_r = 1$ . Write  $\tilde{\mathbf{Y}} = (Y_1, \dots, Y_{c-1})^\top$  for the first  $c - 1$  counts and

$$\eta_r = \log \frac{\pi_r}{\pi_c}, \quad r = 1, \dots, c - 1.$$

Then the probability mass function is

$$f(\mathbf{y}) = h(\mathbf{y}) \exp\{\tilde{\mathbf{y}}^\top \boldsymbol{\eta} - m b(\boldsymbol{\eta})\}, \quad b(\boldsymbol{\eta}) = \log\left(1 + \sum_{s=1}^{c-1} e^{\eta_s}\right), \quad (36.1.1)$$

with  $h(\mathbf{y}) = m!/(y_1! \cdots y_c!)$ . The map  $\boldsymbol{\eta} \mapsto \boldsymbol{\pi}$  is a bijection from  $\mathbb{R}^{c-1}$  onto the interior of the probability simplex, with inverse

$$\pi_r = \frac{e^{\eta_r}}{1 + \sum_{s < c} e^{\eta_s}} \quad (r < c), \quad \pi_c = \frac{1}{1 + \sum_{s < c} e^{\eta_s}}, \quad (36.1.2)$$

and the first two moments are

$$\mathbb{E}(\tilde{\mathbf{Y}}) = m \nabla b(\boldsymbol{\eta}) = m \tilde{\boldsymbol{\pi}}, \quad \text{Cov}(\tilde{\mathbf{Y}}) = m \nabla^2 b(\boldsymbol{\eta}) = m(\text{diag}(\tilde{\boldsymbol{\pi}}) - \tilde{\boldsymbol{\pi}} \tilde{\boldsymbol{\pi}}^\top), \quad (36.1.3)$$

where  $\tilde{\boldsymbol{\pi}} = (\pi_1, \dots, \pi_{c-1})^\top$ .

*Proof.* Since  $\sum_r y_r = m$ ,

$$\prod_{r=1}^c \pi_r^{y_r} = \pi_c^m \prod_{r < c} \left(\frac{\pi_r}{\pi_c}\right)^{y_r} = \exp\left\{\sum_{r < c} y_r \eta_r + m \log \pi_c\right\},$$

and  $\log \pi_c = -b(\boldsymbol{\eta})$  by the definition of  $\eta_r$ , which gives Eq. 36.1.1. Solving  $\eta_r = \log \pi_r - \log \pi_c$  with  $\sum_r \pi_r = 1$  gives Eq. 36.1.2, and every  $\boldsymbol{\pi}$  with positive entries arises from exactly one  $\boldsymbol{\eta}$ , so the map is a bijection.

For the moments, sum [Eq. 36.1.1](#) over the support. By the multinomial theorem,

$$M(\boldsymbol{\eta}) = \sum_{\mathbf{y}} h(\mathbf{y}) \exp(\tilde{\mathbf{y}}^\top \boldsymbol{\eta}) = \left(1 + \sum_{s < c} e^{\eta_s}\right)^m = e^{mb(\boldsymbol{\eta})},$$

a finite sum, which may therefore be differentiated term by term without any regularity condition. Since  $\log M = mb$  is the cumulant generating function of  $\tilde{\mathbf{Y}}$  shifted to  $\boldsymbol{\eta}$ , its first two derivatives are  $E(\tilde{\mathbf{Y}}) = m\nabla b$  and  $\text{Cov}(\tilde{\mathbf{Y}}) = m\nabla^2 b$ . Finally

$$\frac{\partial b}{\partial \eta_r} = \frac{e^{\eta_r}}{1 + \sum_{s < c} e^{\eta_s}} = \pi_r, \quad \frac{\partial^2 b}{\partial \eta_r \partial \eta_s} = \pi_r(\delta_{rs} - \pi_s),$$

where  $\delta_{rs}$  is one if  $r = s$  and zero otherwise. ■

*Remark* (What is new and what is not). [Eq. 36.1.1](#) has the shape of [Definition 34.1.1](#) with the scalar natural parameter replaced by a vector and the dispersion  $\phi/w_i$  by  $1/m$ : the number of trials plays the role of a prior weight. The identity “mean is the gradient of the cumulant function, variance its second derivative” is [Proposition 34.1.2](#), proved there by the same argument. New is only that  $\nabla^2 b$  is a matrix, and a singular one once the baseline coordinate is put back:  $\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top$  kills  $\mathbf{1}$ , because the  $c$  counts sum to the fixed total  $m$ .

## 36.1.2 The model

### Definition 36.1.1 — Baseline-category logit model

Let  $\mathbf{Y}_1, \dots, \mathbf{Y}_n$  be independent with  $\mathbf{Y}_i \sim \text{Mult}(m_i, \boldsymbol{\pi}_i)$ , and let  $\mathbf{x}_{(i)} \in \mathbb{R}^p$  be the  $i$ th row of a model matrix  $\mathbf{X}$  whose first column is  $\mathbf{1}$ . The **baseline-category logit model** with baseline  $c$  is

$$\log \frac{\pi_{ir}}{\pi_{ic}} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r, \quad r = 1, \dots, c-1, \quad (36.1.4)$$

with unknown  $\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_{c-1} \in \mathbb{R}^p$ . Equivalently, writing  $\boldsymbol{\beta}_c = \mathbf{0}$ ,  $\pi_{ir} = \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r) / \sum_{s=1}^c \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}_s)$  for every  $r$ .

This is a generalized linear model in the sense of [Definition 34.2.1](#), with the multinomial random component of [Lemma 36.1.1](#) and the *canonical* link, since [Eq. 36.1.4](#) sets  $\eta_{ir}$  itself equal to a linear predictor. The one departure from [Chapter 34](#) is that each observation carries a vector of natural parameters, so the model’s coefficient vector is the stack  $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^\top, \dots, \boldsymbol{\beta}_{c-1}^\top)^\top$  of length  $(c-1)p$ . Every regressor buys  $c-1$  coefficients: the price of treating the categories as unordered names.

## 36.1.3 The likelihood equations



**Theorem 36.1.2 — Score, information and concavity**

Under [Definition 36.1.1](#) the log-likelihood, as a function of  $\boldsymbol{\beta} = (\boldsymbol{\beta}_1^\top, \dots, \boldsymbol{\beta}_{c-1}^\top)^\top$ , is

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \left\{ \sum_{r < c} y_{ir} \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r - m_i b(\boldsymbol{\eta}_i) \right\}, \quad \eta_{ir} = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r.$$

Then:

- (a) the score has blocks  $\partial \ell / \partial \boldsymbol{\beta}_r = \sum_i (y_{ir} - m_i \pi_{ir}) \mathbf{x}_{(i)} = \mathbf{X}^\top (\mathbf{y}_r - \boldsymbol{\mu}_r)$ , where  $\boldsymbol{\mu}_r$  is the vector with entries  $m_i \pi_{ir}$ ;
- (b) the observed and expected information coincide and have blocks  $\mathbf{X}^\top \mathbf{W}_{rs} \mathbf{X}$  with  $\mathbf{W}_{rs} = \text{diag}\{m_i \pi_{ir} (\delta_{rs} - \pi_{is})\}_{i=1}^n$ ;
- (c)  $\ell$  is concave, and strictly concave when  $\text{rank}(\mathbf{X}) = p$ ; in that case it has at most one maximizer, and the maximizer, if it exists, is the unique solution of the likelihood equations  $\mathbf{X}^\top (\mathbf{y}_r - \boldsymbol{\mu}_r) = \mathbf{0}$  for  $r = 1, \dots, c-1$ .

*Proof.*

- (a) Differentiate:  $\partial b(\boldsymbol{\eta}_i) / \partial \eta_{ir} = \pi_{ir}$  by [Lemma 36.1.1](#), and  $\partial \eta_{ir} / \partial \boldsymbol{\beta}_r = \mathbf{x}_{(i)}$ .
- (b) Differentiating again, and using  $\partial^2 b / \partial \eta_{ir} \partial \eta_{is} = \pi_{ir} (\delta_{rs} - \pi_{is})$ ,

$$-\frac{\partial^2 \ell}{\partial \boldsymbol{\beta}_r \partial \boldsymbol{\beta}_s^\top} = \sum_i m_i \pi_{ir} (\delta_{rs} - \pi_{is}) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top.$$

The data enter only through the score, so the second derivative is nonrandom and observed information equals expected information. This is the canonical-link simplification of [Theorem 34.3.1](#).

- (c) Take  $\mathbf{a} = (\mathbf{a}_1^\top, \dots, \mathbf{a}_{c-1}^\top)^\top$  and put  $t_{ir} = \mathbf{x}_{(i)}^\top \mathbf{a}_r$  for  $r < c$  and  $t_{ic} = 0$ . By (b),

$$\mathbf{a}^\top (-\nabla^2 \ell) \mathbf{a} = \sum_i m_i \left\{ \sum_{r=1}^c \pi_{ir} t_{ir}^2 - \left( \sum_{r=1}^c \pi_{ir} t_{ir} \right)^2 \right\},$$

the inner brace being the variance of the random variable that takes the value  $t_{ir}$  with probability  $\pi_{ir}$ . A variance is nonnegative, so  $\ell$  is concave. It vanishes only if  $t_{i1} = \dots = t_{ic}$ , and since  $t_{ic} = 0$  this forces  $\mathbf{x}_{(i)}^\top \mathbf{a}_r = 0$  for every  $i$  and every  $r < c$ , that is  $\mathbf{X} \mathbf{a}_r = \mathbf{0}$ . When  $\text{rank}(\mathbf{X}) = p$  this gives  $\mathbf{a} = \mathbf{0}$ , so the Hessian is negative definite everywhere and  $\ell$  is strictly concave. A strictly concave function has at most one stationary point, and a stationary point is its maximum. ■

*Idea.* The likelihood equations say  $\mathbf{X}^\top \mathbf{y}_r = \mathbf{X}^\top \hat{\boldsymbol{\mu}}_r$  for each non-baseline category: *fitted counts reproduce observed counts in every direction spanned by a regressor*, so with an intercept each category's fitted total is its observed total. This is the canonical-link moment matching of [Theorem 34.3.1](#), one copy per logit.

Because the information is the expected information, Newton–Raphson and Fisher scoring are

the same algorithm, and by [Theorem 34.4.1](#) each step is a weighted least squares fit, with block weights  $\mathbf{W}_i = m_i(\text{diag}(\tilde{\boldsymbol{\pi}}_i) - \tilde{\boldsymbol{\pi}}_i\tilde{\boldsymbol{\pi}}_i^\top)$  and working response  $\boldsymbol{\eta}_i + \mathbf{W}_i^{-1}(\mathbf{y}_i - m_i\boldsymbol{\pi}_i)_{1:c-1}$ . The only change from [Chapter 34](#) is that the weights are  $(c-1) \times (c-1)$  blocks rather than scalars, so each step is a generalized rather than a diagonally weighted least squares problem ([Theorem 21.3.1](#)).

*Warning* (Existence). Concavity guarantees uniqueness, not existence. As for binary data, the maximum is attained unless the categories can be separated. Take  $m_i = 1$  and  $\text{rank}(\mathbf{X}) = p$  as in [Theorem 36.1.2\(c\)](#), with observation  $i$  in category  $y_i$ . If there are vectors  $\mathbf{a}_1, \dots, \mathbf{a}_c$ , not all equal, with  $\mathbf{x}_{(i)}^\top(\mathbf{a}_{y_i} - \mathbf{a}_r) \geq 0$  for every  $i$  and  $r$ , then differentiating  $\ell$  along that direction gives  $\sum_i \sum_r \pi_{ir} \mathbf{x}_{(i)}^\top(\mathbf{a}_{y_i} - \mathbf{a}_r) \geq 0$  at *every* point of the ray, the fitted probabilities being positive throughout. By [Theorem 36.1.2\(c\)](#) the restriction of  $\ell$  to the ray is strictly concave, so its derivative is strictly decreasing; were it zero somewhere it would be negative beyond, contradicting the display. It is therefore positive throughout, and the supremum is approached only at infinity. (Without the rank condition the derivative can vanish along the whole ray, which forces every  $\mathbf{x}_{(i)}$  orthogonal to every difference  $\mathbf{a}_s - \mathbf{a}_t$ ; the maximum is then attained, but not unique.) The binary case, with its proof and its remedies, is [Theorem 35.3.1](#); the multinomial statement is Albert and Anderson (1984). With many categories and small cells separation is common, and the remedies are the same: a penalty as in [Definition 27.2.1](#), or the equivalent proper prior of [Proposition 30.1.4](#) — see [Chapter 27](#) and [Chapter 30](#).

## 36.1.4 Reading the coefficients

Two facts about [Eq. 36.1.4](#) are easy to state and easy to misuse.

### Proposition 36.1.3 — Changing the baseline

Let  $\hat{\boldsymbol{\beta}}_1, \dots, \hat{\boldsymbol{\beta}}_{c-1}$  maximize the likelihood of [Definition 36.1.1](#), and set  $\hat{\boldsymbol{\beta}}_c = \mathbf{0}$ . Fix any  $s \in \{1, \dots, c\}$ . Then:

- (a) for every pair of categories  $r, r'$ ,  $\log(\pi_{ir}/\pi_{ir'}) = \mathbf{x}_{(i)}^\top(\boldsymbol{\beta}_r - \boldsymbol{\beta}_{r'})$ ; in particular a model with baseline  $s$  is the same model, with coefficients  $\boldsymbol{\beta}_r^{(s)} = \boldsymbol{\beta}_r - \boldsymbol{\beta}_s$ ;
- (b) refitting the model with baseline  $s$  gives  $\hat{\boldsymbol{\beta}}_r^{(s)} = \hat{\boldsymbol{\beta}}_r - \hat{\boldsymbol{\beta}}_s$ , the same maximized log-likelihood and the same fitted probabilities;
- (c) the estimated covariance matrix of  $\hat{\boldsymbol{\beta}}^{(s)}$  obtained from the refitted model equals  $\mathbf{L} \hat{\text{Cov}}(\hat{\boldsymbol{\beta}}) \mathbf{L}^\top$ , where  $\mathbf{L}$  is the linear map of (a); so standard errors of pairwise contrasts do not depend on which baseline was used.

*Proof.*

- (a) Subtract two instances of [Eq. 36.1.4](#); the baseline cancels because  $\log(\pi_{ir}/\pi_{ir'}) = \log(\pi_{ir}/\pi_{ic}) - \log(\pi_{ir'}/\pi_{ic})$ .
- (b) The map  $\mathbf{L} : \boldsymbol{\beta} \mapsto \boldsymbol{\beta}^{(s)}$  defined by  $\boldsymbol{\beta}_r^{(s)} = \boldsymbol{\beta}_r - \boldsymbol{\beta}_s$  for  $r \neq s$  is linear, and it is invertible: from  $\boldsymbol{\beta}^{(s)}$  one recovers  $\boldsymbol{\beta}_r = \boldsymbol{\beta}_r^{(s)} + \boldsymbol{\beta}_s$  with  $\boldsymbol{\beta}_s^{(s)} = \mathbf{0}$ . By (a) the two parameterizations give exactly the same family of probability vectors  $\boldsymbol{\pi}_i$ , so the two likelihoods are the same function composed with  $\mathbf{L}$ . A reparameterization by a bijection carries maximizers to maximizers and leaves the maximum value and every fitted quantity unchanged.

- (c) Under a linear reparameterization  $\boldsymbol{\psi} = \mathbf{L}\boldsymbol{\beta}$  the information transforms as  $\mathbf{J}^{(s)} = \mathbf{L}^{-\top} \mathbf{J} \mathbf{L}^{-1}$ , by the chain rule applied to the score. Hence  $(\mathbf{J}^{(s)})^{-1} = \mathbf{L} \mathbf{J}^{-1} \mathbf{L}^{\top}$ , which is exactly the delta-method covariance of  $\mathbf{L}\hat{\boldsymbol{\beta}}$  — and the delta method is exact here because  $\mathbf{L}$  is linear.

■

So the coefficients are log odds ratios:  $\hat{\beta}_{rj}$  is the estimated change in  $\log(\pi_r/\pi_c)$  per unit of regressor  $j$ , and differences of coefficients are log odds ratios between two non-baseline categories. The baseline is a choice of *display*, not of model. What is not invariant is the reading of a single coefficient, and here is the trap.

### Proposition 36.1.4 — A coefficient is not a derivative of a probability

In Definition 36.1.1, with  $\boldsymbol{\beta}_c = \mathbf{0}$  and  $x_j$  a continuous regressor,

$$\frac{\partial \pi_{ir}}{\partial x_{ij}} = \pi_{ir} \left( \beta_{rj} - \sum_{s=1}^c \pi_{is} \beta_{sj} \right). \quad (36.1.5)$$

So the  $c$  derivatives sum to zero, the derivative for category  $r$  is positive exactly when  $\beta_{rj}$  exceeds the probability-weighted average  $\bar{\beta}_j(\mathbf{x}_{(i)}) = \sum_s \pi_{is} \beta_{sj}$ , and the baseline, whose coefficient is zero by construction, has a strictly positive derivative whenever  $\bar{\beta}_j(\mathbf{x}_{(i)}) < 0$ .

*Proof.* Write  $\pi_{ir} = e^{\eta_{ir}} / \sum_s e^{\eta_{is}}$  with  $\partial \eta_{is} / \partial x_{ij} = \beta_{sj}$ . The quotient rule gives

$$\frac{\partial \pi_{ir}}{\partial x_{ij}} = \pi_{ir} \beta_{rj} - \pi_{ir} \frac{\sum_s e^{\eta_{is}} \beta_{sj}}{\sum_s e^{\eta_{is}}} = \pi_{ir} \left( \beta_{rj} - \sum_s \pi_{is} \beta_{sj} \right).$$

Summing over  $r$  gives  $\bar{\beta}_j - \bar{\beta}_j = 0$ . The remaining statements read off the sign of the bracket, with  $\beta_{cj} = 0$ .

■

Since  $\bar{\beta}_j(\mathbf{x})$  moves with  $\mathbf{x}$ , a fitted probability can rise over one range of the regressor and fall over another, whatever the sign of its coefficient. Read coefficients as odds ratios; read effects on probabilities off the fitted curves.

## 36.1.5 Travel mode choice

### Example. (Choosing how to travel)

Two hundred and ten travellers on the Sydney–Melbourne corridor each chose one of four modes: air (58), train (63), bus (30) or car (59). The regressors are the household income in units of \$10,000 and the size of the travelling party. Take car as the baseline. Fitting Definition 36.1.1 by Newton–Raphson from  $\boldsymbol{\beta} = \mathbf{0}$  takes 7 steps to full double

precision, and gives the following coefficients, with standard errors from the inverse information in parentheses.

| logit        | intercept       | income (per<br>\$10,000) | party size       |
|--------------|-----------------|--------------------------|------------------|
| air vs car   | 0.9435 (0.5498) | 0.0354 (0.1030)          | −0.6006 (0.1992) |
| train vs car | 2.4938 (0.5357) | −0.5731 (0.1184)         | −0.3098 (0.1956) |
| bus vs car   | 1.9780 (0.6717) | −0.3033 (0.1322)         | −0.9404 (0.3245) |

Income does not distinguish air from car ( $z = 0.34$ ) but moves people strongly away from the ground alternatives: each extra \$10,000 of household income multiplies the odds of train rather than car by  $e^{-0.5731} = 0.5638$ . Read through [Proposition 36.1.3](#), the same fit raises the odds of air over train by  $e^{0.6085}$  per \$10,000. Party size pushes travellers towards the car, most sharply away from the bus. The likelihood ratio statistic against the intercept-only model is  $2(-253.3408 + 283.7588) = 60.836$  on six degrees of freedom.

```
import numpy as np
import statsmodels.api as sm

MODES = ["air", "train", "bus", "car"]          # car is category 4, the baseline

raw = sm.datasets.modechoice.load_pandas().data
chosen = raw[raw["choice"] == 1].sort_values("individual").reset_index(drop=True)
y = chosen["mode"].astype(int).to_numpy()      # 1 air, 2 train, 3 bus, 4 car
n, c = len(y), 4
X = np.column_stack([np.ones(n), chosen["hinc"] / 10.0, chosen["psize"]])
Y = np.zeros((n, c))
Y[np.arange(n), y - 1] = 1.0                  # indicator matrix, baseline last

print("chosen mode counts:", Y.sum(axis=0))
```

The fit itself is [Theorem 36.1.2](#) written out: build the probabilities, form the score and the block information, solve, repeat.

```
def probs(X, B):
    """Fitted probabilities of the baseline-category logit model; B is p x (c-1)."""
    eta = np.column_stack([X @ B, np.zeros(len(X))]) # baseline has eta = 0
    eta = eta - eta.max(axis=1, keepdims=True)       # stabilize the
    exponentials
    E = np.exp(eta)
    return E / E.sum(axis=1, keepdims=True)

def loglik(X, Y, B):
    return float(np.sum(Y * np.log(probs(X, B))))

def mnlogit(X, Y, tol=1e-10, maxit=50):
    """Newton-Raphson (= Fisher scoring, the link is canonical) for baseline logits.
    """
    n, p = X.shape
    c = Y.shape[1]
    B = np.zeros((p, c - 1))
```

```

for it in range(1, maxit + 1):
    P = probs(X, B)
    U = (X.T @ (Y[:, :c - 1] - P[:, :c - 1])).T.ravel()      # score, stacked
    by category
    J = np.zeros(((c - 1) * p, (c - 1) * p))                # expected =
    observed information
    for r in range(c - 1):
        for s in range(c - 1):
            w = P[:, r] * ((r == s) - P[:, s])
            J[r * p:(r + 1) * p, s * p:(s + 1) * p] = X.T @ (w[:, None] * X)
        step = np.linalg.solve(J, U)
        B = B + step.reshape(c - 1, p).T
        if np.max(np.abs(step)) < tol:
            break
    return B, J, it

B, J, iters = mnlogit(X, Y)
se = np.sqrt(np.diag(np.linalg.inv(J))).reshape(c - 1, X.shape[1]).T

for r in range(c - 1):
    print(f"{MODES[r]:>6s} vs car: " + " ".join(f"{b:7.4f} ({s:.4f})" for b, s in
        zip(B[:, r], se[:, r])))
print(f"log-likelihood {loglik(X, Y, B):.4f} after {iters} Newton steps")

```

The script behind this section checks the output against the `MNLogit` routine of `statsmodels`, coefficient by coefficient. It also carries out the substitution of [Proposition 36.1.3](#), refitting with train as the baseline and confirming that subtracting the train column from the fitted coefficient matrix reproduces the refit and leaves every fitted probability unchanged.

```

# changing the baseline: subtract the train column (category 2) from every column
Btrain = B - B[:, [1]]
Btrain = np.delete(np.column_stack([Btrain, -B[:, 1]]), 1, axis=1)      # drop train
    , add car
Ytrain = Y[:, [0, 2, 3, 1]]      # put train
    last
Bcheck, _, _ = mnlogit(X, Ytrain)

print("air vs train, refitted :", np.round(Bcheck[:, 0], 6))
print("air vs train, subtracted:", np.round(Btrain[:, 0], 6))
print("fitted probabilities identical:", np.allclose(probs(X, Bcheck[:, [0, 3, 1,
    2]], probs(X, B)))

```

[Figure 36.1.1](#) plots the fitted probabilities against income, and the bus curve in the left panel is [Proposition 36.1.4](#) made visible: the coefficient of income in the bus logit is  $-0.3033$ , yet the fitted probability of the bus *rises* over the first twenty thousand dollars before falling. At an income of \$10,000 with a party of one the fitted probabilities are (0.1528, 0.5243, 0.2182, 0.1047) for air, train, bus and car, with derivatives per \$10,000 of (0.0606,  $-0.1111$ , 0.0126, 0.0378); the car, whose coefficient is fixed at zero, has the second largest of the four, behind air alone.

```

# the derivative of a fitted probability is not the coefficient
Bfull = np.column_stack([B, np.zeros(X.shape[1])])      # append the baseline's
    column of zeros
x0 = np.array([1.0, 1.0, 1.0])      # income $10,000, party
    of one
p0 = probs(x0[None, :], B)[0]

```

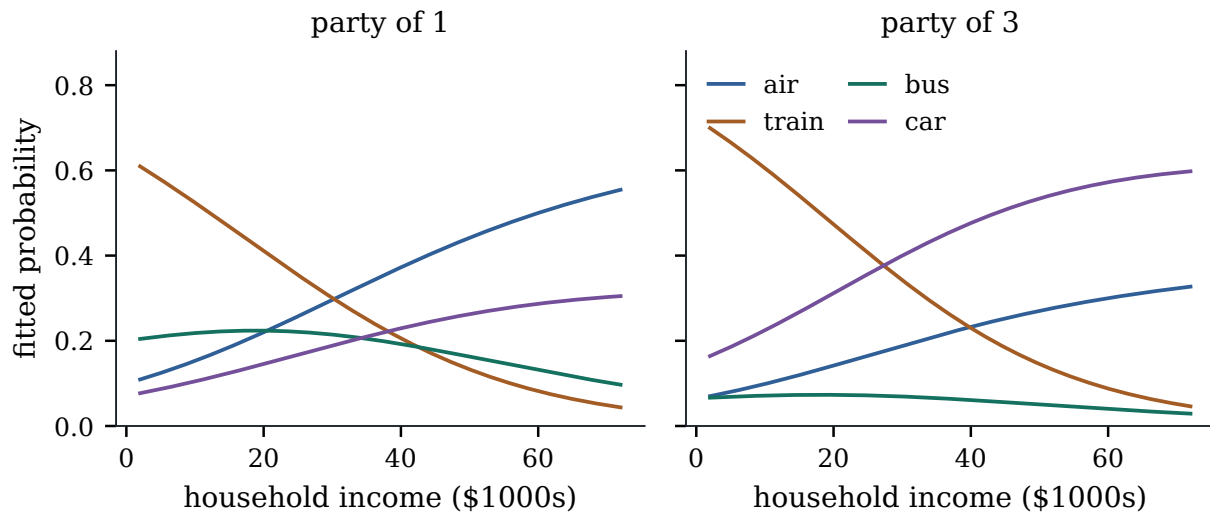


Figure 36.1.1: Fitted mode probabilities from the baseline-category logit model, against household income, for a traveller alone (left) and in a party of three (right). Coefficients are log odds ratios against the car; the curves show what those coefficients do to the probabilities themselves.

```
dp0 = p0 * (Bfull[1] - p0 @ Bfull[1]) # d pi_r / d (income in
    $10,000s)

print("fitted probabilities", np.round(p0, 4))
print("derivatives", np.round(dp0, 4))
print("income coefficients", np.round(Bfull[1], 4))
```

*Remark* (Fitting one logit at a time). It is tempting to estimate  $\beta_r$  by discarding every observation outside  $\{r, c\}$  and running an ordinary binary logistic regression. By [Eq. 36.1.4](#),

$$P(Y_i = r \mid Y_i \in \{r, c\}, \mathbf{x}_{(i)}) = \frac{\pi_{ir}}{\pi_{ir} + \pi_{ic}} = \frac{e^{\mathbf{x}_{(i)}^\top \beta_r}}{1 + e^{\mathbf{x}_{(i)}^\top \beta_r}},$$

so the conditional model is exactly the binary logistic model of [Definition 35.1.1](#) with the same  $\beta_r$ , and the individualized fit is consistent. It is not efficient, and the  $c - 1$  fits are correlated in a way the separate outputs do not report. Begg and Gray (1984) quantified the loss, small when the baseline is a large category and severe otherwise.

## 36.1.6 Inference

The argument of [Theorem 34.5.1](#) carries over verbatim in block notation, although the theorem there is stated for a scalar response: by [Theorem 36.1.2](#) the score is again a sum of independent mean-zero vectors, the information is again nonrandom, and the global concavity its proof leans on is [Theorem 36.1.2\(c\)](#). Under the analogues of its conditions — categories and regressors fixed, sample size growing, the leverages of the stacked design tending to zero, the true parameter interior — the maximum likelihood estimate is consistent and asymptotically normal with covariance the inverse information; nested models are compared by the analysis of deviance of [Theorem 34.5.5](#).

Two cautions are specific to categorical responses. The deviance is *not* a goodness-of-fit statistic for ungrouped data: when every  $m_i = 1$  the saturated model has as many parameters as obser-

vations and the  $\chi^2$  approximation fails, as [Theorem 34.5.5](#) says in general and [Proposition 35.4.1](#) for binary data. And a regressor is not one coefficient here but  $c - 1$ , so “does income matter?” has  $c - 1$  degrees of freedom; testing the coefficients separately and reporting the smallest  $p$ -value is the multiplicity error of [Chapter 13](#) in a new costume.

## 36.1.7 Exercises

### 36.1.7.1 A. Check your understanding

#### Exercise A1.

A response has  $c = 5$  categories and the model matrix has  $p = 4$  columns including the intercept. How many free coefficients has the baseline-category logit model? How many has the saturated model for  $n$  ungrouped observations? How many degrees of freedom does the test that one regressor may be dropped have?

*Solution.* The model has  $(c - 1)p = 16$  free coefficients. The saturated model for ungrouped data gives each observation its own probability vector,  $n(c - 1)$  free parameters. Dropping one regressor removes one coefficient from each of the  $c - 1$  logits, so the test has  $c - 1 = 4$  degrees of freedom.

### 36.1.7.2 B. Practice

#### Exercise B1.

From the fitted coefficients of [Example \(Choosing how to travel\)](#), compute the estimated log odds ratio of bus against train per \$10,000 of income, and the estimated odds ratio of air against bus per extra person in the party. State which of these depend on the choice of baseline.

*Solution.* By [Proposition 36.1.3](#) the bus-against-train income coefficient is  $-0.3033 - (-0.5731) = 0.2698$ , so the odds of bus rather than train rise by a factor  $e^{0.2698}$  per \$10,000. For party size, air against bus is  $-0.6006 - (-0.9404) = 0.3398$ , an odds ratio of  $e^{0.3398}$ . Neither depends on the baseline: by [Proposition 36.1.3\(a\)](#) a difference of coefficients is the pairwise log odds ratio  $\log(\pi_{ir}/\pi_{ir'})$  per unit of the regressor, which the model specifies without reference to any baseline.

#### Exercise B2.

Show directly that the  $c \times c$  matrix  $\Sigma(\boldsymbol{\pi}) = \text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top$  is nonnegative definite with  $\Sigma\mathbf{1} = \mathbf{0}$ , and that its restriction to the first  $c - 1$  coordinates is positive definite when every  $\pi_r > 0$ .

*Solution.* For any  $\mathbf{t}$ ,  $\mathbf{t}^\top \Sigma \mathbf{t} = \sum_r \pi_r t_r^2 - (\sum_r \pi_r t_r)^2$  is the variance of a random variable taking the value  $t_r$  with probability  $\pi_r$ , hence nonnegative, and it vanishes iff all  $t_r$  with  $\pi_r > 0$  are equal. Taking  $\mathbf{t} = \mathbf{1}$  gives  $\Sigma\mathbf{1} = \mathbf{0}$ . If  $\mathbf{t}$  has  $t_c = 0$  and  $\mathbf{t}^\top \Sigma \mathbf{t} = 0$ , then all  $t_r$  are equal to  $t_c = 0$ ; so the leading  $(c - 1) \times (c - 1)$  block is positive definite.

## 36.1.7.3 C. Going deeper

**Exercise C1.**

The **adjacent-category logit model** for an ordered response specifies  $\log(\pi_{i,r+1}/\pi_{ir}) = \theta_r + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  for  $r = 1, \dots, c-1$ , with  $\mathbf{x}_{(i)}$  not containing an intercept. Show that it is the baseline-category model [Definition 36.1.1](#) with slope vectors  $\boldsymbol{\beta}_r = (c-r)\boldsymbol{\beta}^*$  for a single vector  $\boldsymbol{\beta}^*$ , plus a free intercept for each  $r$ : a submodel obtained by  $(c-2)p$  linear restrictions, fitted with the same software after recoding the model matrix.

*Solution.* Summing the adjacent logits from  $r$  to  $c-1$  gives  $\log(\pi_{ic}/\pi_{ir}) = \sum_{s=r}^{c-1} \theta_s + (c-r)\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , so  $\log(\pi_{ir}/\pi_{ic}) = -\sum_{s \geq r} \theta_s - (c-r)\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . This is [Eq. 36.1.4](#) with a free intercept for each  $r$  and slope vector proportional to  $\boldsymbol{\beta}^* = -\boldsymbol{\beta}$  with the known factor  $c-r$ . Fitting is a baseline-category fit on the model matrix whose slope block for logit  $r$  is  $(c-r)\mathbf{X}$ .

**Exercise C2.**

Show that in [Definition 36.1.1](#) the ratio  $\pi_{ir}/\pi_{ir'}$  does not involve any third category — the *independence of irrelevant alternatives*. Construct a three-category example in which this is substantively absurd, and say what the assumption forces about the fitted probabilities when a new category is added that is nearly a copy of an existing one.

## 36.2 Cumulative logit models

Suppose the  $c$  categories are ordered: none, mild, severe; strongly disagree through strongly agree; extremely liberal through extremely conservative. The baseline-category model of [Section 36.1](#) fits such data perfectly well — it fits *any* categorical response — but throws the ordering away. It spends  $(c-1)p$  coefficients where a model that knows the order can often spend  $p$ , and its fit is unchanged if the category labels are shuffled. That last property is the diagnosis: a likelihood invariant to permuting the categories cannot be using the order.

### 36.2.1 Cutting the scale

The device that uses the ordering is to *cut*: for each  $r$  between 1 and  $c-1$ , collapse the response into the binary event  $\{Y_i \leq r\}$  and model its probability. Write

$$\gamma_{ir} = P(Y_i \leq r \mid \mathbf{x}_{(i)}) = \pi_{i1} + \dots + \pi_{ir}, \quad r = 1, \dots, c-1, \quad (36.2.1)$$

for the **cumulative probabilities**, with  $\gamma_{i0} = 0$  and  $\gamma_{ic} = 1$ . The category probabilities are recovered by differencing,  $\pi_{ir} = \gamma_{ir} - \gamma_{i,r-1}$ , so modelling the  $c-1$  cumulative probabilities is modelling the whole distribution.



**Definition 36.2.1 — Cumulative link model**

Let  $G$  be a continuous, strictly increasing distribution function on  $\mathbb{R}$  and let  $\mathbf{x}_{(i)}$  be a vector of  $p$  regressors *without* an intercept. The **cumulative link model** is

$$G^{-1}(\gamma_{ir}) = \theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r, \quad r = 1, \dots, c-1, \quad (36.2.2)$$

with **cutpoints**  $\theta_1, \dots, \theta_{c-1}$  and slope vectors  $\boldsymbol{\beta}_1, \dots, \boldsymbol{\beta}_{c-1}$ . With  $G = \Lambda$ , the standard logistic distribution function  $\Lambda(z) = e^z / (1 + e^z)$ , [Eq. 36.2.2](#) is the **cumulative logit model**

$$\log \frac{P(Y_i \leq r \mid \mathbf{x}_{(i)})}{P(Y_i > r \mid \mathbf{x}_{(i)})} = \theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r;$$

with  $G = \Phi$  it is the **cumulative probit model**, and with  $G(z) = 1 - \exp(-e^z)$  the **cumulative complementary log-log model**. The special case  $\boldsymbol{\beta}_1 = \dots = \boldsymbol{\beta}_{c-1} = \boldsymbol{\beta}$  is the **proportional odds model** when  $G = \Lambda$ , and is the subject of [Section 36.3](#).

Two conventions deserve comment. The intercept is not in  $\mathbf{x}_{(i)}$  because the cutpoints already play that role: shifting every  $\theta_r$  and a hypothetical intercept by the same constant would change nothing, so the two cannot both be free. And the minus sign before  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r$  is worth the confusion it causes once, because it makes a *positive* coefficient mean “larger values of this regressor push the response towards the higher categories”. The cutpoints are unconstrained parameters only if the model is coherent, which is the one real difficulty of this family.

**36.2.2 The ordering constraint****Proposition 36.2.1 — Cutpoints must stay in order**

[Eq. 36.2.2](#) defines a probability distribution over the  $c$  categories at  $\mathbf{x}_{(i)}$  if and only if

$$\theta_1 - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_1 < \theta_2 - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_2 < \dots < \theta_{c-1} - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_{c-1}. \quad (36.2.3)$$

If the slopes are common,  $\boldsymbol{\beta}_r = \boldsymbol{\beta}$  for all  $r$ , this reduces to  $\theta_1 < \dots < \theta_{c-1}$ , a constraint on the parameters alone. If the slopes differ, the constraint involves  $\mathbf{x}_{(i)}$ , and for any two cuts with  $\boldsymbol{\beta}_r \neq \boldsymbol{\beta}_{r+1}$  there are values of  $\mathbf{x}$  at which it fails.

*Proof.* The category probabilities are  $\pi_{ir} = \gamma_{ir} - \gamma_{i,r-1}$ , so they are all positive exactly when  $\gamma_{i1} < \dots < \gamma_{i,c-1}$ , and they automatically sum to one. Since  $G$  is strictly increasing, this is [Eq. 36.2.3](#). With common slopes the terms  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  cancel from every inequality. If  $\mathbf{d} = \boldsymbol{\beta}_{r+1} - \boldsymbol{\beta}_r \neq \mathbf{0}$ , the  $r$ th inequality reads  $\mathbf{x}^\top \mathbf{d} < \theta_{r+1} - \theta_r$ , a half-space; any  $\mathbf{x}$  far enough in the direction  $\mathbf{d}$  violates it. ■

*Warning* (What the constraint costs). The proportional odds model is safe: the constraint holds for every  $\mathbf{x}$  at once, and by [Exercise \(B1\)](#) it is satisfied automatically at the maximum likelihood estimate. A model with cut-specific slopes is *not* a model over

the whole of  $\mathbb{R}^p$ , only on the region where Eq. 36.2.3 holds; one fits it, checks that the fitted probabilities are positive wherever one intends to predict, and reports the check. For a concrete failure take  $c = 3$ , one regressor,  $\theta_1 = 0$ ,  $\beta_1 = 1$ ,  $\theta_2 = 1$ ,  $\beta_2 = 2$ : at  $x = 0$  the linear predictors are 0 and 1, in order; at  $x = 2$  they are  $-2$  and  $-3$ , out of order, and the fitted probability of the middle category is negative.

### 36.2.3 Fitting

The cumulative link model is not a generalized linear model in the strict sense of Definition 34.2.1: the response is a vector and the parameters are shared across its  $c - 1$  coordinates in a way no single link expresses. The likelihood machinery survives intact, because the response is still multinomial and the mean is still a smooth function of the parameters.

#### Proposition 36.2.2 — Score and expected information for a smooth mean

Let  $\mathbf{Y}_i \sim \text{Mult}(m_i, \boldsymbol{\pi}_i)$  independently,  $i = 1, \dots, n$ , with  $\boldsymbol{\pi}_i = \boldsymbol{\pi}_i(\boldsymbol{\psi})$  a smooth function of a parameter  $\boldsymbol{\psi} \in \mathbb{R}^q$  and every  $\pi_{ir} > 0$ . Write  $\tilde{\boldsymbol{\pi}}_i$  for the first  $c - 1$  probabilities,  $\mathbf{D}_i = \partial \tilde{\boldsymbol{\pi}}_i / \partial \boldsymbol{\psi}^\top$  and  $\mathbf{V}_i = \text{diag}(\tilde{\boldsymbol{\pi}}_i) - \tilde{\boldsymbol{\pi}}_i \tilde{\boldsymbol{\pi}}_i^\top$ . Then the score and the expected information are

$$\mathbf{u}(\boldsymbol{\psi}) = \sum_{i=1}^n \mathbf{D}_i^\top \mathbf{V}_i^{-1} (\tilde{\mathbf{y}}_i - m_i \tilde{\boldsymbol{\pi}}_i), \quad \mathbf{J}(\boldsymbol{\psi}) = \sum_{i=1}^n m_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i. \quad (36.2.4)$$

*Proof.* The inverse of  $\mathbf{V}_i$  is  $\text{diag}(1/\tilde{\boldsymbol{\pi}}_i) + \pi_{ic}^{-1} \mathbf{1}\mathbf{1}^\top$ , as multiplying out and using  $\mathbf{1}^\top \tilde{\boldsymbol{\pi}}_i = 1 - \pi_{ic}$  shows. Also  $\partial \ell / \partial \boldsymbol{\psi} = \sum_i \sum_{r < c} (y_{ir} / \pi_{ir}) \partial \pi_{ir} / \partial \boldsymbol{\psi}$ . Fix  $i$ , drop it from the notation, and substitute the inverse:

$$\mathbf{D}^\top \mathbf{V}^{-1} (\tilde{\mathbf{y}} - m \tilde{\boldsymbol{\pi}}) = \sum_{r < c} \frac{\partial \pi_r}{\partial \boldsymbol{\psi}} \frac{y_r - m \pi_r}{\pi_r} + \left( \sum_{r < c} \frac{\partial \pi_r}{\partial \boldsymbol{\psi}} \right) \frac{\sum_{s < c} (y_s - m \pi_s)}{\pi_c}.$$

Because the probabilities sum to one,  $\sum_{r < c} \partial \pi_r / \partial \boldsymbol{\psi} = -\partial \pi_c / \partial \boldsymbol{\psi}$ , and because the counts sum to  $m$ ,  $\sum_{s < c} (y_s - m \pi_s) = -(y_c - m \pi_c)$ . The second term is therefore  $(\partial \pi_c / \partial \boldsymbol{\psi})(y_c - m \pi_c) / \pi_c$ , and the two terms combine into  $\sum_{r < c} (\partial \pi_r / \partial \boldsymbol{\psi})(y_r - m \pi_r) / \pi_r$ , which equals  $\sum_{r < c} (y_r / \pi_r) \partial \pi_r / \partial \boldsymbol{\psi}$  since  $\sum_r \partial \pi_r / \partial \boldsymbol{\psi} = \mathbf{0}$ . Summing over  $i$  gives the score. For the information,  $\text{Cov}(\tilde{\mathbf{Y}}_i) = m_i \mathbf{V}_i$  by Lemma 36.1.1, and the summands of the score are independent with mean zero, so  $\text{Cov}(\mathbf{u}) = \sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} (m_i \mathbf{V}_i) \mathbf{V}_i^{-1} \mathbf{D}_i = \sum_i m_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i$ . ■

For Eq. 36.2.2 the derivatives are immediate: writing  $\eta_{ir} = \theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r$  and  $g = G'$  for the density,

$$\frac{\partial \gamma_{ir}}{\partial \theta_s} = g(\eta_{ir}) \delta_{rs}, \quad \frac{\partial \gamma_{ir}}{\partial \boldsymbol{\beta}_s} = -g(\eta_{ir}) \mathbf{x}_{(i)} \delta_{rs},$$

and  $\mathbf{D}_i$  is obtained by differencing these rows. Fisher scoring  $\boldsymbol{\psi} \leftarrow \boldsymbol{\psi} + \mathbf{J}^{-1} \mathbf{u}$  converges quickly from ordered starting cutpoints and zero slopes. Because Eq. 36.2.4 has the form “derivative of the mean, weighted by the inverse covariance, times the residual”, each step is again a weighted

least squares fit — the block-weighted version of [Theorem 34.4.1](#) and [Proposition 20.6.1](#) — with  $\mathbf{V}_i$  in the place of scalar weights.

*Remark* (Why the fitting is well behaved). Suppose the link density  $g = G'$  is log-concave, as it is for the logistic, the normal and the extreme-value distribution behind the complementary log-log link. Then

$$G(a) - G(b) = \int g(u) \mathbf{1}\{b < u \leq a\} du$$

integrates over  $u$  a function of  $(u, a, b)$  that is log-concave on  $\mathbb{R}^3$ , being a log-concave function of  $u$  times the indicator of a convex set; a marginal of a log-concave function is log-concave, so  $\log\{G(a) - G(b)\}$  is concave on  $\{b < a\}$ . The observation in category  $r$  contributes  $\log\{G(\eta_{ir}) - G(\eta_{i,r-1})\}$  with  $\eta_{ir}$  affine in the parameters, so the log-likelihood is concave on the region where [Eq. 36.2.3](#) holds at the observed  $\mathbf{x}_{(i)}$  — an intersection of half-spaces, hence convex. That is Pratt (1981) and Burrige (1981). Concavity excludes local maxima that are not global, which is what makes scoring from any ordered start dependable; by itself it gives neither existence nor uniqueness, and with cut-specific slopes the region it holds on is cut out by the data.

*Remark* (Asymptotics, and what they rest on). This is not the canonical-link generalized linear model [Theorem 34.5.1](#) is stated for, so that theorem cannot be quoted here. What holds is the standard theory of maximum likelihood in a smooth parametric model: suppose the model is correctly specified and identifiable,  $c$  and  $p$  are fixed,  $\boldsymbol{\psi}^0$  is interior to the region where [Eq. 36.2.3](#) holds,  $G$  is twice continuously differentiable with  $g > 0$ , the regressors are bounded, and  $n^{-1}\mathbf{J}(\boldsymbol{\psi}) \rightarrow \mathbf{J}_\infty$  positive definite uniformly near  $\boldsymbol{\psi}^0$ . Then with probability tending to one the likelihood equations have a consistent solution, and  $\sqrt{n}(\hat{\boldsymbol{\psi}} - \boldsymbol{\psi}^0)$  converges in distribution to  $N_q(\mathbf{0}, \mathbf{J}_\infty^{-1})$ , with the three statistics of [Proposition 34.5.3](#) behaving as they do there.

We do not prove it; treat it as imported. Its first two steps are those of [Theorem 34.5.1](#) — a Lyapunov limit theorem for the score, which [Proposition 36.2.2](#) exhibits as a sum of independent mean-zero terms, and a uniform quadratic expansion on compacts — but not its third, which used the global concavity of a canonical-link generalized linear model. For the classical treatment see van der Vaart (1998, chapter 5) or Lehmann and Casella (1998, chapter 6). Nothing here is exact in finite samples.

## 36.2.4 Where people place themselves

### Example. (A seven-point scale)

The 1996 American National Election Study asked 944 respondents to place themselves on a seven-point scale from extremely liberal (1) to extremely conservative (7); the counts are 16, 103, 147, 256, 170, 218 and 34. With the respondent's age in decades, a seven-point education scale and a twenty-four-point household income scale as regressors, Fisher scoring for the common-slope cumulative logit model converges in 11 steps to

|               | age (decades) | education | income |
|---------------|---------------|-----------|--------|
| $\hat{\beta}$ | 0.0942        | −0.1249   | 0.0276 |

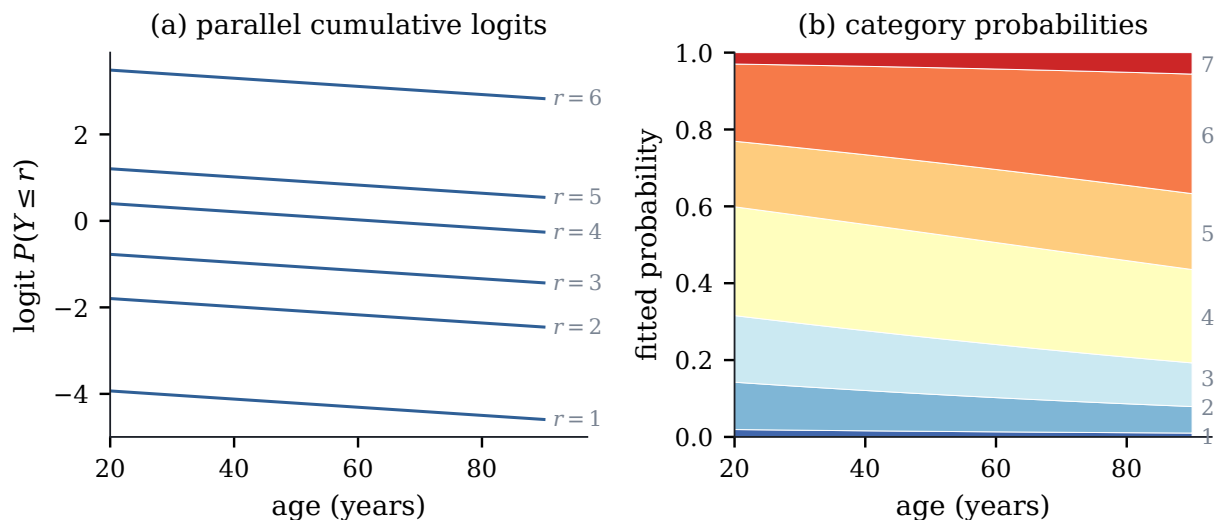


Figure 36.2.1: The fitted cumulative logit model for self-placement, as a function of age at education 4 and income 17. (a) The six cumulative logits are parallel, because the model gives them a common slope. (b) The same fit as category probabilities.

|                |        |        |        |
|----------------|--------|--------|--------|
| standard error | 0.0357 | 0.0395 | 0.0104 |
| $z$            | 2.637  | -3.166 | 2.641  |

with cutpoints  $\hat{\theta} = (-3.7744, -1.6378, -0.6159, 0.5580, 1.3625, 3.6442)$ , in order as [Proposition 36.2.1](#) requires, and maximized log-likelihood  $-1613.8558$ . Older and richer respondents place themselves further to the right, more educated respondents further to the left: each decade of age multiplies the odds of a higher placement by 1.0988, each step up the education scale by 0.8826. Whether one should believe a single common slope is the question of [Section 36.3](#), and for these data the answer turns out to be no.

[Figure 36.2.1](#) shows what the fit says. Panel (a) plots the six fitted cumulative logits against age: exactly parallel lines, which is what a common slope means. Panel (b) differences them into the seven category probabilities, whose bands shift smoothly to the right as age increases and can never cross — the ordering constraint in picture form.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

NAMES = ["age (decades)", "education", "income"]

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].astype(int).to_numpy()           # 1 = extremely liberal ... 7 =
                                                    extremely conservative
X = np.column_stack([anes["age"] / 10.0, anes["educ"], anes["income"]])
n, p = X.shape
c = 7
Y = np.zeros((n, c))
Y[np.arange(n), y - 1] = 1.0

print("category counts:", Y.sum(axis=0))
```

The fitting routine below is [Eq. 36.2.4](#) written out, with any subset of the regressors allowed

cut-specific slopes through the argument `free`. Empty, it is the proportional odds fit; the other settings are for [Section 36.3](#).

```
def cum_gamma(psi, X, c, free):
    """Cumulative probabilities  $\Lambda(\theta_r - x^T \beta_r)$  and their gradients.

     $\psi = [\theta \ (c-1), \beta \ (p), \delta \ ((c-2) \times \text{len}(\text{free}))]$ ; the columns of  $X$ 
    listed
    in `free` get cut-specific slopes  $\beta_r = \beta + \delta_r$ , with  $\delta_1 = 0$ .
    """
    n, p = X.shape
    q, nf, fl = len(psi), len(free), list(free)
    theta, beta = psi[:c - 1], psi[c - 1:c - 1 + p]
    delta = psi[c - 1 + p:].reshape(c - 2, nf)
    G = np.empty((n, c - 1))
    dG = np.zeros((n, c - 1, q))
    for r in range(c - 1):
        beta_r = beta.copy()
        if r >= 1 and nf:
            beta_r[fl] = beta_r[fl] + delta[r - 1]
        g = 1.0 / (1.0 + np.exp(-(theta[r] - X @ beta_r)))
        lam = (g * (1.0 - g))[:, None] # the logistic density at the
    cut
        G[:, r] = g
        dG[:, r, r] = lam[:, 0]
        dG[:, r, c - 1:c - 1 + p] = -lam * X
        if r >= 1 and nf:
            off = c - 1 + p + (r - 1) * nf
            dG[:, r, off:off + nf] = -lam * X[:, fl]
    return G, dG

def cum_probs(psi, X, c, free):
    """Category probabilities  $\pi_{ir}$  and their derivatives  $D[i, r, k] = d \pi_{ir} / d \psi_k$ ."""
    G, dG = cum_gamma(psi, X, c, free)
    P = np.diff(np.column_stack([np.zeros(len(X)), G, np.ones(len(X))]), axis=1)
    D = np.concatenate([dG[:, :1, :], np.diff(dG, axis=1)], axis=1)
    return P, D

def cum_score(psi, X, Y, c, free):
    """Score and expected information of the multinomial likelihood."""
    P, D = cum_probs(psi, X, c, free)
    Pm = P[:, :c - 1]
    Sigma = np.einsum("ir,rs->irs", Pm, np.eye(c - 1)) - np.einsum("ir,is->irs", Pm,
        Pm)
    A = np.linalg.solve(Sigma, D) #  $\Sigma_i^{-1} D_i$ , one
    observation at a time
    U = np.einsum("irk,ir->k", A, Y[:, :c - 1] - Pm)
    J = np.einsum("irk,irl->kl", D, A)
    return U, J

def cum_loglik(psi, X, Y, c, free):
    P, _ = cum_probs(psi, X, c, free)
    return float(np.sum(Y * np.log(np.clip(P, 1e-300, None))))
```

```

def cum_fit(X, Y, c, free=(), tol=1e-10, maxit=100):
    """Fisher scoring for the cumulative logit model."""
    n, p = X.shape
    psi = np.zeros((c - 1) + p + (c - 2) * len(free))
    psi[:c - 1] = np.linspace(-1.5, 1.5, c - 1)    # start from ordered cutpoints,
    no slopes
    for it in range(1, maxit + 1):
        U, J = cum_score(psi, X, Y, c, free)
        step = np.linalg.solve(J, U)
        psi = psi + step
        if np.max(np.abs(step)) < tol:
            break
    return psi, J, it

psi, J, iters = cum_fit(X, Y, c)
theta, beta = psi[:c - 1], psi[c - 1:]
se = np.sqrt(np.diag(np.linalg.inv(J)))
ll_po = cum_loglik(psi, X, Y, c, ())

print("cutpoints", np.round(theta, 4))
for j, name in enumerate(NAMES):
    print(f"{name:>16s}  {beta[j]:8.4f}  ({se[c - 1 + j]:.4f})  z = {beta[j] / se[c - 1 + j]:6.3f}")
print(f"log-likelihood {ll_po:.4f} after {iters} Fisher scoring steps")

```

The script checks the fit, cutpoint by cutpoint, against the `OrderedModel` routine of `statsmodels`.

### 36.2.5 Choosing the link

The three links of [Definition 36.2.1](#) are close over the middle of the probability range and differ in the tails, as in the binary case ([Proposition 35.2.1](#)). Two things are specific to ordinal data. First, the *direction* of the scale is arbitrary. Relabelling the categories in reverse order maps the logit and probit models to themselves with  $\beta$  and the cutpoints negated ([Exercise \(A1\)](#)), but turns the complementary log-log model into a different one, the log-log model; with an asymmetric link the analysis depends on which end of the scale was called “1”, and the choice must be justified.

Second, the complementary log-log link has an interpretation the others lack. With  $G(z) = 1 - \exp(-e^z)$  and common slopes, [Eq. 36.2.2](#) gives

$$P(Y_i > r \mid \mathbf{x}_{(i)}) = \exp\{-e^{\theta_r} e^{-\mathbf{x}_{(i)}^\top \beta}\} = [P(Y > r \mid \mathbf{0})]^{\exp(-\mathbf{x}_{(i)}^\top \beta)}, \quad (36.2.5)$$

so the regressors act multiplicatively on the cumulative hazard of the underlying scale: the grouped-data version of the proportional hazards model of Cox (1972). If the ordered categories are intervals of a survival time, reach for it.

## 36.2.6 Exercises

### 36.2.6.1 A. Check your understanding

#### Exercise A1.

Verify the claim that reversing the category order maps the cumulative logit model to itself with  $\boldsymbol{\beta}$  and the cutpoints negated, and find what it does to the complementary log-log model.

*Solution.* Let  $Y' = c + 1 - Y$ . Then  $\{Y' \leq r\} = \{Y \geq c + 1 - r\} = \{Y > c - r\}$ , so  $\gamma'_{ir} = 1 - \gamma_{i,c-r}$ . Since  $\Lambda^{-1}(1 - u) = -\Lambda^{-1}(u)$ ,  $\Lambda^{-1}(\gamma'_{ir}) = -\theta_{c-r} + \mathbf{x}^\top \boldsymbol{\beta}$ , which is [Eq. 36.2.2](#) with cutpoints  $-\theta_{c-r}$  (still increasing in  $r$ ) and slope  $-\boldsymbol{\beta}$ . The same works for  $\Phi$ . For  $G(z) = 1 - \exp(-e^z)$  the identity  $G^{-1}(1 - u) = -G^{-1}(u)$  fails, and  $G^{-1}(1 - u) = \log(-\log u)$  defines the log-log link, a different family.

### 36.2.6.2 B. Practice

#### Exercise B1.

Show that in the proportional odds model the maximum likelihood estimate automatically satisfies  $\hat{\theta}_1 < \cdots < \hat{\theta}_{c-1}$ , provided every category is observed, so the ordering constraint never binds. (Hint: what is the likelihood if two cutpoints coincide or cross?)

*Solution.* The likelihood is defined only where [Eq. 36.2.3](#) holds, and  $\pi_{ir} \rightarrow 0$  for some  $i$  and  $r$  as the boundary  $\theta_r = \theta_{r+1}$  is approached. If category  $r$  is observed at least once, some  $y_{ir} > 0$  and  $\ell \rightarrow -\infty$  there. Since  $\ell$  is continuous on the open region and tends to  $-\infty$  at the part of its boundary where two cutpoints meet, any maximizer lies in the interior, where the cutpoints are strictly ordered.

#### Exercise B2.

The cumulative logit model is unchanged if the latent scale is multiplied by a positive constant  $a$  and shifted by  $b$ , provided  $\theta_r \mapsto a\theta_r + b$  and  $\boldsymbol{\beta} \mapsto a\boldsymbol{\beta}$ . Deduce that only *ratios* of coefficients, and the standardized distances between cutpoints, are free of the arbitrary scale — and that a coefficient from a logit fit is not comparable with a coefficient from a probit fit without rescaling by roughly 1.8, the standard deviation of the logistic distribution.

#### Exercise B3.

Refit [Example \(A seven-point scale\)](#) after collapsing the seven categories into three (1–3, 4, 5–7) and again after dichotomizing at the middle category. Compare the three sets of slopes and their standard errors, and say which comparison is a check on the model and which is a statement about information.

*Solution.* The estimates from the three fits are  $(0.0942, -0.1249, 0.0276)$ ,  $(0.1350, -0.1148, 0.0343)$  and  $(0.1229, -0.0311, 0.0398)$ , with standard errors that grow as categories are merged. That they estimate the *same* parameters is the collapsing invariance proved in [Section](#)

36.3 — a check on the model, since it holds only if the proportional odds model is true. That the standard errors grow is a statement about information: collapsing throws data away.

### 36.2.6.3 C. Going deeper

#### Exercise C1.

Suppose an underlying continuous response  $U_i$  follows the linear model  $U_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i$  with  $\varepsilon_i$  having distribution function  $G$ , but  $U_i$  is recorded only as the interval  $(\theta_{r-1}, \theta_r]$  it falls in, with *known* cutpoints. Write down the log-likelihood, show it is the cumulative link model with the cutpoints held fixed, and say what is gained by knowing them (Exercise (B2)).

## 36.3 Proportional odds and its checks

Setting  $\boldsymbol{\beta}_1 = \cdots = \boldsymbol{\beta}_{c-1} = \boldsymbol{\beta}$  in Eq. 36.2.2 turns  $(c-1)p$  slopes into  $p$ . With the logistic link the result is the model that carries the name of its most conspicuous consequence.

### Theorem 36.3.1 — The proportional odds model

Let

$$\log \frac{P(Y_i \leq r \mid \mathbf{x}_{(i)})}{P(Y_i > r \mid \mathbf{x}_{(i)})} = \theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \quad r = 1, \dots, c-1. \quad (36.3.1)$$

Then:

- (a) (*latent variable*) Eq. 36.3.1 holds exactly when there is a latent response  $U_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \varepsilon_i$ , with  $\varepsilon_i$  independent of  $\mathbf{x}_{(i)}$  and standard logistic, and  $Y_i = r$  precisely when  $\theta_{r-1} < U_i \leq \theta_r$ , where  $-\infty = \theta_0 < \theta_1 < \cdots < \theta_{c-1} < \theta_c = \infty$ . Replacing the logistic distribution by any continuous, strictly increasing  $G$  gives the general cumulative link model of Definition 36.2.1 with common slopes;
- (b) (*collapsing*) let  $0 = a_0 < a_1 < \cdots < a_k = c$  be integers and define the coarsened response  $Y_i^* = j$  when  $a_{j-1} < Y_i \leq a_j$ . Then  $Y_i^*$  follows the proportional odds model with the *same*  $\boldsymbol{\beta}$  and cutpoints  $\theta_j^* = \theta_{a_j}$ ,  $j = 1, \dots, k-1$ . In particular dichotomizing at any cut  $r$  gives a binary logistic model with slope  $\boldsymbol{\beta}$ ;
- (c) (*coherence*) the ordering constraint Eq. 36.2.3 reduces to  $\theta_1 < \cdots < \theta_{c-1}$  and so holds at every  $\mathbf{x}$  at once;
- (d) (*interpretation*) for any two regressor vectors and every  $r$ ,

$$\frac{\text{odds}(Y > r \mid \mathbf{x}_2)}{\text{odds}(Y > r \mid \mathbf{x}_1)} = \exp\{(\mathbf{x}_2 - \mathbf{x}_1)^\top \boldsymbol{\beta}\}, \quad (36.3.2)$$

a single odds ratio that does not depend on  $r$ ; and if  $(\mathbf{x}_2 - \mathbf{x}_1)^\top \boldsymbol{\beta} > 0$  then the response at  $\mathbf{x}_2$  is stochastically larger than the response at  $\mathbf{x}_1$ .



*Proof.*

- (a) If such a latent variable exists, then  $P(Y_i \leq r) = P(U_i \leq \theta_r) = P(\varepsilon_i \leq \theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}) = \Lambda(\theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ , which is Eq. 36.3.1 after taking logits, since  $\log\{\Lambda(z)/(1 - \Lambda(z))\} = z$ . Conversely, given Eq. 36.3.1, define  $U_i$  by the displayed construction with  $\varepsilon_i$  standard logistic; the computation just made shows that the induced distribution of  $Y_i$  is the one specified by the model, and the cutpoints are increasing by (c).
- (b) The events are nested:  $\{Y_i^* \leq j\} = \{Y_i \leq a_j\}$ , directly from the definition. Hence  $P(Y_i^* \leq j \mid \mathbf{x}_{(i)}) = \gamma_{i,a_j}$ , and applying the logit to both sides,  $\log\{\gamma_{i,a_j}/(1 - \gamma_{i,a_j})\} = \theta_{a_j} - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . This is Eq. 36.3.1 for  $Y^*$  with cutpoints  $\theta_{a_j}$ , which are increasing because the  $\theta_r$  are, and with  $\boldsymbol{\beta}$  untouched. Taking  $k = 2$ ,  $a_1 = r$  gives the binary statement.
- (c) Substitute  $\boldsymbol{\beta}_r = \boldsymbol{\beta}$  into Eq. 36.2.3: the terms  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  cancel, leaving  $\theta_1 < \dots < \theta_{c-1}$ , a condition on the parameters alone. This is the case already recorded in Proposition 36.2.1.
- (d) From Eq. 36.3.1,  $\text{odds}(Y \leq r \mid \mathbf{x}) = \exp(\theta_r - \mathbf{x}^\top \boldsymbol{\beta})$ , so  $\text{odds}(Y > r \mid \mathbf{x}) = \exp(\mathbf{x}^\top \boldsymbol{\beta} - \theta_r)$  and the ratio is  $\exp\{(\mathbf{x}_2 - \mathbf{x}_1)^\top \boldsymbol{\beta}\}$ ; the cutpoint cancels, which is the whole content of Eq. 36.3.2. If  $(\mathbf{x}_2 - \mathbf{x}_1)^\top \boldsymbol{\beta} > 0$  the ratio exceeds one for every  $r$ , so  $P(Y > r \mid \mathbf{x}_2) > P(Y > r \mid \mathbf{x}_1)$  for every  $r$ : stochastic dominance.

■

*Idea.* Part (b) is why the model is worth having: the slope means the same thing however finely the scale was cut, so studies using three categories and seven are comparable. It is also the model's exposed flank, since *every* dichotomization must give that slope.

*Warning* (What proportional odds does not say). Eq. 36.3.2 is about *cumulative* odds. It does not say that a unit of  $x$  has the same effect on every category probability: differencing  $\pi_{ir} = G(\eta_{ir}) - G(\eta_{i,r-1})$  gives  $\partial \pi_{ir} / \partial \mathbf{x} = \{g(\eta_{i,r-1}) - g(\eta_{ir})\} \boldsymbol{\beta}$ , whose sign changes as  $r$  moves across the distribution. Nor does it say that the latent variable of part (a) exists in any physical sense: the construction is a *device for generating the model*, always available once the model holds, and it carries no empirical content. Claims that the latent scale is “really” attitude, utility or severity are claims about the world, and reading  $\boldsymbol{\beta}$  causally needs the whole apparatus of Chapter 25.

### 36.3.1 Checking the assumption

The restriction is  $(c - 2)p$  linear constraints on the unrestricted cumulative model, so the three statistics of Proposition 34.5.3 are all available. The likelihood ratio and Wald tests need the unrestricted model, which by Proposition 36.2.1 may not be fittable; the score test needs only the restricted fit, which always is.

#### Proposition 36.3.2 — Testing and diagnosing the proportional odds restriction

Let  $A \subseteq \{1, \dots, p\}$  index the regressors whose slopes are allowed to vary, and embed Eq. 36.3.1 in the cumulative logit model  $\text{logit } \gamma_{ir} = \theta_r - \mathbf{x}_{(i)}^\top (\boldsymbol{\beta} + \boldsymbol{\delta}_r)$ , where  $\boldsymbol{\delta}_1 = \mathbf{0}$

and  $\delta_r$  is supported on  $A$  for  $r \geq 2$ . Write  $\boldsymbol{\psi} = (\boldsymbol{\theta}, \boldsymbol{\beta}, \boldsymbol{\delta})$  and let  $\hat{\boldsymbol{\psi}}_0$  be the maximum likelihood estimate under  $\boldsymbol{\delta} = \mathbf{0}$ . Then, under the regularity conditions recorded in Section 36.2 and if Eq. 36.3.1 holds:

- (a) the score statistic  $S_{\text{sc}} = \mathbf{u}(\hat{\boldsymbol{\psi}}_0)^\top \mathbf{J}(\hat{\boldsymbol{\psi}}_0)^{-1} \mathbf{u}(\hat{\boldsymbol{\psi}}_0)$ , with  $\mathbf{u}$  and  $\mathbf{J}$  from Eq. 36.2.4, is computable from the restricted fit alone, equals  $\mathbf{u}_\delta^\top [\mathbf{J}^{-1}]_{\delta\delta} \mathbf{u}_\delta$ , and converges in distribution to  $\chi^2$  with  $(c-2)|A|$  degrees of freedom;
- (b) the likelihood ratio and Wald statistics for the same hypothesis have the same limit and are asymptotically equivalent to  $S_{\text{sc}}$ ;
- (c) the  $c-1$  separate binary logistic regressions of  $\mathbf{1}\{Y_i \leq r\}$  on  $\mathbf{x}_{(i)}$  estimate the same  $\boldsymbol{\beta}$ , so plotting their slopes with confidence intervals shows where any departure lies. The  $c-1$  fits are dependent, so the plot is a diagnostic and not a test.

*Proof.*

- (a) By definition of a maximum, the components of  $\mathbf{u}(\hat{\boldsymbol{\psi}}_0)$  in the  $(\boldsymbol{\theta}, \boldsymbol{\beta})$  block vanish: those are the likelihood equations of the restricted model, which is the full model with  $\boldsymbol{\delta}$  held at zero. Writing  $\mathbf{u} = (\mathbf{0}^\top, \mathbf{u}_\delta^\top)^\top$  and partitioning  $\mathbf{J}^{-1}$  conformably gives the stated reduction. That  $\mathbf{u}^\top \mathbf{J}^{-1} \mathbf{u}$  has the  $\chi^2$  limit with degrees of freedom the number of restrictions is the score-test half of Proposition 34.5.3, which for a generalized linear model is proved there and holds here by the same argument applied to the likelihood of Proposition 36.2.2, under the conditions imported in Section 36.2; the restrictions number  $(c-2)|A|$ , one vector  $\delta_r$  of length  $|A|$  for each of the cuts  $r = 2, \dots, c-1$ .
- (b) The asymptotic equivalence of the three statistics is Proposition 34.5.3, and it carries over under the same imported conditions; the hypothesis is a linear restriction on a parameter interior to the parameter space.
- (c) Dichotomizing at cut  $r$  gives, by Theorem 36.3.1(b), the binary logistic model  $\text{logit } P(Y \leq r) = \theta_r - \mathbf{x}^\top \boldsymbol{\beta}$ , with intercept  $\theta_r$  and the same  $\boldsymbol{\beta}$ . Each fit is therefore a correctly specified binary logistic regression under the null and its estimate is consistent for  $\boldsymbol{\beta}$  by Theorem 34.5.1. The fits use overlapping data, so their errors are correlated, which is why they cannot simply be pooled or differenced into a test.

■

Brant (1990) turns (c) into a test by estimating the joint covariance of the  $c-1$  slope estimates — they solve stacked estimating equations, so a sandwich-type covariance in the spirit of Theorem 21.4.2 is available — and forming a Wald statistic for their equality. It tests the same null hypothesis and has the same chi-squared limit. It is *not* the score statistic, it is built from different estimating equations, and the two can differ appreciably in small samples.

*Warning* (The test is not the question). With  $n$  in the thousands the score test rejects departures too small to matter; with  $n$  in the dozens it misses departures that would change every conclusion. The plot in (c) answers the useful question — *how much* do the slopes move, and where. Report both.

## 36.3.2 Self-placement, checked

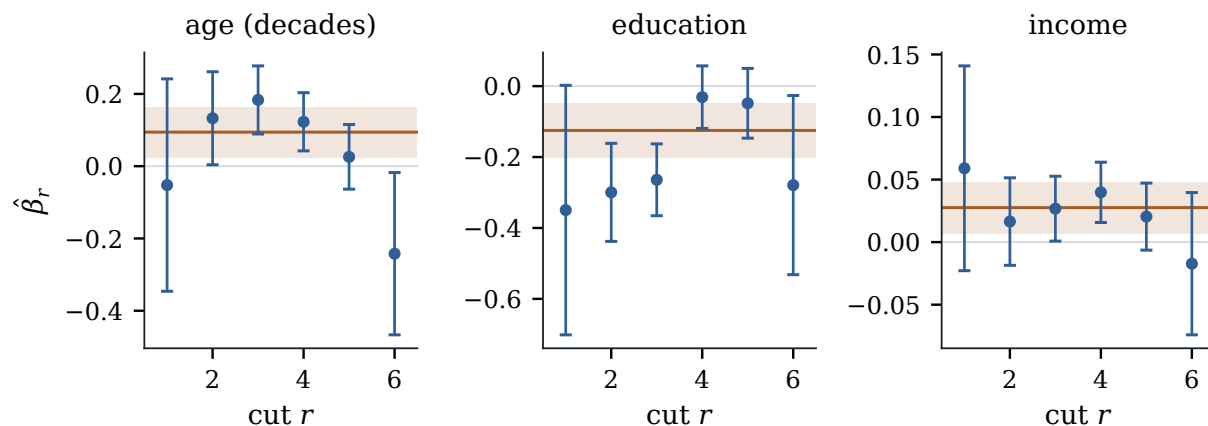


Figure 36.3.1: Slopes from the six separate binary logistic fits of  $\mathbf{1}\{Y \leq r\}$  on the three regressors, with 95% intervals; the horizontal line and band are the proportional odds estimate and its interval. Under proportional odds the six points scatter about the line (Proposition 36.3.2). Education does not.

#### Example. (The proportional odds model does not hold)

The common odds ratio of Theorem 36.3.1(d) is worth having only if it is there. For Example (A seven-point scale), let all three slopes vary across the six cuts. The score statistic of Proposition 36.3.2 is 67.667 on 15 degrees of freedom,  $p = 1.16 \times 10^{-8}$ ; the likelihood ratio statistic against the fitted unrestricted model is 67.420, agreeing to within half a percent as Proposition 36.3.2(b) predicts. One regressor at a time, on  $c - 2 = 5$  degrees of freedom:

| regressor | score statistic | $p$                   | likelihood ratio |
|-----------|-----------------|-----------------------|------------------|
| age       | 17.542          | $3.58 \times 10^{-3}$ | 18.367           |
| education | 41.258          | $8.32 \times 10^{-8}$ | 40.190           |
| income    | 20.056          | $1.22 \times 10^{-3}$ | 20.155           |

Education is the main offender. Figure 36.3.1 shows why: its slope is near  $-0.3$  at the three lowest cuts, near 0 at cuts 4 and 5, and back near  $-0.3$  at cut 6. Education sharply separates the liberal end of the scale from everything above it, says almost nothing about position within the upper middle, and then separates the extreme conservative category again. That is a fact about the data, and a single number cannot express it.

The cells on this page continue those of Section 36.2; this first one repeats the data and the fitting routine so that the page runs on its own.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

NAMES = ["age (decades)", "education", "income"]

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].astype(int).to_numpy()          # 1 = extremely liberal ... 7 =
                                                    extremely conservative
X = np.column_stack([anes["age"] / 10.0, anes["educ"], anes["income"]])
```

```

n, p = X.shape
c = 7
Y = np.zeros((n, c))
Y[np.arange(n), y - 1] = 1.0

def cum_gamma(psi, X, c, free):
    n, p = X.shape
    q, nf, fl = len(psi), len(free), list(free)
    theta, beta = psi[:c - 1], psi[c - 1:c - 1 + p]
    delta = psi[c - 1 + p:].reshape(c - 2, nf)
    G = np.empty((n, c - 1))
    dG = np.zeros((n, c - 1, q))
    for r in range(c - 1):
        beta_r = beta.copy()
        if r >= 1 and nf:
            beta_r[fl] = beta_r[fl] + delta[r - 1]
        g = 1.0 / (1.0 + np.exp(-(theta[r] - X @ beta_r)))
        lam = (g * (1.0 - g))[:, None]          # the logistic density at the
    cut
    G[:, r] = g
    dG[:, r, r] = lam[:, 0]
    dG[:, r, c - 1:c - 1 + p] = -lam * X
    if r >= 1 and nf:
        off = c - 1 + p + (r - 1) * nf
        dG[:, r, off:off + nf] = -lam * X[:, fl]
    return G, dG

def cum_probs(psi, X, c, free):
    G, dG = cum_gamma(psi, X, c, free)
    P = np.diff(np.column_stack([np.zeros(len(X)), G, np.ones(len(X))]), axis=1)
    D = np.concatenate([dG[:, :1, :], np.diff(dG, axis=1)], axis=1)
    return P, D

def cum_score(psi, X, Y, c, free):
    P, D = cum_probs(psi, X, c, free)
    Pm = P[:, :c - 1]
    Sigma = np.einsum("ir,rs->irs", Pm, np.eye(c - 1)) - np.einsum("ir,is->irs", Pm,
        Pm)
    A = np.linalg.solve(Sigma, D)              # Sigma_i^{-1} D_i, one
    observation at a time
    U = np.einsum("irk,ir->k", A, Y[:, :c - 1] - Pm)
    J = np.einsum("irk,irl->kl", D, A)
    return U, J

def cum_loglik(psi, X, Y, c, free):
    P, _ = cum_probs(psi, X, c, free)
    return float(np.sum(Y * np.log(np.clip(P, 1e-300, None))))

def cum_fit(X, Y, c, free=(), tol=1e-10, maxit=100):
    n, p = X.shape
    psi = np.zeros((c - 1) + p + (c - 2) * len(free))
    psi[:c - 1] = np.linspace(-1.5, 1.5, c - 1)    # start from ordered cutpoints,
    no slopes

```

```

for it in range(1, maxit + 1):
    U, J = cum_score(psi, X, Y, c, free)
    step = np.linalg.solve(J, U)
    psi = psi + step
    if np.max(np.abs(step)) < tol:
        break
return psi, J, it

psi, J, iters = cum_fit(X, Y, c)
theta, beta = psi[:c - 1], psi[c - 1:]
se = np.sqrt(np.diag(np.linalg.inv(J)))
ll_po = cum_loglik(psi, X, Y, c, ())
print(f"log-likelihood {ll_po:.4f} after {iters} Fisher scoring steps")

# the same slopes estimated one cut at a time: c - 1 separate binary logistic fits
Xc = sm.add_constant(X)
cut_beta = np.empty((c - 1, p))
cut_se = np.empty((c - 1, p))
for r in range(1, c):
    fit_r = sm.Logit((y <= r).astype(float), Xc).fit(dis=0)
    cut_beta[r - 1] = -np.asarray(fit_r.params)[1:] # logit P(Y <= r) = theta_r -
    x^T beta_r
    cut_se[r - 1] = np.asarray(fit_r.bse)[1:]

for r in range(c - 1):
    print(f"cut {r + 1}: " + " ".join(f"{b:7.4f} ({s:.4f})" for b, s in zip(
        cut_beta[r], cut_se[r])))

```

The score test needs no second fit: evaluate the score and information of the *larger* model at the restricted estimate, with the extra coefficients zero.

```

# score test of proportional odds: all three slopes free to vary across cuts
free = (0, 1, 2)
psi_free = np.zeros((c - 1) + p + (c - 2) * len(free))
psi_free[:c - 1 + p] = psi # the proportional odds fit,
    with delta = 0
U, J_free = cum_score(psi_free, X, Y, c, free)
score_stat = float(U @ np.linalg.solve(J_free, U))
df = (c - 2) * len(free)

psi_np, _, _ = cum_fit(X, Y, c, free=free)
ll_np = cum_loglik(psi_np, X, Y, c, free)
lr_stat = 2 * (ll_np - ll_po)

print(f"score {score_stat:.3f} on {df} df, p = {stats.chi2.sf(score_stat, df):.3g}")
print(f"LR {lr_stat:.3f} on {df} df, p = {stats.chi2.sf(lr_stat, df):.3g}")

```

### 36.3.3 When it fails

Rejecting proportional odds is information, not a disaster. The options, roughly in order of how much they give up.

**Partial proportional odds.** Free the slopes of the offending regressors only, as in [Proposition 36.3.2](#), keeping the restriction for the rest: the model of Peterson and Harrell (1990). It

costs the coherence guarantee, so [Proposition 36.2.1](#) must now be checked at every  $\mathbf{x}$  of interest. Freeing education alone raises the log-likelihood to  $-1593.7610$ , a likelihood ratio statistic of 40.190 on five degrees of freedom ( $p = 1.37 \times 10^{-7}$ ), and lowers Akaike’s criterion ([Definition 29.2.2](#)) from 3245.71 to 3215.52; the fully unrestricted model reaches 3208.29 with ten more parameters. The fitted education slopes by cut are  $-0.2683$ ,  $-0.2988$ ,  $-0.2523$ ,  $-0.0206$ ,  $-0.0498$  and  $-0.2970$ , and every fitted probability stays positive.

```
# partial proportional odds: education alone gets cut-specific slopes
psi_pp, J_pp, _ = cum_fit(X, Y, c, free=(1,))
ll_pp = cum_loglik(psi_pp, X, Y, c, (1,))
educ_slopes = np.concatenate([[psi_pp[c - 1 + 1]], psi_pp[c - 1 + 1] + psi_pp[c - 1
    + p:]])

aic = lambda ll, k: -2 * ll + 2 * k
print("education slope by cut:", np.round(educ_slopes, 4))
print("AIC proportional odds %.2f | partial %.2f | unrestricted %.2f"
      % (aic(ll_po, c - 1 + p), aic(ll_pp, len(psi_pp)), aic(ll_np, len(psi_np))))
```

**A different link.** Non-proportionality is sometimes link misspecification in disguise: the three links disagree in the tails ([Proposition 35.2.1](#)), so data can satisfy a cumulative probit or complementary log-log model with common slopes while violating the logit version. Trying the other two costs nothing.

**A scale effect.** Let the latent variable of [Theorem 36.3.1\(a\)](#) have a dispersion that depends on the regressors,  $\log \sigma(\mathbf{x}) = \mathbf{x}^\top \boldsymbol{\gamma}$ . The cumulative logits become  $(\theta_r - \mathbf{x}^\top \boldsymbol{\beta})e^{-\mathbf{x}^\top \boldsymbol{\gamma}}$ , non-parallel in a particular way: they fan out rather than crossing arbitrarily ([Exercise \(B2\)](#)). This costs  $p$  parameters instead of  $(c - 2)p$ , and when it fits, “the effect is the same but the responses are more spread out here” is a far more useful statement than a table of slopes.

**A different family.** Data that violate proportional odds badly may satisfy a sequential model ([Section 36.4](#)) with common slopes; failing that, the baseline-category model abandons the ordering but is never wrong about it.

**Report the cuts.** Sometimes no single number describes the effect, and the honest answer is [Proposition 36.3.2\(c\)](#). What is *not* a remedy is keeping the proportional odds fit and repairing its standard errors with a sandwich estimator: [Theorem 21.4.2](#) fixes the variance of an estimator whose target is still well defined, and here the target itself has evaporated.

## 36.3.4 Exercises

### 36.3.4.1 A. Check your understanding

#### Exercise A1.

A proportional odds model has  $c = 4$  and  $p = 3$ . How many parameters does it have? How many does the model with cut-specific slopes have, and what are the degrees of freedom of the score test of [Proposition 36.3.2](#) with  $A = \{1, 2, 3\}$  and with  $A = \{2\}$ ?

#### Exercise A2.

In a proportional odds fit the coefficient of a treatment indicator is  $\hat{\beta} = 0.7$ . Write down three correct sentences and one common incorrect one.

*Solution.* Correct: the odds that a treated subject exceeds any fixed level are  $e^{0.7} \approx 2$  times the odds for a control; the same factor applies at every level; the treated response is stochastically larger. Incorrect: “treatment doubles the probability of each category” — the effect on a category probability is a difference of two logistic densities times  $\hat{\beta}$ , and changes sign across the scale.

### 36.3.4.2 B. Practice

#### Exercise B1.

Using [Theorem 36.3.1\(b\)](#), predict what happens to the estimated slopes when the seven categories of [Example \(A seven-point scale\)](#) are collapsed to three, and compare with the actual refit in [Exercise \(B3\)](#). Does the comparison support or undermine the proportional odds assumption here?

*Solution.* Under the model the collapsed fit estimates the same  $\beta$ , so the two should differ only by sampling error, the collapsed one being less precise. The education slope moves from  $-0.1249$  to  $-0.1148$ , small against its standard error of  $0.0416$ , and the age slope from  $0.0942$  to  $0.1350$ , about one standard error: mild evidence at worst. Collapsing to three categories hides the failure that the cut-by-cut plot makes obvious, so a check should use all the cuts.

#### Exercise B2.

Derive the cumulative logits of the scale-effect model  $U_i = \mathbf{x}_{(i)}^\top \beta + \sigma(\mathbf{x}_{(i)})\varepsilon_i$ ,  $\log \sigma(\mathbf{x}) = \mathbf{x}^\top \gamma$ , with  $\varepsilon_i$  standard logistic. Show that it contains [Eq. 36.3.1](#) as the case  $\gamma = \mathbf{0}$ , and that its cumulative logits never cross as functions of  $\mathbf{x}$ .

*Solution.*  $P(Y_i \leq r) = P(\varepsilon_i \leq (\theta_r - \mathbf{x}^\top \beta)e^{-\mathbf{x}^\top \gamma})$ , so  $\text{logit } \gamma_{ir} = (\theta_r - \mathbf{x}^\top \beta)e^{-\mathbf{x}^\top \gamma}$ , which is [Eq. 36.3.1](#) when  $\gamma = \mathbf{0}$ . For two cuts  $r < s$  the difference of the cumulative logits is  $(\theta_r - \theta_s)e^{-\mathbf{x}^\top \gamma}$ , which has the sign of  $\theta_r - \theta_s$  for every  $\mathbf{x}$ , since the exponential is positive: the curves fan out but never cross, so the ordering constraint of [Proposition 36.2.1](#) holds everywhere.

### 36.3.4.3 C. Going deeper

#### Exercise C1.

Show that if the proportional odds model holds and  $\beta_j > 0$ , then  $E\{h(Y) \mid \mathbf{x}\}$  is increasing in  $x_j$  for *every* nondecreasing function  $h$ . Deduce that the sign of  $\beta_j$  is a statement that survives any recoding of the categories into numbers, as long as the recoding respects the order.

*Solution.* By [Theorem 36.3.1\(d\)](#), increasing  $x_j$  makes the response stochastically larger. Stochastic dominance is equivalent to  $E\{h(Y) \mid \mathbf{x}_2\} \geq E\{h(Y) \mid \mathbf{x}_1\}$  for every nondecreasing  $h$ : write  $E h(Y) = h(1) + \sum_{r=1}^{c-1} \{h(r+1) - h(r)\} P(Y > r)$ , a sum of nonnegative

coefficients times the survivor probabilities, each of which increases. Since any order-respecting scoring of the categories is such an  $h$ , the sign of  $\beta_j$  cannot be reversed by rescoreing.

### Exercise C2.

Let [Theorem 36.3.1](#) hold with a single binary regressor, let  $Y_1$  and  $Y_2$  be independent responses from the two groups, and let  $U_1, U_2$  be the latent variables of part (a) of that theorem.

- Show that  $P(U_2 > U_1) = E\{\Lambda(\beta - \varepsilon)\}$  with  $\varepsilon$  standard logistic, so that it is a function of  $\beta$  alone and strictly increasing in it.
- Show that the concordance probability of the *observed* responses,  $P(Y_2 > Y_1) + \frac{1}{2}P(Y_2 = Y_1)$ , is increasing in  $\beta$  for fixed cutpoints. (Hint: write it as  $E\{h(Y_2)\}$  for a nondecreasing  $h$  that depends only on the distribution of  $Y_1$ .) Then evaluate it at  $\beta = 1$  for  $c = 3$  with cutpoints  $(-0.5, 0.5)$  and again with  $(-3, 3)$ , and conclude that, unlike the odds ratio  $e^\beta$ , it is not free of the cutpoints.
- The sample version of (b) is the Mann–Whitney statistic divided by  $n_1 n_2$ , with ties counted half. Say in what sense the proportional odds model is a regression version of the Wilcoxon–Mann–Whitney test, and in what sense it is not.

*Solution.*

- $U_2 - U_1 = \beta + \varepsilon_2 - \varepsilon_1$ , so  $P(U_2 > U_1) = P(\varepsilon_1 < \beta + \varepsilon_2) = E\{\Lambda(\beta + \varepsilon_2)\}$ , which is  $E\{\Lambda(\beta - \varepsilon)\}$  by symmetry of the logistic law. No cutpoint appears, and  $\Lambda$  is strictly increasing, so this increases strictly with  $\beta$ ; at  $\beta = 1$  it is 0.6613.
- Conditioning on  $Y_1$  gives  $E\{h(Y_2)\}$  with  $h(b) = P(Y_1 < b) + \frac{1}{2}P(Y_1 = b)$ , nondecreasing in  $b$  and depending only on the distribution of  $Y_1$  — which does not involve  $\beta$ , group one being the reference. Raising  $\beta$  makes  $Y_2$  stochastically larger by [Theorem 36.3.1\(d\)](#), and  $E\{h(Y_2)\}$  then increases, by the equivalence used in [Exercise \(C1\)](#). It is not a function of  $\beta$  alone: at  $\beta = 1$  the cutpoints  $(-0.5, 0.5)$  give 0.6370 and  $(-3, 3)$  give 0.5482, while  $e^\beta$  is the same in both. Widely spaced cutpoints push mass into the middle category, so ties are common and the concordance probability is pulled towards  $\frac{1}{2}$ .
- It is a regression version in that the two-group comparison is the same stochastic-ordering comparison, extended to several regressors and to adjustment. It is not, in two ways: the rank test is distribution free where the model imposes proportional odds at every cut, a real restriction by [Example \(The proportional odds model does not hold\)](#); and what the model reports is  $\beta$ , not the concordance probability, which by (b) cannot be recovered from  $\beta$  without the cutpoints.

## 36.4 Sequential models

The cumulative models of the last two sections treat an ordered response as a coarsened continuum. Some ordered responses are not like that. A student reaches a master's degree only by passing through a bachelor's; an unemployment spell reaches its second year only by surviving its first. There the categories are *stages*, reached in order, and each stage is a decision taken by



those who got that far. A model built on that picture asks, for each stage, a question conditional on having reached it.

### Definition 36.4.1 — Sequential model

With  $\mathbf{Y}_i \sim \text{Mult}(m_i, \boldsymbol{\pi}_i)$  as in Definition 36.1.1, let

$$\delta_{ir} = P(Y_i = r \mid Y_i \geq r, \mathbf{x}_{(i)}), \quad r = 1, \dots, c-1, \quad (36.4.1)$$

be the **continuation ratios**, or discrete stopping probabilities. The **sequential model** with link  $F^{-1}$  is

$$\delta_{ir} = F(\theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_r), \quad r = 1, \dots, c-1, \quad (36.4.2)$$

for a continuous, strictly increasing distribution function  $F$ . With  $F = \Lambda$  and common slopes  $\boldsymbol{\beta}_r = \boldsymbol{\beta}$  it is the **continuation-ratio logit model**. The category probabilities are

$$\pi_{ir} = \delta_{ir} \prod_{s < r} (1 - \delta_{is}) \quad (r < c), \quad \pi_{ic} = \prod_{s < c} (1 - \delta_{is}). \quad (36.4.3)$$

Two features are visible at once. The sequential model is *not* symmetric in the direction of the scale: reversing the categories reverses which conditional probabilities are modelled. And the cutpoints are completely free, since Eq. 36.4.3 delivers a probability distribution for *any*  $\theta_r$  and  $\boldsymbol{\beta}_r$ . There is no analogue of the ordering constraint Eq. 36.2.3, so cut-specific slopes cost nothing in coherence here, where in a cumulative model they cost the model's existence.

## 36.4.1 The likelihood falls apart

### Theorem 36.4.1 — Factorization of the sequential likelihood

Let  $R_{ir} = \sum_{s > r} y_{is}$  be the number of subjects at covariate pattern  $i$  still “at risk” at stage  $r$ , so that  $R_{i1} = m_i$  and  $R_{ir} = y_{ir} + R_{i,r+1}$ . Under Definition 36.4.1,

$$\prod_{r=1}^c \pi_{ir}^{y_{ir}} = \prod_{r=1}^{c-1} \delta_{ir}^{y_{ir}} (1 - \delta_{ir})^{R_{i,r+1}}. \quad (36.4.4)$$

Consequently:

- (a) the log-likelihood is  $\ell = \sum_{r=1}^{c-1} \ell_r$ , where  $\ell_r(\theta_r, \boldsymbol{\beta}_r) = \sum_i \{y_{ir} \log \delta_{ir} + R_{i,r+1} \log(1 - \delta_{ir})\}$  is the log-likelihood of a binary regression with  $R_{ir}$  trials and  $y_{ir}$  successes;
- (b) if the parameters  $(\theta_r, \boldsymbol{\beta}_r)$  of different stages are unrestricted and distinct, the maximum likelihood estimate is obtained by  $c-1$  *separate* binary regressions, one on the subjects at risk at each stage; the information matrix is block diagonal, and the estimates for different stages are asymptotically independent;

- (c)  $Y_{ir} \mid R_{ir} \sim \text{Bin}(R_{ir}, \delta_{ir})$ , and these conditional distributions determine the multinomial law.

*Proof.* Count exponents on the two sides of Eq. 36.4.4. On the left, substituting Eq. 36.4.3, the factor  $\delta_{ir}$  appears only in  $\pi_{ir}$  for  $r < c$ , with exponent  $y_{ir}$ . The factor  $(1 - \delta_{is})$  appears in  $\pi_{ir}$  for every  $r > s$  and in  $\pi_{ic}$ , so its total exponent is  $\sum_{r>s} y_{ir} = R_{i,s+1}$ . That is exactly Eq. 36.4.4.

- (a) Take logarithms of Eq. 36.4.4 and sum over  $i$ ; by Eq. 36.4.2 the  $r$ th term involves only  $(\theta_r, \boldsymbol{\beta}_r)$ , and since  $R_{ir} = y_{ir} + R_{i,r+1}$ , the  $r$ th term is the binomial log-likelihood with  $R_{ir}$  trials,  $y_{ir}$  successes and success probability  $\delta_{ir}$ , up to a term free of the parameters.
- (b) A sum of functions of disjoint groups of parameters is maximized by maximizing each separately, and its second derivative matrix is block diagonal with the  $r$ th block  $\partial^2 \ell_r / \partial(\theta_r, \boldsymbol{\beta}_r)^{\otimes 2}$ . Taking expectations, the information is block diagonal, so by Theorem 34.5.1 the joint limiting distribution of the stacked estimate is normal with block diagonal covariance: the stages are asymptotically independent.
- (c) Write the multinomial mass function as  $P(\mathbf{Y}_i = \mathbf{y}_i) = h(\mathbf{y}_i) \prod_r \pi_{ir}^{y_{ir}}$  and use Eq. 36.4.4. The combinatorial factor  $h$  likewise factors as  $\prod_{r<c} \binom{R_{ir}}{y_{ir}}$ , because choosing which subjects stop at each stage in turn is the same as choosing the whole allocation. So the mass function is the product of  $c - 1$  binomial mass functions with the stated parameters. ■

*Idea.* A sequential model is barely a multinomial model at all: it is  $c - 1$  binary regressions with a shared parameterization. Separation (Theorem 35.3.1), the links (Proposition 35.2.1), residuals (Proposition 35.4.1) and case-control sampling of Theorem 35.5.1 apply stage by stage, with no new theory.

### Corollary 36.4.2 — One regression on expanded data

Let the slopes be common,  $\boldsymbol{\beta}_r = \boldsymbol{\beta}$ , and the data ungrouped ( $m_i = 1$ ). Build an expanded data set with one row for each pair  $(i, r)$  with  $r \leq \min(y_i, c - 1)$ , response  $\mathbf{1}\{y_i = r\}$ , and regressors  $(\mathbf{e}_r, -\mathbf{x}_{(i)})$ , where  $\mathbf{e}_r$  is the  $r$ th coordinate vector of length  $c - 1$ . Then an ordinary binary regression with link  $F^{-1}$  on the expanded data has, as a function of  $(\boldsymbol{\theta}, \boldsymbol{\beta})$ , the log-likelihood of the sequential model; the two fits, their standard errors and their tests coincide.

*Proof.* With  $m_i = 1$ ,  $y_{ir} = \mathbf{1}\{y_i = r\}$  and  $R_{i,r+1} = \mathbf{1}\{y_i > r\}$ . The  $(i, r)$  term of Eq. 36.4.4 is  $\delta_{ir}$  when  $y_i = r$ ,  $1 - \delta_{ir}$  when  $y_i > r$ , and 1 when  $y_i < r$ , so the rows with  $r > \min(y_i, c - 1)$  contribute nothing and may be dropped. The surviving rows contribute exactly the Bernoulli likelihood with success probability  $\delta_{ir} = F(\theta_r - \mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ , which is the model with linear predictor  $(\mathbf{e}_r, -\mathbf{x}_{(i)})^\top (\boldsymbol{\theta}^\top, \boldsymbol{\beta}^\top)^\top$ . ■

The expanded data set has  $\sum_i \min(y_i, c - 1)$  rows, one per stage each subject reached. That rows from the same subject are not independent looks alarming and is not a problem: Theorem 36.4.1

says the likelihood is *exactly* the product over rows — a factorization, not an approximation.

## 36.4.2 The complementary log-log link joins the two families

The cumulative and sequential families are genuinely different — except at one link, where they are the same.

### Proposition 36.4.3 — Sequential and cumulative complementary log-log models agree

Take  $F(z) = 1 - \exp(-e^z)$  and common slopes. If the sequential model Eq. 36.4.2 holds with cutpoints  $\tilde{\theta}_1, \dots, \tilde{\theta}_{c-1}$ , then the cumulative complementary log-log model Eq. 36.2.2 holds with the same  $\beta$  and cutpoints

$$\theta_r = \log\left(\sum_{s \leq r} e^{\tilde{\theta}_s}\right), \quad r = 1, \dots, c-1, \quad (36.4.5)$$

which are automatically increasing. The map  $\tilde{\theta} \mapsto \theta$  is a bijection onto the increasing sequences, so the two models are the same family, and the maximum likelihood fits agree.

*Proof.* Since  $\{Y_i \geq r\} = \{Y_i > r-1\}$ ,

$$P(Y_i > r \mid Y_i > r-1) = 1 - \delta_{ir} = \exp\{-e^{\tilde{\theta}_r} e^{-\mathbf{x}_{(i)}^\top \beta}\}.$$

Multiplying these for  $s = 1, \dots, r$  telescopes to  $P(Y_i > r) = \exp\{-H_r e^{-\mathbf{x}_{(i)}^\top \beta}\}$  with  $H_r = \sum_{s \leq r} e^{\tilde{\theta}_s}$ , so  $\log\{-\log P(Y_i > r)\} = \log H_r - \mathbf{x}_{(i)}^\top \beta$ , which is Eq. 36.2.2 with  $G(z) = 1 - \exp(-e^z)$  and  $\theta_r = \log H_r$ . The  $H_r$  are strictly increasing because  $e^{\tilde{\theta}_s} > 0$ , and  $\tilde{\theta}_r = \log(H_r - H_{r-1})$  recovers the sequential cutpoints from any increasing  $\theta$ , with  $H_0 = 0$ ; so the map is a bijection. Two parameterizations of the same set of distributions have the same maximized likelihood. ■

This is the observation of Läärä and Matthews (1985), and it explains why the complementary log-log link is the natural one for stage data: by Eq. 36.2.5 the same model is a grouped proportional hazards model, with  $\delta_{ir}$  the discrete hazard of stopping at stage  $r$ . No such identity holds for the logit link, and Figure 36.4.1 shows how far apart the two families then are.

## 36.4.3 The two families on the same data

### Example. (Self-placement, fitted sequentially)

The self-placement data of Example (A seven-point scale) are not a stage process — nobody becomes extremely conservative by first passing through “extremely liberal” —

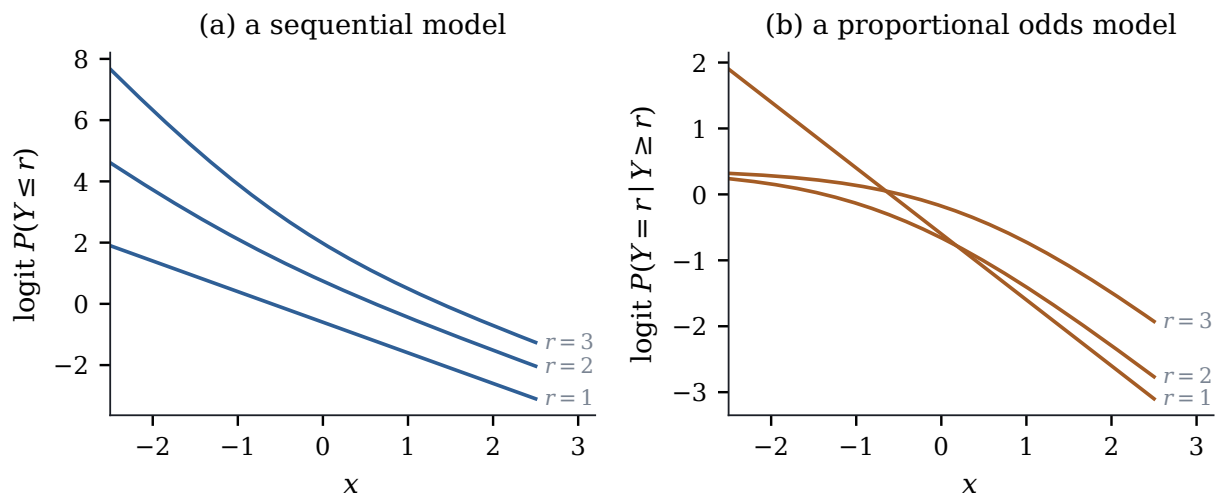


Figure 36.4.1: Each family is simple in its own coordinates and curved in the other's. (a) Cumulative logits implied by a sequential logit model with common slopes. (b) Continuation-ratio logits implied by a proportional odds model. Parallel straight lines in one panel would mean the other family contained this one.

but the sequential model is still a parameterization of the seven category probabilities, and comparing the fits shows what each family buys.

The at-risk counts are 944, 928, 825, 678, 422 and 252, and the expanded data set has 4049 rows. One binary logistic regression on it, as in [Corollary 36.4.2](#), gives cut-points  $(-3.9705, -1.9870, -1.4252, -0.3784, -0.2744, 1.9824)$  and slopes 0.0454 (age, per decade),  $-0.0866$  (education) and 0.0174 (income), with standard errors 0.0270, 0.0303 and 0.0079. The signs agree with the proportional odds fit and the magnitudes are smaller, as one should expect: these are effects on a single stage transition, not on the whole scale. The multinomial log-likelihood is  $-1618.2981$ , which the script confirms equals the log-likelihood the binary regression reports — [Theorem 36.4.1](#) to machine precision.

Akaike's criterion is 3254.60 for the sequential model against 3245.71 for proportional odds ([Definition 29.2.2](#)): with nine parameters each, the cumulative model describes these data better, as a scale of opinion rather than a ladder of stages should. Letting every stage have its own slopes raises the log-likelihood to  $-1578.3794$  and lowers the criterion to 3204.76, just below the unrestricted cumulative model's 3208.29. Once the common-slope restriction goes, the two families have the same  $(c-1)(1+p)$  parameters and fit these data about equally well.

```
import numpy as np
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
y = anes["selfLR"].astype(int).to_numpy()
X = np.column_stack([anes["age"] / 10.0, anes["educ"], anes["income"]])
n, p = X.shape
c = 7

def seq_probs(theta, beta, X, F=lambda z: 1.0 / (1.0 + np.exp(-z))):
    """Category probabilities of the sequential model with  $P(Y = r | Y \geq r) = F(\theta_r - x'\beta)$ ."""
```

```

delta = F(theta[None, :] - (X @ beta)[:, None])
surv = np.column_stack([np.ones(len(X)), np.cumprod(1.0 - delta, axis=1)])
return np.column_stack([delta * surv[:, :-1], surv[:, -1]])

def seq_loglik(theta, beta, X, y, F=lambda z: 1.0 / (1.0 + np.exp(-z))):
    P = seq_probs(theta, beta, X, F)
    return float(np.sum(np.log(P[np.arange(len(y)), y - 1])))

```

With cut-specific slopes the fit is [Theorem 36.4.1\(b\)](#): one logistic regression per stage, on the subjects still at risk.

```

# unrestricted sequential model: one binary logistic fit per stage, on those still
at risk
stage_beta, stage_ll = [], 0.0
for r in range(1, c):
    at_risk = y >= r
    Z = np.column_stack([np.ones(at_risk.sum()), -X[at_risk]])
    fit_r = sm.Logit((y[at_risk] == r).astype(float), Z).fit(dis=0)
    stage_beta.append(np.asarray(fit_r.params))
    stage_ll += fit_r.llf
    print(f"stage {r}: at risk {at_risk.sum():4d}, stop {int((y == r).sum()):4d}, "
          f"theta {fit_r.params[0]:7.4f}, beta {np.round(fit_r.params[1:], 4)}")

print(f"sum of the {c - 1} binary log-likelihoods: {stage_ll:.4f}")

```

With a common slope the fit is [Corollary 36.4.2](#): one logistic regression on the expanded data.

```

# common slope: one binary logistic regression on the expanded data set
rows_X, rows_y = [], []
for i in range(n):
    for r in range(1, min(y[i], c - 1) + 1):          # stages this respondent reached
        stage = np.zeros(c - 1)
        stage[r - 1] = 1.0
        rows_X.append(np.concatenate([stage, -X[i]]))
        rows_y.append(float(y[i] == r))
Xe, ye = np.array(rows_X), np.array(rows_y)

fit_seq = sm.Logit(ye, Xe).fit(dis=0)
theta_seq, beta_seq = fit_seq.params[:c - 1], fit_seq.params[c - 1:]
se_seq = fit_seq.bse[c - 1:]
ll_seq = seq_loglik(theta_seq, beta_seq, X, y)

print(f"expanded data: {len(ye)} rows from {n} respondents")
print("cutpoints", np.round(theta_seq, 4))
print("slopes", np.round(beta_seq, 4), "se", np.round(se_seq, 4))
print(f"binary log-likelihood {fit_seq.llf:.4f} = multinomial log-likelihood {ll_seq:.4f}")

```

Changing the link to the complementary log-log turns the same expanded regression into a cumulative model, by [Proposition 36.4.3](#). The script computes the cumulative cutpoints from [Eq. 36.4.5](#) and confirms that the two ways of assembling the category probabilities agree to within  $3 \times 10^{-16}$ .

```

# Laara and Matthews: with a complementary log-log link the sequential model IS a
cumulative model

```

```

c11 = lambda z: 1.0 - np.exp(-np.exp(z))
fit_c11 = sm.GLM(ye, Xe, family=sm.families.Binomial(sm.families.links.CLogLog())).
    fit()
theta_t, beta_c = fit_c11.params[:c - 1], fit_c11.params[c - 1:]
theta_cum = np.log(np.cumsum(np.exp(theta_t)))           # the cumulative
    model's cutpoints

P_step = seq_probs(theta_t, beta_c, X, F=c11)             # built stage by
    stage
gamma_cum = c11(theta_cum[None, :] - (X @ beta_c)[: , None]) # built from
    cumulative probabilities
P_cum = np.diff(np.column_stack([np.zeros(n), gamma_cum, np.ones(n)]), axis=1)

print("sequential cutpoints ", np.round(theta_t, 4))
print("cumulative cutpoints ", np.round(theta_cum, 4))
print("largest difference between the two sets of fitted probabilities:",
    float(np.max(np.abs(P_step - P_cum))))

```

### Example. (When the mechanism really is sequential)

Generate 4000 observations from a four-stage sequential logit model with one standard normal regressor, cutpoints  $(-0.6, 0, 0.5)$  and slope  $\beta = 1$ ; the category counts come out as 1487, 1094, 693 and 726. The sequential fit recovers the slope as 0.9761, standard error 0.0309. The proportional odds fit of the same data, with the same number of parameters, returns 1.1352, which is not estimating  $\beta$  at all, and Akaike's criterion is 9444.56 against 9515.66. A gap of seventy is not a close call.

The moral is not that one family is better — reverse the simulation and the verdict reverses with it — but that the choice is a modelling decision about *how the response came about*, on which the data have plenty to say.

```

# a mechanism that really is sequential, fitted by both models
def simulate_sequential(n, theta, beta, rng):
    X = rng.normal(size=(n, len(beta)))
    delta = 1.0 / (1.0 + np.exp(-(theta[None, :] - (X @ beta)[: , None])))
    stop = rng.random(delta.shape) < delta
    y = np.where(stop.any(axis=1), stop.argmax(axis=1) + 1, len(theta) + 1)
    return X, y

rng = np.random.default_rng(20360436)
theta0, beta0 = np.array([-0.6, 0.0, 0.5]), np.array([1.0])
Xs, ys = simulate_sequential(4000, theta0, beta0, rng)
print("simulated counts:", np.bincount(ys)[1:])

```

## 36.4.4 Choosing a family

Three families are now available: the cumulative model ([Definition 36.2.1](#)), the sequential model ([Definition 36.4.1](#)) and, through [Exercise \(C1\)](#), adjacent-category logits. They are non-nested, have the same dimension in their common-slope forms, and often fit real data about equally well, so a criterion such as AIC will decline to choose. The mechanism should choose.

- **Cumulative** when the categories are a graded measurement of something continuous: severity, agreement, quality. Its slope is invariant to how the scale was cut ([Theorem 36.3.1](#)), which makes results comparable across studies.
- **Sequential** when the categories are stages that must be passed through, especially for durations and attainments. Its coefficients describe one transition at a time, and [Theorem 36.4.1](#) makes cut-specific slopes cheap.
- **Adjacent-category** when local comparisons are the meaningful ones: category  $r + 1$  rather than  $r$ , among those in one of the two.

None of them addresses overdispersion relative to the multinomial. [Chapter 38](#) treats quasi-likelihood and overdispersion, and [Chapter 40](#) the random-effects models that repeated categorical measurements need.

## 36.4.5 Exercises

### 36.4.5.1 A. Check your understanding

#### Exercise A1.

How many parameters has the sequential model with common slopes? With stage-specific slopes? How many rows has the expanded data set of [Corollary 36.4.2](#) when every subject falls in the last category?

*Solution.* Common slopes:  $(c - 1) + p$ . Stage-specific:  $(c - 1)(1 + p)$ . If every  $y_i = c$  then each subject contributes  $c - 1$  rows, giving  $n(c - 1)$ ; every response in the expanded data is zero, and the fit does not exist — a separation of the kind in [Theorem 35.3.1](#).

#### Exercise A2.

Show by a three-category example that reversing the order of the categories changes the sequential model, but not the cumulative one ([Exercise \(A1\)](#)).

### 36.4.5.2 B. Practice

#### Exercise B1.

Interpret  $\delta_{ir}$  as a discrete hazard and  $\prod_{s \leq r} (1 - \delta_{is})$  as a survivor function. Show that under [Proposition 36.4.3](#) the model says  $P(Y_i > r \mid \mathbf{x}_{(i)}) = S_0(r)^{\exp(-\mathbf{x}_{(i)}^\top \boldsymbol{\beta})}$  for a baseline survivor function  $S_0$ , and identify  $S_0$ .

*Solution.* From the proof of [Proposition 36.4.3](#),  $P(Y_i > r) = \exp\{-H_r e^{-\mathbf{x}^\top \boldsymbol{\beta}}\}$ , so with  $S_0(r) = e^{-H_r}$  we get  $P(Y_i > r) = S_0(r)^{\exp(-\mathbf{x}^\top \boldsymbol{\beta})}$ . Here  $S_0$  is the survivor function at  $\mathbf{x} = \mathbf{0}$  and  $H_r = \sum_{s \leq r} e^{\hat{\theta}_s}$  is the baseline cumulative hazard, which is exactly the grouped-data statement [Eq. 36.2.5](#).

### 36.4.5.3 C. Going deeper

**Exercise C1.**

Let  $T_i$  be a discrete survival time taking values  $1, 2, \dots$ , possibly right-censored at a known time  $c_i$ . Write the likelihood in terms of the discrete hazards  $\lambda_{ir} = P(T_i = r \mid T_i \geq r)$ , and show that it is again the expanded binary likelihood of [Corollary 36.4.2](#), now with subject-specific numbers of rows. What does censoring correspond to in [Theorem 36.4.1](#)?

*Solution.* The contribution of an uncensored subject is  $\lambda_{i,t_i} \prod_{s < t_i} (1 - \lambda_{is})$  and of a censored one  $\prod_{s \leq c_i} (1 - \lambda_{is})$ . Both are products of Bernoulli terms over the periods the subject was at risk, so the expanded binary regression fits them, with a censored subject contributing only zero responses. In [Theorem 36.4.1](#) censoring is the case of a subject who is removed from the risk set without stopping: it reduces  $R_{ir}$  for later  $r$  without adding to any  $y_{ir}$ . This is why continuation-ratio models and discrete-time survival models are the same subject.

**Exercise C2.**

Does the sequential model have a collapsing invariance like [Theorem 36.3.1\(b\)](#)? Show that merging the *last* two categories leaves the model and the remaining parameters intact, but merging any other adjacent pair does not.

## 36.5 Summary and Notes

### 36.5.1 Summary

*Idea.*

1. The multinomial distribution is an exponential family with natural parameters the baseline-category logits, cumulant  $b(\boldsymbol{\eta}) = \log(1 + \sum_s e^{\eta_s})$ , mean  $m\nabla b$  and covariance  $m(\text{diag}(\boldsymbol{\pi}) - \boldsymbol{\pi}\boldsymbol{\pi}^\top)$  ([Lemma 36.1.1](#)). The chapter then chooses which  $c - 1$  logits to make linear.
2. The baseline-category logit model ([Definition 36.1.1](#)) is the canonical-link generalized linear model for that family: score  $\mathbf{X}^\top(\mathbf{y}_r - \boldsymbol{\mu}_r)$  per category, observed information equal to expected, and a log-likelihood strictly concave when  $\text{rank}(\mathbf{X}) = p$ , so the maximum is unique when it exists ([Theorem 36.1.2](#)).
3. Changing the baseline is a linear reparameterization, leaving estimates, fitted values, the maximized likelihood and the standard errors of contrasts unchanged ([Proposition 36.1.3](#)). A coefficient is a log odds ratio against the baseline, not a derivative of a probability ([Proposition 36.1.4](#)).
4. For an ordered response, modelling  $P(Y \leq r)$  uses the ordering ([Definition 36.2.1](#)). Fitted probabilities are positive only where the linear predictors stay in order: automatic with common slopes, a real restriction without them ([Proposition 36.2.1](#)). Score and expected information are  $\sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1}(\tilde{\mathbf{y}}_i - m_i \tilde{\boldsymbol{\pi}}_i)$  and  $\sum_i m_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i$  ([Proposition 36.2.2](#)), and the log-likelihood is concave when the link density is log-concave.



5. The proportional odds model has a latent-variable derivation, slopes that survive any collapsing of adjacent categories, cutpoints that never need constraining, and one odds ratio applying at every cut ([Theorem 36.3.1](#)). The collapsing invariance is also how it is checked.
6. That restriction is tested by a score statistic computed from the restricted fit alone, on  $(c - 2)|A|$  degrees of freedom, and diagnosed by plotting the slopes of the  $c - 1$  dichotomized fits ([Proposition 36.3.2](#)). When it fails: partial proportional odds, a different link, a scale effect, a different family, or an honest table of slopes.
7. A sequential model ([Definition 36.4.1](#)) models the probability of stopping at each stage. Its likelihood factors into  $c - 1$  binomial likelihoods, so with stage-specific slopes it is  $c - 1$  separate binary regressions with asymptotically independent estimates ([Theorem 36.4.1](#)), and with a common slope one binary regression on expanded data ([Corollary 36.4.2](#)). No ordering constraint is needed.
8. With the complementary log-log link the two ordinal families coincide, cutpoints related by a cumulative sum on the exponential scale ([Proposition 36.4.3](#)): the grouped-data proportional hazards model. With the logit link they differ, and the data can tell them apart.

## 36.5.2 Notes and sources

**Coverage.** The topics follow Agresti (2015, chapter 6) and Fahrmeir, Kneib, Lang and Marx (2021, §6.2–6.3); the treatment, proofs, examples and exercises are our own.

**Baseline-category logits.** The model grew up independently in biostatistics and econometrics. McFadden (1974) derived it from random utility maximization, which is the source of the name *multinomial logit* and of the discrete-choice literature; the underlying axiom is Luce’s (1959), and the independence of irrelevant alternatives in [Exercise \(C2\)](#) is its most famous consequence and problem. McFadden’s conditional logit model, in which regressors vary by category, is the standard extension, not treated here. Existence of the maximum likelihood estimate is settled by Albert and Anderson (1984); [Theorem 35.3.1](#) is the binary case. The individualized fits of the remark in [Section 36.1](#) are analysed by Begg and Gray (1984).

**Cumulative models.** Modelling cumulative probabilities with a common slope goes back to Walker and Duncan (1967). McCullagh (1980) named the proportional odds model and gave the latent-variable derivation, the general cumulative link family and the invariance properties of [Theorem 36.3.1](#); McCullagh and Nelder (1989, chapter 5) is a compressed version. Anderson (1984) introduced a different ordinal family, the *stereotype* model, which estimates category scores rather than assuming them. That the log-likelihood of a cumulative link model is concave whenever the link density is log-concave was proved by Pratt (1981) and, for grouped data specifically, by Burrridge (1981); it is the reason Fisher scoring is as docile here as it is. The asymptotics of [Section 36.2](#) are imported rather than proved, this model not being a generalized linear model; the standard accounts are van der Vaart (1998) and Lehmann and Casella (1998). Agresti (2010) is the book-length treatment, and covers the scale-effect models of [Section 36.3](#) and the connection with rank statistics in [Exercise \(C2\)](#).

**Checking proportional odds.** The score test of [Proposition 36.3.2](#) is the one most software reports; Brant (1990) gave the Wald version based on the separate dichotomized fits, which is where the habit of plotting them comes from; it tests the same hypothesis as the score statistic but is a different statistic. Partial proportional odds models are due to Peterson and Harrell (1990), who also noted that such fits can produce negative probabilities — the failure of

**Proposition 36.2.1.**

**Sequential models.** Continuation-ratio logits go back to Fienberg (1980); the sequential formulation used here, with a link and a linear predictor at each stage, is Tutz’s (1991). Fahrmeir and Tutz (2001) and Tutz (2012) give the textbook treatments, the latter with a taxonomy of ordinal models built out of binary ones. The factorization of [Theorem 36.4.1](#) is the algebra that underlies discrete-time survival analysis, where the stages are time intervals and the continuation ratios are hazards; the continuous-time ancestor is Cox (1972), and [Proposition 36.4.3](#), due to Läärä and Matthews (1985), connects the two.

**Data.** Both examples use public-domain data shipped with statsmodels: the travel-mode data of [Example \(Choosing how to travel\)](#) come from the online archive of Greene’s econometrics text, and [Example \(A seven-point scale\)](#) uses an extract of the 1996 American National Election Study, whose education and income variables are ordered scales treated here as numerical scores. Treating them as factors changes the coefficients but no conclusion about proportional odds. Neither example is a causal statement, for the reasons in [Chapter 25](#).

**What is missing.** Responses clustered, repeated or overdispersed relative to [Lemma 36.1.1](#) need more than this chapter offers: [Chapter 38](#) takes up quasi-likelihood and overdispersion, [Chapter 40](#) random effects and estimating equations, and [Chapter 39](#) Bayesian fitting, where the latent variable of [Theorem 36.3.1\(a\)](#) becomes a data augmentation scheme.

### 36.5.3 References

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# Chapter 37

## Counts

A count has no upper bound, cannot be negative, and is nearly always more variable when it is large than when it is small. The linear model fits none of these descriptions and no transformation repairs all of them at once. The generalized linear model does better: put the linear predictor on the log scale, so that the mean is multiplicative, and let the distribution supply a variance that grows with the mean. That is the Poisson log-linear model, the starting point of this chapter.

It is only a starting point. Real counts are almost always more variable than the Poisson allows, often have far more zeros, and are sometimes not a sample of independent events at all but a contingency table in disguise. Each complaint has a model attached to it: exposure and rates, the exact connection between Poisson counts and multinomial proportions, the negative binomial, and the two standard ways to model an excess of zeros.

**What you need.** [Chapter 34](#) (the generalized linear model: [Definition 34.2.1](#), the likelihood equations [Theorem 34.3.1](#), iteratively reweighted least squares [Theorem 34.4.1](#), the deviance [Definition 34.5.1](#) and [Theorem 34.5.5](#), and the three test statistics [Proposition 34.5.3](#)). [Chapter 35](#) supplies the separation theorem [Theorem 35.3.1](#), whose Poisson counterpart is [Proposition 37.1.3](#), and [Chapter 36](#) the baseline-category logit model [Definition 36.1.1](#), which [Section 37.3](#) shows is a log-linear model in disguise. From earlier parts: the two-way layout and its interaction ([Definition 16.2.1](#) and [Proposition 16.2.1](#)), variance-stabilizing transformations ([Theorem 22.4.1](#)), the sandwich covariance estimator ([Theorem 21.4.2](#)), the likelihood ratio test on the boundary of the parameter space ([Theorem 32.5.2](#)), and model selection by AIC and BIC ([Definition 29.2.2](#)).

### 37.0.1 Roadmap

- [Poisson regression](#)
- [Offsets and rates](#)
- [Poisson and multinomial](#)
- [Negative binomial models](#)
- [Zero inflation](#)
- [Summary and notes](#)

## 37.1 Poisson regression

Counts are nonnegative integers, their spread grows with their level, and a regressor acts on them multiplicatively. No transformation satisfies all three at once ([Chapter 22](#)): the square root of a Poisson count has nearly constant variance but a mean no longer linear in the regressors, the logarithm a nearly linear mean but a variance that still moves ([Exercise \(B2\)](#)), and neither is defined at zero. The generalized linear model separates the requirements: the link takes the mean, the distribution takes the variance, and nothing is done to the data.

### 37.1.1 The Poisson as an exponential dispersion family

Write the Poisson mass function in the form [Definition 34.1.1](#) asks for. For  $y = 0, 1, 2, \dots$  and  $\mu > 0$ ,

$$\Pr(Y = y) = \frac{e^{-\mu} \mu^y}{y!} = \exp\{y \log \mu - \mu - \log y!\},$$

so the natural parameter is  $\theta = \log \mu$ , the cumulant function is  $b(\theta) = e^\theta$ , the dispersion is  $\phi = 1$  and the weight is  $w = 1$ . By [Proposition 34.1.2](#),

$$E(Y) = b'(\theta) = e^\theta = \mu, \quad \text{Var}(Y) = \phi b''(\theta) = e^\theta = \mu,$$

so the variance function is  $V(\mu) = \mu$ . The Poisson has no free dispersion parameter: once the mean is fixed, so is the variance. That is the most consequential fact in this chapter, and [Sections 37.4](#) and [37.5](#) are about what to do when the data disagree with it. The canonical link  $\theta = \log \mu$  also makes the mean multiplicative and keeps it positive at every linear predictor: for once convenience and interpretation point the same way.

#### Definition 37.1.1 — The Poisson log-linear model

Let  $Y_1, \dots, Y_n$  be independent, with  $Y_i$  Poisson distributed with mean  $\mu_i > 0$ , and let  $\mathbf{x}_{(i)}$  be the  $i$ th row of a model matrix  $\mathbf{X}$  of full column rank  $p$ . The **Poisson log-linear model** is the generalized linear model ([Definition 34.2.1](#)) with this random component, the log link and the linear predictor  $\eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ :

$$\log \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \quad \text{equivalently} \quad \mu_i = \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}) = e^{\beta_1 x_{i1}} \dots e^{\beta_p x_{ip}}. \quad (37.1.1)$$

The second form carries the interpretation. Increasing the  $j$ th regressor by one unit, the others held anywhere, multiplies the mean by  $e^{\beta_j}$ , the **rate ratio** of that regressor. It does not depend on the levels of the others, which is the log-scale version of “no interaction”.

*Remark* (The identity link). Nothing forces the log link. The model  $\mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  gives regressors additive effects on the count, which is sometimes the question asked: an intervention that prevents three events a year, not one that cuts events by ten per cent. The price is a constrained parameter space, since  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta} > 0$  is needed for every  $i$  and the constraint can bind at the maximum ([Exercise \(C1\)](#)).

## 37.1.2 The likelihood equations

### Theorem 37.1.1 — The Poisson likelihood equations

In the model of [Definition 37.1.1](#) the log-likelihood is

$$\ell(\boldsymbol{\beta}) = \sum_{i=1}^n \{y_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta} - \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})\} - \sum_{i=1}^n \log y_i!, \quad (37.1.2)$$

and, writing  $\boldsymbol{\mu} = \boldsymbol{\mu}(\boldsymbol{\beta})$  for the vector of means and  $\mathbf{W} = \text{diag}(\mu_1, \dots, \mu_n)$ :

- (a) the score is  $\partial\ell/\partial\boldsymbol{\beta} = \mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu})$ , so the likelihood equations are  $\mathbf{X}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$ ;
- (b) the observed and expected information coincide and equal  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ ;
- (c)  $\ell$  is concave, and strictly concave when  $\text{rank}(\mathbf{X}) = p$ , so a stationary point is the unique maximum.

*Proof.* The Poisson has  $\phi = 1$ ,  $w_i = 1$ ,  $V(\mu) = \mu$  and, under the canonical link,  $\partial\mu_i/\partial\eta_i = \mu_i$ . Substituting these in [Theorem 34.3.1](#)(d) gives the score  $\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu})$  and observed information equal to expected information  $\mathbf{X}^\top \text{diag}\{w_i V(\mu_i)\} \mathbf{X} = \mathbf{X}^\top \mathbf{W} \mathbf{X}$ , which are (a) and (b). For (c), that Hessian is nonnegative definite because  $\mathbf{a}^\top \mathbf{X}^\top \mathbf{W} \mathbf{X} \mathbf{a} = \sum_i \mu_i (\mathbf{x}_{(i)}^\top \mathbf{a})^2 \geq 0$ , and the sum vanishes only if  $\mathbf{X} \mathbf{a} = \mathbf{0}$ , since every  $\mu_i > 0$ . Under full column rank that forces  $\mathbf{a} = \mathbf{0}$ , so  $\ell$  is strictly concave. ■

The general score of [Theorem 34.3.1](#) carries a factor  $(\partial\mu_i/\partial\eta_i)/V(\mu_i)$ , here  $\mu_i/\mu_i = 1$ ; what is left has the form of the normal equations  $\mathbf{X}^\top(\mathbf{y} - \hat{\mathbf{Y}}) = \mathbf{0}$ , with the fitted mean vector in place of the projection  $\mathbf{M}\mathbf{y}$ . The residual  $\mathbf{y} - \hat{\boldsymbol{\mu}}$  is orthogonal to every column of  $\mathbf{X}$ , although  $\hat{\boldsymbol{\mu}}$  is not a linear function of  $\mathbf{y}$  and there is no projection in sight ([Exercise \(B2\)](#)).

### Theorem 37.1.2 — Fitted values reproduce observed margins

Let  $\hat{\boldsymbol{\mu}}$  solve the likelihood equations of [Theorem 37.1.1](#). Then  $\mathbf{x}_j^\top \hat{\boldsymbol{\mu}} = \mathbf{x}_j^\top \mathbf{y}$  for every column  $\mathbf{x}_j$  of  $\mathbf{X}$ , and more generally  $\mathbf{a}^\top \hat{\boldsymbol{\mu}} = \mathbf{a}^\top \mathbf{y}$  for every  $\mathbf{a} \in \mathcal{C}(\mathbf{X})$ . In particular:

- (a) if the model has an intercept, the fitted counts sum to the observed total,  $\sum_i \hat{\mu}_i = \sum_i y_i$ ;
- (b) if  $\mathbf{X}$  contains the indicator of a group  $G$ , the fitted counts of that group sum to its observed total,  $\sum_{i \in G} \hat{\mu}_i = \sum_{i \in G} y_i$ ;
- (c) if  $\mathbf{X}$  contains a quantitative regressor  $\mathbf{x}_j$ , the fitted counts reproduce the observed  $\mathbf{x}_j$ -weighted total.

*Proof.* The column identities are [Corollary 34.3.2](#) with  $w_i = 1$ : the likelihood equations say  $\mathbf{X}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$ , that is,  $\mathbf{x}_j^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = 0$  for each  $j$ . The extension to  $\mathcal{C}(\mathbf{X})$  is linearity: any  $\mathbf{a} \in \mathcal{C}(\mathbf{X})$  is  $\mathbf{X}\mathbf{c}$  for some  $\mathbf{c}$ , and  $\mathbf{a}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{c}^\top \mathbf{X}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = 0$ . Parts (a), (b) and

(c) are this statement for  $\mathbf{a} = \mathbf{1}$ , for the group indicator, and for  $\mathbf{x}_j$ .

■

This is what makes log-linear models for contingency tables behave so tidily; [Section 37.3](#) uses it throughout.

### 37.1.3 When the maximum likelihood estimate exists

Strict concavity says that a maximum, if attained, is unique; not that one is attained. The log-likelihood [Eq. 37.1.2](#) can increase along a ray for ever, and then the fitting algorithm diverges. The condition below is the Poisson counterpart of separation in logistic regression ([Theorem 35.3.1](#)).

#### Proposition 37.1.3 — Existence of the maximum likelihood estimate

Let  $\text{rank}(\mathbf{X}) = p$ . The log-likelihood [Eq. 37.1.2](#) attains its maximum at a finite  $\hat{\boldsymbol{\beta}}$  if and only if there is **no** vector  $\mathbf{d} \neq \mathbf{0}$  with

$$\mathbf{x}_{(i)}^\top \mathbf{d} \leq 0 \text{ for every } i, \quad \text{and} \quad \mathbf{x}_{(i)}^\top \mathbf{d} = 0 \text{ whenever } y_i > 0. \quad (37.1.3)$$

*Proof.* Write  $u_i = \mathbf{x}_{(i)}^\top \mathbf{d}$  and, for a fixed  $\boldsymbol{\beta}_0$ ,

$$\ell(\boldsymbol{\beta}_0 + t\mathbf{d}) = \sum_i y_i \eta_i^0 + t \sum_i y_i u_i - \sum_i e^{\eta_i^0} e^{tu_i} - \sum_i \log y_i!,$$

where  $\eta_i^0 = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}_0$ . Suppose such a  $\mathbf{d}$  exists. Then  $\sum_i y_i u_i = 0$ , because every term with  $y_i > 0$  has  $u_i = 0$ . Since every  $u_i \leq 0$ , each  $e^{tu_i}$  is nonincreasing in  $t$ , so  $\ell(\boldsymbol{\beta}_0 + t\mathbf{d})$  is nondecreasing in  $t$ ; and it is strictly increasing unless every  $u_i = 0$ , which full column rank excludes. So the supremum is approached only as  $t \rightarrow \infty$  and is not attained.

Conversely, suppose no such  $\mathbf{d}$  exists, and fix a unit vector  $\mathbf{d}$ . Either some  $u_i > 0$ , and  $\ell(\boldsymbol{\beta}_0 + t\mathbf{d}) \rightarrow -\infty$  because an exponential beats a linear term; or every  $u_i \leq 0$ , in which case some  $i$  with  $y_i > 0$  has  $u_i < 0$ , so  $\sum_i y_i u_i < 0$  while  $-\sum_i e^{\eta_i^0} e^{tu_i} \leq 0$ , and again  $\ell \rightarrow -\infty$ . So there is a  $t_{\mathbf{d}}$  beyond which  $\ell(\boldsymbol{\beta}_0 + t\mathbf{d}) < \ell(\boldsymbol{\beta}_0) - 1$ . Continuity alone does not make that threshold uniform in  $\mathbf{d}$ ; concavity does. By part (c) the difference quotient  $\{\ell(\boldsymbol{\beta}_0 + t\mathbf{d}) - \ell(\boldsymbol{\beta}_0)\}/t$  is nonincreasing in  $t$ , so once it falls below  $-1/t_{\mathbf{d}}$  it stays below, and by continuity it does so on a neighbourhood of  $\mathbf{d}$ . Finitely many such neighbourhoods cover the unit sphere, giving a single  $t^*$  outside whose ball  $\ell < \ell(\boldsymbol{\beta}_0) - 1$ ; and a continuous function on that closed ball attains its maximum.

■

The condition fails when a regressor is the indicator of a set of observations whose counts are all zero: take  $\mathbf{d}$  to be minus that coordinate direction. The coefficient runs to  $-\infty$  and the algorithm reports a huge negative number with an enormous standard error. A ridge or lasso

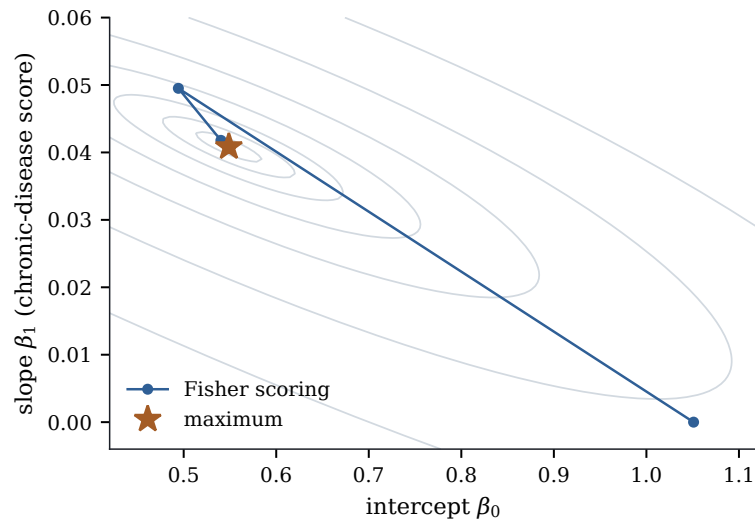


Figure 37.1.1: The log-likelihood of a Poisson log-linear model with an intercept and one regressor, for the physician-visit data used below in this section, with the Fisher scoring path from the start  $\beta = (\log \bar{y}, 0)^\top$ . Contours are drawn at fixed drops below the maximum.

penalty (Chapter 27, Chapter 30) makes the penalized log-likelihood coercive and restores a finite maximizer.

### 37.1.4 Fitting

By Theorem 34.4.1, Fisher scoring is iteratively reweighted least squares: at the current  $\beta$  form the working response and weights

$$z_i = \eta_i + \frac{y_i - \mu_i}{\mu_i}, \quad w_i = \mu_i, \quad (37.1.4)$$

and set  $\beta \leftarrow (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{z}$ . Because the expected and observed information agree (Theorem 37.1.1(b)), Fisher scoring and Newton–Raphson coincide here, and by concavity the iteration converges from any start whenever a maximum exists. The same weighted least squares step appears in Proposition 20.6.1 and Theorem 21.3.1; only the weights differ. Figure 37.1.1 draws the log-likelihood surface of a two-regressor fit with the path from a poor start: the contours are closed and nested, as concavity requires, and the first step already lands near the maximum.

### 37.1.5 Interpretation and inference

The estimate  $\hat{\beta}$  is consistent and asymptotically normal under the conditions of Theorem 34.5.1, with asymptotic covariance  $(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$  at the estimate. Exponentiating the ends of the Wald interval  $\hat{\beta}_j \pm z_{1-\alpha/2} \text{se}(\hat{\beta}_j)$  gives an interval for the rate ratio  $e^{\beta_j}$ , better than a delta method applied to  $e^{\hat{\beta}_j}$ : the log scale is where the normal approximation is accurate, and the transformed interval stays positive.



**Example. (Physician visits in a health insurance experiment)**

The RAND Health Insurance Experiment sample distributed with `statsmodels` records 20190 person-years. The response is the number of outpatient physician visits in the year, with mean 2.860 and a variance 7.093 times as large; [Example \(Physician visits are counts\)](#) used the same data to show how badly a normal linear model describes it.

Fit [Eq. 37.1.1](#) with an intercept, the logarithm of one plus the coinsurance rate (`lncoins`), an indicator of an individual-deductible plan (`idp`), a physical-limitation index, a chronic-disease score, and three indicators of self-rated health (good, fair, poor, against excellent). Fisher scoring converges in 7 steps. The coefficient of `lncoins` is  $-0.0712$ , standard error 0.0022: a rate ratio of 0.9313 per unit, Wald interval (0.9272, 0.9354). Moving across its whole range of 4.615, from free care to the highest coinsurance plan, multiplies expected visits by 0.720. The disease score is not a count of diseases but a continuous index running to 58.6; each unit of it multiplies expected visits by 1.0353, so moving from its lower quartile 6.90 to its upper quartile 13.73 multiplies them by 1.267. Poor self-rated health multiplies them by 1.2441 against excellent health. The fitted counts sum to the observed total, and within each health group to that group's total, as [Theorem 37.1.2](#) requires.

```
import numpy as np
import statsmodels.api as sm

data = sm.datasets.randhie.load_pandas().data          # 20190 person-years, public
            domain
y = data["mdvis"].to_numpy(float)                      # outpatient physician visits
            in the year
names = ["intercept", "lncoins", "idp", "physlm", "disea", "hlthg", "hlthf", "hlthp"
        ]
X = np.column_stack([np.ones(len(y))] + [data[v].to_numpy(float) for v in names
            [1:]])

def poisson_fit(X, y, offset=0.0, tol=1e-11, maxit=50):
    """Fisher scoring for log mu = X beta + offset; returns beta and its iterates.
    """
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(y.mean())
    path = [beta]
    for _ in range(maxit):
        mu = np.exp(X @ beta + offset)                # current fitted means
        z = X @ beta + (y - mu) / mu                  # working response
        XW = X * mu[:, None]                          # weights w_i = mu_i
        step = np.linalg.solve(X.T @ XW, XW.T @ z)
        converged = np.max(np.abs(step - beta)) < tol
        beta = step
        path.append(beta)
    if converged:
        break
    return beta, np.array(path)

beta, path = poisson_fit(X, y)
mu = np.exp(X @ beta)
se = np.sqrt(np.diag(np.linalg.inv(X.T @ (X * mu[:, None]))))
```

```
print("largest entry of X^T (y - mu):", np.abs(X.T @ (y - mu)).max())
for name, b, s in zip(names, beta, se):
    print(f"{name:10s} {b:8.4f} ({s:.4f})    rate ratio {np.exp(b):.4f}")
```

The deviance of the Poisson fit ([Definition 34.5.1](#)) is

$$D = 2 \sum_{i=1}^n \left\{ y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i - \hat{\mu}_i) \right\}, \quad (37.1.5)$$

with the convention  $0 \log 0 = 0$ . If the model has an intercept the second term sums to zero by [Theorem 37.1.2\(a\)](#), and the deviance reduces to  $2 \sum_i y_i \log(y_i/\hat{\mu}_i)$ , the form usually written  $G^2$ ; because  $\phi = 1$  the deviance and the scaled deviance coincide.

Differences of deviances between nested models behave in the usual way: by [Theorem 34.5.5](#) the drop is asymptotically  $\chi^2$  on the difference in dimension, and it is the likelihood ratio statistic of [Proposition 34.5.3](#). The deviance of a *single* model is a different matter: its  $\chi^2$  calibration needs the fitted means to grow, which happens for grouped data with large cells and not for ungrouped records with small counts. In [Example \(Physician visits in a health insurance experiment\)](#) the deviance is 84569.7 on 20182 degrees of freedom and the Pearson statistic is 127526.5, ratios of 4.190 and 6.319. No formal test is available, but ratios of this size are a loud signal of misfit.

```
dev = 2 * np.sum(np.where(y > 0, y * np.log(np.where(y > 0, y, 1) / mu), 0.0) - (y -
    mu))
pearson = np.sum((y - mu) ** 2 / mu)
df = len(y) - X.shape[1]
print(f"deviance {dev:.1f} on {df} df    (ratio {dev / df:.3f})")
print(f"Pearson {pearson:.1f} on {df} df    (ratio {pearson / df:.3f})")
print("fitted total", mu.sum(), " observed total", y.sum())
```

Both statistics are sums of squared residuals: the Pearson residual  $r_i^P = (y_i - \hat{\mu}_i)/\sqrt{\hat{\mu}_i}$  and the deviance residual  $r_i^D = \text{sign}(y_i - \hat{\mu}_i)\sqrt{d_i}$ , with  $d_i$  the  $i$ th term of [Eq. 37.1.5](#). These are the count analogues of [Definition 20.1.1](#); both are badly skewed when  $\hat{\mu}_i$  is small ([Exercise \(B3\)](#)), so a normal probability plot of them tells one little. Leverage and influence transfer from [Chapter 20](#) through the weights: the hat matrix of the last weighted least squares step of [Proposition 20.6.1](#) plays the role of  $\mathbf{M}$ , and a one-step Cook's distance ([Definition 20.4.1](#)) built from it is the standard influence measure for a generalized linear model.

*Idea.* The Poisson log-linear model puts the linear predictor on the log scale, so its coefficients are log rate ratios, and it ties the variance to the mean, so it has no spare parameter for extra variability. The rest of the chapter follows from taking one or other of those decisions seriously. A deviance far larger than its degrees of freedom has three possible causes — a wrong mean model, a wrong variance function, or dependent observations — and reaching for an overdispersion parameter before checking the mean model is a common and expensive mistake, because an inflated variance hides exactly the lack of fit that would have told you what to add.

## 37.1.6 Exercises

### 37.1.6.1 A. Check your understanding

**Exercise A1.**

A Poisson log-linear model for the number of insurance claims has coefficient 0.182 on an indicator of urban residence. State the effect on the expected number of claims in words, and give the multiplicative effect of the indicator on the *odds* that a policy has at least one claim under the same model. Why is the second quantity not simply  $e^{0.182}$ ?

**Exercise A2.**

Show directly from [Theorem 37.1.1](#) that in the model with an intercept only, the maximum likelihood estimate of the common mean is  $\bar{y}$ , and that the deviance equals  $2 \sum_i y_i \log(y_i/\bar{y})$ .

*Solution.* With  $\mathbf{X} = \mathbf{1}$  the likelihood equation is  $\sum_i (y_i - e^\beta) = 0$ , so  $e^{\hat{\beta}} = \bar{y}$ . Substituting into [Eq. 37.1.5](#) and using  $\sum_i (y_i - \bar{y}) = 0$  leaves  $2 \sum_i y_i \log(y_i/\bar{y})$ .

**37.1.6.2 B. Practice****Exercise B1.**

Counts  $y_{ij}$ ,  $j = 1, \dots, n_i$ , are observed in  $c$  groups, and the model is  $\log \mu_{ij} = \beta_i$ , one parameter per group. Show that  $\hat{\mu}_{ij} = \bar{y}_i$ , that  $\widehat{\text{Var}}(\hat{\beta}_i) = 1/(n_i \bar{y}_i)$ , and that a Wald interval for  $\mu_h/\mu_i$  is  $\exp\{\hat{\beta}_h - \hat{\beta}_i \pm z\sqrt{1/(n_h \bar{y}_h) + 1/(n_i \bar{y}_i)}\}$ .

*Solution.* [Theorem 37.1.2\(b\)](#) applied to the group indicators gives  $n_i \hat{\mu}_i = \sum_j y_{ij}$ . The information  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$  is diagonal with entries  $n_i \mu_i$ , so the estimates are independent with variances  $1/(n_i \mu_i)$ ; the variance of their difference is the sum, and exponentiating a Wald interval for  $\beta_h - \beta_i$  gives the stated interval.

**Exercise B2.**

In the setting of [Exercise \(B1\)](#), show that the likelihood ratio statistic for  $H_0 : \mu_1 = \dots = \mu_c$  is  $2 \sum_i n_i \bar{y}_i \log(\bar{y}_i/\bar{y})$ , where  $\bar{y}$  is the overall mean, and state the conditions under which its  $\chi^2(c-1)$  calibration is reasonable.

**Exercise B3.**

For a Poisson fit the Pearson residual is  $r_i^P = (y_i - \hat{\mu}_i)/\sqrt{\hat{\mu}_i}$ . Show that it is strongly skewed when  $\mu_i$  is small, and that the Anscombe residual  $\frac{3}{2}(y_i^{2/3} - \hat{\mu}_i^{2/3})/\hat{\mu}_i^{1/6}$  has variance about one, by applying the delta method to  $Y^{2/3}$ .

**37.1.6.3 C. Going deeper****Exercise C1.**

Consider a Poisson model with the identity link,  $\mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . Show that the log-likelihood

is concave on the open set where every  $\mu_i > 0$ , but that the maximum can occur on its boundary, where some  $\hat{\mu}_i = 0$ ; what happens then to the standard errors an algorithm that ignores the constraint reports? Repeat for the additive *rate* model  $\mu_i = t_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  of [Section 37.2](#), whose coefficients are rate *differences*: show that its likelihood equations are  $\sum_i t_i (y_i - \mu_i) \mathbf{x}_{(i)} / \mu_i = \mathbf{0}$ , and say why an epidemiologist asking how many events an intervention prevents wants it.

### Exercise C2.

Give a data set with  $n = 4$ ,  $p = 2$  for which the condition of [Proposition 37.1.3](#) fails. Then add a ridge penalty  $\lambda \|\boldsymbol{\beta}\|^2$  ([Definition 27.2.1](#)) and show that the penalized log-likelihood has a finite maximizer for every  $\lambda > 0$ .

*Solution.* Take  $\mathbf{X}$  with intercept and a regressor  $x = (0, 0, 1, 1)^\top$  and counts  $\mathbf{y} = (3, 5, 0, 0)^\top$ . Then  $\mathbf{d} = (0, -1)^\top$  has  $\mathbf{x}_{(i)}^\top \mathbf{d} \leq 0$  everywhere and  $= 0$  at the two observations with  $y_i > 0$ , so the condition fails: the slope runs to  $-\infty$  while the intercept converges to  $\log 4$ . Adding  $-\lambda \|\boldsymbol{\beta}\|^2$  makes the objective tend to  $-\infty$  in every direction, since the log-likelihood grows at most linearly along any ray on which it grows at all, so a finite maximizer exists; it is unique by strict concavity of the penalized objective.

## 37.2 Offsets and rates

A count on its own is rarely interesting. Twelve accidents mean one thing on a road carrying a thousand vehicles a day and another on one carrying a hundred thousand. What is comparable is the **rate**: the count divided by the exposure that produced it. The Poisson log-linear model handles rates without new machinery, by moving a known quantity into the linear predictor and refusing to give it a coefficient.

### 37.2.1 Exposure

Let  $t_i > 0$  be the **exposure** of observation  $i$ : person-years of follow-up, vehicle-kilometres, population size, area surveyed. The natural assumption is that the expected count is proportional to exposure,

$$\mu_i = t_i \lambda_i,$$

where  $\lambda_i$  is the expected number of events per unit of exposure, the quantity of interest. Modelling  $\lambda_i$  log-linearly,  $\log \lambda_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , gives

$$\log \mu_i = \log t_i + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}, \quad \mu_i = t_i \exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}). \quad (37.2.1)$$

The term  $\log t_i$  is an **offset**: a column of the linear predictor whose coefficient is known to be one rather than estimated. Coefficients are now log rate ratios,  $e^{\beta_j}$  multiplying the rate per unit of exposure rather than the raw count. This is *not* a regression of the observed rates  $y_i/t_i$ , which would throw away the information that a rate based on a thousand person-years is better determined than one based on five: the response stays the count and the exposure enters as a known constant.

### Proposition 37.2.1 — Offsets

Consider the model Eq. 37.2.1 with  $\text{rank}(\mathbf{X}) = p$ , and write  $\boldsymbol{\mu} = \boldsymbol{\mu}(\boldsymbol{\beta})$  for the mean vector it defines. Then:

- (a) the log-likelihood is  $\ell(\boldsymbol{\beta}) = \sum_i \{y_i(\log t_i + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}) - t_i e^{\mathbf{x}_{(i)}^\top \boldsymbol{\beta}}\} - \sum_i \log y_i!$ ; its score is  $\mathbf{X}^\top (\mathbf{y} - \boldsymbol{\mu})$  and its information is  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$  with  $\mathbf{W} = \text{diag}(\mu_1, \dots, \mu_n)$ , so Theorem 37.1.1 holds verbatim with the offset in place;
- (b) consequently Theorem 37.1.2 holds verbatim: the fitted counts, not the fitted rates, reproduce the observed margins of the model matrix, and with an intercept  $\sum_i \hat{\mu}_i = \sum_i y_i$ ;
- (c) (**grouping**) suppose the rows of  $\mathbf{X}$  take only  $G$  distinct values  $\mathbf{x}_{(g)}$ , with  $n_g$  rows equal to the  $g$ th, all with exposure  $t_i = 1$ . Let  $s_g$  be the total count of group  $g$ . Then fitting Eq. 37.1.1 to the  $n$  individual records and fitting Eq. 37.2.1 to the  $G$  totals  $s_g$  with exposures  $t_g = n_g$  give the same  $\hat{\boldsymbol{\beta}}$ , the same information matrix and the same deviance differences between nested models;
- (d) (**a free exposure coefficient**) the model  $\log \mu_i = \psi \log t_i + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  contains Eq. 37.2.1 as the submodel  $\psi = 1$ . Its fit coincides with the offset fit if and only if  $\hat{\psi} = 1$ , and the likelihood ratio statistic for  $H_0 : \psi = 1$  is asymptotically  $\chi^2(1)$ .

*Proof.*

- (a) Substituting  $\mu_i = t_i e^{\mathbf{x}_{(i)}^\top \boldsymbol{\beta}}$  into the Poisson log-likelihood gives the stated expression, and  $\partial \mu_i / \partial \boldsymbol{\beta} = \mu_i \mathbf{x}_{(i)}$  exactly as before, because the factor  $t_i$  is a constant. The derivative of  $\sum_i y_i \log t_i$  in  $\boldsymbol{\beta}$  is zero. Everything in the proof of Theorem 37.1.1 then goes through with the new  $\mu_i$ , which is (a); and (b) follows from (a) as before.
- (b) Within group  $g$  all the means equal  $\mu_g = e^{\mathbf{x}_{(g)}^\top \boldsymbol{\beta}}$ , so

$$\ell_{\text{ind}}(\boldsymbol{\beta}) = \sum_{g=1}^G \{s_g \log \mu_g - n_g \mu_g\} - \sum_i \log y_i!$$

The grouped model has means  $n_g \mu_g$ , so its log-likelihood is

$$\ell_{\text{grp}}(\boldsymbol{\beta}) = \sum_{g=1}^G \{s_g (\log n_g + \log \mu_g) - n_g \mu_g\} - \sum_g \log s_g!$$

The two differ by a quantity free of  $\boldsymbol{\beta}$ . Log-likelihoods differing by a constant have the same maximizer and the same derivatives of every order, hence the same information; and differences of deviances are differences of log-likelihoods, so the constant cancels there too.

- (c) Setting  $\psi = 1$  recovers Eq. 37.2.1, so the models are nested with one extra parameter. The larger model's log-likelihood is strictly concave in  $(\boldsymbol{\beta}^\top, \psi)^\top$  whenever  $[\mathbf{X}, \log \mathbf{t}]$  has full column rank, by Theorem 37.1.1(c) applied to that model matrix, so its maximizer is unique; it therefore agrees with the offset fit exactly when  $\hat{\psi} = 1$ . The distribution statement is Theorem 34.5.5 for one degree of freedom.

■

Part (c) says that grouping records with identical regressors and carrying the group size as an exposure is *exactly* a bookkeeping operation. It is a property of the Poisson, not of generalized linear models in general, and it comes from sums of independent Poissons being Poisson. The use is both practical — twenty thousand records collapse to a few dozen — and theoretical: the grouped cells may have large counts, so the deviance of a single fitted model is calibrated by [Theorem 34.5.5](#) even though it is not for the individual records.

**Example. (Twenty thousand records in thirty-two rows)**

Take the physician-visit data of [Example \(Physician visits in a health insurance experiment\)](#) and keep only categorical regressors: the individual-deductible indicator, the three self-rated health indicators, and the chronic-disease score banded into four groups. There are 32 distinct covariate patterns, holding between 9 and 3832 person-years. Collapse the 20190 records to these rows, recording the total visits and the number of person-years in each, and fit the rate model with offset the log of the person-years. The coefficients agree with the person-level fit to fourteen decimal places, as [Proposition 37.2.1\(c\)](#) promises.

Read as rates, a person-year in excellent health, in the lowest disease band and on a plan without an individual deductible, is expected to produce 1.9758 visits; an individual deductible multiplies that by 0.8247, poor self-rated health by 1.6875, and the highest disease band by 2.2621.

The grouped deviance is 665.60 on 24 degrees of freedom, and now the  $\chi^2$  calibration applies, since the smallest fitted cell mean is 24.7. The model is rejected. Where does the misfit sit? The fitted model has main effects only, so its 24 residual degrees of freedom are the interactions among the three factors; adding every two-factor interaction among them, 23 parameters in all, drops the deviance to 106.87 on 9 degrees of freedom, a fall of 558.7 on 15. Five sixths of the misfit is therefore mean model, not variance function. What survives the repair — 106.87 on nine degrees of freedom, still far beyond any plausible  $\chi^2$  — is the Poisson variance function, which [Section 37.4](#) takes up.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

def scoring(X, y, offset=0.0, tol=1e-11, maxit=60):
    """Fisher scoring for a Poisson log-linear model with a fixed offset."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(max(y.mean(), 1e-6))
    for _ in range(maxit):
        mu = np.exp(X @ beta + offset)
        z = X @ beta + (y - mu) / mu           # working response
        XW = X * mu[:, None]                  # weights w_i = mu_i
        step = np.linalg.solve(X.T @ XW, XW.T @ z)
        if np.max(np.abs(step - beta)) < tol:
            return step
        beta = step
    return beta

data = sm.datasets.randhie.load_pandas().data.copy()
data["band"] = pd.cut(data["disea"], [-0.1, 6, 10, 15, 60], labels=[0, 1, 2, 3])
```

```

cells = ["idp", "hlthg", "hlthf", "hlthp", "band"]  # all categorical
y = data["mdvis"].to_numpy(float)

def design(frame):
    """Intercept, the plan and health indicators, and three disease-band indicators.
    """
    columns = [np.ones(len(frame))] + [frame[v].to_numpy(float)
                                         for v in ["idp", "hlthg", "hlthf", "hlthp"]]
    band = frame["band"].to_numpy(int)
    return np.column_stack(columns + [(band == k).astype(float) for k in (1, 2, 3)])

grouped = data.groupby(cells, as_index=False, observed=True).agg(
    visits=("mdvis", "sum"), person_years=("mdvis", "size"))
s = grouped["visits"].to_numpy(float)  # total visits in the group
t = grouped["person_years"].to_numpy(float)  # exposure: person-years
X, Xg = design(data), design(grouped)

beta_person = scoring(X, y)  # one row per person-year
beta_group = scoring(Xg, s, offset=np.log(t))  # one row per pattern, offset
log t

print("groups:", len(s), " largest coefficient difference:",
      np.abs(beta_person - beta_group).max())
print("fitted rate, free-plan person-years in excellent health:",
      np.exp(beta_group[0]))

```

## 37.2.2 What a free exposure coefficient buys and costs

Why fix  $\psi = 1$  instead of estimating it, as [Proposition 37.2.1\(d\)](#) allows? It depends on where the exposure came from.

When exposure is a bookkeeping device — the grouping of part (c), where the individual records are the real data — proportionality is a theorem, not an assumption, and a  $\hat{\psi}$  far from one is a symptom of misspecification elsewhere, because  $\log n_g$  is free to proxy for anything correlated with group size. In [Example \(Twenty thousand records in thirty-two rows\)](#) it is  $\hat{\psi} = 1.2266$ , standard error 0.0146, interval (1.1980, 1.2553), with likelihood ratio statistic 249.019 against  $\psi = 1$ . A person-year cannot contribute more than proportionally to its own visits; the statistic is detecting the same lack of fit as the deviance of 665.60.

When exposure genuinely varies across units — different follow-up times, different populations — proportionality is a substantive assumption, and there is a real trade-off, which a simulation makes concrete.

### Example. (Three ways to handle an exposure)

Counts are generated for  $m = 300$  units from  $\log \mu_i = 0.4 + 0.5x_i + \psi \log t_i$ , with  $x_i$  standard normal and the exposure  $t_i$  correlated with it. Three analyses are run on each of 2000 data sets — the correct offset, a free coefficient on  $\log t$ , and no exposure term — and [Figure 37.2.1](#) shows the sampling distributions of the estimated slope of  $x$ , whose



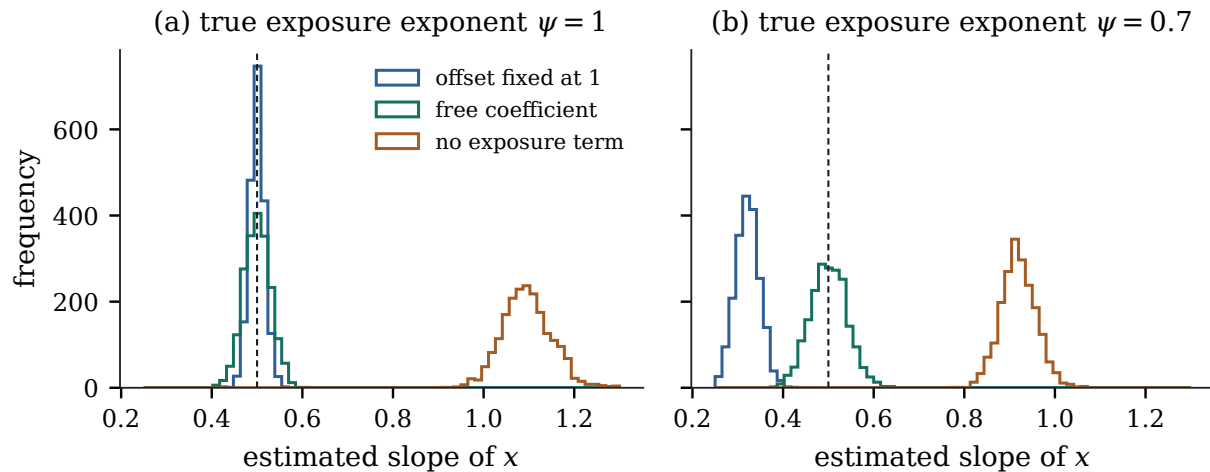


Figure 37.2.1: Sampling distributions of the estimated slope of  $x$  over 2000 simulated data sets, under three treatments of the exposure, when the true exposure exponent is (a) one and (b) 0.7. The dashed line marks the true slope.

true value is 0.5.

When the truth is  $\psi = 1$ , both the offset and the free-coefficient model are unbiased, with means 0.500 and 0.500, but the free coefficient has standard deviation 0.029 against 0.017, a loss of precision by a factor of 1.772: estimating a parameter you already know costs information. Dropping the exposure is a disaster, the slope averaging 1.096, because  $x$  is left to explain the variation in exposure as well as in the rate.

When the truth is  $\psi = 0.7$ , the free-coefficient model is still centred at 0.501, while the offset model, forcing a wrong proportionality, is centred at 0.322 — a bias of more than six of its own standard deviations. Fixing a coefficient you have got wrong costs more than estimating one you knew.

*Warning.* The third analysis fails by ordinary omitted variable bias:  $\log t$  belongs in the linear predictor, and leaving it out loads its effect onto whatever is correlated with it. What is special about an offset is only that its coefficient is known, so it is not a variable to be selected and must never be dropped by a stepwise procedure or a lasso.

### 37.2.3 Rates elsewhere

Two further uses of the same device are worth naming.

**Piecewise-constant hazards.** Split each subject's follow-up into intervals on which the hazard is constant, and let  $d_{ik}$  be subject  $i$ 's events in interval  $k$  and  $t_{ik}$  the time at risk. Treating the  $d_{ik}$  as independent Poisson counts with means  $t_{ik} \exp(\alpha_k + \mathbf{x}_{(i)}^\top \boldsymbol{\beta})$  gives a proportional-hazards model with a step-function baseline, fitted as an ordinary Poisson regression with offset  $\log t_{ik}$  and one parameter per interval ([Exercise \(B1\)](#)); the connection is due to Holford and to Laird and Olivier.

**Standardization.** A standardized morbidity ratio divides the observed count by an expected count  $E_i$  from external reference rates; taking  $\log E_i$  as the offset makes  $\exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$  the ratio itself.



Fixing the exposure coefficient is right when theory or bookkeeping says the count is proportional to the exposure, and costs nothing; estimating it is right when proportionality is in doubt, and costs precision. What is never right is leaving the exposure out.

## 37.2.4 Exercises

### 37.2.4.1 A. Check your understanding

#### Exercise A1.

A rate model uses person-years as the exposure. A colleague re-expresses exposure in person-months, multiplying every  $t_i$  by twelve. Which coefficients change and by how much? Which fitted values change?

#### Exercise A2.

Explain why regressing  $\log(y_i/t_i)$  on  $\mathbf{x}_{(i)}$  by least squares is not equivalent to fitting [Eq. 37.2.1](#), giving two distinct reasons.

### 37.2.4.2 B. Practice

#### Exercise B1.

Suppose subject  $i$  is followed until an event or censoring, with a hazard that is constant at  $h_{ik} = \exp(\alpha_k + \mathbf{x}_{(i)}^\top \boldsymbol{\beta})$  on interval  $k$ . Write  $d_{ik} \in \{0, 1\}$  for the event indicator and  $t_{ik}$  for the time at risk in that interval. Show that the likelihood contribution of subject  $i$  is  $\prod_k h_{ik}^{d_{ik}} e^{-h_{ik} t_{ik}}$ , and hence that maximizing it is the same as fitting a Poisson log-linear model to the  $d_{ik}$  with offset  $\log t_{ik}$ .

*Solution.* With a constant hazard  $h$  on an interval, the probability of surviving a time  $t$  is  $e^{-ht}$  and the density of an event at the end is  $he^{-ht}$ , so the contribution of interval  $k$  is  $h_{ik}^{d_{ik}} e^{-h_{ik} t_{ik}}$ , and the contributions multiply. Its logarithm is  $\sum_{i,k} \{d_{ik} \log h_{ik} - h_{ik} t_{ik}\}$ . The Poisson log-likelihood for counts  $d_{ik}$  with means  $t_{ik} h_{ik}$  differs from it by  $\sum_{i,k} d_{ik} \log t_{ik}$ , a constant in the parameters, so the maximizers agree. The  $d_{ik}$  need not actually be Poisson; only the likelihoods coincide.

#### Exercise B2.

In the setting of [Example \(Three ways to handle an exposure\)](#), suppose the analyst tests  $H_0 : \psi = 1$  and uses the offset model if the test does not reject. Explain why the resulting estimator of the slope of  $x$  has a sampling distribution that is neither of the two shown in [Figure 37.2.1](#), and relate the phenomenon to the pre-test problem [Example \(Coverage after a pretest\)](#).

### 37.2.4.3 C. Going deeper

**Exercise C1.**

Extend [Proposition 37.2.1\(c\)](#) to the case where the individual records already carry exposures  $t_i$  that differ within a group. Show that the grouped fit with exposure  $\sum_{i \in g} t_i$  reproduces the individual fit, and identify exactly what has to be constant within a group.

*Solution.* What must be constant within a group is the row  $\mathbf{x}_{(i)}$ , not the exposure. If  $\mathbf{x}_{(i)} = \mathbf{x}_{(g)}$  for all  $i \in g$  then  $\mu_i = t_i \mu_g^*$  with  $\mu_g^* = e^{\mathbf{x}_{(g)}^\top \boldsymbol{\beta}}$ , so  $\sum_{i \in g} \mu_i = (\sum_{i \in g} t_i) \mu_g^*$  and the log-likelihood is  $\sum_g \{s_g \log \mu_g^* - (\sum_{i \in g} t_i) \mu_g^*\} + \sum_i y_i \log t_i$ , whose last term is free of  $\boldsymbol{\beta}$ . Sums of independent Poissons are Poisson, so the grouped data have exactly the claimed distribution.

**Exercise C2.**

Let the truth be  $\log \mu_i = \psi \log t_i + \beta_0 + \beta_1 x_i$  with  $\psi \neq 1$ , and suppose the offset model is fitted. Show that the limiting value of  $\hat{\beta}_1$  solves  $E\{\mathbf{x} t (t^{\psi-1} e^{\beta_0 + \beta_1 x} - e^{\beta_0^* + \beta_1^* x})\} = \mathbf{0}$ , and argue that  $\beta_1^* = \beta_1$  when  $t^{\psi-1}$  is independent of  $x$  under the weighting by  $t$ .

## 37.3 Poisson and multinomial

A contingency table is a table of counts, so a Poisson log-linear model can be fitted to it. But the counts usually come from a survey with a fixed sample size, or from strata with fixed totals, and then they are not independent Poisson variables at all: they are multinomial. The two descriptions lead to the same fit. That is why software for Poisson regression is also software for contingency tables, and it is the bridge to the baseline-category logit models of [Chapter 36](#).

### 37.3.1 Conditioning on the total

#### Theorem 37.3.1 — Poisson counts conditioned on their total

- (a) Let  $Y_1, \dots, Y_c$  be independent,  $Y_j$  Poisson with mean  $\mu_j > 0$ , and  $N = \sum_j Y_j$ . Then  $N$  is Poisson with mean  $\mu_+ = \sum_j \mu_j$ , and conditionally on  $N = n$  the vector  $(Y_1, \dots, Y_c)$  is multinomial with  $n$  trials and probabilities  $\pi_j = \mu_j / \mu_+$ .
- (b) Suppose the cells are indexed by a stratum  $s = 1, \dots, S$  and a category  $j = 1, \dots, c$ , the counts  $Y_{sj}$  are independent Poisson with means  $\mu_{sj}$ , and the log-linear model is

$$\log \mu_{sj} = \alpha_s + \mathbf{z}_{sj}^\top \boldsymbol{\gamma}, \quad (37.3.1)$$

with  $\boldsymbol{\alpha} \in \mathbb{R}^S$  unrestricted: that is, the model contains a free parameter for every stratum. Then the Poisson log-likelihood splits as

$$\ell(\boldsymbol{\alpha}, \boldsymbol{\gamma}) = \ell_1(\mu_{1+}, \dots, \mu_{S+}) + \ell_2(\boldsymbol{\gamma}) + \text{constant},$$

where  $\ell_1$  is the log-likelihood of  $S$  independent Poisson totals and  $\ell_2$  is the log-likelihood of  $S$  independent multinomial samples with category probabilities

$$\pi_{sj}(\boldsymbol{\gamma}) = \frac{\exp(\mathbf{z}_{sj}^\top \boldsymbol{\gamma})}{\sum_{l=1}^c \exp(\mathbf{z}_{sl}^\top \boldsymbol{\gamma})}, \quad (37.3.2)$$

and the two pieces have no parameters in common.

- (c) Consequently the maximum likelihood estimate  $\hat{\boldsymbol{\gamma}}$  from the Poisson fit is the maximum likelihood estimate from the multinomial fit; the fitted cell counts are  $\hat{\mu}_{sj} = n_s \hat{\pi}_{sj}$  with  $n_s = \sum_j y_{sj}$ ; the observed information for  $\boldsymbol{\gamma}$  is the same in the two models; and deviance differences between models that all contain the stratum parameters agree. The stratum parameters  $\hat{\alpha}_s$  themselves have no multinomial counterpart: they are determined by  $\hat{\mu}_{s+} = n_s$ .

*Proof.*

- (a) Sums of independent Poisson variables are Poisson with the sum of the means (convolve the mass functions, or multiply moment generating functions), so  $\Pr(N = n) = e^{-\mu_+} \mu_+^n / n!$ . For nonnegative integers  $y_j$  with  $\sum_j y_j = n$ ,

$$\begin{aligned} \Pr(Y_1 = y_1, \dots, Y_c = y_c \mid N = n) &= \frac{\prod_j e^{-\mu_j} \mu_j^{y_j} / y_j!}{e^{-\mu_+} \mu_+^n / n!} \\ &= \frac{n!}{\prod_j y_j!} \prod_j \pi_j^{y_j}, \end{aligned}$$

because  $\sum_j \mu_j = \mu_+$  makes the exponential factors cancel and  $\prod_j \mu_j^{y_j} = \mu_+^n \prod_j \pi_j^{y_j}$ . That is the multinomial mass function.

- (b) Write  $\mu_{s+} = \sum_j \mu_{sj}$  and  $\pi_{sj} = \mu_{sj} / \mu_{s+}$ . Under Eq. 37.3.1,  $\mu_{s+} = e^{\alpha_s} \sum_l \exp(\mathbf{z}_{sl}^\top \boldsymbol{\gamma})$ , and the ratio  $\pi_{sj}$  is Eq. 37.3.2, free of  $\boldsymbol{\alpha}$ . Substituting  $\mu_{sj} = \mu_{s+} \pi_{sj}$  in the Poisson log-likelihood,

$$\begin{aligned} \ell &= \sum_{s,j} \{y_{sj} \log \mu_{sj} - \mu_{sj}\} - \sum_{s,j} \log y_{sj}! \\ &= \sum_s \{n_s \log \mu_{s+} - \mu_{s+} - \log n_s!\} \\ &\quad + \sum_s \left\{ \log n_s! - \sum_j \log y_{sj}! + \sum_j y_{sj} \log \pi_{sj} \right\}, \end{aligned}$$

which is the stated split, the first sum being  $\ell_1$  and the second  $\ell_2$  plus the multinomial coefficients. For fixed  $\boldsymbol{\gamma}$ , the map  $\boldsymbol{\alpha} \mapsto (\mu_{1+}, \dots, \mu_{S+})$  is a bijection from  $\mathbb{R}^S$  onto  $(0, \infty)^S$ , so the parameters  $(\mu_{1+}, \dots, \mu_{S+})$  and  $\boldsymbol{\gamma}$  vary independently.

- (c) The two pieces share no parameters and their ranges are unconstrained, so the joint maximum is attained by maximizing each separately:  $\ell_1$  gives  $\hat{\mu}_{s+} = n_s$  and  $\ell_2$  is the multinomial problem, giving  $\hat{\boldsymbol{\gamma}}$  and  $\hat{\mu}_{sj} = n_s \hat{\pi}_{sj}$ . The information for  $\boldsymbol{\gamma}$  is minus the second derivative of  $\ell_2$ , and differences of maximized log-likelihoods between models sharing the stratum parameters lose the common  $\ell_1$ .

■

Part (c) is the practical statement: *fit the table as independent Poisson counts, and provided the*

model contains a free parameter for every total the design fixed, you get the multinomial answer. The converse matters as much: if the design fixed the row totals and the model omits the row parameters, the two fits differ and the Poisson one answers a question the design cannot. The proof used  $s$  only to group cells, so any partition into blocks with fixed totals works the same way.

## 37.3.2 Independence in a two-way table

Take  $S = r$  rows and  $c$  columns, and let  $\mathbf{z}_{ij}$  consist of the column indicators alone, so that Eq. 37.3.1 reads

$$\log \mu_{ij} = \lambda + \lambda_i^R + \lambda_j^C, \quad (37.3.3)$$

the log-linear model with row and column main effects and no interaction. The parameters are redundant by two, and any of the usual side conditions removes the redundancy without changing the fit (Theorem 6.7.1).

### Corollary 37.3.2 — The independence model

Under Eq. 37.3.3 the conditional distribution of the column given the row does not depend on the row, so the two classifications are independent. The maximum likelihood fitted values are

$$\hat{\mu}_{ij} = \frac{y_{i+} y_{+j}}{n}, \quad y_{i+} = \sum_j y_{ij}, \quad y_{+j} = \sum_i y_{ij}, \quad n = y_{++}, \quad (37.3.4)$$

the deviance is  $G^2 = 2 \sum_{i,j} y_{ij} \log(y_{ij}/\hat{\mu}_{ij})$ , the Pearson statistic is  $X^2 = \sum_{i,j} (y_{ij} - \hat{\mu}_{ij})^2 / \hat{\mu}_{ij}$ , and both have  $(r-1)(c-1)$  residual degrees of freedom — and, when the expected counts are large, both are asymptotically  $\chi^2$  on that many degrees of freedom (Theorem 34.5.5).

*Proof.* With  $\mathbf{z}_{ij}$  free of  $i$ , Eq. 37.3.2 gives  $\pi_{ij} = \pi_j$ , which is independence. By Theorem 37.1.2(b) applied to the row and to the column indicators,  $\hat{\mu}_{i+} = y_{i+}$  and  $\hat{\mu}_{+j} = y_{+j}$ . Under Eq. 37.3.3,  $\hat{\mu}_{ij} = \hat{\mu}_{i+} \hat{\mu}_{+j} / \hat{\mu}_{++}$ , because  $\hat{\mu}_{ij} = e^{\lambda} a_i b_j$  factorizes and summing gives  $\hat{\mu}_{i+} \hat{\mu}_{+j} = e^{2\lambda} a_i b_j \sum_l b_l \sum_k a_k = \hat{\mu}_{ij} \hat{\mu}_{++}$ . Substituting the matched margins gives the fit Eq. 37.3.4. The model has  $1 + (r-1) + (c-1)$  free parameters against  $rc$  cells, leaving  $rc - r - c + 1 = (r-1)(c-1)$ . The deviance formula is Eq. 37.1.5 with the second term vanishing by Theorem 37.1.2(a). ■

The fitted values Eq. 37.3.4 are the expected counts of the elementary chi-squared test for independence, available since Pearson (1900); what the log-linear framework adds is everything around it.

### 37.3.3 Association is interaction on the log scale

Adding an interaction term to [Eq. 37.3.3](#) gives the saturated model

$$\log \mu_{ij} = \lambda + \lambda_i^R + \lambda_j^C + \lambda_{ij}^{RC}, \quad (37.3.5)$$

with  $rc$  free parameters and  $\hat{\mu}_{ij} = y_{ij}$ .

#### Proposition 37.3.3 — Interaction parameters are log odds ratios

In [Eq. 37.3.5](#), for any rows  $i \neq k$  and columns  $j \neq l$ ,

$$\lambda_{ij}^{RC} + \lambda_{kl}^{RC} - \lambda_{il}^{RC} - \lambda_{kj}^{RC} = \log \frac{\mu_{ij}\mu_{kl}}{\mu_{il}\mu_{kj}},$$

which does not depend on the side conditions ([Theorem 8.4.2](#)) used to identify the parameters. Under the multinomial reading of [Theorem 37.3.1](#) the right-hand side is the log odds ratio  $\log\{\pi_{ij}\pi_{kl}/(\pi_{il}\pi_{kj})\}$ . The model [Eq. 37.3.3](#) holds if and only if every such quantity is zero.

*Proof.* In the four terms of [Eq. 37.3.5](#) the constant, the row effects and the column effects each appear twice with a plus and twice with a minus sign, so they cancel, leaving the stated combination equal to  $\log \mu_{ij} + \log \mu_{kl} - \log \mu_{il} - \log \mu_{kj}$ . The cancellation uses nothing about the parameterization, so any two parameterizations of the same fitted means agree. Dividing every  $\mu$  by  $\mu_{++}$  leaves the ratio unchanged, giving the multinomial form; and all such quantities vanish exactly when  $\log \mu_{ij}$  is additive in  $i$  and  $j$ , which is [Eq. 37.3.3](#). ■

The combination in [Proposition 37.3.3](#) is the tetrad difference of [Definition 16.2.1](#) applied to  $\log \mu_{ij}$  instead of to a cell mean, so the apparatus of [Chapter 16](#) transfers: interaction contrasts survive reparameterization ([Proposition 16.2.1](#)) and main effects do not. The difference is that there the scale was chosen for convenience and could be changed by a transformation ([Proposition 16.2.2](#)), whereas here the log scale is forced by the link and “no interaction” means “independence”.

#### Example. (Education and party identification)

The 1996 American National Election Studies sample distributed with `statsmodels` records 944 respondents. Cross-classify them by education in four levels (no diploma, high school graduate, some college, degree) and by party identification collapsed to three, with column totals 380, 239 and 325.

Fitting the independence model [Corollary 37.3.2](#) gives  $G^2 = 20.541$  and  $X^2 = 19.338$  on six degrees of freedom. The smallest expected count is 16.46, so the  $\chi^2$  calibration is safe, and the  $p$ -value is 0.0022: independence is rejected. The largest Pearson residual  $(y_{ij} - \hat{\mu}_{ij})/\sqrt{\hat{\mu}_{ij}}$  is  $-2.62$ , at Republicans without a diploma, where 10 were observed against 22.38 expected; the largest positive one,  $2.31$ , is at Democrats in the same row.

The alternative need not be saturated. Education is ordered, so give it scores  $u_i = 1, 2, 3, 4$  and fit  $\log \mu_{ij} = \lambda + \lambda_i^R + \lambda_j^C + u_i \delta_j$ , spending two parameters instead of six on the association. Its deviance is 7.367 on four degrees of freedom,  $p$ -value 0.1177: this model fits, and the drop from independence is 13.174 on two degrees of freedom,  $p = 0.00138$ .

Now read the same three fits as multinomial logit models for party given education (Definition 36.1.1), with Democrat as the baseline: independence becomes the intercept-only logit model, the score model a logit model linear in  $u_i$ , the saturated model a logit model with a free effect per education level. Each pair gives identical fitted counts, and the association parameters  $\delta_j$  are exactly the education slopes of the logit model, 0.1513 for Independent against Democrat and 0.2748 for Republican against Democrat: each step up the education scale multiplies the odds of Republican rather than Democrat by 1.3163. In the saturated model the interaction contrast comparing the degree row with the no-diploma row and the Republican column with the Democrat column is 1.3658, so those odds are 3.919 times as large among respondents with a degree as among those without a diploma.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm

anes = sm.datasets.anes96.load_pandas().data
party = pd.cut(anes["PID"], [-0.5, 1.5, 4.5, 6.5],
               labels=["Democrat", "Independent", "Republican"])
education = pd.cut(anes["educ"], [0.5, 2.5, 3.5, 4.5, 7.5],
                  labels=["no diploma", "high school", "some college", "degree"])
table = pd.crosstab(education, party)
print(table)
```

```
def poisson_mle(X, y, tol=1e-12, maxit=80):
    """Fisher scoring for log mu = X beta."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(y.mean())
    for _ in range(maxit):
        mu = np.exp(X @ beta)
        XW = X * mu[:, None]
        step = np.linalg.solve(X.T @ XW, XW.T @ (X @ beta + (y - mu) / mu))
        if np.max(np.abs(step - beta)) < tol:
            return step
        beta = step
    return beta

def dummies(levels, k):
    """Indicator columns for levels 1, ..., k-1 of a factor taking the values 0..k-1."""
    return np.column_stack([(levels == j).astype(float) for j in range(1, k)])

counts = table.to_numpy(float)
r, c = counts.shape
score = np.arange(1.0, r + 1) # education score u_i = 1, 2, 3, 4
rows, cols = np.divmod(np.arange(counts.size), c)
y = counts.ravel()
main = np.column_stack([np.ones(counts.size), dummies(rows, r), dummies(cols, c)])
```

```

linear = np.column_stack([main, score[rows][:, None] * dummies(cols, c)])
saturated = np.column_stack([main] + [dummies(rows, r)[:, [i]] * dummies(cols, c)[:,
    [j]]
                                for i in range(r - 1) for j in range(c - 1)])

fits = {}
for name, X in [("independence", main), ("linear", linear), ("saturated", saturated)
    ]:
    beta = poisson_mle(X, y)
    fits[name] = (beta, np.exp(X @ beta).reshape(counts.shape))

print("independence fit:\n", np.round(fits["independence"][1], 2))
print("row totals match:", np.allclose(fits["independence"][1].sum(1), counts.sum(1)
    ))
print("column totals match:", np.allclose(fits["independence"][1].sum(0), counts.sum
    (0)))

```

```

from scipy import stats

def deviance(observed, fitted):
    return 2 * np.sum(observed * np.log(observed / fitted))

for name, dfree in [("independence", 6), ("linear", 4)]:
    dev = deviance(counts, fits[name][1])
    pear = np.sum((counts - fits[name][1]) ** 2 / fits[name][1])
    print(f"{name:13s} G2 = {dev:6.3f} X2 = {pear:6.3f} df = {dfree} "
        f"p = {stats.chi2.sf(dev, dfree):.4f}")

```

The next listing fits the three multinomial logit models directly, by Fisher scoring on the multinomial likelihood, and confirms that each reproduces the fitted counts of its log-linear partner.

```

def multinomial_logit(counts, design, tol=1e-12, maxit=100):
    """Baseline-category logits:  $\log(\pi_{ik} / \pi_{i0}) = \text{design}_{i0} \cdot \gamma_k$ ,  $k = 1..c-1$ ."""
    r, c = counts.shape
    q = design.shape[1]
    gamma = np.zeros((c - 1) * q)
    total = counts.sum(1)
    for _ in range(maxit):
        eta = np.column_stack([np.zeros(r)] +
                                [design @ gamma[k * q:(k + 1) * q] for k in range(c -
1)])
        pi = np.exp(eta) / np.exp(eta).sum(1, keepdims=True)
        mu = total[:, None] * pi
        u = np.concatenate([design.T @ (counts[:, k + 1] - mu[:, k + 1])
                                for k in range(c - 1)])
        info = np.zeros((len(gamma), len(gamma)))
        for k in range(c - 1):
            for l in range(c - 1):
                w = total * pi[:, k + 1] * ((k == l) - pi[:, l + 1])
                info[k * q:(k + 1) * q, l * q:(l + 1) * q] = design.T @ (design * w
[:, None])
        step = np.linalg.solve(info, u)
        gamma = gamma + step
        if np.max(np.abs(step)) < tol:

```

```

        break
    return gamma, mu

designs = {"independence": np.ones((r, 1)),
          "linear": np.column_stack([np.ones(r), score]),
          "saturated": np.column_stack([np.ones(r), dummies(np.arange(r), r)])}
for name, design in designs.items():
    gamma, mu = multinomial_logit(counts, design)
    print(f"{name:13s} largest difference in fitted counts: "
          f"{np.abs(mu - fits[name][1]).max():.2e}")

```

### 37.3.4 More than two classifications

With three classifications the same ideas give a family of models, one for each independence structure. Writing the saturated model as

$$\log \mu_{ijk} = \lambda + \lambda_i^A + \lambda_j^B + \lambda_k^C \\ + \lambda_{ij}^{AB} + \lambda_{ik}^{AC} + \lambda_{jk}^{BC} + \lambda_{ijk}^{ABC},$$

the standard submodels delete terms, always a higher-order term before any lower-order term it contains (a **hierarchical** model). Dropping all two-factor terms gives mutual independence; keeping  $\lambda^{AC}$  and  $\lambda^{BC}$  makes  $A$  and  $B$  conditionally independent given  $C$ ; dropping only  $\lambda^{ABC}$  gives **homogeneous association**, in which the  $AB$  odds ratios are the same at every level of  $C$  ([Exercise \(C2\)](#), [Exercise \(B2\)](#)).

[Theorem 37.1.2](#) makes the fitting transparent: the likelihood equations equate observed and fitted counts in exactly the margins spanned by the model's terms. For mutual independence those are the one-way margins and the fit is a product of proportions; for homogeneous association they are all three two-way margins, which has no closed form and is what iterative fitting is for.

*Remark* (Graphical models). Draw a vertex for each classification and an edge for each two-factor term the model retains. For the hierarchical models whose structure that graph captures, conditional independence given a separating set can be read off the picture: these are the **graphical log-linear models** of Darroch, Lauritzen and Speed (1980), the discrete ancestors of graphical models generally. They also say when collapsing over a variable leaves an association unchanged, which is never safe to assume: the  $AB$  association survives collapsing over  $C$  only when  $C$  is conditionally independent of  $A$ , or of  $B$ , given the other (Bishop, Fienberg and Holland, 1975). Otherwise marginal and conditional associations may differ in size or in sign, as [Section 14.5](#) found for partial correlations.

### 37.3.5 Exercises

#### 37.3.5.1 A. Check your understanding

##### Exercise A1.

A three-way table is  $3 \times 4 \times 2$ . How many cells does it have, how many free parameters



does the homogeneous association model have, and what are the residual degrees of freedom?

### Exercise A2.

Verify from [Theorem 37.3.1\(a\)](#) that the conditional distribution of  $Y_1$  given  $N = n$  is binomial with  $n$  trials and probability  $\mu_1/\mu_+$ , and use it to explain why a Poisson model for two counts is the same as a binomial model for one of them given the total.

### 37.3.5.2 B. Practice

#### Exercise B1.

For the  $2 \times 2$  table with entries  $y_{11}, y_{12}, y_{21}, y_{22}$ , compute the independence fit [Eq. 37.3.4](#) by hand and show that  $y_{ij} - \hat{\mu}_{ij} = \pm(y_{11}y_{22} - y_{12}y_{21})/n$  with alternating signs. Deduce that  $X^2 = n(y_{11}y_{22} - y_{12}y_{21})^2 / (y_{1+}y_{2+}y_{+1}y_{+2})$ .

#### Exercise B2.

For the conditional independence model  $\log \mu_{ijk} = \lambda + \lambda_i^A + \lambda_j^B + \lambda_k^C + \lambda_{ik}^{AC} + \lambda_{jk}^{BC}$ , show that the likelihood equations equate the  $AC$  and  $BC$  margins, that  $\hat{\mu}_{ijk} = y_{i+k}y_{+jk}/y_{++k}$ , and that the residual degrees of freedom are  $(r-1)(c-1)\ell$  for an  $r \times c \times \ell$  table.

*Solution.* The terms in the model span the indicators of the  $AC$  and  $BC$  cells, so [Theorem 37.1.2](#) equates those margins. The model says  $\hat{\mu}_{ijk} = f_{ik}g_{jk}$ , so within each level  $k$  it is the independence model of [Corollary 37.3.2](#) applied to the  $k$ th layer, whose fit is  $y_{i+k}y_{+jk}/y_{++k}$ . Each layer contributes  $(r-1)(c-1)$  residual degrees of freedom, and there are  $\ell$  layers.

#### Exercise B3.

In a  $2 \times c \times \ell$  table with  $A$  binary, show that the homogeneous association model implies that the logit of  $\Pr(A = 1 \mid B = j, C = k)$  is additive in  $j$  and  $k$ , and identify the logistic regression that corresponds to it. Which log-linear terms determine the logistic intercept, which the slopes, and which do not appear at all?

*Solution.* The logit is  $\log(\mu_{1jk}/\mu_{2jk})$ . Substituting the model, the terms  $\lambda, \lambda_j^B, \lambda_k^C$  and  $\lambda_{jk}^{BC}$  cancel, leaving  $(\lambda_1^A - \lambda_2^A) + (\lambda_{1j}^{AB} - \lambda_{2j}^{AB}) + (\lambda_{1k}^{AC} - \lambda_{2k}^{AC})$ , which is additive: it is the main-effects logistic regression of  $A$  on  $B$  and  $C$ . The  $A$  main effect gives the intercept and the  $AB, AC$  terms the slopes;  $\lambda^{BC}$ , the association between the explanatory factors, does not appear.

### 37.3.5.3 C. Going deeper

#### Exercise C1.

The model of [Example \(Education and party identification\)](#) with row scores,  $\log \mu_{ij} =$

$\lambda + \lambda_i^R + \lambda_j^C + u_i \delta_j$ , is called a row-effects model. Show that its local log odds ratios  $\log\{\mu_{ij}\mu_{i+1,j+1}/(\mu_{i,j+1}\mu_{i+1,j})\}$  equal  $(u_{i+1} - u_i)(\delta_{j+1} - \delta_j)$ . Then show that giving *both* classifications scores, with association term  $\theta u_i v_j$ , makes every local log odds ratio  $\theta(u_{i+1} - u_i)(v_{j+1} - v_j)$  — the **uniform association** model — and count its parameters.

### Exercise C2.

For a three-way table, show that the likelihood equations of the mutual independence model are  $\hat{\mu}_{i++} = y_{i++}$ ,  $\hat{\mu}_{+j+} = y_{+j+}$  and  $\hat{\mu}_{++k} = y_{++k}$ , and solve them to get  $\hat{\mu}_{ijk} = y_{i++}y_{+j+}y_{++k}/n^2$ . Explain why the homogeneous association model has no such closed form.

## 37.4 Negative binomial models

The Poisson model spends its parameters on the mean and none on the variance. Whether that is enough is an empirical question, and the answer is usually no.

### 37.4.1 Where overdispersion comes from

Call a count model **overdispersed** relative to the Poisson if  $\text{Var}(Y_i) > \text{E}(Y_i)$ . Leave aside the wrong mean model, which [Example \(Twenty thousand records in thirty-two rows\)](#) ruled out first and which no dispersion parameter repairs. Two mechanisms remain: **unobserved heterogeneity**, the units differing in ways the regressors do not capture, which the calculation below shows always inflates the variance; and **dependence between events**, one accident making the next more likely, so that the count is not Poisson even conditionally. This section models the first.

#### Lemma 37.4.1 — The negative binomial as a gamma mixture of Poissons

Let  $\Lambda$  have the gamma distribution with shape  $\kappa > 0$  and rate  $\kappa/\mu$ , so that  $\text{E } \Lambda = \mu$  and  $\text{Var } \Lambda = \mu^2/\kappa$ , and let  $Y \mid \Lambda = \lambda$  be Poisson with mean  $\lambda$ . Then:

- (a) more generally, for any mixing distribution with  $\text{E } \Lambda = \mu$  and  $\text{Var } \Lambda = \tau^2$ ,  $\text{E } Y = \mu$  and  $\text{Var } Y = \mu + \tau^2$ ;
- (b) with the gamma mixing distribution above, the marginal mass function of  $Y$  is

$$\Pr(Y = y) = \frac{\Gamma(y + \kappa)}{\Gamma(\kappa) y!} \left(\frac{\kappa}{\kappa + \mu}\right)^\kappa \left(\frac{\mu}{\kappa + \mu}\right)^y, \quad y = 0, 1, 2, \dots \quad (37.4.1)$$

with  $\text{E } Y = \mu$  and  $\text{Var } Y = \mu + \mu^2/\kappa$ ;

- (c) as  $\kappa \rightarrow \infty$  with  $\mu$  fixed, [Eq. 37.4.1](#) converges to the Poisson mass function with mean  $\mu$ .

*Proof.*

- (a)  $EY = E\Lambda = \mu$ , and by the law of total variance, using  $\text{Var}(Y | \Lambda) = \Lambda$ ,  $\text{Var } Y = E\Lambda + \text{Var } \Lambda = \mu + \tau^2$ .
- (b) The gamma density is  $f(\lambda) = \{(\kappa/\mu)^\kappa / \Gamma(\kappa)\} \lambda^{\kappa-1} e^{-\kappa\lambda/\mu}$ . Therefore

$$\begin{aligned} \Pr(Y = y) &= \int_0^\infty \frac{e^{-\lambda} \lambda^y}{y!} \frac{(\kappa/\mu)^\kappa}{\Gamma(\kappa)} \lambda^{\kappa-1} e^{-\kappa\lambda/\mu} d\lambda \\ &= \frac{(\kappa/\mu)^\kappa}{y! \Gamma(\kappa)} \int_0^\infty \lambda^{y+\kappa-1} e^{-\lambda(1+\kappa/\mu)} d\lambda. \end{aligned}$$

The integral is  $\Gamma(y + \kappa)(1 + \kappa/\mu)^{-(y+\kappa)}$ , so

$$\begin{aligned} \Pr(Y = y) &= \frac{\Gamma(y + \kappa)}{\Gamma(\kappa)y!} \left(\frac{\kappa}{\mu}\right)^\kappa \left(\frac{\mu}{\mu + \kappa}\right)^{y+\kappa} \\ &= \frac{\Gamma(y + \kappa)}{\Gamma(\kappa)y!} \left(\frac{\kappa}{\kappa + \mu}\right)^\kappa \left(\frac{\mu}{\kappa + \mu}\right)^y, \end{aligned}$$

which is [Eq. 37.4.1](#). The moments follow from (a) with  $\tau^2 = \mu^2/\kappa$ .

- (c) Write  $\alpha = 1/\kappa$  and use  $\Gamma(y + \kappa)/\Gamma(\kappa) = \prod_{k=0}^{y-1} (\kappa + k) = \kappa^y \prod_{k=0}^{y-1} (1 + \alpha k)$ . Then [Eq. 37.4.1](#) becomes

$$\Pr(Y = y) = \frac{\mu^y}{y!} \left\{ \prod_{k=0}^{y-1} (1 + \alpha k) \right\} (1 + \alpha\mu)^{-1/\alpha-y}. \quad (37.4.2)$$

As  $\alpha \downarrow 0$  the product tends to one and  $(1 + \alpha\mu)^{-1/\alpha} \rightarrow e^{-\mu}$ , leaving  $e^{-\mu} \mu^y / y!$ .

■

Part (a) says the inflation is unavoidable: *any* unmodelled variation in the mean adds to the variance, whatever its shape. The gamma is chosen in (b) for its closed form, not because heterogeneity is really gamma distributed; the log-normal gives no closed form, and is the natural choice once random effects enter the picture ([Chapter 40](#)).

### Definition 37.4.1 — Negative binomial regression

Let  $Y_1, \dots, Y_n$  be independent with the mass function [Eq. 37.4.1](#), means  $\mu_i$  and shapes  $\kappa_i$ , and let  $\log \mu_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ .

- (a) The **NB2** model takes  $\kappa_i = \kappa$  constant, so that

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i^2}{\kappa}. \quad (37.4.3)$$

The variance is quadratic in the mean, and the coefficient of variation of  $\Lambda_i$  is the constant  $\kappa^{-1/2}$ .

- (b) The **NB1** model takes  $\kappa_i = \alpha\mu_i$  proportional to the mean, so that

$$\text{Var}(Y_i) = \mu_i + \frac{\mu_i}{\alpha} = \phi \mu_i, \quad \phi = 1 + \alpha^{-1}. \quad (37.4.4)$$

The variance is linear in the mean, with a constant inflation factor.

The parameter  $\alpha = 1/\kappa$  in the NB2 model is often called the **dispersion parameter**;

$\alpha = 0$  is the Poisson.

These are genuinely different models, not reparameterizations of one: NB2 says units differ multiplicatively by a fixed relative amount, NB1 that the excess variance is a fixed multiple of the mean.

## 37.4.2 Estimation and inference

### Proposition 37.4.2 — Fitting the negative binomial

In the NB2 model of Definition 37.4.1 with  $\kappa$  treated as a parameter:

- (a) for fixed  $\kappa$  the distribution Eq. 37.4.1 is an exponential dispersion family (Definition 34.1.1) with natural parameter  $\theta = \log\{\mu/(\mu + \kappa)\}$ , cumulant function  $b(\theta) = -\kappa \log(1 - e^\theta)$ , dispersion  $\phi = 1$  and variance function  $V(\mu) = \mu + \mu^2/\kappa$ ;
- (b) with the log link, the score for  $\beta$  is

$$\frac{\partial \ell}{\partial \beta} = \sum_{i=1}^n \frac{y_i - \mu_i}{1 + \mu_i/\kappa} \mathbf{x}_{(i)}, \quad (37.4.5)$$

so that Fisher scoring is iteratively reweighted least squares with weights  $w_i = \mu_i/(1 + \mu_i/\kappa)$ , and the expected information for  $\beta$  is  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$ ;

- (c)  $\beta$  and  $\kappa$  are orthogonal:  $E\{\partial^2 \ell / \partial \beta \partial \kappa\} = \mathbf{0}$ . Hence the asymptotic distribution of  $\hat{\beta}$  is the same whether  $\kappa$  is known or estimated, and the two can be updated alternately;
- (d) the score for  $\kappa$  is  $\sum_i \{\psi(y_i + \kappa) - \psi(\kappa) + \log \frac{\kappa}{\kappa + \mu_i} + 1 - \frac{\kappa + y_i}{\kappa + \mu_i}\}$ , where  $\psi$  is the digamma function; it has no closed-form root and is solved numerically;
- (e) the Poisson model is the boundary case  $\alpha = 1/\kappa = 0$ . The likelihood ratio statistic  $L$  for  $H_0 : \alpha = 0$  is therefore *not* asymptotically  $\chi^2(1)$ ; under the conditions of Self and Liang (1987) its limiting distribution is  $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$ , so the  $p$ -value is  $\Pr\{N(0, 1) > \sqrt{L}\}$ , half the naive one;
- (f) in the NB1 model of Definition 37.4.1(b), with  $\kappa_i = \alpha \mu_i$  and a log link, the score for  $\beta$  at fixed  $\alpha$  is

$$\frac{\partial \ell}{\partial \beta} = \alpha \sum_{i=1}^n \mu_i \left\{ \psi(y_i + \alpha \mu_i) - \psi(\alpha \mu_i) + \log \frac{\alpha}{1 + \alpha} \right\} \mathbf{x}_{(i)}. \quad (37.4.6)$$

The residual  $y_i - \mu_i$  does not appear, so this is not a weighted least squares equation at all. The quasi-Poisson model of Chapter 38 shares the NB1 variance function  $V(\mu) = \phi \mu$  but has the Poisson estimating equation  $\mathbf{X}^\top (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$ , so the two give different  $\hat{\beta}$  from the same data (Exercise (B3)): a variance function does not determine a fit. Overdispersion is the rule rather than the exception, and modelling it changes the coefficients little and the standard errors a great deal; which of the three devices to use turns on whether a likelihood is needed, as it is for AIC and for Section 37.5.

*Proof.*

- (a) Take logarithms in [Eq. 37.4.1](#):

$$\log \Pr(Y = y) = y \log \frac{\mu}{\mu + \kappa} + \kappa \log \frac{\kappa}{\kappa + \mu} + \log \frac{\Gamma(y + \kappa)}{\Gamma(\kappa) y!}.$$

With  $\theta = \log\{\mu/(\mu + \kappa)\}$ , which is a strictly increasing bijection from  $(0, \infty)$  onto  $(-\infty, 0)$ , we have  $\mu = \kappa e^\theta / (1 - e^\theta)$  and  $\kappa/(\kappa + \mu) = 1 - e^\theta$ , so the second term is  $\kappa \log(1 - e^\theta) = -b(\theta)$ . The last term does not involve  $\mu$ . Then  $b'(\theta) = \kappa e^\theta / (1 - e^\theta) = \mu$  and  $b''(\theta) = \kappa e^\theta / (1 - e^\theta)^2 = \mu(1 + \mu/\kappa)$ , which by [Proposition 34.1.2](#) are the mean and the variance.

- (b) By [Theorem 34.3.1](#) the score is  $\sum_i \mathbf{x}_{(i)}(y_i - \mu_i)(\partial \mu_i / \partial \eta_i) / V(\mu_i)$ . With the log link  $\partial \mu_i / \partial \eta_i = \mu_i$ , and  $\mu_i / V(\mu_i) = \mu_i / \{\mu_i(1 + \mu_i/\kappa)\} = 1/(1 + \mu_i/\kappa)$ , giving the score [Eq. 37.4.5](#). The general IRLS weight of [Theorem 34.4.1](#) is  $(\partial \mu_i / \partial \eta_i)^2 / V(\mu_i) = \mu_i / (1 + \mu_i/\kappa)$ .
- (c) Differentiating [Eq. 37.4.5](#), written as  $\sum_i \mathbf{x}_{(i)} \kappa (y_i - \mu_i) / (\mu_i + \kappa)$ , with respect to  $\kappa$  at fixed  $\boldsymbol{\beta}$  (so that  $\mu_i$  is fixed) gives

$$\begin{aligned} \frac{\partial^2 \ell}{\partial \boldsymbol{\beta} \partial \kappa} &= \sum_i \mathbf{x}_{(i)}(y_i - \mu_i) \frac{\partial}{\partial \kappa} \frac{\kappa}{\mu_i + \kappa} \\ &= \sum_i \mathbf{x}_{(i)}(y_i - \mu_i) \frac{\mu_i}{(\mu_i + \kappa)^2}. \end{aligned}$$

Every term is a constant times  $y_i - \mu_i$ , which has mean zero, so the expectation vanishes. The information matrix is therefore block diagonal, and the leading block of its inverse is the inverse of the  $\boldsymbol{\beta}$  block.

- (d) Differentiate the logarithm of [Eq. 37.4.1](#) in  $\kappa$  at fixed  $\mu$ , using  $d \log \Gamma(x) / dx = \psi(x)$ , and collect terms.
- (e) With  $\kappa_i = \alpha \mu_i$  the two ratios in [Eq. 37.4.1](#) become constants,  $\kappa_i / (\kappa_i + \mu_i) = \alpha / (1 + \alpha)$  and  $\mu_i / (\mu_i + \kappa_i) = 1 / (1 + \alpha)$ , so

$$\ell = \sum_i \left\{ \log \frac{\Gamma(y_i + \alpha \mu_i)}{\Gamma(\alpha \mu_i) y_i!} + \alpha \mu_i \log \frac{\alpha}{1 + \alpha} - y_i \log(1 + \alpha) \right\}.$$

Only  $\alpha \mu_i$  depends on  $\boldsymbol{\beta}$ ; differentiating with  $\partial \mu_i / \partial \boldsymbol{\beta} = \mu_i \mathbf{x}_{(i)}$  gives [Eq. 37.4.6](#).

- (f) By [Lemma 37.4.1\(c\)](#) the Poisson is the limit  $\alpha \downarrow 0$ , and  $\alpha \geq 0$  is required for [Eq. 37.4.1](#) to be a distribution, so the null value is on the boundary. The expansion behind the  $\chi^2(1)$  limit assumes an interior maximum; here the unconstrained maximizer of the profile likelihood falls below zero about half the time, and  $L = 0$  on that event. [Theorem 32.5.2](#) computes the analogous mixture exactly in a balanced variance-components problem; the general statement is Self and Liang's. The upper tail of  $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$  at  $L$  is  $\frac{1}{2} \Pr\{\chi^2(1) > L\} = \Pr\{N(0, 1) > \sqrt{L}\}$ .

■

Part (c) makes the alternating algorithm honest as well as convenient: iterate the weighted least squares step of (b) at the current  $\kappa$ , then solve (d) at the current  $\boldsymbol{\beta}$ , and the standard errors computed as though  $\kappa$  were known are asymptotically correct.

### 37.4.3 A score test for overdispersion

The likelihood ratio test in [Proposition 37.4.2\(e\)](#) needs the negative binomial fit. A score test needs only the Poisson fit, which makes it the natural diagnostic to compute alongside any Poisson regression.

#### Theorem 37.4.3 — Score test against NB2 alternatives

Let  $\hat{\mu}_i$  be the fitted means of a Poisson log-linear model with  $\text{rank}(\mathbf{X}) = p$ , and consider the NB2 family [Eq. 37.4.2](#) indexed by  $\alpha \geq 0$ . Then the score for  $\alpha$  at  $\alpha = 0$  is

$$\left. \frac{\partial \ell}{\partial \alpha} \right|_{\alpha=0} = \frac{1}{2} \sum_{i=1}^n \{ (y_i - \mu_i)^2 - y_i \}, \quad (37.4.7)$$

its variance under the Poisson model is  $\frac{1}{2} \sum_i \mu_i^2$ , and its covariance with the score for  $\beta$  is zero. Consequently

$$S = \frac{\sum_{i=1}^n \{ (y_i - \hat{\mu}_i)^2 - y_i \}}{\sqrt{2 \sum_{i=1}^n \hat{\mu}_i^2}} \quad (37.4.8)$$

is asymptotically standard normal under the Poisson model, and the test rejects for large positive  $S$ .

*Proof.* Take logarithms in [Eq. 37.4.2](#):

$$\ell_i(\alpha) = y_i \log \mu_i - \log y_i! + \sum_{k=0}^{y_i-1} \log(1 + \alpha k) - \left( \frac{1}{\alpha} + y_i \right) \log(1 + \alpha \mu_i).$$

Expand in  $\alpha$ . The sum contributes  $\alpha \sum_{k=0}^{y_i-1} k + O(\alpha^2) = \alpha y_i(y_i - 1)/2 + O(\alpha^2)$ . For the last term,  $\log(1 + \alpha \mu_i) = \alpha \mu_i - \alpha^2 \mu_i^2/2 + O(\alpha^3)$ , so

$$\left( \frac{1}{\alpha} + y_i \right) \log(1 + \alpha \mu_i) = \mu_i + \alpha \left( y_i \mu_i - \frac{\mu_i^2}{2} \right) + O(\alpha^2).$$

Hence

$$\begin{aligned} \ell_i(\alpha) &= y_i \log \mu_i - \mu_i - \log y_i! \\ &\quad + \alpha \{ y_i(y_i - 1)/2 - y_i \mu_i + \mu_i^2/2 \} + O(\alpha^2). \end{aligned}$$

The leading term is the Poisson log-likelihood and the coefficient of  $\alpha$  is  $\frac{1}{2} \{ y_i^2 - y_i - 2y_i \mu_i + \mu_i^2 \} = \frac{1}{2} \{ (y_i - \mu_i)^2 - y_i \}$ , which gives [Eq. 37.4.7](#).

For the variance, put  $Z_i = Y_i - \mu_i$  and  $A_i = (Y_i - \mu_i)^2 - Y_i = Z_i^2 - Z_i - \mu_i$ . For a Poisson variable,  $E Z_i = 0$ ,  $E Z_i^2 = \mu_i$ ,  $E Z_i^3 = \mu_i$  and  $E Z_i^4 = \mu_i + 3\mu_i^2$ . Therefore  $E A_i = \mu_i - 0 - \mu_i = 0$  and

$$\begin{aligned} \text{Var } A_i &= E(Z_i^2 - Z_i - \mu_i)^2 \\ &= E Z_i^4 - 2 E Z_i^3 - 2\mu_i E Z_i^2 + E Z_i^2 + 2\mu_i E Z_i + \mu_i^2 \\ &= 2\mu_i^2. \end{aligned}$$

The  $A_i$  are independent, so the variance of Eq. 37.4.7 is  $\frac{1}{4} \sum_i 2\mu_i^2 = \frac{1}{2} \sum_i \mu_i^2$ .

For the covariance, differentiate Eq. 37.4.7 in  $\beta$ , using  $\partial\mu_i/\partial\beta = \mu_i \mathbf{x}_{(i)}$ :

$$\frac{\partial}{\partial\beta} \frac{1}{2} \{(y_i - \mu_i)^2 - y_i\} = -\mu_i(y_i - \mu_i) \mathbf{x}_{(i)},$$

whose expectation is zero. The information between  $\alpha$  and  $\beta$  therefore vanishes at  $\alpha = 0$ , so the usual adjustment for an estimated nuisance parameter is zero and replacing  $\beta$  by  $\hat{\beta}$  does not change the asymptotic variance of the score to first order. Dividing by the standard deviation, with  $\hat{\mu}_i$  for  $\mu_i$ , gives Eq. 37.4.8. For the limit, the  $A_i$  are independent with mean zero and  $E A_i^4 = O(\max(\mu_i, 1)^4)$ , a fourth-moment calculation like the one above. So if the  $\mu_i$  are bounded away from zero and

$$\frac{\sum_{i=1}^n \mu_i^4}{(\sum_{i=1}^n \mu_i^2)^2} \rightarrow 0,$$

Lyapunov's condition holds with  $\delta = 2$  and the standardized sum is asymptotically standard normal. That covers the case in which the fitted means stay in a fixed compact interval inside  $(0, \infty)$ ; Dean and Lawless (1989) treat the general one. ■

The statistic is one-sided by construction: a count *less* variable than the Poisson is possible but not negative binomial, so a large negative  $S$  points somewhere else (Exercise (C1)).

### 37.4.4 The physician visits again

#### Example. (Negative binomial fits to the visit counts)

Refit the model of Example (Physician visits in a health insurance experiment) with the NB2 distribution, alternating the weighted least squares step of Proposition 37.4.2(b) with a one-dimensional solve for  $\kappa$ . The fit gives  $\hat{\kappa} = 0.7637$ , that is  $\hat{\alpha} = 1.3094$ ; at the overall mean of 2.860 visits the model's variance is 13.574 against the Poisson's 2.860. The maximized log-likelihood rises from  $-62737.33$  to  $-43468.52$ , so the likelihood ratio statistic for  $H_0 : \alpha = 0$  is 38537.6; the boundary correction of Proposition 37.4.2(e) halves a  $p$ -value whose logarithm is about  $-19275$ . The score statistic Eq. 37.4.8, from the Poisson fit alone, is 523.0.

What changes is inference. For the individual-deductible indicator the point estimate moves only from  $-0.2648$  to  $-0.2475$ , but its standard error moves from 0.0102 to 0.0218, a factor of 2.125. Two other repairs agree: scaling the Poisson standard errors by  $\sqrt{\hat{\phi}}$ , with  $\hat{\phi} = 6.319$  the Pearson statistic over its degrees of freedom, gives 0.0257, and the sandwich estimator of Theorem 21.4.2 gives 0.0268. A Poisson analysis would have declared effects significant on the strength of a variance assumption the data reject.

Fitting NB1 instead gives log-likelihood  $-43380.59$  with  $\hat{\phi} = 4.766$ , a better fit than NB2 by AIC: 86779.2 against 86955.0, with the Poisson far behind at 125490.7. Figure 37.4.1 shows why the comparison is close. Sorting the person-years by fitted mean into twenty-five bins, the bin variances follow the NB2 curve at the high end, while in the crowded low range, where most of the data and most of the likelihood sit, NB1 is closer.

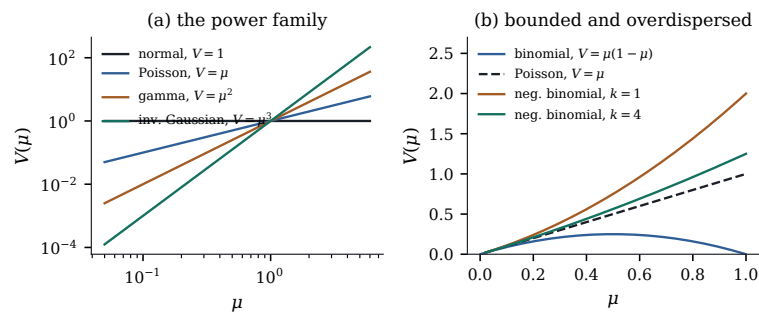


Figure 37.4.1: Sample variances of the physician-visit counts within twenty-five bins of the fitted Poisson mean, against the three fitted variance functions. The Poisson line is far below every bin.

AIC weighs the whole likelihood and the figure the upper tail; when they disagree, choose by the use the model will be put to.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats

data = sm.datasets.randhie.load_pandas().data
y = data["mdvis"].to_numpy(float)
names = ["intercept", "lncoins", "idp", "physlm", "disea", "hlthg", "hlthf", "hlthp"]
X = np.column_stack([np.ones(len(y))] + [data[v].to_numpy(float) for v in names
[1:]])
n, p = X.shape

beta_pois = np.zeros(p)
beta_pois[0] = np.log(y.mean())
for _ in range(50):
    mu = np.exp(X @ beta_pois)
    XW = X * mu[:, None]
    beta_pois = np.linalg.solve(X.T @ XW, XW.T @ (X @ beta_pois + (y - mu) / mu))
mu_pois = np.exp(X @ beta_pois)
info_pois = X.T @ (X * mu_pois[:, None])
se_pois = np.sqrt(np.diag(np.linalg.inv(info_pois)))
log_factorial = special.gammaln(y + 1)
loglik_pois = np.sum(y * np.log(mu_pois) - mu_pois - log_factorial)
print(f"Poisson log-likelihood {loglik_pois:.2f}")
```

```
def nb_loglik(beta, kappa, X, y):
    """Negative binomial log-likelihood: mean exp(X beta), shape kappa (NB2 if
    scalar)."""
    mu = np.exp(X @ beta)
    return np.sum(special.gammaln(y + kappa) - special.gammaln(kappa)
                  - special.gammaln(y + 1)
                  + kappa * np.log(kappa / (kappa + mu))
                  + y * np.log(mu / (mu + kappa)))

def kappa_score(kappa, mu, y):
    """Derivative of the log-likelihood in the shape parameter."""
    return np.sum(special.digamma(y + kappa) - special.digamma(kappa)
                  + np.log(kappa / (kappa + mu)) + 1 - (kappa + y) / (kappa + mu))
```



```

def fit_nb2(X, y, tol=1e-11, maxit=200):
    """Alternate IRLS in beta with a one-dimensional search in kappa."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(y.mean())
    kappa = 1.0
    for _ in range(maxit):
        mu = np.exp(X @ beta)
        w = mu / (1 + mu / kappa)                # NB2 working weights, log link
        XW = X * w[:, None]
        step = np.linalg.solve(X.T @ XW, XW.T @ (X @ beta + (y - mu) / mu))
        new_kappa = optimize.brentq(kappa_score, 1e-4, 1e4,
                                   args=(np.exp(X @ step), y), xtol=1e-12)
        done = max(np.abs(step - beta).max(), abs(new_kappa - kappa)) < tol
        beta, kappa = step, new_kappa
    if done:
        break
    return beta, kappa, nb_loglik(beta, kappa, X, y)

beta_nb2, kappa, loglik_nb2 = fit_nb2(X, y)
mu_nb2 = np.exp(X @ beta_nb2)
weights = mu_nb2 / (1 + mu_nb2 / kappa)          # IRLS weights for NB2 with a log
link
se_nb2 = np.sqrt(np.diag(np.linalg.inv(X.T @ (X * weights[:, None]))))

print(f"kappa = {kappa:.4f}    log-likelihood {loglik_nb2:.2f}")
for name, b, s, sp in zip(names, beta_nb2, se_nb2, se_pois):
    print(f"{name:10s} {b:8.4f}    se {s:.4f}    (Poisson se {sp:.4f})")

```

```

lr = 2 * (loglik_nb2 - loglik_pois)
# the null puts 1/kappa on the boundary, so the null distribution of L is the
mixture
# (1/2) chi2(0) + (1/2) chi2(1) and the p-value is the normal tail at sqrt(L)
log_p = stats.norm.logsf(np.sqrt(lr))

score = np.sum((y - mu_pois) ** 2 - y) / np.sqrt(2 * np.sum(mu_pois ** 2))
print(f"likelihood ratio {lr:.1f}, log p-value {log_p:.0f}")
print(f"score statistic {score:.1f}, log p-value {stats.norm.logsf(score):.0f}")

```

## 37.4.5 Exercises

### 37.4.5.1 A. Check your understanding

#### Exercise A1.

A negative binomial NB2 fit reports  $\hat{\kappa} = 2$ . By what factor does the model inflate the variance over the Poisson at a fitted mean of 1, of 10, and of 100? Repeat for an NB1 fit with  $\hat{\phi} = 3$ , and comment.

**Exercise A2.**

A colleague reports a likelihood ratio statistic of 2.9 for Poisson against NB2 and, comparing with  $\chi^2(1)$ , obtains  $p = 0.089$  and concludes there is no overdispersion. What is the correct  $p$ -value, and does the conclusion change at the five per cent level?

**37.4.5.2 B. Practice****Exercise B1.**

For an intercept-only Poisson model,  $\hat{\mu}_i = \bar{y}$  for every  $i$ . Show that the score statistic [Eq. 37.4.8](#) reduces to  $\sqrt{n/2} \{s_n^2/\bar{y} - 1\}$  with  $s_n^2 = n^{-1} \sum_i (y_i - \bar{y})^2$ , so that it compares the sample variance with the sample mean. Evaluate it for  $n = 100$ ,  $\bar{y} = 3$ ,  $s_n^2 = 5$ .

*Solution.* With  $\hat{\mu}_i = \bar{y}$  the numerator is  $\sum_i (y_i - \bar{y})^2 - n\bar{y} = n(s_n^2 - \bar{y})$  and the denominator is  $\sqrt{2n\bar{y}^2} = \bar{y}\sqrt{2n}$ , so the ratio is  $\sqrt{n/2}(s_n^2 - \bar{y})/\bar{y}$ . For the numbers given this is  $\sqrt{50} \times (5 - 3)/3$ , about 4.7: strong evidence of overdispersion.

**Exercise B2.**

Show that if  $Y$  is NB2 with mean  $\mu$  and shape  $\kappa$ , then  $\Pr(Y = 0) = (1 + \mu/\kappa)^{-\kappa}$ , that this exceeds  $e^{-\mu}$  for every finite  $\kappa$ , and that it tends to one as  $\kappa \rightarrow 0$  with  $\mu$  fixed. Interpret the last statement.

*Solution.* Setting  $y = 0$  in [Eq. 37.4.1](#) leaves  $\{\kappa/(\kappa + \mu)\}^\kappa$ , which is  $(1 + \mu/\kappa)^{-\kappa}$ . Since  $\log(1 + x) < x$  for  $x > 0$ ,  $\kappa \log(1 + \mu/\kappa) < \mu$ , so the probability exceeds  $e^{-\mu}$ . As  $\kappa \rightarrow 0$ ,  $\kappa \log(1 + \mu/\kappa) \rightarrow 0$  and the probability tends to one: extreme heterogeneity concentrates the distribution on zero with occasional very large counts, which is why the negative binomial can on its own account for a great many zeros.

**Exercise B3.**

Show that the quasi-Poisson estimating equation for  $\beta$  is the *same* as the Poisson one, so that  $\hat{\beta}$  is unchanged, while the NB2 equation [Eq. 37.4.5](#) downweights observations with large fitted means. Which model would you expect to be more affected by a single very large count?

**Exercise B4.**

Show that the deviance ([Definition 34.5.1](#)) of the NB2 model with  $\kappa$  known is

$$D = 2 \sum_{i=1}^n \left\{ y_i \log \frac{y_i}{\hat{\mu}_i} - (y_i + \kappa) \log \frac{y_i + \kappa}{\hat{\mu}_i + \kappa} \right\},$$

and that it tends to the Poisson deviance [Eq. 37.1.5](#) as  $\kappa \rightarrow \infty$ . Why does an analysis of deviance based on it need  $\kappa$  to be held fixed across the models compared?

*Solution.* Dropping the terms free of  $\mu$ , the log-likelihood contribution of observation  $i$  is  $y_i \log\{\mu_i/(\mu_i + \kappa)\} + \kappa \log\{\kappa/(\mu_i + \kappa)\}$ . The saturated model puts  $\mu_i = y_i$ , so twice the difference is

$$2\left\{y_i \log \frac{y_i}{\hat{\mu}_i} + (y_i + \kappa) \log \frac{\hat{\mu}_i + \kappa}{y_i + \kappa}\right\},$$

which is the stated formula. For the limit write  $(y_i + \kappa) \log\{1 + (y_i - \hat{\mu}_i)/(\hat{\mu}_i + \kappa)\} = (y_i - \hat{\mu}_i) + O(\kappa^{-1})$ . The deviance depends on  $\kappa$ , so two models fitted with different  $\hat{\kappa}$  have deviances on different scales and their difference is not a likelihood ratio statistic; compare their maximized log-likelihoods instead.

### 37.4.5.3 C. Going deeper

#### Exercise C1.

Construct a count distribution with mean  $\mu$  and variance strictly less than  $\mu$ , and explain why no member of the negative binomial family has this property. What mechanisms produce underdispersed counts in practice, and what would you fit?

*Solution.* The binomial with  $m$  trials and probability  $\pi$  has mean  $m\pi$  and variance  $m\pi(1 - \pi) < m\pi$ , while by [Lemma 37.4.1\(a\)](#) every Poisson mixture has variance at least its mean. Underdispersion arises when events are regular rather than random — inspections on a schedule, quotas — and the natural models are the binomial, or a Conway–Maxwell–Poisson family with a dispersion parameter free on both sides of one.

#### Exercise C2.

[Proposition 37.4.2\(a\)](#) identifies the canonical link of the NB2 family as  $\theta = \log\{\mu/(\mu + \kappa)\}$ . Work out the likelihood equations under this link, show that they take the form  $\mathbf{X}^\top(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$ , and explain why the log link is nevertheless the standard choice.

## 37.5 Zero inflation

Many count data sets have far more zeros than a Poisson or negative binomial model with the right mean can produce. The usual explanation is that zeros come from two places: some units are not at risk at all — they do not smoke, do not drive, are not eligible — and contribute a zero with certainty, while the rest contribute one only when the count happens to be zero. This section writes that story down as a model, does the same for its rival, and asks how the two can be told apart.

### 37.5.1 Two ways to make extra zeros

#### Definition 37.5.1 — Zero-inflated and hurdle models

Let  $f(y; \lambda)$  be a count distribution with parameter  $\lambda$ , such as the Poisson or the NB2 family of [Definition 37.4.1](#), and let  $\mathbf{z}_{(i)}$  and  $\mathbf{x}_{(i)}$  be rows of two model matrices, which

may be equal, overlapping or disjoint.

- (a) The **zero-inflated** model gives  $Y_i$  the mass function

$$\Pr(Y_i = y) = \begin{cases} \pi_i + (1 - \pi_i)f(0; \lambda_i), & y = 0, \\ (1 - \pi_i)f(y; \lambda_i), & y = 1, 2, \dots, \end{cases} \quad (37.5.1)$$

with  $\text{logit } \pi_i = \mathbf{z}_{(i)}^\top \boldsymbol{\gamma}$  and  $\log \lambda_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ . Equivalently, a latent indicator  $Z_i$  is one with probability  $\pi_i$ , and  $Y_i = 0$  when  $Z_i = 1$  while  $Y_i \sim f(\cdot; \lambda_i)$  when  $Z_i = 0$ . Taking  $f$  Poisson gives the **zero-inflated Poisson** (ZIP), and  $f$  negative binomial the **zero-inflated negative binomial** (ZINB).

- (b) The **hurdle** model gives  $Y_i$  the mass function

$$\Pr(Y_i = y) = \begin{cases} 1 - \rho_i, & y = 0, \\ \rho_i \frac{f(y; \lambda_i)}{1 - f(0; \lambda_i)}, & y = 1, 2, \dots, \end{cases} \quad (37.5.2)$$

with  $\text{logit } \rho_i = \mathbf{z}_{(i)}^\top \boldsymbol{\gamma}$  and  $\log \lambda_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ : a binary model decides whether the count is positive, and a **zero-truncated** version of  $f$  decides how large it is.

These are different stories. The zero-inflated model says there are two kinds of unit and a zero does not reveal which; the hurdle model says there is one kind of unit and two decisions, whether to cross the threshold and how far to go once across. Which is right is a subject-matter question, and the data distinguish them only indirectly.

### Proposition 37.5.1 — Properties of the two models

In the notation of Definition 37.5.1, write  $\mu_i = E(Y_i)$  and  $\mathbf{1}_i = \mathbf{1}\{y_i > 0\}$ . The zero-inflated log-likelihood is

$$\begin{aligned} & \sum_{i: y_i=0} \log\{\pi_i + (1 - \pi_i)f(0; \lambda_i)\} \\ & + \sum_{i: y_i>0} [\log(1 - \pi_i) + \log f(y_i; \lambda_i)], \end{aligned}$$

and the hurdle one is

$$\begin{aligned} & \sum_i [\mathbf{1}_i \log \rho_i + (1 - \mathbf{1}_i) \log(1 - \rho_i)] \\ & + \sum_{i: y_i>0} [\log f(y_i; \lambda_i) - \log\{1 - f(0; \lambda_i)\}]. \end{aligned}$$

Then:

- (a) for the zero-inflated model with  $f$  Poisson,

$$\mu_i = (1 - \pi_i)\lambda_i, \quad \text{Var}(Y_i) = \mu_i(1 + \pi_i\lambda_i), \quad (37.5.3)$$

so a zero-inflated Poisson is always overdispersed relative to its own mean unless  $\pi_i = 0$ ;

- (b) the hurdle log-likelihood separates,  $\ell(\boldsymbol{\gamma}, \boldsymbol{\beta}, \cdot) = \ell_{\text{binary}}(\boldsymbol{\gamma}) + \ell_{\text{positive}}(\boldsymbol{\beta}, \cdot)$ , so the two parts can be fitted independently, and their estimates are asymptotically independent; the zero-inflated log-likelihood does not separate;
- (c) the hurdle model can represent **zero deflation**,  $\rho_i > 1 - f(0; \lambda_i)$ , while [Eq. 37.5.1](#) needs  $\pi_i \geq 0$  and can only add zeros; and, for the hurdle model, if  $\mathbf{z}_{(i)}$  contains an intercept then its fitted probabilities of a zero average to the observed proportion of zeros, whatever else is in the binary part. The zero-inflated model satisfies no such identity;
- (d) the zero-inflated model contains  $f$  itself as the limiting case  $\pi_i \equiv 0$ , which lies outside the range where the usual expansion is valid, so a likelihood ratio test of  $f$  against its zero-inflated version is *not* asymptotically  $\chi^2$ ; nor is the Vuong statistic of [Eq. 37.5.5](#) valid for such a comparison.

*Proof.*

- (a) With the latent representation,  $E(Y_i) = E\{E(Y_i | Z_i)\} = (1 - \pi_i)\lambda_i = \mu_i$ . Since  $E(Y_i | Z_i) = \lambda_i(1 - Z_i)$  has variance  $\lambda_i^2\pi_i(1 - \pi_i)$ , the law of total variance gives

$$\begin{aligned}\text{Var}(Y_i) &= E\{\text{Var}(Y_i | Z_i)\} + \text{Var}\{E(Y_i | Z_i)\} \\ &= (1 - \pi_i)\lambda_i + \lambda_i^2\pi_i(1 - \pi_i).\end{aligned}$$

Factoring out  $(1 - \pi_i)\lambda_i = \mu_i$  gives  $\text{Var}(Y_i) = \mu_i(1 + \pi_i\lambda_i)$ , which exceeds  $\mu_i$  whenever  $\pi_i > 0$ .

- (b) With  $\mathbf{1}_i = \mathbf{1}\{y_i > 0\}$ , [Eq. 37.5.2](#) gives

$$\begin{aligned}\ell &= \sum_i \left[ \mathbf{1}_i \log \rho_i + (1 - \mathbf{1}_i) \log(1 - \rho_i) \right] \\ &\quad + \sum_{i: y_i > 0} \left[ \log f(y_i; \lambda_i) - \log\{1 - f(0; \lambda_i)\} \right].\end{aligned}$$

The first sum involves only  $\boldsymbol{\gamma}$  and the second only  $\boldsymbol{\beta}$  and any shape parameter of  $f$ , with unconstrained parameter spaces, so the joint maximum is the pair of separate maxima and the information is block diagonal. For [Eq. 37.5.1](#) the contribution of a zero is  $\log\{\pi_i + (1 - \pi_i)f(0; \lambda_i)\}$ , which involves both parameter vectors inside one logarithm and does not separate.

- (c) In [Eq. 37.5.2](#),  $\rho_i$  is a free probability, so  $\Pr(Y_i = 0) = 1 - \rho_i$  can take any value in  $(0, 1)$ ; in [Eq. 37.5.1](#),  $\Pr(Y_i = 0) = \pi_i + (1 - \pi_i)f(0; \lambda_i) \geq f(0; \lambda_i)$  for  $\pi_i \in [0, 1]$ . The first sum in the display in (b) is the log-likelihood of a logistic regression of  $\mathbf{1}_i$  on  $\mathbf{z}_{(i)}$ , whose likelihood equations are  $\sum_i (\mathbf{1}_i - \hat{\rho}_i)\mathbf{z}_{(i)} = \mathbf{0}$  by [Theorem 34.3.1](#) with the canonical link. If  $\mathbf{z}_{(i)}$  contains an intercept, one of those equations reads  $\sum_i \hat{\rho}_i = \sum_i \mathbf{1}_i$ , so  $n^{-1} \sum_i (1 - \hat{\rho}_i)$  is the observed proportion of zeros. The zero-inflated likelihood equation for the inflation intercept is instead  $\sum_i \hat{\pi}_i = \sum_i \hat{w}_i$ , with  $\hat{w}_i$  the posterior weights [Eq. 37.5.4](#), which says nothing about  $\sum_i \widehat{\Pr}(Y_i = 0)$ .
- (d) Setting  $\pi_i \equiv 0$  in [Eq. 37.5.1](#) recovers  $f$ , but how the  $\chi^2$  argument fails depends on the parameterization. If inflation is a single probability  $\pi$ , the null sits on the boundary of  $[0, 1]$  and [Proposition 37.4.2\(e\)](#) applies verbatim. If  $\text{logit } \pi_i = \mathbf{z}_{(i)}^\top \boldsymbol{\gamma}$ , the null is reached only as the intercept tends to  $-\infty$ , and under it the remaining coefficients of  $\boldsymbol{\gamma}$  are unidentified — a harder non-regularity, where no fixed mixture of  $\chi^2$  laws is the limit. Either way the calibration fails. The Vuong statistic assumes

strictly non-nested models, so that the mean of its summands is nonzero under the null; when they are nested that mean is zero and the limit is not normal.

■

Part (a) matters when choosing among the four combinations. A zero-inflated Poisson is already overdispersed, so excess zeros with no other anomaly can be described either by a ZIP or by a negative binomial; what a ZIP cannot do is produce a long right tail without also producing extra zeros. Zero inflation on a negative binomial addresses zeros and tail separately, which is why ZINB is the workhorse.

## 37.5.2 Fitting

The hurdle model needs no new machinery: fit a logistic regression to  $\mathbf{1}\{y_i > 0\}$ , then maximize  $\sum_{y_i > 0} [\log f(y_i; \lambda_i) - \log\{1 - f(0; \lambda_i)\}]$  numerically over the positive observations. The zero-inflated model does not separate, but its latent representation invites the EM algorithm, which is what Lambert used. The E step computes the posterior probability that observation  $i$  came from the degenerate component,

$$w_i = \Pr(Z_i = 1 \mid Y_i = y_i) = \begin{cases} \frac{\pi_i}{\pi_i + (1 - \pi_i)f(0; \lambda_i)}, & y_i = 0, \\ 0, & y_i > 0, \end{cases} \quad (37.5.4)$$

by Bayes's rule. The M step maximizes the expected complete-data log-likelihood, which *does* separate: a logistic regression of the fractional response  $w_i$  on  $\mathbf{Z}$ , and a count regression of  $y_i$  on  $\mathbf{X}$  with prior weights  $1 - w_i$ . Both are weighted fits already in hand. The M step need only *increase* that expected log-likelihood, not maximize it, so a single weighted least squares step for  $\beta$  and a single Newton step for the shape suffice; either way the observed-data log-likelihood increases at every iteration ([Exercise \(B4\)](#)).

## 37.5.3 An example

### Example. (Five models for the physician visits)

A proportion 0.3124 of the 20190 person-years record no physician visit; the Poisson model of [Example \(Physician visits in a health insurance experiment\)](#) predicts 0.0809.

**Table 37.5.1.** Five count models for the physician-visit data, all with the same eight regressors; the zero-inflated models use them in both parts. The observed proportion of zeros is 0.3124.

| model     | parameters | log-likelihood | AIC      | BIC      | fitted $\Pr(Y = 0)$ |
|-----------|------------|----------------|----------|----------|---------------------|
| Poisson   | 8          | −62737.3       | 125490.7 | 125554.0 | 0.0809              |
| NB2       | 9          | −43468.5       | 86955.0  | 87026.3  | 0.3200              |
| ZIP       | 16         | −54978.6       | 109989.1 | 110115.8 | 0.3125              |
| ZINB      | 17         | −43391.8       | 86817.5  | 86952.0  | 0.3255              |
| hurdle NB | 17         | −43244.6       | 86523.2  | 86657.7  | 0.3124              |

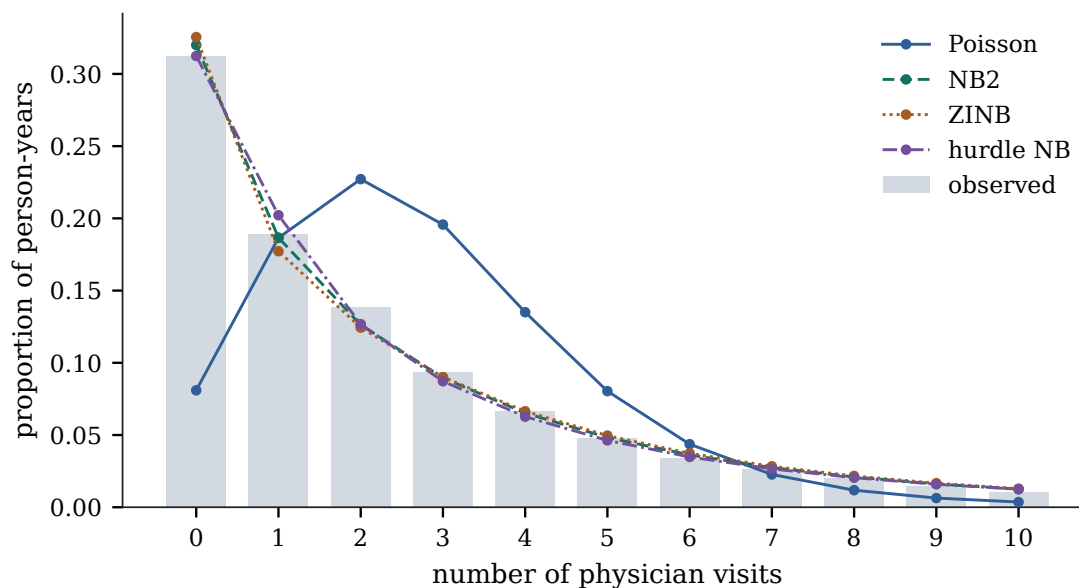


Figure 37.5.1: Observed distribution of physician visits (bars) and the fitted marginal distributions  $n^{-1} \sum_i \Pr(Y_i = k)$  under four models. The Poisson is hopeless at zero and too heavy in the middle; the other three are close to each other and to the data.

Three things stand out. First, zero inflation alone does not save the Poisson: the ZIP reproduces the zeros almost exactly, at 0.3125, and still trails the plain negative binomial by more than eleven thousand in log-likelihood, because the trouble here is the whole shape of the distribution and not only its zeros. Second, the negative binomial *already* matches the zeros, at 0.3200 against 0.3124, as [Exercise \(B2\)](#) predicts: heterogeneity in the rate produces excess zeros as a by-product, and adding zero inflation gains 153.5 in twice the log-likelihood for eight parameters while slightly *overshooting*, at 0.3255. Third, the hurdle model wins on both AIC and BIC, and by construction reproduces the zero proportion exactly.

The hurdle model wins because of the positive counts, not the zeros. Splitting each log-likelihood at  $y = 0$  shows it: on the zeros the hurdle model scores  $-6966.1$  against the ZINB's  $-6757.5$ , a loss of 208.6; on the positive counts  $-36278.5$  against  $-36634.2$ , a gain of 355.7, for a net 147.2. Above two visits the two are indistinguishable in [Figure 37.5.1](#); the difference is in how mass is arranged over the small positive counts, and the hurdle model buys that freedom by giving the truncated component its own shape,  $\hat{\kappa} = 0.557$  against 0.838 in the ZINB. Estimated zero inflation in the ZINB averages only 0.0475: it too spends its extra parameters re-shaping the low counts, which is what a hurdle model does directly.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats

data = sm.datasets.randhie.load_pandas().data
y = data["mdvis"].to_numpy(float)
names = ["lncoins", "idp", "physlm", "disea", "hlthg", "hlthf", "hlthp"]
X = np.column_stack([np.ones(len(y))] + [data[v].to_numpy(float) for v in names])
n, p = X.shape

def poisson_irls(X, y, w=None, start=None, tol=1e-11, maxit=200):
```

```

"""Weighted Poisson regression with a log link; w are prior weights."""
w = np.ones(len(y)) if w is None else w
beta = np.zeros(X.shape[1]) if start is None else start.copy()
if start is None:
    beta[0] = np.log(max((w * y).sum() / w.sum(), 1e-8))
for _ in range(maxit):
    mu = np.exp(X @ beta)
    XW = X * (w * mu)[:, None]
    step = np.linalg.solve(X.T @ XW, XW.T @ (X @ beta + (y - mu) / mu))
    if np.max(np.abs(step - beta)) < tol:
        return step
    beta = step
return beta

def logistic_irls(X, r, tol=1e-11, maxit=200):
    """Logistic regression with a response r in [0, 1] (a fractional response is
    allowed)."""
    gamma = np.zeros(X.shape[1])
    for _ in range(maxit):
        pi = 1 / (1 + np.exp(-X @ gamma))
        v = np.clip(pi * (1 - pi), 1e-10, None)
        XW = X * v[:, None]
        step = np.linalg.solve(X.T @ XW, XW.T @ (X @ gamma + (r - pi) / v))
        if np.max(np.abs(step - gamma)) < tol:
            return step
        gamma = step
    return gamma

def nb_logpmf(y, mu, kappa):
    """log P(Y = y) for the negative binomial with mean mu and shape kappa."""
    return (special.gammaln(y + kappa) - special.gammaln(kappa) - special.gammaln(y
    + 1)
            + kappa * np.log(kappa / (kappa + mu)) + y * np.log(mu / (mu + kappa)))

def poisson_logpmf(y, mu):
    return y * np.log(mu) - mu - special.gammaln(y + 1)

def nb_step(X, y, beta, kappa, w):
    """One IRLS step in beta and one Newton step in log kappa, at prior weights w.
    """
    mu = np.exp(X @ beta)
    XW = X * (w * mu / (1 + mu / kappa))[:, None] # NB2 working weights, log
    link
    beta = np.linalg.solve(X.T @ XW, XW.T @ (X @ beta + (y - mu) / mu))
    mu = np.exp(X @ beta)
    s = np.sum(w * (special.digamma(y + kappa) - special.digamma(kappa)
                    + np.log(kappa / (kappa + mu)) + 1 - (kappa + y) / (kappa + mu))
    )
    d = np.sum(w * (special.polygamma(1, y + kappa) - special.polygamma(1, kappa)
                    + 1 / kappa - 1 / (kappa + mu) - (mu - y) / (kappa + mu) ** 2))
    return beta, float(np.clip(np.exp(np.log(kappa) - s / (kappa * d)), 1e-4, 1e4))

def nb_fit(X, y, w=None, start=None, tol=1e-10, maxit=500):

```



```

"""Weighted NB2 maximum likelihood: iterate nb_step to convergence."""
w = np.ones(len(y)) if w is None else w
beta, kappa = (poisson_irls(X, y, w), 1.0) if start is None else start
for _ in range(maxit):
    new_beta, new_kappa = nb_step(X, y, beta, kappa, w)
    done = max(np.abs(new_beta - beta).max(), abs(new_kappa - kappa)) < tol
    beta, kappa = new_beta, new_kappa
    if done:
        break
return beta, kappa

beta_pois = poisson_irls(X, y)
loglik_pois = poisson_logpmf(y, np.exp(X @ beta_pois)).sum()
beta_nb, kappa_nb = nb_fit(X, y)
loglik_nb = nb_logpmf(y, np.exp(X @ beta_nb), kappa_nb).sum()
print(f"Poisson {loglik_pois:.1f}    NB2 {loglik_nb:.1f} (kappa {kappa_nb:.3f})")

```

```

def zero_inflated_em(X, Z, y, negbin=False, tol=1e-10, maxit=2000):
    """EM for a zero-inflated Poisson or NB2 model; returns (gamma, beta, kappa, loglik).

    gamma indexes the logistic model for the degenerate component, beta the log-linear
    model for the count component. The latent indicator is z_i = 1 if observation i
    comes from the degenerate component at zero.
    """

    zero = y == 0
    gamma = np.zeros(Z.shape[1])
    beta, kappa = (poisson_irls(X, y), np.inf)
    if negbin:
        beta, kappa = nb_fit(X, y)
    previous = -np.inf
    for _ in range(maxit):
        mu = np.exp(X @ beta)
        pi = 1 / (1 + np.exp(-Z @ gamma))
        log_count = nb_logpmf(y, mu, kappa) if negbin else poisson_logpmf(y, mu)
        # E step: the posterior probability that a zero came from the degenerate
part
        w = np.where(zero, pi / (pi + (1 - pi) * np.exp(log_count)), 0.0)
        # M step: a fractional logistic regression and a weighted count regression
        gamma = logistic_irls(Z, w)
        if negbin:
            beta, kappa = nb_step(X, y, beta, kappa, 1 - w)    # one step is enough
        else:
            beta = poisson_irls(X, y, 1 - w, start=beta)
            pi = 1 / (1 + np.exp(-Z @ gamma))
            mu = np.exp(X @ beta)
            log_count = nb_logpmf(y, mu, kappa) if negbin else poisson_logpmf(y, mu)
            loglik = np.sum(np.where(zero,
                                    np.log(pi + (1 - pi) * np.exp(log_count)),
                                    np.log(1 - pi) + log_count))
            if loglik - previous < tol * abs(loglik):
                break
            previous = loglik
    return gamma, beta, kappa, loglik

```

```
gamma_zip, beta_zip, _, loglik_zip = zero_inflated_em(X, X, y)
gamma_zinb, beta_zinb, kappa_zinb, loglik_zinb = zero_inflated_em(X, X, y, negbin=
    True)
print(f"ZIP log-likelihood {loglik_zip:.2f}")
print(f"ZINB log-likelihood {loglik_zinb:.2f} kappa {kappa_zinb:.3f}")
```

```
def truncated_nb_loglik(theta, X, y):
    """Log-likelihood of the zero-truncated NB2 model, parameters (beta, log kappa).
    """
    mu = np.exp(np.clip(X @ theta[:-1], -20.0, 20.0))
    k = np.exp(theta[-1])
    log_p0 = k * np.log(k / (k + mu)) # log P(Y = 0) before
    truncation
    return np.sum(nb_logpmf(y, mu, k) - np.log(-np.expm1(log_p0)))

def hurdle_nb(X, y):
    """Logistic model for y > 0, zero-truncated NB2 for the positive counts."""
    gamma = logistic_irls(X, (y > 0).astype(float))
    pi = 1 / (1 + np.exp(-X @ gamma))
    loglik_zero = np.sum(np.where(y > 0, np.log(pi), np.log1p(-pi)))

    Xp, yp = X[y > 0], y[y > 0]
    beta0, kappa0 = nb_fit(Xp, yp) # an untruncated fit as the
    start
    out = optimize.minimize(lambda t: -truncated_nb_loglik(t, Xp, yp),
        np.append(beta0, np.log(kappa0)), method="BFGS",
        options={"gtol": 1e-5, "maxiter": 800})
    return gamma, out.x[:-1], np.exp(out.x[-1]), loglik_zero - out.fun

gamma_h, beta_h, kappa_h, loglik_hurdle = hurdle_nb(X, y)
print(f"hurdle log-likelihood {loglik_hurdle:.2f} kappa {kappa_h:.3f}")
```

### 37.5.4 Telling them apart

Table 37.5.1 compares fits, not mechanisms, and settles no question about which story is right. It is worth being clear about what the data can and cannot decide.

**Identification.** When  $\mathbf{z}_{(i)}$  and  $\mathbf{x}_{(i)}$  are the same, the two parts compete to explain the same zeros and the likelihood is often very flat: a larger  $\pi_i$  with a larger  $\lambda_i$  fits almost as well as a smaller  $\pi_i$  with a smaller one. The parameters are identified in principle, since the positive counts pin down  $\lambda_i$ , but standard errors can be large, the two parts' coefficients strongly correlated, and the fit sensitive to the start. Restricting  $\mathbf{z}_{(i)}$  to a few variables, or to an intercept, helps and is easier to defend.

**The Vuong test.** For two non-nested models fitted to the same data, Vuong (1989) proposed comparing the pointwise log-likelihood ratios  $m_i = \log f_1(y_i; \hat{\theta}_1) - \log f_2(y_i; \hat{\theta}_2)$  with

$$V = \frac{\sqrt{n} \bar{m}}{s_m}, \quad s_m^2 = \frac{1}{n-1} \sum_{i=1}^n (m_i - \bar{m})^2, \quad (37.5.5)$$

which is asymptotically standard normal when the two models are equally close to the truth, so

that large  $|V|$  favours one or the other by its sign. For the physician visits, ZINB against the hurdle model gives  $V = -7.23$ : taken at face value it prefers the hurdle model, agreeing with AIC.

The test is often misused, routinely applied to compare a count model with its own zero-inflated version — exactly the nested case the theory excludes. Under the null  $\bar{m}$  and  $s_m$  both tend to zero, the ratio has no normal limit, and the reported  $p$ -value is meaningless. In [Example \(Five models for the physician visits\)](#) that comparison gives  $V = 5.90$ , a number with no calibration; Wilson (2015) documents the practice. A further difficulty, quantified by Schennach and Wilhelm (2017), is poor size even in the non-nested case when the two models are close.

```
mu_p, mu_nb = np.exp(X @ beta_pois), np.exp(X @ beta_nb)
pi_zinb = 1 / (1 + np.exp(-X @ gamma_zinb))
mu_zinb = np.exp(X @ beta_zinb)
pi_h = 1 / (1 + np.exp(-X @ gamma_h))
mu_h = np.exp(X @ beta_h)
p0_h = np.exp(kappa_h * np.log(kappa_h / (kappa_h + mu_h)))

def vuong(log_f1, log_f2):
    """Vuong's statistic for model 1 against model 2 (not valid for nested models).
    """
    m = log_f1 - log_f2
    return np.sqrt(len(m)) * m.mean() / m.std(ddof=1)

log_zinb = np.where(y == 0, np.log(pi_zinb + (1 - pi_zinb) * np.exp(nb_logpmf(0.0,
    mu_zinb, kappa_zinb))),
    np.log1p(-pi_zinb) + nb_logpmf(y, mu_zinb, kappa_zinb))
log_hurdle = np.where(y == 0, np.log1p(-pi_h),
    np.log(pi_h) + nb_logpmf(y, mu_h, kappa_h) - np.log1p(-p0_h))
log_nb = nb_logpmf(y, mu_nb, kappa_nb)

print(f"ZINB against NB2 (nested, so not valid):  V = {vuong(log_zinb, log_nb):6.2f}")
print(f"ZINB against the hurdle model:          V = {vuong(log_zinb, log_hurdle):6.2f}")
```

**What does distinguish them.** By [Proposition 37.5.1\(c\)](#) the two differ in what they can say about the positive counts and about zero deflation, and those are where to look: comparing fitted and observed frequencies at  $y = 1, 2, 3$ , as in [Figure 37.5.1](#), is more informative than any single statistic.

*Warning.* Excess zeros are a symptom, not a diagnosis. Before reaching for a two-part model, check that the mean model is right, that the overdispersion is not heterogeneity a negative binomial would absorb, and that the zeros are not an artefact of the recording process: a question never asked, a period of non-observation, a value truncated on entry. The physician-visit data make the point. Of the two models, the hurdle likelihood factorizes and the zero-inflated one does not, which makes the hurdle easier to fit and to interpret; the zero-inflated model has the better story when there really are two kinds of unit.

## 37.5.5 Exercises

### 37.5.5.1 A. Check your understanding

#### Exercise A1.

A ZIP model has  $\pi = 0.3$  and  $\lambda = 4$ . Compute  $EY$ ,  $\text{Var } Y$ ,  $\Pr(Y = 0)$  and the ratio of variance to mean, and compare with a Poisson having the same mean.

### 37.5.5.2 B. Practice

#### Exercise B1.

Let  $Y$  be Poisson with mean  $\lambda$ , truncated to  $\{1, 2, \dots\}$ . Show that  $EY = \lambda/(1 - e^{-\lambda})$  and  $\text{Var } Y = EY\{1 + \lambda - EY\}$ , and deduce that the truncated distribution is underdispersed relative to its own mean.

*Solution.* Write  $c = 1 - e^{-\lambda}$ . The  $y = 0$  term contributes nothing to either moment, so  $EY = \lambda/c$  and  $E\{Y(Y-1)\} = \lambda^2/c$ , giving  $\text{Var } Y = \lambda^2/c + \lambda/c - \lambda^2/c^2 = EY(1 + \lambda - EY)$ . Since  $EY = \lambda/c > \lambda$ , the bracket is less than one.

#### Exercise B2.

Derive the mean and variance of a ZINB with inflation probability  $\pi$ , count mean  $\lambda$  and shape  $\kappa$ . Show that for a given mean and variance there is a curve of  $(\pi, \lambda, \kappa)$  values giving the same first two moments, and explain what this implies for the precision of the estimates.

*Solution.*  $EY = (1 - \pi)\lambda$  and, by the law of total variance,  $\text{Var } Y = (1 - \pi)(\lambda + \lambda^2/\kappa) + \pi(1 - \pi)\lambda^2 = EY\{1 + \lambda(\pi + 1/\kappa)\}$ . The mean fixes  $(1 - \pi)\lambda$  and the variance  $\lambda(\pi + 1/\kappa)$ : two equations in three unknowns, so a one-dimensional family shares the first two moments. Only higher moments separate them, so the likelihood is flat along it and the estimates are strongly correlated with large standard errors.

#### Exercise B3.

Fit a ZINB to the physician-visit data using only an intercept in the inflation part, and compare its AIC with the model of [Example \(Five models for the physician visits\)](#) that uses all the regressors in both parts. Which does AIC prefer, and what does the comparison say about the eight inflation coefficients?

#### Exercise B4.

Show that the EM step described around [Eq. 37.5.4](#) increases the observed-data log-likelihood at every iteration, by writing the observed-data log-likelihood as the expected complete-data log-likelihood minus an entropy term and using Jensen's inequality.

## 37.5.5.3 C. Going deeper

**Exercise C1.**

Take  $f$  Poisson,  $\mathbf{z}_{(i)} = \mathbf{x}_{(i)}$ , and a hurdle model whose binary part uses the complementary log–log link,  $\log\{-\log(1 - \rho_i)\} = \mathbf{z}_{(i)}^\top \boldsymbol{\gamma}$ , in place of the logit of Definition 37.5.1(b). Show that this model contains the plain Poisson as the submodel  $\boldsymbol{\gamma} = \boldsymbol{\beta}$ , so that a likelihood ratio test of the Poisson against it is a standard interior test on  $p$  restrictions, unlike the zero-inflated comparison of Proposition 37.5.1(d). Then show that the same is not true if the binary part uses the logit.

*Solution.* The hurdle model reduces to  $f$  when  $\rho_i = 1 - f(0; \lambda_i)$  for every  $i$ . For the Poisson that is  $1 - e^{-\lambda_i}$ , so  $-\log(1 - \rho_i) = \lambda_i$  and the complementary log–log link makes the constraint read  $\mathbf{z}_{(i)}^\top \boldsymbol{\gamma} = \log \lambda_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$ , that is  $\boldsymbol{\gamma} = \boldsymbol{\beta}$ : a linear restriction of  $p$  dimensions in the interior of  $\mathbb{R}^{2p}$ , so the usual  $\chi^2(p)$  limit applies. With the logit the constraint is  $\mathbf{x}_{(i)}^\top \boldsymbol{\gamma} = \log(e^{\lambda_i} - 1)$ , and since  $t \mapsto \log(e^t - 1)$  is not affine, the vector on the right lies in  $\mathcal{C}(\mathbf{X})$  only in degenerate cases such as  $\mathbf{X}\boldsymbol{\beta}$  constant. The Poisson is then not a smooth submodel at all and no likelihood ratio test of one against the other exists: a model “containing” another is not the same as nesting it.

**Exercise C2.**

In a zero-inflated model  $\beta_j$  is a log rate ratio for the count component, not for the response. Derive  $\partial \log E(Y_i) / \partial x_{ij}$  when  $x_j$  appears in both parts, and explain why reporting  $e^{\hat{\beta}_j}$  as “the effect on the expected count” is wrong.

*Solution.*  $E(Y_i) = (1 - \pi_i)\lambda_i$ , so  $\log E(Y_i) = \log(1 - \pi_i) + \mathbf{x}_{(i)}^\top \boldsymbol{\beta}$  and

$$\frac{\partial \log E(Y_i)}{\partial x_{ij}} = \beta_j - \frac{1}{1 - \pi_i} \frac{\partial \pi_i}{\partial x_{ij}} = \beta_j - \pi_i \gamma_j,$$

using  $\partial \pi_i / \partial x_{ij} = \pi_i(1 - \pi_i)\gamma_j$  for the logit link. The effect on the expected response therefore mixes the two parts and depends on  $\pi_i$ , hence on all the covariates;  $e^{\hat{\beta}_j}$  is the rate ratio only within the non-degenerate component, a conditional statement about a latent class nobody observes.

## 37.6 Summary and Notes

### 37.6.1 Summary

*Idea.*

1. The Poisson is the exponential dispersion family with  $V(\mu) = \mu$  and no free dispersion; its canonical link is the logarithm, so coefficients are log rate ratios (Definition 37.1.1).
2. The likelihood equations are  $\mathbf{X}^\top(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \mathbf{0}$ , the information is  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$  with

$\mathbf{W} = \text{diag}(\hat{\mu}_i)$ , and the log-likelihood is strictly concave at full column rank ([Theorem 37.1.1](#)). Fitted counts reproduce the observed margins spanned by the model matrix ([Theorem 37.1.2](#)).

3. The maximum likelihood estimate exists exactly when no direction  $\mathbf{d}$  satisfies [Eq. 37.1.3](#) ([Proposition 37.1.3](#)); a ridge penalty always restores one.
4. Exposure enters as an offset, a column with coefficient fixed at one. Grouping records with identical regressors and carrying group size as exposure leaves the fit unchanged; estimating the exposure coefficient costs precision, fixing a wrong one costs bias ([Proposition 37.2.1](#)).
5. Independent Poisson counts conditioned on their total are multinomial, and the two fits agree whenever the model has a free parameter for every total the design fixed ([Theorem 37.3.1](#)). Independence is the no-interaction model ([Corollary 37.3.2](#)), and interaction parameters are log odds ratios ([Proposition 37.3.3](#)).
6. A gamma mixture of Poissons is negative binomial, with variance  $\mu + \mu^2/\kappa$  ([Lemma 37.4.1](#), [Definition 37.4.1](#)); any mixing distribution inflates the variance. At fixed  $\kappa$  the NB2 model is a generalized linear model,  $\boldsymbol{\beta}$  and  $\kappa$  are orthogonal, and the test of  $\alpha = 1/\kappa = 0$  is on the boundary, so its null distribution is  $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$ ; part (f) gives the NB1 score and its relation to quasi-Poisson ([Proposition 37.4.2](#)). The score test [Eq. 37.4.8](#) needs only the Poisson fit ([Theorem 37.4.3](#)).
7. Zero-inflated models add a mass at zero; hurdle models cut the distribution at zero and fit the two pieces separately, which factorizes the likelihood ([Definition 37.5.1](#), [Proposition 37.5.1](#)). The Vuong statistic [Eq. 37.5.5](#) is not valid for the nested comparison it is usually applied to.

## 37.6.2 Notes and sources

**Poisson regression.** The log-linear model for counts predates the generalized linear model, but Nelder and Wedderburn (1972) put it in its modern setting, and McCullagh and Nelder (1989) is the standard reference for the deviance, residuals and fitting algorithm. Agresti (2015, chapter 7) and Fahrmeir et al. (2021, section 5.2) are the sources against which this chapter's coverage was checked; Cameron and Trivedi (2013) and Winkelmann (2008) are book-length econometric treatments. The existence condition [Proposition 37.1.3](#) is the Poisson analogue of the separation conditions of [Chapter 35](#), proved here from scratch. The Anscombe residual of [Exercise \(B3\)](#) is from Anscombe (1948).

**Rates.** Treating survival data with piecewise-constant hazards as Poisson counts with an offset is due to Holford (1980) and Laird and Olivier (1981); it is the reason that much of the early software for survival analysis was software for log-linear models.

**Contingency tables.** The Poisson–multinomial correspondence and the maximum likelihood theory of hierarchical log-linear models are in Birch (1963) and, comprehensively, Bishop, Fienberg and Holland (1975). Graphical log-linear models are due to Darroch, Lauritzen and Speed (1980). The chi-squared test for independence is Pearson (1900); [Corollary 37.3.2](#) is its likelihood-based restatement.

**The negative binomial.** The gamma mixture derivation goes back to Greenwood and Yule (1920), who used it for accident proneness. The NB1 and NB2 parameterizations and their different estimating equations are discussed by Cameron and Trivedi (2013, section 3.3) and Hilbe (2011). The score test [Eq. 37.4.8](#) is due to Dean and Lawless (1989); the derivation here, by expanding the log-likelihood in  $\alpha$ , is the usual one. The boundary problem is the setting of

Self and Liang (1987), invoked by [Proposition 37.4.2\(e\)](#); [Theorem 32.5.2](#) computes a comparable case exactly.

**Zeros.** The zero-inflated Poisson model and its EM algorithm are Lambert (1992), written for defect counts in manufacturing; hurdle models are Mullahy (1986). Vuong (1989) introduced the model-selection statistic, for strictly non-nested models; Wilson (2015) documents its misuse for zero-inflation testing, and Schennach and Wilhelm (2017) analyse its size and propose a repair. The definitions and likelihoods of the two model families are [Definition 37.5.1](#) and the preamble to [Proposition 37.5.1](#). [Chapter 38](#) returns to overdispersion without a likelihood, and [Chapter 40](#) to counts that are not independent.

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# Chapter 38

## Quasi-Likelihood and Overdispersion

A generalized linear model asks for a great deal and gives a great deal back. Naming the family fixes the whole distribution of the response, and with it the exact relation between its mean and its variance — the part of the bargain the data most often refuse. Counts of accidents, litters of affected pups, votes in households, durations of strikes are nearly always more variable than the Poisson or the binomial allows, and no choice of link repairs it. This chapter takes the least drastic way out. The likelihood equations of [Chapter 34](#) use the density only through the mean and the variance function, so one may write down those two and estimate from the equations alone. The estimate is unchanged, its consistency survives a wrong variance function, and its standard errors can be read off the data rather than the model. What is given up is the likelihood, and with it the likelihood ratio test, the Akaike criterion, and every statement about a probability.

**What you need.** [Chapter 34](#) throughout: the model ([Definition 34.2.1](#)), the variance function ([Definition 34.1.2](#)), the score and the information ([Theorem 34.3.1](#)), iteratively reweighted least squares ([Theorem 34.4.1](#)), the deviance ([Definition 34.5.1](#), [Theorem 34.5.5](#)) and the asymptotics ([Theorem 34.5.1](#), [Proposition 34.5.2](#)). [Section 21.4](#) for the sandwich estimators ([Definition 21.4.1](#), [Theorem 21.4.2](#)) and [Section 21.3](#) for weighted least squares ([Theorem 21.3.1](#)). [Chapter 37](#) for the negative binomial models ([Definition 37.4.1](#), [Proposition 37.4.2](#)) and the score test for overdispersion ([Theorem 37.4.3](#)). The optimality proof of [Section 38.2](#) uses the projection inequality of [Chapter 6](#); [Chapter 20](#), [Chapter 23](#), [Chapter 29](#) and [Chapter 33](#) are used in single sections.

### 38.0.1 Roadmap

- [Variance functions](#)
- [Quasi-likelihood estimating equations](#)
- [Sandwich covariance estimation](#)
- [Overdispersion](#)
- [Summary and notes](#)

## 38.1 Variance functions

Fitting a generalized linear model needs a family, a link and a linear predictor. [Section 34.3](#) noticed that the family enters the likelihood equations

$$\sum_{i=1}^n x_{ij} \frac{w_i h'(\eta_i)}{V(\mu_i)} (y_i - \mu_i) = 0, \quad j = 1, \dots, p, \quad (38.1.1)$$

only through the variance function  $V$ : two densities with the same  $V$  produce the same estimate. The apparatus of [Definition 34.1.1](#) — a natural parameter, a cumulant function, a density that integrates to one — *derives* [Eq. 38.1.1](#) and then disappears from it. This section puts the variance function first, and the model becomes a mean and a variance and nothing else.

### 38.1.1 Specifying the second moment only

#### Definition 38.1.1 — A variance specification

Let  $Y_1, \dots, Y_n$  be independent, let  $w_1, \dots, w_n > 0$  be known weights, let  $g$  be a strictly monotone differentiable link with inverse  $h$ , and let  $V$  be a positive, continuously differentiable function on an open interval  $\mathcal{M} \subseteq \mathbb{R}$ . A **variance specification** is the pair of assumptions

$$\mathbb{E}(Y_i) = \mu_i = h(\mathbf{x}_i^\top \boldsymbol{\beta}), \quad \text{Var}(Y_i) = \frac{\phi V(\mu_i)}{w_i}, \quad (38.1.2)$$

with  $\boldsymbol{\beta} \in \mathbb{R}^p$  and  $\phi > 0$  unknown. The function  $V$  is the **variance function** and  $\phi$  the **dispersion**. No distribution is assumed for  $Y_i$  beyond these two moments.

The definition is weaker than [Definition 34.2.1](#) in saying nothing about higher moments, the support, or whether any distribution with these moments exists, and stronger in leaving  $\phi$  free even for a  $V$  that came from a family in which it was fixed. The specifications in constant use are

| name           | $V(\mu)$             | $w_i$ | $\phi$ | $\mathcal{M}$ |
|----------------|----------------------|-------|--------|---------------|
| quasi-normal   | 1                    | 1     | free   | $\mathbb{R}$  |
| quasi-Poisson  | $\mu$                | 1     | free   | $(0, \infty)$ |
| quasi-binomial | $\mu(1 - \mu)$       | $m_i$ | free   | $(0, 1)$      |
| quasi-gamma    | $\mu^2$              | 1     | free   | $(0, \infty)$ |
| power          | $\mu^\xi$            | 1     | free   | $(0, \infty)$ |
| NB2            | $\mu + \mu^2/\kappa$ | 1     | 1      | $(0, \infty)$ |

where  $y_i$  is a group proportion in the quasi-binomial row and  $m_i$  the number of trials behind it. A value  $\phi > 1$  is **overdispersion** relative to the family that supplied  $V$ , and  $\phi < 1$  **underdispersion**. The first four rows are variance functions of [Definition 34.1.2](#) with the dispersion set free; the last is that of [Definition 37.4.1\(a\)](#), for which a genuine likelihood exists.

*Remark* (When is a variance specification a distribution?). For  $\phi$  free and  $V(\mu) = \mu^\xi$ , Jorgensen's classification of the Tweedie families ([Exercise \(C2\)](#)) settles whether some exponential dispersion family has exactly these moments: one exists for  $\xi \leq 0$  and for  $\xi \geq 1$ , and for no  $\xi$  strictly between. Existence is not usefulness. At  $\xi = 1$  the Tweedie family with dispersion  $\phi$  is  $\phi W$  with  $W$  Poisson of mean  $\mu/\phi$ : the right two moments, but on the multiples of  $\phi$  rather than the integers. So no *exponential dispersion family* on the counts has mean  $\mu$  and variance  $\phi\mu$  for one  $\phi$  and all  $\mu$ . A family of counts with exactly those two moments does exist — the NB1 model of [Definition 37.4.1\(b\)](#) — but it is not an exponential dispersion family, and its likelihood equations are not [Eq. 38.1.1](#); that is the point of the third subsection of this section.

*Warning.* The quasi-binomial specification is empty when  $m_i = 1$ . An ungrouped binary response has  $Y_i^2 = Y_i$ , so  $\text{Var}(Y_i) = \mu_i(1 - \mu_i)$  exactly for every distribution on  $\{0, 1\}$  with mean  $\mu_i$ , and there is no room for a  $\phi$ . A dispersion estimate far from 1 in an ungrouped logistic fit is a statement about the mean model — an omitted term, a wrong link, a few gross outliers — never about the variance, the second-moment counterpart of [Theorem 34.5.5\(c\)](#).

## 38.1.2 Two ways to inflate a binomial

Where a dispersion is available it can take more than one shape, and the shapes are distinguishable. Suppose observation  $i$  averages  $m_i$  dependent binary trials.

### Lemma 38.1.1 — Exchangeable trials

Let  $Z_1, \dots, Z_m$  be binary variables with common mean  $\pi$  and  $\text{Cov}(Z_s, Z_t) = \rho\pi(1 - \pi)$  for all  $s \neq t$ , so that  $\rho$  is their common correlation, and let  $\bar{Z} = m^{-1} \sum_t Z_t$ . Then

$$\mathbb{E}(\bar{Z}) = \pi, \quad \text{Var}(\bar{Z}) = \left\{1 + \rho(m - 1)\right\} \frac{\pi(1 - \pi)}{m}, \quad (38.1.3)$$

and  $\rho \geq -1/(m - 1)$ , with  $\rho \geq 0$  whenever the trials are conditionally independent and identically distributed given some unobserved variable.

*Proof.* Each  $Z_t$  has variance  $\pi(1 - \pi)$ , so the stated covariance is  $\rho$  times it and  $\text{Var}(\bar{Z}) = m^{-2}\{m\pi(1 - \pi) + m(m - 1)\rho\pi(1 - \pi)\}$ , which is [Eq. 38.1.3](#); nonnegativity gives  $1 + \rho(m - 1) \geq 0$ . If the  $Z_t$  are independent given  $U$  with  $\Pr(Z_t = 1 \mid U) = \Pi$  the same for each  $t$ , the law of total covariance gives  $\text{Cov}(Z_s, Z_t) = \text{Var}(\Pi) \geq 0$ . ■

The variance specification suggested by [Eq. 38.1.3](#) is

$$V_\rho(\mu) = \left\{1 + \rho(m_i - 1)\right\} \mu(1 - \mu), \quad w_i = m_i, \quad \phi = 1,$$

in which the inflation grows with the group size, against the quasi-binomial  $V(\mu) = \mu(1 - \mu)$  with  $w_i = m_i$  and a constant  $\phi$ . The two agree at one group size and nowhere else ([Figure 38.1.1](#)): at  $\phi = 1.8$  and  $\rho = 0.05$  they cross at  $m = 17.0$  and differ by a factor 1.639 at  $m = 40$ .

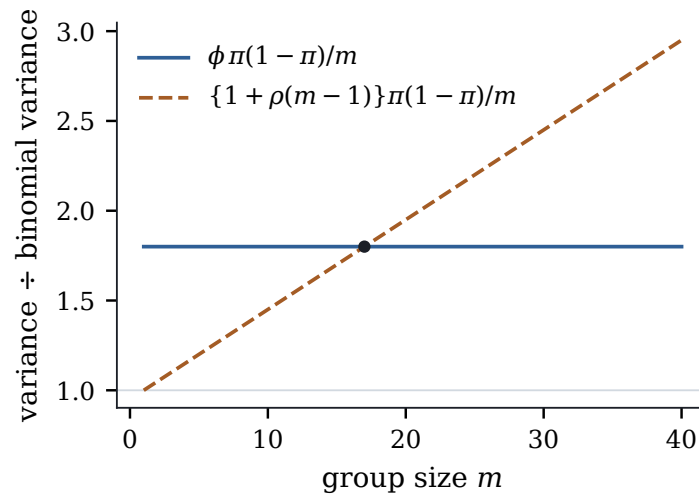


Figure 38.1.1: Two variance functions for a proportion from  $m$  trials, as multiples of  $\pi(1 - \pi)/m$ . A constant dispersion inflates every group equally; exchangeable trials inflate large groups more, by Eq. 38.1.3.

Spread-out group sizes separate them, by a plot of squared Pearson residuals against  $m_i$ ; equal ones do not.

```
import numpy as np

m_grid = np.arange(1, 41)
phi, rho = 1.8, 0.05
inflated = np.full_like(m_grid, phi, dtype=float)      #  $V(p_i) = \phi p_i(1-p_i)/m$ 
exchangeable = 1 + rho * (m_grid - 1)                  #  $V(p_i) = \{1 + \rho(m-1)\} p_i(1-p_i)/m$ 
crossing = 1 + (phi - 1) / rho                          # the one group size where they agree
print(f"the two variance functions agree at m = {crossing:.1f}")
print(f"at m = 40 they differ by a factor {exchangeable[-1] / phi:.3f}")
```

The second specification is not an invention of the quasi-likelihood literature: the **beta-binomial** distribution, a binomial mixed over a beta success probability, has exactly this variance (Exercise (B1)), so a mixture and a correlation leave the same trace in the second moment.

### 38.1.3 A variance function does not determine a fit

A variance function settles the estimating equations, which is less than settling the fit: two models can share a variance function and disagree about  $\hat{\beta}$ , because the second model's likelihood equations need not be Eq. 38.1.1. The NB1 model of Definition 37.4.1(b) has variance  $\phi\mu_i$  with  $\phi = 1 + \alpha^{-1}$ , the quasi-Poisson variance function exactly; but its score Eq. 37.4.6 does not contain the residual  $y_i - \mu_i$  at all, and the two estimates differ in general (Exercise (B3)), as Section 38.4 shows in numbers. A *model* is a set of distributions; a *variance specification* is two moment assumptions plus the rule Eq. 38.1.1 for turning them into an estimate.

## 38.1.4 What the missing density costs

Dropping the density is not free. There is no maximized likelihood, so [Proposition 34.5.3](#) loses one of its three members and [Section 38.3](#) must supply substitutes; no Akaike or Bayesian criterion, since both are  $-2\ell(\hat{\beta})$  plus a penalty ([Definition 29.2.2](#)), though cross-validation ([Definition 29.3.1](#)) survives; and no predictive distribution, so a fitted quasi-Poisson model cannot say what fraction of units will record zero, simulate a data set, or be checked by a predictive comparison, where a negative binomial or beta-binomial model does all three. What survives is the estimate, its consistency, its asymptotic normality, and a covariance estimate that does not need the variance function to be right — the subject of the next two sections.

## 38.1.5 Exercises

### 38.1.5.1 A. Check your understanding

#### Exercise A1.

For each of the following, say whether a dispersion parameter  $\phi \neq 1$  is meaningful, and why: (a) counts of insurance claims per policy; (b) the proportion of a litter of 12 pups showing a defect; (c) an indicator of whether a patient survived; (d) the proportion of 500 sampled voters supporting a candidate, the voters sampled in clusters of 10 households.

*Solution.*

- (a) Yes: nothing ties a count's variance to its mean, and heterogeneity between policies inflates it ([Lemma 37.4.1\(a\)](#)). (b) Yes, with  $w_i = 12$ : correlation between littermates, or variation between litters, inflates the variance by [Lemma 38.1.1](#). (c) No: a binary response has variance  $\mu(1 - \mu)$  whatever else is true. (d) Yes, but the group size is the cluster, not 500; using  $m = 500$  in [Eq. 38.1.3](#) would overstate the inflation.

#### Exercise A2.

Two analysts fit grouped binomial data with group sizes between 5 and 60, one using a constant dispersion  $\phi$  and one using the exchangeable form [Eq. 38.1.3](#) with correlation  $\rho$ . They report  $\hat{\phi} = 2.5$  and  $\hat{\rho} = 0.04$ . At which group size do the two fitted variances agree? For which groups does the first analyst's model claim the larger variance?

*Solution.* Setting  $1 + \rho(m - 1) = \phi$  gives  $m = 1 + (\phi - 1)/\rho = 1 + 1.5/0.04 = 38.5$ ; below it the constant-dispersion model claims the larger variance, above it the smaller.

### 38.1.5.2 B. Practice

#### Exercise B1.

Let  $\Pi$  have a beta distribution with parameters  $a, b > 0$ , so that  $E\Pi = \mu = a/(a + b)$  and  $\text{Var } \Pi = \mu(1 - \mu)/(a + b + 1)$ , and let  $m\bar{Y}$  be binomial  $(m, \Pi)$  given  $\Pi$ . Show that

$E\bar{Y} = \mu$  and

$$\text{Var}(\bar{Y}) = \{1 + \rho(m-1)\} \frac{\mu(1-\mu)}{m}, \quad \rho = \frac{1}{a+b+1},$$

so the beta-binomial has the variance function of [Eq. 38.1.3](#). Deduce that the beta-binomial and an exchangeable-trials model with the same  $\rho$  cannot be told apart by first and second moments.

*Solution.* The conditional mean is  $\Pi$  and the conditional variance  $\Pi(1-\Pi)/m$ , so  $E\bar{Y} = \mu$  and, by the law of total variance and  $E\Pi^2 = \text{Var}\Pi + \mu^2$ ,

$$\text{Var}(\bar{Y}) = \frac{E\{\Pi(1-\Pi)\}}{m} + \text{Var}\Pi = \frac{\mu(1-\mu) - \text{Var}\Pi}{m} + \text{Var}\Pi = \frac{\mu(1-\mu)}{m} \{1 + (m-1)\rho\}$$

with  $\rho = \text{Var}\Pi / \{\mu(1-\mu)\} = 1/(a+b+1)$ . The two specifications therefore give identical first and second moments for every  $m$ .

### Exercise B2.

Under the power specification  $\text{Var}(Y) = \phi\mu^\xi$ , find the coefficient of variation  $\sqrt{\text{Var}(Y)}/E(Y)$  as a function of  $\mu$ . For which  $\xi$  is it constant? Which of the specifications in the table of this section have the property that a multiplicative change of units,  $Y \mapsto cY$ , leaves the specification in the same family with a new  $\phi$ ?

### 38.1.5.3 C. Going deeper

#### Exercise C1.

Let  $Y \mid \Lambda$  have mean  $\Lambda$  and variance  $a\Lambda^\xi$ , and let  $\Lambda$  have mean  $\mu$  and finite variance  $\tau^2$ . Show that the marginal variance is  $a E(\Lambda^\xi) + \tau^2$ , equal to  $a\mu + \tau^2$  when  $\xi = 1$ . For  $\xi = 2$ , use Jensen's inequality to show that it is at least  $a\mu^2 + \tau^2$ , so a mixture of quasi-gamma responses is overdispersed relative to the quasi-gamma specification with the same  $a$ .

## 38.2 Quasi-likelihood estimating equations

Estimation here starts from [Eq. 38.1.1](#) and not from a density. There are two useful ways to read it: as the derivative of a function that is not quite a log-likelihood but close enough to carry the usual arguments, and as the best member of a large family of estimating equations. This section develops both, and they meet in the middle.

### 38.2.1 The quasi-likelihood function

#### Definition 38.2.1 — Quasi-likelihood

Under [Definition 38.1.1](#), the **quasi-likelihood** of a mean  $\mu \in \mathcal{M}$  given a response  $y$  in

the closure of  $\mathcal{M}$ , with weight  $w$ , is

$$Q(\mu; y) = \frac{w}{\phi} \int_y^\mu \frac{y-t}{V(t)} dt, \quad (38.2.1)$$

assumed finite, the integral being read as a limit at an endpoint of  $\mathcal{M}$  where it converges — as it does at  $y = 0$  for  $V(t) = t$  and at  $y \in \{0, 1\}$  for  $V(t) = t(1-t)$ . The quasi-likelihood of the model is  $Q(\boldsymbol{\beta}) = \sum_{i=1}^n Q(\mu_i; y_i)$  with  $\mu_i = h(\mathbf{x}_i^\top \boldsymbol{\beta})$  and weight  $w_i$ .

The definition is designed to make one identity true by construction, and everything else follows from it.

### Proposition 38.2.1 — What the quasi-likelihood does and does not share with a log-likelihood

Write  $Q' = \partial Q / \partial \mu$ . Then, for every variance specification:

- (a)  $Q'(\mu; y) = w(y - \mu) / \{\phi V(\mu)\}$ , and  $Q(\mu; y) \leq 0$  with equality only at  $\mu = y$ : the quasi-likelihood is maximized at the observation, as a saturated fit would be;
- (b)  $E\{Q'(\mu; Y)\} = 0$  and  $\text{Var}\{Q'(\mu; Y)\} = -E\{\partial^2 Q / \partial \mu^2\} = w / \{\phi V(\mu)\}$ , the two Bartlett identities;
- (c) if  $(V, \phi, w)$  are those of an exponential dispersion family (Definition 34.1.1), then  $Q(\mu; y) = \ell(\mu; y) - \ell(y; y)$ , so  $Q$  is the log-likelihood measured from its saturated value, and  $-2\phi Q(\boldsymbol{\beta}) = D(\mathbf{y}, \boldsymbol{\mu})$  is the deviance of Eq. 34.5.4.

*Proof.*

- (a) Differentiating Eq. 38.2.1 in its upper limit gives the stated  $Q'$ . Because  $V > 0$ , the integrand  $(y-t)/V(t)$  is positive for  $t < y$  and negative for  $t > y$ , so for  $\mu > y$  the integral is of a negative function and for  $\mu < y$  it is minus the integral of a positive one: both give  $Q \leq 0$ , with equality only for an empty interval.
- (b)  $E(Y - \mu) = 0$  gives the first identity, and  $\text{Var}(Y) = \phi V(\mu)/w$  gives  $\text{Var}(Q') = w / \{\phi V(\mu)\}$ . Differentiating  $Q'$  again,

$$\frac{\partial^2 Q}{\partial \mu^2} = -\frac{w}{\phi} \cdot \frac{V(\mu) + (y - \mu)V'(\mu)}{V(\mu)^2},$$

whose expectation is  $-w / \{\phi V(\mu)\}$ , since the term carrying  $y - \mu$  has mean zero.

- (c) For an exponential dispersion family the chain rule used in the proof of Theorem 34.3.1 gives  $\partial \ell / \partial \mu = w(y - \mu) / \{\phi V(\mu)\}$ , which is  $Q'$ ; two functions of  $\mu$  with the same derivative differ by a constant, and both sides vanish at  $\mu = y$ . Comparing with Eq. 34.5.4 gives  $D = -2\phi Q$ . ■

The converse of (c) fails:  $\exp Q$  is in general not a density in  $y$  (Exercise (C1)). Part (c) is nevertheless why the objects here look familiar — everything the exponential family contributed to Definition 34.5.1 is already in the integral Eq. 38.2.1 — and for a variance function from no family Eq. 38.2.1 still measures the discrepancy between a fit and an observation, the **quasi-**

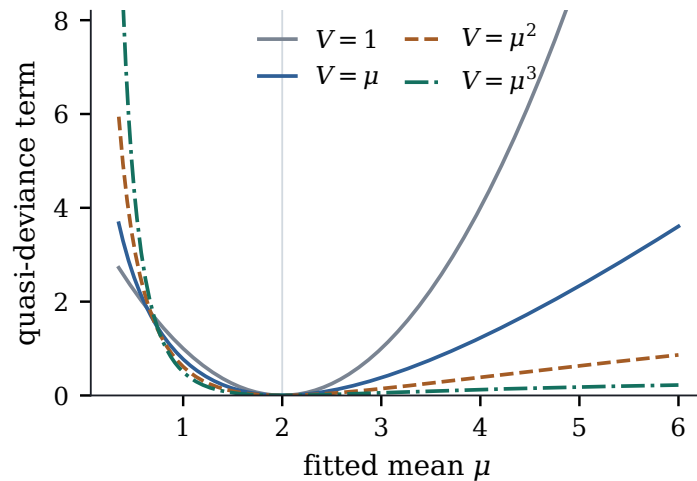


Figure 38.2.1: One term of the quasi-deviance Eq. 38.2.2 at  $y = 2$ ,  $w = 1$ , against the fitted mean, for four power variance functions. Only the constant one penalizes symmetrically.

### deviance

$$D(\mathbf{y}, \boldsymbol{\mu}) = -2\phi Q(\boldsymbol{\beta}) = 2 \sum_{i=1}^n w_i \int_{\mu_i}^{y_i} \frac{y_i - t}{V(t)} dt \geq 0, \quad (38.2.2)$$

which does not involve  $\phi$  and vanishes only at  $\boldsymbol{\mu} = \mathbf{y}$ . Figure 38.2.1 draws one term for four power variance functions: only  $V \equiv 1$  penalizes symmetrically, and under  $V(\mu) = \mu$  a fitted mean one unit too small costs 2.043 times as much as one unit too large, against 4.254 for  $V(\mu) = \mu^2$  and 9.000 for  $V(\mu) = \mu^3$ . A variance function that grows with the mean is a statement about which errors matter.

```
import numpy as np
from scipy import integrate

def quasi_deviance_term(y, mu, V):
    """2 * integral from mu to y of (y - t)/V(t) dt: one observation's quasi-
    deviance."""
    value, _ = integrate.quad(lambda t: (y - t) / V(t), mu, y)
    return 2 * value

y0 = 2.0
closed_form = {
    "V = 1": (lambda m: (y0 - m) ** 2, lambda t: 1.0),
    "V = mu": (lambda m: 2 * (y0 * np.log(y0 / m) - (y0 - m)), lambda t: t),
    "V = mu^2": (lambda m: 2 * (-np.log(y0 / m) + (y0 - m) / m), lambda t: t ** 2),
    "V = mu^3": (lambda m: (y0 - m) ** 2 / (y0 * m ** 2), lambda t: t ** 3),
}

for name, (formula, V) in closed_form.items():
    print(f"{name:9s} at mu = 0.8: {quasi_deviance_term(y0, 0.8, V):.6f}"
          f" closed form {formula(0.8):.6f}")
```

## 38.2.2 The quasi-score



**Theorem 38.2.2 — The quasi-score**

Let  $\mathbf{U}(\boldsymbol{\beta}) = \partial Q / \partial \boldsymbol{\beta}$  be the **quasi-score**, and keep the notation  $\mathbf{W} = \text{diag}\{w_i h'(\eta_i)^2 / V(\mu_i)\}$  and  $\mathbf{D} = \text{diag}\{h'(\eta_i)\}$  of [Theorem 34.3.1](#). Write  $\mathbf{A}_n = \phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X}$ . Then:

- (a)  $\mathbf{U}(\boldsymbol{\beta}) = \phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1}(\mathbf{y} - \boldsymbol{\mu})$ , which is [Eq. 34.3.1](#) exactly. The estimating equations  $\mathbf{U}(\hat{\boldsymbol{\beta}}) = \mathbf{0}$  are [Eq. 38.1.1](#), they do not involve  $\phi$ , and they are solved by the iteratively reweighted least squares of [Theorem 34.4.1](#);
- (b) if the mean model is correct then  $E \mathbf{U}(\boldsymbol{\beta}^0) = \mathbf{0}$ , whatever the true variances are: the quasi-score is an unbiased estimating function;
- (c)  $-E\{\partial \mathbf{U} / \partial \boldsymbol{\beta}^\top\} = \mathbf{A}_n$ , again whatever the true variances are;
- (d) if in addition the variance specification [Eq. 38.1.2](#) is correct, then  $\text{Cov}\{\mathbf{U}(\boldsymbol{\beta}^0)\} = \mathbf{A}_n$  as well: sensitivity and variability agree, which is the second Bartlett identity for  $Q$ ;
- (e) if instead  $\text{Var}(Y_i) = \sigma_i^2$  for arbitrary  $\sigma_i^2$ , then

$$\text{Cov}\{\mathbf{U}(\boldsymbol{\beta}^0)\} = \mathbf{B}_n = \frac{1}{\phi^2} \sum_{i=1}^n \left\{ \frac{w_i h'(\eta_i)}{V(\mu_i)} \right\}^2 \sigma_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top, \quad (38.2.3)$$

which equals  $\mathbf{A}_n$  if and only if  $\sum_i \{w_i h'_i / V_i\}^2 \{\sigma_i^2 - \phi V_i / w_i\} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top = \mathbf{0}$ .

*Proof.*

- (a) By [Proposition 38.2.1\(a\)](#) and the chain rule,

$$\frac{\partial Q}{\partial \beta_j} = \sum_{i=1}^n Q'(\mu_i; y_i) \frac{d\mu_i}{d\eta_i} x_{ij} = \frac{1}{\phi} \sum_{i=1}^n x_{ij} \frac{w_i h'(\eta_i)}{V(\mu_i)} (y_i - \mu_i),$$

which is the coordinate form of [Eq. 34.3.1](#). The map from  $(V, h, \mathbf{w}, \phi)$  to it is the one computed in [Theorem 34.3.1\(a\)](#), so the matrix form, the absence of  $\phi$  and the iteration of [Theorem 34.4.1](#) transfer unchanged; none of the three used the density.

- (b) Every term carries the factor  $y_i - \mu_i$ , which has mean zero when  $\mu_i = h(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^0)$ . No assumption about  $\text{Var}(Y_i)$  is used.
- (c) Write  $a(\eta) = h'(\eta) / V\{h(\eta)\}$ , so that  $\mathbf{U} = \phi^{-1} \sum_i w_i a(\eta_i) (y_i - \mu_i) \mathbf{x}_{(i)}$ . Differentiating,

$$\frac{\partial \mathbf{U}}{\partial \boldsymbol{\beta}^\top} = \frac{1}{\phi} \sum_{i=1}^n w_i \{a'(\eta_i) (y_i - \mu_i) - a(\eta_i) h'(\eta_i)\} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top,$$

the computation of [Theorem 34.3.1\(c\)](#). Expectation annihilates the first term whatever the variances, and  $w_i a(\eta_i) h'(\eta_i) = \mathbf{W}_{ii}$  leaves  $-\mathbf{A}_n$ .

- (d) and (e). The summands of  $\mathbf{U}$  are independent with mean zero, so  $\text{Cov}(\mathbf{U}) = \phi^{-2} \sum_i w_i^2 a(\eta_i)^2 \text{Var}(Y_i) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ , which is [Eq. 38.2.3](#). Substituting  $\sigma_i^2 = \phi V(\mu_i) / w_i$  turns the  $i$ th coefficient into  $\phi^{-2} w_i^2 \{h'_i / V_i\}^2 \phi V_i / w_i = \phi^{-1} w_i h_i'^2 / V_i$ , giving  $\mathbf{A}_n$ . The stated condition is the difference of the two sums.

■

*Remark* (Existence and uniqueness). Nothing above guarantees that [Eq. 38.1.1](#) has a solution, or only one. Under the canonical link of the variance function,  $g'(\mu) = 1/V(\mu)$ ,

the proof of (c) has  $a \equiv 1$  and  $a' \equiv 0$ , so the Jacobian is  $-\mathbf{A}_n$ , negative definite whenever  $\text{rank}(\mathbf{X}) = p$ ;  $Q$  is then strictly concave and a root, if one exists, is the unique maximizer, by [Theorem 34.3.3](#) with  $b''$  replaced by  $V$ . For other links  $Q$  can have several stationary points, as the warning after [Theorem 34.3.3](#) showed.

### Example. (Strike durations)

Kennan's data record the length in days of 62 strikes in US manufacturing, with a measure of unanticipated industrial production at the time each strike began. Take a log link and  $V(\mu) = \mu$ . The production coefficient is  $-7.680$ ; replacing  $V(\mu) = \mu$  by  $V(\mu) = 7.5\mu$  returns the same number to ten decimal places, since  $\phi$  cancels from [Eq. 38.1.1](#), while  $V(\mu) = \mu^2$  moves it to  $-9.353$ , a quadratic variance function downweighting the long strikes.

The mean duration is 42.66 days and the sample variance 2102.7, nearly fifty times as large: a Poisson likelihood is no serious description of these data, and the question is how much of the Poisson machinery survives anyway.

```
import numpy as np
import statsmodels.api as sm

data = sm.datasets.strikes.load_pandas().data
y = data["duration"].to_numpy(float)      # length of the strike, in days
x = data["iprod"].to_numpy(float)         # unanticipated industrial production
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape

def quasi_irls(X, y, V, steps=60):
    """Solve the quasi-score equations for a log link and variance function V."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(y.mean())
    for _ in range(steps):
        eta = X @ beta
        mu = np.exp(eta)
        W = mu ** 2 / V(mu)                # working weight (dmu/deta)^2 / V(mu)
        z = eta + (y - mu) / mu            # working response
        beta = np.linalg.solve((X.T * W) @ X, (X.T * W) @ z)
    return beta

beta_lin = quasi_irls(X, y, lambda m: m)      # V(mu) = mu
beta_scaled = quasi_irls(X, y, lambda m: 7.5 * m) # V(mu) = phi mu: same
equations
beta_sq = quasi_irls(X, y, lambda m: m ** 2)  # V(mu) = mu^2: different
equations
print(f"V = mu      {beta_lin}")
print(f"V = 7.5 mu   {beta_scaled}")
print(f"V = mu^2     {beta_sq}")
```

### 38.2.3 Optimality among linear estimating functions

The second view asks what else might have been used. Any function of the data and the parameter with mean zero at  $\beta^0$  is an **unbiased estimating function**, and setting it to zero defines an estimator; the quasi-score is the best of a natural class.

#### Proposition 38.2.3 — Godambe optimality

Assume Eq. 38.1.2, and consider **linear estimating functions**

$$\mathbf{g}(\beta) = \sum_{i=1}^n \mathbf{a}_i(\beta) \{y_i - \mu_i(\beta)\} = \mathbf{C}(\beta) (\mathbf{y} - \boldsymbol{\mu}),$$

where  $\mathbf{C} = [\mathbf{a}_1, \dots, \mathbf{a}_n]$  is  $p \times n$ , differentiable in  $\beta$ , and does not depend on  $\mathbf{y}$ . Let  $\mathbf{S} = -E\{\partial \mathbf{g} / \partial \beta^\top\}$  and suppose  $\mathbf{S}$  is nonsingular. Define the **Godambe information**

$$\mathcal{J}(\mathbf{g}) = \mathbf{S}^\top \text{Cov}\{\mathbf{g}(\beta^0)\}^{-1} \mathbf{S}.$$

- (a) Every such  $\mathbf{g}$  is unbiased, and  $\mathbf{S} = \mathbf{C} \mathbf{D} \mathbf{X}$ .
- (b)  $\mathcal{J}(\mathbf{g}) \preceq \phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X} = \mathcal{J}(\mathbf{U})$  in the order of nonnegative definiteness, for every linear estimating function  $\mathbf{g}$ .
- (c) Equality holds if and only if  $\mathbf{C}^\top = \boldsymbol{\Sigma}^{-1} \mathbf{D} \mathbf{X} \mathbf{N}$  for a nonsingular  $p \times p$  matrix  $\mathbf{N}$ , with  $\boldsymbol{\Sigma} = \text{Cov}(\mathbf{Y})$ : the coefficients are those of the quasi-score up to one common nonsingular transformation, and  $\mathbf{g}$  then has the same roots as  $\mathbf{U}$ .

*Proof.*

- (a) Unbiasedness follows from  $E(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}$ . Differentiating,

$$\frac{\partial \mathbf{g}}{\partial \beta^\top} = \sum_{i=1}^n \frac{\partial \mathbf{a}_i}{\partial \beta^\top} (y_i - \mu_i) - \mathbf{C} \frac{\partial \boldsymbol{\mu}}{\partial \beta^\top},$$

read entry by entry. Its expectation vanishes because  $\mathbf{a}_i$  does not depend on  $\mathbf{y}$ , and  $\partial \boldsymbol{\mu} / \partial \beta^\top = \mathbf{D} \mathbf{X}$ , so  $\mathbf{S} = \mathbf{C} \mathbf{D} \mathbf{X}$ .

- (b) and (c). Under Eq. 38.1.2,  $\boldsymbol{\Sigma} = \text{diag}\{\phi V(\mu_i)/w_i\}$  is positive definite. Set

$$\mathbf{F} = \boldsymbol{\Sigma}^{1/2} \mathbf{C}^\top, \quad \mathbf{K} = \boldsymbol{\Sigma}^{-1/2} \mathbf{D} \mathbf{X},$$

both  $n \times p$ . Then  $\text{Cov}(\mathbf{g}) = \mathbf{C} \boldsymbol{\Sigma} \mathbf{C}^\top = \mathbf{F}^\top \mathbf{F}$  and  $\mathbf{S} = \mathbf{C} \mathbf{D} \mathbf{X} = \mathbf{F}^\top \mathbf{K}$ , so

$$\mathcal{J}(\mathbf{g}) = \mathbf{K}^\top \mathbf{F} (\mathbf{F}^\top \mathbf{F})^{-1} \mathbf{F}^\top \mathbf{K} = \mathbf{K}^\top \mathbf{P} \mathbf{K},$$

where  $\mathbf{P} = \mathbf{F} (\mathbf{F}^\top \mathbf{F})^{-1} \mathbf{F}^\top$  is the orthogonal projection onto  $\mathcal{C}(\mathbf{F})$  (Theorem 6.3.7), and  $\mathbf{F}^\top \mathbf{F}$  is nonsingular because  $\mathbf{S}$  is. As  $\mathbf{I} - \mathbf{P}$  is symmetric idempotent,  $\mathbf{v}^\top (\mathbf{I} - \mathbf{P}) \mathbf{v} = \|(\mathbf{I} - \mathbf{P}) \mathbf{v}\|^2 \geq 0$ , so  $\mathbf{P} \preceq \mathbf{I}$  and  $\mathcal{J}(\mathbf{g}) \preceq \mathbf{K}^\top \mathbf{K}$ . Finally

$$\mathbf{K}^\top \mathbf{K} = \mathbf{X}^\top \mathbf{D} \boldsymbol{\Sigma}^{-1} \mathbf{D} \mathbf{X} = \sum_{i=1}^n \frac{w_i h'(\eta_i)^2}{\phi V(\mu_i)} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top = \frac{1}{\phi} \mathbf{X}^\top \mathbf{W} \mathbf{X},$$

and [Theorem 38.2.2\(a\),\(d\)](#) identify this as  $\mathcal{J}(\mathbf{U})$ , since for the quasi-score sensitivity and variability are both  $\mathbf{A}_n$ .

Equality in (b) means  $\mathbf{K}^\top(\mathbf{I} - \mathbf{P})\mathbf{K} = \mathbf{0}$ , that is  $\mathcal{C}(\mathbf{K}) \subseteq \mathcal{C}(\mathbf{F})$ . Both matrices have  $p$  columns and, because  $\mathbf{F}^\top\mathbf{K}$  is nonsingular, rank  $p$ ; so their column spaces coincide and  $\mathbf{K} = \mathbf{F}\mathbf{N}$  for a nonsingular  $\mathbf{N}$ . Written out this is  $\mathbf{C}^\top = \boldsymbol{\Sigma}^{-1}\mathbf{D}\mathbf{X}\mathbf{N}^{-1}$ , whose  $i$ th row is a fixed multiple of  $w_i h'(\eta_i) \mathbf{x}_{(i)}^\top / V(\mu_i)$ . Then  $\mathbf{g} = \mathbf{N}^{-\top}\mathbf{U}$ , because  $\mathbf{X}^\top\mathbf{D}\boldsymbol{\Sigma}^{-1}(\mathbf{y} - \boldsymbol{\mu})$  already carries the factor  $\phi^{-1}$  that  $\mathbf{U}$  has; so  $\mathbf{g}$  vanishes exactly where  $\mathbf{U}$  does. ■

*Warning.* The optimality is *within the class of linear estimating functions*. A known true density would have a score nonlinear in  $\mathbf{y}$  with a larger Godambe information — the NB1 score [Eq. 37.4.6](#) is an instance — so “best linear” is the honest reading, as in [Theorem 7.2.1](#). Optimality also assumes [Eq. 38.1.2](#); under a wrong variance function the quasi-score stays unbiased but is no longer optimal, and [Section 38.3](#) prices the loss.

## 38.2.4 Exercises

### 38.2.4.1 A. Check your understanding

#### Exercise A1.

Compute  $\phi w^{-1}Q(\mu; y)$  from [Eq. 38.2.1](#) for  $V(\mu) = 1$ ,  $V(\mu) = \mu$ ,  $V(\mu) = \mu(1 - \mu)$  and  $V(\mu) = \mu^2$ , and check that in each case it is the log-likelihood kernel of the corresponding family, up to a function of  $y$  alone.

*Solution.* For  $V = 1$ ,  $\int_y^\mu (y - t) dt = -(y - \mu)^2/2$ . For  $V(t) = t$  the integrand is  $y/t - 1$ , giving the Poisson kernel  $y \log \mu - \mu$ . For  $V(t) = t(1 - t)$ , partial fractions give  $y/t - (1 - y)/(1 - t)$ , whose integral is the binomial kernel  $y \log \mu + (1 - y) \log(1 - \mu)$ . For  $V(t) = t^2$  the integrand is  $y/t^2 - 1/t$ , giving the gamma kernel  $-y/\mu - \log \mu$ .

#### Exercise A2.

Show that the NB2 variance function  $V(\mu) = \mu + \mu^2/\kappa$  with  $\kappa$  known gives  $\phi w^{-1}Q(\mu; y) = y \log\{\mu/(\mu + \kappa)\} + \kappa \log\{\kappa/(\mu + \kappa)\}$  up to a function of  $y$ , which is the negative binomial log-likelihood kernel of [Eq. 37.4.1](#). NB2 with  $\kappa$  known is an exponential dispersion family, so [Proposition 38.2.1\(c\)](#) applies; explain why the *quasi-Poisson* specification  $V(\mu) = \mu$  with  $\phi$  free does not enjoy the same conclusion, and relate this to the remark of [Section 38.1](#).

### 38.2.4.2 B. Practice

#### Exercise B1.

Take  $n$  independent responses with  $E Y_i = \mu$  (one unknown parameter, no regressors) and  $\text{Var } Y_i = \phi V(\mu)/w_i$ . Show that the quasi-score estimate of [Theorem 38.2.2](#) is the weighted mean  $\sum_i w_i y_i / \sum_i w_i$ , and verify [Proposition 38.2.3](#) directly: among estimating

functions  $\sum_i a_i(y_i - \mu)$  with constants  $a_i$ , the Godambe information is maximized by  $a_i \propto w_i$ .

*Solution.* Parameterizing by  $\mu$  makes  $h' \equiv 1$ , so Eq. 38.1.1 reads  $\sum_i w_i(y_i - \mu)/V(\mu) = 0$  and  $\hat{\mu} = \sum_i w_i y_i / \sum_i w_i$ . For a general  $\mathbf{a}$  the sensitivity is  $S = \sum_i a_i$  and the variability is  $\phi V(\mu) \sum_i a_i^2 / w_i$ , so

$$J = \frac{(\sum_i a_i)^2}{\phi V(\mu) \sum_i a_i^2 / w_i} \leq \frac{\sum_i w_i}{\phi V(\mu)}$$

by the Cauchy–Schwarz inequality applied to  $\sum_i a_i = \sum_i (a_i / \sqrt{w_i}) \sqrt{w_i}$ , with equality if and only if  $a_i \propto w_i$ . The bound is  $\phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X}$  with  $\mathbf{X} = \mathbf{1}$ .

### Exercise B2.

Suppose the link satisfies  $g'(\mu) = 1/V(\mu)$ , the canonical link of the variance function. Show that Eq. 38.1.1 becomes  $\sum_i w_i x_{ij}(y_i - \hat{\mu}_i) = 0$  for every  $j$ , so that the fit reproduces the weighted column totals of  $\mathbf{X}$  exactly, as in Corollary 34.3.2. Check the condition for  $V(\mu) = \mu$  with a log link and for  $V(\mu) = \mu(1 - \mu)$  with a logit link, and find the canonical link of  $V(\mu) = \mu^2$ .

*Solution.* Since  $h$  is the inverse of  $g$ ,  $h'(\eta) = 1/g'(\mu) = V(\mu)$ , so the coefficient  $w_i h'(\eta_i)/V(\mu_i)$  in Eq. 38.1.1 is  $w_i$ . For  $V(\mu) = \mu$  the log link has  $g'(\mu) = 1/\mu$ ; for  $V(\mu) = \mu(1 - \mu)$  the logit link has  $g'(\mu) = 1/\{\mu(1 - \mu)\}$ ; for  $V(\mu) = \mu^2$  the condition  $g'(\mu) = \mu^{-2}$  gives  $g(\mu) = -1/\mu$ , the canonical link of the gamma family. The conclusion is the quasi-likelihood form of Corollary 34.3.2, proved there for exponential dispersion families but needing only Eq. 38.1.1.

### 38.2.4.3 C. Going deeper

#### Exercise C1.

Take the quasi-binomial specification  $V(t) = t(1 - t)$  on  $(0, 1)$  with  $w = 1$  and any  $\phi > 0$ . (a) Show that  $y \mapsto \exp\{Q(\mu; y)\}$  is integrable over  $(0, 1)$ , so normalizes to a density  $f$ . (b) Show that a distribution on  $[0, 1]$  with mean  $m$  has variance at most  $m(1 - m)$ , with equality only on  $\{0, 1\}$ ; deduce that  $f$  has variance strictly below  $m(1 - m)$  at its own mean  $m$ , never  $\phi m(1 - m)$  with  $\phi \geq 1$ . (c) Show that  $f$  is symmetric about  $1/2$  when  $\mu = 1/2$ , and check numerically that at  $\mu = 0.3$ ,  $\phi = 1$  it has mean 0.4408 and variance 0.0687, against  $\mu = 0.3$  and  $\phi\mu(1 - \mu) = 0.21$ . Conclude that integrability of  $\exp Q$  does not make a variance specification a family.

*Solution.*

- (a)  $Q \leq 0$  by Proposition 38.2.1(a), so  $\exp Q \leq 1$  on a bounded interval.
- (b) For  $Y \in [0, 1]$ ,  $E\{Y(1 - Y)\} \geq 0$  gives  $EY^2 \leq m$  and hence  $\text{Var } Y \leq m - m^2$ ; equality forces  $Y(1 - Y) = 0$  almost surely, which a density on the open interval excludes.
- (c) By Exercise (A1) the exponent is  $\phi^{-1}\{y \log \mu + (1 - y) \log(1 - \mu) - y \log y - (1 - y) \log(1 - y)\}$ , unchanged by  $y \mapsto 1 - y$  when  $\mu = 1/2$ . Elsewhere neither moment is reproduced:  $\exp Q$  normalizes without the result being a distribution with the

moments of [Eq. 38.1.2](#).

## 38.3 Sandwich covariance estimation

[Theorem 38.2.2](#) left an asymmetry: the estimating equations are unbiased as soon as the mean model is right, while the identity that turns  $\mathbf{A}_n = \phi^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{X}$  into a covariance needs the variance model too. When the latter fails, the covariance of  $\hat{\boldsymbol{\beta}}$  becomes a product of three matrices instead of one, and the middle factor can be estimated from the residuals even though its  $n$  ingredients cannot.

### 38.3.1 The asymptotic distribution

Throughout  $\boldsymbol{\beta}^0$  is the true parameter,  $\eta_i^0 = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}^0$ ,  $\mu_i^0 = h(\eta_i^0)$ , and  $\sigma_i^2 = \text{Var}(Y_i)$  is *not* assumed equal to  $\phi V(\mu_i^0)/w_i$ . Write

$$a(\eta) = \frac{h'(\eta)}{V\{h(\eta)\}}, \quad \mathbf{A}_n = \frac{1}{\phi} \mathbf{X}^\top \mathbf{W} \mathbf{X} = \frac{1}{\phi} \sum_{i=1}^n w_i a(\eta_i^0) h'(\eta_i^0) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top,$$

and let  $\mathbf{B}_n$  be the matrix [Eq. 38.2.3](#). The conditions are

- (Q1) the  $Y_i$  are independent with  $E Y_i = \mu_i^0$  for a  $\boldsymbol{\beta}^0$  interior to the admissible set,  $p$  is fixed and  $\text{rank}(\mathbf{X}) = p$ ;
- (Q2)  $\|\mathbf{x}_{(i)}\| \leq C$  and  $0 < w_{\min} \leq w_i \leq w_{\max} < \infty$ ;
- (Q3) on a neighbourhood  $\mathcal{N}$  of  $\boldsymbol{\beta}^0$ ,  $h$  and  $a$  are continuously differentiable with  $h'$ ,  $a$ ,  $a'$  bounded and  $h'$ ,  $a$  bounded away from zero, uniformly in  $i$  and  $n$ ;
- (Q4)  $\sigma_i^2 \geq \sigma_{\min}^2 > 0$ ,  $E(Y_i - \mu_i^0)^4 \leq K$ , and  $n^{-1} \mathbf{A}_n \rightarrow \bar{\mathbf{A}}$ ,  $n^{-1} \mathbf{B}_n \rightarrow \bar{\mathbf{B}}$  with both limits positive definite;
- (Q5) with probability tending to one [Eq. 38.1.1](#) has a root  $\hat{\boldsymbol{\beta}} \in \mathcal{N}$  with  $\hat{\boldsymbol{\beta}} \rightarrow \boldsymbol{\beta}^0$  in probability.

#### Theorem 38.3.1 — Consistency, normality and the sandwich

Assume (Q1)–(Q5). Then:

(a)

$$\mathbf{B}_n^{-1/2} \mathbf{A}_n (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0) \xrightarrow{d} N_p(\mathbf{0}, \mathbf{I}), \quad (38.3.1)$$

so that  $\hat{\boldsymbol{\beta}}$  behaves as if its covariance were the **sandwich**  $\mathbf{A}_n^{-1} \mathbf{B}_n \mathbf{A}_n^{-1}$ , which involves the distribution of  $\mathbf{Y}$  only through its first two moments.

- (b) If the variance specification [Eq. 38.1.2](#) is correct, then  $\mathbf{B}_n = \mathbf{A}_n$  by [Theorem 38.2.2\(d\)](#), and [Eq. 38.3.1](#) becomes  $\mathbf{A}_n^{1/2} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0) \rightarrow N_p(\mathbf{0}, \mathbf{I})$ : the **model-based** covariance  $\mathbf{A}_n^{-1} = \phi (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$  is correct, and by [Proposition 38.2.3](#) no linear estimating function has a larger Godambe information — hence, whenever (Q1)–(Q5) hold for that estimating function too, none has a smaller asymptotic covariance.

- (c) Let  $\hat{c}_i = w_i h'(\hat{\eta}_i)/V(\hat{\mu}_i)$  and define the **empirical sandwich**

$$\hat{\Sigma}_{\text{rob}} = (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \left\{ \sum_{i=1}^n \hat{c}_i^2 (y_i - \hat{\mu}_i)^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right\} (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}, \quad (38.3.2)$$

in which  $\phi$  cancels. Then  $n\hat{\Sigma}_{\text{rob}} - n\mathbf{A}_n^{-1}\mathbf{B}_n\mathbf{A}_n^{-1} \rightarrow \mathbf{0}$  in probability, and Eq. 38.3.1 holds with  $\hat{\Sigma}_{\text{rob}}^{-1/2}(\hat{\beta} - \beta^0)$  in place of the left-hand side.

*Proof. Step 1: the quasi-score is asymptotically normal.* By Theorem 38.2.2(a) and (e),  $\mathbf{U}_n(\beta^0) = \phi^{-1} \sum_i w_i a(\eta_i^0) \xi_i \mathbf{x}_{(i)}$  with  $\xi_i = Y_i - \mu_i^0$  independent and mean zero and  $\text{Cov}\{\mathbf{U}_n(\beta^0)\} = \mathbf{B}_n$ . Fix a unit vector  $\mathbf{v}$  and write  $\mathbf{v}^\top \mathbf{B}_n^{-1/2} \mathbf{U}_n = \sum_i d_{ni} \xi_i$  with  $d_{ni} = \phi^{-1} w_i a(\eta_i^0) \mathbf{v}^\top \mathbf{B}_n^{-1/2} \mathbf{x}_{(i)}$ , so that  $\sum_i d_{ni}^2 \sigma_i^2 = 1$ . By (Q4),  $\sigma_i^2 \leq K^{1/2}$  and hence  $\sum_i d_{ni}^2 \geq K^{-1/2}$ , while (Q2)–(Q4) give  $d_{ni}^2 \leq \phi^{-2} w_{\max}^2 a_{\max}^2 C^2 \lambda_{\max}(\mathbf{B}_n^{-1}) = O(n^{-1})$ , because  $\lambda_{\min}(n^{-1}\mathbf{B}_n) \rightarrow \lambda_{\min}(\mathbf{B}) > 0$ . So  $\max_i d_{ni}^2 / \sum_i d_{ni}^2 \rightarrow 0$ , which is Eq. 21.2.2, and the fourth moments are bounded: Lemma 21.2.1(b) with  $\delta = 2$  and the Cramér–Wold device give  $\mathbf{B}_n^{-1/2} \mathbf{U}_n(\beta^0) \rightarrow N_p(\mathbf{0}, \mathbf{I})$ .

*Step 2: the expansion.* By (Q3),  $\beta \mapsto \mathbf{U}_n(\beta)$  is continuously differentiable on  $\mathcal{N}$ , so on  $\{\hat{\beta} \in \mathcal{N}\}$ ,

$$\mathbf{0} = \mathbf{U}_n(\hat{\beta}) = \mathbf{U}_n(\beta^0) + \bar{\mathbf{J}}_n (\hat{\beta} - \beta^0),$$

where

$$\bar{\mathbf{J}}_n = \int_0^1 \frac{\partial \mathbf{U}_n}{\partial \beta^\top} (\beta^0 + t(\hat{\beta} - \beta^0)) dt.$$

*Step 3: the Jacobian.* By the proof of Theorem 38.2.2(c),

$$\frac{\partial \mathbf{U}_n}{\partial \beta^\top}(\beta) = \frac{1}{\phi} \sum_{i=1}^n w_i \{a'(\eta_i) \{y_i - \mu_i(\beta)\} - a(\eta_i) h'(\eta_i)\} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top,$$

with  $\eta_i = \mathbf{x}_{(i)}^\top \beta$ ; take  $\|\beta - \beta^0\| \leq \delta$ , so that every  $\eta_i$  moves by at most  $C\delta$ . The second term is nonrandom and equals  $-\mathbf{A}_n$  at  $\beta^0$ , and since  $ah'$  is uniformly continuous by (Q3) the two differ, after division by  $n$ , by at most  $\phi^{-1} w_{\max} C^2 \omega(C\delta)$  with  $\omega$  the modulus of continuity. In the first term write  $y_i - \mu_i(\beta) = \xi_i + \{\mu_i^0 - \mu_i(\beta)\}$ : the displacement is at most  $h'_{\max} C\delta$ , contributing  $O(\delta)$  to  $n^{-1}$  times the norm, while  $n^{-1} \phi^{-1} \sum_i w_i a'(\eta_i) \xi_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$  has entries that average independent mean-zero variables of bounded variance, hence  $o_p(1)$  by Chebyshev. By (Q5)  $\delta$  may tend to zero, so  $-n^{-1} \bar{\mathbf{J}}_n \rightarrow \bar{\mathbf{A}}$  in probability, with  $\bar{\mathbf{A}}$  nonsingular.

*Step 4: assembling.* With probability tending to one  $\bar{\mathbf{J}}_n$  is invertible and  $\hat{\beta} - \beta^0 = (-\bar{\mathbf{J}}_n)^{-1} \mathbf{U}_n(\beta^0)$ , so

$$\mathbf{B}_n^{-1/2} \mathbf{A}_n (\hat{\beta} - \beta^0) = \left( \mathbf{B}_n^{-1/2} \mathbf{A}_n (-\bar{\mathbf{J}}_n)^{-1} \mathbf{A}_n^{-1} \mathbf{B}_n^{1/2} \right) \mathbf{B}_n^{-1/2} \mathbf{U}_n(\beta^0).$$

By (Q4) and Step 3 the bracket tends in probability to  $\mathbf{I}$ , since  $n^{-1} \mathbf{A}_n$ ,  $n^{-1} \mathbf{B}_n$ , their square roots and inverses all converge to fixed nonsingular limits. Slutsky and Step 1 give Eq. 38.3.1, and (b) is the substitution  $\mathbf{B}_n = \mathbf{A}_n$ .

*Step 5: the empirical middle.* Write  $c_i = w_i a(\eta_i^0)$ , so the middle of Eq. 38.3.2 at the true parameter is  $\sum_i c_i^2 \xi_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$  and  $\phi^2 \mathbf{B}_n = \sum_i c_i^2 \sigma_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ . Each entry of  $n^{-1} \sum_i c_i^2 (\xi_i^2 - \sigma_i^2) \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$  is an average of independent mean-zero variables whose variances are bounded by  $c_{\max}^4 C^4 K$ , so it is  $o_p(1)$  by Chebyshev; this is Step 2 of the proof of Theorem 21.4.2.



Replacing  $\xi_i$  by  $y_i - \hat{\mu}_i$  and  $c_i$  by  $\hat{c}_i$  costs a further  $o_p(1)$ , both displacements being bounded uniformly in  $i$  by a constant times  $\|\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0\|$ , so Step 3 of that proof applies verbatim. The same reasoning gives  $n^{-1}\mathbf{X}^\top \hat{\mathbf{W}}\mathbf{X} \rightarrow \phi \mathbf{\bar{A}}$ , and combining the three factors proves (c). ■

*Remark* (What (Q5) hides, and when it is free). Assuming a consistent root assumes something real: the equations can have no solution, or several. Under the canonical link of the variance function  $Q$  is strictly concave, and the argument for [Theorem 34.5.1](#) gives existence and consistency without (Q5) as soon as  $\lambda_{\min}(\mathbf{A}_n) \rightarrow \infty$ ; for a general link the conditions of Fahrmeir and Kaufmann (1985), quoted in [Proposition 34.5.2](#), do the same and need only unbiasedness. Conditions (Q2)–(Q4) are the quasi-likelihood forms of the bounded-leverage and moment conditions of [Theorem 21.4.2](#), and fail in the same places.

## 38.3.2 Model-based or robust

The two covariance estimates answer different questions. Under the log link with  $V(\mu) = \mu$  and  $w_i = 1$ , where  $\mathbf{W}_{ii} = \mu_i$  and  $a(\eta_i) = 1$ ,

$$\mathbf{A}_n^{-1} \mathbf{B}_n \mathbf{A}_n^{-1} = \left( \sum_i \mu_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right)^{-1} \left( \sum_i \sigma_i^2 \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right) \left( \sum_i \mu_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top \right)^{-1},$$

against the model-based  $\phi(\sum_i \mu_i \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top)^{-1}$ . Only the *shape* of the dependence on  $\mu$  matters, since a constant factor is absorbed into  $\phi$ : a quasi-Poisson analysis is right about the standard errors whenever the true variance is proportional to the mean, whatever the constant, and wrong for any other shape.

**Example.** (Coverage when the variance function is right and when it is wrong)

A quasi-Poisson model  $\log \mu_i = \beta_0 + \beta_1 x_i$  was fitted to data simulated with the same mean and two true variances,  $\text{Var}(Y_i) = \mu_i + 2\mu_i^2$  (quadratic) and  $\text{Var}(Y_i) = 9\mu_i$  (linear), over 3000 replicates at each of seven sample sizes. [Figure 38.3.1](#) records the coverage of nominal 95% intervals for  $\beta_1$  from four standard errors: Poisson with  $\phi = 1$ , model-based scaled by  $\hat{\phi}$ , the empirical sandwich [Eq. 38.3.2](#), and a leverage-corrected sandwich described below.

Ignoring the dispersion is fatal in both panels and does not improve with  $n$ : 0.334 at  $n = 20$  and 0.323 at  $n = 1280$  in the quadratic case, 0.513 and 0.482 in the linear case. A linear true variance makes the specification correct and the  $\hat{\phi}$ -scaled interval already good at  $n = 20$ , covering 0.919 against the sandwich's 0.880; a quadratic one stops it improving, at 0.914 for  $n = 1280$ , while the sandwich climbs to 0.941, buying correctness in the limit at the price of noise in small samples, as in [Example \(Coverage in a skewed design\)](#).

```
import numpy as np

BETA = np.array([1.5, 1.0])
```



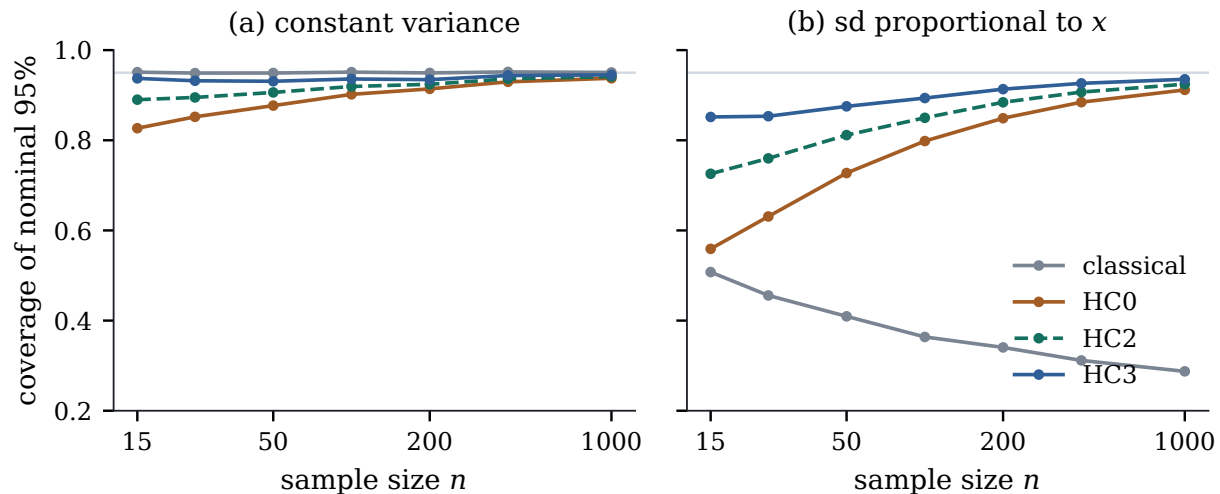


Figure 38.3.1: Coverage of nominal 95% intervals for the slope of a quasi-Poisson fit with a correct mean model. In (a) the true variance is quadratic in the mean, so the working variance function has the wrong shape; in (b) it is proportional to the mean.

```

HALF = 2.0          # the regressor runs over [-HALF, HALF]

def quasi_poisson(X, y, steps=25):
    """IRLS for the quasi-score equations with a log link and  $V(\mu) = \mu$ ."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(max(y.mean(), 0.5))
    for _ in range(steps):
        mu = np.exp(X @ beta)
        z = X @ beta + (y - mu) / mu
        beta = np.linalg.solve((X.T * mu) @ X, (X.T * mu) @ z)
    return beta, np.exp(X @ beta)

def slope_variances(X, y):
    """The four estimates of  $\text{Var}(b_1)$ : naive, phi-scaled, sandwich, leverage-corrected."""
    beta, mu = quasi_poisson(X, y)
    n, p = X.shape
    bread = np.linalg.inv((X.T * mu) @ X)
    root = np.sqrt(mu)[: , None] * X
    hat = np.sum(root * (root @ bread), axis=1)
    r2 = (y - mu) ** 2
    phi = np.sum(r2 / mu) / (n - p)
    meat = lambda w: bread @ ((X * w[: , None]).T @ X) @ bread
    return beta[1], np.array([bread[1, 1], phi * bread[1, 1],
                              meat(r2)[1, 1], meat(r2 / (1 - hat) ** 2)[1, 1]])

def coverage(n, quadratic, reps, rng, kappa=0.5, phi=9.0):
    """Coverage of nominal 95% normal intervals for  $b_1$  over `reps` simulated data sets."""
    x = np.linspace(-HALF, HALF, n)
    X = np.column_stack([np.ones(n), x])
    mu = np.exp(X @ BETA)
    shape = np.full(n, kappa) if quadratic else mu / (phi - 1.0)
    hit = np.zeros(4)
    for _ in range(reps):
        y = rng.negative_binomial(shape, shape / (shape + mu)).astype(float)
        b1, var = slope_variances(X, y)

```

```

        hit += np.abs(b1 - BETA[1]) <= 1.96 * np.sqrt(var)
    return hit / reps

rng = np.random.default_rng(3821)
names = ["Poisson", "phi-scaled", "sandwich", "corrected"]
for n in (25, 100, 400):
    quad = coverage(n, True, 300, rng)
    lin = coverage(n, False, 300, rng)
    print(f"n = {n:4d}    quadratic variance " + " ".join(
        f"{k} {v:.3f}" for k, v in zip(names, quad)))
    print(f"                linear variance      " + " ".join(
        f"{k} {v:.3f}" for k, v in zip(names, lin)))

```

### 38.3.3 The dispersion, the quasi-deviance and tests

The model-based covariance needs  $\phi$ , and so does any  $F$  test; neither  $\hat{\beta}$  nor the sandwich Eq. 38.3.2 does. Since  $\phi$  cancels from Eq. 38.1.1 and scales  $\mathbf{A}_n$  by  $\phi^{-1}$ , it is orthogonal to  $\beta$ : estimating it multiplies the asymptotic covariance of  $\hat{\beta}$  by a consistent factor rather than inflating it.

#### Proposition 38.3.2 — Estimating the dispersion, and quasi-deviance tests

Assume Eq. 38.1.2 with the mean model correct.

- (a)  $E[w_i(Y_i - \mu_i^0)^2 / V(\mu_i^0)] = \phi$  for every  $i$ , so the **Pearson estimator**  $\hat{\phi}_P = X^2 / (n - p)$  with  $X^2 = \sum_i w_i(y_i - \hat{\mu}_i)^2 / V(\hat{\mu}_i)$  — the statistic Eq. 34.5.5 — is consistent for  $\phi$  under (Q1)–(Q5). The divisor  $n - p$  is an analogy with  $s^2$  (Theorem 5.6.1), exact only in the normal linear model.
- (b) The **deviance estimator**  $\hat{\phi}_D = D(\mathbf{y}, \hat{\mu}) / (n - p)$ , with  $D$  the quasi-deviance Eq. 38.2.2, is *not* in general consistent for  $\phi$ . The two agree only so far as a third-order remainder is negligible: large binomial groups and large Poisson means, not small counts. Only  $\hat{\phi}_P$  has expectation exactly  $\phi$  at the true mean.
- (c) For nested mean models  $\mathcal{C}(\mathbf{X}_0) \subseteq \mathcal{C}(\mathbf{X})$  with  $\text{rank}(\mathbf{X}) - \text{rank}(\mathbf{X}_0) = q$ , fitted with the same  $V$ , the **quasi- $F$  statistic** is

$$F = \frac{\{D(\mathbf{y}, \tilde{\mu}) - D(\mathbf{y}, \hat{\mu})\} / q}{\hat{\phi}_P}. \quad (38.3.3)$$

Under the reduced model and (Q1)–(Q5),  $qF \rightarrow \chi^2(q)$ , so referring  $F$  to  $F(q, n - p)$  is asymptotically correct and is the usual small-sample convention. If the variance specification is wrong the limit fails and the Wald statistic built from Eq. 38.3.2 should be used instead.

- (d) No convention makes  $-2Q(\hat{\beta})$  a log-likelihood, so Definition 29.2.2 does not apply. The criterion QAIC =  $D(\mathbf{y}, \hat{\mu}) / \hat{\phi} + 2p$ , with  $\hat{\phi}$  fixed at its value in the largest model, is in common use, but it is a convention, not a theorem. Cross-validation (Definition 29.3.1) needs no likelihood, and the analogue for correlated responses, the QIC of Chapter 40, rests on the same convention.

*Proof.*

- (a) Under Eq. 38.1.2,  $E(Y_i - \mu_i^0)^2 = \phi V(\mu_i^0)/w_i$ , which is the identity. The  $T_i = w_i(Y_i - \mu_i^0)^2/V(\mu_i^0)$  are independent with mean  $\phi$  and bounded variances, so  $n^{-1} \sum_i T_i \rightarrow \phi$  by Chebyshev; replacing  $\mu_i^0$  by  $\hat{\mu}_i$  costs  $o_p(1)$ , by Step 5 of the proof of Theorem 38.3.1 with  $\mathbf{x}_{(i)}\mathbf{x}_{(i)}^\top$  replaced by 1, and  $n/(n-p) \rightarrow 1$ .
- (b) Each term of Eq. 38.2.2 is  $2w_i \int_{\hat{\mu}_i}^{y_i} (y_i - t)V(t)^{-1} dt$ . Expanding  $1/V(t)$  about  $t = y_i$  and integrating gives  $w_i(y_i - \hat{\mu}_i)^2/V(y_i) + O(w_i|y_i - \hat{\mu}_i|^3)$ , the calculation recorded after Eq. 34.5.5: a Pearson term with  $V(y_i)$  in place of  $V(\hat{\mu}_i)$ . The two estimators therefore differ by the remainder and by the gap between those denominators, both negligible when  $|y_i - \hat{\mu}_i|$  is small relative to  $\mu_i$  and neither otherwise; there is no exact unbiasedness to appeal to, since  $E d_i \neq \phi$  in general.
- (c) By Eq. 38.2.2 the difference is  $2\phi\{Q(\hat{\beta}) - Q(\tilde{\beta})\}$ . The proof of Proposition 34.5.3 used the log-likelihood only through the quadratic expansion Eq. 34.5.3, whose ingredients are asymptotic normality of the score and the identity between its variance and minus the expected Hessian. Under Eq. 38.1.2, Step 1 of the proof of Theorem 38.3.1 supplies the first and Theorem 38.2.2(d) the second, with  $\mathbf{A}_n$  in both roles, so that argument goes through word for word with  $Q$  in place of  $\ell$ . Hence  $\{D(\mathbf{y}, \tilde{\mu}) - D(\mathbf{y}, \hat{\mu})\}/\phi \rightarrow \chi^2(q)$ ; part (a) lets  $\hat{\phi}_P$  replace  $\phi$ , and  $qF(q, n-p) \rightarrow \chi^2(q)$  makes the  $F$  reference asymptotically equivalent.
- (d) is a definition and a warning, not a claim.

■

### Example. (Four standard errors for the strike data)

Continue Example (Strike durations). The Pearson statistic is 2393.5 on 60 degrees of freedom, so  $\hat{\phi}_P = 39.89$ , against  $\hat{\phi}_D = 37.22$  from the quasi-deviance. The four standard errors for the production coefficient  $-7.680$  are

| covariance                  | standard error | $t$    |
|-----------------------------|----------------|--------|
| Poisson, $\phi = 1$         | 0.406          | -18.92 |
| model-based, $\hat{\phi}_P$ | 2.564          | -3.00  |
| empirical sandwich          | 2.573          | -2.98  |
| leverage-corrected sandwich | 2.896          | -2.65  |

The Poisson standard error is too small by a factor of about six. More interesting are the second and third rows: the sandwich exceeds the model-based value by a factor of only 1.004, which is evidence *for* the quasi-Poisson variance function, so the model-based standard error is valid and slightly more efficient. The fourth row divides each squared residual by  $(1 - h_{ii})^2$ , with  $h_{ii}$  the leverages of the weighted hat matrix of Definition 34.5.2; the largest is 0.192, enough to matter at  $n = 62$ .

```
import numpy as np
import statsmodels.api as sm

data = sm.datasets.strikes.load_pandas().data
y = data["duration"].to_numpy(float)           # length of the strike, in days
```

```

x = data["iproduct"].to_numpy(float)          # unanticipated industrial production
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape

def quasi_irls(X, y, V, steps=60):
    """Solve the quasi-score equations for a log link and variance function V."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(y.mean())
    for _ in range(steps):
        eta = X @ beta
        mu = np.exp(eta)
        W = mu ** 2 / V(mu)                    # working weight (dmu/deta)^2 / V(mu)
        z = eta + (y - mu) / mu                # working response
        beta = np.linalg.solve((X.T * W) @ X, (X.T * W) @ z)
    return beta

beta_lin = quasi_irls(X, y, lambda m: m)        # V(mu) = mu
mu = np.exp(X @ beta_lin)
pearson = np.sum((y - mu) ** 2 / mu)           # X^2 with V(mu) = mu
phi_pearson = pearson / (n - p)
W = mu                                         # log link with V(mu) = mu: W_ii =
    mu_i
bread = np.linalg.inv((X.T * W) @ X)           # (X' W X)^{-1}
root = np.sqrt(W)[: , None] * X
hat = np.sum(root * (root @ bread), axis=1)    # leverages of the weighted hat matrix
resid = y - mu

cov_model = bread                             # phi = 1: the Poisson standard errors
cov_quasi = phi_pearson * bread                # phi estimated from the same fit
cov_sand = bread @ ((X * resid[: , None] ** 2).T @ X) @ bread
cov_corr = bread @ ((X * (resid ** 2 / (1 - hat) ** 2)[: , None]).T @ X) @ bread

for name, C in [("Poisson", cov_model), ("quasi", cov_quasi),
                ("sandwich", cov_sand), ("corrected", cov_corr)]:
    se = np.sqrt(np.diag(C))
    print(f"{name:10s} slope {beta_lin[1]:8.3f}   se {se[1]:6.3f}   t {beta_lin[1] /
        se[1]:7.2f}")
print(f"largest leverage {hat.max():.4f}, sum {hat.sum():.4f}")

```

### 38.3.4 Corrections, clusters and resampling

Three qualifications finish the section.

**Small samples.** The empirical sandwich is biased downwards, for the reason [Proposition 21.4.1\(a\)](#) identified in the linear model, and the remedy is the same: divide the  $i$ th squared residual by  $1 - h_{ii}$  or by  $(1 - h_{ii})^2$ , with  $h_{ii}$  the diagonal of  $\hat{\mathbf{W}}^{1/2} \mathbf{X} (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \mathbf{X}^\top \hat{\mathbf{W}}^{1/2}$ , giving the analogues of HC2 and HC3 of [Definition 21.4.1](#). The second is the “corrected” line of [Example \(Coverage when the variance function is right and when it is wrong\)](#), better than the plain sandwich at  $n = 80$  in the quadratic panel, 0.906 against 0.891; Kauermann and Carroll (2001) give the leading bias term. The middle of [Eq. 38.3.2](#) is itself an average with very unequal weights, so it behaves like a variance estimate on far fewer than  $n - p$  degrees of freedom ([Example \(One household carries the standard error\)](#)).

**Clusters.** If the observations come in  $G$  independent groups with arbitrary within-group de-

pendence, the middle of Eq. 38.3.2 becomes

$$\sum_{g=1}^G \left( \sum_{i \in g} \hat{c}_i (y_i - \hat{\mu}_i) \mathbf{x}_{(i)} \right) \left( \sum_{i \in g} \hat{c}_i (y_i - \hat{\mu}_i) \mathbf{x}_{(i)} \right)^\top,$$

the cluster-robust form Eq. 33.2.3 with the quasi-score's coefficients in place of the rows of  $\mathbf{X}$ . Its consistency needs  $G \rightarrow \infty$ , by Step 5 above with the cluster in place of the observation. The quasi-score is then no longer optimal, because the within-group correlations are ignored; Chapter 40 puts a generalized estimating equation in its place.

**Resampling.** The bootstrap of Chapter 23 applies unchanged: whole observations (Proposition 23.1.5), the wild bootstrap (Definition 23.2.1), or whole clusters (Exercise (C1)) for small  $G$ .

## 38.3.5 Exercises

### 38.3.5.1 A. Check your understanding

#### Exercise A1.

Under a log link and  $V(\mu) = \mu$  with  $w_i \equiv 1$ , suppose the true variances are  $\sigma_i^2 = \tau \mu_i$  for an unknown  $\tau > 0$ . Show that the sandwich and the model-based covariance have the same probability limit, and identify what  $\hat{\phi}_P$  estimates. Repeat for  $\sigma_i^2 = \tau \mu_i^2$  and show that the two then differ unless all  $\mu_i$  are equal.

#### Exercise A2.

Verify from Eq. 38.3.2 that the empirical sandwich does not change if  $V$  is replaced by  $cV$  for a constant  $c > 0$ , and explain why this is to be expected.

### 38.3.5.2 B. Practice

#### Exercise B1.

Take an intercept-only quasi-Poisson model,  $\log \mu_i = \beta$  for all  $i$ . Show that  $\hat{\mu}_i = \bar{y}$ , that the model-based variance of  $\hat{\beta}$  is  $\hat{\phi}_P / (n\bar{y})$ , and that the sandwich variance is  $\sum_i (y_i - \bar{y})^2 / (n\bar{y})^2$ . Show that the two agree exactly when  $\hat{\phi}_P$  is computed with divisor  $n$  rather than  $n - p$ .

#### Exercise B2.

Let  $\text{Var}(Y_i) = \sigma_i^2$  and consider the quasi-score built from a working variance function  $V$ . Using Proposition 38.2.3 with the *true* variances, show that  $\mathbf{A}_n^{-1} \mathbf{B}_n \mathbf{A}_n^{-1}$  — the asymptotic covariance of Theorem 38.3.1(a) — is never smaller, in the nonnegative definite order, than the inverse Godambe information  $(\sum_i h'(\eta_i)^2 \sigma_i^{-2} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top)^{-1}$  of the best linear estimating function, and that the two agree exactly when  $\sigma_i^2 \propto V(\mu_i)/w_i$ .

*Solution.* Repeat the proof of [Proposition 38.2.3](#) with  $\Sigma = \text{diag}(\sigma_i^2)$  in place of  $\text{diag}\{\phi V(\mu_i)/w_i\}$ ; only its last display used the specification. The quasi-score built from  $V$  is a linear estimating function with  $\mathbf{a}_i = w_i h'(\eta_i) \mathbf{x}_{(i)} / \{\phi V(\mu_i)\}$ , so by [Theorem 38.2.2\(c\),\(e\)](#) its Godambe information is  $\mathbf{A}_n \mathbf{B}_n^{-1} \mathbf{A}_n$ , at most the maximum  $\mathbf{K}^\top \mathbf{K} = \sum_i h'(\eta_i)^2 \sigma_i^{-2} \mathbf{x}_{(i)} \mathbf{x}_{(i)}^\top$ ; inverting reverses the order. Equality needs coefficients proportional to  $h'(\eta_i) \mathbf{x}_{(i)} / \sigma_i^2$ , that is  $w_i / V(\mu_i) \propto 1 / \sigma_i^2$ .

### 38.3.5.3 C. Going deeper

#### Exercise C1.

A large discrepancy between  $\hat{\phi}_P(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$  and [Eq. 38.3.2](#) is evidence that the variance specification is wrong. Propose a test statistic from the difference of the two matrices, say  $\text{tr}\{(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X}) \hat{\Sigma}_{\text{rob}}\} - p \hat{\phi}_P$ , argue informally that its mean is about zero when the specification is correct, and say what stands in the way of a usable null distribution. (This is White's information matrix test; compare [Exercise \(B3\)](#).)

## 38.4 Overdispersion

The word has been used loosely so far. This section says where the extra variability comes from, how to see it, what ignoring it costs, and which repair to make.

### 38.4.1 Where it comes from

#### Proposition 38.4.1 — Sources of overdispersion, and the price of ignoring it

- (a) *Unobserved heterogeneity.* Let  $Y \mid \Lambda$  have mean  $\Lambda$  and variance  $V^*(\Lambda)$ , and let  $\Lambda$  have mean  $\mu$  and variance  $\tau^2$ , with  $0 < \tau^2 < \infty$  and  $E\{V^*(\Lambda)\} < \infty$ . Then  $EY = \mu$  and  $\text{Var } Y = E\{V^*(\Lambda)\} + \tau^2$ , which for a Poisson kernel  $V^*(\lambda) = \lambda$  is  $\mu + \tau^2 > \mu$ : any unmodelled variation in the mean overdisperses.
- (b) *Clustering.* If the response is an average of  $m$  units with pairwise correlation  $\rho > 0$ , its variance is inflated by  $1 + \rho(m-1)$ , by [Lemma 38.1.1](#); the same factor applies to a sum of  $m$  correlated indicators.
- (c) *Contagion.* Let  $Y = \sum_{j=1}^N S_j$ , where  $N$  is Poisson with mean  $\nu$  and the  $S_j$  are independent copies of a positive integer-valued  $S$  with  $ES^2 < \infty$ , independent of  $N$ : events arrive in clusters. Then  $EY = \nu ES$  and  $\text{Var } Y = \nu ES^2$ , so

$$\frac{\text{Var } Y}{EY} = \frac{ES^2}{ES} \geq ES \geq 1, \quad (38.4.1)$$

with equality throughout if and only if  $S \equiv 1$ , that is unless the events arrive singly.

- (d) *Neither of the above.* A wrong mean model, an omitted interaction or a few gross outliers all inflate  $X^2$ , and no dispersion parameter repairs them; for ungrouped

binary data a large  $X^2$  is always a statement about the mean model.

- (e) *The price.* Suppose the mean model is correct and the true variances are  $\sigma_i^2 = \phi_0 V(\mu_i^0)/w_i$  with  $\phi_0 \geq 1$  unknown, while inference uses  $\phi = 1$ . Then under (Q1)–(Q5) any statistic of [Proposition 34.5.3](#) for a  $q$ -dimensional hypothesis converges to  $\phi_0 \chi^2(q)$ , so a nominal level- $\alpha$  test has asymptotic size

$$\Pr\{\chi^2(q) > \chi_{1-\alpha}^2(q)/\phi_0\} \geq \alpha, \quad (38.4.2)$$

increasing in  $\phi_0$ , and tending to 1 as  $q \rightarrow \infty$  for every fixed  $\phi_0 > 1$ . Standard errors are too small by the factor  $\phi_0^{1/2}$ .

*Proof.*

- (a) The law of total variance gives  $\text{Var } Y = E\{V^*(\Lambda)\} + \tau^2$ , and  $E \Lambda = \mu$ ; this is [Lemma 37.4.1](#)(a) with the mixing distribution left arbitrary.  
 (b) is [Lemma 38.1.1](#).  
 (c) Conditioning on  $N$  gives  $E(Y | N) = N E S$  and  $\text{Var}(Y | N) = N \text{Var } S$ , so  $E Y = \nu E S$  and

$$\begin{aligned} \text{Var } Y &= E(N) \text{Var } S + \text{Var}(N)(E S)^2 \\ &= \nu \{\text{Var } S + (E S)^2\} = \nu E S^2, \end{aligned}$$

using  $\text{Var } N = E N = \nu$ . Dividing gives  $\text{Var } Y / E Y = E S^2 / E S$ ; the first inequality in [Eq. 38.4.1](#) is  $E S^2 \geq (E S)^2$ , equality only for degenerate  $S$ , and the second is  $E S \geq 1$ . Equality in both forces  $S \equiv 1$ , and then  $Y = N$  is Poisson.

- (d) The binary claim is the warning of [Section 38.1](#); the rest records what  $X^2$  measures and is not a mathematical assertion.  
 (e) By [Theorem 38.3.1](#)(a) with  $\mathbf{B}_n = \phi_0 \mathbf{A}_n$  — which is [Theorem 38.2.2](#)(e) at  $\sigma_i^2 = \phi_0 V_i/w_i$  with the working  $\phi = 1$  —  $\mathbf{A}_n^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0) \rightarrow N_p(\mathbf{0}, \phi_0 \mathbf{I})$ . The Wald statistic computed as though the covariance were  $\mathbf{A}_n^{-1}$  is  $\|(\mathbf{I} - \mathbf{P}_n)\mathbf{A}_n^{1/2}(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}^0)\|^2$  for a projection  $\mathbf{P}_n$  of rank  $p - q$ , as in the proof of [Proposition 34.5.3](#), hence converges to  $\phi_0$  times a  $\chi^2(q)$  variable; the other two differ from it by  $o_p(1)$  by the same proof, since [Proposition 38.3.2](#)(c) extends the expansion to  $Q$ . The size [Eq. 38.4.2](#) follows and exceeds  $\alpha$  because  $\phi_0 > 1$  lowers the threshold; the limit as  $q \rightarrow \infty$  is [Exercise \(A1\)](#). ■

These sizes are worse than intuition suggests. At  $\phi_0 = 2$  a nominal 5% test of one coefficient rejects with probability 0.166 and of six coefficients with probability 0.391; at  $\phi_0 = 5$  the figures are 0.381 and 0.866. A dispersion of five is not unusual — the strike data of [Example \(Strike durations\)](#) have  $\hat{\phi}_P = 39.89$ .

```
import numpy as np
from scipy import stats

def naive_size(phi, q):
    """Size of a nominal 5% chi-squared test when the statistic is inflated by phi.
    """
    return stats.chi2.sf(stats.chi2.ppf(0.95, q) / phi, q)

print("phi    q = 1    q = 3    q = 6")
```

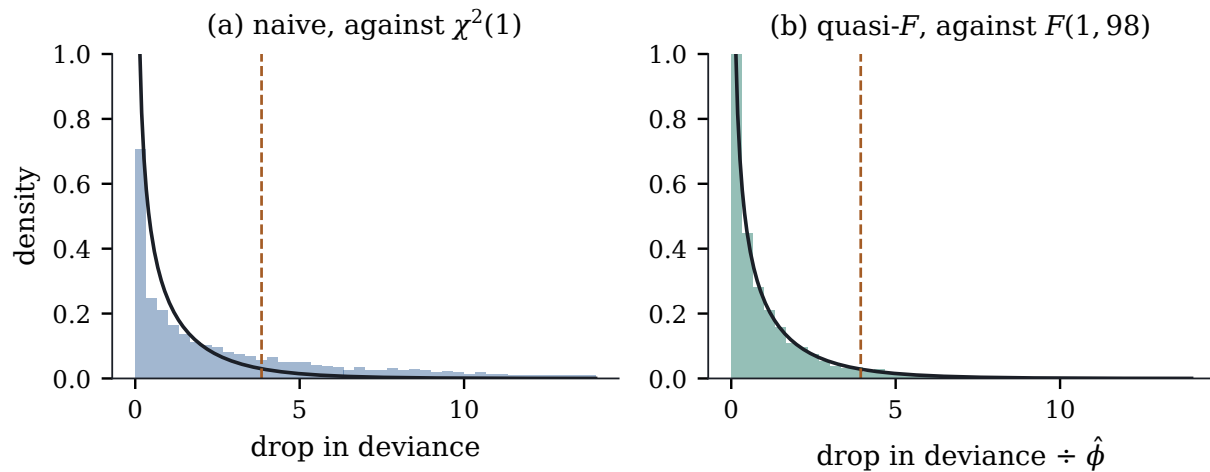


Figure 38.4.1: Simulated null distributions of the drop in deviance for one coefficient, on data whose variance is four times their mean: (a) the raw statistic against  $\chi^2(1)$ ; (b) the statistic divided by the Pearson dispersion, against  $F(1, 98)$ . The dashed line is the upper 5% point.

```
for phi in (1.5, 2.0, 3.0, 5.0, 10.0):
    print(f"{phi:4.1f} " + " ".join(f"{naive_size(phi,q):7.3f}" for q in (1, 3, 6)
    ))
```

#### Example. (A test that rejects six times too often)

Counts were simulated with a constant mean 6.0, a variance 4.0 times the mean and a true slope of zero, and a Poisson log-linear model was fitted to each of 5000 data sets of size 100. Referring the drop in deviance to  $\chi^2(1)$  rejects at the nominal 5% level in 0.322 of the replicates — Eq. 38.4.2 predicts 0.327 at  $\phi_0 = 4$  — with the statistic averaging 3.90 against a  $\chi^2(1)$  mean of one. Dividing by  $\hat{\phi}_P$  and referring to  $F(1, 98)$  rejects in 0.051, and the sandwich Wald test Eq. 38.3.2 in 0.058 (Figure 38.4.1).

```
def poisson_fit(X, y, steps=20):
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(max(y.mean(), 0.5))
    for _ in range(steps):
        mu = np.exp(X @ beta)
        z = X @ beta + (y - mu) / mu
        beta = np.linalg.solve((X.T * mu) @ X, (X.T * mu) @ z)
    return beta, np.exp(X @ beta)

def poisson_deviance(y, mu):
    return 2 * np.sum(np.where(y > 0, y * np.log(np.maximum(y, 1e-300) / mu), 0.0) -
    (y - mu))

def null_statistics(n, phi, mu0, reps, rng):
    """Three statistics for the slope, on data whose true slope is zero."""
    x = np.linspace(-1.0, 1.0, n)
    X = np.column_stack([np.ones(n), x])
    shape = np.full(n, mu0 / (phi - 1.0)) # negative binomial with
    variance phi*mu
    out = np.empty((reps, 3))
    for r in range(reps):
        y = rng.negative_binomial(shape, shape / (shape + mu0)).astype(float)
```



```

    beta, mu = poisson_fit(X, y)
    beta0, mu_null = poisson_fit(X[:, :1], y)
    drop = poisson_deviance(y, mu_null) - poisson_deviance(y, mu)
    phi_hat = np.sum((y - mu) ** 2 / mu) / (n - 2)
    bread = np.linalg.inv((X.T * mu) @ X)
    sand = bread @ ((X * ((y - mu) ** 2)[:, None]).T @ X) @ bread
    out[r] = [drop, drop / phi_hat, beta[1] ** 2 / sand[1, 1]]
return out

N, PHI, MU0 = 100, 4.0, 6.0
rng = np.random.default_rng(3841)
stat = null_statistics(N, PHI, MU0, 400, rng)
rates = [np.mean(stat[:, 0] > stats.chi2.ppf(0.95, 1)),
         np.mean(stat[:, 1] > stats.f.ppf(0.95, 1, N - 2)),
         np.mean(stat[:, 2] > stats.chi2.ppf(0.95, 1))]
print(f"rejection rates at the 5% level: naive {rates[0]:.3f},"
      f" quasi-F {rates[1]:.3f}, sandwich Wald {rates[2]:.3f}")

```

### 38.4.2 Seeing it

Four diagnostics, in the order in which they should be used. **Check the mean model first:** an omitted term can remove an apparent dispersion entirely, as [Example \(Twenty thousand records in thirty-two rows\)](#) showed, and only a mean model that has survived [Definition 34.5.2](#) and the plots of [Chapter 20](#) makes a dispersion parameter mean what it says. **Compare  $\hat{\phi}_P = X^2/(n-p)$  with one,** remembering that it is calibrated only where [Theorem 34.5.5\(b\)](#) applies, noisy for ungrouped counts with small means, and meaningless for ungrouped binary data. **Use a score test:** for counts, [Theorem 37.4.3](#) tests the Poisson against NB2 alternatives from the Poisson fit alone, with a one-sided alternative. **Compare the two covariances:** a large discrepancy between  $\hat{\phi}_P(\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1}$  and the sandwich [Eq. 38.3.2](#) says the *shape* of the variance function is wrong and not merely its scale, while a small one, as in [Example \(Four standard errors for the strike data\)](#), is evidence that the shape is right ([Exercise \(C1\)](#)). A plot of squared Pearson residuals against fitted means, and against group sizes where they vary, shows the same in more detail.

### 38.4.3 Choosing a repair

Four devices are in use, differing in what they assume and what they give.

| device                        | assumes                              | gives                                         | does not give                |
|-------------------------------|--------------------------------------|-----------------------------------------------|------------------------------|
| quasi model with $\hat{\phi}$ | mean and variance shape              | $\hat{\beta}$ , $F$ tests, model-based errors | likelihood, AIC, predictions |
| quasi model with the sandwich | mean model only                      | $\hat{\beta}$ , Wald tests                    | efficiency, small samples    |
| NB2, NB1 or beta-binomial     | a full distribution                  | likelihood, AIC, predictions                  | robustness to that choice    |
| random effect                 | a full distribution and a mixing law | conditional coefficients, prediction of units | the marginal coefficients    |

Putting a dispersion parameter into a quasi model changes the standard errors and leaves  $\hat{\beta}$  exactly where it was ([Theorem 38.2.2\(a\)](#)); changing to a different likelihood with the same variance function can move it by more than a standard error ([Example \(Four analyses of the strike data\)](#)). The fourth device changes the estimand rather than the estimate: a coefficient conditional on a unit's own random effect answers a different question from one describing a population average, a distinction Chapter 40 takes up and does not treat as small.

#### Example. (Four analyses of the strike data)

Continuing [Example \(Strike durations\)](#) and [Example \(Four standard errors for the strike data\)](#), the production coefficient under four treatments of the extra variability is

| fit                                    | coefficient | standard error |
|----------------------------------------|-------------|----------------|
| Poisson, $\phi = 1$                    | -7.680      | 0.406          |
| quasi-Poisson, $V = \phi\mu$           | -7.680      | 2.564          |
| NB1, $\text{Var} = \phi\mu$            | -4.934      | 1.983          |
| NB2, $\text{Var} = \mu + \mu^2/\kappa$ | -9.327      | 2.931          |

The NB1 fit has  $\hat{\phi} = 42.78$ , close to the quasi-Poisson  $\hat{\phi}_P = 39.89$ , and *the same variance function*; yet its coefficient differs by more than a standard error, because its estimating equation [Eq. 37.4.6](#) is not [Eq. 38.1.1](#) — the warning of [Section 38.1](#) in numbers. The NB2 fit, with  $\hat{\alpha} = 1/\hat{\kappa} = 0.954$ , gives -9.327 against the -9.353 of the quasi model with  $V(\mu) = \mu^2$ : at these fitted means the quadratic term of the NB2 variance dominates, so the two nearly coincide.

The tests agree on the conclusion and not on its strength. The drop in deviance is 350.9, overwhelming as  $\chi^2(1)$ ; the quasi- $F$  statistic [Eq. 38.3.3](#) is 8.80 on (1, 60) degrees of freedom,  $p = 0.0043$ , and the score test for overdispersion [Eq. 37.4.8](#) is 224.5. Unanticipated production predicts shorter strikes, at about the evidence a  $t$  statistic of three provides; the Poisson analysis overstates that by orders of magnitude.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.strikes.load_pandas().data
y = data["duration"].to_numpy(float)      # length of the strike, in days
x = data["iprod"].to_numpy(float)         # unanticipated industrial production
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape

def quasi_irls(X, y, V, steps=60):
    """Solve the quasi-score equations for a log link and variance function V."""
    beta = np.zeros(X.shape[1])
    beta[0] = np.log(y.mean())
    for _ in range(steps):
        eta = X @ beta
        mu = np.exp(eta)
        W = mu ** 2 / V(mu)                # working weight (dmu/deta)^2 / V(mu)
        z = eta + (y - mu) / mu            # working response
        beta = np.linalg.solve((X.T * W) @ X, (X.T * W) @ z)
    return beta
```

```

beta_lin = quasi_irls(X, y, lambda m: m)          #  $V(\mu) = \mu$ 
mu = np.exp(X @ beta_lin)
deviance = 2 * np.sum(y * np.log(y / mu) - (y - mu))    # quasi-deviance,  $V(\mu)$ 
                ) = mu
pearson = np.sum((y - mu) ** 2 / mu)                  #  $X^2$  with  $V(\mu) = \mu$ 
phi_pearson = pearson / (n - p)
beta_null = quasi_irls(X[:, :1], y, lambda m: m)
mu_null = np.exp(X[:, :1] @ beta_null)
deviance_null = 2 * np.sum(y * np.log(y / mu_null) - (y - mu_null))
naive_chisq = deviance_null - deviance                # the Poisson likelihood
                ratio
quasi_f = (deviance_null - deviance) / phi_pearson    # referred to  $F(1, n - p)$ 
print(f"naive chi-squared {naive_chisq:.1f}, p = {stats.chi2.sf(naive_chisq, 1):.2e}")
print(f"quasi F {quasi_f:.2f} on (1, {n - p}) df, p = {stats.f.sf(quasi_f, 1, n - p):.4f}")

```

*Idea.* The quasi-score does not know about the variance at all ([Theorem 38.2.2\(b\)](#)), so a wrong variance costs efficiency and calibration but not consistency. Whenever a colleague reports that “adding a dispersion parameter did not change the results”, the coefficients were being looked at and the standard errors were not.

## 38.4.4 Exercises

### 38.4.4.1 A. Check your understanding

#### Exercise A1.

Show that for fixed  $\phi_0 > 1$  the size [Eq. 38.4.2](#) tends to 1 as  $q \rightarrow \infty$ . (Use  $\chi^2(q)/q \rightarrow 1$  in probability, together with  $\chi^2_{1-\alpha}(q)/q \rightarrow 1$ , and compare with the event  $\chi^2(q)/q > \chi^2_{1-\alpha}(q)/(q\phi_0)$ .)

*Solution.* By the law of large numbers  $\chi^2(q)/q \rightarrow 1$  in probability, and the same concentration gives  $\chi^2_{1-\alpha}(q)/q = c_q \rightarrow 1$ . The event in [Eq. 38.4.2](#) is  $\{\chi^2(q)/q > c_q/\phi_0\}$ , whose threshold tends to  $\phi_0^{-1} < 1$ , so its probability tends to 1; at  $q = 1$  it is the finite  $\Pr\{\chi^2(1) > 3.841/\phi_0\}$ . The listing of this section shows the size increasing with  $q$  over the range tabulated, but that is a computation, not a proof.

#### Exercise A2.

A Poisson regression of accident counts on three covariates gives  $X^2/(n-p) = 3.2$ . Give three distinct explanations, and for each one say what you would compute next.

### 38.4.4.2 B. Practice

#### Exercise B1.

In [Proposition 38.4.1\(c\)](#), let  $S$  be geometric on  $\{1, 2, \dots\}$  with mean  $1/\theta$ . Compute  $\text{Var } Y / \text{E } Y$  and show that it equals  $(2 - \theta)/\theta$ . For which  $\theta$  does the compound Poisson

have the variance-to-mean ratio of a negative binomial with  $\kappa = \mu$ ?

### Exercise B2.

Grouped binomial data have group sizes  $m_i$  spread between 2 and 50. Show that under the constant-dispersion specification the Pearson terms  $m_i(y_i - \hat{\mu}_i)^2 / \{\hat{\mu}_i(1 - \hat{\mu}_i)\}$  have expectation  $\phi$  free of  $m_i$ , while under [Eq. 38.1.3](#) they have expectation  $1 + \rho(m_i - 1)$ . Describe the plot this suggests and what each specification predicts for it.

*Solution.* Under  $\text{Var}(Y_i) = \phi\mu_i(1 - \mu_i)/m_i$  the Pearson term at the true mean has expectation  $\phi$ ; under [Eq. 38.1.3](#) the same calculation gives  $1 + \rho(m_i - 1)$ , linear in  $m_i$  with slope  $\rho$ . So plot squared Pearson residuals against  $m_i$  with a smooth: flat supports a constant dispersion, increasing the exchangeable form.

### Exercise B3.

Both the NB1 model and the quasi-Poisson specification have  $\text{Var}(Y_i) = \phi\mu_i$ . Using [Eq. 37.4.6](#) and [Eq. 38.1.1](#), explain why a single very large count influences the two fits differently, and predict which of the two coefficients in [Example \(Four analyses of the strike data\)](#) would move more if the longest strike were deleted.

*Solution.* In the quasi-Poisson equation the log link cancels  $\mu_i$  against  $V(\mu_i)$ , so observation  $i$  enters through the raw residual  $y_i - \mu_i$ : the longest strike, a count of 216 against a fitted mean of 58.99, contributes 157.0. The NB1 score [Eq. 37.4.6](#) contains  $y_i$  only inside  $\psi(y_i + \alpha\mu_i)$ , which grows like  $\log y_i$ , so the NB1 fit is far less sensitive to it and the quasi-Poisson coefficient should move more when it is removed.

## 38.5 Summary and Notes

### 38.5.1 Summary

*Idea.*

1. A **variance specification** ([Definition 38.1.1](#)) gives the mean and the variance of each response and nothing else:  $EY_i = h(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$  and  $\text{Var}(Y_i) = \phi V(\mu_i)/w_i$ , with  $\phi$  free even for a variance function borrowed from a family that fixed it. No distribution need exist with these moments, and for ungrouped binary data none does with  $\phi \neq 1$ . Overdispersion of a group proportion has two shapes, a constant multiple of the binomial variance and the factor  $1 + \rho(m - 1)$  of exchangeable trials ([Lemma 38.1.1](#)), agreeing at one group size only.
2. A variance function fixes the estimating equations, not the fit: the NB1 model shares the quasi-Poisson variance function and gives a different  $\hat{\boldsymbol{\beta}}$  ([Example \(Four analyses of the strike data\)](#)).
3. The **quasi-likelihood**  $Q(\mu; y) = w\phi^{-1} \int_y^\mu (y - t)V(t)^{-1}dt$  of [Definition 38.2.1](#) has the score, the two Bartlett identities and the saturated maximum of a log-

- likelihood, and equals the log-likelihood minus its saturated value for an exponential dispersion family ([Proposition 38.2.1](#)); its  $-2\phi$  multiple is the **quasi-deviance**.
4. The **quasi-score** is the score [Eq. 34.3.1](#), so the estimate is the usual iteratively reweighted least squares fit ([Theorem 38.2.2\(a\)](#)), unbiased as soon as the mean model is right whatever the variances; the identity between its variance and its sensitivity needs the variance model too ([Theorem 38.2.2\(b\)–\(e\)](#)).
  5. Among **linear estimating functions**  $\mathbf{C}(\boldsymbol{\beta})(\mathbf{y} - \boldsymbol{\mu})$  the quasi-score maximizes the Godambe information, and nothing else does except its nonsingular linear transformations ([Proposition 38.2.3](#)). The proof is Gauss–Markov with an orthogonal projection in place of the completion of the square.
  6. Under (Q1)–(Q5),  $\hat{\boldsymbol{\beta}}$  is asymptotically normal with the **sandwich** covariance  $\mathbf{A}_n^{-1} \mathbf{B}_n \mathbf{A}_n^{-1}$ , whose middle is estimated from the residuals ([Theorem 38.3.1](#)); a correct variance specification makes  $\mathbf{B}_n = \mathbf{A}_n$  and collapses it to the model-based  $\phi(\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1}$ , then valid and more efficient.
  7. The dispersion is estimated by  $X^2/(n - p)$ , of expectation exactly  $\phi$  at the true mean, while the deviance version is not in general consistent ([Proposition 38.3.2\(a\),\(b\)](#)). Nested models are compared by the quasi- $F$  statistic [Eq. 38.3.3](#), which needs a correct variance function; the Akaike criterion has no analogue here beyond a convention ([Proposition 38.3.2\(c\),\(d\)](#)).
  8. Overdispersion comes from unobserved heterogeneity, clustering and contagion, each computable exactly ([Proposition 38.4.1\(a\)–\(c\)](#)). Ignoring a dispersion  $\phi_0$  turns a nominal 5% test of one coefficient into one of size  $\Pr\{\chi^2(1) > 3.841/\phi_0\}$ , and of six coefficients into something far worse ([Eq. 38.4.2](#)). The repairs assume different things — a mean and a variance shape, a full distribution, a mixing law — and the last changes the estimand.

## 38.5.2 Notes and sources

**Wedderburn’s paper.** Quasi-likelihood is due to Wedderburn (1974), written two years after the generalized linear model itself (Nelder and Wedderburn 1972) and in the same spirit: notice which part of the structure the algorithm uses, and keep only that. The definition [Eq. 38.2.1](#), the two Bartlett identities and the table of quasi-likelihoods are all in that paper, as is the name. McCullagh and Nelder (1989, chapter 9) is the standard extended treatment; their chapter 10 develops the joint model for the mean and the dispersion, which this chapter does not cover. Multiplying Poisson or binomial standard errors by  $\sqrt{X^2/(n - p)}$  predates the theory — Finney (1947) used it for probit analysis — and Wedderburn said what it estimates.

**Estimating functions and optimality.** The general theory of unbiased estimating functions is Godambe (1960), who showed the likelihood score optimal among them in the sense used in [Proposition 38.2.3](#); Godambe and Heyde (1987) survey the subject. The projection proof given here makes the analogy exact: “optimal among linear estimating functions” stands to “optimal among all” as “best linear unbiased” stands to “minimum variance unbiased” in [Theorem 7.2.1](#). McCullagh (1983) proved the asymptotics under conditions close to (Q1)–(Q5).

**The sandwich.** Its history is traced in [Section 21.4](#): Eicker (1963), Huber (1967) and White (1980) in three literatures at once. The generalized-linear-model version, with the working weights in the bread, is in Royall (1986), who argued for it as the default; Liang and Zeger (1986) made it the foundation of generalized estimating equations. Its small-sample bias is analysed by Kauermann and Carroll (2001), and the leverage corrections of [Example \(Coverage](#)

when the variance function is right and when it is wrong) are the HC2 and HC3 of MacKinnon and White (1985).

**Binomial overdispersion.** The exchangeable-trials calculation Eq. 38.1.3 is the design effect of cluster sampling (Kish 1965), which Chapter 33 met as  $1 + (m - 1)\rho$ . The beta-binomial distribution is due to Skellam (1948), and Williams (1982) proposed fitting the variance function  $\{1 + \rho(m_i - 1)\}\mu(1 - \mu)/m_i$  by alternating a quasi-likelihood step with a moment equation setting  $X^2$  equal to its degrees of freedom.

**Which device to use.** Ver Hoef and Boveng (2007) is a clear applied statement of the quasi-Poisson against negative binomial question, showing that the choice is between the NB1 and NB2 variance functions and matters most where the fitted means are most spread out — what Example (Four analyses of the strike data) shows in miniature. Cameron and Trivedi (2013) covers the quasi-likelihood, negative binomial and sandwich routes together. The “QAIC” convention of Proposition 38.3.2(d) is from Lebreton, Burnham, Clobert and Anderson (1992), widely used in ecology and resting on no theorem.

**The data.** The strike durations of Example (Strike durations) are the subset of Kennan’s (1985) contract-strike series distributed with statsmodels, from Bureau of Labor Statistics records. Kennan fitted duration models of a different kind; the Poisson mean model here illustrates the methods of this chapter and is not an account of strike durations.

**What comes next.** Chapter 39 returns to models with likelihoods, taking up model building, diagnostics and the Bayesian treatment. Chapter 40 extends the estimating-equation idea to correlated responses, where the working variance becomes a working *correlation matrix*. Chapter 41 asks what happens when some responses are not observed at all.

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# Chapter 39

## GLMs in Practice; the Bayesian Thread

Chapters 34 to 38 built the generalized linear model and its inference. This chapter asks two practical questions the theory leaves open. The first is how one arrives at a particular model at all: which family, which link, which scale for the covariates, which interactions, and how candidates are compared. The second is how the model is criticized once fitted, since it has three separable pieces — a mean, a link and a variance function — and a plot that condemns the whole fit is much less useful than one naming the piece at fault.

The rest of the chapter takes up the Bayesian thread, running through the book since [Section 7.6](#). For the normal linear model the posterior was available in closed form and the flat prior reproduced least squares exactly. None of that survives the move to a generalized linear model, for one identifiable reason: the log-likelihood is no longer a quadratic. What replaces it is approximation — the Laplace expansion, which also explains where the BIC comes from — and simulation, in the form of Metropolis–Hastings and Gibbs samplers written out in numpy. The chapter ends where the Bayesian and penalized-likelihood threads meet: a normal prior is ridge regression, a Laplace prior the lasso at its mode, and a merely weakly informative prior is enough to make an estimate exist where maximum likelihood has none.

**What you need.** [Chapter 34](#) throughout ([Definition 34.2.1](#), [Definition 34.2.2](#), [Definition 34.1.2](#), [Theorem 34.3.1](#), [Theorem 34.4.1](#), [Definition 34.5.1](#), [Theorem 34.5.5](#), [Theorem 34.5.1](#), [Definition 34.5.2](#)); [Chapter 35](#) for separation ([Theorem 35.3.1](#)), Firth’s penalty ([Proposition 35.3.2](#)) and the deviance for binary data ([Proposition 35.4.1](#)); [Chapter 37](#) for offsets ([Proposition 37.2.1](#)); [Chapter 38](#) for dispersion ([Proposition 38.3.2](#)) and overdispersion ([Proposition 38.4.1](#)). From earlier parts: [Chapter 20](#) ([Definition 20.4.1](#), [Theorem 20.3.1](#), [Proposition 20.1.3](#)); [Chapter 22](#) for [Proposition 22.5.1](#) and [Theorem 22.3.2](#); the parametric bootstrap of [Proposition 23.3.3](#); [Chapter 29](#) ([Definition 29.2.2](#), [Proposition 29.2.4](#), [Definition 29.3.1](#), [Proposition 29.5.1](#)); and penalties as priors ([Chapter 27](#), [Chapter 30](#): [Definition 27.2.1](#), [Definition 27.4.1](#), [Proposition 30.1.4](#)). The Bayesian thread so far: [Section 7.6](#) for [Theorem 7.6.2](#) and [Corollary 7.6.4](#), [Section 12.6](#) for [Definition 12.6.1](#), and [Section 32.6](#) for [Proposition 32.6.1](#).

### 39.0.1 Roadmap

- [Building a model](#)
- [Diagnostics](#)
- [Bayesian thread IV: what carries over](#)
- [Bayesian generalized linear models and MCMC](#)



- [Priors as regularizers](#)
- [Summary and notes](#)

## 39.1 Building a model

A generalized linear model has three parts that can be chosen separately: a response family, which fixes the variance function  $V(\mu)$ ; a link  $g$ ; and a linear predictor  $\eta = \mathbf{x}^\top \boldsymbol{\beta}$ . [Chapter 34](#) fits whichever combination is handed to it; this section is about how the combination is chosen, and how honest one can be afterwards.

### 39.1.1 The family follows the variance

Two things are known about a response before any model is fitted: what values it can take, and how its spread changes with its level. The first rules out most families at once — a count is not negative, a proportion is bounded, a duration is positive. The second separates those that remain, because within the exponential dispersion family the variance function determines the family ([Exercise \(B2\)](#)). For the families of [Definition 34.1.2](#) the variance is a power of the mean,

$$\text{Var}(Y_i) = \phi \mu_i^\zeta / w_i, \quad (39.1.1)$$

with  $\zeta = 0$  for the normal, 1 for the Poisson, 2 for the gamma and 3 for the inverse Gaussian; in between are the Tweedie families of [Exercise \(C2\)](#). So “which family?” is largely the question “which  $\zeta$ ?”, and with replicates it can be answered before any model is fitted. A second choice arises at once: the scale on which the systematic part is additive. Transforming the response and transforming the mean are different operations ([Exercise \(B1\)](#)), and models on different response scales are not comparable by their likelihoods unless the change of variable is accounted for. The proposition collects the three facts that make these choices more than a matter of taste.

#### Proposition 39.1.1 — Three facts about building a generalized linear model

- (*Reading the variance function off replicates.*) Suppose the observations fall into  $G \geq 2$  groups, group  $g$  holding  $m_g \geq 2$  independent replicates with mean  $\mu_g > 0$ , variance  $\phi \mu_g^\zeta$  and finite fourth moment. Let  $\bar{y}_g$  be the mean of the replicates in group  $g$ , let  $s_g^2$  be their variance with divisor  $m_g - 1$ , and let  $\hat{\zeta}$  be the slope of the weighted least squares regression of  $\log s_g^2$  on  $\log \bar{y}_g$  with any fixed positive weights  $c_g$ . If the  $\log \mu_g$  are not all equal, then  $\hat{\zeta} \rightarrow \zeta$  in probability as  $\min_g m_g \rightarrow \infty$  with  $G$ , the  $\mu_g$  and  $\phi$  fixed.
- (*Additivity is a property of the link, not of the data.*) Let  $g(\mu(x_1, x_2)) = a(x_1) + b(x_2)$  on  $I_1 \times I_2$ , with  $a$  and  $b$  continuous and nonconstant, and let  $J$  be the interval of values taken by  $a + b$ . Let  $\tilde{g}$  be a second link and  $s = \tilde{g} \circ g^{-1}$ , twice differentiable on  $J$ . Then  $\tilde{g}(\mu(x_1, x_2))$  is again a sum of a function of  $x_1$  and a function of  $x_2$  if and only if  $s$  is affine on  $J$ .
- (*Comparing models across response scales.*) Let  $t$  be strictly increasing and continuously differentiable with  $t' > 0$ , and let a model for the transformed responses

$\tilde{y}_i = t(y_i)$  have maximized log-likelihood  $\tilde{\ell}$  and  $d$  estimated parameters. As a model for  $\mathbf{y}$  it has maximized log-likelihood  $\tilde{\ell} + \sum_i \log t'(y_i)$ , so its AIC on the scale of  $\mathbf{y}$  is

$$\text{AIC} = -2\tilde{\ell} + 2d - 2 \sum_{i=1}^n \log t'(y_i). \quad (39.1.2)$$

*Proof.*

- (a) With  $m_g$  replicates of finite fourth moment,  $\bar{y}_g \rightarrow \mu_g$  and  $s_g^2 \rightarrow \phi \mu_g^\zeta$  in probability as  $m_g \rightarrow \infty$ , the second because  $\text{Var}(s_g^2) \rightarrow 0$  when the fourth moment is finite. Both limits are positive, and log is continuous there, so  $L_g = \log \bar{y}_g \rightarrow \log \mu_g$  and  $V_g = \log s_g^2 \rightarrow \log \phi + \zeta \log \mu_g$  in probability. The weighted slope

$$\hat{\zeta} = \frac{\sum_g c_g (L_g - \bar{L}_c)(V_g - \bar{V}_c)}{\sum_g c_g (L_g - \bar{L}_c)^2}, \quad \bar{L}_c = \frac{\sum_g c_g L_g}{\sum_g c_g},$$

is a continuous function of  $(L_1, V_1, \dots, L_G, V_G)$  wherever the denominator is nonzero. At the limit point the denominator is  $\sum_g c_g (\log \mu_g - \overline{\log \mu_c})^2 > 0$ , because the  $\log \mu_g$  are not all equal, and the numerator is  $\zeta$  times the same quantity. The continuous mapping theorem gives  $\hat{\zeta} \rightarrow \zeta$ .

- (b) If  $s(u) = \alpha + \beta u$  then  $\tilde{g}(\mu) = \alpha + \beta a(x_1) + \beta b(x_2)$ , a sum. Conversely suppose  $\tilde{g}(\mu(x_1, x_2)) = A(x_1) + B(x_2)$ . If  $a(x_1) = a(x'_1)$  then  $\mu(x_1, x_2) = \mu(x'_1, x_2)$  for every  $x_2$ , so  $A(x_1) = A(x'_1)$ ; hence  $A^*(u) = A(x_1)$  is well defined on the range  $U$  of  $a$ , and likewise  $B^*$  on the range  $V$  of  $b$ . Both are nondegenerate intervals, being continuous images of intervals under nonconstant maps, and

$$s(u + v) = A^*(u) + B^*(v), \quad u \in U, v \in V.$$

Fix  $u_0, v_0$  interior to  $U, V$ . For any other such  $u, v$  the four terms cancel:

$$s(u + v) - s(u_0 + v) - s(u + v_0) + s(u_0 + v_0) = 0.$$

Differentiating in  $u$  and then in  $v$  gives  $s''(u + v) = 0$ . As  $u + v$  ranges over an open subinterval of  $J$  whose closure is  $J$ , and  $s''$  is continuous,  $s'' \equiv 0$ :  $s$  is affine.

- (c) If  $\tilde{Y} = t(Y)$  has density  $f_{\tilde{Y}}$ , then  $Y$  has density  $f_Y(y) = f_{\tilde{Y}}(t(y)) t'(y)$ , because  $t$  is a strictly increasing differentiable bijection onto its image. Summing logs over independent observations,  $\ell(\cdot; \mathbf{y}) = \tilde{\ell}(\cdot; \tilde{\mathbf{y}}) + \sum_i \log t'(y_i)$  for every parameter value, so the maximizers coincide and the maximized values differ by the stated constant. Substituting into  $\text{AIC} = -2\ell + 2d$  gives [Eq. 39.1.2](#). ■

Part (c) is worth stating because the omitted term is enormous. The AIC that software prints for a normal linear model fitted to  $\log \mathbf{y}$  compares it with other models *for*  $\log \mathbf{y}$ ; against a gamma model for  $\mathbf{y}$  the comparison is meaningless until  $2 \sum_i \log y_i$ , a sum of order  $n$ , is added back.

*Warning.* The same applies to cross-validation, to the deviance and to any other criterion built on a likelihood. A predictive score is a score *for a particular quantity on a particular scale*; predicting  $\log y$  well and predicting  $y$  well are different achievements, and the

second is usually what was wanted.

## 39.1.2 A worked choice

### Example. (How long does a strike last)

The durations of 62 contract strikes in United States manufacturing, from one day to 216, are recorded with a measure of unanticipated industrial production at the time. [Example \(Strike durations\)](#) read the same durations as counts and asked only for a variance function; here they are read as what they are, positive and continuous, and the question is which family to use. The response is strongly right-skewed, so a normal linear model is out; which of the positive families fits the mean–variance relationship?

The covariate takes only 9 distinct values, so the data arrive already in replicate groups, of sizes 2 to 18. The weighted regression of [Proposition 39.1.1\(a\)](#) gives  $\hat{\zeta} = 1.606$  with a standard error of 0.434, computed from the exact normal-theory variance  $\text{Var}(\log s_g^2) = \psi'\{(m_g - 1)/2\}$  — the familiar  $2/(m_g - 1)$  is only its large-group limit, and understates it badly for the two groups of size two. Panel (a) of [Figure 39.1.1](#) shows the nine points with the fitted line and the Poisson and gamma references:  $\zeta = 2$  is 0.9 standard errors away and  $\zeta = 1$  is 1.4, which is weak evidence for the gamma and against anything Poisson-like.

Fit three models with a log link and the same linear predictor, maximizing each over its dispersion parameter as well as over  $\beta$ , and count  $d = 3$  for all three. AIC on the scale of the durations is 585.4 for the gamma, 590.9 for the log-normal and 603.3 for the inverse Gaussian. The log-normal figure is the AIC of the least squares fit to  $\log \mathbf{y}$ , 206.8, plus the Jacobian term  $2 \sum_i \log y_i = 384.1$  of [Eq. 39.1.2](#). Without that term the log-normal appears to beat the gamma by 378.7: not a close call, and in the wrong direction, because the uncorrected number answers a different question. Leave-one-out cross-validation with the log score puts the three in the same order, at  $-4.7192$ ,  $-4.7713$  and  $-4.8664$  per observation.

The gamma slope is to the last decimal the estimate that [Example \(Strike durations\)](#) obtained from  $V(\mu) = \mu^2$  alone: the variance function fixes the estimating equations, and choosing a family on top of it buys a likelihood, not a different fit. The maximum likelihood shape of the gamma fit is  $\hat{\nu} = 1.003$ , so the durations are close to exponential given the covariate, which is what  $\zeta = 2$  with a coefficient of variation near one means. (Software reports instead the moment estimate from the Pearson dispersion  $\hat{\phi} = 0.941$ , namely 1.063; the two agree here, and for the inverse Gaussian they do not.) The fits also disagree about what they estimate: at the mean covariate value the gamma model puts the *mean* duration at 39.4 days, while the log-normal model puts the *median* at 22.2 and, after the retransformation correction of [Proposition 22.5.1](#), the mean at 46.7.

```
import numpy as np
import statsmodels.api as sm
strikes = sm.datasets.strikes.load_pandas().data      # 62 contract strikes, public
    domain
y = strikes["duration"].to_numpy(float)              # days
x = strikes["iprod"].to_numpy(float)                 # unanticipated industrial
    production
```

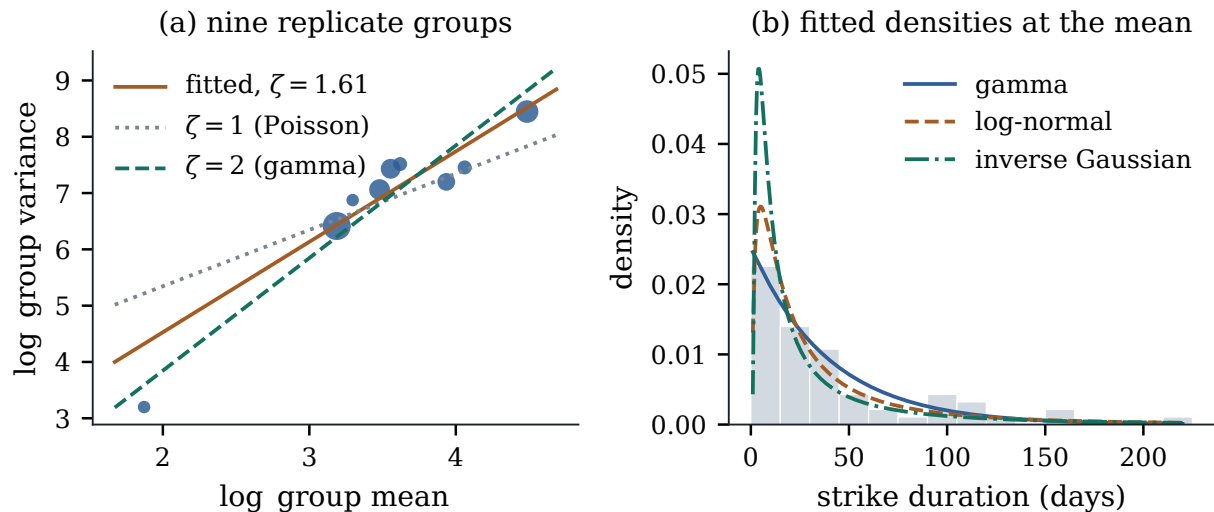


Figure 39.1.1: Choosing a family for the strike durations. (a) Log group variance against log group mean, point size proportional to group size; the fitted slope estimates  $\zeta$  in Eq. 39.1.1, and the reference lines have the Poisson and gamma slopes. (b) The three fitted densities at the mean covariate value, over a histogram of the durations.

```
groups = strikes.groupby("iproduct")["duration"].agg(["count", "mean", "var"])
groups = groups[groups["count"] >= 2]
w = groups["count"].to_numpy() - 1.0
L = np.column_stack([np.ones(len(groups)), np.log(groups["mean"])]])
zeta_fit = np.linalg.solve(L.T @ (w[:, None] * L), L.T @ (w * np.log(groups["var"])))

print(groups.round(2))
print(f"variance-function power zeta = {zeta_fit[1]:.3f}")
```

```
from scipy import optimize, special, stats
X = np.column_stack([np.ones(len(y)), x])
n, p = X.shape
d = p + 1
gam = sm.GLM(y, X, family=sm.families.Gamma(sm.families.links.Log())).fit()
igauss = sm.GLM(y, X, family=sm.families.InverseGaussian(sm.families.links.Log())).fit()
lognorm = sm.OLS(np.log(y), X).fit()

def gamma_loglik(yy, mu, nu):
    return np.sum(stats.gamma.logpdf(yy, nu, scale=mu / nu))

def gamma_shape(yy, mu):
    """Maximum likelihood shape: the root of log nu - digamma(nu) + mean log(y/mu)."""
    c = np.mean(np.log(yy / mu))
    return optimize.brentq(lambda v: np.log(v) - special.digamma(v) + c, 1e-4, 1e4)
```

```

def igauss_lambda(yy, mu):
    """Maximum likelihood precision of the inverse Gaussian, in closed form."""
    return 1.0 / np.mean((yy - mu) ** 2 / (mu**2 * yy))

nu_hat = gamma_shape(y, gam.fittedvalues)
lam_hat = igauss_lambda(y, igauss.fittedvalues)
ll_gam = gamma_loglik(y, gam.fittedvalues, nu_hat)
ll_ig = np.sum(stats.invgauss.logpdf(y, igauss.fittedvalues / lam_hat, scale=lam_hat
    ))
sigma_ml = np.sqrt(lognorm.ssr / n) # the maximizing sigma,
    divisor n
ll_ln = np.sum(stats.norm.logpdf(np.log(y), lognorm.fittedvalues, sigma_ml) - np.log
    (y))

for name, ll in [("gamma, log link", ll_gam), ("inverse Gaussian", ll_ig),
    ("log-normal (on the y scale)", ll_ln)]:
    print(f"{name:28s} loglik {ll:9.2f} AIC {-2 * ll + 2 * d:9.2f}")
print(f"gamma: slope {gam.params[1]:8.3f}, maximum likelihood shape {nu_hat:.3f}")

```

### 39.1.3 Links, scales and interactions

With the family fixed, the link decides what “additive” means. The canonical link ([Definition 34.2.2](#)) keeps the log-likelihood concave ([Theorem 34.3.3](#)) and makes observed and expected information agree ([Theorem 34.3.1\(d\)](#)); neither is a reason to prefer it when its range constrains  $\beta$ . The gamma’s canonical link is the reciprocal, which forces  $\mathbf{x}^\top \beta > 0$  and makes every coefficient a statement about  $1/\mu$ ; the log link has no constraint, keeps the concavity ([Exercise \(C1\)](#)), and makes each coefficient a multiplicative effect on the mean. Almost all gamma regressions use the log link, for interpretation rather than fit.

For the covariates, [Theorem 22.3.2](#) transfers verbatim with the linear predictor in place of the mean: to ask whether  $x_j$  should enter as  $x_j^\lambda$ , add  $x_j \log x_j$  and test its coefficient. Centring matters for a different reason. Under a nonlinear link the intercept is the linear predictor at  $\mathbf{x} = \mathbf{0}$ , a point no observation resembles when the covariates are uncentred; centring each at a representative value makes  $e^{\hat{\beta}_0}$  the fitted mean of a typical case.

Interactions are where the link stops being cosmetic. A two-factor Poisson model with a log link and no interaction says the factors multiply; with an identity link it says they add; by [Proposition 39.1.1\(b\)](#) at most one of these is the additive model of [Definition 16.2.1](#). “No interaction” is therefore never a property of the response alone, and removing an interaction by changing the link is a legitimate simplification, provided the link is reported as part of the finding and was not chosen by looking at the interaction test ([Proposition 29.5.1](#)). On the log scale an interaction coefficient is a ratio of ratios: with binary  $x_1, x_2$  and  $\log \mu = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_{12} x_1 x_2$ ,  $e^{\beta_{12}}$  is the factor by which the effect of  $x_1$  changes when  $x_2$  switches on; on the logit scale it is a ratio of odds ratios, harder to think about because odds ratios do not collapse over omitted covariates ([Example \(The odds ratio does not collapse\)](#)). Offsets need no new theory ([Proposition 37.2.1](#)): a known exposure enters with coefficient fixed at one, and the model describes a rate.

### 39.1.4 Comparing and choosing

Because a generalized linear model is fitted by maximum likelihood, everything in [Chapter 29](#) applies: AIC is  $-2\ell(\hat{\beta}) + 2d$  with  $d = p$  when  $\phi$  is known and  $p + 1$  when it is estimated, BIC replaces 2 by  $\log n$  ([Definition 29.2.2](#)), and their different targets ([Proposition 29.2.4](#)) are unchanged. Two cautions are specific here. First, by [Proposition 39.1.1\(c\)](#) the likelihoods compared must be for the same random variable, and each must be maximized over the dispersion as well as over  $\beta$ . Second, a quasi-likelihood model has no likelihood, so no AIC; the substitute is the quasi-likelihood information criterion built from the independence-model quasi-likelihood and a sandwich penalty, which [Chapter 40](#) defines. The dispersion estimate is not a likelihood quantity either ([Proposition 38.3.2](#)), which is why an overdispersed fit is compared by quasi- $F$  statistics.

Cross-validation needs a loss, and the natural one is the log score: hold out an observation, fit to the rest, record the predictive log density of what was held out. That is [Definition 29.3.1](#) with the deviance as loss, so [Proposition 29.3.1](#) applies. For a  $\phi = 1$  family the log score and the deviance rank models identically; with a dispersion parameter they need not, because the deviance ignores the term in  $\phi$ .

*Idea.* Three questions, three tools. *Which family?* The support and the mean–variance relationship. *Which link?* What should be additive, checked afterwards ([Section 39.2](#)). *Which terms?* A criterion, plus honesty about what the search cost.

### 39.1.5 What the search costs

Every warning of [Section 29.5](#) carries over unchanged, because none used normality: after a search the standard errors are too small, the  $p$ -values too small and the intervals too short ([Proposition 29.5.1](#)). Two remarks are specific here. A drop in deviance is approximately  $\chi^2(q)$  under the smaller model ([Theorem 34.5.5](#)), so a maximum over many candidate terms is not, and the largest always looks significant; and with small  $n$  or sparse cells that approximation is itself poor ([Example \(The Hauck–Donner effect\)](#), [Proposition 35.4.1](#)), so the nominal levels being abused were never right to begin with. Sample splitting ([Proposition 29.5.2](#)) and simultaneous inference over the whole search ([Proposition 29.5.3](#)) are the honest ways out.

### 39.1.6 Exercises

#### 39.1.6.1 A. Check your understanding

##### Exercise A1.

A colleague fits a normal linear model to  $\sqrt{y}$  and a Poisson log-linear model to  $y$ , and compares the two AIC values printed by the software. What is the correction term of [Proposition 39.1.1\(c\)](#) here, and in which direction does omitting it push the comparison?

*Solution.* With  $t(y) = \sqrt{y}$ ,  $t'(y) = 1/(2\sqrt{y})$ , so the correction term is

$$-2 \sum_i \log t'(y_i) = 2 \sum_i \log(2\sqrt{y_i}) = 2n \log 2 + \sum_i \log y_i,$$

and the AIC of the square-root model on the scale of  $\mathbf{y}$  is its printed value plus that amount. It is positive whenever  $\prod_i 4y_i > 1$ , as for counts of any reasonable size it is, so the printed comparison flatters the square-root model. The Poisson model needs no correction, being already a model for  $\mathbf{y}$  — though it is a mass function rather than a density, which makes the comparison dubious anyway.

### Exercise A2.

A two-by-two table of counts has fitted means  $\mu_{11} = 10$ ,  $\mu_{12} = 20$ ,  $\mu_{21} = 15$  and  $\mu_{22} = 30$ . Is there an interaction on the log scale? On the identity scale? Reconcile the two answers with [Proposition 39.1.1\(b\)](#).

*Solution.* On the log scale,  $\log \mu_{11} + \log \mu_{22} - \log \mu_{12} - \log \mu_{21} = \log(10 \cdot 30 / (20 \cdot 15)) = \log 1 = 0$ : no interaction. On the identity scale,  $10 + 30 - 20 - 15 = 5 \neq 0$ : there is one. The two links differ by the map  $s = \exp$ , which is not affine, so [Proposition 39.1.1\(b\)](#) says there is no reason for both to be additive, and here only one is.

### 39.1.6.2 B. Practice

#### Exercise B1.

The canonical link of the gamma family is  $g(\mu) = -1/\mu$  ([Example \(The gamma and its constant coefficient of variation\)](#)). Show that with this link and a model matrix containing an intercept, the parameter space  $\{\boldsymbol{\beta} : \mathbf{x}_{(i)}^\top \boldsymbol{\beta} < 0 \text{ for all } i\}$  is an open convex cone, and that the maximum likelihood estimate may fail to exist by running to its boundary. Contrast with the log link.

*Solution.* The set is an intersection of  $n$  open half-spaces, hence open and convex, and a cone because each constraint is homogeneous. The log-likelihood tends to  $-\infty$  as any  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta} \rightarrow 0^-$ , since  $\mu_i \rightarrow \infty$ ; but the supremum may be approached only as  $\|\boldsymbol{\beta}\| \rightarrow \infty$  inside the cone, in which case no maximizer exists, exactly as in [Theorem 35.3.1](#). With the log link the parameter space is all of  $\mathbb{R}^p$  and the log-likelihood is concave ([Exercise \(C1\)](#)), so the only obstacle is the analogue of separation.

#### Exercise B2.

Show that for a family with  $\phi = 1$  the leave-one-out log score and the leave-one-out deviance differ by a constant that does not depend on the model, so the two rank models identically. Show that this fails when  $\phi$  is estimated.

### 39.1.6.3 C. Going deeper

#### Exercise C1.

Extend [Proposition 39.1.1\(b\)](#) to three covariates: if  $g(\mu) = a_1(x_1) + a_2(x_2) + a_3(x_3)$  with all three nonconstant and continuous, and  $\tilde{g}(\mu)$  is also a sum of three functions of the separate covariates, must  $s = \tilde{g} \circ g^{-1}$  be affine? Does the conclusion change if only two of the three are required to separate?



**Exercise C2.**

Let the true link belong to the power family  $g_\lambda(\mu) = (\mu^\lambda - 1)/\lambda$ , which has  $g_0 = \log$ , and let the model actually fitted use the log link. Following the constructed-variable argument of [Proposition 22.2.4](#), expand  $g_\lambda$  about  $\lambda = 0$  to show that a one degree of freedom test of  $\lambda = 0$  is obtained by adding a multiple of  $\hat{\eta}^2$  to the linear predictor, and identify the constructed variable and the multiple exactly.

## 39.2 Diagnostics

A fitted generalized linear model can be wrong in three ways: the linear predictor may omit a term or use a covariate on the wrong scale, the link may be wrong, or the variance function may be wrong. A single number — the deviance, say — cannot tell these apart, and for ungrouped binary data it cannot even tell that something is wrong ([Proposition 35.4.1\(b\)](#)). What follows is a set of plots and tests, each aimed at one of the three, and a worked case in which they give different answers.

### 39.2.1 Residuals in practice

The three residual types of [Definition 34.5.2](#) differ in what they standardize away. Their common first-order behaviour is the generalized linear version of [Eq. 20.1.2](#).

#### Proposition 39.2.1 — What the standard diagnostics measure

Let a generalized linear model be fitted by maximum likelihood, let  $\hat{\mathbf{W}} = \text{diag}\{w_i h'(\hat{\eta}_i)^2 / V(\hat{\mu}_i)\}$  be the working weights,  $\mathbf{H} = \hat{\mathbf{W}}^{1/2} \mathbf{X} (\mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X})^{-1} \mathbf{X}^\top \hat{\mathbf{W}}^{1/2}$  the weighted hat matrix and  $h_{ii}$  its diagonal.

- (a) (*Moments of the Pearson residuals.*) To first order in the expansion behind [Theorem 34.5.1](#),  $E(r_i^P) \approx 0$  and  $\text{Var}(r_i^P) \approx 1 - h_{ii}$ , and the standardized residual  $r_i^P / \sqrt{1 - h_{ii}}$  has approximate variance one. Moreover  $\sum_i h_{ii} = p$ , so  $h_{ii}$  averages  $p/n$ .
- (b) (*Level of a simulated envelope.*) Let  $|r|_{(1)} \leq \dots \leq |r|_{(n)}$  be the ordered absolute residuals, and let  $B$  independent data sets be generated from the fitted model and refitted, giving vectors of ordered absolute residuals whose  $k$ th entries are  $R_k^{(1)}, \dots, R_k^{(B)}$ . If the model is correct and the fitted parameters are the true ones, and the  $B + 1$  values  $|r|_{(k)}, R_k^{(1)}, \dots, R_k^{(B)}$  are exchangeable with no ties, then

$$\Pr\{|r|_{(k)} < \min_b R_k^{(b)} \text{ or } |r|_{(k)} > \max_b R_k^{(b)}\} = \frac{2}{B+1} \quad (39.2.1)$$

for each fixed  $k$ . The probability that *some*  $k$  leaves the band is larger than  $2/(B+1)$ , and no better than the union bound  $\min\{1, 2n/(B+1)\}$  without further argument.

- (c) (*Influence.*) The one-step deletion results of [Theorem 20.3.1](#) hold with  $\mathbf{X}$  replaced by  $\hat{\mathbf{W}}^{1/2} \mathbf{X}$  and  $\mathbf{y}$  by the working response  $\mathbf{z}$  of [Eq. 34.4.1](#), giving the Cook-type



measure  $C_i = (r_i^P)^2 h_{ii} / \{p(1 - h_{ii})^2\}$  of [Definition 20.4.1](#). The exact deletion identities do not hold, because removing a case changes the weights as well.

- (d) (*Checking the link.*) Let  $\hat{\eta}_i$  be the fitted linear predictor and refit the model with  $\hat{\eta}_i^2$  added as a covariate. Under the fitted link the extra coefficient is zero, so the deviance drop is asymptotically  $\chi^2(1)$ ; a significant drop says the link is wrong, not which link is right.
- (e) (*Checking a covariate: two plots.*) The partial residual for covariate  $j$  on the link scale is  $u_{ij} = \hat{\beta}_j x_{ij} + (y_i - \hat{\mu}_i)/h'(\hat{\eta}_i)$ . Its weighted least squares regression on  $[\mathbf{1}, \mathbf{x}_j]$  with weights  $\hat{\mathbf{W}}$  has slope  $\hat{\beta}_j$  exactly, so a smooth of  $u_{ij}$  against  $x_{ij}$  that departs from a straight line indicates a wrong scale for  $x_j$ , as in [Proposition 20.1.3](#). The added-variable plot is the other half of the pair: let  $\tilde{\mathbf{z}}$  and  $\tilde{\mathbf{x}}_j$  be the residuals of the working response  $\mathbf{z}$  and of  $\mathbf{x}_j$  after  $\hat{\mathbf{W}}$ -weighted regression on the remaining columns  $\mathbf{X}_{-j}$ . Then the  $\hat{\mathbf{W}}$ -weighted regression of  $\tilde{\mathbf{z}}$  on  $\tilde{\mathbf{x}}_j$  has slope exactly  $\hat{\beta}_j$ , so curvature or a single dominating point in that plot is a statement about  $\hat{\beta}_j$  itself.
- (f) (*Checking the variance function.*) If the assumed variance function is correct then  $E|r_i^P| \approx \sqrt{2/\pi}$  for every  $i$ , so a plot of  $|r_i^P|$  against  $\hat{\eta}_i$  should be level. A systematic trend indicates a variance function of the wrong shape; a level plot at the wrong height indicates a dispersion  $\phi \neq 1$ .

*Proof.*

- (a) The expansion used in [Theorem 34.5.1](#) gives, to first order,  $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} \approx (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu})$  and hence  $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} \approx \mathbf{D} \mathbf{X} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})$ , where  $\mathbf{D} = \text{diag}(d\mu_i/d\eta_i)$ . Put  $\mathbf{u} = \mathbf{W}^{1/2} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu})$ , whose entries are the Pearson residuals evaluated at the true parameter, with  $\text{Cov}(\mathbf{u}) = \mathbf{I}$  by the definition of  $\mathbf{W}$ . Then

$$\begin{aligned} \mathbf{W}^{1/2} \mathbf{D}^{-1} (\mathbf{y} - \hat{\boldsymbol{\mu}}) &\approx \mathbf{u} - \mathbf{W}^{1/2} \mathbf{X} (\mathbf{X}^\top \mathbf{W} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{W}^{1/2} \mathbf{u} \\ &= (\mathbf{I} - \mathbf{H}) \mathbf{u}, \end{aligned}$$

so  $E(r_i^P) \approx 0$  and  $\text{Var}(r_i^P) \approx (\mathbf{I} - \mathbf{H})_{ii} = 1 - h_{ii}$ . The trace identity is [Proposition 6.3.4](#) applied to  $\mathbf{H}$ , which is a projection matrix.

- (b) The event that  $|r|_{(k)}$  is the strict minimum or the strict maximum of the  $B + 1$  exchangeable values has probability  $2/(B + 1)$ , because each of the  $B + 1$  values is equally likely to be the smallest and equally likely to be the largest, and with no ties these two events are disjoint for  $B \geq 1$ . The final sentence is the usual multiplicity statement: the events for different  $k$  are highly dependent, so the probability that some  $k$  leaves the band lies between  $2/(B + 1)$  and  $\min\{1, 2n/(B + 1)\}$  by the union bound, and is not  $2/(B + 1)$ .
- (c) [Theorem 20.3.1](#) is an algebraic identity about a least squares fit. At convergence the generalized linear fit is a weighted least squares fit of  $\mathbf{z}$  on  $\mathbf{X}$  with weights  $\hat{\mathbf{W}}$  ([Theorem 34.4.1](#)), so the identity applies to that problem, which is what the substitution says. It is only one step of the algorithm, since refitting without case  $i$  changes  $\hat{\boldsymbol{\mu}}$  and therefore  $\hat{\mathbf{W}}$ .
- (d) The enlarged model contains the fitted one at coefficient zero and adds one parameter, so [Theorem 34.5.5](#) gives the  $\chi^2(1)$  limit under the smaller model. Note that  $\hat{\eta}_i$  is itself estimated; the effect of that is of smaller order and is ignored, as in [Proposition 22.2.4](#).
- (e) The working residual is  $(y_i - \hat{\mu}_i)/h'(\hat{\eta}_i) = z_i - \hat{\eta}_i$  by [Eq. 34.4.1](#). The weighted least squares fit of  $\mathbf{z}$  on  $\mathbf{X}$  has residual vector  $\mathbf{z} - \hat{\boldsymbol{\eta}}$  orthogonal to  $\mathbf{X}$  in the  $\hat{\mathbf{W}}$  inner

product, so  $\mathbf{u}_j = \hat{\beta}_j \mathbf{x}_j + (\mathbf{z} - \hat{\boldsymbol{\eta}})$  regressed on  $[\mathbf{1}, \mathbf{x}_j]$  in that inner product returns  $\hat{\beta}_j$  exactly, by Proposition 20.1.3(a) applied in the  $\hat{\mathbf{W}}$  metric (Theorem 6.9.1). For the added-variable statement, at convergence the fit is the weighted least squares regression of  $\mathbf{z}$  on  $\mathbf{X}$  with weights  $\hat{\mathbf{W}}$  (Theorem 34.4.1), which is an ordinary least squares problem in the  $\hat{\mathbf{W}}$  inner product; Theorem 6.6.2 applied in that inner product gives the claim.

- (f) By (a),  $r_i^P$  has approximate mean zero and variance  $1 - h_{ii} \approx 1$  when the leverages are small, and is approximately normal under the conditions of Theorem 34.5.1. For  $Z \sim N(0, 1)$ ,  $E|Z| = \sqrt{2/\pi}$ . If instead  $\text{Var}(Y_i) = \phi V(\mu_i)/w_i$  with  $\phi \neq 1$ , every  $r_i^P$  is inflated by  $\sqrt{\phi}$  and the plot is level at  $\sqrt{2\phi/\pi}$ .

■

*Remark* (Why absolute residuals). A quantile plot of signed residuals asks two questions at once, and for a discrete response the answer to the first is dominated by the discreteness. Plotting  $|r|_{(k)}$  against  $\Phi^{-1}\{(k + n - 1/8)/(2n + 1/2)\}$ , the expected order statistics of  $|Z|$ , asks only about magnitude, which is what an outlier or an inflated dispersion changes. The envelope turns the picture into a parametric bootstrap test (Proposition 23.3.3), of only approximate level because the data sets are generated at  $\hat{\boldsymbol{\beta}}$ .

## 39.2.2 A model that fails, and why

### Example. (Days of television news)

The 1996 American National Election Studies survey asked its 944 respondents on how many of the past seven days they had watched the television news. Treating the answer as a binomial count out of seven and fitting a logit model in age, education and income gives a deviance of 4447.3 on 940 degrees of freedom and a Pearson statistic of 3598.2, so  $\hat{\phi} = 3.828$ : the model is not close to fitting.

Panel (a) of Figure 39.2.1 is the half-normal plot with an envelope from  $B = 19$  simulated data sets, refitted each time. Every one of the 944 ordered residuals lies above the band — not a little too large in the tail but too large everywhere, the signature of a dispersion problem rather than of outliers.

The targeted checks of Proposition 39.2.1 say where the trouble is not. Adding  $\hat{\eta}^2$  drops the deviance by 0.31 on one degree of freedom,  $p = 0.580$ : the logit link is not the problem. Adding a quadratic in age drops it by 0.03, and panel (a) of Figure 39.2.2 shows the partial residual for age with a smooth that is mildly concave but close to the fitted line, so the scale of age is not the problem either. The leverages are all tiny — the largest 0.0144 against an average of 0.0042 — and the largest Cook-type distance is 0.0970.

What is left is the variance function, and panel (b) makes the case. The local means of  $|r^P|$  in ten equal bins of the linear predictor run from 1.25 to 1.91, against a reference height of  $\sqrt{2/\pi} \approx 0.80$ , and they drift upward: the regression of  $\log(r^P)^2$  on the linear predictor has slope 0.381 with  $t = 4.21$ , so the excess variance is not even constant. The binomial model treats a respondent's seven days as seven independent trials with a common probability, and they are nothing of the kind: 161 respondents watched on

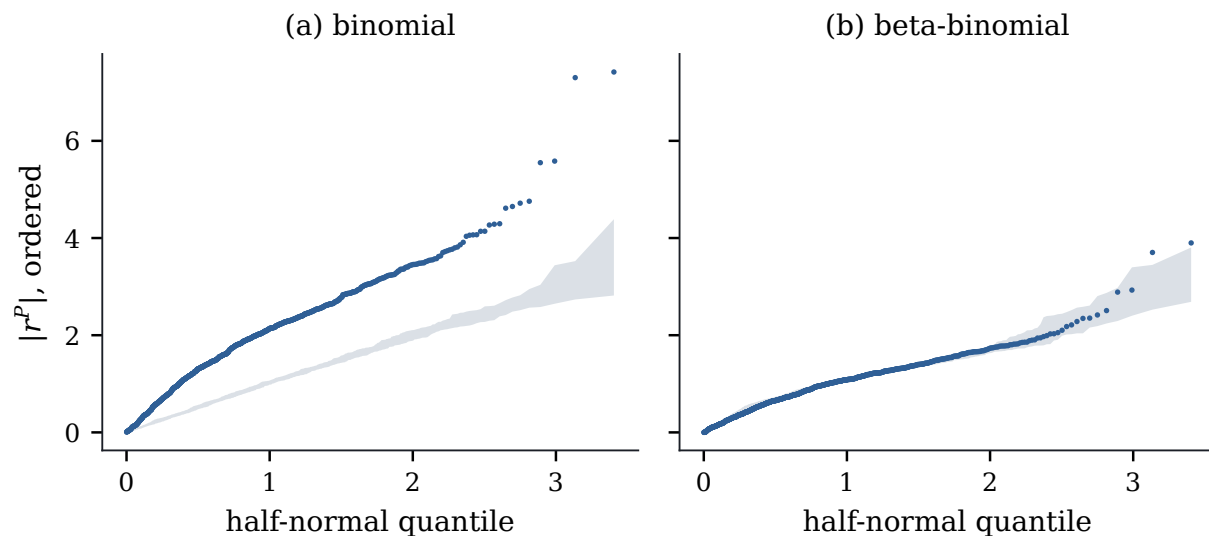


Figure 39.2.1: Half-normal plots of the absolute Pearson residuals with an envelope from 19 simulated data sets, refitted each time. (a) The binomial model: every point lies above the envelope. (b) The beta-binomial model: the upper tail is contained, the lower one is not.

none of the seven days and 288 on all seven. Habits are not coin flips.

The repair is a variance function with a second parameter. A beta-binomial model, in which each respondent has a latent probability from a beta distribution with mean  $\pi_i$  and precision  $s$ , gives  $\hat{s} = 1.072$ , a within-respondent correlation of  $\rho = 1/(1+s) = 0.483$ , and the variance  $7\pi_i(1-\pi_i)(7+s)/(1+s)$ , about 3.90 times the binomial one, against the  $\hat{\phi} = 3.828$  estimated directly. Panel (b) of Figure 39.2.1 repeats the half-normal plot under that model: only 0.2 per cent of the ordered residuals now lie above the envelope, although 22.8 per cent still fall below it, against a pointwise level of 5 per cent on that side — a residual misfit at the bottom of the scale that the beta-binomial does not remove.

The consequences for inference are what Chapter 38 predicts. The age coefficient barely moves, from 0.0424 to 0.0405; its standard error moves from 0.0018 to 0.0033, and the quasi-binomial standard error — the binomial one multiplied by  $\sqrt{\hat{\phi}}$  — is 0.0034, which agrees with the beta-binomial one to two figures. The education coefficient is the casualty: 0.0233 with standard error 0.0178 under the binomial, already unconvincing at  $t = 1.3$ , becomes  $-0.0013$  with standard error 0.0343 — an effect of zero with three times the uncertainty.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats
anes = sm.datasets.anes96.load_pandas().data
m = 7.0 # days in a week
y = anes["TVnews"].to_numpy(float) # successes out of seven
X = np.column_stack([np.ones(len(y)), anes["age"], anes["educ"], anes["income"]])
n, p = X.shape

fit = sm.GLM(np.column_stack([y, m - y]), X, family=sm.families.Binomial()).fit()
mu = fit.fittedvalues * m # fitted expected days
eta = fit.predict(which="linear")
pearson = (y - mu) / np.sqrt(mu * (1 - mu / m)) # r^P, with phi = 1
```

```
print(f"deviance {fit.deviance:8.1f} on {n - p} degrees of freedom")
print(f"Pearson {np.sum(pearson**2):8.1f}; dispersion estimate "
      f"{np.sum(pearson**2) / (n - p):.3f}")
```

```
B = 19 # simulated data sets

def halfnormal_quantiles(k):
    """The horizontal axis: expected half-normal order statistics."""
    i = np.arange(1, k + 1)
    return stats.norm.ppf((i + k - 0.125) / (2 * k + 0.5))

def envelope(draw, refit_abs_resid, seed):
    """Smallest and largest of B ordered |residual| vectors from the fitted model.
    """
    rng = np.random.default_rng(seed)
    out = np.array([np.sort(refit_abs_resid(draw(rng))) for _ in range(B)])
    return out.min(axis=0), out.max(axis=0)

def binomial_abs_resid(ysim):
    f = sm.GLM(np.column_stack([ysim, m - ysim]), X, family=sm.families.Binomial()).
    fit()
    mu_s = f.fittedvalues * m
    return np.abs((ysim - mu_s) / np.sqrt(mu_s * (1 - mu_s / m)))

q = halfnormal_quantiles(n)
lo, hi = envelope(lambda rng: rng.binomial(int(m), fit.fittedvalues).astype(float),
                  binomial_abs_resid, seed=3901)
ordered = np.sort(np.abs(pearson))
outside, above = np.mean((ordered > hi) | (ordered < lo)), np.mean(ordered > hi)
print(f"binomial model: {100 * outside:.1f}% of the ordered residuals leave the
      envelope, "
      f"{100 * above:.1f}% above it")
```

```
# (i) is the link right? Add the squared linear predictor and test it.
fit_link = sm.GLM(np.column_stack([y, m - y]), np.column_stack([X, eta**2]),
                  family=sm.families.Binomial()).fit()
lr_link = fit.deviance - fit_link.deviance

# (ii) is the age term right? A partial residual, on the link scale.
working = (y - mu) / (mu * (1 - mu / m)) # (y - mu) dEta/dMu
partial_age = fit.params[1] * anes["age"].to_numpy() + working

# (iii) is the variance function right? Regress log (r^P)^2 on the linear predictor
lin = sm.OLS(np.log(pearson**2 + 1e-12), sm.add_constant(eta)).fit()

print(f"link test: deviance drop {lr_link:.2f} on 1 d.f., p = {stats.chi2.sf(lr_link
, 1):.3f}")
print(f"slope of log (r^P)^2 on the linear predictor: {lin.params[1]:.3f} "
      f"(t = {lin.tvalues[1]:.2f})")
```

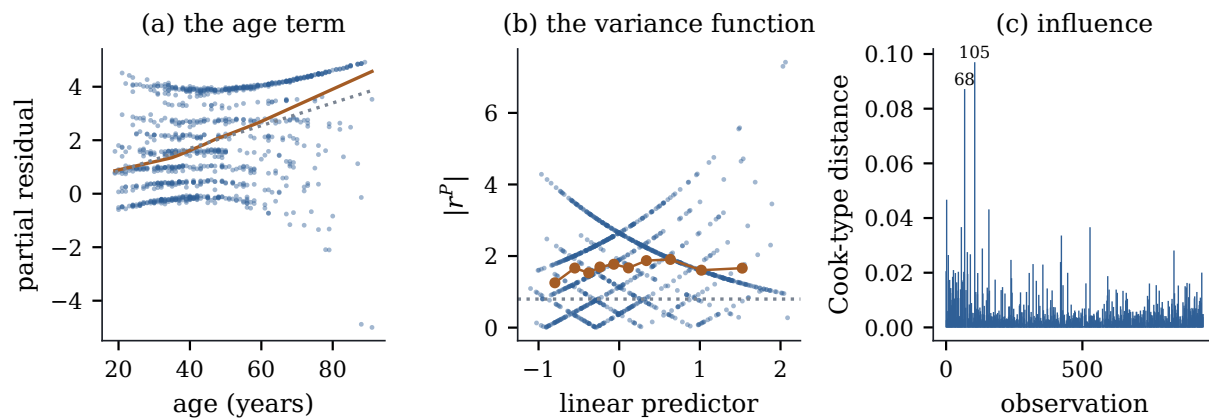


Figure 39.2.2: Three targeted checks of the binomial fit. (a) Partial residual for age, with a smooth (solid) and the fitted linear term (dotted). (b) Absolute Pearson residuals against the linear predictor, with their means in ten equal bins and the reference height  $\sqrt{2/\pi}$ . (c) Cook-type distances. Only (b) shows a problem.

```

scale = np.array([1.0, X[:, 1].std(), X[:, 2].std(), X[:, 3].std()]) # keep BFGS
conditioned

def betabinom_fit(ysim, start):
    """Maximize the beta-binomial likelihood: logit mean X beta, precision s."""
    def negll(theta):
        pi = special.expit((X / scale) @ theta[:p])
        s = np.exp(theta[p])
        v = -np.sum(special.betaln(ysim + s * pi, m - ysim + s * (1 - pi))
                    - special.betaln(s * pi, s * (1 - pi)))
        return v if np.isfinite(v) else 1e12

    theta = np.append(start[:p] * scale, start[p])
    for _ in range(3): # restart until the
        gradient is flat
        opt = optimize.minimize(negll, theta, method="BFGS")
        theta = opt.x
    return np.append(theta[:p] / scale, theta[p]), opt

theta_bb, opt = betabinom_fit(y, np.append(fit.params, 0.1))
beta_bb, s_bb = theta_bb[:p], np.exp(theta_bb[p])
rho = 1.0 / (1.0 + s_bb) # correlation within the
week
print(f"beta-binomial: s = {s_bb:.3f}, within-respondent correlation {rho:.3f}")
for j, name in enumerate(["intercept", "age", "educ", "income"]):
    print(f" {name:10s} binomial {fit.params[j]:8.4f} beta-binomial {beta_bb[j]:8.4f}")

```

### 39.2.3 Which repair

Example (Days of television news) had three candidate repairs and took the third; the choice among them is the subject of Proposition 38.4.1. A **quasi-likelihood** fit keeps the mean model and the shape of the variance function and lets  $\phi$  float (Definition 38.1.1): it changes no coeffi-

cient, because the estimating equations are the same, and multiplies every standard error by  $\sqrt{\hat{\phi}}$ . A **mixture** model, such as the beta-binomial here or the negative binomial of [Definition 37.4.1](#), names a mechanism and is a genuine likelihood, so it has an AIC and a likelihood ratio test, and it generally does move the coefficients. A **random effect** in the linear predictor is the third route, developed in [Chapter 40](#); it changes the meaning of the coefficients as well as their values ([Proposition 40.1.1](#)).

*Idea.* Read diagnostics for a generalized linear model as three separate questions. Is the mean right? Is the link right? Is the variance function right? A plot that fails is useful only when it belongs to one of the three.

## 39.2.4 Exercises

### 39.2.4.1 A. Check your understanding

#### Exercise A1.

[Proposition 39.2.1](#)(b) gives the pointwise level  $2/(B+1)$ . What is  $B$  for a level of 5 per cent? If instead the envelope is taken to be the second smallest and second largest of  $B$  simulated values, what is the pointwise level?

*Solution.*  $2/(B+1) = 0.05$  gives  $B = 39$ . With the second order statistics as limits the observed value must be among the two smallest or two largest of  $B+1$ , which has probability  $4/(B+1)$ , so  $B = 79$  is needed: a less jagged envelope at the cost of more simulation.

#### Exercise A2.

The link test of [Proposition 39.2.1](#)(d) came out non-significant in [Example \(Days of television news\)](#). Is that evidence that the logit link is correct? What would a significant result have shown?

### 39.2.4.2 B. Practice

#### Exercise B1.

Show that in a normal linear model with  $\phi = \sigma^2$  and the identity link, [Proposition 39.2.1](#)(a) reduces to the exact statement  $\text{Cov}(\hat{\boldsymbol{\varepsilon}}) = \sigma^2(\mathbf{I} - \mathbf{M})$  of [Eq. 20.1.2](#), and that the standardization by  $\sqrt{1 - h_{ii}}$  is then exact rather than approximate.

*Solution.* With the identity link and constant variance  $h'(\eta) = 1$ ,  $V(\mu) = 1$ ,  $w_i = 1$ , so  $\mathbf{W} = \mathbf{D} = \mathbf{I}$  and  $\mathbf{H} = \mathbf{X}(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top = \mathbf{M}$ . The expansion in the proof becomes an identity,  $\hat{\boldsymbol{\mu}} - \boldsymbol{\mu} = \mathbf{M}(\mathbf{y} - \boldsymbol{\mu})$ , whence  $\text{Cov}(\mathbf{y} - \hat{\boldsymbol{\mu}}) = \sigma^2(\mathbf{I} - \mathbf{M})$  and  $r_i^P = (y_i - \hat{\mu}_i)/\sigma$  has variance exactly  $1 - h_{ii}$ .

#### Exercise B2.

Let  $Y \mid \Pi \sim \text{Binomial}(m, \Pi)$  with  $\Pi \sim \text{Beta}(s\pi, s(1 - \pi))$ . Show that  $E(Y) = m\pi$  and

$\text{Var}(Y) = m\pi(1 - \pi)(m + s)/(1 + s)$ , and that the ratio to the binomial variance is  $1 + (m - 1)\rho$  with  $\rho = 1/(1 + s)$ . Evaluate at  $m = 7$  and the fitted  $\hat{s} = 1.072$ .

*Solution.*  $E\Pi = \pi$  and  $\text{Var}\Pi = \pi(1 - \pi)/(1 + s)$  for that beta distribution. By the tower property  $EY = mE\Pi = m\pi$ , and

$$\text{Var}Y = E\{m\Pi(1 - \Pi)\} + \text{Var}(m\Pi) = m\{\pi - E\Pi^2\} + m^2\text{Var}\Pi.$$

With  $E\Pi^2 = \text{Var}\Pi + \pi^2$  this is  $m\pi(1 - \pi) - m\text{Var}\Pi + m^2\text{Var}\Pi = m\pi(1 - \pi)\{1 + (m - 1)/(1 + s)\}$ , which is the stated formula since  $1 + (m - 1)/(1 + s) = (m + s)/(1 + s)$ . The ratio to  $m\pi(1 - \pi)$  is  $1 + (m - 1)\rho$ . At  $m = 7$  and  $s = 1.072$ ,  $\rho = 0.483$  and the ratio is  $1 + 6 \times 0.483 = 3.90$ , close to the  $\hat{\phi} = 3.828$  that [Example \(Days of television news\)](#) estimates directly.

### 39.2.4.3 C. Going deeper

#### Exercise C1.

[Proposition 39.2.1\(b\)](#) says the simultaneous level of an envelope exceeds the pointwise one. Construct a simultaneous band from the same  $B$  simulations, by computing for each simulated data set the maximum over  $k$  of the standardized deviation  $\{R_k^{(b)} - \bar{R}_k\}/\text{sd}_k$  and using its  $(1 - \alpha)$  quantile. Show that the resulting band has simultaneous level at most  $\alpha$  under the same exchangeability assumption, and say what it costs.

## 39.3 Bayesian thread IV: what carries over

The Bayesian thread has three instalments behind it. [Section 7.6](#) put a normal-inverse-gamma prior on  $(\boldsymbol{\beta}, \sigma^2)$  in the normal linear model and found the posterior in closed form ([Theorem 7.6.2](#)): precisions add, the posterior mean is a precision-weighted average of the prior mean and the least squares estimate ([Proposition 7.6.3](#)), and the improper prior  $\pi(\boldsymbol{\beta}, \sigma^2) \propto 1/\sigma^2$  reproduces least squares and its  $t$  intervals exactly ([Corollary 7.6.4](#)). [Section 12.6](#) summarized that posterior by credible sets ([Definition 12.6.1](#)), and [Section 32.6](#) read a random effect as a prior, showed that the BLUP is a posterior mean and REML a marginal posterior mode ([Proposition 32.6.1](#)), and ran a Gibbs sampler to do what the closed form could not. This short section says which of those survive the move to a generalized linear model, and why the others do not.

### 39.3.1 What survives

Three things carry over unchanged, being definitions rather than computations. **The posterior is still prior times likelihood:** Bayes' theorem does not know what family  $\mathbf{y}$  came from. **Credible sets still mean what they meant:** [Definition 12.6.1](#) is a statement about a posterior distribution, not about a normal one. **A random effect is still a prior:** the generalized linear mixed models of [Chapter 40](#) are [Eq. 32.6.1](#) with a non-normal first level, and the empirical Bayes reading ([Proposition 27.5.4](#)) survives with them.



### 39.3.2 What does not, and why

Everything that made [Section 7.6](#) *closed form* rests on one fact: the normal log-likelihood is a quadratic in  $\beta$ , so prior times likelihood is again the exponential of a quadratic and completing the square finishes the job. Nothing weaker will do.

#### Proposition 39.3.1 — Why the conjugate machinery stops at the normal family

Let  $\mathbf{y}$  follow the exponential dispersion family of [Definition 34.2.1](#) with the canonical link, so that  $\theta_i = \mathbf{x}_{(i)}^\top \beta$  and  $\ell(\beta) = \phi^{-1} \sum_i w_i \{y_i \theta_i - b(\theta_i)\} + c$ . Suppose  $b$  is twice continuously differentiable on an open interval  $\Theta$ . Then the following are equivalent.

- (a) For every  $n$ , every  $\mathbf{X}$  with  $\text{rank}(\mathbf{X}) = p$  and every  $\mathbf{y}$ , the function  $\beta \mapsto \ell(\beta)$  is a quadratic polynomial on  $\{\beta : \mathbf{x}_{(i)}^\top \beta \in \Theta\}$ .
- (b)  $b''$  is constant on  $\Theta$ .
- (c) The variance function is constant:  $V(\mu) = b''(\theta)$  does not depend on  $\mu$ , so the family is normal ([Exercise \(B2\)](#)).

Consequently, for every family except the normal, a normal prior on  $\beta$  gives a posterior that is not normal and has no closed-form normalizing constant.

*Proof.* The Hessian of  $\ell$  is  $-\phi^{-1} \mathbf{X}^\top \text{diag}\{w_i b''(\mathbf{x}_{(i)}^\top \beta)\} \mathbf{X}$  by [Theorem 34.3.1\(d\)](#). If (b) holds this is constant in  $\beta$ , and a function with a constant Hessian is a quadratic polynomial, giving (a). Conversely, if (a) holds, take  $p = 1$ ,  $n = 1$ ,  $w_1 = 1$ ,  $x_{11} = 1$ : the Hessian is the scalar  $-b''(\beta)/\phi$ , which must then be constant over  $\Theta$ , which is (b). The equivalence of (b) and (c) is the definition  $V(\mu) = b''(\theta)$  of [Definition 34.1.2](#) with  $\mu = b'(\theta)$ , which makes  $\theta \mapsto \mu$  a bijection:  $b'' \equiv \sigma^2$  gives  $b(\theta) = \sigma^2 \theta^2/2$  up to an affine term, the normal cumulant function.

For the last sentence, with  $\pi$  the  $N_p(\mathbf{m}, \Sigma)$  density the log posterior is  $\ell(\beta) - \frac{1}{2}(\beta - \mathbf{m})^\top \Sigma^{-1}(\beta - \mathbf{m})$  up to a constant; were it the log of a normal density it would be a quadratic in  $\beta$ , hence so would  $\ell$ , which forces the normal family. ■

So the failure is not a want of cleverness. It is the same failure that made the likelihood equations of [Theorem 34.3.1](#) nonlinear and forced iteratively reweighted least squares on the frequentist side: the score is not linear in  $\beta$ , so the posterior is not normal, and the integrals defining the posterior mean, the posterior variance and the marginal likelihood

$$m(\mathbf{y}) = \int L(\beta; \mathbf{y}) \pi(\beta) d\beta \tag{39.3.1}$$

have no closed form.



### 39.3.3 Conjugate priors do exist

It is worth being precise, because “generalized linear models have no conjugate priors” is false as stated. Diaconis and Ylvisaker (1979) characterized the conjugate priors of an exponential family; the canonical-link likelihood is an exponential family in  $\boldsymbol{\beta}$ , and their construction gives the density

$$\pi(\boldsymbol{\beta} \mid \mathbf{m}_0, \kappa_0) \propto \exp\left[\kappa_0^{-1}\{\mathbf{m}_0^\top \mathbf{X}\boldsymbol{\beta} - \sum_i v_i b(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})\}\right] \quad (39.3.2)$$

in the form developed for generalized linear models by Bedrick, Christensen and Johnson (1996) and Chen and Ibrahim (2003), with a pseudo-response vector  $\mathbf{m}_0$  and a prior weight  $v_i$  on each observation. It has the shape of the likelihood, so the posterior belongs to the same family with  $\mathbf{m}_0$  and the  $v_i$  updated by addition — the “precisions add” structure of Eq. 7.6.2, the prior read as  $\sum_i v_i$  observations at  $\mathbf{m}_0$ . What is missing is everything one wants afterwards: outside the normal family, and outside degenerate designs such as the intercept-only Poisson model of Exercise (B1), Eq. 39.3.2 has no closed-form normalizing constant, so neither have the posterior mean, the variance, the marginal likelihood or a credible interval. In practice the normal prior of the next section — not conjugate, but interpretable — is what is used.

*Idea.* Everything in the Bayesian thread that was a *definition* survives; everything that was a *closed form* does not. The next two sections replace the closed form by an approximation accurate when  $n$  is large and a simulation accurate when it is run long enough.

### 39.3.4 Exercises

#### 39.3.4.1 A. Check your understanding

##### Exercise A1.

Verify directly, from the form of the Poisson log-likelihood with a log link, that it is not a quadratic in  $\boldsymbol{\beta}$  for  $p = 1$ , and identify the term that spoils it.

*Solution.* With  $\mu_i = e^{x_i\beta}$  the log-likelihood is  $\sum_i \{y_i x_i \beta - e^{x_i \beta}\}$  up to a constant. The first term is linear; the second has second derivative  $-\sum_i x_i^2 e^{x_i \beta}$ , which depends on  $\beta$ . That is the cumulant term  $b(\theta) = e^\theta$ , whose  $b''$  is not constant, as Proposition 39.3.1(b) requires.

##### Exercise A2.

An equal-tailed credible interval for  $\beta_j$  transforms to an equal-tailed credible interval for  $e^{\beta_j}$  by exponentiating its endpoints. Is the same true of a highest-posterior-density interval? Explain.

#### 39.3.4.2 B. Practice

**Exercise B1.**

Take the Poisson family with a log link,  $p = 1$  and  $x_{i1} = 1$  for all  $i$ , so the model has a single parameter  $\beta$  and  $\mu = e^\beta$ . Write out Eq. 39.3.2 explicitly, show that it is the density of  $\beta = \log \mu$  when  $\mu$  has a gamma distribution, and identify the shape and rate in terms of  $\mathbf{m}_0$  and the  $v_i$ . Why does the closed form appear here and not in general?

*Solution.* With  $b(\theta) = e^\theta$ ,  $\theta = \beta$  and  $\kappa_0 = 1$ , Eq. 39.3.2 is  $\pi(\beta) \propto \exp\{a\beta - ve^\beta\}$  with  $a = \mathbf{1}^\top \mathbf{m}_0$  and  $v = \sum_i v_i$ . Substituting  $\mu = e^\beta$ , so  $d\beta = d\mu/\mu$ , the density of  $\mu$  is proportional to  $\mu^{a-1}e^{-v\mu}$ : a gamma with shape  $a$  and rate  $v$ , normalizing constant  $v^a/\Gamma(a)$ . That constant is available because the single linear predictor lets the integral factor into one dimension. With  $p > 1$  the exponent  $\mathbf{m}_0^\top \mathbf{X}\boldsymbol{\beta} - \sum_i v_i e^{\mathbf{x}_{(i)}^\top \boldsymbol{\beta}}$  is a sum of exponentials of different linear forms, and no change of variable makes it a product of gamma integrals.

**39.3.4.3 C. Going deeper****Exercise C1.**

Proposition 39.3.1 assumed the canonical link. Show that with a non-canonical link the conclusion still holds in the following sense: if the log-likelihood is a quadratic in  $\boldsymbol{\beta}$  for every design and every response, then the working weights of Theorem 34.4.1 do not depend on  $\boldsymbol{\beta}$  and the mean is affine in  $\boldsymbol{\beta}$ . Deduce that the model is the normal linear model up to a reparameterization.

## 39.4 Bayesian generalized linear models and MCMC

Proposition 39.3.1 closed off the closed form. What is left is a posterior known only up to a constant, and two ways to work with it: approximate it by a normal density at its mode, accurate to order  $n^{-1}$  and the origin of the BIC; or sample from it, accurate to whatever the Monte Carlo error allows. This section does both, with the samplers written out rather than delegated.

### 39.4.1 The posterior

#### Definition 39.4.1 — The Bayesian generalized linear model

Let  $\mathbf{y}$  follow the generalized linear model of Definition 34.2.1 with link  $g$ , dispersion  $\phi$  and log-likelihood  $\ell(\boldsymbol{\beta})$ , and let  $\pi$  be a prior density on  $\mathbb{R}^p$ . The **posterior density** of  $\boldsymbol{\beta}$  is

$$\pi(\boldsymbol{\beta} \mid \mathbf{y}) = \frac{\exp\{\ell(\boldsymbol{\beta})\} \pi(\boldsymbol{\beta})}{m(\mathbf{y})}, \quad m(\mathbf{y}) = \int_{\mathbb{R}^p} \exp\{\ell(\mathbf{b})\} \pi(\mathbf{b}) d\mathbf{b}, \quad (39.4.1)$$

whenever  $0 < m(\mathbf{y}) < \infty$ ; the posterior is then **proper**. The **maximum a posteriori** estimate  $\hat{\boldsymbol{\beta}}_{\text{MAP}}$  maximizes Eq. 39.4.1, equivalently maximizes the penalized log-likelihood  $\ell(\boldsymbol{\beta}) + \log \pi(\boldsymbol{\beta})$ . The default choice throughout this chapter is the independent normal prior  $\boldsymbol{\beta} \sim N_p(\mathbf{m}, \boldsymbol{\Sigma})$ , for which

$$\log \pi(\boldsymbol{\beta} \mid \mathbf{y}) = \ell(\boldsymbol{\beta}) - \frac{1}{2}(\boldsymbol{\beta} - \mathbf{m})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\beta} - \mathbf{m}) + \text{const.} \quad (39.4.2)$$

Because Eq. 39.4.2 is concave whenever  $\ell$  is (Theorem 34.3.3),  $\hat{\boldsymbol{\beta}}_{\text{MAP}}$  is found by the Fisher scoring of Theorem 34.4.1 with the score replaced by  $\mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1}(\mathbf{y} - \boldsymbol{\mu})/\phi - \boldsymbol{\Sigma}^{-1}(\boldsymbol{\beta} - \mathbf{m})$  and the information by  $\mathbf{X}^\top \mathbf{W} \mathbf{X}/\phi + \boldsymbol{\Sigma}^{-1}$ : one extra term in each. What the prior buys is not only shrinkage but existence.

### Proposition 39.4.1 — When a flat prior gives a posterior

Let  $\pi$  be a prior on  $\mathbb{R}^p$ .

- (a) If  $\pi$  is proper and the likelihood is bounded, the posterior is proper. Both the logistic and the Poisson likelihoods are bounded.
- (b) Take the improper prior  $\pi \propto 1$  and the ungrouped logistic model with  $\text{rank}(\mathbf{X}) = p$ . Then the posterior is proper if and only if the data are in overlap in the sense of Definition 35.3.1, that is if and only if the maximum likelihood estimate exists (Theorem 35.3.1).

*Proof.*

- (a) If  $L \leq L_{\max} < \infty$  then  $m(\mathbf{y}) \leq L_{\max} \int \pi = L_{\max} < \infty$ , and  $m(\mathbf{y}) > 0$  because the integrand is positive on a set of positive measure. The logistic likelihood is a product of probabilities, so  $L \leq 1$ ; the Poisson likelihood  $\prod_i \mu_i^{y_i} e^{-\mu_i} / y_i!$ , maximized freely over  $\boldsymbol{\mu} \in (0, \infty)^n$  at  $\mu_i = y_i$ , is bounded by  $\prod_i y_i^{y_i} e^{-y_i} / y_i!$  (with  $0^0 = 1$ ).
- (b) Write  $s_i = 2y_i - 1$  and use Eq. 35.3.1:  $-\ell(\boldsymbol{\beta}) = \sum_i \log\{1 + \exp(-s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta})\}$ .

*Overlap implies propriety.* Since  $\log(1 + e^{-t}) \geq \max(0, -t) = t_-$ ,

$$-\ell(\boldsymbol{\beta}) \geq \sum_{i=1}^n (s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta})_- =: c(\boldsymbol{\beta}),$$

with  $c$  continuous and homogeneous of degree one. Under overlap  $c(\mathbf{b}) > 0$  for every  $\mathbf{b} \neq \mathbf{0}$ , since  $c(\mathbf{b}) = 0$  forces  $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$  for all  $i$ , which overlap permits only at the origin. The unit sphere is compact, so  $\delta = \min_{\|\mathbf{b}\|=1} c(\mathbf{b}) > 0$  and, by homogeneity,  $c(\boldsymbol{\beta}) \geq \delta \|\boldsymbol{\beta}\|$ . Hence  $\exp\{\ell(\boldsymbol{\beta})\} \leq e^{-\delta \|\boldsymbol{\beta}\|}$ , integrable over  $\mathbb{R}^p$ .

*Separation implies impropriety.* Let  $\mathbf{b} \neq \mathbf{0}$  satisfy  $s_i \mathbf{x}_{(i)}^\top \mathbf{b} \geq 0$  for all  $i$ , put  $S = \{i : \mathbf{x}_{(i)}^\top \mathbf{b} = 0\}$ ,  $\varrho = \min_{i \notin S} s_i \mathbf{x}_{(i)}^\top \mathbf{b} > 0$ ,  $\omega = \max_i \|\mathbf{x}_{(i)}\|$ , and fix  $\rho > 0$ . For  $\boldsymbol{\beta} = t\mathbf{b} + \mathbf{u}$  with  $t \geq t_0 := 2\omega\rho/\varrho$  and  $\|\mathbf{u}\| \leq \rho$ , every  $i \notin S$  has  $s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta} \geq t\varrho - \omega\rho \geq 0$  and every  $i \in S$  has  $s_i \mathbf{x}_{(i)}^\top \boldsymbol{\beta} = s_i \mathbf{x}_{(i)}^\top \mathbf{u} \geq -\omega\rho$ . So  $-\ell(\boldsymbol{\beta}) \leq n \log\{1 + e^{\omega\rho}\}$  on that whole tube, giving  $\exp\{\ell(\boldsymbol{\beta})\} \geq \{1 + e^{\omega\rho}\}^{-n}$  there; and the tube has infinite Lebesgue measure, containing a half-infinite cylinder about the ray. Hence  $m(\mathbf{y}) = \infty$ .

*Warning.* The second part is not a curiosity. Separation is common in small samples and with rare outcomes, and a “noninformative” Bayesian analysis of separated binary data summarizes a measure of infinite mass: whatever the sampler prints is an artefact of where it wandered. [Example \(The separated subgroup, three priors\)](#) makes the divergence visible.

## 39.4.2 The Laplace approximation

For large  $n$  the integral in [Eq. 39.4.1](#) is dominated by a shrinking neighbourhood of the mode, where the integrand looks normal. Write  $q_n(\boldsymbol{\beta}) = \ell(\boldsymbol{\beta}) + \log \pi(\boldsymbol{\beta})$  for the unnormalized log posterior,  $\hat{\boldsymbol{\beta}}_n$  for its maximizer and  $\mathbf{H}_n = -\nabla^2 q_n(\hat{\boldsymbol{\beta}}_n)$ .

### Theorem 39.4.2 — Laplace approximation

Let  $\rho_n = n^{-3/8}$  and  $B_n = \{\boldsymbol{\beta} : \|\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_n\| \leq \rho_n\}$ , and suppose there are constants  $0 < c \leq C < \infty$  and  $a, K > 0$ , free of  $n$ , with:

- (L1)  $q_n$  is four times continuously differentiable on  $B_n$ , with  $\nabla q_n(\hat{\boldsymbol{\beta}}_n) = \mathbf{0}$  and  $\mathbf{H}_n$  positive definite;
- (L2)  $cn \leq \lambda_{\min}(\mathbf{H}_n) \leq \lambda_{\max}(\mathbf{H}_n) \leq Cn$ , and every third and fourth partial derivative of  $q_n$  is bounded in absolute value by  $Cn$  on  $B_n$ ;
- (L3)  $\int_{\mathbb{R}^p \setminus B_n} \exp\{q_n(\boldsymbol{\beta}) - q_n(\hat{\boldsymbol{\beta}}_n)\} d\boldsymbol{\beta} \leq K \exp(-an^{1/4})$ .

Then

$$\int_{\mathbb{R}^p} e^{q_n(\boldsymbol{\beta})} d\boldsymbol{\beta} = (2\pi)^{p/2} \det(\mathbf{H}_n)^{-1/2} e^{q_n(\hat{\boldsymbol{\beta}}_n)} \{1 + O(n^{-1})\}. \quad (39.4.3)$$

*Proof.* Write  $I_n$  for the integral and  $J_n = I_n \det(\mathbf{H}_n)^{1/2} e^{-q_n(\hat{\boldsymbol{\beta}}_n)}$ , so the claim is  $J_n = (2\pi)^{p/2} \{1 + O(n^{-1})\}$ . Substitute  $\boldsymbol{\beta} = \hat{\boldsymbol{\beta}}_n + \mathbf{H}_n^{-1/2} \mathbf{u}$ , Jacobian  $\det(\mathbf{H}_n)^{-1/2}$ :

$$J_n = \int_{\mathbb{R}^p} \exp\{q_n(\hat{\boldsymbol{\beta}}_n + \mathbf{H}_n^{-1/2} \mathbf{u}) - q_n(\hat{\boldsymbol{\beta}}_n)\} d\mathbf{u}.$$

Because  $\nabla q_n(\hat{\boldsymbol{\beta}}_n) = \mathbf{0}$ , Taylor’s theorem on  $B_n$  gives

$$q_n(\hat{\boldsymbol{\beta}}_n + \mathbf{H}_n^{-1/2} \mathbf{u}) - q_n(\hat{\boldsymbol{\beta}}_n) = -\frac{1}{2} \|\mathbf{u}\|^2 + A_n(\mathbf{u}) + r_n(\mathbf{u}),$$

where  $A_n$  is the cubic form built from the third derivatives at  $\hat{\boldsymbol{\beta}}_n$  and  $r_n$  is the fourth-order remainder. By (L2) the derivatives are bounded by  $Cn$  and  $\|\mathbf{H}_n^{-1/2} \mathbf{u}\| \leq (cn)^{-1/2} \|\mathbf{u}\|$ , so

$$|A_n(\mathbf{u})| \leq c_3 n^{-1/2} \|\mathbf{u}\|^3, \quad |r_n(\mathbf{u})| \leq c_4 n^{-1} \|\mathbf{u}\|^4, \quad (39.4.4)$$

with  $c_3 = p^3 C / (6c^{3/2})$  and  $c_4 = p^4 C / (24c^2)$ , valid whenever  $\hat{\boldsymbol{\beta}}_n + \mathbf{H}_n^{-1/2} \mathbf{u} \in B_n$ .

Let  $S_n = \{\mathbf{u} : \|\mathbf{u}\| \leq \sqrt{c}n^{1/8}\}$  and  $U_n$  the image of  $B_n$ . Every  $\mathbf{u} \in S_n$  has  $\|\mathbf{H}_n^{-1/2}\mathbf{u}\| \leq (cn)^{-1/2}\|\mathbf{u}\| \leq n^{-3/8} = \rho_n$ , so  $S_n \subseteq U_n$ ; and every  $\mathbf{u} \in U_n$  has  $\|\mathbf{u}\| \leq \lambda_{\max}(\mathbf{H}_n)^{1/2}\rho_n \leq \sqrt{C}n^{1/8}$ . Split  $J_n$  over  $S_n$ , over  $U_n \setminus S_n$  and outside  $U_n$ .

*Outside  $U_n$ .* Changing back to  $\boldsymbol{\beta}$  and using (L3) with  $\det(\mathbf{H}_n)^{1/2} \leq (Cn)^{p/2}$  bounds that piece by  $(Cn)^{p/2}Ke^{-an^{1/4}}$ , which is  $o(n^{-1})$ .

*Between  $S_n$  and  $U_n$ .* There  $\|\mathbf{u}\| \leq \sqrt{C}n^{1/8}$ , so Eq. 39.4.4 gives  $|A_n| + |r_n| \leq c_3C^{3/2}n^{-1/8} + c_4C^2n^{-1/2} \leq 1$  for large  $n$ ; the integrand is then at most  $e^{1-\|\mathbf{u}\|^2/2}$ , whose integral over  $\|\mathbf{u}\| > \sqrt{c}n^{1/8}$  is  $O(e^{-cn^{1/4}/4}) = o(n^{-1})$ .

*Inside  $S_n$ .* Put  $R_n = A_n + r_n$ , so  $|R_n| \leq 1$  there for large  $n$ . From  $|e^x - 1 - x - \frac{1}{2}x^2| \leq \frac{1}{6}|x|^3e^{|x|}$  for  $|x| \leq 1$ ,

$$\begin{aligned} \int_{S_n} e^{-\|\mathbf{u}\|^2/2} e^{R_n} d\mathbf{u} &= \int_{S_n} e^{-\|\mathbf{u}\|^2/2} \left\{ 1 + R_n + \frac{1}{2}R_n^2 \right\} d\mathbf{u} + \mathcal{E}_n, \\ |\mathcal{E}_n| &\leq \frac{\epsilon}{6} \int_{S_n} e^{-\|\mathbf{u}\|^2/2} |R_n|^3 d\mathbf{u}. \end{aligned}$$

Four pieces. First,  $\int_{S_n} e^{-\|\mathbf{u}\|^2/2} d\mathbf{u} = (2\pi)^{p/2} + O(e^{-cn^{1/4}/2})$ . Second,  $A_n$  is homogeneous of degree three, hence odd, and  $S_n$  is a ball about the origin, so  $\int_{S_n} e^{-\|\mathbf{u}\|^2/2} A_n(\mathbf{u}) d\mathbf{u} = 0$  exactly. Third,  $\int_{S_n} e^{-\|\mathbf{u}\|^2/2} |r_n| d\mathbf{u} \leq c_4n^{-1} \int e^{-\|\mathbf{u}\|^2/2} \|\mathbf{u}\|^4 d\mathbf{u} = O(n^{-1})$ . Fourth, by Eq. 39.4.4  $R_n^2 \leq 2c_3^2n^{-1}\|\mathbf{u}\|^6 + 2c_4^2n^{-2}\|\mathbf{u}\|^8$ , of Gaussian integral  $O(n^{-1})$ , and  $|R_n|^3 = O(n^{-3/2}\|\mathbf{u}\|^9) + O(n^{-3}\|\mathbf{u}\|^{12})$ , of Gaussian integral  $O(n^{-3/2})$ . Adding the three regions gives  $J_n = (2\pi)^{p/2} + O(n^{-1})$ , which is Eq. 39.4.3. ■

*Remark* (When the conditions hold). For a generalized linear model with a proper, positive, twice continuously differentiable prior and the conditions (G1)–(G4) of Theorem 34.5.1, (L1) and (L2) hold with probability tending to one, because  $\mathbf{H}_n = \mathbf{X}^\top \hat{\mathbf{W}} \mathbf{X} / \phi + O(1)$  and (G4) makes its smallest eigenvalue grow linearly. Condition (L3) says the posterior concentrates; verifying it in general is a separate argument this book does not give. Kass and Raftery (1995, section 4) discuss the regularity conditions and the accuracy of the approximation.

### Corollary 39.4.3 — Where the BIC comes from

Under the conditions of Theorem 39.4.2, suppose in addition that  $\mathbf{H}_n/n \rightarrow \bar{\mathbf{H}}$  positive definite, that the prior is continuous and positive at the limit of  $\hat{\boldsymbol{\beta}}_n$ , and that  $\hat{\boldsymbol{\beta}}$  maximizes  $\ell$  with  $\ell(\hat{\boldsymbol{\beta}}) - \ell(\hat{\boldsymbol{\beta}}_n) = O(n^{-1})$ . Then

$$-2 \log m(\mathbf{y}) = -2\ell(\hat{\boldsymbol{\beta}}) + p \log n + O(1), \quad (39.4.5)$$

where the  $O(1)$  term is  $\log \det(\mathbf{H}_n/n) - p \log(2\pi) - 2 \log \pi(\hat{\boldsymbol{\beta}}_n) + o(1)$ . The leading two terms are the BIC of Definition 29.2.2.

*Proof.* Take logarithms in Eq. 39.4.3:

$$2 \log m(\mathbf{y}) = 2\ell(\hat{\boldsymbol{\beta}}_n) + 2 \log \pi(\hat{\boldsymbol{\beta}}_n) + p \log(2\pi) - \log \det \mathbf{H}_n + O(n^{-1}).$$

Write  $\log \det \mathbf{H}_n = p \log n + \log \det(\mathbf{H}_n/n)$  and replace  $\ell(\hat{\boldsymbol{\beta}}_n)$  by  $\ell(\hat{\boldsymbol{\beta}})$  at a cost of  $O(n^{-1})$ ; rearranging gives Eq. 39.4.5 with the stated remainder, which converges because  $\mathbf{H}_n/n \rightarrow \bar{\mathbf{H}}$ .

■

The remainder in Eq. 39.4.5 is  $O(1)$ , not  $o(1)$ : the BIC stays a bounded distance from  $-2 \log m(\mathbf{y})$ , and that distance carries the whole influence of the prior, so a BIC difference is only a crude approximation to a log Bayes factor (Proposition 29.2.4). What makes it useful is that a difference in dimension contributes  $\log n$ , which grows, while the neglected term does not. A third criterion belongs to the same family: the deviance information criterion of Spiegelhalter, Best, Carlin and van der Linde (2002) replaces  $p$  by an effective number of parameters estimated from posterior draws, which suits a model whose parameters are themselves shrunk by a prior.

### Example. (How good is the approximation)

Take a logistic regression with an intercept and one covariate, a  $N(0, 5^2)$  prior on each coefficient, and data simulated at  $\boldsymbol{\beta} = (-0.4, 1.2)^\top$ . Adaptive Gauss–Hermite quadrature on a tensor grid centred at the mode computes Eq. 39.4.1 to machine accuracy, so the Laplace error can be measured exactly.

At  $n = 25$  the Laplace marginal likelihood is too small by 0.10700, or eleven per cent; the relative error falls to 0.001114 at  $n = 1600$ . Regressing the log error on  $\log n$  over seven sample sizes gives a slope of  $-1.06$ , and over those seven sizes the product  $n \times \text{error}$  stays between 1.530 and 2.675: the  $O(n^{-1})$  of Eq. 39.4.3, over a decade and a half of sample sizes (Figure 39.4.1, panel (b)).

Panel (a) shows what that error looks like at  $n = 25$ : the marginal posterior of the slope is visibly right-skewed, mean 2.3291 against mode 1.9271, with total variation distance 0.1827 to the normal approximation. Reporting the mode with the curvature as a standard error would put the estimate in the wrong place and the interval on the wrong side. The corollary is visible too: over the same seven sizes the gap between the BIC and  $-2 \log m(\mathbf{y})$  stays between  $-3.41$  and  $-1.61$ , an  $O(1)$  that does not grow.

```
import numpy as np
from scipy import special, stats
TAU = 5.0                                     # prior standard deviation, N(0, TAU
                                           ~2 I)
BETA0 = np.array([-0.4, 1.2])                 # the truth used to simulate

def simulate(n, seed):
    rng = np.random.default_rng(seed)
    X = np.column_stack([np.ones(n), rng.uniform(-2, 2, n)])
    y = rng.binomial(1, special.expit(X @ BETA0)).astype(float)
    return X, y

def log_post(beta, X, y):
    """Log likelihood plus log prior, up to a constant free of beta."""
    eta = X @ beta
    return np.sum(y * eta - np.logaddexp(0.0, eta)) - np.sum(beta**2) / (2 * TAU**2)
```

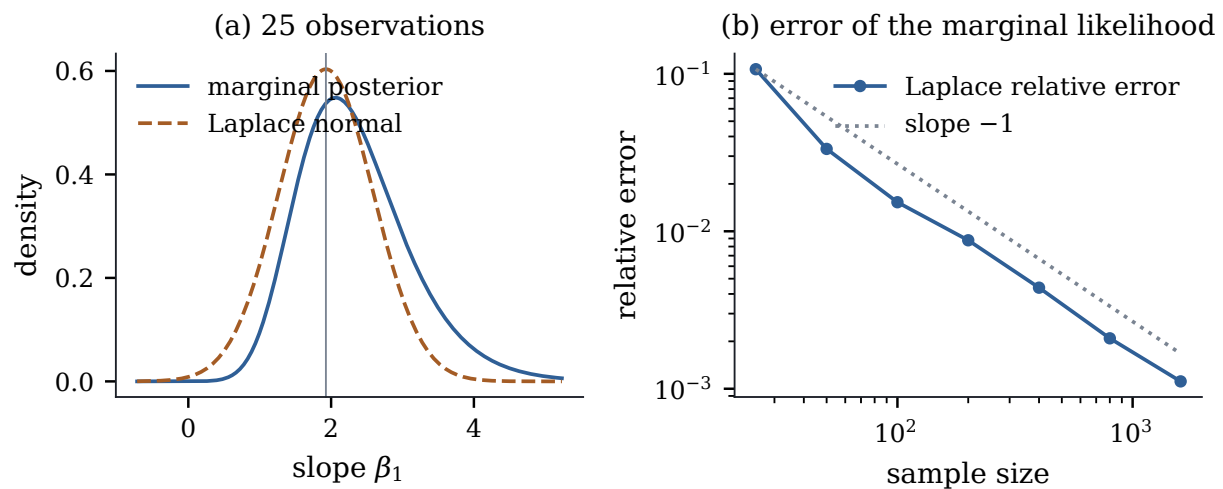


Figure 39.4.1: The Laplace approximation in a two-parameter logistic regression. (a) The marginal posterior of the slope at 25 observations against its normal approximation; the vertical line is the mode. (b) Relative error of the Laplace marginal likelihood against sample size, on log scales, with a reference line of slope  $-1$ .

```
def mode_and_hessian(X, y):
    """Posterior mode and the negative Hessian there, by Newton's method."""
    beta = np.zeros(X.shape[1])
    for _ in range(100):
        pi = special.expit(X @ beta)
        H = X.T @ ((pi * (1 - pi))[:, None] * X) + np.eye(len(beta)) / TAU**2
        g = X.T @ (y - pi) - beta / TAU**2
        beta = beta + np.linalg.solve(H, g)
    return beta, H

def log_marginal_laplace(X, y):
    """log of the Laplace value of the integral of exp(log_post)."""
    beta, H = mode_and_hessian(X, y)
    p = len(beta)
    return (log_post(beta, X, y) + p / 2 * np.log(2 * np.pi)
            - 0.5 * np.linalg.slogdet(H)[1] - p / 2 * np.log(2 * np.pi * TAU**2))

def log_marginal_quadrature(X, y, nodes=60):
    """The same integral by adaptive Gauss-Hermite quadrature: the reference value.
    """
    beta, H = mode_and_hessian(X, y)
    p = len(beta)
    L = np.linalg.cholesky(np.linalg.inv(H))
    z, w = np.polynomial.hermite_e.hermegauss(nodes) # weight exp(-z^2/2)
    Z = np.stack(np.meshgrid(z, z, indexing="ij"), -1).reshape(-1, p)
    logw = np.log(w)[:, None] + np.log(w)[None, :]
    pts = beta + Z @ L.T
    vals = np.array([log_post(b, X, y) for b in pts]) + (Z**2).sum(1) / 2 + logw.
    ravel()
    return (special.logsumexp(vals) + np.linalg.slogdet(L)[1]
            - p / 2 * np.log(2 * np.pi * TAU**2))
```

```
sizes = [25, 50, 100, 200, 400, 800, 1600]
```

```

rel_err = []
for k, n in enumerate(sizes):
    X, y = simulate(n, 3910 + k)
    a, b = log_marginal_laplace(X, y), log_marginal_quadrature(X, y)
    rel_err.append(abs(np.expm1(a - b)))
rel_err = np.array(rel_err)
slope = np.polyfit(np.log(sizes), np.log(rel_err), 1)[0]
print("n, relative error of the Laplace marginal likelihood:")
for n, e in zip(sizes, rel_err):
    print(f" {n:5d} {e:.3e}  n * error = {n * e:.3f}")
print(f"log-log slope {slope:.3f}")

```

Two refinements are quoted without proof. For a positive smooth  $f$ , applying [Theorem 39.4.2](#) to  $\exp\{q_n + \log f\}$  in the numerator of  $E\{f(\boldsymbol{\beta}) \mid \mathbf{y}\} = \int f e^{q_n} / \int e^{q_n}$  raises the accuracy of the ratio to  $O(n^{-2})$ , the leading errors cancelling (Tierney and Kadane, 1986); and the Bernstein–von Mises theorem says the posterior of  $\mathbf{H}_n^{1/2}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}_n)$  converges in total variation to  $N_p(\mathbf{0}, \mathbf{I})$ , so a credible set is asymptotically a confidence set (van der Vaart, 1998, chapter 10).

### 39.4.3 Sampling the posterior

When  $n$  is small, or the posterior skewed as in [Example \(How good is the approximation\)](#), an approximation at the mode is not enough. The alternative is a Markov chain whose stationary distribution is the posterior.

#### Definition 39.4.2 — Metropolis–Hastings, Gibbs, and the two diagnostics

Let  $\pi(\cdot \mid \mathbf{y})$  be a density known up to a constant.

- (a) The **Metropolis–Hastings** chain with proposal density  $q(\cdot \mid \cdot)$  moves from  $\boldsymbol{\beta}^{(t)}$  by drawing  $\boldsymbol{\beta}^* \sim q(\cdot \mid \boldsymbol{\beta}^{(t)})$  and setting  $\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^*$  with probability

$$\alpha(\boldsymbol{\beta}^* \mid \boldsymbol{\beta}^{(t)}) = \min\left\{1, \frac{\pi(\boldsymbol{\beta}^* \mid \mathbf{y}) q(\boldsymbol{\beta}^{(t)} \mid \boldsymbol{\beta}^*)}{\pi(\boldsymbol{\beta}^{(t)} \mid \mathbf{y}) q(\boldsymbol{\beta}^* \mid \boldsymbol{\beta}^{(t)})}\right\}, \quad (39.4.6)$$

and  $\boldsymbol{\beta}^{(t+1)} = \boldsymbol{\beta}^{(t)}$  otherwise. The normalizing constant cancels. A symmetric proposal,  $q(\mathbf{b} \mid \mathbf{b}') = q(\mathbf{b}' \mid \mathbf{b})$ , gives the **random-walk Metropolis** algorithm, in which  $\alpha$  is the ratio of posterior densities capped at one.

- (b) The **Gibbs sampler** partitions  $\boldsymbol{\psi} = (\boldsymbol{\psi}_1, \dots, \boldsymbol{\psi}_K)$  and replaces each block in turn by a draw from its **full conditional**  $\pi(\boldsymbol{\psi}_k \mid \boldsymbol{\psi}_{-k}, \mathbf{y})$ . It is the special case of (a) in which every proposal is accepted.
- (c) For  $c$  chains of length  $N$ , split each in half and let  $W$  be the mean of the  $2c$  within-sequence variances and  $B$  the variance of the  $2c$  sequence means. The **potential scale reduction** is

$$\hat{R} = \left\{ \frac{(N/2 - 1)W/(N/2) + B}{W} \right\}^{1/2}. \quad (39.4.7)$$

- (d) With  $\rho_t$  the lag- $t$  autocorrelation of the stationary chain, the **effective sample size** of  $cN$  draws is  $\text{ESS} = cN/(1 + 2 \sum_{t \geq 1} \rho_t)$ , the sum truncated by Geyer’s rule in the form used here: add the autocorrelations in consecutive pairs  $\rho_{2k-1} + \rho_{2k}$ ,



$k = 1, 2, \dots$ , and stop at the first pair that is negative.

### Proposition 39.4.4 — The chain has the right stationary distribution

Let  $\pi$  be a density on  $\mathbb{R}^p$  and  $q(\cdot | \cdot)$  a transition density with  $q(\mathbf{b}' | \mathbf{b}) > 0$  whenever  $q(\mathbf{b} | \mathbf{b}') > 0$ . The Metropolis–Hastings kernel of Eq. 39.4.6 satisfies detailed balance with respect to  $\pi$ :

$$\pi(\mathbf{b}) q(\mathbf{b}' | \mathbf{b}) \alpha(\mathbf{b}' | \mathbf{b}) = \pi(\mathbf{b}') q(\mathbf{b} | \mathbf{b}') \alpha(\mathbf{b} | \mathbf{b}') \quad \text{for all } \mathbf{b} \neq \mathbf{b}',$$

and consequently  $\pi$  is a stationary distribution of the chain.

*Proof.* Fix  $\mathbf{b} \neq \mathbf{b}'$  and write  $u = \pi(\mathbf{b})q(\mathbf{b}' | \mathbf{b})$ ,  $v = \pi(\mathbf{b}')q(\mathbf{b} | \mathbf{b}')$ . By Eq. 39.4.6,  $\alpha(\mathbf{b}' | \mathbf{b}) = \min(1, v/u)$  and  $\alpha(\mathbf{b} | \mathbf{b}') = \min(1, u/v)$ , so both sides of the identity equal  $\min(u, v)$ . For stationarity let  $P(\mathbf{b}, \cdot)$  be the kernel; for any measurable  $A$ ,

$$\int P(\mathbf{b}, A) \pi(\mathbf{b}) d\mathbf{b} = \int_A \int \pi(\mathbf{b}) q(\mathbf{b}' | \mathbf{b}) \alpha(\mathbf{b}' | \mathbf{b}) d\mathbf{b} d\mathbf{b}' + \int_A \pi(\mathbf{b}') \varrho(\mathbf{b}') d\mathbf{b}',$$

where  $\varrho(\mathbf{b}') = 1 - \int q(\mathbf{b}'' | \mathbf{b}') \alpha(\mathbf{b}'' | \mathbf{b}') d\mathbf{b}''$  is the probability of staying put. By detailed balance the inner integral of the first term is  $\pi(\mathbf{b}') \int q(\mathbf{b} | \mathbf{b}') \alpha(\mathbf{b} | \mathbf{b}') d\mathbf{b} = \pi(\mathbf{b}') \{1 - \varrho(\mathbf{b}')\}$ , so the two terms add to  $\int_A \pi(\mathbf{b}') d\mathbf{b}'$ . ■

Detailed balance is all that is checked here; convergence to  $\pi$  from an arbitrary start needs irreducibility and aperiodicity, and a central limit theorem for the ergodic averages more still (Tierney, 1994). The one design decision is the proposal. A covariance  $\varsigma^2 \mathbf{\Sigma}_{\text{prop}}$  mixes well only when  $\mathbf{\Sigma}_{\text{prop}}$  resembles the posterior covariance, and the Laplace matrix  $\mathbf{H}_n^{-1}$  of Theorem 39.4.2 is the obvious candidate, at the cost of one Newton solve. Roberts, Gelman and Gilks (1997) showed that in a high-dimensional limit the optimal scale is  $\varsigma = 2.38/\sqrt{p}$ , with acceptance rate near 0.234 — derived for a product target, used everywhere as a rule of thumb.

## 39.4.4 Data augmentation for the probit model

A Gibbs sampler needs full conditionals that can be drawn from, and Proposition 39.3.1 says the conditional of  $\boldsymbol{\beta}$  is not normal. Albert and Chib (1993) observed that enlarging the parameter space restores normality, through the latent variable representation of the probit model (Definition 35.2.1).

### Proposition 39.4.5 — The Albert–Chib sampler

Consider the ungrouped probit model  $\Pr(Y_i = 1) = \Phi(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$  with prior  $\boldsymbol{\beta} \sim N_p(\mathbf{m}, \mathbf{\Sigma})$ . Introduce latent variables  $Z_i | \boldsymbol{\beta} \sim N(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}, 1)$ , independent, and set  $Y_i = \mathbf{1}\{Z_i > 0\}$ . Then:

- (a) the marginal distribution of  $\mathbf{Y}$  given  $\boldsymbol{\beta}$  is the probit model, so the posterior of  $\boldsymbol{\beta}$

is unchanged;

- (b) given  $\mathbf{z}$  and  $\mathbf{y}$ ,  $\boldsymbol{\beta} \sim N_p(\mathbf{V}(\mathbf{X}^\top \mathbf{z} + \boldsymbol{\Sigma}^{-1} \mathbf{m}), \mathbf{V})$  with  $\mathbf{V} = (\mathbf{X}^\top \mathbf{X} + \boldsymbol{\Sigma}^{-1})^{-1}$ ;
- (c) given  $\boldsymbol{\beta}$  and  $\mathbf{y}$ , the  $Z_i$  are independent,  $Z_i$  being  $N(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}, 1)$  truncated to  $(0, \infty)$  if  $y_i = 1$  and to  $(-\infty, 0)$  if  $y_i = 0$ .

*Proof.*

- (a)  $\Pr(Y_i = 1 \mid \boldsymbol{\beta}) = \Pr(Z_i > 0 \mid \boldsymbol{\beta}) = \Phi(\mathbf{x}_{(i)}^\top \boldsymbol{\beta})$ , and the  $Z_i$  are independent given  $\boldsymbol{\beta}$ , hence so are the  $Y_i$ .
- (b) The joint density of  $(\boldsymbol{\beta}, \mathbf{z}, \mathbf{y})$  is proportional to

$$\exp\left\{-\frac{1}{2}(\boldsymbol{\beta} - \mathbf{m})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{\beta} - \mathbf{m}) - \frac{1}{2}\|\mathbf{z} - \mathbf{X}\boldsymbol{\beta}\|^2\right\} \prod_i \mathbf{1}\{\text{sign}(z_i) \text{ matches } y_i\}.$$

With  $\mathbf{z}$  fixed the indicator is constant in  $\boldsymbol{\beta}$ , and the exponent is that of the conjugate normal update of [Exercise \(A2\)](#) with  $\sigma^2 = 1$ , response  $\mathbf{z}$  and prior precision  $\boldsymbol{\Sigma}^{-1}$ ; completing the square as in the proof of [Theorem 7.6.2](#) gives the stated normal law.

- (c) With  $\boldsymbol{\beta}$  fixed, the joint density factors over  $i$  into  $\phi(z_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta})$  times the indicator, which is the stated truncated normal density up to its normalizing constant.

■

The price is  $n$  extra draws per sweep; the gain is that every step is accepted and nothing needs tuning. The same device works for the logit link by a different route — the logistic distribution is a scale mixture of normals (Holmes and Held, 2006), the binomial likelihood a mixture over Pólya–gamma variables (Polson, Scott and Windle, 2013) — and, more intricately, for multinomial and ordinal models.

#### Example. (The 1996 survey, sampled)

Return to the logistic regression of [Example \(Voting intention in the 1996 American election\)](#): the intended vote of 944 respondents on party identification, age in decades, education and income. Put an independent  $N(0, 10^2)$  prior on each of the 5 coefficients — nearly flat on this scale — and run four random-walk Metropolis chains of 8000 draws from starts 8 posterior standard deviations away in every coordinate, with the Laplace covariance as the proposal shape and scale  $2.38/\sqrt{5}$ . The acceptance rate is 0.298, and discarding the first half of each chain leaves 16000 draws.

Panel (a) of [Figure 39.4.2](#) shows the four chains for the party coefficient: two of them climb far above the mode before turning back, and all four are indistinguishable once the first few hundred iterations are past. Panel (b) applies the whole recipe — discard the first half, then split what is left — at each run length, and plots  $\hat{R} - 1$  on a log scale. It starts at 5.63 with 200 draws per chain, is below 1.05 by 1000 and below 1.02 by 1400, and its own Monte Carlo variability keeps it brushing the 1.01 line until about 6000; at the full eight thousand the largest value over the five coefficients is 1.0066. Effective sample size is the binding constraint: 1079, or 0.067 effective draws per iteration.

The posterior mean of that coefficient is 1.2300 with posterior standard deviation 0.0705, against the maximum likelihood 1.2188 and standard error 0.0709: with  $n = 944$  and a prior this weak the two analyses agree to the second decimal, as [Theorem 39.4.2](#) and the

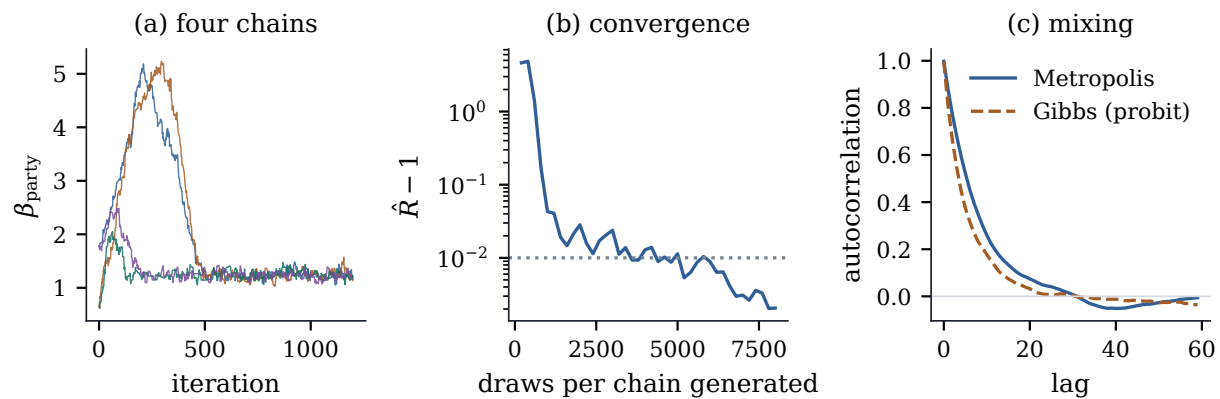


Figure 39.4.2: Diagnostics for the samplers of [Example \(The 1996 survey, sampled\)](#). (a) The first 1200 draws of four Metropolis chains for the party-identification coefficient, from starts eight posterior standard deviations away in every coordinate. (b)  $\hat{R} - 1$ , on a log scale, against draws generated per chain, recomputed from scratch at each length after discarding the first half; the reference line is  $\hat{R} = 1.01$ . (c) Autocorrelation functions of the Metropolis and Albert–Chib chains.

Bernstein–von Mises theorem say they must. What the simulation buys is the ability to ask for any posterior summary at all.

The probit version, sampled by [Proposition 39.4.5](#) with 4000 draws per chain, gives posterior mean 0.6780 against the maximum likelihood 0.6733; the ratio of the logistic coefficient to it is 1.814, against the  $\pi/\sqrt{3} = 1.8138$  of [Proposition 35.2.2](#). Its effective sample size per iteration is 0.092, half again as good and with no tuning, though each iteration costs  $n$  truncated normal draws. Panel (c) compares the two autocorrelation functions.

```
import numpy as np
import statsmodels.api as sm
from scipy import special, stats
anes = sm.datasets.anes96.load_pandas().data
y = anes["vote"].to_numpy(float)           # 1 if the respondent
                                           # expected to vote Dole
X = np.column_stack([np.ones(len(y)), anes["PID"], anes["age"] / 10.0,
                     anes["educ"], anes["income"]])
n, p = X.shape
TAU = 10.0                                # prior: beta ~ N(0, TAU^2
                                           # I)

mle = sm.GLM(y, X, family=sm.families.Binomial()).fit()
w = mle.fittedvalues * (1 - mle.fittedvalues) # the working weights at
                                           # the mode
V_prop = np.linalg.inv(X.T @ (w[:, None] * X) + np.eye(p) / TAU**2) # Laplace
                                           # covariance
CHOL = np.linalg.cholesky(V_prop)
print("maximum likelihood:", mle.params.round(4))

def log_post(beta):
    eta = X @ beta
    return np.sum(y * eta - np.logaddexp(0.0, eta)) - np.sum(beta**2) / (2 * TAU**2)
```

```

def metropolis(n_draws, seed, start, step=2.38 / np.sqrt(5)):
    """Random-walk Metropolis with the Laplace covariance as the proposal shape."""
    rng = np.random.default_rng(seed)
    beta, lp = start.copy(), log_post(start)
    out, accepted = np.empty((n_draws, p)), 0
    for t in range(n_draws):
        prop = beta + step * (CHOL @ rng.standard_normal(p))
        lp_prop = log_post(prop)
        if np.log(rng.random()) < lp_prop - lp:          # the acceptance
            probability
            beta, lp, accepted = prop, lp_prop, accepted + 1
        out[t] = beta
    return out, accepted / n_draws

def split_rhat(draws):
    """Gelman-Rubin statistic on the two halves of every chain."""
    c, N = draws.shape
    s = draws.reshape(2 * c, N // 2)
    means, variances = s.mean(axis=1), s.var(axis=1, ddof=1)
    B, W = means.var(ddof=1), variances.mean()
    return np.sqrt(((N // 2 - 1) / (N // 2) * W + B) / W)

def ess(draws):
    """Effective sample size, with Geyer's initial positive sequence."""
    c, N = draws.shape
    centred = draws - draws.mean(axis=1, keepdims=True)
    var = draws.var(ddof=1)
    rho = np.array([np.mean([np.dot(x[:N - t], x[t:]) / N for x in centred]) / var
                     for t in range(1, min(N, 400))])
    pairs = rho[: 2 * (len(rho) // 2)].reshape(-1, 2).sum(axis=1)
    k = np.argmax(pairs < 0) if np.any(pairs < 0) else len(pairs)
    return c * N / (1 + 2 * pairs[:k].sum())

short, short_rates = zip(*[metropolis(1500, 3960 + c, mle.params) for c in range(2)
])
short = np.array(short)[: , 750: , :]
print(f"two short chains: acceptance {np.mean(short_rates):.3f}, "
      f"party id mean {short[: , :, 1].mean():.4f}, R-hat {split_rhat(short[: , :, 1])
      :.4f}, "
      f"ESS {ess(short[: , :, 1]):.0f}")

```

```

def albert_chib(n_draws, seed, start):
    """Gibbs for the probit model: latent normals, then a normal draw for beta."""
    rng = np.random.default_rng(seed)
    V = np.linalg.inv(X.T @ X + np.eye(p) / TAU**2)
    L = np.linalg.cholesky(V)
    beta, out = start.copy(), np.empty((n_draws, p))
    for t in range(n_draws):
        eta = X @ beta
        lower = stats.norm.cdf(-eta)                      # P(latent < 0)
        u = np.where(y > 0, lower + (1 - lower) * rng.random(n), lower * rng.random(
n))
        z = eta + stats.norm.ppf(np.clip(u, 1e-12, 1 - 1e-12))
        mean = V @ (X.T @ z)
        beta = mean + L @ rng.standard_normal(p)

```

```

        out[t] = beta
    return out

gibbs_short = np.array([albert_chib(600, 3970 + c, np.zeros(p)) for c in range(2)])
[:, 300:, :]
print(f"short Gibbs run: probit party id mean {gibbs_short[:, :, 1].mean():.4f}, "
      f"R-hat {split_rhat(gibbs_short[:, :, 1]):.4f}")

```

### 39.4.5 Checking the fit from the posterior

#### Definition 39.4.3 — Posterior predictive distribution and check

The **posterior predictive** distribution of a replicate data set  $\mathbf{Y}^{\text{rep}}$  is

$$p(\mathbf{y}^{\text{rep}} | \mathbf{y}) = \int f(\mathbf{y}^{\text{rep}} | \boldsymbol{\beta}) \pi(\boldsymbol{\beta} | \mathbf{y}) d\boldsymbol{\beta}, \quad (39.4.8)$$

which is sampled by drawing  $\boldsymbol{\beta}$  from the posterior and then  $\mathbf{y}^{\text{rep}}$  from the model. For a statistic  $T$  the **posterior predictive  $p$ -value** is  $\Pr\{T(\mathbf{Y}^{\text{rep}}) \geq T(\mathbf{y}) | \mathbf{y}\}$ , or the lower-tail version  $\Pr\{T(\mathbf{Y}^{\text{rep}}) \leq T(\mathbf{y}) | \mathbf{y}\}$  when small values of  $T$  are the departure of interest, estimated by the proportion of replicates on the relevant side of the observed value.

The check is only as good as the statistic. One the model was fitted to match — the total number of successes in a model with an intercept — has a  $p$ -value near 1/2 by construction; a useful statistic is one the model does *not* control, such as a subgroup total, a count of zeros or a maximum.

*Warning.* The data are used twice: once to form the posterior and again to compute  $T(\mathbf{y})$ . The resulting  $p$ -value is conservative — its distribution under the model is concentrated towards 1/2 rather than uniform — so a middling value is weak evidence of nothing while an extreme one is strong. Bayarri and Berger (2000) quantify the conservatism; Meng (1994) and Gelman, Meng and Stern (1996) give the general framework, including discrepancies  $T(\mathbf{y}, \boldsymbol{\beta})$  that depend on the parameter too.

#### Example. (Where the survey model fails)

The model of [Example \(The 1996 survey, sampled\)](#) treats party identification as a single linear term. Take  $T$  to be the number of Dole votes among the 108 respondents who called themselves independent Democrats — a subgroup the model has no parameter for. Observed: 7; across four thousand posterior predictive replicates the mean is 14.4, and only 0.034 of them are as low as seven ([Figure 39.4.3](#), panel (a)). Given the conservatism just described, that is a real signal.

It is not one a quadratic term would catch: adding  $\text{PID}^2$  drops the deviance by 0.019, while treating party identification as a seven-level factor drops it by 12.68 on five degrees of freedom,  $p = 0.027$  — the same finding from the likelihood side. Panel (b) shows why: at average covariates the fitted curve passes well above the observed proportion at

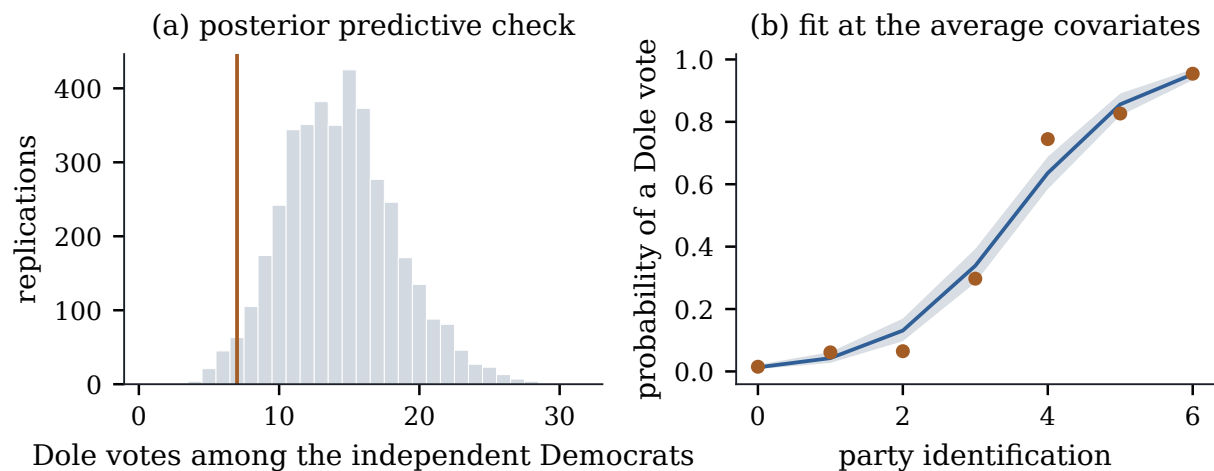


Figure 39.4.3: (a) Posterior predictive distribution of the number of Dole votes among the 108 independent Democrats, with the observed value marked. (b) Fitted probability of a Dole vote against party identification at the average of the other covariates, with a 95 per cent pointwise credible band and the raw category proportions (unadjusted).

party identification two and below it at four. The monotone logistic curve is not flexible enough near the middle of this scale.

```
group = anes["PID"].to_numpy() == 2                                # the independent
    Democrats
observed = y[group].sum()
short4 = np.array([metropolis(2000, 3980 + c, mle.params)[0] for c in range(4)])[:,
    1000:, :]
draws_short = short4.reshape(-1, p)
rng2 = np.random.default_rng(3990)
idx2 = rng2.choice(len(draws_short), 2000, replace=False)
rep_short = np.array([rng2.binomial(1, special.expit(X @ draws_short[i]))[group].sum()
    ()
    for i in idx2])
print(f"short run: {int(observed)} of {int(group.sum())} against {rep_short.mean()
    :.1f} "
    f"replicated; posterior predictive p-value {np.mean(rep_short <= observed):.3f
    }")
```

## 39.4.6 Exercises

### 39.4.6.1 A. Check your understanding

#### Exercise A1.

Show that for a symmetric proposal the acceptance probability [Eq. 39.4.6](#) reduces to  $\min\{1, \exp(q_n(\boldsymbol{\beta}^*) - q_n(\boldsymbol{\beta}^{(t)}))\}$ , and that an uphill move is always accepted.

*Solution.* Symmetry gives  $q(\boldsymbol{\beta}^{(t)} \mid \boldsymbol{\beta}^*) = q(\boldsymbol{\beta}^* \mid \boldsymbol{\beta}^{(t)})$ , so those factors cancel and  $\alpha = \min\{1, \pi(\boldsymbol{\beta}^* \mid \mathbf{y})/\pi(\boldsymbol{\beta}^{(t)} \mid \mathbf{y})\}$ . The normalizing constant cancels too, leaving the

ratio of the unnormalized densities, which is  $\exp\{q_n(\boldsymbol{\beta}^*) - q_n(\boldsymbol{\beta}^{(t)})\}$ . If  $q_n(\boldsymbol{\beta}^*) \geq q_n(\boldsymbol{\beta}^{(t)})$  the ratio is at least one and  $\alpha = 1$ .

### Exercise A2.

A chain is run once, from a start far from the posterior mode, and  $\hat{R}$  computed from its two halves is 1.00. Is that evidence of convergence? What does the definition in Eq. 39.4.7 require?

*Solution.* No.  $\hat{R}$  near one says only that the pieces compared agree; a single chain stuck in one region has two halves that agree perfectly. The definition compares  $2c$  sequences, and the value of the diagnostic comes from starting the  $c$  chains *over-dispersed relative to the posterior*, so that disagreement is detectable. A single chain also cannot reveal a second mode it never visited.

### 39.4.6.2 B. Practice

#### Exercise B1.

Take  $q_n(\beta) = n g(\beta)$  for a fixed smooth  $g$  with a unique interior maximum at  $\beta^*$  and  $g''(\beta^*) < 0$ , and suppose  $g$  is bounded away from its maximum outside any neighbourhood of  $\beta^*$ . Verify (L1)–(L3) of Theorem 39.4.2 and write out the approximation explicitly.

#### Exercise B2.

Corollary 39.4.3 says the prior enters only the  $O(1)$  remainder. Show that if the prior is itself allowed to depend on  $n$  — say  $\boldsymbol{\beta} \sim N_p(\mathbf{0}, n\boldsymbol{\Sigma}_0)$  with  $\boldsymbol{\Sigma}_0$  fixed, so that the prior is made vaguer as data accumulate — then the remainder grows like  $p \log n$ , and say what that does to the BIC.

*Solution.* With  $\pi$  the  $N_p(\mathbf{0}, n\boldsymbol{\Sigma}_0)$  density,  $-2 \log \pi(\hat{\boldsymbol{\beta}}_n) = p \log(2\pi) + p \log n + \log \det \boldsymbol{\Sigma}_0 + \hat{\boldsymbol{\beta}}_n^\top \boldsymbol{\Sigma}_0^{-1} \hat{\boldsymbol{\beta}}_n / n$ , so the term  $-2 \log \pi(\hat{\boldsymbol{\beta}}_n)$  in the remainder grows like  $p \log n$  instead of staying bounded. The expansion then reads  $-2 \log m(\mathbf{y}) = -2\ell(\hat{\boldsymbol{\beta}}) + 2p \log n + O(1)$ : a prior whose spread grows with  $n$  doubles the dimension penalty. This is Lindley's paradox in miniature — a prior that is made vaguer as data accumulate penalizes extra parameters more and more heavily — and it is why the BIC's derivation assumes a fixed prior.

#### Exercise B3.

A chain of 16 000 draws has effective sample size 1000. A colleague thins it by keeping every sixteenth draw, obtaining 1000 nearly independent draws, and says nothing has been lost. Is that right? What is thinning good for?

### 39.4.6.3 C. Going deeper

**Exercise C1.**

Show that the Gibbs update of a block  $\boldsymbol{\psi}_k$  is a Metropolis–Hastings step with proposal  $q(\boldsymbol{\psi}^* \mid \boldsymbol{\psi}) = \pi(\boldsymbol{\psi}_k^* \mid \boldsymbol{\psi}_{-k}, \mathbf{y})$  and that its acceptance probability is one. Deduce that [Proposition 39.4.4](#) covers the Gibbs sampler, and explain why a Gibbs sampler can nevertheless fail to be irreducible.

## 39.5 Priors as regularizers

[Proposition 30.1.4](#) already said that every penalty is a prior and every prior a penalty: the maximizer of a penalized log-likelihood is the mode of a posterior whose log prior is minus the penalty. This section takes that identity in the other direction and asks what a prior *does* in a generalized linear model. The answer is more than shrinkage: a prior carrying almost no information can still be the difference between an estimate and no estimate at all.

### 39.5.1 Two correspondences and a warning about scale

#### Proposition 39.5.1 — What a prior does

Let  $\ell$  be the log-likelihood of a generalized linear model with  $\text{rank}(\mathbf{X}) = p$  and a concave log-likelihood ([Theorem 34.3.3](#)), and write  $\boldsymbol{\beta} = (\beta_0, \boldsymbol{\gamma}^\top)^\top$  with  $\beta_0$  an intercept.

- (a) (*Normal prior, ridge penalty.*) Under  $\gamma_j \sim N(0, \tau^2)$  independently and a flat prior on  $\beta_0$ , the posterior mode maximizes  $\ell(\boldsymbol{\beta}) - \frac{1}{2}\lambda\|\boldsymbol{\gamma}\|^2$  with  $\lambda = 1/\tau^2$ . If  $\ell$  is bounded above, as it is for the binomial and the Poisson, and the response is not constant — for the binomial, both outcomes occur — the mode exists and is unique whether or not the data are separated. The hypothesis on the response cannot be dropped: with a flat prior on  $\beta_0$  the objective is not coercive in  $\beta_0$  when every  $y_i$  is 1. Penalizing the intercept as well makes existence unconditional.
- (b) (*Laplace prior, lasso penalty.*) Under  $\gamma_j$  independent with density  $(2b)^{-1}e^{-|\gamma_j|/b}$  and a flat prior on  $\beta_0$ , the posterior mode maximizes  $\ell(\boldsymbol{\beta}) - \lambda \sum_j |\gamma_j|$  with  $\lambda = 1/b$ , and is therefore the lasso-penalized estimate of [Definition 27.4.1](#), with exact zeros. The *posterior*, by contrast, is never sparse: whenever it is proper,  $\Pr(\gamma_j = 0 \mid \mathbf{y}) = 0$  for every  $j$ .
- (c) (*What a normal prior says about a probability.*) In a binary model with logit link, suppose the linear predictor at some covariate value has the prior  $\eta \sim N(0, \sigma^2)$ . The induced prior density of  $\pi = e^\eta/(1 + e^\eta)$  on  $(0, 1)$  is unimodal with mode  $1/2$  if  $\sigma^2 \leq 2$ , and has exactly two modes, symmetric about  $1/2$ , with a local minimum at  $1/2$ , if  $\sigma^2 > 2$ .
- (d) (*Existence.*) Under the prior of (a), or under independent Cauchy priors on the  $\gamma_j$  with a flat prior on  $\beta_0$ , the logistic posterior mode exists for every data set in which both outcomes occur, separated or not; under (a) it is also unique. Under any proper prior the posterior itself is proper ([Proposition 39.4.1\(a\)](#)), so posterior means and credible intervals exist whether or not the maximum likelihood estimate



does.

*Proof.*

- (a) The log posterior is  $\ell(\boldsymbol{\beta}) - \|\boldsymbol{\gamma}\|^2/(2\tau^2)$  up to a constant, which is the stated penalized log-likelihood with  $\lambda = 1/\tau^2$ ; this is [Proposition 30.1.4](#) for the normal prior. Existence and uniqueness for the logistic model with the intercept left out of the penalty are [Exercise \(B1\)](#), under exactly the stated hypothesis; the same argument — a concave  $\ell$  bounded above, coercive in  $\beta_0$  at fixed  $\boldsymbol{\gamma}$ , plus a strictly concave coercive quadratic — applies to any family whose log-likelihood is concave and bounded above. When  $\boldsymbol{\gamma}$  is penalized and  $\beta_0$  is not, and every  $y_i$  equals 1, the supremum 0 is approached only along  $\boldsymbol{\gamma} = \mathbf{0}$ ,  $\beta_0 \rightarrow +\infty$  and is not attained.
- (b) The log prior is  $-\sum_j |\gamma_j|/b$  up to a constant, giving the penalty with  $\lambda = 1/b$ ; this is [Proposition 30.1.4](#) for the Laplace prior. For the second claim, the unnormalized posterior density is positive and continuous on  $\mathbb{R}^p$ , so when it integrates the marginal posterior of each  $\gamma_j$  is absolutely continuous and  $\Pr(\gamma_j = 0 \mid \mathbf{y}) = 0$ . The sparsity of the lasso is a property of the *mode* of a density with a kink, not of the distribution; [Example \(Posterior mode and posterior mean under a Laplace prior\)](#) makes the same point in the orthonormal linear model. Note what is *not* claimed: the posterior mean can be exactly zero, as it is whenever the posterior happens to be symmetric in  $\gamma_j$  about the origin.
- (c) Write  $t = \log\{\pi/(1-\pi)\}$ , so  $dt/d\pi = 1/\{\pi(1-\pi)\} > 0$ . The induced density satisfies

$$f(\pi) \propto \exp\left(-\frac{t^2}{2\sigma^2}\right) \frac{1}{\pi(1-\pi)},$$

with a constant of proportionality free of  $\pi$ . Since  $t \mapsto \pi$  is strictly increasing, the stationary points of  $f$  in  $\pi$  are the stationary points of  $\psi(t) = \log f(\pi(t))$  in  $t$ . Using  $\pi(1-\pi) = e^t/(1+e^t)^2$ ,

$$\begin{aligned}\psi(t) &= -\frac{t^2}{2\sigma^2} + 2\log(1+e^t) - t + \text{const}, \\ \psi'(t) &= -\frac{t}{\sigma^2} + \frac{2e^t}{1+e^t} - 1 = \tanh(t/2) - \frac{t}{\sigma^2}.\end{aligned}$$

Call the last expression  $G(t)$ ; it is odd and  $G(0) = 0$ . Now  $G'(t) = \frac{1}{2} \text{sech}^2(t/2) - 1/\sigma^2$ , which is even, strictly decreasing in  $|t|$ , and has maximum  $\frac{1}{2} - 1/\sigma^2$  at  $t = 0$ . If  $\sigma^2 \leq 2$  then  $G' \leq 0$  everywhere, with equality only at  $t = 0$  in the boundary case, so  $G$  is strictly decreasing and  $t = 0$  is its only root:  $f$  has one stationary point. It is a maximum because  $f(\pi) \rightarrow 0$  as  $\pi \rightarrow 0$  or  $\pi \rightarrow 1$ , which follows from  $\exp\{-t^2/(2\sigma^2) + |t|\} \rightarrow 0$ .

If  $\sigma^2 > 2$  then  $G'(0) > 0$  and  $G'$  has exactly two zeros, so  $G$  has at most three roots; it has at least three, because  $G(0) = 0$ ,  $G > 0$  just to the right of zero while  $G(t) \rightarrow -\infty$  as  $t \rightarrow \infty$ , and by oddness on the left. So there are roots  $t_- < 0 = t_0 < t_+$  with  $t_- = -t_+$ . Since  $f \rightarrow 0$  at both ends and  $f$  is continuous and positive, the outer two are maxima and the middle one, at  $\pi = 1/2$ , is a minimum.

- (d) With the normal prior the objective is coercive and strictly concave, as in (a). With independent Cauchy priors of scales  $s_j$ , the log posterior is  $\ell(\boldsymbol{\beta}) - \sum_j \log\{1 + (\gamma_j/s_j)^2\}$  up to a constant. For the logistic model  $\ell \leq 0$ , and the penalty tends to  $-\infty$  as  $\|\boldsymbol{\gamma}\| \rightarrow \infty$ ; if  $\boldsymbol{\gamma}$  stays bounded while  $|\beta_0| \rightarrow \infty$  then  $\ell \rightarrow -\infty$ , provided both outcomes occur, by the argument of [Exercise \(B1\)](#). So the objective is

coercive and continuous and attains its maximum on a compact sublevel set. It need not be unique: the Cauchy log density is not concave. The last sentence is [Proposition 39.4.1\(a\)](#).

■

Part (c) is the warning. A prior that is “noninformative” about a coefficient is a strong statement about a probability: a  $N(0, 10^2)$  prior on the linear predictor puts half of its mass on probabilities below 0.0012 or above  $1 - 0.0012$ , saying before any data that the event is nearly certain or nearly impossible and that the analyst does not know which. Panel (a) of [Figure 39.5.1](#) shows the three shapes: a gentle hump at  $\sigma = 1$ , flat at the boundary  $\sigma = \sqrt{2}$ , a deep U at  $\sigma = 4$ .

This is why the default priors recommended for logistic regression are much tighter than “vague”. Gelman, Jakulin, Pittau and Su (2008) rescale every covariate to mean zero and standard deviation 1/2, then put an independent Cauchy prior of scale 2.5 on each coefficient and 10 on the intercept. On that scale a coefficient of 5 multiplies the odds by  $e^{2.5} \approx 12$  for a one standard deviation change, beyond anything social or biomedical data produce; the Cauchy’s heavy tail nonetheless leaves a genuinely large coefficient nearly alone, which a normal prior of the same scale would not.

## 39.5.2 Priors as penalties, checked

*Remark* (Scaling is not optional). Both correspondences put the same  $\lambda$  on every coefficient, which is a statement about the units of the covariates: multiplying  $x_j$  by ten divides  $\gamma_j$  by ten and changes its penalty by a factor of a hundred. Every penalized or weakly-informative fit therefore standardizes first ([Proposition 27.2.1](#), [Proposition 30.2.3](#)). A prior of fixed scale is meaningful only once the covariates have one.

On the 944 respondents of [Example \(Voting intention in the 1996 American election\)](#), with the four covariates standardized, maximizing  $\ell(\boldsymbol{\beta}) - \frac{1}{2}\lambda\|\boldsymbol{\gamma}\|^2$  by Newton’s method and maximizing  $\ell(\boldsymbol{\beta}) - \lambda\sum_j |\gamma_j|$  by coordinate descent gives the two paths. At  $\lambda = 64$ , which is a normal prior with  $\tau = 1/8$ , the ridge estimates of the four slopes are 1.465, 0.105, 0.032 and 0.153; the lasso estimates are 1.773 and three exact zeros. Both have shrunk the party-identification coefficient from its unpenalized value of 2.769 on this standardized scale, the ridge further than the lasso, because the lasso has removed the competitors and the party term absorbs what they explained.

```
import numpy as np
import statsmodels.api as sm
from scipy import optimize, special, stats
anes = sm.datasets.anes96.load_pandas().data
Xf = np.column_stack([anes["PID"], anes["age"], anes["educ"], anes["income"]])
Xf = (Xf - Xf.mean(0)) / Xf.std(0)                                # standardize, as any
                           penalty requires
Xf = np.column_stack([np.ones(len(Xf)), Xf])
yf = anes["vote"].to_numpy(float)

def map_estimate(penalty, lam, steps=30):
    """Posterior mode: maximize the log-likelihood minus lam times a penalty on the
    slopes.

    A normal prior gives the ridge penalty and is solved by Newton's method; a
```

```

Laplace
prior gives the lasso penalty and is solved by coordinate descent on the working
least squares problem of each Newton step.
"""
beta = np.zeros(Xf.shape[1])
for _ in range(steps):
    pi = special.expit(Xf @ beta)
    w = np.maximum(pi * (1 - pi), 1e-8)
    z = Xf @ beta + (yf - pi) / w                                # the working response
    if penalty == "ridge":
        P = np.diag([0.0] + [1.0] * (Xf.shape[1] - 1))
        beta = np.linalg.solve(Xf.T @ (w[:, None] * Xf) + lam * P, Xf.T @ (w * z
    ))
        continue
    for _ in range(40):                                          # coordinate descent for the
        lasso
        for j in range(Xf.shape[1]):
            r = z - Xf @ beta + Xf[:, j] * beta[j]
            num, den = np.sum(w * Xf[:, j] * r), np.sum(w * Xf[:, j] ** 2)
            beta[j] = (num / den if j == 0
                        else np.sign(num) * max(abs(num) - lam, 0.0) / den)
    return beta

unpenalized = map_estimate("ridge", 0.0)                        # the maximum likelihood fit
lams = np.array([1.0, 4.0, 16.0, 64.0, 256.0])
ridge_path = np.array([map_estimate("ridge", l) for l in lams])
lasso_path = np.array([map_estimate("lasso", l) for l in lams])
print(f"unpenalized slopes: {np.round(unpenalized[1:], 3)}")
print("normal prior with sd tau = 1/sqrt(lam) -> ridge; Laplace prior with b = 1/
    lam -> lasso")
for k, l in enumerate(lams):
    print(f" lam {l:6.1f} ridge {np.round(ridge_path[k, 1:], 3)} "
          f"lasso {np.round(lasso_path[k, 1:], 3)}")

```

The Bayesian reading does not stop at the mode. Park and Casella (2008) noted that the Laplace prior is a scale mixture of normals —  $\gamma_j \mid \tau_j^2 \sim N(0, \tau_j^2)$  with  $\tau_j^2$  exponential — which turns the lasso into a hierarchical model with normal full conditionals, and so into a Gibbs sampler whose output is the whole posterior rather than a sparse point. Priors designed to be *more* selective make the mixing distribution heavy-tailed at both ends: a spike-and-slab prior mixes a point mass at zero with a diffuse component, and the horseshoe of Carvalho, Polson and Scott (2010) takes  $\tau_j$  half-Cauchy, so small coefficients are shrunk almost to zero and large ones almost not at all. The other penalties of [Chapter 30](#) carry over by the same substitution, penalized least squares becoming penalized log-likelihood: the elastic net of [Definition 30.2.1](#), the group lasso of [Definition 30.3.1](#) for a factor's dummy variables, and the componentwise boosting of [Definition 30.4.1](#) with the negative log-likelihood as loss and one Newton step per round.

### 39.5.3 A prior that rescues an estimate

**Example.** (The separated subgroup, three priors)

[Example \(A separated subgroup\)](#) fitted the intended vote on party identification among the 13 respondents whose schooling stopped at or before the eighth grade, and found

complete separation: no maximum likelihood estimate exists, and software reports whatever coefficient its iteration limit produced. With only two parameters the posterior can be computed by quadrature, under any prior at all — over the whole plane, by the substitution  $\beta = s \tan u$ , since under separation a Cauchy posterior has a polynomial tail and any fixed box would discard a large part of it.

Under the flat prior the posterior is not a distribution at all, and the arithmetic shows it: integrating the likelihood over a square box centred at the origin gives 8.7 for half-width ten and 778.6 for half-width eighty, each doubling multiplying the mass by at least 4.13. That is the area of the separating cone growing, as in the proof of [Proposition 39.4.1\(b\)](#), and no amount of sampling will reveal it — a chain started near the data's own scale looks well behaved while wandering up a ridge of infinite mass.

An independent  $N(0, 2.5^2)$  prior on the slope and  $N(0, 10^2)$  on the intercept gives a proper posterior with mean 3.077, median 2.931 and a central 95 per cent credible interval of (1.046, 5.922). Independent Cauchy priors of the same scales give mean 6.091, median 4.001 and interval (1.127, 22.995). Both exclude zero decisively. Both lower endpoints agree to within a tenth, and the upper endpoints differ by a factor of 3.9: the data bound the effect from below, and everything above that is prior. Panel (b) of [Figure 39.5.1](#) shows the two posteriors.

Set these beside the frequentist answers of [Section 35.3](#). The ordinary likelihood ratio gives the one-sided interval  $(0.902, \infty)$ : the most honest statement available without a prior, and the least useful. Firth's penalty, the mode under the Jeffreys prior ([Proposition 35.3.2](#)), gives 1.362 with the penalized likelihood ratio interval (0.364, 4.045) — smaller than either posterior median, and a mode rather than a mean. The Jeffreys penalty falls off exponentially along the separating ray ([Proposition 35.3.2](#)) while a Cauchy of scale 2.5 falls off only polynomially, so it bites hardest on exactly the direction the data do not constrain. There is no way to choose among these numbers from the data. Thirteen observations that separate perfectly contain a bound and nothing more, and every method reports that bound plus whatever it assumed. The discipline is to say which assumption produced which number, and to notice that the flat prior is not the assumption-free option but the one whose assumption is inconsistent.

```
sub = anes[anes["educ"] == 1]                                # schooling stopped at or
    before grade eight
Xs = np.column_stack([np.ones(len(sub)), sub["PID"]])
ys = sub["vote"].to_numpy(float)

def loglik(b0, b1):
    """Logistic log-likelihood at one intercept and a whole grid of slopes."""
    eta = b0 + np.outer(b1, Xs[:, 1])
    return np.sum(ys * eta - np.logaddexp(0.0, eta), axis=1)

def log_prior(b0, b1, kind, scale=2.5):
    if kind == "flat":
        return 0.0 * b0
    if kind == "normal":
        return -(b0 / 10.0) ** 2 / 2 - (b1 / scale) ** 2 / 2
    return -np.log1p((b0 / 10.0) ** 2) - np.log1p((b1 / scale) ** 2) # Cauchy

def posterior_of_slope(kind, k=1200, s0=30.0, s1=6.0):
```

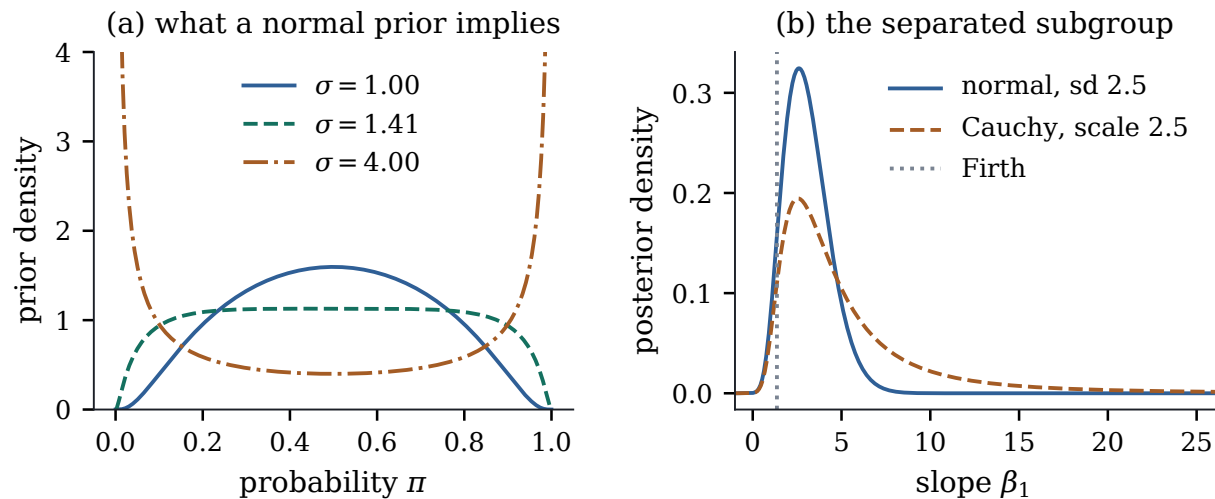


Figure 39.5.1: (a) The prior induced on a probability by a  $N(0, \sigma^2)$  prior on the logit; the middle curve is the boundary case  $\sigma^2 = 2$  of Proposition 39.5.1(c). (b) Posterior densities of the party-identification slope in the separated subgroup of Example (A separated subgroup) under a normal and a Cauchy prior, with Firth's estimate marked.

```

"""Marginal posterior of the slope, by quadrature over the whole plane."""
t, wt = np.polynomial.legendre.leggauss(k)
u, wu = t * np.pi / 2, wt * np.pi / 2
b0, b1 = s0 * np.tan(u), s1 * np.tan(u)
j0, j1 = s0 / np.cos(u) ** 2, s1 / np.cos(u) ** 2
logq = np.array([loglik(a, b1) for a in b0]) + log_prior(b0[:, None], b1[None,
:], kind)
q = np.exp(logq - logq.max())
marg = ((j0 * wu)[:, None] * q).sum(axis=0) # integrate out the intercept
dens = marg / np.sum(marg * j1 * wu) # density with respect to b1
return u, b1, dens, j1, wu

def summarize(u, b1, dens, j1, wu, probs=(0.025, 0.5, 0.975)):
    """Posterior mean and quantiles of the slope from the quadrature grid."""
    f = dens * j1 # the density in the u
    coordinate
    cdf = np.concatenate([[0.0], np.cumsum((f[1:] + f[:-1]) / 2 * np.diff(u))])
    return np.sum(b1 * f * wu), np.interp(probs, cdf / cdf[-1], b1)

post_n, post_c = posterior_of_slope("normal"), posterior_of_slope("cauchy")
mean_n, (lo_n, med_n, hi_n) = summarize(*post_n)
mean_c, (lo_c, med_c, hi_c) = summarize(*post_c)
print(f"posterior mean of the slope: normal {mean_n:.3f}, Cauchy {mean_c:.3f}")

```

*Idea.* A penalty is a prior and a prior a penalty, but the two readings answer different questions. The penalty reading asks what estimate to report and tunes  $\lambda$  by cross-validation; the prior reading asks what is believed before the data, and its scale must be defensible on the scale of the parameter — for a binary response, as a statement about probabilities.

## 39.5.4 Exercises

### 39.5.4.1 A. Check your understanding

#### Exercise A1.

A colleague reports a ridge-penalized logistic fit with  $\lambda = 100$  on standardized covariates. What normal prior does that correspond to, and what odds ratio for a one standard deviation change does the prior regard as, say, two standard deviations from zero?

*Solution.*  $\lambda = 1/\tau^2$  gives  $\tau = 0.1$ . Two prior standard deviations is  $\gamma_j = 0.2$ , an odds ratio of  $e^{0.2} = 1.22$ . The prior therefore says that a one standard deviation change in any covariate almost certainly multiplies the odds by less than about 1.22: a strong belief, not a weak one, which may be appropriate in a high-dimensional screening problem and absurd in a four-covariate model.

#### Exercise A2.

Explain in words why the induced prior of [Proposition 39.5.1\(c\)](#) becomes U-shaped for large  $\sigma$ , without using the calculus of the proof.

### 39.5.4.2 B. Practice

#### Exercise B1.

Show that a Cauchy prior with scale  $s$  is a scale mixture of normals:  $\gamma \mid \lambda \sim N(0, \lambda)$  with  $\lambda$  having an inverse gamma distribution with shape  $1/2$  and rate  $s^2/2$ . Explain how this turns a Cauchy prior into a Gibbs sampler with normal full conditionals in the probit model.

*Solution.* If  $\lambda \sim \text{IG}(1/2, s^2/2)$  then the marginal density of  $\gamma$  is

$$\begin{aligned} & \int_0^\infty \frac{1}{\sqrt{2\pi\lambda}} e^{-\gamma^2/(2\lambda)} \frac{(s^2/2)^{1/2}}{\Gamma(1/2)} \lambda^{-3/2} e^{-s^2/(2\lambda)} d\lambda \\ & \propto \int_0^\infty \lambda^{-2} e^{-(\gamma^2+s^2)/(2\lambda)} d\lambda \propto \frac{1}{\gamma^2 + s^2}, \end{aligned}$$

the Cauchy density of scale  $s$ , the last step by the substitution  $v = 1/\lambda$ . In the Albert–Chib scheme of [Proposition 39.4.5](#), conditioning on the  $\lambda_j$  as well as on  $\mathbf{z}$  makes the prior on  $\boldsymbol{\gamma}$  normal with covariance  $\text{diag}(\lambda_j)$ , so the full conditional of  $\boldsymbol{\beta}$  is normal, and that of each  $\lambda_j$  is inverse gamma with shape 1 and rate  $(\gamma_j^2 + s^2)/2$ : every step is a standard draw.

#### Exercise B2.

For  $\sigma = 10$ , compute the prior probability that  $\pi = e^\eta/(1+e^\eta) < 0.01$  when  $\eta \sim N(0, \sigma^2)$ , and compare with  $\sigma = 1$ . Relate the answers to [Proposition 39.5.1\(c\)](#).

**Exercise B3.**

**Proposition 35.3.2** adds  $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{W} \mathbf{X})$  to the log-likelihood. Show that this is the log of the Jeffreys prior density for the logistic model, so Firth's estimate is a posterior mode. Why does that mode exist even under separation, when the maximum likelihood estimate does not?

*Solution.* The Jeffreys prior is  $\propto \det\{\mathcal{I}(\boldsymbol{\beta})\}^{1/2}$  for the expected information  $\mathcal{I}$ , which for the logistic model is  $\mathbf{X}^\top \mathbf{W} \mathbf{X}$  with  $\mathbf{W} = \text{diag}\{\pi_i(1 - \pi_i)\}$  at  $\boldsymbol{\beta}$  (**Proposition 35.1.3**); its logarithm is  $\frac{1}{2} \log \det(\mathbf{X}^\top \mathbf{W} \mathbf{X})$  up to a constant, so adding it to  $\ell$  gives Firth's penalized log-likelihood and its maximizer is the posterior mode under that prior. It exists under separation for the reason **Proposition 35.3.2** proves: along a separating ray  $t\mathbf{b}$  the weights decay like  $e^{-t\delta}$ , so the log prior falls linearly in  $t$  while the log-likelihood, already at most zero, has nothing left to gain. The likelihood alone is flat along that ray; the prior supplies the curvature.

**39.5.4.3 C. Going deeper****Exercise C1.**

Redo **Proposition 39.5.1(c)** for the probit link: with  $\eta \sim N(0, \sigma^2)$  and  $\pi = \Phi(\eta)$ , show that the induced density of  $\pi$  is unimodal when  $\sigma^2 \leq 1$  and bimodal when  $\sigma^2 > 1$ , and say why the threshold differs from the logit's.

## 39.6 Summary and Notes

### 39.6.1 Summary

*Idea.*

1. A response family is chosen from the support and from the mean–variance relationship; with replicates, the variance-function power is estimated by regressing log group variance on log group mean (**Proposition 39.1.1(a)**).
2. Additivity belongs to the link, not to the data: a mean additive on one link scale is additive on another only when the two differ by an affine map (**Proposition 39.1.1(b)**).
3. Likelihood criteria compare models for the same random variable. A model fitted to  $t(\mathbf{y})$  needs the correction  $-2 \sum_i \log t'(y_i)$ , itself of order  $n$ , before its AIC can stand beside one for  $\mathbf{y}$  (**Proposition 39.1.1(c)**, **Example (How long does a strike last)**).
4. Diagnostics should be aimed. Pearson residuals have approximate variance  $1 - h_{ii}$ ; a simulated envelope has pointwise level  $2/(B + 1)$ ; the link is tested by adding  $\hat{\eta}^2$ ; a covariate's scale by partial-residual and added-variable plots; the variance function by the height and trend of  $|r^P|$  (**Proposition 39.2.1**). In **Example (Days of television news)** only the last is wrong.
5. The log-likelihood is quadratic in  $\boldsymbol{\beta}$  for every design exactly when the family is



normal ([Proposition 39.3.1](#)): in the Bayesian thread every definition survives, every closed form does not.

6. The Bayesian generalized linear model is [Definition 39.4.1](#). A proper prior with a bounded likelihood gives a proper posterior; a flat prior on a logistic model gives one exactly when the data are in overlap, that is exactly when the maximum likelihood estimate exists ([Proposition 39.4.1](#)).
7. Under local smoothness, a growing Hessian and a concentration condition, the Laplace approximation to the marginal likelihood has relative error  $O(n^{-1})$  ([Theorem 39.4.2](#)); its logarithm is the BIC with an  $O(1)$  remainder carrying the whole influence of the prior ([Corollary 39.4.3](#)).
8. Metropolis–Hastings and Gibbs leave the posterior invariant by detailed balance ([Proposition 39.4.4](#), [Definition 39.4.2](#)); Albert–Chib augmentation makes every probit full conditional normal or truncated normal ([Proposition 39.4.5](#)). Convergence is judged by  $\hat{R}$ , mixing by effective sample size ([Example \(The 1996 survey, sampled\)](#)).
9. Posterior predictive checks compare the data with replicates from [Eq. 39.4.8](#); using the data twice makes them conservative, so an extreme  $p$ -value is evidence and a middling one is not ([Definition 39.4.3](#), [Example \(Where the survey model fails\)](#)).
10. A normal prior is ridge, a Laplace prior the lasso at its mode, and a normal prior on a logit is bimodal on the probability scale exactly when  $\sigma^2 > 2$  ([Proposition 39.5.1](#)). A proper prior makes the mode exist under separation, where maximum likelihood has none ([Example \(The separated subgroup, three priors\)](#)).

## 39.6.2 Notes and sources

**Model building.** The three-part structure — random component, systematic component, link — is Nelder and Wedderburn’s (1972); McCullagh and Nelder (1989) remains the standard account of how the pieces are chosen, their chapter 8 for the log link being preferred to the canonical reciprocal in gamma models and their chapter 11 for mean–variance diagnostics. Fahrmeir, Kneib, Lang and Marx (2021, §5.3) develop the log-normal and gamma alternatives for nonnegative responses. The Jacobian correction of [Proposition 39.1.1\(c\)](#) is elementary, widely known, and as widely omitted from software output.

**Diagnostics.** Pregibon (1981) transferred the linear-model diagnostics of [Chapter 20](#) to generalized linear models by the substitution  $\mathbf{X} \mapsto \hat{\mathbf{W}}^{1/2}\mathbf{X}$ , and Pregibon (1980) introduced the goodness-of-link test of [Proposition 39.2.1\(d\)](#). The half-normal plot with a simulated envelope is Atkinson’s (1981, 1985), and the exchangeability argument behind [Eq. 39.2.1](#) is the one that justifies Monte Carlo tests generally. Cook and Weisberg (1983) read a trend in absolute residuals as evidence about the variance function.

**The Bayesian thread.** Lindley and Smith (1972) is the origin of the hierarchical linear model. Diaconis and Ylvisaker (1979) characterized the conjugate priors of an exponential family; the generalized linear form [Eq. 39.3.2](#), with a pseudo-response and per-observation weights, is Bedrick, Christensen and Johnson (1996) and Chen and Ibrahim (2003). Rencher and Schaalje (2008, chapter 11) and Agresti (2015, chapter 10) are the source treatments closest to this chapter, and Gelman, Carlin, Stern, Dunson, Vehtari and Rubin (2013) the standard modern reference for its last two sections.

**Laplace and the BIC.** Laplace’s method is two centuries old; its use for posterior moments



and marginal likelihoods is Tierney and Kadane's (1986). Schwarz (1978) derived the BIC, and Kass and Raftery (1995) made the link between Bayes factors, the Laplace approximation and the BIC standard; their section 4 discusses the regularity conditions that [Theorem 39.4.2](#) assumes, and the set used here — a shrinking ball, bounded scaled derivatives, a concentration bound — is convenient rather than weakest. The deviance information criterion is Spiegelhalter, Best, Carlin and van der Linde (2002); the Bernstein–von Mises theorem in the form quoted is van der Vaart (1998, chapter 10).

**MCMC.** The algorithm of [Eq. 39.4.6](#) is Metropolis, Rosenbluth, Rosenbluth, Teller and Teller (1953) as generalized by Hastings (1970); Tierney (1994) gave the rigorous account for continuous state spaces and Robert and Casella (2004) the textbook treatment. The Gibbs sampler is Geman and Geman (1984), brought into mainstream statistics by Gelfand and Smith (1990). Optimal scaling is Roberts, Gelman and Gilks (1997), and the iteratively reweighted proposal follows Gamerman (1997). Data augmentation for the probit model is Albert and Chib (1993); for the logit, Holmes and Held (2006) use a scale mixture and Polson, Scott and Windle (2013) the Pólya–gamma identity now in general use. [Eq. 39.4.7](#) is Gelman and Rubin (1992) and the truncation rule for the effective sample size is Geyer's (1992). Posterior predictive checking is Rubin (1984), Meng (1994) and Gelman, Meng and Stern (1996), its  $p$ -values analysed by Bayarri and Berger (2000). Hamiltonian Monte Carlo, which is what modern software actually runs, is omitted because its justification needs machinery this book does not develop.

**Priors as penalties.** The ridge–normal correspondence is as old as ridge regression ([Theorem 27.2.2](#)); the lasso–Laplace one was noted by Tibshirani (1996) and turned into a sampler by Park and Casella (2008). Gelman, Jakulin, Pittau and Su (2008) proposed the Cauchy default prior for logistic regression with the rescaling to standard deviation 1/2, and gave the separation argument that [Proposition 39.5.1\(d\)](#) formalizes. That Firth's (1993) penalty is the Jeffreys (1946) prior was observed by Firth himself; Heinze and Schemper (2002) made it the standard remedy. The horseshoe is Carvalho, Polson and Scott (2010). The propriety argument of [Proposition 39.4.1\(b\)](#) is the Bayesian face of [Theorem 35.3.1](#).

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# Chapter 40

## GLMMs and Generalized Estimating Equations

Part VII replaced independent errors by a covariance matrix with structure; Part VIII has so far replaced the normal response by an exponential dispersion family. This chapter does both at once, and the combination is not the sum of its parts. In a linear model the mean is unaffected by what happens to the covariance, so a random effect changes the standard errors and leaves the coefficients alone. Once the link is nonlinear that is false: averaging a curve over a random intercept does not give the same curve back, so the model has *two* sets of coefficients — one for a single cluster, one for the population — and they are not equal.

Two families of methods follow from that fork. If the interest is in what happens within a cluster, and a full probability model is acceptable, the answer is the generalized linear mixed model: random effects inside the linear predictor and a likelihood obtained by integrating them out, numerically, with different approximations giving visibly different answers. If the interest is in the population average, and no model for the dependence is wanted, the answer is generalized estimating equations: solve a quasi-score built from a *working* correlation that is allowed to be wrong, and price the estimate with a sandwich. The second route buys robustness by giving up the likelihood, and with it the deviance, the random-effect predictions, and any tolerance for data that go missing for the wrong reasons.

**What you need.** Chapter 34 throughout (the generalized linear model [Definition 34.2.1](#) and [Definition 34.2.2](#), the likelihood equations [Theorem 34.3.1](#), iteratively reweighted least squares [Theorem 34.4.1](#), the asymptotics [Theorem 34.5.1](#)), Chapter 35 (the logistic model [Definition 35.1.1](#), non-collapsibility [Example \(The odds ratio does not collapse\)](#) and case-control sampling [Theorem 35.5.1](#)), Chapter 36 and Chapter 37 (Poisson regression [Definition 37.1.1](#) and the negative binomial [Definition 37.4.1](#)). From Part VII: the linear mixed model [Definition 32.2.1](#), best linear unbiased prediction [Theorem 32.3.1](#) and its shrinkage reading, REML [Definition 32.4.1](#), the boundary problem [Theorem 32.5.2](#), the covariance models [Definition 33.4.1](#) and the working-covariance result [Proposition 33.4.1](#). From Part V: the sandwich covariance [Theorem 21.4.2](#). Chapter 29 (AIC and BIC, [Definition 29.2.2](#)) and Chapter 25 (what a coefficient means) are used in places. Chapter 38 supplies the quasi-likelihood machinery this chapter generalizes, and Chapter 41 the missingness conditions that [Section 40.4](#) needs.

## 40.0.1 Roadmap

- [Marginal versus conditional models](#)
- [Generalized linear mixed models](#)
- [Integrated likelihood approximations](#)
- [Generalized estimating equations](#)
- [Working correlation](#)
- [Summary and notes](#)

# 40.1 Marginal versus conditional models

A clinician asks what a treatment would do to *this* patient's odds of recovery; a health minister asks what it would do to the recovery rate of the *whole population*. In a linear model the two questions have the same answer, because averaging a straight line over patients returns the same straight line. As soon as the link is nonlinear they do not, and no amount of data removes the difference: it is a property of the model, not of the sample. This section sets up the two targets and proves that the population-averaged coefficient of a logistic model is strictly smaller in magnitude than the individual one.

## 40.1.1 Two coefficients, one model

Throughout this chapter the data are **clustered**: units  $i = 1, \dots, m$  each carry  $n_i$  observations  $Y_{i1}, \dots, Y_{in_i}$ , with  $n = \sum_i n_i$  — a patient observed repeatedly, a school with its pupils, a firm across years. The simplest model that makes observations within a cluster dependent gives each cluster its own level  $U_i$ , with the link  $g$  of [Chapter 34](#):

$$g\{E(Y_{ij} | U_i)\} = \alpha + \beta x_{ij} + U_i, \quad U_i \sim N(0, \tau^2), \quad (40.1.1)$$

with  $Y_{i1}, \dots, Y_{in_i}$  conditionally independent given  $U_i$ . Here  $\beta$  is a **conditional**, or *subject-specific*, coefficient: it compares two covariate values *within one cluster*, with that cluster's level held wherever it happens to be. Write it  $\beta^C$  when the distinction matters.

The other target concerns the response averaged over clusters. Write  $h = g^{-1}$  and

$$m(x) = E\{h(\alpha + \beta x + U)\}, \quad U \sim N(0, \tau^2), \quad (40.1.2)$$

for the **marginal mean function**: the mean response at  $x$  in a cluster drawn at random. The **marginal**, or *population-averaged*, coefficients are the ones a model with the same link would report if fitted to the population: the solution  $(\alpha^M, \beta^M)$  of

$$E \left[ \begin{pmatrix} 1 \\ X \end{pmatrix} \{m(X) - h(\alpha^M + \beta^M X)\} \right] = \mathbf{0}, \quad (40.1.3)$$

the expectation being over the covariate distribution, and assumed to have a unique solution — which it does whenever  $E\{(1, X)^\top(1, X)\}$  is nonsingular and the marginal mean function stays away from the ends of the range of  $h$ . This is the population version of the likelihood

equations [Eq. 34.3.1](#) with the dispersion and the prior weights dropped; for a canonical link the remaining factor  $h'(\eta)/V(\mu)$  is identically one, so for the logit, log and identity links of this section  $(\alpha^M, \beta^M)$  is exactly what an ordinary generalized linear model fitted to the clustered data estimates consistently — a point [Section 40.4](#) returns to. For a binary covariate [Eq. 40.1.3](#) is solved by matching the two means exactly,  $h(\alpha^M) = m(0)$  and  $h(\alpha^M + \beta^M) = m(1)$ , so that  $\beta^M$  is the log odds ratio between the two marginal probabilities.

*Warning.* Neither coefficient is “the” right one and neither is an approximation to the other. They are answers to different questions, and a paper that reports one while discussing the other has made a substantive error, not a numerical one.

## 40.1.2 The attenuation inequality

### Proposition 40.1.1 — Marginal and conditional coefficients

Let  $U$  have mean zero and let  $h$  be the inverse link of the model [Eq. 40.1.1](#).

- (a) **Identity link.** If  $h(\eta) = \eta$  then  $m(x) = \alpha + \beta x$ , so  $\alpha^M = \alpha$  and  $\beta^M = \beta$  for every distribution of  $U$  with mean zero. The two targets coincide.
- (b) **Log link.** If  $h(\eta) = e^\eta$  and  $E e^U < \infty$ , then  $m(x) = E(e^U) e^{\alpha + \beta x}$ , so  $\beta^M = \beta$  and  $\alpha^M = \alpha + \log E e^U$ . For  $U \sim N(0, \tau^2)$  the shift is  $\tau^2/2$ . Only the intercept moves.
- (c) **Probit link.** If  $h = \Phi$  and  $U \sim N(0, \tau^2)$  then

$$m(x) = \Phi\left(\frac{\alpha + \beta x}{\sqrt{1 + \tau^2}}\right), \quad (40.1.4)$$

so the marginal mean function is again a probit, with  $\alpha^M = \alpha/\sqrt{1 + \tau^2}$  and  $\beta^M = \beta/\sqrt{1 + \tau^2}$ .

- (d) **Logit link.** Let  $h(\eta) = e^\eta/(1 + e^\eta)$ , let  $U$  be any random variable with  $0 < \text{Var}\{h(\theta + U)\} < \infty$  for every  $\theta$ , and put  $P_\theta = h(\theta + U)$  and  $F(\theta) = g\{E P_\theta\}$ , the marginal linear predictor. Then

$$F'(\theta) = 1 - \frac{\text{Var}(P_\theta)}{E(P_\theta) E(1 - P_\theta)} \in (0, 1). \quad (40.1.5)$$

Consequently, if  $\beta \neq 0$ , then for any  $x_1 < x_2$  the marginal log odds ratio is strictly attenuated:

$$0 < \text{sign}(\beta) \{g m(x_2) - g m(x_1)\} < |\beta| (x_2 - x_1).$$

In particular a binary covariate has  $0 < |\beta^M| < |\beta^C|$ .

*Proof.*

- (a)  $E(\alpha + \beta x + U) = \alpha + \beta x$ , and this is exactly [Eq. 40.1.3](#) with  $h$  the identity.
- (b)  $E e^{\alpha + \beta x + U} = e^{\alpha + \beta x} E e^U$ . For a normal  $U$ , [Theorem 3.2.1](#) gives  $E e^U = e^{\tau^2/2}$ . The right-hand side is again of the form  $\exp(\alpha^M + \beta^M x)$ , so it solves [Eq. 40.1.3](#) exactly.
- (c) Let  $Z \sim N(0, 1)$  be independent of  $U$  and write  $\eta = \alpha + \beta x$ . Then

$$E \Phi(\eta + U) = E \Pr(Z \leq \eta + U \mid U) = \Pr(Z - U \leq \eta),$$

and  $Z - U \sim N(0, 1 + \tau^2)$ , which is [Eq. 40.1.4](#). Again the result is of the assumed

form, so it solves [Eq. 40.1.3](#).

- (d) Write  $P_\theta = h(\theta + U)$  and  $\bar{p}(\theta) = E P_\theta$ . The logistic inverse link satisfies  $h' = h(1-h)$ , so differentiating under the expectation (legitimate because  $0 \leq h' \leq 1/4$ , so the difference quotients are dominated) gives  $\bar{p}'(\theta) = E\{P_\theta(1 - P_\theta)\}$ . Since  $F(\theta) = \log \bar{p}(\theta) - \log\{1 - \bar{p}(\theta)\}$ ,

$$F'(\theta) = \frac{\bar{p}'(\theta)}{\bar{p}(\theta)\{1 - \bar{p}(\theta)\}} = \frac{E(P_\theta) - E(P_\theta^2)}{E(P_\theta) E(1 - P_\theta)}.$$

The denominator is  $E(P_\theta) - \{E(P_\theta)\}^2$ , so numerator minus denominator equals  $\{E(P_\theta)\}^2 - E(P_\theta^2) = -\text{Var}(P_\theta)$ , which gives [Eq. 40.1.5](#). The ratio is positive because  $P_\theta \in (0, 1)$  makes  $E\{P_\theta(1 - P_\theta)\} > 0$ , and it is strictly less than one because  $\text{Var}(P_\theta) > 0$ . Integrating  $F'$  from  $\alpha + \beta x_1$  to  $\alpha + \beta x_2$ , an interval of positive length because  $\beta \neq 0$ , gives the stated inequality; for a binary covariate the left-hand side is  $\beta^M$  by the remark after [Eq. 40.1.3](#). ■

Part (d) needs no normality of  $U$ , no particular covariate and no asymptotics; it is an identity about the logistic function. Its factor  $F'(\theta)$  is one minus the ratio of the variance of the individual probabilities to the variance a single Bernoulli variable with the average probability would have, so the more the clusters differ, the flatter the population curve.

*Idea.* A nonlinear link and a random effect do not commute. Averaging  $h(\eta + U)$  over  $U$  gives a *different function* of  $\eta$ , and for the logit and probit it is a flatter one. The linear and log links are the exceptions, because for them averaging changes at most an additive or multiplicative constant.

### 40.1.3 How large is the difference?

The probit factor in [Eq. 40.1.4](#) is exact; the logit has no closed form, but the two links are so nearly proportional that a good approximation comes from pretending it does. Matching a standard logistic distribution to a normal one scaled by  $c = 16\sqrt{3}/(15\pi) = 0.5881$  and applying [Eq. 40.1.4](#) gives the classical factor

$$\beta^M \approx \frac{\beta^C}{\sqrt{1 + c^2\tau^2}}, \quad c = \frac{16\sqrt{3}}{15\pi}, \quad (40.1.6)$$

due to Zeger, Liang and Albert (1988). It is an approximation, not a theorem.

#### Example. (Attenuation in a logistic random-intercept model)

Take  $\alpha = 0$ ,  $\beta^C = 1$  and a covariate  $X \sim N(0, 1)$ , and solve [Eq. 40.1.3](#) numerically. With  $\tau = 0.5$  the marginal slope is 0.9502; with  $\tau = 1$  it is 0.8379; with  $\tau = 2$  it is 0.6142. The approximation [Eq. 40.1.6](#) gives 0.8620 and 0.6477 in the last two cases, too large by about three and five per cent. The probit factor at  $\tau = 1$  is 0.7071, far smaller — but that is a different model, with a different meaning for  $\beta^C$ . The covariate distribution matters little: at  $\tau = 1$  a binary covariate gives a marginal log odds ratio of 0.8318. [Figure 40.1.1\(a\)](#) shows why the approximation works at all — the average of

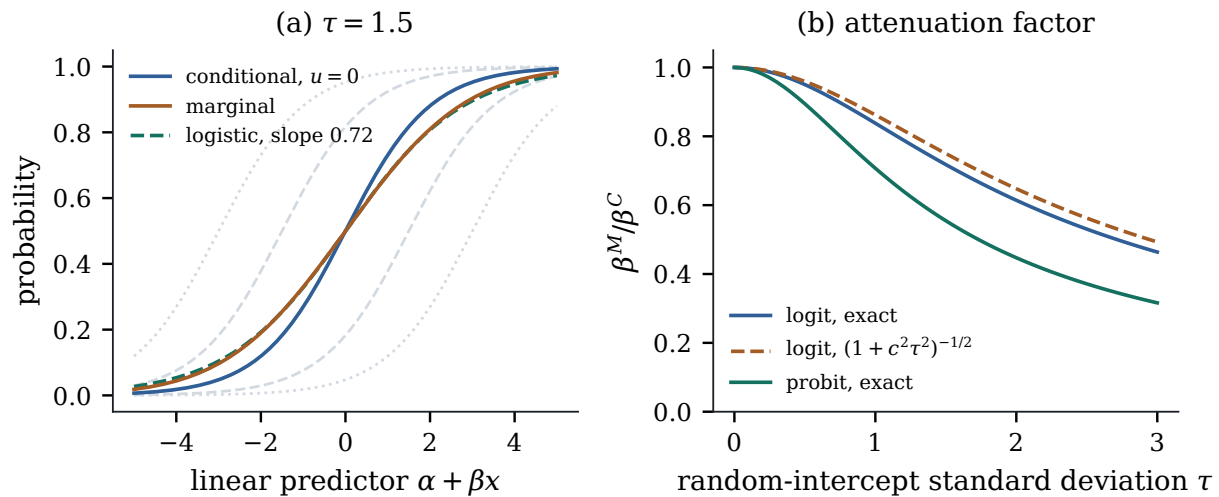


Figure 40.1.1: Attenuation in a random-intercept binary model. (a) With  $\tau = 1.5$ : five conditional logistic curves (grey), their average, and the logistic curve solving Eq. 40.1.3. (b) The ratio  $\beta^M / \beta^C$  against  $\tau$  for the logit (exact and approximate) and for the probit.

the conditional curves is almost exactly a flatter logistic curve — and panel (b) tracks  $\beta^M / \beta^C$  as the clusters become more heterogeneous.

```
import numpy as np
from numpy.polynomial.hermite_e import hermegauss
from scipy.optimize import root
from scipy.stats import norm

def expit(z):
    return 1.0 / (1.0 + np.exp(-z))

# nodes and weights that integrate g(U) over U ~ N(0, 1): sum_k w_k g(z_k)
z_node, z_w = hermegauss(80)
z_w = z_w / np.sqrt(2 * np.pi)

def marginal_curve(eta, tau, link="logit"):
    """E[H(eta + U)] for U ~ N(0, tau^2), by Gauss-Hermite quadrature."""
    H = expit if link == "logit" else norm.cdf
    grid = np.atleast_1d(eta)[..., None] + tau * z_node
    return np.sum(z_w * H(grid), axis=-1)

def beta_marginal(alpha, beta, tau, link="logit"):
    """Population-averaged coefficients: the root of E[(1, x){m(x) - H(a + b x)}] = 0
    when x ~ N(0, 1), with m the marginal curve of the conditional model."""
    H = expit if link == "logit" else norm.cdf
    m = marginal_curve(alpha + beta * z_node, tau, link)

    def score(par):
        resid = m - H(par[0] + par[1] * z_node)
        return [np.sum(z_w * resid), np.sum(z_w * z_node * resid)]

    sol = root(score, [alpha, beta], tol=1e-13)
    return sol.x

C_LOGIT = 16 * np.sqrt(3) / (15 * np.pi)
```



```
def approx_factor(tau, link="logit"):
    c = C_LOGIT if link == "logit" else 1.0
    return 1.0 / np.sqrt(1.0 + c**2 * tau**2)

for t in [0.5, 1.0, 2.0]:
    print(f"tau = {t}:  logit {beta_marginal(0.0, 1.0, t)[1]:.4f}"
          f"    approximation {approx_factor(t):.4f}"
          f"    probit {1 / np.sqrt(1 + t**2):.4f}")
```

### 40.1.4 Attenuation is not bias

Attenuation is not bias: both coefficients are correct answers to different questions. The phenomenon is the **non-collapsibility** of the odds ratio, met in [Section 35.5](#) as [Example \(The odds ratio does not collapse\)](#) — averaging  $2 \times 2$  tables that all share one odds ratio produces a table whose odds ratio is closer to one, with no confounding whatever. Risk differences and risk ratios do not behave this way, which is parts (a) and (b).

It follows that the conditional coefficient depends on what else is in the model: adding a covariate that explains part of the cluster heterogeneity lowers  $\tau^2$  and, by [Eq. 40.1.5](#), moves the conditional and marginal coefficients of the *other* covariates closer together. That is not evidence of confounding. Nor is either coefficient a causal quantity by itself: randomization ([Theorem 25.2.3](#)) licenses a causal reading of the marginal contrast between treated and untreated, but does not turn the marginal log odds ratio into the conditional one, and does not license a conditional reading of  $\beta^C$  unless the random effect really is a property of the cluster that the intervention leaves alone.

Which target, then? The conditional coefficient answers questions about a change *within a unit* — a dose for a patient, a policy for a school — and is comparable across studies with different amounts of unmeasured heterogeneity, provided the random-effect distribution is right. The marginal coefficient answers questions about a population total or rate, and is safer when the cluster structure is a nuisance, because it assumes nothing about the cluster effects. For the identity link the distinction disappears, which is why [Chapter 32](#) never raised it: in [Definition 32.2.1](#) the marginal model [Eq. 32.2.2](#) has the same  $\mathbf{X}\boldsymbol{\beta}$  as the conditional one.

### 40.1.5 Exercises

#### 40.1.5.1 A. Check your understanding

##### Exercise A1.

A random-intercept logistic model for whether a pupil passes an examination has  $\hat{\beta}^C = 0.60$  for one extra hour of study per week and  $\hat{\tau} = 1.2$ . Use [Eq. 40.1.6](#) to say roughly what a population-averaged logistic model would report, and state in one sentence each what the two numbers mean.

*Solution.* The factor is  $(1 + 0.5881^2 \times 1.2^2)^{-1/2}$ . Now  $0.5881^2 = 0.3459$  and  $1.2^2 = 1.44$ , so the factor is  $(1 + 0.4981)^{-1/2} = 1.4981^{-1/2} \approx 0.817$ , giving  $\hat{\beta}^M \approx 0.49$ . The first says: for a given pupil, an extra hour of study multiplies the odds of passing by  $e^{0.60} \approx 1.82$ . The second says: comparing two randomly chosen pupils from the population who differ

by an hour of study, the odds ratio is about  $e^{0.49} \approx 1.63$ .

### Exercise A2.

A Poisson random-intercept model with a log link is fitted and reports  $\hat{\alpha} = -1.2$ ,  $\hat{\beta} = 0.35$ ,  $\hat{\tau}^2 = 0.8$ . What does a population-averaged log-linear model estimate? Which of the three numbers changes?

#### 40.1.5.2 B. Practice

### Exercise B1.

Use [Eq. 40.1.5](#) to show that in a logistic random-intercept model with  $\Pr(Y = 1 | U) = h(U)$  and  $U$  symmetric about 0, the attenuation factor at  $\theta = 0$  equals  $1 - 4 \text{Var}\{h(U)\}$ . Evaluate it for  $U \sim N(0, 1)$ , for which  $\text{Var}\{h(U)\} = 0.0434$ , and compare with the ratio 0.8379 of [Example \(Attenuation in a logistic random-intercept model\)](#).

*Solution.* By symmetry  $Eh(U) = 1/2$ , so the denominator of [Eq. 40.1.5](#) is  $1/4$  and  $F'(0) = 1 - 4 \text{Var}\{h(U)\}$ . With  $\text{Var}\{h(U)\} = 0.0434$  this is  $1 - 4 \times 0.0434 = 0.8264$ . This is the *local* attenuation at  $\theta = 0$ , and it is the strongest: as  $|\theta|$  grows,  $P_\theta$  concentrates near 0 or 1, its variance vanishes faster than  $E P_\theta E(1 - P_\theta)$  and  $F' \rightarrow 1$ . The global ratio  $\beta^M/\beta^C = 0.8379$  of [Example \(Attenuation in a logistic random-intercept model\)](#) is accordingly a little larger, because [Eq. 40.1.3](#) averages  $F'$  over the whole range of  $X$ .

### Exercise B2.

Extend [Proposition 40.1.1\(c\)](#) to a probit model with a random *slope*:  $\Pr(Y = 1 | U_0, U_1) = \Phi(\alpha + \beta x + U_0 + U_1 x)$  with  $(U_0, U_1) \sim N_2(\mathbf{0}, \mathbf{G})$ . Show that the marginal curve is  $\Phi\{(\alpha + \beta x)/s(x)\}$  with  $s(x)^2 = 1 + (1, x)\mathbf{G}(1, x)^\top$ , and explain why the marginal model is now not a probit model in  $x$  at all.

*Solution.* Exactly as in the proof of (c), with  $Z = U_0 + U_1 x$  normal with mean zero and variance  $1 + g_{00} + 2g_{01}x + g_{11}x^2 = s(x)^2$ . The marginal linear predictor  $(\alpha + \beta x)/s(x)$  is a ratio of a linear function to the square root of a quadratic, which is linear in  $x$  only if  $g_{11} = g_{01} = 0$ . With a random slope the attenuation is different at different  $x$ , so no single marginal coefficient describes it.

## 40.2 Generalized linear mixed models

Replace the scalar  $U_i$  of [Section 40.1](#) by a vector of random effects with a covariance matrix, and the single covariate by a model matrix, and the result is the generalized linear mixed model: the linear mixed model of [Chapter 32](#) with a link function and an exponential dispersion family in place of the normal error.

That one change costs more than it looks. In [Definition 32.2.1](#) the marginal distribution was still normal, with covariance  $\mathbf{V} = \mathbf{ZGZ}^\top + \mathbf{R}$ , so everything Part VII knew about generalized

least squares applied unchanged. Here it is an integral with no closed form, and the marginal moments have to be computed one family at a time.

## 40.2.1 The model

### Definition 40.2.1 — Generalized linear mixed model

Let  $\mathbf{u} \sim N_q\{\mathbf{0}, \mathbf{G}(\boldsymbol{\theta})\}$  be a vector of **random effects** with covariance depending on unknown variance components  $\boldsymbol{\theta}$ . Given  $\mathbf{u}$ , let the responses  $Y_1, \dots, Y_n$  be independent, each from the exponential dispersion family [Definition 34.1.1](#) with prior weight  $w_i$ , dispersion  $\phi$  and variance function  $V$ , and with conditional mean

$$\mu_i(\mathbf{u}) = E(Y_i | \mathbf{u}) = h(\eta_i), \quad \eta_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta} + \mathbf{z}_{(i)}^\top \mathbf{u}, \quad (40.2.1)$$

so that  $\text{Var}(Y_i | \mathbf{u}) = \phi V\{\mu_i(\mathbf{u})\}/w_i$ . In matrix form  $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}$  with  $\mathbf{X}$  of size  $n \times p$  and  $\mathbf{Z}$  of size  $n \times q$ , both known. The parameters are the **fixed effects**  $\boldsymbol{\beta}$ , the variance components  $\boldsymbol{\theta}$  and, if it is not fixed by the family, the dispersion  $\phi$ .

Setting  $\mathbf{Z} = \mathbf{0}$  recovers the generalized linear model [Definition 34.2.1](#) exactly; taking the family normal and  $h$  the identity, with  $\mathbf{R} = \phi\mathbf{I}$ , recovers [Definition 32.2.1](#). Two structures cover most applications.

**Clustered data.** Units  $i = 1, \dots, m$  with  $n_i$  observations each, one random effect vector  $\mathbf{u}_i \sim N_{q_0}(\mathbf{0}, \mathbf{G}_0)$  per unit, independent across units. Then  $\mathbf{Z}$  is block diagonal,  $q = mq_0$ , and both the likelihood and the marginal covariance break into  $m$  independent pieces. The **random-intercept** model takes  $q_0 = 1$  and  $\mathbf{z}_{(ij)} = 1$ :

$$g(\mu_{ij}) = \mathbf{x}_{(ij)}^\top \boldsymbol{\beta} + u_i, \quad u_i \stackrel{\text{iid}}{\sim} N(0, \tau^2). \quad (40.2.2)$$

**Crossed or nested effects.** Pupils in classes in schools; raters and items. Here  $\mathbf{G}$  is block diagonal in the *effects* rather than the units and the likelihood does not factorize. [Chapter 32](#) described the linear versions ([Example \(Nested and crossed random factors\)](#)); the difficulty a link function adds is computational, and severe, because the integral in [Section 40.3](#) is then genuinely high-dimensional.

*Remark* (Two extensions, and why they are rarer). A cluster-level random effect can be put into the linear predictor of a baseline-category model [Definition 36.1.1](#) or of a cumulative-logit model [Definition 36.2.1](#), giving mixed models for nominal and ordinal responses; in the ordinal case the effect shifts every cutpoint together, leaving the proportional-odds structure intact. Conditional independence given  $\mathbf{u}_i$  can also be relaxed, by letting the random effects themselves follow an AR(1) process within the cluster — the mixed-model counterpart of the non-diagonal working correlation of [Section 40.5](#). That is rare, because it raises each cluster's integral from dimension  $q_0$  to  $n_i$ .

### Example. (A random-intercept logistic model)

$m$  physicians each treat  $n_i$  patients;  $Y_{ij} = 1$  if patient  $j$  of physician  $i$  recovers. With the

logistic link of [Definition 35.1.1](#), logit  $\Pr(Y_{ij} = 1 \mid u_i) = \mathbf{x}_{(ij)}^\top \boldsymbol{\beta} + u_i$  and  $u_i \sim N(0, \tau^2)$ , the coefficient of a patient-level covariate compares two patients *of the same physician*, and  $\tau$  measures how much physicians differ. A useful summary is the **median odds ratio**  $\exp(\sqrt{2}\tau z_{0.75})$ , the median odds ratio between two randomly chosen physicians treating identical patients ([Exercise \(B1\)](#)). Replacing  $u_i$  by  $u_{0i} + u_{1i}x_{ij}$  gives a random-slope model, at the price of three parameters in  $\mathbf{G}_0$  and, by [Exercise \(B2\)](#), no single marginal coefficient at all.

### Example. (The Poisson–lognormal model)

With  $Y_i \mid u_i$  Poisson ([Definition 37.1.1](#)) of mean  $\exp(\mathbf{x}_{(i)}^\top \boldsymbol{\beta} + u_i)$  and  $u_i \sim N(0, \tau^2)$ , the conditional mean is multiplied by the lognormal variable  $e^{u_i}$ . Unlike the binary case this model is identified with a *single* observation per unit, because the Poisson variance is determined by its mean and any excess must come from  $u_i$ : it is an overdispersion model in the sense of [Chapter 38](#) ([Proposition 38.4.1](#)), and a direct competitor to the negative binomial of [Definition 37.4.1](#), which arises the same way with a gamma mixing variable ([Lemma 37.4.1](#)).

## 40.2.2 The moments a random effect induces

### Proposition 40.2.1 — Marginal moments of a GLMM

In [Definition 40.2.1](#) write  $\mu_i(\mathbf{u}) = h(\eta_i)$  and  $\mu_i^M = E\{\mu_i(\mathbf{u})\}$ .

$$(a) \quad E(Y_i) = \mu_i^M,$$

$$\text{Var}(Y_i) = \frac{\phi}{w_i} E[V\{\mu_i(\mathbf{u})\}] + \text{Var}\{\mu_i(\mathbf{u})\}, \quad (40.2.3)$$

and  $\text{Cov}(Y_i, Y_k) = \text{Cov}\{\mu_i(\mathbf{u}), \mu_k(\mathbf{u})\}$  for  $i \neq k$ . The marginal variance always exceeds the average conditional variance.

(b) **Binomial.** If  $Y_i \mid \mathbf{u} \sim \text{Bin}\{N_i, \pi_i(\mathbf{u})\}$  and  $\bar{\pi}_i = E\pi_i(\mathbf{u})$ , then  $E(Y_i) = N_i\bar{\pi}_i$  and

$$\text{Var}(Y_i) = N_i\bar{\pi}_i(1 - \bar{\pi}_i) \left\{ 1 + (N_i - 1) \frac{\text{Var}\{\pi_i(\mathbf{u})\}}{\bar{\pi}_i(1 - \bar{\pi}_i)} \right\}. \quad (40.2.4)$$

The overdispersion factor is the beta-binomial one, and it equals 1 when  $N_i = 1$ : a *single* Bernoulli response carries no trace of the random effect.

(c) **Poisson with a log link**,  $\phi = 1$ , and a random intercept  $u_i \sim N(0, \tau^2)$  shared by the observations of a cluster. Then  $\mu_{ij}^M = \exp(\mathbf{x}_{(ij)}^\top \boldsymbol{\beta} + \tau^2/2)$ ,

$$\begin{aligned} \text{Var}(Y_{ij}) &= \mu_{ij}^M + (\mu_{ij}^M)^2 (e^{\tau^2} - 1), \\ \text{Cov}(Y_{ij}, Y_{ik}) &= \mu_{ij}^M \mu_{ik}^M (e^{\tau^2} - 1). \end{aligned} \quad (40.2.5)$$

The marginal variance function is *exactly* the NB2 function  $V(\mu) = \mu + \mu^2/\kappa$  of [Definition 37.4.1](#), with  $\kappa = 1/(e^{\tau^2} - 1)$  — although the marginal distribution is not negative binomial and has no closed form.

*Proof.*

- (a) Condition on  $\mathbf{u}$  and use the variance decomposition  $\text{Var}(Y) = \text{E}\{\text{Var}(Y | \mathbf{u})\} + \text{Var}\{\text{E}(Y | \mathbf{u})\}$  and, for  $i \neq k$ ,  $\text{Cov}(Y_i, Y_k) = \text{E}\{\text{Cov}(Y_i, Y_k | \mathbf{u})\} + \text{Cov}\{\text{E}(Y_i | \mathbf{u}), \text{E}(Y_k | \mathbf{u})\}$ , whose first term vanishes by conditional independence. The final claim is  $\text{Var}\{\mu_i(\mathbf{u})\} \geq 0$ .
- (b) With  $\pi = \pi_i(\mathbf{u})$ , Eq. 40.2.3 gives  $\text{Var}(Y_i) = N_i \text{E}\{\pi(1 - \pi)\} + N_i^2 \text{Var}(\pi)$ . Now  $\text{E}\{\pi(1 - \pi)\} = \bar{\pi}_i - \text{E}(\pi^2) = \bar{\pi}_i(1 - \bar{\pi}_i) - \text{Var}(\pi)$ , so  $\text{Var}(Y_i) = N_i \bar{\pi}_i(1 - \bar{\pi}_i) + N_i(N_i - 1) \text{Var}(\pi)$ , which is Eq. 40.2.4.
- (c) For a lognormal factor,  $\text{E} e^{su} = e^{s^2 \tau^2 / 2}$ . Hence  $\mu_{ij}^M = e^{\eta_{ij}^0} \text{E} e^{u_i} = e^{\eta_{ij}^0 + \tau^2 / 2}$ , writing  $\eta_{ij}^0 = \mathbf{x}_{(ij)}^\top \boldsymbol{\beta}$ . By Eq. 40.2.3 with  $V(\mu) = \mu$  and  $\phi = w_{ij} = 1$ ,

$$\text{Var}(Y_{ij}) = \mu_{ij}^M + e^{2\eta_{ij}^0} (\text{E} e^{2u_i} - (\text{E} e^{u_i})^2) = \mu_{ij}^M + e^{2\eta_{ij}^0} (e^{2\tau^2} - e^{\tau^2}),$$

and  $e^{2\eta^0} (e^{2\tau^2} - e^{\tau^2}) = (e^{\eta^0 + \tau^2 / 2})^2 (e^{\tau^2} - 1)$ . The covariance is the same computation with  $\eta_{ij}^0 + \eta_{ik}^0$  in the exponent. Matching  $\mu + \mu^2(e^{\tau^2} - 1)$  with  $\mu + \mu^2/\kappa$  gives  $\kappa$ .

■

Part (b) explains why a random-intercept model for *ungrouped* binary data needs repeated observations: with  $N_i = 1$  the marginal distribution is Bernoulli whatever  $\tau$  is, and  $\tau$  is identified only through the within-cluster *correlation*. Part (c) explains why the Poisson–lognormal and the negative binomial are so hard to tell apart: they agree in their first two moments exactly and differ only in shape, so a likelihood ratio between them compares tails, not overdispersion. The listing checks Eq. 40.2.5 by simulation at  $\tau = 0.6$ , where the matching NB2 shape is  $\kappa = 1/(e^{0.36} - 1) = 2.3077$ .

```
import numpy as np

rng = np.random.default_rng(40_002)
tau, eta1, eta2 = 0.6, 0.4, -0.2
u = rng.normal(0, tau, 1_000_000)
y1 = rng.poisson(np.exp(eta1 + u))
y2 = rng.poisson(np.exp(eta2 + u))
mu1, mu2 = np.exp(eta1 + tau**2 / 2), np.exp(eta2 + tau**2 / 2)

print(f"E Y1      : {y1.mean():.4f} (theory {mu1:.4f})")
print(f"Var Y1     : {y1.var():.4f} (theory {mu1 + mu1**2 * (np.exp(tau**2) - 1):.4f})")
print(f"Cov(Y1,Y2): {np.cov(y1, y2)[0, 1]:.4f} (theory {mu1 * mu2 * (np.exp(tau**2) - 1):.4f})")
```

### 40.2.3 What a random intercept cannot do

**Lemma 40.2.2** — A shared random intercept induces nonnegative correlation

Let  $U$  be a random variable and  $f, g : \mathbb{R} \rightarrow \mathbb{R}$  both nondecreasing, with  $\text{E} f(U)^2$  and  $\text{E} g(U)^2$  finite. Then  $\text{Cov}\{f(U), g(U)\} \geq 0$ . Consequently, in a random-intercept

GLMM Eq. 40.2.2 with any inverse link  $h$  that is nondecreasing,

$$\text{Cov}(Y_{ij}, Y_{ik}) = \text{Cov}\{h(\eta_{ij}^0 + u_i), h(\eta_{ik}^0 + u_i)\} \geq 0 \quad (j \neq k).$$

*Proof.* Let  $U'$  be an independent copy of  $U$ . Because both functions are nondecreasing,  $\{f(U) - f(U')\}\{g(U) - g(U')\} \geq 0$  pointwise. Taking expectations and expanding,

$$0 \leq \mathbb{E}[\{f(U) - f(U')\}\{g(U) - g(U')\}] = 2[\mathbb{E}\{f(U)g(U)\} - \mathbb{E}f(U)\mathbb{E}g(U)] = 2\text{Cov}\{f(U), g(U)\}.$$

The second claim is Proposition 40.2.1(a) with  $f(u) = h(\eta_{ij}^0 + u)$  and  $g(u) = h(\eta_{ik}^0 + u)$ , both nondecreasing. ■

Every link in this book has a nondecreasing inverse, so the consequence is a real restriction: **a model whose only random effect is a cluster-level intercept cannot describe negatively correlated responses.** Such data exist. If exactly one of a traveller's four options is chosen, or exactly one of a litter survives a fixed food supply, the responses compete and their correlation is negative by construction. Section 40.4 analyses a data set of that kind, and the estimating-equation approach has no difficulty with it precisely because it never commits to a mechanism for the dependence.

## 40.2.4 Estimation, inference and prediction

Fitting is the subject of Section 40.3. The random effects cannot be estimated by maximizing a joint likelihood, because their number grows with the sample size — the incidental-parameter trap of Example (Many means, one variance). They are integrated out, giving the **integrated likelihood**

$$L(\boldsymbol{\beta}, \boldsymbol{\theta}, \phi) = \int_{\mathbb{R}^q} \prod_{i=1}^n f(y_i | \mathbf{u}; \boldsymbol{\beta}, \phi) p(\mathbf{u}; \boldsymbol{\theta}) d\mathbf{u}, \quad (40.2.6)$$

which is maximized numerically. Inference for  $\boldsymbol{\beta}$  is then the usual likelihood inference — Wald intervals from the inverse observed information, likelihood ratio tests between nested mean models — as  $m \rightarrow \infty$  with cluster sizes bounded. That claim is imported, not proved here: Theorem 34.5.1 covers a generalized linear model, not a GLMM, and the conditions under which the integrated likelihood behaves like an ordinary one are set out by McCulloch, Searle and Neuhaus (2008, chapter 14). Two warnings carry over from Part VII unchanged. First, a test of  $\tau^2 = 0$ , or of any variance component on the boundary of its range, is *not* a  $\chi^2(1)$  test: the limiting null distribution is the mixture  $\frac{1}{2}\chi^2(0) + \frac{1}{2}\chi^2(1)$  under the conditions of Theorem 32.5.2, so the naive  $p$ -value is twice what it should be. Second, REML has no clean analogue, because Definition 32.4.1 was built on the distribution of  $\mathbf{K}^\top \mathbf{Y}$  for  $\mathcal{C}(\mathbf{K}) = \mathcal{C}(\mathbf{X})^\perp$ , a construction that needs the normal linear model; the approximate versions in use inherit the bias of the working linear model they are built on.

In the linear mixed model Theorem 32.3.1 produced a linear predictor of the random effects, and Eq. 32.3.3 showed it shrinks the cluster's own estimate towards zero. Neither linearity nor an explicit formula survives the link function, but the shrinkage does.

### Proposition 40.2.3 — Empirical Bayes prediction and shrinkage

Consider Eq. 40.2.2 with  $\beta$ ,  $\tau^2$  and  $\phi$  known, write  $\ell_i(u) = \sum_j \log f(y_{ij} | u)$  for cluster  $i$ 's conditional log-likelihood, and assume  $\ell_i$  is concave in  $u$  — as it is for a canonical link by Theorem 34.3.3, and for the gamma with a log link by direct differentiation.

- (a) The posterior density of  $u_i$  given  $\mathbf{y}_i$ , proportional to  $\exp\{\ell_i(u) - u^2/(2\tau^2)\}$ , is log-concave and its mode  $\hat{u}_i$  is unique.
- (b) If  $\ell_i$  attains its maximum at a finite  $\hat{a}_i \neq 0$ , then  $\hat{u}_i$  lies strictly between 0 and  $\hat{a}_i$ : the prediction has the same sign as the cluster's own estimate and is strictly smaller in magnitude. If  $\ell_i$  has no finite maximizer — as when every response in a binary cluster is 0 — then  $\hat{u}_i$  is finite all the same.
- (c) Replacing  $\ell_i$  by its quadratic expansion at  $\hat{a}_i$  gives  $\hat{u}_i \approx \{\tau^2/(\tau^2 + I_i^{-1})\} \hat{a}_i$  with  $I_i = -\ell_i''(\hat{a}_i)$ . In the two cases used in this chapter  $I_i^{-1} = \phi/n_i$  exactly, which is Eq. 32.3.3: clusters with few observations are shrunk harder, and all clusters are shrunk harder when  $\tau^2$  is small relative to the noise.

*Proof.*

- (a) Adding the strictly concave  $-u^2/(2\tau^2)$  to the concave  $\ell_i$  keeps it concave and makes it strictly so, and the sum tends to  $-\infty$  in both directions, so a unique maximizer exists.
- (b) Write  $\psi(u) = \ell_i(u) - u^2/(2\tau^2)$  and suppose  $\hat{a}_i > 0$ . Then  $\ell_i'(\hat{a}_i) = 0$  gives  $\psi'(\hat{a}_i) = -\hat{a}_i/\tau^2 < 0$ , while  $\psi'(0) = \ell_i'(0) > 0$  because  $\ell_i$  is concave with maximum to the right of 0. A continuous derivative positive at 0 and negative at  $\hat{a}_i$  vanishes strictly between them, and by strict concavity that zero is the unique maximizer; the case  $\hat{a}_i < 0$  is symmetric. If  $\ell_i$  increases everywhere,  $\psi'$  is still eventually negative because of the penalty, so  $\hat{u}_i$  is finite.
- (c) With  $\ell_i(u) \approx \ell_i(\hat{a}_i) - \frac{1}{2}I_i(u - \hat{a}_i)^2$ , the penalized maximizer is  $\hat{u}_i = I_i\hat{a}_i/(I_i + \tau^{-2}) = \{\tau^2/(\tau^2 + I_i^{-1})\}\hat{a}_i$ . For a canonical link  $I_i = \sum_j w_{ij}V(\hat{\mu}_{ij})/\phi$ , so  $I_i^{-1} = \phi/(n_i\bar{V}_i)$  with  $\bar{V}_i$  the average of the  $w_{ij}V(\hat{\mu}_{ij})$ ; in the normal case  $V \equiv 1$ ,  $w \equiv 1$  and  $\bar{V}_i = 1$ . For the gamma with a log link  $\ell_i''(u) = -\nu \sum_j y_{ij}e^{-\eta_{ij}}$  with  $\nu = 1/\phi$ , and  $\ell_i'(\hat{a}_i) = 0$  forces  $\sum_j y_{ij}e^{-\eta_{ij}} = n_i$  there, so again  $I_i = n_i/\phi$  exactly. ■

*Remark* (The predictions are not estimates).  $\hat{u}_i$  predicts a random variable, so its error is a prediction error variance and the interval covering  $u_i$  is a *prediction* interval (Theorem 12.3.1), exactly as in Theorem 32.3.1(b). The usual intervals ignore the estimation of  $\beta$  and  $\theta$  and are therefore too short when  $m$  is small.

## 40.2.5 An investment panel with a multiplicative mean

**Example.** (Grunfeld's firms, again)

Chapter 32 fitted a linear mixed model to Grunfeld's panel of 11 US firms observed in



each of 20 years ([Example \(Grunfeld's panel with a random firm effect\)](#)). Investment there ranges over three orders of magnitude and is far more variable for large firms than for small ones, which a constant-variance model cannot represent; a gamma response with a log link can, because the mean is multiplicative and the conditional coefficient of variation is constant. With  $Y_{ij}$  the gross investment of firm  $i$  in year  $j$ ,

$Y_{ij} \mid u_i \sim \text{Gamma}$  with  $\log E(Y_{ij} \mid u_i) = \beta_0 + \beta_1 \log(\text{value}_{ij}) + \beta_2 \log(\text{capital}_{ij}) + u_i$ ,  $u_i \sim N(0, \tau^2)$ , and  $\text{Var}(Y_{ij} \mid u_i) = \phi E(Y_{ij} \mid u_i)^2$ . Fitting by adaptive Gauss–Hermite quadrature ([Section 40.3](#)) gives

|            | $\beta_1$ (value) | $\beta_2$ (capital) | $\phi$ | $\tau$ |
|------------|-------------------|---------------------|--------|--------|
| marginal   | 0.8421            | 0.3386              | 0.2644 | —      |
| gamma GLM  |                   |                     |        |        |
| gamma GLMM | 0.6825            | 0.2315              | 0.0930 | 0.5188 |

The firm-to-firm standard deviation  $\hat{\tau} = 0.5188$  is large: firms a standard deviation apart on the log scale differ by a factor of  $e^{0.5188} \approx 1.68$  in investment at the same value and capital stock. The conditional coefficient of variation is  $\sqrt{\hat{\phi}} = 0.3049$ . For a gamma with shape  $1/\hat{\phi}$  the conditional variance of  $\log Y$  is the trigamma function of the shape, 0.0974, so the fraction of the variance of  $\log Y$  attributable to the firm is  $\hat{\tau}^2/(\hat{\tau}^2 + 0.0974) = 0.734$ .

The fitted conditional intercept is  $-1.3032$ ; by [Proposition 40.1.1\(b\)](#) the log link causes no attenuation of the slopes, and the marginal intercept is  $-1.3032 + \hat{\tau}^2/2 = -1.1686$ . The difference between the two rows is therefore *not* the marginal–conditional distinction. It is the older within-versus-between problem of [Section 6.6](#): the independence fit lets differences between firms contribute to the slopes, while the mixed model assigns most of them to  $u_i$  and estimates the slopes mainly from movement within a firm. A gap of this size accuses the mean model, and [Section 40.5](#) returns to it with a diagnostic and two candidate explanations, [Exercise \(C1\)](#) with a remedy.

With 20 years per firm and  $\hat{\phi} = 0.0930$  small there is almost nothing to shrink: the empirical slope of  $\hat{u}_i$  on  $\hat{a}_i$  is 0.9830, and [Proposition 40.2.3\(c\)](#) predicts 0.9830. Using only each firm's first three years, the same parameters give 0.8951. Shrinkage measures how much a cluster's own data are worth, and is not a fixed property of the model.

## 40.2.6 Exercises

### 40.2.6.1 A. Check your understanding

#### Exercise A1.

For each of the following, say whether the random-intercept variance  $\tau^2$  is identified, and why: (a) one Bernoulli response per cluster; (b) one binomial response with  $N_i = 40$  trials per cluster; (c) one Poisson count per cluster; (d) five Bernoulli responses per cluster.



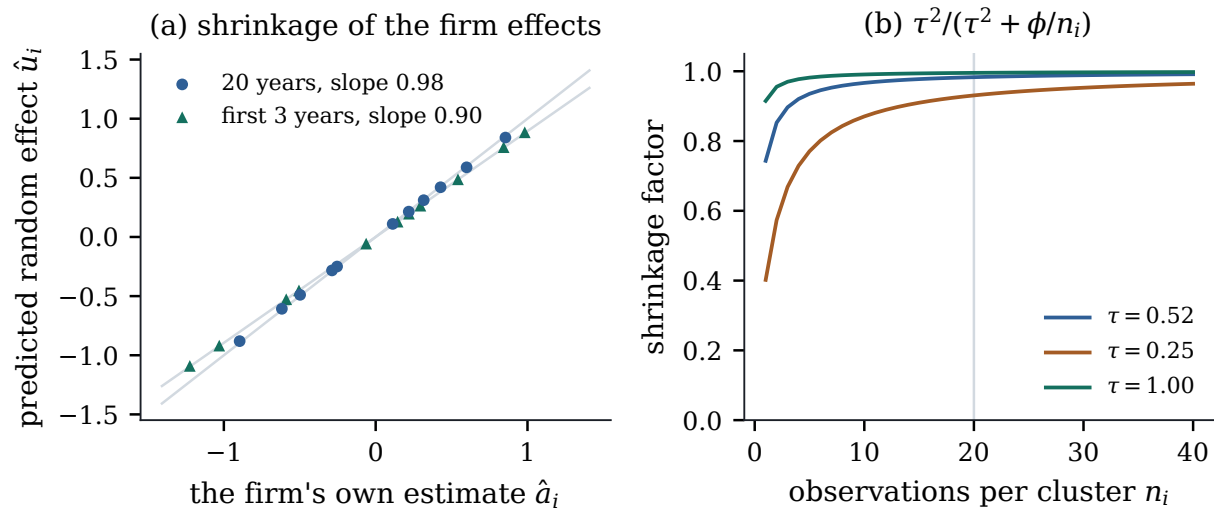


Figure 40.2.1: Shrinkage of the firm effects in the gamma GLMM of [Example \(Grunfeld's firms, again\)](#). (a)  $\hat{u}_i$  against the firm's own estimate  $\hat{a}_i$ , for 20 years (circles) and for the first 3 only (triangles); the diagonal is no shrinkage. (b) The factor  $\tau^2/(\tau^2 + \phi/n_i)$  of [Proposition 40.2.3\(c\)](#) at  $\hat{\phi}$ .

### Exercise A2.

A Poisson–lognormal model is fitted and gives  $\hat{\tau}^2 = 0.5$ . What NB2 shape parameter  $\kappa$  would produce the same marginal variance function? Would you expect the two models to give noticeably different fitted means?

### 40.2.6.2 B. Practice

#### Exercise B1.

In [Example \(A random-intercept logistic model\)](#), let two physicians be drawn at random and let OR be the odds ratio between them for identical patients, taken in the direction that makes it at least one. Show that the median of OR is  $\exp(\sqrt{2}\tau z_{0.75})$ , where  $z_{0.75} = 0.6745$  is the upper quartile of the standard normal. Evaluate it for  $\tau = 1$ .

*Solution.* The log odds ratio between physicians  $i$  and  $k$  is  $u_i - u_k$ , which is  $N(0, 2\tau^2)$ . Taking it in the positive direction gives  $|u_i - u_k|$ , whose median is the upper quartile of  $N(0, 2\tau^2)$ , namely  $\sqrt{2}\tau z_{0.75}$ . Exponentiating gives the claim. For  $\tau = 1$ ,  $\sqrt{2} \times 0.6745 = 0.9539$  and  $e^{0.9539} \approx 2.6$ : a typical pair of physicians differs by a factor of about 2.6 in the odds of recovery, for identical patients.

#### Exercise B2.

Give a two-level model with a *vector* random effect whose induced within-cluster correlation is negative, and explain why [Lemma 40.2.2](#) does not apply. (Hint: let  $\mathbf{z}_{(ij)}$  differ in sign across  $j$ .)

*Solution.* Take  $n_i = 2$ ,  $q_0 = 1$ ,  $z_{i1} = +1$  and  $z_{i2} = -1$ , so  $\eta_{i1} = \eta_{i1}^0 + u_i$  and  $\eta_{i2} = \eta_{i2}^0 - u_i$ . Then  $f(u) = h(\eta_{i1}^0 + u)$  is nondecreasing and  $g(u) = h(\eta_{i2}^0 - u)$  is nonincreasing, so the

association inequality applies with  $-g$  and gives  $\text{Cov}\{f(U), g(U)\} \leq 0$ . The lemma assumed both functions nondecreasing, which is automatic only when every  $z_{(ij)}$  has the same sign — as for an intercept.

### 40.2.6.3 C. Going deeper

#### Exercise C1.

In [Example \(Grunfeld's firms, again\)](#) the mixed-model slopes differ markedly from the independence ones. Add the firm means  $\overline{\log \text{value}}_i$  and  $\overline{\log \text{capital}}_i$  to the model as extra fixed effects, so that the original covariates carry only within-firm variation. Explain why this makes the random effect orthogonal to the covariates in the population, refit, and compare. What test does the difference between the two specifications suggest?

## 40.3 Integrated likelihood approximations

Everything the last section promised rests on the integral [Eq. 40.2.6](#), which has no closed form except in the normal linear case. Fitting a GLMM is therefore a numerical problem before it is a statistical one, and the choice of method is not innocent: three methods in common use can differ in the second significant figure of  $\hat{\beta}$  and in the first of  $\hat{\tau}$ .

### 40.3.1 Why it is hard, and what makes it easier

For the normal linear mixed model  $\mathbf{Y} \mid \mathbf{u}$  and  $\mathbf{u}$  are both normal, so [Eq. 40.2.6](#) is a Gaussian integral and gives [Eq. 32.2.2](#) in closed form. Nothing of the sort happens otherwise: a Poisson likelihood in  $e^u$  against a normal density in  $u$  is not a recognizable integral in any dimension. What *does* help is independence. If the random effects are attached to clusters that are independent of each other, the  $q$ -dimensional integral splits into  $m$  integrals of dimension  $q_0$  each, and with a random intercept  $q_0 = 1$ . A one-dimensional integral repeated  $m$  times is tractable; a  $q$ -dimensional one with  $q$  in the hundreds is not.

#### Theorem 40.3.1 — The integrated likelihood and its approximations

In [Definition 40.2.1](#) write  $\psi_i(\mathbf{u}_i) = \log f(\mathbf{y}_i \mid \mathbf{u}_i; \boldsymbol{\beta}, \phi) + \log p(\mathbf{u}_i; \boldsymbol{\theta})$ .

- (a) **Factorization.** If the random effects split into independent blocks  $\mathbf{u}_1, \dots, \mathbf{u}_m$  with  $\mathbf{u}_i$  entering only the observations of cluster  $i$ , then

$$L(\boldsymbol{\beta}, \boldsymbol{\theta}, \phi) = \prod_{i=1}^m \int_{\mathbb{R}^{q_0}} \exp\{\psi_i(\mathbf{u}_i)\} d\mathbf{u}_i, \quad (40.3.1)$$

a product of  $m$  integrals of dimension  $q_0$ .

- (b) **Laplace approximation.** Fix  $i$  and take  $q_0 = 1$ . Let  $\psi_i$  be four times continuously differentiable with a unique interior maximizer  $\hat{u}_i$ , and  $c_i = -\psi_i''(\hat{u}_i) > 0$ . Then, formally,

$$\int e^{\psi_i(u)} du = \sqrt{\frac{2\pi}{c_i}} e^{\psi_i(\hat{u}_i)} \{1 + \delta_i + \dots\}, \quad (40.3.2)$$

$$\delta_i = \frac{\psi_i^{(4)}(\hat{u}_i)}{8c_i^2} + \frac{5\{\psi_i'''(\hat{u}_i)\}^2}{24c_i^3}.$$

Suppose  $\psi_i = n_i k_i + s_i$ , where  $k_i$  does not depend on  $n_i$  and has a nondegenerate maximum and  $s_i$  has derivatives bounded uniformly in  $n_i$ . (In a GLMM  $n_i k_i$  is the conditional log-likelihood and  $s_i$  the log prior density of  $\mathbf{u}_i$ , which does not grow with  $n_i$ .) Then both terms of  $\delta_i$  are  $O(n_i^{-1})$ : the Laplace approximation has *relative error*  $O(n_i^{-1})$  and the error in the log-likelihood is  $O(\sum_i n_i^{-1})$ .

- (c) **Adaptive Gauss–Hermite quadrature.** Let  $(z_k, w_k)_{k=1}^K$  be the nodes and weights of  $K$ -point Gauss–Hermite quadrature, so that  $\int e^{-z^2} P(z) dz = \sum_k w_k P(z_k)$  for every polynomial  $P$  of degree at most  $2K-1$ . With  $\hat{\sigma}_i = c_i^{-1/2}$ , the substitution  $u = \hat{u}_i + \sqrt{2} \hat{\sigma}_i z$  gives the approximation

$$\int e^{\psi_i(u)} du \approx \sqrt{2} \hat{\sigma}_i \sum_{k=1}^K w_k e^{z_k^2} e^{\psi_i(\hat{u}_i + \sqrt{2} \hat{\sigma}_i z_k)}. \quad (40.3.3)$$

It is exact whenever  $e^{\psi_i(\hat{u}_i + \sqrt{2} \hat{\sigma}_i z) + z^2}$  is a polynomial of degree at most  $2K-1$  in  $z$ , and  $K=1$  reproduces Eq. 40.3.2 without its correction terms.

- (d) **Penalized quasi-likelihood** avoids the integral altogether, by a penalized maximization that is cheaper than either of the above and carries a bias that does not vanish. It is stated and proved separately as Proposition 40.3.2.

*Proof.*

- (a) Under the stated independence the integrand factorizes and Fubini's theorem applies, every factor being nonnegative.
- (b) Substitute  $u = \hat{u}_i + s/\sqrt{c_i}$  and expand  $\psi_i$  about  $\hat{u}_i$ , using  $\psi_i'(\hat{u}_i) = 0$ :

$$\psi_i(u) = \psi_i(\hat{u}_i) - \frac{1}{2}s^2 + a_3 s^3 + a_4 s^4 + \dots,$$

$$a_3 = \frac{\psi_i'''(\hat{u}_i)}{6c_i^{3/2}}, \quad a_4 = \frac{\psi_i^{(4)}(\hat{u}_i)}{24c_i^2}.$$

Hence  $\int e^{\psi_i} du = c_i^{-1/2} e^{\psi_i(\hat{u}_i)} \int e^{-s^2/2} e^{a_3 s^3 + a_4 s^4 + \dots} ds$ . Expanding the second exponential as  $1 + (a_3 s^3 + a_4 s^4) + \frac{1}{2} a_3^2 s^6 + \dots$  and using the standard normal moments  $ES^3 = 0$ ,  $ES^4 = 3$ ,  $ES^6 = 15$  gives the factor  $\sqrt{2\pi} \{1 + 3a_4 + \frac{15}{2} a_3^2 + \dots\}$ , which is Eq. 40.3.2. If  $\psi_i = n_i k_i + s_i$  then  $c_i = n_i |k_i''| + O(1)$ ,  $\psi_i''' = n_i k_i''' + O(1)$  and  $\psi_i^{(4)} = n_i k_i^{(4)} + O(1)$ , so  $a_4 = O(n_i^{-1})$  and  $a_3^2 = O(n_i^{-1})$ .

- (c) The substitution is a change of variable, and  $e^{\psi_i(u)} = e^{-z^2} \{e^{z^2} e^{\psi_i(\hat{u}_i + \sqrt{2} \hat{\sigma}_i z)}\}$ , so the quadrature rule applies to the term in braces. For  $K=1$  the node is  $z_1 = 0$  and the weight is  $w_1 = \sqrt{\pi}$ , giving  $\sqrt{2} \hat{\sigma}_i \sqrt{\pi} e^{\psi_i(\hat{u}_i)} = \sqrt{2\pi/c_i} e^{\psi_i(\hat{u}_i)}$ . ■

*Remark* (What “formally” means in (b)). The expansion in the proof interchanges an infinite series with an integral over the whole line, which the argument does not justify, and the remainder can also be affected by the behaviour of  $\psi_i$  far from its mode. The

rigorous statement — that the expansion is asymptotic in  $n_i$ , with an explicit remainder, under conditions on the derivatives of  $k_i$  and on the tails — is classical, and is given in the form used here by Tierney and Kadane (1986) and Barndorff-Nielsen and Cox (1989, chapter 3). [Chapter 39](#) proves the companion statement for the posterior and the marginal likelihood ([Theorem 39.4.2](#)). We take the error order as given; [Example \(Three methods on clustered binary data\)](#) checks it numerically.

Two readings of (b). First, the accuracy of the Laplace approximation is governed by the observations *per cluster*, not by the sample size: a study with 10 000 patients seen twice each is the hard case, and no number of extra patients repairs it. Second, the relevant  $n_i$  is an *effective* one — a cluster of twenty Bernoulli responses says much less about its own  $u_i$  than a cluster of twenty gamma responses — which is why the same approximation behaves so differently in [Example \(Three methods on clustered binary data\)](#) and [Example \(Grunfeld's firms, again\)](#).

Part (c) matters because plain Gauss–Hermite quadrature places its nodes where the *prior* is large. When the data are informative the posterior of  $u_i$  sits elsewhere and is much narrower, so most nodes fall where the integrand is negligible. The adaptive version puts them at each cluster's own mode and curvature — the same  $\hat{u}_i$  and  $\hat{\sigma}_i$  that [Proposition 40.2.3](#) predicts the random effect with — at the cost of one Newton iteration per cluster per likelihood evaluation.

```
import numpy as np
from numpy.polynomial.hermite import hermgauss
from scipy.optimize import minimize, minimize_scalar

BETA = np.array([-0.5, 1.0])
TAU = 1.0

def expit(z):
    return 1.0 / (1.0 + np.exp(-z))

def cluster_mode(y, X, beta, tau):
    """Mode and curvature of log f(y_i | u) + log p(u), cluster by cluster."""
    eta0, u = X @ beta, np.zeros(y.shape[0])
    for _ in range(100):
        mu = expit(eta0 + u[:, None])
        step = ((y - mu).sum(1) - u / tau**2) / (-(mu * (1 - mu)).sum(1) - 1 / tau**2)
        u = u - step
        if np.max(np.abs(step)) < 1e-12:
            break
    mu = expit(eta0 + u[:, None])
    return u, 1.0 / np.sqrt((mu * (1 - mu)).sum(1) + 1 / tau**2)

def loglik(y, X, beta, tau, K, adapt=True):
    """Integrated log-likelihood by K-node Gauss-Hermite quadrature.

    Adaptive: centre and scale the nodes at each cluster's own mode and curvature.
    K = 1 with adapt=True is exactly the Laplace approximation."""
    node, w = hermgauss(K)
    if adapt:
        centre, scale = cluster_mode(y, X, beta, tau)
    else:
        centre, scale = np.zeros(y.shape[0]), np.full(y.shape[0], tau)
    pts = centre[:, None] + np.sqrt(2) * scale[:, None] * node # (m, K)
    eta = (X @ beta)[:, :, None] + pts[:, None, :] # (m, n, K)
    )
```

```

cond = (y[:, :, None] * eta - np.logaddexp(0.0, eta)).sum(1) # (m, K)
prior = -0.5 * (pts / tau) ** 2 - np.log(tau) - 0.5 * np.log(2 * np.pi)
terms = np.log(w) + node**2 + cond + prior
return float(np.sum(np.log(np.sqrt(2) * scale) + np.logaddexp.reduce(terms, axis
=1)))

def fit_ml(y, X, K, start=None):
    p = X.shape[2]
    obj = lambda par: -loglik(y, X, par[:p], np.exp(par[p]), K)
    x0 = np.r_[np.zeros(p), 0.0] if start is None else start
    r = minimize(obj, x0, method="Nelder-Mead",
                 options=dict(maxiter=8000, maxfev=8000, xatol=1e-8, fatol=1e-10))
    return r.x[:p], float(np.exp(r.x[p])), -r.fun

M_demo, N_demo = 60, 5
rng_demo = np.random.default_rng(40_007)
x_demo = rng_demo.normal(size=(M_demo, N_demo))
X_demo = np.stack([np.ones((M_demo, N_demo)), x_demo], axis=2)
u_demo = rng_demo.normal(0, TAU, M_demo)
y_demo = (rng_demo.uniform(size=(M_demo, N_demo))
           < expit(X_demo @ BETA + u_demo[:, None])).astype(float)

for K in [1, 5, 15]:
    b, t_hat, ll = fit_ml(y_demo, X_demo, K)
    print(f"AGHQ K = {K:2d}:  beta = {b.round(4)}    tau = {t_hat:.4f}    loglik = {ll
:.4f}")

```

## 40.3.2 Penalized quasi-likelihood

The oldest practical method avoids the integral by treating the random effects as parameters with a penalty.

### Proposition 40.3.2 — Penalized quasi-likelihood

In [Definition 40.2.1](#) let  $\ell(\boldsymbol{\beta}, \mathbf{u})$  be the conditional log-likelihood of the data given  $\mathbf{u}$ , and consider the **penalized quasi-likelihood**

$$\ell^{\text{pen}}(\boldsymbol{\beta}, \mathbf{u}) = \ell(\boldsymbol{\beta}, \mathbf{u}) - \frac{1}{2} \mathbf{u}^\top \mathbf{G}^{-1} \mathbf{u}. \quad (40.3.4)$$

- (a) For fixed  $\boldsymbol{\theta}$ , the stationary equations of [Eq. 40.3.4](#) are

$$\mathbf{X}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}) = \mathbf{0}, \quad \mathbf{Z}^\top \mathbf{W} \mathbf{D}^{-1} (\mathbf{y} - \boldsymbol{\mu}) = \phi \mathbf{G}^{-1} \mathbf{u},$$

with  $\mathbf{W}$  and  $\mathbf{D}$  the working weights and derivatives of [Theorem 34.3.1](#) evaluated at  $\boldsymbol{\eta} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}$ . Equivalently, with the working response  $\mathbf{z} = \boldsymbol{\eta} + \mathbf{D}^{-1}(\mathbf{y} - \boldsymbol{\mu})$  of [Eq. 34.4.1](#), they are Henderson's mixed-model equations ([Theorem 32.3.2](#)) for the linear mixed model  $\mathbf{z} \mid \mathbf{u} \sim N_n(\mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\mathbf{u}, \phi \mathbf{W}^{-1})$ ,  $\mathbf{u} \sim N_q(\mathbf{0}, \mathbf{G})$ .

- (b) Iterating — update  $(\boldsymbol{\beta}, \mathbf{u})$  by those equations, then update  $\boldsymbol{\theta}$  by maximum likelihood or REML in the working linear mixed model, and repeat — is the Breslow–Clayton PQL algorithm. It is therefore the Laplace approximation [Eq. 40.3.2](#) with

the further simplification that  $\mathbf{W}$  is treated as fixed when  $\boldsymbol{\theta}$  is updated, so that  $c_i$  contributes no derivative terms.

- (c) The resulting  $\hat{\boldsymbol{\beta}}$  and  $\hat{\boldsymbol{\theta}}$  are biased towards zero, and the bias does **not** vanish as  $m \rightarrow \infty$  with cluster sizes fixed: PQL is inconsistent. It is worst for binary responses with small  $n_i$  and large  $\tau$ , and negligible for responses that carry much information per observation.

*Proof.*

- (a) Differentiating  $\ell$  with respect to  $\boldsymbol{\beta}$  and  $\mathbf{u}$  reproduces Eq. 34.3.1 with model matrix  $[\mathbf{X}, \mathbf{Z}]$ , because  $\mathbf{u}$  enters the linear predictor exactly as  $\boldsymbol{\beta}$  does; the penalty contributes  $-\mathbf{G}^{-1}\mathbf{u}$  to the second block and nothing to the first. Multiplying by  $\phi$  gives the displayed pair. For the second form,  $\mathbf{W}\mathbf{D}^{-1}(\mathbf{y} - \boldsymbol{\mu}) = \mathbf{W}(\mathbf{z} - \boldsymbol{\eta})$  by the definition of  $\mathbf{z}$ , so the equations read  $\mathbf{X}^\top \mathbf{W}(\mathbf{z} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}) = \mathbf{0}$  and  $\mathbf{Z}^\top \mathbf{W}(\mathbf{z} - \mathbf{X}\boldsymbol{\beta} - \mathbf{Z}\mathbf{u}) = \phi \mathbf{G}^{-1}\mathbf{u}$ , which are Eq. 32.3.5 with  $\mathbf{R} = \phi \mathbf{W}^{-1}$ .
- (b) is the observation that maximizing Eq. 40.3.4 over  $\mathbf{u}$  locates  $\hat{\mathbf{u}}$  at the mode of the integrand of Eq. 40.2.6, which is the first ingredient of Eq. 40.3.2; the remaining ingredient,  $\log|c_i|$ , is what the  $\boldsymbol{\theta}$  update drops.
- (c) We do not prove this. The asymptotic bias expansions, and corrections built from them, are given by Breslow and Lin (1995) and Lin and Breslow (1996); [Example \(Three methods on clustered binary data\)](#) measures the bias by simulation.

■

*Remark* (PQL is empirical Bayes). Since  $\exp\{\ell^{\text{pen}}\}$  is proportional to the joint density of  $\mathbf{y}$  and  $\mathbf{u}$ , the PQL estimate of  $\mathbf{u}$  is the posterior mode of Proposition 40.2.3 with  $\boldsymbol{\beta}$  and  $\boldsymbol{\theta}$  replaced by estimates: an empirical Bayes predictor, and nothing about it is wrong. What is wrong is using the same objective to estimate  $\boldsymbol{\theta}$ , because it has been maximized over  $\mathbf{u}$  rather than integrated over it, and the two differ by exactly the  $\log c_i$  term that carries the information about  $\tau$ .

### 40.3.3 Monte Carlo alternatives

When  $q_0$  exceeds two or three, a product grid needs  $K^{q_0}$  nodes and stops being practical. **Monte Carlo EM** treats  $\mathbf{u}$  as missing data: the E step's expectation over the posterior of  $\mathbf{u}$  is replaced by an average over Markov chain Monte Carlo draws, and the M step is an ordinary weighted GLM fit. EM's monotonicity survives only up to Monte Carlo error, so the number of draws must grow with the iterations; Chapter 41 develops the algorithm and proves that the exact version never decreases the likelihood. **Simulated maximum likelihood** estimates each factor of Eq. 40.3.1 by importance sampling with the Laplace approximation as proposal — Eq. 40.3.3 with random nodes. **Fully Bayesian inference** samples  $\boldsymbol{\beta}$ ,  $\mathbf{u}$  and  $\boldsymbol{\theta}$  together, as Section 32.6 did for the linear mixed model (Proposition 32.6.1); without conjugacy the full conditionals need Metropolis–Hastings steps or data augmentation, which Chapter 39 builds (Definition 39.4.2).

### 40.3.4 How different are the answers?

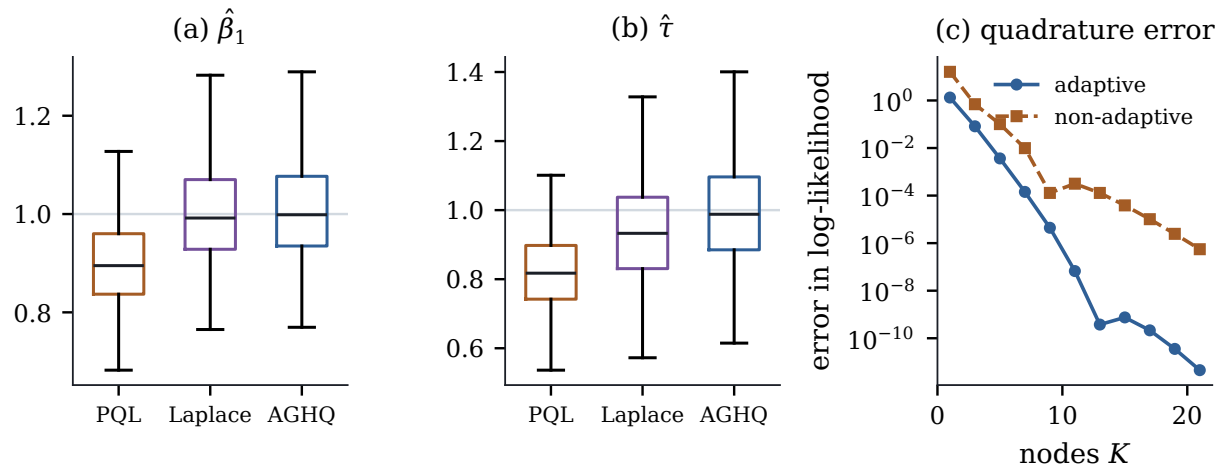


Figure 40.3.1: Approximations to the integrated likelihood, on clustered binary data with  $m = 150$ ,  $n_i = 5$  and  $\tau = 1$ . (a), (b) Sampling distributions of  $\hat{\beta}_1$  and  $\hat{\tau}$  over 200 replications under three methods; the horizontal line is the truth. (c) Error in the integrated log-likelihood against the number of quadrature nodes, adaptive and plain.

#### Example. (Three methods on clustered binary data)

Simulate 150 clusters of 5 binary observations from a random-intercept logistic model with  $\boldsymbol{\beta} = (-0.5, 1.0)^\top$ , a covariate varying within the cluster, and  $\tau = 1$ . Over 200 replications the average estimates are

|                                    | $\hat{\beta}_1$ (truth 1.0) | $\hat{\tau}$ (truth 1.0) |
|------------------------------------|-----------------------------|--------------------------|
| PQL                                | 0.8970                      | 0.8168                   |
| Laplace                            | 1.0022                      | 0.9353                   |
| adaptive Gauss–Hermite,<br>$K = 9$ | 1.0087                      | 0.9909                   |

PQL attenuates the slope by 10.3 per cent and the standard deviation by 18.3 per cent. The Laplace approximation is nearly unbiased for the slope but still shrinks  $\hat{\tau}$  by about six per cent, which is consistent with the  $O(n_i^{-1})$  error of [Theorem 40.3.1\(b\)](#) at  $n_i = 5$ ; the direction and size of the induced bias are measured here, not derived, since part (b) bounds the error in the likelihood and says nothing about where its maximizer moves. Nine adaptive nodes leave a bias of about one per cent, which is the ordinary finite-sample bias of maximum likelihood rather than a numerical error.

Panel (c) of [Figure 40.3.1](#) separates the numerical question from the statistical one. At a fixed parameter value, the error in the integrated log-likelihood with  $K = 5$  is  $3.65136 \times 10^{-3}$  for the adaptive rule and  $1.01404 \times 10^{-1}$  for the plain one — a factor of about 28 — and the adaptive rule is down to  $3.759 \times 10^{-10}$  by  $K = 13$ , while the plain one is still above  $10^{-6}$ , at  $2.460 \times 10^{-6}$ , when  $K = 19$ .

```
def weighted_lmm(z, X, W):
    """REML fit of a weighted random-intercept linear mixed model:  $z_{ij} = x_{ij}'b + u_i + e_{ij}$ 
    with  $\text{Var}(e_{ij}) = 1/W_{ij}$  and  $\text{Var}(u_i) = t2$ . Returns  $b$ ,  $t2$  and the predicted  $u$ ."""
    def solve(t2):
```



```

s, WX = W.sum(1), W[... , None] * X
den = 1 + t2 * s
XtVX = (np.einsum("mnp,mnq->pq", WX, X)
        - t2 * np.einsum("mp,mq->pq", WX.sum(1), WX.sum(1) / den[:, None]))
XtVz = (np.einsum("mnp,mn->p", WX, z)
        - t2 * np.einsum("mp,m->p", WX.sum(1), (W * z).sum(1) / den))
return np.linalg.solve(XtVX, XtVz), XtVX, den

def neg_reml(log_t2):
    t2 = np.exp(log_t2)
    b, XtVX, den = solve(t2)
    r = z - X @ b
    q = np.sum(W * r * r) - t2 * np.sum((W * r).sum(1) ** 2 / den)
    ld = np.sum(np.log(den)) - np.sum(np.log(W))
    return 0.5 * (ld + np.linalg.slogdet(XtVX)[1] + q)

opt = minimize_scalar(neg_reml, bounds=(-12.0, 6.0), method="bounded",
                      options=dict(xatol=1e-10))
t2 = float(np.exp(opt.x))
b, _, den = solve(t2)
r = z - X @ b
return b, t2, t2 * (W * r).sum(1) / den

def fit_pql(y, X, maxit=300):
    """Breslow-Clayton penalized quasi-likelihood: a linear mixed model on the
    working
    response, refitted until it stops changing."""
    beta, u = np.zeros(X.shape[2]), np.zeros(y.shape[0])
    for _ in range(maxit):
        eta = X @ beta + u[:, None]
        mu = expit(eta)
        v = np.clip(mu * (1 - mu), 1e-8, None)          # working weights
        z = eta + (y - mu) / v                          # working response
        b_new, t2, u_new = weighted_lmm(z, X, v)
        done = max(np.max(np.abs(b_new - beta)), np.max(np.abs(u_new - u))) < 1e-10
        beta, u = b_new, u_new
    if done:
        break
    return beta, float(np.sqrt(t2)), u

b_pql, t_pql, _ = fit_pql(y_demo, X_demo)
print(f"PQL          : beta = {b_pql.round(4)}    tau = {t_pql:.4f}")

```

The contrast with [Example \(Grunfeld’s firms, again\)](#) is instructive. That fit has  $n_i = 20$  continuous responses per firm, and refitting it with one adaptive node instead of nine changes no parameter by more than  $5 \times 10^{-4}$  and the maximized log-likelihood, which is  $-937.100$ , by less than 0.01. What separates the two is not the software but the information each cluster carries about its own random effect, exactly as [Theorem 40.3.1\(b\)](#) says.

*Idea.* Report which approximation was used. “A random-intercept logistic model was fitted” is not a complete description when PQL and adaptive quadrature can differ by a tenth of the coefficient. With binary responses in small clusters, use quadrature with enough nodes that the answer stops moving, and say how many.



### 40.3.5 Exercises

#### 40.3.5.1 A. Check your understanding

##### Exercise A1.

A GLMM with two correlated random effects per cluster is fitted by non-adaptive Gauss–Hermite quadrature on a product grid. How many likelihood evaluations per cluster does  $K = 20$  cost? What if there are four random effects per cluster? What does this say about crossed random effects, where  $q_0$  is effectively the whole vector?

##### Exercise A2.

For each of the following, say whether PQL is likely to be adequate and why: (a) counts with means in the hundreds, ten per cluster; (b) binary responses, three per cluster, with a large estimated  $\tau$ ; (c) grouped binomial responses with  $N_i = 200$  trials, one per cluster.

#### 40.3.5.2 B. Practice

##### Exercise B1.

Verify [Theorem 40.3.1\(c\)](#) for  $K = 1$  directly: show that the one-point Gauss–Hermite rule has node  $z_1 = 0$  and weight  $w_1 = \sqrt{\pi}$ , and that [Eq. 40.3.3](#) then reduces to [Eq. 40.3.2](#) without its correction terms. Then explain why the *non*-adaptive rule with  $K = 1$  gives something quite different, and say what.

*Solution.* The one-point rule must integrate  $e^{-z^2}P(z)$  exactly for  $P$  of degree  $\leq 1$ . Taking  $P = 1$  gives  $w_1 = \int e^{-z^2} dz = \sqrt{\pi}$ ; taking  $P = z$  gives  $w_1 z_1 = \int z e^{-z^2} dz = 0$ , so  $z_1 = 0$ . Substituting in [Eq. 40.3.3](#) gives  $\sqrt{2}\hat{\sigma}_i\sqrt{\pi}e^{\psi_i(\hat{u}_i)}$ , which is  $\sqrt{2\pi/c_i}e^{\psi_i(\hat{u}_i)}$ . The non-adaptive rule puts its single node at  $u = 0$  with scale  $\tau$ , giving  $\sqrt{2\pi}\tau e^{\psi_i(0)}$ : the integrand evaluated at the *prior* mode with the prior's width, which is the right answer only when the data say nothing.

##### Exercise B2.

For a cluster of  $n$  Bernoulli responses with common success probability  $\pi$  and a random intercept, show that  $c_i = n\pi(1 - \pi) + \tau^{-2}$ . Compare with the gamma case of [Example \(Grunfeld's firms, again\)](#), where  $c_i = n/\phi + \tau^{-2}$ , and use [Theorem 40.3.1\(b\)](#) to explain the difference between the two fits in terms of an effective cluster size.

*Solution.* For the logit,  $\psi_i''(u) = -\sum_j \mu_{ij}(1 - \mu_{ij}) - \tau^{-2}$ , which at a common  $\pi$  is  $-n\pi(1 - \pi) - \tau^{-2}$ . For the gamma with a log link,  $\psi_i''(u) = -\nu \sum_j y_{ij}e^{-\eta_{ij}} - \tau^{-2}$ , which at the mode is about  $-n/\phi - \tau^{-2}$  since  $\nu = 1/\phi$  and  $y_{ij}e^{-\eta_{ij}} \approx 1$ . The Laplace error is governed by  $c_i\tau^2$ , so the effective cluster size is  $n\pi(1 - \pi)\tau^2$  against  $n\tau^2/\phi$ . With  $n = 5$ ,  $\pi = 1/2$ ,  $\tau = 1$  the first is 1.25; with  $n = 20$ ,  $\phi = 0.0930$ ,  $\tau = 0.52$  the second is about 58. The two fits are in completely different regimes.

## 40.4 Generalized estimating equations

Suppose the question is the marginal one and the cluster structure a nuisance. Then a generalized linear mixed model asks for more than is needed: a distribution for the random effects, a conditional distribution for the responses, and an intractable integral. Everything the question needs is in the *mean*, and [Chapter 38](#) showed that a mean and a variance function suffice to build a usable estimating equation without a likelihood.

Carrying that across to clustered data gives the method of Liang and Zeger (1986), the most widely used in this chapter and the most widely misunderstood. It estimates a marginal coefficient consistently even when the correlation structure it assumes is wrong. It does not estimate a conditional coefficient at all; it produces no likelihood, so most of the model-comparison machinery of the book does not apply; and its robustness to a wrong correlation does not extend to data that go missing for the wrong reasons.

### 40.4.1 The estimating equation

Clusters  $i = 1, \dots, m$  are independent, with response vectors  $\mathbf{Y}_i$  of length  $n_i$  and model matrices  $\mathbf{X}_i$  of size  $n_i \times p$ . The model constrains the marginal mean only:

$$E(\mathbf{Y}_i) = \boldsymbol{\mu}_i(\boldsymbol{\beta}), \quad \mu_{ij}(\boldsymbol{\beta}) = h(\mathbf{x}_{(ij)}^\top \boldsymbol{\beta}), \quad (40.4.1)$$

with  $h$  the inverse of a link as in [Definition 34.2.2](#). Nothing is assumed about the joint distribution of  $\mathbf{Y}_i$ , and in particular nothing about its covariance — although a guess at that covariance is used, deliberately.

#### Definition 40.4.1 — Generalized estimating equations

Let  $V$  be a variance function and  $\phi$  a dispersion, and write

$$\begin{aligned} \mathbf{A}_i(\boldsymbol{\beta}) &= \text{diag}\{V(\mu_{i1}), \dots, V(\mu_{in_i})\}, \\ \mathbf{D}_i(\boldsymbol{\beta}) &= \frac{\partial \boldsymbol{\mu}_i}{\partial \boldsymbol{\beta}^\top} = \text{diag}\{h'(\eta_{ij})\} \mathbf{X}_i. \end{aligned}$$

Let  $\mathbf{R}_i(\boldsymbol{\alpha})$  be a **working correlation matrix**: a correlation matrix for the  $n_i$  observations of cluster  $i$ , depending on a finite parameter  $\boldsymbol{\alpha}$ , chosen by the analyst and *not* assumed correct. The **working covariance** is

$$\mathbf{V}_i(\boldsymbol{\beta}, \boldsymbol{\alpha}, \phi) = \phi \mathbf{A}_i^{1/2} \mathbf{R}_i(\boldsymbol{\alpha}) \mathbf{A}_i^{1/2}, \quad (40.4.2)$$

and the **generalized estimating equation** is

$$\mathbf{U}_m(\boldsymbol{\beta}; \boldsymbol{\alpha}, \phi) = \sum_{i=1}^m \mathbf{D}_i(\boldsymbol{\beta})^\top \mathbf{V}_i^{-1} \{\mathbf{Y}_i - \boldsymbol{\mu}_i(\boldsymbol{\beta})\} = \mathbf{0}. \quad (40.4.3)$$

In practice  $\boldsymbol{\alpha}$  and  $\phi$  are replaced by moment estimates computed from the Pearson

residuals  $r_{ij} = (y_{ij} - \hat{\mu}_{ij})/\sqrt{V(\hat{\mu}_{ij})}$  at the current  $\hat{\beta}$ , and Eq. 40.4.3 is re-solved, until nothing changes.

Within this chapter  $\mathbf{R}_i$  denotes the working correlation matrix, not the error covariance  $\mathbf{R}$  of Part VII.

*Remark* (Three familiar special cases). With  $\mathbf{R}_i = \mathbf{I}$  and one observation per cluster, Eq. 40.4.3 is exactly the likelihood equation Eq. 34.3.1, with  $\mathbf{W}\mathbf{D}^{-1}$  written out. With  $\mathbf{R}_i = \mathbf{I}$  and several observations per cluster it is still that equation, applied as if the observations were independent: the same  $\hat{\beta}$ , different standard errors. And with a general  $\mathbf{R}_i$  but one cluster, it is the quasi-score of Chapter 38 (Theorem 38.2.2) with a full covariance matrix — which for the identity link and  $V \equiv 1$  is the generalized least squares of Proposition 33.4.1.

Solving Eq. 40.4.3 is Fisher scoring. Write  $\mathbf{S}_i = \text{diag}\{h'(\eta_{ij})/\sqrt{V(\mu_{ij})}\}$ ; then  $\mathbf{D}_i^\top \mathbf{V}_i^{-1} = \phi^{-1} \mathbf{X}_i^\top \mathbf{S}_i \mathbf{R}_i^{-1} \mathbf{A}_i^{-1/2}$ , so the update is

$$\hat{\beta} \leftarrow \hat{\beta} + \left( \sum_i \mathbf{X}_i^\top \mathbf{S}_i \mathbf{R}_i^{-1} \mathbf{S}_i \mathbf{X}_i \right)^{-1} \sum_i \mathbf{X}_i^\top \mathbf{S}_i \mathbf{R}_i^{-1} \mathbf{r}_i, \quad (40.4.4)$$

in which  $\phi$  cancels. It is the iteratively reweighted least squares of Theorem 34.4.1 with  $\mathbf{R}_i^{-1}$  inserted between the weights, and it converges just as quickly.

## 40.4.2 What the estimate does and does not have

### Theorem 40.4.1 — Consistency, asymptotic normality and the sandwich

Let the clusters be independent, let  $\beta^0$  be the true value in Eq. 40.4.1, and let  $\Sigma_i^0 = \text{Cov}(\mathbf{Y}_i)$  be arbitrary and unknown. Assume

- (E1)  $n_i \leq n_{\max}$  and  $\|\mathbf{x}_{(ij)}\| \leq C$  for all  $i, j$ ;
- (E2)  $h$  is twice continuously differentiable with  $h'$  bounded,  $V$  is continuous and bounded away from 0 on the relevant range, and  $\beta^0$  is interior to a compact parameter set;
- (E3) the eigenvalues of  $\mathbf{R}_i(\alpha)$  are bounded above and away from zero, uniformly in  $i$  and in  $\alpha$  near its limit;
- (E4)  $E\|\mathbf{Y}_i\|^4 \leq K$  for all  $i$ ;
- (E5)  $\hat{\alpha}$  and  $\hat{\phi}$  are  $\sqrt{m}$ -consistent for some limits  $\alpha^*, \phi^*$ , which need not make  $\mathbf{V}_i$  equal to  $\Sigma_i^0$ ;
- (E6) with  $\mathbf{V}_i$  evaluated at  $(\beta^0, \alpha^*, \phi^*)$ ,

$$\mathbf{A}_m = \frac{1}{m} \sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i \rightarrow \mathbf{A},$$

$$\mathbf{B}_m = \frac{1}{m} \sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \Sigma_i^0 \mathbf{V}_i^{-1} \mathbf{D}_i \rightarrow \mathbf{B},$$

both positive definite.

Then:

- (a) **Unbiasedness.** For every fixed  $\alpha$  and  $\phi$ ,  $E\{\mathbf{U}_m(\beta^0; \alpha, \phi)\} = \mathbf{0}$ . The estimating function is unbiased whatever the working correlation is.
- (b) **Consistency and asymptotic normality.** With probability tending to one, Eq. 40.4.3 has a root  $\hat{\beta}_m$  with  $\hat{\beta}_m \rightarrow \beta^0$  in probability, and

$$\sqrt{m}(\hat{\beta}_m - \beta^0) \xrightarrow{d} N_p(\mathbf{0}, \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}). \quad (40.4.5)$$

- (c) **The sandwich estimator.** Let  $\mathbf{r}_i = \mathbf{Y}_i - \mu_i(\hat{\beta}_m)$  and

$$\begin{aligned} \hat{\mathbf{U}}_R &= \hat{\mathbf{A}}_m^{-1} \left( \sum_i \hat{\mathbf{D}}_i^\top \hat{\mathbf{V}}_i^{-1} \mathbf{r}_i \mathbf{r}_i^\top \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i \right) \hat{\mathbf{A}}_m^{-1}, \\ \hat{\mathbf{A}}_m &= \sum_i \hat{\mathbf{D}}_i^\top \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i, \end{aligned} \quad (40.4.6)$$

with hats denoting evaluation at  $\hat{\beta}_m, \hat{\alpha}, \hat{\phi}$ ;  $\hat{\mathbf{A}}_m$  is  $m$  times the sample version of the  $\mathbf{A}_m$  of (E6). Then  $m \hat{\mathbf{U}}_R \rightarrow \mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$  in probability, and the Wald statistic of a hypothesis  $\mathbf{A}^\top \beta$  of rank  $q$ , formed with  $\hat{\mathbf{U}}_R$ , tends to  $\chi^2(q)$ .

- (d) **When the model-based covariance is valid.** If the working covariance is correct,  $\mathbf{V}_i = \Sigma_i^0$ , then  $\mathbf{B} = \mathbf{A}$  and Eq. 40.4.5 reduces to  $\mathbf{A}^{-1}$ , which the **model-based** estimator  $\hat{\mathbf{U}}_M = (\sum_i \hat{\mathbf{D}}_i^\top \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i)^{-1}$  estimates consistently. Otherwise  $\hat{\mathbf{U}}_M$  is inconsistent, in either direction.

*Proof.*

- (a) At  $\beta = \beta^0$  the matrices  $\mathbf{D}_i$  and  $\mathbf{V}_i$  are deterministic — they depend on the data only through  $\beta$ , which is held fixed — so  $E \mathbf{U}_m = \sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} E\{\mathbf{Y}_i - \mu_i(\beta^0)\} = \mathbf{0}$  by Eq. 40.4.1. Nothing about  $\mathbf{R}_i$  enters, and that is the whole source of the method's robustness.
- (b) and (c). Expand  $\mathbf{U}_m(\beta; \alpha)$  about  $\beta^0$  at  $\alpha = \hat{\alpha}$ . Since  $\mathbf{U}_m$  is vector-valued, no single intermediate point serves, so we use the integral form of Taylor's theorem:

$$\begin{aligned} \mathbf{0} &= \mathbf{U}_m(\hat{\beta}_m; \hat{\alpha}) = \mathbf{U}_m(\beta^0; \hat{\alpha}) + \bar{\mathbf{M}}_m(\hat{\beta}_m - \beta^0), \\ \bar{\mathbf{M}}_m &= \int_0^1 \frac{\partial \mathbf{U}_m}{\partial \beta^\top}(\beta^0 + t(\hat{\beta}_m - \beta^0)) dt. \end{aligned}$$

Three facts give the result.

*The derivative.*  $\partial \mathbf{U}_m / \partial \beta^\top = -\sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i + \mathbf{T}_m$ , where  $\mathbf{T}_m$  collects the terms in which the derivative falls on  $\mathbf{D}_i^\top \mathbf{V}_i^{-1}$  and therefore still multiplies  $\mathbf{Y}_i - \mu_i$ . Each summand of  $\mathbf{T}_m$  has mean zero at  $\beta^0$  and bounded variance by (E1), (E2) and (E4), so  $m^{-1} \mathbf{T}_m \rightarrow \mathbf{0}$  in probability by the weak law for independent summands; and  $m^{-1} \sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i \rightarrow \mathbf{A}$  by (E6). Hence  $m^{-1} \partial \mathbf{U}_m / \partial \beta^\top \rightarrow -\mathbf{A}$ , uniformly for  $\beta$  in a neighbourhood of  $\beta^0$  under (E2); the convergence being uniform over that neighbourhood,  $m^{-1} \bar{\mathbf{M}}_m \rightarrow -\mathbf{A}$  as well, once  $\hat{\beta}_m$  is in it.

*The score.*  $m^{-1/2} \mathbf{U}_m(\beta^0; \alpha^*)$  is  $m^{-1/2}$  times a sum of  $m$  independent mean-zero vectors  $\mathbf{s}_i = \mathbf{D}_i^\top \mathbf{V}_i^{-1}(\mathbf{Y}_i - \mu_i)$  with  $\sum_i \text{Cov}(\mathbf{s}_i) = m\mathbf{B}_m$ . Conditions (E1)–(E4) bound  $E\|\mathbf{s}_i\|^4$  uniformly, which gives the Lyapunov condition, so the Lindeberg–Feller theorem applies and  $m^{-1/2} \mathbf{U}_m \rightarrow N_p(\mathbf{0}, \mathbf{B})$ .

*Estimating  $\alpha$  does not matter.* Differentiating Eq. 40.4.3 with respect to  $\alpha$  leaves the factor  $\mathbf{Y}_i - \boldsymbol{\mu}_i$  untouched, so  $\partial \mathbf{U}_m / \partial \alpha^\top$  at  $\beta^0$  is again a sum of mean-zero terms and  $m^{-1} \partial \mathbf{U}_m / \partial \alpha^\top \rightarrow \mathbf{0}$ . Therefore  $m^{-1/2} \mathbf{U}_m(\beta^0; \hat{\alpha}) = m^{-1/2} \mathbf{U}_m(\beta^0; \alpha^*) + \{m^{-1} \partial \mathbf{U}_m / \partial \alpha^\top\} \sqrt{m}(\hat{\alpha} - \alpha^*) = m^{-1/2} \mathbf{U}_m(\beta^0; \alpha^*) + o_p(1)$  by (E5).

Combining,  $\sqrt{m}(\hat{\beta}_m - \beta^0) = \mathbf{A}^{-1} m^{-1/2} \mathbf{U}_m(\beta^0) + o_p(1)$ , which is Eq. 40.4.5. For (c),  $\mathbf{r}_i \mathbf{r}_i^\top$  has mean  $\Sigma_i^0 + O(m^{-1})$  at  $\hat{\beta}_m$ , and the argument of Theorem 21.4.2 with the cluster in place of the observation — the version already used for Proposition 33.4.1(b) — gives  $m^{-1} \sum_i \hat{\mathbf{D}}_i^\top \hat{\mathbf{V}}_i^{-1} \mathbf{r}_i \mathbf{r}_i^\top \hat{\mathbf{V}}_i^{-1} \hat{\mathbf{D}}_i \rightarrow \mathbf{B}$ . The Wald limit follows from Eq. 40.4.5 and the continuous mapping theorem.

(d) If  $\mathbf{V}_i = \Sigma_i^0$  then  $\mathbf{D}_i^\top \mathbf{V}_i^{-1} \Sigma_i^0 \mathbf{V}_i^{-1} \mathbf{D}_i = \mathbf{D}_i^\top \mathbf{V}_i^{-1} \mathbf{D}_i$  for every  $i$ , so  $\mathbf{B}_m = \mathbf{A}_m$ . ■

*Remark* (What is proved and what is assumed). The existence of a consistent root and the uniform control of the derivative are the two places where the argument leans on standard machinery rather than its own steam; both go exactly as for Theorem 34.5.1, using the concavity-free argument for estimating equations in Liang and Zeger (1986). Condition (E5) is a real assumption: moment estimators of  $\alpha$  from Pearson residuals are  $\sqrt{m}$ -consistent under (E1)–(E4), but an unstructured  $\mathbf{R}$  estimated from few clusters is not. Note the notation:  $\mathbf{U}_m$  is the estimating *function*, while  $\hat{\mathbf{U}}_R$  and  $\hat{\mathbf{U}}_M$  are *covariance estimators*, as in Chapter 21.

*Warning* (The mean model must hold for the whole cluster). Eq. 40.4.1 constrains  $E(\mathbf{Y}_i)$ , the mean of the whole cluster vector, and part (a) uses it in that form; what one usually has in mind is the weaker  $E(Y_{ij} | \mathbf{x}_{(ij)}) = h(\mathbf{x}_{(ij)}^\top \boldsymbol{\beta})$ . A *non-diagonal*  $\mathbf{R}_i$  mixes observations across occasions, so unbiasedness then needs the **full-covariate conditional mean** condition  $E(Y_{ij} | \mathbf{X}_i) = h(\mathbf{x}_{(ij)}^\top \boldsymbol{\beta})$ : the mean at occasion  $j$  must not depend on the covariates at other occasions. With time-varying covariates and feedback — this year’s investment changing next year’s market value, a dose adjusted in response to last month’s reading — that fails even though the occasion-by-occasion mean model is correct, and a non-diagonal working correlation is then *inconsistent* while independence GEE is not (Pepe and Anderson 1994; Diggle, Heagerty, Liang and Zeger 2002, chapter 12).

*Idea.* The asymptotics are in  $m$ , the number of clusters; the total sample size is irrelevant. A trial with ten thousand patients in six hospitals has six independent pieces of information about anything that varies between hospitals, and the sandwich cannot manufacture more — the lesson of Section 33.2.

### 40.4.3 Travel-mode choice

#### Example. (Four options per traveller)

Each of 210 travellers between Australian cities chose one of four modes — air, train, bus, car — and the data record, for every traveller and every mode, the terminal waiting time and a generalized cost. Take  $Y_{ij} = 1$  if traveller  $i$  chose mode  $j$ , fit a logistic model with mode indicators, waiting time and cost, and treat each traveller as a cluster of  $n_i = 4$ .

The sample deliberately over-represents the rarer modes, so by [Theorem 35.5.1](#) the mode indicators and the intercept are shifted by the sampling design and are not population quantities; the within-traveller coefficients are unaffected, and standard errors are the point here in any case.

The four records of a traveller are anything but independent: exactly one is a 1. An independence working correlation gives the same coefficients as ordinary logistic regression, so the question is only what the standard errors should be. The ratio of the sandwich standard error to the model-based one ranges from 0.694 to 1.558 across the six coefficients: for terminal waiting time, whose coefficient is  $-1.0693$  per ten minutes, the sandwich standard error is 1.558 times larger (0.1698 against 0.1090); for generalized cost, whose coefficient is  $-0.0098$  per ten dollars, it is 0.694 times as large (0.0138 against 0.0199).

The sandwich is not a device for making standard errors bigger. Negative within-cluster correlation makes *within-cluster contrasts* better determined than independence would suggest, and cost is such a contrast; waiting time is confounded with the mode indicators, which the traveller's single choice constrains, and is therefore worse determined. Two further observations. The Pearson dispersion estimate is 5.155, which looks like extreme overdispersion and means nothing: Bernoulli responses carry no information about a dispersion parameter ([Proposition 40.2.1](#)(b) with  $N_i = 1$ ). And the exchangeable working correlation estimate is  $-0.3015$ , close to the smallest value  $-1/(n_i - 1) = -1/3$  an exchangeable matrix of size four can have: there the smallest eigenvalue of  $\mathbf{R}$  is  $1 + 3\hat{\alpha} = 0.0956$ . An almost singular structure is a bad working covariance, and independence is the sensible choice.

```
import numpy as np

def expit(z):
    return 1.0 / (1.0 + np.exp(-z))

def working_correlation(kind, r, p, scale_fixed=False):
    """Moment estimate of the working correlation from the Pearson residuals r (m by
    n)."""
    m, n = r.shape
    phi = 1.0 if scale_fixed else np.sum(r**2) / (m * n - p)
    if kind == "independence":
        return np.eye(n), 0.0, phi
    if kind == "exchangeable":
        off = (np.sum(r.sum(1)**2) - np.sum(r**2)) / 2
        a = off / (phi * (m * n * (n - 1) / 2 - p))
        return np.eye(n) + a * (1 - np.eye(n)), a, phi
    if kind == "ar1":
        a = np.sum(r[:, :-1] * r[:, 1:]) / (phi * (m * (n - 1) - p))
        lag = np.abs(np.arange(n)[:, None] - np.arange(n))
        return a**lag, a, phi
    if kind == "unstructured":
        R = (r.T @ r) / (phi * m)
        R = R / np.sqrt(np.outer(np.diag(R), np.diag(R)))
        return R, np.nan, phi
    raise ValueError(kind)

def gee(y, X, kind="independence", family="binomial", scale_fixed=False, maxit=200):
    """Solve  $\sum_i D_i' V_i^{-1} \{y_i - \mu_i\} = 0$  for balanced clusters y (m, n), X
    (m, n, p).
```

```

With  $S_i = \text{diag}(\mu'_{ij} / \sqrt{V(\mu_{ij})})$  and Pearson residuals  $r_i$ , the Fisher-
scoring
step is  $\beta \leftarrow \beta + (\sum_i X_i' S_i R^{-1} S_i X_i)^{-1} \sum_i X_i' S_i R^{-1} r_i$ .""
m, n, p = X.shape
beta, R, alpha, phi = np.zeros(p), np.eye(n), 0.0, 1.0
for _ in range(maxit):
    eta = X @ beta
    if family == "binomial":
        mu = expit(eta)
        dmu, V = mu * (1 - mu), mu * (1 - mu)
    else:
        # log link,  $V(\mu) = \mu^2 (\gamma)$ 
        mu = np.exp(eta)
        dmu, V = mu, mu**2
    r = (y - mu) / np.sqrt(V)
    R, alpha, phi = working_correlation(kind, r, p, scale_fixed)
    Rinv = np.linalg.inv(R)
    S = dmu / np.sqrt(V)
    SX = S[..., None] * X
    A = np.einsum("mjp,jk,mkq->pq", SX, Rinv, SX)
    g = np.einsum("mjp,jk,mk->p", SX, Rinv, r)
    step = np.linalg.solve(A, g)
    beta = beta + step
    if np.max(np.abs(step)) < 1e-11:
        break
u = np.einsum("mjp,jk,mk->mp", SX, Rinv, r)
Ainv = np.linalg.inv(A)
robust = Ainv @ (u.T @ u) @ Ainv
naive = phi * Ainv
covariance
return dict(beta=beta, naive=naive, robust=robust, alpha=alpha, phi=phi, R=R,
            A=A, Ainv=Ainv, S=S, r=r, Rinv=Rinv, X=X)

```

```

import pandas as pd
import statsmodels.api as sm

d = sm.datasets.modechoice.load_pandas().data.sort_values(["individual", "mode"])
m, n = d["individual"].nunique(), 4
# 210 travellers, 4 modes
each
y = d["choice"].to_numpy().reshape(m, n)
D = pd.get_dummies(d["mode"].astype(int), drop_first=True).to_numpy(dtype=float)
cols = [np.ones(m * n)] + [D[:, j] for j in range(3)] + [d["ttme"] / 10, d["gc"] / 10]
X = np.stack([np.asarray(c, dtype=float).reshape(m, n) for c in cols], axis=2)

# a Bernoulli response carries no information about a dispersion parameter, so phi = 1
fit = gee(y, X, "independence", scale_fixed=True)
se_naive, se_robust = np.sqrt(np.diag(fit["naive"])), np.sqrt(np.diag(fit["robust"]))
names = ["intercept", "train", "bus", "car", "ttme/10", "gc/10"]
for k, name in enumerate(names):
    print(f"{name:10s} {fit['beta'][k]:8.4f} naive {se_naive[k]:.4f} robust {se_robust[k]:.4f}")
print("exchangeable alpha:", round(gee(y, X, 'exchangeable', scale_fixed=True)['alpha'], 4))

```



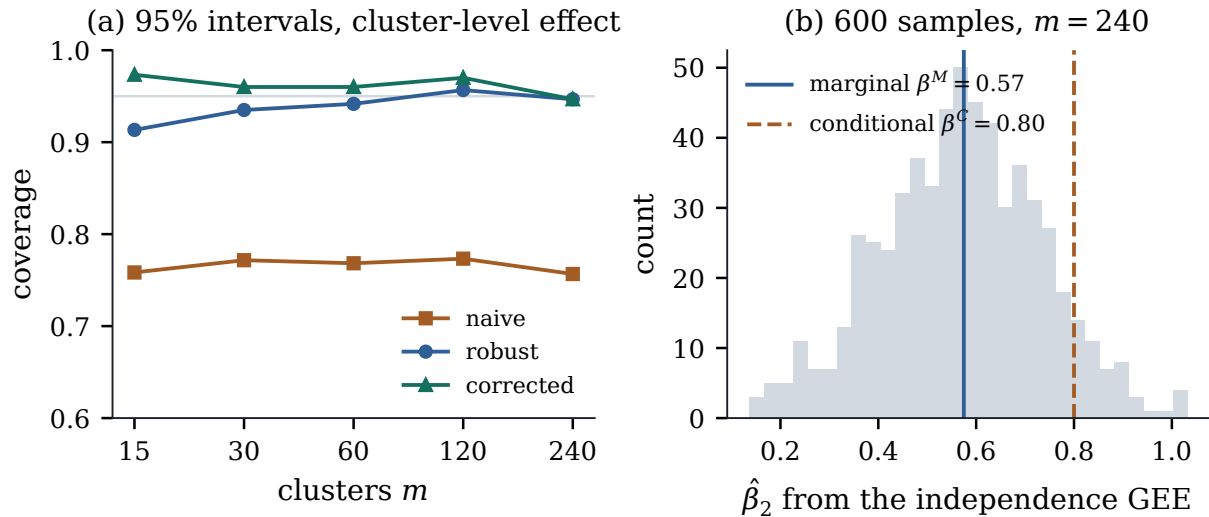


Figure 40.4.1: GEE with a wrong working correlation. (a) Coverage of nominal 95% intervals for the marginal coefficient of a cluster-level covariate, against the number of clusters, for three covariance estimators. (b) The sampling distribution of  $\hat{\beta}_2$  at  $m = 240$ , against the marginal and conditional values.

#### 40.4.4 Small samples, and a simulation

The sandwich Eq. 40.4.6 replaces  $\Sigma_i^0$  by  $\mathbf{r}_i \mathbf{r}_i^\top$ , and the residuals are too small because  $\hat{\beta}_m$  was fitted to them; the bias is of order  $p/m$  and downward, exactly as for Theorem 21.4.2 and Exercise (C2). Two standard repairs, both used in Example (Coverage when the working correlation is wrong):

- **Bias-corrected residuals.** Mancl and DeRouen (2001) replace  $\mathbf{r}_i$  by  $(\mathbf{I} - \mathbf{H}_i)^{-1} \mathbf{r}_i$ , with  $\mathbf{H}_i = \mathbf{D}_i (\sum_k \mathbf{D}_k^\top \mathbf{V}_k^{-1} \mathbf{D}_k)^{-1} \mathbf{D}_i^\top \mathbf{V}_i^{-1}$  cluster  $i$ 's leverage block — the clustered analogue of the HC3 correction of Definition 21.4.1. Kauermann and Carroll (2001) use  $(\mathbf{I} - \mathbf{H}_i)^{-1/2}$ , which corrects less.
- **A  $t$  reference.** Replace the normal quantile by  $t_{m-p}$ . With  $m = 15$  and  $p = 3$  this widens a 95% interval by  $t_{12,0.975}/z_{0.975} = 2.179/1.960$ , about eleven per cent.

##### Example. (Coverage when the working correlation is wrong)

Generate clusters of  $n_i = 8$  binary responses from a random-intercept logistic model with  $\tau = 1.5$ , one covariate varying within the cluster and one fixed for the whole cluster (a treatment assigned to clusters), and fit the marginal model by GEE with an *independence* working correlation — wrong by construction. The target is the marginal cluster-level coefficient, which Eq. 40.1.3 puts at 0.5748 when the conditional coefficient is 0.8. Over 600 replications at  $m = 240$  the mean estimate is 0.5681: consistent for the marginal coefficient, and not for the conditional one.

The three intervals behave quite differently. The model-based interval covers 0.757 of the time and does not improve with  $m$ : it is wrong, not merely imprecise. The sandwich covers 0.913 at  $m = 15$  and 0.947 at  $m = 240$ , and the bias-corrected sandwich with a  $t$  reference covers 0.973 at  $m = 15$  — slightly conservative, which at that sample size is the right side to err on.



## 40.4.5 What GEE does not give

*No likelihood.* Eq. 40.4.3 is not the derivative of anything in general, so there is no deviance, no likelihood ratio test, and no AIC or BIC. Nested mean models are compared by Wald tests built on Eq. 40.4.6 or by score-type tests built on Eq. 40.4.3 under the null; selection uses QIC, the quasi-likelihood analogue of AIC (Section 40.5). Reporting a “deviance” from GEE output is a category error.

*No random effects and no distribution.* There is nothing to predict, so if the object of interest is a cluster — which hospital is doing badly, which firm is unusually profitable — GEE has no answer and Proposition 40.2.3 has no analogue; and because only the mean is modelled, the fit cannot be simulated from, cannot give a prediction interval, and cannot be checked by an envelope or posterior predictive method.

*Few clusters.* Everything above is asymptotic in  $m$ . Below about forty clusters the sandwich needs correcting, and below about fifteen no correction reliably saves it; a likelihood-based method with a covariance model from Definition 33.4.1 is then the better bet.

## 40.4.6 Missing data

The robustness of Eq. 40.4.3 has a sharp limit. If some responses are not observed, write  $R_{ij} = 1$  when  $Y_{ij}$  is recorded, so that the estimating function is really  $\sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1} (\mathbf{Y}_i - \boldsymbol{\mu}_i)$  with  $\mathbf{D}_i = \text{diag}(R_{ij})$ . Part (a) of Theorem 40.4.1 needs  $E\{\mathbf{D}_i(\mathbf{Y}_i - \boldsymbol{\mu}_i)\} = \mathbf{0}$ , a genuine restriction on the missingness.

If the data are **missing completely at random**,  $\mathbf{D}_i$  is independent of  $\mathbf{Y}_i$ , the expectation factorizes, and everything above holds unchanged; the same is true if missingness depends only on covariates that are in the model and fully observed, because the mean model is then still correct conditionally on being observed. But if missingness depends on *previously observed responses* — a patient who was doing badly last visit is more likely to drop out — then  $\mathbf{D}_i$  and  $\mathbf{Y}_i - \boldsymbol{\mu}_i$  are dependent, the estimating function is biased and GEE is inconsistent. This is the **missing at random** case of Definition 41.1.1, and it is exactly the case in which a likelihood-based analysis remains valid.

That asymmetry is the strongest practical argument for the mixed-model route in longitudinal studies with dropout, and it is not widely enough known. The repair is to weight each cluster’s contribution by the inverse of its estimated probability of being observed — the weighted GEE of Robins, Rotnitzky and Zhao (1995), which trades the assumption about the correlation for one about the missingness model. Chapter 41 develops the mechanisms, the ignorability theorem and the alternatives.

## 40.4.7 Exercises

### 40.4.7.1 A. Check your understanding

#### Exercise A1.

Show from Eq. 40.4.3 that with  $\mathbf{R}_i = \mathbf{I}$  the estimate  $\hat{\boldsymbol{\beta}}$  does not depend on  $\phi$  or on the cluster structure at all, and equals the ordinary generalized linear model estimate.

Which quantities *do* depend on the clustering?

### Exercise A2.

A trial randomizes clinics to a new protocol and measures a binary outcome on patients within clinics. The investigators want to report “the effect of the protocol on a patient’s odds of recovery”. Which method should they use, and what would a GEE analysis report instead?

## 40.4.7.2 B. Practice

### Exercise B1.

Take  $m$  clusters of size 2, the identity link,  $V \equiv 1$ ,  $\phi = 1$ , and an exchangeable working correlation with parameter  $\alpha$ . Show that Eq. 40.4.3 reduces to weighted least squares with weight matrix  $\mathbf{R}^{-1}$ ,  $\mathbf{R} = (1 - \alpha)\mathbf{I} + \alpha\mathbf{J}$ , in each cluster, and that when  $\mathbf{x}_{(i1)} = \mathbf{x}_{(i2)}$  for every  $i$  the estimate does not depend on  $\alpha$  at all. Interpret.

*Solution.* The estimating equation is  $\sum_i \mathbf{X}_i^\top \mathbf{R}^{-1}(\mathbf{y}_i - \mathbf{X}_i \boldsymbol{\beta}) = \mathbf{0}$ , which is the normal equation of generalized least squares with covariance  $\mathbf{R}$ . If the two rows of  $\mathbf{X}_i$  are equal,  $\mathbf{X}_i = \mathbf{1}\mathbf{x}_{(i)}^\top$  and  $\mathbf{X}_i^\top \mathbf{R}^{-1} \mathbf{X}_i = (\mathbf{1}^\top \mathbf{R}^{-1} \mathbf{1})\mathbf{x}_{(i)}\mathbf{x}_{(i)}^\top$ ,  $\mathbf{X}_i^\top \mathbf{R}^{-1} \mathbf{y}_i = \mathbf{x}_{(i)} \mathbf{1}^\top \mathbf{R}^{-1} \mathbf{y}_i$ . For an exchangeable  $\mathbf{R}$ ,  $\mathbf{R}^{-1} \mathbf{1} = \{1/(1 + \alpha)\}\mathbf{1}$ , so both sides carry the same scalar factor  $1/(1 + \alpha)$  and it cancels. A cluster-constant covariate sees only the cluster mean, and reweighting a mean by an exchangeable covariance does nothing.

### Exercise B2.

Let  $\mathbf{G}_i$  be any  $n_i \times p$  matrix depending on  $\boldsymbol{\beta}$  but not on  $\mathbf{Y}_i$ . Show that  $\sum_i \mathbf{G}_i^\top (\mathbf{Y}_i - \boldsymbol{\mu}_i)$  is an unbiased estimating function and that Eq. 40.4.3 is the member with  $\mathbf{G}_i = \mathbf{V}_i^{-1} \mathbf{D}_i$ . Which member is optimal, in the sense of Proposition 38.2.3, and why is it not available?

*Solution.* Unbiasedness is immediate from  $E(\mathbf{Y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$ . The Godambe-optimal choice within this linear class is  $\mathbf{G}_i = (\boldsymbol{\Sigma}_i^0)^{-1} \mathbf{D}_i$ , the *true* covariance in place of the working one; Proposition 38.2.3 is the same statement with  $n_i = 1$ . It is unavailable because  $\boldsymbol{\Sigma}_i^0$  is unknown, which is precisely what makes a working correlation necessary.

## 40.4.7.3 C. Going deeper

### Exercise C1.

Show that if  $\mathbf{V}_i = \boldsymbol{\Sigma}_i^0$  then  $\mathbf{A}^{-1} \mathbf{B} \mathbf{A}^{-1} = \mathbf{A}^{-1}$  is the smallest asymptotic covariance attainable in the class of Exercise (B2), in the nonnegative-definite ordering. (Hint: for any  $\mathbf{G}_i$ , compare  $(\sum \mathbf{G}_i^\top \mathbf{D}_i)^{-1} (\sum \mathbf{G}_i^\top \boldsymbol{\Sigma}_i^0 \mathbf{G}_i) (\sum \mathbf{D}_i^\top \mathbf{G}_i)^{-1}$  with  $\mathbf{A}^{-1}$  using a Cauchy–Schwarz argument in the inner product  $\langle \mathbf{a}, \mathbf{b} \rangle = \sum_i \mathbf{a}_i^\top \boldsymbol{\Sigma}_i^0 \mathbf{b}_i$ .)

## 40.5 Working correlation

[Theorem 40.4.1](#) says the working correlation may be wrong without costing consistency. It does not say the choice is free. A wrong one costs *efficiency*, changes the small-sample behaviour of the sandwich, and — because it changes the estimate — can be turned into a diagnostic for the mean model.

### 40.5.1 The catalogue

The structures in use are those of [Definition 33.4.1](#), written as correlation matrices and estimated by moments rather than by likelihood. For a cluster with  $n_i$  observations at occasions  $j = 1, \dots, n_i$ , and Pearson residuals  $r_{ij} = (y_{ij} - \hat{\mu}_{ij}) / \sqrt{V(\hat{\mu}_{ij})}$  with  $\hat{\phi} = \sum_{ij} r_{ij}^2 / (n - p)$ :

| Structure                  | $(\mathbf{R}_i)_{jk}, j \neq k$               | Parameters   |
|----------------------------|-----------------------------------------------|--------------|
| independence               | 0                                             | 0            |
| exchangeable               | $\alpha$                                      | 1            |
| first-order autoregressive | $\alpha^{ j-k }$                              | 1            |
| $q$ -dependent             | $\alpha_{ j-k }$ if $ j - k  \leq q$ , else 0 | $q$          |
| unstructured               | $\alpha_{jk}$                                 | $n(n - 1)/2$ |

The parameters are estimated by matching averages of residual products. Writing  $N_p = \sum_i \binom{n_i}{2}$  for the within-cluster pairs and  $N_a = \sum_i (n_i - 1)$  for the adjacent ones,

$$\hat{\alpha}_{\text{exch}} = \frac{1}{\hat{\phi} (N_p - p)} \sum_i \sum_{j < k} r_{ij} r_{ik}, \quad \hat{\alpha}_{\text{ar}} = \frac{1}{\hat{\phi} (N_a - p)} \sum_i \sum_{j < n_i} r_{ij} r_{i,j+1},$$

and  $\hat{\alpha}_{jk} = \sum_i r_{ij} r_{ik} / (\hat{\phi} m)$ , rescaled to have unit diagonal, for the unstructured case; the  $q$ -dependent estimates are lag-by-lag averages of the same kind.

The design narrows the list at once. The unstructured matrix needs a common set of occasions across clusters and costs  $n(n - 1)/2$  parameters; the autoregressive structure needs an ordering, so it is for longitudinal data and not for pupils in a school; the exchangeable structure is the one a random intercept induces ([Proposition 40.2.1](#)), and is the default for clustered data with no natural order. And every estimated  $\mathbf{R}_i$  must be a valid correlation matrix, which the moment estimates above do not guarantee — [Example \(Four options per traveller\)](#) showed the exchangeable estimate landing almost exactly on the boundary.

```
import numpy as np

KINDS = ["independence", "exchangeable", "ar1", "unstructured"]

def correlation(kind, r, p, phi):
    """Moment estimates of the working correlation from Pearson residuals r (m by n)
    ."""
    m, n = r.shape
    if kind == "independence":
        return np.eye(n)
    if kind == "exchangeable":
        off = (np.sum(r.sum(1) ** 2) - np.sum(r**2)) / 2
        a = off / (phi * (m * n * (n - 1) / 2 - p))
```

```

        return np.eye(n) + a * (1 - np.eye(n))
    if kind == "ar1":
        a = np.sum(r[:, :-1] * r[:, 1:]) / (phi * (m * (n - 1) - p))
        return a ** np.abs(np.arange(n)[:, None] - np.arange(n))
    if kind == "unstructured":
        R = (r.T @ r) / (phi * m)
        return R / np.sqrt(np.outer(np.diag(R), np.diag(R)))
    raise ValueError(kind)

def gee(y, X, kind, family="poisson", maxit=100):
    m, n, p = X.shape
    beta = np.zeros(p)
    beta[0] = np.log(max(y.mean(), 1e-3)) if family != "binomial" else 0.0
    for _ in range(maxit):
        eta = X @ beta
        mu = np.exp(eta)
        dmu, V = (mu, mu) if family == "poisson" else (mu, mu**2)
        r = (y - mu) / np.sqrt(V)
        phi = np.sum(r**2) / (m * n - p)
        R = correlation(kind, r, p, phi)
        Rinv = np.linalg.pinv(R)
        SX = (dmu / np.sqrt(V))[..., None] * X
        A = np.einsum("mjp,jk,mkq->pq", SX, Rinv, SX)
        step = np.linalg.solve(A, np.einsum("mjp,jk,mk->p", SX, Rinv, r))
        beta = beta + step
        if np.max(np.abs(step)) < 1e-10:
            break
    u = np.einsum("mjp,jk,mk->mp", SX, Rinv, r)
    Ainv = np.linalg.inv(A)
    return beta, Ainv @ (u.T @ u) @ Ainv, phi, R

```

## 40.5.2 Efficiency, and where it comes from

### Proposition 40.5.1 — Choosing the working correlation

In the setting of [Theorem 40.4.1](#):

- (a) **Efficiency.** The asymptotic covariance  $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$  of [Eq. 40.4.5](#) is minimized, in the nonnegative-definite ordering and over all unbiased estimating functions linear in  $\mathbf{Y}_i - \boldsymbol{\mu}_i$ , by  $\mathbf{V}_i = \boldsymbol{\Sigma}_i^0$  ([Exercise \(C1\)](#)). Any other working covariance costs efficiency, but the cost depends on the covariate: it is substantial for a covariate that varies *within* clusters and negligible for one that is constant within a cluster, for which the independence choice is in fact hard to beat.
- (b) **Small samples.** The sandwich [Eq. 40.4.6](#) is biased downwards by a term of order  $p/m$ , so Wald tests built on it reject too often when  $m$  is small; the bias-corrected forms of [Section 40.4](#) and a  $t_{m-p}$  reference repair most of it. The bias grows with the number of parameters in  $\mathbf{R}_i$ , because  $\hat{\boldsymbol{\alpha}}$  is then noisier and condition (E5) of [Theorem 40.4.1](#) is stretched.
- (c) **QIC.** Since there is no likelihood, AIC is unavailable. Pan's **quasi-likelihood information criterion** replaces the log-likelihood by the quasi-likelihood  $Q(\boldsymbol{\mu}; \mathbf{y}) = \sum_{ij} \int_{y_{ij}}^{\mu_{ij}} (y_{ij} - t) / \{\phi V(t)\} dt$  of [Definition 38.2.1](#) computed under the *independence* working correlation at the fitted  $\hat{\boldsymbol{\mu}}$ , with  $\phi$  the Pearson estimate of [Propo-](#)

sition 38.3.2 held fixed across the models compared, and replaces the parameter count by a sandwich-based trace:

$$\text{QIC} = -2Q(\hat{\boldsymbol{\mu}}; \mathbf{y}) + 2 \text{tr}(\hat{\boldsymbol{\Omega}}_I \hat{\mathbf{U}}_R), \quad (40.5.1)$$

where  $\hat{\boldsymbol{\Omega}}_I = \sum_i \hat{\mathbf{D}}_i^\top \hat{\mathbf{A}}_i^{-1} \hat{\mathbf{D}}_i / \phi$  is the independence-model information and  $\hat{\mathbf{U}}_R$  is Eq. 40.4.6. If the working model is correct the trace is  $p$  and Eq. 40.5.1 is the Akaike form Definition 29.2.2. It is a sound criterion for choosing the *mean model*, and a weakly justified one for choosing  $\mathbf{R}_i$ .

*Proof.*

- (a) is Exercise (C1); the covariate-dependence is illustrated in Example (How much efficiency is at stake) and explained after it. (b) is the cluster version of the bias calculation in Exercise (C2), applied to  $\hat{\mathbf{D}}_i^\top \hat{\mathbf{V}}_i^{-1} \mathbf{r}_i$  in place of  $\mathbf{X}_g^\top \hat{\boldsymbol{\epsilon}}_g$ . (c) is a definition; the claim that the trace equals  $p$  when the working model is correct is Theorem 40.4.1(d), which gives  $\hat{\mathbf{U}}_R \rightarrow \mathbf{A}^{-1}$ , together with  $\boldsymbol{\Omega}_I = \mathbf{A}$  when  $\mathbf{R}_i = \mathbf{I}$  is correct. ■

*Remark* (Why QIC is weak as a chooser of correlation). Eq. 40.5.1 measures fit by a quasi-likelihood that ignores the correlation altogether, and penalizes by a trace that is smallest when the sandwich is closest to the model-based covariance. Changing  $\mathbf{R}_i$  therefore moves QIC through the penalty almost alone, and rewards whichever structure makes the sandwich small — not whichever is right. Pan (2001) proposed it primarily for the mean model, and Hin and Wang (2009) give a variant (CIC) that drops the fit term for correlation selection. Neither has the standing AIC has for likelihoods.

### Example. (How much efficiency is at stake)

Simulate 60 clusters of 6 Poisson counts with a log link, one covariate varying within the cluster and one constant within it, and generate the dependence in two ways: a shared random intercept, which makes the true correlation exchangeable, and a latent Gaussian AR(1) process with  $\rho = 0.8$ . Fit by GEE with each of the four working correlations and record the sampling variance of each coefficient over 400 replications.

For the **within-cluster** covariate under exchangeable truth, the efficiency of the exchangeable working correlation relative to independence is 1.789; AR(1) and unstructured reach 1.583 and 1.547. Under AR(1) truth the figures are 1.282 for AR(1), 1.183 for exchangeable and 1.160 for unstructured. Matching the truth helps, mismatching it still helps, and the gain is worth having without being dramatic.

For the **cluster-level** covariate the ranking reverses: under exchangeable truth the relative efficiencies are 0.889, 0.938 and 0.834, all below one, so independence is best of the four. The reason is Exercise (B1): a covariate constant within a cluster is estimated from cluster means, on which an exchangeable reweighting has no effect at all, so all a nontrivial  $\mathbf{R}_i$  contributes is the noise in  $\hat{\boldsymbol{\alpha}}$ .

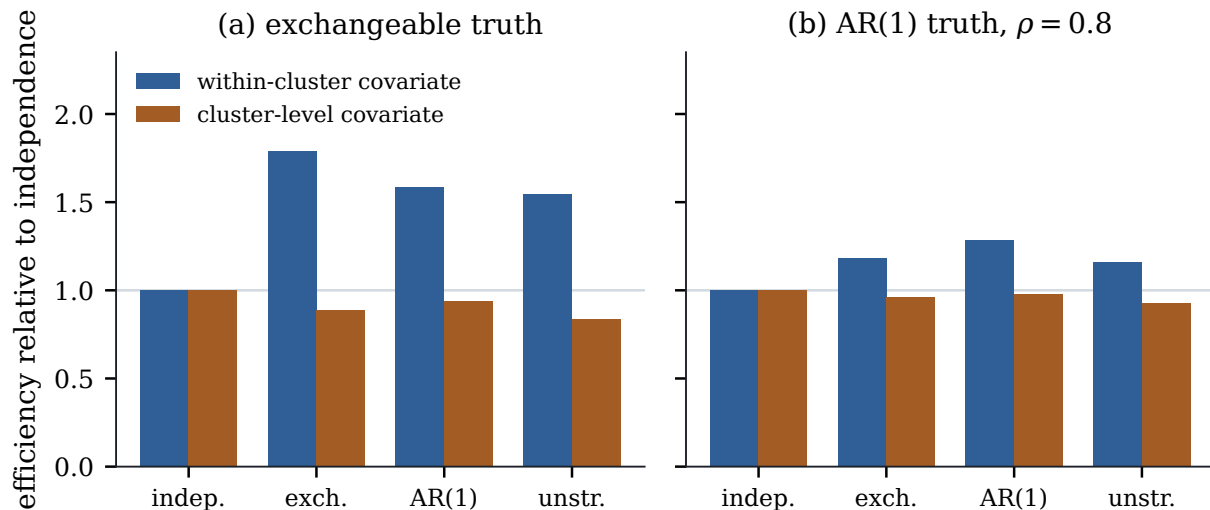


Figure 40.5.1: Efficiency of four working correlations relative to the independence choice, for a within-cluster and a cluster-level covariate, over 400 replications with  $m = 60$  clusters of  $n_i = 6$  Poisson counts. (a) The true dependence is exchangeable (a shared random intercept). (b) The true dependence is a latent AR(1) process with  $\rho = 0.8$ . Bars above one mean more efficient than independence.

### 40.5.3 A diagnostic hiding in the efficiency

By [Theorem 40.4.1\(a\)](#), every working correlation estimates the *same*  $\beta^0$ , provided the mean model [Eq. 40.4.1](#) is right. Two estimates of the same quantity differing by more than their sampling error therefore accuse the mean model, not the correlation. This is the Hausman idea, and it has real diagnostic value because the independence and exchangeable estimates weight within- and between-cluster variation differently: independence is dominated by between-cluster comparisons when clusters are large, and a nontrivial  $\mathbf{R}_i$  shifts weight towards within-cluster ones.

#### Example. (Grunfeld's panel by GEE)

Fit the marginal version of [Example \(Grunfeld's firms, again\)](#) — gamma variance function  $V(\mu) = \mu^2$ , log link, investment on log value and log capital — to the 11 firms by GEE. The AR(1) working correlation estimates a lag-one correlation of 0.894, as one expects of annual investment figures. Yet QIC, with the dispersion held at the independence estimate of the larger model, ranks the four structures

| Structure    | QIC, value and capital | QIC, value only |
|--------------|------------------------|-----------------|
| independence | 8450.1                 | 8657.6          |
| exchangeable | 8524.2                 | 8647.4          |
| AR(1)        | 8574.1                 | 8657.8          |
| unstructured | 8754.7                 | 8704.1          |

and prefers independence. As a comparison of *mean* models the table is informative: at three of the four structures the model with capital wins, by between about 80 and 210 QIC units, and that is the comparison QIC was designed for. Under the unstructured correlation the ranking reverses, which with 11 clusters and a  $20 \times 20$  matrix to estimate is a comment on the structure rather than on the model. As a comparison of structures the column mostly reports that the sandwich is smallest under independence — the remark above in numerical form.

The estimates themselves are the more interesting output. The independence fit gives  $\hat{\beta}_{\text{value}} = 0.8421$  and  $\hat{\beta}_{\text{capital}} = 0.3386$ ; the exchangeable fit gives 0.7214 and 0.2113. The gap in the first coefficient is more than its sandwich standard error of 0.1030, and the two gaps point the same way. By [Theorem 40.4.1\(a\)](#) that accuses the mean model, and two standard explanations fit equally well: firm effects correlated with the covariates, as [Example \(Grunfeld's firms, again\)](#) found, or feedback from investment to the following years' market value, which violates the full-covariate conditional mean condition and makes the exchangeable fit — not the independence one — the inconsistent member of the pair. Either way the remedy is to separate the two sources of variation. Adding the firm means of log value and log capital, as [Exercise \(C1\)](#) asks, makes the independence and exchangeable estimates agree to machine precision, at within-firm slopes 0.6142 and 0.2298, both well below the independence figures above.

```
import statsmodels.api as sm

d = sm.datasets.grunfeld.load_pandas().data.sort_values(["firm", "year"])
m, n = d["firm"].nunique(), 20
y = d["invest"].to_numpy().reshape(m, n)
lv = np.log(d["value"].to_numpy()).reshape(m, n)
lc = np.log(d["capital"].to_numpy()).reshape(m, n)

def qic(y, X, kind, phi):
    """Pan's QIC: the quasi-likelihood of the independence model at the fitted mean,
    plus a
    penalty built from the sandwich. For  $V(\mu) = \mu^2$  the quasi-likelihood is  $-y/\mu - \log \mu$ .
    The dispersion  $\phi$  is held fixed across the models compared, as in Mallows's  $C_p$ .
    """
    beta, robust, _, _ = gee(y, X, kind, family="gamma")
    mu = np.exp(X @ beta)
    Q = np.sum(-y / mu - np.log(mu)) / phi
    omega = np.einsum("mjp,mjq->pq", X, X) / phi      # independence information, log
    link
    return -2 * Q + 2 * np.trace(omega @ robust)

X_full = np.stack([np.ones((m, n)), lv, lc], axis=2)
X_small = np.stack([np.ones((m, n)), lv], axis=2)
phi_ref = gee(y, X_full, "independence", family="gamma")[2]      # from the largest
model

for name, XX in [("value + capital", X_full), ("value only", X_small)]:
    for kind in KINDS:
        print(f"{name:16s} {kind:14s} QIC {qic(y, XX, kind, phi_ref):10.2f}")
```



## 40.5.4 Practical advice

- **Count the clusters first.** Everything here that uses a sandwich needs  $m$  large. Below about forty clusters use a bias correction and a  $t$  reference; below about fifteen, prefer a likelihood-based model with a covariance structure chosen by REML ([Proposition 33.4.1](#)) and accept the assumptions.
- **Match the structure to the design, not to the data.** Exchangeable for unordered clusters, autoregressive for equally spaced longitudinal data, unstructured only when the occasions are common to all clusters and  $m$  is large.
- **Use independence when the covariates are time-varying, or the covariate of interest is cluster-level.** In the first case a non-diagonal  $\mathbf{R}_i$  needs the full-covariate conditional mean condition of [Section 40.4](#), which feedback destroys; in the second it loses nothing ([Example \(How much efficiency is at stake\)](#)). Independence needs neither, estimates nothing extra, and cannot produce an invalid  $\mathbf{R}_i$ .
- **Compare structures to check the mean model.** A large difference between the independence and exchangeable estimates is a warning about  $\mathbf{X}$ , not about  $\mathbf{R}_i$ .
- **Say which covariance you report**, and never put a GEE coefficient and a GLMM coefficient in the same table as if they estimated the same thing. The first costs coverage ([Example \(Coverage when the working correlation is wrong\)](#)); the second is the commonest error in this literature.

*Remark* (Beyond correlations). For binary responses a correlation is an awkward measure of dependence, because the Fréchet bounds on the joint cell probability confine it to an interval that depends on the two means, while a pairwise odds ratio is free in  $(0, \infty)$  whatever the means are. **Alternating logistic regressions**, due to Carey, Zeger and Diggle (1993), exploit this: they replace  $\mathbf{R}_i$  by a model for the pairwise odds ratios and alternate between the mean equation [Eq. 40.4.3](#) and a logistic regression for the association, whose parameters are unconstrained and interpretable in their own right.

## 40.5.5 Exercises

### 40.5.5.1 A. Check your understanding

#### Exercise A1.

For each design, name the working correlation you would use and say why: (a) pupils within 300 schools, class size varying; (b) five annual measurements on each of 500 patients; (c) left and right eyes of 80 patients; (d) irregularly timed clinic visits, between two and eleven per patient, 1200 patients.

#### Exercise A2.

In [Example \(Grunfeld's panel by GEE\)](#), QIC prefers independence among the working correlations but the AR(1) parameter is estimated at 0.894. Are these two statements in conflict? Explain what each of them is measuring.

### 40.5.5.2 B. Practice



**Exercise B1.**

Show that the exchangeable correlation matrix  $\mathbf{R} = (1 - \alpha)\mathbf{I} + \alpha\mathbf{1}\mathbf{1}^\top$  of size  $n$  has eigenvalues  $1 + (n - 1)\alpha$  (once) and  $1 - \alpha$  ( $n - 1$  times), so that it is positive definite exactly when  $-1/(n - 1) < \alpha < 1$ . Verify the value  $1 + 3\hat{\alpha} = 0.0956$  quoted in [Example \(Four options per traveller\)](#) from  $\hat{\alpha} = -0.3015$ , and say what a practitioner should do when a moment estimate falls outside the range.

*Solution.*  $\mathbf{R} = (1 - \alpha)\mathbf{I} + \alpha\mathbf{1}\mathbf{1}^\top$ ;  $\mathbf{1}$  is an eigenvector with eigenvalue  $(1 - \alpha) + n\alpha = 1 + (n - 1)\alpha$ , and any vector orthogonal to  $\mathbf{1}$  has eigenvalue  $1 - \alpha$ . Positivity of both gives the range. With  $\hat{\alpha} = -0.3015$  and  $n = 4$ ,  $1 + 3(-0.3015) = 1 - 0.9045 = 0.0955$ , which matches the quoted 0.0956 to rounding. Outside the range the working covariance is not a covariance at all and the update [Eq. 40.4.4](#) can diverge; the usual remedies are to shrink  $\hat{\alpha}$  towards zero, to project  $\mathbf{R}$  onto the nearest correlation matrix, or — better — to use the independence structure, which is never invalid.

**Exercise B2.**

Formalize the diagnostic of [Example \(Grunfeld's panel by GEE\)](#). Let  $\hat{\boldsymbol{\beta}}_I$  and  $\hat{\boldsymbol{\beta}}_E$  be the independence and exchangeable GEE estimates. Explain why  $\hat{\boldsymbol{\beta}}_I - \hat{\boldsymbol{\beta}}_E$  has mean zero to first order when the mean model is correct, and why its covariance is *not* the difference of the two covariances in general. What resampling scheme would give a usable reference distribution?

*Solution.* Both estimates satisfy [Eq. 40.4.5](#) with the same  $\boldsymbol{\beta}^0$ , so their difference is  $o_p(1)$  and, after scaling, asymptotically normal with mean zero. The classical Hausman simplification  $\text{Cov}(\hat{\boldsymbol{\beta}}_1 - \hat{\boldsymbol{\beta}}_2) = \text{Cov}(\hat{\boldsymbol{\beta}}_2) - \text{Cov}(\hat{\boldsymbol{\beta}}_1)$  needs one of the two to be *efficient*, which neither is here, so the covariance must be obtained from the joint asymptotic distribution of the two estimating functions or by resampling. The cluster bootstrap of [Exercise \(C1\)](#) — resample whole clusters with replacement and refit both — gives the reference distribution directly.

## 40.6 Summary and Notes

### 40.6.1 Summary

*Idea.*

1. A nonlinear link and a random effect do not commute, so a clustered model has two sets of coefficients. The identity and log links are the exceptions; for the probit the marginal coefficient is  $\beta^C / \sqrt{1 + \tau^2}$  exactly, and for the logit it is strictly attenuated with local factor  $1 - \text{Var}(P_\theta) / \{E P_\theta E(1 - P_\theta)\}$  ([Proposition 40.1.1](#)).
2. The generalized linear mixed model puts random effects in the linear predictor of a generalized linear model ([Definition 40.2.1](#)). Its marginal moments follow by conditioning: a binomial response acquires the beta-binomial overdispersion factor, and a Poisson–lognormal response acquires *exactly* the NB2 variance function

([Proposition 40.2.1](#)).

3. A cluster-level random intercept can only induce nonnegative within-cluster correlation, because two nondecreasing functions of one random variable have nonnegative covariance ([Lemma 40.2.2](#)). Data with competition inside a cluster need a different tool.
4. Predicting a random effect in a GLMM is empirical Bayes rather than BLUP, but the shrinkage survives: the posterior mode lies strictly between zero and the cluster's own estimate, with the familiar factor  $\tau^2/(\tau^2 + I_i^{-1})$  to first order ([Proposition 40.2.3](#)).
5. The integrated likelihood factorizes over independent clusters. The Laplace approximation has relative error  $O(n_i^{-1})$  — governed by the observations *per cluster*, not by the sample size — and adaptive Gauss–Hermite quadrature with  $K = 1$  node is exactly the Laplace approximation ([Theorem 40.3.1](#)).
6. Penalized quasi-likelihood is Henderson's mixed-model equations applied to the working response, and is inconsistent: it attenuates both  $\hat{\beta}$  and  $\hat{\theta}$ , badly for binary data in small clusters ([Proposition 40.3.2](#), [Example \(Three methods on clustered binary data\)](#)).
7. Generalized estimating equations solve  $\sum_i \mathbf{D}_i^\top \mathbf{V}_i^{-1}(\mathbf{Y}_i - \boldsymbol{\mu}_i) = \mathbf{0}$  with a working correlation the analyst chooses and does not have to believe ([Definition 40.4.1](#)). The estimating function is unbiased for every working correlation, so  $\hat{\beta}$  is consistent for the marginal coefficient and asymptotically normal with the sandwich covariance  $\mathbf{A}^{-1}\mathbf{B}\mathbf{A}^{-1}$  ([Theorem 40.4.1](#)).
8. GEE gives no likelihood, no random-effect predictions and no distribution; a non-diagonal working correlation also needs the mean model to hold conditionally on the *whole* covariate history, which feedback destroys; and it fails under missingness that depends on previously observed responses — the one case in which a likelihood-based analysis still works ([Section 40.4](#)).
9. A wrong working correlation costs efficiency for covariates that vary within clusters and nothing at all for covariates fixed within a cluster; the sandwich needs a small-sample correction below about forty clusters; and QIC, the quasi-likelihood analogue of AIC, is sound for mean models and weak for correlation structures ([Proposition 40.5.1](#)).
10. Because every working correlation targets the same  $\beta^0$ , a large gap between two GEE fits of the same mean model is evidence against the *mean model* ([Example \(Grunfeld's panel by GEE\)](#)).

## 40.6.2 Notes and sources

**Coverage.** The topics follow Agresti (2015, sections 9.4–9.6) and Fahrmeir, Kneib, Lang and Marx (2021, sections 7.5–7.6), but put the marginal–conditional distinction first, before either estimation method, because that is where the choice between the two traditions is actually made. Book-length treatments are Diggle, Heagerty, Liang and Zeger (2002) for longitudinal data, McCulloch, Searle and Neuhaus (2008) for the mixed-model theory, and Molenberghs and Verbeke (2005) for discrete longitudinal responses.

**Marginal versus conditional.** The distinction was made precise for binary data by Neuhaus, Kalbfleisch and Hauck (1991), still the clearest statement of what each coefficient means and when the two agree. The approximation factor  $(1 + c^2\tau^2)^{-1/2}$  with  $c = 16\sqrt{3}/(15\pi)$  is from

Zeger, Liang and Albert (1988); the constant comes from matching the logistic and normal distributions and predates its use here by a long way. The proof of [Proposition 40.1.1\(d\)](#) given above, through the derivative identity [Eq. 40.1.5](#), seems to be the shortest route to the strict inequality, and needs no assumption on the random-effect distribution beyond non-degeneracy. The median odds ratio of [Exercise \(B1\)](#) was popularized in epidemiology by Larsen and Merlo (2005).

**Fitting.** Breslow and Clayton (1993) introduced penalized quasi-likelihood under that name and set the pattern for GLMM software for a decade; Breslow and Lin (1995) and Lin and Breslow (1996) quantified its bias and proposed corrections. Adaptive Gauss–Hermite quadrature for this problem is due to Liu and Pierce (1994) and, for nonlinear mixed models, Pinheiro and Bates (1995), who also observed that one adaptive node reproduces the Laplace approximation. The Laplace expansion itself is classical; the form used in [Theorem 40.3.1\(b\)](#) is the one in Tierney and Kadane (1986) and Barndorff-Nielsen and Cox (1989, chapter 3). The asymptotics of the integrated likelihood quoted in [Section 40.2](#) are in McCulloch, Searle and Neuhaus (2008, chapter 14). Monte Carlo EM for this problem is Wei and Tanner (1990), with automated sample-size rules in Booth and Hobert (1999); the MCMC route with iteratively reweighted proposals is Gamerman (1997).

**Estimating equations.** Liang and Zeger (1986) and Zeger and Liang (1986) introduced GEE, and the pair is worth reading together: the first gives the theory, the second the practice. The conditions and the expansion in [Theorem 40.4.1](#) follow theirs, with the cluster sandwich argument taken from [Theorem 21.4.2](#) as [Section 33.2](#) already arranged it. The warning about time-varying covariates and the full-covariate conditional mean is due to Pepe and Anderson (1994), and is treated at length in Diggle, Heagerty, Liang and Zeger (2002, chapter 12). Small-sample corrections to the sandwich are due to Mancl and DeRouen (2001), Kauermann and Carroll (2001) and Fay and Graubard (2001); the first is used in [Example \(Coverage when the working correlation is wrong\)](#). QIC is Pan (2001); the criticism of it as a selector of working correlations, and the CIC variant, are in Hin and Wang (2009). Alternating logistic regressions are Carey, Zeger and Diggle (1993). The failure of GEE under missingness that depends on observed responses, and the inverse-probability-weighted repair, are from Robins, Rotnitzky and Zhao (1995).

**What is new here.** [Lemma 40.2.2](#) and its consequence — that a cluster-level random intercept cannot produce negative within-cluster correlation, so that the travel-mode data of [Example \(Four options per traveller\)](#) are outside the reach of any random-intercept GLMM — is elementary, and the association inequality behind it is standard, but we have not seen the two put together as a reason to prefer estimating equations. The efficiency comparison of [Example \(How much efficiency is at stake\)](#) separates within-cluster from cluster-level covariates, making the often-repeated advice “the working correlation hardly matters” precise: it hardly matters for cluster-level covariates and matters by a factor of nearly two for within-cluster ones.

**Data and code.** Grunfeld’s investment panel is the public-domain eleven-firm version distributed with statsmodels, also used in [Chapter 32](#) and [Chapter 33](#); its provenance is Grunfeld (1958), with the version history discussed by Kleiber and Zeileis (2010). The travel-mode choice data are the public-domain sample of 210 non-business trips between Sydney, Canberra and Melbourne from a 1987 intercity mode-choice study, also distributed with statsmodels; the sample over-represents the rarer modes, as [Example \(Four options per traveller\)](#) notes. The code in `code/ch40/` is written from the formulas in this chapter and checked against statsmodels where an equivalent fit exists; the simulations use fixed seeds.

### 40.6.3 References

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# Chapter 41

## Missing Data

Every data set in this book so far has arrived complete. Real ones do not. A respondent skips the question about income, a patient stops coming to the clinic, a sensor fails for three weeks, a laboratory loses a sample. The holes are rarely scattered at random, and the software's silent default — drop every record with a hole in it — is a decision, not a neutral act.

This chapter treats the holes as data. The pattern of what is missing has a distribution of its own, and the subject turns on what that distribution may depend on. If it depends on nothing, almost any analysis survives. If it depends only on what was recorded, likelihood and Bayesian inference survive and most shortcuts do not. If it depends on what was *not* recorded, nothing in the observed data can tell you so. Those three cases organize what follows: when dropping incomplete records is safe, why single imputation lies about precision, what EM is and what it does not give you, how multiple imputation restores the uncertainty imputation destroys, and how to report a conclusion resting on an untestable assumption.

**What you need.** [Chapter 5](#) and [Chapter 6](#) (the linear model and least squares), [Section 7.3](#) (maximum likelihood [Theorem 7.3.1](#), the score and information [Proposition 7.4.7](#)), [Section 7.6](#) and [Section 12.6](#) (the conjugate Bayesian linear model [Theorem 7.6.2](#) and credible sets [Definition 12.6.1](#)), [Chapter 11](#) (the likelihood ratio test [Theorem 11.4.1](#)). [Section 18.6](#) ([Theorem 18.6.1](#), filling in a lost plot of a designed experiment) is the special case this chapter generalizes. [Chapter 23](#) supplies the bootstrap ([Definition 23.1.1](#), [Definition 23.5.1](#)), which is one of the two honest routes to a standard error after EM; [Chapter 25](#) the habit of asking what a coefficient is meant to answer; and [Chapter 34](#) the generalized linear model ([Definition 34.2.1](#), [Theorem 34.3.1](#)), since nothing here is special to a normal response.

### 41.0.1 Roadmap

- [Missingness mechanisms](#)
- [Complete cases and quick repairs](#)
- [Likelihood under missingness at random](#)
- [The EM algorithm](#)
- [Multiple imputation](#)
- [Sensitivity analysis](#)
- [Summary and notes](#)

## 41.1 Missingness mechanisms

A data set with holes carries two kinds of information: the values that were recorded, and the pattern of which values were recorded at all. The second decides whether any analysis of the first means what it appears to. This section gives the pattern a probability model.

### 41.1.1 Notation

Write the **complete data** as an array  $\mathbf{D}$  with  $n$  rows and  $q$  columns, whose  $i$ th row  $\mathbf{d}_{(i)}$  holds the variables of unit  $i$ . Response and covariates sit in the same array: which is the response is a question for the analysis, not for the mechanism. Alongside it put the **missingness indicator** array  $\mathbf{R}$  of the same shape, with  $R_{ij} = 1$  when  $D_{ij}$  is recorded;  $\mathbf{R}$  is free for this use here, as  $\mathbf{H}$  was in Chapters 5 and 10.

The array  $\mathbf{R}$  is always completely observed, which is what makes it data. Split  $\mathbf{D}$  into recorded entries  $\mathbf{D}_{\text{obs}}$  and unrecorded ones  $\mathbf{D}_{\text{mis}}$ . The analyst holds  $(\mathbf{D}_{\text{obs}}, \mathbf{R})$ ; every model here describes  $(\mathbf{D}, \mathbf{R})$  by the factorization

$$f(\mathbf{D}, \mathbf{R} \mid \boldsymbol{\theta}, \boldsymbol{\psi}) = f(\mathbf{D} \mid \boldsymbol{\theta}) f(\mathbf{R} \mid \mathbf{D}, \boldsymbol{\psi}). \quad (41.1.1)$$

Here  $\boldsymbol{\theta}$  is the parameter the analyst cares about — a regression coefficient lives in it — and the second factor is the **missingness mechanism**. Conditioning it on the *complete* data is the essential move: it lets the probability that a value is recorded depend on the value itself, which is exactly the case that causes trouble. This is the **selection** factorization; [Section 41.6](#) uses the other one.

### 41.1.2 Three classes

**Definition 41.1.1** — **Missing completely at random, at random, not at random**

Let  $\mathbf{D}$  and  $\mathbf{R}$  have the joint distribution [Eq. 41.1.1](#), and fix the realized values  $\mathbf{R} = \mathbf{r}$  and  $\mathbf{D}_{\text{obs}} = \mathbf{a}$ . The mechanism is

- (a) **missing completely at random** (MCAR) if  $f(\mathbf{r} \mid \mathbf{D}, \boldsymbol{\psi}) = f(\mathbf{r} \mid \boldsymbol{\psi})$  for every  $\mathbf{D}$  and  $\boldsymbol{\psi}$ : the pattern is independent of the data, recorded or not;
- (b) **missing at random** (MAR) if  $f(\mathbf{r} \mid \mathbf{D}, \boldsymbol{\psi})$  takes the same value for every  $\mathbf{D}$  whose observed part is  $\mathbf{a}$ , and every  $\boldsymbol{\psi}$ , so that it may be written  $f(\mathbf{r} \mid \mathbf{a}, \boldsymbol{\psi})$ ;
- (c) **missing not at random** (MNAR) otherwise.

MCAR implies MAR, and the two buy different things. Under MCAR the recorded units are a random subsample, so *every* analysis of them estimates what it would have estimated with no losses, at the price of a smaller sample. MAR says less: the recorded units are a biased subsample, but the bias is a function of recorded variables, so a method using those variables correctly can undo it.

*Warning.* The names have misled generations of readers. “Missing at random” does not mean the holes are haphazard; it means they are haphazard *within each configuration of the recorded values* — a strong assumption in modest language. “Missing completely at random” is the haphazard one.

Definition [Definition 41.1.1](#) states MAR at the realized  $(\mathbf{r}, \mathbf{a})$ ; requiring it at every pattern is stronger, and the distinction is not pedantry. A Bayesian, conditioning on the data in hand, needs only the realized version; a frequentist, averaging over data sets that did not occur, needs the stronger one (Seaman, Galati, Jackson and Carlin, 2013). This chapter states the realized version and flags where the stronger one is wanted.

### 41.1.3 What each class looks like

#### Example. (A survey that asks about income)

A survey records age  $x$  for everyone and income  $y$  for those who answer. Three stories. *Completely at random:* a printing fault left the income question off one batch of forms, assembled without regard to anything, so  $\Pr(R = 0 \mid x, y)$  is constant. *At random:* younger respondents skip the question more often, and among respondents of a given age, answering has nothing to do with earnings, so  $\Pr(R = 0 \mid x, y) = \Pr(R = 0 \mid x)$ . *Not at random:* high earners skip the question at every age, so  $\Pr(R = 0 \mid x, y)$  depends on  $y$  even given  $x$ . The three are mutually exclusive descriptions of the world, and [Proposition 41.1.1](#) says the observed data cannot tell them apart.

So “the mechanism” is a property not of a variable but of the pair (variable, recorded variables). If the survey also recorded occupation, and high earners skip only because they hold occupations whose members skip, the third story becomes MAR once occupation joins  $\mathbf{D}_{\text{obs}}$ . Enlarging the recorded data can turn MNAR into MAR; it can never do the reverse.

#### Example. (A record with real gaps)

The weekly averages of the continuous Mauna Loa record of atmospheric carbon dioxide run from March 1958 to December 2001 and cover 2284 weeks, of which 59 carry no value — 2.6% of the record, in 22 runs, the longest of 18 consecutive weeks.

They are not spread evenly. Before 1970, 8.6% of weeks are missing; from 1970 on, 0.4%, and a logistic regression of  $1 - R_i$  on calendar time multiplies the odds of a gap by 0.192 per decade. The mechanism depends on a variable recorded for every week, so the record is not MCAR.

Whether it is MAR given time is a question about the instrument, not about the numbers. The published rule is that a day contributes only if the analyser ran steadily for at least six hours, and a week with no such day contributes nothing. So the question is whether the analyser’s failure to deliver six steady hours is related to the concentration it would have read — plausibly not, if the failures are electrical or logistical; plausibly so, if unsettled air both disturbs the instrument and moves the reading. Nothing in [Figure 41.1.1](#) can settle that.

```
import numpy as np
import statsmodels.api as sm
```



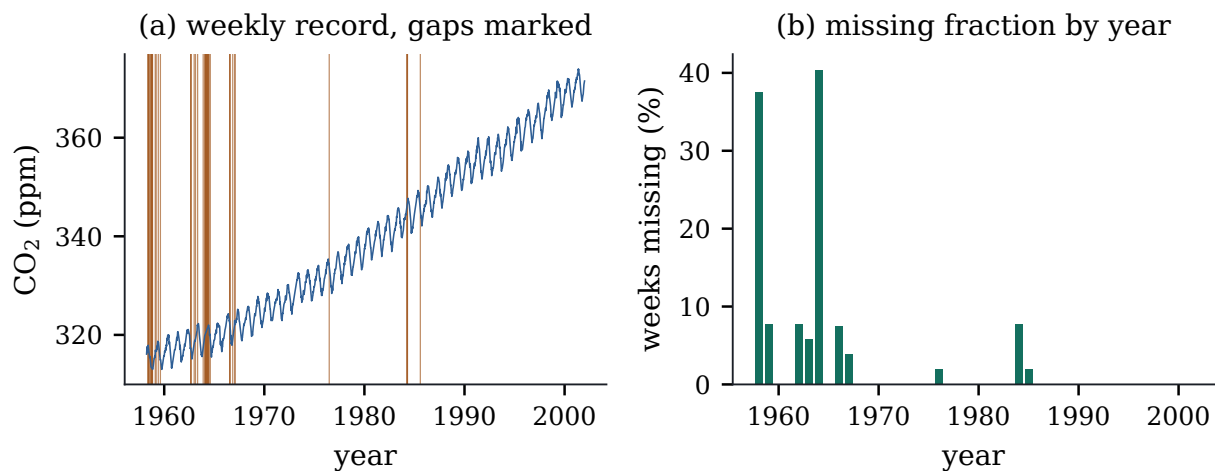


Figure 41.1.1: The weekly Mauna Loa record. (a) The series, with a vertical rule at each missing week. (b) The fraction of weeks missing in each calendar year. The mechanism depends strongly on an observed variable, calendar time, and so is not missing completely at random.

```
co2 = sm.datasets.co2.load_pandas().data["co2"]           # one value per week, some
absent
t = co2.index.year + (co2.index.dayofyear - 0.5) / 365.25
r = (~co2.isna()).to_numpy().astype(float)               # 1 = recorded, 0 = missing
n, n_mis = len(r), int((1 - r).sum())
print(f"{n} weeks, {n_mis} missing ({100 * n_mis / n:.1f}%)")

era = (t >= 1970).astype(float)                           # one observed covariate: is
it 1970 or later?
rate_early = 1 - r[era == 0].mean()
rate_late = 1 - r[era == 1].mean()
print(f"missing before 1970: {100 * rate_early:.1f}%   from 1970 on: {100 *
rate_late:.1f}%")
```

One feature of **R** matters for computation rather than validity. A pattern is **monotone** when the columns can be ordered so that  $R_{ij} = 0$  forces  $R_{ik} = 0$  for  $k > j$ : once a unit drops out it never returns, as in longitudinal dropout. Such patterns are far easier to handle, the joint density factoring into a chain of conditionals each fitted on a nested subsample ([Exercise \(B1\)](#)).

#### 41.1.4 Why the class cannot be read off the data

##### Proposition 41.1.1 — The observed data do not identify the mechanism

Let  $\mathbf{d} = (d_1, d_2)$  be a unit's record, with  $d_1$  always recorded and  $d_2$  recorded exactly when  $R = 1$ . The distribution of the observed data is determined by the three pieces  $f(d_1)$ ,  $\Pr(R = 1 \mid d_1)$  and  $f(d_2 \mid d_1, R = 1)$ , and determines nothing else. Consequently:

- (a) for every conditional density  $g(d_2 \mid d_1)$  there is exactly one joint distribution of  $(\mathbf{d}, R)$  with the given observed-data distribution and with  $f(d_2 \mid d_1, R = 0) = g(d_2 \mid d_1)$ ;

- (b) one of these choices,  $g = f(\cdot \mid d_1, R = 1)$ , is MAR; every other is MNAR; and all assign the observed data the same likelihood.

*Proof.* The observed data are  $(d_1, R)$ , with  $d_2$  when  $R = 1$ . Their density is  $f(d_1) \Pr(R = 0 \mid d_1)$  on  $R = 0$  and  $f(d_1) \Pr(R = 1 \mid d_1) f(d_2 \mid d_1, R = 1)$  on  $R = 1$ ; only the three listed pieces enter.

- (a) Given the three pieces and a candidate  $g$ , define

$$\begin{aligned} f(d_1, d_2, R = 1) &= f(d_1) \Pr(R = 1 \mid d_1) f(d_2 \mid d_1, R = 1), \\ f(d_1, d_2, R = 0) &= f(d_1) \Pr(R = 0 \mid d_1) g(d_2 \mid d_1). \end{aligned}$$

This integrates to one, has the given observed-data distribution, and has conditional density  $g$  for  $d_2$  given  $(d_1, R = 0)$ . Conversely any joint distribution with those properties has these two densities, so it is unique.

- (b) Abbreviate  $p_r = \Pr(R = r \mid d_1)$  and  $f_1 = f(d_2 \mid d_1, R = 1)$ . Under the constructed distribution,

$$\Pr(R = 1 \mid d_1, d_2) = \frac{p_1 f_1}{p_1 f_1 + p_0 g(d_2 \mid d_1)},$$

which is free of  $d_2$  for all  $d_1$  precisely when  $g$  equals  $f(\cdot \mid d_1, R = 1)$ . Every member of the family assigns the observed data the same likelihood by construction, so no statistic computed from those data can prefer one to another. ■

*Idea.* The missing values are missing. No cleverness recovers the conditional distribution of what was not recorded given what was; a choice of that distribution *is* the assumption, and different choices fit the observed data identically. MAR is one such choice: convenient, often defensible, never verifiable.

Two consequences. A test of MCAR against MAR is possible, that comparison involving recorded variables only, and Little (1988) gives one; a test of MAR against MNAR is not. And a sensitivity analysis is not an optional extra but the only way to report what the data do not determine (Section 41.6).

Finally, the classes of Definition 41.1.1 describe the world; whether a procedure works is a property of the *pair*. Under MCAR nearly every method is valid, including dropping incomplete records. Under MAR with distinct parameters, likelihood and Bayesian inference with a correct model for  $\mathbf{D}$  are valid by Theorem 41.3.1, while dropping incomplete records usually is not (Proposition 41.2.1). And under a mechanism depending on a regression's covariates but not on its response, dropping incomplete records is valid *for that regression*, even though the recorded units are a badly biased sample for any other purpose. That third case is the most useful and least well known, and the next section starts there.

## 41.1.5 Exercises

### 41.1.5.1 A. Check your understanding

**Exercise A1.**

Classify each mechanism as MCAR, MAR or MNAR with respect to the recorded variables named.

- (a) A blood assay is run on a random one-third of the samples chosen by a random number generator; age, sex and diagnosis are recorded for all.
- (b) In a weight-loss trial, participants who have gained weight since the last visit stop attending. Weight at every previous visit is recorded.
- (c) A scale reads “over range” and records nothing whenever the true weight exceeds 150 kg.
- (d) A questionnaire item is missing for everyone interviewed by one interviewer, whose identity is recorded.

*Solution.*

- (a) MCAR. (b) MNAR for the current visit’s weight, since whether it is recorded depends on that weight itself; it would be MAR if attendance depended only on the previous, recorded visits. (c) MNAR, and of an extreme kind: missingness is a deterministic function of the unrecorded value. (d) MAR given the interviewer, and not MCAR; without the interviewer’s identity recorded it would be MNAR *given the recorded variables*, which is what [Definition 41.1.1](#) measures.

**Exercise A2.**

Show that MCAR implies MAR, and explain why no enlargement of  $\mathbf{D}_{\text{obs}}$  can turn a MAR mechanism into an MNAR one.

**41.1.5.2 B. Practice****Exercise B1.**

Let  $\mathbf{d} = (d_1, \dots, d_q)$  have a monotone pattern, so that the recorded entries are  $d_1, \dots, d_K$  for some  $K \in \{0, \dots, q\}$ . Write the joint density of  $(\mathbf{d}, K)$  using the chain rule  $f(\mathbf{d}) = \prod_j f(d_j \mid d_1, \dots, d_{j-1})$ , and show that MAR is equivalent to  $\Pr(K = k \mid \mathbf{d})$  depending on  $\mathbf{d}$  only through  $d_1, \dots, d_k$ .

*Solution.* Definition [Definition 41.1.1](#) asks that  $\Pr(K = k \mid \mathbf{d})$  be the same for all  $\mathbf{d}$  agreeing in their first  $k$  coordinates, which is the stated condition. Multiplying the chain-rule factorization by it gives

$$f(\mathbf{d}, K = k) = \left\{ \prod_{j \leq k} f(d_j \mid d_{<j}) \right\} \Pr(K = k \mid d_1, \dots, d_k) \left\{ \prod_{j > k} f(d_j \mid d_{<j}) \right\},$$

and integrating out  $d_{k+1}, \dots, d_q$  removes the last brace entirely. That is the factorization [Section 41.3](#) exploits.

**Exercise B2.**

Two variables  $d_1$  (always recorded) and  $d_2$  (recorded when  $R = 1$ ). Propose a statistic with a known null distribution under MCAR and power against “the mechanism depends on  $d_1$ ”, and explain why no statistic can have power against “the mechanism depends on  $d_2$  given  $d_1$ ”.

**Exercise B3.**

Let  $\mathbf{d} = (d_1, d_2)$  be bivariate normal with mean  $(\mu_1, \mu_2)$  and covariance  $\Sigma$ , and let  $d_2$  be recorded with probability  $\Phi(a + bd_1)$ . Show that the mechanism is MAR, compute  $E(d_2 \mid R = 1)$ , and confirm that it differs from  $\mu_2$  unless  $b = 0$  or  $\sigma_{12} = 0$ . Which class does the mechanism fall in if the probability is instead  $\Phi(a + bd_2)$ ?

*Solution.* The probability depends only on the recorded  $d_1$ , so the mechanism is MAR. Writing  $d_2 = \mu_2 + \beta(d_1 - \mu_1) + e$  with  $\beta = \sigma_{12}/\sigma_{11}$  and  $e$  independent of  $d_1$ ,  $E(d_2 \mid R = 1) = \mu_2 + \beta\{E(d_1 \mid R = 1) - \mu_1\}$ , and  $E(d_1 \mid R = 1) \neq \mu_1$  whenever  $b \neq 0$ , since the selection probability is monotone in  $d_1$ . With  $\Phi(a + bd_2)$  the mechanism depends on the possibly unrecorded  $d_2$ , so it is MNAR for  $b \neq 0$ .

**41.1.5.3 C. Going deeper****Exercise C1.**

Let the units be independent, each with  $d_1 \in \{0, 1\}$  always recorded, and let  $d_2$  be recorded with probability  $p_0$  when  $d_1 = 0$  and with probability  $q(d_2)$  when  $d_1 = 1$ , where  $q$  is not constant. Show that [Definition 41.1.1\(b\)](#) fails at some patterns, but that on any sample in which no unit has  $d_1 = 1$  the realized-value condition holds exactly. What does this say about a frequentist argument, which averages over samples that did not occur, compared with a Bayesian one, which conditions on the sample in hand?

## 41.2 Complete cases and quick repairs

Confronted with holes, software does one of two things without being asked: it deletes the rows that have any, or it fills them and proceeds as though nothing had happened. Both are defensible in circumstances that can be stated precisely, and disastrous outside them. This section states them.

### 41.2.1 When deleting rows is safe

Write  $C_i = 1$  when unit  $i$ 's record is complete in every variable the analysis uses. A **complete-case analysis** applies the analysis to the subsample  $\{i : C_i = 1\}$ , and its behaviour is governed by one thing: the conditional distribution of  $\mathbf{d}_{(i)}$  given  $C_i = 1$ .

**Proposition 41.2.1 — Validity of a complete-case analysis**

Let the rows  $(\mathbf{d}_{(i)}, C_i)$  be independent and identically distributed, with  $\Pr(C_i = 1) > 0$ .

- (a) (*General condition.*) Let a procedure estimate a functional  $T$  consistently at every law in a class containing both  $f$ , the law of  $\mathbf{d}_{(i)}$ , and  $f_{C=1}$ , its conditional law given  $C_i = 1$ . Applied to the complete cases the procedure stays consistent for  $T(f)$  if and only if  $T(f_{C=1}) = T(f)$ . Under MCAR  $f_{C=1} = f$ , and every such functional is preserved.
- (b) (*Regression with selection on the covariates.*) Partition  $\mathbf{d}_{(i)} = (y_i, \mathbf{x}_{(i)})$  and suppose

$$\Pr(C_i = 1 \mid y_i, \mathbf{x}_{(i)}) = \Pr(C_i = 1 \mid \mathbf{x}_{(i)}) \quad \text{almost surely.} \quad (41.2.1)$$

Then the conditional law of  $y_i$  given  $\mathbf{x}_{(i)}$  is the same among the complete cases as in the population. So if a model specifies that conditional law correctly, the complete-case least squares estimate in a linear model is unbiased for  $\boldsymbol{\beta}$  conditionally on the selected covariates; and for a generalized linear model (Definition 34.2.1) the complete-case maximum likelihood estimate is consistent and asymptotically normal, with the usual model-based covariance of Theorem 34.5.1 computed from the complete cases alone, provided their number grows and their covariate second moments converge to a nonsingular limit.

- (c) (*Otherwise.*) If Eq. 41.2.1 fails, write  $\pi(y, \mathbf{x}) = \Pr(C = 1 \mid y, \mathbf{x})$ . The complete-case regression function is

$$E(y \mid \mathbf{x}, C = 1) = E(y \mid \mathbf{x}) + \frac{\text{Cov}\{y, \pi(y, \mathbf{x}) \mid \mathbf{x}\}}{E\{\pi(y, \mathbf{x}) \mid \mathbf{x}\}}, \quad (41.2.2)$$

so the complete cases follow the intended regression function exactly when  $\pi$  is conditionally uncorrelated with the response at every  $\mathbf{x}$ .

*Proof.*

- (a) The complete cases are an independent sample from  $f_{C=1}$ , and their number grows without bound since  $\Pr(C = 1) > 0$ . By the assumed consistency of the procedure at  $f_{C=1}$ , it converges to  $T(f_{C=1})$ , which is the target  $T(f)$  if and only if the two agree. Under MCAR,  $C$  is independent of  $\mathbf{d}$ , hence  $f_{C=1} = f$ .
- (b) By Bayes's rule,

$$f(y \mid \mathbf{x}, C = 1) = \frac{\Pr(C = 1 \mid y, \mathbf{x}) f(y \mid \mathbf{x})}{\Pr(C = 1 \mid \mathbf{x})}, \quad (41.2.3)$$

and under Eq. 41.2.1 the two selection probabilities cancel, leaving  $f(y \mid \mathbf{x})$ . The complete cases are therefore a sample from a joint law whose conditional of  $y$  given  $\mathbf{x}$  is the modelled one, with only the marginal law of  $\mathbf{x}$  distorted. Every likelihood built from  $f(y \mid \mathbf{x}; \boldsymbol{\beta})$  alone conditions on the covariates and is unaffected by that distortion.

(c) Multiply [Eq. 41.2.3](#) by  $y$  and integrate:

$$\begin{aligned} E(y \mid \mathbf{x}, C = 1) &= \frac{E\{y \pi(y, \mathbf{x}) \mid \mathbf{x}\}}{E\{\pi(y, \mathbf{x}) \mid \mathbf{x}\}} \\ &= E(y \mid \mathbf{x}) + \frac{\text{Cov}\{y, \pi(y, \mathbf{x}) \mid \mathbf{x}\}}{E\{\pi(y, \mathbf{x}) \mid \mathbf{x}\}}, \end{aligned}$$

using  $E(y\pi) = E(y)E(\pi) + \text{Cov}(y, \pi)$  throughout at fixed  $\mathbf{x}$ .

■

Part (b) is the most useful positive result in the subject and is routinely overlooked. *Missingness may depend on the covariates as strongly as it likes, in any functional form, without harming a correctly specified regression.* The complete cases are a biased sample of units — their covariate distribution is wrong, so any marginal summary of them is wrong — and yet the fitted conditional mean is right. This is selection on a regressor, as in [Chapter 25](#); the binary-response version is [Exercise \(B1\)](#).

**Warning.** Part (b) needs the conditional *model* to be right. If the true regression is curved and the fitted one is a straight line, selecting on  $\mathbf{x}$  changes which part of the covariate range dominates the fit, and the complete-case slope moves with it. Selection on the covariates is harmless for a correct model and can be very harmful for a wrong one.

### Example. (Four mechanisms, one regression)

Take  $y_i = 1 + 0.8x_i + 0.9w_i + \varepsilon_i$  with  $x_i, w_i, \varepsilon_i$  independent standard normal and  $n = 300$ , and lose the response under four mechanisms: completely at random; at random on  $x$ ; at random on  $x$  and  $w$ ; and not at random, on  $y$ .

**Table 41.2.1.** Mean complete-case estimate of  $\beta_1 = 0.8$  over 2000 replicates, for two analyses under four mechanisms.

| mechanism          | complete cases | fit $y \sim x$ | fit $y \sim x + w$ |
|--------------------|----------------|----------------|--------------------|
| MCAR               | 194            | 0.801          | 0.801              |
| MAR on $x$         | 181            | 0.804          | 0.802              |
| MAR on $x$ and $w$ | 178            | 0.637          | 0.800              |
| MNAR on $y$        | 178            | 0.593          | 0.669              |

The first two rows are [Proposition 41.2.1\(b\)](#): whatever the missingness does with  $x$ , the regression of  $y$  on  $x$  is untouched. The third is the warning attached to it. The mechanism is MAR, using only recorded variables, yet the short regression omitting  $w$  is badly biased, because selecting on  $w$  selects on the part of  $y$  that the short model calls error; adding  $w$  repairs it completely. The fourth row cannot be repaired by any choice of regressors, because the selection is on  $y$  itself.

```
import numpy as np

rng = np.random.default_rng(4101)
n, reps = 300, 2000
beta = np.array([1.0, 0.8, 0.9])
```

# intercept, coefficient of  $x$ ,

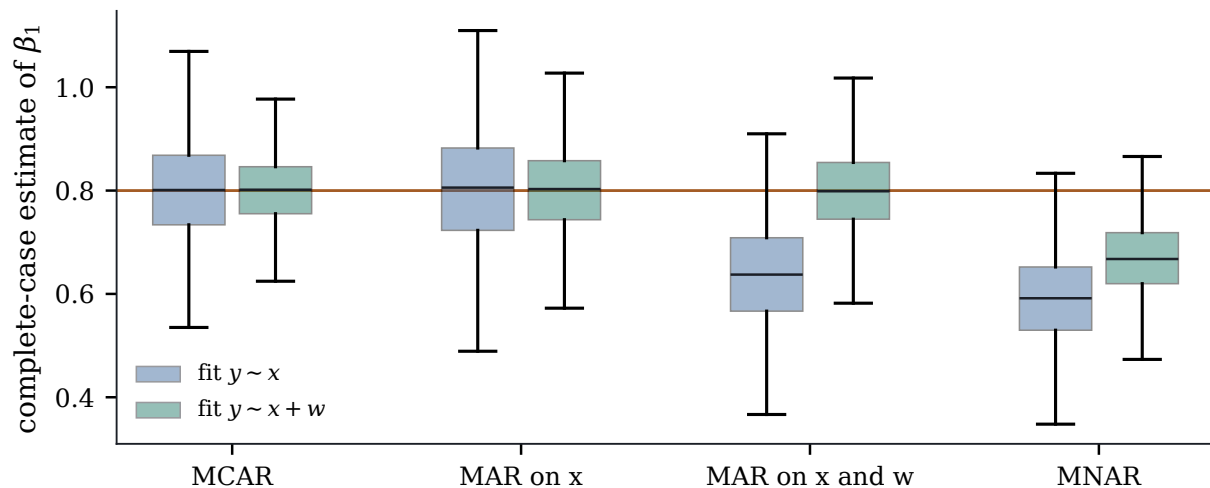


Figure 41.2.1: Complete-case estimates of  $\beta_1 = 0.8$  over 2000 replicates under four mechanisms, for the regression of  $y$  on  $x$  alone and on  $x$  and  $w$ . The horizontal line is the true value.

```

of w

def one_run(mechanism, rng):
    x, w = rng.normal(size=n), rng.normal(size=n)
    y = beta[0] + beta[1] * x + beta[2] * w + rng.normal(size=n)
    if mechanism == "MCAR":
        eta = np.full(n, -0.6)
    elif mechanism == "MAR on x":
        eta = -0.6 + 1.6 * x
        # depends on a regressor only
    elif mechanism == "MAR on x and w":
        eta = -0.6 + 1.2 * x + 1.5 * w
        # depends on observed
        # variables only
    else:
        eta = -0.6 + 1.2 * (y - beta[0])
        # MNAR
        # depends on the value itself
    missing = rng.uniform(size=n) < 1 / (1 + np.exp(-eta))
    obs = ~missing
    short = np.column_stack([np.ones(obs.sum()), x[obs]])
    long = np.column_stack([np.ones(obs.sum()), x[obs], w[obs]])
    # y on x
    # y on x and w
    b_short, *_ = np.linalg.lstsq(short, y[obs], rcond=None)
    b_long, *_ = np.linalg.lstsq(long, y[obs], rcond=None)
    return b_short[1], b_long[1], obs.sum()

names = ["MCAR", "MAR on x", "MAR on x and w", "MNAR"]
out = {m: np.array([one_run(m, rng) for _ in range(reps)]) for m in names}
for m in names:
    s, l, k = out[m].T
    print(f"{m:16s} complete cases {k.mean():5.0f}   y ~ x: {s.mean():.4f}"
          f"      y ~ x + w: {l.mean():.4f}")

```

*Idea.* Ask what the missingness depends on, then put those variables into the analysis. If everything the mechanism uses is in the model and the model is right, the complete cases are enough. If the mechanism uses the response, or a variable the model leaves out, they are not.

## 41.2.2 Available cases

A tempting middle course computes each summary from whichever records carry the variables it needs: a mean from everyone who reported that variable, a covariance from everyone who reported both. This **available-case** analysis, or pairwise deletion, uses more data, at the price of a covariance matrix whose entries come from different subsamples and need not be jointly realizable.

### Example. (A correlation matrix with a negative eigenvalue)

Twelve records on three variables, in three patterns of four:  $d_3$  absent in the first,  $d_2$  in the second,  $d_1$  in the third, with numbers chosen so that the available pairs give  $r_{12} = 0.9$ ,  $r_{13} = 0.9$  and  $r_{23} = -0.9$ . Each is an ordinary correlation on its own four points; together they are impossible, since by [Proposition 2.4.2](#) the third must lie between  $0.81 - \sqrt{0.19 \times 0.19} = 0.62$  and  $0.81 + 0.19 = 1$ . The matrix has smallest eigenvalue  $-0.800$ , so any procedure needing it to be nonnegative definite — a Cholesky factor, a GLS weight matrix, a normal likelihood — fails, even though the matrix itself is perfectly invertible.

```
nan = np.nan
u = np.array([-1.5, -0.5, 0.5, 1.5])
d = np.array([0.5, -1.5, 1.5, -0.5])          # orthogonal to u, same length
pair = np.column_stack([u, 0.9 * u + np.sqrt(1 - 0.81) * d])      #
# correlation +0.9
flip = np.column_stack([u, -(0.9 * u + np.sqrt(1 - 0.81) * d)])    #
# correlation -0.9
D = np.vstack([
    np.column_stack([pair, np.full(4, nan)]),          # pattern A: z3 absent
    np.column_stack([pair[:, 0], np.full(4, nan), pair[:, 1]]),    # pattern B
    # z2 absent
    np.column_stack([np.full(4, nan), flip]),          # pattern C: z1 absent
])
R = np.ones((3, 3))
for i in range(3):
    for j in range(i + 1, 3):
        both = ~np.isnan(D[:, i]) & ~np.isnan(D[:, j])
        R[i, j] = R[j, i] = np.corrcoef(D[both, i], D[both, j])[0, 1]
print("available-case correlation matrix\n", R)
print("eigenvalues", np.round(np.linalg.eigvalsh(R), 3))
```

Under MCAR each entry is consistent and the difficulty is a finite-sample one; under MAR the entries are not consistent at all. Available-case analysis has uses in exploratory work and none in inference.

## 41.2.3 Single imputation

The other reflex is to fill the holes and carry on. **Mean imputation** replaces every missing entry of a variable by the mean of its recorded entries, leaving a pile of entries exactly at the sample mean and a spread that is too small.



### Proposition 41.2.2 — Mean imputation deflates the variance

Let  $z_1, \dots, z_n$  be independent and identically distributed with finite variance  $\sigma^2$ , of which  $k$  are recorded and  $n - k$  missing, the mechanism being completely at random; let  $\tilde{z}_i$  equal  $z_i$  when recorded and  $\bar{z}_{\text{obs}}$  otherwise. Then the sample variance of the filled column satisfies

$$\mathbb{E}\left\{\frac{1}{n-1} \sum_{i=1}^n (\tilde{z}_i - \bar{\tilde{z}})^2 \mid k\right\} = \frac{k-1}{n-1} \sigma^2, \quad (41.2.4)$$

so the estimate of  $\sigma^2$  is too small by the factor  $(k-1)/(n-1) \approx k/n$ , the proportion recorded.

*Proof.* The filled column's mean is  $\bar{\tilde{z}} = \bar{z}_{\text{obs}}$ , because the  $n - k$  inserted values are all equal to it. The imputed entries therefore contribute nothing to  $\sum_i (\tilde{z}_i - \bar{\tilde{z}})^2$ , which reduces to  $\sum_{i \text{ obs}} (z_i - \bar{z}_{\text{obs}})^2$ . Given  $k$ , and under MCAR, that sum has expectation  $(k-1)\sigma^2$ . Dividing by  $n-1$  gives Eq. 41.2.4. ■

With  $n = 300$  and 30% of values missing the factor is 0.699, and the simulation gives 0.700 as the average filled-in sample variance of a standard normal column. Covariances with fully recorded variables are deflated by the same mechanism, so correlations shrink towards zero and coefficients with them.

**Regression imputation** replaces a missing  $z_i$  by its predicted value from a regression on the recorded variables. It fixes the mean but not the spread, since the imputed points lie exactly on the fitted surface. In the same simulation, filling 30% of a variable whose correlation with the predictor is 0.6 raises the sample correlation to 0.667 and lowers the standard deviation to 0.899. Adding a draw from the estimated residual distribution — **stochastic regression imputation** — repairs both, and is the first honest imputation on this list. What it does not repair is the *uncertainty*: the analysis treats  $n$  values as measured when  $n - k$  were invented, and every standard error is too small. Section 41.5 measures the damage — a nominal 95% interval covering 75.9% of the time in the example there — and repairs it.

*Warning.* Never treat a single set of imputed values as data. The estimate may be fine; the standard error, the  $t$  statistic and the interval are not, because the imputed values are correlated with the imputation model's own estimates. The same warning appeared in Section 18.6 for the filled-in plot of a designed experiment, where Theorem 18.6.1(e) gives the exact correction available in that special case.

## 41.2.4 Weighting

Survey statisticians treat unit nonresponse differently. If unit  $i$  is complete with probability  $\pi_i = \Pr(C_i = 1 \mid \mathbf{d}_{(i)})$ , then for any estimating function  $\mathbf{g}$  with  $\mathbb{E}\{\mathbf{g}(\mathbf{d}; \boldsymbol{\beta})\} = \mathbf{0}$  at the true  $\boldsymbol{\beta}$ , the **inverse-probability weighted** equations

$$\sum_{i=1}^n \frac{C_i}{\pi_i} \mathbf{g}(\mathbf{d}_{(i)}; \boldsymbol{\beta}) = \mathbf{0} \quad (41.2.5)$$

are unbiased, because  $E(C_i/\pi_i \mid \mathbf{d}_{(i)}) = 1$ . Under MAR,  $\pi_i$  is a function of recorded variables and can be estimated, typically by a logistic regression of  $C_i$  on them, and the estimator is then consistent when that model is right. This is Horvitz and Thompson's (1952) idea transplanted from sampling to nonresponse, and its weakness is visible in the formula: a unit with a small estimated  $\pi_i$  receives a huge weight, so the estimator is least stable where the data are thinnest. Augmenting Eq. 41.2.5 with a term built from a model for  $E(\mathbf{g} \mid \text{recorded})$  gives the **doubly robust** estimators of Robins, Rotnitzky and Zhao (1994), consistent if either model is right. Weighting is the natural language for unit nonresponse, where there is nothing to impute; likelihood and imputation are the language for item nonresponse, where there is.

## 41.2.5 Exercises

### 41.2.5.1 A. Check your understanding

#### Exercise A1.

Under the mechanism of Proposition 41.2.1(b), show that the complete-case sample mean of  $y$  is generally not consistent for  $E(y)$ , even though the complete-case regression coefficients are consistent. Reconcile the two statements.

*Solution.* Selection changes the law of  $\mathbf{x}$ , and  $E(y) = E\{E(y \mid \mathbf{x})\}$  averages the conditional mean over that law; the complete cases average it over the selected law instead. The regression coefficients describe  $E(y \mid \mathbf{x})$ , which selection leaves alone, so both statements hold at once: the conditional model is right and the marginal average of it is wrong.

### 41.2.5.2 B. Practice

#### Exercise B1.

In Eq. 41.2.2 take  $y \mid \mathbf{x} \sim N(\mathbf{x}^\top \boldsymbol{\beta}, \sigma^2)$  and  $\pi(y, \mathbf{x}) = \Phi(a + by)$ . Show that

$$E(y \mid \mathbf{x}, C = 1) = \mathbf{x}^\top \boldsymbol{\beta} + \frac{b\sigma^2}{\sqrt{1 + b^2\sigma^2}} \frac{\phi(c)}{\Phi(c)}, \quad c = \frac{a + b\mathbf{x}^\top \boldsymbol{\beta}}{\sqrt{1 + b^2\sigma^2}},$$

and deduce that the complete-case regression function is nonlinear in  $\mathbf{x}$  whenever  $b \neq 0$ .

*Solution.* With  $y = \mathbf{x}^\top \boldsymbol{\beta} + \sigma e$ ,  $e$  standard normal,  $E\{\Phi(a + by)\} = \Phi(c)$  by the standard convolution identity, and  $E\{e \Phi(a + by)\} = b\sigma\phi(c)/\sqrt{1 + b^2\sigma^2}$ , obtained by writing  $\Phi(a + by) = \Pr(U \leq a + by)$  for an independent standard normal  $U$  and integrating by parts (equivalently, by Stein's identity  $E\{e h(e)\} = E\{h'(e)\}$ ). Dividing the second by the first and substituting into Eq. 41.2.2 gives the display. The added term depends on  $\mathbf{x}$  through  $c$ , and the ratio  $\phi/\Phi$  is not affine, so the complete-case mean is not linear in  $\mathbf{x}$ ; in particular no single shift of the intercept absorbs it.

**Exercise B2.**

Verify [Eq. 41.2.4](#) by simulation for  $n = 300$  and three missing fractions. Then show that the sample covariance of the filled column with a fully recorded variable  $w$  carries the *same* factor  $(k-1)/(n-1)$ , and explain why the fraction recorded  $k/n$  is only the limiting form of it.

*Solution.* The imputed entries satisfy  $\tilde{z}_i - \bar{\tilde{z}} = 0$ , so  $\sum_i (\tilde{z}_i - \bar{\tilde{z}})(w_i - \bar{w}) = \sum_{\text{obs}} (z_i - \bar{z}_{\text{obs}})w_i$ . Given  $k$  and under MCAR its expectation is  $k\sigma_{zw} - \sigma_{zw} = (k-1)\sigma_{zw}$ , the second term coming from  $E(\bar{z}_{\text{obs}} \sum_{\text{obs}} w_i) = \sigma_{zw}$ . Dividing by  $n-1$  gives  $\{(k-1)/(n-1)\}\sigma_{zw}$ , the factor of [Eq. 41.2.4](#). The two candidates differ only through the divisors used for the sample moments, and  $(k-1)/(n-1) \rightarrow k/n$  as both grow:  $k/n$  is the population-level statement,  $(k-1)/(n-1)$  the exact finite-sample one, and there is no second phenomenon hiding in the gap.

**Exercise B3.**

Prove that [Eq. 41.2.5](#) is an unbiased estimating equation when  $\pi_i > 0$  for every unit, and show by example that the conclusion fails if  $\pi_i = 0$  for a set of covariate values with positive probability. What does that failure correspond to in [Proposition 41.2.1](#)?

**41.2.5.3 C. Going deeper****Exercise C1.**

Let  $E(y | x) = \beta_0 + \beta_1 x + \beta_2 x^2$  with  $\beta_2 \neq 0$ , and let the analyst fit a straight line. Show that even under [Eq. 41.2.1](#) the complete-case slope generally differs from the full-data slope, and identify the functional of the covariate distribution that the straight-line slope estimates (see [Theorem 2.5.2](#) and [Theorem 6.11.1](#)). Under what selection would the two agree?

*Solution.* The straight-line least squares coefficient converges to the best linear predictor coefficient  $\text{Cov}(y, x) / \text{Var}(x)$  computed under whatever law  $x$  has. With  $E(y | x) = \beta_0 + \beta_1 x + \beta_2 x^2$  this is  $\beta_1 + \beta_2 \text{Cov}(x^2, x) / \text{Var}(x)$ , which depends on the third and second central moments of  $x$ . Selection changes those moments, hence the limit, unless it leaves  $\text{Cov}(x^2, x) / \text{Var}(x)$  unchanged — for instance under MCAR, or under a selection probability that is an even function of  $x$  about its mean when the law of  $x$  is symmetric.

## 41.3 Likelihood under missingness at random

A complete-case analysis throws away recorded values: a unit whose income is missing still told us its age. Likelihood throws nothing away. It asks what the *observed* data were likely to be, and the observed data include the partial records. This section writes that likelihood down, proves the theorem that makes it usable, and says what the theorem costs.

### 41.3.1 The observed-data likelihood

The full model is [Eq. 41.1.1](#): a density for the complete array and a mechanism. Since  $\mathbf{D}_{\text{mis}}$  was never seen, the likelihood of what was seen integrates it out,

$$L_{\text{full}}(\boldsymbol{\theta}, \boldsymbol{\psi}) = \int f(\mathbf{D}_{\text{obs}}, \mathbf{D}_{\text{mis}} \mid \boldsymbol{\theta}) f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \mathbf{D}_{\text{mis}}, \boldsymbol{\psi}) d\mathbf{D}_{\text{mis}}, \quad (41.3.1)$$

a function of both parameters. Beside it stands the much simpler

$$L(\boldsymbol{\theta}) = f(\mathbf{D}_{\text{obs}} \mid \boldsymbol{\theta}) = \int f(\mathbf{D}_{\text{obs}}, \mathbf{D}_{\text{mis}} \mid \boldsymbol{\theta}) d\mathbf{D}_{\text{mis}}, \quad (41.3.2)$$

which ignores the mechanism entirely. The **observed-data likelihood** [Eq. 41.3.2](#) is what software computes when told that some entries are absent. The question is when using it is the same as using [Eq. 41.3.1](#).

#### Theorem 41.3.1 — Ignorability

Suppose that

- (i) the mechanism is MAR at the realized  $(\mathbf{R}, \mathbf{D}_{\text{obs}})$  in the sense of [Definition 41.1.1\(b\)](#);
- (ii) the parameters are **distinct**: the parameter space of  $(\boldsymbol{\theta}, \boldsymbol{\psi})$  is a product set  $\Theta \times \Psi$ , so that knowing one places no restriction on the other.

Then the full likelihood factors,

$$L_{\text{full}}(\boldsymbol{\theta}, \boldsymbol{\psi}) = L(\boldsymbol{\theta}) \cdot f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi}), \quad (41.3.3)$$

and consequently:

- (a) the maximizer of  $L_{\text{full}}$  in  $\boldsymbol{\theta}$  is the maximizer of  $L$ , and likelihood ratio statistics for hypotheses about  $\boldsymbol{\theta}$  alone are the same under both;
- (b) if the prior is  $p(\boldsymbol{\theta}, \boldsymbol{\psi}) = p(\boldsymbol{\theta})p(\boldsymbol{\psi})$ , the posterior of  $\boldsymbol{\theta}$  given  $(\mathbf{D}_{\text{obs}}, \mathbf{R})$  equals its posterior given  $\mathbf{D}_{\text{obs}}$  alone, computed from [Eq. 41.3.2](#);
- (c) if in addition the MAR condition holds at every  $(\mathbf{R}, \mathbf{D}_{\text{obs}})$ , the observed-data score has mean zero and the usual maximum likelihood asymptotics ([Theorem 7.3.1](#)) apply to  $L$  under the ordinary regularity conditions.

A mechanism satisfying (i) and (ii) is called **ignorable**.

*Proof.* Under (i) the factor  $f(\mathbf{R} \mid \mathbf{D}, \boldsymbol{\psi})$  in [Eq. 41.3.1](#) takes the same value for every  $\mathbf{D}_{\text{mis}}$ , namely  $f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi})$ , so it comes out of the integral:

$$\begin{aligned} L_{\text{full}}(\boldsymbol{\theta}, \boldsymbol{\psi}) &= f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi}) \int f(\mathbf{D}_{\text{obs}}, \mathbf{D}_{\text{mis}} \mid \boldsymbol{\theta}) d\mathbf{D}_{\text{mis}} \\ &= f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi}) L(\boldsymbol{\theta}), \end{aligned}$$

which is [Eq. 41.3.3](#).

- (a) By (ii),  $\boldsymbol{\theta}$  ranges over  $\Theta$  whatever  $\boldsymbol{\psi}$  is, so  $\max_{\boldsymbol{\psi}} L_{\text{full}} = L(\boldsymbol{\theta}) \max_{\boldsymbol{\psi}} f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi})$ , and the second factor does not involve  $\boldsymbol{\theta}$ . Maximizing over  $\boldsymbol{\theta}$  therefore maximizes  $L$ . For a hypothesis  $\boldsymbol{\theta} \in \Theta_0$ , both the numerator and the denominator of the likelihood ratio carry the same second factor, which cancels.
- (b) The posterior is proportional to  $p(\boldsymbol{\theta})p(\boldsymbol{\psi})L_{\text{full}}$ . By Eq. 41.3.3 this is  $\{p(\boldsymbol{\theta})L(\boldsymbol{\theta})\} \cdot \{p(\boldsymbol{\psi})f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi})\}$ , a product of a function of  $\boldsymbol{\theta}$  and a function of  $\boldsymbol{\psi}$ . Integrating out  $\boldsymbol{\psi}$  multiplies the first bracket by a constant, so the marginal posterior of  $\boldsymbol{\theta}$  is proportional to  $p(\boldsymbol{\theta})L(\boldsymbol{\theta})$ .
- (c) If (i) holds at every realization, then at every realization the joint density of the observed data  $(\mathbf{D}_{\text{obs}}, \mathbf{R})$  is exactly  $L(\boldsymbol{\theta}) f(\mathbf{R} \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\psi})$ , whose logarithm is  $\log L(\boldsymbol{\theta})$  plus a term free of  $\boldsymbol{\theta}$ . So  $\partial \log L / \partial \boldsymbol{\theta}$  is the  $\boldsymbol{\theta}$  component of the score of a correctly specified density for the data actually seen, and has mean zero under the ordinary regularity conditions of Theorem 7.3.1.

■

*Remark* (Where each hypothesis is used). Condition (i) alone gives Eq. 41.3.3 and part (b): a Bayesian who conditions on the data in hand needs nothing more. Condition (ii) is what makes the factorization useful to a frequentist, who maximizes; without it the two factors share parameters and neither can be maximized alone. Part (c) needs the everywhere version of MAR, because a sampling distribution averages over data sets that did not occur. Textbooks usually state “MAR plus distinctness” without separating these roles; the separation is due to Seaman, Galati, Jackson and Carlin (2013).

*Idea.* Under MAR with distinct parameters, the analyst may forget the mechanism and maximize the likelihood of what was recorded. That is the entire licence for the methods of the next two sections — and it is a licence to use a likelihood that is usually an integral, not a product.

## 41.3.2 What the licence costs

Ignorability licenses ignoring the mechanism, not ignoring the missing values. Three consequences follow.

**A model is now needed for variables previously conditioned on.** In a complete data set, least squares never models the distribution of  $\mathbf{X}$ : the likelihood conditions on it. When a covariate is missing, Eq. 41.3.2 integrates over it, and that integral needs a density. Missing responses cost nothing extra; missing covariates cost a model for the covariates.

**When only the response is missing, the partial records add nothing.** If every covariate is recorded, the integral in Eq. 41.3.2 over a missing  $y_i$  is one, and the observed-data likelihood is exactly the complete-case likelihood. This is Proposition 41.2.1(b) again, as a statement about likelihoods, and it is the situation of Theorem 18.6.1.

**The likelihood is no longer a product of simple factors.** Each incomplete record contributes a marginal density, obtained by integrating the complete-data density over the entries it lacks. For the multivariate normal those integrals are marginals of a normal; for almost anything else they have no closed form. Section 41.4 is the standard response.

For a **monotone** pattern the integrals are free: integrating out a record’s unrecorded tail deletes

exactly its conditional factors, so the likelihood becomes a chain of conditional models on nested subsamples, each maximized on its own; that is [Exercise \(B1\)](#) of [Section 41.1](#). The bivariate case is the subject in miniature. With  $u$  always recorded and  $v$  recorded on the first  $k$  units, the estimate of  $\mu_v$  is  $\bar{v}_{(k)} + \hat{\beta}\{\bar{u}_{(n)} - \bar{u}_{(k)}\}$  by [Exercise \(B1\)](#) below: the complete-case mean of  $v$ , corrected by the shift in the mean of  $u$ , using the  $n - k$  records that carry a  $u$  and no  $v$ . Those records were useless to the complete-case analysis and are decisive here — and they repair the damage only to the extent that  $u$  predicts  $v$ , a remark that returns, with teeth, in [Section 41.5](#). Anderson (1957) solved the general monotone multivariate normal case this way.

### 41.3.3 What changes under MNAR

Drop the MAR assumption and [Eq. 41.3.3](#) fails: the integral in [Eq. 41.3.1](#) cannot release the mechanism, and the two parameters are entangled. The mechanism must then be modelled, and the model is not checkable, since [Proposition 41.1.1](#) says that varying the assumed conditional distribution of the missing values traces out a family with identical observed-data likelihoods. Whatever identification such a model achieves comes from functional-form assumptions rather than from the data, and the likelihood surface is often nearly flat in the directions that matter. [Section 41.6](#) takes this up as a problem of reporting rather than of estimation.

### 41.3.4 Exercises

#### 41.3.4.1 A. Check your understanding

##### Exercise A1.

Give a mechanism under which MAR holds but the parameters are not distinct, and show that [Eq. 41.3.3](#) still holds while part (a) of [Theorem 41.3.1](#) can fail. (Hint: let the probability of recording  $y_i$  be a known function of  $\beta$ .)

*Solution.* Take  $\Pr(R_i = 1 \mid \mathbf{x}_{(i)}) = \Phi(\mathbf{x}_{(i)}^\top \beta)$ , with no free  $\psi$  at all. The mechanism depends only on recorded covariates, so MAR holds and the factorization [Eq. 41.3.3](#) is valid, but the second factor now depends on  $\beta$ . Maximizing the product is not the same as maximizing  $L(\beta)$ , and the indicators carry information about  $\beta$ : the parameter spaces are not a product set.

#### 41.3.4.2 B. Practice

##### Exercise B1.

Let  $(u_i, v_i)$  be a bivariate normal sample with  $u$  recorded for all  $n$  units and  $v$  for the first  $k$ , the mechanism depending on  $u$  but not on  $v$ . Reparameterizing by  $(\mu_u, \sigma_u^2)$  and the regression parameters of  $v$  on  $u$ , derive the maximum likelihood estimates and the formula for  $\hat{\mu}_v$  quoted above. Extend to three variables with a monotone pattern:  $u$  recorded  $n$  times,  $v$  the first  $k_2$ ,  $z$  the first  $k_3 \leq k_2$ .

*Solution.* Factor as  $f(u)f(v \mid u)$ ; by [Exercise \(B1\)](#) an unrecorded  $v_i$  loses its conditional factor, so  $L = \prod_{i \leq n} f(u_i) \prod_{i \leq k} f(v_i \mid u_i)$ . The two parameter groups are variation independent, so each product is maximized alone: a normal sample of size  $n$ , and a normal

linear model on  $k$  observations ([Theorem 7.3.1](#)). Since  $\mu_v = \alpha + \beta\mu_u$  and  $\hat{\alpha} = \bar{v}_{(k)} - \hat{\beta}\bar{u}_{(k)}$ , and since a maximum likelihood estimate transforms with the parameterization,  $\hat{\mu}_v$  is as stated. For three variables the same argument on  $f(u)f(v | u)f(z | u, v)$  gives a univariate fit on all  $n$ , the regression of  $v$  on  $u$  over the first  $k_2$ , and the regression of  $z$  on  $(u, v)$  over the first  $k_3$ .

### Exercise B2.

A linear model has a single covariate  $x$  that is sometimes missing, with  $x \sim N(\mu, \tau^2)$  and  $y | x \sim N(\beta_0 + \beta_1 x, \sigma^2)$ . Write the observed-data likelihood contribution of a record with  $y$  recorded and  $x$  missing, and show that it is the density of a normal with mean  $\beta_0 + \beta_1 \mu$  and variance  $\beta_1^2 \tau^2 + \sigma^2$ . Why does this contribution carry information about  $\beta_1$  even though it contains no  $x$ ?

#### 41.3.4.3 C. Going deeper

### Exercise C1.

Show that [Theorem 18.6.1](#) is the special case of [Theorem 41.3.1](#) in which the missing entries are responses, the mechanism depends only on the design, and the complete-data model is the normal linear model. Which part of [Theorem 18.6.1](#) corresponds to [Eq. 41.3.2](#), and which to the warning that the degrees of freedom must be reduced?

## 41.4 The EM algorithm

Maximizing the observed-data likelihood [Eq. 41.3.2](#) directly means differentiating an integral. EM avoids that: it alternates between filling in the missing pieces of the *complete-data log-likelihood* and maximizing the filled-in version, and it never decreases the observed-data likelihood. The iteration of [Theorem 18.6.1\(c\)](#) — fill each lost plot with its fitted value, refit, repeat — is the oldest instance in this book.

### 41.4.1 The two steps

Let  $\ell(\boldsymbol{\theta}) = \log f(\mathbf{D}_{\text{obs}} | \boldsymbol{\theta})$  and  $\ell_c(\boldsymbol{\theta}) = \log f(\mathbf{D} | \boldsymbol{\theta})$  be the observed-data and complete-data log-likelihoods, the second a function of data we do not have. The mechanism is ignorable throughout, so [Theorem 41.3.1](#) licenses working with  $\ell$ .

#### Definition 41.4.1 — EM

Given a current value  $\boldsymbol{\theta}^{(s)}$ , define

$$Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(s)}) = E\{\ell_c(\boldsymbol{\theta}) | \mathbf{D}_{\text{obs}}, \boldsymbol{\theta}^{(s)}\}, \quad (41.4.1)$$

the expectation being over  $\mathbf{D}_{\text{mis}}$  under the conditional density  $f(\mathbf{D}_{\text{mis}} | \mathbf{D}_{\text{obs}}, \boldsymbol{\theta}^{(s)})$ . The

**E step** computes  $Q(\cdot | \boldsymbol{\theta}^{(s)})$ ; the **M step** sets  $\boldsymbol{\theta}^{(s+1)}$  to a maximizer of it. If the M step only *increases*  $Q$ , the scheme is called **generalized EM**.

The E step is not “fill in the missing values”; it fills in whatever functions of them the complete-data log-likelihood contains. When  $\ell_c$  is linear in a set of sufficient statistics, as for every exponential family (Definition 34.1.1), the E step reduces to filling in those statistics (Exercise (B2)) — which is the same as filling in the missing values only when the statistics are linear in them. A normal variance, whose statistic contains  $\sum_i y_i^2$ , is the standing counterexample, and only the second reading is EM.

## 41.4.2 Monotonicity

### Theorem 41.4.1 — The EM sequence never decreases the likelihood

Let  $\boldsymbol{\theta}^{(s+1)}$  satisfy  $Q(\boldsymbol{\theta}^{(s+1)} | \boldsymbol{\theta}^{(s)}) \geq Q(\boldsymbol{\theta}^{(s)} | \boldsymbol{\theta}^{(s)})$ . Then

- (a)  $\ell(\boldsymbol{\theta}^{(s+1)}) \geq \ell(\boldsymbol{\theta}^{(s)})$ , with equality only if  $Q(\boldsymbol{\theta}^{(s+1)} | \boldsymbol{\theta}^{(s)}) = Q(\boldsymbol{\theta}^{(s)} | \boldsymbol{\theta}^{(s)})$  and the two conditional densities of  $\mathbf{D}_{\text{mis}}$  agree almost everywhere;
- (b) if  $\ell$  is bounded above,  $\ell(\boldsymbol{\theta}^{(s)})$  converges;
- (c) if  $Q(\cdot | \cdot)$  is differentiable in its first argument and  $\boldsymbol{\theta}^{(s+1)}$  is an interior maximizer, then at any fixed point  $\boldsymbol{\theta}^*$  of the iteration the observed-data score vanishes:  $\partial \ell(\boldsymbol{\theta}) / \partial \boldsymbol{\theta} |_{\boldsymbol{\theta}^*} = \mathbf{0}$ .

*Proof.* Write  $g(\mathbf{D}_{\text{mis}} | \boldsymbol{\theta}) = f(\mathbf{D}_{\text{mis}} | \mathbf{D}_{\text{obs}}, \boldsymbol{\theta})$  for the conditional density of the missing part. Since  $f(\mathbf{D} | \boldsymbol{\theta}) = f(\mathbf{D}_{\text{obs}} | \boldsymbol{\theta}) g(\mathbf{D}_{\text{mis}} | \boldsymbol{\theta})$ , taking logarithms and rearranging gives, for every  $\mathbf{D}_{\text{mis}}$ ,

$$\ell(\boldsymbol{\theta}) = \ell_c(\boldsymbol{\theta}) - \log g(\mathbf{D}_{\text{mis}} | \boldsymbol{\theta}).$$

The left side does not depend on  $\mathbf{D}_{\text{mis}}$ , so taking the expectation of both sides under  $g(\cdot | \boldsymbol{\theta}^{(s)})$  leaves it unchanged and yields the decomposition

$$\begin{aligned} \ell(\boldsymbol{\theta}) &= Q(\boldsymbol{\theta} | \boldsymbol{\theta}^{(s)}) - H(\boldsymbol{\theta} | \boldsymbol{\theta}^{(s)}), \quad \text{where} \\ H(\boldsymbol{\theta} | \boldsymbol{\theta}^{(s)}) &= E\{\log g(\mathbf{D}_{\text{mis}} | \boldsymbol{\theta}) | \mathbf{D}_{\text{obs}}, \boldsymbol{\theta}^{(s)}\}. \end{aligned} \tag{41.4.2}$$

- (a) Subtract Eq. 41.4.2 at  $\boldsymbol{\theta}^{(s)}$  from the same at  $\boldsymbol{\theta}^{(s+1)}$ :

$$\begin{aligned} \ell(\boldsymbol{\theta}^{(s+1)}) - \ell(\boldsymbol{\theta}^{(s)}) &= \{Q(\boldsymbol{\theta}^{(s+1)} | \boldsymbol{\theta}^{(s)}) - Q(\boldsymbol{\theta}^{(s)} | \boldsymbol{\theta}^{(s)})\} \\ &\quad - \{H(\boldsymbol{\theta}^{(s+1)} | \boldsymbol{\theta}^{(s)}) - H(\boldsymbol{\theta}^{(s)} | \boldsymbol{\theta}^{(s)})\}. \end{aligned}$$

The first brace is nonnegative by hypothesis. For the second, write  $g_s = g(\cdot | \boldsymbol{\theta}^{(s)})$  and  $g_{s+1} = g(\cdot | \boldsymbol{\theta}^{(s+1)})$ ; then

$$\begin{aligned} H(\boldsymbol{\theta}^{(s+1)} | \boldsymbol{\theta}^{(s)}) - H(\boldsymbol{\theta}^{(s)} | \boldsymbol{\theta}^{(s)}) &= \int g_s \log \frac{g_{s+1}}{g_s} \\ &\leq \log \int g_s \frac{g_{s+1}}{g_s} = \log 1 = 0, \end{aligned}$$



by Jensen's inequality applied to the concave logarithm. So the difference of log-likelihoods is a nonnegative number minus a nonpositive one. Equality throughout forces equality in Jensen's inequality, which for the strictly concave logarithm requires  $g_{s+1}/g_s$  to be constant almost everywhere under  $g_s$ , hence  $g_{s+1} = g_s$ , and forces the first brace to vanish.

- (b) A nondecreasing sequence bounded above converges.
- (c) At a fixed point,  $\boldsymbol{\theta}^*$  maximizes  $Q(\cdot \mid \boldsymbol{\theta}^*)$ , so  $\partial Q(\boldsymbol{\theta} \mid \boldsymbol{\theta}^*)/\partial \boldsymbol{\theta}$  vanishes at  $\boldsymbol{\theta} = \boldsymbol{\theta}^*$ . The function  $\boldsymbol{\theta} \mapsto H(\boldsymbol{\theta} \mid \boldsymbol{\theta}^*)$  is maximized at  $\boldsymbol{\theta} = \boldsymbol{\theta}^*$  by the Jensen argument just given, so its derivative vanishes there too. Differentiating Eq. 41.4.2 at  $\boldsymbol{\theta}^*$  gives  $\partial \ell / \partial \boldsymbol{\theta} = \mathbf{0} - \mathbf{0}$ .

■

*Warning.* Part (c) says fixed points are stationary points, not maxima: EM can converge to a saddle point or a local maximum, and with several modes the limit depends on the start. Wu (1983) gives conditions under which the limit points are stationary and under which the sequence converges; none rules out a local maximum. Start from several values.

### 41.4.3 Missing responses in a linear model

The oldest case is the one where every step is explicit, and it settles a question left open in Section 18.6.

#### Proposition 41.4.2 — EM for a normal linear model with missing responses

Let  $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$  with  $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2 \mathbf{I})$  and  $\text{rank}(\mathbf{X}) = p$ ; let the responses of a set of  $m$  rows be unrecorded, the covariates of those rows being known, with an ignorable mechanism, and suppose the observed rows  $\mathbf{X}_o$  have rank  $p$ . Write  $\text{SSE}_o$  for the residual sum of squares of the least squares fit on the observed rows. Then:

- (a) the E step replaces each missing  $y_i$  by  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^{(s)}$  in the cross-product terms of  $\ell_c$ , and each  $y_i^2$  by  $(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^{(s)})^2 + \sigma^{2(s)}$ ;
- (b) the M step is  $\boldsymbol{\beta}^{(s+1)} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}^{(s)}$  with  $\mathbf{y}^{(s)}$  the completed vector, and  $\sigma^{2(s+1)} = n^{-1} \{ \|\mathbf{y}^{(s)} - \mathbf{X}\boldsymbol{\beta}^{(s+1)}\|^2 + m\sigma^{2(s)} \}$ ;
- (c) the unique fixed point is  $\boldsymbol{\beta}^* = \hat{\boldsymbol{\beta}}_o$ , the least squares estimate from the observed rows, and

$$\sigma^{2*} = \frac{\text{SSE}_o}{n - m}, \quad (41.4.3)$$

which is the observed-data maximum likelihood estimate, based on  $n - m$  observations.

*Proof.*

- (a) The complete-data log-likelihood  $-\frac{1}{2}n \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_i (y_i - \mathbf{x}_{(i)}^\top \boldsymbol{\beta})^2$  is quadratic

in the  $y_i$ , so its conditional expectation needs their first two conditional moments. Given the observed data an unrecorded  $y_i$  is a draw from its own model, so those are  $\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^{(s)}$  and  $(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^{(s)})^2 + \sigma^{2(s)}$ .

(b) With those substitutions,

$$Q = -\frac{1}{2}n \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \left\{ \|\mathbf{y}^{(s)} - \mathbf{X}\boldsymbol{\beta}\|^2 + m\sigma^{2(s)} \right\},$$

because each missing row contributes  $E(y_i^2) - 2\mathbf{x}_{(i)}^\top \boldsymbol{\beta} E(y_i) + (\mathbf{x}_{(i)}^\top \boldsymbol{\beta})^2 = (\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^{(s)} - \mathbf{x}_{(i)}^\top \boldsymbol{\beta})^2 + \sigma^{2(s)}$ . Maximizing in  $\boldsymbol{\beta}$  is ordinary least squares on  $\mathbf{y}^{(s)}$ ; maximizing in  $\sigma^2$  gives the stated average.

(c) At a fixed point the completed vector has  $y_i = \mathbf{x}_{(i)}^\top \boldsymbol{\beta}^*$  at the missing rows, so those rows have zero residual and contribute nothing to the least squares criterion; hence  $\boldsymbol{\beta}^*$  minimizes  $\|\mathbf{y}_o - \mathbf{X}_o \mathbf{b}\|^2$  and equals  $\hat{\boldsymbol{\beta}}_o$ , which is [Theorem 18.6.1\(a\)](#). Its residual sum of squares is  $\text{SSE}_o$ , so the variance equation reads  $n\sigma^{2*} = \text{SSE}_o + m\sigma^{2*}$ , giving [Eq. 41.4.3](#). Uniqueness of  $\boldsymbol{\beta}^*$  follows from [Theorem 18.6.1\(b\)](#), and [Eq. 41.4.3](#) then determines  $\sigma^{2*}$ . ■

The term  $m\sigma^{2(s)}$  is the whole point. Without it — fill the gaps with fitted values and hand the completed data to a regression routine — the variance comes out as  $\text{SSE}_o/n$ . That is the error [Theorem 18.6.1\(e\)](#) warns against, now derived rather than asserted: *the filled-in values are not data, and the likelihood knows it.*

### Example. (The Mauna Loa gaps)

Fit the trend-plus-harmonics model of [Section 5.3](#) to all 2284 weeks, with the 59 missing ones handled by EM. The coefficients agree with the complete-case fit to machine precision, as [Proposition 41.4.2\(c\)](#) requires, and the error standard deviation is 0.7992 ppm; filling in the gaps and treating them as data gives 0.7889 ppm, too small by the factor  $\sqrt{n/(n-m)} = 1.0132$ . The discrepancy is small here only because the gaps are few.

```
import numpy as np
import statsmodels.api as sm

co2 = sm.datasets.co2.load_pandas().data["co2"]
t = (co2.index.year + (co2.index.dayofyear - 0.5) / 365.25).to_numpy()
tc = t - 1980.0
X = np.column_stack([np.ones(len(t)), tc, tc**2,
                     np.cos(2 * np.pi * t), np.sin(2 * np.pi * t),
                     np.cos(4 * np.pi * t), np.sin(4 * np.pi * t)])

y = co2.to_numpy()
obs = ~np.isnan(y)
n, m, p = len(y), int((~obs).sum()), X.shape[1]

beta, sigma2 = np.zeros(p), 1.0 # any starting value will do
for sweep in range(60):
    filled = np.where(obs, y, X @ beta) # E step: gaps get their
    fitted values
    beta, *_ = np.linalg.lstsq(X, filled, rcond=None) # M step for the coefficients
```

```

sse = float(np.sum((filled - X @ beta) ** 2))
sigma2 = (sse + m * sigma2) / n          # M step for the variance: add
                                         m*sigma^2
print(f"EM after {sweep + 1} sweeps: sigma-hat = {np.sqrt(sigma2):.4f} ppm")

beta_cc, *_ = np.linalg.lstsq(X[obs], y[obs], rcond=None)
sse_o = float(np.sum((y[obs] - X[obs] @ beta_cc) ** 2))
print(f"observed-data MLE: {np.sqrt(sse_o / (n - m)):.4f} ppm")
print(f"treating the fills as data: {np.sqrt(sse_o / n):.4f} ppm")

```

### 41.4.4 A missing covariate

When a covariate is missing, the conditional distribution the E step needs runs *backwards*: what does a recorded response say about an unrecorded covariate? A model for the covariate is now unavoidable.

#### Proposition 41.4.3 — EM with a missing covariate

Let  $x_i \sim N(\mu, \tau^2)$  independently, and  $y_i | x_i \sim N(\beta_0 + \beta_1 x_i, \sigma^2)$ , with  $\theta = (\beta_0, \beta_1, \sigma^2, \mu, \tau^2)$ . Suppose  $y_i$  is always recorded and  $x_i$  is missing for a subset, with an ignorable mechanism. Then for a record with  $x_i$  missing the E step uses

$$E(x_i | y_i) = \mu + \frac{\beta_1 \tau^2}{\beta_1^2 \tau^2 + \sigma^2} (y_i - \beta_0 - \beta_1 \mu), \quad \text{Var}(x_i | y_i) = \tau^2 - \frac{\beta_1^2 \tau^4}{\beta_1^2 \tau^2 + \sigma^2}, \quad (41.4.4)$$

both evaluated at  $\theta^{(s)}$ , and the M step is the complete-data maximum likelihood fit with the sufficient statistics  $\sum_i x_i$ ,  $\sum_i x_i^2$  and  $\sum_i x_i y_i$  replaced by  $\sum_i E(x_i | y_i)$ ,  $\sum_i E(x_i^2 | y_i)$  and  $\sum_i y_i E(x_i | y_i)$ .

*Proof.* The pair  $(x_i, y_i)$  is bivariate normal, with  $E y_i = \beta_0 + \beta_1 \mu$ ,  $\text{Var } y_i = \beta_1^2 \tau^2 + \sigma^2$  and  $\text{Cov}(x_i, y_i) = \beta_1 \tau^2$ . The conditional moments Eq. 41.4.4 are the standard normal conditional mean and variance (Theorem 3.5.2). The complete-data log-likelihood is

$$-\frac{n}{2} \log \sigma^2 - \frac{\sum_i (y_i - \beta_0 - \beta_1 x_i)^2}{2\sigma^2} - \frac{n}{2} \log \tau^2 - \frac{\sum_i (x_i - \mu)^2}{2\tau^2} + \text{const},$$

which is linear in the three listed statistics together with  $\sum_i y_i$  and  $\sum_i y_i^2$ ; taking conditional expectations therefore replaces exactly those statistics, and the maximizer is the complete-data one evaluated at them. ■

Note what the E step does *not* do: it does not impute  $x_i$  by  $\mu$ , nor from other covariates alone. It uses the response. Imputing a covariate without the response is among the commonest errors in practice, and it attenuates  $\hat{\beta}_1$  towards zero, for the reason Chapter 24 gives.

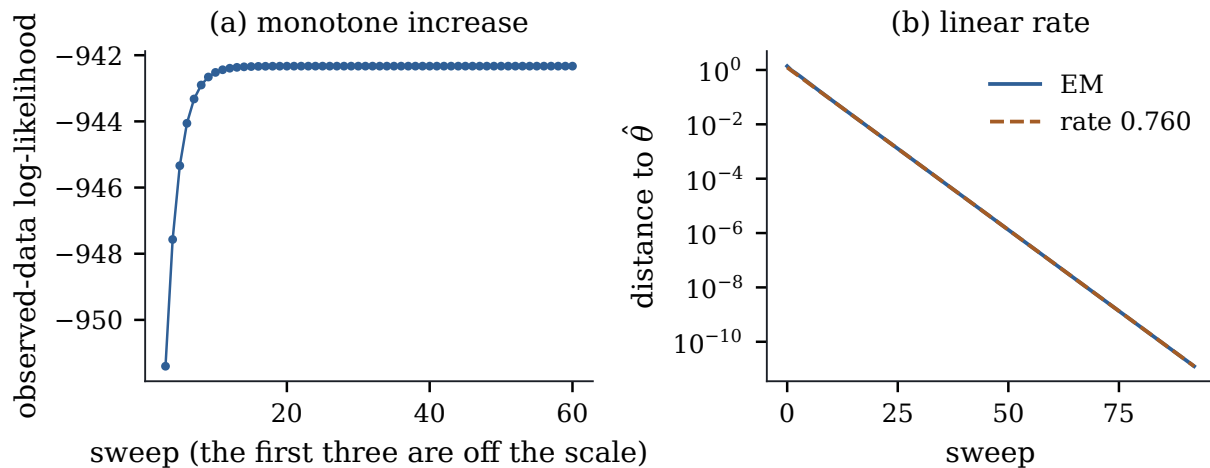


Figure 41.4.1: EM for a linear model with a missing covariate. (a) The observed-data log-likelihood increases at every sweep. (b) The distance from the current iterate to the fixed point decays geometrically at the rate predicted by the fraction of missing information.

#### Example. (An EM fit with a missing covariate)

Simulate  $N = 400$  records with  $\beta_1 = 0.8$ , and let the covariate be missing more often when the response is large; this loses 169 covariate values. The mechanism is MAR — it uses only the recorded  $y$  — so likelihood is valid and the complete-case fit is not. EM gives  $\hat{\beta}_1 = 0.825$ ; the complete cases give 0.620, attenuated by a quarter.

Figure 41.4.1 shows the two faces of the algorithm. The observed-data log-likelihood climbs monotonically, as Theorem 41.4.1(a) requires, and the distance to the fixed point falls by the constant factor 0.760 per sweep, matching to three decimals the largest eigenvalue of  $\mathcal{I}_c^{-1}\mathcal{I}_m$  below. Reaching eight decimal places takes 68 sweeps — slow by the standards of Newton’s method, which is the price of never differentiating an integral.

```
rng = np.random.default_rng(4104)
N = 400
theta_true = dict(b0=1.0, b1=0.8, s2=1.0, mu=0.5, t2=1.0)
x = theta_true["mu"] + np.sqrt(theta_true["t2"]) * rng.normal(size=N)
y2 = theta_true["b0"] + theta_true["b1"] * x + np.sqrt(theta_true["s2"]) * rng.
    normal(size=N)
seen = rng.uniform(size=N) > 1 / (1 + np.exp(-(-0.4 + 2.0 * (y2 - y2.mean()))))
print(f"{N - seen.sum()} of {N} covariates missing, more often when y is large")

def estep(th, y, seen):
    """Conditional mean and second moment of the covariate given the response."""
    b0, b1, s2, mu, t2 = th
    vy = b1**2 * t2 + s2 # marginal variance of y
    cond_mean = mu + (b1 * t2 / vy) * (y - b0 - b1 * mu)
    cond_var = t2 - (b1 * t2) ** 2 / vy
    ex = np.where(seen, x, cond_mean)
    ex2 = np.where(seen, x**2, cond_mean**2 + cond_var)
    return ex, ex2

def mstep(ex, ex2, y):
    mu, n_ = ex.mean(), len(y)
```

```

t2 = ex2.mean() - mu**2
sxx = ex2.sum() - n_ * mu**2
sxy = (ex * y).sum() - n_ * mu * y.mean()
b1 = sxy / sxx
b0 = y.mean() - b1 * mu
s2 = ((y**2).sum() - 2 * b0 * y.sum() - 2 * b1 * (ex * y).sum()
      + n_ * b0**2 + 2 * b0 * b1 * ex.sum() + b1**2 * ex2.sum()) / n_
return np.array([b0, b1, s2, mu, t2])

def loglik(th, y, seen):
    """Observed-data log-likelihood: joint for complete rows, marginal for the rest.
    """
    b0, b1, s2, mu, t2 = th
    ll = -0.5 * np.sum(np.log(2 * np.pi * s2) + (y[seen] - b0 - b1 * x[seen])**2 /
                      s2)
    ll += -0.5 * np.sum(np.log(2 * np.pi * t2) + (x[seen] - mu)**2 / t2)
    vy = b1**2 * t2 + s2
    ll += -0.5 * np.sum(np.log(2 * np.pi * vy) + (y[~seen] - b0 - b1 * mu)**2 / vy)
    return ll

th = np.array([0.0, 0.0, 1.0, 0.0, 1.0])
path, iterates = [loglik(th, y2, seen)], [th]
for sweep in range(200):
    ex, ex2 = estep(th, y2, seen)
    th = mstep(ex, ex2, y2)
    path.append(loglik(th, y2, seen))
    iterates.append(th)
print("EM   b0, b1 =", np.round(th[:2], 4))

cc = np.polyfit(x[seen], y2[seen], 1)[::-1] # complete cases only
print("complete cases:", np.round(cc, 4))

```

### 41.4.5 Rate of convergence, and the standard errors EM does not give

Both the speed of EM and its principal defect come from the same matrix. Define the **complete-data information**  $\mathcal{I}_c = -\partial^2 Q(\theta | \theta^*) / \partial \theta \partial \theta^\top$  at  $\theta = \theta^*$ , the **observed information**  $\mathcal{I}_o = -\partial^2 \ell / \partial \theta \partial \theta^\top$  there, and their difference  $\mathcal{I}_m = \mathcal{I}_c - \mathcal{I}_o$ , the **missing information**.

#### Proposition 41.4.4 — The missing-information principle

At any  $\theta$ , with  $\mathbf{S}_c$  the complete-data score,

$$\frac{\partial \ell}{\partial \theta} = E(\mathbf{S}_c | \mathbf{D}_{\text{obs}}, \theta), \quad \mathcal{I}_o = E(-\partial \mathbf{S}_c / \partial \theta^\top | \mathbf{D}_{\text{obs}}, \theta) - \text{Var}(\mathbf{S}_c | \mathbf{D}_{\text{obs}}, \theta). \quad (41.4.5)$$

Consequently  $\mathcal{I}_m = \text{Var}(\mathbf{S}_c | \mathbf{D}_{\text{obs}}, \theta)$  is nonnegative definite: the observed information

is never larger than the expected complete-data information.

*Proof.* Differentiate the identity  $\ell(\boldsymbol{\theta}) = \ell_c(\boldsymbol{\theta}) - \log g(\mathbf{D}_{\text{mis}} \mid \boldsymbol{\theta})$  from the proof of [Theorem 41.4.1](#) and take the conditional expectation under  $g(\cdot \mid \boldsymbol{\theta})$ . The conditional score of  $g$  has mean zero, being the score of a density in its own parameter, so the first identity follows. Differentiating again,

$$-\frac{\partial^2 \ell}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top} = -\frac{\partial^2 \ell_c}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top} + \frac{\partial^2 \log g}{\partial \boldsymbol{\theta} \partial \boldsymbol{\theta}^\top},$$

and taking the conditional expectation turns the last term into minus the information of  $g$ , which by the usual information identity equals minus the variance of its score. Since the score of  $g$  is  $\mathbf{S}_c - \partial \ell / \partial \boldsymbol{\theta}$  and the second term is a constant given the observed data, that variance is  $\text{Var}(\mathbf{S}_c \mid \mathbf{D}_{\text{obs}})$ . ■

Identity [Eq. 41.4.5](#) is Louis’s formula (Louis, 1982); “observed = complete minus missing” is the missing-information principle of Orchard and Woodbury (1972). Two consequences:

**The rate.** Near a fixed point the EM map is approximately linear with Jacobian  $\mathbf{I} - \mathcal{I}_c^{-1} \mathcal{I}_o = \mathcal{I}_c^{-1} \mathcal{I}_m$ , so the iterates converge linearly at its largest eigenvalue — the largest **fraction of missing information** (Dempster, Laird and Rubin, 1977). In [Example \(An EM fit with a missing covariate\)](#) that eigenvalue is 0.760 and the measured rate is 0.760; the log-likelihood gap, quadratic near the maximum, falls at its square, 0.577 per sweep. In [Theorem 18.6.1\(c\)](#) the same matrix is  $\mathbf{M}_{mm}$ , the submatrix of the hat matrix belonging to the lost plots.

**The standard errors.** The matrix EM manipulates is  $\mathcal{I}_c$ , which is *too large*, so inverting it understates the variance by the fraction of missing information. Three honest routes: [Eq. 41.4.5](#); numerical differentiation of  $\ell$ , as this section’s script does; or the bootstrap, resampling records and rerunning EM ([Definition 23.5.1](#)). In [Example \(An EM fit with a missing covariate\)](#) these give 0.0611 and 0.0605 (bootstrap, 300 resamples), against 0.0512 from the complete-data information — an understatement of 16%.

*Warning.* The variance of an EM estimate is never the inverse of the matrix the M step inverts. If a routine reports standard errors after EM without saying how it got them, assume it has made this mistake.

## 41.4.6 Exercises

### 41.4.6.1 A. Check your understanding

#### Exercise A1.

The EM algorithm for the zero-inflated model of [Section 37.5](#) treats the component label as the missing datum. Identify  $\mathbf{D}_{\text{mis}}$ ,  $Q$  and the M step there, and check that [Exercise \(B4\)](#) is the special case of [Theorem 41.4.1\(a\)](#).

### 41.4.6.2 B. Practice

**Exercise B1.**

In [Proposition 41.4.2](#), suppose one carries out the iteration but *omits* the term  $m\sigma^{2(s)}$  from the M step for the variance. Show that the coefficients are unaffected and that the variance converges to  $\text{SSE}_o/n$ . Relate the discrepancy to [Theorem 18.6.1\(e\)](#).

*Solution.* The coefficient update does not involve  $\sigma^2$  at all, so  $\boldsymbol{\beta}^{(s)}$  follows the same sequence and converges to  $\hat{\boldsymbol{\beta}}_o$ ; at that point the completed residual sum of squares is  $\text{SSE}_o$ . The truncated variance update is  $\sigma^{2(s+1)} = \text{SSE}_o/n$  once the coefficients have settled. That is the complete-data formula applied to  $n$  “observations”,  $m$  of which have been manufactured to fit exactly; reducing the divisor by  $m$  is the degrees-of-freedom correction of [Theorem 18.6.1\(e\)](#).

**Exercise B2.**

Let the complete-data model be an exponential dispersion family ([Definition 34.1.1](#)) with sufficient statistic  $\mathbf{t}(\mathbf{D})$ . Show that the E step reduces to computing  $E\{\mathbf{t}(\mathbf{D}) \mid \mathbf{D}_{\text{obs}}, \boldsymbol{\theta}^{(s)}\}$  and the M step to solving the complete-data likelihood equations with  $\mathbf{t}$  replaced by that expectation. Why does the normal variance in [Proposition 41.4.2](#) still require the extra term  $m\sigma^{2(s)}$ ?

*Solution.* The complete-data log-likelihood is  $\boldsymbol{\theta}^\top \mathbf{t}(\mathbf{D}) - A(\boldsymbol{\theta})$  plus a term free of  $\boldsymbol{\theta}$ , so it is affine in  $\mathbf{t}$  and its conditional expectation replaces  $\mathbf{t}$  by  $E(\mathbf{t} \mid \cdot)$ . Setting the derivative to zero gives  $E_{\boldsymbol{\theta}} \mathbf{t} = E(\mathbf{t} \mid \mathbf{D}_{\text{obs}})$ . The normal case is no exception: the sufficient statistics include  $\sum_i y_i^2$ , and its conditional expectation over a missing row is  $(\mathbf{x}_{(i)}^\top \boldsymbol{\beta}^{(s)})^2 + \sigma^{2(s)}$ , which is where the extra term comes from. The trap is to substitute the conditional *mean of the data* rather than the conditional mean of the *statistic*.

**41.4.6.3 C. Going deeper****Exercise C1.**

Construct a two-component normal mixture with missing labels — the classical EM problem — for which the likelihood is unbounded, and show that EM started near a degenerate configuration drives one variance to zero while the likelihood increases at every step. Reconcile this with [Theorem 41.4.1\(b\)](#).

## 41.5 Multiple imputation

Filling a hole with a plausible value is not wrong; filling it with *one* plausible value and then forgetting that it was a guess is. Multiple imputation keeps the convenience of a complete rectangular data set and restores the uncertainty a single filling destroys, by filling several times and letting the variation between fillings speak.

## 41.5.1 The idea

Rubin's proposal (1978, 1987) has three parts.

1. **Impute.** Draw  $M$  independent completions  $\mathbf{D}^{(1)}, \dots, \mathbf{D}^{(M)}$ , each replacing  $\mathbf{D}_{\text{mis}}$  by a draw from its posterior predictive distribution given  $\mathbf{D}_{\text{obs}}$  under an **imputation model**.
2. **Analyse.** Apply the intended complete-data analysis to each completion, getting estimates  $\hat{Q}_1, \dots, \hat{Q}_M$  and complete-data variance estimates  $U_1, \dots, U_M$ .
3. **Combine.** Pool them by [Theorem 41.5.1](#).

Step 1 is the only novelty. Drawing from the posterior predictive distribution means drawing the imputation model's *parameters* as well as the values; an imputation that conditions on a point estimate treats it as known, which is one of the ways the procedure fails.

### Definition 41.5.1 — Multiple imputation

Let  $\gamma$  parameterize a model for the conditional distribution of  $\mathbf{D}_{\text{mis}}$  given  $\mathbf{D}_{\text{obs}}$ , with prior  $p(\gamma)$ . An imputation is **proper** in Rubin's sense if the  $M$  completions are independent draws from

$$f(\mathbf{D}_{\text{mis}} \mid \mathbf{D}_{\text{obs}}) = \int f(\mathbf{D}_{\text{mis}} \mid \mathbf{D}_{\text{obs}}, \gamma) p(\gamma \mid \mathbf{D}_{\text{obs}}) d\gamma, \quad (41.5.1)$$

the posterior predictive distribution. Three standard constructions:

- (a) **Regression imputation with a parameter draw.** For a continuous variable modelled as normal given the others, draw  $\sigma^{2*} = \text{SSE}/\chi_{k-q}^2$  from its posterior under a flat prior, then  $\gamma^* \sim N(\hat{\gamma}, \sigma^{2*}(\mathbf{Z}_o^\top \mathbf{Z}_o)^{-1})$ , then impute  $\mathbf{z}^\top \gamma^* + \sigma^* \varepsilon$  with  $\varepsilon$  standard normal. This is [Theorem 7.6.2](#) with the noninformative limit [Corollary 7.6.4](#).
- (b) **Predictive mean matching.** Compute the predicted value for the record with the gap, find the  $d$  recorded records with the closest predicted values, and impute one of their *observed* values at random. Every imputed value then actually occurred, so the marginal distribution is respected even when the normal model is wrong, at the cost of a less transparent theory.
- (c) **Chained equations.** With gaps in several variables, cycle through them, imputing each in turn from a regression on the current versions of all the others, for several sweeps, and keep the completion at the end of a sweep.

Construction (c), also called fully conditional specification or MICE (van Buuren, 2018), is what makes multiple imputation usable on real files, where dozens of variables have gaps and no one will write a joint density for all of them. It is a Gibbs-like sweep (compare [Example \(The proficiency study, four ways\)](#)) carrying a risk the Gibbs sampler does not: the specified conditionals need not correspond to *any* joint distribution, and then the sweep has no stationary distribution and the result depends on the order of the variables. The dependence is usually slight, and the flexibility is worth the untidiness, but it should be known.



## 41.5.2 Rubin's rules

### Theorem 41.5.1 — Combining M analyses

Let  $Q$  be the estimand and suppose the complete-data analysis is calibrated in the sense that, given the complete data  $\mathbf{D}$ , it reports  $\hat{Q}(\mathbf{D})$  and  $U(\mathbf{D})$  with

$$E(Q | \mathbf{D}) = \hat{Q}(\mathbf{D}), \quad \text{Var}(Q | \mathbf{D}) = U(\mathbf{D}). \quad (41.5.2)$$

Let  $\mathbf{D}^{(1)}, \dots, \mathbf{D}^{(M)}$  be independent draws from Eq. 41.5.1, generated by the analyst and therefore conditionally independent of  $Q$  given  $\mathbf{D}_{\text{obs}}$ ; put  $\hat{Q}_\ell = \hat{Q}(\mathbf{D}^{(\ell)})$  and  $U_\ell = U(\mathbf{D}^{(\ell)})$ , and define

$$\bar{Q} = \frac{1}{M} \sum_{\ell} \hat{Q}_\ell, \quad \bar{U} = \frac{1}{M} \sum_{\ell} U_\ell, \quad B = \frac{1}{M-1} \sum_{\ell} (\hat{Q}_\ell - \bar{Q})^2. \quad (41.5.3)$$

Then, with all expectations conditional on  $\mathbf{D}_{\text{obs}}$ :

- (a) (*Decomposition.*)  $\text{Var}(Q | \mathbf{D}_{\text{obs}}) = E(U | \mathbf{D}_{\text{obs}}) + \text{Var}(\hat{Q} | \mathbf{D}_{\text{obs}})$ , and  $\bar{U}$  and  $B$  are unbiased for the two terms;
- (b) (*Total variance.*)  $\bar{Q}$  is unbiased for  $E(Q | \mathbf{D}_{\text{obs}})$ , and

$$T = \bar{U} + \left(1 + \frac{1}{M}\right)B \quad (41.5.4)$$

is unbiased for the variance of  $Q - \bar{Q}$  given  $\mathbf{D}_{\text{obs}}$ ;

- (c) (*Degrees of freedom.*) treating  $\bar{U}$  as fixed and  $(M-1)B / \text{Var}(\hat{Q} | \mathbf{D}_{\text{obs}})$  as  $\chi_{M-1}^2$ , the variance of  $T$  matches that of a scaled  $\chi_\nu^2$  variable with

$$\nu = (M-1) \left\{ \frac{T}{(1 + M^{-1})B} \right\}^2 = \frac{M-1}{\gamma^2}, \quad \gamma = \frac{(1 + M^{-1})B}{T}, \quad (41.5.5)$$

and the interval  $\bar{Q} \pm t_{\nu, 1-\alpha/2} \sqrt{T}$  is the standard multiple-imputation interval;  $\gamma$  is the estimated **fraction of missing information**;

- (d) (*Efficiency.*) write  $T_\infty = E(U) + \text{Var}(\hat{Q})$  for the total variance at  $M = \infty$  and  $\gamma_\infty = \text{Var}(\hat{Q})/T_\infty$  for the limiting fraction of missing information. Then the target of  $T$  exceeds  $T_\infty$  by the factor  $1 + \gamma_\infty/M$ , so using  $M$  imputations rather than infinitely many inflates the standard error by  $(1 + \gamma_\infty/M)^{1/2}$ .

*Proof.* Throughout, condition on  $\mathbf{D}_{\text{obs}}$  and write  $E$  and  $\text{Var}$  for the conditional operators.

- (a) The conditional variance decomposition applied to  $Q$  with the intermediate conditioning variable  $\mathbf{D}$  gives  $\text{Var}(Q) = E\{\text{Var}(Q | \mathbf{D})\} + \text{Var}\{E(Q | \mathbf{D})\}$ , which is  $E(U) + \text{Var}(\hat{Q})$  by Eq. 41.5.2. The completions are draws of  $\mathbf{D}$  from its conditional law, so  $E\bar{U} = E(U)$ , and the sample variance of independent draws is unbiased for their variance, so  $E B = \text{Var}(\hat{Q})$ .
- (b)  $E\bar{Q} = E\hat{Q} = E\{E(Q | \mathbf{D})\} = EQ$ , so  $\bar{Q}$  is unbiased for the posterior mean. Because the completions are conditionally independent of  $Q$  given  $\mathbf{D}_{\text{obs}}$ , so is  $\bar{Q}$ ,

and therefore

$$\begin{aligned}\text{Var}(Q - \bar{Q}) &= \text{Var}(Q) + \text{Var}(\bar{Q}) = \{E(U) + \text{Var}(\hat{Q})\} + \frac{\text{Var}(\hat{Q})}{M} \\ &= E(U) + \left(1 + \frac{1}{M}\right) \text{Var}(\hat{Q}),\end{aligned}$$

using (a) for the first bracket and independence of the  $M$  draws for the second term. By (a) again,  $\bar{U}$  and  $(1 + M^{-1})B$  are unbiased for the two pieces, so  $T$  is unbiased for the whole.

- (c) Under the stated approximation  $T = \bar{U} + (1 + M^{-1})B$  has variance  $(1 + M^{-1})^2 \text{Var}(B)$ , which is  $2(1 + M^{-1})^2 \{\text{Var}(\hat{Q})\}^2 / (M - 1)$ . A scaled  $\chi^2_\nu$  variable with mean  $T$  has variance  $2T^2/\nu$ . Equating and replacing  $\text{Var}(\hat{Q})$  by its estimate  $B$ ,

$$\frac{2T^2}{\nu} = \frac{2(1 + M^{-1})^2 B^2}{M - 1} \implies \nu = (M - 1) \frac{T^2}{(1 + M^{-1})^2 B^2},$$

which is [Eq. 41.5.5](#).

- (d) By (b) the target of  $T$  is

$$E(U) + (1 + M^{-1}) \text{Var}(\hat{Q}) = T_\infty + \frac{\text{Var}(\hat{Q})}{M} = T_\infty \left(1 + \frac{\gamma_\infty}{M}\right).$$

Taking square roots gives the inflation factor.

■

*Remark* (What the proof assumes). The calibration [Eq. 41.5.2](#) is a Bayesian statement: the complete-data analysis behaves as a posterior mean and variance would. Rubin's rules are therefore exactly right when imputation and analysis come from the *same* joint model, and approximately right otherwise. The degrees of freedom in part (c) can exceed the complete-data degrees of freedom, absurdly, when  $M$  is large and the sample small; Barnard and Rubin (1999) give the corrected formula.

*Idea.* The between-imputation variance  $B$  is the price of not knowing the missing values, and  $\bar{U}$  is the price of having a finite sample. Adding them is the whole of multiple imputation. The factor  $1 + 1/M$  is the price of using finitely many imputations to measure  $B$ .

### 41.5.3 How many imputations

Part (d) is why the classical advice was “three to five imputations”. With  $\gamma_\infty = 0.3$  and  $M = 5$  the standard error is inflated by 1.0296, three per cent; with  $\gamma_\infty = 0.5$  the inflation is 1.0488 at  $M = 5$  and 1.0124 at  $M = 20$ .

The advice has shifted upward since, for two reasons the theorem does not address. The degrees of freedom [Eq. 41.5.5](#) grow like  $M - 1$ , so a small  $M$  gives a fat-tailed reference distribution and wide intervals even when the point estimate is fine; and  $\bar{Q}$  is itself a Monte Carlo average, so a second analyst with a different seed gets a different number — uncomfortable when that number is a  $p$ -value near a threshold. Imputations are cheap, and the curve in [Figure 41.5.1\(b\)](#) is flat past  $M = 20$  for every plausible  $\gamma$ .

## 41.5.4 Coverage in practice

### Example. (Five analyses of one problem)

Take  $n = 200$  records with two correlated covariates and  $y = 1 + 0.5x_1 + 0.6x_2 + \varepsilon$ , and lose  $x_2$  with a probability depending on  $x_1$  and  $y$  — both recorded, so MAR — leaving about 106 complete records. Over 2000 replicates, for the coefficient of  $x_2$ :

**Table 41.5.1.** Five analyses of the same MAR problem. The estimand is  $\beta_2 = 0.6$ ; “mean se” is the average reported standard error and “sd” the actual spread of the estimates over 2000 replicates.

| analysis                      | bias   | mean se | sd    | coverage |
|-------------------------------|--------|---------|-------|----------|
| complete cases                | -0.151 | 0.101   | 0.102 | 67.2%    |
| mean imputation               | -0.319 | 0.125   | 0.108 | 25.6%    |
| single imputation             | -0.010 | 0.082   | 0.138 | 75.9%    |
| multiple imputation, $M = 5$  | -0.013 | 0.118   | 0.119 | 95.0%    |
| multiple imputation, $M = 20$ | -0.013 | 0.115   | 0.114 | 94.5%    |

Each row fails in its own way. The complete cases are biased, the missingness depending on the response. Mean imputation is far worse, and its reported standard error is *larger* than its actual spread. A single proper imputation removes the bias but reports a standard error three fifths of the truth, and covers 75.9% of the time. Multiple imputation with five draws gets both right, with an average estimated fraction of missing information 0.463 and median degrees of freedom 18.3.

```
import numpy as np

n = 200
rho = 0.5
beta_true = np.array([1.0, 0.5, 0.6])           # intercept, x1, x2

def make_data(rng):
    x1 = rng.normal(size=n)
    x2 = rho * x1 + np.sqrt(1 - rho**2) * rng.normal(size=n)
    y = beta_true[0] + beta_true[1] * x1 + beta_true[2] * x2 + rng.normal(size=n)
    eta = -0.3 + 0.8 * x1 + 2.0 * (y - y.mean())    # depends on observed values
    # only: MAR
    seen = rng.uniform(size=n) > 1 / (1 + np.exp(-eta))
    return x1, x2, y, seen

def fit(X, y):
    """Least squares with the coefficient of x2 and its squared standard error."""
    b, *_ = np.linalg.lstsq(X, y, rcond=None)
```

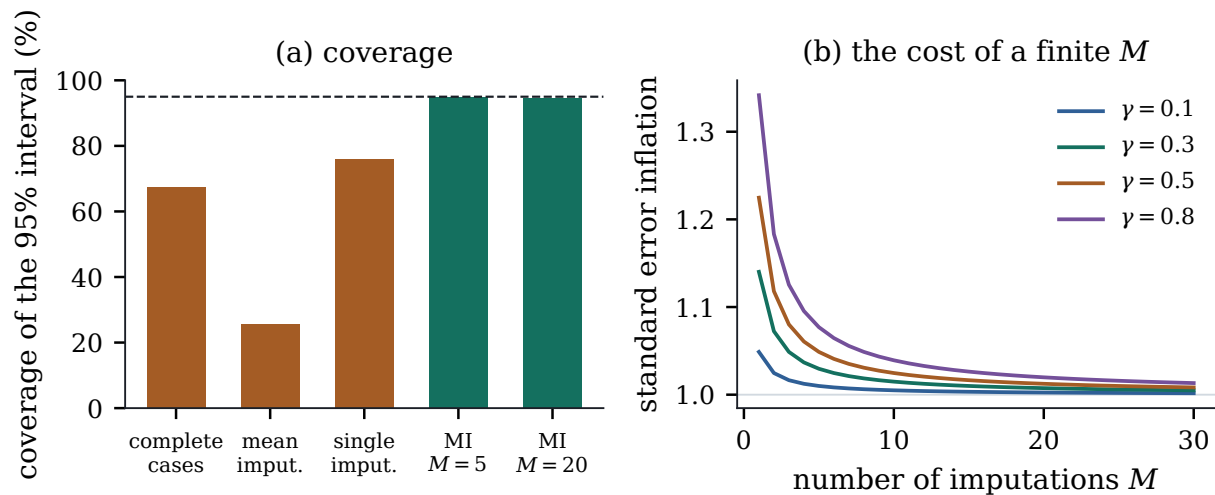


Figure 41.5.1: (a) Coverage of nominal 95% intervals for  $\beta_2$  under five analyses of the same missing-at-random problem. (b) The factor by which a finite number of imputations inflates the standard error, as a function of  $M$ , for four values of the fraction of missing information.

```

dof = len(y) - X.shape[1]
s2 = np.sum((y - X @ b) ** 2) / dof
v = s2 * np.linalg.inv(X.T @ X)[2, 2]
return b[2], v, dof

def impute_once(x1, x2, y, seen, rng):
    """One draw from the posterior predictive distribution of the missing x2."""
    Z = np.column_stack([np.ones(n), x1, y]) # imputation model: x2 on x1
    and y
    Zo, k = Z[seen], int(seen.sum())
    g, *_ = np.linalg.lstsq(Zo, x2[seen], rcond=None)
    sse = float(np.sum((x2[seen] - Zo @ g) ** 2))
    s2_star = sse / rng.chisquare(k - Z.shape[1]) # draw the variance ...
    cov = s2_star * np.linalg.inv(Zo.T @ Zo)
    g_star = rng.multivariate_normal(g, cov) # ... then the coefficients
    draw = Z @ g_star + np.sqrt(s2_star) * rng.normal(size=n)
    return np.where(seen, x2, draw) # observed values are kept

def rubin(estimates, variances):
    """Combine M analyses: the average, the total variance and the degrees of
    freedom."""
    M = len(estimates)
    q_bar, u_bar = np.mean(estimates), np.mean(variances)
    b = np.var(estimates, ddof=1)
    total = u_bar + (1 + 1 / M) * b
    gamma = (1 + 1 / M) * b / total # fraction of missing
    information
    nu = (M - 1) / gamma**2
    return q_bar, total, nu

demo_rng = np.random.default_rng(4105)
x1d, x2d, yd, seend = make_data(demo_rng) # one data set, x2 missing at
random
qs, us = [], []

```

```

for _ in range(5):                                     # M = 5 completions of it
    xi = impute_once(x1d, x2d, yd, seend, demo_rng)
    qi, ui, _ = fit(np.column_stack([np.ones(n), x1d, xi]), yd)
    qs.append(qi)
    us.append(ui)
q_bar, total, nu = rubin(qs, us)
print(f"M = 5: estimate {q_bar:.3f}, total variance {total:.4f}, "
      f"degrees of freedom {nu:.1f}")

```

Chained equations handle gaps in more than one variable: ten sweeps of the two conditional regressions below recover the coefficient of  $x_2$  when both covariates are incomplete.

```

rho = 0.5
beta_true = np.array([1.0, 0.5, 0.6])                 # intercept, x1, x2

def make_data(rng):
    x1 = rng.normal(size=n)
    x2 = rho * x1 + np.sqrt(1 - rho**2) * rng.normal(size=n)
    y = beta_true[0] + beta_true[1] * x1 + beta_true[2] * x2 + rng.normal(size=n)
    eta = -0.3 + 0.8 * x1 + 2.0 * (y - y.mean())        # depends on observed values
    only: MAR
    seen = rng.uniform(size=n) > 1 / (1 + np.exp(-eta))
    return x1, x2, y, seen

rng2 = np.random.default_rng(4107)
x1, x2, y, seen2 = make_data(rng2)
seen1 = rng2.uniform(size=n) > 0.25                  # x1 now has gaps of its own
x1_work = np.where(seen1, x1, x1[seen1].mean())       # a crude start
x2_work = np.where(seen2, x2, x2[seen2].mean())
for sweep in range(10):                              # one sweep = one pass over
    the variables
    Z = np.column_stack([np.ones(n), x2_work, y])     # x1 given the others
    g, *_ = np.linalg.lstsq(Z[seen1], x1[seen1], rcond=None)
    s = np.std(x1[seen1] - Z[seen1] @ g, ddof=3)
    x1_work = np.where(seen1, x1, Z @ g + s * rng2.normal(size=n))
    Z = np.column_stack([np.ones(n), x1_work, y])     # x2 given the others
    g, *_ = np.linalg.lstsq(Z[seen2], x2[seen2], rcond=None)
    s = np.std(x2[seen2] - Z[seen2] @ g, ddof=3)
    x2_work = np.where(seen2, x2, Z @ g + s * rng2.normal(size=n))
b_chained, *_ = np.linalg.lstsq(np.column_stack([np.ones(n), x1_work, x2_work]), y,
rcond=None)
print("after chained equations:", np.round(b_chained, 3))

```

### 41.5.5 Congeniality

Multiple imputation splits the work between two models: the imputer's and the analyst's. Nothing forces them to agree, and Meng (1994) showed that the consequences depend on which way they disagree.

**The imputation model omits something the analysis uses.** If the analyst fits an interaction and the imputer imputed one of its covariates from a model without it, the imputed values carry no interaction and the analyst's interaction is biased towards zero. This is the serious

failure, and easy to commit: the imputer usually works first, for a whole file, without knowing which analyses will be run.

**The imputation model includes something the analysis does not.** Then the imputations are richer than the analysis needs; estimates are usually fine and Rubin's  $T$  tends to be *too large*, so the procedure is conservative rather than wrong.

Meng calls a pair of models **congenial** when both are implied by a single Bayesian model, and shows that Rubin's variance estimator is consistent under congeniality and can be biased either way without it. Two rules follow: put into the imputation model every variable the analysis will use, including the response and every transformation and interaction to be fitted; and when imputer and analyst are different people, err towards a rich imputation model.

*Warning.* Omitting the response from the imputation model is the single commonest mistake. It looks cautious — “I do not want to build the answer into the data” — and it is exactly backwards: it imputes covariates as though they were unrelated to the response, which biases every coefficient towards zero. The response belongs in the imputation model.

## 41.5.6 Exercises

### 41.5.6.1 A. Check your understanding

#### Exercise A1.

Set  $M = 1$  in [Theorem 41.5.1](#) and explain why the formulas break down. Which of the three quantities in [Eq. 41.5.3](#) cannot be computed, and what does that say about single imputation?

*Solution.* With  $M = 1$  the between-imputation sum of squares has no degrees of freedom:  $B$  is  $0/0$ . So the part of the uncertainty that comes from not knowing the missing values cannot be estimated at all from one imputation, which is exactly why a single imputation reports  $\bar{U}$  alone and understates the variance.

#### Exercise A2.

From [Eq. 41.5.5](#), compute  $\nu$  for  $M = 5$  and  $\gamma \in \{0.1, 0.3, 0.5\}$ . At which value of  $\gamma$  does the multiple-imputation reference distribution become so heavy-tailed that the interval is more than 20% wider than a normal-based one?

*Solution.* Equation [Eq. 41.5.5](#) gives  $\nu = (M - 1)/\gamma^2$ , so at  $M = 5$  the three values are  $4/0.01 = 400$ ,  $4/0.09 = 44.4$  and  $4/0.25 = 16$ . The interval is wider than the normal-based one by the ratio of the  $t$  quantile to 1.96, and a table shows that ratio passing 1.2 at about seven degrees of freedom, that is at  $\gamma = \sqrt{4/7.2} \approx 0.75$ . Only a problem in which three quarters of the information is missing costs that much at  $M = 5$  — and raising  $M$  is the cheap repair.

### 41.5.6.2 B. Practice

**Exercise B1.**

Construct a simulation in which the analysis fits  $y$  on  $x_1, x_2$  and their interaction, while  $x_2$  is imputed from a model without the interaction. Measure the bias in the interaction coefficient and its coverage. Then repeat with the interaction included in the imputation model.

**41.5.6.3 C. Going deeper****Exercise C1.**

Show that when the imputation model and the analysis model are the same Bayesian model,  $\bar{Q}$  and  $T$  converge as  $M \rightarrow \infty$  to the exact posterior mean and variance of  $Q$  given  $\mathbf{D}_{\text{obs}}$ . Deduce that multiple imputation is, in the congenial case, a Monte Carlo scheme for a posterior that could also be sampled directly — and say why one might still prefer the imputation route.

## 41.6 Sensitivity analysis

Everything in the last three sections rested on missingness at random, and [Proposition 41.1.1](#) says the data cannot check it. That is a reason neither to despair nor to pretend, but to report how much the conclusion depends on the assumption — which is what a sensitivity analysis does.

### 41.6.1 Two factorizations

The selection factorization [Eq. 41.1.1](#) splits the joint law into a model for the data and a model for who responds. The other order gives the **pattern-mixture** factorization

$$f(\mathbf{D}, \mathbf{R} \mid \boldsymbol{\theta}_1, \boldsymbol{\theta}_2) = f(\mathbf{D} \mid \mathbf{R}, \boldsymbol{\theta}_1) f(\mathbf{R} \mid \boldsymbol{\theta}_2), \quad (41.6.1)$$

a mixture, over patterns, of a different distribution of the data within each pattern. The two describe the same object and differ only in what is easy to say.

#### Proposition 41.6.1 — Selection and pattern-mixture models

Consider a single variable  $v$  recorded when  $R = 1$ , alongside always recorded variables  $\mathbf{u}$ .

- (a) Every selection model  $f(v \mid \mathbf{u}) \Pr(R = 1 \mid v, \mathbf{u})$  corresponds to the pattern-mixture model with

$$f(v \mid \mathbf{u}, R = r) \propto f(v \mid \mathbf{u}) \Pr(R = r \mid v, \mathbf{u}),$$

$$\Pr(R = r \mid \mathbf{u}) = \int f(v \mid \mathbf{u}) \Pr(R = r \mid v, \mathbf{u}) dv,$$

and conversely, so neither formulation is more general.

- (b) The observed data identify  $f(v \mid \mathbf{u}, R = 1)$  and  $\Pr(R = 1 \mid \mathbf{u})$ , and nothing about  $f(v \mid \mathbf{u}, R = 0)$ . The marginal

$$f(v \mid \mathbf{u}) = \Pr(R = 1 \mid \mathbf{u}) f(v \mid \mathbf{u}, R = 1) + \Pr(R = 0 \mid \mathbf{u}) f(v \mid \mathbf{u}, R = 0) \quad (41.6.2)$$

is therefore identified only after the analyst supplies the last factor. Missing at random is the supply  $f(v \mid \mathbf{u}, R = 0) = f(v \mid \mathbf{u}, R = 1)$ .

- (c) Let the mechanism be  $\Pr(R = 1 \mid v, \mathbf{u}) = h\{a(\mathbf{u}) + cv\}$ , with  $h$  continuous, strictly increasing and valued in  $(0, 1)$  and  $a(\cdot)$  an unrestricted function. Then  $c$  is not identified: for every  $c$  at which the tilt [Eq. 41.6.3](#) below is integrable and its integral covers the required value, there are an  $a(\cdot)$  and an  $f(v \mid \mathbf{u}, R = 0)$  reproducing the observed-data distribution exactly. Any standard error reported for  $c$  comes from restricting  $a(\cdot)$  — to  $a + \mathbf{u}^\top \mathbf{b}$ , say — and from the assumed forms of  $h$  and  $f(v \mid \mathbf{u})$ , not from the data.

*Proof.*

- (a) Both expressions are the joint density  $f(v, R = r \mid \mathbf{u})$  written in the two orders, by the definition of conditional probability; the displayed formulas are Bayes's rule in each direction, and the correspondence is one-to-one wherever the denominators are positive.
- (b) The observed data consist of  $(\mathbf{u}, R)$  for all units and  $v$  for those with  $R = 1$ . Its distribution is determined by the law of  $\mathbf{u}$ ,  $\Pr(R = 1 \mid \mathbf{u})$  and  $f(v \mid \mathbf{u}, R = 1)$ , and by nothing else, exactly as in [Proposition 41.1.1](#). Equation [Eq. 41.6.2](#) is the law of total probability. Setting the unidentified factor equal to the identified one is the MAR choice, since then  $\Pr(R = 1 \mid v, \mathbf{u})$  computed by Bayes's rule is free of  $v$ .
- (c) Fix  $c$ , write  $\pi(\mathbf{u}) = \Pr(R = 1 \mid \mathbf{u})$  and  $f_1(v \mid \mathbf{u}) = f(v \mid \mathbf{u}, R = 1)$  for the two identified pieces, and abbreviate  $h_c(v) = h\{a(\mathbf{u}) + cv\}$ . Bayes's rule applied to (a) forces the unrecorded conditional to be the recorded one *tilted* by the odds of not responding:

$$f(v \mid \mathbf{u}, R = 0) = \frac{f_1(v \mid \mathbf{u})}{K(\mathbf{u})} \cdot \frac{1 - h_c(v)}{h_c(v)}, \quad K(\mathbf{u}) = \int f_1 \frac{1 - h_c}{h_c} dv. \quad (41.6.3)$$

Take that as the definition and assemble  $f(v \mid \mathbf{u})$  from [Eq. 41.6.2](#). Then  $f(v \mid \mathbf{u})h_c(v) = \pi f_1 h_c + \{(1 - \pi)/K\} f_1 (1 - h_c)$ , which collapses to  $\pi(\mathbf{u}) f_1(v \mid \mathbf{u})$  — free of  $h_c$  — exactly when  $K(\mathbf{u}) = \{1 - \pi(\mathbf{u})\}/\pi(\mathbf{u})$ . Integrating over  $v$  then returns  $\Pr(R = 1 \mid \mathbf{u}) = \pi(\mathbf{u})$ , and dividing returns  $f(v \mid \mathbf{u}, R = 1) = f_1(v \mid \mathbf{u})$ : both identified pieces are reproduced, for this  $c$ .

It remains to solve  $K(\mathbf{u}) = \{1 - \pi(\mathbf{u})\}/\pi(\mathbf{u})$  for the intercept  $a(\mathbf{u})$ , one scalar equation per  $\mathbf{u}$ . Since  $(1 - h_c)/h_c$  decreases pointwise in  $a(\mathbf{u})$ ,  $K$  is continuous and strictly decreasing, with  $K \rightarrow 0$  as  $a(\mathbf{u}) \rightarrow \infty$ ; a solution exists whenever the target lies in its range. Both provisos bite. For  $h = \Phi$  and  $f_1$  normal with standard deviation  $s$ , the integrand's logarithm behaves like  $(c^2 - s^{-2})v^2/2$  as  $v \rightarrow -\infty$ , so  $K$  is finite exactly when  $|c|s < 1$ , and then runs over all of  $(0, \infty)$ ; for larger  $|c|$  no such pattern-mixture model exists and the construction is unavailable. Within the range, the observed-data likelihood profiled over the remaining parameters is constant in  $c$ , so  $c$  is not identified. Restricting  $a(\mathbf{u})$  to  $a + \mathbf{u}^\top \mathbf{b}$  leaves one equation per  $\mathbf{u}$  and only  $1 + \dim \mathbf{u}$  unknowns; the profile is then no longer flat, and the curvature — the reported standard error —



comes from that restriction, not from the data.

■

Part (c) is worth dwelling on. Fit a Heckman-style selection model — a normal outcome equation and a probit selection equation, as in Heckman (1979) — and the software returns an estimate of the selection coefficient with a standard error; part (c) says that number is an artefact of the assumed functional forms. With an **exclusion restriction**, a variable entering the selection equation and not the outcome equation, the parameter becomes identified by more than functional form; without one it does not. Reporting such a fit as though the data had spoken is the central dishonesty of MNAR modelling, and the reason this section prefers to display the untestable part rather than estimate it.

*Idea.* Under MNAR nothing is estimated that was not assumed. The useful question is not “what is the answer?” but “how far must the assumption move before the answer changes?”

## 41.6.2 Delta-adjustment and tipping points

The pattern-mixture form suggests an honest device. Impute under MAR, then shift the imputed values by a fixed amount  $\delta$  before analysing them: that is, assume

$$E(v \mid \mathbf{u}, R = 0) = E(v \mid \mathbf{u}, R = 1) - \delta. \quad (41.6.4)$$

Here  $\delta$  is a *sensitivity parameter*: not estimated, but set. Zero is MAR, and a positive  $\delta$  says the unrecorded values are worse than their recorded counterparts with the same covariates. Because [Eq. 41.6.4](#) constrains only the unidentified factor of [Eq. 41.6.2](#), every  $\delta$  fits the observed records equally well — in the example below all give the observed-data log-likelihood  $-373.62$  — so the choice is a statement about the world, not about the data. Repeating the analysis over a grid of  $\delta$  and reporting the curve is a **tipping-point analysis**: the value  $\delta^*$  at which the conclusion reverses is the smallest departure from MAR that would overturn it.

### Example. (How far must the assumption move?)

A two-arm study of 400 units records a baseline measurement for everyone and an outcome for those who stay. Dropout is likelier in the treated arm and at high baseline values, and 147 outcomes are lost — 83 treated and 64 control. Multiple imputation under MAR gives a treatment effect of 0.475 with 95% interval (0.199, 0.750).

Now suppose the treated units who dropped out would have done worse than the MAR imputation says, by  $\delta$ . The estimate falls steadily and the lower limit reaches zero at  $\delta^* = 0.482$ , which is 0.452 of a residual standard deviation. The finding therefore survives the assumption that treated dropouts were up to about half a standard deviation worse than comparable stayers, and not more. Whether that is reassuring is a question for whoever knows why people dropped out.

```
import numpy as np
from scipy import stats
```

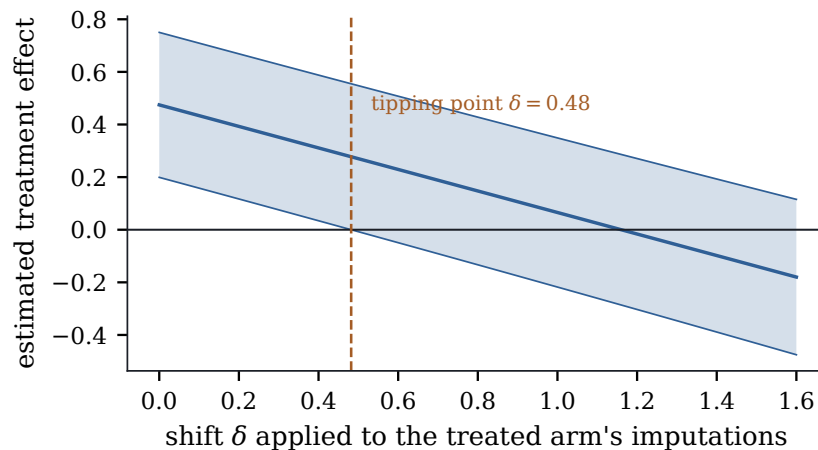


Figure 41.6.1: Estimated treatment effect and its 95% interval as the imputed outcomes of the treated arm are shifted down by  $\delta$ . Every  $\delta$  fits the recorded outcomes equally well. The tipping point is where the lower limit reaches zero.

```
rng = np.random.default_rng(4108)
n, tau_true = 400, 0.6
arm = np.repeat([0, 1], n // 2)
base = rng.normal(size=n)
y = 0.5 + tau_true * arm + 0.6 * base + rng.normal(size=n)
eta = -1.0 + 0.7 * arm + 0.8 * base
seen = rng.uniform(size=n) > 1 / (1 + np.exp(-eta))
print(f"{n - seen.sum()} of {n} outcomes lost:"
      f" {int((~seen & (arm == 1)).sum())} treated, {int((~seen & (arm == 0)).sum())} control")
```

```
Z = np.column_stack([np.ones(n), arm, base])

def mi_estimate(delta, M, rng):
    """Multiple imputation under MAR, then shift the treated arm's imputations by -delta."""
    Zo, k = Z[seen], int(seen.sum())
    g, *_ = np.linalg.lstsq(Zo, y[seen], rcond=None)
    sse = float(np.sum((y[seen] - Zo @ g) ** 2))
    qs, us = [], []
    for _ in range(M):
        s2 = sse / rng.chisquare(k - Z.shape[1])
        g_star = rng.multivariate_normal(g, s2 * np.linalg.inv(Zo.T @ Zo))
        draw = Z @ g_star + np.sqrt(s2) * rng.normal(size=n) - delta * arm
        y_full = np.where(seen, y, draw)
        b, *_ = np.linalg.lstsq(Z, y_full, rcond=None)
        resid = y_full - Z @ b
        v = (resid @ resid / (n - 3)) * np.linalg.inv(Z.T @ Z)[1, 1]
        qs.append(b[1])
        us.append(v)
    M_ = len(qs)
    q_bar, u_bar, b_var = np.mean(qs), np.mean(us), np.var(qs, ddof=1)
    total = u_bar + (1 + 1 / M_) * b_var
    gamma = (1 + 1 / M_) * b_var / total
    return q_bar, total, (M_ - 1) / gamma**2
```

```

deltas = np.linspace(0.0, 1.6, 33)
est, lo, hi = [], [], []
for d in deltas:
    q, t, nu = mi_estimate(d, 50, np.random.default_rng(4109))
    half = stats.t.ppf(0.975, nu) * np.sqrt(t)
    est.append(q)
    lo.append(q - half)
    hi.append(q + half)
est, lo, hi = map(np.array, (est, lo, hi))
tipping = float(np.interp(0.0, -lo, deltas))          # where the lower limit
reaches zero
print(f"delta = 0: effect {est[0]:.3f}, 95% interval ({lo[0]:.3f}, {hi[0]:.3f})")
print(f"tipping point: delta = {tipping:.3f}")

```

Variations abound. The shift may be applied to one arm or both, on the response scale or a link scale, and may depend on covariates or grow with time since dropout; in trials it is often replaced by **jump to reference**, in which a treated dropout is imputed as though from the control arm, or **copy increments in reference**, in which only the subsequent changes are borrowed. Each is a particular choice of the unidentified factor in [Eq. 41.6.2](#).

### 41.6.3 Honest reporting

The technical content of this section is thin; the discipline is not. A report should say how much is missing and where, by variable and by pattern; why, as far as the reasons recorded at collection time reveal, those being the only evidence about the mechanism that will ever exist; what was assumed, naming the variables MAR is conditional on, since “missing values were imputed” states no assumption at all; what the method was, including whether the imputation model contained the response; and how far the conclusion travels, as a sensitivity curve or at least a tipping point. That last item is where regulators have converged: the National Research Council (2010) report recommends designing to prevent missingness and pre-specifying both the primary analysis and the sensitivity analyses, rather than choosing them after seeing the result.

*Warning.* A sensitivity analysis chosen after seeing the primary result is not a sensitivity analysis; it is a selection procedure, with the properties [Chapter 29](#) describes ([Proposition 29.5.1](#)). Fix the grid of  $\delta$  and the summary to be reported before the analysis is run.

### 41.6.4 Exercises

#### 41.6.4.1 A. Check your understanding

##### Exercise A1.

Write the bivariate normal of [Exercise \(B3\)](#) in both factorizations, selection and pattern-mixture, and identify which parameters are estimable from the observed data in each. Why is the pattern-mixture form better suited to a sensitivity analysis?

**Exercise A2.**

In [Example \(How far must the assumption move?\)](#) the tipping point is 0.482 on the outcome scale. Express it as a multiple of the estimated treatment effect and as a multiple of the residual standard deviation, and say which you would put in a report and why.

**41.6.4.2 B. Practice****Exercise B1.**

For the model of [Example \(How far must the assumption move?\)](#), show analytically that the treatment effect estimated after a  $\delta$ -shift of the treated arm's imputations is  $\hat{\tau}(\delta) = \hat{\tau}(0) - \delta p_1$ , where  $p_1$  is the fraction of treated units whose outcome is missing, provided the analysis model contains the arm indicator. Check the slope against the figure.

*Solution.* With the arm indicator in the model and a completed data set, the estimated effect is a difference of arm means adjusted for the baseline. Shifting the imputed outcomes of the treated arm down by  $\delta$  lowers that arm's adjusted mean by  $\delta$  times the fraction of its units that were imputed, and leaves the control arm alone; the baseline adjustment is unaffected because the shift is constant within the arm. Hence  $\hat{\tau}(\delta) = \hat{\tau}(0) - \delta p_1$ . With 83 of 200 treated units missing,  $p_1 \approx 0.42$ , and the slope of the curve in [Figure 41.6.1](#) matches.

**41.6.4.3 C. Going deeper****Exercise C1.**

For a monotone pattern with three time points, state the **complete-case missing value** restriction — the unidentified conditional for a unit that dropped out after time  $j$  equals the corresponding conditional among units seen at all three times — and the **neighbouring-case** alternative. Show that both are particular choices of the unidentified factor in [Eq. 41.6.2](#), and that the MAR choice lies between them in a sense you should make precise.

**Exercise C2.**

Suppose the response is bounded in  $[0, 1]$ . Compute the **worst-case bounds** for its mean: impute every missing value by 0, then by 1. Show that the width of the resulting interval is the fraction missing, that it requires no assumption at all, and that it is usually too wide to be useful. Discuss where such bounds nevertheless earn their place.

*Solution.* If a fraction  $\lambda$  of values is missing, the mean of the completed data ranges over an interval of width  $\lambda$  as the imputations range over  $[0, 1]$ , and every value in that interval is attained by some choice, so the bounds are sharp and assumption-free — they are the Manski-style bounds for a bounded outcome. They are useless when  $\lambda$  is large compared with the effect of interest, which is the usual case. They earn their place as a first line in a report: if the worst-case bound already excludes the null, no assumption

about the mechanism is needed at all, and that is worth knowing before any modelling begins.

## 41.7 Summary and Notes

### 41.7.1 Summary

*Idea.*

1. The pattern of missingness is data. Writing the joint law as a model for the complete array times a mechanism [Eq. 41.1.1](#) separates what the analyst wants from how the holes arose, and sorts mechanisms into MCAR, MAR and MNAR ([Definition 41.1.1](#)).
2. Nothing in the observed data distinguishes MAR from MNAR: every choice of the conditional distribution of the unrecorded values given the recorded ones gives the same observed-data likelihood ([Proposition 41.1.1](#)). MCAR against MAR is testable; MAR against MNAR is not.
3. A complete-case analysis estimates functionals of the law given completeness. It is valid under MCAR, and — the useful case — valid for a correctly specified regression whenever the missingness depends on the covariates but not on the response ([Proposition 41.2.1](#)). It is not generally valid under MAR.
4. Mean imputation deflates the variance by the fraction recorded ([Proposition 41.2.2](#)); regression imputation without noise inflates correlations; any single imputation understates every standard error; available-case correlation matrices need not be nonnegative definite.
5. Under MAR with distinct parameters the mechanism is ignorable: likelihood, likelihood ratio tests and Bayesian posteriors may be computed from the observed-data likelihood alone ([Theorem 41.3.1](#)). The licence costs a model for any variable that can go missing, including covariates. For a monotone pattern that likelihood factors into conditional models on nested subsamples ([Exercise \(B1\)](#) of [Section 41.1](#) and [Exercise \(B1\)](#) of [Section 41.3](#)).
6. EM maximizes the observed-data likelihood by alternating a conditional expectation of the complete-data log-likelihood with a complete-data fit ([Definition 41.4.1](#)). Every sweep increases the observed-data likelihood ([Theorem 41.4.1](#)), and fixed points are stationary points, not necessarily maxima.
7. For missing responses in a linear model, EM returns the observed-data least squares estimate and  $\text{SSE}_o/(n - m)$  ([Proposition 41.4.2](#)), deriving the degrees-of-freedom correction that [Theorem 18.6.1\(e\)](#) asserted. For a missing covariate the E step must condition on the response ([Proposition 41.4.3](#)).
8. Observed information equals expected complete-data information minus the variance of the complete-data score ([Proposition 41.4.4](#)). That difference sets both EM's rate of convergence and the amount by which the complete-data information understates the variance; standard errors come from Louis's formula, numerical differentiation or the bootstrap, never from the matrix the M step inverts.
9. Multiple imputation draws  $M$  completions from the posterior predictive distribution ([Definition 41.5.1](#)), analyses each, and combines by  $\bar{Q}$  and  $T = \bar{U} + (1 + M^{-1})B$

on  $(M - 1)/\gamma^2$  degrees of freedom ([Theorem 41.5.1](#)). A finite  $M$  inflates the standard error by  $(1 + \gamma_\infty/M)^{1/2}$ . The imputation model must contain everything the analysis will use, above all the response.

10. Under MNAR the selection and pattern-mixture factorizations describe the same unidentified object ([Proposition 41.6.1](#)). A delta-adjustment names the unidentified part, every value of it fits the observed data equally well, and the tipping point reports how far the assumption must move before the conclusion does.

## 41.7.2 Notes and sources

**The framework.** The classification into MCAR, MAR and MNAR, the selection factorization and the ignorability theorem are due to Rubin (1976); the book-length development is Little and Rubin (2019), whose organization this chapter follows, the proofs, examples and numbers here being the book's own. Schafer (1997) treats the multivariate normal and categorical cases, including the monotone factorizations of [Section 41.3](#), whose closed form is Anderson's (1957). The realized-value and everywhere versions of MAR are separated by Seaman, Galati, Jackson and Carlin (2013); the test of MCAR is Little (1988), and the general form of [Proposition 41.1.1](#) is Molenberghs, Beunckens, Sotito and Kenward (2008).

**Complete cases.** Part (b) of [Proposition 41.2.1](#) has been rediscovered many times; the clearest statements are Little (1992) and White and Carlin (2010), who also note that complete-case analysis can beat multiple imputation for efficiency when the mechanism depends only on co-variables. Inverse-probability weighting descends from Horvitz and Thompson (1952), the doubly robust augmentation from Robins, Rotnitzky and Zhao (1994).

**EM.** The algorithm was named and given its general form by Dempster, Laird and Rubin (1977), who also identified the rate of convergence as the largest fraction of missing information; special cases are older, the iteration of [Theorem 18.6.1\(c\)](#) being the missing-plot scheme of Healy and Westmacott (1956), built on Yates's (1933) formula. The missing-information principle is Orchard and Woodbury (1972), and [Eq. 41.4.5](#) is Louis's (1982). Wu (1983) corrected the convergence claims of the original paper; Meng and Rubin (1991) extract the observed information from the EM iterations themselves, and McLachlan and Krishnan (2008) is the book-length treatment.

**Multiple imputation.** The idea is Rubin's (1978) and the definitive statement is Rubin (1987), where the combining rules, the degrees of freedom and the efficiency calculation of [Theorem 41.5.1](#) all appear; the small-sample degrees of freedom are Barnard and Rubin (1999). Congeniality is Meng's (1994), whose discussion in the same issue remains the best entry point to when Rubin's variance estimator is consistent. Chained equations were developed by Van Buuren, Boshuizen and Knook (1999) and by Raghunathan, Lepkowski, Van Hoewyk and Solenberger (2001); van Buuren (2018) is the practical handbook.

**MNAR and sensitivity.** Selection models in the econometric tradition begin with Heckman (1979), pattern-mixture models with Little (1993), who introduced the complete-case missing-value restriction of [Exercise \(C1\)](#). The non-identifiability in [Proposition 41.6.1\(c\)](#) is why Manski's (1990) assumption-free bounds, the subject of [Exercise \(C2\)](#), remain worth computing. Delta-adjustment and tipping-point analyses became standard in clinical trials after the National Research Council (2010) report, which is short and unusually clear about the difference between preventing missing data and repairing it. The reference-based schemes named in [Section 41.6](#) are due to Carpenter, Roger and Kenward (2013).

**Where this goes next.** Nothing here is special to a normal response: ignorability, EM and Rubin’s rules apply verbatim to the generalized linear models of Chapters 34 to 37, with harder integrals and an E step usually done by simulation. The estimating-equation methods of Chapter 40 stay consistent when the missingness is completely at random or depends only on modelled, fully observed covariates, and become inconsistent when it depends on previously observed responses — the MAR case in which likelihood remains valid — because the estimating function  $E\{R_i(Y_i - \mu_i)\}$  is then no longer zero. The repair is inverse-probability weighting.

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